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# Foreword

This Technical Specification (TS) has been produced by ETSI 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

x the first digit:

1 presented to TSG for information;

2 presented to TSG for approval;

3 or greater indicates TSG approved document under change control.

y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

z the third digit is incremented when editorial only changes have been incorporated in the document.

# Introduction

The present document is part 3 of a multi-parts TS:

3GPP TS 36.521-1 [10]: Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) conformance specification Radio transmission and reception; Part 1: Conformance Testing.

3GPP TS 36.521-2 [23]: Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) conformance specification Radio transmission and reception; Part 2: Implementation Conformance Statement (ICS).

3GPP TS 36.521-3: Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) conformance specification Radio transmission and reception; Part 3: Radio Resource Management (RRM) conformance testing.

# 1 Scope

The present document specifies the measurement procedures for the conformance test of the user equipment (UE) that contain requirements for support of RRM (Radio Resource Management) as part of the 3G Long Term Evolution (3G LTE).

The requirements are listed in different clauses only if the corresponding parameters deviate. More generally, tests are only applicable to those mobiles that are intended to support the appropriate functionality. To indicate the circumstances in which tests apply, this is noted in the "*test applicability*" part of the test.

For example only Release 8 and later UE declared to support LTE shall be tested for this functionality. In the event that for some tests different conditions apply for different releases, this is indicated within the text of the test itself.

# 2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document.*

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 36.101: "E-UTRA UE radio transmission and reception".

[3] ITU-R Recommendation SM.329-10, "Unwanted emissions in the spurious domain".

[4] 3GPP TS 36.133: "E-UTRA requirements for support of radio resource management".

[5] 3GPP TS 36.331: "E-UTRA Radio Resource Control (RRC): protocol specification".

[6] 3GPP TS 36.304: "E-UTRA UE procedures in idle mode".

[7] 3GPP TS 36.508: "Common test environments for User Equipment (UE)".

[8] 3GPP TS 36.213: "E-UTRA Physical layer procedures".

[9] 3GPP TS 36.211: "E-UTRA Physical Channels and Modulation".

[10] 3GPP TS 36.521-1: "User Equipment (UE) conformance specification Radio transmission and reception. Part 1: Conformance Testing".

[11] 3GPP TS 36.321: "E-UTRA Medium Access Control (MAC): protocol specification".

[12] 3GPP TS 36.214: "E-UTRA Physical layer - Measurements".

[13] 3GPP TS 45.010: "Radio subsystem synchronization".

[14] 3GPP TS 36.306: "E-UTRA UE radio access capabilities".

[15] 3GPP TS 45.008: "Radio subsystem link control".

[16] 3GPP TS 45.005: "Radio transmission and reception"

[17] 3GPP2 C.S0033-B: "Recommended Minimum Performance Standards for cdma2000 High Rate Packet Data Access Terminal".

[18] 3GPP2 C.S0024-B: "cdma2000 High Rate Packet Data Air Interface Specification".

[19] 3GPP2 C.S0011-C: “Recommended Minimum Performance Standards for cdma2000 Spread Spectrum Mobile Stations”.

[20] 3GPP TR 36.903: "Derivation of test tolerances for Radio Resource Management (RRM) conformance tests ".

[21] 3GPP TS 25.133: "Requirements for Support of Radio Resource Management (FDD)".

[22] 3GPP TS 25.123: "Requirements for Support of Radio Resource Management (TDD)".

[23] 3GPP TS 36.521-2: "Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) conformance specification; Radio transmission and reception; Part 2: Implementation Conformance Statement (ICS)".

[24] 3GPP TS 34.108: “UTRA Common test environments for User Equipment (UE)".

[25] 3GPP TS 36.521-3 Release 10: "User Equipment (UE) conformance specification; Radio transmission and reception; Part 3: Radio Resource Management (RRM) conformance testing".

[26] 3GPP TS 36.321: “Evolved Universal Terrestrial Radio Access (E-UTRA); Medium Access Control (MAC) protocol specification”.

[27] 3GPP TS 37.571-1: User Equipment (UE) conformance specification for UE positioning; Part 1: Conformance test specification.

[28] 3GPP TS 36.423: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); X2 Application Protocol (X2AP)".

[29] 3GPP TS 36.104: "Evolved Universal Terrestrial Radio Access (E-UTRA); Base Station (BS) radio transmission and reception".

[30] 3GPP TS 36.300: "Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Universal Terrestrial Radio Access Network (E-UTRAN); Overall description; Stage 2".

# 3 Definitions, symbols and abbreviations

## 3.1 Definitions

For the purposes of the present document, the terms and definitions given in TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

**Asynchronous Dual Connectivity:** As defined in TS 36.300 [30].

**Carrier aggregation:** Aggregation of two or more component carriers in order to support wider transmission bandwidths.

**Carrier aggregation band:** A set of one or more operating bands across which multiple carriers are aggregated with a specific set of technical requirements.

**Carrier aggregation bandwidth class:** A class defined by the aggregated transmission bandwidth configuration and maximum number of component carriers supported by a UE.

**Carrier aggregation configuration:** A combination of CA operating band(s) and CA bandwidth class(es) supported by a UE.

**Channel edge:** The lowest and highest frequency of the carrier, separated by the channel bandwidth.

**Channel bandwidth:** The RF bandwidth supporting a single E-UTRA RF carrier with the transmission bandwidth configured in the uplink or downlink of a cell. The channel bandwidth is measured in MHz and is used as a reference for transmitter and receiver RF requirements.

**Contiguous carriers:** A set of two or more carriers configure in a spectrum block where there are no RF requirements based on co-existence for un-coordinated operation within the spectrum block.

**Dual Connectivity:** As defined in TS 36.300 [30].

**High operating band:** an operating band with a higher downlink frequency with respect to another, low, operating band.

**Inter-band carrier aggregation:** Carrier aggregation of component carriers in different operating bands.

NOTE: Carriers aggregated in each band can be contiguous or non-contiguous

**Intra-band carrier aggregation:** Contiguous carriers aggregated in the same operating band.

**Intra-band non-contiguous carrier aggregation:** Non-contiguous carriers aggregated in the same operating band.

**Low operating band:** an operating band with a lower downlink frequency with respect to another, high, operating band.

**Master Cell Group:** As defined in TS 36.300 [30].

**Master eNB:** As defined in TS 36.300 [30].

**Maximum throughput:** The maximum achievable throughput for a reference measurement channel.

**Maximum Output Power:** The mean power level per carrier of UE measured at the antenna connector in a specified reference condition.

**Mean power:** When applied to E-UTRA transmission this is the power measured in the operating system bandwidth of the carrier. The period of measurement shall be at least one subframe (1ms) unless otherwise stated.

**NB-IoT:** NB-IoT allows access to network services via E-UTRA with a channel bandwidth limited to 200 kHz.

**NB-IoT stand-alone operation**: A NB-IoT is operating standalone when it utilizes its own spectrum, for example the spectrum used by GERAN systems as a replacement of one or more GSM carriers, as well as scattered spectrum for potential IoT deployment.

**NB-IoT guard band operation:** NB-IoT is operating in guard band when it utilizes the unused resource block(s) within a E-UTRA carrier's guard-band

**NB-IoT in-band operation:** NB-IoT is operating in-band when it utilizes the resource block(s) within a normal E-UTRA carrier

**Normal Performance Group:** For UE which supports Increased UE carrier monitoring UTRA or E-UTRA the group of inter-frequency carriers or inter-RAT carriers is divided into two groups. The group which has a better delay performance compared to the other group is refered to as the normal performance group

**Occupied bandwidth:** The width of a frequency band such that, below the lower and above the upper frequency limits, the mean powers emitted are each equal to a specified percentage $\beta/2$ of the total mean power of a given emission.

**Output power:** The mean power of one carrier of the UE, delivered to a load with resistance equal to the nominal load impedance of the transmitter.

**Primary Cell:** As defined in TS 36.331 [5] subclause 3.1.

**Primary SCell:** As defined in TS 36.300 [30].

**Primary Secondary Timing Advance Group**: Timing Advance Group containing the PSCell.

**Primary Timing Advance Group**: Timing Advance Group containing the PCell.

**Reduced Performance Group:** For UE which supports Increased UE carrier monitoring UTRA or E-UTRA the group of inter-frequency carriers or inter-RAT carriers is divided into two groups. The group which has worse delay performance compared to the other group is refered to as the reduced performance group

**Secondary Cell:** As defined in TS 36.331 [5] subclause 3.1.

**Serving Cell:** As defined in TS 36.331 [5] subclause 3.1.

**Secondary Cell Group:** As defined in TS 36.300 [30].

**Secondary Timing Advance Group**: As defined in TS 36.331 [5].

**Synchronous Dual Connectivity:** As defined in TS 36.300 [25].

**TDD configuration with CA:** The same uplink-downlink and special subframe configurations [9] in the PCell and SCell are assumed unless otherwise stated.

**TDD configuration with inter-frequency:** The same uplink-downlink and special subframe configurations [9] in all the cells on the serving and inter-frequency carriers are assumed unless otherwise stated.

**Throughput:** The number of payload bits successfully received per second for a reference measurement channel in a specified reference condition.

**Timing Advance Group**: As defined in TS 36.331 [5].

**Transmission bandwidth:** Bandwidth of an instantaneous transmission from a UE or BS, measured in Resource Block units.

**Transmission bandwidth configuration:** The highest transmission bandwidth allowed for uplink or downlink in a given channel bandwidth, measured in Resource Block units.

**UE category 0 applicability:** The requirements for a UE category 0 are derived assuming UE category 0 [14] and a single antenna receiver.

**UE category M1 applicability:** The requirements for UE category M1 are derived assuming: DL Category M1 and Uplink Category M1, operation in any LTE system bandwidth but with a channel bandwidth of 1.4 MHz and transmission bandwidth of 6 PRBs in downlink and uplink, and a single antenna receiver. DL UE category M1 and UL UE category M1 are defined in TS 36.306 [14]. The requirements for CEMode A shall apply provided the UE category M1 is configured with CEMode A, SCH $\hat{E}s/Iot \geq -6$ dB and CRS $\hat{E}s/Iot \geq -6$ dB. The CEMode A and the number of repetition levels for different physical channels are defined in TS 36.213 [8]. The requirements for CEMode B shall apply provided the UE category M1 is configured with CEMode B, SCH $\hat{E}s/Iot \geq -15$ dB and CRS $\hat{E}s/Iot \geq -15$ dB. The CEMode B and the number of repetition levels for different physical channels are defined in TS 36.213 [8].

**UE category NB1 applicability:** The requirements for UE category NB1 are derived assuming UE category NB1[14] and a single antenna receiver.

## 3.2 Symbols

For the purposes of the present document, the following symbols apply:

| | |
|---|---|
| $BW_{Channel}$ | Channel bandwidth, defined in TS 36.101 subclause 3.2 |
| $BW_{Channel_CA}$ | Aggregated channel bandwidth, expressed in MHz, defined in TS 36.101 subclause 3.2. |
| CPICH_Ec | Average energy per PN chip for the CPICH |
| CPICH_Ec/Io | The ratio of the received energy per PN chip for the CPICH to the total received power spectral density at the UE antenna connector. |
| Ec | Average energy per PN chip |
| $\hat{E}s$ | Received energy per RE (power normalized to the subcarrier spacing) during the useful part of the symbol, i.e. excluding the cyclic prefix, at the UE antenna connector |
| Io | The total received power density, including signal and interference, as measured at the UE antenna connector. |
| Ioc | The power spectral density (integrated in a noise bandwidth equal to the chip rate and normalized to the chip rate) of a band limited noise source (simulating interference from cells, which are not defined in a test procedure) as measured at the UE antenna connector. |
| Iot | The received power spectral density of the total noise and interference for a certain RE (power integrated over the RE and normalized to the subcarrier spacing) as measured at the UE antenna connector |
| $N_{oc}$ | The power spectral density of a white noise source (average power per RE normalised to the subcarrier spacing), simulating interference from cells that are not defined in a test procedure, as measured at the UE antenna connector |
| $n_{PRB}$ | Physical Resource Block number as defined in subclause 3.1 in 3GPP TS 36.211. |

| | |
|---|---|
| $P_{CMAX}$ | Configured UE transmitted power as defined in subclause 6.2.5 in 3GPP TS 36.101. |
| $Q_{out_Cat\ M1}$ | Defined in TS36.133 clause 7.19.2 |
| $Q_{in_Cat\ M1}$ | Defined in TS36.133 clause 7.19.2 |
| S | Defined in TS 36.304, subclause 5.2.3.2 for E-UTRAN |
| SCH_Ec/Ior | The ratio of the transmit energy per PN chip of the SCH to the total transmit power spectral density at the UTRA Node B antenna connector |
| SCH_RP | Received (linear) average power of the resource elements that carry E-UTRA synchronisation signal, measured at the UE antenna connector |
| $S_{ServingCcell}$ | Defined in TS 36.304 |
| Sintersearch | Defined in TS 25.304, subclause 5.2.6.1.5 |
| Sintrasearch | Defined in TS 25.304, subclause 5.2.6.1.5 for UTRAN and in TS 36.304, subclause 5.2.4.7 for E-UTRAN |
| Snonintrasearch | Defined in TS 36.304, subclause 5.2.4.7 |
| SsearchRAT | Defined in TS 25.304, subclause 5.2.6.1.5 |
| $Thresh_{x,\ high}$ | Defined in TS 36.304, subclause 5.2.4.7 |
| $Thresh_{x,\ low}$ | Defined in TS 36.304, subclause 5.2.4.7 |
| $Thresh_{serving,\ low}$ | Defined in TS 36.304, subclause 5.2.4.7 |
| $T_{RE\text{-}ESTABLISH\text{-}REQ}$ | The RRC Re-establishment delay requirement, the time between the moment when erroneous CRCs are applied, to when the UE starts to send preambles on the PRACH. |
| Treselection | Defined in TS 25.304, subclause 5.2.6.1.5 |
| $Treselection_{RAT}$ | Defined in TS 36.304, subclause 5.2.4.7 |
| $Treselection_{EUTRAN}$ | Defined in TS 36.304, subclause 5.2.4.7 |
| $Treselection_{UTRAN}$ | Defined in TS 36.304, subclause 5.2.4.7 |
| $Treselection_{GERAN}$ | Defined in TS 36.304, subclause 5.2.4.7 |
| $T_S$ | Basic time unit, defined in TS 36.211, clause 4 |

## 3.3 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

| | |
|---|---|
| 1x RTT | CDMA2000 1x Radio Transmission Technology |
| ABS | Almost Blank Subframe |
| ARQ | Automatic Repeat Request |
| AWGN | Additive White Gaussian Noise |
| BCCH | Broadcast Control Channel |
| BCH | Broadcast Channel |
| BS | Base Station |
| BSIC | Base transceiver Station Identity Code |
| CA | Carrier Aggregation |
| CC | Component Carriers |
| CCCH SDU | Common Control Channel SDU |
| CCTrCH | Coded Composite Transport Channel |
| CFN | Connection Frame Number |
| CPICH | Common Pilot Channel |
| CPICH Ec/No | CPICH received energy per chip divided by the power density in the band |
| CRS | Cell-specific Reference Signals |
| C-RNTI | Cell RNTI |
| CQI | Channel Quality Indicator |
| CSI | Channel-State Information |
| CSI-RS | CSI Reference Signal |
| DC | Dual Connectivity |
| DCI | Downlink Control Information |
| DCCH | Dedicated Control Channel |
| DL | Downlink |
| DPCH | Dedicated Physical Channel |
| DPCCH | Dedicated Physical Control Channel |
| DRX | Discontinuous Reception |
| DTX | Discontinuous Transmission |

| | |
|---|---|
| DwPTS | Downlink Pilot Time-Slot |
| EARFCN | E-UTRA Absolute Radio Frequency Channel Number |
| EPRE | Energy Per Resource Element |
| eICIC | Enhanced Inter-Cell Interference Coordination |
| E-UTRA | Evolved UMTS Terrestrial Radio Access |
| E-UTRAN | Evolved UMTS Terrestrial Radio Access Network |
| FDD | Frequency Division Duplex |
| FeICIC | Further Enhanced Inter-Cell Interference Coordination |
| FGI | Feature Group Indicator |
| FRC | Fixed Reference Channel |
| GSM | Global System for Mobile communication |
| HARQ | Hybrid Automatic Repeat Request |
| HD-FDD | Half- Duplex FDD |
| HO | Handover |
| HRPD | High Rate Packet Data |
| ICIC | Inter-Cell Interference Coordination |
| MAC | Medium Access Control |
| MCG | Master Cell Group |
| MeNB | Master eNB |
| MBSFN | Multimedia Broadcast multicast service Single Frequency Network |
| MBSFN ABS | MBSFN Almost Blank Subframe |
| MDT | Minimization of Drive Tests |
| MIB | Master Information Block |
| OCNG | OFDMA Channel Noise Generator |
| OFDM | Orthogonal Frequency Division Multiplexing |
| OFDMA | Orthogonal Frequency Division Multiple Access |
| OTDOA | Observed Time Difference of Arrival |
| PBCH | Physical Broadcast Channel |
| PCC | Primary Component Carrier |
| PCCH | Paging Control Channel |
| P-CCPCH | Primary Common Control Physical Channel |
| PCell | Primary Cell |
| PCFICH | Physical Control Format Indicator Channel |
| PDCCH | Physical Downlink Control Channel |
| PDSCH | Physical Downlink Shared Channel |
| PHICH | Physical Hybrid ARQ Indicator Channel |
| PLMN | Public Land Mobile Network |
| PMI | Precoding Matrix Indicator |
| PRACH | Physical Random Access Channel |
| PSCell | Primary SCell |
| PSS | Primary Synchronization Signal |
| PSS_RA | PSS-to-EPRE ratio for the channel PSS |
| psTAG | Primary Secondary Timing Advance Group |
| pTAG | Primary Timing Advance Group |
| PUCCH | Physical Uplink Control Channel |
| PUSCH | Physical Uplink Shared Channel |
| RACH | Random Access Channel |
| RAT | Radio Access Channel |
| REFSENS | Reference Sensitivity power level |
| RLC | Radio Link Control |
| RMC | Reference Measurement Channel |
| r.m.s | Root Mean Square |
| RNC | Radio Network Controller |
| RNTI | Radio Network Temporary Identifier |
| RRC | Radio Resource Control |
| RRM | Radio Resource Management |
| RSRP | Reference Signal Received Power |
| RSRQ | Reference Signal Received Quality |
| RSSI | Received Signal Strength Indicator |
| SCC | Secondary Component Carrier |
| SCH | Synchronization Channel |
| SCell | Secondary Cell |

| | |
|---|---|
| SDU | Service Data Unit |
| SFN | System Frame Number |
| SNR | Signal-to-Noise Ratio |
| SON | Self Organizing Network |
| SRS | Sounding Reference Signal |
| SSS | Secondary Synchronization Signal |
| SSS_RA | SSS-to-RS EPRE ratio for the channel SSS |
| sTAG | Secondary Timing Advance Group |
| TAG | Timing Advance Group |
| TDD | Time Division Duplex |
| TTI | Transmission Time Interval |
| UE | User Equipment |
| UL | Uplink |
| UMTS | Universal Mobile Telecommunications System |
| UpPTS | Uplink Pilot Time-Slot |
| UTRA | UMTS Terrestrial Radio Access |
| UTRAN | UMTS Terrestrial Radio Access Network |
| WB-RSRQ | Wide Bandwidth RSRQ |

# 3.4 Void

# 3.5 Additional notation

## 3.5.1 Groups of bands

The intention with the band grouping below is to increase the readability of the specification.

**Table 3.5.1-1: E-UTRA band groups**

| Group | E-UTRA FDD | | E-UTRA TDD | |
|---|---|---|---|---|
| | Band group notation | Operating bands | Band group notation | Operating bands |
| A | FDD_A | 1, 4, 6, 10, 11, 18, 19, 21, 23, 24, 32 Note 4 | TDD_A | 33, 34, 35, 36, 37, 38, 39, 40, 45 |
| B | FDD_B | 65, 66 Note 5 | TDD_B | - |
| C | FDD_C | 9, 30 | TDD_C | 42, 43 |
| D | FDD_D | 28 | TDD_D | - |
| E | FDD_E | 2, 5, 7, 27 | TDD_E | 41, 44 |
| F | FDD_F | 26 Note 3 | TDD_F | - |
| G | FDD_G | 3, 8, 12, 13, 14, 17, 20, 22, 29 Note 2 | TDD_G | - |
| H | FDD_H | 25 | TDD_H | - |
| I | FDD_I | - | TDD_I | - |
| J | FDD_J | - | TDD_J | - |
| K | FDD_K | - | TDD_K | - |
| L | FDD_L | - | TDD_L | - |
| M | FDD_M | - | TDD_M | - |
| N | FDD_N | 31 | TDD_N | - |

NOTE 1: The bands within the same group have the same Io conditions in a corresponding requirement in this specification.
NOTE 2: Band 29 is used only for E-UTRA carrier aggregation with other E-UTRA bands.
NOTE 3: The minimum Io condition for Band 26 is reduced by 0.5 dB when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz.
NOTE 4: Band 32 is used only for E-UTRA carrier aggregation with other E-UTRA bands.
NOTE 5: The range 2180-2200 MHz of the DL operating band 66 is restricted to E-UTRA operation when carrier aggregation is configured.

# 3A Requirements for support of RRM

## 3A.1 General

Radio Resource Management (RRM) ensures the efficient use of the available radio resources and also provides mechanisms that enable E-UTRAN to meet radio resource related requirements. The requirements that are tested include:

- Idle mode, the cell re-selection algorithms that are controlled by the setting of parameters (thresholds and hysteresis values) that define the best cell and/or determine when the UE should select a new cell
- The configuration of the UE measurement and reporting procedures that are transmitted via dedicated signalling in connected mode
- Connected mode, the mobility of radio connections that has to be supported
- Handover decisions that may be based on UE or eNB measurements
- Inter-RAT RRM, the management of radio resources in connection with inter-RAT mobility, e.g. Inter-RAT handover

Inter frequency and inter-RAT test cases are performed without frequency overlapping between cells required in the test.

- For bands with bandwidth not accommodating all the E-UTRAN cells required in the test without frequency overlapping, inter band testing shall be done according to subclause 3A.3.5. If the UE does not support the combination given in subclause 3A.3.5, the relevant tests are applicable only to the bands with the necessary bandwidth.
- In case when frequency overlapping occurs due to the frequency channel selection defined for the test (i.e. Cell number as per Annex E), other frequency channels which avoid the frequency overlapping shall be selected. If no suitable selection is found the test is not applicable for the affected band.

For test cases in clauses 4, 5, 6, 7, 8, 9 the initial conditions of the downlink physical channels signal levels and downlink physical channels required are specified in Annex C.0.

Unless otherwise mentioned, in those test case where delay is the test criteria, if HARQ or even RLC retransmission happens and is detected by SS, the extra delay due to retransmission shall be excluded in the delay calculation before making a pass or fail judgement on the UE in the test.

## 3A.2 Requirements Classification for Statistical Testing

The test requirements are expressed as absolute requirements with a single value stating the requirement or expressed as a success rate. The statistical nature depends on the type of test requirement. Some have large statistical variations, while others are not statistical in nature at all. When testing a parameter with a statistical nature, a confidence level is set. This establishes the probability that a Device Under Test (DUT) passing the test actually meets the test requirement and determines how many times a test have to be repeated and what the pass and fail criteria is. The statistical significance shall be set according to Annex G.

## 3A.3 RRM Test Configurations

The cell configuration of cells described in the test cases shall be set according to TS 36.508 [7] section 4.4.7.

### 3A.3.1 UE with Single or Multiple Antenna Connector

For testing a UE with more than one E-UTRA antenna connector, the connection diagram configurations are described in TS 36.508 [7] Annex A for the case of 2 E-UTRA RX antennas. For UEs with more than one E-UTRA antenna connector the fading of the signals and the AWGN signals applied to each receiver antenna connector shall be

uncorrelated. The levels of the test signal applied to each of the antenna connectors shall be as defined in the respective test cases.

For testing a UE with a single E-UTRA antenna connector, the connection diagram configurations are not described in TS 36.508 [7] Annex A. If the E-UTRA UE supports only single RX antenna, the RX diversity connector in the diagram is not applicable.

### 3A.3.2 Test configuration for Inter band test cases

It is allowed to use separate AWGN generators for the different bands in interband test cases, although the connection diagrams in 36.508 Annex A display one wideband AWGN generator per DUT antenna connector. When interband test cases are also inter RAT, then it is necessary to use separated AWGN generators per RAT because of different noise density in different RATs. This is displayed accordingly in the connections in 36.508 Annex A.

### 3A.3.3 Test configuration for Inter RAT test cases

The DUT may employ common antennas for different RATs or separated ones, leading to different connections. The diagrams in 36.508 Annex A display only the connections with common antennas for different RATs without excluding the separate case. Note that in case of separate antennas, also separate AWGNs and faders are necessary, if applicable.

The non E-UTRA RATs are undefined with respect to the RX antenna configuration. The diagrams in 36.508 Annex A display RX diversity with 2 antennas for the non E-UTRA RATs without excluding the single antenna case. If the non E-UTRA RAT support only single RX antenna, the RX diversity connector in the diagram is not applicable.

For UEs with more than one non-E-UTRA antenna connector the fading of the signals and the AWGN signals applied to each receiver antenna connector shall be uncorrelated. The levels of the test signal applied to each of the antenna connectors shall be as defined in the respective test cases.

With respect to the non E-UTRAN cell and AWGN, the diagrams in 36.508 Annex A are fully equipped, even if for certain RATs (e.g. GSM) the AWGN generator may be not applicable.

### 3A.3.4 UE with Multiband Capability

The Radio Resource Management performance of a UE in sections 4 – 8 is considered to be independent from all bands. Therefore, the required performance in the respective test cases can be verified in one of the bands supported by the UE, with the exception of inter-band testing requirements in clause 3A.1. The test cases in section 9 are considered to be band dependant and are therefore applicable in all of the supported bands in the UE.

### 3A.3.5 E-UTRAN operating band configuration

Inter-band configuration is not affecting the test purpose since the minimum requirements are valid regardless of band. Band combinations defined in table 3A.3.5-1 shall be used for testing.

For 8.20.1 and 8.20.2, there are three E-UTRA cells, Cell 1(PCell), Cell 2(neighbor cell) and Cell 3(SCell), all on different frequencies. The default cell configuration is to test on Cell1 and Cell2 on the same operating band and Cell3 on the different operating band. If the operating band of Cell 1 and Cell2 is overlapped (the operating band is less than 20MHz) and if the operating band of Cell3 has more than or equal to 20MHz, then Cell2 and Cell3 on the same band and Cell1 on the different band to test. Otherwise the test is not applicable.

**Table 3A.3.5-1: Inter-band configuration**

| Band under test | Additional band |
|---|---|
| 31 | 7 |
| 5,12,13,14,17, 30 | 4 |
| [23] | TBD |
| 11,18,19,21,27 | 1 |
| [34],[37] | TBD |
| Note 1: The band under test should contain the inter-frequency (neighbour) cell.<br>Note 2: The additional band should contain the serving cell of the test. If more than one inter-frequency cell is needed, that cell should be on the additional band.<br>Note 3: The bands and cells referred in this table are E-UTRAN bands and cells only.Note 4: Bands 5, 11, 23, 37 only need inter-band configuration in test cases where 3 cells are required. | |

## 3A.3.6 CA configuration

For CA testing, unless otherwise stated, the logical carriers PCC/SCCs are mapped on physical frequencies as defined in Tables 3A.3.6-1.

**Table 3A.3.6-1: PCC/SCC frequency mapping**

| CA Configuration | PCC-SCC mapping | Notes |
|---|---|---|
| Intra-band CA | CC1-CC2 | 1 |
| Inter-band CA (CA_x-y) | Bx-By<br>(if not supported by the UE, then By-Bx) | 2 |
| NOTE 1: Notation CCi-CCj means PCC on component carrier CCi and SCC on component carrier CCj, with CCi/j frequencies defined in the corresponding intra-band contiguous / non-contiguous CA band in TS36.508.<br>NOTE 2: Notation Bi–Bj means PCC on component Band i and SCC on component Band j, with single Band i/j frequencies defined in TS36.508. | | |

## 3A.3.7 HD-FDD category 0 testing

For testing of category 0 HD-FDD UE in FDD operating bands, the SS shall ensure downlink transmission is only configured in downlink subframes and that UL grants are given to ensure UE is transmitting only in uplink subframes throughout the test. The subframes reserved for uplink and downlink respectively are specified in the corresponding RMC definitions in Annex A.

# 3A.4 Antenna Configuration

Unless otherwise specified, E-UTRA FDD or E-UTRA TDD cells in all RRM Test cases in AWGN propagation condition are configured with Antenna Configuration 1x2.

## 3A.4.1 Antenna connection for 4 Rx capable UEs

### 3A.4.1.1 Introduction

All tests in sections 4 to 9 are specified for UEs supporting either category 0 (1RX) or 2RX. In this section, the antenna connection method for applying 2RX tests to UEs supporting 4RX antenna ports is specified. No tests cases are currently specified in section 4-9 which are applicable only to 4RX antenna ports, so 4RX capable UEs are always tested by reusing tests which were originally specified for 2RX UEs.

## 3A.4.1.2 Principle of testing

### 3A.4.1.2.1 Single carrier tests

For 4RX capable UEs supporting at least one 2RX band, the, all single carrier tests specified in section 4 to 8 shall be tested on any band where 2RX is supported with the antenna connection specified in 3A.4.1.2.3. For single carrier tests specified in section 9, all tests shall be tested with the antenna connction specified in 3A.4.1.2.3 for bands where 2RX is supported, and the antenna connction specified in 3A.4.1.2.4 for bands where 4RX is supported.

For 4RX capable UEs which do not support any 2RX band, all tests specified in sections 4 to 9 shall be tested using the antenna connection specified in section 3A.4.1.2.4. For radio link monitoring tests, the SNR levels are modified according to table 3A.4.1.2.1-1 and table 3A.4.1.2.1-2.

**Table 3A.4.1.2.1-1 Modified parameters for RLM out of sync testing with 4 RX antenna connection**

| Test case | SNR during T3 (dB) | | | |
|---|---|---|---|---|
| | Test 1 | Test 2 | Test 3 | Test 4 |
| 7.3.1_1 | -17 | -17 | -15 | -15.7 |
| 7.3.3_1 | -16.6 | -16.7 | -14.8 | -15.4 |
| 7.3.5_1 | -15.7 | -17 | N/A | N/A |
| 7.3.7_1 | -15.4 | -16.6 | N/A | N/A |
| 7.3.9_1 (cell 1) | -15.9 | N/A | N/A | N/A |
| 7.3.10_1 (cell 1) | -15.9 | N/A | N/A | N/A |
| 7.3.13_1 (cell 1) | -15.9 | N/A | N/A | N/A |
| 7.3.14_1 (cell 1) | -15.9 | N/A | N/A | N/A |
| 7.3.17_1 (cell 1) | -17.2 | N/A | N/A | N/A |
| 7.3.18_1 (cell 1) | -17.2 | N/A | N/A | N/A |
| 7.3.23_1 | -15.7 | N/A | N/A | N/A |

**Table 3A.4.1.2.1-1 Modified parameters for RLM out of sync single carrier testing with 4 RX antenna connection**

| Test case | SNR during T3 (dB) | | SNR during T4 (dB) | |
|---|---|---|---|---|
| | Test 1 | Test 2 | Test 1 | Test 2 |
| 7.3.2_1 | -14 | -15.7 | -9.9 | -10.8 |
| 7.3.4_1 | -14.8 | -15.4 | -9.9 | -10.8 |
| 7.3.6_1 | -17 | N/A | -12.2 | N/A |
| 7.3.8_1 | -16.6 | N/A | -12.6 | N/A |
| 7.3.11_1 (cell 1) | -15.9 | N/A | -10.8 | N/A |
| 7.3.12_1 (cell 1) | -15.9 | N/A | -10.8 | N/A |
| 7.3.15_1 (cell 1) | -15.9 | N/A | -10.8 | N/A |
| 7.3.16_1 (cell 1) | -15.9 | N/A | -10.8 | N/A |
| 7.3.19_1 (cell 1) | -17.2 | N/A | -12.1 | N/A |
| 7.3.20_1 (cell 1) | -17.2 | N/A | -12.1 | N/A |
| 7.3.21_1 (cell 1) | -17.2 | N/A | -12.1 | N/A |
| 7.3.22_1 (cell 1) | -17.2 | N/A | -12.1 | N/A |
| 7.3.24_1 | -15.7 | N/A | -10.8 | N/A |
| 7.3.25_1 | -15.7 | N/A | -10.8 | N/A |

### 3A.4.1.2.2 Carrier aggregation and Dual connectivity tests

All carrier aggregation and dual connectivity tests are performed using the antenna connection in section 3A.4.1.2.3 for the PCell antenna connection if the PCell is on a band where 2RX is supported or the antenna connection in 3A.4.1.2.4 for the PCell antenna connection if the PCell is on a band where 4RX is supported.

All carrier aggregation and dual connectivity tests are performed using the antenna connection in section 3A.4.1.2.3 for the SCell or PSCell antenna connection if an SCell or PSCell is on band where 2RX is supported or the testing procedure in 3A.4.1.2.4 for the SCell or PSCell antenna connection if an SCell or PSCell is on a band where 4RX is supported.

For dual connectivity radio link monitoring tests with the PSCell on a band where 4RX is supported, the PSCell SNR levels are modified according to table 3A.4.1.2.2 -1 and table 3A.4.1.2.2 -2.

**Table 3A.4.1.2.2-1 Modified parameters for dual connectivity RLM out of sync testing with 4 RX antenna connection**

| Test case | SNR during T3 (dB) |
|---|---|
| 7.3.38_1 (cell 2) | -15.7 (5MHz)<br>-15.7 (10MHz)<br>-16.3 (20 MHz) |
| 7.3.39_1 (cell 2) | -15.7 (5MHz)<br>-15.7 (10MHz)<br>-16.3 (20 MHz) |
| 7.3.40_1 (cell 2) | -15.4 (5MHz)<br>-15.4 (10MHz)<br>-16.1 (20MHz) |

**Table 3A.4.1.2.2-2 Modified parameters for RLM out of sync testing with 4 RX antenna connection**

| Test case | SNR during T3 (dB) | SNR during T4 (dB) |
|---|---|---|
| 7.3.41_1 (cell 2) | -15.7 (5 Mhz)<br>-17.0 (10 Mhz)<br>-17.0 (20 Mhz) | -10.8 (5MHz)<br>-12.2 (10MHz)<br>-12.2 (20 MHz) |
| 7.3.42_1 (cell 2) | -15.7 (5 Mhz)<br>-17.0 (10 Mhz)<br>-17.0 (20 Mhz) | -10.8 (5MHz)<br>-12.2 (10MHz)<br>-12.2 (20 MHz) |
| 7.3.43_1 (cell 2) | -16.6 (5MHz)<br>-16.6 (10 MHz)<br>-16.6 (20 MHz) | -12.6 (5MHz)<br>-12.6 (10 MHz)<br>-12.6 (20 MHz) |

#### 3A.4.1.2.3 Antenna connection for bands where 2RX is supported

For bands where 2RX is supported, it is left to the UE declaration and AP configuration to decide which 2 of the 4 Rx ports are connected with data source from system simulator. The remaning 2 Rx ports shall be connected with zero input**.** No test parameters or requirements are modified.

#### 3A.4.1.2.4 Antenna connection for bands where 4RX is supported

For bands where 4RX is supporetd, all 4 Rx are connected with data source from system simulator**.** The system simulator shall provide independent noise and fading (low correlation) for each antenna port. Except for the modifications to radio link monitoring theresholds described in sections 3A.4.1.2.1 and 3A.4.1.2.2, no test parameters or requirements are modified.

## 3A.5 Test Cases for Synchronous and Asynchronous Dual Connectivity

### 3A.5.1 Introduction

This clause defines a principle which is applicable to test cases verifying RRM requirements for dual connectivity (DC) operation in synchronous and asynchronous scenarios.

### 3A.5.2 Principle of Testing

Test cases may be defined in both synchronous DC and asynchronous DC scenarios to verify the same RRM requirement.

If test cases are defined in both synchronous and asynchronous DC scenarios to verify the same RRM requirement then the UE capable of both synchronous and asynchronous DC operations needs to be tested with one of the tests in either synchronous or asynchronous DC scenarios.

# 4 E-UTRAN RRC_IDLE State Mobility

After the UE has switched on and a PLMN has been selected, the cell selection process takes place. This process allows the UE to select a suitable cell where to camp on in order to access available services. In this process the UE can use stored information (*Stored information cell selection*) or not (*Initial cell selection*).

When the UE is in either Camped Normally state or Camped on Any Cell state on a cell, the UE attempts to detect, synchronise, and monitor intra-frequency, inter-frequency and inter-RAT cells indicated by the serving cell, the cell re-selection evaluation process takes place. This process allows the UE to select a more suitable cell and camp on it. In this process the UE measurement activity is controlled by measurement rules defined in TS 36.304 [6] clause 5.2.4.2, allowing the UE to limit its measurement activity.

## 4.1 E-UTRAN Cell Selection

Editor's note: There are currently no tests defined for E-UTRAN cell selection.

## 4.2 E-UTRAN Cell Re-Selection

### 4.2.1 E-UTRAN FDD - FDD cell re-selection intra frequency case

#### 4.2.1.1 Test purpose

To verify that when the current and target cell operates on the same carrier frequency the UE is able to search and measure cells to meet the intra-frequency cell re-selection requirements.

#### 4.2.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

#### 4.2.1.3 Minimum conformance requirements

The cell re-selection delay shall be less than $T_{evaluate\ E\text{-}UTRAN_Intra} + T_{SI\text{-}EUTRA}$ in RRC_IDLE state.

The UE shall be able to identify new intra-frequency cells and perform RSRP measurement of identified intra-frequency cells without an explicit intra-frequency neighbour list containing physical layer cell identities.

The UE shall be able to evaluate whether a newly detectable intra-frequency cell meets the re-selection criteria defined in TS 36.304 [6] within $T_{detect,\ EUTRAN_Intra}$ as defined in table 4.2.2.3-1 of TS 36.133 [4] clause 4.2.2.3 when Treselection = 0 provided that the cell is at least 3 dB better ranked.

The UE shall measure RSRP at least every $T_{measure,EUTRAN_Intra}$ as defined in table 4.2.2.3-1 of TS 36.133 [4] clause 4.2.2.3 for intra-frequency cells that are identified and measured according to the measurement rules.

The UE shall filter RSRP measurement of each measured intra-frequency cell using at least 2 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least $T_{measure,EUTRAN_Intra} / 2$.

For an intra-frequency cell that has been already detected, but that has not been re-selected to, the filtering shall be such that the UE shall be capable of evaluating that the intra-frequency cell has met the re-selection criterion defined in TS

36.304 [6] within $T_{evaluateFDD,Intra}$ as defined in table 4.2.2.3-1 of TS 36.133 [4] clause 4.2.2.3 when Treselection = 0 provided that the cell is at least 3 dB better ranked. When evaluating cells for re-selection, the side conditions are RSRP and SCh apply to both serving and non-serving intra-frequency cells.

If Treselection timer has a non-zero value and the intra-frequency cell is better ranked than the serving cell, the UE shall evaluate this intra-frequency cell for the Treselection time. If this cell remains better ranked within this duration, then the UE shall re-select that cell.

The UE shall evaluate the intra-frequency cell re-selection criteria as defined in TS 36.304 [6] at least every DRX cycle. The DRX cycle length is 1.28 seconds. When a non-zero value of Treselection is used, the UE shall only perform re-selection on an evaluation which occurs simultaneously to, or later than the expiry of the Treselection timer.

At intra-frequency cell re-selection, the UE shall monitor the downlink of serving cell for paging reception until the UE is capable to start monitoring downlink channels of the target intra-frequency cell for paging reception. The interruption time shall not exceed $T_{SI-EUTRA}$ + 50 ms. $T_{SI-EUTRA}$ is the time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks as defined in TS 36.331 [5] for a E-UTRAN cell.

For idle mode cell re-selection purposes, the UE shall be capable of monitoring at least:

- Intra-frequency carrier

In addition to the requirements defined in TS 36.133 [4] clause 4.2.2.9 a UE in RRC_IDLE state shall be capable of monitoring a total of at least 8 carrier frequency layers, which includes serving layer, comprising of any defined in TS 36.133 [4] clause 4.2.2.9 combination of E-UTRA FDD, E-UTRA TDD, UTRA FDD, UTRA TDD, GSM (one GSM layer corresponds to 32 cells), cdma2000 1x and HRPD layers.

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.3 and A.4.2.1.

## 4.2.1.4 Test description

### 4.2.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. The general test parameter settings are set up according to Table 4.2.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are as defined in clause 4.2.1.4.3

5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 4.2.1.4.1-1: General Test Parameters for E-UTRAN FDD-FDD intra cell re-selection test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2 | |
| T2 end condition | Active cell | | Cell2 | |
| | Neighbour cells | | Cell1 | |
| Final condition | Visited cell | | Cell1 | |
| E-UTRA RF Channel Number | | | 1 | Only one FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| Time offset between cells | | | 3 ms | Asynchronous cells<br>3ms or 92160*Ts |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| PRACH configuration | | | 4 | As specified in table 5.7.1-2 in 3GPP TS 36.211 [9] |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | >7 | During T1, Cell 2 shall be powered off, and during the off time the physical cell identity shall be changed, The intention is to ensure that Cell 2 has not been detected by the UE prior to the start of period T2 |
| T2 | | s | 40 | T2 need to be defined so that cell re-selection reaction time is taken into account. |
| T3 | | s | 15 | T3 need to be defined so that cell re-selection reaction time is taken into account. |

### 4.2.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The UE is requested to monitor the neighbouring cell on one E-UTRA FDD carrier. In the test there are three successive time periods, with time duration of T1, T2 and T3 respectively. Only Cell 1 is already identified by the UE prior to the start of the test, i.e. Cell 2 is not identified by the UE prior to the start of the test. Cell 1 and Cell 2 belong to different tracking areas. Furthermore, UE has not registered with network for the tracking area containing Cell 2.

In the following test procedure “UE responds” means “UE starts transmitting preamble on PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure according to TS 36.508 [7] clause 4.5A.2”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2. Set Cell 2 physical cell identity = initial cell 2 physical cell identity.

2. Set the parameters according to T1 in Table 4.2.1.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for one iteration of the test procedure loop.

4. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 4.2.1.5-1.

5. The SS waits for random access requests information from the UE to perform cell re-selection to a newly detectable cell, Cell 2.

6. If the UE responds on the newly detectable cell, Cell 2 during time duration T2 within 34 seconds from the beginning of time period T2, then count a success for the event “Re-select newly detected Cell 2”. Otherwise count a fail for the event “Re-select newly detected Cell 2”.

7. If the UE has re-selected Cell 2 within T2, after the re-selection or when T2 expires, continue with step 8. Otherwise, if T2 expires and the UE has not yet re-selected Cell 2, skip to step 12.

8. The SS shall switch the power setting from T2 to T3 as specified in Table 4.2.1.5-1.

9. The SS waits for random access requests information from the UE to perform cell re-selection to an already detected cell, Cell 1.

10. If the UE responds on the already detected cell, Cell 1 during time duration T3 within 8 seconds from the beginning of time period T3, then count a success for the event “Re-select already detected Cell 1”. Otherwise count a fail for the event “Re-select already detected Cell 1”.

11. If the UE has re-selected Cell 1 within T3, after the re-selection or when T3 expires, skip to step 13. Otherwise, if T3 expires and the UE has not yet re-selected Cell 1, continue with step 12.

12. Switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.

13. Repeat step 2-12 until a test verdict has been achieved.
    Each of the events “Re-select newly detected Cell 2” and “Re-select already detected Cell 1” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. Different events may require different times for a verdict.
    If both events pass, the test passes. If one event fails, the test fails.

#### 4.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 4.2.1.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra cell re-selection test case**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.1-1<br>Table H.2.1-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

### 4.2.1.5 Test requirement

Tables 4.2.1.4.1-1 and 4.2.1.5-1 defines the primary level settings including test tolerances for E-UTRAN FDD-FDD intra-frequency cell re-selection test case.

**Table 4.2.1.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD-FDD intra frequency cell re-selection test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.2 (OP.2 FDD) | | OP.2 FDD | | | OP.2 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | | | | | | | |
| PSS_RA | | | | | | | |
| SSS_RA | | | | | | | |
| PCFICH_RB | | | | | | | |
| PHICH_RA | | | | | | | |
| PHICH_RB | | | | | | | |
| PDCCH_RA | | | | | | | |
| PDCCH_RB | | | | | | | |
| PDSCH_RA | | | | | | | |
| PDSCH_RB | | | | | | | |
| OCNG_RA Note 1 | | | | | | | |
| OCNG_RB Note 1 | | | | | | | |
| Qrxlevmin | dBm | -140 | -140 | -140 | -140 | -140 | -140 |
| Pcompensation | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $Qhyst_s$ | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $Qoffset_{s,n}$ | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| Cell_selection_and_re selection_quality_measurement | | RSRP | | | RSRP | | |
| $\hat{E}_s / I_{ot}$ | dB | 16.00 | -3.55 | 3.24 | -infinity | 3.24 | -3.55 |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 16.00 | 13.00 | 16.45 | -infinity | 16.45 | 13.00 |
| RSRP Note 3 | dBm/15 kHz | -82.00 | -85.00 | -81.55 | -infinity | -81.55 | -85.00 |
| Treselection | s | 0 | 0 | 0 | 0 | 0 | 0 |
| Sintrasearch | dB | Not sent | | | Not sent | | |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The cell re-selection delay to a newly detectable cell is defined as the time from the beginning of time period T2, to the moment when the UE camps on Cell 2, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 2.

The cell re-selection delay to a newly detectable cell test requirement in this case is expressed as:

Cell re-selection delay to a newly detectable cell = $T_{detect,E\text{-}UTRAN_Intra} + T_{SI\text{-}EUTRA}$

$T_{detect,E\text{-}UTRAN_Intra}$ = 32 s; as specified in TS 36.133 [4] clause 4.2.2.3.

$T_{SI\text{-}EUTRA}$ = 1280 ms; maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell is assumed in this test; as specified in TS 36.133 [4] clause 4.2.2.3

The cell re-selection delay to a newly detectable cell shall be less than a total of 34 seconds in this test case (note: this gives a total of 33.28 seconds but the test allows 34 seconds).

The cell re-selection delay to an already detected cell is defined as the time from the beginning of time period T3, to the moment when the UE camps on Cell 1, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 1.

The cell re-selection delay to an already detected cell test requirement in this case is expressed as:

Cell re-selection to an already detected cell delay = $T_{evaluate,E-UTRAN_Intra} + T_{SI-EUTRA}$

$T_{evaluate,E-UTRAN_Intra}$ = 6.40 s; as specified in TS 36.133 [4] clause 4.2.2.3.

$T_{SI-EUTRA}$ = 1280 ms; maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell is assumed in this test; as specified in TS 36.133 [4] clause 4.2.2.3

The cell re-selection delay to an already detected cell shall be less than a total of 8 seconds in this test case (note: this gives a total of 7.68 seconds but the test allows 8 seconds).

For the test to pass, both events above shall pass.

The statistical pass/ fail decisions are done separated for each event. For an event to pass, the total number of successful loops shall be more than 90% of the cases with a confidence level of 95%.

## 4.2.2 E-UTRAN TDD - TDD cell re-selection intra frequency case

### 4.2.2.1 Test purpose

To verify that when the current and target cell operates on the same carrier frequency the UE is able to search and measure cells to meet the intra-frequency cell re-selection requirements.

### 4.2.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

### 4.2.2.3 Minimum conformance requirements

The cell re-selection delay shall be less than $T_{evaluate,E-UTRAN_Intra} + T_{SI-EUTRA}$ in RRC_IDLE state.

The UE shall be able to identify new intra-frequency cells and perform RSRP measurement of identified intra-frequency cells without an explicit intra-frequency neighbour list containing physical layer cell identities.

The UE shall be able to evaluate whether a newly detectable intra-frequency cell meets the re-selection criteria defined in TS 36.304 [6] within $T_{detect,EUTRAN_Intra}$ as defined in table 4.2.2.3-1 of TS 36.133 [4] clause 4.2.2.3 when Treselection = 0.

The UE shall measure RSRP at least every $T_{measure,EUTRAN_Intra}$ as defined in table 4.2.2.3-1 of TS 36.133 [4] clause 4.2.2.3 for intra-frequency cells that are identified and measured according to the measurement rules.

The UE shall filter RSRP measurements of each measured intra-frequency cell using at least 2 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least $T_{measure,EUTRAN_Intra}$ / 2.

For an intra-frequency cell that has been already detected, but that has not been re-selected to, the filtering shall be such that the UE shall be capable of evaluating that the intra-frequency cell has met the re-selection criterion defined in TS 36.304 [6] within $T_{evaluate,E-UTRAN_Intra}$ as defined in table 4.2.2.3-1 of TS 36.133 [4] clause 4.2.2.3 when Treselection = 0 provided that the cell is at least 3 dB better ranked. When evaluating cells for reselection, the side conditions for RSRP and SCH apply to both serving and non-serving intra-frequency cells.

If Treselection timer has a non-zero value and the intra-frequency cell is better ranked than the serving cell, the UE shall evaluate this intra-frequency cell for the Treselection time. If this cell remains better ranked within this duration, then the UE shall re-select that cell.

The UE shall evaluate the intra-frequency cell re-selection criteria as defined in TS 36.304 [6] at least every DRX cycle. The DRX cycle length is 1.28 seconds.

At intra-frequency cell re-selection, the UE shall monitor the downlink of serving cell for paging reception until the UE is capable to start monitoring downlink channels of the target intra-frequency cell for paging reception. The interruption time shall not exceed $T_{SI-EUTRA}$ + 50 ms. $T_{SI-EUTRA}$ is the time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks as defined in TS 36.331 [5] for a E-UTRAN cell.

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.3 and A.4.2.2.

## 4.2.2.4 Test description

### 4.2.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.
2. The general test parameter settings are set up according to Table 4.2.2.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.2.2.4.3.
5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for registration with the power level set according to Table clause C.0 and C.1 for this test.

**Table 4.2.2.4.1-1: General Test Parameters for E-UTRAN TDD-TDD intra cell re-selection test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2 | |
| T2 end condition | Active cell | | Cell2 | |
| | Neighbour cells | | Cell1 | |
| Final condition | Visited cell | | Cell1 | |
| E-UTRA RF Channel Number | | | 1 | Only one TDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| Time offset between cells | | µs | 3 | Synchronous cells<br>3µs or 92*Ts |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in 3GPP TS 36.211 |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in 3GPP TS 36.211 |
| PRACH configuration index | | | 53 | As specified in table 5.7.1-3 in 3GPP TS 36.211 |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | >7 | During T1, Cell 2 shall be powered off, and during the off time the physical cell identity shall be changed, The intention is to ensure that Cell 2 has not been detected by the UE prior to the start of period T2 |
| T2 | | s | 40 | T2 need to be defined so that cell re-selection reaction time is taken into account. |
| T3 | | s | 15 | T3 need to be defined so that cell re-selection reaction time is taken into account. |

### 4.2.2.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The UE is requested to monitor the neighbouring cell on one E-UTRA TDD carrier. In the test there are three successive time periods, with time duration of T1, T2, and T3 respectively. Only Cell 1 is already identified by the UE prior to the start of the test. Cell 1 and Cell 2 belong to different tracking areas. Furthermore, UE has not registered with network for the tracking area containing Cell 2.

In the following test procedure “UE responds” means “UE starts transmitting preamble on PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure according to TS 36.508 [7] clause 4.5A.2”

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2. Set Cell 2 physical cell identity = initial cell 2 physical cell identity.

2. Set the parameters according to T1 in Table 4.2.2.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for one iteration of the test procedure loop.

4. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 4.2.2.5-1.

5. The SS waits for random access requests information from the UE to perform cell re-selection to a newly detectable cell, Cell 2.

6. If the UE responds on the newly detectable cell, Cell 2, during time duration T2 within 34 seconds from the beginning of time period T2, then count a success for the event “Re-select newly detected Cell 2”. Otherwise count a fail for the event “Re-select newly detected Cell 2”.

7. If the UE has re-selected Cell 2 within T2, after the re-selection or when T2 expires, continue with step 8. Otherwise, if T2 expires and the UE has not yet re-selected Cell 2, skip to step 12.

8. The SS shall switch the power setting from T2 to T3 as specified in Table 4.2.2.5-1.

9. The SS waits for random access requests information from the UE to perform cell re-selection to an already detected cell, Cell 1.

10. If the UE responds on the already detected cell, Cell 1, during time duration T3 within 8 seconds from the beginning of time period T3, then count a success for the event “Re-select already detected Cell 1”. Otherwise count a fail for the event “Re-select already detected Cell 1”.

11. If the UE has re-selected Cell 1 within T3, after the re-selection or when T3 expires, skip to step 13. Otherwise, if T3 expires and the UE has not yet re-selected Cell 1, continue with step 12.

12. Switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.

13. Repeat step 2-12 until a test verdict has been achieved.
    Each of the events “Re-select newly detected Cell 2” and “Re-select already detected Cell 1” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. Different events may require different times for a verdict.
    If both events pass, the test passes. If one event fails, the test fails.

#### 4.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 4.2.2.4.3-1: Common Exception messages for E-UTRAN TDD-TDD intra cell re-selection test case**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.1-1<br>Table H.2.1-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-2 |

### 4.2.2.5 Test requirement

Tables 4.2.2.4.1-1, and 4.2.2.5-1 defines the primary level settings including test tolerances for E-UTRAN TDD-TDD intra frequency cell re-selection test case.

**Table 4.2.2.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD-TDD intra frequency cell re-selection test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Pattern defined in D.2.2 (OP.2 TDD) | | OP.2 TDD | | | OP.2 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | | | | | | | |
| PSS_RA | | | | | | | |
| SSS_RA | | | | | | | |
| PCFICH_RB | | | | | | | |
| PHICH_RA | | | | | | | |
| PHICH_RB | | | | | | | |
| PDCCH_RA | | | | | | | |
| PDCCH_RB | | | | | | | |
| PDSCH_RA | | | | | | | |
| PDSCH_RB | | | | | | | |
| OCNG_RA Note 1 | | | | | | | |
| OCNG_RB Note 1 | | | | | | | |
| Qrxlevmin | dBm | -140 | | | -140 | | |
| Pcompensation | dB | 0 | | | 0 | | |
| $Qhyst_s$ | dB | 0 | | | 0 | | |
| $Qoffset_{s,n}$ | dB | 0 | | | 0 | | |
| Cell_selection_and_reselection_quality_measurement | | RSRP | | | RSRP | | |
| $\hat{E}_s / I_{ot}$ | dB | 16.00 | -3.55 | 3.24 | -infinity | 3.24 | -3.55 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 16.00 | 13.00 | 16.45 | -infinity | 16.45 | 13.00 |
| RSRP | dBm/15 kHz | -82.00 | -85.00 | -81.55 | -infinity | -81.55 | -85.00 |
| Treselection | s | 0 | 0 | 0 | 0 | 0 | 0 |
| Sintrasearch | dB | Not sent | | | Not sent | | |
| Propagation Condition | | AWGN | | | | | |
| Note: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | | | | | |

The cell re-selection delay to a newly detectable cell is defined as the time from the beginning of time period T2, to the moment when the UE camps on Cell 2, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 2.

The cell re-selection delay to a newly detectable cell test requirement in this case is expressed as:

Cell re-selection delay to a newly detectable cell = $T_{detect,EUTRAN_Intra} + T_{SI\text{-}EUTRA}$

$T_{detect,EUTRAN_Intra}$ = 32 s; as specified in TS 36.133 [4] clause 4.2.2.3.

$T_{SI\text{-}EUTRA}$ = 1280 ms; maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell is assumed in this test; as specified in TS 36.133 [4] clause 4.2.2.7

The cell re-selection delay to a newly detectable cell shall be less than a total of 34 seconds in this test case (note: this gives a total of 33.28 seconds but the test allows 34 seconds).

The cell re-selection delay to an already detected cell is defined as the time from the beginning of time period T3, to the moment when the UE camps on Cell 1 and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 1.

The cell re-selection delay to an already detected cell test requirement in this case is expressed as:

Cell re-selection to an already detected cell delay = $T_{evaluate,E\text{-}UTRAN_Intra} + T_{SI\text{-}EUTRA}$

$T_{evaluate,E\text{-}UTRAN_Intra}$ = 6.40 s; as specified in TS 36.133 [4] clause 4.2.2.3

$T_{SI\text{-}EUTRA}$ = 1280 ms; as specified in TS 36.133 [4] clause 4.2.2.7

The cell re-selection delay to an already detected cell shall be less than a total of 8 seconds in this test case (note: this gives a total of 7.68 seconds but the test allows 8 seconds).

For the test to pass, both events above shall pass.

The statistical pass/ fail decisions are done separated for each event. For an event to pass, the total number of successful loops shall be more than 90% of the cases with a confidence level of 95%.

## 4.2.3 E-UTRAN FDD - FDD cell re-selection inter frequency case

### 4.2.3.1 Test purpose

To verify that when the neighbour cell operates on a different carrier frequency, compared to the current cell the UE is able to search and measure cells to meet the inter-frequency cell re-selection requirements.

### 4.2.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

### 4.2.3.3 Minimum conformance requirements

The cell re-selection delay shall be less than $T_{evaluate,E\text{-}UTRAN_Inter} + T_{SI\text{-}EUTRA}$ in RRC_IDLE state.

The UE shall be able to identify new inter-frequency cells and perform RSRP measurements of identified inter-frequency cells if carrier frequency information is provided by the serving cell, even if no explicit neighbour list with physical layer cell identities is provided.

If the $S_{ServingCell}$ of the E-UTRA serving cell is greater than $S_{nonintrasearch}$ then the UE shall search for inter-frequency layers of higher priority at least every $T_{higher_priority_search}$ where $T_{higher_priority_search}$ is described in TS 36.133 [4] clause 4.2.2 as $T_{higher_priority_search}$ = (60 * $N_{layers}$) seconds, where the parameter $N_{layers}$ is the total number of configured higher priority E-UTRA carrier frequencies.

If the $S_{ServingCell}$ of the E-UTRA serving cell is less than or equal to $S_{nonintrasearch}$ then the UE shall search for and measure inter-frequency layers of higher, equal or lower priority in preparation for possible re-selection. In this scenario, the minimum rate at which the UE is required to search for and measure higher priority layers shall be the same as that defined below for lower or equal priority inter-frequency layers.

The UE shall be able to evaluate whether a newly detectable lower or equal priority inter-frequency cell meets the re-selection criteria defined in TS 36.304 [6] within $K_{carrier}$ * $T_{detect,EUTRAN_Inter}$ (as defined in table 4.2.2.4-1 of TS 36.133 [4] clause 4.2.2.4) if at least carrier frequency information is provided for inter-frequency neighbour cells by the serving cells when $Treselection_{EUTRAN}$ = 0 provides that the re-selection criteria is met by a margin of at least 5 dB for re-selection based on ranking or 6 dB for re-selection based on absolute priorities. The parameter $K_{carrier}$ is the number of E-UTRA inter-frequency carriers indicated by the serving cell.

When higher priority cells are found by the higher priority search, they shall be measured at least every $T_{measure,EUTRAN_Inter}$. If re-selection to any higher priority cell is not triggered within ($T_{evaluateFDD,\ Inter}$ + $Treselection_{EUTRAN}$) after it is found in a higher priority search, the UE is not required to continue making measurements of the cell to evaluate the ongoing possibility of re-selection. If the UE detects on an E-UTRA carrier a cell whose physical identity is indicated as not allowed for that carrier in the measurement control system information of the serving cell, the UE is not required to perform measurements on that cell.

The UE shall measure RSRP at least every $K_{carrier}$ * $T_{measure,EUTRAN_Inter}$ DRX cycle as defined in table 4.2.2.4-1 of TS 36.133 [4] clause 4.2.2.4 for identified lower or equal priority inter-frequency cells. If the UE detects on an E-UTRA carrier a cell whose physical identity is indicated as not allowed for that carrier in the measurement control system information of the serving cell, the UE is not required to perform measurements on that cell.

The UE shall filter RSRP measurements of each measured higher, lower and equal priority inter-frequency cell using at least 2 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least $T_{measure,EUTRAN_Inter}$ / 2.

The UE shall not consider an E-UTRA neighbour cell in cell re-selection, if it is indicated as not allowed in the measurement control system information of the serving cell.

For an inter-frequency cell that has been already detected, but that has not been re-selected to, the filtering shall be such that the UE shall be capable of evaluating that the inter-frequency cell has met re-selection criterion defined in TS 36.304 [6] within $K_{carrier}$ * $T_{evaluate,E-UTRAN_Inter}$ as defined in table 4.2.2.4-1 of TS 36.133 [4] clause 4.2.2.4 when $Treselection_{EUTRAN}$ = 0 provides that the re-selection criteria is met by a margin of at least 5 dB for re-selection based on ranking or 6 dB for re-selection based on absolute priorities. When evaluating cells for re-selection, the side conditions for RSRP and SCH apply to both serving and inter-frequency cells.

If $Treselection_{EUTRAN}$ timer has a non-zero value and the inter-frequency cell is better ranked than the serving cell, the UE shall evaluate this inter-frequency cell for the $Treselection_{EUTRAN}$ time. If this cell remains better ranked within this duration, then the UE shall re-select that cell.

The UE shall evaluate the inter-frequency cell re-selection criteria as defined in TS 36.304 [6] at least every DRX cycle. The DRX cycle length is 1.28 seconds. When a non-zero value of $Treselection_{EUTRAN}$ is used, the UE shall only perform re-selection on an evaluation which occurs simultaneously to, or later than the expiry of the $Treselection_{EUTRAN}$ timer.

At inter-frequency cell re-selection, the UE shall monitor the downlink of serving cell for paging reception until the UE is capable to start monitoring downlink channels of the target inter-frequency cell for paging reception. The interruption time shall not exceed $T_{SI-EUTRA}$ + 50 ms. $T_{SI-EUTRA}$ is the time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks as defined in TS 36.331 [5] for a E-UTRAN cell.

For idle mode cell re-selection purposes, the UE shall be capable of monitoring at least:

- Depending on UE capability, 3 FDD E-UTRA inter-frequency carriers

In addition to the requirements defined in TS 36.133 [4] clause 4.2.2.9 a UE in RRC_IDLE state shall be capable of monitoring a total of at least 8 carrier frequency layers, which includes serving layer, comprising of any defined in TS 36.133 [4] clause 4.2.2.9 combination of E-UTRA FDD, E-UTRA TDD, UTRA FDD, UTRA TDD, GSM (one GSM layer corresponds to 32 cells) , cdma2000 1x and HRPD layers.

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.4 and A.4.2.3.

## 4.2.3.4 Test description

### 4.2.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.
2. The general test parameter settings are set up according to Table 4.2.3.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.2.3.4.3.
5. There are two E-UTRA FDD carriers and two cells specified in the test. Cell 2 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 4.2.3.4.1-1: General Test Parameters for E-UTRAN FDD-FDD inter frequency cell re-selection test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell2 | UE shall be forced to cell 2 in the initialisation phase, so that reselection to cell 1 occurs during the first T1 phase |
| T1 end condition | Active cell | | Cell1 | UE shall perform reselection to cell 1 during T1 |
| | Neighbour cell | | Cell2 | |
| Final condition | Active cell | | Cell2 | UE shall perform reselection to cell 2 during T3 |
| E-UTRA RF Channel Number | | | 1, 2 | Two FDD carrier frequencies are used. |
| Time offset between cells | | ms | 3 | Asynchronous cells<br>3ms or 92160*Ts |
| PRACH configuration | | | 4 | As specified in table 5.7.1-2 in TS 36.211 [9] |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 | T1 need to be defined so that cell re-selection reaction time is taken into account. |
| T2 | | s | >7 | During T2, cell 2 shall be powered off, and during the off time the physical cell identity shall be changed, The intention is to ensure that cell 2 has not been detected by the UE prior to the start of period T3. |
| T3 | | s | 75 | T3 need to be defined so that cell re-selection reaction time is taken into account. |

### 4.2.3.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The UE is requested to monitor the neighbouring cell on one of the E-UTRA FDD carriers. In the test there are three successive time periods, with time duration of T1, T2 and T3 respectively. Both Cell 1 and Cell 2 are already identified by the UE prior to the start of the test. Cell 1 and Cell 2 belong to different tracking areas and Cell 2 is of higher priority than Cell 1. Furthermore, UE has not registered with network for the tracking area containing Cell 1.

In the following test procedure “UE responds” means “UE starts transmitting preamble on PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure according to TS 36.508 [7] clause 4.5A.2”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2.
2. Set the parameters according to duration T0 in Table 4.2.3.5-1
3. Set the parameters according to duration T1 in Table 4.2.3.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
4. The SS waits for random access requests information from the UE to perform cell re-selection on the lower priority cell, Cell 1.
5. If the UE responds on lower priority cell, Cell 1 during time duration T1 within 8 seconds from the beginning of time period T1, then count a success for the event “Re-select lower priority Cell 1”. Otherwise count a fail for the event “Re-select lower priority Cell 1”.
6. If the UE has re-selected Cell 1 within T1, after the re-selection or when T1 expires, continue with step 7. Otherwise, if T1 expires and the UE has not yet re-selected Cell 1, skip to step 12.
7. The SS shall switch the power setting from T1 to T2 as specified in Table 4.2.3.5-2. During time duration T2, Cell 2 shall be powered OFF and the physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) shall be changed to ensure Cell 2 is not detected by the UE.
8. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 4.2.3.5-2.
9. The SS waits for random access requests information from the UE to perform cell re-selection on the higher priority cell, Cell 2.

10. If the UE responds on higher priority cell, Cell 2 during time duration T3 within 68 seconds from the beginning of time period T3, then count a success for the event "Re-select higher priority Cell 2". Otherwise count a fail for the event "Re-select higher priority Cell 2".

11. If the UE has re-selected Cell 2 within T3, after the re-selection or when T3 expires, skip to step 13. Otherwise, if T3 expires and the UE has not yet re-selected Cell 2, continue with step 12.

12. Switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 2.
    Set the parameters according to duration T0 in Table 4.2.3.5-1.

13. Repeat step 3-12 until a test verdict has been achieved.
    Each of the events "Re-select lower priority Cell 1" and "Re-select higher priority Cell 2" is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. Different events may require different times for a verdict.
    If both events pass, the test passes. If one event fails, the test fails.

#### 4.2.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 4.2.3.4.3-1: Common Exception messages for E-UTRAN FDD-FDD inter frequency cell re-selection test case**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.2-1<br>Table H.2.2-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

### 4.2.3.5 Test requirement

Tables 4.2.3.4.1-1, 4.2.3.5-1 and 4.2.3.5-2 define the primary level settings including test tolerances for E-UTRAN FDD-FDD inter-frequency cell re-selection test case. Note that the time period for T0 is system implementation dependent.

**Table 4.2.3.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD-FDD inter frequency cell re-selection test case**

| Parameter | Unit | Cell 1 | Cell 2 |
|---|---|---|---|
| | | T0 | |
| E-UTRA RF Channel number | | 1 | 2 |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.1.2 (OP.2 FDD) | | OP.2 FDD | |
| PBCH_RA | dB | 0 | 0 |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |

| | | | |
|---|---|---|---|
| Qrxlevmin | dBm | -140 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -99.35 | |
| RSRP Note 3 | dBm/15 KHz | -103.05 | -84.95 |
| $\hat{E}_s / I_{ot}$ | dB | -3.70 | 14.40 |
| $\hat{E}_s / N_{oc}$ | dB | -3.70 | 14.40 |
| $Treselection_{EUTRAN}$ | S | 0 | 0 |
| Snonintrasearch | dB | 50 | Not sent |
| $Thresh_{x, high}$ | dB | 48 | 48 |
| $Thresh_{serving, low}$ | dB | 44 | 44 |
| $Thresh_{x, low}$ | dB | 50 | 50 |
| Propagation Condition | | AWGN | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.<br>Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | |

**Table 4.2.3.5-2: Cell Specific Test requirement Parameters for E-UTRAN FDD-FDD inter frequency cell re-selection test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.2 (OP.2 FDD) | | OP.2 FDD | | | OP.2 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| Qrxlevmin | dBm | -140 | | | -140 | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -99.35 | | | | | |
| RSRP Note 3 | dBm/15 KHz | -82.95 | -82.95 | -82.95 | -103.05 | -infinity | -84.95 |
| $\hat{E}_s / I_{ot}$ | dB | 6.40 | 16.40 | 16.40 | -3.70 | -infinity | 14.40 |
| $\hat{E}_s / N_{oc}$ | dB | 16.40 | 16.40 | 16.40 | -3.70 | -infinity | 14.40 |
| $Treselection_{EUTRAN}$ | S | 0 | | | 0 | | |
| Snonintrasearch | dB | 50 | | | Not sent | | |
| $Thresh_{x, high}$ | dB | 48 | | | 48 | | |
| $Thresh_{serving, low}$ | dB | 44 | | | 44 | | |
| $Thresh_{x, low}$ | dB | 50 | | | 50 | | |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The cell re-selection delay to lower priority is defined as the time from the beginning of time period T1, to the moment when the UE camps on Cell 1, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 1.

The cell re-selection delay to lower priority test requirement in this case is expressed as:

Cell re-selection delay to lower priority = $T_{evaluate,E\text{-}UTRAN_Inter} + T_{SI\text{-}EUTRA}$

$T_{evaluate,E\text{-}UTRAN_Inter}$ = 6.40 s; as specified in TS 36.133 [4] clause 4.2.2.4

$T_{SI\text{-}EUTRA}$ = 1280 ms; as specified in TS 36.133 [4] clause 4.2.2.4

The cell re-selection delay to lower priority shall be less than a total of 8 seconds in this test case (note: this gives a total of 7.68 seconds but the test allows 8 seconds).

The cell re-selection delay to higher priority is defined as the time from the beginning of time period T3, to the moment when the UE camps on Cell 2 and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 2.

The cell re-selection delay to higher priority test requirement in this case is expressed as:

Cell re-selection delay to higher priority = $T_{higher_priority_search}$ + $T_{evaluate,E\text{-}UTRAN_Inter}$ + $T_{SI\text{-}EUTRA}$

$T_{higher_priority_search}$ = 60 s; as specified in TS 36.133 [4] clause 4.2.2

$T_{evaluate,E\text{-}UTRAN_Inter}$ = 6.40 s; as specified in TS 36.133 [4] clause 4.2.2.4

$T_{SI\text{-}EUTRA}$ = 1280 ms; as specified in TS 36.133 [4] clause 4.2.2.4

The cell re-selection delay to higher priority shall be less than a total of 68 seconds in this test case (note: this gives a total of 67.68 seconds but the test allows 68 seconds).

For the test to pass, both events above shall pass.

The statistical pass/ fail decisions are done separately for each event. For an event to pass, the total number of successful loops shall be more than 90% of the cases with a confidence level of 95%.

## 4.2.4 E-UTRAN FDD - TDD cell re-selection inter frequency case

### 4.2.4.1 Test purpose

To verify that when the neighbour cell operates on a different carrier frequency and mode, compared to the current cell the UE is able to search and measure cells to meet the inter-frequency cell re-selection requirements.

### 4.2.4.2 Test applicability

This test applies to all types of E-UTRA UE supporting FDD and TDD release 9 and forward.

### 4.2.4.3 Minimum conformance requirements

The UE shall be able to identify new inter-frequency cells and perform RSRP or RSRQ measurements of identified inter-frequency cells if carrier frequency information is provided by the serving cell, even if no explicit neighbour list with physical layer cell identities is provided.

If Srxlev > $S_{nonIntraSearchP}$ and Squal > $S_{nonIntraSearchQ}$ then the UE shall search for inter-frequency layers of higher priority at least every $T_{higher_priority_search}$ where $T_{higher_priority_search}$ is described in section 4.2.2.

If Srxlev ≤ $S_{nonIntraSearchP}$ or Squal ≤ $S_{nonIntraSearchQ}$ then the UE shall search for and measure inter-frequency layers of higher, equal or lower priority in preparation for possible reselection. In this scenario, the minimum rate at which the UE is required to search for and measure higher priority layers shall be the same as that defined below.

The UE shall be able to evaluate whether a newly detectable inter-frequency cell meets the reselection criteria defined in TS36.304 [6] within $K_{carrier}$ * $T_{detect,EUTRAN_Inter}$ if at least carrier frequency information is provided for inter-frequency neighbour cells by the serving cells when $T_{reselection}$ = 0 provided that the reselection criteria is met by a margin of at least 5dB for reselections based on ranking or 6dB for RSRP reselections based on absolute priorities or 4dB for RSRQ reselections based on absolute priorities.

The parameter $K_{carrier}$ is the number of E-UTRA inter-frequency carriers indicated by the serving cell. An inter-frequency cell is considered to be detectable according to RSRP, RSRP Ês/Iot, SCH_RP and SCH Ês/Iot defined in Annex I.1.2 for a corresponding Band.

When higher priority cells are found by the higher priority search, they shall be measured at least every $T_{measure,E\text{-}UTRAN_Inter}$. If, after detecting a cell in a higher priority search, it is determined that reselection has not occurred then the UE is not required to continuously measure the detected cell to evaluate the ongoing possibility of reselection. However, the minimum measurement filtering requirements specified later in this section shall still be met by the UE before it makes any determination that it may stop measuring the cell. If the UE detects on a E-UTRA carrier a cell whose physical identity is indicated as not allowed for that carrier in the measurement control system information of the serving cell, the UE is not required to perform measurements on that cell.

The UE shall measure RSRP or RSRQ at least every $K_{carrier}$ * $T_{measure,EUTRAN_Inter}$ (see table 4.2.4.3-1) for identified lower or equal priority inter-frequency cells. If the UE detects on a E-UTRA carrier a cell whose physical identity is indicated as not allowed for that carrier in the measurement control system information of the serving cell, the UE is not required to perform measurements on that cell.

The UE shall filter RSRP or RSRQ measurements of each measured higher, lower and equal priority inter-frequency cell using at least 2 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least $T_{measure,EUTRAN_Inter}/2$.

The UE shall not consider a E-UTRA neighbour cell in cell reselection, if it is indicated as not allowed in the measurement control system information of the serving cell.

For an inter-frequency cell that has been already detected, but that has not been reselected to, the filtering shall be such that the UE shall be capable of evaluating that the inter-frequency cell has met reselection criterion defined TS 36.304 [6] within $K_{carrier} * T_{evaluate,E-UTRAN_Inter}$ when $T_{reselection} = 0$ as specified in table 4.2.4.3-1 provided that the reselection criteria is met by a margin of at least 5dB for reselections based on ranking or 6dB for RSRP reselections based on absolute priorities or 4dB for RSRQ reselections based on absolute priorities. When evaluating cells for reselection, the side conditions for RSRP and SCH apply to both serving and inter-frequency cells.

If $T_{reselection}$ timer has a non zero value and the inter-frequency cell is better ranked than the serving cell, the UE shall evaluate this inter-frequency cell for the $T_{reselection}$ time. If this cell remains better ranked within this duration, then the UE shall reselect that cell.

**Table 4.2.4.3-1 : $T_{detect,EUTRAN_Inter}$, $T_{measure,EUTRAN_Inter}$ and $T_{evaluate,E-UTRAN_Inter}$**

| DRX cycle length [s] | $T_{detect,EUTRAN_Inter}$ [s] (number of DRX cycles) | $T_{measure,EUTRAN_Inter}$ [s] (number of DRX cycles) | $T_{evaluate,E-UTRAN_Inter}$ [s] (number of DRX cycles) |
|---|---|---|---|
| 0.32 | 11.52 (36) | 1.28 (4) | 5.12 (16) |
| 0.64 | 17.92 (28) | 1.28 (2) | 5.12 (8) |
| 1.28 | 32(25) | 1.28 (1) | 6.4 (5) |
| 2.56 | 58.88 (23) | 2.56 (1) | 7.68 (3) |

For higher priority cells, a UE may optionally use a shorter value for $T_{measureE-UTRA_Inter}$, which shall not be less than Max(0.64 s, one DRX cycle).

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.4 and A.4.2.4.

## 4.2.4.4 Test description

### 4.2.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.
2. The general test parameter settings are set up according to Table 4.2.4.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.2.4.4.3.
5. There are two E-UTRA carriers and two cells specified in the test. One E-UTRA FDD Cell 1 is the neighbour cell on the FDD carrier frequency, One E-UTRA TDD Cell 2 is the cell on the TDD carrier frequency and is used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 4.2.4.4.1-1: General Test Parameters for E-UTRAN FDD-TDD inter frequency cell re-selection test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell2 | UE shall be forced to cell 2 in the initialisation phase, so that reselection to cell 1 occurs during the first T1 phase |
| T1 end condition | Active cell | | Cell1 | UE shall perform reselection to cell 1 during T1 |
| | Neighbour cell | | Cell2 | |
| Final condition | Active cell | | Cell2 | UE shall perform reselection to cell 2 during T3 |
| Cell 1 E-UTRA RF Channel Number | | | 1 | One FDD carrier frequency is used. And Cell 1 is on RF channel number 1. |
| Cell 2 E-UTRA RF Channel Number | | | 2 | One TDD carrier frequencies is used. And Cell 2 is on RF channel number 2. |
| Time offset between cells | | | 3 ms | Asynchronous cells |
| E-UTRA FDD PRACH configuration | | | 4 | As specified in table 5.7.1-2 in 3GPP TS 36.211 |
| E-UTRA TDD PRACH configuration | | | 53 | As specified in table 5.7.1-3 in 3GPP TS 36.211 |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in 3GPP TS 36.211 |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in 3GPP TS 36.211 |
| E-UTRA FDD Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| E-UTRA TDD Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 | T1 need to be defined so that cell re-selection reaction time is taken into account. |
| T2 | | s | >7 | During T2, cell 2 shall be powered off, and during the off time the physical cell identity shall be changed, The intention is to ensure that cell 2 has not been detected by the UE prior to the start of period T3. |
| T3 | | s | 75 | T3 need to be defined so that cell re-selection reaction time is taken into account. |

### 4.2.4.4.2 Test procedure

The test scenario comprises of 1 E-UTRA FDD Cell 1 and 1 E-UTRA TDD Cell 2. The UE is requested to monitor the neighbouring cell on the E-UTRA TDD carriers. In the test there are three successive time periods, with time duration of T1, T2 and T3 respectively. Both Cell 1 and Cell 2 are already identified by the UE prior to the start of the test. Cell 1 and Cell 2 belong to different tracking areas and Cell 2 is of higher priority than Cell 1. Furthermore, UE has not registered with network for the tracking area containing Cell 1.

In the following test procedure “UE responds” means “UE starts transmitting preamble on PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure according to TS 36.508 [7] clause 4.5A.2”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2.
2. Set the parameters according to duration T0 in Table 4.2.4.5-1
3. Set the parameters according to duration T1 in Table 4.2.4.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
4. The SS waits for random access requests information from the UE to perform cell re-selection on the lower priority cell, Cell 1.
5. If the UE responds on lower priority cell, Cell 1 during time duration T1 within 8 seconds from the beginning of time period T1, then count a success for the event “Re-select lower priority Cell 1”. Otherwise count a fail for the event “Re-select lower priority Cell 1”.

6. If the UE has re-selected Cell 1 within T1, after the re-selection or when T1 expires, continue with step 7. Otherwise, if T1 expires and the UE has not yet re-selected Cell 1, skip to step 12.

7. The SS shall switch the power setting from T1 to T2 as specified in Table 4.2.4.5-2. During time duration T2, Cell 2 shall be powered OFF and the physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) shall be changed to ensure Cell 2 is not detected by the UE.

8. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 4.2.4.5-2.

9. The SS waits for random access requests information from the UE to perform cell re-selection on the higher priority cell, Cell 2.

10. If the UE responds on higher priority cell, Cell 2 during time duration T3 within 68 seconds from the beginning of time period T3, then count a success for the event “Re-select higher priority Cell 2”. Otherwise count a fail for the event “Re-select higher priority Cell 2”.

11. If the UE has re-selected Cell 2 within T3, after the re-selection or when T3 expires, skip to step 13. Otherwise, if T3 expires and the UE has not yet re-selected Cell 2, continue with step 12.

12. Switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 2.
    Set the parameters according to duration T0 in Table 4.2.4.5-1.

13. Repeat step 3-12 until a test verdict has been achieved.
    Each of the events “Re-select lower priority Cell 1” and “Re-select higher priority Cell 2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. Different events may require different times for a verdict.
    If both events pass, the test passes. If one event fails, the test fails.

#### 4.2.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 4.2.4.4.3-1: Common Exception messages for E-UTRAN FDD-TDD inter frequency cell re-selection test case**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.2-1<br>Table H.2.2-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1<br>Table H.3.2-2 |

### 4.2.4.5 Test requirement

Tables 4.2.4.4.1-1, 4.2.4.5-1 and 4.2.4.5-2 define the primary level settings including test tolerances for E-UTRAN FDD-TDD inter-frequency cell re-selection test case. Note that the time period for T0 is system implementation dependent.

**Table 4.2.4.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD-TDD inter frequency cell re-selection test case in AWGN**

| Parameter | Unit | Cell 1 | Cell 2 |
|---|---|---|---|
| | | T0 | |
| E-UTRA RF Channel number | | 1 | 2 |
| $BW_{channel}$ | MHz | 10 | 10 |
| OCNG Patterns defined in D.1.2 (OP.2 FDD) and D.2.2 (OP.2 TDD) | | OP.2 FDD | OP.2 TDD |
| PBCH_RA | dB | 0 | 0 |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| Qrxlevmin | dBm | -140 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -99.35 | |
| RSRP Note 3 | dBm/15 KHz | -103.05 | -84.95 |
| $\hat{E}_s / I_{ot}$ | dB | -3.70 | 14.4 |
| $\hat{E}_s / N_{oc}$ | dB | -3.70 | 14.4 |
| $Treselection_{EUTRAN}$ | S | 0 | 0 |
| Snonintrasearch | dB | 50 | Not sent |
| $Thresh_{x, high}$ | dB | 48 | 48 |
| $Thresh_{serving, low}$ | dB | 44 | 44 |
| $Thresh_{x, low}$ | dB | 50 | 50 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 4.2.4.5-2: Cell Specific Test requirement Parameters for E-UTRAN FDD-TDD inter frequency cell re-selection test case in AWGN**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.2 (OP.2 FDD) and D.2.2 (OP.2 TDD) | | OP.2 FDD | | | OP.2 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| Qrxlevmin | dBm | -140 | | | -140 | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -99.35 | | | | | |
| RSRP Note 3 | dBm/15 KHz | -82.95 | -82.95 | -82.95 | -103.05 | -infinity | -84.95 |
| $\hat{E}_s / I_{ot}$ | dB | 16.40 | 16.40 | 16.40 | -3.70 | -infinity | 14.40 |
| $\hat{E}_s / N_{oc}$ | dB | 16.40 | 16.40 | 16.40 | -3.70 | -infinity | 14.40 |
| $Treselection_{EUTRAN}$ | s | 0 | | | 0 | | |
| Snonintrasearch | dB | 50 | | | Not sent | | |
| $Thresh_{x, high}$ | dB | 48 | | | 48 | | |
| $Thresh_{serving, low}$ | dB | 44 | | | 44 | | |
| $Thresh_{x, low}$ | dB | 50 | | | 50 | | |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The cell re-selection delay to lower priority is defined as the time from the beginning of time period T1, to the moment when the UE camps on Cell 1, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 1.

The cell re-selection delay to lower priority test requirement in this case is expressed as:

Cell re-selection delay to lower priority = $T_{evaluate,E\text{-}UTRAN_Inter} + T_{SI\text{-}EUTRA}$

$T_{evaluate,E\text{-}UTRAN_Inter}$ = 6.40 s; as specified in TS 36.133 [4] clause 4.2.2.4

$T_{SI\text{-}EUTRA}$ = 1280 ms; as specified in TS 36.133 [4] clause 4.2.2.4

The cell re-selection delay to lower priority shall be less than a total of 8 seconds in this test case (note: this gives a total of 7.68 seconds but the test allows 8 seconds).

The cell re-selection delay to higher priority is defined as the time from the beginning of time period T3, to the moment when the UE camps on Cell 2 and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 2.

The cell re-selection delay to higher priority test requirement in this case is expressed as:

Cell re-selection delay to higher priority = $T_{higher_priority_search} + T_{evaluate,E\text{-}UTRAN_Inter} + T_{SI\text{-}EUTRA}$

$T_{higher_priority_search}$ = 60 s; as specified in TS 36.133 [4] clause 4.2.2

$T_{evaluate,E\text{-}UTRAN_Inter}$ = 6.40 s; as specified in TS 36.133 [4] clause 4.2.2.4

$T_{SI\text{-}EUTRA}$ = 1280 ms; as specified in TS 36.133 [4] clause 4.2.2.4

The cell re-selection delay to higher priority shall be less than a total of 68 seconds in this test case (note: this gives a total of 67.68 seconds but the test allows 68 seconds).

For the test to pass, both events above shall pass.

The statistical pass/ fail decisions are done separately for each event. For an event to pass, the total number of successful loops shall be more than 90% of the cases with a confidence level of 95%.

## 4.2.5 E-UTRAN TDD - FDD cell re-selection inter frequency case

### 4.2.5.1 Test purpose

To verify that when the neighbour cell operates on a different carrier frequency and mode, compared to the current cell the UE is able to search and measure cells to meet the inter-frequency cell re-selection requirements.

### 4.2.5.2 Test applicability

This test applies to all types of E-UTRA UE supporting FDD and TDD release 9 and forward.

### 4.2.5.3 Minimum conformance requirements

The UE shall be able to identify new inter-frequency cells and perform RSRP or RSRQ measurements of identified inter-frequency cells if carrier frequency information is provided by the serving cell, even if no explicit neighbour list with physical layer cell identities is provided.

If Srxlev > $S_{nonIntraSearchP}$ and Squal > $S_{nonIntraSearchQ}$ then the UE shall search for inter-frequency layers of higher priority at least every $T_{higher_priority_search}$ where $T_{higher_priority_search}$ is described in section 4.2.2.

If Srxlev ≤ $S_{nonIntraSearchP}$ or Squal ≤ $S_{nonIntraSearchQ}$ then the UE shall search for and measure inter-frequency layers of higher, equal or lower priority in preparation for possible reselection. In this scenario, the minimum rate at which the UE is required to search for and measure higher priority layers shall be the same as that defined below.

The UE shall be able to evaluate whether a newly detectable inter-frequency cell meets the reselection criteria defined in TS36.304 [6] within $K_{carrier} * T_{detect,EUTRAN_Inter}$ if at least carrier frequency information is provided for inter-frequency neighbour cells by the serving cells when $T_{reselection}$ = 0 provided that the reselection criteria is met by a margin of at least 5dB for reselections based on ranking or 6dB for RSRP reselections based on absolute priorities or 4dB for RSRQ reselections based on absolute priorities.

The parameter $K_{carrier}$ is the number of E-UTRA inter-frequency carriers indicated by the serving cell. An inter-frequency cell is considered to be detectable according to RSRP, RSRP Ês/Iot, SCH_RP and SCH Ês/Iot defined in Annex I.1.2 for a corresponding Band.

When higher priority cells are found by the higher priority search, they shall be measured at least every $T_{measure,E\text{-}UTRAN_Inter}$. If, after detecting a cell in a higher priority search, it is determined that reselection has not occurred then the UE is not required to continuously measure the detected cell to evaluate the ongoing possibility of reselection. However, the minimum measurement filtering requirements specified later in this section shall still be met by the UE before it makes any determination that it may stop measuring the cell. If the UE detects on a E-UTRA carrier a cell whose physical identity is indicated as not allowed for that carrier in the measurement control system information of the serving cell, the UE is not required to perform measurements on that cell.

The UE shall measure RSRP or RSRQ at least every $K_{carrier} * T_{measure,EUTRAN_Inter}$ (see table 4.2.5.3-1) for identified lower or equal priority inter-frequency cells. If the UE detects on a E-UTRA carrier a cell whose physical identity is indicated

as not allowed for that carrier in the measurement control system information of the serving cell, the UE is not required to perform measurements on that cell.

The UE shall filter RSRP or RSRQ measurements of each measured higher, lower and equal priority inter-frequency cell using at least 2 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least $T_{measure,EUTRAN_Inter}/2$.

The UE shall not consider a E-UTRA neighbour cell in cell reselection, if it is indicated as not allowed in the measurement control system information of the serving cell.

For an inter-frequency cell that has been already detected, but that has not been reselected to, the filtering shall be such that the UE shall be capable of evaluating that the inter-frequency cell has met reselection criterion defined TS 36.304 [6] within $K_{carrier} * T_{evaluate,E-UTRAN_Inter}$ when $T_{reselection} = 0$ as specified in table 4.2.5.3-1 provided that the reselection criteria is met by a margin of at least 5dB for reselections based on ranking or 6dB for RSRP reselections based on absolute priorities or 4dB for RSRQ reselections based on absolute priorities. When evaluating cells for reselection, the side conditions for RSRP and SCH apply to both serving and inter-frequency cells.

If $T_{reselection}$ timer has a non zero value and the inter-frequency cell is better ranked than the serving cell, the UE shall evaluate this inter-frequency cell for the $T_{reselection}$ time. If this cell remains better ranked within this duration, then the UE shall reselect that cell.

**Table 4.2.5.3-1 : $T_{detect,EUTRAN_Inter}$, $T_{measure,EUTRAN_Inter}$ and $T_{evaluate,E-UTRAN_Inter}$**

| DRX cycle length [s] | $T_{detect,EUTRAN_Inter}$ [s] (number of DRX cycles) | $T_{measure,EUTRAN_Inter}$ [s] (number of DRX cycles) | $T_{evaluate,E-UTRAN_Inter}$ [s] (number of DRX cycles) |
|---|---|---|---|
| 0.32 | 11.52 (36) | 1.28 (4) | 5.12 (16) |
| 0.64 | 17.92 (28) | 1.28 (2) | 5.12 (8) |
| 1.28 | 32(25) | 1.28 (1) | 6.4 (5) |
| 2.56 | 58.88 (23) | 2.56 (1) | 7.68 (3) |

For higher priority cells, a UE may optionally use a shorter value for $T_{measureE-UTRA_Inter}$, which shall not be less than Max(0.64 s, one DRX cycle).

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.4 and A.4.2.5.

## 4.2.5.4 Test description

### 4.2.5.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.
2. The general test parameter settings are set up according to Table 4.2.5.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.2.5.4.3.
5. There are two E-UTRA carriers and two cells specified in the test. One E-UTRA TDD Cell 1 is the neighbour cell on the TDD carrier frequency, One E-UTRA FDD Cell 2 is the cell on the FDD carrier frequency and is used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 4.2.5.4.1-1: General Test Parameters for E-UTRAN TDD-FDD inter frequency cell re-selection test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell2 | UE shall be forced to cell 2 in the initialisation phase, so that reselection to cell 1 occurs during the first T1 phase |
| T1 end condition | Active cell | | Cell1 | UE shall perform reselection to cell 1 during T1 |
| | Neighbour cell | | Cell2 | |
| Final condition | Active cell | | Cell2 | UE shall perform reselection to cell 2 during T3 |
| Cell 1 E-UTRA RF Channel Number | | | 1 | One TDD carrier frequency is used. And Cell 1 is on RF channel number 1. |
| Cell 2 E-UTRA RF Channel Number | | | 2 | One FDD carrier frequencies is used. And Cell 2 is on RF channel number 2. |
| Time offset between cells | | | 3 ms | Asynchronous cells |
| E-UTRA TDD PRACH configuration | | | 53 | As specified in table 5.7.1-3 in 3GPP TS 36.211 |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in 3GPP TS 36.211 |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in 3GPP TS 36.211 |
| E-UTRA FDD PRACH configuration | | | 4 | As specified in table 5.7.1-2 in 3GPP TS 36.211 |
| E-UTRA FDD Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| E-UTRA TDD Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 | T1 need to be defined so that cell re-selection reaction time is taken into account. |
| T2 | | s | >7 | During T2, cell 2 shall be powered off, and during the off time the physical cell identity shall be changed, The intention is to ensure that cell 2 has not been detected by the UE prior to the start of period T3. |
| T3 | | s | 75 | T3 need to be defined so that cell re-selection reaction time is taken into account. |

### 4.2.5.4.2 Test procedure

The test scenario comprises of 1 E-UTRA TDD Cell 1 and 1 E-UTRA FDD Cell 2. The UE is requested to monitor the neighbouring cell on the E-UTRA FDD carrier. In the test there are three successive time periods, with time duration of T1, T2 and T3 respectively. Both Cell 1 and Cell 2 are already identified by the UE prior to the start of the test. Cell 1 and Cell 2 belong to different tracking areas and Cell 2 is of higher priority than Cell 1. Furthermore, UE has not registered with network for the tracking area containing Cell 1.

In the following test procedure “UE responds” means “UE starts transmitting preamble on PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure according to TS 36.508 [7] clause 4.5A.2”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2.
2. Set the parameters according to duration T0 in Table 4.2.5.5-1.
3. Set the parameters according to duration T1 in Table 4.2.5.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
4. The SS waits for random access requests information from the UE to perform cell re-selection on the lower priority cell, Cell 1.
5. If the UE responds on lower priority cell, Cell 1 during time duration T1 within 8 seconds from the beginning of time period T1, then count a success for the event “Re-select lower priority Cell 1”. Otherwise count a fail for the event “Re-select lower priority Cell 1”.

6. If the UE has re-selected Cell 1 within T1, after the re-selection or when T1 expires, continue with step 7. Otherwise, if T1 expires and the UE has not yet re-selected Cell 1, skip to step 12.

7. The SS shall switch the power setting from T1 to T2 as specified in Table 4.2.5.5-2. During time duration T2, Cell 2 shall be powered OFF and the physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) shall be changed to ensure Cell 2 is not detected by the UE.

8. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 4.2.5.5-2.

9. The SS waits for random access requests information from the UE to perform cell re-selection on the higher priority cell, Cell 2.

10. If the UE responds on higher priority cell, Cell 2 during time duration T3 within 68 seconds from the beginning of time period T3, then count a success for the event "Re-select higher priority Cell 2". Otherwise count a fail for the event "Re-select higher priority Cell 2".

11. If the UE has re-selected Cell 2 within T3, after the re-selection or when T3 expires, skip to step 13. Otherwise, if T3 expires and the UE has not yet re-selected Cell 2, continue with step 12.

12. Switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 2.
    Set the parameters according to duration T0 in Table 4.2.5.5-1.

13. Repeat step 3-12 until a test verdict has been achieved.
    Each of the events "Re-select lower priority Cell 1" and "Re-select higher priority Cell 2" is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. Different events may require different times for a verdict.
    If both events pass, the test passes. If one event fails, the test fails.

#### 4.2.5.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 4.2.5.4.3-1: Common Exception messages for E-UTRAN TDD-FDD inter frequency cell re-selection test case**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.2-1<br>Table H.2.2-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1<br>Table H.3.2-2 |

### 4.2.5.5 Test requirement

Tables 4.2.5.4.1-1, 4.2.5.5-1 and 4.2.5.5-2 define the primary level settings including test tolerances for E-UTRAN TDD-FDD inter-frequency cell re-selection test case. Note that the time period for T0 is system implementation dependent.

**Table 4.2.5.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD-FDD inter frequency cell re-selection test case**

| Parameter | Unit | Cell 1 | Cell 2 |
|---|---|---|---|
| | | T0 | |
| E-UTRA RF Channel number | | 1 | 2 |
| $BW_{channel}$ | MHz | 10 | 10 |
| OCNG Patterns defined in D.1.2 (OP.2 FDD) and D.2.2 (OP.2 TDD) | | OP.2 TDD | OP.2 FDD |
| PBCH_RA | dB | 0 | 0 |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| Qrxlevmin | dBm | -140 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -99.35 | |
| RSRP Note 3 | dBm/15 KHz | -103.05 | -84.95 |
| $\hat{E}_s / I_{ot}$ | dB | -3.70 | 14.40 |
| $\hat{E}_s / N_{oc}$ | dB | -3.70 | 14.40 |
| $Treselection_{EUTRAN}$ | S | 0 | 0 |
| Snonintrasearch | dB | 50 | Not sent |
| $Thresh_{x, high}$ | dB | 48 | 48 |
| $Thresh_{serving, low}$ | dB | 44 | 44 |
| $Thresh_{x, low}$ | dB | 50 | 50 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 4.2.5.5-2: Cell Specific Test requirement Parameters for E-UTRAN TDD-FDD inter frequency cell re-selection test case in AWGN**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.2 (OP.2 FDD) and D.2.2 (OP.2 TDD) | | OP.2 TDD | | | OP.2 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| Qrxlevmin | dBm | -140 | | | -140 | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -99.35 | | | | | |
| RSRP Note 3 | dBm/15 KHz | -82.95 | -82.95 | -82.95 | -103.05 | -infinity | -84.95 |
| $\hat{E}_s / I_{ot}$ | dB | 16.40 | 16.40 | 16.40 | -3.70 | -infinity | 14.40 |
| $\hat{E}_s / N_{oc}$ | dB | 16.40 | 16.40 | 16.40 | -3.70 | -infinity | 14.40 |
| $Treselection_{EUTRAN}$ | s | 0 | | | 0 | | |
| Snonintrasearch | dB | 50 | | | Not sent | | |
| $Thresh_{x, high}$ | dB | 48 | | | 48 | | |
| $Thresh_{serving, low}$ | dB | 44 | | | 44 | | |
| $Thresh_{x, low}$ | dB | 50 | | | 50 | | |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The cell re-selection delay to lower priority is defined as the time from the beginning of time period T1, to the moment when the UE camps on Cell 1, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 1.

The cell re-selection delay to lower priority test requirement in this case is expressed as:

Cell re-selection delay to lower priority = $T_{evaluate,E\text{-}UTRAN_Inter} + T_{SI\text{-}EUTRA}$

$T_{evaluate,E\text{-}UTRAN_Inter}$ = 6.40 s; as specified in TS 36.133 [4] clause 4.2.2.4

$T_{SI\text{-}EUTRA}$ = 1280 ms; as specified in TS 36.133 [4] clause 4.2.2.4

The cell re-selection delay to lower priority shall be less than a total of 8 seconds in this test case (note: this gives a total of 7.68 seconds but the test allows 8 seconds).

The cell re-selection delay to higher priority is defined as the time from the beginning of time period T3, to the moment when the UE camps on Cell 2 and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 2.

The cell re-selection delay to higher priority test requirement in this case is expressed as:

Cell re-selection delay to higher priority = $T_{higher_priority_search}$ + $T_{evaluate,E\text{-}UTRAN_Inter}$ + $T_{SI\text{-}EUTRA}$

$T_{higher_priority_search}$ = 60 s; as specified in TS 36.133 [4] clause 4.2.2

$T_{evaluate,E\text{-}UTRAN_Inter}$ = 6.40 s; as specified in TS 36.133 [4] clause 4.2.2.4

$T_{SI\text{-}EUTRA}$ = 1280 ms; as specified in TS 36.133 [4] clause 4.2.2.4

The cell re-selection delay to higher priority shall be less than a total of 68 seconds in this test case (note: this gives a total of 67.68 seconds but the test allows 68 seconds).

For the test to pass, both events above shall pass.

The statistical pass/ fail decisions are done separately for each event. For an event to pass, the total number of successful loops shall be more than 90% of the cases with a confidence level of 95%.

## 4.2.6 E-UTRAN TDD - TDD cell re-selection inter frequency case

### 4.2.6.1 Test purpose

To verify that when the neighbour cell operates on a different carrier frequency, compared to the current cell the UE is able to search and measure cells to meet the inter-frequency cell re-selection requirements.

### 4.2.6.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

### 4.2.6.3 Minimum conformance requirements

The cell re-selection delay shall be less than $T_{evaluate,E\text{-}UTRAN_Inter}$ + $T_{SI\text{-}EUTRA}$ in RRC_IDLE state.

The UE shall be able to identify new inter-frequency cells and perform RSRP measurements of identified inter-frequency cells if carrier frequency information is provided by the serving cell, even if no explicit neighbour list with physical layer cell identities is provided.

If the $S_{ServingCell}$ of the E-UTRA serving cell is greater than $S_{nonintrasearch}$ then the UE shall search for inter-frequency layers of higher priority at least every $T_{higher_priority_search}$ where $T_{higher_priority_search}$ is described in section 4.2.2 of TS 36.133 [4].

If the $S_{ServingCell}$ of the E-UTRA serving cell is less than or equal to $S_{nonintrasearch}$ then the UE shall search for and measure inter-frequency layers of higher, equal or lower priority in preparation for possible re-selection. In this scenario, the minimum rate at which the UE is required to search for and measure higher priority layers shall be the same as that defined below for a lower or equal priority inter-frequency layers.

The UE shall be able to evaluate whether a newly detectable lower or equal priority inter-frequency cell meets the re-selection criteria defined in TS 36.304 [6] within $K_{carrier}$ * $T_{detect,EUTRAN_Inter}$ (as defined in table 4.2.2.4-1 of TS 36.133 [4] clause 4.2.2.4) if at least carrier frequency information is provided for inter-frequency neighbour cells by the serving cells when $Treselection_{EUTRAN}$ = 0 provided that the reselection criteria is met by a margin of at least 5dB for reselections based on ranking or 6dB for reselections based on absolute priorities. The parameter $K_{carrier}$ is the number of E-UTRA inter-frequency carriers indicated by the serving cell.

When higher priority cells are found by the higher priority search, they shall be measured at least every $T_{measure,EUTRAN_Inter}$. If, after it is found in a higher priority search, it is determined that reselection has not occurred then the UE is not required to continuously measure the detected cell to evaluate the ongoing possibility of re-selection. However, the minimum measurement filtering requirements specified later in this section shall still be met by the UE before it makes any determination that it may stop measuring the cell. If the UE detects on a E-UTRA carrier a cell whose physical identity is indicated as not allowed for that carrier in the measurement control system information of the serving cell, the UE is not required to perform measurements on that cell.

The UE shall measure RSRP at least every $K_{carrier}$ * $T_{measure,EUTRAN_Inter}$ as defined in table 4.2.2.4-1 of TS 36.133 [4] clause 4.2.2.4 for identified lower or equal priority inter-frequency cells. If the UE detects on an E-UTRA carrier a cell

whose physical identity is indicated as not allowed for that carrier in the measurement control system information of the serving cell, the UE is not required to perform measurements on that cell.

The UE shall filter RSRP measurements of each measured higher, lower and equal priority inter-frequency cell using at least 2 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least $T_{measure,EUTRAN_Inter} / 2$.

The UE shall not consider an E-UTRA neighbour cell in cell re-selection, if it is indicated as not allowed in the measurement control system information of the serving cell.

For an inter-frequency cell that has been already detected, but that has not been re-selected to, the filtering shall be such that the UE shall be capable of evaluating that the inter-frequency cell has met re-selection criterion defined in TS 36.304 [6] within $K_{carrier} * T_{evaluate,E-UTRAN_Inter}$ as defined in table 4.2.2.4-1 of TS 36.133 [4] clause 4.2.2.4 when $Treselection_{EUTRAN} = 0$ provides that the re-selection criteria is met by a margin of at least 5dB for re-selection based on ranking or 6dB for re-selection based on absolute priorities. When evaluating cells for reselection, the side conditions for RSRP and SCH apply to both serving and inter-frequency cells.

If $Treselection_{EUTRAN}$ timer has a non-zero value and the inter-frequency cell is better ranked than the serving cell, the UE shall evaluate this inter-frequency cell for the $Treselection_{EUTRAN}$ time. If this cell remains better ranked within this duration, then the UE shall re-select that cell.

The UE shall evaluate the inter-frequency cell re-selection criteria as defined in TS 36.304 [6] at least every DRX cycle. The DRX cycle length is 1.28 seconds.

At inter-frequency cell re-selection, the UE shall monitor the downlink of serving cell for paging reception until the UE is capable to start monitoring downlink channels of the target inter-frequency cell for paging reception. The interruption time shall not exceed $T_{SI-EUTRA}$ + 50 ms. $T_{SI-EUTRA}$ is the time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks as defined in TS 36.331 [5] for a E-UTRAN cell.

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.4 and A.4.2.6.

## 4.2.6.4 Test description

### 4.2.6.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (eNodeB emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.
2. The general test parameter settings are set up according to Table 4.2.6.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.2.6.4.3.
5. There are two E-UTRA TDD carriers and two cells specified in the test. Cell 2 is the cell used for registration with the power level set according to clause C.0 and C.1for this test.

**Table 4.2.6.4.1-1: General Test Parameters for E-UTRAN TDD-TDD inter frequency cell re-selection test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell2 | UE shall be forced to cell 2 in the initialisation phase, so that reselection to cell 1 occurs during the first T1 phase |
| T1 end condition | Active cell | | Cell1 | UE shall perform reselection to cell 1 during T1 |
| | Neighbour cell | | Cell2 | |
| Final condition | Active cell | | Cell2 | UE shall perform reselection to cell 2 during T3 |
| E-UTRA RF Channel Number | | | 1, 2 | Two TDD carrier frequencies are used. |
| Time offset between cells | | µs | 3 | Synchronous cells<br>3µs or 92*Ts |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in 3GPP TS 36.211 |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in 3GPP TS 36.211 |
| PRACH configuration index | | | 53 | As specified in table 5.7.1-3 in 3GPP TS 36.211 |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 | T1 need to be defined so that cell re-selection reaction time is taken into account. |
| T2 | | s | >7 | During T2, cell 2 shall be powered off, and during the off time the physical cell identity shall be changed, The intention is to ensure that cell 2 has not been detected by the UE prior to the start of period T3. |
| T3 | | s | 75 | T3 need to be defined so that cell re-selection reaction time is taken into account. |

### 4.2.6.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The UE is requested to monitor the neighbouring cell on one E-UTRA TDD carrier. In the test there are three successive time periods, with time duration of T1, T2, and T3 respectively. Both Cell 1 and Cell 2 are already identified by the UE prior to the start of the test. Cell 1 and Cell 2 belong to different tracking areas and Cell 2 is of higher priority than Cell 1. Furthermore, UE has not registered with network for the tracking area containing Cell 1.

In the following test procedure “UE responds” means “UE starts transmitting preamble on PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure according to TS 36.508 [7] clause 4.5A.2”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2.

2. Set the parameters according to duration T0 in Table 4.2.6.5-1

3. Set the parameters according to duration T1 in Table 4.2.6.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

4. The SS waits for random access requests information from the UE to perform cell re-selection procedure on the lower priority cell, Cell 1.

5. If the UE responds on the lower priority cell, Cell 1, during time duration T1 within 8 seconds from the beginning of time period T1, then count a success for the event “Re-select lower priority Cell 1”. Otherwise count a fail for the event “Re-select lower priority Cell 1”.

6. If the UE has re-selected Cell 1 within T1, after the re-selection or when T1 expires, continue with step 7. Otherwise, if T1 expires and the UE has not yet re-selected Cell 1, skip to step 12.

7. The SS shall switch the power setting from T1 to T2 as specified in Table 4.2.6.5-2. During time duration T2, Cell 2 shall be powered OFF and change Cell 2 physical cell identity to ((current cell 2 physical cell identity + 1) mod 14 + 2) to ensure Cell 2 is not detected by the UE.

8. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 4.2.6.5-2.

9. The SS waits for random access requests information from the UE to perform cell re-selection procedure on the higher priority cell, Cell 2.

10. If the UE responds on higher priority cell, Cell 2, during time duration T3 within 68 seconds from the beginning of time period T3, then count a success for the event “Re-select higher priority Cell 2”. Otherwise count a fail for the event “Re-select higher priority Cell 2”.

11. If the UE has re-selected Cell 2 within T3, after the re-selection or when T3 expires, skip to step 13.
Otherwise, if T3 expires and the UE has not yet re-selected Cell 2, continue with step 12.

12. Switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 2.
Set the parameters according to duration T0 in Table 4.2.6.5-1.

13. Repeat step 3-12 until a test verdict has been achieved.
Each of the events “Re-select lower priority Cell 1” and “Re-select higher priority Cell 2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. Different events may require different times for a verdict.
If both events pass, the test passes. If one event fails, the test fails.

### 4.2.6.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.4.3.3 and 4.6.3 with the following exceptions:

**Table 4.2.6.4.3-1: Common Exception messages for E-UTRAN TDD-TDD inter frequency cell re-selection test case**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.2-1<br>Table H.2.2-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-2 |

## 4.2.6.5 Test requirement

Tables 4.2.6.4.1-1, 4.2.6.5-1 and 4.2.6.5-2 defines the primary level settings including test tolerances for E-UTRAN TDD-TDD inter frequency cell re-selection test case. Note that the time period for T0 is system implementation dependent.

**Table 4.2.6.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD-TDD inter frequency cell re-selection test case**

| Parameter | Unit | Cell 1 | Cell 2 |
|---|---|---|---|
| | | T0 | |
| E-UTRA RF Channel number | | 1 | 2 |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.2.2 (OP.2 TDD) | | OP.2 TDD | |
| PBCH_RA | dB | 0 | 0 |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA[Note 1] | dB | | |
| OCNG_RB[Note 1] | dB | | |

| | | | |
|---|---|---|---|
| Qrxlevmin | dBm | -140 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -99.35 | |
| RSRP Note 3 | dBm/15 KHz | -103.05 | -84.95 |
| $\hat{E}_s / I_{ot}$ | dB | -3.70 | 14.40 |
| $\hat{E}_s / N_{oc}$ | dB | -3.70 | 14.40 |
| $Treselection_{EUTRAN}$ | S | 0 | 0 |
| Snonintrasearch | dB | 50 | Not sent |
| $Thresh_{x, high}$ | dB | 48 | 48 |
| $Thresh_{serving, low}$ | dB | 44 | 44 |
| $Thresh_{x, low}$ | dB | 50 | 50 |
| Propagation Condition | | AWGN | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.<br>Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | |

**Table 4.2.6.5-2: Cell Specific Test requirement Parameters for E-UTRAN TDD-TDD inter frequency cell re-selection test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Pattern defined in D.2.2 (OP.2 TDD) | | OP.2 TDD | | | OP.2 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Qrxlevmin | dBm | -140 | | | -140 | | |
| $N_{oc}$ | dBm/15 kHz | -99.35 | | | | | |
| RSRP | dBm/15 KHz | -82.95 | -82.95 | -82.95 | -103.05 | -infinity | -84.95 |
| $\hat{E}_s / I_{ot}$ | dB | 16.40 | 16.40 | 16.40 | -3.70 | -infinity | 14.40 |
| $\hat{E}_s / N_{oc}$ | dB | 16.40 | 16.40 | 16.40 | -3.70 | -infinity | 14.40 |
| $Treselection_{EUTRAN}$ | S | 0 | | | 0 | | |
| Snonintrasearch | dB | 50 | | | Not sent | | |
| $Thresh_{x, high}$ | dB | 48 | | | 48 | | |
| $Thresh_{serving, low}$ | dB | 44 | | | 44 | | |
| $Thresh_{x, low}$ | dB | 50 | | | 50 | | |
| Propagation Condition | | AWGN | | | | | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.<br>Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | | | | | |

The cell re-selection delay to higher priority is defined as the time from the beginning of time period T3, to the moment when the UE camps on Cell 2 and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 2.

The cell re-selection delay to higher priority test requirement in this case is expressed as:

Cell re-selection delay to higher priority = $T_{higher_priority_search} + T_{evaluate,E\text{-}UTRAN_Inter} + T_{SI\text{-}EUTRA}$

$T_{higher_priority_search}$ = 60 s; as specified in TS 36.133 [4] clause 4.2.2

$T_{evaluate,E\text{-}UTRAN_Inter}$ = 6.40 s; as specified in TS 36.133 [4] clause 4.2.2.4

$T_{SI\text{-}EUTRA}$ = 1280 ms; as specified in TS 36.133 [4] clause 4.2.2.7

The cell re-selection delay to higher priority shall be less than a total of 68 seconds in this test case (note: this gives a total of 67.68 seconds but the test allows 68 seconds).

The cell re-selection delay to lower priority is defined as the time from the beginning of time period T1, to the moment when the UE camps on Cell 1 and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 1.The cell re-selection delay to lower priority test requirement in this case is expressed as:

Cell re-selection delay to lower priority = $T_{evaluate,E\text{-}UTRAN_Inter} + T_{SI\text{-}EUTRA}$

$T_{evaluate,E\text{-}UTRAN_Inter}$ = 6.40 s; as specified in TS 36.133 [4] clause 4.2.2.4

$T_{SI\text{-}EUTRA}$ = 1280 ms; as specified in TS 36.133 [4] clause 4.2.2.7

The cell re-selection delay to lower priority shall be less than a total of 8 seconds in this test case (note: this gives a total of 7.68 seconds but the test allows 8 seconds).

For the test to pass, both events above shall pass.

The statistical pass/ fail decisions are done separately for each event. For an event to pass, the total number of successful loops shall be more than 90% of the cases with a confidence level of 95%.

## 4.2.7 E-UTRAN FDD – FDD Inter frequency case in the existence of non-allowed CSG cell

### 4.2.7.1 Test purpose

To verify that when the neighbour cell operates on a different carrier frequency and there is the interference from non-allowed CSG cell and the layers have equal priority, compared to the current cell the UE is able to search and measure cells to meet the inter-frequency cell re-selection requirements.

### 4.2.7.2 Test applicability

This test applies to all types of E-UTRA UE supporting FDD release 9 and forward.

### 4.2.7.3 Minimum conformance requirements

The cell re-selection delay shall be less than $T_{evaluate,E\text{-}UTRAN_Inter} + T_{SI\text{-}EUTRA}$ in RRC_IDLE state.

The UE shall be able to identify new inter-frequency cells and perform RSRQ measurements of identified inter-frequency cells if carrier frequency information is provided by the serving cell, even if no explicit neighbour list with physical layer cell identities is provided.

If Srxlev > $S_{nonIntraSearchP}$ and Squal > $S_{nonIntraSearchQ}$ then the UE shall search for inter-frequency layers of higher priority at least every $T_{higher_priority_search}$ where $T_{higher_priority_search}$ is described in TS 36.133 [4] clause 4.2.2 as $T_{higher_priority_search}$ = (60 * $N_{layers}$) seconds, where the parameter $N_{layers}$ is the total number of configured higher priority E-UTRA carrier frequencies.

If Srxlev ≤ $S_{nonIntraSearchP}$ or Squal ≤ $S_{nonIntraSearchQ}$ then the UE shall search for and measure inter-frequency layers of higher, equal or lower priority in preparation for possible re-selection. In this scenario, the minimum rate at which the UE is required to search for and measure higher priority layers shall be the same as that defined below for lower or equal priority inter-frequency layers.

The UE shall be able to evaluate whether a newly detectable inter-frequency cell meets the re-selection criteria defined in TS 36.304 [6] within $K_{carrier}$ * $T_{detect,EUTRAN_Inter}$ (as defined in table 4.2.2.4-1 of TS 36.133 [4] clause 4.2.2.4) if at least carrier frequency information is provided for inter-frequency neighbour cells by the serving cells when $Treselection_{EUTRAN}$ = 0 provides that the re-selection criteria is met by a margin of at least 5 dB for re-selection based on ranking or 6 dB for re-selection based on absolute priorities or 4dB for RSRQ reselections based on absolute priorities. The parameter $K_{carrier}$ is the number of E-UTRA inter-frequency carriers indicated by the serving cell.

When higher priority cells are found by the higher priority search, they shall be measured at least every $T_{measure,EUTRAN_Inter}$. If re-selection to any higher priority cell is not triggered within ($T_{evaluateFDD,\ Inter}$ + $Treselection_{EUTRAN}$) after it is found in a higher priority search, the UE is not required to continue making measurements of the cell to evaluate the ongoing possibility of re-selection. If the UE detects on an E-UTRA carrier a cell whose physical identity is indicated as not allowed for that carrier in the measurement control system information of the serving cell, the UE is not required to perform measurements on that cell.

The UE shall measure RSRQ at least every $K_{carrier}$ * $T_{measure,EUTRAN_Inter}$ DRX cycle as defined in table 4.2.2.4-1 of TS 36.133 [4] clause 4.2.2.4 for identified lower or equal priority inter-frequency cells. If the UE detects on an E-UTRA carrier a cell whose physical identity is indicated as not allowed for that carrier in the measurement control system information of the serving cell, the UE is not required to perform measurements on that cell.

The UE shall filter RSRQ measurements of each measured higher, lower and equal priority inter-frequency cell using at least 2 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least $T_{measure,EUTRAN_Inter}$ / 2.

The UE shall not consider an E-UTRA neighbour cell in cell re-selection, if it is indicated as not allowed in the measurement control system information of the serving cell.

For an inter-frequency cell that has been already detected, but that has not been re-selected to, the filtering shall be such that the UE shall be capable of evaluating that the inter-frequency cell has met re-selection criterion defined in TS 36.304 [6] within $K_{carrier}$ * $T_{evaluate,E\text{-}UTRAN_Inter}$ as defined in table 4.2.2.4-1 of TS 36.133 [4] clause 4.2.2.4 when $Treselection_{EUTRAN}$ = 0 provides that the re-selection criteria is met by a margin of at least 5 dB for re-selection based on ranking or 6 dB for re-selection based on absolute priorities or 4dB for RSRQ reselections based on absolute priorities.

When evaluating cells for re-selection, the side conditions for RSRP and SCH apply to both serving and inter-frequency cells.

If $Treselection_{EUTRAN}$ timer has a non-zero value and the inter-frequency cell is better ranked than the serving cell, the UE shall evaluate this inter-frequency cell for the $Treselection_{EUTRAN}$ time. If this cell remains better ranked within this duration, then the UE shall re-select that cell.

The UE shall evaluate the inter-frequency cell re-selection criteria as defined in TS 36.304 [6] at least every DRX cycle. The DRX cycle length is 1.28 seconds. When a non-zero value of $Treselection_{EUTRAN}$ is used, the UE shall only perform re-selection on an evaluation which occurs simultaneously to, or later than the expiry of the $Treselection_{EUTRAN}$ timer.

At inter-frequency cell re-selection, the UE shall monitor the downlink of serving cell for paging reception until the UE is capable to start monitoring downlink channels of the target inter-frequency cell for paging reception. The interruption time shall not exceed $T_{SI-EUTRA}$ + 50 ms. $T_{SI-EUTRA}$ is the time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks as defined in TS 36.331 [5] for a E-UTRAN cell.

For idle mode cell re-selection purposes, the UE shall be capable of monitoring at least:

- Depending on UE capability, 3 FDD E-UTRA inter-frequency carriers

In addition to the requirements defined in TS 36.133 [4] clause 4.2.2.9 a UE in RRC_IDLE state shall be capable of monitoring a total of at least 8 carrier frequency layers, which includes serving layer, comprising of any defined in TS 36.133 [4] clause 4.2.2.9 combination of E-UTRA FDD, E-UTRA TDD, UTRA FDD, UTRA TDD, GSM (one GSM layer corresponds to 32 cells) , cdma2000 1x and HRPD layers.

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.4 and A.4.2.7.

## 4.2.7.4 Test description

### 4.2.7.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

The UE's allowed CSG list is empty.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.27.
2. The general test parameter settings are set up according to Table 4.2.7.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.2.7.4.3.
5. There are two E-UTRA FDD carriers and two cells specified in the test and one non-allowed E-UTRA FDD CSG cell. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test and Cell 3 is the cell used for adding interference from non-allowed CSG cell.

**Table 4.2.7.4.1-1: General test parameters for FDD-FDD inter frequency cell re-selection test case with non-allowed CSG cell**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | UE shall be forced to cell 1 in the initialisation phase |
| Final condition | Active cell | | Cell2 | UE shall perform reselection to cell 2 during T2 |
| E-UTRA RF Channel Number | | | 1, 2 | Two FDD carrier frequencies are used. |
| Time offset between cells | | ms | 3 | Asynchronous cells |
| PRACH configuration | | | 4 | As specified in table 5.7.1-2 in TS 36.211 |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 | T1 need to be defined so that the non-allowed CSG cell is identified. |
| T2 | | s | 40 | T2 need to be defined so that cell re-selection reaction time is taken into account. |
| T3 | | s | 15 | T3 need to be defined so that whether cell re-selection would not occur is insured. |

### 4.2.7.4.2 Test procedure

The test consists of one active cell, one neighbour cell and one non-allowed CSG cell. The UE is requested to monitor the neighbouring cell on one of the E-UTRA FDD carriers. In the test there are four successive time periods, with time duration of T0, T1, T2 and T3 respectively. Cell 1 is already identified by the UE prior to the start of the test. Cell 1 and Cell 2 belong to different tracking areas and Cell 3 is a non-allowed CSG cell. Furthermore, UE has not registered with network for the tracking area containing Cell 2.

In the following test procedure “UE responds” means “UE starts transmitting preamble on PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure according to TS 36.508 [7] clause 4.5A.2”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2. Set Cell 2 physical cell identity = initial cell 2 physical cell identity.

2. Set the parameters according to duration T0 in Table 4.2.7.5-1. If the UE is already camped in Cell 1, wait until T0 expires and skip to step 5.

3. The SS waits for random access requests information from the UE to perform cell re-selection on Cell 1.

4. If the UE has re-selected Cell 1 within T0, after the re-selection or when T0 expires, continue with step 5. Otherwise, if T0 expires and the UE has not yet re-selected Cell 1, skip to step 12.

5. Set the parameters according to duration T1 in Table 4.2.7.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

6. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for one iteration of the test procedure loop.

7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 4.2.7.5-2. T2 starts.

8. The SS waits for random access requests information from the UE to perform cell re-selection to a newly detectable cell, Cell 2.

9. If the UE has responded on the newly detected Cell 2 during time duration within 34 seconds from the beginning of time period T2 count a success for the event “Re-select newly detected Cell 2”, and after the re-selection, continue with step 10.
   Otherwise, if the UE has not re-selected Cell 2 count a fail for the event “Re-select newly detected Cell 2”, and skip to step 12.

10. After the re-selection or when T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 4.2.7.5-2. T3 starts.

11. If the UE has not re-selected Cell 1 when T3 expires, count a success for the event “Hold out on Cell 2”, and skip to step 13. Otherwise, if the UE responds on the known cell, Cell 1 during time duration T3, count a fail for the event “Hold out on Cell 2”, and continue with step 12.

12. Switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.

13. Repeat step 2-12 until a test verdict has been achieved.
    Each of the events “Re-select newly detected Cell 2” and “Hold out on Cell 2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. Different events may require different times for a verdict.
    If both events pass, the test passes. If one event fails, the test fails.

#### 4.2.7.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 4.2.7.4.3-1: Common Exception messages for E-UTRAN FDD – FDD Inter frequency case in the existence of non-allowed CSG cell**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

**Table 4.2.7.4.3-2: *SystemInformationBlockType1*: E-UTRAN inter frequency cell re-selection case in the existence of non-allowed CSG cell**

| Derivation Path: TS 36.508 [7] clause 4.4.3.2, Table 4.4.3.2-3 SystemInformationBlockType1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType1 ::= SEQUENCE { | | | |
| cellAccessRelatedInfo SEQUENCE { | | | |
| csg-Indication | TRUE | Cell 3 | |
| csg-Identity | Physical Cell ID of the Cell 3 | Cell 3 | |
| } | | | |
| nonCriticalExtension SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| cellSelectionInfo-v920 SEQUENCE { | | | |
| q-QualMin-r9 | -20 (-20dB) | Cell 1, 2, 3 | |
| q-QualMinOffset-r9 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 4.2.7.4.3-3: *SystemInformationBlockType3*: E-UTRAN inter frequency cell re-selection case in the existence of non-allowed CSG cell**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType3 ::= SEQUENCE { | | | |
| intraFreqCellReselectionInfo SEQUENCE { | | | |
| lateNonCriticalExtension { | | | |
| s-IntraSearch-v920 SEQUENCE {} | Not present | | |
| s-NonIntraSearch-v920 SEQUENCE { | | | |
| s-NonIntraSearchP-r9 | 25 | Cell 1 | |
| s-NonIntraSearchQ-r9 | 10 | Cell 1 | |
| } | | | |
| q-QualMin-r9 | -20 (-20dB) | Cell 1, 2, 3 | |
| threshServingLowQ-r9 | Not present | | |
| } | | | |
| } | | | |

**Table 4.2.7.4.3-4: *SystemInformationBlockType4*: E-UTRAN inter frequency cell re-selection case in the existence of non-allowed CSG cell**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-3 SystemInformationBlockType4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType4 ::= SEQUENCE { | | | |
| intraFreqNeighCellList SEQUENCE (SIZE (1..maxCellIntra)) OF SEQUENCE { | | | |
| physCellId | Physical Cell ID of the Cell 3 | Cell 1 | |
| | Physical Cell ID of the Cell 1 | Cell 3 | |
| } | | | |
| csg-PhysCellIdRange SEQUENCE { | | | |
| start | Physical Cell ID of the cell on which this SIB is transmitted | Cell 3 | |
| range | Not present | | |
| } | | | |
| } | | | |

**Table 4.2.7.4.3-5: *SystemInformationBlockType5*: E-UTRAN inter frequency cell re-selection case in the existence of non-allowed CSG cell**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-4 SystemInformationBlockType5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType5 ::= SEQUENCE { | | | |
| interFreqCarrierFreqList SEQUENCE (SIZE (1..maxFreq)) OF SEQUENCE { | | | |
| dl-CarrierFreq | Frequency of the Cell 2 | Cell 1, 3 | |
| | Frequency of the Cell 1 | Cell 2 | |
| q-RxLevMin | -70 (-140 dBm) | Cell 1, 2, 3 | |
| interFreqNeighCellList[*n*] SEQUENCE (SIZE (1..maxCellInter)) OF SEQUENCE { | | Cell 2 | |
| physCellId | Physical Cell ID of the Cell 1 | | |
| q-OffsetCell | dB0 | | |
| } | | | |
| q-QualMin-r9 | -20 (-20dB) | Cell 1, 2, 3 | |
| threshX-Q-r9 SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |

**Table 4.2.7.4.3-6: *SystemInformationBlockType9*: E-UTRAN inter frequency cell re-selection case in the existence of non-allowed CSG cell**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-8 SystemInformationBlockType9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType9 ::= SEQUENCE { | | | |
| hnb-Name | "3gppTest" | Cell 3 | |
| } | | | |

## 4.2.7.5 Test requirement

Tables 4.2.7.4.1-1, 4.2.7.5-1 and 4.2.7.5-2 defines the primary level settings including test tolerances for E-UTRAN FDD – FDD Inter frequency case in the existence of non-allowed CSG cell test case. Note that the time period for T0 is system implementation dependent.

**Table 4.2.7.5-1: Cell specific Test requirement parameters for FDD-FDD inter frequency cell re-selection test case with non-allowed CSG cell**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| | | T0 | | |
| E-UTRA RF Channel number | | 1 | 2 | 1 |
| $BW_{channel}$ | MHz | 10 | 10 | 10 |
| OCNG Patterns defined in D.1.2 | | OP.2 FDD | OP.2 FDD | OP.2 FDD |
| PBCH_RA | dB | 0 | 0 | 0 |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| Qrxlevmin | dBm | -140 | -140 | -140 |
| Qqualmin | dB | -20 | -20 | -20 |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | |
| RSRP Note 3 | dBm/15 kHz | -85.00 | -101.00 | -Infinity |
| RSRQ Note 3 | dB | -11.00 | -15.56 | -Infinity |
| $\hat{E}_s / I_{ot}$ | dB | 13.00 | -3.00 | -Infinity |
| $\hat{E}_s / N_{oc}$ | dB | 13.00 | -3.00 | -Infinity |
| Treselection | s | 0 | 0 | 0 |
| SnonintrasearchQ | dB | 10 | Not sent | Not sent |
| Propagation Condition | | AWGN | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and RSRQ and Es/Iot levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 4.2.7.5-2: Cell specific test parameters for FDD-FDD inter frequency cell re-selection test case with non-allowed CSG cell**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3(Non-allowed CSG cell) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.2 (OP.2 FDD) | | OP.2 FDD | | | OP.2 FDD | | | OP.2 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | |
| Qrxlevmin | dBm | -140 | | | -140 | | | -140 | | |
| Qqualmin | dB | -20 | | | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | | | | | | | |
| RSRP Note 3 | dBm/15 kHz | -90.20 | -90.0 | -85.00 | -Infinity | -85.00 | -90.00 | -90.00 | -85.00 | -60.00 |
| RSRQ Note 3 | dB | -14.24 | -17.15 | -35.81 | -Infinity | -11.00 | -11.43 | -14.04 | -12.15 | -10.81 |
| $\hat{E}_s / I_{ot}$ | dB | -0.84 | -5.21 | -25.00 | -Infinity | 13.00 | 8.00 | -0.47 | 4.36 | 24.79 |
| $\hat{E}_s / N_{oc}$ | dB | 7.80 | 8.00 | 13.00 | -Infinity | 13.00 | 8.00 | 8.00 | 13.00 | 38.00 |
| Treselection | s | 0 | | | 0 | | | 0 | | |
| SnonintrasearchQ | dB | 10 | | | Not sent | | | Not sent | | |
| Propagation Condition | | AWGN | | | | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and RSRQ levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The cell reselection delay to a newly detectable cell is defined as the time from the beginning of time period T2, to the moment when the UE camps on Cell 2, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 2.

The cell re-selection delay to a newly detectable cell test requirement in this case is expressed as:

Cell re-selection delay to a newly detectable cell = $T_{evaluate, E-UTRAN_Inter} + T_{SI-EUTRA}$

$T_{evaluate, E-UTRAN_Inter}$ = 32 s; as specified in TS 36.133 [4] clause 4.2.2.4.

$T_{SI-EUTRA}$ = 1280 ms; as specified in TS 36.133 [4] clause 4.2.2.4.

The cell re-selection delay to a newly detectable cell shall be less than a total of 34 seconds in this test case (note: this gives a total of 33.28 s, but the test allows 34 seconds.)

The probability of re-selection from Cell 1 to Cell 2 during T2 observed during testing shall be at least 90%.

The probability of re-selection from Cell 2 to Cell 1 during T3 observed during testing shall be less than 10%.

For the test to pass, both events above shall pass with a confidence level of 95%.

## 4.2.8 E-UTRAN TDD – TDD Inter frequency case in the existence of non-allowed CSG cell

### 4.2.8.1 Test purpose

To verify that when the neighbour cell operates on a different carrier frequency and there is the interference from non-allowed CSG cell and the layers have equal priority, compared to the current cell the UE is able to search and measure cells to meet the inter-frequency cell re-selection requirements.

### 4.2.8.2 Test applicability

This test applies to all types of E-UTRA UE supporting TDD release 9 and forward.

### 4.2.8.3 Minimum conformance requirements

The cell re-selection delay shall be less than $T_{evaluate,E-UTRAN_Inter} + T_{SI-EUTRA}$ in RRC_IDLE state.

The UE shall be able to identify new inter-frequency cells and perform RSRQ measurements of identified inter-frequency cells if carrier frequency information is provided by the serving cell, even if no explicit neighbour list with physical layer cell identities is provided.

If Srxlev > $S_{nonIntraSearchP}$ and Squal > $S_{nonIntraSearchQ}$ then the UE shall search for inter-frequency layers of higher priority at least every $T_{higher_priority_search}$ where $T_{higher_priority_search}$ is described in section 4.2.2 of TS 36.133 [4].

If Srxlev ≤ $S_{nonIntraSearchP}$ or Squal ≤ $S_{nonIntraSearchQ}$ then the UE shall search for and measure inter-frequency layers of higher, equal or lower priority in preparation for possible re-selection. In this scenario, the minimum rate at which the UE is required to search for and measure higher priority layers shall be the same as that defined below for a lower or equal priority inter-frequency layers.

The UE shall be able to evaluate whether a newly detectable inter-frequency cell meets the re-selection criteria defined in TS 36.304 [6] within $K_{carrier} * T_{detect,EUTRAN_Inter}$ (as defined in table 4.2.2.4-1 of TS 36.133 [4] clause 4.2.2.4) if at least carrier frequency information is provided for inter-frequency neighbour cells by the serving cells when $Treselection_{EUTRAN}$ = 0 provided that the reselection criteria is met by a margin of at least 5dB for reselections based on ranking or 6dB for reselections based on absolute priorities or 4dB for RSRQ reselections based on absolute priorities. The parameter $K_{carrier}$ is the number of E-UTRA inter-frequency carriers indicated by the serving cell.

When higher priority cells are found by the higher priority search, they shall be measured at least every $T_{measure,EUTRAN_Inter}$. If, after it is found in a higher priority search, it is determined that reselection has not occurred then the UE is not required to continuously measure the detected cell to evaluate the ongoing possibility of re-selection. However, the minimum measurement filtering requirements specified later in this section shall still be met by the UE before it makes any determination that it may stop measuring the cell. If the UE detects on a E-UTRA carrier a cell whose physical identity is indicated as not allowed for that carrier in the measurement control system information of the serving cell, the UE is not required to perform measurements on that cell.

The UE shall measure RSRQ at least every $K_{carrier} * T_{measure,EUTRAN_Inter}$ as defined in table 4.2.2.4-1 of TS 36.133 [4] clause 4.2.2.4 for identified lower or equal priority inter-frequency cells. If the UE detects on an E-UTRA carrier a cell whose physical identity is indicated as not allowed for that carrier in the measurement control system information of the serving cell, the UE is not required to perform measurements on that cell.

The UE shall filter RSRQ measurements of each measured higher, lower and equal priority inter-frequency cell using at least 2 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least $T_{measure,EUTRAN_Inter} / 2$.

The UE shall not consider an E-UTRA neighbour cell in cell re-selection, if it is indicated as not allowed in the measurement control system information of the serving cell.

For an inter-frequency cell that has been already detected, but that has not been re-selected to, the filtering shall be such that the UE shall be capable of evaluating that the inter-frequency cell has met re-selection criterion defined in TS 36.304 [6] within $K_{carrier} * T_{evaluate,E-UTRAN_Inter}$ as defined in table 4.2.2.4-1 of TS 36.133 [4] clause 4.2.2.4 when $Treselection_{EUTRAN} = 0$ provides that the re-selection criteria is met by a margin of at least 5dB for re-selection based on ranking or 6dB for re-selection based on absolute priorities or 4dB for RSRQ reselections based on absolute priorities. When evaluating cells for reselection, the side conditions for RSRP and SCH apply to both serving and inter-frequency cells.

If $Treselection_{EUTRAN}$ timer has a non-zero value and the inter-frequency cell is better ranked than the serving cell, the UE shall evaluate this inter-frequency cell for the $Treselection_{EUTRAN}$ time. If this cell remains better ranked within this duration, then the UE shall re-select that cell.

The UE shall evaluate the inter-frequency cell re-selection criteria as defined in TS 36.304 [6] at least every DRX cycle. The DRX cycle length is 1.28 seconds.

At inter-frequency cell re-selection, the UE shall monitor the downlink of serving cell for paging reception until the UE is capable to start monitoring downlink channels of the target inter-frequency cell for paging reception. The interruption time shall not exceed $T_{SI-EUTRA}$ + 50 ms. $T_{SI-EUTRA}$ is the time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks as defined in TS 36.331 [5] for a E-UTRAN cell.

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.4 and A.4.2.8.

## 4.2.8.4 Test description

### 4.2.8.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

The UE's allowed CSG list is empty.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.27.
2. The general test parameter settings are set up according to Table 4.2.8.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.2.8.4.3.
5. There are two E-UTRA TDD carriers and two cells specified in the test and one non-allowed E-UTRA TDD CSG cell. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test and Cell 3 is the cell used for adding interference from non-allowed CSG cell.

**Table 4.2.8.4.1-1: General test parameters for TDD-TDD inter frequency cell re-selection test case with non-allowed CSG cell**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | UE shall be forced to cell 1 in the initialisation phase |
| Final condition | Active cell | | Cell2 | UE shall perform reselection to cell 2 during T2 |
| E-UTRA RF Channel Number | | | 1, 2 | Two TDD carrier frequencies are used. |
| Time offset between cells | | µs | 3 | Synchronous cells |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in 3GPP TS 36.211 |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in 3GPP TS 36.211 |
| PRACH configuration | | | 53 | As specified in table 5.7.1-3 in TS 36.211 |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | 15 | T1 need to be defined so that the non-allowed CSG cell is identified. |
| T2 | | s | 40 | T2 need to be defined so that cell re-selection reaction time is taken into account. |
| T3 | | s | 15 | T3 need to be defined so that whether cell re-selection would not occur is insured. |

### 4.2.8.4.2 Test procedure

The test consists of one active cell, one neighbour cell and one now-allowed CSG cell. The UE is requested to monitor the neighbouring cell on one of the E-UTRA TDD carriers. In the test there are four successive time periods, with time duration of T0, T1, T2 and T3 respectively. Cell 1is already identified by the UE prior to the start of the test. Cell 1 and Cell 2 belong to different tracking areas and Cell 3 is a non-allowed CSG cell. Furthermore, UE has not registered with network for the tracking area containing Cell 2.

In the following test procedure “UE responds” means “UE starts transmitting preamble on PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure according to TS 36.508 [7] clause 4.5A.2”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2. Set Cell 2 physical cell identity = initial cell 2 physical cell identity.

2. Set the parameters according to duration T0 in Table 4.2.8.5-1. If the UE is already camped in Cell 1, wait until T0 expires and skip to step 5.

3. The SS waits for random access requests information from the UE to perform cell re-selection on Cell 1.

4. If the UE has re-selected Cell 1 within T0, after the re-selection or when T0 expires, continue with step 5. Otherwise, if T0 expires and the UE has not yet re-selected Cell 1, skip to step 12.

5. Set the parameters according to duration T1 in Table 4.2.8.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

6. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for one iteration of the test procedure loop.

7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 4.2.8.5-2. T2 starts.

8. The SS waits for random access requests information from the UE to perform cell re-selection to a newly detectable cell, Cell 2.

9. If the UE has responded on the newly detected Cell 2 during time duration within 34 seconds from the beginning of time period T2 count a success for the event “Re-select newly detected Cell 2”, and after the re-selection, continue with step 10.
   Otherwise, if the UE has not re-selected Cell 2 count a fail for the event “Re-select newly detected Cell 2”, and skip to step 12.

10. After the re-selection or when T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 4.2.8.5-2. T3 starts.

11. If the UE has not re-selected Cell 1 when T3 expires, count a success for the event “Hold out on Cell 2”, and skip to step 13. Otherwise, if the UE responds on the known cell, Cell 1 during time duration T3, count a fail for the event “Hold out on Cell 2”, and continue with step 12.

12. Switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.

13. Repeat step 2-12 until a test verdict has been achieved.
    Each of the events “Re-select newly detected Cell 2” and “Hold out on Cell 2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. Different events may require different times for a verdict.
    If both events pass, the test passes. If one event fails, the test fails.

#### 4.2.8.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 4.2.8.4.3-1: Common Exception messages for E-UTRAN TDD – TDD Inter frequency case in the existence of non-allowed CSG cell**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.2-2 |

**Table 4.2.8.4.3-2: *SystemInformationBlockType1*: E-UTRAN TDD – TDD Inter frequency case in the existence of non-allowed CSG cell**

| Derivation Path: TS 36.508 [7] clause 4.4.3.2, Table 4.4.3.2-3 SystemInformationBlockType1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType1 ::= SEQUENCE { | | | |
| cellAccessRelatedInfo SEQUENCE { | | | |
| csg-Indication | TRUE | Cell 3 | |
| csg-Identity | Physical Cell ID of the Cell 3 | Cell 3 | |
| } | | | |
| nonCriticalExtension SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| cellSelectionInfo-v920 SEQUENCE { | | | |
| q-QualMin-r9 | -20 (-20dB) | Cell 1, 2, 3 | |
| q-QualMinOffset-r9 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 4.2.8.4.3-3: *SystemInformationBlockType3*: E-UTRAN TDD – TDD Inter frequency case in the existence of non-allowed CSG cell**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType3 ::= SEQUENCE { | | | |
| intraFreqCellReselectionInfo SEQUENCE { | | | |
| lateNonCriticalExtension { | | | |
| s-IntraSearch-v920 SEQUENCE {} | Not present | | |
| s-NonIntraSearch-v920 SEQUENCE { | | | |
| s-NonIntraSearchP-r9 | 25 | Cell 1 | |
| s-NonIntraSearchQ-r9 | 10 | Cell 1 | |
| } | | | |
| q-QualMin-r9 | -20 (-20dB) | Cell 1, 2, 3 | |
| threshServingLowQ-r9 | Not present | | |
| } | | | |
| } | | | |

**Table 4.2.8.4.3-4: *SystemInformationBlockType4*: E-UTRAN TDD – TDD Inter frequency case in the existence of non-allowed CSG cell**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-3 SystemInformationBlockType4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType4 ::= SEQUENCE { | | | |
| intraFreqNeighCellList SEQUENCE (SIZE (1..maxCellIntra)) OF SEQUENCE { | | | |
| physCellId | Physical Cell ID of the Cell 3 | Cell 1 | |
| | Physical Cell ID of the Cell 1 | Cell 3 | |
| } | | | |
| csg-PhysCellIdRange SEQUENCE { | | | |
| start | Physical Cell ID of the cell on which this SIB is transmitted | Cell 3 | |
| range | Not present | | |
| } | | | |
| } | | | |

**Table 4.2.8.4.3-5: *SystemInformationBlockType5*: E-UTRAN TDD – TDD Inter frequency case in the existence of non-allowed CSG cell**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-4 SystemInformationBlockType5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType5 ::= SEQUENCE { | | | |
| interFreqCarrierFreqList SEQUENCE (SIZE (1..maxFreq)) OF SEQUENCE { | | | |
| dl-CarrierFreq | Frequency of the Cell 2 | Cell 1, 3 | |
| | Frequency of the Cell 1 | Cell 2 | |
| q-RxLevMin | -70 (-140 dBm) | Cell 1, 2, 3 | |
| interFreqNeighCellList[*n*] SEQUENCE (SIZE (1..maxCellInter)) OF SEQUENCE { | | Cell 2 | |
| physCellId | Physical Cell ID of the Cell 1 | | |
| q-OffsetCell | dB0 | | |
| } | | | |
| q-QualMin-r9 | -20 (-20dB) | Cell 1, 2, 3 | |
| threshX-Q-r9 SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |

**Table 4.2.8.4.3-6: *SystemInformationBlockType9*: E-UTRAN TDD – TDD Inter frequency case in the existence of non-allowed CSG cell**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-8 SystemInformationBlockType9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType9 ::= SEQUENCE { | | | |
| hnb-Name | "3gppTest" | Cell 3 | |
| } | | | |

## 4.2.8.5 Test requirement

Tables 4.2.8.4.1-1, 4.2.8.5-1 and 4.2.8.5-2 defines the primary level settings including test tolerances for E-UTRAN TDD – TDD Inter frequency case in the existence of non-allowed CSG cell test case. Note that the time period for T0 is system implementation dependent.

**Table 4.2.8.5-1: Cell specific Test requirement parameters for FDD-FDD inter frequency cell re-selection test case with non-allowed CSG cell**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|
| | | T0 | | |
| E-UTRA RF Channel number | | 1 | 2 | 1 |
| $BW_{channel}$ | MHz | 10 | 10 | 10 |
| OCNG Patterns defined in D.2.2 | | OP.2 TDD | OP.2 TDD | OP.2 TDD |
| PBCH_RA | dB | 0 | 0 | 0 |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| Qrxlevmin | dBm | -140 | -140 | -140 |
| Qqualmin | dB | -20 | -20 | -20 |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | |
| RSRP Note 3 | dBm/15 kHz | -85.00 | -101.00 | -Infinity |
| RSRQ Note 3 | dB | -11.00 | -15.56 | -Infinity |
| $\hat{E}_s / I_{ot}$ | dB | 13.00 | -3.00 | -Infinity |
| $\hat{E}_s / N_{oc}$ | dB | 13.00 | -3.00 | -Infinity |
| Treselection | s | 0 | 0 | 0 |
| SnonintrasearchQ | dB | 10 | Not sent | Not sent |
| Propagation Condition | | AWGN | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and RSRQ and Es/Iot levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 4.2.8.5-2: Cell specific test parameters for TDD-TDD inter frequency cell re-selection test case with non-allowed CSG cell**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 (Non-allowed CSG cell) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | | 10 | | |
| OCNG Pattern defined in D..2.2 (OP.2 TDD) | | OP.2 TDD | | | OP.2 TDD | | | OP.2 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | |
| Qrxlevmin | dBm | -140 | | | -140 | | | -140 | | |
| Qqualmin | dB | -20 | | | | | | | | |
| $N_{oc}$ Note 2 | dBm/ 15kHz | -98 | | | | | | | | |
| RSRP Note 3 | dBm/ 15kHz | -90.20 | -90.00 | -85.00 | -Infinity | -85.00 | -90.00 | -90.00 | -85.00 | -60.00 |
| RSRQ Note 3 | dB | -14.24 | -17.15 | -35.81 | -Infinity | -11.00 | -11.43 | -14.04 | -12.15 | -10.81 |
| $\hat{E}_s / I_{ot}$ | dB | -0.84 | -5.21 | -25.00 | -Infinity | 13.00 | 8.00 | -0.47 | 4.36 | 24.79 |
| $\hat{E}_s / N_{oc}$ | dB | 7.80 | 8.00 | 13.00 | -Infinity | 13.00 | 8.00 | 8.00 | 13.00 | 38.00 |
| Treselection | S | 0 | | | 0 | | | 0 | | |
| SnonintrasearchQ | dB | 10 | | | Not sent | | | Not sent | | |
| Propagation Condition | | AWGN | | | | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and RSRQ levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The cell reselection delay to a newly detectable cell is defined as the time from the beginning of time period T2, to the moment when the UE camps on Cell 2, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 2.

The cell re-selection delay to a newly detectable cell test requirement in this case is expressed as:

Cell re-selection delay to a newly detectable cell = $T_{evaluate,\ E\text{-}UTRAN_Intra} + T_{SI\text{-}EUTRA}$

$T_{evaluate,E\text{-}UTRAN_Inter}$ = 32 s; as specified in TS 36.133 [4] clause 4.2.2.4.

$T_{SI\text{-}EUTRA}$ = 1280 ms; as specified in TS 36.133 [4] clause 4.2.2.4.

The cell re-selection delay to a newly detectable cell shall be less than a total of 34 seconds in this test case (note: this gives a total of 33.28 s, but the test allows 34 seconds.)

The probability of re-selection from Cell 1 to Cell 2 during T2 observed during testing shall be at least 90%.

The probability of re-selection from Cell 2 to Cell 1 during T3 observed during testing shall be less than 10%.

For the test to pass, both events above shall pass with a confidence level of 95%.

## 4.2.9 E-UTRAN FDD - FDD intra frequency cell re-selection case for 5MHz bandwidth

### 4.2.9.1 Test purpose

Same test purpose as in clause 4.2.1.1.

### 4.2.9.2 Test applicability

This test applies to all types of E-UTRA FDD UE Release 8 and forward and only support E-UTRA Band 31.

### 4.2.9.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 4.2.1.3 with the following exceptions:

- Instead of A.4.2.1 → use A.4.2.9.

### 4.2.9.4 Test description

#### 4.2.9.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.
2. The general test parameter settings are set up according to Table 4.2.9.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.2.9.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 4.2.9.4.1-1: General Test Parameters for E-UTRAN FDD-FDD intra cell re-selection test case for 5MHz bandwidth**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 | |
| Note 1: See Table 4.2.1.4.1-1 for the other parameters.<br>Note 2: This is according to the principle defined in TS 36.133 [4] section A.3.7.2. | | | |

#### 4.2.9.4.2 Test procedure

Same test procedure as in clause 4.2.1.4.2 with the following exceptions:

- Instead of Table 4.2.1.5-1 → use Table 4.2.9.5-1.

#### 4.2.9.4.3 Message contents

Same message contents as in clause 4.2.1.4.3.

### 4.2.9.5 Test requirement

Same test requirement as in clause 4.2.1.5 with the following exceptions:

Tables 4.2.9.4.1-1 and 4.2.9.5-1 defines the primary level settings including test tolerances for E-UTRAN FDD-FDD intra-frequency cell re-selection test case.

**Table 4.2.9.5-1: Cell specific test parameters for FDD intra frequency cell reselection test case in AWGN for 5MHz**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| $BW_{channel}$ | MHz | 5 | | | 5 | | |
| OCNG Patterns defined in D.1.16 (OP.16 FDD) | | OP.16 FDD | | | OP.16 FDD | | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: See Table 4.2.1.5-1 for the other parameters. | | | | | | | |

## 4.2.12 E-UTRAN FDD – FDD Intra frequency case for Cat-M1 UE in normal coverage

### 4.2.12.1 Test purpose

To verify for a category M1 UE that when the current and target cell operates on the same carrier frequency in normal coverage the UE is able to search and measure cells to meet the respective intra-frequency cell re-selection requirements.

### 4.2.12.2 Test applicability

This test applies to all types of E-UTRA FD-FDD UE release 13 and forward of UE category M1.

### 4.2.12.3 Minimum conformance requirements

The cell re-selection delay shall be less than $T_{evaluate\ E\text{-}UTRAN_Intra} + T_{SI\text{-}EUTRA}$ in RRC_IDLE state.

The CAT-M1 UE shall be able to identify new intra-frequency cells and perform RSRP measurement of identified intra-frequency cells without an explicit intra-frequency neighbour list containing physical layer cell identities.

The CAT-M1 UE shall be able to evaluate whether a newly detectable intra-frequency cell meets the re-selection criteria defined in TS 36.304 [6] within $T_{detect,\ EUTRAN_Intra}$ as defined in table 4.2.2.3-1 of TS 36.133 [4] clause 4.2.2.3 when Treselection = 0 provided that the cell is at least 3 dB better ranked.

The CAT-M1 UE shall measure RSRP at least every $T_{measure,EUTRAN_Intra}$ as defined in table 4.2.2.3-1 of TS 36.133 [4] clause 4.2.2.3 for intra-frequency cells that are identified and measured according to the measurement rules.

The CAT-M1 UE shall filter RSRP measurement of each measured intra-frequency cell using at least 2 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least $T_{measure,EUTRAN_Intra} / 2$.

For an intra-frequency cell that has been already detected, but that has not been re-selected to, the filtering shall be such that the CAT-M1 UE shall be capable of evaluating that the intra-frequency cell has met the re-selection criterion defined in TS 36.304 [6] within $T_{evaluateFDD,Intra}$ as defined in table 4.2.2.3-1 of TS 36.133 [4] clause 4.2.2.3 when Treselection = 0 provided that the cell is at least 3 dB better ranked. When evaluating cells for re-selection, the side conditions are RSRP and SCh apply to both serving and non-serving intra-frequency cells.

If Treselection timer has a non-zero value and the intra-frequency cell is better ranked than the serving cell, the CAT-M1 UE shall evaluate this intra-frequency cell for the Treselection time. If this cell remains better ranked within this duration, then the CAT-M1 UE shall re-select that cell.

The CAT-M1 UE shall evaluate the intra-frequency cell re-selection criteria as defined in TS 36.304 [6] at least every DRX cycle. The DRX cycle length is 1.28 seconds. When a non-zero value of Treselection is used, the CAT-M1 UE shall only perform re-selection on an evaluation which occurs simultaneously to, or later than the expiry of the Treselection timer.

At intra-frequency cell re-selection, the CAT-M1 UE shall monitor the downlink of serving cell for paging reception until the CAT-M1 UE is capable to start monitoring downlink channels of the target intra-frequency cell for paging reception. The interruption time shall not exceed $T_{SI-EUTRA}$ + 50 ms. $T_{SI-EUTRA}$ is the time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks as defined in TS 36.331 [5] for a E-UTRAN cell.

For idle mode cell re-selection purposes, the CAT-M1 UE shall be capable of monitoring at least:

- Intra-frequency carrier

In addition to the requirements defined in TS 36.133 [4] clause 4.2.2.9 a CAT-M1 UE in RRC_IDLE state shall be capable of monitoring a total of at least 8 carrier frequency layers, which includes serving layer, comprising of any defined in TS 36.133 [4] clause 4.2.2.9 combination of E-UTRA FDD, E-UTRA TDD, UTRA FDD, UTRA TDD, GSM (one GSM layer corresponds to 32 cells), cdma2000 1x and HRPD layers.

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.3 and A.4.2.12.

## 4.2.12.4 Test description

### 4.2.12.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.
2. The general test parameter settings are set up according to Table 4.2.12.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.2.12.4.3.
5. There is one E-UTRA FD-FDD cell (Cell 1) and one E-UTRA FDD cell (Cell 2) specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 4.2.12.4.1-1: General test parameters for FDD intra frequency cell reselection test case for Cat-M1 UE in normal coverage**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2 | |
| T2 end condition | Active cell | | Cell2 | |
| | Neighbour cells | | Cell1 | |
| Final condition | Visited cell | | Cell1 | |
| E-UTRA RF Channel Number | | | 1 | Only one FDD carrier frequency is used. |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| PRACH Configuration | | | PRACH_2CE | Refer to A.9 |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | >7 | During T1, Cell 2 shall be powered off, and during the off time the physical cell identity shall be changed, The intention is to ensure that Cell 2 has not been detected by the UE prior to the start of period T2 |
| T2 | | s | 40 | T2 need to be defined so that cell re-selection reaction time is taken into account. |
| T3 | | s | 15 | T3 need to be defined so that cell re-selection reaction time is taken into account. |

### 4.2.12.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The CAT-M1 UE is requested to monitor the neighbouring cell on one E-UTRA FDD carrier. In the test there are three successive time periods, with time duration of T1, T2 and T3 respectively. Only Cell 1 is already identified by the CAT-M1 UE prior to the start of the test, i.e. Cell 2 is not identified by the UE prior to the start of the test. Cell 1 and Cell 2 belong to different tracking areas. Furthermore, the CAT-M1 UE has not registered with network for the tracking area containing Cell 2.

In the following test procedure “UE responds” means “UE starts transmitting preamble on PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure according to TS 36.508 [7] clause 4.5A.2”.

1. Ensure the CAT-M1 UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2. Set Cell 2 physical cell identity = initial cell 2 physical cell identity.

2. Set the parameters according to T1 in Table 4.2.12.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for one iteration of the test procedure loop.

4. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 4.2.12.5-1.

5. The SS waits for random access requests information from the UE to perform cell re-selection to a newly detectable cell, Cell 2.

6. If the CAT-M1 UE responds on the newly detectable cell, Cell 2 during time duration T2 within 34 seconds from the beginning of time period T2, then count a success for the event “Re-select newly detected Cell 2”. Otherwise count a fail for the event “Re-select newly detected Cell 2”.

7. If the CAT-M1 UE has re-selected Cell 2 within T2, after the re-selection or when T2 expires, continue with step 8. Otherwise, if T2 expires and the CAT-M1 UE has not yet re-selected Cell 2, skip to step 12.

8. The SS shall switch the power setting from T2 to T3 as specified in Table 4.2.12.5-1.

9. The SS waits for random access requests information from the CAT-M1 UE to perform cell re-selection to an already detected cell, Cell 1.

10. If the CAT-M1 UE responds on the already detected cell, Cell 1 during time duration T3 within 8 seconds from the beginning of time period T3, then count a success for the event “Re-select already detected Cell 1”. Otherwise count a fail for the event “Re-select already detected Cell 1”.

11. If the CAT-M1 UE has re-selected Cell 1 within T3, after the re-selection or when T3 expires, skip to step 13. Otherwise, if T3 expires and the CAT-M1 UE has not yet re-selected Cell 1, continue with step 12.

12. Switch off and on the CAT-M1 UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.

13. Repeat step 2-12 until a test verdict has been achieved.
    Each of the events “Re-select newly detected Cell 2” and “Re-select already detected Cell 1” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. Different events may require different times for a verdict.
    If both events pass, the test passes. If one event fails, the test fails.

#### 4.2.12.4.3 Message contents

Message contents are according to 3GPP TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 4.2.12.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra cell re-selection test case**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.1-1<br>Table H.2.1-2 |

**Table 4.2.12.4.3-2: *PRACH-Config-DEFAULT*: E-UTRAN FDD cell re-selection and handover**

| Derivation Path: 3GPP TS 36.508 [7] clause 4.6.3, Table 4.6.3-7A PRACH-Config-v1310-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| prach-ParametersListCE-r13 SEQUENCE { | | | |
| prach-ConfigIndex-r13[1] | 4 | | |
| } | | | |

### 4.2.12.5 Test requirement

Tables 4.2.12.4.1-1 and 4.2.12.5-1 defines the primary level settings including test tolerances for E-UTRAN FDD – FDD Intra frequency case for Cat-M1 cell re-selection test case.

**Table 4.2.12.5-1: Cell specific test parameters for FDD intra frequency cell reselection test case for Cat-M1 UE in normal coverage**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | |
| OCNG Patterns | | OP.21 FDD | | | OP.21 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| MPDCCH_RA | dB | | | | | | |
| MPDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| Qrxlevmin | dBm | -140 | -140 | -140 | -140 | -140 | -140 |
| Pcompensation | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $Qhyst_s$ | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $Qoffset_{s,n}$ | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| Cell_selection_and_reselection_quality_measurement | | RSRP | | | RSRP | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 16 | 13 | 16.45 | -infinity | 16.45 | 13 |
| $\hat{E}_s / I_{ot}$ | dB | 16 | -3.55 | 3.24 | -infinity | 3.24 | -3.55 |
| RSRP Note3 | dBm/15 kHz | -82 | -85 | -81.55 | -infinity | -81.55 | -85 |
| Treselection | S | 0 | 0 | 0 | 0 | 0 | 0 |
| Sintrasearch | dB | Not sent | | | Not sent | | |
| Propagation Condition | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x1 | | | 1x1 | | |
| Timing offset to Cell 1 | ms | - | | | 3 | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The cell reselection delay to a newly detectable cell is defined as the time from the beginning of time period T2, to the moment when the UE camps on Cell 2, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 2.

The cell re-selection delay to a newly detectable cell shall be less than 34 s.

The cell reselection delay to an already detected cell is defined as the time from the beginning of time period T3, to the moment when the UE camps on Cell 1, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 1.

The cell re-selection delay to an already detected cell shall be less than 8 s.

The rate of correct cell reselections observed during repeated tests shall be at least 90%.

NOTE: The cell re-selection delay to a newly detectable cell can be expressed as: $T_{detect,EUTRAN_Intra} + T_{SI}$, and to an already detected cell can be expressed as: $T_{evaluate,EUTRAN_Intra} + T_{SI}$,

Where:

$T_{detect,EUTRAN_Intra}$ See Table 4.2.2.3-1 in TS 36.133 [4] clause 4.2.2.3

$T_{evaluate,EUTRAN_Intra}$ See Table 4.2.2.3-1 in TS 36.133 [4] clause 4.2.2.3

$T_{SI}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell; 1280 ms is assumed in this test case.

This gives a total of 33.28 s, allow 34 s for the cell re-selection delay to a newly detectable cell and 7.68 s, allow 8 s for the cell re-selection delay to an already detected cell in the test case.

## 4.2.13 E-UTRAN HD – FDD Intra frequency case for Cat-M1 UE in normal coverage

### 4.2.13.1 Test purpose

To verify for a category M1 UE that when the current and target cell operates on the same carrier frequency in normal coverage the UE is able to search and measure cells to meet the respective intra-frequency cell re-selection requirements.

### 4.2.13.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 13 and forward of UE category M1.

### 4.2.13.3 Minimum conformance requirements

The cell re-selection delay shall be less than $T_{evaluate\ E\text{-}UTRAN_Intra} + T_{SI\text{-}EUTRA}$ in RRC_IDLE state.

The CAT-M1 UE shall be able to identify new intra-frequency cells and perform RSRP measurement of identified intra-frequency cells without an explicit intra-frequency neighbour list containing physical layer cell identities.

The CAT-M1 UE shall be able to evaluate whether a newly detectable intra-frequency cell meets the re-selection criteria defined in TS 36.304 [6] within $T_{detect,\ EUTRAN_Intra}$ as defined in table 4.2.2.3-1 of TS 36.133 [4] clause 4.2.2.3 when Treselection = 0 provided that the cell is at least 3 dB better ranked.

The CAT-M1 UE shall measure RSRP at least every $T_{measure,EUTRAN_Intra}$ as defined in table 4.2.2.3-1 of TS 36.133 [4] clause 4.2.2.3 for intra-frequency cells that are identified and measured according to the measurement rules.

The CAT-M1 UE shall filter RSRP measurement of each measured intra-frequency cell using at least 2 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least $T_{measure,EUTRAN_Intra} / 2$.

For an intra-frequency cell that has been already detected, but that has not been re-selected to, the filtering shall be such that the CAT-M1 UE shall be capable of evaluating that the intra-frequency cell has met the re-selection criterion defined in TS 36.304 [6] within $T_{evaluateFDD,Intra}$ as defined in table 4.2.2.3-1 of TS 36.133 [4] clause 4.2.2.3 when Treselection = 0 provided that the cell is at least 3 dB better ranked. When evaluating cells for re-selection, the side conditions are RSRP and SCh apply to both serving and non-serving intra-frequency cells.

If Treselection timer has a non-zero value and the intra-frequency cell is better ranked than the serving cell, the CAT-M1 UE shall evaluate this intra-frequency cell for the Treselection time. If this cell remains better ranked within this duration, then the CAT-M1 UE shall re-select that cell.

The CAT-M1 UE shall evaluate the intra-frequency cell re-selection criteria as defined in TS 36.304 [6] at least every DRX cycle. The DRX cycle length is 1.28 seconds. When a non-zero value of Treselection is used, the CAT-M1 UE shall only perform re-selection on an evaluation which occurs simultaneously to, or later than the expiry of the Treselection timer.

At intra-frequency cell re-selection, the CAT-M1 UE shall monitor the downlink of serving cell for paging reception until the CAT-M1 UE is capable to start monitoring downlink channels of the target intra-frequency cell for paging reception. The interruption time shall not exceed $T_{SI\text{-}EUTRA}$ + 50 ms. $T_{SI\text{-}EUTRA}$ is the time required for receiving all the

relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks as defined in TS 36.331 [5] for a E-UTRAN cell.

For idle mode cell re-selection purposes, the CAT-M1 UE shall be capable of monitoring at least:

- Intra-frequency carrier

In addition to the requirements defined in TS 36.133 [4] clause 4.2.2.9 a CAT-M1 UE in RRC_IDLE state shall be capable of monitoring a total of at least 8 carrier frequency layers, which includes serving layer, comprising of any defined in TS 36.133 [4] clause 4.2.2.9 combination of E-UTRA FDD, E-UTRA TDD, UTRA FDD, UTRA TDD, GSM (one GSM layer corresponds to 32 cells), cdma2000 1x and HRPD layers.

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.3 and A.4.2.13.

## 4.2.13.4 Test description

### 4.2.13.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.
2. The general test parameter settings are set up according to Table 4.2.12.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.2.12.4.3
5. There is one E-UTRA HD-FDD cell (Cell 1) and one E-UTRA FDD cell (Cell 2) specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 4.2.13.4.1-1: General test parameters for HD-FDD intra frequency cell reselection test case for Cat-M1 UE in normal coverage**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2 | |
| T2 end condition | Active cell | | Cell2 | |
| | Neighbour cells | | Cell1 | |
| Final condition | Visited cell | | Cell1 | |
| E-UTRA RF Channel Number | | | 1 | Only one carrier frequency is used. |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| PRACH Configuration | | | PRACH_2CE | Refer to A.9 |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | >7 | During T1, Cell 2 shall be powered off, and during the off time the physical cell identity shall be changed, The intention is to ensure that Cell 2 has not been detected by the UE prior to the start of period T2 |
| T2 | | s | 40 | T2 need to be defined so that cell re-selection reaction time is taken into account. |
| T3 | | s | 15 | T3 need to be defined so that cell re-selection reaction time is taken into account. |

#### 4.2.13.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The CAT-M1 UE is requested to monitor the neighbouring cell on one E-UTRA FDD carrier. In the test there are three successive time periods, with time duration of T1, T2 and T3 respectively. Only Cell 1 is already identified by the CAT-M1 UE prior to the start of the test, i.e. Cell 2 is not identified by the UE prior to the start of the test. Cell 1 and Cell 2 belong to different tracking areas. Furthermore, the CAT-M1 UE has not registered with network for the tracking area containing Cell 2.

In the following test procedure "UE responds" means "UE starts transmitting preamble on PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure according to TS 36.508 [7] clause 4.5A.2".

1. Ensure the CAT-M1 UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2. Set Cell 2 physical cell identity = initial cell 2 physical cell identity.

2. Set the parameters according to T1 in Table 4.2.13.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for one iteration of the test procedure loop.

4. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 4.2.13.5-1.

5. The SS waits for random access requests information from the UE to perform cell re-selection to a newly detectable cell, Cell 2.

6. If the CAT-M1 UE responds on the newly detectable cell, Cell 2 during time duration T2 within 34 seconds from the beginning of time period T2, then count a success for the event "Re-select newly detected Cell 2". Otherwise count a fail for the event "Re-select newly detected Cell 2".

7. If the CAT-M1 UE has re-selected Cell 2 within T2, after the re-selection or when T2 expires, continue with step 8. Otherwise, if T2 expires and the CAT-M1 UE has not yet re-selected Cell 2, skip to step 12.

8. The SS shall switch the power setting from T2 to T3 as specified in Table 4.2.13.5-1.

9. The SS waits for random access requests information from the CAT-M1 UE to perform cell re-selection to an already detected cell, Cell 1.

10. If the CAT-M1 UE responds on the already detected cell, Cell 1 during time duration T3 within 8 seconds from the beginning of time period T3, then count a success for the event "Re-select already detected Cell 1". Otherwise count a fail for the event "Re-select already detected Cell 1".

11. If the CAT-M1 UE has re-selected Cell 1 within T3, after the re-selection or when T3 expires, skip to step 13. Otherwise, if T3 expires and the CAT-M1 UE has not yet re-selected Cell 1, continue with step 12.

12. Switch off and on the CAT-M1 UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.

13. Repeat step 2-12 until a test verdict has been achieved.
    Each of the events "Re-select newly detected Cell 2" and "Re-select already detected Cell 1" is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. Different events may require different times for a verdict.
    If both events pass, the test passes. If one event fails, the test fails.

#### 4.2.13.4.3 Message contents

Message contents are according to 3GPP TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 4.2.12.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra cell re-selection test case**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.1-1<br>Table H.2.1-2 |

**Table 4.2.12.4.3-2: *PRACH-Config-DEFAULT*: E-UTRAN FDD cell re-selection and handover**

| Derivation Path:3GPP TS 36.508 [7] clause 4.6.3, Table 4.6.3-7A PRACH-Config-v1310-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| prach-ParametersListCE-r13 SEQUENCE { | | | |
| prach-ConfigIndex-r13[1] | 4 | | |
| } | | | |

## 4.2.14 E-UTRAN TDD – TDD Intra frequency case for Cat-M1 UE in normal coverage

### 4.2.14.1 Test purpose

To verify for a category M1 UE that when the current and target cell operates on the same carrier frequency in normal coverage the UE is able to search and measure cells to meet the respective intra-frequency cell re-selection requirements.

### 4.2.14.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 13 and forward of UE category M1.

### 4.2.14.3 Minimum conformance requirements

The cell re-selection delay shall be less than $T_{evaluate,E\text{-}UTRAN_Intra} + T_{SI\text{-}EUTRA}$ in RRC_IDLE state.

The Cat-M1 UE shall be able to identify new intra-frequency cells and perform RSRP measurement of identified intra-frequency cells without an explicit intra-frequency neighbour list containing physical layer cell identities.

The Cat-M1 UE shall be able to evaluate whether a newly detectable intra-frequency cell meets the re-selection criteria defined in TS 36.304 [6] within $T_{detect,EUTRAN_Intra}$ as defined in table 4.2.2.3-1 of TS 36.133 [4] clause 4.2.2.3 when Treselection = 0.

The Cat-M1 UE shall measure RSRP at least every $T_{measure,EUTRAN_Intra}$ as defined in table 4.2.2.3-1 of TS 36.133 [4] clause 4.2.2.3 for intra-frequency cells that are identified and measured according to the measurement rules.

The Cat-M1 UE shall filter RSRP measurements of each measured intra-frequency cell using at least 2 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least $T_{measure,EUTRAN_Intra} / 2$.

For an intra-frequency cell that has been already detected, but that has not been re-selected to, the filtering shall be such that the Cat-M1 UE shall be capable of evaluating that the intra-frequency cell has met the re-selection criterion defined in TS 36.304 [6] within $T_{evaluate,E\text{-}UTRAN_Intra}$ as defined in table 4.2.2.3-1 of TS 36.133 [4] clause 4.2.2.3 when Treselection = 0 provided that the cell is at least 3 dB better ranked. When evaluating cells for reselection, the side conditions for RSRP and SCH apply to both serving and non-serving intra-frequency cells.

If Treselection timer has a non-zero value and the intra-frequency cell is better ranked than the serving cell, the Cat-M1 UE shall evaluate this intra-frequency cell for the Treselection time. If this cell remains better ranked within this duration, then the Cat-M1 UE shall re-select that cell.

The Cat-M1 UE shall evaluate the intra-frequency cell re-selection criteria as defined in TS 36.304 [6] at least every DRX cycle. The DRX cycle length is 1.28 seconds.

At intra-frequency cell re-selection, the UE shall monitor the downlink of serving cell for paging reception until the UE is capable to start monitoring downlink channels of the target intra-frequency cell for paging reception. The interruption time shall not exceed $T_{SI-EUTRA}$ + 50 ms. $T_{SI-EUTRA}$ is the time required for receiving all the relevant system information data according to the reception procedure and the RRC procedure delay of system information blocks as defined in TS 36.331 [5] for a E-UTRAN cell.

For idle mode cell re-selection purposes, the Cat-M1 UE shall be capable of monitoring at least:

- Intra-frequency carrier

In addition to the requirements defined in TS 36.133 [4] clause 4.2.2.9 a Cat-M1 UE in RRC_IDLE state shall be capable of monitoring a total of at least 8 carrier frequency layers, which includes serving layer, comprising of any defined in TS 36.133 [4] clause 4.2.2.9 combination of E-UTRA FDD, E-UTRA TDD, UTRA FDD, UTRA TDD, GSM (one GSM layer corresponds to 32 cells), cdma2000 1x and HRPD layers.

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.3 and A.4.2.14.

## 4.2.14.4 Test description

### 4.2.14.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20(using only main UE Tx/Rx antenna).
2. The general test parameter settings are set up according to Table 4.2.14.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.2.14.4.3.
5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for registration with the power level set according to Table clause C.0 and C.1 for this test.

**Table 4.2.14.4.1-1: General test parameters for TDD intra frequency cell re-selection test case for Cat-M1 UE**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2 | |
| T2 end condition | Active cell | | Cell2 | |
| | Neighbour cells | | Cell1 | |
| Final condition | Visited cell | | Cell1 | |
| E-UTRA RF Channel Number | | | 1 | Only one TDD carrier frequency is used. |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211 |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211 |
| PRACH Configuration | | | PRACH 2CE | Refer to A.9 |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | >7 | During T1, Cell 2 shall be powered off, and during the off time the physical cell identity shall be changed, The intention is to ensure that Cell 2 has not been detected by the UE prior to the start of period T2 |
| T2 | | s | 40 | T2 need to be defined so that cell re-selection reaction time is taken into account. |
| T3 | | s | 15 | T3 need to be defined so that cell re-selection reaction time is taken into account. |

### 4.2.14.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The UE is requested to monitor the neighbouring cell on one E-UTRA TDD carrier. In the test there are three successive time periods, with time duration of T1, T2, and T3 respectively. Only Cell 1 is already identified by the UE prior to the start of the test. Cell 1 and Cell 2 belong to different tracking areas. Furthermore, UE has not registered with network for the tracking area containing Cell 2.

In the following test procedure “UE responds” means “UE starts transmitting preamble on PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure according to TS 36.508 [7] clause 4.5A.2”

1. Ensure the UE is in State 2A-RF-CE according to TS 36.508 [7] clause 7.2A.2. Set Cell 2 physical cell identity = initial cell 2 physical cell identity.

2. Set the parameters according to T1 in Table 4.2.14.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for one iteration of the test procedure loop.

4. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 4.2.14.5-1.

5. The SS waits for random access requests information from the UE to perform cell re-selection to a newly detectable cell, Cell 2.

6. If the Cat-M1 UE responds on the newly detectable cell, Cell 2, during time duration T2 within 34 seconds from the beginning of time period T2, then count a success for the event “Re-select newly detected Cell 2”. Otherwise count a fail for the event “Re-select newly detected Cell 2”.

7. If the Cat-M1 UE has re-selected Cell 2 within T2, after the re-selection or when T2 expires, continue with step 8.
   Otherwise, if T2 expires and the Cat-M1 UE has not yet re-selected Cell 2, skip to step 12.

8. The SS shall switch the power setting from T2 to T3 as specified in Table 4.2.14.5-1.

9. The SS waits for random access requests information from the UE to perform cell re-selection to an already detected cell, Cell 1.

10. If the Cat-M1 UE responds on the already detected cell, Cell 1, during time duration T3 within 8 seconds from the beginning of time period T3, then count a success for the event “Re-select already detected Cell 1”. Otherwise count a fail for the event “Re-select already detected Cell 1”.

11. If the Cat-M1 UE has re-selected Cell 1 within T3, after the re-selection or when T3 expires, skip to step 13. Otherwise, if T3 expires and the Cat-M1 UE has not yet re-selected Cell 1, continue with step 12.

12. Switch off and on the UE and ensure the UE is in State 2A-RF-CE according to TS 36.508 [7] clause 7.2A.2A in Cell 1.

13. Repeat step 2-12 until a test verdict has been achieved.
    Each of the events “Re-select newly detected Cell 2” and “Re-select already detected Cell 1” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. Different events may require different times for a verdict.
    If both events pass, the test passes. If one event fails, the test fails.

#### 4.2.14.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 4.2.14.4.3-1: Common Exception messages for E-UTRAN TDD-TDD intra cell re-selection test case**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.1-1<br>Table H.2.1-2 |

**Table 4.2.14.4.3-2: *PRACH-Config-v1310-DEFAULT*: E-UTRAN TDD cell re-selection and handover**

| Derivation Path: 36.508 clause 4.6.3 Table 4.6.3-7A | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PRACH-Config-v1310-DEFAULT ::= SEQUENCE { | | | |
| rsrp-ThresholdsPrachInfoList-r13 SEQUENCE (SIZE(1..3)) OF { | 1 entry | | |
| RSRP-Range[1] | 42 | -99dBm | |
| } | | | |
| PRACH-Config-v1310-DEFAULT ::= SEQUENCE { | | | |
| prach-ParametersListCE-r13 SEQUENCE (SIZE(1..maxCE-Level-r13)) OF SEQUENCE { | 2 entries | | |
| PRACH-ParametersCE-r13[1] SEQUENCE { | | | |
| prach-ConfigIndex-r13[1] | 53 | INTEGER (0..63) | TDD |
| prach-FreqOffset-r13[1] | Same as prach-FrequencyOffset | | |
| prach-StartingSubframe-r13[1] | sf2 | | |
| maxNumPreambleAttemptCE-r13[1] | n5 | | |
| numRepetitionPerPreambleAttempt-r13[1] | n1 | | |
| mpdcch-NarrowbandsToMonitor-r13 SEQUENCE (SIZE(1..2)) OF INTEGER (1..maxAvailNarrowBands-r13)[1] { | 1 entry | | |
| mpdcch-NarrowbandsToMonitor-r13[1] | 1 | | |
| } | | | |
| mpdcch-NumRepetition-RA-r13[1] | r1 | | |
| prach-HoppingConfig-r13[1] | off | | |
| } | | | |
| prach-ConfigIndex-r13[2] | 53 | INTEGER (0..63) | TDD |
| prach-FreqOffset-r13[2] | Same as prach-FrequencyOffset | | |
| prach-StartingSubframe-r13[2] | sf128 | | |
| maxNumPreambleAttemptCE-r13[2] | n5 | | |
| numRepetitionPerPreambleAttempt-r13[2] | n1 | | |
| mpdcch-NarrowbandsToMonitor-r13 SEQUENCE (SIZE(1..2)) OF INTEGER (1..maxAvailNarrowBands-r13)[2] { | 1 entry | | |
| mpdcch-NarrowbandsToMonitor-r13[1] | 1 | | |
| } | | | |
| mpdcch-NumRepetition-RA-r13[2] | r4 | | |
| prach-HoppingConfig-r13[2] | off | | |
| } | | | |
| } | | | |
| } | | | |

## 4.2.14.5 Test requirement

Tables 4.2.14.4.1-1, and 4.2.14.5-1 defines the primary level settings including test tolerances for E-UTRAN TDD-TDD intra frequency case for Cat-M1 cell re-selection test case.

**Table 4.2.14.5-1: Cell specific test parameters for TDD intra frequency cell re-selection test case for Cat-M1 UE in AWGN in normal coverage**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | |
| OCNG Patterns | | OP.2 TDD | | | OP.2 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| MPDCCH_RA | dB | | | | | | |
| MPDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| Qrxlevmin | dBm | -140 | -140 | -140 | -140 | -140 | -140 |
| Pcompensation | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $Qhyst_s$ | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $Qoffset_{s,n}$ | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| Cell_selection_and_reselection_quality_measurement | | RSRP | | | RSRP | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 16.00 | 13.00 | 16.45 | -infinity | 16.45 | 13.00 |
| $\hat{E}_s / I_{ot}$ | dB | 16.00 | -3.55 | 3.24 | -infinity | 3.24 | -3.55 |
| RSRP Note3 | dBm/15 kHz | -82.00 | -85.00 | -81.55 | -infinity | -81.55 | -85.00 |
| Treselection | s | 0 | 0 | 0 | 0 | 0 | 0 |
| Sintrasearch | dB | Not sent | | | Not sent | | |
| Propagation Condition | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x1 | | | 1x1 | | |
| Timing offset to Cell 1 | μs | - | | | 3 | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The cell reselection delay to a newly detectable cell is defined as the time from the beginning of time period T2, to the moment when the UE camps on Cell 2, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 2.

The cell re-selection delay to a newly detectable cell shall be less than 34 s.

The cell reselection delay to an already detected cell is defined as the time from the beginning of time period T3, to the moment when the UE camps on cell 1, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on cell 1.

The cell re-selection delay shall be less than 8 s.

The rate of correct cell reselections observed during repeated tests shall be at least 90%.

NOTE: The cell re-selection delay to a newly detectable cell can be expressed as: $T_{detect,EUTRAN_Intra} + T_{SI\text{-}EUTRA}$, and to an already detected cell can be expressed as: $T_{evaluate,\ E\text{-}UTRAN_\ intra} + T_{SI\text{-}EUTRA}$,

Where:

$T_{detect,EUTRAN_Intra}$ See Table 4.2.2.3-1 in clause 4.2.2.3 of TS 36.133 [4]

$T_{evaluate,EUTRAN_intra}$ See Table 4.2.2.3-1 in clause 4.2.2.3 of TS 36.133 [4]

$T_{SI-EUTRA}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell; 1280 ms is assumed in this test case.

This gives a total of 33.28 s, allow 34 s for the cell re-selection delay to a newly detectable cell and 7.68 s, allow 8 s for the cell re-selection delay to an already detected cell in the test case.

### 4.2.13.5 Test requirement

Tables 4.2.13.4.1-1 and 4.2.13.5-1 defines the primary level settings including test tolerances for E-UTRAN HD FDD – FDD Intra frequency case for Cat-M1 cell re-selection test case.

**Table 4.2.13.5-1: Cell specific test parameters for HD-FDD intra frequency cell reselection test case for Cat-M1 UE in normal coverage**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | |
| OCNG Patterns | | OP.21 FDD | | | OP.21 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| MPDCCH_RA | dB | | | | | | |
| MPDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| Qrxlevmin | dBm | -140 | -140 | -140 | -140 | -140 | -140 |
| Pcompensation | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $Qhyst_s$ | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $Qoffset_{s,n}$ | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| Cell_selection_and_reselection_quality_measurement | | RSRP | | | RSRP | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 16 | 13 | 16.45 | -infinity | 16.45 | 13 |
| $\hat{E}_s / I_{ot}$ | dB | 16 | -3.55 | 3.24 | -infinity | 3.24 | -3.55 |
| RSRP Note3 | dBm/15 kHz | -82 | -85 | -81.55 | -infinity | -81.55 | -85 |
| Treselection | s | 0 | 0 | 0 | 0 | 0 | 0 |
| Sintrasearch | dB | Not sent | | | Not sent | | |
| Propagation Condition | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x1 | | | 1x1 | | |
| Timing offset to Cell 1 | ms | - | | | 3 | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The cell reselection delay to a newly detectable cell is defined as the time from the beginning of time period T2, to the moment when the UE camps on Cell 2, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 2.

The cell re-selection delay to a newly detectable cell shall be less than 34 s.

The cell reselection delay to an already detected cell is defined as the time from the beginning of time period T3, to the moment when the UE camps on Cell 1, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 1.

The cell re-selection delay to an already detected cell shall be less than 8 s.

The rate of correct cell reselections observed during repeated tests shall be at least 90%.

NOTE: The cell re-selection delay to a newly detectable cell can be expressed as: $T_{detect,EUTRAN_Intra} + T_{SI}$, and to an already detected cell can be expressed as: $T_{evaluate,EUTRAN_Intra} + T_{SI}$,

Where:

$T_{detect,EUTRAN_Intra}$ See Table 4.2.2.3-1 in TS 36.133 [4] clause 4.2.2.3

$T_{evaluate,EUTRAN_Intra}$ See Table 4.2.2.3-1 in TS 36.133 [4] clause 4.2.2.3

$T_{SI}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell; 1280 ms is assumed in this test case.

This gives a total of 33.28 s, allow 34 s for the cell re-selection delay to a newly detectable cell and 7.68 s, allow 8 s for the cell re-selection delay to an already detected cell in the test case.

## 4.2.15 E-UTRAN FDD – FDD Intra frequency case for Cat-M1 UE in enhanced coverage

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- *Core requirement are being futher investigated as marked in TS36.133.*
- *The Test tolerances and Test system uncertainties applicable to this test are undefined.*
- *Annex E of Cell Configuration table is undefined.*
- *Message content is TBD.*

### 4.2.15.1 Test purpose

To verify for a category M1 UE that when the current and target cell operates on the same carrier frequency in enhanced coverage the UE is able to search and measure cells to meet the respective intra-frequency cell re-selection requirements.

### 4.2.15.2 Test applicability

This test applies to all types of E-UTRA FD-FDD UE release 13 and forward of UE category M1. Applicability requires support CE Mode B.

### 4.2.15.3 Minimum conformance requirements

The cell re-selection delay shall be less than $T_{evaluate,\ E\text{-}UTRAN_intra_EC} + T_{SI\text{-}EUTRA}$ in RRC_IDLE state.

The requirements for enhanced coverage in idle mode shall apply provided the UE category M1 with the radio condition that SCH Ês/Iot ≥ -15 dB and CRS Ês/Iot ≥ -15 dB. The CAT-M1 UE is considered to be in enhanced coverage area of neighbor cell according to RSRP, RSRP Ês/Iot, SCH_RP and SCH Ês/Iot of the neighbor cell defined in Annex B.1.3 of TS36.133[4] for a corresponding Band.

The CAT-M1 UE shall be able to identify new intra-frequency cells and perform RSRP measurements of identified intra-frequency cells without an explicit intra-frequency neighbour list containing physical layer cell identities. The UE shall not cause any interruption to the paging reception and acquisition of SI while performing measurement on serving or any neighbor cells.

The CAT-M1 UE shall be able to evaluate whether a newly detectable intra-frequency cell meets the reselection criteria defined in TS36.304[6] within $T_{detect,EUTRAN_Intra_EC}$ when that Treselection= 0 .

The CAT-M1 UE shall measure RSRP at least every $T_{measure,EUTRAN_Intra_EC}$ for intra-frequency cells that are identified and measured according to the measurement rules.

The CAT-M1 UE shall filter RSRP measurements of each measured intra-frequency cell using at least 4 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least $T_{measure,EUTRAN_Intra_EC}/2$.

The CAT-M1 UE shall not consider an E-UTRA neighbour cell in cell reselection, if it is indicated as not allowed in the measurement control system information of the serving cell.

For an intra-frequency cell that has been already detected, but that has not been reselected to, the filtering shall be such that the UE shall be capable of evaluating that the intra-frequency cell has met reselection criterion defined in TS36.304[6] within $T_{evaluate,E-UTRAN_intra_EC}$ when $T_{reselection} = 0$, provided that the cell is at least 3dB better ranked and additional conditions Table 4.2.15.3-2 are met. When evaluating cells for reselection, the side conditions for RSRP and SCH apply to both serving and non-serving intra-frequency cells.

If $T_{reselection}$ timer has a non zero value and the intra-frequency cell is better ranked than the serving cell, the UE shall evaluate this intra-frequency cell for the $T_{reselection}$ time. If this cell remains better ranked within this duration, then the UE shall reselect that cell.

For UE not configured with eDRX_IDLE cycle, $T_{detect,EUTRAN_Intra_EC}$, $T_{measure,EUTRAN_Intra_EC}$ and $T_{evaluate,\ E\text{-}UTRAN_intra_EC}$ are specified in Table 4.2.15.3-1 provided that additional conditions Table 4.2.15.3-2 are met.

**Table 4.2.15.3-1 : $T_{detect,EUTRAN_Intra_EC}$, $T_{measure,EUTRAN_Intra_EC}$ and $T_{evaluate,\ E\text{-}UTRAN_intra_EC}$**

| DRX cycle length [s] | $T_{detect,EUTRAN_Intra_EC}$ [s] (number of DRX cycles) | $T_{measure,EUTRAN_Intra_EC}$ [s] (number of DRX cycles) | $T_{evaluate,E\text{-}UTRAN_intra_EC}$ [s] (number of DRX cycles) |
|---|---|---|---|
| 0.32 | 330.24 (1032) | 1.28 (4) | 10.24 (32) |
| 0.64 | 330.24 (516) | 1.28 (2) | 10.24 (16) |
| 1.28 | 524.8 (410) | 1.28 (1) | 12.8 (10) |
| 2.56 | 1039.36 (406) | 2.56 (1) | 15.36 (6) |

**Table 4.2.15.3-2: Conditions on SCH Ês/Iot of identified and of the neighbour cell**

| SCH Ês/Iot of already identified cell including serving cell: Q1 | Neighbouring cell SCH Ês/Iot: Q2 | $T_{detect,EUTRAN_Intra,\ ca\ tM1}$ (s) |
|---|---|---|
| -15≤Q1<-6 | -15≤ Q2 < -6 | As defined in Table Table 4.2.2.11.2-1 |
| -15≤Q1<-6 | Q2≥-6 | Requirements in 4.2.2 apply |
| Q1≥ -6 | Q2≥-6 | Requirements in 4.2.2 apply |

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.11 and A.4.2.15.

## 4.2.15.4 Test description

### 4.2.15.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20 using only main UE Tx/Rx antenna..
2. The general test parameter settings are set up according to Table 4.2.15.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.2.15.4.3.
5. There is one E-UTRA FD-FDD cell (Cell 1) and one E-UTRA FDD cell (Cell 2) specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 4.2.15.4.1-1: General test parameters for FDD intra frequency cell reselection test case for Cat-M1 UE in enhanced coverage**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2 | |
| T2 end condition | Active cell | | Cell2 | |
| | Neighbour cells | | Cell1 | |
| Final condition | Visited cell | | Cell1 | |
| E-UTRA RF Channel Number | | | 1 | Only one FDD carrier frequency is used. |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| PRACH Parameters | | | PRACH_4CE | Refer to A.9 |
| DRX cycle length | | s | 0.64 | The value shall be used for all cells in the test. |
| T1 | | s | >7 | During T1, Cell 2 shall be powered off, and during the off time the physical cell identity shall be changed, The intention is to ensure that Cell 2 has not been detected by the UE prior to the start of period T2 |
| T2 | | s | 340 | T2 need to be defined so that cell re-selection reaction time is taken into account. |
| T3 | | s | 15 | T3 need to be defined so that cell re-selection reaction time is taken into account. |

### 4.2.15.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The CAT-M1 UE is requested to monitor the neighbouring cell on one E-UTRA FDD carrier. In the test there are three successive time periods, with time duration of T1, T2 and T3 respectively. Only Cell 1 is already identified by the CAT-M1 UE prior to the start of the test, i.e. Cell 2 is not identified by the UE prior to the start of the test. Cell 1 and Cell 2 belong to different tracking areas. Furthermore, the CAT-M1 UE has not registered with network for the tracking area containing Cell 2.

In the following test procedure “UE responds” means “UE starts transmitting preamble on PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure according to TS 36.508 [7] clause 4.5A.2”.

1. Ensure the CAT-M1 UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2. Set Cell 2 physical cell identity = initial cell 2 physical cell identity.

2. Set the parameters according to T1 in Table 4.2.15.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for one iteration of the test procedure loop.

4. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 4.2.15.5-1.

5. The SS waits for random access requests information from the UE to perform cell re-selection to a newly detectable cell, Cell 2.

6. If the CAT-M1 UE responds on the newly detectable cell, Cell 2 during time duration T2 within 332 seconds from the beginning of time period T2, then count a success for the event “Re-select newly detected Cell 2”. Otherwise count a fail for the event “Re-select newly detected Cell 2”.

7. If the CAT-M1 UE has re-selected Cell 2 within T2, after the re-selection or when T2 expires, continue with step 8. Otherwise, if T2 expires and the CAT-M1 UE has not yet re-selected Cell 2, skip to step 12.

8. The SS shall switch the power setting from T2 to T3 as specified in Table 4.2.15.5-1.

9. The SS waits for random access requests information from the CAT-M1 UE to perform cell re-selection to an already detected cell, Cell 1.

10. If the CAT-M1 UE responds on the already detected cell, Cell 1 during time duration T3 within 12 seconds from the beginning of time period T3, then count a success for the event “Re-select already detected Cell 1”. Otherwise count a fail for the event “Re-select already detected Cell 1”.

11. If the CAT-M1 UE has re-selected Cell 1 within T3, after the re-selection or when T3 expires, skip to step 13. Otherwise, if T3 expires and the CAT-M1 UE has not yet re-selected Cell 1, continue with step 12.

12. Switch off and on the CAT-M1 UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.

13. Repeat step 2-12 until a test verdict has been achieved.
    Each of the events “Re-select newly detected Cell 2” and “Re-select already detected Cell 1” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. Different events may require different times for a verdict.
    If both events pass, the test passes. If one event fails, the test fails.

#### 4.2.15.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

TBD

### 4.2.15.5 Test requirement

Tables 4.2.15.4.1-1 and 4.2.15.5-1 defines the primary level settings including test tolerances for E-UTRAN FDD – FDD Intra frequency case for Cat-M1 cell re-selection test case.

**Table 4.2.15.5-1: Cell specific test parameters for FDD intra frequency cell reselection test case for Cat-M1 UE in enhanced coverage**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | |
| OCNG Patterns defined in clause D.1.2 | | OP.2 FDD | | | OP.2 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| MPDCCH_RA | dB | | | | | | |
| MPDCCH_PB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| Qrxlevmin | dBm | -140 | -140 | -140 | -140 | -140 | -140 |
| Pcompensation | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $Qhyst_s$ | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $Qoffset_{s,n}$ | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| Cell_selection_and_reselection_quality_measurement | | RSRP | | | RSRP | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | -8+TT | -11+TT | -8+TT | -infinity | -8+TT | -11+TT |
| $\hat{E}_s / I_{ot}$ | dB | -8+TT | -11.64+TT | -8.33+TT | -infinity | -8.33+TT | -11.64+TT |
| RSRP Note3 | dBm/15 kHz | -106+TT | -109+TT | -106+TT | -infinity | -106+TT | -109+TT |
| Treselection | s | 0 | 0 | 0 | 0 | 0 | 0 |
| Sintrasearch | dB | Not sent | | | Not sent | | |
| Propagation Condition | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x1 | | | 1x1 | | |
| Timing offset to Cell 1 | ms | - | | | 3 | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The cell reselection delay to a newly detectable cell is defined as the time from the beginning of time period T2, to the moment when the UE camps on Cell 2, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 2.

The cell re-selection delay to a newly detectable cell shall be less than 332 s.

The cell reselection delay to an already detected cell is defined as the time from the beginning of time period T3, to the moment when the UE camps on Cell 1, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 1.

The cell re-selection delay to an already detected cell shall be less than 12 s.

The rate of correct cell reselections observed during repeated tests shall be at least 90%.

NOTE: The cell re-selection delay to a newly detectable cell can be expressed as: $T_{detect,EUTRAN_Intra_EC} + T_{SI}$, and to an already detected cell can be expressed as: $T_{evaluate,\ E\text{-}UTRAN_intra_EC} + T_{SI}$.

Where:

$T_{detect,EUTRAN_Intra_EC}$ See Table 4.2.15.3-1

$T_{evaluate,\ E\text{-}UTRAN_intra_EC}$ See Table 4.2.15.3-11

$T_{SI}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell; 1280 ms is assumed in this test case.

This gives a total of 331.52 s, allow 332 s for the cell re-selection delay to a newly detectable cell and 11.52 s, allow 12 s for the cell re-selection delay to an already detected cell in the test case.

## 4.2.16 E-UTRAN HD – FDD Intra frequency case for Cat-M1 UE in enhanced coverage

Editor's notes: This test case is incomplete. The following items are missing or incomplete:

- Test system uncertainties are not yet defined
- Test Tolerances are not yet defined
- TS 36.133 core requirements are being futher investigated and based on outcome of the investigation the corresponding test requirements and test parameters can be updated

### 4.2.16.1 Test purpose

To verify for a category M1 UE in enhanced coverage that the UE is able to search and measure cells to meet the respective HD-FDD intra-frequency cell re-selection requirements.

### 4.2.16.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 13 and forward of UE category M1 that support operating in CEModeB.

### 4.2.16.3 Minimum conformance requirements

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.11 and A.4.2.16.

### 4.2.16.4 Test description

#### 4.2.16.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.
2. The general test parameter settings are set up according to Table 4.2.12.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.2.12.4.3
5. There is one E-UTRA HD-FDD cell (Cell 1) and one E-UTRA FDD cell (Cell 2) specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 4.2.16.4.1-1: General test parameters for HD-FDD intra frequency cell reselection test case for Cat-M1 UE in enhanced coverage**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2 | |
| T2 end condition | Active cell | | Cell2 | |
| | Neighbour cells | | Cell1 | |
| Final condition | Visited cell | | Cell1 | |
| E-UTRA RF Channel Number | | | 1 | Only one carrier frequency is used. |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| PRACH Parameters | | | PRACH_4CE | Refer to A.3.16 in TS 36.133 [4] |
| DRX cycle length | | s | 0.64 | The value shall be used for all cells in the test. |
| T1 | | s | >7 | During T1, Cell 2 shall be powered off, and during the off time the physical cell identity shall be changed, The intention is to ensure that Cell 2 has not been detected by the UE prior to the start of period T2 |
| T2 | | s | 340 | T2 need to be defined so that cell re-selection reaction time is taken into account. |
| T3 | | s | 15 | T3 need to be defined so that cell re-selection reaction time is taken into account. |

### 4.2.16.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The CAT-M1 UE is requested to monitor the neighbouring cell on one E-UTRA FDD carrier. In the test there are three successive time periods, with time duration of T1, T2 and T3 respectively. Only Cell 1 is already identified by the CAT-M1 UE prior to the start of the test, i.e. Cell 2 is not identified by the UE prior to the start of the test. Cell 1 and Cell 2 belong to different tracking areas. Furthermore,the CAT-M1 UE has not registered with network for the tracking area containing Cell 2.

In the following test procedure “UE responds” means “UE starts transmitting preamble on PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure according to TS 36.508 [7] clause 4.5A.2”.

1. Ensure the CAT-M1 UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2. Set Cell 2 physical cell identity = initial cell 2 physical cell identity.

2. Set the parameters according to T1 in Table 4.2.16.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for one iteration of the test procedure loop.

4. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 4.2.16.5-1.

5. The SS waits for random access requests information from the UE to perform cell re-selection to a newly detectable cell, Cell 2.

6. If the CAT-M1 UE responds on the newly detectable cell, Cell 2 during time duration T2 within 332 seconds from the beginning of time period T2, then count a success for the event “Re-select newly detected Cell 2”. Otherwise count a fail for the event “Re-select newly detected Cell 2”.

7. If the CAT-M1 UE has re-selected Cell 2 within T2, after the re-selection or when T2 expires, continue with step 8. Otherwise, if T2 expires and the CAT-M1 UE has not yet re-selected Cell 2, skip to step 12.

8. The SS shall switch the power setting from T2 to T3 as specified in Table 4.2.16.5-1.

9. The SS waits for random access requests information from the CAT-M1 UE to perform cell re-selection to an already detected cell, Cell 1.

10. If the CAT-M1 UE responds on the already detected cell, Cell 1 during time duration T3 within 12 seconds from the beginning of time period T3, then count a success for the event “Re-select already detected Cell 1”. Otherwise count a fail for the event “Re-select already detected Cell 1”.

11. If the CAT-M1 UE has re-selected Cell 1 within T3, after the re-selection or when T3 expires, skip to step 13. Otherwise, if T3 expires and the CAT-M1 UE has not yet re-selected Cell 1, continue with step 12.

12. Switch off and on the CAT-M1 UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.

13. Repeat step 2-12 until a test verdict has been achieved.
    Each of the events “Re-select newly detected Cell 2” and “Re-select already detected Cell 1” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. Different events may require different times for a verdict.
    If both events pass, the test passes. If one event fails, the test fails.

### 4.2.16.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 4.2.12.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra cell re-selection test case**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.1-1<br>Table H.2.1-2 |

**Table 4.2.12.4.3-2: *PRACH-Config-DEFAULT*: E-UTRAN FDD cell re-selection and handover**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-7A PRACH-Config-v1310-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| prach-ParametersListCE-r13 SEQUENCE { | | | |
| prach-ConfigIndex-r13[1] | 4 | | |
| } | | | |

## 4.2.16.5 Test requirement

Tables 4.2.16.4.1-1 and 4.2.16.5-1 defines the primary level settings including test tolerances for E-UTRAN HD FDD – FDD Intra frequency case for Cat-M1 cell re-selection test case.

**Table 4.2.16.5-1: Cell specific test parameters for HD-FDD intra frequency cell reselection test case for Cat-M1 UE in enhanced coverage**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | |
| OCNG Patterns | | OP.2 FDD | | | OP.2 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| MPDCCH_RA | dB | | | | | | |
| MPDCCH_PB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| Qrxlevmin | dBm | -140 | -140 | -140 | -140 | -140 | -140 |
| Pcompensation | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $Qhyst_s$ | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $Qoffset_{s,n}$ | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| Cell_selection_and_reselection_quality_measurement | | RSRP | | | RSRP | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | -8 + TT | -11 + TT | -8 + TT | -infinity | -8 + TT | -11 + TT |
| $\hat{E}_s / I_{ot}$ | dB | -8 + TT | -11.64 | -8.33 + TT | -infinity | -8.33 + TT | -11.64 + TT |
| RSRP Note3 | dBm/15 kHz | -106 + TT | -109 + TT | -106 + TT | -infinity | -106 + TT | -109 + TT |
| Treselection | s | 0 | 0 | 0 | 0 | 0 | 0 |
| Sintrasearch | dB | Not sent | | | Not sent | | |
| Propagation Condition | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x1 | | | 1x1 | | |
| Timing offset to Cell 1 | ms | - | | | 3 | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The cell reselection delay to a newly detectable cell is defined as the time from the beginning of time period T2, to the moment when the UE camps on Cell 2, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 2.

The cell re-selection delay to a newly detectable cell shall be less than 332 s.

The cell reselection delay to an already detected cell is defined as the time from the beginning of time period T3, to the moment when the UE camps on Cell 1, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 1.

The cell re-selection delay to an already detected cell shall be less than 12 s.

The rate of correct cell reselections observed during repeated tests shall be at least 90%.

NOTE: The cell re-selection delay to a newly detectable cell can be expressed as: $T_{detect,EUTRAN_Intra} + T_{SI}$, and to an already detected cell can be expressed as: $T_{evaluate,EUTRAN_Intra} + T_{SI}$,

Where:

$T_{detect,EUTRAN_Intra}$ See Table 4.2.2.11-1 in TS 36.133 [4] clause 4.2.2.11

$T_{evaluate,EUTRAN_Intra}$ See Table 4.2.2.11-1 in TS 36.133 [4] clause 4.2.2.11

$T_{SI}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell; 1280 ms is assumed in this test case.

This gives a total of 331.52 s, allow 332 s for the cell re-selection delay to a newly detectable cell and 11.52 s, allow 12 s for the cell re-selection delay to an already detected cell in the test case.

## 4.2.17 E-UTRAN TDD – TDD Intra frequency case for Cat-M1 UE in enhanced coverage

### 4.2.17.1 Test purpose

To verify for a category M1 UE in enhanced coverage that the UE is able to search and measure cells to meet the respective TDD-TDD intra-frequency cell re-selection requirements.

### 4.2.17.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 13 and forward of UE category M1 that support operating in CEModeB.

### 4.2.17.3 Minimum conformance requirements

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.11 and A.4.2.17.

### 4.2.17.4 Test description

#### 4.2.17.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20(using only main UE Tx/Rx antenna).
2. The general test parameter settings are set up according to Table 4.2.17.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.2.17.4.3.
5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for registration with the power level set according to Table clause C.0 and C.1 for this test.

**Table 4.2.17.4.1-1: General test parameters for TDD intra frequency cell re-selection test case for Cat-M1 UE**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | |
| | Neighbour cells | | Cell2 | |
| T2 end condition | Active cell | | Cell2 | |
| | Neighbour cells | | Cell1 | |
| Final condition | Visited cell | | Cell1 | |
| E-UTRA RF Channel Number | | | 1 | Only one TDD carrier frequency is used. |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211 |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211 |
| PRACH Parameters | | | PRACH_4CE | Refer to A.3.16 in TS 36.133 [4] |
| DRX cycle length | | s | 0.64 | The value shall be used for all cells in the test. |
| T1 | | s | >7 | During T1, Cell 2 shall be powered off, and during the off time the physical cell identity shall be changed, The intention is to ensure that Cell 2 has not been detected by the UE prior to the start of period T2 |
| T2 | | s | 340 | T2 need to be defined so that cell re-selection reaction time is taken into account. |
| T3 | | s | 15 | T3 need to be defined so that cell re-selection reaction time is taken into account. |

### 4.2.17.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The CAT-M1 UE is requested to monitor the neighbouring cell on one E-UTRA TDD carrier. In the test there are three successive time periods, with time duration of T1, T2, and T3 respectively. Only Cell 1 is already identified by the CAT-M1 UE prior to the start of the test. Cell 1 and Cell 2 belong to different tracking areas. Furthermore, CAT-M1 UE has not registered with network for the tracking area containing Cell 2.

In the following test procedure “UE responds” means “UE starts transmitting preamble on PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure according to TS 36.508 [7] clause 4.5A.2”

1. Ensure the CAT-M1 UE is in State 2A-RF-CE according to TS 36.508 [7] clause 7.2A.2. Set Cell 2 physical cell identity = initial cell 2 physical cell identity.

2. Set the parameters according to T1 in Table 4.2.17.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for one iteration of the test procedure loop.

4. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 4.2.17.5-1.

5. The SS waits for random access requests information from the CAT-M1 UE to perform cell re-selection to a newly detectable cell, Cell 2.

6. If the Cat-M1 UE responds on the newly detectable cell, Cell 2, during time duration T2 within 332 seconds from the beginning of time period T2, then count a success for the event “Re-select newly detected Cell 2”. Otherwise count a fail for the event “Re-select newly detected Cell 2”.

7. If the Cat-M1 UE has re-selected Cell 2 within T2, after the re-selection or when T2 expires, continue with step 8.
   Otherwise, if T2 expires and the Cat-M1 UE has not yet re-selected Cell 2, skip to step 12.

8. The SS shall switch the power setting from T2 to T3 as specified in Table 4.2.17.5-1.

9. The SS waits for random access requests information from the CAT-M1 UE to perform cell re-selection to an already detected cell, Cell 1.

10. If the Cat-M1 UE responds on the already detected cell, Cell 1, during time duration T3 within 12 seconds from the beginning of time period T3, then count a success for the event “Re-select already detected Cell 1”. Otherwise count a fail for the event “Re-select already detected Cell 1”.

11. If the Cat-M1 UE has re-selected Cell 1 within T3, after the re-selection or when T3 expires, skip to step 13. Otherwise, if T3 expires and the Cat-M1 UE has not yet re-selected Cell 1, continue with step 12.

12. Switch off and on the CAT-M1 UE and ensure the UE is in State 2A-RF-CE according to TS 36.508 [7] clause 7.2A.2A in Cell 1.

13. Repeat step 2-12 until a test verdict has been achieved.
    Each of the events “Re-select newly detected Cell 2” and “Re-select already detected Cell 1” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. Different events may require different times for a verdict.
    If both events pass, the test passes. If one event fails, the test fails.

#### 4.2.17.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 4.2.17.4.3-1: Common Exception messages for E-UTRAN TDD-TDD intra cell re-selection test case**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.1-1<br>Table H.2.1-2 |

**Table 4.2.17.4.3-2: *PRACH-Config-v1310-DEFAULT*: E-UTRAN TDD cell re-selection and handover**

| Derivation Path: 36.508 clause 4.6.3 Table 4.6.3-7A | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PRACH-Config-v1310-DEFAULT ::= SEQUENCE { | | | |
| rsrp-ThresholdsPrachInfoList-r13 SEQUENCE (SIZE(1..3)) OF { | 1 entry | | |
| RSRP-Range[1] | 42 | -99dBm | |
| } | | | |
| PRACH-Config-v1310-DEFAULT ::= SEQUENCE { | | | |
| prach-ParametersListCE-r13 SEQUENCE (SIZE(1..maxCE-Level-r13)) OF SEQUENCE { | 2 entries | | |
| PRACH-ParametersCE-r13[1] SEQUENCE { | | | |
| prach-ConfigIndex-r13[1] | 53 | INTEGER (0..63) | TDD |
| prach-FreqOffset-r13[1] | Same as prach-FrequencyOffset | | |
| prach-StartingSubframe-r13[1] | sf2 | | |
| maxNumPreambleAttemptCE-r13[1] | n5 | | |
| numRepetitionPerPreambleAttempt-r13[1] | n1 | | |
| mpdcch-NarrowbandsToMonitor-r13 SEQUENCE (SIZE(1..2)) OF INTEGER (1..maxAvailNarrowBands-r13)[1] { | 1 entry | | |
| mpdcch-NarrowbandsToMonitor-r13[1] | 1 | | |
| } | | | |
| mpdcch-NumRepetition-RA-r13[1] | r1 | | |
| prach-HoppingConfig-r13[1] | off | | |
| } | | | |
| prach-ConfigIndex-r13[2] | 53 | INTEGER (0..63) | TDD |
| prach-FreqOffset-r13[2] | Same as prach-FrequencyOffset | | |
| prach-StartingSubframe-r13[2] | sf128 | | |
| maxNumPreambleAttemptCE-r13[2] | n5 | | |
| numRepetitionPerPreambleAttempt-r13[2] | n1 | | |
| mpdcch-NarrowbandsToMonitor-r13 SEQUENCE (SIZE(1..2)) OF INTEGER (1..maxAvailNarrowBands-r13)[2] { | 1 entry | | |
| mpdcch-NarrowbandsToMonitor-r13[1] | 1 | | |
| } | | | |
| mpdcch-NumRepetition-RA-r13[2] | r4 | | |
| prach-HoppingConfig-r13[2] | off | | |
| } | | | |
| } | | | |
| } | | | |

## 4.2.17.5 Test requirement

Tables 4.2.17.4.1-1, and 4.2.17.5-1 defines the primary level settings including test tolerances for E-UTRAN TDD-TDD intra frequency case for Cat-M1 cell re-selection test case.

**Table 4.2.17.5-1: Cell specific test parameters for TDD intra frequency cell re-selection test case for Cat-M1 UE in AWGN in enhanced coverage**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | |
| OCNG Patterns | | OP.2 TDD | | | OP.2 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| MPDCCH_RA | dB | | | | | | |
| MPDCCH_PB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| Qrxlevmin | dBm | -140 | -140 | -140 | -140 | -140 | -140 |
| Pcompensation | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $Qhyst_s$ | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $Qoffset_{s,n}$ | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| Cell_selection_and_reselection_quality_measurement | | RSRP | | | RSRP | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | -8 + TT | -11 + TT | -8 + TT | -infinity | -8 + TT | -11 + TT |
| $\hat{E}_s / I_{ot}$ | dB | -8 + TT | -11.64 + TT | -8.33 + TT | -infinity | -8.33 + TT | -11.64 + TT |
| RSRP Note3 | dBm/15 kHz | -106 + TT | -109 + TT | -106 + TT | -infinity | -106 + TT | -109 + TT |
| Treselection | s | 0 | 0 | 0 | 0 | 0 | 0 |
| Sintrasearch | dB | Not sent | | | Not sent | | |
| Propagation Condition | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x1 | | | 1x1 | | |
| Timing offset to Cell 1 | µs | - | | | 3 | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The cell reselection delay to a newly detectable cell is defined as the time from the beginning of time period T2, to the moment when the UE camps on Cell 2, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 2.

The cell re-selection delay to a newly detectable cell shall be less than 332 s.

The cell reselection delay to an already detected cell is defined as the time from the beginning of time period T3, to the moment when the UE camps on Cell 1, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on Cell 1.

The cell re-selection delay to an already detected cell shall be less than 12 s.

The rate of correct cell reselections observed during repeated tests shall be at least 90%.

NOTE: The cell re-selection delay to a newly detectable cell can be expressed as: $T_{detect,EUTRAN_Intra} + T_{SI}$, and to an already detected cell can be expressed as: $T_{evaluate,EUTRAN_Intra} + T_{SI}$,

Where:

$T_{detect,EUTRAN_Intra}$ See Table 4.2.2.11-1 in TS 36.133 [4] clause 4.2.2.11

$T_{evaluate,EUTRAN_Intra}$ See Table 4.2.2.11-1 in TS 36.133 [4] clause 4.2.2.11

$T_{SI}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell; 1280 ms is assumed in this test case.

This gives a total of 331.52 s, allow 332 s for the cell re-selection delay to a newly detectable cell and 11.52 s, allow 12 s for the cell re-selection delay to an already detected cell in the test case.

# 4.3 E-UTRAN to UTRAN Cell Re-Selection

## 4.3.1 E-UTRAN FDD - UTRAN FDD cell re-selection

### 4.3.1.1 E-UTRA FDD - UTRAN FDD cell reselection: UTRA FDD is of higher priority

#### 4.3.1.1.1 Test purpose

To verify that the UE is able to search and measure neighbouring UTRA FDD cells and compare to the E-UTRA serving cell to meet the inter-RAT cell re-selection requirements when the UTRA is of higher priority.

#### 4.3.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support UTRA FDD.

#### 4.3.1.1.3 Minimum conformance requirements

When the measurement rules indicate that UTRA FDD cells are to be measured, the UE shall measure CPICH Ec/Io and CPICH RSCP of detected UTRA FDD cells in the neighbour frequency list at the minimum measurement rate. The parameter $N_{UTRA_carrier}$ is the number of carriers used in the neighbour frequency list. The UE shall filter CPICH Ec/Io and CPICH RSCP measurements of each measured UTRA FDD cell using at least 2 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least half the minimum specified measurement period.

If the $S_{ServingCell}$ of the E-UTRA serving cell is greater than $S_{nonintrasearch}$ then the UE shall search for inter-RAT layers of higher priority at least every $T_{higher_priority_search}$ where $T_{higher_priority_search}$ is described in TS 36.133 [4] clause 4.2.2 as $T_{higher_priority_search} = (60 * N_{layers})$ seconds, where the parameter $N_{layers}$ is the total number of configured higher priority carrier frequencies.

If the $S_{ServingCell}$ of the E-UTRA serving cell is less than or equal to $S_{nonintrasearch}$ then the UE shall search for and measure inter-RAT layers of higher or lower priority in preparation for possible re-selection. In this scenario, the minimum rate at which the UE is required to search for and measure higher priority inter-RAT layers shall be the same as that defined below for lower priority RATs.

The UE shall evaluate whether newly detectable UTRA FDD cells have met the re-selection criteria in TS 36.304 [6] within time $N_{UTRA_carrier} * T_{detectUTRA_FDD}$ (as defined in table 4.2.2.5.1-1 of TS 36.133 [4] clause 4.2.2.5.1) when the $S_{ServingCell}$ of the E-UTRA serving cell is less than $S_{nonintrasearch}$ when $Treselection_{RAT} = 0$ provided that the re-selection criteria is met by a margin of at least 6 dB.

Cells which have been detected shall be measured at least every $N_{UTRA_carrier} * T_{measureUTRA_FDD}$ when the $S_{ServingCell}$ of the E-UTRA serving cell is less than $S_{nonintrasearch}$.

When higher priority UTRA FDD cells are found by the higher priority search, they shall be measured at least every $T_{measureUTRA_FDD}$. If re-selection to any higher priority cell is not triggered within ($T_{evaluateUTRA_FDD} + Treselection_{RAT}$) after it is found in a higher priority search, the UE is not required to continue making measurements of the cell to evaluate the ongoing possibility of re-selection.

For a cell that has been already detected, but that has not been re-selected to, the filtering shall be such that the UE shall be capable of evaluating that an already identified UTRA FDD cell has met re-selection criterion defined in TS 36.304 [6] within $N_{UTRA_carrier}$ * $T_{evaluateUTRA_FDD}$ when $Treselection_{RAT}$ = 0 as specified in table 4.2.2.5.1-1 of TS 36.133 [4] clause 4.2.2.5.1 provided that the reselection criteria is met by a margin of at least 6dB.

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.5.1 and A.4.3.1.1.

### 4.3.1.1.4 Test description

#### 4.3.1.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.22.
2. The general test parameter settings are set up according to Table 4.3.1.1.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.3.1.1.4.3.
5. There is one E-UTRA FDD cell and one UTRA FDD cell specified in the test. Cell 1 is the cell used for registration according to TS 36.508 [7] clause 7.2A.2 with the power level set according to Annex C.0 and C.1.

**Table 4.3.1.1.4.1-1: General test parameters for E-UTRA FDD- higher priority UTRA FDD inter RAT cell re-selection test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell 1 | UE is on cell 1 in the initialisation phase, so that reselection to cell 2 occurs during T2 |
| T2 end condition | Active cell | | Cell 2 | UE shall perform reselection to cell 2 during T2 |
| | Neighbour cell | | Cell 1 | |
| T3 end condition | Active cell | | Cell 1 | UE shall perform reselection to cell 1 during T3 |
| | Neighbour cell | | Cell 2 | |
| E-UTRA PRACH configuration | | | 4 | As specified in table 5.7.1-2 in TS 36.211 [9] |
| E_UTRA Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | >20 | During T1, cell 2 shall be powered off, and during the off time the primary scrambling code shall be changed, The intention is to ensure that cell 2 has not been detected by the UE prior to the start of period T2. |
| T2 | | s | 85 | T2 needs to be defined so that cell re-selection reaction time is taken into account |
| T3 | | s | 25 | T3 needs to be defined so that cell re-selection reaction time is taken into account. |

#### 4.3.1.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. E-UTRA cell 1 is already identified by the UE prior to the start of the test. Before T1 the UE is camped on to cell 1. During T1, cell 2 shall be powered off, and during the off time the scrambling code shall be changed. At the start of T2 cell 2 becomes stronger than $Thresh_{x_high}$, the UE is expected to detect cell 2, send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Routing Area Update procedure on cell 2. At the start of T3 cell 2 becomes weaker than $Thresh_{serving_low}$, and the UE reselects to Cell 1.

In the following test procedure “UE responds” means “UE starts transmitting preamble on PRACH for sending the RRC CONNECTION REQUEST message to perform a Routing Area Update procedure”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2.
2. Set the parameters according to duration T1 in Table 4.3.1.1.5-1 and 4.3.1.1.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. During T1, Cell 2 shall be powered off and the SS shall set Cell 2 primary scrambling code = ((current cell 2 primary scrambling code – 50) mod 200 + 100).
4. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 4.3.1.1.5-1 and 4.3.1.1.5-2. The SS waits for random access requests information from the UE to perform cell re-selection procedure on the higher priority cell, Cell 2.
5. If the UE responds on Cell 2 within 81s from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
6. If the UE has re-selected Cell 2 within T2, after the re-selection or when T2 expires, continue with step 7. Otherwise, if T2 expires and the UE has not yet re-selected Cell 2, skip to step 10.
7. The SS shall switch the power setting from T2 to T3 as specified in Table 4.3.1.1.5-1 and 4.3.1.1.5-2.
8. The SS waits for random access requests information from the UE to perform cell re-selection procedure on the lower priority cell, Cell 1.
9. If the UE has re-selected Cell 1 within T3, after the re-selection or when T3 expires, skip to step 11. Otherwise, if T3 expires and the UE has not yet re-selected Cell 1, continue with step 10.
10. Switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.
11. Repeat step 2-10 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

#### 4.3.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 and TS 34.108 [24] clause 6.1.0b with the following exceptions:

**Table 4.3.1.1.4.3-1: Common Exception messages**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.3-1<br>Table H.2.3-2<br>Table H.2.3-13 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

**Table 4.3.1.1.4.3-2: System Information Block type 19: Inter-RAT E-UTRAN FDD - UTRA FDD is of higher priority cell re-selection**

| Derivation Path: 36.508 [7] clause 4.4.4.1 Table 4.4.4.1-1: System Information Block type 19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SysInfoType19 ::= SEQUENCE { | | | |
| utra-PriorityInfoList SEQUENCE { | | | |
| utra-ServingCell SEQUENCE { | | | |
| priority | 5 | | |
| s-PrioritySearch1 | 31 (62dB) | | |
| s-PrioritySearch2 | 0 | default value is 0 | |
| threshServingLow | 18 (36dB) | | |
| } | | | |
| eutra-FrequencyAndPriorityInfoList SEQUENCE (SIZE (1..maxNumEUTRAFreqs)) OF SEQUENCE | | *n* denotes the index of the entry | |
| earfcn | | Downlink EARFCN of Cell 1 | |
| measurementBandwidth | 50 | Enumerated(6, 15, 25, 50, 75, 100) | |
| priority | 4 | | |
| qRxLevMinEUTRA | -70 (-140 dBm) | | |
| threshXhigh | 2 (4 dB) | | |
| threshXlow | 25 (50 dB) | | |
| eutra-blackListedCellList | Not present | | |
| eutraDetection | TRUE | | |
| } | | | |
| nonCriticalExtensions SEQUENCE {} | Not present | | |
| } | | | |

### 4.3.1.1.5 Test requirement

Tables 4.3.1.1.4-1, 4.3.1.1.5-1 and 4.3.1.1.5-2 define the primary level settings including test tolerances for inter-RAT cell re-selection E-UTRA FDD to UTRA FDD test case (UTRA is of higher priority).

**Table 4.3.1.1.5-1: Cell specific Test Parameters for Cell 1(E-UTRA FDD)**

| Parameter | Unit | Cell 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| OCNG Patterns defined in D.1.2 (OP.2 FDD) | | OP.2 FDD | | |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| Qqualmin for UTRA neighbour cell | dB | -20 | | |
| Qrxlevmin for UTRA neighbour cell | dBm | -115 | | |
| Qrxlevmin | dBm | -140 | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | |
| RSRP | dBm/15 KHz | -82.95 | -82.95 | -82.95 |
| $\hat{E}_s / I_{ot}$ | dB | 15.05 | 15.05 | 15.05 |
| $\hat{E}_s / N_{oc}$ | dB | 15.05 | 15.05 | 15.05 |
| $Treselection_{EUTRAN}$ | s | 0 | | |
| Snonintrasearch | dB | 50 | | |
| $Thresh_{x, high}$ (Note 2) | dB | 40 | | |
| Propagation Condition | | AWGN | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2 : This refers to the value of $Thresh_{x, high}$ which is included in E-UTRA system information, and is a threshold for the UTRA target cell

**Table 4.3.1.1.5-2: Cell specific test parameters for Cell 2 (higher priority UTRA FDD)**

| Parameter | Unit | Cell 2 (UTRA) | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| UTRA RF Channel Number | | Channel 2 | | |
| CPICH_Ec/Ior | dB | -10 | | |
| PCCPCH_Ec/Ior | dB | -12 | | |
| SCH_Ec/Ior | dB | -12 | | |
| PICH_Ec/Ior | dB | -15 | | |
| OCNS_Ec/Ior | dB | -0.941 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -∞ | 11.90 | -5.70 |
| $I_{oc}$ | dBm/3,84 MHz | -70.10 | | |
| CPICH_Ec/Io | dB | -∞ | -10.27 | -16.74 |
| CPICH_RSCP | dBm | -∞ | -68.20 | -85.80 |
| Propagation Condition | | AWGN | | |
| Qqualmin | dB | -20 | | |
| Qrxlevmin | dBm | -115 | | |
| QrxlevminEUTRA | dBm | -140 | | |
| UE_TXPWR_MAX_RACH | dBm | 21 | | |
| Treselection | s | 0 | | |
| Sprioritysearch1 | dB | 62 | | |
| Sprioritysearch2 | dB | 0 | | |
| $Thresh_{serving, low}$ | dB | 36 | | |
| $Thresh_{x, low}$ (Note 1) | dB | 50 | | |
| Note 1: This refers to the value of $Thresh_{x, low}$ which is included in UTRA system information, and is a threshold for the E-UTRA target cell. | | | | |

The cell reselection delay to higher priority is defined as the time from the beginning of time period T2, to the moment when the UE camps on cell 2, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message on cell 2.

The cell re-selection delay to higher priority shall be less than 81 s.

NOTE: The cell re-selection delay to higher priority cell can be expressed as: $T_{higher_priority_search} + T_{evaluateUTRA_FDD} + T_{SI\text{-}UTRA}$

Where:

$T_{higher_priority_search}$ See section 4.2.2 of TS 36.133 [4]; 60s is assumed in this test case

$T_{evaluateUTRA\text{-}FDD}$ See Table 4.2.2.5.1-1 of TS 36.133 [4] clause 4.2.2.5.1

$T_{SI\text{-}UTRA}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell; 1280 ms is assumed in this test case.

This gives a total of 80.48 s for higher priority cell search, allow 81 s for higher priority cell reselection in the test case.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 4.3.1.2 E-UTRAN FDD - UTRAN FDD cell re-selection: UTRA FDD is of lower priority

### 4.3.1.2.1 Test purpose

To verify that the UE is able to search and measure neighbouring UTRA FDD cells and compare to the E-UTRA serving cell to meet the inter-RAT cell re-selection requirements when the UTRA is of lower priority.

### 4.3.1.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support UTRA FDD.

#### 4.3.1.2.3 Minimum conformance requirements

When the measurement rules indicate that UTRA FDD cells are to be measured, the UE shall measure CPICH Ec/Io and CPICH RSCP of detected UTRA FDD cells in the neighbour frequency list at the minimum measurement rate. The parameter $N_{UTRA_carrier_FDD}$ is the number of carriers used in the neighbour frequency list. The UE shall filter CPICH Ec/Io and CPICH RSCP measurements of each measured UTRA FDD cell using at least 2 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least half the minimum specified measurement period.

If the $S_{ServingCell}$ of the E-UTRA serving cell is greater than $S_{nonintrasearch}$then the UE shall search for inter-RAT layers of higher priority at least every $T_{higher_priority_search}$ where $T_{higher_priority_search}$ is described in TS 36.133 [4] clause 4.2.2 as $T_{higher_priority_search}$ = (60 * $N_{layers}$) seconds, where the parameter $N_{layers}$ is the total number of configured higher priority carrier frequencies.

If the $S_{ServingCell}$ of the E-UTRA serving cell is less than or equal to $S_{nonintrasearch}$ then the UE shall search for and measure inter-RAT layers of higher or lower priority in preparation for possible re-selection. In this scenario, the minimum rate at which the UE is required to search for and measure higher priority inter-RAT layers shall be the same as that defined below for lower priority RATs.

The UE shall evaluate whether newly detectable UTRA FDD cells have met the re-selection criteria in TS 36.304 [6] within time $N_{UTRA_carrier_FDD}$ * $T_{detectUTRA_FDD}$ (as defined in table 4.2.2.5.1-1 of TS 36.133 [4] clause 4.2.2.5.1) when the $S_{ServingCell}$ of the E-UTRA serving cell is less than $S_{nonintrasearch}$ when $Treselection_{RAT}$ = 0 provided that the re-selection criteria is met by a margin of at least 6 dB.

Cells which have been detected shall be measured at least every $N_{UTRA_carrier}$ * $T_{measureUTRA_FDD}$ when the $S_{ServingCell}$ of the E-UTRA serving cell is less than $S_{nonintrasearch}$.

When higher priority UTRA FDD cells are found by the higher priority search, they shall be measured at least every $T_{measureUTRA_FDD.}$ If, after detecting a cell in a higher priority search, it is determined that reselection has not occurred then the UE is not required to continuously measure the detected cell to evaluate the ongoing possibility of re-selection. However, the minimum measurement filtering requirements specified later in this section shall still be met by the UE before it makes any determination that it may stop measuring the cell.

For a cell that has been already detected, but that has not been re-selected to, the filtering shall be such that the UE shall be capable of evaluating that an already identified UTRA FDD cell has met re-selection criterion defined in TS 36.304 [6] within $N_{UTRA_carrier}$ * $T_{evaluateUTRA_FDD}$ when $Treselection_{RAT}$ = 0 as specified in table 4.2.2.5.1-1 of TS 36.133 [4] clause 4.2.2.5.1 provided that the reselection criteria is met by a margin of at least 6dB.

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.5.1 and A.4.3.1.2.

#### 4.3.1.2.4 Test description

##### 4.3.1.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.22.
2. The general test parameter settings are set up according to Table 4.3.1.2.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.3.1.2.4.3.
5. There is one E-UTRA FDD cell and one UTRA FDD cell specified in the test. Cell 1 is the cell used for registration according to TS 36.508 [7] clause 7.2A.2 with the power level set according to Annex C.0 and C.1.

**Table 4.3.1.2.4.1-1: General test parameters for EUTRA FDD- lower priority UTRA FDD inter RAT cell re-selection test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | E-UTRAN cell |
| T1 end condition | Active cells | | Cell1 | UE shall perform reselection to cell 1 during T1 for subsequent iterations of the test |
| | Neighbour cell | | Cell2 | |
| T2 end condition | Active cell | | Cell2 | UE shall perform reselection to cell 2 during T2 |
| | Neighbour cell | | Cell1 | |
| E-UTRA PRACH configuration | | | 4 | As specified in table 5.7.1-2 in TS 36.211 [9] |
| E_UTRA Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | 85 | T1 need to be defined so that cell re-selection reaction time is taken into account. |
| T2 | | s | 25 | T2 need to be defined so that cell re-selection reaction time is taken into account. |

### 4.3.1.2.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The UE is requested to monitor the neighbouring cell on one UTRA FDD carrier. In the test there are two successive time periods, with time duration of T1 and T2 respectively. Both Cell 1 and Cell 2 are already identified by the UE prior to the start of the test. At T1 the UE is camped on to Cell 1. Cell 2 is of lower priority than Cell 1. Cell 1 and Cell 2 belong to different tracking areas. Furthermore, UE has not registered with network for the tracking area containing Cell 2.

In the following test procedure “UE responds” means “UE starts transmitting preamble on PRACH for sending the RRC CONNECTION REQUEST message to perform a Routing Area Update procedure”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2.
2. Set the parameters according to T1 in Table 4.3.1.2.5-1 and 4.3.1.2.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts. If the UE is already camped in Cell 1, wait until T1expires and skip to step 5.
3. The SS waits for random access requests information from the UE to perform cell re-selection on Cell 1.
4. If the UE has re-selected Cell 1 within T1, after the re-selection or when T1 expires, continue with step 5. Otherwise, if T1 expires and the UE has not yet re-selected Cell 1, skip to step 9.
5. The SS shall switch the power setting from T1 to T2 as specified in Table 4.3.1.2.5-1 and 4.3.1.2.5-2.
6. The SS waits for random access requests information from the UE to perform cell re-selection on Cell 2.
7. If the UE responds on Cell 2 during time duration T2 within 21 seconds from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
8. If the UE has re-selected Cell 2 within T2, after the re-selection or when T2 expires, skip to step 10. Otherwise, if T2 expires and the UE has not yet re-selected Cell 2, continue with step 9.
9. Switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.
10. Repeat step 2-9 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

### 4.3.1.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 and TS 34.108 [24] clause 6.1.0b with the following exceptions:

**Table 4.3.1.2.4.3-1: Common Exception messages**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.3-5<br>Table H.2.3-6<br>Table H.2.3-13 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

**Table 4.3.1.2.4.3-2: System Information Block type 19: Inter-RAT E-UTRAN FDD - UTRA FDD is of lower priority cell re-selection**

| Derivation Path: 36.508 [7] clause 4.4.4.1 Table 4.4.4.1-1: System Information Block type 19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SysInfoType19 ::= SEQUENCE { | | | |
| utra-PriorityInfoList SEQUENCE { | | | |
| utra-ServingCell SEQUENCE { | | | |
| priority | 3 | | |
| s-PrioritySearch1 | 21 (42dB) | | |
| s-PrioritySearch2 | 0 | default value is 0 | |
| threshServingLow | 0 (0dB) | | |
| } | | | |
| eutra-FrequencyAndPriorityInfoList SEQUENCE (SIZE (1..maxNumEUTRAFreqs)) OF SEQUENCE | | *n* denotes the index of the entry | |
| earfcn | | Downlink EARFCN of Cell 1 | |
| measurementBandwidth | 50 | Enumerated(6, 15, 25, 50, 75, 100) | |
| priority | 4 | | |
| qRxLevMinEUTRA | -70 (-140 dBm) | | |
| threshXhigh | 24 (48 dB) | | |
| threshXlow | 1 (2 dB) | Default value | |
| eutra-blackListedCellList | Not present | | |
| eutraDetection | TRUE | | |
| } | | | |
| nonCriticalExtensions SEQUENCE {} | Not present | | |
| } | | | |

## 4.3.1.2.5 Test requirement

Tables 4.3.1.2.4.1-1, 4.3.1.2.5-1 and 4.3.1.2.5-2 define the primary level settings including test tolerances for E-UTRAN FDD- UTRAN FDD inter-RAT cell re-selection test case which UTRA is of lower priority.

**Table 4.3.1.2.5-1: Cell specific test parameters for cell 1 (E-UTRA)**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.1.2 (OP.2 FDD) | | OP.2 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| Qqualmin for UTRA neighbour cell | dB | -20 | |
| Qrxlevmin for UTRA neighbour cell | dBm | -115 | |
| Qrxlevmin | dBm | -140 | |
| $N_{oc}$ | dBm/15 kHz | -99.35 | |
| RSRP | dBm/15 KHz | -84.95 | -103.05 |
| $\hat{E}_s / I_{ot}$ | dB | 14.40 | -3.70 |
| $\hat{E}_s / N_{oc}$ | dB | 14.40 | -3.70 |
| $Treselection_{EUTRAN}$ | s | 0 | |
| Snonintrasearch | dB | Not sent | |
| $Thresh_{serving, low}$ | dB | 44 | |
| $Thresh_{x, low}$ (Note 2) | dB | 42 | |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: This refers to the value of $Thresh_{x, low}$ which is included in E-UTRA system information, and is a threshold for the UTRA target cell.

**Table 4.3.1.2.5-2: Cell specific test parameters for Cell 2(Lower priority UTRA FDD)**

| Parameter | Unit | Cell 2 (UTRA) | |
|---|---|---|---|
| | | T1 | T2 |
| UTRA RF Channel Number | | Channel 2 | |
| CPICH_Ec/Ior | dB | -10 | |
| PCCPCH_Ec/Ior | dB | -12 | |
| SCH_Ec/Ior | dB | -12 | |
| PICH_Ec/Ior | dB | -15 | |
| OCNS_Ec/Ior | dB | -0.941 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 13.80 | 13.80 |
| $I_{oc}$ | dBm/3,84 MHz | -70 | |
| CPICH_Ec/Io | dB | -10.18 | -10.18 |
| CPICH_RSCP | dBm | -66.20 | -66.20 |
| Propagation Condition | | AWGN | |
| Qqualmin | dB | -20 | |
| Qrxlevmin | dBm | -115 | |
| QrxlevminEUTRA | dBm | -140 | |
| UE_TXPWR_MAX_RACH | dBm | 21 | |
| Treselection | s | 0 | |
| Sprioritysearch1 | dB | 42 | |
| Sprioritysearch2 | dB | 0 | |
| $Thresh_{x, high}$ (Note 1) | dB | 48 | |
| Note 1: This refers to the value of $Thresh_{x, high}$ which is included in UTRA system information, and is a threshold for the E-UTRA target cell. | | | |

The cell reselection delay to lower priority is defined as the time from the beginning of time period T2, to the moment when the UE camps on cell 2, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message on cell 2.

The cell re-selection delay to lower priority shall be less than 21 s.

NOTE: The cell re-selection delay to lower priority cell can be expressed as: $T_{evaluateUTRA_FDD} + T_{SI-UTRA}$

Where:

$T_{evaluateUTRA-FDD}$ See Table 4.2.2.5.1-1 of TS 36.133 [4] clause 4.2.2.5.1

$T_{SI-UTRA}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell; 1280 ms is assumed in this test case.

This gives a total of 20.48 s for lower priority cell reselection, allow 21 s.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 4.3.1.3 E-UTRAN FDD - UTRAN FDD cell re-selection in fading propagation conditions: UTRA FDD is of lower priority

### 4.3.1.3.1 Test purpose

To verify that the UE is able to search and measure neighbouring UTRA FDD cells and compare to the E-UTRA serving cell to meet the inter-RAT cell re-selection requirements when the UTRA is of lower priority. The E-UTRA cell is in fading propagation conditions and the UTRA cell is in AWGN propagation conditions.

### 4.3.1.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support UTRA FDD.

#### 4.3.1.3.3 Minimum conformance requirements

When the measurement rules indicate that UTRA FDD cells are to be measured, the UE shall measure CPICH Ec/Io and CPICH RSCP of detected UTRA FDD cells in the neighbour frequency list at the minimum measurement rate. The parameter $N_{UTRA_carrier_FDD}$ is the number of carriers used in the neighbour frequency list. The UE shall filter CPICH Ec/Io and CPICH RSCP measurements of each measured UTRA FDD cell using at least 2 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least half the minimum specified measurement period.

If the $S_{ServingCell}$ of the E-UTRA serving cell is greater than $S_{nonintrasearch}$ then the UE shall search for inter-RAT layers of higher priority at least every $T_{higher_priority_search}$ where $T_{higher_priority_search}$ is described in TS 36.133 [4] clause 4.2.2 as $T_{higher_priority_search}$ = (60 * $N_{layers}$) seconds, where the parameters $N_{layers}$ is the total number of configured higher priority carrier frequencies.

If the $S_{ServingCell}$ of the E-UTRA serving cell is less than or equal to $S_{nonintrasearch}$ then the UE shall search for and measure inter-RAT layers of higher or lower priority in preparation for possible re-selection. In this scenario, the minimum rate at which the UE is required to search for and measure higher priority inter-RAT layers shall be the same as that defined below for lower priority RATs.

The UE shall evaluate whether newly detectable UTRA FDD cells have met the re-selection criteria in TS 36.304 [6] within time $N_{UTRA_carrier_FDD}$ * $T_{detectUTRA_FDD}$ (as defined in table 4.2.2.5.1-1 of TS 36.133 [4] clause 4.2.2.5.1) when the $S_{ServingCell}$ of the E-UTRA serving cell is greater than $S_{nonintrasearch}$ when $Treselection_{RAT}$ = 0 provided that the re-selection criteria is met by a margin of at least 6 dB.

Cells which have been detected shall be measured at least every $N_{UTRA_carrier}$ * $T_{measureUTRA_FDD}$ when the $S_{ServingCell}$ of the E-UTRA serving cell is less than $S_{nonintrasearch}$.

When higher priority UTRA FDD cells are found by the higher priority search, they shall be measured at least every $T_{measureUTRA_FDD}$. If, after it detecting a cell in a higher priority search, it is determined that reselection has not occurred then the UE is not required to continuously measure the detected cell to evaluate the ongoing possibility of re-selection. However, the minimum measurement filtering requirements specified later in this section shall still be met by the UE before it makes any determination that it may stop measuring the cell.

For a cell that has been already detected, but that has not been re-selected to, the filtering shall be such that the UE shall be capable of evaluating that an already identified UTRA FDD cell has met re-selection criterion defined in TS 36.304 [6] within $N_{UTRA_carrier}$ * $T_{evaluateUTRA_FDD}$ when $Treselection_{RAT}$ = 0 as specified in table 4.2.2.5.1-1 of TS 36.133 [4] clause 4.2.2.5.1 provided that the reselection criteria is met by a margin of at least 6dB.

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.5.1 and A.4.3.1.3.

#### 4.3.1.3.4 Test description

##### 4.3.1.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.26.
2. The general test parameter settings are set up according to Table 4.3.1.3.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.3.1.3.4.3.
5. There is one E-UTRA FDD cell and one UTRA FDD cell specified in the test. Cell 1 is the cell used for registration according to TS 36.508 [7] clause 7.2A.2 with the power level set according to Annex C.0 and C.1.

**Table 4.3.1.3.4.1-1: General test parameters for EUTRA FDD- lower priority UTRA FDD inter RAT cell re-selection test case in fading conditions**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | E-UTRAN Cell. |
| T1 end condition | Active cells | | Cell1 | UE shall perform reselection to cell 1 during T1 for subsequent iterations of the test. |
| | Neighbour cell | | Cell2 | |
| T3 end condition | Active cell | | Cell2 | UE shall perform reselection to cell 2 during T3 |
| | Neighbour cell | | Cell1 | |
| E-UTRA PRACH configuration | | | 4 | As specified in table 5.7.1-2 in TS 36.211 [9] |
| E_UTRA Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | <85 | T1 need to be defined so that cell re-selection reaction time is taken into account. T1 is terminated when the UE starts to send preambles to cell 1 |
| T2 | | s | 64 | The start of T2 is defined as the time when the UE starts to send PRACH preambles to cell 1 |
| T3 | | s | <25 | T3 need to be defined so that cell re-selection reaction time is taken into account. T3 is terminated when the UE starts to send preambles to cell 2 |
| T4 | | s | 64 | The start of T4 is defined as the time when the UE starts to send PRACH preambles to cell 2 |

### 4.3.1.3.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The UE is requested to monitor the neighbouring cell on one UTRA FDD carrier. In the test there are four successive time periods, with time duration of T1, T2, T3 and T4 respectively. Both Cell 1 and Cell 2 are already identified by the UE prior to the start of the test. At T1 the UE is camped on to Cell 1. Cell 2 is of lower priority than Cell 1. Cell 1 and Cell 2 belong to different tracking areas. Furthermore, UE has not registered with network for the tracking area containing Cell 2. Time duration T2 and T4 are not used for cell re-selection in the test.

In the following test procedure, “UE responds on Cell 1” means “UE starts transmitting preamble on PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure according to TS 36.508 [7] clause 4.5A.2” and “UE responds on Cell 2” means “UE starts transmitting preamble on PRACH for sending the RRC CONNECTION REQUEST message to perform a Routing Area Update procedure”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2.
2. Set the parameters according to duration T0 in Table 4.3.1.3.5-1 and 4.3.1.3.5-2.
3. Set the parameters according to T2 in Table 4.3.1.3.5-3 and 4.3.1.3.5-4. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T2 starts.
4. The SS monitors for possible random access request information from the UE to perform cell re-selection on Cell 2.
5. If the UE responds on Cell 2 within T2, then count a fail for the event “Hold out on Cell 1” and skip to step 10. Otherwise, count a success for the event “Hold out on Cell 1” and after T2 expires continue with step 6.
6. The SS shall switch the power setting from T2 to T3 as specified in Table 4.3.1.3.5-3 and 4.3.1.3.5-4.
7. The SS waits for random access request information from the UE to perform cell re-selection on Cell 2.
8. If the UE responds on Cell 2 during time duration T3 within 21 seconds from the beginning of time period T3, then count a success for the event “Re-select lower priority Cell 2”.
   Otherwise, count a fail for the event “Re-select lower priority Cell 2”.

9. If the UE responds on Cell 2 within T3, at the moment of the request-reception continue with step 10. Otherwise, after T3 expires skip to step 17.

10. The SS shall switch the power setting from T3 to T4 as specified in Table 4.3.1.3.5-3 and 4.3.1.3.5-4.

11. The SS monitors for possible random access request information from the UE to perform cell re-selection on Cell 1.

12. If the UE responds on Cell 1 within T4, then count a fail for the event “Hold out on Cell 2” and skip to step 17. Otherwise, count a success for the event “Hold out on Cell 2” and after T4 expires continue with step 13.

13. The SS shall switch the power setting from T4 to T1 as specified in Table 4.3.1.3.5-3 and 4.3.1.3.5-4.

14. The SS waits for random access requests information from the UE to perform cell re-selection on Cell 1.

15. If the UE responds on Cell 1 within T1, at the moment of the request-reception skip to step 17. Otherwise, after T1 expires continue with step 16.

16. Switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1. Set the parameters according to duration T0 in Table 4.3.1.3.5-1 and 4.3.1.3.5-2.

17. Repeat step 3-16 until a test verdict has been achieved.
    Each of the events “Hold out on Cell 1”, “Re-select lower priority Cell 2” and “Hold out on Cell 2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. Different events may require different times for a verdict.
    If both events pass, the test passes. If one event fails, the test fails.

#### 4.3.1.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 and TS 34.108 [24] clause 6.1.0b with the following exceptions:

**Table 4.3.1.3.4.3-1: Common Exception messages**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.3-5<br>Table H.2.3-6<br>Table H.2.3-13 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

**Table 4.3.1.3.4.3-2: System Information Block type 19: Inter-RAT E-UTRAN FDD - UTRA FDD is of lower priority cell re-selection in fading conditions**

| Derivation Path: 36.508 [7] clause 4.4.4.1 Table 4.4.4.1-1: System Information Block type 19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SysInfoType19 ::= SEQUENCE { | | | |
| utra-PriorityInfoList SEQUENCE { | | | |
| utra-ServingCell SEQUENCE { | | | |
| priority | 3 | | |
| s-PrioritySearch1 | 21 (42dB) | | |
| s-PrioritySearch2 | 0 | default value is 0 | |
| threshServingLow | 0 (0dB) | | |
| } | | | |
| eutra-FrequencyAndPriorityInfoList SEQUENCE (SIZE (1..maxNumEUTRAFreqs)) OF SEQUENCE | | *n* denotes the index of the entry | |
| earfcn | | Downlink EARFCN of Cell 1 | |
| measurementBandwidth | 50 | Enumerated(6, 15, 25, 50, 75, 100) | |
| priority | 4 | | |
| qRxLevMinEUTRA | -70 (-140 dBm) | | |
| threshXhigh | 22 (44 dB) | | |
| threshXlow | 1 (2 dB) | Default value | |
| eutra-blackListedCellList | Not present | | |
| eutraDetection | TRUE | | |
| } | | | |
| nonCriticalExtensions SEQUENCE {} | Not present | | |
| } | | | |

### 4.3.1.3.5 Test requirement

Tables 4.3.1.3.4.1-1, 4.3.1.3.5-1, 4.3.1.3.5-2, 4.3.1.3.5-3 and 4.3.1.3.5-4 define the primary level settings including test tolerances for E-UTRAN FDD- UTRAN FDD inter-RAT cell re-selection test case which UTRA is of lower priority. Note that the time period for T0 is system implementation dependent.

**Table 4.3.1.3.5-1: Cell specific test parameters for Cell 1 (Lower priority E-UTRA FDD)**

| Parameter | Unit | Cell 1 |
|---|---|---|
| | | T0 |
| E-UTRA RF Channel number | | 1 |
| $BW_{channel}$ | MHz | 10 |
| Antenna Configuration | | 1x2 |
| OCNG Patterns defined in D.1.2 (OP.2 FDD) | | OP.2 FDD |
| PSS_RA | dB | 0 |
| SSS_RA | dB | 0 |
| PCFICH_RB | dB | 0 |
| PHICH_RA | dB | 0 |
| PHICH_RB | dB | 0 |
| PDCCH_RA | dB | 0 |
| PDCCH_RB | dB | 0 |
| PDSCH_RA | dB | 0 |
| PDSCH_RB | dB | 0 |
| OCNG_RA Note 1 | dB | 0 |
| OCNG_RB Note 1 | dB | 0 |
| Qqualmin for UTRA neighbour cell | dB | -20 |
| Qrxlevmin for UTRA neighbour cell | dBm | -115 |
| Qrxlevmin | dBm | -140 |
| $N_{oc}$ | dBm/15 kHz | -104 |
| RSRP | dBm/15 KHz | -82 |
| $\hat{E}_s / I_{ot}$ | dB | 22 |
| $\hat{E}_s / N_{oc}$ | dB | 22 |
| $Treselection_{EUTRAN}$ | s | 0 |
| Snonintrasearch | dB | Not sent |
| $Thresh_{serving, low}$ | dB | 44 |
| $Thresh_{x, low}$ (Note 2) | dB | 42 |
| Propagation Condition | | AWGN |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: This refers to the value of $Thresh_{x, low}$ which is included in E-UTRA system information, and is a threshold for the UTRA target cell.

**Table 4.3.1.3.5-2: Cell specific test parameters for Cell 2 (Lower priority UTRA FDD)**

| Parameter | Unit | Cell 2 (UTRA) |
|---|---|---|
| | | T0 |
| UTRA RF Channel Number | | Channel 2 |
| CPICH_Ec/Ior | dB | -10 |
| PCCPCH_Ec/Ior | dB | -12 |
| SCH_Ec/Ior | dB | -12 |
| PICH_Ec/Ior | dB | -15 |
| OCNS_Ec/Ior | dB | -0.941 |
| $\hat{I}_{or}/I_{oc}$ | dB | 13.80 |
| $I_{oc}$ | dBm/3,84 MHz | -70 |
| CPICH_Ec/Io | dB | -10.18 |
| CPICH_RSCP | dBm | -66.20 |
| Propagation Condition | | AWGN |
| Qqualmin | dB | -20 |
| Qrxlevmin | dBm | -115 |
| QrxlevminEUTRA | dBm | -140 |
| UE_TXPWR_MAX_RACH | dBm | 21 |
| Treselection | s | 0 |
| Sprioritysearch1 | dB | 42 |
| Sprioritysearch2 | dB | 0 |
| $Thresh_{x, high}$ (Note 1) | dB | 44 |
| Note 1: This refers to the value of $Thresh_{x, high}$ which is included in UTRA system information, and is a threshold for the E-UTRA target cell | | |

**Table 4.3.1.3.5-3: Cell specific test parameters for cell 1 (E-UTRA)**

| Parameter | Unit | Cell 1 | | | |
|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 |
| E-UTRA RF Channel number | | 1 | | | |
| $BW_{channel}$ | MHz | 10 | | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | |
| OCNG Patterns defined in D.1.2 (OP.2 FDD) | | OP.2 FDD | | | |
| PSS_RA | dB | 0 | | | |
| SSS_RA | dB | 0 | | | |
| PCFICH_RB | dB | 0 | | | |
| PHICH_RA | dB | 0 | | | |
| PHICH_RB | dB | 0 | | | |
| PDCCH_RA | dB | 0 | | | |
| PDCCH_RB | dB | 0 | | | |
| PDSCH_RA | dB | 0 | | | |
| PDSCH_RB | dB | 0 | | | |
| OCNG_RA Note 1 | dB | 0 | | | |
| OCNG_RB Note 1 | dB | 0 | | | |
| Qqualmin for UTRA neighbour cell | dB | -20 | | | |
| Qrxlevmin for UTRA neighbour cell | dBm | -115 | | | |
| Qrxlevmin | dBm | -140 | | | |
| $N_{oc}$ | dBm/15 kHz | -104 | | | |
| RSRP | dBm/15 KHz | -82 | -82 | -107 | -107 |
| $\hat{E}_s / I_{ot}$ | dB | 22 | 22 | -3 | -3 |
| $\hat{E}_s / N_{oc}$ | dB | 22 | 22 | -3 | -3 |
| $Treselection_{EUTRAN}$ | s | 0 | | | |
| Snonintrasearch | dB | Not sent | | | |
| $Thresh_{serving, low}$ | dB | 44 | | | |
| $Thresh_{x, low}$ (Note 2) | dB | 42 | | | |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total t
spectral density is achieved for all OFDM symbols.

Note 2: This refers to the value of $Thresh_{x, low}$ which is included in E-UTRA system inform
threshold for the UTRA target cell.

**Table 4.3.1.3.5-4: Cell specific test parameters for Cell 2 (Lower priority UTRA FDD)**

| Parameter | Unit | Cell 2 (UTRA) | | | |
|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 |
| UTRA RF Channel Number | | Channel 2 | | | |
| CPICH_Ec/Ior | dB | -10 | | | |
| PCCPCH_Ec/Ior | dB | -12 | | | |
| SCH_Ec/Ior | dB | -12 | | | |
| PICH_Ec/Ior | dB | -15 | | | |
| OCNS_Ec/Ior | dB | -0.941 | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 13.80 | 13.80 | 13.80 | 13.80 |
| $I_{oc}$ | dBm/3,84 MHz | -70 | | | |
| CPICH_Ec/Io | dB | -10.18 | -10.18 | -10.18 | -10.18 |
| CPICH_RSCP | dBm | -66.20 | -66.20 | -66.20 | -66.20 |
| Propagation Condition | | AWGN | | | |
| Qqualmin | dB | -20 | | | |
| Qrxlevmin | dBm | -115 | | | |
| QrxlevminEUTRA | dBm | -140 | | | |
| UE_TXPWR_MAX_RACH | dBm | 21 | | | |
| Treselection | s | 0 | | | |
| Sprioritysearch1 | dB | 42 | | | |
| Sprioritysearch2 | dB | 0 | | | |
| $Thresh_{x, high}$ (Note 1) | dB | 44 | | | |
| Note 1: This refers to the value of $Thresh_{x, high}$ which is included in UTRA system information, and is a threshold for the E-UTRA target cell | | | | | |

The probability of re-selection from Cell 1 to Cell 2 during T2 observed during testing shall be less than 10%.

The probability of re-selection from Cell 2 to Cell 1 during T4 observed during testing shall be less than 10%.

The cell reselection delay to lower priority is defined as the time from the beginning of time period T3, to the moment when the UE camps on Cell 2 and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message on Cell 2. In order to evaluate re-selection delay, the SS first needs to verify that the UE is camped on Cell 1 at the start of T3.

The cell re-selection delay to lower priority shall be less than 21 s.

NOTE: The cell re-selection delay to lower priority cell can be expressed as: $T_{evaluateUTRA_FDD} + T_{SI-UTRA}$

Where:

$T_{evaluateUTRA-FDD}$ See Table 4.2.2.5.1-1 of TS 36.133 [4] clause 4.2.2.5.1

$T_{SI-UTRA}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell; 1280 ms is assumed in this test case.

This gives a total of 20.48 s for lower priority cell reselection, allow 21 s.

For the test to pass, all the events above shall pass.

The statistical pass/ fail decisions are done separated for each event. For an event to pass, the total number of successful loops shall be more than 90% of the cases with a confidence level of 95%.

## 4.3.1.4 E-UTRAN FDD - UTRAN FDD cell re-selection: UTRA FDD is of lower priority for 5MHz bandwidth

### 4.3.1.4.1 Test purpose

Same test purpose as in clause 4.3.1.2.1.

#### 4.3.1.4.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support only E-UTRA Band 31 and UTRA FDD.

#### 4.3.1.4.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 4.3.1.2 with the following exceptions:

- Instead of A.4.3.1.2 → use A.4.3.1.4.

#### 4.3.1.4.4 Test description

##### 4.3.1.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.22.
2. The general test parameter settings are set up according to Table 4.3.1.4.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.3.1.4.4.3.
5. There is one E-UTRA FDD cell and one UTRA FDD cell specified in the test. Cell 1 is the cell used for registration according to TS 36.508 [7] clause 7.2A.2 with the power level set according to Annex C.0 and C.1.

**Table 4.3.1.4.4.1-1: General test parameters for EUTRA FDD- lower priority UTRA FDD inter RAT cell re-selection test case for 5MHz bandwidth**

Same as Table 4.3.1.2.4.1-1.

##### 4.3.1.4.4.2 Test procedure

Same test procedure as in clause 4.3.1.2.4.2 with the following exceptions:

- Instead of Table 4.3.1.2.5-1 → use Table 4.3.1.4.5-1.

##### 4.3.1.4.4.3 Message contents

Same message contents as 4.3.1.2.4.3 with the following exceptions:

**Table 4.3.1.4.4.3-1: System Information Block type 19: Inter-RAT E-UTRAN FDD - UTRA FDD is of lower priority cell re-selection for 5MHz bandwidth**

| Derivation Path: 36.508 [7] clause 4.4.4.1 Table 4.4.4.1-1: System Information Block type 19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SysInfoType19 ::= SEQUENCE { | | | |
| utra-PriorityInfoList SEQUENCE { | | | |
| utra-ServingCell SEQUENCE { | | | |
| priority | 3 | | |
| s-PrioritySearch1 | 21 (42dB) | | |
| s-PrioritySearch2 | 0 | default value is 0 | |
| threshServingLow | 0 (0dB) | | |
| } | | | |
| eutra-FrequencyAndPriorityInfoList SEQUENCE (SIZE (1..maxNumEUTRAFreqs)) OF SEQUENCE | | *n* denotes the index of the entry | |
| earfcn | | Downlink EARFCN of Cell 1 | |
| measurementBandwidth | 25 | Enumerated(6, 15, 25, 50, 75, 100) | |
| priority | 4 | | |
| qRxLevMinEUTRA | -70 (-140 dBm) | | |
| threshXhigh | 24 (48 dB) | | |
| threshXlow | 1 (2 dB) | Default value | |
| eutra-blackListedCellList | Not present | | |
| eutraDetection | TRUE | | |
| } | | | |
| nonCriticalExtensions SEQUENCE {} | Not present | | |
| } | | | |

### 4.3.1.4.5 Test requirement

Same test requirement as in clause 4.3.1.2.5 with the following exceptions:

Tables 4.3.1.4.4.1-1, 4.3.1.4.5-1 and 4.3.1.4.5-2 define the primary level settings including test tolerances for E-UTRAN FDD- UTRAN FDD inter-RAT cell re-selection test case which UTRA is of lower priority.

**Table 4.3.1.4.5-1: Cell specific test parameters for cell 1 (E-UTRA) for 5MHz bandwidth**

| **Parameter** | **Unit** | **Cell 1** | |
|---|---|---|---|
| | | **T1** | **T2** |
| $BW_{channel}$ | MHz | 5 | |
| OCNG Patterns defined in D.1.16 (OP.16 FDD) | | OP.16 FDD | |
| Note 1: See Table 4.3.1.2.5-1 for the other parameters. | | | |

## 4.3.2 E-UTRAN FDD - UTRAN TDD cell re-selection

### 4.3.2.1 Test purpose

To verify that the UE is able to search and measure neighbouring UTRAN TDD cells and compare to the E-UTRA serving cell to meet the inter-RAT cell re-selection requirements when the UTRA cell is of lower priority.

### 4.3.2.2 Test applicability

This test applies to all types of release 8 and forward E-UTRA FDD UEs that support release 9 and forward UTRA TDD.

### 4.3.2.3 Minimum conformance requirements

#### 4.3.2.3.1 3.84Mcps TDD option

There are no requirements so this is not tested.

#### 4.3.2.3.2 1.28Mcps TDD option

When the measurement rules indicate that UTRA TDD cells are to be measured, the UE shall measure P-CCPCH RSCP of detected UTRA TDD cells in the neighbour frequency list at the minimum measurement rate specified in this section. The parameter $N_{UTRA_carrier_TDD}$ is the number of carriers in the neighbour frequency list. The UE shall filter P-CCPCH RSCP measurements of each measured UTRA TDD cell using at least 2 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least half the minimum specified measurement period. P-CCPCH RSCP of UTRAN TDD cells shall not be filtered over a longer period than that specified in TS 36.133 [4] table 4.2.2.5.2-1.

If the $S_{ServingCell}$of the E-UTRA serving cell is greater than $S_{nonintrasearch}$then the UE shall search for inter-RAT layers of higher priority at least every $T_{higher_priority_search}$ where $T_{higher_priority_search}$ is described in section 36.133 [4] clauses 4.2.2.

If the $S_{ServingCell}$of the E-UTRA serving cell is less than or equal to $S_{nonintrasearch}$then the UE shall search for and measure inter-RAT layers of higher, equal or lower priority in preparation for possible reselection. In this scenario, the minimum rate at which the UE is required to search for and measure higher priority inter-RAT layers shall be the same as that defined below for lower priority RATs.

The UE shall evaluate whether newly detectable UTRA TDD cells have met the reselection criteria in TS 36.304 within time ($N_{UTRA_carrier_TDD}$) * $T_{detectUTRA_TDD}$ when the $S_{ServingCell}$ of the E-UTRA serving cell (or other cells on the same frequency layer) is less than $S_{nonintrasearch}$ when $T_{reselection}$ = 0 provided that the reselection criteria is met by a margin of at least 6dB.

Cells which have been detected shall be measured at least every ($N_{UTRA_carrier_TDD}$) * $T_{measureUTRA_TDD}$ when the $S_{ServingCell}$ of the E-UTRA serving cell is less than $S_{nonintrasearch}$.

When higher priority UTRA TDD cells are found by the higher priority search, they shall be measured at least every $T_{measure,UTRA_TDD}$. If, after detecting a cell in a higher priority search, it is determined that reselection has not occurred then the UE is not required to continuously measure the detected cell to evaluate the ongoing possibility of reselection. However, the minimum measurement filtering requirements specified later in this section shall still be met by the UE before it makes any determination that it may stop measuring the cell.

For a cell that has been already detected, but that has not been reselected to, the filtering shall be such that the UE shall be capable of evaluating that an already identified UTRA TDD cell has met reselection criterion defined in TS 36.304 [6] within $N_{UTRA_carrier_TDD}$ *$T_{evaluateUTRA_TDD}$ when $T_{reselection}$ = 0 provided that the reselection criteria is met by a margin of at least 6 dB.

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.5.2 and A.4.3.2.

#### 4.3.2.3.3 7.68Mcps TDD option

There are no requirements so this is not tested.

### 4.3.2.4 Test description

#### 4.3.2.4.1 3.84Mcps TDD option

There are no requirements so this is not tested.

#### 4.3.2.4.2 1.28Mcps TDD option

##### 4.3.2.4.2.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.

2. The general test parameter settings are set up according to Table 4.3.2.4.2.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are as defined in clause 4.3.2.4.2.3.

5. There is one E-UTRA FDD cell and one UTRA TDD cell specified in the test. Cell 1 is the cell used for registration according to TS 36.508 [7] clause 7.2A.2 with the power level set according to Annex C.0 and C.1.

**Table 4.3.2.4.2.1-1: General test parameters for E-UTRA FDD to UTRA (1.28 Mcps TDD OPTION) Cell Re-selection**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell 1 | E-UTRA FDD cell |
| T1 end condition | Active cell | | Cell1 | UE shall perform reselection to cell 1 during T1 for subsequent iterations of the test. |
| | Neighbour cell | | Cell2 | 1.28 Mcps TDD OPTION cell |
| T2 end condition | Active cell | | Cell2 | UE shall perform reselection to cell 2 during T2 |
| | Neighbour cell | | Cell1 | E-UTRA FDD cell |
| CP length of cell 1 | | | normal | |
| E-UTRA PRACH configuration | | | 4 | As specified in table 5.7.1-2 in TS 36.211 |
| Time offset between cells | | | 3 ms | Asynchronous cells<br>3ms or 92160*Ts |
| Access Barring Information | | - | Not sent | No additional delays in random access procedure. |
| Treselection | | s | 0 | |
| DRX cycle length | | s | 1,28 | |
| HCS | | | Not used | |
| T1 | | s | 85 | T1 need to be defined so that cell re-selection reaction time is taken into account. |
| T2 | | s | 25 | |

### 4.3.2.4.2.2 Test procedure

The test consists of one active cell and one neighbour cell. In the test there are two successive time periods, with time duration of T1 and T2 respectively. Both Cell 1 and Cell 2 are already identified by the UE prior to the test. Cell 1 and Cell 2 belong to different tracking areas. The UTRA TDD layer is configured at a lower priority than the E-UTRA FDD layer.

The ranking of the cells shall be made according to the cell reselection criteria specified in TS 36.304.

In the following test procedure “UE responds” means “UE starts transmitting the SYNCH-UL sequence in the UpPTS for sending RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2.

2. Set the parameters according to T1 in Table 4.3.2.5.2-1 and 4.3.2.5.2-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts. If the UE is already camped in Cell 1, wait until T1expires and skip to step 5.

3. The SS waits for random access requests information from the UE to perform cell re-selection on the higher priority cell, Cell 1.

4. If the UE has re-selected Cell 1 within T1, after the re-selection or when T1 expires, continue with step 5. Otherwise, if T1 expires and the UE has not yet re-selected Cell 1, skip to step 9.

5. The SS shall switch the power setting from T1 to T2 as specified in Table 4.3.2.5.2-1 and 4.3.2.5.2-2.
6. The SS waits for random access requests information from the UE to perform cell re-selection on the lower priority cell, Cell 2.
7. If the UE responds on lower priority cell, Cell 2 during time duration T2 within 21 seconds from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
8. If the UE has re-selected Cell 2 within T2, after the re-selection or when T2 expires, skip to step 10. Otherwise, if T2 expires and the UE has not yet re-selected Cell 2, continue with step 9.
9. Switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.
10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 4.3.2.4.2.3 Message contents

Message contents are according TS 36.508 [7] clause 4.6 and TS 34.108 [24] clause 6.1.0b with the following exceptions:

**Table 4.3.2.4.2.3-1: Common Exception messages**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.3-7<br>Table H.2.3-8<br>Table H.2.3-14 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

**Table 4.3.2.4.2.3-2: System Information Block type 19: E-UTRA FDD- lower priority UTRA TDD inter RAT cell re-selection test case**

| Derivation Path: 36.508 clause 4.4.4.1 Table 4.4.4.1-1: System Information Block type 19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SysInfoType19 ::= SEQUENCE { | | | |
| utra-PriorityInfoList SEQUENCE { | | | |
| utra-ServingCell SEQUENCE { | | | |
| priority | 3 | | |
| s-PrioritySearch1 | 0 (0dB) | | |
| s-PrioritySearch2 | Not present | default value is 0 | |
| threshServingLow | 0 (0dB) | | |
| } | | | |
| eutra-FrequencyAndPriorityInfoList SEQUENCE (SIZE (1..maxNumEUTRAFreqs)) OF SEQUENCE | | *n* denotes the index of the entry | |
| earfcn | | Downlink EARFCN of Cell 1 | |
| measurementBandwidth | 50 | Enumerated(6, 15, 25, 50, 75, 100) | |
| priority | 4 | | |
| qRxLevMinEUTRA | -70 (-140 dBm) | | |
| threshXhigh | 23 (46 dB) | | |
| threshXlow | 1 (2 dB) | Default value | |
| eutra-blackListedCellList | Not present | | |
| eutraDetection | TRUE | | |
| } | | | |
| nonCriticalExtensions SEQUENCE {} | Not present | | |
| } | | | |

### 4.3.2.4.3 7.68 Mcps TDD option

There are no requirements so this is not tested.

### 4.3.2.5 Test requirement

#### 4.3.2.5.1 3.84Mcps TDD option

There are no requirements so this is not tested.

#### 4.3.2.5.2 1.28Mcps TDD option

Tables 4.3.2.4.2-1, 4.3.2.5.2-1 and 4.3.2.5.2-2 defines the primary level settings including test tolerances for E-UTRAN FDD to UTRA TDD cell re-selection test case.

**Table 4.3.2.5.2-1: Cell specific test parameters for cell re-selection E-UTRA FDD to UTRA TDD test case (cell 1)**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| PBCH_RA | dB | 0 | 0 |
| PBCH_RB | dB | | |
| PSS_RB | dB | | |
| SSS_RB | dB | | |
| PCFICH_PA | dB | | |
| PHICH_PA | dB | | |
| PHICH_PB | dB | | |
| PDCCH_PA | dB | | |
| PDCCH_PB | dB | | |
| PDSCH_PA | dB | | |
| PDSCH_PB | dB | | |
| OCNG_RA Note1 | dB | | |
| OCNG_RB Note1 | dB | | |
| $Q_{rxlevmin}$ | dBm/15kHz | -140 | -140 |
| $N_{oc}$ | dBm/15kHz | -98 | |
| RSRP | dBm/15kHz | -86.9 | -101.1 |
| $\hat{E}_s / I_{ot}$ | dB | 11.1 | -3.1 |
| $S_{nonintrasearch}$ | dB | Not sent | |
| $Thresh_{serving, low}$ | dB | 46 (-94dBm) | |
| $Thresh_{x, low}$ (Note2) | dB | 24 (-79dBm) | |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: This refers to the value of Threshx, low which is included in E-UTRA system information, and is a threshold for the UTRA TDD target cell.

**Table 4.3.2.5.2-2: Cell specific test parameters for cell re-selection E-UTRA FDD to UTRA TDD test case (cell 2)**

| Parameter | Unit | Cell 2 (UTRA) | | | |
|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPTS | |
| | | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number (Note1) | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 |
| OCNS_Ec/Ior | dB | -3 | -3 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 11 | 11 | 11 | 11 |
| $I_{oc}$ | dBm/1.28 MHz | -80 | | | |
| PCCPCH RSCP | dBm | -72 | -72 | n.a. | n.a. |
| Propagation Condition | | AWGN | | | |
| $Q_{rxlevmin}$ | dBm | -103 | | | |
| $Qoffset1_{s,n}$ | dB | C1, C2: 0 | | | |
| $Qhyst1_{s}$ | dB | 0 | | | |
| $Thresh_{x, high}$ (Note2) | dB | 46 (-94dBm) | | | |
| Note1: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number.<br>Note 2: This refers to the value of $Thresh_{x, high}$ which is included in UTRA system information, and is a threshold for the E-UTRA target cell. | | | | | |

The cell reselection delay to lower priority is defined as the time from the beginning of time period T2, to the moment when the UE camps on cell 2, and starts to send the SYNCH-UL sequence in the UpPTS for sending the RRC CONNECTION REQUEST message on cell 2.

The cell re-selection delay to lower priority shall be less than 21 s.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

NOTE: The cell re-selection delay to lower priority cell can be expressed as: $T_{evaluateUTRA_TDD} + T_{SI-UTRA}$

Where:

$T_{evaluateUTRA_TDD}$ 19.2s, as specified in TS 36.133 [4] table 4.2.2.5.2-1

$T_{SI-UTRA}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell; 1280 ms is assumed in this test case.

This gives a total of 20.48 s, allow 21 s for lower priority cell reselection in the test case.

#### 4.3.2.5.3 7.68 Mcps TDD option

There are no requirements so this is not tested.

## 4.3.3 E-UTRAN TDD - UTRAN FDD cell re-selection

### 4.3.3.1 Test purpose

To verify that the UE is able to search and measure neighbouring UTRA FDD cells and compare to the E-UTRA serving cell to meet the inter-RAT cell re-selection requirements when the UTRA is of lower priority.

### 4.3.3.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward that support UTRA FDD.

### 4.3.3.3 Minimum conformance requirements

When the measurement rules indicate that UTRA FDD cells are to be measured, the UE shall measure CPICH Ec/Io and CPICH RSCP of detected UTRA FDD cells in the neighbour frequency list at the minimum measurement rate. The parameter $N_{UTRA_carrier_FDD}$ is the number of carriers in the neighbour frequency list. The UE shall filter CPICH Ec/Io and CPICH RSCP measurements of each measured UTRA FDD cell using at least 2 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least half the minimum specified measurement period.

If the $S_{ServingCell}$ of the E-UTRA serving cell is greater than $S_{nonintrasearch}$then the UE shall search for inter-RAT layers of higher priority at least every $T_{higher_priority_search}$ where $T_{higher_priority_search}$ is described in TS 36.133 [4] clause 4.2.2 as $T_{higher_priority_search}$ = (60 * $N_{layers}$) seconds, where the parameter $N_{layers}$ is the total number of configured higher priority carrier frequencies.

If the $S_{ServingCell}$ of the E-UTRA serving cell is less than or equal to $S_{nonintrasearch}$ then the UE shall search for and measure inter-RAT layers of higher or lower priority in preparation for possible re-selection. In this scenario, the minimum rate at which the UE is required to search for and measure higher priority inter-RAT layers shall be the same as that defined below for lower priority RATs.

The UE shall evaluate whether newly detectable UTRA FDD cells have met the re-selection criteria in TS 36.304 [6] within time $N_{UTRA_carrier_FDD}$ * $T_{detectUTRA_FDD}$ (as defined in table 4.2.2.5.1-1 of TS 36.133 [4] clause 4.2.2.5.1) when the $S_{ServingCell}$ of the E-UTRA serving cell is greater than $S_{nonintrasearch}$ when $Treselection_{RAT}$ = 0 provided that the re-selection criteria is met by a margin of at least 6 dB.

Cells which have been detected shall be measured at least every $N_{UTRA_carrier}$ * $T_{measureUTRA_FDD}$ when the $S_{ServingCell}$ of the E-UTRA serving cell is less than $S_{nonintrasearch}$.

When higher priority UTRA FDD cells are found by the higher priority search, they shall be measured at least every $T_{measureUTRA_FDD.}$ If re-selection to any higher priority cell is not triggered within ($T_{evaluateUTRA_FDD}$ + $Treselection_{RAT}$) after it is found in a higher priority search, the UE is not required to continue making measurements of the cell to evaluate the ongoing possibility of re-selection.

For a cell that has been already detected, but that has not been re-selected to, the filtering shall be such that the UE shall be capable of evaluating that an already identified UTRA FDD cell has met re-selection criterion defined in TS 36.304 [6] within $N_{UTRA_carrier}$ * $T_{evaluateUTRA_FDD}$ as defined in table 4.2.2.5.1-1 of TS 36.133 [4] clause 4.2.2.5.1 when $Treselection_{RAT}$ = 0.

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.5.1 and A.4.3.3.

### 4.3.3.4 Test description

#### 4.3.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.22.
2. The general test parameter settings are set up according to Table 4.3.3.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.3.3.4.3.
5. There is one E-UTRA TDD cell and one UTRA FDD cell specified in the test. Cell 1 is the cell used for registration according to TS 36.508 [7] clause 7.2A.2 with the power level set according to Annex C.0 and C.1.

**Table 4.3.3.4.1-1: General test parameters for EUTRA TDD- lower priority UTRA FDD inter RAT cell re-selection test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | E-UTRAN cell |
| T1 end condition | Active cells | | Cell1 | UE shall perform reselection to cell 1 during T1 for subsequent iterations of the test |
| | Neighbour cell | | Cell2 | |
| T2 end condition | Active cell | | Cell2 | UE shall perform reselection to cell 2 during T2 |
| | Neighbour cell | | Cell1 | |
| E-UTRA PRACH configuration | | | 53 | As specified in table 5.7.1-2 in TS 36.211 |
| Uplink-downlink configuration of cell 1 | | | 1 | As specified in table 4.2.2 in TS 36.211 |
| Special subframe configuration of cell 1 | | | 6 | As specified in table 4.2.1 in TS 36.211 |
| E_UTRA Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | 85 | T1 need to be defined so that cell re-selection reaction time is taken into account. |
| T2 | | s | 25 | T2 need to be defined so that cell re-selection reaction time is taken into account. |

### 4.3.3.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The UE is requested to monitor the neighbouring cell on one UTRA FDD carrier. In the test there are two successive time periods, with time duration of T1 and T2 respectively. Both Cell 1 and Cell 2 are already identified by the UE prior to the start of the test. Cell 2 is of lower priority than Cell 1.

In the following test procedure “UE responds” means “UE starts transmitting preamble on PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2.
2. Set the parameters according to T1 in Table 4.3.3.5-1 and 4.3.3.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts. If the UE is already camped in Cell 1, wait until T1 expires and skip to step 5.
3. The SS waits for random access requests information from the UE to perform cell re-selection on Cell 1.
4. If the UE has re-selected Cell 1 within T1, after the re-selection or when T1 expires, continue with step 5. Otherwise, if T1 expires and the UE has not yet re-selected Cell 1, skip to step 9.
5. The SS shall switch the power setting from T1 to T2 as specified in Table 4.3.3.5-1 and 4.3.3.5-2.
6. The SS waits for random access requests information from the UE to perform cell re-selection on Cell 2.
7. If the UE responds on Cell 2 during time duration T2 within 21 seconds from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
8. If the UE has re-selected Cell 2 within T2, after the re-selection or when T2 expires, skip to step 10. Otherwise, if T2 expires and the UE has not yet re-selected Cell 2, continue with step 9.
9. Switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.
10. Repeat step 2-9 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

### 4.3.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 and TS 34.108 [24] clause 6.1.0b with the following exceptions:

**Table 4.3.3.4.3-1: Common Exception messages**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.3-5<br>Table H.2.3-6<br>Table H.2.3-13 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-2 |

**Table 4.3.3.4.3-2: System Information Block type 19: EUTRA TDD- lower priority UTRA FDD inter RAT cell re-selection test case**

| Derivation Path: 36.508 clause 4.4.4.1 Table 4.4.4.1-1: System Information Block type 19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SysInfoType19 ::= SEQUENCE { | | | |
| utra-PriorityInfoList SEQUENCE { | | | |
| utra-ServingCell SEQUENCE { | | | |
| priority | 3 | | |
| s-PrioritySearch1 | 21 (42dB) | | |
| s-PrioritySearch2 | 0 | default value is 0 | |
| threshServingLow | 0 (0dB) | | |
| } | | | |
| eutra-FrequencyAndPriorityInfoList SEQUENCE (SIZE (1..maxNumEUTRAFreqs)) OF SEQUENCE | | *n* denotes the index of the entry | |
| earfcn | | Downlink EARFCN of Cell 1 | |
| measurementBandwidth | 50 | Enumerated(6, 15, 25, 50, 75, 100) | |
| priority | 4 | | |
| qRxLevMinEUTRA | -70 (-140 dBm) | | |
| threshXhigh | 24 (48 dB) | | |
| threshXlow | 1 (2 dB) | Default value | |
| eutra-blackListedCellList | Not present | | |
| eutraDetection | TRUE | | |
| } | | | |
| nonCriticalExtensions SEQUENCE {} | Not present | | |
| } | | | |

## 4.3.3.5 Test requirement

Tables 4.3.3.4.1-1, 4.3.3.5-1 and 4.3.3.5-2 define the primary level settings including test tolerances for E-UTRAN TDD- UTRAN FDD intra frequency cell re-selection test case which UTRA is of lower priority.

**Table 4.3.3.5-1: Cell specific test parameters for Cell 1(E-UTRA TDD)**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.2.1 (OP.2 TDD) | | OP.2 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| Qqualmin for UTRA neighbour cell | dB | -20 | |
| Qrxlevmin for UTRA neighbour cell | dBm | -115 | |
| Qrxlevmin | dBm | -140 | |
| $N_{oc}$ | dBm/15 kHz | -99.35 | |
| RSRP | dBm/15 KHz | -84.95 | -103.05 |
| $\hat{E}_s / I_{ot}$ | dB | 14.40 | -3.70 |
| $\hat{E}_s / N_{oc}$ | dB | 14.40 | -3.70 |
| $Treselection_{EUTRAN}$ | s | 0 | |
| Snonintrasearch | dB | Not sent | |
| $Thresh_{serving, low}$ | dB | 44 | |
| $Thresh_{x, low}$ (Note 2) | dB | 42 | |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: This refers to the value of $Thresh_{x, low}$ which is included in E-UTRA system information, and is a threshold for the UTRA target cell.

**Table 4.3.3.5-2: Cell specific test parameters for Cell 2(Lower priority UTRA FDD)**

| Parameter | Unit | Cell 2 (UTRA) | |
|---|---|---|---|
| | | T1 | T2 |
| UTRA RF Channel Number | | Channel 2 | |
| CPICH_Ec/Ior | dB | -10 | |
| PCCPCH_Ec/Ior | dB | -12 | |
| SCH_Ec/Ior | dB | -12 | |
| PICH_Ec/Ior | dB | -15 | |
| OCNS_Ec/Ior | dB | -0.941 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 13.80 | 13.80 |
| $I_{oc}$ | dBm/3,84 MHz | -70 | |
| CPICH_Ec/Io | dB | -10.18 | -10.18 |
| CPICH_RSCP | dBm | -66.20 | -66.20 |
| Propagation Condition | | AWGN | |
| Qqualmin | dB | -20 | |
| Qrxlevmin | dBm | -115 | |
| QrxlevminEUTRA | dBm | -140 | |
| UE_TXPWR_MAX_RACH | dBm | 21 | |
| Treselection | S | 0 | |
| Sprioritysearch1 | dB | 42 | |
| Sprioritysearch2 | dB | 0 | |
| $Thresh_{x, high}$ (Note 1) | dB | 48 | |
| Note: This refers to the value of $Thresh_{x, high}$ which is included in UTRA system information, and is a threshold for the E-UTRA target cell. | | | |

The cell reselection delay to lower priority is defined as the time from the beginning of time period T2, to the moment when the UE camps on cell 2, and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST message on cell 2.

The cell re-selection delay to lower priority test requirement in this case is expressed as:

Cell re-selection delay to lower priority = $T_{evaluateUTRA_FDD} + T_{SI-EUTRA}$

$T_{evaluateUTRA_FDD}$ = 19.2 s; as specified in TS 36.133 [4] clause 4.2.2.5

$T_{SI-EUTRA}$ = 1280 ms; as specified in TS 36.133 [4] clause 4.2.2.4

The cell re-selection delay to lower priority shall be less than a total of 21 seconds in this test case (note: this gives a total of 20.48 seconds but the test allows 21 seconds).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 4.3.4 E-UTRAN TDD - UTRAN TDD cell re-selection

### 4.3.4.1 E-UTRA TDD - UTRAN TDD cell re-selection: UTRA is of higher priority

#### 4.3.4.1.1 Test purpose

To verify that the UE is able to search and measure neighbouring UTRA TDD cells and compare to the E-UTRA serving cell to meet the inter-RAT cell re-selection requirements when the UTRA is of higher priority.

#### 4.3.4.1.2 Test applicability

This test applies to all types of release 8 and forward E-UTRA TDD UEs that support release 9 and forward UTRA TDD.

#### 4.3.4.1.3 Minimum conformance requirements

When the measurement rules indicate that UTRA TDD cells are to be measured, the UE shall measure P-CCPCH RSCP of detected UTRA TDD cells in the neighbour frequency list at the minimum measurement rate. The parameter $N_{UTRA_carrier_TDD}$ is the number of carriers in the neighbour frequency list. The UE shall filter P-CCPCH RSCP measurements of each measured UTRA TDD cell using at least 2 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least half the minimum specified measurement period.

If the $S_{ServingCell}$ of the E-UTRA serving cell is greater than $S_{nonintrasearch}$ then the UE shall search for inter-RAT layers of higher priority at least every $T_{higher_priority_search}$ where $T_{higher_priority_search}$ is described in TS 36.133 [4] clause 4.2.2 as $T_{higher_priority_search}$ = (60 * $N_{layers}$) seconds, where the parameter $N_{layers}$ is the total number of configured higher priority carrier frequencies.

If the $S_{ServingCell}$ of the E-UTRA serving cell is less than or equal to $S_{nonintrasearch}$ then the UE shall search for and measure inter-RAT layers of higher, or lower priority in preparation for possible re-selection. In this scenario, the minimum rate at which the UE is required to search for and measure higher priority inter-RAT layers shall be the same as that defined below for lower priority RATs.

The UE shall evaluate whether newly detectable UTRA TDD cells have met the re-selection criteria in TS 36.304 [6] within time ($N_{UTRA_carrier_TDD}$) * $T_{detectUTRA_TDD}$ (as defined in table 4.2.2.5.2-1 of TS 36.133 [4] clause 4.2.2.5.2) when the $S_{ServingCell}$ of the E-UTRA serving cell is less than $S_{nonintrasearch}$ when $Treselection_{RAT}$ = 0 provided that the reselection criteria is met by a margin of at least 6dB.

Cells which have been detected shall be measured at least every ($N_{UTRA_carrier_TDD}$) * $T_{measureUTRA_TDD}$ when the $S_{ServingCell}$ of the E-UTRA serving cell is less than $S_{nonintrasearch}$.

When higher priority UTRA TDD cells are found by the higher priority search, they shall be measured at least every $T_{measureUTRA_TDD.}$ If, after detecting a cell in a higher priority search, it is determined that reselection has not occurred then the UE is not required to continuously measure the detected cell to evaluate the ongoing possibility of re-selection. However, the minimum measurement filtering requirements specified later in this section shall still be met by the UE before it makes any determination that it may stop measuring the cell.

For a cell that has been already detected, but that has not been re-selected to, the filtering shall be such that the UE shall be capable of evaluating that an already identified UTRA TDD cell has met re-selection criterion defined in TS 36.304 [6] within $N_{UTRA_carrier_TDD}$ * $T_{evaluateUTRA_TDD}$ when $Treselection_{RAT}$ = 0.as specified in table 4.2.2.5.2-1 of TS 36.133 [4] clause 4.2.2.5.2 provided that the reselection criteria is met by a margin of at least 6dB.

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.5.2 and A.4.3.4.1.

#### 4.3.4.1.4 Test description

##### 4.3.4.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B/eNodeB emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.
2. The general test parameter settings are set up according to Table 4.3.4.1.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.3.4.1.4.3.
5. There is one E-UTRA TDD cell and one UTRA TDD cell specified in the test. Cell 1 is the cell used for registration according to TS 36.508 [7] clause 7.2A.2 with the power level set according to Annex C.0 and C.1.

**Table 4.3.4.1.4.1-1: General test parameters for E-UTRAN to UTRAN (1.28 Mcps TDD OPTION) Cell Re-selection**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell 1 | UE is on cell 1 in the initialisation phase, so that reselection to cell 2 occurs during the first T2 phase |
| T2 end condition | Active cell | | Cell2 | UE shall perform reselection to cell 2 during T2 |
| | Neighbour cell | | Cell1 | |
| T3 end condition | Active cell | | Cell1 | UE shall perform reselection to cell 1 during T3 |
| | Neighbour cell | | Cell2 | |
| Uplink-downlink configuration of cell 1 | | | 1 | As specified in table 4.2.2 in TS 36.211 [9] |
| Special subframe configuration of cell 1 | | | 6 | As specified in table 4.2.1 in TS 36.211 [9] |
| PRACH configuration of cell 1 | | | 53 | As specified in table 5.7.1-3 in TS 36.211 [9] |
| CP length of cell 1 | | | Normal | |
| Time offset between cells | | | 3 ms | Asynchronous cells<br>3ms or 92160*Ts |
| Access Barring Information | | - | Not sent | No additional delays in random access procedure. |
| Treselection | | s | 0 | |
| DRX cycle length | | s | 1,28 | |
| HCS | | | Not used | |
| T1 | | s | >20 | During T1, cell 2 shall be powered off, and during the off time the primary scrambling code shall be changed, The intention is to ensure that cell 2 has not been detected by the UE prior to the start of period T2. |
| T2 | | s | 85 | T2 needs to be defined so that cell re-selection reaction time is taken into account. |
| T3 | | s | 25 | T3 needs to be defined so that cell re-selection reaction time is taken into account. |

### 4.3.4.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. E-UTRA cell 1 is already identified by the UE prior to the start of the test. Before T1 the UE is camped on to cell 1. During T1, cell 2 shall be powered off, and during the off time the cell id shall be changed. At starting T2 cell 2 becomes stronger than $Thresh_{x_high}$, the UE is expected to detect cell 2, send preambles on the PRACH for sending the RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure on cell 2. At the start of T3, cell 2 becomes weaker than $Thresh_{serving_low}$, and the UE reselects to Cell 1.

In the following test procedure “UE responds” means “UE starts transmitting the SYNCH-UL sequence in the UpPTS for sending RRC CONNECTION REQUEST message to perform a Routing Area Update procedure”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2.

2. Set the parameters according to duration T1 in Table 4.3.4.1.5-1and 4.3.4.1.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. During T1, cell 2 shall be powered off and the SS shall set Cell 2 cell parameter id = (current cell 2 cell parameter id +4) mod 16.

4. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 4.3.4.1.5-1 and 4.3.4.1.5-2. The SS waits for random access requests information from the UE to perform cell re-selection procedure on the higher priority cell, Cell 2.

5. If the UE responds on Cell 2 within 81s from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

6. If the UE has re-selected Cell 2 within T2, after the re-selection or when T2 expires, continue with step 7. Otherwise, if T2 expires and the UE has not yet re-selected Cell 2, skip to step 10.

7. The SS shall switch the power setting from T2 to T3 as specified in Table 4.3.4.1.5-1 and 4.3.4.1.5-2.

8. The SS waits for random access requests information from the UE to perform cell re-selection procedure on the lower priority cell, Cell 1.

9. If the UE has re-selected Cell 1 within T3, after the re-selection or when T3 expires, skip to step 11. Otherwise, if T3 expires and the UE has not yet re-selected Cell 1, continue with step 10.

10. Switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.

11. Repeat step 2-10 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

#### 4.3.4.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 and TS 34.108 [24] clause 6.1.0b with the following exceptions:

**Table 4.3.4.1.4.3-1: Common Exception messages**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.3-3<br>Table H.2.3-4<br>Table H.2.3-14 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-2 |

**Table 4.3.4.1.4.3-2: System Information Block type 19: E-UTRA TDD- higher priority UTRA TDD inter RAT cell re-selection test case**

| Derivation Path: 36.508 clause 4.4.4.1 Table 4.4.4.1-1: System Information Block type 19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SysInfoType19 ::= SEQUENCE { | | | |
| utra-PriorityInfoList SEQUENCE { | | | |
| utra-ServingCell SEQUENCE { | | | |
| priority | 5 | | |
| s-PrioritySearch1 | 31 (62dB) | | |
| s-PrioritySearch2 | Not present | default value is 0 | |
| threshServingLow | 12 (24dB) | | |
| } | | | |
| eutra-FrequencyAndPriorityInfoList SEQUENCE (SIZE (1..maxNumEUTRAFreqs)) OF SEQUENCE | | *n* denotes the index of the entry | |
| earfcn | | Downlink EARFCN of Cell 1 | |
| measurementBandwidth | 50 | Enumerated(6, 15, 25, 50, 75, 100) | |
| priority | 4 | | |
| qRxLevMinEUTRA | -70 (-140 dBm) | | |
| threshXhigh | 2 (4 dB) | | |
| threshXlow | 23 (46 dB) | Default value | |
| eutra-blackListedCellList | Not present | | |
| eutraDetection | TRUE | | |
| } | | | |
| nonCriticalExtensions SEQUENCE {} | Not present | | |
| } | | | |

#### 4.3.4.1.5 Test requirement

Tables 4.3.4.1.4-1, 4.3.4.1.5-1 and 4.3.4.1.5-2 define the primary level settings including test tolerances for cell re-selection E-UTRA TDD to UTRA TDD test case (UTRA is of higher priority).

**Table 4.3.4.1.5-1: Cell specific test requirement parameters for cell re-selection E-UTRA TDD to UTRA TDD test case (cell 1)**

| Parameter | Unit | Cell 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| OCNG Patterns defined in D.2.2 (OP.2 TDD) | | OP.2 TDD | | |
| PBCH_RA | dB | 0 | 0 | 0 |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $Q_{rxlevmin}$ | dBm | -140 | -140 | -140 |
| $N_{oc}$ | dBm/15kHz | -98 | | |
| RSRP | dBm/15kHz | -86.9 | -86.9 | -86.9 |
| $\hat{E}_s / I_{ot}$ | dB | 11.1 | 11.1 | 11.1 |
| $Thresh_{x, high}$ (Note2) | dB | 24(-79dBm) | | |
| $S_{nonintrasearch}$ | dB | 46 | | |
| Propagation Condition | | AWGN | | |

Note 1: OCNG shall be used such that cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: This refers to the value of $Thresh_{x, high}$ which is included in E-UTRA system information, and is a threshold for the UTRA target cell.

**Table 4.3.4.1.5-2: Cell specific test requirement parameters for cell re-selection E-UTRA TDD to UTRA TDD test case (cell 2)**

| Parameter | Unit | Cell 2 (UTRA) | | | | | |
|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | | DwPTS | | |
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| UTRA RF Channel Number (Note1) | | Channel 2 | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | -3 | | | |
| DwPCH_Ec/Ior | dB | | | | 0 | 0 | 0 |
| OCNS_Ec/Ior | dB | -3 | -3 | -3 | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -inf | 11 | -3 | -inf | 11 | -3 |
| $I_{oc}$ | dBm/1.28 MHz | -80 | | | | | |
| PCCPCH RSCP | dBm | -inf | -72 | -86 | n.a. | | |
| Propagation Condition | | AWGN | | | | | |
| $Q_{rxlevmin}$ | dBm | -103 | | | | | |
| $Qoffset_{s,n}$ | dB | C1, C2: 0 | | | | | |
| $Qhyst_{s}$ | dB | 0 | | | | | |
| $S_{nonintrasearch}$ | dB | Not sent | | | | | |
| $Thresh_{serving, low}$ | dB | 24 (-79dBm) | | | | | |
| $Thresh_{x, low}$ (Note2) | dB | 46 (-94dBm) | | | | | |
| Note 1: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number.<br>Note 2: This refers to the value of Threshx, low which is included in UTRA system information, and is a threshold for the E-UTRA target cell. | | | | | | | |

The cell reselection delay to higher priority is defined as the time from the beginning of time period T2, to the moment when the UE camps on cell 2, and starts to send the SYNCH-UL sequence in the UpPTS for sending the RRC CONNECTION REQUEST message on cell 2.

The cell re-selection delay to higher priority shall be less than 81 s.

The rate of correct cell reselections observed during repeated tests shall be at least 90%.

NOTE: The cell re-selection delay to higher priority cell can be expressed as: $T_{higher_priority_search} + T_{evaluateUTRA_TDD} + T_{SI_UTRA}$,

Where:

$T_{higher_priority_search}$ 60s, See TS 36.133 [4] section 4.2.2.5

$T_{evaluateUTRA_TDD}$ 19.2s, See TS 36.133 [4] Table 4.2.2.5.2-1

$T_{SI_UTRA}$ Maximum repetition period of relevant system info blocks that need to be received by the UE to camp on a cell; 1280 ms is assumed in this test case.

This gives a total of 80.48 s, allow 81 s for higher priority cell reselection in the test case.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 4.3.4.2 E-UTRAN TDD - UTRAN TDD cell re-selection: UTRA is of lower priority

### 4.3.4.2.1 Test purpose

To verify that the UE is able to search and measure neighbouring UTRA TDD cells and compare to the E-UTRA serving cell to meet the inter-RAT cell re-selection requirements when the UTRA is of lower priority.

#### 4.3.4.2.2 Test applicability

This test applies to all types of release 8 and forward E-UTRA TDD UEs that support release 9 and forward UTRA TDD.

#### 4.3.4.2.3 Minimum conformance requirements

When the measurement rules indicate that UTRA TDD cells are to be measured, the UE shall measure P-CCPCH RSCP of detected UTRA TDD cells in the neighbour frequency list at the minimum measurement rate. The parameter $N_{UTRA_carrier_TDD}$ is the number of carriers in the neighbour frequency list. The UE shall filter P-CCPCH RSCP measurements of each measured UTRA TDD cell using at least 2 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least half the minimum specified measurement period.

If the $S_{ServingCell}$ of the E-UTRA serving cell is greater than $S_{nonintrasearch}$then the UE shall search for inter-RAT layers of higher priority at least every $T_{higher_priority_search}$ where $T_{higher_priority_search}$ is described in TS 36.133 [4] clause 4.2.2 as $T_{higher_priority_search}$ = (60 * $N_{layers}$) seconds, where the parameter $N_{layers}$ is the total number of configured higher priority carrier frequencies.

If the $S_{ServingCell}$ of the E-UTRA serving cell is less than or equal to $S_{nonintrasearch}$ then the UE shall search for and measure inter-RAT layers of higher, or lower priority in preparation for possible re-selection. In this scenario, the minimum rate at which the UE is required to search for and measure such layers is not reduced and shall be the same as that defined below for lower priority RATs.

The UE shall evaluate whether newly detectable UTRA TDD cells have met the re-selection criteria in TS 36.304 [6] within time ($N_{UTRA_carrier_TDD}$) * $T_{detectUTRA_TDD}$ (as defined in table 4.2.2.5.2-1 of TS 36.133 [4] clause 4.2.2.5.2) when the $S_{ServingCell}$ of the E-UTRA serving cell is less than $S_{nonintrasearch}$ when $Treselection_{RAT}$ = 0 provided that the reselection criteria is met by a margin of at least 6dB.

Cells which have been detected shall be measured at least every ($N_{UTRA_carrier_TDD}$) * $T_{measureUTRA_TDD}$ when the $S_{ServingCell}$ of the E-UTRA serving cell is less than $S_{nonintrasearch}$.

When higher priority UTRA TDD cells are found by the higher priority search, they shall be measured at least every $T_{measureUTRA_TDD.}$ If, after detecting a cell in a higher priority search, it is determined that reselection has not occurred then the UE is not required to continuously measure the detected cell to evaluate the ongoing possibility of re-selection. However, the minimum measurement filtering requirements specified later in this section shall still be met by the UE before it makes any determination that it may stop measuring the cell.

For a cell that has been already detected, but that has not been re-selected to, the filtering shall be such that the UE shall be capable of evaluating that an already identified UTRA TDD cell has met re-selection criterion defined in TS 36.304 [6] within $N_{UTRA_carrier_TDD}$ * $T_{evaluateUTRA_TDD}$ when $Treselection_{RAT}$ = 0 as specified in table 4.2.2.5.2-1of TS36.133 [4] provided that the reselection criteria is met by a margin of at least 6dB.

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.5.2 and A.4.3.4.2.

#### 4.3.4.2.4 Test description

##### 4.3.4.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B/eNodeB emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.
2. The general test parameter settings are set up according to Table 4.3.4.2.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.3.4.2.4.3.

5. There is one E-UTRA TDD cell and one UTRA TDD cell specified in the test. Cell 1 is the cell used for registration according to TS 36.508 [7] clause 7.2A.2 with the power level set according to Annex C.0 and C.1.

**Table 4.3.4.2.4.1-1: General test parameters for E-UTRAN to UTRAN (1.28 Mcps TDD OPTION) Cell Re-selection**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell 1 | E-UTRAN TDD Cell |
| T1 end condition | Active cell | | Cell1 | UE shall perform reselection to cell 1 during T1 for subsequent iterations of the test. |
| | Neighbour cell | | Cell2 | 1.28 Mcps TDD OPTION cell |
| T2 end condition | Active cell | | Cell2 | UE shall perform reselection to cell 2 during T2 |
| | Neighbour cell | | Cell1 | E-UTRA TDD cell |
| Uplink-downlink configuration of cell 1 | | | 1 | As specified in table 4.2.2 in TS 36.211 [9] |
| Special subframe configuration of cell 1 | | | 6 | As specified in table 4.2.1 in TS 36.211 [9] |
| PRACH configuration of cell 1 | | | 53 | As specified in table 5.7.1-3 in TS 36.211 [9] |
| CP length of cell 1 | | | Normal | |
| Time offset between cells | | | 3 ms | Asynchronous cells<br>3ms or 92160*Ts |
| Access Barring Information | | - | Not sent | No additional delays in random access procedure. |
| Treselection | | s | 0 | |
| DRX cycle length | | s | 1,28 | |
| HCS | | | Not used | |
| T1 | | s | 85 | |
| T2 | | s | 25 | |

### 4.3.4.2.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The UE is requested to monitor the neighbouring cell on one UTRA TDD carrier. In the test there are two successive time periods, with time duration of T1 and T2 respectively. Both Cell 1 and Cell 2 are already identified by the UE prior to the start of the test. At T1 the UE is camped on to Cell 1. Cell 1 and Cell 2 belong to different tracking areas. Furthermore, UE has not registered with network for the tracking area containing Cell 2.

In the following test procedure “UE responds” means “UE starts transmitting the SYNCH-UL sequence in the UpPTS for sending RRC CONNECTION REQUEST message to perform a Routing Area Update procedure”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2.

2. Set the parameters according to T1 in Table 4.3.4.2.5-1 and 4.3.4.2.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts. If the UE is already camped in Cell 1, wait until T1 expires and skip to step 5.

3. The SS waits for random access requests information from the UE to perform cell re-selection on Cell 1.

4. If the UE has re-selected Cell 1 within T1, after the re-selection or when T1 expires, continue with step 5. Otherwise, if T1 expires and the UE has not yet re-selected Cell 1, skip to step 9.

5. The SS shall switch the power setting from T1 to T2 as specified in Table 4.3.4.2.5-1 and 4.3.4.2.5-2.

6. The SS waits for random access requests information from the UE to perform cell re-selection on Cell 2.

7. If the UE responds on Cell 2 during time duration T2 within 21 seconds from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

8. If the UE has re-selected Cell 2 within T2, after the re-selection or when T2 expires, skip to step 10. Otherwise, if T2 expires and the UE has not yet re-selected Cell 2, continue with step 9.

9. Switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.

10. Repeat step 2-9 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

#### 4.3.4.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 and TS 34.108 [24] clause 6.1.0b with the following exceptions:

**Table 4.3.4.2.4.3-1: Common Exception messages**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.3-7<br>Table H.2.3-8<br>Table H.2.3-14 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-2 |

**Table 4.3.4.2.4.3-2: System Information Block type 19: E-UTRA FDD- lower priority UTRA TDD inter RAT cell re-selection test case**

| Derivation Path: 36.508 clause 4.4.4.1 Table 4.4.4.1-1: System Information Block type 19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SysInfoType19 ::= SEQUENCE { | | | |
| utra-PriorityInfoList SEQUENCE { | | | |
| utra-ServingCell SEQUENCE { | | | |
| priority | 3 | | |
| s-PrioritySearch1 | 0 (0dB) | | |
| s-PrioritySearch2 | Not present | default value is 0 | |
| threshServingLow | 0 (0dB) | | |
| } | | | |
| eutra-FrequencyAndPriorityInfoList SEQUENCE (SIZE (1..maxNumEUTRAFreqs)) OF SEQUENCE | | *n* denotes the index of the entry | |
| earfcn | | Downlink EARFCN of Cell 1 | |
| measurementBandwidth | 50 | Enumerated(6, 15, 25, 50, 75, 100) | |
| priority | 4 | | |
| qRxLevMinEUTRA | -70 (-140 dBm) | | |
| threshXhigh | 23 (46 dB) | | |
| threshXlow | 1 (2 dB) | Default value | |
| eutra-blackListedCellList | Not present | | |
| eutraDetection | TRUE | | |
| } | | | |
| nonCriticalExtensions SEQUENCE {} | Not present | | |
| } | | | |

### 4.3.4.2.5 Test requirement

Tables 4.3.4.2.4.1-1, 4.3.4.2.5-1 and 4.3.4.2.5-2 define the primary level settings including test tolerances for E-UTRAN TDD- UTRAN TDD intra frequency cell re-selection test case which UTRA is of lower priority.

**Table 4.3.4.2.5-1: Cell specific test requirement parameters for cell re-selection E-UTRA TDD to UTRA TDD test case (cell 1)**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.2.2 (OP.2 TDD) | | OP.2 TDD | |
| PBCH_RA | dB | 0 | 0 |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA(Note1) | dB | | |
| OCNG_RB(Note1) | dB | | |
| $Q_{rxlevmin}$ | dBm | -140 | -140 |
| $N_{oc}$ | dBm/15kHz | -98 | |
| RSRP | dBm/15kHz | -86.9 | -101.1 |
| $\hat{E}_s / I_{ot}$ | dB | 11.1 | -3.1 |
| $S_{nonintrasearch}$ | dB | Not sent | |
| $Thresh_{serving, low}$ | dB | 46 (-94dBm) | |
| $Thresh_{x, low (Note2)}$ | dB | 24 (-79dBm) | |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: This refers to the value of Threshx, low which is included in E-UTRA system information, and is a threshold for the UTRA target cell

**Table 4.3.4.2.5-2: Cell specific test requirement parameters for cell re-selection E-UTRA TDD to UTRA TDD test case (cell 2)**

| Parameter | Unit | Cell 2 (UTRA) | | | |
|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPTS | |
| | | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number (Note1) | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 |
| OCNS_Ec/Ior | dB | -3 | -3 | | |
| $\hat{I}_{or} / I_{oc}$ | dB | 11 | 11 | 11 | 11 |
| $I_{oc}$ | dBm/1.28 MHz | -80 | | | |
| PCCPCH RSCP | dBm | -72 | -72 | n.a. | n.a. |
| Propagation Condition | | AWGN | | | |
| $Q_{rxlevmin}$ | dBm | -103 | | | |
| $Qoffset_{s,n}$ | dB | C1, C2: 0 | | | |
| $Qhyst_s$ | dB | 0 | | | |
| $Thresh_{x, high}$ (Note2) | dB | 46 (-94dBm) | | | |

Note 1: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number.

Note 2: This refers to the value of $Thresh_{x, high}$ which is included in UTRA system information and is a threshold for the E-UTRA target cell.

The cell re-selection delay to lower priority is defined as the time from the beginning of time period T2, to the moment when the UE camps on Cell 2 and starts to send the SYNCH-UL sequence in the UpPTS for sending the RRC CONNECTION REQUEST message to perform a Routing Area Update procedure on Cell 2.

The cell re-selection delay to lower priority shall be less than 21 s.

NOTE: The cell re-selection delay to lower priority cell can be expressed as: $T_{evaluateUTRA_TDD} + T_{SI_UTRA}$,

Where:

$T_{evaluateUTRA_TDD}$ 19.2s, See Table 4.2.2.5.2-1 of TS 36.133 [4] clause 4.2.2.5.1

$T_{SI_UTRA}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell; 1280 ms is assumed in this test case.

This gives a total of 20.48 s, allow 21 s for lower priority cell reselection in the test case. For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

### 4.3.4.3 EUTRA TDD-UTRA TDD cell reselection in fading propagation conditions: UTRA TDD is of lower priority

#### 4.3.4.3.1 Test purpose

To verify that the UE is able to search and measure neighbouring UTRA TDD cells and compare to the E-UTRA serving cell to meet the inter-RAT cell re-selection requirements when the UTRA is of lower priority. The E-UTRA cell is in fading propagation conditions and the UTRA cell is in AWGN propagation conditions.

#### 4.3.4.3.2 Test applicability

This test applies to all types of release 8 and forward E-UTRA TDD UEs that support release 9 and forward UTRA TDD.

#### 4.3.4.3.3 Minimum conformance requirements

When the measurement rules indicate that UTRA TDD cells are to be measured, the UE shall measure P-CCPCH RSCP of detected UTRA TDD cells in the neighbour frequency list at the minimum measurement rate specified in this section. The parameter $N_{UTRA_carrier_TDD}$ is the number of carriers in the neighbour frequency list. The UE shall filter P-CCPCH RSCP measurements of each measured UTRA TDD cell using at least 2 measurements. Within the set of measurements used for the filtering, at least two measurements shall be spaced by at least half the minimum specified measurement period. P-CCPCH RSCP of UTRAN TDD cells shall not be filtered over a longer period than that specified in table 4.3.4.3.3-1.

The UE shall evaluate whether newly detectable UTRA TDD cells have met the reselection criteria in TS 36.304 [6] within time $(N_{UTRA_carrier_TDD}) * T_{detectUTRA_TDD}$ when the $S_{ServingCell}$ of the E-UTRA serving cell is less than $S_{nonintrasearch}$ when $T_{reselection} = 0$ provided that the reselection criteria is met by a margin of at least 6dB.

Cells which have been detected shall be measured at least every $(N_{UTRA_carrier_TDD}) * T_{measureUTRA_TDD}$ when the $S_{ServingCell}$ of the E-UTRA serving cell is less than $S_{nonintrasearch}$.

When higher priority UTRA TDD cells are found by the higher priority search, they shall be measured at least every $T_{measure,UTRA_TDD}$. If, after detecting a cell in a higher priority search, it is determined that reselection has not occurred then the UE is not required to continuously measure the detected cell to evaluate the ongoing possibility of reselection. However, the minimum measurement filtering requirements specified later in this section shall still be met by the UE before it makes any determination that it may stop measuring the cell.

For a cell that has been already detected, but that has not been reselected to, the filtering shall be such that the UE shall be capable of evaluating that an already identified UTRA TDD cell has met reselection criterion defined in TS 36.304 [6] within $N_{UTRA_carrier_TDD} * T_{evaluateUTRA_TDD}$ when $T_{reselection} = 0$ as specified in table 4.3.4.3.3-1 provided that the reselection criteria is met by a margin of at least 6dB.

**Table 4.3.4.3.3-1: $T_{detectUTRA_TDD}$, $T_{measureUTRA_TDD}$ and $T_{evaluateUTRA_TDD}$**

| DRX cycle length [s] | $T_{detectUTRA_TDD}$ [s] | $T_{measureUTRA_TDD}$ [s] (number of DRX cycles) | $T_{evaluateUTRA_TDD}$ [s] (number of DRX cycles) |
|---|---|---|---|
| 0.32 | 30 | 5.12 (16) | 15.36 (48) |
| 0.64 | | 5.12 (8) | 15.36 (24) |
| 1.28 | | 6.4(5) | 19.2 (15) |
| 2.56 | 60 | 7.68 (3) | 23.04 (9) |

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.5.2 and A.4.3.4.3.

## 4.3.4.3.4 Test description

### 4.3.4.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.26.
2. The general test parameter settings are set up according to Table 4.3.4.3.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.3.4.3.4.3.
5. There is one E-UTRA TDD cell and one UTRA TDD cell specified in the test. Cell 1 is the cell used for registration according to TS 36.508 [7] clause 7.2A.2 with the power level set according to Annex C.0 and C.1.

**Table 4.3.4.3.4.1-1: General test parameters for EUTRA TDD- lower priority UTRA TDD inter RAT cell re-selection test case in fading conditions**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | E-UTRAN cell |
| T1 end condition | Active cells | | Cell1 | UE shall perform reselection to cell 1 during T1 for subsequent iterations of the test |
| | Neighbour cell | | Cell2 | |
| T3 end condition | Active cell | | Cell2 | UE shall perform reselection to cell 2 during T3 |
| | Neighbour cell | | Cell1 | |
| E-UTRA PRACH configuration | | | 53 | As specified in table 5.7.1-3 in TS 36.211 [9] |
| Uplink-downlink configuration of cell 1 | | | 1 | As specified in table 4.2.2 in TS 36.211 [9] |
| Special subframe configuration of cell 1 | | | 6 | As specified in table 4.2.1 in TS 36.211 [9] |
| E_UTRA Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | <85 | T1 need to be defined so that cell re-selection reaction time is taken into account. T1 is terminated when the UE starts to send preambles to cell 1 |
| T2 | | s | 64 | The start of T2 is defined as the time when the UE starts to send PRACH preambles to cell 1 |
| T3 | | s | <25 | T3 need to be defined so that cell re-selection reaction time is taken into account. T3 is terminated when the UE starts to send PRACH preambles to cell 2 |
| T4 | | s | 64 | The start of T4 is defined as the time when the UE starts to send PRACH preambles to cell 2 |

### 4.3.4.3.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The UE is requested to monitor the neighbouring cell on one UTRA TDD carrier. In the test there are four successive time periods, with time duration of T1, T2, T3 and T4 respectively. Both E-UTRA Cell 1 and UTRA Cell 2 are already identified by the UE prior to the start of the test. At T1 the UE is camped on to Cell 1. Cell 2 is of lower priority than Cell 1. Cell 1 and Cell 2 belong to different tracking areas. Furthermore, UE has not registered with network for the tracking area containing Cell 2. Time duration T2 and T4 are not used for cell re-selection in the test.

In the following test procedure, “UE responds on Cell 1” means “UE starts transmitting preamble on PRACH for sending RRC CONNECTION REQUEST message to perform a Tracking Area Update procedure according to TS 36.508 [7] clause 4.5A.2” and “UE responds on Cell 2” means “UE starts transmitting the SYNCH-UL sequence in the UpPTS for sending RRC CONNECTION REQUEST message to perform a Routing Area Update procedure”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2.
2. Set the parameters according to duration T0 in Table 4.3.4.3.5-1 and 4.3.4.3.5-2.
3. Set the parameters according to T2 in Table 4.3.4.3.5-3 and 4.3.4.3.5-4. Propagation conditions are set according to Annex B clause B.1.1 and B.2.2. T2 starts.
4. The SS monitors for possible random access request information from the UE to perform cell re-selection on Cell 2.
5. If the UE responds on Cell 2 within T2, then count a fail for the event “Hold out on Cell 1” and skip to step 10. Otherwise, count a success for the event “Hold out on Cell 1” and after T2 expires continue with step 6.
6. The SS shall switch the power setting from T2 to T3 as specified in Table 4.3.4.3.5-3 and 4.3.4.3.5-4.
7. The SS waits for random access requests information from the UE to perform cell re-selection on Cell 2.

8. If the UE responds on Cell 2 during time duration T3 within 21 seconds from the beginning of time period T3, then count a success for the event "Re-select lower priority Cell 2".
   Otherwise, count a fail for the event "Re-select lower priority Cell 2".

9. If the UE responds on Cell 2 within T3, at the moment of the request-reception continue with step 10.
   Otherwise, after T3 expires skip to step 17.

10. The SS shall switch the power setting from T3 to T4 as specified in Table 4.3.4.3.5-3 and 4.3.4.3.5-4.

11. The SS monitors for possible random access request information from the UE to perform cell re-selection on Cell 1.

12. If the UE responds on Cell 1 within T4, then count a fail for the event "Hold out on Cell 2" and skip to step 17.
    Otherwise, count a success for the event "Hold out on Cell 2" and after T4 expires continue with step 13.

13. The SS shall switch the power setting from T4 to T1 as specified in Table 4.3.4.3.5-3 and 4.3.4.3.5-4.

14. The SS waits for random access requests information from the UE to perform cell re-selection on Cell 1.

15. If the UE responds on Cell 1 within T1, at the moment of the request-reception skip to step 17.
    Otherwise, after T1 expires continue with step 16.

16. Switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.
    Set the parameters according to duration T0 in Table 4.3.4.3.5-1 and 4.3.4.3.5-2.

17. Repeat step 3-16 until a test verdict has been achieved.
    Each of the events "Hold out on Cell 1", "Re-select lower priority Cell 2" and "Hold out on Cell 2" is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. Different events may require different times for a verdict.
    If both events pass, the test passes. If one event fails, the test fails.

#### 4.3.4.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 and TS 34.108 [24] clause 6.1.0b with the following exceptions:

**Table 4.3.4.3.4.3-1: Common Exception messages**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.3-5<br>Table H.2.3-8 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-2 |

**Table 4.3.4.3.4.3-2: System Information Block type 3 (1.28 Mcps TDD): Inter-RAT E-UTRA TDD - UTRA TDD is of lower priority cell re-selection**

| Derivation Path: 34.108 clause 6.1.0b | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| - SIB4 Indicator | TRUE | | |
| - Cell identity | 0000 0000 0000 0000<br>0000 0000 0001B | | |
| - Cell selection and re-selection info | | | |
| - Mapping info | Not present | | |
| - Cell selection and reselection quality measure | (no data) | | |
| - CHOICE mode | TDD | | |
| - Sintrasearch | 10 dB | | |
| - Sintersearch | 10 dB | | |
| - SsearchHCS | Not present | | |
| - RAT List | Not present | | |
| - Qrxlevmin | -103 dBm | | |
| - Qhyst1s | 0 dB | | |
| - Treselections | 0 seconds | | |

| | | | |
|---|---|---|---|
| - HCS Serving cell information<br>- Maximum allowed UL TX power | Not present<br>21dBm | | |

**Table 4.3.4.3.4.3-3: System Information Block type 19: Inter-RAT E-UTRA TDD - UTRA TDD is of lower priority cell re-selection**

| Derivation Path: 36.508 clause 4.4.4.1 Table 4.4.4.1-1: System Information Block type 19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SysInfoType19 ::= SEQUENCE { | | | |
| utra-PriorityInfoList SEQUENCE { | | | |
| utra-ServingCell SEQUENCE { | | | |
| priority | 3 | | |
| s-PrioritySearch1 | 0 (0dB) | | |
| s-PrioritySearch2 | Not present | default value is 0 | |
| threshServingLow | 0 (0dB) | | |
| } | | | |
| eutra-FrequencyAndPriorityInfoList SEQUENCE (SIZE (1..maxNumEUTRAFreqs)) OF SEQUENCE | | *n* denotes the index of the entry | |
| earfcn | | Downlink EARFCN of Cell 1 | |
| measurementBandwidth | 50 | Enumerated(6, 15, 25, 50, 75, 100) | |
| priority | 4 | | |
| qRxLevMinEUTRA | -70 (-140 dBm) | | |
| threshXhigh | 22 (44 dB) | | |
| threshXlow | 1 (2 dB) | Default value | |
| eutra-blackListedCellList | Not present | | |
| eutraDetection | TRUE | | |
| } | | | |
| nonCriticalExtensions SEQUENCE {} | Not present | | |
| } | | | |

## 4.3.4.3.5 Test requirement

Tables 4.3.4.3.4.1-1, 4.3.4.3.5-1, 4.3.4.3.5-2, 4.3.4.3.5-3 and 4.3.4.3.5-4 define the primary level settings including test tolerances for E-UTRA TDD- UTRA TDD inter-RAT cell re-selection test case which UTRA is of lower priority.

**Table 4.3.4.3.5-1: Cell specific test parameters for Cell 1 (E-UTRA TDD)**

| Parameter | Unit | Cell 1 |
|---|---|---|
| | | T0 |
| E-UTRA RF Channel number | | 1 |
| $BW_{channel}$ | MHz | 10 |
| Antenna Configuration | | 1x2 |
| OCNG Patterns defined in D.2.2 (OP.2 TDD) | | OP.2 TDD |
| PSS_RA | dB | 0 |
| SSS_RA | dB | 0 |
| PCFICH_RB | dB | 0 |
| PHICH_RA | dB | 0 |
| PHICH_RB | dB | 0 |
| PDCCH_RA | dB | 0 |
| PDCCH_RB | dB | 0 |
| PDSCH_RA | dB | 0 |
| PDSCH_RB | dB | 0 |
| OCNG_RA Note 1 | dB | 0 |
| OCNG_RB Note 1 | dB | 0 |
| Qrxlevmin for UTRA neighbour cell | dBm | -103 |
| Qrxlevmin | dBm | -140 |
| $N_{oc}$ | dBm/15 kHz | -104 |
| RSRP | dBm/15 kHz | -82 |
| $\hat{E}_s / I_{ot}$ | dB | 22 |
| $\hat{E}_s / N_{oc}$ | dB | 22 |
| $Treselection_{EUTRAN}$ | s | 0 |
| Snonintrasearch | dB | Not sent |
| $Thresh_{serving, low}$ | dB | 44 |
| $Thresh_{x, low}$ (Note 2) | dB | 24 |
| Propagation Condition | | AWGN |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: This refers to the value of $Thresh_{x, low}$ which is included in E-UTRA system information, and is a threshold for the UTRA target cell.

**Table 4.3.4.3.5-2: Cell specific test parameters for Cell 2 (UTRA TDD)**

| Parameter | Unit | Cell 2 (UTRA) |
|---|---|---|
| | | T0 |
| Timeslot Number | | 0 |
| UTRA RF Channel Number (Note 1) | | Channel 2 |
| PCCPCH_Ec/Ior | dB | -3 |
| DwPCH_Ec/Ior | dB | |
| OCNS_Ec/Ior | dB | -3 |
| $\hat{I}_{or}/I_{oc}$ | dB | 13 |
| $I_{oc}$ | dBm/1.28 MHz | -80 |
| PCCPCH RSCP | dBm | -70 |
| Propagation Condition | | AWGN |
| Qrxlevmin | dBm | -103 |
| QrxlevminEUTRA | dBm | -140 |
| UE_TXPWR_MAX_RACH | dBm | 21 |
| Treselection | s | 0 |
| $Thresh_{x, high}$ (Note 2) | dB | 44 |

Note1: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number.

Note 2: This refers to the value of $Thresh_{x, high}$ which is included in UTRA system information, and is a threshold for the E-UTRA target cell

**Table 4.3.4.3.5-3: Cell specific test parameters for cell 1 (E-UTRA)**

| Parameter | Unit | Cell 1 | | | |
|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 |
| E-UTRA RF Channel number | | 1 | | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | |
| $BW_{channel}$ | MHz | 10 | | | |
| OCNG Patterns defined in D.2.2 (OP.2 TDD) | | OP.2 TDD | | | |
| PSS_RA | dB | 0 | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| Qrxlevmin for UTRA neighbour cell | dBm | -103 | | | |
| Qrxlevmin | dBm | -140 | | | |
| $N_{oc}$ | dBm/15 kHz | -104 | | | |
| RSRP | dBm/15 KHz | -82 | -82 | -107 | -107 |
| $\hat{E}_s / I_{ot}$ | dB | 22 | 22 | -3 | -3 |
| $\hat{E}_s / N_{oc}$ | dB | 22 | 22 | -3 | -3 |
| $Treselection_{EUTRAN}$ | s | 0 | | | |
| Snonintrasearch | dB | Not sent | | | |
| $Thresh_{serving, low}$ | dB | 44 | | | |
| $Thresh_{x, low}$ (Note 2) | dB | 24 | | | |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: This refers to the value of $Thresh_{x, low}$ which is included in E-UTRA system information, and is a threshold for the UTRA target cell.

**Table 4.3.4.3.5-4: Cell specific test parameters for cell 2 (UTRA)**

| Parameter | Unit | Cell 2 (UTRA) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | | | DwPTS | | | |
| | | T1 | T2 | T3 | T4 | T1 | T2 | T3 | T4 |
| UTRA RF Channel Number (Note1) | | Channel 2 | | | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | | | | |
| DwPCH_Ec/Ior | dB | | | | | 0 | | | |
| OCNS_Ec/Ior | dB | -3 | | | | | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | 13 | 13 | 13 | 13 | 13 | 13 | 13 | 13 |
| $I_{oc}$ | dBm/1.28 MHz | -80 | | | | | | | |
| PCCPCH RSCP | dBm | -70 | -70 | -70 | -70 | n.a. | n.a. | n.a. | n.a. |
| Propagation Condition | | AWGN | | | | | | | |
| Qrxlevmin | dBm | -103 | | | | | | | |
| $Qrxlevmin_{EUTRA}$ | dBm | -140 | | | | | | | |
| UE_TXPWR_MAX_RACH | dBm | 21 | | | | | | | |
| Treselection | s | 0 | | | | | | | |
| $Thresh_{x, high}$ (Note2) | dB | 44 | | | | | | | |
| Note1: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number.<br>Note2: This refers to the value of $Thresh_{x, high}$ which is included in UTRA system information, and is a threshold for the E-UTRA target cell | | | | | | | | | |

The probability of reselection from cell 1 to cell 2 during T2 observed during testing shall be less than 10%

The probability of reselection from cell 2 to cell 1 during T4 observed during testing shall be less than 10%

The cell reselection delay to lower priority is defined as the time from the beginning of time period T3, to the moment when the UE camps on cell 2, and starts to send the SYNCH-UL sequence in the UpPTS for sending the RRC CONNECTION REQUEST message on cell 2. In order to evaluate reselection delay, the system simulator first needs to verify that the UE is camped on cell 1 at the start of T3

The cell re-selection delay to lower priority cell can be expressed as:

Cell re-selection delay to lower priority = $T_{evaluateUTRA_TDD} + T_{SI\text{-}UTRA}$

$T_{evaluateUTRA_TDD}$ 19.2s, as specified in TS 36.133 [4] Table 4.2.2.5.2-1

$T_{SI\text{-}UTRA}$ Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell; 1280 ms is assumed in this test case.

The cell re-selection delay to lower priority shall be less than a total of 21 seconds in this test case (note: this gives a total of 20.48 seconds but the test allows 21 seconds).

For the test to pass, all the events above shall pass.

The statistical pass/ fail decisions are done separated for each event. For an event to pass, the total number of successful loops shall be more than 90% of the cases with a confidence level of 95%.

# 4.4 E-UTRAN to GSM Cell Re-Selection

## 4.4.1 E-UTRAN FDD - GSM cell re-selection

### 4.4.1.1 Test purpose

To verify that the UE is able to search and measure neighbouring GSM cells and compare to the E-UTRA serving cell to meet the inter-RAT cell re-selection requirements.

### 4.4.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support GSM.

### 4.4.1.3 Minimum conformance requirements

The cell re-selection delay shall be less than 4 * $T_{measure, GSM}$ + $T_{BCCH}$ in RRC_IDLE state.

When the measurement rules defined in TS 36.304 [6] indicates that E-UTRAN inter-frequencies or inter-RAT frequency cells are to be measured, the UE shall measure the signal level of the GSM BCCH carriers if the GSM BCCH carriers are indicated in the measurement control system information of the serving cell, GSM BCCH carriers of lower priority than the serving cell shall be measured at least every $T_{measure,GSM}$ as defined in table 4.2.2.5.3-1 of TS 36.133 [4] clause 4.2.2.5.3.

When higher priority GSM BCCH carriers are found by the higher priority search, they shall be measured at least every $T_{measure,GSM}$, and the UE shall decode the BSIC of the GSM BCCH carrier. If, after detecting a cell in a higher priority search, it is determined that re-selection has not occurred then the UE is not required to continuously measure the detected cell to evaluate the ongoing possibility of re-selection, or to continuously verify the BSIC of the GSM BCCH carrier every 30s.

However, the minimum measurement filtering requirements specified shall still be met by the UE before it makes any determination that it may stop measuring the cell.

The UE shall maintain a running average of 4 measurements for each GSM BCCH carrier. The measurement samples for each cell shall be as far as possible uniformly distributed over the averaging period.

If continuous GSM measurements are required by the measurement rules in TS 36.304 [6], the UE shall attempt to verify the BSIC at least every 30 seconds for each of the 4 strongest GSM BCCH carriers. If a change of BSIC is detected for one GSM cell then that GSM BCCH carrier shall be treated as a new GSM neighbour cell. If the UE detects on a BCCH carrier a BSIC which is indicated as not allowed for that carrier in the measurement control system information of the serving cell, the UE is not required to perform BSIC re-confirmation for that cell.

The UE shall not consider the GSM BCCH carrier in cell re-selection, if the UE cannot demodulate the BSIC of that GSM BCCH carrier. Additionally, the UE shall not consider a GSM neighbour cell in cell re-selection, if it is indicated as not allowed in the measurement control system information of the serving cell.

The UE shall evaluate the inter-RAT cell re-selection criteria as defined in TS 36.304 [6] at least every DRX cycle. The DRX cycle length is 1.28 seconds. When a non-zero value of $Treselection_{EUTRAN}$ is used, the UE shall only perform re-selection on an evaluation which occurs simultaneously to, or later than the expiry of the $Treselection_{EUTRAN}$ timer.

At inter-RAT cell re-selection, the UE shall monitor the downlink of serving cell for paging reception until the UE is capable to start monitoring downlink channels for paging reception of the target inter-frequency cell. For E-UTRAN to GSM cell re-selection the interruption time must not exceed $T_{BCCH}$ + 50 ms. $T_{BCCH}$ is the maximum time allowed to read BCCH data from a GSM cell as defined in TS 45.008 [15] clause 6.2.

For idle mode cell re-selection purposes, the UE shall be capable of monitoring at least:

- Depending on UE capability, 3 FDD E-UTRA inter-frequency carriers, and
- Depending on UE capability, 32 GSM carriers

In addition to the requirements defined in TS 36.133 [4] clause 4.2.2.9 a UE in RRC_IDLE state shall be capable of monitoring a total of at least 8 carrier frequency layers, which includes serving layer, comprising of any defined in TS 36.133 [4] clause 4.2.2.9 combination of E-UTRA FDD, E-UTRA TDD, UTRA FDD, UTRA TDD, GSM (one GSM layer corresponds to 32 cells) , cdma2000 1x and HRPD layers.

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.5.3 and A.4.4.1.

### 4.4.1.4 Test description

#### 4.4.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. The general test parameter settings are set up according to Table 4.4.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are as defined in clause 4.4.1.4.3.

5. There is one E-UTRA FDD cell and one GSM cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 4.4.1.4.1-1: General Test Parameters for E-UTRAN FDD - GSM cell re-selection test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | UE shall be forced to cell 1 in the initialisation phase and shall be able to detect and monitor the 4 strongest GSM BCCH carriers in T1. Cell 1 is an E-UTRA FDD cell. |
| Final condition | Neighbour cell | | Cell2 | UE shall perform reselection to cell 2 during T2. Cell 2 is a GSM cell. |
| E-UTRA RF Channel Number | | | 1 | 1 E-UTRA FDD carrier frequency |
| GSM ARFCN | | | 1 | 12 GSM BCCH carriers are used |
| PRACH configuration | | | 4 | As specified in table 5.7.1-2 in TS 36.211 [9] |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| CP length of cell 1 | | | Normal | |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| T1 | | s | 35 | T1 need to be defined so that cell re-selection reaction time is taken into account. |
| T2 | | s | 35 | T2 need to be defined so that the higher layer search periodicity and cell re-selection reaction time are taken into account. |
| Propagation channel | | | AWGN | |

### 4.4.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The UE is requested to monitor the neighbouring cells on one E-UTRA FDD carrier and twelve GSM cells. In the test there are two successive time periods, with time duration of T1 and T2 respectively. Cell 1 (E-UTRA FDD cell) and Cell 2 (GSM cell) shall belong to different Location Areas. During initialization before the start of the test, the UE is camped on Cell 1. By the end of T1, the UE has identified BSIC on the GSM BCCH carrier of Cell 2 but the signal levels do not meet the re-selection criterion during T1. At the start of T2, the signal levels change such that Cell 2 meets the re-selection criterion. The GSM layer is configured at a lower priority than the serving E-UTRA FDD layer.

In the following test procedure “UE responds” means “UE starts transmitting RR Channel Request message for location update procedure”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2.

2. Set the parameters according to T1 in Table’s 4.4.1.5-1 and 4.4.1.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts. If the UE is already camped in Cell 1, wait until T1 expires and skip to step 5.

3. The SS waits for random access requests information from the UE to perform cell re-selection on Cell 1.

4. If the UE has re-selected Cell 1 within T1, after the re-selection or when T1 expires, continue with step 5. Otherwise, if T1 expires and the UE has not yet re-selected Cell 1, skip to step 9.

5. The SS shall switch the power setting from T1 to T2 as specified in Table’s 4.4.1.5-1 and 4.4.1.5-2.

6. The SS waits for location update information from the UE to perform cell re-selection on Cell 2.

7. If the UE responds on Cell 2 during time duration T2 within 27.9 seconds from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

8. If the UE has re-selected Cell 2 within T2, after the re-selection or when T2 expires, skip to step 10. Otherwise, if T2 expires and the UE has not yet re-selected Cell 2, continue with step 9.

9. Switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.

10. Repeat step 2-9 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

#### 4.4.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 4.4.1.4.3-1: Common Exception messages for E-UTRAN FDD - GSM cell re-selection test case**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.3-9<br>Table H.2.3-10 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

All GSM cell messages indicated shall use the same content as described in the default message content in TS 45.008 [15] clause 9 for Rel-4 and later releases, with the exceptions above and as specified in Tables 4.4.1.5-1 and 4.4.1.5-2.

### 4.4.1.5 Test requirement

Tables 4.4.1.4.1-1, 4.4.1.5-1 and 4.4.1.5-2 defines the primary level settings including test tolerances for E-UTRAN FDD to GSM cell re-selection test case.

**Table 4.4.1.5-1: Cell Specific Test requirement Parameters for Cell 1 E-UTRAN FDD cell**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.1.2 (OP.2 FDD) | | OP.2 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA[Note 1] | dB | | |
| OCNG_RB[Note 1] | dB | | |

| | | | |
|---|---|---|---|
| Qrxlevmin | dBm | -140 | |
| $N_{oc}$ | dBm/15 kHz | -99.4 | |
| RSRP | dBm/15 KHz | -88.9 | -103.1 |
| $\hat{E}_s/I_{ot}$ | dB | 10.5 | -3.70 |
| $\hat{E}_s/N_{oc}$ | dB | 10.5 | -3.70 |
| $Treselection_{EUTRAN}$ | s | 0 | |
| Snonintrasearch | dB | Not sent | |
| $Thresh_{serving, low}$ | dB | 44 | |
| $Thresh_{x, low}$ Note 2 | dB | 24 | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | |
| Note 2: This refers to $Thresh_{x, low}$ which is included in E-UTRA system information, and is a threshold for GSM target cell. | | | |

**Table 4.4.1.5-2: Cell Specific Test requirement Parameters for Cell 2 GSM cell**

| Parameter | Unit | Cell 2 (GSM) | |
|---|---|---|---|
| | | T1 | T2 |
| Absolute RF Channel Number | | ARFCN 1 | |
| RXLEV | dBm | -90.00 | -75.00 |
| RXLEV_ACCESS_MIN | dBm | -105 | |
| MS_TXPWR_MAX_CCH | dBm | 24 | |

The cell re-selection delay is defined as the time from the beginning of time period T2, to the moment when the UE camps on Cell 2, and starts to send the RR Channel Request message for location update to Cell 2.

The cell re-selection delay test requirement in this case is expressed as:

Cell re-selection delay = 4 * $T_{measureGSM}$ + $T_{BCCH}$

$T_{measureGSM}$ = 6.40 s; as specified in TS 36.133 [4] clause 4.2.2.3

$T_{BCCH}$ = 1.9 s; maximum time allowed to read the BCCH data from GSM cell, when being synchronized to a BCCH carrier; as specified in TS 45.008 [15] clause 6.2

The cell re-selection delay shall be less than a total of 27.9 seconds in this test case (note: this gives a total of 26 seconds for the $T_{measureGSM}$ calculation plus 1.9 s for $T_{BCCH}$ but the test allows 27.9 seconds).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 4.4.2 E-UTRAN TDD - GSM cell re-selection

### 4.4.2.1 Test purpose

To verify that the UE is able to search and measure neighbouring GSM cells and compare to the E-UTRA serving cell to meet the inter-RAT cell re-selection requirements.

### 4.4.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward that support GSM.

### 4.4.2.3 Minimum conformance requirements

If the $S_{ServingCell}$ of the E-UTRA serving cell (or other cells on the same frequency layer) is greater than $S_{nonintrasearch}$ then:

- The UE may not search for, or measure GSM cells if the priority of GSM is equal to, or lower than the serving cell.

- The UE shall search for and measure GSM cells if the priority of GSM is higher than the serving cell. The minimum rate at which the UE is required to search for and measure such layers may be reduced in this scenario to maintain UE battery life.

If the $S_{ServingCell}$ of the E-UTRA serving cell is less than or equal to $S_{nonintrasearch}$ then the UE shall measure, according to the measurement rules defined in TS 36.304 [6] at least every $T_{measure,GSM}$ as defined in table 4.2.2.5.3-1 of TS 36.133 [4] clause 4.2.2.5.3:

- If a detailed neighbour cell list is provided, the signal level of the GSM BCCH carrier of each GSM neighbour cell indicated in the measurement control system information of the serving cell; or

- If only BCCH carriers are provided, the signal level of the GSM BCCH carriers indicated in the measurement control system information of the serving cell

If the RSRP of the E-UTRA serving cell is greater than $S_{nonintrasearch}$ then the UE shall search for GSM BCCH carrier at least every $T_{higher_priority_search}$ where $T_{higher_priority_search}$ is described in TS 36.133 [4] clause 4.2.2 as $T_{higher_priority_search}$ = (60 * $N_{layers}$) seconds, where the parameter $N_{layers}$ is the total number of configured higher priority carrier frequencies. When higher priority GSM BCCH carriers are found by the higher priority search, they shall be measured at least every $T_{measure,GSM}$, and the UE shall decode the BSIC of the GSM BCCH carrier. If re-selection to any higher priority cell is not triggered within (4 * $T_{measure_GSM}$ + $Treselection_{RAT}$) after it has been found in a higher priority search, the UE is not required to continue make measurements of the BCCH carrier to evaluate the ongoing possibility of re-selection, or to continuously verify the BSIC of the GSM BCCH carrier every 30s.

The UE shall maintain a running average of 4 measurements for each GSM BCCH carrier. The measurement samples for each cell shall be as far as possible uniformly distributed over the averaging period.

If continuous GSM measurements are required by the measurement rules in TS 36.304 [6], the UE shall attempt to verify the BSIC at least every 30 seconds for each of the 4 strongest GSM BCCH carriers. If a change of BSIC is detected for one GSM cell then that GSM BCCH carrier shall be treated as a new GSM neighbour cell. If the UE detects on a BCCH carrier a BSIC which is indicated as not allowed for that carrier in the measurement control system information of the serving cell, the UE is not required to perform BSIC re-confirmation for that cell.

The UE shall not consider the GSM BCCH carrier in cell re-selection, if the UE cannot demodulate the BSIC of that GSM BCCH carrier. Additionally, the UE shall not consider a GSM neighbour cell in cell re-selection, if it is indicated as not allowed in the measurement control system information of the serving cell.

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.5.3 and A.4.4.2.

## 4.4.2.4 Test description

### 4.4.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. The general test parameter settings are set up according to Table 4.4.2.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are as defined in clause 4.4.2.4.3.

5. There is one E-UTRA TDD cell and one GSM cell specified in the test. Cell 1 (E-UTRA TDD cell) is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 4.4.2.4.1-1: General test parameters for E-UTRA TDD GSM cell re-selection test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell1 | UE shall be forced to cell 1 in the initialisation phase and shall be able to detect and monitor the 4 strongest GSM BCCH carriers in T1. Cell 1 is an E-UTRA TDD cell. |
| Final condition | Neighbour cell | | Cell2 | UE shall perform reselection to cell 2 during T2. Cell 2 is a GSM cell. |
| E-UTRA RF Channel Number | | | 1 | 1 E-UTRA TDD carrier frequency |
| GSM ARFCN | | | 1 | 12 GSM BCCH carriers are used |
| Uplink-downlink configuration of cell 1 | | | 1 | As specified in table 4.2.2 in TS 36.211 |
| Special subframe configuration for cell 1 | | | 6 | As specified in table 4.2.1 in TS 36.211 |
| PRACH configuration for cell 1 | | | 53 | As specified in table 5.7.1-3 in TS 36.211 |
| CP length of cell 1 | | | Normal | |
| Access Barring Information | | | Not Sent | No additional delays in random access procedure. |
| DRX cycle length | | s | 1.28 | The value shall be used for all cells in the test. |
| Propagation channel | | | AWGN | |
| T1 | | s | 35 | T1 need to be defined so that cell re-selection reaction time is taken into account. |
| T2 | | s | 35 | T2 need to be defined so that the higher layer search periodicity and cell re-selection reaction time are taken into account. |

### 4.4.2.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The UE is requested to monitor the neighbouring cell on one E-UTRA TDD carrier and twelve GSM cells. In the test there are two successive time periods, with time duration of T1 and T2 respectively. Cell 1 (E-UTRA TDD cell) and Cell 2 (GSM cell) belong to different Location Areas. By the end of T1, the UE has identified BSIC on the GSM BCCH carrier of Cell 2 but the signal levels do not meet the re-selection criterion. At the start of T2, the signal levels change such that Cell 2 meets the re-selection criterion. The GSM layer is configured at a lower priority than the serving E-UTRA TDD layer.

In the following test procedure “UE responds” means “UE starts transmitting RR Channel Request message for location update procedure”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2.
2. Set the parameters according to T1 in Table’s 4.4.2.5-1 and 4.4.2.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts. If the UE is already camped in Cell 1, wait until T1 expires and skip to step 5.
3. The SS waits for random access requests information from the UE to perform cell re-selection on Cell 1.
4. If the UE has re-selected Cell 1 within T1, after the re-selection or when T1 expires, continue with step 5. Otherwise, if T1 expires and the UE has not yet re-selected Cell 1, skip to step 9.
5. The SS shall switch the power setting from T1 to T2 as specified in Table’s 4.4.2.5-1 and 4.4.2.5-2.
6. The SS waits for location update information from the UE to perform cell re-selection on Cell 2.
7. If the UE responds on Cell 2 during time duration T2 within 27.9 seconds from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
8. If the UE has re-selected Cell 2 within T2, after the re-selection or when T2 expires, skip to step 10. Otherwise, if T2 expires and the UE has not yet re-selected Cell 2, continue with step 9.
9. Switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.
10. Repeat step 2-9 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

#### 4.4.2.4.3 Message contents

Message contents are according to TS 36.508 [4] clause 4.6 with the following exceptions:

**Table 4.4.2.4.3-1: Common Exception messages for E-UTRAN TDD - GSM cell re-selection test case**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.3-9<br>Table H.2.3-10 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-2 |

All GSM cell messages indicated shall use the same content as described in the default message content in TS 45.008 [15] clause 9 for Rel-4 and later releases, with the exceptions above and as specified in Tables 4.4.2.5-1 and 4.4.2.5-2.

### 4.4.2.5 Test requirement

Tables 4.4.2.4.1-1, 4.4.2.5-1 and 4.4.2.5-2 defines the primary level settings including test tolerances for E-UTRAN TDD to GSM cell re-selection test case.

**Table 4.4.2.5-1: Cell Specific Test requirement Parameters for Cell 1 E-UTRAN TDD cell**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.2.2 (OP.2 TDD) | | OP.2 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $Q_{rxlevmin}$ | dBm | -140 | |
| $N_{oc}$ | dBm/15 kHz | -99.4 | |
| RSRP | dBm/15 KHz | -88.9 | -103.1 |
| $\hat{E}_s / I_{ot}$ | dB | 10.5 | -3.70 |
| $\hat{E}_s / N_{oc}$ | dB | 10.5 | -3.70 |
| $T_{reselectionEUTRAN}$ | s | 0 | |
| $S_{nonintrasearch}$ | dB | Not sent | |
| $Thresh_{serving, low}$ | dB | 44 | |
| $Thresh_{x, low}$ (Note 2) | dB | 24 | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: This refers to $Thresh_{x, low}$ which is included in E-UTRA system information, and is a threshold for GSM target cell. | | | |

**Table 4.4.2.5-2: Cell Specific Test requirement Parameters for Cell 2 GSM cell**

| Parameter | Unit | Cell 2 (GSM) | |
|---|---|---|---|
| | | T1 | T2 |
| Absolute RF Channel Number | | ARFCN 1 | |
| RXLEV | dBm | -90.00 | -75.00 |
| RXLEV_ACCESS_MIN | dBm | -105 | |
| MS_TXPWR_MAX_CCH | dBm | 24 | |

The cell re-selection delay is defined as the time from the beginning of time period T2, to the moment when the UE camps on Cell 2, and starts to send the RR Channel Request message for location update to Cell 2.

The cell re-selection delay test requirement in this case is expressed as:

Cell re-selection delay = 4 * TmeasureGSM + $T_{BCCH}$

TmeasureGSM = 6.4 s; as specified in TS 36.133 [4] Table 4.2.2.5.3-1 in clause 4.2.2.5.3

$T_{BCCH}$ = 1.9 s; the maximum time allowed to read the BCCH data, when being synchronized to a BCCH carrier as specified in TS 45.008 [15] clause 6.2

This gives a total of 25.6 s + $T_{BCCH}$, allow 26 s + $T_{BCCH}$ in the test case.

The cell re-selection delay shall be less than a total of 27.9 seconds in this test case (note: this gives a total of 26 seconds for the $T_{measureGSM}$ calculation plus 1.9 s for $T_{BCCH}$ but the test allows 27.9 seconds).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 4.5 E-UTRAN to HRPD Cell Re-Selection

## 4.5.1 E-UTRAN FDD - HRPD Cell re-selection

### 4.5.1.1 E-UTRAN FDD - HRPD Cell Reselection: HRPD is of Lower Priority

#### 4.5.1.1.1 Test purpose

To verify that the UE is able to search and measure neighbouring HRPD cells and compare to the E-UTRA serving cell to meet the inter-RAT cell re-selection requirements.

#### 4.5.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support HRPD.

#### 4.5.1.1.3 Minimum conformance requirements

In order to perform measurement and cell reselection to HRPD cell, the UE shall acquire the timing of HRPD cells.

When the measurement rules indicate that HRPD cells are to be measured, the UE shall measure CDMA2000 HRPD Pilot Strength of HRPD cells in the neighbour cell list at the minimum measurement rate specified in this section.

The parameter ‘Number of HRPD Neighbour Frequency’, which is transmitted on E-UTRAN BCCH, is the number of carriers used for all HRPD cells in the neighbour cell list.

When the RSRP of the E-UTRA serving cell (or other cells on the same frequency layer) is lower than ‘HRPD Start Measuring E-UTRAN Rx Power Strength Threshold’ and HRPD is of lower priority than the currently selected E-UTRAN frequency layer, the UE shall measure CDMA2000 HRPD Pilot Strength of the HRPD cells at least every (Number of HRPD Neighbour Frequency)*$T_{measureHRPD}$. In case HRPD is of higher priority than the currently selected E-

UTRAN frequency layer the UE shall measure HRPD cells at least every (Number of HRPD Neighbour Frequency)* $T_{higher_proirty_search}T_{higher_priority_measure}$. The parameter $T_{higher_proirty_search}$ $T_{higher_priority_measure}$ is defined in section 4.2.2 of TS 36.133 [4].

The UE shall be capable of evaluating that the HRPD cell has met cell reselection criterion defined in TS 36.304 [6] within $T_{evaluateHRPD}$.

Table 4.2.2.5.4-1 of TS 36.133 [4] clause 4.2.2.5.4 gives values of $T_{measureHRPD}$ and $T_{evaluateHRPD}$

### 4.5.1.1.4 Test description

#### 4.5.1.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] Clause 4.3.1.

1. Connect the SS (nodeB emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.22
2. The general test parameter settings are set up according to Table 4.5.1.1.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.5.1.1.4.3
5. There is one E-UTRA FDD cell and one HRPD cell specified in the test. Cell 1(E-UTRA FDD cell) is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table4.5.1.1.4.1-1: General Test Parameters for E-UTRAN FDD - lower priority HRPD Cell Re-selection**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell 1 | E-UTRAN FDD cell |
| | Neighbour cell | | Cell 2 | HRPD cell |
| Final condition | Active cell | | Cell 2 | HRPD cell is selecting during T2 |
| DRX cycle length | | s | 1.28 | |
| E-UTRA FDD RF Channel Number | | | 1 | Only one FDD carrier frequency is used. |
| E-UTRA FDD Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| HRPD RF Channel Number | | | 1 | Only one HRPD carrier frequency is used. |
| E-UTRA FDD PRACH configuration | | | 4 | As specified in table 5.7.1-2 in TS 36.211 |
| E_UTRA FDD Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| T1 | | s | 30 | |
| T2 | | s | 30 | |

#### 4.5.1.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The UE is requested to monitor the neighbouring cell on one HRPD cell. In the test there are two successive time periods, with time duration of T1 and T2 respectively.

Both E-UTRAN FDD cell 1 and HRPD cell 2 are already identified by the UE prior to the start of the test. At T1 the UE is camped on to cell 1. Cell2 is of lower priority than cell 1. Cell 1 and Cell 2 belong to different tracking areas.

In the following test procedure “UE responds” means “UE starts transmitting access probe preambles on the Access Channel”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.

2. Set the parameters according to T1 in Tables 4.5.1.1.5-1 and 4.5.1.1.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts

3. When T1 expires the SS shall switch the power setting from T1 to T2 as specified in Tables 4.5.1.1.5-1 and 4.5.1.1.5-2.

4. The SS waits for probe preambles on the Access Channel on cell 2 from the UE to perform cell re-selection on Cell 2.

5. If the UE responds on Cell 2 during time duration T2 within 21 seconds from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

6. After T2 expires, switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.

7. Repeat steps 2-6 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

#### 4.5.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 and 4.4.7.1 with the following exceptions:

**Table 4.5.1.1.4.3-1: Common Exception messages**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.3-11<br>Table H.2.3-12 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

### 4.5.1.1.5 Test requirement

Tables 4.5.1.1.5-1 and 4.5.1.1.5-2 define the primary level settings including test tolerances for E-UTRAN FDD - HRPD cell re-selection test (HRPD cell is of lower priority).

**Table 4.5.1.1.5-1: Cell Specific Test Parameters for E-UTRAN FDD (Cell # 1)**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.1.2 (OP.2 FDD) | | OP.2 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA[Note 1] | dB | | |
| OCNG_RB[Note 1] | dB | | |

| | | | |
|---|---|---|---|
| $N_{oc}$ | dBm/15 kHz | -99.4 | |
| RSRP | dBm/15 KHz | -88.9 | -103.1 |
| $\hat{E}_s / I_{ot}$ | dB | 10.5 | -3.7 |
| $\hat{E}_s / N_{oc}$ | dB | 10.5 | -3.7 |
| $Treselection_{EUTRAN}$ | S | 0 | |
| Snonintrasearch | dB | Not sent | |
| cellReselectionPriority | - | 1 | |
| Qrxlevmin | dBm | -140 | |
| Qrxlevminoffset | dB | 0 | |
| Pcompensation | dB | 0 | |
| $S_{ServingCell}$ | dB | 51.1 | 36.9 |
| $Thresh_{serving, low}$ | dB | 44 | |
| Propagation Condition | | AWGN | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | |

**Table 4.5.1.1.5-2: Cell Specific Test Parameters for HRPD (cell # 2)**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | **T1** | **T2** |
| HRPD RF Channel Number | | 1 | |
| $\frac{Control \quad E_b}{N_t}$ (38.4 kbps) | dB | 21 | |
| $\frac{Control \quad E_b}{N_t}$ (76.8 kbps) | dB | 18 | |
| $\hat{I}_{or} / I_{oc}$ | dB | 0 | 0 |
| $I_{oc}$ | dBm/ 1.2288 MHz | -55 | |
| CDMA2000 HRPD Pilot Strength | dB | -3 | -3 |
| Propagation Condition | | AWGN | |
| $S_{nonServingCell,x}$ | | -6 | |
| Treselection | s | 0 | |
| hrpd-CellReselectionPriority | - | 0 | |
| $Thresh_{x, low}$ | | -14 | |

The cell reselection delay to lower priority is defined as the time from the beginning of time period T2, to the moment when the UE camps on cell 2 and starts to send access probe preambles on the Access Channel on cell 2.

The cell re-selection delay to the lower priority cell 2 shall be less than 21 s.

The cell re-selection delay to lower priority cell can be expressed as: $T_{evaluateHRPD} + T_{SI-HRPD}$

Where:

$T_{evaluatHRPD}$ 19.2 s for 1.28 s DRX cycle as specified in TS 36.133 [4] Clause 4.2.2.5 Table 4.2.2.5.4-1

$T_{SI-HRPD}$ Maximum repetition period of relevant system information blocks that need to be received by the UE to camp on cell 2; 1280 ms is assumed in this test case.

This gives a total of 20.48 s for the lower priority cell reselection, allow 21 s in the test case.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 4.5.2 E-UTRAN TDD - HRPD Cell re-selection

### 4.5.2.1 E-UTRAN TDD - HRPD Cell Reselection: HRPD is of Lower Priority

#### 4.5.2.1.1 Test purpose

To verify that the UE is able to search and measure neighbouring HRPD cells and compare to the E-UTRA serving cell to meet the inter-RAT cell re-selection requirements.

#### 4.5.2.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward that support HRPD.

#### 4.5.2.1.3 Minimum conformance requirements

In order to perform measurement and cell reselection to HRPD cell, the UE shall acquire the timing of HRPD cells.

When the measurement rules indicate that HRPD cells are to be measured, the UE shall measure CDMA2000 HRPD Pilot Strength of HRPD cells in the neighbour cell list at the minimum measurement rate specified in this section.

The parameter 'Number of HRPD Neighbour Frequency', which is transmitted on E-UTRAN BCCH, is the number of carriers used for all HRPD cells in the neighbour cell list.

When the RSRP of the E-UTRA serving cell (or other cells on the same frequency layer) is lower than 'HRPD Start Measuring E-UTRAN Rx Power Strength Threshold' and HRPD is of lower priority than the currently selected E-UTRAN frequency layer, the UE shall measure CDMA2000 HRPD Pilot Strength of the HRPD cells at least every (Number of HRPD Neighbour Frequency)*$T_{measureHRPD}$. In case HRPD is of higher priority than the currently selected E-UTRAN frequency layer the UE shall measure HRPD cells at least every (Number of HRPD Neighbour Frequency)* $T_{higher_proirty_search}T_{higher_priority_measure}$. The parameter $T_{higher_proirty_search}$ $T_{higher_priority_measure}$ is defined in section 4.2.2 of TS 36.133 [4].

The UE shall be capable of evaluating that the HRPD cell has met cell reselection criterion defined in TS 36.304 [6] within $T_{evaluateHRPD}$.

Table 4.2.2.5.4-1 of TS 36.133 [4] clause 4.2.2.5.4 gives values of $T_{measureHRPD}$ and $T_{evaluateHRPD}$

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.5.4 and A.4.5.2.1.

#### 4.5.2.1.4 Test description

##### 4.5.2.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] Clause 4.3.1.

1. Connect the SS (nodeB emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.22
2. The general test parameter settings are set up according to Table 4.5.2.1.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.5.2.1.4.3
5. There is one E-UTRA TDD cell and one HRPD cell specified in the test. Cell 1(E-UTRA TDD cell) is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 4.5.2.1.4.1-1: General Test Parameters for E-UTRAN TDD - lower priority HRPD Cell Re-selection**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell 1 | E-UTRAN TDD cell |
| | Neighbour cell | | Cell 2 | HRPD cell |
| Final condition | Active cell | | Cell 2 | HRPD cell is selecting during T2 |
| Uplink-downlink configuration of cell 1 | | | 1 | As specified in table 4.2.2 in TS 36.211 |
| Special subframe configuration of cell 1 | | | 6 | As specified in table 4.2.1 in TS 36.211 |
| CP length of cell 1 | | | Normal | |
| DRX cycle length | | s | 1.28 | |
| E-UTRA TDD RF Channel Number | | | 1 | Only one TDD carrier frequency is used. |
| E-UTRA TDD Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| HRPD RF Channel Number | | | 1 | Only one HRPD carrier frequency is used. |
| E-UTRA TDD PRACH configuration of cell 1 | | | 53 | As specified in table 4.7.1-3 in TS 36.211 |
| E_UTRA TDD Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| T1 | | s | 30 | |
| T2 | | s | 30 | |

### 4.5.2.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The UE is requested to monitor the neighbouring cell on one HRPD cell. In the test there are two successive time periods, with time duration of T1 and T2 respectively.

Both E-UTRAN TDD cell 1 and HRPD cell 2 are already identified by the UE prior to the start of the test. At T1 the UE is camped on to cell 1. Cell2 is of lower priority than Cell 1. Cell 1 and Cell 2 belong to different tracking areas.

In the following test procedure “UE responds” means “UE starts transmitting access probe preambles on the Access Channel”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.
2. Set the parameters according to T1 in Tables 4.5.2.1.5-1 and 4.5.2.1.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts
3. When T1 expires the SS shall switch the power setting from T1 to T2 as specified in Tables 4.5.2.1.5-1 and 4.5.1.1.5-2.
4. The SS waits for probe preambles on the Access Channel on cell 2 from the UE to perform cell re-selection on Cell 2.
5. If the UE responds on Cell 2 during time duration T2 within 21 seconds from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
6. After T2 expires, switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.
7. Repeat steps 2-6 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

### 4.5.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 and 4.4.7.1 with the following exceptions:

**Table 4.5.2.1.4.3-1: Common Exception messages**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.3-11<br>Table H.2.3-12 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-2 |

### 4.5.2.1.5 Test requirement

Tables 4.5.2.1.5-1 and 4.5.2.1.5-2 define the primary level settings including test tolerances for E-UTRAN TDD - HRPD cell re-selection test (HRPD cell is of lower priority).

**Table 4.5.2.1.1-2: Cell Specific Test Parameters for E-UTRAN TDD (Cell # 1)**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.2.2 (OP.2 TDD) | | OP.2 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $N_{oc}$ | dBm/15 kHz | -99.4 | |
| RSRP | dBm/15 KHz | -88.9 | -103.1 |
| $\hat{E}_s / I_{ot}$ | dB | 10.5 | -3.7 |
| $\hat{E}_s / N_{oc}$ | dB | 10.5 | -3.7 |
| $Treselection_{EUTRAN}$ | S | 0 | |
| Snonintrasearch | dB | Not sent | |
| cellReselectionPriority | - | 1 | |
| Qrxlevmin | dBm | -140 | |
| Qrxlevminoffset | dB | 0 | |
| Pcompensation | dB | 0 | |
| $S_{ServingCell}$ | dB | 51.1 | 36.9 |
| $Thresh_{serving, low}$ | dB | 44 | |
| Propagation Condition | | AWGN | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | |

**Table 4.5.2.1.1-3: Cell Specific Test Parameters for HRPD (cell # 2)**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| HRPD RF Channel Number | | 1 | |
| $\frac{\text{Control } E_b}{N_t}$ (38.4 kbps) | dB | 21 | |
| $\frac{\text{Control } E_b}{N_t}$ (76.8 kbps) | dB | 18 | |
| $\hat{I}_{or}/I_{oc}$ | dB | 0 | 0 |
| $I_{oc}$ | dBm/ 1.2288 MHz | -55 | |
| CDMA2000 HRPD Pilot Strength | dB | -3 | -3 |
| Propagation Condition | | AWGN | |
| $S_{nonServingCell,x}$ | | -6 | |
| Treselection | s | 0 | |
| hrpd-CellReselectionPriority | - | 0 | |
| $Thresh_{x, low}$ | | -14 | |

The cell reselection delay to lower priority is defined as the time from the beginning of time period T2, to the moment when the UE camps on cell 2 and starts to send access probe preambles on the Access Channel on cell 2.

The cell re-selection delay to the lower priority cell 2 shall be less than 21 s.

The cell re-selection delay to lower priority cell can be expressed as: $T_{evaluateHRPD} + T_{SI-HRPD}$

Where:

$T_{evaluatHRPD}$ 19.2 s for 1.28 s DRX cycle as specified in TS 36.133 [4] Clause 4.2.2.5 Table 4.2.2.5.4-1

$T_{SI-HRPD}$ Maximum repetition period of relevant system information blocks that need to be received by the UE to camp on cell 2; 1280 ms is assumed in this test case.

This gives a total of 20.48 s for the lower priority cell reselection, allow 21 s in the test case.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 4.6 E-UTRAN to cdma2000 1xRTT Cell Re-Selection

## 4.6.1 E-UTRAN FDD - cdma2000 1xRTT Cell re-selection

### 4.6.1.1 E-UTRAN FDD - cdma2000 1x Cell Reselection: cdma2000 1X is of Lower Priority

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

The intra-frequency cell reselection criteria related to scaling of measurement rules parameters need to be specified when parameters are finalized

The intra-frequency cell reselection criteria related to exact scaling parameters for different mobility states are undefined

Measurement bandwidth (current assumption is 6RB) is undefined

The "out of service" criteria is undefined

The transmission scheme (1Tx or 2Tx) undefined

The Message contents are undefined

The Test system uncertainties applicable to this test are undefined, including uncertainties above 3GHz

Test tolerances have not yet been applied to the wanted and interfering signal levels

#### 4.6.1.1.1 Test purpose

To verify that the UE is able to search and measure neighbouring cdma2000 1xRTT cells and compare to the E-UTRA serving cell to meet the inter-RAT cell re-selection requirements when the cdma2000 1x is of lower priority.

#### 4.6.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support cdma2000 1xRTT.

#### 4.6.1.1.3 Minimum conformance requirements

In order to perform measurement and cell reselection to cdma2000 1X cell, the UE shall acquire the timing of cdma2000 1X cells.

When the measurement rules indicate that cdma2000 1X cells are to be measured, the UE shall measure cdma2000 1x RTT Pilot Strength of cdma2000 1X cells in the neighbour cell list at the minimum measurement rate specified in this section.

The parameter 'Number of CDMA2000 1X Neighbour Frequency', which is transmitted on E-UTRAN BCCH, is the number of carriers used for all cdma2000 1X cells in the neighbour cell list.

When the RSRP of the E-UTRA serving cell (or other cells on the same frequency layer) is lower than 'CDMA2000 1X Start Measuring E-UTRAN Rx Power Strength Threshold' and cdma2000 1X is of lower priority than the currently selected E-UTRAN frequency layer, the UE shall measure Pilot Ec/Io of the CDMA2000 1X cells at least every (Number of CDMA2000 1X Neighbour Frequency)*$T_{measureCDMA2000\ 1X}$. In case cdma2000 1X is of higher priority than the currently selected E-UTRAN frequency layer, the UE shall measure cdma2000 1X cells at least every (Number of CDMA2000 1X Neighbour Frequency)* $T_{higher_proirty_search}T_{higher_priority_measure}$. The parameter $T_{higher_proirty_search}$ $T_{higher_priority_measure}$ is defined in section 4.2.2 of TS 36.133 [4].

The UE shall be capable of evaluating that the cdma2000 1X cell has met cell reselection criterion defined in TS 36.304 [6] within $T_{evaluateCDMA2000\ 1X}$.

Table 4.2.2.5.5-1 of TS 36.133 [4] clause 4.2.2.5.5 gives values of $T_{measureCDMA2000\ 1X}$ and $T_{evaluateCDMA2000\ 1X.}$

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.5.5 and A.4.6.1.1.

#### 4.6.1.1.4 Test description

##### 4.6.1.1.4.1 Initial conditions

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] Clause 4.3.1.

1. Connect the SS (nodeB emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.22
2. The general test parameter settings are set up according to Table 4.6.1.1.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.6.1.1.4.3
5. There is one E-UTRA FDD cell and one CDMA2000 1xRTT cell specified in the test. Cell 1(E-UTRA FDD cell) is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 4.6.1.1.4.1-1: General Test Parameters for E-UTRAN FDD - lower priority cdma2000 1X Cell Re-selection**

| Parameter | Unit | Value | Comment |
|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell 1 | E-UTRAN FDD cell |
| | Neighbour cell | | Cell 2 | cdma2000 1X cell |
| Final condition | Active cell | | Cell 2 | cdma2000 1X cell is selecting during T2 |
| DRX cycle length | | s | 1.28 | |
| E-UTRA FDD RF Channel Number | | | 1 | Only one FDD carrier frequency is used. |
| E-UTRA FDD Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| cdma2000 1X RF Channel Number | | | 1 | Only one cdma2000 1X carrier frequency is used. |
| E-UTRA FDD PRACH configuration | | | 4 | As specified in table 5.7.1-2 in TS 36.211 [9] |
| E_UTRA FDD Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| T1 | | s | 30 | |
| T2 | | s | 30 | |

#### 4.6.1.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The UE is requested to monitor the neighbouring cell on one CDMA200 1xRTT cell. In the test there are two successive time periods, with time duration of T1 and T2 respectively.

Both E-UTRAN FDD cell 1 and CDMA2000 1xRTT cell 2 are already identified by the UE prior to the start of the test. At T1 the UE is camped on to cell 1. Cell2 is of lower priority than cell 1. Cell 1 and Cell 2 belong to different tracking areas.

In the following test procedure “UE responds” means “UE starts transmitting access probe preambles on the Access Channel”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.
2. Set the parameters according to T1 in Tables 4.6.1.1.5-1 and 4.6.1.1.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Tables 4.6.1.1.5-1 and 4.6.1.1.5-2.
4. The SS waits for access probe preambles on the Access Channel on cell 2 from the UE to perform cell re-selection on Cell 2.
5. If the UE responds on Cell 2 during time duration T2 within 21 seconds from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
6. After T2 expires, switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.
7. Repeat steps 2-6 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

#### 4.6.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 4.6.1.1.4.3-1: Common Exception messages**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.3-15<br>Table H.2.3-16 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

#### 4.6.1.1.5 Test requirements

Tables 4.6.1.1.5-1 and 4.6.1.1.5-2 define the primary level settings including test tolerances for E-UTRAN FDD - cdma2000 1xRTT cell re-selection test (cdma2000 1x cell is of lower priority).

**Table 4.6.1.1.5-1: Cell Specific Test Parameters for E-UTRAN FDD (Cell # 1)**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.1.2 (OP.2 FDD) | | OP.2 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | |
| RSRP Note 3 | dBm/15 KHz | -89+ TT | -100+ TT |
| $\hat{E}_s / I_{ot}$ | dB | 9+ TT | -2+ TT |
| $\hat{E}_s / N_{oc}$ | dB | 9 | -2 |
| $Treselection_{EUTRAN}$ | S | 0 | |
| Snonintrasearch | dB | Not sent | |
| cellReselectionPriority | - | 1 | |
| Qrxlevmin | dBm | -140 | |
| Qrxlevminoffset | dB | 0 | |
| Pcompensation | dB | 0 | |
| $S_{ServingCell}$ | dB | 51 | 40 |
| $Thresh_{serving, low}$ | dB | 43 | |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 4.6.1.1.5-2: Cell Specific Test Parameters for cdma2000 1X (cell # 2)**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| cdma2000 1X RF Channel Number | | 1 | |
| $\frac{\text{Pilot } E_c}{I_{or}}$ | dB | [-7] | |
| $\frac{\text{Sync } E_c}{I_{or}}$ | dB | [-16] | |
| $\frac{\text{Paging } E_c}{I_{or}}$ (4.8 kbps) | dB | [-12] | |
| $\hat{I}_{or}/I_{oc}$ | dB | [0] + TT | [0] + TT |
| $I_{oc}$ | dBm/ 1.2288 MHz | -55 | |
| CDMA2000 1xRTT Pilot Strength | dB | [-10] + TT | [-10] + TT |
| Propagation Condition | | AWGN | |
| $S_{nonServingCell,x}$ | | [-20] | |
| Treselection | s | 0 | |
| oneXRTT-CellReselectionPriority | - | 0 | |
| $Thresh_{x, low}$ | | [-28] | |

The cell reselection delay to lower priority is defined as the time from the beginning of time period T2, to the moment when the UE camps on cell 2 and starts to send access probe preambles on the Access Channel on cell 2.

The cell re-selection delay to the lower priority cell 2 shall be less than 21 s.

The cell re-selection delay to lower priority cell can be expressed as: $T_{evaluatecdma2000\ 1X} + T_{SI\text{-}cdma2000\ 1X}$

Where:

$T_{evaluatcdma2000\ 1X}$ = 19.2 s for 1.28 s DRX cycle as specified in TS 36.133 [4] Clause 4.2.2.5 Table 4.2.2.5.5-1

$T_{SI\text{-}cdma2000\ 1X}$ Maximum repetition period of relevant system information blocks that need to be received by the UE to camp on cell 2; 1280 ms is assumed in this test case.

This gives a total of 20.48 s for the lower priority cell reselection, allow 21 s in the test case.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 4.6.2 E-UTRAN TDD - cdma2000 1xRTT Cell re-selection

### 4.6.2.1 E-UTRAN TDD - cdma2000 1x Cell Reselection: cdma2000 1X is of Lower Priority

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

The Test system uncertainties applicable to this test are undefined, including uncertainties above 3GHz

Test tolerances have not yet been applied to the wanted and interfering signal levels

#### 4.6.2.1.1 Test purpose

To verify that the UE is able to search and measure neighbouring cdma2000 1xRTT cells and compare to the E-UTRA serving cell to meet the inter-RAT cell re-selection requirements when the cdma2000 1x is of lower priority.

#### 4.6.2.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward that support cdma2000 1xRTT.

#### 4.6.2.1.3 Minimum conformance requirements

In order to perform measurement and cell reselection to cdma2000 1X cell, the UE shall acquire the timing of cdma2000 1X cells.

When the measurement rules indicate that cdma2000 1X cells are to be measured, the UE shall measure cdma2000 1x RTT Pilot Strength of cdma2000 1X cells in the neighbour cell list at the minimum measurement rate specified in this section.

The parameter 'Number of CDMA2000 1X Neighbour Frequency', which is transmitted on E-UTRAN BCCH, is the number of carriers used for all cdma2000 1X cells in the neighbour cell list.

When the RSRP of the E-UTRA serving cell (or other cells on the same frequency layer) is lower than 'CDMA2000 1X Start Measuring E-UTRAN Rx Power Strength Threshold' and cdma2000 1X is of lower priority than the currently selected E-UTRAN frequency layer, the UE shall measure Pilot Ec/Io of the CDMA2000 1X cells at least every (Number of CDMA2000 1X Neighbour Frequency)*$T_{measureCDMA2000\ 1X}$. In case cdma2000 1X is of higher priority than the currently selected E-UTRAN frequency layer, the UE shall measure cdma2000 1X cells at least every (Number of CDMA2000 1X Neighbour Frequency)* $T_{higher_proirty_search}T_{higher_priority_measure}$. The parameter $T_{higher_proirty_search}$ $T_{higher_priority_measure}$ is defined in section 4.2.2 of TS 36.133 [4].

The UE shall be capable of evaluating that the cdma2000 1X cell has met cell reselection criterion defined in TS 36.304 [6] within $T_{evaluateCDMA2000\ 1X}$.

Table 4.2.2.5.5-1 of TS 36.133 [4] clause 4.2.2.5.5 gives values of $T_{measureCDMA2000\ 1X}$ and $T_{evaluateCDMA2000\ 1X.}$

The normative reference for this requirement is TS 36.133 [4] clause 4.2.2.5.5 and A.4.6.2.1.

#### 4.6.2.1.4 Test description

##### 4.6.2.1.4.1 Initial conditions

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] Clause 4.3.1.

1. Connect the SS (nodeB emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.22
2. The general test parameter settings are set up according to Table 4.6.2.1.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 4.6.2.1.4.3.
5. There is one E-UTRA TDD cell and one CDMA2000 1xRTT cell specified in the test. Cell 1(E-UTRA TDD cell) is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 4.6.2.1.4.1-1: General Test Parameters for E-UTRAN TDD - lower priority cdma2000 1X Cell Re-selection**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active cell | | Cell 1 | E-UTRAN TDD cell |
| | Neighbour cell | | Cell 2 | cdma2000 1X cell |
| Final condition | Active cell | | Cell 2 | cdma2000 1X cell is selecting during T2 |
| DRX cycle length | | s | 1.28 | |
| E-UTRA TDD RF Channel Number | | | 1 | Only one TDD carrier frequency is used. |
| E-UTRA TDD Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| cdma2000 1X RF Channel Number | | | 1 | Only one cdma2000 1X carrier frequency is used. |
| E-UTRA TDD PRACH configuration | | | 53 | As specified in table 5.7.1-3 in TS 36.211 |

| | | | |
|---|---|---|---|
| Uplink-downlink configuration of cell 1 | | 1 | As specified in table 4.2.2 in TS 36.211 |
| Special subframe configuration of cell 1 | | 6 | As specified in table 4.2.1 in TS 36.211 |
| E_UTRA TDD Access Barring Information | - | Not Sent | No additional delays in random access procedure. |
| T1 | s | 30 | |
| T2 | s | 30 | |

#### 4.6.2.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The UE is requested to monitor the neighbouring cell on one CDMA200 1xRTT cell. In the test there are two successive time periods, with time duration of T1 and T2 respectively.

Both E-UTRAN TDD cell 1 and CDMA2000 1xRTT cell 2 are already identified by the UE prior to the start of the test. At T1 the UE is camped on to cell 1. Cell2 is of lower priority than cell 1. Cell 1 and Cell 2 belong to different tracking areas.

In the following test procedure “UE responds” means “UE starts transmitting access probe preambles on the Access Channel”.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.
2. Set the parameters according to T1 in Tables 4.6.2.1.5-1 and 4.6.2.1.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Tables 4.6.2.1.5-1 and 4.6.2.1.5-2.
4. The SS waits for access probe preambles on the Access Channel on cell 2 from the UE to perform cell re-selection on Cell 2.
5. If the UE responds on Cell 2 during time duration T2 within 21 seconds from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
6. After T2 expires, switch off and on the UE and ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2 in Cell 1.
7. Repeat steps 2-6 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

#### 4.6.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 4.6.2.1.4.3-1: Common Exception messages**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.3-15<br>Table H.2.3-16 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-2 |

### 4.6.2.1.5 Test requirements

Tables 4.6.2.1.5-1 and 4.6.2.1.5-2 define the primary level settings including test tolerances for E-UTRAN TDD - cdma2000 1xRTT cell re-selection test (cdma2000 1x cell is of lower priority).

**Table 4.6.2.1.5-1: Cell Specific Test Parameters for E-UTRAN TDD (Cell # 1)**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.2.2 (OP.2 TDD) | | OP.2 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | |
| RSRP Note 3 | dBm/15 KHz | -89+TT | -102+TT |
| $\hat{E}_s / I_{ot}$ | dB | 9+TT | -4+TT |
| $\hat{E}_s / N_{oc}$ | dB | 9 | -4 |
| $Treselection_{EUTRAN}$ | S | 0 | |
| Snonintrasearch | dB | Not sent | |
| cellReselectionPriority | - | 1 | |
| Qrxlevmin | dBm | -140 | |
| Qrxlevminoffset | dB | 0 | |
| Pcompensation | dB | 0 | |
| $\mathbf{S_{ServingCell}}$ | dB | 51 | 38 |
| $Thresh_{serving, low}$ | dB | 44 | |
| Propagation Condition | | AWGN | |

Note 1: CNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: SRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 4.6.2.1.5-2: Cell Specific Test Parameters for cdma2000 1X (cell # 2)**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| cdma2000 1X RF Channel Number | | 1 | |
| $\frac{\text{Pilot } E_c}{I_{or}}$ | dB | [-7] | |
| $\frac{\text{Sync } E_c}{I_{or}}$ | dB | [-16] | |
| $\frac{\text{Paging } E_c}{I_{or}}$ (4.8 kbps) | dB | [-12] | |
| $\hat{I}_{or}/I_{oc}$ | dB | [0] + TT | [0] + TT |
| $I_{oc}$ | dBm/ 1.2288 MHz | -55 | |
| CDMA2000 1xRTT Pilot Strength | dB | [-10] + TT | [-10] + TT |
| Propagation Condition | | AWGN | |
| $S_{nonServingCell,x}$ | | [-20] | |
| Treselection | s | 0 | |
| oneXRTT-CellReselectionPriority | - | 0 | |
| $Thresh_{x, low}$ | | [-28] | |

The cell reselection delay to lower priority is defined as the time from the beginning of time period T2, to the moment when the UE camps on cell 2 and starts to send access probe preambles on the Access Channel on cell 2.

The cell re-selection delay to the lower priority cell 2 shall be less than 21 s.

The cell re-selection delay to lower priority cell can be expressed as: $T_{evaluatecdma2000\ 1X} + T_{SI\text{-}cdma2000\ 1X}$

Where:

$T_{evaluatcdma2000\ 1X}$ = 19.2 s for 1.28 s DRX cycle as specified in TS 36.133 [4] Clause 4.2.2.5 Table 4.2.2.5.5-1

$T_{SI\text{-}cdma2000\ 1X}$ Maximum repetition period of relevant system information blocks that need to be received by the UE to camp on cell 2; 1280 ms is assumed in this test case.

This gives a total of 20.48 s for the lower priority cell reselection, allow 21 s in the test case.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 5 E-UTRAN RRC_CONNECTED State Mobility

When the UE is in RRC_CONNECTED state on a cell, network-controlled UE-assisted handovers are performed. The UE makes measurements of attributes of the serving and neighbour cells to enable the handover process. This process allows the UE to transfer a connection between the UE and current cell to target cell.

SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Annex A1. The SS sends downlink MAC padding bits on the DL RMC.

Due to the undefined UE behaviour regarding the sending of HARQ-ACK after receiving a RRC message triggering an handover (acc. 3GPP TS 36.331 [5] Subclause 5.3.5.4), the SS behaviour when waiting for the appropriate HARQ acknowledgement should be as follows:

- Reception of an HARQ-ACK will cause no HARQ delay exclusion (acc. subclause 3A.1).
- Reception of an HARQ-NACK will cause HARQ retransmission and HARQ delay exclusion (acc. subclause 3A.1).
- UE-DTX (as observed by SS) will cause HARQ retransmission, but no HARQ delay exclusion (acc. subclause 3A.1).

Uplink for E-UTRA cell(s) is configured according to Annex A.3.

# 5.1 E-UTRAN Handover

## 5.1.1 E-UTRAN FDD-FDD Handover intra frequency case

### 5.1.1.1 Test purpose

To verify the UE's ability to perform handover in RRC_CONNECTED state when an intra-frequency handover is commanded by meeting the UE maximum RRC procedure delay and interruption time requirements.

### 5.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

### 5.1.1.3 Minimum conformance requirements

The handover delay $D_{handover}$ shall be less than maximum RRC procedure delay + $T_{interrupt}$ in RRC_CONNECTED state.

When the UE receive a RRC message implying handover the UE shall be ready to start the transmission of the new uplink PRACH channel within $D_{handover}$ seconds from the end of the last TTI containing the RRC command.

Where:

$D_{handover}$ equals the maximum RRC procedure delay defined in TS 36.331 [5] clause 11.2 plus the interruption time stated in TS 36.133 [4] clause 5.1.2.1.2.

The interruption time is the time between end of the last TTI containing the RRC command on the old PDSCH and the time the UE starts transmission of the new PRACH, excluding the RRC procedure delay. This requirement applies when UE is not required to perform any synchronisation procedure before transmitting on the new PRACH.

When intra-frequency handover is commanded, the interruption time shall be less than $T_{interrupt}$. The $T_{interrupt}$ equation is defined as:

$$T_{interrupt} = T_{search} + T_{IU} + 20 \text{ ms}$$

Where:

$T_{search}$ is the time required to search the target cell when the target cell is not already known when the handover command is received by the UE. If the target cell is known, then $T_{search}$ = 0 ms. If the target cell is unknown and signal quality is sufficient for successful cell detection on the first attempt, then $T_{search}$ = 80 ms. Regardless of whether DRX is in use by the UE, $T_{search}$ shall still be based on non-DRX target cell search times.

$T_{IU}$ is the interruption uncertainty in acquiring the first available PRACH occasion in new cell. $T_{IU}$ can be up to 30 ms.

NOTE: The actual value of $T_{IU}$ shall depend upon the PRACH configuration used in the target cell.

In the interruption requirement a cell is know if it has been meeting the relevant cell identification requirement during the last 5 seconds otherwise it is unknown. Relevant cell identification requirements are described in TS 36.133 [4] clause 8.1.2.2.1 for intra-frequency handover.

The normative reference for this requirement is TS 36.133 [4] clause 5.1.2.1 and A.5.1.1

### 5.1.1.4 Test description

#### 5.1.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. The general test parameter settings are set up according to Table 5.1.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 5.1.1.4.3.

5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.1.1.4.1-1: General Test Parameters for E-UTRAN FDD-FDD intra frequency handover test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 FDD | As specified in clause A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in clause A.2.1 |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbouring cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF Channel Number | | | 1 | Only one FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| A3-Offset | | dB | 0 | |
| Hysteresis | | dB | 0 | |
| Time To Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | | OFF |
| CP length | | | Normal | |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| PRACH configuration | | | 4 | As specified in table 5.7.1-2 in TS 36.211 [9] |
| Time offset between cells | | ms | 3 | Asynchronous cells<br>3ms or 92160*Ts |
| T1 | | s | 5 | |
| T2 | | s | ≤5 | |
| T3 | | s | 1 | |

### 5.1.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and send a measurement report. T3 is defined as the end of the last TTI containing the RRC message implying handover.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

2. Set the parameters according to T1 in Table 5.1.1.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. The SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 5.1.1.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3.

7. SS shall transmit an RRCConnectionReconfiguration message defined in Table H.3.2-3 implying handover to Cell 2.

8. The start of T3 is the instant when the last TTI containing the RRC Connection reconfiguration message implying handover is sent to the UE, at that instant the SS shall switch the power setting from T2 to T3 as specified in Table 5.1.1.1.5-1.

9. The UE shall transmit RRCConnectionReconfigurationComplete message.

10. If the UE transmits the uplink PRACH channel to Cell 2 less than 50 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

11. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

13. Repeat step 2-12 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 5.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 5.1.1.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra frequency handover test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7<br>Table H.3.2-1<br>Table H.3.2-3 |

**Table 5.1.1.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD intra frequency handover test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 5.1.1.4.3-3: MeasResults: Additional E-UTRAN FDD-FDD intra frequency handover test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCellsCHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 5.1.1.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD intra frequency handover test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 5.1.1.4.3-5: *RACH-ConfigCommon-DEFAULT*: Additional E-UTRAN FDD-FDD intra frequency handover test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-12 RACH-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RACH-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| powerRampingParameters SEQUENCE { | | | |
| preambleInitialReceivedTargetPower | dBm-90 | | |
| } | | | |
| } | | | |

## 5.1.1.5 Test requirement

Tables 5.1.1.4.1-1 and 5.1.1.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-FDD intra frequency handover test case.

**Table 5.1.1.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD-FDD intra frequency handover test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.1 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD | OP.1 FDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 8 | -3.8 | -3.8 | -Infinity | 2.86 | 2.86 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 8 | 8 | 8 | - Infinity | 11.5 | 11.5 |
| RSRP Note 3 | dBm/15 KHz | -90 | -90 | -90 | - Infinity | -86.5 | -86.5 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE start to transmit the PRACH to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

$T_{interrupt} = T_{search} + T_{IU} + 20$ ms

$T_{search} = 0$, since Cell 2 is known prior to the test

$T_{IU} = 15$ ms, since 10 ms due to uncertainty in frame and 5 ms additional delay due to PRACH transmission occasion

Maximum RRC procedure delay = 15 ms as defined in TS 36.331 [5] clause 11.2

The handover delay $D_{handover}$ shall be less than a total of 50 ms in this test case (note: this gives a total of 15 ms for maximum RRC procedure delay plus 35 ms for $T_{interrupt}$).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.1.2 E-UTRAN TDD-TDD Handover intra frequency case

### 5.1.2.1 Test purpose

To verify the UE's ability to perform handover in RRC_CONNECTED state when an intra-frequency handover is commanded by meeting the UE maximum RRC procedure delay and interruption time requirements.

### 5.1.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

### 5.1.2.3 Minimum conformance requirements

The handover delay $D_{handover}$ shall be less than maximum RRC procedure delay + $T_{interrupt}$ in RRC_CONNECTED state.

When the UE receives a RRC message implying handover, the UE shall be ready to start the transmission of the new uplink PRACH channel within $D_{handover}$ seconds from the end of the last TTI containing the RRC command.

Where:

$D_{handover}$ equals the maximum RRC procedure delay defined in TS 36.331 [5] clause 11.2 plus the interruption time stated in TS 36.133 [4] clause 5.2.2.4.2.

The interruption time is the time between end of the last TTI containing the RRC command on the old PDSCH and the time the UE starts transmission of the new PRACH, excluding the RRC procedure delay. This requirement applies when UE is not required to perform any synchronisation procedure before transmitting on the new PRACH.

When intra-frequency handover is commanded, the interruption time shall be less than $T_{interrupt}$. The $T_{interrupt}$ equation is defined as:

$$T_{interrupt} = T_{search} + T_{IU} + 20 \text{ ms}$$

Where:

$T_{search}$ is the time required to search the target cell when the target cell is not already known when the handover command is received by the UE. If the target cell is known, then $T_{search}$ = 0 ms. If the target cell is unknown and signal quality is sufficient for successful cell detection on the first attempt, then $T_{search}$ = 80 ms. Regardless of whether DRX is in use by the UE, $T_{search}$ shall still be based on non-DRX target cell search times.

$T_{IU}$ is the interruption uncertainty in acquiring the first available PRACH occasion in new cell. $T_{IU}$ can be up to 30ms.

NOTE: The actual value of $T_{IU}$ shall depend upon the PRACH configuration used in the target cell.

In the interruption requirement a cell is know if it has been meeting the relevant cell identification requirement during the last 5 seconds otherwise it is unknown. Relevant cell identification requirements are described in TS 36.133 [4] clause 8.1.2.2.2 for intra-frequency handover.

The normative reference for this requirement is TS 36.133 [4] clause 5.2.2.4 and A.5.1.2.

### 5.1.2.4 Test description

#### 5.1.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. The general test parameter settings are set up according to Table 5.1.2.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 5.1.2.4.3.

5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.1.2.4.1-1: General Test Parameters for E-UTRAN TDD/TDD Intra Frequency Handover case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 TDD | As specified in Annex A.1.2 |
| PCFICH/PDCCHPHICH parameters | | | DL Reference Measurement Channel R.6 TDD | As specified in Annex A.2.2 |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbouring cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF Channel Number | | | 1 | Only one TDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| A3-Offset | | dB | 0 | |
| Hysteresis | | dB | 0 | |
| Time To Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | | OFF |
| CP length | | | Normal | |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211 |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211 |
| PRACH configuration index | | | 53 | As specified in table 5.7.1-3 in TS 36.211 |
| Time offset between cells | | µs | 3 | Synchronous cells<br>3µs or 92*Ts |
| T1 | | s | 5 | |
| T2 | | s | ≤5 | |
| T3 | | s | 1 | |

### 5.1.2.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and send a measurement report. The start of T3 is defined as the end of the last TTI containing the RRC message implying handover.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

2. Set the parameters according to T1 in Table 5.1.2.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 5.1.2.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3.

7. SS shall transmit an RRCConnectionReconfiguration message defined in Table H.3.2-3 implying handover to Cell 2.

8. The start of T3 is the instant when the last TTI containing the RRC Connection reconfiguration message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T2 to T3 as specified in Table 5.1.2.1.5-1.

9. The UE shall transmit RRCConnectionReconfigurationComplete message.

10. If the UE transmits the uplink PRACH channel to Cell 2 less than 50ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

11. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

13. Repeat step 2-12 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 5.1.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 5.1.2.4.3-1: Common Exception messages for E-UTRAN TDD-TDD intra frequency handover test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7<br>Table H.3.2-2<br>Table H.3.2-3 |

**Table 5.1.2.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD-TDD intra frequency handover test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 5.1.2.4.3-3: MeasResults: Additional E-UTRAN TDD-TDD intra frequency handover test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 5.1.2.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN TDD-TDD intra frequency handover test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of the Cell 2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 5.1.2.4.3-5: *RACH-ConfigCommon-DEFAULT*: Additional E-UTRAN TDD-TDD intra frequency handover test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-12 RACH-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RACH-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| powerRampingParameters SEQUENCE { | | | |
| preambleInitialReceivedTargetPower | dBm-98 | | |
| } | | | |

## 5.1.2.5 Test requirement

Tables 5.1.2.4.1-1 and 5.1.2.5-1 define the primary level settings including test tolerances for E-UTRAN TDD/TDD Intra Frequency Handover test.

**Table 5.1.2.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD/TDD Intra Frequency Handover case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | OP.1 TDD | OP.2 TDD | OP.2 TDD | OP.2 TDD | OP.1 TDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA[Note] | dB | | | | | | |
| OCNG_RB[Note] | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 8 | -3.8 | -3.8 | -Infinity | 2.86 | 2.86 |
| $N_{oc}$ | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 8 | 8 | 8 | - Infinity | 11.5 | 11.5 |
| RSRP | dBm/15 KHz | -90 | -90 | -90 | - Infinity | -86.5 | -86.5 |
| Propagation Condition | | AWGN | | | | | |
| Note: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | | | | | |

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE start to transmit the PRACH to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

The maximum RRC procedure delay test requirement in this case is 15ms, as specified in TS 36.331 [5] clause 11.2.

The $T_{interrupt}$ test requirement in this case is 35 ms expressed as:

$T_{interrupt} = T_{search} + T_{IU} + 20$ ms

$T_{search}$ = 0, since cell 2 is known prior to the test

$T_{IU}$ = 15 ms, since 10 ms due to uncertainty in frame and 5 ms additional delay due to PRACH transmission occasion

The handover delay $D_{handover}$ shall be less than a total of 50 ms in this test case (note: this gives a total of 15 ms for maximum RRC procedure delay plus 35 ms for $T_{interrupt}$).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.1.3 E-UTRAN FDD-FDD Handover inter frequency case

### 5.1.3.1 Test purpose

To verify the UE's ability to perform handover in RRC_CONNECTED state when an inter-frequency handover is commanded by meeting the UE maximum RRC procedure delay and interruption time requirements.

### 5.1.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward. Applicability requires support for FGI bits 5, 13, and 25.

### 5.1.3.3 Minimum conformance requirements

The handover delay $D_{handover}$ shall be less than maximum RRC procedure delay + $T_{interrupt}$ in RRC_CONNECTED state.

When the UE receive a RRC message implying handover the UE shall be ready to start the transmission of the new uplink PRACH channel within $D_{handover}$ seconds from the end of the last TTI containing the RRC command.

Where:

$D_{handover}$ equals the maximum RRC procedure delay defined in TS 36.331 [5] clause 11.2 plus the interruption time stated in TS 36.133 [4] clause 5.1.2.1.2.

The interruption time is the time between end of the last TTI containing the RRC command on the old PDSCH and the time the UE starts transmission of the new PRACH, excluding the RRC procedure delay. This requirement applies when UE is not required to perform any synchronisation procedure before transmitting on the new PRACH.

When inter-frequency handover is commanded, the interruption time shall be less than $T_{interrupt}$. The $T_{interrupt}$ equation is defined as:

$$T_{interrupt} = T_{search} + T_{IU} + 20 \text{ ms}$$

Where:

$T_{search}$ is the time required to search the target cell when the target cell is not already known when the handover command is received by the UE. If the target cell is known, then $T_{search} = 0$ ms. If the target cell is unknown and signal quality is sufficient for successful cell detection on the first attempt, then $T_{search} = 80$ ms. Regardless of whether DRX is in use by the UE, $T_{search}$ shall still be based on non-DRX target cell search times.

$T_{IU}$ is the interruption uncertainty in acquiring the first available PRACH occasion in the new cell. $T_{IU}$ can be up to 30 ms.

NOTE: The actual value of $T_{IU}$ shall depend upon the PRACH configuration used in the target cell.

In the interruption requirement a cell is know if it has been meeting the relevant cell identification requirement during the last 5 seconds otherwise it is unknown. Relevant cell identification requirements are described in TS 36.133 [4] clause 8.1.2.3.1 for inter-frequency handover.

The normative reference for this requirement is TS 36.133 [4] clause 5.1.2.1 and A.5.2.1.

### 5.1.3.4 Test description

#### 5.1.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. The general test parameter settings are set up according to Table 5.1.3.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 5.1.3.4.3.

5. There are two E-UTRA FDD carriers and one cell on each carrier specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.1.3.4.1-1: General Test Parameters for E-UTRAN FDD-FDD inter frequency handover test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 FDD | As specified in clause A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in clause A.2.1 |
| Initial conditions | Active cell | | Cell 1 | Cell 1 is on RF channel number 1 |
| | Neighbour cell | | Cell 2 | Cell 2 is on RF channel number 2 |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF channel number | | | 1, 2 | Two FDD carriers are used |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| A3-Offset | | dB | -4 | |
| Hysteresis | | dB | 0 | |
| TimeToTrigger | | Ms | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | DRX_L | As specified in Table 5.1.3.5-2 |
| PRACH configuration | | | 4 | As specified in table 5.7.1-2 in 3GPP TS 36.211 [9] |
| Access Barring Information | | - | Not sent | No additional delays in random access procedure |
| Time offset between cells | | 3 | ms | Asynchronous cells<br>3ms or 92160*Ts |
| Gap pattern configuration Id | | | 0 | As specified in Table 8.1.2.1-1 in 3GPP TS 36.133 [4] started before T2 starts |
| T1 | | s | 5 | |
| T2 | | s | ≤5 | |
| T3 | | s | 1 | |

### 5.1.3.4.2 Test procedure

The test consists of one active cell and one neighbour cell. PDCCHs indicating new transmissions shall be sent continuously to ensure that the UE would not enter the DRX state. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and send a measurement report. T3 is defined as the end of the last TTI containing the RRC message implying handover.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

2. Set the parameters according to T1 in Table 5.1.3.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 5.1.3.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3.

7. SS shall transmit an RRCConnectionReconfiguration message defined in Table H.3.2-3 implying handover to Cell 2.

8. The start of T3 is the instant when the last TTI containing the RRC connection reconfiguration message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T2 to T3 as specified in Table 5.1.3.5-1.

9. The UE shall transmit RRCConnectionReconfigurationComplete message.

10. If the UE transmits the uplink PRACH channel to Cell 2 less than 50 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

11. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

13. Repeat step 2-12 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 5.1.3.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 5.1.3.4.3-1: Common Exception messages for E-UTRAN FDD-FDD inter-frequency handover test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-3<br>Table H.3.1-7<br>Table H.3.2-1<br>Table H.3.2-3<br>Table H.3.6-2 |

**Table 5.1.3.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD inter frequency handover test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -8 (-4 dB) | -4 is actual value in dB (-8 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 5.1.3.4.3-3: *MeasResults*: Additional E-UTRAN FDD-FDD inter frequency handover test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 5.1.3.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD inter frequency handover test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 5.1.3.4.3-5: *RACH-ConfigCommon-DEFAULT*: Additional E-UTRAN FDD-FDD inter frequency handover test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-12 RACH-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RACH-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| powerRampingParameters SEQUENCE { | | | |
| preambleInitialReceivedTargetPower | dBm-90 | | |
| } | | | |
| } | | | |

## 5.1.3.5 Test requirement

Tables 5.1.3.4.1-1, 5.1.3.5-1, and 5.1.3.5-2 define the primary level settings including test tolerances for E-UTRAN FDD-FDD inter frequency handover test case.

**Table 5.1.3.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD-FDD inter frequency handover test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.1 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD | OP.1 FDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | 4 | -Infinity | 7.50 | 7.50 |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | 4 | -Infinity | 7.50 | 7.50 |
| RSRP Note 3 | dBm/15 KHz | -94 | -94 | -94 | -Infinity | -90.5 | -90.5 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 5.1.3.5-2: Reference DRX-Configuration to be used in E-UTRAN FDD-FDD inter frequency handover test case**

| Parameter | Value | Comments |
|---|---|---|
| Reference configuration | DRX_L | As defined in 4.8.2.1.5 in TS 36.508 [7.] |
| onDurationTimer | psf6 | |
| drx-InactivityTimer | psf1920 | |
| drx-RetransmissionTimer | psf16 | |
| longDRX-CycleStartOffset | sf1280, 0 | |
| shortDRX | disabled | |
| Note: For further information see section 6.3.2 in 3GPP TS 36.331 [5]. | | |

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE start to transmit the PRACH to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

$T_{interrupt} = T_{search} + T_{IU} + 20$ ms

$T_{search} = 0$, since Cell 2 is known prior to the test

$T_{IU}$ = 15 ms, since 10 ms due to uncertainty in frame and 5 ms additional delay due to PRACH transmission occasion

Maximum RRC procedure delay = 15 ms as defined in TS 36.331 [5] clause 11.2

The handover delay $D_{handover}$ shall be less than a total of 50 ms in this test case (note: this gives a total of 15 ms for maximum RRC procedure delay plus 35 ms for $T_{interrupt}$).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.1.4 E-UTRAN TDD-TDD Handover inter frequency case

### 5.1.4.1 Test purpose

To verify the UE's ability to perform handover in RRC_CONNECTED state when an inter-frequency handover is commanded by meeting the UE maximum RRC procedure delay and interruption time requirements.

### 5.1.4.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward. Applicability requires support for FGI bits 5, 13, and 25.

### 5.1.4.3 Minimum conformance requirements

The handover delay $D_{handover}$ shall be less than maximum RRC procedure delay + $T_{interrupt}$ in RRC_CONNECTED state.

When the UE receives a RRC message implying handover, the UE shall be ready to start the transmission of the new uplink PRACH channel within $D_{handover}$ seconds from the end of the last TTI containing the RRC command.

Where:

$D_{handover}$ equals the maximum RRC procedure delay defined in TS 36.331 [5] clause 11.2 plus the interruption time stated in TS 36.133 [4] clause 5.2.2.4.2.

The interruption time is the time between end of the last TTI containing the RRC command on the old PDSCH and the time the UE starts transmission of the new PRACH, excluding the RRC procedure delay. This requirement applies when UE is not required to perform any synchronisation procedure before transmitting on the new PRACH.

When inter-frequency handover is commanded, the interruption time shall be less than $T_{interrupt}$. The $T_{interrupt}$ equation is defined as:

$$T_{interrupt} = T_{search} + T_{IU} + 20 \text{ ms}$$

Where:

$T_{search}$ is the time required to search the target cell when the target cell is not already known when the handover command is received by the UE. If the target cell is known, then $T_{search}$ = 0 ms. If the target cell is unknown and signal quality is sufficient for successful cell detection on the first attempt, then $T_{search}$ = 80 ms. Regardless of whether DRX is in use by the UE, $T_{search}$ shall still be based on non-DRX target cell search times.

$T_{IU}$ is the interruption uncertainty in acquiring the first available PRACH occasion in new cell. $T_{IU}$ can be up to 30ms.

NOTE: The actual value of $T_{IU}$ shall depend upon the PRACH configuration used in the target cell.

The normative reference for this requirement is TS 36.133 [4] clause 5.2.2.4 and A.5.1.4

## 5.1.4.4 Test description

### 5.1.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.
2. The general test parameter settings are set up according to Table 5.1.4.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are defined in clause 5.1.4.4.3.
5. There are two E-UTRA TDD carriers and one cell on each carrier specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.1.4.4.1-1: General Test Parameters for E-UTRAN TDD-TDD inter frequency handover test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 TDD | As specified in clause A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 TDD | As specified in clause A.2.2 |
| Initial conditions | Active cell | | Cell 1 | Cell 1 is on RF channel number 1 |
| | Neighbour cell | | Cell 2 | Cell 2 is on RF channel number 2 |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF channel number | | | 1, 2 | Two TDD carriers are used |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| A3-Offset | | dB | -4 | |
| Hysteresis | | dB | 0 | |
| TimeToTrigger | | Ms | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | DRX_L | As specified in Table 5.1.4.5-2 |
| CP length | | | Normal | |
| PRACH configuration | | | 53 | As specified in table 5.7.1-3 in TS 36.211 |
| Access Barring Information | | - | Not sent | No additional delays in random access procedure |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211 |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211 |
| Time offset between cells | | µs | 3 | Synchronous cells<br>3µs or 92*Ts |
| Gap pattern configuration Id | | | 1 | As specified in Table 8.1.2.1-1 in 3GPP TS 36.133 [4] started before T2 starts |
| T1 | | s | 5 | |
| T2 | | s | ≤5 | |
| T3 | | s | 1 | |

**Table 5.1.4.4.1-2: Void**

### 5.1.4.4.2 Test procedure

The test consists of one active cell and one neighbour cell. PDCCHs indicating new transmissions shall be sent continuously to ensure that the UE would not enter the DRX state. The test consists of three successive time periods,

with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and send a measurement report. The start of T3 is defined as the end of the last TTI containing the RRC message implying handover.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 5.1.4.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 5.1.4.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3.
7. SS shall transmit an RRCConnectionReconfiguration message defined in Table H.3.2-3 implying handover to Cell 2.
8. The start of T3 is the instant when the last TTI containing the RRC connection reconfiguration message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T2 to T3 as specified in Table 5.1.4.5-1.
9. The UE shall transmit RRCConnectionReconfigurationComplete message.
10. If the UE transmits the uplink PRACH channel to Cell 2 less than 50 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
11. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
13. Repeat step 2-12 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 5.1.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 5.1.4.4.3-1: Common Exception messages for E-UTRAN TDD-TDD inter-frequency handover test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-3<br>Table H.3.1-7<br>Table H.3.2-2<br>Table H.3.2-3<br>Table H.3.6-2 |

**Table 5.1.4.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD-TDD inter frequency handover test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -8 (-4 dB) | -4 is actual value in dB (-8 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 5.1.4.4.3-3: *MeasResults*: Additional E-UTRAN TDD-TDD inter frequency handover test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 5.1.4.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN TDD-TDD inter frequency handover test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of the Cell 2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 5.1.4.4.3-5: *RACH-ConfigCommon-DEFAULT*: Additional E-UTRAN TDD-TDD inter frequency handover test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-12 RACH-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RACH-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| powerRampingParameters SEQUENCE { | | | |
| preambleInitialReceivedTargetPower | dBm-98 | | |
| } | | | |
| } | | | |

## 5.1.4.5 Test requirement

Tables 5.1.4.4.1-1 and 5.1.4.5-1 define the primary level settings including test tolerances for E-UTRAN TDD/TDD Inter Frequency Handover test.

**Table 5.1.4.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD/TDD Inter Frequency Handover case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in TS 36.133 [4] D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | OP.1 TDD | OP.2 TDD | OP.2 TDD | OP.2 TDD | OP.1 TDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA[Note] | dB | | | | | | |
| OCNG_RB[Note] | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | 4 | -Infinity | 7.50 | 7.50 |
| $N_{oc}$ | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | 4 | -Infinity | 7.50 | 7.50 |
| RSRP | dBm/15 KHz | -94 | -94 | -94 | - Infinity | -90.5 | -90.5 |
| Propagation Condition | | AWGN | | | | | |
| Note: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | | | | | |

**Table 5.1.4.5-2: Reference DRX-Configuration to be used in E-UTRAN TDD-TDD inter frequency handover test case**

| Parameter | Value | Comments |
|---|---|---|
| Reference configuration | DRX_L | As defined in 4.8.2.1.5 in TS 36.508 [7.] |
| onDurationTimer | psf6 | |
| drx-InactivityTimer | psf1920 | |
| drx-RetransmissionTimer | psf16 | |
| longDRX-CycleStartOffset | sf1280, 0 | |
| shortDRX | disabled | |
| Note: For further information see section 6.3.2 in 3GPP TS 36.331 [5]. | | |

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE start to transmit the PRACH to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

The maximum RRC procedure delay test requirement in this case is 15ms, as specified in TS 36.331 [5] clause 11.2.

The $T_{interrupt}$ test requirement in this case is 35 ms expressed as:

$T_{interrupt} = T_{search} + T_{IU} + 20$ ms

$T_{search} = 0$, since cell 2 is known prior to the test

$T_{IU} = 15$ ms, since 10 ms due to uncertainty in frame and 5 ms additional delay due to PRACH transmission occasion

The handover delay $D_{handover}$ shall be less than a total of 50 ms in this test case.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.1.5 E-UTRAN FDD-FDD inter frequency Handover: unknown target cell

### 5.1.5.1 Test purpose

To verify the UE's ability to perform handover in RRC_CONNECTED state when an inter-frequency handover: unknown target cell is commanded by meeting the handover to an unknown target cell delay requirements.

### 5.1.5.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward. Applicability requires support for FGI bits 13, and 25.

### 5.1.5.3 Minimum conformance requirements

Procedure delays for all procedures that can command a handover are specified in TS 36.331 [5].

When the UE receive a RRC message implying handover the UE shall be ready to start the transmission of the new uplink PRACH channel within $D_{handover}$ seconds from the end of the last TTI containing the RRC command.

Where:

$D_{handover}$ equals the maximum RRC procedure delay defined in TS 36.331 [5] clause 11.2 plus the interruption time stated in TS 36.133 [4] clause 5.1.2.1.2.

The interruption time is the time between end of the last TTI containing the RRC command on the old PDSCH and the time the UE starts transmission of the new PRACH, excluding the RRC procedure delay. This requirement applies when UE is not required to perform any synchronisation procedure before transmitting on the new PRACH.

When inter-frequency handover is commanded, the interruption time shall be less than $T_{interrupt}$. The $T_{interrupt}$ equation is defined as:

$$T_{interrupt} = T_{search} + T_{IU} + 20 \text{ ms}$$

Where:

$T_{search}$ is the time required to search the target cell when the target cell is not already known when the handover command is received by the UE. If the target cell is known, then $T_{search} = 0$ ms. If the target cell is unknown and signal quality is sufficient for successful cell detection on the first attempt, then Tsearch = 80 ms. Regardless of whether DRX is in use by the UE, $T_{search}$ shall still be based on non-DRX target cell search times.

$T_{IU}$ is the interruption uncertainty in acquiring the first available PRACH occasion in the new cell. $T_{IU}$ can be up to 30 ms.

NOTE: The actual value of $T_{IU}$ shall depend upon the PRACH configuration used in the target cell.

In the interruption requirement a cell is known if it has been meeting the relevant cell identification requirement during the last 5 seconds otherwise it is unknown. Relevant cell identification requirements are described in TS 36.133 [4] clause 8.1.2.3.1 for inter-frequency handover.

The normative reference for this requirement is TS 36.133 [4] clause 5.1.2.1 and A.5.1.5.

## 5.1.5.4 Test description

### 5.1.5.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.
2. The general test parameter settings are set up according to Table 5.1.5.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are defined in clause 5.1.5.4.3.
5. There are two E-UTRA FDD carriers and one cell on each carrier specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.1.5.4.1-1: General Test Parameters for E-UTRAN FDD-FDD inter frequency handover: unknown target cell test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Initial conditions | Active cell | | Cell 1 | Cell 1 is on RF channel number 1 |
| | Neighbouring cell | | Cell 2 | Cell 2 is on RF channel number 2 |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF channel number | | | 1, 2 | Two FDD carriers are used |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| DRX | | | OFF | Non-DRX test |
| PRACH configuration | | | 4 | As specified in table 5.7.1-2 in 3GPP TS 36.211 [9] |
| Access Barring Information | | - | Not sent | No additional delays in random access procedure |
| Time offset between cells | | | 3 ms | Asynchronous cells<br>3ms or 92160*Ts |
| T1 | | s | ≤5 | |
| T2 | | s | 1 | |

### 5.1.5.4.2 Test procedure

The test scenario comprises of two E-UTRA FDD carriers and one cell on each carrier as given in tables 5.1.5.4.1-1 and 5.1.5.5-1. No gap patterns are configured in the test case. The test consists of two successive time periods, with time durations of T1and T2 respectively. At the start of time duration T1, the UE does not have any timing information of Cell 2. A RRC message implying handover shall be sent to the UE during period T1. The start of T2 is the instant when the last TTI containing the RRC message implying handover is sent to the UE. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and start to transmit the PRACH to Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 5.1.5.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message implying handover to Cell 2.
4. The start of T2 is the instant when the last TTI containing the RRC connection reconfiguration message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T1 to T2 as specified in Table 5.1.5.5-1. T2 starts.
5. The UE shall transmit RRCConnectionReconfigurationComplete message.
6. If the UE transmits the uplink PRACH channel to Cell 2 less than 130 ms from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
7. After T2 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell..
8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
9. Repeat step 2-8 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 5.1.5.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 5.1.5.4.3-1: Common Exception messages for E-UTRAN FDD-FDD inter-frequency handover: unknown target cell test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1<br>Table H.3.2-3 |

**Table 5.1.5.4.3-2: *RACH-ConfigCommon-DEFAULT*: Additional E-UTRAN FDD-FDD inter frequency handover: unknown target cell test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-12 RACH-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RACH-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| powerRampingParameters SEQUENCE { | | | |
| preambleInitialReceivedTargetPower | dBm-90 | | |
| } | | | |
| } | | | |

## 5.1.5.5 Test requirement

Tables 5.1.5.4.1-1 and 5.1.5.5-1define the primary level settings including test tolerances for E-UTRAN FDD-FDD inter frequency handover: unknown target cell test.

**Table 5.1.5.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD-FDD inter frequency handover: unknown target cell test case**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.2 FDD | OP.2 FDD | OP.1 FDD |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA[Note 1] | dB | | | | |
| OCNG_RB[Note 1] | dB | | | | |

| | | | | | |
|---|---|---|---|---|---|
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | -Infinity | 7 |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | -Infinity | 7 |
| RSRP Note 3 | dBm/15 KHz | -94 | -94 | -Infinity | -91 |
| Propagation Condition | | AWGN | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The handover to an unknown target cell delay is defined as the time from the beginning of time period T2, to the moment the UE start to transmit the PRACH to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$ (note: the target cell is unknown)

$T_{interrupt} = T_{search} + T_{IU} + 20$ ms

$T_{search}$ = 80, since Cell 2 is unknown prior to the test

$T_{IU}$ = 15 ms, since 10 ms due to uncertainty in frame and 5 ms additional delay due to PRACH transmission occasion

Maximum RRC procedure delay = 15 ms as defined in TS 36.331 [5] clause 11.2

The handover delay $D_{handover}$ to an unknown target cell shall be less than a total of 130 ms in this test case (note: this gives a total of 15 ms for maximum RRC procedure delay plus 115 ms for $T_{interrupt}$).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.1.6 E-UTRAN TDD-TDD inter frequency handover: unknown target cell

### 5.1.6.1 Test purpose

To verify the UE's ability to perform handover in RRC_CONNECTED state when the target cell is unknown and an inter-frequency handover is commanded by meeting the UE maximum RRC procedure delay and interruption time requirements.

### 5.1.6.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward. Applicability requires support for FGI bits 13, and 25.

### 5.1.6.3 Minimum conformance requirements

Procedure delays for all procedures that can command a handover are specified in 3GPP TS 36.331 [5].

When the UE receive a RRC message implying handover the UE shall be ready to start the transmission of the new uplink PRACH channel within $D_{handover}$ seconds from the end of the last TTI containing the RRC command.

Where:

$D_{handover}$ equals the maximum RRC procedure delay defined in TS 36.331 [5] clause 11.2 plus the interruption time stated in TS 36.133 [4] clause 5.2.2.4.2.

The interruption time is the time between end of the last TTI containing the RRC command on the old PDSCH and the time the UE starts transmission of the new PRACH, excluding the RRC procedure delay. This requirement applies when UE is not required to perform any synchronisation procedure before transmitting on the new PRACH.

When inter-frequency handover is commanded, the interruption time shall be less than $T_{interrupt}$.

$$T_{interrupt} = T_{search} + T_{IU} + 20 \text{ ms}$$

Where:

$T_{search}$ is the time required to search the target cell when the target cell is not already known when the handover command is received by the UE. If the target cell is known, then $T_{search}$ = 0 ms. If the target cell is unknown and signal quality is sufficient for successful cell detection on the first attempt, then $T_{search}$ = 80 ms. Regardless of whether DRX is in use by the UE, $T_{search}$ shall still be based on non-DRX target cell search times.

$T_{IU}$ is the interruption uncertainty in acquiring the first available PRACH occasion in new cell. $T_{IU}$ can be up to 30ms.

NOTE: The actual value of $T_{IU}$ shall depend upon the PRACH configuration used in the target cell.

In the interruption requirement a cell is known if it has been meeting the relevant cell identification requirement during the last 5 seconds otherwise it is unknown. Relevant cell identification requirements are described in TS 36.133 [4] clause 8.1.2.3.4 for inter-frequency handover.

The normative reference for this requirement is TS 36.133 [4] clause 5.2.2.4 and A.5.1.6

## 5.1.6.4 Test description

### 5.1.6.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.
2. The general test parameter settings are set up according to Table 5.1.6.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are defined in clause 5.1.6.4.3.
5. There are two E-UTRA TDD carriers and one cell on each carrier specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.1.6.4.1-1: General test parameters for E-UTRAN TDD-TDD inter frequency handover test case when the target cell is unknown**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCHPHICH parameters | | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbouring cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF Channel Number | | | 1, 2 | Two TDD carriers |
| DRX | | | OFF | Non-DRX test |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211 |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211 |
| PRACH configuration index | | | 53 | As specified in table 5.7.1-3 in TS 36.211 |
| Time offset between cells | | μs | 3 | Synchronous cells<br>3μs or 92*Ts |
| Gap pattern configuration | | | - | No gap pattern configured |
| T1 | | s | ≤5 | |
| T2 | | s | 1 | |

### 5.1.6.4.2 Test procedure

The test scenario comprises of two E-UTRA TDD carriers and one cell on each carrier as given in tables 5.1.6.4.1-1 and 5.1.6.5-1. No gap patterns are configured in the test case. The test consists of two successive time periods, with time durations of T1 and T2 respectively. At the start of time duration T1, the UE does not have any timing information of Cell 2. Starting T2, Cell 2 becomes detectable and the UE receives a RRC handover command from the network. The start of T2 is the instant when the last TTI containing the RRC message implying handover is sent to the UE.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 5.1.6.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message implying handover to Cell 2.
4. The start of T2 is the instant when the last TTI containing the RRC connection reconfiguration message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T1 to T2 as specified in Table 5.1.6.5-1. T2 starts.
5. The UE shall transmit RRCConnectionReconfigurationComplete message.
6. If the UE transmits the uplink PRACH channel to Cell 2 less than 130 ms from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
7. After T2 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
9. Repeat step 2-8 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 5.1.6.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 5.1.6.4.3-1: Common Exception messages for E-UTRAN TDD-TDD inter frequency handover unknown target cell test requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.2-2<br>Table H.3.2-3 |

**Table 5.1.6.4.3-2: *RACH-ConfigCommon-DEFAULT*: Additional E-UTRAN TDD-TDD inter frequency handover: unknown target cell test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-12 RACH-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RACH-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| powerRampingParameters SEQUENCE { | | | |
| preambleInitialReceivedTargetPower | dBm-98 | | |
| } | | | |
| } | | | |

## 5.1.6.5 Test requirement

Tables 5.1.6.4.1-1 and 5.1.6.5-1 define the primary level settings including test tolerances for E-UTRAN TDD-TDD inter frequency handover test case when the target cell is unknown.

**Table 5.1.6.5-1: Cell specific test parameters for E-UTRAN TDD-TDD inter frequency handover test case when the target cell is unknown**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| OCNG Patterns defined in D.2.1(OP.1 TDD) and in D.2.2(OP.2 TDD) | | OP.1 TDD | OP.2 TDD | OP.2 TDD | OP.1 TDD |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RANote 1 | dB | | | | |
| OCNG_RBNote 1 | dB | | | | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 | -Infinity | -93 |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | -Infinity | 5 |
| SCH_RP Note 4 | dBm/15 kHz | -94 | -94 | -Infinity | -93 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | -Infinity | 5 |
| Propagation Condition | | AWGN | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The handover delay is defined as the time from the beginning of time period T2, to the moment the UE start to transmit the PRACH to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

The maximum RRC procedure delay test requirement in this case is 15ms, as specified in TS 36.331 [5] clause 11.2.

The $T_{interrupt}$ test requirement in this case is expressed as:

$T_{interrupt} = T_{search} + T_{IU}$ + 20 ms (note: the target cell is unknown)

$T_{search}$ = 80 ms, since Cell 2 is unknown prior to the test

$T_{IU}$ = 15 ms, since 10 ms due to uncertainty in frame and 5 ms additional delay due to PRACH transmission occasion

The handover delay $D_{handover}$ to an unknown target cell shall be less than a total of 130 ms in this test case (note: this gives a total of 15 ms for maximum RRC procedure delay plus 115 ms for $T_{interrupt}$).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.1.7 E-UTRAN FDD-TDD Handover inter frequency case

### 5.1.7.1 Test purpose

To verify the UE's ability to perform handover in RRC_CONNECTED state when an E-UTRAN FDD-TDD inter-frequency handover is commanded by meeting the UE maximum RRC procedure delay and interruption time requirements.

### 5.1.7.2 Test applicability

This test applies to all types of E-UTRA UE supporting FDD and TDD release 9 and forward. Applicability requires support for FGI bits 5, 25, and 30.

### 5.1.7.3 Minimum conformance requirements

Procedure delays for all procedures that can command a handover are specified in 3GPP TS 36.331 [5].

When the UE receives a RRC message implying handover, the UE shall be ready to start the transmission of the new uplink PRACH channel within $D_{handover}$ seconds from the end of the last TTI containing the RRC command.

Where:

$D_{handover}$ equals the maximum RRC procedure delay to be defined in section 11.2 in 3GPP TS36.331 [5] plus the interruption time stated in TS 36.133 [4] section 5.2.2.4.2.

The interruption time is the time between end of the last TTI containing the RRC command on the old PDSCH and the time the UE starts transmission of the new PRACH, excluding the RRC procedure delay. This requirement applies when UE is not required to perform any synchronisation procedure before transmitting on the new PRACH.

When inter-frequency handover is commanded, the interruption time shall be less than $T_{interrupt}$

$$T_{interrupt} = T_{search} + T_{IU} + 20 \text{ ms}$$

Where

$T_{search}$ is the time required to search the target cell when the target cell is not already known when the handover command is received by the UE. If the target cell is known, then $T_{search}$ = 0 ms. If the target cell is unknown and signal quality is sufficient for successful cell detection on the first attempt, then $T_{search}$ = 80 ms. Regardless of whether DRX is in use by the UE, $T_{search}$ shall still be based on non-DRX target cell search times.

$T_{IU}$ is the interruption uncertainty in acquiring the first available PRACH occasion in the new cell. $T_{IU}$ can be up to 30 ms.

NOTE: The actual value of $T_{IU}$ shall depend upon the PRACH configuration used in the target cell.

In the interruption requirement a cell is known if it has been meeting the relevant cell identification requirement during the last 5 seconds otherwise it is unknown. Relevant cell identification requirements are described in TS 36.133 [4] Section 8.1.2.3.4 for inter-frequency handover.

The normative reference for this requirement is TS 36.133 [4] clause 5.2.2.4 and A.5.1.7

### 5.1.7.4 Test description

#### 5.1.7.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.
2. The general test parameter settings are set up according to Table 5.1.7.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are defined in clause 5.1.7.4.3.
5. There are two E-UTRA carriers and one E-UTRA FDD Cell 1 and E-UTRA TDD Cell 2 on each carrier specified in the test. E-UTRA FDD Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.1.7.4.1-1: General Test Parameters for E-UTRAN FDD-TDD inter frequency handover test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Cell 1 PDSCH parameters | | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| Cell 1 PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Cell 2 PDSCH parameters | | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| Cell 2 PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Gap Pattern Id | | | 0 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbour cell | | Cell 2 | |
| Final conditions | Active cell | | Cell 2 | |
| Cell 1 E-UTRA RF channel number | | | 1 | One FDD carrier is used |
| Cell 2 E-UTRA RF channel number | | | 2 | One TDD carrier is used |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| A3-Offset | | dB | -4 | |
| Hysteresis | | dB | 0 | |
| Time to Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | |
| DRX | | | DRX_L | As specified in 3GPP TS 36.133 [4] section A.3.3 |
| CP length | | | Normal | |
| E-UTRA TDD Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in 3GPP TS 36.211. Applicable to cell 2. |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in 3GPP TS 36.211. Applicable to cell 2 |
| E-UTRA TDD PRACH configuration | | | 53 | As specified in table 5.7.1-3 in 3GPP TS 36.211 |
| Time offset between cells | | | 3 ms | Asynchronous cells |
| T1 | | s | 5 | |
| T2 | | s | ≤5 | |
| T3 | | s | 1 | |

### 5.1.7.4.2 Test procedure

The test scenario comprises of one E-UTRA FDD cell and one E-UTRA TDD cell. PDCCHs indicating new transmissions shall be sent continuously to ensure that the UE would not enter the DRX state. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and

send a measurement report. The start of T3 is defined as the end of the last TTI containing the RRC message implying handover.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 5.1.7.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 5.1.7.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3.
7. SS shall transmit an RRCConnectionReconfiguration message defined in Table H.3.2-3 implying handover to Cell 2.
8. The start of T3 is the instant when the last TTI containing the RRC connection reconfiguration message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T2 to T3 as specified in Table 5.1.7.5-1.
9. The UE shall transmit RRCConnectionReconfigurationComplete message.
10. If the UE transmits the uplink PRACH channel to Cell 2 less than 50 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
11. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
13. Repeat step 2-12 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 5.1.7.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 5.1.7.4.3-1: Common Exception messages for E-UTRAN FDD-TDD inter-frequency handover test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-3<br>Table H.3.1-7<br>Table H.3.2-2<br>Table H.3.2-3<br>Table H.3.6-2 |

**Table 5.1.7.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-TDD inter frequency handover test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -8 (-4 dB) | -4 is actual value in dB (-8 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 5.1.7.4.3-3: *MeasResults*: Additional E-UTRAN FDD-TDD inter frequency handover test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 5.1.7.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN FDD-TDD inter frequency handover test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of the Cell 2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 5.1.7.4.3-5: *RACH-ConfigCommon-DEFAULT*: Additional E-UTRAN FDD-TDD inter frequency handover test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-12 RACH-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RACH-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| powerRampingParameters SEQUENCE { | | | |
| preambleInitialReceivedTargetPower | dBm-98 | | |
| } | | | |
| } | | | |

## 5.1.7.5 Test requirement

Tables 5.1.7.4.1-1 and 5.1.7.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-TDD Inter Frequency Handover test.

**Table 5.1.7.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD (cell #1) in E-UTRAN FDD-TDD Inter Frequency Handover test case**

| Parameter | Unit | Cell 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $\hat{E}_s / I_{ot}$ | dB | 4.00 | 4.00 | 4.00 |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | |
| $\hat{E}_s / N_{oc}$ | dB | 4.00 | 4.00 | 4.00 |
| RSRP Note 3 | dBm/15 KHz | -94.00 | -94.00 | -94.00 |
| Propagation Condition | AWGN | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameter themselves.

**Table 5.1.7.5-2: Cell specific test parameters for E-UTRAN TDD (cell #2) in E-UTRAN FDD-TDD Inter frequency handover test case**

| Parameter | Unit | Cell 2 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel number | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.2 TDD | OP.2 TDD | OP.1 TDD |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $\hat{E}_s / I_{ot}$ | dB | -Infinity | 7.50 | 7.50 |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | |
| $\hat{E}_s / N_{oc}$ | dB | -Infinity | 7.50 | 7.50 |
| RSRP Note 3 | dBm/15 KHz | -Infinity | -90.50 | -90.50 |
| Propagation Condition | AWGN | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameter themselves.

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE start to transmit the PRACH to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

The maximum RRC procedure delay test requirement in this case is 15ms, as specified in TS 36.331 [5] clause 11.2.

The $T_{interrupt}$ test requirement in this case is 35 ms expressed as:

$T_{interrupt} = T_{search} + T_{IU} + 20$ ms

$T_{search} = 0$, since cell 2 is known prior to the test

$T_{IU} = 15$ ms, since 10 ms due to uncertainty in frame and 5 ms additional delay due to PRACH transmission occasion

The handover delay $D_{handover}$ shall be less than a total of 50 ms in this test case.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.1.8 E-UTRAN TDD-FDD Handover inter frequency case

### 5.1.8.1 Test purpose

To verify the UE's ability to perform handover in RRC_CONNECTED state when an E-UTRAN TDD-FDD inter-frequency handover is commanded by meeting the UE maximum RRC procedure delay and interruption time requirements.

### 5.1.8.2 Test applicability

This test applies to all types of E-UTRA UE supporting FDD and TDD release 9 and forward. Applicability requires support for FGI bits 5, 25, and 30.

### 5.1.8.3 Minimum conformance requirements

Procedure delays for all procedures that can command a handover are specified in 3GPP TS 36.331 [5].

When the UE receives a RRC message implying handover the UE shall be ready to start the transmission of the new uplink PRACH channel within $D_{handover}$ seconds from the end of the last TTI containing the RRC command.

Where:

$D_{handover}$ equals the maximum RRC procedure delay to be defined in section 11.2 in 3GPP TS 36.331 [5] plus the interruption time stated in TS 36.133 [4] section 5.1.2.1.2.

The interruption time is the time between end of the last TTI containing the RRC command on the old PDSCH and the time the UE starts transmission of the new PRACH, excluding the RRC procedure delay. This requirement applies when UE is not required to perform any synchronisation procedure before transmitting on the new PRACH.

When inter-frequency handover is commanded, the interruption time shall be less than $T_{interrupt}$

$$T_{interrupt} = T_{search} + T_{IU} + 20 \text{ ms}$$

Where:

$T_{search}$ is the time required to search the target cell when the target cell is not already known when the handover command is received by the UE. If the target cell is known, then $T_{search}$ = 0 ms. If the target cell is unknown and signal quality is sufficient for successful cell detection on the first attempt, then $T_{search}$ = 80 ms. Regardless of whether DRX is in use by the UE, $T_{search}$ shall still be based on non-DRX target cell search times.

$T_{IU}$ is the interruption uncertainty in acquiring the first available PRACH occasion in the new cell. $T_{IU}$ can be up to 30 ms.

NOTE: The actual value of $T_{IU}$ shall depend upon the PRACH configuration used in the target cell.

In the interruption requirement a cell is known if it has been meeting the relevant cell identification requirement during the last 5 seconds otherwise it is unknown. Relevant cell identification requirements are described in TS 36.133 [4] Section 8.1.2.3.1 for inter-frequency handover.

The normative reference for this requirement is TS 36.133 [4] clause 5.1.2.1 and A.5.1.8.

### 5.1.8.4 Test description

#### 5.1.8.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. The general test parameter settings are set up according to Table 5.1.8.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 5.1.8.4.3.

5. There are two E-UTRA carriers and one E-UTRA TDD Cell 1 and one E-UTRA FDD Cell 2 on each carrier specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.1.8.4.1-1: General Test Parameters for E-UTRAN TDD-FDD inter frequency handover test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Cell 1 PDSCH parameters | | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| Cell 1 PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Cell 2 PDSCH parameters | | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| Cell 2 PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Initial conditions | Active cell | | Cell 1 | Cell 1 is on RF channel number 1 |
| | Neighbouring cell | | Cell 2 | Cell 2 is on RF channel number 2 |
| Final condition | Active cell | | Cell 2 | |
| Cell 1 E-UTRA RF channel number | | | 1 | One TDD carrier is used |
| Cell 2 E-UTRA RF channel number | | | 2 | One FDD carrier is used |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| A3-Offset | | dB | -4 | |
| Hysteresis | | dB | 0 | |
| TimeToTrigger | | s | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | DRX_L | As specified in TS 36.133 [4] section A.3.3 |
| E-UTRA FDD PRACH configuration | | | 4 | As specified in table 5.7.1-2 in 3GPP TS 36.211 |
| E-UTRA FDD Access Barring Information | | - | Not sent | No additional delays in random access procedure |
| Time offset between cells | | | 3 ms | Asynchronous cells |
| Gap pattern configuration Id | | | 0 | As specified in Table 8.1.2.1-1 started before T2 starts |
| T1 | | s | 5 | |
| T2 | | s | ≤5 | |
| T3 | | s | 1 | |

### 5.1.8.4.2 Test procedure

The test consists of one active cell and one neighbour cell. PDCCHs indicating new transmissions should be sent continuously to ensure that the UE would not enter the DRX state. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and send a measurement report. The start of T3 is defined as the end of the last TTI containing the RRC message implying handover.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

2. Set the parameters according to T1 in Table 5.1.8.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 5.1.8.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3.
7. SS shall transmit an RRCConnectionReconfiguration message defined in Table H.3.2-3 implying handover to Cell 2.
8. The start of T3 is the instant when the last TTI containing the RRC connection reconfiguration message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T2 to T3 as specified in Table 5.1.8.5-1.
9. The UE shall transmit RRCConnectionReconfigurationComplete message.
10. If the UE transmits the uplink PRACH to Cell 2 less than 50 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
11. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
13. Repeat step 2-12 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 5.1.8.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 5.1.8.4.3-1: Common Exception messages for E-UTRAN TDD-FDD inter-frequency handover test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-3<br>Table H.3.1-7<br>Table H.3.2-1<br>Table H.3.2-3<br>Table H.3.6-2 |

**Table 5.1.8.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD-FDD inter frequency handover test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -8 (-4 dB) | -4 is actual value in dB (-8 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 5.1.8.4.3-3: *MeasResults*: Additional E-UTRAN TDD-FDD inter frequency handover test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 5.1.8.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN TDD-FDD inter frequency handover test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of the Cell 2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 5.1.8.4.3-5: *RACH-ConfigCommon-DEFAULT*: Additional E-UTRAN TDD-FDD inter frequency handover test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-12 RACH-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RACH-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| powerRampingParameters SEQUENCE { | | | |
| preambleInitialReceivedTargetPower | dBm-90 | | |
| } | | | |
| } | | | |

## 5.1.8.5 Test requirement

Tables 5.1.8.4.1-1, 5.1.8.5-1, and 5.1.8.5-2 define the primary level settings including test tolerances for E-UTRAN TDD-FDD inter frequency handover test case.

**Table 5.1.8.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD (cell #1) in TDD-FDD Inter frequency handover test case**

| Parameter | Unit | Cell 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $\hat{E}_s / I_{ot}$ | dB | 4.00 | 4.00 | 4.00 |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | |
| $\hat{E}_s / N_{oc}$ | dB | 4.00 | 4.00 | 4.00 |
| RSRP Note 3 | dBm/15 KHz | -94.00 | -94.00 | -94.00 |
| Propagation Condition | | AWGN | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameter themselves.

**Table 5.1.8.5-2: Cell specific test parameters for E-UTRAN FDD (cell #2) in E-UTRAN TDD-FDD Inter frequency handover test case**

| Parameter | Unit | Cell 2 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel number | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.2 FDD | OP.2 FDD | OP.1 FDD |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $\hat{E}_s / I_{ot}$ | dB | -Infinity | 7.50 | 7.50 |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | |
| $\hat{E}_s / N_{oc}$ | dB | -Infinity | 7.50 | 7.50 |
| RSRP Note 3 | dBm/15 KHz | -Infinity | -90.50 | -90.50 |
| Propagation Condition | AWGN | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameter themselves.

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE start to transmit the PRACH to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

$T_{interrupt} = T_{search} + T_{IU} + 20$ ms

$T_{search} = 0$, since Cell 2 is known prior to the test

$T_{IU} = 15$ ms, since 10 ms due to uncertainty in frame and 5 ms additional delay due to PRACH transmission occasion

Maximum RRC procedure delay = 15 ms as defined in TS 36.331 [5] clause 11.2

The handover delay $D_{handover}$ shall be less than a total of 50 ms in this test case (note: this gives a total of 15 ms for maximum RRC procedure delay plus 35 ms for $T_{interrupt}$).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.1.9 E-UTRAN FDD-FDD Intra frequency handover for 5MHz bandwidth

### 5.1.9.1 Test purpose

Same test purpose as in clause 5.1.1.

### 5.1.9.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that only support E-UTRA Band 31.

### 5.1.9.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 5.1.1.3 with the following exceptions:

- Instead of A.5.1.1 → use A.5.1.9.

### 5.1.9.4 Test description

#### 5.1.9.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.
2. The general test parameter settings are set up according to Table 5.1.9.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are defined in clause 5.1.9.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.1.9.4.1-1: General Test Parameters for E-UTRAN FDD-FDD intra frequency handover test case for 5MHz bandwidth**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.5 FDD | As specified in clause A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.11 FDD | As specified in clause A.2.1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 | |
| Note 1: See Table 5.1.1.4.1-1 for other general test parameters.<br>Note 2: This test is performed according to the principle defined in TS 36.133 [4] section A.3.7.2 | | | |

#### 5.1.9.4.2 Test procedure

Same test procedure as in clause 5.1.1.4.2 with the following exceptions:

- Instead of Table 5.1.1.5-1 → use Table 5.1.9.5-1.

#### 5.1.9.4.3 Message contents

Same message contents as in clause 5.1.1.4.3.

### 5.1.9.5 Test requirement

Same test requirement as in clause 5.1.1.5 with the following exceptions:

Tables 5.1.9.4.1-1 and 5.1.9.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-FDD intra frequency handover test case for 5MHz bandwidth.

**Table 5.1.9.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD-FDD intra frequency handover test case for 5MHz bandwidth**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| $BW_{channel}$ | MHz | 5 | | | 5 | | |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and in D.1.16 (OP.16 FDD) | | OP.15 FDD | OP.15 FDD | OP.16 FDD | OP.16 FDD | OP.16 FDD | OP.15 FDD |
| Note: See Table 5.1.1.5-1 for other cell-specific test parameters. | | | | | | | |

## 5.1.10 E-UTRAN FDD-FDD Handover intra frequency handover for UE category 0

### 5.1.10.1 Test purpose

To verify the UE's ability to perform handover in RRC_CONNECTED state when an intra-frequency handover is commanded by meeting the UE maximum RRC procedure delay and interruption time requirements.

### 5.1.10.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward of UE Category 0.

### 5.1.10.3 Minimum conformance requirements

The handover delay $D_{handover}$ shall be less than maximum RRC procedure delay + $T_{interrupt}$ in RRC_CONNECTED state.

When the UE receive a RRC message implying handover the UE shall be ready to start the transmission of the new uplink PRACH channel within $D_{handover}$ seconds from the end of the last TTI containing the RRC command.

Where:

> $D_{handover}$ equals the maximum RRC procedure delay defined in TS 36.331 [5] clause 11.2 plus the interruption time stated in TS 36.133 [4] clause 5.1.2.1.2.

The interruption time is the time between end of the last TTI containing the RRC command on the old PDSCH and the time the UE starts transmission of the new PRACH, excluding the RRC procedure delay. This requirement applies when UE is not required to perform any synchronisation procedure before transmitting on the new PRACH.

When intra-frequency handover is commanded, the interruption time shall be less than $T_{interrupt}$. The $T_{interrupt}$ equation is defined as:

$$T_{interrupt} = T_{search} + T_{IU} + 20 \text{ ms}$$

Where:

> $T_{search}$ is the time required to search the target cell when the target cell is not already known when the handover command is received by the UE. If the target cell is known, then $T_{search} = 0$ ms. If the target cell is unknown and signal quality is sufficient for successful cell detection on the first attempt, then $T_{search} = 80$ ms. Regardless of whether DRX is in use by the UE, $T_{search}$ shall still be based on non-DRX target cell search times.

$T_{IU}$ is the interruption uncertainty in acquiring the first available PRACH occasion in new cell. $T_{IU}$ can be up to 30 ms.

NOTE: The actual value of $T_{IU}$ shall depend upon the PRACH configuration used in the target cell.

In the interruption requirement a cell is know if it has been meeting the relevant cell identification requirement during the last 5 seconds otherwise it is unknown. Relevant cell identification requirements are described in TS 36.133 [4] clause 8.1.2.2.1 for intra-frequency handover.

The normative reference for this requirement is TS 36.133 [4] clause 5.1.2.1 and A.5.1.10

## 5.1.10.4 Test description

### 5.1.10.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources (without using the faders) to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20 using only main Tx/Rx antenna.
2. The general test parameter settings are set up according to Table 5.1.10.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are defined in clause 5.1.10.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.1.10.4.1-1: General Test Parameters for E-UTRAN FDD-FDD intra frequency handover for UE category 0 test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.13 FDD | As specified in clause A.1.3 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in clause A.2.1 |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbouring cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF Channel Number | | | 1 | Only one FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| A3-Offset | | dB | 0 | |
| Hysteresis | | dB | 0 | |
| Time To Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | | OFF |
| CP length | | | Normal | |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| PRACH configuration | | | 4 | As specified in table 5.7.1-2 in TS 36.211 [9] |
| Time offset between cells | | ms | 3 | Asynchronous cells<br>3ms or 92160*Ts |
| T1 | | s | 5 | |
| T2 | | s | ≤5 | |
| T3 | | s | 1 | |

#### 5.1.10.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and send a measurement report. T3 is defined as the end of the last TTI containing the RRC message implying handover.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 5.1.10.5-1. Propagation conditions are set according to Annex B clause B.2. T1 starts.
3. The SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 5.1.10.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3.
7. SS shall transmit an RRCConnectionReconfiguration message defined in Table H.3.2-3 implying handover to Cell 2.
8. The start of T3 is the instant when the last TTI containing the RRC Connection reconfiguration message implying handover is sent to the UE, at that instant the SS shall switch the power setting from T2 to T3 as specified in Table 5.1.10.5-1.
9. The UE shall transmit RRCConnectionReconfigurationComplete message.
10. If the UE transmits the uplink PRACH channel to Cell 2 less than 50 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
11. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
13. Repeat step 2-12 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 5.1.10.4.3 Message contents

Same message content as in clause 5.1.1.4.3.

### 5.1.10.5 Test requirement

Tables 5.1.10.4.1-1 and 5.1.10.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-FDD intra frequency handover test case.

**Table 5.1.10.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD-FDD intra frequency handover for UE category 0 test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.1 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD | OP.1 FDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 8 | -3.8 | -3.8 | -Infinity | 2.86 | 2.86 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 8 | 8 | 8 | - Infinity | 11.5 | 11.5 |
| RSRP Note 3 | dBm/15 KHz | -90 | -90 | -90 | - Infinity | -86.5 | -86.5 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE start to transmit the PRACH to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

$T_{interrupt} = T_{search} + T_{IU} + 20$ ms

$T_{search} = 0$, since Cell 2 is known prior to the test

$T_{IU}$ = 15 ms, since 10 ms due to uncertainty in frame and 5 ms additional delay due to PRACH transmission occasion

Maximum RRC procedure delay = 15 ms as defined in TS 36.331 [5] clause 11.2

The handover delay $D_{handover}$ shall be less than a total of 50 ms in this test case (note: this gives a total of 15 ms for maximum RRC procedure delay plus 35 ms for $T_{interrupt}$).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.1.11 E-UTRAN HD-FDD Handover intra frequency handover for UE category 0

### 5.1.11.1 Test purpose

To verify the UE's ability to perform handover in RRC_CONNECTED state when an intra-frequency handover is commanded by meeting the UE maximum RRC procedure delay and interruption time requirements.

### 5.1.11.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward of UE Category 0.

### 5.1.11.3 Minimum conformance requirements

Procedure delays for all procedures that can command a handover are specified in TS 36.331 [5].

When the UE receives a RRC message implying handover the UE shall be ready to start the transmission of the new uplink PRACH channel within $D_{handover}$ seconds from the end of the last TTI containing the RRC command.

Where:

$D_{handover}$ equals the maximum RRC procedure delay to be defined in clause 11.2 in TS 36.331 [5] plus the interruption time stated in TS 36.133 [4] clause 5.1.2.1.2.

The interruption time is the time between end of the last TTI containing the RRC command on the old PDSCH and the time the UE starts transmission of the new PRACH, excluding the RRC procedure delay. This requirement applies when UE is not required to perform any synchronisation procedure before transmitting on the new PRACH.

When intra-frequency handover is commanded, the interruption time shall be less than $T_{interrupt}$

$$T_{interrupt} = T_{search} + T_{IU} + 20 \text{ ms}$$

Where:

$T_{search}$ is the time required to search the target cell when the target cell is not already known when the handover command is received by the UE. If the target cell is known, then $T_{search}$ = 0 ms. If the target cell is unknown and signal quality is sufficient for successful cell detection on the first attempt, then $T_{search}$ = 80 ms. Regardless of whether DRX is in use by the UE, $T_{search}$ shall still be based on non-DRX target cell search times.

$T_{IU}$ is the interruption uncertainty in acquiring the first available PRACH occasion in the new cell. $T_{IU}$ can be up to 30 ms.

NOTE: The actual value of $T_{IU}$ shall depend upon the PRACH configuration used in the target cell.

In the interruption requirement a cell is known if it has been meeting the relevant cell identification requirement during the last 5 seconds otherwise it is unknown. Relevant cell identification requirements are described in TS 36.133 [4] Clause 8.1.2.2.1 for intra-frequency handover.

The normative reference for this requirement is TS 36.133 [4] clause 5.2.2.5 and A.5.1.11.

### 5.1.11.4 Test description

#### 5.1.11.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources (without using the faders) to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20 using only main Tx/Rx antenna.

2. The general test parameter settings are set up according to Table 5.1.11.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 5.1.11.4.3.

5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.1.11.4.1-1: General Test Parameters for E-UTRAN HD-FDD intra frequency handover for UE category 0 test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.1 HD-FDD | As specified in clause A.1.4 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.3 HD-FDD | As specified in clause A.2.3 |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbouring cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF Channel Number | | | 1 | Only one FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| A3-Offset | | dB | 0 | |
| Hysteresis | | dB | 0 | |
| Time To Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | | OFF |
| CP length | | | Normal | |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| PRACH configuration | | | 4 | As specified in table 5.7.1-2 in TS 36.211 [9] |
| Time offset between cells | | ms | 3 | Asynchronous cells<br>3ms or 92160*Ts |
| T1 | | s | 5 | |
| T2 | | s | ≤5 | |
| T3 | | s | 1 | |

### 5.1.11.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and send a measurement report. T3 is defined as the end of the last TTI containing the RRC message implying handover.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

2. Set the parameters according to T1 in Table 5.1.11.5-1. Propagation conditions are set according to Annex B clause B.1. T1 starts.

3. The SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 5.1.11.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3.

7. SS shall transmit an RRCConnectionReconfiguration message defined in Table [H.3.2-3] implying handover to Cell 2.

8. The start of T3 is the instant when the last TTI containing the RRC Connection reconfiguration message implying handover is sent to the UE, at that instant the SS shall switch the power setting from T2 to T3 as specified in Table 5.1.11.5-1.

9. The UE shall transmit RRCConnectionReconfigurationComplete message.

10. If the UE transmits the uplink PRACH channel to Cell 2 less than 50 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

11. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

13. Repeat step 2-12 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 5.1.11.4.3 Message contents

Same message content as in clause 5.1.1.4.3.

### 5.1.11.5 Test requirement

Tables 5.1.11.4.1-1 and 5.1.11.5-1 define the primary level settings including test tolerances for E-UTRAN HD-FDD-HD-FDD intra frequency handover test case.

**Table 5.1.11.5-1: Cell Specific Test requirement Parameters for E-UTRAN HD-FDD intra frequency handover for UE category 0 test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.1 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD | OP.1 FDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 8 | -3.8 | -3.8 | -Infinity | 2.86 | 2.86+TT |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 8 | 8 | 8 | - Infinity | 11.5 | 11.5 |
| RSRP Note 3 | dBm/15 KHz | -90 | -90 | -90 | - Infinity | -86.5 | -86.5 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE start to transmit the PRACH to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

$T_{interrupt} = T_{search} + T_{IU} + 20$ ms

$T_{search} = 0$, since Cell 2 is known prior to the test

$T_{IU} = 15$ ms, since 10 ms due to uncertainty in frame and 5 ms additional delay due to PRACH transmission occasion

Maximum RRC procedure delay = 15 ms as defined in TS 36.331 [5] clause 11.2

The handover delay $D_{handover}$ shall be less than a total of 50 ms in this test case (note: this gives a total of 15 ms for maximum RRC procedure delay plus 35 ms for $T_{interrupt}$).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.1.12 E-UTRAN TDD-TDD Handover intra frequency handover for UE category 0

### 5.1.12.1 Test purpose

To verify the UE's ability to perform handover in RRC_CONNECTED state when an intra-frequency handover is commanded by meeting the UE maximum RRC procedure delay and interruption time requirements.

### 5.1.12.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward of UE Category 0.

### 5.1.12.3 Minimum conformance requirements

The handover delay $D_{handover}$ shall be less than maximum RRC procedure delay + $T_{interrupt}$ in RRC_CONNECTED state.

When the UE receives a RRC message implying handover, the UE shall be ready to start the transmission of the new uplink PRACH channel within $D_{handover}$ seconds from the end of the last TTI containing the RRC command.

Where:

$D_{handover}$ equals the maximum RRC procedure delay defined in TS 36.331 [5] clause 11.2 plus the interruption time stated in TS 36.133 [4] clause 5.2.2.4.2.

The interruption time is the time between end of the last TTI containing the RRC command on the old PDSCH and the time the UE starts transmission of the new PRACH, excluding the RRC procedure delay. This requirement applies when UE is not required to perform any synchronisation procedure before transmitting on the new PRACH.

When intra-frequency handover is commanded, the interruption time shall be less than $T_{interrupt}$. The $T_{interrupt}$ equation is defined as:

$$T_{interrupt} = T_{search} + T_{IU} + 20 \text{ ms}$$

Where:

$T_{search}$ is the time required to search the target cell when the target cell is not already known when the handover command is received by the UE. If the target cell is known, then $T_{search}$ = 0 ms. If the target cell is unknown and signal quality is sufficient for successful cell detection on the first attempt, then $T_{search}$ = 80 ms. Regardless of whether DRX is in use by the UE, $T_{search}$ shall still be based on non-DRX target cell search times.

$T_{IU}$ is the interruption uncertainty in acquiring the first available PRACH occasion in new cell. $T_{IU}$ can be up to 30ms.

NOTE: The actual value of $T_{IU}$ shall depend upon the PRACH configuration used in the target cell.

In the interruption requirement a cell is know if it has been meeting the relevant cell identification requirement during the last 5 seconds otherwise it is unknown. Relevant cell identification requirements are described in TS 36.133 [4] clause 8.1.2.2.2 for intra-frequency handover.

The normative reference for this requirement is TS 36.133 [4] clause 5.2.2.4 and A.5.1.12.

### 5.1.12.4 Test description

#### 5.1.12.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources (without using the faders) to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20 using only main Tx/Rx antenna.

2. The general test parameter settings are set up according to Table 5.1.12.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 5.1.12.4.3.

5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.1.12.4.1-1: General Test Parameters for E-UTRAN TDD-TDD intra frequency handover for UE category 0 test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.12 TDD | As specified in clause A.1.5 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 TDD | As specified in clause A.2.2 |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbouring cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF Channel Number | | | 1 | Only one TDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| A3-Offset | | dB | 0 | |
| Hysteresis | | dB | 0 | |
| Time To Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | | OFF |
| CP length | | | Normal | |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211 [9] |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211 [9] |
| PRACH configuration | | | 53 | As specified in table 5.7.1-3 in TS 36.211 [9] |
| Time offset between cells | | S | 3 μs | Synchronous cells |
| T1 | | S | 5 | |
| T2 | | S | ≤5 | |
| T3 | | S | 1 | |

### 5.1.12.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and send a measurement report. T3 is defined as the end of the last TTI containing the RRC message implying handover.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

2. Set the parameters according to T1 in Table 5.1.12.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. The SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 5.1.12.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3.

7. SS shall transmit an RRCConnectionReconfiguration message defined in Table H.3.2-3 implying handover to Cell 2.

8. The start of T3 is the instant when the last TTI containing the RRC Connection reconfiguration message implying handover is sent to the UE, at that instant the SS shall switch the power setting from T2 to T3 as specified in Table 5.1.12.1.5-1.

9. The UE shall transmit RRCConnectionReconfigurationComplete message.

10. If the UE transmits the uplink PRACH channel to Cell 2 less than 50 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

11. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

13. Repeat step 2-12 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 5.1.12.4.3 Message contents

Same message content as in clause 5.1.2.4.3.

### 5.1.12.5 Test requirement

Tables 5.1.12.4.1-1 and 5.1.12.5-1 define the primary level settings including test tolerances for E-UTRAN TDD-TDD intra frequency handover test case.

**Table 5.1.12.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD-TDD intra frequency handover for UE category 0 test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | OP.1 TDD | OP.2 TDD | OP.2 TDD | OP.2 TDD | OP.1 TDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 8 | -3.8 | -3.8 | -Infinity | 2.86 | 2.86 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 8 | 8 | 8 | - Infinity | 11.5 | 11.5 |
| RSRP Note 3 | dBm/15 KHz | -90 | -90 | -90 | - Infinity | -86.5 | -86.5 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE start to transmit the PRACH to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

$T_{interrupt} = T_{search} + T_{IU} + 20$ ms

$T_{search} = 0$, since Cell 2 is known prior to the test

$T_{IU} = 15$ ms, since 10 ms due to uncertainty in frame and 5 ms additional delay due to PRACH transmission occasion

Maximum RRC procedure delay = 15 ms as defined in TS 36.331 [5] clause 11.2

The handover delay $D_{handover}$ shall be less than a total of 50 ms in this test case (note: this gives a total of 15 ms for maximum RRC procedure delay plus 35 ms for $T_{interrupt}$).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.1.13 E-UTRAN FDD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

*- Some message contents and exceptions are undefined*

*- $T_{interrupt}$ is updated taking $T_{MIB}$ in to account at RAN4#81bis. However, specific $T_{MIB}$ value is still under discussion in RAN4*

### 5.1.13.1 Test purpose

To verify a category M1 UE's ability to perform handover in RRC_CONNECTED state when an intra-frequency handover is commanded by meeting the UE maximum RRC procedure delay and interruption time requirements.

### 5.1.13.2 Test applicability

This test applies to all types of E-UTRA FD-FDD UE release 13 and forward of UE category M1.

### 5.1.13.3 Minimum conformance requirements

The handover delay shall be less than maximum RRC procedure delay + $T_{interrupt}$ in RRC_CONNECTED state.

When the UE receive a RRC message implying handover the UE shall be ready to start the transmission of the new uplink PRACH channel within $D_{handover}$ seconds from the end of the last TTI containing the RRC command.

Where:

$D_{handover}$ equals the maximum RRC procedure delay defined in TS 36.331 [5] clause 11.2 plus the interruption time stated in TS 36.133 [4] clause 5.5.2.1.2.

The interruption time is the time between the end of the last TTI containing the RRC command on the old PDSCH and the moment the UE has transmitted all repetitions of PRACH in the target cell, excluding the RRC procedure delay. This requirement applies when UE is not required to perform any synchronisation procedure before transmitting on the new PRACH.

When intra-frequency handover is commanded, the interruption time shall be less than $T_{interrupt}$. The $T_{interrupt}$ equation is defined as:

$$T_{interrupt} = T_{search} + T_{MIB} + T_{IU} + 20 \text{ ms}$$

Where:

$T_{search}$ is the time required to search the target cell when the target cell is not already known when the handover command is received by the UE. If the target cell is known, then $T_{search}$ = 0 ms. If the target cell is unknown and signal quality is sufficient for successful cell detection on the first attempt, then $T_{search}$ = 80 ms. Regardless of whether DRX is in use by the UE, $T_{search}$ shall still be based on non-DRX target cell search times.

$T_{MIB}$ is the time required for acquiring the MIB information of the target cell.

$T_{IU}$ is the time required to complete the transmission of PRACH in the target cell. The actual value of $T_{IU}$ shall depend upon the uncertainity in acquiring the first available PRACH occasion based on the PRACH configuration used in the target cell and the PRACH coverage enhancement level used by the UE for sending the random access preamble to the target cell.

In the interruption requirement a cell is known if it has been meeting the relevant cell identification requirement for a time duration equal or longer than the time duration required for the cell identification. Otherwise, it is unknown. The time duration required for the cell identification is specified in relevant intra-frequency cell identification requirements as described in TS 36.133 [4] clause 8.13.2.1 for CEModeA.

The normative reference for this requirement is TS 36.133 [4] clause 5.5.2.1 and A.5.1.13.

## 5.1.13.4 Test description

### 5.1.13.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connector as shown in TS 36.508 [7] Annex A figure A.20 using only main Tx/Rx antenna.

2. The general test parameter settings are set up according to Table 5.1.13.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 5.1.13.4.3.

5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.1.13.4.1-1: General test parameters for E-UTRAN FDD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbouring cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF Channel Number | | | 1 | Only one FDD carrier frequency is used. |
| A3-Offset | | dB | 0 | |
| Hysteresis | | dB | 0 | |
| Time To Trigger | | s | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | | OFF |
| CP length | | | Normal | |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| PRACH configuration | | | PRACH_4CE | As specified in A.9 |
| T1 | | s | 5 | |
| T2 | | s | ≤5 | |
| T3 | | s | 1 | |
| Gap pattern ID | | | 1 | |

### 5.1.13.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and send a measurement report. T3 is defined as the end of the last TTI containing the RRC message implying handover.

1. Ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3AA. Cell 1 is the active cell.

2. Set the parameters according to T1 in Table 5.1.13.5-1. Propagation conditions are set according to Annex B clause B.1. T1 starts.

3. The SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 5.1.13.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3.

7. SS shall transmit an RRCConnectionReconfiguration message defined in Table H.3.2-3 implying handover to Cell 2.

8. The start of T3 is the instant when the last TTI containing the RRC Connection reconfiguration message implying handover is sent to the UE, at that instant the SS shall switch the power setting from T2 to T3 as specified in Table 5.1.13.5-1.

9. The UE shall transmit RRCConnectionReconfigurationComplete message.

10. If the UE transmits the uplink PRACH channel to Cell 2 less than 50 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

11. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3AA. Cell 1 is the active cell.

12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

13. Repeat steps 2-12 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 5.1.13.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the condition CEModeA with the following exceptions:

**Table 5.1.13.4.3-1: Common Exception messages for E-UTRAN FDD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7<br>Table H.3.2-3 |

**Table 5.1.13.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 5.1.13.4.3-3:*MeasGapConfig-DP2*: Additional E-UTRAN FDD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA**

| Derivation Path: 36.508, clause 4.6.6 Table 4.6.6-1B | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasGapConfig-GP2 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| gapOffset CHOICE { | | | |
| gp1 | 0 | TGRP = 80 ms | Gap pattern ID = 1 |
| } | | | |
| } | | | |
| } | | | |

**Table 5.1.13.4.3-4: *MeasResults*: Additional Additional E-UTRAN FDD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 5.1.13.4.3-5: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 5.1.13.4.3-6: *PRACH-Config-v1310-DEFAULT*: Additional E-UTRAN FDD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA**

| Derivation Path: 36.508 clause 4.6.3 Table 4.6.3-7A | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PRACH-Config-v1310-DEFAULT ::= SEQUENCE { | | | |
| rsrp-ThresholdsPrachInfoList-r13 SEQUENCE (SIZE(1..3)) OF { | 3 entries | | |
| RSRP-Range[1] | 34 | -107dBm | |
| RSRP-Range[2] | 42 | -99dBm | |
| RSRP-Range[3] | 49 | -92dBm | |
| } | | | |
| PRACH-Config-v1310-DEFAULT ::= SEQUENCE { | | | |
| prach-ParametersListCE-r13 SEQUENCE (SIZE(1..maxCE-Level-r13)) OF SEQUENCE { | 4 entries | | |
| PRACH-ParametersCE-r13[1] SEQUENCE { | | | |
| prach-ConfigIndex-r13[1] | 4 | INTEGER (0..63) | FDD |
| prach-FreqOffset-r13[1] | Same as prach-FrequencyOffset | | |
| prach-StartingSubframe-r13[1] | sf2 | | |
| maxNumPreambleAttemptCE-r13[1] | n5 | | |
| numRepetitionPerPreambleAttempt-r13[1] | n1 | | |
| mpdcch-NarrowbandsToMonitor-r13 SEQUENCE (SIZE(1..2)) OF INTEGER (1..maxAvailNarrowBands-r13)[1] { | 1 entry | | |
| mpdcch-NarrowbandsToMonitor-r13[1] | 1 | | |
| } | | | |
| mpdcch-NumRepetition-RA-r13[1] | r1 | | |
| prach-HoppingConfig-r13[1] | off | | |
| } | | | |
| PRACH-ParametersCE-r13[2] SEQUENCE { | | | |
| prach-ConfigIndex-r13[2] | 4 | INTEGER (0..63) | FDD |
| prach-FreqOffset-r13[2] | Same as prach-FrequencyOffset | | |
| prach-StartingSubframe-r13[2] | sf2 | | |
| maxNumPreambleAttemptCE-r13[2] | n5 | | |
| numRepetitionPerPreambleAttempt-r13[2] | n32 | | |
| mpdcch-NarrowbandsToMonitor-r13 SEQUENCE (SIZE(1..2)) OF INTEGER (1..maxAvailNarrowBands-r13)[2] { | 1 entry | | |
| mpdcch-NarrowbandsToMonitor-r13[1] | 1 | | |
| } | | | |
| mpdcch-NumRepetition-RA-r13[2] | r4 | | |
| prach-HoppingConfig-r13[2] | off | | |
| } | | | |
| PRACH-ParametersCE-r13[3] SEQUENCE { | | | |
| prach-ConfigIndex-r13[3] | 4 | INTEGER (0..63) | FDD |
| prach-FreqOffset-r13[3] | Same as prach-FrequencyOffset | | |
| prach-StartingSubframe-r13[3] | sf2 | | |
| maxNumPreambleAttemptCE-r13[3] | n5 | | |
| numRepetitionPerPreambleAttempt-r13[3] | n64 | | |
| mpdcch-NarrowbandsToMonitor-r13 SEQUENCE (SIZE(1..2)) OF INTEGER (1..maxAvailNarrowBands-r13)[3] { | 1 entry | | |
| mpdcch-NarrowbandsToMonitor-r13[1] | 1 | | |
| } | | | |
| mpdcch-NumRepetition-RA-r13[3] | r16 | | |
| prach-HoppingConfig-r13[3] | off | | |
| } | | | |
| PRACH-ParametersCE-r13[4] SEQUENCE { | | | |
| prach-ConfigIndex-r13[4] | 4 | INTEGER (0..63) | FDD |
| prach-FreqOffset-r13[4] | Same as prach-FrequencyOffset | | |
| prach-StartingSubframe-r13[4] | sf2 | | |
| maxNumPreambleAttemptCE-r13[4] | n5 | | |
| numRepetitionPerPreambleAttempt-r13[4] | n128 | | |

| | | | |
|---|---|---|---|
| mpdcch-NarrowbandsToMonitor-r13 SEQUENCE (SIZE(1..2)) OF INTEGER (1..maxAvailNarrowBands-r13)[4] { | 1 entry | | |
| mpdcch-NarrowbandsToMonitor-r13[1] | 1 | | |
| } | | | |
| mpdcch-NumRepetition-RA-r13[4] | r64 | | |
| prach-HoppingConfig-r13[4] | off | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.1.13.4.3-7: *EPDCCH-Config-r11-DEFAULT*: Additional E-UTRAN FDD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| EPDCCH-Config-r11 ::= SEQUENCE{ | | | |
| config-r11{ | | | |
| setup SEQUENCE { | | | |
| startSymbol-r11 | 2 | | |
| setConfigToReleaseList-r11 | Not present | | |
| setConfigToAddModList-r11 SEQUENCE { | 1 entry | | |
| setConfigId-r11[1] | 0 | | |
| transmissionType-r11[1] | distributed | | |
| resourceBlockAssignment-r11[1] SEQUENCE{ | | | |
| numberPRB-Pairs-r11 | n4 | | |
| resourceBlockAssignment-r11 | 1001 | | |
| } | | | |
| dmrs-ScramblingSequenceInt-r11[1] | 0 | | |
| pucch-ResourceStartOffset-r11[1] | 0 | | |
| re-MappingQCL-ConfigListId-r11[1] | 0 | | |
| numberPRB-Pairs-v1310 | Not present | | |
| mpdcch-config-r13 CHOICE { | | | |
| setup SEQUENCE { | | | |
| csi-NumRepetitionCE-r13 | v4 | | |
| mpdcch-pdsch-HoppingConfig-r13 | on | | |
| mpdcch-StartSF-UESS-r13 CHOICE { | | | |
| fdd-r13 | v1 | | FDD |
| } | | | |
| mpdcch-NumRepetition-r13 | r8 | | |
| mpdcch-Narrowband-r13 | 1 | Narrowband index 0 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.1.13.4.3-8: *SystemInformationBlockType1-BR-r13* : Additional E-UTRAN FDD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType1-BR-r13 ::= SEQUENCE { | | | |
| freqHoppingParametersDL-r13 SEQUENCE { | | | |
| mpdcch-pdsch-HoppingNB-r13 | nb2 | | |
| interval-DLHoppingConfigCommonModeA-r13 CHOICE { | | | |
| interval-FDD-r13 | int4 | | FDD |
| } | | | |
| mpdcch-pdsch-HoppingOffset-r13 | [FFS] | INTEGER (1..16 (maxAvailNarrowBands-r13)) | |
| } | | | |
| } | | | |

## 5.1.13.5 Test requirement

Tables 5.1.13.4.1-1 and 5.1.13.5-1 define the primary level settings including test tolerances.

**Table 5.1.13.5-1: Cell specific test parameters for FDD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | |
| PDSCH Reference Measurement Channel defined in clause A.8.1 | | R.20 FDD | | | - | | |
| MPDCCH Reference Channel defined in clause A.7.1 | | R.16 FDD | | | R.16 FDD | | |
| PCFICH/PDCCH/PHICH Reference Channel defined in clause A.2.1 | | R.6 FDD | | | R.6 FDD | | |
| OCNG Patterns defined in D.1.21 | | OP.21 FDD | | | OP.21 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| MPDCCH_RA | dB | | | | | | |
| MPDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 8 | 8 | 8 | -Infinity | 11.5 | 11.5 |
| $\hat{E}_s / I_{ot}$ Note 3 | dB | 8 | -3.80 | -3.80 | -Infinity | 2.86 | 2.86 |
| RSRP Note 3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -86.5 | -86.5 |
| Io Note 3 | dBm/9MHz | -61.58 | -56.91 | -56.91 | Specified in columns for Cell 1 | | |
| Propagation Condition | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x1 | | | 1x1 | | |
| Timing offset to Cell 1 | | - | | | 3ms or 92160*Ts Asynchronous cells | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: Es/Iot, RSRP and Io level has been derived from other parameters for information purpose. They are not settable parameters themselves.

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE starts to transmit the PRACH to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

$T_{interrupt} = T_{search} + T_{MIB} + T_{IU} + 20$ ms

$T_{search} = 0$, since Cell 2 is known prior to the test

$T_{MIB}$ = [FFS]

$T_{IU}$ = 15 ms, since 10 ms due to uncertainty in frame and 5 ms additional delay due to PRACH transmission occasion

Maximum RRC procedure delay = 15 ms as defined in TS 36.331 [5] clause 11.2

The handover delay shall be less than a total of [FFS] ms in this test case (note: derived from 15 ms for maximum RRC procedure delay plus [FFS] ms for $T_{interrupt}$).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.1.14 E-UTRAN HD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

*- Some message contents and exceptions in 5.1.13 to which this test case refers back are undefined*

*- $T_{interrupt}$ is updated taking $T_{MIB}$ in to account at RAN4#81bis. However, specific $T_{MIB}$ value is still under discussion in RAN4*

### 5.1.14.1 Test purpose

To verify a category M1 UE's ability to perform handover in RRC_CONNECTED state when an intra-frequency handover is commanded by meeting the UE maximum RRC procedure delay and interruption time requirements.

### 5.1.14.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 13 and forward of UE category M1.

### 5.1.14.3 Minimum conformance requirements

The handover delay shall be less than maximum RRC procedure delay + $T_{interrupt}$ in RRC_CONNECTED state.

When the UE receive a RRC message implying handover the UE shall be ready to start the transmission of the new uplink PRACH channel within $D_{handover}$ seconds from the end of the last TTI containing the RRC command.

Where:

$D_{handover}$ equals the maximum RRC procedure delay defined in TS 36.331 [5] clause 11.2 plus the interruption time stated in TS 36.133 [4] clause 5.5.2.1.2.

The interruption time is the time between the end of the last TTI containing the RRC command on the old PDSCH and the moment the UE has transmitted all repetitions of PRACH in the target cell, excluding the RRC procedure delay. This requirement applies when UE is not required to perform any synchronisation procedure before transmitting on the new PRACH.

When intra-frequency handover is commanded, the interruption time shall be less than $T_{interrupt}$. The $T_{interrupt}$ equation is defined as:

$T_{interrupt} = T_{search} + T_{MIB} + T_{IU} + 20$ ms

Where:

$T_{search}$ is the time required to search the target cell when the target cell is not already known when the handover command is received by the UE. If the target cell is known, then $T_{search} = 0$ ms. If the target cell is unknown and signal quality is sufficient for successful cell detection on the first attempt, then $T_{search} = 80$ ms. Regardless of whether DRX is in use by the UE, $T_{search}$ shall still be based on non-DRX target cell search times.

$T_{MIB}$ is the time required for acquiring the MIB information of the target cell.

$T_{IU}$ is the time required to complete the transmission of PRACH in the target cell. The actual value of $T_{IU}$ shall depend upon the uncertainity in acquiring the first available PRACH occasion based on the PRACH configuration used in the target cell and the PRACH coverage enhancement level used by the UE for sending the random access preamble to the target cell.

In the interruption requirement a cell is known if it has been meeting the relevant cell identification requirement for a time duration equal or longer than the time duration required for the cell identification. Otherwise, it is unknown. The time duration required for the cell identification is specified in relevant intra-frequency cell identification requirements as described in TS 36.133 [4] clause 8.13.2.1 for CEModeA.

The normative reference for this requirement is TS 36.133 [4] clause 5.5.2.2 and A.5.1.14.

## 5.1.14.4 Test description

### 5.1.14.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connector as shown in TS 36.508 [7] Annex A figure A.20 using only main Tx/Rx antenna.
2. The general test parameter settings are set up according to Table 5.1.14.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are defined in clause 5.1.14.4.3.
5. There is one E-UTRA HD-FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.1.14.4.1-1: General test parameters for E-UTRAN HD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbouring cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF Channel Number | | | 1 | Only one HD-FDD carrier frequency is used. |
| A3-Offset | | dB | 0 | |
| Hysteresis | | dB | 0 | |
| Time To Trigger | | s | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | | OFF |
| CP length | | | Normal | |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| PRACH configuration | | | PRACH_4CE | As specified in A.9 |
| T1 | | s | 5 | |
| T2 | | s | ≤5 | |
| T3 | | s | 1 | |
| Gap pattern ID | | | 1 | |

### 5.1.14.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information

of Cell 2. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and send a measurement report. T3 is defined as the end of the last TTI containing the RRC message implying handover.

1. Ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3AA. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 5.1.14.5-1. Propagation conditions are set according to Annex B clause B.1. T1 starts.
3. The SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 5.1.14.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3.
7. SS shall transmit an RRCConnectionReconfiguration message defined in Table H.3.2-3 implying handover to Cell 2.
8. The start of T3 is the instant when the last TTI containing the RRC Connection reconfiguration message implying handover is sent to the UE, at that instant the SS shall switch the power setting from T2 to T3 as specified in Table 5.1.14.5-1.
9. The UE shall transmit RRCConnectionReconfigurationComplete message.
10. If the UE transmits the uplink PRACH channel to Cell 2 less than 50 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
11. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3AA. Cell 1 is the active cell.
12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
13. Repeat steps 2-12 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 5.1.14.4.3 Message contents

Same message contents as in clause 5.1.13.4.3.

### 5.1.14.5 Test requirement

Tables 5.1.14.4.1-1 and 5.1.14.5-1 define the primary level settings including test tolerances.

**Table 5.1.14.5-1: Cell specific test parameters for E-UTRAN HD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | |
| PDSCH Reference Measurement Channel defined in clause A.8.2 | | R.10 HD-FDD | | | - | | |
| MPDCCH Reference Channel defined in clause A.7.2 | | R.6 HD-FDD | | | R.6 HD-FDD | | |
| PCFICH/PDCCH/PHICH Reference Channel defined in clause A.2.3 | | R.3 HD-FDD | | | R.3 HD-FDD | | |
| OCNG Patterns defined in D.1.21 | | OP.21 FDD | | | OP.21 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| MPDCCH_RA | dB | | | | | | |
| MPDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ Note 3 | dB | 8 | 8 | 8 | -Infinity | 11.5 | 11.5 |
| $\hat{E}_s / I_{ot}$ | dB | 8 | -3.80 | -3.80 | -Infinity | 2.86 | 2.86 |
| RSRP Note 3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -86.5 | -86.5 |
| Io Note 3 | dBm/9MHz | -61.58 | -56.91 | -56.91 | Specified in columns for Cell 1 | | |
| Propagation Condition | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x1 | | | 1x1 | | |
| Timing offset to Cell 1 | | - | | | 3ms or 92160*Ts Asynchronous cells | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: Es/Iot, RSRP and Io level has been derived from other parameters for information purpose. They are not settable parameters themselves

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE starts to transmit the PRACH to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

$T_{interrupt} = T_{search} + T_{MIB} + T_{IU} + 20$ ms

$T_{search} = 0$, since Cell 2 is known prior to the test

$T_{MIB}$ = [FFS]

$T_{IU}$ = 15 ms, since 10 ms due to uncertainty in frame and 5 ms additional delay due to PRACH transmission occasion

Maximum RRC procedure delay = 15 ms as defined in TS 36.331 [5] clause 11.2

The handover delay shall be less than a total of [FFS] ms in this test case (note: derived from 15 ms for maximum RRC procedure delay plus [FFS] ms for $T_{interrupt}$).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.1.15 E-UTRAN TDD Intra frequency handover for Cat-M1 UEs in CEModeA

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

*- Some message contents and exceptions are undefined*

*- $T_{interrupt}$ is updated taking $T_{MIB}$ in to account at RAN4#81bis. However, specific $T_{MIB}$ value is still under discussion in RAN4*

### 5.1.15.1 Test purpose

To verify a category M1 UE's ability to perform handover in RRC_CONNECTED state when an intra-frequency handover is commanded by meeting the UE maximum RRC procedure delay and interruption time requirements.

### 5.1.15.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 13 and forward of UE category M1.

### 5.1.15.3 Minimum conformance requirements

The handover delay shall be less than maximum RRC procedure delay + $T_{interrupt}$ in RRC_CONNECTED state.

When the UE receive a RRC message implying handover the UE shall be ready to start the transmission of the new uplink PRACH channel within $D_{handover}$ seconds from the end of the last TTI containing the RRC command.

Where:

$D_{handover}$ equals the maximum RRC procedure delay defined in TS 36.331 [5] clause 11.2 plus the interruption time stated in TS 36.133 [4] clause 5.5.2.1.2.

The interruption time is the time between end of the last TTI containing the RRC command on the old PDSCH and the moment the UE has transmitted all repetitions of PRACH in the target cell, excluding the RRC procedure delay. This requirement applies when UE is not required to perform any synchronisation procedure before transmitting on the new PRACH.

When intra-frequency handover is commanded, the interruption time shall be less than $T_{interrupt}$. The $T_{interrupt}$ equation is defined as:

$T_{interrupt} = T_{search} + T_{MIB} + T_{IU} + 20$ ms

Where:

$T_{search}$ is the time required to search the target cell when the target cell is not already known when the handover command is received by the UE. If the target cell is known, then $T_{search} = 0$ ms. If the target cell is unknown and signal quality is sufficient for successful cell detection on the first attempt, then $T_{search} = 80$ ms. Regardless of whether DRX is in use by the UE, $T_{search}$ shall still be based on non-DRX target cell search times.

$T_{MIB}$ is the time required for acquiring the MIB information of the target cell.

$T_{IU}$ is the time required to complete the transmission of PRACH in the target cell. The actual value of $T_{IU}$ shall depend upon the uncertainity in acquiring the first available PRACH occasion based on the PRACH configuration used in the target cell and the PRACH coverage enhancement level used by the UE for sending the random access preamble to the target cell. In the interruption requirement a cell is known if it has been meeting the relevant cell identification requirement for a time duration equal or longer than the time duration required for the cell identification. Otherwise, it is unknown. The time duration required for the cell identification is specified in relevant intra-frequency cell identification requirements as described in TS 36.133 [4] clause 8.13.2.1 for CEModeA.

The normative reference for this requirement is TS 36.133 [4] clause 5.5.2.3 and A.5.1.15.

## 5.1.15.4 Test description

### 5.1.15.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connector as shown in TS 36.508 [7] Annex A figure A.20 using only main Tx/Rx antenna.

2. The general test parameter settings are set up according to Table 5.1.15.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 5.1.15.4.3.

5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.1.15.4.1-1: General test parameters for E-UTRAN TDD Intra frequency handover for Cat-M1 UEs in CEModeA test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbouring cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF Channel Number | | | 1 | Only one TDD carrier frequency is used. |
| A3-Offset | | dB | 0 | |
| Hysteresis | | dB | 0 | |
| Time To Trigger | | s | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | | OFF |
| CP length | | | Normal | |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211 |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211 |
| PRACH configuration index | | | PRACH_4CE | As specified in A.9 |
| T1 | | s | 5 | |
| T2 | | s | ≤5 | |
| T3 | | s | 1 | |
| Gap pattern ID | | | 1 | |

### 5.1.15.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and send a measurement report. T3 is defined as the end of the last TTI containing the RRC message implying handover.

1. Ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3AA. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 5.1.15.5-1. Propagation conditions are set according to Annex B clause B.1. T1 starts.
3. The SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 5.1.15.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3.
7. SS shall transmit an RRCConnectionReconfiguration message defined in Table H.3.2-3 implying handover to Cell 2.
8. The start of T3 is the instant when the last TTI containing the RRC Connection reconfiguration message implying handover is sent to the UE, at that instant the SS shall switch the power setting from T2 to T3 as specified in Table 5.1.15.5-1.
9. The UE shall transmit RRCConnectionReconfigurationComplete message.
10. If the UE transmits the uplink PRACH channel to Cell 2 less than 50 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
11. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3AA. Cell 1 is the active cell.
12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
13. Repeat steps 2-12 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 5.1.15.4.3 Message contents

Same message contents as in clause 5.1.13.4.3 with following exceptions:

- Instead of Table 5.1.13.4.3-6 → use Table 5.1.15.4.3-1.
- Instead of Table 5.1.13.4.3-7 → use Table 5.1.15.4.3-2.
- Instead of Table 5.1.13.4.3-8 → use Table 5.1.15.4.3-3.

**Table 5.1.15.4.3-1: *PRACH-Config-v1310-DEFAULT*: Additional E-UTRAN FDD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA**

| Derivation Path: 36.508 clause 4.6.3 Table 4.6.3-7A | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PRACH-Config-v1310-DEFAULT ::= SEQUENCE { | | | |
| rsrp-ThresholdsPrachInfoList-r13 SEQUENCE (SIZE(1..3)) OF { | 3 entries | | |
| RSRP-Range[1] | 34 | -107dBm | |
| RSRP-Range[2] | 42 | -99dBm | |
| RSRP-Range[3] | 49 | -92dBm | |
| } | | | |
| PRACH-Config-v1310-DEFAULT ::= SEQUENCE { | | | |
| prach-ParametersListCE-r13 SEQUENCE (SIZE(1..maxCE-Level-r13)) OF SEQUENCE { | 4 entries | | |
| PRACH-ParametersCE-r13[1] SEQUENCE { | | | |
| prach-ConfigIndex-r13[1] | 53 | INTEGER (0..63) | TDD |
| prach-FreqOffset-r13[1] | Same as prach-FrequencyOffset | | |
| prach-StartingSubframe-r13[1] | sf2 | | |
| maxNumPreambleAttemptCE-r13[1] | n5 | | |
| numRepetitionPerPreambleAttempt-r13[1] | n1 | | |
| mpdcch-NarrowbandsToMonitor-r13 SEQUENCE (SIZE(1..2)) OF INTEGER (1..maxAvailNarrowBands-r13)[1] { | 1 entry | | |
| mpdcch-NarrowbandsToMonitor-r13[1] | 1 | | |
| } | | | |
| mpdcch-NumRepetition-RA-r13[1] | r1 | | |
| prach-HoppingConfig-r13[1] | off | | |
| } | | | |
| PRACH-ParametersCE-r13[2] SEQUENCE { | | | |
| prach-ConfigIndex-r13[2] | 53 | INTEGER (0..63) | TDD |
| prach-FreqOffset-r13[2] | Same as prach-FrequencyOffset | | |
| prach-StartingSubframe-r13[2] | sf2 | | |
| maxNumPreambleAttemptCE-r13[2] | n5 | | |
| numRepetitionPerPreambleAttempt-r13[2] | n32 | | |
| mpdcch-NarrowbandsToMonitor-r13 SEQUENCE (SIZE(1..2)) OF INTEGER (1..maxAvailNarrowBands-r13)[2] { | 1 entry | | |
| mpdcch-NarrowbandsToMonitor-r13[1] | 1 | | |
| } | | | |
| mpdcch-NumRepetition-RA-r13[2] | r4 | | |
| prach-HoppingConfig-r13[2] | off | | |
| } | | | |
| PRACH-ParametersCE-r13[3] SEQUENCE { | | | |
| prach-ConfigIndex-r13[3] | 53 | INTEGER (0..63) | TDD |
| prach-FreqOffset-r13[3] | Same as prach-FrequencyOffset | | |
| prach-StartingSubframe-r13[3] | sf2 | | |
| maxNumPreambleAttemptCE-r13[3] | n5 | | |
| numRepetitionPerPreambleAttempt-r13[3] | n64 | | |
| mpdcch-NarrowbandsToMonitor-r13 SEQUENCE (SIZE(1..2)) OF INTEGER (1..maxAvailNarrowBands-r13)[3] { | 1 entry | | |
| mpdcch-NarrowbandsToMonitor-r13[1] | 1 | | |
| } | | | |
| mpdcch-NumRepetition-RA-r13[3] | r4 | | |
| prach-HoppingConfig-r13[3] | off | | |
| } | | | |
| PRACH-ParametersCE-r13[4] SEQUENCE { | | | |
| prach-ConfigIndex-r13[4] | 53 | INTEGER (0..63) | TDD |
| prach-FreqOffset-r13[4] | Same as prach-FrequencyOffset | | |
| prach-StartingSubframe-r13[4] | sf2 | | |
| maxNumPreambleAttemptCE-r13[4] | n5 | | |
| numRepetitionPerPreambleAttempt-r13[4] | n128 | | |

| | | | |
|---|---|---|---|
| mpdcch-NarrowbandsToMonitor-r13 SEQUENCE (SIZE(1..2)) OF INTEGER (1..maxAvailNarrowBands-r13)[4] { | 1 entry | | |
| mpdcch-NarrowbandsToMonitor-r13[1] | 1 | | |
| } | | | |
| mpdcch-NumRepetition-RA-r13[4] | r64 | | |
| prach-HoppingConfig-r13[4] | off | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.1.15.4.3-2: *EPDCCH-Config-r11-DEFAULT*: Additional E-UTRAN FDD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| EPDCCH-Config-r11 ::= SEQUENCE{ | | | |
| config-r11{ | | | |
| setup SEQUENCE { | | | |
| startSymbol-r11 | 2 | | |
| setConfigToReleaseList-r11 | Not present | | |
| setConfigToAddModList-r11 SEQUENCE { | 1 entry | | |
| setConfigId-r11[1] | 0 | | |
| transmissionType-r11[1] | distributed | | |
| resourceBlockAssignment-r11[1] SEQUENCE{ | | | |
| numberPRB-Pairs-r11 | n4 | | |
| resourceBlockAssignment-r11 | 1001 | | |
| } | | | |
| dmrs-ScramblingSequenceInt-r11[1] | 0 | | |
| pucch-ResourceStartOffset-r11[1] | 0 | | |
| re-MappingQCL-ConfigListId-r11[1] | 0 | | |
| numberPRB-Pairs-v1310 | Not present | | |
| mpdcch-config-r13 CHOICE { | | | |
| setup SEQUENCE { | | | |
| csi-NumRepetitionCE-r13 | v4 | | |
| mpdcch-pdsch-HoppingConfig-r13 | on | | |
| mpdcch-StartSF-UESS-r13 CHOICE { | | | |
| tdd-r13 | v1 | | TDD |
| } | | | |
| mpdcch-NumRepetition-r13 | r8 | | |
| mpdcch-Narrowband-r13 | 1 | Narrowband index 0 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.1.15.4.3-3: *SystemInformationBlockType1-BR-r13* : Additional E-UTRAN FDD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType1-BR-r13 ::= SEQUENCE { | | | |
| freqHoppingParametersDL-r13 SEQUENCE { | | | |
| mpdcch-pdsch-HoppingNB-r13 | nb2 | | |
| interval-DLHoppingConfigCommonModeA-r13 CHOICE { | | | |
| interval-TDD-r13 | Int10 | | TDD |
| } | | | |
| mpdcch-pdsch-HoppingOffset-r13 | [FFS] | INTEGER (1..16 (maxAvailNarrowBands-r13)) | |
| } | | | |
| } | | | |

## 5.1.15.5 Test requirement

Tables 5.1.15.4.1-1 and 5.1.15.5-1 define the primary level settings including test tolerances.

**Table 5.1.15.5-1: Cell specific test parameters for E-UTRAN TDD Intra frequency handover for Cat-M1 UEs in CEModeA test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | |
| PDSCH Reference Measurement Channel defined in clause A.8.3 | | R.16 TDD | | | - | | |
| MPDCCH Reference Channel defined in clause A.7.3 | | R.14 TDD | | | R.14 TDD | | |
| PCFICH/PDCCH/PHICH Reference Channel defined in clause A.2.2 | | R.6 TDD | | | R.6 TDD | | |
| OCNG Patterns defined in D.2.11 | | OP.11 TDD | | | OP.11 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| MPDCCH_RA | dB | | | | | | |
| MPDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 8 | 8 | 8 | -Infinity | 11.5 | 11.5 |
| $\hat{E}_s / I_{ot}$ Note 3 | dB | 8 | -3.80 | -3.80 | -Infinity | 2.86 | 2.86 |
| RSRP Note 3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -86.5 | -86.5 |
| Io Note 3 | dBm/9MHz | -61.58 | -56.91 | -56.91 | Specified in columns for Cell 1 | | |
| Propagation Condition | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x1 | | | 1x1 | | |
| Timing offset to Cell 1 | | - | | | 3 μs or 92*Ts Synchronous cells | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: Es/Iot, RSRP and Io level has been derived from other parameters for information purpose. They are not settable parameters themselves

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE starts to transmit the PRACH to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

$T_{interrupt} = T_{search} + T_{MIB} + T_{IU} + 20$ ms

$T_{search} = 0$, since Cell 2 is known prior to the test

$T_{MIB}$ = [FFS]

$T_{IU}$ = 15 ms, since 10 ms due to uncertainty in frame and 5 ms additional delay due to PRACH transmission occasion

Maximum RRC procedure delay = 15 ms as defined in TS 36.331 [5] clause 11.2

The handover delay shall be less than a total of [FFS] ms in this test case (note: derived from 15 ms for maximum RRC procedure delay plus [FFS] ms for $T_{interrupt}$).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 5.2 Handover from E-UTRAN to other RATs

## 5.2.1 E-UTRAN FDD - UTRAN FDD handover

### 5.2.1.1 Test purpose

To verify the UE's ability to transfer a connection between the UE and E-UTRAN to UTRAN in RRC_CONNECTED state by meeting the UE RRC procedure delay and interruption time requirements.

### 5.2.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support UTRA FDD. Applicability requires support for FGI bits 8, and 22.

### 5.2.1.3 Minimum conformance requirements

When the UE receives a RRC message implying handover to UTRAN the UE shall be ready to start the transmission of the new UTRA uplink DPCCH within $D_{handover}$ seconds from the end of the last E-UTRAN TTI containing the RRC MOBILITY FROM E-UTRA command.

Where:

$D_{handover}$ equals the RRC procedure delay, which is 50 ms plus the interruption time stated in TS 36.133 [4] clause 5.3.1.1.2.

The interruption time is the time between the end of the last TTI containing the RRC command on the E-UTRAN PDSCH and the time the UE starts transmission on the uplink DPCCH in UTRAN FDD, excluding the RRC procedure delay. The interruption time depends on whether the target cell is known to the UE or not. The target cell is known if it has been measured by the UE during the last 5 seconds otherwise it is unknown. The UE shall always perform a UTRA synchronisation procedure as part of the handover procedure.

If the target cell is known the interruption time shall be less than $T_{interrupt1}$.

$$T_{interrupt1} = T_{IU} + T_{sync} + 50 + 10*F_{max} + T_{MC} \text{ ms}$$

If the target cell is unknown the interruption time shall be less than $T_{interrupt2}$.

$$T_{interrupt2} = T_{IU} + T_{sync} + 150 + 10*F_{max} + T_{MC} \text{ ms}$$

This requirement shall be met, provided that there is one target cell in the MOBILITY FROM E-UTRA command. Performance requirements for E-UTRA to UTRA soft handover are not specified. When UE is connected to an E-UTRA cell, UTRA SFN timing measurements are not reported. This implies that the timing of the DPCH of the UTRA target cells in the active set cannot be configured by UTRAN to guarantee that all target cells fall within the UE reception window of $T_0$ +/- 148 chips.

Where:

$T_{IU}$ is the interruption uncertainty when changing the timing from the E-UTRAN to the new UTRAN cell. $T_{IU}$ can be up to one UTRA frame (10 ms).

$F_{max}$ denotes the maximum number of radio frames within the transmission time intervals of all transport channels that are multiplexed into the same CCTrCH on the UTRA target cell. If HS-PDSCH is configured in the UTRA target cell, $F_{max}$ is 4 radio frames.

$T_{sync}$ is the time required for measuring the downlink DPCCH channel as stated in TS 25.214 [12] clause 4.3.1.2. In case higher layers indicate the usage of a post-verification period $T_{sync}$ = 0 ms. Otherwise $T_{sync}$ = 40 ms.

$T_{MC}$ is 0 ms if a single UTRA cell is configured as the handover target, otherwise 20 ms if handover to UTRA with 1, 2 or 3 UTRA carriers with secondary HS-PDSCH is configured.

The phase reference is the UTRA primary CPICH.

The requirements assume that N312 has the smallest possible value i.e. only one in-sync is required.

The normative reference for this requirement is TS 36.133 [4] clause 5.3.1 and A.5.2.1.

### 5.2.1.4 Test description

#### 5.2.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.22.
2. The general test parameter settings are set up according to Table 5.2.1.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are defined in clause 5.2.1.4.3.
5. There is one E-UTRA FDD serving cell and one UTRA FDD cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.2.1.4.1-1: General Test Parameters for E-UTRAN FDD - UTRAN FDD handover test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Initial conditions | Active cell | | Cell 1 | E-UTRAN cell |
| | Neighbouring cell | | Cell 2 | UTRAN cell |
| Final condition | Active cell | | Cell 2 | UTRAN cell |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| Gap Pattern Id | | | 0 | As specified in TS 36.133 [4] Table 8.1.2.1-1 started before T2 starts |
| E-UTRAN FDD measurement quantity | | | RSRP | |
| Inter-RAT (UTRAN FDD) measurement quantity | | | CPICH Ec/N0 | |
| b2-Threshold1 | | dBm | -90 | Absolute E-UTRAN RSRP threshold for event B2 |
| b2-Threshold2-UTRA | | dB | -18 | Absolute UTRAN CPICH Ec/Iothreshold for event B2 |
| Hysteresis | | dB | 0 | |
| TimeToTrigger | | dB | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | OFF | Non-DRX test |
| Access Barring Information | | - | Not sent | No additional delays in random access procedure |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 10 | |
| UTRA RF Channel Number | | | 1 | One UTRA FDD carrier frequency is used. |
| Monitored UTRA FDD cell list size | | | 12 | UTRA cells on UTRA RF channel 1 provided in the cell before T2. |
| Post-verification period | | | False | |
| T1 | | s | 5 | |
| T2 | | s | ≤5 | |
| T3 | | s | 1 | |

### 5.2.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE does not have any timing information of Cell 2. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and send a measurement report. Gap pattern configuration with id #0 as specified in Table 8.1.2.1-1 is configured before T2 begins to enable inter-RAT frequency monitoring. The RRC message implying handover shall be sent to the UE during T2, after the UE has reported Event B2. The start of T3 is the instant when the last TTI containing the RRC message implying handover is sent to the UE. The handover message shall contain cell 2 as the target cell.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table's 5.2.1.5-1 and 5.2.1.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. The SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 5.2.1.5-1 and 5.2.1.5-2.
6. UE shall transmit a MeasurementReport message triggered by Event B2.

7. SS shall transmit a MobilityFromEUTRACommand message implying handover to Cell 2.

8. The start of T3 is the instant when the last TTI containing the RRC message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T2 to T3 as specified in Tables 5.2.1.5-1 and 5.2.1.5-2.

9. If the UE transmits the UL DPCCH to Cell 2 less than 190 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure testes is increased by one.

10. After T3 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

11. The SS shall set Cell 2 primary scrambling code = ((current cell 2 primary scrambling code - 50) mod 200 + 100) for next iteration of the test procedure loop.

12. Repeat step 2-11 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 5.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 and 4.7B with the following exceptions:

**Table 5.2.1.4.3-1: Common Exception messages for E-UTRAN FDD - UTRAN FDD handover**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-5<br>Table H.3.1-7<br>Table H.3.3-1<br>Table H.3.3-3 |

**Table 5.2.1.4.3-2: *ReportConfigInterRAT-B2-UTRA*: Additional E-UTRAN FDD - UTRAN FDD handover**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-8 ReportConfigInterRAT-B2(EUTRA-Thres, UTRA-Thres) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B2-UTRA(EUTRA-Thres, UTRA-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB2 SEQUENCE { | | | |
| b2-Threshold1 CHOICE { | | | |
| threshold-RSRP | 50(-90 dBm) | -90 dBm EUTRA-Thres is actual threshold value in dBm (50 - 140 dBm) | |
| } | | | |
| b2-Threshold2 CHOICE { | | | |
| b2-Threshold2-UTRA CHOICE { | | | |
| thresholdUTRA-EcN0 | 13 (-18 dB) | -18 dB is actual UTRA-Thres is actual Ec/Io value in dB ((13 - 49)/2 dB) | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

**Table 5.2.1.4.3-3: *MeasResults*: Additional E-UTRAN FDD - UTRAN FDD handover**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultsServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultsNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 5.2.1.4.3-4: *MeasResultListUTRA*: Additional E-UTRAN FDD - UTRAN FDD handover**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| fdd | PhysCellIdUTRA-FDD | This is the typical value range used in UTRAN FDD tests. | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-EcN0 | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 5.2.1.4.3-5: HANDOVER TO UTRAN COMMAND**

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| Downlink information common for all radio links | | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | Initialize | | |
| - Default DPCH Offset Value | Arbitrary set to value 0..306688 by step of 512 | | |

## 5.2.1.5 Test requirement

Tables 5.2.1.4.1-1, 5.2.1.5-1 and 5.2.1.5-2 define the primary level settings including test tolerances for E-UTRAN FDD - UTRAN FDD handover test.

**Table 5.2.1.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD to UTRAN FDD handover test case (Cell 1)**

| Parameter | Unit | Cell 1 (E-UTRA) | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA[Note] | dB | | | |
| OCNG_RB[Note] | dB | | | |

| | | | | |
|---|---|---|---|---|
| $\hat{E}_s / I_{ot}$ | dB | -1.10 | -1.10 | -1.10 |
| $\hat{E}_s / N_{oc}$ | | -1.10 | -1.10 | -1.10 |
| $N_{oc}$ | dBm/15 kHz | -98 | | |
| RSRP Note 2 | dBm/15 KHz | -99.10 | -99.10 | -99.10 |
| Io Note 2 | dBm/9 MHz | –67.72 | -67.72 | -67.72 |
| Propagation Condition | | AWGN | | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | | |

**Table 5.2.1.5-2: Cell Specific Test requirement Parameters for Cell 2 UTRAN FDD cell**

| Parameter | Unit | Cell 2 (UTRA) | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| CPICH_Ec/Ior | dB | -10 | | |
| PCCPCH_Ec/Ior | dB | -12 | | |
| SCH_Ec/Ior | dB | -12 | | |
| PICH_Ec/Ior | dB | -15 | | |
| DCH_Ec/Ior | dB | N/A | N/A | Note 1 |
| OCNS_Ec/Ior | dB | -0.941 | -0.941 | Note 2 |
| $\hat{I}_{or} / I_{oc}$ | dB | -infinity | -1.8 | -1.8 |
| $I_{oc}$ | dBm/3,84 MHz | -70 | -70 | -70 |
| CPICH_Ec/Io | dB | -infinity | -14 | -14 |
| Propagation Condition | | AWGN | | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$ | | | | |

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE start to transmit the UL DPCCH to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt1}$ (note: the target cell is known)

$T_{interrupt1} = T_{IU} + T_{sync} + 50 + 10*F_{max} + T_{MC}$ ms

$T_{IU}$ = 10 ms; $T_{IU}$ can be up to one UTRA frame (10 ms).

$F_{max}$ = 4 radio frames; The maximum radio frames within the transmission time intervals to fit into DCCH with 40 ms TTI

$T_{sync}$ = 40 ms; In case higher layers indicate the usage of a post-verification period $T_{sync}$ = 0 ms. Otherwise $T_{sync}$ = 40 ms

$T_{MC}$ = 0 ms if a single UTRA cell is configured as the handover target. Otherwise $T_{MC}$ = 20 ms

Maximum RRC procedure delay = 50 ms as defined in TS 36.331 [5] clause 11.2

The handover delay $D_{handover}$ shall be less than a total of 190 ms in this test case (note: this gives a total of 50 ms for maximum RRC procedure delay plus 140 ms for $T_{interrupt1}$).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.2.2 E-UTRAN TDD - UTRAN FDD handover

### 5.2.2.1 Test purpose

To verify the UE's ability to transfer a connection between the UE and E-UTRAN to UTRAN in RRC_CONNECTED state by meeting the UE RRC procedure delay and interruption time requirements.

### 5.2.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward that support UTRA FDD. Applicability requires support for FGI bits 8, and 22.

### 5.2.2.3 Minimum conformance requirements

The handover delay $D_{handover}$ shall be less than maximum RRC procedure delay + $T_{interrupt}$ in RRC_CONNECTED state.

When the UE receives a RRC message implying handover to UTRAN the UE shall be ready to start the transmission of the new UTRA uplink DPCCH within $D_{handover}$ seconds from the end of the last E-UTRAN TTI containing the RRC MOBILITY FROM E-UTRA command.

Where:

$D_{handover}$ equals the RRC procedure delay, which is 50ms plus the interruption time stated in TS 36.133 [4] clause 5.3.1.1.2.

The interruption time is the time between the end of the last TTI containing the RRC command on the E-UTRAN PDSCH and the time the UE starts transmission on the uplink DPCCH in UTRAN FDD, excluding the RRC procedure delay. The interruption time depends on whether the target cell is known to the UE or not. The target cell is known if it has been measured by the UE during the last 5 seconds otherwise it is unknown. The UE shall always perform a UTRA synchronisation procedure as part of the handover procedure.

If the target cell is known the interruption time shall be less than $T_{interrupt1}$. The $T_{interrupt1}$ equation is defined as:

$$T_{interrupt1} = T_{IU} + T_{sync} + 50 + 10*F_{max} + T_{MC} \text{ ms}$$

If the target cell is unknown the interruption time shall be less than $T_{interrupt2}$. The $T_{interrupt2}$ equation is defined as:

$$T_{interrupt2} = T_{IU} + T_{sync} + 150 + 10*F_{max} + T_{MC} \text{ ms}$$

This requirement shall be met, provided that there is one target cell in the MOBILITY FROM E-UTRA command. When UE is connected to an E-UTRA cell, UTRA SFN timing measurements are not reported. This implies that the timing of the DPCH of the UTRA target cells in the active set cannot be configured by UTRAN to guarantee that all target cells fall within the UE reception window of $T_0$ +/- 148 chips.

Where:

$T_{IU}$ is the interruption uncertainty when changing the timing from the E-UTRAN to the new UTRAN cell. $T_{IU}$ can be up to one UTRA frame (10 ms).

$F_{max}$ denotes the maximum number of radio frames within the transmission time intervals of all transport channels that are multiplexed into the same CCTrCH on the UTRA target cell. If HS-PDSCH is configured in the UTRA target cell, $F_{max}$ is 4 radio frames.

$T_{sync}$ is the time required for measuring the downlink DPCCH channel as stated in TS 25.214 [12] clause 4.3.1.2. In case higher layers indicate the usage of a post-verification period $T_{sync}$ = 0 ms. Otherwise $T_{sync}$ = 40 ms.

$T_{MC}$ is 0 ms if a single UTRA cell is configured as the handover target, otherwise 20 ms if handover to UTRA with 1, 2 or 3 UTRA carriers with secondary HS-PDSCH is configured.

The phase reference is the UTRA primary CPICH.

The requirements assume that N312 has the smallest possible value i.e. only one "in_sync" is required.

The normative reference for this requirement is TS 36.133 [4] clause 5.3.1 and A.5.2.2.

### 5.2.2.4 Test description

#### 5.2.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.22.
2. The general test parameter settings are set up according to Table 5.2.2.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are defined in clause 5.2.2.4.3.
5. There is one E-UTRA TDD serving cell and one UTRA FDD cell specified in the test. Cell 1 (E-UTRA TDD cell) is the cell used for call setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.2.2.4.1-1: General test parameters for E-UTRAN TDD-UTRAN FDD handover**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters (E-UTRAN TDD) | | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN TDD) | | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Initial conditions | Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| | Neighbour cell | | Cell 2 | Cell 2 is on UTRA RF channel number 1. |
| Final conditions | Active cell | | Cell 2 | |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in 3GPP TS 36.211. Applicable to cell 1. |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in 3GPP TS 36.211. Applicable to cell 1 |
| E-UTRAN TDD measurement quantity | | | RSRP | |
| Inter-RAT (UTRA FDD) measurement quantity | | | CPICH Ec/Io | |
| b2-Threshold1 | | dBm | -90 | Absolute E-UTRAN RSRP threshold for event B2 |
| b2-Threshold2-UTRA | | dB | -18 | UTRAN FDD CPICH Ec/Io threshold for event B2 |
| Hysteresis | | dB | 0 | |
| DRX | | | OFF | No DRX configured. |
| Time to Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | |
| CP length | | | Normal | Applicable to cell 1 |
| Gap pattern configuration Id | | | 0 | As specified in Table 8.1.2.1-1; to start before T2 starts |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| UTRA RF Channel Number | | | 1 | One UTRA FDD carrier frequency is used. |
| Monitored UTRA FDD cell list size | | | 12 | UTRA cells on UTRA RF channel 1 provided in the cell list before T2. |
| Post-verification period | | | False | Post verification is not used. |
| T1 | | s | 5 | |
| T2 | | s | ≤5 | |
| T3 | | s | 1 | |

### 5.2.2.4.2 Test procedure

The test consists of one E-UTRAN TDD cell and one neighbour UTRAN FDD cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE does not have any timing information of Cell 2. Gap pattern configuration with id #0 as specified in Table 8.1.2.1-1 is configured before the start of T2 to enable the monitoring of UTRAN FDD. A neighbouring cell list, including the UTRA cell (Cell 2) is sent to the UE before T2 starts. During the time T2, Cell 2 becomes detectable and the UE is expected to detect and send a measurement report. The RRC message implying handover shall be sent to the UE during T2, after the UE has reported Event B2. The start of T3 is the instant when the last TTI containing the RRC message implying handover is sent to the UE. The handover message shall contain cell 2 as the target cell.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

2. Set the parameters according to T1 in Table 5.2.2.5-1 and 5.2.2.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 5.2.2.5-1 and 5.2.2.5-2.
6. UE shall transmit a MeasurementReport message triggered by Event B2.
7. SS shall transmit a MobilityFromEUTRACommand message implying handover to Cell 2.
8. The start of T3 is the instant when the last TTI containing the RRC message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T2 to T3 as specified in Table 5.2.2.5-1 and 5.2.2.5-2.
9. If the UE transmits the Uplink DPCCH channel to Cell 2 less than 190 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure testes is increased by one.
10. After T3 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
11. The SS shall set Cell 2 primary scrambling code = ((current cell 2 primary scrambling code – 50) mod 200 + 100) for next iteration of the test procedure loop.
12. Repeat step 2-11 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 5.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 and 4.7B with the following exceptions:

**Table 5.2.2.4.3-1: Common Exception messages for E-UTRAN TDD - UTRAN FDD handover**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-5<br>Table H.3.1-7<br>Table H.3.3-1<br>Table H.3.3-3 |

**Table 5.2.2.4.3-2: *ReportConfigInterRAT-B2-UTRA*: Additional E-UTRAN TDD - UTRAN FDD handover**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-8 ReportConfigInterRAT-B2(EUTRA-Thres, UTRA-Thres)- | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B2-UTRA(EUTRA-Thres, UTRA-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB2 SEQUENCE { | | | |
| b2-Threshold1 CHOICE { | | | |
| Threshold-RSRP | 50 (-90dBm) | -90 dBm EUTRA-Thres is actual threshold value in dBm (50 - 140 dBm) | |
| } | | | |
| b2-Threshold2 CHOICE { | | | |
| b2-Threshold2-UTRA CHOICE { | | | |
| thresholdUTRA-EcN0 | 13 (-18dB) | -18 dB is actual UTRA-Thres is actual Ec/NO value in dB ((13 - 49)/2 dB) | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

**Table 5.2.2.4.3-3: *MeasResults*: Additional E-UTRAN TDD - UTRAN FDD handover**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultsServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultsNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 5.2.2.4.3-4: *MeasResultListUTRA*: Additional E-UTRAN TDD - UTRAN FDD handover**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| fdd | PhysCellIdUTRA-FDD | This is the typical value range used in UTRAN FDD tests. | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-EcN0 | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 5.2.2.4.3-5: HANDOVER TO UTRAN COMMAND**

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| Downlink information common for all radio links | | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | Initialize | | |
| - Default DPCH Offset Value | Arbitrary set to value 0..306688 by step of 512 | | |

## 5.2.2.5 Test requirement

Tables 5.2.2.4.1-1, 5.2.2.5-1 and 5.2.2.5-2 define the primary level settings including test tolerances for E-UTRAN TDD - UTRAN FDD handover test.

**Table 5.2.2.5-1: Cell specific test parameters for E-UTRAN TDD (cell 1) for handover to UTRAN FDD (cell # 2)**

| Parameter | Unit | Cell 1 (E-UTRAN) | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.1 TDD) | | OP.1 TDD | | OP.2 TDD |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA[Note] | | | | |
| OCNG_RB[Note] | | | | |

| | | | | |
|---|---|---|---|---|
| RSRP | dBm/15 kHz | -99.10 | -99.10 | -99.10 |
| $\hat{E}_s / I_{oc}$ | dB | -1.10 | -1.10 | -1.10 |
| $\hat{E}_s / N_{oc}$ | dB | -1.10 | -1.10 | -1.10 |
| $N_{oc}$ | dBm/15 kHz | -98 | | |
| Propagation Condition | | AWGN | | |
| Note: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | | |

**Table 5.2.2.5-2: Cell specific test parameters for UTRAN FDD (cell # 2) for handover from E-UTRAN TDD cell (cell #1)**

| Parameter | Unit | Cell 2 (UTRA) | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| CPICH_Ec/Ior | dB | -10 | | |
| PCCPCH_Ec/Ior | dB | -12 | | |
| SCH_Ec/Ior | dB | -12 | | |
| PICH_Ec/Ior | dB | -15 | | |
| DPCH_Ec/Ior | dB | N/A | N/A | Note 1 |
| OCNS_Ec/Ior | dB | -0.941 | -0.941 | Note 2 |
| $\hat{I}_{or} / I_{oc}$ | dB | -infinity | -1.8 | -1.8 |
| $I_{oc}$ | dBm/3.84 MHz | -70 | | |
| CPICH_Ec/Io | dB | -infinity | -14 | -14 |
| Propagation Condition | | AWGN | | |
| Note 1: The DPCH level is controlled by the power control loop<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$ | | | | |

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE start to transmit the UL DPCCH to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt1}$ (note: the target cell is known)

$T_{interrupt1} = T_{IU} + T_{sync} + 50 + 10*F_{max} + T_{MC}$ ms

$T_{IU}$ = 10 ms; $T_{IU}$ can be up to one UTRA frame (10 ms).

$F_{max}$ = 4 radio frames; The maximum radio frames within the transmission time intervals to fit into DCCH with 40 ms TTI

$T_{sync}$ = 40 ms; In case higher layers indicate the usage of a post-verification period $T_{sync}$ = 0 ms. Otherwise $T_{sync}$ = 40 ms

$T_{MC}$ = 0 ms if a single UTRA cell is configured as the handover target. Otherwise $T_{MC}$ = 20 ms

Maximum RRC procedure delay = 50 ms as defined in TS 36.331 [5] clause 11.2

The handover delay $D_{handover}$ shall be less than a total of 190 ms in this test case (note: this gives a total of 50ms for maximum RRC procedure delay plus 140 ms for Tinterrupt1).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.2.3 E-UTRAN FDD - GSM handover

### 5.2.3.1 Test purpose

To verify the UE's ability to transfer a connection between the UE and E-UTRAN to GSM in RRC_CONNECTED state by meeting the UE handover delay and interruption time requirements.

### 5.2.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support GSM and inter-RAT PS handover to GERAN. Applicability requires support for FGI bits 9, 15 and 23.

### 5.2.3.3 Minimum conformance requirements

The handover delay given in table 5.2.3.3-1 and interruption time given in table 5.2.3.3-2 requirements for the case where the UE has not synchronised to the GSM cell before receiving the RRC MOBILITY FROM E-UTRA command are valid when the signal quality of the GSM cell is sufficient for successful synchronisation with one attempt. If the UE is unable to synchronise to the GSM cell on the first attempt, it shall continue to search for synchronisation information for up to 800 ms duration. If after 800 ms the UE has not synchronised to the GSM cell it shall follow the handover failure procedure specified in TS 36.331 [5].

When the UE receives a RRC MOBILITY FROM E-UTRA command the UE shall be ready to transmit (as specified in 3GPP TS 45.010 [13]) on the channel of the new RAT within the value defined in TS 36.133 [4] clause 5.3.3.2.1 and shown in table 5.2.3.3-1 from the end of the last TTI containing the RRC command.

The UE shall process the RRC procedures for the MOBILITY FROM E-UTRA command within 50 ms, which is noted as RRC procedure delay.

**Table 5.2.3.3-1: E-UTRAN/GSM handover - handover delay**

| UE synchronisation status | handover delay [ms] |
|---|---|
| The UE has synchronised to the GSM cell before the RRC MOBILITY FROM E-UTRA COMMAND is received | 90 |
| The UE has not synchronised to the GSM cell before RRC the MOBILITY FROM E-UTRA COMMAND is received | 190 |

The interruption time is the time between the end of the TTI containing the RRC command on the E-UTRAN PDSCH and the time the UE starts transmission on the uplink channel in GSM, excluding the RRC procedure delay. The interruption time depends on whether the UE has synchronized to the target GSM cell or not and shall be less than the value defined in TS 36.133 [4] clause 5.3.3.2.1 and shown in table 5.2.3.3-2.

**Table 5.2.3.3-2: E-UTRAN/GSM handover - interruption time**

| Synchronisation status | Interruption time [ms] |
|---|---|
| The UE has synchronised to the GSM cell before the RRC MOBILITY FROM E-UTRA COMMAND is received | 40 |
| The UE has not synchronised to the GSM cell before the RRC MOBILITY FROM E-UTRA COMMAND is received | 140 |

The normative reference for this requirement is TS 36.133 [4] clause 5.3.3 and A.5.2.3.

## 5.2.3.4 Test description

### 5.2.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 5.2.3.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 5.2.3.4.3.
5. There is one E-UTRA FDD serving cell and one GSM cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.2.3.4.1-1: General Test Parameters for E-UTRAN FDD - GSM handover test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Gap Pattern Id | | | 1 | As specified in TS 36.133 [4] section 8.1.2.1. |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbour cell | | Cell 2 | |
| Final conditions | Active cell | | Cell 2 | |
| Inter-RAT measurement quantity | | | GSM Carrier RSSI | |
| Threshold other system | | dBm | -80 | Absolute GSM carrier RSSI threshold for event B1. |
| Hysteresis | | dB | 0 | |
| Time to Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | | OFF |
| T1 | | s | 20 | |
| T2 | | s | 7 | |
| T3 | | s | 1 | |

### 5.2.3.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. The RRC message implying handover to Cell 2 shall be sent to the UE during T2, after the UE has reported Event B1. T3 is defined as the end of the last E-UTRAN TTI containing the RRC message implying handover. The requirements are also applicable for a UE not requiring measurement gap, in which case no measurement gap pattern is sent.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table's 5.2.3.5-1 and 5.2.3.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 5.2.3.5-1 and 5.2.3.5-2. T2 starts.
6. UE shall transmit a MeasurementReport message triggered by Event B1.
7. SS shall transmit a MobilityFromEUTRACommand message implying handover to Cell 2.
8. The start of T3 is the instant when the last TTI containing the RRC message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T2 to T3 as specified in Table 5.2.3.5-2. T3 starts.
9. If the UE sends access bursts on the new DCCH to Cell 2 less than 100 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
10. After T3 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
11. The SS shall set a different BSIC on Cell 2, as the previous timing information of Cell 2 is invalid in the UE for the next iteration of the test procedure loop.
12. Repeat step 2-11 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 5.2.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 5.2.3.4.3-1: Common Exception messages for E-UTRAN FDD - GSM handover**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-6<br>Table H.3.1-7<br>Table H.3.3-2<br>Table H.3.3-3 |

**Table 5.2.3.4.3-2: *SystemInformationBlockType7*: Additional E-UTRAN FDD - GSM handover**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-3 SystemInformationBlockType7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| commonInfo SEQUENCE { | | | |
| p-MaxGERAN | 33 (33 dBm) | | GSM 400 & GSM 900 & GSM 850 & GSM 700 |
| | 30 (30 dBm) | | DCS 1800 & PCS 1900 |
| } | | | |

**Table 5.2.3.4.3-3: *ReportConfigInterRAT-B1-GERAN*: Additional E-UTRAN FDD - GSM handover**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7A ReportConfigInterRAT-B1-GERAN(GERAN-Thres) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-GERAN(GERAN-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| b1-thresholdGERAN | 30 (-80 dBm) | -80 is actual value in dBm (30 - 110 dBm) | |

**Table 5.2.3.4.3-4: *MeasResults*: Additional E-UTRAN FDD - GSM handover**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListGERAN | MeasResultListGERAN | | |
| } | | | |
| } | | | |

**Table 5.2.3.4.3-5: *MeasResultListGERAN:* Additional E-UTRAN FDD - GSM handover**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListGERAN ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| MeasResultGERAN SEQUENCE { | | | |
| carrierFreq | CarrierFreqGERAN | Contains the carrier frequency of the target GERAN cell | |
| physCellId | PhysCellIdGERAN | Contains the Base Station Identity Code (BSIC) | |
| } | | | |
| measResult SEQUENCE { | | | |
| rssi | | Set according to specific test | |
| } | | | |
| } | | | |

## 5.2.3.5 Test requirement

Tables 5.2.3.4.1-1, 5.2.3.5-1 and 5.2.3.5-2 defines the primary level settings including test tolerances for E-UTRAN FDD to GSM handover test case.

**Table 5.2.3.5-1: Cell Specific Test requirement Parameters for Cell 1 E-UTRAN FDD cell**

| **Parameter** | **Unit** | **Cell 1** | |
|---|---|---|---|
| | | **T1, T2** | **T3** |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.2 FDD |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note1 | dB | | |
| OCNG_RB Note1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 (AWGN) | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | |

| | | |
|---|---|---|
| RSRP Note 3 | dBm/15kHz | -94 |
| Propagation Condition | | AWGN |
| Note 1: OCNG shall be used such that cell 1 is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.<br>Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | |

**Table 5.2.3.5-2: Cell Specific Test requirement Parameters for Cell 2 GSM cell**

| Parameter | Unit | Cell 2 (GSM) | |
|---|---|---|---|
| | | T1 | T2, T3 |
| Absolute RF Channel Number | | ARFCN 1 | |
| RXLEV | dBm | -85 | -75 |

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE start to send access bursts on the new DCCH to Cell 2.

The handover delay $T_{Handover\ delay}$ test requirement in this case is expressed as:

Handover delay $T_{Handover\ delay}$ = handover delay + $T_{offset}$ + $T_{UL}$

Handover delay = 90 ms; this is based on handover delay value as defined in Table 5.2.3.3.-1

$T_{offset}$ = 4.65 ms; GSM timing uncertainty between the time from when the UE is ready to transmit until the start of the next timeslot in GSM 26 multiframe structure

$T_{UL}$ = 4.65 ms; the time the UE has to wait in case the next timeslot is an idle frame or a SACCH frame

The handover delay $T_{Handover\ delay}$ shall be less than a total of 100 ms in this test case (note: this gives a total of 99.29 ms but the test allows 100 ms).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.2.4 E-UTRAN TDD - UTRAN TDD handover

### 5.2.4.1 Test purpose

To verify the UE's ability to transfer a connection between the UE and E-UTRAN to UTRAN in RRC_CONNECTED state by meeting the UE RRC procedure delay and interruption time requirements.

### 5.2.4.2 Test applicability

This test applies to all types of release 8 and forward E-UTRA TDD UEs that support release 9 and forward UTRA TDD. Applicability requires support for FGI bits 8, and 22.

### 5.2.4.3 Minimum conformance requirements

When the UE receives a RRC message implying E-UTRAN/UTRAN TDD handover the UE shall be ready to start the transmission of the new uplink DPCH or the SYNC-UL within $D_{handover}$ seconds from the end of the last TTI containing the RRC MOBILITY FROM E-UTRA command.

Where:

$D_{handover}$ equals the RRC procedure performance value plus the interruption time stated in TS 36.133 [4] clause 5.3.2.2.2.

The interruption time is the time between the end of the last TTI containing the RRC command on the E-UTRAN PDSCH and the time the UE starts transmission on the uplink DPCH or the SYNC-UL in UTRAN TDD, excluding the RRC procedure delay. The interruption time depends on whether the target cell is known to the UE or not.

The UE shall always perform a UTRA synchronisation procedure as part of the handover procedure.

If the target cell has been measured by the UE during the last 5 seconds, the interruption time shall be less than $T_{interrupt1}$

$$T_{interrupt1} = T_{offset} + T_{UL} + 30 * F_{SFN} + 20 + 10 * F_{max} \text{ ms}$$

If the target cell has not been measured by the UE during the last 5 seconds, the interruption time shall be less than $T_{interrupt2}$

$$T_{interrupt2} = T_{offset} + T_{UL} + 30 * F_{SFN} + 180 + 10 * F_{max} \text{ ms}$$

Where:

$T_{offset}$ Equal to 10 ms, the frame timing uncertainty between the old cell and the target cell and the time that can elapse until the appearance of a Beacon channel

$T_{UL}$ Equal to 10 ms, the time that can elapse until the appearance of the UL timeslot in the target cell

$F_{SFN}$ Equal to 1 if SFN decoding is required and equal to 0 otherwise

$F_{max}$ denotes the maximum number of radio frames within the transmission time intervals of all transport channels that are multiplexed into the same CCTrCH.

The interruption time requirements for an unknown target cell shall apply only if the signal quality of the unknown target cell is sufficient for successful synchronisation with one attempt.

The normative reference for this requirement is TS 36.133 [4] clause 5.3.2 and A.5.2.4.

## 5.2.4.4 Test description

### 5.2.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.22.
2. The general test parameter settings are set up according to Table 5.2.4.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are defined in clause 5.2.4.4.3.
5. There is one E-UTRA TDD serving cell and one UTRA TDD cell specified in the test. Cell 1 (E-UTRA TDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.2.4.4.1-1: General test parameters for E-UTRA TDD to UTRA (1.28 Mcps TDD OPTION) handover test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Initial conditions | Active cell | | Cell 1 | E-UTRA TDD cell |
| | Neighbour cell | | Cell 2 | UTRA 1.28Mcps TDD Cell |
| Final conditions | Active cell | | Cell 2 | |
| Gap Pattern Id | | | 0 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| Uplink-downlink configuration of cell 1 | | | 1 | As specified in table 4.2.2 in TS 36.211 [9] |
| Special subframe configuration of cell 1 | | | 6 | As specified in table 4.2.1 in TS 36.211 [9] |
| CP length of cell 1 | | | Normal | |
| Time offset between cells | | | 3 ms | Asynchronous cells<br>3ms or 92160*Ts |
| Access Barring Information | | | Not Sent | No additional delays in random access procedure. |
| Assigned Sub-Channel Number | | | 1 | No additional delays in random access procedure due to ASC. |
| Hysteresis | | dB | 0 | |
| Time To Trigger | | dB | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | OFF | |
| Ofn | | dB | 0 | |
| Hys | | dB | 0 | |
| Thresh1 | | dBm | -93 | E-UTRA event B2 threshold |
| Thresh2 | | dBm | -80 | UTRA event B2 threshold |
| T1 | | s | 5 | |
| T2 | | s | ≤10 | |
| T3 | | s | 1 | |

### 5.2.4.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. A neighbouring cell list, including the UTRA cell (Cell 2) is sent to the UE before T2 starts. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and send a measurement report. T3 is defined as the end of the last TTI containing the RRC message implying handover.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table's 5.2.4.5-1 and 5.2.4.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. The SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 5.2.4.5-1 and 5.2.4.5-2. T2 starts.
6. UE shall transmit a MeasurementReport message triggered by Event B2.
7. SS shall transmit a MobilityFromEUTRACommand message implying handover to Cell 2.
8. The start of T3 is the instant when the last TTI containing the RRC MobilityFromEUTRACommand message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T2 to T3 as specified in Tables 5.2.4.5-1 and 5.2.4.5-2. T3 starts.

9. If the UE transmits the UL to Cell 2 less than 120 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure testes is increased by one.

10. After T3 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

11. The SS shall set Cell 2 cell parameter id =(current cell 2 cell parameter id +4) mod 16 for next iteration of the test procedure loop.

12. Repeat step 2-11 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 5.2.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 and 4.7B with the following exceptions:

**Table 5.2.4.4.3-1: Common Exception messages for E-UTRA TDD to UTRA TDD cell handover**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-5<br>Table H.3.1-7<br>Table H.3.3-1<br>Table H.3.3-3 |

**Table 5.2.4.4.3-2: *ReportConfigInterRAT-B2-UTRA*: Additional E-UTRAN TDD - UTRAN TDD handover**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-8 ReportConfigInterRAT-B2(EUTRA-Thres, UTRA-Thres) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B2-UTRA(EUTRA-Thres, UTRA-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB2 SEQUENCE { | | | |
| b2-Threshold1 CHOICE { | | | |
| threshold-RSRP | 47 (-93 dBm) | -93 dBm EUTRA-Thres is actual threshold value in dBm (47 - 140 dBm) | |
| } | | | |
| b2-Threshold2 CHOICE { | | | |
| b2-Threshold2 UTRA CHOICE { | | | |
| utra-RSCP | 35 (-80 dB) | -80 dB is actual UTRA-Thres is actual RSCP value in dB (35 - 115 dBm) | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

**Table 5.2.4.4.3-3: *MeasResults*: Additional E-UTRAN TDD - UTRAN TDD handover**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultsServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultsNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 5.2.4.4.3-4: *MeasResultListUTRA*: Additional E-UTRAN TDD - UTRAN TDD handover**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| tdd | PhysCellIdUTRA-TDD | This is the typical value range used in UTRAN TDD tests. | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-RSCP | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 5.2.4.4.3-5: HANDOVER TO UTRAN COMMAND**

| **Information Element** | **Value/remark** | **Comment** | **Condition** |
|---|---|---|---|
| CHOICE channel requirement | Uplink DPCH info | | |
| - CHOICE mode | TDD | | |
| - Uplink Timing Advance Control | | | |
| - CHOICE Timing Advance | enabled | | |
| - CHOICE TDD option | 1.28 Mcps | | |
| - Uplink synchronization parameters | | | |
| - Uplink synchronization step size | 1 | | |
| - Uplink synchronization frequency | 1 | | |
| - Synchronization parameters | | | |
| - SYNC_UL codes bitmap | "11111111" | | |
| - FPACH info | | | |
| - Timeslot number | 0 | | |
| - Channelisation code | (16/15) | | |
| - Midamble Shift and burst type | | | |
| - CHOICE TDD option | 1.28 Mcps TDD /REL-4/ | | |
| - Midamble Allocation Mode | Default midamble | | |
| - Midamble configuration | 4 (k=8) | | |
| - Midamble Shift | Not present | | |
| - WT | 4 | | |
| | | | |
| - $PRX_{UpPCHdes}$ | 15(-105dBm) | | |
| - SYNC_UL procedure | Not present | | |
| - Activation time | Not Present | | |
| Downlink information common for all radio links | | | |

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | Initialize | | |
| - Default DPCH Offset Value | Arbitrary set to value 0..306688 by step of 512 | | |
| -Downlink information for each radio links | | | |
| - CHOICE mode | TDD | | |
| - Primary CCPCH info | | | |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | TDD | | |
| - Activation time | Not Present | | |

## 5.2.4.5 Test requirement

Tables 5.2.4.4.1-1, 5.2.4.5-1 and 5.2.4.5-2 define the primary level settings including test tolerances for E-UTRAN TDD - UTRAN TDD handover test.

**Table 5.2.4.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD to UTRA TDD handover test case (Cell 1)**

| Parameter | Unit | Cell 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | | OP.2 TDD |
| PBCH_RA | dB | 0 | 0 | 0 |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $\hat{E}_s / I_{ot}$ | dB | 15.10 | -3 | -3 |
| $\hat{E}_s / N_{oc}$ | dB | 15.10 | -3 | -3 |
| $N_{oc}$ | dBm/15kHz | -99.05 | | |
| RSRP | dBm/15kHz | -83,95 | -102.05 | -102.05 |
| SCH_RP | dBm/15 kHz | -83.95 | -102.05 | -102.05 |
| Io Note 2 | dBm/9MHz | -56.04 | -69.50 | -69.50 |
| Propagation Condition | | AWGN | | |

Note 1: OCNG shall be used such that cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 5.2.4.5-2: Cell Specific Test requirement Parameters for Cell 2 UTRAN TDD cell**

| Parameter | Unit | Cell 2 (UTRA) | | | | | |
|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | | DwPTS | | |
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| UTRA RF Channel Number Note 1 | | 2 | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | | |
| DwPCH_Ec/Ior | dB | | | | 0 | | |
| OCNS_Ec/Ior | dB | -3 | | | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -3 | 12.6 | 12.6 | -3 | 12.6 | 12.6 |
| $I_{oc}$ | dBm/1.28 MHz | -80.8 | | | | | |
| PCCPCH RSCP | dBm | -86.8 | -71.2 | -71.2 | n.a. | | |
| Io Note 2 | dBm/1.28 MHz | -79.04 | -67.97 | -67.97 | -79.04 | -67.97 | -67.97 |
| Propagation Condition | | AWGN | | | | | |
| Note 1: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number.<br>Note 2: PCCPCH_RSCP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves | | | | | | | |

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE start to transmit the SYNCH-UL sequence in the UpPTS to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

$D_{handover}$ = maximum RRC procedure delay + $T_{interrupt1}$

$T_{interrupt1}= T_{offset}+T_{UL}+30*F_{SFN}+20 +10*F_{max}$ ms

$T_{offse}$= 10 ms; The frame timing uncertainty between the old cell and the target cell and the time that can elapse until the appearance of a Beacon channel

$T_{UL}$ = 10 ms; The time that can elapse until the appearance of the UL timeslot in the target cell

$F_{SFN}$ = 1; Equal to 1 if SFN decoding is required and equal to 0 otherwise.

$F_{max}$ = 0; The maximum number of radio frames within the transmission time intervals of all transport channels that are multiplexed into the same CCTrCH.

Maximum RRC procedure delay = 50 ms as defined in TS 36.331 [5] clause 11.2.

The handover delay $D_{handover}$ shall be less than a total of 120 ms in this test case.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.2.5 E-UTRAN FDD - UTRAN TDD handover

### 5.2.5.1 Test purpose

To verify the UE's ability to transfer a connection between the UE and E-UTRAN FDD to UTRAN TDD in RRC_CONNECTED state by meeting the UE RRC procedure delay and interruption time requirements.

### 5.2.5.2 Test applicability

This test applies to all types of release 8 and forward E-UTRA FDD UEs that support release 9 and forward UTRA TDD. Applicability requires support for FGI bits 8, and 22.

### 5.2.5.3 Minimum conformance requirements

#### 5.2.5.3.1 void

#### 5.2.5.3.2 1.28Mcps TDD option

When the UE receives a RRC message implying E-UTRAN/UTRAN TDD handover the UE shall be ready to start the transmission of the new uplink DPCH or the SYNC-UL within $D_{handover}$ seconds from the end of the last TTI containing the RRC MOBILITY FROM E-UTRA command.

Where:

$D_{handover}$ equals the RRC procedure delay, which is 50 ms plus the interruption time stated in TS 36.133 [4] clause 5.3.2.2.2.

The interruption time is the time between the end of the last TTI containing the RRC command on the E-UTRAN PDSCH and the time the UE starts transmission on the uplink DPCH or the SYNC-UL in UTRAN TDD, excluding the RRC procedure delay. The interruption time depends on whether the target cell is known to the UE or not.

The UE shall always perform a UTRA synchronisation procedure as part of the handover procedure.

If the target cell has been measured by the UE during the last 5 seconds, the interruption time shall be less than $T_{interrupt1}$.

$$T_{interrupt1} = T_{offset} + T_{UL} + 30 * F_{SFN} + 20 + 10 * F_{max} \text{ ms}$$

If the target cell has not been measured by the UE during the last 5 seconds, the interruption time shall be less than $T_{interrupt2}$.

$$T_{interrupt2} = T_{offset} + T_{UL} + 30 * F_{SFN} + 180 + 10 * F_{max} \text{ ms}$$

Where:

| | |
|---|---|
| $T_{offset}$ | Equal to 10 ms, the frame timing uncertainty between the old cell and the target cell and the time that can elapse until the appearance of a Beacon channel |
| $T_{UL}$ | Equal to 10 ms, the time that can elapse until the appearance of the UL timeslot in the target cell |
| $F_{SFN}$ | Equal to 1 if SFN decoding is required and equal to 0 otherwise |
| $F_{max}$ | denotes the maximum number of radio frames within the transmission time intervals of all transport channels that are multiplexed into the same CCTrCH. |

The interruption time requirements for an unknown target cell shall apply only if the signal quality of the unknown target cell is sufficient for successful synchronisation with one attempt.

The normative reference for this requirement is TS 36.133 [4] clause 5.3.2 and A.5.2.5.

#### 5.2.5.3.2 void

### 5.2.5.4 Test description

#### 5.2.5.4.1 void

#### 5.2.5.4.2 1.28Mcps TDD option

5.2.5.4.2.1Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.22.

2. The general test parameter settings are set up according to Table 5.2.5.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 5.2.5.4.3.

5. There is one E-UTRA FDD serving cell and one UTRATDD cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.2.5.4.2.1-1: General Test Parameters for E-UTRAN FDD - UTRAN TDD handover test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Initial conditions | Active cell | | Cell 1 | E-UTRA FDD cell |
| | Neighbour cell | | Cell 2 | UTRA 1.28Mcps TDD Cell |
| Final conditions | Active cell | | Cell 2 | |
| Gap Pattern Id | | | 1 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| E-UTRAN FDD measurement quantity | | | RSRP | |
| UTRAN TDD measurement quantity | | | P-CCPCH RSCP | |
| CP length of cell 1 | | | Normal | |
| Access Barring Information | | | Not Sent | No additional delays in random access procedure. |
| Assigned Sub-Channel Number | | | 1 | No additional delays in random access procedure due to ASC. |
| Hysteresis | | dB | 0 | |
| Time To Trigger | | dB | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | OFF | |
| Ofn | | dB | 0 | |
| Hys | | dB | 0 | |
| Thresh1 | | dBm | -93 | Absolute E-UTRAN RSRP threshold for event B2 |
| Thresh2 | | dBm | -80 | Absolute UTRAN P-CCPCH RSCP threshold for event B2 |
| T1 | | s | 5 | |
| T2 | | s | ≤ 10 | |
| T3 | | s | 1 | |

### 5.2.5.4.2.2 Test procedure

The test scenario comprises of two cells, E-UTRA TDD cell1 and UTRA TDD cell2. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE does not have any timing information of Cell 2. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and send a measurement report. Gap pattern configuration with id #1 as specified in TS 36.133 [4] Table 8.1.2.1-1 is configured before T2 begins to enable inter-RAT frequency monitoring.

A RRC message implying handover shall be sent to the UE during period T2, after the UE has reported Event B2. The start of T3 is the instant when the last TTI containing the RRC message implying handover is sent to the UE. The handover message shall contain cell 2 as the target cell.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

2. Set the parameters according to T1 in Table's 5.2.5.5.2-1 and 5.2.5.5.2-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 5.2.5.5.2.-1 and 5.2.5.5.2-2.
6. UE shall transmit a MeasurementReport message triggered by Event B2.
7. SS shall transmit a MobilityFromEUTRACommand message implying handover to Cell 2.
8. The start of T3 is the instant when the last TTI containing the RRC message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T2 to T3 as specified in Tables 5.2.5.5-1 and 5.2.5.5-2.
9. If the UE transmits the UL DPCH Cell 2 less than 120 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure testes is increased by one.
10. After T3 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
11. SS shall change set cell 2 cell parameter id =(current cell 2 cell parameter id +4) mod 16.
12. Repeat step 2-11 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 5.2.5.4.2.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 5.2.5.4.2.3-1: Common Exception messages for E-UTRAN FDD - UTRAN TDD handover**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-5<br>Table H.3.1-7<br>Table H.3.3-1<br>Table H.3.3-3 |

**Table 5.2.5.4.2.3-2: *ReportConfigInterRAT-B2-UTRA*: Additional E-UTRAN FDD - UTRAN TDD handover**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B2-UTRA(EUTRA-Thres, UTRA-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB2 SEQUENCE { | | | |
| b2-Threshold1 CHOICE { | | | |
| threshold-RSRP | 47 (-93 dBm) | -93 dBm EUTRA-Thres is actual threshold value in dBm ( 46 - 140 dBm) | |
| } | | | |
| b2-Threshold2 CHOICE { | | | |
| b2-Threshold2UTRA CHOICE { | | | |
| utra-RSCP | 35 (-80 dBm) | -80 dBm is actual UTRA-Thres is actual RSCP value in dBm (36-115dBm) | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

**Table 5.2.5.4.2.3-3: *MeasResults*: Additional E-UTRAN FDD - UTRAN FDD handover**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 5.2.5.4.2.3-4: *MeasResultListUTRA*: Additional E-UTRAN FDD - UTRAN TDD handover**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| tdd | PhysCellIdUTRA-TDD | | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-RSCP | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 5.2.5.4.2.3-5: HANDOVER TO UTRAN COMMAND**

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| CHOICE channel requirement | Uplink DPCH info | | |
| - CHOICE mode | TDD | | |
| - Uplink Timing Advance Control | | | |
| - CHOICE Timing Advance | enabled | | |
| - CHOICE TDD option | 1.28 Mcps | | |
| - Uplink synchronization parameters | | | |
| - Uplink synchronization step size | 1 | | |
| - Uplink synchronization frequency | 1 | | |
| - Synchronization parameters | | | |
| - SYNC_UL codes bitmap | "11111111" | | |
| - FPACH info | | | |
| - Timeslot number | 0 | | |
| - Channelisation code | (16/15) | | |
| - Midamble Shift and burst type | | | |
| - CHOICE TDD option | 1.28 Mcps TDD /REL-4/ | | |
| - Midamble Allocation Mode | Default midamble | | |
| - Midamble configuration | 4 (k=8) | | |
| - Midamble Shift | Not present | | |
| - WT | 4 | | |
| | | | |
| - $PRX_{UpPCHdes}$ | 15(-105dBm) | | |
| - SYNC_UL procedure | Not present | | |
| - Activation time | Not present | | |
| Downlink information common for all radio links | | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | Initialize | | |
| - Default DPCH Offset Value | Arbitrary set to value 0..306688 by step of 512 | | |
| -Downlink information for each radio links | | | |
| - CHOICE mode | TDD | | |
| - Primary CCPCH info | | | |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | TDD | | |
| - Activation time | Not Present | | |

### 5.2.5.4.3 7.68 Mcps TDD option

Editor's note: FFS

## 5.2.5.5 Test requirement

### 5.2.5.5.1 3.84Mcps TDD option

Editor's note: FFS

### 5.2.5.5.2 1.28Mcps TDD option

Tables 5.2.5.4.2.1-1, 5.2.5.5.2-1 and 5.2.5.5.2-2 define the primary level settings including test tolerances for E-UTRAN FDD - UTRAN TDD handover test.

**Table 5.2.5.5-1: Cell Specific Test requirement Parameters for Cell 1 E-UTRAN FDD cell**

| Parameter | Unit | Cell 1 (E-UTRA) | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $\hat{E}_s / N_{oc}$ | dB | 11 + TT | -3 + TT | -3+ TT |
| $N_{oc}$ | dBm/15 kHz | -98 | | |
| $\hat{E}_s / I_{ot}$ | dB | 11 + TT | -3 + TT | -3+ TT |
| RSRP | dBm/15 KHz | -87 + TT | -101 + TT | -101+ TT |
| Propagation Condition | | AWGN | | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | | |

**Table 5.2.5.5-2: Cell Specific Test requirement Parameters for Cell 2 UTRAN TDD cell**

| Parameter | Unit | Cell 2 (UTRA) | | | | | |
|---|---|---|---|---|---|---|---|
| Timeslot Number | | 0 | | | DwPTS | | |
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| UTRA RF Channel Number* | | Channel 2 | | | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | | | |
| DwPCH_Ec/Ior | dB | | | | 0 | | |
| OCNS_Ec/Ior | dB | -3 | | | | | |
| $\hat{I}_{or} / I_{oc}$ | dB | -3 TT | 11 TT | 11 TT | -3 TT | 11 TT | 11 TT |
| $I_{oc}$ | dBm/1.28 MHz | -80 | | | | | |
| PCCPCH RSCP | dBm | -86 TT | -72 TT | -72 TT | n.a. | | |
| Propagation Condition | | AWGN | | | | | |
| * Note: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number. | | | | | | | |

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE start to transmit the SYNCH-UL sequence in the UpPTS to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

$T_{interrupt1}$= $T_{offset}$+$T_{UL}$+30*$F_{SFN}$+20 +10*$F_{max}$ ms

$T_{offse}$= 10 ms; The frame timing uncertainty between the old cell and the target cell and the time that can elapse until the appearance of a Beacon channel

$T_{UL}$ = 10 ms; The time that can elapse until the appearance of the UL timeslot in the target cell

$F_{SFN}$ = 1; Equal to 1 if SFN decoding is required and equal to 0 otherwise.

$F_{max}$ = 0; The maximum number of radio frames within the transmission time intervals of all transport channels that are multiplexed into the same CCTrCH.

Maximum RRC procedure delay = 50 ms as defined in TS 36.331 [5] clause 11.2

The handover delay $D_{handover}$ shall be less than a total of 120 ms in this test case.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

#### 5.2.5.5.2 7.68 Mcps TDD option

Editor's note: FFS

## 5.2.6 E-UTRA TDD - GSM handover

### 5.2.6.1 Test purpose

To verify the UE's ability to transfer a connection between the UE and E-UTRAN to GSM in RRC_CONNECTED state by meeting the UE handover delay and interruption time requirements.

### 5.2.6.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward that support GSM and inter-RAT PS handover to GERAN. Applicability requires support for FGI bits 9, 15 and 23.

### 5.2.6.3 Minimum conformance requirements

The handover delay $T_{Handover}$ delay shall be less than handover delay + $T_{offset}$ + $T_{UL}$ in RRC_CONNECTED state.

The handover delay given in table 5.2.6.3-1 and interruption time given in table 5.2.6.3-2 requirements for the case where the UE has not synchronised to the GSM cell before receiving the RRC MOBILITY FROM E-UTRA command are valid when the signal quality of the GSM cell is sufficient for successful synchronisation with one attempt. If the UE is unable to synchronise to the GSM cell on the first attempt, it shall continue to search for synchronisation information for up to 800 ms duration. If after 800 ms the UE has not synchronised to the GSM cell it shall follow the handover failure procedure specified in TS 36.331 [5].

When the UE receives a RRC MOBILITY FROM E-UTRA command the UE shall be ready to transmit (as specified in 3GPP TS 45.010 [13]) on the channel of the new RAT within the value defined in TS 36.133 [4] clause 5.3.3.2.1 and shown in table 5.2.6.3-1 from the end of the last TTI containing the RRC command.

The UE shall process the RRC procedures for the MOBILITY FROM E-UTRA command within 50 ms, which is noted as RRC procedure delay.

**Table 5.2.6.3-1: E-UTRAN/GSM handover - handover delay**

| UE synchronisation status | handover delay [ms] |
|---|---|
| The UE has synchronised to the GSM cell before the RRC MOBILITY FROM E-UTRA COMMAND is received | 90 |
| The UE has not synchronised to the GSM cell before RRC the MOBILITY FROM E-UTRA COMMAND is received | 190 |

The interruption time, i.e. the time between the end of the last TTI containing a transport block on the old channel and the time the UE is ready to transmit on the new channel, shall be less than the value defined in TS 36.133 [4] clause 5.3.3.2.2 and shown in table 5.2.6.3-2:

**Table 5.2.6.3-2: E-UTRAN/GSM handover - interruption time**

| Synchronisation status | Interruption time [ms] |
|---|---|
| The UE has synchronised to the GSM cell before the RRC MOBILITY FROM E-UTRA COMMAND is received | 40 |
| The UE has not synchronised to the GSM cell before the RRC MOBILITY FROM E-UTRA COMMAND is received | 140 |

The normative reference for this requirement is TS 36.133 [4] clause 5.3.3 and A.5.2.6.

## 5.2.6.4 Test description

### 5.2.6.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 5.2.6.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 5.2.6.4.3.
5. There is one E-UTRA TDD serving cell and one GSM cell specified in the test. Cell 1 (E-UTRA TDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.2.6.4.1-1: General Test Parameters for E-UTRAN TDD - GSM handover test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Gap Pattern Id | | | 1 | As specified in TS 36.133 [4] section 8.1.2.1. |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbour cell | | Cell 2 | |
| Final conditions | Active cell | | Cell 2 | |
| Uplink-downlink configuration of cell 1 | | | 1 | As specified in table 4.2.2 in TS 36.211 [9] |
| Special subframe configuration of cell 1 | | | 6 | As specified in table 4.2.1 in TS 36.211 [9] |
| CP length of cell 1 | | | Normal | |
| Inter-RAT measurement quantity | | | GSM Carrier RSSI | |
| E-UTRA RF Channel Number | | | 1 | E-UTRA RF Channel Number |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | E-UTRA Channel Bandwidth ($BW_{channel}$) |
| Threshold other system | | dBm | -80 | Absolute GSM carrier RSSI threshold for event B1. |
| Hysteresis | | dB | 0 | |
| Time to Trigger | | ms | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | | OFF |
| T1 | | s | 20 | |
| T2 | | s | 7 | |
| T3 | | s | 1 | |

### 5.2.6.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. The RRC message implying handover to Cell 2 shall be sent to the UE during T2, after the UE has reported Event B1. T3 is defined as the end of the last E-UTRAN TTI containing the RRC message implying handover. The requirements are also applicable for a UE not requiring measurement gap, in which case no measurement gap pattern is sent.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table's 5.2.6.5-1 and 5.2.6.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 5.2.6.5-1 and 5.2.6.5-2. T2 starts.
6. UE shall transmit a MeasurementReport message triggered by Event B1.
7. SS shall transmit a MobilityFromEUTRACommand message implying handover to Cell 2.
8. The start of T3 is the instant when the last TTI containing the RRC message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T2 to T3 as specified in Table 5.2.6.5-2. T3 starts.
9. If the UE sends access bursts on the new DCCH to Cell 2 less than 100 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure testes is increased by one.
10. After T3 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

11. The SS shall set a different BSIC on Cell 2, as the previous timing information of Cell 2 is invalid in the UE for the next iteration of the test procedure loop.

12. Repeat step 2-11 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 5.2.6.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 5.2.6.4.3-1: Common Exception messages for E-UTRAN TDD - GSM handover**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-6<br>Table H.3.1-7<br>Table H.3.3-2<br>Table H.3.3-3 |

**Table 5.2.6.4.3-2: *SystemInformationBlockType7*: Additional E-UTRAN TDD - GSM handover**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-3 SystemInformationBlockType7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| commonInfo SEQUENCE { | | | |
| p-MaxGERAN | 33 (33 dBm) | | GSM 400 & GSM 900 & GSM 850 & GSM 700 |
| | 30 (30 dBm) | | DCS 1800 & PCS 1900 |
| } | | | |

**Table 5.2.6.4.3-3: ReportConfigInterRAT-B1-GERAN: Additional E-UTRAN TDD - GSM handover**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-GERAN(GERAN-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| B1-Threshold-GERAN CHOICE { | | | |
| thresholdGERAN | 30 (-80 dBm) | -80 is actual value in dBm (30 - 110 dBm) | |

**Table 5.2.6.4.3-4: MeasResults: Additional E-UTRAN TDD - GSM handover**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListGERAN | MeasResultListGERAN | | |
| } | | | |
| } | | | |

**Table 5.2.6.4.3-5: MeasResultListGERAN: Additional E-UTRAN TDD - GSM handover**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListGERAN ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| MeasResultGERAN SEQUENCE { | | | |
| CarrierFreq | CarrierFreqGERAN | Contains the carrier frequency of the target GERAN cell | |
| physCellId | PhysCellIdGERAN | Contains the Base Station Identity Code (BSIC) and is used %% | |
| } | | | |
| Cgi-Info SEQUENCE { | | | |
| cellGlobalId | CellGlobalIdGERAN | | |
| routingAreaCode | Not present | | |
| } | | | |
| measResult SEQUENCE { | | | |
| Rssi | | Set according to specific test | |
| } | | | |
| } | | | |

## 5.2.6.5 Test requirement

Tables 5.2.6.4.1-1, 5.2.6.5-1 and 5.2.6.5-2 defines the primary level settings including test tolerances for E-UTRAN TDD to GSM handover test case.

**Table 5.2.6.5-1: Cell Specific Test requirement Parameters for Cell 1 E-UTRAN TDD cell**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1, T2 | T3 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | OP.2 TDD |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note1 | dB | | |
| OCNG_RB Note1 | dB | | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 (AWGN) | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | |
| RSRP Note 3 | dBm/15kHz | -94 | |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that cell 1 is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 5.2.6.5-2: Cell Specific Test requirement Parameters for Cell 2 GSM cell**

| Parameter | Unit | Cell 2 (GSM) | |
|---|---|---|---|
| | | T1 | T2, T3 |
| Absolute RF Channel Number | | ARFCN 1 | |
| RXLEV | dBm | -85 | -75 |

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE start to send access bursts on the new DCCH to Cell 2.

The handover delay $T_{Handover\ delay}$ test requirement in this case is expressed as:

Handover delay $T_{Handover\ delay}$ = handover delay + $T_{offset}$ + $T_{UL}$

Handover delay = 90 ms; this is based on handover delay value as defined in Table 5.2.6.3.-1

$T_{offset}$ = 4.65 ms; GSM timing uncertainty between the time from when the UE is ready to transmit until the start of the next timeslot in GSM 26 multiframe structure

$T_{UL}$ = 4.65 ms; the time the UE has to wait in case the next timeslot is an idle frame or a SACCH frame

The handover delay $T_{Handover\ delay}$ shall be less than a total of 100 ms in this test case (note: this gives a total of 99.3 ms but the test allows 100 ms).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.2.7 E-UTRAN FDD - UTRAN FDD handover: unknown target cell

### 5.2.7.1 Test purpose

To verify the UE's ability to transfer a connection between the UE and E-UTRAN to UTRAN handover: unknown target cell in RRC_CONNECTED state by meeting the handover to an unknown target cell delay requirements.

### 5.2.7.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support UTRA FDD. Applicability requires support for FGI bits 8, and 22.

### 5.2.7.3 Minimum conformance requirements

When the UE receives a RRC message implying handover to UTRAN the UE shall be ready to start the transmission of the new UTRA uplink DPCCH within $D_{handover}$ seconds from the end of the last E-UTRAN TTI containing the RRC MOBILITY FROM E-UTRA command.

Where:

$D_{handover}$ equals the RRC procedure delay, which is 50 ms plus the interruption time stated in TS 36.133 [4] clause 5.3.1.1.2.

The interruption time is the time between the end of the last TTI containing the RRC command on the E-UTRAN PDSCH and the time the UE starts transmission on the uplink DPCCH in UTRAN FDD, excluding the RRC procedure delay. The interruption time depends on whether the target cell is known to the UE or not.

The target cell is known if it has been measured by the UE during the last 5 seconds otherwise it is unknown. The UE shall always perform a UTRA synchronisation procedure as part of the handover procedure.

If the target cell is known the interruption time shall be less than $T_{interrupt1}$.

$$T_{interrupt1} = T_{IU} + T_{sync} + 50 + 10*F_{max} + T_{MC} \text{ ms}$$

If the target cell is unknown the interruption time shall be less than $T_{interrupt2}$.

$$T_{interrupt2} = T_{IU} + T_{sync} + 150 + 10*F_{max} + T_{MC} \text{ ms}$$

This requirement shall be met, provided that there is one target cell in the MOBILITY FROM E-UTRA command. When UE is connected to an E-UTRA cell, UTRA SFN timing measurements are not reported. This implies that the timing of the DPCH of the UTRA target cells in the active set cannot be configured by UTRAN to guarantee that all target cells fall within the UE reception window of $T_0$ +/- 148 chips.

Where:

$T_{IU}$ is the interruption uncertainty when changing the timing from the E-UTRAN to the new UTRAN cell. $T_{IU}$ can be up to one UTRA frame (10 ms).

$F_{max}$ denotes the maximum number of radio frames within the transmission time intervals of all transport channels that are multiplexed into the same CCTrCH on the UTRA target cell. If HS-PDSCH is configured in the UTRA target cell, $F_{max}$ is 4 radio frames.

$T_{sync}$ is the time required for measuring the downlink DPCCH channel as stated in TS 25.214 [12] clause 4.3.1.2. In case higher layers indicate the usage of a post-verification period $T_{sync}$ = 0 ms. Otherwise $T_{sync}$ = 40 ms.

$T_{MC}$ is 0 ms if a single UTRA cell is configured as the handover target, otherwise 20 ms if handover to UTRA with 1, 2 or 3 UTRA carriers with secondary HS-PDSCH is configured.

The phase reference is the primary CPICH.

The requirements assume that N312 has the smallest possible value i.e. only one in-sync is required.

The normative reference for this requirement is TS 36.133 [4] clause 5.3.1 and A.5.2.7.

## 5.2.7.4 Test description

### 5.2.7.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.22.
2. The general test parameter settings are set up according to Table 5.2.7.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are defined in clause 5.2.7.4.3.
5. There is one E-UTRA FDD serving cell and one UTRA FDD cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.2.7.4.1-1: General Test Parameters for E-UTRAN FDD - UTRAN FDD handover: unknown target cell test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Initial conditions | Active cell | | Cell 1 | E-UTRAN cell |
| | Neighbouring cell | | Cell 2 | UTRAN cell |
| Final condition | Active cell | | Cell 2 | UTRAN cell |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| E-UTRAN FDD measurement quantity | | | RSRP | |
| Inter-RAT (UTRAN FDD) measurement quantity | | | CPICH Ec/N0 | |
| DRX | | | OFF | Non-DRX test |
| Access Barring Information | | - | Not sent | No additional delays in random access procedure |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 10 | |
| UTRA RF Channel Number | | | 1 | One UTRA FDD carrier frequency is used. |
| Monitored UTRA FDD cell list size | | | 12 | UTRA cells on UTRA RF channel 1 provided in the cell before T2. |
| Post-verification period | | | False | |
| T1 | | s | ≤5 | |
| T2 | | s | 1 | |

### 5.2.7.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of two successive time periods, with time durations of T1and T2 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. A RRC message implying handover shall be sent to the UE during period T1. The start of T2 is the instant when the last TTI containing the RRC message implying handover is sent to the UE. Starting T2, Cell 2 becomes detectable and the UE receives a RRC handover command from the network.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table's 5.2.7.5-1 and 5.2.7.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. SS shall transmit a MobilityFromEUTRACommand message implying handover to Cell 2.
4. The start of T2 is the instant when the last TTI containing the RRC MobilityFromEUTRACommand message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T1 to T2 as specified in Tables 5.2.7.5-1 and 5.2.7.5-2. T2 starts.
5. If the UE transmits the UL DPCCH to Cell 2 less than 290 ms from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
6. After T2 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
7. The SS shall set Cell 2 primary scrambling code = ((current cell 2 primary scrambling code - 50) mod 200 + 100) for next iteration of the test procedure loop.
8. Repeat step 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 5.2.7.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 5.2.7.4.3-1: Common Exception messages for E-UTRAN FDD - UTRAN FDD handover: unknown target cell test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.3-1<br>Table H.3.3-3 |

**Table 5.2.7.4.3-2: HANDOVER TO UTRAN COMMAND**

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| Downlink information common for all radio links | | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | Initialize | | |
| - Default DPCH Offset Value | Arbitrary set to value 0..306688 by step of 512 | | |

## 5.2.7.5 Test requirement

Tables 5.2.7.4.1-1, 5.2.7.5-1 and 5.2.7.5-2 define the primary level settings including test tolerances for E-UTRAN FDD - UTRAN FDD handover: unknown target cell test.

**Table 5.2.7.5-1: Cell Specific Test requirement Parameters for Cell 1 in E-UTRAN FDD - UTRAN FDD handover: unknown target cell test**

| Parameter | Unit | Cell 1 (E-UTRA) | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.2 FDD |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 0 | 0 |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 0 | 0 |
| RSRP Note 3 | dBm/15 KHz | -98 | -98 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 5.2.7.5-2: Cell Specific Test requirement Parameters for Cell 2 in E-UTRAN - UTRAN FDD handover: unknown target cell test**

| Parameter | Unit | Cell 2 (UTRA) | |
|---|---|---|---|
| | | T1 | T2 |
| CPICH_Ec/Ior | dB | -10 | |
| PCCPCH_Ec/Ior | dB | -12 | |
| SCH_Ec/Ior | dB | -12 | |
| PICH_Ec/Ior | dB | -15 | |
| DCH_Ec/Ior | dB | Note 1 | |
| OCNS_Ec/Ior | dB | Note 2 | |
| $\hat{I}_{or} / I_{oc}$ | dB | -infinity | -1.8 |
| $I_{oc}$ | dBm/3,84 MHz | -70 | -70 |
| CPICH_Ec/Io | dB | -infinity | -14 |
| Propagation Condition | AWGN | | |

Note 1: The DPCH level is controlled by the power control loop

Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$

The handover to an unknown target cell delay is defined as the time from the beginning of time period T2, to the moment the UE start to transmit the UL DPCCH to Cell 2.

The handover to an unknown target cell delay test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt2}$ (note: the target cell is unknown)

$T_{interrupt2} = T_{IU} + T_{sync} + 150 + 10*F_{max} + T_{MC}$ ms

$T_{IU}$ = 10 ms; $T_{IU}$ can be up to one UTRA frame (10 ms).

$F_{max}$ = 4 radio frames; The maximum radio frames within the transmission time intervals to fit into DCCH with 40 ms TTI

$T_{sync}$ = 40 ms; In case higher layers indicate the usage of a post-verification period $T_{sync}$ = 0 ms. Otherwise $T_{sync}$ = 40 ms

$T_{MC}$ = 0 ms if a single UTRA cell is configured as the handover target. Otherwise $T_{MC}$ = 20 ms

Maximum RRC procedure delay = 50 ms as defined in TS 36.331 [5] clause 11.2

The handover to an unknown target cell delay shall be less than a total of 290 ms in this test case (note: this gives a total of 50 ms for maximum RRC procedure delay plus 240 ms for $T_{interrupt2}$).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.2.8 E-UTRAN FDD - GSM handover: unknown target cell

### 5.2.8.1 Test purpose

To verify the UE's ability to transfer a connection between the UE and E-UTRAN to GSM handover: unknown target cell in RRC_CONNECTED state by meeting the handover to an unknown target cell delay requirements.

### 5.2.8.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support GSM and inter-RAT PS handover to GERAN. Applicability requires support for FGI bits 9, and 23.

### 5.2.8.3 Minimum conformance requirements

The handover delay $T_{Handover\ delay}$ shall be less than handover delay + $T_{offset}$ + $T_{UL}$ in RRC_CONNECTED state.

The handover delay given in table 5.2.8.3-1 and interruption time given in table 5.2.8.3-2 requirements for the case where the UE has not synchronised to the GSM cell before receiving the RRC MOBILITY FROM E-UTRA command are valid when the signal quality of the GSM cell is sufficient for successful synchronisation with one attempt. If the UE is unable to synchronise to the GSM cell on the first attempt, it shall continue to search for synchronisation information for up to 800 ms duration. If after 800 ms the UE has not synchronised to the GSM cell it shall follow the handover failure procedure specified in TS 36.331 [5].

When the UE receives a RRC MOBILITY FROM E-UTRA command the UE shall be ready to transmit (as specified in 3GPP TS 45.010 [13]) on the channel of the new RAT within the value defined in TS 36.133 [4] clause 5.3.3.2.1 and shown in table 5.2.8.3-1 from the end of the last TTI containing the RRC command.

The UE shall process the RRC procedures for the MOBILITY FROM E-UTRA command within 50 ms, which is noted as RRC procedure delay.

**Table 5.2.8.3-1: E-UTRAN/GSM handover - handover delay**

| UE synchronisation status | handover delay [ms] |
|---|---|
| The UE has synchronised to the GSM cell before the RRC MOBILITY FROM E-UTRA COMMAND is received | 90 |
| The UE has not synchronised to the GSM cell before RRC the MOBILITY FROM E-UTRA COMMAND is received | 190 |

The interruption time is the time between the end of the TTI containing the RRC command on the E-UTRAN PDSCH and the time the UE starts transmission on the uplink channel in GSM, excluding the RRC procedure delay. The interruption time depends on whether the UE has synchronized to the target GSM cell or not and shall be less than the value defined in TS 36.133 [4] clause 5.3.3.2.1 and shown in table 5.2.8.3-2.

**Table 5.2.8.3-2: E-UTRAN/GSM handover - interruption time**

| Synchronisation status | Interruption time [ms] |
|---|---|
| The UE has synchronised to the GSM cell before the RRC MOBILITY FROM E-UTRA COMMAND is received | 40 |
| The UE has not synchronised to the GSM cell before the RRC MOBILITY FROM E-UTRA COMMAND is received | 140 |

The normative reference for this requirement is TS 36.133 [4] clause 5.3.3 and A.5.2.8.

## 5.2.8.4 Test description

### 5.2.8.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 5.2.8.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 5.2.8.4.3.
5. There is one E-UTRA FDD serving cell and one GSM cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.2.8.4.1-1: General Test Parameters for E-UTRAN FDD - GSM handover: unknown target cell test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Gap Pattern Id | | | None | No measurement gaps shall be provided. |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbour cell | | Cell 2 | |
| Final conditions | Active cell | | Cell 2 | |
| DRX | | | OFF | No DRX configured |
| T1 | | s | ≤7 | |
| T2 | | s | 1 | |

## 5.2.8.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of two successive time periods, with time durations of T1 and T2 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. A RRC message implying handover shall be sent to the UE during period T1. The start of T2 is the instant when the last TTI containing the RRC message implying handover is sent to the UE. Starting T2, Cell 2 becomes detectable and the UE receives a RRC handover command from the network.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table's 5.2.8.5-1 and 5.2.8.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. The neighbour cell shall broadcast its own BSIC, SS shall transmit a MobilityFromEUTRACommand message implying handover to Cell 2.
4. The start of T2 is the instant when the last TTI containing the RRC MobilityFromEUTRACommand message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T1 to T2 as specified in Table 5.2.8.5-1 and 5.2.8.5-2. T2 starts.
5. If the UE sends access bursts on the new DCCH to Cell 2 less than 200 ms from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure testes is increased by one.
6. After T2 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
7. The SS shall set a different BSIC on Cell 2, as the previous timing information of Cell 2 is invalid in the UE for the next iteration of the test procedure loop.
8. Repeat step 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

## 5.2.8.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 5.2.8.4.3-1: Common Exception messages for E-UTRAN FDD - GSM handover: unknown target cell test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.3-2<br>Table H.3.3-3 |

**Table 5.2.8.4.3-2: *SystemInformationBlockType7*: Additional E-UTRAN FDD - GSM handover: unknown target cell test requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-3 SystemInformationBlockType7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| commonInfo SEQUENCE { | | | |
| p-MaxGERAN | 33 (33 dBm) | | GSM 400 & GSM 900 & GSM 850 & GSM 700 |
| | 30 (30 dBm) | | DCS 1800 & PCS 1900 |
| } | | | |

## 5.2.8.5 Test requirement

Tables 5.2.8.4.1-1, 5.2.8.5-1 and 5.2.8.5-2 defines the primary level settings including test tolerances for E-UTRAN FDD to GSM handover: unknown target cell test.

**Table 5.2.8.5-1: Cell Specific Test requirement Parameters for Cell 1 in E-UTRAN FDD - GSM handover: unknown target cell test**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.2 FDD |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note1 | dB | | |
| OCNG_RB Note1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | |
| RSRP Note 3 | dBm/15 kHz | -94 | |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that cell 1 is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 5.2.8.5-2: Cell Specific Test requirement Parameters for Cell 2 in E-UTRAN FDD - GSM handover: unknown target cell test**

| Parameter | Unit | Cell 2 (GSM) | |
|---|---|---|---|
| | | T1 | T2 |
| Absolute RF Channel Number | | ARFCN 1 | |
| RXLEV | dBm | -Infinity | -75 |

The handover to an unknown target cell delay is defined as the time from the beginning of time period T2, to the moment the UE start to send access bursts on the new DCCH to Cell 2.

The handover to an unknown target cell delay test requirement in this case is expressed as:

Handover delay $T_{Handover\ delay}$ = handover delay + $T_{offset}$ + $T_{UL}$

Handover delay = 190 ms; this is based on handover delay value as defined in Table 5.2.3.3.-1

$T_{offset}$ = 4.65 ms; GSM timing uncertainty between the time from when the UE is ready to transmit until the start of the next timeslot in GSM 26 multiframe structure

$T_{UL}$ = 4.65 ms; the time the UE has to wait in case the next timeslot is an idle frame or a SACCH frame

The handover to an unknown target cell delay shall be less than a total of 200 ms in this test case (note: this gives a total of 199.3 ms but the test allows 200 ms).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.2.9 E-UTRAN TDD - GSM handover: unknown target cell

### 5.2.9.1 Test purpose

To verify the UE's ability to transfer a connection between the UE and E-UTRAN TDD to GSM in RRC_CONNECTED state by meeting the UE RRC procedure delay and interruption time requirements when the target cell is unknown.

### 5.2.9.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward that support GSM and inter-RAT PS handover to GERAN. Applicability requires support for FGI bits 9, and 23.

### 5.2.9.3 Minimum conformance requirements

The handover delay given in table 5.2.9.3-1 and interruption time given in table 5.2.9.3-2 requirements for the case where the UE has not synchronised to the GSM cell before receiving the RRC MOBILITY FROM E-UTRA command are valid when the signal quality of the GSM cell is sufficient for successful synchronisation with one attempt. If the UE is unable to synchronise to the GSM cell on the first attempt, it shall continue to search for synchronisation information for up to 800 ms duration. If after 800 ms the UE has not synchronised to the GSM cell it shall follow the handover failure procedure specified in TS 36.331 [5].

When the UE receives a RRC MOBILITY FROM E-UTRA command the UE shall be ready to transmit (as specified in 3GPP TS 45.010 [13]) on the channel of the new RAT within the value defined in TS 36.133 [4] clause 5.3.3.2.1 and shown in table 5.2.9.3-1 from the end of the last TTI containing the RRC command.

The UE shall process the RRC procedures for the MOBILITY FROM E-UTRA command within 50 ms, which is noted as RRC procedure delay.

**Table 5.2.9.3-1: E-UTRAN/GSM handover - handover delay**

| UE synchronisation status | handover delay [ms] |
|---|---|
| The UE has synchronised to the GSM cell before the RRC MOBILITY FROM E-UTRA COMMAND is received | 90 |
| The UE has not synchronised to the GSM cell before RRC the MOBILITY FROM E-UTRA COMMAND is received | 190 |

The interruption time is the time between the end of the TTI containing the RRC command on the E-UTRAN PDSCH and the time the UE starts transmission on the uplink channel in GSM, excluding the RRC procedure delay. The interruption time depends on whether the UE has synchronized to the target GSM cell or not and shall be less than the value defined in TS 36.133 [4] clause 5.3.3.2.2 and shown in table 5.2.9.3-2.

**Table 5.2.9.3-2: E-UTRAN/GSM handover - interruption time**

| Synchronisation status | Interruption time [ms] |
|---|---|
| The UE has synchronised to the GSM cell before the RRC MOBILITY FROM E-UTRA COMMAND is received | 40 |
| The UE has not synchronised to the GSM cell before the RRC MOBILITY FROM E-UTRA COMMAND is received | 140 |

The normative reference for this requirement is TS 36.133 [4] clause 5.3.3 and A.5.2.9.

## 5.2.9.4 Test description

### 5.2.9.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.14.
2. The general test parameter settings are set up according to Table 5.2.9.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 5.2.9.4.3.
5. There is one E-UTRA TDD serving cell and one GSM cell specified in the test. Cell 1 (E-UTRA TDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.2.9.4.1-1: General Test Parameters for E-UTRAN TDD to GSM handover test case; unknown target cell**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Gap Pattern Id | | | None | No measurement gaps shall be provided. |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbour cell | | Cell 2 | |
| Final conditions | Active cell | | Cell 2 | |
| DRX | | | OFF | No DRX configured |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in 3GPP TS 36.211 [8] |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in 3GPP TS 36.211 [8] |
| T1 | | s | 7 | |
| T2 | | s | 1 | |

### 5.2.9.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of two successive time periods, with time duration of T1, T2 respectively. At the start of time duration T1, the UE will not have any timing information of cell 2. Starting T2, cell 2 becomes detectable and the UE receives a RRC handover command from the network. The start of T2 is the instant when the last TTI containing the RRC message implying handover is sent to the UE. No Gap pattern configuration shall be used.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table's 5.2.9.5-1 and 5.2.9.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. The neighbour cell shall broadcast its own BSIC, SS shall transmit a MobilityFromEUTRACommand message implying handover to Cell 2.
4. The start of T2 is the instant when the last TTI containing the RRC MobilityFromEUTRACommand message implying handover is sent to the UE, At that instant the SS shall switch the power settings from T1 to T2 as specified in Table 5.2.9.5-1 and 5.2.9.5-2. T2 starts.
5. If the UE sends access bursts on the new DCCH to Cell 2 less than 200 ms from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
6. After T2 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
7. The SS shall set a different BSIC on Cell 2, as the previous timing information of Cell 2 is invalid in the UE for the next iteration of the test procedure loop.
8. Repeat step 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 5.2.9.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 5.2.9.4.3-1: Common Exception messages for E-UTRAN TDD - GSM handover: unknown target cell test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.3-2<br>Table H.3.3-3 |

**Table 5.2.9.4.3-2: *SystemInformationBlockType7*: Additional E-UTRAN TDD - GSM handover: unknown target cell test requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-3 SystemInformationBlockType7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| commonInfo SEQUENCE { | | | |
| p-MaxGERAN | 33 (33 dBm) | | GSM 400 & GSM 900 & GSM 850 & GSM 700 |
| | 30 (30 dBm) | | DCS 1800 & PCS 1900 |
| } | | | |

## 5.2.9.5 Test requirement

Tables 5.2.9.4.1-1, 5.2.9.5-1 and 5.2.9.5-2 defines the primary level settings including test tolerances for E-UTRAN TDD to GSM handover test case when the target cell is unknown.

**Table 5.2.9.5-1: Cell Specific Test requirement Parameters for Cell 1 E-UTRAN TDD GSM handover: unknown target cell test**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2(OP.2 TDD) | | OP.1 TDD | OP.2 TDD |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note1 | dB | | |
| OCNG_RB Note1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | |

| | | |
|---|---|---|
| RSRP Note 3 | dBm/15 kHz | -94 |
| Propagation Condition | | AWGN |
| Note 1: OCNG shall be used such that cell 1 is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.<br>Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | |

**Table 5.2.9.5-2: Cell Specific Test requirement Parameters for Cell 2 E-UTRAN TDD – GSM handover: unknown target cell test**

| Parameter | Unit | Cell 2 (GSM) | |
|---|---|---|---|
| | | T1 | T2 |
| Absolute RF Channel Number | | ARFCN 1 | |
| RXLEV | dBm | -Infinity | -75 |

The handover delay is defined as the time from the beginning of time period T2, to the moment the UE start to send access bursts on the new DCCH to Cell 2.

The handover delay $T_{Handover\ delay}$ test requirement in this case is expressed as:

Handover delay $T_{Handover\ delay}$ = handover delay + $T_{offset}$ + $T_{UL}$

Handover delay = 190 ms; this is based on handover delay value as defined in Table 5.2.9.3.-1

$T_{offset}$ = 4.65 ms; GSM timing uncertainty between the time from when the UE is ready to transmit until the start of the next timeslot in GSM 26 multiframe structure

$T_{UL}$ = 4.65 ms; the time the UE has to wait in case the next timeslot is an idle frame or a SACCH frame

The handover delay $T_{Handover\ delay}$ shall be less than a total of 200 ms in this test case (note: this gives a total of 199.3 ms but the test allows 200 ms).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.2.10 E-UTRAN TDD - UTRAN TDD handover: unknown target cell

### 5.2.10.1 Test purpose

To verify the UE's ability to transfer a connection between the UE and E-UTRAN TDD to UTRAN TDD in RRC_CONNECTED state by meeting the UE RRC procedure delay and interruption time requirements when the target cell is unknown.

### 5.2.10.2 Test applicability

This test applies to all types of release 8 and forward E-UTRA TDD UEs that support release 9 and forward UTRA TDD. Applicability requires support for FGI bits 8, and 22.

### 5.2.10.3 Minimum conformance requirements

When the UE receives a RRC message implying E-UTRAN/UTRAN TDD handover the UE shall be ready to start the transmission of the new uplink DPCH or the SYNC-UL within $D_{handover}$ seconds from the end of the last TTI containing the RRC MOBILITY FROM E-UTRA command.

Where:

$D_{handover}$ equals the RRC procedure delay, which is 50 ms plus the interruption time stated in TS 36.133 [4] clause 5.3.2.2.2.

The interruption time is the time between the end of the last TTI containing the RRC command on the E-UTRAN PDSCH and the time the UE starts transmission on the uplink DPCH or the SYNC-UL in UTRAN TDD, excluding the RRC procedure delay. The interruption time depends on whether the target cell is known to the UE or not.

The UE shall always perform a UTRA synchronisation procedure as part of the handover procedure.

If the target cell has been measured by the UE during the last 5 seconds, the interruption time shall be less than $T_{interrupt1}$

$$T_{interrupt1} = T_{offset} + T_{UL} + 30 * F_{SFN} + 20 + 10 * F_{max} \text{ ms}$$

If the target cell has not been measured by the UE during the last 5 seconds, the interruption time shall be less than $T_{interrupt2}$

$$T_{interrupt2} = T_{offset} + T_{UL} + 30 * F_{SFN} + 180 + 10 * F_{max} \text{ ms}$$

Where:

$T_{offset}$ Equal to 10 ms, the frame timing uncertainty between the old cell and the target cell and the time that can elapse until the appearance of a Beacon channel

$T_{UL}$ Equal to 10 ms, the time that can elapse until the appearance of the UL timeslot in the target cell

$F_{SFN}$ Equal to 1 if SFN decoding is required and equal to 0 otherwise

$F_{max}$ denotes the maximum number of radio frames within the transmission time intervals of all transport channels that are multiplexed into the same CCTrCH.

The interruption time requirements for an unknown target cell shall apply only if the signal quality of the unknown target cell is sufficient for successful synchronisation with one attempt.

The normative reference for this requirement is TS 36.133 [4] clause 5.3.2 and A.5.2.10.

## 5.2.10.4 Test description

### 5.2.10.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.22.
2. The general test parameter settings are set up according to Table 5.2.10.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are defined in clause 5.2.10.4.3.
5. There is one E-UTRA TDD serving cell and one UTRA TDD cell specified in the test. Cell 1 (E-UTRA TDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.2.10.4.1-1: General test parameters for E-UTRA TDD to unknown UTRA (1.28 Mcps TDD OPTION) handover test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Initial conditions | Active cell | | Cell 1 | E-UTRAN TDD cell |
| | Neighbour cell | | Cell 2 | UTRA 1.28Mcps TDD cell |
| Final conditions | Active cell | | Cell 2 | UTRA 1.28Mcps TDD cell |
| CP length of cell 1 | | | Normal | |
| Uplink-downlink configuration of cell 1 | | | 1 | As specified in table 4.2.2 in TS 36.211 [9] |
| Special subframe configuration of cell 1 | | | 6 | As specified in table 4.2.1 in TS 36.211 [9] |
| Time offset between cells | | | 3 ms | Asynchronous cells<br>3ms or 92160*Ts |
| Access Barring Information | | | Not Sent | No additional delays in random access procedure. |
| Assigned Sub-Channel Number | | | 1 | No additional delays in random access procedure due to ASC. |
| TimeToTrigger | | dB | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | OFF | |
| T1 | | s | 5 | During T1, cell 2 shall be powered off, and during the off time the physical layer cell identity shall be changed. |
| T2 | | s | 1 | |

### 5.2.10.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, a RRC message implying handover to UTRA 1.28Mcps TDD cell shall be sent to the UE including activation time "now". The end of the last TTI containing handover message is the beginning of T2 duration.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

2. Set the parameters according to T1 in Tables 5.2.10.5-1 and 5.2.10.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. The SS shall transmit a MobilityFromEUTRACommand message implying handover to Cell 2.

4. The start of T2 is the instant when the last TTI containing the RRC MobilityFromEUTRACommand message implying handover is sent to the UE, At that instant the SS shall switch the power settings from T1 to T2 as specified in Tables 5.2.10.5-1 and 5.2.10.5-2. T2 starts.

5. If the UE transmits the UL to Cell 2 less than 280ms from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

6. After T2 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

7. The SS shall set Cell 2 cell parameter id =(current cell 2 cell parameter id +4) mod 16 for next iteration of the test procedure loop.

8. Repeat step 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 5.2.10.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 5.2.10.4.3-1: Common Exception messages for E-UTRA TDD to unknown UTRA TDD cell handover**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.3-1<br>Table H.3.3-3 |

**Table 5.2.10.4.3-2: HANDOVER TO UTRAN COMMAND**

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| CHOICE channel requirement | Uplink DPCH info | | |
| - CHOICE mode | TDD | | |
| - Uplink Timing Advance Control | | | |
| - CHOICE Timing Advance | enabled | | |
| - CHOICE TDD option | 1.28 Mcps | | |
| - Uplink synchronization parameters | | | |
| - Uplink synchronization step size | 1 | | |
| - Uplink synchronization frequency | 1 | | |
| - Synchronization parameters | | | |
| - SYNC_UL codes bitmap | "11111111" | | |
| - FPACH info | | | |
| - Timeslot number | 0 | | |
| - Channelisation code | (16/15) | | |
| - Midamble Shift and burst type | | | |
| - CHOICE TDD option | 1.28 Mcps TDD /REL-4/ | | |
| - Midamble Allocation Mode | Default midamble | | |
| - Midamble configuration | 4 (k=8) | | |
| - Midamble Shift | Not present | | |
| - WT | 4 | | |
| | | | |
| - $PRX_{UpPCHdes}$ | 15(-105dBm) | | |
| - SYNC_UL procedure | Not present | | |
| - Activation time | Not Present | | |
| Downlink information common for all radio links | | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | Initialize | | |
| - Default DPCH Offset Value | Arbitrary set to value 0..306688 by step of 512 | | |
| -Downlink information for each radio links | | | |
| - CHOICE mode | TDD | | |
| - Primary CCPCH info | | | |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | TDD | | |
| - Activation time | Not Present | | |

## 5.2.10.5 Test requirement

Tables 5.2.10.4.1-1, 5.2.10.5-1 and 5.2.10.5-2 define the primary level settings including test tolerances for E-UTRAN TDD to unknown UTRAN TDD cell handover test.

**Table 5.2.10.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD to unknown UTRAN TDD cell handover test case (cell1)**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| BWchannel | MHz | 10 | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2(OP.2 TDD) | | OP.1 TDD | OP.2 TDD |
| PBCH_RA | dB | 0 | 0 |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA (Note 1) | dB | | |
| OCNG_RB (Note 1) | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 3 | 3 |
| $\hat{E}_s / N_{oc}$ | dB | 3 | 3 |
| $N_{oc}$ | dBm/15kHz | -98 | |
| RSRP | dBm/15kHz | -95 | -95 |
| SCH_RP | dBm/15 kHz | -95 | -95 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 5.2.10.5-2: Cell Specific Test requirement Parameters for Cell 2 UTRAN TDD cell**

| Parameter | Unit | Cell 2 (UTRA) | | | |
|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPTS | |
| | | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number[Note1] | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | |
| DwPCH_Ec/Ior | dB | | | 0 | |
| OCNS_Ec/Ior | dB | -3 | | | |
| $\hat{I}_{or} / I_{oc}$ | dB | -infinity | 13 | -infinity | 13 |
| $I_{oc}$ | dBm/1.28 MHz | -80 | | | |
| PCCPCH RSCP | dBm | -infinity | -70 | n.a. | |
| Propagation Condition | | AWGN | | | |

Note1: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number.

Note2: P-CCPCH RSCP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The handover delay is defined as the time from the beginning of time period T2, to the moment the UE start to transmit SYNCH-UL sequence in the UpPTS to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

$D_{handover}$ = maximum RRC procedure delay + $T_{interrupt2}$

$T_{interrupt2}$= $T_{offset}$+$T_{UL}$+30*$F_{SFN}$+180 +10*$F_{max}$ ms

$T_{offset}$ = 10 ms; The frame timing uncertainty between the old cell and the target cell and the time that can elapse until the appearance of a Beacon channel

$T_{UL}$ = 10 ms; The time that can elapse until the appearance of the UL timeslot in the target cell

$F_{SFN}$ = 1; Equal to 1 if SFN decoding is required and equal to 0 otherwise.

$F_{max}$ = 0; The maximum number of radio frames within the transmission time intervals of all transport channels that are multiplexed into the same CCTrCH.

Maximum RRC procedure delay = 50 ms as defined in TS 36.331 [5] clause 11.2.

The handover delay $D_{handover}$ shall be less than a total of 280 ms in this test case.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.2.11 E-UTRAN FDD - UTRAN FDD handover for 5MHz Bandwidth

### 5.2.11.1 Test purpose

Same test purpose as in clause 5.2.1.1.

### 5.2.11.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support UTRA FDD and only Band 31. Applicability requires support for FGI bits 8, and 22.

### 5.2.11.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 5.2.1.3 with the following exceptions:

- Instead of A.5.2.1 → use A.5.2.11.

### 5.2.11.4 Test description

#### 5.2.11.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.22.
2. The general test parameter settings are set up according to Table 5.2.11.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are defined in clause 5.2.11.4.3.
5. There is one E-UTRA FDD serving cell and one UTRA FDD cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.2.11.4.1-1: General Test Parameters for E-UTRAN FDD - UTRAN FDD handover test case for 5MHz bandwidth**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.5 FDD | As specified in clause A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.11 FDD | As specified in clause A.2.1 |
| E-UTRA Channel Bandwidth (BWchannel) | MHz | 5 | |
| Note 1: See Table 5.2.1.4.1-1 for other general test parameters.<br>Note 2: This test is according to the principle defined in TS 36.133 [4] section A.3.7.2. | | | |

### 5.2.11.4.2 Test procedure

Same test procedure as in clause 5.2.1.4.2 with the following exceptions:

- Instead of Table 5.2.1.5-1 → use Table 5.2.11.5-1.

### 5.2.11.4.3 Message contents

Same message contents as in clause 5.2.1.4.3.

### 5.2.11.5 Test requirement

Same test requirement as in clause 5.2.1.5 with the following exceptions:

Tables 5.2.11.4.1-1, 5.2.11.5-1 and 5.2.1.5-2 define the primary level settings including test tolerances for E-UTRAN FDD - UTRAN FDD handover test.

**Table 5.2.11.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD to UTRAN FDD handover test case (Cell 1)**

| Parameter | Unit | Cell 1 (E-UTRA) | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| $BW_{channel}$ | MHz | 10 | | |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and in D.1.16 (OP.16 FDD) | | OP.15 FDD | OP.15 FDD | OP.16 **FDD** |
| Io Note 2 | dBm/4.5 MHz | -70.73 | -70.73 | -70.73 |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 3: See Table 5.2.1.5-1 for other cell specific test parameters. | | | | |

# 5.3 Handover from E-UTRAN to non-3GPP RATs

## 5.3.1 E-UTRAN FDD - HRPD handover

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined: Protocol procedure for the test state is under discussion.

- The Test system uncertainties applicable above 3GHz are undefined
- The Test Tolerances and Test Requirements applicable above 3GHz are undefined

### 5.3.1.1 Test purpose

To verify the UE's ability to transfer a connection between the UE and E-UTRAN to HRPD in RRC_CONNECTED state by meeting the UE RRC procedure delay and interruption time requirements.

### 5.3.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support HRPD. Applicability requires support for FGI bits 12, and 26.

### 5.3.1.3 Minimum conformance requirements

The handover delay $D_{handover}$ shall be less than maximum RRC procedure delay (which is 50 ms) + $T_{interrupt}$ in RRC_CONNECTED state.

When the UE receives a RRC message implying handover to HRPD, the UE shall be ready to start the transmission of the new reverse control channel in HRPD within $D_{handover}$ from the end of the last E-UTRAN TTI containing the RRC command.

The interruption time is the time between the end of the last TTI containing the RRC command on the E-UTRAN PDSCH and the time the UE starts transmission of the reverse control channel in HRPD, excluding the RRC procedure delay. The interruption time depends on whether the target cell is known to the UE or not.

An HRPD cell is known if it has been measured by the UE during the last 5 seconds otherwise it is unknown. Under the reference conditions specified in sub-clause 6.6 of [17], the interruption time shall be less than $T_{interrupt}$

$$T_{interrupt} = T_{IU} + 40 + 10*KC*SW_K + 10*OC*SW_O \text{ ms}$$

Where:

$T_{IU}$ It is the interruption uncertainty when changing the timing from the E-UTRAN to the new HRPD cell. $T_{IU}$ can be up to one HRPD frame (26.66 ms).

$SW_K$ is $SW_K = \left\lceil \frac{srch_win_k}{60} \right\rceil$ where srch_win_k is the number of HRPD chips indicated by the search window for known target HRPD cells in the message

$SW_O$ is $SW_O = \left\lceil \frac{srch_win_o}{60} \right\rceil$ where srch_win_o is the number of HRPD chips indicated by the search window for unknown target HRPD cells in the message

KC It is the number of known target HRPD cells in the message, and

OC It is the number of unknown target HRPD cells in the message.

Note: An additional delay in the interruption time may occur due to the reverse link silence interval [18], which is specific to HRPD.

The normative reference for this requirement is TS 36.133 [4] clause 5.4.1 and A.5.3.1.

### 5.3.1.4 Test description

#### 5.3.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.22.

2. The general test parameter settings are set up according to Table 5.3.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 5.3.1.4.3.

5. There is one E-UTRA FDD serving cell and one HRPD cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.3.1.4.1-1: General test parameters for E-UTRAN FDD to HRPD handover test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Initial conditions | Active cell | | Cell 1 | E-UTRAN FDD cell |
| | Neighbouring cell | | Cell 2 | HRPD cell |
| Final condition | Active cell | | Cell 2 | HRPD cell |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| Gap Pattern Id | | | 0 | As specified in TS 36.133 [4] Table 8.1.2.1-1 started before T2 starts |
| E-UTRAN FDD measurement quantity | | | RSRP | |
| Inter-RAT (HRPD) measurement quantity | | | CDMA2000 HRPD Pilot Strength | |
| b2-Threshold1 | | dBm | -90 | Absolute E-UTRAN RSRP threshold for event B2 |
| b2-Threshold2-CDMA2000 | | dB | -7 | Absolute 'CDMA2000 HRPD Pilot Strength' threshold for event B2 |
| Hysteresis | | dB | 0 | |
| TimeToTrigger | | dB | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | OFF | Non-DRX test |
| Access Barring Information | | - | Not sent | No additional delays in random access procedure |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 10 | |
| HRPD RF Channel Number | | | 1 | One HRPD carrier frequency is used. |
| HRPD neighbour cell list size | | | 8 | HRPD cells on HRPD RF channel 1 provided in the cell list before T2. |
| cdma2000-SearchWindowSize | | | 8 (60 PN chips) | Search window size as defined in section 6.3.5 in 3GPP TS 36.331 [5] |
| T1 | | s | 5 | |
| T2 | | s | ≤10 | |
| T3 | | s | 1 | |

### 5.3.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and send a measurement report. The RRC message implying handover shall be sent to the UE during T2, after the UE has reported Event B2. T3 is defined as the end of the last E-UTRAN TTI containing the RRC message implying handover.

1. Ensure the UE is in [State 3B-RF] according to TS 36.508 [7] clause [7.2A.4]. Cell 1 is the active cell.

2. Set the parameters according to T1 in Table's 5.3.1.5-1 and 5.3.1.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. The neighbour cell shall broadcast its own PN offset and the measurement cell list of Cell 1 shall contain the PN offset of Cell 2. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 5.3.1.5-1 and 5.3.1.5-2.

6. UE shall transmit a MeasurementReport message triggered by Event B2.

7. The SS transmits a *HandoverFromEUTRAPreparationRequest* on Cell 1.

8. The UE transmit tunnelled HRPD *Connection Request* and *Route Update* messages contained in an *ULHandoverPreparationTransfer* message on Cell 1.

9. SS shall transmit a MobilityFromEUTRACommand message implying handover to Cell 2. The tunnelled HRPD *Traffic Channel Assignment, HRPD Silence Parameters* and *HRPD Open Loop Parameters* messages are contained in *MobilityFromEUTRACommand.*

10. The start of T3 is the instant when the last TTI containing the RRC message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T2 to T3 as specified in Tables 5.3.1.5-1 and 5.3.1.5-2.

11. If the UE starts to transmit the Reverse Control Channel to Cell 2 in less than 127 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure testes is increased by one.

12. After T3 expires, the UE shall be switched off. Then ensure the UE is in State 3B according to TS 36.508 [7] clause 4.5.3B. Cell 1 is the active cell.

13. The SS shall set a different PN Offset on Cell 2 so that the previous timing information of Cell 2 is invalid in the UE for the next iteration of the test procedure loop.

14. Repeat steps 2-13 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 5.3.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 and 4.4.7.1 with the following exceptions.

**Table 5.3.1.4.3-1: Common Exception messages for E-UTRAN FDD - HRPD handover**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.3.1-8<br>Table H.3.3-3<br>Table H.3.3-4 |

**Table 5.3.1.4.2-2: MeasObjectCDMA2000: Additional E-UTRAN FDD - HRPD handover**

| Derivation Path: 36.508, Table 4.6.6-1C | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectCDMA2000-GENERIC ::= SEQUENCE { | | | |
| cdma2000-Type | TypeHRPD | | |
| carrierFreq SEQUENCE { | | | |
| bandClass | Band Class of frequency under test | | |
| Arfcn | f14 | | |
| } | | | |
| searchWindowSize | 8 | | |
| offsetFreq | db0 | | |
| cellsToRemoveList | Not present | | |
| cellsToAddModList CHOICE { | | | |
| cellsToAddModListCDMA2000 ::= SEQUENCE (SIZE (1.. maxCellMeas)) OF SEQUENCE { | | | |
| cellIndex [1] | 1 | | |
| physCellId [1] | PN offset of Cell 2 | | |
| cellIndex [2] | 2 | | |
| physCellId [2] | PN offset of Cell 2 + 4 | | |
| cellIndex [3] | 3 | | |
| physCellId [3] | PN offset of Cell 2 + 8 | | |
| cellIndex [4] | 4 | | |
| physCellId [4] | PN offset of Cell 2 + 12 | | |
| cellIndex [5] | 5 | | |
| physCellId [5] | PN offset of Cell 2 + 16 | | |
| cellIndex [6] | 6 | | |
| physCellId [6] | PN offset of Cell 2 + 20 | | |
| cellIndex [7] | 7 | | |
| physCellId [7] | PN offset of Cell 2 + 24 | | |
| cellIndex [8] | 8 | | |
| physCellId [8] | PN offset of Cell 2 + 28 | | |
| } | | | |
| } | | | |
| cellForWhichToReportCGI | Not present | | |
| } | | | |

**Table 5.3.1.4.3-3: *ReportConfigInterRAT-B2-CDMA2000*: Additional E-UTRAN FDD - HRPD handover**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7C ReportConfigInterRAT-B2-CDMA2000(EUTRA-Thres, CDMA2000-Thres) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B2-CDMA2000(EUTRA-Thres, CDMA2000-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB2 SEQUENCE { | | | |
| b2-Threshold1 CHOICE { | | | |
| threshold-RSRP | 50(-90 dBm) | -90 dBm EUTRA-Thres is actual threshold value in dBm ( 50 - 140 dBm) | |
| } | | | |
| b2-Threshold2 CHOICE { | | | |
| b2-Threshold2CDMA2000 | 14 (-7 dB) | Integer (0..63) | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| maxReportCells | 1 | | |
| reportInterval | ms2048 | | |
| reportAmount | r1 | | |
| } | | | |

**Table 5.3.1.4.3-4: *MeasuredResults*: Additional E-UTRAN FDD - HRPD handover**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasuredResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultsServing | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| neighbouringMeasResults CHOICE { | | | |
| measResultsCDMA2000 | MeasResultsCDMA2000 | | |
| } | | | |
| } | | | |

**Table 5.3.1.4.3-5: *MeasResultListCDMA2000*: Additional E-UTRAN FDD - HRPD handover**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsCDMA2000 ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity of Cell 2 | | |
| cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| pilotPnPhase | | Set according to specific test | |
| pilotStrength | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 5.3.1.4.3-6: *HandoverFromEUTRAPreparationRequest***

| Derivation Path: 36.508 Table 4.6.1-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| HandoverFromEUTRAPreparationRequest ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| handoverFromEUTRAPreparationRequest-r8 SEQUENCE { | | | |
| cdma2000-Type | typeHRPD | | |
| Rand | Not present | | |
| mobilityParameters | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.1.4.3-7: *ULHandoverPreparationTransfer***

| Derivation Path: 36.508 Table 4.6.1-24 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULHandoverPreparationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| ulHandoverPreparationTransfer-r8 SEQUENCE { | | | |
| cdma2000-Type | typeHRPD | | |
| Meid | Not present | | |
| dedicatedInfo | Set according to Table 8.4.5.4.3.3-7 | HRPD Connection Request and Route Update | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.1.4.3-8: *HRPD Connection Request***

| **Information Element** | **Value/remark** | **Comment** | **Condition** |
|---|---|---|---|
| SAPState | '0'B | SAP Header | |

| | | | |
|---|---|---|---|
| SessionConfigurationToken | 16 bits, Set by UE | | |
| ConnectionLayerFormat | 1 bit, Set by UE | | |
| ATI Record | 34 bits, Set based on UATI assigned to UE | | |
| Reserved | '0000'B | | |
| MessageID | '0000001' | Connection Request | |
| TransactionID | Any allowed value | 8 bit field | |
| RequestReason | '0000' | Access Terminal Initiated | |

**Table 5.3.1.4.3-9: *HRPD Route Update***

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| SAPState | '0'B | SAP Header | |
| SessionConfigurationToken | 16 bits, Set by UE | | |
| ConnectionLayerFormat | 1 bit, Set by UE | | |
| ATI Record | 34 bits, Set based on UATI assigned to UE | | |
| Reserved | '0000'B | | |
| MessageID | '0000000'B | Route Update | |
| MessageSequence | 8 bits, Set by UE | | |
| ReferencePilotPN | 9 bits, Set by UE | | |
| ReferencePilotStrength | 6 bits, Set by UE | | |
| ReferenceKeep | '1'B | | |
| NumPilots | '0000'B | | |
| CompatibleReserved | '0'B | | |
| ReferencePilotChannelIncluded | '1'B | | |
| ReferencePilotChannel | 24 bits, Set by UE | | |
| ReferencePilotArrivalIncluded | '1'B | | |
| ReferencePilotArrival | 15 bits, Set by UE | | |
| Reserved | 0-7 bits, Set all 0s by UE | | |

**Table 5.3.1.4.3-10: *HRPD Traffic Channel Assignment***

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| SAPState | '1'B | SAP Header | |
| SessionConfigurationToken | '0'B | | |
| ConnectionLayerFormat | 1 bit, Set by SS | | |
| ATI Record | 34 bits, Set based on UATI assigned to UE | | |
| Reserved | '0000'B | | |
| | | | |
| MessageID | '00000001'B | Traffic Channel Assignment | |
| MessageSequence | Set by SS | 8 bit field | |
| ChannelIncluded | '1'B | Channel record included | |
| Channel | '00000000000000000111010'B | channel record for Cell 15 | |
| FrameOffset | '1010'B | frame offset for Cell 15 | |
| DRCLength | '01'B | DRCLength for Cell 15 | |
| DRCChannelGainBase | '111101'B | ratio of the power level of the DRC Channel (when it is transmitted) to the power level of the Reverse Traffic Pilot Channel expressed as 2's complement value in units of 0.5 dB | |
| ACKChannelGain | '000110'B | ratio of the power level of the Ack Channel (when it is transmitted) to the power level of the Reverse Traffic Pilot Channel expressed as 2's complement value in units of 0.5 dB | |
| NumPilots | '1'B | | |
| PilotPN | '000110010'B | PN Offset of target sector (Cell 15) | |
| SofterHandoff | '0'B | Set to '0'since only 1 pilot included in message | |
| MACIndexLSBs | Set by SS | 6 least significant bits of the MACIndex assigned to UE | |
| DRCCover | '001'B | index of the DRC cover associated with target sector (Cell 15) | |
| RABLength | '01'B | 2 bit field | |
| RABOffset | '010'B | 3 bit field | |

**Table 5.3.1.4.3-11: *HRPD Silence Parameters***

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| SAPState | '0'B | SAP Header | |
| SessionConfigurationToken | 16 bits, Set by UE | | |
| ConnectionLayerFormat | 1 bit, Set by SS | | |
| ATI Record | 34 bits, Set based on UATI assigned to UE | | |
| Reserved | '0000'B | | |
| MessageID | '00000010'B | | |
| ReverseLinkSilenceDuration | 2 bits, Set by SS | | |
| ReverseLinkSilencePeriod | 2 bits, Set by SS | | |
| Reserved | 0-7 bits, Set all 0s by SS | | |

**Table 5.3.1.4.3-12: *HRPD Open Loop Parameters***

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| SAPState | '0'B | SAP Header | |
| SessionConfigurationToken | 16 bits, Set by UE | | |
| ConnectionLayerFormat | 1 bit, Set by SS | | |
| ATI Record | 34 bits, Set based on UATI assigned to UE | | |
| Reserved | '0000'B | | |
| MessageID | '00000111'B | | |
| NumPilots | '0001'B | | |
| PilotPN | 9 bits, Set by SS | | |
| OpenLoopAdjust | 8 bits, Set by SS | | |
| InitialAdjust | 5 bits, Set by SS | | |
| PilotStrengthIncluded | 1 bit, Set by SS | | |
| PilotStrengthNominal | 3 bits, Set by SS | | |
| PilotStrengthCorrectionMin | 3 bits, Set by SS | | |
| PilotStrengthCorrectionMax | 3 bits, Set by SS | | |
| Reserved | 0-7 bits, Set all 0s by SS | | |

## 5.3.1.5 Test requirement

Tables 5.3.1.4.1-1, 5.3.1.5-1 and 5.3.1.5-2 define the primary level settings including test tolerances for E-UTRAN FDD - HRPD handover test.

**Table 5.3.1.5-1: Cell Specific Test requirement Parameters for Cell 1 E-UTRAN FDD cell**

| Parameter | Unit | Cell 1 (E-UTRA) | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |

| | | | | |
|---|---|---|---|---|
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 (AWGN) | | |
| RSRP Note 3 | dBm/15 KHz | -98.80 | -98.80 | -98.80 |
| $\hat{E}_s / N_{oc}$ | dB | -0.8 | -0.8 | -0.8 |
| $\hat{E}_s / I_{ot}$ | dB | -0.8 | -0.8 | -0.8 |
| Propagation Condition | | AWGN | | |
| Note 1: OCNG shall be used such that cell 1 is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | | |
| Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. | | | | |
| Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | | |

**Table 5.3.1.5-2: Cell Specific Test requirement Parameters for Cell 2 HRPD cell**

| Parameter | Unit | Cell 2 (HRPD) | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| $\frac{\text{Control } E_b}{N_t}$ (38.4 kbps) | dB | 21 | | |
| $\frac{\text{Control } E_b}{N_t}$ (76.8 kbps) | dB | 18 | | |
| $\hat{I}_{or} / I_{oc}$ | dB | -infinity | 0 | 0 |
| $I_{oc}$ | dBm/1.2288 MHz | -55 | | |
| CDMA2000 HRPD Pilot Strength | dB | -infinity | -3 | -3 |
| Propagation Condition | | AWGN | | |

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE start to transmit the reverse control channel in HRPD to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

$T_{interrupt} = T_{IU} + 40 + 10*KC*SW_K + 10*OC*SW_O$ ms

$T_{IU}$ = 26.66 ms; $T_{IU}$ can be up to one HRPD frame (26.66 ms).

$SW_K = 1$; $SW_K = \left\lceil \frac{\text{srch_win_k}}{60} \right\rceil$ where srch_win_k is the number of HRPD chips (60) indicated by the search window for known target HRPD cells in the message

KC = 1; 1 known cell; HRPD cell is identified during T2 and is therefore known before T3

OC = 0; OC is the number of unknown target HRPD cells (0).

Maximum RRC procedure delay = 50 ms as defined in TS 36.133 [4].

The handover delay $D_{handover}$ shall be less than a total of 127 ms in this test case (note: this gives a total of 50 ms for maximum RRC procedure delay plus 76.66 ms for Tinterrupt - allow 127 ms in the test).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.3.2 E-UTRAN FDD - cdma2000 1xRTT handover

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

The Mobility From EUTRA Command message parameters are undefined

InterRAT-Target and InterRAT-Message field description is FFS

The Message contents are undefined

The Test system uncertainties applicable to this test are undefined

The Test tolerances applicable to this test are undefined

### 5.3.2.1 Test purpose

To verify the UE's ability to transfer a connection between the UE and E-UTRAN to cdma2000 1xRTT in RRC_CONNECTED state by meeting the UE RRC procedure delay and interruption time requirements.

### 5.3.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support cdma2000 1xRTT. Applicability requires support for FGI bits 11, and 24.

### 5.3.2.3 Minimum conformance requirements

The handover delay $D_{handover}$ shall be less than maximum RRC procedure delay (which is 130 ms) + $T_{interrupt}$ in RRC_CONNECTED state.

When the UE receives a RRC message implying handover to cdma2000 1xRTT, the UE shall be ready to start the transmission of the new reverse control channel in cdma2000 1xRTT within $D_{handover}$ from the end of the last E-UTRAN TTI containing the RRC command.

The interruption time is the time between the end of the last TTI containing the RRC command on the E-UTRAN PDSCH and the time the UE starts transmission of the reverse control channel in cdma2000 1xRTT, excluding the RRC procedure delay. The interruption time depends on whether the target cell is known to the UE or not.

A cdma2000 1xRTT cell is known if it has been measured by the UE during the last 5 seconds otherwise it is unknown. Under the reference conditions specified in sub-clause 4.2.1 of [19], the interruption time shall be less than $T_{interrupt}$:

$$T_{interrupt} = T_{IU} + [40] + [10]*KC*SW_K + [10]*OC*SW_O \text{ ms}$$

Where:

$T_{IU}$ It is the interruption uncertainty when changing the timing from the E-UTRAN to the new cdma2000 1xRTT cell. $T_{IU}$ can be up to one cdma2000 1xRTT frame (20 ms).

$SW_K$ is $SW_K = \left\lceil \frac{srch_win_k}{60} \right\rceil$ where srch_win_k is the number of cdma2000 1xRTT chips indicated by the search window for known target cdma2000 1xRTT cells in the message

$SW_O$ is $SW_O = \left\lceil \frac{srch_win_o}{60} \right\rceil$ where srch_win_o is the number of cdma2000 1xRTT chips indicated by the search window for unknown target cdma2000 1xRTT cells in the message

KC It is the number of known target cdma2000 1xRTT cells in the message, and

OC It is the number of unknown target cdma2000 1xRTT cells in the message.

The normative reference for this requirement is TS 36.133 [4] clause 5.4.2 and A.5.3.2.

### 5.3.2.4 Test description

#### 5.3.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.22.
2. The general test parameter settings are set up according to Table 5.3.2.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are defined in clause 5.3.2.4.3.
5. There is one E-UTRA FDD serving cell and one cdma2000 1xRTT cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.3.2.4.1-1: General test parameters for E-UTRAN FDD to cdma2000 1xRTT handover test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Initial conditions | Active cell | | Cell 1 | E-UTRAN FDD cell |
| | Neighbouring cell | | Cell 2 | cdma2000 1X cell |
| Final condition | Active cell | | Cell 2 | cdma2000 1X cell |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| Gap Pattern Id | | | 0 | As specified in TS 36.133 [4] Table 8.1.2.1-1 started before T2 starts |
| E-UTRAN FDD measurement quantity | | | RSRP | |
| Inter-RAT (cdma2000 1X) measurement quantity | | | CDMA2000 1xRTT Pilot Strength | |
| b2-Threshold1 | | dBm | -90 | Absolute E-UTRAN RSRP threshold for event B2 |
| b2-Threshold2-CDMA2000 | | dB | -14 | Absolute 'CDMA2000 1xRTT Pilot Strength' threshold for event B2 |
| Hysteresis | | dB | 0 | |
| TimeToTrigger | | dB | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | OFF | Non-DRX test |
| Access Barring Information | | - | Not sent | No additional delays in random access procedure |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 10 | |
| cdma2000 1X RF Channel Number | | | 1 | One HRPD carrier frequency is used. |
| cdma2000 1X neighbour cell list size | | | 8 | cdma2000 1X cells on cdma2000 1X RF channel 1 provided in the cell list before T2. |
| cdma2000-SearchWindowSize | | | 8 (60 PN chips) | Search window size as defined in section 6.3.5 in 3GPP TS 36.331 [5] |
| T1 | | s | 5 | |
| T2 | | s | ≤10 | |
| T3 | | s | 1 | |

#### 5.3.2.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and send a measurement report. The RRC message implying handover shall be sent to the UE during T2, after the UE has reported Event B2. T3 is defined as the end of the last E-UTRAN TTI containing the RRC message implying handover.

1. Ensure the UE is in FFS according to TS 36.508 [7] clause FFS. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table's 5.3.2.5-1 and 5.3.2.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. The neighbour cell shall broadcast its own PN offset and the measurement cell list of Cell 1 shall contain the PN offset of Cell 2. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 5.3.2.5-1 and 5.3.2.5-2.
6. UE shall transmit a MeasurementReport message triggered by Event B2.
7. SS shall transmit a MobilityFromEUTRACommand message implying handover to Cell 2.
8. The start of T3 is the instant when the last TTI containing the RRC message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T2 to T3 as specified in Tables 5.3.2.5-1 and 5.3.2.5-2.
9. If the UE starts to transmit the Reverse Control Channel to Cell 2 in less than 200 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure testes is increased by one.
10. After T3 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
11. The SS shall set a different PN Offset on Cell 2 so that the previous timing information of Cell 2 is invalid in the UE for the next iteration of the test procedure loop.
12. Repeat steps 2-11 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 5.3.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions [FFS]

### 5.3.2.5 Test requirement

Tables 5.3.2.4.1-1, 5.3.2.5-1 and 5.3.2.5-2 define the primary level settings including test tolerances for E-UTRAN FDD - cdma2000 1xRTT handover test.

**Table 5.3.2.5-1: Cell Specific Test requirement Parameters for Cell 1 E-UTRAN FDD cell**

| Parameter | Unit | Cell 1 (E-UTRA) | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 (AWGN) | | |
| RSRP Note 3 | dBm/15 KHz | -98 + TT | -98 + TT | -98 + TT |
| $\hat{E}_s / N_{oc}$ | dB | 0 + TT | 0 + TT | 0 +TT |
| $\hat{E}_s / I_{ot}$ | dB | 0 + TT | 0 + TT | 0 + TT |
| Propagation Condition | | AWGN | | |

Note 1: OCNG shall be used such that cell 1 is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 5.3.2.5-2: Cell Specific Test requirement Parameters for Cell 2 cdma2000 1xRTT cell**

| Parameter | Unit | Cell 2 (cdma2000 1X) | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| $\frac{\text{Pilot } E_c}{I_{or}}$ | dB | -7 | | |
| $\frac{\text{Sync } E_c}{I_{or}}$ | dB | -16 | | |
| $\frac{\text{Paging } E_c}{I_{or}}$ (4.8 kbps) | dB | -12 | | |
| $\hat{I}_{or} / I_{oc}$ | dB | -infinity | 0 + TT | 0 + TT |
| $I_{oc}$ | dBm/1.2288 MHz | -55 | | |
| CDMA2000 1xRTT Pilot Strength | dB | -infinity | -10 + TT | -10 + TT |
| Propagation Condition | | AWGN | | |

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE start to transmit the reverse control channel in cdma2000 1xRTT to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

$T_{interrupt} = T_{IU} + [40] + [10]*KC*SW_K + [10]*OC*SW_O$ ms

$T_{IU}$ = 20 ms; $T_{IU}$ can be up to one cdma2000 1xRTT frame (20 ms).

$SW_K = 1$; $SW_K = \left\lceil \frac{srch_win_k}{60} \right\rceil$ where srch_win_k is the number of cdma2000 1xRTT chips (60) indicated by the search window for known target cdma2000 1xRTT cells in the message

KC = 1; 1 known cell; cdma2000 1xRTT cell is identified during T2 and is therefore known before T3

OC = 0; OC is the number of unknown target cdma2000 1xRTT cells (0).

Maximum RRC procedure delay = 130 ms as defined in TS 36.133 [4].

The handover delay $D_{handover}$ shall be less than a total of 200 ms in this test case (note: this gives a total of 130 ms for maximum RRC procedure delay plus 70 ms for Tinterrupt).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.3.3 E-UTRAN FDD - HRPD handover: unknown target cell

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

The MobilityFromEUTRACommand message parameters are undefined

targetRAT-Type and targetRAT-MessageContainer field descriptions are FFS

The Message contents are undefined

The Test system uncertainties applicable to this test are undefined

The Test tolerances applicable to this test are undefined

### 5.3.3.1 Test purpose

To verify the UE's ability to transfer a connection between the UE and E-UTRAN to HRPD handover: unknown target cell in RRC_CONNECTED state by meeting the handover to an unknown target cell delay requirements.

### 5.3.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support HRPD. Applicability requires support for FGI bits 12, and 26.

### 5.3.3.3 Minimum conformance requirements

The handover delay $D_{handover}$ shall be less than maximum RRC procedure delay (which is 50 ms) + $T_{interrupt}$ in RRC_CONNECTED state.

When the UE receives a RRC message implying handover to HRPD, the UE shall be ready to start the transmission of the new reverse control channel in HRPD within $D_{handover}$ from the end of the last E-UTRAN TTI containing the RRC command.

The interruption time is the time between the end of the last TTI containing the RRC command on the E-UTRAN PDSCH and the time the UE starts transmission of the reverse control channel in HRPD, excluding the RRC procedure delay. The interruption time depends on whether the target cell is known to the UE or not.

An HRPD cell is known if it has been measured by the UE during the last 5 seconds otherwise it is unknown. Under the reference conditions specified in sub-clause 6.6 of [17], the interruption time shall be less than $T_{interrupt}$

$$T_{interrupt} = T_{IU} + [40] + [10]*KC*SW_K + [10]*OC*SW_O \text{ ms}$$

Where:

$T_{IU}$ It is the interruption uncertainty when changing the timing from the E-UTRAN to the new HRPD cell. $T_{IU}$ can be up to one HRPD frame (26.66 ms).

$SW_K$ is $SW_K = \left\lceil \frac{srch_win_k}{60} \right\rceil$ where srch_win_k is the number of HRPD chips indicated by the search window for known target HRPD cells in the message

$SW_O$ is $SW_O = \left\lceil \frac{srch_win_o}{60} \right\rceil$ where srch_win_o is the number of HRPD chips indicated by the search window for unknown target HRPD cells in the message

KC It is the number of known target HRPD cells in the message, and

OC It is the number of unknown target HRPD cells in the message.

NOTE: An additional delay in the interruption time may occur due to the reverse link silence interval [18], which is specific to HRPD.

The normative reference for this requirement is TS 36.133 [4] clause 5.4.1 and A.5.3.3.

### 5.3.3.4 Test description

#### 5.3.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.22.
2. The general test parameter settings are set up according to Table 5.3.3.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are defined in clause 5.3.3.4.3.
5. There is one E-UTRA FDD serving cell and one HRPD cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.3.3.4.1-1: General test parameters for E-UTRAN FDD to HRPD handover: unknown target cell test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Initial conditions | Active cell | | Cell 1 | E-UTRAN FDD cell |
| | Neighbouring cell | | Cell 2 | HRPD cell |
| Final condition | Active cell | | Cell 2 | HRPD cell |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| DRX | | | OFF | Non-DRX test |
| Access Barring Information | | - | Not sent | No additional delays in random access procedure |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 10 | |
| HRPD RF Channel Number | | | 1 | One HRPD carrier frequency is used. |
| cdma2000-SearchWindowSize | | | 8 (60 PN chips) | Search window size as defined in section 6.3.5 in 3GPP TS 36.331 |
| T1 | | s | ≤5 | |
| T2 | | s | 1 | |

### 5.3.3.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of two successive time periods, with time durations of T1 and T2 respectively. At the start of time duration T1, the UE does not have any timing information of Cell 2. A RRC message implying handover shall be sent to the UE during period T1. The start of T2 is the instant when the last TTI containing the RRC message implying handover is sent to the UE. Starting at T2, Cell 2 becomes detectable and the UE receives a RRC handover command from the network.

1. Ensure the UE is in [State 3B-RF] according to TS 36.508 [7] clause [7.2A.4]. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table's 5.3.3.5-1 and 5.3.3.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. The SS shall transmit the message containing Information Element systemTimeInfo as defined in TS 36.331 [5] clause 6.3.1. The neighbour cell shall broadcast its own PN offset. The SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. The SS shall transmit a MobilityFromEUTRACommand message implying handover to Cell 2.
6. The start of T2 is the instant when the last TTI containing the RRC message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T1 to T2 as specified in Table's 5.3.3.5-1 and 5.3.3.5-2.
7. If the UE starts to transmit the Reverse Control Channel to Cell 2 in less than 127 ms from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure testes is increased by one.
8. After T2 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
9. The SS shall set a different PN Offset on Cell 2 so that the previous timing information of Cell 2 is invalid in the UE for the next iteration of the test procedure loop.
10. Repeat steps 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 5.3.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions [FFS]

### 5.3.3.5 Test requirement

Tables 5.3.3.4.1-1, 5.3.3.5-1 and 5.3.3.5-2 define the primary level settings including test tolerances for E-UTRAN FDD - HRPD handover: unknown target cell test.

**Table 5.3.3.5-1: Cell Specific Test requirement Parameters for Cell 1 E-UTRAN FDD to HRPD handover: unknown target cell test case**

| Parameter | Unit | Cell 1 (E-UTRAN FDD) | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | |
| RSRP Note 3 | dBm/15 kHz | -98 + TT | -98 + TT |
| $\hat{E}_s / N_{oc}$ | dB | 0 + TT | 0 + TT |
| $\hat{E}_s / I_{ot}$ | dB | 0 + TT | 0 + TT |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 5.3.3.5-2: Cell Specific Test requirement Parameters for Cell 2 E-UTRAN FDD to HRPD handover: unknown target cell test case**

| Parameter | Unit | Cell 2 (HRPD) | |
|---|---|---|---|
| | | T1 | T2 |
| $\frac{\text{Control } E_b}{N_t}$ (38.4 kbps) | dB | 21 | |
| $\frac{\text{Control } E_b}{N_t}$ (76.8 kbps) | dB | 18 | |
| $\hat{I}_{or}/I_{oc}$ | dB | -infinity | 0 + TT |
| $I_{oc}$ | dBm/1.2288 MHz | -55 | |
| CDMA2000 HRPD Pilot Strength | dB | -infinity | -3 + TT |
| Propagation Condition | | AWGN | |

The handover to an unknown target cell delay is defined as the time from the beginning of time period T2, to the moment the UE start to transmit the reverse control channel in HRPD to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

$T_{interrupt} = T_{IU} + [40] + [10]*KC*SW_K + [10]*OC*SW_O$ ms

$T_{IU}$ = 26.66 ms; $T_{IU}$ can be up to one HRPD frame (26.66 ms).

$SW_O = 1$; $SW_O = \left\lceil \frac{srch_win_o}{60} \right\rceil$ where srch_win_o is the number of HRPD chips (60) indicated by the search window for unknown target HRPD cells in the message

KC = 0; KC is the number of known target HRPD cells (0).

OC = 1; OC is the number of unknown target HRPD cells (1).

Maximum RRC procedure delay = 50 ms as defined in TS 36.133 [4].

The handover delay $D_{handover}$ shall be less than a total of 127 ms in this test case (note: this gives a total of 50 ms for maximum RRC procedure delay plus 76.66 ms for Tinterrupt - allow 127 ms in the test).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.3.4 E-UTRAN FDD - cdma2000 1xRTT handover: unknown target cell

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

The Mobility From EUTRA Command message parameters are undefined

targetRAT-Type and targetRAT-MessageContainer field descriptions are FFS

The Message contents are undefined

The Test system uncertainties applicable to this test are undefined

The Test tolerances applicable to this test are undefined

### 5.3.4.1 Test purpose

To verify the UE's ability to transfer a connection between the UE and E-UTRAN to cdma2000 1xRTT handover: unknown target cell in RRC_CONNECTED state by meeting the handover to an unknown target cell delay requirements.

### 5.3.4.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support cdma2000 1xRTT. Applicability requires support for FGI bits 11, and 24.

### 5.3.4.3 Minimum conformance requirements

The handover delay $D_{handover}$ shall be less than maximum RRC procedure delay (which is 130 ms) + $T_{interrupt}$ in RRC_CONNECTED state.

When the UE receives a RRC message implying handover to cdma2000 1xRTT, the UE shall be ready to start the transmission of the new reverse control channel in cdma2000 1xRTT within $D_{handover}$ from the end of the last E-UTRAN TTI containing the RRC command.

The interruption time is the time between the end of the last TTI containing the RRC command on the E-UTRAN PDSCH and the time the UE starts transmission of the reverse control channel in cdma2000 1xRTT, excluding the RRC procedure delay. The interruption time depends on whether the target cell is known to the UE or not.

A cdma2000 1xRTT cell is known if it has been measured by the UE during the last 5 seconds otherwise it is unknown. Under the reference conditions specified in sub-clause 4.2.1 of [19], the interruption time shall be less than $T_{interrupt}$:

$$T_{interrupt} = T_{IU} + [40] + [10]*KC*SW_K + [10]*OC*SW_O \text{ ms}$$

Where:

$T_{IU}$ It is the interruption uncertainty when changing the timing from the E-UTRAN to the new cdma2000 1xRTT cell. $T_{IU}$ can be up to one cdma2000 1xRTT frame (20 ms).

$SW_K$ is $SW_K = \left\lceil \frac{srch_win_k}{60} \right\rceil$ where srch_win_k is the number of cdma2000 1xRTT chips indicated by the search window for known target cdma2000 1xRTT cells in the message

$SW_O$ is $SW_O = \left\lceil \frac{srch_win_o}{60} \right\rceil$ where srch_win_o is the number of cdma2000 1xRTT chips indicated by the search window for unknown target cdma2000 1xRTT cells in the message

KC It is the number of known target cdma2000 1xRTT cells in the message, and

OC It is the number of unknown target cdma2000 1xRTT cells in the message.

The normative reference for this requirement is TS 36.133 [4] clause 5.4.2 and A.5.3.4.

### 5.3.4.4 Test description

#### 5.3.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.22.
2. The general test parameter settings are set up according to Table 5.3.4.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are defined in clause 5.3.4.4.3.

5. There is one E-UTRA FDD serving cell and one cdma2000 1xRTT cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.3.4.4.1-1: General test parameters for E-UTRAN FDD to cdma2000 1xRTT handover: unknown target cell test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Initial conditions | Active cell | | Cell 1 | E-UTRAN FDD cell |
| | Neighbouring cell | | Cell 2 | cdma2000 1X cell |
| Final condition | Active cell | | Cell 2 | cdma2000 1X cell |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| DRX | | | OFF | Non-DRX test |
| Access Barring Information | | - | Not sent | No additional delays in random access procedure |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 10 | |
| cdma2000 1X RF Channel Number | | | 1 | One HRPD carrier frequency is used. |
| cdma2000-SearchWindowSize | | | 8 (60 PN chips) | Search window size as defined in section 6.3.5 in 3GPP TS 36.331 |
| T1 | | s | ≤5 | |
| T2 | | s | 1 | |

### 5.3.4.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of two successive time periods, with time durations of T1 and T2 respectively. At the start of time duration T1, the UE does not have any timing information of Cell 2. A RRC message implying handover shall be sent to the UE during period T1. The start of T2 is the instant when the last TTI containing the RRC message implying handover is sent to the UE. Starting at T2, Cell 2 becomes detectable and the UE receives a RRC handover command from the network.

1. Ensure the UE is in FFS according to TS 36.508 [7] clause FFS. Cell 1 is the active cell.

2. Set the parameters according to T1 in Table's 5.3.4.5-1 and 5.3.4.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. The SS shall transmit the message containing Information Element systemTimeInfo as defined in TS 36.331 [5] clause 6.3.1. The neighbour cell shall broadcast its own PN offset. The SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. The SS shall transmit a MobilityFromEUTRACommand message implying handover to Cell 2.

6. The start of T2 is the instant when the last TTI containing the RRC message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T1 to T2 as specified in Table's 5.3.4.5-1 and 5.3.4.5-2.

7. If the UE starts to transmit the Reverse Control Channel to Cell 2 in less than 200 ms from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure testes is increased by one.

8. After T2 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

9. The SS shall set a different PN Offset on Cell 2 so that the previous timing information of Cell 2 is invalid in the UE for the next iteration of the test procedure loop.

10. Repeat steps 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 5.3.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions [FFS]

### 5.3.4.5 Test requirement

Tables 5.3.4.4.1-1, 5.3.4.5-1 and 5.3.4.5-2 define the primary level settings including test tolerances for E-UTRAN FDD - cdma2000 1xRTT handover: unknown target cell test.

**Table 5.3.4.5-1: Cell Specific Test requirement Parameters for Cell 1 E-UTRAN FDD to cdma2000 1xRTT handover: unknown target cell test case**

| Parameter | Unit | Cell 1 (E-UTRAN FDD) | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | |
| RSRP Note 3 | dBm/15 kHz | -98 + TT | -98 + TT |
| $\hat{E}_s / N_{oc}$ | dB | 0 + TT | 0 + TT |
| $\hat{E}_s / I_{ot}$ | dB | 0 + TT | 0 + TT |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 5.3.4.5-2: Cell Specific Test requirement Parameters for Cell 2 E-UTRAN FDD to cdma2000 1xRTT handover: unknown target cell test case**

| Parameter | Unit | Cell 2 (cdma2000 1X) | |
|---|---|---|---|
| | | T1 | T2 |
| $\frac{\text{Pilot } E_c}{I_{or}}$ | dB | -7 | |
| $\frac{\text{Sync } E_c}{I_{or}}$ | dB | -16 | |
| $\frac{\text{Paging } E_c}{I_{or}}$ (4.8 kbps) | dB | -12 | |
| $\hat{I}_{or}/I_{oc}$ | dB | -infinity | 0 + TT |
| $I_{oc}$ | dBm/1.2288 MHz | -55 | |
| CDMA2000 1xRTT Pilot Strength | dB | -infinity | -10 + TT |
| Propagation Condition | | AWGN | |

The handover to an unknown target cell delay is defined as the time from the beginning of time period T2, to the moment the UE start to transmit the reverse control channel in cdma2000 1xRTT to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

$T_{interrupt} = T_{IU} + [40] + [10]*KC*SW_K + [10]*OC*SW_O$ ms

$T_{IU}$ = 20 ms; $T_{IU}$ can be up to one cdma2000 1xRTT frame (20 ms).

$SW_O = 1$; $SW_O = \left\lceil \frac{\text{srch_win_o}}{60} \right\rceil$ where srch_win_o is the number of cdma2000 1xRTT chips (60) indicated by the search window for unknown target cdma2000 1xRTT cells in the message

KC = 0; KC is the number of known target cdma2000 1xRTT cells (0).

OC = 1; OC is the number of unknown target cdma2000 1xRTT cells (1).

Maximum RRC procedure delay = 130 ms as defined in TS 36.133 [4].

The handover delay $D_{handover}$ shall be less than a total of 200 ms in this test case (note: this gives a total of 130 ms for maximum RRC procedure delay plus 70 ms for Tinterrupt).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.3.5 E-UTRAN TDD - HRPD handover

Editor's note: This Test case is incomplete for frequencies above 3GHz

- The Test system uncertainties applicable above 3GHz are undefined
- The Test Tolerances and Test Requirements applicable above 3GHz are undefined

### 5.3.5.1 Test purpose

To verify the UE's ability to transfer a connection between the UE and E-UTRAN to HRPD in RRC_CONNECTED state by meeting the UE RRC procedure delay and interruption time requirements.

### 5.3.5.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward that support HRPD. Applicability requires support for FGI bits 12, and 26.

### 5.3.5.3 Minimum conformance requirements

The handover delay $D_{handover}$ shall be less than maximum RRC procedure delay (which is 50 ms) + $T_{interrupt}$ in RRC_CONNECTED state.

When the UE receives a RRC message implying handover to HRPD, the UE shall be ready to start the transmission of the new reverse control channel in HRPD within $D_{handover}$ from the end of the last E-UTRAN TTI containing the RRC command.

The interruption time is the time between the end of the last TTI containing the RRC command on the E-UTRAN PDSCH and the time the UE starts transmission of the reverse control channel in HRPD, excluding the RRC procedure delay. The interruption time depends on whether the target cell is known to the UE or not.

An HRPD cell is known if it has been measured by the UE during the last 5 seconds otherwise it is unknown. Under the reference conditions specified in sub-clause 6.6 of [17], the interruption time shall be less than $T_{interrupt}$

$$T_{interrupt} = T_{IU} + 40 + 10*KC*SW_K + 10*OC*SW_O \text{ ms}$$

Where:

$T_{IU}$ It is the interruption uncertainty when changing the timing from the E-UTRAN to the new HRPD cell. $T_{IU}$ can be up to one HRPD frame (26.66 ms).

$SW_K$ is $SW_K = \left\lceil \frac{srch_win_k}{60} \right\rceil$ where srch_win_k is the number of HRPD chips indicated by the search window for known target HRPD cells in the message

$SW_O$ is $SW_O = \left\lceil \frac{srch_win_o}{60} \right\rceil$ where srch_win_o is the number of HRPD chips indicated by the search window for unknown target HRPD cells in the message

KC It is the number of known target HRPD cells in the message, and

OC It is the number of unknown target HRPD cells in the message.

Note: An additional delay in the interruption time may occur due to the reverse link silence interval [18], which is specific to HRPD.

The normative reference for this requirement is TS 36.133 [4] clause 5.4.1 and A.5.3.5.

### 5.3.5.4 Test description

#### 5.3.5.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.22.
2. The general test parameter settings are set up according to Table 5.3.5.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 5.3.5.4.3.

5. There is one E-UTRA TDD serving cell and one HRPD cell specified in the test. Cell 1 (E-UTRA TDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.3.5.4.1-1: General test parameters for E-UTRAN TDD to HRPD handover test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Initial conditions | Active cell | | Cell 1 | E-UTRAN TDD cell |
| | Neighbouring cell | | Cell 2 | HRPD cell |
| Final condition | Active cell | | Cell 2 | HRPD cell |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| Gap Pattern Id | | | 0 | As specified in TS 36.133 [4] Table 8.1.2.1-1 started before T2 starts |
| E-UTRAN TDD measurement quantity | | | RSRP | |
| Inter-RAT (HRPD) measurement quantity | | | CDMA2000 HRPD Pilot Strength | |
| b2-Threshold1 | | dBm | -90 | Absolute E-UTRAN RSRP threshold for event B2 |
| b2-Threshold2-CDMA2000 | | dB | -7 | Absolute 'CDMA2000 HRPD Pilot Strength' threshold for event B2 |
| Hysteresis | | dB | 0 | |
| TimeToTrigger | | s | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | OFF | Non-DRX test |
| Access Barring Information | | - | Not sent | No additional delays in random access procedure |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 10 | |
| Uplink-downlink configuration of cell 1 | | | 1 | As specified in table 4.2.2 in TS 36.211 |
| Special subframe configuration of cell 1 | | | 6 | As specified in table 4.2.1 in TS 36.211 |
| HRPD RF Channel Number | | | 1 | One HRPD carrier frequency is used. |
| HRPD neighbour cell list size | | | 8 | HRPD cells on HRPD RF channel 1 provided in the cell list before T2. |
| cdma2000-SearchWindowSize | | | 8 (60 PN chips) | Search window size as defined in section 6.3.5 in 3GPP TS 36.331 |
| T1 | | s | 5 | |
| T2 | | s | ≤10 | |
| T3 | | s | 1 | |

### 5.3.5.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and send a measurement report. The RRC message implying handover shall be sent to the UE during T2, after the UE has reported Event B2. T3 is defined as the end of the last E-UTRAN TTI containing the RRC message implying handover.

1. Ensure the UE is in [State 3B-RF] according to TS 36.508 [7] clause 7.2A.4. Cell 1 is the active cell.

2. Set the parameters according to T1 in Table's 5.3.5.4.4-1 and 5.3.5.4.4-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. The neighbour cell shall broadcast its own PN offset and the measurement cell list of Cell 1 shall contain the PN offset of Cell 2. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 5.3.5.4.4-1 and 5.3.5.4.4-2.

6. The UE shall transmit a MeasurementReport message triggered by Event B2.

7. The SS transmits a *HandoverFromEUTRAPreparationRequest* on Cell 1.

8. The UE transmits tunnelled HRPD *Connection Request* and *Route Update* messages contained in an *ULHandoverPreparationTransfer* message on Cell 1.

9. SS shall transmit a MobilityFromEUTRACommand message implying handover to Cell 2. The tunnelled HRPD *Traffic Channel Assignment, HRPD Silence Parameters* and *HRPD Open Loop Parameters* messages are contained in *MobilityFromEUTRACommand.*

10. The start of T3 is the instant when the last TTI containing the RRC message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T2 to T3 as specified in Tables 5.3.5.4.4-1 and 5.3.5.4.4-2.

11. If the UE starts to transmit the Reverse Control Channel to Cell 2 in less than 127 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure testes is increased by one.

12. After T3 expires, the UE shall be switched off. Then ensure the UE is in State 3B-RF according to TS 36.508 [7] clause 7.2A.4. Cell 1 is the active cell.

13. The SS shall set a different PN Offset on Cell 2 so that the previous timing information of Cell 2 is invalid in the UE for the next iteration of the test procedure loop.

14. Repeat steps 2-13 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 5.3.5.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 and 4.4.7.1 with the following exceptions.

**Table 5.3.5.4.3-1: Common Exception messages for E-UTRAN TDD - HRPD handover**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.3.1-8 |

**Table 5.3.5.4.3-2: MeasObjectCDMA2000: Additional E-UTRAN TDD - HRPD handover**

| Derivation Path: 36.508, Table 4.6.6-1C | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectCDMA2000-GENERIC ::= SEQUENCE { | | | |
| cdma2000-Type | TypeHRPD | | |
| carrierFreq SEQUENCE { | | | |
| bandClass | Band Class of frequency under test | | |
| Arfcn | f14 | | |
| } | | | |
| searchWindowSize | 8 | | |
| offsetFreq | db0 | | |
| cellsToRemoveList | Not present | | |
| cellsToAddModList CHOICE { | | | |
| cellsToAddModListCDMA2000 ::= SEQUENCE (SIZE (1.. maxCellMeas)) OF SEQUENCE { | | | |
| cellIndex [1] | 1 | | |
| physCellId [1] | PN offset of Cell 2 | | |
| cellIndex [2] | 2 | | |
| physCellId [2] | PN offset of Cell 2 + 4 | | |
| cellIndex [3] | 3 | | |
| physCellId [3] | PN offset of Cell 2 + 8 | | |
| cellIndex [4] | 4 | | |
| physCellId [4] | PN offset of Cell 2 + 12 | | |
| cellIndex [5] | 5 | | |
| physCellId [5] | PN offset of Cell 2 + 16 | | |
| cellIndex [6] | 6 | | |
| physCellId [6] | PN offset of Cell 2 + 20 | | |
| cellIndex [7] | 7 | | |
| physCellId [7] | PN offset of Cell 2 + 24 | | |
| cellIndex [8] | 8 | | |
| physCellId [8] | PN offset of Cell 2 + 28 | | |
| } | | | |
| } | | | |
| cellForWhichToReportCGI | Not present | | |
| } | | | |

**Table 5.3.5.4.3-3: *ReportConfigInterRAT-B2-CDMA2000*: Additional E-UTRAN TDD - HRPD handover**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7C ReportConfigInterRAT-B2-CDMA2000(EUTRA-Thres, CDMA2000-Thres) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B2-CDMA2000(EUTRA-Thres, CDMA2000-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB2 SEQUENCE { | | | |
| b2-Threshold1 CHOICE { | | | |
| threshold-RSRP | 50(-90 dBm) | -90 dBm EUTRA-Thres is actual threshold value in dBm ( 50 - 140 dBm) | |
| } | | | |
| b2-Threshold2 CHOICE { | | | |
| b2-Threshold2CDMA2000 | 14 (-7 dB) | Integer (0..63) | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| maxReportCells | 1 | | |
| reportInterval | ms2048 | | |
| reportAmount | r1 | | |
| } | | | |

**Table 5.3.5.4.3-4: *MeasuredResults*: Additional E-UTRAN TDD - HRPD handover**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasuredResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultsCDMA2000 | MeasResultsCDMA2000 | | |
| } | | | |
| } | | | |

**Table 5.3.5.4.3-5: *MeasResultsCDMA2000*: Additional E-UTRAN TDD - HRPD handover**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsCDMA2000 ::= SEQUENCE { | | | |
| preRegistrationStatusHRPD | TRUE | | |
| measResultListCDMA2000 ::=SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity of Cell 2 | | |
| cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| pilotStrength | | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.5.4.3-6: *HandoverFromEUTRAPreparationRequest***

| Derivation Path: 36.508 Table 4.6.1-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| HandoverFromEUTRAPreparationRequest ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| handoverFromEUTRAPreparationRequest-r8 SEQUENCE { | | | |
| cdma2000-Type | typeHRPD | | |
| Rand | Not present | | |
| mobilityParameters | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.5.4.3-7: *ULHandoverPreparationTransfer***

| Derivation Path: 36.508 Table 4.6.1-24 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULHandoverPreparationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| ulHandoverPreparationTransfer-r8 SEQUENCE { | | | |
| cdma2000-Type | typeHRPD | | |
| Meid | Not present | | |
| dedicatedInfo | Set according to Table 5.3.5.4.3-8 | HRPD Connection Request and Route Update | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.5.4.3-8: *dedicatedInfo* in *ULHandoverPreparationTransfer* (Table 5.3.5.4.3-7)**

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| SAPState | '0'B | SAP Header | |
| SessionConfigurationToken | 16 bits, Set by UE | | |
| ConnectionLayerFormat | 1 bit, Set by UE | | |
| ATI Record | 34 bits, Set based on UATI assigned to UE | | |
| Reserved | '0000'B | | |
| Length | Length of *HRPD Route Update* message (Table 5.3.5.4.3-10) + length of StreamHeader + length of SLPHeader + length of SNPHeader below, Set by SS | Connection Layer Header | |
| StreamHeader | '00'B | Stream Layer header. Stream 0 is assigned to the Default Signalling Application | |
| SLPHeader | Set by the UE | Signalling Link Protocol SLP-D and SLP-F headers. | |
| SNPHeader | '00001110' | Signalling Network Protocol header. InConfigurationProtocol=0, Type=Route Update. | |
| SessionLayerPacket | *HRPD Route Update* message (Table 5.3.5.4.3-10) | | |
| Length | Length of *HRPD Connection Request* message (Table 5.3.5.4.3-9) + length of StreamHeader + length of SLPHeader + length of SNPHeader below, Set by SS | Connection Layer Header | |
| StreamHeader | '00'B | Stream Layer header. Stream 0 is assigned to the Default Signalling Application | |
| SLPHeader | Set by the UE | Signalling Link Protocol SLP-D and SLP-F headers. | |
| SNPHeader | '00001100' | Signalling Network Protocol header. InConfigurationProtocol=0, Type=Idle State. | |
| SessionLayerPacket | *HRPD Connection Request* message (Table 5.3.5.4.3-9) | | |

**Table 5.3.5.4.3-9: *HRPD Connection Request* (Table 5.3.5.4.3-8)**

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| MessageID | '0000001' | Connection Request | |
| TransactionID | Any allowed value | 8 bit field | |
| RequestReason | '0000' | Access Terminal Initiated | |

**Table 5.3.5.4.3-10: *HRPD Route Update* (Table 5.3.5.4.3-8)**

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| MessageID | '0000000'B | Route Update | |
| MessageSequence | 8 bits, Set by UE | | |
| ReferencePilotPN | 9 bits, Set by UE | | |
| ReferencePilotStrength | 6 bits, Set by UE | | |
| ReferenceKeep | '1'B | | |
| NumPilots | '0000'B | | |
| CompatibleReserved | '0'B | | |
| ReferencePilotChannelIncluded | '1'B | | |
| ReferencePilotChannel | 24 bits, Set by UE | | |
| ReferencePilotArrivalIncluded | '1'B | | |
| ReferencePilotArrival | 15 bits, Set by UE | | |
| Reserved | 0-7 bits, Set all 0s by UE | | |

**Table 5.3.5.4.3-11: *MobilityFromEUTRACommand* (step 9)**

| Derivation Path: 36.508, Table 4.6.1-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MobilityFromEUTRACommand ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| mobilityFromEUTRACommand-r8 SEQUENCE { | | | |
| cs-FallbackIndicator | False | | |
| purpose CHOICE{ | | | |
| handover SEQUENCE { | | | |
| targetRAT-Type | cdma2000-HRPD | | |
| targetRAT-MessageContainer | Set according to Table 5.3.5.4.3-12 | HRPD Silence Parameters and HRPD Open Loop Parameters, HRPD Traffic Channel Assignment | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.5.4.3-12: *targetRAT-MessageContainer* in *MobilityFromEUTRACommand* (Table 5.3.5.4.3-11)**

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| SAPState | '1'B | SAP Header | |
| SessionConfigurationToken | '0'B | | |
| ConnectionLayerFormat | 1 bit, Set by SS | | |
| ATI Record | 34 bits, Set based on UATI assigned to UE | | |
| Reserved | '0000'B | | |
| Length | Length of *HRPD Silence Parameters* message (Table 5.3.5.4.3-14) + length of StreamHeader + length of SLPHeader + length of SNPHeader below, Set by SS | Connection Layer Header (Note 1) | |
| StreamHeader | '00'B | Stream Layer header. Stream 0 is assigned to the Default Signalling Application | |
| SLPHeader | Set by the SS | Signalling Link Protocol SLP-D and SLP-F headers. | |
| SNPHeader | '00001111' | Signalling Network Protocol header. InConfigurationProtocol=0, Type=Overhead Messages. | |
| SessionLayerPacket | *HRPD Silence Parameters* message (Table 5.3.5.4.3-14) | (Note 1) | |
| Length | Length of *HRPD Open Loop Parameters* message (Table 5.3.5.4.3-15) + length of StreamHeader + length of SLPHeader + length of SNPHeader below, Set by SS | Connection Layer Header (Note 1) | |
| StreamHeader | '00'B | Stream Layer header. Stream 0 is assigned to the Default Signalling Application | |
| SLPHeader | Set by the SS | Signalling Link Protocol SLP-D and SLP-F headers. | |
| SNPHeader | '00000100' | Signalling Network Protocol header. InConfigurationProtocol=0, Type=Reverse Traffic Channel MAC. | |
| SessionLayerPacket | *HRPD Open Loop Parameters* message (Table 5.3.5.4.3-15) | (Note 1) | |
| Length | Length of *HRPD Traffic Channel Assignment* message (Table 5.3.5.4.3-13) + length of StreamHeader + length of SLPHeader + length of SNPHeader below, Set by SS | Connection Layer Header | |

| | | | |
|---|---|---|---|
| StreamHeader | '00'B | Stream Layer header. Stream 0 is assigned to the Default Signalling Application | |
| SLPHeader | Set by the SS | Signalling Link Protocol SLP-D and SLP-F headers. | |
| SNPHeader | '00001110' | Signalling Network Protocol header. InConfigurationProtocol=0, Type=Route Update. | |
| SessionLayerPacket | *HRPD Traffic Channel Assignment* message (Table 5.3.5.4.3-13) | | |

NOTE 1: *HRPD Silence Parameters* message and *HRPD Open Loop Parameters* message can be sent in any order.

**Table 5.3.5.4.3-13: *HRPD Traffic Channel Assignment* (Table 5.3.5.4.3-12)**

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| MessageID | '00000001'B | Traffic Channel Assignment | |
| MessageSequence | Set by SS | 8 bit field | |
| ChannelIncluded | '1'B | Channel record included | |
| Channel | '0000000000000000011 11010'B | channel record for Cell 15 | |
| FrameOffset | '1010'B | frame offset for Cell 15 | |
| DRCLength | '01'B | DRCLength for Cell 15 | |
| DRCChannelGainBase | '111101'B | ratio of the power level of the DRC Channel (when it is transmitted) to the power level of the Reverse Traffic Pilot Channel expressed as 2's complement value in units of 0.5 dB | |
| ACKChannelGain | '000110'B | ratio of the power level of the Ack Channel (when it is transmitted) to the power level of the Reverse Traffic Pilot Channel expressed as 2's complement value in units of 0.5 dB | |
| NumPilots | '1'B | | |
| PilotPN | '000110010'B | PN Offset of target sector (Cell 15) | |
| SofterHandoff | '0'B | Set to '0'since only 1 pilot included in message | |
| MACIndexLSBs | Set by SS | 6 least significant bits of the MACIndex assigned to UE | |
| DRCCover | '001'B | index of the DRC cover associated with target sector (Cell 15) | |
| RABLength | '01'B | 2 bit field | |
| RABOffset | '010'B | 3 bit field | |

**Table 5.3.5.4.3-14: *HRPD Silence Parameters* (Table 5.3.5.4.3-12)**

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| MessageID | '00000010'B | | |
| ReverseLinkSilenceDuration | 2 bits, Set by SS | | |
| ReverseLinkSilencePeriod | 2 bits, Set by SS | | |
| Reserved | 0-7 bits, Set all 0s by SS | | |

**Table 5.3.5.4.3-15: *HRPD Open Loop Parameters* (Table 5.3.5.4.3-12)**

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| MessageID | '00000111'B | | |
| NumPilots | '0001'B | | |
| PilotPN | 9 bits, Set by SS | | |
| OpenLoopAdjust | 8 bits, Set by SS | | |
| InitialAdjust | 5 bits, Set by SS | | |
| PilotStrengthIncluded | 1 bit, Set by SS | | |
| PilotStrengthNominal | 3 bits, Set by SS | | |
| PilotStrengthCorrectionMin | 3 bits, Set by SS | | |
| PilotStrengthCorrectionMax | 3 bits, Set by SS | | |
| Reserved | 0-7 bits, Set all 0s by SS | | |

## 5.3.5.5 Test requirement

Tables 5.3.5.4.1-1, 5.3.5.5-1 and 5.3.5.5-2 define the primary level settings including test tolerances for E-UTRAN TDD - HRPD handover.

**Table 5.3.5.5-1: Cell specific test parameters for E-UTRAN TDD cell#1 for handover to HRPD cell # 2**

| Parameter | Unit | Cell 1 (E-UTRA) | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | | OP.2 TDD |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | |
| RSRP Note 3 | dBm/15 KHz | -98.8 | -98.8 | -98.8 |
| $\hat{E}_s / N_{oc}$ | dB | -0.8 | -0.8 | -0.8 |
| $\hat{E}_s / I_{ot}$ | dB | -0.8 | -0.8 | -0.8 |
| Propagation Condition | | AWGN | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 5.3.5.5-2: Cell specific test parameters for HRPD (cell # 2) for handover from E-UTRAN TDD cell (cell #1)**

| Parameter | Unit | Cell 2 (HRPD) | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| $\frac{\text{Control } E_b}{N_t}$ (38.4 kbps) | dB | 21 | | |
| $\frac{\text{Control } E_b}{N_t}$ (76.8 kbps) | dB | 18 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -infinity | 0 | 0 |
| $I_{oc}$ | dBm/1.2288 MHz | -55 | | |
| CDMA2000 HRPD Pilot Strength | dB | -infinity | -3 | -3 |
| Propagation Condition | | AWGN | | |

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE start to transmit the reverse control channel in HRPD to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

$T_{interrupt} = T_{IU} + 40 + 10*KC*SW_K + 10*OC*SW_O$ ms

$T_{IU}$ = 26.66 ms; $T_{IU}$ can be up to one HRPD frame (26.66 ms).

$SW_K = 1$; $SW_K = \left\lceil \frac{srch_win_k}{60} \right\rceil$ where srch_win_k is the number of HRPD chips (60) indicated by the search window for known target HRPD cells in the message

KC = 1; 1 known cell; HRPD cell is identified during T2 and is therefore known before T3

OC = 0; OC is the number of unknown target HRPD cells (0).

Maximum RRC procedure delay = 50 ms as defined in TS 36.133 [4].

The handover delay $D_{handover}$ shall be less than a total of 127 ms in this test case (note: this gives a total of 50 ms for maximum RRC procedure delay plus 76.66 ms for Tinterrupt - allow 127 ms in the test).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 5.3.6 E-UTRAN TDD - cdma2000 1xRTT handover

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

The Test system uncertainties applicable to this test are undefined

The Test tolerances applicable to this test are undefined

### 5.3.6.1 Test purpose

To verify the UE's ability to transfer a connection between the UE and E-UTRAN to cdma2000 1xRTT in RRC_CONNECTED state by meeting the UE RRC procedure delay and interruption time requirements.

### 5.3.6.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward that support cdma2000 1xRTT. Applicability requires support for FGI bits 11 and 24.

### 5.3.6.3 Minimum conformance requirements

The handover delay $D_{handover}$ shall be less than maximum RRC procedure delay (which is 130 ms) + $T_{interrupt}$ in RRC_CONNECTED state.

When the UE receives a RRC message implying handover to cdma2000 1xRTT, the UE shall be ready to start the transmission of the new reverse control channel in cdma2000 1xRTT within $D_{handover}$ from the end of the last E-UTRAN TTI containing the RRC command.

The interruption time is the time between the end of the last TTI containing the RRC command on the E-UTRAN PDSCH and the time the UE starts transmission of the reverse control channel in cdma2000 1xRTT, excluding the RRC procedure delay. The interruption time depends on whether the target cell is known to the UE or not.

A cdma2000 1xRTT cell is known if it has been measured by the UE during the last 5 seconds otherwise it is unknown. Under the reference conditions specified in sub-clause 4.2.1 of [19], the interruption time shall be less than $T_{interrupt}$:

$$T_{interrupt} = T_{IU} + 40 + 10*KC*SW_K + 10*OC*SW_O \text{ ms}$$

Where:

$T_{IU}$ It is the interruption uncertainty when changing the timing from the E-UTRAN to the new cdma2000 1xRTT cell. $T_{IU}$ can be up to one cdma2000 1xRTT frame (20 ms).

$SW_K$ is $SW_K = \left\lceil \frac{srch_win_k}{60} \right\rceil$ where srch_win_k is the number of cdma2000 1xRTT chips indicated by the search window for known target cdma2000 1xRTT cells in the message

$SW_O$ is $SW_O = \left\lceil \frac{srch_win_o}{60} \right\rceil$ where srch_win_o is the number of cdma2000 1xRTT chips indicated by the search window for unknown target cdma2000 1xRTT cells in the message

KC It is the number of known target cdma2000 1xRTT cells in the message, and

OC It is the number of unknown target cdma2000 1xRTT cells in the message.

The normative reference for this requirement is TS 36.133 [4] clause 5.4.2 and A.5.3.6.

### 5.3.6.4 Test description

#### 5.3.6.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.22.
2. The general test parameter settings are set up according to Table 5.3.6.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are defined in clause 5.3.6.4.3.
5. There is one E-UTRA TDD serving cell and one cdma2000 1xRTT cell specified in the test. Cell 1 (E-UTRA TDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 5.3.6.4.1-1: General test parameters for E-UTRAN TDD to cdma2000 1xRTT handover test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Initial conditions | Active cell | | Cell 1 | E-UTRAN TDD cell |
| | Neighbouring cell | | Cell 2 | cdma2000 1X cell |
| Final condition | Active cell | | Cell 2 | cdma2000 1X cell |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| Gap Pattern Id | | | 0 | As specified in TS 36.133 [4]Table 8.1.2.1-1 started before T2 starts |
| E-UTRAN TDD measurement quantity | | | RSRP | |
| Inter-RAT (cdma2000 1X) measurement quantity | | | CDMA2000 1xRTT Pilot Strength | |
| b2-Threshold1 | | dBm | -90 | Absolute E-UTRAN RSRP threshold for event B2 |
| b2-Threshold2-CDMA2000 | | dB | -14 | Absolute 'CDMA2000 1xRTT Pilot Strength' threshold for event B2 |
| Hysteresis | | dB | 0 | |
| TimeToTrigger | | S | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | OFF | Non-DRX test |
| Access Barring Information | | - | Not sent | No additional delays in random access procedure |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 10 | |
| cdma2000 1X RF Channel Number | | | 1 | One cdma2000 1X carrier frequency is used. |
| cdma2000 1X neighbour cell list size | | | 8 | cdma2000 1X cells on cdma2000 1X RF channel 1 provided in the cell list before T2. |
| cdma2000-SearchWindowSize | | | 8 (60 PN chips) | Search window size as defined in section 6.3.5 in 3GPP TS 36.331 [5] |
| T1 | | S | 5 | |
| T2 | | S | ≤10 | |
| T3 | | S | 1 | |

### 5.3.6.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE may not have any timing information of Cell 2. Starting T2, Cell 2 becomes detectable and the UE is expected to detect and send a measurement report. The RRC message implying handover shall be sent to the UE during T2, after the UE has reported Event B2. T3 is defined as the end of the last E-UTRAN TTI containing the RRC message implying handover.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

2. Set the parameters according to T1 in Tables 5.3.6.5-1 and 5.3.6.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. The neighbour cell shall broadcast its own PN offset and the measurement cell list of Cell 1 shall contain the PN offset of Cell 2. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Tables 5.3.6.5-1 and 5.3.6.5-2.

6. The UE shall transmit a MeasurementReport message triggered by Event B2.

7. The SS transmits a *HandoverFromEUTRAPreparationRequest* on Cell 1.

8. The UE transmits tunnelled *1xRTT GCSNA Encapsulated Page Response* messages contained in an *ULHandoverPreparationTransfer* message on Cell 1.

9. The SS transmits a tunnelled *1xRTT GCSNA Encapsulated Handoff Direction* message contained in a *MobilityFromEUTRACommand* on Cell1 to order the UE to perform inter RAT to Cell 2.

10. The start of T3 is the instant when the last TTI containing the RRC message implying handover is sent to the UE, at that instant the SS shall switch the power settings from T2 to T3 as specified in Tables 5.3.6.5-1 and 5.3.6.5-2.

11. If the UE starts to transmit the Reverse Control Channel to Cell 2 in less than 200 ms from the beginning of time period T3 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

12. After T3 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

13. The SS shall set a different PN Offset on Cell 2 so that the previous timing information of Cell 2 is invalid in the UE for the next iteration of the test procedure loop.

14. Repeat steps 2-13 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 5.3.6.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions.

**Table 5.3.6.4.3-1: Common Exception messages for E-UTRAN TDD - cdma2000 1xRTT handover**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7 |

**Table 5.3.6.4.3-2: *MeasConfig* (Table 5.3.6.4.3-1)**

| Derivation Path: 36.508, Table 4.6.6-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig ::= SEQUENCE { | | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entries | | |
| measObjectId[1] | IdMeasObject-f17 | | |
| measObject[1] | MeasObjectCDMA2000-GENERIC | | |
| measObjectId[2] | IdMeasObject-f1 | | |
| measObject[2] | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId)) OF SEQUENCE { | 1 entry | | |
| reportConfigId[1] | IdReportConfig-B2-CDMA2000 | | |
| reportConfig[1] | ReportConfigInterRAT-B2-CDMA2000(-90, -14) | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) OF SEQUENCE { | 1 entry | | |
| measId[1] | 1 | | |
| measObjectId[1] | IdMeasObject-f17 | | |
| reportConfigId[1] | IdReportConfig-B2-CDMA2000 | | |
| } | | | |
| measGapConfig CHOICE { | | | |
| Setup SEQUENCE { | | | |
| gapOffset CHOICE { | | | |
| gp0 | 60 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.6.4.3-3: *MeasObjectCDMA2000*: Additional E-UTRAN TDD - cdma2000 1xRTT handover**

| Derivation Path: 36.508, Table 4.6.6-1C | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectCDMA2000-GENERIC ::= SEQUENCE { | | | |
| cdma2000-Type | type1XRTT | | |
| carrierFreq SEQUENCE { | | | |
| bandClass | Band Class of frequency under test | | |
| Arfcn | f14 | | |
| } | | | |
| searchWindowSize | 8 | | |
| offsetFreq | db0 | | |
| cellsToRemoveList | Not present | | |
| cellsToAddModList CHOICE { | | | |
| cellsToAddModListCDMA2000 ::= SEQUENCE (SIZE (1.. maxCellMeas)) OF SEQUENCE { | | | |
| cellIndex [1] | 1 | | |
| physCellId [1] | PN offset of Cell 2 | | |
| cellIndex [2] | 2 | | |
| physCellId [2] | PN offset of Cell 2 + 4 | | |
| cellIndex [3] | 3 | | |
| physCellId [3] | PN offset of Cell 2 + 8 | | |
| cellIndex [4] | 4 | | |
| physCellId [4] | PN offset of Cell 2 + 12 | | |
| cellIndex [5] | 5 | | |
| physCellId [5] | PN offset of Cell 2 + 16 | | |
| cellIndex [6] | 6 | | |
| physCellId [6] | PN offset of Cell 2 + 20 | | |
| cellIndex [7] | 7 | | |
| physCellId [7] | PN offset of Cell 2 + 24 | | |
| cellIndex [8] | 8 | | |
| physCellId [8] | PN offset of Cell 2 + 28 | | |
| } | | | |
| } | | | |
| cellForWhichToReportCGI | Not present | | |
| } | | | |

**Table 5.3.6.4.3-4: *ReportConfigInterRAT-B2-CDMA2000*: Additional E-UTRAN TDD - cdma2000 1xRTT handover**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7C ReportConfigInterRAT-B2-CDMA2000(EUTRA-Thres, CDMA2000-Thres) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B2-CDMA2000(EUTRA-Thres, CDMA2000-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB2 SEQUENCE { | | | |
| b2-Threshold1 CHOICE { | | | |
| threshold-RSRP | 50(-90 dBm) | -90 dBm EUTRA-Thres is actual threshold value in dBm ( 50 - 140 dBm) | |
| } | | | |
| b2-Threshold2 CHOICE { | | | |
| b2-Threshold2CDMA2000 | 28 (-14 dB) | Integer (0..63) | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| maxReportCells | 1 | | |
| reportInterval | ms2048 | | |
| reportAmount | r1 | | |
| } | | | |

**Table 5.3.6.4.3-5: *MeasuredResults*: Additional E-UTRAN TDD - cdma2000 1xRTT handover**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasuredResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultsCDMA2000 | MeasResultsCDMA2000 | | |
| } | | | |
| } | | | |

**Table 5.3.6.4.3-6: *MeasResultsCDMA2000*: Additional E-UTRAN TDD - cdma2000 1xRTT handover**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsCDMA2000 ::= SEQUENCE { | | | |
| preRegistrationStatusHRPD | FALSE | | |
| measResultListCDMA2000 ::=SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity of Cell 2 | | |
| cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| pilotStrength | | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.6.4.3-7: *HandoverFromEUTRAPreparationRequest***

| Derivation Path: 36.508 Table 4.6.1-4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| HandoverFromEUTRAPreparationRequest ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| handoverFromEUTRAPreparationRequest-r8 SEQUENCE { | | | |
| cdma2000-Type | type1XRTT | | |
| Rand | Set by SS | Random Challenge Data as broadcast on Cell 2 | |
| mobilityParameters | Set according to 36.508 Table 4.5.2C.4-6 | CDMA2000Parameters | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.6.4.3-8: *ULHandoverPreparationTransfer***

| Derivation Path: 36.508 Table 4.6.1-24 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ULHandoverPreparationTransfer ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| ulHandoverPreparationTransfer-r8 SEQUENCE { | | | |
| cdma2000-Type | type1XRTT | | |
| Meid | UE's meid | | |
| dedicatedInfo | Set according to Table 5.3.6.4.3-8 | 1xRTT GCSNA Encapsulated Page Response message | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.6.4.3-9: *1xRTT GCSNA Encapsulated Page Response* (Table 5.3.6.4.3-7)**

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| MessageID | ‘00000001’B | GCSNA1xCircuitService message | |
| GCSNAOption | ‘00000001’B | | |
| AlternativeGCSNAOption_INCL | ‘0’B | | |
| IWSIDIncl | ‘0’B | | |
| AckRequired | ‘0’B | | |
| StopDupDetect | ‘0’B | | |
| MessageSequence | 6 bits, Set by UE | | |
| NumTLACEncapsulated1xL3PDU | ‘00’B | | |
| Reserved | ‘0000’B | | |
| 1xLogicalChannel | ‘0’B | | |
| 1xProtocolRevision | ‘00000110’B | | |
| MsgType | ‘00000101’B | Page Response message | |
| NumTLACHeaderRecords | ‘0001’B | | |
| TLACHeaderRecordType | ‘0000’B | | |
| TLACHeaderRecordLength | 8 bits, Set by UE | | |
| MSID_TYPE | 3 bits, Set by UE | Should be matched with PREF_MSID_TYPE | |
| MSID_LEN | 4 bits, Set by UE | | |
| MSID | Variable, Set by UE | | |
| Reserved | ‘0000000’B | | |
| 1xL3PDULength | 16 bits, Set by UE | | |
| MOB_TERM | ‘1’B | | |
| SLOT_CYCLE_INDEX | ’010’B | | |
| MOB_P_REV | 8 bits, Set by UE | | |
| SCM | 8 bits, Set by UE | | |
| REQUEST_MODE | ’001’B | | |
| SERVICE_OPTION | 16 bits, Set by UE | | |
| PM | ’0’B | | |
| NAR_AN_CAP | ’0’B | | |
| ENCRYPTION_SUPPORTED | ’0000’B | | |
| NUM_ALT_SO | ‘000’B | | |
| UZID_INCL | ’0’B | | |
| CH_IND | ’0’B | | |
| OTD_SUPPORTED | ’0000’B | | |
| QPCH_SUPPORTED | ’0’B | | |
| ENHANCED_RC | ’0’B | | |
| FOR_RC_PREF | ’0000’B | | |
| REV_RC_PREF | ’0’B | | |
| FCH_SUPPORTED | ’0’B | | |
| FCH Capability Type-specific fields | Variable | | |
| DCCH_SUPPORTED | ’1’B | | |
| REV_FCH_GATING_REQ | ’0’B | | |

**Table 5.3.6.4.3-10: *MobilityFromEUTRACommand***

| Derivation Path: 36.508, Table 4.6.1-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MobilityFromEUTRACommand ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| mobilityFromEUTRACommand-r9 SEQUENCE { | | | |
| cs-FallbackIndicator | False | | |
| purpose CHOICE{ | | | |
| e-CSFB-r9 SEQUENCE { | | | |
| messageContCDMA2000-1XRTT-r9 | Set according to Table 5.3.6.4.3-10 | 1xRTT GCSNA Encapsulated Handoff Direction message | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.6.4.3-11: *1xRTT GCSNA Encapsulated Handoff Direction* (Table5.3.6.4.3-9)**

| Information Element | Value/remark | Comment | Condition |
|---|---|---|---|
| MessageID | ‘00000001’B | | |
| GCSNAOption | ‘00000001’B | | |
| AlternativeGCSNAOption_INCL | ‘0’B | | |
| IWSIDIncl | ‘0’B | | |
| AckRequired | ‘0’B | | |
| StopDupDetect | ‘0’B | | |
| MessageSequence | Set by SS | | |
| NumTLACEncapsulated1xL3PDU | ‘00’B | | |
| Reserved | ‘0000’B | | |
| 1xLogicalChannel | ‘1’B | | |
| 1xProtocolRevision | ‘00000110’B | | |
| MsgType | ‘00100010’B | Universal Handoff Direction message | |
| NumTLACHeaderRecords | ‘0000’B | | |
| Reserved | ‘000’B | | |
| 1xL3PDULength | 16 bits, Set by SS | | |
| USE_TIME | ’0’B | | |
| ACTION_TIME | ’000000’B | | |
| HDM_SEQ | 2 bits, Set by SS | | |
| PARMS_INCL | ’1’B | | |
| P_REV | ‘00000110’B | | |
| SERV_NEG_TYPE | ’1’B | | |
| SEARCH_INCLUDED | ’1’B | | |
| SRCH_WIN_A | ’1000’B | | |
| SRCH_WIN_N | ’1001’B | | |
| SRCH_WIN_R | ’1011’B | | |
| T_ADD | ’010100’B | | |
| T_DROP | ’011110’B | | |
| T_COMP | ’1010’B | | |
| T_TDROP | ’0100’B | | |
| SOFT_SLOPE | ’000000’B | | |
| ADD_INTERCEPT | ’000000’B | | |
| DROP_INTERCEPT | ’000000’B | | |
| EXTRA_PARMS | ’1’B | | |
| PACKET_ZONE_ID | ’00000000’B | | |
| FRAME_OFFSET | 4 bits, Set by SS | | |
| PRIVATE_LCM | ’0’B | | |
| RESET_L2 | ’1’B | | |
| RESET_FPC | ’1’B | | |
| ENCRYPT_MODE | ’00’B | | |
| NOM_PWR_EXT | ’0’B | | |
| NOM_PWR | ’0000’B | | |
| RLGAIN_TRAFFIC_PILOT | ’000000’B | | |
| DEFAULT_RLAG | ’1’B | | |
| NUM_PREAMBLE | ’000’B | | |
| BAND_CLASS | 5 bits, Set by SS | | |
| CDMA_FREQ | 11 bits, Set by SS | | |
| RETURN_IF_HANDOFF_FAIL | ’0’B | | |
| PERIODIC_SEARCH | ’0’B | | |
| SCR_INCLUDED | ’1’B | | |
| NNSCR_INCLUDED | ’1’B | | |
| USE_PWR_CNTL_STEP | ’0’B | | |
| CLEAR_RETRY_DELAY | ’0’B | | |
| SCH_INCL | ’1’B | | |
| FPC_SUBCHAN_GAIN | ’01010’B | | |
| USE_PC_TIME | ’0’B | | |
| CH_IND | ’101’B | | |
| ACTIVE_SET_REC_LEN | 8 bits, Set by SS | | |
| NUM_PILOTS | ’001’B | | |
| SRCH_OFFSET_INCL | ’1’B | | |
| PILOT_PN | ’000000000’B | | |
| SRCH_OFFSET | ’010’B | | |
| ADD_PILOT_REC_INCL | ’0’B | | |

| | | | |
|---|---|---|---|
| PWR_COMB_IND | '0'B | | |
| CODE_CHAN_FCH | 11 bits, Set by SS | | |
| QOF_MASK_ID_FCH | '00'B | | |
| RESERVED | 0-7 bits | | |
| REV_FCH_GATING_MODE | '0'B | | |

### 5.3.6.5 Test requirement

Tables 5.3.6.4.1-1, 5.3.6.5-1 and 5.3.6.5-2 define the primary level settings including test tolerances for E-UTRAN TDD - cdma2000 1xRTT handover test.

**Table 5.3.6.5-1: Cell Specific Test requirement Parameters for Cell 1 E-UTRAN TDD cell**

| Parameter | Unit | Cell 1 (E-UTRA) | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | | OP.2 TDD |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | |
| RSRP Note 3 | dBm/15 KHz | -98+TT | -98+TT | -98+TT |
| $\hat{E}_s / N_{oc}$ | dB | 0+TT | 0+TT | 0+TT |
| $\hat{E}_s / I_{ot}$ | dB | 0+TT | 0+TT | 0+TT |
| Propagation Condition | | AWGN | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 5.3.6.5-2: Cell Specific Test requirement Parameters for Cell 2 cdma2000 1xRTT cell**

| Parameter | Unit | Cell 2 (cdma2000 1X) | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| $\frac{\text{Pilot } E_c}{I_{or}}$ | dB | -7 | | |
| $\frac{\text{Sync } E_c}{I_{or}}$ | dB | -16 | | |
| $\frac{\text{Paging } E_c}{I_{or}}$ (4.8 kbps) | dB | -12 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -infinity | 0+TT | 0+TT |
| $I_{oc}$ | dBm/1.2288 MHz | -55 | | |
| CDMA2000 1xRTT Pilot Strength | dB | -infinity | -10+TT | -10+TT |
| Propagation Condition | | AWGN | | |

The handover delay is defined as the time from the beginning of time period T3, to the moment the UE start to transmit the reverse control channel in cdma2000 1xRTT to Cell 2.

The handover delay $D_{handover}$ test requirement in this case is expressed as:

Handover delay $D_{handover}$ = maximum RRC procedure delay + $T_{interrupt}$

$T_{interrupt} = T_{IU} + 40 + 10*KC*SW_K + 10*OC*SW_O$ ms

$T_{IU}$ = 20 ms; $T_{IU}$ can be up to one cdma2000 1xRTT frame (20 ms).

$SW_K = 1$; $SW_K = \left\lceil \frac{\text{srch_win_k}}{60} \right\rceil$ where srch_win_k is the number of cdma2000 1xRTT chips (60) indicated by the search window for known target cdma2000 1xRTT cells in the message

KC = 1; 1 known cell; cdma2000 1xRTT cell is identified during T2 and is therefore known before T3

OC = 0; OC is the number of unknown target cdma2000 1xRTT cells (0).

Maximum RRC procedure delay = 130 ms as defined in TS 36.133 [4].

The handover delay $D_{handover}$ shall be less than a total of 200 ms in this test case (note: this gives a total of 130 ms for maximum RRC procedure delay plus 70 ms for $T_{interrupt}$).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 6 RRC Connection Mobility Control

When the UE is in RRC_CONNECTED, for which security has been activated, initiate the RRC re-establishment procedure in order to continue the RRC connection, the RRC re-establishment process takes place. In this process the UE initiates the procedure when one of the following conditions is met: upon re-entry of the service area after having detected radio link failure, upon handover failure or when lower layers detect problems as defined in TS 36.331 [5] clause 5.3.7.2. After selecting the best cell the UE send a 'RRC Connection Re-establishment Request message' to the System Simulator as defined in TS 36.331 [5] clause 5.3.7. The connection re-establishment succeeds only if the concerned cell is prepared i.e. has a valid UE context within the specified UE re-establishment delay period.

When the random access procedure is initiated by a PDCCH order or by the MAC sublayer itself, the random access process takes place. This process allows the PDCCH order or RRC optionally to indicate a random access preamble and PRACH resource as defined in TS 36.321 [11] clause 5.1. In this process from the physical layer perspective, the L1 random access procedure encompasses the transmission of random access preamble and random access response as defined in TS 36.213 [8] clause 6.1. The random access procedure is used when establishing the L1 communication between the UE and E-UTRAN.

SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Annex A1. The SS sends downlink MAC padding bits on the DL RMC.

Uplink is configured according to Annex A.3. This applies only for Re-establishment tests (subclause 6.1).

# 6.1 RRC Re-establishment

## 6.1.1 E-UTRAN FDD Intra-frequency RRC Re-establishment

### 6.1.1.1 Test purpose

To verify that the UE is able to send a RRC Connection Re-establishment Request message to the System Simulator within the specified re-establishment delay limits from the moment it detects a loss in RRC connection.

### 6.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

### 6.1.1.3 Minimum conformance requirements

In RRC connected mode the UE shall be capable of sending *RRCConnectionReestablishmentRequest* message within $T_{re\text{-}establish_delay}$ seconds from the moment it detects a loss in RRC connection. The total RRC connection delay ($T_{re\text{-}establish_delay}$) shall be less than:

$$T_{re\text{-}esta}blish_delay = TUL_grant + T_{UE_re\text{-}establish_delay}$$

$T_{UL_grant}$: It is the time required to acquire and process uplink grant from the target cell. The uplink grant is required to transmit *RRCConnectionReestablishmentRequest* message.

The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) is the time between the moments when any of the conditions requiring RRC re-establishment as defined in section 5.3.7 in TS 36.331 [5] is detected by the UE to the time when the UE sends PRACH to the target cell. The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) requirement shall be less than:

$$TUE\text{-}re\text{-}establish_delay = 50\ ms + Nfreq * Tsearch + TSI + TPRACH$$

$T_{search}$: It is the time required by the UE to search the target cell.

$T_{search}$ = It is [100] ms if the target cell is known by the UE; the target cell is known if it has been measured by the UE in the last 5 seconds.

$T_{search}$ = It is 800 ms if the target cell is unknown by the UE; the target cell is unknown if it has not been measured by the UE in the last 5 seconds.

$T_{SI}$ = It is the time required for receiving all the relevant system information according to the reception procedure and the RRC procedure delay of system information blocks defined in TS 36.331 [5] for E-UTRAN cell.

$T_{PRACH}$ = The additional delay caused by the random access procedure; it will be at least 10 ms due to random access occasion and there might be additional delay due to ramping procedure.

$N_{freq}$: It is the total number of E-UTRA frequencies to be monitored for RRC re-establishment; $N_{freq}$ = 1 if the target cell is known.

There is no requirement if the target cell does not contain the UE context.

The normative reference for this requirement is TS 36.133 [4] clause 6.1.2.1 and A.6.1.1.

### 6.1.1.4 Test description

#### 6.1.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.20.
2. The parameter settings for the cells are set up according to Table 6.1.1.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 6.1.1.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.1.1.4.1-1: General test parameters for E-UTRAN FDD intra-frequency RRC Re-establishment test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbouring cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF Channel Number | | | 1 | Only one FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| N310 | | - | 1 | Maximum consecutive out-of-sync indications from lower layers |
| N311 | | - | 1 | Minimum consecutive in-sync indications from lower layers |
| T310 | | ms | 0 | Radio link failure timer; T310 is disabled |
| T311 | | ms | 3000 | RRC re-establishment timer |
| DRX | | | OFF | |
| CP length | | | Normal | |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| PRACH configuration index | | | 4 | As specified in table 5.7.1-2 in TS 36.211 |
| Time offset between cells | | ms | 3 | Asynchronous cells<br>3ms or 92160*Ts |
| T1 | | s | 5 | |
| T2 | | ms | 200 | |
| T3 | | s | 3 | |

#### 6.1.1.4.2 Test procedure

The test consists of 3 successive time periods, with time duration of T1, T2 and T3 respectively. At the start of time period T2, cell 1, which is the active cell, is deactivated. The time period T3 starts after the occurrence of the radio link failure.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 6.1.1.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 6.1.1.5-1. T2 starts
6. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 6.1.1.5-1. T3 starts
7. If the UE starts to send PRACH preambles to cell 2 for sending the *RRCConnectionReestablishmentRequest* message to cell 2 within 1.5 s from the beginning of time period T3, then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
8. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

### 6.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.1.1.4.3-1: Common Exception messages for E-UTRAN FDD Intra-frequency RRC Re-establishment**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.5-1 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

**Table 6.1.1.4.3-2: MeasConfig-DEFAULT: Additional E-UTRAN intra frequency RRC Re-establishment requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModifyList | Not present | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModifyList | Not present | | |
| quantityConfig | Not present | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| hrpd-PreRegistrationInfo | Not present | | |
| speedDependentParameters | Not present | | |
| } | | | |

## 6.1.1.5 Test requirement

Table 6.1.1.5-1 defines the primary level settings including test tolerances for E-UTRAN FDD Intra-frequency RRC Re-establishment test case.

**Table 6.1.1.5-1: Cell specific test parameters for E-UTRAN FDD intra-frequency RRC Re-establishment test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.1 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD | OP.1 FDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 1.54 | -Infinity | -Infinity | -3.79 | 4 | 4 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 7 | -Infinity | -Infinity | 4 | 4 | 4 |
| RSRP Note 3 | dBm/15 KHz | -91 | -Infinity | -Infinity | -94 | -94 | -94 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The RRC re-establishment delay is defined as the time from the start of time period T3, to the moment when the UE starts to send PRACH preambles to cell 2 for sending the *RRCConnectionReestablishmentRequest* message to cell 2.

The RRC re-establishment delay to a known E-UTRA FDD intra frequency cell shall be less than 1.5 s.

The rate of correct RRC re-establishments observed during repeated tests shall be at least 90%.

NOTE: The RRC re-establishment delay in the test is derived from the following expression:

$$T_{re\text{-}establish_delay} = T_{UL_grant} + T_{UE_re\text{-}establish_delay}.$$

Where:

$T_{UL_grant}$ = It is the time required to acquire and process uplink grant from the target cell. The PRACH reception at the system simulator is used as a trigger for the completion of the test; hence $T_{UL_grant}$ is not used.

$$TUE_re\text{-}establish_delay = 50\ ms + Nfreq * Tsearch + TSI + TPRACH$$

$N_{freq} = 1$

$T_{search} = 100$ ms

$T_{SI} = 1280$ ms; it is the time required for receiving all the relevant system information as defined in TS 36.331 [5] for the target E-UTRAN FDD cell.

$T_{PRACH}$ = 15 ms; it is the additional delay caused by the random access procedure.

This gives a total of 1445 ms, within the allowance of 1.5 s in the test case.

## 6.1.2 E-UTRAN FDD Inter-frequency RRC Re-establishment

### 6.1.2.1 Test purpose

To verify that the UE is able to send a RRC Connection Re-establishment Request message to the System Simulator within the specified re-establishment delay limits from the moment it detects a loss in RRC connection.

### 6.1.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward. Applicability requires support for FGI bit 25.

### 6.1.2.3 Minimum conformance requirements

In RRC connected mode the UE shall be capable of sending *RRCConnectionReestablishmentRequest* message within $T_{re\text{-}establish_delay}$ seconds from the moment it detects a loss in RRC connection. The total RRC connection delay ($T_{re\text{-}establish_delay}$) shall be less than:

$$T_{re\text{-}establish_delay} = T_{UL_grant} + T_{UE_re\text{-}establish_delay}$$

$T_{UL_grant}$: It is the time required to acquire and process uplink grant from the target cell. The uplink grant is required to transmit *RRCConnectionReestablishmentRequest* message.

The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) is the time between the moments when any of the conditions requiring RRC re-establishment as defined in section 5.3.7 in TS 36.331 [5] is detected by the UE to the time when the UE sends PRACH to the target cell. The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) requirement shall be less than:

$$TUE\text{-}re\text{-}establish_delay = 50\ ms + Nfreq * Tsearch + TSI + TPRACH$$

$T_{search}$: It is the time required by the UE to search the target cell.

$T_{search}$ = It is [100] ms if the target cell is known by the UE; the target cell is known if it has been measured by the UE in the last 5 seconds.

$T_{search}$ = It is 800 ms if the target cell is unknown by the UE; the target cell is unknown if it has not been measured by the UE in the last 5 seconds.

$T_{SI}$ = It is the time required for receiving all the relevant system information according to the reception procedure and the RRC procedure delay of system information blocks defined in TS 36.331 [5] for E-UTRAN cell.

$T_{PRACH}$ = The additional delay caused by the random access procedure; it will be at least 10 ms due to random access occasion and there might be additional delay due to ramping procedure.

$N_{freq}$: It is the total number of E-UTRA frequencies to be monitored for RRC re-establishment; $N_{freq}$ = 1 if the target cell is known.

There is no requirement if the target cell does not contain the UE context.

The normative reference for this requirement is TS 36.133 [4] clause 6.1.2.1 and A.6.1.2.

### 6.1.2.4 Test description

#### 6.1.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.14.
2. The parameter settings for the cells are set up according to Table 6.1.2.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 6.1.2.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.1.2.4.1-1: General test parameters for E-UTRAN FDD inter-frequency RRC Re-establishment test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbouring cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF Channel Number (cell 1) | | | 1 | |
| E-UTRA RF Channel Number (cell 2) | | | 2 | |
| E-UTRA FDD inter-frequency carrier list size | | | 1 | 2 E-UTRA FDD carrier frequencies in total: 1 intra-frequency and 1 inter-frequency |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| N310 | | - | 1 | Maximum consecutive out-of-sync indications from lower layers |
| N311 | | - | 1 | Minimum consecutive in-sync indications from lower layers |
| T310 | | ms | 0 | Radio link failure timer; T310 is disabled |
| T311 | | ms | 5000 | RRC re-establishment timer |
| DRX | | | OFF | |
| CP length | | | Normal | |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| PRACH configuration index | | | 4 | As specified in table 5.7.1-2 in TS 36.211 |
| Time offset between cells | | ms | 3 | Asynchronous cells<br>3ms or 92160*Ts |
| T1 | | s | 5 | |
| T2 | | ms | 200 | |
| T3 | | s | 5 | |

### 6.1.2.4.2 Test procedure

The test consists of 3 successive time periods, with time duration of T1, T2 and T3 respectively. At the start of time period T2, cell 1, which is the active cell, is deactivated. The time period T3 starts after the occurrence of the radio link failure.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 6.1.2.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 6.1.2.5-1. T2 starts

6. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 6.1.2.5-1. T3 starts

7. If the UE starts to send PRACH preambles to cell 2 for sending the *RRCConnectionReestablishmentRequest* message to cell 2 within 3 s from the beginning of time period T3, then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

8. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

### 6.1.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.1.2.4.3-1: Common Exception messages for E-UTRAN FDD Inter-frequency RRC Re-establishment**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.5-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

**Table 6.1.2.4.3-2: MeasConfig-DEFAULT: Additional E-UTRAN inter frequency RRC Re-establishment requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE OF { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | serving frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | inter frequency | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModifyList | Not present | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModifyList | Not present | | |
| quantityConfig | Not present | | |
| measGapConfig | MeasGapConfig-GP1 | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

### 6.1.2.5 Test requirement

Table 6.1.2.5-1 defines the primary level settings including test tolerances for E-UTRAN FDD Inter-frequency RRC Re-establishment test case.

**Table 6.1.2.5-1: Cell specific test parameters for E-UTRAN FDD inter-frequency RRC Re-establishment test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.1 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD | OP.1 FDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | -Infinity | -Infinity | -Infinity | -Infinity | 7 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | -Infinity | -Infinity | - Infinity | - Infinity | 7 |
| RSRP Note 3 | dBm/15 KHz | -94 | -Infinity | -Infinity | - Infinity | -Infinity | -91 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The RRC re-establishment delay is defined as the time from the start of time period T3, to the moment when the UE starts to send PRACH preambles to cell 2 for sending the *RRCConnectionReestablishmentRequest* message to cell 2.

The RRC re-establishment delay to an unknown E-UTRA FDD inter frequency cell shall be less than 3 s.

The rate of correct RRC re-establishments observed during repeated tests shall be at least 90%.

NOTE: The RRC re-establishment delay in the test is derived from the following expression:

$$T_{re\text{-}establish_delay} = T_{UL_grant} + T_{UE_re\text{-}establish_delay}.$$

Where:

$T_{UL_grant}$ = It is the time required to acquire and process uplink grant from the target cell. The PRACH reception at the system simulator is used as a trigger for the completion of the test; hence $T_{UL_grant}$ is not used.

TUE_re-establish_delay = 50 ms + Nfreq* Tsearch + TSI + TPRACH

$N_{freq}$ = 2

$T_{search}$ = 800 ms

$T_{SI}$ = 1280 ms; it is the time required for receiving all the relevant system information as defined in TS 36.331 [5] for the target E-UTRAN FDD cell.

$T_{PRACH}$ = 15 ms; it is the additional delay caused by the random access procedure.

This gives a total of 2945 ms, within the allowance of 3 s in the test case.

## 6.1.3 E-UTRAN TDD Intra-frequency RRC Re-establishment

### 6.1.3.1 Test purpose

To verify that the UE is able to send a RRC Connection Re-establishment Request message to the System Simulator within the specified re-establishment delay limits from the moment it detects a loss in RRC connection.

### 6.1.3.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

### 6.1.3.3 Minimum conformance requirements

In RRC connected mode the UE shall be capable of sending *RRCConnectionReestablishmentRequest* message within $T_{re\text{-}establish_delay}$ seconds from the moment it detects a loss in RRC connection. The total RRC connection delay ($T_{re\text{-}establish_delay}$) shall be less than:

$$T_{re\text{-}establish_delay} = T_{UL_grant} + T_{UE_re\text{-}establish_delay}$$

$T_{UL_grant}$: It is the time required to acquire and process uplink grant from the target cell. The uplink grant is required to transmit *RRCConnectionReestablishmentRequest* message.

The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) is the time between the moments when any of the conditions requiring RRC re-establishment as defined in section 5.3.7 in TS 36.331 [5] is detected by the UE to the time when the UE sends PRACH to the target cell. The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) requirement shall be less than:

$$TUE\text{-}re\text{-}establish_delay = 50\ ms + Nfreq * Tsearch + TSI + T_{PRACH}$$

$T_{search}$ is the time required by the UE to search the target cell.

$T_{search}$ is [100] ms if the target cell is known by the UE; the target cell is known if it has been measured by the UE in the last 5 seconds.

$T_{search}$ is 800 ms if the target cell is unknown by the UE; the target cell is unknown if it has not been measured by the UE in the last 5 seconds.

$T_{SI}$ is the time required for receiving all the relevant system information according to the reception procedure and the RRC procedure delay of system information blocks defined in TS 36.331 [5] for E-UTRAN cell.

$T_{PRACH}$ is the additional delay caused by the random access procedure; it will be at least 10 ms due to random access occasion and there might be additional delay due to ramping procedure.

$N_{freq}$ is the total number of E-UTRA frequencies to be monitored for RRC re-establishment; $N_{freq}$ = 1 if the target cell is known.

There is no requirement if the target cell does not contain the UE context.

The normative reference for this requirement is TS 36.133 [4] clause 6.1.2.1 and A.6.1.3.

### 6.1.3.4 Test description

#### 6.1.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.20.
2. The parameter settings for the cells are set up according to Table 6.1.3.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 6.1.3.4.3.
5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.1.3.4.1-1: General test parameters for E-UTRAN TDD intra-frequency RRC Re-establishment test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbouring cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF Channel Number | | | 1 | Only one TDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| N310 | | - | 1 | Maximum consecutive out-of-sync indications from lower layers |
| N311 | | - | 1 | Minimum consecutive in-sync indications from lower layers |
| T310 | | ms | 0 | Radio link failure timer; T310 is disabled |
| T311 | | ms | 3000 | RRC re-establishment timer |
| DRX | | | OFF | |
| CP length | | | Normal | |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211 |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211 |
| PRACH configuration index | | | 53 | As specified in table 5.7.1-3 in TS 36.211 |
| Time offset between cells | | µs | 3 | Synchronous cells<br>3µs or 92*Ts |
| T1 | | s | 5 | |
| T2 | | ms | 200 | |
| T3 | | s | 3 | |

#### 6.1.3.4.2 Test procedure

The test consists of 3 successive time periods, with time duration of T1, T2 and T3 respectively. At the start of time period T2, cell 1, which is the active cell, is deactivated. The time period T3 starts after the occurrence of the radio link failure.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 6.1.3.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 6.1.3.5-1. T2 starts
6. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 6.1.3.5-1. T3 starts
7. If the UE starts to send PRACH preambles to cell 2 within 1.5 s from the beginning of time period T3. then the number of successful tests is increased by one.
8. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

### 6.1.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.1.3.4.3-1: Common Exception messages for E-UTRAN intra frequency RRC Re-establishment requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.5-1 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-2 |

**Table 6.1.3.4.3-2: MeasConfig-DEFAULT: Additional E-UTRAN intra frequency RRC Re-establishment requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModifyList | Not present | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModifyList | Not present | | |
| quantityConfig | Not present | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| hrpd-PreRegistrationInfo | Not present | | |
| speedDependentParameters | Not present | | |
| } | | | |

### 6.1.3.5 Test requirement

Table 6.1.3.5-1 defines the primary level settings including test tolerances for E-UTRAN TDD Intra-frequency RRC Re-establishment test case.

**Table 6.1.3.5-1: Cell specific test parameters for E-UTRAN TDD intra-frequency RRC Re-establishment test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | OP.1 TDD | OP.2 TDD | OP.2 TDD | OP.2 TDD | OP.1 TDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 1.54 | -Infinity | -Infinity | -3.79 | 4 | 4 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 7 | -Infinity | -Infinity | 4 | 4 | 4 |
| RSRP Note 3 | dBm/15 KHz | -91 | -Infinity | -Infinity | -94 | -94 | -94 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The RRC re-establishment delay is defined as the time from the start of time period T3, to the moment when the UE starts to send PRACH preambles to cell 2 for sending the *RRCConnectionReestablishmentRequest* message to cell 2.

The RRC re-establishment delay to a known E-UTRA TDD intra frequency cell shall be less than 1.5 s.

The rate of correct RRC re-establishments observed during repeated tests shall be at least 90%.

NOTE: The RRC re-establishment delay in the test is derived from the following expression:

$$T_{re\text{-}establish_delay} = T_{UL_grant} + T_{UE_re\text{-}establish_delay}.$$

Where:

$T_{UL_grant}$ = It is the time required to acquire and process uplink grant from the target cell. The PRACH reception at the system simulator is used as a trigger for the completion of the test; hence $T_{UL_grant}$ is not used.

$$T_{UE_re\text{-}establish_delay} = 50\ ms + N_{freq} * T_{search} + T_{SI} + T_{PRACH}$$

$N_{freq} = 1$

$T_{search}$ = 100 ms

$T_{SI}$ = 1280 ms; it is the time required for receiving all the relevant system information as defined in TS 36.331 [5] for the target E-UTRAN TDD cell.

$T_{PRACH}$ = 15 ms; it is the additional delay caused by the random access procedure.

This gives a total of 1445 ms, within the allowance of 1.5 s in the test case.

## 6.1.4 E-UTRAN TDD Inter-frequency RRC Re-establishment

### 6.1.4.1 Test purpose

To verify that the UE is able to send a RRC Connection Re-establishment Request message to the System Simulator within the specified re-establishment delay limits from the moment it detects a loss in RRC connection.

### 6.1.4.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward. Applicability requires support for FGI bit 25.

### 6.1.4.3 Minimum conformance requirements

In RRC connected mode the UE shall be capable of sending *RRCConnectionReestablishmentRequest* message within $T_{re\text{-}establish_delay}$ seconds from the moment it detects a loss in RRC connection. The total RRC connection delay ($T_{re\text{-}establish_delay}$) shall be less than:

$$T_{re\text{-}establish_delay} = T_{UL_grant} + T_{UE_re\text{-}establish_delay}$$

$T_{UL_grant}$: It is the time required to acquire and process uplink grant from the target cell. The uplink grant is required to transmit *RRCConnectionReestablishmentRequest* message.

The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) is the time between the moments when any of the conditions requiring RRC re-establishment as defined in section 5.3.7 in TS 36.331 [5] is detected by the UE to the time when the UE sends PRACH to the target cell. The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) requirement shall be less than:

$$TUE\text{-}re\text{-}establish_delay = 50\ ms + Nfreq * Tsearch + TSI + T_{PRACH}$$

$T_{search}$: It is the time required by the UE to search the target cell.

$T_{search}$ = It is [100] ms if the target cell is known by the UE; the target cell is known if it has been measured by the UE in the last 5 seconds.

$T_{search}$ = It is 800 ms if the target cell is unknown by the UE; the target cell is unknown if it has not been measured by the UE in the last 5 seconds.

$T_{SI}$ = It is the time required for receiving all the relevant system information according to the reception procedure and the RRC procedure delay of system information blocks defined in TS 36.331 [5] for E-UTRAN cell.

$T_{PRACH}$ = The additional delay caused by the random access procedure; it will be at least 10 ms due to random access occasion and there might be additional delay due to ramping procedure.

$N_{freq}$: It is the total number of E-UTRA frequencies to be monitored for RRC re-establishment; $N_{freq}$ = 1 if the target cell is known.

There is no requirement if the target cell does not contain the UE context.

The normative reference for this requirement is TS 36.133 [4] clause 6.1.2.1 and A.6.1.4.

### 6.1.4.4 Test description

#### 6.1.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.14.
2. The parameter settings for the cells are set up according to Table 6.1.4.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 6.1.4.4.3.
5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.1.4.4.1-1: General test parameters for E-UTRAN TDD inter-frequency RRC Re-establishment test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbouring cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF Channel Number (cell 1) | | | 1 | |
| E-UTRA RF Channel Number (cell 2) | | | 2 | |
| E-UTRA TDD inter-frequency carrier list size | | | 1 | 2 E-UTRA TDD carrier frequencies in total: 1 intra-frequency and 1 inter-frequency |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| N310 | | - | 1 | Maximum consecutive out-of-sync indications from lower layers |
| N311 | | - | 1 | Minimum consecutive in-sync indications from lower layers |
| T310 | | ms | 0 | Radio link failure timer; T310 is disabled |
| T311 | | ms | 5000 | RRC re-establishment timer |
| DRX | | | OFF | |
| CP length | | | Normal | |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211 |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211 |
| PRACH configuration index | | | 53 | As specified in table 5.7.1-3 in TS 36.211 |
| Time offset between cells | | µs | 3 | Synchronous cells<br>3µs or 92*Ts |
| T1 | | s | 5 | |
| T2 | | ms | 200 | |
| T3 | | s | 5 | |

#### 6.1.4.4.2 Test procedure

The test consists of 3 successive time periods, with time duration of T1, T2 and T3 respectively. At the start of time period T2, cell 1, which is the active cell, is deactivated. The time period T3 starts after the occurrence of the radio link failure.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 6.1.3.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 6.1.3.5-1. T2 starts.
6. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 6.1.3.5-1. T3 starts.
7. If the UE starts to send PRACH preambles to cell 2 within 3s from the beginning of time period T3. then the number of successful tests is increased by one.
8. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

#### 6.1.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.1.4.4.3-1: Common Exception messages for E-UTRAN inter frequency RRC Re-establishment requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.5-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-2 |

**Table 6.1.4.4.3-2: MeasConfig-DEFAULT: Additional E-UTRAN inter frequency RRC Re-establishment requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE OF { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | serving frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | inter frequency | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModifyList | Not present | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModifyList | Not present | | |
| quantityConfig | Not present | | |
| measGapConfig | MeasGapConfig-GP1 | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedDependentParameters | Not present | | |
| } | | | |

## 6.1.4.5 Test requirement

Table 6.1.4.5-1 defines the primary level settings including test tolerances for E-UTRAN TDD Inter-frequency RRC Re-establishment test case.

**Table 6.1.4.5-1: Cell specific test parameters for E-UTRAN TDD inter-frequency RRC Re-establishment test case**

| **Parameter** | **Unit** | **Cell 1** | | | **Cell 2** | | |
|---|---|---|---|---|---|---|---|
| | | **T1** | **T2** | **T3** | **T1** | **T2** | **T3** |
| E-UTRA RF Channel Number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in A.2.1 (OP.1 TDD) and in A.2.2 (OP.2 TDD) | | OP.1 TDD | OP.1 TDD | OP.2 TDD | OP.2 TDD | OP.2 TDD | OP.1 TDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $\hat{E}_s / I_{ot}$ | dB | 4 | -Infinity | -Infinity | -Infinity | -Infinity | 7 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | -Infinity | -Infinity | - Infinity | - Infinity | 7 |
| RSRP Note 3 | dBm/15 KHz | -94 | -Infinity | -Infinity | - Infinity | -Infinity | -91 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The RRC re-establishment delay is defined as the time from the start of time period T3, to the moment when the UE starts to send PRACH preambles to cell 2 for sending the *RRCConnectionReestablishmentRequest* message to cell 2.

The RRC re-establishment delay to an unknown E-UTRA TDD inter frequency cell shall be less than 3 s.

The rate of correct RRC re-establishments observed during repeated tests shall be at least 90%.

NOTE: The RRC re-establishment delay in the test is derived from the following expression:

$$T_{re\text{-}establish_delay} = T_{UL_grant} + T_{UE_re\text{-}establish_delay}.$$

Where:

$T_{UL_grant}$ = It is the time required to acquire and process uplink grant from the target cell. The PRACH reception at the system simulator is used as a trigger for the completion of the test; hence $T_{UL_grant}$ is not used.

$T_{UE_re\text{-}establish_delay}$ = 50 ms + $N_{freq}$* $T_{search}$ + $T_{SI}$ + $T_{PRACH}$

$N_{freq}$ = 2

$T_{search}$ = 800 ms

$T_{SI}$ = 1280 ms; it is the time required for receiving all the relevant system information as defined in TS 36.331 [5] for the target E-UTRAN TDD cell.

$T_{PRACH}$ = 15 ms; it is the additional delay caused by the random access procedure.

This gives a total of 2945 ms, within the allowance of 3 s in the test case.

## 6.1.5 E-UTRAN FDD Intra-frequency RRC Re-establishment for 5MHz Bandwidth

### 6.1.5.1 Test purpose

Same test purpose as clause 6.1.1.1.

### 6.1.5.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that only support E-UTRA Band 31.

### 6.1.5.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 6.1.1.3 with the following exceptions:

- Instead of A.6.1.1 → use A.6.1.5.

### 6.1.5.4 Test description

#### 6.1.5.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.20.
2. The parameter settings for the cells are set up according to Table 6.1.5.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 6.1.5.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.1.5.4.1-1: General test parameters for E-UTRAN FDD intra-frequency RRC Re-establishment test case for 5MHz bandwidth**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.5 FDD | As specified in clause A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.11 FDD | As specified in clause A.2.1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 | |
| Note 1: See Table 6.1.1.4.1-1 for the other parameters.<br>Note 2: This test is according to the principle defined in TS 36.133 [4] section A.3.7.2. | | | |

#### 6.1.5.4.2 Test procedure

Same test procedure as in clause 6.1.1.4.2 with the following exceptions:

- Instead of Table 6.1.1.5-1 → use Table 6.1.5.5-1.

#### 6.1.5.4.3 Message contents

Same message contents as defined in clause 6.1.1.4.3.

### 6.1.5.5 Test requirement

Same test requirement as in clause 6.1.1.5 with the following exceptions:

Table 6.1.5.5-1 defines the primary level settings including test tolerances for E-UTRAN FDD Intra-frequency RRC Re-establishment test case.

**Table 6.1.5.5-1: Cell specific test parameters for E-UTRAN FDD intra-frequency RRC Re-establishment test case for 5MHz bandwidth**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| $BW_{channel}$ | MHz | 5 | | | 5 | | |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | OP.15 FDD | OP.15 FDD | OP.16 FDD | OP.16 FDD | OP.16 FDD | OP.15 FDD |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: See Table 6.1.1.5-1 for the other parameters. | | | | | | | |

## 6.1.6 E-UTRAN FD-FDD Intra-frequency RRC Re-establishment for UE category 0

### 6.1.6.1 Test purpose

To verify that the UE is able to send a RRC Connection Re-establishment Request message to the System Simulator within the specified re-establishment delay limits from the moment it detects a loss in RRC connection.

### 6.1.6.2 Test applicability

This test applies to all types of E-UTRA FD-FDD UE release 12 and forward of UE category 0.

### 6.1.6.3 Minimum conformance requirements

In RRC connected mode the UE shall be capable of sending *RRCConnectionReestablishmentRequest* message within $T_{re\text{-}establish_delay}$ seconds from the moment it detects a loss in RRC connection. The total RRC connection delay ($T_{re\text{-}establish_delay}$) shall be less than:

$$T_{re\text{-}esta}blish_delay = TUL_grant + T_{UE_re\text{-}establish_delay}$$

$T_{UL_grant}$: It is the time required to acquire and process uplink grant from the target cell. The uplink grant is required to transmit *RRCConnectionReestablishmentRequest* message.

The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) is the time between the moments when any of the conditions requiring RRC re-establishment as defined in section 5.3.7 in TS 36.331 [5] is detected by the UE to the time when the UE sends PRACH to the target cell. The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) requirement shall be less than:

$$TUE\text{-}re\text{-}establish_delay = 50\ ms + Nfreq*Tsearch + TSI + TPRACH$$

$T_{search}$ = It is the time required by the UE to search the target cell.

$T_{search}$ = It is [100] ms if the target cell is known by the UE; the target cell is known if it has been measured by the UE in the last 5 seconds.

$T_{search}$ = It is 800 ms if the target cell is unknown by the UE; the target cell is unknown if it has not been measured by the UE in the last 5 seconds.

$T_{SI}$ = It is the time required for receiving all the relevant system information according to the reception procedure and the RRC procedure delay of system information blocks defined in TS 36.331 [5] for E-UTRAN cell.

$T_{PRACH}$ = The additional delay caused by the random access procedure; it will be at least 10 ms due to random access occasion and there might be additional delay due to ramping procedure.

$N_{freq}$: It is the total number of E-UTRA frequencies to be monitored for RRC re-establishment; $N_{freq}$ = 1 if the target cell is known.

There is no requirement if the target cell does not contain the UE context.

The normative reference for this requirement is TS 36.133 [4] clause 6.1.2.1 and A.6.1.6.

## 6.1.6.4 Test description

### 6.1.6.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.20(using only main UE Tx/Rx antenna).
2. The parameter settings for the cells are set up according to Table 6.1.6.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 6.1.6.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.1.6.4.1-1: General test parameters for E-UTRAN FDD intra-frequency RRC Re-establishment test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.13 FDD | As specified in clause A.1.3 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in clause A.2.1 |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbouring cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF Channel Number | | | 1 | Only one FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| N310 | | - | 1 | Maximum consecutive out-of-sync indications from lower layers |
| N311 | | - | 1 | Minimum consecutive in-sync indications from lower layers |
| T310 | | ms | 0 | Radio link failure timer; T310 is disabled |
| T311 | | ms | 3000 | RRC re-establishment timer |
| DRX | | | OFF | |
| CP length | | | Normal | |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| PRACH configuration index | | | 4 | As specified in table 5.7.1-2 in TS 36.211 |
| Time offset between cells | | ms | 3 | Asynchronous cells |
| T1 | | s | 5 | |
| T2 | | ms | 200 | |
| T3 | | s | 3 | |

### 6.1.6.4.2 Test procedure

The test consists of 3 successive time periods, with time duration of T1, T2 and T3 respectively. At the start of time period T2, cell 1, which is the active cell, is deactivated. The time period T3 starts after the occurrence of the radio link failure.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

2. Set the parameters according to T1 in Table 6.1.6.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 6.1.6.5-1. T2 starts
6. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 6.1.6.5-1. T3 starts
7. If the UE starts to send PRACH preambles to cell 2 for sending the *RRCConnectionReestablishmentRequest* message to cell 2 within 1.5 s from the beginning of time period T3, then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
8. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

### 6.1.6.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.1.6.4.3-1: Common Exception messages for E-UTRAN FDD Intra-frequency RRC Re-establishment**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.5-1 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

**Table 6.1.6.4.3-2: MeasConfig-DEFAULT: Additional E-UTRAN intra frequency FDD RRC Re-establishment requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModifyList | Not present | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModifyList | Not present | | |
| quantityConfig | Not present | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| hrpd-PreRegistrationInfo | Not present | | |
| speedDependentParameters | Not present | | |
| } | | | |

## 6.1.6.5 Test requirement

Table 6.1.6.5-1 defines the primary level settings including test tolerances for E-UTRAN FDD Intra-frequency RRC Re-establishment test case.

**Table 6.1.6.5-1: Cell specific test parameters for E-UTRAN FDD intra-frequency RRC Re-establishment test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2(OP.2 FDD) | | OP.1 FDD | OP.1 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD | OP.1 FDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 1.54 | -Infinity | -Infinity | -3.79 | 4 | 4 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 7 | -Infinity | -Infinity | 4 | 4 | 4 |
| RSRP Note 3 | dBm/15 KHz | -91 | -Infinity | -Infinity | -94 | -94 | -94 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The RRC re-establishment delay is defined as the time from the start of time period T3, to the moment when the UE starts to send PRACH preambles to cell 2 for sending the *RRCConnectionReestablishmentRequest* message to cell 2.

The RRC re-establishment delay to a known E-UTRA FDD intra frequency cell shall be less than 1.5 s.

The rate of correct RRC re-establishments observed during repeated tests shall be at least 90%.

NOTE: The RRC re-establishment delay in the test is derived from the following expression:

$$T_{re\text{-}establish_delay} = T_{UL_grant} + T_{UE_re\text{-}establish_delay}.$$

Where:

$T_{UL_grant}$ = It is the time required to acquire and process uplink grant from the target cell. The PRACH reception at the system simulator is used as a trigger for the completion of the test; hence $T_{UL_grant}$ is not used.

TUE_re-establish_delay = 50 ms + Nfreq* Tsearch + TSI + TPRACH

$N_{freq}$ = 1

$T_{search}$ = 100 ms

$T_{SI}$ = 1280 ms; it is the time required for receiving all the relevant system information as defined in TS 36.331 [5] for the target E-UTRAN FDD cell.

$T_{PRACH}$ = 15 ms; it is the additional delay caused by the random access procedure.

This gives a total of 1445 ms, within the allowance of 1.5 s in the test case.

## 6.1.7 E-UTRAN HD-FDD Intra-frequency RRC Re-establishment for UE category 0

### 6.1.7.1 Test purpose

To verify that the UE is able to send a RRC Connection Re-establishment Request message to the System Simulator within the specified re-establishment delay limits from the moment it detects a loss in RRC connection.

### 6.1.7.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 12 and forward of UE category 0.

### 6.1.7.3 Minimum conformance requirements

In RRC connected mode the UE shall be capable of sending *RRCConnectionReestablishmentRequest* message within $T_{re\text{-}establish_delay}$ seconds from the moment it detects a loss in RRC connection. The total RRC connection delay ($T_{re\text{-}establish_delay}$) shall be less than:

$$T_{re\text{-}establish_delay} = TUL_grant + T_{UE_re\text{-}establish_delay}$$

$T_{UL_grant}$: It is the time required to acquire and process uplink grant from the target cell. The uplink grant is required to transmit *RRCConnectionReestablishmentRequest* message.

The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) is the time between the moments when any of the conditions requiring RRC re-establishment as defined in section 5.3.7 in TS 36.331 [5] is detected by the UE to the time when the UE sends PRACH to the target cell. The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) requirement shall be less than:

$$TUE\text{-}re\text{-}establish_delay = 50\ ms + Nfreq * Tsearch + TSI + TPRACH$$

$T_{search}$: It is the time required by the UE to search the target cell.

$T_{search}$ = It is [100] ms if the target cell is known by the UE; the target cell is known if it has been measured by the UE in the last 5 seconds.

$T_{search}$ = It is 800 ms if the target cell is unknown by the UE; the target cell is unknown if it has not been measured by the UE in the last 5 seconds.

$T_{SI}$ = It is the time required for receiving all the relevant system information according to the reception procedure and the RRC procedure delay of system information blocks defined in TS 36.331 [5] for E-UTRAN cell.

$T_{PRACH}$ = The additional delay caused by the random access procedure; it will be at least 10 ms due to random access occasion and there might be additional delay due to ramping procedure.

$N_{freq}$: It is the total number of E-UTRA frequencies to be monitored for RRC re-establishment; $N_{freq}$ = 1 if the target cell is known.

There is no requirement if the target cell does not contain the UE context.

The normative reference for this requirement is TS 36.133 [4] clause 6.1.2.1 and A.6.1.7.

### 6.1.7.4 Test description

#### 6.1.7.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.20(using only main UE Tx/Rx antenna).
2. The parameter settings for the cells are set up according to Table 6.1.7.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 6.1.7.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.1.7.4.1-1: General test parameters for E-UTRAN HD-FDD intra-frequency RRC Re-establishment test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.1 HD-FDD | As specified in clause A.1.4 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.3 HD-FDD | As specified in clause A.2.3 |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbouring cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF Channel Number | | | 1 | Only one FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| N310 | | - | 1 | Maximum consecutive out-of-sync indications from lower layers |
| N311 | | - | 1 | Minimum consecutive in-sync indications from lower layers |
| T310 | | ms | 0 | Radio link failure timer; T310 is disabled |
| T311 | | ms | 3000 | RRC re-establishment timer |
| DRX | | | OFF | |
| CP length | | | Normal | |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| PRACH configuration index | | | 4 | As specified in table 5.7.1-2 in TS 36.211 |
| Time offset between cells | | ms | 3 | Asynchronous cells |
| T1 | | s | 5 | |
| T2 | | ms | 200 | |
| T3 | | s | 3 | |

### 6.1.7.4.2 Test procedure

The test consists of 3 successive time periods, with time duration of T1, T2 and T3 respectively. At the start of time period T2, cell 1, which is the active cell, is deactivated. The time period T3 starts after the occurrence of the radio link failure.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 6.1.7.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 6.1.7.5-1. T2 starts
6. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 6.1.7.5-1. T3 starts

7. If the UE starts to send PRACH preambles to cell 2 for sending the *RRCConnectionReestablishmentRequest* message to cell 2 within 1.5 s from the beginning of time period T3, then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

8. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

#### 6.1.7.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.1.7.4.3-1: Common Exception messages for E-UTRAN FDD Intra-frequency RRC Re-establishment**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.5-1 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

**Table 6.1.7.4.3-2: MeasConfig-DEFAULT: Additional E-UTRAN FDD intra frequency RRC Re-establishment requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModifyList | Not present | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModifyList | Not present | | |
| quantityConfig | Not present | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| hrpd-PreRegistrationInfo | Not present | | |
| speedDependentParameters | Not present | | |
| } | | | |

### 6.1.7.5 Test requirement

Table 6.1.7.5-1 defines the primary level settings including test tolerances for E-UTRAN HD-FDD Intra-frequency RRC Re-establishment test case.

**Table 6.1.7.5-1: Cell specific test parameters for E-UTRAN HD-FDD intra-frequency RRC Re-establishment test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.1(OP.1 FDD) and in D.1.2(OP.2 FDD) | | OP.1 FDD | OP.1 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD | OP.1 FDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 1.54 | -Infinity | -Infinity | -3.79 | 4 | 4 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 7 | -Infinity | -Infinity | 4 | 4 | 4 |
| RSRP Note 3 | dBm/15 KHz | -91 | -Infinity | -Infinity | -94 | -94 | -94 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The RRC re-establishment delay is defined as the time from the start of time period T3, to the moment when the UE starts to send PRACH preambles to cell 2 for sending the *RRCConnectionReestablishmentRequest* message to cell 2.

The RRC re-establishment delay to a known E-UTRA FDD intra frequency cell shall be less than 1.5 s.

The rate of correct RRC re-establishments observed during repeated tests shall be at least 90%.

NOTE: The RRC re-establishment delay in the test is derived from the following expression:

$$T_{re\text{-}establish_delay} = T_{UL_grant} + T_{UE_re\text{-}establish_delay}.$$

Where:

$T_{UL_grant}$ = It is the time required to acquire and process uplink grant from the target cell. The PRACH reception at the system simulator is used as a trigger for the completion of the test; hence $T_{UL_grant}$ is not used.

TUE_re-establish_delay = 50 ms + Nfreq* Tsearch + TSI + TPRACH

$N_{freq}$ = 1

$T_{search}$ = 100 ms

$T_{SI}$ = 1280 ms; it is the time required for receiving all the relevant system information as defined in TS 36.331 [5] for the target E-UTRAN FDD cell.

$T_{PRACH}$ = 15 ms; it is the additional delay caused by the random access procedure.

This gives a total of 1445 ms, within the allowance of 1.5 s in the test case.

## 6.1.8 E-UTRAN TDD Intra-frequency RRC Re-establishment for UE category 0

### 6.1.8.1 Test purpose

To verify that the category 0 UE is able to send a RRC Connection Re-establishment Request message to the System Simulator within the specified re-establishment delay limits from the moment it detects a loss in RRC connection.

### 6.1.8.2 Test applicability

This test applies to all type of E-UTRA TDD UE release 12 and forward of UE category 0.

### 6.1.8.3 Minimum conformance requirements

In RRC connected mode the UE shall be capable of sending *RRCConnectionReestablishmentRequest* message within $T_{re\text{-}establish_delay}$ seconds from the moment it detects a loss in RRC connection. The total RRC connection delay ($T_{re\text{-}establish_delay}$) shall be less than:

$$T_{re\text{-}establish_delay} = T_{UL_grant} + T_{UE_re\text{-}establish_delay}$$

$T_{UL_grant}$: It is the time required to acquire and process uplink grant from the target PCell. The uplink grant is required to transmit *RRCConnectionReestablishmentRequest* message.

The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) is the time between the moments when any of the conditions requiring RRC re-establishment as defined in clause 5.3.7 in TS 36.331 [2] is detected by the UE to the time when the UE sends PRACH to the target PCell. The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) requirement shall be less than:

$$T_{UE\text{-}re\text{-}establish_delay} = 50 \text{ ms} + N_{freq} * Tsearch + T_{SI} + T_{PRACH}$$

- $T_{search}$: It is the time required by the UE to search the target PCell.

  $T_{search}$ = It is 100 ms if the target PCell is known by the UE; the target PCell is known if it has been measured by the UE in the last 5 seconds.

  $T_{search}$ = It is 800 ms if the target PCell is unknown by the UE; the target PCell is unknown if it has not been measured by the UE in the last 5 seconds.

  $T_{SI}$ = It is the time required for receiving all the relevant system information according to the reception procedure and the RRC procedure delay of system information blocks defined in TS 36.331 [2] for the target PCell.

  $T_{PRACH}$ = The additional delay caused by the random access procedure; it will be at least 10 ms due to random access occasion and there might be additional delay due to ramping procedure.

  $N_{freq}$: It is the total number of E-UTRA frequencies to be monitored for RRC re-establishment; $N_{freq}$ = 1 if the target PCell is known.

There is no requirement if the target cell does not contain the UE context.

The normative reference for this requirement is TS 36.133 [4] clause 6.1.2.1 and A.6.1.8.

### 6.1.8.4 Test description

#### 6.1.8.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.20 using only main UE Tx/Rx antenna.
2. The parameter settings for the cells are set up according to Table 6.1.8.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 6.1.8.4.3.
5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.1.8.4.1-1: General test parameters for E-UTRAN TDD intra-frequency RRC Re-establishment test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.12 TDD | As specified in clause A.1.5 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 TDD | As specified in clause A.2.2 |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbouring cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF Channel Number | | | 1 | Only one TDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| N310 | | - | 1 | Maximum consecutive out-of-sync indications from lower layers |
| N311 | | - | 1 | Minimum consecutive in-sync indications from lower layers |
| T310 | | ms | 0 | Radio link failure timer; T310 is disabled |
| T311 | | ms | 3000 | RRC re-establishment timer |
| DRX | | | OFF | |
| CP length | | | Normal | |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211 |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211 |
| PRACH configuration index | | | 53 | As specified in table 5.7.1-3 in TS 36.211 |
| Time offset between cells | | μs | 3 | Synchronous cells |
| T1 | | s | 5 | |
| T2 | | ms | 200 | |
| T3 | | s | 3 | |

### 6.1.8.4.2 Test procedure

The test consists of 3 successive time periods, with time duration of T1, T2 and T3 respectively. At the start of time period T2, cell 1, which is the active cell, is deactivated. The time period T3 starts after the occurrence of the radio link failure.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 6.1.8.5-1. Propagation conditions are set according to Annex B clause B.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 6.1.8.5-1. T2 starts
6. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 6.1.8.5-1. T3 starts
7. If the UE starts to send PRACH preambles to cell 2 within 1.5 s from the beginning of time period T3. then the number of successful tests is increased by one.
8. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

### 6.1.8.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.1.8.4.3-1: Common Exception messages for E-UTRAN intra frequency RRC Re-establishment requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.5-1 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-2 |

**Table 6.1.8.4.3-2: MeasConfig-DEFAULT: Additional E-UTRAN intra frequency RRC Re-establishment requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModifyList | Not present | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModifyList | Not present | | |
| quantityConfig | Not present | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| hrpd-PreRegistrationInfo | Not present | | |
| speedDependentParameters | Not present | | |
| } | | | |

## 6.1.8.5 Test requirement

Table 6.1.8.5-1 and Table 6.1.8.4.1-1 defines the primary level settings including test tolerances for E-UTRAN TDD Intra-frequency RRC Re-establishment test case.

**Table 6.1.8.5-1: Cell specific test parameters for E-UTRAN TDD intra-frequency RRC Re-establishment test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | OP.1 TDD | OP.2 TDD | OP.2 TDD | OP.2 TDD | OP.1 TDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 1.54 | -Infinity | -Infinity | -3.79 | 4 | 4 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 7 | -Infinity | -Infinity | 4 | 4 | 4 |
| RSRP Note 3 | dBm/15 KHz | -91 | -Infinity | -Infinity | -94 | -94 | -94 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The RRC re-establishment delay is defined as the time from the start of time period T3, to the moment when the UE starts to send PRACH preambles to cell 2 for sending the *RRCConnectionReestablishmentRequest* message to cell 2.

The RRC re-establishment delay to a known E-UTRA TDD intra frequency cell shall be less than 1.5 s.

The rate of correct RRC re-establishments observed during repeated tests shall be at least 90%.

NOTE: The RRC re-establishment delay in the test is derived from the following expression:

$$T_{re\text{-}establish_delay} = T_{UL_grant} + T_{UE_re\text{-}establish_delay}.$$

Where:

$T_{UL_grant}$ = It is the time required to acquire and process uplink grant from the target cell. The PRACH reception at the system simulator is used as a trigger for the completion of the test; hence $T_{UL_grant}$ is not used.

$T_{UE_re\text{-}establish_delay} = 50\ ms + N_{freq} * T_{search} + T_{SI} + T_{PRACH}$

$N_{freq} = 1$

$T_{search} = 100\ ms$

$T_{SI}$ = 1280 ms; it is the time required for receiving all the relevant system information as defined in TS 36.331 for the target E-UTRAN TDD cell.

$T_{PRACH}$ = 15 ms; it is the additional delay caused by the random access procedure.

This gives a total of 1445 ms, allow 1.5 s in the test case.

## 6.1.9 E-UTRAN FD-FDD Intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA

### 6.1.9.1 Test purpose

To verify that the UE is able to send a RRC Connection Re-establishment Request message to the System Simulator within the specified re-establishment delay limits from the moment it detects a loss in RRC connection.

### 6.1.9.2 Test applicability

This test applies to all types of E-UTRA FD-FDD UE release 13 and forward of UE category M1.

### 6.1.9.3 Minimum conformance requirements

In RRC connected mode the UE shall be capable of sending *RRCConnectionReestablishmentRequest* message within $T_{re\text{-}establish_delay}$ seconds from the moment it detects a loss in RRC connection. The total RRC connection delay ($T_{re\text{-}establish_delay}$) shall be less than:

$$T_{re\text{-}esta}blish_delay = TUL_grant + T_{UE_re\text{-}establish_delay}$$

$T_{UL_grant}$: It is the time required to acquire and process uplink grant from the target cell. The uplink grant is required to transmit *RRCConnectionReestablishmentRequest* message.

The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) is the time between the moments when any of the conditions requiring RRC re-establishment as defined in section 5.3.7 in TS 36.331 [5] is detected by the UE to the time when the UE sends PRACH to the target cell. The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) requirement shall be less than:

$$T_{UE\text{-}re\text{-}establish_delay} = 50 \text{ ms} + N_{freq} * Tsearch + T_{SI} + T_{PRACH}$$

$T_{search}$ = It is the time required by the UE to search the target cell.

$T_{search}$ = It is [100] ms if the target cell is known by the UE; the target cell is known if it has been measured by the UE in the last 5 seconds.

$T_{search}$ = It is 800 ms if the target cell is unknown by the UE; the target cell is unknown if it has not been measured by the UE in the last 5 seconds.

$T_{SI}$ = It is the time required for receiving all the relevant system information according to the reception procedure and the RRC procedure delay of system information blocks defined in TS 36.331 [5] for E-UTRAN cell.

$T_{PRACH}$ = The additional delay caused by the random access procedure; it will be at least 10 ms due to random access occasion and there might be additional delay due to ramping procedure.

$N_{freq}$: It is the total number of E-UTRA frequencies to be monitored for RRC re-establishment; $N_{freq}$ = 1 if the target cell is known.

There is no requirement if the target cell does not contain the UE context.

The normative reference for this requirement is TS 36.133 [4] clause 6.1.2.1 and A.6.1.9.

### 6.1.9.4 Test description

#### 6.1.9.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.20(using only main UE Tx/Rx antenna).
2. The parameter settings for the cells are set up according to Table 6.1.9.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 6.1.9.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.1.9.4.1-1: General test parameters for E-UTRAN FD-FDD intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.20 FDD | As specified in clause A.8.1 |
| MPDCCH parameters | | | DL Reference Measurement Channel R.16 FDD | As specified in clause A.7.1 |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbouring cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF Channel Number | | | 1 | Only one FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| PRACH Configuration | | | PRACH_2CE | As specified in A.9 |
| N310 | | - | 1 | Maximum consecutive out-of-sync indications from lower layers |
| N311 | | - | 1 | Minimum consecutive in-sync indications from lower layers |
| T310 | | ms | 0 | Radio link failure timer; T310 is disabled |
| T311 | | ms | 3000 | RRC re-establishment timer |
| DRX | | | OFF | |
| CP length | | | Normal | |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| PRACH configuration index | | | 4 | As specified in table 5.7.1-2 in TS 36.211 |
| Time offset between cells | | ms | 3 | Asynchronous cells |
| T1 | | s | 5 | |
| T2 | | ms | 200 | |
| T3 | | s | 3 | |

### 6.1.9.4.2 Test procedure

The test consists of 3 successive time periods, with time duration of T1, T2 and T3 respectively. At the start of time period T2, cell 1, which is the active cell, is deactivated. The time period T3 starts after the occurrence of the radio link failure.

1. Ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3AA. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 6.1.9.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 6.1.9.5-1. T2 starts
6. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 6.1.9.5-1. T3 starts

7. If the UE starts to send PRACH preambles to cell 2 for sending the *RRCConnectionReestablishmentRequest* message to cell 2 within 1.5 s from the beginning of time period T3, then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

8. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3AA. Cell 1 is the active cell.

9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

#### 6.1.9.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the condition CEModeA with the following exceptions:

**Table 6.1.9.4.3-1: Common Exception messages for E-UTRAN FD-FDD Intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.5-1 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

**Table 6.1.9.4.3-2: MeasConfig-DEFAULT: Additional E-UTRAN intra frequency FD-FDD RRC Re-establishment requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModifyList | Not present | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModifyList | Not present | | |
| quantityConfig | Not present | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| hrpd-PreRegistrationInfo | Not present | | |
| speedDependentParameters | Not present | | |
| } | | | |

### 6.1.9.5 Test requirement

Table 6.1.9.5-1 defines the primary level settings including test tolerances for E-UTRAN FDD Intra-frequency RRC Re-establishment test case.

**Table 6.1.9.5-1: Cell specific test parameters for E-UTRAN FD-FDD intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.21 (OP.21 FDD) and in D.1.2 (OP.2 FDD) | | OP.21 FDD | OP.21 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD | OP.21 FDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| MPDCCH_RA | dB | | | | | | |
| MPDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 1.54 | -Infinity | -Infinity | -3.79 | 4 | 4 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 7 | -Infinity | -Infinity | 4 | 4 | 4 |
| RSRP Note 3 | dBm/15 KHz | -91 | -Infinity | -Infinity | -94 | -94 | -94 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The RRC re-establishment delay is defined as the time from the start of time period T3, to the moment when the UE starts to send PRACH preambles to cell 2 for sending the *RRCConnectionReestablishmentRequest* message to cell 2.

The RRC re-establishment delay to a known E-UTRA FDD intra frequency cell shall be less than 1.5 s.

The rate of correct RRC re-establishments observed during repeated tests shall be at least 90%.

NOTE: The RRC re-establishment delay in the test is derived from the following expression:

$$T_{re\text{-}establish_delay} = T_{UL_grant} + T_{UE_re\text{-}establish_delay}.$$

Where:

$T_{UL_grant}$ = It is the time required to acquire and process uplink grant from the target cell. The PRACH reception at the system simulator is used as a trigger for the completion of the test; hence $T_{UL_grant}$ is not used.

TUE_re-establish_delay = 50 ms + Nfreq* Tsearch + TSI + TPRACH

$N_{freq}$ = 1

$T_{search}$ = 100 ms

$T_{SI}$ = 1280 ms; it is the time required for receiving all the relevant system information as defined in TS 36.331 [5] for the target E-UTRAN FDD cell.

$T_{PRACH}$ = 15 ms; it is the additional delay caused by the random access procedure.

This gives a total of 1445 ms, within the allowance of 1.5 s in the test case.

## 6.1.10 E-UTRAN HD-FDD Intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA

### 6.1.10.1 Test purpose

To verify that the UE is able to send a RRC Connection Re-establishment Request message to the System Simulator within the specified re-establishment delay limits from the moment it detects a loss in RRC connection.

### 6.1.10.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 13 and forward of UE category M1.

### 6.1.10.3 Minimum conformance requirements

In RRC connected mode the UE shall be capable of sending *RRCConnectionReestablishmentRequest* message within $T_{re\text{-}establish_delay}$ seconds from the moment it detects a loss in RRC connection. The total RRC connection delay ($T_{re\text{-}establish_delay}$) shall be less than:

$$T_{re\text{-}establish_delay} = TUL_grant + T_{UE_re\text{-}establish_delay}$$

$T_{UL_grant}$: It is the time required to acquire and process uplink grant from the target cell. The uplink grant is required to transmit *RRCConnectionReestablishmentRequest* message.

The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) is the time between the moments when any of the conditions requiring RRC re-establishment as defined in section 5.3.7 in TS 36.331 [5] is detected by the UE to the time when the UE sends PRACH to the target cell. The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) requirement shall be less than:

$$TUE\text{-}re\text{-}establish_delay = 50\ ms + Nfreq * Tsearch + TSI + TPRACH$$

$T_{search}$: It is the time required by the UE to search the target cell.

$T_{search}$ = It is [100] ms if the target cell is known by the UE; the target cell is known if it has been measured by the UE in the last 5 seconds.

$T_{search}$ = It is 800 ms if the target cell is unknown by the UE; the target cell is unknown if it has not been measured by the UE in the last 5 seconds.

$T_{SI}$ = It is the time required for receiving all the relevant system information according to the reception procedure and the RRC procedure delay of system information blocks defined in TS 36.331 [5] for E-UTRAN cell.

$T_{PRACH}$ = The additional delay caused by the random access procedure; it will be at least 10 ms due to random access occasion and there might be additional delay due to ramping procedure.

$N_{freq}$: It is the total number of E-UTRA frequencies to be monitored for RRC re-establishment; $N_{freq}$ = 1 if the target cell is known.

There is no requirement if the target cell does not contain the UE context.

The normative reference for this requirement is TS 36.133 [4] clause 6.1.2.1 and A.6.1.10.

### 6.1.10.4 Test description

#### 6.1.10.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.20(using only main UE Tx/Rx antenna).
2. The parameter settings for the cells are set up according to Table 6.1.10.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 6.1.10.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.1.10.4.1-1: General test parameters for E-UTRAN HD-FDD intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.10 HD-FDD | As specified in clause A.8.2 |
| MPDCCH parameters | | | DL Reference Measurement Channel R.6 HD-FDD | As specified in clause A.7.2 |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbouring cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF Channel Number | | | 1 | Only one FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| PRACH Configuration | | | PRACH_2CE | As specified in A.9 |
| N310 | | - | 1 | Maximum consecutive out-of-sync indications from lower layers |
| N311 | | - | 1 | Minimum consecutive in-sync indications from lower layers |
| T310 | | ms | 0 | Radio link failure timer; T310 is disabled |
| T311 | | ms | 3000 | RRC re-establishment timer |
| DRX | | | OFF | |
| CP length | | | Normal | |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| PRACH configuration index | | | 4 | As specified in table 5.7.1-2 in TS 36.211 |
| Time offset between cells | | ms | 3 | Asynchronous cells |
| T1 | | s | 5 | |
| T2 | | ms | 200 | |
| T3 | | s | 3 | |

### 6.1.10.4.2 Test procedure

The test consists of 3 successive time periods, with time duration of T1, T2 and T3 respectively. At the start of time period T2, cell 1, which is the active cell, is deactivated. The time period T3 starts after the occurrence of the radio link failure.

1. Ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3AA. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 6.1.10.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 6.1.10.5-1. T2 starts
6. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 6.1.10.5-1. T3 starts

7. If the UE starts to send PRACH preambles to cell 2 for sending the *RRCConnectionReestablishmentRequest* message to cell 2 within 1.5 s from the beginning of time period T3, then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

8. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3AA. Cell 1 is the active cell.

9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

### 6.1.10.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the condition CEModeA with the following exceptions:

**Table 6.1.10.4.3-1: Common Exception messages for E-UTRAN HD-FDD Intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.5-1 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

**Table 6.1.10.4.3-2: MeasConfig-DEFAULT: Additional E-UTRAN HD-FDD intra frequency RRC Re-establishment requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModifyList | Not present | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModifyList | Not present | | |
| quantityConfig | Not present | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| hrpd-PreRegistrationInfo | Not present | | |
| speedDependentParameters | Not present | | |
| } | | | |

## 6.1.10.5 Test requirement

Table 6.1.10.5-1 defines the primary level settings including test tolerances for E-UTRAN HD-FDD Intra-frequency RRC Re-establishment test case.

**Table 6.1.10.5-1: Cell specific test parameters for E-UTRAN HD-FDD intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.21 (OP.21 FDD) and in D.1.2 (OP.2 FDD) | | OP.21 FDD | OP.21 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD | OP.21 FDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| MPDCCH_RA | dB | | | | | | |
| MPDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 1.54 | -Infinity | -Infinity | -3.79 | 4 | 4 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 7 | -Infinity | -Infinity | 4 | 4 | 4 |
| RSRP Note 3 | dBm/15 KHz | -91 | -Infinity | -Infinity | -94 | -94 | -94 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The RRC re-establishment delay is defined as the time from the start of time period T3, to the moment when the UE starts to send PRACH preambles to cell 2 for sending the *RRCConnectionReestablishmentRequest* message to cell 2.

The RRC re-establishment delay to a known E-UTRA FDD intra frequency cell shall be less than 1.5 s.

The rate of correct RRC re-establishments observed during repeated tests shall be at least 90%.

NOTE: The RRC re-establishment delay in the test is derived from the following expression:

$$T_{re\text{-}establish_delay} = T_{UL_grant} + T_{UE_re\text{-}establish_delay}.$$

Where:

$T_{UL_grant}$ = It is the time required to acquire and process uplink grant from the target cell. The PRACH reception at the system simulator is used as a trigger for the completion of the test; hence $T_{UL_grant}$ is not used.

TUE_re-establish_delay = 50 ms + Nfreq* Tsearch + TSI + TPRACH

$N_{freq}$ = 1

$T_{search}$ = 100 ms

$T_{SI}$ = 1280 ms; it is the time required for receiving all the relevant system information as defined in TS 36.331 [5] for the target E-UTRAN FDD cell.

$T_{PRACH}$ = 15 ms; it is the additional delay caused by the random access procedure.

This gives a total of 1445 ms, within the allowance of 1.5 s in the test case.

## 6.1.11 E-UTRAN TDD Intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA

### 6.1.11.1 Test purpose

To verify that the UE is able to send a RRC Connection Re-establishment Request message to the System Simulator within the specified re-establishment delay limits from the moment it detects a loss in RRC connection.

### 6.1.11.2 Test applicability

This test applies to all type of E-UTRA TDD UE release 13 and forward of UE category M1.

### 6.1.11.3 Minimum conformance requirements

In RRC connected mode the UE shall be capable of sending *RRCConnectionReestablishmentRequest* message within $T_{re\text{-}establish_delay}$ seconds from the moment it detects a loss in RRC connection. The total RRC connection delay ($T_{re\text{-}establish_delay}$) shall be less than:

$$T_{re\text{-}establish_delay} = T_{UL_grant} + T_{UE_re\text{-}establish_delay}$$

$T_{UL_grant}$: It is the time required to acquire and process uplink grant from the target PCell. The uplink grant is required to transmit *RRCConnectionReestablishmentRequest* message.

The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) is the time between the moments when any of the conditions requiring RRC re-establishment as defined in clause 5.3.7 in TS 36.331 [2] is detected by the UE to the time when the UE sends PRACH to the target PCell. The UE re-establishment delay ($T_{UE_re\text{-}establish_delay}$) requirement shall be less than:

$$T_{UE\text{-}re\text{-}establish_delay} = 50\ ms + N_{freq} * Tsearch + T_{SI} + T_{PRACH}$$

$T_{search}$: It is the time required by the UE to search the target PCell.

$T_{search}$ = It is 100 ms if the target PCell is known by the UE; the target PCell is known if it has been measured by the UE in the last 5 seconds.

$T_{search}$ = It is 800 ms if the target PCell is unknown by the UE; the target PCell is unknown if it has not been measured by the UE in the last 5 seconds.

$T_{SI}$ = It is the time required for receiving all the relevant system information according to the reception procedure and the RRC procedure delay of system information blocks defined in TS 36.331 [2] for the target PCell.

$T_{PRACH}$ = The additional delay caused by the random access procedure; it will be at least 10 ms due to random access occasion and there might be additional delay due to ramping procedure.

$N_{freq}$: It is the total number of E-UTRA frequencies to be monitored for RRC re-establishment; $N_{freq}$ = 1 if the target PCell is known.

There is no requirement if the target cell does not contain the UE context.

The normative reference for this requirement is TS 36.133 [4] clause 6.1.2.1 and A.6.1.11.

### 6.1.11.4 Test description

#### 6.1.11.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.20 (using only main UE Tx/Rx antenna).
2. The parameter settings for the cells are set up according to Table 6.1.11.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 6.1.11.4.3.
5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.1.11.4.1-1: General test parameters for E-UTRAN TDD intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA test case**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.16 TDD | As specified in clause A.8.3 |
| MPDCCH parameters | | | DL Reference Measurement Channel R.14 TDD | As specified in clause A.7.3 |
| Initial conditions | Active cell | | Cell 1 | |
| | Neighbouring cell | | Cell 2 | |
| Final condition | Active cell | | Cell 2 | |
| E-UTRA RF Channel Number | | | 1 | Only one TDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| PRACH Configuration | | | PRACH_2CE | As specified in A.9 |
| N310 | | - | 1 | Maximum consecutive out-of-sync indications from lower layers |
| N311 | | - | 1 | Minimum consecutive in-sync indications from lower layers |
| T310 | | ms | 0 | Radio link failure timer; T310 is disabled |
| T311 | | ms | 3000 | RRC re-establishment timer |
| DRX | | | OFF | |
| CP length | | | Normal | |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211 |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211 |
| PRACH configuration index | | | 53 | As specified in table 5.7.1-3 in TS 36.211 |
| Time offset between cells | | µs | 3 | Synchronous cells |
| T1 | | s | 5 | |
| T2 | | ms | 200 | |
| T3 | | s | 3 | |

### 6.1.11.4.2 Test procedure

The test consists of 3 successive time periods, with time duration of T1, T2 and T3 respectively. At the start of time period T2, cell 1, which is the active cell, is deactivated. The time period T3 starts after the occurrence of the radio link failure.

1. 10. Ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3AA.Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 6.1.11.5-1. Propagation conditions are set according to Annex B clause B.1. T1 starts.
3. SS shall transmit an *RRCConnectionReconfiguration* message.
4. The UE shall transmit *RRCConnectionReconfigurationComplete* message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 6.1.11.5-1. T2 starts

6. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 6.1.11.5-1. T3 starts

7. If the UE starts to send PRACH preambles to cell 2 for sending the *RRCConnectionReestablishmentRequest* message to cell 2 within 1.5 s from the beginning of time period T3, then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

8. After T3 expires, cause UE handover back to Cell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3AA. Cell 1 is the active cell.

9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

### 6.1.11.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 6 with the condition CEModeA with the following exceptions:

**Table 6.1.11.4.3-1: Common Exception messages for E-UTRAN TDD Intra-frequency RRC Re-establishment requirement for Cat-M1 UE in CEModeA**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.5-1 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-2 |

**Table 6.1.11.4.3-2: MeasConfig-DEFAULT: Additional E-UTRAN intra frequency RRC Re-establishment requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModifyList | Not present | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModifyList | Not present | | |
| quantityConfig | Not present | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| hrpd-PreRegistrationInfo | Not present | | |
| speedDependentParameters | Not present | | |
| } | | | |

## 6.1.11.5 Test requirement

Table 6.1.11.5-1 and Table 6.1.11.4.1-1 defines the primary level settings including test tolerances for E-UTRAN TDD Intra-frequency RRC Re-establishment test case.

**Table 6.1.11.5-1: Cell specific test parameters for E-UTRAN TDD intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.2.11 (OP.11 TDD) and in D.2.2 (OP.2 TDD) | | OP.11 TDD | OP.11 TDD | OP.2 TDD | OP.2 TDD | OP.2 TDD | OP.11 TDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| MPDCCH_RA | dB | | | | | | |
| MPDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 1.54 | -Infinity | -Infinity | -3.79 | 4 | 4 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 7 | -Infinity | -Infinity | 4 | 4 | 4 |
| RSRP Note 3 | dBm/15 KHz | -91 | -Infinity | -Infinity | -94 | -94 | -94 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The RRC re-establishment delay is defined as the time from the start of time period T3, to the moment when the UE starts to send PRACH preambles to cell 2 for sending the *RRCConnectionReestablishmentRequest* message to cell 2.

The RRC re-establishment delay to a known E-UTRA TDD intra frequency cell shall be less than 1.5 s.

The rate of correct RRC re-establishments observed during repeated tests shall be at least 90%.

NOTE: The RRC re-establishment delay in the test is derived from the following expression:

$$T_{re\text{-}establish_delay} = T_{UL_grant} + T_{UE_re\text{-}establish_delay}.$$

Where:

$T_{UL_grant}$ = It is the time required to acquire and process uplink grant from the target cell. The PRACH reception at the system simulator is used as a trigger for the completion of the test; hence $T_{UL_grant}$ is not used.

$T_{UE_re\text{-}establish_delay} = 50 \text{ ms} + N_{freq} * T_{search} + T_{SI} + T_{PRACH}$

$N_{freq} = 1$

$T_{search} = 100$ ms

$T_{SI}$ = 1280 ms; it is the time required for receiving all the relevant system information as defined in TS 36.331 for the target E-UTRAN TDD cell.

$T_{PRACH}$ = 15 ms; it is the additional delay caused by the random access procedure.

This gives a total of 1445 ms, allow 1.5 s in the test case.

## 6.1.15 HD-FDD Intra-frequency RRC Re-establishment for category NB1 UE in In-Band mode under normal coverage

Editor's notes: This clause is incomplete, the following items are TBD

- Test frequencies is TBD.
- Test system uncertainty and tolerance is undefined.
- Connection diagram is TBD.
- Test procedure is incomplete.
- Test requirement of TS36.133 included TBD and bracket.
- Test configuration for eCell, PRB, number of E-UTRA Cell used is pending confirmation with RAN4.

### 6.1.15.1 Test purpose

The purpose is to verify that the NB-IoT FDD intra-frequency RRC re-establishment delay is within the specified limits from the moment it detects a loss in RRC connection.

### 6.1.15.2 Test applicability

This test case applies to all types of UE release 13 and forward of UE category NB1. Applicability requires support for User Plane CIoT EPS optimization.

### 6.1.15.3 Minimum conformance requirements

In RRC connected mode the UE shall be capable of sending *RRCConnectionReestablishmentRequest* message within $T_{re\text{-}establish_delay_NB\text{-}IoT}$ seconds from the moment it detects a loss in RRC connection. The total RRC connection delay ($T_{re\text{-}establish_delay_NB\text{-}IoT}$) shall be less than:

$$T_{re\text{-}establish_delay_NB\text{-}IoT} = T_{UL_grant} + T_{UE_re\text{-}establish_delay_NB\text{-}IoT}$$

$T_{UL_grant}$: It is the time required to acquire and process uplink grant from the target cell. The uplink grant is required to transmit *RRCConnectionReestablishmentRequest* message.

The UE re-establishment delay ($T_{UE_re\text{-}establish_delay_NB\text{-}IoT}$) is the time between the moments when any of the conditions requiring RRC re-establishment as defined in clause 5.3.7 in TS 36.331 [2] is detected by the UE to the time when the UE sends NPRACH preamble to the target cell. The UE re-establishment delay ($T_{UE_re\text{-}establish_delay_NB\text{-}IoT}$) requirement shall be less than:

$$T_{UE\text{-}re\text{-}establish_delay_NB\text{-}IoT} = 100\ ms + N_{NB\text{-}Iot\text{-}freq} * Tsearch_NB\text{-}IoT + T_{SI_NB\text{-}IoT} + T_{NPRACH_NB\text{-}IoT}$$

$T_{search_NB\text{-}IoT}$: It is the time required by the UE to search the target cell:

If the target cell is known, then $T_{search}$ = 0 ms. then $T_{search}$ = 0 ms. If the target cell is unknown and signal quality is sufficient for successful cell detection on the first attempt, then $T_{search}$ = 80 ms. Otherwise,

$T_{search}$ = [1400] ms if Q ≥ -6dB or, $T_{search}$ = [14800] ms if -15dB ≤ Q < -6dB.

Note: Q is the NSCH Ês/Iot of target cell. A cell is unknown if it has not been measured by the UE in the last 5 seconds. Otherwise it is known.

$T_{SI_NB\text{-}IoT}$: It is the time required for receiving all the relevant system information according to the reception procedure and the RRC procedure delay of system information blocks defined in TS 36.331 [2] for the target cell.

$T_{NPRACH_NB\text{-}IoT}$: The additional delay caused by the random access procedure; it will be at least 40 ms due to random access occasion and there might be additional delay due to ramping procedure.

$N_{NB\text{-}Iot\text{-}freq}$: It is the total number of NB-IoT frequencies to be monitored for RRC re-establishment; $N_{NB\text{-}Iot\text{-}freq} = 1$ if the target cell is known.

There is no requirement if the target cell does not contain the UE context.

The normative reference for this requirement is TS 36.133 [4] clause 6.5 and A.6.1.15.

## 6.1.15.4 Test description

### 6.1.15.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 8.1.1.

Frequencies to be tested: TBD.

Channel Bandwidth to be tested: eCell is 10 MHz as defined in TS 36.508 [7] clause 4.3.1, nCell is 200kHz as defined in TS 36.508 [7] clause 8.1.3.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure [TBD].
2. The parameter settings for the cells are set up according to Table 6.1.15.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 6.1.15.4.3.
5. There is one NB-IoT carrier and one E-UTRA FDD carrier and three cells specified in the test. nCell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.1.15.4.1-1: General test parameters for HD-FDD Intra-frequency RRC Re-establishment for UE category NB1 in In-Band mode under normal coverage**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| NB-IOT operational mode | | | In-band | |
| Initial condition | Active cell | | nCell1 | |
| | Neighbour cells | | eCell1, nCell2 | |
| Final condition | Active cell | | nCell2 | |
| E-UTRA RF Channel Number | | | 1 | One carrier frequency is used for eCell. |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| NPRACH Configuration | | | 1 | Refer to Table A.10.3-1 |
| N310 | | - | 1 | Maximum consecutive out-of-sync indications from lower layers |
| N311 | | - | 1 | Minimum consecutive in-sync indications from lower layers |
| T310 | | ms | 0 | Radio link failure timer; T310 is disabled |
| T311 | | ms | TBD | RRC re-establishment timer |
| DRX | | | OFF | |
| T1 | | s | 5 | |
| T2 | | ms | [400] | |
| T3 | | s | [10] | |

### 6.1.15.4.2 Test procedure

The test consists of 3 successive time periods, with time duration of T1, T2 and T3 respectively. At the start of time period T2, nCell 1, which is the active cell, is deactivated. The time period T3 starts after the occurrence of the radio link failure.

1. Ensure the UE is in State 2A-NB with UP CIoT Optimisation according to TS 36.508 [7] clause 8.1.5. nCell 1 is the active cell.
2. Set the parameters according to T1 in Table 6.1.15.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 6.1.15.5-1. T2 starts.
4. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 6.1.15.5-1. T3 starts.
5. If the UE starts to send NPRACH preambles to nCell 2 for sending the *RRCConnectionReestablishmentRequest-NB* message to nCell 2 within [TBD ms] from the beginning of time period T3, then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
6. After T3 expires, switch off the UE. Then ensure the UE is in State 2A-NB with UP CIoT Optimisation according to TS 36.508 [7] clause 8.1.5. nCell 1 is the active cell.
7. Repeat step 2-6 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

#### 6.1.15.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 8.1.6.

### 6.1.15.5 Test requirement

Table 6.1.15.5-1 and Table 6.1.15.5-2 defines the primary level settings including test tolerances for HD-FDD Intra-frequency RRC Re-establishment test case.

**Table 6.1.15.5-1: nCell 1, nCell 2 specific test parameters for HD-FDD Intra-frequency RRC Re-establishment for category NB1 UE in In-Band mode under normal coverage**

| | | nCell 1 | | | nCell 2 | | |
|---|---|---|---|---|---|---|---|
| | | **T1** | **T2** | **T3** | **T1** | **T2** | **T3** |
| $BW_{channel}$ | kHz | 200 | | | 200 | | |
| PRB location within eCell | - | 30 | | | 30 | | |
| NPDSCH parameters defined in clause A.10.2 | | R.14 HD-FDD | | | R.14 HD-FDD | | |
| NPDCCH parameters defined in clause A.10.1 | | R.26 HD-FDD | | | R.26 HD-FDD | | |
| NOCNG Patterns defined in clause D.3 | - | NOP.1 FDD | | | NOP.1 FDD | | |
| NPBCH_RA | dB | 0 | | | 0 | | |
| NPBCH_RB | dB | | | | | | |
| NPSS_RA | dB | | | | | | |
| NSSS_RA | dB | | | | | | |
| NPDCCH_RA | dB | | | | | | |
| NPDCCH_RB | dB | | | | | | |
| NPDSCH_RA | dB | | | | | | |
| NPDSCH_RB | dB | | | | | | |
| NOCNG_RA Note 1 | dB | | | | | | |
| NOCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ | dBm/15 kHz | Specified in Table 6.1.15.5-2 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 7+TT | -Infinity | -Infinity | 4+TT | 4+TT | 4+TT |
| $\hat{E}_s / I_{ot}$ Note2 | dB | 1.54+TT | -Infinity | -Infinity | -3.79+TT | 4+TT | 4+TT |
| NRSRP Note2 | dBm/15 kHz | -91+TT | -Infinity | -Infinity | -94+TT | -94+TT | -94+TT |
| Propagation Condition | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x1 | | | 1x1 | | |
| Timing offset to nCell 1 | ms | - | | | 3 | | |

Note 1: NOCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Es/Iot and NRSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 6.1.15.5-2: eCell 1 specific test parameters for HD-FDD Intra-frequency RRC Re-establishment for category NB1 UE in In-Band mode under normal coverage**

| | | eCell 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| $BW_{channel}$ | MHz | 10 | | |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| [PCFICH_RB] | dB | | | |
| [PDCCH_RA] | dB | | | |
| [PDCCH_RB] | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | |
| $\hat{E}_s / N_{oc}$ | dB | 4+TT | 4+TT | 4+TT |
| $\hat{E}_s / I_{ot}$ Note2 | dB | 4+TT | 4+TT | 4+TT |
| RSRP Note2 | dBm/15 kHz | -94+TT | -94+TT | -94+TT |
| Propagation Condition | | AWGN | | |
| Antenna Configuration | | 1x1 | | |

Note 1: OCNG shall be used such that the eCell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power $N_{oc}$.

Note 3: Es/Iot and RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The RRC re-establishment delay is defined as the time from the start of time period T3, to the moment when the UE starts to send NPRACH preambles to ncell 2 for sending the *RRCConnectionReestablishmentRequest-NB* message to ncell 2.

The RRC re-establishment delay to a unknown NB-IoT FDD intra frequency cell shall be less than TBD s.

The rate of correct RRC re-establishments observed during repeated tests shall be at least 90%.

NOTE: The RRC re-establishment delay in the test is derived from the following expression:

$$T_{re\text{-}establish_delay} = T_{UL_grant} + T_{UE\text{-}re\text{-}establish_delay_NB\text{-}IoT}.$$

Where:

$T_{UL_grant}$ = It is the time required to acquire and process uplink grant from the target cell. The NPRACH reception at the system simulator is used as a trigger for the completion of the test; hence $T_{UL_grant}$ is not used.

$T_{UE\text{-}re\text{-}establish_delay_NB\text{-}IoT} = 100\ ms + N_{NB\text{-}Iot\text{-}freq} * T_{search_NB\text{-}IoT} + T_{SI_NB\text{-}IoT} + T_{NPRACH_NB\text{-}IoT}$

$N_{NB\text{-}Iot\text{-}freq} = 1$

$T_{search_NB\text{-}IoT}$ = [1400] ms

$T_{SI_NB\text{-}IoT}$ = TBD ms; it is the time required for receiving all the relevant system information as defined in TS 36.331 for the target NB-IoT FDD cell.

$T_{NPRACH_NB\text{-}IoT}$ = [80] ms; it is the additional delay caused by the random access procedure.

## 6.1.16 HD-FDD Inter-frequency RRC Re-establishment for category NB1 UE in In-Band mode under enhanced coverage

Editor's notes: This clause is incomplete, the following items are TBD

- Test frequencies is TBD.
- Test system uncertainty and tolerance is TBD
- Connection diagram is TBD
- Test configuration for nCell PRB and number of E-UTRAN Cell is pending confirmation by RAN4.
- Test requirement and procedure still bracket.

### 6.1.16.1 Test purpose

The purpose is to verify that the NB-IoT FDD inter-frequency RRC re-establishment delay is within the specified limits from the moment it detects a loss in RRC connection.

### 6.1.16.2 Test applicability

This test case applies to all types of UE release 13 and forward of UE category NB1. Applicability requires support for user plane CIoT EPS optimization.

### 6.1.16.3 Minimum conformance requirements

In RRC connected mode the UE shall be capable of sending *RRCConnectionReestablishmentRequest* message within $T_{re\text{-}establish_delay_NB\text{-}IoT}$ seconds from the moment it detects a loss in RRC connection. The total RRC connection delay ($T_{re\text{-}establish_delay_NB\text{-}IoT}$) shall be less than:

$$T_{re\text{-}establish_delay_NB\text{-}IoT} = T_{UL_grant} + T_{UE_re\text{-}establish_delay_NB\text{-}IoT}$$

$T_{UL_grant}$: It is the time required to acquire and process uplink grant from the target cell. The uplink grant is required to transmit *RRCConnectionReestablishmentRequest* message.

The UE re-establishment delay ($T_{UE_re\text{-}establish_delay_NB\text{-}IoT}$) is the time between the moments when any of the conditions requiring RRC re-establishment as defined in clause 5.3.7 in TS 36.331 [2] is detected by the UE to the time when the UE sends NPRACH preamble to the target cell. The UE re-establishment delay ($T_{UE_re\text{-}establish_delay_NB\text{-}IoT}$) requirement shall be less than:

$$T_{UE\text{-}re\text{-}establish_delay_NB\text{-}IoT} = 100\ ms + N_{NB\text{-}Iot\text{-}freq} * Tsearch_NB\text{-}IoT + T_{SI_NB\text{-}IoT} + T_{NPRACH_NB\text{-}IoT}$$

$T_{search_NB\text{-}IoT}$: It is the time required by the UE to search the target cell:

If the target cell is known, then $T_{search}$ = 0 ms. then $T_{search}$ = 0 ms. If the target cell is unknown and signal quality is sufficient for successful cell detection on the first attempt, then $T_{search}$ = 80 ms. Otherwise, $T_{search}$ = [1400] ms if Q ≥ -6dB or, $T_{search}$ = [14800] ms if -15dB ≤ Q < -6dB.

Note: Q is the NSCH Ês/Iot of target cell. A cell is unknown if it has not been measured by the UE in the last 5 seconds. Otherwise it is known.

$T_{SI_NB\text{-}IoT}$: It is the time required for receiving all the relevant system information according to the reception procedure and the RRC procedure delay of system information blocks defined in TS 36.331 [2] for the target cell.

$T_{NPRACH_NB\text{-}IoT}$: The additional delay caused by the random access procedure; it will be at least 40 ms due to random access occasion and there might be additional delay due to ramping procedure.

$N_{NB\text{-}Iot\text{-}freq}$: It is the total number of NB-IoT frequencies to be monitored for RRC re-establishment; $N_{NB\text{-}Iot\text{-}freq} = 1$ if the target cell is known.

There is no requirement if the target cell does not contain the UE context.

The normative reference for this requirement is TS 36.133 [4] clause 6.5 and A.6.1.16.

## 6.1.16.4 Test description

### 6.1.16.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 8.1.1.

Frequencies to be tested: TBD.

Channel Bandwidth to be tested: eCell is 10 MHz as defined in TS 36.508 [7] clause 4.3.1, nCell is 200kHz as defined in TS 36.508 [7] clause 8.1.3.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure [TBD].
2. The parameter settings for the cells are set up according to Table 6.1.16.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 6.1.16.4.3.
5. There is one NB-IoT carrier and one E-UTRA FDD carrier and three cells specified in the test. nCell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.1.16.4.1-1: General test parameters for HD-FDD Inter-frequency RRC Re-establishment for category NB1 UE in In-Band mode under enhanced coverage**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| NB-IOT operational mode | | | In-band | |
| Initial condition | Active cell | | nCell1 | |
| | Neighbour cells | | eCell1, nCell2 | |
| Final condition | Active cell | | nCell2 | |
| E-UTRA RF Channel Number | | | 1 | One carrier frequency is used for eCell. |
| Access Barring Information | | - | Not Sent | No additional delays in random access procedure. |
| NPRACH Configuration | | | 1 | Refer to Table A.10.3-1 |
| N310 | | - | 1 | Maximum consecutive out-of-sync indications from lower layers |
| N311 | | - | 1 | Minimum consecutive in-sync indications from lower layers |
| T310 | | ms | 0 | Radio link failure timer; T310 is disabled |
| T311 | | ms | TBD | RRC re-establishment timer |
| DRX | | | OFF | |
| T1 | | s | 5 | |
| T2 | | ms | [800] | |
| T3 | | s | [10] | |

### 6.1.16.4.2 Test procedure

The test consists of 3 successive time periods, with time duration of T1, T2 and T3 respectively. At the start of time period T2, nCell 1, which is the active cell, is deactivated. The time period T3 starts after the occurrence of the radio link failure.

1. Ensure the UE is in State 2A-NB with UP CIoT Optimisation according to TS 36.508 [7] clause 8.1.5. nCell 1 is the active cell.

2. Set the parameters according to T1 in Table 6.1.16.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 6.1.16.5-1. T2 starts
4. When T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 6.1.16.5-1. T3 starts
5. If the UE starts to send NPRACH preambles to nCell2 for sending the*RRCConnectionReestablishmentRequest-NB* message to nCell 2 within [TBD ms] from the beginning of time period T3, then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
6. After T3 expires, cause UE handover back to nCell 1 (if the handover fails, switch off the UE) or switch off the UE. Then ensure the UE is in State 2A-NB with UP CIoT Optimisation according to TS 36.508 [7] clause 8.1.5. nCell 1 is the active cell.
7. Repeat step 2-6 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

#### 6.1.16.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 8.1.6.

### 6.1.16.5 Test requirement

Table 6.1.16.5-1 and 6.1.16.5-2 defines the primary level settings including test tolerances for HD- FDD Inter-frequency RRC Re-establishment test case.

**Table 6.1.16.5-1: nCell 1, nCell 2 specific test parameters for HD-FDD Inter-frequency RRC Re-establishment for category NB1 UE in In-Band mode under enhanced coverage**

| | | nCell 1 | | | nCell 2 | | |
|---|---|---|---|---|---|---|---|
| | | **T1** | **T2** | **T3** | **T1** | **T2** | **T3** |
| $BW_{channel}$ | kHz | 200 | | | 200 | | |
| PRB location within eCell | - | [30] | | | [30] | | |
| NPDSCH parameters defined in clause A.10.2 | | R.14 HD-FDD | | | R.14 HD-FDD | | |
| NPDCCH parameters defined in clause A.10.1 | | R.26 HD-FDD | | | R.26 HD-FDD | | |
| NOCNG Patterns defined in clause D.3 | - | NOP.1 FDD | | | NOP.1 FDD | | |
| NPBCH_RA | dB | 0 | | | 0 | | |
| NPBCH_RB | dB | | | | | | |
| NPSS_RA | dB | | | | | | |
| NSSS_RA | dB | | | | | | |
| NPDCCH_RA | dB | | | | | | |
| NPDCCH_RB | dB | | | | | | |
| NPDSCH_RA | dB | | | | | | |
| NPDSCH_RB | dB | | | | | | |
| NOCNG_RA Note 1 | dB | | | | | | |
| NOCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ | dBm/15 kHz | Specified in Table A.6.1.16.1-3 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 7+TT | -Infinity | -Infinity | -Infinity | -12.6+TT | -12.6+TT |
| $\hat{E}_s / I_{ot}$ Note2 | dB | 1.54+TT | -Infinity | -Infinity | -Infinity | -12.6+TT | -12.6+TT |
| NRSRP Note2 | dBm/15 kHz | -91+TT | -Infinity | -Infinity | -Infinity | -110.6+TT | -110.6+TT |
| Propagation Condition | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x1 | | | 1x1 | | |
| Timing offset to nCell 1 | ms | - | | | 3 | | |

Note 1: NOCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Es/Iot and NRSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 6.1.16.5-2: eCell 1 specific test parameters for HD-FDD Inter-frequency RRC Re-establishment for category NB1 UE in In-Band mode under enhanced coverage**

| | | eCell 1 | | |
|---|---|---|---|---|
| | | **T1** | **T2** | **T3** |
| $BW_{channel}$ | MHz | 10 | | |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| [PCFICH_RB] | dB | | | |
| [PDCCH_RA] | dB | | | |
| [PDCCH_RB] | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $N_{oc}$ | dBm/15 kHz | [-98]+TT | | |
| $\hat{E}_s / N_{oc}$ | dB | [-12.6] +TT | [-12.6] +TT | [-12.6] +TT |
| $\hat{E}_s / I_{ot}$ Note2 | dB | [-12.6] +TT | [-12.6] +TT | [-12.6] +TT |
| RSRP Note2 | dBm/15 kHz | [-110.6] +TT | [-110.6] +TT | [-110.6] +TT |
| Propagation Condition | | AWGN | | |
| Antenna Configuration | | 1x1 | | |

Note 1: OCNG shall be used such that the eCell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power $N_{oc}$.

Note 3: Es/Iot and RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The RRC re-establishment delay is defined as the time from the start of time period T3, to the moment when the UE starts to send NPRACH preambles to cell 2 for sending the *RRCConnectionReestablishmentRequest-NB* message to ncell 2.

The RRC re-establishment delay to a unknown NB-IoT FDD inter frequency cell shall be less than TBD s.

The rate of correct RRC re-establishments observed during repeated tests shall be at least 90%.

NOTE: The RRC re-establishment delay in the test is derived from the following expression:

$$T_{re\text{-}establish_delay} = T_{UL_grant} + T_{UE\text{-}re\text{-}establish_delay_NB\text{-}IoT}.$$

Where:

$T_{UL_grant}$ = It is the time required to acquire and process uplink grant from the target cell. The NPRACH reception at the system simulator is used as a trigger for the completion of the test; hence $T_{UL_grant}$ is not used.

$T_{UE\text{-}re\text{-}establish_delay_NB\text{-}IoT} = 100\ ms + N_{NB\text{-}Iot\text{-}freq} * T_{search_NB\text{-}IoT} + T_{SI_NB\text{-}IoT} + T_{NPRACH_NB\text{-}IoT}$

$N_{NB\text{-}Iot\text{-}freq} = 2$

$T_{search_NB\text{-}IoT}$ = [14800] ms

$T_{SI_NB\text{-}IoT}$ = TBD ms; it is the time required for receiving all the relevant system information as defined in TS 36.331 for the target NB-IoT FDD cell.

$T_{NPRACH_NB\text{-}IoT}$ = [80] ms; it is the additional delay caused by the random access procedure.

# 6.2 Random Access

## 6.2.1 E-UTRAN FDD - Contention Based Random Access Test

### 6.2.1.1 Test purpose

To verify that the UE behaviour of the random access procedure is according to the E-UTRAN FDD contention based random access requirements in an AWGN channel model and that the PRACH power settings and timing are within the specified limits.

### 6.2.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

### 6.2.1.3 Minimum conformance requirements

The random access procedure is used when establishing the layer 1 communication between the UE and E-UTRAN. The random access is as defined in TS 36.213 [8] clause 6 and the control of the RACH transmission is as defined in TS 36.321 [11] clause 5.1.

The UE shall have capability to calculate PRACH transmission power according to the PRACH power formula as defined in TS 36.213 [8] clause 6.1 and apply this power level at the first preamble or additional preambles. The absolute power applied to the first preamble shall have an accuracy as defined in table 6.3.5.1.1-1 of TS 36.101 [2] clause 6.3.5.1.1. The relative power applied to additional preambles shall have an accuracy as specified in table 6.3.5.2.1-1 of TS 36.101 [2] clause 6.3.5.2.1.

The UE shall indicate a Random Access problem to upper layers if the maximum number of preambles transmission counter has been reached.

The UE may stop monitoring for Random Access Response(s) and shall transmit the msg3 if the Random Access Response contains a Random Access Preamble identifier corresponding to the transmitted Random Access Preamble.

The UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if all received Random Access Responses contain Random Access Preamble identifiers that do not match the transmitted Random Access Preamble.

The UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if no Random Access Response is received within the RA Response window defined in clause 5.1.4 TS 36.321.

The UE shall re-transmit the msg3 upon the reception of a NACK on msg3.

The UE shall send ACK if the Contention Resolution is successful.

The UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires unless the received message includes a UE Contention Resolution Identity MAC control element and the UE Contention Resolution Identity included in the MAC control element matches the CCCH SDU transmitted in the uplink message.

The UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if the Contention Resolution Timer expires.

The normative reference for this requirement is TS 36.133 [4] clause 6.2.2.1 and A.6.2.1.

### 6.2.1.4 Test description

#### 6.2.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18.

2. Propagation conditions are set according to Annex B clause B.0.

3. There is one E-UTRA FDD carrier and one cell specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

#### 6.2.1.4.2 Test procedure

The test consists of a single cell. The UE has a downlink PDSCH allocated centred on the centre sub-carrier. The E-UTRAN shall not explicitly signal a Random Access Preamble ID to the UE.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2.

2. Set the parameters according to Tables 6.2.1.5-1 and 6.2.1.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. The UE shall establish a connection setup with SS, the random access procedure within the connection setup is used in the test.

4. Test 1: Correct behaviour when receiving random access response reception

   4.1. In Test 1, the UE shall send the preamble to the SS. In response to the first 4 preambles, the SS shall transmit a random access response not corresponding to the transmitted random access preamble.

   4.2. The UE shall consider the random access response reception not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires if all received random access response contain random access preamble identifiers that do not match the transmitted random access preamble.

   4.3. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after 5 preambles have been received by the SS.

   4.4. The UE shall consider this random access response reception successful and transmit the msg3.

   4.5. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.1.5-3 and 6.2.1.5-5. Measure the relative power and timing applied to additional preambles (last 4 preambles) and it shall not exceed the values specified in Tables 6.2.1.5-4 and 6.2.1.5-5.

5. Test 2: Correct behaviour when not receiving random access response reception

   5.1. Repeat step 1-3.

   5.2. In Test 2, the UE shall send the preamble to the SS. The SS shall not respond to the first 4 preambles.

   5.3. The UE shall consider the random access response reception not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires if no random access response is received within the RA Response window of 10 sub-frames.

   5.4. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after 5 preambles have been received by the SS.

   5.5. The UE shall consider this random access response reception successful and transmit the msg3.

   5.6. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.1.5-3 and 6.2.1.5-5. Measure the relative power and timing applied to additional preambles (last 4 preambles) and it shall not exceed the values specified in Tables 6.2.1.5-4 and 6.2.1.5-5.

6. Test 3: Correct behaviour when receiving a NACK on msg3

   6.1. Repeat step 1-3.

6.2. In Test 3, the UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

6.3. The UE shall consider this random access response reception successful and transmit the msg3.

6.4. The SS shall send NACK to all UE msg3 following a successful random access response.

6.5. The UE shall consider this contention resolution not successful then re-transmit the msg3 upon the reception of a NACK on msg3 until the maximum number of HARQ retransmission is reached.

7. Test 4: Correct behaviour when receiving an incorrect message over Temporary C-RNTI

7.1. Repeat step 1-3.

7.2. In Test 4, the UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

7.3. The UE shall consider this random access response reception successful and transmit the msg3.

7.4. The SS shall send a message addressed to the Temporary C-RNTI with a UE contention resolution identity included in the MAC control element not matching the CCCH SDU transmitted in msg3 uplink message.

7.5. The UE shall consider the contention resolution not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires.

8. Test 5: Correct behaviour when receiving a correct message over Temporary C-RNTI

8.1. In Test 5, the SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

8.2. The UE shall consider this random access response reception successful and transmit the msg3.

8.3. The SS shall send a message addressed to the Temporary C-RNTI with a UE contention resolution identity included in the MAC control element matching the CCCH SDU transmitted in the msg3 uplink message.

8.4. The UE shall send ACK and consider the contention resolution successful and the random access procedure successfully completed.

9. Test 6: Correct behaviour when contention resolution timer expires

9.1. Repeat step 1-3.

9.2. In Test 6, the UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

9.3. The UE shall consider this random access response reception successful and transmit the msg3.

9.4. The SS shall send an ACK for msg3 but not send msg4 allowing the contention resolution timer to expire.

9.5. The UE shall consider the contention resolution not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires.

9.6. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

9.7. The UE shall consider this random access response reception successful and transmit the msg3.

#### 6.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.2.1.4.3-1: Common Exception messages for E-UTRAN FDD - Contention Based Random Access test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | H.2.6-1<br>H.2.6-2<br>H.2.6-3 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

**Table 6.2.1.4.3-2: *UplinkPowerControlCommon-DEFAULT*: Additional E-UTRAN FDD - Contention Based Random Access test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-25 UplinkPowerControlCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlCommon-DEFAULT ::= SEQUENCE { | | | |
| p0-NominalPUSCH | -101 (-101 dBm) | | |
| } | | | |

## 6.2.1.5 Test requirement

Tables 6.2.1.5-1 and 6.2.1.5-2 define the primary level settings for E-UTRAN FDD - contention based random access test. Table 6.2.1.5-5 defines the uplink timing error limit including test tolerances.

**Table 6.2.1.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD - Contention Based Random Access test**

| Parameter | Unit | Value | Comments |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern Note 1 | | OP.1/2 FDD Note 1 | As defined in D.1.1/2. |
| PDSCH parameters Note 4 | | DL Reference Measurement Channel R.0 FDD Note 4 | As defined in A.1.1. |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As defined in A.1.2. |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |

| | | | |
|---|---|---|---|
| $\hat{E}_s / I_{ot}$ | dB | 3 | |
| $N_{oc}$ | dBm/15 KHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 3 | |
| Io Note 2 | dBm/9 MHz | -65.5 | |
| RSRP Note 3 | dBm/15 KHz | -95 | |
| referenceSignalPower | dBm/15 KHz | -5 | As defined in clause 6.3.2 in 3GPP TS 36.331 [5]. |
| Configured UE transmitted power ( $P_{CMAX}$ ) | dBm | 23 | As defined in clause 6.2.5 in 3GPP TS 36.101 [2]. |
| PRACH Configuration Index | - | 4 | As defined in table 5.7.1-2 in 3GPP TS 36.211 [9]. |
| Back off Parameter Index | - | 2 | As defined in table 7.2-1 in 3GPP TS 36.321 [11]. |
| Propagation Condition | - | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. The OCNG pattern is chosen during the test according to the presence of a DL reference measurement channel.
Note 2: Io level has been derived from other parameters for information purpose. It is not a settable parameter.
Note 3: RSRP level has been derived from other parameters for information purposes. It is not a settable parameter.
Note 4: The DL PDSCH reference measurement channel is used in the test only when a downlink transmission dedicated to the UE under test is required.

**Table 6.2.1.5-2: RACH-Configuration parameters for E-UTRAN FDD - Contention Based Random Access test**

| Field | Value | Comment |
|---|---|---|
| powerRampingStep | dB2 | |
| preambleInitialReceivedTargetPower | dBm-120 | |
| preambleTransMax | n6 | |
| ra-ResponseWindowSize | sf10 | 10 sub-frames |
| mac-ContentionResolutionTimer | sf48 | 48 sub-frames |
| maxHARQ-Msg3Tx | 4 | |
| Note: For further information see Section 6.3.2 in 3GPP TS 36.331 [5]. | | |

Test 1: Correct behaviour when receiving random access response reception

- The power of the first preamble shall be -30 dBm to within the accuracy specified in Table 6.2.1.5-3.
- The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.1.5-4.
- The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.1.5-5.

Test 2: Correct behaviour when not receiving random access response reception-

- The power of the first preamble shall be -30 dBm to within the accuracy specified in Table 6.2.1.5-3.
- The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.1.5-4.
- The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.1.5-5.

Test 3: Correct behaviour when receiving a NACK on msg3

- The UE shall re-transmit the msg3 upon the reception of a NACK on msg3 until the maximum number of HARQ retransmission is reached.

Test 4: Correct behaviour when receiving an incorrect message over Temporary C-RNTI

- The UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires unless the received message includes a UE contention resolution identity MAC control element

and the UE contention resolution identity included in the MAC control element matches the CCCH SDU transmitted in the uplink message.

Test 5: Correct behaviour when receiving a correct message over Temporary C-RNTI

- The UE shall send ACK if the contention resolution is successful.

Test 6: Correct behaviour when contention resolution timer expires

- The UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if the contention resolution timer expires.

**Table 6.2.1.5-3: Absolute power tolerance for E-UTRAN FDD - Contention Based Random Access test**

| Tolerance | |
|---|---|
| **Normal Conditions** | **Extreme Conditions** |
| ± 10.5 dB | ± 13.5 dB |

**Table 6.2.1.5-4: Relative power tolerance for E-UTRAN FDD - Contention Based Random Access test**

| power step size (Up or down) | PRACH | |
|---|---|---|
| | **Normal Conditions** | **Extreme Conditions** |
| ΔP [dB] | [dB] | [dB] |
| 2 ≤ ΔP < 3 | ± 3.7 | ± 5.7 |
| Note 1: For extreme conditions an additional ± 2.0 dB relaxation is allowed for PRACH allocations | | |

**Table 6.2.1.5-5: Test requirements for $T_e$ Timing Error Limit for E-UTRAN FDD – Contention Based Random Access test**

| Downlink Bandwidth (MHz) | $T_e$ |
|---|---|
| ≥3 | $15*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

## 6.2.2 E-UTRAN FDD - Non-Contention Based Random Access Test

### 6.2.2.1 Test purpose

To verify that the UE behaviour of the random access procedure is according to the E-UTRAN FDD non-contention based random access requirements in an AWGN channel model and that the PRACH power settings and timing are within the specified limits.

### 6.2.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

### 6.2.2.3 Minimum conformance requirements

The random access procedure is used when establishing the layer 1 communication between the UE and E-UTRAN. The random access is as defined in TS 36.213 [8] clause 6 and the control of the RACH transmission is as defined in TS 36.321 [11] clause 5.1.

The UE shall have capability to calculate PRACH transmission power according to the PRACH power formula as defined in TS 36.213 [8] clause 6.1 and apply this power level at the first preamble or additional preambles. The absolute power applied to the first preamble shall have an accuracy as defined in table 6.3.5.1.1-1 of TS 36.101 [2] clause 6.3.5.1.1. The relative power applied to additional preambles shall have an accuracy as specified in table 6.3.5.2.1-1 of TS 36.101 [2] clause 6.3.5.2.1.

The UE shall indicate a Random Access problem to upper layers if the maximum number of preambles transmission counter has been reached.

The UE may stop monitoring for Random Access Response(s) if the Random Access Response contains a Random Access Preamble identifier corresponding to the transmitted Random Access Preamble.

The UE shall re-transmit the preamble with the calculated PRACH transmission power if all received Random Access Responses contain Random Access Preamble identifiers that do not match the transmitted Random Access Preamble.

The UE shall re-transmit the preamble with the calculated PRACH transmission power.

The normative reference for this requirement is TS 36.133 [4] clause 6.2.2.2 and A.6.2.2.

## 6.2.2.4 Test description

### 6.2.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18.

2. Propagation conditions are set according to Annex B clause B.0.

3. There is one E-UTRA FDD carrier and one cell specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

### 6.2.2.4.2 Test procedure

The test consists of a single cell. The UE has a downlink PDSCH allocated centred on the centre sub-carrier. The E-UTRAN shall signal a Random Access Preamble ID to the UE via a PDCCH order.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Tables 6.2.2.5-1 and 6.2.2.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. The SS shall signal a Random Access Preamble ID via a PDCCH order to the UE and initiate a Non-contention based Random Access procedure.

4. Test 1: Correct behaviour when receiving Random Access Response

   4.1 In Test 1, the UE shall send the signalled preamble to the SS. In response to the first 4 preambles, the SS shall transmit a random access response not corresponding to the transmitted random access preamble.

   4.2. The UE shall consider the random access response reception not successful then re-transmit the preamble with the calculated PRACH transmission power if all received random access response contain random access preamble identifiers that do not match the transmitted random access preamble.

   4.3. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after 5 preambles have been received by the SS.

   4.4. The UE shall consider this random access response reception successful.

   4.5. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.2.5-3 and 6.2.2.5-5. Measure the relative power and timing applied to additional preambles (last 4 preambles) and it shall not exceed the values specified in Tables 6.2.2.5-4 and 6.2.2.5-5.

5. Test 2: Correct behaviour when not receiving Random Access Response

   5.1. Repeat step 1-3.

5.2. In Test 2, the UE shall send the signalled preamble to the SS. The SS shall not respond to the first 4 preamble.

5.3. The UE shall consider the random access response reception not successful then re-transmit the preamble with the calculated PRACH transmission power.

5.4. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after 5 preambles have been received by the SS.

5.5. The UE shall consider this random access response reception successful.

5.6. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.2.5-3 and 6.2.2.5-5. Measure the relative power and timing applied to additional preambles (last 4 preambles) and it shall not exceed the values specified in Tables 6.2.2.5-4 and 6.2.2.5-5.

### 6.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.2.2.4.3-1: Common Exception messages for E-UTRAN FDD - Non-Contention Based Random Access test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | H.2.6-1<br>H.2.6-2<br>H.2.6-3 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

**Table 6.2.2.4.3-2: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN FDD - Non-Contention Based Random Access test requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | CQI-ReportConfig-DEFAULT | | RBC |
| soundingRS-LU-ConfigDedicated | Not present | | RBC |

**Table 6.2.2.4.3-3: *MAC-MainConfig-RBC*: Additional E-UTRAN FDD - Non-Contention Based Random Access test requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| timeAlignmentTimerDedicated | Infinity | | |

**Table 6.2.2.4.3-4: *UplinkPowerControlCommon-DEFAULT*: Additional E-UTRAN FDD - Non-Contention Based Random Access test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-25 UplinkPowerControlCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlCommon-DEFAULT ::= SEQUENCE { | | | |
| p0-NominalPUSCH | -101 (-101 dBm) | | |
| } | | | |

### 6.2.2.5 Test requirement

Tables 6.2.2.5-1 and 6.2.2.5-2 define the primary level settings for E-UTRAN FDD - non-contention based random access test. Table 6.2.2.5-5 defines the uplink timing error limit including test tolerances.

**Table 6.2.2.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD - Non-Contention Based Random Access test**

| Parameter | Unit | Value | Comments |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern | | OP.1 FDD | As defined in D.1.1. |
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As defined in A.1.1. |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As defined in A.1.2. |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 3 | |
| $N_{oc}$ | dBm/15 KHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 3 | |
| Io Note 2 | dBm/9 MHz | -65.5 | |
| RSRP Note 3 | dBm/15 KHz | -95 | |
| referenceSignalPower | dBm/15 KHz | -5 | As defined in clause 6.3.2 in 3GPP TS 36.331 [5]. |
| Configured UE transmitted power ( $P_{CMAX}$ ) | dBm | 23 | As defined in clause 6.2.5 in 3GPP TS 36.101 [2]. |
| PRACH Configuration Index | - | 4 | As defined in table 5.7.1-2 in 3GPP TS 36.211 [9]. |
| Back off Parameter Index | - | 2 | As defined in table 7.2-1 in 3GPP TS 36.321 [11]. |
| Propagation Condition | - | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Io level has been derived from other parameters for information purpose. It is not a settable parameter.
Note 3: RSRP level has been derived from other parameters for information purposes. It is not a settable parameter.

**Table 6.2.2.5-2: RACH-Configuration parameters for E-UTRAN FDD - Non-Contention Based Random Access test**

| Field | Value | Comment |
|---|---|---|
| powerRampingStep | dB2 | |
| preambleInitialReceivedTargetPower | dBm-120 | |
| preambleTransMax | n6 | |
| Ra-ResponseWindowSize | sf10 | 10 sub-frames |
| Note: For further information see Section 6.3.2 in 3GPP TS 36.331 [5]. | | |

Test 1: Correct behaviour when receiving Random Access Response

- The power of the first preamble shall be -30 dBm to within the accuracy specified Table 6.2.2.5-3.
- The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.2.5-4.
- The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.2.5-5.

Test 2: Correct behaviour when not receiving Random Access Response

- The power of the first preamble shall be -30 dBm to within the accuracy specified in Table 6.2.2.5-3..
- The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.2.5-4.
- The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.2.5-5.

**Table 6.2.2.5-3: Absolute power tolerance for E-UTRAN FDD - Non-Contention Based Random Access test**

| Tolerance | |
|---|---|
| **Normal Conditions** | **Extreme Conditions** |
| ± 10.5 dB | ± 13.5 dB |

**Table 6.2.2.5-4: Relative power tolerance for E-UTRAN FDD - Non-Contention Based Random Access test**

| power step size (Up or down) | PRACH | |
|---|---|---|
| | **Normal Conditions** | **Extreme Conditions** |
| ΔP [dB] | [dB] | [dB] |
| 2 ≤ ΔP < 3 | ± 3.7 | ± 5.7 |
| Note 1: For extreme conditions an additional ± 2.0 dB relaxation is allowed for PRACH allocations | | |

**Table 6.2.2.5-5: Test requirements for Te Timing Error Limit for E-UTRAN FDD – Non-Contention Based Random Access test**

| Downlink Bandwidth (MHz) | $T_e$ |
|---|---|
| ≥3 | $15*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

## 6.2.3 E-UTRAN TDD - Contention Based Random Access Test

### 6.2.3.1 Test purpose

To verify that the UE behaviour of the random access procedure is according to the E-UTRAN TDD contention based random access requirements in an AWGN channel model and that the PRACH power settings and timing are within the specified limits.

### 6.2.3.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

### 6.2.3.3 Minimum conformance requirements

The random access procedure is used when establishing the layer 1 communication between the UE and E-UTRAN. The random access is as defined in TS 36.213 [8] clause 6 and the control of the RACH transmission is as defined in TS 36.321 [11] clause 5.1.

The UE shall have capability to calculate PRACH transmission power according to the PRACH power formula as defined in TS 36.213 [8] clause 6.1 and apply this power level at the first preamble or additional preambles. The absolute power applied to the first preamble shall have an accuracy as defined in table 6.3.5.1.1-1 of TS 36.101 [2] clause 6.3.5.1.1. The relative power applied to additional preambles shall have an accuracy as specified in table 6.3.5.2.1-1 of TS 36.101 [2] clause 6.3.5.2.1.

The UE shall indicate a Random Access problem to upper layers if the maximum number of preambles transmission counter has been reached.

The UE may stop monitoring for Random Access Response(s) and shall transmit the msg3 if the Random Access Response contains a Random Access Preamble identifier corresponding to the transmitted Random Access Preamble.

The UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if all received Random Access Responses contain Random Access Preamble identifiers that do not match the transmitted Random Access Preamble.

The UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if no Random Access Response is received within the RA Response window defined in clause 5.1.4 TS 36.321.

The UE shall re-transmit the msg3 upon the reception of a NACK on msg3.

The UE shall send ACK if the Contention Resolution is successful.

The UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires unless the received message includes a UE Contention Resolution Identity MAC control element and the UE Contention Resolution Identity included in the MAC control element matches the CCCH SDU transmitted in the uplink message.

The UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if the Contention Resolution Timer expires.

The normative reference for this requirement is TS 36.133 [4] clause 6.2.2.1 and A.6.2.3.

## 6.2.3.4 Test description

### 6.2.3.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18.
2. Propagation conditions are set according to Annex B clause B.0.
3. There is one E-UTRA TDD carrier and one cell specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

### 6.2.3.4.2 Test procedure

The test consists of a single cell. The UE has a downlink PDSCH allocated on the centre sub-carrier. The E-UTRAN shall not explicitly signal a Random Access Preamble ID to the UE.

1. Ensure the UE is in State 2A-RF according to TS 36.508 [7] clause 7.2A.2.
2. Set the parameters according to Tables 6.2.3.5-1 and 6.2.3.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
3. The UE set up a connection with SS, and the random access procedure used in the connection setup is used in the test.
4. Test 1: Correct behaviour when receiving random access response reception

4.1 In Test 1, the UE shall send the preamble to the SS. In response to the first 4 preambles, the SS shall transmit a random access response not corresponding to the transmitted random access preamble.

4.2 The UE shall consider the random access response reception not successful then re- select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires if all received random access response contain random access preamble identifiers that do not match the transmitted random access preamble.

4.3 The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after 5 preambles have been received by the SS.

4.4 The UE shall consider this random access response reception successful and transmit the msg3.

4.5 Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.3.5-3 and 6.2.3.5-5. Measure the relative power and timing applied to additional preambles (last 4 preambles) and it shall not exceed the values specified in Tables 6.2.3.5-4 and 6.2.3.5-5.

5. Test 2: Correct behaviour when *not* receiving random access response reception

5.1 Repeat step 1-3.

5.2 In Test 2, the UE shall send the preamble to the SS. The SS shall not respond to the first 4 preambles.

5.3 The UE shall consider the random access response reception not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires if no random access response is received within the RA Response window of 10 sub-frames.

5.4 The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after 5 preambles have been received by the SS.

5.5 The UE shall consider this random access response reception successful and transmit the msg3.

5.6 Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.3.5-3and 6.2.3.5-5. Measure the relative power and timing applied to additional preambles (last 4 preambles) and it shall not exceed the values specified in Tables 6.2.3.5-4 and 6.2.3.5-5.

6. Test 3: Correct behaviour when receiving a NACK on msg3

6.1 Repeat step 1-3.

6.2 In Test 3, the UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preambles have been received by the SS.

6.3 The UE shall consider this random access response reception successful and transmit the msg3.

6.4 The SS shall send NACK all UE msg3 following a successful random access response.

6.5 The UE shall consider this contention resolution not successful then re-transmit the msg3 upon the reception of a NACK on msg3 until the maximum number of HARQ retransmission is reached.

7. Test 4: Correct behaviour when receiving an incorrect message over Temporary C-RNTI

7.1 Repeat step 1-3.

7.2 In Test 4, the UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after first preambles have been received by the SS.

7.3 The UE shall consider this random access response reception successful and transmit the msg3.

7.4 The SS shall send a message addressed to the Temporary C-RNTI with a UE contention resolution identity included in the MAC control element not matching the CCCH SDU transmitted in msg3 uplink message.

7.5 The UE shall consider the contention resolution not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires.

8. Test 5: Correct behaviour when receiving a correct message over Temporary C-RNTI

   8.1 Repeat step 1-3

   8.2 In Test 5, the SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preambles have been received by the SS.

   8.3 The UE shall consider this random access response reception successful and transmit the msg3.

   8.4 The SS shall send a message addressed to the Temporary C-RNTI with a UE contention resolution identity included in the MAC control element matching the CCCH SDU transmitted in the msg3 uplink message.

   8.5 The UE shall send ACK and consider the contention resolution successful and the random access procedure successfully completed.

9. Test 6: Correct behaviour when contention resolution timer expires

   9.1 Repeat step 1-3.

   9.2 In Test 6, the UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble have been received by the SS.

   9.3 The UE shall consider this random access response reception successful and transmit the msg3.

   9.4 The SS shall send an ACK for msg3 but not send msg4 allowing the contention resolution timer to expire.

   9.5 The UE shall consider the contention resolution not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires.

#### 6.2.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.2.3.4.3-1: Common Exception messages for E-UTRAN TDD -Contention Based Random Access test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | H.2.6-1<br>H.2.6-2<br>H.2.6-3 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-2 |

**Table 6.2.3.4.3-2: *UplinkPowerControlCommon-DEFAULT*: Additional E-UTRAN TDD - Contention Based Random Access test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-25 UplinkPowerControlCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlCommon-DEFAULT ::= SEQUENCE { | | | |
| p0-NominalPUSCH | -101 (-101 dBm) | | |
| } | | | |

### 6.2.3.5 Test requirement

Tables 6.2.3.5-1 and 6.2.3.5-2 define the primary level settings for E-UTRAN TDD - contention based random access test. Table 6.2.3.5-5 defines the uplink timing error limit including test tolerances.

**Table 6.2.3.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD - Contention Based Random Access test**

| Parameter | Unit | Value | Comments |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern Note 1 | | OP.1/2 TDD Note 1 | As defined in D.2.1/2. |
| PDSCH parameters Note 4 | | DL Reference Measurement Channel R.0 TDD Note 4 | As defined in A.1.2. |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As defined in A.2.2. |
| Special subframe configuration | - | 6 | As specified in table 4.2-1 in 3GPP TS 36.211 [9]. |
| Uplink-downlink configuration | - | 1 | As specified in table 4.2-2 in 3GPP TS 36.211 [9]. |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 3 | |
| $N_{oc}$ | dBm/15 KHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 3 | |
| Io Note 2 | dBm/9 MHz | -65.5 | |
| RSRP Note 3 | dBm/15 KHz | -95 | |
| referenceSignalPower | dBm/15 KHz | -5 | As defined in clause 6.3.2 in 3GPP TS 36.331 [5]. |
| Configured UE transmitted power ( $P_{CMAX}$ ) | dBm | 23 | As defined in clause 6.2.5 in 3GPP TS 36.101 [2]. |
| PRACH Configuration Index | - | 53 | As defined in table 5.7.1-3 in 3GPP TS 36.211 [9]. |
| Back off Parameter Index | - | 2 | As defined in table 7.2-1 in 3GPP TS 36.321 [11]. |
| Propagation Condition | - | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. The OCNG pattern is chosen during the test according to the presence of a DL reference measurement channel.
Note 2: Io level has been derived from other parameters for information purpose. It is not a settable parameter.
Note 3: RSRP level has been derived from other parameters for information purposes. It is not a settable parameter.
Note 4: The DL PDSCH reference measurement channel is used in the test only when a downlink transmission dedicated to the DUT is required.

**Table 6.2.3.5-2: RACH-Configuration parameters for E-UTRAN TDD - Contention Based Random Access test**

| Field | Value | Comment |
|---|---|---|
| numberOfRA-Preambles | n52 | |
| sizeOfRA-PreamblesGroupA | n52 | No group B. |
| powerRampingStep | dB2 | |
| preambleInitialReceivedTargetPower | dBm-120 | |
| preambleTransMax | n6 | |
| ra-ResponseWindowSize | sf10 | 10 sub-frames |
| mac-ContentionResolutionTimer | sf48 | 48 sub-frames |
| maxHARQ-Msg3Tx | 4 | |
| Note: For further information see Section 6.3.2 in 3GPP TS 36.331 [5]. | | |

Test 1: Correct behaviour when receiving random access response reception

- The power of the first preamble shall be -22 dBm to within the accuracy specified in Table 6.2.3.5-3.
- The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.3.5-4.
- The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.3.5-5.

Test 2: Correct behaviour when not receiving random access response reception

- The power of the first preamble shall be -22 dBm to within the accuracy specified in Table 6.2.3.5-3.
- The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.3.5-4.
- The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.3.5-5.

Test 3: Correct behaviour when receiving a NACK on msg3

- The UE shall re-transmit the msg3 upon the reception of a NACK on msg3 until the maximum number of HARQ retransmission is reached.

Test 4: Correct behaviour when receiving an incorrect message over Temporary C-RNTI

- The UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires unless the received message includes a UE contention resolution identity MAC control element and the UE contention resolution identity included in the MAC control element matches the CCCH SDU transmitted in the uplink message.

Test 5: Correct behaviour when receiving a correct message over Temporary C-RNTI

- The UE shall send ACK if the contention resolution is successful.

Test 6: Correct behaviour when contention resolution timer expires

- The UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if the contention resolution timer expires.

**Table 6.2.3.5-3: Absolute power tolerance for E-UTRAN TDD - Contention Based Random Access test**

| Tolerance | |
|---|---|
| **Normal Conditions** | **Extreme Conditions** |
| ± 10.5 dB | ± 13.5 dB |

**Table 6.2.3.5-4: Relative power tolerance for E-UTRAN TDD - Contention Based Random Access test**

| power step size (Up or down) | PRACH | |
|---|---|---|
| | Normal Conditions | Extreme Conditions |
| ΔP [dB] | [dB] | [dB] |
| 2 ≤ ΔP < 3 | ± 3.7 | ± 5.7 |
| Note 1: For extreme conditions an additional ± 2.0 dB relaxation is allowed for PRACH allocations | | |

**Table 6.2.3.5-5: Test requirements for $T_e$ Timing Error Limit for E-UTRAN TDD – Contention Based Random Access test**

| Downlink Bandwidth (MHz) | $T_e$ |
|---|---|
| ≥3 | $15*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

## 6.2.4 E-UTRAN TDD - Non-Contention Based Random Access Test

### 6.2.4.1 Test purpose

To verify that the UE behaviour of the random access procedure is according to the E-UTRAN TDD non-contention based random access requirements in an AWGN model and that the PRACH power settings and timing are within the specified limits.

### 6.2.4.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

### 6.2.4.3 Minimum conformance requirements

The random access procedure is used when establishing the layer 1 communication between the UE and E-UTRAN. The random access is as defined in TS 36.213 [8] clause 6 and the control of the RACH transmission is as defined in TS 36.321 [11] clause 5.1.

The UE shall have capability to calculate PRACH transmission power according to the PRACH power formula as defined in TS 36.213 [8] clause 6.1 and apply this power level at the first preamble or additional preambles. The absolute power applied to the first preamble shall have an accuracy as defined in table 6.3.5.1.1-1 of TS 36.101 [2] clause 6.3.5.1.1. The relative power applied to additional preambles shall have an accuracy as specified in table 6.3.5.2.1-1 of TS 36.101 [2] clause 6.3.5.2.1.

The UE shall indicate a Random Access problem to upper layers if the maximum number of preambles transmission counter has been reached.

The UE may stop monitoring for Random Access Response(s) if the Random Access Response contains a Random Access Preamble identifier corresponding to the transmitted Random Access Preamble.

The UE shall re-transmit the preamble with the calculated PRACH transmission power if all received Random Access Responses contain Random Access Preamble identifiers that do not match the transmitted Random Access Preamble.

The UE shall re-transmit the preamble with the calculated PRACH transmission power if no Random Access Response is received within the RA response window.

The normative reference for this requirement is TS 36.133 [4] clause 6.2.2.2 and A.6.2.4.

### 6.2.4.4 Test description

#### 6.2.4.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18.

2. Propagation conditions are set according to Annex B clause B.0.

3. There is one E-UTRA TDD carrier and one cell specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

#### 6.2.4.4.2 Test procedure

The test consists of a single cell. The UE has a downlink PDSCH allocated centred on the centre sub-carrier. The E-UTRAN shall signal a Random Access Preamble ID to the UE via a PDCCH order.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Tables 6.2.4.5-1 and 6.2.4.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. The SS shall signal a Random Access Preamble ID via a PDCCH order to the UE and initiate a Non-contention based Random Access procedure.

4. Test 1: Correct behaviour when receiving Random Access Response

4.1. In Test 1, the UE shall send the signalled preamble to the SS. In response to the first 4 preambles, the SS shall transmit a random access response not corresponding to the transmitted random access preamble.

4.2 The UE shall consider the random access response reception not successful then re-transmit the preamble with the calculated PRACH transmission power if all received random access response contain random access preamble identifiers that do not match the transmitted random access preamble.

4.3 The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after 5 preambles have been received by the SS.

4.4 The UE shall consider this random access response reception successful.

4.5 Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.4.5-3 and 6.2.4.5-5. Measure the relative power and timing applied to additional preambles (last 4 preambles) and it shall not exceed the values specified in Tables 6.2.4.5-4 and 6.2.4.5-5.

5. Test 2: Correct behaviour when *not* receiving Random Access Response

5.1 Repeat step 1-3.

5.2 In Test 2, the UE shall send the signalled preamble to the SS. The SS shall not respond to the first 4 preambles.

5.3 The UE shall consider the random access response reception not successful then re-transmit the preamble with the calculated PRACH transmission power.

5.4 The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after 5 preambles have been received by the SS.

5.5 The UE shall consider this random access response reception successful.

5.6 Measure the power and timing of the first preamble and it shall not exceed the values specified in clause Tables 6.2.4.5-3 and 6.2.4.5-5. Measure the relative power and timing applied to additional preambles (last 4 preambles) and it shall not exceed the values specified in Tables 6.2.4.5-4 and 6.2.4.5-5.

### 6.2.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.2.4.4.3-1: Common Exception messages for E-UTRAN TDD - Non-Contention Based Random Access test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | H.2.6-1<br>H.2.6-2<br>H.2.6-3 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-2 |

**Table 6.2.4.4.3-2: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN TDD - Non-Contention Based Random Access test requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | CQI-ReportConfig-DEFAULT | | RBC |
| soundingRS-LU-ConfigDedicated | Not present | | RBC |
| } | | | |

**Table 6.2.4.4.3-3: *MAC-MainConfig-RBC*: Additional E-UTRAN TDD - Non-Contention Based Random Access test requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| } | | | |

**Table 6.2.4.4.3-4: *UplinkPowerControlCommon-DEFAULT*: Additional E-UTRAN TDD - Non-Contention Based Random Access test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-25 UplinkPowerControlCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlCommon-DEFAULT ::= SEQUENCE { | | | |
| p0-NominalPUSCH | -101 (-101 dBm) | | |
| } | | | |

## 6.2.4.5 Test requirement

Tables 6.2.4.5-1 and 6.2.4.5-2 define the primary level settings for E-UTRAN TDD - non-contention based random access test. Table 6.2.4.5-5 defines the uplink timing error limit including test tolerances.

**Table 6.2.4.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD - Non-Contention Based Random Access test**

| Parameter | Unit | Value | Comments |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern | | OP.1 TDD | As defined in D.2.1. |
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As defined in A.1.2. |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As defined in A.2.2. |
| Special subframe configuration | - | 6 | As specified in table 4.2-1 in 3GPP TS 36.211 [9]. |
| Uplink-downlink configuration | - | 1 | As specified in table 4.2-2 in 3GPP TS 36.211 [9]. |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 3 | |
| $N_{oc}$ | dBm/15 KHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 3 | |
| Io Note 2 | dBm/9 MHz | -65.5 | |
| RSRP Note 3 | dBm/15 KHz | -95 | |
| referenceSignalPower | dBm/15 KHz | -5 | As defined in clause 6.3.2 in 3GPP TS 36.331 [5]. |
| Configured UE transmitted power ( $P_{CMAX}$ ) | dBm | 23 | As defined in clause 6.2.5 in 3GPP TS 36.101 [2]. |
| PRACH Configuration Index | - | 53 | As defined in table 5.7.1-3 in 3GPP TS 36.211 [9]. |
| Back off Parameter Index | - | 2 | As defined in table 7.2-1 in 3GPP TS 36.321 [11]. |
| Propagation Condition | - | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Io level has been derived from other parameters for information purpose. It is not a settable parameter.
Note 3: RSRP level has been derived from other parameters for information purposes. It is not a settable parameter.

**Table 6.2.4.5-2: RACH-Configuration parameters for E-UTRAN TDD – Non-Contention Based Random Access test**

| Field | Value | Comment |
|---|---|---|
| powerRampingStep | dB2 | |
| preambleInitialReceivedTargetPower | dBm-120 | |
| preambleTransMax | n6 | |
| Ra-ResponseWindowSize | sf10 | 10 sub-frames |
| Note: For further information see Section 6.3.2 in 3GPP TS 36.331 [5]. | | |

Test 1: Correct behaviour when receiving Random Access Response

- The power of the first preamble shall be -22 dBm to within the accuracy specified in Table 6.2.4.5-3.
- The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.4.5-4.
- The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.4.5-5.

Test 2: Correct behaviour when *not* receiving Random Access Response

- The power of the first preamble shall be -22 dBm to within the accuracy specified in Table 6.2.4.5-3.
- The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.4.5-4.
- The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.4.5-5.

**Table 6.2.4.5-3: Absolute power tolerance for E-UTRAN TDD - Non-Contention Based Random Access test**

| Tolerance | |
|---|---|
| **Normal Conditions** | **Extreme Conditions** |
| ± 10.5 dB | ± 13.5 dB |

**Table 6.2.4.5-4: Relative power tolerance for E-UTRAN TDD - Non-Contention Based Random Access test**

| power step size (Up or down) | PRACH | |
|---|---|---|
| | **Normal Conditions** | **Extreme Conditions** |
| ΔP [dB] | [dB] | [dB] |
| 2 ≤ ΔP < 3 | ± 3.7 | ± 5.7 |
| Note 1: For extreme conditions an additional ± 2.0 dB relaxation is allowed for PRACH allocations | | |

**Table 6.2.4.5-5: Test requirements for $T_e$ Timing Error Limit for E-UTRAN TDD – Non-Contention Based Random Access test**

| Downlink Bandwidth (MHz) | $T_e$ |
|---|---|
| ≥3 | $15*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

## 6.2.5 E-UTRAN FDD - Contention Based Random Access Test for 5MHz Bandwidth

### 6.2.5.1 Test purpose

Same test purpose as defined in clause 6.2.1.1.

### 6.2.5.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that only support Band 31.

### 6.2.5.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 6.2.1.3 with the following exceptions:

- Instead of A.6.2.1 → use A.6.2.5.

### 6.2.5.4 Test description

#### 6.2.5.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18.
2. Propagation conditions are set according to Annex B clause B.0.
3. There is one E-UTRA FDD carrier and one cell specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

#### 6.2.5.4.2 Test procedure

Same test procedure as defined in clause 6.2.1.4.2 with the following exceptions:

- Instead of Table 6.2.1.5-1 → use Table 6.2.5.5-1.

#### 6.2.5.4.3 Message contents

Same message contents as defined in clause 6.2.1.4.3.

### 6.2.5.5 Test requirement

Same test requirement as defined in clause 6.2.1.5 with the following exceptions:

Tables 6.2.5.5-1 and 6.2.1.5-2 define the primary level settings for E-UTRAN FDD - contention based random access test. Table 6.2.1.5-5 defines the uplink timing error limit including test tolerances.

**Table 6.2.5.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD - Contention Based Random Access test for 5MHz bandwidth**

| Parameter | Unit | Value | Comments |
|---|---|---|---|
| $BW_{channel}$ | MHz | 5 | |
| OCNG Pattern Note 1 | | OP.15/16 FDD Note 1 | As defined in D.1.15/16. |
| PDSCH parameters Note 2 | | DL Reference Measurement Channel R.5 FDD Note 2 | As defined in A.1.1. |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.11 FDD | As defined in A.2.1. |
| Io Note 2 | dBm/4.5 MHz | **-68.5** | |
| Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. The OCNG pattern is chosen during the test according to the presence of a DL reference measurement channel.<br>Note 2: The DL PDSCH reference measurement channel is used in the test only when a downlink transmission dedicated to the UE under test is required.<br>Note 3: See Table 6.2.1.5-1 for the other parameters.<br>Note 4: This test is according to the principle defined in TS 36.133 [4] section A.3.7.2. | | | |

## 6.2.6 E-UTRAN FDD - Non-Contention Based Random Access Test for 5MHz Bandwidth

### 6.2.6.1 Test purpose

Same test purpose as defined in clause 6.2.2.1.

### 6.2.6.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that only support band 31.

### 6.2.6.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 6.2.2.3 with the following exceptions:

- Instead of A.6.2.2 → use A.6.2.6.

### 6.2.6.4 Test description

#### 6.2.6.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18.
2. Propagation conditions are set according to Annex B clause B.0.
3. There is one E-UTRA FDD carrier and one cell specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

#### 6.2.6.4.2 Test procedure

Same test procedure as defined in clause 6.2.2.4.2 with the following exceptions:

- Instead of Table 6.2.2.5-1 → use Table 6.2.6.5-1.

#### 6.2.6.4.3 Message contents

Same message contents as defined in clause 6.2.2.4.3.

### 6.2.6.5 Test requirement

Same test requirement as defined in clause 6.2.2.5 with the following exceptions:

Tables 6.2.6.5-1 and 6.2.2.5-2 define the primary level settings for E-UTRAN FDD - non-contention based random access test for 5MHz bandwidth. Table 6.2.2.5-5 defines the uplink timing error limit including test tolerances.

**Table 6.2.6.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD - Non-Contention Based Random Access test for 5MHz bandwidth**

| Parameter | Unit | Value | Comments |
|---|---|---|---|
| $BW_{channel}$ | MHz | 5 | |
| OCNG Pattern Note 1 | | OP.15 FDD Note 1 | As defined in D.1.15. |
| PDSCH parameters Note 2 | | DL Reference Measurement Channel R.5 FDD Note 2 | As defined in A.1.1. |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.11 FDD | As defined in A.2.1. |
| Io Note 2 | dBm/4.5 MHz | -68.5 | |
| Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: Io level has been derived from other parameters for information purpose. It is not a settable parameter.<br>Note 3: See Table 6.2.2.5-1 for the other parameters.<br>Note 4: This test is according to the principle defined in TS 36.133 [4] section A.3.7.2. | | | |

## 6.2.7 E-UTRAN FDD – Non-Contention Based Random Access Test for SCell in sTAG

### 6.2.7.1 Test purpose

To verify that the UE behaviour of the random access procedure, for the SCell, is according to the E-UTRAN FDD non-contention based random access requirements in an AWGN model and that the PRACH power settings and timing, for the SCell, are within the specified limits.

### 6.2.7.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 11 and forward that supports Uplink Carrier Aggregation and multiple timing advances.

### 6.2.7.3 Minimum conformance requirements

The random access procedure is used when establishing the layer 1 communication between the UE and E-UTRAN. The random access is specified in clause 6 of TS 36.213 [8] and the control of the RACH transmission is specified in clause 5.1 of TS 36.321 [11]. Contention based random access procedures can only be carried out on PCell, while non-contention based random access procedures can be carried out on both PCell and an activated SCell.

The UE shall have capability to calculate PRACH transmission power according to the PRACH power formula defined in TS 36.213 [8] and apply this power level at the first preamble or additional preambles. The absolute power applied to the first preamble shall have an accuracy as specified in table 6.3.5.1.1-1 of TS 36.101 [2]. The relative power applied to additional preambles shall have an accuracy as specified in table 6.3.5.2.1-1 of 36.101 [2].

The UE shall indicate a Random Access problem to upper layers if the maximum number of preamble transmission counter has been reached for the random access procedure on PCell as specified in clause 5.1.4 in TS 36.321 [11].

The UE shall stop preamble transmission if maximum number of preamble transmission counter has been reached for the random access procedure on an activated Scell as specified in clause 5.1.4 in TS 36.321 [11].

The UE may stop monitoring for Random Access Response(s) if the Random Access Response contains a Random Access Preamble identifier corresponding to the transmitted Random Access Preamble.

The UE shall re-transmit the preamble with the calculated PRACH transmission power if all received Random Access Responses contain Random Access Preamble identifiers that do not match the transmitted Random Access Preamble.

The UE shall re-transmit the preamble with the calculated PRACH transmission power.

The normative reference for this requirement is TS 36.133 [4] clause 6.2.2 and A.6.2.7.

## 6.2.7.4 Test description

### 6.2.7.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz for both PCell and SCell as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.42 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier. PCell and SCell are different timing advance group. Cell 1 is in the primary Timing Advance Group (pTAG) and Cell 2 is in the secondary Timing Advance Group (sTAG).Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test, Cell 2 shall be powered OFF.

### 6.2.7.4.2 Test procedure

This test consists of two cells. Cell 1 is PCell and Cell 2 is SCell. PCell and SCell are different timing advance group. Cell 1 is in the primary Timing Advance Group (pTAG) and Cell 2 is in the secondary Timing Advance Group (sTAG). The purpose of the PCell is to allow the SCell to be configured and to handle the Random Access Response which takes place on PCell.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCell according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4 with UL CA.

4. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).

5. Set the parameters according to Tables 6.2.7.5-1 and 6.2.7.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. The SS shall signal, on Cell 2, the SCell, a Random Access Preamble ID via a PDCCH order to the UE and initiate a Non-contention based Random Access procedure.

7. Test 1: Correct behaviour when receiving Random Access Response

   7.1. In Test 1, the UE shall send, on Cell 2, the SCell, the signalled preamble to the SS. In response to the first 4 preambles, the SS shall transmit on PCell Cell 1 a random access response not corresponding to the transmitted random access preamble.

   7.2. The UE shall consider the random access response reception not successful then re-transmit the preamble on SCell Cell 2 with the calculated PRACH transmission power.

   7.3. The SS shall transmit on PCell Cell 1 a random access response containing a random access preamble identifier corresponding to the transmitted random access after 5 preambles have been received by the SS.

   7.4. The UE shall consider this random access response reception successful.

   7.5. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.7.5-3 and 6.2.7.5-5. Measure the relative power and timing applied to additional preambles (last 4 preambles) and it shall not exceed the values specified in Tables 6.2.7.5-4 and 6.2.7.5-5.

8. Test 2: Correct behaviour when not receiving Random Access Response

   8.1. Repeat step 1-6.

8.2. In Test 2, the UE shall send, on Cell 2, the SCell, the signalled preamble to the SS. The SS shall not respond to the first 4 preambles.

8.3. The UE shall consider the random access response reception not successful then re-transmit the preamble with the calculated PRACH transmission power.

8.4. The SS shall transmit on PCell Cell 1 a random access response containing a random access preamble identifier corresponding to the transmitted random access after 5 preambles have been received by the SS.

8.5. The UE shall consider this random access response reception successful.

8.6. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.7.5-3 and 6.2.7.5-5. Measure the relative power and timing applied to additional preambles (last 4 preambles) and it shall not exceed the values specified in Tables 6.2.7.5-4 and 6.2.7.5-5.

9. Test 3: Correct behaviour when maximum number of preamble transmission counter has been reached

9.1. Repeat step 1-6.

9.2. In Test 3, the UE shall send, on Cell 2, the SCell, the signalled preamble to the SS. In response to the first 5 preambles, the SS shall transmit on PCell Cell 1 a random access response not corresponding to the transmitted random access preamble.

9.3. The UE shall consider the random access response reception not successful then re-transmit the preamble on SCell Cell 2 with the calculated PRACH transmission power.

9.4. The SS shall still transmit on PCell Cell 1 a random access response containing a random access preamble identifier not corresponding to the transmitted random access after 5 preambles have been received by the SS.

9.5. The UE shall consider this random access response reception not successful and stop to transmit the preamble on SCell Cell 2.

9.6. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.7.5-3 and 6.2.7.5-5. Measure the relative power and timing applied to additional preambles (last 5 preambles) and it shall not exceed the values specified in Tables 6.2.7.5-4 and 6.2.7.5-5.

### 6.2.7.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.2.7.4.3-1: Common Exception messages for E-UTRAN FDD - Non-Contention Based Random Access for Scell test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | H.2.6-1<br>H.2.6-2<br>H.2.6-3 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

**Table 6.2.7.4.3-2: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN FDD - Non-Contention Based Random Access test for Scell requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | CQI-ReportConfig-DEFAULT | | RBC |
| soundingRS-LU-ConfigDedicated | Not present | | RBC |

**Table 6.2.7.4.3-3: *MAC-MainConfig-RBC*: Additional E-UTRAN FDD - Non-Contention Based Random Access test for Scell requirement**

| Derivation Path: 36.508, Table 4.8.2.1.5-1, condition SCell_AddMod | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| stag-ToAddModList-r11 SEQUENCE (SIZE (1..maxSTAG-r11)) OF SEQUENCE { | Not present | | |
| stag-Id-r11 | 1 | | |
| timeAlignmentTimerSTAG-r11 | infinity | | |
| } | | | |
| } | | | |

**Table 6.2.7.4.3-4: *UplinkPowerControlCommon-DEFAULT*: Additional E-UTRAN FDD - Non-Contention Based Random Access test for Scell requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-25 UplinkPowerControlCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlCommon-DEFAULT ::= SEQUENCE { | | | |
| p0-NominalPUSCH | -101 (-101 dBm) | | |
| } | | | |

**Table 6.2.7.4.3-5: *SCellToAddMod-r10:* Additional E-UTRAN FDD - Non-Contention Based Random Access test for Scell requirement**

| Derivation Path: 36.508, Table 4.6.3-19D | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SCellToAddMod-r10 ::= SEQUENCE { | | | |
| sCellIndex-r10 | 1 | | |
| cellIdentification-r10 SEQUENCE { | | | |
| physCellId-r10 | PhysicalCellIdentity of Cell 2 | | |
| dl-CarrierFreq-r10 | Same downlink EARFCN as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 6.2.7.4.3-6: *RadioResourceConfigCommonSCell-r10:* Additional E-UTRAN FDD - Non-Contention Based Random Access test for Scell requirement**

| Derivation Path: 36.508, Table 4.6.3-13A, condition UL_CA | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigCommonSCell-r10-DEFAULT ::= SEQUENCE { | | | |
| rach-ConfigCommonSCell-r11 SEQUENCE { | | | |
| powerRampingParameters-r11 SEQUENCE { | | | |
| powerRampingStep | dB2 | | |
| preambleInitialReceivedTargetPower | dBm-120 | | |
| } | | | |
| ra-SupervisionInfo-r11 SEQUENCE { | | | |
| preambleTransMax-r11 | n6 | | |
| } | | | |
| } | | | |
| prach-ConfigSCell-r11 SEQUENCE { | | | |
| rootSequenceIndex | See TS 36.508 [18] clause 4.4.2, Table 4.4.2-1A and clause 6.3.2.2 Table 6.3.2.2-2 | See table 5.7.2-4 in TS 36.211 [41] for PRACH format 0-3 | FDD |
| prach-ConfigInfo SEQUENCE { | | | |
| prach-ConfigIndex | 4 | | FDD |
| highSpeedFlag | FALSE | | |
| zeroCorrelationZoneConfig | | | |
| prach-FreqOffset | See TS 36.508 [18] clause 4.6.8 | Channel-bandwidth-dependent parameter | |
| } | | | |
| } | | | |
| uplinkPowerControlCommonSCell-v1130 SEQUENCE | Not present | | |
| } | | | |

**Table 6.2.7.4.3-7: *RadioResourceConfigDedicatedSCell-r10*: Additional E-UTRAN FDD - Non-Contention Based Random Access test for Scell requirement**

| Derivation Path: 36.508, Table 4.6.3-19AA | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicatedSCell-r10 ::= SEQUENCE { | | | |
| physicalConfigDedicatedSCell-r10 | PhysicalConfigDedicated SCell-r10-DEFAULT | | |
| mac-MainConfigSCell-r11 SEQUENCE { | | | |
| stag-Id-r11 | 1 | | |
| } | | | |
| } | | | |

## 6.2.7.5 Test requirement

Tables 6.2.7.5-1 and 6.2.7.5-2 define the primary level settings for E-UTRAN FDD - non-contention based random access test for SCell. Table 6.2.7.5-5 defines the uplink timing error limit including test tolerances.

**Table 6.2.7.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD - Non-Contention Based Random Access test for SCell**

| Parameter | Unit | Cell 1 | Cell 2 | Comments |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | 2 | |
| $BW_{channel}$ | MHz | 10 | 10 | |
| Active PCell | | Cell 1 | | Primary cell of RF channel number 1. |
| Active SCell | | | Cell 2 | Secondary cell of RF channel number 2. |
| TAG configuration | | pTAG | sTAG | pTAG+sTAG configures Cell 1 and Cell 2 to separate TAGs |
| OCNG Pattern | | OP.1 FDD | OP.1 FDD | As defined in D.1.1. |
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | DL Reference Measurement Channel R.0 FDD | As defined in A.1.1. |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | DL Reference Measurement Channel R.6 FDD | As defined in A.1.2. |
| PBCH_RA | dB | 0 | 0 | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $\hat{E}_s / I_{ot}$ | dB | 3 | 3 | |
| $N_{oc}$ | dBm/15 KHz | -98 | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 3 | 3 | |
| Io Note 2 | dBm/9 MHz | -65.5 | -65.5 | |
| RSRP Note 3 | dBm/15 KHz | -95 | -95 | |
| referenceSignalPower | dBm/15 KHz | -5 | -5 | As defined in clause 6.3.2 in TS 36.331 [5]. |
| Configured UE transmitted power ( $P_{CMAX,c}$ ) | dBm | 23 | 23 | As defined in clause 6.2.5 in TS 36.101 [2]. |
| PRACH Configuration Index | - | 4 | 4 | As defined in table 5.7.1-2 in TS 36.211 [9]. |
| Backoff Parameter Index | - | 2 | 2 | As defined in table 7.2-1 in TS 36.321 [11]. |
| Propagation Condition | - | AWGN | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Io level has been derived from other parameters for information purpose. It is not a settable parameter.
Note 3: RSRP level has been derived from other parameters for information purposes. It is not a settable parameter.

**Table 6.2.7.5-2: RACH-Configuration parameters for E-UTRAN FDD - Non-Contention Based Random Access test for SCell**

| Field | Value | Comment |
|---|---|---|
| powerRampingStep | dB2 | |
| preambleInitialReceivedTargetPower | dBm-120 | |
| preambleTransMax | n6 | |
| ra-ResponseWindowSize | sf10 | 10 sub-frames |
| Note: For further information see Clause 6.3.2 in TS 36.331 [5]. | | |

Test 1: Correct behaviour when receiving Random Access Response

- The power of the first preamble shall be -30 dBm to within the accuracy specified Table 6.2.7.5-3.
- The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.7.5-4.
- The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.7.5-5.

Test 2: Correct behaviour when not receiving Random Access Response

- The power of the first preamble shall be -30 dBm to within the accuracy specified in Table 6.2.7.5-3.
- The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.7.5-4.
- The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.7.5-5.

Test 3: Correct behaviour when maximum number of preamble transmission counter has been reached

- The power of the first preamble shall be -30 dBm to within the accuracy specified Table 6.2.7.5-3.
- The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.7.5-4.
- The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.7.5-5.

**Table 6.2.7.5-3: Absolute power tolerance for E-UTRAN FDD - Non-Contention Based Random Access test for SCell**

| Conditions | Tolerance |
|---|---|
| Normal | ± 10.5 dB |
| Extreme | ± 13.5 dB |

**Table 6.2.7.5-4: Relative power tolerance for E-UTRAN FDD - Non-Contention Based Random Access test for SCell**

| power step size (Up or down) | PRACH | |
|---|---|---|
| | Normal Conditions | Extreme Conditions |
| ΔP [dB] | [dB] | [dB] |
| 2 ≤ ΔP < 3 | ± 3.7 | ± 5.7 |
| Note 1: For extreme conditions an additional ± 2.0 dB relaxation is allowed for PRACH allocations | | |

**Table 6.2.7.5-5: Test requirements for Te Timing Error Limit for E-UTRAN FDD – Non-Contention Based Random Access test for SCell**

| Downlink Bandwidth (MHz) | $T_e$ |
|---|---|
| ≥3 | 15*Ts |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

## 6.2.8 E-UTRAN TDD – Non-Contention Based Random Access Test for SCell in sTAG

### 6.2.8.1 Test purpose

To verify that the UE behaviour of the random access procedure, for the SCell, is according to the E-UTRAN TDD non-contention based random access requirements in an AWGN model and that the PRACH power settings and timing, for the SCell, are within the specified limits.

### 6.2.8.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 11 and forward that supports Uplink Carrier Aggregation and multiple timing advances.

### 6.2.8.3 Minimum conformance requirements

The random access procedure is used when establishing the layer 1 communication between the UE and E-UTRAN. The random access is specified in clause 6 of TS 36.213 [8] and the control of the RACH transmission is specified in clause 5.1 of TS 36.321 [11]. Contention based random access procedures can only be carried out on PCell, while non-contention based random access procedures can be carried out on both PCell and an activated SCell.

The UE shall have capability to calculate PRACH transmission power according to the PRACH power formula defined in TS 36.213 [8] and apply this power level at the first preamble or additional preambles. The absolute power applied to the first preamble shall have an accuracy as specified in table 6.3.5.1.1-1 of TS 36.101 [2]. The relative power applied to additional preambles shall have an accuracy as specified in table 6.3.5.2.1-1 of 36.101 [2].

The UE shall indicate a Random Access problem to upper layers if the maximum number of preamble transmission counter has been reached for the random access procedure on PCell as specified in clause 5.1.4 in TS 36.321 [11].

The UE shall stop preamble transmission if maximum number of preamble transmission counter has been reached for the random access procedure on an activated Scell as specified in clause 5.1.4 in TS 36.321 [11].

The UE may stop monitoring for Random Access Response(s) if the Random Access Response contains a Random Access Preamble identifier corresponding to the transmitted Random Access Preamble.

The UE shall re-transmit the preamble with the calculated PRACH transmission power if all received Random Access Responses contain Random Access Preamble identifiers that do not match the transmitted Random Access Preamble.

The UE shall re-transmit the preamble with the calculated PRACH transmission power.

The normative reference for this requirement is TS 36.133 [4] clause 6.2.2 and A.6.2.8.

### 6.2.8.4 Test description

#### 6.2.8.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz for both PCell and SCell as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.42 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier. PCell and SCell are different timing advance group. Cell 1 is in the primary Timing Advance Group (pTAG) and Cell 2 is in the secondary Timing Advance Group (sTAG).Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test, Cell 2 shall be powered OFF.

#### 6.2.8.4.2 Test procedure

This test consists of two cells. Cell 1 is PCell and Cell 2 is SCell. PCell and SCell are different timing advance group. Cell 1 is in the primary Timing Advance Group (pTAG) and Cell 2 is in the secondary Timing Advance Group (sTAG). The purpose of the PCell is to allow the SCell to be configured and to handle the Random Access Response which takes place on PCell.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCell according to Annex C.0 and C.1 for all downlink physical channels.
3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4 with UL CA.
4. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).
5. Set the parameters according to Tables 6.2.8.5-1 and 6.2.8.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
6. The SS shall signal, on Cell 2, the SCell, a Random Access Preamble ID via a PDCCH order to the UE and initiate a Non-contention based Random Access procedure.
7. Test 1: Correct behaviour when receiving Random Access Response

   7.1. In Test 1, the UE shall send, on Cell 2, the SCell, the signalled preamble to the SS. In response to the first 4 preambles, the SS shall transmit on PCell Cell 1 a random access response not corresponding to the transmitted random access preamble.

   7.2. The UE shall consider the random access response reception not successful then re-transmit the preamble on SCell Cell 2 with the calculated PRACH transmission power.

   7.3. The SS shall transmit on PCell Cell 1 a random access response containing a random access preamble identifier corresponding to the transmitted random access after 5 preambles have been received by the SS.

   7.4. The UE shall consider this random access response reception successful.

   7.5. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.8.5-3 and 6.2.8.5-5. Measure the relative power and timing applied to additional preambles (last 4 preambles) and it shall not exceed the values specified in Tables 6.2.8.5-4 and 6.2.8.5-5.

8. Test 2: Correct behaviour when not receiving Random Access Response

   8.1. Repeat step 1-6.

   8.2. In Test 2, the UE shall send, on Cell 2, the SCell, the signalled preamble to the SS. The SS shall not respond to the first 4 preambles.

   8.3. The UE shall consider the random access response reception not successful then re-transmit the preamble with the calculated PRACH transmission power.

   8.4. The SS shall transmit on PCell Cell 1 a random access response containing a random access preamble identifier corresponding to the transmitted random access after 5 preambles have been received by the SS.

   8.5. The UE shall consider this random access response reception successful.

   8.6. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.8.5-3 and 6.2.8.5-5. Measure the relative power and timing applied to additional preambles (last 4 preambles) and it shall not exceed the values specified in Tables 6.2.8.5-4 and 6.2.8.5-5.

9. Test 3: Correct behaviour when maximum number of preamble transmission counter has been reached

   9.1. Repeat step 1-6.

   9.2. In Test 3, the UE shall send, on Cell 2, the SCell, the signalled preamble to the SS. In response to the first 5 preambles, the SS shall transmit on PCell Cell 1 a random access response not corresponding to the transmitted random access preamble.

9.3. The UE shall consider the random access response reception not successful then re-transmit the preamble on SCell Cell 2 with the calculated PRACH transmission power.

9.4. The SS shall still transmit on PCell Cell 1 a random access response containing a random access preamble identifier not corresponding to the transmitted random access after 5 preambles have been received by the SS.

9.5. The UE shall consider this random access response reception not successful and stop to transmit the preamble on SCell Cell 2.

9.6. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.8.5-3 and 6.2.8.5-5. Measure the relative power and timing applied to additional preambles (last 5 preambles) and it shall not exceed the values specified in Tables 6.2.8.5-4 and 6.2.8.5-5.

#### 6.2.8.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.2.8.4.3-1: Common Exception messages for E-UTRAN TDD - Non-Contention Based Random Access for Scell test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | H.2.6-1<br>H.2.6-2<br>H.2.6-3 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

**Table 6.2.8.4.3-2: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN TDD - Non-Contention Based Random Access test for Scell requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | CQI-ReportConfig-DEFAULT | | RBC |
| soundingRS-LU-ConfigDedicated | Not present | | RBC |

**Table 6.2.8.4.3-3: *MAC-MainConfig-RBC*: Additional E-UTRAN TDD - Non-Contention Based Random Access test for Scell requirement**

| Derivation Path: 36.508, Table 4.8.2.1.5-1, condition SCell_AddMod | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| stag-ToAddModList-r11 SEQUENCE (SIZE (1..maxSTAG-r11)) OF SEQUENCE { | Not present | | |
| stag-Id-r11 | 1 | | |
| timeAlignmentTimerSTAG-r11 | infinity | | |
| } | | | |
| } | | | |

**Table 6.2.8.4.3-4: *UplinkPowerControlCommon-DEFAULT*: Additional E-UTRAN TDD - Non-Contention Based Random Access test for Scell requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-25 UplinkPowerControlCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlCommon-DEFAULT ::= SEQUENCE { | | | |
| p0-NominalPUSCH | -101 (-101 dBm) | | |
| } | | | |

**Table 6.2.8.4.3-5: *SCellToAddMod-r10:* Additional E-UTRAN TDD - Non-Contention Based Random Access test for Scell requirement**

| Derivation Path: 36.508, Table 4.6.3-19D | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SCellToAddMod-r10 ::= SEQUENCE { | | | |
| sCellIndex-r10 | 1 | | |
| cellIdentification-r10 SEQUENCE { | | | |
| physCellId-r10 | PhysicalCellIdentity of Cell 2 | | |
| dl-CarrierFreq-r10 | Same downlink EARFCN as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 6.2.8.4.3-6: *RadioResourceConfigCommonSCell-r10:* Additional E-UTRAN TDD - Non-Contention Based Random Access test for Scell requirement**

| Derivation Path: 36.508, Table 4.6.3-13A, condition UL_CA | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigCommonSCell-r10-DEFAULT ::= SEQUENCE { | | | |
| rach-ConfigCommonSCell-r11 SEQUENCE { | | | |
| powerRampingParameters-r11 SEQUENCE { | | | |
| powerRampingStep | dB2 | | |
| preambleInitialReceivedTargetPower | dBm-120 | | |
| } | | | |
| ra-SupervisionInfo-r11 SEQUENCE { | | | |
| preambleTransMax-r11 | n6 | | |
| } | | | |
| } | | | |
| prach-ConfigSCell-r11 SEQUENCE { | | | |
| rootSequenceIndex | See TS 36.508 [18] clause 4.4.2, Table 4.4.2-1A and clause 6.3.2.2 Table 6.3.2.2-2 | See table 5.7.2-5 in TS 36.211 [41] for PRACH format 4 | |
| prach-ConfigInfo SEQUENCE { | | | |
| prach-ConfigIndex | 53 | | |
| highSpeedFlag | FALSE | | |
| zeroCorrelationZoneConfig | | | |
| prach-FreqOffset | See TS 36.508 [18] clause 4.6.8 | Channel-bandwidth-dependent parameter | |
| } | | | |
| } | | | |
| uplinkPowerControlCommonSCell-v1130 SEQUENCE | Not present | | |
| } | | | |

**Table 6.2.8.4.3-7: *RadioResourceConfigDedicatedSCell-r10*: Additional E-UTRAN TDD - Non-Contention Based Random Access test for Scell requirement**

| Derivation Path: 36.508, Table 4.6.3-19AA | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicatedSCell-r10 ::= SEQUENCE { | | | |
| physicalConfigDedicatedSCell-r10 | PhysicalConfigDedicated SCell-r10-DEFAULT | | |
| mac-MainConfigSCell-r11 SEQUENCE { | | | |
| stag-Id-r11 | 1 | | |
| } | | | |
| } | | | |

## 6.2.8.5 Test requirement

Tables 6.2.8.5-1 and 6.2.8.5-2 define the primary level settings for E-UTRAN TDD - non-contention based random access test for SCell. Table 6.2.8.5-5 defines the uplink timing error limit including test tolerances.

**Table 6.2.8.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD - Non-Contention Based Random Access test for SCell**

| Parameter | Unit | Cell 1 | Cell 2 | Comments |
|---|---|---|---|---|
| E-UTRA RF Channel Number | - | 1 | 1 | |
| $BW_{channel}$ | MHz | 10 | 10 | |
| Active PCell | | Cell 1 | | Primary cell of RF channel number 1. |
| Active SCell | | | Cell 2 | Secondary cell of RF channel number 2. |
| TAG configuration | | pTAG | sTAG | pTAG+sTAG configures Cell 1 and Cell 2 to separate TAGs |
| OCNG Pattern | - | OP.1 TDD | OP.1 TDD | As defined in D.2.1. |
| PDSCH parameters | - | DL Reference Measurement Channel R.0 TDD | DL Reference Measurement Channel R.0 TDD | As defined in A.1.2. |
| PCFICH/PDCCH/PHICH parameters | - | DL Reference Measurement Channel R.6 TDD | DL Reference Measurement Channel R.6 TDD | As defined in A.2.2. |
| Special subframe configuration | - | 6 | 6 | As specified in table 4.2-1 in TS 36.211 [9]. |
| Uplink-downlink configuration | - | 1 | 1 | As specified in table 4.2-2 in TS 36.211 [9]. |
| PBCH_RA | dB | 0 | 0 | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $\hat{E}_s / I_{ot}$ | dB | 3 | 3 | |
| $N_{oc}$ | dBm/15 KHz | -98 | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 3 | 3 | |
| Io Note 2 | dBm/9 MHz | -65.5 | -65.5 | |
| RSRP Note 3 | dBm/15 KHz | -95 | -95 | |
| referenceSignalPower | dBm/15 KHz | -5 | -5 | As defined in clause 6.3.2 in TS 36.331 [5]. |
| Configured UE transmitted power ( $P_{CMAX,c}$ ) | dBm | 23 | 23 | As defined in clause 6.2.5 in TS 36.101 [2]. |
| PRACH Configuration Index | - | 53 | 53 | As defined in table 5.7.1-3 in TS 36.211 [9]. |
| Backoff Parameter Index | - | 2 | 2 | As defined in table 7.2-1 in TS 36.321 [11]. |
| Propagation Condition | - | AWGN | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Io level has been derived from other parameters for information purpose. It is not a settable parameter.
Note 3: RSRP level has been derived from other parameters for information purposes. It is not a settable parameter.

**Table 6.2.8.5-2: RACH-Configuration parameters for E-UTRAN TDD - Non-Contention Based Random Access test for SCell**

| Field | Value | Comment |
|---|---|---|
| powerRampingStep | dB2 | |
| preambleInitialReceivedTargetPower | dBm-120 | |
| preambleTransMax | n6 | |
| ra-ResponseWindowSize | sf10 | 10 sub-frames |
| Note: For further information see Clause 6.3.2 in TS 36.331 [5]. | | |

Test 1: Correct behaviour when receiving Random Access Response

- The power of the first preamble shall be -22 dBm to within the accuracy specified Table 6.2.8.5-3.
- The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.8.5-4.
- The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.8.5-5.

Test 2: Correct behaviour when not receiving Random Access Response

- The power of the first preamble shall be -22 dBm to within the accuracy specified in Table 6.2.8.5-3.
- The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.8.5-4.
- The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.8.5-5.

Test 3: Correct behaviour when maximum number of preamble transmission counter has been reached

- The power of the first preamble shall be -30 dBm to within the accuracy specified Table 6.2.8.5-3.
- The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.8.5-4.
- The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.8.5-5.

**Table 6.2.8.5-3: Absolute power tolerance for E-UTRAN TDD - Non-Contention Based Random Access test for SCell**

| Conditions | Tolerance |
|---|---|
| Normal | ± 10.5 dB |
| Extreme | ± 13.5 dB |

**Table 6.2.8.5-4: Relative power tolerance for E-UTRAN TDD - Non-Contention Based Random Access test for SCell**

| power step size (Up or down) | PRACH | |
|---|---|---|
| | Normal Conditions | Extreme Conditions |
| ΔP [dB] | [dB] | [dB] |
| 2 ≤ ΔP < 3 | ± 3.7 | ± 5.7 |
| Note 1: For extreme conditions an additional ± 2.0 dB relaxation is allowed for PRACH allocations | | |

**Table 6.2.8.5-5: Test requirements for Te Timing Error Limit for E-UTRAN TDD – Non-Contention Based Random Access test for SCell**

| Downlink Bandwidth (MHz) | $T_e$ |
|---|---|
| ≥3 | 15*Ts |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

# 6.2.9

# 6.2.10 E-UTRAN FDD - Contention based random access test for Cat-M1 UEs in normal coverage

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- The Test system uncertainties applicable to this test are undefined

- The Test tolerances applicable to this test are undefined

- Annex E needs to be updated

- Some requirements in Test 1 and Test 2 are still within square brackets in TS36.133

- prach-StartingSubframe is smaller than numRepetitionPerPreambleAttempt for Level 2 and Level 3 in Table 6.2.10.5-3, which violates TS 36.331 spec. It needs to be corrected in TS 36.133.

## 6.2.10.1 Test purpose

The purpose of this test is to verify whether the behaviour of the random access procedure of a Cat-M1 UE in Normal Coverage is according to the E-UTRAN FDD contention based random access requirements specified in TS 36.133 [4] Clause 6.2.2, Clause 6.2.3 and Clause 7.1.2 in an AWGN channel model, whether the PRACH power settings and timing are within the specified limits and that the UE determines properly the enhanced coverage level based on the RSRP measurement and the configured criterion in RSRP-ThresholdsPrach as defined in TS 36.331 [5].

## 6.2.10.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 13 and forward of UE category M1.

## 6.2.10.3 Minimum conformance requirements

The random access procedure is used when establishing the layer 1 communication between the UE and E-UTRAN. The random access is specified in clause 6 of TS 36.213[8] and the control of the RACH transmission is specified in clause 5.1 of TS 36.321[11]. Contention based random access procedures can only be carried out on PCell and PSCell.

The CAT-M1 UE shall have capability to calculate PRACH transmission power according to the PRACH power formula as defined in TS 36.213 [8] clause 6.1 and apply this power level at the first preamble or additional preambles. The absolute power applied to the first preamble shall have the accuracy as defined in Table 6.3.5.1.1-1 of TS 36.101 [2]. The relative power applied to additional preambles shall have the accuracy as specified in Table 6.3.5.2.1-1 of TS 36.101 [2].

The CAT-M1 UE shall indicate a Random Access problem to upper layers if the maximum number of preambles transmission counter has been reached for the random access procedure on PCell or PSCell as specified in clause 5.1.4 in TS 36.321 [11].

The CAT-M1 UE shall stop preamble transmission if maximum number of preamble transmission counter has been reached for the random access procedure on an activated Scell as specified in clause 5.1.4 in TS 36.321 [11].

The CAT-M1 UE may stop monitoring for Random Access Response(s) and shall transmit the msg3 if the Random Access Response contains a Random Access Preamble identifier corresponding to the transmitted Random Access Preamble.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if all received Random Access Responses contain Random Access Preamble identifiers that do not match the transmitted Random Access Preamble.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if no Random Access Response is received within the RA Response window defined in clause 5.1.4 TS 36.321 [11].

The Cat-M1 UE shall re-transmit the msg3 upon the reception of a NACK on msg3.

The Cat-M1 UE shall send ACK if the Contention Resolution is successful.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires unless the received message includes a UE Contention Resolution Identity MAC control element and the Cat-M1 UE Contention Resolution Identity included in the MAC control element matches the CCCH SDU transmitted in the uplink message.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if the Contention Resolution Timer expires.

In addition to the requirements defined in TS 36.133 [4] clause 6.2.1 and 6.2.2, the Cat-M1 UE shall also execute the random access procedure defined in clause 5.1 in TS 36.321 [11] using the PRACH configuration contained in *PRACH-ConfigSIB* in TS 36.331 [5]:

- Determine the enhanced coverage level based on the RSRP measurement and the configured criterion (*RSRP-ThresholdsPrach* in TS 36.331 [5]) as defined in section 5.1.1, TS 36.321 [11];
- Select PRACH resources [5] configured for the corresponding enhanced coverage level as determined in the previous step;
- Transmits or re- transmits PRACH preamble using the selected PRACH resources and PRACH configuration.

The normative reference for this requirement is TS 36.133 [4] clause 6.2.2.1, 6.2.3, 7.1.2 and A.6.2.10.

The test parameters are given in Tables 6.2.10.5-1, 6.2.10.5-2 and 6.2.10.5-3.

## 6.2.10.4 Test description

### 6.2.10.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18 using only main UE Tx/Rx antenna.
2. Propagation conditions are set according to Annex B clause B.0.
3. There is one E-UTRA FDD carrier and one cell specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

### 6.2.10.4.2 Test procedures

The test consists of a single cell. The UE has a downlink PDSCH allocated centred on the centre sub-carrier. The EUTRAN shall not explicitly signal a Random Access Preamble via dedicated signalling in the downlink to the Cat-M1 UE.

1. Ensure the CAT-M1 UE is in State 2A-RF-CE
2. Set the parameters according to Tables 6.2.10.5-1, 6.2.10.5-2 and 6.2.10.5-3 as appropriate where CE level 0 shall be used. Propagation conditions are set according to Annex B clause B.1.1.
3. The CAT-M1 UE shall establish a connection setup with SS, the random access procedure within the connection setup is used in the test.
4. Test 1: Correct behaviour when receiving random access response

   4.1. In Test 1, the CAT-M1 UE shall send the preamble to the SS. In response to the first [4] preambles, the SS shall transmit a random access response not corresponding to the transmitted random access preamble.

   4.2. The CAT-M1 UE shall consider the random access response reception not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires if all

received random access response contain random access preamble identifiers that do not match the transmitted random access preamble.

4.3. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after [5] preambles have been received by the SS.

4.4. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

4.5. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.10.5-4 and 6.2.10.5-6. The power of the first preamble shall be -30 dBm. Measure the relative power and timing applied to additional preambles and it shall not exceed the values specified in Table 6.2.10.5-5. The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.10.5-6.

5. Test 2: Correct behaviour when not receiving random access response

5.1. Repeat step 1-3.

5.2. In Test 2, the CAT-M1 UE shall send the preamble to the SS. The SS shall not respond to the first [4] preambles.

5.3. The CAT-M1 UE shall consider the random access response reception not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires if no random access response is received within the RA Response window of 20 sub-frames.

5.4. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after [5] preambles have been received by the SS.

5.5. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

5.6. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.10.5-4 and 6.2.10.5-6. The power of the first preamble shall be -30 dBm. Measure the relative power and timing applied to additional preambles and it shall not exceed the values specified in Table 6.2.10.5-5. The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.10.5-6.

6. Test 3: Correct behaviour when receiving a NACK on msg3

6.1. Repeat step 1-3.

6.2. In Test 3, the CAT-M1 UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

6.3. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

6.4. The SS shall send NACK to all UE msg3 following a successful random access response.

6.5. The CAT-M1 UE shall consider this contention resolution not successful then re-transmit the msg3 upon the reception of a NACK on msg3 until the maximum number of HARQ retransmission is reached.

7. Test 4: Correct behaviour when receiving an incorrect message over Temporary C-RNTI

7.1. Repeat step 1-3.

7.2. In Test 4, the CAT-M1 UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

7.3. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

7.4. The SS shall send a message addressed to the Temporary C-RNTI with a UE contention resolution identity included in the MAC control element not matching the CCCH SDU transmitted in msg3 uplink message.

7.5. The CAT-M1 UE shall consider the contention resolution not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires.

8. Test 5: Correct behaviour when receiving a correct message over Temporary C-RNTI

8.1. In Test 5, the SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

8.2. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

8.3. The SS shall send a message addressed to the Temporary C-RNTI with a UE contention resolution identity included in the MAC control element matching the CCCH SDU transmitted in the msg3 uplink message.

8.4. The CAT-M1 UE shall send ACK and consider the contention resolution successful and the random access procedure successfully completed.

9. Test 6: Correct behaviour when contention resolution timer expires

9.1. Repeat step 1-3.

9.2. In Test 6, the CAT-M1 UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

9.3. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

9.4. The SS shall send an ACK for msg3 but not send msg4 allowing the contention resolution timer to expire.

9.5. The CAT-M1 UE shall consider the contention resolution not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires.

9.6. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

9.7. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

10. Test 7: PRACH Resource Selection

10.1. Repeat step 1-3.

10.2. In Test 7, the CAT-M1 UE shall send the preamble to the SS.

10.3. The CAT-M1 UE shall determines the enhanced coverage level 0 based on the RSRP measurement and the configured criterion (*RSRP-ThresholdsPrach* in TS 36.331 [5]) as defined in section 5.1.1, TS 36,321 [11].

10.4. The CAT-M1 UE shall selects PRACH resources configured for the corresponding enhanced coverage level 0 as determined in the previous step.

10.5. The CAT-M1 UE shall transmit or re- transmits PRACH preamble using the selected PRACH resources, PRACH configuration and calculated PRACH transmission power for the corresponding coverage level 0.

### 6.2.10.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.2.10.4.3-1: Common Exception messages for E-UTRAN FDD - Contention Based Random Access test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.6-1<br>Table H.2.6-2<br>Table H.2.6-3 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

**Table 6.2.10.4.3-2: *UplinkPowerControlCommon-DEFAULT*: Additional E-UTRAN FDD – Contention Based Random Access test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-25 UplinkPowerControlCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlCommon-DEFAULT ::= SEQUENCE { | | | |
| p0-NominalPUSCH | -101 (-101 dBm) | | |
| } | | | |

## 6.2.10.5 Test requirement

Tables 6.2.10.5-1, 6.2.10.5-2 and 6.2.10.5-3 define the primary level settings for E-UTRAN FDD - contention based random access test. Table 6.2.10.5-6 defines the uplink timing error limit including test tolerances.

For the tests specified in section 6.2.10.4.2, the power of the first preamble shall be -30 dBm to within the accuracy specified in Table 6.2.10.5-4. The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.10.5-5. The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.10.5-6.

**Table 6.2.10.5-1: General test parameters for FDD contention based random access test**

| Parameter | Unit | Value | Comments |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern Note 1 | | OP.21 FDD | As defined in TS 36.133 [4] clause A.3.2.1.21. |
| PDSCH parameters Note 2 | | R.20 FDD | As defined in TS 36.133 [4] clause A.3.1.4.1 |
| MPDCCH parameters Note 2 | | R.16 FDD | As defined in TS 36.133 [4] clause A.3.1.3.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As defined in TS 36.133 [4] clause A.3.1.2.1 |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| MPDCCH_RA | dB | | |
| MPDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $N_{oc}$ | dBm/15 KHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 3+TT | |
| $\hat{E}_s / I_{ot}$ Note 3 | dB | 3+TT | |
| RSRP Note 3 | dBm/15 KHz | -95+TT | |
| Io Note 3 | dBm/9 MHz | -65.5+TT | |
| Propagation Condition | - | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. The OCNG pattern is chosen during the test according to the presence of a DL reference measurement channel.

Note 2: The PDSCH and MPDCCH reference measurement channels are used in the test only when a downlink transmission dedicated to the UE under test is required.

Note 3: Es/Iot, RSRP and Io level has been derived from other parameters for information purpose. They are not settable parameters themselves.

**Table 6.2.10.5-2: RACH-Configuration parameters for FDD contention based random access test**

| Field | Value | | | | Comment |
|---|---|---|---|---|---|
| **Parameters not per CE Levels** | | | | | |
| powerRampingStep | dB2 | | | | |
| preambleInitialReceivedTargetPower | dBm-120 | | | | |
| preambleTransMax | n6 | | | | |
| maxHARQ-Msg3Tx | 4 | | | | |
| rar-HoppingConfig | Off | | | | |
| **Parameters per CE Levels** | | | | | |
| ***CE Level*** | ***Level 0*** | ***Level 1*** | ***Level 2*** | ***Level 3*** | |
| ra-ResponseWindowSize (per CE) | sf20 | sf80 | sf180 | sf320 | |
| mac-ContentionResolutionTimer (per CE) | sf80 | sf120 | sf200 | sf480 | |
| PreambleMappingInfo {firstPreamble, lastPreamble} | {0, 9} | {10,19} | {20,29} | {30,39} | |
| Note: For further information see Clause 6.3.2 in TS 36.331 [5]. | | | | | |

**Table 6.2.10.5-3: PRACH-Configuration parameters for FDD contention based random access test**

| Field | Value | | | | Comment |
|---|---|---|---|---|---|
| **Parameters not per CE Levels** | | | | | |
| rsrp-ThresholdsPrach | {24, 31, 38} | | | | Corresponding to {-116, -109, -102} dBm as defined in TS36.133 [4] clause 9.1.21.5 |
| mpdcch-startSF-CSS-RA | v1 | | | | |
| referenceSignalPower | -5 dBm/15 KHz | | | | As defined in clause 6.3.2 in TS 36.331 [5]. |
| maxHARQ-Msg3Tx | 4 | | | | As defined in Table 5.7.1-2 in TS 36.211 [9] |
| Backoff Parameter Index | 2 | | | | As defined in Table 7.2-1 in TS 36.321 [11] |
| Configured UE transmitted power ( $P_{CMAX}$ ) | 23 dBm | | | | As defined in clause 6.2.5 in TS 36.101 [2] |
| **Parameters per PRACH CE Levels** | | | | | |
| ***CE Level*** | ***Level 0*** | ***Level 1*** | ***Level 2*** | ***Level 3*** | |
| prach-ConfigIndex | 4 | 4 | 4 | 4 | As defined in Table 5.7.1-2 in TS 36.211 [9] |
| prach-FreqOffset | 0 | 0 | 0 | 0 | |
| prach-StartingSubframe | sf2 | sf4 | Sf8 | sf16 | |
| maxNumPreambleAttempt | n3 | n5 | n7 | n10 | |
| numRepetitionPerPreambleAttempt | n1 | n4 | n16 | n64 | |
| mpdcch-NarrowbandsToMonitor | 2 | 2 | 2 | 2 | |
| mpdcch-NumRepetition-RA | r8 | r8 | r128 | r128 | |
| prach-HoppingConfig | Off | Off | Off | Off | |
| Note 1: See Clause 6.3.2 in TS 36.331 [5] for further information on the parameters in this table. | | | | | |

**Table 6.2.10.5-4: Absolute power tolerance for E-UTRAN FDD - Contention Based Random Access test**

| Tolerance | |
|---|---|
| **Normal Conditions** | **Extreme Conditions** |
| ± 9 dB+TT | ± 12 dB+TT |
| Note: TS 36.133 [4] clause A.6.2.10.2 and TS 36.101 [2] clause 6.3.5.1.1. | |

**Table 6.2.10.5-5: Relative power tolerance for E-UTRAN FDD - Contention Based Random Access test**

| Power Step Size (Up or Down) | PRACH | |
|---|---|---|
| | Normal Conditions | Extreme Conditions |
| ΔP [dB] | [dB] | [dB] |
| 2 ≤ ΔP < 3 | ± 3.0+TT | ± 5.0+TT |
| Note 1: TS 36.133 [4] clause A.6.2.10.2 and TS 36.101 [2] clause 6.3.5.2.1.<br>Note 2: For extreme conditions an additional ± 2.0 dB relaxation is allowed for PRACH allocations. | | |

**Table 6.2.10.5-6: Test requirements for $T_e$ Timing Error Limit for E-UTRAN FDD – Contention Based Random Access test**

| Downlink Bandwidth (MHz) | $T_e$ |
|---|---|
| 1.4 | 24*$T_S$ |
| ≥ 3 | 12*$T_S$ |
| Note 1: TS 36.133 [4] clauses A.6.2.10.2 and clause 7.1.2.<br>Note 2: $T_S$ is the basic timing unit defined in TS 36.211 [9]. | |

## 6.2.11 E-UTRAN HD-FDD - Contention based random access test for Cat-M1 UEs in normal coverage

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- The Test system uncertainties applicable to this test are undefined.

- The Test tolerances applicable to this test are undefined.

- Annex E needs to be updated.

- Some requirements in Test 1 and Test 2 are still within square brackets in TS36.133.

### 6.2.11.1 Test purpose

The purpose of this test is to verify whether the behaviour of the random access procedure of a Cat-M1 UE in Normal Coverage is according to the E-UTRAN HD-FDD contention based random access requirements specified in TS 36.133 [4] Clause 6.2.2, Clause 6.2.3 and Clause 7.1.2 in an AWGN channel model, whether the PRACH power settings and timing are within the specified limits and that the UE determines properly the enhanced coverage level based on the RSRP measurement and the configured criterion in RSRP-ThresholdsPrach as defined in TS 36.331 [5].

### 6.2.11.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 13 and forward of UE category M1.

### 6.2.11.3 Minimum conformance requirements

The random access procedure is used when establishing the layer 1 communication between the UE and E-UTRAN. The random access is specified in clause 6 of TS 36.213[8] and the control of the RACH transmission is specified in clause 5.1 of TS 36.321[11]. Contention based random access procedures can only be carried out on PCell and PSCell.

The CAT-M1 UE shall have capability to calculate PRACH transmission power according to the PRACH power formula as defined in TS 36.213 [8] clause 6.1 and apply this power level at the first preamble or additional preambles. The absolute power applied to the first preamble shall have the accuracy as defined in Table 6.3.5.1.1-1 of TS 36.101 [2]. The relative power applied to additional preambles shall have the accuracy as specified in Table 6.3.5.2.1-1 of TS 36.101 [2].

The CAT-M1 UE shall indicate a Random Access problem to upper layers if the maximum number of preambles transmission counter has been reached for the random access procedure on PCell or PSCell as specified in clause 5.1.4 in TS 36.321 [11].

The CAT-M1 UE shall stop preamble transmission if maximum number of preamble transmission counter has been reached for the random access procedure on an activated Scell as specified in clause 5.1.4 in TS 36.321 [11].

The CAT-M1 UE may stop monitoring for Random Access Response(s) and shall transmit the msg3 if the Random Access Response contains a Random Access Preamble identifier corresponding to the transmitted Random Access Preamble.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if all received Random Access Responses contain Random Access Preamble identifiers that do not match the transmitted Random Access Preamble.

The CAT-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if no Random Access Response is received within the RA Response window defined in clause 5.1.4 TS 36.321 [11].

The CAT-M1 UE shall re-transmit the msg3 upon the reception of a NACK on msg3.

The Cat-M1 UE shall send ACK if the Contention Resolution is successful.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires unless the received message includes a UE Contention Resolution Identity MAC control element and the Cat-M1 UE Contention Resolution Identity included in the MAC control element matches the CCCH SDU transmitted in the uplink message.

The CAT-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if the Contention Resolution Timer expires.

In addition to the requirements defined in TS 36.133 [4] clause 6.2.1 and 6.2.2, the Cat-M1 UE shall also execute the random access procedure defined in clause 5.1 in TS 36.321 [11] using the PRACH configuration contained in *PRACH-ConfigSIB* in TS 36.331 [5]:

- Determine the enhanced coverage level based on the RSRP measurement and the configured criterion (*RSRP-ThresholdsPrach* in TS 36.331 [5]) as defined in section 5.1.1, TS 36.321 [11];
- Select PRACH resources [5] configured for the corresponding enhanced coverage level as determined in the previous step;
- Transmits or re- transmits PRACH preamble using the selected PRACH resources and PRACH configuration.

The normative reference for this requirement is TS 36.133 [4] clause 6.2.2.1, 6.2.3, 7.1.2 and A.6.2.10.

The test parameters are given in Tables 6.2.11.5-1, 6.2.11.5-2 and 6.2.11.5-3.

## 6.2.11.4 Test description

### 6.2.11.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18 using only main UE Tx/Rx antenna.
2. Propagation conditions are set according to Annex B clause B.0.
3. There is one E-UTRA HD-FDD carrier and one cell specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

#### 6.2.11.4.2 Test procedures

The test consists of a single cell. The UE has a downlink PDSCH allocated centred on the centre sub-carrier. The EUTRAN shall not explicitly signal a Random Access Preamble via dedicated signalling in the downlink to the Cat-M1 UE.

1. Ensure the CAT-M1 UE is in State 2A-RF-CE according to TS 36.508 [7] clause 7.2A.2.

2. Set the parameters according to Tables 6.2.11.5-1, 6.2.11.5-2 and 6.2.11.5-3 as appropriate, where the CE level 0 shall be used in this test. Propagation conditions are set according to Annex B clause B.1.1.

3. The CAT-M1 UE shall establish a connection setup with SS, the random access procedure within the connection setup is used in the test.

4. Test 1: Correct behaviour when receiving random access response

   4.1. In Test 1, the CAT-M1 UE shall send the preamble to the SS. In response to the first [4] preambles, the SS shall transmit a random access response not corresponding to the transmitted random access preamble.

   4.2. The CAT-M1 UE shall consider the random access response reception not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires if all received random access response contain random access preamble identifiers that do not match the transmitted random access preamble.

   4.3. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after [5] preambles have been received by the SS.

   4.4. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

   4.5. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.11.5-4 and 6.2.11.5-6. The power of the first preamble shall be -30 dBm. Measure the relative power and timing applied to additional preambles and it shall not exceed the values specified in Table 6.2.11.5-5. The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.11.5-6.

5. Test 2: Correct behaviour when not receiving random access response

   5.1. Repeat step 1-3.

   5.2. In Test 2, the CAT-M1 UE shall send the preamble to the SS. The SS shall not respond to the first [4] preambles.

   5.3. The CAT-M1 UE shall consider the random access response reception not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires if no random access response is received within the RA Response window of 20 sub-frames.

   5.4. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after [5] preambles have been received by the SS.

   5.5. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

   5.6. Measure the power and timing of the first preamble and it shall not exceed the values specified Tables 6.2.11.5-4 and 6.2.11.5-6. The power of the first preamble shall be -30 dBm. Measure the relative power and timing applied to additional preambles and it shall not exceed the values specified in Table 6.2.11.5-5. The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.11.5-6.

6. Test 3: Correct behaviour when receiving a NACK on msg3

   6.1. Repeat step 1-3.

   6.2. In Test 3, the CAT-M1 UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

   6.3. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

   6.4. The SS shall send NACK to all UE msg3 following a successful random access response.

6.5. The CAT-M1 UE shall consider this contention resolution not successful then re-transmit the msg3 upon the reception of a NACK on msg3 until the maximum number of HARQ retransmission is reached.

7. Test 4: Correct behaviour when receiving an incorrect message over Temporary C-RNTI

7.1. Repeat step 1-3.

7.2. In Test 4, the CAT-M1 UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

7.3. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

7.4. The SS shall send a message addressed to the Temporary C-RNTI with a UE contention resolution identity included in the MAC control element not matching the CCCH SDU transmitted in msg3 uplink message.

7.5. The CAT-M1 UE shall consider the contention resolution not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires.

8. Test 5: Correct behaviour when receiving a correct message over Temporary C-RNTI

8.1. In Test 5, the SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

8.2. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

8.3. The SS shall send a message addressed to the Temporary C-RNTI with a UE contention resolution identity included in the MAC control element matching the CCCH SDU transmitted in the msg3 uplink message.

8.4. The CAT-M1 UE shall send ACK and consider the contention resolution successful and the random access procedure successfully completed.

9. Test 6: Correct behaviour when contention resolution timer expires

9.1. Repeat step 1-3.

9.2. In Test 6, the CAT-M1 UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

9.3. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

9.4. The SS shall send an ACK for msg3 but not send msg4 allowing the contention resolution timer to expire.

9.5. The CAT-M1 UE shall consider the contention resolution not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires.

9.6. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

9.7. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

10. Test 7: PRACH Resource Selection

10.1. Repeat step 1-3.

10.1. In Test 7, the CAT-M1 UE shall send the preamble to the SS.

10.3. The CAT-M1 UE shall determines the enhanced coverage level 0 based on the RSRP measurement and the configured criterion (*RSRP-ThresholdsPrach* in TS 36.331 [5]) as defined in section 5.1.1, TS 36,321 [11].

10.4. The CAT-M1 UE shall selects PRACH resources configured for the corresponding enhanced coverage level 0 as determined in the previous step.

10.5. The CAT-M1 UE shall transmit or re- transmits PRACH preamble using the selected PRACH resources, PRACH configuration and calculated PRACH transmission power for the corresponding coverage level 0.

### 6.2.11.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.2.11.4.3-1: Common Exception messages for E-UTRAN HD-FDD - Contention Based Random Access test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.6-1<br>Table H.2.6-2<br>Table H.2.6-3 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

**Table 6.2.11.4.3-2: *UplinkPowerControlCommon-DEFAULT*: Additional E-UTRAN HD-FDD – Contention Based Random Access test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-25 UplinkPowerControlCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlCommon-DEFAULT ::= SEQUENCE { | | | |
| p0-NominalPUSCH | -101 (-101 dBm) | | |
| } | | | |

## 6.2.11.5 Test requirement

Tables 6.2.11.5-1, 6.2.11.5-2 and 6.2.11.5-3 define the primary level settings for E-UTRAN HD-FDD - contention based random access test. Table 6.2.11.5-6 defines the uplink timing error limit including test tolerances.

For the tests specified in section 6.2.11.4.2, the power of the first preamble shall be -30 dBm to within the accuracy specified in Table 6.2.11.5-4. The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.11.5-5. The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.11.5-6.

**Table 6.2.11.5-1: General test parameters for HD-FDD contention based random access test**

| Parameter | Unit | Value | Comments |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern Note 1 | | OP.21 FDD | As defined in TS 36.133 [4] clause A.3.2.1.21. |
| PDSCH parameters Note 2 | | R.10 HD-FDD | As defined in TS 36.133 [4] clause A.3.1.4.2 |
| MPDCCH parameters Note 2 | | R.6 HD-FDD | As defined in TS 36.133 [4] clause A.3.1.3.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.3 HD-FDD | As defined in TS 36.133 [4] clause A.3.1.2.3 |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| MPDCCH_RA | dB | | |
| MPDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $N_{oc}$ | dBm/15 KHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 3+TT | |
| $\hat{E}_s / I_{ot}$ Note 3 | dB | 3+TT | |
| RSRP Note 3 | dBm/15 KHz | -95+TT | |
| Io Note 3 | dBm/9 MHz | -65.5+TT | |
| Propagation Condition | - | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. The OCNG pattern is chosen during the test according to the presence of a DL reference measurement channel.

Note 2: The PDSCH and MPDCCH reference measurement channels are used in the test only when a downlink transmission dedicated to the UE under test is required.

Note 3: Es/Iot, RSRP and Io level has been derived from other parameters for information purpose. They are not settable parameters themselves.

**Table 6.2.11.5-2: RACH-Configuration parameters for FDD contention based random access test**

| Field | Value | | | | Comment |
|---|---|---|---|---|---|
| **Parameters not per CE Levels** | | | | | |
| powerRampingStep | dB2 | | | | |
| preambleInitialReceivedTargetPower | dBm-120 | | | | |
| preambleTransMax | n6 | | | | |
| maxHARQ-Msg3Tx | 4 | | | | |
| rar-HoppingConfig | Off | | | | |
| **Parameters per CE Levels** | | | | | |
| ***CE Level*** | ***Level 0*** | ***Level 1*** | ***Level 2*** | ***Level 3*** | |
| ra-ResponseWindowSize (per CE) | sf20 | sf80 | sf180 | sf320 | |
| mac-ContentionResolutionTimer (per CE) | sf80 | sf120 | sf200 | sf480 | |
| PreambleMappingInfo {firstPreamble, lastPreamble} | {0, 9} | {10,19} | {20,29} | {30,39} | |
| Note: For further information see Clause 6.3.2 in TS 36.331 [5]. | | | | | |

**Table 6.2.11.5-3: PRACH-Configuration parameters for FDD contention based random access test**

| Field | Value | | | | Comment |
|---|---|---|---|---|---|
| **Parameters not per CE Levels** | | | | | |
| rsrp-ThresholdsPrach | {24, 31, 38} | | | | Corresponding to {-116, -109, -102} dBm as defined in TS36.133 [4] clause 9.1.21.5 |
| mpdcch-startSF-CSS-RA | v1 | | | | |
| referenceSignalPower | -5 dBm/15 KHz | | | | As defined in clause 6.3.2 in TS 36.331 [5]. |
| maxHARQ-Msg3Tx | 4 | | | | As defined in Table 5.7.1-2 in TS 36.211 [9] |
| Backoff Parameter Index | 2 | | | | As defined in Table 7.2-1 in TS 36.321 [11] |
| Configured UE transmitted power ( $P_{CMAX}$ ) | 23 dBm | | | | As defined in clause 6.2.5 in TS 36.101 [2] |
| **Parameters per PRACH CE Levels** | | | | | |
| ***CE Level*** | ***Level 0*** | ***Level 1*** | ***Level 2*** | ***Level 3*** | |
| prach-ConfigIndex | 4 | 4 | 4 | 4 | As defined in Table 5.7.1-2 in TS 36.211 [9] |
| prach-FreqOffset | 0 | 0 | 0 | 0 | |
| prach-StartingSubframe | sf2 | sf4 | Sf8 | sf16 | |
| maxNumPreambleAttempt | n3 | n5 | n7 | n10 | |
| numRepetitionPerPreambleAttempt | n1 | n4 | n16 | n64 | |
| mpdcch-NarrowbandsToMonitor | 2 | 2 | 2 | 2 | |
| mpdcch-NumRepetition-RA | r8 | r8 | r128 | r128 | |
| prach-HoppingConfig | Off | Off | Off | Off | |
| Note 1: See Clause 6.3.2 in TS 36.331 [5] for further information on the parameters in this table. | | | | | |

**Table 6.2.11.5-4: Absolute power tolerance for E-UTRAN FDD - Contention Based Random Access test**

| Tolerance | |
|---|---|
| **Normal Conditions** | **Extreme Conditions** |
| ± 9 dB+TT | ± 12 dB+TT |
| Note: TS 36.133 [4] clause A.6.2.11.2 and TS 36.101 [2] clause 6.3.5.1.1. | |

**Table 6.2.11.5-5: Relative power tolerance for E-UTRAN FDD - Contention Based Random Access test**

| Power Step Size (Up or Down) | PRACH | |
|---|---|---|
| | Normal Conditions | Extreme Conditions |
| ΔP [dB] | [dB] | [dB] |
| 2 ≤ ΔP < 3 | ± 3.0+TT | ± 5.0+TT |
| Note 1: TS 36.133 [4] clause A.6.2.11.2 and TS 36.101 [2] clause 6.3.5.2.1.<br>Note 2: For extreme conditions an additional ± 2.0 dB relaxation is allowed for PRACH allocations. | | |

**Table 6.2.11.5-6: Test requirements for $T_e$ Timing Error Limit for E-UTRAN FDD – Contention Based Random Access test**

| Downlink Bandwidth (MHz) | $T_e$ |
|---|---|
| 1.4 | 24*$T_S$ |
| ≥ 3 | 12*$T_S$ |
| Note 1: TS 36.133 [4] clauses A.6.2.11.2 and clause 7.1.2.<br>Note 2: $T_S$ is the basic timing unit defined in TS 36.211 [9]. | |

## 6.2.12 E-UTRAN TDD Contention Based Random Access Test for Cat-M1 UEs in Normal Coverage

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- Some requirements in Test 1 and Test 2 are still within square brackets in TS36.133

### 6.2.12.1 Test purpose

To verify whether the behavior of the random access procedure of a Cat-M1 UE in Normal Coverage is according to the E-UTRAN TDD contention based random access requirements specified in TS 36.133 [4] Clause 6.2.2, Clause 6.2.3 and Clause 7.1.2 in an AWGN channel model, whether the PRACH power settings and timing are within the specified limits and that the UE determines properly the enhanced coverage level based on the RSRP measurement and the configured criterion in RSRP-ThresholdsPrach as defined in TS 36.331 [5].

### 6.2.12.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 13 and forward of UE category M1.

### 6.2.12.3 Minimum conformance requirements

The random access procedure is used when establishing the layer 1 communication between the UE and E-UTRAN. The random access is specified in clause 6 of TS 36.213[8] and the control of the RACH transmission is specified in clause 5.1 of TS 36.321[11]. Contention based random access procedures can only be carried out on PCell and PSCell.

The Cat-M1 UE shall have capability to calculate PRACH transmission power according to the PRACH power formula as defined in TS 36.213 [8] clause 6.1 and apply this power level at the first preamble or additional preambles. The absolute power applied to the first preamble shall have an accuracy as defined in table 6.3.5.1.1-1 of TS 36.101 [2]. The relative power applied to additional preambles shall have an accuracy as specified in table 6.3.5.2.1-1 of TS 36.101 [2].

The Cat-M1 UE shall indicate a Random Access problem to upper layers if the maximum number of preambles transmission counter has been reached for the random access procedure on PCell or PSCell as specified in clause 5.1.4 in TS 36.321 [11].

The Cat-M1 UE shall stop preamble transmission if maximum number of preamble transmission counter has been reached for the random access procedure on an activated Scell as specified in clause 5.1.4 in TS 36.321 [11].

The Cat-M1 UE may stop monitoring for Random Access Response(s) and shall transmit the msg3 if the Random Access Response contains a Random Access Preamble identifier corresponding to the transmitted Random Access Preamble.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if all received Random Access Responses contain Random Access Preamble identifiers that do not match the transmitted Random Access Preamble.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if no Random Access Response is received within the RA Response window defined in clause 5.1.4 TS 36.321 [11].

The Cat-M1 UE shall re-transmit the msg3 upon the reception of a NACK on msg3.

The Cat-M1 UE shall send ACK if the Contention Resolution is successful.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires unless the received message includes a UE Contention Resolution Identity MAC control element and the Cat-M1 UE Contention Resolution Identity included in the MAC control element matches the CCCH SDU transmitted in the uplink message.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if the Contention Resolution Timer expires.

In addition to the requirements defined in TS 36.133 [4] clause 6.2.1 and 6.2.2, the Cat-M1 UE shall also execute the random access procedure defined in clause 5.1 in TS 36.321 [11] using the PRACH configuration contained in *PRACH-ConfigSIB* in TS 36.331 [5]:

- Determine the enhanced coverage level based on the RSRP measurement and the configured criterion (*RSRP-ThresholdsPrach* in TS 36.331 [5]) as defined in section 5.1.1, TS 36.321 [11];
- Select PRACH resources [5] configured for the corresponding enhanced coverage level as determined in the previous step;
- Transmits or re- transmits PRACH preamble using the selected PRACH resources and PRACH configuration.

The normative reference for this requirement is TS 36.133 [4] clause 6.2.2.1, 6.2.3, 7.1.2 and A.6.2.12.

The test parameters are given in tables 6.2.12.5-1, 6.2.12.5-2 and 6.2.12.5-3.

## 6.2.12.4 Test description

### 6.2.12.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18(using only main UE Tx/Rx antenna).
2. Propagation conditions are set according to Annex B clause B.0.
3. There is one E-UTRA TDD carrier and one cell specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

### 6.2.12.4.2 Test procedure

The test consists of a single cell. The UE has a downlink PDSCH allocated on the centre sub-carrier. The E-UTRAN shall not explicitly signal a Random Access Preamble via dedicated signalling in the downlikn to the Cat-M1 UE.

1. Ensure the Cat-M1 UE is in State 2A-RF-CE according to TS 36.508 [7] clause 7.2A.2.

2. Set the parameters according to Tables 6.2.12.5-1, 6.2.12.5-2 and 6.2.12.5-3 as appropriate where CE level 0 shall be used. Propagation conditions are set according to Annex B clause B.1.1.

3. The Cat-M1 UE shall establish a connection setup with SS, the random access procedure within the connection setup is used in the test.

4. Test 1: Correct behaviour when receiving random access response

   4.1 In Test 1, the Cat-M1 UE shall send the preamble to the SS. In response to the first [4] preambles, the SS shall transmit a random access response not corresponding to the transmitted random access preamble.

   4.2 The Cat-M1 UE shall consider the random access response reception not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires if all received random access response contain random access preamble identifiers that do not match the transmitted random access preamble.

   4.3 The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after [5] preambles have been received by the SS.

   4.4 The Cat-M1 UE shall consider this random access response reception successful and transmit the msg3.

   4.5. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.12.5-4 and 6.2.12.5-6. The power of the first preamble shall be -30 dBm. Measure the relative power and timing applied to additional preambles and it shall not exceed the values specified in Table 6.2.12.5-5. The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.12.5-6.

5. Test 2: Correct behaviour when not receiving random access response reception

   5.1 Repeat step 1-3.

   5.2 In Test 2, the Cat-M1 UE shall send the preamble to the SS. The SS shall not respond to the first [4] preambles.

   5.3 The Cat-M1 UE shall consider the random access response reception not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires if no random access response is received within the RA Response window of 20 sub-frames.

   5.4 The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after [5] preambles have been received by the SS.

   5.5 The UE shall consider this random access response reception successful and transmit the msg3.

   5.6. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.12.5-4 and 6.2.12.5-6. The power of the first preamble shall be -30 dBm. Measure the relative power and timing applied to additional preambles and it shall not exceed the values specified in Table 6.2.12.5-5. The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.12.5-6.

6. Test 3: Correct behaviour when receiving a NACK on msg3

   6.1 Repeat step 1-3.

   6.2 In Test 3, the Cat-M1 UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preambles have been received by the SS.

   6.3 The Cat-M1 UE shall consider this random access response reception successful and transmit the msg3.

   6.4 The SS shall send NACK all UE msg3 following a successful random access response.

   6.5 The Cat-M1 UE shall consider this contention resolution not successful then re-transmit the msg3 upon the reception of a NACK on msg3 until the maximum number of HARQ retransmission is reached.

7. Test 4: Correct behaviour when receiving an incorrect message over Temporary C-RNTI

   7.1 Repeat step 1-3.

7.2 In Test 4, the Cat-M1 UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after first preambles have been received by the SS.

7.3 The Cat-M1 UE shall consider this random access response reception successful and transmit the msg3.

7.4 The SS shall send a message addressed to the Temporary C-RNTI with a UE contention resolution identity included in the MAC control element not matching the CCCH SDU transmitted in msg3 uplink message.

7.5 The Cat-M1 UE shall consider the contention resolution not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires.

8. Test 5: Correct behaviour when receiving a correct message over Temporary C-RNTI

8.1 Repeat step 1-3

8.2 In Test 5, the SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preambles have been received by the SS.

8.3 The Cat-M1 UE shall consider this random access response reception successful and transmit the msg3.

8.4 The SS shall send a message addressed to the Temporary C-RNTI with a UE contention resolution identity included in the MAC control element matching the CCCH SDU transmitted in the msg3 uplink message.

8.5 The Cat-M1 UE shall send ACK and consider the contention resolution successful and the random access procedure successfully completed.

9. Test 6: Correct behaviour when contention resolution timer expires

9.1 Repeat step 1-3.

9.2 In Test 6, the Cat-M1 UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble have been received by the SS.

9.3 The Cat-M1 UE shall consider this random access response reception successful and transmit the msg3.

9.4 The SS shall send an ACK for msg3 but not send msg4 allowing the contention resolution timer to expire.

9.5 The Cat-M1 UE shall consider the contention resolution not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires.

9.6. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

9.7. The Cat-M1 UE shall consider this random access response reception successful and transmit the msg3.

10. Test 7: PRACH Resource Selection

10.1. Repeat step 1-3.

10.2. In Test 7, the Cat-M1 UE shall send the preamble to the SS.

10.3. The Cat-M1 UE shall determines the enhanced coverage level 0 based on the RSRP measurement and the configured criterion (RSRP-ThresholdsPrach in TS 36.331 [5]) as defined in section 5.1.1, TS 36,321 [11].

10.4. The Cat-M1 UE shall selects PRACH resources configured for the corresponding enhanced coverage level 0 as determined in the previous step.

10.5. The Cat-M1 UE shall transmit or re-transmits PRACH preamble using the selected PRACH resources, PRACH configuration and calculated PRACH transmission power for the corresponding coverage level 0.

#### 6.2.12.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.2.12.4.3-1: Common Exception messages for E-UTRAN TDD -Contention Based Random Access test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | H.2.6-1<br>H.2.6-2<br>H.2.6-3 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-2 |

**Table 6.2.12.4.3-2: *UplinkPowerControlCommon-DEFAULT*: Additional E-UTRAN TDD - Contention Based Random Access test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-25 UplinkPowerControlCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlCommon-DEFAULT ::= SEQUENCE { | | | |
| p0-NominalPUSCH | -101 (-101 dBm) | | |
| } | | | |

**Table 6.2.12.4.3-3: *RACH-ConfigCommon-DEFAULT*: E-UTRAN TDD - Contention Based Random Access test requirement**

| Derivation Path: 36.508 clause 4.6.3 Table 4.6.3-12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RACH-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| preambleInfo SEQUENCE { | | | |
| preamblesGroupAConfig SEQUENCE { | | | |
| sizeOfRA-PreamblesGroupA | n52 | No group B. | |
| } | | | |
| } | | | |
| } | | | |
| RACH-CE-LevelInfoList-r13 ::= SEQUENCE (SIZE (1..maxCE-Level-r13)) OF SEQUENCE { | | | |
| preambleMappingInfo-r13[1] SEQUENCE { | | | |
| firstPreamble-r13 | 0 | | |
| lastPreamble-r13 | 9 | | |
| } | | | |
| preambleMappingInfo-r13[2] SEQUENCE { | | | |
| firstPreamble-r13 | 10 | | |
| lastPreamble-r13 | 19 | | |
| } | | | |
| } | | | |

**Table 6.2.12.4.3-4: *PRACH-Config-v1310-DEFAULT*: E-UTRAN TDD - Contention Based Random Access test requirement**

| Derivation Path: 36.508 clause 4.6.3 Table 4.6.3-7A | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PRACH-Config-v1310-DEFAULT ::= SEQUENCE { | | | |
| rsrp-ThresholdsPrachInfoList-r13 SEQUENCE (SIZE(1..3)) OF { | 3 entry | | |
| RSRP-Range[1] | 24 | -116 dBm | |
| RSRP-Range[2] | 31 | -109 dBm | |
| RSRP-Range[3] | 38 | -102 dBm | |
| } | | | |
| PRACH-Config-v1310-DEFAULT ::= SEQUENCE { | | | |
| prach-ParametersListCE-r13 SEQUENCE (SIZE(1..maxCE-Level-r13)) OF SEQUENCE { | 2 entries | | |
| PRACH-ParametersCE-r13[1] SEQUENCE { | | | |
| prach-ConfigIndex-r13[1] | 53 | INTEGER (0..63) | TDD |
| prach-FreqOffset-r13[1] | Same as prach-FrequencyOffset | | |
| prach-StartingSubframe-r13[1] | sf2 | | |
| maxNumPreambleAttemptCE-r13[1] | n3 | | |
| numRepetitionPerPreambleAttempt-r13[1] | n1 | | |
| mpdcch-NarrowbandsToMonitor-r13 SEQUENCE (SIZE(1..2)) OF INTEGER (1..maxAvailNarrowBands-r13)[1] { | 1 entry | | |
| mpdcch-NarrowbandsToMonitor-r13[1] | 2 | | |
| } | | | |
| mpdcch-NumRepetition-RA-r13[1] | r8 | | |
| prach-HoppingConfig-r13[1] | off | | |
| } | | | |
| prach-ConfigIndex-r13[2] | 53 | INTEGER (0..63) | TDD |
| prach-FreqOffset-r13[2] | Same as prach-FrequencyOffset | | |
| prach-StartingSubframe-r13[2] | sf4 | | |
| maxNumPreambleAttemptCE-r13[2] | n5 | | |
| numRepetitionPerPreambleAttempt-r13[2] | n4 | | |
| mpdcch-NarrowbandsToMonitor-r13 SEQUENCE (SIZE(1..2)) OF INTEGER (1..maxAvailNarrowBands-r13)[2] { | 1 entry | | |
| mpdcch-NarrowbandsToMonitor-r13[1] | 2 | | |
| } | | | |
| mpdcch-NumRepetition-RA-r13[2] | r8 | | |
| prach-HoppingConfig-r13[2] | off | | |
| } | | | |
| } | | | |
| } | | | |

## 6.2.12.5 Test requirement

Tables 6.2.12.5-1, 6.2.12.5-2 and 6.2.12.5-3 define the primary level settings for E-UTRAN TDD - contention based random access test. Table 6.2.12.5-6 defines the uplink timing error limit including test tolerances.

For the tests specified in section 6.2.12.4.2, the power of the first preamble shall be -30 dBm to within the accuracy specified in Table 6.2.12.5-4. The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.12.5-5. The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.12.5-6.

**Table 6.2.12.5-1: Cell Specific Test requirement parameters for TDD contention based random access test**

| Parameter | Unit | Value | Comments |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern Note 1 | | OP.11 TDD | As defined in TS 36.133 [4] clause A.3.2.2.11. |
| PDSCH parameters Note 2 | | R.16 TDD | As defined in TS 36.133 [4] clause A.3.1.4.3 |
| MPDCCH parameters Note 2 | | R.14 TDD | As defined in TS 36.133 [4] clause A.3.1.3.3 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As defined in TS 36.133 [4] clause A.3.1.2.2 |
| Special subframe configuration | - | 6 | As specified in table 4.2-1 in TS 36.211 [11]. |
| Uplink-downlink configuration | - | 1 | As specified in table 4.2-2 in TS 36.211 [11]. |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| MPDCCH_RA | dB | | |
| MPDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $N_{oc}$ | dBm/15 KHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 3 | |
| $\hat{E}_s / I_{ot}$ Note 3 | dB | 3 | |
| RSRP Note 3 | dBm/15 KHz | -95 | |
| Io Note 3 | dBm/9 MHz | -65.5 | |
| Propagation Condition | - | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. The OCNG pattern is chosen during the test according to the presence of a DL reference measurement channel.

Note 2: The PDSCH and MPDCCH reference measurement channels are used in the test only when a downlink transmission dedicated to the UE under test is required.

Note 3: Es/Iot, RSRP and Io level has been derived from other parameters for information purpose. They are not settable parameters themselves.

**Table 6.2.12.5-2: RACH-Configuration parameters for TDD contention based random access test**

| Field | Value | | | | Comment |
|---|---|---|---|---|---|
| **Parameters not per CE Levels** | | | | | |
| numberOfRA-Preambles | n52 | | | | |
| sizeOfRA-PreamblesGroupA | n52 | | | | No group B. |
| powerRampingStep | dB2 | | | | |
| preambleInitialReceivedTargetPower | dBm-120 | | | | |
| preambleTransMax | n6 | | | | |
| maxHARQ-Msg3Tx | 4 | | | | |
| rar-HoppingConfig | Off | | | | |
| **Parameters per CE Levels** | | | | | |
| ***CE Level*** | ***Level 0*** | ***Level 1*** | ***Level 2*** | ***Level 3*** | |
| ra-ResponseWindowSize (per CE) | sf20 | sf80 | sf180 | sf320 | |
| mac-ContentionResolutionTimer (per CE) | sf80 | sf120 | sf200 | sf480 | |
| PreambleMappingInfo {firstPreamble, lastPreamble} | {0, 9} | {10,19} | {20,29} | {30,39} | |
| Note 1: For further information see Clause 6.3.2 in TS 36.331 [5]. | | | | | |

**Table 6.2.12.5-3: PRACH-Configuration parameters for TDD contention based random access test**

| Field | Value | | | | Comment |
|---|---|---|---|---|---|
| **Parameters not per CE Levels** | | | | | |
| rsrp-ThresholdsPrach | {24, 31, 38} | | | | Corresponding to {-116, -109, -102} dBm as defined in Section 9.1.21.5 |
| mpdcch-startSF-CSS-RA | v1 | | | | |
| referenceSignalPower | -5 dBm/15 KHz | | | | As defined in clause 6.3.2 in TS 36.331 [5]. |
| maxHARQ-Msg3Tx | 4 | | | | As defined in table 5.7.1-2 in TS 36.211 [9] |
| Backoff Parameter Index | 2 | | | | As defined in table 7.2-1 in TS 36.321 [11] |
| Configured UE transmitted power ($P_{CMAX}$) | 23 dBm | | | | As defined in clause 6.2.5 in TS 36.101 [2] |
| **Parameters per PRACH CE Levels** | | | | | |
| **CE Level** | **Level 0** | **Level 1** | **Level 2** | **Level 3** | |
| prach-ConfigIndex | 53 | 53 | 53 | 53 | As defined in table 5.7.1-3 in TS 36.211 [9] |
| prach-FreqOffset | 0 | 0 | 0 | 0 | |
| prach-StartingSubframe | sf2 | sf4 | sf8 | sf16 | |
| maxNumPreambleAttempt | n3 | n5 | n7 | n10 | |
| numRepetitionPerPreambleAttempt | n1 | n4 | n16 | n64 | |
| mpdcch-NarrowbandsToMonitor | 2 | 2 | 2 | 2 | |
| mpdcch-NumRepetition-RA | r8 | r8 | r128 | r128 | |
| prach-HoppingConfig | Off | Off | Off | Off | |
| Note 1: See Clause 6.3.2 in TS 36.331 [5] for further information on the parameters in this table. | | | | | |

**Table 6.2.12.5-4: Absolute power tolerance for E-UTRAN TDD - Contention Based Random Access test**

| Tolerance | |
|---|---|
| **Normal Conditions** | **Extreme Conditions** |
| ± 10.5 dB | ± 13.5 dB |
| Note: TS 36.133 [4] clause A.6.2.12.2 and TS 36.101 [2] clause 6.3.5.1.1 | |

**Table 6.2.12.5-5: Relative power tolerance for E-UTRAN TDD - Contention Based Random Access test**

| Power Step Size (Up or Down) | PRACH | |
|---|---|---|
| | Normal Conditions | Extreme Conditions |
| ΔP [dB] | [dB] | [dB] |
| $2 \leq \Delta P < 3$ | ± 3.7 | ± 5.7 |
| Note 1: TS 36.133 [4] clause A.6.2.12.2 and TS 36.101 [2] clause 6.3.5.2.1<br>Note 2: For extreme conditions an additional ± 2.0 dB relaxation is allowed for PRACH allocations | | |

**Table 6.2.12.5-6: Test requirements for $T_e$ Timing Error Limit for E-UTRAN TDD – Contention Based Random Access test**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| 1.4 | $27*T_S$ |
| ≥3 | $15*T_S$ |
| Note 1: TS 36.133 [4] clauses A.6.2.12.2 and clause 7.1.2<br>Note 2: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

## 6.2.13 E-UTRAN FDD - Contention Based Random Access Test for Cat-M1 UEs in Enhanced Coverage

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

- The Test system uncertainties applicable to this test are undefined (Cat-M1 power control uncertainty and TT requirements in TS 36.521-1 Aug 2016 clause 6.3.5EA.1 on are all undefined)

- The Test tolerances applicable to this test are undefined

- Annex E needs to be updated

- Some requirements in Test 1 and Test 2 and timing error limit are still within square brackets in TS36.133

- FFSs in TS 36.508 [7] clause 4.6.3, Table 4.6.3-7A PRACH-Config-v1310-DEFAULT and Table 4.6.3-12 PRACH-ConfigCommon-DEFAUL have not been updated. If the entries there are inconsistent with the parameters settings in this section, then more tables need to be added in Message contents section for exceptions.

### 6.2.13.1 Test purpose

The purpose of this test is to verify whether the behavior of the random access procedure of a Cat-M1 UE in Enhanced Coverage is according to the E-UTRAN FDD contention based random access requirements specified in TS 36.133 [4] Clause 6.2.2, Clause 6.2.3 and Clause 7.1.2 in an AWGN channel model, whether the PRACH power settings and timing are within the specified limits, and whether the UE determines properly the enhanced coverage level based on the RSRP measurement and the configured criterion in RSRP-ThresholdsPrach as defined in TS 36.331 [5].

### 6.2.13.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 13 and forward of UE category M1.

### 6.2.13.3 Minimum conformance requirements

The random access procedure is used when establishing the layer 1 communication between the UE and E-UTRAN. The random access is specified in clause 6 of TS 36.213[8] and the control of the RACH transmission is specified in clause 5.1 of TS 36.321[11]. Contention based random access procedures can only be carried out on PCell and PSCell.

The CAT-M1 UE shall have capability to calculate PRACH transmission power according to the PRACH power formula as defined in TS 36.213 [8] clause 6.1 and apply this power level at the first preamble or additional preambles. The absolute power applied to the first preamble shall have the accuracy as defined in Table 6.3.5.1.1-1 of TS 36.101

[2]. The relative power applied to additional preambles shall have the accuracy as specified in Table 6.3.5.2.1-1 of TS 36.101 [2].

The CAT-M1 UE shall indicate a Random Access problem to upper layers if the maximum number of preambles transmission counter has been reached for the random access procedure on PCell or PSCell as specified in clause 5.1.4 in TS 36.321 [11].

The CAT-M1 UE shall stop preamble transmission if maximum number of preamble transmission counter has been reached for the random access procedure on an activated Scell as specified in clause 5.1.4 in TS 36.321 [11].

The CAT-M1 UE may stop monitoring for Random Access Response(s) and shall transmit the msg3 if the Random Access Response contains a Random Access Preamble identifier corresponding to the transmitted Random Access Preamble.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if all received Random Access Responses contain Random Access Preamble identifiers that do not match the transmitted Random Access Preamble.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if no Random Access Response is received within the RA Response window defined in clause 5.1.4 TS 36.321 [11].

The Cat-M1 UE shall re-transmit the msg3 upon the reception of a NACK on msg3.

The Cat-M1 UE shall send ACK if the Contention Resolution is successful.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires unless the received message includes a UE Contention Resolution Identity MAC control element and the Cat-M1 UE Contention Resolution Identity included in the MAC control element matches the CCCH SDU transmitted in the uplink message.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if the Contention Resolution Timer expires.

Per TS 36.133 [4] clause 6.2.3, in addition to the requirements defined in TS 36.133 [4] clause 6.2.1 and 6.2.2, the Cat-M1 UE shall also execute the random access procedure defined in clause 5.1 in TS 36.321 [11] using the PRACH configuration contained in *PRACH-ConfigSIB* in TS 36.331 [5]:

- Determine the enhanced coverage level based on the RSRP measurement and the configured criterion (*RSRP-ThresholdsPrach* in TS 36.331 [5]) as defined in section 5.1.1, TS 36.321 [11];
- Select PRACH resources (TS 36.331 [5]) configured for the corresponding enhanced coverage level as determined in the previous step and;
- Transmits or re- transmits PRACH preamble using the selected PRACH resources and PRACH configuration.

The normative reference for this requirement is TS 36.133 [4] clause 6.2.2.1, 6.2.3, 7.1.2 and A.6.2.13.

The test parameters are given in Tables 6.2.13.5-1, 6.2.13.5-2 and 6.2.13.5-3.

## 6.2.13.4 Test description

### 6.2.13.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18 using only main UE Tx/Rx antenna.
2. Propagation conditions are set according to Annex B clause B.0.

3. There is one E-UTRA FDD carrier and one cell specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

### 6.2.13.4.2 Test procedures

The test consists of a single cell. The UE has a downlink PDSCH allocated centered on the centre sub-carrier. The EUTRAN shall not explicitly signal a Random Access Preamble via dedicated signaling in the downlink to the Cat-M1 UE.

1. Ensure the CAT-M1 UE is in State 2A-RF-CE

2. Set the parameters according to Tables 6.2.13.5-1, 6.2.13.5-2 and 6.2.13.5-3 as appropriate where CE level 1 shall be used. Propagation conditions are set according to Annex B clause B.1.1.

3. The CAT-M1 UE shall establish a connection setup with SS, the random access procedure within the connection setup is used in the test.

4. Test 1: Correct behavior when receiving random access response

   4.1. In Test 1, the CAT-M1 UE shall send the preamble to the SS. In response to the first [4] preambles, the SS shall transmit a random access response not corresponding to the transmitted random access preamble.

   4.2. The CAT-M1 UE shall consider the random access response reception not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires if all received random access response contain random access preamble identifiers that do not match the transmitted random access preamble.

   4.3. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after [5] preambles (the preamble may be transmitted multiple times in each attempt) have been received by the SS.

   4.4. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

   4.5. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.13.5-4 and 6.2.13.5-6. The power of the first preamble shall be -30 dBm. Measure the relative power and timing applied to additional preambles and it shall not exceed the values specified in Table 6.2.13.5-5. The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.13.5-6.

5. Test 2: Correct behavior when not receiving random access response

   5.1. Repeat step 1-3.

   5.2. In Test 2, the CAT-M1 UE shall send the preamble to the SS. The SS shall not respond to the first [4] preambles.

   5.3. The CAT-M1 UE shall consider the random access response reception not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires if no random access response is received within the RA Response window of 80 sub-frames.

   5.4. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after [5] preambles have been received by the SS.

   5.5. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

   5.6. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.13.5-4 and 6.2.13.5-6. The power of the first preamble shall be -30 dBm. Measure the relative power and timing applied to additional preambles and it shall not exceed the values specified in Table 6.2.13.5-5. The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.13.5-6.

6. Test 3: Correct behavior when receiving a NACK on msg3

   6.1. Repeat step 1-3.

   6.2. In Test 3, the CAT-M1 UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

6.3. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

6.4. The SS shall send NACK to all UE msg3 following a successful random access response.

6.5. The CAT-M1 UE shall consider this contention resolution not successful then re-transmit the msg3 upon the reception of a NACK on msg3 until the maximum number of HARQ retransmission is reached.

7. Test 4: Correct behavior when receiving an incorrect message over Temporary C-RNTI

7.1. Repeat step 1-3.

7.2. In Test 4, the CAT-M1 UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

7.3. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

7.4. The SS shall send a message addressed to the Temporary C-RNTI with a UE contention resolution identity included in the MAC control element not matching the CCCH SDU transmitted in msg3 uplink message.

7.5. The CAT-M1 UE shall consider the contention resolution not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires unless the received message includes a UE contention resolution identity MAC control element and the UE contention resolution identity included in the MAC control element matches the CCCH SDU transmitted in msg3 uplink message.

8. Test 5: Correct behavior when receiving a correct message over Temporary C-RNTI

8.1. In Test 5, the SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

8.2. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

8.3. The SS shall send a message addressed to the Temporary C-RNTI with a UE contention resolution identity included in the MAC control element matching the CCCH SDU transmitted in the msg3 uplink message.

8.4. The CAT-M1 UE shall send ACK and consider the contention resolution successful and the random access procedure successfully completed.

9. Test 6: Correct behavior when contention resolution timer expires

9.1. Repeat step 1-3.

9.2. In Test 6, the CAT-M1 UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

9.3. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

9.4. The SS shall send an ACK for msg3 but not send msg4 allowing the contention resolution timer to expire.

9.5. The CAT-M1 UE shall consider the contention resolution not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires.

9.6. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

9.7. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

10. Test 7: PRACH Resource Selection

10.1. Repeat step 1-3.

10.2. In Test 7, the CAT-M1 UE shall send the preamble to the SS.

10.3. The CAT-M1 UE shall determines the enhanced coverage level 1 based on the RSRP measurement and the configured criterion (*RSRP-ThresholdsPrach* in TS 36.331 [5]) as defined in section 5.1.1, TS 36,321 [11].

10.4. The CAT-M1 UE shall selects PRACH resources configured for the corresponding enhanced coverage level 1 as determined in the previous step.

10.5. The CAT-M1 UE shall transmit or re- transmits PRACH preamble using the selected PRACH resources, PRACH configuration and calculated PRACH transmission power for the corresponding coverage level 1.

### 6.2.13.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.2.13.4.3-1: Common Exception messages for E-UTRAN FDD - Contention Based Random Access test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.6-1<br>Table H.2.6-2<br>Table H.2.6-3 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

**Table 6.2.13.4.3-2: *UplinkPowerControlCommon-DEFAULT*: Additional E-UTRAN FDD – Contention Based Random Access test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-25 UplinkPowerControlCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlCommon-DEFAULT ::= SEQUENCE { | | | |
| p0-NominalPUSCH | -101 (-101 dBm) | | |
| } | | | |

## 6.2.13.5 Test requirement

Tables 6.2.13.5-1, 6.2.13.5-2 and 6.2.13.5-3 define the primary level settings for E-UTRAN FDD - contention based random access test. Table 6.2.13.5-6 defines the uplink timing error limit including test tolerances.

For the tests specified in section 6.2.13.4.2, the power of the first preamble shall be -30 dBm to within the accuracy specified in Table 6.2.13.5-4. The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.13.5-5. The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.13.5-6.

**Table 6.2.13.5-1: General test parameters for FDD contention based random access test**

| Parameter | Unit | Value | Comments |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern Note 1 | | OP.21 FDD | As defined in TS 36.133 [4] clause A.3.2.1.21. |
| PDSCH parameters Note 2 | | R.22 FDD | As defined in TS 36.133 [4] clause A.3.1.4.4 |
| MPDCCH parameters Note 2 | | R.18 FDD | As defined in TS 36.133 [4] clause A.3.1.3.4 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As defined in TS 36.133 [4] clause A.3.1.2.1 |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| MPDCCH_RA | dB | | |
| MPDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $N_{oc}$ | dBm/15 KHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | -12+TT | |
| $\hat{E}_s / I_{ot}$ Note 3 | dB | -12+TT | |
| RSRP Note 3 | dBm/15 KHz | -110+TT | |
| Io Note 3 | dBm/9 MHz | -70+TT | |
| Propagation Condition | - | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. The OCNG pattern is chosen during the test according to the presence of a DL reference measurement channel.

Note 2: The PDSCH and MPDCCH reference measurement channels are used in the test only when a downlink transmission dedicated to the UE under test is required.

Note 3: Es/Iot, RSRP and Io level has been derived from other parameters for information purpose. They are not settable parameters themselves.

**Table 6.2.13.5-2: RACH-Configuration parameters for FDD contention based random access test**

| Field | Value | | | | Comment |
|---|---|---|---|---|---|
| **Parameters not per CE Levels** | | | | | |
| powerRampingStep | dB2 | | | | |
| preambleInitialReceivedTargetPower | dBm-120 | | | | |
| preambleTransMax | n6 | | | | |
| maxHARQ-Msg3Tx | 4 | | | | |
| rar-HoppingConfig | Off | | | | |
| **Parameters per CE Levels** | | | | | |
| ***CE Level*** | ***Level 0*** | ***Level 1*** | ***Level 2*** | ***Level 3*** | |
| ra-ResponseWindowSize (per CE) | sf20 | sf80 | sf180 | sf320 | |
| mac-ContentionResolutionTimer (per CE) | sf80 | sf120 | sf200 | sf480 | |
| PreambleMappingInfo {firstPreamble, lastPreamble} | {0, 9} | {10,19} | {20,29} | {30,39} | |
| Note: For further information see Clause 6.3.2 in TS 36.331 [5]. | | | | | |

**Table 6.2.13.5-3: PRACH-Configuration parameters for FDD contention based random access test**

| Field | Value | | | | Comment |
|---|---|---|---|---|---|
| **Parameters not per CE Levels** | | | | | |
| rsrp-ThresholdsPrach | {16, 26, 36} | | | | Corresponding to {-124, -114, -104} dBm as defined in TS36.133 [4] clause 9.1.21.5 |
| mpdcch-startSF-CSS-RA | v1 | | | | |
| referenceSignalPower | -5 dBm/15 KHz | | | | As defined in clause 6.3.2 in TS 36.331 [5]. |
| maxHARQ-Msg3Tx | 4 | | | | As defined in Table 5.7.1-2 in TS 36.211 [9] |
| Backoff Parameter Index | 2 | | | | As defined in Table 7.2-1 in TS 36.321 [11] |
| Configured UE transmitted power ( $P_{CMAX}$ ) | 23 dBm | | | | As defined in clause 6.2.5 in TS 36.101 [2] |
| **Parameters per PRACH CE Levels** | | | | | |
| ***CE Level*** | ***Level 0*** | ***Level 1*** | ***Level 2*** | ***Level 3*** | |
| prach-ConfigIndex | 4 | 4 | 4 | 4 | As defined in Table 5.7.1-2 in TS 36.211 [9] |
| prach-FreqOffset | 0 | 0 | 0 | 0 | |
| prach-StartingSubframe | sf2 | sf4 | Sf8 | sf16 | |
| maxNumPreambleAttempt | n3 | n5 | n7 | n10 | |
| numRepetitionPerPreambleAttempt | n1 | n4 | n16 | n64 | |
| mpdcch-NarrowbandsToMonitor | 2 | 2 | 2 | 2 | |
| mpdcch-NumRepetition-RA | r8 | r8 | r128 | r128 | |
| prach-HoppingConfig | Off | Off | Off | Off | |
| Note 1: See Clause 6.3.2 in TS 36.331 [5] for further information on the parameters in this table. | | | | | |

**Table 6.2.13.5-4: Absolute power tolerance for E-UTRAN FDD - Contention Based Random Access test**

| Tolerance | |
|---|---|
| **Normal Conditions** | **Extreme Conditions** |
| ±9+TT dB | ± 12+TT dB |
| Note: TS 36.133 [4] clause A.6.2.13.2 and TS 36.101 [2] clause 6.3.5.1.1 | |

**Table 6.2.13.5-5: Relative power tolerance for E-UTRAN FDD - Contention Based Random Access test**

| Power Step Size (Up or Down) | PRACH | |
|---|---|---|
| | Normal Conditions | Extreme Conditions |
| ΔP [dB] | [dB] | [dB] |
| 2 ≤ ΔP < 3 | ± 3.0+TT | ± 5.0+TT |
| Note 1: TS 36.133 [4] clause A.6.2.13.2 and TS 36.101 [2] clause 6.3.5.2.1<br>Note 2: For extreme conditions an additional ± 2.0 dB relaxation is allowed for PRACH allocations | | |

**Table 6.2.13.5-6: Test requirements for $T_e$ Timing Error Limit for E-UTRAN FDD – Contention Based Random Access test**

| CEMode | $T_e$ |
|---|---|
| A | 30*$T_S$ |
| B | [52.5]*$T_S$ |
| Note 1: TS 36.133 [4] clauses A.6.2.13.2, 7.24 and clause 7.1.2<br>Note 2: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

## 6.2.14 E-UTRAN HD-FDD - Contention Based Random Access Test for Cat-M1 UEs in Enhanced Coverage

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

- Annex E needs to be updated

- The Test system uncertainties applicable to this test are undefined

- The Test tolerances applicable to this test are undefined

- Some requirements in Test 1 and Test 2 and timing error limit are still within square brackets in TS36.133

- FFSs in TS 36.508 [7] clause 4.6.3, Table 4.6.3-7A PRACH-Config-v1310-DEFAULT and Table 4.6.3-12 PRACH-ConfigCommon-DEFAUL have not been updated. If the entries there are inconsistent with the parameters settings in this section, then more tables need to be added in Message contents section for exceptions.

### 6.2.14.1 Test purpose

The purpose of this test is to verify whether the behavior of the random access procedure of a Cat-M1 UE in Enhanced Coverage is according to the E-UTRAN HD-FDD contention based random access requirements specified in TS 36.133 [4] Clause 6.2.2, Clause 6.2.3 and Clause 7.1.2 in an AWGN channel model, whether the PRACH power settings and timing are within the specified limits, and whether the UE determines properly the enhanced coverage level based on the RSRP measurement and the configured criterion in RSRP-ThresholdsPrach as defined in TS 36.331 [5].

### 6.2.14.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 13 and forward of UE category M1.

### 6.2.14.3 Minimum conformance requirements

The random access procedure is used when establishing the layer 1 communication between the UE and E-UTRAN. The random access is specified in clause 6 of TS 36.213[8] and the control of the RACH transmission is specified in clause 5.1 of TS 36.321[11]. Contention based random access procedures can only be carried out on PCell and PSCell.

The CAT-M1 UE shall have capability to calculate PRACH transmission power according to the PRACH power formula as defined in TS 36.213 [8] clause 6.1 and apply this power level at the first preamble or additional preambles.

The absolute power applied to the first preamble shall have the accuracy as defined in Table 6.3.5.1.1-1 of TS 36.101 [2]. The relative power applied to additional preambles shall have the accuracy as specified in Table 6.3.5.2.1-1 of TS 36.101 [2].

The CAT-M1 UE shall indicate a Random Access problem to upper layers if the maximum number of preambles transmission counter has been reached for the random access procedure on PCell or PSCell as specified in clause 5.1.4 in TS 36.321 [11].

The CAT-M1 UE shall stop preamble transmission if maximum number of preamble transmission counter has been reached for the random access procedure on an activated Scell as specified in clause 5.1.4 in TS 36.321 [11].

The CAT-M1 UE may stop monitoring for Random Access Response(s) and shall transmit the msg3 if the Random Access Response contains a Random Access Preamble identifier corresponding to the transmitted Random Access Preamble.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if all received Random Access Responses contain Random Access Preamble identifiers that do not match the transmitted Random Access Preamble.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if no Random Access Response is received within the RA Response window defined in clause 5.1.4 TS 36.321 [11].

The Cat-M1 UE shall re-transmit the msg3 upon the reception of a NACK on msg3.

The Cat-M1 UE shall send ACK if the Contention Resolution is successful.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires unless the received message includes a UE Contention Resolution Identity MAC control element and the Cat-M1 UE Contention Resolution Identity included in the MAC control element matches the CCCH SDU transmitted in the uplink message.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if the Contention Resolution Timer expires.

Per TS 36.133 [4] clause 6.2.3, in addition to the requirements defined in TS 36.133 [4] clause 6.2.1 and 6.2.2, the Cat-M1 UE shall also execute the random access procedure defined in clause 5.1 in TS 36.321 [11] using the PRACH configuration contained in *PRACH-ConfigSIB* in TS 36.331 [5]:

- Determine the enhanced coverage level based on the RSRP measurement and the configured criterion (*RSRP-ThresholdsPrach* in TS 36.331 [5]) as defined in section 5.1.1, TS 36.321 [11];
- Select PRACH resources (TS 36.331 [5]) configured for the corresponding enhanced coverage level as determined in the previous step;
- Transmits or re- transmits PRACH preamble using the selected PRACH resources and PRACH configuration.

The normative reference for this requirement is TS 36.133 [4] clause 6.2.2.1, 6.2.3, 7.1.2 and A.6.2.14.

The test parameters are given in Tables 6.2.14.5-1, 6.2.14.5-2 and 6.2.14.5-3.

## 6.2.14.4 Test description

### 6.2.14.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18 using only main UE Tx/Rx antenna.
2. Propagation conditions are set according to Annex B clause B.0.

3. There is one E-UTRA HD-FDD carrier and one cell specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

### 6.2.14.4.2 Test procedures

The test consists of a single cell. The UE has a downlink PDSCH allocated centered on the centre sub-carrier. The EUTRAN shall not explicitly signal a Random Access Preamble via dedicated signaling in the downlink to the Cat-M1 UE.

1. Ensure the CAT-M1 UE is in State 2A-RF-CE

2. Set the parameters according to Tables 6.2.14.5-1, 6.2.14.5-2 and 6.2.14.5-3 as appropriate where CE level 1 shall be used. Propagation conditions are set according to Annex B clause B.1.1.

3. The CAT-M1 UE shall establish a connection setup with SS, the random access procedure within the connection setup is used in the test.

4. Test 1: Correct behavior when receiving random access response

   4.1. In Test 1, the CAT-M1 UE shall send the preamble to the SS. In response to the first [4] preambles, the SS shall transmit a random access response not corresponding to the transmitted random access preamble.

   4.2. The CAT-M1 UE shall consider the random access response reception not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires if all received random access response contain random access preamble identifiers that do not match the transmitted random access preamble.

   4.3. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after [5] preambles (the preamble may be transmitted multiple times in each attempt) have been received by the SS.

   4.4. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

   4.5. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.14.5-4 and 6.2.14.5-6. The power of the first preamble shall be -30 dBm. Measure the relative power and timing applied to additional preambles and it shall not exceed the values specified in Table 6.2.14.5-5. The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.14.5-6.

5. Test 2: Correct behavior when not receiving random access response

   5.1. Repeat step 1-3.

   5.2. In Test 2, the CAT-M1 UE shall send the preamble to the SS. The SS shall not respond to the first [4] preambles.

   5.3. The CAT-M1 UE shall consider the random access response reception not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires if no random access response is received within the RA Response window of 80 sub-frames.

   5.4. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after [5] preambles have been received by the SS.

   5.5. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

   5.6. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.14.5-4 and 6.2.14.5-6. The power of the first preamble shall be -30 dBm. Measure the relative power and timing applied to additional preambles and it shall not exceed the values specified in Table 6.2.14.5-5. The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.14.5-6.

6. Test 3: Correct behavior when receiving a NACK on msg3

   6.1. Repeat step 1-3.

   6.2. In Test 3, the CAT-M1 UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

6.3. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

6.4. The SS shall send NACK to all UE msg3 following a successful random access response.

6.5. The CAT-M1 UE shall consider this contention resolution not successful then re-transmit the msg3 upon the reception of a NACK on msg3 until the maximum number of HARQ retransmission is reached.

7. Test 4: Correct behavior when receiving an incorrect message over Temporary C-RNTI

7.1. Repeat step 1-3.

7.2. In Test 4, the CAT-M1 UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

7.3. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

7.4. The SS shall send a message addressed to the Temporary C-RNTI with a UE contention resolution identity included in the MAC control element not matching the CCCH SDU transmitted in msg3 uplink message.

7.5. The CAT-M1 UE shall consider the contention resolution not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires unless the received message includes a UE contention resolution identity MAC control element and the UE contention resolution identity included in the MAC control element matches the CCCH SDU transmitted in msg3 uplink message.

8. Test 5: Correct behavior when receiving a correct message over Temporary C-RNTI

8.1. In Test 5, the SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

8.2. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

8.3. The SS shall send a message addressed to the Temporary C-RNTI with a UE contention resolution identity included in the MAC control element matching the CCCH SDU transmitted in the msg3 uplink message.

8.4. The CAT-M1 UE shall send ACK and consider the contention resolution successful and the random access procedure successfully completed.

9. Test 6: Correct behavior when contention resolution timer expires

9.1. Repeat step 1-3.

9.2. In Test 6, the CAT-M1 UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

9.3. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

9.4. The SS shall send an ACK for msg3 but not send msg4 allowing the contention resolution timer to expire.

9.5. The CAT-M1 UE shall consider the contention resolution not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires.

9.6. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

9.7. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

10. Test 7: PRACH Resource Selection

10.1. Repeat step 1-3.

10.2. In Test 7, the CAT-M1 UE shall send the preamble to the SS.

10.3. The CAT-M1 UE shall determines the enhanced coverage level 1 based on the RSRP measurement and the configured criterion (*RSRP-ThresholdsPrach* in TS 36.331 [5]) as defined in section 5.1.1, TS 36,321 [11].

10.4. The CAT-M1 UE shall selects PRACH resources configured for the corresponding enhanced coverage level 1 as determined in the previous step.

10.5. The CAT-M1 UE shall transmit or re- transmits PRACH preamble using the selected PRACH resources, PRACH configuration and calculated PRACH transmission power for the corresponding coverage level 1.

#### 6.2.14.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.2.14.4.3-1: Common Exception messages for E-UTRAN HD-FDD - Contention Based Random Access test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.6-1<br>Table H.2.6-2<br>Table H.2.6-3 |
| Default RRC messages and information elements contents exceptions | Table H.3.2-1 |

**Table 6.2.14.4.3-2: *UplinkPowerControlCommon-DEFAULT*: Additional E-UTRAN HD-FDD - Contention Based Random Access test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-25 UplinkPowerControlCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlCommon-DEFAULT ::= SEQUENCE { | | | |
| p0-NominalPUSCH | -101 (-101 dBm) | | |
| } | | | |

### 6.2.14.5 Test requirement

Tables 6.2.14.5-1, 6.2.14.5-2 and 6.2.14.5-3 define the primary level settings for E-UTRAN HD-FDD - contention based random access test. Table 6.2.14.5-6 defines the uplink timing error limit including test tolerances.

For the tests specified in section 6.2.14.4.2, the power of the first preamble shall be -30 dBm to within the accuracy specified in Table 6.2.14.5-4. The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.14.5-5. The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.14.5-6.

**Table 6.2.14.5-1: General test parameters for HD-FDD contention based random access test**

| Parameter | Unit | Value | Comments |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern Note 1 | | OP.21 FDD | As defined in TS 36.133 [4] clause A.3.2.1.21. |
| PDSCH parameters Note 2 | | R.12 HD-FDD | As defined in TS 36.133 [4] clause A.3.1.4.5 |
| MPDCCH parameters Note 2 | | R.8 HD-FDD | As defined in TS 36.133 [4] clause A.3.1.3.5 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.3 HD-FDD | As defined in TS 36.133 [4] clause A.3.1.2.3 |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| MPDCCH_RA | dB | | |
| MPDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $N_{oc}$ | dBm/15 KHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | -12+TT | |
| $\hat{E}_s / I_{ot}$ Note 3 | dB | -12+TT | |
| RSRP Note 3 | dBm/15 KHz | -110+TT | |
| Io Note 3 | dBm/9 MHz | -70+TT | |
| Propagation Condition | - | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. The OCNG pattern is chosen during the test according to the presence of a DL reference measurement channel.

Note 2: The PDSCH and MPDCCH reference measurement channels are used in the test only when a downlink transmission dedicated to the UE under test is required.

Note 3: Es/Iot, RSRP and Io level has been derived from other parameters for information purpose. They are not settable parameters themselves.

**Table 6.2.14.5-2: RACH-Configuration parameters for HD-FDD contention based random access test**

| Field | Value | | | | Comment |
|---|---|---|---|---|---|
| **Parameters not per CE Levels** | | | | | |
| powerRampingStep | dB2 | | | | |
| preambleInitialReceivedTargetPower | dBm-120 | | | | |
| preambleTransMax | n6 | | | | |
| maxHARQ-Msg3Tx | 4 | | | | |
| rar-HoppingConfig | Off | | | | |
| **Parameters per CE Levels** | | | | | |
| ***CE Level*** | ***Level 0*** | ***Level 1*** | ***Level 2*** | ***Level 3*** | |
| ra-ResponseWindowSize (per CE) | sf20 | sf80 | sf180 | sf320 | |
| mac-ContentionResolutionTimer (per CE) | sf80 | sf120 | sf200 | sf480 | |
| PreambleMappingInfo {firstPreamble, lastPreamble} | {0, 9} | {10,19} | {20,29} | {30,39} | |
| Note: For further information see Clause 6.3.2 in TS 36.331 [5]. | | | | | |

**Table 6.2.14.5-3: PRACH-Configuration parameters for HD-FDD contention based random access test**

| Field | Value | | | | Comment |
|---|---|---|---|---|---|
| **Parameters not per CE Levels** | | | | | |
| rsrp-ThresholdsPrach | {16, 26, 36} | | | | Corresponding to {-124, -114, -104} dBm as defined in TS36.133 [4] clause 9.1.21.5 |
| mpdcch-startSF-CSS-RA | v1 | | | | |
| referenceSignalPower | -5 dBm/15 KHz | | | | As defined in clause 6.3.2 in TS 36.331 [5]. |
| maxHARQ-Msg3Tx | 4 | | | | As defined in Table 5.7.1-2 in TS 36.211 [9] |
| Backoff Parameter Index | 2 | | | | As defined in Table 7.2-1 in TS 36.321 [11] |
| Configured UE transmitted power ( $P_{CMAX}$ ) | 23 dBm | | | | As defined in clause 6.2.5 in TS 36.101 [2] |
| **Parameters per PRACH CE Levels** | | | | | |
| ***CE Level*** | ***Level 0*** | ***Level 1*** | ***Level 2*** | ***Level 3*** | |
| prach-ConfigIndex | 4 | 4 | 4 | 4 | As defined in Table 5.7.1-2 in TS 36.211 [9] |
| prach-FreqOffset | 0 | 0 | 0 | 0 | |
| prach-StartingSubframe | sf2 | sf4 | Sf8 | sf16 | |
| maxNumPreambleAttempt | n3 | n5 | n7 | n10 | |
| numRepetitionPerPreambleAttempt | n1 | n4 | n16 | n64 | |
| mpdcch-NarrowbandsToMonitor | 2 | 2 | 2 | 2 | |
| mpdcch-NumRepetition-RA | r8 | r8 | r128 | r128 | |
| prach-HoppingConfig | Off | Off | Off | Off | |
| Note 1: See Clause 6.3.2 in TS 36.331 [5] for further information on the parameters in this table. | | | | | |

**Table 6.2.14.5-4: Absolute power tolerance for E-UTRAN HD-FDD - Contention Based Random Access test**

| Tolerance | |
|---|---|
| **Normal Conditions** | **Extreme Conditions** |
| ±9+TT dB | ± 12+TT dB |
| Note: TS 36.133 [4] clause A.6.2.14.2 and TS 36.101 [2] clause 6.3.5.1.1 | |

**Table 6.2.14.5-5: Relative power tolerance for E-UTRAN HD-FDD - Contention Based Random Access test**

| Power Step Size (Up or Down) | PRACH | |
|---|---|---|
| | Normal Conditions | Extreme Conditions |
| ΔP [dB] | [dB] | [dB] |
| 2 ≤ ΔP < 3 | ± 3.0+TT | ± 5.0+TT |
| Note 1: TS 36.133 [4] clause A.6.2.14.2 and TS 36.101 [2] clause 6.3.5.2.1<br>Note 2: For extreme conditions an additional ± 2.0 dB relaxation is allowed for PRACH allocations | | |

**Table 6.2.14.5-6: Test requirements for $T_e$ Timing Error Limit for E-UTRAN HD-FDD - Contention Based Random Access test**

| CEMode | $T_e$ |
|---|---|
| A | 30*$T_S$ |
| B | [52.5]*$T_S$ |
| Note 1: TS 36.133 [4] clauses A.6.2.14.2, 7.24 and clause 7.1.2<br>Note 2: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

## 6.2.15 E-UTRAN TDD - Contention Based Random Access Test for Cat-M1 UEs in Enhanced Coverage

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

- The Test system uncertainties applicable to this test are undefined (Cat-M1 power control uncertainty and TT requirements in TS 36.521-1 Aug 2016 clause 6.3.5EA.1 on are all undefined)

- The Test tolerances applicable to this test are undefined

- Annex E needs to be updated

- Some requirements in Test 1 and Test 2 and timing error limit are still within square brackets in TS36.133

- FFSs in TS 36.508 [7] clause 4.6.3, Table 4.6.3-7A PRACH-Config-v1310-DEFAULT and Table 4.6.3-12 PRACH-ConfigCommon-DEFAUL have not been updated. If the entries there are inconsistent with the parameters settings in this section, then more tables need to be added in Message contents section for exceptions.

### 6.2.15.1 Test purpose

The purpose of this test is to verify whether the behavior of the random access procedure of a Cat-M1 UE in Enhanced Coverage is according to the E-UTRAN TDD contention based random access requirements specified in TS 36.133 [4] Clause 6.2.2, Clause 6.2.3 and Clause 7.1.2 in an AWGN channel model, whether the PRACH power settings and timing are within the specified limits, and whether the UE determines properly the enhanced coverage level based on the RSRP measurement and the configured criterion in RSRP-ThresholdsPrach as defined in TS 36.331 [5].

### 6.2.15.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 13 and forward of UE category M1.

### 6.2.15.3 Minimum conformance requirements

The random access procedure is used when establishing the layer 1 communication between the UE and E-UTRAN. The random access is specified in clause 6 of TS 36.213[8] and the control of the RACH transmission is specified in clause 5.1 of TS 36.321[11]. Contention based random access procedures can only be carried out on PCell and PSCell.

The CAT-M1 UE shall have capability to calculate PRACH transmission power according to the PRACH power formula as defined in TS 36.213 [8] clause 6.1 and apply this power level at the first preamble or additional preambles. The absolute power applied to the first preamble shall have the accuracy as defined in Table 6.3.5.1.1-1 of TS 36.101 [2]. The relative power applied to additional preambles shall have the accuracy as specified in Table 6.3.5.2.1-1 of TS 36.101 [2].

The CAT-M1 UE shall indicate a Random Access problem to upper layers if the maximum number of preambles transmission counter has been reached for the random access procedure on PCell or PSCell as specified in clause 5.1.4 in TS 36.321 [11].

The CAT-M1 UE shall stop preamble transmission if maximum number of preamble transmission counter has been reached for the random access procedure on an activated Scell as specified in clause 5.1.4 in TS 36.321 [11].

The CAT-M1 UE may stop monitoring for Random Access Response(s) and shall transmit the msg3 if the Random Access Response contains a Random Access Preamble identifier corresponding to the transmitted Random Access Preamble.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if all received Random Access Responses contain Random Access Preamble identifiers that do not match the transmitted Random Access Preamble.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if no Random Access Response is received within the RA Response window defined in clause 5.1.4 TS 36.321 [11].

The Cat-M1 UE shall re-transmit the msg3 upon the reception of a NACK on msg3.

The Cat-M1 UE shall send ACK if the Contention Resolution is successful.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires unless the received message includes a UE Contention Resolution Identity MAC control element and the Cat-M1 UE Contention Resolution Identity included in the MAC control element matches the CCCH SDU transmitted in the uplink message.

The Cat-M1 UE shall re-select a preamble and transmit with the calculated PRACH transmission power when the back off time expires if the Contention Resolution Timer expires.

Per TS 36.133 [4] clause 6.2.3, in addition to the requirements defined in TS 36.133 [4] clause 6.2.1 and 6.2.2, the Cat-M1 UE shall also execute the random access procedure defined in clause 5.1 in TS 36.321 [11] using the PRACH configuration contained in *PRACH-ConfigSIB* in TS 36.331 [5]:

- Determine the enhanced coverage level based on the RSRP measurement and the configured criterion (*RSRP-ThresholdsPrach* in TS 36.331 [5]) as defined in section 5.1.1, TS 36.321 [11];
- Select PRACH resources (TS 36.331 [5]) configured for the corresponding enhanced coverage level as determined in the previous step and;
- Transmits or re- transmits PRACH preamble using the selected PRACH resources and PRACH configuration.

The normative reference for this requirement is TS 36.133 [4] clause 6.2.2.1, 6.2.3, 7.1.2 and A.6.2.15.

The test parameters are given in Tables 6.2.15.5-1, 6.2.15.5-2 and 6.2.15.5-3.

## 6.2.15.4 Test description

### 6.2.15.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18 using only main UE Tx/Rx antenna.

2. Propagation conditions are set according to Annex B clause B.0.

3. There is one E-UTRA TDD carrier and one cell specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

### 6.2.15.4.2 Test procedures

The test consists of a single cell. The UE has a downlink PDSCH allocated centered on the centre sub-carrier. The EUTRAN shall not explicitly signal a Random Access Preamble via dedicated signaling in the downlink to the Cat-M1 UE.

1. Ensure the CAT-M1 UE is in State 2A-RF-CE

2. Set the parameters according to Tables 6.2.15.5-1, 6.2.15.5-2 and 6.2.15.5-3 as appropriate where CE level 1 shall be used. Propagation conditions are set according to Annex B clause B.1.1.

3. The CAT-M1 UE shall establish a connection setup with SS, the random access procedure within the connection setup is used in the test.

4. Test 1: Correct behavior when receiving random access response

   4.1. In Test 1, the CAT-M1 UE shall send the preamble to the SS. In response to the first [4] preambles, the SS shall transmit a random access response not corresponding to the transmitted random access preamble.

   4.2. The CAT-M1 UE shall consider the random access response reception not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires if all received random access response contain random access preamble identifiers that do not match the transmitted random access preamble.

   4.3. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after [5] preambles (the preamble may be transmitted multiple times in each attempt) have been received by the SS.

   4.4. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

   4.5. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.15.5-4 and 6.2.15.5-6. The power of the first preamble shall be -30 dBm. Measure the relative power and timing applied to additional preambles and it shall not exceed the values specified in Table 6.2.15.5-5. The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.15.5-6.

5. Test 2: Correct behavior when not receiving random access response

   5.1. Repeat step 1-3.

   5.2. In Test 2, the CAT-M1 UE shall send the preamble to the SS. The SS shall not respond to the first [4] preambles.

   5.3. The CAT-M1 UE shall consider the random access response reception not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires if no random access response is received within the RA Response window of 80 sub-frames.

   5.4. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after [5] preambles have been received by the SS.

   5.5. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

   5.6. Measure the power and timing of the first preamble and it shall not exceed the values specified in Tables 6.2.15.5-4 and 6.2.15.5-6. The power of the first preamble shall be -30 dBm. Measure the relative power and timing applied to additional preambles and it shall not exceed the values specified in Table 6.2.15.5-5. The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.15.5-6.

6. Test 3: Correct behavior when receiving a NACK on msg3

   6.1. Repeat step 1-3.

6.2. In Test 3, the CAT-M1 UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

6.3. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

6.4. The SS shall send NACK to all UE msg3 following a successful random access response.

6.5. The CAT-M1 UE shall consider this contention resolution not successful then re-transmit the msg3 upon the reception of a NACK on msg3 until the maximum number of HARQ retransmission is reached.

7. Test 4: Correct behavior when receiving an incorrect message over Temporary C-RNTI

7.1. Repeat step 1-3.

7.2. In Test 4, the CAT-M1 UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

7.3. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

7.4. The SS shall send a message addressed to the Temporary C-RNTI with a UE contention resolution identity included in the MAC control element not matching the CCCH SDU transmitted in msg3 uplink message.

7.5. The CAT-M1 UE shall consider the contention resolution not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires unless the received message includes a UE contention resolution identity MAC control element and the UE contention resolution identity included in the MAC control element matches the CCCH SDU transmitted in msg3 uplink message.

8. Test 5: Correct behavior when receiving a correct message over Temporary C-RNTI

8.1. Repeat step 1-3

8.2. In Test 5, the SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

8.3. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

8.4. The SS shall send a message addressed to the Temporary C-RNTI with a UE contention resolution identity included in the MAC control element matching the CCCH SDU transmitted in the msg3 uplink message.

8.5. The CAT-M1 UE shall send ACK and consider the contention resolution successful and the random access procedure successfully completed.

9. Test 6: Correct behavior when contention resolution timer expires

9.1. Repeat step 1-3.

9.2. In Test 6, the CAT-M1 UE shall send the preamble to the SS. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

9.3. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

9.4. The SS shall send an ACK for msg3 but not send msg4 allowing the contention resolution timer to expire.

9.5. The CAT-M1 UE shall consider the contention resolution not successful then re-select a preamble and transmit with the calculated PRACH transmission power when the backoff time expires.

9.6. The SS shall transmit a random access response containing a random access preamble identifier corresponding to the transmitted random access after the first preamble has been received by the SS.

9.7. The CAT-M1 UE shall consider this random access response reception successful and transmit the msg3.

10. Test 7: PRACH Resource Selection

10.1. Repeat step 1-3.

10.2. In Test 7, the CAT-M1 UE shall send the preamble to the SS.

10.3. The CAT-M1 UE shall determines the enhanced coverage level 1 based on the RSRP measurement and the configured criterion (*RSRP-ThresholdsPrach* in TS 36.331 [5]) as defined in section 5.1.1, TS 36,321 [11].

10.4. The CAT-M1 UE shall selects PRACH resources configured for the corresponding enhanced coverage level 1 as determined in the previous step.

10.5. The CAT-M1 UE shall transmit or re- transmits PRACH preamble using the selected PRACH resources, PRACH configuration and calculated PRACH transmission power for the corresponding coverage level 1.

### 6.2.15.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.2.15.4.3-1: Common Exception messages for E-UTRAN TDD - Contention Based Random Access test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.6-1<br>Table H.2.6-2<br>Table H.2.6-3 |

**Table 6.2.15.4.3-2: *UplinkPowerControlCommon-DEFAULT*: Additional E-UTRAN TDD – Contention Based Random Access test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-25 UplinkPowerControlCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlCommon-DEFAULT ::= SEQUENCE { | | | |
| p0-NominalPUSCH | -101 (-101 dBm) | | |
| } | | | |

## 6.2.15.5 Test requirement

Tables 6.2.15.5-1, 6.2.15.5-2 and 6.2.15.5-3 define the primary level settings for E-UTRAN TDD - contention based random access test. Table 6.2.15.5-6 defines the uplink timing error limit including test tolerances.

For the tests specified in section 6.2.15.4.2, the power of the first preamble shall be -30 dBm to within the accuracy specified in Table 6.2.15.5-4. The relative power for preamble ramping step shall be 2 dB to within the accuracy specified in Table 6.2.15.5-5. The transmit timing of all PRACH transmissions shall be within the accuracy specified in Table 6.2.15.5-6.

**Table 6.2.15.5-1: General test parameters for TDD contention based random access test**

| Parameter | Unit | Value | Comments |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern Note 1 | | OP.11 TDD | As defined in TS 36.133 [4] clause A.3.2.2.11. |
| PDSCH parameters Note 2 | | R.18 TDD | As defined in TS 36.133 [4] clause A.3.1.4.6 |
| MPDCCH parameters Note 2 | | R.16 TDD | As defined in TS 36.133 [4] clause A.3.1.3.6 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As defined in TS 36.133 [4] clause A.3.1.2.2 |
| Special subframe configuration | - | 6 | As specified in table 4.2-1 in TS 36.211 [9]. |
| Uplink-downlink configuration | - | 1 | As specified in table 4.2-2 in TS 36.211 [9]. |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| MPDCCH_RA | dB | | |
| MPDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $N_{oc}$ | dBm/15 KHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | -12+TT | |
| $\hat{E}_s / I_{ot}$ Note 3 | dB | -12+TT | |
| RSRP Note 3 | dBm/15 KHz | -110+TT | |
| Io Note 3 | dBm/9 MHz | -70+TT | |
| Propagation Condition | - | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. The OCNG pattern is chosen during the test according to the presence of a DL reference measurement channel.

Note 2: The PDSCH and MPDCCH reference measurement channels are used in the test only when a downlink transmission dedicated to the UE under test is required.

Note 3: Es/Iot, RSRP and Io level has been derived from other parameters for information purpose. They are not settable parameters themselves.

**Table 6.2.15.5-2: RACH-Configuration parameters for TDD contention based random access test**

| Field | Value | | | | Comment |
|---|---|---|---|---|---|
| **Parameters not per CE Levels** | | | | | |
| numberOfRA-Preambles | n52 | | | | |
| sizeOfRA-PreamblesGroupA | n52 | | | | No group B. |
| powerRampingStep | dB2 | | | | |
| preambleInitialReceivedTargetPower | dBm-120 | | | | |
| preambleTransMax | n6 | | | | |
| maxHARQ-Msg3Tx | 4 | | | | |
| rar-HoppingConfig | Off | | | | |
| **Parameters per CE Levels** | | | | | |
| ***CE Level*** | ***Level 0*** | ***Level 1*** | ***Level 2*** | ***Level 3*** | |
| ra-ResponseWindowSize (per CE) | sf20 | sf80 | sf180 | sf320 | |
| mac-ContentionResolutionTimer (per CE) | sf80 | sf120 | sf200 | sf480 | |
| PreambleMappingInfo {firstPreamble, lastPreamble} | {0, 9} | {10,19} | {20,29} | {30,39} | |
| Note: For further information see Clause 6.3.2 in TS 36.331 [5]. | | | | | |

**Table 6.2.15.5-3: PRACH-Configuration parameters for TDD contention based random access test**

| Field | Value | | | | Comment |
|---|---|---|---|---|---|
| **Parameters not per CE Levels** | | | | | |
| rsrp-ThresholdsPrach | {16, 26, 36} | | | | Corresponding to {-124, -114, -104} dBm as defined in TS36.133 [4] clause 9.1.21.5 |
| mpdcch-startSF-CSS-RA | v1 | | | | |
| referenceSignalPower | -5 dBm/15 KHz | | | | As defined in clause 6.3.2 in TS 36.331 [5]. |
| maxHARQ-Msg3Tx | 4 | | | | As defined in Table 5.7.1-2 in TS 36.211 [9] |
| Backoff Parameter Index | 2 | | | | As defined in Table 7.2-1 in TS 36.321 [11] |
| Configured UE transmitted power ($P_{CMAX}$) | 23 dBm | | | | As defined in clause 6.2.5 in TS 36.101 [2] |
| **Parameters per PRACH CE Levels** | | | | | |
| ***CE Level*** | ***Level 0*** | ***Level 1*** | ***Level 2*** | ***Level 3*** | |
| prach-ConfigIndex | 4 | 4 | 4 | 4 | As defined in Table 5.7.1-2 in TS 36.211 [9] |
| prach-FreqOffset | 0 | 0 | 0 | 0 | |
| prach-StartingSubframe | sf2 | sf4 | Sf8 | sf16 | |
| maxNumPreambleAttempt | n3 | n5 | n7 | n10 | |
| numRepetitionPerPreambleAttempt | n1 | n4 | n16 | n64 | |
| mpdcch-NarrowbandsToMonitor | 2 | 2 | 2 | 2 | |
| mpdcch-NumRepetition-RA | r8 | r8 | r128 | r128 | |
| prach-HoppingConfig | Off | Off | Off | Off | |
| Note 1: See Clause 6.3.2 in TS 36.331 [5] for further information on the parameters in this table. | | | | | |

**Table 6.2.15.5-4: Absolute power tolerance for E-UTRAN TDD - Contention Based Random Access test**

| Tolerance | |
|---|---|
| **Normal Conditions** | **Extreme Conditions** |
| ±9+TT dB | ± 12+TT dB |
| Note: TS 36.133 [4] clause A.6.2.15.2 and TS 36.101 [2] clause 6.3.5.1.1 | |

**Table 6.2.15.5-5: Relative power tolerance for E-UTRAN TDD - Contention Based Random Access test**

| Power Step Size (Up or Down) | PRACH | |
|---|---|---|
| | Normal Conditions | Extreme Conditions |
| ΔP [dB] | [dB] | [dB] |
| 2 ≤ ΔP < 3 | ± 3.0+TT | ± 5.0+TT |
| Note 1: TS 36.133 [4] clause A.6.2.15.2 and TS 36.101 [2] clause 6.3.5.2.1<br>Note 2: For extreme conditions an additional ± 2.0 dB relaxation is allowed for PRACH allocations | | |

**Table 6.2.15.5-6: Test requirements for $T_e$ Timing Error Limit for E-UTRAN TDD – Contention Based Random Access test**

| CEMode | $T_e$ |
|---|---|
| A | 30*$T_S$ |
| B | [52.5]*$T_S$ |
| Note 1: TS 36.133 [4] clauses A.6.2.15.2, 7.24 and clause 7.1.2<br>Note 2: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

# 6.3 RRC Connection Release with Redirection

## 6.3.1 Redirection from E-UTRAN FDD to UTRAN FDD

### 6.3.1.1 Test purpose

The purpose of this test is to verify that the UE makes correct performing the RRC connection release with redirection to the target UTRAN FDD cell. This test will partly verify the RRC connection release with redirection to UTRAN FDD cell requirements in section 6.3.2.1.

### 6.3.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward that support UTRA FDD.

### 6.3.1.3 Minimum conformance requirements

The UE shall be capable of performing the RRC connection release with redirection to the target UTRAN FDD cell within $T_{connection_release_redirect_UTRA\ FDD}$.

The time delay ($T_{connection_release_redirect_UTRA\ FDD}$) is the time between the end of the last TTI containing the RRC command, "*RRCConnectionRelease*" as defined in section 5.3.8 in TS 36.331 [5] on the E-UTRAN PDSCH and the time the UE starts to send random access to the target UTRA FDD cell. The time delay ($T_{connection_release_redirect_UTRA\ FDD}$) shall be less than:

$$T_{connection_release_redirect_UTRA\ FDD} = T_{RRC_procedure_delay} + T_{identify\text{-}UTRA\ FDD} + T_{SI\text{-}UTRA\ FDD} + T_{RA}$$

The target UTRA FDD cell shall be considered detectable when:

- CPICH Ec/Io $\geq$ -15 dB,
- SCH_Ec/Io $\geq$ -15 dB for at least one channel tap and SCH_Ec/Ior is equally divided between primary synchronisation code and secondary synchronisation code.

$T_{RRC_procedure_delay}$: It is the RRC procedure for processing the received message "*RRCConnectionRelease*". It shall be less than 110 ms.

$T_{identify\text{-}UTRA\ FDD}$: It is the time to identify the target UTRA FDD cell. It shall be less than 500 ms.

$T_{SI-UTRA\ FDD}$: It is the time required for acquiring all the relevant system information of the target UTRA FDD cell. This time depends upon whether the UE is provided with the relevant system information of the target UTRA FDD cell or not by the E-UTRAN before the RRC connection is released.

$T_{RA}$: It is the delay caused due to the random access procedure when sending random access to the target UTRA FDD cell.

The normative reference for this requirement is TS 36.133 [4] clause 6.3.2.1 and A.6.3.1.

## 6.3.1.4 Test description

### 6.3.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.22.

2. The general test parameter settings are set up according to Table 6.3.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are as defined in clause 6.3.1.4.3.

5. There is one E-UTRA FDD cell and one UTRA FDD cell specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.3.1.4.1-1: General test parameters for RRC Connection Release with Redirection from E-UTRAN FDD to UTRAN FDD under AWGN propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on UTRA RF channel number 1. |
| CP length | | Normal | Applicable to cell 1 |
| E-UTRA RF Channel Number | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| UTRA RF Channel Number | | 1 | One UTRA FDD carrier frequency is used. |
| Inter-RAT (UTRA FDD) measurement quantity | | CPICH Ec/Io | |
| Filter coefficient | | 0 | L3 filtering is not used. |
| DRX | | OFF | |
| UTRA FDD cell list size | | 16 | UTRA cells on UTRA RF channel 1 provided in the "*RRCConnectionRelease*" message from the E-UTRAN |
| T1 | s | ≤5 | |
| T2 | s | 1 | |

### 6.3.1.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. The test consists of two successive time periods, with time duration of T1, and T2 respectively. The "*RRCConnectionRelease*" message containing the relevant system information of Cell 2 shall be sent to the UE during period T1 and the start of T2 is the instant when the last TTI containing the RRC message is sent to the UE. Prior to time duration T2, the UE shall not have any timing information of Cell 2. Cell 2 is powered up at the beginning of the T2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 6.3.1.5-1 and 6.3.1.5-2. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. SS shall transmit an *RRCConnectionRelease* containing the relevant system information of Cell 2 during period T1.
4. The SS shall start T2 timer when the last TTI containing the *RRCConnectionRelease* message is sent to UE.
5. When T2 starts, the SS shall switch the power setting from T1 to T2 as specified in Table 6.3.1.5-1 for Cell 1, and Table 6.3.1.5-2 for Cell 2.
6. If the UE transmits the PRACH to Cell 2 less than 650 ms from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
7. After T2 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.Cell 1 is the active cell.
8. The SS shall set Cell 2 primary scrambling code = ((current cell 2 primary scrambling code - 50) mod 200 + 100) for next iteration of the test procedure loop.
9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

### 6.3.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.3.1.4.3-1: *RRCConnectionRelease*: Additional Redirection from E-UTRAN FDD to UTRAN FDD test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| rrcConnectionRelease-r8 SEQUENCE { | | | |
| releaseCause | other | | |
| redirectedCarrierInfo CHOICE { | | | |
| utra-FDD | Cell 2 Carrier Information | | |
| } | | | |
| idleModeMobilityControlInfo | Not present | | |
| nonCriticalExtension SEQUENCE { | | | |
| lateNonCriticalExtension | Not present | | |
| nonCriticalExtension { | | | |
| cellInfoList-r9 CHOICE { | | | |
| utra-FDD-r9 | CellInfoListUTRA-FDD-r9 | | |
| } | | | |
| nonCriticalExtension | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 6.3.1.4.3-2: *CellInfoListUTRA-FDD-r9*: Additional Redirection from E-UTRAN FDD to UTRAN FDD test requirement (step3)**

| Derivation Path: TS 36.331 [5] clause 6.6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellInfoListUTRA-FDD-r9 ::= SEQUENCE (SIZE (1..16)) OF SEQUENCE { | | | |
| physCellId-r9 | Cell 2 physical cell identity | | |
| utra-BCCH-Container-r9 | Cell 2 Relevant System Information | | |
| } | | | |

## 6.3.1.5 Test requirement

Tables 6.3.1.4.1-1, 6.3.1.5-1 and 6.3.1.5-2 define the primary level settings including test tolerances for Redirection from E-UTRAN FDD to UTRAN FDD test.

**Table 6.3.1.5-1: Cell Specific Test requirement Parameters for cell #1 E-UTRAN FDD to UTRAN FDD RRC release with redirection under AWGN propagation conditions**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 6.3.1.5-2: Cell Specific Test Parameters for cell #2 E-UTRAN FDD to UTRAN FDD RRC release with redirection under AWGN propagation conditions**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| UTRA RF Channel Number | | 1 | |
| CPICH_Ec/Ior | dB | -10 | |
| PCCPCH_Ec/Ior | dB | -12 | |
| SCH_Ec/Ior | dB | -12 | |
| PICH_Ec/Ior | dB | -15 | |
| DPCH_Ec/Ior | dB | N/A | |
| OCNS | | -0.941 | |
| $\hat{I}_{or}/I_{oc}$ | dB | -∞ | 0.42 |
| $I_{oc}$ | dBm/3.84 MHz | -70 | |
| CPICH_Ec/Io[Note 3] | dB | -∞ | -12.81 |
| Propagation Condition | | AWGN | |

Note 1: The DPCH level is controlled by the power control loop.
Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$
Note 3: This gives an SCH Ec/Io of -15dB

The UE shall start to transmit random access to Cell 2 less than 650 ms from the beginning of time period T2.

The rate of correct RRC connection release redirection to UTRAN FDD observed during repeated tests shall be at least 90%.

The overall delays measured test requirement can be expressed as

Test Requirement = $T_{RRC_procedure_delay} + T_{identify\text{-}UTRA\ FDD} + T_{SI\text{-}UTRA\ FDD} + T_{RA}$

$T_{RRC_procedure_delay}$ = 110 ms

$T_{identify\text{-}UTRA\ FDD}$ = 500 ms

$T_{SI\text{-}UTRA\ FDD}$ =the time required for acquiring all the relevant system information of the target UTRA FDD cell. This time depends upon whether the UE is provided with the relevant system information of the target UTRA FDD cell or not by the E-UTRAN before the RRC connection is released. 0 ms is assumed in this test case.

$T_{RA}$ = The additional delay caused by the random access procedure. 40 ms is assumed in this test case.

This gives a total of 650 ms for overall delays measured.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 6.3.2 Redirection from E-UTRAN TDD to UTRAN FDD

### 6.3.2.1 Test purpose

The purpose of this test is to verify that the UE performs the RRC connection release with redirection to the target UTRAN FDD cell. This test will partly verify the RRC connection release with redirection to UTRAN FDD cell requirements.

### 6.3.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward that support UTRA FDD.

### 6.3.2.3 Minimum conformance requirements

The UE shall be capable of performing the RRC connection release with redirection to the target UTRAN FDD cell within $T_{connection_release_redirect_UTRA\ FDD}$.

The time delay ($T_{connection_release_redirect_UTRA\ FDD}$) is the time between the end of the last TTI containing the RRC command, "*RRCConnectionRelease*" as defined in section 5.3.8 in TS 36.331 [5] on the E-UTRAN PDSCH and the time the UE starts to send random access to the target UTRA FDD cell. The time delay ($T_{connection_release_redirect_UTRA\ FDD}$) shall be less than:

$$T_{connection_release_redirect_UTRA\ FDD} = T_{RRC_procedure_delay} + T_{identify\text{-}UTRA\ FDD} + T_{SI\text{-}UTRA\ FDD} + T_{RA}$$

The target UTRA FDD cell shall be considered detectable when:

- CPICH Ec/Io $\geq$ -15 dB,
- SCH_Ec/Io $\geq$ -15 dB for at least one channel tap and SCH_Ec/Ior is equally divided between primary synchronisation code and secondary synchronisation code.

$T_{RRC_procedure_delay}$: It is the RRC procedure for processing the received message "*RRCConnectionRelease*". It shall be less than 110 ms.

$T_{identify\text{-}UTRA\ FDD}$: It is the time to identify the target UTRA FDD cell. It shall be less than 500 ms.

$T_{SI\text{-}UTRA\ FDD}$: It is the time required for acquiring all the relevant system information of the target UTRA FDD cell. This time depends upon whether the UE is provided with the relevant system information of the target UTRA FDD cell or not by the E-UTRAN before the RRC connection is released.

$T_{RA}$: It is the delay caused due to the random access procedure when sending random access to the target UTRA FDD cell.

The normative reference for this requirement is TS 36.133 [4] clause 6.3.2.1 and A.6.3.2.

### 6.3.2.4 Test description

#### 6.3.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.22.
2. The general test parameter settings are set up according to Table 6.3.2.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 6.3.2.4.3.
5. There is one E-UTRA TDD cell and one UTRA FDD cell specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.3.2.4.1-1: General test parameters for RRC Connection Release with Redirection from E-UTRAN TDD to UTRAN FDD under AWGN propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on UTRA RF channel number 1. |
| CP length | | Normal | Applicable to cell 1 |
| E-UTRA RF Channel Number | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211 |
| Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in TS 36.211 |
| UTRA RF Channel Number | | 1 | One UTRA FDD carrier frequency is used. |
| Inter-RAT (UTRA FDD) measurement quantity | | CPICH Ec/Io | |
| Filter coefficient | | 0 | L3 filtering is not used. |
| DRX | | OFF | |
| UTRA FDD cell list size | | 16 | UTRA cells on UTRA RF channel 1 provided in the "*RRCConnectionRelease*" message from the E-UTRAN |
| T1 | s | ≤5 | |
| T2 | s | 1 | |

### 6.3.2.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. The test consists of two successive time periods, with time duration of T1, and T2 respectively. The "*RRCConnectionRelease*" message containing the relevant system information of Cell 2 shall be sent to the UE during period T1 and the start of T2 is the instant when the last TTI containing the RRC message is sent to the UE. Prior to time duration T2, the UE shall not have any timing information of Cell 2. Cell 2 is powered up at the beginning of the T2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 6.3.2.5-1 and 6.3.2.5-2. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. SS shall transmit an *RRCConnectionRelease* containing the relevant system information of Cell 2 during period T1.
4. The SS shall start T2 timer when the last TTI containing the *RRCConnectionRelease* message is sent to UE.
5. When T2 starts, the SS shall switch the power setting from T1 to T2 as specified in Table 6.3.2.5-1 for Cell 1, and Table 6.3.2.5-2 for Cell 2.
6. If the UE transmits the PRACH to Cell 2 less than 650 ms from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
7. After T2 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.Cell 1 is the active cell.
8. The SS shall set Cell 2 primary scrambling code = ((current cell 2 primary scrambling code - 50) mod 200 + 100) for next iteration of the test procedure loop.
9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

### 6.3.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.3.2.4.3-1: *RRCConnectionRelease*: Additional Redirection from E-UTRAN TDD to UTRAN FDD test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| rrcConnectionRelease-r8 SEQUENCE { | | | |
| releaseCause | other | | |
| redirectedCarrierInfo CHOICE { | | | |
| utra-FDD | Cell 2 Carrier Information | | |
| } | | | |
| idleModeMobilityControlInfo | Not present | | |
| nonCriticalExtension SEQUENCE { | | | |
| lateNonCriticalExtension | Not present | | |
| nonCriticalExtension { | | | |
| cellInfoList-r9 CHOICE { | | | |
| utra-FDD-r9 | CellInfoListUTRA-FDD-r9 | | |
| } | | | |
| nonCriticalExtension | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 6.3.2.4.3-2: *CellInfoListUTRA-FDD-r9*: Additional Redirection from E-UTRAN TDD to UTRAN FDD test requirement (step3)**

| Derivation Path: TS 36.331 [5] clause 6.6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellInfoListUTRA-FDD-r9 ::= SEQUENCE (SIZE (1..16)) OF SEQUENCE { | | | |
| physCellId-r9 | Cell 2 physical cell identity | | |
| utra-BCCH-Container-r9 | Cell 2 Relevant System Information | | |
| } | | | |

## 6.3.2.5 Test requirement

Tables 6.3.2.4.1-1, 6.3.2.5-1 and 6.3.2.5-2 define the primary level settings including test tolerances for Redirection from E-UTRAN TDD to UTRAN FDD test.

**Table 6.3.2.5-1: Cell Specific Test requirement Parameters for cell #1 E-UTRAN TDD to UTRAN FDD RRC release with redirection under AWGN propagation conditions**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 6.3.2.5-2: Cell Specific Test Parameters for cell #2 E-UTRAN TDD to UTRAN FDD RRC release with redirection under AWGN propagation conditions**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| UTRA RF Channel Number | | 1 | |
| CPICH_Ec/Ior | dB | -10 | |
| PCCPCH_Ec/Ior | dB | -12 | |
| SCH_Ec/Ior | dB | -12 | |
| PICH_Ec/Ior | dB | -15 | |
| DPCH_Ec/Ior | dB | N/A | |
| OCNS | | -0.941 | |
| $\hat{I}_{or} / I_{oc}$ | dB | -∞ | 0.42 |
| $I_{oc}$ | dBm/3.84 MHz | -70 | |
| CPICH_Ec/Io Note 3 | dB | -∞ | -12.81 |
| Propagation Condition | | AWGN | |

Note 1: The DPCH level is controlled by the power control loop.
Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.
Note 3: This gives an SCH Ec/Io of -15dB

The UE shall start to transmit random access to Cell 2 less than 650 ms from the beginning of time period T2.

The rate of correct RRC connection release redirection to UTRAN FDD observed during repeated tests shall be at least 90%.

The overall delays measured test requirement can be expressed as

Test Requirement = $T_{RRC_procedure_delay}$ + $T_{identify\text{-}UTRA\ FDD}$ + $T_{SI\text{-}UTRA\ FDD}$ + $T_{RA}$

$T_{RRC_procedure_delay}$ = 110 ms

$T_{identify\text{-}UTRA\ FDD}$ = 500 ms

$T_{SI\text{-}UTRA\ FDD}$ =the time required for acquiring all the relevant system information of the target UTRA FDD cell. This time depends upon whether the UE is provided with the relevant system information of the target UTRA FDD cell or not by the E-UTRAN before the RRC connection is released. 0 ms is assumed in this test case.

$T_{RA}$ = The additional delay caused by the random access procedure. 40 ms is assumed in this test case.

This gives a total of 650 ms for overall delays measured.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 6.3.3 Redirection from E-UTRAN FDD to GERAN when System Information is provided

### 6.3.3.1 Test Purpose

To verify that the UE is able to performs the RRC connection release with redirection from the E-UTRA FDD to the target GERAN cell to meet the RRC connection release with redirection to GERAN requirements.

### 6.3.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward that support GERAN.

### 6.3.3.3 Minimum conformance requirements

The UE shall be capable of performing the RRC connection release with redirection to the target GERAN cell within $T_{connection_release_redirect_GERAN}$.

The time delay ($T_{connection_release_redirect_GERAN}$) is the time between the end of the last TTI containing the RRC command, "*RRCConnectionRelease*" as defined in section 5.3.8 in TS36.331 [5] on the E-UTRAN PDSCH and the time the UE starts to send random access to the target GERAN cell. The time delay ($T_{connection_release_redirect_GERAN}$) shall be less than:

$$T_{connection_release_redirect_GERAN} = T_{RRC_procedure_delay} + T_{identify\text{-}GERAN} + T_{SI\text{-}GERAN} + T_{RA}$$

The target GERAN cell shall be considered detectable when the UE receives the GERAN cell at levels down to 10 dB + the reference sensitivity level or reference interference levels as specified in TS 45.005 [16].

$T_{RRC_procedure_delay}$: It is the RRC procedure for processing the received message "*RRCConnectionRelease*". It shall be less than 110 ms.

$T_{identify\text{-}UTRA\ GERAN}$: It is the time to identify the BSIC of the target GERAN cell. It shall be less than 1 second.

$T_{SI\text{-}UTRA\ GERAN}$: It is the time required for acquiring all the relevant system information of the target GERAN cell. This time depends upon whether the UE is provided with the relevant system information of the target GERAN cell or not by the E-UTRAN before the RRC connection is released.

$T_{RA}$: It is the delay caused due to the random access procedure when sending random access burst to the target GERAN cell.

The normative reference for this requirement is TS 36.133 [4] clause 6.3.2.2 and A.6.3.3.

### 6.3.3.4 Test description

#### 6.3.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.14.
2. The parameter settings for the cells are set up according to Table 6.3.3.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 6.3.3.4.3.
5. There is one E-UTRA FDD cell and one GERAN cell specified in the test. Cell 1(E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.3.3.4.1-1: General test parameters for RRC connection release with redirection from E-UTRAN FDD to GERAN in AWGN**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on Absolute RF Channel Number 1 (GSM cell) |
| CP length | | Normal | Applicable to cell 1 |
| E-UTRA RF Channel Number | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| DRX | | OFF | |
| Monitored GSM cell list size | | 6 GSM neighbour including ARFCN 1 | GSM cells are provided in the "*RRCConnectionRelease*" message. |
| T1 | s | 5 | |
| T2 | s | 2 | |

#### 6.3.3.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. The test consists of two successive time periods, with time duration of T1, and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2. The start of T2 is the instant when the last TTI containing the RRC message, "*RRCConnectionRelease*", is received by the UE from cell 1. The "*RRCConnectionRelease*" message shall contain all the relevant system information of cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 6.3.3.5-1 and 6.3.3.5-2. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. SS shall transmit an *RRCConnectionRelease* containing the relevant system information of Cell 2 during period T1.
4. The SS shall start T2 timer when the last TTI containing the *RRCConnectionRelease* message is sent to UE.
5. When T2 starts, the SS shall switch the power setting from T1 to T2 as specified in Table 6.3.3.5-1 for Cell 1, and Table 6.3.3.5-2 for Cell 2.

6. If the UE transmits the random access burst on RACH of the target GERAN Cell 2 less than 1120 ms from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

7. After T2 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

8. The SS shall set a different BSIC on Cell 2, as the previous timing information of Cell 2 is invalid in the UE for the next iteration of the test procedure loop.

9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

#### 6.3.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.3.3.4.3-1: *RRCConnectionRelease*: Additional Redirection from E-UTRAN FDD to GERAN test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| rrcConnectionRelease-r8 SEQUENCE { | | | |
| releaseCause | other | | |
| redirectedCarrierInfo CHOICE { | | | |
| geran | Cell 2 Carrier Information | | |
| } | | | |
| idleModeMobilityControlInfo | Not present | | |
| nonCriticalExtension SEQUENCE { | | | |
| lateNonCriticalExtension | Not present | | |
| nonCriticalExtension { | | | |
| cellInfoList-r9 CHOICE { | | | |
| geran-r9 | CellInfoListGERAN-r9 | | |
| } | | | |
| nonCriticalExtension | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 6.3.3.4.3-2: *CellInfoListGERAN-r9*: Additional Redirection from E-UTRAN FDD to GERAN test requirement (step3)**

| Derivation Path: TS 36.331 [5] clause 6.6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellInfoListGERAN-r9 ::= SEQUENCE (SIZE (1..6)) OF SEQUENCE { | | | |
| physCellId-r9 | Cell 2 physical cell identity | | |
| carrierFreq-r9 | Cell 2 carrier frequency | | |
| systemInformation-r9 | Cell 2 relevant system information | | |
| } | | | |

### 6.3.3.5 Test requirement

Tables 6.3.3.4.1-1, 6.3.3.5-1 and 6.3.3.5-2 define the primary level settings including test tolerances for Redirection from E-UTRAN FDD to GERAN test.

**Table 6.3.3.5-1: Cell specific test parameters for E-UTRA FDD cell (cell #1) for RRC connection release with redirection from E-UTRAN FDD to GERAN in AWGN**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| $N_{oc}$ | dBm/15 kHz | -98 | |
| RSRP | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 6.3.3.5-2: Cell specific test parameters for GERAN cell (cell #2) for RRC connection release with redirection from E-UTRAN FDD to GERAN in AWGN**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| Absolute RF Channel Number | | ARFNC 1 | |
| RXLEV | dBm | -Infinity | -75 |
| GSM BSIC | | N/A | Valid |

The UE shall begin to send access bursts on RACH of the target GERAN cell (cell #2) less than 1120 ms from the beginning of time period T2.

The rate of correct "RRC connection release with redirection to GERAN" observed during repeated tests shall be at least 90%.

The overall delays measured test requirement can be expressed as:

$T_{connection_release_redirect_GERAN} = T_{RRC_procedure_delay} + T_{identify\text{-}GERAN} + T_{SI\text{-}GERAN} + T_{RA}$

$T_{RRC_procedure_delay}$ = 110 ms, which is the time for processing the received message "*RRCConnectionRelease*.

$T_{identify\text{-}GERAN}$ = 1000 ms, which is the time for identifying the target GERAN cell.

$T_{SI\text{-}GERAN}$ = 0; UE does not have to read the system information of the GERAN cell since all relevant SI is provided to the UE in the "RRCConnectionRelease" message.

$T_{RA}$ = 10 ms, which is about 2 GSM frames (2*4.65 ms) to account for the GSM timing uncertainty.

This gives a total of 1120 ms for overall delays measured.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 6.3.4 Redirection from E-UTRAN TDD to GERAN when System Information is provided

### 6.3.4.1 Test Purpose

To verify that the UE is able to performs the RRC connection release with redirection from the E-UTRA TDD to the target GERAN cell to meet the RRC connection release with redirection to GERAN requirements.

### 6.3.4.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward that support GERAN.

### 6.3.4.3 Minimum conformance requirements

The UE shall be capable of performing the RRC connection release with redirection to the target GERAN cell within $T_{connection_release_redirect_GERAN}$.

The time delay ($T_{connection_release_redirect_GERAN}$) is the time between the end of the last TTI containing the RRC command, "*RRCConnectionRelease*" as defined in section 5.3.8 in TS36.331 [5] on the E-UTRAN PDSCH and the time the UE starts to send random access to the target GERAN cell. The time delay ($T_{connection_release_redirect_GERAN}$) shall be less than:

$$T_{connection_release_redirect_GERAN} = T_{RRC_procedure_delay} + T_{identify\text{-}GERAN} + T_{SI\text{-}GERAN} + T_{RA}$$

The target GERAN cell shall be considered detectable when the UE receives the GERAN cell at levels down to 10 dB + the reference sensitivity level or reference interference levels as specified in TS 45.005 [16].

$T_{RRC_procedure_delay}$: It is the RRC procedure for processing the received message "*RRCConnectionRelease*". It shall be less than 110 ms.

$T_{identify\text{-}UTRA\ GERAN}$: It is the time to identify the BSIC of the target GERAN cell. It shall be less than 1 second.

$T_{SI\text{-}UTRA\ GERAN}$: It is the time required for acquiring all the relevant system information of the target GERAN cell. This time depends upon whether the UE is provided with the relevant system information of the target GERAN cell or not by the E-UTRAN before the RRC connection is released.

$T_{RA}$: It is the delay caused due to the random access procedure when sending random access burst to the target GERAN cell.

The normative reference for this requirement is TS 36.133 [4] clause 6.3.2.2 and A.6.3.4.

### 6.3.4.4 Test description

#### 6.3.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.14.

2. The parameter settings for the cells are set up according to Table 6.3.4.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are as defined in clause 6.3.4.4.3.

5. There is one E-UTRA TDD cell and one GERAN cell specified in the test. Cell 1(E-UTRA TDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.3.4.4.1-1: General test parameters for RRC connection release with redirection from E-UTRAN TDD to GERAN in AWGN**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on Absolute RF Channel Number 1 (GSM cell) |
| CP length | | Normal | Applicable to cell 1 |
| Special subframe configuration | | 6 | As specified in table 4.2.1 in TS 36.211. The same configuration applies to all cells. |
| Uplink-downlink configuration | | 1 | |
| E-UTRA RF Channel Number | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| DRX | | OFF | |
| Monitored GSM cell list size | | 6 GSM neighbour including ARFCN 1 | GSM cells are provided in the "*RRCConnectionRelease*" message. |
| T1 | s | 5 | |
| T2 | s | 2 | |

### 6.3.4.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. The test consists of two successive time periods, with time duration of T1, and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2. The start of T2 is the instant when the last TTI containing the RRC message, "*RRCConnectionRelease*", is received by the UE from cell 1. The "*RRCConnectionRelease*" message shall contain all the relevant system information of cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

2. Set the parameters according to T1 in Table 6.3.4.5-1 and 6.3.4.5-2. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.

3. SS shall transmit an *RRCConnectionRelease* containing the relevant system information of Cell 2 during period T1.

4. The SS shall start T2 timer when the last TTI containing the *RRCConnectionRelease* message is sent to UE.

5. When T2 starts, the SS shall switch the power setting from T1 to T2 as specified in Table 6.3.4.5-1 for Cell 1, and Table 6.3.4.5-2 for Cell 2.

6. If the UE transmits the random access burst on RACH of the target GERAN Cell 2 less than 1120 ms from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

7. After T2 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

8. The SS shall set a different BSIC on Cell 2, as the previous timing information of Cell 2 is invalid in the UE for the next iteration of the test procedure loop.

9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

#### 6.3.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.3.4.4.3-1: *RRCConnectionRelease*: Additional Redirection from E-UTRAN TDD to GERAN test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| rrcConnectionRelease-r8 SEQUENCE { | | | |
| releaseCause | other | | |
| RedirectedCarrierInfo CHOICE { | | | |
| geran | Cell 2 Carrier Information | | |
| } | | | |
| idleModeMobilityControlInfo | Not present | | |
| nonCriticalExtension SEQUENCE { | | | |
| lateNonCriticalExtension | Not present | | |
| nonCriticalExtension { | | | |
| cellInfoList-r9 CHOICE { | | | |
| geran-r9 | CellInfoListGERAN-r9 | | |
| } | | | |
| nonCriticalExtension | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 6.3.4.4.3-2: *CellInfoListGERAN-r9*: Additional Redirection from E-UTRAN TDD to GERAN test requirement (step3)**

| Derivation Path: TS 36.331 [5] clause 6.6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellInfoListGERAN-r9 ::= SEQUENCE (SIZE (1..6)) OF SEQUENCE { | | | |
| physCellId-r9 | Cell 2 physical cell identity | | |
| carrierFreq-r9 | Cell 2 carrier frequency | | |
| systemInformation-r9 | Cell 2 relevant system information | | |
| } | | | |

### 6.3.4.5 Test requirement

Tables 6.3.4.4.1-1, 6.3.4.5-1 and 6.3.4.5-2 define the primary level settings including test tolerances for Redirection from E-UTRAN TDD to GERAN test.

**Table 6.3.4.5-1: Cell specific test parameters for E-UTRA TDD cell (cell #1) for RRC connection release with redirection from E-UTRAN TDD to GERAN in AWGN**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| $N_{oc}$ | dBm/15 kHz | -98 | |
| RSRP | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 6.4.3.5-2: Cell specific test parameters for GERAN cell (cell #2) for RRC connection release with redirection from E-UTRAN TDD to GERAN in AWGN**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| Absolute RF Channel Number | | ARFNC 1 | |
| RXLEV | dBm | -Infinity | -75 |
| GSM BSIC | | N/A | Valid |

The UE shall begin to send access bursts on RACH of the target GERAN cell (cell #2) less than 1120 ms from the beginning of time period T2.

The rate of correct "RRC connection release with redirection to GERAN" observed during repeated tests shall be at least 90%.

The overall delays measured test requirement can be expressed as:

$T_{connection_release_redirect_GERAN} = T_{RRC_procedure_delay} + T_{identify\text{-}GERAN} + T_{SI\text{-}GERAN} + T_{RA}$

$T_{RRC_procedure_delay}$ = 110 ms, which is the time for processing the received message "*RRCConnectionRelease*.

$T_{identify\text{-}GERAN}$ = 1000 ms, which is the time for identifying the target GERAN cell.

$T_{SI\text{-}GERAN}$ = 0; UE does not have to read the system information of the GERAN cell since all relevant SI is provided to the UE in the "RRCConnectionRelease" message.

$T_{RA}$ = 10 ms, which is about 2 GSM frames (2*4.65 ms) to account for the GSM timing uncertainty.

This gives a total of 1120 ms for overall delays measured.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 6.3.5 E-UTRA TDD RRC connection release redirection to UTRA TDD

### 6.3.5.1 Test purpose

The purpose of this test is to verify that the UE performs the RRC connection release with redirection from the E-UTRA TDD to the target UTRA TDD cell within $T_{connection_release_redirect_UTRA\ TDD}$. This test will partly verify the RRC connection release with redirection to UTRA TDD requirements in TS 36.133 [4] section 6.3.2.3.

### 6.3.5.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward that support UTRA TDD.

### 6.3.5.3 Minimum conformance requirements

The UE shall be capable of performing the RRC connection release with redirection to the target UTRAN TDD cell within $T_{connection_release_redirect_UTRA\ TDD}$.

The time delay ($T_{connection_release_redirect_UTRA\ TDD}$) is the time between the end of the last TTI containing the RRC command, "*RRCConnectionRelease*" as defined in section 5.3.8 in TS 36.331 [5] on the E-UTRAN PDSCH and the time the UE starts to send random access to the target UTRA TDD cell. The time delay ($T_{connection_release_redirect_UTRA\ TDD}$) shall be less than:

$$T_{connection_release_redirect_UTRA\ TDD} = T_{RRC_procedure_delay} + T_{identify\text{-}UTRA\ TDD} + N_{redirect\text{-}UTRA\ TDD} + T_{SI\text{-}UTRA\ TDD} + T_{RA}$$

The target UTRA TDD cell shall be considered detectable when:

- P-CCPCH Ec/Io $\geq$ -6 dB,
- DwPCH_Ec/Io $\geq$ -1 dB.

$T_{RRC_procedure_delay}$: It is the RRC procedure for processing the received message "*RRCConnectionRelease*". It shall be less than 110 ms.

$T_{identify\text{-}UTRA\ TDD}$: It is the time to identify the target UTRA TDD cell. It shall be less than 500 ms.

$T_{SI\text{-}UTRA\ TDD}$: It is the time required for acquiring all the relevant system information of the target UTRA TDD cell. This time depends upon whether the UE is provided with the relevant system information of the target UTRA TDD cell or not by the E-UTRAN before the RRC connection is released.

$T_{RA}$: It is the delay caused due to the random access procedure when sending random access to the target UTRA TDD cell.

$N_{redirect\text{-}UTRA\ TDD}$: It is the total number of target UTRA TDD frequencies included in RedirectedCarrierInfo in "*RRCConnectionRelease*" message. It can be up to 4 UTRA TDD frequencies.

The normative reference for this requirement is TS 36.133 [4] clause 6.3.2.3 and A.6.3.5.

### 6.3.5.4 Test description

#### 6.3.5.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.

2. The general test parameter settings are set up according to Table 6.3.5.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 6.3.5.4.3.

5. There is one E-UTRA TDD serving cell and one UTRA TDD cell specified in the test. Cell 1 (E-UTRA TDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.3.5.4.1-1: General test parameters for E-UTRA TDD RRC connection release redirection to UTRA TDD**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on UTRA RF channel number 1. |
| Time offset between cells | ms | 3 | Asynchronous cells |
| E-UTRA RF Channel Number | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| CP length | | Normal | Applicable to cell 1 |
| Special subframe configuration of cell 1 | | 6 | As specified in table 4.2.1 in TS 36.211 |
| Uplink-downlink configuration of cell 1 | | 1 | As specified in table 4.2-2 in TS 36.211. |
| UTRA RF Channel Number | | 1 | One UTRA TDD carrier frequency is used. |
| Filter coefficient | | 0 | L3 filtering is not used. |
| DRX | | OFF | |
| Monitored UTRA TDD cell list size | | 16 | UTRA cells on UTRA RF channel 1 provided in the "*RRCConnectionRelease*" message from the E-UTRAN |
| T1 | s | 5 | |
| T2 | s | 1 | |

### 6.3.5.4.2 Test procedure

The test consists of two successive time periods, with time duration of T1, and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2. The "*RRCConnectionRelease*" message containing the relevant system information of Cell 2 shall be sent to the UE during period T1 and the start of T2 is the instant when the last TTI containing the RRC message, "*RRCConnectionRelease*", is received by the UE from cell 1. The "*RRCConnectionRelease*" message shall contain all the relevant system information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

2. Set the parameters according to T1 in Table 6.3.5.5-1 and 6.3.5.5-2. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.

3. SS shall transmit an *RRCConnectionRelease* containing the relevant system information of Cell 2 during period T1.

4. The SS shall start T2 timer when the last TTI containing the *RRCConnectionRelease* message is sent to UE.

5. When T2 starts, the SS shall switch the power setting from T1 to T2 as specified in Table 6.3.5.5-1for Cell 1, and Table 6.3.5.5-2 for Cell 2.

6. If the UE transmits the SYNCH-UL sequence in the UpPTS to Cell 2 less than 650 ms from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

7. After T2 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

8. The SS shall set Cell 2 cell parameter id = (current cell 2 cell parameter id +4) mod 16 for next iteration of the test procedure loop.

9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

#### 6.3.5.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.3.5.4.3-1: *RRCConnectionRelease*: E-UTRA TDD RRC connection release redirection to UTRA TDD test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| rrcConnectionRelease-r8 SEQUENCE { | | | |
| releaseCause | other | | |
| redirectedCarrierInfo CHOICE { | | | |
| utra-TDD | Cell 2 Carrier Information | | |
| } | | | |
| idleModeMobilityControlInfo | Not present | | |
| nonCriticalExtension SEQUENCE { | | | |
| lateNonCriticalExtension | Not present | | |
| nonCriticalExtension { | | | |
| cellInfoList-r9 CHOICE { | | | |
| utra-TDD-r9 | CellInfoListUTRA-TDD-r9 | | |
| } | | | |
| nonCriticalExtension | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 6.3.5.4.3-2: *CellInfoListUTRA-TDD-r9*: E-UTRA TDD RRC connection release redirection to UTRA TDD test requirement (step3)**

| Derivation Path: TS 36.331 [5] clause 6.6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellInfoListUTRA-TDD-r9 ::= SEQUENCE (SIZE (1..16)) OF SEQUENCE { | | | |
| physCellId-r9 | Cell 2 physical cell identity | | |
| utra-BCCH-Container-r9 | Cell 2 Relevant System Information | | |
| } | | | |

### 6.3.5.5 Test requirement

Tables 6.3.5.4.1-1, 6.3.5.5-1 and 6.3.5.5-2 define the primary level settings including test tolerances for Redirection from E-UTRAN TDD to UTRAN TDD test.

**Table 6.3.5.5-1: Cell specific test parameters for cell 1 in E-UTRA TDD RRC connection release redirection to UTRA TDD test**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 6.3.5.5-2: Cell specific test parameters for cell 2 in E-UTRA TDD RRC connection release redirection to UTRA TDD test**

| Parameter | Unit | Cell 2 (UTRA TDD) | | | |
|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPTS | |
| | | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number Note1 | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -4.77 | -4.77 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 |
| OCNS_Ec/Ior Note2 | dB | -1.76 | -1.76 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -inf | 8 | -inf | 8 |
| $I_{oc}$ | dBm/1.28 MHz | -80 | | | |
| PCCPCH RSCP Note3 | dBm | -inf | -76.77 | n.a. | n.a. |
| PCCPCH_Ec/Io Note3 | dB | -inf | -5.41 | n.a. | n.a. |
| DwPCH_Ec/Io Note3 | dB | n.a. | n.a. | -inf | -0.64 |
| Propagation Condition | | AWGN | | | |

Note 1: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number.

Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.

Note 3: P-CCPCH RSRP, PCCPCH_Ec/Io and DwPCH_Ec/Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall start to transmit the SYNCH-UL sequence in the UpPTS to Cell 2 less than 650 ms from the beginning of time period T2.

The rate of correct RRC connection release redirection to UTRA TDD observed during repeated tests shall be at least 90%.

The time delay can be expressed as: $T_{RRC_procedure_delay} + T_{identify\text{-}UTRA\ TDD} + N_{redirect\text{-}UTRA\ TDD} + T_{SI\text{-}UTRA\ TDD} + T_{RA}$, where:

$T_{RRC_procedure_delay}$ = 110 ms, which is specified in TS 36.133 [4] section 6.3.2.3.

$T_{identify\text{-}UTRA\ TDD}$ = 500 ms; which is defined in TS 36.133 [4] section 6.3.2.3.

$N_{redirect\text{-}UTRA\ TDD}$ = 0. It is the total number of target UTRA TDD frequencies included in RedirectedCarrierInfo in "*RRCConnectionRelease*" message. It can be up to 4 UTRA TDD frequencies.

$T_{SI\text{-}UTRA\ TDD}$ = 0 ms, UE does not have to read the system information of the UTRAN TDD since all relevant SI is provided to the UE in the "*RRCConnectionRelease*" message.

$T_{RA}$ = 40ms. This is the additional delay caused by the random access procedure

It gives a total delay of 650 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 6.3.6 E-UTRA FDD RRC connection release redirection to UTRA TDD

### 6.3.6.1 Test purpose

The purpose of this test is to verify that the UE performs the RRC connection release with redirection from the E-UTRA FDD to the target UTRA TDD cell within $T_{connection_release_redirect_UTRA\ TDD}$. This test will partly verify the RRC connection release with redirection to UTRA TDD requirements in TS 36.133 [4] section 6.3.2.3.

### 6.3.6.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward that support UTRA TDD.

### 6.3.6.3 Minimum conformance requirements

The UE shall be capable of performing the RRC connection release with redirection to the target UTRAN TDD cell within $T_{connection_release_redirect_UTRA\ TDD}$.

The time delay ($T_{connection_release_redirect_UTRA\ TDD}$) is the time between the end of the last TTI containing the RRC command, "*RRCConnectionRelease*" as defined in section 5.3.8 in TS 36.331 [5] on the E-UTRAN PDSCH and the time the UE starts to send random access to the target UTRA TDD cell. The time delay ($T_{connection_release_redirect_UTRA\ TDD}$) shall be less than:

$$T_{connection_release_redirect_UTRA\ TDD} = T_{RRC_procedure_delay} + T_{identify\text{-}UTRA\ TDD} + N_{redirect\text{-}UTRA\ TDD} + T_{SI\text{-}UTRA\ TDD} + T_{RA}$$

The target UTRA TDD cell shall be considered detectable when:

- P-CCPCH Ec/Io $\geq$ -6 dB,
- DwPCH_Ec/Io $\geq$ -1 dB.

$T_{RRC_procedure_delay}$: It is the RRC procedure for processing the received message "*RRCConnectionRelease*". It shall be less than 110 ms.

$T_{identify\text{-}UTRA\ TDD}$: It is the time to identify the target UTRA TDD cell. It shall be less than 500 ms.

$T_{SI\text{-}UTRA\ TDD}$: It is the time required for acquiring all the relevant system information of the target UTRA TDD cell. This time depends upon whether the UE is provided with the relevant system information of the target UTRA TDD cell or not by the E-UTRAN before the RRC connection is released.

$T_{RA}$: It is the delay caused due to the random access procedure when sending random access to the target UTRA TDD cell.

$N_{redirect\text{-}UTRA\ TDD}$: It is the total number of target UTRA TDD frequencies included in RedirectedCarrierInfo in "*RRCConnectionRelease*" message. It can be up to 4 UTRA TDD frequencies.

The normative reference for this requirement is TS 36.133 [4] clause 6.3.2.3 and A.6.3.6.

## 6.3.6.4 Test description

### 6.3.6.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.23.
2. The general test parameter settings are set up according to Table 6.3.6.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 6.3.6.4.3.
5. There is one E-UTRA FDD serving cell and one UTRA TDD cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.3.6.4.1-1: General test parameters for E-UTRA FDD RRC connection release redirection to UTRA TDD**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on UTRA RF channel number 1. |
| CP length | | Normal | Applicable to cell 1 |
| Time offset between cells | ms | 3 | Asynchronous cells |
| E-UTRA RF Channel Number | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Filter coefficient | | 0 | L3 filtering is not used. |
| DRX | | OFF | |
| Monitored UTRA TDD cell list size | | 16 | UTRA cells on UTRA RF channel 1 provided in the "*RRCConnectionRelease*" message from the E-UTRAN |
| T1 | s | 5 | |
| T2 | s | 1 | |

### 6.3.6.4.2 Test procedure

The test consists of two successive time periods, with time duration of T1, and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2. The "*RRCConnectionRelease*" message containing the relevant system information of Cell 2 shall be sent to the UE during period T1 and the start of T2 is the instant when the

last TTI containing the RRC message, "*RRCConnectionRelease*", is received by the UE from cell 1. The "*RRCConnectionRelease*" message shall contain all the relevant system information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 6.3.6.5-1 and 6.3.6.5-2. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. SS shall transmit an *RRCConnectionRelease* containing the relevant system information of Cell 2 during period T1.
4. The SS shall start T2 timer when the last TTI containing the *RRCConnectionRelease* message is sent to UE.
5. When T2 starts, the SS shall switch the power setting from T1 to T2 as specified in Table 6.3.6.5-1 for Cell 1, and Table 6.3.6.5-2 for Cell 2.
6. If the UE transmits the SYNCH-UL sequence in the UpPTS to Cell 2 less than 650 ms from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
7. After T2 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
8. The SS shall set Cell 2 cell parameter id = (current cell 2 cell parameter id +4) mod 16 for next iteration of the test procedure loop.
9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

### 6.3.6.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.3.6.4.3-1: *RRCConnectionRelease*: E-UTRA FDD RRC connection release redirection to UTRA TDD test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| rrcConnectionRelease-r8 SEQUENCE { | | | |
| releaseCause | other | | |
| redirectedCarrierInfo CHOICE { | | | |
| utra-TDD | Cell 2 Carrier Information | | |
| } | | | |
| idleModeMobilityControlInfo | Not present | | |
| nonCriticalExtension SEQUENCE { | | | |
| lateNonCriticalExtension | Not present | | |
| nonCriticalExtension { | | | |
| cellInfoList-r9 CHOICE { | | | |
| utra-TDD-r9 | CellInfoListUTRA-TDD-r9 | | |
| } | | | |
| nonCriticalExtension | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 6.3.6.4.3-2: *CellInfoListUTRA-TDD-r9*: E-UTRA FDD RRC connection release redirection to UTRA TDD test requirement (step3)**

| Derivation Path: TS 36.331 [5] clause 6.6.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellInfoListUTRA-TDD-r9 ::= SEQUENCE (SIZE (1..16)) OF SEQUENCE { | | | |
| physCellId-r9 | Cell 2 physical cell identity | | |
| utra-BCCH-Container-r9 | Cell 2 Relevant System Information | | |
| } | | | |

## 6.3.6.5 Test requirement

Tables 6.3.6.4.1-1, 6.3.6.5-1 and 6.3.6.5-2 define the primary level settings including test tolerances for Redirection from E-UTRAN FDD to UTRAN TDD test.

**Table 6.3.6.1-2: Cell specific test parameters for cell 1 in E-UTRA FDD RRC connection release redirection to UTRA TDD test**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.2.1 (OP.1 FDD) | | OP.1 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 6.3.6.1-3: Cell specific test parameters for cell 2 in E-UTRA FDD RRC connection release redirection to UTRA TDD test**

| Parameter | Unit | Cell 2 (UTRA TDD) | | | |
|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPTS | |
| | | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number Note1 | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -4.77 | -4.77 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 |
| OCNS_Ec/Ior Note2 | dB | -1.76 | -1.76 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -inf | 8 | -inf | 8 |
| $I_{oc}$ | dBm/1.28 MHz | -80 | | | |
| PCCPCH RSCP Note3 | dBm | -inf | -76.77 | n.a. | n.a. |
| PCCPCH_Ec/Io Note3 | dB | -inf | -5.41 | n.a. | n.a. |
| DwPCH_Ec/Io Note3 | dB | n.a. | n.a. | -inf | -0.64 |
| Propagation Condition | | AWGN | | | |

Note 1: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number.
Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.
Note 3: P-CCPCH RSRP, PCCPCH_Ec/Io and DwPCH_Ec/Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall start to transmit the SYNCH-UL sequence in the UpPTS to Cell 2 less than 650 ms from the beginning of time period T2.

The rate of correct RRC connection release redirection to UTRA TDD observed during repeated tests shall be at least 90%.

The time delay can be expressed as: $T_{RRC_procedure_delay} + T_{identify\text{-}UTRA\ TDD} + N_{redirect\text{-}UTRA\ TDD} + T_{SI\text{-}UTRA\ TDD} + T_{RA}$, where:

$T_{RRC_procedure_delay}$ = 110 ms, which is specified in TS 36.133 [4] section 6.3.2.3.

$T_{identify\text{-}UTRA\ TDD}$ = 500 ms; which is defined in TS 36.133 [4] section 6.3.2.3.

$N_{redirect\text{-}UTRA\ TDD}$ = 0. It is the total number of target UTRA TDD frequencies included in RedirectedCarrierInfo in "*RRCConnectionRelease*" message. It can be up to 4 UTRA TDD frequencies.

$T_{SI\text{-}UTRA\ TDD}$ = 0 ms, UE does not have to read the system information of the UTRAN TDD since all relevant SI is provided to the UE in the "*RRCConnectionRelease*" message.

$T_{RA}$ = 40ms. This is the additional delay caused by the random access procedure.

This gives a total delay of 650 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 6.3.7 E-UTRA TDD RRC connection release redirection to UTRA TDD without SI provided

### 6.3.7.1 Test purpose

The purpose of this test is to verify that the UE performs the RRC connection release with redirection from the E-UTRA TDD to the target UTRA TDD cell within $T_{connection_release_redirect_UTRA\ TDD}$. This test will partly verify the RRC connection release with redirection to UTRA TDD requirements in TS 36.133 [4] section 6.3.2.3.

### 6.3.7.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward that support UTRA TDD.

### 6.3.7.3 Minimum conformance requirements

The UE shall be capable of performing the RRC connection release with redirection to the target UTRAN TDD cell within $T_{connection_release_redirect_UTRA\ TDD}$.

The time delay ($T_{connection_release_redirect_UTRA\ TDD}$) is the time between the end of the last TTI containing the RRC command, "*RRCConnectionRelease*" as defined in section 5.3.8 in TS 36.331 [5] on the E-UTRAN PDSCH and the time the UE starts to send random access to the target UTRA TDD cell. The time delay ($T_{connection_release_redirect_UTRA\ TDD}$) shall be less than:

$$T_{connection_release_redirect_UTRA\ TDD} = T_{RRC_procedure_delay} + T_{identify\text{-}UTRA\ TDD} + N_{redirect\text{-}UTRA\ TDD} + T_{SI\text{-}UTRA\ TDD} + T_{RA}$$

The target UTRA TDD cell shall be considered detectable when:

- P-CCPCH Ec/Io $\geq$ -6 dB,
- DwPCH_Ec/Io $\geq$ -1 dB.

$T_{RRC_procedure_delay}$: It is the RRC procedure for processing the received message "*RRCConnectionRelease*". It shall be less than 110 ms.

$T_{identify\text{-}UTRA\ TDD}$: It is the time to identify the target UTRA TDD cell. It shall be less than 500 ms.

$T_{SI\text{-}UTRA\ TDD}$: It is the time required for acquiring all the relevant system information of the target UTRA TDD cell. This time depends upon whether the UE is provided with the relevant system information of the target UTRA TDD cell or not by the E-UTRAN before the RRC connection is released.

$T_{RA}$: It is the delay caused due to the random access procedure when sending random access to the target UTRA TDD cell.

$N_{redirect\text{-}UTRA\ TDD}$: It is the total number of target UTRA TDD frequencies included in RedirectedCarrierInfo in "*RRCConnectionRelease*" message. It can be up to 4 UTRA TDD frequencies.

The normative reference for this requirement is TS 36.133 [4] clause 6.3.2.3 and A.6.3.7.

### 6.3.7.4 Test description

#### 6.3.7.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.
2. The general test parameter settings are set up according to Table 6.3.7.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 6.3.7.4.3.
5. There is one E-UTRA TDD serving cell and one UTRA TDD cell specified in the test. Cell 1 (E-UTRA TDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.3.7.4.1-1: General test parameters for E-UTRA TDD RRC connection release redirection to UTRA TDD without SI provided**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on UTRA RF channel number 1. |
| Time offset between cells | ms | 3 | Asynchronous cells |
| E-UTRA RF Channel Number | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| CP length | | Normal | Applicable to cell 1 |
| Uplink-downlink configuration of cell 1 | | 1 | As specified in table 4.2.2 in TS 36.211 |
| Special subframe configuration of cell 1 | | 6 | As specified in table 4.2.1 in TS 36.211 |
| UTRA RF Channel Number | | 1 | One UTRA TDD carrier frequency is used. |
| Filter coefficient | | 0 | L3 filtering is not used. |
| DRX | | OFF | |
| Monitored UTRA TDD cell list size | | none | No explicit neighbour list is provided to the UE |
| T1 | s | 5 | |
| T2 | s | 2 | |

### 6.3.7.4.2 Test procedure

The test consists of two successive time periods, with time duration of T1, and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2. The "*RRCConnectionRelease*" message not containing the relevant system information of Cell 2 shall be sent to the UE during period T1 and the start of T2 is the instant when the last TTI containing the RRC message, "*RRCConnectionRelease*", is received by the UE from Cell 1. The "*RRCConnectionRelease*" message shall not contain all the relevant system information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

2. Set the parameters according to T1 in Table 6.3.7.5-1 and 6.3.7.5-2. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.

3. The SS shall transmit an *RRCConnectionRelease* not containing the relevant system information of Cell 2 during period T1.

4. The SS shall start T2 timer when the last TTI containing the *RRCConnectionRelease* message is sent to the UE.

5. When T2 starts, the SS shall switch the power setting from T1 to T2 as specified in Table 6.3.7.5-1for Cell 1, and Table 6.3.7.5-2 for Cell 2.

6. If the UE transmits the SYNCH-UL sequence in the UpPTS to Cell 2 less than 1930 ms from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

7. After T2 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

8. The SS shall set Cell 2 cell parameter id = (current cell 2 cell parameter id +4) mod 16 for next iteration of the test procedure loop.

9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

#### 6.3.7.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.3.7.4.3-1: *RRCConnectionRelease*: E-UTRA TDD RRC connection release redirection to UTRA TDD without SI provided test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| rrcConnectionRelease-r8 SEQUENCE { | | | |
| releaseCause | other | | |
| redirectedCarrierInfo CHOICE { | | | |
| utra-TDD | Cell 2 Carrier Information | | |
| } | | | |
| idleModeMobilityControlInfo | Not present | | |
| nonCriticalExtension SEQUENCE { | | | |
| lateNonCriticalExtension | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 6.3.7.5 Test requirement

Tables 6.3.7.4.1-1, 6.3.7.5-1 and 6.3.7.5-2 define the primary level settings including test tolerances for Redirection from E-UTRAN TDD to UTRAN TDD test without SI provided.

**Table 6.3.7.5-1: Cell specific test parameters for cell 1 in E-UTRA TDD RRC connection release redirection to UTRA TDD test without SI provided**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 6.3.7.5-2: Cell specific test parameters for cell 2 in E-UTRA TDD RRC connection release redirection to UTRA TDD test without SI provided**

| Parameter | Unit | Cell 2 (UTRA TDD) | | | |
|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPTS | |
| | | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number[Note1] | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -4.77 | -4.77 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 |
| OCNS_Ec/Ior[Note2] | dB | -1.76 | -1.76 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -inf | 8 | -inf | 8 |
| $I_{oc}$ | dBm/1.28 MHz | -80 | | | |
| PCCPCH RSCP [Note3] | dBm | -inf | -76.77 | n.a. | n.a. |
| PCCPCH_Ec/Io [Note3] | dB | -inf | -5.41 | n.a. | n.a. |
| DwPCH_Ec/Io [Note3] | dB | n.a. | n.a. | -inf | -0.64 |
| Propagation Condition | | AWGN | | | |

Note 1: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number.
Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.
Note 3: P-CCPCH RSRP, PCCPCH_Ec/Io and DwPCH_Ec/Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall start to transmit the SYNCH-UL sequence in the UpPTS to Cell 2 less than 1930 ms from the beginning of time period T2.

The time delay can be expressed as: $T_{RRC_procedure_delay} + T_{identify\text{-}UTRA\ TDD} + N_{redirect\text{-}UTRA\ TDD} + T_{SI\text{-}UTRA\ TDD} + T_{RA}$, where:

$T_{RRC_procedure_delay}$ = 110 ms; It is the RRC procedure for processing the received message "*RRCConnectionRelease*".

$T_{identify\text{-}UTRA\ TDD}$ = 500 ms; It is the time to identify the target UTRA TDD cell.

$N_{redirect\text{-}UTRA\ TDD}$ = 0. It is the total number of target UTRA TDD frequencies included in RedirectedCarrierInfo in "*RRCConnectionRelease*" message. It can be up to 4 UTRA TDD frequencies.

$T_{SI\text{-}UTRA\ TDD}$: Maximum repetition period of relevant system info blocks that need to be received by the UE during RRC connection release redirection to UTRA TDD cell. 1280 ms is assumed in this test case.

$T_{RA}$ = 40ms, this is the additional delay caused by the random access procedure.

This gives a total delay of 1930 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 6.3.8 E-UTRA FDD RRC connection release redirection to UTRA TDD without SI provided

### 6.3.8.1 Test purpose

The purpose of this test is to verify that the UE performs the RRC connection release with redirection from the E-UTRA FDD to the target UTRA TDD cell within $T_{connection_release_redirect_UTRA\ TDD}$. This test will partly verify the RRC connection release with redirection to UTRA TDD requirements in TS 36.133 [4] section 6.3.2.3.

### 6.3.8.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward that support UTRA TDD.

### 6.3.8.3 Minimum conformance requirements

The UE shall be capable of performing the RRC connection release with redirection to the target UTRAN TDD cell within $T_{connection_release_redirect_UTRA\ TDD}$.

The time delay ($T_{connection_release_redirect_UTRA\ TDD}$) is the time between the end of the last TTI containing the RRC command, "*RRCConnectionRelease*" as defined in section 5.3.8 in TS 36.331 [5] on the E-UTRAN PDSCH and the time the UE starts to send random access to the target UTRA TDD cell. The time delay ($T_{connection_release_redirect_UTRA\ TDD}$) shall be less than:

$$T_{connection_release_redirect_UTRA\ TDD} = T_{RRC_procedure_delay} + T_{identify\text{-}UTRA\ TDD} + N_{redirect\text{-}UTRA\ TDD} + T_{SI\text{-}UTRA\ TDD} + T_{RA}$$

The target UTRA TDD cell shall be considered detectable when:

- P-CCPCH Ec/Io $\geq$ -6 dB,
- DwPCH_Ec/Io $\geq$ -1 dB.

$T_{RRC_procedure_delay}$: It is the RRC procedure for processing the received message "*RRCConnectionRelease*". It shall be less than 110 ms.

$T_{identify\text{-}UTRA\ TDD}$: It is the time to identify the target UTRA TDD cell. It shall be less than 500 ms.

$T_{SI\text{-}UTRA\ TDD}$: It is the time required for acquiring all the relevant system information of the target UTRA TDD cell. This time depends upon whether the UE is provided with the relevant system information of the target UTRA TDD cell or not by the E-UTRAN before the RRC connection is released.

$T_{RA}$: It is the delay caused due to the random access procedure when sending random access to the target UTRA TDD cell.

$N_{redirect\text{-}UTRA\ TDD}$: It is the total number of target UTRA TDD frequencies included in RedirectedCarrierInfo in "*RRCConnectionRelease*" message. It can be up to 4 UTRA TDD frequencies.

The normative reference for this requirement is TS 36.133 [4] clause 6.3.2.3 and A.6.3.8.

### 6.3.8.4 Test description

#### 6.3.8.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.
2. The general test parameter settings are set up according to Table 6.3.8.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 6.3.8.4.3.
5. There is one E-UTRA FDD serving cell and one UTRA TDD cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.3.8.4.1-1: General test parameters for E-UTRA FDD RRC connection release redirection to UTRA TDD without SI provided**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on UTRA RF channel number 1. |
| Time offset between cells | ms | 3 | Asynchronous cells |
| E-UTRA RF Channel Number | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| CP length | | Normal | Applicable to cell 1 |
| UTRA RF Channel Number | | 1 | One UTRA TDD carrier frequency is used. |
| Filter coefficient | | 0 | L3 filtering is not used. |
| DRX | | OFF | |
| Monitored UTRA TDD cell list size | | none | No explicit neighbour list is provided to the UE |
| T1 | S | 5 | |
| T2 | S | 2 | |

### 6.3.8.4.2 Test procedure

The test consists of two successive time periods, with time duration of T1, and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2. The "*RRCConnectionRelease*" message not containing the relevant system information of Cell 2 shall be sent to the UE during period T1 and the start of T2 is the instant when the last TTI containing the RRC message, "*RRCConnectionRelease*", is received by the UE from cell 1. The "*RRCConnectionRelease*" message shall not contain all the relevant system information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 6.3.8.5-1 and 6.3.8.5-2. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. The SS shall transmit an *RRCConnectionRelease* not containing the relevant system information of Cell 2 during time period T1.
4. The SS shall start T2 timer when the last TTI containing the *RRCConnectionRelease* message is sent to the UE.
5. When T2 starts, the SS shall switch the power setting from T1 to T2 as specified in Table 6.3.8.5-1 for Cell 1 and Table 6.3.8.5-2 for Cell 2.
6. If the UE transmits the SYNCH-UL sequence in the UpPTS to Cell 2 less than 1930 ms from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
7. After T2 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
8. The SS shall set Cell 2 cell parameter id = (current cell 2 cell parameter id +4) mod 16 for next iteration of the test procedure loop.
9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

### 6.3.8.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.3.8.4.3-1: *RRCConnectionRelease*: E-UTRA FDD RRC connection release redirection to UTRA TDD test without SI provided requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| rrcConnectionRelease-r8 SEQUENCE { | | | |
| releaseCause | other | | |
| redirectedCarrierInfo CHOICE { | | | |
| utra-TDD | Cell 2 Carrier Information | | |
| } | | | |
| idleModeMobilityControlInfo | Not present | | |
| nonCriticalExtension SEQUENCE { | | | |
| lateNonCriticalExtension | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 6.3.8.5 Test requirement

Tables 6.3.8.4.1-1, 6.3.8.5-1 and 6.3.8.5-2 define the primary level settings including test tolerances for Redirection from E-UTRAN FDD to UTRAN TDD test.

**Table 6.3.8.5-1: Cell specific test parameters for cell 1 in E-UTRA FDD RRC connection release redirection to UTRA TDD test without SI provided**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 6.3.8.5-2: Cell specific test parameters for cell 2 in E-UTRA FDD RRC connection release redirection to UTRA TDD test without SI provided**

| Parameter | Unit | Cell 2 (UTRA TDD) | | | |
|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPTS | |
| | | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number Note1 | | Channel 1 | | | |
| PCCPCH_Ec/Ior | dB | -4.77 | -4.77 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 |
| OCNS_Ec/Ior Note2 | dB | -1.76 | -1.76 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -inf | 8 | -inf | 8 |
| $I_{oc}$ | dBm/1.28 MHz | -80 | | | |
| PCCPCH RSCP Note3 | dBm | -inf | -76.77 | n.a. | n.a. |
| PCCPCH_Ec/Io Note3 | dB | -inf | -5.41 | n.a. | n.a. |
| DwPCH_Ec/Io Note3 | dB | n.a. | n.a. | -inf | -0.64 |
| Propagation Condition | | AWGN | | | |

Note 1: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number.

Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.

Note 3: P-CCPCH RSRP, PCCPCH_Ec/Io and DwPCH_Ec/Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall start to transmit the SYNCH-UL sequence in the UpPTS to Cell 2 less than 1930 ms from the beginning of time period T2.

The time delay can be expressed as: $T_{RRC_procedure_delay} + T_{identify\text{-}UTRA\ TDD} + N_{redirect\text{-}UTRA\ TDD} + T_{SI\text{-}UTRA\ TDD} + T_{RA}$, where:

$T_{RRC_procedure_delay}$ = 110 ms; It is the RRC procedure for processing the received message "*RRCConnectionRelease*".

$T_{identify\text{-}UTRA\ TDD}$ = 500 ms; It is the time to identify the target UTRA TDD cell.

$N_{redirect\text{-}UTRA\ TDD}$ = 0. It is the total number of target UTRA TDD frequencies included in RedirectedCarrierInfo in "*RRCConnectionRelease*" message. It can be up to 4 UTRA TDD frequencies.

$T_{SI\text{-}UTRA\ TDD}$: Maximum repetition period of relevant system info blocks that need to be received by the UE during RRC connection release redirection to UTRA TDD cell. 1280 ms is assumed in this test case.

$T_{RA}$ = 40ms, this is the additional delay caused by the random access procedure.

This gives a total delay of 1930 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 6.3.9 Redirection from E-UTRAN FDD to UTRAN FDD without System Information

### 6.3.9.1 Test purpose

The purpose of this test is to verify that the UE makes correct performing the RRC connection release with redirection to the target UTRAN FDD cell without system information. This test will partly verify the RRC connection release with redirection to UTRAN FDD cell requirements in section 6.3.2.1.

### 6.3.9.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward that support UTRA FDD.

### 6.3.9.3 Minimum conformance requirements

The UE shall be capable of performing the RRC connection release with redirection to the target UTRAN FDD cell within $T_{connection_release_redirect_UTRA\ FDD}$.

The time delay ($T_{connection_release_redirect_UTRA\ FDD}$) is the time between the end of the last TTI containing the RRC command, "*RRCConnectionRelease*" as defined in section 5.3.8 in TS 36.331 [5] on the E-UTRAN PDSCH and the time the UE starts to send random access to the target UTRA FDD cell. The time delay ($T_{connection_release_redirect_UTRA\ FDD}$) shall be less than:

$$T_{connection_release_redirect_UTRA\ FDD} = T_{RRC_procedure_delay} + T_{identify\text{-}UTRA\ FDD} + T_{SI\text{-}UTRA\ FDD} + T_{RA}$$

The target UTRA FDD cell shall be considered detectable when:

- CPICH Ec/Io $\geq$ -15 dB,
- SCH_Ec/Io $\geq$ -15 dB for at least one channel tap and SCH_Ec/Ior is equally divided between primary synchronisation code and secondary synchronisation code.

$T_{RRC_procedure_delay}$: It is the RRC procedure for processing the received message "*RRCConnectionRelease*". It shall be less than 110 ms.

$T_{identify\text{-}UTRA\ FDD}$: It is the time to identify the target UTRA FDD cell. It shall be less than 500 ms.

$T_{SI\text{-}UTRA\ FDD}$: It is the time required for acquiring all the relevant system information of the target UTRA FDD cell. This time depends upon whether the UE is provided with the relevant system information of the target UTRA FDD cell or not by the E-UTRAN before the RRC connection is released.

$T_{RA}$: It is the delay caused due to the random access procedure when sending random access to the target UTRA FDD cell.

The normative reference for this requirement is TS 36.133 [4] clause 6.3.2.1 and A.6.3.1.

### 6.3.9.4 Test description

#### 6.3.9.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.22.
2. The general test parameter settings are set up according to Table 6.3.9.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 6.3.9.4.3.
5. There is one E-UTRA FDD cell and one UTRA FDD cell specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.3.9.4.1-1: General test parameters for RRC Connection Release with Redirection from E-UTRAN FDD to UTRAN FDD under AWGN propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on UTRA RF channel number 1. |
| CP length | | Normal | Applicable to cell 1 |
| E-UTRA RF Channel Number | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| UTRA RF Channel Number | | 1 | One UTRA FDD carrier frequency is used. |
| Inter-RAT (UTRA FDD) measurement quantity | | CPICH Ec/Io | |
| Filter coefficient | | 0 | L3 filtering is not used. |
| DRX | | OFF | |
| UTRA FDD cell list size | | None | No explicit neighbour list is provided to the UE |
| T1 | s | ≤5 | |
| T2 | s | 2 | |

### 6.3.9.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. The test consists of two successive time periods, with time duration of T1, and T2 respectively. The "*RRCConnectionRelease*" message not containing any system information of Cell 2 shall be sent to the UE during period T1 and the start of T2 is the instant when the last TTI containing the RRC message is sent to the UE. Prior to time duration T2, the UE shall not have any timing information of Cell 2. Cell 2 is powered up at the beginning of the T2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 6.3.9.5-1 and 6.3.9.5-2. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. SS shall transmit an *RRCConnectionRelease* not containing the relevant system information of Cell 2 during period T1.
4. The SS shall start T2 timer when the last TTI containing the *RRCConnectionRelease* message is sent to UE.
5. When T2 starts, the SS shall switch the power setting from T1 to T2 as specified in Table 6.3.9.5-1 for Cell 1, and Table 6.3.9.5-2 for Cell 2.
6. If the UE transmits the PRACH to Cell 2 less than 1930 ms from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
7. After T2 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
8. The SS shall set Cell 2 primary scrambling code = ((current cell 2 primary scrambling code - 50) mod 200 + 100) for next iteration of the test procedure loop.
9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

### 6.3.9.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.3.9.4.3-1: *RRCConnectionRelease*: Additional Redirection from E-UTRAN FDD to UTRAN FDD test requirement without system information (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| rrcConnectionRelease-r8 SEQUENCE { | | | |
| releaseCause | other | | |
| redirectedCarrierInfo CHOICE { | | | |
| utra-FDD | Cell 2 Carrier Information | | |
| } | | | |
| idleModeMobilityControlInfo | Not present | | |
| nonCriticalExtension SEQUENCE { | | | |
| lateNonCriticalExtension | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 6.3.9.5 Test requirement

Tables 6.3.9.4.1-1, 6.3.9.5-1 and 6.3.9.5-2 define the primary level settings including test tolerances for Redirection from E-UTRAN FDD to UTRAN FDD without system information test.

**Table 6.3.9.5-1: Cell Specific Test requirement Parameters for cell #1 E-UTRAN FDD to UTRAN FDD RRC release with redirection under AWGN propagation conditions**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 6.3.9.5-2: Cell Specific Test Parameters for cell #2 E-UTRAN FDD to UTRAN FDD RRC release with redirection under AWGN propagation conditions**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| UTRA RF Channel Number | | 1 | |
| CPICH_Ec/Ior | dB | -10 | |
| PCCPCH_Ec/Ior | dB | -12 | |
| SCH_Ec/Ior | dB | -12 | |
| PICH_Ec/Ior | dB | -15 | |
| DPCH_Ec/Ior | dB | N/A | |
| OCNS | | -0.941 | |
| $\hat{I}_{or} / I_{oc}$ | dB | -∞ | 0.42 |
| $I_{oc}$ | dBm/3.84 MHz | -70 | |
| CPICH_Ec/Io Note 3 | dB | -∞ | 12.81 |
| Propagation Condition | | AWGN | |

Note 1: The DPCH level is controlled by the power control loop.
Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$
Note 3: This gives an SCH Ec/Io of -15dB

The UE shall start to transmit random access to Cell 2 less than 1930 ms from the beginning of time period T2.

The rate of correct RRC connection release redirection to UTRAN FDD observed during repeated tests shall be at least 90%.

The overall delays measured test requirement can be expressed as

Test Requirement = $T_{RRC_procedure_delay}$ + $T_{identify\text{-}UTRA\ FDD}$ + $T_{SI\text{-}UTRA\ FDD}$ + $T_{RA}$

$T_{RRC_procedure_delay}$ = 110 ms

$T_{identify\text{-}UTRA\ FDD}$ = 500 ms

$T_{SI\text{-}UTRA\ FDD}$ =the time required for acquiring all the relevant system information of the target UTRA FDD cell. This time depends upon whether the UE is provided with the relevant system information of the target UTRA FDD cell or not by the E-UTRAN before the RRC connection is released. 1280 ms is assumed in this test case.

$T_{RA}$ = The additional delay caused by the random access procedure. 40 ms is assumed in this test case.

This gives a total of 1930 ms for overall delays measured.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 6.3.10 Redirection from E-UTRAN FDD to GERAN when System Information is not provided

### 6.3.10.1 Test Purpose

To verify that the UE is able to performs the RRC connection release with redirection from the E-UTRA FDD to the target GERAN cell when system information is not provided to meet the RRC connection release with redirection to GERAN requirements.

### 6.3.10.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward that support GERAN.

### 6.3.10.3 Minimum conformance requirements

The UE shall be capable of performing the RRC connection release with redirection to the target GERAN cell within $T_{connection_release_redirect_GERAN}$.

The time delay ($T_{connection_release_redirect_GERAN}$) is the time between the end of the last TTI containing the RRC command, "*RRCConnectionRelease*" as defined in section 5.3.8 in TS36.331 [5] on the E-UTRAN PDSCH and the time the UE starts to send random access to the target GERAN cell. The time delay ($T_{connection_release_redirect_GERAN}$) shall be less than:

$$T_{connection_release_redirect_GERAN} = T_{RRC_procedure_delay} + T_{identify\text{-}GERAN} + T_{SI\text{-}GERAN} + T_{RA}$$

The target GERAN cell shall be considered detectable when the UE receives the GERAN cell at levels down to 10 dB + the reference sensitivity level or reference interference levels as specified in TS 45.005 [16].

$T_{RRC_procedure_delay}$: It is the RRC procedure for processing the received message "*RRCConnectionRelease*". It shall be less than 110 ms.

$T_{identify\text{-}UTRA\ GERAN}$: It is the time to identify the BSIC of the target GERAN cell. It shall be less than 1 second.

$T_{SI\text{-}UTRA\ GERAN}$: It is the time required for acquiring all the relevant system information of the target GERAN cell. This time depends upon whether the UE is provided with the relevant system information of the target GERAN cell or not by the E-UTRAN before the RRC connection is released.

$T_{RA}$: It is the delay caused due to the random access procedure when sending random access burst to the target GERAN cell.

The normative reference for this requirement is TS 36.133 [4] clause 6.3.2.2 and A.6.3.10.

## 6.3.10.4 Test description

### 6.3.10.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.22.
2. The parameter settings for the cells are set up according to Table 6.3.10.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 6.3.10.4.3.
5. There is one E-UTRA FDD cell and one GERAN cell specified in the test. Cell 1(E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.3.10.4.1-1: General test parameters for RRC connection release with redirection from E-UTRAN FDD to GERAN in AWGN**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on Absolute RF Channel Number 1 (GSM cell) |
| CP length | | Normal | Applicable to cell 1 |
| E-UTRA RF Channel Number | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| DRX | | OFF | |
| Monitored GSM cell list size | | 6 GSM neighbour including ARFCN 1 | Only the list of GERAN carrier frequencies is provided in the "*RRCConnectionRelease*" message. |
| T1 | s | ≤5 | |
| T2 | s | 4 | |

### 6.3.10.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. The test consists of two successive time periods, with time duration of T1, and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2. The start of T2 is the instant when the last TTI containing the RRC message, "*RRCConnectionRelease*", is received by the UE from cell 1. The "*RRCConnectionRelease*" message shall not contain the relevant system information of cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 6.3.10.5-1 and 6.3.10.5-2. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. SS shall transmit an *RRCConnectionRelease* without the relevant system information of Cell 2 during period T1.
4. The SS shall start T2 timer when the last TTI containing the *RRCConnectionRelease* message is sent to UE.
5. When T2 starts, the SS shall switch the power setting from T1 to T2 as specified in Table 6.3.10.5-1 for Cell 1, and Table 6.3.10.5-2 for Cell 2.

6. If the UE transmits the random access burst on RACH of the target GERAN Cell 2 less than 3020 ms from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

7. After T2 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

8. The SS shall set a different BSIC on Cell 2, as the previous timing information of Cell 2 is invalid in the UE for the next iteration of the test procedure loop.

9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

#### 6.3.10.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.3.10.4.3-1: *RRCConnectionRelease*: Additional Redirection from E-UTRAN FDD to GERAN test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| rrcConnectionRelease-r8 SEQUENCE { | | | |
| releaseCause | other | | |
| redirectedCarrierInfo CHOICE { | | | |
| geran | Cell 2 Carrier Information | | |
| } | | | |
| idleModeMobilityControlInfo | Not present | | |
| nonCriticalExtension SEQUENCE { | | | |
| lateNonCriticalExtension | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 6.3.10.5 Test requirement

Tables 6.3.10.4.1-1, 6.3.10.5-1 and 6.3.10.5-2 define the primary level settings including test tolerances for Redirection from E-UTRAN FDD to GERAN test.

**Table 6.3.10.5-1: Cell specific test parameters for E-UTRA FDD cell (cell #1) for RRC connection release with redirection from E-UTRAN FDD to GERAN in AWGN**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| $N_{oc}$ | dBm/15 kHz | -98 | |
| RSRP | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 6.3.10.5-2: Cell specific test parameters for GERAN cell (cell #2) for RRC connection release with redirection from E-UTRAN FDD to GERAN in AWGN**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| Absolute RF Channel Number | | ARFNC 1 | |
| RXLEV | dBm | -Infinity | -75 |
| GSM BSIC | | N/A | Valid |

The UE shall begin to send access bursts on RACH of the target GERAN cell (cell #2) less than 3020 ms from the beginning of time period T2.

The overall delays measured test requirement can be expressed as:

$T_{connection_release_redirect_GERAN} = T_{RRC_procedure_delay} + T_{identify\text{-}GERAN} + T_{SI\text{-}GERAN} + T_{RA}$

$T_{RRC_procedure_delay}$ = 110 ms, which is the time for processing the received message "*RRCConnectionRelease*.

$T_{identify\text{-}GERAN}$ = 1000 ms, which is the time for identifying the target GERAN cell.

$T_{SI\text{-}GERAN}$ = 1900 ms; which is the maximum time allowed to read BCCH data from the target GERAN cell.

$T_{RA}$ = 10 ms, which is about 2 GSM frames (2*4.65 ms) to account for the GSM timing uncertainty.

This gives a total of 3020 ms for overall delays measured.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 6.3.11 Redirection from E-UTRAN TDD to GERAN when System Information is not provided

### 6.3.11.1 Test Purpose

To verify that the UE is able to performs the RRC connection release with redirection from the E-UTRA TDD to the target GERAN cell when system information is not provided to meet the RRC connection release with redirection to GERAN requirements.

### 6.3.11.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward that support GERAN.

### 6.3.11.3 Minimum conformance requirements

The UE shall be capable of performing the RRC connection release with redirection to the target GERAN cell within $T_{connection_release_redirect_GERAN}$.

The time delay ($T_{connection_release_redirect_GERAN}$) is the time between the end of the last TTI containing the RRC command, "*RRCConnectionRelease*" as defined in section 5.3.8 in TS36.331 [5] on the E-UTRAN PDSCH and the time the UE starts to send random access to the target GERAN cell. The time delay ($T_{connection_release_redirect_GERAN}$) shall be less than:

$$T_{connection_release_redirect_GERAN} = T_{RRC_procedure_delay} + T_{identify\text{-}GERAN} + T_{SI\text{-}GERAN} + T_{RA}$$

The target GERAN cell shall be considered detectable when the UE receives the GERAN cell at levels down to 10 dB + the reference sensitivity level or reference interference levels as specified in TS 45.005 [16].

$T_{RRC_procedure_delay}$: It is the RRC procedure for processing the received message "*RRCConnectionRelease*". It shall be less than 110 ms.

$T_{identify\text{-}UTRA\ GERAN}$: It is the time to identify the BSIC of the target GERAN cell. It shall be less than 1 second.

$T_{SI\text{-}UTRA\ GERAN}$: It is the time required for acquiring all the relevant system information of the target GERAN cell. This time depends upon whether the UE is provided with the relevant system information of the target GERAN cell or not by the E-UTRAN before the RRC connection is released.

$T_{RA}$: It is the delay caused due to the random access procedure when sending random access burst to the target GERAN cell.

The normative reference for this requirement is TS 36.133 [4] clause 6.3.2.2 and A.6.3.11.

### 6.3.11.4 Test description

#### 6.3.11.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.14.
2. The parameter settings for the cells are set up according to Table 6.3.11.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 6.3.11.4.3.

5. There is one E-UTRA TDD cell and one GERAN cell specified in the test. Cell 1(E-UTRA TDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.3.11.4.1-1: General test parameters for RRC connection release with redirection from E-UTRAN TDD to GERAN in AWGN**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2. |
| Active | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on Absolute RF Channel Number 1 (GSM cell) |
| CP length | | Normal | Applicable to cell 1 |
| Special subframe configuration | | 6 | As specified in table 4.2.1 in TS 36.211. The same configuration applies to all cells. |
| Uplink-downlink configuration | | 1 | |
| E-UTRA RF Channel Number | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| DRX | | OFF | |
| Monitored GSM cell list size | | 6 GSM neighbour including ARFCN 1 | Only the list of GERAN carrier frequencies is provided in the "*RRCConnectionRelease*" message. |
| T1 | s | ≤5 | |
| T2 | s | 4 | |

### 6.3.11.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. The test consists of two successive time periods, with time duration of T1, and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2. The start of T2 is the instant when the last TTI containing the RRC message, "*RRCConnectionRelease*", is received by the UE from cell 1. The "*RRCConnectionRelease*" message shall not contain the relevant system information of cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.
2. Set the parameters according to T1 in Table 6.3.11.5-1 and 6.3.11.5-2. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. The SS shall transmit an *RRCConnectionRelease* not containing the relevant system information of Cell 2 during time period T1.
4. The SS shall start T2 timer when the last TTI containing the *RRCConnectionRelease* message is sent to the UE.
5. When T2 starts, the SS shall switch the power setting from T1 to T2 as specified in Table 6.3.11.5-1 for Cell 1 and Table 6.3.11.5-2 for Cell 2.
6. If the UE transmits the random access burst on RACH of the target GERAN Cell 2 less than 3020 ms from the beginning of time period T2 then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
7. After T2 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Cell 1 is the active cell.
8. The SS shall set a different BSIC on Cell 2, as the previous timing information of Cell 2 is invalid in the UE for the next iteration of the test procedure loop.
9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

### 6.3.11.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.3.11.4.3-1: *RRCConnectionRelease*: Additional Redirection from E-UTRAN FDD to GERAN test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| rrcConnectionRelease-r8 SEQUENCE { | | | |
| releaseCause | other | | |
| redirectedCarrierInfo CHOICE { | | | |
| geran | Cell 2 Carrier Information | | |
| } | | | |
| idleModeMobilityControlInfo | Not present | | |
| nonCriticalExtension SEQUENCE { | | | |
| lateNonCriticalExtension | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 6.3.11.5 Test requirement

Tables 6.3.11.4.1-1, 6.3.11.5-1 and 6.3.11.5-2 define the primary level settings including test tolerances for Redirection from E-UTRAN TDD to GERAN test.

**Table 6.3.11.5-1: Cell specific test parameters for E-UTRA TDD cell (cell #1) for RRC connection release with redirection from E-UTRAN TDD to GERAN in AWGN**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| $N_{oc}$ | dBm/15 kHz | -98 | |
| RSRP | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 6.3.11.5-2: Cell specific test parameters for GERAN cell (cell #2) for RRC connection release with redirection from E-UTRAN TDD to GERAN in AWGN**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| Absolute RF Channel Number | | ARFNC 1 | |
| RXLEV | dBm | -Infinity | -75 |
| **GSM BSIC** | | N/A | Valid |

The UE shall begin to send access bursts on RACH of the target GERAN cell (cell #2) less than 3020 ms from the beginning of time period T2.

The overall delays measured test requirement can be expressed as:

$T_{connection_release_redirect_GERAN} = T_{RRC_procedure_delay} + T_{identify\text{-}GERAN} + T_{SI\text{-}GERAN} + T_{RA}$

$T_{RRC_procedure_delay}$ = 110 ms, which is the time for processing the received message "*RRCConnectionRelease*.

$T_{identify\text{-}GERAN}$ = 1000 ms, which is the time for identifying the target GERAN cell.

$T_{SI\text{-}GERAN}$ = 1900 ms; which is the maximum time allowed to read BCCH data from the target GERAN cell.

$T_{RA}$ = 10 ms, which is about 2 GSM frames (2*4.65 ms) to account for the GSM timing uncertainty.

This gives a total of 3020 ms for overall delays measured.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 6.3.12 E-UTRAN TDD RRC connection release redirection to UTRAN FDD without SI provided

### 6.3.12.1 Test purpose

The purpose of this test is to verify that the UE performs the RRC connection release with redirection from the E-UTRAN TDD to the target UTRAN FDD cell within $T_{connection_release_redirect_UTRAN\ FDD}$. This test will partly verify the RRC connection release with redirection to UTRAN FDD requirements.

### 6.3.12.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward that support UTRA FDD.

### 6.3.12.3 Minimum conformance requirements

The UE shall be capable of performing the RRC connection release with redirection to the target UTRAN FDD cell within $T_{connection_release_redirect_UTRA\ FDD}$.

The time delay ($T_{connection_release_redirect_UTRA\ FDD}$) is the time between the end of the last TTI containing the RRC command, "*RRCConnectionRelease*" as defined in section 5.3.8 in TS 36.331 [5] on the E-UTRAN PDSCH and the time the UE starts to send random access to the target UTRA FDD cell. The time delay ($T_{connection_release_redirect_UTRA\ FDD}$) shall be less than:

$$T_{connection_release_redirect_UTRA\ FDD} = T_{RRC_procedure_delay} + T_{identify\text{-}UTRA\ FDD} + T_{SI\text{-}UTRA\ FDD} + T_{RA}$$

The target UTRA FDD cell shall be considered detectable when:

- CPICH Ec/Io $\geq$ -15 dB,
- SCH_Ec/Io $\geq$ -15 dB for at least one channel tap and SCH_Ec/Ior is equally divided between primary synchronisation code and secondary synchronisation code.

$T_{RRC_procedure_delay}$: It is the RRC procedure for processing the received message "*RRCConnectionRelease*". It shall be less than 110 ms.

$T_{identify\text{-}UTRA\ FDD}$: It is the time to identify the target UTRA FDD cell. It shall be less than 500 ms.

$T_{SI\text{-}UTRA\ FDD}$: It is the time required for acquiring all the relevant system information of the target UTRA FDD cell. This time depends upon whether the UE is provided with the relevant system information of the target UTRA FDD cell or not by the E-UTRAN before the RRC connection is released.

$T_{RA}$: It is the delay caused due to the random access procedure when sending random access to the target UTRA FDD cell.

The normative reference for this requirement is TS 36.133 [4] clause 6.3.2.1 and A.6.3.12.

## 6.3.12.4 Test description

### 6.3.12.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.22.
2. The general test parameter settings are set up according to Table 6.3.12.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are as defined in clause 6.3.12.4.3.
5. There is one E-UTRA TDD cell and one UTRA FDD cell specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 6.3.12.4.1-1: General test parameters for E-UTRAN TDD RRC connection release redirection to UTRAN FDD without SI provided**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRAN RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on UTRAN RF channel number 1. |
| E-UTRAN RF Channel Number | | 1 | One E-UTRAN TDD carrier frequency is used. |
| E-UTRAN Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Uplink-downlink configuration of cell 1 | | 1 | As specified in table 4.2.2 in TS 36.211 |
| Special subframe configuration of cell 1 | | 6 | As specified in table 4.2.1 in TS 36.211 |
| CP length | | Normal | Applicable to cell 1 |
| UTRAN RF Channel Number | | 1 | One UTRAN TDD carrier frequency is used. |
| Filter coefficient | | 0 | L3 filtering is not used. |
| DRX | | OFF | |
| Monitored UTRAN FDD cell list size | | None | No explicit neighbour list is provided to the UE |
| T1 | s | ≤5 | |
| T2 | s | 2 | |

### 6.3.12.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. The test consists of two successive time periods, with time duration of T1, and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2. Cell 2 is powered up at the beginning of the T2. The "*RRCConnectionRelease*" message not containing any system information of Cell 2 shall be sent to the UE during period T1. The start of T2 is the instant when the last TTI containing the RRC message, "*RRCConnectionRelease*", is received by the UE from Cell 1.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

2. Set the parameters according to T1 in Table 6.3.12.5-1 and 6.3.12.5-2. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.

3. The SS shall transmit an *RRCConnectionRelease* not containing the relevant system information of Cell 2 during period T1.

4. The SS shall start T2 timer when the last TTI containing the *RRCConnectionRelease* message is sent to the UE.

5. When T2 starts, the SS shall switch the power setting from T1 to T2 as specified in Table 6.3.12.5-1 for Cell 1, and Table 6.3.12.5-2 for Cell 2.

6. If the UE transmits the PRACH to Cell 2 less than 1930 ms from the beginning of time period T2 then the number of successful tests are increased by one. Otherwise, the number of failure tests is increased by one.

7. After T2 expires, the UE shall be switched off. Then ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

8. The SS shall set Cell 2 primary scrambling code = ((current cell 2 primary scrambling code - 50) mod 200 + 100) for next iteration of the test procedure loop.

9. Repeat step 2-8 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved.

#### 6.3.12.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 6.3.12.4.3-1: *RRCConnectionRelease*: Additional E-UTRAN TDD RRC connection release redirection to UTRAN FDD without SI provided test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| rrcConnectionRelease-r8 SEQUENCE { | | | |
| releaseCause | other | | |
| redirectedCarrierInfo CHOICE { | | | |
| utra-FDD | Cell 2 Carrier Information | | |
| } | | | |
| idleModeMobilityControlInfo | Not present | | |
| nonCriticalExtension SEQUENCE { | | | |
| lateNonCriticalExtension | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 6.3.12.5 Test requirement

Tables 6.3.12.4.1-1, 6.3.12.5-1 and 6.3.12.5-2 define the primary level settings including test tolerances for Redirection from E-UTRAN TDD to UTRAN FDD test.

**Table 6.3.12.5-1: Cell Specific Test requirement Parameters for cell #1 E-UTRAN TDD RRC connection release redirection to UTRAN FDD without SI provided under AWGN propagation conditions**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 6.3.12.5-2: Cell Specific Test Parameters for cell #2 E-UTRAN TDD RRC connection release redirection to UTRAN FDD without SI provided under AWGN propagation conditions**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| UTRA RF Channel Number | | 1 | |
| CPICH_Ec/Ior | dB | -10 | |
| PCCPCH_Ec/Ior | dB | -12+TT | |
| SCH_Ec/Ior | dB | -12 | |
| PICH_Ec/Ior | dB | -15 | |
| DPCH_Ec/Ior | dB | N/A | |
| OCNS | | -0.941 | |
| $\hat{I}_{or} / I_{oc}$ | dB | -∞ | 0.42 |
| $I_{oc}$ | dBm/3.84 MHz | -70 | |
| CPICH_Ec/Io Note 3 | dB | -∞ | -12.81 |
| Propagation Condition | | AWGN | |

Note 1: The DPCH level is controlled by the power control loop.
Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.
Note 3: This gives an SCH Ec/Io of -15dB

The UE shall start to send random access to the target UTRAN FDD cell (Cell 2) less than 1930 ms from the beginning of time period T2.

The time delay can be expressed as: $T_{RRC_procedure_delay} + T_{identify\text{-}UTRAN\ FDD} + T_{SI\text{-}UTRAN\ FDD} + T_{RA}$, where:

$T_{RRC_procedure_delay}$ = 110 ms, which is specified in TS36.133 [4] section 6.3.2.1.

$T_{identify\text{-}UTRAN\ FDD}$ = 500 ms; which is defined in TS36.133 [4] section 6.3.2.1.

$T_{SI\text{-}UTRAN\ FDD}$: Maximum repetition period of relevant system info blocks that need to be received by the UE during RRC connection release redirection to UTRAN FDD cell. 1280 ms is assumed in this test case.

$T_{RA}$ = 40ms, this is the additional delay caused by the random access procedure.

This gives a total delay of 1930 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 7 Timing and Signalling Characteristics

The timing requirements are applicable for the uplink physical channels and signals specified in TS 36.211 [9] clause 5 (for uplink physical channels) as defined.

SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Annex A1. The SS sends downlink MAC padding bits on the DL RMC.

Uplink is configured according to Annex A.3.

## 7.1 UE Transmit Timing

### 7.1.1 E-UTRAN FDD - UE Transmit Timing Accuracy

#### 7.1.1.1 Test purpose

To verify the UE have the capability to follow the frame timing change of the connected System Simulator. The method used is that the UE initial transmit timing accuracy, the maximum amount of timing change in one adjustment, and the minimum and maximum adjustment rate are within the specified limits based on the requirements.

#### 7.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward. Applicability requires support for FGI bit 5.

#### 7.1.1.3 Minimum conformance requirements

The UE initial transmission timing error shall be less than or equal to $\pm T_e$ where the timing error limit value $T_e$ is defined in TS 36.133 [4] clause 7.1.2 and shown in table 7.1.1.3-1. This requirement applies when it is the first transmission in a DRX cycle for PUCCH, PUSCH and SRS or it is the PRACH transmission. The uplink frame transmission takes place $(N_{\mathrm{TA}} + N_{\mathrm{TA\ offset}}) \times T_s$ before the reception of the first detected path (in time) of the corresponding downlink frame from the reference cell. The reference point for the UE initial transmit timing control requirement shall be the downlink timing minus $(N_{\mathrm{TA_Ref}} + N_{\mathrm{TA\ offset}}) \times T_s$.

where:

$N_{TA}$ is $0 \leq N_{TA} \leq 20512$

$N_{TA_Ref}$ is 0 for PRACH; $(N_{\mathrm{TA_Ref}} + N_{\mathrm{TA\ offset}})$ (in $T_s$ units) for other channels is the difference between UE transmission timing and the downlink timing immediately after when the last timing advance in TS 36.133 [4] clause 7.3 was applied. $N_{\mathrm{TA_Ref}}(N_{\mathrm{TA_Ref}} + N_{\mathrm{TA\ offset}})$ (in $T_s$ units) for other channels is not changed until next timing advance is received.

$N_{TA\ offset}$ is 0 for frame structure type 1 as defined in TS 36.211 [9] clause 8.1.$T_S$ denotes the basic time unit. The size of various fields in the time domain is expressed as a number of time units $T_S$ = 1/(15000 x 2048) seconds.

**Table 7.1.1.3-1: $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| 1.4 | 24*$T_S$ |
| ≥3 | 12*$T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

When it is not the first transmission in a DRX cycle or there is no DRX cycle, and when it is the transmission for PUCCH, PUSCH and SRS transmission, the UE shall be capable of changing the transmission timing according to the received downlink frame except when the timing advance in TS 36.133 [4] clause 7.3 is applied. When the transmission

timing error between the UE and the reference timing exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$. The reference timing shall be $(N_{TA_Ref} + N_{TA\,offset}) \times T_s$ before the downlink timing.

All adjustments made to the UE uplink timing shall follow these rules:

1) The maximum amount of the magnitude of the timing change in one adjustment shall be $T_q$.

2) The minimum aggregate adjustment rate shall be 7 * × $T_S$ per second.

3) The maximum aggregate adjustment rate shall be $T_q$ per 200 ms.

Where the maximum autonomous time adjustment step $T_q$ is defined in TS 36.133 [4] clause 7.1.2 and shown in table 7.1.1.3-2.

**Table 7.1.1.3-2: $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| 1.4 | $17.5*T_S$ |
| 3 | $9.5*T_S$ |
| 5 | $5.5*T_S$ |
| ≥10 | $3.5*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

The normative reference for this requirement is TS 36.133 [4] clause 7.1 and A.7.1.1.

NOTE 1: Both the allowed UE frequency error and the UE time adjustment quantisation can affect the result of the test, and it is not possible to test the UE Transmit Timing Accuracy in isolation from these core requirements. They are therefore taken into account when setting the Test Tolerances and the Test Requirements

## 7.1.1.4 Test description

### 7.1.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz and 1.4 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18.

2. Propagation conditions are set according to Annex B clause B.0.

3. There is one E-UTRA FDD carrier and one cell specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

### 7.1.1.4.2 Test procedure

The test consists of a single cell. The transmit timing accuracy is verified related to the downlink frame timing of Cell 1. The downlink timing of Cell 1 is changed and the changes in UE transmit timing are observed. The transmit timing is verified by the UE transmitting SRS used as a measurement reference facilitating the SS timing estimation.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Test 1, Test 2, or Test 3 in Tables 7.1.1.5-1, 7.1.1.5-2 and 7.1.1.5-3 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. The SS shall check that the UE transmit timing offset is within the limits specified in Table 7.1.1.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1.

6. The SS adjusts the downlink timing for Cell 1 to a delay of $+64 \times T_S$ (approximately +2 μs) for ≥3 MHz downlink bandwidth (Test 1, Test 2) and a delay of $+128 \times T_S$ (approximately +4 μs) for 1.4MHz downlink bandwidth (Test 3) compared to that in step 5. For Test 2, the downlink timing adjustment shall be done at (40 ± 15 )ms before the next DRX On duration period.

7. Step 7 applies for Test 1 and Test 3, but is omitted for Test 2. The test system samples the UE Transmit Timing once per SRS transmission (periodicity 2 subframes). To check Rule 1, the SS shall check that the maximum time adjustment step size $T_q$ between one SRS transmission (subframe n) to next (subframe n+2) is within Rule 1 as specified in clause 7.1.1.5. To check that the minimum adjustment rate is within Rule 2 as specified in clause 7.1.1.5, the SS shall measure the change in SRS transmission timing over a 1.002s sliding window, with step size 2ms. To check that the maximum adjustment rate is within Rule 3 as specified in clause 7.1.1.5, the SS shall measure the change in SRS transmission timing over a 198ms sliding window, with step size 2ms. The three rules apply until the UE transmit timing offset is within the limits specified in Table 7.1.1.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1.

8. In case for Test 1 and Test 3, the SS shall check that the UE transmit timing offset stays within the limits specified in Table 7.1.1.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1. In case for Test 2, the SS shall check that the UE transmit timing offset stays within the limits specified in Table 7.1.1.5-4 For that purpose, the SRS transmission that takes place in the first DRX ON duration after downlink timing of Cell 1 is changed is used..

9. The SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

10. After the RRC connection release, the SS switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

11. Repeat step 1-10 for each sub-test in Tables 7.1.1.5-1, 7.1.1.5-2 and 7.1.1.5-3 as appropriate.

### 7.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.1.1.4.3-1: Common Exception messages for UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.4-2 |

**Table 7.1.1.4.3-2: *RRCConnectionReconfiguration*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.331 [5] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| radioResourceConfigDedicated SEQUENCE { | | | |
| mac-MainConfig CHOICE { | | | |
| explicitValue | MAC-MainConfig-RBC | | |
| } | | | |
| physicalConfigDedicated | PhysicalConfigDedicated - DEFAULT | | RBC |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.1.1.4.3-3: *SoundingRS-RL-ConfigCommon-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | | Set according to specific test; bw5 for Test 1 and Test 2 and bw7 for Test 3 | |
| srs-SubframeConfig | | Set according to specific test; sc1 for Test 1 and Test 3 and sc3 for Test 2 | FDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | Not present | | FDD |
| } | | | |

**Table 7.1.1.4.3-4: *SoudingRS-UL-ConfigDedicated-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | bw0 used with no frequency hopping. bw3 used with frequency hopping | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | | Set according to specific test; 0 for Test 1 and 77 for Test 2 and 0 for Test 3 | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 7.1.1.4.3-5: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement for Test 2**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | DRX_S |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf80 typical value in simulations | |
| sf80 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.1.1.4.3-6: *CQI-ReportConfig-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm30 | This IE should be omitted for test 3 | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| release | NULL | | |
| } | | | |

**Table 7.1.1.4.3-7: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| } | | | |

## 7.1.1.5 Test requirement

Tables 7.1.1.5-1, 7.1.1.5-2 and 7.1.1.5-3 define the primary settings including test tolerances for UE transmit timing accuracy for E-UTRAN FDD test.

Tables 7.1.1.5-4, 7.1.1.5-5 and 7.1.1.5-6 define the rules for adjustments made to the UE uplink timing include Test Tolerances.

**Table 7.1.1.5-1: Cell Specific Test requirement Parameters for UE transmit timing accuracy for E-UTRAN FDD test case**

| Parameter | Unit | Value | | |
|---|---|---|---|---|
| | | Test 1 | Test 2 | Test 3 |
| E-UTRA RF Channel Number | | 1 | 1 | 1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | 10 | 1.4 |
| DRX cycle | ms | OFF | 80[Note5] | OFF |
| PDCCH/PCFICH/PHICH Reference measurement channel[Note1] | | R.6 FDD | R.6 FDD | R.8 FDD |
| OCNG Pattern[Note2] | | OP.2 FDD | OP.2 FDD | OP.4 FDD |
| PBCH_RA | dB | 0 | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| OCNG_RA[Note3] | | | | |
| OCNG_RB[Note3] | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | -98 | -98 |
| $\hat{E}_s / I_{ot}$ | dB | 3.30 | 3.30 | 3.30 |
| $\hat{E}_s / N_{oc}$ | dB | 3.30 | 3.30 | 3.30 |
| Io[Note4] | dBm/9 MHz | -65.25 | -65.25 | N/A |
| | dBm/1.08 MHz | N/A | N/A | -74.46 |
| Propagation condition | - | AWGN | AWGN | AWGN |

Note 1: For the reference measurement channels, see section A.2.1.
Note 2: For the OCNG pattern, see section D.1.2.
Note 3: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: Io level has been derived from other parameters for information purpose. It is not a settable parameter.
Note 5: DRX related parameters are defined in Table 7.1.1.5-3.

**Table 7.1.1.5-2: Sounding Reference Signal Configuration to be used in UE transmit timing accuracy for E-UTRAN FDD test case**

| Field | Test 1 | Test 2 | Test 3 | Comment |
|---|---|---|---|---|
| | Value | | | |
| srs-BandwidthConfig | bw5 | bw5 | bw7 | |
| srs-SubframeConfig | sc1 | sc3 | sc1 | |
| ackNackSRS-SimultaneousTransmission | FALSE | FALSE | FALSE | |
| srsMaxUpPts | N/A | N/A | N/A | Not applicable for FDD |
| srs-Bandwidth | 0 | 0 | 0 | No hopping |
| srs-HoppingBandwidth | hbw0 | hbw0 | hbw0 | |
| freqDomainPosition | 0 | 0 | 0 | |
| duration | TRUE | TRUE | TRUE | Indefinite duration |
| srs-ConfigIndex | 0 | 77 | 0 | SRS periodicity of 2ms and 80 ms for Test 1 and 2, respectively. |
| transmissionComb | 0 | 0 | 0 | |
| cyclicShift | cs0 | cs0 | cs0 | No cyclic shift |
| Note: For further information see section 6.3.2 in 3GPP TS 36.331 [5]. | | | | |

**Table 7.1.1.5-3: DRX Configuration to be used in UE transmit timing accuracy for E-UTRAN FDD test case**

| Field | Test2 | Comment |
|---|---|---|
| | Value | |
| onDurationTimer | psf1 | |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf80 | |
| shortDRX | disable | |
| Note: For further information see section 6.3.2 in 3GPP TS 36.331 [5]. | | |

The UE transmit timing offset shall be within the requirements in Table 7.1.1.5-4.

The reference point for the UE initial transmit timing control test requirement shall be the downlink timing minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$.

**Table 7.1.1.5-4: Test requirement for $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| 1.4 | $27*T_S$ |
| ≥3 | $15*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

The UE shall be capable of changing the transmission timing according to the received downlink frame. When the transmission timing error between the UE and the reference cell exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$.

All adjustments made to the UE uplink timing shall follow these rules:

1) For tests 1 and 3, the maximum adjustment step size $T_q$ shall be within the requirements in Table 7.1.1.5-5

2) For test 1, the minimum aggregate adjustment rate shall be $3.4 \times T_S$ per 1.002 s.

3) For tests 1 and 3, the maximum aggregate adjustment rate shall be within the requirements in Table 7.1.1.5-6.

**Table 7.1.1.5-5: Test requirement for $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| 1.4 | $18*T_S$ |
| ≥10 | $4*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

**Table 7.1.1.5-6: Test requirement for maximum aggregate adjustment rate per 198 ms**

| Downlink Bandwidth (MHz) | Maximum adjustment per 198 ms |
|---|---|
| 1.4 | $18.6*T_S$ |
| ≥10 | $4.6*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

An illustration of the measurement principle is shown in Figure 7.1.1.5-1.

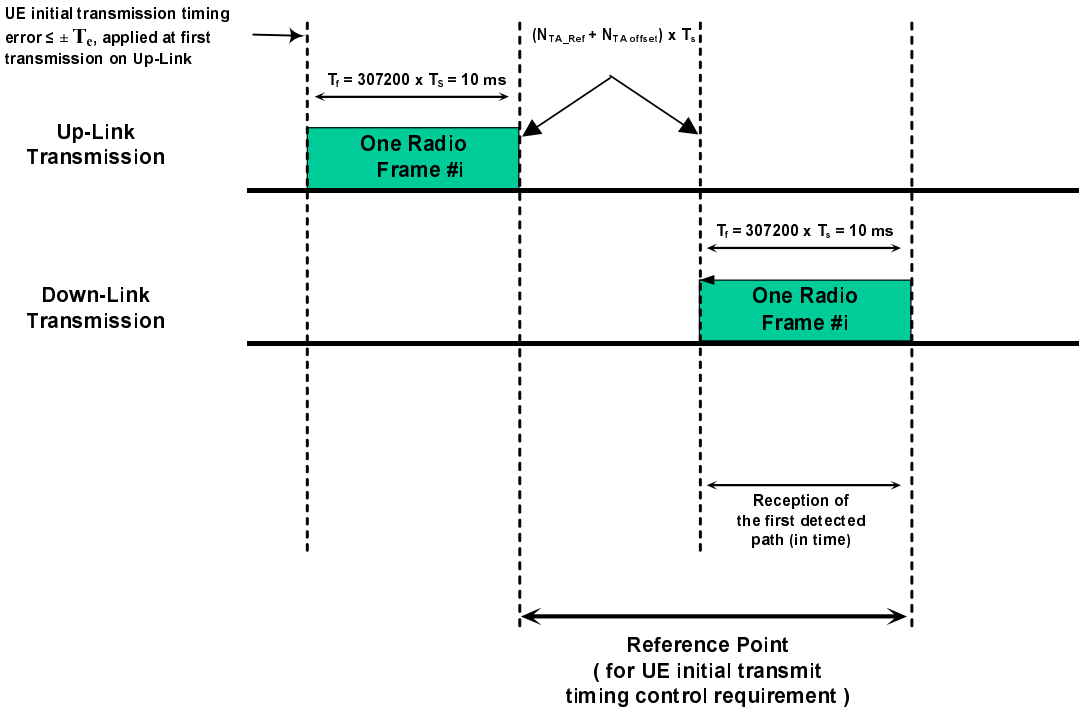


**Figure 7.1.1.5-1: Illustration of measurement principle**

# 7.1.1_1 E-UTRAN FDD - UE Transmit Timing Accuracy (Non DRx UE)

## 7.1.1_1.1 Test purpose

Same test purpose as in clause 7.1.1.1

## 7.1.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD release 8 UE's not supporting FGI bit 5.

## 7.1.1_1.3 Minimum conformance requirements

Same minimum conformance requirement as 7.1.1.3

## 7.1.1_1.4 Test description

### 7.1.1_1.4.1 Initial conditions

Same initial condition as in clause 7.1.1.4.1

### 7.1.1_1.4.2 Test procedure

The test consists of a single cell. The transmit timing accuracy is verified related to the downlink frame timing of Cell 1. The downlink timing of Cell 1 is changed and the changes in UE transmit timing are observed. The transmit timing is verified by the UE transmitting SRS used as a measurement reference facilitating the SS timing estimation.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Test 1 or Test 3 in Tables 7.1.1_1.5-1 and 7.1.1_1.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. The SS shall check that the UE transmit timing offset is within the limits specified in Table 7.1.1_1.5-3 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1.
6. The SS adjusts the downlink timing for Cell 1 to a delay of $+64 \times T_S$ (approximately +2 μs) for ≥3 MHz downlink bandwidth (Test 1) and a delay of $+128 \times T_S$ (approximately +4 μs) for 1.4MHz downlink bandwidth (Test 3) compared to that in step 5.
7. The test system samples the UE Transmit Timing once per SRS transmission (periodicity 2 subframes). To check Rule 1, the SS shall check that the maximum time adjustment step size $T_q$ between one SRS transmission (subframe n) to next (subframe n+2) is within Rule 1 as specified in clause 7.1.1_1.5. To check that the minimum adjustment rate is within Rule 2 as specified in clause 7.1.1_1.5, the SS shall measure the change in SRS transmission timing over a 1.002s sliding window, with step size 2ms. To check that the maximum adjustment rate is within Rule 3 as specified in clause 7.1.1_1.5, the SS shall measure the change in SRS transmission timing over a 198ms sliding window, with step size 2ms. The three rules apply until the UE transmit timing offset is within the limits specified in Table 7.1.1_1.5-3 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1.
8. The SS shall check that the UE transmit timing offset stays within the limits specified in Table 7.1.1_1.5-3 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1.
9. The SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
10. After the RRC connection release, the SS switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
11. Repeat step 1-10 for each sub-test in Tables 7.1.1_1.5-1 and 7.1.1_1.5-2 as appropriate.

### 7.1.1_1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with exceptions listed in Tables 7.1.1.4.3-1, 7.1.1.4.3-2, 7.1.1.4.3-6, and 7.1.1.4.3-7 of section 7.1.1.4.3 and the following:

**Table 7.1.1_1.4.3-1: *SoundingRS-RL-ConfigCommon-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | | Set according to specific test; bw5 for Test 1 and bw7 for Test 3 | |
| srs-SubframeConfig | | Set according to specific test; sc1 for Test 1 and Test 3 | FDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | Not present | | FDD |
| } | | | |

**Table 7.1.1_1.4.3-2: *SoudingRS-UL-ConfigDedicated-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | bw0 used with no frequency hopping. bw3 used with frequency hopping | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| Duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | | Set according to specific test; 0 for Test 1 and Test 3 | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

## 7.1.1_1.5 Test requirement

Tables 7.1.1_1.5-1 and 7.1.1_1.5-2 define the primary settings including test tolerances for UE transmit timing accuracy for E-UTRAN FDD test.

Tables 7.1.1_1.5-3, 7.1.1_1.5-4 and 7.1.1_1.5-5 define the rules for adjustments made to the UE uplink timing including Test Tolerances.

**Table 7.1.1_1.5-1: Cell Specific Test requirement Parameters for UE transmit timing accuracy for E-UTRAN FDD test case**

| Parameter | Unit | Value | | |
|---|---|---|---|---|
| | | Test 1 | Test 2 | Test 3 |
| E-UTRA RF Channel Number | | 1 | N/A | 1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | N/A | 1.4 |
| DRX cycle | ms | OFF | N/A | OFF |
| PDCCH/PCFICH/PHICH Reference measurement channel[Note1] | | R.6 FDD | N/A | R.8 FDD |
| OCNG Pattern[Note2] | | OP.2 FDD | N/A | OP.4 FDD |
| PBCH_RA | dB | 0 | N/A | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| OCNG_RA[Note3] | | | | |
| OCNG_RB[Note3] | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | N/A | -98 |
| $\hat{E}_s / I_{ot}$ | dB | 3.30 | N/A | 3.30 |
| $\hat{E}_s / N_{oc}$ | dB | 3.30 | N/A | 3.30 |
| Io[Note4] | dBm/9 MHz | -65.25 | N/A | N/A |
| | dBm/1.08 MHz | N/A | N/A | -74.46 |
| Propagation condition | - | AWGN | N/A | AWGN |

Note 1: For the reference measurement channels, see section A.2.1.
Note 2: For the OCNG pattern, see section D.1.2.
Note 3: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: Io level has been derived from other parameters for information purpose. It is not a settable parameter.

**Table 7.1.1_1.5-2: Sounding Reference Signal Configuration to be used in UE transmit timing accuracy for E-UTRAN FDD test case**

| Field | Test 1 | Test 2 | Test 3 | Comment |
|---|---|---|---|---|
| | Value | | | |
| srs-BandwidthConfig | bw5 | N/A | bw7 | |
| srs-SubframeConfig | sc1 | N/A | sc1 | |
| ackNackSRS-SimultaneousTransmission | FALSE | N/A | FALSE | |
| srsMaxUpPts | N/A | N/A | N/A | Not applicable for FDD |
| srs-Bandwidth | 0 | N/A | 0 | No hopping |
| srs-HoppingBandwidth | hbw0 | N/A | hbw0 | |
| freqDomainPosition | 0 | N/A | 0 | |
| Duration | TRUE | N/A | TRUE | Indefinite duration |
| srs-ConfigIndex | 0 | N/A | 0 | SRS periodicity of 2ms for Test 1 and Test 3, respectively. |
| transmissionComb | 0 | N/A | 0 | |
| cyclicShift | cs0 | N/A | cs0 | No cyclic shift |
| Note: For further information see section 6.3.2 in 3GPP TS 36.331 [5]. | | | | |

The UE transmit timing offset shall be within the requirements in Table 7.1.1_1.5-3.

The reference point for the UE initial transmit timing control test requirement shall be the downlink timing minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$.

**Table 7.1.1_1.5-3: Test requirement for $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| 1.4 | $27*T_S$ |
| ≥3 | $15*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

The UE shall be capable of changing the transmission timing according to the received downlink frame. When the transmission timing error between the UE and the reference cell exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$.

All adjustments made to the UE uplink timing shall follow these rules:

1) The maximum adjustment step size $T_q$ shall be within the requirements in Table 7.1.1_1.5-4.

2) For test 1, the minimum aggregate adjustment rate shall be $3.4 \times T_S$ per second.

3) The maximum aggregate adjustment rate shall be within the requirements in Table 7.1.1_1.5-5.

**Table 7.1.1_1.5-4: Test requirement for $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| 1.4 | $18*T_S$ |
| ≥10 | $4*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

**Table 7.1.1_1.5-5: Test requirement for maximum aggregate adjustment rate per 200 ms**

| Downlink Bandwidth (MHz) | Maximum adjustment per 200 ms |
|---|---|
| 1.4 | $18.6*T_S$ |
| ≥10 | $4.6*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

An illustration of the measurement principle is shown in Figure 7.1.1.5-1 in section 7.1.1.5.

## 7.1.2 E-UTRAN TDD - UE Transmit Timing Accuracy

### 7.1.2.1 Test purpose

To verify the UE have the capability to follow the frame timing change of the connected System Simulator. The method used is that the UE initial transmit timing accuracy, the maximum amount of timing change in one adjustment, and the minimum and maximum adjustment rate are within the specified limits based on the requirements.

### 7.1.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward. Applicability requires support for FGI bit 5.

### 7.1.2.3 Minimum conformance requirements

The UE initial transmission timing error shall be less than or equal to $\pm T_e$ where the timing error limit value $T_e$ as defined in table 7.1.2-1 of TS 36.133 [4] clause 7.1.2. This requirement applies when it is the first transmission in a DRX cycle for PUCCH, PUSCH and SRS or it is the PRACH transmission. The uplink frame transmission takes place $(N_{\mathrm{TA}} + N_{\mathrm{TA\,offset}}) \times T_s$ before the reception of the first detected path (in time) of the corresponding downlink frame from the reference cell. The reference point for the UE initial transmit timing control requirement shall be the downlink timing minus $(N_{\mathrm{TA_Ref}} + N_{\mathrm{TA\,offset}}) \times T_s$.

where:

$N_{TA}$ is $0 \le N_{TA} \le 20512$

$N_{TA_Ref}$ is 0 for PRACH; $N_{(N_{\mathrm{TA_Ref}} + N_{\mathrm{TA\,offset}})}$ (in $T_s$ units) for other channels is the difference between UE transmission timing and the downlink timing immediately after when the last timing advance in TS 36.133 [4] clause 7.3 was applied. $N_{TA_Ref}$ in $T_s$ units) for other channels is not changed until next timing advance is received.

$N_{TA\,offset}$ is 624 for frame structure type 2 as defined in TS 36.211 [9] clause 8.1. $T_S$ denotes the basic time unit. The size of various fields in the time domain is expressed as a number of time units $T_S = 1/(15000 \times 2048)$ seconds.

When it is not the first transmission in a DRX cycle or there is no DRX cycle, and when it is the transmission for PUCCH, PUSCH and SRS transmission, the UE shall be capable of changing the transmission timing according to the received downlink frame except when the timing advance in TS 36.133 [4] clause 7.3 is applied. When the transmission timing error between the UE and the reference timing exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$. The reference timing shall be $(N_{\mathrm{TA_Ref}} + N_{\mathrm{TA\,offset}}) \times T_s$ before the downlink timing.

All adjustments made to the UE uplink timing $(N_{\mathrm{TA}} + N_{\mathrm{TA\,offset}}) \times T_s$ shall follow these rules:

1) The maximum amount of the magnitude of the timing change in one adjustment shall be $T_q$
2) The minimum aggregate adjustment rate shall be $7 \times T_S$ per second.
3) The maximum aggregate adjustment rate shall be $T_q$ per 200 ms.

Where the maximum timing error value $T_e$ is specified in table 7.1.2.3-1 and maximum autonomous time adjustment step $T_q$ is specified in table 7.1.2.3-2.

The normative reference for this requirement is TS 36.133 [4] clause 7.1 and A.7.1.2.

**Table 7.1.2.3-1: $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| 1.4 | $24*T_S$ |
| ≥3 | $12*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 | |

**Table 7.1.2.3-2: $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| 1.4 | $17.5*T_S$ |
| 3 | $9.5*T_S$ |
| 5 | $5.5*T_S$ |
| ≥10 | $3.5*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 | |

NOTE 1: Both the allowed UE frequency error and the UE time adjustment quantisation can affect the result of the test, and it is not possible to test the UE Transmit Timing Accuracy in isolation from these core requirements. They are therefore taken into account when setting the Test Tolerances and the Test Requirements

## 7.1.2.4 Test description

### 7.1.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and

Channel Bandwidth to be tested: 10 MHz and 1.4 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noises source to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18.

2. Propagation conditions are set according to Annex B clause B.0.

3. There is one E-UTRA TDD carrier and one cell specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

### 7.1.2.4.2 Test procedure

The test consists of a single cell. The transmit timing accuracy is verified related to the downlink frame timing of Cell 1. The downlink timing of Cell 1 is changed and the changes in UE transmit timing are observed. The transmit timing is verified by the UE transmitting SRS used as a measurement reference facilitating the SS timing estimation.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Test 1, Test 2, or Test 3 in Tables 7.1.2.5-1 and 7.1.2.5-2 and 7.1.2.5-3 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. The SS shall check that the UE transmit timing offset is $624\times T_S$ to within the $T_e$ limits specified in Table 7.1.2.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1.

6. The SS adjusts the downlink timing for Cell 1 to a delay of $+64 \times T_S$ (approximately +2 μs) for ≥3 MHz downlink bandwidth (Test 1, Test 2) and a delay of $+128\times T_S$ (approximately +4 μs) for 1.4MHz downlink (Test 3) bandwidth compared to that in step 5. For Test 2, the downlink timing adjustment shall be done at (40 ± 15 )ms before the next DRX On duration period.

7. Step 7 applies for Test 1 and Test 3, but is omitted for Test 2. The test system samples the UE Transmit Timing once per SRS transmission (periodicity 10 subframes). To check Rule 1, the SS shall check that the maximum time adjustment step size $T_q$ between one SRS transmission (subframe n) to next (subframe n+10) is within Rule 1 as specified in clause 7.1.2.5. To check that the minimum adjustment rate is within Rule 2 as specified in clause 7.1.2.5, the SS shall measure the change in SRS transmission timing over a 1.01s sliding window, with step size 10ms. To check that the maximum adjustment rate is within Rule 3 as specified in clause 7.1.2.5, the SS shall measure the change in SRS transmission timing over a 190ms sliding window, with step size 10ms. The three rules apply until the UE transmit timing offset is $(624 \times T_S)$ to within the limits specified in Table 7.1.2.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1.

8. In case for Test 1 and Test 3, the SS shall check that the UE transmit timing offset stays at $624\times T_S$ to within the $T_e$ limits specified in Table 7.1.2.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1. In case for Test 2, the SS shall check that the UE transmit timing offset stays within the limits specified in Table 7.1.2.5-4. For that purpose, the SRS transmission that takes place in the first DRX ON duration after downlink timing of Cell 1 is changed is used.

9. The SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

10. After the RRC connection release, the SS switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

11. Repeat step 1-10 for each sub-test in Tables 7.1.2.5-1, 7.1.2.5-2 and 7.1.2.5-3 as appropriate.

### 7.1.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.1.2.4.3-1: Common Exception messages for UE transmit timing accuracy for E-UTRAN TDD test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.4-2 |

**Table 7.1.2.4.3-2: *RRCConnectionReconfiguration*: Additional UE transmit timing accuracy for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.331 [5] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| radioResourceConfigDedicated SEQUENCE { | | | |
| mac-MainConfig CHOICE { | | | |
| explicitValue | MAC-MainConfig-RBC | | |
| } | | | |
| physicalConfigDedicated | PhysicalConfigDedicated - DEFAULT | | RBC |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.1.2.4.3-3: *SoundingRS-UL-ConfigCommon-DEFAULT*: Additional UE transmit timing for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [5] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRsUl-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | | Set according to specific test; bw5 for Test 1 and Test 2 , bw7 for Test 3 | |
| srs-SubframeConfig | sc3 | | TDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | Not present | | |
| } | | | |

**Table 7.1.2.4.3-4: *SoundingRS-UL-ConfigDedicated-DEFAULT*: Additional UE transmit timing for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRSUL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| enable SEQUENCE { | | | |
| srsBandwidth | bw0 | bw0 used with no frequency hopping. bw3 used with frequency hopping | |
| srsHoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | indefinite duration | |
| srs-ConfigIndex | | Set according to specific test; 15 for Test 1, Test3 and 85 for Test 2 | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 7.1.2.4.4-5: MAC-MainConfig-RBC: Additional UE transmit timing for E-UTRAN TDD test requirement for Test 2**

| Derivation Path: TS 36.508 [7] clause 4.8.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf80 typical value in simulations | |
| sf80 | 1 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.1.2.4.3-6: *CQI-ReportConfig-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm30 | This IE should be omitted for test 3 | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| release | NULL | | |
| } | | | |

**Table 7.1.2.4.3-7: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| } | | | |

## 7.1.2.5 Test requirement

Tables 7.1.2.5-1, 7.1.2.5-2 and 7.1.2.5-3 define the primary settings including test tolerances for UE transmit timing for E-UTRAN TDD test.

Tables 7.1.2.5-4, 7.1.2.5-5 and 7.1.2.5-6 define the rules for adjustments made to the UE uplink timing include Test Tolerances.

**Table 7.1.2.5-1: Test Parameters for UE Transmit Timing Accuracy Tests for E-UTRAN TDD**

| Parameter | Unit | Value | | |
|---|---|---|---|---|
| | | Test 1 | Test 2 | Test 3 |
| E-UTRA RF Channel Number | | 1 | 1 | 1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | 10 | 1.4 |
| Special subframe configuration[Note1] | | 6 | 6 | 6 |
| Uplink-downlink configuration[Note2] | | 1 | 1 | 1 |
| DRX cycle | ms | OFF | 80[Note7] | OFF |
| PDCCH/PCFICH/PHICH Reference measurement channel[Note3] | | R.6 TDD | R.6 TDD | R.8 TDD |
| OCNG Pattern[Note4] | | OP.2 TDD | OP.2 TDD | OP.4 TDD |
| PBCH_RA | dB | 0 | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| OCNG_RA[Note5] | | | | |
| OCNG_RB[Note5] | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | -98 | -98 |
| $\hat{E}_s / I_{ot}$ | dB | 3.30 | 3.30 | 3.30 |
| $\hat{E}_s / N_{oc}$ | dB | 3.30 | 3.30 | 3.30 |
| Io[Note6] | dBm/9 MHz | -65.25 | -65.25 | N/A |
| | dBm/1.08 MHz | N/A | N/A | -74.46 |
| Propagation condition | - | AWGN | AWGN | AWGN |

Note 1: For the special subframe configuration see table 4.2-1 in 3GPP TS 36.211 [9]
Note 2: For the uplink-downlink configuration see table 4.2-2 in 3GPP TS 36.211 [9]
Note 3: For the reference measurement channels, see section A.2.2.
Note 4 For the OCNG pattern, see section D.2.2(for 10MHz) and D.2.4(for 1.4MHz).
Note 5: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 6: Io level has been derived from other parameters for information purpose. It is not a settable parameter.
Note 7: DRX related parameters are defined in Table 7.1.2.5-3.

**Table 7.1.2.5-2: Sounding Reference Signal Configuration to be used in UE Transmit Timing Accuracy Tests for E-UTRAN TDD**

| Field | Test 1 | Test 2 | Test 3 | Comment |
|---|---|---|---|---|
| | Value | | | |
| srsBandwidthConfiguration | bw5 | bw5 | bw7 | |
| srsSubframeConfiguration | sc3 | sc3 | sc3 | Once every 5 subframes |
| ackNackSrsSimultaneousTransmission | FALSE | FALSE | FALSE | |
| srsMaxUpPTS | FALSE | FALSE | FALSE | |
| srsBandwidth | 0 | 0 | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | hbw0 | hbw0 | |
| frequencyDomainPosition | 0 | 0 | 0 | |
| duration | TRUE | TRUE | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 15 | 85 | 15 | SRS periodicity of 10 and 80 ms for Test 1 and 2, respectively. |
| transmissionComb | 0 | 0 | 0 | |
| cyclicShift | cs0 | cs0 | cs0 | No cyclic shift |
| Note: For further information see section 6.3.2 in 3GPP TS 36.331 [5]. | | | | |

**Table 7.1.2.5-3: DRX Configuration to be used in UE Transmit Timing Accuracy Test 2 for E-UTRAN TDD**

| Field | Test2 | Comment |
|---|---|---|
| | Value | |
| onDurationTimer | psf1 | |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf80 | |
| shortDRX | disable | |
| Note: For further information see section 6.3.2 in 3GPP TS 36.331 [5]. | | |

The UE transmit timing offset shall be within the requirements in Table 7.1.2.5-4.

The reference point for the UE initial transmit timing control test requirement shall be the downlink timing minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$ seconds.

**Table 7.1.2.5-4: Test requirement for $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| 1.4 | $27*T_S$ |
| ≥3 | $15*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

The UE shall be capable of changing the transmission timing according to the received downlink frame. When the transmission timing error between the UE and the reference cell exceeds $\pm T_e$ seconds the UE is required to adjust its timing to within $\pm T_e$ seconds.

All adjustments made to the UE uplink timing shall follow these rules:

1) For tests 1 and 3, the maximum adjustment step size $T_q$ shall be within the requirements in Table 7.1.2.5-5.

2) For test 1, the minimum aggregate adjustment rate shall be $3.4 \times T_S$ per 1.01 s.

3) For tests 1 and 3, the maximum aggregate adjustment rate shall be within the requirements in Table 7.1.2.5-6.

**Table 7.1.2.5-5: Test requirement for $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| 1.4 | $18*T_S$ |
| ≥10 | $4*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

**Table 7.1.2.5-6: Test requirement for maximum aggregate adjustment rate per 190 ms**

| Downlink Bandwidth (MHz) | Maximum adjustment per 190 ms |
|---|---|
| 1.4 | $18.6*T_S$ |
| ≥10 | $4.6*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

An illustration of the measurement principle is shown in Figure 7.1.2.5-4.

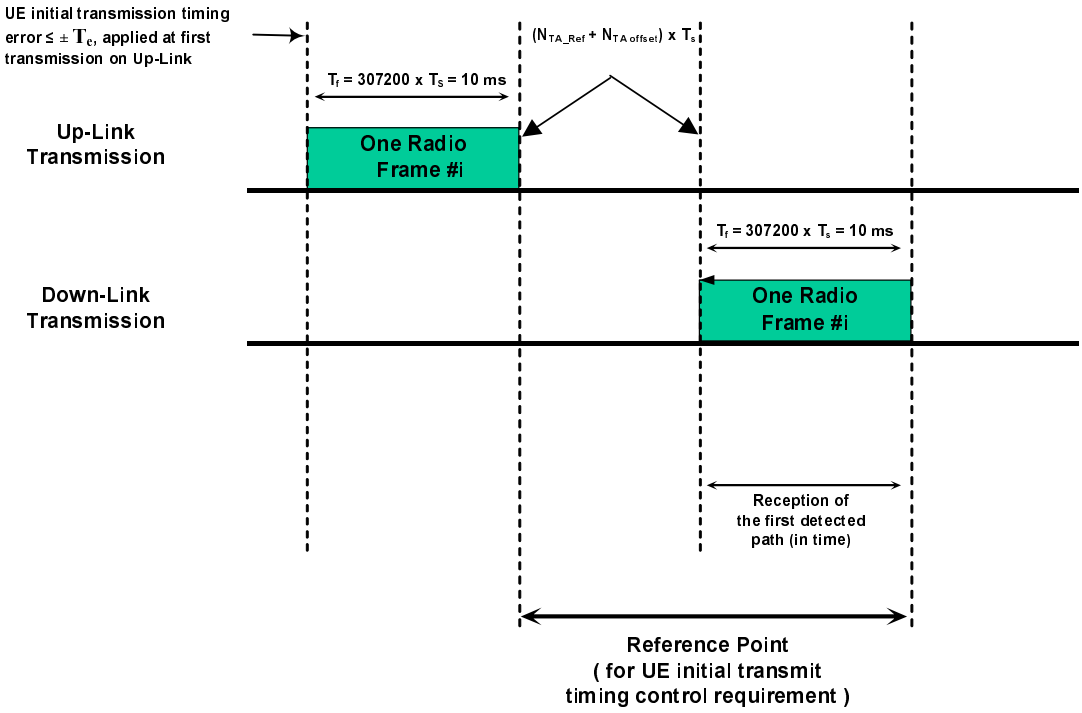


**Figure 7.1.2.5-4: Illustration of measurement principle**

# 7.1.2_1 E-UTRAN TDD - UE Transmit Timing Accuracy (Non DRx UE)

## 7.1.2_1.1 Test purpose

Same test purpose as 7.1.2.1

## 7.1.2_1.2 Test applicability

This test applies to all types of E-UTRA TDD release 8 UE's not supporting FGI bit 5.

## 7.1.2_1.3 Minimum conformance requirements

Same minimum conformance requirement as 7.1.2.3

## 7.1.2_1.4 Test description

### 7.1.2_1.4.1 Initial conditions

Same initial condition as 7.1.2.4.1

### 7.1.2_1.4.2 Test procedure

The test consists of a single cell. The transmit timing accuracy is verified related to the downlink frame timing of Cell 1. The downlink timing of Cell 1 is changed and the changes in UE transmit timing are observed. The transmit timing is verified by the UE transmitting SRS used as a measurement reference facilitating the SS timing estimation.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Test 1 or Test 3 in Tables 7.1.2_1.5-1 and 7.1.2_1.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. The SS shall check that the UE transmit timing offset is 624× $T_S$ to within the $T_e$ limits specified in Table 7.1.2_1.5-3 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1.
6. The SS adjusts the downlink timing for Cell 1 to a delay of +64 × $T_S$ (approximately +2 μs) for ≥3 MHz downlink bandwidth (Test 1) and a delay of +128× $T_S$ (approximately +4 μs) for 1.4MHz downlink (Test 3) bandwidth compared to that in step 5.
7. The test system samples the UE Transmit Timing once per SRS transmission (periodicity 10 subframes). To check Rule 1, the SS shall check that the maximum time adjustment step size $T_q$ between one SRS transmission (subframe n) to next (subframe n+10) is within Rule 1 as specified in clause 7.1.2_1.5. To check that the minimum adjustment rate is within Rule 2 as specified in clause 7.1.2_1.5, the SS shall measure the change in SRS transmission timing over a 1.01s sliding window, with step size 10ms. To check that the maximum adjustment rate is within Rule 3 as specified in clause 7.1.2_1.5, the SS shall measure the change in SRS transmission timing over a 190ms sliding window, with step size 10ms. The three rules apply until the UE transmit timing offset is（624 × $T_S$) to within the limits specified in Table 7.1.2_1.5-3 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1.
8. The SS shall check that the UE transmit timing offset stays at 624× $T_S$ to within the $T_e$ limits specified in Table 7.1.2_1.5-3 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1.
9. The SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
10. After the RRC connection release, the SS switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
11. Repeat step 1-10 for each sub-test in Tables 7.1.2_1.5-1 and 7.1.2_1.5-2 as appropriate.

### 7.1.2_1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with exceptions listed in Tables 7.1.2.4.3-1, 7.1.2.4.3-2, 7.1.2.4.3-6, and 7.1.2.4.3-7 of section 7.1.2.4.3 and the following

**Table 7.1.2_1.4.3-1: *SoundingRS-UL-ConfigCommon-DEFAULT*: Additional UE transmit timing for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRsUl-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | | Set according to specific test; bw5 for Test 1, bw7 for Test 3 | |
| srs-SubframeConfig | sc3 | | TDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | Not present | | |
| } | | | |

**Table 7.1.2_1.4.3-2: *SoundingRS-UL-ConfigDedicated-DEFAULT*: Additional UE transmit timing for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRSUL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| enable SEQUENCE { | | | |
| srsBandwidth | bw0 | bw0 used with no frequency hopping. bw3 used with frequency hopping | |
| srsHoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | indefinite duration | |
| srs-ConfigIndex | | Set according to specific test; 15 for Test 1 and Test 3 | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

## 7.1.2_1.5 Test requirement

Tables 7.1.2_1.5-1 and 7.1.2_1.5-2 define the primary settings including test tolerances for UE transmit timing for E-UTRAN TDD test.

Tables 7.1.2_1.5-3, 7.1.2_1.5-4 and 7.1.2_1.5-5 define the rules for adjustments made to the UE uplink timing including Test Tolerances.

**Table 7.1.2_1.5-1: Test Parameters for UE Transmit Timing Accuracy Tests for E-UTRAN TDD**

| Parameter | Unit | Value | | |
|---|---|---|---|---|
| | | Test 1 | Test 2 | Test 3 |
| E-UTRA RF Channel Number | | 1 | N/A | 1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | N/A | 1.4 |
| Special subframe configuration[Note1] | | 6 | N/A | 6 |
| Uplink-downlink configuration[Note2] | | 1 | N/A | 1 |
| DRX cycle | ms | OFF | N/A | OFF |
| PDCCH/PCFICH/PHICH Reference measurement channel[Note3] | | R.6 TDD | N/A | R.8 TDD |
| OCNG Pattern[Note4] | | OP.2 TDD | N/A | OP.4 TDD |
| PBCH_RA | dB | 0 | N/A | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| OCNG_RA[Note3] | | | | |
| OCNG_RB[Note3] | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | N/A | -98 |
| $\hat{E}_s / I_{ot}$ | dB | 3.30 | N/A | 3.30 |
| $\hat{E}_s / N_{oc}$ | dB | 3.30 | N/A | 3.30 |
| Io[Note6] | dBm/9 MHz | -65.25 | N/A | N/A |
| | dBm/1.08 MHz | N/A | N/A | -74.46 |
| Propagation condition | - | AWGN | N/A | AWGN |

Note 1: For the special subframe configuration see table 4.2-1 in 3GPP TS 36.211 [9]
Note 2: For the uplink-downlink configuration see table 4.2-2 in 3GPP TS 36.211 [9]
Note 3: For the reference measurement channels, see section A.2.2.
Note 4 For the OCNG pattern, see section D.2.2(for 10MHz) and D.2.4(for 1.4MHz).
Note 5: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 6: Io level has been derived from other parameters for information purpose. It is not a settable parameter.

**Table 7.1.2_1.5-2: Sounding Reference Signal Configuration to be used in UE Transmit Timing Accuracy Tests for E-UTRAN TDD**

| Field | Test 1 | Test 2 | Test3 | Comment |
|---|---|---|---|---|
| | Value | | | |
| srsBandwidthConfiguration | bw5 | N/A | bw7 | |
| srsSubframeConfiguration | sc3 | N/A | sc3 | Once every 5 subframes |
| ackNackSrsSimultaneousTransmission | FALSE | N/A | FALSE | |
| srsMaxUpPTS | FALSE | N/A | FALSE | |
| srsBandwidth | 0 | N/A | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | N/A | hbw0 | |
| frequencyDomainPosition | 0 | N/A | 0 | |
| duration | TRUE | N/A | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 15 | N/A | 15 | SRS periodicity of 10 ms for Test 1 and Test 3, respectively. |
| transmissionComb | 0 | N/A | 0 | |
| cyclicShift | cs0 | N/A | cs0 | No cyclic shift |
| Note: For further information see section 6.3.2 in 3GPP TS 36.331 [5]. | | | | |

The UE transmit timing offset shall be within the requirements in Table 7.1.2_1.5-3.

The reference point for the UE initial transmit timing control test requirement shall be the downlink timing minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$ seconds.

**Table 7.1.2_1.5-3: Test requirement for $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| 1.4 | $27*T_S$ |
| ≥3 | $15*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

The UE shall be capable of changing the transmission timing according to the received downlink frame. When the transmission timing error between the UE and the reference cell exceeds $\pm T_e$ seconds the UE is required to adjust its timing to within $\pm T_e$ seconds.

All adjustments made to the UE uplink timing shall follow these rules:

1) The maximum adjustment step size $T_q$ shall be within the requirements in Table 7.1.2_1.5-4.

2) For test 1, the minimum aggregate adjustment rate shall be $3.4 \times T_S$ per 1.01 s.

3) The maximum aggregate adjustment rate shall be within the requirements in Table 7.1.2_1.5-5.

**Table 7.1.2_1.5-4: Test requirement for $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| 1.4 | $18*T_S$ |
| ≥10 | $4*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

**Table 7.1.2_1.5-5: Test requirement for maximum aggregate adjustment rate per 190 ms**

| Downlink Bandwidth (MHz) | Maximum adjustment per 190 ms |
|---|---|
| 1.4 | $18.6*T_S$ |
| ≥10 | $4.6*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

An illustration of the measurement principle is shown in Figure 7.1.2.5-4 in section 7.1.2.5.

## 7.1.3 E-UTRAN FDD – UE Transmit Timing Accuracy Tests for SCell

### 7.1.3.1 Test purpose

To verify the UE have the capability to follow the frame timing change of the connected System Simulator. The method used is that the UE initial transmit timing accuracy, the maximum amount of timing change in one adjustment, and the minimum and maximum adjustment rate are within the specified limits based on the requirements.

### 7.1.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 11 that supports Uplink Carrier Aggregation. Applicability requires support for FGI bit 5.

### 7.1.3.3 Minimum conformance requirements

The UE initial transmission timing error shall be less than or equal to $\pm T_e$ where the timing error limit value $T_e$ is specified in Table 7.1.2-1. This requirement applies when it is the first transmission in a DRX cycle for PUCCH, PUSCH and SRS or it is the PRACH transmission. The reference point for the UE initial transmit timing control requirement shall be the downlink timing of the reference cell minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$. The downlink timing is defined as the time when the first detected path (in time) of the corresponding downlink frame is received from the reference cell. $N_{\text{TA_Ref}}$ for PRACH is defined as 0. $(N_{\text{TA_Ref}} + N_{\text{TA offset}})$ (in $T_s$ units) for other channels is the difference between UE transmission timing and the Downlink timing immediately after when the last timing advance in clause 7.3 was applied. $N_{\text{TA_Ref}}$ for other channels is not changed until next timing advance is received.

**Table 7.1.3.3-1: $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| 1.4 | $24*T_S$ |
| ≥3 | $12*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211. | |

When it is not the first transmission in a DRX cycle or there is no DRX cycle, and when it is the transmission for PUCCH, PUSCH and SRS transmission, the UE shall be capable of changing the transmission timing according to the received downlink frame of the reference cell except when the timing advance in clause 7.3 is applied. When the transmission timing error between the UE and the reference timing exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$. The reference timing shall be $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$ before the downlink timing of the reference cell. All adjustments made to the UE uplink timing shall follow these rules:

1) The maximum amount of the magnitude of the timing change in one adjustment shall be $T_q$ seconds.

2) The minimum aggregate adjustment rate shall be $7*T_S$ per second.

3) The maximum aggregate adjustment rate shall be $T_q$ per 200ms.

where the maximum autonomous time adjustment step $T_q$ is specified in Table 7.1.2-2.

**Table 7.1.3.3-2: $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| 1.4 | $17.5*T_S$ |
| 3 | $9.5*T_S$ |
| 5 | $5.5*T_S$ |
| ≥10 | $3.5*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 | |

The normative reference for this requirement is TS 36.133 [4] clause 7.1.2 and A.7.1.3.

## 7.1.3.4 Test description

### 7.1.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.45.
2. Propagation conditions are set according to Annex B clause B.0.
3. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 shall be powered OFF.

### 7.1.3.4.2 Test procedure

The test consists of two cells, PCell and SCell. The transmit timing accuracy is verified related to the downlink frame timing of PCell (Cell 1). The downlink timing of Cell 1 is changed and the changes in UE transmit timing of both PCell and SCell are observed. The transmit timing is verified by the UE transmitting SRS (Sounding Reference Symbols) used as a measurement reference facilitating the SS timing estimation.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCell according to Annex C.0 and C.1 for all downlink physical channels.
3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4 with UL CA.
4. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).
5. Set the parameters according to Test 1 and Test 2 in Tables 7.1.3.5-1, 7.1.3.5-2 and 7.1.3.5-3 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
6. The SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
7. The UE shall transmit RRCConnectionReconfigurationComplete message.
8. The SS shall check that the UE transmit timing offsets of both PCell and SCell are within the limits specified in Table 7.1.3.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of PCell (Cell 1).
9. The SS adjusts the downlink timing for PCell (Cell 1) to a delay of $+64 \times T_S$ (approximately +2 µs) compared to that in step 8. For Test 2, the downlink timing adjustment shall be done at (40 ± 15) ms before the next DRX On duration period.
10. Step 10 applies for Test 1, but is omitted for Test 2. The test system samples the UE Transmit Timing once per SRS transmission (periodicity 2 subframes). To check Rule 1, the SS shall check that the maximum time adjustment step size $T_q$ between one SRS transmission (subframe n) to next (subframe n+2) is within Rule 1 as specified in clause 7.1.3.5. To check that the minimum adjustment rate is within Rule 2 as specified in clause 7.1.3.5, the SS shall measure the change in SRS transmission timing over a 1.002s sliding window, with step size 2ms. To check that the maximum adjustment rate is within Rule 3 as specified in clause 7.1.3.5, the SS shall measure the change in SRS transmission timing over a 198ms sliding window, with step size 2ms. The three rules apply until the UE transmit timing offsets of both PCell and SCell are within the limits specified in Table 7.1.3.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of PCell (Cell 1).
11. In case for Test 1, the SS shall check that the UE transmit timing offsets of both PCell and SCell stay within the limits specified in Table 7.1.3.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of PCell (Cell 1). In case for Test 2, the SS shall check that the UE transmit timing offsets of both PCell and SCell stay within the limits specified in Table 7.1.3.5-4. For that purpose, the SRS transmission that takes place in the first DRX ON duration after downlink timing of PCell (Cell 1) is changed is used..

12. The SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

13. After the RRC connection release, the SS switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 1-13 for each sub-test in Tables 7.1.3.5-1, 7.1.3.5-2 and 7.1.3.5-3 as appropriate.

### 7.1.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.1.3.4.3-1: Common Exception messages for UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.4-2 |

**Table 7.1.3.4.3-2: *RRCConnectionReconfiguration*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.331 [5] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| radioResourceConfigDedicated SEQUENCE { | | | |
| mac-MainConfig CHOICE { | | | |
| explicitValue | MAC-MainConfig-RBC | | |
| } | | | |
| physicalConfigDedicated | PhysicalConfigDedicated - DEFAULT | | RBC |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.1.3.4.3-3: *PhysicalConfigDedicated- DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT with condition RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated-v1020 SEQUENCE { | | | |
| srs-AntennaPort-r10 | an1 | | |
| } | | | |

**Table 7.1.3.4.3-4: *SoundingRS-RL-ConfigCommon-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw5 | | |
| srs-SubframeConfig | | Set according to specific test; sc1 for Test 1 and sc3 for Test 2 | FDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | Not present | | FDD |
| } | | | |

**Table 7.1.3.4.3-5: *SoudingRS-UL-ConfigDedicated-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | bw0 used with no frequency hopping. | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | | Set according to specific test; 0 for Test 1 and 77 for Test 2 | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 7.1.3.4.3-6: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement for Test 2**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | DRX_S |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf80 typical value in simulations | |
| sf80 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.1.3.4.3-7: *CQI-ReportConfig-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm30 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| release | NULL | | |
| } | | | |

**Table 7.1.3.4.3-8: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| } | | | |

## 7.1.3.5 Test requirement

Tables 7.1.3.5-1, 7.1.3.5-2 and 7.1.3.5-3 define the primary settings including test tolerances for UE transmit timing accuracy for E-UTRAN FDD test.

Tables 7.1.3.5-4, 7.1.3.5-5 and 7.1.3.5-6 define the rules for adjustments made to the UE uplink timing include Test Tolerances.

**Table 7.1.3.5-1: Cell Specific Test requirement Parameters for UE transmit timing accuracy for E-UTRAN FDD test case**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | Test 1 | Test 2 | Test 1 | Test 2 |
| E-UTRA RF Channel Number | | 1 | 1 | 2 | 2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 20 | 20 | 20 | 20 |
| Active PCell | | Cell 1 | Cell 1 | | |
| Active SCell | | | | Cell 2 | Cell 2 |
| TAG configuration | | pTAG | pTAG | pTAG | pTAG |
| DRX cycle | ms | OFF | 80[Note5] | OFF | 80[Note5] |
| PDCCH/PCFICH/PHICH Reference measurement channel[Note1] | | R.10 FDD | R.10 FDD | R.10 FDD | R.10 FDD |
| OCNG Pattern[Note2] | | OP.12 FDD | OP.12 FDD | OP.12 FDD | OP.12 FDD |
| PBCH_RA | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| OCNG_RA[Note3] | | | | | |
| OCNG_RB[Note3] | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | -98 | -98 | -98 |
| $\hat{E}_s / I_{ot}$ | dB | 3.30 | 3.30 | 3.30 | 3.30 |
| $\hat{E}_s / N_{oc}$ | dB | 3.30 | 3.30 | 3.30 | 3.30 |
| Io[Note4] | dBm/18 MHz | -62.24 | -62.24 | -62.24 | -62.24 |
| Propagation condition | - | AWGN | AWGN | AWGN | AWGN |

Note 1: For the reference measurement channels, see clause A.2.1.
Note 2: For the OCNG pattern, see clause D.1.12.
Note 3: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: Io level has been derived from other parameters for information purpose. It is not a settable parameter.
Note 5: DRX related parameters are defined in Table 7.1.3.5-3.

**Table 7.1.3.5-2: Sounding Reference Symbol Configuration to be used in UE transmit timing accuracy for E-UTRAN FDD test case**

| Field | Cell 1 | | Cell 2 | | Comment |
|---|---|---|---|---|---|
| | Test 1 | Test 2 | Test 1 | Test 2 | |
| srsBandwidthConfiguration | bw5 | bw5 | bw5 | bw5 | |
| srsSubframeConfiguration | sc1 | sc3 | sc1 | sc3 | |
| ackNackSrsSimultaneousTransmission | FALSE | FALSE | FALSE | FALSE | |
| srsMaxUpPTS | N/A | N/A | N/A | N/A | Not applicable for FDD |
| srsBandwidth | 0 | 0 | 0 | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | hbw0 | hbw0 | hbw0 | |
| frequencyDomainPosition | 0 | 0 | 0 | 0 | |
| duration | TRUE | TRUE | TRUE | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 0 | 77 | 0 | 77 | SRS periodicity of 2ms and 80 ms for Test 1 and 2, respectively. |
| transmissionComb | 0 | 0 | 0 | 0 | |
| cyclicShift | cs0 | cs0 | cs0 | cs0 | No cyclic shift |
| srsAntennaPort | an1 | an1 | an1 | an1 | Number of SRS antenna ports |
| NOTE: For further information see clause 6.3.2 in TS 36.331 [5]. | | | | | |

**Table 7.1.3.5-3: drx-Configuration to be used in UE Transmit Timing Accuracy Test 2 for E-UTRAN FDD**

| Field | Cell 1 | Cell 2 | Comment |
|---|---|---|---|
| onDurationTimer | psf1 | psf1 | |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf80 | sf80 | |
| shortDRX | disable | disable | |
| NOTE: For further information see clause 6.3.2 in TS 36.331 [5]. | | | |

The UE transmit timing offset shall be within the requirements in Table 7.1.3.5-4.

The reference point for the UE initial transmit timing control test requirement shall be the downlink timing minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$.

**Table 7.1.3.5-4: Test requirement for $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| ≥3 | $15*T_S$ |
| NOTE: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

The UE shall be capable of changing the transmission timing according to the received downlink frame. When the transmission timing error between the UE and the reference cell exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$.

All adjustments made to the UE uplink timing shall follow these rules:

1) For tests 1, the maximum adjustment step size $T_q$ shall be within the requirements in Table 7.1.3.5-5

2) For test 1, the minimum aggregate adjustment rate shall be $3.4 \times T_S$ per 1.002 s.

3) For tests 1, the maximum aggregate adjustment rate shall be within the requirements in Table 7.1.3.5-6.

**Table 7.1.3.5-5: Test requirement for $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| ≥10 | 4*$T_S$ |
| NOTE: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

**Table 7.1.3.5-6: Test requirement for maximum aggregate adjustment rate per 198 ms**

| Downlink Bandwidth (MHz) | Maximum adjustment per 198 ms |
|---|---|
| ≥10 | 4.6*$T_S$ |
| NOTE: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

# 7.1.3_1 E-UTRAN FDD – UE Transmit Timing Accuracy Tests for SCell (Rel-12 and forward)

## 7.1.3_1.1 Test purpose

To verify the UE have the capability to follow the frame timing change of the connected System Simulator. The method used is that the UE initial transmit timing accuracy, the maximum amount of timing change in one adjustment, and the minimum and maximum adjustment rate are within the specified limits based on the requirements.

## 7.1.3_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward that supports Uplink Carrier Aggregation. Applicability requires support for FGI bit 5.

## 7.1.3_1.3 Minimum conformance requirements

The UE initial transmission timing error shall be less than or equal to $\pm T_e$ where the timing error limit value $T_e$ is specified in Table 7.1.2-1. This requirement applies when it is the first transmission in a DRX cycle for PUCCH, PUSCH and SRS or it is the PRACH transmission. The reference point for the UE initial transmit timing control requirement shall be the downlink timing of the reference cell minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$. The downlink timing is defined as the time when the first detected path (in time) of the corresponding downlink frame is received from the reference cell. $N_{\text{TA_Ref}}$ for PRACH is defined as 0. $(N_{\text{TA_Ref}} + N_{\text{TA offset}})$ (in $T_s$ units) for other channels is the difference between UE transmission timing and the Downlink timing immediately after when the last timing advance in clause 7.3 was applied. $N_{\text{TA_Ref}}$ for other channels is not changed until next timing advance is received.

**Table 7.1.3_1.3-1: $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| 1.4 | 24*$T_S$ |
| ≥3 | 12*$T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211. | |

When it is not the first transmission in a DRX cycle or there is no DRX cycle, and when it is the transmission for PUCCH, PUSCH and SRS transmission, the UE shall be capable of changing the transmission timing according to the received downlink frame of the reference cell except when the timing advance in clause 7.3 is applied. When the transmission timing error between the UE and the reference timing exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$. The reference timing shall be $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$ before the downlink timing of the reference cell. All adjustments made to the UE uplink timing shall follow these rules:

1. The maximum amount of the magnitude of the timing change in one adjustment shall be $T_q$ seconds.
2. The minimum aggregate adjustment rate shall be 7*$T_S$ per second.
3. The maximum aggregate adjustment rate shall be $T_q$ per 200ms.

where the maximum autonomous time adjustment step $T_q$ is specified in Table 7.1.2-2.

**Table 7.1.3_1.3-2: $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| 1.4 | $17.5*T_S$ |
| 3 | $9.5*T_S$ |
| 5 | $5.5*T_S$ |
| ≥10 | $3.5*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 | |

The normative reference for this requirement is TS 36.133 [4] clause 7.1.2 and A.7.1.3.

## 7.1.3_1.4 Test description

### 7.1.3_1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.45.
2. Propagation conditions are set according to Annex B clause B.0.
3. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 shall be powered OFF.

### 7.1.3_1.4.2 Test procedure

The test consists of two cells, PCell and SCell. The transmit timing accuracy is verified related to the downlink frame timing of PCell (Cell 1). The downlink timing of Cell 1 is changed and the changes in UE transmit timing of both PCell and SCell are observed. The transmit timing is verified by the UE transmitting SRS (Sounding Reference Symbols) used as a measurement reference facilitating the SS timing estimation.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCell according to Annex C.0 and C.1 for all downlink physical channels except PHICH.
3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4 with UL CA.
4. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).
5. Set the parameters according to Test 1, Test 2 and Test 3 in Tables 7.1.3_1.5-1, 7.1.3_1.5-2 and 7.1.3_1.5-3 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
6. The SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
7. The UE shall transmit RRCConnectionReconfigurationComplete message.
8. The SS shall check that the UE transmit timing offsets of both PCell and SCell are within the limits specified in Table 7.1.3_1.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of PCell (Cell 1).
9. The SS adjusts the downlink timing for PCell (Cell 1) to a delay of $+64 \times T_S$ (approximately +2 μs) (for Test 1 and Test 2) or $+32 \times T_S$ (for Test 3) compared to that in step 8. For Test 2, the downlink timing adjustment shall be done at (40 ± 15) ms before the next DRX On duration period. For Test 3, the downlink timing adjustment shall be done at (320 ± 15) ms before the next DRX On duration period.

10. Step 10 applies for Test 1, but is omitted for Test 2 and Test 3. The test system samples the UE Transmit Timing once per SRS transmission (periodicity 2 subframes). To check Rule 1, the SS shall check that the maximum time adjustment step size $T_q$ between one SRS transmission (subframe n) to next (subframe n+2) is within Rule 1 as specified in clause 7.1.3_1.5. To check that the minimum adjustment rate is within Rule 2 as specified in clause 7.1.3.5, the SS shall measure the change in SRS transmission timing over a 1.002s sliding window, with step size 2ms. To check that the maximum adjustment rate is within Rule 3 as specified in clause 7.1.3_1.5, the SS shall measure the change in SRS transmission timing over a 198ms sliding window, with step size 2ms. The three rules apply until the UE transmit timing offsets of both PCell and SCell are within the limits specified in Table 7.1.3_1.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of PCell (Cell 1).

11. In case for Test 1, the SS shall check that the UE transmit timing offsets of both PCell and SCell stay within the limits specified in Table 7.1.3_1.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of PCell (Cell 1). In case for Test 2 and Test 3, the SS shall check that the UE transmit timing offsets of both PCell and SCell stay within the limits specified in Table 7.1.3_1.5-4. For that purpose, the SRS transmission that takes place in the first DRX ON duration after downlink timing of PCell (Cell 1) is changed is used.

12. The SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

13. After the RRC connection release, the SS switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 1-13 for each sub-test in Tables 7.1.3_1.5-1, 7.1.3_1.5-2 and 7.1.3_1.5-3 as appropriate.

### 7.1.3_1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.1.3_1.4.3-1: Common Exception messages for UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.4-2 |

**Table 7.1.3_1.4.3-2: *RRCConnectionReconfiguration*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.331 [5] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| radioResourceConfigDedicated SEQUENCE { | | | |
| mac-MainConfig CHOICE { | | | |
| explicitValue | MAC-MainConfig-RBC | | |
| } | | | |
| physicalConfigDedicated | PhysicalConfigDedicated - DEFAULT | | RBC |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.1.3_1.4.3-3: *PhysicalConfigDedicated- DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT with condition RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated-v1020 SEQUENCE { | | | |
| srs-AntennaPort-r10 | an1 | | |
| } | | | |

**Table 7.1.3_1.4.3-4: *SoundingRS-RL-ConfigCommon-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw5 | | |
| srs-SubframeConfig | | Set according to specific test; sc1 for Test 1 and sc3 for Test 2 and Test 3 | FDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | Not present | | FDD |
| } | | | |

**Table 7.1.3_1.4.3-5: *SoudingRS-UL-ConfigDedicated-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | bw0 used with no frequency hopping. | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | | Set according to specific test; 0 for Test 1, 77 for Test 2 and 317 for Test 3 | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 7.1.3_1.4.3-6: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement for Test 2 and Test 3**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | DRX_S |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf80 typical value in simulations | |
| sf80 | 0 | Test 2 | |
| sf640 | 0 | Test 3 | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.1.3_1.4.3-7: *CQI-ReportConfig-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm30 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| release | NULL | | |
| } | | | |

**Table 7.1.3_1.4.3-8: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| } | | | |

## 7.1.3_1.5 Test requirement

Tables 7.1.3_1.5-1, 7.1.3_1.5-2 and 7.1.3_1.5-3 define the primary settings including test tolerances for UE transmit timing accuracy for E-UTRAN FDD test.

Tables 7.1.3_1.5-4, 7.1.3_1.5-5 and 7.1.3_1.5-6 define the rules for adjustments made to the UE uplink timing include Test Tolerances.

**Table 7.1.3_1.5-1: Cell Specific Test requirement Parameters for UE transmit timing accuracy for E-UTRAN FDD test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | Test 1 | Test 2 | Test 3 | Test 1 | Test 2 | Test 3 |
| E-UTRA RF Channel Number | | 1 | 1 | 1 | 2 | 2 | 2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 20 | 20 | 20 | 20 | 20 | 20 |
| Active PCell | | Cell 1 | Cell 1 | Cell 1 | | | |
| Active SCell | | | | | Cell 2 | Cell 2 | Cell 2 |
| TAG configuration | | pTAG | pTAG | pTAG | pTAG | pTAG | pTAG |
| DRX cycle | ms | N/A | 80Note5 | 640Note5 | N/A | 80Note5 | 640Note5 |
| PDCCH/PCFICH/PHICH Reference measurement channelNote1 | | R.10 FDD | R.10 FDD | R.10 FDD | R.10 FDD | R.10 FDD | R.10 FDD |
| OCNG PatternNote2 | | OP.12 FDD | OP.12 FDD | OP.12 FDD | OP.12 FDD | OP.12 FDD | OP.12 FDD |
| PBCH_RA | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | |
| PSS_RA | | | | | | | |
| SSS_RA | | | | | | | |
| PCFICH_RB | | | | | | | |
| PHICH_RA | | | | | | | |
| PHICH_RB | | | | | | | |
| PDCCH_RA | | | | | | | |
| PDCCH_RB | | | | | | | |
| OCNG_RANote3 | | | | | | | |
| OCNG_RBNote3 | | | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | -98 | -98 | -98 | -98 | -98 |
| $\hat{E}_s / I_{ot}$ | dB | 3.30 | 3.30 | 3.30 | 3.30 | 3.30 | 3.30 |
| $\hat{E}_s / N_{oc}$ | dB | 3.30 | 3.30 | 3.30 | 3.30 | 3.30 | 3.30 |
| IoNote4 | dBm/18 MHz | -62.24 | -62.24 | -62.24 | -62.24 | -62.24 | -62.24 |
| Propagation condition | - | AWGN | AWGN | AWGN | AWGN | AWGN | AWGN |

Note 1: For the reference measurement channels, see clause A.2.1.
Note 2: For the OCNG pattern, see clause D.1.12.
Note 3: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: Io level has been derived from other parameters for information purpose. It is not a settable parameter.
Note 5: DRX related parameters are defined in Table 7.1.3_1.5-3.

**Table 7.1.3_1.5-2: Sounding Reference Symbol Configuration to be used in UE transmit timing accuracy for E-UTRAN FDD test case**

| Field | Cell 1 | | | Cell 2 | | | Comment |
|---|---|---|---|---|---|---|---|
| | Test 1 | Test 2 | Test 3 | Test 1 | Test 2 | Test 3 | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| srsBandwidthConfiguration | bw5 | bw5 | bw5 | bw5 | bw5 | bw5 | |
| srsSubframeConfiguration | sc1 | sc3 | sc3 | sc1 | sc3 | sc3 | |
| ackNackSrsSimultaneousTransmission | FALSE | FALSE | FALSE | FALSE | FALSE | FALSE | |
| srsMaxUpPTS | N/A | N/A | N/A | N/A | N/A | N/A | Not applicable for FDD |
| srsBandwidth | 0 | 0 | 0 | 0 | 0 | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | hbw0 | hbw0 | hbw0 | hbw0 | hbw0 | |
| frequencyDomainPosition | 0 | 0 | 0 | 0 | 0 | 0 | |
| duration | TRUE | TRUE | TRUE | TRUE | TRUE | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 0 | 77 | 317 | 0 | 77 | 317 | SRS periodicity of 2ms, 80 ms and 320ms for Test 1, 2 and 3, respectively. |
| transmissionComb | 0 | 0 | 0 | 0 | 0 | 0 | |
| cyclicShift | cs0 | cs0 | cs0 | cs0 | cs0 | cs0 | No cyclic shift |
| srsAntennaPort | an1 | an1 | an1 | an1 | an1 | an1 | Number of SRS antenna ports |
| NOTE: For further information see clause 6.3.2 in TS 36.331 [5]. | | | | | | | |

**Table 7.1.3_1.5-3: drx-Configuration to be used in UE Transmit Timing Accuracy Test 2 and Test 3 for E-UTRAN FDD**

| Field | Test 2 | | Test 3 | | Comment |
|---|---|---|---|---|---|
| | Cell 1 | Cell 2 | Cell 1 | Cell 2 | |
| onDurationTimer | psf1 | psf1 | psf1 | psf1 | |
| drx-InactivityTimer | psf1 | psf1 | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf80 | sf80 | Sf640 | Sf640 | |
| shortDRX | disable | disable | disable | Disable | |
| NOTE: For further information see clause 6.3.2 in TS 36.331 [5]. | | | | | |

The UE transmit timing offset shall be within the requirements in Table 7.1.3_1.5-4.

The reference point for the UE initial transmit timing control test requirement shall be the downlink timing minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$.

**Table 7.1.3_1.5-4: Test requirement for $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| ≥3 | $15*T_S$ |
| NOTE: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

The UE shall be capable of changing the transmission timing according to the received downlink frame. When the transmission timing error between the UE and the reference cell exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$.

All adjustments made to the UE uplink timing shall follow these rules:

1. For tests 1, the maximum adjustment step size $T_q$ shall be within the requirements in Table 7.1.3_1.5-5
2. For test 1, the minimum aggregate adjustment rate shall be $3.4 \times T_S$ per 1.002 s.
3. For tests 1, the maximum aggregate adjustment rate shall be within the requirements in Table 7.1.3_1.5-6.

**Table 7.1.3_1.5-5: Test requirement for $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| ≥10 | $4*T_S$ |
| NOTE: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

**Table 7.1.3_1.5-6: Test requirement for maximum aggregate adjustment rate per 198 ms**

| Downlink Bandwidth (MHz) | Maximum adjustment per 198 ms |
|---|---|
| ≥10 | $4.6*T_S$ |
| NOTE: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

## 7.1.4 E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell

### 7.1.4.1 Test purpose

To verify the UE have the capability to follow the frame timing change of the connected System Simulator. The method used is that the UE initial transmit timing accuracy, the maximum amount of timing change in one adjustment, and the minimum and maximum adjustment rate are within the specified limits based on the requirements.

### 7.1.4.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 11 that supports Uplink Carrier Aggregation. Applicability requires support for FGI bit 5.

### 7.1.4.3 Minimum conformance requirements

The UE initial transmission timing error shall be less than or equal to $\pm T_e$ where the timing error limit value $T_e$ is specified in Table 7.1.2-1. This requirement applies when it is the first transmission in a DRX cycle for PUCCH, PUSCH and SRS or it is the PRACH transmission. The reference point for the UE initial transmit timing control requirement shall be the downlink timing of the reference cell minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$. The downlink timing is defined as the time when the first detected path (in time) of the corresponding downlink frame is received from the reference cell. $N_{\text{TA_Ref}}$ for PRACH is defined as 0. $(N_{\text{TA_Ref}} + N_{\text{TA offset}})$ (in $T_s$ units) for other channels is the difference between UE transmission timing and the Downlink timing immediately after when the last timing advance in clause 7.3 was applied. $N_{\text{TA_Ref}}$ for other channels is not changed until next timing advance is received.

**Table 7.1.4.3-1: $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| 1.4 | $24*T_S$ |
| ≥3 | $12*T_S$ |
| Note: TS is the basic timing unit defined in TS 36.211 | |

When it is not the first transmission in a DRX cycle or there is no DRX cycle, and when it is the transmission for PUCCH, PUSCH and SRS transmission, the UE shall be capable of changing the transmission timing according to the received downlink frame of the reference cell except when the timing advance in clause 7.3 is applied. When the transmission timing error between the UE and the reference timing exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$. The reference timing shall be $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$ before the downlink timing of the reference cell. All adjustments made to the UE uplink timing shall follow these rules:

1) The maximum amount of the magnitude of the timing change in one adjustment shall be $T_q$ seconds.

2) The minimum aggregate adjustment rate shall be $7*T_S$ per second.

3) The maximum aggregate adjustment rate shall be $T_q$ per 200ms.

where the maximum autonomous time adjustment step $T_q$ is specified in Table 7.1.2-2.

**Table 7.1.4.3-2: $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| 1.4 | $17.5*T_S$ |
| 3 | $9.5*T_S$ |
| 5 | $5.5*T_S$ |
| ≥10 | $3.5*T_S$ |
| Note: TS is the basic timing unit defined in TS 36.211 | |

The normative reference for this requirement is TS 36.133 [4] clause 7.1.2 and A.7.1.4

## 7.1.4.4 Test description

### 7.1.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.45.

2. Propagation conditions are set according to Annex B clause B.0.

3. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 shall be powered OFF.

### 7.1.4.4.2 Test procedure

The test consists of two cells, PCell and SCell. The transmit timing accuracy is verified related to the downlink frame timing of PCell (Cell 1). The downlink timing of Cell 1 is changed and the changes in UE transmit timing of both PCell and SCell are observed. The transmit timing is verified by the UE transmitting SRS (Sounding Reference Symbols) used as a measurement reference facilitating the SS timing estimation.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCell according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4 with UL CA.

4. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).

5. Set the parameters according to Test 1 and Test 2 in Tables 7.1.4.5-1, 7.1.4.5-2 and 7.1.4.5-3 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. The SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. The SS shall check that the UE transmit timing offsets of both PCell and SCell are $624\times T_S$ to within the $T_e$ limits specified in Table 7.1.4.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of PCell (Cell 1).

9. The SS adjusts the downlink timing for PCell (Cell 1) to a delay of $+64 \times T_S$ (approximately +2 µs) compared to that in step 8. For Test 2, the downlink timing adjustment shall be done at (40 ± 15) ms before the next DRX On duration period.

10. Step 10 applies for Test 1, but is omitted for Test 2. The test system samples the UE Transmit Timing once per SRS transmission (periodicity 10 subframes). To check Rule 1, the SS shall check that the maximum time adjustment step size $T_q$ between one SRS transmission (subframe n) to next (subframe n+10) is within Rule 1 as

specified in clause 7.1.4.5. To check that the minimum adjustment rate is within Rule 2 as specified in clause 7.1.4.5, the SS shall measure the change in SRS transmission timing over a 1.01s sliding window, with step size 10ms. To check that the maximum adjustment rate is within Rule 3 as specified in clause 7.1.4.5, the SS shall measure the change in SRS transmission timing over a 190ms sliding window, with step size 10ms. The three rules apply until the UE transmit timing offsets of both PCell and SCell are（$624 \times T_S$) to within the limits specified in Table 7.1.4.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of PCell (Cell 1).

11. In case for Test 1, the SS shall check that the UE transmit timing offsets of both PCell and SCell stay at（$624 \times T_S$) to within the $T_e$ limits specified in Table 7.1.4.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of PCell (Cell 1). In case for Test 2, the SS shall check that the UE transmit timing offsets of both PCell and SCell stay within the limits specified in Table 7.1.4.5-4. For that purpose, the SRS transmission that takes place in the first DRX ON duration after downlink timing of Cell 1 is changed is used.

12. The SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

13. After the RRC connection release, the SS switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 1-13 for each sub-test in Tables 7.1.4.5-1, 7.1.4.5-2 and 7.1.4.5-3 as appropriate.

### 7.1.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.1.4.4.3-1: Common Exception messages for UE transmit timing accuracy for E-UTRAN TDD test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.4-2 |

**Table 7.1.4.4.3-2: *RRCConnectionReconfiguration*: Additional UE transmit timing accuracy for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.331 [5] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| radioResourceConfigDedicated SEQUENCE { | | | |
| mac-MainConfig CHOICE { | | | |
| explicitValue | MAC-MainConfig-RBC | | |
| } | | | |
| physicalConfigDedicated | PhysicalConfigDedicated - DEFAULT | | RBC |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.1.4.4.3-3: *PhysicalConfigDedicated- DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT with condition RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated-v1020 SEQUENCE { | | | |
| srs-AntennaPort-r10 | an1 | | |
| } | | | |

**Table 7.1.4.4.3-4: *SoundingRS-RL-ConfigCommon-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw5 | | |
| srs-SubframeConfig | sc3 | Once every 5 subframes | TDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srs-MaxUpPts | Not present | | |
| } | | | |

**Table 7.1.4.4.3-5: *SoudingRS-UL-ConfigDedicated-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | bw0 used with no frequency hopping. | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | Set according to specific test; 15 for Test 1 and 85 for Test 2 | SRS periodicity of 10 and 80 ms for Test 1 and 2, respectively. | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 7.1.4.4.3-6: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN TDD test requirement for Test 2**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | DRX_S |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf80 typical value in simulations | |
| sf80 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.1.4.4.3-7: *CQI-ReportConfig-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm30 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| release | NULL | | |
| } | | | |

**Table 7.1.4.4.3-8: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| } | | | |

## 7.1.4.5 Test requirement

Tables 7.1.4.5-1, 7.1.4.5-2 and 7.1.4.5-3 define the primary settings including test tolerances for UE transmit timing accuracy for E-UTRAN TDD test.

Tables 7.1.4.5-4, 7.1.4.5-5 and 7.1.4.5-6 define the rules for adjustments made to the UE uplink timing include Test Tolerances.

**Table 7.1.4.5-1: Cell Specific Test requirement Parameters for UE transmit timing accuracy for E-UTRAN TDD test case**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | Test 1 | Test 2 | Test 1 | Test 2 |
| E-UTRA RF Channel Number | | 1 | 1 | 2 | 2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 20 | 20 | 20 | 20 |
| E-UTRA RF Channel Number | | 1 | 1 | 2 | 2 |
| Active PCell | | Cell 1 | Cell 1 | | |
| Active Scell | | | | Cell 2 | Cell 2 |
| TAG configuration | | pTAG | pTAG | pTAG | pTAG |
| Special subframe configuration[Note1] | | 6 | 6 | 6 | 6 |
| Uplink-downlink configuration[Note2] | | 1 | 1 | 1 | 1 |
| DRX cycle | ms | OFF | 80[Note7] | OFF | 80[Note7] |
| PDCCH/PCFICH/PHICH Reference measurement channel[Note3] | | R.10 TDD | R.10 TDD | R.10 TDD | R.10 TDD |
| OCNG Pattern[Note4] | | OP.8 TDD | OP.8 TDD | OP.8 TDD | OP.8 TDD |
| PBCH_RA | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | 0 | 0 | | 0 |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| OCNG_RA[Note5] | | | | | |
| OCNG_RB[Note5] | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | -98 | -98 | -98 |
| $\hat{E}_s / I_{ot}$ | dB | 3.30 | 3.30 | 3.30 | 3.30 |
| $\hat{E}_s / N_{oc}$ | dB | 3.30 | 3.30 | 3.30 | 3.30 |
| Io[Note6] | dBm/18 MHz | -62.24 | -62.24 | -62.24 | -62.24 |
| Propagation condition | - | AWGN | AWGN | AWGN | AWGN |

Note 1: For the special subframe configuration see table 4.2-1 in TS 36.211 [9].
Note 2: For the uplink-downlink configuration see table 4.2-2 in TS 36.211 [9].
Note 3: For the reference measurement channels, see clause A.2.2.
Note 4: For the OCNG pattern, see clause D.2.8.
Note 5: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 6: Io level has been derived from other parameters for information purpose. It is not a settable parameter.
Note 7: DRX related parameters are defined in Table 7.1.4.5-3.

**Table 7.1.4.5-2: Sounding Reference Symbol Configuration to be used in UE transmit timing accuracy for E-UTRAN TDD test case**

| Field | Cell 1 | | Cell 2 | | Comment |
|---|---|---|---|---|---|
| | Test 1 | Test 1 | Test 1 | Test 2 | Tset3 |
| srsBandwidthConfiguration | bw5 | bw5 | bw5 | bw5 | |
| srsSubframeConfiguration | sc3 | sc3 | sc3 | sc3 | Once every 5 subframes |
| ackNackSrsSimultaneousTransmission | FALSE | FALSE | FALSE | FALSE | |
| srsMaxUpPTS | FALSE | FALSE | FALSE | FALSE | |
| srsBandwidth | 0 | 0 | 0 | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | hbw0 | hbw0 | hbw0 | |
| frequencyDomainPosition | 0 | 0 | 0 | 0 | |
| duration | TRUE | TRUE | TRUE | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 15 | 85 | 15 | 85 | SRS periodicity of 10 and 80 ms for Test 1 and 2, respectively. |
| transmissionComb | 0 | 0 | 0 | 0 | |
| cyclicShift | cs0 | cs0 | cs0 | cs0 | No cyclic shift |
| srsAntennaPort | an1 | an1 | an1 | an1 | Number of SRS antenna ports |
| Note: For further information see clause 6.3.2 in TS 36.331 [5]. | | | | | |

**Table 7.1.4.5-3: drx-Configuration to be used in UE Transmit Timing Accuracy Test 2 for E-UTRAN TDD**

| Field | Cell 1 | Cell 2 | Comment |
|---|---|---|---|
| onDurationTimer | psf1 | psf1 | |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf80 | sf80 | |
| shortDRX | disable | disable | |
| Note: For further information see clause 6.3.2 in TS 36.331 [5]. | | | |

The UE transmit timing offset shall be within the requirements in Table 7.1.4.5-4.

The reference point for the UE initial transmit timing control test requirement shall be the downlink timing minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$.

**Table 7.1.4.5-4: Test requirement for $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| ≥3 | $15*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

The UE shall be capable of changing the transmission timing according to the received downlink frame. When the transmission timing error between the UE and the reference cell exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$.

All adjustments made to the UE uplink timing shall follow these rules:

1) For tests 1, the maximum adjustment step size $T_q$ shall be within the requirements in Table 7.1.4.5-5

2) For test 1, the minimum aggregate adjustment rate shall be $3.4 \times T_S$ per 1.01 s.

3) For tests 1, the maximum aggregate adjustment rate shall be within the requirements in Table 7.1.4.5-6.

**Table 7.1.4.5-5: Test requirement for $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| ≥10 | 4*$T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

**Table 7.1.4.5-6: Test requirement for maximum aggregate adjustment rate per 190 ms**

| Downlink Bandwidth (MHz) | Maximum adjustment per 190 ms |
|---|---|
| ≥10 | 4.6*$T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

# 7.1.4_1 E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell (Rel-12 and forward)

## 7.1.4_1.1 Test purpose

To verify the UE have the capability to follow the frame timing change of the connected System Simulator. The method used is that the UE initial transmit timing accuracy, the maximum amount of timing change in one adjustment, and the minimum and maximum adjustment rate are within the specified limits based on the requirements.

## 7.1.4_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward that supports Uplink Carrier Aggregation. Applicability requires support for FGI bit 5.

## 7.1.4_1.3 Minimum conformance requirements

The UE initial transmission timing error shall be less than or equal to $\pm T_e$ where the timing error limit value $T_e$ is specified in Table 7.1.2-1. This requirement applies when it is the first transmission in a DRX cycle for PUCCH, PUSCH and SRS or it is the PRACH transmission. The reference point for the UE initial transmit timing control requirement shall be the downlink timing of the reference cell minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$. The downlink timing is defined as the time when the first detected path (in time) of the corresponding downlink frame is received from the reference cell. $N_{\text{TA_Ref}}$ for PRACH is defined as 0. $(N_{\text{TA_Ref}} + N_{\text{TA offset}})$ (in $T_s$ units) for other channels is the difference between UE transmission timing and the Downlink timing immediately after when the last timing advance in clause 7.3 was applied. $N_{\text{TA_Ref}}$ for other channels is not changed until next timing advance is received.

**Table 7.1.4_1.3-1: $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| 1.4 | 24*$T_S$ |
| ≥3 | 12*$T_S$ |
| Note: TS is the basic timing unit defined in TS 36.211 | |

When it is not the first transmission in a DRX cycle or there is no DRX cycle, and when it is the transmission for PUCCH, PUSCH and SRS transmission, the UE shall be capable of changing the transmission timing according to the received downlink frame of the reference cell except when the timing advance in clause 7.3 is applied. When the transmission timing error between the UE and the reference timing exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$. The reference timing shall be $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$ before the downlink timing of the reference cell. All adjustments made to the UE uplink timing shall follow these rules:

1. The maximum amount of the magnitude of the timing change in one adjustment shall be $T_q$ seconds.
2. The minimum aggregate adjustment rate shall be 7*$T_S$ per second.
3. The maximum aggregate adjustment rate shall be $T_q$ per 200ms.

where the maximum autonomous time adjustment step $T_q$ is specified in Table 7.1.4_1.3-2.

**Table 7.1.4_1.3-2: $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| 1.4 | $17.5*T_S$ |
| 3 | $9.5*T_S$ |
| 5 | $5.5*T_S$ |
| ≥10 | $3.5*T_S$ |
| Note: TS is the basic timing unit defined in TS 36.211 | |

The normative reference for this requirement is TS 36.133 [4] clause 7.1.2 and A.7.1.4

## 7.1.4_1.4 Test description

### 7.1.4_1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.45.
2. Propagation conditions are set according to Annex B clause B.0.
3. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 shall be powered OFF.

### 7.1.4_1.4.2 Test procedure

The test consists of two cells, PCell and SCell. The transmit timing accuracy is verified related to the downlink frame timing of PCell (Cell 1). The downlink timing of Cell 1 is changed and the changes in UE transmit timing of both PCell and SCell are observed. The transmit timing is verified by the UE transmitting SRS (Sounding Reference Symbols) used as a measurement reference facilitating the SS timing estimation.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCell according to Annex C.0 and C.1 for all downlink physical channels except PHICH.
3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4 with UL CA.
4. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).
5. Set the parameters according to Test 1, Test 2 and Test 3 in Tables 7.1.4_1.5-1, 7.1.4_1.5-2 and 7.1.4_1.5-3 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
6. The SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
7. The UE shall transmit RRCConnectionReconfigurationComplete message.
8. The SS shall check that the UE transmit timing offsets of both PCell and SCell are $624\times T_S$ to within the $T_e$ limits specified in Table 7.1.4.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of PCell (Cell 1).
9. The SS adjusts the downlink timing for PCell (Cell 1) to a delay of $+64 \times T_S$ (approximately +2 µs) (for Test 1 and Test 2) or $+32\times T_S$ (for Test 3) compared to that in step 8. For Test 2, the downlink timing adjustment shall be done at (40 ± 15) ms before the next DRX On duration period. For Test 3, the downlink timing adjustment shall be done at (320 ± 15) ms before the next DRX On duration period.

10. Step 10 applies for Test 1, but is omitted for Test 2 and Test 3. The test system samples the UE Transmit Timing once per SRS transmission (periodicity 10 subframes). To check Rule 1, the SS shall check that the maximum time adjustment step size $T_q$ between one SRS transmission (subframe n) to next (subframe n+10) is within Rule 1 as specified in clause 7.1.4.5. To check that the minimum adjustment rate is within Rule 2 as specified in clause 7.1.4.5, the SS shall measure the change in SRS transmission timing over a 1.01s sliding window, with step size 10ms. To check that the maximum adjustment rate is within Rule 3 as specified in clause 7.1.4_1.5, the SS shall measure the change in SRS transmission timing over a 190ms sliding window, with step size 10ms. The three rules apply until the UE transmit timing offsets of both PCell and SCell are $(624 \times T_S)$ to within the limits specified in Table 7.1.4_1.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of PCell (Cell 1).

11. In case for Test 1, the SS shall check that the UE transmit timing offsets of both PCell and SCell stay at $(624 \times T_S)$ to within the $T_e$ limits specified in Table 7.1.4_1.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of PCell (Cell 1). In case for Test 2 and Test 3, the SS shall check that the UE transmit timing offsets of both PCell and SCell stay within the limits specified in Table 7.1.4_1.5-4. For that purpose, the SRS transmission that takes place in the first DRX ON duration after downlink timing of Cell 1 is changed is used.

12. The SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

13. After the RRC connection release, the SS switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 1-13 for each sub-test in Tables 7.1.4_1.5-1, 7.1.4_1.5-2 and 7.1.4_1.5-3 as appropriate.

### 7.1.4_1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.1.4_1.4.3-1: Common Exception messages for UE transmit timing accuracy for E-UTRAN TDD test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.4-2 |

**Table 7.1.4_1.4.3-2: *RRCConnectionReconfiguration*: Additional UE transmit timing accuracy for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.331 [5] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| radioResourceConfigDedicated SEQUENCE { | | | |
| mac-MainConfig CHOICE { | | | |
| explicitValue | MAC-MainConfig-RBC | | |
| } | | | |
| physicalConfigDedicated | PhysicalConfigDedicated - DEFAULT | | RBC |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.1.4_1.4.3-3: *PhysicalConfigDedicated- DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT with condition RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated-v1020 SEQUENCE { | | | |
| srs-AntennaPort-r10 | an1 | | |
| } | | | |

**Table 7.1.4_1.4.3-4: *SoundingRS-RL-ConfigCommon-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw5 | | |
| srs-SubframeConfig | sc3 | Once every 5 subframes | TDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srs-MaxUpPts | Not present | | |
| } | | | |

**Table 7.1.4_1.4.3-5: *SoudingRS-UL-ConfigDedicated-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | bw0 used with no frequency hopping. | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | Set according to specific test; 15 for Test 1, 85 for Test 2 and 325 for Test 3 | SRS periodicity of 10, 80 ms and 320ms for Test 1, 2 and 3,, respectively. | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 7.1.4_1.4.3-6: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN TDD test requirement for Test 2**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | DRX_S |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf80 typical value in simulations | |
| sf80 | 0 | Test 2 | |
| Sf640 | 0 | Test 3 | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.1.4_1.4.3-7: *CQI-ReportConfig-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm30 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| release | NULL | | |
| } | | | |

**Table 7.1.4_1.4.3-8: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN TDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| } | | | |

## 7.1.4_1.5 Test requirement

Tables 7.1.4_1.5-1, 7.1.4_1.5-2 and 7.1.4_1.5-3 define the primary settings including test tolerances for UE transmit timing accuracy for E-UTRAN TDD test.

Tables 7.1.4_1.5-4, 7.1.4_1.5-5 and 7.1.4_1.5-6 define the rules for adjustments made to the UE uplink timing include Test Tolerances.

**Table 7.1.4_1.5-1: Cell Specific Test requirement Parameters for UE transmit timing accuracy for E-UTRAN TDD test case**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | Test 1 | Test 2 | Test 3 | Test 1 | Test 2 | Test 3 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | 1 | 1 | 2 | 2 | 2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 20 | 20 | 20 | 20 | 20 | 20 |
| E-UTRA RF Channel Number | | 1 | 1 | 1 | 2 | 2 | 2 |
| Active PCell | | Cell 1 | Cell 1 | Cell 1 | | | |
| Active SCell | | | | | Cell 2 | Cell 2 | Cell 2 |
| TAG configuration | | pTAG | pTAG | pTAG | pTAG | pTAG | pTAG |
| Special subframe configuration[Note1] | | 6 | 6 | 6 | 6 | 6 | 6 |
| Uplink-downlink configuration[Note2] | | 1 | 1 | 1 | 1 | 1 | 1 |
| DRX cycle | ms | OFF | 80[Note7] | 640[Note7] | OFF | 80[Note7] | 640[Note7] |
| PDCCH/PCFICH/PHICH Reference measurement channel[Note3] | | R.10 TDD | R.10 TDD | R.10 TDD | R.10 TDD | R.10 TDD | R.10 TDD |
| OCNG Pattern[Note4] | | OP.8 TDD | OP.8 TDD | OP.8 TDD | OP.8 TDD | OP.8 TDD | OP.8 TDD |
| PBCH_RA | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | 0 | 0 | 0 | | 0 | 0 |
| PSS_RA | | | | | | | |
| SSS_RA | | | | | | | |
| PCFICH_RB | | | | | | | |
| PHICH_RA | | | | | | | |
| PHICH_RB | | | | | | | |
| PDCCH_RA | | | | | | | |
| PDCCH_RB | | | | | | | |
| OCNG_RA[Note5] | | | | | | | |
| OCNG_RB[Note5] | | | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | -98 | -98 | -98 | -98 | -98 |
| $\hat{E}_s / I_{ot}$ | dB | 3.30 | 3.30 | 3.30 | 3.30 | 3.30 | 3.30 |
| $\hat{E}_s / N_{oc}$ | dB | 3.30 | 3.30 | 3.30 | 3.30 | 3.30 | 3.30 |
| Io[Note6] | dBm/18 MHz | -62.24 | -62.24 | -62.24 | -62.24 | -62.24 | -62.24 |
| Propagation condition | - | AWGN | AWGN | AWGN | AWGN | AWGN | AWGN |

Note 1: For the special subframe configuration see table 4.2-1 in TS 36.211 [9].
Note 2: For the uplink-downlink configuration see table 4.2-2 in TS 36.211 [9].
Note 3: For the reference measurement channels, see clause A.2.2.
Note 4: For the OCNG pattern, see clause D.2.8.
Note 5: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 6: Io level has been derived from other parameters for information purpose. It is not a settable parameter.
Note 7: DRX related parameters are defined in Table 7.1.4_1.5-3.

**Table 7.1.4_1.5-2: Sounding Reference Symbol Configuration to be used in UE transmit timing accuracy for E-UTRAN TDD test case**

| Field | Cell 1 | | | Cell 2 | | | Comment |
|---|---|---|---|---|---|---|---|
| | Test 1 | Test 2 | Test 3 | Test 1 | Test 2 | Test 3 | |
| | Tset1 | Tset2 | Tset3 | Tset1 | Tset2 | Tset3 | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| srsBandwidthConfiguration | bw5 | bw5 | bw5 | bw5 | bw5 | bw5 | |
| srsSubframeConfiguration | sc3 | sc3 | sc3 | sc3 | sc3 | sc3 | Once every 5 subframes |
| ackNackSrsSimultaneous Transmission | FALSE | FALSE | FALSE | FALSE | FALSE | FALSE | |
| srsMaxUpPTS | FALSE | FALSE | FALSE | FALSE | FALSE | FALSE | |
| srsBandwidth | 0 | 0 | 0 | 0 | 0 | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | hbw0 | hbw0 | hbw0 | hbw0 | hbw0 | |
| frequencyDomainPosition | 0 | 0 | 0 | 0 | 0 | 0 | |
| duration | TRUE | TRUE | TRUE | TRUE | TRUE | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 15 | 85 | 325 | 15 | 85 | 325 | SRS periodicity of 10, 80 ms and 320ms for Test 1, 2 and 3, respectively. |
| transmissionComb | 0 | 0 | 0 | 0 | 0 | 0 | |
| cyclicShift | cs0 | cs0 | cs0 | cs0 | cs0 | cs0 | No cyclic shift |
| srsAntennaPort | an1 | an1 | an1 | an1 | an1 | an1 | Number of SRS antenna ports |
| Note: For further information see clause 6.3.2 in TS 36.331 [5]. | | | | | | | |

**Table 7.1.4_1.5-3: drx-Configuration to be used in UE Transmit Timing Accuracy Test 2 and Test 3 for E-UTRAN TDD**

| Field | Test 2 | | Test 3 | | Comment |
|---|---|---|---|---|---|
| | Cell 1 | Cell 2 | Cell 1 | Cell 2 | |
| onDurationTimer | psf1 | psf1 | psf1 | psf1 | |
| drx-InactivityTimer | psf1 | psf1 | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf80 | sf80 | Sf640 | Sf640 | |
| shortDRX | disable | disable | disable | disable | |
| Note: For further information see clause 6.3.2 in TS 36.331 [5]. | | | | | |

The UE transmit timing offset shall be within the requirements in Table 7.1.4_1.5-4.

The reference point for the UE initial transmit timing control test requirement shall be the downlink timing minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$.

**Table 7.1.4_1.5-4: Test requirement for $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| ≥3 | $15*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

The UE shall be capable of changing the transmission timing according to the received downlink frame. When the transmission timing error between the UE and the reference cell exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$.

All adjustments made to the UE uplink timing shall follow these rules:

1. For tests 1, the maximum adjustment step size $T_q$ shall be within the requirements in Table 7.1.4_1.5-5.
2. For test 1, the minimum aggregate adjustment rate shall be $3.4 \times T_S$ per 1.01 s.
3. For tests 1, the maximum aggregate adjustment rate shall be within the requirements in Table 7.1.4_1.5-6.

**Table 7.1.4_1.5-5: Test requirement for $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| ≥10 | $4*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

**Table 7.1.4_1.5-6: Test requirement for maximum aggregate adjustment rate per 190 ms**

| Downlink Bandwidth (MHz) | Maximum adjustment per 190 ms |
|---|---|
| ≥10 | $4.6*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

# 7.1.4A E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell for 20 MHz +10 MHz bandwidth

## 7.1.4A.1 Test purpose

Same test purpose as defined in clause 7.1.4.1.

## 7.1.4A.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 11 and forward that supports Uplink Carrier Aggregation. Applicability requires support for FGI bit 5.

## 7.1.4A.3 Minimum conformance requirements

Same minimum conformance requirements as defined in clause 7.1.4.3 with the following exception:

- The normative reference for this requirement is TS 36.133 [4] clause 7.1.2 and A.7.1.4A.

## 7.1.4A.4 Test description

### 7.1.4A.4.1 Initial conditions

- Same initial conditions as in clause 7.1.4.4.1 with the following exceptions: Channel Bandwidth to be tested: 20 MHz for Cell 1 on the PCC, 10MHz for Cell 2 on the SCC

### 7.1.4A.4.2 Test procedure

Same test procedure as in clause 7.1.4.4.2

### 7.1.4A.4.3 Message contents

Same message contents as in clause 7.1.4.4.3

## 7.1.4A.5 Test requirement

Table 7.1.4A.5-1, 7.1.4A.5-2 and 7.1.4A.5-3 define the primary settings including test tolerances for UE transmit timing accuracy for E-UTRAN TDD test. The listed parameter values in Tables 7.1.4A.5-1 will replace the values of corresponding parameters in Table 7.1.4.5-1.

Table 7.1.4A.5-4, 7.1.4A.5-5 and 7.1.4A.5-6 define the rules for adjustments made to the UE uplink timing include Test Tolerances.

**Table 7.1.4A.5-1: Test Parameters for UE Transmit Timing Accuracy Tests for E-UTRAN TDD for 20 MHz +10 MHz**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | Test 1 | Test 2 | Test 1 | Test 2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 20 | 20 | 10 | 10 |
| PDCCH/PCFICH/PHICH Reference measurement channel Note3 | | R.10 TDD | R.10 TDD | R.6 TDD | R.6 TDD |
| OCNG Pattern Note4 | | OP.8 TDD | OP.8 TDD | OP.2 TDD | OP.2 TDD |
| Io Note6 | dBm/18 MHz | -62.24 | -62.24 | N/A | N/A |
| | dBm/9 MHz | N/A | N/A | -65.25 | -65.25 |

**Table 7.1.4A.5-2: Sounding Reference Symbol Configuration to be used in UE transmit timing accuracy for E-UTRAN TDD test case for 20 MHz +10 MHz**

| Field | Cell 1 | | Cell 2 | | Comment |
|---|---|---|---|---|---|
| | Test 1 | Test 2 | Test 1 | Test 2 | Tset3 |
| srsBandwidthConfiguration | bw5 | bw5 | bw5 | bw5 | |
| srsSubframeConfiguration | sc3 | sc3 | sc3 | sc3 | Once every 5 subframes |
| ackNackSrsSimultaneousTransmission | FALSE | FALSE | FALSE | FALSE | |
| srsMaxUpPTS | FALSE | FALSE | FALSE | FALSE | |
| srsBandwidth | 0 | 0 | 0 | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | hbw0 | hbw0 | hbw0 | |
| frequencyDomainPosition | 0 | 0 | 0 | 0 | |
| duration | TRUE | TRUE | TRUE | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 15 | 85 | 15 | 85 | SRS periodicity of 10 and 80 ms for Test 1 and 2, respectively. |
| transmissionComb | 0 | 0 | 0 | 0 | |
| cyclicShift | cs0 | cs0 | cs0 | cs0 | No cyclic shift |
| srsAntennaPort | an1 | an1 | an1 | an1 | Number of SRS antenna ports |
| Note: For further information see clause 6.3.2 in TS 36.331 [5]. | | | | | |

**Table 7.1.4A.5-3: drx-Configuration to be used in UE Transmit Timing Accuracy Test 2 for E-UTRAN TDD for 20 MHz +10 MHz**

| Field | Cell 1 | Cell 2 | Comment |
|---|---|---|---|
| onDurationTimer | psf1 | psf1 | |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf80 | sf80 | |
| shortDRX | disable | disable | |
| Note: For further information see clause 6.3.2 in TS 36.331 [5]. | | | |

The UE transmit timing offset shall be within the requirements in Table 7.1.4A.5-4.

The reference point for the UE initial transmit timing control test requirement shall be the downlink timing minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$.

**Table 7.1.4A.5-4: Test requirement for $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| ≥3 | $15*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

The UE shall be capable of changing the transmission timing according to the received downlink frame. When the transmission timing error between the UE and the reference cell exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$.

All adjustments made to the UE uplink timing shall follow these rules:

1) For tests 1, the maximum adjustment step size $T_q$ shall be within the requirements in Table 7.1.4A.5-5.

2) For test 1, the minimum aggregate adjustment rate shall be $3.4 \times T_S$ per 1.01 s.

3) For tests 1, the maximum aggregate adjustment rate shall be within the requirements in Table 7.1.4A.5-6.

**Table 7.1.4A.5-5: Test requirement for $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| ≥10 | $4*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

**Table 7.1.4A.5-6: Test requirement for maximum aggregate adjustment rate per 190 ms**

| Downlink Bandwidth (MHz) | Maximum adjustment per 190 ms |
|---|---|
| ≥10 | $4.6*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

## 7.1.5 E-UTRAN FDD - UE Transmit Timing Accuracy Tests for 5MHz Bandwidth

### 7.1.5.1 Test purpose

Same test purpose as defined in clause 7.1.1.1.

### 7.1.5.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that only support band 31.

### 7.1.5.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 7.1.1.3 with the following exceptions:

- Instead of A.7.1.1 → use A.7.1.5.

### 7.1.5.4 Test description

#### 7.1.5.4.1 Initial conditions

Same initial conditions as defined in 7.1.1.4.1 except that the Channel Bandwidth to be tested is 5MHz as defined in TS 36.508 [7] clause 4.3.1.

#### 7.1.5.4.2 Test procedure

The test consists of a single cell. The transmit timing accuracy is verified related to the downlink frame timing of Cell 1. The downlink timing of Cell 1 is changed and the changes in UE transmit timing are observed. The transmit timing is

verified by the UE transmitting SRS (Sounding Reference Symbols) used as a measurement reference facilitating the SS timing estimation.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Test 1 in Tables 7.1.5.5-1 and 7.1.5.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. The SS shall check that the UE transmit timing offset is within the limits specified in Table 7.1.5.5-3 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1.
6. The SS adjusts the downlink timing for Cell 1 to a delay of $+64 \times T_S$ (approximately +2 μs).
7. The test system samples the UE Transmit Timing once per SRS transmission (periodicity 2 subframes). To check Rule 1, the SS shall check that the maximum time adjustment step size $T_q$ between one SRS transmission (subframe n) to next (subframe n+2) is within Rule 1 as specified in clause 7.1.5.5. To check that the minimum adjustment rate is within Rule 2 as specified in clause 7.1.5.5, the SS shall measure the change in SRS transmission timing over a 1.572s sliding window, with step size 2ms. To check that the maximum adjustment rate is within Rule 3 as specified in clause 7.1.5.5, the SS shall measure the change in SRS transmission timing over a 198ms sliding window, with step size 2ms. The three rules apply until the UE transmit timing offset is within the limits specified in Table 7.1.5.5-3 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1.
8. The SS shall check that the UE transmit timing offset stays within the limits specified in Table 7.1.5.5-3 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1.

### 7.1.5.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.1.5.4.3-1: Common Exception messages for UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.4-2 |

**Table 7.1.5.4.3-2: *RRCConnectionReconfiguration*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.331 [5] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| radioResourceConfigDedicated SEQUENCE { | | | |
| mac-MainConfig CHOICE { | | | |
| explicitValue | MAC-MainConfig-RBC | | |
| } | | | |
| physicalConfigDedicated | PhysicalConfigDedicated - DEFAULT | | RBC |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.1.5.4.3-3: *SoundingRS-RL-ConfigCommon-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw5 | | |
| srs-SubframeConfig | sc1 | | |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | Not present | | |
| } | | | |

**Table 7.1.5.4.3-4: *SoudingRS-UL-ConfigDedicated-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | bw0 used with no frequency hopping. bw3 used with frequency hopping | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | 0 | | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 7.1.5.4.3-5: *CQI-ReportConfig-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm30 | This IE should be omitted for test 3 | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| release | NULL | | |
| } | | | |

**Table 7.1.5.4.3-6: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| } | | | |

## 7.1.5.5 Test requirement

Tables 7.1.5.5-1 and 7.1.5.5-2 define the primary settings including test tolerances for UE transmit timing accuracy for E-UTRAN FDD test.

Tables 7.1.5.5-3, 7.1.5.5-4 and 7.1.5.5-5 define the rules for adjustments made to the UE uplink timing include Test Tolerances.

**Table 7.1.5.5-1: Cell Specific Test requirement Parameters for UE transmit timing accuracy for E-UTRAN FDD test case for 5MHz bandwidth**

| **Parameter** | **Unit** | **Value** |
|---|---|---|
| | | **Test 1** |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 |
| PDCCH/PCFICH/PHICH Reference measurement channel[Note1] | | R.11 FDD |
| OCNG Pattern[Note2] | | OP.16 FDD |
| Io[Note4] | dBm/4.5 MHz | -68.27 |
| Note 1: For the reference measurement channels, see clause A.2.1.<br>Note 2: For the OCNG pattern, see clause D.1.16.<br>Note 3: See Table 7.1.1.5-1 for the other parameters.<br>Note 4: This test is according to the principle defined in TS 36.133 [4] section A.3.7.2. | | |

**Table 7.1.5.5-2: Sounding Reference Symbol Configuration to be used in UE transmit timing accuracy for E-UTRAN FDD test case for 5MHz bandwidth**

| Field | Test 1 | Comment |
|---|---|---|
| | Value | |
| srs-BandwidthConfig | bw5 | |
| srs-SubframeConfig | sc1 | |
| ackNackSRS-SimultaneousTransmission | FALSE | |
| srsMaxUpPts | N/A | Not applicable for FDD |
| srs-Bandwidth | 0 | No hopping |
| srs-HoppingBandwidth | hbw0 | |
| freqDomainPosition | 0 | |
| duration | TRUE | Indefinite duration |
| srs-ConfigIndex | 0 | SRS periodicity of 2ms and 80 ms for Test 1 and 2, respectively. |
| transmissionComb | 0 | |
| cyclicShift | cs0 | No cyclic shift |

The UE transmit timing offset shall be within the requirements in Table 7.1.5.5-3.

The reference point for the UE initial transmit timing control test requirement shall be the downlink timing minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$.

**Table 7.1.5.5-3: Test requirement for $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| ≥3 | $15*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9]. | |

The UE shall be capable of changing the transmission timing according to the received downlink frame. When the transmission timing error between the UE and the reference cell exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$.

All adjustments made to the UE uplink timing shall follow these rules:

1) The maximum adjustment step size $T_q$ shall be within the requirements in Table 7.1.5.5-4

2) The minimum aggregate adjustment rate shall be $5.7 \times T_S$ per 1.572 s.

3) The maximum aggregate adjustment rate shall be within the requirements in Table 7.1.5.5-5.

**Table 7.1.5.5-4: Test requirement for $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| 5 | 6.0*Ts |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

**Table 7.1.5.5-5: Test requirement for maximum aggregate adjustment rate per 198 ms**

| Downlink Bandwidth (MHz) | Maximum adjustment per 198 ms |
|---|---|
| 5 | 6.6*Ts |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

An illustration of the measurement principle is shown in Figure

# 7.1.6 E-UTRAN FDD – UE Transmit Timing Accuracy Tests for SCell in sTAG

## 7.1.6.1 Test purpose

To verify that the UE is capable of following the frame timing change of the connected System simulator and that the UE initial transmit timing accuracy, maximum amount of timing change in one adjustment, minimum and maximum adjustment rate are within the specified limits for SCell in sTAG based on the requirements.

## 7.1.6.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 11 and forward that supports Uplink Carrier Aggregation and multiple timing advances and support of FGI bit 5.

## 7.1.6.3 Minimum conformance requirements

The UE initial transmission timing error shall be less than or equal to $\pm T_e$ where the timing error limit value $T_e$ is specified in Table 7.1.6.3-1. This requirement applies when it is the first transmission in a DRX cycle for PUCCH, PUSCH and SRS or it is the PRACH transmission. The reference point for the UE initial transmit timing control requirement shall be the downlink timing of the reference cell minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$. The downlink timing is defined as the time when the first detected path (in time) of the corresponding downlink frame is received from the reference cell. $N_{\text{TA_Ref}}$ for PRACH is defined as 0. $(N_{\text{TA_Ref}} + N_{\text{TA offset}})$ (in $T_s$ units) for other channels is the difference between UE transmission timing and the Downlink timing immediately after when the last timing advance in clause 7.3 was applied. $N_{\text{TA_Ref}}$ for other channels is not changed until next timing advance is received.

**Table 7.1.6.3-1: $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| 1.4 | $24*T_S$ |
| ≥3 | $12*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 | |

When it is not the first transmission in a DRX cycle or there is no DRX cycle, and when it is the transmission for PUCCH, PUSCH and SRS transmission, the UE shall be capable of changing the transmission timing according to the received downlink frame of the reference cell except when the timing advance in clause 7.3 is applied. When the transmission timing error between the UE and the reference timing exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$, as long as, for the UE configured with a pTAG and an sTAG, the transmission timing difference between TAGs does not exceed the maximum transmission timing difference (i.e., 32.47us) after such adjustment. If the transmission timing difference is bigger than the maximum transmission timing difference (i.e., 32.47us) UE may stop adjustment. The reference timing shall be $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$ before the downlink timing of the reference cell. All adjustments made to the UE uplink timing shall follow these rules:

1) The maximum amount of the magnitude of the timing change in one adjustment shall be $T_q$ seconds.

2) The minimum aggregate adjustment rate shall be $7*T_S$ per second.

3) The maximum aggregate adjustment rate shall be $T_q$ per 200ms.

where the maximum autonomous time adjustment step $T_q$ is specified in Table 7.1.6.3-2.

**Table 7.1.6.3-2: $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| 1.4 | $17.5*T_S$ |
| 3 | $9.5*T_S$ |
| 5 | $5.5*T_S$ |
| ≥10 | $3.5*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 | |

The normative reference for this requirement is TS 36.133 [4] clause 7.1.2 and A.7.1.6.

## 7.1.6.4 Test description

### 7.1.6.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz for both PCell and SCell as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.45.

2. Propagation conditions are set according to Annex B clause B.0.

3. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier. PCell is in the Primary Timing Advance Group (pTAG) and SCell is in the secondary Timing Advance Group (sTAG). Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 shall be powered OFF.

### 7.1.6.4.2 Test procedure

The test consists of two cells, PCell and SCell. PCell and SCell are in the Primary Timing Advance Group (pTAG) and the secondary Timing Advance Group (sTAG) respectively. The transmit timing accuracy is verified related to the downlink frame timing of SCell (Cell 2) in sTAG. The downlink timing of Cell 2 is changed and the change in UE transmit timing of SCell is observed. The transmit timing is verified by the UE transmitting SRS (Sounding Reference Symbols) used as a measurement reference facilitating the SS timing estimation.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCell according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4 with UL CA.

4. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).

5. Set the parameters according to Test 1 and Test 2 in Tables 7.1.6.5-1, 7.1.6.5-2 and 7.1.6.5-3 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. The SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. The SS shall check that the UE transmit timing offsets of SCell in sTAG are within the limit specified in Table 7.1.6.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of the activated SCell (Cell 2).

9. The SS adjusts the downlink transmit timing for the activated SCell (Cell 2) to a delay of $+64 \times T_S$ (approximately +2 μs) compared to that in step 8. For Test 2, the downlink timing adjustment shall be done at (40 ± 15 )ms before the next DRX On duration period.

10. Step 10 applies for Test 1, but is omitted for Test 2. The test system samples the UE Transmit Timing once per SRS transmission (periodicity 2 subframes). To check Rule 1, the SS shall check that the maximum time adjustment step size $T_q$ between one SRS transmission (subframe n) to next (subframe n+2) is within Rule 1 as specified in clause 7.1.6.5. To check that the minimum adjustment rate is within Rule 2 as specified in clause 7.1.6.5, the SS shall measure the change in SRS transmission timing over a 1.002s sliding window, with step size 2ms. To check that the maximum adjustment rate is within Rule 3 as specified in clause 7.1.6.5, the SS shall measure the change in SRS transmission timing over a 198ms sliding window, with step size 2ms. The three rules apply until the UE transmit timing offsets of SCell are within the limit specified in Table 7.1.6.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of the activated SCell (Cell 2).

11. In case for Test 1, the SS shall check that the UE transmit timing offsets of the SCell in sTAG stay within the limit specified in Table 7.1.6.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of the activated SCell (Cell 2). In case for Test 2, the SS shall check that the UE transmit timing offsets of the activated SCell (Cell 2) stay within the limit specified in Table 7.1.6.5-4 with respect to the first timing adjustment after downlink frame of the activated SCell (Cell 2) is changed. For that purpose, the SRS transmission that takes place in the first DRX ON duration after downlink timing of the activated SCell (Cell 2) is changed is used.

12. The SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

13. After the RRC connection release, the SS switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 1-13 for each sub-test in Tables 7.1.6.5-1, 7.1.6.5-2 and 7.1.6.5-3 as appropriate.

### 7.1.6.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.1.6.4.3-1: *RRCConnectionReconfiguration*: Additional E-UTRAN FDD UE transmit timing accuracy test for SCell in sTAG requirement**

| Derivation Path: Clause 4.6.1 Table 4.6.1-8, condition SCell_AddMod |
|---|

**Table 7.1.6.4.3-2: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN FDD test for SCell in sTAG requirement for Test 1**

| Derivation Path: 36.508, Table 4.8.2.1.5-1, condition SCell_AddMod | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| stag-ToAddModList-r11 SEQUENCE (SIZE (1..maxSTAG-r11)) OF SEQUENCE { | Not present | | |
| stag-Id-r11 | 1 | | |
| timeAlignmentTimerSTAG-r11 | infinity | | |
| } | | | |
| } | | | |

**Table 7.1.6.4.3-3: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN FDD test for SCell in sTAG requirement for Test 2**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC, condition SCell_AddMod | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | DRX_S |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf80 typical value in simulations | |
| sf80 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| stag-ToAddModList-r11 SEQUENCE (SIZE (1..maxSTAG-r11)) OF SEQUENCE { | Not present | | |
| stag-Id-r11 | 1 | | |
| timeAlignmentTimerSTAG-r11 | infinity | | |
| } | | | |
| } | | | |

**Table 7.1.6.4.3-4: *SCellToAddMod-r10:* Additional UE transmit timing accuracy for E-UTRAN FDD test for SCell in sTAG requirement**

| Derivation Path: 36.508, Table 4.6.3-19D | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SCellToAddMod-r10 ::= SEQUENCE { | | | |
| sCellIndex-r10 | 1 | | |
| cellIdentification-r10 SEQUENCE { | | | |
| physCellId-r10 | PhysicalCellIdentity of Cell 2 | | |
| dl-CarrierFreq-r10 | Same downlink EARFCN as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 7.1.6.4.3-5: *RadioResourceConfigCommonSCell-r10:* Additional UE transmit timing accuracy for E-UTRAN FDD test for SCell in sTAG requirement**

| Derivation Path: 36.508, Table 4.6.3-13A, condition UL_CA | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigCommonSCell-r10-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigCommon-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw5 | | |
| srs-SubframeConfig | sc3 | | |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srs-MaxUpPts | Not Present | | |
| } | | | |
| } | | | |
| uplinkPowerControlCommonSCell-v1130 SEQUENCE | Not present | | |
| } | | | |

**Table 7.1.6.4.3-6: *RadioResourceConfigDedicatedSCell-r10:* Additional UE transmit timing accuracy for E-UTRAN FDD test for SCell in sTAG requirement**

| Derivation Path: 36.508, Table 4.6.3-19AA | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicatedSCell-r10 ::= SEQUENCE { | | | |
| physicalConfigDedicatedSCell-r10 | PhysicalConfigDedicated SCell-r10-DEFAULT | | |
| mac-MainConfigSCell-r11 SEQUENCE { | | | |
| stag-Id-r11 | 1 | | |
| } | | | |
| } | | | |

**Table 7.1.6.4.3-7: *PhysicalConfigDedicatedSCell-r10-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD test for SCell in sTAG requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-6A PhysicalConfigDedicatedSCell-r10-DEFAULT with UL_CA | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicatedSCell-r10-DEFAULT ::= SEQUENCE { | | | |
| ul-Configuration-r10 SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | bw0 used with no frequency hopping. | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | | Set according to specific test; 15 for Test 1 and 85 for Test 2 | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |
| soundingRS-UL-ConfigDedicated-v1020 SEQUENCE { | | | |
| srs-AntennaPort-r10 | an1 | | |
| } | | | |
| } | | | |
| } | | | |

## 7.1.6.5 Test requirement

Tables 7.1.6.5-1, 7.1.6.5-2 and 7.1.6.5-3 define the primary settings including test tolerances for UE transmit timing accuracy for SCell in sTAG for E-UTRAN FDD test.

Tables 7.1.6.5-4, 7.1.6.5-5 and 7.1.6.5-6 define the rules for adjustments made to the UE uplink timing include Test Tolerances.

**Table 7.1.6.5-1: Cell Specific Test requirement Parameters for UE transmit timing accuracy for SCell in sTAG for E-UTRAN FDD test case**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | Test 1 | Test 2 | Test 1 | Test 2 |
| E-UTRA RF Channel Number | | 1 | 1 | 2 | 2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | 10 | 10 | 10 |
| Active PCell | | Cell 1 | Cell 1 | | |
| Active SCell | | | | Cell 2 | Cell 2 |
| TAG configuration | | pTAG | pTAG | sTAG | sTAG |
| DRX cycle | ms | OFF | 80[Note5] | OFF | 80[Note5] |
| PDCCH/PCFICH/PHICH Reference measurement channel[Note1] | | R.6 FDD | R.6 FDD | R.6 FDD | R.6 FDD |
| OCNG Pattern[Note2] | | OP.2 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD |
| PBCH_RA | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| OCNG_RA[Note3] | | | | | |
| OCNG_RB[Note3] | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | -98 | -98 | -98 |
| $\hat{E}_s / I_{ot}$ | dB | 3.30 | 3.30 | 3.30 | 3.30 |
| $\hat{E}_s / N_{oc}$ | dB | 3.30 | 3.30 | 3.30 | 3.30 |
| Io[Note4] | dBm/9 MHz | -65.25 | -65.25 | -65.25 | -65.25 |
| Propagation condition | - | AWGN | AWGN | AWGN | AWGN |

Note 1: For the reference measurement channels, see clause A.2.1.
Note 2: For the OCNG pattern, see clause D.1.2.
Note 3: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: Io level has been derived from other parameters for information purpose. It is not a settable parameter.
Note 5: DRX related parameters are defined in Table 7.1.6.5-3.

**Table 7.1.6.5-2: Sounding Reference Symbol Configuration to be used in UE transmit timing accuracy for SCell in sTAG for E-UTRAN FDD test case**

| Field | Cell 1 | | Cell 2 | | Comment |
|---|---|---|---|---|---|
| | Test 1 | Test 2 | Test 1 | Test 2 | |
| srsBandwidthConfiguration | bw5 | bw5 | bw5 | bw5 | |
| srsSubframeConfiguration | sc3 | sc3 | sc3 | sc3 | Once every 5 subframes |
| ackNackSrsSimultaneousTransmission | FALSE | FALSE | FALSE | FALSE | |
| srsMaxUpPTS | FALSE | FALSE | FALSE | FALSE | |
| srsBandwidth | 0 | 0 | 0 | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | hbw0 | hbw0 | hbw0 | |
| frequencyDomainPosition | 0 | 0 | 0 | 0 | |
| duration | TRUE | TRUE | TRUE | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 15 | 85 | 15 | 85 | SRS periodicity of 10 and 80 ms for Test 1 and 2, respectively. |
| transmissionComb | 0 | 0 | 0 | 0 | |
| cyclicShift | cs0 | cs0 | cs0 | cs0 | No cyclic shift |
| srsAntennaPort | an1 | an1 | an1 | an1 | Number of SRS antenna ports |
| Note: For further information see clause 6.3.2 in TS 36.331 [5]. | | | | | |

**Table 7.1.6.5-3: drx-Configuration to be used in UE Transmit Timing Accuracy Test 2 for SCell in sTAG for E-UTRAN FDD**

| Field | Cell 1 | Cell 2 | Comment |
|---|---|---|---|
| onDurationTimer | psf1 | psf1 | |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf80 | sf80 | |
| shortDRX | disable | disable | |
| Note: For further information see clause 6.3.2 in TS 36.331 [5]. | | | |

After the SCell (Cell 2) is activated, the UE transmit timing offsets for SCell in sTAG shall be within the requirements in Table 7.1.6.5-4.

The reference point for the UE initial transmit timing control test requirement shall be the downlink timing of the activated SCell minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$.

**Table 7.1.6.5-4: Test requirement for $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| ≥3 | $15*T_S$ |
| NOTE: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

The UE shall be capable of changing the transmission timing according to the received downlink frame of the reference cell except when the timing advance in clause 7.3 TS 36.133 [4] is applied. When the transmission timing error between the UE and the reference timing exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$.

The reference timing shall be $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$ before the downlink timing of the reference cell. All adjustments made to the UE uplink timing shall follow these rules:

1) For test 1, the maximum adjustment step size $T_q$ shall be within the requirements in Table 7.1.6.5-5.

2) For test 1, the minimum aggregate adjustment rate shall be $3.4 \times T_S$ per 1.002 s.

3) For test 1, the maximum aggregate adjustment rate shall be within the requirements in Table 7.1.6.5-6.

**Table 7.1.6.5-5: Test requirement for $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| ≥10 | 4.0*$T_S$ |
| NOTE: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

**Table 7.1.6.5-6: Test requirement for maximum aggregate adjustment rate per 198 ms**

| Downlink Bandwidth (MHz) | Maximum adjustment per 198 ms |
|---|---|
| ≥10 | 4.6*$T_S$ |
| NOTE: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

## 7.1.7 E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell in sTAG

### 7.1.7.1 Test purpose

To verify that the UE is capable of following the frame timing change of the connected System simulator and that the UE initial transmit timing accuracy, maximum amount of timing change in one adjustment, minimum and maximum adjustment rate are within the specified limits for SCell in sTAG based on the requirements.

### 7.1.7.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 11 and forward that supports Uplink Carrier Aggregation. Applicability requires support for FGI bit 5.

### 7.1.7.3 Minimum conformance requirements

The UE initial transmission timing error shall be less than or equal to $\pm T_e$ where the timing error limit value $T_e$ is specified in Table 7.1.7.3-1. This requirement applies when it is the first transmission in a DRX cycle for PUCCH, PUSCH and SRS or it is the PRACH transmission. The reference point for the UE initial transmit timing control requirement shall be the downlink timing of the reference cell minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$. The downlink timing is defined as the time when the first detected path (in time) of the corresponding downlink frame is received from the reference cell. $N_{\text{TA_Ref}}$ for PRACH is defined as 0. $(N_{\text{TA_Ref}} + N_{\text{TA offset}})$ (in $T_s$ units) for other channels is the difference between UE transmission timing and the Downlink timing immediately after when the last timing advance in clause 7.3 was applied. $N_{\text{TA_Ref}}$ for other channels is not changed until next timing advance is received.

**Table 7.1.7.3-1: $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| 1.4 | 24*$T_S$ |
| ≥3 | 12*$T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 | |

When it is not the first transmission in a DRX cycle or there is no DRX cycle, and when it is the transmission for PUCCH, PUSCH and SRS transmission, the UE shall be capable of changing the transmission timing according to the received downlink frame of the reference cell except when the timing advance in clause 7.3 is applied. When the transmission timing error between the UE and the reference timing exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$, as long as, for the UE configured with a pTAG and an sTAG, the transmission timing difference between TAGs does not exceed the maximum transmission timing difference (i.e., 32.47us) after such adjustment. If the transmission timing difference is bigger than the maximum transmission timing difference (i.e., 32.47us) UE may stop adjustment. The reference timing shall be $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$ before the downlink timing of the reference cell. All adjustments made to the UE uplink timing shall follow these rules:

1) The maximum amount of the magnitude of the timing change in one adjustment shall be $T_q$ seconds.

2) The minimum aggregate adjustment rate shall be 7*$T_S$ per second.

3) The maximum aggregate adjustment rate shall be $T_q$ per 200ms.

where the maximum autonomous time adjustment step $T_q$ is specified in Table 7.1.7.3-2.

**Table 7.1.7.3-2: $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_q$ |
|---|---|
| 1.4 | 17.5*$T_S$ |
| 3 | 9.5*$T_S$ |
| 5 | 5.5*$T_S$ |
| ≥10 | 3.5*$T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 | |

The normative reference for this requirement is TS 36.133 [4] clause 7.1.2 and A.7.1.7.

## 7.1.7.4 Test description

### 7.1.7.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz for both PCell and SCell as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.45.
2. Propagation conditions are set according to Annex B clause B.0.
3. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier. PCell is in the Primary Timing Advance Group (pTAG) and SCell is in the secondary Timing Advance Group (sTAG). Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 shall be powered OFF.

### 7.1.7.4.2 Test procedure

The test consists of two cells, PCell and SCell. PCell and SCell are in the Primary Timing Advance Group (pTAG) and the secondary Timing Advance Group (sTAG) respectively. The transmit timing accuracy is verified related to the downlink frame timing of SCell (Cell 2) in sTAG. The downlink timing of Cell 2 is changed and the change in UE transmit timing of SCell is observed. The transmit timing is verified by the UE transmitting SRS (Sounding Reference Symbols) used as a measurement reference facilitating the SS timing estimation.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCell according to Annex C.0 and C.1 for all downlink physical channels.
3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4 with UL CA.
4. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).
5. Set the parameters according to Test 1 and Test 2 in Tables 7.1.7.5-1, 7.1.7.5-2 and 7.1.7.5-3 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
6. The SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
7. The UE shall transmit RRCConnectionReconfigurationComplete message.
8. The SS shall check that the UE transmit timing offsets of SCell in sTAG are $624 \times T_S$ to within the $T_e$ limit specified in Table 7.1.7.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of the activated SCell (Cell 2).

9. The SS adjusts the downlink transmit timing for the activated SCell (Cell 2) to a delay of +64 × $T_S$ (approximately +2 μs) compared to that in step 8. For Test 2, the downlink timing adjustment shall be done at (40 ± 15 )ms before the next DRX On duration period.

10. Step 10 applies for Test 1, but is omitted for Test 2. The test system samples the UE Transmit Timing once per SRS transmission (periodicity 10 subframes). To check Rule 1, the SS shall check that the maximum time adjustment step size $T_q$ between one SRS transmission (subframe n) to next (subframe n+10) is within Rule 1 as specified in clause 7.1.7.5. To check that the minimum adjustment rate is within Rule 2 as specified in clause 7.1.7.5, the SS shall measure the change in SRS transmission timing over a 1.01s sliding window, with step size 10ms. To check that the maximum adjustment rate is within Rule 3 as specified in clause 7.1.7.5, the SS shall measure the change in SRS transmission timing over a 190ms sliding window, with step size 10ms. The three rules apply until the UE transmit timing offsets of SCell are（624 × $T_S$) to within the limit specified in Table 7.1.7.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of the activated SCell (Cell 2).

11. In case for Test 1, the SS shall check that the UE transmit timing offsets of the SCell in sTAG stay at（624 × $T_S$) to within the $T_e$ limit specified in Table 7.1.7.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of the activated SCell (Cell 2). In case for Test 2, the SS shall check that the UE transmit timing offsets of the activated SCell (Cell 2) stay within the limit specified in Table 7.1.7.5-4 with respect to the first timing adjustment after downlink frame of the activated SCell (Cell 2) is changed. For that purpose, the SRS transmission that takes place in the first DRX ON duration after downlink timing of the activated SCell (Cell 2) is changed is used.

12. The SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

13. After the RRC connection release, the SS switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 1-13 for each sub-test in Tables 7.1.7.5-1, 7.1.7.5-2 and 7.1.7.5-3 as appropriate.

### 7.1.7.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.1.7.4.3-1: *RRCConnectionReconfiguration*: Additional E-UTRAN TDD UE transmit timing accuracy test for SCell in sTAG requirement**

| Derivation Path: Clause 4.6.1 Table 4.6.1-8, condition SCell_AddMod |
|---|

**Table 7.1.7.4.3-2: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN TDD test for SCell in sTAG requirement for Test 1**

| Derivation Path: 36.508, Table 4.8.2.1.5-1, condition SCell_AddMod | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| stag-ToAddModList-r11 SEQUENCE (SIZE (1..maxSTAG-r11)) OF SEQUENCE { | Not present | | |
| stag-Id-r11 | 1 | | |
| timeAlignmentTimerSTAG-r11 | infinity | | |
| } | | | |
| } | | | |

**Table 7.1.7.4.3-3: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN TDD test for SCell in sTAG requirement for Test 2**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC, condition SCell_AddMod | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | DRX_S |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf80 typical value in simulations | |
| sf80 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| stag-ToAddModList-r11 SEQUENCE (SIZE (1..maxSTAG-r11)) OF SEQUENCE { | Not present | | |
| stag-Id-r11 | 1 | | |
| timeAlignmentTimerSTAG-r11 | infinity | | |
| } | | | |
| } | | | |

**Table 7.1.7.4.3-4: *SCellToAddMod-r10:* Additional UE transmit timing accuracy for E-UTRAN TDD test for SCell in sTAG requirement**

| Derivation Path: 36.508, Table 4.6.3-19D | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SCellToAddMod-r10 ::= SEQUENCE { | | | |
| sCellIndex-r10 | 1 | | |
| cellIdentification-r10 SEQUENCE { | | | |
| physCellId-r10 | PhysicalCellIdentity of Cell 2 | | |
| dl-CarrierFreq-r10 | Same downlink EARFCN as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 7.1.7.4.3-5: *RadioResourceConfigCommonSCell-r10:* Additional UE transmit timing accuracy for E-UTRAN TDD test for SCell in sTAG requirement**

| Derivation Path: 36.508, Table 4.6.3-13A, condition UL_CA | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigCommonSCell-r10-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigCommon-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw5 | | |
| srs-SubframeConfig | sc3 | | |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srs-MaxUpPts | Not Present | | |
| } | | | |
| } | | | |
| uplinkPowerControlCommonSCell-v1130 SEQUENCE | Not present | | |
| } | | | |

**Table 7.1.7.4.3-6: *RadioResourceConfigDedicatedSCell-r10:* Additional UE transmit timing accuracy for E-UTRAN TDD test for SCell in sTAG requirement**

| Derivation Path: 36.508, Table 4.6.3-19AA | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicatedSCell-r10 ::= SEQUENCE { | | | |
| physicalConfigDedicatedSCell-r10 | PhysicalConfigDedicated SCell-r10-DEFAULT | | |
| mac-MainConfigSCell-r11 SEQUENCE { | | | |
| stag-Id-r11 | 1 | | |
| } | | | |
| } | | | |

**Table 7.1.7.4.3-7: *PhysicalConfigDedicatedSCell-r10-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN TDD test for SCell in sTAG requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-6A PhysicalConfigDedicatedSCell-r10-DEFAULT with UL_CA | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicatedSCell-r10-DEFAULT ::= SEQUENCE { | | | |
| ul-Configuration-r10 SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | bw0 used with no frequency hopping. | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | | Set according to specific test; 15 for Test 1 and 85 for Test 2 | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |
| soundingRS-UL-ConfigDedicated-v1020 SEQUENCE { | | | |
| srs-AntennaPort-r10 | an1 | | |
| } | | | |
| } | | | |
| } | | | |

## 7.1.7.5 Test requirement

Tables 7.1.7.5-1, 7.1.7.5-2 and 7.1.7.5-3 define the primary settings including test tolerances for UE transmit timing accuracy for SCell in sTAG for E-UTRAN TDD test.

Tables 7.1.7.5-4, 7.1.7.5-5 and 7.1.7.5-6 define the rules for adjustments made to the UE uplink timing include Test Tolerances.

**Table 7.1.7.5-1: Cell Specific Test requirement Parameters for UE transmit timing accuracy for SCell in sTAG for E-UTRAN TDD test case**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | Test 1 | Test 2 | Test 1 | Test 2 |
| E-UTRA RF Channel Number | | 1 | 1 | 2 | 2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | 10 | 10 | 10 |
| E-UTRA RF Channel Number | | 1 | 1 | 2 | 2 |
| Active PCell | | Cell 1 | Cell 1 | | |
| Active SCell | | | | Cell 2 | Cell 2 |
| TAG configuration | | pTAG | pTAG | sTAG | sTAG |
| Special subframe configuration[Note1] | | 6 | 6 | 6 | 6 |
| Uplink-downlink configuration[Note2] | | 1 | 1 | 1 | 1 |
| DRX cycle | ms | OFF | 80[Note7] | OFF | 80[Note7] |
| PDCCH/PCFICH/PHICH Reference measurement channel[Note3] | | R.6 TDD | R.6 TDD | R.6 TDD | R.6 TDD |
| OCNG Pattern[Note4] | | OP.2 TDD | OP.2 TDD | OP.2 TDD | OP.2 TDD |
| PBCH_RA | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | 0 | 0 | | 0 |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| OCNG_RA[Note5] | | | | | |
| OCNG_RB[Note5] | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | -98 | -98 | -98 |
| $\hat{E}_s / I_{ot}$ | dB | 3.30 | 3.30 | 3.30 | 3.30 |
| $\hat{E}_s / N_{oc}$ | dB | 3.30 | 3.30 | 3.30 | 3.30 |
| Io[Note6] | dBm/9 MHz | --65.25 | --65.25 | --65.25 | --65.25 |
| Propagation condition | - | AWGN | AWGN | AWGN | AWGN |

Note 1: For the special subframe configuration see table 4.2-1 in TS 36.211 [9].
Note 2: For the uplink-downlink configuration see table 4.2-2 in TS 36.211 [9].
Note 3: For the reference measurement channels, see clause A.2.2.
Note 4: For the OCNG pattern, see clause D.2.2.
Note 5: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 6: Io level has been derived from other parameters for information purpose. It is not a settable parameter.
Note 7: DRX related parameters are defined in Table 7.1.7.5-3.

**Table 7.1.7.5-2: Sounding Reference Symbol Configuration to be used in UE transmit timing accuracy for SCell in sTAG for E-UTRAN TDD test case**

| Field | Cell 1 | | Cell 2 | | Comment |
|---|---|---|---|---|---|
| | Test 1 | Test 2 | Test 1 | Test 2 | |
| srsBandwidthConfiguration | bw5 | bw5 | bw5 | bw5 | |
| srsSubframeConfiguration | sc3 | sc3 | sc3 | sc3 | Once every 5 subframes |
| ackNackSrsSimultaneousTransmission | FALSE | FALSE | FALSE | FALSE | |
| srsMaxUpPTS | FALSE | FALSE | FALSE | FALSE | |
| srsBandwidth | 0 | 0 | 0 | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | hbw0 | hbw0 | hbw0 | |
| frequencyDomainPosition | 0 | 0 | 0 | 0 | |
| duration | TRUE | TRUE | TRUE | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 15 | 85 | 15 | 85 | SRS periodicity of 10 and 80 ms for Test 1 and 2, respectively. |
| transmissionComb | 0 | 0 | 0 | 0 | |
| cyclicShift | cs0 | cs0 | cs0 | cs0 | No cyclic shift |
| srsAntennaPort | an1 | an1 | an1 | an1 | Number of SRS antenna ports |
| Note: For further information see clause 6.3.2 in TS 36.331 [5]. | | | | | |

**Table 7.1.7.5-3: drx-Configuration to be used in UE Transmit Timing Accuracy Test 2 for SCell in sTAG for E-UTRAN TDD**

| Field | Cell 1 | Cell 2 | Comment |
|---|---|---|---|
| onDurationTimer | psf1 | psf1 | |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf80 | sf80 | |
| shortDRX | disable | disable | |
| Note: For further information see clause 6.3.2 in TS 36.331 [5]. | | | |

After the SCell (Cell 2) is activated, the UE transmit timing offsets for SCell in sTAG shall be within the requirements in Table 7.1.7.5-4.

The reference point for the UE initial transmit timing control test requirement shall be the downlink timing of the activated SCell minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$.

**Table 7.1.7.5-4: Test requirement for $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| ≥3 | $15*T_S$ |
| NOTE: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

The UE shall be capable of changing the transmission timing according to the received downlink frame of the reference cell except when the timing advance in clause 7.3 TS 36.133 [4] is applied. When the transmission timing error between the UE and the reference timing exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$.

The reference timing shall be $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$ before the downlink timing of the reference cell. All adjustments made to the UE uplink timing shall follow these rules:

1) For test 1, the maximum adjustment step size $T_q$ shall be within the requirements in Table 7.1.7.5-5

2) For test 1, the minimum aggregate adjustment rate shall be $3.4 \times T_S$ per 1.01 s.

3) For test 1, the maximum aggregate adjustment rate shall be within the requirements in Table 7.1.7.5-6.

**Table 7.1.7.5-5: Test requirement for $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| ≥10 | $4.0*T_S$ |
| NOTE: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

**Table 7.1.7.5-6: Test requirement for maximum aggregate adjustment rate per 190 ms**

| Downlink Bandwidth (MHz) | Maximum adjustment per 190 ms |
|---|---|
| ≥10 | $4.6*T_S$ |
| NOTE: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

## 7.1.7A E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell in sTAG for 20MHz +20MHz bandwidth

### 7.1.7A.1 Test purpose

Same test purpose as defined in clause 7.1.7.1.

### 7.1.7A.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 11 and forward that supports Uplink Carrier Aggregation and multiple timing advances. Applicability requires support for FGI bit 5.

### 7.1.7A.3 Minimum conformance requirements

Same minimum conformance requirements as defined in clause 7.1.7.3 with the following exception:

- The normative reference for this requirement is TS 36.133 [4] clause 7.1.2 and A.7.1.7A.

### 7.1.7A.4 Test description

#### 7.1.7A.4.1 Initial conditions

Same initial conditions as in clause 7.1.7.4.1 with the following exceptions:

- Channel Bandwidth to be tested: 20 MHz for both PCell and SCell

#### 7.1.7A.4.2 Test procedure

Same test procedure as in clause 7.1.7.4.2

#### 7.1.7A.4.3 Message contents

Same message contents as in clause 7.1.7.4.3

### 7.1.7A.5 Test requirement

Table 7.1.7A.5-1, 7.1.7A.5-2 and 7.1.7A.5-3 define the primary settings including test tolerances for UE transmit timing accuracy for SCell in sTAG for E-UTRAN TDD test. The listed parameter values in Tables 7.1.7A.5-1 will replace the values of corresponding parameters in Table 7.1.7.5-1 other parameters keep the same.

Table 7.1.7A.5-4, 7.1.7A.5-5 and 7.1.7A.5-6 define the rules for adjustments made to the UE uplink timing include Test Tolerances.

**Table 7.1.7A.5-1: Cell Specific Test requirement Parameters for UE transmit timing accuracy for SCell in sTAG for 20MHz +20MHz**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | Test 1 | Test 2 | Test 1 | Test 2 |
| E-UTRA RF Channel Number | | 1 | 1 | 2 | 2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 20 | 20 | 20 | 20 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2 | | R.10 TDD | R.10 TDD | R.10 TDD | R.10 TDD |
| OCNG Pattern defined in D.2.8 (OP.8 TDD) | | OP.8 TDD | OP.8 TDD | OP.8 TDD | OP.8 TDD |
| Io Note1 | dBm/18 MHz | -62.24 | -62.24 | -62.24 | -62.24 |
| Note 1: Io level has been derived from other parameters for information purpose. It is not a settable parameter. | | | | | |

**Table 7.1.7A.5-2: Sounding Reference Symbol Configuration to be used in UE transmit timing accuracy for SCell in sTAG for 20MHz +20MHz**

| Field | Cell 1 | | Cell 2 | | Comment |
|---|---|---|---|---|---|
| | Test 1 | Test 2 | Test 1 | Test 2 | |
| srsBandwidthConfiguration | bw5 | bw5 | bw5 | bw5 | |
| srsSubframeConfiguration | sc3 | sc3 | sc3 | sc3 | Once every 5 subframes |
| ackNackSrsSimultaneousTransmission | FALSE | FALSE | FALSE | FALSE | |
| srsMaxUpPTS | FALSE | FALSE | FALSE | FALSE | |
| srsBandwidth | 0 | 0 | 0 | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | hbw0 | hbw0 | hbw0 | |
| frequencyDomainPosition | 0 | 0 | 0 | 0 | |
| duration | TRUE | TRUE | TRUE | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 15 | 85 | 15 | 85 | SRS periodicity of 10 and 80 ms for Test 1 and 2, respectively. |
| transmissionComb | 0 | 0 | 0 | 0 | |
| cyclicShift | cs0 | cs0 | cs0 | cs0 | No cyclic shift |
| srsAntennaPort | an1 | an1 | an1 | an1 | Number of SRS antenna ports |
| Note: For further information see clause 6.3.2 in TS 36.331 [5]. | | | | | |

**Table 7.1.7A.5-3: drx-Configuration to be used in UE Transmit Timing Accuracy Test 2 for SCell in sTAG for 20MHz +20MHz**

| Field | Cell 1 | Cell 2 | Comment |
|---|---|---|---|
| onDurationTimer | psf1 | psf1 | |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf80 | sf80 | |
| shortDRX | disable | disable | |
| Note: For further information see clause 6.3.2 in TS 36.331 [5]. | | | |

After the SCell (Cell 2) is activated, the UE transmit timing offsets for SCell in sTAG shall be within the requirements in Table 7.1.7A.5-4.

The reference point for the UE initial transmit timing control test requirement shall be the downlink timing of the activated SCell minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$.

**Table 7.1.7A.5-4: Test requirement for $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| ≥3 | $15*T_S$ |
| NOTE: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

The UE shall be capable of changing the transmission timing according to the received downlink frame of the reference cell except when the timing advance in clause 7.3 TS 36.133 [4] is applied. When the transmission timing error between the UE and the reference timing exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$.

The reference timing shall be $(N_{\mathrm{TA_Ref}} + N_{\mathrm{TA\,offset}}) \times T_s$ before the downlink timing of the reference cell. All adjustments made to the UE uplink timing shall follow these rules:

1. For test 1, the maximum adjustment step size $T_q$ shall be within the requirements in Table 7.1.7A.5-5.
2. For test 1, the minimum aggregate adjustment rate shall be $3.4 \times T_S$ per 1.01 s.
3. For test 1, the maximum aggregate adjustment rate shall be within the requirements in Table 7.1.7A.5-6.

**Table 7.1.7A.5-5: Test requirement for $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| ≥10 | $4.0*T_S$ |
| NOTE: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

**Table 7.1.7A.5-6: Test requirement for maximum aggregate adjustment rate per 190 ms**

| Downlink Bandwidth (MHz) | Maximum adjustment per 190 ms |
|---|---|
| ≥10 | $4.6*T_S$ |
| NOTE: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

# 7.1.7B E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell in sTAG for 20MHz +10MHz bandwidth

## 7.1.7B.1 Test purpose

Same test purpose as defined in clause 7.1.7.1.

## 7.1.7B.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 11 and forward that supports Uplink Carrier Aggregation and multiple timing advances. Applicability requires support for FGI bit 5.

## 7.1.7B.3 Minimum conformance requirements

Same minimum conformance requirements as defined in clause 7.1.7.3 with the following exception:

- The normative reference for this requirement is TS 36.133 [4] clause 7.1.2 and A.7.1.7B.

## 7.1.7B.4 Test description

### 7.1.7B.4.1 Initial conditions

Same initial conditions as in clause 7.1.7.4.1 with the following exceptions:

- Channel Bandwidth to be tested: 20 MHz for PCell and 10 MHz for SCell

### 7.1.7B.4.2 Test procedure

Same test procedure as in clause 7.1.7.4.2.

### 7.1.7B.4.3 Message contents

Same message contents as in clause 7.1.7.4.3.

## 7.1.7B.5 Test requirement

Table 7.1.7B.5-1, 7.1.7B.5-2 and 7.1.7B.5-3 define the primary settings including test tolerances for UE transmit timing accuracy for SCell in sTAG for E-UTRAN TDD test. The listed parameter values in Tables 7.1.7B.5-1 will replace the values of corresponding parameters in Table 7.1.7.5-1 other parameters keep the same.

Table 7.1.7B.5-4, 7.1.7B.5-5 and 7.1.7B.5-6 define the rules for adjustments made to the UE uplink timing include Test Tolerances.

**Table 7.1.7B.5-1: Cell Specific Test requirement Parameters for UE transmit timing accuracy for SCell in sTAG for 20MHz +10MHz**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | Test 1 | Test 2 | Test 1 | Test 2 |
| E-UTRA RF Channel Number | | 1 | 1 | 2 | 2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 20 | 20 | 10 | 10 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2 | | R.10 TDD | R.10 TDD | R.6 TDD | R.6 TDD |
| OCNG Pattern defined in D.2.2 (OP.2 TDD) and in D.2.8 (OP.8 TDD) | | OP.8 TDD | OP.8 TDD | OP.2 TDD | OP.2 TDD |
| $Io^{Note1}$ | dBm/18 MHz | -62.24 | -62.24 | - | - |
| | dBm/9 MHz | - | - | -65.25 | -65.25 |
| Note 1: Io level has been derived from other parameters for information purpose. It is not a settable parameter. | | | | | |

**Table 7.1.7B.5-2: Sounding Reference Symbol Configuration to be used in UE transmit timing accuracy for SCell in sTAG for 20MHz +10MHz**

| Field | Cell 1 | | Cell 2 | | Comment |
|---|---|---|---|---|---|
| | Test 1 | Test 2 | Test 1 | Test 2 | |
| srsBandwidthConfiguration | bw5 | bw5 | bw5 | bw5 | |
| srsSubframeConfiguration | sc3 | sc3 | sc3 | sc3 | Once every 5 subframes |
| ackNackSrsSimultaneousTransmission | FALSE | FALSE | FALSE | FALSE | |
| srsMaxUpPTS | FALSE | FALSE | FALSE | FALSE | |
| srsBandwidth | 0 | 0 | 0 | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | hbw0 | hbw0 | hbw0 | |
| frequencyDomainPosition | 0 | 0 | 0 | 0 | |
| duration | TRUE | TRUE | TRUE | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 15 | 85 | 15 | 85 | SRS periodicity of 10 and 80 ms for Test 1 and 2, respectively. |
| transmissionComb | 0 | 0 | 0 | 0 | |
| cyclicShift | cs0 | cs0 | cs0 | cs0 | No cyclic shift |
| srsAntennaPort | an1 | an1 | an1 | an1 | Number of SRS antenna ports |
| Note: For further information see clause 6.3.2 in TS 36.331 [5]. | | | | | |

**Table 7.1.7B.5-3: drx-Configuration to be used in UE Transmit Timing Accuracy Test 2 for SCell in sTAG for 20MHz +10MHz**

| Field | Cell 1 | Cell 2 | Comment |
|---|---|---|---|
| onDurationTimer | psf1 | psf1 | |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf80 | sf80 | |
| shortDRX | disable | disable | |
| Note: For further information see clause 6.3.2 in TS 36.331 [5]. | | | |

After the SCell (Cell 2) is activated, the UE transmit timing offsets for SCell in sTAG shall be within the requirements in Table 7.1.7B.5-4.

The reference point for the UE initial transmit timing control test requirement shall be the downlink timing of the activated SCell minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$.

**Table 7.1.7B.5-4: Test requirement for $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| ≥3 | $15*T_S$ |
| NOTE: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

The UE shall be capable of changing the transmission timing according to the received downlink frame of the reference cell except when the timing advance in clause 7.3 TS 36.133 [4] is applied. When the transmission timing error between the UE and the reference timing exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$.

The reference timing shall be $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$ before the downlink timing of the reference cell. All adjustments made to the UE uplink timing shall follow these rules:

1. For test 1, the maximum adjustment step size $T_q$ shall be within the requirements in Table 7.1.7B.5-5
2. For test 1, the minimum aggregate adjustment rate shall be $3.4 \times T_S$ per 1.01 s.

3. For test 1, the maximum aggregate adjustment rate shall be within the requirements in Table 7.1.7B.5-6.

**Table 7.1.7B.5-5: Test requirement for $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| ≥10 | 4.0*$T_S$ |
| NOTE: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

**Table 7.1.7B.5-6: Test requirement for maximum aggregate adjustment rate per 190 ms**

| Downlink Bandwidth (MHz) | Maximum adjustment per 190 ms |
|---|---|
| ≥10 | 4.6*$T_S$ |
| NOTE: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

## 7.1.8 Void

## 7.1.9 Void

## 7.1.10 E-UTRAN FDD – UE Transmit Timing Accuracy Tests for Cat-M1 UE in CEModeA

### 7.1.10.1 Test purpose

To verify that the Cat-M1 UE in CEModeA have the capability to follow the frame timing change of the connected System Simulator. The method used is that the UE initial transmit timing accuracy, the maximum amount of timing change in one adjustment, and the minimum and maximum adjustment rate are within the specified limits based on the requirements.

### 7.1.10.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 13 and forward of UE category M1. Applicability requires support for FGI bit 5.

### 7.1.10.3 Minimum conformance requirements

The UE initial transmission timing error shall be less than or equal to $\pm T_e$ where the timing error limit value $T_e$ is defined in TS 36.133 [4] clause 7.1.2 and shown in table 7.1.10.3-1. This requirement applies when it is the first transmission in a DRX, eDRX_CONN cycle, or the first transmission in a repetition period (R>1) for PUCCH, PUSCH and SRS or it is the PRACH transmission. The reference point for the UE initial transmit timing control requirement shall be the downlink timing of the reference cell minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$. The downlink timing is defined as the time when the first detected path (in time) of the corresponding downlink frame is received from the reference cell. $N_{\text{TA_Ref}}$ for PRACH is defined as 0. $(N_{\text{TA_Ref}} + N_{\text{TA offset}})$ (in $T_s$ units) for other channels is the difference between UE transmission timing and the Downlink timing immediately after when the last timing advance in clause 7.3 was applied. $N_{\text{TA_Ref}}$ for other channels is not changed until next timing advance is received.

**Table 7.1.10.3-1: $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| 1.4 | 24*$T_S$ |
| ≥3 | 12*$T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

Cat-M1 UEs configured with CEModeA shall follow the requirement $T_e$ defined for 1.4MHz in Table 7.1.10.3-1 regardless the downlink carrier bandwidth *(Editor's note: It is FFS for the Cat-M1 UEs configured with CEModeB).*

When it is not the first transmission in a DRX or eDRX_CONN cycle or there is no DRX or no eDRX_CONN cycle, and when it is the transmission for PUCCH, PUSCH and SRS transmission, the UE shall, when no repetitions are configured on the uplink or the repetition period is R=1, be capable of changing the transmission timing according to the received downlink frame of the reference cell except when the timing advance in clause 7.3 is applied.

When in a TAG the transmission timing error between the UE and the reference timing exceeds $\pm T_e$, or in a sTAG the UE changes the downlink SCell for deriving the UE transmit timing for cells in the sTAG configured with one or two uplinks, the UE is required to adjust its timing to within $\pm T_e$ in that TAG, as long as,

- the UE is configured with a pTAG and one or two sTAG, the transmission timing difference between TAGs does not exceed the maximum transmission timing difference (i.e., 32.47us) after such adjustment, or
- the UE is configured with synchronous dual connectivity, the transmission timing difference between pTAG and psTAG does not exceed the maximum transmission timing difference (i.e., 35.21us) after such adjustment.

If the transmission timing difference after such adjustment is bigger than the maximum transmission timing difference UE may stop adjustment in this TAG. The reference timing shall be $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$ before the downlink timing of the reference cell. All adjustments made to the UE uplink timing under the above mentioned scenarios shall follow these rules:

1) The maximum amount of the magnitude of the timing change in one adjustment shall be $T_q$ seconds.
2) The minimum aggregate adjustment rate shall be 7 * $T_S$ per second.
3) The maximum aggregate adjustment rate shall be $T_q$ per 200 ms.

Where the maximum autonomous time adjustment step $T_q$ is defined in TS 36.133 [4] clause 7.1.2 and shown in table 7.1.10.3-2.

**Table 7.1.10.3-2: $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| 1.4 | 17.5*$T_S$ |
| 3 | 9.5*$T_S$ |
| 5 | 5.5*$T_S$ |
| ≥10 | 3.5*$T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

Cat-M1 UEs configured with CEModeA shall follow the requirement $T_q$ defined for 1.4MHz in Table 7.1.2-2 regardless the downlink carrier bandwidth *(Editor's note: It is FFS for the Cat-M1 UEs configured with CEModeB).*

When a repetition period is configured on the uplink, the UE shall not adjust the uplink transmission timing autonomously during an ongoing repetition period for which R>1.

The normative reference for this requirement is TS 36.133 [4] clause 7.1 and A.7.1.10.

NOTE 1: Both the allowed UE frequency error and the UE time adjustment quantisation can affect the result of the test, and it is not possible to test the UE Transmit Timing Accuracy in isolation from these core requirements. They are therefore taken into account when setting the Test Tolerances and the Test Requirements.

## 7.1.10.4 Test description

### 7.1.10.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18(using only main UE Tx/Rx antenna).

2. Propagation conditions are set according to Annex B clause B.0.

3. There is one E-UTRA FDD carrier and one cell specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

### 7.1.10.4.2 Test procedure

The test consists of a single cell. The transmit timing accuracy is verified related to the downlink frame timing of Cell 1. The downlink timing of Cell 1 is changed and the changes in UE transmit timing are observed. The transmit timing is verified by the UE transmitting SRS used as a measurement reference facilitating the SS timing estimation.

1. Ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3AA.

2. Set the parameters according to Test 1, Test 2, or Test 3 in Tables 7.1.10.5-1, 7.1.10.5-2 and 7.1.10.5-3 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. The SS shall check that the UE transmit timing offset is within the limits specified in Table 7.1.10.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1.

6. The SS adjusts the downlink timing for Cell 1 to a delay of $+64 \times T_S$ (approximately +2 μs) for Test 1 and Test2 and to and a delay of $+32 \times T_S$ for Test 3 compared to that in step 5. For Test 2 and Test 3, the downlink timing adjustment shall be done at (40 ± 15 )ms before the next DRX On duration period.

7. Step 7 applies for Test 1, but is omitted for Test 2 and Test 3. The test system samples the UE Transmit Timing once per SRS transmission (periodicity 2 subframes). To check Rule 1, the SS shall check that the maximum time adjustment step size $T_q$ between one SRS transmission (subframe n) to next (subframe n+2) is within Rule 1 as specified in clause 7.1.10.5. To check that the minimum adjustment rate is within Rule 2 as specified in clause 7.1.1.5, the SS shall measure the change in SRS transmission timing over a 1.002s sliding window, with step size 2ms. To check that the maximum adjustment rate is within Rule 3 as specified in clause 7.1.10.5, the SS shall measure the change in SRS transmission timing over a 198ms sliding window, with step size 2ms. The 0three rules apply until the UE transmit timing offset is within the limits specified in Table 7.1.10.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1.

8. In case for Test 1 and Test 3, the SS shall check that the UE transmit timing offset stays within the limits specified in Table 7.1.10.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1. In case for Test 2 and Test 3, the SS shall check that the UE transmit timing offset stays within the limits specified in Table 7.1.10.5-4 For that purpose, the SRS transmission that takes place in the first DRX ON duration after downlink timing of Cell 1 is changed is used.

9. The SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

10. After the RRC connection release, the SS switches off and on the UE and ensures the UE is in State 3A-RF –CE according to TS 36.508 [7] clause 7.2A.3AA.

11. Repeat step 1-10 for each sub-test in Tables 7.1.10.5-1, 7.1.10.5-2 and 7.1.10.5-3 as appropriate.

### 7.1.10.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the condition CEModeA with the following exceptions:

**Table 7.1.10.4.3-1: Common Exception messages for UE transmit timing accuracy for E-UTRAN FDD Cat-M1 UE in CEModeA test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.4-2 |

**Table 7.1.10.4.3-2: *RRCConnectionReconfiguration*: Additional UE transmit timing accuracy for E-UTRAN FDD Cat-M1 UE in CEModeA test requirement**

| Derivation Path: TS 36.331 [5] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| radioResourceConfigDedicated SEQUENCE { | | | |
| mac-MainConfig CHOICE { | | | |
| explicitValue | MAC-MainConfig-RBC | | |
| } | | | |
| physicalConfigDedicated | PhysicalConfigDedicated - DEFAULT | | RBC |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.1.10.4.3-3: *SoundingRS-RL-ConfigCommon-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD Cat-M1 UE in CEModeA test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw5 | | |
| srs-SubframeConfig | | Set according to specific test; sc1 for Test 1 and sc3 for Test 2 and Test 3 | FDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | Not present | | FDD |
| } | | | |

**Table 7.1.10.4.3-4: *SoudingRS-UL-ConfigDedicated-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | bw0 used with no frequency hopping. bw3 used with frequency hopping | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | | Set according to specific test; 0 for Test 1 and 77 for Test 2 and 317 for Test 3 | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 7.1.10.4.3-5: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN FDD Cat-M1 UE in CEModeA test requirement for Test 2 and Test 3**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | DRX_S |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | Set according to specific test; sf80 for Test 2 and sf640 for Test 3 | |
| sf80 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.1.10.4.3-6: *CQI-ReportConfig-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN FDD Cat-M1 UE in CEModeA test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm20 | Allowed value for CEModeA configured UE | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| release | NULL | | |
| } | | | |

**Table 7.1.10.4.3-7: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| } | | | |

## 7.1.10.5 Test requirement

Tables 7.1.10.5-1, 7.1.10.5-2 and 7.1.10.5-3 define the primary settings including test tolerances for UE transmit timing accuracy for E-UTRAN FDD test.

Tables 7.1.10.5-4, 7.1.10.5-5 and 7.1.10.5-6 define the rules for adjustments made to the UE uplink timing include Test Tolerances.

**Table 7.1.10.5-1: Cell Specific Test requirement Parameters for UE transmit timing accuracy for E-UTRAN FDD for Cat-M1 UE in CEModeA test case**

| Parameter | Unit | Value | | |
|---|---|---|---|---|
| | | Test 1 | Test 2 | Test 3 |
| E-UTRA RF Channel Number | | 1 | 1 | 1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | 10 | 10 |
| DRX cycle | ms | N/A | 80[Note5] | 640[Note5] |
| PRACH Configuration | | PRACH_2CE As specified in A.9 | | |
| MPCCH Reference measurement channel[Note1] | | R.16 FDD | R.16 FDD | R.16 FDD |
| OCNG Pattern[Note2] | | OP.21 FDD | OP.21 FDD | OP.21 FDD |
| PBCH_RA | dB | 0 | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| MPDCCH_RA | | | | |
| MPDCCH_RB | | | | |
| OCNG_RA[Note3] | | | | |
| OCNG_RB[Note3] | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | -98 | -98 |
| $\hat{E}_s / I_{ot}$ | dB | 3.30 | 3.30 | 3.30 |
| $\hat{E}_s / N_{oc}$ | dB | 3.30 | 3.30 | 3+.30 |
| Io[Note4] | dBm/9 MHz | -65.25 | -65.25 | -65.25 |
| | dBm/1.08 MHz | N/A | N/A | N/A |
| Propagation condition | - | AWGN | AWGN | AWGN |

Note 1: For the reference measurement channels, see clause A.7.1.
Note 2: For the OCNG pattern, see clause D.1.21.
Note 3: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: Io level has been derived from other parameters for information purpose. It is not a settable parameter.
Note 5: DRX related parameters are defined in Table 7.1.10.5-3.

**Table 7.1.10.5-2: Sounding Reference Signal Configuration to be used in UE transmit timing accuracy for E-UTRAN FDD for Cat-M1 UE in CEModeA test case**

| Field | Value | | | Comment |
|---|---|---|---|---|
| | Test 1 | Test 2 | Test 3 | |
| srsBandwidthConfiguration | bw5 | bw5 | bw5 | |
| srsSubframeConfiguration | sc1 | sc3 | sc3 | |
| ackNackSrsSimultaneousTransmission | FALSE | FALSE | FALSE | |
| srsMaxUpPTS | N/A | N/A | N/A | Not applicable for FDD |
| srsBandwidth | 0 | 0 | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | hbw0 | hbw0 | |
| frequencyDomainPosition | 0 | 0 | 0 | |
| duration | TRUE | TRUE | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 0 | 77 | 317 | SRS periodicity of 2ms, 80 ms and 320ms for Test 1, 2 and 4, respectively. |
| transmissionComb | 0 | 0 | 0 | |
| cyclicShift | cs0 | cs0 | cs0 | No cyclic shift |
| SRS-AntennaPort | an1 | | | Number of antenna ports used for SRS transmission |

Note 1: For further information see clause 6.3.2 in TS 36.331[5].

**Table 7.1.10.5-3: DRX Configuration to be used in UE transmit timing accuracy for E-UTRAN FDD for Cat-M1 UE in CEModeA test case**

| Field | Value | | Comment |
|---|---|---|---|
| | Test 2 | Test 3 | |
| onDurationTimer | psf1 | psf1 | |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf80 | sf640 | |
| shortDRX | disable | disable | |
| Note 1: For further information see clause 6.3.2 in TS 36.331[5]. | | | |

The following sequence of events shall be used to verify that the requirements are met.

For the 10MHz channel bandwidth, the test sequence shall be carried out in RRC_CONNECTED for both non-DRX (for Test1) and DRX with a cycle length of 80 ms or a cycle length of 640 ms (Tests 2 and , respectively):

a) After a connection is set up with the cell, the test system shall verify that the UE transmit timing offset is within $N_{TA}\times T_S \pm 15\times T_S$ with respect to the first detected path (in time) of the corresponding downlink frame of cell 1.

b) The test system adjusts the downlink transmit timing for the cell by $+64\times T_S$ (for Test 1 and Test 2) or $+32\times T_S$ (for Test 3) compared to that in (a).

c) The test system shall verify that for Test 1 the adjustment step size and the adjustment rate shall be according to the requirements in clause 7.1.2 until the UE transmit timing offset is within $N_{TA}\times T_S \pm 15\times T_S$ with respect to the first detected path (in time) of the corresponding downlink frame of cell 1. Skip this step for Test 2 and Test 3.

d) The test system shall verify that the UE transmit timing offset stays within $N_{TA}\times T_S \pm 15\times T_S$ with respect to the first detected path (in time) of the corresponding downlink frame of cell 1. For test 2 and test 3 the UE transmit timing offset shall be verified for the first transmission in the DRX cycle immediately after DL timing adjustment.

The UE transmit timing offset shall be within the requirements in Table 7.1.10.5-4.

The reference point for the UE initial transmit timing control test requirement shall be the downlink timing minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}})\times T_s$.

**Table 7.1.10.5-4: Test requirement for $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| ≥3 | $15*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9]. | |

The UE shall be capable of changing the transmission timing according to the received downlink frame. When the transmission timing error between the UE and the reference cell exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$.

All adjustments made to the UE uplink timing shall follow these rules:

1) The maximum adjustment step size $T_q$ shall be within the requirements in Table 7.1.10.5-5

2) The minimum aggregate adjustment rate shall be $3.4*T_S$ per second (To be adjusted after TT analysis).

3) The maximum aggregate adjustment rate shall be within the requirements in Table 7.1.10.5-6.

**Table 7.1.10.5-5: Test requirement for $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| ≥10 | $4*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9]. | |

**Table 7.1.10.5-6: Test requirement for maximum aggregate adjustment rate per 198 ms**

| Downlink Bandwidth (MHz) | Maximum adjustment per 198 ms |
|---|---|
| ≥10 | 4.6*$T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9]. | |

## 7.1.11 E-UTRAN HD-FDD – UE Transmit Timing Accuracy Tests for Cat-M1 UE in CEModeA

### 7.1.11.1 Test purpose

To verify that the Cat-M1 UE in CEModeA have the capability to follow the frame timing change of the connected System Simulator. The method used is that the UE initial transmit timing accuracy, the maximum amount of timing change in one adjustment, and the minimum and maximum adjustment rate are within the specified limits based on the requirements.

### 7.1.11.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 13 and forward of UE category M1. Applicability requires support for FGI bit 5.

### 7.1.11.3 Minimum conformance requirements

The UE initial transmission timing error shall be less than or equal to ±$T_e$ where the timing error limit value $T_e$ is defined in TS 36.133 [4] clause 7.1.2 and shown in table 7.1.11.3-1. This requirement applies when it is the first transmission in a DRX, eDRX_CONN cycle, or the first transmission in a repetition period (R>1) for PUCCH, PUSCH and SRS or it is the PRACH transmission. The reference point for the UE initial transmit timing control requirement shall be the downlink timing of the reference cell minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$. The downlink timing is defined as the time when the first detected path (in time) of the corresponding downlink frame is received from the reference cell. $N_{\text{TA_Ref}}$ for PRACH is defined as 0. $(N_{\text{TA_Ref}} + N_{\text{TA offset}})$ (in $T_s$ units) for other channels is the difference between UE transmission timing and the Downlink timing immediately after when the last timing advance in clause 7.3 was applied. $N_{\text{TA_Ref}}$ for other channels is not changed until next timing advance is received.

**Table 7.1.11.3-1: $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| 1.4 | 24*$T_S$ |
| ≥3 | 12*$T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

Cat-M1 UEs configured with CEModeA shall follow the requirement $T_e$ defined for 1.4MHz in Table 7.1.11.3-1 regardless the downlink carrier bandwidth *(Editor's note: It is FFS for the Cat-M1 UEs configured with CEModeB).*

When it is not the first transmission in a DRX or eDRX_CONN cycle or there is no DRX or no eDRX_CONN cycle, and when it is the transmission for PUCCH, PUSCH and SRS transmission, the UE shall, when no repetitions are configured on the uplink or the repetition period is R=1, be capable of changing the transmission timing according to the received downlink frame of the reference cell except when the timing advance in clause 7.3 is applied.

When in a TAG the transmission timing error between the UE and the reference timing exceeds ±$T_e$, or in a sTAG the UE changes the downlink SCell for deriving the UE transmit timing for cells in the sTAG configured with one or two uplinks, the UE is required to adjust its timing to within ±$T_e$ in that TAG, as long as,

- the UE is configured with a pTAG and one or two sTAG, the transmission timing difference between TAGs does not exceed the maximum transmission timing difference (i.e., 32.47us) after such adjustment, or

- the UE is configured with synchronous dual connectivity, the transmission timing difference between pTAG and psTAG does not exceed the maximum transmission timing difference (i.e., 35.21us) after such adjustment.

If the transmission timing difference after such adjustment is bigger than the maximum transmission timing difference UE may stop adjustment in this TAG. The reference timing shall be $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$ before the downlink timing of the reference cell. All adjustments made to the UE uplink timing under the above mentioned scenarios shall follow these rules:

1) The maximum amount of the magnitude of the timing change in one adjustment shall be $T_q$ seconds.

2) The minimum aggregate adjustment rate shall be 7 * $T_S$ per second.

3) The maximum aggregate adjustment rate shall be $T_q$ per 200 ms.

Where the maximum autonomous time adjustment step $T_q$ is defined in TS 36.133 [4] clause 7.1.2 and shown in table 7.1.11.3-2.

**Table 7.1.11.3-2: $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| 1.4 | 17.5*$T_S$ |
| 3 | 9.5*$T_S$ |
| 5 | 5.5*$T_S$ |
| ≥10 | 3.5*$T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9] | |

Cat-M1 UEs configured with CEModeA shall follow the requirement $T_q$ defined for 1.4MHz in Table 7.1.2-2 regardless the downlink carrier bandwidth *(Editor's note: It is FFS for the Cat-M1 UEs configured with CEModeB).*

*When a repetition period is configured on the uplink, the UE shall not adjust the uplink transmission timing autonomously during an ongoing repetition period for which R>1.*

The normative reference for this requirement is TS 36.133 [4] clause 7.1 and A.7.1.11.

NOTE 1: Both the allowed UE frequency error and the UE time adjustment quantisation can affect the result of the test, and it is not possible to test the UE Transmit Timing Accuracy in isolation from these core requirements. They are therefore taken into account when setting the Test Tolerances and the Test Requirements

## 7.1.11.4 Test description

### 7.1.11.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18(using only main UE Tx/Rx antenna).

2. Propagation conditions are set according to Annex B clause B.0.

3. There is one E-UTRA HD-FDD carrier and one cell specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

### 7.1.11.4.2 Test procedure

The test consists of a single cell. The transmit timing accuracy is verified related to the downlink frame timing of Cell 1. The downlink timing of Cell 1 is changed and the changes in UE transmit timing are observed. The transmit timing is verified by the UE transmitting SRS used as a measurement reference facilitating the SS timing estimation.

1. Ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3AA.

2. Set the parameters according to Test 1, Test 2, or Test 3 in Tables 7.1.11.5-1, 7.1.11.5-2 and 7.1.11.5-3 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. The SS shall check that the UE transmit timing offset is within the limits specified in Table 7.1.11.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1.

6. The SS adjusts the downlink timing for Cell 1 to a delay of $+64 \times T_S$ (approximately +2 μs) for Test 1 and Test2 and to and a delay of $+32 \times T_S$ for Test 3 compared to that in step 5. For Test 2 and Test 3, the downlink timing adjustment shall be done at (40 ± 15 )ms before the next DRX On duration period.

7. Step 7 applies for Test 1, but is omitted for Test 2 and Test 3. The test system samples the UE Transmit Timing once per SRS transmission (periodicity 2 subframes). To check Rule 1, the SS shall check that the maximum time adjustment step size $T_q$ between one SRS transmission (subframe n) to next (subframe n+2) is within Rule 1 as specified in clause 7.1.11.5. To check that the minimum adjustment rate is within Rule 2 as specified in clause 7.1.11.5, the SS shall measure the change in SRS transmission timing over a 1.002s sliding window, with step size 2ms. To check that the maximum adjustment rate is within Rule 3 as specified in clause 7.1.11.5, the SS shall measure the change in SRS transmission timing over a 198ms sliding window, with step size 2ms. The three rules apply until the UE transmit timing offset is within the limits specified in Table 7.1.11.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1.

8. In case for Test 1 and Test 3, the SS shall check that the UE transmit timing offset stays within the limits specified in Table 7.1.11.5-4 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1. In case for Test 2 and Test 3, the SS shall check that the UE transmit timing offset stays within the limits specified in Table 7.1.11.5-4 For that purpose, the SRS transmission that takes place in the first DRX ON duration after downlink timing of Cell 1 is changed is used.

9. The SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

10. After the RRC connection release, the SS switches off and on the UE and ensures the UE is in State 3A-RF –CE according to TS 36.508 [7] clause 7.2A.3AA.

11. Repeat step 1-10 for each sub-test in Tables 7.1.11.5-1, 7.1.11.5-2 and 7.1.11.5-3 as appropriate.

### 7.1.11.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the condition CEModeA with the following exceptions:

**Table 7.1.11.4.3-1: Common Exception messages for UE transmit timing accuracy for E-UTRAN HD-FDD Cat-M1 UE in CEModeA test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.4-2 |

**Table 7.1.11.4.3-2: *RRCConnectionReconfiguration*: Additional UE transmit timing accuracy for E-UTRAN HD-FDD Cat-M1 UE in CEModeA test requirement**

| Derivation Path: TS 36.331 [5] clause 6.2.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| radioResourceConfigDedicated SEQUENCE { | | | |
| mac-MainConfig CHOICE { | | | |
| explicitValue | MAC-MainConfig-RBC | | |
| } | | | |
| physicalConfigDedicated | PhysicalConfigDedicated - DEFAULT | | RBC |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.1.11.4.3-3: *SoundingRS-RL-ConfigCommon-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN HD-FDD Cat-M1 UE in CEModeA test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw5 | | |
| srs-SubframeConfig | | Set according to specific test; sc1 for Test 1 and sc3 for Test 2 and Test 3 | FDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | Not present | | FDD |
| } | | | |

**Table 7.1.11.4.3-4: *SoudingRS-UL-ConfigDedicated-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN HD-FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | bw0 used with no frequency hopping. bw3 used with frequency hopping | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | | Set according to specific test; 0 for Test 1 and 77 for Test 2 and 317 for Test 3 | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 7.1.11.4.3-5: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN HD-FDD Cat-M1 UE in CEModeA test requirement for Test 2 and Test 3**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | DRX_S |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | Set according to specific test; sf80 for Test 2 and sf640 for Test 3 | |
| sf80 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.1.11.4.3-6: *CQI-ReportConfig-DEFAULT*: Additional UE transmit timing accuracy for E-UTRAN HD-FDD Cat-M1 UE in CEModeA test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm20 | Allowed value for CEModeA configured UE | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| release | NULL | | |
| } | | | |

**Table 7.1.11.4.3-7: *MAC-MainConfig-RBC*: Additional UE transmit timing accuracy for E-UTRAN HD-FDD test requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| } | | | |

## 7.1.11.5 Test requirement

Tables 7.1.11.5-1, 7.1.11.5-2 and 7.1.11.5-3 define the primary settings including test tolerances for UE transmit timing accuracy for E-UTRAN HD-FDD test.

Tables 7.1.11.5-4, 7.1.11.5-5 and 7.1.11.5-6 define the rules for adjustments made to the UE uplink timing include Test Tolerances.

**Table 7.1.11.5-1: Cell Specific Test requirement Parameters for UE transmit timing accuracy for E-UTRAN HD-FDD for Cat-M1 UE in CEModeA test case**

| Parameter | Unit | Value | | |
|---|---|---|---|---|
| | | Test 1 | Test 2 | Test 3 |
| E-UTRA RF Channel Number | | 1 | 1 | 1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | 10 | 10 |
| DRX cycle | ms | N/A | 80[Note5] | 640[Note5] |
| PRACH Configuration | | PRACH_2CE As specified in A.9 | | |
| MPCCH Reference measurement channel[Note1] | | R.16 HD-FDD | R.16 HD-FDD | R.16 HD-FDD |
| OCNG Pattern[Note2] | | OP.21 FDD | OP.21 FDD | OP.21 FDD |
| PBCH_RA | dB | 0 | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| MPDCCH_RA | | | | |
| MPDCCH_RB | | | | |
| OCNG_RA[Note3] | | | | |
| OCNG_RB[Note3] | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | -98 | -98 |
| $\hat{E}_s / I_{ot}$ | dB | 3.30 | 3.30 | 3.30 |
| $\hat{E}_s / N_{oc}$ | dB | 3.30 | 3.30 | 3.30 |
| Io[Note4] | dBm/9 MHz | -65.25 | -65.25 | -65.25 |
| | dBm/1.08 MHz | N/A | N/A | N/A |
| Propagation condition | - | AWGN | AWGN | AWGN |

Note 1: For the reference measurement channels, see clause A.7.2.
Note 2: For the OCNG pattern, see clause D.1.21.
Note 3: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: Io level has been derived from other parameters for information purpose. It is not a settable parameter.
Note 5: DRX related parameters are defined in Table 7.1.11.5-3.

**Table 7.1.11.5-2: Sounding Reference Signal Configuration to be used in UE transmit timing accuracy for E-UTRAN HD-FDD for Cat-M1 UE in CEModeA test case**

| Field | Value | | | Comment |
|---|---|---|---|---|
| | Test 1 | Test 2 | Test 3 | |
| srsBandwidthConfiguration | bw5 | bw5 | bw5 | |
| srsSubframeConfiguration | sc1 | sc3 | sc3 | |
| ackNackSrsSimultaneousTransmission | FALSE | FALSE | FALSE | |
| srsMaxUpPTS | N/A | N/A | N/A | Not applicable for FDD |
| srsBandwidth | 0 | 0 | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | hbw0 | hbw0 | |
| frequencyDomainPosition | 0 | 0 | 0 | |
| duration | TRUE | TRUE | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 0 | 77 | 317 | SRS periodicity of 2ms, 80 ms and 320ms for Test 1, 2 and 4, respectively. |
| transmissionComb | 0 | 0 | 0 | |
| cyclicShift | cs0 | cs0 | cs0 | No cyclic shift |
| SRS-AntennaPort | an1 | | | Number of antenna ports used for SRS transmission |

Note 1: For further information see clause 6.3.2 in TS 36.331[5].

**Table 7.1.11.5-3: DRX Configuration to be used in UE transmit timing accuracy for E-UTRAN FDD for Cat-M1 UE in CEModeA test case**

| Field | Value | | Comment |
|---|---|---|---|
| | Test 2 | Test 3 | |
| onDurationTimer | psf1 | psf1 | |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf80 | sf640 | |
| shortDRX | disable | disable | |
| Note 1: For further information see clause 6.3.2 in TS 36.331[5]. | | | |

The following sequence of events shall be used to verify that the requirements are met.

For the 10MHz channel bandwidth, the test sequence shall be carried out in RRC_CONNECTED for both non-DRX (for Test1) and DRX with a cycle length of 80 ms or a cycle length of 640 ms (Tests 2 and , respectively):

a) After a connection is set up with the cell, the test system shall verify that the UE transmit timing offset is within $N_{TA}\times T_S \pm 15\times T_S$ with respect to the first detected path (in time) of the corresponding downlink frame of cell 1.

b) The test system adjusts the downlink transmit timing for the cell by $+64\times T_S$ (for Test 1 and Test 2) or $+32\times T_S$ (for Test 3) compared to that in (a).

c) The test system shall verify that for Test 1 the adjustment step size and the adjustment rate shall be according to the requirements in clause 7.1.2 until the UE transmit timing offset is within $N_{TA}\times T_S \pm 15\times T_S$ with respect to the first detected path (in time) of the corresponding downlink frame of cell 1. Skip this step for Test 2 and Test 3.

d) The test system shall verify that the UE transmit timing offset stays within $N_{TA}\times T_S \pm 15\times T_S$ with respect to the first detected path (in time) of the corresponding downlink frame of cell 1. For test 2 and test 3 the UE transmit timing offset shall be verified for the first transmission in the DRX cycle immediately after DL timing adjustment.

The UE transmit timing offset shall be within the requirements in Table 7.1.11.5-4.

The reference point for the UE initial transmit timing control test requirement shall be the downlink timing minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}})\times T_s$.

**Table 7.1.11.5-4: Test requirement for $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| ≥3 | $15*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9]. | |

The UE shall be capable of changing the transmission timing according to the received downlink frame. When the transmission timing error between the UE and the reference cell exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$.

All adjustments made to the UE uplink timing shall follow these rules:

1) The maximum adjustment step size $T_q$ shall be within the requirements in Table 7.1.11.5-5

2) The minimum aggregate adjustment rate shall be $3.4*T_S$ per second (To be adjusted after TT analysis).

3) The maximum aggregate adjustment rate shall be within the requirements in Table 7.1.11.5-6.

**Table 7.1.11.5-5: Test requirement for $T_q$ Maximum Autonomous Time Adjustment Step**

| Downlink Bandwidth (MHz) | $T_{q_}$ |
|---|---|
| ≥10 | $4*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9]. | |

**Table 7.1.11.5-6: Test requirement for maximum aggregate adjustment rate per 198 ms**

| Downlink Bandwidth (MHz) | Maximum adjustment per 198 ms |
|---|---|
| ≥10 | $4.6*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9]. | |

# 7.2 UE Timing Advance

## 7.2.1 E-UTRAN FDD - UE Timing Advance Adjustment Accuracy

### 7.2.1.1 Test purpose

To verify the UE in RRC_CONNECTED state adjusts the timing of its transmissions with accuracy by meeting the E-UTRAN FDD timing advance adjustment requirements in an AWGN model.

### 7.2.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

### 7.2.1.3 Minimum conformance requirements

The timing advance is initiated from E-UTRAN with MAC message that implies and adjustment of the timing advance according to TS 36.321 [11] clause 5.2.

The UE shall adjust the timing of its uplink transmission timing at sub-frame *n*+6 for a timing advancement command received in sub-frame *n*.

The UE shall adjust the timing of its transmission with a relative accuracy better than or equal to $\pm 4 \times T_S$ seconds to the signalled timing advance value compared to the timing of preceding uplink transmission. The timing advance command is expressed in multiple of $16 \times T_S$ and is relative to the current uplink timing.

In case of random access response, 11-bit timing advance command, $T_A$, indicates $N_{TA}$ values by index values of $T_A$ = 0, 1, 2, ..., 1282, where an amount of the time alignment is given by $N_{TA} = T_A \times 16$ according to TS 36.213 [8] clause 4.2.3. $N_{TA}$ is defined in TS 36.211 [9].

In other cases, 6-bit timing advance command, $T_A$, indicates adjustment of the current $N_{TA}$ value, $N_{TA,old}$, to the new $N_{TA}$ value, $N_{TA,new}$, by index values of $T_A$ = 0, 1, 2,..., 63, where $N_{TA,new} = N_{TA,old} + (T_A - 31) \times 16$ according to TS 36.213 [8] clause 4.2.3. Adjustment of $N_{TA}$ value by a positive or a negative amount indicates advancing or delaying the uplink transmission timing by a given amount respectively.

For UE timers, the UE shall comply with the timer accuracies according to TS 36.133 [4] table 7.2.2-1. The requirements are only related to the actual timing measurements internally in the UE.

The normative reference for this requirement is TS 36.133 [4] clause 7.3 and A.7.2.1.

### 7.2.1.4 Test description

#### 7.2.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18.

2. The general test parameter settings are set according to Table 7.2.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. There is one E-UTRA FDD carrier and one cell specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.2.1.4.1-1: General Test Parameters for E-UTRAN FDD - UE timing advance adjustment accuracy test case**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Timing Advance Command ($T_A$) value during T1 | | 31 | $N_{TA}$ = 0 for the purpose of establishing a reference value from which the timing advance adjustment accuracy can be measured during T2 |
| Timing Advance Command ($T_A$) value during T2 | | 39 | $N_{TA}$ = 128 |
| DRX | | OFF | |
| T1 | s | 5 | |
| T2 | s | 5 | |

### 7.2.1.4.2 Test procedure

The test consists of a single cell. The test consists of two successive time periods, with time durations of T1 and T2 respectively. In each time period, timing advance commands are sent to the UE and SRS are sent from the UE and received by the SS. By measuring the reception of the SRS, the transmit timing, and hence the timing advance adjustment accuracy, can be measured. The UE Time Alignment Timer (timeAlignmentTimer IE), defined in TS 36.321 [11] clause 5.2, shall be configured so that it does not expire in the duration of the test.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Tables 7.2.1.5-1 and 7.2.1.5-2. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. SS shall transmit one message with a timing advance command, $T_A$. The timing advance command, $T_A$, shall be set to 31 which indicate adjustment of the current $N_{TA}$ value. The timing advance adjustment during T1 shall be $N_{TA} = 0$.

6. The UE shall transmit SRS and adjust its uplink timing at the beginning of sub-frame n+6 for a timing advance command $T_A$ received in sub-frame n. Sub-frame n is the sub-frame where the UE received the time advance command.

7. When T1 expires, the SS shall switch the timing advance command ($T_A$) from T1 to T2 as specified in Table 7.2.1.5-1.

8. SS shall transmit a sequence of messages with timing advance command, $T_A$. The timing advance command, $T_A$, shall be set to 39 which indicate adjustment of the current $N_{TA}$ value. The timing advance adjustment during T2 shall be $N_{TA} = 128$.

9. The UE shall transmit SRS and adjust its uplink timing at the beginning of sub-frame n+6 for a timing advance command $T_A$ received in sub-frame n. Sub-frame n is the sub-frame where the UE received the time advance command.

10. The result from the SRS and adjustment of the timing advance in step 9) is used to measure that the UE adjusts the timing of its transmission with a relative accuracy better than or equal to $\pm 4.5 \times T_S$ to the signalled timing advance value compared to the timing of preceding uplink transmission.

11. If the UE adjust the timing of its transmission within a relative accuracy greater than or equal to ± 4.5 × $T_S$ to the signalled timing advance value compared to the timing of preceding uplink transmission then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

12. The SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

13. After the RRC connection release, the SS:

    - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),

    or

    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 2-13 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 7.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.2.1.4.3-1: Common Exception messages for E-UTRAN FDD - UE timing advance adjustment accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.4-2 |

**Table 7.2.1.4.3-2: *SoundingRS-UL-ConfigCommon-DEFAULT*: Additional E-UTRAN FDD - UE timing advance adjustment accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw5 | Channel-bandwidth-dependent parameter | |
| srs-SubframeConfig | sc3 | | FDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | Not present | | FDD |
| } | | | |
| } | | | |

**Table 7.2.1.4.3-3: *SoundingRSUL-ConfigDedicated-DEFAULT*: Additional E-UTRAN FDD - UE timing advance adjustment accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | bw0 used with no frequency hopping. bw3 used with frequency hopping | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | 7 | SRS periodicity of 10 | FDD |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 7.2.1.4.3-4: *MAC-MainConfig-RBC*: Additional E-UTRAN FDD - UE timing advance adjustment accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| dl-SCH-Config SEQUENCE {} | Not present | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| timeAlignmentTimerDedicated | Infinity | | |

## 7.2.1.5 Test requirement

Tables 7.2.1.4.1-1, 7.2.1.5-1 and 7.2.1.5-2 define the primary level settings for E-UTRAN FDD - UE timing advance adjustment accuracy test.

**Table 7.2.1.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD - UE timing advance adjustment accuracy test case**

| Parameter | Unit | Value | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note1 | dB | | |
| OCNG_RB Note1 | dB | | |
| Timing Advance Command ($T_A$) | | 31 | 39 |
| $\hat{E}_s / I_{ot}$ | dB | 3 | |
| $N_{oc}$ | dBm/15 KHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 3 | |
| Io Note2 | dBm/9 MHz | -65.5 | |
| Propagation Condition | | AWGN | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: Io level has been derived from other parameters for information purpose. It is not a settable parameter. | | | |

**Table 7.2.1.5-2: Sounding Reference Signal Configuration to be used in E-UTRAN FDD - UE timing advance adjustment accuracy test case**

| Field | Value | Comment |
|---|---|---|
| srs-BandwidthConfig | bw5 | |
| srs-SubframeConfig | sc3 | Once every 5 subframes |
| ackNackSRS-SimultaneousTransmission | FALSE | |
| srsMaxUpPts | N/A | Not applicable for E-UTRAN FDD |
| srs-Bandwidth | 0 | No hopping |
| srs-HoppingBandwidth | hbw0 | |
| freqDomainPosition | 0 | |
| duration | TRUE | Indefinite duration |
| srs-ConfigIndex | 7 | SRS periodicity of 10. |
| transmissionComb | 0 | |
| cyclicShift | cs0 | No cyclic shift |
| Note: For further information see section 6.3.2 in 3GPP TS 36.331 [15]. | | |

The UE shall apply the signalled Timing Advance value to the transmission timing at the designated activation time i.e. 6 sub-frames after the reception of the timing advance command.

The UE shall adjust the timing of its transmission with a relative accuracy better than or equal to $\pm 4.5 \times T_S$ seconds to the signalled timing advance value compared to the timing of preceding uplink transmission. This requirement includes test tolerances.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 7.2.2 E-UTRAN TDD - UE Timing Advance Adjustment Accuracy

### 7.2.2.1 Test purpose

The purpose of the test is to verify E-UTRAN TDD Timing Advance adjustment accuracy requirements, in an AWGN model.

### 7.2.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

### 7.2.2.3 Minimum conformance requirements

The timing advance is initiated from E-UTRAN with MAC message that implies and adjustment of the timing advance according to TS 36.321 [11] clause 5.2.

The UE shall adjust the timing of its uplink transmission timing at sub-frame *n*+6 for a timing advancement command received in sub-frame *n*.

The UE shall adjust the timing of its transmission with a relative accuracy better than or equal to $\pm 4 \times T_S$ seconds to the signalled timing advance value compared to the timing of preceding uplink transmission. The timing advance command is expressed in multiple of $16 \times T_S$ and is relative to the current uplink timing.

In case of random access response, 11-bit timing advance command, $T_A$, indicates $N_{TA}$ values by index values of $T_A$ = 0, 1, 2, ..., 1282, where an amount of the time alignment is given by $N_{TA} = T_A \times 16$ according to TS 36.213 [8] clause 4.2.3. $N_{TA}$ is defined in TS 36.211 [9].

In other cases, 6-bit timing advance command, $T_A$, indicates adjustment of the current $N_{TA}$ value, $N_{TA,old}$, to the new $N_{TA}$ value, $N_{TA,new}$, by index values of $T_A$ = 0, 1, 2,..., 63, where $N_{TA,new} = N_{TA,old} + (T_A - 31) \times 16$ according to TS 36.213 [8] clause 4.2.3. Adjustment of $N_{TA}$ value by a positive or a negative amount indicates advancing or delaying the uplink transmission timing by a given amount respectively.

For UE timers, the UE shall comply with the timer accuracies according to TS 36.133 [4] table 7.2.2-1. The requirements are only related to the actual timing measurements internally in the UE.

The normative reference for this requirement is TS 36.133 [4] clause 7.3 and A.7.2.2.

### 7.2.2.4 Test description

#### 7.2.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noises source to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18.
2. The general test parameter settings are set according to Table 7.2.2.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. There is one E-UTRA TDD carrier and one cell specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.2.2.4-1 General Test Parameters for E-UTRAN TDD Timing Advance Accuracy Test**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Timing Advance Command ($T_A$) value during T1 | | 31 | $N_{TA}$ = 0 for the purpose of establishing a reference value from which the timing advance adjustment accuracy can be measured during T2 |
| Timing Advance Command ($T_A$) value during T2 | | 39 | $N_{TA}$ = 128 |
| DRX | | OFF | |
| T1 | s | 5 | |
| T2 | s | 5 | |

### 7.2.2.4.2 Test procedure

The test consists of two successive time periods, with time duration of T1 and T2 respectively. In each time period, timing advance commands are sent to the UE and Sounding Reference Signals (SRS), are sent from the UE and received by the test equipment. By measuring the reception of the SRS, the transmit timing, and hence the timing advance adjustment accuracy, can be measured. The UE Time Alignment Timer (timeAlignmentTimer IE), defined in TS 36.321 [11] clause 5.2, shall be configured so that it does not expire in the duration of the test.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Tables 7.2.2.5-1, 7.2.2.5-2 and 7.2.2.5-3. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. SS shall transmit one message with a timing advance command, $T_A$. The timing advance command, $T_A$, shall be set to 31 which indicate adjustment of the current $N_{TA}$ value. The timing advance adjustment during T1 shall be $N_{TA}$ = 0.
6. The UE shall transmit SRS and adjust its uplink timing from the beginning of sub-frame n+6 for a timing advance command $T_A$ received in sub-frame n. Sub-frame n is the sub-frame where the UE receive the time advance command.
7. When T1 expires, the SS shall switch the timing advance command ($T_A$) from T1 to T2 as specified in Table 7.2.1.5-1.
8. SS shall transmit a sequence of messages with timing advance command, $T_A$. The timing advance command, $T_A$, shall be set to 39 which indicate adjustment of the current $N_{TA}$ value. The timing advance adjustment during T2 shall be $N_{TA}$ = 128.
9. The UE shall transmit SRS and adjust its uplink timing from the beginning of sub-frame n+6 for a timing advance command $T_A$ received in sub-frame n. Sub-frame n is the sub-frame where the UE receive the time advance command.
10. The timing of the first SRS transmission after sub-frame n+6 and adjustment of the timing advance in step 9) is used to measure that the UE adjusts the timing of its transmission with a relative accuracy better than or equal to $\pm 4.5 \times T_S$ to the signalled timing advance value compared to the timing of preceding uplink transmission.
11. If the UE adjust the timing of its transmission within a relative accuracy greater than or equal to $\pm 4.5 \times T_S$ to the signalled timing advance value compared to the timing of preceding uplink transmission then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.
12. The SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

13. After the RRC connection release, the SS:

    - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),

    or

    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 2-13 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 7.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.2.2.4.3-1: Common Exception messages for E-UTRAN TDD - UE timing advance adjustment accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.4-2 |

**Table 7.2.2.4.3-2: *SoundingRS-UL-ConfigCommon-DEFAULT*: Additional UE transmit timing for E-UTRAN TDD test requirement**

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRSUL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw5 | Channel-bandwidth-dependent parameter | |
| srs-SubframeConfig | sc3 | | TDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | Not present | | |
| } | | | |

**Table 7.2.2.4.3-3: *SoundingRS-UL-ConfigDedicated-DEFAULT*: Additional UE transmit timing for E-UTRAN TDD test requirement**

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRSUL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| enable SEQUENCE { | | | |
| srsBandwidth | bw0 | bw0 used with no frequency hopping. bw3 used with frequency hopping | |
| srsHoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | 15 | SRS periodicity of 10 | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 7.2.1.4.3-4: *MAC-MainConfig-RBC*: Additional E-UTRAN TDD - UE timing advance adjustment accuracy test requirement**

| Derivation Path: 36.508 clause 4.8.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| dl-SCH-Config SEQUENCE {} | Not present | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |

## 7.2.2.5 Test requirement

Tables 7.2.2.4.1-1, 7.2.2.5-1 and 7.2.2.5-2 define the primary level settings for E-UTRAN TDD - UE timing advance adjustment accuracy test.

**Table 7.2.2.5-1 Cell specific Test Parameters for E-UTRAN TDD Timing Advance Accuracy Test**

| Parameter | Unit | Value | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| Special subframe configuration[Note1] | | 6 | |
| Uplink-downlink configuration[Note2] | | 1 | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA[Note3] | dB | | |
| OCNG_RB[Note3] | dB | | |
| Timing Advance Command ($T_A$) | | 31 | 39 |
| $\hat{E}_s / I_{ot}$ | dB | 3 | |
| $N_{oc}$ | dBm/15 KHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 3 | |
| Io[Note4] | dBm/9 MHz | -65.5 | |
| Propagation Condition | | AWGN | |

Note 1: For the special subframe configuration see table 4.2-1 in 3GPP TS 36.211.
Note 2: For the uplink-downlink configuration see table 4.2-2 in 3GPP TS 36.211.
Note 3: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: Io level has been derived from other parameters for information purpose. It is not a settable parameter.

**Table 7.2.2.5-2:Sounding Reference Signal Configuration for E-UTRAN TDD Transmit Timing Accuracy Test**

| Field | Value | Comment |
|---|---|---|
| srsBandwidthConfiguration | bw5 | |
| srsSubframeConfiguration | sc3 | Once every 5 subframes |
| ackNackSrsSimultaneousTransmission | FALSE | |
| srsMaxUpPTS | N/A | |
| srsBandwidth | bw0 | No hopping |
| srsHoppingBandwidth | hbw0 | |
| frequencyDomainPosition | 0 | |
| Duration | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 15 | SRS periodicity of 10ms. |
| transmissionComb | 0 | |
| cyclicShift | cs0 | No cyclic shift |
| Note: For further information see section 6.3.2 in 3GPP TS 36.331. | | |

The UE shall apply the signalled Timing Advance value to the transmission timing at the designated activation time i.e. 6 sub frames after the reception of the timing advance command.

The UE shall adjust the timing of its transmissions with a relative accuracy better than or equal to $\pm 4.5 \times T_S$ seconds to the signalled timing advance value compared to the timing of preceding uplink transmission. This requirement includes test tolerances.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 7.2.3 E-UTRAN FDD - UE Timing Advance Adjustment Accuracy Test for 5MHz Bandwidth

### 7.2.3.1 Test purpose

Same test purpose as defined in clause 7.2.1.1.

### 7.2.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that only support E-UTRA Band 31.

### 7.2.3.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 7.2.1.3 with the following exceptions:

- Instead of A.7.2.1 → use A.7.2.3.

### 7.2.3.4 Test description

#### 7.2.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18.
2. The general test parameter settings are set according to Table 7.2.3.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. There is one E-UTRA FDD carrier and one cell specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.2.3.4.1-1: General Test Parameters for E-UTRAN FDD - UE timing advance adjustment accuracy test case**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.5 FDD | As specified in clause A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.11 FDD | As specified in clause A.2.1 |
| Note 1: For the reference measurement channels, see clause A.1.1 and A.2.1.<br>Note 2: See Table 7.2.1.4.1-1 for the ther parameters.<br>Note 3: This test is according to the principle defined in TS 36.133 [4] section A.3.7.2 | | | |

#### 7.2.3.4.2 Test procedure

Same test procedure as defined in clause 7.2.1.4.2 with the following exceptions:

- Instead of Table 7.2.1.5-1 → use Table 7.2.3.5-1.

#### 7.2.3.4.3 Message contents

Same message contents as defined in clause 7.2.1.4.3.

### 7.2.3.5 Test requirement

Same test requirement as defined in clause 7.2.1.5 with the following exceptions:

Tables 7.2.3.4.1-1, 7.2.3.5-1 and 7.2.1.5-2 define the primary level settings for E-UTRAN FDD - UE timing advance adjustment accuracy test for 5MHz bandwidth.

**Table 7.2.3.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD - UE timing advance adjustment accuracy for 5MHz bandwidth test case**

| Parameter | Unit | Value | |
|---|---|---|---|
| | | T1 | T2 |
| $BW_{channel}$ | MHz | 5 | |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) | | OP.15 FDD | |
| $Io^{Note2}$ | dBm/4.5 MHz | -68.5 | |
| Note 1: For the reference measurement channels, see clause A.2.1.<br>Note 2: See Table 7.2.1.5-1 for the other parameters. | | | |

## 7.2.4 E-UTRAN FDD - UE Timing Advance Adjustment Accuracy Test for SCell in sTAG

### 7.2.4.1 Test purpose

To verify the UE in RRC_CONNECTED state adjusts the timing of its transmissions with accuracy by meeting the E-UTRAN FDD timing advance adjustment accuracy requirements in an AWGN model.

### 7.2.4.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 11 and forward that supports Uplink Carrier Aggregation and multiple timing advances.

### 7.2.4.3 Minimum conformance requirements

The timing advance is initiated from E-UTRAN with MAC message that implies and adjustment of the timing advance according to TS 36.321 [11] clause 5.2.

The UE shall adjust the timing of its uplink transmission timing at sub-frame *n*+6 for a timing advancement command received in sub-frame *n*.

The UE shall adjust the timing of its transmission with a relative accuracy better than or equal to $\pm 4 \times T_S$ seconds to the signalled timing advance value compared to the timing of preceding uplink transmission. The timing advance command is expressed in multiple of $16 \times T_S$ and is relative to the current uplink timing.

In case of random access response, 11-bit timing advance command, $T_A$, for a TAG indicates $N_{TA}$ values by index values of $T_A = 0, 1, 2, ..., 1282$, where an amount of the time alignment for the TAG is given by $N_{TA} = T_A \times 16$ according to TS 36.213 [8] clause 4.2.3. $N_{TA}$ is defined in TS 36.211 [9].

In other cases, 6-bit timing advance command, $T_A$, for a TAG indicates adjustment of the current $N_{TA}$ value, $N_{TA,old}$, to the new $N_{TA}$ value, $N_{TA,new}$, by index values of $T_A = 0, 1, 2,..., 63$, where $N_{TA,new} = N_{TA,old} + (T_A - 31) \times 16$ according to TS 36.213 [8] clause 4.2.3. Adjustment of $N_{TA}$ value by a positive or a negative amount indicates advancing or delaying the uplink transmission timing for the TAG by a given amount respectively.

For UE timers, the UE shall comply with the timer accuracies according to TS 36.133 [4] table 7.2.2-1. The requirements are only related to the actual timing measurements internally in the UE.

The normative reference for this requirement is TS 36.133 [4] clause 7.3 and A.7.2.4.

## 7.2.4.4 Test description

### 7.2.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz for both PCell and SCell as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.45.
2. The general test parameter settings are set according to Table 7.2.4.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Two Cells are used. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier. PCell is in the Primary Timing Advance Group (pTAG) and SCell is in the secondary Timing Advance Group (sTAG). Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 shall be powered OFF.

**Table 7.2.4.4.1-1: General Test Parameters for E-UTRAN FDD - UE timing advance adjustment accuracy test for SCell in sTAG**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Timing Advance Command ($T_A$) value during T1 | | 31 | $N_{TA}$ = 0 for the purpose of establishing a reference value from which the timing advance adjustment accuracy can be measured during T2 |
| Timing Advance Command ($T_A$) value during T2 | | 39 | $N_{TA}$ = 128 |
| DRX | | OFF | |
| T1 | s | 5 | |
| T2 | s | 5 | |

### 7.2.4.4.2 Test procedure

The test consists of two cells. The test consists of two successive time periods, with time durations of T1 and T2 respectively. In each time period, timing advance commands for sTAG are sent to the UE and SRS are sent from the UE and received by the SS. By measuring the reception of the SRS, the transmit timing, and hence the timing advance adjustment accuracy, can be measured for SCell in sTAG. The UE Time Alignment Timer (timeAlignmentTimer IE), defined in TS 36.321 [11] clause 5.2, shall be configured so that it does not expire in the duration of the test.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCell according to Annex C.0 and C.1 for all downlink physical channels.
3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4 with UL CA.
4. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).
5. Set the parameters according to T1 in Tables 7.2.4.5-1 and 7.2.4.5-2. Propagation conditions are set according to Annex B clause B.1.1.
6. SS shall transmit an RRCConnectionReconfiguration message.
7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. SS shall transmit one message with a Timing Advance Command MAC Control Element for sTAG, $T_A$. The timing advance command, $T_A$, shall be set to 31 which indicate zero adjustment of the current $N_{TA}$ value for sTAG. The timing advance adjustment during T1 shall be $N_{TA} = 0$.

9. The UE shall transmit SRS and adjust its uplink timing at the beginning of sub-frame n+6 for a timing advance command $T_A$ received in sub-frame n. Sub-frame n is the sub-frame where the UE received the time advance command.

10. When T1 expires, the SS shall switch the timing advance command ($T_A$) from T1 to T2 as specified in Table 7.2.4.5-1.

11. SS shall transmit a sequence of messages with Timing Advance Command MAC Control Element for sTAG, $T_A$. The timing advance command, $T_A$, shall be set to 39 which indicate adjustment of the current $N_{TA}$ value for sTAG. The timing advance adjustment during T2 shall be $N_{TA} = 128$.

12. The UE shall transmit SRS and adjust its uplink timing at the beginning of sub-frame n+6 for a timing advance command $T_A$ received in sub-frame n. Sub-frame n is the sub-frame where the UE received the time advance command.

13. The result from the SRS and adjustment of the timing advance in step 12) is used to measure that the UE adjusts the timing of its transmission with a relative accuracy better than or equal to $\pm 4.0 \times T_S$ +TT to the signalled timing advance value compared to the timing of preceding uplink transmission.

14. If the UE adjust the timing of its transmission within a relative accuracy greater than or equal to $\pm 4.5 \times T_S$ to the signalled timing advance value compared to the timing of preceding uplink transmission then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

15. The SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

16. After the RRC connection release, the SS:

    - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),

    or

    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

17. Repeat step 2-16 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.2.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.2.4.4.3-1: *RRCConnectionReconfiguration*: Additional E-UTRAN FDD - UE timing advance adjustment accuracy test for SCell in sTAG requirement**

| Derivation Path: Clause 4.6.1 Table 4.6.1-8, condition SCell_AddMod |
|---|

**Table7.2.4.4.3-2: *MAC-MainConfig-RBC*: Additional E-UTRAN FDD - UE timing advance adjustment accuracy test for SCell in sTAG requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC, condition SCell_AddMod | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| stag-ToAddModList-r11 SEQUENCE (SIZE (1..maxSTAG-r11)) OF SEQUENCE { | Not present | | |
| stag-Id-r11 | 1 | | |
| timeAlignmentTimerSTAG-r11 | infinity | | |
| } | | | |
| } | | | |

**Table 7.2.4.4.3-3: *SCellToAddMod-r10:* Additional E-UTRAN FDD - UE timing advance adjustment accuracy test for SCell in sTAG requirement**

| Derivation Path: 36.508, Table 4.6.3-19D | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SCellToAddMod-r10 ::= SEQUENCE { | | | |
| sCellIndex-r10 | 1 | | |
| cellIdentification-r10 SEQUENCE { | | | |
| physCellId-r10 | PhysicalCellIdentity of Cell 2 | | |
| dl-CarrierFreq-r10 | Same downlink EARFCN as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 7.2.4.4.3-4: *RadioResourceConfigCommonSCell-r10:* Additional E-UTRAN FDD - UE timing advance adjustment accuracy test for SCell in sTAG requirement**

| Derivation Path: 36.508, Table 4.6.3-13A, condition UL_CA | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigCommonSCell-r10-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigCommon-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw5 | | |
| srs-SubframeConfig | sc3 | | |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srs-MaxUpPts | Not Present | | |
| } | | | |
| } | | | |
| uplinkPowerControlCommonSCell-v1130 SEQUENCE | Not present | | |
| } | | | |

**Table 7.2.4.4.3-5: *RadioResourceConfigDedicatedSCell-r10:* Additional E-UTRAN FDD - UE timing advance adjustment accuracy test for SCell in sTAG requirement**

| Derivation Path: 36.508, Table 4.6.3-19AA | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicatedSCell-r10 ::= SEQUENCE { | | | |
| physicalConfigDedicatedSCell-r10 | PhysicalConfigDedicated SCell-r10-DEFAULT | | |
| mac-MainConfigSCell-r11 SEQUENCE { | | | |
| stag-Id-r11 | 1 | | |
| } | | | |
| } | | | |

**Table 7.2.4.4.3-6: *PhysicalConfigDedicatedSCell-r10-DEFAULT*: Additional E-UTRAN FDD - UE timing advance adjustment accuracy test for SCell in sTAG requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-6A PhysicalConfigDedicatedSCell-r10-DEFAULT with UL_CA | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicatedSCell-r10-DEFAULT ::= SEQUENCE { | | | |
| ul-Configuration-r10 SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | bw0 used with no frequency hopping. | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | 7 | | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |
| soundingRS-UL-ConfigDedicated-v1020 SEQUENCE { | | | |
| srs-AntennaPort-r10 | an1 | | |
| } | | | |
| } | | | |
| } | | | |

## 7.2.4.5 Test requirement

Tables 7.2.4.4.1-1, 7.2.4.5-1 and 7.2.4.5-2 define the primary level settings for E-UTRAN FDD - UE timing advance adjustment accuracy test for SCell in sTAG.

**Table 7.2.4.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD - UE timing advance adjustment accuracy test for SCell in sTAG**

| Parameter | Unit | Value | | | |
|---|---|---|---|---|---|
| | | Cell1 | | Cell2 | |
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Active PCell | | Cell1 | Cell1 | | |
| Active SCell | | | | Cell2 | Cell2 |
| TAG configuration | | pTAG | pTAG | sTAG | sTAG |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | | OP.1 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA[Note1] | dB | | | | |
| OCNG_RB[Note1] | dB | | | | |
| Timing Advance Command ($T_A$) | | / | / | 31 | 39 |
| $\hat{E}_s / I_{ot}$ | dB | 3 | | 3 | |
| $N_{oc}$ | dBm/15 KHz | -98 | | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 3 | | 3 | |
| Io[Note2] | dBm/9 MHz | -65.5 | | -65.5 | |
| Propagation Condition | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Io level has been derived from other parameters for information purpose. It is not a settable parameter.

**Table 7.2.4.5-2: Sounding Reference Signal Configuration to be used in E-UTRAN FDD - UE timing advance adjustment accuracy test for SCell in sTAG**

| Field | Value | Comment |
|---|---|---|
| srsBandwidthConfiguration | bw5 | |
| srsSubframeConfiguration | sc3 | Once every 5 subframes |
| ackNackSrsSimultaneousTransmission | FALSE | |
| srsMaxUpPTS | N/A | Not applicable for E-UTRAN FDD |
| srsBandwidth | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | |
| frequencyDomainPosition | 0 | |
| Duration | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 7 | SRS periodicity of 10. |
| transmissionComb | 0 | |
| cyclicShift | cs0 | No cyclic shift |
| SRS-AntennaPort | an1 | Number of antenna ports used for SRS transmission |
| Note: For further information see clause 6.3.2 in TS 36.331 [5]. | | |

The UE shall apply the signalled Timing Advance value to the transmission timing at the designated activation time i.e. 6 sub-frames after the reception of the timing advance command.

The UE shall adjust the timing of its transmission with a relative accuracy better than or equal to $\pm 4.5 \times T_S$ seconds to the signalled timing advance value compared to the timing of preceding uplink transmission. This requirement includes test tolerances.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 7.2.5 E-UTRAN TDD - UE Timing Advance Adjustment Accuracy Test for SCell in sTAG

### 7.2.5.1 Test purpose

To verify the UE in RRC_CONNECTED state adjusts the timing of its transmissions with accuracy by meeting the E-UTRAN TDD timing advance adjustment accuracy requirements in an AWGN model.

### 7.2.5.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 11 and forward that supports Uplink Carrier Aggregation and multiple timing advances.

### 7.2.5.3 Minimum conformance requirements

The timing advance is initiated from E-UTRAN with MAC message that implies and adjustment of the timing advance according to TS 36.321 [11] clause 5.2.

The UE shall adjust the timing of its uplink transmission timing at sub-frame *n*+6 for a timing advancement command received in sub-frame *n*.

The UE shall adjust the timing of its transmission with a relative accuracy better than or equal to $\pm 4 \times T_S$ seconds to the signalled timing advance value compared to the timing of preceding uplink transmission. The timing advance command is expressed in multiple of $16 \times T_S$ and is relative to the current uplink timing.

In case of random access response, 11-bit timing advance command, $T_A$, for a TAG indicates $N_{TA}$ values by index values of $T_A = 0, 1, 2, ..., 1282$, where an amount of the time alignment for the TAG is given by $N_{TA} = T_A \times 16$ according to TS 36.213 [8] clause 4.2.3. $N_{TA}$ is defined in TS 36.211 [9].

In other cases, 6-bit timing advance command, $T_A$, for a TAG indicates adjustment of the current $N_{TA}$ value, $N_{TA,old}$, to the new $N_{TA}$ value, $N_{TA,new}$, by index values of $T_A = 0, 1, 2,..., 63$, where $N_{TA,new} = N_{TA,old} + (T_A - 31) \times 16$ according to TS 36.213 [8] clause 4.2.3. Adjustment of $N_{TA}$ value by a positive or a negative amount indicates advancing or delaying the uplink transmission timing for the TAG by a given amount respectively.

For UE timers, the UE shall comply with the timer accuracies according to TS 36.133 [4] table 7.2.2-1. The requirements are only related to the actual timing measurements internally in the UE.

The normative reference for this requirement is TS 36.133 [4] clause 7.3 and A.7.2.5.

### 7.2.5.4 Test description

#### 7.2.5.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz for both PCell and SCell as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.45.

2. The general test parameter settings are set according to Table 7.2.5.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Two Cells are used. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier. PCell is in the Primary Timing Advance Group (pTAG) and SCell is in the secondary Timing Advance Group (sTAG). Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 shall be powered OFF.

**Table 7.2.5.4.1-1: General Test Parameters for E-UTRAN TDD - UE timing advance adjustment accuracy test for SCell in sTAG**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Timing Advance Command ($T_A$) value during T1 | | 31 | $N_{TA}$ = 0 for the purpose of establishing a reference value from which the timing advance adjustment accuracy can be measured during T2 |
| Timing Advance Command ($T_A$) value during T2 | | 39 | $N_{TA}$ = 128 |
| DRX | | OFF | |
| T1 | s | 5 | |
| T2 | s | 5 | |

### 7.2.5.4.2 Test procedure

The test consists of two cells. The test consists of two successive time periods, with time durations of T1 and T2 respectively. In each time period, timing advance commands for sTAG are sent to the UE and SRS are sent from the UE and received by the SS. By measuring the reception of the SRS, the transmit timing, and hence the timing advance adjustment accuracy, can be measured for SCell in sTAG. The UE Time Alignment Timer (timeAlignmentTimer IE), defined in TS 36.321 [11] clause 5.2, shall be configured so that it does not expire in the duration of the test.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCell according to Annex C.0 and C.1 for all downlink physical channels.
3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4 with UL CA.
4. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).
5. Set the parameters according to T1 in Tables 7.2.5.5-1 and 7.2.5.5-2. Propagation conditions are set according to Annex B clause B.1.1.
6. SS shall transmit an RRCConnectionReconfiguration message.
7. The UE shall transmit RRCConnectionReconfigurationComplete message.
8. SS shall transmit one message with a Timing Advance Command MAC Control Element for sTAG, $T_A$. The timing advance command, $T_A$, shall be set to 31 which indicate zero adjustment of the current $N_{TA}$ value for sTAG. The timing advance adjustment during T1 shall be $N_{TA}$ = 0.
9. The UE shall transmit SRS and adjust its uplink timing at the beginning of sub-frame n+6 for a timing advance command $T_A$ received in sub-frame n. Sub-frame n is the sub-frame where the UE received the time advance command.
10. When T1 expires, the SS shall switch the timing advance command ($T_A$) from T1 to T2 as specified in Table 7.2.5.5-1.
11. SS shall transmit a sequence of messages with Timing Advance Command MAC Control Element for sTAG, $T_A$. The timing advance command, $T_A$, shall be set to 39 which indicate adjustment of the current $N_{TA}$ value for sTAG. The timing advance adjustment during T2 shall be $N_{TA}$ = 128.

12. The UE shall transmit SRS and adjust its uplink timing at the beginning of sub-frame n+6 for a timing advance command $T_A$ received in sub-frame n. Sub-frame n is the sub-frame where the UE received the time advance command.

13. The result from the SRS and adjustment of the timing advance in step 12) is used to measure that the UE adjusts the timing of its transmission with a relative accuracy better than or equal to $\pm 4.5 \times T_S$ to the signalled timing advance value compared to the timing of preceding uplink transmission.

14. If the UE adjust the timing of its transmission within a relative accuracy greater than or equal to $\pm 4.5 \times T_S$ to the signalled timing advance value compared to the timing of preceding uplink transmission then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

15. The SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

16. After the RRC connection release, the SS:

    - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),

    or

    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

17. Repeat step 2-16 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 7.2.5.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.2.5.4.3-1: *RRCConnectionReconfiguration*: Additional E-UTRAN TDD - UE timing advance adjustment accuracy test for SCell in sTAG requirement**

| Derivation Path: Clause 4.6.1 Table 4.6.1-8, condition SCell_AddMod |
|---|

**Table7.2.5.4.3-2: *MAC-MainConfig-RBC*: Additional E-UTRAN TDD - UE timing advance adjustment accuracy test for SCell in sTAG requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC, condition SCell_AddMod | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| stag-ToAddModList-r11 SEQUENCE (SIZE (1..maxSTAG-r11)) OF SEQUENCE { | Not present | | |
| stag-Id-r11 | 1 | | |
| timeAlignmentTimerSTAG-r11 | infinity | | |
| } | | | |
| } | | | |

**Table 7.2.5.4.3-3: *SCellToAddMod-r10:* Additional E-UTRAN TDD - UE timing advance adjustment accuracy test for SCell in sTAG requirement**

| Derivation Path: 36.508, Table 4.6.3-19D | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SCellToAddMod-r10 ::= SEQUENCE { | | | |
| sCellIndex-r10 | 1 | | |
| cellIdentification-r10 SEQUENCE { | | | |
| physCellId-r10 | PhysicalCellIdentity of Cell 2 | | |
| dl-CarrierFreq-r10 | Same downlink EARFCN as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 7.2.5.4.3-4: *RadioResourceConfigCommonSCell-r10:* Additional E-UTRAN TDD - UE timing advance adjustment accuracy test for SCell in sTAG requirement**

| Derivation Path: 36.508, Table 4.6.3-13A, condition UL_CA | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigCommonSCell-r10-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigCommon-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw5 | | |
| srs-SubframeConfig | sc3 | | |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srs-MaxUpPts | Not Present | | |
| } | | | |
| } | | | |
| uplinkPowerControlCommonSCell-v1130 SEQUENCE | Not present | | |
| } | | | |

**Table 7.2.5.4.3-5: *RadioResourceConfigDedicatedSCell-r10:* Additional E-UTRAN TDD - UE timing advance adjustment accuracy test for SCell in sTAG requirement**

| Derivation Path: 36.508, Table 4.6.3-19AA | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicatedSCell-r10 ::= SEQUENCE { | | | |
| physicalConfigDedicatedSCell-r10 | PhysicalConfigDedicated SCell-r10-DEFAULT | | |
| mac-MainConfigSCell-r11 SEQUENCE { | | | |
| stag-Id-r11 | 1 | | |
| } | | | |
| } | | | |

**Table 7.2.5.4.3-6: *PhysicalConfigDedicatedSCell-r10-DEFAULT*: Additional E-UTRAN TDD - UE timing advance adjustment accuracy test for SCell in sTAG requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-6A PhysicalConfigDedicatedSCell-r10-DEFAULT with UL_CA | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicatedSCell-r10-DEFAULT ::= SEQUENCE { | | | |
| ul-Configuration-r10 SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | bw0 used with no frequency hopping. | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | 15 | | |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |
| soundingRS-UL-ConfigDedicated-v1020 SEQUENCE { | | | |
| srs-AntennaPort-r10 | an1 | | |
| } | | | |
| } | | | |
| } | | | |

## 7.2.5.5 Test requirement

Tables 7.2.5.4.1-1, 7.2.5.5-1 and 7.2.5.5-2 define the primary level settings for E-UTRAN TDD - UE timing advance adjustment accuracy test for SCell in sTAG.

**Table 7.2.5.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD - UE timing advance adjustment accuracy test for SCell in sTAG**

| Parameter | Unit | Value | | | |
|---|---|---|---|---|---|
| | | Cell 1 | | Cell 2 | |
| | | T1 | T2 | T1 | T2 |

| | | | |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | 2 |
| $BW_{channel}$ | MHz | 10 | 10 |
| Active PCell | | Cell1 | |
| Active SCell | | | Cell2 |
| TAG configuration | | pTAG | sTAG |
| Special subframe configuration[Note1] | | 6 | 6 |
| Uplink-downlink configuration[Note2] | | 1 | 1 |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | OP.1 TDD |
| PBCH_RA | dB | 0 | 0 |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA[Note3] | dB | | |
| OCNG_RB[Note3] | dB | | |
| Timing Advance Command ($T_A$) | | | 31 / 39 |
| $\hat{E}_s / I_{ot}$ | dB | 3 | 3 |
| $N_{oc}$ | dBm/15 KHz | -98 | -98 |
| $\hat{E}_s / N_{oc}$ | dB | 3 | 3 |
| Io[Note4] | dBm/9 MHz | -65.5 | -65.5 |
| Propagation Condition | | AWGN | AWGN |

Note 1: For the special subframe configuration see table 4.2-1 in TS 36.211.
Note 2: For the uplink-downlink configuration see table 4.2-2 in TS 36.211.
Note 3: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: Io level has been derived from other parameters for information purpose. It is not a settable parameter.

**Table 7.2.5.5-2: Sounding Reference Signal Configuration to be used in E-UTRAN TDD - UE timing advance adjustment accuracy test for SCell in sTAG**

| Field | Value | Comment |
|---|---|---|
| srsBandwidthConfiguration | bw5 | |
| srsSubframeConfiguration | sc3 | Once every 5 subframes |
| ackNackSrsSimultaneousTransmission | FALSE | |
| srsMaxUpPTS | N/A | |
| srsBandwidth | bw0 | No hopping |
| srsHoppingBandwidth | hbw0 | |
| frequencyDomainPosition | 0 | |
| Duration | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 15 | SRS periodicity of 10ms. |
| transmissionComb | 0 | |
| cyclicShift | cs0 | No cyclic shift |
| SRS-AntennaPort | an1 | Number of antenna ports used for SRS transmission |
| Note: For further information see clause 6.3.2 in TS 36.331 [5]. | | |

The UE shall apply the signalled Timing Advance value to the transmission timing at the designated activation time i.e. 6 sub-frames after the reception of the timing advance command.

The UE shall adjust the timing of its transmission with a relative accuracy better than or equal to $\pm 4.5 \times T_S$ seconds to the signalled timing advance value compared to the timing of preceding uplink transmission. This requirement includes test tolerances.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 7.2.5A E-UTRAN TDD - UE Timing Advance Adjustment Accuracy Test for SCell in sTAG for 20MHz +20MHz bandwidth

### 7.2.5A.1 Test purpose

Same test purpose as defined in clause 7.2.5.1.

### 7.2.5A.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 11 and forward that supports Uplink Carrier Aggregation and multiple timing advances.

### 7.2.5A.3 Minimum conformance requirements

Same minimum conformance requirements as defined in clause 7.2.5.3 with the following exception:

- The normative reference for this requirement is TS 36.133 [4] clause 7.3 and A.7.2.5A.

### 7.2.5A.4 Test description

#### 7.2.5A.4.1 Initial conditions

Same initial conditions as in clause 7.2.5.4.1 with the following exceptions:

- Channel Bandwidth to be tested: 20 MHz for both PCell and SCell
- Instead of table 7.2.5.4.1-1, use table 7.2.5A.4.1-1.

**Table 7.2.5A.4.1-1: General Test Parameters for E-UTRAN TDD - UE timing advance adjustment accuracy test for SCell in sTAG for 20 MHz +20 MHz**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.3 TDD | As specified in section A.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.10 TDD | As specified in section A.2 |
| Timing Advance Command ($T_A$) value during T1 | | 31 | $N_{TA}$ = 0 for the purpose of establishing a reference value from which the timing advance adjustment accuracy can be measured during T2 |
| Timing Advance Command ($T_A$) value during T2 | | 39 | $N_{TA}$ = 128 |
| DRX | | OFF | |
| T1 | s | 5 | |
| T2 | s | 5 | |

#### 7.2.5A.4.2 Test procedure

Same test procedure as in clause 7.2.5.4.2

### 7.2.5A.4.3 Message contents

Same message contents as in clause 7.2.5.4.3

## 7.2.5A.5 Test requirement

Table 7.2.5A.4.1-1, 7.2.5A.5-1 and 7.2.5A.5-2 define the primary level settings. The listed parameter values in Table 7.2.5A.5-1 will replace the values of corresponding parameters in Table 7.2.5.5-1 other parameters keep the same.

**Table 7.2.5A.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD - UE timing advance adjustment accuracy test for SCell in sTAG for 20 MHz +20 MHz**

| Parameter | Unit | Value | | | |
|---|---|---|---|---|---|
| | | Cell 1 | | Cell 2 | |
| | | T1 | T2 | T1 | T2 |
| $BW_{channel}$ | MHz | 20 | | 20 | |
| OCNG Patterns defined in D.2.7 | | OP.7 TDD | | OP.7 TDD | |
| $Io^{Note4}$ | dBm/18 MHz | -62.5 | | -62.5 | |

**Table 7.2.5A.5-2: Sounding Reference Signal Configuration to be used in E-UTRAN TDD - UE timing advance adjustment accuracy test for SCell in sTAG for 20 MHz +20 MHz**

| Field | Value | Comment |
|---|---|---|
| srsBandwidthConfiguration | bw5 | |
| srsSubframeConfiguration | sc3 | Once every 5 subframes |
| ackNackSrsSimultaneousTransmission | FALSE | |
| srsMaxUpPTS | N/A | |
| srsBandwidth | bw0 | No hopping |
| srsHoppingBandwidth | hbw0 | |
| frequencyDomainPosition | 0 | |
| Duration | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 15 | SRS periodicity of 10ms. |
| transmissionComb | 0 | |
| cyclicShift | cs0 | No cyclic shift |
| SRS-AntennaPort | an1 | Number of antenna ports used for SRS transmission |
| Note: For further information see clause 6.3.2 in TS 36.331 [5]. | | |

The UE shall apply the signalled Timing Advance value to the transmission timing at the designated activation time i.e. 6 sub-frames after the reception of the timing advance command.

The UE shall adjust the timing of its transmission with a relative accuracy better than or equal to $\pm 4.5 \times T_S$ seconds to the signalled timing advance value compared to the timing of preceding uplink transmission. This requirement includes test tolerances.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 7.2.5B E-UTRAN TDD - UE Timing Advance Adjustment Accuracy Test for SCell in sTAG for 20MHz +10MHz bandwidth

## 7.2.5B.1 Test purpose

Same test purpose as defined in clause 7.2.5.1.

## 7.2.5B.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 11 and forward that supports Uplink Carrier Aggregation and multiple timing advances.

## 7.2.5B.3 Minimum conformance requirements

Same minimum conformance requirements as defined in clause 7.2.5.3 with the following exception:

- The normative reference for this requirement is TS 36.133 [4] clause 7.3 and A.7.2.5B.

## 7.2.5B.4 Test description

### 7.2.5B.4.1 Initial conditions

Same initial conditions as in clause 7.2.5.4.1 with the following exceptions:

- Channel Bandwidth to be tested: 20 MHz for PCell and 10 MHz for SCell
- Instead of table 7.2.5.4.1-1, use table 7.2.5B.4.1-1.

**Table 7.2.5B.4.1-1: General Test Parameters for E-UTRAN TDD - UE timing advance adjustment accuracy test for SCell in sTAG for 20 MHz +10 MHz**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | For Cell 1: DL Reference Measurement Channel R.3 TDD<br>For Cell 2: DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | For Cell 1: DL Reference Measurement Channel R.10 TDD<br>For Cell 2: DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Timing Advance Command ($T_A$) value during T1 | | 31 | $N_{TA}$ = 0 for the purpose of establishing a reference value from which the timing advance adjustment accuracy can be measured during T2 |
| Timing Advance Command ($T_A$) value during T2 | | 39 | $N_{TA}$ = 128 |
| DRX | | OFF | |
| T1 | s | 5 | |
| T2 | s | 5 | |

### 7.2.5B.4.2 Test procedure

Same test procedure as in clause 7.2.5.4.2

### 7.2.5B.4.3 Message contents

Same message contents as in clause 7.2.5.4.3

## 7.2.5B.5 Test requirement

Table 7.2.5B.4.1-1, 7.2.5B.5-1 and 7.2.5B.5-2 define the primary level settings. The listed parameter values in Table 7.2.5B.5-1 will replace the values of corresponding parameters in Table 7.2.5.5-1 other parameters keep the same.

**Table 7.2.5B.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD - UE timing advance adjustment accuracy test for SCell in sTAG for 20 MHz +10 MHz**

| Parameter | Unit | Value | | | |
|---|---|---|---|---|---|
| | | Cell 1 | | Cell 2 | |
| | | T1 | T2 | T1 | T2 |

| $BW_{channel}$ | MHz | 20 | 10 |
|---|---|---|---|
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.7 (OP.7 TDD) | | OP.7 TDD | OP.1 TDD |
| $Io^{Note4}$ | dBm/18 MHz | -62.5 | N/A |
| | dBm/9 MHz | N/A | -65.5 |

**Table 7.2.5B.5-2: Sounding Reference Signal Configuration to be used in E-UTRAN TDD - UE timing advance adjustment accuracy test for SCell in sTAG for 20 MHz +10 MHz**

| Field | Value | Comment |
|---|---|---|
| srsBandwidthConfiguration | bw5 | |
| srsSubframeConfiguration | sc3 | Once every 5 subframes |
| ackNackSrsSimultaneousTransmission | FALSE | |
| srsMaxUpPTS | N/A | |
| srsBandwidth | bw0 | No hopping |
| srsHoppingBandwidth | hbw0 | |
| frequencyDomainPosition | 0 | |
| Duration | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 15 | SRS periodicity of 10ms. |
| transmissionComb | 0 | |
| cyclicShift | cs0 | No cyclic shift |
| SRS-AntennaPort | an1 | Number of antenna ports used for SRS transmission |
| Note: For further information see clause 6.3.2 in TS 36.331 [5]. | | |

The UE shall apply the signalled Timing Advance value to the transmission timing at the designated activation time i.e. 6 sub-frames after the reception of the timing advance command.

The UE shall adjust the timing of its transmission with a relative accuracy better than or equal to $\pm 4.5 \times T_S$ seconds to the signalled timing advance value compared to the timing of preceding uplink transmission. This requirement includes test tolerances.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 7.2.6 E-UTRAN FDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA

### 7.2.6.1 Test purpose

To verify the UE in RRC_CONNECTED state adjusts the timing of its transmissions with accuracy by meeting the E-UTRAN FDD timing advance adjustment accuracy requirements for Cat-M1 UE in CEModeA in an AWGN model.

### 7.2.6.2 Test applicability

This test applies to all types of E-UTRA FD-FDD UE Release 13 and forward of UE category M1.

### 7.2.6.3 Minimum conformance requirements

The timing advance is initiated from E-UTRAN with MAC message that implies and adjustment of the timing advance according to TS 36.321 [11] clause 5.2.

The UE shall adjust the timing of its uplink transmission timing at sub-frame *n*+6 for a timing advancement command received in sub-frame *n*.

The UE shall adjust the timing of its transmission with a relative accuracy better than or equal to $\pm 4 \times T_S$ seconds to the signalled timing advance value compared to the timing of preceding uplink transmission. The timing advance command is expressed in multiple of $16 \times T_S$ and is relative to the current uplink timing.

In case of random access response, 11-bit timing advance command, $T_A$, for a cell indicates $N_{TA}$ values by index values of $T_A$ = 0, 1, 2, ..., 1282, where an amount of the time alignment for the cell is given by $N_{TA} = T_A \times 16$ according to TS 36.213 [8] clause 4.2.3. $N_{TA}$ is defined in TS 36.211 [9].

In other cases, 6-bit timing advance command, $T_A$, indicates adjustment of the current $N_{TA}$ value, $N_{TA,old}$, to the new $N_{TA}$ value, $N_{TA,new}$, by index values of $T_A$ = 0, 1, 2,..., 63, where $N_{TA,new} = N_{TA,old} + (T_A - 31) \times 16$ according to TS 36.213 [8] clause 4.2.3. Adjustment of $N_{TA}$ value by a positive or a negative amount indicates advancing or delaying the uplink transmission timing by a given amount respectively.

For UE timers, the UE shall comply with the timer accuracies according to TS 36.133 [4] table 7.2.2-1. The requirements are only related to the actual timing measurements internally in the UE.

The normative reference for this requirement is TS 36.133 [4] clause 7.3.2.2 and A.7.2.6.

## 7.2.6.4 Test description

### 7.2.6.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.18 using only main UE Tx/Rx antenna.
2. The general test parameter settings are set according to Table 7.2.6.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. There is one E-UTRA FDD carrier and one cell specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.2.6.4.1-1: General Test Parameters for E-UTRAN FDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA**

| | | | |
|---|---|---|---|
| PDSCH parameters: DL Reference Measurement Channel | | R.20 FDD | As specified in clause A.8.1 |
| MPDCCH parameters: DL Reference Measurement Channel | | R.16 FDD | As specified in clause A.7.1 |
| Timing Advance Command ($T_A$) value during T1 | | 31 | $N_{TA}$ = 0 for the purpose of establishing a reference value from which the timing advance adjustment accuracy can be measured during T2 |
| Timing Advance Command ($T_A$) value during T2 | | 39 | $N_{TA}$ = 128 |
| DRX | | OFF | |
| T1 | s | 5 | |
| T2 | s | 5 | |

### 7.2.6.4.2 Test procedure

The test consists of a single cell. The test consists of two successive time periods, with time durations of T1 and T2 respectively. In each time period, timing advance commands are sent to the UE and SRS are sent from the UE and received by the SS. By measuring the reception of the SRS, the transmit timing, and hence the timing advance adjustment accuracy, can be measured. The UE Time Alignment Timer (*timeAlignmentTimer* IE), defined in TS 36.321 [11] clause 5.2, shall be configured so that it does not expire in the duration of the test.

1. Ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3AA.

2. Set the parameters according to T1 in Tables 7.2.6.5-1 and 7.2.6.5-2. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an *RRCConnectionReconfiguration* message.

4. The UE shall transmit *RRCConnectionReconfigurationComplete* message.

5. SS shall transmit one message with a Timing Advance Command, $T_A$. The timing advance command, $T_A$, shall be set to 31 which indicate adjustment of the current $N_{TA}$ value. The timing advance adjustment during T1 shall be $N_{TA} = 0$.

6. The UE shall transmit SRS and adjust its uplink timing at the beginning of sub-frame n+6 for a timing advance command $T_A$ received in sub-frame n. Sub-frame n is the sub-frame where the UE received the time advance command.

7. When T1 expires, the SS shall switch the timing advance command ($T_A$) from T1 to T2 as specified in Table 7.2.6.5-1.

8. SS shall transmit a sequence of messages with Timing Advance Command, $T_A$. The timing advance command, $T_A$, shall be set to 39 which indicate adjustment of the current $N_{TA}$ value. The timing advance adjustment during T2 shall be $N_{TA} = 128$.

9. The UE shall transmit SRS and adjust its uplink timing at the beginning of sub-frame n+6 for a timing advance command $T_A$ received in sub-frame n. Sub-frame n is the sub-frame where the UE received the time advance command.

10. The result from the SRS and adjustment of the timing advance in step 9) is used to measure that the UE adjusts the timing of its transmission with a relative accuracy better than or equal to $\pm 4.5 \times T_S$ to the signalled timing advance value compared to the timing of preceding uplink transmission.

11. If the UE adjust the timing of its transmission within a relative accuracy greater than or equal to $\pm 4.5 \times T_S$ to the signalled timing advance value compared to the timing of preceding uplink transmission then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

12. The SS shall transmit *RRCConnectionRelease* message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

13. After the RRC connection release, the SS:

    - transmits in Cell 1 a Paging message (including *PagingRecord* with *ue-Identity*) for the UE and ensures the UE is in State 3A-CE according to TS 36.508 [7] clause 4.5.3AB (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3AA),

or

    - switches off and on the UE and ensures the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3AA.

14. Repeat step 2-13 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.2.6.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.2.6.4.3-1: Common Exception messages for E-UTRAN FDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.4-2 |

**Table 7.2.6.4.3-2: *SoundingRS-UL-ConfigCommon-DEFAULT*: Additional E-UTRAN FDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw5 | Channel-bandwidth-dependent parameter | |
| srs-SubframeConfig | sc3 | | FDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | Not present | | FDD |
| } | | | |
| } | | | |

**Table 7.2.6.4.3-3: *SoundingRSUL-ConfigDedicated-DEFAULT*: Additional E-UTRAN FDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw0 | bw0 used with no frequency hopping. bw3 used with frequency hopping | |
| srs-HoppingBandwidth | hbw0 | | |
| freqDomainPosition | 0 | | |
| duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | 7 | SRS periodicity of 10 | FDD |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 7.2.6.4.3-4: *MAC-MainConfig-RBC*: Additional E-UTRAN FDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| dl-SCH-Config SEQUENCE {} | Not present | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| timeAlignmentTimerDedicated | Infinity | | |

## 7.2.6.5 Test requirement

Tables 7.2.6.4.1-1, 7.2.6.5-1 and 7.2.6.5-2 define the primary level settings for E-UTRAN FDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA.

**Table 7.2.6.5-1: Cell specific Test Parameters for E-UTRAN FDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA**

| Parameter | Unit | Value | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| PDSCH parameters: DL Reference Measurement Channel defined in clause A.8.1 | | R.20 FDD | |
| MPDCCH parameters: DL Reference Measurement Channel defined in clause A.7.1 | | R.16 FDD | |
| OCNG Patterns defined in clause D.1.21 | | OP.21 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| MPDCCH_RA | dB | | |
| MPDCCH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note1 | dB | | |
| OCNG_RB Note1 | dB | | |
| Timing Advance Command ($T_A$) | | 31 | 39 |
| $\hat{E}_s / I_{ot}$ | dB | 3 | |
| $N_{oc}$ | dBm/15 KHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 3 | |
| Io Note2 | dBm/9 MHz | -65.5 | |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that cells is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Io level has been derived from other parameters for information purpose. It is not a settable parameter.

**Table 7.2.6.5-2: Sounding Reference Symbol Configuration to be used in E-UTRAN FDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA**

| Field | Value | Comment |
|---|---|---|
| srsBandwidthConfiguration | Bw5 | |
| srsSubframeConfiguration | sc3 | Once every 5 subframes |
| ackNackSrsSimultaneousTransmission | FALSE | |
| srsMaxUpPTS | N/A | Not applicable for E-UTRAN FDD |
| srsBandwidth | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | |
| frequencyDomainPosition | 0 | |
| Duration | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 7 | SRS periodicity of 10. |
| transmissionComb | 0 | |
| cyclicShift | cs0 | No cyclic shift |
| SRS-AntennaPort | an1 | Number of antenna ports used for SRS transmission |

Note 1: For further information see clause 6.3.2 in TS 36.331.

The UE shall apply the signalled Timing Advance value to the transmission timing at the designated activation time i.e. 6 sub-frames after the reception of the timing advance command.

The UE shall adjust the timing of its transmission with a relative accuracy better than or equal to $\pm 4.5 \times T_S$ seconds to the signalled timing advance value compared to the timing of preceding uplink transmission. This requirement includes test tolerances.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 7.2.7 E-UTRAN HD-FDD UE Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA

### 7.2.7.1 Test purpose

To verify the UE in RRC_CONNECTED state adjusts the timing of its transmissions with accuracy by meeting the E-UTRAN HD-FDD UE timing advance adjustment accuracy requirements for Cat-M1 UE in CEModeA in an AWGN model.

### 7.2.7.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE Release 13 and forward of UE category M1.

### 7.2.7.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 7.2.6.

The normative reference for this requirement is TS 36.133 [4] clause 7.3.2.2 and A.7.2.7.

### 7.2.7.4 Test description

#### 7.2.7.4.1 Initial conditions

Same initial conditions as in clause 7.2.6.4.1 with following exceptions:

- Instead of Table 7.2.6.4.1-1 → use Table 7.2.7.4.1-1.

**Table 7.2.7.4.1-1: General Test Parameters for E-UTRAN HD-FDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters: DL Reference Measurement Channel | | R.10 HD-FDD | As specified in clause A.8.2 |
| MPDCCH parameters: DL Reference Measurement Channel | | R.6 HD-FDD | As specified in clause A.7.2 |
| Timing Advance Command ($T_A$) value during T1 | | 31 | $N_{TA}$ = 0 for the purpose of establishing a reference value from which the timing advance adjustment accuracy can be measured during T2 |
| Timing Advance Command ($T_A$) value during T2 | | 39 | $N_{TA}$ = 128 |
| DRX | | OFF | |
| T1 | s | 5 | |
| T2 | s | 5 | |

#### 7.2.7.4.2 Test procedure

Same test procedure as in clause 7.2.6.4.2 with following exceptions:

- Instead of Tables 7.2.6.5-1 → use Table 7.2.7.5-1.
- Instead of Table 7.2.6.5-2 → use Table 7.2.7.5-1.

#### 7.2.7.4.3 Message contents

Same message contents as in clause 7.2.6.4.3.

### 7.2.7.5 Test requirement

Tables 7.2.7.4.1-1, 7.2.7.5-1 and 7.2.7.5-2 define the primary level settings for E-UTRAN HD-FDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA.

**Table 7.2.7.5-1: Cell specific Test Parameters for E-UTRAN HD-FDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA**

| Parameter | Unit | Value | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| PDSCH parameters:<br>DL Reference Measurement Channel defined in clause A.8.2 | | R.10 HD-FDD | |
| MPDCCH parameters:<br>DL Reference Measurement Channel defined in clause A.7.2 | | R.6 HD-FDD | |
| OCNG Patterns defined in clause D.1.21 | | OP.21 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| MPDCCH_RA | dB | | |
| MPDCCH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note1 | dB | | |
| OCNG_RB Note1 | dB | | |
| Timing Advance Command ($T_A$) | | 31 | 39 |
| $\hat{E}_s / I_{ot}$ | dB | 3 | |
| $N_{oc}$ | dBm/15 KHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 3 | |
| Io Note2 | dBm/9 MHz | -65.5 | |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Io level has been derived from other parameters for information purpose. It is not a settable parameter.

**Table 7.2.7.5-2: Sounding Reference Symbol Configuration to be used in E-UTRAN HD-FDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA**

| Field | Value | Comment |
|---|---|---|
| srsBandwidthConfiguration | Bw5 | |
| srsSubframeConfiguration | sc3 | Once every 5 subframes |
| ackNackSrsSimultaneousTransmission | FALSE | |
| srsMaxUpPTS | N/A | |
| srsBandwidth | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | |
| frequencyDomainPosition | 0 | |
| Duration | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 7 | SRS periodicity of 10. |
| transmissionComb | 0 | |
| cyclicShift | cs0 | No cyclic shift |
| SRS-AntennaPort | an1 | Number of antenna ports used for SRS transmission |
| Note 1: For further information see clause 6.3.2 in TS 36.331. | | |

The UE shall apply the signalled Timing Advance value to the transmission timing at the designated activation time i.e. 6 sub-frames after the reception of the timing advance command.

The UE shall adjust the timing of its transmission with a relative accuracy better than or equal to $\pm 4.5 \times T_S$ seconds to the signalled timing advance value compared to the timing of preceding uplink transmission. This requirement includes test tolerances.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 7.2.8 E-UTRAN TDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA

### 7.2.8.1 Test purpose

To verify the UE in RRC_CONNECTED state adjusts the timing of its transmissions with accuracy by meeting the E-UTRAN TDD timing advance adjustment accuracy requirements for Cat-M1 UE in CEModeA in an AWGN model.

### 7.2.8.2 Test applicability

This test applies to all types of E-UTRA TDD UE Release 13 and forward of UE category M1.

### 7.2.8.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 7.2.6.

The normative reference for this requirement is TS 36.133 [4] clause 7.3.2.2 and A.7.2.8.

### 7.2.8.4 Test description

#### 7.2.8.4.1 Initial conditions

Same initial conditions as in clause 7.2.6.4.1 with following exceptions:

- Instead of Table 7.2.6.4.1-1 → use Table 7.2.8.4.1-1.

**Table 7.2.8.4.1-1: General Test Parameters for E-UTRAN TDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA**

| | | | |
|---|---|---|---|
| PDSCH parameters: DL Reference Measurement Channel | | R.16 TDD | As specified in clause A.8.3 |
| MPDCCH parameters: DL Reference Measurement Channel | | R.14 TDD | As specified in clause A.7.3 |
| Timing Advance Command ($T_A$) value during T1 | | 31 | $N_{TA}$ = 0 for the purpose of establishing a reference value from which the timing advance adjustment accuracy can be measured during T2 |
| Timing Advance Command ($T_A$) value during T2 | | 39 | $N_{TA}$ = 128 |
| DRX | | OFF | |
| T1 | s | 5 | |
| T2 | s | 5 | |

### 7.2.8.4.2 Test procedure

Same test procedure as in clause 7.2.6.4.2 with following exceptions:

- Instead of Tables 7.2.6.5-1 → use Table 7.2.8.5-1.
- Instead of Table 7.2.6.5-2 → use Table 7.2.8.5-1.

### 7.2.8.4.3 Message contents

Same message contents as in clause 7.2.6.4.3 with condition TDD.

## 7.2.8.5 Test requirement

Tables 7.2.8.4.1-1, 7.2.8.5-1 and 7.2.8.5-2 define the primary level settings for E-UTRAN TDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA.

**Table 7.2.8.5-1: Cell specific Test Parameters for E-UTRAN TDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA**

| Parameter | Unit | Value | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| PDSCH parameters: DL Reference Measurement Channel defined in clause A.8.3 | | R.16 TDD | |
| MPDCCH parameters: DL Reference Measurement Channel defined in clause A.7.3 | | R.14 TDD | |
| OCNG Patterns defined in clause D.2.11 | | OP.11 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| MPDCCH_RA | dB | | |
| MPDCCH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA[Note1] | dB | | |
| OCNG_RB[Note1] | dB | | |
| Timing Advance Command ($T_A$) | | 31 | 39 |
| $\hat{E}_s / I_{ot}$ | dB | 3 | |
| $N_{oc}$ | dBm/15 KHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 3 | |
| Io[Note2] | dBm/9 MHz | -65.5 | |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that cells is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Io level has been derived from other parameters for information purpose. It is not a settable parameter.

**Table 7.2.8.5-2: Sounding Reference Symbol Configuration to be used in E-UTRAN TDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA**

| Field | Value | Comment |
|---|---|---|
| srsBandwidthConfiguration | Bw5 | |
| srsSubframeConfiguration | sc3 | Once every 5 subframes |
| ackNackSrsSimultaneousTransmission | FALSE | |
| srsMaxUpPTS | N/A | Not applicable for E-UTRAN TDD |
| srsBandwidth | 0 | No hopping |
| srsHoppingBandwidth | hbw0 | |
| frequencyDomainPosition | 0 | |
| Duration | TRUE | Indefinite duration |
| Srs-ConfigurationIndex | 15 | SRS periodicity of 10. |
| transmissionComb | 0 | |
| cyclicShift | cs0 | No cyclic shift |
| SRS-AntennaPort | an1 | Number of antenna ports used for SRS transmission |

Note 1: For further information see clause 6.3.2 in TS 36.331.

The UE shall apply the signalled Timing Advance value to the transmission timing at the designated activation time i.e. 6 sub-frames after the reception of the timing advance command.

The UE shall adjust the timing of its transmission with a relative accuracy better than or equal to ±4.5 × $T_S$ seconds to the signalled timing advance value compared to the timing of preceding uplink transmission. This requirement includes test tolerances.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 7.2.9 HD-FDD UE Timing Advance Adjustment Accuracy Test for Category NB1 UE in Standalone Mode under Enhance Coverage

Editor's notes: This clause is incomplete, the following items are TBD

- Test frequencies is FFS.
- Test system uncertainty and tolerance is undefined.
- Connection diagram is TBD.

### A.7.2.9.1 Test Purpose

The purpose of the test is to verify E-UTRAN Timing Advance adjustment accuracy requirements for category NB1 UE in enhanced coverage in an AWGN model.

### 7.2.9.2 Test applicability

This test applies to all types of HD-FDD UE release 13 and forward of UE Category NB1.

### 7.2.9.3 Minimum conformance requirements

The timing advance is initiated from E-UTRAN with MAC message that implies and adjustment of the timing advance, see TS 36.321 [17] clause 5.2.

UE shall adjust the timing of its uplink transmission timing at sub-frame *n*+12 for a timing advance command received in sub-frame *n*. In case repetitions are used on the downlink, sub-frame *n* refers to the last subframe in the repetition period in which the message containing the MAC control information was received. The UE shall not apply a TA command during an uplink repetition period.

The UE shall adjust the timing of its transmissions with a relative accuracy better than or equal to ±13.33* $T_S$ seconds to the signalled timing advance value compared to the timing of preceding uplink transmission. The timing advance command is expressed in multiples of 16* $T_S$ and is relative to the current uplink timing.

In case of random access response, 11-bit timing advance command, $T_A$, indicates $N_{TA}$ values by index values of $T_A$ = 0, 1, 2, ..., 1282, where an amount of the time alignment is given by $N_{TA} = T_A \times 16$ according to TS 36.213 [8] clause 16.1.2. $N_{TA}$ is defined in TS 36.211 [9].

In other cases, 6-bit timing advance command, $T_A$, indicates adjustment of the current $N_{TA}$ value, $N_{TA,old}$, to the new $N_{TA}$ value, $N_{TA,new}$, by index values of $T_A$ = 0, 1, 2,..., 63, where $N_{TA,new} = N_{TA,old} + (T_A - 31) \times 16$ according to TS 36.213 [8] clause 16.1.2. Adjustment of $N_{TA}$ value by a positive or a negative amount indicates advancing or delaying the uplink transmission timing by a given amount respectively.

For UE timers, the UE shall comply with the timer accuracies according to TS 36.133 [4] table 7.21.2-1. The requirements are only related to the actual timing measurements internally in the UE.

The normative reference for this requirement is TS 36.133 [4] clause 7.22 and A.7.2.9.

## 7.2.9.4 Test description

### 7.2.9.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 8.1.1.

Frequencies to be tested: FFS.

Channel Bandwidth to be tested: 200 KHz.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure [TBD].
2. The general test parameter settings are set according to Table 7.2.9.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. There is one E-UTRA FDD carrier and one cell specified in the test. nCell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.2.9.4.1-1: General Test Parameters for E-UTRAN Timing Advance Accuracy Test for Category NB1 UE in Standalone Mode under Enhanced Coverage**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| NB-IoT operational mode | | | Standalone | |
| CP Length | | | Normal | |
| Timing Advance Command ($T_A$) value during T1 | | | 31 | $N_{TA}$ = 0 for the purpose of establishing a reference value from which the timing advance adjustment accuracy can be measured during T2 |
| Timing Advance Command ($T_A$) value during T2 | | | 39 | $N_{TA}$ = 128 |
| Number of repetitons | NPDCCH | | 128 | |
| | NPUSCH | | 32 | |
| DRX | | | OFF | |
| T1 | | s | 5 | |
| T2 | | s | 5 | |

### 7.2.9.4.2 Test procedure

The test consists of a single cell. The test consists of two successive time periods, with time duration of T1 and T2 respectively. In each time period, timing advance commands are sent to the UE and the UE is scheduled in uplink subframes to transmit NPUSCH, which is received by the test equipment. By measuring the reception of the NPUSCH, the transmit timing, and hence the timing advance adjustment accuracy, can be measured.The UE Time Alignment Timer (timeAlignmentTimer IE), defined in TS 36.321 [11] clause 5.2, shall be configured so that it does not expire in the duration of the test.very

1. Ensure the UE is in idle State 2A-NB with CP CIoT Optimisation according to TS 36.508 [7] clause 8.1.5.
2. Set the parameters according to T1 in Tables 7.2.9.5-1. Propagation conditions are set according to Annex B clause B.1.1.
3. SS sends uplink scheduling information via NPDCCH DCI format N0 for C_RNTI to schedule uplink subframes according to TS 36.508 [7] clause 8.1.3.6, Table 8.1.3.6.1.1-1.
4. SS shall transmit one message with a timing advance command, $T_A$. The timing advance command, $T_A$, shall be set to 31 which indicate adjustment of the current $N_{TA}$ value. The timing advance adjustment during T1 shall be $N_{TA} = 0$.
5. The UE shall transmit NPUSCH and adjust its uplink timing at the beginning of sub-frame n+40 for a timing advance command $T_A$ received in sub-frame n. Sub-frame n is the last subframe in the repetition period of NPDCCH in which the timing advance command is received by the UE.

6. When T1 expires, the SS shall switch the timing advance command ($T_A$) from T1 to T2 as specified in Table 7.2.9.5-1.

7. SS shall transmit a sequence of messages with timing advance command, $T_A$. The timing advance command, $T_A$, shall be set to 39 which indicate adjustment of the current $N_{TA}$ value. The timing advance adjustment during T2 shall be $N_{TA}$ = 128.

8. The UE shall transmit NPUSCH and adjust its uplink timing at the beginning of sub-frame n+40 for a timing advance command $T_A$ received in sub-frame n. Sub-frame n is the last subframe in the repetition period of NPDCCH in which the timing advance command is received by the UE.

9. The result from the NPUSCH and adjustment of the timing advance in step 8) is used to measure that the UE adjusts the timing of its transmission with a relative accuracy better than or equal to $\pm 13.33 \times T_S$ to the signalled timing advance value compared to the timing of preceding uplink transmission.

10. If the UE adjust the timing of its transmission within a relative accuracy greater than or equal to $\pm 13.33 \times T_S$ to the signalled timing advance value compared to the timing of preceding uplink transmission then the number of successful tests is increased by one. Otherwise, the number of failure tests is increased by one.

11. The SS shall transmit RRCConnectionRelease-NB message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

12. After the RRC connection release, the SS:

    - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 2A-NB with CP CIoT Optimisation according to TS 36.508 [7] clause 8.1.5 (if the paging fails, switches off and on the UE and ensures the UE is in State 2A-NB with CP CIoT Optimisation according to TS 36.508 [7] clause 8.1.5),

    or

    - switches off and on the UE and ensures the UE is in State 2A-NB with CP CIoT Optimisation according to TS 36.508 [7] clause 8.1.5.

14. Repeat step 2-13 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.2.9.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 8.1.6 with the following exceptions:

**Table 7.2.9.4.3-4: *MAC-MainConfig-NB*: Additional HD-FDD - UE timing advance adjustment accuracy test requirement**

| Derivation Path: TS 36.508 clause 8.1.8, Table 8.1.8.2.1.5-1: MAC-MainConfig-NB-SRB | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-NB-SRB ::= SEQUENCE { | | | |
| drx-Config-r13 CHOICE { | | | |
| release | NULL | | |
| } | | | |
| timeAlignmentTimerDedicated-r13 | Infinity | | |
| } | | | |
| } | | | |

## 7.2.9.5 Test requirement

Tables 7.2.9.4.1-1, 7.2.9.5-1 and 7.2.9.5-2 define the primary level settings for HD-FDD UE timing advance adjustment accuracy test.

**Table 7.2.9.5-1: Cell specific Test Parameters for E-UTRAN Timing Advance Accuracy Test for Category NB1 UE in Standalone Mode under Enhanced Coverage**

| Parameter | Unit | Value | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | kHz | 200 | |
| NPDSCH parameters: DL Reference Measurement Channel defined in A.1.6.4 | | R.20 HD-FDD | |
| NPDCCH parameters: DL Reference Measurement Channel defined in A.2.4.4 | | R.32 HD-FDD | |
| NOCNG Patterns defined in D.3 | | NOP.3 FDD | |
| NPBCH_RA | dB | 0 | |
| NPBCH_RB | dB | | |
| NPSS_RA | dB | | |
| NSSS_RA | dB | | |
| NPDCCH_RA | dB | | |
| NPDCCH_RB | dB | | |
| NPDSCH_RA | dB | | |
| NPDSCH_RB | dB | | |
| NOCNG_RA[Note1] | dB | | |
| NOCNG_RB[Note1] | dB | | |
| Timing Advance Command ($T_A$) | | 31 | 39 |
| $\hat{E}_s / I_{ot}$ | dB | -12+TT | |
| $N_{oc}$ | dBm/15 KHz | -88+TT | |
| $\hat{E}_s / N_{oc}$ | dB | -12+TT | |
| Io[Note2] | dBm/ 180 KHz | -76.9+TT | |
| Antenna Configuration | | 1x1 | |
| Propagation Condition | | AWGN | |
| Note 1: OCNG shall be used such that the cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: Io level has been derived from other parameters for information purpose. It is not a settable parameter. | | | |

The UE shall apply the signalled Timing Advance value to the transmission timing at subframe $n$+40, where subframe $n$ is the last subframe in the repetition period of NPDCCH in which the timing advance command is received by the UE.

NOTE: In the test requirement the timing advance adjustment delay of 40 subfames include:

- NPUSCH repetition period equal to 32 subframes and
- Uplink delay of 8 subframes. This is the time between the last subframe in the repetition period of NPDCCH and the first subframe in the repetition period of the corresponding NPUSCH.

The UE shall adjust the timing of its transmission with a relative accuracy better than or equal to $\pm 13.33 \times T_S$ seconds to the signalled timing advance value compared to the timing of preceding uplink transmission. This requirement includes test tolerances.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 7.3 Radio Link Monitoring

## 7.3.1 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync

### 7.3.1.1 Test purpose

The purpose of this test is to verify that the UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the serving cell. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in TS 36.133 section 7.6.

### 7.3.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

### 7.3.1.3 Minimum conformance requirements

When the downlink radio link quality estimated over the last 200 ms period becomes worse than the threshold $Q_{out}$, Layer 1 of the UE shall send an out-of-sync indication to the higher layers within 200 ms $Q_{out}$ evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations shall be performed as specified in section 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least 10 ms.

The transmitter power shall be turned off within 40 ms after expiry of T310 timer as specified in TS 36.331 [5] section 5.3.11.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.1.

### 7.3.1.4 Test description

The test consists of four subtests with one cell configured; the difference between the subtests is the number of transmitter antennas and the propagation channel. Each subtest consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.1.4-1 shows the three different time durations and the corresponding variation of the downlink SNR in the active cell to emulate out-of-sync states.

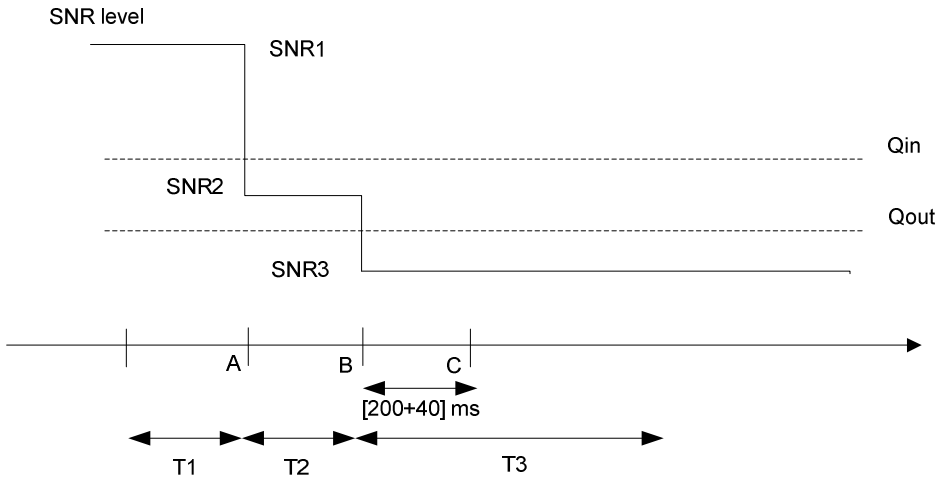


**Figure 7.3.1.4-1: SNR variation for out-of-sync testing**

#### 7.3.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as stated below.
   For subtest 1:As Shown in TS 36.508 [7] Annex A figure A.9 (without using the faders).
   For subtest 2: As Shown in TS 36.508 [7] Annex A figure A.10 (without using the faders).
   For subtest 3: As Shown in TS 36.508 [7] Annex A figure A.9
   For subtest 4: As Shown in TS 36.508 [7] Annex A figure A.10

2. The general test parameter settings for the different subtests are set up according to Table 7.3.1.4-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 7.3.1.4.3

5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.1.4.1-1: General test parameters for E-UTRAN FDD out-of-sync testing**

| Parameter | | Unit | Value | | | | Comment |
|---|---|---|---|---|---|---|---|
| | | | Test 1 | Test 2 | Test 3 | Test 4 | |
| PCFICH/PDCCH/PHICH parameters | | | R.6 FDD | R.7 FDD | R.6 FDD | R.7 FDD | As specified in section A.2.1<br>None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.2 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD | As specified in section D.1.2. |
| Active cell | | | Cell 1 | Cell 1 | Cell 1 | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| CP length | | | Normal | Normal | Normal | Normal | |
| E-UTRA RF Channel Number | | | 1 | 1 | 1 | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | 10 | 10 | 10 | |
| Correlation Matrix and Antenna Configuration | | | 1x2 | 2x2 | 1x2 Low | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.521-1 [10] Annex B.2.3.2 |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | 1A | 1A | 1A | As defined in section 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | 2 | 2 | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 in section 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | 8 | 8 | 8 | |
| | $\rho_A$, $\rho_B$ | | 0 | -3 | 0 | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | 1 | 4 | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 4 | 1 | 4 | 1 | |
| DRX | | | OFF | OFF | OFF | OFF | |
| Layer 3 filtering | | | Enabled | Enabled | Enabled | Enabled | Counters:<br>N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | 0 | 0 | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | 1000 | 1000 | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | PUCCH 1-0 | PUCCH 1-0 | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 2 | 2 | 2 | 2 | Minimum CQI reporting periodicity |
| Propagation channel | | | AWGN | AWGN | ETU 70 Hz | ETU 70 Hz | |
| T1 | | s | 1 | 1 | 1 | 1 | |
| T2 | | s | 0.4 | 0.4 | 0.4 | 0.4 | |
| T3 | | s | 0.5 | 0.5 | 0.5 | 0.5 | |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | | | | |

#### 7.3.1.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.1.5-1 for subtest 1 and 2 and according to T1 in Table and 7.3.1.5-2 for subtest 3 and 4. Propagation conditions are set according to Annex B clause B.1.1 for subtests 1 and 2 and according to B.2.2 for subtests 3 and 4. T1 starts.
3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.1.5-1 for subtests 1 and 2 and according to T2 in Table 7.3.1.5-2 for subtests 3 and 4. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.1.5-1 for subtests 1 and 2 and according to T3 in Table 7.3.1.5-2 for subtests 3 and 4. T3 starts.
5. If the SS:

   a) detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B

   and

   b) does not detect any uplink power higher than -48.5 dBm from time point C (240 ms after the start of T3) until T3 expires,

   the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
6. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.1.5-1 for subtests 1 and 2 and Table 7.3.1.5-2 for subtests 3 and 4.
7. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
8. Repeat steps 2-7 for all subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 7.3.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.1.4.3-1: Common Exception messages for E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.1.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.1.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync test 1 and 3 requirements**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| } | | | |

**Table 7.3.1.4.3-4: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync test 2 and 4 requirements**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.1.4.3-5: *MAC-MainConfig-RBC*: E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

### 7.3.1.5 Test requirement

**Table 7.3.1.5-1: Cell specific test parameters for E-UTRAN FDD (cell # 1) for out-of-sync radio link monitoring tests # 1 and # 2**

| Parameter | Unit | Test 1 | | | Test 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| Antenna Configuration | | 1x2 | | | 2x2 | | |
| OCNG Pattern defined in D.1 (FDD) | | OP.2 FDD | | | OP2 FDD | | |
| $\rho_A$, $\rho_B$ | | 0 | | | -3 | | |
| PCFICH_RB | dB | 4 | | | 1 | | |
| PDCCH_RA | dB | 0 | | | -3 | | |
| PDCCH_RB | dB | 0 | | | -3 | | |
| PBCH_RA | dB | 0 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| SNR Note 6 | dB | -4.1 | -8.9 | -14.1 | -4.3 | -8.9 | -14.1 |
| | | | | | | | |
| | | | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | | -98 | | |
| Propagation condition | | AWGN | | | AWGN | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.1.4-1.

**Table 7.3.1.5-2: Cell specific test parameters for E-UTRAN FDD (cell # 1) for out-of-sync radio link monitoring tests # 3 and # 4**

| Parameter | Unit | Test 3 | | | Test 4 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | 2x2 Low | | |
| OCNG Pattern defined in D.1 (FDD) | | OP.2 FDD | | | OP.2 FDD | | |
| $\rho_A$, $\rho_B$ | | 0 | | | -3 | | |
| PCFICH_RB | dB | 4 | | | 1 | | |
| PDCCH_RA | dB | 0 | | | -3 | | |
| PDCCH_RB | dB | 0 | | | -3 | | |
| PBCH_RA | dB | 0 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| SNR Note 6 | dB | -0.6 | -4.7 | -12.3 | -1.4 | -5.3 | -13.1 |
| | | | | | | | |
| | | | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | | -98 | | |
| Propagation condition | | ETU 70 Hz | | | ETU 70 Hz | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.1.4-1.

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal no later than time point C (240ms after the start of time duration T3).

The uplink signal is verified on the basis of the UE output power:
- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal
- UE output power equal to or less than Transmit OFF power -48.5 dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

# 7.3.1_1 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync with 4 Rx antenna ports

Editor's note: This Test case is incomplete

- The Test Tolerances and Test Requirements applicable are undefined
- Test case are not present in TS36.521-2

## 7.3.1_1.1 Test purpose

Same as 7.3.1.1.

## 7.3.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward that supports 4Rx on all its FDD operating bands.

## 7.3.1_1.3 Minimum conformance requirements

Same as 7.3.1.3.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.1.

## 7.3.1_1.4 Test description

Same as 7.3.1.4.

### 7.3.1_1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as stated below.
   For subtest 1:As Shown in TS 36.508 [7] Annex A figure A.75 (without using the faders).
   For subtest 2: As Shown in TS 36.508 [7] Annex A figure A.75 (without using the faders).
   For subtest 3: As Shown in TS 36.508 [7] Annex A figure A.76
   For subtest 4: As Shown in TS 36.508 [7] Annex A figure A.76
2. The general test parameter settings for the different subtests are set up according to Table 7.3.1_1.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.1_1.4.3.1
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.1_1.4.1-1: General test parameters for E-UTRAN FDD out-of-sync testing**

See Table 7.3.1.4.1-1 with the following changes:

- Instead of antenna configuration 1x2 use 1x4, instead of 2x2 use 2x4
- Instead of antenna configuration 1x2 low use 1x4 low, instead of 2x2 low use 2x4 low

### 7.3.1_1.4.2 Test procedure

Same Test Procedure as in 7.3.1.4.2 with the following changes:

- Instead of Table 7.3.1.5-1 use Table 7.3.1_1.5-1
- Instead of Table 7.3.1.5-2 use Table 7.3.1_1.5-2

### 7.3.1_1.4.3 Message contents

Same as 7.3.1.4.3.

## 7.3.1_1.5 Test requirement

**Table 7.3.1_1.5-1: Cell specific test parameters for E-UTRAN FDD (cell # 1) for out-of-sync radio link monitoring tests # 1 and # 2**

| Parameter | Unit | Test 1 | | | Test 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| Antenna Configuration | | 1x4 | | | 2x4 | | |
| OCNG Pattern defined in D.1 (FDD) | | OP.2 FDD | | | OP2 FDD | | |
| $\rho_A$, $\rho_B$ | | 0 | | | -3 | | |
| PCFICH_RB | dB | 4 | | | 1 | | |
| PDCCH_RA | dB | 0 | | | -3 | | |
| PDCCH_RB | dB | 0 | | | -3 | | |
| PBCH_RA | dB | 0 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| SNR Note 6, Note 7 | dB | -4.1 | -8.9 | -17+TT | -4.3 | -8.9 | -17+TT |
| | | | | | | | |
| | | | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | | -98 | | |
| Propagation condition | | AWGN | | | AWGN | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.1.4-1.
Note 7: The SNR is adjusted for 4Rx as defined in 3A.4.1.2.1

**Table 7.3.1_1.5-2: Cell specific test parameters for E-UTRAN FDD (cell # 1) for out-of-sync radio link monitoring tests # 3 and # 4**

| Parameter | Unit | Test 3 | | | Test 4 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| Correlation Matrix and Antenna Configuration | | 1x4 Low | | | 2x4 Low | | |
| OCNG Pattern defined in D.1 (FDD) | | OP.2 FDD | | | OP.2 FDD | | |
| $\rho_A$, $\rho_B$ | | 0 | | | -3 | | |
| PCFICH_RB | dB | 4 | | | 1 | | |
| PDCCH_RA | dB | 0 | | | -3 | | |
| PDCCH_RB | dB | 0 | | | -3 | | |
| PBCH_RA | dB | 0 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| SNR Note 6, Note 7 | dB | -0.6 | -4.7 | -15+TT | -1.4 | -5.3 | -15.7+TT |
| | | | | | | | |
| | | | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | | -98 | | |
| Propagation condition | | ETU 70 Hz | | | ETU 70 Hz | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.1.4-1.
Note 7: The SNR is adjusted for 4Rx as defined in 3A.4.1.2.1

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal no later than time point C (240ms after the start of time duration T3).

The uplink signal is verified on the basis of the UE output power:
- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal
- UE output power equal to or less than Transmit OFF power -48.5 dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.2 E-UTRAN FDD Radio Link Monitoring Test for In-sync

### 7.3.2.1 Test purpose

The purpose of this test is to verify that the UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the serving cell. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in TS 36.133 section 7.6.

### 7.3.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

### 7.3.2.3 Minimum conformance requirements

When the downlink radio link quality estimated over the last 100 ms period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send an in-sync indication to the higher layers within 100 ms $Q_{in}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations shall be performed as specified in section 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least 10 ms.

The transmitter power shall be turned off within 40 ms after expiry of T310 timer as specified in TS 36.331 [5] section 5.3.11.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.2.

### 7.3.2.4 Test description

The test consists of two subtests with one cell configured; the difference between the subtests is the number of transmitter antennas. Each subtest consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.2.4-1 shows the different time durations and the corresponding variation of the downlink SNR in the active cell to emulate in-sync states.

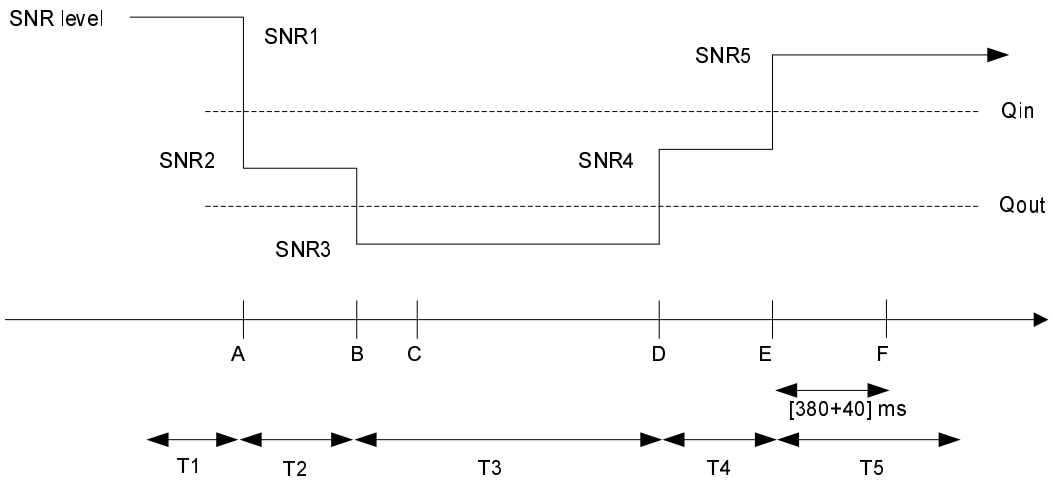


**Figure 7.3.2.4-1: SNR variation for in-sync testing**

#### 7.3.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. For subtest 1 (one transmitter antenna): Connect the SS (node B emulator) and faders, to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.9.
   For subtest 2 (two transmitter antennas): Connect the SS (node B emulator) and faders, to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10.

2. The general test parameter settings for the different subtest are set up according to Table 7.3.2.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 7.3.2.4.3.

5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.2.4.1-1: General test parameters for E-UTRAN FDD in-sync testing**

| Parameter | | Unit | Value | | Comment |
|---|---|---|---|---|---|
| | | | Test 1 | Test 2 | |
| PCFICH/PDCCH/PHICH parameters | | | R.6 FDD | R.7 FDD | As specified in section A.2.1<br>None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.2 FDD | OP.2 FDD | As specified in section D.1.2. |
| Active cell | | | Cell 1 | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| CP length | | | Normal | Normal | |
| E-UTRA RF Channel Number | | | 1 | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | 10 | |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.521-1 [10] Annex B.2.3.2 |
| In sync transmission parameters (Note 1) | DCI format | | 1C | 1C | As defined in section 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | 2 | In sync threshold $Q_{in}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 section and Table 7.6.1-2 respectively. |
| | Aggregation level | CCE | 4 | 4 | |
| | $\rho_A$, $\rho_B$ | | 0 | -3 | |
| | Ratio of PDCCH to RS EPRE | | 0 | -3 | |
| | Ratio of PCFICH to RS EPRE | | 4 | 1 | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | 1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 section 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | 8 | |
| | $\rho_A$, $\rho_B$ | | 0 | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 4 | 1 | |
| DRX | | | OFF | OFF | |
| Layer 3 filtering | | | Enabled | Enabled | Counters:<br>N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 2 | 2 | Minimum CQI reporting periodicity |
| Propagation channel | | | ETU 70 Hz | ETU 70 Hz | |
| T1 | | s | 0.5 | 0.5 | |
| T2 | | s | 0.4 | 0.4 | |
| T3 | | s | 1.36 | 1.36 | |
| T4 | | s | 0.4 | 0.4 | |
| T5 | | s | 1 | 1 | |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | | |

#### 7.3.2.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.2.5-1 for subtests 1 and 2 respectively. Propagation conditions are set according to Annex B.2.2. T1 starts.
3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.2.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.2.5-1. T3 starts.
5. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.2.5-1. T4 starts.
6. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.2.5-1. T5 starts.
7. If the SS detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (420 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
8. After T5 expires, repeat steps 2-7 for both subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 7.3.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.2.4.3-1: Common Exception messages for E-UTRAN FDD Radio Link Monitoring Test for in-sync**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.2.4.3-2 : *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring Test for in-sync**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.2.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring Test for in-sync test 1 requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| } | | | |

**Table 7.3.2.4.3-4: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring Test for in-sync test 2 requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.2.4.3-5: *MAC-MainConfig-RBC*: E-UTRAN FDD Radio Link Monitoring Test for in-sync**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

### 7.3.2.5 Test requirement

**Table 7.3.2.5-1: Cell specific test parameters for E-UTRAN FDD (cell # 1) for in-sync radio link monitoring tests # 1 and # 2**

| Parameter | Unit | Test 1 | | | | | Test 2 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | 10 | | | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | | | 2x2 Low | | | | |
| OCNG Pattern defined in D.1 (FDD) | | OP.2 FDD | | | | | OP.2 FDD | | | | |
| $\rho_A$, $\rho_B$ | | 0 | | | | | -3 | | | | |
| PCFICH_RB | dB | 4 | | | | | 1 | | | | |
| PDCCH_RA | dB | 0 | | | | | -3 | | | | |
| PDCCH_RB | dB | 0 | | | | | -3 | | | | |
| PBCH_RA | dB | 0 | | | | | -3 | | | | |
| PBCH_RB | dB | | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | | |
| SNR Note 6 | dB | -0.6 | -4.7 | -12.3 | -7.2 | -0.6 | -1.4 | -5.3 | -13.1 | -8.2 | -1.4 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | | -98 | | | | |
| Propagation condition | | ETU 70 Hz | | | | | ETU 70 Hz | | | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.2.4-1

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (420 ms after the start of time duration T5) the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:
- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.2_1 E-UTRAN FDD Radio Link Monitoring Test for In-sync with 4 Rx antenna ports

Editor's note: This Test case is incomplete

- The Test Tolerances and Test Requirements applicable are undefined

- Test case are not present in TS36.521-2

### 7.3.2_1.1 Test purpose

Same as 7.3.2.1.

### 7.3.2_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward that supports 4Rx on all its FDD operating bands.

### 7.3.2_1.3 Minimum conformance requirements

Same as 7.3.2.3.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.2.

### 7.3.2_1.4 Test description

Same as 7.3.2.4.

#### 7.3.2_1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. For subtest 1 (one transmitter antenna): Connect the SS (node B emulator) and faders, to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.75.
   For subtest 2 (two transmitter antennas): Connect the SS (node B emulator) and faders, to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.76.
2. The general test parameter settings for the different subtest are set up according to Table 7.3.2_1.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.2_1.4.3.
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.2_1.4.1-1: General test parameters for E-UTRAN FDD in-sync testing**

See Table 7.3.2.4.1 with the following changes:

- Instead of antenna configuration 1x2 low use 1x4 low, instead of 2x2 low use 2x4 low

#### 7.3.2_1.4.2 Test procedure

Same Test Procedure as in 7.3.2.4.2 with the following changes:

- Instead of Table 7.3.2.5-1 use Table 7.3.2_1.5-1

#### 7.3.2_1.4.3 Message contents

Same as 7.3.2.4.3.

## 7.3.2_1.5 Test requirement

**Table 7.3.2_1.5-1: Cell specific test parameters for E-UTRAN FDD (cell # 1) for in-sync radio link monitoring tests # 1 and # 2**

| Parameter | Unit | Test 1 | | | | | Test 2 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | 10 | | | | |
| Correlation Matrix and Antenna Configuration | | 1x4 Low | | | | | 2x4 Low | | | | |
| OCNG Pattern defined in D.1 (FDD) | | OP.2 FDD | | | | | OP.2 FDD | | | | |
| $\rho_A$, $\rho_B$ | | 0 | | | | | -3 | | | | |
| PCFICH_RB | dB | 4 | | | | | 1 | | | | |
| PDCCH_RA | dB | 0 | | | | | -3 | | | | |
| PDCCH_RB | dB | 0 | | | | | -3 | | | | |
| PBCH_RA | dB | 0 | | | | | -3 | | | | |
| PBCH_RB | dB | | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | | |
| SNR Note 6 Note 7 | dB | -0.6 | -4.7 | -14+TT | -9.9+TT | -0.6 | -1.4 | -5.3 | -15.7+TT | -10.8+TT | -1.4 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | | -98 | | | | |
| Propagation condition | | ETU 70 Hz | | | | | ETU 70 Hz | | | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.2.4-1
Note 7: The SNR is adjusted for 4Rx as defined in 3A.4.1.2.1

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (420 ms after the start of time duration T5) the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:
- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.3 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync

### 7.3.3.1 Test purpose

The purpose of this test is to verify that the UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the serving cell. This test will partly verify the E-UTRAN TDD radio link monitoring requirements in TS 36.133 section 7.6.

### 7.3.3.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

### 7.3.3.3 Minimum conformance requirements

When the downlink radio link quality estimated over the last 200 ms period becomes worse than the threshold $Q_{out}$, Layer 1 of the UE shall send an out-of-sync indication to the higher layers within 200 ms $Q_{out}$ evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations shall be performed as specified in section 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least 10 ms.

The transmitter power shall be turned off within 40 ms after expiry of T310 timer as specified in TS 36.331 [5] section 5.3.11.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.3.

### 7.3.3.4 Test description

The test consists of four subtests with one cell configured; the difference between the subtests is the number of transmitter antennas and the propagation channel. Each subtest consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.3.4-1 shows the three different time durations and the corresponding variation of the downlink SNR in the active cell to emulate out-of-sync states.

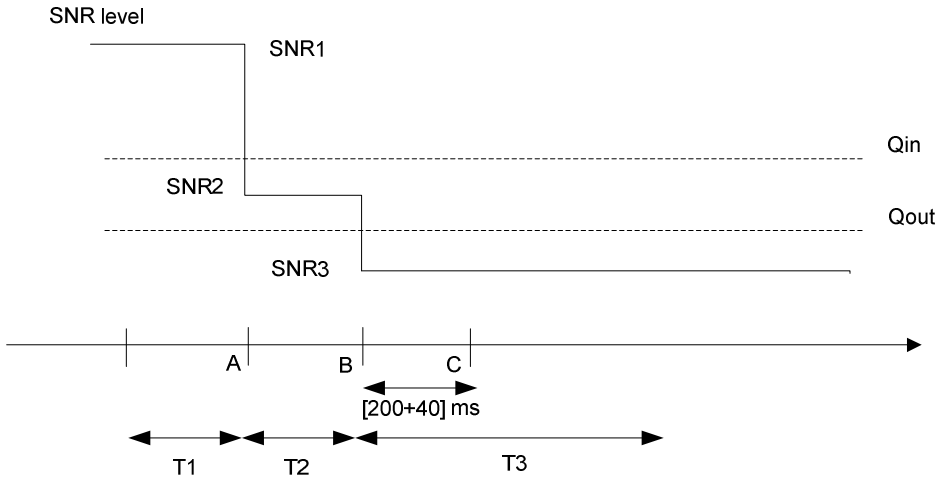


**Figure 7.3.3.4-1: SNR variation for out-of-sync testing**

#### 7.3.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as stated below.
   For subtest 1: As Shown in TS 36.508 [7] Annex A figure A.9 (without using the faders).
   For subtest 2: As Shown in TS 36.508 [7] Annex A figure A.10 (without using the faders).
   For subtest 3: As Shown in TS 36.508 [7] Annex A figure A.9
   For subtest 4: As Shown in TS 36.508 [7] Annex A figure A.10

2. The general test parameter settings for the different subtests are set up according to Table 7.3.3.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 7.3.3.4.3.

5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.3.4.1-1: General test parameters for E-UTRAN TDD out-of-sync testing**

| Parameter | | Unit | Value | | | | Comment |
|---|---|---|---|---|---|---|---|
| | | | Test 1 | Test 2 | Test 3 | Test 4 | |
| PCFICH/PDCCH/PHICH parameters | | | R.6 TDD | R.7 TDD | R.6 TDD | R.7 TDD | As specified in section A.2.1<br>None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.2 TDD | OP.2 TDD | OP.2 TDD | OP.2 TDD | As specified in section D.2.2. |
| Active cell | | | Cell 1 | Cell 1 | Cell 1 | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| CP length | | | Normal | Normal | Normal | Normal | |
| E-UTRA RF Channel Number | | | 1 | 1 | 1 | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | 10 | 10 | 10 | |
| Correlation Matrix and Antenna Configuration | | | 1x2 | 2x2 | 1x2 Low | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.521-1 [10] Annex B.2.3.2 |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | 1A | 1A | 1A | As defined in section 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | 2 | 2 | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 in section 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | 8 | 8 | 8 | |
| | $\rho_A$, $\rho_B$ | | 0 | -3 | 0 | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | 1 | 4 | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 4 | 1 | 4 | 1 | |
| DRX | | | OFF | OFF | OFF | OFF | |
| Layer 3 filtering | | | Enabled | Enabled | Enabled | Enabled | Counters:<br>N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | 0 | 0 | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | 1000 | 1000 | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | PUCCH 1-0 | PUCCH 1-0 | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 1 | 1 | 1 | 1 | Minimum CQI reporting periodicity |
| Propagation channel | | | AWGN | AWGN | ETU 70 Hz | ETU 70 Hz | |
| T1 | | s | 1 | 1 | 1 | 1 | |
| T2 | | s | 0.4 | 0.4 | 0.4 | 0.4 | |
| T3 | | s | 0.5 | 0.5 | 0.5 | 0.5 | |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel | | | | | | | |

#### 7.3.3.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.3.5-1 for subtests 1 and 2 and according to T1 in Table and 7.3.3.5-2 for subtest 3 and 4. Propagation conditions are set according to Annex B clause B.1.1 for subtests 1 and 2 and according to B.2.2 for subtests 3 and 4. T1 starts.
3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.3.5-1 for subtests 1 and 2 and according to T2 in Table 7.3.3.5-2 for subtests 3 and 4. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.3.5-1 for subtests 1 and 2 and according to T3 in Table 7.3.3.5-2 for subtests 3 and 4. T3 starts.
5. If the SS:

   a) detects uplink power equal to or higher than -39 dBm in each uplink subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B

   and

   b) does not detect any uplink power higher than -48.5 dBm from time point C (240 ms after the start of T3) until T3 expires

   the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
6. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.3.5-1 for subtests 1 and 2 and Table 7.3.3.5-2 for subtests 3 and 4.
7. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
8. Repeat steps 2-7 for all subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 7.3.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.3.4.3-1: Common Exception messages for E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.3.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1C in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.3.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync test 1 and 3 requirements**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| } | | | |

**Table 7.3.3.4.3-4: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync test 2 and 4 requirements**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.3.4.3-5: *MAC-MainConfig-RBC*: E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

### 7.3.3.5 Test requirement

**Table 7.3.3.5-1: Cell specific test parameters for E-UTRAN TDD (cell # 1) for out-of-sync radio link monitoring tests # 1 and # 2**

| Parameter | Unit | Test 1 | | | Test 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| Antenna Configuration | | 1x2 | | | 2x2 | | |
| Special subframe configuration[Note1] | | 6 | | | 6 | | |
| Uplink-downlink configuration[Note2] | | 1 | | | 1 | | |
| OCNG Pattern defined in D.2 (TDD) | | OP.2 TDD | | | OP.2 TDD | | |
| $\rho_A$, $\rho_B$ | | 0 | | | -3 | | |
| PCFICH_RB | dB | 4 | | | 1 | | |
| PDCCH_RA | dB | 0 | | | -3 | | |
| PDCCH_RB | dB | 0 | | | -3 | | |
| PBCH_RA | dB | 0 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA[Note 3] | dB | | | | | | |
| OCNG_RB[Note 3] | dB | | | | | | |
| SNR [Note 8] | dB | -4.5 | -8.5 | -13.7 | -4.6 | -8.6 | -13.8 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | -98 | | |
| Propagation condition | | AWGN | | | AWGN | | |

Note 1: For the special subframe configuration see table 4.2-1 in 3GPP TS 36.211.
Note 2: For the uplink-downlink configuration see table 4.2-2 in 3GPP TS 36.211.
Note 3: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 5: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 6: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 7: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 8: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.3.4-1.

**Table 7.3.3.5-2: Cell specific test parameters for E-UTRAN TDD (cell # 1) for out-of-sync radio link monitoring tests # 3 and # 4**

| Parameter | Unit | Test 3 | | | Test 4 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | 2x2 Low | | |
| Special subframe configuration Note1 | | 6 | | | 6 | | |
| Uplink-downlink configuration Note2 | | 1 | | | 1 | | |
| OCNG Pattern defined in D.2 (TDD) | | OP.2 TDD | | | OP.2 TDD | | |
| $\rho_A$, $\rho_B$ | | 0 | | | -3 | | |
| PCFICH_RB | dB | 4 | | | 1 | | |
| PDCCH_RA | dB | 0 | | | -3 | | |
| PDCCH_RB | dB | 0 | | | -3 | | |
| PBCH_RA | dB | 0 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 3 | dB | | | | | | |
| OCNG_RB Note 3 | dB | | | | | | |
| SNR Note 8 | dB | -0.6 | -4.5 | -12.1 | -1.4 | -5.0 | -12-8 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | -98 | | |
| Propagation condition | | ETU 70 Hz | | | ETU 70 Hz | | |

Note 1: For the special subframe configuration see table 4.2-1 in 3GPP TS 36.211.
Note 2: For the uplink-downlink configuration see table 4.2-2 in 3GPP TS 36.211.
Note 3: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 5: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 6: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 7: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 8: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.3.4-1.

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least in all uplink subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal no later than time point C (240ms after the start of the time duration T3).

The uplink signal is verified on the basis of the UE output power:
- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal
- UE output power equal to or less than Transmit OFF power -48.5 dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.4 E-UTRAN TDD Radio Link Monitoring Test for In-sync

### 7.3.4.1 Test purpose

The purpose of this test is to verify that the UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the serving cell. This test will partly verify the E-UTRAN TDD radio link monitoring requirements in TS 36.133 section 7.6.

### 7.3.4.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

### 7.3.4.3 Minimum conformance requirements

When the downlink radio link quality estimated over the last 100 ms period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send an in-sync indication to the higher layers within 100 ms $Q_{in}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations shall be performed as specified in section 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least 10 ms.

The transmitter power shall be turned off within 40 ms after expiry of T310 timer as specified in TS 36.331 [5] section 5.3.11.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.4.

### 7.3.4.4 Test description

The test consists of 2 subtests with one cell configured; the difference between the subtests is the number of transmitter antennas. Each subtest consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.4.4-1 shows the different time durations and the corresponding variation of the downlink SNR in the active cell to emulate in-sync states.

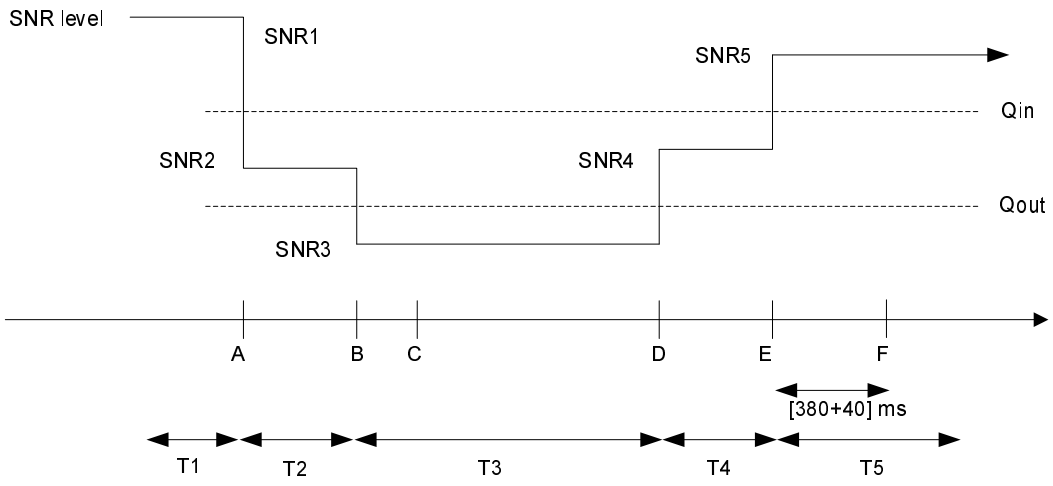


**Figure 7.3.4.4-1: SNR variation for in-sync testing**

#### 7.3.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. For subtest 1 (one transmitter antenna): Connect the SS (node B emulator) and faders, to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.9.
   For subtest 2 (two transmitter antennas): Connect the SS (node B emulator) and faders, to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10.

2. The general test parameter settings for the different subtest are set up according to Table 7.3.4.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 7.3.4.4.3.

6. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.4.4.1-1: General test parameters for E-UTRAN TDD in-sync testing**

| Parameter | | Unit | Value | | Comment |
|---|---|---|---|---|---|
| | | | Test 1 | Test 2 | |
| PCFICH/PDCCH/PHICH parameters | | | R.6 TDD | R.7 TDD | As specified in section A.2.2<br>None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.2 TDD | OP.2 TDD | As specified in section D.2.2. |
| Active cell | | | Cell 1 | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| CP length | | | Normal | Normal | |
| E-UTRA RF Channel Number | | | 1 | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | 10 | |
| Correlation Matrix and Antenna Configuration | | | 1x2 Low | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.521-1 [10] Annex B.2.3.2 |
| n sync transmission parameters (Note 1) | DCI format | | 1C | 1C | As defined in section 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | 2 | In sync threshold $Q_{in}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 section and Table 7.6.1-2 respectively. |
| | Aggregation level | CCE | 4 | 4 | |
| | $\rho_A$, $\rho_B$ | | 0 | -3 | |
| | Ratio of PDCCH to RS EPRE | | 0 | -3 | |
| | Ratio of PCFICH to RS EPRE | | 4 | 1 | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | 1A | As defined in section 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 section 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | 8 | |
| | $\rho_A$, $\rho_B$ | | 0 | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 4 | 1 | |
| DRX | | | OFF | OFF | |
| Layer 3 filtering | | | Enabled | Enabled | Counters:<br>N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 1 | 1 | Minimum CQI reporting periodicity |
| Propagation channel | | | ETU 70 Hz | ETU 70 Hz | |
| T1 | | s | 0.5 | 0.5 | |
| T2 | | s | 0.4 | 0.4 | |
| T3 | | s | 1.46 | 1.46 | |
| T4 | | s | 0.4 | 0.4 | |
| T5 | | s | 1 | 1 | |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | | |

#### 7.3.4.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.4.5-1 for subtest 1 and 2 respectively. Propagation conditions are set according to Annex B.2.2. T1 starts.
3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.4.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.4.5-1. T3 starts.
5. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.4.5-1. T4 starts.
6. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.4.5-1. T5 starts.
7. If the SS detects uplink power equal to or higher than -39 dBm in each uplink subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during from time point A to time point F (420 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
8. After T5 expires, repeat steps 2-7 for both subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 7.3.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.4.4.3-1: Common Exception messages for E-UTRAN TDD Radio Link Monitoring Test for in-sync**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.4.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for In-sync**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1C in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|

| | |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.4.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for in-sync test 1 requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| } | | | |

**Table 7.3.4.4.3-4: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for in-sync test 2 requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.4.4.3-5: *MAC-MainConfig-RBC*: E-UTRAN TDD Radio Link Monitoring Test for in-sync**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

### 7.3.4.5 Test requirement

**Table 7.3.4.5-1: Cell specific test parameters for E-UTRAN TDD (cell # 1) for In-sync radio link monitoring tests # 1 and # 2**

| Parameter | Unit | Test 1 | | | | | Test 2 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | 10 | | | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | | | 2x2 Low | | | | |
| Special subframe configuration Note1 | | 6 | | | | | 6 | | | | |
| Uplink-downlink configuration Note2 | | 1 | | | | | 1 | | | | |
| OCNG Pattern defined in D.2 (TDD) | | OP.2 TDD | | | | | OP.2 TDD | | | | |
| $\rho_A$, $\rho_B$ | | 0 | | | | | -3 | | | | |
| PCFICH_RB | dB | 4 | | | | | 1 | | | | |
| PDCCH_RA | dB | 0 | | | | | -3 | | | | |
| PDCCH_RB | dB | 0 | | | | | -3 | | | | |
| PBCH_RA | dB | 0 | | | | | -3 | | | | |
| PBCH_RB | dB | | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | | |
| OCNG_RA Note 3 | dB | | | | | | | | | | |
| OCNG_RB Note 3 | dB | | | | | | | | | | |
| SNR Note 8 | dB | -0.6 | -4.5 | -12.1 | -7.2 | -0.6 | -1.4 | -5.0 | -12.8 | -8.2 | -1.4 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | | -98 | | | | |
| Propagation condition | | ETU 70 Hz | | | | | ETU 70 Hz | | | | |

Note 1: For the special subframe configuration see table 4.2-1 in 3GPP TS 36.211.
Note 2: For the uplink-downlink configuration see table 4.2-2 in 3GPP TS 36.211.
Note 3: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 5: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 6: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 7: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 8: The SNR in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.4.4-1.

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (420 ms after the start of time duration T5) the UE shall transmit uplink signal at least in all uplink subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:
- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.5 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync in DRX

### 7.3.5.1 Test purpose

The purpose of this test is to verify that the UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the serving cell when DRX is used. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in TS 36.133 section 7.6.

### 7.3.5.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward. Applicability requires support for FGI bit 5.

### 7.3.5.3 Minimum conformance requirements

When DRX is used the $Q_{out}$ evaluation period ($T_{Evaluate_Q_{out_DRX}}$) and the $Q_{in}$ evaluation period ($T_{Evaluate_Q_{in_DRX}}$) is specified in TS 36.133 [4] Table 7.6.2.2-1 will be used.

When the downlink radio link quality estimated over the last $T_{Evaluate_Q_{out_DRX}}$ [s] period becomes worse than the threshold $Q_{out}$, Layer 1 of the UE shall send out-of-sync indication to the higher layers within $T_{Evaluate_Q_{out_DRX}}$ [s] evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

When the downlink radio link quality estimated over the last $T_{Evaluate_Q_{in_DRX}}$ [s] period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send in-sync indications to the higher layers within $T_{Evaluate_Q_{in_DRX}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations shall be performed as specified in section 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10 ms, DRX_cycle_length).

Upon start of T310 timer as specified in section 5.3.11 in TS 36.331, the UE shall monitor the link for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry of T310 timer.

The transmitter power shall be turned off within 40 ms after expiry of T310 counter as specified in TS 36.331 [5] section 5.3.11.

**Table 7.3.5.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX}}$ and $T_{Evaluate_Q_{in_DRX}}$ (s) (DRX cycles) |
|---|---|
| ≤0.04 | Note (20) |
| 0.04 < DRX cycle ≤ 0.64 | Note (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note (5) |
| Note: Evaluation period length in time depends on the length of the DRX cycle in use | |

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.5.

### 7.3.5.4 Test description

The test consists of two subtests with one cell configured; the difference between the subtests is the DRX cycle length, number of transmit antennas and the propagation conditions. Each subtest consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.5.4-1 shows the three different time durations and the corresponding variation of the downlink SNR in the active cell to emulate out-of-sync states.

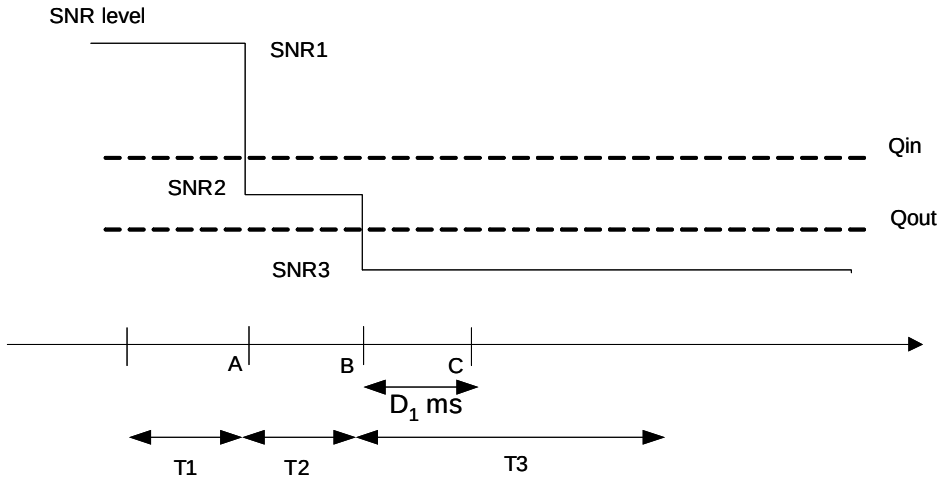


**Figure 7.3.5.4-1: SNR variation for out-of-sync testing in DRX**

### 7.3.5.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as stated below.
   For subtest 1: As Shown in TS 36.508 [7] Annex A figure A.10
   For subtest 2: As Shown in TS 36.508 [7] Annex A figure A.9 (without using the faders)

2. The general test parameter settings for the different subtests are set up according to Table 7.3.5.4-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 7.3.5.4.3

5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.5.4.1-1: General test parameters for E-UTRAN FDD out-of-sync in DRX testing**

| Parameter | | Unit | Value | | Comment |
|---|---|---|---|---|---|
| | | | Test 1 | Test 2 | |
| PCFICH/PDCCH/PHICH parameters | | | R.7 FDD | R.6 FDD | As specified in section A.2.1. None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.2 FDD | OP.2 FDD | As specified in section D.1.2. |
| Active cell | | | Cell 1 | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| CP length | | | Normal | Normal | |
| E-UTRA RF Channel Number | | | 1 | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | 10 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | 1x2 | Correlation Matrix and Antenna Configuration are defined in TS 36.521-1 [10] Annex B.2.3.2 |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | 1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] in section 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | 0 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | 4 | |
| DRX cycle | | ms | 40 | 1280 | See Table 7.3.5.5-2 |
| Layer 3 filtering | | | Enabled | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213 [8]. |
| CQI reporting periodicity | | ms | 2 | 2 | Minimum CQI reporting periodicity |
| Propagation channel | | | ETU 70 Hz | AWGN | . |
| T1 | | s | 4 | 32 | |
| T2 | | s | 1.6 | 12.8 | |
| T3 | | s | 1.8 | 13 | |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | | |

### 7.3.5.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 7.3.5.5-1 for subtest 1 and 2. Propagation conditions are set according to Annex B clause B.2.2 for subtests 1 and according to B.1.1 for subtests 2. T1 starts.

3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.5.5-1 for subtests 1 and 2. T2 starts.

4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.5.5-1 for subtests 1 and 2. T3 starts.

5. If the SS:

   a) detects uplink power equal to or higher than -39 dBm in the On-duration part of every DRX cycle in the subframe according the configured CQI reporting mode (PUCCH 1-0) during the period from time point A to time point B

   and

   b) For subtest 1: does not detect any uplink power higher than -48.5 dBm from time point C (900 ms after the start of T3) until T3 expires,
   For subtest 2: does not detect any uplink power higher than -48.5 dBm from time point C (6500 ms after the start of T3) until T3 expires,

   the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.

6. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.5.5-1 for subtests 1 and 2.

7. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

8. Repeat steps 2-7 for all subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.5.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.5.4.3-1: Common Exception messages for E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync in DRX**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.5.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync in DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.5.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN FDD Radio Link Monitoring test for Out-of-sync in DRX, subtest #1 requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf40 typical value in real network for real-time services. | |
| sf40 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.5.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN FDD Radio Link Monitoring test for Out-of-sync in DRX, subtest #2 requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf1280 typical value in real network for best-effort services. | |
| sf1280 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.5.4.3-5: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN FDD Radio Link Monitoring test for Out-of-sync in DRX, subtest #1 and #2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.5.4.3-6: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring test for Out-of-sync in DRX, subtest #1 requirements**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.5.4.3-7: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD Radio Link Monitoring test for Out-of-sync in DRX, subtest #1 and #2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

### 7.3.5.5 Test requirement

**Table 7.3.5.5-1: Cell specific test parameters for E-UTRAN FDD (cell # 1) for out-of-sync radio link monitoring subtests #1 and # 2**

| Parameter | Unit | Test 1 | | | Test 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | 1x2 | | |
| OCNG Pattern defined in D.1 (FDD) | | OP.2 FDD | | | OP.2 FDD | | |
| $\rho_A$, $\rho_B$ | | -3 | | | 0 | | |
| PCFICH_RB | dB | 1 | | | 4 | | |
| PDCCH_RA | dB | -3 | | | 0 | | |
| PDCCH_RB | dB | -3 | | | 0 | | |
| PBCH_RA | dB | -3 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA[Note1] | dB | | | | | | |
| OCNG_RB[Note1] | dB | | | | | | |
| SNR[Note 6] | dB | -1.4 | -5.3 | -13.1 | -4.1 | -8.9 | -14.1 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | -98 | | |
| Propagation condition | | ETU 70 Hz | | | AWGN | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.5.4-1.

**Table 7.3.5.5-2: DRX-Configuration for E-UTRAN FDD Radio Link Monitoring out-of-sync in DRX test**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf2 | psf2 | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf40 | sf1280 | |
| shortDRX | disable | disable | |

**Table 7.3.5.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD Radio Link Monitoring out-of-sync in DRX test**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | infinity | infinity | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| sr-ConfigIndex | 0 | 0 | For further information see section 6.3.2 in 3GPP TS 36.331 and section10.1 in 3GPP TS 36.213 [8]. |

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

In subtest 1 and subtest 2 during the period from time point A to time point B the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the subframe according the configured CQI reporting mode (PUCCH 1-0).

In subtest 1 the UE shall stop transmitting uplink signal no later than time point C (duration D1 = 900 ms after the start of time duration T3).

In subtest 2 the UE shall stop transmitting uplink signal no later than time point C (duration D1 = 6500 ms after the start of time duration T3).

The uplink signal is verified on the basis of the UE output power:
- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal
- UE output power equal to or less than Transmit OFF power -48.5 dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.6 E-UTRAN FDD Radio Link Monitoring Test for In-sync in DRX

### 7.3.6.1 Test purpose

The purpose of this test is to verify that the UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the serving cell when DRX is used. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in TS 36.133 section 7.6.

### 7.3.6.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward. Applicability requires support for FGI bit 5.

### 7.3.6.3 Minimum conformance requirements

When DRX is used the $Q_{out}$ evaluation period ($T_{Evaluate_Q_{out_DRX}}$) and the $Q_{in}$ evaluation period ($T_{Evaluate_Q_{in_DRX}}$) is specified in TS 36.133 [4] Table 7.6.2.2-1 will be used.

When the downlink radio link quality estimated over the last $T_{Evaluate_Q_{out_DRX}}$ [s] period becomes worse than the threshold $Q_{out}$, Layer 1 of the UE shall send out-of-sync indication to the higher layers within $T_{Evaluate_Q_{out_DRX}}$ [s] evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

When the downlink radio link quality estimated over the last $T_{Evaluate_Q_{in_DRX}}$ [s] period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send in-sync indications to the higher layers within $T_{Evaluate_Q_{in_DRX}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations shall be performed as specified in section 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10 ms, DRX_cycle_length).

Upon start of T310 timer as specified in section 5.3.11 in TS 36.331, the UE shall monitor the link for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry of T310 timer.

The transmitter power shall be turned off within 40 ms after expiry of T310 counter as specified in TS 36.331 [5] section 5.3.11.

**Table 7.3.6.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX}}$ and $T_{Evaluate_Q_{in_DRX}}$ (s) (DRX cycles) |
|---|---|
| ≤0.04 | Note (20) |
| 0.04 < DRX cycle ≤ 0.64 | Note (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note (5) |
| Note: Evaluation period length in time depends on the length of the DRX cycle in use | |

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.6.

## 7.3.6.4 Test description

The test consists of one subtest with one cell configured. The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.6.4-1 shows the different time durations and the corresponding variation of the downlink SNR in the active cell to emulate in-sync states.

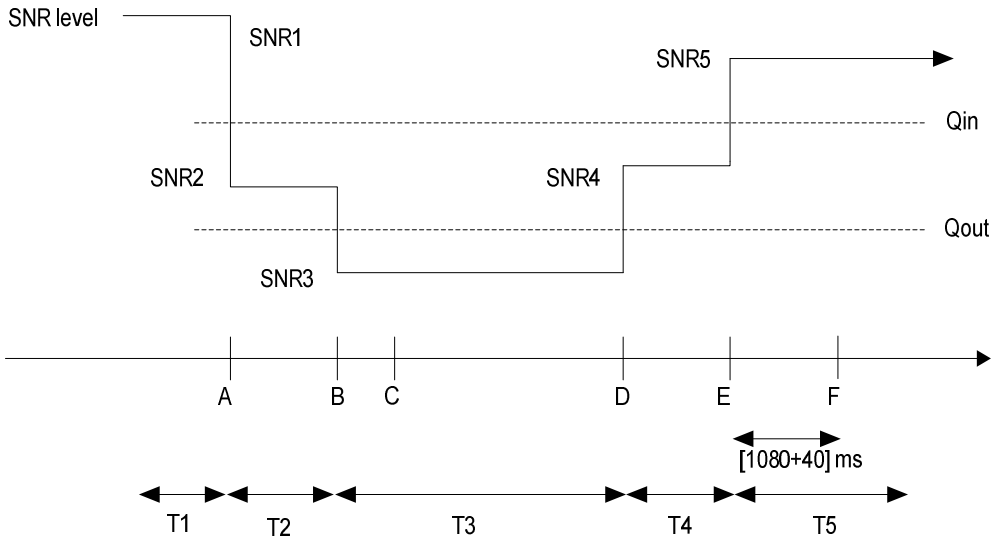


**Figure 7.3.6.4-1: SNR variation for in-sync testing in DRX**

### 7.3.6.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.18.

2. The general test parameter settings for the test is set up according to Table 7.3.6.4-1.

3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.6.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.6.4.1-1: General test parameters for E-UTRAN FDD in-sync in DRX testing**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | | R.6 FDD | As specified in section A.2.1. None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.2 FDD | As specified in section D.1.2. |
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| CP length | | | Normal | |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| Configuration | | | 1x2 | |
| In sync transmission parameters (Note 1) | DCI format | | 1C | As defined in section 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | In sync threshold $Q_{in}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] in section and Table 7.6.1-2 respectively. |
| | Aggregation level | CCE | 4 | |
| | $\rho_A$, $\rho_B$ | | 0 | |
| | Ratio of PDCCH to RS EPRE | | 0 | |
| | Ratio of PCFICH to RS EPRE | | 4 | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] in section 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | 0 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 4 | |
| DRX cycle | | ms | 40 | See Table 7.3.6.5-2 |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213 [8]. |
| CQI reporting periodicity | | ms | 2 | Minimum CQI reporting periodicity |
| Propagation channel | | | AWGN | |
| T1 | | s | 4 | |
| T2 | | s | 1.6 | |
| T3 | | s | 1.46 | |
| T4 | | s | 0.4 | |
| T5 | | s | 4 | |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

## 7.3.6.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.6.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.6.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.6.5-1. T3 starts.
5. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.6.5-1. T4 starts.
6. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.6.5-1. T5 starts.
7. If the SS detects uplink power equal to or higher than -39 dBm in the On-duration part of every DRX cycle in the subframe according the configured CQI reporting mode (PUCCH 1-0) during the period from time point A to time point F (1120 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
8. Repeat steps 2-7 for all subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.6.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.6.4.3-1: Common Exception messages for E-UTRAN FDD Radio Link Monitoring Test for In-sync in DRX**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.6.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring Test for In-sync in DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.6.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN FDD Radio Link Monitoring test for In-sync in DRX requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf40 typical value in real network for real-time services. | |
| sf40 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.6.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN FDD Radio Link Monitoring test for In-sync in DRX requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.6.4.3-5: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD Radio Link Monitoring test for In-sync in DRX requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

### 7.3.6.5 Test requirement

**Table 7.3.6.5-1: Cell specific test parameters for E-UTRAN FDD (cell # 1) for In-sync radio link monitoring test**

| Parameter | Unit | Test 1 | | | | |
|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | |
| Antenna Configuration | | 1x2 | | | | |
| OCNG Pattern defined in D.1 (FDD) | | OP.2 FDD | | | | |
| $\rho_A$, $\rho_B$ | | 0 | | | | |
| PCFICH_RB | dB | 4 | | | | |
| PDCCH_RA | dB | 0 | | | | |
| PDCCH_RB | dB | 0 | | | | |
| PBCH_RA | dB | 0 | | | | |
| PBCH_RB | dB | | | | | |
| PSS_RA | dB | | | | | |
| SSS_RA | dB | | | | | |
| PHICH_RA | dB | | | | | |
| PHICH_RB | dB | | | | | |
| PDSCH_RA | dB | | | | | |
| PDSCH_RB | dB | | | | | |
| OCNG_RA Note1 | dB | | | | | |
| OCNG_R B Note1 | dB | | | | | |
| SNR Note 6 | dB | -4.1 | -8.9 | -14.1 | -9.3 | -4.1 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | |
| Propagation condition | | AWGN | | | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.6.4-1.

**Table 7.3.6.5-2: DRX-Configuration for E-UTRAN FDD Radio Link Monitoring in-sync in DRX test**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf2 | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf40 | |
| shortDRX | disable | |

**Table 7.3.6.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD Radio Link Monitoring in-sync in DRX test**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| sr-ConfigIndex | 0 | For further information see section 6.3.2 in 3GPP TS 36.331 [8] and section10.1 in 3GPP TS 36.213 [8]. |

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (1120 ms after the start of time duration T5) the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the subframe according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:
- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.7 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync in DRX

### 7.3.7.1 Test purpose

The purpose of this test is to verify that the UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the serving cell when DRX is used. This test will partly verify the E-UTRAN TDD radio link monitoring requirements in TS 36.133 section 7.6.

### 7.3.7.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward. Applicability requires support for FGI bit 5.

### 7.3.7.3 Minimum conformance requirements

When DRX is used the $Q_{out}$ evaluation period ($T_{Evaluate_Q_{out_DRX}}$) and the $Q_{in}$ evaluation period ($T_{Evaluate_Q_{in_DRX}}$) is specified in TS 36.133 [4] Table 7.6.2.2-1 will be used.

When the downlink radio link quality estimated over the last $T_{Evaluate_Q_{out_DRX}}$ [s] period becomes worse than the threshold $Q_{out}$, Layer 1 of the UE shall send out-of-sync indication to the higher layers within $T_{Evaluate_Q_{out_DRX}}$ [s] evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331.

When the downlink radio link quality estimated over the last $T_{Evaluate_Q_{in_DRX}}$ [s] period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send in-sync indications to the higher layers within $T_{Evaluate_Q_{in_DRX}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331.

The out-of-sync and in-sync evaluations shall be performed as specified in section 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10 ms, DRX_cycle_length).

Upon start of T310 timer as specified in section 5.3.11 in TS 36.331 [5], the UE shall monitor the link for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry of T310 timer.

The transmitter power shall be turned off within 40 ms after expiry of T310 counter as specified in TS 36.331 [5] section 5.3.11.

**Table 7.3.7.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX}}$ and $T_{Evaluate_Q_{in_DRX}}$ (s) (DRX cycles) |
|---|---|
| ≤0.04 | Note (20) |
| 0.04 < DRX cycle ≤ 0.64 | Note (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note (5) |
| Note: Evaluation period length in time depends on the length of the DRX cycle in use | |

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.7.

### 7.3.7.4 Test description

The test consists of two subtests with one cell configured; the difference between the subtests is the DRX cycle length, number of transmit antennas and the propagation conditions. Each subtest consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.7.4-1 shows the three different time durations and the corresponding variation of the downlink SNR in the active cell to emulate out-of-sync states.

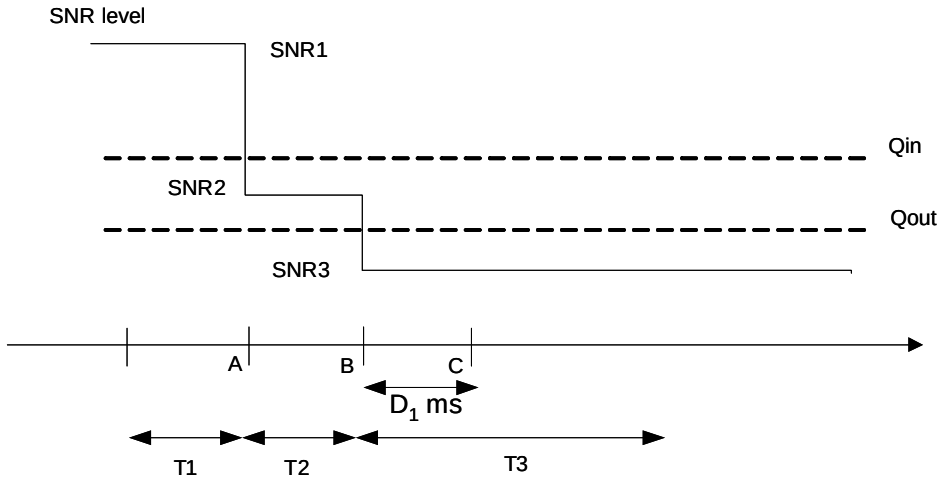


**Figure 7.3.7.4-1: SNR variation for out-of-sync testing in DRX**

#### 7.3.7.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as stated below.
   For subtest 1: As Shown in TS 36.508 [7] Annex A figure A.10
   For subtest 2: As Shown in TS 36.508 [7] Annex A figure A.9 (without using the faders)
2. The general test parameter settings for the different subtests are set up according to Table 7.3.7.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.7.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.7.4.1-1: General test parameters for E-UTRAN TDD out-of-sync in DRX testing**

| Parameter | | Unit | Value | | Comment |
|---|---|---|---|---|---|
| | | | Test 1 | Test 2 | |
| PCFICH/PDCCH/PHICH parameters | | | R.7 TDD | R.6 TDD | As specified in section A.2.2. None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.2 TDD | OP.2 TDD | As specified in section D.2.2. |
| Active cell | | | Cell 1 | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| CP length | | | Normal | Normal | |
| E-UTRA RF Channel Number | | | 1 | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | 10 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | 1x2 | Correlation Matrix and Antenna Configuration are defined in TS 36.521-1 [10] Annex B.2.3.2 |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | 1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] in section 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | 0 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | 4 | |
| DRX cycle | | ms | 40 | 1280 | See Table 7.3.7.5-2 |
| Layer 3 filtering | | | Enabled | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213 [8]. |
| CQI reporting periodicity | | ms | 1 | 1 | Minimum CQI reporting periodicity |
| Propagation channel | | | ETU 70 Hz | AWGN | . |
| T1 | | s | 4 | 32 | |
| T2 | | s | 1.6 | 12.8 | |
| T3 | | s | 1.8 | 13 | |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | | |

### 7.3.7.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 7.3.7.5-1 for subtest 1 and 2. Propagation conditions are set according to Annex B clause B.2.2 for subtests 1 and according to B.1.1 for subtests 2. T1 starts.

3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.7.5-1 for subtests 1 and 2. T2 starts.

4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.7.5-1 for subtests 1 and 2. T3 starts.

5. If the SS:

   a) detects uplink power equal to or higher than -39 dBm in the On-duration part of every DRX cycle in the uplink subframe according the configured CQI reporting mode (PUCCH 1-0) during the period from time point A to time point B

   and

   b) For subtest 1: does not detect any uplink power higher than -48.5 dBm from time point C (900 ms after the start of T3) until T3 expires,
   For subtest 2: does not detect any uplink power higher than -48.5 dBm from time point C (6500 ms after the start of T3) until T3 expires,

   the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.

6. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.7.5-1 for subtests 1 and 2.

7. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

8. Repeat steps 2-7 for all subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.7.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.7.4.3-1: Common Exception messages for E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync in DRX**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.7.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync in DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1C in TS 36.213 [8]) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213 [8]) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.7.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN TDD Radio Link Monitoring test for Out-of-sync in DRX, subtest #1 requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf40 typical value in real network for real-time services. | |
| sf40 | 2 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.7.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN TDD Radio Link Monitoring test for Out-of-sync in DRX, subtest #2 requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_DRX_L |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf1280 typical value in real network for best-effort services. | |
| sf1280 | 2 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.7.4.3-5: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN TDD Radio Link Monitoring test for Out-of-sync in DRX, subtest #1 and #2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.7.4.3-6: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring test for Out-of-sync in DRX, subtest #1 requirements**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.7.4.3-7: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN TDD Radio Link Monitoring test for Out-of-sync in DRX, subtest #1 and #2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 2 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

### 7.3.7.5 Test requirement

**Table 7.3.7.5-1: Cell specific test parameters for E-UTRAN TDD (cell # 1) for out-of-sync radio link monitoring subtests #1 and # 2**

| Parameter | Unit | Test 1 | | | Test 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | 1x2 | | |
| Special subframe configuration Note1 | | 6 | | | 6 | | |
| Uplink-downlink configuration Note2 | | 1 | | | 1 | | |
| OCNG Pattern defined in D.2 (TDD) | | OP.2 TDD | | | OP.2 TDD | | |
| $\rho_A$, $\rho_B$ | | -3 | | | 0 | | |
| PCFICH_RB | dB | 1 | | | 4 | | |
| PDCCH_RA | dB | -3 | | | 0 | | |
| PDCCH_RB | dB | -3 | | | 0 | | |
| PBCH_RA | dB | -3 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note1 | dB | | | | | | |
| OCNG_RB Note1 | dB | | | | | | |
| SNR Note 8 | dB | -1.4 | -5.0 | -12.8 | -4.5 | -8.5 | -13.7 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | -98 | | |
| Propagation condition | | ETU 70 Hz | | | AWGN | | |

Note 1: For the special subframe configuration see table 4.2-1 in 3GPP TS 36.211 [9].
Note 2: For the uplink-downlink configuration see table 4.2-2 in 3GPP TS 36.211 [9].
Note 3: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 5: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 6: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 7: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 8: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.7.4-1.

**Table 7.3.7.5-2: DRX-Configuration for E-UTRAN TDD Radio Link Monitoring out-of-sync in DRX test**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf2 | psf2 | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf40 | sf1280 | |
| shortDRX | disable | Disable | |

**Table 7.3.7.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN TDD Radio Link Monitoring out-of-sync in DRX test**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | infinity | infinity | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| sr-ConfigIndex | 2 | 2 | For further information see section 6.3.2 in 3GPP TS 36.331 [5] and section10.1 in 3GPP TS 36.213 [8]. |

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

In subtest 1 and subtest 2 during the period from time point A to time point B the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the uplink subframe according to the configured CQI reporting mode (PUCCH 1-0).

In subtest 1 the UE shall stop transmitting uplink signal no later than time point C (duration D1 = 900 ms after the start of time duration T3).

In subtest 2 the UE shall stop transmitting uplink signal no later than time point C (duration D1 = 6500 ms after the start of time duration T3).

The uplink signal is verified on the basis of the UE output power:
- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal
- UE output power equal to or less than Transmit OFF power -48.5 dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.8 E-UTRAN TDD Radio Link Monitoring Test for In-sync in DRX

### 7.3.8.1 Test purpose

The purpose of this test is to verify that the UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the serving cell when DRX is used. This test will partly verify the E-UTRAN TDD radio link monitoring requirements in TS 36.133 section 7.6.

### 7.3.8.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward. Applicability requires support for FGI bit 5.

### 7.3.8.3 Minimum conformance requirements

When DRX is used the $Q_{out}$ evaluation period ($T_{Evaluate_Q_{out_DRX}}$) and the $Q_{in}$ evaluation period ($T_{Evaluate_Q_{in_DRX}}$) is specified in TS 36.133 [4] Table 7.6.2.2-1 will be used.

When the downlink radio link quality estimated over the last $T_{Evaluate_Q_{out_DRX}}$ [s] period becomes worse than the threshold $Q_{out}$, Layer 1 of the UE shall send out-of-sync indication to the higher layers within $T_{Evaluate_Q_{out_DRX}}$ [s] evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

When the downlink radio link quality estimated over the last $T_{Evaluate_Q_{in_DRX}}$ [s] period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send in-sync indications to the higher layers within $T_{Evaluate_Q_{in_DRX}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331.

The out-of-sync and in-sync evaluations shall be performed as specified in section 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10 ms, DRX_cycle_length).

Upon start of T310 timer as specified in section 5.3.11 in TS 36.331 [5], the UE shall monitor the link for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry of T310 timer.

The transmitter power shall be turned off within 40 ms after expiry of T310 counter as specified in TS 36.331 [5] section 5.3.11.

**Table 7.3.8.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX}}$ and $T_{Evaluate_Q_{in_DRX}}$ (s) (DRX cycles) |
|---|---|
| ≤0.04 | Note (20) |
| 0.04 < DRX cycle ≤ 0.64 | Note (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note (5) |
| Note: Evaluation period length in time depends on the length of the DRX cycle in use | |

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.8.

## 7.3.8.4 Test description

The test consists of one subtest with one cell configured. The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.8.4-1 shows the different time durations and the corresponding variation of the downlink SNR in the active cell to emulate in-sync states.

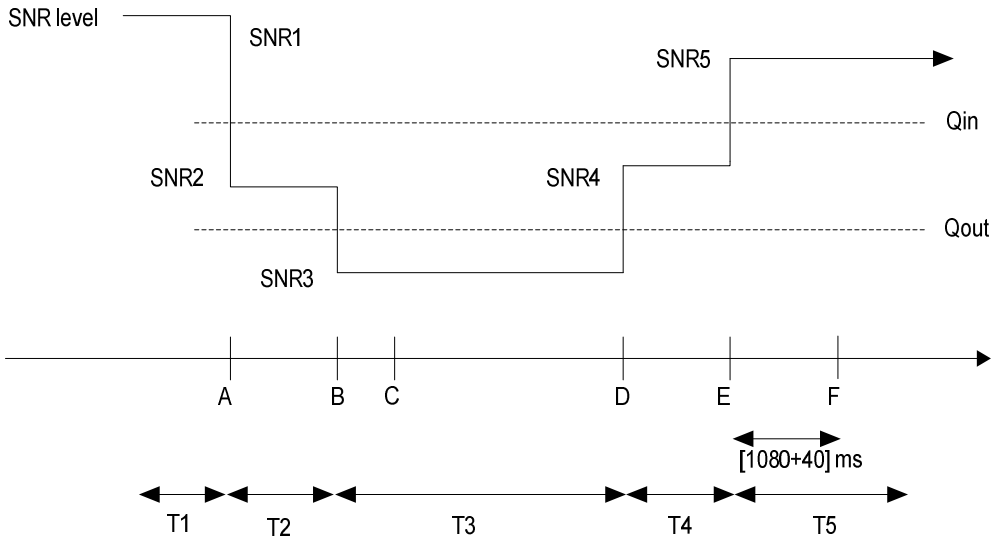


**Figure 7.3.8.4-1: SNR variation for in-sync testing in DRX**

### 7.3.8.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1 Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.18.

2. The general test parameter settings for the test is set up according to Table 7.3.8.4-1.

3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.8.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.8.4.1-1: General test parameters for E-UTRAN TDD in-sync in DRX testing**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | | R.6 TDD | As specified in section A.2.2. None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.2 TDD | As specified in section D.2.2. |
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| CP length | | | Normal | |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| Antenna Configuration | | | 1x2 | |
| In sync transmission parameters (Note 1) | DCI format | | 1C | As defined in section 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | In sync threshold $Q_{in}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] in section and Table 7.6.1-2 respectively. |
| | Aggregation level | CCE | 4 | |
| | $\rho_A$, $\rho_B$ | | 0 | |
| | Ratio of PDCCH to RS EPRE | | 0 | |
| | Ratio of PCFICH to RS EPRE | | 4 | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] in section 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | 0 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 4 | |
| DRX cycle | | ms | 40 | See Table 7.3.8.5-2 |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213 [8]. |
| CQI reporting periodicity | | ms | 2 | Minimum CQI reporting periodicity |
| Propagation channel | | | AWGN | |
| T1 | | s | 4 | |
| T2 | | s | 1.6 | |
| T3 | | s | 1.46 | |
| T4 | | s | 0.4 | |
| T5 | | s | 4 | |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

## 7.3.8.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.8.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.
3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.8.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.8.5-1. T3 starts.
5. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.8.5-1. T4 starts.
6. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.8.5-1. T5 starts.
7. If the SS detects uplink power equal to or higher than -39 dBm in the On-duration part of every DRX cycle in the uplink subframe according the configured CQI reporting mode (PUCCH 1-0) during the period from time point A to time point F (1120 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
8. Repeat steps 2-7 for all subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.8.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.8.4.3-1: Common Exception messages for E-UTRAN TDD Radio Link Monitoring Test for In-sync in DRX**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.8.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for In-sync in DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1C in TS 36.213 [8]) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213 [8]) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.8.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN TDD Radio Link Monitoring test for In-sync in DRX requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf40 typical value in real network for real-time services. | |
| sf40 | 2 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.8.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN TDD Radio Link Monitoring test for In-sync in DRX requirement**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.8.4.3-5: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN TDD Radio Link Monitoring test for In-sync in DRX requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 2 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

## 7.3.8.5 Test requirement

**Table 7.3.8.5-1: Cell specific test parameters for E-UTRAN TDD (cell # 1) for In-sync radio link monitoring test**

| Parameter | Unit | Test 1 | | | | |
|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | |
| Antenna Configuration | | 1x2 | | | | |
| Special subframe configuration[Note1] | | 6 | | | | |
| Uplink-downlink configuration[Note2] | | 1 | | | | |
| OCNG Pattern defined in D.2 (TDD) | | OP.2 TDD | | | | |
| $\rho_A$, $\rho_B$ | | 0 | | | | |
| PCFICH_RB | dB | 4 | | | | |
| PDCCH_RA | dB | 0 | | | | |
| PDCCH_RB | dB | 0 | | | | |
| PBCH_RA | dB | 0 | | | | |
| PBCH_RB | dB | | | | | |
| PSS_RA | dB | | | | | |
| SSS_RA | dB | | | | | |
| PHICH_RA | dB | | | | | |
| PHICH_RB | dB | | | | | |
| PDSCH_RA | dB | | | | | |
| PDSCH_RB | dB | | | | | |
| OCNG_RA[Note1] | dB | | | | | |
| OCNG_R B[Note1] | dB | | | | | |
| SNR [Note 8] | dB | -4.5 | -8.5 | -13.7 | -9.7 | -4.5 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | |
| Propagation condition | | AWGN | | | | |

Note 1: For the special subframe configuration see table 4.2-1 in 3GPP TS 36.211 [9].
Note 2: For the uplink-downlink configuration see table 4.2-2 in 3GPP TS 36.211 [9].
Note 3: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 5: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 6: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 7: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 8: The SNR in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.8.4-1.

**Table 7.3.8.5-2: DRX-Configuration for E-UTRAN TDD Radio Link Monitoring in-sync in DRX test**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf2 | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf40 | |
| shortDRX | disable | |

**Table 7.3.8.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN TDD Radio Link Monitoring in-sync in DRX test**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| sr-ConfigIndex | 2 | For further information see section 6.3.2 in 3GPP TS 36.331 [5] and section10.1 in 3GPP TS 36.213 [8]. |

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (1120 ms after the start of time duration T5) the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the uplink subframe according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:
- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.9 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with Non MBSFN ABS (eICIC)

### 7.3.9.1 Test purpose

To verify that the UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the serving cell under time domain measurement resource restriction.

### 7.3.9.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward. Applicability requires support for FGI bit 115.

### 7.3.9.3 Minimum conformance requirements

When the downlink radio link quality of the PCell estimated over the last 200 ms period becomes worse than the threshold Qout, Layer 1 of the UE shall send an out-of-sync indication for the PCell to the higher layers within 200 ms Qout evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in TS 36.213 [8] clause 4.2.1. Two successive indications from Layer 1 shall be separated by at least 10 ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in TS 36.331 [5] clause 5.3.11.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2.1 and A.7.3.9.

### 7.3.9.4 Test description

There are two E-UTRA cells in the test on the same RF channel. Cell 1 is the serving victim cell and Cell 2 is the neighbour aggressor cell. Non-MBSFN ABS pattern is configured in the aggressor cell (Cell 2). The UE is configured with a time domain measurement restriction pattern for performing serving cell (Cell 1) measurements. The UE is also configured for periodic CQI reporting in PUCCH 1-0 mode.

The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.9.4-1 shows the variation of the downlink SNR in the serving cell (Cell 1) to emulate out-of-sync states.

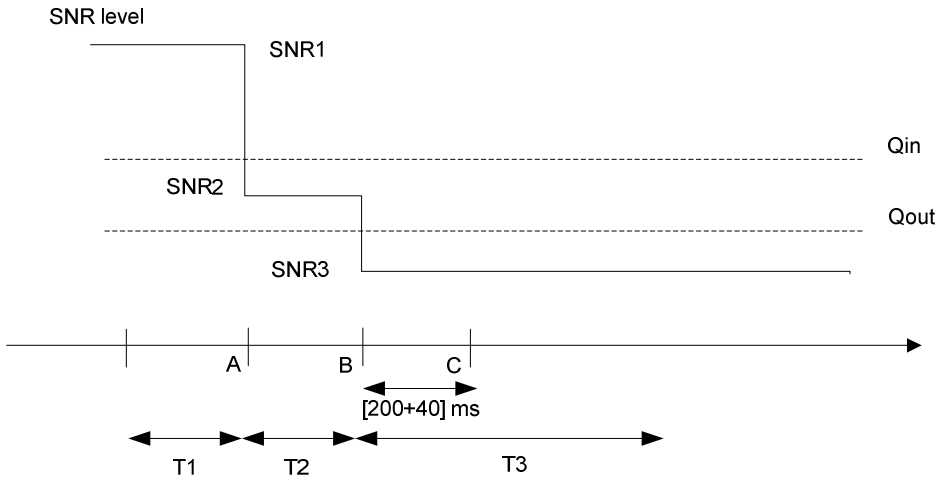


**Figure 7.3.9.4-1 SNR variation in the serving cell for out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS**

### 7.3.9.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), AWGN noise sources and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.40
2. The general test parameter settings for the different subtests are set up according to Table 7.3.9.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.9.4.3
5. At this stage only Cell 1 is switched on. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1.

**Table 7.3.9.4.1-1: General test parameters for E-UTRAN FDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | | R.9.FDD | As specified in section A.2.1.<br>None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.6 FDD | As specified in section D.1.6. |
| Serving cell (PCell) | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| Neighbour cell | | | Cell 2 | Aggressor cell on E-UTRA RF channel number 1 |
| Neighbour cell ABS configuration | | | Non-MBSFN ABS | As defined in Table C.3.1.1.2-1 |
| CP length | | | Normal | |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.521-1 [10] Annex B.2.3.2 |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 3 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] section 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX | | | OFF | |
| Layer 3 filtering | | | Enabled | Counters:: N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 2 | Minimum CQI reporting periodicity |
| Time offset between cells | | μs | 3 | Synchronous cells |
| T1 | | s | 1 | |
| T2 | | s | 0.4 | |
| T3 | | s | 0.5 | |
| Physical cell ID PCI | | | $(PCI_{cell1}-PCI_{cell2})$mod3 != 0 | Cell IDs are chosen such that CRS from cells 1 and 2 do not overlap in frequency |
| ABS pattern | | | '10000000100000001000000010000000010000000' | FDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 2. The first/leftmost bit corresponds to the PCell subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (40) divided by 10. No MBSFN subframes are configured in the ABS subframes. |
| Time domain measurement resource restriction pattern | | | '10000000100000001000000010000000010000000' | Time domain measurement resource restriction pattern for serving cell measurement signalled to the UE in message measSubframePattern PCell -r10 as defined in TS 36.331, clause 6.3.2. |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel | | | | |

#### 7.3.9.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to Cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

Non-MBSFN ABS pattern is configured in the aggressor cell (Cell 2). The UE is configured by higher layers with a time domain measurement restriction pattern for performing serving cell (Cell 1) measurements. The patterns shall be configured prior to the start of T1.

1. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters for both Cell 1 and Cell 2 according to T1 in Table 7.3.9.5-1. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.
3. When T1 expires the SS shall change the SNR values to T2 as specified in Table 7.3.9.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR values to T3 as specified in Table 7.3.9.5-1. T3 starts.
5. If the SS:

   a) detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B

   and

   b) does not detect any uplink power higher than -48.5 dBm from time point C (240 ms after the start of T3) until T3 expires,

   the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
6. When T3 expires the SS shall change the SNR values to T1 as specified in Table 7.3.9.5-1.
7. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.
8. Repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 7.3.9.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.9.4.3-1: Common Exception messages for E-UTRAN FDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1<br>Table H.2.7-1<br>Table H.2.7-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.9.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.9.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.9.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN FDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

**Table 7.3.9.4.3-5: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN FDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '10000000100000001000 00001000000010000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.9.4.3-6: *SystemInformationBlockType1 :* Additional E-UTRAN FDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.2 Table 4.4.3.2-3 SystemInformationBlockType1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType1 ::= SEQUENCE { | | | |
| cellSelectionInfo SEQUENCE { | | | |
| q-RxLevMin | -35 (-70 dBm) | | Cell 2 |
| } | | | |

## 7.3.9.5 Test requirements

**Table 7.3.9.5-1: Cell specific test requirement parameters for E-UTRAN FDD for out-of-sync radio link monitoring under time domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | 2x2 Low | | |
| OCNG Pattern defined in D.1.6 (FDD) | | OP.6 FDD | | | OP.6 FDD | | |
| $\rho_A$, $\rho_B$ | | -3 | | | -3 | | |
| PCFICH_RB | dB | 1 | | | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.2-1. | | |
| PDCCH_RA | dB | 1 | | | | | |
| PDCCH_RB | dB | 1 | | | | | |
| PBCH_RA | dB | -3 | | | | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note1 | dB | | | | | | |
| OCNG_RB Note1 | dB | | | | | | |
| SNR Note 6 | dB | -0.4 | -4.5 | -13.3 | 4.8 | 4.8 | 5.2 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | -98 | | |
| Propagation condition | | ETU 30 | | | ETU 30 | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal quality, signal-to-interference-plus-noise ratio, on the CRS REs.
Note 6: The SNR in the restricted measurement subframes during time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.9.4-1.

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal no later than time point C (240 ms after the start of time duration T3).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal
- UE output power equal to or less than Transmit OFF power -48.5 dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.10 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with Non MBSFN ABS (eICIC)

### 7.3.10.1 Test purpose

To verify that the UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the serving cell under time domain measurement resource restriction.

### 7.3.10.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward. Applicability requires support for FGI bit 115.

### 7.3.10.3 Minimum conformance requirements

When the downlink radio link quality of the PCell estimated over the last 200 ms period becomes worse than the threshold Qout, Layer 1 of the UE shall send an out-of-sync indication for the PCell to the higher layers within 200 ms Qout evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in TS 36.213 [8] clause 4.2.1. Two successive indications from Layer 1 shall be separated by at least 10 ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in TS 36.331 [5] clause 5.3.11.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2.1 and A.7.3.10.

### 7.3.10.4 Test description

There are two E-UTRA cells in the test on the same RF channel. Cell 1 is the serving victim cell and Cell 2 is the neighbour aggressor cell. Non-MBSFN ABS pattern is configured in the aggressor cell (Cell 2). The UE is configured with a time domain measurement restriction pattern for performing serving cell (Cell 1) measurements. The UE is also configured for periodic CQI reporting in PUCCH 1-0 mode.

The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.10.4-1 shows the variation of the downlink SNR in the serving cell (Cell 1) to emulate out-of-sync states.

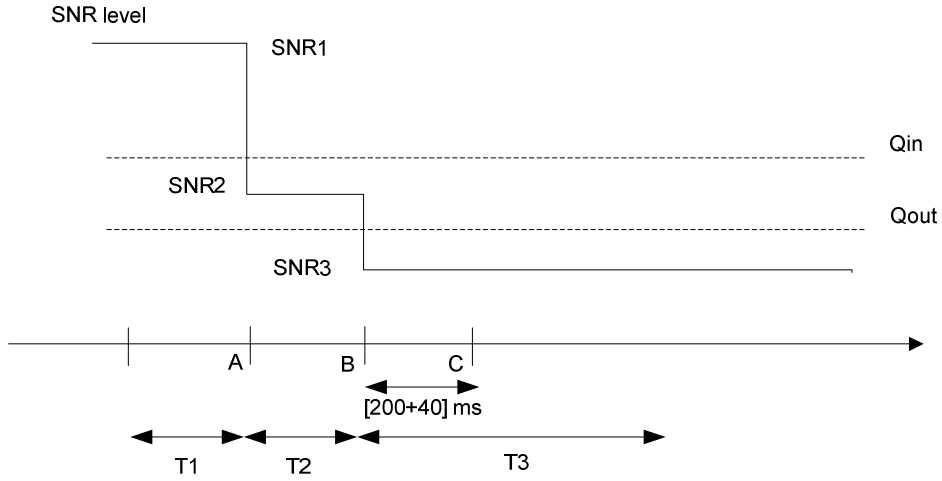


**Figure 7.3.10.4-1 SNR variation in the serving cell for out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS**

### 7.3.10.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), AWGN noise sources and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.40
2. The general test parameter settings for the different subtests are set up according to Table 7.3.10.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.10.4.3
5. At this stage only Cell 1 is switched on. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1.

**Table 7.3.10.4.1-1: General test parameters for E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | | R.9 TDD | As specified in section A.2.2. None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.2 TDD | As specified in section D.2.2. |
| Serving cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| Neighbour cell | | | Cell 2 | Cell 2 is the aggressor cell on E-UTRA RF channel number 1 |
| Neighbour cell ABS configuration | | | Non-MBSFN ABS | As defined in Table C.3.1.1.2-1 |
| CP length | | | Normal | |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.521-1 [10] Annex B.2.3.2 |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 3 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] section 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| Physical cell ID PCI | | | $(PCI_{cell1}-PCI_{cell2})mod3 != 0$ | Cell IDs are chosen such that CRS from cells 1 and 2 do not overlap in frequency. |
| ABS pattern | | | 10000000001000000000 | TDD ABS Pattern Info IE is configured in Cell 2 as defined in section 9.2.54 in TS 36.423 [28]. The first/leftmost bit corresponds to the subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (20) divided by 10. No MBSFN subframes are configured in the ABS subframes. |
| Time domain measurement resource restriction pattern | | | 10000000001000000000 | MeasSubframePattern IE is configured in UE for serving cell measurement as defined in section 6.3.6 in TS 36.331. |
| DRX | | | OFF | |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 1 | Minimum CQI reporting periodicity |
| Time offset between cells | | µs | 3 | |
| Propagation channel | | | ETU30 | |
| T1 | | s | 1 | |
| T2 | | s | 0.4 | |
| T3 | | s | 0.5 | |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

### 7.3.10.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to Cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms.

Non-MBSFN ABS pattern is configured in the aggressor cell (Cell 2). The UE is configured by higher layers with a time domain measurement restriction pattern for performing serving cell (Cell 1) measurements. The patterns shall be configured prior to the start of T1.

1. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters for both Cell 1 and Cell 2 according to T1 in Table 7.3.10.5-1. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.
3. When T1 expires the SS shall change the SNR values to T2 as specified in Table 7.3.10.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR values to T3 as specified in Table 7.3.10.5-1. T3 starts.
5. If the SS:

   a) detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B

   and

   b) does not detect any uplink power higher than -48.5 dBm from time point C (240 ms after the start of T3) until T3 expires,

   the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
6. When T3 expires the SS shall change the SNR values to T1 as specified in Table 7.3.10.5-1.
7. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.
8. Repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.10.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.10.4.3-1: Common Exception messages for E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.10.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1C in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.10.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.10.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

**Table 7.3.10.4.3-5: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 | | | |
| subframeConfig1-5-r10 | '10000000001000000000' | BIT STRING (SIZE (20)) | |
| | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.10.4.3-6: *SystemInformationBlockType1 :* Additional E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.2 Table 4.4.3.2-3 SystemInformationBlockType1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType1 ::= SEQUENCE { | | | |
| cellSelectionInfo SEQUENCE { | | | |
| q-RxLevMin | -35 (-70 dBm) | | Cell 2 |
| } | | | |

## 7.3.10.5 Test requirements

**Table 7.3.10.5-1: Cell specific test requirement parameters for E-UTRAN TDD for out-of-sync radio link monitoring under time domain measurement resource restriction with non-MBSFN ABS (eICIC)**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | 2x2 Low | | |
| Special subframe configuration[Note1] | | 6 | | | 6 | | |
| Uplink-downlink configuration[Note2] | | 1 | | | 1 | | |
| OCNG Pattern defined in D.2.2 (TDD) | | OP.2 TDD | | | OP.2 TDD | | |
| $\rho_A$, $\rho_B$ | | -3 | | | -3 | | |
| PCFICH_RB | dB | 1 | | | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.2-1. | | |
| PDCCH_RA | dB | 1 | | | | | |
| PDCCH_RB | dB | 1 | | | | | |
| PBCH_RA | dB | -3 | | | | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA[Note 3] | dB | | | | | | |
| OCNG_RB[Note 3] | dB | | | | | | |
| SNR [Note 8] | dB | -0.4 | -4.5 | -13.3 | 4.8 | 4.8 | 5.2 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | -98 | | |
| Propagation condition | | ETU30 | | | ETU30 | | |

Note 1: For the special subframe configuration see table 4.2-1 in 3GPP TS 36.211.
Note 2: For the uplink-downlink configuration see table 4.2-2 in 3GPP TS 36.211.
Note 3: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 5: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 6: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 7: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 8: The SNR in time periods T1, T2 and T3 of active cell is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.10.4-1.

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least in all uplink subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal no later than time point C (240 ms after the start of the time duration T3).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal

- UE output power equal to or less than Transmit OFF power -48.5 dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.11 E-UTRAN FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with Non MBSFN ABS (eICIC)

### 7.3.11.1 Test purpose

To verify that the UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the serving cell.

### 7.3.11.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward. Applicability requires support for FGI bit 115.

### 7.3.11.3 Minimum conformance requirements

When the downlink radio link quality of the PCell estimated over the last 100 ms period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 100 ms $Q_{in}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in TS 36.213 [8] clause 4.2.1. Two successive indications from Layer 1 shall be separated by at least 10 ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in TS 36.331 [5] clause 5.3.11.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2.1 and A.7.3.11.

### 7.3.11.4 Test description

There are two E-UTRA cells in the test on the same RF channel. Cell 1 is the serving victim cell and Cell 2 is the neighbour aggressor cell. Non-MBSFN ABS pattern is configured in the aggressor cell (Cell 2). The UE is configured with a time domain measurement restriction pattern for performing serving cell (Cell 1) measurements. The UE is also configured for periodic CQI reporting in PUCCH 1-0 mode.

The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.11.4-1 shows the variation of the downlink SNR in the serving cell (Cell 1) to emulate in-sync states.

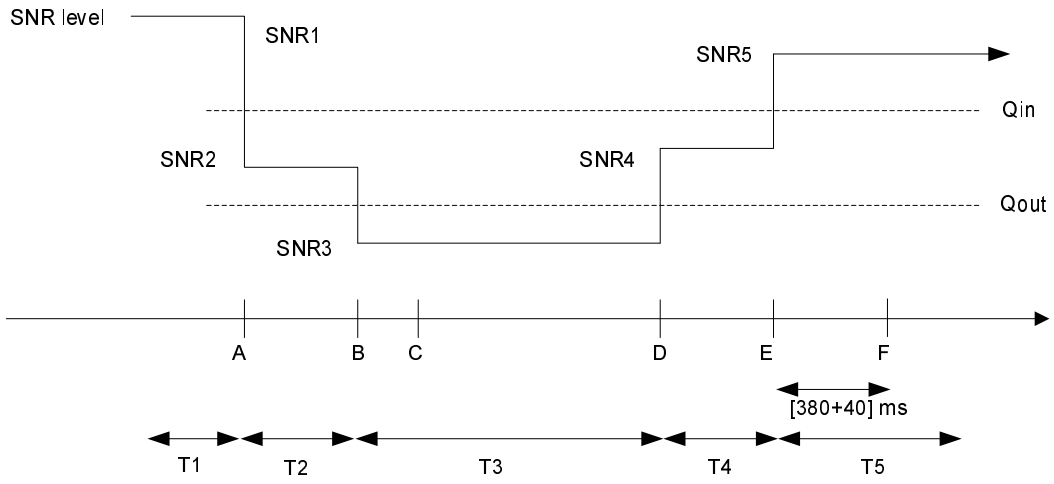


**Figure 7.3.11.4-1 SNR variation in the serving cell for in-sync testing under time domain measurement resource restriction with non-MBSFN ABS**

### 7.3.11.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), AWGN noise sources and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.40
2. The general test parameter settings for the different subtests are set up according to Table 7.3.11.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.11.4.3
5. At this stage only Cell 1 is switched on. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1.

**Table 7.3.11.4.1-1: General test parameters for E-UTRAN FDD in-sync testing under time domain measurement resource restriction (eICIC)**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | | R.9 FDD | As specified in section A.2.1. None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.6 FDD | As specified in section D.1.6. |
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| Neighbour cell | | | Cell 2 | Cell 2 is on E-UTRA RF channel number 1; Cell 2 generates interference over restricted resources. |
| Neighbour cell ABS configuration | | | Non-MBSFN ABS | As defined in Table C.3.1.1.2-2 |
| CP length | | | Normal | |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 10 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.521-1 [10] Annex B.2.3.2 |
| In sync transmission parameters for the active cell (Note 1) | DCI format | | 1C | As defined in section 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 3 | In sync threshold Qin and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] section 7.6.1 and Table 7.6.1-2 respectively. |
| | Aggregation level | CCE | 4 | |
| | ρA, ρB | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | -3 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| Out of sync transmission parameters for active cell (Note 1) | DCI format | | 1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 3 | Out of sync threshold Qout and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] section 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | ρA, ρB | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX | | | OFF | |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 2 | Minimum CQI reporting periodicity |
| Time offset between cells | | μs | 3 | |
| Propagation channel | | | ETU30 | |
| T1 | | s | 0.5 | |
| T2 | | s | 0.4 | |

| | | | |
|---|---|---|---|
| T3 | s | 1.46 | |
| T4 | s | 0.4 | |
| T5 | s | 1 | |
| Physical cell ID PCI | | ($PCI_{cell1}$-$PCI_{cell2}$)mod3 != 0 | Cell IDs are chosen such that CRS from cells 1 and 2 do not overlap in frequency |
| ABS pattern | | '10000000100000001000000100000010000' | FDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 2. The first/leftmost bit corresponds to the PCell subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (40) divided by 10. No MBSFN subframes are configured in the ABS subframes. |
| Time domain measurement resource restriction pattern | | '10000000100000001000000100000010000' | Time domain measurement pattern for serving cell measurement signalled to the UE in message measSubframePattern PCell - r10 as defined in TS 36.331, clause 6.3.2. |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | |

### 7.3.11.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to Cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

Non-MBSFN ABS pattern is configured in the aggressor cell (Cell 2). The UE is configured by higher layers with a time domain measurement restriction pattern for performing serving cell (Cell 1) measurements. The patterns shall be configured prior to the start of T1.

1. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters for both Cell 1 and Cell 2 according to T1 in Table 7.3.11.5-1. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.
3. When T1 expires the SS shall change the SNR values to T2 as specified in Table 7.3.11.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR values to T3 as specified in Table 7.3.11.5-1. T3 starts.
5. When T3 expires the SS shall change the SNR values to T4 as specified in Table 7.3.11.5-1. T4 starts.
6. When T4 expires the SS shall change the SNR values to T5 as specified in Table 7.3.11.5-1. T5 starts.
7. If the SS detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (420 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
8. After T5 expires, repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.11.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.11.4.3-1: Common Exception messages for E-UTRAN FDD in-sync testing under time domain measurement resource restriction (eICIC)**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2<br>Table H.2.7-1<br>Table H.2.7-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.11.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FDD in-sync testing under time domain measurement resource restriction (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**TTable 7.3.11.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FDD in-sync testing under time domain measurement resource restriction (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.11.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN FDD in-sync testing under time domain measurement resource restriction (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

**Table 7.3.11.4.3-5: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN FDD in-sync testing under time domain measurement resource restriction (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '10000000100000001000 00001000000010000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| } | | | |
| } | | | |

## 7.3.11.5 Test requirements

**Table 7.3.11.5-1: Cell specific test requirement parameters for E-UTRAN FDD for in-sync radio link monitoring under time domain measurement resource restriction (eICIC)**

| Parameter | Unit | Cell 1 | | | | | Cell 2 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | 10 | | | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | | | 2x2 Low | | | | |
| PCFICH/PDCCH/PHICH parameters | | R.9 FDD | | | | | R.9 FDD | | | | |
| Number of Control OFDM symbols | | 3 | | | | | 3 | | | | |
| OCNG Pattern defined D.1.6 (FDD) | | OP.6 FDD | | | | | OP.6 FDD | | | | |
| $\rho_A$, $\rho_B$ | | -3 | | | | | -3 | | | | |
| PCFICH_RB | dB | 1 | | | | | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.2-2. | | | | |
| PDCCH_RA | dB | -3 | | | | | | | | | |
| PDCCH_RB | dB | -3 | | | | | | | | | |
| PBCH_RA | dB | -3 | | | | | | | | | |
| PBCH_RB | dB | | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | | |
| SNR Note 6 | dB | -0.4 | -4.5 | -13.3 | -8.2 | -0.4 | 4.8 | 4.8 | 5.2 | 5.2 | 4.8 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | | -98 | | | | |
| Propagation condition | | ETU30 | | | | | ETU30 | | | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2, T3, T4 and T5 of the active cell is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.11.4-1.

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (420 ms after the start of time duration T5) the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.12 E-UTRAN TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with Non MBSFN ABS (eICIC)

### 7.3.12.1 Test purpose

To verify that the UE properly detects in sync for the purpose of monitoring downlink radio link quality of the serving cell.

### 7.3.12.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward. Applicability requires support for FGI bit 115.

### 7.3.12.3 Minimum conformance requirements

When the downlink radio link quality of the PCell estimated over the last 100 ms period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 100 ms $Q_{in}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in TS 36.213 [8] clause 4.2.1. Two successive indications from Layer 1 shall be separated by at least 10 ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in TS 36.331 [5] clause 5.3.11.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2.1 and A.7.3.11.

### 7.3.12.4 Test description

There are two E-UTRA cells in the test on the same RF channel. Cell 1 is the serving victim cell and Cell 2 is the neighbour aggressor cell. Non-MBSFN ABS pattern is configured in the aggressor cell (Cell 2). The UE is configured with a time domain measurement restriction pattern for performing serving cell (Cell 1) measurements. The UE is also configured for periodic CQI reporting in PUCCH 1-0 mode.

The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.12.4-1 shows the variation of the downlink SNR in the serving cell (Cell 1) to emulate in-sync states.

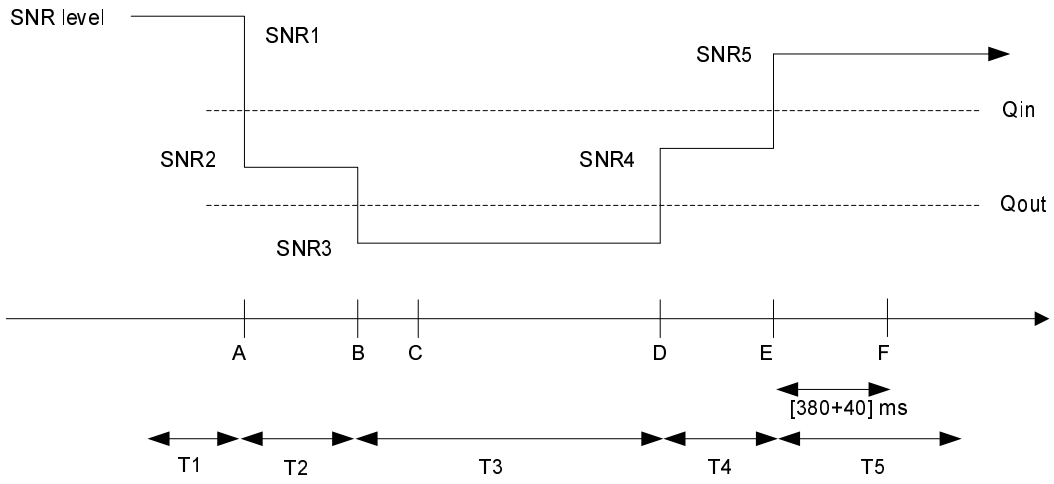


**Figure 7.3.12.4-1 SNR variation in the serving cell for in-sync testing under time domain measurement resource restriction with non-MBSFN ABS**

### 7.3.12.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), AWGN noise sources and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.40
2. The general test parameter settings for the different subtests are set up according to Table 7.3.12.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.12.4.3
5. At this stage only Cell 1 is switched on. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1.

**Table 7.3.12.4.1-1: General test parameters for E-UTRAN TDD in-sync testing under time domain measurement resource restriction (eICIC)**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | | R.9 TDD | As specified in section A.2.2. None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.2 TDD | As specified in section D.2.2. |
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| Neighbour cell | | | Cell 2 | Cell 2 is on E-UTRA RF channel number 1; Cell 2 generates interference over restricted resources. |
| Neighbour cell ABS configuration | | | Non-MBSFN ABS | As defined in Table C.3.1.1.2-2 |
| CP length | | | Normal | |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 10 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.521-1 [10] Annex B.2.3.2 |
| In sync transmission parameters for the active cell (Note 1) | DCI format | | 1C | As defined in section 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 3 | In sync threshold Qin and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] section 7.6.1 and Table 7.6.1-2 respectively. |
| | Aggregation level | CCE | 4 | |
| | ρA, ρB | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | -3 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| Out of sync transmission parameters for active cell (Note 1) | DCI format | | 1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 3 | Out of sync threshold Qout and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] section 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | ρA, ρB | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX | | | OFF | |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 1 | Minimum CQI reporting periodicity |
| Time offset between cells | | μs | 3 | |
| Propagation channel | | | ETU30 | |
| T1 | | s | 0.5 | |
| T2 | | s | 0.4 | |

| | | | |
|---|---|---|---|
| T3 | s | 1.46 | |
| T4 | s | 0.4 | |
| T5 | s | 1 | |
| Physical cell ID PCI | | ($PCI_{cell1}$-$PCI_{cell2}$ )mod3 != 0 | Cell IDs are chosen such that CRS from cells 1 and 2 do not overlap in frequency |
| ABS pattern | | '1000000000 1000000000' | TDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 2. The first/leftmost bit corresponds to the subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (20) divided by 10. No MBSFN subframes are configured in the ABS subframes. |
| Time domain measurement resource restriction pattern | | '1000000000 1000000000' | Time domain measurement pattern for serving cell measurement signalled to the UE in message measSubframePattern PCell - r10 as defined in TS 36.331, clause 6.3.2. Configured in Cell 1. |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | |

### 7.3.12.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to Cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms.

Non-MBSFN ABS pattern is configured in the aggressor cell (Cell 2). The UE is configured by higher layers with a time domain measurement restriction pattern for performing serving cell (Cell 1) measurements. The patterns shall be configured prior to the start of T1.

1. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters for both Cell 1 and Cell 2 according to T1 in Table 7.3.12.5-1. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.
3. When T1 expires the SS shall change the SNR values to T2 as specified in Table 7.3.12.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR values to T3 as specified in Table 7.3.12.5-1. T3 starts.
5. When T3 expires the SS shall change the SNR values to T4 as specified in Table 7.3.12.5-1. T4 starts.
6. When T4 expires the SS shall change the SNR values to T5 as specified in Table 7.3.12.5-1. T5 starts.
7. If the SS detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (420 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
8. After T5 expires, repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.12.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.12.4.3-1: Common Exception messages for E-UTRAN TDD in-sync testing under time domain measurement resource restriction (eICIC)**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.12.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD in-sync testing under time domain measurement resource restriction (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1C in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.12.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD in-sync testing under time domain measurement resource restriction (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.12.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN TDD in-sync testing under time domain measurement resource restriction (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

**Table 7.3.12.4.3-5: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN TDD in-sync testing under time domain measurement resource restriction (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 | | | |
| subframeConfig1-5-r10 | '10000000001000000000' | BIT STRING (SIZE (20)) | |
| | | | |
| } | | | |
| } | | | |
| } | | | |

## 7.3.12.5 Test requirements

**Table 7.3.12.5-1: Cell specific test requirement parameters for E-UTRAN TDD for in-sync radio link monitoring under time domain measurement resource restriction (eICIC)**

| Parameter | Unit | Cell 1 | | | | | Cell 2 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | 10 | | | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | | | 2x2 Low | | | | |
| Special subframe configuration Note1 | | 6 | | | | | 6 | | | | |
| Uplink-downlink configuration Note2 | | 1 | | | | | 1 | | | | |
| PCFICH/PDCCH/PHICH parameters | | R.9 TDD | | | | | R.9 TDD | | | | |
| Number of Control OFDM symbols | | 3 | | | | | 3 | | | | |
| OCNG Pattern defined in D.2.2 (TDD) | | OP.2 TDD | | | | | OP.2 TDD | | | | |
| $\rho_A$, $\rho_B$ | | -3 | | | | | -3 | | | | |
| PCFICH_RB | dB | 1 | | | | | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.2-2. | | | | |
| PDCCH_RA | dB | -3 | | | | | | | | | |
| PDCCH_RB | dB | -3 | | | | | | | | | |
| PBCH_RA | dB | -3 | | | | | | | | | |
| PBCH_RB | dB | | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | | |
| OCNG_RA Note 3 | dB | | | | | | | | | | |
| OCNG_RB Note 3 | dB | | | | | | | | | | |
| SNR Note 8 | dB | -0.4 | -4.5 | -13.3 | -8.2 | -0.4 | 4.8] | 4.8 | 5.2 | 5.2 | 4.8 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | | -98 | | | | |
| Propagation condition | | ETU30 | | | | | ETU30 | | | | |

Note 1: For the special subframe configuration see table 4.2-1 in 3GPP TS 36.211.
Note 2: For the uplink-downlink configuration see table 4.2-2 in 3GPP TS 36.211.
Note 3: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 5: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 6: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 7: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 8: The SNR in time periods T1, T2, T3, T4 and T5 of active cell is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.12.4-1.

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (420 ms after the start of time duration T5) the UE shall transmit uplink signal at least in all uplink subframes configured for CQI transmission according to the configured CQI mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.13 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)

### 7.3.13.1 Test purpose

To verify that the UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the serving cell under time domain measurement resource restriction.

### 7.3.13.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward. Applicability requires support for FGI bit 115.

### 7.3.13.3 Minimum conformance requirements

When the downlink radio link quality of the PCell estimated over the last 200 ms period becomes worse than the threshold Qout, Layer 1 of the UE shall send an out-of-sync indication for the PCell to the higher layers within 200 ms Qout evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in TS 36.213 [8] clause 4.2.1. Two successive indications from Layer 1 shall be separated by at least 10 ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in TS 36.331 [5] clause 5.3.11.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2.1 and A.7.3.13.

### 7.3.13.4 Test description

There are two E-UTRA cells in the test on the same RF channel. Cell 1 is the serving victim cell and Cell 2 is the neighbour aggressor cell. MBSFN ABS pattern is configured in the aggressor cell (Cell 2). The UE is configured with a time domain measurement restriction pattern for performing serving cell (Cell 1) measurements. The UE is also configured for periodic CQI reporting in PUCCH 1-0 mode.

The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.13.4-1 shows the variation of the downlink SNR in the serving cell (Cell 1) to emulate out-of-sync states.

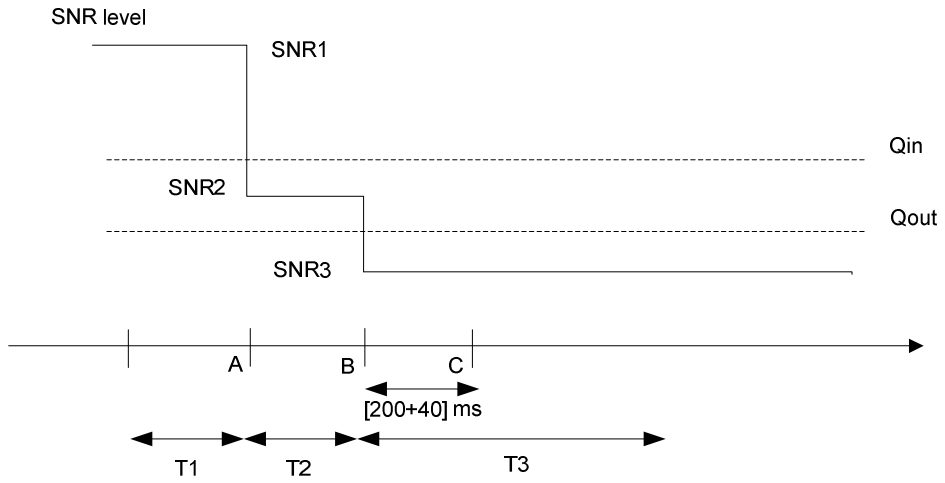


**Figure 7.3.13.4-1: SNR variation in the serving cell for out-of-sync testing under time domain measurement resource restriction with MBSFN ABS**

#### 7.3.13.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), AWGN noise sources and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.40.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.13.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.13.4.3.
5. At this stage only Cell 1 is switched on. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1.

**Table 7.3.13.4.1-1: General test parameters for E-UTRAN FDD out-of-sync testing under time domain measurement resource restriction with MBSFN ABS (eICIC)**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | | R.9.FDD | As specified in section A.2.1<br>None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.6 FDD for the serving cell (Cell 1)<br>OP.9 FDD for the neighbour cell (Cell 2) | As specified in section D.1.6 and D.1.9 respectively |
| Serving cell (PCell) | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| Neighbour cell | | | Cell 2 | Aggressor cell on E-UTRA RF channel number 1 |
| Neighbour cell ABS configuration | | | MBSFN ABS | As defined in Table C.3.1.2.2-1 |
| CP length | | | Normal | |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.521-1 [10] Annex B.2.3.2 |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 3 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] section 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX | | | OFF | |
| Layer 3 filtering | | | Enabled | Counters:: N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 2 | Minimum CQI reporting periodicity |
| Time offset between cells | | µs | 3 | Synchronous cells |
| T1 | | s | 1 | |
| T2 | | s | 0.4 | |
| T3 | | s | 0.5 | |
| Physical cell ID PCI | | | ($PCI_{cell1}$-$PCI_{cell2}$ ) mod 3 = 0, $PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell IDs are chosen such that CRS from cells 1 and 2 overlap in frequency |
| ABS pattern | | | '01000000100000001000000000010000001000000' | FDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 2. The first/leftmost bit corresponds to the PCell subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (40) divided by 10. MBSFN subframes are configured in the ABS subframes. |
| Time domain measurement resource restriction pattern | | | '01000000100000001000000000010000001000000' | Time domain measurement resource restriction pattern for serving cell measurement signalled to the UE in message measSubframePattern-Serv-r10 as defined in TS 36.331, clause 6.3.2. |
| Note: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

### 7.3.13.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to Cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

MBSFN ABS pattern is configured in the aggressor cell (Cell 2). The UE is configured by higher layers with a time domain measurement restriction pattern for performing serving cell (Cell 1) measurements. The patterns shall be configured prior to the start of T1.

1. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters for both Cell 1 and Cell 2 according to T1 in Table 7.3.13.5-1. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.

3. When T1 expires the SS shall change the SNR values to T2 as specified in Table 7.3.13.5-1. T2 starts.

4. When T2 expires the SS shall change the SNR values to T3 as specified in Table 7.3.13.5-1. T3 starts.

5. If the SS:
   a) detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B
   and
   b) does not detect any uplink power higher than -48.5 dBm from time point C (240 ms after the start of T3) until T3 expires,
   the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.

6. When T3 expires the SS shall change the SNR values to T1 as specified in Table 7.3.13.5-1.

7. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.

8. Repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.13.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.13.4.3-1: Common Exception messages for E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.13.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.13.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.13.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

**Table 7.3.13.4.3-5: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '01000000100000001000 00000010000001000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.13.4.3-6: *SystemInformationBlockType2*: Additional E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 SystemInformationBlockType2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfiguration | Not present | | Cell 1 |
| } | | | |

**Table 7.3.13.4.3-7: *SystemInformationBlockType3*: Additional E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1 |
| | '01'B (No MBSFN subframes are present in all neighbour cells) | | Cell 2 |

**Table 7.3.13.4.3-8: *SystemInformationBlockType1 :* Additional E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.2 Table 4.4.3.2-3 SystemInformationBlockType1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType1 ::= SEQUENCE { | | | |
| cellSelectionInfo SEQUENCE { | | | |
| q-RxLevMin | -35 (-70 dBm) | | Cell 2 |
| } | | | |

## 7.3.13.5 Test requirements

**Table 7.3.13.5-1: Cell specific test requirement parameters for E-UTRAN FDD for out-of-sync radio link monitoring under time domain measurement resource restriction with MBSFN ABS (eICIC)**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | 2x2 Low | | |
| OCNG Pattern defined in D.1.6 (FDD) and D.1.9 (FDD) | | OP.6 FDD | | | OP.9 FDD | | |
| $\rho_A$, $\rho_B$ | | -3 | | | -3 | | |
| PCFICH_RB | dB | 1 | | | Non-ABS and ABS subframe channel powers defined in Table C.3.1.2.2-1. | | |
| PDCCH_RA | dB | 1 | | | | | |
| PDCCH_RB | dB | 1 | | | | | |
| PBCH_RA | dB | -3 | | | | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note1 | dB | | | | | | |
| OCNG_RB Note1 | dB | | | | | | |
| SNR Note 6 | dB | -0.4 | -4.5 | -13.3 | 4.8 | 4.8 | 5.2 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | -98 | | |
| Propagation condition | | ETU 30 Hz | | | ETU 30 Hz | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal quality, signal-to-interference-plus-noise ratio, on the CRS REs.
Note 6: The SNR in the restricted measurement subframes during time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.13.4-1.

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal no later than time point C (240 ms after the start of time duration T3).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal.

- UE output power equal to or less than Transmit OFF power -48.5 dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.14 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)

### 7.3.14.1 Test purpose

To verify that the UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the serving cell under time domain measurement resource restriction.

### 7.3.14.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward. Applicability requires support for FGI bit 115.

### 7.3.14.3 Minimum conformance requirements

When the downlink radio link quality of the PCell estimated over the last 200 ms period becomes worse than the threshold Qout, Layer 1 of the UE shall send an out-of-sync indication for the PCell to the higher layers within 200 ms Qout evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in TS 36.213 [8] clause 4.2.1. Two successive indications from Layer 1 shall be separated by at least 10 ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in TS 36.331 [5] clause 5.3.11.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2.1 and A.7.3.14.

### 7.3.14.4 Test description

There are two E-UTRA cells in the test on the same RF channel. Cell 1 is the serving victim cell and Cell 2 is the neighbour aggressor cell. MBSFN ABS pattern is configured in the aggressor cell (Cell 2). The UE is configured with a time domain measurement restriction pattern for performing serving cell (Cell 1) measurements. The UE is also configured for periodic CQI reporting in PUCCH 1-0 mode.

The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.14.4-1 shows the variation of the downlink SNR in the serving cell (Cell 1) to emulate out-of-sync states.

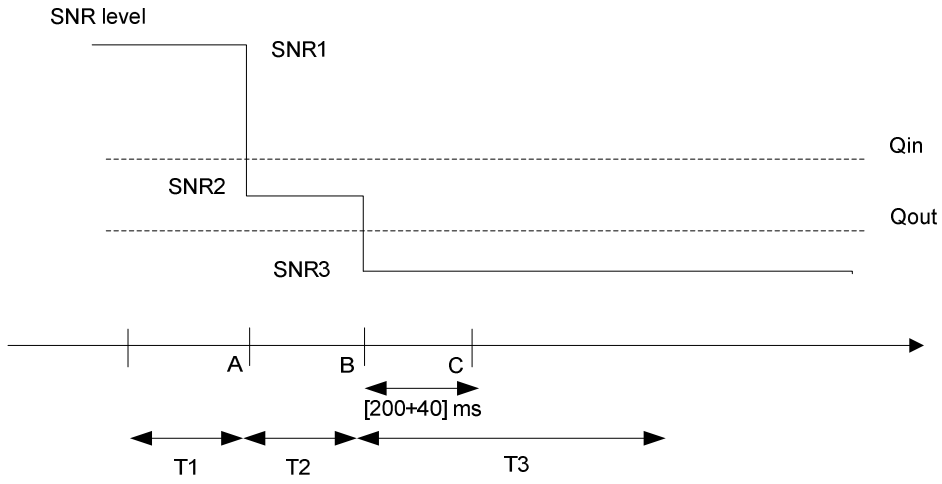


**Figure 7.3.14.4-1: SNR variation in the serving cell for out-of-sync testing under time domain measurement resource restriction with MBSFN ABS**

#### 7.3.14.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), AWGN noise sources and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.40.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.14.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.14.4.3.
5. At this stage only Cell 1 is switched on. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1.

**Table 7.3.14.4.1-1: General test parameters for E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with MBSFN ABS (eICIC)**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | | R.9.TDD | As specified in clause A.2.2.<br>None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.2 TDD for the serving cell (Cell 1)<br>OP.6 TDD for the neighbour cell (Cell 2) | As specified in clause D.2.2 and D.2.6 respectively |
| Serving cell (PCell) | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| Neighbour cell | | | Cell 2 | Aggressor cell on E-UTRA RF channel number 1 |
| Neighbour cell ABS configuration | | | MBSFN ABS | As defined in Table C.3.1.2.2-1 |
| CP length | | | Normal | |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.101 [5] Annex B.2.3.2 |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 3 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] clause 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX | | | OFF | |
| Layer 3 filtering | | | Enabled | Counters:: N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 1 | Minimum CQI reporting periodicity |
| Time offset between cells | | µs | 3 | Synchronous cells |
| T1 | | s | 1 | |
| T2 | | s | 0.4 | |
| T3 | | s | 0.5 | |
| Physical cell ID PCI | | | ($PCI_{cell1}$-$PCI_{cell2}$ ) mod 3 = 0, $PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell IDs are chosen such that CRS from cells 1 and 2 overlap in frequency |
| ABS pattern | | | '00001000000000100000' | MBSFN ABS. TDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 2.<br>The first/leftmost bit corresponds to the PCell subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (20) divided by 10. All ABS subframes are MBSFN subframes. |
| Time domain measurement resource restriction pattern | | | '00001000000000100000' | Time-domain measurement resource restriction pattern for serving cell measurements signalled to the UE in message measSubframePattern-Serv-r10 as defined in TS 36.331, clause 6.3.2. |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

### 7.3.14.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to Cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms.

MBSFN ABS pattern is configured in the aggressor cell (Cell 2). The UE is configured by higher layers with a time domain measurement restriction pattern for performing serving cell (Cell 1) measurements. The patterns shall be configured prior to the start of T1.

1. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters for both Cell 1 and Cell 2 according to T1 in Table 7.3.14.5-1. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.

3. When T1 expires the SS shall change the SNR values to T2 as specified in Table 7.3.14.5-1. T2 starts.

4. When T2 expires the SS shall change the SNR values to T3 as specified in Table 7.3.14.5-1. T3 starts.

5. If the SS:
   a) detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B
   and
   b) does not detect any uplink power higher than -48.5 dBm from time point C (240 ms after the start of T3) until T3 expires,
   the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.

6. When T3 expires the SS shall change the SNR values to T1 as specified in Table 7.3.14.5-1.

7. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.

8. Repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.14.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.14.4.3-1: Common Exception messages for E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with MBSFN ABS (eICIC)**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.14.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1C in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.14.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.14.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

**Table 7.3.14.4.3-5: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 | | | |
| subframeConfig1-5-r10 | '00001000000000100000' | BIT STRING (SIZE (20)) | |
| | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.14.4.3-6: *SystemInformationBlockType2*: Additional E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| mbsfn-SubframeConfigList ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF SEQUENCE { | | | |
| radioframeAllocationPeriod | n1 | Every radio frame is with MBSFN subframe | Cell 2 |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| oneFrame | '010000'B | Subframe 4 is used for MBSFN | |
| } | | | |
| } | | | |

**Table 7.3.14.4.3-7: *SystemInformationBlockType3*: Additional E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1 |
| | '01'B (No MBSFN subframes are present in all neighbour cells) | | Cell 2 |

**Table 7.3.14.4.3-8: *SystemInformationBlockType1 :* Additional E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.2 Table 4.4.3.2-3 SystemInformationBlockType1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType1 ::= SEQUENCE { | | | |
| cellSelectionInfo SEQUENCE { | | | |
| q-RxLevMin | -35 (-70 dBm) | | Cell 2 |
| } | | | |

## 7.3.14.5 Test requirements

**Table 7.3.14.5-1: Cell specific test requirement parameters for E-UTRAN TDD for out-of-sync radio link monitoring under time domain measurement resource restriction with MBSFN ABS (eICIC)**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| Special subframe configuration[Note1] | | 6 | | | 6 | | |
| Uplink-downlink configuration[Note2] | | 1 | | | 1 | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | 2x2 Low | | |
| OCNG Pattern defined in D.2.2 (TDD) and D.2.6 (TDD) | | OP.2 TDD | | | OP.6 TDD | | |
| $\rho_A$, $\rho_B$ | | -3 | | | -3 | | |
| PCFICH_RB | dB | 1 | | | Non-ABS and ABS subframe channel powers defined in Table C.3.1.2.2-1. | | |
| PDCCH_RA | dB | 1 | | | | | |
| PDCCH_RB | dB | 1 | | | | | |
| PBCH_RA | dB | -3 | | | | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA[Note3] | dB | | | | | | |
| OCNG_RB[Note3] | dB | | | | | | |
| SNR [Note 7,8] | dB | -0.4 | -4.5 | -13.3 | 4.8 | 4.8 | 5.2 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | -98 | | |
| Propagation condition | | ETU 30 Hz | | | ETU 30 Hz | | |

Note 1: For the special subframe configuration see table 4.2-1 in TS 36.211.
Note 2: For the uplink-downlink subframe configuration see table 4.2-2 in TS 36.211.
Note 3: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 5: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 6: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 7: SNR levels correspond to the signal quality, signal-to-interference-plus-noise ratio, on the CRS REs.
Note 8: The SNR in the restricted measurement subframes during time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in Figure 7.3.14.4-1.

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least in all uplink subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal no later than time point C (240 ms after the start of the time duration T3).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal.

- UE output power equal to or less than Transmit OFF power -48.5 dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.15 E-UTRAN FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)

### 7.3.15.1 Test purpose

To verify that the UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the serving cell.

### 7.3.15.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward. Applicability requires support for FGI bit 115.

### 7.3.15.3 Minimum conformance requirements

When the downlink radio link quality of the PCell estimated over the last 100 ms period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 100 ms $Q_{in}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in TS 36.213 [8] clause 4.2.1. Two successive indications from Layer 1 shall be separated by at least 10 ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in TS 36.331 [5] clause 5.3.11.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2.1 and A.7.3.15.

### 7.3.15.4 Test description

There are two E-UTRA cells in the test on the same RF channel. Cell 1 is the serving victim cell and Cell 2 is the neighbour aggressor cell. MBSFN ABS pattern is configured in the aggressor cell (Cell 2). The UE is configured with a time domain measurement restriction pattern for performing serving cell (Cell 1) measurements. The UE is also configured for periodic CQI reporting in PUCCH 1-0 mode.

The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.15.4-1 shows the variation of the downlink SNR in the serving cell (Cell 1) to emulate in-sync states.

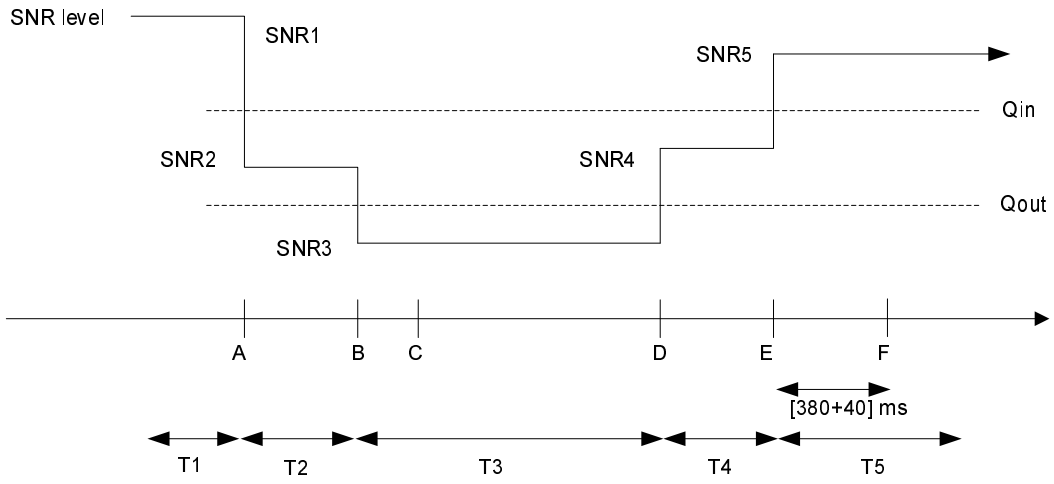


**Figure 7.3.15.4-1: SNR variation in the serving cell for in-sync testing under time domain measurement resource restriction with MBSFN ABS**

### 7.3.15.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), AWGN noise sources and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.40.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.15.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.15.4.3.
5. At this stage only Cell 1 is switched on. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1.

**Table 7.3.15.4.1-1: General test parameters for E-UTRAN FDD in-sync testing under time domain measurement resource restriction with MBSFN ABS (eICIC)**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | | R.9 FDD | As specified in section A.2.1. None of the PDCCH are intended for the UE under test. |
| Serving cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| Neighbour cell | | | Cell 2 | Cell 2 is the aggressor cell on E-UTRA RF channel number 1 |
| OCNG parameters for Cell 1 | | | OP.6 FDD | As specified in section D.1.6. |
| OCNG parameters for Cell 2 | | | OP.9 FDD | As specified in section D.1.9. |
| CP length | | | Normal | |
| Neighbour cell ABS configuration | | | MBSFN ABS | As defined in Table C.3.1.2.2-2 |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 10 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.521-1 [10] Annex B.2.3.2 |
| In sync transmission parameters (Note 1) | DCI format | | 1C | As defined in section 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 3 | In sync threshold Qin and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] section 7.6.1 and Table 7.6.1-2 respectively. |
| | Aggregation level | CCE | 4 | |
| | ρA, ρB | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | -3 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 3 | Out of sync threshold Qout and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] section 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | ρA, ρB | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| Physical cell ID PCI | | | $(PCI_{cell1}-PCI_{cell2}) \bmod 3 = 0$, PCIcell1 not equal to PCIcell2 | Cell IDs are chosen such that CRS from cells 1 and 2 overlap in frequency. |
| ABS pattern | | | 01000000100000001000000000010000000001000000 | FDD ABS Pattern Info IE is configured in Cell 2 as defined in section 9.2.54 in TS 36.423 [28]. The first/leftmost bit corresponds to the subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (40) divided by 10. All ABS subframes are MBSFN subframes. |
| Time domain measurement | | | 01000000100000001000 | MeasSubframePattern IE is |

| | | | |
|---|---|---|---|
| resource restriction pattern | | 0000010000001000000 | configured in UE for serving cell measurement as defined in section 6.3.6 in TS 36.331. |
| DRX | | OFF | |
| Layer 3 filtering | | Enabled | Counters:<br>N310 = 1; N311 = 1 |
| T310 timer | ms | 2000 | T310 is enabled |
| T311 timer | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | ms | 2 | Minimum CQI reporting periodicity |
| Time offset between cells | µs | 3 | |
| Propagation channel | | ETU30 | |
| T1 | s | 0.5 | |
| T2 | s | 0.4 | |
| T3 | s | 1.46 | |
| T4 | s | 0.4 | |
| T5 | s | 1 | |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | |

### 7.3.15.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to Cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

MBSFN ABS pattern is configured in the aggressor cell (Cell 2). The UE is configured by higher layers with a time domain measurement restriction pattern for performing serving cell (Cell 1) measurements. The patterns shall be configured prior to the start of T1.

1. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters for both Cell 1 and Cell 2 according to T1 in Table 7.3.15.5-1. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.

3. When T1 expires the SS shall change the SNR values to T2 as specified in Table 7.3.15.5-1. T2 starts.

4. When T2 expires the SS shall change the SNR values to T3 as specified in Table 7.3.15.5-1. T3 starts.

5. When T3 expires the SS shall change the SNR values to T4 as specified in Table 7.3.15.5-1. T4 starts.

6. When T4 expires the SS shall change the SNR values to T5 as specified in Table 7.3.15.5-1. T5 starts.

7. If the SS detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (420 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.

8. After T5 expires, repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.15.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.15.4.3-1: Common Exception messages for E-UTRAN FDD Radio Link Monitoring Test for in-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.15.4.3-2 : *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring Test for in-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.15.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring Test for in-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.15.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN FDD Radio Link Monitoring Test for in-sync**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

**Table 7.3.15.4.3-5: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional E-UTRAN FDD Radio Link Monitoring Test for in-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '01000000100000001000 00000010000001000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.15.4.3-6: *SystemInformationBlockType2*: Additional E-UTRAN FDD Radio Link Monitoring Test for in-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 SystemInformationBlockType2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfiguration | Not present | | Cell 1 |
| } | | | |

**Table 7.3.15.4.3-7: *SystemInformationBlockType3*: Additional E-UTRAN FDD Radio Link Monitoring Test for in-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1 |
| | '01'B (No MBSFN subframes are present in all neighbour cells) | | Cell 2 |

## 7.3.15.5 Test requirements

**Table 7.3.15.5-1: Cell specific test requirement parameters for E-UTRAN FDD for in-sync radio link monitoring under time domain measurement resource restriction with MBSFN ABS (eICIC)**

| Parameter | Unit | Cell 1 | | | | | Cell 2 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | 10 | | | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | | | 2x2 Low | | | | |
| OCNG Pattern defined in D.1.6 (FDD) and D.1.9 (FDD) | | OP.6 FDD | | | | | OP.9 FDD | | | | |
| $\rho_A$, $\rho_B$ | | -3 | | | | | -3 | | | | |
| PCFICH_RB | dB | 1 | | | | | Non-ABS and ABS subframe channel powers defined in Table C.3.1.2.2-2. | | | | |
| PDCCH_RA | dB | -3 | | | | | | | | | |
| PDCCH_RB | dB | -3 | | | | | | | | | |
| PBCH_RA | dB | -3 | | | | | | | | | |
| PBCH_RB | dB | | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | | |
| SNR Note 6 | dB | -0.4 | -4.5 | -13.3 | -8.2 | -0.4 | 4.8 | 4.8 | 5.2 | 5.2 | 4.8 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | | -98 | | | | |
| Propagation condition | | ETU30 | | | | | ETU30 | | | | |

Note 1: OCNG shall be used such that the resources in Cell 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2, T3, T4 and T5 of the active cell is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.15.4-1.

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (420 ms after the start of time duration T5) the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal.

The rate of correct events observed during repeated tests shall be at least 90%.

# 7.3.16 E-UTRAN TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)

## 7.3.16.1 Test purpose

To verify that the UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the serving cell under time domain measurement resource restriction.

## 7.3.16.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward. Applicability requires support for FGI bit 115.

## 7.3.16.3 Minimum conformance requirements

When the downlink radio link quality of the PCell estimated over the last 100 ms period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 100 ms $Q_{in}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in TS 36.213 [8] clause 4.2.1. Two successive indications from Layer 1 shall be separated by at least 10 ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in TS 36.331 [5] clause 5.3.11.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2.1 and A.7.3.16.

## 7.3.16.4 Test description

There are two E-UTRA cells in the test on the same RF channel. Cell 1 is the serving victim cell and Cell 2 is the neighbour aggressor cell. MBSFN ABS pattern is configured in the aggressor cell (Cell 2). The UE is configured with a time domain measurement restriction pattern for performing serving cell (Cell 1) measurements. The UE is also configured for periodic CQI reporting in PUCCH 1-0 mode.

The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.16.4-1 shows the variation of the downlink SNR in the serving cell (Cell 1) to emulate in-sync states.

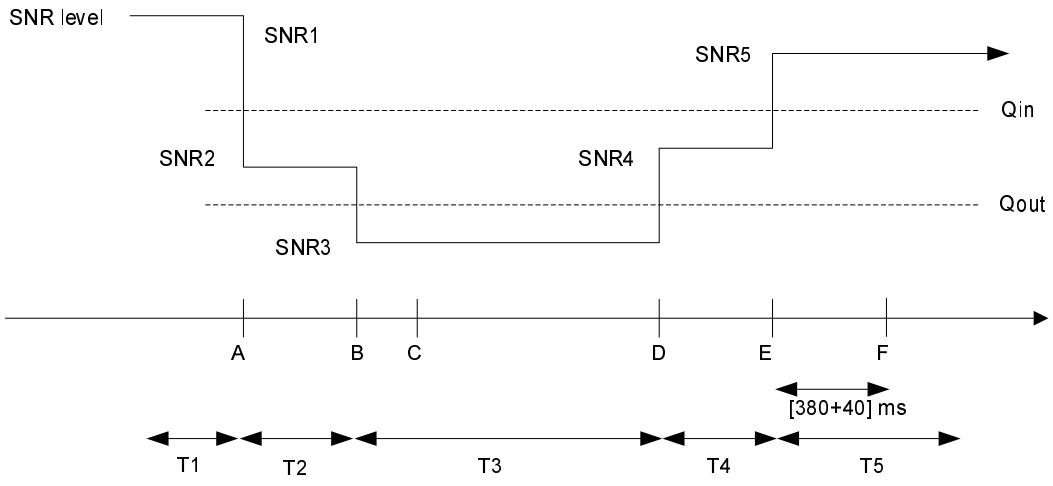


**Figure 7.3.16.4-1: SNR variation in the serving cell for in-sync testing under time domain measurement resource restriction with MBSFN ABS**

#### 7.3.16.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), AWGN noise sources and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.40.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.16.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.16.4.3.
5. At this stage only Cell 1 is switched on. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1.

**Table 7.3.16.4.1-1: General test parameters for E-UTRAN TDD in-sync testing under time domain measurement resource restriction with MBSFN ABS**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | | R.9 TDD | As specified in clause A.2.2. None of the PDCCH are intended for the UE under test |
| Serving cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| Neighbour cell | | | Cell 2 | Cell 2 is the aggressor cell on E-UTRA RF channel number 1 |
| Neighbour cell ABS configuration | | | MBSFN ABS | As defined in Table C.3.1.2.2-2 |
| OCNG parameters for Cell 1 | | | OP.2 TDD | As specified in clause D.2.2. |
| OCNG parameters for Cell 2 | | | OP.6 TDD | As specified in clause D.2.6. |
| CP length | | | Normal | |
| Neighbour cell ABS configuration | | | MBSFN ABS | |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 10 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.101 [5] Annex B.2.3.2 |
| In sync transmission parameters (Note 1) | DCI format | | 1C | As defined in clause 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 3 | In sync threshold Qin and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] clause 7.6.1 and Table 7.6.1-2 respectively. |
| | Aggregation level | CCE | 4 | |
| | ρA, ρB | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | -3 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 3 | Out of sync threshold Qout and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] clause 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | ρA, ρB | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| Physical cell ID PCI | | | ($PCI_{cell1}$-$PCI_{cell2}$ ) mod 3 = 0, PCIcell1 not equal to PCIcell2 | Cell IDs are chosen such that CRS from cells 1 and 2 overlap in frequency. |
| ABS pattern | | | 00001000000000100000 | TDD ABS Pattern Info IE is configured in Cell 2 as defined in clause 9.2.54 in TS 36.423 [28]. The first/leftmost bit corresponds to the subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (20) divided by 10. All ABS subframes are MBSFN subframes. |
| Time domain measurement | | | 00001000000000100000 | MeasSubframePattern IE is |

| | | | |
|---|---|---|---|
| resource restriction pattern | | | configured in UE for serving cell measurement as defined in clause 6.3.6 in TS 36.331. |
| DRX | | OFF | |
| Layer 3 filtering | | Enabled | Counters:<br>N310 = 1; N311 = 1 |
| T310 timer | ms | 2000 | T310 is enabled |
| T311 timer | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | ms | 1 | Minimum CQI reporting periodicity |
| Time offset between cells | µs | 3 | |
| Propagation channel | | ETU30 | |
| T1 | s | 0.5 | |
| T2 | s | 0.4 | |
| T3 | s | 1.46 | |
| T4 | s | 0.4 | |
| T5 | s | 1 | |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | |

### 7.3.16.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to Cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms.

MBSFN ABS pattern is configured in the aggressor cell (Cell 2). The UE is configured by higher layers with a time domain measurement restriction pattern for performing serving cell (Cell 1) measurements. The patterns shall be configured prior to the start of T1.

1. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters for both Cell 1 and Cell 2 according to T1 in Table 7.3.16.5-1. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.
3. When T1 expires the SS shall change the SNR values to T2 as specified in Table 7.3.16.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR values to T3 as specified in Table 7.3.16.5-1. T3 starts.
5. When T3 expires the SS shall change the SNR values to T4 as specified in Table 7.3.16.5-1. T4 starts.
6. When T4 expires the SS shall change the SNR values to T5 as specified in Table 7.3.16.5-1. T5 starts.
7. If the SS detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (420 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
8. After T5 expires, repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.16.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.16.4.3-1: Common Exception messages for E-UTRAN TDD in-sync testing under time domain measurement resource restriction (eICIC)**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.16.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD in-sync testing under time domain measurement resource restriction (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1C in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.16.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD in-sync testing under time domain measurement resource restriction (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.16.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN TDD in-sync testing under time domain measurement resource restriction (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

**Table 7.3.16.4.3-5: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN TDD in-sync testing under time domain measurement resource restriction (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 | | | |
| subframeConfig1-5-r10 | '00001000000000100000' | BIT STRING (SIZE (20)) | |
| | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.16.4.3-6: *SystemInformationBlockType2*: Additional E-UTRAN TDD Radio Link Monitoring Test for in-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| mbsfn-SubframeConfigList ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF SEQUENCE { | | | |
| radioframeAllocationPeriod | n1 | Every radio frame is with MBSFN subframe | Cell 2 |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| oneFrame | '010000'B | Subframe 4 is used for MBSFN | |
| } | | | |
| } | | | |

**Table 7.3.16.4.3-7: *SystemInformationBlockType3*: Additional E-UTRAN TDD Radio Link Monitoring Test for in-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1 |
| | '01'B (No MBSFN subframes are present in all neighbour cells) | | Cell 2 |

## 7.3.16.5 Test requirements

**Table 7.3.16.5-1: Cell specific test parameters for E-UTRAN TDD for in-sync radio link monitoring under time domain measurement resource restriction with MBSFN ABS**

| Parameter | Unit | Cell 1 | | | | | Cell 2 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | 10 | | | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | | | 2x2 Low | | | | |
| Special subframe configuration[Note1] | | 6 | | | | | 6 | | | | |
| Uplink-downlink configuration[Note2] | | 1 | | | | | 1 | | | | |
| OCNG Pattern defined in D.2.2 (TDD) and D.2.6 (TDD) | | OP.2 TDD | | | | | OP.6 TDD | | | | |
| $\rho_A$, $\rho_B$ | | -3 | | | | | -3 | | | | |
| PCFICH_RB | dB | 1 | | | | | Non-ABS and ABS subframe channel powers defined in Table C.3.1.2.2-2. | | | | |
| PDCCH_RA | dB | -3 | | | | | | | | | |
| PDCCH_RB | dB | -3 | | | | | | | | | |
| PBCH_RA | dB | -3 | | | | | | | | | |
| PBCH_RB | dB | | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | | |
| OCNG_RA[Note 1] | dB | | | | | | | | | | |
| OCNG_RB[Note 1] | dB | | | | | | | | | | |
| SNR [Note 8] | dB | -0.4 | -4.5 | -13.3 | -8.2 | -0.4 | 4.8 | 4.8 | 5.2 | 5.2 | 4.8 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | | -98 | | | | |
| Propagation condition | | ETU30 | | | | | ETU30 | | | | |

Note 1: For the special subframe configuration see table 4.2-1 in TS 36.211.
Note 2: For the uplink-downlink configuration see table 4.2-2 in TS 36.211.
Note 3: OCNG shall be used such that the resources in Cell 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 5: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 6: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 7: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 8: The SNR in time periods T1, T2, T3, T4 and T5 of the active cell is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.16.4-1.

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (420 ms after the start of time duration T5) the UE shall transmit uplink signal at least in all uplink subframes configured for CQI transmission according to the configured CQI mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.17 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non MBSFN ABS (feICIC)

### 7.3.17.1 Test purpose

To verify that the UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the PCell under time domain measurement resource restriction with CRS assistance information.

### 7.3.17.2 Test applicability

This test applies to all types of E-UTRA FDD and CRS interference handling UE release 11 and forward. Applicability requires support of FGI bit 115.

### 7.3.17.3 Minimum conformance requirements

When the downlink radio link quality of the PCell estimated over the last 200 ms period becomes worse than the threshold Qout, Layer 1 of the UE shall send an out-of-sync indication for the PCell to the higher layers within 200 ms Qout evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in TS 36.213 [8] clause 4.2.1. Two successive indications from Layer 1 shall be separated by at least 10 ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in TS 36.331 [5] clause 5.3.11.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2.1 and A.7.3.17.

### 7.3.17.4 Test description

There are three E-UTRA cells in the test on the same RF channel. There are three active cells in the test: Cell 1 is the PCell cell and Cell 2 and Cell 3 are the neighbour cells. Non-MBSFN ABS pattern is configured in both Cell 2 and Cell 3 in this test. The UE is configured by higher layers with a time domain measurement restriction pattern for performing PCell measurements. The UE is also configured for periodic CQI reporting in PUCCH 1-0 mode with periodicity of 2 ms.

The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.17.4-1 shows the variation of the downlink SNR in the serving cell (Cell 1) to emulate out-of-sync states.

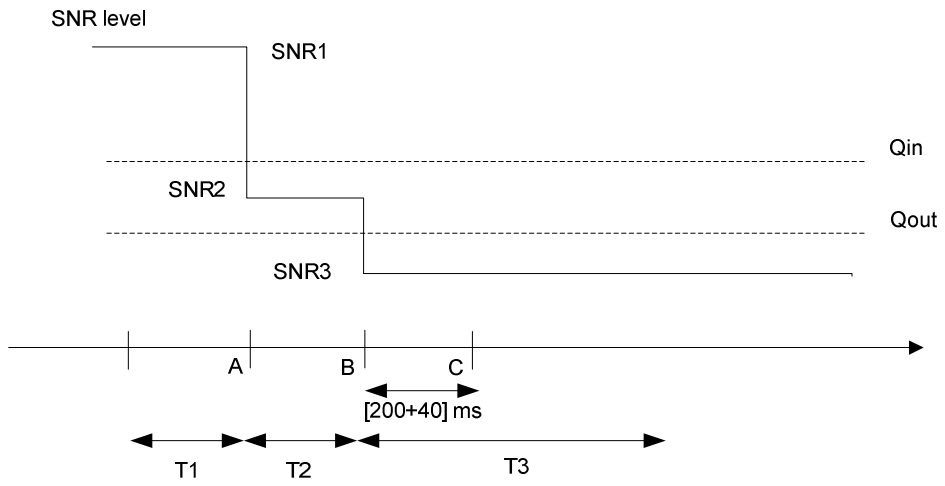


**Figure 7.3.17.4-1: SNR variation in the serving cell for out-of-sync testing under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

#### 7.3.17.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), AWGN noise sources and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.48.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.17.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.17.4.3
5. At this stage only Cell 1 is switched on. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1.

**Table 7.3.17.4.1-1: General test parameters for E-UTRAN FDD out-of-sync testing under time domain measurement resource restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC)**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | | R.7 FDD | As specified in clause A.2.1<br>None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.6 FDD | As specified in section D.1.6. |
| PCell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| Neighbor cells | | | Cell 2 and Cell 3 | Both of aggressor cells on E-UTRA RF channel number 1 |
| Neighbor cell ABS configuration | | | Non-MBSFN ABS | As defined in Table C.3.1.1.2-1 |
| CP length | | | Normal | |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.521-1 [10] Annex B.2.3.2 |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] clause 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX | | | OFF | |
| Layer 3 filtering | | | Enabled | Counters:: N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 2 | Minimum CQI reporting periodicity |
| Time offset between cells | | μs | Cell 2 time offset with respect to Cell 1: 3<br>Cell 3 time offset with respect to Cell 1: 2 | Three synchronous cells |
| Frequency shift between cells | | Hz | Cell 2 frequency shift with respect to Cell 1: 300<br>Cell 3 frequency shift with respect to Cell 1: -100 | |
| T1 | | s | 1 | |
| T2 | | s | 0.4 | |
| T3 | | s | 0.5 | |
| Physical cell IDs | | | $(PCI_{cell1}-PCI_{cell2})mod3 = 0$<br>$(PCI_{cell1}-PCI_{cell3})mod3!= 0$<br><br>$PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell PCIs are selected so that all conditions are met |
| ABS pattern | | | '10000000100000001000000010000000010000000' | FDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. The first/leftmost bit corresponds to the PCell subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (40) divided by 10. No MBSFN subframes are configured in the ABS subframes.<br>Configured in both Cell 2 and Cell 3 prior to the start of T1. |

| | | | | |
|---|---|---|---|---|
| Time domain measurement resource restriction pattern | | | '10000000100000001000 00001000000010000000' | Time domain measurement resource restriction pattern for serving cell measurement signalled to the UE in message measSubframePatternPCell-r10 as defined in TS 36.331, clause 6.3.2. |
| CRS assistance information | physCellId | | see PCI conditions above | The CRS assistance information is provided for Cell 2 and Cell 3 in *CRS-AssistanceInfo*. It includes a single *MBSFN-SubframeConfig* element with subframe allocation *oneFrame*='000000' |
| | antennaPortsCount | | an2 | |
| | mbsfn-SubframeConfigList | | *oneFrame* = '000000' | |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel | | | | |

### 7.3.17.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to Cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

Non-MBSFN ABS pattern is configured in both Cell 2 and Cell 3 in this test. The UE is configured by higher layers with a time domain measurement restriction pattern for performing PCell (Cell 1) measurements. The Non-MBSFN ABS pattern, the time domain measurement resource restriction pattern and the CRS assistance information shall be configured prior to the start of T1.

1. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3 with exceptions listed in 7.2A.6.

2. Set the parameters for Cell 1, Cell 2 and Cell 3 according to T1 in Table 7.3.17.5-1. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.

3. When T1 expires the SS shall change the SNR values to T2 as specified in Table 7.3.17.5-1. T2 starts.

4. When T2 expires the SS shall change the SNR values to T3 as specified in Table 7.3.17.5-1. T3 starts.

5. If the SS:
   a) detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B

   and

   b) does not detect any uplink power higher than -48.5 dBm from time point C (240 ms after the start of T3) until T3 expires,
   the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.

6. When T3 expires the SS shall change the SNR values to T1 as specified in Table 7.3.17.5-1.

7. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.

8. Repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.17.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 7.2A.6 with the following exceptions:

**Table 7.3.17.4.3-1: Common Exception messages for E-UTRAN FDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (feICIC)**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1<br>Table H.2.7-1<br>Table H.2.7-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.17.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.17.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.17.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN FDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

**Table 7.3.17.4.3-5: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN FDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '10000000100000001000 00001000000010000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| } | | | |
| NeighCellsCRS-Info-r11 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| CRS-AssistancedInfoList-r11 ::= SEQUENCE (SIZE (1..maxCellReport)) OF CRS-AssistanceInfo-r11 { | | 2 Entries | |
| CRS-AssistanceInfo-r11 :: = SEQUENCE { | | | |
| physCellId-r11 | ($PCI_{cell1}$-$PCI_{cell2}$) mod3 = 0<br><br>$PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell PCIs are selected so that both conditions are met for Cell2 | |
| antennaPortsCount-r11 | an2 | | |
| mbsfn-SubframeConfigList-r11 ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF MBSFN-SubframeConfig { | | | |
| MBSFN-SubframeConfig:: = SEQUENCE { | | | |
| radioframeAllocationPeriod | n1 | | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| oneFrame | '000000' | The CRS assistance information is provided for Cell2 in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation oneFrame='000000' | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| CRS-AssistanceInfo-r11 :: = SEQUENCE { | | | |
| physCellId-r11 | ($PCI_{cell1}$-$PCI_{cell3}$) mod3! = 0 | Cell PCIs are selected so that both conditions are met for Cell3 | |
| antennaPortsCount-r11 | an2 | | |
| mbsfn-SubframeConfigList-r11 ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF MBSFN-SubframeConfig { | | | |
| MBSFN-SubframeConfig:: = SEQUENCE { | | | |
| radioframeAllocationPeriod | n1 | | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| oneFrame | '000000' | The CRS assistance information is provided for for | |

| | | | |
|---|---|---|---|
| | | Cell3 in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation oneFrame='000000' | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.17.4.3-6: *SystemInformationBlockType1 :* Additional E-UTRAN FDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.2 Table 4.4.3.2-3 SystemInformationBlockType1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType1 ::= SEQUENCE { | | | |
| cellSelectionInfo SEQUENCE { | | | |
| q-RxLevMin | -35 (-70 dBm) | | Cell 2, Cell 3 |
| } | | | |

### 7.3.17.5 Test requirements

**Table 7.3.17.5-1: Cell specific test parameters for E-UTRAN FDD for out-of-sync radio link monitoring under time domain measurement resource restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC)**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | | 10 | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | 2x2 Low | | | 2x2 Low | | |
| OCNG Pattern defined in D.1.6 (FDD) | | OP.6 FDD | | | OP.6 FDD | | | OP.6 FDD | | |
| $\rho_A$, $\rho_B$ | | -3 | | | -3 | | | -3 | | |
| PCFICH_RB | dB | 1 | | | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.2-1. | | | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.2-1 | | |
| PDCCH_RA | dB | 1 | | | | | | | | |
| PDCCH_RB | dB | 1 | | | | | | | | |
| PBCH_RA | dB | -3 | | | | | | | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA[Note1] | dB | | | | | | | | | |
| OCNG_RB[Note1] | dB | | | | | | | | | |
| SNR [Note 6] | dB | -0.6 | -4.3 | -14.6 | 3.8 | 3.8 | 4.2 | 1.8 | 1.8 | 2.2 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | -98 | | | -98 | | |
| Propagation condition | | ETU 30 Hz | | | ETU 30 Hz | | | ETU 30 Hz | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal quality, signal-to-interference-plus-noise ratio, on the CRS REs.
Note 6: The SNR in the restricted measurement subframes during time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.17.4-1.

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal no later than time point C (240 ms after the start of time duration T3).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal

- UE output power equal to or less than Transmit OFF power -48.5 dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.18 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non MBSFN ABS (feICIC)

### 7.3.18.1 Test purpose

To verify that the UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the PCell under time domain measurement resource restriction with CRS assistance information.

### 7.3.18.2 Test applicability

This test applies to all types of E-UTRA TDD and CRS interference handling UE release 11 and forward. Applicability requires support of FGI bit 115.

### 7.3.18.3 Minimum conformance requirements

When the downlink radio link quality of the PCell estimated over the last 200 ms period becomes worse than the threshold Qout, Layer 1 of the UE shall send an out-of-sync indication for the PCell to the higher layers within 200 ms Qout evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in TS 36.213 [8] clause 4.2.1. Two successive indications from Layer 1 shall be separated by at least 10 ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in TS 36.331 [5] clause 5.3.11.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2.1 and A.7.3.18.

### 7.3.18.4 Test description

There are three E-UTRA cells in the test on the same RF channel. There are three active cells in the test: Cell 1 is the PCell cell and Cell 2 and Cell 3 are the neighbour cells. Non-MBSFN ABS pattern is configured in both Cell 2 and Cell 3 in this test. The UE is configured by higher layers with a time domain measurement restriction pattern for performing PCell measurements. The UE is also configured for periodic CQI reporting in PUCCH 1-0 mode with periodicity of 2 ms. The non-MBSFN ABS pattern, the time domain measurement resource restriction pattern and the CRS assistance information shall be configured prior to the start of T1.

The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.18.4-1 shows the variation of the downlink SNR in the serving cell (Cell 1) to emulate out-of-sync states.

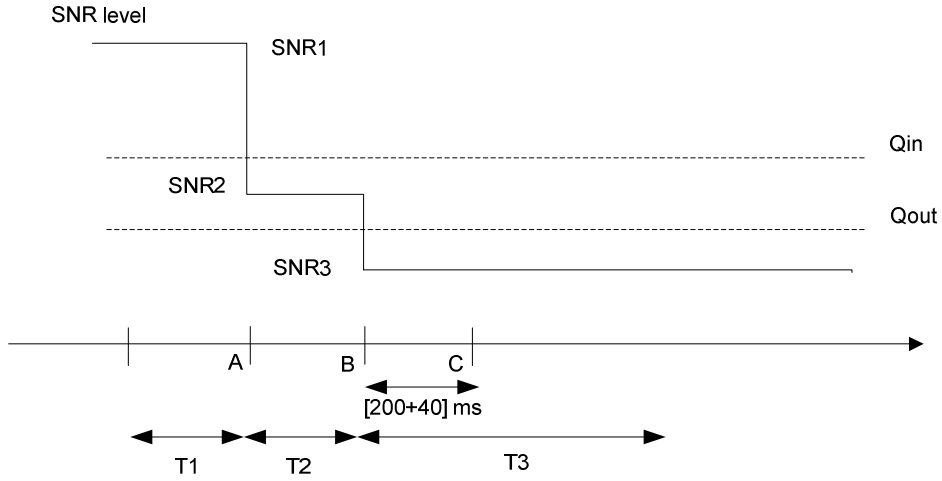


**Figure 7.3.18.4-1:SNR variation in the serving cell for out-of-sync testing under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

### 7.3.18.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), AWGN noise sources and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.48.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.18.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.18.4.3.
5. At this stage only Cell 1 is switched on. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1.

**Table 7.3.18.4.1-1: General test parameters for E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC)**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | | R.7.TDD | As specified in clause A.2.1<br>None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.2 TDD | As specified in clause D.2.2 |
| PCell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| Neighbor cells | | | Cell 2 and Cell 3 | Both of aggressor cells on E-UTRA RF channel number 1 |
| Neighbor cell ABS configuration | | | Non-MBSFN ABS | As defined in Table C.3.1.1.2-1 |
| CP length | | | Normal | |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.521-1 [10] Annex B.2.3.2 |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] clause 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX | | | OFF | |
| Layer 3 filtering | | | Enabled | Counters:: N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 1 | Minimum CQI reporting periodicity |
| Time offset between cells | | μs | Cell 2 time offset with respect to Cell 1: 3<br>Cell 3 time offset with respect to Cell 1: 2 | Three synchronous cells |
| Frequency shift between cells | | Hz | Cell 2 frequency shift with respect to Cell 1: 300<br>Cell 3 frequency shift with respect to Cell 1: -100 | |
| T1 | | s | 1 | |
| T2 | | s | 0.4 | |
| T3 | | s | 0.5 | |
| Physical cell IDs | | | $(PCI_{cell1}-PCI_{cell2})mod3 = 0$<br>$(PCI_{cell1}-PCI_{cell3})mod3!= 0$<br><br>$PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell PCIs are selected so that all conditions are met |
| ABS pattern | | | '00001000000000100000' | TDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. The first/leftmost bit corresponds to the PCell subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (20) divided by 10. No MBSFN subframes are configured in the ABS subframes.<br>Configured in both Cell 2 and Cell 3 prior to the start of T1. |

| | | | | |
|---|---|---|---|---|
| Time domain measurement resource restriction pattern | | | '0000100000000000100000' | Time domain measurement resource restriction pattern for serving cell measurement signalled to the UE in message measSubframePatternPCell-r10 as defined in TS 36.331, clause 6.3.2. |
| CRS assistance information | physCellId | | see PCI conditions above | The CRS assistance information is provided for Cell 2 only in *CRS-AssistanceInfo*. It includes a single *MBSFN-SubframeConfig* element with subframe allocation *oneFrame*='000000' |
| | antennaPortsCount | | an2 | |
| | mbsfn-SubframeConfigList | | *oneFrame* = '000000' | |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel | | | | |

### 7.3.18.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to Cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

Non-MBSFN ABS pattern is configured in both Cell 2 and Cell 3 in this test. The UE is configured by higher layers with a time domain measurement restriction pattern for performing PCell (Cell 1) measurements. The Non-MBSFN ABS pattern, the time domain measurement resource restriction pattern and the CRS assistance information shall be configured prior to the start of T1.

1. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3 with exceptions listed in 7.2A.6.
2. Set the parameters for Cell 1, Cell 2 and Cell 3 according to T1 in Table 7.3.18.5-1. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.
3. When T1 expires the SS shall change the SNR values to T2 as specified in Table 7.3.18.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR values to T3 as specified in Table 7.3.18.5-1. T3 starts.
5. If the SS:

   a) detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B

   and

   b) does not detect any uplink power higher than -48.5 dBm from time point C (240 ms after the start of T3) until T3 expires,

   the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
6. When T3 expires the SS shall change the SNR values to T1 as specified in Table 7.3.18.5-1.
7. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.
8. Repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.18.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 7.2A.6 with the following exceptions:

**Table 7.3.18.4.3-1: Common Exception messages for E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (feICIC)**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1<br>Table H.2.7-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.18.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1C in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.18.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.18.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

**Table 7.3.18.4.3-5: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 | | | |
| subframeConfig1-5-r10 | '00001000000000100000' | BIT STRING (SIZE (20)) | |
| | | | |
| } | | | |
| } | | | |
| NeighCellsCRS-Info-r11 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| CRS-AssistancedInfoList-r11 ::= SEQUENCE (SIZE (1..maxCellReport)) OF CRS-AssistanceInfo-r11 { | | 2 Entries | |
| CRS-AssistanceInfo-r11 :: = SEQUENCE { | | | |
| physCellId-r11 | ($PCI_{cell1}$-$PCI_{cell2}$) mod3 = 0 $PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell PCIs are selected so that both conditions are met for Cell2 | |
| antennaPortsCount-r11 | an2 | | |
| mbsfn-SubframeConfigList-r11 ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF MBSFN-SubframeConfig { | | | |
| MBSFN-SubframeConfig:: = SEQUENCE { | | | |
| radioframeAllocationPeriod | n1 | | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| oneFrame | '000000' | The CRS assistance information is provided for Cell2 in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation oneFrame='000000' | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| CRS-AssistanceInfo-r11 :: = SEQUENCE { | | | |
| physCellId-r11 | ($PCI_{cell1}$-$PCI_{cell3}$) mod3! = 0 | Cell PCIs are selected so that both conditions are met for Cell3 | |
| antennaPortsCount-r11 | an2 | | |
| mbsfn-SubframeConfigList-r11 ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF MBSFN-SubframeConfig { | | | |
| MBSFN-SubframeConfig:: = SEQUENCE { | | | |
| radioframeAllocationPeriod | n1 | | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| oneFrame | '000000' | The CRS assistance information is | |

| | | | |
|---|---|---|---|
| | | provided for for Cell3 in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation oneFrame='000000' | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.18.4.3-6: *SystemInformationBlockType1 :* Additional E-UTRAN TDD out-of-sync testing under time domain measurement resource restriction with non-MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.2 Table 4.4.3.2-3 SystemInformationBlockType1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType1 ::= SEQUENCE { | | | |
| cellSelectionInfo SEQUENCE { | | | |
| q-RxLevMin | -35 (-70 dBm) | | Cell 2, Cell 3 |
| } | | | |

## 7.3.18.5 Test requirements

**Table 7.3.18.5-1: Cell specific test requirement parameters for E-UTRAN TDD for out-of-sync radio link monitoring under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS (feICIC)**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | | 10 | | |
| Special subframe configuration[Note1] | | 6 | | | 6 | | | 6 | | |
| Uplink-downlink configuration[Note2] | | 1 | | | 1 | | | 1 | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | 2x2 Low | | | 2x2 Low | | |
| OCNG Pattern defined in D.2.2 (TDD) | | OP.2 TDD | | | OP.2 TDD | | | OP.2 TDD | | |
| $\rho_A$, $\rho_B$ | | -3 | | | -3 | | | -3 | | |
| PCFICH_RB | dB | 1 | | | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.2-1. | | | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.2-1. | | |
| PDCCH_RA | dB | 1 | | | | | | | | |
| PDCCH_RB | dB | 1 | | | | | | | | |
| PBCH_RA | dB | -3 | | | | | | | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA[Note1] | dB | | | | | | | | | |
| OCNG_RB[Note1] | dB | | | | | | | | | |
| SNR [Note 6] | dB | -0.6 | -4.3 | -14.6 | 3.8 | 3.8 | 4.2 | 1.8 | 1.8 | 2.2 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | -98 | | | -98 | | |
| Propagation condition | | ETU 30 Hz | | | ETU 30 Hz | | | ETU 30 Hz | | |

Note 1: For the special subframe configuration see table 4.2-1 in TS 36.211.
Note 2: For the uplink-downlink subframe configuration see table 4.2-2 in TS 36.211.
Note 3: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 5: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 6: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 7: SNR levels correspond to the signal quality, signal-to-interference-plus-noise ratio, on the CRS REs.
Note 8: The SNR in the restricted measurement subframes during time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in Figure 7.3.18.4-1.

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least in all uplink subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal no later than time point C (240 ms after the start of the time duration T3).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal

- UE output power equal to or less than Transmit OFF power -48.5 dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.19 E-UTRAN FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non MBSFN ABS (feICIC)

### 7.3.19.1 Test purpose

To verify that the UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the PCell under time domain measurement resource restriction with CRS assistance information.

### 7.3.19.2 Test applicability

This test applies to all types of E-UTRA FDD and CRS interference handling UE release 11 and forward. Applicability requires support of FGI bit 115.

### 7.3.19.3 Minimum conformance requirements

When the downlink radio link quality of the PCell estimated over the last 100 ms period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 100 ms $Q_{in}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in TS 36.213 [8] clause 4.2.1. Two successive indications from Layer 1 shall be separated by at least 10 ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in TS 36.331 [5] clause 5.3.11.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2.1 and A.7.3.19.

### 7.3.19.4 Test description

There are three E-UTRA cells in the test on the same RF channel. There are three active cells in the test: Cell 1 is the PCell cell and Cell 2 and Cell 3 are the neighbour cells. Non-MBSFN ABS pattern is configured in both Cell 2 and Cell 3 in this test. The UE is configured by higher layers with a time domain measurement restriction pattern for performing PCell measurements. The UE is also configured for periodic CQI reporting in PUCCH 1-0 mode with periodicity of 2 ms.

The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.19.4-1 shows the variation of the downlink SNR in the active cell (Cell 1) to emulate in-sync state. Prior to the start of the time duration T1, the UE shall be fully synchronized to Cell 1. The Non-MBSFN ABS pattern, the time domain measurement resource restriction pattern and the CRS assistance information shall be configured prior to the start of T1.

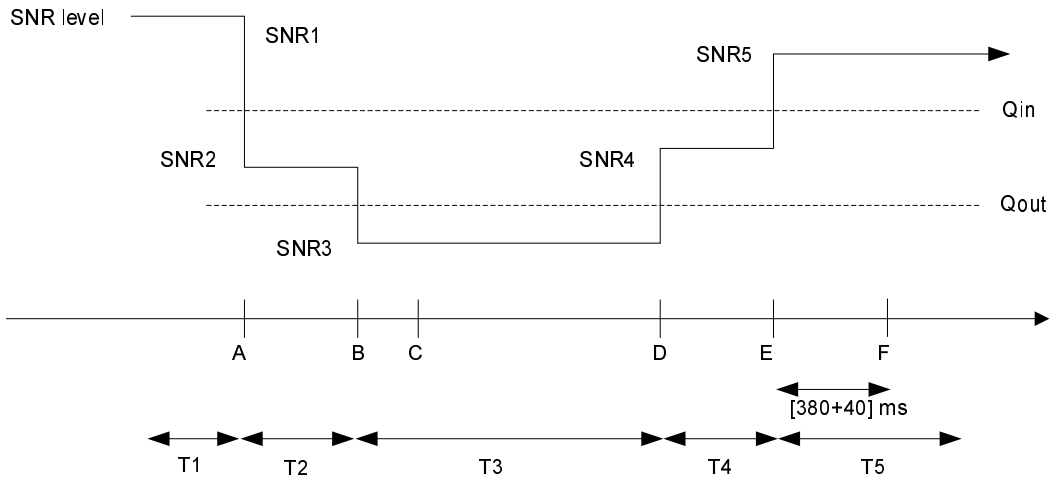


**Figure 7.3.19.4-1: SNR variation in the serving cell for in-sync testing under time domain measurement resource restriction with non-MBSFN ABS**

### 7.3.19.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), AWGN noise sources and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.48.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.19.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.19.4.3
5. At this stage only Cell 1 is switched on. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1.

**Table 7.3.19.4.1-1: General test parameters for E-UTRAN FDD in-sync radio link monitoring test (feICIC)**

| Parameter | | Unit | Value | | | Comment |
|---|---|---|---|---|---|---|
| | | | Test 1 | | | |
| | | | Cell 1 | Cell 2 | Cell 3 | |
| PCFICH/PDCCH/PHICH parameters | | | R.9 FDD | R.9 FDD | R.9 FDD | As specified in section A.2.1. None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.6 FDD | OP.6 FDD | OP.6 FDD | As specified in section D.1.6 |
| Active cell | | | PCell | Neighbor Cell | Neighbor Cell | Cell 1, Cell 2 and Cell 3 are on E-UTRA RF channel number 1 |
| CP length | | | Normal | Normal | Normal | |
| E-UTRA RF Channel Number | | | 1 | 1 | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | 10 | 10 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | 2x2 Low | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.521-1 [10] Annex B.2.3.2 |
| Neighbor Cell ABS configuration | | | N/A | Non-MBSFN ABS | | As defined in Table C.3.1.1.2-2. |
| ABS Pattern | | | N/A | 1000000010 0000001000 0000100000 0010000000 | 1000000010 0000001000 0000100000 0010000000 | FDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. The first/leftmost bit corresponds to the PCell subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (40) divided by 10. No MBSFN subframes are configured in the ABS subframes. Configured in both Cell 2 and Cell 3 prior to the start of T1. |
| Time domain measurement resource restriction pattern | | | '1000000010000000100000000010000000010000000' | N/A | N/A | Time domain measurement resource restriction pattern for serving cell measurement signalled to the UE in message measSubframePatternPCell-r10 as defined in TS 36.331, clause 6.3.2. |
| CRS assistant information | physCellId | | N/A | see PCI conditions below | see PCI conditions below | The CRS assistance information is provided for Cell 2 and Cell 3 in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation oneFrame='000000' |
| | antennaPortsCount | | | an2 | an2 | |
| | mbsfn-SubframeConfigList | | | oneFrame = '000000' | oneFrame = '000000' | |
| Time offset between cells (With respect to Cell 1) | | us | 0 | 3 | 2 | |
| Frequency shift between cells (With respect to Cell 1) | | Hz | 0 | 300 | -100 | |
| Physical Cell ID | | | $PCI_{cell1}$ | ($PCI_{cell1}$-$PCI_{cell2}$) mod3 = 0 | ($PCI_{cell1}$-$PCI_{cell3}$) mod3! = 0 | Cell PCIs are selected so that all conditions are met |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | $PCI_{cell1}$ not equal to $PCI_{cell2}$ | | |
| In sync transmission parameters (Note 1) | | DCI format | 1C | 1C | 1C | |
| In sync transmission parameters (Note 1) | Number of Control OFDM symbols | | 3 | 3 | 3 | As defined in section 5.3.3.1.4 in TS 36.212 |
| | Aggregation level | CCE | 4 | 4 | 4 | In sync threshold $Q_{in}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] section 7.6.1 and Table 7.6.1-2 respectively. |
| | $\rho_A$, $\rho_B$ | | -3 | -3 | -3 | |
| | Ratio of PDCCH to RS EPRE | | -3 | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.2-2. | | |
| | Ratio of PCFICH to RS EPRE | | 1 | | | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | 1A | 1A | |
| | Number of Control OFDM symbols | | 3 | 3 | 3 | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Aggregation level | CCE | 8 | 8 | 8 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] section 7.6.1 and Table 7.6.1-1 respectively. |
| | $\rho_A$, $\rho_B$ | | -3 | -3 | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.2-2. | | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | | | |
| DRX | | | OFF | OFF | OFF | |
| Layer 3 filtering | | | Enabled | Disable | Disable | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | N/A | | T310 is enabled |
| T311 timer | | ms | 1000 | | | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | | | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 2 | | | Minimum CQI reporting periodicity |
| T1 | | s | 0.5 | N/A | | |
| T2 | | s | 0.4 | | | |
| T3 | | s | 1.46 | | | |
| T4 | | s | 0.4 | | | |
| T5 | | s | 1 | | | |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | | | |

### 7.3.19.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to Cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

Non-MBSFN ABS pattern is configured in both Cell 2 and Cell 3 in this test. The UE is configured by higher layers with a time domain measurement restriction pattern for performing PCell (Cell 1) measurements. The Non-MBSFN ABS pattern, the time domain measurement resource restriction pattern and the CRS assistance information shall be configured prior to the start of T1.

1. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3 with exceptions listed in 7.2A.6.

2. Set the parameters for Cell 1, Cell 2, and Cell 3 according to T1 in Table 7.3.19.5-1. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.
3. When T1 expires the SS shall change the SNR values to T2 as specified in Table 7.3.19.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR values to T3 as specified in Table 7.3.19.5-1. T3 starts.
5. When T3 expires the SS shall change the SNR values to T4 as specified in Table 7.3.19.5-1. T4 starts.
6. When T4 expires the SS shall change the SNR values to T5 as specified in Table 7.3.19.5-1. T5 starts.
7. If the SS detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (420 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
8. After T5 expires, repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.19.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 7.2A.6 with the following exceptions:

**Table 7.3.19.4.3-1: Common Exception messages for E-UTRAN FDD in-sync testing under time domain measurement resource restriction (feICIC)**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2<br>Table H.2.7-1<br>Table H.2.7-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.19.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FDD in-sync testing under time domain measurement resource restriction (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.19.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FDD in-sync testing under time domain measurement resource restriction (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.19.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN FDD in-sync testing under time domain measurement resource restriction (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

**Table 7.3.19.4.3-5: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN FDD in-sync testing under time domain measurement resource restriction (feICIC)**

| Derivation Path: TS 36.508 [7] clause 5.2A.5 Table 5.2A.5.1.1-2 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '10000000100000001000 00001000000010000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| } | | | |
| NeighCellsCRS-Info-r11 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| CRS-AssistancedInfoList-r11 ::= SEQUENCE (SIZE (1..maxCellReport)) OF CRS-AssistanceInfo-r11 { | | 2 Entries | |
| CRS-AssistancedInfo-r11 ::= SEQUENCE { | | | |
| physCellId-r11 | ($PCI_{cell1}$-$PCI_{cell2}$) mod3 = 0<br><br>$PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell PCIs are selected so that both conditions are met for Cell2 | |
| antennaPortsCount-r11 | an2 | | |
| mbsfn-SubframeConfigList-r11 ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF MBSFN-SubframeConfig { | | | |
| MBSFN-SubframeConfig:: = SEQUENCE { | | | |
| radioframeAllocationPeriod | n1 | | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| oneFrame | '000000' | The CRS assistance information is provided for Cell2 in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation oneFrame='000000' | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| CRS-AssistanceInfo-r11 :: = SEQUENCE { | | | |
| physCellId-r11 | ($PCI_{cell1}$-$PCI_{cell3}$) mod3! = 0 | Cell PCIs are selected so that both conditions are met for Cell3 | |
| antennaPortsCount-r11 | an2 | | |
| mbsfn-SubframeConfigList-r11 ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF MBSFN-SubframeConfig { | | | |
| MBSFN-SubframeConfig:: = SEQUENCE { | | | |
| radioframeAllocationPeriod | n1 | | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| oneFrame | '000000' | The CRS assistance information is | |

| | | | |
|---|---|---|---|
| | | provided for Cell 3 in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation oneFrame='000000' | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 7.3.19.5 Test requirements

**Table 7.3.19.5-1: Cell specific test requirement parameters for E-UTRAN FDD for in-sync radio link monitoring (feICIC)**

| Parameter | Unit | Test 1 | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Cell1 | | | | | Cell2 | | | | | Cell3 | | | | |
| | | T1 | T2 | T3 | T4 | T5 | T1 | T2 | T3 | T4 | T5 | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | | 1 | | | | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | 10 | | | | | 10 | | | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | | | 2x2 Low | | | | | 2x2 Low | | | | |
| PCFICH/PDCCH/ PHICH parameters | | R.9 FDD | | | | | R.9 FDD | | | | | R.9 FDD | | | | |
| OCNG Pattern defined in D.1.6 (FDD) | | OP.6 FDD | | | | | OP.6 FDD | | | | | OP.6 FDD | | | | |
| $\rho_A$, $\rho_B$ | | -3 | | | | | -3 | | | | | -3 | | | | |
| PCFICH_RB | dB | 1 | | | | | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.2-2. | | | | | | | | | |
| PDCCH_RA | dB | -3 | | | | | | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | | | | | | | |
| PBCH_RA | dB | | | | | | | | | | | | | | | |
| PBCH_RB | dB | | | | | | | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | | | | | | | |
| SNR Note 6 | dB | -0.6 | -4.3 | -14.6 | -9.5 | -0.6 | 3.8 | 3.8 | 4.2 | 4.2 | 3.8 | 1.8 | 1.8 | 2.2 | 2.2 | 1.8 |
| $N_{oc}$ | dBm/ 15 kHz | -98 | | | | | -98 | | | | | -98 | | | | |
| Propagation condition | Hz | ETU 30 | | | | | ETU 30 | | | | | ETU 30 | | | | |

Note 1: OCNG shall be used such that the resources in cell 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.19.4-1.

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (420 ms after the start of time duration T5) the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.20 E-UTRAN TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non MBSFN ABS (feICIC)

### 7.3.20.1 Test purpose

To verify that the UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the Pcell under time domain measurement resource restriction with CRS assistance information.

### 7.3.20.2 Test applicability

This test applies to all types of E-UTRA TDD and CRS interference handling UE release 11 and forward. Applicability requires support of FGI bit 115.

### 7.3.20.3 Minimum conformance requirements

When the downlink radio link quality of the PCell estimated over the last 100 ms period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 100 ms $Q_{in}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in TS 36.213 [8] clause 4.2.1. Two successive indications from Layer 1 shall be separated by at least 10 ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in TS 36.331 [5] clause 5.3.11.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2.1 and A.7.3.20.

### 7.3.20.4 Test description

There are three E-UTRA cells in the test on the same RF channel. There are three active cells in the test: Cell 1 is the PCell cell and Cell 2 and Cell 3 are the neighbour cells. Non-MBSFN ABS pattern is configured in both Cell 2 and Cell 3 in this test. The UE is configured by higher layers with a time domain measurement restriction pattern for performing PCell measurements. The UE is also configured for periodic CQI reporting in PUCCH 1-0 mode with periodicity of 2 ms.

The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.20.4-1 shows the variation of the downlink SNR in the active cell (Cell 1) to emulate in-sync state. Prior to the start of the time duration T1, the UE shall be fully synchronized to Cell 1. The Non-MBSFN ABS pattern, the time domain measurement resource restriction pattern and the CRS assistance information shall be configured prior to the start of T1.

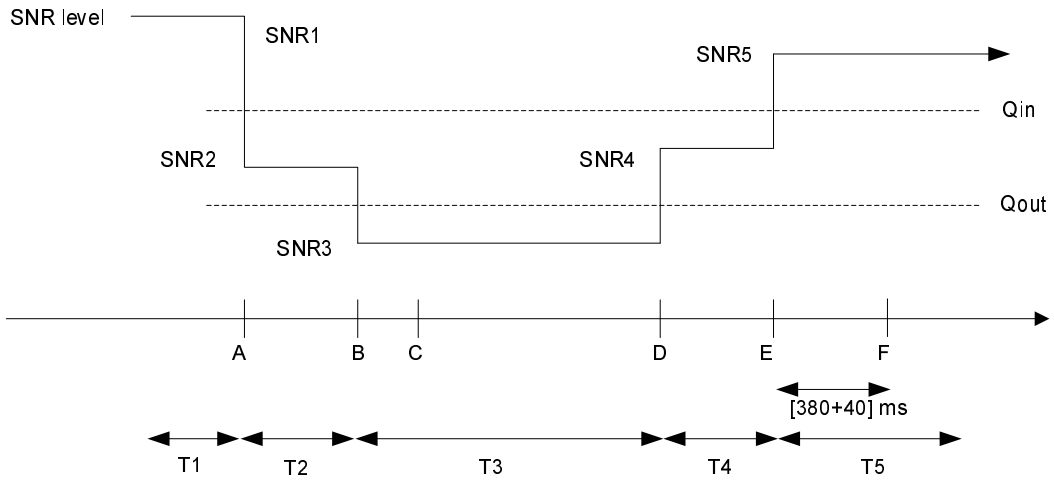


**Figure 7.3.20.4-1: SNR variation in the serving cell for in-sync testing under time domain measurement resource restriction with non-MBSFN ABS**

### 7.3.20.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), AWGN noise sources and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.48.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.20.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.20.4.3
5. At this stage only Cell 1 is switched on. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1.

**Table 7.3.20.4.1-1: General test parameters for E-UTRAN TDD in-sync radio link monitoring test (feICIC)**

| Parameter | | Unit | Value | | | Comment |
|---|---|---|---|---|---|---|
| | | | Test 1 | | | |
| | | | Cell 1 | Cell 2 | Cell 3 | |
| PCFICH/PDCCH/PHICH parameters | | | R.9 TDD | R.9 TDD | R.9 TDD | As specified in section A.2.2. None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.2 TDD | OP.2 TDD | OP.2 TDD | As specified in section D.2.2. |
| Active cell | | | PCell | Neighbor Cell | Neighbor Cell | Cell 1, Cell 2 and Cell 3 are on E-UTRA RF channel number 1 |
| CP length | | | Normal | Normal | Normal | |
| E-UTRA RF Channel Number | | | 1 | 1 | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | 10 | 10 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | 2x2 Low | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.521-1 [10] Annex B.2.3.2 |
| Neighbor Cell ABS configuration | | | N/A | Non-MBSFN ABS | | As defined in Table C.3.1.1.2-2 |
| ABS Pattern | | | N/A | '00001000000000100000' | '00001000000000100000' | TDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 2 and Cell 3. The first/leftmost bit corresponds to the PCell subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (20) divided by 10. No MBSFN subframes are configured in the ABS subframes. |
| Time domain measurement resource restriction pattern | | | '00001000000000100000' | N/A | N/A | Time domain measurement resource restriction pattern for serving cell measurement signalled to the UE in message measSubframePatternPCell-r10 as defined in TS 36.331, clause 6.3.2. |
| CRS assistant information | physCellId | | N/A | see PCI conditions below | see PCI conditions below | The CRS assistance information is provided for Cell 2 and Cell 3 in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation oneFrame='000000' |
| | antennaPortsCount | | | an2 | an2 | |
| | mbsfn-SubframeConfigList | | | oneFrame = '000000' | oneFrame = '000000' | |
| Time offset from Cell 1 | | us | 0 | 3 | 2 | |
| Frequency offset | | Hz | 0 | 300 | -100 | |
| Physical Cell ID | | | $PCI_{cell1}$ | ($PCI_{cell1}$-$PCI_{cell2}$) mod3 = 0<br><br>$PCI_{cell1}$ not equal to | ($PCI_{cell1}$-$PCI_{cell3}$) mod3 != 0 | Cell PCIs are selected so that all conditions are met |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | $PCI_{cell2}$ | | |
| In sync transmission parameters (Note 1) | DCI format | | 1C | 1C | 1C | As defined in section 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 3 | 3 | 3 | In sync threshold $Q_{in}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] section 7.6.1 and Table 7.6.1-2 respectively. |
| | Aggregation level | CCE | 4 | 4 | 4 | |
| | $\rho_A$, $\rho_B$ | | -3 | -3 | -3 | |
| | Ratio of PDCCH to RS EPRE | | -3 | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.2-2. | | |
| | Ratio of PCFICH to RS EPRE | | 1 | | | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | 1A | 1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 3 | 3 | 3 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] section 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | 8 | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | -3 | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.2-2. | | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | | | |
| DRX | | | OFF | OFF | OFF | |
| Layer 3 filtering | | | Enabled | Disable | Disable | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | N/A | | T310 is enabled |
| T311 timer | | ms | 1000 | | | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | | | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 1 | | | Minimum CQI reporting periodicity |
| T1 | | s | 0.5 | N/A | | |
| T2 | | s | 0.4 | | | |
| T3 | | s | 1.46 | | | |
| T4 | | s | 0.4 | | | |
| T5 | | s | 1 | | | |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | | | |

### 7.3.20.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to Cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

Non-MBSFN ABS pattern is configured in both Cell 2 and Cell 3 in this test. The UE is configured by higher layers with a time domain measurement restriction pattern for performing PCell (Cell 1) measurements. The Non-MBSFN ABS pattern, the time domain measurement resource restriction pattern and the CRS assistance information shall be configured prior to the start of T1.

1. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters for Cell 1, Cell 2, and Cell 3 according to T1 in Table 7.3.20.5-1. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.

3. When T1 expires the SS shall change the SNR values to T2 as specified in Table 7.3.20.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR values to T3 as specified in Table 7.3.20.5-1. T3 starts.
5. When T3 expires the SS shall change the SNR values to T4 as specified in Table 7.3.20.5-1. T4 starts.
6. When T4 expires the SS shall change the SNR values to T5 as specified in Table 7.3.20.5-1. T5 starts.
7. If the SS detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (420 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
8. After T5 expires, repeat steps 2-10 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.20.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 7.2A.6 with the following exceptions:

**Table 7.3.20.4.3-1: Common Exception messages for E-UTRAN TDD in-sync testing under time domain measurement resource restriction (feICIC)**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.20.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD in-sync testing under time domain measurement resource restriction (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1C in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.20.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD in-sync testing under time domain measurement resource restriction (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.20.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN TDD in-sync testing under time domain measurement resource restriction (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

**Table 7.3.20.4.3-5: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN TDD in-sync testing under time domain measurement resource restriction (feICIC)**

| Derivation Path: TS 36.508 [7] clause 5.2A.5, Table 5.2A.5.1.1-2 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 | | | |
| subframeConfig1-5-r10 | '00001000000000100000' | BIT STRING (SIZE (20)) | |
| | | | |
| } | | | |
| } | | | |
| NeighCellsCRS-Info-r11 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| CRS-AssistancedInfoList-r11 ::= SEQUENCE (SIZE (1..maxCellReport)) OF CRS-AssistanceInfo-r11 { | | 2 Entries | |
| CRS-AssistancedInfo-r11 ::= SEQUENCE { | | | |
| physCellId-r11 | ($PCI_{cell1}$-$PCI_{cell2}$) mod3 = 0 $PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell PCIs are selected so that both conditions are met for Cell2 | |
| antennaPortsCount-r11 | an2 | | |
| mbsfn-SubframeConfigList-r11 ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF MBSFN-SubframeConfig { | | | |
| MBSFN-SubframeConfig:: = SEQUENCE { | | | |
| radioframeAllocationPeriod | n1 | | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| oneFrame | '000000' | The CRS assistance information is provided for Cell2 in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation oneFrame='000000' | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| CRS-AssistanceInfo-r11 :: = SEQUENCE { | | | |
| physCellId-r11 | ($PCI_{cell1}$-$PCI_{cell3}$) mod3! = 0 | Cell PCIs are selected so that both conditions are met for Cell3 | |
| antennaPortsCount-r11 | an2 | | |
| mbsfn-SubframeConfigList-r11 ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF MBSFN-SubframeConfig { | | | |
| MBSFN-SubframeConfig:: = SEQUENCE { | | | |
| radioframeAllocationPeriod | n1 | | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| oneFrame | '000000' | The CRS assistance information is | |

| | | | |
|---|---|---|---|
| | | provided for Cell 3 in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation oneFrame='000000' | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 7.3.20.5 Test requirements

**Table 7.3.20.5-1: Cell specific test requirement parameters for E-UTRAN TDD for in-sync radio link monitoring (feICIC)**

| Parameter | Unit | Test 1 | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Cell1 | | | | | Cell2 | | | | | Cell3 | | | | |
| | | T1 | T2 | T3 | T4 | T5 | T1 | T2 | T3 | T4 | T5 | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | | 1 | | | | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | 10 | | | | | 10 | | | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | | | 2x2 Low | | | | | 2x2 Low | | | | |
| Special subframe configuration Note 1 | | 6 | | | | | 6 | | | | | 6 | | | | |
| Uplink-downlink configuration Note 2 | | 1 | | | | | 1 | | | | | 1 | | | | |
| PCFICH/PDCCH/ PHICH parameters | | R.9 TDD | | | | | R.9 TDD | | | | | R.9 TDD | | | | |
| OCNG Pattern defined in D.2.2 (TDD) | | OP.2 TDD | | | | | OP.2 TDD | | | | | OP.2 TDD | | | | |
| $\rho_A$, $\rho_B$ | | -3 | | | | | -3 | | | | | -3 | | | | |
| PCFICH_RB | dB | 1 | | | | | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.2-2. | | | | | | | | | |
| PDCCH_RA | dB | -3 | | | | | | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | | | | | | | |
| PBCH_RA | dB | | | | | | | | | | | | | | | |
| PBCH_RB | dB | | | | | | | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | | | | | | | |
| OCNG_RA Note 3 | dB | | | | | | | | | | | | | | | |
| OCNG_RB Note 3 | dB | | | | | | | | | | | | | | | |
| SNR Note 8 | dB | -0.6 | | | | | 3.8 | 3.8 | 4.2 | 4.2 | 3.8 | 1.8 | 1.8 | 2.2 | 2.2 | 1.8 |
| $N_{oc}$ | dBm/ 15 kHz | -98 | | | | | -98 | | | | | -98 | | | | |
| Propagation condition | Hz | ETU 30 | | | | | ETU 30 | | | | | ETU 30 | | | | |

Note 1: For the special subframe configuration see table 4.2-1 in 3GPP TS 36.211.
Note 2: For the uplink-downlink configuration see table 4.2-2 in 3GPP TS 36.211.
Note 3: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 5: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 6: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 7: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 8: The SNR in time periods T1, T2, T3, T4 and T5 of active cell is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.20.4-1.

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (420 ms after the start of time duration T5) the UE shall transmit uplink signal at least in all uplink subframes configured for CQI transmission according to the configured CQI mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.21 E-UTRAN FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and MBSFN ABS (feICIC)

### 7.3.21.1 Test purpose

To verify that the UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the serving cell under time domain measurement resource restriction with CRS assistance information.

### 7.3.21.2 Test applicability

This test applies to all types of E-UTRA FDD and CRS interference handling UE release 11 and forward. Applicability requires support of FGI bit 115.

### 7.3.21.3 Minimum conformance requirements

When the downlink radio link quality of the PCell estimated over the last 100 ms period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 100 ms $Q_{in}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in TS 36.213 [8] clause 4.2.1. Two successive indications from Layer 1 shall be separated by at least 10 ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in TS 36.331 [5] clause 5.3.11.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2.1 and A.7.3.21.

### 7.3.21.4 Test description

There are three E-UTRA cells in the test on the same RF channel. There are three active cells in the test: Cell 1 is the PCell cell and Cell 2 and Cell 3 are the neighbour cells. MBSFN ABS pattern is configured in both Cell 2 and Cell 3 in this test. The UE is configured by higher layers with a time domain measurement restriction pattern for performing PCell measurements. The UE is also configured for periodic CQI reporting in PUCCH 1-0 mode with periodicity of 2 ms.

The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.21.4-1 shows the variation of the downlink SNR in the active cell (Cell 1) to emulate in-sync state. Prior to the start of the time duration T1, the UE shall be fully synchronized to Cell 1. The MBSFN ABS pattern, the time domain measurement resource restriction pattern and the CRS assistance information shall be configured prior to the start of T1.

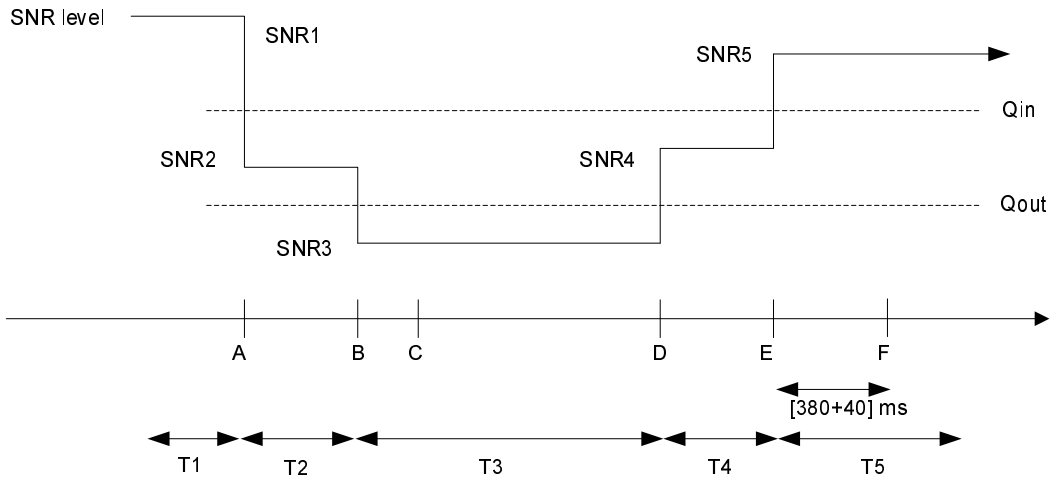


**Figure 7.3.21.4-1: SNR variation in the serving cell for in-sync testing under time domain measurement resource restriction with MBSFN ABS**

### 7.3.21.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), AWGN noise sources and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.48.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.21.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.21.4.3.
5. At this stage only Cell 1 is switched on. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1.

**Table 7.3.21.4.1-1: General test parameters for E-UTRAN FDD in-sync testing radio link monitoring test (feICIC)**

| Parameter | | Unit | Value | | | Comment |
|---|---|---|---|---|---|---|
| | | | Test 1 | | | |
| | | | Cell 1 | Cell 2 | Cell 3 | |
| PCFICH/PDCCH/PHICH parameters | | | R.9 FDD | R.9 FDD | R.9 FDD | As specified in section A.2.1. None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.6 FDD | OP.9 FDD | OP.9 FDD | As specified in section D.1.6 and D.1.9. |
| Active cell | | | PCell | Neighbor Cell | Neighbor Cell | Cell 1, Cell 2 and Cell 3 are on E-UTRA RF channel number 1 |
| CP length | | | Normal | Normal | Normal | |
| E-UTRA RF Channel Number | | | 1 | 1 | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | 10 | 10 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | 2x2 Low | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.521-1 [10] Annex B.2.3.2 |
| Neighbor Cell ABS configuration | | | N/A | MBSFN ABS | | As defined in Table C.3.1.2.2-2. |
| ABS Pattern | | | N/A | 0100000010 0000001000 0000001000 0001000000 | 0100000010 0000001000 0000001000 0001000000 | FDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. The first/leftmost bit corresponds to the PCell subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (40) divided by 10. MBSFN subframes are configured in the ABS subframes configured in Cell 2 and Cell 3 prior to the start of T1. |
| Time domain measurement resource restriction pattern | | | '0100000010000000100000000100000100000' | N/A | N/A | Time domain measurement resource restriction pattern for serving cell measurement signalled to the UE in message measSubframePatternPCell-r10 as defined in TS 36.331, clause 6.3.2. |
| CRS assistance information | physCellId | | N/A | see PCI conditions below | see PCI conditions below | The CRS assistance information is provided for Cell 2 and Cell 3 in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation fourFrames = '10000100010000010000 1000' |
| | antennaPortsCount | | | an2 | an2 | |
| | mbsfn-SubframeConfigList | | | fourFrames = '10000100010000010000' | fourFrames = '10000100010000010000' | |
| Time offset between cells (With respect to Cell 1) | | us | 0 | 3 | 2 | |
| Frequency shift between cells (With respect to Cell 1) | | Hz | 0 | 300 | -100 | |
| Physical Cell ID | | | $PCI_{cell1}$ | ($PCI_{cell1}$-$PCI_{cell2}$) | ($PCI_{cell1}$-$PCI_{cell3}$) | Cell PCIs are selected so that all conditions are met |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | mod3 = 0<br><br>$PCI_{cell1}$ not equal to $PCI_{cell2}$ | mod3 != 0 | |
| In sync transmission parameters (Note 1) | | DCI format | 1C | 1C | 1C | |
| In sync transmission parameters (Note 1) | Number of Control OFDM symbols | | 3 | 3 | 3 | As defined in section 5.3.3.1.4 in TS 36.212 |
| | Aggregation level | CCE | 4 | 4 | 4 | In sync threshold $Q_{in}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] section 7.6.1 and Table 7.6.1-2 respectively. |
| | $\rho_A$, $\rho_B$ | | -3 | -3 | -3 | |
| | Ratio of PDCCH to RS EPRE | | -3 | Non-ABS and ABS subframe channel powers defined in Table C.3.1.2.2-2. | | |
| | Ratio of PCFICH to RS EPRE | | 1 | | | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | 1A | 1A | |
| | Number of Control OFDM symbols | | 3 | 3 | 3 | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Aggregation level | CCE | 8 | 8 | 8 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] section 7.6.1 and Table 7.6.1-1 respectively. |
| | $\rho_A$, $\rho_B$ | | -3 | -3 | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | Non-ABS and ABS subframe channel powers defined in Table C.3.1.2.2-2. | | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | | | |
| DRX | | | OFF | OFF | OFF | |
| Layer 3 filtering | | | Enabled | Disable | Disable | Counters:<br>N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | N/A | | T310 is enabled |
| T311 timer | | ms | 1000 | | | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | | | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 2 | | | Minimum CQI reporting periodicity |
| T1 | | s | 0.5 | N/A | | |
| T2 | | s | 0.4 | | | |
| T3 | | s | 1.46 | | | |
| T4 | | s | 0.4 | | | |
| T5 | | s | 1 | | | |
| NOTE 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | | | |

### 7.3.21.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to Cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

MBSFN ABS pattern is configured in both Cell 2 and Cell 3 in this test. The UE is configured by higher layers with a time domain measurement restriction pattern for performing PCell (Cell 1) measurements. The MBSFN ABS pattern, the time domain measurement resource restriction pattern and the CRS assistance information shall be configured prior to the start of T1.

1. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3 with exceptions listed in 7.2A.6.

2. Set the parameters for Cell 1, Cell 2, and Cell 3 according to T1 in Table 7.3.21.5-1. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.
3. When T1 expires the SS shall change the SNR values to T2 as specified in Table 7.3.21.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR values to T3 as specified in Table 7.3.21.5-1. T3 starts.
5. When T3 expires the SS shall change the SNR values to T4 as specified in Table 7.3.21.5-1. T4 starts.
6. When T4 expires the SS shall change the SNR values to T5 as specified in Table 7.3.21.5-1. T5 starts.
7. If the SS detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (420 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
8. After T5 expires, repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.21.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 7.2A.6 with the following exceptions:

**Table 7.3.21.4.3-1: Common Exception messages for E-UTRAN FDD Radio Link Monitoring Test for in-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (feICIC)**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.21.4.3-2 : *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring Test for in-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (feICIC) requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.21.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring Test for in-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (felCIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.21.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN FDD Radio Link Monitoring Test for in-sync**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

**Table 7.3.21.4.3-5: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional E-UTRAN FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [7] clause 5.2A.5, Table 5.2A.5.1.1-2 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '01000000100000001000 00000010000001000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| } | | | |
| NeighCellsCRS-Info-r11 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| CRS-AssistancedInfoList-r11 ::= SEQUENCE (SIZE (1..maxCellReport)) OF CRS-AssistanceInfo-r11 { | | 2 Entries | |
| CRS-AssistancedInfo-r11 ::= SEQUENCE { | | | |
| physCellId-r11 | ($PCI_{cell1}$-$PCI_{cell2}$) mod3 = 0<br><br>$PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell PCIs are selected so that both conditions are met for Cell2 | |
| antennaPortsCount-r11 | an2 | | |
| mbsfn-SubframeConfigList-r11 ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF MBSFN-SubframeConfig { | | | |
| MBSFN-SubframeConfig:: = SEQUENCE { | | | |
| radioframeAllocationPeriod | n4 | | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| fourFrames | '10000100010000010000 01000' | The CRS assistance information is provided for Cell 2 in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation fourFrams=' 10000100010000 0100001000' | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| CRS-AssistanceInfo-r11 :: = SEQUENCE { | | | |
| physCellId-r11 | ($PCI_{cell1}$-$PCI_{cell3}$) mod3! = 0 | Cell PCIs are selected so that both conditions are met for Cell3 | |
| antennaPortsCount-r11 | an2 | | |
| mbsfn-SubframeConfigList-r11 ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF MBSFN-SubframeConfig { | | | |
| MBSFN-SubframeConfig:: = SEQUENCE { | | | |
| radioframeAllocationPeriod | n4 | | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| fourFrames | '10000100010000010000 | The CRS | |

| | | | |
|---|---|---|---|
| | 1000' | assistance information is provided for Cell 3 in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation fourFrams=' 10000100010000 0100001000' | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.21.4.3-6: *SystemInformationBlockType2*: Additional E-UTRAN FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 SystemInformationBlockType2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfiguration | Not present | | Cell 1 |
| mbsfn-SubframeConfigList ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF SEQUENCE { | | | Cell 2, Cell 3 |
| radioframeAllocationPeriod | n4 | | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| fourFrames | '10000100010000010000 1000'B | | |
| } | | | |
| } | | | |

**Table 7.3.21.4.3-7: *SystemInformationBlockType3*: Additional E-UTRAN FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1, Cell 2, Cell 3 |

## 7.3.21.5 Test requirements

**Table 7.3.21.5-1: Cell specific test parameters for E-UTRAN TDD in-sync radio link monitoring test (feICIC)**

| Parameter | Unit | Test 1 | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Cell1 | | | | | Cell2 | | | | | Cell3 | | | | |
| | | T1 | T2 | T3 | T4 | T5 | T1 | T2 | T3 | T4 | T5 | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | | 1 | | | | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | 10 | | | | | 10 | | | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | | | 2x2 Low | | | | | 2x2 Low | | | | |
| PCFICH/PDCCH/ PHICH parameters | | R.9 FDD | | | | | R.9 FDD | | | | | R.9 FDD | | | | |
| OCNG Pattern defined in D.1.6 and D.1.9 (FDD) | | OP.6 FDD | | | | | OP.9 FDD | | | | | OP.9 FDD | | | | |
| $\rho_A$, $\rho_B$ | | -3 | | | | | -3 | | | | | -3 | | | | |
| PCFICH_RB | dB | 1 | | | | | Non-ABS and ABS subframe channel powers defined in Table C.3.1.2.2-2. | | | | | | | | | |
| PDCCH_RA | dB | -3 | | | | | | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | | | | | | | |
| PBCH_RA | dB | | | | | | | | | | | | | | | |
| PBCH_RB | dB | | | | | | | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | | | | | | | |
| SNR Note 6 | dB | -0.6 | | | | | 3.8 | 3.8 | 4.2 | 4.2 | 3.8 | 1.8 | 1.8 | 2.2 | 2.2 | 1.8 |
| $N_{oc}$ | dBm/ 15 kHz | -98 | | | | | -98 | | | | | -98 | | | | |
| Propagation condition | Hz | ETU 30 | | | | | ETU 30 | | | | | ETU 30 | | | | |

NOTE 1: OCNG shall be used such that the resources in cell 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
NOTE 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
NOTE 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
NOTE 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
NOTE 6: The SNR in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.21.4-1.

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (420 ms after the start of time duration T5) the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.22 E-UTRAN TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and MBSFN ABS (feICIC)

### 7.3.22.1 Test purpose

To verify that the UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the serving cell under time domain measurement resource restriction with CRS assistance information.

### 7.3.22.2 Test applicability

This test applies to all types of E-UTRA TDD and CRS interference handling UE release 11 and forward. Applicability requires support of FGI bit 115.

### 7.3.22.3 Minimum conformance requirements

When the downlink radio link quality of the PCell estimated over the last 100 ms period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 100 ms $Q_{in}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in TS 36.213 [8] clause 4.2.1. Two successive indications from Layer 1 shall be separated by at least 10 ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in TS 36.331 [5] clause 5.3.11.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2.1 and A.7.3.22.

### 7.3.22.4 Test description

There are three E-UTRA cells in the test on the same RF channel. There are three active cells in the test: Cell 1 is the PCell cell and Cell 2 and Cell 3 are the neighbour cells. MBSFN ABS pattern is configured in both Cell 2 and Cell 3 in this test. The UE is configured by higher layers with a time domain measurement restriction pattern for performing PCell measurements. The UE is also configured for periodic CQI reporting in PUCCH 1-0 mode with periodicity of 2 ms.

The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.22.4-1 shows the variation of the downlink SNR in the active cell (Cell 1) to emulate in-sync state. Prior to the start of the time duration T1, the UE shall be fully synchronized to Cell 1. The MBSFN ABS pattern, the time domain measurement resource restriction pattern and the CRS assistance information shall be configured prior to the start of T1

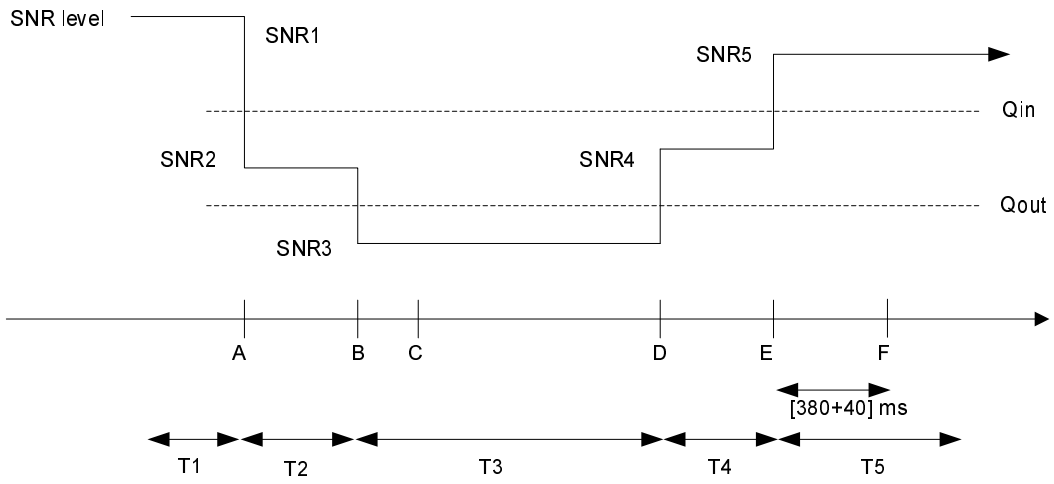


**Figure 7.3.22.4-1: SNR variation in the serving cell for in-sync testing under time domain measurement resource restriction with MBSFN ABS**

### 7.3.22.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), AWGN noise sources and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.48.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.22.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.22.4.3.
5. At this stage only Cell 1 is switched on. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1.

**Table 7.3.22.4.1-1: General test parameters for E-UTRAN TDD in-sync radio link monitoring test**

| rameter | Unit | Value | | | Com |
|---|---|---|---|---|---|
| | | Test 1 | | | |
| | | Cell 1 | Cell 2 | Cell 3 | |
| H/PHICH | | R.9 TDD | R.9 TDD | R.9 TDD | As specified in s None of the PD intended for the |
| ters | | OP.2 TDD | OP.6 TDD | OP.6 TDD | As specified in s and D.2.6. |
| | | PCell | Neighbor Cell | Neighbor Cell | Cell 1, Cell 2 an E-UTRA RF cha |
| | | Normal | Normal | Normal | |
| annel Number | | 1 | 1 | 1 | One E-UTRA TI frequency is use |
| el Bandwidth | MHz | 10 | 10 | 10 | |
| rix and Antenna | | 2x2 Low | 2x2 Low | 2x2 Low | Correlation Mat Antenna Config defined in TS 36 Annex B.2.3.2 |
| BS configuration | | N/A | MBSFN ABS | | As defined in Ta |
| | | N/A | 00001000000000100000 | 00001000000000100000 | TDD ABS Patte defined in TS 36 clause 9.2.54. C Cell 2 and Cell 3 The first/leftmos corresponds to subframe #0 of satisfying SFN r where x is the s string (20) divid MBSFN subfran configured in th subframes. |
| easurement tion pattern | | '00001000000000100000' | N/A | N/A | Time domain m resource restric serving cell mea signalled to the measSubframel r10 as defined i clause 6.3.2. |
| physCellId | | N/A | see PCI conditions below | see PCI conditions below | The CRS assist information is pr 2 and Cell 3 in C AssistanceInfo. single MBSFN-SubframeConfig subframe alloca fourFrames = '010000100001 |
| antennaPortsCount | | | an2 | an2 | |
| mbsfn-SubframeConfigList | | | fourFrames = '01000010000100001000000' | fourFrames = '01000010000100001000000' | |
| n Cell 1 | us | 0 | 3 | 2 | |
| et | Hz | 0 | 300 | -100 | |
| ) | | $PCI_{cell1}$ | ($PCI_{cell1}$-$PCI_{cell2}$ ) mod3 = 0 $PCI_{cell1}$ not equal to $PCI_{cell2}$ | ($PCI_{cell1}$-$PCI_{cell3}$ ) mod3 != 0 | Cell PCIs are se all conditions ar |
| DCI format | | 1C | 1C | 1C | As defined in se in TS 36.212 |
| Number of Control OFDM symbols | | 2 | 2 | 2 | In sync threshol corresponding h PDCCH/PCFICH parameters are TS 36.133 [4] se and Table 7.6.1 |
| Aggregation level | CCE | 4 | 4 | 4 | |
| $\rho_A$, $\rho_B$ | | -3 | -3 | -3 | |
| Ratio of PDCCH to RS EPRE | | -3 | Non-ABS and ABS subframe channel powers defined in Table C.3.1.2.2-2. | | |
| Ratio of PCFICH to RS EPRE | | 1 | | | |
| DCI format | | 1A | 1A | 1A | As defined in se |

| | | | | | |
|---|---|---|---|---|---|
| | | | | | in TS 36.212 |
| Number of Control OFDM symbols | | 2 | 2 | 2 | Out of sync thre the correspondi |
| Aggregation level | CCE | 8 | 8 | 8 | PDCCH/PCFICI |
| $\rho_A$, $\rho_B$ | | -3 | -3 | -3 | parameters are |
| Ratio of PDCCH to RS EPRE | dB | 1 | Non-ABS and ABS subframe channel powers defined in Table C.3.1.2.2-2. | | TS 36.133 [4] s and Table 7.6.1 |
| Ratio of PCFICH to RS EPRE | dB | 1 | | | |
| | | OFF | OFF | OFF | |
| | | Enabled | Disable | Disable | Counters: N310 = 1; N311 |
| | ms | 2000 | N/A | | T310 is enabled |
| | ms | 1000 | | | T311 is enabled |
| porting mode | | PUCCH 1-0 | | | As defined in Ta TS 36.213 |
| eriodicity | ms | 1 | | | Minimum CQI re periodicity |
| | s | 0.5 | N/A | | |
| | s | 0.4 | | | |
| | s | 1.46 | | | |
| | s | 0.4 | | | |
| | s | 1 | | | |

CH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Chan

### 7.3.22.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to Cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

MBSFN ABS pattern is configured in both Cell 2 and Cell 3 in this test. The UE is configured by higher layers with a time domain measurement restriction pattern for performing PCell (Cell 1) measurements. The MBSFN ABS pattern, the time domain measurement resource restriction pattern and the CRS assistance information shall be configured prior to the start of T1.

1. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3 with exceptions listed in 7.2A.6.
2. Set the parameters for Cell 1, Cell 2, and Cell 3 according to T1 in Table 7.3.22.5-1. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.
3. When T1 expires the SS shall change the SNR values to T2 as specified in Table 7.3.22.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR values to T3 as specified in Table 7.3.22.5-1. T3 starts.
5. When T3 expires the SS shall change the SNR values to T4 as specified in Table 7.3.22.5-1. T4 starts.
6. When T4 expires the SS shall change the SNR values to T5 as specified in Table 7.3.22.5-1. T5 starts.
7. If the SS detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (420 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
8. After T5 expires, repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.22.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 7.2A.6 with the following exceptions:

**Table 7.3.22.4.3-1: Common Exception messages for E-UTRAN TDD Radio Link Monitoring Test for in-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (feICIC)**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.22.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for in-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (feICIC) requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.22.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for in-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.22.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN TDD Radio Link Monitoring Test for in-sync**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

**Table 7.3.22.4.3-5: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional E-UTRAN TDD Radio Link Monitoring Test for in-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [7] clause 5.2A.5, Table 5.2A.5.1.1-2 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 | '00001000000000100000' | BIT STRING (SIZE (20)) | |
| } | | | |
| } | | | |
| NeighCellsCRS-Info-r11 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| CRS-AssistancedInfoList-r11 ::= SEQUENCE (SIZE (1..maxCellReport)) OF CRS-AssistanceInfo-r11 { | | 2 Entries | |
| CRS-AssistancedInfo-r11 ::= SEQUENCE { | | | |
| physCellId-r11 | ($PCI_{cell1}$-$PCI_{cell2}$) mod3 = 0<br>$PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell PCIs are selected so that both conditions are met for Cell2 | |
| antennaPortsCount-r11 | an2 | | |
| mbsfn-SubframeConfigList-r11 ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF MBSFN-SubframeConfig { | | | |
| MBSFN-SubframeConfig:: = SEQUENCE { | | | |
| radioframeAllocationPeriod | n4 | | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| fourFrames | '01000010000100000000' | The CRS assistance information is provided for Cell 2 in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation fourFrams= '01000010000100 0010000000' | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| CRS-AssistanceInfo-r11 :: = SEQUENCE { | | | |
| physCellId-r11 | ($PCI_{cell1}$-$PCI_{cell3}$) mod3! = 0 | Cell PCIs are selected so that both conditions are met for Cell3 | |
| antennaPortsCount-r11 | an2 | | |
| mbsfn-SubframeConfigList-r11 ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF MBSFN-SubframeConfig { | | | |
| MBSFN-SubframeConfig:: = SEQUENCE { | | | |
| radioframeAllocationPeriod | n4 | | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| fourFrames | '01000010000100001000 0000' | The CRS assistance information is | |

| | | | |
|---|---|---|---|
| | | provided for Cell 3 in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation fourFrams= '01000010000100 0010000000' | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.22.4.3-6: *SystemInformationBlockType2*: Additional E-UTRAN TDD Radio Link Monitoring Test for in-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 SystemInformationBlockType2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfigList | Not present | | Cell 1 |
| mbsfn-SubframeConfigList ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF SEQUENCE { | | | Cell 2, Cell 3 |
| radioframeAllocationPeriod | n4 | | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| fourFrames | '01000010000100001000 0000'B | | |
| } | | | |
| } | | | |

**Table 7.3.22.4.3-7: *SystemInformationBlockType3*: Additional E-UTRAN TDD Radio Link Monitoring Test for in-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1, Cell 2, Cell 3 |

### 7.3.22.5 Test requirements

**Table 7.3.22.5-1: Cell specific test parameters for E-UTRAN TDD in-sync radio link monitoring test (feICIC)**

| Parameter | Unit | Test 1 | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Cell1 | | | | | Cell2 | | | | | Cell3 | | | | |
| | | T1 | T2 | T3 | T4 | T5 | T1 | T2 | T3 | T4 | T5 | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | | 1 | | | | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | | 10 | | | | | 10 | | | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | | | 2x2 Low | | | | | 2x2 Low | | | | |
| Special subframe configuration Note 1 | | 6 | | | | | 6 | | | | | 6 | | | | |
| Uplink-downlink configuration Note 2 | | 1 | | | | | 1 | | | | | 1 | | | | |
| PCFICH/PDCCH/ PHICH parameters | | R.9 TDD | | | | | R.9 TDD | | | | | R.9 TDD | | | | |
| OCNG Pattern defined in D.2.2 and D.2.6 (TDD) | | OP.2 TDD | | | | | OP.6 TDD | | | | | OP.6 TDD | | | | |
| $\rho_A$, $\rho_B$ | | -3 | | | | | -3 | | | | | -3 | | | | |
| PCFICH_RB | dB | 1 | | | | | Non-ABS and ABS subframe channel powers defined in Table C.3.1.2.2-2 | | | | | | | | | |
| PDCCH_RA | dB | -3 | | | | | | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | | | | | | | |
| PBCH_RA | dB | | | | | | | | | | | | | | | |
| PBCH_RB | dB | | | | | | | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | | | | | | | |
| OCNG_RA Note 3 | dB | | | | | | | | | | | | | | | |
| OCNG_RB Note 3 | dB | | | | | | | | | | | | | | | |
| SNR Note 8 | dB | -0.6 | -4.3 | -14.6 | -9.5 | -0.6 | 3.8 | 3.8 | 4.2 | 4.2 | 3.8 | 1.8 | 1.8 | 2.2 | 2.2 | 1.8 |
| $N_{oc}$ | dBm/ 15 kHz | -98 | | | | | -98 | | | | | -98 | | | | |
| Propagation condition | Hz | ETU 30 | | | | | ETU 30 | | | | | ETU 30 | | | | |

Note 1: For the special subframe configuration see table 4.2-1 in 3GPP TS 36.211.
Note 2: For the uplink-downlink configuration see table 4.2-2 in 3GPP TS 36.211.
Note 3: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 5: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 6: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 7: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 8: The SNR in time periods T1, T2, T3, T4 and T5 of active cell is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.22.4-1.

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (420 ms after the start of time duration T5) the UE shall transmit uplink signal at least in all uplink subframes configured for CQI transmission according to the configured CQI mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.23 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync for 5MHz Bandwidth

### 7.3.23.1 Test purpose

Same test purpose as defined in clause 7.3.1.1.

### 7.3.23.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that only support E-UTRA Band 31.

### 7.3.23.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 7.3.1.3 with the following exceptions:

- Instead of A.7.3.1 → use A.7.3.23.

### 7.3.23.4 Test description

The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.23.4-1 shows the three different time durations and the corresponding variation of the downlink SNR in the active cell to emulate out-of-sync states.

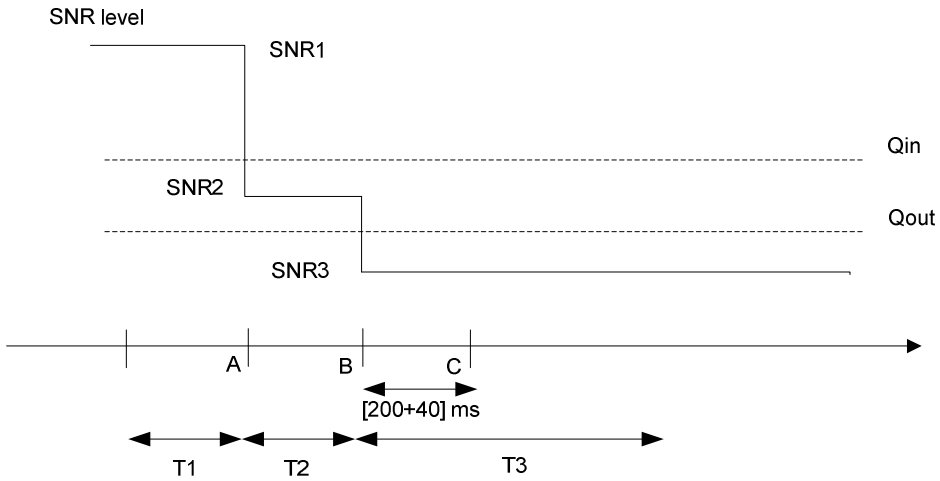


**Figure 7.3.23.4-1: SNR variation for out-of-sync testing**

#### 7.3.23.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10.

2. The general test parameter settings for the different subtests are set up according to Table 7.3.23.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.23.4.3.
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.23.4.1-1: General test parameters for E-UTRAN FDD out-of-sync testing for 5MHz bandwidth**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| | | | Test 4 | |
| PCFICH/PDCCH/PHICH parameters | | | R.12 FDD | As specified in clause A.2.1. None of the PDCCH are intended for the UE under test |
| OCNG parameters | | | OP.16 FDD | As specified in clause D.1.16. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 5 | |
| Out of sync transmission parameters (Note 1) | Number of Control OFDM Symbols | | 3 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] clause 7.6.1 and Table 7.6.1-1 respectively. |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |
| Note 2: See Table 7.3.1.4.1-1 for other general test parameters. | | | | |
| Note 3: This test is according to the principle defined in TS 36.133 [4] section A.3.7.2. | | | | |

### 7.3.23.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.23.5-1. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.
3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.23.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.23.5-1. T3 starts.
5. If the SS:

   a) detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B

   and

   b) does not detect any uplink power higher than -48.5 dBm from time point C (240 ms after the start of T3) until T3 expires,

   the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
6. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.23.5-1.
7. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
8. Repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.23.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.23.4.3-1: Common Exception messages for E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync for 5MHz bandwidth**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.2.4-1 |

**Table 7.3.23.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync for 5MHz bandwidth**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.23.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync for 5MHz bandwidth requirements**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.23.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync for 5MHz bandwidth**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

### 7.3.23.5 Test requirement

Same test requirement as defined in clause 7.3.1.5 with the following exceptions:

Tables 7.3.1.4.1-1 and 7.3.23.5-1 define the primary level settings for E-UTRAN FDD (cell # 1) for out-of-sync radio link monitoring for 5MHz bandwidth.

**Table 7.3.23.5-1: Cell specific test parameters for E-UTRAN FDD for out-of-sync radio link monitoring test #4 under 5MHz Bandwidth**

| Parameter | Unit | Test 4 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| $BW_{channel}$ | MHz | 5 | | |
| OCNG Pattern defined in A.3.2.1.16 (FDD) | | OP.16 FDD | | |
| SNR Note 6 | dB | -1.4 | -4.8 | -13.1 |
| Note 1: See Table 7.3.1.5-2 for other cell specific test parameters. | | | | |

## 7.3.24 E-UTRAN FDD Radio Link Monitoring Test for In-sync for 5MHz Bandwidth

### 7.3.24.1 Test purpose

Same test purpose as defined in clause 7.3.2.1.

### 7.3.24.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that only support Band 31.

### 7.3.24.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 7.3.1.3 with the following exceptions:

- Instead of A.7.3.2 → use A.7.3.24.

### 7.3.24.4 Test description

The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.24.4-1 shows the different time durations and the corresponding variation of the downlink SNR in the active cell to emulate in-sync states.

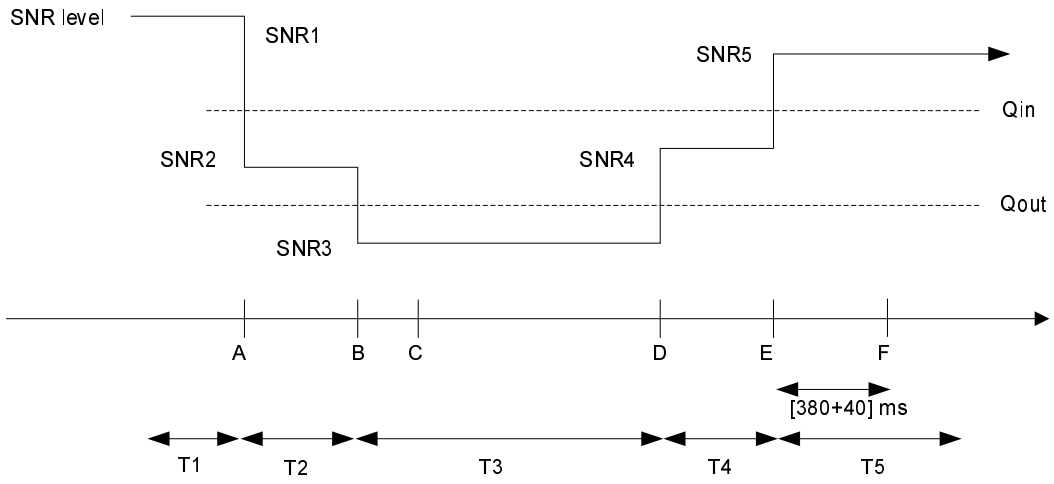


**Figure 7.3.24.4-1: SNR variation for in-sync testing**

### 7.3.24.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders, to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10.
2. The general test parameter settings for the different subtest are set up according to Table 7.3.24.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.24.4.3.
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.24.4.1-1: General test parameters for E-UTRAN FDD in-sync for 5MHz bandwidth testing**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | | R.12 FDD | As specified in clause A.2.1. None of the PDCCH are intended for the UE under test |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 5 | |
| In sync transmission parameters (Note 1) | Number of Control OFDM symbols | | 3 | In sync threshold Qin and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] clause 7.6.1 and Table 7.6.1-2 respectively. |
| Out of sync transmission parameters (Note 1) | Number of Control OFDM symbols | | 3 | Out of sync threshold Qout and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] clause 7.6.1 and Table 7.6.1-1 respectively. |
| Note 1: See Table 7.3.2.4.1-1 for other general test 2 parameters.<br>Note 2: This test is performed according to the principle defined in TS 36.133 [4] section A.3.7.2. | | | | |

### 7.3.24.4.2 Test procedure

Same test procedure as defined in clause 7.3.2.4.2 with only Test 2 applied and with the following exceptions:

- Instead of Table 7.3.2.5-1 → use Table 7.3.24.5-1.

### 7.3.24.4.3 Message contents

Same message contents as defined in clause 7.3.2.4.3 with only Test 2 applied.

## 7.3.24.5 Test requirement

Same test requirement as defined in clause 7.3.2.5 with the following exceptions:

Tables 7.3.24.4.1-1 and 7.3.24.5-1 define the primary level settings for E-UTRAN FDD (cell # 1) for in-sync radio link monitoring test for 5MHz bandwidth.

**Table 7.3.24.5-1: Cell specific test parameters for E-UTRAN FDD (cell # 1) for in-sync radio link monitoring test for 5MHz bandwidth**

| Parameter | Unit | T1 | T2 | T3 | T4 | T5 |
|---|---|---|---|---|---|---|
| $BW_{channel}$ | MHz | 5 | | | | |
| OCNG Pattern defined in D.1.16 (FDD) | | OP.16 FDD | | | | |
| SNR | dB | -1.4 | -4.8 | -13.1 | -8.2 | -1.4 |
| Propagation condition | | ETU 70 Hz | | | | |
| Note 1: See Table 7.3.2.5-1 for other general test 2 parameters. | | | | | | |

# 7.3.25 E-UTRAN FDD Radio Link Monitoring Test for In-sync in DRX for 5MHz Bandwidth

## 7.3.25.1 Test purpose

Same test purpose as defined in clause 7.3.6.1.

## 7.3.25.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that only support E-UTRA Band 31. Applicability requires support for FGI bit 5.

## 7.3.25.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 7.3.6.3 with the following exceptions:

- Instead of A.7.3.6 → use A.7.3.25.

## 7.3.25.4 Test description

Same test description as defined in clause 7.3.6.4.

### 7.3.25.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.18.
2. The general test parameter settings for the test are set up according to Table 7.3.25.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.25.4.3.
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.25.4.1-1: General test parameters for E-UTRAN FDD in-sync in DRX for 5MHz bandwidth testing**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PCFICH/PDCCH/PHICH parameters | | | R.12 FDD | As specified in clause A.2.1. None of the PDCCH are intended for the UE under test |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 5 | |
| In sync transmission parameters( Note 1) | Number of Control OFDM symbols | | 3 | In sync threshold $Q_{in}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] clause 7.6.1 and Table 7.6.1-2 respectively. |
| Out of sync transmission parameters( Note 1) | Number of Control OFDM symbols | | 3 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] clause 7.6.1 and Table 7.6.1-1 respectively. |
| Note 1: See Table 7.3.6.4.1-1 for other general test parameters.<br>Note 2: This test is performed according to the principle defined in TS 36.133 [4] section A.3.7.2. | | | | |

### 7.3.25.4.2 Test procedure

Same test procedure as defined in clause 7.3.6.4.2 with the following exceptions:

- Instead of Table 7.3.6.5-1 → use Table 7.3.25.5-1.

### 7.3.25.4.3 Message contents

Same message contents as defined in clause 7.3.6.4.3.

## 7.3.25.5 Test requirement

Same test requirement as defined in clause 7.3.6.5 with the following exceptions:

Tables 7.3.25.4.1-1, 7.3.25.5-1, 7.3.6.5-2 and 7.3.6.5-3 define the primary level settings for E-UTRAN FDD (cell # 1) radio link monitoring test for In-sync in DRX for 5MHz bandwidth.

**Table 7.3.25.5-1: Cell specific test parameters for E-UTRAN FDD (cell # 1) for In-sync radio link monitoring in DRX for 5MHz bandwidth test**

| Parameter | Unit | T1 | T2 | T3 | T4 | T5 |
|---|---|---|---|---|---|---|
| $BW_{channel}$ | MHz | 5 | | | | |
| OCNG Pattern defined in D.1.16 (FDD) | | OP.16 FDD | | | | |
| SNR | dB | -1.7 | -5.1 | -12.8 | -7.9 | -1.7 |
| Propagation condition | | AWGN | | | | |
| Note 1: See Table 7.3.6.5-1 for other general test parameters. | | | | | | |

# 7.3.26 E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync for UE category 0

## 7.3.26.1 Test purpose

The purpose of this test is to verify that the FD-FDD category 0 UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the PCell. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in TS 36.133 [4] section 7.11.

## 7.3.26.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward of UE Category 0.

## 7.3.26.3 Minimum conformance requirements

The UE category 0 applicability of the requirements for performing radio link monitoring is defined in Section 3.1.

When the downlink radio link quality of the PCell estimated over the last 200 ms period becomes worse than the threshold $Q_{out_Cat0}$, Layer 1 of the UE shall send an out-of-sync indication for the PCell to the higher layers within 200ms $Q_{out_Cat0}$ evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last 100 ms period becomes better than the threshold $Q_{in_Cat0}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 100 ms $Q_{in_Cat0}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least 10ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

The normative reference for this requirement is TS 36.133 [4] clause 7.11.1, 7.11.2 and A.7.3.26.

## 7.3.26.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.26.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync state. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

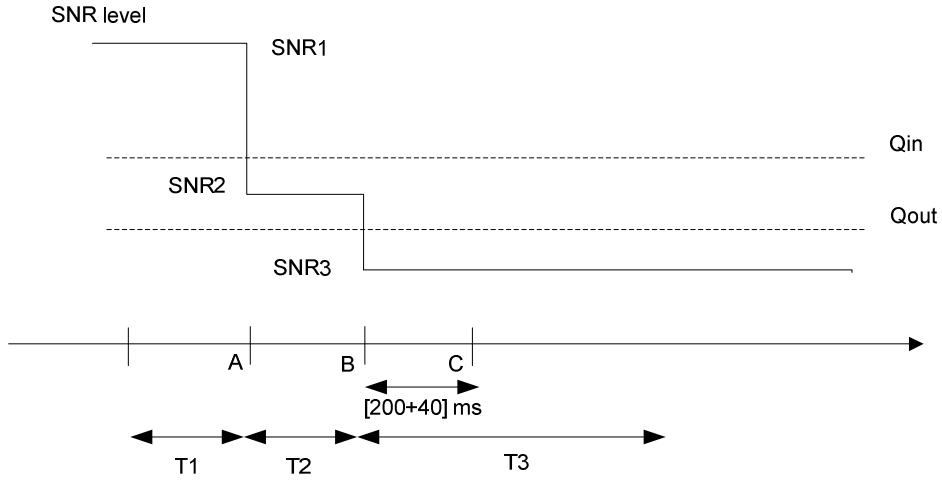


**Figure 7.3.26.4-1: SNR variation for out-of-sync testing**

### 7.3.26.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.26.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.26.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.26.4.1-1: General test parameters for E-UTRAN FD-FDD out-of-sync test for category 0**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| CP length | | | Normal | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] in section 7.11.1 and Table 7.11.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX | | | OFF | |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 2 | Minimum CQI reporting periodicity |
| T1 | | s | 1 | |
| T2 | | s | 0.4 | |
| T3 | | s | 0.5 | |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

## 7.3.26.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.26.5-1. Propagation conditions are set according to Annex B clause B.2. T1 starts.
3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.26.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.26.5-1. T3 starts.
5. If the SS:
   a) detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B

and

   b) does not detect any uplink power higher than -48.5 dBm from time point C (240 ms after the start of T3) until T3 expires,

   the number of successful tests is increased by one.

Otherwise the number of failed tests is increased by one.

6. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.26.5-1.
7. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
8. Repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.26.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.26.4.3-1: Common Exception messages for E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync for UE category 0**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.26.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.26.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| } | | | |

**Table 7.3.26.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

## 7.3.26.5 Test requirement

**Table 7.3.26.5-1: Cell specific test parameters for E-UTRAN FD-FDD (cell # 1) for out-of-sync radio link monitoring for UE category 0**

| Parameter | Unit | Test 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| PCFICH/PDCCH/PHICH parameters defined in A.2.1 | | R.7 FDD | | |
| OCNG Pattern defined in D.1.2 (FDD) | | OP.2 FDD | | |
| $\rho_A$, $\rho_B$ | | -3 | | |
| PCFICH_RB | dB | 1 | | |
| PDCCH_RA | dB | 4 | | |
| PDCCH_RB | dB | 4 | | |
| PBCH_RA | dB | -3 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | |
| SNR Note 6 | dB | -1.2 | -6.0 | -13.8 |
| Propagation condition | | ETU 70Hz | | |
| Correlation Matrix and Antenna Configuration | | 2x1 Low | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.26.4-1.

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal no later than time point C (240ms after the start of time duration T3).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal
- UE output power equal to or less than Transmit OFF power -48.5 dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.27 E-UTRAN FD-FDD Radio Link Monitoring Test for In-sync for UE category 0

### 7.3.27.1 Test purpose

The purpose of this test is to verify that the FD-FDD category 0 UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the PCell. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in TS 36.133 [4] section 7.11.

### 7.3.27.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward of UE category 0.

### 7.3.27.3 Minimum conformance requirements

The UE category 0 applicability of the requirements for performing radio link monitoring is defined in Section 3.1.

When the downlink radio link quality of the PCell estimated over the last 200 ms period becomes worse than the threshold $Q_{out_Cat0}$, Layer 1 of the UE shall send an out-of-sync indication for the PCell to the higher layers within 200ms $Q_{out_Cat0}$ evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last 100 ms period becomes better than the threshold $Q_{in_Cat0}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 100 ms $Q_{in_Cat0}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least 10ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

The normative reference for this requirement is TS 36.133 [4] clause 7.11.1, 7.11.2 and A.7.3.27.

### 7.3.27.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.27.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

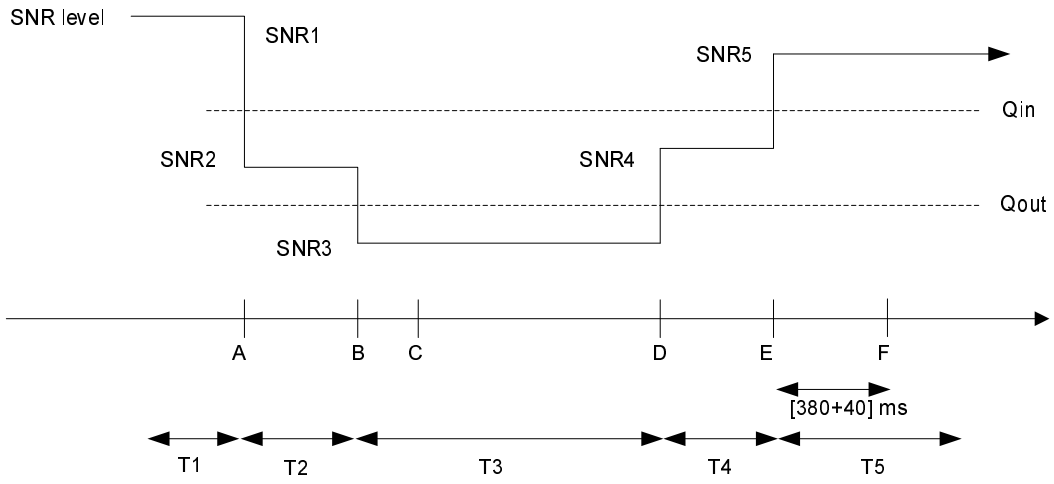


**Figure 7.3.27.4-1: SNR variation for in-sync testing**

### 7.3.27.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.27.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.27.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.27.4.1-1: General test parameters for E-UTRAN FD-FDD Radio Link Monitoring in-sync test for UE category 0**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| CP length | | | Normal | |
| In sync transmission parameters (Note 1) | DCI format | | 1C | As defined in clause 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | In sync threshold $Q_{in}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS36.133 [4] clause 7.11.1 and Table 7.11.1-2 respectively. |
| | Aggregation level | CCE | 4 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS36.133 [4] clause 7.11.1 and Table 7.11.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX | | | OFF | |
| Layer 3 filtering | | | *Enabled* | *Counters: N310 = 1; N311 = 1* |
| T310 timer | | *ms* | *2000* | *T310 is enabled* |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 2 | Minimum CQI reporting periodicity |
| T1 | | s | 0.5 | |
| T2 | | s | 0.4 | |
| T3 | | s | 1.46 | |
| T4 | | s | 0.4 | |
| T5 | | s | 1 | |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

## 7.3.27.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.27.5-1. Propagation conditions are set according to Annex B.2. T1 starts.

3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.27.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.27.5-1. T3 starts.
5. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.27.5-1. T4 starts.
6. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.27.5-1. T5 starts.
7. If the SS detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (420 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
8. After T5 expires, repeat steps 2-7 for both subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.27.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.27.4.3-1: Common Exception messages for E-UTRAN FD-FDD Radio Link Monitoring Test for in-sync for UE category 0**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.27.4.3-2 : *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FD-FDD Radio Link Monitoring Test for in-sync for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.27.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FD-FDD Radio Link Monitoring Test for in-sync for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.27.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN FD-FDD Radio Link Monitoring Test for in-sync for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

## 7.3.27.5 Test requirement

**Table 7.3.27.5-1: Cell specific test parameters for E-UTRAN FD-FDD in-sync Radio Link Monitoring Test for UE category 0**

| Parameter | Unit | Test 1 | | | | |
|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | |
| PCFICH/PDCCH/PHICH parameters defined in clause A.2.1 | | R.7 FDD | | | | |
| OCNG Pattern defined in D.1.2 (FDD) | | OP.2 FDD | | | | |
| $\rho_A$, $\rho_B$ | | -3 | | | | |
| PCFICH_RB | dB | 1 | | | | |
| PDCCH_RA | dB | 1 | | | | |
| PDCCH_RB | dB | 1 | | | | |
| PBCH_RA | dB | -3 | | | | |
| PBCH_RB | dB | | | | | |
| PSS_RA | dB | | | | | |
| SSS_RA | dB | | | | | |
| PHICH_RA | dB | | | | | |
| PHICH_RB | dB | | | | | |
| PDSCH_RA | dB | | | | | |
| PDSCH_RB | dB | | | | | |
| OCNG_RA Note 1 | dB | | | | | |
| OCNG_RB Note 1 | dB | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | |
| SNR Note 6 | dB | -1.2 | -6.0 | -13.8 | -8.0 | -1.2 |
| Propagation condition | | ETU 70Hz | | | | |
| Correlation Matrix and Antenna Configuration | | 2x1 Low | | | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure A.7.3.27.1-1.

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (420 ms after the start of time duration T5) the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

# 7.3.28 E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category 0

## 7.3.28.1 Test purpose

The purpose of this test is to verify that the FD-FDD category 0 UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the serving cell when DRX is used. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in TS 36.133 section 7.11.

## 7.3.28.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward of UE category 0. Applicability requires support for FGI bit 5.

## 7.3.28.3 Minimum conformance requirements

The UE category 0 applicability of the requirements for performing radio link monitoring is defined in Section 3.1.

When DRX is used for FD-FDD and TDD category 0 UEs, the $Q_{out_Cat0}$ evaluation period ($T_{Evaluate_Q_{out_DRX_Cat0}}$) and the $Q_{in_Cat0}$ evaluation period ($T_{Evaluate_Q_{in_DRX_Cat0}}$) is specified in TS36.133 [4] Table 7.11.2.2-1 will be used.

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{out_DRX_Cat0}}$ [s] period becomes worse than the threshold $Q_{out_Cat0}$, Layer 1 of the UE shall send out-of-sync indication for the PCell to the higher layers within $T_{Evaluate_Q_{out_DRX_Cat0}}$ [s] evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{in_DRX_Cat0}}$ [s] period becomes better than the threshold $Q_{in_Cat0}$, Layer 1 of the UE shall send in-sync indications for the PCell to the higher layers within $T_{Evaluate_Q_{in_DRX_Cat0}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10ms, DRX_cycle_length).

Upon start of T310 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.28.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX for FD-FDD and TDD UE category 0**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX_Cat0}}$ and $T_{Evaluate_Q_{in_DRX_Cat0}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in TS36.133 [4] clause 7.11.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note (20) |
| 0.04 < DRX cycle ≤ 0.64 | Note (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note (5) |
| Note: Evaluation period length in time depends on the length of the DRX cycle in use | |

The normative reference for this requirement is TS 36.133 [4] clause 7.11.1, 7.11.2 and A.7.3.28.

## 7.3.28.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.28.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully

synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode PDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to "infinity" so that UL timing alignment is maintained during the test.

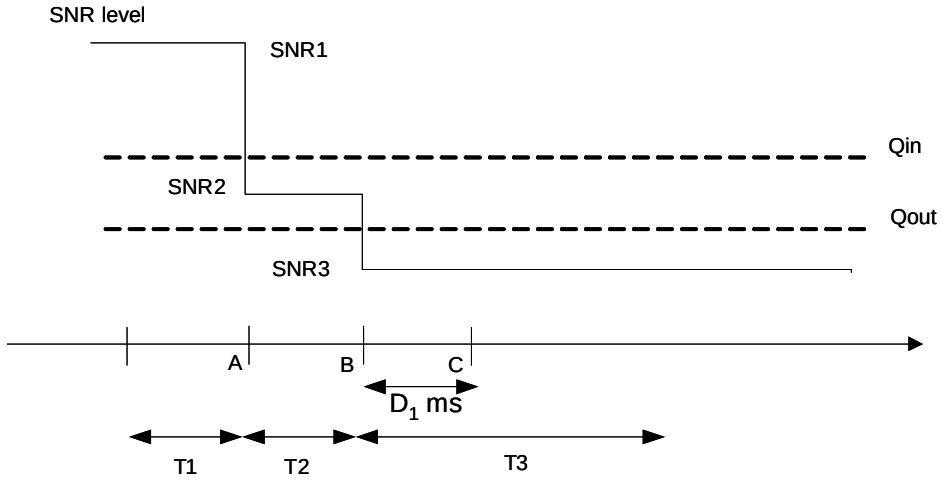


**Figure 7.3.28.4-1: SNR variation for out-of-sync testing in DRX**

### 7.3.28.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 (without using the faders) using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.28.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.28.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.28.4.1-1: General test parameters for E-UTRAN FD-FDD out-of-sync test in DRX for category 0**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 10 | |
| CP length | | | Normal | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | Out of sync threshold Qout and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] clause 7.11.1 and Table 7.11.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX cycle | | ms | 1280 | See Table 7.3.28.5-2 |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 2 | Minimum CQI reporting periodicity |
| T1 | | s | 32 | |
| T2 | | s | 12.8 | |
| T3 | | s | 13 | |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

## 7.3.28.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.28.5-1. Propagation conditions are set according to Annex B clause B.1. T1 starts.
3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.28.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.28.5-1. T3 starts.
5. If the SS:

   a) detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B

and

   b) does not detect any uplink power higher than -48.5 dBm from time point C (6500 ms after the start of T3)

until T3 expires,
the number of successful tests is increased by one.

Otherwise the number of failed tests is increased by one.

6. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.28.5-1.

7. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

8. Repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.28.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.28.4.3-1: Common Exception messages for E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for category 0 UE**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.28.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.28.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN FD-FDD Radio Link Monitoring test for Out-of-sync in DRX for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf1280 typical value in real network for best-effort services. | |
| sf1280 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.28.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN FD-FDD Radio Link Monitoring test for Out-of-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.28.4.3-5: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FD-FDD Radio Link Monitoring test for Out-of-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.28.4.3-6: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FD-FDD Radio Link Monitoring test for Out-of-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

## 7.3.28.5 Test requirement

**Table 7.3.28.5-1: Cell specific test parameters for E-UTRAN FD-FDD (cell # 1) for out-of-sync radio link monitoring tests in DRX for UE category 0**

| Parameter | Unit | Test 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| PCFICH/PDCCH/PHICH parameters defined in clause A.2.1 | | R.7 FDD | | |
| OCNG Pattern defined in D.1.2 (FDD) | | OP.2 FDD | | |
| $\rho_A$, $\rho_B$ | | -3 | | |
| PCFICH_RB | dB | 1 | | |
| PDCCH_RA | dB | 4 | | |
| PDCCH_RB | dB | 4 | | |
| PBCH_RA | dB | -3 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note1 | dB | | | |
| OCNG_RB Note1 | dB | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | |
| SNR Note 6 | dB | -5.5 | -9.4 | -14.6 |
| Propagation condition | | AWGN | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.28.4-1.

**Table 7.3.28.5-2: DRX-Configuration for E-UTRAN FD-FDD out-of-sync test in DRX for UE category 0**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf2 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf1280 | |
| shortDRX | disable | |

**Table 7.3.28.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FD-FDD out-of-sync test in DRX for UE category 0**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213. |

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the subframe according to the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal no later than time point C (duration $D_1$ = 6500 ms after the start of time duration T3.

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal
- UE output power equal to or less than Transmit OFF power -48.5 dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.29 E-UTRAN FD-FDD Radio Link Monitoring Test for In-sync in DRX for UE category 0

### 7.3.29.1 Test purpose

The purpose of this test is to verify that the FD-FDD category 0 UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the PCell when DRX is used. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in TS 36.133 [4] section 7.11.

### 7.3.29.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward of UE category 0. Applicability requires support for FGI bit 5.

### 7.3.29.3 Minimum conformance requirements

The UE category 0 applicability of the requirements for performing radio link monitoring is defined in Section 3.1.

When DRX is used for FD-FDD and TDD category 0 UEs, the $Q_{out_Cat0}$ evaluation period ($T_{Evaluate_Q_{out_DRX_Cat0}}$) and the $Q_{in_Cat0}$ evaluation period ($T_{Evaluate_Q_{in_DRX_Cat0}}$) is specified in TS36.133 [4] Table 7.11.2.2-1 will be used.

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{out_DRX_Cat0}}$ [s] period becomes worse than the threshold $Q_{out_Cat0}$, Layer 1 of the UE shall send out-of-sync indication for the PCell to the higher layers within

$T_{Evaluate_Q_{out_DRX_Cat0}}$ [s] evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{in_DRX_Cat0}}$ [s] period becomes better than the threshold $Q_{in_Cat0}$, Layer 1 of the UE shall send in-sync indications for the PCell to the higher layers within $T_{Evaluate_Q_{in_DRX_Cat0}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10ms, DRX_cycle_length).

Upon start of T310 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.29.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX for FD-FDD and TDD UE category 0**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX_Cat0}}$ and $T_{Evaluate_Q_{in_DRX_Cat0}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in TS36.133 [4] clause 7.11.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note (20) |
| 0.04 < DRX cycle ≤ 0.64 | Note (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note (5) |
| Note: Evaluation period length in time depends on the length of the DRX cycle in use | |

The normative reference for this requirement is TS 36.133 [4] clause 7.11.1, 7.11.2 and A.7.3.29.

## 7.3.29.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.29.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode PDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to “infinity” so that UL timing alignment is maintained during the test.

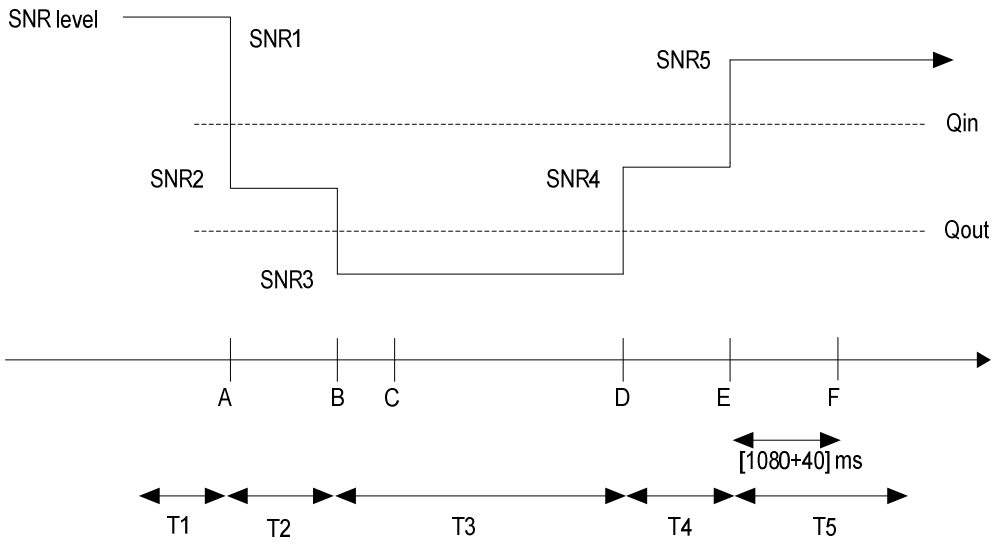


**Figure 7.3.29.4-1: SNR variation for in-sync testing in DRX**

### 7.3.29.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 (without using the faders) using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.29.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.29.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.29.4.1-1: General test parameters for E-UTRAN FD-FDD in-sync test in DRX for category 0**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| CP length | | | Normal | |
| In sync transmission parameters (Note 1) | DCI format | | 1C | As defined in clause 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | In sync threshold $Q_{in}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] clause 7.11.1 and Table 7.11.1-2 respectively. |
| | Aggregation level | CCE | 4 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | | 1 | |
| | Ratio of PCFICH to RS EPRE | | 1 | |

| | | | | |
|---|---|---|---|---|
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] clause 7.11.1 and Table 7.11.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX cycle | | ms | 40 | See Table 7.3.29.5-2 |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 2 | Minimum CQI reporting periodicity |
| T1 | | s | 4 | |
| T2 | | s | 1.6 | |
| T3 | | s | 1.46 | |
| T4 | | s | 0.4 | |
| T5 | | s | 4 | |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

### 7.3.29.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.29.5-1. Propagation conditions are set according to Annex B.1. T1 starts.
3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.29.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.29.5-1. T3 starts.
5. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.29.5-1. T4 starts.
6. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.29.5-1. T5 starts.
7. If the SS detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (1120 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
8. After T5 expires, repeat steps 2-7 for both subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.29.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.29.4.3-1: Common Exception messages for E-UTRAN FD-FDD Radio Link Monitoring Test for In-sync in DRX for category 0 UE**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.29.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FD-FDD Radio Link Monitoring Test for In-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.29.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN FD-FDD Radio Link Monitoring test for In-sync in DRX for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf40 typical value in real network for real-time services. | |
| sf40 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.29.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN FD-FDD Radio Link Monitoring test for In-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.29.4.3-5: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FD-FDD Radio Link Monitoring test for In-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.29.4.3-6: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FD-FDD Radio Link Monitoring test for In-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

## 7.3.29.5 Test requirement

**Table 7.3.29.5-1: Cell specific test parameters for E-UTRAN FD-FDD in-sync Radio Link Monitoring Test in DRX for UE category 0**

| Parameter | Unit | Test 1 | | | | |
|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | |
| PCFICH/PDCCH/PHICH parameters defined in clause A.2.1 | | R.7 FDD | | | | |
| OCNG Pattern defined in D.1.2 (FDD) | | OP.2 FDD | | | | |
| $\rho_A$, $\rho_B$ | | -3 | | | | |
| PCFICH_RB | dB | 1 | | | | |
| PDCCH_RA | dB | 1 | | | | |
| PDCCH_RB | dB | 1 | | | | |
| PBCH_RA | dB | -3 | | | | |
| PBCH_RB | dB | | | | | |
| PSS_RA | dB | | | | | |
| SSS_RA | dB | | | | | |
| PHICH_RA | dB | | | | | |
| PHICH_RB | dB | | | | | |
| PDSCH_RA | dB | | | | | |
| PDSCH_RB | dB | | | | | |
| OCNG_RA Note1 | dB | | | | | |
| OCNG_RB Note1 | dB | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | |
| SNR Note 8 | dB | -5.5 | -9.4 | -14.6 | -10.7 | -5.5 |
| Propagation condition | | AWGN | | | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.29.4-1.

**Table 7.3.29.5-2: DRX-Configuration for E-UTRAN FD-FDD in-sync radio link monitoring tests in DRX for UE category 0**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf2 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf40 | |
| shortDRX | disable | |

**Table 7.3.29.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FD-FDD in-sync radio link monitoring test in DRX for UE category 0**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213. |

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (1120 ms after the start of time duration T5) the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the subframe according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.30 E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync for UE category 0

### 7.3.30.1 Test purpose

The purpose of this test is to verify that the HD-FDD category 0 UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the PCell. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in TS 36.133 [4] section 7.11.

### 7.3.30.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 12 and forward of UE Category 0.

### 7.3.30.3 Minimum conformance requirements

The UE category 0 applicability of the requirements for performing radio link monitoring is defined in Section 3.1.

When no DRX is used: The HD-FDD category 0 UE shall meet all applicable requirements specified in TS.36.133 [4] clause 7.11.2.1 under the following conditions

- at least 1 DL subframe per radio frame of PCell is available at the UE during $Q_{in_Cat0}$ and $Q_{out_Cat0}$ evaluation periods.

When the downlink radio link quality of the PCell estimated over the last 200 ms period becomes worse than the threshold $Q_{out_Cat0}$, Layer 1 of the UE shall send an out-of-sync indication for the PCell to the higher layers within

200ms $Q_{out_Cat0}$ evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last 100 ms period becomes better than the threshold $Q_{in_Cat0}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 100 ms $Q_{in_Cat0}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least 10ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

The normative reference for this requirement is TS 36.133 [4] clause 7.11.1, 7.11.2, 7.11.3 and A.7.3.30.

### 7.3.30.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.30.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync state. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 10 ms.

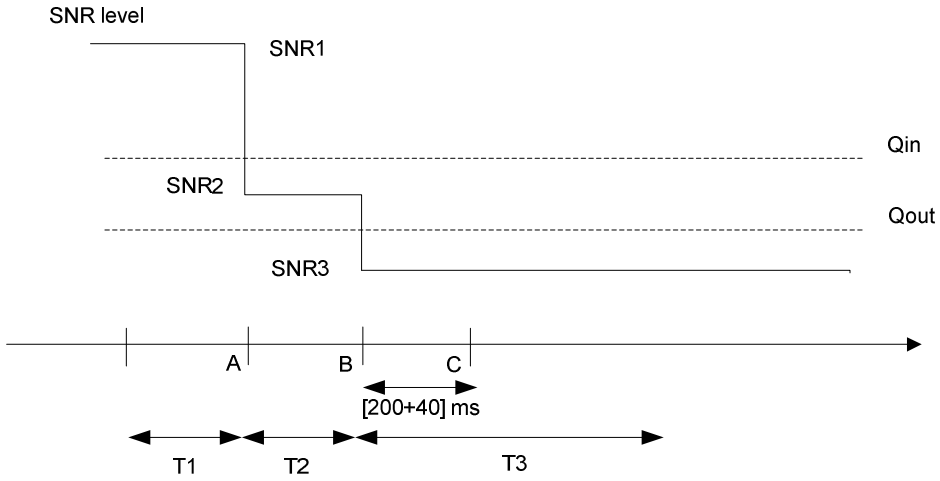


**Figure 7.3.30.4-1: SNR variation for out-of-sync testing**

#### 7.3.30.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.30.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.30.4.3.
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.30.4.1-1: General test parameters for E-UTRAN HD-FDD out-of-sync test for category 0**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| CP length | | | Normal | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] in section 7.11.1 and Table 7.11.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX | | | OFF | |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 10 | Minimum CQI reporting periodicity |
| T1 | | s | 1 | |
| T2 | | s | 0.4 | |
| T3 | | s | 0.5 | |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

## 7.3.30.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of [TBD] ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 7.3.30.5-1. Propagation conditions are set according to Annex B clause B.2. T1 starts.

3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.30.5-1. T2 starts.

4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.30.5-1. T3 starts.

5. If the SS:
   a) detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B

and

   b) does not detect any uplink power higher than -48.5 dBm from time point C (240 ms after the start of T3) until T3 expires,
   the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.

6. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.30.5-1.

7. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

8. Repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.30.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.30.4.3-1: Common Exception messages for E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync for UE category 0**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.30.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 7 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.30.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| } | | | |

**Table 7.3.30.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

## 7.3.30.5 Test requirement

**Table 7.3.30.5-1: Cell specific test parameters for E-UTRAN HD-FDD (cell # 1) for out-of-sync radio link monitoring for UE category 0**

| Parameter | Unit | Test 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| PCFICH/PDCCH/PHICH parameters defined in A.2.3 | | R.4 HD-FDD | | |
| OCNG Pattern defined in D.1.2 (FDD) | | OP.2 FDD | | |
| $\rho_A$, $\rho_B$ | | -3 | | |
| PCFICH_RB | dB | 1 | | |
| PDCCH_RA | dB | 4 | | |
| PDCCH_RB | dB | 4 | | |
| PBCH_RA | dB | -3 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | |
| SNR Note 6 | dB | -0.3 | -5.1 | -12.9 |
| Propagation condition | | ETU 70Hz | | |
| Correlation Matrix and Antenna Configuration | | 2x1 Low | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.30.4-1.

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal no later than time point C (240ms after the start of time duration T3).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal.

- UE output power equal to or less than Transmit OFF power -48.5 dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.31 E-UTRAN HD-FDD Radio Link Monitoring Test for In-sync for UE category 0

### 7.3.31.1 Test purpose

The purpose of this test is to verify that the HD-FDD category 0 UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the PCell. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in TS 36.133 [4] section 7.11.

### 7.3.31.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 12 and forward of UE category 0.

### 7.3.31.3 Minimum conformance requirements

The UE category 0 applicability of the requirements for performing radio link monitoring is defined in Section 3.1.

When no DRX is used: The HD-FDD category 0 UE shall meet all applicable requirements specified in TS.36.133 [4] clause 7.11.2.1 under the following conditions:

- at least 1 DL subframe per radio frame of PCell is available at the UE during $Q_{in_Cat0}$ and $Q_{out_Cat0}$ evaluation periods.

When the downlink radio link quality of the PCell estimated over the last 200 ms period becomes worse than the threshold $Q_{out_Cat0}$, Layer 1 of the UE shall send an out-of-sync indication for the PCell to the higher layers within 200ms $Q_{out_Cat0}$ evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last 100 ms period becomes better than the threshold $Q_{in_Cat0}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 100 ms $Q_{in_Cat0}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least 10ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

The normative reference for this requirement is TS 36.133 [4] clause 7.11.1, 7.11.2, 7.11.3 and A.7.3.31.

### 7.3.31.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.31.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 10 ms.

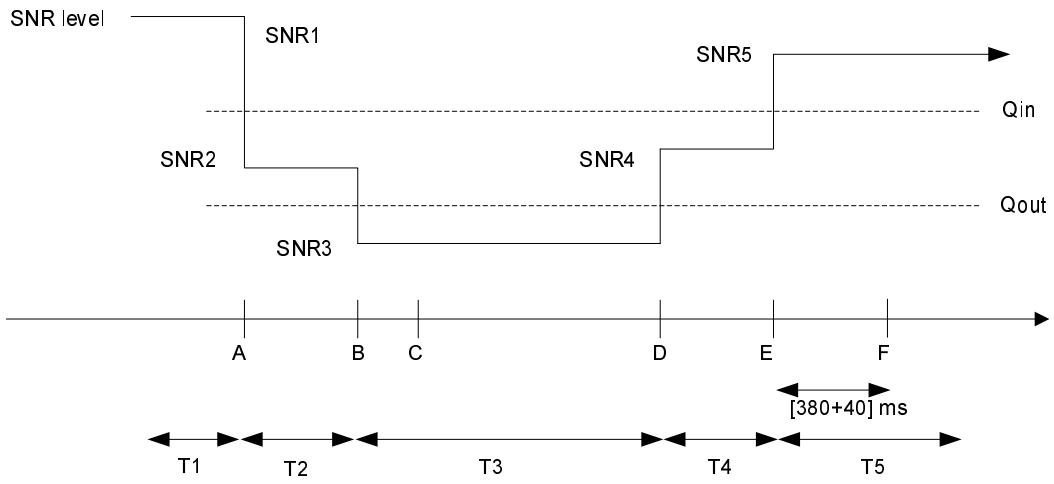


**Figure 7.3.31.4-1: SNR variation for in-sync testing**

### 7.3.31.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.31.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.31.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.31.4.1-1: General test parameters for E-UTRAN HD-FDD Radio Link Monitoring in-sync test for UE category 0**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| CP length | | | Normal | |
| In sync transmis-sion parameters (Note 1) | DCI format | | 1C | As defined in clause 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | In sync threshold $Q_{in}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS36.133 [4] clause 7.11.1 and Table 7.11.1-2 respectively. |
| | Aggregation level | CCE | 4 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| Out of sync transmis-sion parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS36.133 [4] clause 7.11.1 and Table 7.11.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX | | | OFF | |
| Layer 3 filtering | | | *Enabled* | *Counters: N310 = 1; N311 = 1* |
| T310 timer | | *ms* | *2000* | *T310 is enabled* |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 10 | Minimum CQI reporting periodicity |
| T1 | | s | 0.5 | |
| T2 | | s | 0.4 | |
| T3 | | s | 1.46 | |
| T4 | | s | 0.4 | |
| T5 | | s | 1 | |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

## 7.3.31.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of [TBD] ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.31.5-1. Propagation conditions are set according to Annex B.2. T1 starts.

3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.31.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.31.5-1. T3 starts.
5. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.31.5-1. T4 starts.
6. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.31.5-1. T5 starts.
7. If the SS detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (420 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
8. After T5 expires, repeat steps 2-7 for both subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.31.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.31.4.3-1: Common Exception messages for E-UTRAN HD-FDD Radio Link Monitoring Test for in-sync for UE category 0**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.31.4.3-2 : *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN HD-FDD Radio Link Monitoring Test for in-sync for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 7 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.31.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN HD-FDD Radio Link Monitoring Test for in-sync for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.31.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN HD-FDD Radio Link Monitoring Test for in-sync for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

## 7.3.31.5 Test requirement

**Table 7.3.31.5-1: Cell specific test parameters for E-UTRAN HD-FDD in-sync Radio Link Monitoring Test for UE category 0**

| Parameter | Unit | Test 1 | | | | |
|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 |

| | | | | | | |
|---|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | |
| PCFICH/PDCCH/PHICH parameters defined in clause A.2.3 | | R.4 HD-FDD | | | | |
| OCNG Pattern defined in D.1.2 (FDD) | | OP.2 FDD | | | | |
| $\rho_A$, $\rho_B$ | | -3 | | | | |
| PCFICH_RB | dB | 1 | | | | |
| PDCCH_RA | dB | 1 | | | | |
| PDCCH_RB | dB | 1 | | | | |
| PBCH_RA | dB | -3 | | | | |
| PBCH_RB | dB | | | | | |
| PSS_RA | dB | | | | | |
| SSS_RA | dB | | | | | |
| PHICH_RA | dB | | | | | |
| PHICH_RB | dB | | | | | |
| PDSCH_RA | dB | | | | | |
| PDSCH_RB | dB | | | | | |
| OCNG_RA Note 1 | dB | | | | | |
| OCNG_RB Note 1 | dB | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | |
| SNR Note 6 | dB | -0.3 | -5.1 | -12.9 | -7.1 | -0.3 |
| Propagation condition | | ETU 70Hz | | | | |
| Correlation Matrix and Antenna Configuration | | 2x1 Low | | | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure A.7.3.27.1-1.

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (420 ms after the start of time duration T5) the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.32 E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category 0

### 7.3.32.1 Test purpose

The purpose of this test is to verify that the HD-FDD category 0 UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the serving cell when DRX is used. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in TS 36.133 section 7.11.

### 7.3.32.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 12 and forward of UE category 0. Applicability requires support for FGI bit 5.

### 7.3.32.3 Minimum conformance requirements

The UE category 0 applicability of the requirements for performing radio link monitoring is defined in Section 3.1.

When DRX is used for HD-FDD category 0 UEs, the $Q_{out}$ evaluation period ($T_{Evaluate_Q_{out_DRX_Cat0}}$) and the $Q_{in}$ evaluation period ($T_{Evaluate_Q_{in_DRX_Cat0}}$) specified in Table 7.3.32.3-1 will be used.

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{out_DRX_Cat0}}$ [s] period becomes worse than the threshold $Q_{out_Cat0}$, Layer 1 of the UE shall send out-of-sync indication for the PCell to the higher layers within $T_{Evaluate_Q_{out_DRX_Cat0}}$ [s] evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{in_DRX_Cat0}}$ [s] period becomes better than the threshold $Q_{in_Cat0}$, Layer 1 of the UE shall send in-sync indications for the PCell to the higher layers within $T_{Evaluate_Q_{in_DRX_Cat0}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10ms, DRX_cycle_length).

Upon start of T310 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.32.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX for HD-FDD UE category 0**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX}}$ and $T_{Evaluate_Q_{in_DRX}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in TS36.133 clause 7.11.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note (40) |
| 0.04 < DRX cycle ≤ 0.16 | Note (20) |
| 0.16 < DRX cycle ≤ 0.64 | Note (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note (5) |
| Note: Evaluation period length in time depends on the length of the DRX cycle in use | |

The normative reference for this requirement is TS 36.133 [4] clause 7.11.1, 7.11.2, 7.11.3 and A.7.3.32.

### 7.3.32.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.32.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 5 ms. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode PDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to “infinity” so that UL timing alignment is maintained during the test.

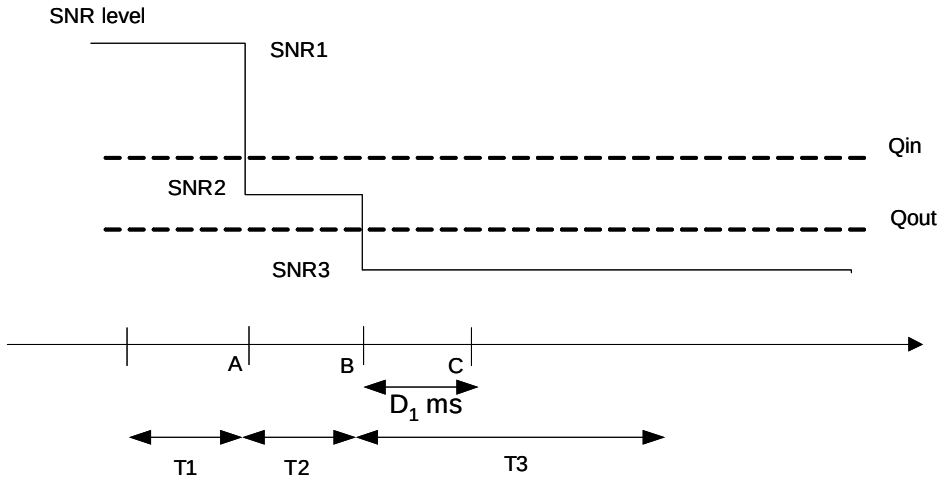


**Figure 7.3.32.4-1: SNR variation for out-of-sync testing in DRX**

### 7.3.32.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 (without using the faders) using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.32.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.32.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.32.4.1-1: General test parameters for E-UTRAN HD-FDD out-of-sync test in DRX for category 0**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 10 | |
| CP length | | | Normal | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | Out of sync threshold Qout and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] clause 7.11.1 and Table 7.11.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX cycle | | ms | 1280 | See Table 7.3.32.4-2 |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 5 | Minimum CQI reporting periodicity |
| T1 | | s | 32 | |
| T2 | | s | 12.8 | |
| T3 | | s | 13 | |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

## 7.3.32.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 5 ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.32.5-1. Propagation conditions are set according to Annex B clause B.1. T1 starts.
3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.32.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.32.5-1. T3 starts.
5. If the SS:
   a) detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B

and

   b) does not detect any uplink power higher than -48.5 dBm from time point C (6500 ms after the start of T3) until T3 expires,
   the number of successful tests is increased by one.

Otherwise the number of failed tests is increased by one.

6. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.32.5-1.
7. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
8. Repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.32.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.32.4.3-1: Common Exception messages for E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for category 0 UE**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.32.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 2 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.32.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN HD-FDD Radio Link Monitoring test for Out-of-sync in DRX for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf1280 typical value in real network for best-effort services. | |
| sf1280 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.32.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN HD-FDD Radio Link Monitoring test for Out-of-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.32.4.3-5: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN HD-FDD Radio Link Monitoring test for Out-of-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.32.4.3-6: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN HD-FDD Radio Link Monitoring test for Out-of-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

## 7.3.32.5 Test requirement

**Table 7.3.32.5-1: Cell specific test parameters for E-UTRAN HD-FDD (cell # 1) for out-of-sync radio link monitoring tests in DRX for UE category 0**

| Parameter | Unit | Test 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| PCFICH/PDCCH/PHICH parameters defined in clause A.2.3 | | R.4 HD-FDD | | |
| OCNG Pattern defined in D.1.2 (FDD) | | OP.2 FDD | | |
| $\rho_A$, $\rho_B$ | | -3 | | |
| PCFICH_RB | dB | 1 | | |
| PDCCH_RA | dB | 4 | | |
| PDCCH_RB | dB | 4 | | |
| PBCH_RA | dB | -3 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note1 | dB | | | |
| OCNG_RB Note1 | dB | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | |
| SNR Note 6 | dB | -4.8 | -8.9 | -14.1 |
| Propagation condition | | AWGN | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.28.4-1.

**Table 7.3.32.5-2: DRX-Configuration for E-UTRAN HD-FDD out-of-sync test in DRX for UE category 0**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | Psf5 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf1280 | |
| shortDRX | disable | |

**Table 7.3.32.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN HD-FDD out-of-sync test in DRX for UE category 0**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213. |

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the subframe according to the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal no later than time point C (duration $D_1$ = 6500 ms after the start of time duration T3.

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal
- UE output power equal to or less than Transmit OFF power -48.5 dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.33 E-UTRAN HD-FDD Radio Link Monitoring Test for In-sync in DRX for UE category 0

### 7.3.33.1 Test purpose

The purpose of this test is to verify that the HD-FDD category 0 UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the PCell when DRX is used. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in TS 36.133 [4] section 7.11.

### 7.3.33.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 12 and forward of UE category 0. Applicability requires support for FGI bit 5.

### 7.3.33.3 Minimum conformance requirements

The UE category 0 applicability of the requirements for performing radio link monitoring is defined in Section 3.1.

When DRX is used for HD-FDD category 0 UEs, the $Q_{out}$ evaluation period ($T_{Evaluate_Q_{out_DRX_Cat0}}$) and the $Q_{in}$ evaluation period ($T_{Evaluate_Q_{in_DRX_Cat0}}$) specified in Table 7.3.33.3-1 will be used.

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{out_DRX_Cat0}}$ [s] period becomes worse than the threshold $Q_{out_Cat0}$, Layer 1 of the UE shall send out-of-sync indication for the PCell to the higher layers within

$T_{Evaluate_Q_{out_DRX_Cat0}}$ [s] evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{in_DRX_Cat0}}$ [s] period becomes better than the threshold $Q_{in_Cat0}$, Layer 1 of the UE shall send in-sync indications for the PCell to the higher layers within $T_{Evaluate_Q_{in_DRX_Cat0}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10ms, DRX_cycle_length).

Upon start of T310 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.33.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX for HD-FDD UE category 0**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX}}$ and $T_{Evaluate_Q_{in_DRX}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in TS36.133 clause 7.11.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note (40) |
| 0.04 < DRX cycle ≤ 0.16 | Note (20) |
| 0.16 < DRX cycle ≤ 0.64 | Note (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note (5) |
| Note: Evaluation period length in time depends on the length of the DRX cycle in use | |

The normative reference for this requirement is TS 36.133 [4] clause 7.11.1, 7.11.2, 7.11.3 and A.7.3.33.

## 7.3.33.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.29.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 5 ms. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode PDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to “infinity” so that UL timing alignment is maintained during the test.

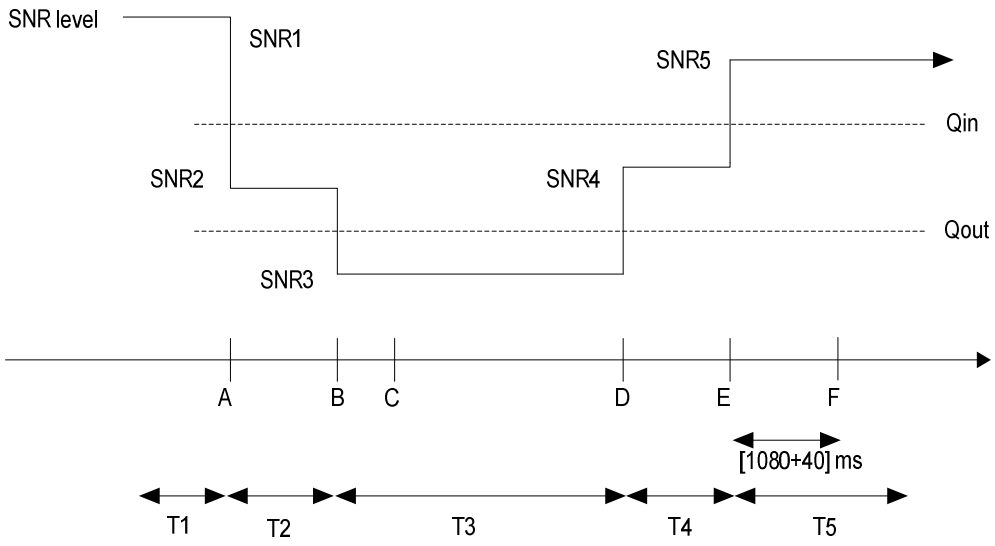


**Figure 7.3.33.1-1: SNR variation for in-sync testing in DRX**

### 7.3.33.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 (without using the faders) using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.33.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.33.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.33.4.1-1: General test parameters for E-UTRAN HD-FDD in-sync test in DRX for category 0**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| CP length | | | Normal | |
| In sync transmission parameters (Note 1) | DCI format | | 1C | As defined in clause 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | In sync threshold $Q_{in}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] clause 7.11.1 and Table 7.11.1-2 respectively. |
| | Aggregation level | CCE | 4 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | | 1 | |
| | Ratio of PCFICH to RS EPRE | | 1 | |

| | | | | |
|---|---|---|---|---|
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] clause 7.11.1 and Table 7.11.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX cycle | | ms | 40 | See Table 7.3.29.1-3 |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 5 | Minimum CQI reporting periodicity |
| T1 | | s | 4 | |
| T2 | | s | 1.6 | |
| T3 | | s | 1.46 | |
| T4 | | s | 0.4 | |
| T5 | | s | 4 | |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

### 7.3.33.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 5 ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.33.5-1. Propagation conditions are set according to Annex B.1. T1 starts.
3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.33.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.33.5-1. T3 starts.
5. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.33.5-1. T4 starts.
6. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.33.5-1. T5 starts.
7. If the SS detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (1120 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
8. After T5 expires, repeat steps 2-7 for both subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.33.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.33.4.3-1: Common Exception messages for E-UTRAN HD-FDD Radio Link Monitoring Test for In-sync in DRX for category 0 UE**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.33.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN HD-FDD Radio Link Monitoring Test for In-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 2 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.33.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN HD-FDD Radio Link Monitoring test for In-sync in DRX for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | Psf5 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf40 typical value in real network for real-time services. | |
| sf40 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.33.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN HD-FDD Radio Link Monitoring test for In-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.33.4.3-5: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN HD-FDD Radio Link Monitoring test for In-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.33.4.3-6: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN HD-FDD Radio Link Monitoring test for In-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

## 7.3.33.5 Test requirement

**Table 7.3.33.5-1: Cell specific test parameters for E-UTRAN HD-FDD (cell # 1) for In-sync radio link monitoring tests in DRX for UE category 0**

| Parameter | Unit | Test 1 | | | | |
|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | |
| PCFICH/PDCCH/PHICH parameters specified in clause A.2.3 | | R.4 HD-FDD | | | | |
| OCNG Pattern defined in D.1.2 (FDD) | | OP.2 FDD | | | | |
| $\rho_A$, $\rho_B$ | | -3 | | | | |
| PCFICH_RB | dB | 1 | | | | |
| PDCCH_RA | dB | 1 | | | | |
| PDCCH_RB | dB | 1 | | | | |
| PBCH_RA | dB | -3 | | | | |
| PBCH_RB | dB | | | | | |
| PSS_RA | dB | | | | | |
| SSS_RA | dB | | | | | |
| PHICH_RA | dB | | | | | |
| PHICH_RB | dB | | | | | |
| PDSCH_RA | dB | | | | | |
| PDSCH_RB | dB | | | | | |
| OCNG_RA Note1 | dB | | | | | |
| OCNG_RB Note1 | dB | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | |
| SNR Note 6 | dB | -4.8 | -8.9 | -14.1 | -10.0 | -4.8 |
| Propagation condition | | AWGN | | | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure A.7.3.33.1-1.

**Table 7.3.33.5-2: DRX-Configuration for E-UTRAN HD-FDD in-sync radio link monitoring tests in DRX for UE category 0**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf5 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf40 | |
| shortDRX | disable | |

**Table 7.3.33.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN HD-FDD in-sync radio link monitoring tests in DRX for UE category 0**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213. |

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (1120 ms after the start of time duration T5) the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the subframe according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.34 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync for UE category 0

### 7.3.34.1 Test purpose

The purpose of this test is to verify that the TDD category 0 UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the PCell. This test will partly verify the E-UTRAN TDD radio link monitoring requirements in TS 36.133 [4] section 7.11.

### 7.3.34.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward of UE Category 0.

### 7.3.34.3 Minimum conformance requirements

The UE category 0 applicability of the requirements for performing radio link monitoring is defined in Section 3.1.

When the downlink radio link quality of the PCell estimated over the last 200 ms period becomes worse than the threshold $Q_{out_Cat0}$, Layer 1 of the UE shall send an out-of-sync indication for the PCell to the higher layers within 200ms $Q_{out_Cat0}$ evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last 100 ms period becomes better than the threshold $Q_{in_Cat0}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 100 ms $Q_{in_Cat0}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least 10ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

The normative reference for this requirement is TS 36.133 [4] clause 7.11.1, 7.11.2 and A.7.3.34.

### 7.3.34.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.26.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync state. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

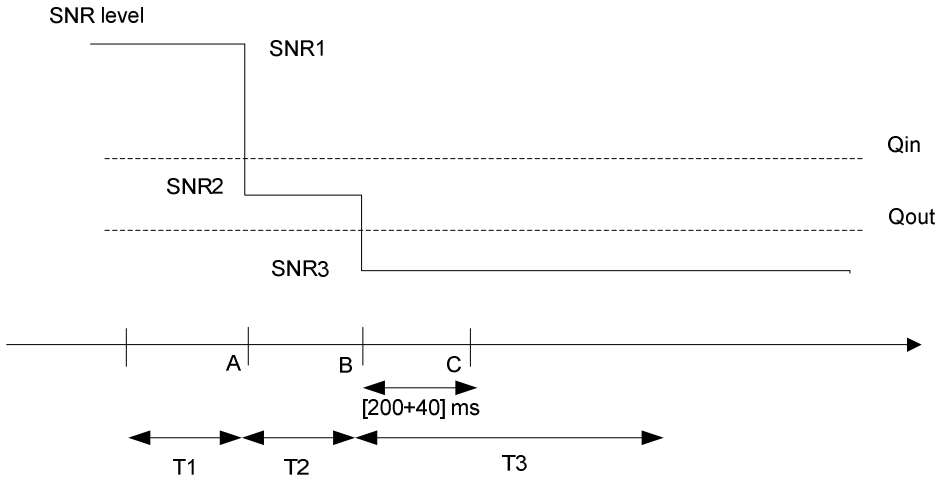


**Figure 7.3.34.4-1: SNR variation for out-of-sync testing**

#### 7.3.34.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.34.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.34.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.34.4.1-1: General test parameters for E-UTRAN TDD out-of-sync test for category 0**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| CP length | | | Normal | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] in section 7.11.1 and Table 7.11.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX | | | OFF | |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 1 | Minimum CQI reporting periodicity |
| T1 | | s | 1 | |
| T2 | | s | 0.4 | |
| T3 | | s | 0.5 | |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

## 7.3.34.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.34.5-1. Propagation conditions are set according to Annex B clause B.1. T1 starts.
3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.34.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.34.5-1. T3 starts.
5. If the SS:
   a) detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B

and

   b) does not detect any uplink power higher than -48.5 dBm from time point C (240 ms after the start of T3) until T3 expires,
   the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.

6. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.34.5-1.

7. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

8. Repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.34.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.34.4.3-1: Common Exception messages for E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync for UE category 0**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.34.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.34.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.34.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

## 7.3.34.5 Test requirement

**Table 7.3.34.5-1: Cell specific test parameters for E-UTRAN TDD (cell # 1) for out-of-sync radio link monitoring for UE category 0**

| Parameter | Unit | Test 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| Special subframe configuration Note 1 | | 6 | | |
| Uplink-downlink configuration Note 1 | | 1 | | |
| PCFICH/PDCCH/PHICH parameters defined in A.2.1 | | R.7 TDD | | |
| OCNG Pattern defined in D.1.2 (TDD) | | OP.2 TDD | | |
| $\rho_A$, $\rho_B$ | | -3 | | |
| PCFICH_RB | dB | 1 | | |
| PDCCH_RA | dB | 4 | | |
| PDCCH_RB | dB | 4 | | |
| PBCH_RA | dB | -3 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | |
| SNR Note 6 | dB | -0.7 | -5.0 | -12.8 |
| Propagation condition | | ETU 70Hz | | |
| Correlation Matrix and Antenna Configuration | | 2x1 Low | | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 4: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 5: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 6: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 7: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.34.4-1.

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal no later than time point C (240ms after the start of time duration T3).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal
- UE output power equal to or less than Transmit OFF power -48.5 dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.35 E-UTRAN TDD Radio Link Monitoring Test for In-sync for UE category 0

### 7.3.35.1 Test purpose

The purpose of this test is to verify that the TDD category 0 UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the PCell. This test will partly verify the E-UTRAN TDD radio link monitoring requirements in TS 36.133 [4] section 7.11.

### 7.3.35.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward of UE category 0.

### 7.3.35.3 Minimum conformance requirements

The UE category 0 applicability of the requirements for performing radio link monitoring is defined in Section 3.1.

When the downlink radio link quality of the PCell estimated over the last 200 ms period becomes worse than the threshold $Q_{out_Cat0}$, Layer 1 of the UE shall send an out-of-sync indication for the PCell to the higher layers within 200ms $Q_{out_Cat0}$ evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last 100 ms period becomes better than the threshold $Q_{in_Cat0}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 100 ms $Q_{in_Cat0}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least 10ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

The normative reference for this requirement is TS 36.133 [4] clause 7.11.1, 7.11.2 and A.7.3.35.

### 7.3.35.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.35.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms.

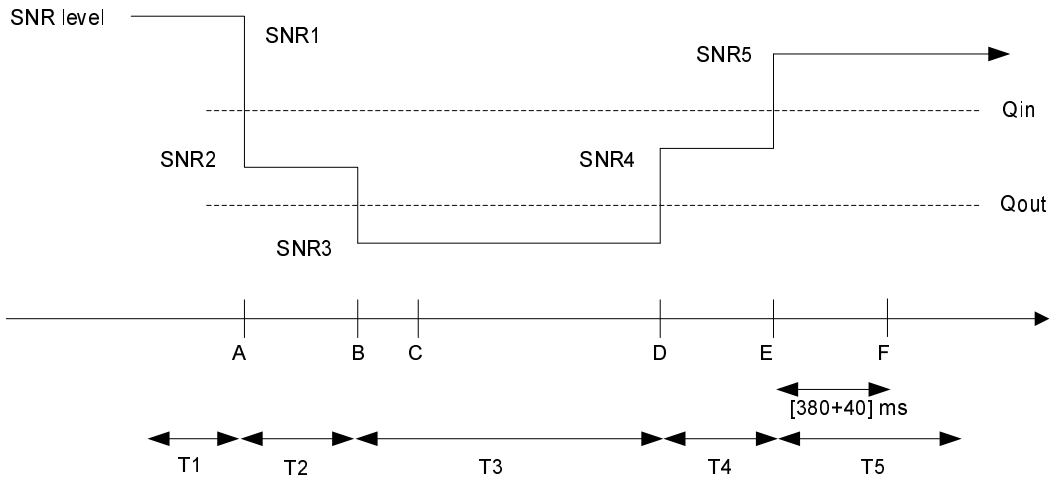


**Figure 7.3.35.4-1: SNR variation for in-sync testing**

### 7.3.35.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.35.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.35.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.35.4.1-1: General test parameters for E-UTRAN TDD Radio Link Monitoring in-sync test for UE category 0**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| CP length | | | Normal | |
| In sync transmis-sion parameters (Note 1) | DCI format | | 1C | As defined in clause 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | In sync threshold $Q_{in}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS36.133 [4] clause 7.11.1 and Table 7.11.1-2 respectively. |
| | Aggregation level | CCE | 4 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| Out of sync transmis-sion parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS36.133 [4] clause 7.11.1 and Table 7.11.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX | | | OFF | |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 1 | Minimum CQI reporting periodicity |
| T1 | | s | 0.5 | |
| T2 | | s | 0.4 | |
| T3 | | s | 1.46 | |
| T4 | | s | 0.4 | |
| T5 | | s | 1 | |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

## 7.3.35.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 7.3.35.5-1. Propagation conditions are set according to Annex B.2. T1 starts.

3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.35.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.35.5-1. T3 starts.
5. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.35.5-1. T4 starts.
6. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.35.5-1. T5 starts.
7. If the SS detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (420 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
8. After T5 expires, repeat steps 2-7 for both subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.35.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.35.4.3-1: Common Exception messages for E-UTRAN TDD Radio Link Monitoring Test for in-sync for UE category 0**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.35.4.3-2 : *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for in-sync for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.35.4.3-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for in-sync for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.35.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN TDD Radio Link Monitoring Test for in-sync for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

## 7.3.35.5 Test requirement

**Table 7.3.35.5-1: Cell specific test parameters for E-UTRAN TDD in-sync Radio Link Monitoring Test for UE category 0**

| Parameter | Unit | Test 1 | | | | |
|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | |
| Special subframe configuration Note 1 | | 6 | | | | |
| Uplink-downlink configuration Note 1 | | 1 | | | | |
| PCFICH/PDCCH/PHICH parameters defined in clause A.2.1 | | R.7 TDD | | | | |
| OCNG Pattern defined in D.1.2 (TDD) | | OP.2 TDD | | | | |
| $\rho_A$, $\rho_B$ | | -3 | | | | |
| PCFICH_RB | dB | 1 | | | | |
| PDCCH_RA | dB | 1 | | | | |
| PDCCH_RB | dB | 1 | | | | |
| PBCH_RA | dB | -3 | | | | |
| PBCH_RB | dB | | | | | |
| PSS_RA | dB | | | | | |
| SSS_RA | dB | | | | | |
| PHICH_RA | dB | | | | | |
| PHICH_RB | dB | | | | | |
| PDSCH_RA | dB | | | | | |
| PDSCH_RB | dB | | | | | |
| OCNG_RA Note 1 | dB | | | | | |
| OCNG_RB Note 1 | dB | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | |
| SNR Note 6 | dB | -0.7 | -5.0 | -12.8 | -7.5 | -0.7 |
| Propagation condition | | ETU 70Hz | | | | |
| Correlation Matrix and Antenna Configuration | | 2x1 Low | | | | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 4: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 5: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 6: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 7: The SNR in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.35.4-1.

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (420 ms after the start of time duration T5) the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.36 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category 0

### 7.3.36.1 Test purpose

The purpose of this test is to verify that the TDD category 0 UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the serving cell when DRX is used. This test will partly verify the E-UTRAN TDD radio link monitoring requirements in TS 36.133 section 7.11.

### 7.3.36.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward of UE category 0. Applicability requires support for FGI bit 5.

### 7.3.36.3 Minimum conformance requirements

The UE category 0 applicability of the requirements for performing radio link monitoring is defined in Section 3.1.

When DRX is used for FD-FDD and TDD category 0 UEs, the $Q_{out_Cat0}$ evaluation period ($T_{Evaluate_Q_{out_DRX_Cat0}}$) and the $Q_{in_Cat0}$ evaluation period ($T_{Evaluate_Q_{in_DRX_Cat0}}$) is specified in TS36.133 [4] Table 7.11.2.2-1 will be used.

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{out_DRX_Cat0}}$ [s] period becomes worse than the threshold $Q_{out_Cat0}$, Layer 1 of the UE shall send out-of-sync indication for the PCell to the higher layers within $T_{Evaluate_Q_{out_DRX_Cat0}}$ [s] evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{in_DRX_Cat0}}$ [s] period becomes better than the threshold $Q_{in_Cat0}$, Layer 1 of the UE shall send in-sync indications for the PCell to the higher layers within $T_{Evaluate_Q_{in_DRX_Cat0}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10ms, DRX_cycle_length).

Upon start of T310 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.36.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX for TDD and TDD UE category 0**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX_Cat0}}$ and $T_{Evaluate_Q_{in_DRX_Cat0}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in TS36.133 [4] clause 7.11.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note (20) |
| 0.04 < DRX cycle ≤ 0.64 | Note (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note (5) |
| Note: Evaluation period length in time depends on the length of the DRX cycle in use | |

The normative reference for this requirement is TS 36.133 [4] clause 7.11.1, 7.11.2 and A.7.3.36.

### 7.3.36.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.36.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode PDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to "infinity" so that UL timing alignment is maintained during the test.

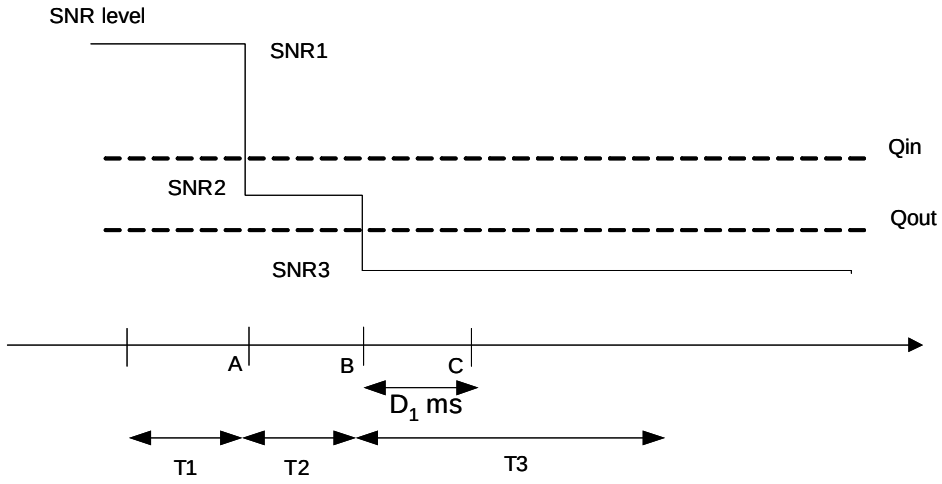


**Figure 7.3.36.4-1: SNR variation for out-of-sync testing in DRX**

#### 7.3.36.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 (without using the faders) using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.28.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.28.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.36.4.1-1: General test parameters for E-UTRAN TDD out-of-sync test in DRX for category 0**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 10 | |
| CP length | | | Normal | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | Out of sync threshold Qout and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] clause 7.11.1 and Table 7.11.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX cycle | | ms | 1280 | See Table 7.3.28.1-3 |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 1 | Minimum CQI reporting periodicity |
| T1 | | s | 32 | |
| T2 | | s | 12.8 | |
| T3 | | s | 13 | |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

## 7.3.36.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.36.5-1. Propagation conditions are set according to Annex B clause B.1. T1 starts.
3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.36.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.36.5-1. T3 starts.
5. If the SS:
   a) detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B

and

   b) does not detect any uplink power higher than -48.5 dBm from time point C (6500 ms after the start of T3) until T3 expires,
   the number of successful tests is increased by one.

Otherwise the number of failed tests is increased by one.

6. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.36.5-1.
7. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
8. Repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.36.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.36.4.3-1: Common Exception messages for E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync in DRX for category 0 UE**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.36.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.36.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN TDD Radio Link Monitoring test for Out-of-sync in DRX for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf1280 typical value in real network for best-effort services. | |
| sf1280 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.36.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN TDD Radio Link Monitoring test for Out-of-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.36.4.3-5: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring test for Out-of-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.36.4.3-6: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN TDD Radio Link Monitoring test for Out-of-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 2 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

## 7.3.36.5 Test requirement

**Table 7.3.36.5-1: Cell specific test parameters for E-UTRAN TDD (cell # 1) for out-of-sync radio link monitoring tests in DRX for UE category 0**

| Parameter | Unit | Test 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| Special subframe configuration Note 1 | | 6 | | |
| Uplink-downlink configuration Note 1 | | 1 | | |
| PCFICH/PDCCH/PHICH parameters defined in clause A.2.1 | | R.7 TDD | | |
| OCNG Pattern defined in D.1.2 (TDD) | | OP.2 TDD | | |
| $\rho_A$, $\rho_B$ | | -3 | | |
| PCFICH_RB | dB | 1 | | |
| PDCCH_RA | dB | 4 | | |
| PDCCH_RB | dB | 4 | | |
| PBCH_RA | dB | -3 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note1 | dB | | | |
| OCNG_RB Note1 | dB | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | |
| SNR Note 6 | dB | -5.0 | -9.0 | -14.2 |
| Propagation condition | | AWGN | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 4: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 5: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 6: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 7: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.36.4-1.

**Table 7.3.36.5-2: DRX-Configuration for E-UTRAN TDD out-of-sync test in DRX for UE category 0**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf2 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf1280 | |
| shortDRX | disable | |

**Table 7.3.36.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN TDD out-of-sync test in DRX for UE category 0**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213. |

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the subframe according to the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal no later than time point C (duration $D_1$ = 6500 ms after the start of time duration T3.

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal.
- UE output power equal to or less than Transmit OFF power -48.5 dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.37 E-UTRAN TDD Radio Link Monitoring Test for In-sync in DRX for UE category 0

### 7.3.37.1 Test purpose

The purpose of this test is to verify that the TDD category 0 UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the PCell when DRX is used. This test will partly verify the E-UTRAN TDD radio link monitoring requirements in TS 36.133 [4] section 7.11.

### 7.3.37.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward of UE category 0. Applicability requires support for FGI bit 5.

### 7.3.37.3 Minimum conformance requirements

The UE category 0 applicability of the requirements for performing radio link monitoring is defined in Section 3.1.

When DRX is used for FD-FDD and TDD category 0 UEs, the $Q_{out_Cat0}$ evaluation period ($T_{Evaluate_Q_{out_DRX_Cat0}}$) and the $Q_{in_Cat0}$ evaluation period ($T_{Evaluate_Q_{in_DRX_Cat0}}$) is specified in TS36.133 [4] Table 7.11.2.2-1 will be used.

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{out_DRX_Cat0}}$ [s] period becomes worse than the threshold $Q_{out_Cat0}$, Layer 1 of the UE shall send out-of-sync indication for the PCell to the higher layers within $T_{Evaluate_Q_{out_DRX_Cat0}}$ [s] evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{in_DRX_Cat0}}$ [s] period becomes better than the threshold $Q_{in_Cat0}$, Layer 1 of the UE shall send in-sync indications for the PCell to the higher layers within $T_{Evaluate_Q_{in_DRX_Cat0}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10ms, DRX_cycle_length).

Upon start of T310 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.37.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX for TDD and TDD UE category 0**

| DRX cycle length (s) | $T_{Evaluate_Q_{out}_DRX_Cat0}$ and $T_{Evaluate_Q_{in}_DRX_Cat0}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in TS36.133 [4] clause 7.11.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note (20) |
| 0.04 < DRX cycle ≤ 0.64 | Note (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note (5) |
| Note: Evaluation period length in time depends on the length of the DRX cycle in use | |

The normative reference for this requirement is TS 36.133 [4] clause 7.11.1, 7.11.2 and A.7.3.37.

## 7.3.37.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.37.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode PDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to "infinity" so that UL timing alignment is maintained during the test.

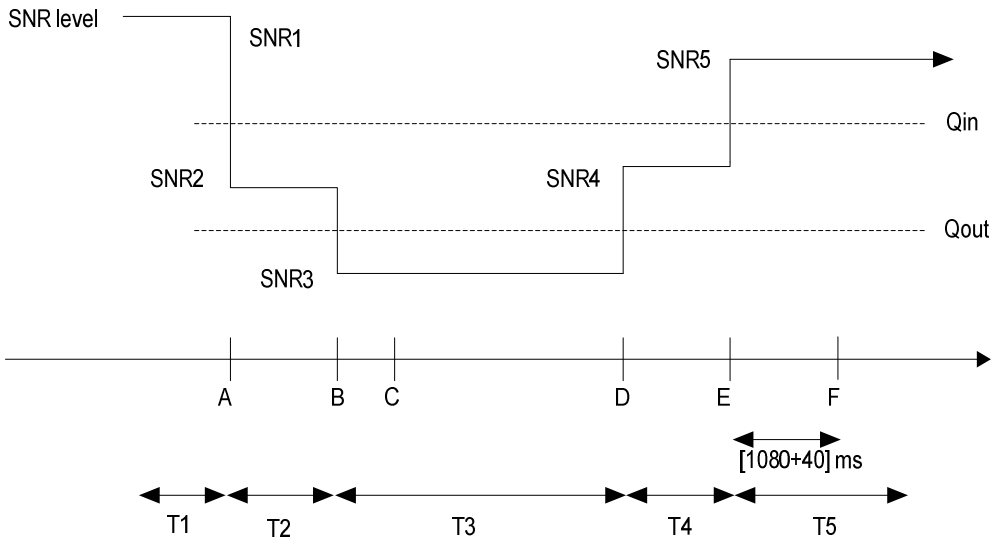


**Figure 7.3.37.4-1: SNR variation for in-sync testing in DRX**

### 7.3.37.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 (without using the faders) using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.37.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.37.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.37.4.1-1: General test parameters for E-UTRAN TDD in-sync test in DRX for category 0**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| E-UTRA RF Channel Number | | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | |
| CP length | | | Normal | |
| In sync transmission parameters (Note 1) | DCI format | | 1C | As defined in clause 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | In sync threshold $Q_{in}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] clause 7.11.1 and Table 7.11.1-2 respectively. |
| | Aggregation level | CCE | 4 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | | 1 | |
| | Ratio of PCFICH to RS EPRE | | 1 | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS 36.133 [4] clause 7.11.1 and Table 7.11.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX cycle | | ms | 40 | See Table 7.3.29.1-3 |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 2 | Minimum CQI reporting periodicity |
| T1 | | s | 4 | |
| T2 | | s | 1.6 | |
| T3 | | s | 1.46 | |
| T4 | | s | 0.4 | |
| T5 | | s | 4 | |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

### 7.3.37.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.37.5-1. Propagation conditions are set according to Annex B.1. T1 starts.
3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.37.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.37.5-1. T3 starts.
5. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.37.5-1. T4 starts.
6. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.37.5-1. T5 starts.
7. If the SS detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (1120 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
8. After T5 expires, repeat steps 2-7 for both subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.37.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.37.4.3-1: Common Exception messages for E-UTRAN TDD Radio Link Monitoring Test for In-sync in DRX for category 0 UE**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.37.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for In-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.37.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN TDD Radio Link Monitoring test for In-sync in DRX for UE category 0**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf40 typical value in real network for real-time services. | |
| sf40 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.37.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN TDD Radio Link Monitoring test for In-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.37.4.3-5: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring test for In-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.37.4.3-6: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN TDD Radio Link Monitoring test for In-sync in DRX for category 0 UE**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 2 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

## 7.3.37.5 Test requirement

**Table 7.3.37.5-1: Cell specific test parameters for E-UTRAN TDD (cell # 1) for in-sync radio link monitoring test in DRX for UE category 0**

| Parameter | Unit | Test 1 | | | | |
|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | |
| Special subframe configuration Note 1 | | 6 | | | | |
| Uplink-downlink configuration Note 1 | | 1 | | | | |
| PCFICH/PDCCH/PHICH parameters defined in section A.2.2 | | R.7 TDD | | | | |
| OCNG Pattern defined in D.1.2 (TDD) | | OP.2 TDD | | | | |
| $\rho_A$, $\rho_B$ | | -3 | | | | |
| PCFICH_RB | dB | 1 | | | | |
| PDCCH_RA | dB | 1 | | | | |
| PDCCH_RB | dB | 1 | | | | |
| PBCH_RA | dB | -3 | | | | |
| PBCH_RB | dB | | | | | |
| PSS_RA | dB | | | | | |
| SSS_RA | dB | | | | | |
| PHICH_RA | dB | | | | | |
| PHICH_RB | dB | | | | | |
| PDSCH_RA | dB | | | | | |
| PDSCH_RB | dB | | | | | |
| OCNG_RA Note2 | dB | | | | | |
| OCNG_RB Note2 | dB | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | |
| SNR Note 7 | dB | -5.0 | -9.0 | -14.2 | -10.2 | -5.0 |
| Propagation condition | | AWGN | | | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | | | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 4: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 5: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 6: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 7: The SNR in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.37.4-1.

**Table 7.3.37.5-2: DRX-Configuration for E-UTRAN TDD in-sync test in DRX for UE category 0**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf2 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | Sf40 | |
| shortDRX | disable | |

**Table 7.3.37.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN TDD in-sync test in DRX for UE category 0**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 2 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213. |

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (1120 ms after the start of time duration T5) the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the subframe according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.38 E-UTRAN FDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC

### 7.3.38.1 Test purpose

The purpose of this test is to verify that the UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the PSCell when DRX is used in synchronous dual connectivity. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in TS36.133[4] clause 7.6.

### 7.3.38.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward that supports Dual Connectivity.

### 7.3.38.3 Minimum conformance requirements

When DRX is used the $Q_{out}$ evaluation period ($T_{Evaluate_Q_{out_DRX}}$) and the $Q_{in}$ evaluation period ($T_{Evaluate_Q_{in_DRX}}$) is specified in TS 36.133[4] Table 7.6.2.2-1 will be used.

When the downlink radio link quality of the PCell or PSCell estimated over the last $T_{Evaluate_Q_{in_DRX}}$ [s] period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send in-sync indications for the PCell or PSCell to the higher layers within $T_{Evaluate_Q_{in_DRX}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations shall be performed as specified in section 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10 ms, DRX_cycle_length).

Upon start of T310 timer or T313 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link of PCell or PSCell for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer or T313 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer and the transmitter power of PSCell if configured shall be turned off within 40 ms after expiry of T313 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.38.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX}}$ and $T_{Evaluate_Q_{in_DRX}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in TS36.133[4] clause 7.6.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note 1 (20) |
| 0.04 < DRX cycle ≤ 0.64 | Note 1 (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note 1 (5) |

Note 1: Evaluation period length in time depends on the length of the DRX cycle in use.
Note 2: MCG's DRX configuration is applied for PCell RLM evaluation and SCG's DRX configuration is applied for PSCell RLM evaluation.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.38.

## 7.3.38.4 Test description

There are two cells, cell 1 is PCell and cell 2 is PSCell, in the test. The test consists of three successive time periods with time duration of T1, T2 and T3 respectively. Figure 7.3.38.4-1 shows the variation of the downlink SNR in the PCell and PSCell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell 2. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode PDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to "infinity" so that UL timing alignment is maintained during the test.

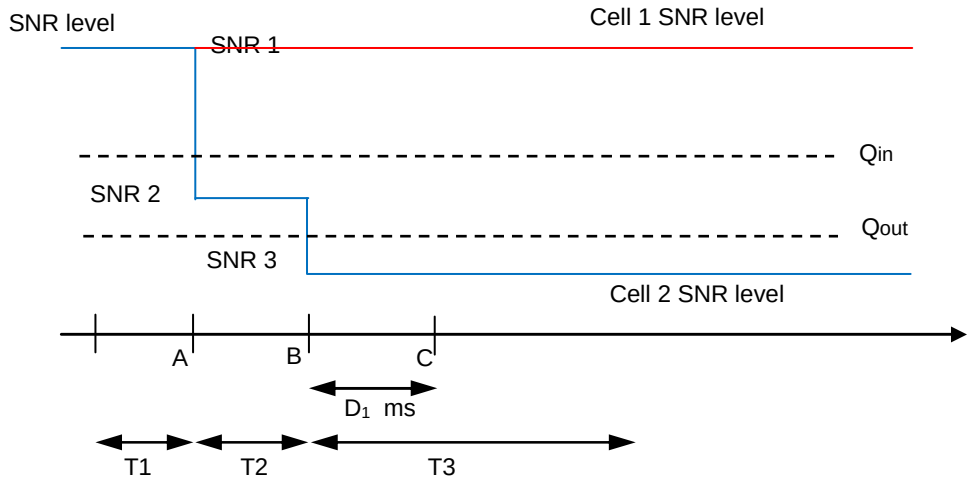


**Figure 7.3.38.4-1 SNR variation for out-of-sync testing in DRX**

### 7.3.38.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 7.3.38.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different DC configurations (per band) as defined in TS 36.521-1 [10] clause 5.4.2.

1. Connect the SS (node B emulator) and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.40 (without connection the AWGN sources) as appropriate.

2. The general test parameter settings for the different subtests are set up according to Table 7.3.38.4-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 7.3.38.4.3.

5. There are two cells specified in this test, Cell 1 and Cell2. Cell 1 is the PCell on RF channel 1and Cell 2 is the PSCell on RF channel 2. Cell1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.38.4.1-1: General test parameters for E-UTRAN FDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1<br>Cell 2 | Cell 1 is PCell on E-UTRA RF channel number 1, and cell 2 is PSCell on E-UTRA RF channel number 2 |
| CP length | | | Normal | |
| E-UTRA RF Channel Number | | | 1, 2 | Two E-UTRA FDD carrier frequencies are used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 5, 10, 20 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.101 [2] Annex B.2.3.2 |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 5MHz: 3<br>10MHz: 2<br>20MHz: 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS36.133[4] clause 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | ρA, ρB | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX cycle on cell 1 | | ms | 640 | See Table 7.3.38.5-2 |
| DRX cycle on cell 2 | | ms | 40 | See Table 7.3.38.5-2 |
| Timing offset between cell 1 and cell 2 | | μs | 33 | For synchronous dual connectivity |
| Layer 3 filtering | | | Enabled | Counters:<br>N310 = 1; N311 = 1; N313 = 1; N314 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| T313 timer | | ms | 0 | T313 is disabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 2 | Minimum CQI reporting periodicity |
| T1 | | s | 4 | |
| T2 | | s | 1.6 | |
| T3 | | s | 1.8 | |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

## 7.3.38.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell 2.

1. Ensure the UE is in State 3A-RF-DC2 if UE supports DC Split DRB bearer, according to TS 36.508 [7] clause 5.2A.2B; else ensure the UE is in state 3A-RF-DC1, according to TS 36.508 [7] clause 5.2A.2A.

2. The SS shall transmit an RRCConnectionReconfiguration message configuring the UE with the message content defined in clause 7.3.38.4.3.

3. The UE shall transmit RRCConnectionReconfigurationComplete message.

4. Set the parameters according to T1 in Table 7.3.38.5-1 as appropriate for BW dependant parameters. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.

5. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.38.5-1. T2 starts.

6. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.38.5-1. T3 starts.

7. If the SS on Cell2:
   a) detects uplink power equal to or higher than -39 dBm in the On-duration part of every DRX cycle in the subframe according the configured CQI reporting mode (PUCCH 1-0) during the period from time point A to time point B
   and
   b) does not detect any uplink power higher than -48.5 dBm from time point C (900 ms after the start of T3) until T3 expires,

   the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.

8. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.38.5-1.

9. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF-DC2 if UE supports DC Split DRB bearer, according to TS 36.508 [7] clause 5.2A.2B; else ensure the UE is in state 3A-RF-DC1, according to TS 36.508 [7] clause 5.2A.2A.

10. Repeat steps 2-9 for all subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.38.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.38.4.3-1: Common Exception messages for E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.38.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.38.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN FDD Radio Link Monitoring test for Out-of-sync in DRX in synchronous DC, for cell 1**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf640 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.38.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN FDD Radio Link Monitoring test for Out-of-sync in DRX in synchronous DC, for cell 2**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf40 typical value in real network for real-time services. | |
| sf40 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.38.4.3-5: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN FDD Radio Link Monitoring test for Out-of-sync in DRX in synchronous DC**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.38.4.3-6: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring test for Out-of-sync in DRX in synchronous DC**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.38.4.3-7: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD Radio Link Monitoring test for Out-of-sync in DRX in synchronous DC**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 20 | 5 MHz channel bandwidth parameter | To be set depending on BW under test |
| | 41 | 10 MHz channel bandwidth parameter | |
| | 84 | 20 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 7.3.38.4.3-8: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN FDD Radio Link Monitoring test for Out-of-sync in DRX in synchronous DC**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| rlf-TimersAndConstantsSCG-r12 CHOICE { | | | |
| setup SEQUENCE { | | | |
| t313-r12 | ms0 | | |
| n313-r12 | n1 | | |
| n314-r12 | n1 | | |
| } | | | |
| } | | | |
| } | | | |

## 7.3.38.5 Test requirement

**Table 7.3.38.5-1: Cell specific test parameters for E-UTRAN FDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 5, 10, 20 | | | 5, 10, 20 | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | 2x2 Low | | |
| PCFICH/PDCCH/PHICH parameters<br>None of the PDCCH are intended for the UE under test | | 5MHz: R.12 FDD<br>10MHz: R.7 FDD<br>20MHz: R.13 FDD | | | 5MHz: R.12 FDD<br>10MHz: R.7 FDD<br>20MHz: R.13 FDD | | |
| OCNG Pattern | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | |
| ρA, ρB | | -3 | | | -3 | | |
| PCFICH_RB | dB | 1 | | | 1 | | |
| PDCCH_RA | dB | 1 | | | 1 | | |
| PDCCH_RB | dB | 1 | | | 1 | | |
| PBCH_RA | dB | -3 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note1 | dB | | | | | | |
| OCNG_RB Note1 | dB | | | | | | |
| SNR Note 6 (5MHz bandwidth) | dB | -1.4 | -1.4 | -1.4 | -1.4 | -4.8 | -13.1 |
| SNR Note 6 (10MHz bandwidth) | dB | -1.4 | -1.4 | -1.4 | -1.4 | -5.3 | -13.1 |
| SNR Note 6 (20MHz bandwidth) | dB | -2.0 | -2.0 | -2.0 | -22.0 | -5.9 | -13.7 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | -98 | | |
| Propagation condition | | ETU 70 Hz | | | ETU 70 Hz | | |
| Time offset to cell1 | μs | - | | | 33 | | |

Note 1: OCNG shall be used such that the resources in cell 1 and cell 2 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2, and T3 are denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.38.4-1.

**Table 7.3.38.5-2: DRX-Configuration for E-UTRAN FDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC**

| Field | Value | | Comment |
|---|---|---|---|
| | Cell 1 | Cell 2 | |
| onDurationTimer | psf2 | psf2 | As specified in clause 6.3.2 in TS 36.331 [5] |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | Sf640 | Sf40 | |
| shortDRX | disable | disable | |

**Table 7.3.38.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331[5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331[5] and section 10.1 in TS 36.213 [8]. |

The UE behaviour during time durations T1, T2, and T3 shall be as follows:

During time durations T1, T2 and T3, the UE shall transmit uplink signal at least in all subframes configured for CQI transmission on Cell1.

During the period from time point A to time point B the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the uplink subframe on Cell2.

The UE shall stop transmitting uplink signal no later than time point C (duration D1 = 900 ms after the start of time duration T3) on PSCell.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.39 E-UTRAN FDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in asynchronous DC

### 7.3.39.1 Test purpose

The purpose of this test is to verify that the UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the PSCell when DRX in asynchronous dual connectivity is used. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in TS36.133[4] clause 7.6.

### 7.3.39.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward that supports asynchronous Dual Connectivity.

### 7.3.39.3 Minimum conformance requirements

When DRX is used the $Q_{out}$ evaluation period ($T_{Evaluate_Q_{out_DRX}}$) and the $Q_{in}$ evaluation period ($T_{Evaluate_Q_{in_DRX}}$) is specified in TS 36.133[4] Table 7.6.2.2-1 will be used.

When the downlink radio link quality of the PCell or PSCell estimated over the last $T_{Evaluate_Q_{in_DRX}}$ [s] period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send in-sync indications for the PCell or PSCell to the higher layers within $T_{Evaluate_Q_{in_DRX}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations shall be performed as specified in section 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10 ms, DRX_cycle_length).

Upon start of T310 timer or T313 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link of PCell or PSCell for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer or T313 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer and the transmitter power of PSCell if configured shall be turned off within 40 ms after expiry of T313 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.39.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX}}$ and $T_{Evaluate_Q_{in_DRX}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in TS36.133[4] clause 7.6.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note 1 (20) |
| 0.04 < DRX cycle ≤ 0.64 | Note 1 (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note 1 (5) |

Note 1: Evaluation period length in time depends on the length of the DRX cycle in use.
Note 2: MCG's DRX configuration is applied for PCell RLM evaluation and SCG's DRX configuration is applied for PSCell RLM evaluation.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.39.

## 7.3.39.4 Test description

There are two cells, cell 1 is PCell in MCG and cell 2 is PSCell in SCG, in the test. Before the test starts the UE is connected to cell 1 on radio channel 1 and to cell 2 on radio channel 2. The test consists of three successive time periods with time duration of T1, T2 and T3 respectively. The downlink SNR in cell 1 keeps constant in the test. Figure 7.3.39.4-1 shows the variation of the downlink SNR in the PCell and PSCell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell 2. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode PDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to "infinity" so that UL timing alignment is maintained during the test.

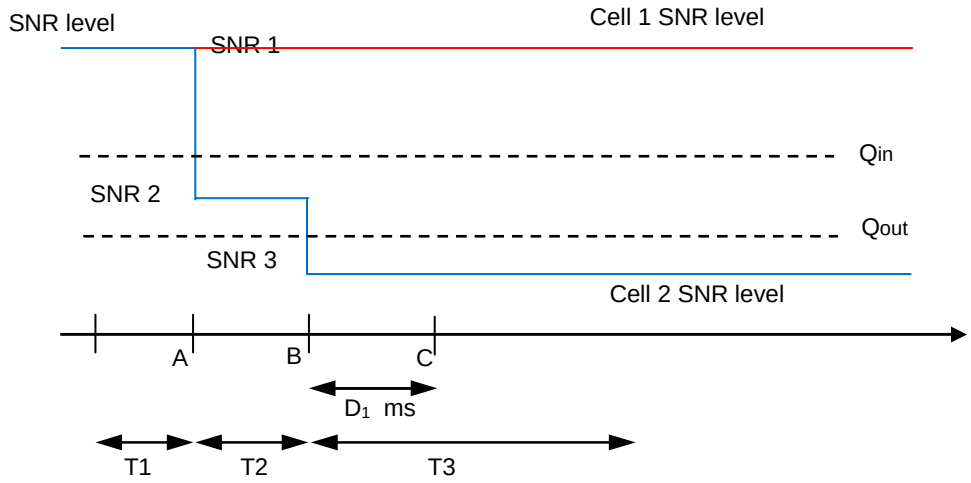


**Figure 7.3.39.4-1: SNR variation for out-of-sync testing in DRX**

### 7.3.39.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 7.3.39.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different DC configurations (per band) as defined in TS 36.521-1 [10] clause 5.4.2.

1. Connect the SS (node B emulator) and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.40 (without connection the AWGN sources) as appropriate.

2. The general test parameter settings for the different subtests are set up according to Table 7.3.39.4-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 7.3.39.4.3

5. There are two cells specified in this test, Cell 1 and Cell2. Cell 1 is the PCell on RF channel 1and Cell 2 is the PSCell on RF channel 2. Cell1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.39.4.1-1: General test parameters for E-UTRAN FDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in asynchronous DC**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1<br>Cell 2 | Cell 1 (PCell) is on E-UTRA RF channel number 1 and cell 2 (PSCell) is on E-UTRA RF channel number 2 |
| CP length | | | Normal | |
| E-UTRA RF Channel Number | | | 1, 2 | Two E-UTRA FDD carrier frequencies are used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 5, 10, 20 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.101 [2] Annex B.2.3.2 |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 5MHz: 3<br>10MHz: 2<br>20MHz: 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS36.133[4] clause 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | ρA, ρB | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX cycle on cell 1 | | ms | 640 | See Table 7.3.39.5-2 |
| DRX cycle on cell 2 | | ms | 40 | See Table 7.3.39.5-2 |
| Layer 3 filtering | | | Enabled | Counters:<br>N310 = 1; N311 = 1; N313 = 1; N314 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| T313 timer | | ms | 0 | T313 is disabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 2 | Minimum CQI reporting periodicity |
| T1 | | s | 4 | |
| T2 | | s | 1.6 | |
| T3 | | s | 1.8 | |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

## 7.3.39.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell 2.

1. Ensure the UE is in State 3A-RF-DC2 if UE supports DC Split DRB bearer, according to TS 36.508 [7] clause 5.2A.2B; else ensure the UE is in state 3A-RF_DC1, according to TS 36.508 [7] clause 5.2A.2A.

2. The SS shall transmit an RRCConnectionReconfiguration message configuring the UE with the message content defined in clause 7.3.39.4.3.

3. The UE shall transmit RRCConnectionReconfigurationComplete message.

4. Set the parameters according to T1 in Table 7.3.39.5-1 as appropriate for BW dependant parameters. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.

5. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.39.5-1. T2 starts.

6. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.39.5-1. T3 starts.

7. If the SS on Cell2:
   a) detects uplink power equal to or higher than -39 dBm in the On-duration part of every DRX cycle in the subframe according the configured CQI reporting mode (PUCCH 1-0) during the period from time point A to time point B
   and
   b) does not detect any uplink power higher than -48.5 dBm from time point C (900 ms after the start of T3) until T3 expires,
   the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.

8. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.39.5-1.

9. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF-DC2 if UE supports DC Split DRB bearer, according to TS 36.508 [7] clause 5.2A.2B; else ensure the UE is in state 3A-RF-DC1, according to TS 36.508 [7] clause 5.2A.2A.

10. Repeat steps 2-9 for all subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.39.4.3 Message contents

Same message contents as defined in clause 7.3.38.4.3.

## 7.3.39.5 Test requirement

**Table 7.3.39.5-1: Cell specific test parameters for E-UTRAN FDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in asynchronous DC**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 5, 10, 20 | | | 5, 10, 20 | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | 2x2 Low | | |
| PCFICH/PDCCH/PHICH parameters<br>None of the PDCCH are intended for the UE under test | | 5MHz: R.12 FDD<br>10MHz: R.7 FDD<br>20MHz: R.13 FDD | | | 5MHz: R.12 FDD<br>10MHz: R.7 FDD<br>20MHz: R.13 FDD | | |
| OCNG Pattern | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | |
| ρA, ρB | | -3 | | | -3 | | |
| PCFICH_RB | dB | 1 | | | 1 | | |
| PDCCH_RA | dB | 1 | | | 1 | | |
| PDCCH_RB | dB | 1 | | | 1 | | |
| PBCH_RA | dB | -3 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note1 | dB | | | | | | |
| OCNG_RB Note1 | dB | | | | | | |
| SNR Note 6 (5MHz bandwidth) | dB | -1.4 | -1.4 | -1.4 | -1.4 | -4.8 | -13.1 |
| SNR Note 6 (10MHz bandwidth) | dB | -1.4 | -1.4 | -1.4 | -1.4 | -5.3 | -13.1 |
| SNR Note 6 (20MHz bandwidth) | dB | -2.0 | -2.0 | -2.0 | -2.0 | -5.9 | -13.7 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | -98 | | |
| Propagation condition | | ETU 70 Hz | | | ETU 70 Hz | | |
| Receive time offset to cell1 Note 7 | μs | - | | | 500 | | |

Note 1: OCNG shall be used such that the resources in cell 1 and cell 2 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2, and T3 are denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.39.4-1.
Note 7: Receive time difference between subframe boundaries of signals received from the two cells at the UE antenna connector including time alignment error between the two cells.

**Table 7.3.39.5-2: DRX-Configuration for E-UTRAN FDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in asynchronous DC**

| Field | Value | | Comment |
|---|---|---|---|
| | Cell 1 | Cell 2 | |
| onDurationTimer | psf2 | psf2 | As specified in clause 6.3.2 in TS 36.331 [5] |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | Sf640 | Sf40 | |
| shortDRX | disable | disable | |

**Table 7.3.39.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in asynchronous DC**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331[5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331[5] and section 10.1 in TS 36.213 [8]. |

The UE behaviour during time durations T1, T2, and T3 shall be as follows:

During time durations T1, T2 and T3, the UE shall transmit uplink on cell 1 at least once every DRX cycle, in the On-duration part of the cycle in the subframe according to the configured CQI reporting mode (PUCCH 1-0).

During the period from time point A to time point B the UE shall transmit uplink signal on cell 2 at least once every DRX cycle, in the On-duration part of the cycle in the subframe according to the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal on cell 2 no later than time point C (duration D1 = 900 ms after the start of time duration T3).

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.40 E-UTRAN TDD-TDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC

### 7.3.40.1 Test purpose

The purpose of this test is to verify that the UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the PSCell when DRX is used. This test will partly verify the E-UTRAN TDD radio link monitoring requirements in TS36.133[4] clause 7.6.

### 7.3.40.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward that supports Dual Connectivity.

### 7.3.40.3 Minimum conformance requirements

When DRX is used the $Q_{out}$ evaluation period ($T_{Evaluate_Q_{out_DRX}}$) and the $Q_{in}$ evaluation period ($T_{Evaluate_Q_{in_DRX}}$) is specified in TS 36.133[4] Table 7.6.2.2-1 will be used.

When the downlink radio link quality of the PCell or PSCell estimated over the last $T_{Evaluate_Q_{in_DRX}}$ [s] period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send in-sync indications for the PCell or PSCell to the higher layers within $T_{Evaluate_Q_{in_DRX}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations shall be performed as specified in section 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10 ms, DRX_cycle_length).

Upon start of T310 timer or T313 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link of PCell or PSCell for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer or T313 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer and the transmitter power of PSCell if configured shall be turned off within 40 ms after expiry of T313 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.40.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX}}$ and $T_{Evaluate_Q_{in_DRX}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in TS36.133[4] clause 7.6.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note 1 (20) |
| 0.04 < DRX cycle ≤ 0.64 | Note 1 (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note 1 (5) |

Note 1: Evaluation period length in time depends on the length of the DRX cycle in use.
Note 2: MCG's DRX configuration is applied for PCell RLM evaluation and SCG's DRX configuration is applied for PSCell RLM evaluation.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.40.

## 7.3.40.4 Test description

There are two cells, cell 1 is PCell and cell 2 is PSCell, in the test. The test consists of three successive time periods with time duration of T1, T2 and T3 respectively. Figure 7.3.40.4-1 shows the variation of the downlink SNR in the PCell and PSCell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell 2. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode PDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to "infinity" so that UL timing alignment is maintained during the test.

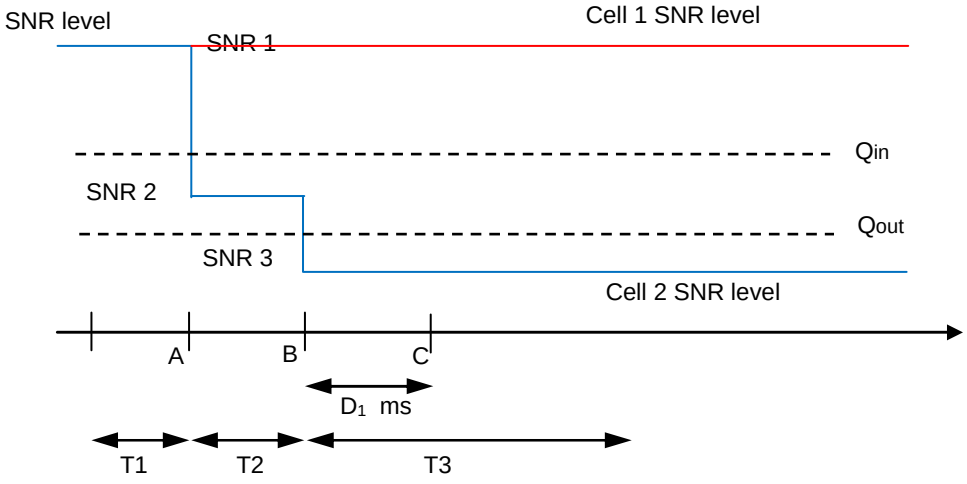


**Figure 7.3.40.4-1 SNR variation for out-of-sync testing in DRX**

### 7.3.40.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 7.3.40.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different DC configurations (per band) as defined in TS 36.521-1 [10] clause 5.4.2.

1. Connect the SS (node B emulator) and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.40 (without connection the AWGN sources) as appropriate.

2. The general test parameter settings for the different subtests are set up according to Table 7.3.40.4-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 7.3.40.4.3.

5. There are two cells specified in this test, Cell 1 and Cell2. Cell 1 is the PCell on RF channel 1and Cell 2 is the PSCell on RF channel 2. Cell1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.40.4.1-1: General test parameters for E-UTRAN TDD-TDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1<br>Cell 2 | Cell 1 is PCell on E-UTRA RF channel number 1, and cell 2 is PSCell on E-UTRA RF channel number 2 |
| CP length | | | Normal | |
| E-UTRA RF Channel Number | | | 1, 2 | Two E-UTRA TDD carrier frequencies are used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 5, 10, 20 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.101 [2] Annex B.2.3.2 |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS36.133[4] clause 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | ρA, ρB | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX cycle on cell 1 | | ms | 640 | See Table 7.3.40.5-2 |
| DRX cycle on cell 2 | | ms | 40 | See Table 7.3.40.5-2 |
| Timing offset between cell 1 and cell 2 | | μs | 33 | For synchronous dual connectivity |
| Layer 3 filtering | | | Enabled | Counters:<br>N310 = 1; N311 = 1; N313 = 1; N314 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| T313 timer | | ms | 0 | T313 is disabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 1 | Minimum CQI reporting periodicity |
| T1 | | s | 4 | |
| T2 | | s | 1.6 | |
| T3 | | s | 1.8 | |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

## 7.3.40.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell2.

1. Ensure the UE is in State 3A-RF-DC2 if UE supports DC Split DRB bearer, according to TS 36.508 [7] clause 5.2A.2B; else ensure the UE is in state 3A-RF_DC1, according to TS 36.508 [7] clause 5.2A.2A.

2. The SS shall transmit an RRCConnectionReconfiguration message configuring the UE with the message content defined in clause 7.3.40.4.3.

3. The UE shall transmit RRCConnectionReconfigurationComplete message.

4. Set the parameters according to T1 in Table 7.3.40.5-1 as appropriate for BW dependant parameters. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.

5. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.40.5-1. T2 starts.

6. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.40.5-1. T3 starts.

7. If the SS on Cell2:
   a) detects uplink power equal to or higher than -39 dBm in the On-duration part of every DRX cycle in the subframe according the configured CQI reporting mode (PUCCH 1-0) during the period from time point A to time point B
   and
   b) does not detect any uplink power higher than -48.5 dBm from time point C (900 ms after the start of T3) until T3 expires,
   The number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.

8. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.40.5-1.

9. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF-DC2 if UE supports DC Split DRB bearer, according to TS 36.508 [7] clause 5.2A.2B; else ensure the UE is in state 3A-RF-DC1, according to TS 36.508 [7] clause 5.2A.2A.

10. Repeat steps 2-9 for all subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.40.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.40.4.3-1: Common Exception messages for E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.40.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1C in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.40.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN TDD Radio Link Monitoring test for Out-of-sync in DRX in synchronous DC, for cell 1**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf640 | 2 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.40.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN FDD Radio Link Monitoring test for Out-of-sync in DRX in synchronous DC, for cell 2**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf40 typical value in real network for real-time services. | |
| sf40 | 2 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.40.4.3-5: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN FDD Radio Link Monitoring test for Out-of-sync in DRX in synchronous DC**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.40.4.3-6: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring test for Out-of-sync in DRX in synchronous DC**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.38.4.3-7: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD Radio Link Monitoring test for Out-of-sync in DRX in synchronous DC**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 20 | 5 MHz channel bandwidth parameter | To be set depending on BW under test |
| | 41 | 10 MHz channel bandwidth parameter | |
| | 84 | 20 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 2 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 7.3.40.4.3-8: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN FDD Radio Link Monitoring test for Out-of-sync in DRX in synchronous DC**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| rlf-TimersAndConstantsSCG-r12 CHOICE { | | | |
| setup SEQUENCE { | | | |
| t313-r12 | ms0 | | |
| n313-r12 | n1 | | |
| n314-r12 | n1 | | |
| } | | | |
| } | | | |
| } | | | |

### 7.3.40.5 Test requirement

**Table 7.3.40.5-1: Cell specific test parameters for E-UTRAN TDD-TDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 5, 10, 20 | | | 5, 10, 20 | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | 2x2 Low | | |
| Special subframe configuration[Note1] | | 6 | | | 6 | | |
| Uplink-downlink configuration[Note2] | | 1 | | | 1 | | |
| PCFICH/PDCCH/PHICH parameters<br>None of the PDCCH are intended for the UE under test | | 5MHz: R.12 TDD<br>10MHz: R.7 TDD<br>20MHz: R.13 TDD | | | 5MHz: R.12 TDD<br>10MHz: R.7 TDD<br>20MHz: R.13 TDD | | |
| OCNG Pattern | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | |
| ρA, ρB | | -3 | | | -3 | | |
| PCFICH_RB | dB | 1 | | | 1 | | |
| PDCCH_RA | dB | 1 | | | 1 | | |
| PDCCH_RB | dB | 1 | | | 1 | | |
| PBCH_RA | dB | -3 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA[Note3] | dB | | | | | | |
| OCNG_RB[Note3] | dB | | | | | | |
| SNR [Note 8] (5MHz bandwidth) | dB | -0.7 | -0.7 | -0.7 | -0.7 | -4.3 | -12.8 |
| SNR [Note 8] (10MHz bandwidth) | dB | -1.4 | -1.4 | -1.4 | -1.4 | -5.0 | -12.8 |
| SNR [Note 8] (20MHz bandwidth) | dB | -2.1 | -2.1 | -2.1 | -2.1 | -5.7 | -13.5 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | -98 | | |
| Propagation condition | | ETU 70 Hz | | | ETU 70 Hz | | |
| Time offset to cell1 | μs | - | | | 33 | | |

Note 1: For the special subframe configuration see table 4.2-1 in TS 36.211.
Note 2: For the uplink-downlink configuration see table 4.2-2 in TS 36.211.
Note 3: OCNG shall be used such that the resources in cell 1 and cell 2 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 5: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 6: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 7: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 8: The SNR in time periods T1, T2, and T3 are denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.40.41-1.

**Table 7.3.40.5-2: DRX-Configuration for E-UTRAN TDD-TDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC**

| Field | Value | | Comment |
|---|---|---|---|
| | Cell 1 | Cell 2 | |
| onDurationTimer | psf2 | psf2 | As specified in clause 6.3.2 in TS 36.331 [5] |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | Sf640 | Sf40 | |
| shortDRX | disable | disable | |

**Table 7.3.40.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN TDD-TDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331[5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331[5] and section 10.1 in TS 36.213 [8]. |

The UE behaviour during time durations T1, T2, and T3 shall be as follows:

During time durations T1, T2 and T3, the UE shall transmit uplink signal at least in all subframes configured for CQI transmission on Cell1.

During the period from time point A to time point B the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the uplink subframe on Cell2.

The UE shall stop transmitting uplink signal no later than time point C (duration D1 = 900 ms after the start of time duration T3) on PSCell.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.41 E-UTRAN FDD-FDD Radio Link Monitoring Test for In-sync in DRX in synchronous dual connectivity

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

*The Test system uncertainties applicable to this test are undefined*

*The Test tolerances applicable to this test are undefined*

### 7.3.41.1 Test purpose

The purpose of this test is to verify that the UE properly detects the out of sync and in sync for the purpose of monitoring downlink radio link quality of the PSCell when DRX is used in dual connectivity. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in TS36.133[4] clause 7.6.

### 7.3.41.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward that supports Dual Connectivity.

### 7.3.41.3 Minimum conformance requirements

When DRX is used the $Q_{out}$ evaluation period ($T_{Evaluate_Q_{out_DRX}}$) and the $Q_{in}$ evaluation period ($T_{Evaluate_Q_{in_DRX}}$) is specified in TS 36.133[4] Table 7.6.2.2-1 will be used.

When the downlink radio link quality of the PCell or PSCell estimated over the last $T_{Evaluate_Q_{in_DRX}}$ [s] period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send in-sync indications for the PCell or PSCell to the higher layers within $T_{Evaluate_Q_{in_DRX}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The In-sync and in-sync evaluations shall be performed as specified in section 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10 ms, DRX_cycle_length).

Upon start of T310 timer or T313 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link of PCell or PSCell for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer or T313 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer and the transmitter power of PSCell if configured shall be turned off within 40 ms after expiry of T313 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.41.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX}}$ and $T_{Evaluate_Q_{in_DRX}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in TS36.133[4] clause 7.6.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note 1 (20) |
| 0.04 < DRX cycle ≤ 0.64 | Note 1 (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note 1 (5) |
| Note 1: Evaluation period length in time depends on the length of the DRX cycle in use.<br>Note 2: MCG's DRX configuration is applied for PCell RLM evaluation and SCG's DRX configuration is applied for PSCell RLM evaluation. | |

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.41.

## 7.3.41.4 Test description

There are two cells, cell 1 is PCell and cell 2 is PSCell, in the test. The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.38.4-1 shows the variation of the downlink SNR in the PSCell to emulate In-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell 2. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms. In the test, DRX configuration for PCell and PSCell is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode PDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to "infinity" so that UL timing alignment is maintained during the test.

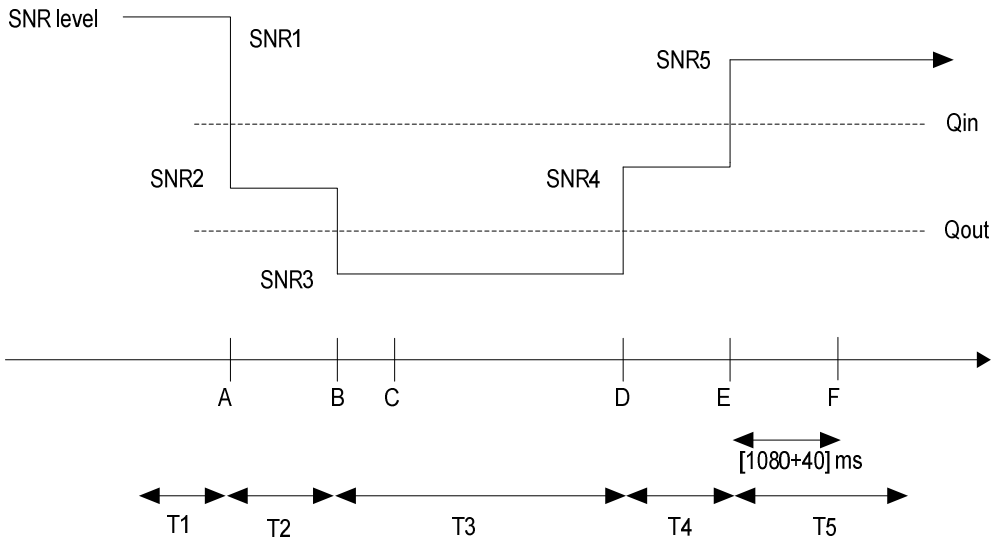


**Figure 7.3.41.4-1: SNR variation of cell 2 (PSCell) for in-sync testing in DRX**

#### 7.3.41.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 7.3.41.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different DC configurations (per band) as defined in TS 36.521-1 [10] clause 5.4.2.

1. Connect the SS (node B emulator) and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.14 as appropriate.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.41.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.41.4.3.
5. There are two cells specified in this test, Cell 1 and Cell2. Cell 1 is the PCell on RF channel 1and Cell 2 is the PSCell on RF channel 2. Cell1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.41.4.1-1: General test parameters for E-UTRAN FDD-FDD Radio Link Monitoring Test for In-sync in DRX in synchronous dual connectivity**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2 | Two E-UTRA FDD carrier frequency are used. |
| Active cell | | | Cell 1, Cell 2 | Cell 1 is PCell on E-UTRA RF channel number 1, and cell 2 is PSCell on E-UTRA RF channel number 2 |
| CP length | | | Normal | |
| In sync transmission parameters (Note 1) | DCI format | | 1C | As defined in clause 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | For 10MHz and 20MHz channel BW |
| | Number of Control OFDM symbols | | 3 | For 5MHz channel BW |
| | Aggregation level | CCE | 4 | In sync threshold $Q_{in}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS36.133[4] clause 7.6.1 and Table 7.6.1-2 respectively. |
| | ρA, ρB | | 0 | |
| | Ratio of PDCCH to RS EPRE | | 0 | |
| | Ratio of PCFICH to RS EPRE | | 4 | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | For 10MHz and 20MHz channel BW |
| | Number of Control OFDM symbols | | 3 | For 5MHz channel BW |
| | Aggregation level | CCE | 8 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS36.133[4] clause 7.6.1 and Table 7.6.1-1 respectively. |
| | ρA, ρB | | 0 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 4 | |
| DRX cycle on Cell 1 | | ms | 640 | See Table 7.3.41.5-2 |
| DRX cycle on Cell 2 | | ms | 40 | See Table A.7.3.41.5-2 |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1, N313 = 1, N314 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| T313 timer | | ms | 2000 | T313 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 2 | Minimum CQI reporting periodicity |
| T1 | | s | 4 | |
| T2 | | s | 1.6 | |
| T3 | | s | 1.46 | |
| T4 | | s | 0.4 | |
| T5 | | s | 4 | |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel.<br>Note 2: Even a UE capable of both synchronous and asynchronous DC operations is required to pass this test case in accordance with the principle defined in section 3A.5 | | | | |

## 7.3.41.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell 2.

1. Ensure the UE is in State 3A-RF-DC2 if UE supports DC Split DRB bearer, according to TS 36.508 [7] clause 5.2A.2B; else ensure the UE is in state 3C-RF, according to TS 36.508 [7] clause 5.2A.2A.

2. The SS shall transmit an RRCConnectionReconfiguration message configuring the UE with the message content defined in clause 7.3.41.4.3.

3. The UE shall transmit RRCConnectionReconfigurationComplete message.
4. Set the parameters according to T1 in Table 7.3.41.5-1 for subtests 1 and 2 respectively. Propagation conditions are set according to Annex B.2.2. T1 starts.
5. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.41.5-1. T2 starts.
6. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.41.5-1. T3 starts.
7. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.41.5-1. T4 starts.
8. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.41.5-1. T5 starts.
9. If the SS on Cell 2 detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (1120 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
10. After T5 expires, repeat steps 2-9 for both subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.41.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.41.4.3-1: Common Exception messages for E-UTRAN FDD Radio Link Monitoring Test for In-sync in DRX in synchronous DC**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.41.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FDD Radio Link Monitoring Test for In-sync in DRX in synchronous DC**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.41.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN FDD Radio Link Monitoring test for In-sync in DRX in synchronous DC, for cell 1**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf640 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.41.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN FDD Radio Link Monitoring test for In-sync in DRX in synchronous DC, for cell 2**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf40 typical value in real network for real-time services. | |
| sf40 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.41.4.3-5: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN FDD Radio Link Monitoring test for In-sync in DRX in synchronous DC**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.41.4.3-6: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD Radio Link Monitoring test for In-sync in DRX in synchronous DC**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 20 | 5 MHz channel bandwidth parameter | To be set depending on BW under test |
| | 41 | 10 MHz channel bandwidth parameter | |
| | 84 | 20 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 7.3.41.4.3-7: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN FDD Radio Link Monitoring test for In-sync in DRX in synchronous DC**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| rlf-TimersAndConstantsSCG-r12 CHOICE { | | | |
| setup SEQUENCE { | | | |
| t313-r12 | ms2000 | | |
| n313-r12 | n1 | | |
| n314-r12 | n1 | | |
| } | | | |
| } | | | |
| } | | | |

## 7.3.41.5 Test requirement

**Table 7.3.41.5-1: Cell specific test parameters for E-UTRAN FDD-FDD Radio Link Monitoring Test for In-sync in DRX in synchronous dual connectivity**

| Parameter | Unit | Cell 1(PCell) | Cell 2 (PSCell) | | | | |
|---|---|---|---|---|---|---|---|
| | | T1 ~ T5 | T1 | T2 | T3 | T4 | T5 |

| E-UTRA RF Channel Number | | | 1 | 2 | | | | |
|---|---|---|---|---|---|---|---|---|
| $BW_{channel}$ | | MHz | 5: $N_{RB,c}$ = 25<br>10: $N_{RB,c}$ = 50<br>20: $N_{RB,c}$ = 100 | 5: $N_{RB,c}$ = 25<br>10: $N_{RB,c}$ = 50<br>20: $N_{RB,c}$ = 100 | | | | |
| PCFICH/PDCCH/PHICH parameters defined in A.2.1 | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | | |
| OCNG Pattern defined in D.1 (FDD) | | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | | | |
| $\rho_A$, $\rho_B$ | | | 0 | 0 | | | | |
| PCFICH_RB | | dB | 4 | 4 | | | | |
| PDCCH_RA | | dB | 0 | 0 | | | | |
| PDCCH_RB | | dB | 0 | 0 | | | | |
| PBCH_RA | | dB | 0 | 0 | | | | |
| PBCH_RB | | dB | | | | | | |
| PSS_RA | | dB | | | | | | |
| SSS_RA | | dB | | | | | | |
| PHICH_RA | | dB | | | | | | |
| PHICH_RB | | dB | | | | | | |
| PDSCH_RA | | dB | | | | | | |
| PDSCH_RB | | dB | | | | | | |
| OCNG_RA[Note1] | | dB | | | | | | |
| OCNG_RB[Note1] | | dB | | | | | | |
| SNR [Note 6] | 5MHz $BW_{channel}$ | dB | -1.7 | -1.7 | -5.1 | -12.8 | -7.9 | -1.7 |
| | 10MHz $BW_{channel}$ | | -4.1 | -4.1 | -8.9 | -14.1 | -9.3 | -4.1 |
| | 20MHz $BW_{channel}$ | | -4.1 | -4.1 | -8.9 | -14.1 | -9.3 | -4.1 |
| $N_{oc}$ | | dBm/15 kHz | -98 | | | | | |
| Propagation condition | | | AWGN | AWGN | | | | |
| Correlation Matrix and Antenna Configuration | | | 1x2 | 1x2 | | | | |
| Receive time offset to cell1 [Note 7] | | μs | - | 33 | | | | |

Note 1: OCNG shall be used such that the resources in cell 1 and cell 2 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR of cell 2 in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.41.4-1.
Note 7: Receive time difference between subframe boundaries of signals received from the two cells at the UE antenna connector including time alignment error between the two cells.

**Table 7.3.41.5-2: DRX-Configuration for E-UTRAN FDD-FDD Radio Link Monitoring Test for In-sync in DRX in synchronous dual connectivity**

| Field | Value | | Comment |
|---|---|---|---|
| | Cell 1 | Cell 2 | |
| onDurationTimer | psf2 | psf2 | As specified in section 6.3.2 in 3GPP TS 36.331[5] |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf640 | sf40 | |
| shortDRX | disable | disable | |

**Table 7.3.41.5-3: TimeAlignmentTimer Configuration for E-UTRAN FDD-FDD Radio Link Monitoring Test for In-sync in DRX in synchronous dual connectivity**

| Field | Value | |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331[5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331[5] and section10.1 in TS 36.213[8]. |

The UE behaviour during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (1120 ms after the start of time duration T5) the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the subframe according to the configured CQI reporting mode (PUCCH 1-0) on PCell and PSCell.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.42 E-UTRAN FDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in asynchronous DC

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

*The Test system uncertainties applicable to this test are undefined*

*The Test tolerances applicable to this test are undefined*

### 7.3.42.1 Test purpose

The purpose of this test is to verify that the UE properly detects the out of sync and in sync for the purpose of monitoring downlink radio link quality of the PSCell when DRX is used in asynchronous dual connectivity. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in clause 7.6.

### 7.3.42.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward that supports asynchronous Dual Connectivity.

### 7.3.42.3 Minimum conformance requirements

When DRX is used the $Q_{out}$ evaluation period ($T_{Evaluate_Q_{out_DRX}}$) and the $Q_{in}$ evaluation period ($T_{Evaluate_Q_{in_DRX}}$) is specified in TS 36.133[4] Table 7.6.2.2-1 will be used.

When the downlink radio link quality of the PCell or PSCell estimated over the last $T_{Evaluate_Q_{in_DRX}}$ [s] period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send in-sync indications for the PCell or PSCell to the higher layers within $T_{Evaluate_Q_{in_DRX}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The In-sync and in-sync evaluations shall be performed as specified in section 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10 ms, DRX_cycle_length).

Upon start of T310 timer or T313 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link of PCell or PSCell for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer or T313 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer and the transmitter power of PSCell if configured shall be turned off within 40 ms after expiry of T313 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.41.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX}}$ and $T_{Evaluate_Q_{in_DRX}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in TS36.133[4] clause 7.6.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note 1 (20) |
| 0.04 < DRX cycle ≤ 0.64 | Note 1 (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note 1 (5) |

Note 1: Evaluation period length in time depends on the length of the DRX cycle in use.
Note 2: MCG's DRX configuration is applied for PCell RLM evaluation and SCG's DRX configuration is applied for PSCell RLM evaluation.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.42.

## 7.3.42.4 Test description

There are two cells in the test. Cell 1 is PCell in MCG and cell 2 is PSCell in SCG. Before the test starts the UE is connected to cell 1 on radio channel 1 and to cell 2 on radio channel 2. The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. The downlink SNR in cell 1 keeps constant in the test. Figure 7.3.42.4-1 shows the variation of the downlink SNR in cell 2 to emulate In-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell 2 in asynchronous dual connectivity. For both cell 1 and cell 2, the UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode PDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to "infinity" so that UL timing alignment is maintained during the test.

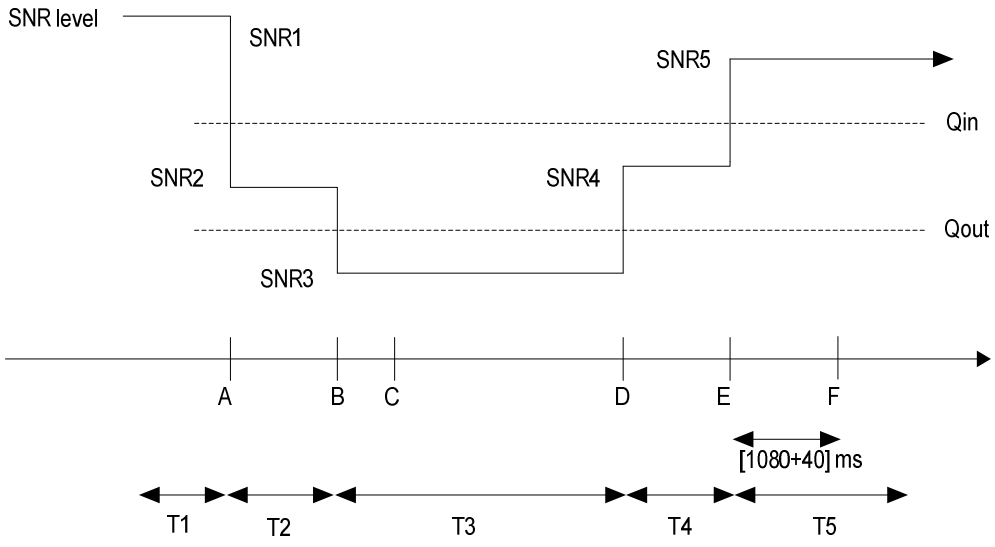


**Figure 7.3.42.4-1: Cell 2 SNR variation for in-sync testing in DRX**

### 7.3.42.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 7.3.42.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different DC configurations (per band) as defined in TS 36.521-1 [10] clause 5.4.2.

1. Connect the SS (node B emulator) and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.14 as appropriate.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.42.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.42.4.3.
5. There are two cells specified in this test, Cell 1 and Cell2. Cell 1 is the PCell on RF channel 1and Cell 2 is the PSCell on RF channel 2. Cell1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.42.4.1-1: General test parameters for E-UTRAN FDD-FDD Radio Link Monitoring in-sync test in DRX dual connectivity**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cells | | | Cell 1 and cell 2 | Cell 1 (PCell) is on E-UTRA RF channel number 1 and cell 2 (PSCell) is on E-UTRA RF channel number 2 |
| CP length | | | Normal | |
| Antenna Configuration | | | 1x2 | |
| In sync transmission parameters (Note 1) | DCI format | | 1C | As defined in clause 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 5MHz: 3<br>10MHz: 2<br>20MHz: 2 | In sync threshold $Q_{in}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS36.133[4] clause and Table 7.6.1-2 respectively. |
| | Aggregation level | CCE | 4 | |
| | $\rho_A$, $\rho_B$ | | 0 | |
| | Ratio of PDCCH to RS EPRE | | 0 | |
| | Ratio of PCFICH to RS EPRE | | 4 | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS36.133[4] clause 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | $\rho_A$, $\rho_B$ | | 0 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 4 | |

| | | | |
|---|---|---|---|
| DRX cycle in cell 1 | ms | 640 | See Table 7.3.42.5-2 |
| DRX cycle in cell 2 | ms | 40 | See Table 7.3.42.5-2 |
| Layer 3 filtering | | Enabled | Counters:<br>N310 = 1; N311 = 1;<br>N313 = 1; N314 = 1 |
| T310 timer | ms | 2000 | T310 is enabled |
| T311 timer | ms | 1000 | T311 is enabled |
| T313 timer | ms | 2000 | T313 is enabled |
| Periodic CQI reporting mode | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | ms | 2 | Minimum CQI reporting periodicity |
| T1 | s | 4 | |
| T2 | s | 1.6 | |
| T3 | s | 1.46 | |
| T4 | s | 0.4 | |
| T5 | s | 4 | |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel.<br>Note 2: The parameters in the table apply to both cell 1 and cell 2 unless defined otherwise.<br>Note 3: Even a UE capable of both synchronous and asynchronous DC operations is required to pass this test case in accordance with the principle defined in section 3A.5. | | | |

### 7.3.42.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell 2.

1. Ensure the UE is in State 3A-RF-DC2 if UE supports DC Split DRB bearer, according to TS 36.508 [7] clause 5.2A.2B; else ensure the UE is in state 3A-RF-DC1, according to TS 36.508 [7] clause 5.2A.2A.
2. The SS shall transmit an RRCConnectionReconfiguration message configuring the UE with the message content defined in clause 7.3.42.4.3.
3. The UE shall transmit RRCConnectionReconfigurationComplete message.
4. Set the parameters according to T1 in Table 7.3.42.5-1 for subtests 1 and 2 respectively. Propagation conditions are set according to Annex B.2.2. T1 starts.
5. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.42.5-1. T2 starts.
6. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.42.5-1. T3 starts.
7. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.42.5-1. T4 starts.
8. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.42.5-1. T5 starts.
9. If the SS on Cell 2 detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (1120 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
10. After T5 expires, repeat steps 2-9 for both subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.42.4.3 Message contents

Same message contents as defined in clause 7.3.41.4.3.

## 7.3.42.5 Test requirement

**Table 7.3.42.5-1: Cell specific test parameters for E-UTRAN FDD-FDD Radio Link Monitoring Test for In-sync in DRX in asynchronous dual connectivity**

| Parameter | | Unit | Cell 1 (PCell) | Cell 2 (PSCell) | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | T1 ~ T5 | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | | 1 | 2 | | | | |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | | MHz | 5, 10, 20 | 5, 10, 20 | | | | |
| PCFICH/PDCCH/PHICH parameters. None of the PDCCH are intended for the UE under test. | | | 5MHz: R.11 FDD 10MHz: R.6 FDD 20MHz: R.10 FDD | 5MHz: R.11 FDD 10MHz: R.6 FDD 20MHz: R.10 FDD | | | | |
| OCNG Pattern | | | 5MHz: OP.16 FDD 10MHz: OP.2 FDD 20MHz: OP.12 FDD | 5MHz: OP.16 FDD 10MHz: OP.2 FDD 20MHz: OP.12 FDD | | | | |
| $\rho_A$, $\rho_B$ | | | 0 | 0 | | | | |
| PCFICH_RB | | dB | 4 | 4 | | | | |
| PDCCH_RA | | dB | 0 | 0 | | | | |
| PDCCH_RB | | dB | 0 | 0 | | | | |
| PBCH_RA | | dB | 0 | 0 | | | | |
| PBCH_RB | | dB | | | | | | |
| PSS_RA | | dB | | | | | | |
| SSS_RA | | dB | | | | | | |
| PHICH_RA | | dB | | | | | | |
| PHICH_RB | | dB | | | | | | |
| PDSCH_RA | | dB | | | | | | |
| PDSCH_RB | | dB | | | | | | |
| OCNG_RA Note1 | | dB | | | | | | |
| OCNG_RB Note1 | | dB | | | | | | |
| SNR Note 6 | 5MHz $BW_{channel}$ | dB | -1.7 | -1.7 | -5.1 | -12.8 | -7.9 | -1.7 |
| | 10MHz $BW_{channel}$ | dB | -4.1 | -4.1 | -8.9 | -14.1 | -9.3 | -4.1 |
| | 20MHz $BW_{channel}$ | dB | -4.1 | -4.1 | -8.9 | -14.1 | -9.3 | -4.1 |
| $N_{oc}$ | | dBm/15 kHz | -98 | | | | | |
| Propagation condition | | | AWGN | | | | | |
| Receive time offset to cell1 Note 7 | | μs | - | 500 | | | | |

Note 1: OCNG shall be used such that the resources in cell 1 and cell 2 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR of cell 2 in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.42.4-1.
Note 7: Receive time difference between subframe boundaries of signals received from the two cells at the UE antenna connector including time alignment error between the two cells..

**Table 7.3.42.1-3: DRX-Configuration for E-UTRAN FDD In-sync tests**

| Field | Cell 1 | Cell 2 | Comment |
|---|---|---|---|
| onDurationTimer | psf2 | psf2 | As specified in clause 6.3.2 in TS 36.331[5] |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf640 | sf40 | |
| shortDRX | disable | disable | |

**Table 7.3.42.1-4: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD In-sync testing**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331[5]. |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331[5] and section10.1 in TS 36.213[8]. |

The UE behaviour during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (1120 ms after the start of time duration T5) the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the subframe according to the configured CQI reporting mode (PUCCH 1-0) on PCell and PSCell.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.43 E-UTRAN TDD-TDD Radio Link Monitoring Test for In-sync in DRX in synchronous dual connectivity

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

*The Test system uncertainties applicable to this test are undefined*

*The Test tolerances applicable to this test are undefined*

### 7.3.43.1 Test purpose

The purpose of this test is to verify that the UE properly detects the out of sync and in sync for the purpose of monitoring downlink radio link quality of the PSCell when DRX is used in synchronous dual connectivity. This test will partly verify the E-UTRAN TDD radio link monitoring requirements in clause 7.6.

### 7.3.43.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward that supports Dual Connectivity.

### 7.3.43.3 Minimum conformance requirements

When DRX is used the $Q_{out}$ evaluation period ($T_{Evaluate_Q_{out_DRX}}$) and the $Q_{in}$ evaluation period ($T_{Evaluate_Q_{in_DRX}}$) is specified in TS 36.133[4] Table 7.6.2.2-1 will be used.

When the downlink radio link quality of the PCell or PSCell estimated over the last $T_{Evaluate_Q_{in_DRX}}$ [s] period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send in-sync indications for the PCell or PSCell to the higher layers within $T_{Evaluate_Q_{in_DRX}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The In-sync and in-sync evaluations shall be performed as specified in section 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10 ms, DRX_cycle_length).

Upon start of T310 timer or T313 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link of PCell or PSCell for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer or T313 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer and the transmitter power of PSCell if configured shall be turned off within 40 ms after expiry of T313 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.41.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX}}$ and $T_{Evaluate_Q_{in_DRX}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in TS36.133[4] clause 7.6.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note 1 (20) |
| 0.04 < DRX cycle ≤ 0.64 | Note 1 (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note 1 (5) |

Note 1: Evaluation period length in time depends on the length of the DRX cycle in use.
Note 2: MCG's DRX configuration is applied for PCell RLM evaluation and SCG's DRX configuration is applied for PSCell RLM evaluation.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.43.

## 7.3.43.4 Test description

There are two cells, cell 1 is PCell and cell 2 is PSCell, in the test. The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.43.4-1 shows the variation of the downlink SNR in the PSCell to emulate In-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell 2. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms. In the test, DRX configuration for PCell and PSCell is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode PDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to "infinity" so that UL timing alignment is maintained during the test.

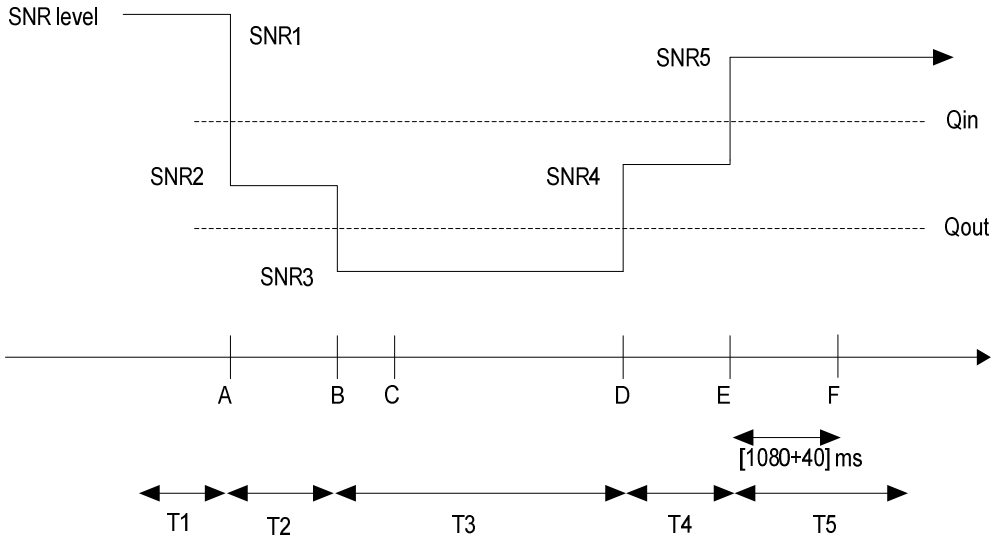


**Figure 7.3.43.4-1: SNR variation of cell 2 (PSCell) for in-sync testing in DRX**

### 7.3.43.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 7.3.43.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different DC configurations (per band) as defined in TS 36.521-1 [10] clause 5.4.2.

1. Connect the SS (node B emulator) and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.14 as appropriate.

2. The general test parameter settings for the different subtests are set up according to Table 7.3.43.4-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 7.3.43.4.3.

5. There are two cells specified in this test, Cell 1 and Cell2. Cell 1 is the PCell on RF channel 1and Cell 2 is the PSCell on RF channel 2. Cell1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.43.4.1-1: General test parameters for E-UTRAN TDD-TDD Radio Link Monitoring Test for In-sync in DRX in synchronous dual connectivity**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2 | Two E-UTRA TDD carrier frequency are used. |
| Active cell | | | Cell 1, Cell 2 | Cell 1 is PCell on E-UTRA RF channel number 1, and cell 2 is PSCell on E-UTRA RF channel number 2 |
| CP length | | | Normal | |
| In sync transmission parameters (Note 1) | DCI format | | 1C | As defined in clause 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | For 10MHz and 20MHz channel BW |
| | Number of Control OFDM symbols | | 3 | For 5MHz channel BW |
| | Aggregation level | CCE | 4 | In sync threshold $Q_{in}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in clause 7.6.1 and Table 7.6.1-2 respectively. |
| | ρA, ρB | | 0 | |
| | Ratio of PDCCH to RS EPRE | | 0 | |
| | Ratio of PCFICH to RS EPRE | | 4 | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | For 10MHz and 20MHz channel BW |
| | Number of Control OFDM symbols | | 3 | For 5MHz channel BW |
| | Aggregation level | CCE | 8 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in clause 7.6.1 and Table 7.6.1-1 respectively. |
| | ρA, ρB | | 0 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 4 | |
| DRX cycle on Cell 1 | | ms | 640 | See Table A.7.3.43.1-3 |
| DRX cycle on Cell 2 | | ms | 40 | See Table A.7.3.43.1-3 |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1, N313 = 1, N314 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| T313 timer | | ms | 2000 | T313 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 1 | Minimum CQI reporting periodicity |
| T1 | | s | 4 | |
| T2 | | s | 1.6 | |
| T3 | | s | 1.46 | |
| T4 | | s | 0.4 | |
| T5 | | s | 4 | |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel.<br>Note 2: Even a UE capable of both synchronous and asynchronous DC operations is required to pass this test case in accordance with the principle defined in section 3A.5. | | | | |

## 7.3.43.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell 2.

1. Ensure the UE is in State 3A-RF-DC2 if UE supports DC Split DRB bearer, according to TS 36.508 [7] clause 5.2A.2B; else ensure the UE is in state 3A-RF-DC1, according to TS 36.508 [7] clause 5.2A.2A.

2. The SS shall transmit an RRCConnectionReconfiguration message configuring the UE with the message content defined in clause 7.3.43.4.3.

3. The UE shall transmit RRCConnectionReconfigurationComplete message.
4. Set the parameters according to T1 in Table 7.3.43.5-1 for subtests 1 and 2 respectively. Propagation conditions are set according to Annex B.2.2. T1 starts.
5. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.43.5-1. T2 starts.
6. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.43.5-1. T3 starts.
7. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.43.5-1. T4 starts.
8. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.43.5-1. T5 starts.
9. If the SS on Cell 2 detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (1120 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
10. After T5 expires, repeat steps 2-9 for both subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.43.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.43.4.3-1: Common Exception messages for E-UTRAN TDD Radio Link Monitoring Test for In-sync in DRX in synchronous DC**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.43.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for In-sync in DRX in synchronous DC**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1C in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.43.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN TDD Radio Link Monitoring test for In-sync in DRX in synchronous DC, for cell 1**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf640 | 2 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.43.4.3-4: *MAC-MainConfig-RBC*: E-UTRAN TDD Radio Link Monitoring test for In-sync in DRX in synchronous DC, for cell 2**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf40 typical value in real network for real-time services. | |
| sf40 | 2 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.43.4.3-5: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN TDD Radio Link Monitoring test for In-sync in DRX in synchronous DC**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.43.4.3-6: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN TDD Radio Link Monitoring test for In-sync in DRX in synchronous DC**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 20 | 5 MHz channel bandwidth parameter | To be set depending on BW under test |
| | 41 | 10 MHz channel bandwidth parameter | |
| | 84 | 20 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 7.3.43.4.3-7: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN TDD Radio Link Monitoring test for In-sync in DRX in synchronous DC**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| rlf-TimersAndConstantsSCG-r12 CHOICE { | | | |
| setup SEQUENCE { | | | |
| t313-r12 | ms2000 | | |
| n313-r12 | n1 | | |
| n314-r12 | n1 | | |
| } | | | |
| } | | | |
| } | | | |

## 7.3.43.5 Test requirement

**Table 7.3.43.5-1: Cell specific test parameters for E-UTRAN TDD-TDD Radio Link Monitoring Test for In-sync in DRX in synchronous dual connectivity**

| Parameter | | Unit | Cell 1(PCell) | Cell 2 (PSCell) | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | T1 ~ T5 | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | | 1 | 2 | | | | |
| $BW_{channel}$ | | MHz | 5: $N_{RB,c}$ = 25<br>10: $N_{RB,c}$ = 50<br>20: $N_{RB,c}$ = 100 | 5: $N_{RB,c}$ = 25<br>10: $N_{RB,c}$ = 50<br>20: $N_{RB,c}$ = 100 | | | | |
| Special subframe configuration Note1 | | | 6 | 6 | | | | |
| Uplink-downlink configuration Note2 | | | 1 | 1 | | | | |
| PCFICH/PDCCH/PHICH parameters defined in A.2.2 | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | | |
| OCNG Pattern defined in D.2 (TDD) | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | | | |
| $\rho_A$, $\rho_B$ | | | 0 | 0 | | | | |
| PCFICH_RB | | dB | 4 | 4 | | | | |
| PDCCH_RA | | dB | 0 | 0 | | | | |
| PDCCH_RB | | dB | 0 | 0 | | | | |
| PBCH_RA | | dB | 0 | 0 | | | | |
| PBCH_RB | | dB | | | | | | |
| PSS_RA | | dB | | | | | | |
| SSS_RA | | dB | | | | | | |
| PHICH_RA | | dB | | | | | | |
| PHICH_RB | | dB | | | | | | |
| PDSCH_RA | | dB | | | | | | |
| PDSCH_RB | | dB | | | | | | |
| OCNG_RA Note1 | | dB | | | | | | |
| OCNG_RB Note1 | | dB | | | | | | |
| SNR Note 6 | 5MHz $BW_{channel}$ | dB | -4.5 | -4.5 | -8.5 | -13.7 | -9.7 | -4.5 |
| | 10MHz $BW_{channel}$ | | -4.5 | -4.5 | -8.5 | -13.7 | -9.7 | -4.5 |
| | 20MHz $BW_{channel}$ | | -4.5 | -4.5 | -8.5 | -13.7 | -9.7 | -4.5 |
| $N_{oc}$ | | dBm/15 kHz | -98 | | | | | |
| Propagation condition | | | AWGN | AWGN | | | | |
| Correlation Matrix and Antenna Configuration | | | 1x2 | 1x2 | | | | |
| Receive time offset to cell1 Note 9 | | μs | - | 33 | | | | |

Note 1: For the special subframe configuration see table 4.2-1 in TS 36.211.
Note 2: For the uplink-downlink configuration see table 4.2-2 in TS 36.211.
Note 3: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 5: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 6: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 7: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 8: The SNR in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure A.7.3.43.1-1.
Note 9: Receive time difference between subframe boundaries of signals received from the two cells at the UE antenna connector including time alignment error between the two cells.

**Table 7.3.43.5-2: DRX-Configuration for E-UTRAN TDD-TDD Radio Link Monitoring Test for In-sync in DRX in synchronous dual connectivity**

| Field | Value | | Comment |
|---|---|---|---|
| | Cell 1 | Cell 2 | |
| onDurationTimer | psf2 | psf2 | As specified in section 6.3.2 in 3GPP TS 36.331[5] |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf640 | sf40 | |
| shortDRX | disable | disable | |

**Table 7.3.43.5-3: TimeAlignmentTimer -Configuration for E-UTRAN TDD-TDD Radio Link Monitoring Test for In-sync in DRX in synchronous dual connectivity**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331[5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331[5] and section10.1 in TS 36.213[8]. |

The UE behaviour during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (1120 ms after the start of time duration T5) the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the subframe according to the configured CQI reporting mode (PUCCH 1-0) on PCell and PSCell.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.44 E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in FDD

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

*The Test system uncertainties applicable to this test are undefined*

*The Test tolerances applicable to this test are undefined*

### 7.3.44.1 Test purpose

The purpose of this test is to verify that the UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the PSCell when DRX is used in synchronous dual connectivity. This test will partly verify the E-UTRA radio link monitoring requirements in TS36.133[4] clause 7.6.

### 7.3.44.2 Test applicability

This test applies to all types of E-UTRA FDD and TDD UE release 12 and forward that supports Dual Connectivity.

### 7.3.44.3 Minimum conformance requirements

When DRX is used the $Q_{out}$ evaluation period ($T_{Evaluate_Q_{out_DRX}}$) and the $Q_{in}$ evaluation period ($T_{Evaluate_Q_{in_DRX}}$) is specified in TS 36.133[4] Table 7.6.2.2-1 will be used.

When the downlink radio link quality of the PCell or PSCell estimated over the last $T_{Evaluate_Q_{in_DRX}}$ [s] period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send in-sync indications for the PCell or PSCell to the higher layers within $T_{Evaluate_Q_{in_DRX}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations shall be performed as specified in section 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10 ms, DRX_cycle_length).

Upon start of T310 timer or T313 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link of PCell or PSCell for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer or T313 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer and the transmitter power of PSCell if configured shall be turned off within 40 ms after expiry of T313 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.44.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX}}$ and $T_{Evaluate_Q_{in_DRX}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in TS36.133[4] clause 7.6.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note 1 (20) |
| 0.04 < DRX cycle ≤ 0.64 | Note 1 (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note 1 (5) |

Note 1: Evaluation period length in time depends on the length of the DRX cycle in use.
Note 2: MCG's DRX configuration is applied for PCell RLM evaluation and SCG's DRX configuration is applied for PSCell RLM evaluation.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.44.

## 7.3.44.4 Test description

There are two cells, cell 1 is PCell and cell 2 is PSCell, in the test. The test consists of three successive time periods with time duration of T1, T2 and T3 respectively. Figure 7.3.44.4-1 shows the variation of the downlink SNR in the PCell and PSCell to emulate out-of-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell 2. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms on cell 1 and 1ms on cell 2. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode PDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to "infinity" so that UL timing alignment is maintained during the test.

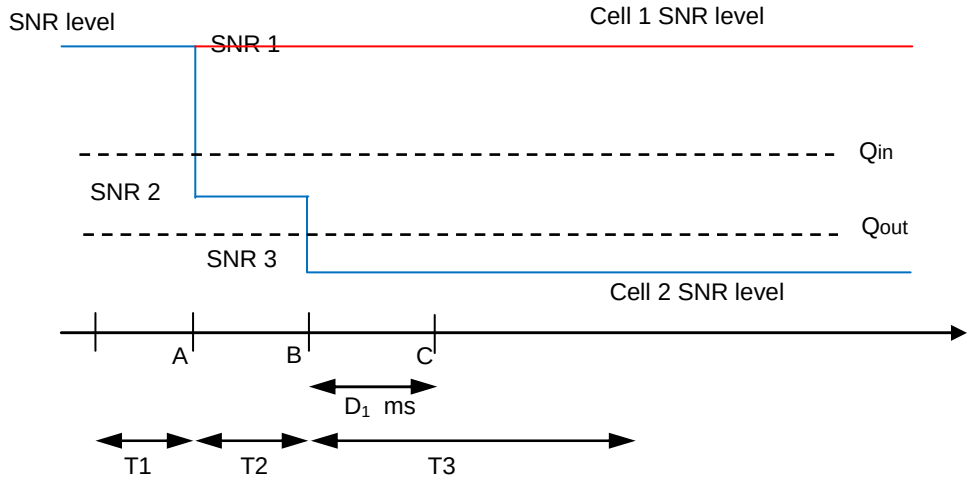


**Figure 7.3.44.4-1 SNR variation for out-of-sync testing in DRX**

### 7.3.44.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 7.3.44.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different DC configurations (per band) as defined in TS 36.521-1 [10] clause 5.4.2.

1. Connect the SS (node B emulator) and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.40 (without connection the AWGN sources) as appropriate.

2. The general test parameter settings for the different subtests are set up according to Table 7.3.44.4-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 7.3.44.4.3.

5. There are two cells specified in this test, Cell 1 and Cell2. Cell 1 is the PCell on RF channel 1and Cell 2 is the PSCell on RF channel 2. Cell1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.44.4.1-1: General test parameters for E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in FDD**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1<br>Cell 2 | Cell 1 is PCell on E-UTRA FDD RF channel number 1, and cell 2 is PSCell on E-UTRA TDD RF channel number 2 |
| CP length | | | Normal | |
| E-UTRA RF Channel Number | | | 1, 2 | One E-UTRA FDD carrier frequency and one E-UTRA TDD carrier frequency are used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 5, 10, 20 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.101 [2] Annex B.2.3.2 |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 5MHz: 3<br>10MHz: 2<br>20MHz: 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS36.133[4] clause 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | ρA, ρB | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX cycle on cell 1 | | ms | 640 | See Table 7.3.44.5-2 |
| DRX cycle on cell 2 | | ms | 40 | See Table 7.3.44.5-2 |
| Timing offset between cell 1 and cell 2 | | μs | 33 | For synchronous dual connectivity |
| Layer 3 filtering | | | Enabled | Counters:<br>N310 = 1; N311 = 1; N313 = 1; N314 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| T313 timer | | ms | 0 | T313 is disabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| T1 | | s | 4 | |
| T2 | | s | 1.6 | |
| T3 | | s | 1.8 | |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

## 7.3.44.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell 2.

1. Ensure the UE is in State 3A-RF-DC2 if UE supports DC Split DRB bearer, according to TS 36.508 [7] clause 5.2A.2B; else ensure the UE is in state 3A-RF-DC1, according to TS 36.508 [7] clause 5.2A.2A.

2. The SS shall transmit an RRCConnectionReconfiguration message configuring the UE with the message content defined in clause 7.3.44.4.3.

3. The UE shall transmit RRCConnectionReconfigurationComplete message.

4. Set the parameters according to T1 in Table 7.3.44.5-1 as appropriate for BW dependant parameters. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.

5. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.44.5-1. T2 starts.

6. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.44.5-1. T3 starts.

7. If the SS on Cell2:
   a) detects uplink power equal to or higher than -39 dBm in the On-duration part of every DRX cycle in the subframe according the configured CQI reporting mode (PUCCH 1-0) during the period from time point A to time point B
   and
   b) does not detect any uplink power higher than -48.5 dBm from time point C (900 ms after the start of T3) until T3 expires,

   the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.

8. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.44.5-1.

9. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF-DC2 if UE supports DC Split DRB bearer, according to TS 36.508 [7] clause 5.2A.2B; else ensure the UE is in state 3A-RF-DC1, according to TS 36.508 [7] clause 5.2A.2A.

10. Repeat steps 2-9 for all subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.44.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.44.4.3-1: Common Exception messages for E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in FDD**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.44.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in FDD, for Cell 1 (FDD)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.44.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in FDD, for cell 1**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf640 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.44.4.3-4: *MAC-MainConfig-RBC* E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in FDD, for cell 2**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf40 typical value in real network for real-time services. | |
| sf40 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.44.4.3-5: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in FDD**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.44.4.3-6: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in FDD**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.44.4.3-7: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in FDD**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 20 | 5 MHz channel bandwidth parameter | To be set depending on BW under test |
| | 41 | 10 MHz channel bandwidth parameter | |
| | 84 | 20 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 7.3.44.4.3-8: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in FDD**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| rlf-TimersAndConstantsSCG-r12 CHOICE { | | | |
| setup SEQUENCE { | | | |
| t313-r12 | ms0 | | |
| n313-r12 | n1 | | |
| n314-r12 | n1 | | |
| } | | | |
| } | | | |
| } | | | |

## 7.3.44.5 Test requirement

**Table 7.3.44.5-1: Cell specific test parameters for E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in FDD**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 5, 10, 20 | | | 5, 10, 20 | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | 2x2 Low | | |
| Special subframe configuration[Note7] | | - | | | 6 | | |
| Uplink-downlink configuration[Note8] | | - | | | 1 | | |
| PCFICH/PDCCH/PHICH parameters<br>None of the PDCCH are intended for the UE under test | | 5MHz: R.12 FDD<br>10MHz: R.7 FDD<br>20MHz: R.13 FDD | | | 5MHz: R.12 TDD<br>10MHz: R.7 TDD<br>20MHz: R.13 TDD | | |
| OCNG Pattern | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | |
| ρA, ρB | | -3 | | | -3 | | |
| PCFICH_RB | dB | 1 | | | 1 | | |
| PDCCH_RA | dB | 1 | | | 1 | | |
| PDCCH_RB | dB | 1 | | | 1 | | |
| PBCH_RA | dB | -3 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA[Note1] | dB | | | | | | |
| OCNG_RB[Note1] | dB | | | | | | |
| SNR [Note 6] (5MHz bandwidth) | dB | -2.3+TT | -2.3+TT | -2.3+TT | -1.6+TT | -5.2+TT | -11.9+TT |
| SNR [Note 6] (10MHz bandwidth) | dB | -2.3+TT | -2.3+TT | -2.3+TT | -2.3+TT | -5.9+TT | -11.9+TT |
| SNR [Note 6] (20MHz bandwidth) | dB | -2.9+TT | -2.9+TT | -2.9+TT | -3.0+TT | -6.6+TT | -12.6+TT |
| $N_{oc}$ | dBm/15 kHz | -98 | | | -98 | | |
| Propagation condition | | ETU 70 Hz | | | ETU 70 Hz | | |
| Time offset to cell1 [Note 9] | μs | - | | | 33 | | |
| CQI reporting periodicity | ms | 2 | | | 1 | | |

Note 1: OCNG shall be used such that the resources in cell 1 and cell 2 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2, and T3 are denoted as SNR1, SNR2 and SNR3 respectively in figure A.7.3.38.1-1.
Note 7: For the special subframe configuration see table 4.2-1 in TS 36.211.
Note 8: For the uplink-downlink configuration see table 4.2-2 in TS 36.211.
Note 9: Receive time difference between subframe boundaries of signals received from the two cells at the UE antenna connector including time alignment error between the two cells.

**Table 7.3.44.5-2: DRX-Configuration for E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in FDD**

| Field | Value | | Comment |
|---|---|---|---|
| | Cell 1 | Cell 2 | |
| onDurationTimer | psf2 | psf2 | As specified in clause 6.3.2 in TS 36.331 [5] |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | Sf640 | Sf40 | |
| shortDRX | disable | disable | |

**Table 7.3.44.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331[5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331[5] and section 10.1 in TS 36.213 [8]. |

The UE behaviour during time durations T1, T2, and T3 shall be as follows:

During time durations T1, T2 and T3, the UE shall transmit uplink signal at least in all subframes configured for CQI transmission on Cell1.

During the period from time point A to time point B the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the uplink subframe on Cell2.

The UE shall stop transmitting uplink signal no later than time point C (duration D1 = 900 ms after the start of time duration T3) on PSCell.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.45 E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in TDD

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

*The Test system uncertainties applicable to this test are undefined*

*The Test tolerances applicable to this test are undefined*

### 7.3.45.1 Test purpose

The purpose of this test is to verify that the UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the PSCell when DRX is used in synchronous dual connectivity. This test will partly verify the E-UTRA radio link monitoring requirements in TS36.133[4] clause 7.6.

### 7.3.45.2 Test applicability

This test applies to all types of E-UTRA FDD and TDD UE release 12 and forward that supports Dual Connectivity.

### 7.3.45.3 Minimum conformance requirements

When DRX is used the $Q_{out}$ evaluation period ($T_{Evaluate_Q_{out_DRX}}$) and the $Q_{in}$ evaluation period ($T_{Evaluate_Q_{in_DRX}}$) is specified in TS 36.133[4] Table 7.6.2.2-1 will be used.

When the downlink radio link quality of the PCell or PSCell estimated over the last $T_{Evaluate_Q_{in_DRX}}$ [s] period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send in-sync indications for the PCell or PSCell to the higher layers within $T_{Evaluate_Q_{in_DRX}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The out-of-sync and in-sync evaluations shall be performed as specified in section 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10 ms, DRX_cycle_length).

Upon start of T310 timer or T313 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link of PCell or PSCell for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer or T313 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer and the transmitter power of PSCell if configured shall be turned off within 40 ms after expiry of T313 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.45.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX}}$ and $T_{Evaluate_Q_{in_DRX}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in TS36.133[4] clause 7.6.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note 1 (20) |
| 0.04 < DRX cycle ≤ 0.64 | Note 1 (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note 1 (5) |

Note 1: Evaluation period length in time depends on the length of the DRX cycle in use.
Note 2: MCG's DRX configuration is applied for PCell RLM evaluation and SCG's DRX configuration is applied for PSCell RLM evaluation.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.45.

## 7.3.45.4 Test description

There are two cells, cell 1 is PCell and cell 2 is PSCell, in the test. The test consists of three successive time periods with time duration of T1, T2 and T3 respectively. Figure 7.3.45.4-1 shows the variation of the downlink SNR in the PCell and PSCell to emulate out-of-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell 2. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms on cell 1 and 1ms on cell 2. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode PDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to "infinity" so that UL timing alignment is maintained during the test.

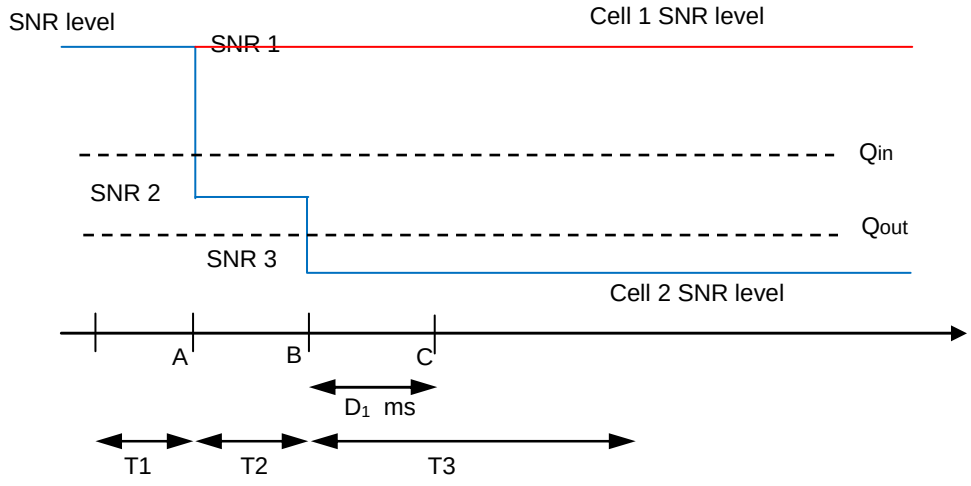


**Figure 7.3.45.4-1 SNR variation for out-of-sync testing in DRX**

### 7.3.45.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 7.3.45.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different DC configurations (per band) as defined in TS 36.521-1 [10] clause 5.4.2.

1. Connect the SS (node B emulator) and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.40 (without connection the AWGN sources) as appropriate.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.45.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.45.4.3.
5. There are two cells specified in this test, Cell 1 and Cell2. Cell 1 is the PCell on RF channel 1and Cell 2 is the PSCell on RF channel 2. Cell1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.45.4.1-1: General test parameters for E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in TDD**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1<br>Cell 2 | Cell 1 is PCell on E-UTRA TDD RF channel number 1, and cell 2 is PSCell on E-UTRA FDD RF channel number 2 |
| CP length | | | Normal | |
| E-UTRA RF Channel Number | | | 1, 2 | One E-UTRA TDD carrier frequency and one E-UTRA FDD carrier frequency are used. |
| E-UTRA Channel Bandwidth (BWchannel) | | MHz | 5, 10, 20 | |
| Correlation Matrix and Antenna Configuration | | | 2x2 Low | Correlation Matrix and Antenna Configuration are defined in TS 36.101 [2] Annex B.2.3.2 |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 5MHz: 3<br>10MHz: 2<br>20MHz: 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS36.133[4] clause 7.6.1 and Table 7.6.1-1 respectively. |
| | Aggregation level | CCE | 8 | |
| | ρA, ρB | | -3 | |
| | Ratio of PDCCH to RS EPRE | dB | 1 | |
| | Ratio of PCFICH to RS EPRE | dB | 1 | |
| DRX cycle on cell 1 | | ms | 640 | See Table 7.3.45.5-2 |
| DRX cycle on cell 2 | | ms | 40 | See Table 7.3.45.5-2 |
| Timing offset between cell 1 and cell 2 | | μs | 33 | For synchronous dual connectivity |
| Layer 3 filtering | | | Enabled | Counters:<br>N310 = 1; N311 = 1; N313 = 1; N314 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| T313 timer | | ms | 0 | T313 is disabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| T1 | | s | 4 | |
| T2 | | s | 1.6 | |
| T3 | | s | 1.8 | |
| Note 1: PDCCH/PCFICH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

### 7.3.45.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell 2.

1. Ensure the UE is in State 3A-RF-DC2 if UE supports DC Split DRB bearer, according to TS 36.508 [7] clause 5.2A.2B; else ensure the UE is in state 3A-RF-DC1, according to TS 36.508 [7] clause 5.2A.2A.

2. The SS shall transmit an RRCConnectionReconfiguration message configuring the UE with the message content defined in clause 7.3.45.4.3.

3. The UE shall transmit RRCConnectionReconfigurationComplete message.

4. Set the parameters according to T1 in Table 7.3.45.5-1 as appropriate for BW dependant parameters. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.

5. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.45.5-1. T2 starts.

6. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.45.5-1. T3 starts.

7. If the SS on Cell2:
   a) detects uplink power equal to or higher than -39 dBm in the On-duration part of every DRX cycle in the subframe according the configured CQI reporting mode (PUCCH 1-0) during the period from time point A to time point B
   and
   b) does not detect any uplink power higher than -48.5 dBm from time point C (900 ms after the start of T3) until T3 expires,

   the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.

8. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.45.5-1.

9. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF-DC2 if UE supports DC Split DRB bearer, according to TS 36.508 [7] clause 5.2A.2B; else ensure the UE is in state 3A-RF-DC1, according to TS 36.508 [7] clause 5.2A.2A.

10. Repeat steps 2-9 for all subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.45.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.45.4.3-1: Common Exception messages for E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in TDD**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.45.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in TDD, for Cell 1 (FDD)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.45.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in TDD, for cell 1**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf640 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.45.4.3-4: *MAC-MainConfig-RBC* E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in TDD, for cell 2**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf40 typical value in real network for real-time services. | |
| sf40 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.45.4.3-5: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in TDD**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.45.4.3-6: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in TDD**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.45.4.3-7: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in TDD**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 20 | 5 MHz channel bandwidth parameter | To be set depending on BW under test |
| | 41 | 10 MHz channel bandwidth parameter | |
| | 84 | 20 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 7.3.45.4.3-8: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in TDD**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| rlf-TimersAndConstantsSCG-r12 CHOICE { | | | |
| setup SEQUENCE { | | | |
| t313-r12 | ms0 | | |
| n313-r12 | n1 | | |
| n314-r12 | n1 | | |
| } | | | |
| } | | | |
| } | | | |

## 7.3.45.5 Test requirement

**Table 7.3.45.5-1: Cell specific test parameters for E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in TDD**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 5, 10, 20 | | | 5, 10, 20 | | |
| Correlation Matrix and Antenna Configuration | | 2x2 Low | | | 2x2 Low | | |
| Special subframe configuration[Note1] | | 6 | | | - | | |
| Uplink-downlink configuration[Note2] | | 1 | | | - | | |
| PCFICH/PDCCH/PHICH parameters<br>None of the PDCCH are intended for the UE under test | | 5MHz: R.12 TDD<br>10MHz: R.7 TDD<br>20MHz: R.13 TDD | | | 5MHz: R.12 FDD<br>10MHz: R.7 FDD<br>20MHz: R.13 FDD | | |
| OCNG Pattern | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | |
| ρA, ρB | | -3 | | | -3 | | |
| PCFICH_RB | dB | 1 | | | 1 | | |
| PDCCH_RA | dB | 1 | | | 1 | | |
| PDCCH_RB | dB | 1 | | | 1 | | |
| PBCH_RA | dB | -3 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA[Note3] | dB | | | | | | |
| OCNG_RB[Note3] | dB | | | | | | |
| SNR [Note 8] (5MHz bandwidth) | dB | -1.6+TT | -1.6+TT | -1.6+TT | -2.3+TT | -5.7+TT | -12.2+TT |
| SNR [Note 8] (10MHz bandwidth) | dB | -2.3+TT | -2.3+TT | -2.3+TT | -2.3+TT | -6.2+TT | -12.2+TT |
| SNR [Note 8] (20MHz bandwidth) | dB | -3.0+TT | -3.0+TT | -3.0+TT | -2.9+TT | -6.8+TT | -12.8+TT |
| $N_{oc}$ | dBm/15 kHz | -98 | | | -98 | | |
| Propagation condition | | ETU 70 Hz | | | ETU 70 Hz | | |
| Time offset to cell1 [Note 9] | μs | - | | | 33 | | |
| CQI reporting periodicity | ms | 1 | | | 2 | | |

Note 1: For the special subframe configuration see table 4.2-1 in TS 36.211.
Note 2: For the uplink-downlink configuration see table 4.2-2 in TS 36.211.
Note 3: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 5: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 6: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 7: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 8: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure A.7.3.40.1-1.
Note 9: Receive time difference between subframe boundaries of signals received from the two cells at the UE antenna connector including time alignment error between the two cells.

**Table 7.3.45.5-2: DRX-Configuration for E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in TDD**

| Field | Value | | Comment |
|---|---|---|---|
| | Cell 1 | Cell 2 | |
| onDurationTimer | psf2 | psf2 | As specified in clause 6.3.2 in TS 36.331 [5] |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | Sf640 | Sf40 | |
| shortDRX | disable | disable | |

**Table 7.3.45.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331[5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331[5] and section 10.1 in TS 36.213 [8]. |

The UE behaviour during time durations T1, T2, and T3 shall be as follows:

During time durations T1, T2 and T3, the UE shall transmit uplink signal at least in all subframes configured for CQI transmission on Cell1.

During the period from time point A to time point B the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the uplink subframe on Cell2.

The UE shall stop transmitting uplink signal no later than time point C (duration D1 = 900 ms after the start of time duration T3) on PSCell.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.46 E-UTRAN TDD-FDD Radio Link Monitoring Test for In-sync in DRX for PSCell in synchronous DC with PCell in FDD

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

*The Test system uncertainties applicable to this test are undefined*

*The Test tolerances applicable to this test are undefined*

### 7.3.46.1 Test purpose

The purpose of this test is to verify that the UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the PSCell when DRX is used in synchronous dual connectivity. This test will partly verify the E-UTRA radio link monitoring requirements in TS36.133[4] clause 7.6.

### 7.3.46.2 Test applicability

This test applies to all types of E-UTRA FDD and TDD UE release 12 and forward that supports Dual Connectivity.

### 7.3.46.3 Minimum conformance requirements

When DRX is used the $Q_{out}$ evaluation period ($T_{Evaluate_Q_{out_DRX}}$) and the $Q_{in}$ evaluation period ($T_{Evaluate_Q_{in_DRX}}$) is specified in TS 36.133[4] Table 7.6.2.2-1 will be used.

When the downlink radio link quality of the PCell or PSCell estimated over the last $T_{Evaluate_Q_{in_DRX}}$ [s] period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send in-sync indications for the PCell or PSCell to the higher layers within $T_{Evaluate_Q_{in_DRX}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The In-sync and in-sync evaluations shall be performed as specified in section 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10 ms, DRX_cycle_length).

Upon start of T310 timer or T313 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link of PCell or PSCell for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer or T313 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer and the transmitter power of PSCell if configured shall be turned off within 40 ms after expiry of T313 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.46.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX}}$ and $T_{Evaluate_Q_{in_DRX}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in TS36.133[4] clause 7.6.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note 1 (20) |
| 0.04 < DRX cycle ≤ 0.64 | Note 1 (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note 1 (5) |
| Note 1: Evaluation period length in time depends on the length of the DRX cycle in use.<br>Note 2: MCG's DRX configuration is applied for PCell RLM evaluation and SCG's DRX configuration is applied for PSCell RLM evaluation. | |

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.46.

## 7.3.46.4 Test description

There are two cells, cell 1 is PCell and cell 2 is PSCell, in the test. The test consists of five successive time periods with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.46.4-1 shows the variation of the downlink SNR in the PCell and PSCell to emulate in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell 2. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms on cell 1 and 1ms on cell 2. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode PDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to "infinity" so that UL timing alignment is maintained during the test.

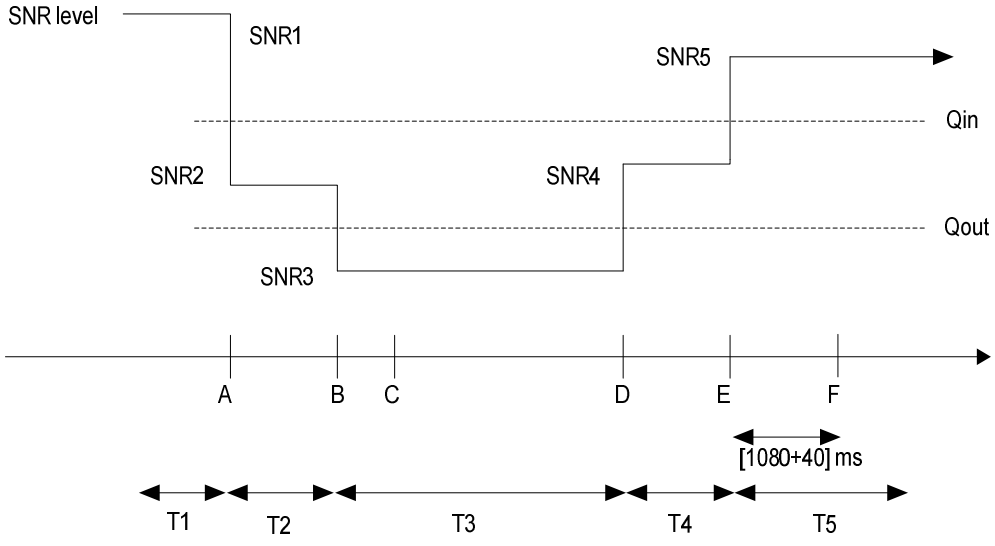


**Figure 7.3.46.4-1 SNR variation of cell 2 (PSCell) for in-sync testing in DRX**

#### 7.3.46.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 7.3.46.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different DC configurations (per band) as defined in TS 36.521-1 [10] clause 5.4.2.

1. Connect the SS (node B emulator) and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.14 as appropriate.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.46.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.46.4.3.
5. There are two cells specified in this test, Cell 1 and Cell2. Cell 1 is the PCell on RF channel 1and Cell 2 is the PSCell on RF channel 2. Cell1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.46.4.1-1: General test parameters for E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in TDD**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2 | One E-UTRA FDD carrier frequency and One E-UTRA TDD carrier frequency are used. |
| Active cell | | | Cell 1, Cell 2 | Cell 1 is PCell on E-UTRA FDD RF channel number 1, and cell 2 is PSCell on E-UTRA TDD RF channel number 2 |
| CP length | | | Normal | |
| In sync transmission parameters (Note 1) | DCI format | | 1C | As defined in clause 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | For 10MHz and 20MHz channel BW |
| | Number of Control OFDM symbols | | 3 | For 5MHz channel BW |
| | Aggregation level | CCE | 4 | In sync threshold $Q_{in}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS36.133[4] clause 7.6.1 and Table 7.6.1-2 respectively. |
| | ρA, ρB | | 0 | |
| | Ratio of PDCCH to RS EPRE | | 0 | |
| | Ratio of PCFICH to RS EPRE | | 4 | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | For 10MHz and 20MHz channel BW |
| | Number of Control OFDM symbols | | 3 | For 5MHz channel BW |
| | Aggregation level | CCE | 8 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS36.133[4] clause 7.6.1 and Table 7.6.1-1 respectively. |
| | ρA, ρB | | 0 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 4 | |
| DRX cycle on Cell 1 | | ms | 640 | See Table 7.3.46.5-2 |
| DRX cycle on Cell 2 | | ms | 40 | See Table 7.3.46.5-2 |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1, N313 = 1, N314 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| T313 timer | | ms | 2000 | T313 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| T1 | | s | 4 | |
| T2 | | s | 1.6 | |
| T3 | | s | 1.46 | |
| T4 | | s | 0.4 | |
| T5 | | s | 4 | |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel.<br>Note 2: Even a UE capable of both synchronous and asynchronous DC operations is required to pass this test case in accordance with the principle defined in section 3A.5. | | | | |

## 7.3.46.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell 2.

1. Ensure the UE is in State 3A-RF-DC2 if UE supports DC Split DRB bearer, according to TS 36.508 [7] clause 5.2A.2B; else ensure the UE is in state 3C-RF, according to TS 36.508 [7] clause 5.2A.2A.

2. The SS shall transmit an RRCConnectionReconfiguration message configuring the UE with the message content defined in clause 7.3.46.4.3.

3. The UE shall transmit RRCConnectionReconfigurationComplete message.
4. Set the parameters according to T1 in Table 7.3.61.5-1 for subtests 1 and 2 respectively. Propagation conditions are set according to Annex B.2.2. T1 starts.
5. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.46.5-1. T2 starts.
6. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.46.5-1. T3 starts.
7. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.46.5-1. T4 starts.
8. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.46.5-1. T5 starts.
9. If the SS on Cell 2 detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (1120 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
10. After T5 expires, repeat steps 2-9 for both subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.46.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.46.4.3-1: Common Exception messages for E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in FDD**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.46.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in FDD**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.46.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in FDD, for cell 1**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf640 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.46.4.3-4: *MAC-MainConfig-RBC* E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in FDD, for cell 2**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf40 typical value in real network for real-time services. | |
| sf40 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.46.4.3-5: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in FDD**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.46.4.3-6: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in FDD**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.46.4.3-7: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in FDD**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 20 | 5 MHz channel bandwidth parameter | To be set depending on BW under test |
| | 41 | 10 MHz channel bandwidth parameter | |
| | 84 | 20 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 7.3.46.4.3-8: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in FDD**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| rlf-TimersAndConstantsSCG-r12 CHOICE { | | | |
| setup SEQUENCE { | | | |
| t313-r12 | ms2000 | | |
| n313-r12 | n1 | | |
| n314-r12 | n1 | | |
| } | | | |
| } | | | |
| } | | | |

## 7.3.46.5 Test requirement

**Table 7.3.46.5-1: Cell specific test parameters for E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in FDD**

| Parameter | | Unit | Cell 1(PCell) | Cell 2 (PSCell) | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | T1 ~ T5 | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | | 1 | 2 | | | | |
| $BW_{channel}$ | | MHz | 5: $N_{RB,c}$ = 25<br>10: $N_{RB,c}$ = 50<br>20: $N_{RB,c}$ = 100 | 5: $N_{RB,c}$ = 25<br>10: $N_{RB,c}$ = 50<br>20: $N_{RB,c}$ = 100 | | | | |
| Special subframe configuration Note1 | | | - | 6 | | | | |
| Uplink-downlink configuration Note2 | | | - | 1 | | | | |
| PCFICH/PDCCH/PHICH parameters defined in A.3.1.2.1 and A.3.1.2.2 | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | | |
| OCNG Pattern defined in A.3.2.1 (FDD) and A.3.2.2 (TDD) | | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | | | |
| $\rho_A$, $\rho_B$ | | | 0 | 0 | | | | |
| PCFICH_RB | | dB | 4 | 4 | | | | |
| PDCCH_RA | | dB | 0 | 0 | | | | |
| PDCCH_RB | | dB | 0 | 0 | | | | |
| PBCH_RA | | dB | 0 | 0 | | | | |
| PBCH_RB | | dB | | | | | | |
| PSS_RA | | dB | | | | | | |
| SSS_RA | | dB | | | | | | |
| PHICH_RA | | dB | | | | | | |
| PHICH_RB | | dB | | | | | | |
| PDSCH_RA | | dB | | | | | | |
| PDSCH_RB | | dB | | | | | | |
| OCNG_RA Note3 | | dB | | | | | | |
| OCNG_RB Note3 | | dB | | | | | | |
| SNR Note 8 | 5MHz $BW_{channel}$ | dB | -2.3+TT | -5.1+TT | -9.1+TT | -13.1+TT | -9.1+TT | -5.1+TT |
| | 10MHz $BW_{channel}$ | | -4.7+TT | -5.1+TT | -9.1+TT | -13.1+TT | -9.1+TT | -5.1+TT |
| | 20MHz $BW_{channel}$ | | -4.7+TT | -5.1+TT | -9.1+TT | -13.1+TT | -9.1+TT | -5.1+TT |
| $N_{oc}$ | | dBm/15 kHz | -98 | | | | | |
| Propagation condition | | | AWGN | AWGN | | | | |
| Correlation Matrix and Antenna Configuration | | | 1x2 | 1x2 | | | | |
| Receive time offset to cell1 Note 9 | | μs | - | 33 | | | | |
| CQI reporting periodicity | | ms | 2 | 1 | | | | |

Note 1: For the special subframe configuration see table 4.2-1 in TS 36.211.
Note 2: For the uplink-downlink configuration see table 4.2-2 in TS 36.211.
Note 3: OCNG shall be used such that the resources in cell 1 and cell 2 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 5: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 6: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 7: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 8: The SNR of Cell 2 in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure A.7.3.46.1-1.
Note 9: Receive time difference between subframe boundaries of signals received from the two cells at the UE antenna connector including time alignment error between the two cells.

**Table 7.3.46.5-2: DRX-Configuration for E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in FDD**

| Field | Value | | Comment |
|---|---|---|---|
| | Cell 1 | Cell 2 | |
| onDurationTimer | psf2 | psf2 | As specified in clause 6.3.2 in TS 36.331 [5] |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | Sf640 | Sf40 | |
| shortDRX | disable | disable | |

**Table 7.3.46.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331[5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331[5] and section 10.1 in TS 36.213 [8]. |

The UE behaviour during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (1120 ms after the start of time duration T5) the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the subframe according to the configured CQI reporting mode (PUCCH 1-0) on PCell and PSCell.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.47 E-UTRAN TDD-FDD Radio Link Monitoring Test for In-sync in DRX for PSCell in synchronous DC with PCell in TDD

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

*The Test system uncertainties applicable to this test are undefined*

*The Test tolerances applicable to this test are undefined*

### 7.3.47.1 Test purpose

The purpose of this test is to verify that the UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the PSCell when DRX is used in synchronous dual connectivity. This test will partly verify the E-UTRA radio link monitoring requirements in TS36.133[4] clause 7.6.

### 7.3.47.2 Test applicability

This test applies to all types of E-UTRA FDD and TDD UE release 12 and forward that supports Dual Connectivity.

### 7.3.47.3 Minimum conformance requirements

When DRX is used the $Q_{out}$ evaluation period ($T_{Evaluate_Q_{out_DRX}}$) and the $Q_{in}$ evaluation period ($T_{Evaluate_Q_{in_DRX}}$) is specified in TS 36.133[4] Table 7.6.2.2-1 will be used.

When the downlink radio link quality of the PCell or PSCell estimated over the last $T_{Evaluate_Q_{in_DRX}}$ [s] period becomes better than the threshold $Q_{in}$, Layer 1 of the UE shall send in-sync indications for the PCell or PSCell to the higher layers within $T_{Evaluate_Q_{in_DRX}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5] clause 5.5.3.2.

The In-sync and in-sync evaluations shall be performed as specified in section 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least max(10 ms, DRX_cycle_length).

Upon start of T310 timer or T313 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link of PCell or PSCell for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer or T313 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer and the transmitter power of PSCell if configured shall be turned off within 40 ms after expiry of T313 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.47.3-1: $Q_{out}$ and $Q_{in}$ Evaluation Period in DRX**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX}}$ and $T_{Evaluate_Q_{in_DRX}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in TS36.133[4] clause 7.6.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note 1 (20) |
| 0.04 < DRX cycle ≤ 0.64 | Note 1 (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note 1 (5) |

Note 1: Evaluation period length in time depends on the length of the DRX cycle in use.
Note 2: MCG's DRX configuration is applied for PCell RLM evaluation and SCG's DRX configuration is applied for PSCell RLM evaluation.

The normative reference for this requirement is TS 36.133 [4] clause 7.6.2 and A.7.3.47.

## 7.3.47.4 Test description

There are two cells, cell 1 is PCell and cell 2 is PSCell, in the test. The test consists of five successive time periods with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.47.4-1 shows the variation of the downlink SNR in the PCell and PSCell to emulate in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell 2. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms on cell 1 and 1ms on cell 2. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode PDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to "infinity" so that UL timing alignment is maintained during the test.

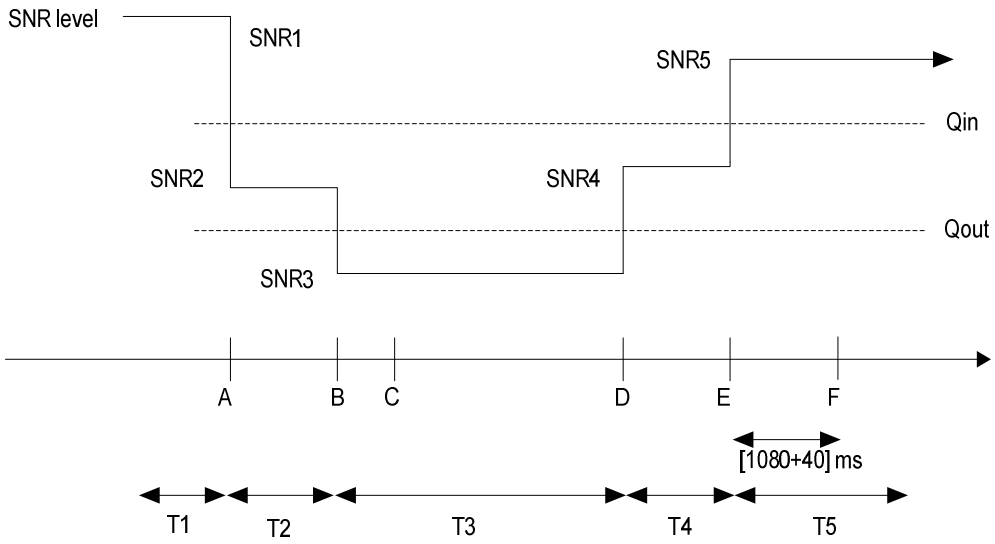


**Figure 7.3.47.4-1 SNR variation of cell 2 (PSCell) for in-sync testing in DRX**

#### 7.3.47.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 7.3.47.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different DC configurations (per band) as defined in TS 36.521-1 [10] clause 5.4.2.

1. Connect the SS (node B emulator) and faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.14 as appropriate.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.47.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.47.4.3.
5. There are two cells specified in this test, Cell 1 and Cell2. Cell 1 is the PCell on RF channel 1and Cell 2 is the PSCell on RF channel 2. Cell1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.47.4.1-1: General test parameters for E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in TDD**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2 | One E-UTRA TDD carrier frequency and One E-UTRA FDD carrier frequency are used. |
| Active cell | | | Cell 1, Cell 2 | Cell 1 is PCell on E-UTRA TDD RF channel number 1, and cell 2 is PSCell on E-UTRA FDD RF channel number 2 |
| CP length | | | Normal | |
| In sync transmission parameters (Note 1) | DCI format | | 1C | As defined in clause 5.3.3.1.4 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | For 10MHz and 20MHz channel BW |
| | Number of Control OFDM symbols | | 3 | For 5MHz channel BW |
| | Aggregation level | CCE | 4 | In sync threshold $Q_{in}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS36.133[4] clause 7.6.1 and Table 7.6.1-2 respectively. |
| | ρA, ρB | | 0 | |
| | Ratio of PDCCH to RS EPRE | | 0 | |
| | Ratio of PCFICH to RS EPRE | | 4 | |
| Out of sync transmission parameters (Note 1) | DCI format | | 1A | As defined in clause 5.3.3.1.3 in TS 36.212 |
| | Number of Control OFDM symbols | | 2 | For 10MHz and 20MHz channel BW |
| | Number of Control OFDM symbols | | 3 | For 5MHz channel BW |
| | Aggregation level | CCE | 8 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical PDCCH/PCFICH transmission parameters are as specified in TS36.133[4] clause 7.6.1 and Table 7.6.1-1 respectively. |
| | ρA, ρB | | 0 | |
| | Ratio of PDCCH to RS EPRE | dB | 4 | |
| | Ratio of PCFICH to RS EPRE | dB | 4 | |
| DRX cycle on Cell 1 | | ms | 640 | See Table 7.3.47.5-2 |
| DRX cycle on Cell 2 | | ms | 40 | See Table 7.3.47.5-2 |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1, N313 = 1, N314 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| T313 timer | | ms | 2000 | T313 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| T1 | | s | 4 | |
| T2 | | s | 1.6 | |
| T3 | | s | 1.47 | |
| T4 | | s | 0.4 | |
| T5 | | s | 4 | |
| Note 1: PDCCH/PCFICH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel.<br>Note 2: Even a UE capable of both synchronous and asynchronous DC operations is required to pass this test case in accordance with the principle defined in section 3A.5. | | | | |

## 7.3.47.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1 and cell 2.

1. Ensure the UE is in State 3A-RF-DC2 if UE supports DC Split DRB bearer, according to TS 36.508 [7] clause 5.2A.2B; else ensure the UE is in state 3C-RF, according to TS 36.508 [7] clause 5.2A.2A.

2. The SS shall transmit an RRCConnectionReconfiguration message configuring the UE with the message content defined in clause 7.3.47.4.3.

3. The UE shall transmit RRCConnectionReconfigurationComplete message.
4. Set the parameters according to T1 in Table 7.3.61.5-1 for subtests 1 and 2 respectively. Propagation conditions are set according to Annex B.2.2. T1 starts.
5. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.47.5-1. T2 starts.
6. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.47.5-1. T3 starts.
7. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.47.5-1. T4 starts.
8. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.47.5-1. T5 starts.
9. If the SS on Cell 2 detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (1120 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
10. After T5 expires, repeat steps 2-9 for both subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.47.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.47.4.3-1: Common Exception messages for E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in TDD**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.47.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in TDD**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.47.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in TDD, for cell 1**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf640 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.47.4.3-4: *MAC-MainConfig-RBC* E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in TDD, for cell 2**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf40 typical value in real network for real-time services. | |
| sf40 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.47.4.3-5: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in TDD**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.47.4.3-6: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in TDD**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | 2TX |
| } | | | |
| } | | | |

**Table 7.3.47.4.3-7: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in TDD**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 20 | 5 MHz channel bandwidth parameter | To be set depending on BW under test |
| | 41 | 10 MHz channel bandwidth parameter | |
| | 84 | 20 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 7.3.47.4.3-8: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in TDD**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| rlf-TimersAndConstantsSCG-r12 CHOICE { | | | |
| setup SEQUENCE { | | | |
| t313-r12 | ms2000 | | |
| n313-r12 | n1 | | |
| n314-r12 | n1 | | |
| } | | | |
| } | | | |
| } | | | |

## 7.3.47.5 Test requirement

**Table 7.3.47.5-1: Cell specific test parameters for E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in TDD**

| Parameter | | Unit | Cell 1(PCell) | Cell 2 (PSCell) | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | T1 ~ T5 | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | | 1 | 2 | | | | |
| $BW_{channel}$ | | MHz | 5: $N_{RB,c}$ = 25<br>10: $N_{RB,c}$ = 50<br>20: $N_{RB,c}$ = 100 | 5: $N_{RB,c}$ = 25<br>10: $N_{RB,c}$ = 50<br>20: $N_{RB,c}$ = 100 | | | | |
| Special subframe configuration[Note1] | | | 6 | - | | | | |
| Uplink-downlink configuration[Note2] | | | 1 | - | | | | |
| PCFICH/PDCCH/PHICH parameters defined in A 2.1 and A. 2.2 | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | | |
| OCNG Pattern defined in D.1 (FDD) and D.2 (TDD) | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | | | |
| $\rho_A$, $\rho_B$ | | | 0 | 0 | | | | |
| PCFICH_RB | | dB | 4 | 4 | | | | |
| PDCCH_RA | | dB | 0 | 0 | | | | |
| PDCCH_RB | | dB | 0 | 0 | | | | |
| PBCH_RA | | dB | 0 | 0 | | | | |
| PBCH_RB | | dB | | | | | | |
| PSS_RA | | dB | | | | | | |
| SSS_RA | | dB | | | | | | |
| PHICH_RA | | dB | | | | | | |
| PHICH_RB | | dB | | | | | | |
| PDSCH_RA | | dB | | | | | | |
| PDSCH_RB | | dB | | | | | | |
| OCNG_RA[Note1] | | dB | | | | | | |
| OCNG_RB[Note1] | | dB | | | | | | |
| SNR [Note 6] | 5MHz $BW_{channel}$ | dB | -5.1+TT | -2.3+TT | -5.7+TT | -12.2+TT | -7.3+TT | -2.3+TT |
| | 10MHz $BW_{channel}$ | | -5.1+TT | -4.7+TT | -9.5+TT | -13.5+TT | -8.7+TT | -4.7+TT |
| | 20MHz $BW_{channel}$ | | -5.1+TT | -4.7+TT | -9.5+TT | -13.5+TT | -8.7+TT | -4.7+TT |
| $N_{oc}$ | | dBm/15 kHz | -98 | | | | | |
| Propagation condition | | | AWGN | AWGN | | | | |
| Correlation Matrix and Antenna Configuration | | | 1x2 | 1x2 | | | | |
| Receive time offset to cell1 [Note 9] | | μs | - | 33 | | | | |
| CQI reporting periodicity | | ms | 1 | 2 | | | | |

Note 1: For the special subframe configuration see table 4.2-1 in TS 36.211.
Note 2: For the uplink-downlink configuration see table 4.2-2 in TS 36.211.
Note 3: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 5: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 6: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 7: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 8: The SNR of Cell 2 in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure A.7.3.47.1-1.
Note 9: Receive time difference between subframe boundaries of signals received from the two cells at the UE antenna connector including time alignment error between the two cells.

**Table 7.3.47.5-2: DRX-Configuration for E-UTRAN TDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC with PCell in TDD**

| Field | Value | | Comment |
|---|---|---|---|
| | Cell 1 | Cell 2 | |
| onDurationTimer | psf2 | psf2 | As specified in clause 6.3.2 in TS 36.331 [5] |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | Sf640 | Sf40 | |
| shortDRX | disable | disable | |

**Table 7.3.47.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in synchronous DC**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331[5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331[5] and section 10.1 in TS 36.213 [8]. |

The UE behaviour during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (1120 ms after the start of time duration T5) the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the subframe according to the configured CQI reporting mode (PUCCH 1-0) on PCell and PSCell.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.48 E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync for Cat-M1 UE in CEMode A

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- Message contents are FFS.

- SNR are TBD in Ran-4.

### 7.3.48.1 Test purpose

The purpose of this test is to verify that the FD-FDD Cat-M1 UE properly detects the out of sync and in sync for the purpose of monitoring downlink radio link quality of the PCell in CEModeA. This test will partly verify the E-UTRAN FDD radio link monitoring requirements for Cat-M1 UE defined in clause 7.19.

### 7.3.48.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 13 and forward of UE category M1.

### 7.3.48.3 Minimum conformance requirements

The UE category CAT-M1 applicability of the requirements for performing radio link monitoring is defined in Section 3.6 of TS 36.133[4].

When the downlink radio link quality of the PCell estimated over the last 400 ms period becomes worse than the threshold $Q_{out_CatM1}$, Layer 1 of the UE shall send an out-of-sync indication for the PCell to the higher layers within 400 ms $Q_{out_CatM1}$ evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last 200 ms period becomes better than the threshold $Q_{in_CatM1}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 200 ms $Q_{in_CatM1}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least 10ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

The normative reference for this requirement is TS 36.133 [4] clause 7.19.1, 7.19.2, 7.19.3 and A.7.3.53.

## 7.3.48.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.48.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode without repetition with a reporting periodicity of 2 ms.

In the test, the RRC parameter *numberPRB-Pairs* is set to 6 and the *RRC parameter mPDCCH-NumRepetition* is set to 8. UE shall successfully complete the RRC reconfiguration accordingly prior to the start of time duration T1.

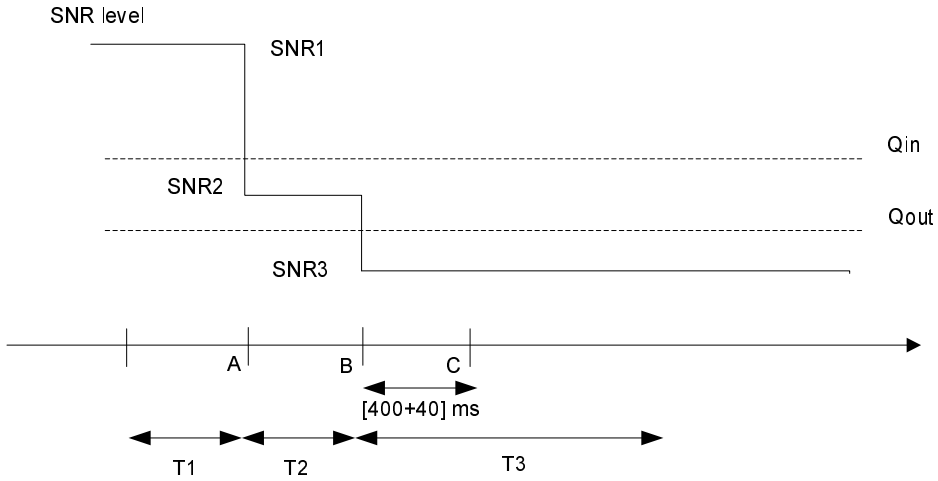


**Figure 7.3.48.4-1: SNR variation for out-of-sync testing**

### 7.3.48.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.48.4.1-1.
3. Propagation conditions are set according to Annex B clause B.2.
4. Message contents are defined in clause 7.3.48.4.3.
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.48.4.1-1: General test parameters for E-UTRAN FD-FDD out-of-sync testing for UE Cat-M1 in CEMode A**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| CP length | | | Normal | |
| Out of sync transmission parameters (Note 1) | DCI format | | 6-1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of OFDM symbols for legacy control channels | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical MPDCCH transmission parameters are as specified in section 7.19.2 and Table 7.19.2-1 respectively in TS 36.133[4] |
| | M-PDCCH aggregation level | eCCE | 24 | |
| | M-PDCCH repetition level | | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| DRX | | | OFF | |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213 [8]. |
| CQI reporting periodicity | | ms | 2 | Minimum CQI reporting periodicity |
| T1 | | s | 2 | |
| T2 | | s | 0.8 | |
| T3 | | s | 1.8 | |
| Note 1: MPDCCH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

## 7.3.48.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

1. Ensure the UE is in State 3A-RF-CE with the condition CE Mode A according to TS 36.508 [7] clause 7.2A.3AA .

2. SS sends a RRC Reconfiguration message configuring M-PDCCH aggregation level and M-PDCCH repetition level as defined in Table 7.3.48.4.1-1.

3. Set the parameters according to T1 in Table 7.3.48.5-1. Propagation conditions are set according to Annex B clause B.1. T1 starts.

4. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.48.5-1. T2 starts.

5. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.48.5-1. T3 starts.

6. If the SS:
   a) detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B

and

b) does not detect any uplink power higher than -48.5 dBm from time point C (440 ms after the start of T3) until T3 expires, the number of successful tests is increased by one.

Otherwise the number of failed tests is increased by one.

7. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.48.5-1.

8. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF-CE with CE ModeA activated according to TS 36.508 [7] clause 7.2A.3AA.

9. Repeat steps 2-8 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.48.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.48.4.3-1: Common Exception messages for E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync for Cat-M1 UE in CEMode A**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.48.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync for CAT_M1 UE**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.48.4.3-3: *EPDCCH-Config-r11-DEFAULT:* Additional E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync for CAT_M1 UE**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2B EPDCCH-Config-r11-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| mpdcch-config-r13 CHOICE { | | | |
| setup SEQUENCE { | | | |
| csi-NumRepetitionCE-r13 | v4 | | |
| mpdcch-pdsch-HoppingConfig-r13 | on | | |
| mpdcch-StartSF-UESS-r13 CHOICE { | | | |
| fdd-r13 | v1 | | |
| } | | | |
| mpdcch-NumRepetition-r13 | r8 | | |
| mpdcch-Narrowband-r13 | 1 | Narrowband index 0 | |
| } | | | |
| } | | | |

## 7.3.48.5 Test requirement

**Table 7.3.48.5-1: Cell specific test parameters for E-UTRAN FD-FDD (cell # 1) for out-of-sync radio link monitoring tests for Cat-M1 in CEMode A**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | | 1 | | |
| $BW_{channel}$ | | MHz | 10 | | |
| MPDCCH parameters as defined in A.3.1.3.1 | | | R.17 FDD Note 7 | | |
| OCNG Pattern defined in A.3.2.1.21 (FDD) | | | OP.21 FDD | | |
| $\rho_A$, $\rho_B$ | | | -3 | | |
| MPDCCH_RA | | dB | 0 | | |
| MPDCCH_RB | | dB | 0 | | |
| PBCH_RA | | dB | -3 | | |
| PBCH_RB | | dB | | | |
| PSS_RA | | dB | | | |
| SSS_RA | | dB | | | |
| OCNG_RA Note 1 | | dB | | | |
| OCNG_RB Note 1 | | dB | | | |
| $N_{oc}$ | | dBm/15 kHz | -98 | | |
| SNR Note 6 | Set 1 | dB | [FFS] | [FFS] | [FFS] |
| | Set 2 | | [FFS] | [FFS] | [FFS] |
| Propagation condition | | | ETU 30Hz | | |
| Correlation Matrix and Antenna Configuration | | | 2x1 Low | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.48.4-1. The SNR values correspond to either Set 1 or Set 2 of the out of sync parameters.
Note 7: Aggregation level and repetition levels specified in 7.3.48.4.1-1 are used for this test .

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal no later than time point C (440 ms after the start of time duration T3).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal

- UE output power equal to or less than Transmit OFF power -48.5 dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.49 E-UTRAN FD-FDD Radio Link Monitoring Test for In-Sync for Cat-M1 UE in CEMode A

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- Message contents are FFS.

- SNR are TBD in Ran-4.

### 7.3.49.1 Test purpose

The purpose of this test is to verify that the FD-FDD Cat-M1 UE properly detects the out of sync and in sync for the purpose of monitoring downlink radio link quality of the PCell in CEModeA. This test will partly verify the E-UTRAN FDD radio link monitoring requirements for Cat-M1 UE defined in clause 7.19.

### 7.3.49.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 13 and forward of UE category M1.

### 7.3.49.3 Minimum conformance requirements

The UE category CAT-M1 applicability of the requirements for performing radio link monitoring is defined in Section 3.6 of TS 36.133[4].

When the downlink radio link quality of the PCell estimated over the last 400 ms period becomes worse than the threshold $Q_{out_CatM1}$, Layer 1 of the UE shall send an out-of-sync indication for the PCell to the higher layers within 400 ms $Q_{out_CatM1}$ evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last 200 ms period becomes better than the threshold $Q_{in_CatM1}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 200 ms $Q_{in_CatM1}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least 10ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

The normative reference for this requirement is TS 36.133 [4] clause 7.19.1, 7.19.2, 7.19.3 and A.7.3.49.

### 7.3.49.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.49.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode without repetition with a reporting periodicity of 2 ms.

In the test, the RRC parameter *numberPRB-Pairs* is set to 4 and the *RRC parameter mPDCCH-NumRepetition* is set to 4. UE shall successfully complete the RRC reconfiguration accordingly prior to the start of time duration T1.

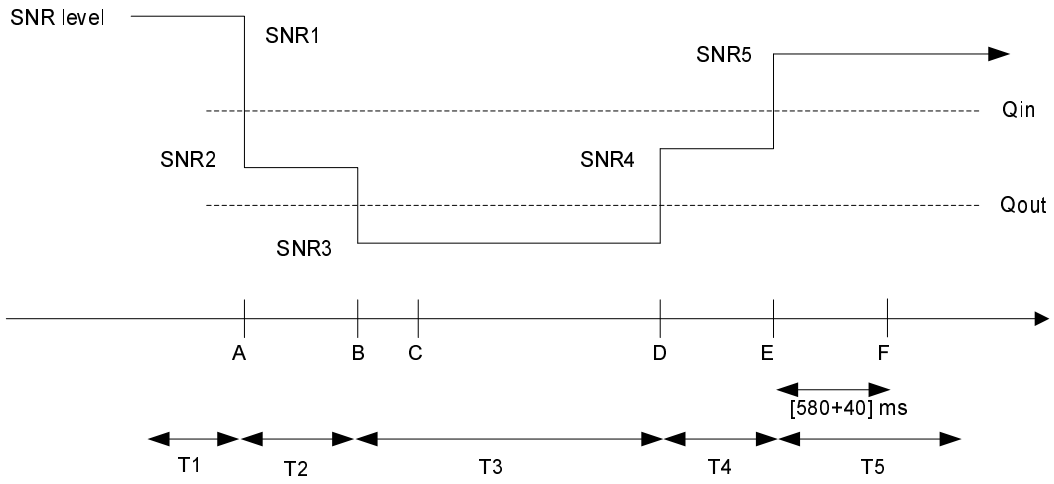


**Figure 7.3.49.4-1: SNR variation for in-sync testing**

### 7.3.49.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.49.4.1-1.
3. Propagation conditions are set according to Annex B clause B.2.
4. Message contents are defined in clause 7.3.49.4.3.
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.49.4.1-1: General test parameters for E-UTRAN FD-FDD in-sync testing for UE Cat-M1 in CEMode A**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| CP length | | | Normal | |
| In sync transmission parameters (Note 1) | DCI format | | 6-1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of OFDM symbols for legacy control channels | | 2 | In sync threshold $Q_{in}$ and the corresponding hypothetical MPDCCH transmission parameters are as specified in section 7.19.2 and Table 7.19.2-1 respectively in TS 36.133[4].. |
| | M-PDCCH aggregation level | eCCE | 4 | |
| | M-PDCCH repetition level | | 2 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| Out of sync transmission parameters (Note 1) | DCI format | | 6-1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of OFDM symbols for legacy control channels | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical MPDCCH transmission parameters are as specified in section 7.19.2 and Table 7.19.2-1 respectively in TS 36.133[4].. |
| | M-PDCCH aggregation level | eCCE | 16 | |
| | M-PDCCH repetition level | | 4 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| DRX | | | OFF | |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213 [8]. |
| CQI reporting periodicity | | ms | 2 | Minimum CQI reporting periodicity |
| T1 | | s | 2 | |
| T2 | | s | 0.8 | |
| T3 | | s | 1.36 | |
| T4 | | s | 0.4 | |
| T5 | | s | 2 | |
| Note 1: MPDCCH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

### 7.3.49.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

1. Ensure the UE is in State 3A-RF-CE with the condition CE Mode A according to TS 36.508 [7] clause 7.2A.3AA.

2. SS sends a RRC Reconfiguration message configuring M-PDCCH aggregation level and M-PDCCH repetition level as defined in Table 7.3.49.4.1-1.

3. Set the parameters according to T1 in Table 7.3.49.5-1. Propagation conditions are set according to Annex B.2. T1 starts.

4. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.49.5-1. T2 starts.
5. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.49.5-1. T3 starts.
6. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.49.5-1. T4 starts.
7. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.49.5-1. T5 starts.
8. If the SS detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (620 ms after the start of time duration T5) the number of successful tests is increased by one.

Otherwise the number of failed tests is increased by one.

9. After T5 expires, repeat steps 2-8 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved

### 7.3.49.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.49.4.3-1: Common Exception messages for E-UTRAN FD-FDD Radio Link Monitoring Test for in-sync for UE category CAT-M1**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.49.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FD-FDD Radio Link Monitoring Test for in-sync for UE category CAT-M1**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.49.4.3-3: *EPDCCH-Config-r11-DEFAULT:* Additional E-UTRAN FD-FDD Radio Link Monitoring Test for in-sync for UE category CAT-M1**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2B EPDCCH-Config-r11-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| numberPRB-Pairs-v1310 CHOICE { | | | |
| setup | n4 | | |
| } | | | |
| mpdcch-config-r13 CHOICE { | | | |
| setup SEQUENCE { | | | |
| csi-NumRepetitionCE-r13 | v4 | | |
| mpdcch-pdsch-HoppingConfig-r13 | on | | |
| mpdcch-StartSF-UESS-r13 CHOICE { | | | |
| fdd-r13 | v1 | | |
| } | | | |
| mpdcch-NumRepetition-r13 | r4 | | |
| mpdcch-Narrowband-r13 | 1 | Narrowband index 0 | |
| } | | | |
| } | | | |

## 7.3.49.5 Test requirement

**Table 7.3.49.5-1: Cell specific test parameters for E-UTRAN FD-FDD (cell # 1) for in-sync radio link monitoring tests for Cat-M1 in CEMode A**

| Parameter | Unit | Test 1 | | | | |
|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | |
| MPDCCH parameters as defined in A.3.1.3.1 | | R.17 FDD | | | | |
| OCNG Pattern defined in A.3.2.1.21 (FDD) | | OP.21 FDD | | | | |
| $\rho_A$, $\rho_B$ | | -3 | | | | |
| MPDCCH_RA | dB | 0 | | | | |
| MPDCCH_RB | dB | 0 | | | | |
| PBCH_RA | dB | -3 | | | | |
| PBCH_RB | dB | | | | | |
| PSS_RA | dB | | | | | |
| SSS_RA | dB | | | | | |
| OCNG_RA Note 1 | dB | | | | | |
| OCNG_RB Note 1 | dB | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | |
| SNR Note 6 | dB | [5.4] +0.9 | [-4.1] +0.9 | [-13.1 ]-0.9 | [-1.6]-0.9 | [5.4] +0.9 |
| Propagation condition | | ETU 30Hz | | | | |
| Correlation Matrix and Antenna Configuration | | 2x1 Low | | | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.49.4.1-1.

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F ([620] ms after the start of time duration T5) the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.50 E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category M1 configured in CEMode A

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined

- The SNRs are TBD

- UE Transmit minimum power and Transmit OFF power are TBD for a carrier frequency above 3GHz

- The Test system uncertainties applicable to this test are undefined

- The Test tolerances applicable to this test are undefined

- Annex E needs to be updated

### 7.3.50.1 Test purpose

The purpose of this test is to verify that the FD-FDD category M1 UE configured in CEMode A properly detects the out of sync for the purpose of monitoring downlink radio link quality of the PCell when DRX is used. This test will partly verify the E-UTRAN FD-FDD radio link monitoring requirements in TS 36.133 Section 7.19 [4].

### 7.3.50.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 13 and forward of UE category M1. Applicability requires support for FGI bit 5.

### 7.3.50.3 Minimum conformance requirements

The UE Cat-M1 applicability of the requirements for performing radio link monitoring is defined in Section 3.1.

The UE shall estimate the downlink radio link quality and compare it to the thresholds $Q_{out_Cat\ M1}$ and $Q_{in_Cat\ M1}$ for the purpose of monitoring downlink radio link quality of the PCell.

When DRX is used for FD-FDD and TDD category M1 UEs, the $Q_{out_CatM1}$ evaluation period ($T_{Evaluate_Q_{out_DRX_CatM1}}$) and the $Q_{in_CatM1}$ evaluation period ($T_{Evaluate_Q_{in_DRX_CatM1}}$) specified in Table 7.3.50.3-1 (TS 36.133 [4] Table 7.19.2.2-1) will be used.

When eDRX_CONN cycle is used for FD-FDD and TDD UE category M1 UEs, the $Q_{out_CatM1}$ evaluation period ($T_{Evaluate_Q_{out_DRX_CatM1}}$) and the $Q_{in_CatM1}$ evaluation period ($T_{Evaluate_Q_{in_DRX_CatM1}}$) specified in Table 7.3.50.3-2 (TS 36.133 [4] Table 7.19.2.2-2) will be used.

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{out_DRX_CatM1}}$ [s] period becomes worse than the threshold $Q_{out_CatM1}$, Layer 1 of the UE shall send out-of-sync indication for the PCell to the higher layers within $T_{Evaluate_Q_{out_DRX_CatM1}}$ [s] evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{in_DRX_CatM1}}$ [s] period becomes better than the threshold $Q_{in_CatM1}$, Layer 1 of the UE shall send in-sync indications for the PCell to the higher layers within $T_{Evaluate_Q_{in_DRX_CatM1}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. When DRX is used, two successive indications from Layer 1 shall be separated by at least max(10ms, DRX_cycle_length). When eDRX_CONN is used, two successive indications from Layer 1 shall be separated by at least max(10 ms, eDRX_CONN cycle length).

Upon start of T310 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.50.3-1: $Q_{out_CatM1}$ and $Q_{in_CatM1}$ Evaluation Period in DRX for FD-FDD and TDD UE category M1**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX_CatM1}}$ and $T_{Evaluate_Q_{in_DRX_CatM1}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in TS36.133 [4] clause 7.19.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note (20) |
| 0.04 < DRX cycle ≤ 0. 64 | Note (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note (5) |
| NOTE: Evaluation period length in time depends on the length of the DRX cycle in use. | |

**Table 7.3.50.3-2: $Q_{out_CatM1}$ and $Q_{in_CatM1}$ evaluation period when eDRX_CONN cycle is configured for FD-FDD and TDD UE category M1**

| eDRX_CONN cycle length (s) | $T_{Evaluate_Q_{out_DRX_CatM1}}$ and $T_{Evaluate_Q_{in_DRX_CatM1}}$ (s) (eDRX_CONN cycles) |
|---|---|
| 2.56 < eDRX_CONN cycle ≤ 10.24 | Note (5) |
| NOTE: Evaluation period length in time depends on the length of the eDRX_CONN cycle in use. | |

The normative reference for this requirement is TS 36.133 [4] clause 7.19.1, 7.19.2 and A.7.3.50.

The test parameters are given in Tables 7.3.50.4.1-1, 7.3.50.5-1, 7.3.50.5-2 and 7.3.50.5-3.

## 7.3.50.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.50.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync states.

There are two sets of values for MPDCCH aggregation level and MPDCCH repetition level, respectively. In each run of the test, the test equipment randomly selects one set of values (Set 1 or Set 2) for both MPDCCH aggregation level and MPDCCH repetition level, and sends the RRC configuration to the UE. MPDCCH aggregation level is determined by RRC parameter *numberPRB-Pairs*, and value 6 and 4 are corresponding to Set 1 and Set 2, respectively. MPDCCH repetition level is determined by RRC parameter *mPDCCH-NumRepetition*, and value 8 and 4 are corresponding to Set 1 and Set 2, respectively. UE shall successfully complete the RRC reconfiguration accordingly prior to the start of time duration T1.

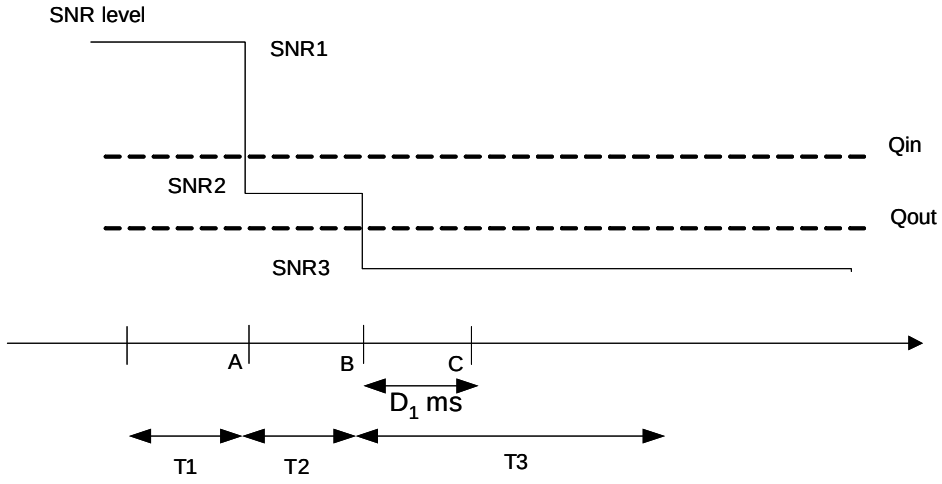


**Figure 7.3.50.4-1: SNR variation for out-of-sync testing in DRX**

### 7.3.50.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A. 10 (without using the faders) using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.50.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0.
4. Message contents are defined in clause 7.3.50.4.3.
5. There is one cell and one E-UTRA FD-FDD carrier specified in the test. Cell 1 is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.50.4.1-1: General test parameters for E-UTRAN FD-FDD out-of-sync tests in DRX for UE category M1 configured in CEMode A**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| CP length | | | Normal | |
| Out of sync transmission parameters (Note 1) | DCI format | | 6-1A | As defined in TS 36.212 [31] |
| | Number of OFDM symbols for legacy control channels | | 2 | Out of sync threshold $Q_{out,\ Cat\ M1}$ and the corresponding hypothetical MPDCCH transmission parameters are as specified in TS 36.133 [4] clause 7.19.2 and Table 7.19.2-1 respectively |
| | MPDCCH aggregation level | eCCE | Set 1: 24<br>Set 2:16 | |
| | MPDCCH repetition level | | Set 1: 8<br>Set 2: 4 | |
| | Ratio of MPDCCH to RS EPRE | | 0 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| DRX cycle | | Ms | 1280 | See Table 7.3.50.5-2 |
| Layer 3 filtering | | | Enabled | Counters:<br>N310 = 1; N311 = 1 |
| T310 timer | | Ms | 0 | T310 is disabled |
| T311 timer | | Ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213 [8]. |
| CQI reporting periodicity | | Ms | 2 | Minimum CQI reporting periodicity |
| T1 | | S | 32 | |
| T2 | | S | 12.8 | |
| T3 | | S | 13 | |
| Note 1: MPDCCH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

### 7.3.50.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms without repetition. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode MPDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to “infinity” so that UL timing alignment is maintained during the test.

1. Ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.50.5-1. Propagation conditions are set according to Annex B clause B.1. T1 starts.
3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.50.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.50.5-1. T3 starts.
5. If the SS a) detects uplink power equal to or higher than [-39] dBm (as defined in TS 36.521-1[10] clause 6.3.2.5 for carrier frequency less than [3]GHz) in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B, and b) does not detect any uplink power higher than [-48.5] dBm (as defined in TS 36.521-1[10] clause 6.3.3.5 for carrier frequency less than [3]GHz) from time point C (6500 ms (D1) after the start of T3) until T3 expires, then the number of successful tests is increased by one. Otherwise the number of failed tests is increased by one.
6. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.50.5-1.
7. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3.
8. Repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.50.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.50.4.3-1: Common Exception messages for E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category M1 configured in CEMode A**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.50.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category M1 configured in CEMode A**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULTT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig- DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213 [8]) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | OFF | (see Table 7.2.2- | |
| 1B and clause 7.2.2 in TS 36.213 [8]) | | | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.50.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN FD-FDD Radio Link Monitoring test for In-sync in DRX UE category M1 configured in CEMode A**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf10 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | Sf1280 typical value in real network for real-time services. | |
| Sf1280 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.50.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN FD-FDD Radio Link Monitoring test for Out-of-sync in UE category M1 configured in CEMode A**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.50.4.3-5: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FD-FDD Radio Link Monitoring test for Out-of-sync in DRX in UE category M1 configured in CEMode A**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | |
| } | | | |
| } | | | |

**Table 7.3.50.4.3-6: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FD-FDD Radio Link Monitoring test for Out-of-sync in DRX for UE category M1 with CE mode A**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

## 7.3.50.5 Test requirement

**Table 7.3.50.5-1: Cell specific test parameters for E-UTRAN FD-FDD (cell # 1) for out-of-sync radio link monitoring tests in DRX for UE category M1 configured in CEMode A**

| Parameter | Unit | Test 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| MPDCCH parameters defined in A.3.1.3 | | R.17 FDD Note 7 | | |
| OCNG Pattern defined in A.3.2.1 (FDD) | | OP.21 FDD | | |
| $\rho_A$, $\rho_B$ | | -3 | | |
| MPDCCH_RA | dB | 0 | | |
| MPDCCH_RB | dB | 0 | | |
| PBCH_RA | dB | -3 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| OCNG_RA Note1 | dB | | | |
| OCNG_RB Note1 | dB | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | |
| SNR Note 6 | dB | TBD | TBD | TBD |
| Propagation condition | | AWGN | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains MPDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.50.4-1.
Note 7: Aggregation level and repetition level specified in Table 7.3.50.4.1-1 are used for this test.

**Table 7.3.50.5-2: DRX-Configuration for E-UTRAN FD-FDD out-of-sync tests for UE category M1 configured in CEMode A**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf10 | As specified in clause 6.3.2 in TS 36.331 [5] |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf1280 | |
| shortDRX | disable | |

**Table 7.3.50.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FD-FDD out-of-sync testing for UE category M1 configured in CEMode A**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331 [5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331 [5] and section 10.1 in TS 36.213 [8]. |

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the subframe according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power [-39] dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5 for f<3GHz) means uplink signal,

- UE output power equal to or less than Transmit OFF power [-48.5] dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5 for f<3GHz) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.51 E-UTRAN FD-FDD Radio Link Monitoring Test for In-sync in DRX for UE Category M1 Configured in CEMode A

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- The SNRs are TBD

- UE Transmit minimum power is TBD for a carrier frequency above 3GHz ([-39 dBm])

- The Test system uncertainties applicable to this test are undefined

- The Test tolerances applicable to this test are undefined

- Annex E needs to be updated

### 7.3.51.1 Test purpose

The purpose of this test is to verify that the FD-FDD category M1 UE configured in CEMode A properly detects the out of sync and in sync for the purpose of monitoring downlink radio link quality of the PCell when DRX is used. This test will partly verify the E-UTRAN FD-FDD radio link monitoring requirements in clause 7.19.

### 7.3.51.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 13 and forward of UE category M1. Applicability requires support for FGI bit 5.

### 7.3.51.3 Minimum conformance requirements

The UE category M1 applicability of the requirements for performing radio link monitoring in subclause 7.19 is defined in Section 3.6.

When DRX is used for FD-FDD and TDD UE category M1 UEs, the $Q_{out_CatM1}$ evaluation period ($T_{Evaluate_Q_{out_DRX_CatM1}}$) and the $Q_{in_CatM1}$ evaluation period ($T_{Evaluate_Q_{in_DRX_CatM1}}$) specified in Table 7. 3.51.3-1 will be used.

When eDRX_CONN cycle is used for FD-FDD and TDD UE category M1 UEs, the $Q_{out_CatM1}$ evaluation period ($T_{Evaluate_Q_{out_DRX_CatM1}}$) and the $Q_{in_CatM1}$ evaluation period ($T_{Evaluate_Q_{in_DRX_CatM1}}$) specified in Table 7. 3.51.3-2 will be used.

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{out_DRX_CatM1}}$ [s] period becomes worse than the threshold $Q_{out_CatM1}$, Layer 1 of the UE shall send out-of-sync indication for the PCell to the higher layers within $T_{Evaluate_Q_{out_DRX_CatM1}}$ [s] evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{in_DRX_CatM1}}$ [s] period becomes better than the threshold $Q_{in_CatM1}$, Layer 1 of the UE shall send in-sync indications for the PCell to the higher layers within $T_{Evaluate_Q_{in_DRX_CatM1}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in [3]. When DRX is used, two successive indications from Layer 1 shall be separated by at least max(10ms, DRX_cycle_length). When eDRX_CONN is used, two successive indications from Layer 1 shall be separated by at least max(10 ms, eDRX_CONN cycle length).

Upon start of T310 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.51.3-1: $Q_{out_CatM1}$ and $Q_{in_CatM1}$ Evaluation Period in DRX for FD-FDD and TDD UE category M1**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX_CatM1}}$ and $T_{Evaluate_Q_{in_DRX_CatM1}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in clause 7.19.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note (20) |
| 0.04 < DRX cycle ≤ 0. 64 | Note (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note (5) |
| NOTE: Evaluation period length in time depends on the length of the DRX cycle in use. | |

**Table 7.3.51.3-2: $Q_{out_CatM1}$ and $Q_{in_CatM1}$ evaluation period when eDRX_CONN cycle is configured for FD-FDD and TDD UE category M1**

| eDRX_CONN cycle length (s) | $T_{Evaluate_Q_{out_DRX_CatM1}}$ and $T_{Evaluate_Q_{in_DRX_CatM1}}$ (s) (eDRX_CONN cycles) |
|---|---|
| 2.56 < eDRX_CONN cycle ≤ 10.24 | Note (5) |
| NOTE: Evaluation period length in time depends on the length of the eDRX_CONN cycle in use. | |

The normative reference for this requirement is TS 36.133 [4] clause 7.19.1, 7.19.2 and A.7.3.51.

### 7.3.51.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.51.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms without repetition. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode MPDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to “infinity” so that UL timing alignment is maintained during the test.

There are two sets of values for MPDCCH aggregation level and MPDCCH repetition level, respectively. In each run of the test, the test equipment randomly selects one set of values (Set 1 or Set 2) for both MPDCCH aggregation level and MPDCCH repetition level, and sends the RRC configuration to the UE. MPDCCH aggregation level is determined by RRC parameter *numberPRB-Pairs*, and value 6 and 4 are corresponding to Set 1 and Set 2, respectively. MPDCCH repetition level is determined by RRC parameter *mPDCCH-NumRepetition*, and value 8 and 4 are corresponding to Set 1 and Set 2, respectively. UE shall successfully complete the RRC reconfiguration accordingly prior to the start of time duration T1.

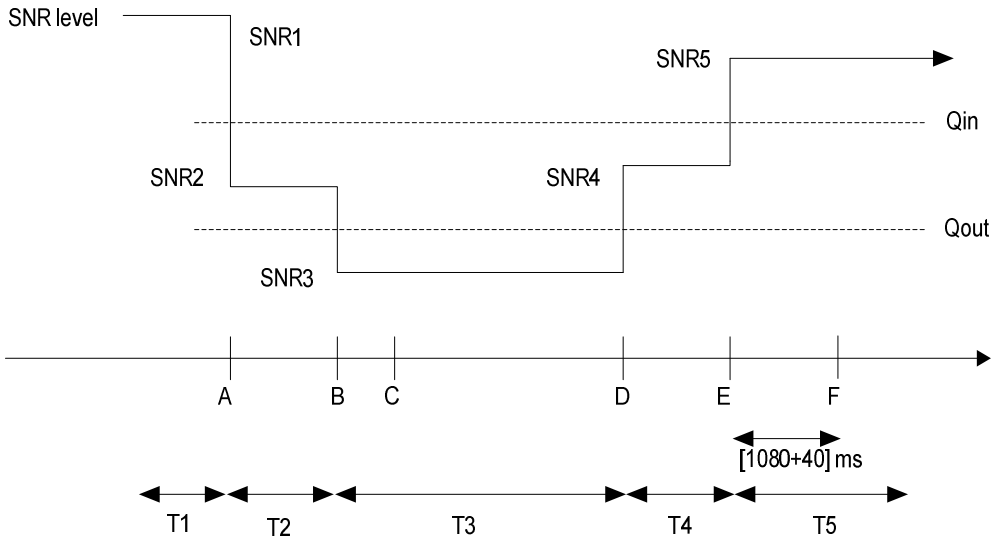


**Figure 7.3.51.4-1: SNR variation for in-sync testing in DRX**

#### 7.3.51.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 (without using the faders) using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.51.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.51.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.51.4.1-1: General test parameters for E-UTRAN FD-FDD in-sync test in DRX for UE category M1 configured in CEMode A**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| CP length | | | Normal | |
| In sync transmission parameters (Note 1) | DCI format | | 6-1A | As defined in TS 36.212 |
| | Number of OFDM symbols for legacy control channels | | 2 | In sync threshold $Q_{in, Cat M1}$ and the corresponding hypothetical MPDCCH transmission parameters are as specified in clause 7.19.2 and Table 7.19.2-1 respectively |
| | MPDCCH aggregation level | eCCE | Set 1: 8<br>Set 2:4 | |
| | MPDCCH repetition level | | Set 1: 4<br>Set 2: 2 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of MPDCCH to RS EPRE | | 0 | |
| Out of sync transmission parameters (Note 1) | DCI format | | 6-1A | As defined in TS 36.212 |
| | Number of OFDM symbols for legacy control channels | | 2 | Out of sync threshold $Q_{out, Cat M1}$ and the corresponding hypothetical MPDCCH transmission parameters are as specified in clause 7.19.2 and Table 7.19.2-1 respectively |
| | MPDCCH aggregation level | eCCE | Set 1: 24<br>Set 2:16 | |
| | MPDCCH repetition level | | Set 1: 8<br>Set 2: 4 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of MPDCCH to RS EPRE | dB | 0 | |
| DRX cycle | | ms | 40 | See Table 7.3.51.5-2 |
| Layer 3 filtering | | | Enabled | Counters:<br>N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213 |
| CQI reporting periodicity | | ms | 2 | Minimum CQI reporting periodicity |
| T1 | | s | 4 | |
| T2 | | s | 1.6 | |
| T3 | | s | 1.46 | |
| T4 | | s | 0.4 | |
| T5 | | s | 4 | |
| Note 1: MPDCCH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

## 7.3.51.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

1. Ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.51.5-1. The SS shall select randomly one set of values (Set 1 or Set 2) for both MPDCCH aggregation level and MPDCCH repetition level. Propagation conditions are set according to Annex B.1. T1 starts.
3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.51.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.51.5-1. T3 starts.
5. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.51.5-1. T4 starts.
6. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.51.5-1. T5 starts.

7. If the SS detects uplink power equal to or higher than [-39 dBm] (as defined in TS 36.521-1 clause 6.3.2.5 for carrier frequency less than [3]GHz) in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (1120 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.

8. After T5 expires, repeat steps 2-7 for both subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.51.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.51.4.3-1: Common Exception messages for E-UTRAN FD-FDD Radio Link Monitoring Test for In-sync in DRX for UE category M1 configured in CEMode A**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.51.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FD-FDD Radio Link Monitoring Test for In-sync in DRX for UE category M1 configured in CEMode A**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | OFF | (see Table 7.2.2-1B and clause 7.2.2 in TS 36.213 [8]) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.51.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN FD-FDD Radio Link Monitoring test for In-sync in DRX for UE category M1 configured in CEMode A**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf10 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf40 typical value in real network for real-time services. | |
| sf40 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.51.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN FD-FDD Radio Link Monitoring test for In-sync in DRX for UE category M1 configured in CEMode A**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.51.4.3-5: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FD-FDD Radio Link Monitoring test for In-sync in DRX for UE category M1 configured in CEMode A**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | |
| } | | | |
| } | | | |

**Table 7.3.51.4.3-6: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FD-FDD Radio Link Monitoring test for In-sync in DRX for UE category M1 configured in CEMode A**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

## 7.3.51.5 Test requirement

**Table 7.3.51.5-1: Cell specific test parameters for E-UTRAN FD-FDD (cell # 1) for in-sync radio link monitoring test in DRX for UE category M1 configured in CEMode A**

| Parameter | Unit | Test 1 | | | | |
|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | |
| MPDCCH parameters defined in A.3.1.3 | | R.17 FDD Note 7 | | | | |
| OCNG Pattern defined in A.3.2.1 (FDD) | | OP.21 FDD | | | | |
| $\rho_A$, $\rho_B$ | | -3 | | | | |
| MPDCCH_RA | dB | 0 | | | | |
| MPDCCH_RB | dB | 0 | | | | |
| PBCH_RA | dB | -3 | | | | |
| PBCH_RB | dB | | | | | |
| PSS_RA | dB | | | | | |
| SSS_RA | dB | | | | | |
| OCNG_RA Note1 | dB | | | | | |
| OCNG_RB Note1 | dB | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | |
| SNR Note 8 | dB | TBD | TBD | TBD | TBD | TBD |
| Propagation condition | | AWGN | | | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains MPDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.51.4-1.
Note 7: Aggregation level and repetition level specified in Table 7.3.51.4.1-1are used for this test.

**Table 7.3.51.5-2: DRX-Configuration for E-UTRAN FD-FDD out-of-sync tests for UE category M1 configured in CEMode A**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf10 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf40 | |
| shortDRX | disable | |

**Table 7.3.51.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FD-FDD out-of-sync testing for UE category M1 configured in CEMode A**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331 and section 10.1 in TS 36.213. |

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (1120 ms after the start of time duration T5) the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the subframe according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power [-39 dBm] (as defined in TS 36.521-1 [10] clause 6.3.2.5 for carrier frequency less than [3] GHz) means uplink signal

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.52 E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category CAT-M1

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- Message contents are FFS.

- SNR are TBD in Ran-4

### 7.3.52.1 Test purpose

The purpose of this test is to verify that the HD-FDD Cat-M1 UE properly detects the out of sync for the purpose of monitoring downlink radio link quality of the serving cell. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in TS 36.133 section 7.19.

### 7.3.52.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 12 and forward of UE category M1.

### 7.3.52.3 Minimum conformance requirements

The UE category CAT-M1 applicability of the requirements for performing radio link monitoring is defined in Section 3.6 of TS 36.133[4].

When the downlink radio link quality of the PCell estimated over the last 400 ms period becomes worse than the threshold $Q_{out_CatM1}$, Layer 1 of the UE shall send an out-of-sync indication for the PCell to the higher layers within 400

ms $Q_{out_CatM1}$ evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last 200 ms period becomes better than the threshold $Q_{in_CatM1}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 200 ms $Q_{in_CatM1}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least 10ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

The normative reference for this requirement is TS 36.133 [4] clause 7.19.1, 7.19.2, 7.19.3 and A.7.3.53.

### 7.3.52.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.52.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode without repetition with a reporting periodicity of 2 ms.

In the test, the RRC parameter *numberPRB-Pairs* is set to 6 and the *RRC parameter mPDCCH-NumRepetition* is set to 8. UE shall successfully complete the RRC reconfiguration accordingly prior to the start of time duration T1.

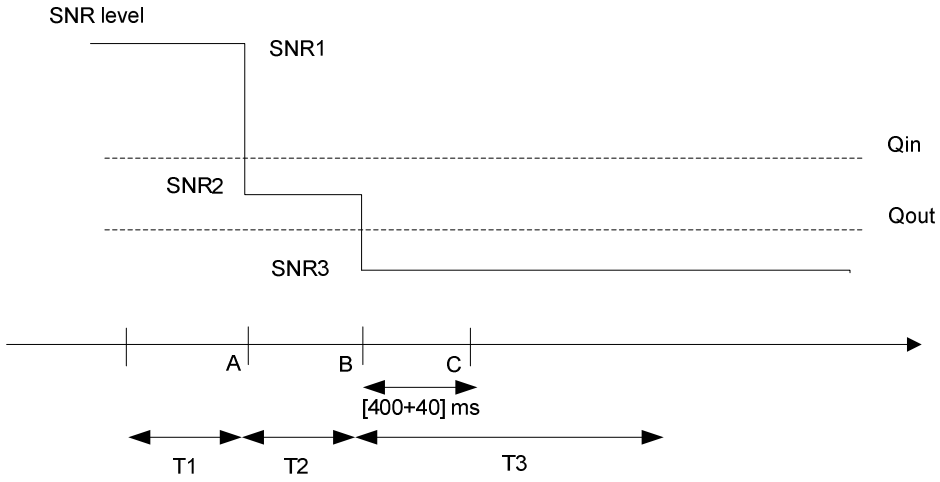


**Figure 7.3.52.4-1: SNR variation for out-of-sync testing**

#### 7.3.52.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.52.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.52.4.3

5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.52.4.1-1: General test parameters for E-UTRAN HD-FDD out-of-sync testing for UE Cat-M1 in CEMode A**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| CP length | | | Normal | |
| Out of sync transmission parameters (Note 1) | DCI format | | 6-1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of OFDM symbols for legacy control channels | | 2 | Out of sync threshold Qout and the corresponding hypothetical MPDCCH transmission parameters are as specified in section 7.19.2 and Table 7.19.2-1 respectively in TS 36.133 [4]. |
| | M-PDCCH aggregation level | eCCE | 24 | |
| | M-PDCCH repletion level | | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| DRX | | | OFF | |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213 [8]. |
| CQI reporting periodicity | | ms | 20 | Minimum CQI reporting periodicity |
| T1 | | s | 2 | |
| T2 | | s | 0.8 | |
| T3 | | s | 1.8 | |
| Note 1: MPDCCH transmission parameters corresponding to the in-sync and out of sync transmission need not be included in the Reference Measurement Channel. | | | | |

## 7.3.52.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 2 ms.

1. Ensure the UE is in State 3A-RF-CE with the condition CE Mode A according to TS 36.508 [7] clause 7.2A.3AA .

2. SS sends a RRC Reconfiguration message configuring M-PDCCH aggregation level and M-PDCCH repetition level as defined in Table 7.3.52.4.1-1.

3. Set the parameters according to T1 in Table 7.3.52.5-1. Propagation conditions are set according to Annex B clause B.1. T1 starts.

4. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.52.5-1. T2 starts.

5. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.52.5-1. T3 starts.

6. If the SS:
   a) detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B

   and

   b) does not detect any uplink power higher than -48.5 dBm from time point C (440 ms after the start of T3) until T3 expires, the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.

7. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.52.5-1.

8. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF-CE with CE ModeA activated according to TS 36.508 [7] clause 7.2A.3AA.

9. Repeat steps 2-8 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.52.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.52.4.3-1: Common Exception messages for E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for category CAT-M1 UE**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.52.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category CAT-M1**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 17 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.52.4.3-3: *EPDCCH-Config-r11-DEFAULT:* Additional E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category CAT-M1**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2B EPDCCH-Config-r11-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| numberPRB-Pairs-v1310 CHOICE { | | | |
| setup | n6 | | |
| } | | | |
| mpdcch-config-r13 CHOICE { | | | |
| setup SEQUENCE { | | | |
| csi-NumRepetitionCE-r13 | v4 | | |
| mpdcch-pdsch-HoppingConfig-r13 | on | | |
| mpdcch-StartSF-UESS-r13 CHOICE { | | | |
| fdd-r13 | v1 | | |
| } | | | |
| mpdcch-NumRepetition-r13 | r8 | | |
| mpdcch-Narrowband-r13 | 1 | Narrowband index 0 | |
| } | | | |
| } | | | |

## 7.3.52.5 Test requirement

**Table 7.3.52.5-1: Cell specific test parameters for E-UTRAN HD-FDD (cell # 1) for out-of-sync radio link monitoring tests for Cat-M1 in CEMode A**

| Parameter | Unit | Test 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| MPDCCH parameters as defined in A.3.1.3.1 | | R.7 HD-FDD | | |
| OCNG Pattern defined in A.3.2.1.21 (FDD) | | OP.21 FDD | | |
| $\rho_A$, $\rho_B$ | | -3 | | |
| MPDCCH_RA | dB | 0 | | |
| MPDCCH_RB | dB | 0 | | |
| PBCH_RA | dB | -3 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | |
| SNR Note 6 | dB | [0.0]+0.9 | [-6.9]+0.9 | [-15.9]-0.9 |
| Propagation condition | | ETU 30Hz | | |
| Correlation Matrix and Antenna Configuration | | 2x1 Low | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure A.7.3.x5.1-1.

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal no later than time point C (440 ms after the start of time duration T3).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal

- UE output power equal to or less than Transmit OFF power -48.5 dBm (as defined in TS 36.521-1 [10] clause 6.3.3.5) means no uplink signal.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 7.3.53 E-UTRAN HD-FDD Radio Link Monitoring Test for In-sync for UE category CAT-M1

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- Message contents are FFS.

- SNR are TBD in Ran-4.

### 7.3.53.1 Test purpose

The purpose of this test is to verify that the HD-FDD Cat-M1 UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the serving cell. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in TS 36.133 [4] section 7.19.

### 7.3.53.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 13 and forward of UE category M1.

### 7.3.53.3 Minimum conformance requirements

The UE category M1 applicability of the requirements for performing radio link monitoring in subclause 7.19 is defined in Section 3.6 of TS 36.133 [4].

When the downlink radio link quality of the PCell estimated over the last 400 ms period becomes worse than the threshold $Q_{out_CatM1}$, Layer 1 of the UE shall send an out-of-sync indication for the PCell to the higher layers within 400 ms $Q_{out_CatM1}$ evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last 200 ms period becomes better than the threshold $Q_{in_CatM1}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 200 ms $Q_{in_CatM1}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least 10ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

The normative reference for this requirement is TS 36.133 [4] clause 7.19.1, 7.19.2, 7.19.3 and A.7.3.53.

### 7.3.53.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.53.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode without repetition with a reporting periodicity of [[FFS]] ms.

In the test, the RRC parameter *numberPRB-Pairs* is set to 4 and the *RRC parameter mPDCCH-NumRepetition* is set to 4. UE shall successfully complete the RRC reconfiguration accordingly prior to the start of time duration T1.

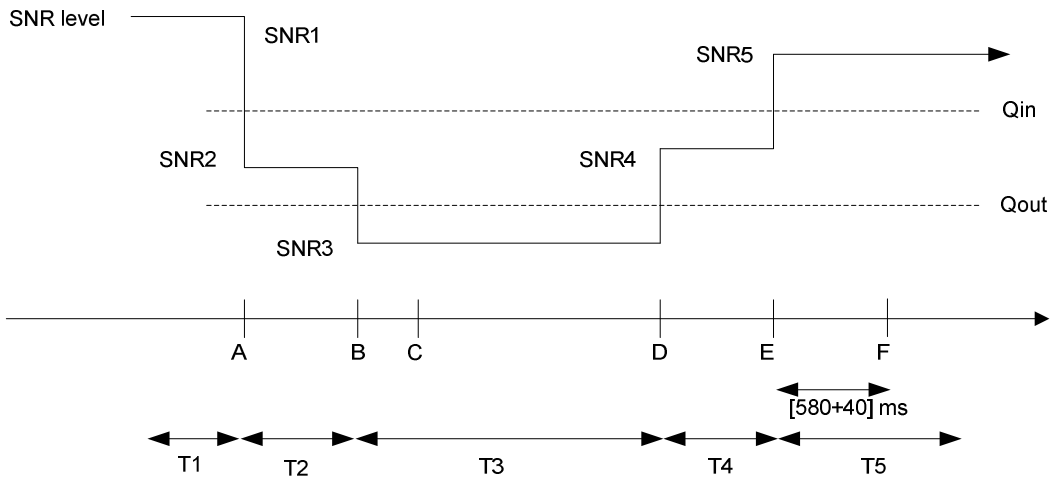


**Figure 7.3.53.4-1: SNR variation for in-sync testing**

### 7.3.53.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.53.4-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.53.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.53.4.1-1: General test parameters for E-UTRAN HD-FDD (cell # 1) for in-sync radio link monitoring tests for Cat-M1 in CEMode A**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| CP length | | | Normal | |
| In sync transmission parameters (Note 1) | DCI format | | 6-1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of OFDM symbols for legacy control channels | | 2 | In sync threshold $Q_{in}$ and the corresponding hypothetical MPDCCH transmission parameters are as specified in section 7.19.2 and Table 7.19.2-1 respectively in TS 36.133 [4].. |
| | M-PDCCH aggregation level | eCCE | 4 | |
| | | | 2 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| Out of sync transmission parameters (Note 1) | DCI format | | 6-1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of OFDM symbols for legacy control channels | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical MPDCCH transmission parameters are as specified in section 7.19.2 and Table 7.19.2-1 respectively in TS 36.133 [4].. |
| | M-PDCCH aggregation level | eCCE | 16 | |
| | M-PDCCH repletion level | | 4 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| DRX | | | OFF | |
| Layer 3 filtering | | | Enabled | Counters:<br>N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213 [8]. |
| CQI reporting periodicity | | ms | 20 | Minimum CQI reporting periodicity |
| T1 | | s | 2 | |
| T2 | | s | 0.8 | |
| T3 | | s | 1.36 | |
| T4 | | s | 0.4 | |
| T5 | | s | 2 | |
| Note 1: MPDCCH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

### 7.3.53.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of [[FFS]] ms.

1. Ensure the UE is in State 3A-RF-CE with the condition CE Mode A according to TS 36.508 [7] clause 7.2A.3AA.

2. SS sends a RRC Reconfiguration message configuring M-PDCCH aggregation level and M-PDCCH repetition level as defined in Table 7.3.53.4.1-1.

3. Set the parameters according to T1 in Table 7.3.53.5-1. Propagation conditions are set according to Annex B.2. T1 starts.

4. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.53.5-1. T2 starts.
5. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.53.5-1. T3 starts.
6. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.53.5-1. T4 starts.
7. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.53.5-1. T5 starts.
8. If the SS detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (620 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
9. After T5 expires, repeat steps 2-8 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.53.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.53.4.3-1: Common Exception messages for E-UTRAN HD-FDD Radio Link Monitoring Test for in-sync for UE category CAT-M1**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.53.4.3-2: *CQI-ReportConfig-DEFAULT:* E-UTRAN HD-FDD Radio Link Monitoring Test for in-sync for UE category CAT-M1**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 17 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.52.4.3-3: *EPDCCH-Config-r11-DEFAULT:* Additional E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category CAT-M1**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2B EPDCCH-Config-r11-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| numberPRB-Pairs-v1310 CHOICE { | | | |
| setup | n4 | | |
| } | | | |
| mpdcch-config-r13 CHOICE { | | | |
| setup SEQUENCE { | | | |
| csi-NumRepetitionCE-r13 | v4 | | |
| mpdcch-pdsch-HoppingConfig-r13 | on | | |
| mpdcch-StartSF-UESS-r13 CHOICE { | | | |
| fdd-r13 | v1 | | |
| } | | | |
| mpdcch-NumRepetition-r13 | r4 | | |
| mpdcch-Narrowband-r13 | 1 | Narrowband index 0 | |
| } | | | |
| } | | | |

### 7.3.53.5 Test requirement

**Table 7.3.53.5-1: Cell specific test parameters for E-UTRAN HD-FDD (cell # 1) for in-sync radio link monitoring tests for Cat-M1 in CEMode A**

| Parameter | Unit | Test 1 | | | | |
|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | |
| MPDCCH parameters as defined in A.3.1.3.1 | | R.7 HD-FDD | | | | |
| OCNG Pattern defined in A.3.2.1.21 (FDD) | | OP.21 FDD | | | | |
| $\rho_A$, $\rho_B$ | | -3 | | | | |
| MPDCCH_RA | dB | 0 | | | | |
| MPDCCH_RB | dB | 0 | | | | |
| PBCH_RA | dB | -3 | | | | |
| PBCH_RB | dB | | | | | |
| PSS_RA | dB | | | | | |
| SSS_RA | dB | | | | | |
| OCNG_RA Note 1 | dB | | | | | |
| OCNG_RB Note 1 | dB | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | |
| SNR Note 6 | dB | [5.4] +0.9 | [- 4.1] +0.9 | [-- 13.1 ] - 0.9 | [ – 1.6] -0.9 | [5.4] +0.9 |
| Propagation condition | | ETU 30Hz | | | | |
| Correlation Matrix and Antenna Configuration | | 2x1 Low | | | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.53.4.1-1.

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F ([620] ms after the start of time duration T5) the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power -39 dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

7.3.54

## 7.3.55 E-UTRAN HD-FDD Radio Link Monitoring Test for In-sync in DRX for UE Category M1 configured in CEMode A

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- The CQI reporting periodicity and SNR are TBD

- UE Transmit minimum power is TBD for a carrier frequency above 3GHz ([-39 dBm])

- The Test system uncertainties applicable to this test are undefined

- The Test tolerances applicable to this test are undefined

- Annex E needs to be updated

### 7.3.55.1 Test purpose

The purpose of this test is to verify that the HD-FDD category M1 UE configured in CEMode A properly detects the out of sync and in sync for the purpose of monitoring downlink radio link quality of the PCell when DRX is used. This test will partly verify the E-UTRAN HD-FDD radio link monitoring requirements in TS 36.133 [4] Section 7.19.

### 7.3.55.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 13 and forward of UE category M1. Applicability requires support for FGI bit 5.

### 7.3.55.3 Minimum conformance requirements

The UE category M1 applicability of the requirements for performing radio link monitoring is defined in Section 3.1.

When DRX is used for HD-FDD category M1 with CE mode A UEs, the $Q_{out_CatM1}$ evaluation period ($T_{Evaluate_Qout_DRX_CatM1}$) and the $Q_{in_CatM1}$ evaluation period ($T_{Evaluate_Qin_DRX_CatM1}$) specified in Table 7.3.55.3-1 (TS 36.133 [4] Table 7.19.3.2-1) will be used.

When eDRX_CONN cycle is used for HD-FDD category M1 with CE mode A UEs, the $Q_{out_CatM1}$ evaluation period ($T_{Evaluate_Qout_DRX_CatM1}$) and the $Q_{in_CatM1}$ evaluation period ($T_{Evaluate_Qin_DRX_CatM1}$) specified in Table 7.3.55.3-2 (TS 36.133 [4] Table 7.19.3.2-2) will be used.

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Qout_DRX_CatM1}$ [s] period becomes worse than the threshold $Q_{out_CatM1}$, Layer 1 of the UE shall send out-of-sync indication for the PCell to the higher layers within $T_{Evaluate_Qout_DRX_CatM1}$ [s] evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Qin_DRX_CatM1}$ [s] period becomes better than the threshold $Q_{in_CatM1}$, Layer 1 of the UE shall send in-sync indications for the PCell to the higher layers within $T_{Evaluate_Qin_DRX_CatM1}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. When DRX is used, two successive indications from Layer 1 shall be separated by at least max(10ms, DRX_cycle_length). When eDRX_CONN is used, two successive indications from Layer 1 shall be separated by at least max(10 ms, eDRX_CONN cycle length).

Upon start of T310 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.55.3-1: $Q_{out_CatM1}$ and $Q_{in_CatM1}$ Evaluation Period in DRX for HD-FDD UE category M1 with CE mode A**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX_CatM1}}$ and $T_{Evaluate_Q_{in_DRX_CatM1}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in clause 7.19.3.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note (40) |
| 0.04 < DRX cycle ≤ 0. 16 | Note (20) |
| 0. 16 < DRX cycle ≤ 0.64 | Note (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note (5) |
| Note: Evaluation period length in time depends on the length of the DRX cycle in use. | |

**Table 7.3.55.3-2: $Q_{out_CatM1}$ and $Q_{in_CatM1}$ evaluation period when eDRX_CONN cycle is configured for HD-FDD UE category M1 with CE mode A**

| eDRX_CONN cycle length (s) | $T_{Evaluate_Q_{out_DRX_CatM1}}$ and $T_{Evaluate_Q_{in_DRX_CatM1}}$ (s) (eDRX_CONN cycles) |
|---|---|
| 2.56 < eDRX_CONN cycle ≤ 10.24 | Note (5) |
| NOTE: Evaluation period length in time depends on the length of the eDRX_CONN cycle in use. | |

The normative reference for this requirement is TS 36.133 [4] clause 7.19.1, 7.19.3 and A.7.3.55.

## 7.3.55.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.55.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of TBD ms without repetition. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode MPDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to “infinity” so that UL timing alignment is maintained during the test.

There are two sets of values for MPDCCH aggregation level and MPDCCH repetition level, respectively. In each run of the test, the test equipment randomly selects one set of values (Set 1 or Set 2) for both MPDCCH aggregation level and MPDCCH repetition level, and sends the RRC configuration to the UE. MPDCCH aggregation level is determined by RRC parameter *numberPRB-Pairs*, and value 6 and 4 are corresponding to Set 1 and Set 2, respectively. MPDCCH repetition level is determined by RRC parameter *mPDCCH-NumRepetition*, and value 8 and 4 are corresponding to Set 1 and Set 2, respectively. UE shall successfully complete the RRC reconfiguration accordingly prior to the start of time duration T1.

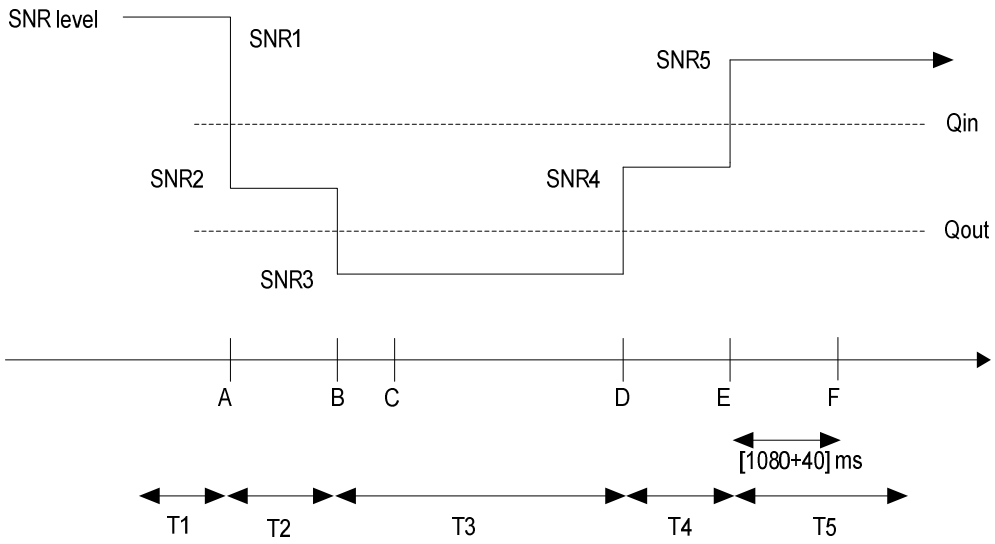


**Figure 7.3.55.4-1: SNR variation for in-sync testing in DRX**

### 7.3.55.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 (without using the faders) using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.55.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 7.3.55.4.3.
5. There is one cell and one E-UTRA HD-FDD carrier specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.55.4.1-1: General test parameters for E-UTRAN HD-FDD in-sync test in DRX for UE category M1 with CE mode A**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| CP length | | | Normal | |
| In sync transmission parameters (Note 1) | DCI format | | 6-1A | As defined in TS 36.212 |
| | Number of OFDM symbols for legacy control channels | | 2 | In sync threshold $Q_{in, Cat M1}$ and the corresponding hypothetical MPDCCH transmission parameters are as specified in clause 7.19.2 and Table 7.19.2-1 respectively |
| | MPDCCH aggregation level | eCCE | Set 1: 8<br>Set 2:4 | |
| | MPDCCH repetition level | | Set 1: 4<br>Set 2: 2 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of MPDCCH to RS EPRE | | 0 | |
| Out of sync transmission parameters (Note 1) | DCI format | | 6-1A | As defined in TS 36.212 |
| | Number of OFDM symbols for legacy control channels | | 2 | Out of sync threshold $Q_{out, Cat M1}$ and the corresponding hypothetical MPDCCH transmission parameters are as specified in clause 7.19.2 and Table 7.19.2-1 respectively |
| | MPDCCH aggregation level | eCCE | Set 1: 24<br>Set 2:16 | |
| | MPDCCH repetition level | | Set 1: 8<br>Set 2: 4 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of MPDCCH to RS EPRE | dB | 0 | |
| DRX cycle | | ms | 40 | See Table 7.3.55.5-2 |
| Layer 3 filtering | | | Enabled | Counters:<br>N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213 |
| CQI reporting periodicity | | ms | TBD | |
| T1 | | s | 4 | |
| T2 | | s | 1.6 | |
| T3 | | s | 1.46 | |
| T4 | | s | 0.4 | |
| T5 | | s | 4 | |
| Note 1: MPDCCH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

## 7.3.55.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of TBD ms.

1. Ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 7.3.55.5-1. The SS shall select randomly one set of values (Set 1 or Set 2) for both MPDCCH aggregation level and MPDCCH repetition level. Propagation conditions are set according to Annex B.1. T1 starts.
3. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.55.5-1. T2 starts.
4. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.55.5-1. T3 starts.
5. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.55.5-1. T4 starts.
6. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.55.5-1. T5 starts.

7. If the SS detects uplink power equal to or higher than [-39] dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (1120 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.

8. After T5 expires, repeat steps 2-7 for both subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.55.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.55.4.3-1: Common Exception messages for E-UTRAN HD-FDD Radio Link Monitoring Test for In-sync in DRX for UE category M1 with CE mode A**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.55.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN HD-FDD Radio Link Monitoring Test for In-sync in DRX for UE category M1 with CE mode A**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | TBD | (see Table 7.2.2-1A in TS 36.213 [8]) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | OFF | (see Table 7.2.2-1B and clause 7.2.2 in TS 36.213 [8]) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.55.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN HD-FDD Radio Link Monitoring test for In-sync in DRX for UE category M1 with CE mode A**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf20 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf40 typical value in real network for real-time services. | |
| sf40 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.55.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN HD-FDD Radio Link Monitoring test for In-sync in DRX for UE category M1 with CE mode A**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.55.4.3-5: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN HD-FDD Radio Link Monitoring test for In-sync in DRX for UE category M1 with CE mode A**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | |
| } | | | |
| } | | | |

**Table 7.3.55.4.3-6: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN HD-FDD Radio Link Monitoring test for In-sync in DRX for UE category M1 with CE mode A**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

## 7.3.55.5 Test requirement

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (1120 ms after the start of time duration T5) the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the subframe according to the configured CQI reporting mode (PUCCH 1-0).

The uplink signal is verified on the basis of the UE output power:

- UE output power equal to or higher than Transmit minimum power [-39] dBm (as defined in TS 36.521-1 [10] clause 6.3.2.5) means uplink signal

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

**Table 7.3.55.5-1: Cell specific test parameters for E-UTRAN HD-FDD (cell # 1) for in-sync radio link monitoring test in DRX for UE category M1 configured in CEMode A**

| Parameter | Unit | Test 1 | | | | |
|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | |
| MPDCCH parameters defined in A.3.1.3 | | R.7 HD-FDD Note 7 | | | | |
| OCNG Pattern defined in A.3.2.1 (FDD) | | OP.21 FDD | | | | |
| $\rho_A$, $\rho_B$ | | -3 | | | | |
| MPDCCH_RA | dB | 0 | | | | |
| MPDCCH_RB | dB | 0 | | | | |
| PBCH_RA | dB | -3 | | | | |
| PBCH_RB | dB | | | | | |
| PSS_RA | dB | | | | | |
| SSS_RA | dB | | | | | |
| OCNG_RA Note1 | dB | | | | | |
| OCNG_RB Note1 | dB | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | |
| SNR Note 8 | dB | TBD | TBD | TBD | TBD | TBD |
| Propagation condition | | AWGN | | | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains MPDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.55.4-1.
Note 7: Aggregation level and repetition level specified in Table 7.3.55.4.1-1 are used for this test.

**Table 7.3.55.5-2: DRX-Configuration for E-UTRAN HD-FDD out-of-sync tests for UE category M1 configured in CEMode A**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf20 | As specified in clause 6.3.2 in TS 36.331 [5] |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf40 | |
| shortDRX | disable | |

**Table 7.3.55.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN HD-FDD in-sync radio link monitoring tests in DRX for UE category M1 with CE mode A**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331 [5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213 [5]. |

## 7.3.56 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync for Cat-M1 UE in CEMode A

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- Test Tolerances need to be completed

- Annex E updates are pending

- Message contents are FFS.

- SNR values are in [] (square brackets)

### 7.3.56.1 Test purpose

The purpose of this test is to verify that the TDD Cat-M1 UE properly detects the out of sync and in sync for the purpose of monitoring downlink radio link quality of the PCell in CEModeA. This test will partly verify the E-UTRAN TDD radio link monitoring requirements for Cat-M1 UE defined in TS 36.133[4] clause 7.19.

### 7.3.56.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 13 and forward of UE category M1.

### 7.3.56.3 Minimum conformance requirements

The UE category CAT-M1 applicability of the requirements for performing radio link monitoring is defined in Section 3.6 of TS 36.133[4].

When the downlink radio link quality of the PCell estimated over the last 400 ms period becomes worse than the threshold $Q_{out_CatM1}$, Layer 1 of the UE shall send an out-of-sync indication for the PCell to the higher layers within 400 ms $Q_{out_CatM1}$ evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last 200 ms period becomes better than the threshold $Q_{in_CatM1}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 200 ms $Q_{in_CatM1}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least 10ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

The normative reference for this requirement is TS 36.133 [4] clause 7.19.1, 7.19.2, 7.19.3 and A.7.3.56.

### 7.3.56.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.56.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms.

UE shall successfully complete the RRC reconfiguration accordingly prior to the start of time duration T1.

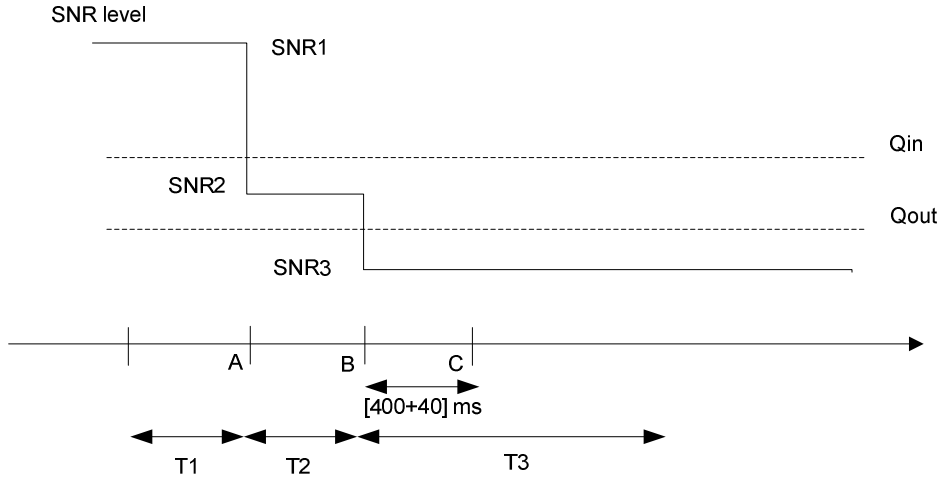


**Figure 7.3.56.4-1: SNR variation for out-of-sync testing**

### 7.3.56.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.56.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0
4. Message contents are defined in clause 7.3.56.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.56.4.1-1: General test parameters for E-UTRAN TDD out-of-sync testing for UE Cat-M1 in CEMode A**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| CP length | | | Normal | |
| Out of sync transmission parameters (Note 1) | DCI format | | 6-1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of OFDM symbols for legacy control channels | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical MPDCCH transmission parameters are as specified in section 7.19.2 and Table 7.19.2-1 respectively. |
| | M-PDCCH aggregation level | eCCE | 24 | |
| | M-PDCCH repetition level | | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| DRX | | | OFF | |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 1 | Minimum CQI reporting periodicity |
| T1 | | s | 2 | |
| T2 | | s | 0.8 | |
| T3 | | s | 1.8 | |
| Note 1: MPDCCH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

## 7.3.56.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms.

1. Ensure the UE is in State 3A-RF-CE with the condition CE Mode A according to TS 36.508 [7] clause 7.2A.3AA .

2. SS sends a RRC Reconfiguration message configuring M-PDCCH aggregation level and M-PDCCH repetition level as defined in Table 7.3.56.4.1-1

3. Set the parameters according to T1 in Table 7.3.56.5-1. Propagation conditions are set according to Annex B clause B.1. T1 starts.

4. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.56.5-1. T2 starts.

5. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.56.5-1. T3 starts.

6. If the SS:
   a) detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B

and

   b) does not detect any uplink power higher than -48.5 dBm from time point C (440 ms after the start of T3) until T3 expires, the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.

7. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.56.5-1.

8. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF-CE with CE ModeA activated according to TS 36.508 [7] clause 7.2A.3AA.

9. Repeat steps 2-8 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.56.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.56.4.3-1: Common Exception messages for E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync for Cat-M1 UE in CEMode A**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table [FFS] |
| Default RRC messages and information elements contents exceptions | |

### 7.3.56.5 Test requirement

**Table 7.3.56.5-1: Cell specific test parameters for E-UTRAN TDD (cell # 1) for out-of-sync radio link monitoring tests for Cat-M1 in CEMode A**

| Parameter | Unit | Test 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| Special subframe configuration Note 1 | | 6 | | |
| Uplink-downlink configuration Note 1 | | 1 | | |
| MPDCCH parameters as defined in A.7.3 | | R.15 TDD Note 8 | | |
| OCNG Pattern defined in D.2.11 (TDD) | | OP.11 TDD | | |
| $\rho_A$, $\rho_B$ | | -3 | | |
| PBCH_RA | dB | -3 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | |
| SNR Note 7 | dB | [-0.5] | [-7.3] | [-16.3] |
| Propagation condition | | ETU 30Hz | | |
| Correlation Matrix and Antenna Configuration | | 2x1 Low | | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 4: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 5: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 6: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 7: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.56.4-1.
Note 8: Aggregation level and repetition levels specified in 7.3.56.4-1 are used for this test.

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal no later than time point C (440 ms after the start of time duration T3).

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.57 E-UTRAN TDD Radio Link Monitoring Test for In-Sync for Cat-M1 UE in CEMode A

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- Test Tolerances need to be completed

- Annex E updates are pending

- Message contents are FFS.

- SNR values are in [] (square bracket)

### 7.3.57.1 Test purpose

The purpose of this test is to verify that the TDD Cat-M1 UE properly detects the out of sync and in sync for the purpose of monitoring downlink radio link quality of the PCell in CEModeA. This test will partly verify the E-UTRAN TDD radio link monitoring requirements for Cat-M1 UE defined in TS 36.133[4] clause 7.19.

### 7.3.57.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 13 and forward of UE category M1.

### 7.3.57.3 Minimum conformance requirements

The UE category CAT-M1 applicability of the requirements for performing radio link monitoring is defined in Section 3.6 of TS 36.133[4].

When the downlink radio link quality of the PCell estimated over the last 400 ms period becomes worse than the threshold $Q_{out_CatM1}$, Layer 1 of the UE shall send an out-of-sync indication for the PCell to the higher layers within 400 ms $Q_{out_CatM1}$ evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last 200 ms period becomes better than the threshold $Q_{in_CatM1}$, Layer 1 of the UE shall send an in-sync indication for the PCell to the higher layers within 200 ms $Q_{in_CatM1}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. Two successive indications from Layer 1 shall be separated by at least 10ms.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

The normative reference for this requirement is TS 36.133 [4] clause 7.19.1, 7.19.2, 7.19.3 and A.7.3.57.

### 7.3.57.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.57.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode without repetition with a reporting periodicity of 1 ms.

UE shall successfully complete the RRC reconfiguration accordingly prior to the start of time duration T1.

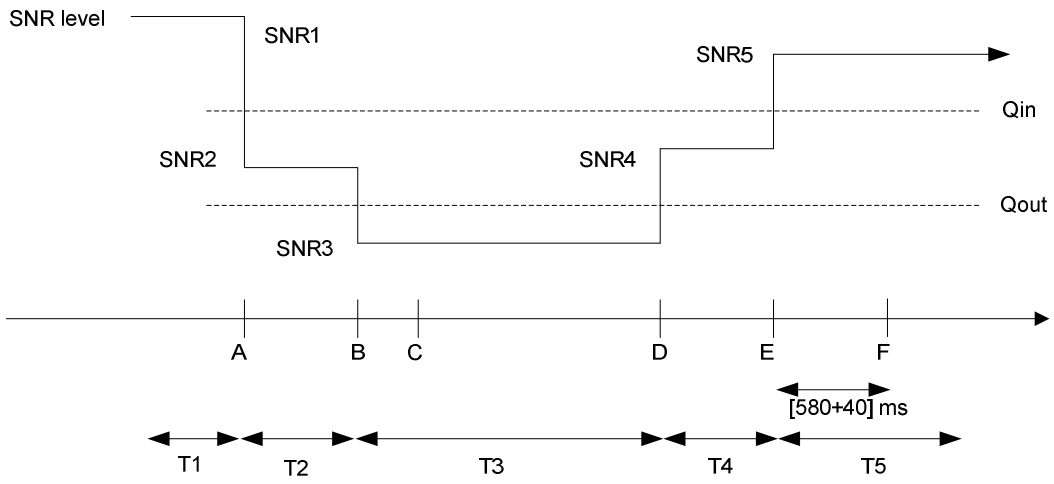


**Figure 7.3.57.4-1: SNR variation for in-sync testing**

### 7.3.57.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.57.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0
4. Message contents are defined in clause 7.3.57.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.57.4.1-1: General test parameters for E-UTRAN TDD in-sync testing for UE Cat-M1 in CEMode A**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| CP length | | | Normal | |
| In sync transmission parameters (Note 1) | DCI format | | 6-1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of OFDM symbols for legacy control channels | | 2 | In sync threshold $Q_{in}$ and the corresponding hypothetical MPDCCH transmission parameters are as specified in section 7.19.2 and Table 7.19.2-1 respectively. |
| | M-PDCCH aggregation level | eCCE | 4 | |
| | | | 2 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| Out of sync transmission parameters (Note 1) | DCI format | | 6-1A | As defined in section 5.3.3.1.3 in TS 36.212 |
| | Number of OFDM symbols for legacy control channels | | 2 | Out of sync threshold $Q_{out}$ and the corresponding hypothetical MPDCCH transmission parameters are as specified in section 7.19.2 and Table 7.19.2-1 respectively. |
| | M-PDCCH aggregation level | eCCE | 16 | |
| | M-PDCCH repetition level | | 4 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| DRX | | | OFF | |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 1 | Minimum CQI reporting periodicity |
| T1 | | s | 2 | |
| T2 | | s | 0.8 | |
| T3 | | s | 1.36 | |
| T4 | | s | 0.4 | |
| T5 | | s | 2 | |
| Note 1: MPDCCH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

### 7.3.57.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode without repetition with a reporting periodicity of 1 ms.

1. Ensure the UE is in State 3A-RF-CE with the condition CE Mode A according to TS 36.508 [7] clause 7.2A.3AA.

2. SS sends a RRC Reconfiguration message configuring M-PDCCH aggregation level and M-PDCCH repetition level as defined in Table 7.3.57.4.1-1

3. Set the parameters according to T1 in Table 7.3.57.5-1. Propagation conditions are set according to Annex B.2. T1 starts.

4. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.57.5-1. T2 starts.

5. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.57.5-1. T3 starts.
6. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.57.5-1. T4 starts.
7. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.57.5-1. T5 starts.
8. If the SS detects uplink power equal to or higher than -39 dBm in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (620 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.
9. After T5 expires, repeat steps 2-8 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 7.3.57.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.57.4.3-1: Common Exception messages for E-UTRAN TDD Radio Link Monitoring Test for in-sync for UE category CAT-M1**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | FFS |
| Default RRC messages and information elements contents exceptions | |

### 7.3.57.5 Test requirement

**Table 7.3.57.5-1: Cell specific test parameters for E-UTRAN TDD (cell # 1) for in-sync radio link monitoring tests for Cat-M1 in CEMode A**

| Parameter | Unit | Test 1 | | | | |
|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T5 |
| E-UTRA RF Channel Number | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | |
| Special subframe configuration Note 1 | | 6 | | | | |
| Uplink-downlink configuration Note 1 | | 1 | | | | |
| MPDCCH parameters as defined in A.7.3. | | R.15 TDD | | | | |
| OCNG Pattern defined in D.2.11 (TDD) | | OP.11 TDD | | | | |
| $\rho_A$, $\rho_B$ | | -3 | | | | |
| MPDCCH_RA | dB | 0 | | | | |
| MPDCCH_RB | dB | 0 | | | | |
| PBCH_RA | dB | -3 | | | | |
| PBCH_RB | dB | | | | | |
| PSS_RA | dB | | | | | |
| SSS_RA | dB | | | | | |
| OCNG_RA Note 2 | dB | | | | | |
| OCNG_RB Note 2 | dB | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | |
| SNR Note 6 | dB | [5.0] | [-4.2] | [-13.2] | [-2.0] | [5.0] |
| Propagation condition | | ETU 30Hz | | | | |
| Correlation Matrix and Antenna Configuration | | 2x1 Low | | | | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 4: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 5: The signal contains PDCCH for UEs other than the device under test as part of OCNG.
Note 6: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 7: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure 7.3.57.4-1.

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F ([680+40] ms after the start of time duration T5) the UE shall transmit uplink signal at least in all subframes configured for CQI transmission according to the configured CQI reporting mode (PUCCH 1-0).

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.58 E- UTRAN TDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category M1 configured in CEMode A

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

- UE Transmit minimum power and Transmit OFF power are TBD for a carrier frequency above 3GHz

- The Test system uncertainties applicable to this test are undefined

- The Test tolerances applicable to this test are undefined

- Annex E needs to be updated

- SNR values are in [] (square bracket)

### 7.3.58.1 Test purpose

The purpose of this test is to verify that the TDD category M1 UE configured in CEMode A properly detects the out of sync for the purpose of monitoring downlink radio link quality of the PCell when DRX is used. This test will partly verify the E-UTRAN TDD radio link monitoring requirements in TS 36.133[4] clause 7.19.

### 7.3.58.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 13 and forward of UE category M1. Applicability requires support for FGI bit 5.

### 7.3.58.3 Minimum conformance requirements

The UE category CAT-M1 applicability of the requirements for performing radio link monitoring in subclause 7.19 of TS 36.133[4] is defined in Section 3.6 of TS 36.133[4].

When DRX is used for FD-FDD and TDD category M1 UEs, the $Q_{out_CatM1}$ evaluation period ($T_{Evaluate_Q_{out_DRX_CatM1}}$) and the $Q_{in_CatM1}$ evaluation period ($T_{Evaluate_Q_{in_DRX_CatM1}}$) specified in Table 7.3.58.3-1 (TS 36.133 [4] Table 7.19.2.2-1) will be used.

When eDRX_CONN cycle is used for FD-FDD and TDD UE category M1 UEs, the $Q_{out_CatM1}$ evaluation period ($T_{Evaluate_Q_{out_DRX_CatM1}}$) and the $Q_{in_CatM1}$ evaluation period ($T_{Evaluate_Q_{in_DRX_CatM1}}$) specified in Table 7.3.58.3-2 (TS 36.133 [4] Table 7.19.2.2-2) will be used.

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{out_DRX_CatM1}}$ [s] period becomes worse than the threshold $Q_{out_CatM1}$, Layer 1 of the UE shall send out-of-sync indication for the PCell to the higher layers within $T_{Evaluate_Q_{out_DRX_CatM1}}$ [s] evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [5].

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{in_DRX_CatM1}}$ [s] period becomes better than the threshold $Q_{in_CatM1}$, Layer 1 of the UE shall send in-sync indications for the PCell to the higher layers within $T_{Evaluate_Q_{in_DRX_CatM1}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [5].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in TS 36.213 [8]. When DRX is used, two successive indications from Layer 1 shall be separated by at least max(10ms, DRX_cycle_length). When eDRX_CONN is used, two successive indications from Layer 1 shall be separated by at least max(10 ms, eDRX_CONN cycle length).

Upon start of T310 timer as specified in clause 5.3.11 in TS 36.331 [5], the UE shall monitor the link for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [5].

**Table 7.3.58.3-1: $Q_{out_CatM1}$ and $Q_{in_CatM1}$ Evaluation Period in DRX for FD-FDD and TDD UE category M1**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX_CatM1}}$ and $T_{Evaluate_Q_{in_DRX_CatM1}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in TS36.133 [4] clause 7.19.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note (20) |
| 0.04 < DRX cycle ≤ 0. 64 | Note (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note (5) |
| NOTE: Evaluation period length in time depends on the length of the DRX cycle in use | |

**Table 7.3.58.3-2: $Q_{out_CatM1}$ and $Q_{in_CatM1}$ evaluation period when eDRX_CONN cycle is configured for FD-FDD and TDD UE category M1**

| eDRX_CONN cycle length (s) | $T_{Evaluate_Q_{out_DRX_CatM1}}$ and $T_{Evaluate_Q_{in_DRX_CatM1}}$ (s) (eDRX_CONN cycles) |
|---|---|
| 2.56 < eDRX_CONN cycle ≤ 10.24 | Note (5) |
| NOTE: Evaluation period length in time depends on the length of the eDRX_CONN cycle in use | |

The normative reference for this requirement is TS 36.133 [4] clause 7.19.1, 7.19.2 and A.7.3.58.

The test parameters are given in Tables 7.3.58.4.1-1, 7.3.58.5-1, 7.3.58.5-2 and 7.3.58.5-3.

## 7.3.58.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. Figure 7.3.58.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms without repetition. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode MPDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to “infinity” so that UL timing alignment is maintained during the test.

UE shall successfully complete the RRC reconfiguration accordingly prior to the start of time duration T1.

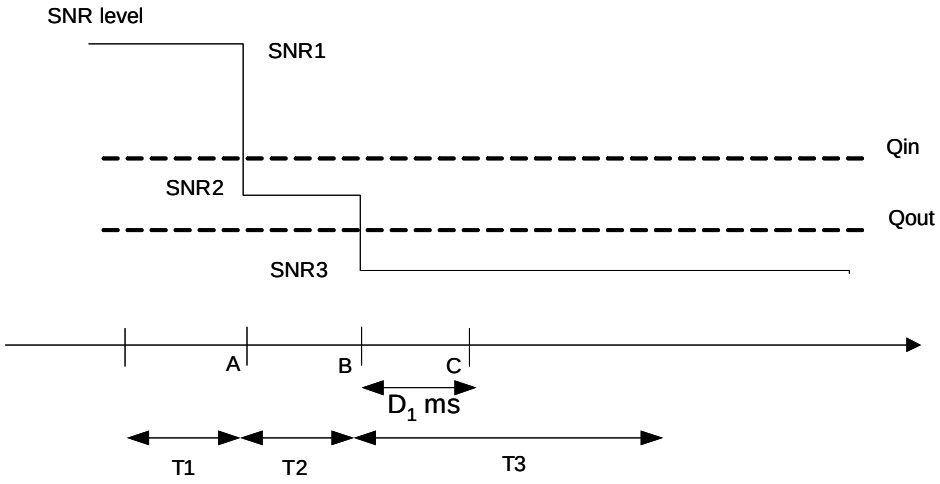


**Figure 7.3.58.4-1: SNR variation for out-of-sync testing in DRX**

### 7.3.58.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.58.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0
4. Message contents are defined in clause 7.3.58.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.58.4.1-1: General test parameters for E-UTRAN TDD out-of-sync tests in DRX for UE category M1 configured in CEMode A**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| CP length | | | Normal | |
| Out of sync transmission parameters (Note 1) | DCI format | | 6-1A | As defined in TS 36.212 |
| | Number of OFDM symbols for legacy control channels | | 2 | Out of sync threshold $Q_{out,\ Cat\ M1}$ and the corresponding hypothetical MPDCCH transmission parameters are as specified in clause 7.19.2 and Table 7.19.2-1 respectively. |
| | MPDCCH aggregation level | eCCE | 16 | |
| | MPDCCH repetition level | | 4 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of MPDCCH to RS EPRE | dB | 0 | |
| DRX cycle | | ms | 1280 | See Table A.7.3.58.1-3 |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 0 | T310 is disabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 1 | Minimum CQI reporting periodicity |
| T1 | | s | 32 | |
| T2 | | s | 12.8 | |
| T3 | | s | 13 | |
| Note 1: MPDCCH corresponding to the out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

### 7.3.58.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms without repetition. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode MPDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to “infinity” so that UL timing alignment is maintained during the test.

1. Ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3.

2. SS sends a RRC Reconfiguration message configuring M-PDCCH aggregation level and M-PDCCH repetition level as defined in Table 7.3.58.4.1-1

3. Set the parameters according to T1 in Table 7.3.58.5-1. Propagation conditions are set according to Annex B clause B.1. T1 starts.

4. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.58.5-1. T2 starts.

5. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.58.5-1. T3 starts.

6. If the SS a) detects uplink power equal to or higher than [-39] dBm (as defined in TS 36.521-1[10] clause 6.3.2.5 for carrier frequency less than [3]GHz) in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point B, and b) does not detect any uplink power higher than [-48.5] dBm (as defined in TS 36.521-1[10] clause 6.3.3.5 for carrier frequency less than [3]GHz) from time point C (6500 ms (D1) after the start of T3) until T3 expires, then the number of successful tests is increased by one. Otherwise the number of failed tests is increased by one.

7. When T3 expires the SS shall change the SNR value to T1 as specified in Table 7.3.58.5-1.

8. If the UE has not re-established the connection in at least 1s, the UE is switched off and then on. Ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3.

9. Repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.58.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.58.4.3-1: Common Exception messages for E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category M1 configured in CEMode A**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-1 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.58.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category M1 configured in CEMode A**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULTT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig- DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213 [8]) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | OFF | (see Table 7.2.2-1B and clause 7.2.2 in TS 36.213 [8]) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|

| | |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.58.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN TDD Radio Link Monitoring test for In-sync in DRX UE category M1 configured in CEMode A**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf10 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | Sf1280 typical value in real network for real-time services. | |
| Sf1280 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.58.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN TDD Radio Link Monitoring test for Out-of-sync in UE category M1 configured in CEMode A**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.58.4.3-5: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring test for Out-of-sync in DRX in UE category M1 configured in CEMode A**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | |
| } | | | |
| } | | | |

**Table 7.3.58.4.3-6: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN TDD Radio Link Monitoring test for Out-of-sync in DRX for UE category M1 with CE mode A**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

## 7.3.58.5 Test requirement

**Table 7.3.58.5-1: Cell specific test parameters for E-UTRAN TDD (cell # 1) for out-of-sync radio link monitoring tests in DRX for UE category M1 configured in CEMode A**

| Parameter | Unit | Test 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| Special subframe configuration | | 6 | | |
| Uplink-downlink configuration | | 1 | | |
| MPDCCH parameters defined in A.7.3 | | R.15 TDD | | |
| OCNG Pattern defined in D.2.11 (TDD) | | OP.11 TDD | | |
| $\rho_A$, $\rho_B$ | | -3 | | |
| MPDCCH_RA | dB | 0 | | |
| MPDCCH_RB | dB | 0 | | |
| PBCH_RA | dB | -3 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| OCNG_RA Note1 | dB | | | |
| OCNG_RB Note1 | dB | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | |
| SNR Note 6 | dB | [-0.1] | [-7.17] | [-15.17] |
| Propagation condition | | AWGN | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains MPDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2 and T3 is denoted as SNR1, SNR2 and SNR3 respectively in figure A.7.3.58.1-1.

**Table 7.3.58.5-2: DRX-Configuration for E-UTRAN TDD out-of-sync tests for UE category M1 configured in CEMode A**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf10 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf1280 | |
| shortDRX | disable | |

**Table 7.3.58.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN TDD out-of-sync testing for UE category M1 configured in CEMode A**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331 and section 10.1 in TS 36.213. |

The UE behaviour in each test during time durations T1, T2 and T3 shall be as follows:

During the period from time point A to time point B the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the subframe according to the configured CQI reporting mode (PUCCH 1-0).

The UE shall stop transmitting uplink signal no later than time point C (duration $D_1$ = 6500 ms after the start of time duration T3.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.59 E- UTRAN TDD Radio Link Monitoring Test for In-sync in DRX for UE category M1 configured in CEMode A

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- UE Transmit minimum power is TBD for a carrier frequency above 3GHz ([-39 dBm])

- The Test system uncertainties applicable to this test are undefined

- The Test tolerances applicable to this test are undefined

- Annex E needs to be updated

- SNR values are in [] (square bracket)

### 7.3.59.1 Test purpose

The purpose of this test is to verify that the TDD category M1 UE configured in CEMode A properly detects the in sync for the purpose of monitoring downlink radio link quality of the PCell when DRX is used. This test will partly verify the E-UTRAN TDD radio link monitoring requirements in TS 36.133[4] clause 7.19.

### 7.3.59.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 13 and forward of UE category M1. Applicability requires support for FGI bit 5.

### 7.3.59.3 Minimum conformance requirements

The UE category CAT-M1 applicability of the requirements for performing radio link monitoring in subclause 7.19 of TS 36.133[4] is defined in Section 3.6 of TS 36.133[4].

When DRX is used for FD-FDD and TDD UE category M1 UEs, the $Q_{out_CatM1}$ evaluation period ($T_{Evaluate_Q_{out_DRX_CatM1}}$) and the $Q_{in_CatM1}$ evaluation period ($T_{Evaluate_Q_{in_DRX_CatM1}}$) specified in Table 7.3.59.3-1 will be used.

When eDRX_CONN cycle is used for FD-FDD and TDD UE category M1 UEs, the $Q_{out_CatM1}$ evaluation period ($T_{Evaluate_Q_{out_DRX_CatM1}}$) and the $Q_{in_CatM1}$ evaluation period ($T_{Evaluate_Q_{in_DRX_CatM1}}$) specified in Table 7.3.59.3-2 will be used.

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{out_DRX_CatM1}}$ [s] period becomes worse than the threshold $Q_{out_CatM1}$, Layer 1 of the UE shall send out-of-sync indication for the PCell to the higher layers within $T_{Evaluate_Q_{out_DRX_CatM1}}$ [s] evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [2].

When the downlink radio link quality of the PCell estimated over the last $T_{Evaluate_Q_{in_DRX_CatM1}}$ [s] period becomes better than the threshold $Q_{in_CatM1}$, Layer 1 of the UE shall send in-sync indications for the PCell to the higher layers within $T_{Evaluate_Q_{in_DRX_CatM1}}$ [s] evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [2].

The out-of-sync and in-sync evaluations of the PCell shall be performed as specified in clause 4.2.1 in [3]. When DRX is used, two successive indications from Layer 1 shall be separated by at least max(10ms, DRX_cycle_length). When eDRX_CONN is used, two successive indications from Layer 1 shall be separated by at least max(10 ms, eDRX_CONN cycle length).

Upon start of T310 timer as specified in clause 5.3.11 in TS 36.331 [2], the UE shall monitor the link for recovery using the evaluation period and Layer 1 indication interval corresponding to the non-DRX mode until the expiry or stop of T310 timer.

The transmitter power of the UE shall be turned off within 40 ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [2].

**Table 7.3.59.3-1: $Q_{out_CatM1}$ and $Q_{in_CatM1}$ Evaluation Period in DRX for FD-FDD and TDD UE category M1**

| DRX cycle length (s) | $T_{Evaluate_Q_{out_DRX_CatM1}}$ and $T_{Evaluate_Q_{in_DRX_CatM1}}$ (s) (DRX cycles) |
|---|---|
| ≤ 0.01 | Non-DRX requirements in clause 7.19.2.1 are applicable. |
| 0.01 < DRX cycle ≤0.04 | Note (20) |
| 0.04 < DRX cycle ≤ 0. 64 | Note (10) |
| 0.64 < DRX cycle ≤ 2.56 | Note (5) |
| NOTE: Evaluation period length in time depends on the length of the DRX cycle in use | |

**Table 7.3.59.3-2: $Q_{out_CatM1}$ and $Q_{in_CatM1}$ evaluation period when eDRX_CONN cycle is configured for FD-FDD and TDD UE category M1**

| eDRX_CONN cycle length (s) | $T_{Evaluate_Q_{out_DRX_CatM1}}$ and $T_{Evaluate_Q_{in_DRX_CatM1}}$ (s) (eDRX_CONN cycles) |
|---|---|
| 2.56 < eDRX_CONN cycle ≤ 10.24 | Note (5) |
| NOTE: Evaluation period length in time depends on the length of the eDRX_CONN cycle in use | |

The normative reference for this requirement is TS 36.133 [4] clause 7.19.1, 7.19.2 and A.7.3.59.

The test parameters are given in Tables 7.3.59.4.1-1, 7.3.59.5-1, 7.3.59.5-2 and 7.3.59.5-3.

### 7.3.59.4 Test description

There is one cell (cell 1), which is the active cell, in the test. The test consists of five successive time periods, with time duration of T1, T2, T3, T4 and T5 respectively. Figure 7.3.59.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms without repetition. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode MPDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to “infinity” so that UL timing alignment is maintained during the test.

UE shall successfully complete the RRC reconfiguration accordingly prior to the start of time duration T1.

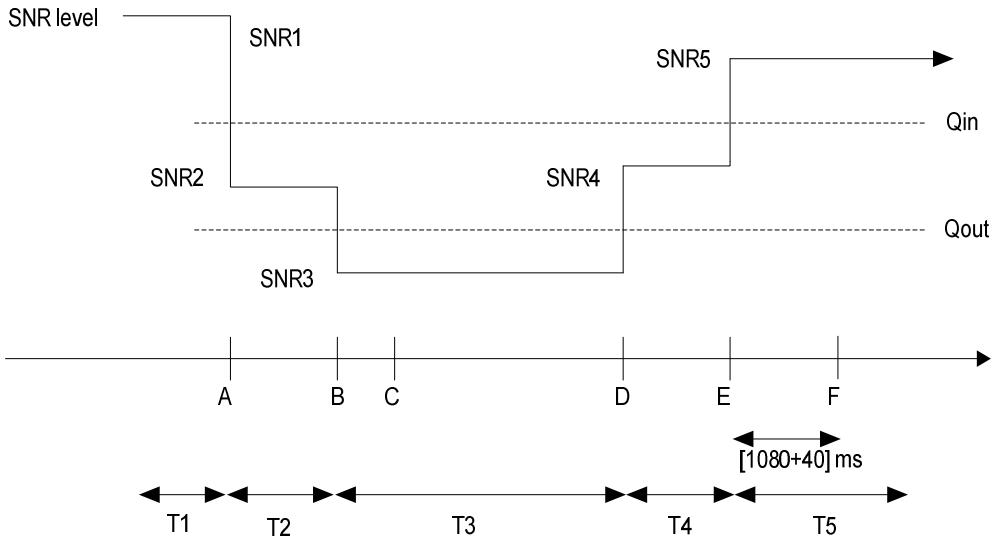


**Figure 7.3.59.4-1: SNR variation for in-sync testing in DRX**

#### 7.3.59.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 using only main Tx/Rx antenna.
2. The general test parameter settings for the different subtests are set up according to Table 7.3.59.4.1-1.
3. Propagation conditions are set according to Annex B clause B.0
4. Message contents are defined in clause 7.3.59.4.3
5. There is one cell specified in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.59.4.1-1: General test parameters for E-UTRAN TDD In-sync tests in DRX for UE category M1 configured in CEMode A**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Active cell | | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1 |
| CP length | | | Normal | |
| In sync transmission parameters (Note 1) | DCI format | | 6-1A | As defined in TS 36.212 |
| | Number of OFDM symbols for legacy control channels | | 2 | In sync threshold $Q_{in, Cat M1}$ and the corresponding hypothetical MPDCCH transmission parameters are as specified in clause 7.19.2 and Table 7.19.2-1 respectively. |
| | MPDCCH aggregation level | eCCE | 8 | |
| | MPDCCH repetition level | | 4 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of MPDCCH to RS EPRE | | 0 | |
| Out of sync transmission parameters (Note 1) | DCI format | | 6-1A | As defined in TS 36.212 |
| | Number of OFDM symbols for legacy control channels | | 2 | Out of sync threshold $Q_{out, Cat M1}$ and the corresponding hypothetical MPDCCH transmission parameters are as specified in clause 7.19.2 and Table 7.19.2-1 respectively. |
| | MPDCCH aggregation level | eCCE | 24 | |
| | MPDCCH repetition level | | 8 | |
| | $\rho_A$, $\rho_B$ | | -3 | |
| | Ratio of MPDCCH to RS EPRE | dB | 0 | |
| DRX cycle | | ms | 40 | See Table A.7.3.59.1-3 |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| Periodic CQI reporting mode | | | PUCCH 1-0 | As defined in table 7.2.2-1 in TS 36.213. |
| CQI reporting periodicity | | ms | 1 | Minimum CQI reporting periodicity |
| T1 | | s | 4 | |
| T2 | | s | 1.6 | |
| T3 | | s | 1.46 | |
| T4 | | s | 0.4 | |
| T5 | | s | 4 | |
| Note 1: MPDCCH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

### 7.3.59.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for periodic CQI reporting in PUCCH 1-0 mode with a reporting periodicity of 1 ms without repetition. In the test, DRX configuration is enabled and DRX inactivity timer has already been expired, i.e. UE tries to decode MPDCCH and to send periodic CQI during the period when On-duration timer is running. Time alignment timers shall be set to "infinity" so that UL timing alignment is maintained during the test.

1. Ensure the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3.
2. SS sends a RRC Reconfiguration message configuring M-PDCCH aggregation level and M-PDCCH repetition level as defined in Table 7.3.59.4.1-1
3. Set the parameters according to T1 in Table 7.3.59.5-1. Propagation conditions are set according to Annex B.1. T1 starts.
4. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.59.5-1. T2 starts.
5. When T2 expires the SS shall change the SNR value to T3 as specified in Table 7.3.59.5-1. T3 starts.

6. When T3 expires the SS shall change the SNR value to T4 as specified in Table 7.3.59.5-1. T4 starts.

7. When T4 expires the SS shall change the SNR value to T5 as specified in Table 7.3.59.5-1. T5 starts.

8. If the SS detects uplink power equal to or higher than [-39 dBm] (as defined in TS 36.521-1 clause 6.3.2.5 for carrier frequency less than [3]GHz) in each subframe configured for CQI transmission (according CQI reporting mode PUCCH 1-0) during the period from time point A to time point F (1120 ms after the start of time duration T5) the number of successful tests is increased by one.

   Otherwise the number of failed tests is increased by one.

9. After T5 expires, repeat steps 2-7 for both subtests until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.59.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 7.3.59.4.3-1: Common Exception messages for E-UTRAN TDD Radio Link Monitoring Test for In-sync in DRX for UE category M1 configured in CEMode A**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.4-2 |
| Default RRC messages and information elements contents exceptions | |

**Table 7.3.59.4.3-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring Test for In-sync in DRX for UE category M1 configured in CEMode A**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULTT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig- DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportPeriodic | ON | | |
| cqi-ReportPeriodic ::= CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213 [8]) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | OFF | (see Table 7.2.2-1B and clause 7.2.2 in TS 36.213 [8]) | |
| simultaneousAckNackAndCQI | FALSE | BOOLEAN | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| CQI_PERIODIC | When periodic CQI reporting should be enabled |

**Table 7.3.59.4.3-3: *MAC-MainConfig-RBC*: E-UTRAN TDD Radio Link Monitoring test for In-sync in DRX UE category M1 configured in CEMode A**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | pc_FeatrGrp_5 AND DRX_S |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf10 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | Sf1280 typical value in real network for real-time services. | |
| Sf1280 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | infinity | | |
| } | | | |

**Table 7.3.59.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN TDD Radio Link Monitoring test for In-sync in UE category M1 configured in CEMode A**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | RBC |
| } | | | |

**Table 7.3.59.4.3-5: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD Radio Link Monitoring test for In-sync in DRX in UE category M1 configured in CEMode A**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| defaultValue | NULL | | |
| } | | | |
| } | | | |

**Table 7.3.59.4.3-6: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN TDD Radio Link Monitoring test for In-sync in DRX for UE category M1 with CE mode A**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

## 7.3.59.5 Test requirement

**Table 7.3.59.5-1: Cell specific test parameters for E-UTRAN TDD (cell # 1) for In-sync radio link monitoring tests in DRX for UE category M1 configured in CEMode A**

| Parameter | Unit | Test 1 | | | | |
|---|---|---|---|---|---|---|
| | | **T1** | **T2** | **T3** | **T4** | **T5** |
| E-UTRA RF Channel Number | | 1 | | | | |
| $BW_{channel}$ | MHz | 10 | | | | |
| Special subframe configuration | | 6 | | | | |
| Uplink-downlink configuration | | 1 | | | | |
| MPDCCH parameters defined in A.7.3 | | R.15 TDD | | | | |
| OCNG Pattern defined in D.2.11 (TDD) | | OP.11 TDD | | | | |
| $\rho_A$, $\rho_B$ | | **-3** | | | | |
| MPDCCH_RA | dB | **0** | | | | |
| MPDCCH_RB | dB | **0** | | | | |
| PBCH_RA | dB | **-3** | | | | |
| PBCH_RB | dB | | | | | |
| PSS_RA | dB | | | | | |
| SSS_RA | dB | | | | | |
| OCNG_RA Note1 | dB | | | | | |
| OCNG_RB Note1 | dB | | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | | | |
| SNR Note 8 | dB | [-5.13] | [-10.13] | [-18.13] | [-11.13] | [-5.13] |
| Propagation condition | | AWGN | | | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | | | |

Note 1: OCNG shall be used such that the resources in cell # 1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains MPDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T2, T3, T4 and T5 is denoted as SNR1, SNR2, SNR3, SNR4 and SNR5 respectively in figure 7.3.59.4.1-1.

**Table 7.3.59.5-2: DRX-Configuration for E-UTRAN TDD out-of-sync tests for UE category M1 configured in CEMode A**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf10 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf40 | |
| shortDRX | disable | |

**Table A.7.3.59.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN TDD out-of-sync testing for UE category M1 configured in CEMode A**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331 and section 10.1 in TS 36.213. |

The UE behaviour in each test during time durations T1, T2, T3, T4 and T5 shall be as follows:

During the period from time point A to time point F (1120 ms after the start of time duration T5) the UE shall transmit uplink signal at least once every DRX cycle, in the On-duration part of the cycle in the subframe according to the configured CQI reporting mode (PUCCH 1-0).

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.EE Radio Link Monitoring Test for In-sync without DRX under Normal Coverage for category NB1 UE

Editor's note: This test case is not complete. Following aspects are either missing or not yet determined:

- Test system uncertainties and test tolerances is TBD
- Message contents are FFS
- Test Requirements are TBD

### 7.3.EE.1 Test Purpose

The purpose of this test is to verify that the HD-FDD category NB UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the PCell when the UE is under normal coverage conditions. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in TS 36.133 [4] section 7.23.

### 7.3.EE.2 Test Applicability

This test case applies to all types of E-UTRA HD-FDD UE release 13 and forward of UE Category NB1.

### 7.3.EE.3 Minimum Conformance Requirements

When the downlink radio link quality of the NB-IoT cell estimated over the last $T_{Evaluate_Q_{out_NB\text{-}IoT}}$ period becomes worse than the threshold $Q_{out_NB\text{-}IoT}$, Layer 1 of the UE shall send an out-of-sync indication for the NB-IoT cell to the higher layers within $T_{Evaluate_Q_{out_NB\text{-}IoT}}$ $Q_{out_NB\text{-}IoT}$ evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [2].

When the downlink radio link quality of the NB-IoT cell estimated over the last $T_{Evaluate_Q_{in_NB\text{-}IoT}}$ period becomes better than the threshold $Q_{in_NB\text{-}IoT}$, Layer 1 of the UE shall send an in-sync indication for the NB-IoT cell to the higher layers

within $T_{Evaluate_Q_{in_NB\text{-}IoT}}$ $Q_{in_NB\text{-}IoT}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [2].

The out-of-sync and in-sync evaluations of the NB-IoT cell shall be performed as specified in clause 4.2.1 in [3]. Two successive indications from Layer 1 shall be separated by at least [10]ms.

The transmitter power of the UE shall be turned off within [40] ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [2]. The following table 7.23.2.1-1 defines the $T_{Evaluate_Q_{out_NB\text{-}IoT}}$ and $T_{Evaluate_Q_{in_NB\text{-}IoT}}$ for each coverage mode.

**Table 7.3.EE.3-1 Mapping of coverage mode and $Q_{out}$ and $Q_{in}$ Evaluation Period in non-DRX for HD-FDD Category NB1 UE**

| Coverage Mode | $T_{Evaluate_Q_{out_NB\text{-}IoT}}$ | $T_{Evaluate_Q_{in_NB\text{-}IoT}}$ |
|---|---|---|
| Normal | [400]ms | [200]ms |
| Enhanced | [4000]ms | [2000]ms |

The normative reference for this requirement is TS 36.133 [4] clause 7.23 and A.7.3.EE

## 7.3.EE.4 Test description

There is one cell (nCell 1), which is the active cell, in the test. The test consists of five successive time periods, with time duration of T1, dT, T2, dT and T3 respectively. Figure 7.3.EE.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to nCell 1. The UE is scheduled in every uplink subframe to transmit NPUSCH, which is received by the test equipment. By measuring the reception of the NPUSCH, detection of out of sync and in sync requirements can be measured. In the test, DRX configuration is disabled, i.e. UE tries to decode NPDCCH and to send NPUSCH during the period when On-duration timer is running. Time alignment timers shall be set to “infinity” so that UL timing alignment is maintained during the test.

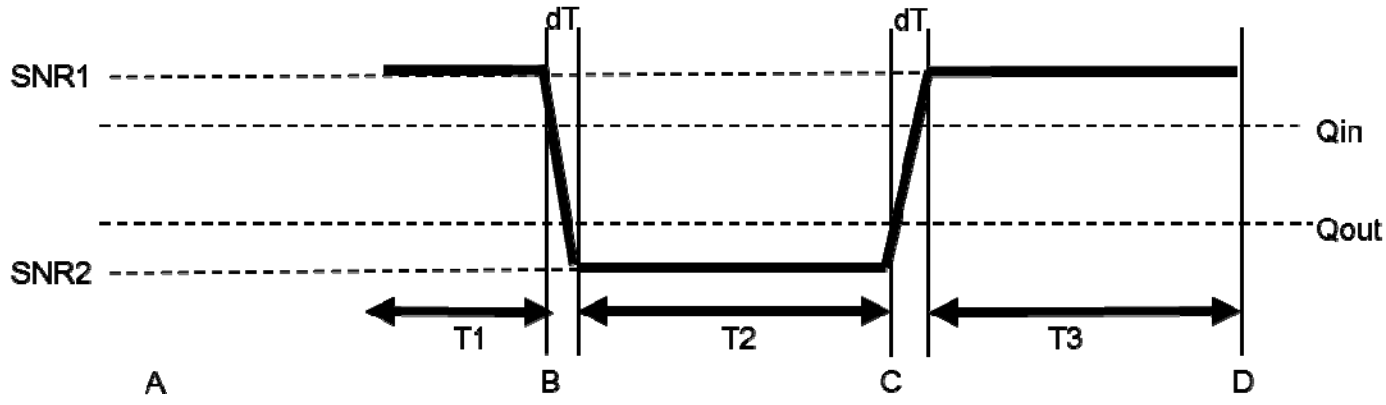


**Figure 7.3.EE.4-1: SNR variation for in-sync testing without DRX**

### 7.3.EE.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to FFS

Channel Bandwidth to be tested: As specified in Table 7.3.EE.4.1-1 and 7.3.EE.4.1-2.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 using only main Tx/Rx antenna.

2. The general test parameter settings for the different subtests are set up according to Table 7.3.EE.4.1-1 and 7.3.EE.4.1-2.

3. Propagation conditions are set according to Annex B clause B.2

4. Message contents are defined in clause 7.3.EE.4.3

5. There is one NB1 cell and one EUTRA Cell specified in this test. nCell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.EE.4.1-1: General test parameters for HD-FDD in-sync test without DRX for UE category NB1 in normal coverage**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| NB-IoT operational mode | | | In-band | |
| Active cell | | | nCell 1 | NB-IOT cell nCell1 is within E-UTRA cell eCell1 |
| CP length | | | Normal | |
| In sync transmission parameters (Note 1) | DCI format | | Format N1 | As defined in TS 36.212[21] |
| | Number of OFDM symbols for legacy control channels | | 3 | In sync threshold $Q_{in, Cat NB1}$ and the corresponding hypothetical NPDCCH transmission parameters are as specified in clause 7.19.2 and Table 7.19.2-1 respectively in TS 36.133[4]. |
| | NPDCCH aggregation level | eCCE | [2] | |
| | NPDCCH repetition level | | [TBD] | |
| | Ratio of NPDSCH to NRS EPRE | | 0 | |
| | Ratio of NPDCCH to NRS EPRE | | 0 | |
| Out of sync transmission parameters (Note 1) | DCI format | | Format N1 | As defined in TS 36.212[21] |
| | Number of OFDM symbols for legacy control channels | | 3 | Out of sync threshold $Q_{out, Cat NB1}$ and the corresponding hypothetical NPDCCH transmission parameters are as specified in clause 7.19.2 and Table 7.19.2-1 respectively in TS 36.133[4]. |
| | NPDCCH aggregation level | eCCE | 2 | |
| | NPDCCH repetition level | | 256 | |
| | Ratio of NPDSCH to NRS EPRE | | 0 | |
| | Ratio of NPDCCH to RS EPRE | dB | 0 | |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| T1 | | s | [4] | |
| dT | | s | [2] | |
| T2 | | s | [4] | |
| dT | | s | [2] | |
| T3 | | s | [4] | |
| Note 1: NPDCCH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

**Table 7.3.EE.4.1-2: nCell 1 specific test parameters for HD-FDD in-sync radio link monitoring test without DRX for UE category NB1 in normal coverage**

| Parameter | Unit | nCell 1 | | | | |
|---|---|---|---|---|---|---|
| | | T1 | dT | T2 | dT | T3 |
| $BW_{channel}$ | kHz | 200 | | | | |
| PRB location within eCell | - | 30 | | | | |
| NPDCCH parameters defined in A.3.1.3 | | R.26 HD-FDD | | | | |
| OCNG Pattern defined in A.3.2.3 | | NOP.1 FDD | | | | |
| NPDSCH_RA | | 0 | | | | |
| NPDSCH_RB | | 0 | | | | |
| NPDCCH_RA | dB | 0 | | | | |
| NPDCCH_RB | dB | 0 | | | | |
| NPBCH_RA | dB | 0 | | | | |
| NPBCH_RB | dB | | | | | |
| NPSS_RA | dB | | | | | |
| NSSS_RA | dB | | | | | |
| NOCNG_RA Note1 | dB | | | | | |
| NOCNG_RB Note1 | dB | | | | | |
| $N_{oc}$ | dBm/15 kHz | Specified in Table 7.3.EE.4.1-3 | | | | |
| SNR Note 5, Note 6 | dB | [6] | Note 7 | [-13] | Note 8 | [6] |
| Propagation condition | | EPA5Hz | | | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | | | |

Note 1: OCNG shall be used such that the resources in ncell1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains NPDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, dT, T2, dT and T3 is denoted as SNR1, SNR2, and SNR1 respectively in figure 7.3.EE.4-1.
Note 7: The Test system shall gradually reduce its transmit power in steps of [≤1]dB every [0.1] seconds till SNR2 can be ensured at the end of dT.
Note 8: The Test system shall gradually increase its transmit power in steps of [≤1]dB every [0.1] seconds till SNR1 can be ensured at the end of dT.

**Table 7.3.EE.4.1-3: eCell 1 specific test parameters for HD-FDD in-sync radio link monitoring test without DRX for UE category NB1 in normal coverage**

| | | eCell 1 |
|---|---|---|
| | | T1-T3 |
| $BW_{channel}$ | MHz | 10 |
| OCNG Pattern | - | [TBD] |
| PBCH_RA | dB | 0 |
| PBCH_RB | dB | |
| PSS_RA | dB | |
| SSS_RA | dB | |
| PCFICH_RB | dB | |
| PDCCH_RA | dB | |
| PDCCH_RB | dB | |
| OCNG_RA Note 1 | dB | |
| OCNG_RB Note 1 | dB | |
| $N_{oc}$ | dBm/15 kHz | -98 |
| $\hat{E}_s / N_{oc}$ | dB | [TBD] |
| Propagation Condition | | EPA5Hz |
| Antenna Configuration | | 2x1 |
| Note 1: OCNG shall be used such that the eCell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power $N_{oc}$. | | |

**Table 7.3.EE.4.1-4: *TimeAlignmentTimer* -Configuration for E-UTRAN HD-FDD out-of-sync testing for UE category NB1 configured in normal coverage**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331 [5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331 [5] and section 10.1 in TS 36.213 [8]. |

## 7.3.EE.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for NPUSCH transmission for every uplink sub-frame.

1. Ensure the UE is in State 2A-NB with CP CIoT optimisation according to TS 36.508 [7] clause 8.1.5.
2. SS sends a RRC Reconfiguration message configuring NPDCCH repetition level defined in FFS.
3. Set the parameters according to T1 in Table 7.3.EE.5-1. Propagation conditions are set according to Annex B clause B.1. T1 starts.
4. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.EE.5-1 in dT s with SNR level changing by FFS dB in FFS S. T2 starts.
5. When T2 expires the SS shall increase the SNR value to T3 as specified in Table 7.3.EE.5-1 in dT s with SNR level changing by FFS dB in FFS s. T3 starts.
6. During T3 SS sends continuous uplink grants and if the SS detects uplink transmission equal to or higher than - FFS in each uplink subframe configured for NPUSCH during the period from time point C+dT to time point D (FFS ms after the start of time duration T3) the number of successful tests is increased by one.
7. Otherwise the number of failed tests is increased by one.

8. After T3 expires, repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved

### 7.3.EE.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 and FFS:

## 7.3.EE.5 Test requirement

**Table 7.3.EE.5.1-1: nCell 1 specific test parameters for HD-FDD in-sync radio link monitoring test without DRX for UE category NB1 in normal coverage**

| Parameter | Unit | nCell 1 | | | | |
|---|---|---|---|---|---|---|
| | | T1 | dT | T2 | dT | T3 |
| $BW_{channel}$ | kHz | 200 | | | | |
| PRB location within eCell | - | 30 | | | | |
| NPDCCH parameters defined in A.3.1.3 | | R.26 HD-FDD | | | | |
| OCNG Pattern defined in A.3.2.3 | | NOP.1 FDD | | | | |
| NPDSCH_RA | | 0 | | | | |
| NPDSCH_RB | | 0 | | | | |
| NPDCCH_RA | dB | 0 | | | | |
| NPDCCH_RB | dB | 0 | | | | |
| NPBCH_RA | dB | 0 | | | | |
| NPBCH_RB | dB | | | | | |
| NPSS_RA | dB | | | | | |
| NSSS_RA | dB | | | | | |
| NOCNG_RA Note1 | dB | | | | | |
| NOCNG_RB Note1 | dB | | | | | |
| $N_{oc}$ | dBm/15 kHz | Specified in Table 7.3.EE.5.1-2 | | | | |
| SNR Note 5, Note 6 | dB | [6]+TT | Note 7 | [-13]+TT | Note 8 | [6] +TT |
| Propagation condition | | EPA5Hz | | | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | | | |

Note 1: OCNG shall be used such that the resources in ncell1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains NPDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, dT, T2, dT and T3 is denoted as SNR1, SNR2, and SNR1 respectively in figure 7.3.EE.4-1.
Note 7: The Test system shall gradually reduce its transmit power in steps of [≤1]dB every [0.1] seconds till SNR2 can be ensured at the end of dT.
Note 8: The Test system shall gradually increase its transmit power in steps of [≤1]dB every [0.1] seconds till SNR1 can be ensured at the end of dT.

**Table 7.3.EE.5.1-2: eCell 1 specific test parameters for HD-FDD in-sync radio link monitoring test without DRX for UE category NB1 in normal coverage**

| | | eCell 1 |
|---|---|---|
| | | T1-T3 |
| $BW_{channel}$ | MHz | 10 |
| OCNG Pattern | - | [TBD] |
| PBCH_RA | dB | 0 |
| PBCH_RB | dB | |
| PSS_RA | dB | |
| SSS_RA | dB | |
| PCFICH_RB | dB | |
| PDCCH_RA | dB | |
| PDCCH_RB | dB | |
| OCNG_RA Note 1 | dB | |
| OCNG_RB Note 1 | dB | |
| $N_{oc}$ | dBm/15 kHz | -98 |
| $\hat{E}_s / N_{oc}$ | dB | [TBD] +TT |
| Propagation Condition | | EPA5Hz |
| Antenna Configuration | | 2x1 |
| Note 1: OCNG shall be used such that the eCell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power $N_{oc}$. | | |

During the period from time point C+dT to time point D, the UE behaviors in each test shall be as follows:

UE under test is expected to decode the uplink grant and switch to uplink and complete the uplink transmission. Thereafter UE switches back to downlink. This is considered a correct event.

The rate of correct events observed during repeated tests shall be at least 90%.

## 7.3.GG Radio Link Monitoring Test for In-sync without DRX under Enhanced Coverage for category NB1 UE

Editor's note: This test case is not complete. Following aspects are either missing or not yet determined:

- Test system uncertainties and test tolerances is TBD
- Message contents are FFS
- Test Requirements are TBD

### 7.3.GG.1 Test Purpose

The purpose of this test is to verify that the HD-FDD category NB UE properly detects the in sync for the purpose of monitoring downlink radio link quality of the PCell when the UE is under normal coverage conditions. This test will partly verify the E-UTRAN FDD radio link monitoring requirements in TS 36.133 [4] section 7.23.

### 7.3.GG.2 Test Applicability

This test case applies to all types of E-UTRA HD-FDD UE release 13 and forward of UE Category NB1.

### 7.3.GG.3 Minimum Conformance Requirements

When the downlink radio link quality of the NB-IoT cell estimated over the last $T_{Evaluate_Q_{out_NB\text{-}IoT}}$ period becomes worse than the threshold $Q_{out_NB\text{-}IoT}$, Layer 1 of the UE shall send an out-of-sync indication for the NB-IoT cell to the

higher layers within $T_{Evaluate_Q_{out_NB\text{-}IoT}}$ $Q_{out_NB\text{-}IoT}$ evaluation period. A Layer 3 filter shall be applied to the out-of-sync indications as specified in TS 36.331 [2].

When the downlink radio link quality of the NB-IoT cell estimated over the last $T_{Evaluate_Q_{in_NB\text{-}IoT}}$ period becomes better than the threshold $Q_{in_NB\text{-}IoT}$, Layer 1 of the UE shall send an in-sync indication for the NB-IoT cell to the higher layers within $T_{Evaluate_Q_{in_NB\text{-}IoT}}$ $Q_{in_NB\text{-}IoT}$ evaluation period. A L3 filter shall be applied to the in-sync indications as specified in TS 36.331 [2].

The out-of-sync and in-sync evaluations of the NB-IoT cell shall be performed as specified in clause 4.2.1 in [3]. Two successive indications from Layer 1 shall be separated by at least [10]ms.

The transmitter power of the UE shall be turned off within [40] ms after expiry of T310 timer as specified in clause 5.3.11 in TS 36.331 [2]. The following table 7.23.2.1-1 defines the $T_{Evaluate_Q_{out_NB\text{-}IoT}}$ and $T_{Evaluate_Q_{in_NB\text{-}IoT}}$ for each coverage mode.

**Table 7.3.GG.3-1: Mapping of coverage mode and $Q_{out}$ and $Q_{in}$ Evaluation Period in non-DRX for HD-FDD Category NB1 UE**

| Coverage Mode | $T_{Evaluate_Q_{out_NB\text{-}IoT}}$ | $T_{Evaluate_Q_{in_NB\text{-}IoT}}$ |
|---|---|---|
| Normal | [400]ms | [200]ms |
| Enhanced | [4000]ms | [2000]ms |

The normative reference for this requirement is TS 36.133 [4] clause 7.23 and A.7.3.GG

## 7.3.GG.4 Test description

There is one cell (nCell 1), which is the active cell, in the test. The test consists of five successive time periods, with time duration of T1, dT, T2, dT and T3 respectively. Figure 7.3.GG.4-1 shows the variation of the downlink SNR in the active cell to emulate out-of-sync and in-sync states. Prior to the start of the time duration T1, the UE shall be fully synchronized to nCell 1. The UE is scheduled in every uplink subframe to transmit NPUSCH, which is received by the test equipment. By measuring the reception of the NPUSCH, detection of out of sync and in sync requirements can be measured. In the test, DRX configuration is disabled, i.e. UE tries to decode NPDCCH and to send NPUSCH during the period when On-duration timer is running. Time alignment timers shall be set to “infinity” so that UL timing alignment is maintained during the test.

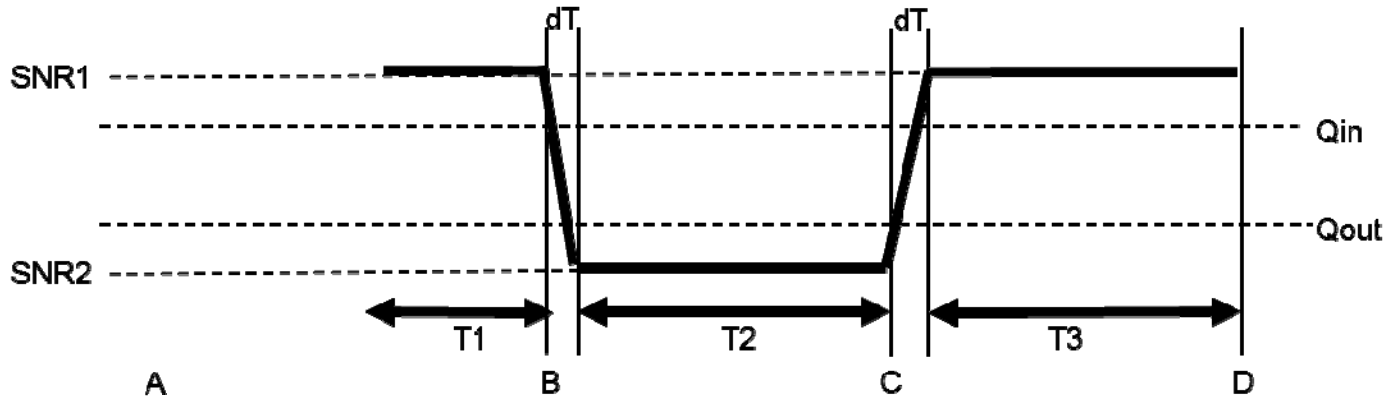


**Figure 7.3.GG.4-1: SNR variation for in-sync testing without DRX**

### 7.3.GG.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to FFS

Channel Bandwidth to be tested: As specified in Table 7.3.GG.4.1-1 and 7.3.GG.4.1-2.

1. Connect the SS (node B emulator) and AWGN noise sources or faders to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.10 using only main Tx/Rx antenna.

2. The general test parameter settings for the different subtests are set up according to Table 7.3.GG.4.1-1 and 7.3.GG.4.1-2.

3. Propagation conditions are set according to Annex B clause B.2

4. Message contents are defined in clause 7.3.GG.4.3

5. There is one NB1 cell and one EUTRA Cell specified in this test. nCell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 7.3.GG.4.1-1: General test parameters for HD-FDD in-sync test without DRX for UE category NB1 in enhanced coverage**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| NB-IoT operational mode | | | In-band | |
| Active cell | | | nCell 1 | NB-IOT cell nCell1 is within E-UTRA cell eCell1 |
| CP length | | | Normal | |
| In sync transmission parameters (Note 1) | DCI format | | Format N1 | As defined in TS 36.212[21] |
| | Number of OFDM symbols for legacy control channels | | 3 | In sync threshold $Q_{in, Cat NB1}$ and the corresponding hypothetical NPDCCH transmission parameters are as specified in clause 7.19.2 and Table 7.19.2-1 respectively in TS 36.133 [4] |
| | NPDCCH aggregation level | eCCE | [2] | |
| | NPDCCH repetition level | | [TBD] | |
| | Ratio of NPDSCH to NRS EPRE | | 0 | |
| | Ratio of NPDCCH to NRS EPRE | | 0 | |
| Out of sync transmission parameters (Note 1) | DCI format | | Format N1 | As defined in TS 36.212[21] |
| | Number of OFDM symbols for legacy control channels | | 3 | Out of sync threshold $Q_{out, Cat NB1}$ and the corresponding hypothetical NPDCCH transmission parameters are as specified in clause 7.19.2 and Table 7.19.2-1 respectively in TS 36.133 [4] |
| | NPDCCH aggregation level | eCCE | 2 | |
| | NPDCCH repetition level | | 256 | |
| | Ratio of NPDSCH to NRS EPRE | | 0 | |
| | Ratio of NPDCCH to RS EPRE | dB | 0 | |
| Layer 3 filtering | | | Enabled | Counters: N310 = 1; N311 = 1 |
| T310 timer | | ms | 2000 | T310 is enabled |
| T311 timer | | ms | 1000 | T311 is enabled |
| T1 | | s | [4] | |
| dT | | s | [2] | |
| T2 | | s | [4] | |
| dT | | s | [2] | |
| T3 | | s | [4] | |
| Note 1: NPDCCH corresponding to the in-sync and out of sync transmission parameters need not be included in the Reference Measurement Channel. | | | | |

**Table 7.3.GG.4.1-2: nCell 1 specific test parameters for HD-FDD in-sync radio link monitoring test without DRX for UE category NB1 in enhanced coverage**

| Parameter | Unit | nCell 1 | | | | |
|---|---|---|---|---|---|---|
| | | T1 | dT | T2 | dT | T3 |
| $BW_{channel}$ | kHz | 200 | | | | |
| PRB location within eCell | - | 30 | | | | |
| NPDCCH parameters defined in A.3.1.3 | | R.26 HD-FDD | | | | |
| OCNG Pattern defined in A.3.2.3 | | NOP.1 FDD | | | | |
| NPDSCH_RA | | 0 | | | | |
| NPDSCH_RB | | 0 | | | | |
| NPDCCH_RA | dB | 0 | | | | |
| NPDCCH_RB | dB | 0 | | | | |
| NPBCH_RA | dB | 0 | | | | |
| NPBCH_RB | dB | | | | | |
| NPSS_RA | dB | | | | | |
| NSSS_RA | dB | | | | | |
| NOCNG_RA Note1 | dB | | | | | |
| NOCNG_RB Note1 | dB | | | | | |
| $N_{oc}$ | dBm/15 kHz | Specified in Table 7.3.GG.4.1-3 | | | | |
| SNR Note 5, Note 6 | dB | [0] | Note 7 | [-20] | Note 8 | [0] |
| Propagation condition | | EPA5Hz | | | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | | | |

Note 1: OCNG shall be used such that the resources in ncell1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains NPDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, T3 and T5 is denoted as SNR1, SNR2, and SNR1 respectively in figure A.7.3.X.1-1.
Note 7: The Test system shall gradually reduce its transmit power in steps of [≤1]dB every [0.1] seconds till SNR2 can be ensured at the end of dT.
Note 8: The Test system shall gradually increase its transmit power in steps of [≤1]dB every [0.1] seconds till SNR1 can be ensured at the end of dT.

**Table 7.3.GG.4.1-3: eCell 1 specific test parameters for HD-FDD in-sync radio link monitoring test without DRX for UE category NB1 in enhanced coverage**

| Parameter | Unit | eCell 1 |
|---|---|---|
| | | T1-T3 |
| $BW_{channel}$ | MHz | 10 |
| OCNG Pattern | - | [TBD] |
| PBCH_RA | dB | 0 |
| PBCH_RB | dB | |
| PSS_RA | dB | |
| SSS_RA | dB | |
| PCFICH_RB | dB | |
| PDCCH_RA | dB | |
| PDCCH_RB | dB | |
| OCNG_RA Note 1 | dB | |
| OCNG_RB Note 1 | dB | |
| $N_{oc}$ | dBm/15 kHz | -98 |
| $\hat{E}_s / N_{oc}$ | dB | [TBD] |
| Propagation Condition | | EPA5Hz |
| Antenna Configuration | | 2x1 |

Note 1: OCNG shall be used such that the eCell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power $N_{oc}$.

**Table 7.3.GG.4.1-4: *TimeAlignmentTimer* -Configuration for E-UTRAN HD-FDD out-of-sync testing for UE category NB1 configured in enhanced coverage**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331 [5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331 [5] and section 10.1 in TS 36.213 [8]. |

## 7.3.GG.4.2 Test procedure

Prior to the start of the time duration T1, the UE shall be fully synchronized to cell 1. The UE shall be configured for NPUSCH transmission for every uplink sub-frame.

1. Ensure the UE is in State 2A-NB with CP CIoT optimisation according to TS 36.508 [7] clause 8.1.5 .
2. SS sends a RRC Reconfiguration message configuring NPDCCH repetition level defined in FFS.
3. Set the parameters according to T1 in Table 7.3.GG.5-1. Propagation conditions are set according to Annex B clause B.1. T1 starts.
4. When T1 expires the SS shall change the SNR value to T2 as specified in Table 7.3.GG.5-1 in dT s with SNR level changing by FFS dB in FFS S. T2 starts.
5. When T2 expires the SS shall increase the SNR value to T3 as specified in Table 7.3.GG.5-1 in dT s with SNR level changing by FFS dB in FFS s. T3 starts.
6. During T3 SS sends continuous uplink grants and if the SS detects uplink transmission equal to or higher than - FFS in each uplink subframe configured for NPUSCH during the period from time point C+dT to time point D (FFS ms after the start of time duration T3) the number of successful tests is increased by one.
7. Otherwise the number of failed tests is increased by one.

8. After T3 expires, repeat steps 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 7.3.GG.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 and FFS:

## 7.3.GG.5 Test requirement

**Table 7.3.GG.5.1-1: nCell 1 specific test parameters for HD-FDD in-sync radio link monitoring test without DRX for UE category NB1 in enhanced coverage**

| Parameter | Unit | nCell 1 | | | | |
|---|---|---|---|---|---|---|
| | | T1 | dT | T2 | dT | T3 |
| $BW_{channel}$ | kHz | 200 | | | | |
| PRB location within eCell | - | 30 | | | | |
| NPDCCH parameters defined in A.3.1.3 | | R.26 HD-FDD | | | | |
| OCNG Pattern defined in A.3.2.3 | | NOP.1 FDD | | | | |
| NPDSCH_RA | | 0 | | | | |
| NPDSCH_RB | | 0 | | | | |
| NPDCCH_RA | dB | 0 | | | | |
| NPDCCH_RB | dB | 0 | | | | |
| NPBCH_RA | dB | 0 | | | | |
| NPBCH_RB | dB | | | | | |
| NPSS_RA | dB | | | | | |
| NSSS_RA | dB | | | | | |
| NOCNG_RA Note1 | dB | | | | | |
| NOCNG_RB Note1 | dB | | | | | |
| $N_{oc}$ | dBm/15 kHz | Specified in Table 7.3.GG.5.1-2 | | | | |
| SNR Note 5, Note 6 | dB | [0]+TT | Note 7 | [-20]+TT | Note 8 | [0] +TT |
| Propagation condition | | EPA5Hz | | | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | | | |

Note 1: OCNG shall be used such that the resources in ncell1 are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The uplink resources for CQI reporting are assigned to the UE prior to the start of time period T1.
Note 3: The timers and layer 3 filtering related parameters are configured prior to the start of time period T1.
Note 4: The signal contains NPDCCH for UEs other than the device under test as part of OCNG.
Note 5: SNR levels correspond to the signal to noise ratio over the cell-specific reference signal REs.
Note 6: The SNR in time periods T1, dT, T2, dT and T3 is denoted as SNR1, SNR2, and SNR1 respectively in figure 7.3.GG.4-1.
Note 7: The Test system shall gradually reduce its transmit power in steps of [≤1]dB every [0.1] seconds till SNR2 can be ensured at the end of dT.
Note 8: The Test system shall gradually increase its transmit power in steps of [≤1]dB every [0.1] seconds till SNR1 can be ensured at the end of dT.

**Table 7.3.GG.5.1-2: eCell 1 specific test parameters for HD-FDD in-sync radio link monitoring test without DRX for UE category NB1 in enhanced coverage**

| Parameter | Unit | eCell 1 |
|---|---|---|
| | | T1-T3 |
| $BW_{channel}$ | MHz | 10 |
| OCNG Pattern | - | [TBD] |
| PBCH_RA | dB | 0 |
| PBCH_RB | dB | |
| PSS_RA | dB | |
| SSS_RA | dB | |
| PCFICH_RB | dB | |
| PDCCH_RA | dB | |
| PDCCH_RB | dB | |
| OCNG_RA Note 1 | dB | |
| OCNG_RB Note 1 | dB | |
| $N_{oc}$ | dBm/15 kHz | -98 |
| $\hat{E}_s / N_{oc}$ | dB | [TBD] +TT |
| Propagation Condition | | EPA5Hz |
| Antenna Configuration | | 2x1 |
| Note 1: OCNG shall be used such that the eCell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power $N_{oc}$. | | |

During the period from time point C+dT to time point D, the UE behaviors in each test shall be as follows:

UE under test is expected to decode the uplink grant and switch to uplink and complete the uplink transmission. Thereafter UE switches back to downlink. This is considered a correct event.

The rate of correct events observed during repeated tests shall be at least 90%.

# 7.4 Void

# 7.5 Proximity-based Services

## 7.5.1 E-UTRAN FDD – UE ProSe Direct Discovery Transmission Timing Accuracy Test

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

- Call procedure of the test case needs to be tweaked

- Test case applicability is undefined.

- Message contents are TBD.

- Connection diagrams are TBD

- PDSCH scheduling and OCNG configuration during discovery is under investigation.

### 7.5.1.1 Test purpose

To verify the timing requirements of a UE for ProSe Direct Discovery transmissions when PCell downlink timing is used as a reference.

### 7.5.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE FFS.

### 7.5.1.3 Minimum conformance requirements

The requirements in this subclause are applicable when the reference timing used for ProSe transmissions is the serving cell (RRC_IDLE) or PCell (RRC_CONNECTED). The sidelink transmissions takes place $(N_{\text{TA,SL}} + N_{\text{TA offset}}) \cdot T_s$ before the reception of the first detected path (in time) of the corresponding downlink frame from the reference cell, where $N_{\text{TA offset}}$ is specified in Section 8.1 of [9]. The value of $N_{\text{TA,SL}}$ differs between sidelink channels and signals, and is either $N_{\text{TA, SL}} = N_{\text{TA}}$ or $N_{\text{TA, SL}} = 0$ as specified in Section 9.9 of [9].

The normative reference for this requirement is TS 36.133 [4] clause 7.16.2.1 and A.7.5.1.

**Table 7.5.1-1: $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| 1.4 | $24*T_S$ |
| ≥3 | $12*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 | |

### 7.5.1.4 Test description

#### 7.5.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure FFS
2. Propagation conditions are set according to Annex B clause B.0.
3. There is one E-UTRA FDD carrier and one cell specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

#### 7.5.1.4.2 Test procedure

The test consists of a single cell. The transmit timing accuracy is verified related to the downlink frame timing of Cell 1. The downlink timing of Cell 1 is changed and the changes in UE transmit timing are observed. The transmit timing is verified by the UE transmitting PSDCH used as a measurement reference facilitating the SS timing estimation.

1. Ensure the UE is in State 4 according to TS 36.508 [7] clause 4.5.4 and UE test loop Mode 4 is activated.
2. Set the parameters according to Test 1 in Tables 7.5.1.5-1, as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
3. The SS shall check that the UE transmit timing offset is within the limits specified in Table 7.5.1.5-2 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1.
4. The SS adjusts the downlink timing for Cell 1 to a delay of $+32 \times T_S$ (approximately +1 μs) compared to that in step 3.
5. The SS shall check that the UE transmit timing offset stays within the limits specified in Table 7.5.1.5-2. For that purpose, the PSDCH transmission that takes place at least after the first DRX ON duration after downlink timing of Cell 1 is changed is used.

6. The SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

#### 7.5.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions FFS.

### 7.5.1.5 Test requirement

Tables 7.5.1.5-1 defines the primary settings including test tolerances for UE transmit timing accuracy for E-UTRAN FDD test.

Tables 7.5.1.5-2, defines the UE uplink timing include Test Tolerances.

**Table 7.5.1.5-1: Test parameters for ProSe Transmission Timing Accuracy test for E-UTRAN FDD**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 | |
| Active cell | | Cell 1 | E-UTRA FDD Cell1 on RF channel number 1 |
| CP length of Cell 1 | | Normal | |
| drx-Configuration | | DRX_P1 | As specified in Table C.3.4.2-1. |
| ProSe Direct Discovery resource pool configuration | | As specified in Table H.6.1.1-1 (Configuration #1) | IE values unless specified otherwise in this test. |
| PDCCH/PCFICH/PHICH Reference measurement channel Note1 | | R.11 FDD | |
| OCNG Pattern Note2 | | OP.16 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | | | |
| PSS_RA | | | |
| SSS_RA | | | |
| PCFICH_RB | | | |
| PHICH_RA | | | |
| PHICH_RB | | | |
| PDCCH_RA | | | |
| PDCCH_RB | | | |
| OCNG_RA Note3 | | | |
| OCNG_RB Note3 | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 3.30 | |
| RSRP Note4 | dBm/15 kHz | -94.7 | |
| SCH_RP Note 4 | dBm/15 kHz | -94.7 | |
| Propagation condition | | AWGN | |

Note 1: For the reference measurement channels, see clause A.1.
Note 2: For the OCNG pattern, see clause D.1.
Note 3: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE transmit timing offset shall be within the requirements in Table 7.5.1.5-2.

The reference point for the UE initial transmit timing control test requirement shall be the downlink timing minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$.

**Table 7.5.1.5-2: Test requirement for $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| ≥3 | $15*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9]. | |

The UE shall be capable of changing the transmission timing according to the received downlink frame. When the transmission timing error between the UE and the reference cell exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$.

## 7.5.2 FFS

## 7.5.3 E-UTRAN FDD – Interruptions due to ProSe Direct Discovery

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

- The Test system uncertainties applicable to this test are undefined
- The Test tolerances applicable to this test are undefined
- Call procedure of the test case needs to be tweaked
- Test case applicability is undefined.
- Message contents are TBD.
- Connection diagrams are TBD
- Annex E updates are pending.
- PDSCH scheduling and OCNG configuration during discovery is under investigation.

### 7.5.3.1 Test purpose

To verify that the UE is able to complete PCELL interruption due to ProSe Direct discovery.

### 7.5.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE FFS.

### 7.5.3.3 Minimum conformance requirements

For this test, the UE is triggered by the test loop function or the upper layers to monitor ProSe Direct Discovery.

The test parameters are given in Table 7.5.3.3-1 and Table 7.5.3.3-2 below. There is one active cell (PCell) in this test and 24 active Sidelink transmissions in this test (with 12 active Sidelink UEs per configured discovery subframe). Two tests (Test 1 and Test 2) are defined to verify interruptions due to synchronous (Test 1) and asynchronous (Test 2) ProSe Direct Discovery.

The tests consist of three successive time periods, with time duration of T1, T2 and T3 respectively.

During T1, the UE is in RRC_IDLE and monitoring the ProSe Direct Discovery announcements from other active Sidelink UEs on the ProSe Direct Discovery resources.

During T2, the test system establishes a RRC connection with the UE. No PDSCH traffic is scheduled for UE during T2, and the UE is expected to transmit *SidelinkUEInformation* indicating *discRxInterest* during T2. On reception of *SidelinkUEInformation*, the test system shall RRC reconfiguration message to the UE and wait for the UE to respond with RRC reconfiguration complete message before transitioning to T3. If the UE does not transmit *SidelinkUEInformation* for up to [2] sec, the test system shall transition to T3.

During T3, the UE is scheduled with PDSCH traffic on PCell downlink. The test system will count the missed ACK/NACKs during T3 to verify the allowed interruptions during ProSe Direct Discovery.

The normative reference for this requirement is TS 36.133 [4] clause 7.16.3.1 and A.7.5.3.

**Table 7.5.3.3-1: Test parameters for interruption due to ProSe Direct Discovery tests**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| E-UTRA RF Channel Number | | 1 | | |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 | | |
| Active cell | | Cell 1 | | E-UTRA FDD Cell1 on RF channel number 1 |
| CP length of Cell 1 | | Normal | | |
| T1 | s | 5.12 | | |
| T2 | s | Up to receiving RRC reconfiguration setup complete from the UE, or up to [2] sec if UE does not transmit *SidelinkUEInformation* during this period. | | |
| T3 | s | 10.24 | | |

**Table 7.5.3.3-2: ProSe Direct Discovery configuration for interruption due to ProSe Direct Discovery tests**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| E-UTRA RF Channel Number | | 1 | | UL carrier frequency |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 | | |
| ProSe Direct Discovery resource pool configuration | | As specified in Table H.6.1.1-1 (Configuration #1) | As specified in Table H.6.1.1-2 (Configuration #2) | IE values unless specified otherwise in this test. |
| Active Sidelink UEs Configuration | | PDP.1.FDD As specified in Table A.6.2-1 | PDP.2.FDD As specified in Table A.6.2-1 | Transmitting ProSe Direct Discovery (Test 1 and 2) and SLSS (for Test 2) |

**Table 7.5.3.3-3: Cell specific test parameters for interruption due to ProSe Direct Discovery tests**

| Parameter | | Unit | Cell 1 | | |
|---|---|---|---|---|---|
| | | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | | 1 | | |
| $BW_{channel}$ | | MHz | 5 | | |
| UE RRC state | | | IDLE | CONNECTED | |
| Paging configuration | defaultPagingCycle | | rf256 | N/A | |
| | nB | | T / 32 | | |
| DRX | | | N/A | OFF | |
| PDSCH Reference measurement channel defined in A.3.1.1.1 Note1 | | | N/A | None | R.5 FDD |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.1 Note1 | | | R.11 FDD | | |
| OCNG Pattern | | | OP.16 FDD | | |
| PBCH_RA | | dB | 0 | | |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA Note 1 | | | | | |
| OCNG_RB Note 1 | | | | | |
| $N_{oc}$ Note2 | | dBm/15 kHz | -98 | | |
| $\hat{E}_s / N_{oc}$ | | dB | 16 | | |
| RSRP Note3 | | dBm/15 kHz | -82 | | |
| SCH_RP Note 3 | | dBm/15 kHz | -82 | | |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

## 7.5.3.4 Test description

### 7.5.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure FFS.

2. Propagation conditions are set according to Annex B clause B.0.

3. There is one E-UTRA FDD carrier and one cell specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

#### 7.5.3.4.2 Test procedure

There is one active cell (PCell) in this test and 24 active Sidelink transmissions in this test (with 12 active Sidelink UEs per configured discovery subframe). Two tests (Test 1 and Test 2) are defined to verify interruptions due to synchronous (Test 1) and asynchronous (Test 2) ProSe Direct Discovery.

1. Ensure the UE is in FFS according to TS 36.508 [7] clause FFS.
2. Set the parameters according to Test 1 time period T1 in Tables 7.5.3.5-1, as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
3. UE is in idle mode in T1 and is monitoring ProSe Direct Discovery announcements from other active Sidelink UEs.
4. On the expiry of T1, TE pages the UE and establishes a RRC Connection with the UE. TE doesn't schedules any PDSCH traffic to the UE.
5. UE is expected to send a Sidelink UEInformation indicating discRxInterest, upon reception, TE is expected to send a RRC Connection reconfiguration message to the UE.
6. Once the UE sends RRC reconfiguration complete message or [2] secs have elapsed after start of T2, start T3.
7. During T3, TE schedules PDSCH traffic on PCell and counts ACK/NACK and record the same.
8. For Test 1, at least 98.75% of all expected ACK/NACKs during T3 shall be transmitted by the ProSe UE. For Test 2, at least 97.5% of all expected ACK/NACKs during T3 shall be transmitted by the ProSe UE.
9. Repeat steps 1 to 8 for Test 2.
10. A UE is considered pass if it meets the test requirements defined in 7.5.3.5 for both Test 1 and Test 2.

#### 7.5.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions FFS.

### 7.5.3.5 Test requirement

Tables 7.5.3.5-1 define the primary level settings including test tolerances for interruptions due to ProSe Direct discovery.

**Table 7.5.3.5-1: Cell specific test requirement parameters for interruption due to ProSe Direct Discovery tests**

| Parameter | | Unit | Cell 1 | | |
|---|---|---|---|---|---|
| | | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | | 1 | | |
| $BW_{channel}$ | | MHz | 5 | | |
| UE RRC state | | | IDLE | CONNECTED | |
| Paging configuration | defaultPagingCycle | | rf256 | N/A | |
| | nB | | T / 32 | | |
| DRX | | | N/A | OFF | |
| PDSCH Reference measurement channel defined in A.3.1.1.1 Note1 | | | N/A | None | R.5 FDD |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.1 Note1 | | | R.11 FDD | | |
| OCNG Pattern | | | OP.16 FDD | | |
| PBCH_RA | | dB | 0 | | |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA Note 1 | | | | | |
| OCNG_RB Note 1 | | | | | |
| $N_{oc}$ Note2 | | dBm/15 kHz | -98 | | |
| $\hat{E}_s / N_{oc}$ | | dB | 16 | | |
| RSRP Note3 | | dBm/15 kHz | -82 +TT | | |
| SCH_RP Note 3 | | dBm/15 kHz | -82 | | |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall be scheduled on PCell continuously during T3.

In Test 1, at least 98.75% of all expected ACK/NACKs during T3 shall be transmitted by the ProSe UE. The missed ACK/NACKs can occur only on subframe ‘n’, if either n±1 subframe is a discovery subframe, or if n-3, or n-5 is a discovery subframe.

NOTE 1: For the test configuration in Table 7.5.3.5-1, the specific subframes where missed ACK/NACKs are allowed are when (subframe mod 320) = 159, 163, 162, 166, corresponding to allowed interruptions on subframe 159 and 162.

In Test 2, at least 97.5% of all expected ACK/NACKs during T3 shall be transmitted by the ProSe UE. The missed ACK/NACK can occur only on subframe ‘n’, if either n±5 subframe is a discovery or SLSS subframe, or if n+1, or n-9 is a discovery or SLSS subframe.

NOTE 2: For the test configuration in Table 7.5.3.5-1, the specific subframes where missed ACK/NACKs are allowed are when (subframe mod 320) = 135, 139, 145, 149, 155, 159, 166, 170, corresponding to allowed interruptions on subframes 135, 145, 155 and 166.

## 7.5.4 E-UTRAN FDD – UE ProSe Direct Communication Transmission Timing Accuracy Test

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

- Call procedure of the test case needs to be tweaked

- Test case applicability is undefined.

- Message contents are TBD.

- Connection diagrams are TBD

- PDSCH scheduling and OCNG configuration during discovery is under investigation.

### 7.5.4.1 Test purpose

To verify the timing requirements of a UE for ProSe Direct Communication transmissions when PCell downlink timing is used as a reference.

### 7.5.4.2 Test applicability

This test applies to all types of E-UTRA FDD UE FFS.

### 7.5.4.3 Minimum conformance requirements

The requirements in this subclause are applicable when the reference timing used for ProSe transmissions is the serving cell (RRC_IDLE) or PCell (RRC_CONNECTED). The sidelink transmissions takes place $(N_{\text{TA,SL}} + N_{\text{TA offset}}) \cdot T_s$ before the reception of the first detected path (in time) of the corresponding downlink frame from the reference cell, where $N_{\text{TA offset}}$ is specified in Section 8.1 of [9]. The value of $N_{\text{TA,SL}}$ differs between sidelink channels and signals, and is either $N_{\text{TA, SL}} = N_{\text{TA}}$ or $N_{\text{TA, SL}} = 0$ as specified in Section 9.9 of [9].

The normative reference for this requirement is TS 36.133 [4] clause 7.16.2.1.1 and A.7.5.4.

**Table 7.5.4-1: $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| 1.4 | 24*$T_S$ |
| ≥3 | 12*$T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 | |

### 7.5.4.4 Test description

#### 7.5.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure FFS

2. Propagation conditions are set according to Annex B clause B.0.

3. There is one E-UTRA FDD carrier and one cell specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

#### 7.5.4.4.2 Test procedure

The test consists of a single cell. The transmit timing accuracy is verified related to the downlink frame timing of Cell 1. The downlink timing of Cell 1 is changed and the changes in UE transmit timing are observed. The transmit timing is verified by the UE transmitting PSDCH used as a measurement reference facilitating the SS timing estimation.

1. Ensure the UE is in State 4 according to TS 36.508 [7] clause 4.5.4 and UE test loop Mode 4 is activated.
2. Set the parameters according to Test 1 in Tables 7.5.4.5-1, as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
3. The SS shall check that the ProSe UE SLSS transmit timing offset is within the limits specified in Table 7.5.4.5-2 with respect to the first detected path (in time) of the corresponding downlink frame of Cell 1.
4. The SS adjusts the downlink timing for Cell 1 to a delay of $+32 \times T_S$ (approximately +1 μs) compared to that in step 3.
5. After waiting for one SLSS period (40ms), the SS shall check that the UE transmit timing offset stays within the limits specified in Table 7.5.4.5-2.
6. The SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

#### 7.5.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions FFS.

### 7.5.4.5 Test requirement

Tables 7.5.4.5-1 defines the primary settings including test tolerances for UE transmit timing accuracy for E-UTRAN FDD test.

Tables 7.5.4.5-2, define the UE uplink timing include Test Tolerances.

**Table 7.5.4.5-1: Test parameters for ProSe Transmission Timing Accuracy test for E-UTRAN FDD**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 or 10 | Note 5 |
| Active cell | | Cell 1 | E-UTRA FDD Cell1 on RF channel number 1 |
| CP length of Cell 1 | | Normal | |
| drx-Configuration | | DRX_P1 | As specified in Table C.3.4.2 |
| ProSe Direct Communication configuration | | As specified in Table H.6.1.2-1: (Configuration #1) | IE values unless specified otherwise in this test. |
| networkControlledSyncTx | | ON | Configured |
| PDCCH/PCFICH/PHICH Reference measurement channel Note1 | | 5 MHz: R.11 FDD<br>10 MHz: R.6 FDD | |
| OCNG Pattern Note2 | | 5 MHz: OP.16 FDD<br>10 MHz: OP.2 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | | | |
| PSS_RA | | | |
| SSS_RA | | | |
| PCFICH_RB | | | |
| PHICH_RA | | | |
| PHICH_RB | | | |
| PDCCH_RA | | | |
| PDCCH_RB | | | |
| OCNG_RA Note3 | | | |
| OCNG_RB Note3 | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 3.30 | |
| RSRP Note4 | dBm/15 kHz | -94.7 | |
| SCH_RP Note 4 | dBm/15 kHz | -94.7 | |
| Propagation condition | | AWGN | |

Note 1: For the reference measurement channels, see clause A.3.1.
Note 2: For the OCNG pattern, see clause A.3.2.
Note 3: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 4: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: This test is according to the principle defined in section A.3.12.3.

The UE transmit timing offset shall be within the requirements in Table 7.5.4.5-2.

The reference point for the UE initial transmit timing control test requirement shall be the downlink timing minus $(N_{\text{TA_Ref}} + N_{\text{TA offset}}) \times T_s$.

**Table 7.5.4.5-2: Test requirement for $T_e$ Timing Error Limit**

| Downlink Bandwidth (MHz) | $T_{e_}$ |
|---|---|
| ≥3 | $15*T_S$ |
| Note: $T_S$ is the basic timing unit defined in TS 36.211 [9]. | |

The UE shall be capable of changing the transmission timing according to the received downlink frame. When the transmission timing error between the UE and the reference cell exceeds $\pm T_e$ the UE is required to adjust its timing to within $\pm T_e$.

# 8 UE Measurements Procedures

When the UE is in RRC_CONNECTED state on a cell, UE reports measurement information in accordance with the measurement configuration as provided by the System Simulator. To initiate a specific measurement, the System Simulator sends a 'RRC Connection Reconfiguration message' to the UE including a measurement ID and type, a command (setup, modify, release), the measurement objects, the measurement quantity, the reporting quantities and the reporting criteria (periodical/event-triggered), after that the measurement reporting process takes place. In this process when the reporting criteria are fulfilled the UE sends a 'Measurement Report message' to the System Simulator including the measurement ID and the results. The reporting criteria that trigger the UE to send a 'Measurement Report message' to the System Simulator is event-triggered as defined in TS 36.331 [5] clause 5.5.3. The measurement reporting succeeds only if the measurement report is sent within the specified measurement reporting delay period.

The reference channels in this section assume transmission of PDSCH with a maximum number of 5 HARQ transmissions unless otherwise specified. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Annex A1. The SS sends downlink MAC padding bits on the DL RMC.

Uplink is configured according to Annex A.3.

## 8.1 E-UTRAN FDD intra frequency measurements

### 8.1.1 E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells

#### 8.1.1.1 Test purpose

To verify the UE's ability to make a correct reporting of an event under fading propagation conditions in asynchronous cells within the E-UTRA FDD-FDD intra frequency cell search requirements.

#### 8.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

#### 8.1.1.3 Minimum conformance requirements

The measurement reporting delay shall be less than $T_{identify_intra}$ in RRC_CONNECTED state.

When no DRX is in use the UE shall be able to identify a new detectable FDD intra frequency cell within

$$T_{identify\ intra} = T_{basic\ identify\ E-UTRA_FDD,\ intra} \cdot \frac{T_{Measurement\ Period,\ Intra}}{T_{Intra}} \quad ms$$

Where:

$T_{basic_identify_E\text{-}UTRA_FDD,\ intra}$ is 800 ms.

$T_{Intra}$: This is the minimum time that is available for intra-frequency measurements, during the measurement period with an arbitrarily chosen timing. Time is assumed to be available for performing intra-frequency measurements whenever the receiver is guaranteed to be active on the intra-frequency carrier.

A cell shall be considered detectable when

- RSRP related side condition given in TS 36.133 [4] clauses 9.1.2.1 and 9.1.2.2 and RSRQ related side conditions given in clause 9.1.5.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.1 for a corresponding Band.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{Measurement_Period\ Intra}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra-frequency measurements is 200 ms. When no measurement gaps are activated, the UE shall be capable of performing RSRP and RSRQ measurements for 8 identified intra-frequency cells, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of 200 ms. When measurement gaps are activated, the UE shall be capable of performing measurements for at least $Y_{measurement\ intra}$ cells, where $Y_{measurement\ intra}$ is defined in the following equation. If the UE has identified more than $Y_{measurement\ intra}$ cells, the UE shall perform measurements at least 8 identified intra-frequency cells but the reporting rate of RSRP and RSRQ measurements of cells from UE physical layer to higher layers may be decreased.

$$Y_{measurement\ intra} = Floor\left\{ X_{basic\ measurement\ FDD} \cdot \frac{T_{Intra}}{T_{Measurement_Period,\ Intra}} \right\} \text{cells}$$

Where:

$X_{basic\ measurement\ FDD}$ = 8 (cells).

$T_{Measurement_Period\ Intra}$ = 200 ms. The measurement period for intra-frequency RSRP and RSRQ measurements.

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clauses 9.1.2.1 and 9.1.2.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in the clause 9.1.5.1

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clauses 9.1.2.1, 9.1.2.2, and 9.1.5.1, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ intra}$ defined in TS 36.133 [4] clause 8.1.2.2.1.2. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra}$ defined in TS 36.133 [4] section 8.1.2.2.1.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected. The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.2.1 and A.8.1.1.

## 8.1.1.4 Test description

### 8.1.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15.

2. The general test parameter settings are set up according to Table 8.1.1.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.1.1.4.3.

5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.1.4.1-1: General Test Parameters for E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in clause A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in clause A.2.1 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | One FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| A3-Offset | dB | -6 | |
| CP length | | Normal | |
| Hysteresis | dB | 0 | |
| Time To Trigger | ms | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | | OFF |
| Time offset between cells | ms | 3 | Asynchronous cells<br>3ms or 92160*Ts |
| T1 | s | 5 | |
| T2 | s | 5 | |

### 8.1.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.1.1.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.1.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 802 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),

or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.1.1.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.1.1.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.1.1.4.3-3: *MeasResults*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServiCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.1.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.1.1.5 Test requirement

Tables 8.1.1.4.1-1 and 8.1.1.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-FDD event triggered reporting under fading propagation conditions in asynchronous cells test.

**Table 8.1.1.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells**

| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | |
|---|---|---|---|---|---|
| | | **T1** | **T2** | **T1** | **T2** |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_PB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_PB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA[Note 1] | dB | | | | |
| OCNG_RB[Note 1] | dB | | | | |

| | | | | | |
|---|---|---|---|---|---|
| $\hat{E}_s / I_{ot}$ | dB | 6.10 | -0.95 | -Infinity | -0.95 |
| $N_{oc}$ Note 3 | dBm/15 KHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | 6.10 | 6.10 | -Infinity | 6.10 |
| RSRP Note 4 | dBm/15 KHz | -91.90 | -91.90 | -Infinity | -91.90 |
| SCH_RP Note 4 | dBm/15 KHz | -91.90 | -91.90 | -Infinity | -91.90 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra}$

$$T_{identify_intra} = T_{basic\ identify\ E-UTRA_FDD,\ intra} \cdot \frac{T_{Measurement\ Period,\ Intra}}{T_{Intra}}$$

$T_{basic_identify_E\text{-}UTRA_FDD,\ intra}$ = 800 ms

$T_{Measurement_Period,Intra}$ = 200 ms

$T_{Intra}$ = 200 ms

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 802 ms in this test case (note: this gives a total of 800 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.2 E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells

### 8.1.2.1 Test purpose

To verify the UE's ability to make a correct reporting of an event under fading propagation conditions in synchronous cells within the E-UTRA FDD-FDD intra frequency cell search requirements.

### 8.1.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward. Applicability requires support for FGI bit 5.

### 8.1.2.3 Minimum conformance requirements

The measurement reporting delay shall be less than $T_{identify_intra}$ in RRC_CONNECTED state.

When no DRX is in use the UE shall be able to identify a new detectable FDD intra frequency cell within

$$T_{identify\ intra} = T_{basic\ identify\ E-UTRA_FDD,\ intra} \cdot \frac{T_{Measurement\ Period,\ Intra}}{T_{Intra}} \quad ms$$

Where:

$T_{basic_identify_E\text{-}UTRA_FDD,\ intra}$ is 800 ms.

$T_{Intra}$: This is the minimum time that is available for intra-frequency measurements, during the measurement period with an arbitrarily chosen timing. Time is assumed to be available for performing intra-frequency measurements whenever the receiver is guaranteed to be active on the intra-frequency carrier.

A cell shall be considered detectable when

- RSRP related side condition given in TS 36.133 [4] clauses 9.1.2.1 and 9.1.2.2 and RSRQ related side conditions given in clause 9.1.5.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.1 for a corresponding Band.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{Measurement_Period\ Intra}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra-frequency measurements is 200 ms. When no measurement gaps are activated, the UE shall be capable of performing RSRP and RSRQ measurements for 8 identified intra-frequency cells, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of 200 ms. When measurement gaps are activated, the UE shall be capable of performing measurements for at least $Y_{measurement\ intra}$ cells, where $Y_{measurement\ intra}$ is defined in the following equation. If the UE has identified more than $Y_{measurement\ intra}$ cells, the UE shall perform measurements at least 8 identified intra-frequency cells but the reporting rate of RSRP and RSRQ measurements of cells from UE physical layer to higher layers may be decreased.

$$Y_{measurement\ intra} = Floor\left\{ X_{basic\ measurement\ FDD} \cdot \frac{T_{Intra}}{T_{Measurement_Period,\ Intra}} \right\} \text{cells}$$

Where:

$X_{basic\ measurement\ FDD}$ = 8 (cells).

$T_{Measurement_Period\ Intra}$ = 200 ms. The measurement period for intra-frequency RSRP and RSRQ measurements.

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clauses 9.1.2.1 and 9.1.2.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in the clause 9.1.5.1.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clauses 9.1.2.1, 9.1.2.2, and 9.1.5.1, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ intra}$ defined in TS 36.133 [4] clause 8.1.2.2.1.2. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra}$ defined in TS 36.133 [4] section 8.1.2.2.1.1 becomes undetectable for a period ≤ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.2.1 and A.8.1.2.

## 8.1.2.4 Test description

### 8.1.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15.

2. The general test parameter settings are set up according to Table 8.1.2.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.1.2.4.3.

5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.2.4.1-1: General Test Parameters for E-UTRAN FDD-FDD event triggered reporting under fading propagation conditions in synchronous cells**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in clause A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in clause A.2.1 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | One FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| A3-Offset | dB | -6 | |
| CP length | | Normal | |
| Hysteresis | dB | 0 | |
| Time To Trigger | ms | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | DRX_L | As specified in Table 8.1.2.5-2 |
| Time offset between cells | µs | 3 | Synchronous cells<br>3µs or 92*Ts |
| T1 | s | 5 | |
| T2 | s | 5 | |

### 8.1.2.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 is used. PDCCHs indicating new transmissions shall be sent continuously to ensure that the UE would not enter the DRX state. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.1.2.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2.T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.2.5-1. T2 starts.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 802 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10.Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.1.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.1.2.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7<br>Table H.3.6-2 |

**Table 8.1.2.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.1.2.4.3-3: *MeasResults*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCell CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.2.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

### 8.1.2.5 Test requirement

Tables 8.1.2.4.1-1, 8.1.2.5-1, and 8.1.2.5-2 define the primary level settings including test tolerances for E-UTRAN FDD-FDD event triggered reporting under fading propagation conditions in synchronous cells test.

**Table 8.1.2.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD-FDD event triggered reporting under fading propagation conditions in synchronous cells**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 6.10 | -0.95 | -Infinity | -0.95 |
| $N_{oc}$ Note 3 | dBm/15 KHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | 6.10 | 6.10 | -Infinity | 6.10 |
| RSRP Note 4 | dBm/15 KHz | -91.90 | -91.90 | -Infinity | -91.90 |
| SCH_RP Note 4 | dBm/15 KHz | -91.90 | -91.90 | -Infinity | -91.90 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.1.2.5-2: Reference DRX-Configuration to be used in E-UTRAN FDD-FDD event triggered reporting under fading propagation conditions in synchronous cells**

| Parameter | Value | Comments |
|---|---|---|
| Reference configuration | DRX_L | As defined in 4.8.2.1.5 in TS 36.508 [7.] |
| onDurationTimer | psf6 | |
| drx-InactivityTimer | psf1920 | |
| drx-RetransmissionTimer | psf16 | |
| longDRX-CycleStartOffset | sf1280, 0 | |
| shortDRX | disabled | |
| Note: For further information see section 6.3.2 in 3GPP TS 36.331 [5]. | | |

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra}$

$$T_{identify_intra} = T_{basic\ identify\ E-UTRA_FDD,\ intra} \cdot \frac{T_{Measurement\ Period,\ Intra}}{T_{Intra}}$$

$T_{basic_identify_E\text{-}UTRA_FDD,\ intra}$= 800 ms

$T_{Measurement_Period,Intra}$ = 200 ms

$T_{Intra}$ = 200 ms

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 802 ms in this test case (note: this gives a total of 800 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.3 E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX

### 8.1.3.1 Test purpose

To verify the UE's ability to make a correct reporting of an event in DRX under fading propagation conditions in synchronous cells within the E-UTRA FDD-FDD intra frequency cell search requirements.

### 8.1.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward. Applicability requires support for FGI bit 5.

### 8.1.3.3 Minimum conformance requirements

When DRX is in use the UE shall be able to identify a new detectable FDD intra frequency cell within $T_{identify_intra}$ as defined in table 8.1.2.2.1.2-1 of TS 36.133 [4] clause 8.1.2.2.1.2.

A cell shall be considered detectable when

- RSRP related side condition given in TS 36.133 [4] clauses 9.1.2.1 and 9.1.2.2 and RSRQ related side conditions given in clause 9.1.5.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.1 for a corresponding Band.

In the RRC_CONNECTED state with DRX cycles of 80 ms or greater the measurement period for intra frequency measurements is $T_{measure_intra}$ as defined in table 8.1.2.2.1.2-2 of TS 36.133 [4] clause 8.1.2.2.1.2. The UE shall be capable of performing RSRP and RSRQ measurement for 8 identified intra frequency cells, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_intra}$.

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clauses 9.1.2.1 and 9.1.2.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in the clause 9.1.5.1.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clauses 9.1.2.1, 9.1.2.2, and 9.1.5.1, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra}$ defined in TS 36.133 [4] clause 8.1.2.2.1.2. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra}$ defined in TS 36.133 [4] section 8.1.2.2.1.1 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.2.1 and A.8.1.3.

## 8.1.3.4 Test description

### 8.1.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15.

2. The general test parameter settings are set up according to Table 8.1.3.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.1.3.4.3.

5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.3.4.1-1: General Test Parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | | As specified in section A.2.1 |
| Active cell | | Cell 1 | | |
| Neighbour cell | | Cell 2 | | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | | One FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| A3-Offset | dB | -6 | | |
| CP length | | Normal | | |
| Hysteresis | dB | 0 | | |
| Time To Trigger | dB | 0 | | |
| Filter coefficient | | 0 | | L3 filtering is not used |
| DRX | | ON | | DRX related parameters are defined in Table 8.1.3.5-2 |
| Time offset between cells | | 3 μs | | Synchronous cells<br>3μs or 92*Ts |
| T1 | s | 5 | | |
| T2 | s | 5 | 30 | |

## 8.1.3.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

In Test 1 when DRX = 40 ms is used, UE needs to be provided at least once every 500 ms with new Timing Advance Command MAC control element to restart the Timer Alignment Timer to keep the UE uplink time alignment. Furthermore, the UE is allocated with PUSCH resource at every DRX cycle. In Test 2 when DRX = 1280 ms is used, the uplink time alignment is not maintained and the UE needs to use RACH to obtain uplink allocation for measurement reporting.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.1.3.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.3.5-1 and 8.1.3.5-2.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 842 ms for Test 1 or less than 26882 ms for Test 2 then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

11. Repeat step 1-10 for each sub-test in Table 8.1.3.4.1-1 as appropriate.

### 8.1.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.1.3.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra frequency event triggered reporting in DRX under fading propagation conditions test 1 and 2 requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-2<br>Table H.3.1-7<br>Table H.3.7-1<br>Table H.3.7-2<br>Table H.3.7-3 |

**Table 8.1.3.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting with in DRX under fading propagation conditions test 1 and 2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.1.3.4.3-3: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting with in DRX under fading propagation conditions test 1 and 2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventide CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.1.3.4.3-4: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting in DRX under fading propagation conditions test 1 and 2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.1.3.4.3-5: *MeasResults*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting in DRX under fading propagation conditions test 1 and 2 requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.3.4.3-6: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting in DRX under fading propagation conditions test 1 and 2 requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| physCellId | PhysCellId | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.1.3.5 Test requirement

Tables 8.1.3.4.1-1, 8.1.3.5-1, 8.1.3.5-2 and 8.1.3.5-3 define the primary level settings including test tolerances for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used test.

**Table 8.1.3.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting in DRX under fading propagation conditions**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 6.10 | -0.95 | -Infinity | -0.95 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | 6.10 | 6.10 | -Infinity | 6.10 |
| RSRP Note 3 | dBm/15 KHz | -91.90 | -91.90 | -Infinity | -91.90 |
| SCH_RP Note 3 | dBm/15 KHz | -91.90 | -91.90 | -Infinity | -91.90 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.1.3.5-2: DRX Configuration to be used for E-UTRAN FDD-FDD intra-frequency event triggered reporting in DRX under fading propagation conditions**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf40 | sf1280 | |
| shortDRX | disable | disable | |

**Table 8.1.3.5-3: TimeAlignmentTimer and sr-ConfigIndex-Configuration to be used for E-UTRAN FDD-FDD intra-frequency event triggered reporting in DRX under fading propagation conditions**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |

| | | | |
|---|---|---|---|
| TimeAlignmentTimer | sf500 | sf500 | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| sr-ConfigIndex | 0 | 0 | For further information see section 6.3.2 in 3GPP TS 36.331 [5] and section10.1 in 3GPP TS 36.213 [8]. |

In Test 1 when DRX cycle length = 40 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2 on PUSCH.

In Test 2 when DRX cycle length = 1280 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE starts to send preambles on the PRACH for Scheduling Request (SR) to obtain allocation to send the measurement report to Cell 2 on PUSCH.

For both tests:

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event A3 measurement report.

The overall delay measured when DRX cycle length is 40 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra}$

$T_{identify_intra}$ = 800 ms. When DRX cycle length is 40 ms then the $T_{identify_intra}$ is 0.8 s.

TTI insertion uncertainty = 2 ms

DRX cycle length = 40 ms

The overall delay measured when DRX cycle length is 40 ms shall be less than a total of 842 ms.

The overall delay measured when DRX cycle length is 1280 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra}$

$T_{identify_intra}$ = 25600 ms. When DRX cycle length is 1280 ms then the $T_{identify_intra}$ is 20 x 1280 ms.

TTI insertion uncertainty = 2 ms

DRX cycle length = 1280 ms

The overall delay measured when DRX cycle length is 1280 ms shall be less than a total of 26882 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.4 Void

## 8.1.5 E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps

### 8.1.5.1 Test purpose

To verify the requirement for identification of a new CGI of E-UTRA cell with autonomous gaps.

### 8.1.5.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward with support of intra-frequency SI acquisition for HO.

### 8.1.5.3 Minimum conformance requirements

No explicit neighbour list is provided to the UE for identifying a new CGI of E-UTRA cell. The UE shall identify and report the CGI when requested by the network for the purpose 'reportCGI'. The UE may make autonomous gaps in downlink reception and uplink transmission for receiving MIB and SIB1 message according to section 5.5.3.1 of TS 36.331 [5]. Note that a UE is not required to use autonomous gap if si-RequestForHO is set to false. If autonomous gaps are used for measurement with the purpose of 'reportCGI', the UE shall be able to identify a new CGI of E-UTRA cell within:

$$T_{identify_CGI,\,intra} = T_{basic_identify_CGI,\,intra} \quad ms$$

Where

$T_{basic_identify_CGI,\,intra}$ = 150 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new CGI of an E-UTRA cell is defined, provided that the E-UTRA cell has been already identified by the UE.

A cell shall be considered identifiable following conditions are fulfilled:

- RSRP related side conditions given in TS 36.133 [4] Section 9.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.2 for a corresponding Band

The requirement for identifying a new CGI of an E-UTRA cell within $T_{basic_identify_CGI,.intra}$ is applicable when no DRX is used as well as when all the DRX cycles specified in 3GPP TS 36.331 [5] are used.

Given that continuous DL data allocation and no DRX is used, and no measurement gaps are configured, the UE shall have more than 60 ACK/NACKs transmitted during the identification of a new CGI of E-UTRA cell.

The ECGI reporting delay occurs due to the delay uncertainty when inserting the ECGI measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. In case DRX is used, the ECGI reporting may be delayed until the next DRX cycle.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.2.3 and A.8.1.5.

### 8.1.5.4 Test description

#### 8.1.5.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. The general test parameter settings are set up according to Table 8.1.5.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.1.5.4.3.

5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.5.4.1-1: General test parameters for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.3 FDD | As specified in clause A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in clause A.2.1 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | Only one FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| CP length | | Normal | |
| A3-Offset | dB | -3 | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| si-RequestForHO | | TRUE | As specified in section 5.5.3.1 in TS 36.331 [5]. |
| Time offset between cells | ms | 3 | Asynchronous cells<br>3ms or 92160*Ts |
| T1 | s | 5 | |
| T2 | s | ≤10 | |
| T3 | s | 5 | |

### 8.1.5.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. PDCCHs indicating new transmissions shall be sent continuously to ensure that the UE would have ACK/NACK sending during identifying a new CGI of E-UTRAN cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE does not have any timing information of cell 2. Starting T2, cell 2 becomes detectable and the UE is expected to detect and send a measurement report.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.1.5.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration with event A3 configured message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.5.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3.

7. SS shall transmit an RRCConnectionReconfiguration message during period T2, The RRC message shall create a measurement report configuration with purpose *reportCGI* and *si-RequestForHO* set to TRUE.

8. The SS shall start T3 timer when the last TTI containing the RRC message implying SI reading is sent to UE.

9. The UE shall transmit RRCConnectionReconfigurationComplete message.

10. UE shall transmit a MeasurementReport message containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3. The UE shall be scheduled continuously throughout the test, and from the start of T3 until 170ms at least 80 ACK/NACK shall be detected as being transmitted by the UE. If the overall delays measured from the beginning of time period T3 is less than 170 ms, and the UE have more than 80 ACK/NACKs transmitted from the start of T3 until 170 ms, then the number of successful tests is increased by one. If the UE fails to report the cell global identifier within the overall delays measured requirement, or the UE have less than 80 ACK/NACKs transmitted from the start of T3 until 170 ms, then the number of failure tests is increased by one.

11. After the SS receive the MeasurementReport message in step 10) or when T3 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

13. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 2-13 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.1.5.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.1.5.4.3-1: Common Exception messages for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.1.5.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -6 (-3 dB) | -3 is actual value in dB (-6 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.5.4.3-3: *MeasResults*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step6)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.5.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step6)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.1.5.4.3-5: *ReportConfigEUTRA-PERIODICAL*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportCGI | | |
| } | | | |
| } | | | |
| reportAmount | 1 | | |
| si-RequestForHO-r9 | setup | | |
| } | | | |

**Table 8.1.5.4.3-6: *MeasResultListEUTRA*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| cgi-Info SEQUENCE{ | | | |
| cellGlobalId SEQUENCE{ | | | |
| plmn-Identity | plmn-Identity | | |
| cellIdentity | cellIdentity | | |
| } | | | |
| trackingAreaCode | | | |
| plmn-IdentityList | | | |
| } | | | |
| } | | | |

**Table 8.1.5.4.3-7: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | |
| cellForWhichToReportCGI | Physical Cell ID of Cell 2 | | |
| } | | | |

**Table 8.1.5.4.3-8: *SystemInformationBlockType2*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-1 SystemInformationBlockType2 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfig | Not present | | Cell 1 |
| } | | | |

**Table 8.1.5.4.3-9: *SystemInformationBlockType3*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-2 SystemInformationBlockType3 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType3 ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1 |
| | '01'B (No MBSFN subframes are present in all neighbour cells) | | Cell 2 |
| } | | | |

**Table 8.1.5.4.3-10: *MeasConfig-DEFAULT*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f1 | | |
| reportConfigId | IdReportConfig-P | | |
| } | | | |
| measGapConfig | Not present | | |
| } | | | |

**Table 8.1.5.4.3-11: *MeasResults*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.5.4.3-12: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | |
| } | | | |

## 8.1.5.5 Test requirement

Tables 8.1.5.4.1-1 and 8.1.5.5-1 define the primary level settings including test tolerances for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test.

**Table 8.1.5.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.10 (OP.10 FDD) and in D.1.2 (OP.2 FDD) | | OP.10 FDD | OP.10 FDD | OP.10 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 8 | -3.3 | -3.3 | -Infinity | 2.36 | 2.36 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 8 | 8 | 8 | -Infinity | 11 | 11 |
| RSRP Note 3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| SCH_RP Note3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall transmit a measurement report containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Test requirement = RRC Procedure delay + $T_{identify_CGI,\ intra}$ + TTI insertion uncertainty

RRC procedure delay=15ms

$$T_{identify_CGI,\ intra} = T_{basic_identify_CGI,\ intra} \quad ms$$

$T_{basic_identify_CGI,\ intra}$ = 150 ms

TTI insertion uncertainty = 2 ms

The overall delays measured is 167 ms, allow 170 ms. The UE shall be scheduled continuously throughout the test, and from the start of T3 until 170 ms at least 80 ACK/NACK shall be detected as being transmitted by the UE.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

NOTE: The overall 80 ACK/NACK number is caused by two parts. Firstly, at least 60 ACK/NACK shall be sent during identifying the cell global identifier of cell 2 according to the requirement in TS 36.133 [4] Section 8.1.2.2.3.1. Secondly, given that continuous DL data allocation, additional 20 ACK/NACK shall be sent from the start of T3 until 170 ms excludes 150 ms for identifying the cell global identifier of cell 2.

## 8.1.6 E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX

### 8.1.6.1 Test purpose

To verify the requirement for identification of a new CGI of E-UTRA cell with autonomous gaps in DRX.

### 8.1.6.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward with support of intra-frequency SI acquisition for HO. Applicability requires support for FGI bit 5

### 8.1.6.3 Minimum conformance requirements

No explicit neighbour list is provided to the UE for identifying a new CGI of E-UTRA cell. The UE shall identify and report the CGI when requested by the network for the purpose 'reportCGI'. The UE may make autonomous gaps in downlink reception and uplink transmission for receiving MIB and SIB1 message according to section 5.5.3.1 of TS 36.331 [5]. Note that a UE is not required to use autonomous gap if si-RequestForHO is set to false. If autonomous gaps are used for measurement with the purpose of 'reportCGI', the UE shall be able to identify a new CGI of E-UTRA cell within:

$$T_{identify_CGI,\,intra} = T_{basic_identify_CGI,\,intra} \quad ms$$

Where

$T_{basic_identify_CGI,\,intra}$ = 150 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new CGI of an E-UTRA cell is defined, provided that the E-UTRA cell has been already identified by the UE.

A cell shall be considered identifiable following conditions are fulfilled:

- RSRP related side conditions given in TS 36.133 [4] Section 9.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.2 for a corresponding Band

The requirement for identifying a new CGI of an E-UTRA cell within $T_{basic_identify_CGI,.intra}$ is applicable when no DRX is used as well as when all the DRX cycles specified in 3GPP TS 36.331 [5] are used.

The ECGI reporting delay occurs due to the delay uncertainty when inserting the ECGI measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. In case DRX is used, the ECGI reporting may be delayed until the next DRX cycle.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.2.3 and A.8.1.6.

### 8.1.6.4 Test description

#### 8.1.6.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. The general test parameter settings are set up according to Table 8.1.6.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.1.6.4.3.

5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.6.4.1-1: General test parameters for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps when DRX is used**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in clause A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in clause A.2.1 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | Only one FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| CP length | | Normal | |
| A3-Offset | dB | -3 | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | ON | DRX related parameters are defined in Table 8.1.6.5-2 |
| si-RequestForHO | | TRUE | As specified in section 5.5.3.1 in TS 36.331 [5]. |
| Time offset between cells | ms | 3 | Asynchronous cells |
| T1 | s | 5 | |
| T2 | s | ≤30 | UE shall report cell within 25.6s (20 DRX cycles) |
| T3 | s | 5 | |

### 8.1.6.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE does not have any timing information of cell 2. Starting T2, cell 2 becomes detectable and the UE is expected to detect and send a measurement report.

In the test UE needs to be provided at least once every 1280ms with new Timing Advance Command MAC control element to restart the Time alignment timer to keep UE uplink time alignment. Furthermore UE sends scheduling request (SR) to obtain allocation to send the measurement report on PUSCH.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.1.6.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration with event A3 configured message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.6.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3.

7. SS shall transmit an RRCConnectionReconfiguration message during period T2, The RRC message shall create a measurement report configuration with purpose *reportCGI* and *si-RequestForHO* set to TRUE.

8. The SS shall start T3 timer when the last TTI containing the RRC message implying SI reading is sent to UE.

9. The UE shall transmit RRCConnectionReconfigurationComplete message.

10. UE shall transmit a measurement report containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3. If the overall delays measured from the beginning of time period T3 is less than 170 ms then the number of successful tests is increased by one. If the UE fails to report the cell global identifier within the overall delays measured requirement then the number of failure tests is increased by one.

11. After the SS receive the MeasurementReport message in step 10) or when T3 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

13. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 2-13 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.1.6.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.1.6.4.3-1: Common Exception messages for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.1.6.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.6.4.3-3: *MAC-MainConfig-RBC*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf1280 typical value in real network for real-time services. | |
| sf1280 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| } | | | |

**Table 8.1.6.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.6, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| … | | | |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | |
| } | | | |

**Table 8.1.6.4.3-5: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.1.6.4.3-6: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -6 (-3 dB) | -3 is actual value in dB (-6 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.6.4.3-7: *MeasResults*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.6.4.3-8: *MeasResultListEUTRA*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step6)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.1.6.4.3-9: *ReportConfigEUTRA-PERIODICAL*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportCGI | | |
| } | | | |
| } | | | |
| reportAmount | 1 | | |
| si-RequestForHO-r9 | setup | | |
| } | | | |

**Table 8.1.6.4.3-10: *MeasResultListEUTRA*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| cgi-Info SEQUENCE{ | | | |
| cellGlobalId SEQUENCE{ | | | |
| plmn-Identity | plmn-Identity | | |
| cellIdentity | cellIdentity | | |
| } | | | |
| trackingAreaCode | | | |
| plmn-IdentityList | | | |
| } | | | |
| } | | | |

**Table 8.1.6.4.3-11: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellForWhichToReportCGI | Physical Cell ID of Cell 2 | | |
| } | | | |

**Table 8.1.6.4.3-12: *MeasConfig-DEFAULT*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f1 | | |
| reportConfigId | IdReportConfig-P | | |
| } | | | |
| measGapConfig | Not present | | |
| } | | | |

**Table 8.1.6.4.3-13: *MeasResults*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

## 8.1.6.5 Test requirement

Tables 8.1.6.4.1-1, 8.1.6.5-1, 8.1.6.5-2 and 8.1.6.5-3 define the primary level settings including test tolerances for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps when DRX is used test.

**Table 8.1.6.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.1 FDD | OP.1 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 8 | -3.3 | -3.3 | -Infinity | 2.36 | 2.36 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 8 | 8 | 8 | -Infinity | 11 | 11 |
| RSRP Note 3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| SCH_RP Note3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.1.6.5-2: DRX configuration for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf1280 | |
| shortDRX | disable | |

**Table 8.1.6.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | Infinity | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| sr-ConfigIndex | 0 | For further information see section 6.3.2 in 3GPP TS 36.331 [5] and section10.1 in 3GPP TS 36.213 [8] |

The UE shall transmit a measurement report containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Test requirement = RRC Procedure delay + $T_{identify_CGI,\,intra}$ + TTI insertion uncertainty

RRC procedure delay=15ms

$$T_{identify_CGI,\,intra} = T_{basic_identify_CGI,\,intra} \quad ms$$

$T_{basic_identify_CGI,\,intra}$ = 150 ms

TTI insertion uncertainty = 2 ms

The overall delay measured is 167 ms, allow 170 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.7 E-UTRAN FDD-FDD Intra-Frequency Event-Triggered Reporting under Time Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC)

### 8.1.7.1 Test purpose

To verify that the UE correctly detects and reports Event A3 (neighbour becomes offset better than PCell) defined in [5] under a time domain measurement resource restriction and non-MBSFN ABS configured in the aggressor cell.

### 8.1.7.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward. Applicability requires support for FGI bit 115.

### 8.1.7.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD intra-frequency cell within

$$T_{identify_\,intra_eICIC} = T_{basic_identify_E\text{-}UTRA_FDD_eICIC,\,intra} \cdot \frac{T_{Measurement_Period_eICIC,\,Intra}}{T_{Intra}} \quad ms$$

where

$T_{basic_identify_E\text{-}UTRA_FDD_eICIC,\,intra}$ is 1000 ms.

$T_{Intra}$ is the minimum time available for intra-frequency measurements, during the measurement period with an arbitrarily chosen timing. The time is assumed to be available for performing intra-frequency measurements whenever the receiver is guaranteed to be active on the intra-frequency carrier.

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] clauses 9.1.2.3 and 9.1.2.4 and RSRQ related side conditions given in TS 36.133 [4] clause 9.1.5.2 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.8 for a corresponding Band.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{Measurement_Period_eICIC,\ Intra}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra-frequency measurements is 200 ms. When no measurement gaps are activated, the UE shall be capable of performing RSRP and RSRQ measurements for 8 identified intra-frequency cells, including also the cells which are not measured in the subframes indicated by the time-domain measurement resource restriction pattern, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of 200 ms. When measurement gaps are activated the UE shall be capable of performing measurements for at least $Y_{measurement_intra_eICIC}$ cells , where $Y_{measurement_intra_eICIC}$ is defined in the following equation. If the UE has identified more than $Y_{measurement_intra_eICIC}$ cells, the UE shall perform measurements of at least 8 identified intra-frequency cells but the reporting rate of RSRP and RSRQ measurements of cells from UE physical layer to higher layers may be decreased.

$$Y_{measurement_intra_eICIC} = Floor\left\{ X_{basic_measurement_FDD_eICIC} \cdot \frac{T_{Intra}}{T_{Measurement_Period_eICIC,\ Intra}} \right\} \text{ cells}$$

where

$X_{basic_measurement_FDD_eICIC}$ = 8 (cells)

$T_{Measurement_Period_eICIC,\ Intra}$ = 200 ms is the measurement period for intra-frequency RSRP and RSRQ measurements.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra_eICIC}$ defined above. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_eICIC}$ defined above becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_eICIC,\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.8.1 and A.8.1.7.

## 8.1.7.4 Test description

### 8.1.7.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.15.

2. The general test parameter settings are set up according to Table 8.1.7.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.1.7.4.3.

5. At this stage only Cell 1 is switched on. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1.

**Table 8.1.7.4.1-1: General test parameters for E-UTRAN FDD-FDD intra-frequency event-triggered reporting under time domain measurement resource restriction with Non-MBSFN ABS (eICIC)**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| PCell | | Cell 1 | Also the aggressor cell. Active in T1 and T2 |
| Neighbour cell | | Cell 2 | Cell to be identified. Active only in T2. |
| PCell ABS configuration | | Non-MBSFN ABS | As defined in Table C.3.1.1.1-1 |
| E-UTRA RF Channel Number | | 1 | One FDD carrier frequency is used |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | For all cells in the test |
| A3-Offset | dB | -11 | |
| Event A3 measurement quantity | | RSRP | |
| CP length | | Normal | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | | OFF |
| Time offset between cells | | 3 µs | Synchronous cells |
| T1 | s | 5 | |
| T2 | s | 5 | |
| Physical cell ID PCI | | ($PCI_{cell1}$ - $PCI_{cell2}$ )mod6 !=0 | Cell PCIs are selected so that the condition is met |
| ABS pattern | | '10000000100000001000000010000000010000000' | FDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 1 during T1.<br>The first/leftmost bit corresponds to the subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (40) divided by 10. No MBSFN subframes are configured in the ABS subframes. |
| Time domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | '10000000100000001000000010000000010000000' | Time domain measurement resource restriction pattern for neighbour cell measurement signalled to the UE in measSubframePattern-Neigh IE in measSubframePatternConfig-Neigh, as defined in TS 36.331, clause 6.3.5.<br>Configured during T1 for Cell 2 measurements. |
| Time domain measurement resource restriction pattern for PCell measurements | | '01000000010000000100000001000000001000000' | Configured during T1 for Cell 1 measurements |

## 8.1.7.4.2 Test procedure

There are two synchronous E-UTRA cells in the test on the same RF channel. Cell 1 is the Pcell and also the aggressor cell, Cell 2 is the neighbour victim cell to be identified. Non-MBSFN ABS pattern is configured for aggressor cell (Cell 1). The UE is configured with a time domain measurement resource restriction pattern for performing E-UTRAN FDD intra-frequency measurements on neighbour cells (Cell 2). The UE is also configured with a time domain measurement resource restriction pattern for PCell (Cell 1) measurements. It is indicated to the UE that event-triggered reporting with Event A3 is used.

The test consists of two successive time periods with time duration of T1 and T2, respectively. During time duration T1, the UE shall not have any timing information on Cell 2. The information for both measurement patterns shall be provided to the UE during T1.

1. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters for both Cell 1 and Cell 2 according to T1 in Table 8.1.7.5-1. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.7.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 1002 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A in Cell 1 according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.1.7.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.1.7.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra-frequency event-triggered reporting under time domain measurement resource restriction with Non-MBSFN ABS (eICIC)**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.7-1<br>Table H.2.7-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.1.7.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD intra-frequency event-triggered reporting under time domain measurement resource restriction with Non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -22 (-11 dB) | -11 is actual value in dB (-22 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.1.7.4.3-3: *MeasResults*: Additional E-UTRAN FDD-FDD intra-frequency event-triggered reporting under time domain measurement resource restriction with Non-MBSFN ABS (eICIC)**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCell CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.7.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD intra-frequency event-triggered reporting under time domain measurement resource restriction with Non-MBSFN ABS (eICIC)**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.1.7.4.3-5: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD-FDD intra-frequency event-triggered reporting under time domain measurement resource restriction with Non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| neighCellConfig | '01'B (No MBSFN subframes are present in all neighbour cells) | | FDD with E-UTRA FDD neighbour cell |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternFDD-r10 | '10000000100000001000 00001000000010000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 2 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.7.4.3-6: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN FDD-FDD intra-frequency event-triggered reporting under time domain measurement resource restriction with Non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '01000000010000000100 00001000000001000000' | BIT STRING (SIZE (40)) | Cell1 |
| } | | | |
| } | | | |
| } | | | |

## 8.1.7.5 Test requirements

**Table 8.1.7.5-1: Cell specific test requirement parameters for E-UTRAN FDD-FDD Intra-Frequency Event-Triggered Reporting under Time Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC)**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.1.5 (OP.5 FDD) and in D.1.6 (OP.6 FDD) | | OP.5 FDD | | OP.6 FDD | |
| PBCH_RA | dB | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1. | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | |
| $(\hat{E}_s / N_{oc})_{meas}$ Note 5 | dB | 1 | 1 | -Infinity | -3.2 |
| $(\hat{E}_s / N_{oc})_{ABS}$ | dB | 1 | 1 | N/A | N/A |
| RSRP Note 4,5 | dBm/15 kHz | -97 | -97 | -Infinity | -101.2 |
| SCH_RP Note 4 | dBm/15 kHz | -97 | -97 | -Infinity | -101.2 |
| CRS $\hat{E}_s / I_{ot}$ | dB | 1 | -0.7 | -Infinity | -3.2 |
| SCH $\hat{E}_s / I_{ot}$ | dB | 1 | -0.7 | -Infinity | -6.74 |
| Propagation Condition | | ETU30 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.

Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 5: RSRP is estimated for Cell 2 during the restricted measurement subframes for neighbour cells. RSRP is estimated for Cell 1 during the PCell restricted subframes.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra_eICIC}$

$$T_{identify_intra_eICIC} = T_{basic_identify_E-UTRA_FDD_eICIC,intra} \cdot \frac{T_{Measurement_Period_eICIC,Intra}}{T_{Intra}} \quad ms$$

$T_{basic_identify_E-UTRA_FDD_eICIC,intra}$ = 1000 ms

$T_{Measurement_Period_eICIC,Intra}$ = 200 ms

$T_{Intra}$ = 200 ms

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 1002 ms in this test case (note: this gives a total of 1000 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.8 E-UTRAN FDD-FDD Intra-Frequency Event-Triggered Reporting under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC)

### 8.1.8.1 Test purpose

To verify that the UE correctly detects and reports Event A3 (neighbour becomes offset better than PCell) defined in [5] under a time domain measurement resource restriction pattern and CRS assistance information, and when non-MBSFN ABS configured in the interfering cells.

### 8.1.8.2 Test applicability

This test applies to all types of E-UTRA FDD and CRS interference handling UE release 11 and forward. Applicability requires support of FGI bit 115.

### 8.1.8.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD intra-frequency cell within

$$T_{identify_intra_FeICIC} = T_{basic_identify_E-UTRA_FDD_FeICIC,intra} \cdot \frac{T_{Measurement_Period_FeICIC,Intra}}{T_{Intra}} \quad ms$$

where

$T_{basic_identify_E-UTRA_FDD_FeICIC,intra}$ is 1000 ms.

$T_{Intra}$ is the minimum time available for intra-frequency measurements, during the measurement period with an arbitrarily chosen timing. The time is assumed to be available for performing intra-frequency measurements whenever the receiver is guaranteed to be active on the intra-frequency carrier.

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] clauses 9.1.2.5 and 9.1.2.6 and RSRQ related side conditions given in TS 36.133 [4] clause 9.1.5.3 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.8 for a corresponding Band.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{Measurement_Period_eICIC,Intra}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra-frequency measurements is 200 ms. When no measurement gaps are activated, the UE shall be capable of performing RSRP and RSRQ measurements for 8 identified intra-frequency cells, including also the cells which are not measured in the subframes indicated by the time-domain measurement resource restriction pattern and the cells indicated in the CRS assistance information, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of 200 ms. When measurement gaps are activated the UE shall be capable of performing measurements for at least $Y_{measurement_intra_FeICIC}$ cells , where $Y_{measurement_intra_FeICIC}$ is defined in the following equation. If the UE has identified more than $Y_{measurement_intra_FeICIC}$ cells, the UE shall perform measurements of at least 8 identified intra-frequency cells but the reporting rate of RSRP and RSRQ measurements of cells from UE physical layer to higher layers may be decreased.

$$Y_{measurement_intra_FeICIC} = Floor\left\{ X_{basic_measurement_FDD_FeICIC} \cdot \frac{T_{Intra}}{T_{Measurement_Period_FeICIC,\ Intra}} \right\} \text{ cells}$$

where

$X_{basic_measurement_FDD_FeICIC}$ = 8 (cells).

$T_{Measurement_Period_FeICIC,\ Intra}$ = 200 ms is the measurement period for intra-frequency RSRP and RSRQ measurements.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra_eICIC}$ defined above. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_FeICIC}$ defined above becomes undetectable for a period ≤ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_FeICIC,\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.8.3 and A.8.1.8.

## 8.1.8.4 Test description

### 8.1.8.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.54.

2. The general test parameter settings are set up according to Table 8.1.8.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.1.8.4.3.

5. In this test, there are three E-UTRA cells, Cell 1, Cell 2 and Cell 3, all on same frequencies. At this stage only Cell 1 is switched on. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1.

**Table 8.1.8.4.1-1: General test parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 FDD | As specified in clause A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in clause A.2.1 |
| PCell | | | Cell 1 | Also a first interfering cell to Cell 3. Active in T1 and T2. |
| Neighbour cells | | | Cell 2 and Cell 3 | Cell 2 is a second interfering cell; Cell 2 is active in T1 and T2.<br>Cell 3 is the cell to be identified; Cell 3 is active only in T2. |
| ABS transmission configuration | | | Non-MBSFN ABS | As defined in Table C.3.1.1.1-1 |
| E-UTRA RF Channel Number | | | 1 | One FDD carrier frequency is used |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | For all cells in the test |
| A3-Offset | | dB | -14 | |
| Event A3 measurement quantity | | | RSRP | |
| CP length | | | Normal | |
| Hysteresis | | dB | 0 | |
| Time To Trigger | | s | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | | OFF |
| Time offset between cells | | µs | Cell 2 offset with respect to Cell 1: 0<br>Cell 3 offset with respect to Cell 1: -2.5 | Three synchronous cells |
| T1 | | S | 5 | |
| T2 | | S | 5 | |
| Physical cell IDs | | | ($PCI_{cell1}$ - $PCI_{cell3}$ )mod6 = 0<br>($PCI_{cell2}$ - $PCI_{cell3}$ )mod6 != 0<br><br>$PCI_{cell1}$ not equal to $PCI_{cell3}$ | Cell PCIs are selected so that all conditions are met |
| ABS pattern | | | '10000000100000001000000010000000010000000' | FDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54.<br>The first/leftmost bit corresponds to the PCell subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (40) divided by 10. No MBSFN subframes are configured in the ABS subframes.<br>Configured in Cell 1 and Cell 2 during T1. |
| Time domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | | '10000000100000001000000010000000010000000' | Time domain measurement resource restriction pattern for neighbour cell measurement signalled to the UE in measSubframePatternNeigh IE in measSubframePatternConfigNeigh, as defined in TS 36.331, clause 6.3.5.<br>Configured during T1 for Cell 3 measurements. The cell list in measSubframeCellList IE shall contain Cell 3 but not Cell 2. |
| Time domain measurement resource restriction pattern for PCell measurements | | | '01000000010000000100000001000000001000000' | Configured during T1 for Cell 1 measurements |
| CRS assistance information | physCellId | | see PCI conditions above | The CRS assistance information is provided for Cell 2 only in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation *one Frame*='000000'. |
| | antennaPortsCount | | 1 | |
| | mbsfn-SubframeConfigList | | *oneFrame* = '000000' | |

### 8.1.8.4.2 Test procedure

The test parameters are given in Tables 8.1.8.4.1-1 and 8.1.8.51-1. It is indicated to the UE in the measurement control information that event-triggered reporting with Event A3 is used. In the test, there are three synchronous cells, Cell 1, Cell 2, and Cell 3, on the same RF channel. Cell 1 is the PCell. Cell 3 is the cell to be identified. A non-MBSFN ABS pattern is configured in each of the Cell 1 and Cell 2 during the entire test. The test consists of two successive time periods with time duration of T1 and T2, respectively. During time duration T1, the UE shall not have any timing information on Cell 3.

The UE is configured by higher layers with a time domain measurement resource restriction pattern for performing E-UTRAN FDD intra-frequency measurements on neighbour cells, namely Cell 3 measurements. The UE is also provided via higher layers with the CRS assistance information for Cell 2. The information for both measurement patterns and the CRS assistance information shall be provided via RRC to the UE during T1.

1. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3 with exceptions listed in 7.2A.6.

2. Set the parameters for Cell 1 and Cell 2 according to T1 in Table 8.1.8.5-1. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 for Cell 1, Cell 2, and Cell 3 as specified in Table 8.1.8.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 1002 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. Set Cell 3 physical cell identity = ((current cell 3 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. Formulas are applied until PCI-s values are found, which fulfill the PCI conditions described in Table 8.1.8.4.1-1.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A in Cell 1 according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.1.8.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 5.2A.5.1 with the following exceptions:

**Table 8.1.8.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra-frequency event triggered reporting under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.7-1<br>Table H.2.7-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7<br>Table H.3.5-5<br>Table H.5.1-1<br>Table H.5.1-2<br>Table H.5.1-3 |

**Table 8.1.8.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -28 (-14 dB) | -14 is actual value in dB (-28 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| maxReportCells | 2 | | |
| } | | | |

**Table 8.1.8.4.3-3: *MeasResults*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCell CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.8.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of Cell 2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| physCellId | PhysCellId of Cell 3 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.1.8.4.3-5: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternFDD-r10 | '10000000100000001000 00001000000010000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 3 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.8.4.3-6: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '01000000010000000100000001000000001000000' | BIT STRING (SIZE (40)) | Cell1 |
| } | | | |
| } | | | |
| } | | | |

## 8.1.8.5 Test requirements

Tables 8.1.8.4.1-1 and 8.1.8.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-FDD intra-frequency event triggered reporting under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS test.

**Table 8.1.8.5-1: Cell specific test parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.1.5 (OP.5 FDD) and in D.1.6 (OP.6 FDD) | | OP.5 FDD | | OP.6 FDD | | N/A | OP.6 FDD |
| PBCH_RA | dB | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1. | | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1 | | N/A | 0 |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | | | |
| $(\hat{E}_s / N_{oc})$ | dB | 4 | 4 | 2 | 2 | -Infinity | -3.20 |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 | -96 | -96 | -Infinity | -101.2 |
| SCH_RP Note 4 | dBm/15 kHz | -94 | -94 | -96 | -96 | -Infinity | -101.2 |
| CRS $\hat{E}_s / I_{ot}$ Note 5 | dB | 4 | 2.30 | 2 | 0.30 | -Infinity | -8.66 |
| SCH $\hat{E}_s / I_{ot}$ | dB | -0.12 | -0.86 | -3.45 | -4.01 | -Infinity | -10.27 |
| Propagation Condition | | ETU30 | | ETU30 | | ETU30 | |

NOTE 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
NOTE 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.
NOTE 4: RSRP, SCH_RP, and $\hat{E}_s / I_{ot}$ levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 5: Applies during the restricted measurement subframes configured for neighbour cell (Cell 3) measurements.

The UE shall send one Event A3 triggered measurement report for Cell 3, with a measurement reporting delay less than 1000 ms from the beginning of time period T2.

The UE shall not send event-triggered measurement reports as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90% for the tested Event A3.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra_FeICIC}$

$$T_{identify_intra_FeICIC} = T_{basic_identify_E\text{-}UTRA_FDD_FeICIC,\ intra} \cdot \frac{T_{Measurement_Period_FeICIC,\ Intra}}{T_{Intra}} \quad ms$$

where

$T_{basic_identify_E\text{-}UTRA_FDD_FeICIC,\ intra}$ is 1000 ms.

$T_{Measurement_Period_FeICIC,\ Intra}$ = 200 ms

$T_{Intra}$ = 200 ms

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 1002 ms in this test case (note: this gives a total of 1000 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.9 E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells for 5MHz Bandwidth

### 8.1.9.1 Test purpose

Same test purpose as defined in clause 8.1.1.1.

### 8.1.9.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that only support E-UTRA Band 31.

### 8.1.9.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.1.1.3 with the following exceptions:

- Instead of A.8.1.1 → use A.8.1.9.

### 8.1.9.4 Test description

#### 8.1.9.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15.
2. The general test parameter settings are set up according to Table 8.1.9.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.1.9.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.9.4.1-1: General Test Parameters for E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells for 5MHz bandwidth**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.5 FDD | As specified in clause A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.11 FDD | As specified in clause A.2.1 |
| Note 1: See Table 8.1.1.4.1-1 for the other parameters.<br>Note 2: This test is according to the principle defined in TS 36.133 [4] section A.3.7.2. | | | |

### 8.1.9.4.2 Test procedure

Same test procedure as defined in clause 8.1.1.4.2 with the following exceptions:

- Instead of Table 8.1.1.5-1 → use Table 8.1.9.5-1.

### 8.1.9.4.3 Message contents

Same message contents as defined in clause 8.1.1.4.3.

## 8.1.9.5 Test requirement

Same test requirement as defined in clause 8.1.1.5 with the following exceptions:

Tables 8.1.9.4.1-1 and 8.1.9.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-FDD event triggered reporting under fading propagation conditions in asynchronous cells for 5MHz bandwidth test.

**Table 8.1.9.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells for 5MHz bandwidth**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| BWchannel | MHz | 5 | | 5 | |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | OP.15 FDD | | OP.16 FDD | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: See Table 8.1.1.5-1 for the other parameters. | | | | | |

# 8.1.10 E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX for 5MHz Bandwidth

## 8.1.10.1 Test purpose

Same test purpose as defined in clause 8.1.3.1.

## 8.1.10.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that only support E-UTRA Band 31. Applicability requires support for FGI bit 5.

## 8.1.10.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.1.3.3 with the following exceptions:

- Instead of A.8.1.3 → use A.8.1.10.

## 8.1.10.4 Test description

### 8.1.10.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15.
2. The general test parameter settings are set up according to Table 8.1.10.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.1.10.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.10.4.1-1: General Test Parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used for 5MHz bandwidth**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| PDSCH parameters | | DL Reference Measurement Channel R.5 FDD | | As specified in clause A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.11 FDD | | As specified in clause A.2.1 |
| Channel Bandwidth (BWchannel) | MHz | 5 | | |
| NOTE 1: See Table 8.1.3.4.1-1 for the other parameters.<br>NOTE 2: This test is according to the principle defined in TS 36.133 [4] section A.3.7.2. | | | | |

### 8.1.10.4.2 Test procedure

Same test procedure as defined in clause 8.1.3.4.2 with the following exceptions:

- Instead of Table 8.1.3.5-1 → use Table 8.1.10.5-1.

### 8.1.10.4.3 Message contents

Same message contents as defined in clause 8.1.3.4.3.

## 8.1.10.5 Test requirement

Same test requirement as defined in clause 8.1.3.5 with the following exceptions:

Tables 8.1.10.4.1-1, 8.1.10.5-1, 8.1.3.5-2 and 8.1.3.5-3 define the primary level settings including test tolerances for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used for 5MHz bandwidth test.

**Table 8.1.10.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting in DRX under fading propagation conditions for 5MHz bandwidth**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| $BW_{channel}$ | MHz | 5 | | 5 | |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and in D.1.16 (OP.16 FDD) | | OP.15 FDD | | OP.16 FDD | |
| NOTE: See Table 8.1.3.5-1 for the other parameters. | | | | | |

## 8.1.11 E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for UE category 0

### 8.1.11.1 Test purpose

To verify the category 0 UE's ability to make a correct reporting of an event under fading propagation conditions in asynchronous cells within the E-UTRA FDD-FDD intra-frequency cell search requirements.

### 8.1.11.2 Test applicability

This test applies to all type of E-UTRA FD-FDD UE release 12 and forward of UE Category 0.

### 8.1.11.3 Minimum conformance requirements

The measurement reporting delay shall be less than $T_{identify_intra_UE\ cat\ 0}$ in RRC_CONNECTED state.

When no DRX is in use the UE shall be able to identify a new detectable FDD intra frequency cell within

$$T_{identify_intra_UE\ cat\ 0} = T_{basic_identify_E\text{-}UTRA_FDD_UE\ cat\ 0} \cdot \frac{T_{Measurement_Period_UE\ cat\ 0,\ Intra}}{T_{Intra}} \quad ms$$

Where:

- $T_{basic_identify_E\text{-}UTRA_FDD_UE\ cat\ 0,\ intra}$ is 1000 ms
- $T_{Intra}$: This is the minimum time that is available for intra frequency measurements, during the measurement period with an arbitrarily chosen timing. Time is assumed to be available for performing intra frequency measurements whenever the receiver is guaranteed to be active on the intra frequency carrier.

A cell shall be considered detectable when

- RSRP related side condition given in TS 36.133 [4] clauses 9.1.13.1 and 9.1.13.2 and RSRQ related side conditions given in TS 36.133 [4] clause 9.1.13.3 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.1 for a corresponding Band.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{Measurement_Period_UE\ cat\ 0\ Intra}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is 400 ms. When no measurement gaps are activated, the low complexity UE shall be capable of performing RSRP and RSRQ measurements for 8 identified-intra-frequency cells, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of 400 ms. When measurement gaps are activated the UE shall be capable of performing measurements for at least $Y_{measurement\ intra_UE\ cat\ 0}$ cells , where $Y_{measurement\ intra_UE\ cat\ 0}$ is defined in the following equation. If the UE has identified more than $Y_{measurement\ intra_UE\ cat\ 0}$ cells, the UE shall perform

measurements of at least 8 identified intra- frequency cells but the reporting rate of RSRP and RSRQ measurements of cells from UE physical layer to higher layers may be decreased.

$$Y_{measurement_intra_UE\ cat\ 0} = Floor\left\{ X_{basic_measurement_FDD_UE\ cat\ 0} \cdot \frac{T_{Intra}}{T_{Measurement_Period_UE\ cat\ 0,\ Intra}} \right\}$$

cells where

- $X_{basic\ measurement\ FDD_UE\ cat\ 0}$ = 8 (cells)
- $T_{Measurement_Period_UE\ cat\ 0,\ Intra}$ = 400 ms. The measurement period for Intra frequency RSRP and RSRQ measurements.

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clauses 9.1.13.1 and 9.1.13.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clause 9.1.13.3.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clauses 9.1.13.1, 9.1.13.2 and 9.1.13.3, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra_UE\ cat\ 0}$ defined in TS 36.133 [4] clause 8.5.2.1.1.1. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_UE\ cat\ 0}$ defined in TS 36.133 [4] clause 8.5.2.1.1.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_UE\ cat\ 0,\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.5.2.1.1.1 and A.8.1.11.

## 8.1.11.4 Test description

### 8.1.11.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15 using only main UE Tx/Rx antenna.
2. The general test parameter settings are set up according to Table 8.1.11.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.1.11.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.11.4.1-1: General test parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.13 FDD | As specified in clause A.1.3 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in clause A.2.1 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | One FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| A3-Offset | dB | -6 | |
| CP length | | Normal | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | | OFF |
| Time offset between cells | | 3 ms | Asynchronous cells |
| T1 | s | 5 | |
| T2 | s | 5 | |

### 8.1.11.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.1.11.5-1. Propagation conditions are set according to Annex B clauses B.1 and B.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.11.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 1002 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.1.11.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.1.11.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.1.11.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.1.11.4.3-3: *MeasResults*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServiCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.11.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.1.11.5 Test requirement

Tables 8.1.11.4.1-1 and 8.1.11.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-FDD event triggered reporting under fading propagation conditions in asynchronous cells test.

**Table 8.1.11.5-1: Cell specific test parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |

| | | | | | |
|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | 2x1 | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1 FDD) and in A.3.2.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_PB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_PB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 6.10 | --0.95 | -Infinity | -0.95 |
| $N_{oc}$ Note 3 | dBm/15 KHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | 6.10 | 6.10 | -Infinity | 6.10 |
| RSRP Note 4 | dBm/15 KHz | -91.9 | -91.9 | -Infinity | -91.9 |
| SCH_RP Note 4 | dBm/15 KHz | -91.9 | -91.9 | -Infinity | -91.9 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra_UE\ cat\ 0}$

$$T_{identify\ intra_UE\ cat\ 0} = T_{basic_identify_E-UTRA_FDD_UE\ cat\ 0} \cdot \frac{T_{Measurement_Period_UE\ cat\ 0,\ Intra}}{T_{Intra}}$$

$T_{basic_identify_E-UTRA_FDD_UE\ cat\ 0}$ = 1000 ms

$T_{Measurement_Period_UE\ cat\ 0,\ Intra}$ = 400 ms

$T_{Intra}$ = 400 ms

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 1002 ms in this test case (note: this gives a total of 1000 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.12 E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for UE category 0

### 8.1.12.1 Test purpose

To verify the category 0 UE's ability to make a correct reporting of an event under fading propagation conditions in synchronous cells within the E-UTRA FDD-FDD intra-frequency cell search requirements.

### 8.1.12.2 Test applicability

This test applies to all types of E-UTRA FD-FDD UE release 12 and forward of UE category 0. Applicability requires support for FGI bit 5.

### 8.1.12.3 Minimum conformance requirements

The measurement reporting delay shall be less than $T_{identify_intra_UE\ cat\ 0}$ in RRC_CONNECTED state.

When no DRX is in use the UE shall be able to identify a new detectable FDD intra frequency cell within

$$T_{identify_intra_UE\ cat\ 0} = T_{basic_identify_E\text{-}UTRA_FDD_UE\ cat\ 0} \cdot \frac{T_{Measurement_Period_UE\ cat\ 0,\ Intra}}{T_{Intra}} \quad ms$$

Where:

$T_{basic_identify_E\text{-}UTRA_FDD_UE\ cat\ 0,\ intra}$ is 1000 ms

$T_{Intra}$ : This is the minimum time that is available for intra frequency measurements, during the measurement period with an arbitrarily chosen timing. Time is assumed to be available for performing intra frequency measurements whenever the receiver is guaranteed to be active on the intra frequency carrier.

A cell shall be considered detectable when

- RSRP related side condition given in TS 36.133 [4] clauses 9.1.13.1 and 9.1.13.2 and RSRQ related side conditions given in TS 36.133 [4] clause 9.1.13.3 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.1 for a corresponding Band.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{Measurement_Period_UE\ cat\ 0\ Intra}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is 400 ms. When no measurement gaps are activated, the low complexity UE shall be capable of performing RSRP and RSRQ measurements for 8 identified-intra-frequency cells, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of 400 ms. When measurement gaps are activated the UE shall be capable of performing measurements for at least $Y_{measurement\ intra_UE\ cat\ 0}$ cells , where $Y_{measurement\ intra_UE\ cat\ 0}$ is defined in the following equation. If the UE has identified more than $Y_{measurement\ intra_UE\ cat\ 0}$ cells, the UE shall perform measurements of at least 8 identified intra- frequency cells but the reporting rate of RSRP and RSRQ measurements of cells from UE physical layer to higher layers may be decreased.

$$Y_{measurement_intra_UE\ cat\ 0} = Floor\left\{ X_{basic_measurement_FDD_UE\ cat\ 0} \cdot \frac{T_{Intra}}{T_{Measurement_Period_UE\ cat\ 0,\ Intra}} \right\}$$

cells where

$X_{basic\ measurement\ FDD_UE\ cat\ 0}$ = 8 (cells)

$T_{Measurement_Period_UE\ cat\ 0,\ Intra}$ = 400 ms. The measurement period for Intra frequency RSRP and RSRQ measurements.

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clauses 9.1.13.1 and 9.1.13.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clause 9.1.13.3.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clauses 9.1.13.1, 9.1.13.2 and 9.1.13.3, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra_UE\ cat\ 0}$ defined in TS 36.133 [4] clause 8.5.2.1.1.1. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_UE\ cat\ 0}$ defined in TS 36.133 [4] clause 8.5.2.1.1.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_UE\ cat\ 0,\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.5.2.1.1.1 and A.8.1.12.

## 8.1.12.4 Test description

### 8.1.12.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15 using only main UE Tx/Rx antenna.
2. The general test parameter settings are set up according to Table 8.1.12.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.1.12.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.12.4.1-1: General test parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.13 FDD | As specified in clause A.1.3 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in clause A.2.1 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | One FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| A3-Offset | dB | -6 | |
| CP length | | Normal | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | DRX_L | As specified in Table 8.1.12.5-2 |
| Time offset between cells | | 3 μs | Synchronous cells |
| T1 | s | 5 | |
| T2 | s | 5 | |

## 8.1.12.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 is used. PDCCHs indicating new transmissions shall be sent continuously to ensure that the UE would not enter the DRX state. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.1.12.5-1. Propagation conditions are set according to Annex B clauses B.1 and B.2. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.12.5-1. T2 starts.
6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 1002 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.
7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.1.12.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.1.12.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7<br>Table H.3.6-2 |

**Table 8.1.12.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.1.12.4.3-3: *MeasResults*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCell CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.12.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.1.12.5 Test requirement

Tables 8.1.12.4.1-1, 8.1.12.5-1 and 8.1.12.5-2 define the primary level settings including test tolerances for E-UTRAN FDD-FDD event triggered reporting under fading propagation conditions in synchronous cells test.

**Table 8.1.12.5-1: Cell specific test parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | 2x1 | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1 FDD) and in A.3.2.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 6.10 | -0.95 | -Infinity | -0.96 |
| $N_{oc}$ Note 3 | dBm/15 KHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | 6.10 | 6.10 | -Infinity | 6.10 |
| RSRP Note 4 | dBm/15 KHz | -91.9 | -91.9 | -Infinity | -91.9 |
| SCH_RP Note 4 | dBm/15 KHz | -91.9 | -91.9 | -Infinity | -91.9 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.1.12.5-2: Reference DRX-Configuration to be used in E-UTRAN FDD-FDD event triggered reporting under fading propagation conditions in synchronous cells**

| Parameter | Value | Comments |
|---|---|---|
| Reference configuration | DRX_L | As defined in 4.8.2.1.5 in TS 36.508 [7] |
| onDurationTimer | psf6 | |
| drx-InactivityTimer | psf1920 | |
| drx-RetransmissionTimer | psf16 | |
| longDRX-CycleStartOffset | sf1280, 0 | |
| shortDRX | disabled | |
| Note: For further information see section 6.3.2 in 3GPP TS 36.331 [5]. | | |

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra_UE\ cat\ 0}$

$$T_{identify\ \ intra_UE\ cat\ 0} = T_{basic_identify_E-UTRA\ _\ FDD_UE\ cat\ 0} \cdot \frac{T_{Measurement_Period_UE\ cat\ 0,\ Intra}}{T_{Intra}}$$

$T_{basic_identify_E-UTRA_FDD_UE\ cat\ 0}$ = 1000 ms

$T_{Measurement_Period_UE\ cat\ 0,\ Intra}$ = 400 ms

$T_{Intra}$ = 400 ms

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 1002 ms in this test case (note: this gives a total of 1000 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.13 E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX for UE category 0

### 8.1.13.1 Test purpose

To verify the category 0 UE's ability to make a correct reporting of an event in DRX under fading propagation conditions in synchronous cells within the E-UTRA FDD-FDD intra-frequency cell search requirements.

### 8.1.13.2 Test applicability

This test applies to all types of E-UTRA FD-FDD UE release 12 and forward of UE category 0. Applicability requires support for FGI bit 5.

### 8.1.13.3 Minimum conformance requirements

When DRX is in use the UE shall be able to identify a new detectable FDD intra frequency cell within $T_{identify_intra_UE\ cat\ 0}$ as shown in Table 8.1.13.3-1.

**Table 8.1.13.3-1: Requirement to identify a newly detectable FDD intra frequency cell**

| DRX cycle length (s) | $T_{identify_intra_UE\ cat\ 0}$ (s) (DRX cycles) |
|---|---|
| ≤0.04 | [1] (Note1) |
| 0.04<DRX-cycle≤0.08 | Note2 (40) |
| 0.128 | 3.2 (25) |
| 0.128<DRX-cycle≤2.56 | Note2(20) |
| Note 1: Number of DRX cycle depends upon the DRX cycle in use.<br>Note 2: Time depends upon the DRX cycle in use. | |

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] clauses 9.1.13.1 and 9.1.13.2 and RSRQ related side conditions given in TS 36.133 [4] clause 9.1.13.3 are fulfilled for a corresponding Band,

- SCH_RP and SCH Ês/Iot according to Annex I.2.1 for a corresponding Band

In the RRC_CONNECTED state the measurement period for intra frequency measurements is $T_{measure_intra_UE\ cat\ 0}$ as shown in Table 8.1.13.3-2. The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified intra-frequency cells, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_intra_UE\ cat\ 0}$.

**Table 8.1.13.3-2: Requirement to measure FDD intra frequency cells**

| DRX cycle length (s) | $T_{measure_intra_UE\ cat\ 0}$ (s) (DRX cycles) |
|---|---|
| ≤0.08 | 0.4 (Note1) |
| 0.08<DRX-cycle≤2.56 | Note2 (5) |
| Note 1: Number of DRX cycle depends upon the DRX cycle in use<br>Note 2: Time depends upon the DRX cycle in use | |

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clauses 9.1.13.1 and 9.1.13.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clause 9.1.13.3.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.1.13.1, 9.1.13.2 and 9.1.13.3, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra,\ UE\ cat\ 0}$ defined in TS 36.133 [4] clause 8.5.2.1.1.2. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_UE\ cat\ 0}$ defined in TS 36.133 [4] clause 8.5.2.1.1.2 becomes undetectable for a period ≤ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_intra_UE\ cat\ 0}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.5.2.1.1.2 and A.8.1.13

## 8.1.13.4 Test description

### 8.1.13.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15 using only main UE Tx/Rx antenna.

2. The general test parameter settings are set up according to Table 8.1.13.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.1.13.4.3.

5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.13.4.1-1: General test parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| PDSCH parameters | | DL Reference Measurement Channel R.13 FDD | | As specified in clause A.1.3 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | | As specified in clause A.2.1 |
| Active cell | | Cell 1 | | |
| Neighbour cell | | Cell 2 | | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | | One FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| A3-Offset | dB | -6 | | |
| CP length | | Normal | | |
| Hysteresis | dB | 0 | | |
| Time To Trigger | s | 0 | | |
| Filter coefficient | | 0 | | L3 filtering is not used |
| DRX | | ON | | DRX related parameters are defined in Table 8.1.13.5-2 |
| Time offset between cells | | 3 μs | | Synchronous cells |
| T1 | s | 5 | | |
| T2 | s | 5 | 30 | |

## 8.1.13.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

In Test 1 when DRX = 40 ms is used, UE needs to be provided at least once every 500 ms with new Timing Advance Command MAC control element to restart the Timer Alignment Timer to keep the UE uplink time alignment. Furthermore, the UE is allocated with PUSCH resource at every DRX cycle. In Test 2 when DRX = 1280 ms is used, the uplink time alignment is not maintained and the UE needs to use RACH to obtain uplink allocation for measurement reporting.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.1.13.5-1. Propagation conditions are set according to Annex B clauses B.1 and B.2. T1 start.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.13.5-1 and 8.1.13.5-2.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 1042 ms for Test 1 or less than 26882 ms for Test 2 then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

11. Repeat step 1-10 for each sub-test in Table 8.1.13.4.1-1 as appropriate.

### 8.1.13.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.1.13.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra frequency event triggered reporting in DRX under fading propagation conditions test 1 and 2 requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-2<br>Table H.3.1-7<br>Table H.3.7-1<br>Table H.3.7-2<br>Table H.3.7-3 |

**Table 8.1.13.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting with in DRX under fading propagation conditions test 1 and 2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.1.13.4.3-3: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting with in DRX under fading propagation conditions test 1 and 2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventide CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.1.13.4.3-4: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting in DRX under fading propagation conditions test 1 and 2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.1.13.4.3-5: *MeasResults*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting in DRX under fading propagation conditions test 1 and 2 requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.13.4.3-6: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting in DRX under fading propagation conditions test 1 and 2 requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| physCellId | PhysCellId | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.1.13.5 Test requirement

Tables 8.1.13.4.1-1, 8.1.13.5-1, 8.1.13.5-2 and 8.1.13.5-3 define the primary level settings including test tolerances for E-UTRAN FDD-FDD event triggered reporting under fading propagation conditions in synchronous cells tests when DRX is used.

**Table 8.1.13.5-1: Cell specific test parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | 2x1 | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1 FDD) and in A.3.2.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 6.10 | -0.95 | -Infinity | -0.95 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | 6.10 | 6.10 | -Infinity | 6.10 |
| RSRP Note 3 | dBm/15 KHz | -91.9 | -91.9 | -Infinity | -91.9 |
| SCH_RP Note 3 | dBm/15 KHz | -91.9 | -91.9 | -Infinity | -91.9 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.1.13.5-2: DRX-Configuration for E-UTRAN FDD-FDD intra-frequency event triggered reporting in DRX under fading propagation conditions in synchronous cells**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf40 | sf1280 | |
| shortDRX | disable | disable | |

**Table 8.1.13.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD-FDD intra-frequency event triggered reporting in DRX under fading propagation conditions in synchronous cells**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | 0 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213. |

In Test 1 when DRX cycle length = 40 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2 on PUSCH.

In Test 2 when DRX cycle length = 1280 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE starts to send preambles on the PRACH for Scheduling Request (SR) to obtain allocation to send the measurement report to Cell 2 on PUSCH.

For both tests:

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event A3 measurement report.

The overall delay measured when DRX cycle length is 40 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra_UE\ cat\ 0}$

$T_{identify_intra_UE\ cat\ 0}$ = 1000 ms. When DRX cycle length is 40 ms then the $T_{identify_intra_UE\ cat\ 0}$ is 1s.

TTI insertion uncertainty = 2 ms

DRX cycle length = 40 ms

The overall delay measured when DRX cycle length is 40 ms shall be less than a total of 1042 ms.

The overall delay measured when DRX cycle length is 1280 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra_UE\ cat\ 0}$

$T_{identify_intra_UE\ cat\ 0}$ = 25600 ms. When DRX cycle length is 1280 ms then the $T_{identify_intra_UE\ cat\ 0}$ is 20 x 1280 ms.

TTI insertion uncertainty = 2 ms

DRX cycle length = 1280 ms

The overall delay measured when DRX cycle length is 1280 ms shall be less than a total of 26882 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.14 E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for UE category 0

Editor's notes: This test is incomplete. The following aspects are either missing or not yet determined:

- TT is TBD

- Test system uncertainties is TBD

### 8.1.14.1 Test purpose

To verify the UE makes correct reporting of an event. This test will partly verify the FDD intra-frequency cell search requirements.

### 8.1.14.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 12 and forward of UE category 0. Applicability requires support for FGI bit 5.

### 8.1.14.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD intra frequency cell within

$$T_{identify\ \ intra_UE\ cat\ 0} = T_{basic_identify_E\text{-}UTRA\ _FDD_UE\ cat\ 0} \cdot \frac{T_{Measurement_Period_UE\ cat\ 0,\ Intra}}{T_{Intra}} \quad ms$$

where

$T_{basic_identify_E\text{-}UTRA_FDD_UE\ cat\ 0,\ intra}$ is 1000 ms

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] 9.1.13.1 and 9.1.13.2 and RSRQ related side conditions given in Clause 9.1.13.3 are fulfilled for a corresponding Band,

- SCH_RP and SCH Ês/Iot according to TS 36.133 [4] Annex B.2.1 for a corresponding Band.

$T_{Intra}$: This is the minimum time that is available for intra frequency measurements, during the measurement period with an arbitrarily chosen timing. Time is assumed to be available for performing intra frequency measurements whenever the receiver is guaranteed to be active on the intra frequency carrier.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{Measurement_Period_\ UE\ cat\ 0\ Intra}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is 400 ms. When no measurement gaps are activated, the low complexity UE shall be capable of performing RSRP and RSRQ measurements for 8 identified-intra-frequency cells, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of 400 ms. When measurement gaps are activated the UE shall be capable of performing measurements for at least $Y_{measurement\ intra_UE\ cat\ 0}$ cells , where $Y_{measurement\ intra_\ UE\ cat\ 0}$ is defined in the following equation. If the UE has identified more than $Y_{measurement\ intra_UE\ cat\ 0}$ cells, the UE shall perform measurements of at least 8 identified intra- frequency cells but the reporting rate of RSRP and RSRQ measurements of cells from UE physical layer to higher layers may be decreased.

$$Y_{measurement_intra_UE\ cat\ 0} = Floor\left\{ X_{basic\ _measurement\ _FDD_UE\ cat\ 0} \cdot \frac{T_{Intra}}{T_{Measurement_Period_UE\ cat\ 0,\ Intra}} \right\}$$

cells where

$X_{basic\ measurement\ FDD_UE\ cat\ 0}$ = 8 (cells)

$T_{Measurement_Period_UE\ cat\ 0,\ Intra}$ = 400 ms. The measurement period for Intra frequency RSRP and RSRQ measurements.

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clauses 9.1.13.1 and 9.1.13.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clause 9.1.13.3.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ intra_UE\ cat\ 0}$ defined in TS 36.133 [4] Clause 8.5.2.1.1.1. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_UE\ cat\ 0}$ defined in TS 36.133 [4] clause 8.5.2.1.1.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_UE\ cat\ 0,\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.5.2.1.2.1 and A.8.1.14.

## 8.1.14.4 Test description

### 8.1.14.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20(using only main UE Tx/Rx antenna).
2. The general test parameter settings are set up according to Table 8.1.14.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.1.14.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.14.4.1-1: General test parameters for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.1 HD-FDD | As specified in clause A.3.1.1.4 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.3 HD-FDD | As specified in clause A.3.1.2.3 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | One FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| A3-Offset | dB | -6 | |
| CP length | | Normal | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | | OFF |
| Time offset between cells | | 3 ms | Asynchronous cells |
| T1 | s | 5 | |
| T2 | s | 5 | |

### 8.1.14.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time duration of T1, and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell..
2. Set the parameters according to T1 in Table 8.1.14.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration with event A3 configured message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.14.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 1002ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.
7. After the SS receives the MeasurementReport message in step 6 or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   -switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.1.14.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.1.14.4.3-1: Common Exception messages for E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.1.14.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.1.14.4.3-3: *MeasResults*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServiCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.14.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.1.14.5 Test requirement

Table 8.1.14.4.1-1 and 8.1.14.5.1-1 define the primary level settings including test tolerances for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells.

**Table 8.1.14.5.1-1: Cell specific test parameters for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells.**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | 2x1 | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1 FDD) and in A.3.2.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_PB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_PB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 4+TT | -1.46+TT | -Infinity | -1.46+TT |
| $N_{oc}$ Note 3 | dBm/15 KHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4+TT | 4+TT | -Infinity | 4+TT |
| RSRP Note 4 | dBm/15 KHz | -94 | -94 | -Infinity | -94 |
| SCH_RP Note 4 | dBm/15 KHz | -94 | -94 | -Infinity | -94 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 1000 ms from the beginning of time period T2.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90%.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra_UE\ cat\ 0}$

$$T_{identify_intra_UE\ cat\ 0} = T_{basic_identify_E\text{-}UTRA\ _\ FDD_UE\ cat\ 0,\ intra} \cdot \frac{T_{Measurement\ _\ Period_UE\ cat\ 0,\ Intra}}{T_{Intra}}$$

$T_{basic_identify_E\text{-}UTRA_FDD_UE\ cat\ 0}$ = 1000 ms

$T_{Measurement_Period_UE\ cat\ 0,\ Intra}$ = 400 ms

$T_{Intra}$ = 400 ms

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 1002 ms in this test case (note: this gives a total of 1000 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.15 E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for UE category 0

Editor's notes: This test is incomplete. The following aspects are either missing or not yet determined:

- TT is TBD

- Test system uncertainties is TBD

### 8.1.15.1 Test purpose

To verify the UE makes correct reporting of an event. This test will partly verify the FDD intra-frequency cell search requirements.

### 8.1.15.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward of UE category 0. Applicability requires support for FGI bit 5.

### 8.1.15.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD intra frequency cell within

$$T_{identify\ \ intra_UE\ cat\ 0} = T_{basic_identify_E\text{-}UTRA\ _\ FDD_UE\ cat\ 0} \cdot \frac{T_{Measurement_Period_UE\ cat\ 0,\ Intra}}{T_{Intra}} \quad ms$$

where

$T_{basic_identify_E\text{-}UTRA_FDD_UE\ cat\ 0,\ intra}$ is 1000 ms.

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] Sections 9.1.13.1 and 9.1.13.2 and RSRQ related side conditions given in TS 36.133 [4] Clause 9.1.13.3 are fulfilled for a corresponding Band,

- SCH_RP and SCH Ês/Iot according to TS 36.133 [4] Annex B.2.1 for a corresponding Band.

$T_{Intra}$ : This is the minimum time that is available for intra frequency measurements, during the measurement period with an arbitrarily chosen timing. Time is assumed to be available for performing intra frequency measurements whenever the receiver is guaranteed to be active on the intra frequency carrier.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{Measurement_Period_UE\ cat\ 0\ Intra}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is 400 ms. When no measurement gaps are activated, the low complexity UE shall be capable of performing RSRP and RSRQ measurements for 8 identified-intra-frequency cells, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of 400 ms. When measurement gaps are activated the UE shall be capable of performing measurements for at least $Y_{measurement\ intra_UE\ cat\ 0}$ cells , where $Y_{measurement\ intra_UE\ cat\ 0}$ is defined in the following equation. If the UE has identified more than $Y_{measurement\ intra_UE\ cat\ 0}$ cells, the UE shall perform measurements of at least 8 identified intra- frequency cells but the reporting rate of RSRP and RSRQ measurements of cells from UE physical layer to higher layers may be decreased.

$$Y_{measurement_intra_UE\ cat\ 0} = Floor\left\{ X_{basic_measurement_FDD_UE\ cat\ 0} \cdot \frac{T_{Intra}}{T_{Measurement_Period_UE\ cat\ 0,\ Intra}} \right\}$$

cells where

$X_{basic\ measurement\ FDD_UE\ cat\ 0}$ = 8 (cells);

$T_{Measurement_Period_UE\ cat\ 0,\ Intra}$ = 400 ms. The measurement period for Intra frequency RSRP and RSRQ measurements.

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clauses 9.1.13.1 and 9.1.13.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clause 9.1.13.3.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ intra_UE\ cat\ 0}$ defined in TS 36.133 [4] Clause 8.5.2.1.1.1. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_UE\ cat\ 0}$ defined in TS 36.133 [4] clause 8.5.2.1.1.1 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_UE\ cat\ 0,\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.5.2.1.2.1 and A.8.1.15.

## 8.1.15.4 Test description

### 8.1.15.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.20 (using only main UE Tx/Rx antenna).

2. The general test parameter settings are set up according to Table 8.1.15.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.1.15.4.3.

5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.15.4.1-1: General test parameters for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.1 HD-FDD | As specified in clause A.3.1.1.4 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.3 HD-FDD | As specified in clause A.3.1.2.3 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | One FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| A3-Offset | dB | -6 | |
| CP length | | Normal | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | DRX_L | As specified in clause A.3.3 |
| Time offset between cells | | 3 μs | Synchronous cells |
| T1 | s | 5 | |
| T2 | s | 5 | |

## 8.1.15.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 is used. PDCCHs indicating new transmissions or retransmissions should be sent continuously to ensure that the UE would not enter the DRX state. The test consists of two successive time periods, with time duration of T1, and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell..

2. Set the parameters according to T1 in Table 8.1.15.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration with event A3 configured message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.15.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 1002 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receives the MeasurementReport message in step 6 or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),

or
-switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.1.15.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.1.15.4.3-1: Common Exception messages for E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7<br>Table H.3.6-2 |

**Table 8.1.15.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.1.15.4.3-3: *MeasResults*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCell CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.15.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.1.15.5 Test requirement

Table 8.1.15.4.1-1 and 8.1.15.5.1-1 define the primary level settings including test tolerances for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells.

**Table 8.1.15.5.1-1: Cell specific test parameters for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | 2x1 | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1 FDD) and in A.3.2.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 4+TT | -1.46+TT | -Infinity | -1.46+TT |
| $N_{oc}$ Note 3 | dBm/15 KHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4+TT | 4+TT | -Infinity | 4+TT |
| RSRP Note 4 | dBm/15 KHz | -94 | -94 | -Infinity | -94 |
| SCH_RP Note 4 | dBm/15 KHz | -94 | -94 | -Infinity | -94 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 1000 ms from the beginning of time period T2.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90%.

The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra_UE\ cat\ 0}$

$$T_{identify_intra_UE\ cat\ 0} = T_{basic_identify_E-UTRA_FDD_UE\ cat\ 0,\ intra} \cdot \frac{T_{Measurement_Period_UE\ cat\ 0,\ Intra}}{T_{Intra}}$$

$T_{basic_identify_E-UTRA_FDD_UE\ cat\ 0}$ = 1000 ms

$T_{Measurement_Period_UE\ cat\ 0,\ Intra}$ = 400 ms

$T_{Intra}$ = 400 ms

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 1002 ms in this test case (note: this gives a total of 1000 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.16 E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX for UE category 0

Editor's notes: This test is incomplete. The following aspects are either missing or not yet determined:

- TT is TBD

- Test system uncertainties is TBD

### 8.1.16.1 Test purpose

To verify the UE makes correct reporting of an event in DRX. The tests will partly verify the FDD-FDD intra-frequency cell search in DRX requirements.

### 8.1.16.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 12 and forward with support of intra-frequency SI acquisition for HO of UE category 0.

### 8.1.16.3 Minimum conformance requirements

When DRX is in use the UE shall be able to identify a new detectable FDD intra frequency cell within $T_{identify_intra_UE\ cat\ 0}$ as shown in table 8.1.16.3-1.

**Table 8.1.16.3-1: Requirement to identify a newly detectable HD-FDD intrafrequency cell**

| DRX cycle length (s) | $T_{identify_intra_UE\ cat\ 0}$ (s) (DRX cycles) |
|---|---|
| ≤0.04 | 1 (Note1) |
| 0.04<DRX-cycle≤0.08 | Note2 (50) |
| 0.128 | 3.2 (32) |
| 0.128<DRX-cycle≤2.56 | Note2(25) |
| Note1: Number of DRX cycle depends upon the DRX cycle in use.<br>Note2: Time depends upon the DRX cycle in use. | |

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] Sections 9.1.13.1 and 9.1.13.2 and RSRQ related side conditions given in TS 36.133 [4] Clause 9.1.13.3 are fulfilled for a corresponding Band,

- SCH_RP and SCH Ês/Iot according to TS 36.133 [4] Annex B.2.1 for a corresponding Band.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is $T_{measure_intra_UE\ cat\ 0}$ as shown in table 8.1.16.3-2. The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified-intra-frequency cells, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_intra_UE\ cat\ 0}$.

**Table 8.1.16.3-2: Requirement to measure HD-FDD intrafrequency cells**

| DRX cycle length (s) | $T_{measure_intra_UE\ cat\ 0}$ (s) (DRX cycles) |
|---|---|
| ≤0.04 | 0.4 (Note1) |
| 0.04<DRX-cycle≤0.16 | Note2 (7) |
| 0.16<DRX-cycle≤2.56 | Note2(5) |
| Note1: Number of DRX cycle depends upon the DRX cycle in use.<br>Note2: Time depends upon the DRX cycle in use. | |

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clauses 9.1.13.1 and 9.1.13.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clauses 9.1.13.3.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.13.1, 9.1.13.2 and 9.1.13.3, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra_UE\ cat\ 0}$ defined in TS 36.133 [4] Clause 8.5.2.1.1.2 When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_UE\ cat\ 0}$ defined in TS 36.133 [4] clause 8.5.2.1.1.2 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_intra_UE\ cat\ 0}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.5.2.1.2.2 and A.8.1.16.

## 8.1.16.4 Test description

### 8.1.16.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.20(using only main UE Tx/Rx antenna).
2. The general test parameter settings are set up according to Table 8.1.16.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.1.16.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.16.4.1-1: General test parameters for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.1 HD-FDD | As specified in clause A.3.1.1.4 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.3 HD-FDD | As specified in clause A.3.1.2.3 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | One FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| A3-Offset | dB | -6 | |
| CP length | | Normal | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | DRX_L | As specified in clause A.3.3 |
| Time offset between cells | | 3 μs | Synchronous cells |
| T1 | s | 5 | |
| T2 | s | 5 | |

## 8.1.16.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. In the measurement control information, it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time duration of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2.

In Test 1 UE needs to be provided at least once every 500ms with new Timing Advance Command MAC control element to restart the Time alignment timer to keep UE uplink time alignment. Furthermore UE is allocated with PUSCH resource at every DRX cycle.

In Test 2 the uplink time alignment is not maintained and UE needs to use RACH to obtain UL allocation for measurement reporting.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.1.16.5-1. Propagation conditions are set according to Annex B clauses B.1 and B.2. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.16.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 1042 ms for Test 1 or less than 26882 ms for Test 2 then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.
7. After the SS receives the MeasurementReport message in step 6 or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),

or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

11. Repeat step 1-10 for each sub-test in Table 8.1.16.4.1-1 as appropriate.

### 8.1.16.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.1.16.4.3-1: Common Exception messages for E-UTRAN HD-FDD intra frequency event triggered reporting in DRX under fading propagation conditions test 1 and 2 requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-2<br>Table H.3.1-7<br>Table H.3.7-1<br>Table H.3.7-2<br>Table H.3.7-3 |

**Table 8.1.16.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting with in DRX under fading propagation conditions test 1 and 2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.1.16.4.3-3: *ReportConfigEUTRA-A3*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting with in DRX under fading propagation conditions test 1 and 2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventide CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.1.16.4.3-4: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN HD-FDD intra-frequency event triggered reporting in DRX under fading propagation conditions test 1 and 2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.1.16.4.3-5: *MeasResults*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting in DRX under fading propagation conditions test 1 and 2 requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.16.4.3-6: *MeasResultListEUTRA*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting in DRX under fading propagation conditions test 1 and 2 requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| physCellId | PhysCellId | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.1.16.5 Test requirement

Table 8.1.16.4.1-1 and A.8.1.16.5.1-1 define the primary level settings including test tolerances for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells.

**Table 8.1.16.5.1-1: Cell specific test parameters for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | 2x1 | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1 FDD) and in A.3.2.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 4+TT | -1.46+TT | -Infinity | -1.46+TT |
| $N_{oc}$ Note 3 | dBm/15 KHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4+TT | 4+TT | -Infinity | 4+TT |
| RSRP Note 4 | dBm/15 KHz | -94 | -94 | -Infinity | -94 |
| SCH_RP Note 4 | dBm/15 KHz | -94 | -94 | -Infinity | -94 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

In Test 1, the UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 1000 ms from the beginning of time period T2. The measurement reporting delay is defined as the time from the beginning of time period T2 to the moment when the UE send the measurement report on PUSCH.

In Test 2, the UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 25600 ms from the beginning of time period T2. The measurement reporting delay is defined as the time from the beginning of time period T2 to the moment when the UE starts to send preambles on the PRACH for scheduling request (SR) to obtain allocation to send the measurement report on PUSCH.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90%.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event A3 measurement report.

The overall delay measured when DRX cycle length is 40 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra_UE\ cat\ 0}$

$T_{identify_intra_UE\ cat\ 0}$ = 1000 ms. When DRX cycle length is 40 ms then the $T_{identify_intra_UE\ cat\ 0}$ is 1s.

TTI insertion uncertainty = 2 ms

DRX cycle length = 40 ms

The overall delay measured when DRX cycle length is 40 ms shall be less than a total of 1042 ms.

The overall delay measured when DRX cycle length is 1280 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra_UE\ cat\ 0}$

$T_{identify_intra_UE\ cat\ 0}$ = 25600 ms. When DRX cycle length is 1280 ms then the $T_{identify_intra_UE\ cat\ 0}$ is 20 x 1280 ms.

TTI insertion uncertainty = 2 ms

DRX cycle length = 1280 ms

The overall delay measured when DRX cycle length is 1280 ms shall be less than a total of 26882 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.17 E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for UE category 0

### 8.1.17.1 Test purpose

To verify the category 0 UE's ability to make a correct reporting of an event under fading propagation conditions in synchronous cells within the E-UTRA TDD-TDD intra-frequency cell search requirements.

### 8.1.17.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward of UE category 0. Applicability requires support for FGI bit 5.

### 8.1.17.3 Minimum conformance requirements

The measurement reporting delay shall be less than $T_{identify_intra_UE\ cat\ 0}$ in RRC_CONNECTED state.

When no DRX is in use the UE shall be able to identify a new detectable TDD intra frequency cell within

$$T_{identify_intra_UE\ cat\ 0} = T_{basic_identify_E-UTRA_TDD_UE\ cat\ 0,\ intra} \cdot \frac{T_{Measurement_Period_UE\ cat\ 0,\ Intra}}{T_{Intra}} \quad ms$$

Where:

$T_{basic_identify_E-UTRA_TDD_UE\ cat\ 0,\ intra}$ is 1000 ms

$T_{Intra}$ : This is the minimum time that is available for intra frequency measurements, during the measurement period with an arbitrarily chosen timing. Time is assumed to be available for performing intra frequency measurements whenever the receiver is guaranteed to be active on the intra frequency carrier.

A cell shall be considered detectable when

- RSRP related side condition given in TS 36.133 [4] clauses 9.1.13.1 and 9.1.13.2 and RSRQ related side conditions given in TS 36.133 [4] clause 9.1.13.3 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.1 for a corresponding Band.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{Measurement_Period_UE\ cat\ 0,\ Intra}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is 400 ms. When no measurement gaps are activated, the UE shall be capable of performing RSRP and RSRQ measurements for 8 identified-intra-frequency cells, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of 400 ms. When measurement gaps are activated the UE shall be capable of performing measurements for at least $Y_{measurement\ intra_UE\ cat\ 0}$ cells , where $Y_{measurement\ intra_UE\ cat\ 0}$ is defined in the following equation. If the UE has identified more than $Y_{measurement\ ,intra_UE\ cat\ 0}$ cells, the UE shall perform measurements of at least 8 identified intra-frequency cells but the reporting rate of RSRP and RSRQ measurements of cells from UE physical layer to higher layers may be decreased.

$$Y_{measurement\ \ intra_UE\ cat\ 0} = Floor\left\{ X_{basic\ measurement\ TDD_UE\ cat\ 0} \cdot \frac{T_{Intra}}{T_{Measurement_Period_UE\ cat\ 0,\ Intra}} \right\}$$

cells where

$X_{basic\ measurement\ TDD_UE\ cat\ 0}$ = 8 (cells)

$T_{Measurement_Period\ intra_UE\ cat\ 0}$ = 400 ms. The measurement period for Intra frequency RSRP and RSRQ measurements.

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clauses 9.1.13.1 and 9.1.13.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clause 9.1.13.3.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clauses 9.1.13.1, 9.1.13.2 and 9.1.13.3, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ intra_UE\ cat\ 0}$ defined in TS 36.133 [4] clauses 8.5.2.1.3.1. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_UE\ cat\ 0}$ defined in TS 36.133 [4] clauses 8.5.2.1.3.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ Intra_UE\ cat\ 0}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.5.2.1.3.1 and A.8.1.17.

## 8.1.17.4 Test description

### 8.1.17.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15 using only main UE Tx/Rx antenna.
2. The general test parameter settings are set up according to Table 8.1.17.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.1.17.4.3.
5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.17.4.1-1: General test parameters for E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.12 TDD | As specified in clause A.1.5 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in clause A.2.2 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | One TDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| A3-Offset | dB | -6 | |
| CP length | | Normal | |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211. The same configuration in both cells |
| Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in TS 36.211. The same configuration in both cells |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | DRX_L | As specified in Table 8.1.17.5-2 |
| Time offset between cells | | 3 µs | Synchronous cells |
| T1 | s | 5 | |
| T2 | s | 5 | |

## 8.1.17.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 is used. PDCCHs indicating new transmissions or retransmissions shall be sent continuously to ensure that the UE would not enter the DRX state. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.1.17.5-1. Propagation conditions are set according to Annex B clauses B.1 and B.2. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.17.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 1002 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receives the MeasurementReport message in step 6 or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.1.17.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.1.17.4.3-1: Common Exception messages for E-UTRAN TDD-TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7<br>Table H.3.6-2 |

**Table 8.1.17.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD-TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.1.17.4.3-3: MeasResults: Additional E-UTRAN TDD-TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | Report Cell 1 | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.17.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN TDD-TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | Report Cell 2 | |
| physCellId | PhysCellId of the Cell 2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

## 8.1.17.5 Test requirement

Tables 8.1.17.4.1-1, 8.1.17.5-1 and 8.1.17.5-2 define the primary level settings including test tolerances for E-UTRAN TDD-TDD event triggered reporting under fading propagation conditions in synchronous cells test.

**Table 8.1.17.5-1: Cell specific test parameters for E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | 2x1 | |
| OCNG Pattern defined in A.3.2.2.1 (OP.1 TDD) and in A.3.2.2.2 (OP.2) | | OP.1 TDD | | OP.2 TDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | |
| RSRP Note 4 | dBm/15 kHz | -91.4 | -91.4 | -Infinity | -91.4 |
| $\hat{E}_s / I_{ot}$ | dB | 6.60 | -0.86 | -Infinity | -0.86 |
| SCH_RP Note 4 | dBm/15 kHz | -91.4 | -91.4 | -Infinity | -91.4 |
| $\hat{E}_s / N_{oc}$ | dB | 6.60 | 6.60 | -Infinity | 6.60 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.1.17.5-2: Reference DRX-Configuration to be used in E-UTRAN TDD-TDD event triggered reporting under fading propagation conditions in synchronous cells**

| Parameter | Value | Comments |
|---|---|---|
| Reference configuration | DRX_L | As defined in 4.8.2.1.5 in TS 36.508 [7] |
| onDurationTimer | psf6 | |
| drx-InactivityTimer | psf1920 | |
| drx-RetransmissionTimer | psf16 | |
| longDRX-CycleStartOffset | sf1280, 0 | |
| shortDRX | disabled | |
| Note: For further information see section 6.3.2 in 3GPP TS 36.331 [5]. | | |

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra_UE\ cat\ 0}$

$$T_{identify_intra_UE\ cat\ 0} = T_{basic_identify_E\text{-}UTRA\ _TDD_UE\ cat\ 0,\ intra} \cdot \frac{T_{Measurement\ _Period_UE\ cat\ 0,\ Intra}}{T_{Intra}}$$

$T_{basic_identify_E\text{-}UTRA_TDD_UE\ cat\ 0}$ = 1000 ms

$T_{Measurement_Period_UE\ cat\ 0,\ Intra}$ = 400 ms

$T_{Intra}$ = 400 ms

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 1002 ms in this test case (note: this gives a total of 1000 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.18 E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX for UE category 0

### 8.1.18.1 Test purpose

To verify the category 0 UE's ability to make a correct reporting of an event under fading propagation conditions in synchronous cells with DRX within the E-UTRA TDD-TDD intra frequency cell search in DRX requirements.

### 8.1.18.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward of UE category 0. Applicability requires support for FGI bit 5.

### 8.1.18.3 Minimum conformance requirements

When DRX is in use the UE shall be able to identify a new detectable TDD intra frequency cell within $T_{identify_intra_UE\ cat\ 0}$ as shown in Table 8.1.18.3-1.

**Table 8.1.18.3-1: Requirement to identify a newly detectable TDD intra frequency cell**

| DRX cycle length (s) | $T_{identify_intra_UE\ cat\ 0}$ (s) (DRX cycles) |
|---|---|
| ≤0.04 | 1 (Note1) |
| 0.04<DRX-cycle≤0.08 | Note2 (40) |
| 0.128 | 3.2 (25) |
| 0.128<DRX-cycle≤2.56 | Note2(20) |
| Note 1: Number of DRX cycle depends upon the DRX cycle in use.<br>Note 2: Time depends upon the DRX cycle in use. | |

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] clause 9.1.13.1 and 9.1.13.2 and RSRQ related side conditions given in TS 36.133 [4] clause 9.1.13.3 are fulfilled for a corresponding Band,

- SCH_RP and SCH Ês/Iot according to Annex I.2.1 for a corresponding Band.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is $T_{measure_intra_UE\ cat\ 0}$ as shown in Table 8.1.18.3-2. The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified intra-frequency cells, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_intra_UE\ cat\ 0}$.

**Table 8.1.18.3-2: Requirement to measure TDD intra frequency cells**

| DRX cycle length (s) | $T_{measure_intra_UE\ cat\ 0}$ (s) (DRX cycles) |
|---|---|
| ≤0.08 | 0.4 (Note1) |
| 0.08<DRX-cycle≤2.56 | Note2 (5) |
| Note 1: Number of DRX cycle depends upon the DRX cycle in use.<br>Note 2: Time depends upon the DRX cycle in use. | |

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clauses 9.1.13.1 and 9.1.13.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clause 9.1.13.3.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.1.13.1, 9.1.13.2 and 9.1.13.3, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra,\ UE\ cat\ 0}$ defined in TS 36.133 [4] clause 8.5.2.1.3.2. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_UE\ cat\ 0}$ defined in TS 36.133 [4] clause 8.5.2.1.3.2 becomes undetectable for a period ≤ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_intra_UE\ cat\ 0}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.5.2.1.3.2 and A.8.1.18

## 8.1.18.4 Test description

### 8.1.18.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15 using only main UE Tx/Rx antenna.

2. The general test parameter settings are set up according to Table 8.1.18.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.1.18.4.3.

5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.18.4.1-1: General test parameters for E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| PDSCH parameters | | DL Reference Measurement Channel R.12 TDD | | As specified in clause A.1.5 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | | As specified in clause A.2.2 |
| Active cell | | Cell 1 | | |
| Neighbour cell | | Cell 2 | | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | | One TDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| A3-Offset | dB | -6 | | |
| CP length | | Normal | | |
| Special subframe configuration | | 6 | | As specified in table 4.2-1 in TS 36.211. The same configuration in both cells |
| Uplink-downlink configuration | | 1 | | As specified in table 4.2-2 in TS 36.211. The same configuration in both cells |
| Hysteresis | dB | 0 | | |
| Time To Trigger | s | 0 | | |
| Filter coefficient | | 0 | | L3 filtering is not used |
| DRX | | ON | | DRX related parameters are defined in Table 8.1.18.5-2 |
| Time offset between cells | | 3 µs | | Synchronous cells |
| T1 | s | 5 | | |
| T2 | s | 5 | 30 | |

## 8.1.18.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

In Test 1 UE needs to be provided at least once every 500ms with new Timing Advance Command MAC control element to restart the Time alignment timer to keep UE uplink time alignment. Furthermore, the UE is allocated with PUSCH resource at every DRX cycle.

In Test 2 the uplink time alignment is not maintained and UE needs to use RACH to obtain UL allocation for measurement reporting.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.1.18.5-1. Propagation conditions are set according to Annex B clauses B.1 and B.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.18.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 1042 ms for Test 1 or less than 26882 ms for Test 2 then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receives the MeasurementReport message in step 6 or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

11. Repeat step 1-10 for each sub-test in Table 8.1.18.4.1-1 as appropriate.

### 8.1.18.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.1.18.4.3-1: Common Exception messages for E-UTRAN TDD-TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.1.18.4.3-2: RRCConnectionReconfiguration: Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | | | |
| | MeasConfig-DEFAULT | | MEAS |
| radioResourceConfigDedicated SEQUENCE { | | | |
| MAC-MainConfig-RBC SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf40 | 9 | For Test 1 | |
| sf1280 | 9 | For Test 2 | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | sf500 | | |
| } | | | |
| PhysicalConfigDedicated SEQUENCE { | | | |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.18.4.3-3: ReportConfigEUTRA-A3: Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.18.4.3-4: SchedulingRequest-Config-DEFAULT: Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20: SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | Channel-bandwidth-dependent parameter | |
| sr-ConfigIndex | 2 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.1.18.4.3-5: MeasResults: Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | Set according to specific test | |
| rsrqResult | INTEGER(0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.18.4.3-6: MeasResultListEUTRA: Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| 1 entry physCellId | PhysCellId | INTEGER (0..503) of Cell 2 | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

## 8.1.18.5 Test requirement

Tables 8.1.18.4.1-1, 8.1.18.5-1, 8.1.18.5-2 and 8.1.18.5-3 define the primary level settings including test tolerances for E-UTRAN TDD-TDD event triggered reporting under fading propagation conditions in synchronous cells tests when DRX is used.

**Table 8.1.18.5-1: Cell specific test parameters for E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | 2x1 | |
| OCNG Pattern defined in A.3.2.2.1 (OP.1 TDD) and in A.3.2.2.2 (OP.2) | | OP.1 TDD | | OP.2 TDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | | |
| RSRP Note 3 | dBm/15 kHz | -91.4 | -91.4 | -Infinity | -91.4 |
| $\hat{E}_s / I_{ot}$ | dB | 6.60 | -0.86 | -Infinity | -0.86 |
| SCH_RP Note 3 | dBm/15 kHz | -91.4 | -91.4 | -Infinity | -91.4 |
| $\hat{E}_s / N_{oc}$ | dB | 6.60 | 6.60 | -Infinity | 6.60 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.1.18.5-2: DRX-Configuration for E-UTRAN TDD-TDD intra-frequency event triggered reporting in DRX under fading propagation conditions in synchronous cells**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf40 | sf1280 | |
| shortDRX | disable | disable | |

**Table 8.1.18.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN TDD-TDD intra-frequency event triggered reporting in DRX under fading propagation conditions in synchronous cells**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 2 | 2 | For further information see clause 6.3.2 in TS 36.331 and clause 10.1 in TS 36.213. |

In Test 1 when DRX cycle length = 40 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2 on PUSCH.

In Test 2 when DRX cycle length = 1280 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE starts to send preambles on the PRACH for Scheduling Request (SR) to obtain allocation to send the measurement report to Cell 2 on PUSCH.

For both tests:

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event A3 measurement report.

The overall delay measured when DRX cycle length is 40 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra_UE\ cat\ 0}$

$T_{identify_intra_UE\ cat\ 0}$ = 1000 ms. When DRX cycle length is 40 ms then the $T_{identify_intra_UE\ cat\ 0}$ is 1s.

TTI insertion uncertainty = 2 ms

DRX cycle length = 40 ms

The overall delay measured when DRX cycle length is 40 ms shall be less than a total of 1042 ms.

The overall delay measured when DRX cycle length is 1280 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra_UE\ cat\ 0}$

$T_{identify_intra_UE\ cat\ 0}$ = 25600 ms. When DRX cycle length is 1280 ms then the $T_{identify_intra_UE\ cat\ 0}$ is 20 x 1280 ms.

TTI insertion uncertainty = 2 ms

DRX cycle length = 1280 ms

The overall delay measured when DRX cycle length is 1280 ms shall be less than a total of 26882 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.19 E-UTRAN FD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps for UE category 0

### 8.1.19.1 Test purpose

To verify the requirement for identification of a new CGI of E-UTRA cell with autonomous gaps for UE category 0.

### 8.1.19.2 Test applicability

This test applies to all types of E-UTRA FD-FDD UE release 12 and forward with support of intra-frequency SI acquisition for HO of UE category 0.

### 8.1.19.3 Minimum conformance requirements

No explicit neighbour list is provided to the UE for identifying a new CGI of E-UTRA cell. The UE shall identify and report the CGI when requested by the network for the purpose 'reportCGI'. The UE may make autonomous gaps in downlink reception and uplink transmission for receiving MIB and SIB1 message according to clause 5.5.3.1 of TS 36.331 [5]. Note that a UE is not required to use autonomous gap if si-RequestForHO is set to false. If autonomous gaps are used for measurement with the purpose of 'reportCGI', regardless of whether DRX is used or not, or whether SCell(s) are configured or not, the UE shall be able to identify a new CGI of E-UTRA cell within:

$$T_{identify_CGI_LC\text{-}UE,\ intra} = T_{basic_identify_CGI_LC\text{-}UE,\ intra} \quad ms$$

Where

$T_{basic_identify_CGI_LC\text{-}UE,\ intra}$ = 190 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new CGI of an E-UTRA cell is defined, provided that the E-UTRA cell has been already identified by the UE.

A cell shall be considered identifiable following conditions are fulfilled:

- RSRP related side conditions given in TS 36.133 [4] Clause 9.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.2 for a corresponding Band

The MIB of an E-UTRA cell whose CGI is identified shall be considered decodable by the UE provided the PBCH demodulation requirements are met according to [TBD].

The requirement for identifying a new CGI of an E-UTRA cell within $T_{basic_identify_CGI_LC\text{-}UE,.intra}$ is applicable when no DRX is used as well as when all the DRX cycles specified in TS 36.331 [5] are used.

Within the time, $T_{identify_CGI_LC\text{-}UE,\ intra}$ $ms$, over which the UE identifies the new CGI of E-UTRA cell, the UE shall transmit at least 92 ACK/NACKs on PCell provided that:

- there is continuous DL data allocation,
- no DRX cycle is used,
- no measurement gaps are configured,
- only one code word is transmitted in each subframe,
- no MBSFN subframes are configured in the PCell or each of activated SCell(s).

The ECGI reporting delay occurs due to the delay uncertainty when inserting the ECGI measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. In case DRX is used, the ECGI reporting may be delayed until the next DRX cycle. If IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.5.2.1.4 and A.8.1.19.

## 8.1.19.4 Test description

### 8.1.19.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20(using only main UE Tx/Rx antenna).
2. The general test parameter settings are set up according to Table 8.1.19.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.1.19.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.19.4.1-1: General test parameters for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps for UE category 0**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.15 FDD | As specified in clause A.1.3 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.7FDD | As specified in clause A.2.1 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | Only one FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| CP length | | Normal | |
| A3-Offset | dB | -3 | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| si-RequestForHO | | TRUE | As specified in clause 5.5.3.1 in TS 36.331. |
| T1 | s | 5 | |
| T2 | s | ≤10 | |
| T3 | s | 5 | |

### 8.1.19.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. PDCCHs indicating new transmissions shall be sent continuously to ensure that the UE would have ACK/NACK sending during identifying a new CGI of E-UTRAN cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE does not have any timing information of cell 2. Starting T2, cell 2 becomes detectable and the UE is expected to detect and send a measurement report.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.1.19.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration with event A3 configured message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.19.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3.

7. SS shall transmit an RRCConnectionReconfiguration message during period T2, The RRC message shall create a measurement report configuration with purpose *reportCGI* and *si-RequestForHO* set to TRUE.

8. The SS shall start T3 timer when the last TTI containing the RRC message implying SI reading is sent to UE.

9. The UE shall transmit RRCConnectionReconfigurationComplete message.

10. UE shall transmit a MeasurementReport message containing the cell global identifier of cell 2 within 210 milliseconds from the start of T3. The UE shall be scheduled continuously throughout the test, and from the start of T3 until 210ms at least 112 ACK/NACK shall be detected as being transmitted by the UE. If the overall delays measured from the beginning of time period T3 is less than 210 ms, and the UE have more than 112 ACK/NACKs transmitted from the start of T3 until 210 ms, then the number of successful tests is increased by one. If the UE fails to report the cell global identifier within the overall delays measured requirement, or the UE have less than 112 ACK/NACKs transmitted from the start of T3 until 210 ms, then the number of failure tests is increased by one.

11. After the SS receive the MeasurementReport message in step 10) or when T3 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

13. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 2-13 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.1.19.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.1.19.4.3-1: Common Exception messages for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.1.19.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -6 (-3 dB) | -3 is actual value in dB (-6 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.19.4.3-3: *MeasResults*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step6)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.19.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step6)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.1.19.4.3-5: *ReportConfigEUTRA-PERIODICAL*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportCGI | | |
| } | | | |
| } | | | |
| reportAmount | r1 | | |
| si-RequestForHO-r9 | setup | | |
| } | | | |

**Table 8.1.19.4.3-6: *MeasResultListEUTRA*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| cgi-Info SEQUENCE{ | | | |
| cellGlobalId SEQUENCE{ | | | |
| plmn-Identity | plmn-Identity | | |
| cellIdentity | cellIdentity | | |
| } | | | |
| trackingAreaCode | | | |
| plmn-IdentityList | | | |
| } | | | |
| } | | | |

**Table 8.1.19.4.3-7: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | |
| cellForWhichToReportCGI | Physical Cell ID of Cell 2 | | |
| } | | | |

**Table 8.1.19.4.3-8: *SystemInformationBlockType2*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-1 SystemInformationBlockType2 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfig | Not present | | Cell 1 |
| } | | | |

**Table 8.1.19.4.3-9: *SystemInformationBlockType3*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-2 SystemInformationBlockType3 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType3 ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1 |
| | '01'B (No MBSFN subframes are present in all neighbour cells) | | Cell 2 |
| } | | | |

**Table 8.1.19.4.3-10: *MeasConfig-DEFAULT*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f1 | | |
| reportConfigId | IdReportConfig-P | | |
| } | | | |
| measGapConfig | Not present | | |
| } | | | |

**Table 8.1.19.4.3-11: *MeasResults*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.19.4.3-12: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | |
| } | | | |

## 8.1.19.5 Test requirement

Tables 8.1.19.4.1-1 and 8.1.19.5-1 define the primary level settings including test tolerances for E-UTRAN FD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps for UE category 0 test.

**Table 8.1.19.5-1: Cell specific test parameters for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps for UE category 0**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | | 2x1 | | |
| OCNG Patterns defined inD.1.10 (OP.10 FDD) and in D.1.2(OP.2 FDD) | | OP.10 FDD | OP.10 FDD | OP.10 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD |
| PBCH_RA | dB | -3 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | 0 | | | 0 | | |
| PHICH_RA | dB | -3 | | | -3 | | |
| PHICH_PB | dB | | | | | | |
| PDCCH_RA | dB | 0 | | | 0 | | |
| PDCCH_PB | dB | | | | | | |
| PDSCH_RA | dB | -3 | | | -3 | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 8 | -3.3 | -3.3 | -Infinity | 2.36 | 2.36 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 8 | 8 | 8 | -Infinity | 11 | 11 |
| RSRP Note 3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| SCH_RP Note3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| Propagation Condition | | AWGN | | | | | |
| Timing offset to Cell 1 | ms | - | | | 3 | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall transmit a measurement report containing the cell global identifier of cell 2 within 210 milliseconds from the start of T3.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Test requirement = RRC Procedure delay + $T_{identify_CGI,\,intra}$ + TTI insertion uncertainty

RRC procedure delay=15ms

$$T_{identify_CGI,\,intra} = T_{basic_identify_CGI,\,intra} \quad ms$$

$T_{basic_identify_CGI,\,intra}$ = 190 ms

TTI insertion uncertainty = 2 ms

The overall delays measured is 207 ms, allow 210 ms. The UE shall be scheduled continuously throughout the test, and from the start of T3 until 210 ms at least 112 ACK/NACK shall be detected as being transmitted by the UE.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

NOTE: The overall 112 ACK/NACK number is caused by two parts. Firstly, at least 92 ACK/NACK shall be sent during identifying the cell global identifier of cell 2 according to the requirement in Clause 8.5.2.1.4. Secondly, given that continuous DL data allocation, additional 20 ACK/NACK shall be sent from the start of T3 until 210 ms excludes 190 ms for identifying the cell global identifier of cell.

## 8.1.20 E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0

### 8.1.20.1 Test purpose

To verify the requirement for identification of a new CGI of E-UTRA cell with autonomous gaps with DRX for UE category 0.

### 8.1.20.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward with support of intra-frequency SI acquisition for HO. Applicability requires support for FGI bit 5 of UE category 0.

### 8.1.20.3 Minimum conformance requirements

No explicit neighbour list is provided to the UE for identifying a new CGI of E-UTRA cell. The UE shall identify and report the CGI when requested by the network for the purpose ‘reportCGI’. The UE may make autonomous gaps in downlink reception and uplink transmission for receiving MIB and SIB1 message according to clause 5.5.3.1 of TS 36.331 [5]. Note that a UE is not required to use autonomous gap if si-RequestForHO is set to false. If autonomous gaps are used for measurement with the purpose of ‘reportCGI’, regardless of whether DRX is used or not, or whether SCell(s) are configured or not, the UE shall be able to identify a new CGI of E-UTRA cell within:

$$T_{identify_CGI_LC\text{-}UE,\,intra} = T_{basic_identify_CGI_LC\text{-}UE,\,intra} \quad ms$$

Where

$T_{basic_identify_CGI_LC\text{-}UE,\,intra}$ = 190 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new CGI of an E-UTRA cell is defined, provided that the E-UTRA cell has been already identified by the UE.

A cell shall be considered identifiable following conditions are fulfilled:

- RSRP related side conditions given in TS 36.133 [4] Section 9.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.2 for a corresponding Band.

The MIB of an E-UTRA cell whose CGI is identified shall be considered decodable by the UE provided the PBCH demodulation requirements are met according to [TBD].

The requirement for identifying a new CGI of an E-UTRA cell within $T_{basic_identify_CGI_LC\text{-}UE,.intra}$ is applicable when no DRX is used as well as when all the DRX cycles specified in TS 36.331 [5] are used.

Within the time, $T_{identify_CGI_LC\text{-}UE,intra}$ $ms$ , over which the UE identifies the new CGI of E-UTRA cell, the UE shall transmit at least 92 ACK/NACKs on PCell provided that:

- there is continuous DL data allocation,
- no DRX cycle is used,
- no measurement gaps are configured,
- only one code word is transmitted in each subframe,
- no MBSFN subframes are configured in the PCell or each of activated SCell(s).

The ECGI reporting delay occurs due to the delay uncertainty when inserting the ECGI measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. In case DRX is used, the ECGI reporting may be delayed until the next DRX cycle. If IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.5.2.1.4 and A.8.1.20.

## 8.1.20.4 Test description

### 8.1.20.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1 Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20(using only main UE Tx/Rx antenna).

2. The general test parameter settings are set up according to Table 8.1.20.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.1.20.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.20.4.1-1: General test parameters for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.14 FDD | As specified in clause A.1.3 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.7 FDD | As specified in clause A.2.1 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | Only one FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| CP length | | Normal | |
| A3-Offset | dB | -3 | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | ON | DRX related parameters are defined in Table A.8.1.20.5-2 |
| si-RequestForHO | | TRUE | As specified in clause 5.5.3.1 in TS 36.331. |
| T1 | s | 5 | |
| T2 | s | ≤30 | UE should report cell within 25.6s (20 DRX cycles) |
| T3 | s | 5 | |

### 8.1.20.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE does not have any timing information of cell 2. Starting T2, cell 2 becomes detectable and the UE is expected to detect and send a measurement report.

In the test UE needs to be provided at least once every 1280ms with new Timing Advance Command MAC control element to restart the Time alignment timer to keep UE uplink time alignment. Furthermore UE sends scheduling request (SR) to obtain allocation to send the measurement report on PUSCH.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.1.20.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration with event A3 configured message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.20.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3.
7. SS shall transmit an RRCConnectionReconfiguration message during period T2, The RRC message shall create a measurement report configuration with purpose *reportCGI* and *si-RequestForHO* set to TRUE.
8. The SS shall start T3 timer when the last TTI containing the RRC message implying SI reading is sent to UE.
9. The UE shall transmit RRCConnectionReconfigurationComplete message.
10. UE shall transmit a measurement report containing the cell global identifier of cell 2 within 210 milliseconds from the start of T3. If the overall delays measured from the beginning of time period T3 is less than 210 ms then the number of successful tests is increased by one. If the UE fails to report the cell global identifier within the overall delays measured requirement then the number of failure tests is increased by one.

11. After the SS receive the MeasurementReport message in step 10) or when T3 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

13. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 2-13 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.1.20.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.1.20.4.3-1: Common Exception messages for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.1.20.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.20.4.3-3: *MAC-MainConfig-RBC*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf1280 typical value in real network for real-time services. | |
| sf1280 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| } | | | |

**Table 8.1.20.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.6, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | |
| } | | | |

**Table 8.1.20.4.3-5: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.1.20.4.3-6: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -6 (-3 dB) | -3 is actual value in dB (-6 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.20.4.3-7: *MeasResults*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.20.4.3-8: *MeasResultListEUTRA*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step6)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.1.20.4.3-9: *ReportConfigEUTRA-PERIODICAL*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportCGI | | |
| } | | | |
| } | | | |
| reportAmount | r1 | | |
| si-RequestForHO-r9 | setup | | |
| } | | | |

**Table 8.1.20.4.3-10: *MeasResultListEUTRA*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| cgi-Info SEQUENCE{ | | | |
| cellGlobalId SEQUENCE{ | | | |
| plmn-Identity | plmn-Identity | | |
| cellIdentity | cellIdentity | | |
| } | | | |
| trackingAreaCode | | | |
| plmn-IdentityList | | | |
| } | | | |
| } | | | |

**Table 8.1.20.4.3-11: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellForWhichToReportCGI | Physical Cell ID of Cell 2 | | |
| } | | | |

**Table 8.1.20.4.3-12: *MeasConfig-DEFAULT*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f1 | | |
| reportConfigId | IdReportConfig-P | | |
| } | | | |
| measGapConfig | Not present | | |
| } | | | |

**Table 8.1.20.4.3-13: *MeasResults*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

## 8.1.20.5 Test requirement

Tables 8.1.20.4.1-1, 8.1.20.5-1, 8.1.20.5-2 and 8.1.20.5-3 define the primary level settings including test tolerances for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0 test.

**Table 8.1.20.5-1: Cell specific test parameters for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | | 2x1 | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.1 FDD | OP.1 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD |
| PBCH_RA | dB | -3 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | 0 | | | 0 | | |
| PHICH_RA | dB | -3 | | | -3 | | |
| PHICH_PB | dB | | | | | | |
| PDCCH_RA | dB | 0 | | | 0 | | |
| PDCCH_PB | dB | | | | | | |
| PDSCH_RA | dB | -3 | | | -3 | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 8 | -3.3 | -3.3 | -Infinity | 2.36 | 2.36 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 8 | 8 | 8 | -Infinity | 11 | 11 |
| RSRP Note 3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| SCH_RP Note3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| Propagation Condition | | AWGN | | | | | |
| Timing offset to Cell 1 | ms | - | | | 3 | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.1.20.5-2: DRX configuration for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf1280 | |
| shortDRX | disable | |

**Table 8.1.20.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | Infinity | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213. |

The UE shall transmit a measurement report containing the cell global identifier of cell 2 within 210 milliseconds from the start of T3.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Test requirement = RRC Procedure delay + $T_{identify_CGI,\,intra}$ + TTI insertion uncertainty

RRC procedure delay=15ms

$$T_{identify_CGI,\,intra} = T_{basic_identify_CGI,\,intra} \quad ms$$

$T_{basic_identify_CGI,\,intra}$ = 190 ms

TTI insertion uncertainty = 2 ms

The overall delay measured is 207 ms, allow 210 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.21 E-UTRAN HD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps for UE category 0

### 8.1.21.1 Test purpose

To verify the requirement for identification of a new CGI of E-UTRA cell with autonomous gaps for UE category 0.

### 8.1.21.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 12 and forward with support of intra-frequency SI acquisition for HO of UE category 0.

### 8.1.21.3 Minimum conformance requirements

No explicit neighbour list is provided to the UE for identifying a new CGI of E-UTRA cell. The UE shall identify and report the CGI when requested by the network for the purpose 'reportCGI'. The UE may make autonomous gaps in downlink reception and uplink transmission for receiving MIB and SIB1 message according to clause 5.5.3.1 of TS 36.331 [5]. Note that a UE is not required to use autonomous gap if si-RequestForHO is set to false. If autonomous gaps are used for measurement with the purpose of 'reportCGI', regardless of whether DRX is used or not, or whether SCell(s) are configured or not, the UE shall be able to identify a new CGI of E-UTRA cell within:

$$T_{identify_CGI_LC\text{-}UE,\,intra} = T_{basic_identify_CGI_LC\text{-}UE,\,intra} \quad ms$$

Where

$T_{basic_identify_CGI_LC\text{-}UE,\ intra}$ = 190 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new CGI of an E-UTRA cell is defined, provided that the E-UTRA cell has been already identified by the UE.

A cell shall be considered identifiable following conditions are fulfilled:

- RSRP related side conditions given in TS 36.133 [4] Clause 9.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.2 for a corresponding Band

The MIB of an E-UTRA cell whose CGI is identified shall be considered decodable by the UE provided the PBCH demodulation requirements are met according to [TBD].

The requirement for identifying a new CGI of an E-UTRA cell within $T_{basic_identify_CGI_LC\text{-}UE,.intra}$ is applicable when no DRX is used as well as when all the DRX cycles specified in TS 36.331 [5] are used.

Within the time, $T_{identify_CGI_LC\text{-}UE,\ intra}$ *ms* , over which the UE identifies the new CGI of E-UTRA cell, the UE shall transmit at least 92 ACK/NACKs on PCell provided that:

- there is continuous DL data allocation,
- no DRX cycle is used,
- no measurement gaps are configured,
- only one code word is transmitted in each subframe,
- no MBSFN subframes are configured in the PCell or each of activated SCell(s).

The ECGI reporting delay occurs due to the delay uncertainty when inserting the ECGI measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. In case DRX is used, the ECGI reporting may be delayed until the next DRX cycle. If IDC autonomous denial is configured, an additional delay can be expected.

For the UE supporting half duplex FDD operation there is no requirement in terms of number of ACK/NACK transmission on PCell.

The normative reference for this requirement is TS 36.133 [4] clause 8.5.2.1.5 and A.8.1.21.

## 8.1.21.4 Test description

### 8.1.21.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20(using only main UE Tx/Rx antenna).
2. The general test parameter settings are set up according to Table 8.1.21.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.1.21.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.21.4.1-1: General test parameters for E-UTRAN HD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps for UE category 0**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.2 HD-FDD | As specified in clause A.1.4 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.4 HD-FDD | As specified in clause A.2.3 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | Only one FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| CP length | | Normal | |
| A3-Offset | dB | -3 | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| si-RequestForHO | | TRUE | As specified in clause 5.5.3.1 in TS 36.331. |
| T1 | s | 5 | |
| T2 | s | ≤10 | |
| T3 | s | 5 | |

### 8.1.21.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. PDCCHs indicating new transmissions shall be sent continuously to ensure that the UE would have ACK/NACK sending during identifying a new CGI of E-UTRAN cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE does not have any timing information of cell 2. Starting T2, cell 2 becomes detectable and the UE is expected to detect and send a measurement report.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.1.21.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration with event A3 configured message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.21.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3.
7. SS shall transmit an RRCConnectionReconfiguration message during period T2, The RRC message shall create a measurement report configuration with purpose *reportCGI* and *si-RequestForHO* set to TRUE.
8. The SS shall start T3 timer when the last TTI containing the RRC message implying SI reading is sent to UE.
9. The UE shall transmit RRCConnectionReconfigurationComplete message.
10. UE shall transmit a MeasurementReport message containing the cell global identifier of cell 2 within 210 milliseconds from the start of T3. The UE shall be scheduled continuously throughout the test, and from the start of T3 until 210ms at least 112 ACK/NACK shall be detected as being transmitted by the UE. If the overall delays measured from the beginning of time period T3 is less than 210 ms, and the UE have more than 112 ACK/NACKs transmitted from the start of T3 until 210 ms, then the number of successful tests is increased by one. If the UE fails to report the cell global identifier within the overall delays measured requirement, or the UE have less than 112 ACK/NACKs transmitted from the start of T3 until 210 ms, then the number of failure tests is increased by one.
11. After the SS receive the MeasurementReport message in step 10) or when T3 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

13. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 2-13 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.1.21.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.1.21.4.3-1: Common Exception messages for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.1.21.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -6 (-3 dB) | -3 is actual value in dB (-6 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.21.4.3-3: *MeasResults*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step6)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.21.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step6)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.1.21.4.3-5: *ReportConfigEUTRA-PERIODICAL*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportCGI | | |
| } | | | |
| } | | | |
| reportAmount | r1 | | |
| si-RequestForHO-r9 | setup | | |
| } | | | |

**Table 8.1.21.4.3-6: *MeasResultListEUTRA*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| cgi-Info SEQUENCE{ | | | |
| cellGlobalId SEQUENCE{ | | | |
| plmn-Identity | plmn-Identity | | |
| cellIdentity | cellIdentity | | |
| } | | | |
| trackingAreaCode | | | |
| plmn-IdentityList | | | |
| } | | | |
| } | | | |

**Table 8.1.21.4.3-7: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | |
| cellForWhichToReportCGI | Physical Cell ID of Cell 2 | | |
| } | | | |

**Table 8.1.21.4.3-8: *SystemInformationBlockType2*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-1 SystemInformationBlockType2 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfig | Not present | | Cell 1 |
| } | | | |

**Table 8.1.21.4.3-9: *SystemInformationBlockType3*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-2 SystemInformationBlockType3 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType3 ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1 |
| | '01'B (No MBSFN subframes are present in all neighbour cells) | | Cell 2 |
| } | | | |

**Table 8.1.21.4.3-10: *MeasConfig-DEFAULT*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f1 | | |
| reportConfigId | IdReportConfig-P | | |
| } | | | |
| measGapConfig | Not present | | |
| } | | | |

**Table 8.1.21.4.3-11: *MeasResults*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.21.4.3-12: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | |
| } | | | |

## 8.1.21.5 Test requirement

Tables 8.1.21.4.1-1 and 8.1.21.5-1 define the primary level settings including test tolerances for E-UTRAN HD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps for UE category 0 test.

**Table 8.1.21.5-1: Cell specific test parameters for E-UTRAN HD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps for UE category 0**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | | 2x1 | | |
| OCNG Patterns defined in D.1.10 (OP.10 FDD) and in D.1.2 (OP.2 FDD) | | OP.10 FDD | OP.10 FDD | OP.10 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD |
| PBCH_RA | dB | -3 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | 0 | | | 0 | | |
| PHICH_RA | dB | -3 | | | -3 | | |
| PHICH_PB | dB | | | | | | |
| PDCCH_RA | dB | 0 | | | 0 | | |
| PDCCH_PB | dB | | | | | | |
| PDSCH_RA | dB | -3 | | | -3 | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 8 | -3.3 | -3.3 | -Infinity | 2.36 | 2.36 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 8 | 8 | 8 | -Infinity | 11 | 11 |
| RSRP Note 3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| SCH_RP Note3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| Propagation Condition | | AWGN | | | | | |
| Timing offset to Cell 1 | ms | - | | | 3 | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall transmit a measurement report containing the cell global identifier of cell 2 within 210 milliseconds from the start of T3.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Test requirement = RRC Procedure delay + $T_{identify_CGI,\,intra}$ + TTI insertion uncertainty

RRC procedure delay=15ms

$$T_{identify_CGI,\,intra} = T_{basic_identify_CGI,\,intra} \quad ms$$

$T_{basic_identify_CGI,\,intra}$ = 190 ms

TTI insertion uncertainty = 2 ms

The overall delays measured is 207 ms, allow 210 ms. The UE shall be scheduled continuously throughout the test, and from the start of T3 until 210 ms at least 112 ACK/NACK shall be detected as being transmitted by the UE.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.22 E-UTRAN HD- FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0

### 8.1.22.1 Test purpose

To verify the requirement for identification of a new CGI of E-UTRA cell with autonomous gaps with DRX for UE category 0.

### 8.1.22.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 12 and forward with support of intra-frequency SI acquisition for HO. Applicability requires support for FGI bit 5 of UE category 0.

### 8.1.22.3 Minimum conformance requirements

No explicit neighbour list is provided to the UE for identifying a new CGI of E-UTRA cell. The UE shall identify and report the CGI when requested by the network for the purpose 'reportCGI'. The UE may make autonomous gaps in downlink reception and uplink transmission for receiving MIB and SIB1 message according to clause 5.5.3.1 of TS 36.331 [5]. Note that a UE is not required to use autonomous gap if si-RequestForHO is set to false. If autonomous gaps are used for measurement with the purpose of 'reportCGI', regardless of whether DRX is used or not, or whether SCell(s) are configured or not, the UE shall be able to identify a new CGI of E-UTRA cell within:

$$T_{identify_CGI_LC\text{-}UE,\ intra} = T_{basic_identify_CGI_LC\text{-}UE,\ intra} \quad ms$$

Where

$T_{basic_identify_CGI_LC\text{-}UE,\ intra}$ = 190 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new CGI of an E-UTRA cell is defined, provided that the E-UTRA cell has been already identified by the UE.

A cell shall be considered identifiable following conditions are fulfilled:

- RSRP related side conditions given in TS 36.133 [4] Section 9.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.2 for a corresponding Band

The MIB of an E-UTRA cell whose CGI is identified shall be considered decodable by the UE provided the PBCH demodulation requirements are met according to [TBD].

The requirement for identifying a new CGI of an E-UTRA cell within $T_{basic_identify_CGI_LC\text{-}UE,.intra}$ is applicable when no DRX is used as well as when all the DRX cycles specified in TS 36.331 [5] are used.

Within the time, $T_{identify_CGI_LC\text{-}UE,\ intra}$ $ms$, over which the UE identifies the new CGI of E-UTRA cell, the UE shall transmit at least 92 ACK/NACKs on PCell provided that:

- there is continuous DL data allocation,
- no DRX cycle is used,
- no measurement gaps are configured,
- only one code word is transmitted in each subframe,
- no MBSFN subframes are configured in the PCell or each of activated SCell(s).

The ECGI reporting delay occurs due to the delay uncertainty when inserting the ECGI measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. In case DRX is used, the ECGI reporting may be delayed until the next DRX cycle. If IDC autonomous denial is configured, an additional delay can be expected.

For the UE supporting half duplex FDD operation there is no requirement in terms of number of ACK/NACK transmission on PCell.

The normative reference for this requirement is TS 36.133 [4] clause 8.5.2.1.5 and A.8.1.22.

## 8.1.22.4 Test description

### 8.1.22.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20(using only main UE Tx/Rx antenna).
2. The general test parameter settings are set up according to Table 8.1.22.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.1.22.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.22.4.1-1: General test parameters for E-UTRAN HD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.2 HD-FDD | As specified in clause A.1.4 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.4 HD-FDD | As specified in clause A.2.3 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | Only one FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| CP length | | Normal | |
| A3-Offset | dB | -3 | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | ON | DRX related parameters are defined in Table A.8.1.6.1-3 |
| si-RequestForHO | | TRUE | As specified in clause 5.5.3.1 in TS 36.331. |
| T1 | s | 5 | |
| T2 | s | ≤30 | UE should report cell within 25.6s (20 DRX cycles) |
| T3 | s | 5 | |

### 8.1.22.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE does not have any timing information of cell 2. Starting T2, cell 2 becomes detectable and the UE is expected to detect and send a measurement report.

In the test UE needs to be provided at least once every 1280ms with new Timing Advance Command MAC control element to restart the Time alignment timer to keep UE uplink time alignment. Furthermore UE sends scheduling request (SR) to obtain allocation to send the measurement report on PUSCH.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.1.22.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration with event A3 configured message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.22.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3.
7. SS shall transmit an RRCConnectionReconfiguration message during period T2, The RRC message shall create a measurement report configuration with purpose *reportCGI* and *si-RequestForHO* set to TRUE.
8. The SS shall start T3 timer when the last TTI containing the RRC message implying SI reading is sent to UE.
9. The UE shall transmit RRCConnectionReconfigurationComplete message.
10. UE shall transmit a measurement report containing the cell global identifier of cell 2 within 210 milliseconds from the start of T3. If the overall delays measured from the beginning of time period T3 is less than 210 ms then the number of successful tests is increased by one. If the UE fails to report the cell global identifier within the overall delays measured requirement then the number of failure tests is increased by one.
11. After the SS receive the MeasurementReport message in step 10) or when T3 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
13. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
14. Repeat step 2-13 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.1.22.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.1.22.4.3-1: Common Exception messages for E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.1.22.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.22.4.3-3: *MAC-MainConfig-RBC*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf1280 typical value in real network for real-time services. | |
| sf1280 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| } | | | |

**Table 8.1.22.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.6, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| … | | | |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | |
| } | | | |

**Table 8.1.22.4.3-5: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.1.22.4.3-6: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -6 (-3 dB) | -3 is actual value in dB (-6 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.22.4.3-7: *MeasResults*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.22.4.3-8: *MeasResultListEUTRA*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step6)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.1.22.4.3-9: *ReportConfigEUTRA-PERIODICAL*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportCGI | | |
| } | | | |
| } | | | |
| reportAmount | 1 | | |
| si-RequestForHO-r9 | setup | | |
| } | | | |

**Table 8.1.22.4.3-10: *MeasResultListEUTRA*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| cgi-Info SEQUENCE{ | | | |
| cellGlobalId SEQUENCE{ | | | |
| plmn-Identity | plmn-Identity | | |
| cellIdentity | cellIdentity | | |
| } | | | |
| trackingAreaCode | | | |
| plmn-IdentityList | | | |
| } | | | |
| } | | | |

**Table 8.1.22.4.3-11: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellForWhichToReportCGI | Physical Cell ID of Cell 2 | | |
| } | | | |

**Table 8.1.22.4.3-12: *MeasConfig-DEFAULT*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f1 | | |
| reportConfigId | IdReportConfig-P | | |
| } | | | |
| measGapConfig | Not present | | |
| } | | | |

**Table 8.1.22.4.3-13: *MeasResults*: Additional E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

## 8.1.22.5 Test requirement

Tables 8.1.22.4.1-1, 8.1.22.5-1, 8.1.22.5-2 and 8.1.22.5-3 define the primary level settings including test tolerances for E-UTRAN HD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0 test.

**Table 8.1.22.5-1: Cell specific test parameters for E-UTRAN HD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | | 2x1 | | |
| OCNG Patterns defined in D.1.1(OP.1 FDD) and in D.1.2(OP.2 FDD) | | OP.1 FDD | OP.1 FDD | OP.1 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD |
| PBCH_RA | dB | -3 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | 0 | | | 0 | | |
| PHICH_RA | dB | -3 | | | -3 | | |
| PHICH_PB | dB | | | | | | |
| PDCCH_RA | dB | 0 | | | 0 | | |
| PDCCH_PB | dB | | | | | | |
| PDSCH_RA | dB | -3 | | | -3 | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 8 | -3.3 | -3.3 | -Infinity | 2.36 | 2.36 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 8 | 8 | 8 | -Infinity | 11 | 11 |
| RSRP Note 3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| SCH_RP Note3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| Propagation Condition | | AWGN | | | | | |
| Timing offset to Cell 1 | ms | - | | | 3 | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.1.22.5-2: DRX configuration for E-UTRAN HD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf1280 | |
| shortDRX | disable | |

**Table 8.1.22.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN HD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | Infinity | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213. |

The UE shall transmit a measurement report containing the cell global identifier of cell 2 within 210 milliseconds from the start of T3.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Test requirement = RRC Procedure delay + $T_{identify_CGI,\ intra}$ + TTI insertion uncertainty

RRC procedure delay=15ms

$$T_{identify_CGI,\ intra} = T_{basic_identify_CGI,\ intra} \quad ms$$

$T_{basic_identify_CGI,\ intra}$ = 190 ms

TTI insertion uncertainty = 2 ms

The overall delay measured is 207 ms, allow 210 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.23 E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeA

Editor's notes: This test case is incomplete. The following items are missing or incomplete:

- Test tolerances and test requirement values
- The cell mapping needs to be defined.
- Table I.2.x-1[TBD] needs to be added in Annex I.

### 8.1.23.1 Test purpose

To verify Cat-M1 UE's ability to make a correct reporting of an event under fading propagation conditions in asynchronous cells within the E-UTRA FDD-FDD intra-frequency cell search requirements for Cat-M1 UE.

### 8.1.23.2 Test applicability

This test applies to all types of E-UTRA FD-FDD UE release 13 and forward of UE category M1.

### 8.1.23.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify and measure a new detectable FDD intra frequency cell according to requirements in Table 8.1.23.3-1 when SCH Ês/Iot >= -6 dB.

**Table 8.1.23.3-1: Requirement on cell identification delay and measurement delay for FDD intra frequency cell**

| Gap pattern ID | Cell identification delay ($T_{identify_intra_UE\ cat\ M1}$) | Measurement delay ($T_{measure_intra_UE\ cat\ M1}$) |
|---|---|---|
| 0 | 1.44 seconds | 480 ms |
| 1 | 2.88 seconds | 960 ms |

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] Clauses 9.1.21.1 and 9.1.21.2 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex Table I.2.x-1[TBD] for a corresponding Band.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{measure_intra_UE\ cat\ M1}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is according to Table 8.1.23.3-1. When measurement gaps are activated the UE shall be capable of performing measurements for at least 6 cells. If the UE has identified more than 6 cells, the UE shall perform measurements but the reporting rate of RSRP measurement of cells from UE physical layer to higher layers may be decreased.

Reported RSRP measurement contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] Clauses 9.1.21.1 and 9.1.21.2.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ intra_UE\ cat\ M1}$ defined in Clause 8.1.23.3. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_UE\ cat\ M1}$ defined in Clause 8.1.23.3 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_UE\ cat\ M1,\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clauses 8.13.2.1.1.1 and A.8.1.23.

## 8.1.23.4 Test description

### 8.1.23.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.15 using only main UE Tx/Rx antenna.
2. The general test parameter settings are set up according to Table 8.1.23.4.1-1.
3. Propagation conditions are set according to Annex B.0.

4. Message contents are defined in clause 8.1.23.4.3.

5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.23.4.1-1: General test parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeA**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | One radio channel is used. |
| Active cell | | | Cell 1 | |
| Neighbour cell | | | Cell 2 | Cell to be identified. |
| CP length | | | Normal | |
| DRX | | | OFF | |
| A3 | Offset | dB | -6 | |
| | Hysteresis | dB | 0 | |
| | Time To Trigger | S | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| Gap pattern ID | | | 1 | |
| T1 | | S | 5 | |
| T2 | | S | 5 | |

### 8.1.23.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information, it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA.

2. Set the parameters according to T1 in Table 8.1.23.5-1. Propagation conditions are set according to Annex B.1 and B.2. T1 starts.

3. The SS shall transmit the RRCConnectionReconfiguration message.

4. The UE shall transmit the RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.23.5-1.

6. The UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than [2880+TT] ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receives the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit the RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A-CE with the condition CEModeA according to TS 36.508 [7] clause 4.5.3AB (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA.

10. Repeat steps 2-9 until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

### 8.1.23.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the condition CEModeA and the following exceptions:

**Table 8.1.23.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeA test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2a<br>Table H.3.1-7 |

**Table 8.1.23.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeA test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.1.23.4.3-3: *MeasResults*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeA test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServiCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.23.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeA test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.1.23.5 Test requirement

Tables 8.1.23.4.1-1 and 8.1.23.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-FDD event triggered reporting under fading propagation conditions in asynchronous cells test.

**Table 8.1.23.5-1: Cell specific test parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeA**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | | |
| $BW_{channel}$ | MHz | 10 | | | |
| PDSCH parameters: DL Reference Measurement Channel | | R.20 FDD | | - | |
| MPDCCH parameters: DL Reference Measurement Channel | | R.16 FDD | | R.16 FDD | |
| OCNG Patterns | | OP.21 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| MPDCCH_RA | dB | | | | |
| MPDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98+TT | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4+TT | 4+TT | -Infinity | 4+TT |
| $\hat{E}_s / I_{ot}$ Note 3 | dB | 4+TT | -1.46+TT | -Infinity | -1.46+TT |
| RSRP Note 3 | dBm/15 KHz | -94+TT | -94+TT | -Infinity | -94+TT |
| SCH_RP Note 3 | dBm/15 KHz | -94+TT | -94 | -Infinity | -94+TT |
| Io Note 3 | dBm/9MHz | -64.76+TT | -62.42+TT | Specified in Cell 1 columns | |
| Propagation Condition | | ETU70 | | ETU70 | |
| Antenna Configuration | | 1x1 | | 1x1 | |
| Timing offset to Cell 1 | ms | - | | 3 | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: Es/Iot, RSRP, SCH_RP and Io have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra_UE\ cat\ M1}$

$T_{identify_intra_UE\ cat\ M1}$ = 2880 ms

TTI insertion uncertainty = TT ms

The overall delays measured shall be less than a total of 2880+TT ms in this test case (note: this gives a total of 2880 ms for measurement reporting delay plus TT ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.24 E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA

Editor's notes: This test case is incomplete. The following items are missing or incomplete:

- Test tolerances and test requirement values
- The cell mapping needs to be defined.
- Table I.2.x-1[TBD] needs to be added in Annex I.

### 8.1.24.1 Test purpose

To verify Cat-M1 UE's ability to make a correct reporting of an event under fading propagation conditions in synchronous cells within the E-UTRA FDD-FDD intra-frequency cell search requirements for Cat-M1 UE.

### 8.1.24.2 Test applicability

This test applies to all types of E-UTRA FD-FDD UE release 13 and forward of UE category M1.

### 8.1.24.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify and measure a new detectable FDD intra frequency cell according to requirements in Table 8.1.24.3-1 when SCH Ês/Iot >= -6 dB.

**Table 8.1.24.3-1: Requirement on cell identification delay and measurement delay for FDD intra frequency cell**

| Gap pattern ID | Cell identification delay ($T_{identify_intra_UE\ cat\ M1}$) | Measurement delay ($T_{measure_intra_UE\ cat\ M1}$) |
|---|---|---|
| 0 | 1.44 seconds | 480 ms |
| 1 | 2.88 seconds | 960 ms |

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] Sections 9.1.21.1 and 9.1.21.2 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex Table I.2.x-1[TBD] for a corresponding Band.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{measure_intra_UE\ cat\ M1}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is according to Table 8.1.24.3-1. When measurement gaps are activated the UE shall be capable of performing measurements for at least 6 cells. If the UE has identified more than 6 cells, the UE shall perform measurements but the reporting rate of RSRP measurement of cells from UE physical layer to higher layers may be decreased.

Reported RSRP measurement contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.21.1 and 9.1.21.2.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ intra_UE\ cat\ M1}$ defined in Clause 8.1.24.3. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_UE\ cat\ M1}$ defined in Clause 8.1.24.3 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_UE\ cat\ M1,\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.13.2.1.1.1 and A.8.1.24.

## 8.1.24.4 Test description

### 8.1.24.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15 using only main UE Tx/Rx antenna.
2. The general test parameter settings are set up according to Table 8.1.24.4.1-1.
3. Propagation conditions are set according to Annex B.0.
4. Message contents are defined in clause 8.1.24.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.24.1-1: General test parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | One radio channel is used. |
| Active cell | | | Cell 1 | |
| Neighbour cell | | | Cell 2 | Cell to be identified. |
| CP length | | | Normal | |
| DRX | | | OFF | |
| A3 | Offset | dB | -6 | |
| | Hysteresis | dB | 0 | |
| | Time To Trigger | S | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| Gap pattern ID | | | 1 | |
| T1 | | S | 5 | |
| T2 | | S | 5 | |

### 8.1.24.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information, it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA.
2. Set the parameters according to T1 in Table 8.1.24.5-1. Propagation conditions are set according to Annex B clauses B.1 and B.2. T1 starts.
3. The SS shall transmit the RRCConnectionReconfiguration message.
4. The UE shall transmit the RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.24.5-1.
6. The UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than [2880+TT] ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.
7. After the SS receives the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit the RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A-CE with the condition CEModeA according to TS 36.508 [7] clause 4.5.3AB (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA.
10. Repeat steps 2-9 until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

### 8.1.24.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the condition CEModeA and the following exceptions:

**Table 8.1.24.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2a<br>Table H.3.1-7 |

**Table 8.1.24.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.1.24.4.3-3: *MeasResults*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA test requirement**

| Derivation Path: 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServiCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.24.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA test requirement**

| Derivation Path: 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.1.24.5 Test requirement

Tables 8.1.24.4.1-1 and 8.1.24.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-FDD event triggered reporting under fading propagation conditions in synchronous cells test.

**Table 8.1.24.5-1: Cell specific test parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | | |
| $BW_{channel}$ | MHz | 10 | | | |
| PDSCH parameters: DL Reference Measurement Channel | | R.20 FDD | | - | |
| MPDCCH parameters: DL Reference Measurement Channel | | R.16 FDD | | R.16 FDD | |
| OCNG Patterns | | OP.21 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| MPDCCH_RA | dB | | | | |
| MPDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4+TT | 4+TT | -Infinity | 4+TT |
| $\hat{E}_s / I_{ot}$ Note 3 | dB | 4+TT | -1.46+TT | -Infinity | -1.46+TT |
| RSRP Note 3 | dBm/15 KHz | -94+TT | -94+TT | -Infinity | -94+TT |
| SCH_RP Note 3 | dBm/15 KHz | -94+TT | -94+TT | -Infinity | -94+TT |
| Io Note 3 | dBm/9MHz | -64.76+TT | -62.42+TT | Specified in Cell 1 columns | |
| Propagation Condition | | ETU70 | | ETU70 | |
| Antenna Configuration | | 1x1 | | 1x1 | |
| Timing offset to Cell 1 | µs | - | | 3 | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: Es/Iot, RSRP, SCH_RP and Io have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra_UE\ cat\ M1}$

$T_{identify_intra_UE\ cat\ M1}$ = 2880 ms

TTI insertion uncertainty = TT ms

The overall delays measured shall be less than a total of 2880+TT ms in this test case (note: this gives a total of 2880 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.25 E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA in DRX

Editor's notes: This test case is incomplete. The following items are missing or incomplete:

- Test tolerances and test requirement values
- The cell mapping needs to be defined.
- Table I.2.x-1[TBD] needs to be added in Annex I.

### 8.1.25.1 Test purpose

To verify Cat-M1 UE's ability to make a correct reporting of an event in DRX under fading propagation conditions in asynchronous cells within the E-UTRA FDD-FDD intra-frequency cell search requirements in DRX for Cat-M1 UE.

### 8.1.25.2 Test applicability

This test applies to all types of E-UTRA FD-FDD UE release 13 and forward of UE category M1. Applicability requires support for FGI bit 5.

### 8.1.25.3 Minimum conformance requirements

When DRX is in use the UE shall be able to identify a new detectable FDD intra frequency cell within $T_{identify_intra_UE\ cat\ M1}$ as shown in Table 8.1.25.3-1.

When eDRX_CONN is in use the UE shall be able to identify a new detectable FDD intra frequency cell within $T_{identify_intra_UE\ cat\ M1}$ as shown in Table 8.1.25.3-2.

**Table 8.1.25.3-1: Requirement to identify a newly detectable FDD intra frequency cell**

| Gap pattern ID | DRX cycle length (s) | $T_{identify_intra_UE\ cat\ M1}$ (s) (DRX cycles) |
|---|---|---|
| 0 | ≤0.04 | 1.44 (Note1) |
| | 0.04<DRX-cycle≤0.08 | Note2 (40) |
| | 0.128 | 3.2 (25) |
| | 0.128<DRX-cycle≤2.56 | Note2(20) |
| 1 | <0.128 | 2.88 (Note1) |
| | 0.128 | 3.2 (25) |
| | 0.128<DRX-cycle≤2.56 | Note2(20) |
| Note1: Number of DRX cycle depends upon the DRX cycle in use.<br>Note2: Time depends upon the DRX cycle in use. | | |

**Table 8.1.25.3-2: Requirement to identify a newly detectable FDD intra frequency cell when eDRX_CONN cycle is used**

| eDRX_CONN cycle length (s) | $T_{identify_intra_UE\ cat\ M1}$ (s) (eDRX_CONN cycles) |
|---|---|
| 2.56<eDRX_CONN cycle≤10.24 | Note (20) |
| NOTE: Time depends upon the eDRX_CONN cycle in use. | |

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] Sections 9.1.21.1 and 9.1.21.2 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex Table I.2.x-1[TBD] for a corresponding Band

In the RRC_CONNECTED state the measurement period for intra frequency measurements is $T_{measure_intra_UE\ cat\ M1}$. When DRX is used, $T_{measure_intra_UE\ cat\ M1}$ is as specified in Table 8.1.25.3-3. When eDRX_CONN is used, $T_{measure_intra_UE\ cat\ M1}$ is as specified in Table 8.1.25.3-4. The UE shall be capable of performing RSRP measurements for 6 identified-intra-frequency cells, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_intra_UE\ cat\ M1}$.

**Table 8.1.25.3-3: Requirement to measure FDD intra frequency cells**

| Gap pattern ID | DRX cycle length (s) | $T_{measure_intra_UE\ cat\ M1}$ (s) (DRX cycles) |
|---|---|---|
| 0 | <0.128 | 0.48 (Note1) |
| | 0.128≤DRX-cycle≤2.56 | Note2 (5) |
| 1 | <0.256 | 0.96 (Note1) |
| | 0.256≤DRX-cycle≤2.56 | Note2 (5) |
| Note1: Number of DRX cycle depends upon the DRX cycle in use.<br>Note2: Time depends upon the DRX cycle in use. | | |

**Table 8.1.25.3-4: Requirement to measure FDD intra frequency cells when eDRX_CONN cycle is used**

| eDRX_CONN cycle length (s) | $T_{measure_intra_UE\ cat\ M1}$ (s) (eDRX_CONN cycles) |
|---|---|
| 2.56<eDRX_CONN cycle≤10.24 | Note (5) |
| NOTE: Time depends upon the eDRX_CONN cycle in use. | |

Reported RSRP measurement contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.21.1 and 9.1.21.2.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra,\ UE\ cat\ M1}$ defined in Clause 8.1.25.3. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_UE\ cat\ M1}$ defined in clause 8.1.25.3 becomes undetectable for a period ≤ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_intra_UE\ cat\ M1}$ provided the timing to that cell has not

changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.13.2.1.1.2 and A.8.1.25.

## 8.1.25.4 Test description

### 8.1.25.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.15 using only main UE Tx/Rx antenna.

2. The general test parameter settings are set up according to Table 8.1.25.4.1-1.

3. Propagation conditions are set according to Annex B.0.

4. Message contents are defined in clause 8.1.25.4.3.

5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.25.4.1-1: General test parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE when DRX is used**

| Parameter | | Unit | Value | | Comment |
|---|---|---|---|---|---|
| | | | Test1 | Test2 | |
| E-UTRA RF Channel Number | | | 1 | 1 | One radio channel is used. |
| Active cell | | | Cell 1 | Cell 1 | |
| Neighbour cell | | | Cell 2 | Cell 2 | Cell to be identified. |
| CP length | | | Normal | Normal | |
| DRX | | | ON | ON | DRX related parameters are defined in Table 8.1.25.5-2 |
| A3 | Offset | dB | -6 | -6 | |
| | Hysteresis | dB | 0 | 0 | |
| | Time To Trigger | s | 0 | 0 | |
| Filter coefficient | | | 0 | 0 | L3 filtering is not used |
| Gap pattern ID | | | 0 | 0 | |
| T1 | | s | 5 | 5 | |
| T2 | | s | 5 | 30 | |

### 8.1.25.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information, it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

In Test 1 when DRX = 40 ms is used, UE needs to be provided at least once every 500 ms with new Timing Advance Command MAC control element to restart the Timer Alignment Timer to keep the UE uplink time alignment. Furthermore, the UE is allocated with PUSCH resource at every DRX cycle. In Test 2 when DRX = 1280 ms is used, the uplink time alignment is not maintained and the UE needs to use RACH to obtain uplink allocation for measurement reporting.

1. Ensure the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA.

2. Set the parameters according to T1 in Table 8.1.25.5-1. Propagation conditions are set according to Annex B.1 and B.2. T1 starts.

3. The SS shall transmit the RRCConnectionReconfiguration message.

4. The UE shall transmit the RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Tables 8.1.25.5-1 and 8.1.25.5-2.

6. The UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than [1480+TT] ms for Test 1 or less than [26880+TT] ms for Test 2 then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receives the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit the RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A-CE with the condition CEModeA according to TS 36.508 [7] clause 4.5.3AB (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA.

10. Repeat steps 2-9 until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved.

11. Repeat steps 1-10 for each sub-test Table 8.1.25.4.1-1 as appropriate.

### 8.1.25.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the condition CEModeA and the following exceptions:

**Table 8.1.25.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CE in DRX test 1 and 2 requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-2a<br>Table H.3.1-7<br>Table H.3.7-1<br>Table H.3.7-2<br>Table H.3.7-3 |

**Table 8.1.25.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CE in DRX test 1 and 2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.1.25.4.3-3: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CE in DRX test 1 and 2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventide CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.1.25.4.3-4: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CE in DRX test 1 and 2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.1.25.4.3-5: *MeasResults*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CE in DRX test 1 and 2 requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.25.4.3-6: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CE in DRX test 1 and 2 requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| physCellId | PhysCellId | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.1.25.5 Test requirement

Tables 8.1.25.4.1-1, 8.1.25.5-1, 8.1.25.5-2, and 8.1.25.5-3 define the primary level settings including test tolerances for E-UTRAN FDD-FDD event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CE in DRX.

**Table 8.1.25.5-1: Cell specific test parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE when DRX is used**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | | |
| $BW_{channel}$ | MHz | 10 | | | |
| PDSCH parameters: DL Reference Measurement Channel | | R.20 FDD | | - | |
| MPDCCH parameters: DL Reference Measurement Channel | | R.16 FDD | | R.16 FDD | |
| OCNG Patterns | | OP.21 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| MPDCCH_RA | dB | | | | |
| MPDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98+TT | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4+TT | 4+TT | -Infinity | 4+TT |
| $\hat{E}_s / I_{ot}$ Note 3 | dB | 4+TT | -1.46+TT | -Infinity | -1.46+TT |
| RSRP Note 3 | dBm/15 KHz | -94+TT | -94+TT | -Infinity | -94+TT |
| SCH_RP Note 3 | dBm/15 KHz | -94+TT | -94+TT | -Infinity | -94+TT |
| Io Note 3 | dBm/9MHz | -64.76+TT | -62.42+TT | Specified in Cell 1 columns | |
| Propagation Condition | | ETU70 | | ETU70 | |
| Antenna Configuration | | 1x1 | | 1x1 | |
| Timing offset to Cell 1 | µs | - | | 3 | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: Es/Iot, RSRP, SCH_RP and Io have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

**Table 8.1.25.5-2: DRX-Configuration for E-UTRAN FDD-FDD intra-frequency event triggered reporting in DRX under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf40 | sf1280 | |
| shortDRX | disable | disable | |

**Table 8.1.25.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD-FDD intra-frequency event triggered reporting in DRX under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | 0 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213. |

In Test 1 when DRX cycle length = 40 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2 on PUSCH.

In Test 2 when DRX cycle length = 1280 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE starts to send preambles on the PRACH for Scheduling Request (SR) to obtain allocation to send the measurement report to Cell 2 on PUSCH.

For both tests:

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event A3 measurement report.

The overall delay measured when DRX cycle length is 40 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra_UE\ cat\ M1}$

$T_{identify_intra_UE\ cat\ M1}$ = 1440 ms. When DRX cycle length is 40 ms then the $T_{identify_intra_UE\ cat\ M1}$ is 1440 ms.

TTI insertion uncertainty = TT ms

DRX cycle length = 40 ms

The overall delay measured when DRX cycle length is 40 ms shall be less than a total of 1480+TT ms.

The overall delay measured when DRX cycle length is 1280 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra_UE\ cat\ M1}$

$T_{identify_intra_UE\ cat\ M1}$ = 25600 ms. When DRX cycle length is 1280 ms then the $T_{identify_intra_UE\ cat\ M1}$ is 20 x 1280 ms.

TTI insertion uncertainty = TT ms

DRX cycle length = 1280 ms

The overall delay measured when DRX cycle length is 1280 ms shall be less than a total of 26880+TT ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.26 E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeA

Editor's notes: This test case is incomplete. The following items are missing or incomplete:

- Test tolerances and test requirement values
- Annex E needs to be updated

### 8.1.26.1 Test purpose

To verify Cat-M1 UE's ability to make a correct reporting of an event under fading propagation conditions in asynchronous cells within the E-UTRA HD-FDD intra-frequency cell search requirements for Cat-M1 UE.

### 8.1.26.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE Release 13 and forward of UE category M1.

### 8.1.26.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify and measure a new detectable FDD intra frequency cell according to requirements in Table 8.1.26.3-1 when SCH Ês/Iot >= -6 dB

**Table 8.1.26.3-1: Requirement on cell identification delay and measurement delay for FDD intrafrequency cell**

| Gap pattern ID | Cell identification delay ($T_{identify_intra_UE\ cat\ M1}$) | Measurement delay ($T_{measure_intra_UE\ cat\ M1}$) |
|---|---|---|
| 0 | 1.44 seconds | 480 ms |
| 1 | 2.88 seconds | 960 ms |

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] Clauses 9.1.21.1 and 9.1.21.2 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to TS 36.133 [4] Table B.2.14-2 for a corresponding Band.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{measure_intra_UE\ cat\ M1}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is according to Table 8.1.26.3-1. When measurement gaps are activated the UE shall be capable of performing measurements for at least 6 cells. If the UE has identified more than 6 cells, the UE shall perform measurements but the reporting rate of RSRP measurement of cells from UE physical layer to higher layers may be decreased.

Reported RSRP measurement contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] Clauses 9.1.21.1 and 9.1.21.2.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$.This measurement reporting delay excludes a delay which caused by no UL resoureces for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ intra_UE\ cat\ M1}$ defined in Clause 8.1.26.3. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_UE\ cat\ M1}$ defined in Clause 8.1.26.3 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_UE\ cat\ M1,\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clauses 8.13.2.1.2.1 and A.8.1.26.

## 8.1.26.4 Test description

### 8.1.26.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (Node B emulator), faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.15 using only main UE Tx/Rx antenna.
2. The general test parameter settings are set up according to Table 8.1.26.4.1-1.
3. Propagation conditions are set according to Annex B.0.
4. Message contents are defined in clause 8.1.26.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.26.4.1-1: General test parameters for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeA**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | One radio channel is used. |
| Active cell | | | Cell 1 | |
| Neighbour cell | | | Cell 2 | Cell to be identified. |
| CP length | | | Normal | |
| DRX | | | OFF | |
| A3 | Offset | dB | -6 | |
| | Hysteresis | dB | 0 | |
| | Time To Trigger | S | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| Gap pattern ID | | | 1 | |
| T1 | | S | 5 | |
| T2 | | S | 5 | |

### 8.1.26.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information, it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA.
2. Set the parameters according to T1 in Table 8.1.26.5-1. Propagation conditions are set according to Annex B.1. T1 starts.

3. The SS shall transmit the RRCConnectionReconfiguration message.
4. The UE shall transmit the RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.26.5-1.
6. The UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than [2882+TT] ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.
7. After the SS receives the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit the RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A-CE with the condition CEModeA according to TS 36.508 [7] clause 4.5.3AB (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA.
10. Repeat steps 2-9 until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

### 8.1.26.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the condition CEModeA and the following exceptions:

**Table 8.1.26.4.3-1: Common Exception messages for E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeA test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2a<br>Table H.3.1-7 |

**Table 8.1.26.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeA test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.1.26.4.3-3: *MeasResults*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeA test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServiCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.26.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeA test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.1.26.5 Test requirement

Tables 8.1.26.4.1-1 and 8.1.26.5-1 define the primary level settings including test tolerances for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells.

**Table 8.1.26.5-1: Cell specific test parameters for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeA**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | | |
| $BW_{channel}$ | MHz | 10 | | | |
| PDSCH parameters: DL Reference Measurement Channel | | R.20 HD-FDD | | - | |
| MPDCCH parameters: DL Reference Measurement Channel | | R.16 HD-FDD | | R.16 HD-FDD | |
| OCNG Patterns | | OP.21 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| MPDCCH_RA | dB | | | | |
| MPDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98+TT | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4+TT | 4+TT | -Infinity | 4+TT |
| $\hat{E}_s / I_{ot}$ Note 3 | dB | 4+TT | -1.46+TT | -Infinity | -1.46+TT |
| RSRP Note 3 | dBm/15 KHz | -94+TT | -94+TT | -Infinity | -94+TT |
| SCH_RP Note 3 | dBm/15 KHz | -94+TT | -94+TT | -Infinity | -94+TT |
| Io Note 3 | dBm/9MHz | -64.76+TT | -62.42+TT | Specified in Cell 1 columns | |
| Propagation Condition | | ETU70 | | ETU70 | |
| Antenna Configuration | | 1x1 | | 1x1 | |
| Timing offset to Cell 1 | ms | - | | 3 | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: Es/Iot, RSRP, SCH_RP and Io have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra_UE\ cat\ M1}$

$T_{identify_intra_UE\ cat\ M1}$ = [2880+TT]

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of [2880+TTI insertion uncertainty+TT] ms in this test case (note: this gives a total of 2880 ms for measurement reporting delay plus TT ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.27 E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA

Editor's notes: This test case is incomplete. The following items are missing or incomplete:

- Test tolerances and test requirement values
- Annex E needs to be updated

### 8.1.27.1 Test purpose

To verify Cat-M1 UE's ability to make a correct reporting of an event under fading propagation conditions in synchronous cells within the E-UTRA HD-FDD intra-frequency cell search requirements for Cat-M1 UE.

### 8.1.27.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE Release 13 and forward of UE category M1.

### 8.1.27.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify and measure a new detectable FDD intra frequency cell according to requirements in Table 8.1.27.3-1 when SCH Ês/Iot >= -6 dB

**Table 8.1.27.3-1: Requirement on cell identification delay and measurement delay for FDD intrafrequency cell**

| Gap pattern ID | Cell identification delay ($T_{identify_intra_UE\ cat\ M1}$) | Measurement delay ($T_{measure_intra_UE\ cat\ M1}$) |
|---|---|---|
| 0 | 1.44 seconds | 480 ms |
| 1 | 2.88 seconds | 960 ms |

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] Sections 9.1.21.1 and 9.1.21.2 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to TS 36.133 [4] Table B.2.14-2 for a corresponding Band.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{measure_intra_UE\ cat\ M1}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is according to Table 8.1.27.3-1. When measurement gaps are activated the UE shall be capable of performing measurements for at least 6 cells. If the UE has identified more than 6 cells, the UE shall perform measurements but the reporting rate of RSRP measurement of cells from UE physical layer to higher layers may be decreased.

Reported RSRP measurement contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.21.1 and 9.1.21.2.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$.This measurement reporting delay excludes a delay which caused by no UL resoureces for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ intra_UE\ cat\ M1}$ defined in Clause 8.1.27.3. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_UE\ cat\ M1}$ defined in Clause 8.1.27.3 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_UE\ cat\ M1,\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.13.2.1.2.1 and A.8.1.27.

## 8.1.27.4 Test description

### 8.1.27.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (Node B emulator) , faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.15 using only main UE Tx/Rx antenna.
2. The general test parameter settings are set up according to Table 8.1.27.4.1-1.
3. Propagation conditions are set according to Annex B.0.
4. Message contents are defined in clause 8.1.27.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.27.1-1: General test parameters for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | One radio channel is used. |
| Active cell | | | Cell 1 | |
| Neighbour cell | | | Cell 2 | Cell to be identified. |
| CP length | | | Normal | |
| DRX | | | DRX_L | As specified in TS 36.133 [4] clause A.3.3 |
| A3 | Offset | dB | -6 | |
| | Hysteresis | dB | 0 | |
| | Time To Trigger | S | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| Gap pattern ID | | | 1 | As specified in TS 36.133 [4] clause 8.1.2.1 |
| T1 | | S | 5 | |
| T2 | | S | 5 | |

### 8.1.27.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information, it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA.
2. Set the parameters according to T1 in Table 8.1.27.5-1. Propagation conditions are set according to Annex B clauses B.1. T1 starts.
3. The SS shall transmit the RRCConnectionReconfiguration message.
4. The UE shall transmit the RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.27.5-1.
6. The UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than [2882+TT] ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.
7. After the SS receives the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit the RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A-CE with the condition CEModeA according to TS 36.508 [7] clause 4.5.3AB (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA.
10. Repeat steps 2-9 until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

### 8.1.27.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the condition CEModeA and the following exceptions:

**Table 8.1.27.4.3-1: Common Exception messages for E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2a<br>Table H.3.1-7<br>Table H.3.6-2 |

**Table 8.1.27.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.1.27.4.3-3: *MeasResults*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA test requirement**

| Derivation Path: 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServiCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.27.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA test requirement**

| Derivation Path: 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.1.27.5 Test requirement

Tables 8.1.27.4.1-1 and 8.1.27.5-1 define the primary level settings including test tolerances for E-UTRAN HD-FDD event triggered reporting under fading propagation conditions in synchronous cells.

**Table 8.1.27.5-1: Cell specific test parameters for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | | |
| $BW_{channel}$ | MHz | 10 | | | |
| PDSCH parameters: DL Reference Measurement Channel | | R.20 FDD | | - | |
| MPDCCH parameters: DL Reference Measurement Channel | | R.16 FDD | | R.16 FDD | |
| OCNG Patterns | | OP.21 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| MPDCCH_RA | dB | | | | |
| MPDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4+TT | 4+TT | -Infinity | 4+TT |
| $\hat{E}_s / I_{ot}$ Note 3 | dB | 4+TT | -1.46+TT | -Infinity | -1.46+TT |
| RSRP Note 3 | dBm/15 KHz | -94+TT | -94+TT | -Infinity | -94+TT |
| SCH_RP Note 3 | dBm/15 KHz | -94+TT | -94+TT | -Infinity | -94+TT |
| Io Note 3 | dBm/9MHz | -64.76+TT | -62.42+TT | Specified in Cell 1 columns | |
| Propagation Condition | | ETU70 | | ETU70 | |
| Antenna Configuration | | 1x1 | | 1x1 | |
| Timing offset to Cell 1 | μs | - | | 3 | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: Es/Iot, RSRP, SCH_RP and Io have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra_UE\ cat\ M1}$

$T_{identify_intra_UE\ cat\ M1}$ = [2880+TT]

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of [2880+TTI insertion uncertainty +TT] ms in this test case (note: this gives a total of 2880 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.28 E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA in DRX

Editor's notes: This test case is incomplete. The following items are missing or incomplete:

- Test tolerances and test requirement values
- Annex E needs to be updated

### 8.1.28.1 Test purpose

To verify Cat-M1 UE's ability to make a correct reporting of an event in DRX under fading propagation conditions in asynchronous cells within the E-UTRA HD-FDD intra-frequency cell search requirements in DRX for Cat-M1 UE.

### 8.1.28.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE Release 13 and forward of UE category M1.

### 8.1.28.3 Minimum conformance requirements

When DRX is in use the UE shall be able to identify a new detectable HD-FDD intra frequency cell within $T_{identify_intra_UE\ cat\ M1}$ as shown in Table 8.1.28.3-1.

When eDRX_CONN is in use the UE shall be able to identify a new detectable HD-FDD intra frequency cell within $T_{identify_intra_UE\ cat\ M1}$ as shown in Table 8.1.28.3-2.

**Table 8.1.28.3-1: Requirement to identify a newly detectable HD-FDD intrafrequency cell**

| Gap pattern ID | DRX cycle length (s) | $T_{identify_intra_UE\ cat\ M1}$ (s) (DRX cycles) |
|---|---|---|
| 0 | ≤0.04 | 1.44 (Note1) |
| | 0.04<DRX-cycle≤0.08 | Note2 (40) |
| | 0.128 | 3.2 (25) |
| | 0.128<DRX-cycle≤2.56 | Note2(20) |
| 1 | ≤0.08 | 2.88 (Note1) |
| | 0.128 | 3.2 (32) |
| | 0.128<DRX-cycle≤2.56 | Note2(25) |
| Note1: Number of DRX cycle depends upon the DRX cycle in use<br>Note2: Time depends upon the DRX cycle in use | | |

**Table 8.1.28.3-2: Requirement to identify a newly detectable HD-FDD intrafrequency cell when eDRX_CONN cycle is used**

| eDRX_CONN cycle length (s) | $T_{identify_intra_UE\ cat\ M1}$ (s) (eDRX_CONN cycles) |
|---|---|
| 2.56<eDRX_CONN cycle≤10.24 | Note (25) |
| NOTE: Time depends upon the eDRX_CONN cycle in use | |

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] Sections 9.1.21.1 and 9.1.21.2 are fulfilled for a corresponding Band,

- SCH_RP and SCH Ês/Iot according to TS 36.133 [4] Table B.2.14-2 for a corresponding Band

In the RRC_CONNECTED state the measurement period for intra frequency measurements is $T_{measure_intra_UE\ cat\ M1}$. When DRX is used, $T_{measure_intra_UE\ cat\ M1}$ is as specified in Table 8.1.28.3-3. When eDRX_CONN is used, $T_{measure_intra_UE\ cat\ M1}$ is as specified in Table 8.1.28.3-4. The UE shall be capable of performing RSRP measurements for 6 identified-intra-frequency cells, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_intra_UE\ cat\ M1}$.

**Table 8.1.28.3-3: Requirement to measure HD-FDD intrafrequency cells**

| Gap pattern ID | DRX cycle length (s) | $T_{measure_intra_UE\ cat\ M1}$ (s) (DRX cycles) |
|---|---|---|
| 0 | <0.08 | 0.48 (Note1) |
| | 0.08≤DRX-cycle≤0.16 | Note2 (7) |
| | 0.16<DRX-cycle≤2.56 | Note2(5) |
| 1 | <0.16 | 0.96 (Note1) |
| | DRX-cycle=0.16 | 1.12 (7) |
| | 0.16<DRX-cycle≤2.56 | Note2(5) |
| Note1: Number of DRX cycle depends upon the DRX cycle in use<br>Note2: Time depends upon the DRX cycle in use | | |

**Table 8.1.28.3-4: Requirement to measure HD-FDD intrafrequency cells when eDRX_CONN cycle is used**

| eDRX_CONN cycle length (s) | $T_{measure_intra_UE\ cat\ M1}$ (s) (eDRX_CONN cycles) |
|---|---|
| 2.56<eDRX_CONN cycle≤10.24 | Note (5) |
| NOTE: Time depends upon the eDRX_CONN cycle in use | |

Reported RSRP measurement contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.21.1 and 9.1.21.2.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$.This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra,\ UE\ cat\ M1}$ defined in Clause 8.1.28.3. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_UE\ cat\ M1}$ defined in clause 8.1.28.3 becomes undetectable for a period ≤ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_intra_UE\ cat\ M1}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.13.2.1.1.2 and A.8.1.28.

## 8.1.28.4 Test description

### 8.1.28.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.15 using only main UE Tx/Rx antenna.
2. The general test parameter settings are set up according to Table 8.1.28.4.1-1.
3. Propagation conditions are set according to Annex B.0.
4. Message contents are defined in clause 8.1.28.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.28.4.1-1: General test parameters for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE when DRX is used**

| Parameter | | Unit | Value | | Comment |
|---|---|---|---|---|---|
| | | | Test1 | Test2 | |
| E-UTRA RF Channel Number | | | 1 | 1 | One radio channel is used. |
| Active cell | | | Cell 1 | Cell 1 | |
| Neighbour cell | | | Cell 2 | Cell 2 | Cell to be identified. |
| CP length | | | Normal | Normal | |
| DRX | | | ON | ON | DRX related parameters are defined in Table 8.1.28.5-2 |
| A3 | Offset | dB | -6 | -6 | |
| | Hysteresis | dB | 0 | 0 | |
| | Time To Trigger | s | 0 | 0 | |
| Filter coefficient | | | 0 | 0 | L3 filtering is not used |
| Gap pattern ID | | | 0 | 0 | As specified in TS 36.133 [4] clause 8.1.2.1. |
| T1 | | s | 5 | 5 | |
| T2 | | s | 5 | 35 | |

### 8.1.28.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information, it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

In Test 1 when DRX = 40 ms is used, UE needs to be provided at least once every 500 ms with new Timing Advance Command MAC control element to restart the Timer Alignment Timer to keep the UE uplink time alignment. Furthermore, the UE is allocated with PUSCH resource at every DRX cycle. In Test 2 when DRX = 1280 ms is used, the uplink time alignment is not maintained and the UE needs to use RACH to obtain uplink allocation for measurement reporting.

1. Ensure the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA.
2. Set the parameters according to T1 in Table 8.1.28.5-1. Propagation conditions are set according to Annex B.1. T1 starts.
3. The SS shall transmit the RRCConnectionReconfiguration message.
4. The UE shall transmit the RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Tables 8.1.28.5-1 and 8.1.28.5-2.
6. The UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than [1480+TT] ms for Test 1 or less than [26882+TT] ms for Test 2 then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.
7. After the SS receives the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit the RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A-CE with the condition CEModeA according to TS 36.508 [7] clause 4.5.3AB (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA.
10. Repeat steps 2-9 until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved.
11. Repeat steps 1-10 for each sub-test Table 8.1.28.4.1-1 as appropriate.

### 8.1.28.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the condition CEModeA and the following exceptions:

**Table 8.1.28.4.3-1: Common Exception messages for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CE in DRX test 1 and 2 requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-2a<br>Table H.3.1-7<br>Table H.3.7-1<br>Table H.3.7-2<br>Table H.3.7-3 |

**Table 8.1.28.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CE in DRX test 1 and 2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.1.28.4.3-3: *ReportConfigEUTRA-A3*: Additional E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CE in DRX test 1 and 2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventide CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.1.28.4.3-4: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CE in DRX test 1 and 2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.1.28.4.3-5: *MeasResults*: Additional E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CE in DRX test 1 and 2 requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.28.4.3-6: *MeasResultListEUTRA*: Additional E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CE in DRX test 1 and 2 requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| physCellId | PhysCellId | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.1.28.5 Test requirement

Tables 8.1.28.4.1-1, 8.1.28.5-1, 8.1.28.5-2, and 8.1.28.5-3 define the primary level settings including test tolerances for E-UTRAN HD-FDD event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CE in DRX.

**Table 8.1.28.5-1: Cell specific test parameters for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE when DRX is used**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | | |
| $BW_{channel}$ | MHz | 10 | | | |
| PDSCH parameters: DL Reference Measurement Channel | | R.10 HD-FDD | | - | |
| MPDCCH parameters: DL Reference Measurement Channel | | R.6 HD-FDD | | R.6 HD-FDD | |
| OCNG Patterns | | OP.21 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| MPDCCH_RA | dB | | | | |
| MPDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4+TT | 4+TT | -Infinity | 4+TT |
| $\hat{E}_s / I_{ot}$ Note 3 | dB | 4+TT | -1.46+TT | -Infinity | -1.46+TT |
| RSRP Note 3 | dBm/15 KHz | -94+TT | -94+TT | -Infinity | -94+TT |
| SCH_RP Note 3 | dBm/15 KHz | -94+TT | -94+TT | -Infinity | -94+TT |
| Io Note 3 | dBm/9MHz | -64.76+TT | -62.42+Tt | Specified in Cell 1 columns | |
| Propagation Condition | | ETU70 | | ETU70 | |
| Antenna Configuration | | 1x1 | | 1x1 | |
| Timing offset to Cell 1 | μs | - | | 3 | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: Es/Iot, RSRP, SCH_RP and Io have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

**Table 8.1.28.5-2: DRX-Configuration for E-UTRAN HD-FDD intra-frequency event triggered reporting in DRX under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf40 | sf1280 | |
| shortDRX | disable | disable | |

**Table 8.1.28.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN HD-FDD intra-frequency event triggered reporting in DRX under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | 0 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213. |

In Test 1 when DRX cycle length = 40 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2 on PUSCH.

In Test 2 when DRX cycle length = 1280 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE starts to send preambles on the PRACH for Scheduling Request (SR) to obtain allocation to send the measurement report to Cell 2 on PUSCH.

For both tests:

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event A3 measurement report.

The overall delay measured when DRX cycle length is 40 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra_UE\ cat\ M1}$

$T_{identify_intra_UE\ cat\ M1}$ = 1440 ms. When DRX cycle length is 40 ms then the $T_{identify_intra_UE\ cat\ M1}$ is 1440 ms.

TTI insertion uncertainty = [TT] ms

DRX cycle length = 40 ms

The overall delay measured when DRX cycle length is 40 ms shall be less than a total of 1480+TT ms.

The overall delay measured when DRX cycle length is 1280 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra_UE\ cat\ M1}$

$T_{identify_intra_UE\ cat\ M1}$ = 25600 ms. When DRX cycle length is 1280 ms then the $T_{identify_intra_UE\ cat\ M1}$ is 20 x 1280 ms.

TTI insertion uncertainty = 2 ms

DRX cycle length = 1280 ms

The overall delay measured when DRX cycle length is 1280 ms shall be less than a total of [26880+ TTI insertion uncertainty +TT] ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.29 E-UTRAN TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- *The Test system uncertainties and Test tolerance applicable to this test are undefined.*

### 8.1.29.1 Test purpose

To verify that the Cat-M1 UE in CEModeA to make a correct reporting of an event under fading propagation conditions in synchronous cells within the E-UTRA TDD intra-frequency cell search requirements.

### 8.1.29.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 13 and forward of UE category M1.

### 8.1.29.3 Minimum conformance requirements

The measurement reporting delay shall be less than $T_{identify_intra_UE\ cat\ M1}$ in RRC_CONNECTED state.

When no DRX is in use, the UE shall be able to identify and measure a new detectable TDD intra frequency cell according to requirements in Table 8.1.29.3-1 when SCH Ês/Iot >= -6 dB

**Table 8.1.29.3-1: Requirement on cell identification delay and measurement delay for TDD intrafrequency cell**

| Gap pattern ID | Cell identification delay ($T_{identify_intra_UE\ cat\ M1}$) | Measurement delay ($T_{measure_intra_UE\ cat\ M1}$) |
|---|---|---|
| 0 | 1.44 seconds | 480 ms |
| 1 | 2.88 seconds | 960 ms |

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] Sections 9.1.21.1 and 9.1.21.2 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.14-1 for a corresponding Band.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{measure_intra_UE\ cat\ M1}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is according to Table 8.1.29.3-1. When measurement gaps are activated the UE shall be capable of performing measurements for at least 6 cells. If the UE has identified more than 6 cells, the UE shall perform measurements but the reporting rate of RSRP measurements of cells from UE physical layer to higher layers may be decreased.

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] sub-clauses 9.1.21.1 and 9.1.21.2.

Reported RSRP measurement contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.21.1 and 9.1.21.2.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$.This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ intra_UE\ cat\ M1}$ defined in TS 36.133 [4] clause 8.13.2.1.3.1. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period Tidentify_intra_UE cat M1 defined in TS 36.133 [4] clause 8.13.2.1.3.1 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than TMeasurement_Period Intra_UE cat M1 provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.13.2.1.3.1 and A.8.1.29.

## 8.1.29.4 Test description

### 8.1.29.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15 using only main UE Tx/Rx antenna.
2. The general test parameter settings are set up according to Table 8.1.29.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.1.29.4.3.
5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.29.4.1-1: General test parameters for E-UTRAN TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA**

| Parameter | | Unit | Value | Comments |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | One radio channel is used for this test. |
| Active cell | | | Cell 1 | |
| Neighbour cell | | | Cell 2 | Cell to be identified. |
| CP length | | | Normal | |
| DRX | | | OFF | |
| A3 | Offset | dB | -6 | |
| | Hysteresis | dB | 0 | |
| | Time To Trigger | s | 0 | |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211. The same configuration in both cells |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211. The same configuration in both cells |
| Filter coefficient | | | 0 | L3 filtering is not used |
| Gap Pattern Id | | | 1 | |
| T1 | | s | 5 | |
| T2 | | s | 5 | |

### 8.1.29.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 is used. MPDCCHs indicating new transmissions or retransmissions shall be sent continuously to ensure that the UE would not enter the DRX state. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF-CE with condition CEMode A according to TS 36.508 [7] clause 7.2A.3AA.
2. Set the parameters according to T1 in Table 8.1.29.5-1. Propagation conditions are set according to Annex B clauses B.1 and B.2. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.29.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 2882 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.
7. After the SS receives the MeasurementReport message in step 6 or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A-CE with the condition CEMode A according to TS 36.508 [7] clause 4.5.3AB (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF-CE according to TS 36.508 [7] clause 7.2A.3AA),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEMode A according to TS 36.508 [7] clause 7.2A.3AA.
10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.1.29.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the condition CEMode A and the following exceptions:

**Table 8.1.29.4.3-1: Common Exception messages for E-UTRAN TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.1.29.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.1.29.4.3-3: MeasResults: Additional E-UTRAN TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA test requirement**

| Derivation Path: 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | Set according to specific test | |
| rsrqResult | INTEGER(0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.29.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA test requirement**

| Derivation Path: 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId | INTEGER (0..503) of Cell 2 | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

## 8.1.29.5 Test requirement

Tables 8.1.29.4.1-1 and 8.1.29.5-1 define the primary level settings including test tolerances for E-UTRAN TDD event triggered reporting under fading propagation conditions in synchronous cells test.

**Table 8.1.29.5-1: Cell specific test parameters for E-UTRAN TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | | |
| $BW_{channel}$ | MHz | 10 | | | |
| PDSCH parameters: DL Reference Measurement Channel | | R.16 TDD | | - | |
| MPDCCH parameters: DL Reference Measurement Channel | | R.14 TDD | | R.14 TDD | |
| OCNG Patterns | | OP.11 TDD | | OP.2 TDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| MPDCCH_RA | dB | | | | |
| MPDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | | |
| $\hat{E}_s/N_{oc}$ | dB | 4+TT | 4+TT | -Infinity | 4+TT |
| $\hat{E}_s/I_{ot}$ Note 3 | dB | 4+TT | -1.46+TT | -Infinity | -1.46+TT |
| RSRP Note 3 | dBm/15 kHz | -94+TT | -94+TT | -Infinity | -94+TT |
| SCH_RP Note 3 | dBm/15 kHz | -94+TT | -94+TT | -Infinity | -94+TT |
| Io Note 3 | dBm/9 MHz | -64.76+TT | -62.42+TT | -64.76 | -62.42+TT |
| Propagation Condition | | ETU 70 | | ETU 70 | |
| Antenna Configuration | | 1x1 | | 1x1 | |
| Time offset to Cell 1 | µs | - | | 3 | |

Note 1 OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: $\hat{E}_s/I_{ot}$, RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra_UE\ cat\ M1}$

$T_{identify_intra_UE\ cat\ M1}$ = 2880 ms

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 2882 ms in this test case (note: this gives a total of 2880 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.30 E-UTRAN TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA in DRX

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- *The Test system uncertainties and Test tolerance applicable to this test are undefined.*

### 8.1.30.1 Test purpose

To verify that the Cat-M1 UE in CEModeA to make a correct reporting of an event under fading propagation conditions in synchronous cells within the E-UTRA TDD intra-frequency cell search requirements in DRX requirements.

### 8.1.30.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 13 and forward of UE category M1. Applicability requires support for FGI bit 5.

### 8.1.30.3 Minimum conformance requirements

When DRX is in use the UE shall be able to identify a new detectable TDD intra frequency cell within $T_{identify_intra_UE\ catM1}$ as shown in table 8.1.30.3-1.

When eDRX_CONN is in use the UE shall be able to identify a new detectable TDD intra frequency cell within $T_{identify_intra_UE\ cat\ M1}$ as shown in table 8.1.30.3-1A.

**Table 8.1.30.3-1: Requirement to identify a newly detectable TDD intrafrequency cell**

| Gap pattern ID | DRX cycle length (s) | $T_{identify_intra_UE\ cat\ M1}$ (s) (DRX cycles) |
|---|---|---|
| 0 | ≤0.04 | 1.44 (Note1) |
| | 0.04<DRX-cycle≤0.08 | Note2 (40) |
| | 0.128 | 3.2 (25) |
| | 0.128<DRX-cycle≤2.56 | Note2(20) |
| 1 | <0.128 | 2.88 (Note1) |
| | 0.128 | 3.2 (25) |
| | 0.128<DRX-cycle≤2.56 | Note2(20) |
| Note1: Number of DRX cycle depends upon the DRX cycle in use<br>Note2: Time depends upon the DRX cycle in use | | |

**Table 8.1.30.3-1A: Requirement to identify a newly detectable TDD intrafrequency cell when eDRX_CONN cycle is used**

| eDRX_CONN cycle length (s) | $T_{identify_intra_UE\ cat\ M1}$ (s) (eDRX_CONN cycles) |
|---|---|
| 2.56<eDRX_CONN cycle≤10.24 | Note (20) |
| NOTE: Time depends upon the eDRX_CONN cycle in use | |

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] Sections 9.1.21.1 and 9.1.21.2 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.14-1 for a corresponding Band.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is $T_{measure_intra_UE\ cat\ M1}$. When DRX is used, $T_{measure_intra_UE\ cat\ M1}$ is as specified in table 8.1.30.3-2. When eDRX_CONN is used, $T_{measure_intra_UE\ cat\ M1}$ is as specified in table 8.1.30.3-2A. The UE shall be capable of performing RSRP measurements for 6 identified intra-frequency cells and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_intra_UE\ cat\ M1}$.

**Table 8.1.30.3-2: Requirement to measure TDD intra frequency cells**

| Gap pattern ID | DRX cycle length (s) | $T_{measure_intra_UE\ cat\ M1}$ (s) (DRX cycles) |
|---|---|---|
| 0 | <0.128 | 0.48 (Note1) |
| | 0.128≤DRX-cycle≤2.56 | Note2 (5) |
| 1 | <0.256 | 0.96 (Note1) |
| | 0.256≤DRX-cycle≤2.56 | Note2 (5) |
| Note1: Number of DRX cycle depends upon the DRX cycle in use<br>Note2: Time depends upon the DRX cycle in use | | |

**Table 8.1.30.3-2A: Requirement to measure TDD intra frequency cells when eDRX_CONN cycle is used**

| eDRX_CONN cycle length (s) | $T_{measure_intra_UE\ cat\ M1}$ (s) (eDRX_CONN cycles) |
|---|---|
| 2.56<eDRX_CONN cycle≤10.24 | Note (5) |
| NOTE: Time depends upon the eDRX_CONN cycle in use. | |

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] sub-clauses 9.1.21.1 and 9.1.21.2.

Reported RSRP measurement contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.21.1 and 9.1.21.2.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$.This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra_UE\ cat\ M1}$ defined in TS 36.133 [4] clause 8.13.2.1.3.2. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_UE\ cat\ M1}$ defined in TS 36.133 [4] clause 8.13.2.1.3.2 becomes undetectable for a period ≤ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_intra_UE\ cat\ M1}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.13.2.1.3.2 and A.8.1.30.

## 8.1.30.4 Test description

### 8.1.30.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15 using only main UE Tx/Rx antenna.
2. The general test parameter settings are set up according to Table 8.1.30.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.1.30.4.3.
5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.30.4.1-1: General test parameters for E-UTRAN TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA when DRX is used**

| Parameter | | Unit | Value | | Comments |
|---|---|---|---|---|---|
| | | | Test 1 | Test 2 | |
| E-UTRA RF Channel Number | | | 1 | | One radio channel is used for both tests. |
| Active cell | | | Cell 1 | | |
| Neighbour cell | | | Cell 2 | | Cell to be identified. |
| CP length | | | Normal | | |
| DRX | | | ON | | DRX related parameters are defined in Table 8.1.30.5-2 |
| A3 | Offset | dB | -6 | | |
| | Hysteresis | dB | 0 | | |
| | Time To Trigger | s | 0 | | |
| Special subframe configuration | | | 6 | | As specified in table 4.2-1 in TS 36.211. The same configuration in both cells |
| Uplink-downlink configuration | | | 1 | | As specified in table 4.2-2 in TS 36.211. The same configuration in both cells |
| Filter coefficient | | | 0 | | L3 filtering is not used |
| Gap Pattern Id | | | 0 | | |
| T1 | | s | 5 | | |
| T2 | | s | 5 | 30 | |

### 8.1.30.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information, it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

In Test 1 when DRX = 40 ms is used, UE needs to be provided at least once every 500 ms with new Timing Advance Command MAC control element to restart the Timer Alignment Timer to keep the UE uplink time alignment. Furthermore, the UE is allocated with PUSCH resource at every DRX cycle. In Test 2 when DRX = 1280 ms is used, the uplink time alignment is not maintained and the UE needs to use RACH to obtain uplink allocation for measurement reporting.

1. Ensure the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA.
2. Set the parameters according to T1 in Table 8.1.30.5-1. Propagation conditions are set according to Annex B.1 and B.2. T1 starts.

3. The SS shall transmit the RRCConnectionReconfiguration message.

4. The UE shall transmit the RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Tables 8.1.30.5-1 and 8.1.30.5-2.

6. The UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 1482 ms for Test 1 or less than 26882 ms for Test 2 then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receives the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit the RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A-CE with the condition CEModeA according to TS 36.508 [7] clause 4.5.3AB (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA.

10. Repeat steps 2-9 until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved.

11. Repeat steps 1-10 for each sub-test Table 8.1.30.4.1-1 as appropriate.

### 8.1.30.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the condition CEModeA and the following exceptions:

**Table 8.1.30.4.3-1: Common Exception messages for E-UTRAN TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA when DRX is used test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.1.30.4.3-2: RRCConnectionReconfiguration: Additional E-UTRAN TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA when DRX is used test requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | | | |
| | MeasConfig-DEFAULT | | MEAS |
| radioResourceConfigDedicated SEQUENCE { | | | |
| MAC-MainConfig-RBC SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf40 | 9 | For Test 1 | |
| sf1280 | 9 | For Test 2 | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | sf500 | | |
| } | | | |
| PhysicalConfigDedicated SEQUENCE { | | | |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.30.4.3-3: ReportConfigEUTRA-A3: Additional E-UTRAN TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA when DRX is used test requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.30.4.3-4: SchedulingRequest-Config-DEFAULT: Additional E-UTRAN TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA when DRX is used test requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20: SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | Channel-bandwidth-dependent parameter | |
| sr-ConfigIndex | 2 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.1.30.4.3-5: MeasResults: Additional E-UTRAN TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA when DRX is used test requirements**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | Set according to specific test | |
| rsrqResult | INTEGER(0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.30.4.3-6: MeasResultListEUTRA: Additional E-UTRAN TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA when DRX is used test requirements**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| 1 entry physCellId | PhysCellId | INTEGER (0..503) of Cell 2 | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

## 8.1.30.5 Test requirement

Tables 8.1.30.4.1-1, 8.1.30.5-1, 8.1.30.5-2 and 8.1.30.5-3 define the primary level settings including test tolerances for E-UTRAN TDD event triggered reporting under fading propagation conditions in synchronous cells tests when DRX is used.

**Table 8.1.30.5-1: Cell specific test parameters for E-UTRAN TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used for Cat-M1 UE in CEModeA**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | | |
| $BW_{channel}$ | MHz | 10 | | | |
| PDSCH parameters: DL Reference Measurement Channel | | R.16 TDD | | - | |
| MPDCCH parameters: DL Reference Measurement Channel | | R.14 TDD | | R.14 TDD | |
| OCNG Patterns | | OP.11 TDD | | OP.2 TDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| MPDCCH_RA | dB | | | | |
| MPDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | | |
| $\hat{E}_s/N_{oc}$ | dB | 4+TT | 4+TT | -Infinity | 4+TT |
| $\hat{E}_s/I_{ot}$ Note 3 | dB | 4+TT | -1.46+TT | -Infinity | -1.46+TT |
| RSRP Note 3 | dBm/15 kHz | -94+TT | -94+TT | -Infinity | -94+TT |
| SCH_RP Note 3 | dBm/15 kHz | -94+TT | -94+TT | -Infinity | -94+TT |
| Io Note 3 | dBm/9 MHz | -64.76+TT | -62.42+TT | -64.76+TT | -62.42+TT |
| Propagation Condition | | ETU 70 | | ETU 70 | |
| Antenna Configuration | | 1x1 | | 1x1 | |
| Time offset to Cell 1 | µs | - | | 3 | |

Note 1 OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: $\hat{E}_s/I_{ot}$, RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

**Table 8.1.30.5-2: DRX-Configuration for E-UTRAN TDD intra-frequency event triggered reporting in DRX under fading propagation conditions in synchronous cells**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf40 | sf1280 | |
| shortDRX | disable | disable | |

**Table 8.1.30.5-3: *TimeAlignmentTimer-Configuration* for E-UTRAN TDD intra-frequency event triggered reporting in DRX under fading propagation conditions in synchronous cells**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 2 | 2 | For further information see clause 6.3.2 in TS 36.331 and clause 10.1 in TS 36.213. |

In Test 1 when DRX cycle length = 40 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2 on PUSCH.

In Test 2 when DRX cycle length = 1280 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE starts to send preambles on the PRACH for Scheduling Request (SR) to obtain allocation to send the measurement report to Cell 2 on PUSCH.

For both tests:

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event A3 measurement report.

The overall delay measured when DRX cycle length is 40 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra_UE\ cat\ M1}$

$T_{identify_intra_UE\ cat\ M1}$ = 1440 ms. When DRX cycle length is 40 ms then the $T_{identify_intra_UE\ cat\ M1}$ is 1440 ms.

TTI insertion uncertainty = 2 ms

DRX cycle length = 40 ms

The overall delay measured when DRX cycle length is 40 ms shall be less than a total of 1482 ms.

The overall delay measured when DRX cycle length is 1280 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra_UE\ cat\ M1}$

$T_{identify_intra_UE\ cat\ M1}$ = 25600 ms. When DRX cycle length is 1280 ms then the $T_{identify_intra_UE\ cat\ M1}$ is 20 x 1280 ms.

TTI insertion uncertainty = 2 ms

DRX cycle length = 1280 ms

The overall delay measured when DRX cycle length is 1280 ms shall be less than a total of 26882 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.33 E-UTRAN HD-FDD Intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeB

Editor's notes: This test case is incomplete. The following items are missing or incomplete:

- Test tolerances, Test system uncertainties and test requirement values
- Annex E needs to be updated

### 8.1.33.1 Test purpose

To verify Cat-M1 UE's ability to make a correct reporting of an event under fading propagation conditions in asynchronous cells within the E-UTRA HD-FDD intra-frequency cell search requirements for Cat-M1 UE.

### 8.1.33.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 13 and forward of UE category M1 that support CE Mode B.

### 8.1.33.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify and measure a new detectable FDD intra frequency cell according to requirements in Table 8.1.33.3-1 provided that additional conditions table 8.1.33.3-2 is met.

**Table 8.1.33.3-1: Requirement on cell identification delay and measurement delay for FDD intrafrequency cell**

| Gap pattern ID | Cell identification delay ($T_{identify_intra_UE\ cat\ M1}$) | Measurement delay ($T_{measure_intra_UE\ cat\ M1}$) |
|---|---|---|
| 0 | 320.8 s | 800 ms |
| 1 | 321.6 s | 1600ms |

**Table 8.1.33.3-2: Conditions on SCH Ês/Iot for cell identification delay and measurement delay for FDD intrafrequency cell**

| SCH Ês/Iot of already identified cell including serving cell: Q1 | Neighouring cell SCH Ês/Iot: Q2 | Cell identification delay ($T_{identify_intra_UE\ cat\ M1}$) | Measurement delay ($T_{measure_intra_UE\ cat\ M1}$) |
|---|---|---|---|
| -15≤Q1<-6 | -15≤ Q2 < -6 | As defined in table 8.1.33.3-1 | As defined in table 8.1.33.3-1 |
| -15≤Q1<-6 | Q2≥-6 | Requirements in TS 36.133 [4] Clauses 8.13.2 apply | Requirements in TS 36.133 [4] Clauses 8.13.2 apply |
| Q1≥ -6 | Q2≥-6 | Requirements in TS 36.133 [4] Clauses 8.13.2 apply | Requirements in TS 36.133 [4] Clauses 8.13.2 apply |

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] Clauses 9.1.21.3 and 9.1.21.4 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to TS 36.133 [4] Table B.2.14-4 for a corresponding Band.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{measure_intra_UE\ cat\ M1}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is according to Table 8.1.33.3-1. When measurement gaps are activated the UE shall be capable of performing measurements for at least 6 cells. If the UE has identified more than 6 cells, the UE shall perform measurements but the reporting rate of RSRP measurements of cells from UE physical layer to higher layers may be decreased.

Reported RSRP measurement contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] Clauses 9.1.21.3 and 9.1.21.4.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$.This measurement reporting delay excludes a delay which caused by no UL resoureces for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ intra_UE\ cat\ M1}$ defined in Clause 8.1.33.3. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_UE\ cat\ M1}$ defined in Clause 8.1.33.3 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_UE\ cat\ M1,\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.13.3.1.2.1 and A.8.1.33.

## 8.1.33.4 Test description

### 8.1.33.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (Node B emulator), faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.15 using only main UE Tx/Rx antenna.
2. The general test parameter settings are set up according to Table 8.1.33.4.1-1.
3. Propagation conditions are set according to Annex B.0.
4. Message contents are defined in clause 8.1.33.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.33.4.1-1: General test parameters for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeB**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | One radio channel is used for this test |
| Active Cell | | | Cell 1 | |
| Neighbour cell | | | Cell 2 | Cell to be identified |
| CP length | | | Normal | |
| DRX | | | OFF | |
| A3 | Offset | dB | -8 | |
| | Hysteresis | dB | 0 | |
| | Time To Trigger | s | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| Gap pattern ID | | | 0 | |
| T1 | | s | 5 | |
| T2 | | s | 325 | |

### 8.1.33.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information, it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF-CE with the condition CEModeB according to TS 36.508 [7] clause 7.2A.3AA.
2. Set the parameters according to T1 in Table 8.1.33.5-1. Propagation conditions are set according to Annex B.1 and B.2. T1 starts.
3. The SS shall transmit the RRCConnectionReconfiguration message.
4. The UE shall transmit the RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.33.5-1.
6. The UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 3210 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.
7. After the SS receives the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit the RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A-CE with the condition CEModeB according to TS 36.508 [7] clause 4.5.3AB (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeB according to TS 36.508 [7] clause 7.2A.3AA),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeB according to TS 36.508 [7] clause 7.2A.3AA.
10. Repeat steps 2-9 until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

### 8.1.33.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the condition CEModeB and the following exceptions:

**Table 8.1.33.4.3-1: Common Exception messages for E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeB test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2a<br>Table H.3.1-7 |

**Table 8.1.33.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeB test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -16 (-8 dB) | -8 is actual value in dB (-16 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.33.4.3-3: *MeasResults*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeB test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.33.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeB test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.1.33.5 Test requirement

Tables 8.1.33.4.1-1 and 8.1.33.5-1 define the primary level settings including test tolerances for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells.

**Table A.8.1.33.5-1: Cell specific test parameters for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeB**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | | |
| $BW_{channel}$ | MHz | 10 | | | |
| PDSCH parameters: DL Reference Measurement Channel | | R.12 HD-FDD | | - | |
| MPDCCH parameters: DL Reference Measurement Channel | | R.8 HD-FDD | | R.8 HD-FDD | |
| OCNG Patterns | | OP.21 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| MPDCCH_RA | dB | | | | |
| MPDCCH_PB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | -12+TT | -12+TT | -infinity | -12+TT |
| $\hat{E}_s / I_{ot}$ Note 3 | dB | -12+TT | -12.27+TT | -infinity | -12.27+TT |
| RSRP Note 3 | dBm/15 kHz | -110+TT | -110+TT | -infinity | -110+TT |
| SCH_RP Note 3 | dBm/15 kHz | -110+TT | -110+TT | -infinity | -110+TT |
| Io Note 3 | dBm/9MHz | -69.95+TT | -69.70+TT | Specified in Cell 1 columns | |
| Propagation Condition | | ETU70 | | ETU70 | |
| Antenna Configuration | | 1x1 | | 1x1 | |
| Timing offset to Cell 1 | ms | - | | 3 | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: Es/Iot, RSRP, SCH_RP and Io have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra_UE\ cat\ M1}$

$T_{identify_intra_UE\ cat\ M1} = 3208$

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 3210 ms in this test case (note: this gives a total of 3208 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.34 E-UTRAN HD-FDD Intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB

Editor's notes: This test case is incomplete. The following items are missing or incomplete:

- Test tolerances, Test system uncertainties and test requirement values
- Annex E needs to be updated

### 8.1.34.1 Test purpose

To verify Cat-M1 UE's ability to make a correct reporting of an event under fading propagation conditions in synchronous cells within the E-UTRA HD-FDD intra-frequency cell search requirements for Cat-M1 UE.

### 8.1.34.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 13 and forward of UE category M1 that support CE Mode B.

### 8.1.34.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify and measure a new detectable FDD intra frequency cell according to requirements in Table 8.1.34.3-1 provided that additional conditions table 8.1.34.3-2 is met.

**Table 8.1.34.3-1: Requirement on cell identification delay and measurement delay for FDD intrafrequency cell**

| Gap pattern ID | Cell identification delay ($T_{identify_intra_UE\ cat\ M1}$) | Measurement delay ($T_{measure_intra_UE\ cat\ M1}$) |
|---|---|---|
| 0 | 320.8 s | 800 ms |
| 1 | 321.6 s | 1600ms |

**Table 8.1.34.3-2: Conditions on SCH Ês/Iot for cell identification delay and measurement delay for FDD intrafrequency cell**

| SCH Ês/Iot of already identified cell including serving cell: Q1 | Neighouring cell SCH Ês/Iot: Q2 | Cell identification delay ($T_{identify_intra_UE\ cat\ M1}$) | Measurement delay ($T_{measure_intra_UE\ cat\ M1}$) |
|---|---|---|---|
| -15≤Q1<-6 | -15≤ Q2 < -6 | As defined in table 8.1.34.3-1 | As defined in table 8.1.34.3-1 |
| -15≤Q1<-6 | Q2≥-6 | Requirements in TS 36.133 [4] Clauses 8.13.2 apply | Requirements in TS 36.133 [4] Clauses 8.13.2 apply |
| Q1≥ -6 | Q2≥-6 | Requirements in TS 36.133 [4] Clauses 8.13.2 apply | Requirements in TS 36.133 [4] Clauses 8.13.2 apply |

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] Clauses 9.1.21.3 and 9.1.21.4 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to TS 36.133 [4] Table B.2.14-4 for a corresponding Band.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{measure_intra_UE\ cat\ M1}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is according to Table 8.1.34.3-1. When measurement gaps are activated the UE shall be capable of performing measurements for at least 6 cells. If the UE has identified more than 6 cells, the UE shall perform measurements but the reporting rate of RSRP measurements of cells from UE physical layer to higher layers may be decreased.

Reported RSRP measurement contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] Clauses 9.1.21.3 and 9.1.21.4.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$.This measurement reporting delay excludes a delay which caused by no UL resoureces for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ intra_UE\ cat\ M1}$ defined in Clause 8.1.34.3. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_UE\ cat\ M1}$ defined in Clause 8.1.34.3 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_UE\ cat\ M1,\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.13.3.1.2.1 and A.8.1.34.

## 8.1.34.4 Test description

### 8.1.34.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (Node B emulator), faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.15 using only main UE Tx/Rx antenna.
2. The general test parameter settings are set up according to Table 8.1.34.4.1-1.
3. Propagation conditions are set according to Annex B.0.
4. Message contents are defined in clause 8.1.34.4.3.
5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.34.4.1-1: General test parameters for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | One radio channel is used for this test |
| Active Cell | | | Cell 1 | |
| Neighbour cell | | | Cell 2 | Cell to be identified |
| CP length | | | Normal | |
| DRX | | | OFF | |
| A3 | Offset | dB | -8 | |
| | Hysteresis | dB | 0 | |
| | Time To Trigger | s | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| Gap pattern ID | | | 0 | |
| T1 | | s | 5 | |
| T2 | | s | 325 | |

## 8.1.34.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information, it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF-CE with the condition CEModeB according to TS 36.508 [7] clause 7.2A.3AA.
2. Set the parameters according to T1 in Table 8.1.34.5-1. Propagation conditions are set according to Annex B.1 and B.2. T1 starts.
3. The SS shall transmit the RRCConnectionReconfiguration message.
4. The UE shall transmit the RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.34.5-1.
6. The UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 3210 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.
7. After the SS receives the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit the RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A-CE with the condition CEModeB according to TS 36.508 [7] clause 4.5.3AB (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeB according to TS 36.508 [7] clause 7.2A.3AA),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeB according to TS 36.508 [7] clause 7.2A.3AA.
10. Repeat steps 2-9 until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

## 8.1.34.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the condition CEModeB and the following exceptions:

**Table 8.1.34.4.3-1: Common Exception messages for E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2a<br>Table H.3.1-7 |

**Table 8.1.34.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -16 (-8 dB) | -8 is actual value in dB (-16 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.34.4.3-3: *MeasResults*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.34.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN HD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.1.34.5 Test requirement

Tables 8.1.34.4.1-1 and 8.1.34.5-1 define the primary level settings including test tolerances for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells.

**Table A.8.1.34.5-1: Cell specific test parameters for E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | | |
| $BW_{channel}$ | MHz | 10 | | | |
| PDSCH parameters: DL Reference Measurement Channel | | R.12 HD-FDD | | - | |
| MPDCCH parameters: DL Reference Measurement Channel | | R.8 HD-FDD | | R.8 HD-FDD | |
| OCNG Patterns | | OP.21 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| MPDCCH_RA | dB | | | | |
| MPDCCH_PB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | -12+TT | -12+TT | -infinity | -12+TT |
| $\hat{E}_s / I_{ot}$ Note 3 | dB | -12+TT | -12.27+TT | -infinity | -12.27+TT |
| RSRP Note 3 | dBm/15 KHz | -110+TT | -110+TT | -infinity | -110+TT |
| SCH_RP Note 3 | dBm/15 KHz | -110+TT | -110+TT | -infinity | -110+TT |
| Io Note 3 | dBm/9MHz | -69.95+TT | -69.70+TT | Specified in Cell 1 columns | |
| Propagation Condition | | ETU70 | | ETU70 | |
| Antenna Configuration | | 1x1 | | 1x1 | |
| Timing offset to Cell 1 | µs | - | | 3 | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: Es/Iot, RSRP, SCH_RP and Io have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra_UE\ cat\ M1}$

$T_{identify_intra_UE\ cat\ M1} = 3208$

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 3210 ms in this test case (note: this gives a total of 3208 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.1.35 E-UTRAN TDD Intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB

Editor's notes: This test case is incomplete. The following items are missing or incomplete:

- Test tolerances, Test system uncertainties and test requirement values
- Annex E needs to be updated

### 8.1.35.1 Test purpose

To verify Cat-M1 UE's ability to make a correct reporting of an event under fading propagation conditions in synchronous cells within the E-UTRA TDD intra-frequency cell search requirements for Cat-M1 UE.

### 8.1.35.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 13 and forward of UE category M1 that support CE Mode B.

### 8.1.35.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify and measure a new detectable TDD intra frequency cell according to requirements in Table 8.1.35.3-1provided that additional conditions Table 8.1.35.3-2 is met.

**Table 8.1.35.3-1: Requirement on cell identification delay and measurement delay for TDD intrafrequency cell**

| Gap pattern ID | Cell identification delay ($T_{identify_intra_UE\ cat\ M1}$) | Measurement delay ($T_{measure_intra_UE\ cat\ M1}$) |
|---|---|---|
| 0 | 320.8 s | 800 ms |
| 1 | 321.6 s | 1600ms |

**Table 8.1.35.3-2: Conditions on SCH Ês/Iot for cell identification delay and measurement delay for TDD intrafrequency cell**

| SCH Ês/Iot of already identified cell including serving cell: Q1 | Neighbouring cell SCH Ês/Iot: Q2 | Cell identification delay ($T_{identify_intra_UE\ cat\ M1}$) | Measurement delay ($T_{measure_intra_UE\ cat\ M1}$) |
|---|---|---|---|
| -15≤Q1<-6 | -15≤ Q2 < -6 | As defined in table 8.1.35.3-1 | As defined in table 8.1.35.3-1 |
| -15≤Q1<-6 | Q2≥-6 | Requirements in TS 36.133 [4] Clauses 8.13.2 apply | Requirements in TS 36.133 [4] Clauses 8.13.2 apply |
| Q1≥ -6 | Q2≥-6 | Requirements in TS 36.133 [4] Clauses 8.13.2 apply | Requirements in TS 36.133 [4] Clauses 8.13.2 apply |

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] Clauses 9.1.21.3 and 9.1.21.4 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to TS 36.133 [4] Table B.2.14-3 for a corresponding Band.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{measure_intra_UE\ cat\ M1}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is according to Table 8.1.35.3-1. When measurement gaps are activated the UE shall be capable of performing measurements for at least 6 cells. If the UE has identified more than 6 cells, the UE shall perform measurements but the reporting rate of RSRP measurements of cells from UE physical layer to higher layers may be decreased.

Reported RSRP measurement contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] Clauses 9.1.21.3 and 9.1.21.4.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$.This measurement reporting delay excludes a delay which caused by no UL resoureces for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ intra_UE\ cat\ M1}$ defined in Clause 8.1.35.3. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_UE\ cat\ M1}$ defined in Clause 8.1.35.3 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_UE\ cat\ M1,\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.13.3.1.3.1 and A.8.1.35.

## 8.1.35.4 Test description

### 8.1.35.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (Node B emulator), faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.15 using only main UE Tx/Rx antenna.
2. The general test parameter settings are set up according to Table 8.1.35.4.1-1.
3. Propagation conditions are set according to Annex B.0.
4. Message contents are defined in clause 8.1.35.4.3.
5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.1.35.4.1-1: General test parameters for E-UTRAN TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB**

| Parameter | | Unit | Value | Comments |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | One radio channel is used for this test. |
| Active cell | | | Cell 1 | |
| Neighbour cell | | | Cell 2 | Cell to be identified. |
| CP length | | | Normal | |
| DRX | | | OFF | |
| A3 | Offset | dB | -8 | |
| | Hysteresis | dB | 0 | |
| | Time To Trigger | s | 0 | |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211[9]. The same configuration in both cells |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211[9]. The same configuration in both cells |
| Filter coefficient | | | 0 | L3 filtering is not used |
| Gap Pattern Id | | | 0 | |
| T1 | | s | 5 | |
| T2 | | s | 325 | |

### 8.1.35.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information, it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF-CE with the condition CEModeB according to TS 36.508 [7] clause 7.2A.3AA.
2. Set the parameters according to T1 in Table 8.1.35.5-1. Propagation conditions are set according to Annex B.1 and B.2. T1 starts.
3. The SS shall transmit the RRCConnectionReconfiguration message.
4. The UE shall transmit the RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.1.35.5-1.
6. The UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 3210 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.
7. After the SS receives the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit the RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A-CE with the condition CEModeB according to TS 36.508 [7] clause 4.5.3AB (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeB according to TS 36.508 [7] clause 7.2A.3AA),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF-CE with the condition CEModeB according to TS 36.508 [7] clause 7.2A.3AA.
10. Repeat steps 2-9 until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

### 8.1.35.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the condition CEModeB and the following exceptions:

**Table 8.1.35.4.3-1: Common Exception messages for E-UTRAN TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2a<br>Table H.3.1-7 |

**Table 8.1.35.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -16 (-8 dB) | -8 is actual value in dB (-16 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.1.35.4.3-3: *MeasResults*: Additional E-UTRAN TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.1.35.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.1.35.5 Test requirement

Tables 8.1.35.4.1-1 and 8.1.35.5-1 define the primary level settings including test tolerances for E-UTRAN TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells.

**Table A.8.1.35.5-1: Cell specific test parameters for E-UTRAN TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | | |
| $BW_{channel}$ | MHz | 10 | | | |
| PDSCH parameters: DL Reference Measurement Channel | | R.18 TDD | | - | |
| MPDCCH parameters: DL Reference Measurement Channel | | R.16 TDD | | R.16 TDD | |
| OCNG Patterns | | OP.11 TDD | | OP.2 TDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| MPDCCH_RA | dB | | | | |
| MPDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | -12+TT | -12+TT | -infinity | -12+TT |
| $\hat{E}_s / I_{ot}$ Note 3 | dB | -12+TT | -12.27+TT | -infinity | -12.27+TT |
| RSRP Note 3 | dBm/15 kHz | -110+TT | -110+TT | -infinity | -110+TT |
| SCH_RP Note 3 | dBm/15 kHz | -110+TT | -110+TT | -infinity | -110+TT |
| Io Note 3 | dBm/9MHz | -69.95+TT | -69.70+TT | Specified in Cell 1 columns | |
| Propagation Condition | | ETU70 | | ETU70 | |
| Antenna Configuration | | 1x1 | | 1x1 | |
| Timing offset to Cell 1 | ms | - | | 3 | |

Note 1 OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: $\hat{E}_s/I_{ot}$, RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra_UE\ cat\ M1}$

$T_{identify_intra_UE\ cat\ M1} = 3208$

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 3210 ms in this test case (note: this gives a total of 3208 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 8.2 E-UTRAN TDD intra frequency measurements

## 8.2.1 E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells

### 8.2.1.1 Test purpose

To verify the UE's ability to make a correct reporting of an event under fading propagation conditions in synchronous cells within the E-UTRA TDD-TDD intra frequency cell search requirements.

### 8.2.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward. Applicability requires support for FGI bit 5.

### 8.2.1.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable TDD intra frequency cell within

$$T_{\text{identify intra}} = T_{\text{basic identify } E-UTRA_\text{TDD, intra}} \cdot \frac{T_{\text{Measurement Period, Intra}}}{T_{\text{Intra}}} \quad ms$$

where

$T_{\text{basic_identify_E-UTRA_TDD, intra}}$ is 800 ms

A cell shall be considered detectable when

- RSRP related side condition given in TS 36.133 [4] Sections 9.1.2.1 and 9.1.2.2 and RSRQ related side conditions given in Section 9.1.5.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.1 for a corresponding Band.

$T_{\text{Intra}}$: This is the minimum time that is available for intra frequency measurements, during the measurement period with an arbitrarily chosen timing. Time is assumed to be available for performing intra frequency measurements whenever the receiver is guaranteed to be active on the intra frequency carrier.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{\text{Measurement_Period Intra}}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is 200 ms. When no measurement gaps are activated, the UE shall be capable of performing RSRP and RSRQ measurements for 8 identified-intra-frequency cells, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of 200 ms. When measurement gaps are activated the UE shall be capable of performing measurements for at least $Y_{\text{measurement intra}}$ cells , where $Y_{\text{measurement intra}}$ is defined in the following equation. If the UE has identified more than $Y_{\text{measurement intra}}$ cells, the UE shall perform measurements at least 8 identified intra-frequency cells but the reporting rate of RSRP and RSRQ measurements of cells from UE physical layer to higher layers may be decreased.

$$Y_{\text{measurement intra}} = Floor\left\{ X_{\text{basic measurement TDD}} \cdot \frac{T_{\text{Intra}}}{T_{\text{Measurement_Period, Intra}}} \right\} \text{cells}$$

where

$X_{basic\ measurement\ TDD}$ = 8 (cells)

$T_{Measurement_Period\ Intra}$ = 200 ms. The measurement period for Intra frequency RSRP and RSRQ measurements.

$T_{Intra}$: This is the minimum time that is available for intra frequency measurements, during the measurement period with an arbitrarily chosen timing. Time is assumed to be available for performing intra frequency measurements whenever the receiver is guaranteed to be active on the intra frequency carrier.

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] sub-clauses 9.1.2.1 and 9.1.2.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in the sub-clause 9.1.5.1.

Reported RSRP and RSRQ measurements contained in periodically triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.2.1, 9.1.2.2, and 9.1.5.1, respectively.

Reported RSRP and RSRQ measurements contained in event triggered periodic measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.2.1, 9.1.2.2, and 9.1.5.1, respectively.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.2.1, 9.1.2.2, and 9.1.5.1, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra}$ defined in TS 36.133 [4] Section 8.1.2.2.2.1 When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra}$ defined in TS 36.133 [4] section 8.1.2.2.1.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.2.2 and A.8.1.2.

## 8.2.1.4 Test description

### 8.2.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15.

2. The general test parameter settings are set up according to Table 8.2.1.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.2.1.4.3.

5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.2.1.4.1-1: General test parameters for E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | One TDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| A3-Offset | dB | -6 | |
| CP length | | Normal | |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211. The same configuration in both cells |
| Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in TS 36.211. The same configuration in both cells |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | DRX_L | As specified in Table 8.2.1.5-2 |
| Time offset between cells | µs | 3 | Synchronous cells<br>3µs or 92*Ts |
| T1 | s | 5 | |
| T2 | s | 5 | |

### 8.2.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 is used. PDCCHs indicating new transmissions or retransmissions shall be sent continuously to ensure that the UE would not enter the DRX state. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.2.1.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2.T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.2.1.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 802 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receives the MeasurementReport message in step 6 or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 8.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.2.1.4.3-1: Common Exception messages for E-UTRAN TDD-TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7<br>Table H.3.6-2 |

**Table 8.2.1.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD-TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.2.1.4.3-3: MeasResults: Additional E-UTRAN TDD-TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | Report Cell 1 | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.2.1.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN TDD-TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | Report Cell 2 | |
| physCellId | PhysCellId of the Cell 2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

## 8.2.1.5 Test requirement

Tables 8.2.1.4.1-1, 8.2.1.5-1, and 8.2.1.5-2 define the primary level settings including test tolerances for E-UTRAN TDD-TDD event triggered reporting under fading propagation conditions in synchronous cells test.

**Table 8.2.1.5-1: Cell specific test parameters for E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Pattern defined in A.2.1 (OP.1 TDD) and in A.2.2 (OP.2) | | OP.1 TDD | | OP.2 TDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ | dBm/15 kHz | -98 | | | |
| RSRP | dBm/15 kHz | -91.40 | -91.40 | -Infinity | -91.40 |
| $\hat{E}_s / I_{ot}$ | dB | 6.60 | -0.86 | -Infinity | -0.86 |
| SCH_RP | dBm/15 kHz | -91.40 | -91.40 | -Infinity | -91.40 |
| $\hat{E}_s / N_{oc}$ | dB | 6.60 | 6.60 | -Infinity | 6.60 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

**Table 8.2.1.5-2: Reference DRX-Configuration to be used in E-UTRAN TDD-TDD event triggered reporting under fading propagation conditions in synchronous cells**

| Parameter | Value | Comments |
|---|---|---|
| Reference configuration | DRX_L | As defined in 4.8.2.1.5 in TS 36.508 [7.] |
| onDurationTimer | psf6 | |
| drx-InactivityTimer | psf1920 | |
| drx-RetransmissionTimer | psf16 | |
| longDRX-CycleStartOffset | sf1280, 0 | |
| shortDRX | disabled | |
| Note: For further information see section 6.3.2 in 3GPP TS 36.331 [5]. | | |

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra}$

$$T_{identify_intra} = T_{basic\ identify\ E\text{–}UTRA_TDD,\ intra} \cdot \frac{T_{Measurement\ Period,\ Intra}}{T_{Intra}}$$

$T_{basic_identify_E\text{-}UTRA_TDD,\ intra}$= 800 ms

$T_{Measurement_Period,Intra}$ = 200 ms

$T_{Intra}$ = 200 ms

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 802 ms in this test case (note: this gives a total of 800 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.2.2 E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX

### 8.2.2.1 Test purpose

To verify the UE's ability to make a correct reporting of an event under fading propagation conditions in synchronous cells with DRX within the E-UTRA TDD-TDD intra frequency cell search in DRX requirements.

### 8.2.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward. Applicability requires support for FGI bit 5.

### 8.2.2.3 Minimum conformance requirements

Note: The state when no DRX is used is assumed to be the one in which the DRX Inactivity Timer is running, and the state when DRX is used is assumed to be otherwise for this performance requirement.

When DRX is in use the UE shall be able to identify a new detectable TDD intra frequency cell within $T_{identify_intra}$ as shown in table 8.2.2.3-1

**Table 8.2.2.3-1: Requirement to identify a newly detectable TDD intra frequency cell**

| DRX cycle length (s) | $T_{identify_intra}$ (s) (DRX cycles) |
|---|---|
| ≤0.04 | 0.8 (Note1) |
| 0.04<DRX-cycle≤0.08 | Note2 (40) |
| 0.08<DRX-cycle≤2.56 | Note2(20) |
| Note1: Number of DRX cycle depends upon the DRX cycle in use<br>Note2: Time depends upon the DRX cycle in use | |

A cell shall be considered detectable when

- RSRP related side condition given in TS 36.133 [4] Sections 9.1.2.1 and 9.1.2.2 and RSRQ related side conditions given in Section 9.1.5.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.1 for a corresponding Band.

In the RRC_CONNECTED state with DRX cycles of 80ms or greater the measurement period for intra frequency measurements is $T_{measure_intra}$ as shown in table 8.2.2.3-2. The UE shall be capable of performing RSRP and RSRQ measurements for TS 45.008 [15] identified-intra-frequency cells and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_intra}$.

**Table 8.2.2.3-2: Requirement to measure TDD intra frequency cells**

| DRX cycle length (s) | $T_{measure_intra}$ (s) (DRX cycles) |
|---|---|
| ≤0.04 | 0.2 (Note1) |
| 0.04<DRX-cycle≤2.56 | Note2 (5) |
| Note1: Number of DRX cycle depends upon the DRX cycle in use.<br>Note2: Time depends upon the DRX cycle in use. | |

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] sub-clauses 9.1.2.1 and 9.1.2.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in the sub-clause 9.1.5.1.

Reported RSRP and RSRQ measurements contained in periodically triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.2.1, 9.1.2.2, and 9.1.5.1, respectively.

Reported RSRP and RSRQ measurements contained in event triggered periodic measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.2.1, 9.1.2.2, and 9.1.5.1, respectively.

The first report in event triggered periodic measurement reporting shall meet the requirements specified in TS 36.133 [4] section 8.1.2.2.2.1.3 Event Triggered Reporting.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.2.1, 9.1.2.2, and 9.1.5.1, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra}$ defined in TS 36.133 [4] Section 8.1.2.2.2 When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra}$ defined in TS 36.133 [4] section 8.1.2.2.1.1 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.2.2 and A.8.2.2.

## 8.2.2.4 Test description

### 8.2.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15.
2. The general test parameter settings are set up according to Table 8.2.2.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.2.2.4.3.
5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.2.2.4.1-1: General test parameters for E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used**

| Parameter | Unit | Value | | Comment |
|---|---|---|---|---|
| | | Test 1 | Test 2 | |
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | | As specified in section A.2.2 |
| Active cell | | Cell 1 | | |
| Neighbour cell | | Cell 2 | | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | | One TDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| A3-Offset | dB | -6 | | |
| CP length | | Normal | | |
| Special subframe configuration | | 6 | | As specified in table 4.2-1 in TS 36.211 [9]. The same configuration in both cells |
| Uplink-downlink configuration | | 1 | | As specified in table 4.2-2 in TS 36.211 [9]. The same configuration in both cells |
| Hysteresis | dB | 0 | | |
| Time To Trigger | s | 0 | | |
| Filter coefficient | | 0 | | L3 filtering is not used |
| DRX | | ON | | DRX related parameters are defined in Table 8.2.2.4-2 |
| Time offset between cells | µs | 3 | | Synchronous cells 3µs or 92*Ts |
| T1 | s | 5 | | |
| T2 | s | 5 | 30 | |

### 8.2.2.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

In Test 1 UE needs to be provided at least once every 500ms with new Timing Advance Command MAC control element to restart the Time alignment timer to keep UE uplink time alignment. Furthermore, the UE is allocated with PUSCH resource at every DRX cycle.

In Test 2 the uplink time alignment is not maintained and UE needs to use RACH to obtain UL allocation for measurement reporting.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.2.2.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.2.2.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 842 ms for Test 1 and 26882 ms for Test 2 then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receives the MeasurementReport message in step 6 or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

11. Repeat step 1-10 for each sub-test in Table 8.2.2.4.1-1 as appropriate.

### 8.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.2.2.4.3-1: Common Exception messages for E-UTRAN TDD-TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.2.2.4.3-2: RRCConnectionReconfiguration: Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | | | |
| | MeasConfig-DEFAULT | | MEAS |
| radioResourceConfigDedicated SEQUENCE { | | | |
| MAC-MainConfig-RBC SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf40 | 9 | For Test 1 | |
| sf1280 | 9 | For Test 2 | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | sf500 | | |
| } | | | |
| PhysicalConfigDedicated SEQUENCE { | | | |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.2.4.3-3: ReportConfigEUTRA-A3: Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.2.4.3-4: SchedulingRequest-Config-DEFAULT: Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20: SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | Channel-bandwidth-dependent parameter | |
| sr-ConfigIndex | 2 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.2.2.4.3-5: MeasResults: Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | Set according to specific test | |
| rsrqResult | INTEGER(0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.2.2.4.3-6: MeasResultListEUTRA: Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| 1 entry physCellId | PhysCellId | INTEGER (0..503) of Cell 2 | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

## 8.2.2.5 Test requirement

Tables 8.2.2.5-1, 8.2.2.5-2 and 8.2.2.5-3 defines the primary level settings including test tolerances for E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used test.

**Table 8.2.2.5-1: Cell specific test parameters for E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used**

| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | |
|---|---|---|---|---|---|
| | | **T1** | **T2** | **T1** | **T2** |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2) | | OP.1 TDD | | OP.2 TDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |

| | | | | | |
|---|---|---|---|---|---|
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | | |
| RSRP Note 3 | dBm/15 kHz | -91.40 | -91.40 | -Infinity | -91.40 |
| $\hat{E}_s / I_{ot}$ | dB | 6.60 | -0.86 | -Infinity | -0.86 |
| SCH_RP Note 3 | dBm/15 kHz | -91.40 | -91.40 | -Infinity | -91.40 |
| $\hat{E}_s / N_{oc}$ | dB | 6.60 | 6.60 | -Infinity | 6.60 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.2.2.5-2: DRX-Configuration for E-UTRAN TDD-TDD intra-frequency event triggered reporting in DRX under fading propagation conditions in synchronous cells**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | Sf40 | sf1280 | |
| shortDRX | disable | disable | |

**Table 8.2.2.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN TDD-TDD intra-frequency event triggered reporting in DRX under fading propagation conditions in synchronous cells**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| sr-ConfigIndex | 2 | 2 | For further information see section 6.3.2 in 3GPP TS 36.331 [5] and section 10.1 in 3GPP TS 36.213 [8]. |

In Test 1 when DRX cycle length = 40 ms is used, the overall delay measured is defined as the time from the beginning of time period T2 to the moment when the UE send one Event A3 triggered measurement report on PUSCH.

In Test 2 when DRX cycle length = 1280 ms is used, the overall delay measured is defined as the time from the beginning of time period T2 to the moment when the UE starts to send preambles on the PRACH for scheduling request (SR) to obtain allocation to send the measurement report on PUSCH.

For both tests:

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event A3 measurement report.

The overall delay measured when DRX cycle length is 40 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra}$

$T_{identify_intra}$ = 800 ms. When DRX cycle length is 40 ms then the $T_{identify_intra}$ is 0.8 s.

TTI insertion uncertainty = 2 ms

DRX cycle length = 40 ms

The overall delay measured when DRX cycle length is 40 ms shall be less than a total of 842 ms.

The overall delay measured when DRX cycle length is 1280 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra}$

$T_{identify_intra}$ = 25600 ms. When DRX cycle length is 1280 ms then the $T_{identify_intra}$ is 20 x 1280 ms.

TTI insertion uncertainty = 2 ms

DRX cycle length = 1280 ms

The overall delay measured when DRX cycle length is 1280 ms shall be less than a total of 26882 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.2.3 E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps

### 8.2.3.1 Test purpose

To verify the requirement for identification of a new CGI of E-UTRA cell with autonomous gaps in TS 36.133 [4] section 8.1.2.2.4.

### 8.2.3.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward with support of intra-frequency SI acquisition for HO.

### 8.2.3.3 Minimum conformance requirements

No explicit neighbour list is provided to the UE for identifying a new CGI of E-UTRA cell. The UE shall identify and report the CGI when requested by the network for the purpose 'reportCGI'. The UE may make autonomous gaps in downlink reception and uplink transmission for receiving MIB and SIB1 message according to section 5.5.3.1 of TS 36.331 [5]. Note that a UE is not required to use autonomous gap if si-RequestForHO is set to false. If autonomous gaps are used for measurement with the purpose of 'reportCGI', the UE shall be able to identify a new CGI of E-UTRA cell within:

$$T_{identify_CGI,\,intra} = T_{basic_identify_CGI,\,intra} \quad ms$$

Where

$T_{basic_identify_CGI,\,intra}$ = 150 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new CGI of an E-UTRA cell is defined, provided that the E-UTRA cell has been already identified by the UE.

A cell shall be considered identifiable following conditions are fulfilled:

- RSRP related side conditions given in TS 36.133 [4] Section 9.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.2 for a corresponding Band

The requirement for identifying a new CGI of an E-UTRA cell within $T_{basic_identify_CGI,.intra}$ is applicable when no DRX is used as well as when all the DRX cycles specified in 3GPP TS 36.331 [5] are used.

Given that continuous DL data allocation and no DRX is used, and no measurement gaps are configured, the UE shall be able to transmit at least the number of ACK/NACKs stated in Table 8.2.3.3-1 during the identification of a new CGI of E-UTRA cell.

**Table 8.2.3.3-1: Requirement on minimum number of ACK/NACKs to transmit during $T_{basic_identify_CGI,\ intra}$**

| UL/DL configuration | Minimum number of transmitted ACK/NACKs |
|---|---|
| 0 | 18 |
| 1 | 35 |
| 2 | 43 |
| 3 | 36 |
| 4 | 39 |
| 5 | 42 |
| 6 | 30 |

The ECGI reporting delay occurs due to the delay uncertainty when inserting the ECGI measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. In case DRX is used, the ECGI reporting may be delayed until the next DRX cycle.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.2.4 and A.8.2.3.

## 8.2.3.4 Test description

### 8.2.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. The general test parameter settings are set up according to Table 8.2.3.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.2.3.4.3.

5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.2.3.4.1-1: General test parameters for E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | Only one TDD carrier frequency is used. |
| Channel Bandwidth (BWchannel) | MHz | 10 | |
| CP length | | Normal | |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211 [9]. The same configuration in both cells |
| Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in TS 36.211 [9]. The same configuration in both cells |
| A3-Offset | dB | -3 | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| si-RequestForHO | | TRUE | As specified in section 5.5.3.1 in TS 36.331 [5]. |
| Time offset between cells | μs | 3 | Synchronous cells |
| T1 | s | 5 | |
| T2 | s | ≤10 | |
| T3 | s | 5 | |

### 8.2.3.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. PDCCHs indicating new transmissions shall be sent continuously to ensure that the UE would have ACK/NACK sending during identifying a new CGI of E-UTRAN cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE does not have any timing information of cell 2. Starting T2, cell 2 becomes detectable and the UE is expected to detect and send a measurement report.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.2.3.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration with event A3 configured message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.2.3.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3.

7. SS shall transmit an RRCConnectionReconfiguration message during period T2, The RRC message shall create a measurement report configuration with purpose reportCGI and si-RequestForHO set to TRUE.

8. The SS shall start T3 timer when the last TTI containing the RRC message implying SI reading is sent to UE.

9. The UE shall transmit RRCConnectionReconfigurationComplete message.

10. UE shall transmit a MeasurementReport message containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3. The UE shall be scheduled continuously throughout the test, and from the start of T3 until 170 ms at least 47 ACK/NACK shall be detected as being transmitted by the UE. If the overall delays measured from the beginning of time period T3 is less than 170 ms, and the UE have more than 47 ACK/NACKs transmitted from the start of T3 until 170 ms, then the number of successful tests is increased by one. If the UE

fails to report the cell global identifier within the overall delays measured requirement, or the UE have less than 47 ACK/NACKs transmitted from the start of T3 until 170 ms, then the number of failure tests is increased by one.

11. After the SS receive the MeasurementReport message in step 10) or when T3 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

13. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 2-13 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.2.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.2.3.4.3-1: Common Exception messages for E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.2.3.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -6 (-3 dB) | -3 is actual value in dB (-6 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.3.4.3-3: *MeasResults*: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.2.3.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step6)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.2.3.4.3-5: ReportConfigEUTRA-PERIODICAL: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportCGI | | |
| } | | | |
| } | | | |
| reportAmount | 1 | | |
| si-RequestForHO-r9 | setup | | |
| } | | | |

**Table 8.2.3.4.3-6: *MeasResultListEUTRA*: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| cgi-Info SEQUENCE{ | | | |
| cellGlobalId SEQUENCE{ | | | |
| plmn-Identity | plmn-Identity | | |
| cellIdentity | cellIdentity | | |
| } | | | |
| trackingAreaCode | | | |
| plmn-IdentityList | | | |
| } | | | |
| } | | | |

**Table 8.2.3.4.3-7: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellForWhichToReportCGI | Physical Cell ID of Cell 2 | | Cell 1 |
| } | | | |

**Table 8.2.3.4.3-8: *MeasConfig-DEFAULT*: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step 7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f1 | | |
| reportConfigId | IdReportConfig-P | | |
| } | | | |
| measGapConfig | Not present | | |
| } | | | |

**Table 8.2.3.4.3-9: *MeasResults*: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

## 8.2.3.5 Test requirement

Tables 8.2.3.4.1-1 and 8.2.3.5-1 define the primary level settings including test tolerances for E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test.

**Table 8.2.3.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | OP.1 TDD | OP.1 TDD | OP.2 TDD | OP.2 TDD | OP.2 TDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 8 | -3.33 | -3.33 | -Infinity | 2.36 | 2.36 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 8 | 8 | 8 | -Infinity | 11 | 11 |
| RSRP Note 3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| SCH_RP Note3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall transmit a measurement report containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Test requirement = RRC Procedure delay + $T_{identify_CGI,\,intra}$ + TTI insertion uncertainty

RRC procedure delay=15ms

$$T_{identify_CGI,\,intra} = T_{basic_identify_CGI,\,intra} \quad ms$$

$T_{basic_identify_CGI,\,intra}$ = 150 ms

TTI insertion uncertainty = 2 ms

The overall delays measured is 167 ms, allow 170 ms. The UE shall be scheduled continuously throughout the test, and from the start of T3 until 170 ms at least 47 ACK/NACK shall be detected as being transmitted by the UE.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

NOTE: The overall 47 ACK/NACK number is caused by two parts. Firstly, at least 35 ACK/NACK shall be sent during identifying the cell global identifier of cell 2 according to the requirement for UL/DL configuration #1 in Table 8.2.3.3-1. Secondly, given that continuous DL data allocation, additional 12 ACK/NACK shall be sent from the start of T3 until 170 ms excludes 150 ms for identifying the cell global identifier of cell 2.

## 8.2.4 E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX

### 8.2.4.1 Test purpose

To verify the requirement for identification of a new CGI of E-UTRA cell with autonomous gaps in DRX.

### 8.2.4.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward with support of intra-frequency SI acquisition for HO. Applicability requires support for FGI bit 5

### 8.2.4.3 Minimum conformance requirements

No explicit neighbour list is provided to the UE for identifying a new CGI of E-UTRA cell. The UE shall identify and report the CGI when requested by the network for the purpose 'reportCGI'. The UE may make autonomous gaps in downlink reception and uplink transmission for receiving MIB and SIB1 message according to section 5.5.3.1 of TS 36.331 [5]. Note that a UE is not required to use autonomous gap if si-RequestForHO is set to false. If autonomous gaps are used for measurement with the purpose of 'reportCGI', the UE shall be able to identify a new CGI of E-UTRA cell within:

$$T_{identify_CGI,\,intra} = T_{basic_identify_CGI,\,intra} \quad ms$$

Where

$T_{basic_identify_CGI,\,intra}$ = 150 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new CGI of an E-UTRA cell is defined, provided that the E-UTRA cell has been already identified by the UE.

A cell shall be considered identifiable following conditions are fulfilled:

- RSRP related side conditions given in TS 36.133 [4] Section 9.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.2 for a corresponding Band

The requirement for identifying a new CGI of an E-UTRA cell within $T_{basic_identify_CGI,.intra}$ is applicable when no DRX is used as well as when all the DRX cycles specified in 3GPP TS 36.331 [5] are used.

The ECGI reporting delay occurs due to the delay uncertainty when inserting the ECGI measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. In case DRX is used, the ECGI reporting may be delayed until the next DRX cycle.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.2.4 and A.8.2.4.

### 8.2.4.4 Test description

#### 8.2.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. The general test parameter settings are set up according to Table 8.2.4.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.2.4.4.3.

5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.2.4.4.1-1: General test parameters for E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps when DRX is used**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | Only one TDD carrier frequency is used. |
| Channel Bandwidth (BWchannel) | MHz | 10 | |
| CP length | | Normal | |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211 [9]. The same configuration in both cells |
| Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in TS 36.211 [9]. The same configuration in both cells |
| A3-Offset | dB | -3 | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | ON | DRX related parameters are defined in Table 8.2.4.5-2 |
| si-RequestForHO | | TRUE | As specified in section 5.5.3.1 in TS 36.331 [5]. |
| Time offset between cells | μs | 3 | Synchronous cells |
| T1 | s | 5 | |
| T2 | s | ≤30 | UE shall report cell within 25.6s (20 DRX cycles) |
| T3 | s | 5 | |

### 8.2.4.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE does not have any timing information of cell 2. Starting T2, cell 2 becomes detectable and the UE is expected to detect and send a measurement report.

In the test UE needs to be provided at least once every 1280ms with new Timing Advance Command MAC control element to restart the Time alignment timer to keep UE uplink time alignment. Furthermore UE sends scheduling request (SR) to obtain allocation to send the measurement report on PUSCH.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.2.4.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration with event A3 configured message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.2.4.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3.

7. SS shall transmit an RRCConnectionReconfiguration message during period T2, The RRC message shall create a measurement report configuration with purpose *reportCGI* and *si-RequestForHO* set to TRUE.

8. The SS shall start T3 timer when the last TTI containing the RRC message implying SI reading is sent to UE.

9. The UE shall transmit RRCConnectionReconfigurationComplete message.

10. UE shall transmit a measurement report containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3. If the overall delays measured from the beginning of time period T3 is less than 170 ms then the number of successful tests is increased by one. If the UE fails to report the cell global identifier within the overall delays measured requirement then the number of failure tests is increased by one.

11. After the SS receive the MeasurementReport message in step 10) or when T3 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

13. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 2-13 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.2.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.2.4.4.3-1: Common Exception messages for E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.2.4.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.4.4.3-3: *MAC-MainConfig-RBC*: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf1280 typical value in real network for real-time services. | |
| sf1280 | 9 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| } | | | |

**Table 8.2.4.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.6, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| … | | | |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | |
| } | | | |

**Table 8.2.4.4.3-5: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 2 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.2.4.4.3-6: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -6 (-3 dB) | -3 is actual value in dB (-6 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.4.4.3-7: *MeasResults*: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.2.4.4.3-8: *MeasResultListEUTRA*: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step6)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.2.4.4.3-9: *ReportConfigEUTRA-PERIODICAL*: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportCGI | | |
| } | | | |
| } | | | |
| reportAmount | 1 | | |
| si-RequestForHO-r9 | setup | | |
| } | | | |

**Table 8.2.4.4.3-10: *MeasResultListEUTRA*: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| cgi-Info SEQUENCE{ | | | |
| cellGlobalId SEQUENCE{ | | | |
| plmn-Identity | plmn-Identity | | |
| cellIdentity | cellIdentity | | |
| } | | | |
| trackingAreaCode | | | |
| plmn-IdentityList | | | |
| } | | | |
| } | | | |

**Table 8.2.4.4.3-11: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellForWhichToReportCGI | Physical Cell ID of Cell 2 | | Cell 1 |
| } | | | |

**Table 8.2.4.4.3-12: *MeasConfig-DEFAULT*: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f1 | | |
| reportConfigId | IdReportConfig-P | | |
| } | | | |
| measGapConfig | Not present | | |
| } | | | |

**Table 8.2.4.4.3-13: *MeasResults*: Additional E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

## 8.2.4.5 Test requirement

Tables 8.2.4.4.1-1, 8.2.4.5-1, 8.2.4.5-2 and 8.2.4.5-3 define the primary level settings including test tolerances for E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps when DRX is used test.

**Table 8.2.4.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | OP.1 TDD | OP.1 TDD | OP.2 TDD | OP.2 TDD | OP.2 TDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 8 | -3.33 | -3.33 | -Infinity | 2.36 | 2.36 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 8 | 8 | 8 | -Infinity | 11 | 11 |
| RSRP Note 3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| SCH_RP Note3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.2.4.5-2: DRX configuration for E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf1280 | |
| shortDRX | disable | |

**Table 8.2.4.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | Infinity | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| sr-ConfigIndex | 2 | For further information see section 6.3.2 in 3GPP TS 36.331 [5] and section10.1 in 3GPP TS 36.213 [8] |

The UE shall transmit a measurement report containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Test requirement = RRC Procedure delay + $T_{identify_CGI,\,intra}$ + TTI insertion uncertainty

RRC procedure delay=15ms

$$T_{identify_CGI,\,intra} = T_{basic_identify_CGI,\,intra} \quad ms$$

$T_{basic_identify_CGI,\,intra}$ = 150 ms

TTI insertion uncertainty = 2 ms

The overall delay measured is 167 ms, allow 170 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.2.5 E-UTRAN TDD-TDD Intra-Frequency Event-Triggered Reporting under Time Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC)

### 8.2.5.1 Test purpose

To verify that the UE correctly detects and reports Event A3 (neighbour becomes offset better than PCell) defined in [5] under a time domain measurement resource restriction and non-MBSFN ABS configured in the aggressor cell.

### 8.2.5.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward. Applicability requires support for FGI bit 115.

### 8.2.5.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable TDD intra-frequency cell within

$$T_{identify_intra_eICIC} = T_{basic_identify_E\text{-}UTRA_TDD_eICIC,\,intra} \cdot \frac{T_{Measurement_Period_eICIC,\,Intra}}{T_{Intra}} \quad ms$$

where

$T_{basic_identify_E\text{-}UTRA_TDD_eICIC,\,intra}$ is 1000 ms.

$T_{Intra}$ is the minimum time available for intra-frequency measurements, during the measurement period with an arbitrarily chosen timing. The time is assumed to be available for performing intra-frequency measurements whenever the receiver is guaranteed to be active on the intra-frequency carrier.

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] clauses 9.1.2.3 and 9.1.2.4 and RSRQ related side conditions given in TS 36.133 [4] clause 9.1.5.2 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex B.2.8 for a corresponding Band.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{Measurement_Period_eICIC,\ Intra}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra-frequency measurements is 200 ms. When no measurement gaps are activated, the UE shall be capable of performing RSRP and RSRQ measurements for 8 identified intra-frequency cells , including also the cells which are not measured in the subframes indicated by the time-domain measurement resource restriction pattern, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of 200 ms. When measurement gaps are activated the UE shall be capable of performing measurements for at least $Y_{measurement_intra_eICIC}$ cells , where $Y_{measurement_intra_eICIC}$ is defined in the following equation. If the UE has identified more than $Y_{measurement_intra_eICIC}$ cells, the UE shall perform measurements of at least 8 identified intra-frequency cells but the reporting rate of RSRP and RSRQ measurements of cells from UE physical layer to higher layers may be decreased.

$$Y_{measurement_intra_eICIC} = Floor\left\{ X_{basic_measurement_TDD_eICIC} \cdot \frac{T_{Intra}}{T_{Measurement_Period_eICIC,\ Intra}} \right\} \text{ cells}$$

where

$X_{basic_measurement_TDD_eICIC}$ = 8 (cells)

$T_{Measurement_Period_eICIC,\ Intra}$ = 200 ms is the measurement period for intra-frequency RSRP and RSRQ measurements.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra_eICIC}$ defined above. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_eICIC}$ defined above becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_intra_eICIC}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.8.2 and A.8.2.5.

## 8.2.5.4 Test description

### 8.2.5.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.15.

2. The general test parameter settings are set up according to Table 8.2.5.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.2.5.4.3.

5. At this stage only Cell 1 is switched on. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1.

**Table 8.2.5.4.1-1: General test parameters for E-UTRAN TDD-TDD intra-frequency event-triggered reporting under time domain measurement resource restriction with Non-MBSFN ABS (eICIC)**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| PCell | | Cell 1 | Also the aggressor cell. Active in T1 and T2 |
| Neighbour cell | | Cell 2 | Cell to be identified. Active only in T2. |
| PCell ABS configuration | | Non-MBSFN ABS | As defined in Table C.3.1.1.1-1 |
| E-UTRA RF Channel Number | | 1 | One TDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| A3-Offset | dB | -11 | |
| Event A3 measurement quantity | | RSRP | |
| CP length | | Normal | |
| Special subframe configuration | | 6 | As specified in Table 4.2-1 in TS 36.211. The same configuration in both cells |
| Uplink-downlink configuration | | 1 | As specified in Table 4.2-2 in TS 36.211. The same configuration in both cells |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | | OFF |
| Time offset between cells | | 3 μs | Synchronous cells |
| T1 | s | 5 | |
| T2 | s | 5 | |
| Physical cell ID PCI | | $(PCI_{cell1} - PCI_{cell2})mod6 != 0$ | Cell PCIs are selected so that the condition is met |
| ABS pattern | | '00000000010000000001' | TDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 1 during T1.<br>The first/leftmost bit corresponds to the subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (20) divided by 10. No MBSFN subframes are configured in the ABS subframes. |
| Time domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | '00000000010000000001' | Time domain measurement resource restriction pattern for neighbour cell measurement signalled to the UE in measSubframePattern-Neigh IE in measSubframePatternConfig-Neigh, as defined in TS 36.331, clause 6.3.5. Configured during T1 for Cell 2 measurements. |
| Time domain measurement resource restriction pattern for PCell measurements | | '10000000001000000000' | Configured during T1 for Cell 1 measurements |

## 8.2.5.4.2 Test procedure

There are two synchronous E-UTRA cells in the test on the same RF channel. Cell 1 is the Pcell and also the aggressor cell, Cell 2 is the neighbour victim cell to be identified. Non-MBSFN ABS pattern is configured for aggressor cell (Cell 1). The UE is configured with a time domain measurement resource restriction pattern for performing E-UTRAN FDD intra-frequency measurements on neighbour cells (Cell 2). The UE is also configured with a time domain measurement resource restriction pattern for PCell (Cell 1) measurements. It is indicated to the UE that event-triggered reporting with Event A3 is used.

The test consists of two successive time periods with time duration of T1 and T2, respectively. During time duration T1, the UE shall not have any timing information on Cell 2. The information for both measurement patterns shall be provided to the UE during T1.

1. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters for both Cell 1 and Cell 2 according to T1 in Table 8.2.5.5-1. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.2.5.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 1002 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A in Cell 1 according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.2.5.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.2.5.4.3-1: Common Exception messages for E-UTRAN TDD-TDD intra-frequency event-triggered reporting under time domain measurement resource restriction with Non-MBSFN ABS (eICIC)**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.7-1 |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.2.5.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD-TDD intra-frequency event-triggered reporting under time domain measurement resource restriction with Non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -22 (-11 dB) | -11 is actual value in dB (-22 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.2.5.4.3-3: *MeasResults*: Additional E-UTRAN TDD-TDD intra-frequency event-triggered reporting under time domain measurement resource restriction with Non-MBSFN ABS (eICIC)**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | Report Cell 1 | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.2.5.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN TDD-TDD intra-frequency event-triggered reporting under time domain measurement resource restriction with Non-MBSFN ABS (eICIC)**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | Report Cell 2 | |
| physCellId | PhysCellId of the Cell 2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 8.2.5.4.3-5: MeasObjectEUTRA-GENERIC: Additional E-UTRAN TDD-TDD intra-frequency event-triggered reporting under time domain measurement resource restriction with Non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternTDD-r10 | | | |
| subframeConfig1-5-r10 | '00000000010000000001' | BIT STRING (SIZE (20)) | Cell1 |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 2 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.5.4.3-6: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN TDD-TDD intra-frequency event-triggered reporting under time domain measurement resource restriction with Non-MBSFN ABS (eICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 | | | |
| subframeConfig1-5-r10 | '10000000001000000000' | BIT STRING (SIZE (20)) | |
| | | | |
| } | | | |
| } | | | |
| } | | | |

## 8.2.5.5 Test requirements

**Table 8.2.5.5-1: Cell specific test requirement parameters for E-UTRAN TDD-TDD Intra-Frequency Event-Triggered Reporting under Time Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC)**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2) | | OP.1 TDD | | OP.2 TDD | |
| PBCH_RA | dB | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1. | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | |
| $(\hat{E}_s / N_{oc})_{meas}$ Note 5 | dB | 1 | 1 | -Infinity | -3.2 |
| $(\hat{E}_s / N_{oc})_{ABS}$ | dB | 1 | 1 | N/A | N/A |
| RSRP Note 4,5 | dBm/15 kHz | -97 | -97 | -Infinity | -101.2 |
| SCH_RP Note 4 | dBm/15 kHz | -97 | -97 | -Infinity | -101.2 |
| CRS $\hat{E}_s / I_{ot}$ | dB | 1 | -0.7 | -Infinity | -6.74 |
| SCH $\hat{E}_s / I_{ot}$ | dB | 1 | 0.7 | -Infinity | -3.2 |
| Propagation Condition | | ETU30 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: RSPP is estimated for Cell 2 during the restricted measurement subframes for neighbour cells. RSPP is estimated for Cell 1 during the PCell restricted subframes.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra_eICIC}$

$$T_{identify_intra_eICIC} = T_{basic_identify_E-UTRA_TDD_eICIC,\ intra} \cdot \frac{T_{Measurement_Period_eICIC,\ Intra}}{T_{Intra}} \quad ms$$

$T_{basic_identify_E-UTRA_TDD_eICIC,\ intra}$ = 1000 ms

$T_{Measurement_Period_eICIC,\ Intra}$ = 200 ms

$T_{Intra}$ = 200 ms

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 1002 ms in this test case (note: this gives a total of 1000 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.2.6 E-UTRAN TDD-TDD Intra-Frequency Event-Triggered Reporting under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC)

### 8.2.6.1 Test purpose

To verify that the UE correctly detects and reports Event A3 (neighbour becomes offset better than PCell) defined in [5] under a time domain measurement resource restriction pattern and CRS assistance information, and when non-MBSFN ABS configured in the interfering cells.

### 8.2.6.2 Test applicability

This test applies to all types of E-UTRA TDD and CRS interference handling UE release 11 and forward. Applicability requires support of FGI bit 115.

### 8.2.6.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD intra-frequency cell within

$$T_{identify_intra_FeICIC} = T_{basic_identify_E-UTRA_TDD_FeICIC,\ intra} \cdot \frac{T_{Measurement_Period_FeICIC,\ Intra}}{T_{Intra}} \quad ms$$

where

- $T_{basic_identify_E-UTRA_TDD_eICIC,\ intra}$ is 1000 ms.
- $T_{Intra}$ is the minimum time available for intra-frequency measurements, during the measurement period with an arbitrarily chosen timing. The time is assumed to be available for performing intra-frequency measurements whenever the receiver is guaranteed to be active on the intra-frequency carrier.

A cell shall be considered detectable when:

- RSRP related side conditions given in TS 36.133 [4] clauses 9.1.2.5 and 9.1.2.6 and RSRQ related side conditions given in TS 36.133 [4] clause 9.1.5.3 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.8 for a corresponding Band.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{Measurement_Period_FeICIC,\ Intra}$. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra-frequency measurements is 200 ms. When no measurement gaps are activated, the UE shall be capable of performing RSRP and RSRQ measurements for 8 identified

intra-frequency cells , including also the cells which are not measured in the subframes indicated by the time-domain measurement resource restriction pattern and the cells indicated in the CRS assistance information, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of 200 ms. When measurement gaps are activated the UE shall be capable of performing measurements for at least $Y_{measurement_intra_FeICIC}$ cells , where $Y_{measurement_intra_FeICIC}$ is defined in the following equation. If the UE has identified more than $Y_{measurement_intra_FeICIC}$ cells, the UE shall perform measurements of at least 8 identified intra-frequency cells but the reporting rate of RSRP and RSRQ measurements of cells from UE physical layer to higher layers may be decreased.

$$Y_{measurement_intra_FeICIC} = Floor\left\{ X_{basic_measurement_TDD_FeICIC} \cdot \frac{T_{Intra}}{T_{Measurement_Period_FeICIC,\ Intra}} \right\} \text{ cells}$$

where

- $X_{basic_measurement_TDD_FeICIC}$ = 8 (cells)
- $T_{Measurement_Period_FeICIC,\ Intra}$ = 200ms is the measurement period for intra-frequency RSRP and RSRQ measurements.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra_FeICIC}$ defined above. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_FeICIC}$ defined above becomes undetectable for a period ≤ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_FeICIC,\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.8.4 and A.8.2.6.

## 8.2.6.4 Test description

### 8.2.6.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.54.
2. The general test parameter settings are set up according to Table 8.2.6.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.2.6.4.3.
5. In this test, there are three E-UTRA cells, Cell 1, Cell 2 and Cell 3, all on same frequencies. At this stage only Cell 1 is switched on. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1.

**Table 8.2.6.4.1-1: General test parameters for E-UTRAN TDD-TDD intra-frequency event triggered reporting under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 TDD | As specified in clause A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 TDD | As specified in clause A.2.2 |
| PCell | | | Cell 1 | Also a first interfering cell to Cell 3. Active in T1 and T2. |
| Neighbour cells | | | Cell 2 and Cell 3 | Cell 2 is a second interfering cell; Cell 2 is active in T1 and T2.<br>Cell 3 is the cell to be identified; Cell 3 is active only in T2. |
| ABS transmission configuration | | | Non-MBSFN ABS | As defined in Table C.3.1.1.1-1 |
| E-UTRA RF Channel Number | | | 1 | One TDD carrier frequency is used |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | For all cells in the test |
| A3-Offset | | dB | -14 | |
| Event A3 measurement quantity | | | RSRP | |
| CP length | | | Normal | |
| Special subframe configuration | | | 6 | As specified in Table 4.2-1 in TS 36.211. The same configuration in both cells |
| Uplink-downlink configuration | | | 1 | As specified in Table 4.2-2 in TS 36.211. The same configuration in both cells |
| Hysteresis | | dB | 0 | |
| Time To Trigger | | s | 0 | |
| Filter coefficient | | | 0 | L3 filtering is not used |
| DRX | | | | OFF |
| Time offset between cells | | µs | Cell 2 offset with respect to Cell 1: 0<br>Cell 3 offset with respect to Cell 1: -2.5 | Three synchronous cells |
| T1 | | s | 5 | |
| T2 | | s | 5 | |
| Physical cell IDs | | | ($PCI_{cell1}$ - $PCI_{cell3}$ )mod6 = 0<br>($PCI_{cell2}$ - $PCI_{cell3}$ )mod6 != 0<br><br>$PCI_{cell1}$ not equal to $PCI_{cell3}$ | Cell PCIs are selected so that all conditions are met |
| ABS pattern | | | ‘00000000010000000001’ | TDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54.<br>The first/leftmost bit corresponds to the PCell subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (20) divided by 10. No MBSFN subframes are configured in the ABS subframes.<br>Provided to the UE for Cell 1 and Cell 2 during T1. |
| Time domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | | ‘00000000010000000001’ | Time domain measurement resource restriction pattern for neighbour cell measurement signalled to the UE in measSubframePatternNeigh IE in measSubframePatternConfigNeigh, as defined in TS 36.331, clause 6.3.5.<br>Provided to the UE during T1 for Cell 3 measurements. The cell list in measSubframeCellList IE shall contain Cell 3 but not Cell 2. |
| Time domain measurement resource restriction pattern for PCell measurements | | | ‘10000000001000000000’ | Configured during T1 for Cell 1 measurements |
| CRS assistance information | physCellId | | see PCI conditions above | The CRS assistance information is provided for Cell 2 only in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig |
| | antennaPortsCount | | 1 | |

| | | | | |
|---|---|---|---|---|
| | mbsfn-SubframeConfigList | | *oneFrame* = '000000' | element with subframe allocation *oneFrame*='000000'. |

### 8.2.6.4.2 Test procedure

The test parameters are given in Tables 8.2.6.4.1-1 and 8.2.6.51-1. It is indicated to the UE in the measurement control information that event-triggered reporting with Event A3 is used. In the test, there are three synchronous cells, Cell 1, Cell 2, and Cell 3, on the same RF channel. Cell 1 is the PCell. Cell 3 is the cell to be identified. A non-MBSFN ABS pattern is configured in each of the Cell 1 and Cell 2 during the entire test. The test consists of two successive time periods with time duration of T1 and T2, respectively. During time duration T1, the UE shall not have any timing information on Cell 3.

The UE is configured by higher layers with a time domain measurement resource restriction pattern for performing E-UTRAN TDD intra-frequency measurements on neighbour cells, namely Cell 3 measurements. The UE is also provided via higher layers with the CRS assistance information for Cell 2. The information for both measurement patterns and the CRS assistance information shall be provided via RRC to the UE during T1.

1. Ensure the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3 with exceptions listed in 7.2A.6.

2. Set the parameters for Cell 1 and Cell 2 according to T1 in Table 8.2.6.5-1. Propagation conditions are set according to Annex B clause B.2.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 for Cell 1, Cell 2, and Cell 3 as specified in Table 8.2.6.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 1002 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. Set Cell 3 physical cell identity = ((current cell 3 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. Formulas are applied until PCI-s values are found, which fulfill the PCI conditions described in Table 8.2.6.4.1-1.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A in Cell 1 according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF in Cell 1 according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.2.6.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 5.2A.5.1 with the following exceptions:

**Table 8.2.6.4.3-1: Common Exception messages for E-UTRAN TDD-TDD intra-frequency event triggered reporting under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.7-1<br>Table H.2.7-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7<br>Table H.3.5-5<br>Table H.5.1-1<br>Table H.5.1-2<br>Table H.5.1-3 |

**Table 8.2.6.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -28 (-14 dB) | -14 is actual value in dB (-28 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| maxReportCells | 2 | | |
| } | | | |

**Table 8.2.6.4.3-3: *MeasResults*: Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | Report Cell 1 | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.2.6.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | Report Cell 2 | |
| physCellId | PhysCellId of Cell 2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| physCellId | PhysCellId of Cell 3 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.2.6.4.3-5: MeasObjectEUTRA-GENERIC: Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternTDD-r10 | | | |
| subframeConfig1-5-r10 | '00000000010000000001' | BIT STRING (SIZE (20)) | Cell1 |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 3 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.6.4.3-6: *RadioResourceConfigDedicated-SRB2-DRB:* Additional E-UTRAN TDD-TDD intra-frequency event-triggered reporting under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 | | | |
| subframeConfig1-5-r10 | '10000000001000000000' | BIT STRING (SIZE (20)) | |
| | | | |
| } | | | |
| } | | | |
| } | | | |

## 8.2.6.5 Test requirements

Tables 8.2.6.4.1-1 and 8.2.6.5-1 define the primary level settings including test tolerances for E-UTRAN TDD-TDD intra-frequency event triggered reporting under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS test.

**Table 8.2.6.5-1: Cell specific test parameters for E-UTRAN TDD-TDD intra-frequency event triggered reporting under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | | OP.2.TDD | | N/A | OP.2 TDD |
| PBCH_RA | dB | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1. | | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1. | | N/A | 0 |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | | | |
| $(\hat{E}_s / N_{oc})$ | dB | 4 | 4 | 2 | 2 | -Infinity | -3.20 |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 | -96 | -96 | -Infinity | -101.2 |
| SCH_RP Note 4 | dBm/15 kHz | -94 | -94 | -96 | -96 | -Infinity | -101.2 |
| CRS $\hat{E}_s / I_{ot}$ Note 5 | dB | 4 | 2.30 | 2 | 0.30 | -Infinity | -8.66 |
| SCH $\hat{E}_s / I_{ot}$ | dB | -0.12 | -0.86 | -3.45 | -4.01 | -Infinity | -10.27 |
| Propagation Condition | | ETU30 | | ETU30 | | ETU30 | |

NOTE 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
NOTE 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.
NOTE 4: RSRP, SCH_RP, and $\hat{E}_s / I_{ot}$ levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 5: Applies during the restricted measurement subframes configured for neighbour cell (Cell 3) measurements.

The UE shall send one Event A3 triggered measurement report for Cell 3, with a measurement reporting delay less than 1000 ms from the beginning of time period T2.

The UE shall not send event-triggered measurement reports as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90% for the tested Event A3.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_intra_FeICIC}$

$$T_{identify_intra_FeICIC} = T_{basic_identify_E\text{-}UTRA_TDD_FeICIC,\ intra} \cdot \frac{T_{Measurement_Period_FeICIC,\ Intra}}{T_{Intra}} \quad ms$$

where

$T_{basic_identify_E\text{-}UTRA_TDD_FeICIC,\ intra}$ is 1000 ms.

$T_{Measurement_Period_FeICIC,\ Intra}$ = 200 ms

$T_{Intra}$ = 200 ms

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 1002 ms in this test case (note: this gives a total of 1000 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.2.7 E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps for UE category 0

### 8.2.7.1 Test purpose

To verify the requirement for identification of a new CGI of E-UTRA cell with autonomous gaps.

### 8.2.7.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward with support of intra-frequency SI acquisition for HO. Applicability requires support for FGI bit 5 of UE category 0.

### 8.2.7.3 Minimum conformance requirements

No explicit neighbour list is provided to the UE for identifying a new CGI of E-UTRA cell. The UE shall identify and report the CGI when requested by the network for the purpose 'reportCGI'. The UE may make autonomous gaps in downlink reception and uplink transmission for receiving MIB and SIB1 messages according to clause 5.5.3.1 of TS 36.331 [2]. Note that a UE is not required to use autonomous gap if si-RequestForHO is set to false. If autonomous gaps are used for measurement with the purpose of 'reportCGI', regardless of whether DRX is used or not, or whether SCell(s) are configured or not, the UE shall be able to identify a new CGI of E-UTRA cell within:

$$T_{identify_CGI_LC\text{-}UE,\ intra} = T_{basic_identify_CGI_LC\text{-}UE,\ intra} \quad ms$$

Where

$T_{basic_identify_CGI_LC\text{-}UE,\ intra}$ = 190 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new CGI of an E-UTRA cell is defined, provided that the E-UTRA cell has been already identified by the UE.

A cell shall be considered identifiable when the following conditions are fulfilled:

- RSRP related side conditions given in TS 36.133 [4] Clause 9.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to TS 36.133 [4] Annex B.2.2 for a corresponding Band

The MIB of an E-UTRA cell whose CGI is identified shall be considered decodable by the UE provided the PBCH demodulation requirements are met according to [TBD].

The requirement for identifying a new CGI of an E-UTRA cell within $T_{basic_identify_CGI_LC\text{-}UE,\ intra}$ is applicable when no DRX is used as well as when all the DRX cycles specified in TS 36.331 [2] are used.

Within the time, $T_{identify_CGI_LC\text{-}UE,\,intra}$ *ms* , over which the UE identifies the new CGI of E-UTRA cell, the UE shall be able to transmit at least the number of ACK/NACKs stated in Table 8.2.7.3-1 on PCell provided that:

- there is continuous DL data allocation,
- no DRX cycle is used,
- no measurement gaps are configured,
- only one code word is transmitted in each subframe,
- no MBSFN subframes are configured in the PCell or each of activated SCell(s).

**Table 8.2.7.3-1: Requirement on minimum number of ACK/NACKs to transmit during $T_{basic_identify_CGI_LC\text{-}UE,\,intra}$**

| UL/DL configuration | Minimum number of transmitted ACK/NACKs |
|---|---|
| 0 (Note 1) | 30 |
| 1 | 54 |
| 2 | 68 |
| 3 | 56 |
| 4 | 61 |
| 5 | 66 |
| 6 | 46 |
| Note 1: The applicability of this requirement is TBD. | |

The ECGI reporting delay occurs due to the delay uncertainty when inserting the ECGI measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. In case DRX is used, the ECGI reporting may be delayed until the next DRX cycle. If IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.5.2.1.6 and A.8.2.7.

## 8.2.7.4 Test description

### 8.2.7.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.20(using only main UE Tx/Rx antenna).
2. The general test parameter settings are set up according to Table 8.2.7.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.2.7.4.3.
5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.2.7.4.1-1: General test parameters for E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.13 TDD | As specified in clause A.3.1.1.5 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.7 TDD | As specified in clause A.3.1.2.2 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | Only one TDD carrier frequency is used. |
| Channel Bandwidth (BWchannel) | MHz | 10 | |
| CP length | | Normal | |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211. The same configuration in both cells |
| Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in TS 36.211. The same configuration in both cells |
| A3-Offset | dB | -3 | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| si-RequestForHO | | TRUE | As specified in clause 5.5.3.1 in TS 36.331. |
| T1 | s | 5 | |
| T2 | s | ≤10 | |
| T3 | s | 5 | |

### 8.2.7.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. PDCCHs indicating new transmissions should be sent continuously to ensure that the UE would have ACK/NACK sending during identifying a new CGI of E-UTRAN cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE does not have any timing information of cell 2. Starting T2, cell 2 becomes detectable and the UE is expected to detect and send a measurement report.

A RRC message implying SI reading shall be sent to the UE during period T2, after the UE has reported Event A3. The RRC message shall create a measurement report configuration with purpose *reportCGI* and *si-RequestForHO* set to TRUE. The start of T3 is the instant when the last TTI containing the RRC message implying SI reading is sent to the UE.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.2.7.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration with event A3 configured message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.2.7.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3.
7. SS shall transmit an RRCConnectionReconfiguration message during period T2, The RRC message shall create a measurement report configuration with purpose *reportCGI* and *si-RequestForHO* set to TRUE.
8. The SS shall start T3 timer when the last TTI containing the RRC message implying SI reading is sent to UE.
9. The UE shall transmit RRCConnectionReconfigurationComplete message.

10. UE shall transmit a measurement report containing the cell global identifier of cell 2 within 210 milliseconds from the start of T3. If the overall delays measured from the beginning of time period T3 is less than 210 ms then the number of successful tests is increased by one. If the UE fails to report the cell global identifier within the overall delays measured requirement then the number of failure tests is increased by one.

11. After the SS receive the MeasurementReport message in step 10) or when T3 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

13. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 2-13 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.2.7.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.2.7.4.3-1: Common Exception messages for E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.2.7.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -6 (-3 dB) | -3 is actual value in dB (-6 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.7.4.3-3: *MeasResults*: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.2.7.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step6)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.2.7.4.3-5: ReportConfigEUTRA-PERIODICAL: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportCGI | | |
| } | | | |
| } | | | |
| reportAmount | 1 | | |
| si-RequestForHO-r9 | setup | | |
| } | | | |

**Table 8.2.7.4.3-6: *MeasResultListEUTRA*: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| cgi-Info SEQUENCE{ | | | |
| cellGlobalId SEQUENCE{ | | | |
| plmn-Identity | plmn-Identity | | |
| cellIdentity | cellIdentity | | |
| } | | | |
| trackingAreaCode | | | |
| plmn-IdentityList | | | |
| } | | | |
| } | | | |

**Table 8.2.7.4.3-7: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellForWhichToReportCGI | Physical Cell ID of Cell 2 | | Cell 1 |
| } | | | |

**Table 8.2.7.4.3-8: *MeasConfig-DEFAULT*: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step 7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f1 | | |
| reportConfigId | IdReportConfig-P | | |
| } | | | |
| measGapConfig | Not present | | |
| } | | | |

**Table 8.2.7.4.3-9: *MeasResults*: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

## 8.2.7.5 Test requirement

Table 8.2.7.4.1-1 and A.8.2.7.5.1-1 define the primary level settings including test tolerances for E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps.

**Table 8.2.7.5.1-1: Cell specific test parameters for E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | | 2x1 | | |
| OCNG Patterns defined in A.3.2.2.1 (OP.1 TDD) and in A.3.2.2.2 (OP.2 TDD) | | OP.1 TDD | OP.1 TDD | OP.1 TDD | OP.2 TDD | OP.2 TDD | OP.2 TDD |
| PBCH_RA | dB | -3 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | 0 | | | 0 | | |
| PHICH_RA | dB | -3 | | | -3 | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | 0 | | | 0 | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | -3 | | | -3 | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 8 | -3.3 | -3.3 | -Infinity | 2.36 | 2.36 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 8 | 8 | 8 | -Infinity | 11 | 11 |
| RSRP Note 3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| SCH_RP Note3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| Propagation Condition | | AWGN | | | | | |
| Timing offset to Cell 1 | μs | - | | | 3 | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall transmit a measurement report containing the cell global identifier of cell 2 within 190 milliseconds from the start of T3.

Test requirement = RRC Procedure delay + $T_{identify_CGI_LC\text{-}UE,\ intra}$ + reporting delay

= 15 + 190 + 2ms from the start of T3

= 207 ms, allow 210 ms.

The UE shall be scheduled continuously throughout the test, and from the start of T3 until 190 ms at least 66 ACK/NACK shall be detected as being transmitted by the UE.

The rate of correct events observed during repeated tests shall be at least 90%.

NOTE: The overall 66 ACK/NACK number is caused by two parts. Firstly, at least 54 ACK/NACK shall be sent during identifying the cell global identifier of cell 2 according to the requirement for UL/DL configuration #1 in Table 8.1.2.2.4.1-1 of Clause 8.1.2.2.4.1. Secondly, given that continuous DL data allocation, additional 12 ACK/NACK shall be sent from the start of T3 until 210 ms excludes 190 ms for identifying the cell global identifier o

## 8.2.8 E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0

### 8.2.8.1 Test purpose

To verify the requirement for identification of a new CGI of E-UTRA cell with autonomous gaps.

### 8.2.8.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward with support of intra-frequency SI acquisition for HO. Applicability requires support for FGI bit 5 of UE category 0.

### 8.2.8.3 Minimum conformance requirements

No explicit neighbour list is provided to the UE for identifying a new CGI of E-UTRA cell. The UE shall identify and report the CGI when requested by the network for the purpose 'reportCGI'. The UE may make autonomous gaps in downlink reception and uplink transmission for receiving MIB and SIB1 messages according to clause 5.5.3.1 of TS 36.331 [2]. Note that a UE is not required to use autonomous gap if si-RequestForHO is set to false. If autonomous gaps are used for measurement with the purpose of 'reportCGI', regardless of whether DRX is used or not, or whether SCell(s) are configured or not, the UE shall be able to identify a new CGI of E-UTRA cell within:

$$T_{identify_CGI_LC\text{-}UE,\ intra} = T_{basic_identify_CGI_LC\text{-}UE,\ intra} \quad ms$$

Where

$T_{basic_identify_CGI_LC\text{-}UE,\ intra}$ = 190 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new CGI of an E-UTRA cell is defined, provided that the E-UTRA cell has been already identified by the UE.

A cell shall be considered identifiable when the following conditions are fulfilled:

- RSRP related side conditions given in TS 36.133 [4] Clause 9.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to TS 36.133 [4] Annex B.2.2 for a corresponding Band.

The MIB of an E-UTRA cell whose CGI is identified shall be considered decodable by the UE provided the PBCH demodulation requirements are met according to [TBD].

The requirement for identifying a new CGI of an E-UTRA cell within $T_{basic_identify_CGI_LC\text{-}UE,\ intra}$ is applicable when no DRX is used as well as when all the DRX cycles specified in TS 36.331 [2] are used.

Within the time, $T_{identify_CGI_LC\text{-}UE,\ intra}$ $ms$, over which the UE identifies the new CGI of E-UTRA cell, the UE shall be able to transmit at least the number of ACK/NACKs stated in Table 8.2.8.3-1 on PCell provided that:

- there is continuous DL data allocation,
- no DRX cycle is used,
- no measurement gaps are configured,
- only one code word is transmitted in each subframe,
- no MBSFN subframes are configured in the PCell or each of activated SCell(s).

**Table 8.2.8.3-1: Requirement on minimum number of ACK/NACKs to transmit during $T_{basic_identify_CGI_LC-UE, intra}$**

| UL/DL configuration | Minimum number of transmitted ACK/NACKs |
|---|---|
| 0 (Note 1) | 30 |
| 1 | 54 |
| 2 | 68 |
| 3 | 56 |
| 4 | 61 |
| 5 | 66 |
| 6 | 46 |
| Note 1: The applicability of this requirement is TBD. | |

The ECGI reporting delay occurs due to the delay uncertainty when inserting the ECGI measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. In case DRX is used, the ECGI reporting may be delayed until the next DRX cycle. If IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.5.2.1.6 and A.8.2.8.

## 8.2.8.4 Test description

### 8.2.8.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.20(using only main UE Tx/Rx antenna).
2. The general test parameter settings are set up according to Table 8.2.8.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.2.8.4.3.
5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.2.8.4.1-1: General test parameters for E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.13 TDD | As specified in clause A.3.1.1.5 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.7 TDD | As specified in clause A.3.1.2.2 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | Only one TDD carrier frequency is used. |
| Channel Bandwidth (BWchannel) | MHz | 10 | |
| CP length | | Normal | |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211. The same configuration in both cells |
| Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in TS 36.211. The same configuration in both cells |
| A3-Offset | dB | -3 | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | ON | DRX related parameters are defined in Table A.8.2.8.1-3 |
| si-RequestForHO | | TRUE | As specified in clause 5.5.3.1 in TS 36.331. |
| T1 | s | 5 | |
| T2 | s | ≤30 | UE should report cell within 25.6s (20 DRX cycles) |
| T3 | s | 5 | |

### 8.2.8.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE does not have any timing information of cell 2. Starting T2, cell 2 becomes detectable and the UE is expected to detect and send a measurement report.

A RRC message implying SI reading shall be sent to the UE during period T2, after the UE has reported Event A3. The RRC message shall create a measurement report configuration with purpose *reportCGI* and *si-RequestForHO* set to TRUE. The start of T3 is the instant when the last TTI containing the RRC message implying SI reading is sent to the UE.

In the test UE needs to be provided at least once every 1280ms with new Timing Advance Command MAC control element to restart the Time alignment timer to keep UE uplink time alignment. Furthermore UE sends scheduling request (SR) to obtain allocation to send the measurement report on PUSCH.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.2.8.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration with event A3 configured message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.2.8.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3.
7. SS shall transmit an RRCConnectionReconfiguration message during period T2, The RRC message shall create a measurement report configuration with purpose *reportCGI* and *si-RequestForHO* set to TRUE.

8. The SS shall start T3 timer when the last TTI containing the RRC message implying SI reading is sent to UE.

9. The UE shall transmit RRCConnectionReconfigurationComplete message.

10. UE shall transmit a measurement report containing the cell global identifier of cell 2 within 210 milliseconds from the start of T3. If the overall delays measured from the beginning of time period T3 is less than 210 ms then the number of successful tests is increased by one. If the UE fails to report the cell global identifier within the overall delays measured requirement then the number of failure tests is increased by one.

11. After the SS receive the MeasurementReport message in step 10) or when T3 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

13. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 2-13 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.2.8.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.2.8.4.3-1: Common Exception messages for E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.2.8.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.8.4.3-3: *MAC-MainConfig-RBC*: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf1280 typical value in real network for real-time services. | |
| sf1280 | 9 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| } | | | |

**Table 8.2.8.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.6, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| … | | | |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | |
| } | | | |

**Table 8.2.8.4.3-5: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 2 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.2.8.4.3-6: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -6 (-3 dB) | -3 is actual value in dB (-6 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.8.4.3-7: *MeasResults*: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.2.8.4.3-8: *MeasResultListEUTRA*: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step6)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.2.8.4.3-9: *ReportConfigEUTRA-PERIODICAL*: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportCGI | | |
| } | | | |
| } | | | |
| reportAmount | 1 | | |
| si-RequestForHO-r9 | setup | | |
| } | | | |

**Table 8.2.8.4.3-10: *MeasResultListEUTRA*: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| cgi-Info SEQUENCE{ | | | |
| cellGlobalId SEQUENCE{ | | | |
| plmn-Identity | plmn-Identity | | |
| cellIdentity | cellIdentity | | |
| } | | | |
| trackingAreaCode | | | |
| plmn-IdentityList | | | |
| } | | | |
| } | | | |

**Table 8.2.8.4.3-11: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellForWhichToReportCGI | Physical Cell ID of Cell 2 | | Cell 1 |
| } | | | |

**Table 8.2.8.4.3-12: *MeasConfig-DEFAULT*: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step7)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f1 | | |
| reportConfigId | IdReportConfig-P | | |
| } | | | |
| measGapConfig | Not present | | |
| } | | | |

**Table 8.2.8.4.3-13: *MeasResults*: Additional E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step10)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

## 8.2.8.5 Test requirement

Tables 8.2.8.4.1-1, 8.2.8.5-1, 8.2.8.5-2 and 8.2.8.5-3 define the primary level settings including test tolerances for Cell specific test parameters for E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX.

**Table 8.2.8.5.1-1: Cell specific test parameters for E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| Correlation Matrix and Antenna Configuration | | 2x1 | | | 2x1 | | |
| OCNG Patterns defined in A.3.2.2.1 (OP.1 TDD) and in A.3.2.2.2 (OP.2 TDD) | | OP.1 TDD | OP.1 TDD | OP.1 TDD | OP.2 TDD | OP.2 TDD | OP.2 TDD |
| PBCH_RA | dB | -3 | | | -3 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | 0 | | | 0 | | |
| PHICH_RA | dB | -3 | | | -3 | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | 0 | | | 0 | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | -3 | | | -3 | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 8 | -3.3 | -3.3 | -Infinity | 2.36+ | 2.36 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 8 | 8 | 8 | -Infinity | 11 | 11 |
| RSRP Note 3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| SCH_RP Note3 | dBm/15 KHz | -90 | -90 | -90 | -Infinity | -87 | -87 |
| Propagation Condition | | AWGN | | | | | |
| Timing offset to Cell 1 | μs | - | | | 3 | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.2.8.5.1-2: DRX configuration for E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf1280 | |
| shortDRX | disable | |

**Table 8.2.8.5.1-3: *TimeAlignmentTimer* -Configuration for E-UTRAN TDD Intra frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | infinity | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 2 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213. |

The UE shall transmit a measurement report containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3.

Test requirement = RRC Procedure delay + $T_{identify_CGI_LC\text{-}UE,\ intra}$ + reporting delay

= 15 + 190 + 2ms from the start of T3

= 207 ms, allow 210 ms.

The rate of correct events observed during repeated tests shall be at least 90%.

# 8.3 E-UTRAN FDD-FDD Inter-frequency Measurements

## 8.3.1 E-UTRAN FDD-FDD Inter-frequency event triggered reporting under fading propagation conditions in asynchronous cells

### 8.3.1.1 Test purpose

To verify the UE's ability to make a correct reporting of an event under fading propagation conditions in asynchronous cells within the E-UTRA FDD inter-frequency cell search requirements. This test will partly verify the FDD-FDD inter-frequency cell search requirements in section 8.1.2.3.

### 8.3.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward. Applicability requires support for FGI bit 25.

### 8.3.1.3 Minimum conformance requirements

When measurement gaps are scheduled the UE shall be able to identify a new FDD inter-frequency within $T_{Identify_Inter}$ according to the following expression:

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new FDD inter-frequency cell is defined.

$N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1 and $T_{Inter1}$ is defined in TS 36.133 [4] section 8.1.2.1.

A cell shall be considered detectable provided following conditions are fulfilled:

- RSRP and RSRP Ês/Iot according to Annex I.2.3 for a corresponding Band,
- other RSRP related side conditions given in Sections 9.1.3.1 and 9.1.3.2 and RSRQ related side conditions given in Sections 9.1.6.1 and 9.1.6.2 are fulfilled,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band.

When measurement gaps are scheduled for FDD inter frequency measurements the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with measurement accuracy as specified in TS 36.133 sub-clauses 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively, with measurement period given by table 8.3.1.3-1.

**Table 8.3.1.3-1: Measurement period and measurement bandwidth**

| Configuration | Physical Layer Measurement period: $T_{Measurement_Period_Inter_FDD}$ [ms] | Measurement bandwidth [RB] |
|---|---|---|
| 0 | 480 x $N_{freq}$ | 6 |
| 1 (Note) | 240 x $N_{freq}$ | 50 |
| Note: This configuration is optional. | | |

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per FDD inter-frequency and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period defined in Table 8.3.1.3-1.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clauses 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_inter}$ defined in TS 36.133 [4] clause 8.1.2.3.1.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra}$ defined in TS 36.133 [4] section 8.1.2.2.1.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.1 and A.8.3.1.

## 8.3.1.4 Test description

### 8.3.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15.

2. The general test parameter settings are set up according to Table 8.3.1.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.3.1.4.3.

5. There are two E-UTRA FDD carriers and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.3.1.4.1-1: General test parameters for E-UTRAN FDD-FDD inter frequency event triggered reporting in fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| E-UTRA RF Channel Number | | 1, 2 | Two FDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active cell | | Cell 1 | Cell 1 is on RF channel number 1 |
| Neighbour cell | | Cell 2 | Cell 2 is on RF channel number 2 |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| A3-Offset | dB | -6 | |
| Hysteresis | dB | 0 | |
| CP length | | Normal | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | OFF |
| Time offset between cells | ms | 3 | Asynchronous cells<br>3ms or 92160*Ts |
| T1 | s | 5 | |
| T2 | s | 5 | |

### 8.3.1.4.2 Test procedure

The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2. Gap pattern configuration is configured before T2 begins to enable inter-frequency monitoring.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.3.1.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.3.1.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the measurement reporting delay from the beginning of time period T2 is less than 3842ms the number of successful tests is increased by one. If the UE fails to report the event within the measurement reporting delay requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.3.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.3.1.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.3.1-9 |

**Table 8.3.1.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.3.1.4.3-3: *MeasResults*: Additional E-UTRAN FDD-FDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.3.1.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

## 8.3.1.5 Test requirement

Tables 8.3.1.4.1-1 and 8.3.1.5-1 define the primary level settings including test tolerances for event triggered reporting under fading propagation conditions in asynchronous inter frequency cells test.

**Table 8.3.1.5-1: Cell specific test requirement parameters for E-UTRAN FDD-FDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells**

| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | |
|---|---|---|---|---|---|
| | | **T1** | **T2** | **T1** | **T2** |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA[Note 1] | dB | | | | |
| OCNG_RB[Note 1] | dB | | | | |

| | | | | | |
|---|---|---|---|---|---|
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 | -Infinity | -90.30 |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | -Infinity | 7.70 |
| SCH_RP | dBm/15 kHz | -94 | -94 | -Infinity | -90.30 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | -Infinity | 7.70 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 3842 ms from the beginning of time period T2.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = 3840 ms

TTI insertion uncertainty = $TTI_{DCCH}$ = 1 ms; $2xTTI_{DCCH}$ = 2 ms

The overall delays measured shall be less than a total of 3842 ms in this test case (note: this gives a total of 3840 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

## 8.3.2 E-UTRAN FDD-FDD Inter-frequency event triggered reporting when DRX is used under fading propagation conditions in asynchronous cells

### 8.3.2.1 Test purpose

To verify the UE's ability to make a correct reporting of an event in DRX under fading propagation conditions in asynchronous cells within the E-UTRA FDD-FDD inter-frequency cell search requirements.

### 8.3.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward. Applicability requires support for FGI bits 5, and 25.

### 8.3.2.3 Minimum conformance requirements

When measurement gaps are scheduled the UE shall be able to identify a new FDD inter-frequency within $T_{Identify_Inter}$ as shown in table 8.3.2.3-1:

**Table 8.3.2.3-1: Requirement to identify a newly detectable FDD inter-frequency cell**

| DRX cycle length (s) | $T_{identify_inter}$ (s) (DRX cycles) | |
|---|---|---|
| | Gap period = 40 ms | Gap period = 80 ms |
| ≤0.16 | Non DRX Requirements in TS 36.133 [4] clause 8.1.2.3.1.1 are applicable | Non DRX Requirements in TS 36.133 [4] clause 8.1.2.3.1.1 are applicable |
| 0.256 | 5.12*$N_{freq}$ (20*$N_{freq}$) | 7.68*$N_{freq}$ (30*$N_{freq}$) |
| 0.32 | 6.4*$N_{freq}$ (20*$N_{freq}$) | 7.68*$N_{freq}$ (24*$N_{freq}$) |
| 0.32 < DRX-cycle ≤ 2.56 | Note (20*$N_{freq}$) | Note (20*$N_{freq}$) |
| Note: Time depends upon the DRX cycle in use | | |

The non DRX requirements in TS 36.133 [4] clause 8.1.2.3.1.1 states that when measurement gaps are scheduled the UE shall be able to identify a new FDD inter-frequency within $T_{Identify\ Inter}$ according to the following expression:

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new FDD inter-frequency cell is defined.

$N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1 and $T_{Inter1}$ is defined in TS 36.133 [4] section 8.1.2.1.

A cell shall be considered detectable provided following conditions are fulfilled:

- RSRP and RSRP Ês/Iot according to Annex I.2.3 for a corresponding Band,
- other RSRP related side conditions given in Sections 9.1.3.1 and 9.1.3.2 and RSRQ related side conditions given in Sections 9.1.6.1 and 9.1.6.2 are fulfilled,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band.

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per FDD inter-frequency and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period defined in Table 8.3.2.3-2.

**Table 8.3.2.3-2: Requirement to measure FDD inter-frequency cells**

| DRX cycle length (s) | $T_{measure_inter}$ (s) (DRX cycles) |
|---|---|
| ≤0.08 | Non DRX Requirements in TS 36.133 [4] clause 8.1.2.3.1.1 are applicable |
| 0.08 < DRX-cycle ≤ 2.56 | Note (5*$N_{freq}$) |
| Note: Time depends upon the DRX cycle in use | |

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clauses 9.1.3.1 and 9.1.3.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in the sub-clauses 9.1.6.1 and 9.1.6.2.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clauses 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x TTI DCCH. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ inter}$ defined in TS 36.133 [4] clause 8.1.2.2.1.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra}$ defined in TS 36.133 [4] section 8.1.2.2.1.1 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.1 and A.8.3.2.

## 8.3.2.4 Test description

### 8.3.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15.

2. The general test parameter settings are set up according to Table 8.3.2.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.3.2.4.3.

5. There are two E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.3.2.4.1-1: General Test Parameters for E-UTRAN FDD-FDD inter-frequency event triggered reporting when DRX is used under fading propagation conditions in asynchronous cells**

| Parameter | Unit | Test 1 | Test 2 | Comment |
|---|---|---|---|---|
| | | Value | | |
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | | As specified in section A.1.1 Note that UE may only be allocated at *On Duration* |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | | As specified in section A.2.1. |
| E-UTRA RF Channel Number | | 1, 2 | | Two FDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| Active cell | | Cell 1 | | Cell 1 is on RF channel number 1 |
| Neighbour cell | | Cell 2 | | Cell 2 is on RF channel number 2 |
| Gap Pattern Id | | 0 | | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| A3-Offset | dB | -6 | | |
| Hysteresis | dB | 0 | | |
| CP length | | Normal | | |
| TimeToTrigger | s | 0 | | |
| Filter coefficient | | 0 | | L3 filtering is not used |
| PRACH configuration | | 4 | | As specified in table 5.7.1-2 in TS 36.211 |
| Access Barring Information | - | Not Sent | | No additional delays in random access procedure. |
| DRX | | ON | | DRX related parameters are defined in Table 8.3.2.5-2 |
| Time offset between cells | | 3 ms | | Asynchronous cells<br>3ms or 92160*Ts |
| T1 | s | 5 | | |
| T2 | s | 5 | 30 | |

### 8.3.2.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

In Test 1 when DRX = 40 ms is used, UE needs to be provided at least once every 500 ms with new Timing Advance Command MAC control element to restart the Timer Alignment Timer to keep the UE uplink time alignment. Furthermore, the UE is allocated with PUSCH resource at every DRX cycle. In Test 2 when DRX = 1280 ms is used, the uplink time alignment is not maintained and the UE needs to use RACH to obtain uplink allocation for measurement reporting.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.3.2.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.3.2.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 3882 ms for Test 1 or less than 26882 ms for Test 2 then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

11. Repeat step 1-10 for each sub-test in Table 8.3.2.4.1-1 as appropriate.

### 8.3.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.3.2.4.3-1: Common Exception messages for E-UTRAN FDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test 1 and 2 requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-7<br>Table H.3.1-9<br>Table H.3.7-1<br>Table H.3.7-2<br>Table H.3.7-3 |

**Table 8.3.2.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test 1 and 2 requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.3.2.4.3-3: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test 1 and 2 requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventide CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.3.2.4.3-4: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test 1 and 2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.3.2.4.3-5: *MeasResults*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test 1 and 2 requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.3.2.4.3-6: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test 1 and 2 requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| physCellId | PhysCellId | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.3.2.4.3-7: *PRACH-Config-DEFAULT*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test 1 and 2 requirements**

| Derivation Path: TS 36.508 [7] clause 7.3.2 Table 7.3.2-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PRACH-Config-DEFAULT ::= SEQUENCE { | | | |
| prach-ConfigIndex | 4 | | |
| } | | | |

## 8.3.2.5 Test requirement

Tables 8.3.2.4.1-1, 8.3.2.5-1, 8.3.2.5-2 and 8.3.2.5-3 define the primary level settings including test tolerances for E-UTRAN FDD-FDD inter-frequency event triggered reporting when DRX is used under fading propagation conditions in asynchronous cells test.

**Table 8.3.2.5-1: Cell Specific Test Parameters for E-UTRAN FDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA[Note 1] | dB | | | | |
| OCNG_RB[Note 1] | dB | | | | |

| | | | | | |
|---|---|---|---|---|---|
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | | |
| RSRP Note 3 | dBm/15 kHz | -94 | -94 | -Infinity | -90.30 |
| $\hat{E}_s/I_{ot}$ | dB | 4 | 4 | -Infinity | 7.70 |
| SCH_RP Note 3 | dBm/15 kHz | -94 | -94 | -Infinity | -90.30 |
| $\hat{E}_s/N_{oc}$ | dB | 4 | 4 | -Infinity | 7.70 |
| Propagation Condition | | ETU70 | | | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.<br>Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | | | |

**Table 8.3.2.5-2: DRX-Configuration to be used in E-UTRAN FDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf40 | sf1280 | |
| shortDRX | disable | disable | |
| Note: For further information see section 6.3.2 in 3GPP TS 36.331 [5]. | | | |

**Table 8.3.2.5-3: *TimeAlignmentTimer* and *sr-ConfigIndex*-Configuration to be used in E-UTRAN FDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | For further information see section 6.3.2 in 3GPP TS 36.331 [5]. |
| sr-ConfigIndex | 0 | 0 | For further information see section 6.3.2 in 3GPP TS 36.331 and section10.1 in 3GPP TS 36.213 [8]. |

In Test 1 when DRX cycle length = 40 ms is used, the overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2 on PUSCH.

In Test 2 when DRX cycle length = 1280 ms is used, the overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE starts to send preambles on the PRACH for Scheduling Request (SR) to obtain allocation to send the measurement report to Cell 2 on PUSCH.

For both tests:

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event A3 measurement report.

The overall delay measured when DRX cycle length is 40 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty+ DRX cycle length

Measurement reporting delay = $T_{Identify_Inter}$

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new FDD inter-frequency cell is defined.

$T_{Inter1}$ = 60 ms. It is defined in table 8.1.2.1-1 of TS 36.133 [4] clause 8.1.2.1.

$N_{freq}$ = 1. It is defined in TS 36.133 [4] clause 8.1.2.1.1.

TTI insertion uncertainty = 2 ms

DRX cycle length = 40 ms

The overall delays measured when DRX cycle length is 40 ms shall be less than a total of 3882 ms.

The overall delay measured when DRX cycle length is 1280 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_inter}$

$T_{identify_inter}$ = 25600 ms. When DRX cycle length is 1280 ms then the $T_{identify_inter}$ is 20 x 1280 ms, as defined in Table 8.3.2.3-1.

TTI insertion uncertainty = 2 ms

DRX cycle length = 1280 ms

The overall delays measured when DRX cycle length is 1280 ms shall be less than a total of 26882 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.3.3 E-UTRAN FDD-FDD Inter frequency event triggered reporting under AWGN propagation conditions in asynchronous cells with DRX when L3 filtering is used

### 8.3.3.1 Test purpose

To verify the UE's ability to make a correct reporting of an event in DRX when L3 filtering is used in AWGN propagation conditions in asynchronous cells within the E-UTRA FDD-FDD inter frequency cell search in DRX requirements and the UE behaviour with the filterCoefficent.

### 8.3.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward. Applicability requires support for FGI bits 5, and 25.

### 8.3.3.3 Minimum conformance requirements

When DRX is in use the UE shall be able to identify a new detectable FDD inter frequency cell within $T_{identify\ inter}$ as defined in table 8.1.2.3.1.2-1 of TS 36.133 [4] clause 8.1.2.3.1.2.

A cell shall be considered detectable when

- RSRP and RSRP Ês/Iot according to Annex I.2.3 for a corresponding Band,

- other RSRP related side condition given in TS 36.133 [4] clauses 9.1.3.1 and 9.1.3.2 and RSRQ related side conditions given in Sections 9.1.6.1 and 9.1.6.2 are fulfilled,

- SCH_RP|$_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band.

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per FDD inter-frequency and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period defined in table 8.3.3.3-1.

**Table 8.3.3.3-1: Requirement to measure FDD inter frequency cells**

| DRX cycle length (s) | $T_{measure_inter}$ (s) (DRX cycles) |
|---|---|
| ≤0.08 | Non DRX Requirements in section 8.1.2.3.1.1 in 3GPP TS 36.133 [4] are applicable |
| 0.08<DRX-cycle≤2.56 | Note (5*$N_{freq}$) |
| Note: Time depends upon the DRX cycle in use | |

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clauses 9.1.3.1 and 9.1.3.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in the sub-clauses 9.1.6.1 and 9.1.6.2.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clauses 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_inter}$ defined in TS 36.133 [4] clause 8.1.2.3.1.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_inter}$ defined in TS 36.133 [4] clause 8.1.2.3.1.1 and then enters or leaves the reporting range, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ Inter}$ defined in TS 36.133 [4] clause 8.1.2.3.1.1 provided the timing to that cell has not changed more than ± 50 Ts while measurement gap has not been available and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.1 and A.8.3.3.

## 8.3.3.4 Test description

### 8.3.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. The general test parameter settings are set up according to Table 8.3.3.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.3.3.4.3.

5. There is one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.3.3.4.1-1: General Test Parameters for E-UTRAN FDD-FDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1, 2 | Two FDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Gap Pattern Id | | 1 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1 |
| Neighbour A3-Offset Ofn | dB | -14 | |
| CP length | | Normal | |
| Hysteresis | dB | 0 | |
| Time To Trigger | dB | 0 | |
| Filter coefficient | | 9 | L3 filtering is used |
| DRX | | ON | DRX related parameters are defined in Table 8.3.3.5-2 |
| Time offset between cells | | 3 ms | Asynchronous cells |
| T1 | s | 30 | |
| T2 | s | 9 | |

### 8.3.3.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 and the filter coefficient is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively.

In the Test when DRX = 1280 ms is used, a non-zero L3 filtering is configured. The uplink time alignment is not maintained and UE needs to use RACH to obtain UL allocation for measurement reporting.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.3.3.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. During T1 the SS continuously checks the absence of an A3 triggered measurement report from the neighbour cell. If the UE does not send this during T1, then count neither success nor fail. If the UE sends such a report, then count a fail. Upon a fail in step 5, neither success nor fail are counted in step 7.

6. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.3.3.5-1 and 8.3.3.5-2.

7. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 7682 ms then the number of “successes” is increased by one, otherwise count a fail.

8. After the SS receive the MeasurementReport message in step 7) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

9. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

10. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

11. Repeat step 2-10 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

12. Repeat step 1-11 for each sub-test in Table 8.3.3.4.1-1 as appropriate.

### 8.3.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.3.3.4.3-1: Common Exception messages for E-UTRAN FDD-FDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used in test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-3<br>Table H.3.7-2<br>Table H.3.7-3 |

**Table 8.3.3.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used in test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.3.3.4.3-3: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used in test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventid CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | 0 (0 dB) | The actual value is IE value * 0.5 dB IE Value INTEGER (-30..30) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | The actual value is IE value * 0.5 dB | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.3.3.4.3-4: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.3.3.4.3-5: *MeaResults*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.3.3.4.3-6: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| physCellId | PhysCellId | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.3.3.4.3-7: *FilterCoefficient*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| filterCoefficient | fc9 | | |

**Table 8.3.3.4.3-8: *MeasObjectEUTRA-GENERIC(Freq)*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC(Freq) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| carrierFreq | Downlink EARFCN for Freq | | |
| allowedmeasBandwidth | The number of the resource blocks for Freq | | |
| presenceAntennaPort1 | FALSE | | |
| neighCellConfig | '10'B (The MBSFN subframe allocations of all neighbour cells are identical to or subsets of that in the serving cells) | | |
| offsetFreq | -14 (dB-14) | -14 dB is actual value in dB (Value dB-14 corresponds to -14 dB) | |
| cellsToRemoveList | Not present | | |
| cellsToAddModList | Not present | | |
| blackCellsToRemoveList | Not present | | |
| blackCellsToAddModList | Not present | | |
| cellForWhichToReportCGI | Not present | | |
| } | | | |

## 8.3.3.5 Test requirement

Tables 8.3.3.4.1-1, 8.3.3.5-1, 8.3.3.5-2 and 8.3.3.5-3 define the primary level settings including test tolerances for E-UTRAN FDD-FDD inter-frequency event triggered reporting under AWGN propagation conditions in asynchronous cells with DRX when L3 filtering is used test.

**Table 8.3.3.5-1: Cell Specific Test Parameters for E-UTRAN FDD-FDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 4.00 | 0.00 | 4.00 | 24.00 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -95.50 | -95.50 | -98.00 | -98.00 |
| $\hat{E}_s / N_{oc}$ | dB | 4.00 | 0.00 | 4.00 | 24.00 |
| RSRP Note 3 | dBm/15 KHz | -91.50 | -95.50 | -94.00 | -74.00 |
| SCH_RP Note 3 | dBm/15 KHz | -91.50 | -95.50 | -94.00 | -74.00 |
| Propagation Condition | | AWGN | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.3.3.5-2: DRX-Configuration to be used in E-UTRAN FDD-FDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in section 6.3.2 in 3GPP TS 36.331 [5]. |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf1280 | |
| shortDRX | disable | |

**Table 8.3.3.5-3: *TimeAlignmentTimer* and *sr-ConfigIndex*-Configuration to be used in E-UTRAN FDD-FDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | sf500 | As specified in section 6.3.2 in 3GPP TS 36.331 [5]. |
| sr-ConfigIndex | 0 | For further information see section 6.3.2 in 3GPP TS 36.331 [5] and section 10.1 in 3GPP TS 36.213 [8]. |

The overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE starts to send preambles on the PRACH for Scheduling Request (SR) to obtain allocation to send the measurement report to Cell 2 on PUSCH.

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event A3 measurement report.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled. The overall delay measured when DRX cycle length is 1280 ms test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{measure_inter}$

$T_{measure_inter}$ = 6400 ms. When DRX cycle length is 1280 ms then the $T_{measure_inter}$ is 5 x 1280 ms, as defined in Table 8.3.3.3-1.

TTI insertion uncertainty = 2 ms

DRX cycle length = 1280 ms

The overall delay measured when DRX cycle length is 1280 ms shall be less than a total of 7682 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.3.4 E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps

### 8.3.4.1 Test purpose

To verify the requirement for identification of a new CGI of E-UTRA cell with autonomous gaps.

### 8.3.4.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward with support of inter-frequency SI acquisition for HO.

### 8.3.4.3 Minimum conformance requirements

No explicit neighbour list is provided to the UE for identifying a new CGI of E-UTRA cell. The UE shall identify and report the CGI when requested by the network for the purpose of 'reportCGI'. The UE may make autonomous gaps in both downlink reception and uplink transmission for receiving MIB and SIB1 message according to section 5.5.3.1 of TS 36.331 [5]. Note that a UE is not required to use autonomous gap if si-RequestForHO is set to false. If autonomous gaps are used for measurement with the purpose of 'reportCGI', regardless of whether DRX is used or not, the UE shall be able to identify a new CGI of E-UTRA cell within:

$$T_{identify_CGI,\,inter} = T_{basic_identify_CGI,\,inter} \quad ms$$

Where

$T_{basic_identify_CGI,\,inter}$ = 150 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new CGI of E-UTRA cell is defined, provided that the E-UTRA cell has been already identified by the UE.

A cell shall be considered identifiable following conditions are fulfilled:

- RSRP related side conditions given in TS 36.133 [4] Section 9.1 are fulfilled for a corresponding Band,
- SCH_RP$|_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band

The requirement for identifying a new CGI of an E-UTRA cell within $T_{basic_identify_CGI,inter}$ is applicable when no DRX is used as well as when all the DRX cycles specified in 3GPP TS 36.331 [5] are used.

Given that continuous DL data allocation and no DRX is used, and no measurement gaps are configured, the UE shall have more than 60 ACK/NACKs transmitted during the identification of a new CGI of E-UTRA cell.

The ECGI reporting delay occurs due to the delay uncertainty when inserting the ECGI measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. In case DRX is used, the ECGI reporting may be delayed until the next DRX cycle.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.5 and A.8.3.4.

## 8.3.4.4 Test description

### 8.3.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.
2. The general test parameter settings are set up according to Table 8.3.4.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.3.4.4.3.
5. There are two E-UTRA FDD carriers and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.3.4.4.1-1: General test parameters for E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.3 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| E-UTRA RF channel number | | 1, 2 | Two FDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active cell | | Cell 1 | Cell 1 is on RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 1 is on RF channel number 2. |
| CP length | | Normal | |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| A3-Offset | dB | -6 | |
| Hysteresis | dB | 0 | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| si-RequestForHO | | TRUE | As specified in section 5.5.3.1 in TS 36.331 [5]. |
| Time offset between cells | ms | 3 | Asynchronous cells |
| T1 | s | 5 | |
| T2 | s | ≤10 | |
| T3 | s | 5 | |

### 8.3.4.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. PDCCHs indicating new transmissions shall be sent continuously to ensure that the UE would have ACK/NACK sending during identifying a new CGI of E-UTRAN cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE does not have any timing information of cell 2. Gap pattern configuration is configured before T2 begins to enable inter-frequency monitoring. Starting T2, cell 2 becomes detectable and the UE is expected to detect and send a measurement report.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.3.4.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration with event A3 configured message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.3.4.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3.
7. The SS shall transmit an RRCConnectionReconfiguration message during period T2. The RRC message shall release MeasGapConfig according to Table 8.3.4.4.3-6.
8. The UE shall transmit RRCConnectionReconfigurationComplete message.
9. SS shall transmit an RRCConnectionReconfiguration message during period T2, The RRC message shall create a measurement report configuration with purpose *reportCGI* and *si-RequestForHO* set to TRUE.
10. The SS shall start T3 timer when the last TTI containing the RRC message implying SI reading is sent to UE.
11. The UE shall transmit RRCConnectionReconfigurationComplete message
12. UE shall transmit a MeasurementReport message containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3. The UE shall be scheduled continuously throughout the test, and from the start of T3 until 170 ms at least 80 ACK/NACK shall be detected as being transmitted by the UE. If the overall delays

measured from the beginning of time period T3 is less than 170 ms, and the UE have more than 80 ACK/NACKs transmitted from the start of T3 until 170 ms, then the number of successful tests is increased by one. If the UE fails to report the cell global identifier within the overall delays measured requirement, or the UE have less than 80 ACK/NACKs transmitted from the start of T3 until 170 ms, then the number of failure tests is increased by one.

13. After the SS receive the MeasurementReport message (in step 12) or when T3 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

14. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

15. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

16. Repeat step 2-15 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.3.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.3.4.4.3-1: Common Exception messages for E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.3.1-9 |

**Table 8.3.4.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.3.4.4.3-3: *MeasResults*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.3.4.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step6)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.3.4.4.3-5: *ReportConfigEUTRA-PERIODICAL*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step9)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportCGI | | |
| } | | | |
| } | | | |
| reportAmount | 1 | | |
| si-RequestForHO-r9 | setup | | |
| } | | | |

**Table 8.3.4.4.3-6: MeasGapConfig- Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step7)**

| Derivation Path: 36.331, clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasGapConfig ::= CHOICE { | | | |
| release | NULL | | |
| } | | | |

**Table 8.3.4.4.3-7: *MeasResultListEUTRA*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step12)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| cgi-Info SEQUENCE{ | | | |
| cellGlobalId SEQUENCE{ | | | |
| plmn-Identity | plmn-Identity | | |
| cellIdentity | cellIdentity | | |
| } | | | |
| trackingAreaCode | | | |
| plmn-IdentityList | | | |
| } | | | |
| } | | | |

**Table 8.3.4.4.3-8: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step9)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(f1) ::= SEQUENCE { | | f1 is the frequency of the serving cell | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | |
| } | | | |
| MeasObjectEUTRA-GENERIC(f2) ::= SEQUENCE { | | f2 is the frequency of the neighbouring cell | |
| neighCellConfig | '01'B (No MBSFN subframes are present in all neighbour cells) | | |
| cellForWhichToReportCGI | Physical Cell ID of Cell 2 | | |
| } | | | |

**Table 8.3.4.4.3-9: *SystemInformationBlockType2*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-1 SystemInformationBlockType2 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfig | Not present | | Cell 1 |
| } | | | |

**Table 8.3.4.4.3-10: *SystemInformationBlockType3*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-2 SystemInformationBlockType3 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType3 ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1 |
| | '01'B (No MBSFN subframes are present in all neighbour cells) | | Cell 2 |
| } | | | |

**Table 8.3.4.4.3-11: *SystemInformationBlockType5*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-3 SystemInformationBlockType5 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType5 ::= SEQUENCE { | | | |
| neighCellConfig[*n*] | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1 |
| | '01'B (No MBSFN subframes are present in all neighbour cells) | | Cell 2 |
| } | | | |

**Table 8.3.4.4.3-12: *MeasConfig-DEFAULT*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| } | | | |

**Table 8.3.4.4.3-13: *MeasConfig-DEFAULT*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step9)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | IdReportConfig-P | | |
| } | | | |
| } | | | |

**Table 8.3.4.4.3-14: *MeasResults*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step12)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.3.4.4.3-15: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(f1) ::= SEQUENCE { | | f1 is the frequency of the serving cell | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | |
| } | | | |
| MeasObjectEUTRA-GENERIC(f2) ::= SEQUENCE { | | f2 is the frequency of the neighbouring cell | |
| neighCellConfig | '01'B (No MBSFN subframes are present in all neighbour cells) | | |
| } | | | |

### 8.3.4.5 Test requirement

Tables 8.3.4.4.1-1 and 8.3.4.5-1 define the primary level settings including test tolerances for E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test.

**Table 8.3.4.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.10 (OP.10 FDD) and in D.1.2 (OP.2 FDD) | | OP.10 FDD | OP.10 FDD | OP.10 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | 4 | -Infinity | 7 | 7 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | 4 | -Infinity | 7 | 7 |
| RSRP Note 3 | dBm/15 KHz | -94 | -94 | -94 | -Infinity | -91 | -91 |
| SCH_RP Note3 | dBm/15 KHz | -94 | -94 | -94 | -Infinity | -91 | -91 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall transmit a measurement report containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Test requirement = RRC Procedure delay + $T_{identify_CGI,\ inter}$ + TTI insertion uncertainty

RRC procedure delay=15ms

$$T_{identify_CGI,\ inter} = T_{basic_identify_CGI,\ inter} \quad ms$$

$T_{basic_identify_CGI,\ inter}$ = 150 ms

TTI insertion uncertainty = 2 ms

The overall delays measured is 167 ms, allow 170 ms. The UE shall be scheduled continuously throughout the test, and from the start of T3 until 170 ms at least 80 ACK/NACK shall be detected as being transmitted by the UE.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

NOTE: The overall 80 ACK/NACK number is caused by two parts. Firstly, at least 60 ACK/NACK shall be sent during identifying the cell global identifier of cell 2 according to the requirement in TS 36.133 [4] Section 8.1.2.2.3.1. Secondly, given that continuous DL data allocation, additional 20 ACK/NACK shall be sent from the start of T3 until 170 ms excludes 150 ms for identifying the cell global identifier of cell 2.

## 8.3.5 E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX

### 8.3.5.1 Test purpose

To verify the requirement for identification of a new CGI of E-UTRA cell with autonomous gaps in DRX.

### 8.3.5.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward with support of inter-frequency SI acquisition for HO. Applicability requires support for FGI bit 5.

### 8.3.5.3 Minimum conformance requirements

No explicit neighbour list is provided to the UE for identifying a new CGI of E-UTRA cell. The UE shall identify and report the CGI when requested by the network for the purpose 'reportCGI'. The UE may make autonomous gaps in downlink reception and uplink transmission for receiving MIB and SIB1 message according to section 5.5.3.1 of TS 36.331 [5]. Note that a UE is not required to use autonomous gap if si-RequestForHO is set to false. If autonomous gaps are used for measurement with the purpose of 'reportCGI', the UE shall be able to identify a new CGI of E-UTRA cell within:

$$T_{identify_CGI,\,inter} = T_{basic_identify_CGI,\,inter} \quad ms$$

Where

$T_{basic_identify_CGI,\,inter}$ = 150 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new CGI of an E-UTRA cell is defined, provided that the E-UTRA cell has been already identified by the UE.

A cell shall be considered identifiable following conditions are fulfilled:

- RSRP related side conditions given in TS 36.133 [4] Section 9.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band

The requirement for identifying a new CGI of an E-UTRA cell within $T_{basic_identify_CGI,\,inter}$ is applicable when no DRX is used as well as when all the DRX cycles specified in 3GPP TS 36.331 [5] are used.

The ECGI reporting delay occurs due to the delay uncertainty when inserting the ECGI measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. In case DRX is used, the ECGI reporting may be delayed until the next DRX cycle.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.5 and A.8.3.5.

### 8.3.5.4 Test description

#### 8.3.5.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. The general test parameter settings are set up according to Table 8.3.5.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.3.5.4.3.

5. There are two E-UTRA FDD carriers and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.3.5.4.1-1: General test parameters for E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps when DRX is used**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| E-UTRA RF channel number | | 1, 2 | Two FDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active cell | | Cell 1 | Cell 1 is on RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 1 is on RF channel number 2. |
| CP length | | Normal | |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| A3-Offset | dB | -6 | |
| Hysteresis | dB | 0 | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | ON | DRX related parameters are defined in Table 8.3.5.5-2 |
| si-RequestForHO | | TRUE | As specified in section 5.5.3.1 in TS 36.331 [5]. |
| Time offset between cells | ms | 3 | Asynchronous cells |
| T1 | s | 5 | |
| T2 | s | ≤30 | UE shall report cell within 25.6s (20 DRX cycles) |
| T3 | s | 5 | |

### 8.3.5.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE does not have any timing information of cell 2. Gap pattern configuration is configured before T2 begins to enable inter-frequency monitoring. Starting T2, cell 2 becomes detectable and the UE is expected to detect and send a measurement report.

In the test UE needs to be provided at least once every 1280ms with new Timing Advance Command MAC control element to restart the Time alignment timer to keep UE uplink time alignment. Furthermore UE sends scheduling request (SR) to obtain allocation to send the measurement report on PUSCH.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.3.5.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration with event A3 configured message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.3.5.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3.
7. The SS shall transmit an RRCConnectionReconfiguration message during period T2. The RRC message shall release MeasGapConfig according to Table 8.3.5.4.3-10.
8. The UE shall transmit RRCConnectionReconfigurationComplete message.
9. SS shall transmit an RRCConnectionReconfiguration message during period T2, The RRC message shall create a measurement report configuration with purpose *reportCGI* and *si-RequestForHO* set to TRUE.
10. The SS shall start T3 timer when the last TTI containing the RRC message implying SI reading is sent to UE.
11. The UE shall transmit RRCConnectionReconfigurationComplete message.
12. UE shall transmit a measurement report containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3. If the overall delays measured from the beginning of time period T3 is less than 170 ms then the number of successful tests is increased by one. If the UE fails to report the cell global identifier within the overall delays measured requirement then the number of failure tests is increased by one.
13. After the SS receive the MeasurementReport message (in step 12) or when T3 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
14. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
15. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
16. Repeat step 2-15 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.3.5.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.3.5.4.3-1: Common Exception messages for E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-7<br>Table H.3.1-9 |

**Table 8.3.5.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.3.5.4.3-3: *MAC-MainConfig-RBC*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf1280 typical value in real network for real-time services. | |
| sf1280 | 9 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| } | | | |

**Table 8.3.5.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.6, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| … | | | |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | |
| } | | | |

**Table 8.3.5.4.3-5: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.3.5.4.3-6: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12(-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.3.5.4.3-7: *MeasResults*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.3.5.4.3-8: *MeasResultListEUTRA*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step6)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.3.5.4.3-9: *ReportConfigEUTRA-PERIODICAL*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step9)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportCGI | | |
| } | | | |
| } | | | |
| reportAmount | 1 | | |
| si-RequestForHO-r9 | setup | | |
| } | | | |

**Table 8.3.5.4.3-10: MeasGapConfig-Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step7)**

| Derivation Path: 36.331, clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasGapConfig ::= CHOICE { | | | |
| release | NULL | | |
| } | | | |

**Table 8.3.5.4.3-11: *MeasResultListEUTRA*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step12)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| cgi-Info SEQUENCE{ | | | |
| cellGlobalId SEQUENCE{ | | | |
| plmn-Identity | plmn-Identity | | |
| cellIdentity | cellIdentity | | |
| } | | | |
| trackingAreaCode | | | |
| plmn-IdentityList | | | |
| } | | | |
| } | | | |

**Table 8.3.5.4.3-12: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step9)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(f2) ::= SEQUENCE { | | f2 is the frequency of the neighbouring cell | |
| cellForWhichToReportCGI | Physical Cell ID of Cell 2 | | |
| } | | | |

**Table 8.3.5.4.3-13: *MeasConfig-DEFAULT*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| } | | | |

**Table 8.3.5.4.3-14: *MeasConfig-DEFAULT*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step9)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | IdReportConfig-P | | |
| } | | | |
| } | | | |

**Table 8.3.5.4.3-15: *MeasResults*: Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step12)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

## 8.3.5.5 Test requirement

Tables 8.3.5.4.1-1, 8.3.5.5-2 and 8.3.5.5-3 define the primary level settings including test tolerances for E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps when DRX is used test.

**Table 8.3.5.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.1 FDD | OP.1 FDD | OP.2 FDD | OP.2 FDD | OP.2 FDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $\hat{E}_s/I_{ot}$ | dB | 4 | 4 | 4 | -Infinity | 7 | 7 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s/N_{oc}$ | dB | 4 | 4 | 4 | -Infinity | 7 | 7 |
| RSRP Note 3 | dBm/15 KHz | -94 | -94 | -94 | -Infinity | -91 | -91 |
| SCH_RP Note3 | dBm/15 KHz | -94 | -94 | -94 | -Infinity | -91 | -91 |
| Propagation Condition | | AWGN | | | | | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | | | | | |
| Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. | | | | | | | |
| Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | | | | | |

**Table 8.3.5.5-2: DRX configuration for E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf1280 | |
| shortDRX | disable | |

**Table 8.3.5.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | Infinity | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| sr-ConfigIndex | 0 | For further information see section 6.3.2 in 3GPP TS 36.331 [5] and section10.1 in 3GPP TS 36.213 [8] |

The UE shall transmit a measurement report containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Test requirement = RRC Procedure delay + $T_{identify_CGI,\ inter}$ + TTI insertion uncertainty

RRC procedure delay=15ms

$$T_{identify_CGI,\ inter} = T_{basic_identify_CGI,\ inter} \quad ms$$

$T_{basic_identify_CGI,\ inter}$ = 150 ms

TTI insertion uncertainty = 2 ms

The overall delay measured is 167 ms, allow 170 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.3.6 E-UTRAN FDD-FDD Inter-frequency event triggered reporting without measurement gaps under AWGN propagation conditions in asynchronous cells

### 8.3.6.1 Test purpose

To verify theUE's ability to make a correct reporting of an event without measurement gaps within the E-UTRA FDD inter-frequency cell search requirements.

### 8.3.6.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward with support of Measurement without gaps. Applicability requires support for FGI bits 25.

### 8.3.6.3 Minimum conformance requirements

When measurement gaps are scheduled, or the UE supports capability of conducting such measurements without gaps, the UE shall be able to identify a new FDD inter-frequency within $T_{Identify_Inter}$ according to the following expression:

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new FDD inter-frequency cell is defined.

$N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1 and $T_{inter1}$ is defined in TS 36.133 [4] section 8.1.2.1

A cell shall be considered detectable provided following conditions are fulfilled:

- RSRP and RSRP Ês/Iot according to Annex B.2.3 for a corresponding Band
- other RSRP related side conditions given in Sections 9.1.3.1 and 9.1.3.2 and RSRQ related side conditions given in TS 36.133 [4] sections 9.1.6.1 and 9.1.6.2 are fulfilled.
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band

When measurement gaps are scheduled for FDD inter frequency measurements, or the UE supports capability of conducting such measurements without gaps, the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with measurement accuracy as specified in TS 36.133 [4] sub-clauses 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively, with measurement period given by table 8.3.6.3-1.

**Table 8.3.6.3-1: Measurement period and measurement bandwidth**

| Configuration | Physical Layer Measurement period: $T_{Measurement_Period_Inter_FDD}$ [ms] | Measurement bandwidth [RB] |
|---|---|---|
| 0 | 480 x $N_{freq}$ | 6 |
| 1 (Note) | 240 x $N_{freq}$ | 50 |
| Note: This configuration is optional | | |

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per FDD inter-frequency for up to 3 FDD inter-frequencies and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period defined in Table 8.3.6.3-1.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify-inter}$ defined in TS 36.133 [4] clause 8.1.2.3.1.1. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_inter}$ defined in TS 36.133 [4] clause 8.1.2.3.1.1 and then triggers the measurement report as per TS 36.331 [2], the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_Inter_FDD}$ defined in TS 36.133 [4] clause 8.1.2.3.1.1 provided the timing to that cell has not changed more than ± 50 Ts while measurement gap has not been available and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.
The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3 and A.8.3.6.

## 8.3.6.4 Test description

### 8.3.6.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. The general test parameter settings are set up according to Table 8.3.6.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.3.6.4.3.

5. There are two E-UTRA FDD carriers and two cells on different carrier frequencies and no gaps are configured in this test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.3.6.4.1-1: General test parameters for E-UTRAN FDD-FDD inter-frequency event triggered reporting without measurement gaps**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.3 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| E-UTRA RF Channel Number | | 1, 2 | Two FDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active PCell | | Cell 1 | Cell 1 is on RF channel number 1 |
| Neighbour cell | | Cell 2 | Cell 2 is on RF channel number 2 |
| A3-Offset | dB | -6 | |
| Hysteresis | dB | 0 | |
| CP length | | Normal | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | OFF |
| Time offset between cells | | 3 ms | Asynchronous cells |
| T1 | s | 5 | |
| T2 | s | 5 | |

### 8.3.6.4.2 Test procedure

The test consists of two successive time periods, with time duration of T1, and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2. PDCCH on the PCell shall be sent continuously to ensure that the UE sends ACK/NACKs throughout the test.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.3.6.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.3.6.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3. If the measurement reporting delay from the beginning of time period T2 is less than 3842ms and at least 85% of all ACK/NACKS transmitted by the UE are detected by the system simulator until the measurement report is received during T2 the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement or less than 85% of all ACK/NACKS transmitted by the UE are detected by the system simulator then the number of failure tests is increased by one.
7. After the SS receives the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.3.6.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.3.6.4.3-1: Common Exception messages for E-UTRAN FDD-FDD inter frequency event triggered reporting without measurement gaps under AWGN propagation conditions in synchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7 |

**Table 8.3.6.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD inter frequency event triggered reporting without measurement gaps under AWGN propagation conditions in asynchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.3.6.4.3-3: *MeasResults*: Additional E-UTRAN FDD-FDD inter frequency event triggered reporting without measurement gaps under AWGN propagation conditions in asynchronous cells test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.3.6.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD inter frequency event triggered reporting without measurement gaps under AWGN propagation conditions in asynchronous cells test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 8.3.6.4.3-5: *MeasConfig-DEFAULT*: Additional E-UTRAN FDD-FDD inter frequency event triggered reporting without measurement gaps under AWGN propagation conditions in asynchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the neighbouring cell(inter frequency cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | inter frequency cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-A3 | | |
| reportConfig | ReportConfigEUTRA-A3 | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-A3 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 8.3.6.4.3-6: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD-FDD inter frequency event triggered reporting without measurement gaps under AWGN propagation conditions in asynchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(f1) ::= SEQUENCE { | | f1 is the frequency of the serving cell | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | |
| } | | | |
| MeasObjectEUTRA-GENERIC(f2) ::= SEQUENCE { | | f2 is the frequency of the neighbouring cell | |
| neighCellConfig | '01'B (No MBSFN subframes are present in all neighbour cells) | | |
| } | | | |

**Table 8.3.6.4.3-7: *SystemInformationBlockType2*: Additional E-UTRAN FDD-FDD inter frequency event triggered reporting without measurement gaps under AWGN propagation conditions in asynchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-1 SystemInformationBlockType2 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfig | Not present | | Cell 1 |
| } | | | |

**Table 8.3.6.4.3-8: *SystemInformationBlockType3*: Additional E-UTRAN FDD-FDD inter frequency event triggered reporting without measurement gaps under AWGN propagation conditions in asynchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-2 SystemInformationBlockType3 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType3 ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1 |
| | '01'B (No MBSFN subframes are present in all neighbour cells) | | Cell 2 |
| } | | | |

**Table 8.3.6.4.3-9: *SystemInformationBlockType5*: Additional E-UTRAN FDD-FDD inter frequency event triggered reporting without measurement gaps under AWGN propagation conditions in asynchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-3 SystemInformationBlockType5 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType5 ::= SEQUENCE { | | | |
| neighCellConfig[*n*] | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1 |
| | '01'B (No MBSFN subframes are present in all neighbour cells) | | Cell 2 |
| } | | | |

## 8.3.6.5 Test requirement

Tables 8.3.6.4.1-1 and 8.3.6.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-FDD Inter-frequency event triggered reporting without measurement gaps under AWGN propagation conditions in asynchronous cells test.

**Table 8.3.6.5-1: Cell specific test parameters for E-UTRAN FDD-FDD inter-frequency event triggered reporting without measurement gaps**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| OCNG Patterns defined in D.1.10 (OP.10 FDD) and in D.1.2 (OP.2 FDD) | | OP.10 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |

| | | | | | |
|---|---|---|---|---|---|
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | | |
| RSRP Note 3 | dBm/15 kHz | -94 | -94 | -Infinity | -91 |
| $\hat{E}_s/I_{ot}$ | dB | 4 | 4 | -Infinity | 7 |
| SCH_RP Note 3 | dBm/15 kHz | -94 | -94 | -Infinity | -91 |
| $\hat{E}_s/N_{oc}$ | dB | 4 | 4 | -Infinity | 7 |
| Propagation Condition | | AWGN | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 3842 ms from the beginning of time period T2.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = 3840 ms

TTI insertion uncertainty = $TTI_{DCCH}$ = 1 ms; 2x$TTI_{DCCH}$ = 2 ms

The overall delays measured shall be less than a total of 3842 ms in this test case (note: this gives a total of 3840 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall send continuous ACK/NACK throughout the test, and from the start of T2 until Event A3 is reported, at least 85% ACK/NACK shall be detected.

For a test to be considered successful requirements on both event detection and percentage of transmitted ACK/NACKs have to be fulfilled simultaneously. The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

# 8.4 E-UTRAN TDD-TDD inter frequency measurements

## 8.4.1 E-UTRAN TDD-TDD Inter-frequency event triggered reporting under fading propagation conditions in synchronous cells

### 8.4.1.1 Test purpose

To verify the UE's ability to make a correct reporting of an event under fading propagation conditions in synchronous cells within the E-UTRA TDD inter-frequency cell search requirements. This test will partly verify the TDD-TDD inter-frequency cell search requirements in section 8.1.2.3.

### 8.4.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward. Applicability requires support for FGI bit 25.

### 8.4.1.3 Minimum conformance requirements

When measurement gaps are scheduled the UE shall be able to identify a new TDD inter-frequency within $T_{Identify_Inter}$ according to the following expression:

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new TDD inter-frequency cell is defined.

$T_{inter1}$ is defined in TS 36.133 [4] section 8.1.2.1

$N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1.

A cell shall be considered detectable provided following conditions are fulfilled:

- RSRP and RSRP Ês/Iot according to Annex I.2.3 for a corresponding Band,
- other RSRP related side conditions given in TS 36.133 [4] Sections 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively are fulfilled.
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band.

When measurement gaps are scheduled for TDD inter frequency measurements the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with measurement accuracy as specified in TS 36.133 sub-clauses 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively, with measurement period ($T_{Measurement_Period_TDD_Inter}$) given by table 8.4.4.1.3-1.

**Table 8.4.1.3-1: $T_{Measurement_Period_TDD_Inter}$ for different configurations**

| Configuration | Measurement bandwidth [RB] | Number of UL/DL sub-frames per half frame (5 ms) | | DwPTS | | $T_{Measurement_Period_TDD_Inter}$ [ms] |
|---|---|---|---|---|---|---|
| | | DL | UL | Normal CP | Extended CP | |
| 0 | 6 | 2 | 2 | $19760 \cdot T_s$ | $20480 \cdot T_s$ | 480 x $N_{freq}$ |
| 1 (Note 1) | 50 | 2 | 2 | $19760 \cdot T_s$ | $20480 \cdot T_s$ | 240 x $N_{freq}$ |
| Note 1: This configuration is optional.<br>Note 2: $T_s$ is defined in 3GPP TS 36.211 [9]. | | | | | | |

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per TDD inter-frequency and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period $T_{Measurement_Period_TDD_Inter.}$

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clauses 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{Identify_Inter}$ defined in TS 36.133 [4] clause 8.1.2.3.2.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra}$ defined in TS 36.133 [4] section 8.1.2.2.1.1 becomes undetectable for a period ≤ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.2 and A.8.4.1.

### 8.4.1.4 Test description

#### 8.4.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15.

2. The general test parameter settings are set up according to Table 8.4.1.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.4.1.4.3.

5. There are two E-UTRA TDD carriers and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.4.1.4.1-1: General test parameters for E-UTRAN TDD-TDD inter frequency event triggered reporting in fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Gap Pattern Id | | 1 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211. The same configuration in both cells |
| Uplink-downlink configuration | | 1 | As specified in 3GPP TS 36.211 section 4.2 Table 4.2-2 |
| CP length | | Normal | |
| E-UTRA RF Channel Number | | 1, 2 | Two TDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active cell | | Cell 1 | Cell 1 is on RF channel number 1 |
| Neighbour cell | | Cell 2 | Cell 2 is on RF channel number 2 |
| A3-Offset | dB | -6 | |
| Hysteresis | dB | 0 | |
| TimeToTrigger | dB | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| Time offset between cells | µs | 3 | Synchronous cells<br>3µs or 92*Ts |
| T1 | s | 5 | |
| T2 | s | 10 | |

#### 8.4.1.4.2 Test procedure

The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2. Gap pattern configuration is configured before T2 begins to enable inter-frequency monitoring.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.4.1.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.4.1.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the measurement reporting delay from the beginning of time period T2 is less than 7682 ms the number of successful tests is increased by one. If the UE fails to report the event within the measurement reporting delay requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit an RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.4.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.4.1.4.3-1: Common Exception messages for Additional E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H3.1-1<br>Table H3.1-3<br>Table H.3.1-7 |

**Table 8.4.1.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.4.1.4.3-3: *MeasResults*: Additional E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.4.1.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| MeasResultEUTRA ::= SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.4.1.5 Test requirement

Tables 8.4.1.4.1-1 and 8.4.1.5-1 define the primary level settings including test tolerances for event triggered reporting under fading propagation conditions in asynchronous inter frequency cells test.

**Table 8.4.1.5-1: Cell specific test parameters for E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in synchronous cells**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2) | | OP.1 TDD | | OP.2 TDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | -Infinity | 7.70 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 | -Infinity | -90.30 |
| SCH_RP | dBm/15 kHz | -94 | -94 | -infinity | -90.30 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | -Infinity | 7.70 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE priori to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 7682 ms from the beginning of time period T2.

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 7682 ms in this test case (note: a total of 7680 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

1) NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

## 8.4.2 E-UTRAN TDD-TDD Inter-frequency event triggered reporting when DRX is used under fading propagation conditions in synchronous cells

### 8.4.2.1 Test purpose

To verify the UE's ability to make a correct reporting of an event under fading propagation conditions in synchronous cells with DRX within the E-UTRA TDD-TDD inter frequency cell search in DRX requirements.

### 8.4.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward. Applicability requires support for FGI bits 5, and 25.

### 8.4.2.3 Minimum conformance requirements

When DRX is in use the UE shall be able to identify a new detectable E-UTRAN TDD inter frequency cell within $T_{identify_inter}$ as shown in table 8.4.2.3-1.

**Table 8.4.2.3-1: Requirement to identify a newly detectable TDD inter frequency cell**

| DRX cycle length (s) | $T_{identify_inter}$ (s) (DRX cycles) | |
|---|---|---|
| | Gap period = 40 ms | Gap period = 80 ms |
| ≤0.16 | Non DRX Requirements in section 8.1.2.3.2.1 are applicable | Non DRX Requirements in section 8.1.2.3.2.1 are applicable |
| 0.256 | 5.12*Nfreq (20*Nfreq) | 7.68*Nfreq (30*Nfreq) |
| 0.32 | 6.4*Nfreq (20*Nfreq) | 7.68*Nfreq (24*Nfreq) |
| 0.32<DRX-cycle≤2.56 | Note (20*Nfreq) | Note (20*Nfreq) |
| Note: Time depends upon the DRX cycle in use | | |

A cell shall be considered detectable provided following conditions are fulfilled:

- RSRP and RSRP Ês/Iot according to Annex I.2.3 for a corresponding Band,
- other RSRP related side conditions given in TS 36.133 [4] Sections 9.1.3.1 and 9.1.3.2 and RSRQ related side conditions given in Sections 9.1.6.1 and 9.1.6.2 are fulfilled.
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band.

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per TDD inter-frequency and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period defined in Table 8.4.2.3-2.

**Table 8.4.2.3-2: Requirement to measure TDD inter frequency cells**

| DRX cycle length (s) | $T_{measure_inter}$ (s) (DRX cycles) |
|---|---|
| ≤0.84 | Non DRX Requirements in section 8.1.2.3.1.1 are applicable |
| 0.08<DRX-cycle≤2.56 | Note ($5*N_{freq}$) |
| Note: Time depends upon the DRX cycle in use | |

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] sub-clauses 9.1.3.1 and 9.1.3.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in the sub-clauses 9.1.6.1 and 9.1.6.2.

Reported RSRP and RSRQ measurements contained in periodically triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively.

Reported RSRP and RSRQ measurements contained in event triggered periodic measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively.

The first report in event triggered periodic measurement reporting shall meet the requirements specified in TS 36.133 [4] section 8.1.2.3.2.2.1.3 Event Triggered Reporting.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{Identify_Inter}$ defined in TS 36.133 [4] Section 8.1.2.3.2.2 When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra}$ defined in TS 36.133 [4] section 8.1.2.2.1.1 becomes undetectable for a period ≤ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.2.2 and A.8.4.2.

## 8.4.2.4 Test description

### 8.4.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15.

2. The general test parameter settings are set up according to Table 8.4.2.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.4.2.4.3.

5. There are two cells on different carrier frequencies and gap pattern configuration # 0 as defined in TS 36.133 [4] Table 8.1.2.1-1 is provided. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.4.2.4.1-1: General test parameters for E-UTRAN TDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions**

| Parameter | Unit | Test 1 | Test 2 | Comment |
|---|---|---|---|---|
| | | Value | | |
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | | As specified in section A.1.2. Note that UE may only be allocated at *On Duration* |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | | As specified in section A.2.2. |
| E-UTRA RF Channel Number | | 1, 2 | | Two TDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| Active cell | | Cell 1 | | Cell 1 is on RF channel number 1 |
| Neighbour cell | | Cell 2 | | Cell 2 is on RF channel number 2 |
| Gap Pattern Id | | 0 | | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| Uplink-downlink configuration | | 1 | | As specified in 3GPP TS 36.211 section 4.2 Table 4.2-2 |
| Special subframe configuration | | 6 | | As specified in table 4.2-1 in TS 36.211. The same configuration in both cells |
| A3-Offset | dB | -6 | | |
| Hysteresis | dB | 0 | | |
| CP length | | Normal | | |
| TimeToTrigger | s | 0 | | |
| Filter coefficient | | 0 | | L3 filtering is not used |
| PRACH configuration | | 4 | | As specified in table 5.7.1-3 in TS 36.211 |
| Access Barring Information | - | Not Sent | | No additional delays in random access procedure. |
| DRX | | ON | | DRX related parameters are defined in Table 8.4.2.4.1-2 |
| Time offset between cells | µs | 3 | | Synchronous cells<br>3µs or 92*Ts |
| T1 | s | 5 | | |
| T2 | s | 5 | 30 | |

## 8.4.2.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

In Test 1 UE needs to be provided at least once every 500ms with new Timing Advance Command MAC control element to restart the Time alignment timer to keep UE uplink time alignment. Furthermore, the UE is allocated with PUSCH resource at every DRX cycle. In Test 2 the uplink time alignment is not maintained and UE needs to use RACH to obtain UL allocation for measurement reporting.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.4.2.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.4.2.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 3882 ms for Test 1 and 26882 ms for Test 2 then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receives the MeasurementReport message in step 6 or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

11. Repeat step 1-10 for each sub-test in Table 8.4.2.4.1-1 as appropriate.

### 8.4.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.4.2.4.3-1: Common Exception messages for Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-7<br>Table H.3.1-9<br>Table H.3.7-1<br>Table H.3.7-2<br>Table H.3.7-3 |

**Table 8.4.2.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | | | |
| | MeasConfig-DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated- HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.4.2.4.3-3: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.4.2.4.3-4: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20: SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | Channel-bandwidth-dependent parameter | |
| sr-ConfigIndex | 2 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.4.2.4.3-5: *MeasResults:* Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | Set according to specific test | |
| rsrqResult | INTEGER(0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.4.2.4.3-6: *MeasResultListEUTRA*: Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 8.4.2.4.3-7: *PRACH-Config-DEFAULT*: Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test requirement**

| Derivation Path: TS 36.508 [7] clause 7.3.2 Table 7.3.2-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PRACH-Config-DEFAULT ::= SEQUENCE { | | | |
| prach-ConfigIndex | 4 | | |
| } | | | |

## 8.4.2.5 Test requirement

Tables 8.4.2.5-1, 8.4.2.5-2 and 8.4.2.5-3 define the primary level settings including test tolerances for E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells when DRX is used test.

**Table 8.4.2.5-1: Cell specific test parameters for E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in synchronous cells**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | | OP.2 TDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |

| | | | | | |
|---|---|---|---|---|---|
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | | |
| RSRP Note 3 | dBm/15 kHz | -94 | -94 | -Infinity | -90.30 |
| $\hat{E}_s/I_{ot}$ | dB | 4 | 4 | -Infinity | 7.70 |
| SCH_RP Note 3 | dBm/15 kHz | -94 | -94 | -Infinity | -90.30 |
| $\hat{E}_s/N_{oc}$ | dB | 4 | 4 | -Infinity | 7.70 |
| Propagation Condition | | ETU70 | | | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.<br>Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | | | |

**Table 8.4.2.5-2: drx-Configuration to be used in E-UTRAN TDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf40 | sf1280 | |
| shortDRX | disable | disable | |

**Table 8.4.2.5-3: *TimeAlignmentTimer* and *sr-ConfigIndex* -Configuration to be used in E-UTRAN TDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | For further information see section 6.3.2 in 3GPP TS 36.331 [5]. |
| sr-ConfigIndex | 2 | 2 | For further information see section 6.3.2 in 3GPP TS 36.331 and 10.1 in 3GPP TS 36.213 [8]. |

In Test1 when DRX cycle length = 40 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment when the UE send one Event A3 triggered measurement report on PUSCH.

In Test 2 when DRX cycle length = 1280 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE starts to send preambles on the PRACH for Scheduling Request (SR) to obtain allocation to send the measurement report to Cell 2 on PUSCH.TTI insertion uncertainty = 2 ms

For both tests:

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event A3 measurement report.

The overall delay measured when DRX cycle length is 40 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{Identify_Inter}$

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new FDD inter-frequency cell is defined.

$T_{Inter1}$ = 60 ms. It is defined in table 8.1.2.1-1 of TS 36.133 [4] clause 8.1.2.1.

$N_{freq}$ = 1. It is defined in TS 36.133 [4] clause 8.1.2.1.1.

TTI insertion uncertainty = 2 ms

DRX cycle length = 40 ms

The overall delay measured when DRX cycle length is 40 ms shall be less than a total of 3882 ms.

The overall delay measured when DRX cycle length is 1280 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_inter}$

$T_{identify_inter}$ = 25600 ms. When DRX cycle length is 1280 ms then the $T_{identify_inter}$ is 20 x 1280 ms, as defined in Table 8.4.2.3-1.

TTI insertion uncertainty = 2 ms

DRX cycle length = 1280 ms

The overall delay measured when DRX cycle length is 1280 ms shall be less than a total of 26882 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.4.3 E-UTRAN TDD-TDD inter-frequency event triggered reporting under AWGN propagation conditions in synchronous cells with DRX when L3 filtering is used

### 8.4.3.1 Test purpose

To verify the UE's ability to make a correct reporting of an event in DRX when L3 filtering is used in AWGN propagation conditions in synchronous cells within the E-UTRA TDD-TDD inter frequency cell search in DRX requirements and the UE behaviour with the filterCoefficent.

### 8.4.3.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward. Applicability requires support for FGI bits 5, and 25.

### 8.4.3.3 Minimum conformance requirements

When DRX is in use the UE shall be able to identify a new detectable E-UTRAN TDD inter frequency cell within $T_{identify_inter}$ as shown in table 8.4.3.3-1

**Table 8.4.3.3-1: Requirement to identify a newly detectable TDD inter frequency cell**

| DRX cycle length (s) | $T_{identify_inter}$ (s) (DRX cycles) | |
|---|---|---|
| | Gap period = 40 ms | Gap period = 80 ms |
| ≤0.16 | Non DRX Requirements in section 8.1.2.3.2.1 are applicable | Non DRX Requirements in section 8.1.2.3.2.1 are applicable |
| 0.256 | 5.12*Nfreq (20*Nfreq) | 7.68*Nfreq (30*Nfreq) |
| 0.32 | 6.4*Nfreq (20*Nfreq) | 7.68*Nfreq (24*Nfreq) |
| 0.32<DRX-cycle≤2.56 | Note (20*Nfreq) | Note (20*Nfreq) |
| Note: Time depends upon the DRX cycle in use | | |

A cell shall be considered detectable provided following conditions are fulfilled:

- RSRP and RSRP Ês/Iot according to Annex I.2.3 for a corresponding Band,
- other RSRP related side conditions given in TS 36.133 [4] Sections 9.1.3.1 and 9.1.3.2 and RSRQ related side conditions given in Sections 9.1.6.1 and 9.1.6.2 are fulfilled,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band.

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per TDD inter-frequency for up to 3 TDD inter-frequencies and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period defined in Table 8.4.3.3-2.

**Table 8.4.3.3-2: Requirement to measure TDD inter frequency cells**

| DRX cycle length (s) | $T_{measure_inter}$ (s) (DRX cycles) |
|---|---|
| ≤0.08 | Non DRX Requirements in section 8.1.2.3.2.1 are applicable |
| 0.08 <DRX-cycle ≤ 2.56 | Note ($5*N_{freq}$) |
| Note: Time depends upon the DRX cycle in use | |

The RSRP measurement accuracy for all measured cells shall be as specified in the TS 36.133 [4] sub-clauses 9.1.3.1 and 9.1.3.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in the sub-clauses 9.1.6.1 and 9.1.6.2.

Reported RSRP measurements contained in periodically triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively.

Reported RSRP and RSRQ measurements contained in event triggered periodic measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively.

The first report in event triggered periodic measurement reporting shall meet the requirements specified in TS 36.133 [4] section 8.1.2.3.2.2.1.3 Event Triggered Reporting.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{Identify_Inter}$ defined in TS 36.133 [4] Section 8.1.2.3.2.2. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{Identify_Inter}$ in TS 36.133 [4] section 8.1.2.3.2.2 and then enters or leaves the reporting range, the event triggered measurement reporting delay shall be less than $T_{measure_inter}$ in TS 36.133 [4] section 8.1.2.3.2.2 provided the timing to that cell has not changed more than ± 50 Ts while measurement gap has not been available and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.2.2 and A.8.4.3.

## 8.4.3.4 Test description

### 8.4.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. The general test parameter settings are set up according to Table 8.4.3.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.4.3.4.3.

5. There is one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.4.3.4.1-1: General test parameters for E-UTRAN TDD-TDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1, 2 | Two TDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Time offset between cells | μs | 3 | synchronous cells |
| Gap Pattern Id | | 1 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1 |
| Uplink-downlink configuration of cells | | 1 | As specified in table 4.2.2 in TS 36.211 |
| Special subframe configuration of cells | | 6 | As specified in table 4.2.1 in TS 36.211 |
| Neighbour A3-Offset Ofn | dB | -14 | |
| CP length | | Normal | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 9 | L3 filtering is used |
| DRX | | ON | DRX related parameters are defined in Table 8.4.3.5-2 |
| T1 | s | 30 | |
| T2 | s | 9 | |

### 8.4.3.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 and the filter coefficient is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively.

In the Test when DRX = 1280 ms is used, a non-zero L3 filtering is configured. The uplink time alignment is not maintained and UE needs to use RACH to obtain UL allocation for measurement reporting.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.4.3.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. During T1 the SS continuously checks the absence of an A3 triggered measurement report from the neighbour cell. If the UE does not send this during T1, then count neither success nor fail. If the UE sends such a report, then count a fail. Upon a fail in step 5, neither success nor fail are counted in step 7.

6. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.4.3.5-1 and 8.4.3.5-2.

7. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 7682 ms then the number of “successes” is increased by one. Otherwise count a fail.

8. After the SS receive the MeasurementReport message in step 7) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

9. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

10. After the RRC connection release, the SS:

- transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),

or

- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

11. Repeat step 2-10 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 8.4.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.4.3.4.3-1: Common Exception messages for E-UTRAN TDD-TDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used in test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-3<br>Table H.3.7-2<br>Table H.3.7-3 |

**Table 8.4.3.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used in test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.4.3.4.3-3: *MeasObjectEUTRA-GENERIC(f2)*: Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC(Freq) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| carrierFreq | Downlink EARFCN for frequency f2 defined in 36.508 | | |
| allowedmeasBandwidth | The number of the resource blocks for frequency f2 | | |
| presenceAntennaPort1 | FALSE | | |
| neighCellConfig | '01'B (No MBSFN subframes are present in all neighbour cells) | | |
| offsetFreq | dB-14 | -14 dB is actual value in dB (Value dB-14 corresponds to -14 dB) | |
| cellsToRemoveList | Not present | | |
| cellsToAddModList | Not present | | |
| blackCellsToRemoveList | Not present | | |
| blackCellsToAddModList | Not present | | |
| cellForWhichToReportCGI | Not present | | |
| } | | | |

**Table 8.4.3.4.3-4: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used in test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventid CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | 0 (0 dB) | The actual value is IE value * 0.5 dB IE Value INTEGER (-30..30) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | The actual value is IE value * 0.5 dB | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.4.3.4.3-5: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 2 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.4.3.4.3-6: *MeaResults*: Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.4.3.4.3-7: *MeasResultListEUTRA*: Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| physCellId | PhysCellId | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.4.3.4.3-8: *FilterCoefficient*: Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| filterCoefficient | fc9 | | |

## 8.4.3.5 Test requirement

Tables 8.4.3.4.1-1, 8.4.3.5-1, 8.4.3.5-2 and 8.4.3.5-3 define the primary level settings including test tolerances for E-UTRAN FDD-FDD inter-frequency event triggered reporting under AWGN propagation conditions in asynchronous cells with DRX when L3 filtering is used test.

**Table 8.4.3.5-1: Cell specific test parameters for E-UTRAN TDD-TDD inter-frequency event triggered reporting under AWGN propagation conditions with DRX when L3 filtering is used**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | | OP.2 TDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 4.00 | 0.00 | 4.00 | 24.00 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -95.50 | -95.50 | -98.00 | -98.00 |
| $\hat{E}_s / N_{oc}$ | dB | 4.00 | 0.00 | 4.00 | 24.00 |
| RSRP Note 3 | dBm/15 KHz | -91.50 | -95.50 | -94.00 | -74.00 |
| SCH_RP Note 3 | dBm/15 KHz | -91.50 | -95.50 | -94.00 | -74.00 |
| Propagation Condition | | AWGN | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.4.3.5-2: DRX-Configuration for E-UTRAN TDD-TDD inter-frequency event triggered reporting in DRX under AWGN propagation conditions with DRX when L3 filtering is used**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in section 6.3.2 in 3GPP TS 36.331 [5]. |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf1280 | |
| shortDRX | disable | |

**Table 8.4.3.5-3: TimeAlignmentTimer -Configuration for E-UTRAN TDD-TDD inter-frequency event triggered reporting in DRX under AWGN propagation conditions with DRX when L3 filtering is used**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | sf500 | As specified in section 6.3.2 in 3GPP TS 36.331 [5]. |
| sr-ConfigIndex | 2 | For further information see section 6.3.2 in 3GPP TS 36.331 [5] and section 10.1 in 3GPP TS 36.213 [8]. |

The overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE starts to send preambles on the PRACH for Scheduling Request (SR) to obtain allocation to send the measurement report to Cell 2 on PUSCH.

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event A3 measurement report.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The overall delay measured when DRX cycle length is 1280 ms test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{measure_inter}$

$T_{measure_inter}$ = 6400 ms. When DRX cycle length is 1280 ms then the $T_{measure_inter}$ is 5 x 1280 ms, as defined in Table 8.4.3.3-1.

TTI insertion uncertainty = 2 ms

DRX cycle length = 1280 ms

The overall delay measured when DRX cycle length is 1280 ms shall be less than a total of 7682 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.4.4 E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps

### 8.4.4.1 Test purpose

To verify the requirement for identification of a new CGI of E-UTRA cell with autonomous gaps.

### 8.4.4.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward with support of inter-frequency SI acquisition for HO.

### 8.4.4.3 Minimum conformance requirements

No explicit neighbour list is provided to the UE for identifying a new CGI of E-UTRA cell. The UE shall identify and report the CGI when requested by the network for the purpose of 'reportCGI'. The UE may make autonomous gaps in both downlink reception and uplink transmission for receiving MIB and SIB1 message according to section 5.5.3.1 of TS 36.331 [5]. Note that a UE is not required to use autonomous gap if si-RequestForHO is set to false. If autonomous gaps are used for measurement with the purpose of 'reportCGI', regardless of whether DRX is used or not, the UE shall be able to identify a new CGI of E-UTRA cell within:

$$T_{identify_CGI,inter} = T_{basic_identify_CGI,inter} \quad ms$$

Where

$T_{basic_identify_CGI,inter}$ = 150 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new CGI of E-UTRA cell is defined, provided that the E-UTRA cell has been already identified by the UE.

A cell shall be considered identifiable following conditions are fulfilled:

- RSRP related side conditions given in TS 36.133 [4] Section 9.1 are fulfilled for a corresponding Band,
- SCH_RP $\geq$ -125 dBm for Bands 33, 34, 35, 36, 37, 38, 39, 40, 42, 43 and SCH Ês/Iot $\geq$ - 4 dB.
- SCH_RP $\geq$ -124 dBm for Band 41 and SCH Ês/Iot $\geq$ - 4 dB.

The requirement for identifying a new CGI of an E-UTRA cell within $T_{basic_identify_CGI,inter}$ is applicable when no DRX is used as well as when all the DRX cycles specified in 3GPP TS 36.331 [5] are used.

Given that continuous DL data allocation and no DRX is used, no measurement gaps are configured, and TDD configuration as in TS 36.133 [4] Table 8.1.2.3.2.1-1 is used, the UE shall have more than 30 ACK/NACKs transmitted during the identification of a new CGI of E-UTRA cell.

The ECGI reporting delay occurs due to the delay uncertainty when inserting the ECGI measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. In case DRX is used, the ECGI reporting may be delayed until the next DRX cycle.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.7 and A.8.4.4.

### 8.4.4.4 Test description

#### 8.4.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.
2. The general test parameter settings are set up according to Table 8.4.4.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.4.4.4.3.
5. There are two E-UTRA TDD carriers and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.4.4.4.1-1: General test parameters for E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| E-UTRA RF channel number | | 1, 2 | Two TDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active cell | | Cell 1 | Cell 1 is on RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 1 is on RF channel number 2. |
| CP length | | Normal | |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211 [9]. The same configuration in both cells |
| Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in TS 36.211 [9]. The same configuration in both cells |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| A3-Offset | dB | -6 | |
| Hysteresis | dB | 0 | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| si-RequestForHO | | TRUE | As specified in section 5.5.3.1 in TS 36.331 [5]. |
| Time offset between cells | µs | 3 | Synchronous cells |
| T1 | s | 5 | |
| T2 | s | ≤10 | |
| T3 | s | 5 | |

### 8.4.4.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. PDCCHs indicating new transmissions should be sent continuously to ensure that the UE would have ACK/NACK sending during identifying a new CGI of E-UTRAN cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE does not have any timing information of cell 2. Gap pattern configuration is configured before T2 begins to enable inter-frequency monitoring. Starting T2, cell 2 becomes detectable and the UE is expected to detect and send a measurement report.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.4.4.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration with event A3 configured message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.4.4.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3.
7. The SS shall transmit an RRCConnectionReconfiguration message during period T2. The RRC message shall release MeasGapConfig according to Table 8.4.4.4.3-6.
8. The UE shall transmit RRCConnectionReconfigurationComplete message.
9. SS shall transmit an RRCConnectionReconfiguration message during period T2, The RRC message shall create a measurement report configuration with purpose *reportCGI* and *si-RequestForHO* set to TRUE.
10. The SS shall start T3 timer when the last TTI containing the RRC message implying SI reading is sent to UE.

11. The UE shall transmit RRCConnectionReconfigurationComplete message.

12. UE shall transmit a MeasurementReport message containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3. The UE shall be scheduled continuously throughout the test, and from the start of T3 until 170 ms at least 42 ACK/NACK shall be detected as being transmitted by the UE. If the overall delays measured from the beginning of time period T3 is less than 170 ms, and the UE have more than 42 ACK/NACKs transmitted from the start of T3 until 170 ms, then the number of successful tests is increased by one. If the UE fails to report the cell global identifier within the overall delays measured requirement, or the UE have less than 42 ACK/NACKs transmitted from the start of T3 until 170 ms, then the number of failure tests is increased by one.

13. After the SS receive the MeasurementReport message (in step 12) or when T3 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

14. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

15. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

16. Repeat step 2-15 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 8.4.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.4.4.4.3-1: Common Exception messages for E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.3.1-9 |

**Table 8.4.4.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.4.4.4.3-3: *MeasResults*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.4.4.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step6)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.4.4.4.3-5: *ReportConfigEUTRA-PERIODICAL*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step9)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportCGI | | |
| } | | | |
| } | | | |
| reportAmount | 1 | | |
| si-RequestForHO-r9 | setup | | |
| } | | | |

**Table 8.4.4.4.3-6: MeasGapConfig- Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step7)**

| Derivation Path: 36.331, clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasGapConfig ::= CHOICE { | | | |
| release | NULL | | |
| | | | |
| } | | | |

**Table 8.4.4.4.3-7: *MeasResultListEUTRA*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step12)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| cgi-Info SEQUENCE{ | | | |
| cellGlobalId SEQUENCE{ | | | |
| plmn-Identity | plmn-Identity | | |
| cellIdentity | cellIdentity | | |
| } | | | |
| trackingAreaCode | | | |
| plmn-IdentityList | | | |
| } | | | |
| } | | | |

**Table 8.4.4.4.3-8: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellForWhichToReportCGI | Physical Cell ID of Cell 2 | | Cell 1 |
| } | | | |

**Table 8.4.4.4.3-9: *MeasConfig-DEFAULT*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| } | | | |

**Table 8.4.4.4.3-10: *MeasConfig-DEFAULT*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step9)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | IdReportConfig-P | | |
| } | | | |
| } | | | |

**Table 8.4.4.4.3-11: *MeasResults*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step12)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

## 8.4.4.5 Test requirement

Tables 8.4.4.4.1-1 and 8.4.4.5-1 define the primary level settings including test tolerances for E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test.

**Table 8.4.4.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | OP.1 TDD | OP.1 TDD | OP.2 TDD | OP.2 TDD | OP.2 TDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | 4 | -Infinity | 7 | 7 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | 4 | -Infinity | 7 | 7 |
| RSRP Note 3 | dBm/15 KHz | -94 | -94 | -94 | -Infinity | -91 | -91 |
| SCH_RP Note3 | dBm/15 KHz | -94 | -94 | -94 | -Infinity | -91 | -91 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall transmit a measurement report containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Test requirement = RRC Procedure delay + $T_{identify_CGI,\ inter}$ + TTI insertion uncertainty

RRC procedure delay=15ms

$$T_{identify_CGI,\ inter} = T_{basic_identify_CGI,\ inter} \quad ms$$

$T_{basic_identify_CGI,\ inter}$ = 150 ms

TTI insertion uncertainty = 2 ms

The overall delays measured is 167 ms, allow 170 ms. The UE shall be scheduled continuously throughout the test, and from the start of T3 until 170 ms at least 42 ACK/NACK shall be detected as being transmitted by the UE.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

NOTE: The overall 42 ACK/NACK number is caused by two parts. Firstly, at least 30 ACK/NACK shall be sent during identifying the cell global identifier of cell 2 according to the requirement in Section TS 36.133 [4] 8.1.2.2.3.1. Secondly, given that continuous DL data allocation, additional 12 ACK/NACK shall be sent from the start of T3 until 170 ms excludes 150 ms for identifying the cell global identifier of cell 2.

## 8.4.5 E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX

### 8.4.5.1 Test purpose

To verify the requirement for identification of a new CGI of E-UTRA cell with autonomous gaps with DRX.

### 8.4.5.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward with support of inter-frequency SI acquisition for HO. Applicability requires support for FGI bit 5.

### 8.4.5.3 Minimum conformance requirements

No explicit neighbour list is provided to the UE for identifying a new CGI of E-UTRA cell. The UE shall identify and report the CGI when requested by the network for the purpose of 'reportCGI'. The UE may make autonomous gaps in both downlink reception and uplink transmission for receiving MIB and SIB1 message according to section 5.5.3.1 of TS 36.331 [5]. Note that a UE is not required to use autonomous gap if si-RequestForHO is set to false. If autonomous gaps are used for measurement with the purpose of 'reportCGI', regardless of whether DRX is used or not, the UE shall be able to identify a new CGI of E-UTRA cell within:

$$T_{identify_CGI,inter} = T_{basic_identify_CGI,inter} \quad ms$$

Where

$T_{basic_identify_CGI,inter}$ = 150 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new CGI of E-UTRA cell is defined, provided that the E-UTRA cell has been already identified by the UE.

A cell shall be considered identifiable following conditions are fulfilled:

- RSRP related side conditions given in TS 36.133 [4] Section 9.1 are fulfilled for a corresponding Band,
- SCH_RP $\geq$ -125 dBm for Bands 33, 34, 35, 36, 37, 38, 39, 40, 42, 43 and SCH Ês/Iot $\geq$ - 4 dB.
- SCH_RP $\geq$ -124 dBm for Band 41 and SCH Ês/Iot $\geq$ - 4 dB.

The requirement for identifying a new CGI of an E-UTRA cell within $T_{basic_identify_CGI,inter}$ is applicable when no DRX is used as well as when all the DRX cycles specified in 3GPP TS 36.331 [5] are used.

The ECGI reporting delay occurs due to the delay uncertainty when inserting the ECGI measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. In case DRX is used, the ECGI reporting may be delayed until the next DRX cycle.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.7 and A.8.4.5.

### 8.4.5.4 Test description

#### 8.4.5.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. The general test parameter settings are set up according to Table 8.4.5.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.4.5.4.3.

5. There are two E-UTRA TDD carriers and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.4.5.4.1-1: General test parameters for E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| E-UTRA RF channel number | | 1, 2 | Two TDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active cell | | Cell 1 | Cell 1 is on RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 1 is on RF channel number 2. |
| CP length | | Normal | |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211 [9]. The same configuration in both cells |
| Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in TS 36.211 [9]. The same configuration in both cells |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| A3-Offset | dB | -6 | |
| Hysteresis | dB | 0 | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | ON | DRX related parameters are defined in Table 8.4.5.5-2 |
| si-RequestForHO | | TRUE | As specified in section 5.5.3.1 in TS 36.331 [5]. |
| Time offset between cells | µs | 3 | Synchronous cells |
| T1 | s | 5 | |
| T2 | s | ≤30 | UE shall report cell within 25.6s (20 DRX cycles) |
| T3 | s | 5 | |

### 8.4.5.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE does not have any timing information of cell 2. Gap pattern configuration is configured before T2 begins to enable inter-frequency monitoring. Starting T2, cell 2 becomes detectable and the UE is expected to detect and send a measurement report.

In the test UE needs to be provided at least once every 1280ms with new Timing Advance Command MAC control element to restart the Time alignment timer to keep UE uplink time alignment. Furthermore UE sends scheduling request (SR) to obtain allocation to send the measurement report on PUSCH.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.4.5.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration with event A3 configured message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.4.5.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3.
7. The SS shall transmit an RRCConnectionReconfiguration message during period T2. The RRC message shall release MeasGapConfig according to Table 8.4.5.4.3-10.
8. The UE shall transmit RRCConnectionReconfigurationComplete message.
9. SS shall transmit an RRCConnectionReconfiguration message during period T2, The RRC message shall create a measurement report configuration with purpose *reportCGI* and *si-RequestForHO* set to TRUE.
10. The SS shall start T3 timer when the last TTI containing the RRC message implying SI reading is sent to UE.
11. The UE shall transmit RRCConnectionReconfigurationComplete message.
12. UE shall transmit a MeasurementReport message containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3. If the overall delays measured from the beginning of time period T3 is less than 170 ms, then the number of successful tests is increased by one. If the UE fails to report the cell global identifier within the overall delays measured requirement, then the number of failure tests is increased by one.
13. After the SS receive the MeasurementReport message (in step 12) or when T3 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
14. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
15. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
16. Repeat step 2-15 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.4.5.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.4.5.4.3-1: Common Exception messages for E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-7<br>Table H.3.1-9 |

**Table 8.4.5.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.4.5.4.3-3: *MAC-MainConfig-RBC*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf1280 typical value in real network for real-time services. | |
| sf1280 | 9 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | Infinity | | |
| } | | | |

**Table 8.4.5.4.3-4: *PhysicalConfigDedicated-DEFAULT*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.6, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| … | | | |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | |
| } | | | |

**Table 8.4.5.4.3-5: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 2 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.4.5.4.3-6: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12(-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.4.5.4.3-7: *MeasResults*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.4.5.4.3-8: *MeasResultListEUTRA*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step6)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.4.5.4.3-9 *ReportConfigEUTRA-PERIODICAL*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step9)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportCGI | | |
| } | | | |
| } | | | |
| reportAmount | 1 | | |
| si-RequestForHO-r9 | setup | | |
| } | | | |

**Table 8.4.5.4.3-10: MeasGapConfig- Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step7)**

| Derivation Path: 36.331, clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasGapConfig ::= CHOICE { | | | |
| release | NULL | | |
| } | | | |

**Table 8.4.5.4.3-11: *MeasResultListEUTRA*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step12)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| cgi-Info SEQUENCE{ | | | |
| cellGlobalId SEQUENCE{ | | | |
| plmn-Identity | plmn-Identity | | |
| cellIdentity | cellIdentity | | |
| } | | | |
| trackingAreaCode | | | |
| plmn-IdentityList | | | |
| } | | | |
| } | | | |

**Table 8.4.5.4.3-12: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellForWhichToReportCGI | Physical Cell ID of Cell 2 | | Cell 1 |
| } | | | |

**Table 8.4.5.4.3-13: *MeasConfig-DEFAULT*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| } | | | |

**Table 8.4.5.4.3-14: *MeasConfig-DEFAULT*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step9)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | IdReportConfig-P | | |
| } | | | |
| } | | | |

**Table 8.4.5.4.3-15: *MeasResults*: Additional E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test requirement (step12)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

### 8.4.5.5 Test requirement

Tables 8.4.5.4.1-1 ,8.4.5.5-1, 8.4.5.5-2 and 8.4.5.5-3 define the primary level settings including test tolerances for E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX test.

**Table 8.4.5.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | OP.1 TDD | OP.1 TDD | OP.2 TDD | OP.2 TDD | OP.2 TDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | 4 | -Infinity | 7 | 7 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | 4 | -Infinity | 7 | 7 |
| RSRP Note 3 | dBm/15 KHz | -94 | -94 | -94 | -Infinity | -91 | -91 |
| SCH_RP Note3 | dBm/15 KHz | -94 | -94 | -94 | -Infinity | -91 | -91 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.4.5.5-2: DRX configuration for E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf1280 | |
| shortDRX | disable | |

**Table 8.4.5.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | Infinity | As specified in section 6.3.2 in 3GPP TS 36.331 [5] |
| sr-ConfigIndex | 2 | For further information see section 6.3.2 in 3GPP TS 36.331 [5] and section10.1 in 3GPP TS 36.213 [8]. |

The UE shall transmit a measurement report containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Test requirement = RRC Procedure delay + $T_{identify_CGI,\ inter}$ + TTI insertion uncertainty

RRC procedure delay=15ms

$$T_{identify_CGI,\ inter} = T_{basic_identify_CGI,\ inter} \quad ms$$

$T_{basic_identify_CGI,\ inter}$ = 150 ms

TTI insertion uncertainty = 2 ms

The overall delays measured is 167 ms, allow 170 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 8.5 E-UTRAN FDD - UTRAN measurements

## 8.5.1 E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions

### 8.5.1.1 Test purpose

To verify the UE's ability to make a correct reporting of an event under fading propagation conditions within the E-UTRA FDD - UTRA FDD cell search requirements.

### 8.5.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support UTRA FDD. Applicability requires support for FGI bits 15 and 22.

### 8.5.1.3 Minimum conformance requirements

The measurement reporting delay shall be less than $T_{identify,\ UTRA_FDD}$ in RRC_CONNECTED state.

When explicit neighbour list is provided and no DRX is used the UE shall be able to identify a new detectable cell within

$$T_{identify,\ UTRA_FDD} = T_{basic_identify_UTRA_FDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq} \quad ms$$

Where:

$T_{basic_identify_UTRA_FDD}$ = 300 ms. This is the time period used in the inter-RAT equation where the maximum allowed time for the UE to identify a new UTRA FDD cell is defined.

$T_{Inter1}$ = 30 ms. This is the minimum available time for inter-RAT measurement during 480 ms period

$N_{Freq}$: This is the number of UTRA carriers being monitored

A cell shall be considered detectable when

- CPICH Ec/Io $\underline{\geq}$ -20 dB,
- SCH_Ec/Io ≥ -17 dB for at least one channel tap and SCH_Ec/Ior is equally divided between primary synchronisation code and secondary synchronisation code. When L3 filtering is used an additional delay can be expected.

When measurement gaps are scheduled for UTRA FDD inter RAT measurement the UE physical layer shall be capable of reporting measurements to higher layers with measurement accuracy as specified in TS 36.133 [4] clause 9.2 of with measurement period given by

$$T_{measurement_UTRA_FDD} = Max\left\{T_{Measurement_Period\ UTRA_FDD}, T_{basic_measurement_UTRA_FDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms$$

Where:

$X_{basic\ measurement\ UTRA_FDD}$ = 6 (cells)

$T_{Measurement_Period\ UTRA_FDD}$ = 480 ms. The period used for calculating the measurement period.

$T_{basic_measurement_UTRA_FDD}$ = 50 ms. This is the time period used in the equation where the maximum allowed time for the UE to identify a new UTRA FDD cell is defined.

$T_{Inter1}$ = 30 ms. This is the minimum available time for inter-RAT measurement during 480 ms period

$N_{Freq}$: This is the number of UTRA carriers being monitored

If the UE does not need measurement gaps to perform UTRA FDD measurements, the measurement period for UTRA FDD measurements is 480 ms.

The UE shall be capable of performing UTRA FDD CPICH measurements for $X_{basic\ measurement\ UTRA_FDD}$ inter-frequency cells per FDD frequency and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{Measurement_UTRA_FDD}$.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between any events that will trigger a measurement report until the UE starts to transmit the measurement report over the Uu interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: twice the TTI of the uplink DCCH. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ UTRA_FDD}$ defined in TS 36.133 [4] clause 8.1.2.4.1.1.1. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.1 and A.8.5.1.

### 8.5.1.4 Test description

#### 8.5.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.23.
2. The general test parameter settings are set up according to Table 8.5.1.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.5.1.4.3.
5. There is one E-UTRA FDD serving cell and one UTRA FDD cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.5.1.4.1-1: General Test Parameters for E-UTRAN FDD- UTRAN FDD event triggered reporting under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Gap Pattern Id | | 1 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on UTRA RF channel number 1. |
| CP length | | Normal | Applicable to Cell 1 |
| E-UTRA RF Channel Number | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| UTRA RF Channel Number | | 1 | One UTRA FDD carrier frequency is used. |
| Inter-RAT (UTRA FDD) measurement quantity | | CPICH Ec/Io | |
| b1-Threshold-UTRA | dB | -18 | CPICH Ec/Io threshold for event B1. |
| Hysteresis | dB | 0 | |
| Time To Trigger | ms | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used. |
| DRX | | OFF | |
| Monitored UTRA FDD cell list size | | 12 | UTRA cells on UTRA RF channel 1 provided in the cell list. |
| T1 | s | 5 | |
| T2 | s | 6 | |

#### 8.5.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event B1 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table's 8.5.1.5-1 and 8.5.1.5-2. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2.. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 8.5.1.5-1 and 8.5.1.5-2.

6. UE shall transmit a MeasurementReport message triggered by Event B1. If the overall delay measured from the beginning of time period T2 is less than 4802 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. The SS shall set Cell 2 primary scrambling code = ((current cell 2 primary scrambling code - 50) mod 200 + 100) for the next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.5.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.5.1.4.3-1: Common Exception messages for E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-4<br>Table H.3.1-7 |

**Table 8.5.1.4.3-3: *ReportConfigInterRAT-B1-UTRA*: Additional E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7B ReportConfigInterRAT-B1(EUTRA-Thres) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-UTRA(UTRA-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| b1-ThresholdUTRA CHOICE { | | | |
| utra-EcN0 | 13 (-18 dB) | -18 dB is actual EcNO value in dB ((13 - 49)/2 dB) | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.5.1.4.3-4: *MeasResults*: Additional E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 8.5.1.4.3-5: *MeasResultListUTRA*: Additional E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| fdd | 250 | PhysCellIdUTRA-FDD INTEGER (0..511) | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-EcN0 | | Set according to specific test | |
| } | | | |
| } | | | |

## 8.5.1.5 Test requirement

Tables 8.5.1.4.1-1, 8.5.1.5-1 and 8.5.1.5-2 define the primary level settings including test tolerances for E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions in synchronous cells test.

**Table 8.5.1.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD (Cell 1) for event triggered reporting of UTRAN FDD cell under fading propagation conditions**

| **Parameter** | **Unit** | **Cell 1** | |
|---|---|---|---|
| | | **T1** | **T2** |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | |
| OCNG Pattern defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| $N_{oc}$ | dBm/15 kHz | -98 | |
| RSRP | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | ETU70 | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2. | | | |

**Table 8.5.1.5-2: Cell Specific Test requirement Parameters for Cell 2 UTRAN FDD event triggered reporting under fading propagation conditions**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| UTRA RF Channel Number | | 1 | |
| CPICH_Ec/Ior | dB | -10 | |
| PCCPCH_Ec/Ior | dB | -12 | |
| SCH_Ec/Ior | dB | -12 | |
| PICH_Ec/Ior | dB | -15 | |
| DPCH_Ec/Ior | dB | N/A | |
| OCNS | | -0.941 | |
| $\hat{I}_{or}/I_{oc}$ | dB | -Infinity | -1.8 |
| $I_{oc}$ | dBm/3.84 MHz | -70 | |
| CPICH_Ec/Io | dB | -Infinity | -14 |
| Propagation Condition | | Case 5 (Note 3) | |
| Note 1: The DPCH level is controlled by the power control loop.<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.<br>Note 3: Case 5 propagation conditions are defined in Annex A of 3GPP TS 25.101. | | | |

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event B1 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify, UTRA_FDD}$

$$T_{identify,\ UTRA_FDD} = T_{basic_identify_UTRA_FDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq} \quad ms$$

$T_{basic_identify_UTRA_FDD}$ = 300 ms

$T_{Inter1}$ = 30 ms

$N_{Freq}$ = 1

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 4802 ms in this test case (note: this gives a total of 4800 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.5.2 E-UTRAN FDD - UTRAN FDD SON ANR cell search reporting under AWGN propagation conditions

### 8.5.2.1 Test purpose

The purpose of this test is to verify that the UE makes correct reporting of the strongest UTRAN cell for SON automatic neighbour relations. This test will partly verify the E-UTRAN FDD - UTRAN FDD cell search requirements for identification of a new UTRA FDD cell for SON given in TS 36.133 [4] section 8.1.2.4.7.1.1.

### 8.5.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support UTRA FDD. Applicability requires support for FGI bits 5, 19 and 22.

### 8.5.2.3 Minimum conformance requirements

When no DRX is used the UE shall be able to identify a new cell within:

$$T_{identify,\ UTRA_FDD} = T_{basic_identify_UTRA_FDD} \cdot \frac{480}{Tinter1} \cdot N_{Freq} \quad ms$$

$T_{basic_identify_UTRA_FDD}$ = 300 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new UTRA FDD cell is defined.

A cell shall be considered identifiable following conditions are fulfilled:

- CPICH Ec/Io $\geq$ -20 dB,
- SCH_Ec/Io $\geq$ -17 dB for at least one channel tap and SCH_Ec/Ior is equally divided between primary synchronisation code and secondary synchronisation code. When L3 filtering is used an additional delay can be expected.

If the UE is unable to identify the UTRA cell for SON within $8*T_{identify,\ UTRA_FDD}$ ms, the UE may stop searching UTRA cells for SON.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.7 and A.8.5.2.

### 8.5.2.4 Test description

#### 8.5.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.
2. The general test parameter settings are set up according to Table 8.5.2.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.5.2.4.3.
5. There is one E-UTRA FDD serving cell and one UTRA FDD cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.5.2.4.1-1: General test parameters for E-UTRAN FDD-UTRAN FDD cell search reporting for SON ANR in AWGN propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1. |
| Gap Pattern Id | | 1 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on UTRA RF channel number 1. |
| CP length | | Normal | Applicable to cell 1 |
| E-UTRA RF Channel Number | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| UTRA RF Channel Number | | 1 | One UTRA FDD carrier frequency is used. |
| Inter-RAT (UTRA FDD) measurement quantity | | CPICH Ec/Io | |
| Filter coefficient | | 0 | L3 filtering is not used. |
| DRX | | OFF | |
| Monitored UTRA FDD cell list size | | None | No explicit neighbour list is provided to the UE |
| T1 | s | >5 | During T1, cell 2 shall be powered off, and during the off time the primary scrambling code shall be changed, The intention is to ensure that cell 2 has not been detected by the UE prior to the start of period T2. |
| T2 | s | 6 | |

### 8.5.2.4.2 Test procedure

The test consists of one active E-UTRAN cell and one neighbour UTRAN cell. In the measurement control information it is indicated to the UE that periodical reporting with the purpose 'reportStrongestCellsForSON' is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Tables 8.5.2.5-1 and 8.5.2.5-2. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.

3. The SS shall set Cell 2 primary scrambling code = ((current cell 2 primary scrambling code - 50) mod 200 + 100) for next iteration of the test procedure loop.

4. SS shall transmit an RRCConnectionReconfiguration message.

5. The UE shall transmit RRCConnectionReconfigurationComplete message.

6. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Tables 8.5.2.5-1 and 8.5.2.5-2.

7. The UE shall transmit a MeasurementReport message containing the primary scrambling code of cell 2. If the overall delays measured from the beginning of time period T2 is less than 4802 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

8. After the SS receive the MeasurementReport message in step 7) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),

or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 8.5.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.5.2.4.3-1: Common Exception messages for E-UTRAN FDD-UTRAN FDD cell search reporting for SON ANR in AWGN propagation conditions**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7 |

**Table 8.5.2.4.3-2: *MeasConfig-DEFAULT*: Additional E-UTRAN FDD-UTRAN FDD cell search reporting for SON ANR in AWGN propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF MeasObjectToAddMod | 2 entry | | |
| MeasObjectToAddMod ::= SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell (E-UTRA Cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | E-UTRA Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod ::= SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the neighbouring cell(UTRA Cell) | |
| measObject CHOICE { | | | |
| MeasObjectUTRA | MeasObjectUTRA-GENERIC(f2) | UTRA Cell | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF ReportConfigToAddMod | 1 entry | | |
| ReportConfigToAddMod ::= SEQUENCE { | | | |
| reportConfigId | idReportConfig | | |
| reportConfig CHOICE { | | | |
| reportConfigInterRAT | ReportConfigInterRAT-SON-UTRA | | |
| } | | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of MeasIdToAddMod | 1 entry | | |
| MeasIdToAddMod ::= SEQUENCE { | | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | MeasGapConfig-GP2 | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 8.5.2.4.3-3: *ReportConfigInterRAT-SON-UTRA*: Additional E-UTRAN FDD-UTRAN FDD cell search reporting for SON ANR in AWGN propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-SON-UTRA::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose ENUMERATED { | reportStrongestCellsForSON | | |
| } | | | |
| } | | | |
| } | | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | 1 | | |
| } | | | |

**Table 8.5.2.4.3-4: *MeasResults*: Additional E-UTRAN FDD-UTRAN FDD cell search reporting for SON ANR in AWGN propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 8.5.2.4.3-5: *MeasResultListUTRA*: Additional E-UTRAN FDD-UTRAN FDD cell search reporting for SON ANR in AWGN propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultUTRA | | | |
| MeasResultUTRA ::= SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| fdd | PhysCellIdUTRA-FDD | The primary scrambling code, as defined in TS 25.331 | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-EcN0 | | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 8.5.2.5 Test requirement

Tables 8.5.2.4.1-1, 8.5.2.5-1 and 8.5.2.5-2 define the primary level settings including test tolerances for UTRAN FDD cell search for SON ANR under AWGN propagation conditions test.

**Table 8.5.2.5-1: Cell specific test parameters for E-UTRAN FDD (cell # 1) for UTRAN FDD cell search for SON ANR under AWGN propagation conditions**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.5.2.5-2: Cell specific test parameters for UTRAN FDD (cell # 2) for UTRAN FDD cell search for SON ANR under AWGN propagation conditions**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| UTRA RF Channel Number | | 1 | |
| CPICH_Ec/Ior | dB | -10 | |
| PCCPCH_Ec/Ior | dB | -12 | |
| SCH_Ec/Ior | dB | -12 | |
| PICH_Ec/Ior | dB | -15 | |
| DPCH_Ec/Ior | dB | N/A | |
| OCNS | | -0.941 | |
| $\hat{I}_{or}/I_{oc}$ | dB | -Infinity | -2.95 |
| $I_{oc}$ | dBm/3.84 MHz | -70 | |
| CPICH_Ec/Io | dB | -Infinity | -14.73 |
| Propagation Condition | | AWGN | |
| Note 1: The DPCH level is controlled by the power control loop.<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$ | | | |

The UE shall send the first measurement report containing the primary scrambling code of cell 2, with a measurement reporting delay less than 4802 ms from the beginning of time period T2.

When no DRX is used the UE shall be able to identify a new cell within:

$$T_{identify,\ UTRA_FDD} = T_{basic_identify_UTRA_FDD} \cdot \frac{480}{Tinter1} \cdot N_{Freq} \quad ms$$

Where:

$T_{basic_identify_UTRA_FDD}$ = 300 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new UTRA FDD cell is defined.

$T_{Inter1}$ = 30 ms. TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 4802 ms in this test case (note: this gives a total of 4800 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The rate of correct measurement reports observed with this delay during repeated tests shall be at least 90% with a confidence level of 95%.

## 8.5.3 E-UTRAN FDD - UTRAN FDD event triggered reporting when DRX is used under fading propagation conditions

### 8.5.3.1 Test purpose

To verify the UE's ability to make a correct reporting of an event in DRX under fading propagation conditions within the E-UTRA FDD - UTRA FDD cell search requirements.

### 8.5.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support UTRA FDD. Applicability requires support for FGI bits 5, 15 and 22.

### 8.5.3.3 Minimum conformance requirements

The measurement reporting delay shall be less than $T_{identify,\ UTRA_FDD}$ in RRC_CONNECTED state.

When explicit neighbour list is provided and DRX is used the UE shall be able to identify a new detectable cell belonging to the neighbour cell list within Tidentify, UTRA_FDD as in table 8.5.3.3-1.

**Table 8.5.3.3-1: Requirements to identify a newly detectable UTRA FDD cell**

| DRX cycle length (s) | $T_{identify_UTRA_FDD}$ (s) (DRX cycles) | |
|---|---|---|
| | Gap period = 40 ms | Gap period = 80 ms |
| ≤0.04 | Non DRX Requirements in TS 36.133 [4] section 8.1.2.4.1.1 are applicable | Non DRX Requirements in TS 36.133 [4] section 8.1.2.4.1.1 are applicable |
| 0.064 | 2.56* Nfreq (40* Nfreq) | 4.8* Nfreq (75* Nfreq) |
| 0.08 | 3.2* Nfreq (40* Nfreq) | 4.8* Nfreq (60* Nfreq) |
| 0.128 | 3.2* Nfreq (25* Nfreq) | 4.8* Nfreq (37.5* Nfreq) |
| 0.16 | 3.2* Nfreq (20* Nfreq) | 4.8* Nfreq (30* Nfreq) |
| 0.16<DRX-cycle≤2.56 | Note (20* Nfreq) | Note (20* Nfreq) |
| Note: Time depends upon the DRX cycle in use | | |

A cell shall be considered detectable when

- CPICH Ec/Io $\geq$ -20 dB,
- SCH_Ec/Io ≥ -17 dB for at least one channel tap and SCH_Ec/Ior is equally divided between primary synchronisation code and secondary synchronisation code. When L3 filtering is used an additional delay can be expected.

The UE shall be capable of performing RSCP and Ec/Io measurements of at least 6 UTRA cells per UTRA FDD carrier and the UE physical layer shall be capable of reporting RSCP and Ec/Io measurements to higher layers with the measurement period defined in table 8.5.3.3-2.

**Table 8.5.3.3-2: Requirements to measure UTRA FDD cells**

| DRX cycle length (s) | $T_{measure_UTRA_FDD}$ (s) (DRX cycles) | |
|---|---|---|
| | **Gap period = 40 ms** | **Gap period = 80 ms** |
| ≤0.04 | Non DRX Requirements in TS 36.133 [4] section 8.1.2.4.1.1 are applicable | Non DRX Requirements in TS 36.133 [4] section 8.1.2.4.1.1 are applicable |
| 0.064 | 0.48* $N_{freq}$ (7.5* $N_{freq}$) | 0.8* $N_{freq}$ (12.5* $N_{freq}$) |
| 0.08 | 0.48* $N_{freq}$ (6* $N_{freq}$) | 0.8* $N_{freq}$ (10* $N_{freq}$) |
| 0.128 | 0.64* $N_{freq}$ (5* $N_{freq}$) | 0.8* $N_{freq}$ (6.25* $N_{freq}$) |
| 0.128<DRX-cycle≤2.56 | Note (5* $N_{freq}$) | Note (5* $N_{freq}$) |
| Note: Time depends upon the DRX cycle in use | | |

The measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clause 9.1.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between any events that will trigger a measurement report until the UE starts to transmit the measurement report over the Uu interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: twice the TTI of the uplink DCCH. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ UTRA_FDD}$ defined in TS 36.133 [4] clause 8.1.2.4.1.1.1. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.1.2 and A.8.5.3.

## 8.5.3.4 Test description

### 8.5.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.23.

2. The general test parameter settings are set up according to Table 8.5.3.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.5.3.4.3.

5. There is one E-UTRA FDD serving cell and one UTRA FDD cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.5.3.4.1-1: General Test Parameters for E-UTRAN FDD- UTRAN FDD event triggered reporting when DRX is used under fading propagation conditions**

| Parameter | Unit | Test 1 | Test 2 | Comment |
|---|---|---|---|---|
| | | Value | | |
| PDSCH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.0 FDD | | As specified in section A.1.1 Note that UE may only be allocated at *On Duration* |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.6 FDD | | As specified in section A.2.1. |
| Gap Pattern Id | | 0 | | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| Active cell | | Cell 1 | | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | | Cell 2 is on UTRA RF channel number 1. |
| CP length | | Normal | | Applicable to cell 1 |
| E-UTRA RF Channel Number | | 1 | | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| UTRA RF Channel Number | | 1 | | One UTRA FDD carrier frequency is used. |
| Inter-RAT (UTRA FDD) measurement quantity | | CPICH Ec/Io | | |
| b1-Threshold-UTRA | dB | -18 | | CPICH Ec/Io threshold for event B1. |
| Hysteresis | dB | 0 | | |
| TimeToTrigger | s | 0 | | |
| Filter coefficient | | 0 | | L3 filtering is not used |
| PRACH configuration | | 4 | | As specified in table 5.7.1-2 in TS 36.211 [9] |
| Access Barring Information | - | Not Sent | | No additional delays in random access procedure. |
| DRX | | ON | | DRX related parameters are defined in Table 8.5.3.5-2 |
| Monitored UTRA FDD cell list size | | 12 | | UTRA cells on UTRA RF channel 1 provided in the cell list. |
| T1 | s | 5 | | |
| T2 | s | 6 | 30 | |

### 8.5.3.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event B1 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

In Test 1 when DRX = 40 ms is used, UE needs to be provided at least once every 500 ms with new Timing Advance Command MAC control element to restart the Timer Alignment Timer to keep the UE uplink time alignment. Furthermore, the UE is allocated with PUSCH resource at every DRX cycle. In Test 2 when DRX = 1280 ms is used, the uplink time alignment is not maintained and the UE needs to use RACH to obtain uplink allocation for measurement reporting.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.5.3.5-1, 8.5.3.5-2, 8.5.3.5-3 and 8.5.3.5-5. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 8.5.3.5-1 and 8.5.3.5-4. T2 starts.

6. UE shall transmit a MeasurementReport message triggered by Event B1. If the overall delays measured from the beginning of time period T2 is less than 2442 ms for Test 1 or less than 26882 ms for Test 2 then the number of

successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. The SS shall set Cell 2 primary scrambling code = ((current cell 2 primary scrambling code - 50) mod 200 + 100) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

11. Repeat step 1-10 for each sub-test in Table 8.3.2.4.1-1 as appropriate.

### 8.5.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.5.3.4.3-1: Common Exception messages for E-UTRAN FDD - UTRAN FDD event triggered reporting when DRX is used under fading propagation conditions**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-4<br>Table H.3.1-7<br>Table H.3.7-1<br>Table H.3.7-2<br>Table H.3.7-3 |

**Table 8.5.3.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN FDD - UTRAN FDD event triggered reporting when DRX is used under fading propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.5.3.4.3-3: *ReportConfigInterRAT-B1-UTRA*: Additional E-UTRAN FDD - UTRAN FDD event triggered reporting when DRX is used under fading propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7B ReportConfigInterRAT-B1(EUTRA-Thres) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-UTRA(UTRA-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| b1-ThresholdUTRA CHOICE { | | | |
| utra-EcN0 | 13 (-18 dB) | -18 dB is actual EcNO value in dB ((13 - 49)/2 dB) | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0(0 dB) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.5.3.4.3-4: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD - UTRAN FDD event triggered reporting when DRX is used under fading propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.5.3.4.3-5: *MeasResults*: Additional E-UTRAN FDD - UTRAN FDD event triggered reporting when DRX is used under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 8.5.3.4.3-6: *MeasResultListUTRA*: Additional E-UTRAN FDD - UTRAN FDD event triggered reporting when DRX is used under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| fdd | 250 | PhysCellIdUTRA-FDD | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-EcN0 | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 8.5.3.4.3-7: *PRACH-Config-DEFAULT*: Additional E-UTRAN FDD - UTRAN FDD event triggered reporting when DRX is used under fading propagation conditions**

| Derivation Path: TS 36.508 [7] clause 7.3.2 Table 7.3.2-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PRACH-Config-DEFAULT ::= SEQUENCE { | | | |
| prach-ConfigIndex | 4 | | |
| } | | | |

## 8.5.3.5 Test requirement

Tables 8.5.3.4.1-1, 8.5.3.5-1 and 8.5.3.5-4 define the primary level settings including test tolerances for E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions in synchronous cells test.

**Table 8.5.3.5-1: Cell Specific Test requirement Parameters for Cell 1 E-UTRAN FDD event triggered reporting when DRX is used under fading propagation conditions**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | |
| OCNG Pattern defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |

| | | | |
|---|---|---|---|
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | |
| RSRP Note 3 | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| Propagation Condition | | ETU70 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.5.3.5-2: DRX-Configuration to be used in E-UTRAN FDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf40 | sf1280 | |
| shortDRX | disable | disable | |
| Note: For further information see section 6.3.2 in 3GPP TS 36.331 [5]. | | | |

**Table 8.5.3.5-3: *TimeAlignmentTimer* and *sr-ConfigIndex*-Configuration to be used in E-UTRAN FDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | For further information see section 6.3.2 in 3GPP TS 36.331 [5]. |
| sr-ConfigIndex | 0 | 0 | For further information see section 6.3.2 in 3GPP TS 36.331 and section10.1 in 3GPP TS 36.213 [8]. |

**Table 8.5.3.5-4: Cell Specific Test requirement Parameters for Cell 2 UTRAN FDD event triggered reporting when DRX is used under fading propagation conditions**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| UTRA RF Channel Number | | 1 | |
| CPICH_Ec/Ior | dB | -10 | |
| PCCPCH_Ec/Ior | dB | -12 | |
| SCH_Ec/Ior | dB | -12 | |
| PICH_Ec/Ior | dB | -15 | |
| DPCH_Ec/Ior | dB | N/A | |
| OCNS | | -0.941 | |
| $\hat{I}_{or}/I_{oc}$ | dB | -Infinity | -1.8 |
| $I_{oc}$ | dBm/3.84 MHz | -70 | |
| CPICH_Ec/Io | dB | -Infinity | -14 |
| Propagation Condition | | Case 5 (Note 3) | |
| Note 1: The DPCH level is controlled by the power control loop.<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.<br>Note 3: Case 5 propagation conditions are defined in Annex A of 3GPP TS 25.101. | | | |

In Test 1 when DRX cycle length = 40 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event B1 triggered measurement report to Cell 2 on PUSCH.

In Test 2 when DRX cycle length = 1280 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE starts to send preambles on the PRACH for Scheduling Request (SR) to obtain allocation to send the measurement report to Cell 2 on PUSCH.

For both tests:

The overall delays measured when DRX cycle length is 40 ms in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event B1 measurement report.

The overall delay measured when DRX cycle length is 40 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify, UTRA_FDD}$=

$$T_{identify,\ UTRA_FDD} = T_{basic_identify_UTRA_FDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq} \quad ms$$

Where:

$T_{basic_identify_UTRA_FDD}$ = 300 ms. It is the time period used in the inter RAT equation where the maximum allowed time for the UE to identify a new UTRA FDD cell is defined.

$T_{Inter1}$ = 60 ms.

$N_{freq}$ = 1. It is defined in TS 36.133 [4] clause 8.1.2.1.1.

TTI insertion uncertainty = 2 ms

DRX cycle length = 40 ms

The overall delay measured when DRX cycle length is 40 ms shall be less than a total of 2442 ms.

The overall delay measured when DRX cycle length is 1280 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_UTRA_FDD}$

$T_{identify_UTRA_FDD}$ = 25600 ms. When DRX cycle length is 1280 ms the $T_{identify_UTRA_FDD}$ is 20 x 1280 ms.

TTI insertion uncertainty = 2 ms

DRX cycle length = 1280 ms

The overall delay measured when DRX cycle length is 1280 ms shall be less than a total of 26882 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.5.4 E-UTRAN FDD - UTRAN FDD enhanced cell identification under AWGN propagation conditions

### 8.5.4.1 Test purpose

The purpose of this test is to verify that the UE makes correct enhanced reporting of UTRAN cells. This test will partly verify the Enhanced UTRA FDD cell identification requirements in section 8.1.2.4.1.1.1a.

### 8.5.4.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward that support UTRA FDD. Applicability requires support for FGI bit 15.

### 8.5.4.3 Minimum conformance requirements

When explicit neighbour list is provided and no DRX is used or when DRX cycle length ≤ 40 ms the UE shall be able to identify a new detectable cell belonging to the monitored set within $T_{identify,\ enhanced_UTRA_FDD}$:

$$T_{identify,\ enhanced_UTRA_FDD} = (T_{basic_identify_enhanced_UTRA_FDD} \cdot \frac{480}{T_{inter1}} + 480)\ N_{Freq} \quad ms$$

A cell shall be considered detectable when:

- CPICH Ec/Io ≥ -15 dB,
- SCH_Ec/Io ≥ -15 dB for at least one channel tap and SCH_Ec/Ior is equally divided between primary synchronisation code and secondary synchronisation code. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.1.1.1a and A.8.5.4.

### 8.5.4.4 Test description

#### 8.5.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.

2. The general test parameter settings are set up according to Table 8.5.4.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.5.4.4.3.

5. There is one E-UTRA FDD serving cell and one UTRA FDD cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 (UTRAN FDD cell) is to be searched. Gap pattern configuration # 0 as defined in TS 36.133 [4] table 8.1.2.1-1 is provided.

**Table 8.5.4.4.1-1: General test parameters for E-UTRAN FDD - UTRAN FDD enhanced cell identification under AWGN propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1. |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on UTRA RF channel number 1. |
| CP length | | Normal | Applicable to cell 1 |
| E-UTRA RF Channel Number | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| UTRA RF Channel Number | | 1 | One UTRA FDD carrier frequency is used. |
| Inter-RAT (UTRA FDD) measurement quantity | | CPICH Ec/Io | |
| b1-Threshold-UTRA | dB | -18 | CPICH Ec/Io threshold for event B1. |
| Hysteresis | dB | 0 | |
| Time To Trigger | ms | 0 | |
| Monitored UTRA FDD cell list size | | 12 | UTRA cells on UTRA RF channel 1 provided in the cell list |
| T1 | s | 5 | |
| T2 | s | 2 | |

### 8.5.4.4.2 Test procedure

The test consists of two successive time periods, with time duration of T1, and T2 respectively. During time period T1, gaps are activated and an interRAT measurement reporting configuration is configured, and linked to a UTRA measurement object corresponding to channel UARFCN 1. Prior to time duration T2, the UE shall not have any timing information of cell 2. Cell 2 is powered up at the beginning of the T2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Tables 8.5.4.5-1 and 8.5.4.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. The SS shall transmit an *RRCConnectionReconfiguration* with Event B1 configured message.

4. The UE shall transmit *RRCConnectionReconfigurationComplete* message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Tables 8.5.4.5-1 and 8.5.4.5-2. T2 starts.

6. UE shall transmit a *MeasurementReport* message triggered by Event B1. If the overall delay measured from the beginning of time period T2 is less than 962 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit *RRCConnectionRelease* message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. The SS shall set Cell 2 primary scrambling code = ((current cell 2 primary scrambling code - 50) mod 200 + 100) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.5.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.5.4.4.3-1: Common Exception messages for E-UTRAN FDD - UTRAN FDD enhanced cell identification under AWGN propagation conditions**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-4<br>Table H.3.1-7 |

**Table 8.5.4.4.3-2: *ReportConfigInterRAT-B1-UTRA*: Additional E-UTRAN FDD - UTRAN FDD enhanced cell identification under AWGN propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6 Table 4.6.6-7B | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-UTRA(UTRA-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| b1-ThresholdUTRA CHOICE { | | | |
| utra-EcN0 | 13 (-18 dB) | UTRA-Thres is actual Ec/No value in dB (UTRA-Thres * 2 + 49) | UTRA-FDD |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0dB) | The actual value is IE value * 0.5 dB. | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.5.4.4.3-3: *MeasResults*: Additional E-UTRAN FDD - UTRAN FDD enhanced cell identification under AWGN propagation conditions**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 8.5.4.4.3-4: *MeasResultListUTRA*: E-UTRAN FDD - UTRAN FDD enhanced cell identification under AWGN propagation conditions**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| fdd | PhysCellIdUTRA-FDD | INTEGER (0..511) | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-EcN0 | | Set according to specific test INTEGER (0..49) | |
| } | | | |
| } | | | |

## 8.5.4.5 Test requirement

Tables 8.5.4.4.1-1, 8.5.4.5-1 and 8.5.4.5-2 define the primary level settings including test tolerances for E-UTRAN FDD - UTRAN FDD enhanced cell identification under AWGN propagation conditions test.

**Table 8.5.4.5-1: Cell specific test parameters for cell #1 E-UTRAN FDD - UTRAN FDD enhanced cell identification under AWGN propagation conditions**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.5.4.5-2: Cell specific test parameters for cell #2 E-UTRAN FDD - UTRAN FDD enhanced cell identification under AWGN propagation conditions**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| UTRA RF Channel Number | | 1 | |
| CPICH_Ec/Ior | dB | -10 | |
| PCCPCH_Ec/Ior | dB | -12 | |
| SCH_Ec/Ior | dB | -12 | |
| PICH_Ec/Ior | dB | -15 | |
| DPCH_Ec/Ior | dB | N/A | |
| OCNS | | -0.941 | |
| $\hat{I}_{or} / I_{oc}$ | dB | - infinity | 0.02 |
| $I_{oc}$ | dBm/3.84 MHz | -70 | |
| CPICH_Ec/Io Note 3 | dB | - infinity | -13 |
| Propagation Condition | | AWGN | |

Note 1: The DPCH level is controlled by the power control loop.
Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.
Note 3: This gives an SCH Ec/Io of -15dB

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send the first measurement report containing the primary scrambling code of cell 2.

The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify,\ enhanced_UTRA_FDD}$

$$T_{identify,\ enhanced_UTRA_FDD} = (T_{basic_identify_enhanced_UTRA_FDD} \cdot \frac{480}{T_{inter1}} + 480)\ N_{Freq} \quad ms$$

$T_{basic_identify_enhanced_UTRA_FDD}$ = 60 ms. This is the time period used in the inter RAT equation in TS 36.133 [4] section 8.1.2.4.1.1.1a where the maximum allowed time for the UE to identify a new UTRA FDD cell is defined.

$T_{Inter1}$ = 60 ms. It is defined in TS 36.133 [4] section 8.1.2.1.

$N_{Freq}$ = 1. It is defined in TS 36.133 [4] clause 8.1.2.1.1.

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 962 ms in this test case (note: this gives a total of 960 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.5.5

FFS

## 8.5.6 E-UTRAN FDD - UTRAN FDD event triggered reporting without measurement gaps under AWGN propagation conditions

### 8.5.6.1 Test purpose

To verify the UE's ability to make a correct reporting of an event without measurement gaps within the E-UTRA FDD - UTRA FDD cell search requirements.

### 8.5.6.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward that support UTRA FDD with support of Measurement without gaps. Applicability requires support for FGI bits 15 and 22.

### 8.5.6.3 Minimum conformance requirements

When explicit neighbour list is provided and no DRX is used, either measurement gaps are scheduled or the UE supports capability of conducting such measurements without gaps, the UE shall be able to identify a new detectable cell belonging to the monitored set within

$$T_{identify,\ UTRA_FDD} = T_{basic_identify_UTRA_FDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq} \quad ms$$

A cell shall be considered detectable when

- CPICH Ec/Io $\geq$ -20 dB,

- SCH_Ec/Io $\geq$ -17 dB for at least one channel tap and SCH_Ec/Ior is equally divided between primary synchronisation code and secondary synchronisation code. When L3 filtering is used an additional delay can be expected.

When measurement gaps are scheduled for UTRA FDD inter RAT measurements, or the UE supports capability of conducting such measurements without gaps, the UE physical layer shall be capable of reporting measurements to higher layers with measurement accuracy as specified in TS 36.133 [4] clause 9.2 with measurement period given by

$$T_{measurement_UTRA_FDD} = Max\left\{T_{Measurement_Period\ UTRA_FDD}, T_{basic_measurement_UTRA_FDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms$$

The UE shall be capable of performing UTRA FDD CPICH measurements for $X_{basic\ measurementUTRA_FDD}$ inter-frequency cells per FDD frequency and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{Measurement_UTRA_FDD}$.

$X_{basic\ measurement\ UTRA_FDD} = 6$

$T_{Measurement_Period\ UTRA_FDD}$ = 480 ms. The period used for calculating the measurement period $T_{measurement_UTRA_FDD}$ for UTRA FDD CPICH measurements.

$T_{basic_identify_UTRA_FDD}$ = 300 ms. This is the time period used in the inter RAT equation in TS 36.133 [4] clause 8.1.2.4.1.1.1 where the maximum allowed time for the UE to identify a new UTRA FDD cell is defined.

$T_{basic_identify_enhanced_UTRA_FDD}$ = 60 ms. This is the time period used in the inter RAT equation in TS 36.133 [4] clause 8.1.2.4.1.1.1a where the maximum allowed time for the UE to identify a new UTRA FDD cell is defined.

$T_{basic_measurement_UTRA_FDD}$ = 50 ms. This is the time period used in the equation for defining the measurement period for inter RAT CPICH measurements.

$N_{freq}$ is defined in clause 8.1.2.1.1 and $T_{inter1}$ is defined in TS 36.133 [4] clause 8.1.2.1

Reported measurements in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as the reporting criteria is not fulfilled.

The measurement reporting delay is defined as the time between any event that will trigger a measurement report until the UE starts to transmit the measurement report over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify,\ UTRA_FDD}$ defined in TS 36.133 [4] clause 8.1.2.4.1.1.1 for the minimum requirements or $T_{identify,\ enhanced_UTRA_FDD}$ defined in TS 36.133 [4] clause 8.1.2.4.1.1.1a for the enhanced requirements When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.1 and A.8.5.6.

## 8.5.6.4 Test description

### 8.5.6.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.

2. The general test parameter settings are set up according to Table 8.5.6.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.5.6.4.3.

5. There is one E-UTRA FDD serving cell and one UTRA FDD cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.5.6.1-1: General test parameters for E-UTRAN FDD-UTRAN FDD event triggered reporting without measurement gaps under AWGN propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.3 FDD | As specified in clause A.1.1. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.6 FDD | As specified in clause A.2.1. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on UTRA RF channel number 1. |
| CP length | | Normal | Applicable to cell 1 |
| E-UTRA RF Channel Number | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| UTRA RF Channel Number | | 1 | One UTRA FDD carrier frequency is used. |
| Inter-RAT (UTRA FDD) measurement quantity | | CPICH Ec/Io | |
| b1-Threshold-UTRA | dB | -18 | CPICH Ec/Io threshold for event B1. |
| Hysteresis | dB | 0 | |
| Time To Trigger | ms | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used. |
| DRX | | OFF | |
| Monitored UTRA FDD cell list size | | 12 | UTRA cells on UTRA RF channel 1 provided in the cell list. |
| T1 | s | 5 | |
| T2 | s | 6 | |

### 8.5.6.4.2 Test procedure

The test consists of one active E-UTRAN cell and one neighbour UTRAN cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event B1 is used. The test consists of two successive time periods, with time duration of T1, and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2. PDCCH on the PCell shall be sent continuously to ensure that the UE sends ACK/NACKs throughout the test.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table's 8.5.6.5-1 and 8.5.6.5-2. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 8.5.6.5-1 and 8.5.6.5-2.

6. UE shall transmit a MeasurementReport message triggered by Event B1. If the overall delay measured from the beginning of time period T2 is less than 4802 ms and at least 85% of all ACK/NACKS transmitted by the UE are detected by the system simulator until the measurement report is received during T2 the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement or less than 85% of all ACK/NACKS transmitted by the UE are detected by the system simulator then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. The SS shall set Cell 2 primary scrambling code = ((current cell 2 primary scrambling code - 50) mod 200 + 100) for the next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.5.6.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.5.6.4.3-1: Common Exception messages for E-UTRAN FDD - UTRAN FDD event triggered reporting without measurement gaps under AWGN propagation conditions**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7 |

**Table 8.5.6.4.3-2: *ReportConfigInterRAT-B1-UTRA*: Additional E-UTRAN FDD - UTRAN FDD event triggered reporting without measurement gaps under AWGN propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7B ReportConfigInterRAT-B1(EUTRA-Thres) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-UTRA(UTRA-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| b1-ThresholdUTRA CHOICE { | | | |
| utra-EcN0 | 13 (-18 dB) | -18 dB is actual EcNO value in dB ((13 - 49)/2 dB) | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.5.6.4.3-3: *MeasResults*: Additional E-UTRAN FDD - UTRAN FDD event triggered reporting without measurement gaps under AWGN propagation conditions**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 8.5.6.4.3-4: *MeasResultListUTRA*: Additional E-UTRAN FDD - UTRAN FDD event triggered reporting without measurement gaps under AWGN propagation conditions**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| fdd | 250 | PhysCellIdUTRA-FDD INTEGER (0..511) | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-EcN0 | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 8.5.6.4.3-5: *MeasConfig-DEFAULT*: Additional E-UTRAN FDD - UTRAN FDD event triggered reporting without measurement gaps under AWGN propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell (E-UTRA Cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | E-UTRA Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the neighbouring cell(UTRA Cell) | |
| measObject CHOICE { | | | |
| measObjectUTRA | MeasObjectUTRA-GENERIC(f2) | UTRA Cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-B1 | | |
| reportConfig | ReportConfigInterRAT-B1-UTRA | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-B1 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | UTRAN |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 8.5.6.4.3-6: *SystemInformationBlockType2*: Additional E-UTRAN FDD - UTRAN FDD event triggered reporting without measurement gaps under AWGN propagation conditions**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-1 SystemInformationBlockType2 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfig | Not present | | Cell 1 |
| } | | | |

### 8.5.6.5 Test requirement

Tables 8.5.6.4.1-1, 8.5.6.5-1 and 8.5.6.5-2 define the primary level settings including test tolerances for E-UTRAN FDD - UTRAN FDD event triggered reporting without measurement gaps under AWGN propagation conditions test.

**Table 8.5.6.5-1: Cell specific test parameters for E-UTRAN FDD (cell # 1) for event triggered reporting of UTRAN FDD cell without measurement gaps under AWGN propagation conditions**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern defined in D.1.10 (OP.10 FDD) | | OP.10 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| $N_{oc}$ | dBm/15 kHz | -98 | |
| RSRP | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2. | | | |

**Table 8.5.6.5-2: Cell specific test parameters for UTRAN FDD (cell # 2) for event triggered reporting of UTRAN FDD cell without measurement gaps under AWGN propagation conditions**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| UTRA RF Channel Number | | 1 | |
| CPICH_Ec/Ior | dB | -10 | |
| PCCPCH_Ec/Ior | dB | -12 | |
| SCH_Ec/Ior | dB | -12 | |
| PICH_Ec/Ior | dB | -15 | |
| DPCH_Ec/Ior | dB | N/A | |
| OCNS | | -0.941 | |
| $\hat{I}_{or} / I_{oc}$ | dB | -Infinity | -1.8 |
| $I_{oc}$ | dBm/3.84 MHz | -70 | |
| CPICH_Ec/Io | dB | -Infinity | -14 |
| Propagation Condition | | AWGN | |
| Note 1: The DPCH level is controlled by the power control loop.<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$. | | | |

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event B1 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify, UTRA_FDD}$

$$T_{identify,\ UTRA_FDD} = T_{basic_identify_UTRA_FDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq} \quad ms$$

$T_{basic_identify_UTRA_FDD}$ = 300 ms

$T_{Inter1}$ = 30 ms

$N_{Freq}$ = 1

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 4802 ms in this test case (note: this gives a total of 4800 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall send continuous ACK/NACK throughout the test, and from the start of T2 until Event A3 is reported, at least 85% ACK/NACK shall be detected.

For a test to be considered successful requirements on both event detection and percentage of transmitted ACK/NACKs have to be fulfilled simultaneously.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.5.7 E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions for 5MHz Bandwidth

### 8.5.7.1 Test purpose

Same test purpose as defined in clause 8.5.1.1.

### 8.5.7.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support UTRA FDD and only support E-UTRA Band 31. Applicability requires support for FGI bits 15 and 22.

### 8.5.7.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.5.1.3 with the following exceptions:

- Instead of A.8.5.1 → use A.8.5.7.

### 8.5.7.4 Test description

#### 8.5.7.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.23.
2. The general test parameter settings are set up according to Table 8.5.7.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.5.7.4.3.
5. There is one E-UTRA FDD serving cell and one UTRA FDD cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.5.7.4.1-1: General Test Parameters for E-UTRAN FDD- UTRAN FDD event triggered reporting under fading propagation conditions for 5MHz bandwidth**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.5 FDD | As specified in clause A.1.1. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.11 FDD | As specified in clause A.2.1. |
| E-UTRA Channel Bandwidth (BWchannel) | MHz | 5 | |
| NOTE 1: See Table 8.5.1.4.1-1 for the other parameters.<br>NOTE 2: This test is according to the principle defined in TS 36.133 [4] section A.3.7.2. | | | |

### 8.5.7.4.2 Test procedure

Same test procedure as defined in clause 8.5.1.4.2 with the following exceptions:

- Instead of Table 8.5.1.5-1 → use Table 8.5.7.5-1.

### 8.5.7.4.3 Message contents

Same message contents as defined in clause 8.5.1.4.3.

## 8.5.7.5 Test requirement

Same test requirement as defined in clause 8.5.1.5 with the following exceptions:

Tables 8.5.7.4.1-1, 8.5.7.5-1 and 8.5.1.5-2 define the primary level settings including test tolerances for E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions in synchronous cells for 5MHz bandwidth test.

**Table 8.5.7.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD (Cell 1) for event triggered reporting of UTRAN FDD cell under fading propagation conditions for 5MHz bandwidth**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| $BW_{channel}$ | MHz | 5 | |
| OCNG Pattern defined in D.1.15 | | OP.15 FDD | |
| NOTE: See Table 8.5.1.5-1 for the other parameters. | | | |

# 8.6 E-UTRAN TDD - UTRAN FDD measurements

## 8.6.1 E-UTRAN TDD - UTRAN FDD event triggered reporting under fading propagation conditions

### 8.6.1.1 Test purpose

To verify the UE's ability to make a correct reporting of an event under fading propagation conditions within the E-UTRA TDD - UTRA FDD cell search requirements.

### 8.6.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward that support UTRA FDD. Applicability requires support for FGI bits 15 and 22.

### 8.6.1.3 Minimum conformance requirements

The measurement reporting delay shall be less than $T_{identify,\ UTRA_FDD}$ in RRC_CONNECTED state.

When explicit neighbour list is provided and no DRX is used the UE shall be able to identify a new detectable cell belonging to the monitored set within

$$T_{identify,\ UTRA_FDD} = T_{basic_identify_UTRA_FDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq} \quad ms$$

Where:

$T_{basic_identify_UTRA_FDD}$ = 300 ms. This is the time period used in the inter-RAT equation where the maximum allowed time for the UE to identify a new UTRA FDD cell is defined.

$T_{Inter1}$ = 30 ms. This is the minimum available time for inter-RAT measurement during 480 ms period

$N_{Freq}$: This is the number of UTRA carriers being monitored

A cell shall be considered detectable when

- CPICH Ec/Io $\geq$ -20 dB,
- SCH_Ec/Io $\geq$ -17 dB for at least one channel tap and SCH_Ec/Ior is equally divided between primary synchronisation code and secondary synchronisation code. When L3 filtering is used an additional delay can be expected.

When measurement gaps are scheduled for UTRA FDD inter RAT measurement the UE physical layer shall be capable of reporting measurements to higher layers with measurement accuracy as specified in TS 36.133 [4] clause 9.2 of with measurement period given by

$$T_{measurement_UTRA_FDD} = Max\left\{T_{Measurement_Period\ UTRA_FDD}, T_{basic_measurement_UTRA_FDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms$$

Where:

$X_{basic\ measurement\ UTRA_FDD}$ = 6 (cells)

$T_{Measurement_Period\ UTRA_FDD}$ = 480 ms. The period used for calculating the measurement period.

$T_{basic_measurement_UTRA_FDD}$ = 50 ms. This is the time period used in the equation where the maximum allowed time for the UE to identify a new UTRA FDD cell is defined.

$T_{Inter1}$ = 30 ms. This is the minimum available time for inter-RAT measurement during 480 ms period

$N_{Freq}$: This is the number of UTRA carriers being monitored

If the UE does not need measurement gaps to perform UTRA FDD measurements, the measurement period for UTRA FDD measurements is 480 ms.

The UE shall be capable of performing UTRA FDD CPICH measurements for $X_{basic\ measurement\ UTRA_FDD}$ inter-frequency cells per FDD frequency and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{Measurement_UTRA_FDD}$.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between any events that will trigger a measurement report until the UE starts to transmit the measurement report over the Uu interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: twice the TTI of the uplink DCCH. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ UTRA_FDD}$ defined in TS 36.133 [4] clause 8.1.2.4.1.1.1. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.2 and A.8.6.1.

## 8.6.1.4 Test description

### 8.6.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.23.

2. The general test parameter settings are set up according to Table 8.6.1.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.6.1.4.3.

5. There is one E-UTRA TDD serving cell and one UTRA FDD cell specified in the test. Cell 1 (E-UTRA TDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.6.1.4.1-1: General test parameters for E-UTRAN TDD-UTRAN FDD event triggered reporting in fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2. |
| Gap Pattern Id | | 1 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on UTRA RF channel number 1. |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in 3GPP TS 36.211. Applicable to cell 1. |
| Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in 3GPP TS 36.211. Applicable to cell 1. |
| CP length | | Normal | Applicable to cell 1. |
| E-UTRA RF Channel Number | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| UTRA RF Channel Number | | 1 | One UTRA FDD carrier frequency is used. |
| Inter-RAT (UTRA FDD) measurement quantity | | CPICH Ec/Io | |
| b1-Threshold-UTRA | dB | -18 | CPICH Ec/Io threshold for event B1. |
| Hysteresis | dB | 0 | |
| Time To Trigger | ms | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used. |
| DRX | | OFF | |
| Monitored UTRA cell list size | | 12 | UTRA cells on UTRA RF channel 1 provided in the cell list. |
| T1 | s | 5 | |
| T2 | s | 6 | |

### 8.6.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event B1 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table's 8.6.1.5-1 and 8.6.1.5-2. Propagation conditions are set according to Annex B clause B.1.1 and B.2.2.. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 8.6.1.5-1 and 8.6.1.5-2.

6. UE shall transmit a MeasurementReport message triggered by Event B1. If the overall delays measured from the beginning of time period T2 is less than 4802 ms then the number of successful tests is increased by one. If the

UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. The SS shall set Cell 2 primary scrambling code = ((current cell 2 primary scrambling code - 50) mod 200 + 100) for the next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.6.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.6.1.4.3-1: Common Exception messages for E-UTRAN TDD - UTRAN FDD event triggered reporting under fading propagation conditions**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-4<br>Table H.3.1-7 |

**Table 8.6.1.4.3-3: *ReportConfigInterRAT-B1-UTRA*: Additional E-UTRAN TDD - UTRAN FDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-UTRA(UTRA-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| b1-ThresholdUTRA CHOICE { | | | |
| utra-EcN0 | 13 (-18dB) | -18 dB is actual EcNO value in dB ((13 - 49)/2 dB) | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0dB) | | |
| } | | | |
| } | | | |

**Table 8.6.1.4.3-4: *MeasResults*: Additional E-UTRAN TDD - UTRAN FDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServiCCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 8.6.1.4.3-5: *MeasResultListUTRA*: Additional E-UTRAN TDD - UTRAN FDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| fdd | UTRA-FDD-CellIdentity | | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-EcN0 | | Set according to specific test | |
| } | | | |
| } | | | |

## 8.6.1.5 Test requirement

Tables 8.6.1.4.1-1, 8.6.1.5-1 and 8.6.1.5-2 define the primary level settings including test tolerances for E-UTRAN TDD - UTRAN FDD event triggered reporting under fading propagation conditions in synchronous cells test.

**Table 8.6.1.5-1: Cell specific test parameters for E-UTRAN TDD (cell # 1) for event triggered reporting of UTRAN FDD cell under fading propagation conditions**

| **Parameter** | **Unit** | **Cell 1** | |
|---|---|---|---|
| | | **T1** | **T2** |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |

| | | | |
|---|---|---|---|
| OCNG_RB[Note 1] | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| $N_{oc}$ | dBm/15 kHz | -98 | |
| RSRP | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | ETU70 | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2. | | | |

**Table 8.6.1.5-2: Cell specific test parameters for UTRAN FDD (cell # 2) for event triggered reporting of UTRAN FDD cell under fading propagation conditions**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| UTRA RF Channel Number | | 1 | |
| CPICH_Ec/Ior | dB | -10 | |
| PCCPCH_Ec/Ior | dB | -12 | |
| SCH_Ec/Ior | dB | -12 | |
| PICH_Ec/Ior | dB | -15 | |
| DPCH_Ec/Ior | dB | N/A | |
| OCNS | | -0.941 | |
| $\hat{I}_{or} / I_{oc}$ | dB | -Infinity | -1.8 |
| $I_{oc}$ | dBm/3.84 MHz | -70 | |
| CPICH_Ec/Io | dB | -Infinity | -14 |
| Propagation Condition | | Case 5 (Note 3) | |
| Note 1: The DPCH level is controlled by the power control loop.<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.<br>Note 3: Case 5 propagation conditions are defined in Annex A of 3GPP TS 25.101. | | | |

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event B1 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2 \times TTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify, UTRA_FDD}$

$$T_{identify,\ UTRA_FDD} = T_{basic_identify_UTRA_FDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq} \quad ms$$

$T_{basic_identify_UTRA_FDD}$ = 300 ms

$T_{Inter1}$ = 30 ms

$N_{Freq}$ = 1

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 4802 ms in this test case (note: this gives a total of 4800 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 8.7 E-UTRAN TDD - UTRAN measurements

## 8.7.1 E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions

Editor's note: It only includes 1.28 Mcps TDD option related requirements

### 8.7.1.1 Test purpose

To verify the UE's ability to make a correct reporting of an event under fading propagation conditions within the E-UTRA TDD - UTRA TDD cell search requirements.

### 8.7.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 that support UTRA TDD. Applicability requires support for FGI bits 15 and 22.

This test applies to all types of E-UTRA TDD UE release 9 and forward that support UTRA TDD and not supporting UTRA FDD. Applicability requires support for FGI bits 15 and 22.

This test applies to all types of E-UTRA TDD UE release 9 and forward that support UTRA TDD. Applicability requires support for FGI bits 15 and 39.

### 8.7.1.3 Minimum conformance requirements

The measurement reporting delay shall be less than $T_{identify,\ UTRA_TDD}$ in RRC_CONNECTED state.

When explicit neighbour list is provided and no DRX is used the UE shall be able to identify a new detectable cell belonging to the monitored set within

$$T_{identify,\ UTRA_TDD} = Max\left\{5000, T_{basic\ identify\ UTRA_TDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms$$

If the UE does not require transmit gap to perform inter-RAT UTRA TDD measurements, the UE shall be able to identify a new detectable inter-RAT UTRA TDD cell belonging to the monitored set within 5000 ms.

A cell shall be considered detectable when

- P-CCPCH Ec/Io $\geq$ -8 dB,
- DwPCH_Ec/Io $\geq$ -5 dB.

When L3 filtering is used an additional delay can be expected.

When measurement gaps are scheduled for UTRA TDD inter RAT measurements the UE physical layer shall be capable of reporting measurements to higher layers with measurement accuracy as specified in TS 36.133 [4] clause 9.3 with measurement period given by

$$T_{measurement\ UTRA_TDD} = Max\left\{T_{Measurement_Period\ UTRA_TDD}, T_{basic\ measurement\ UTRA_TDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms$$

If the UE does not need measurement gaps to perform UTRA TDD measurements, the measurement period for UTRA TDD measurements is 480 ms.

The UE shall be capable of performing UTRA TDD P-CCPCH RSCP measurements for $X_{basic\ measurementUTRA_TDD}$ inter-frequency cells per TDD frequency of the monitored set and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{Measurement_UTRA_TDD}$.

$X_{basic\ measurementUTRA_TDD} = 6$

$T_{Measurement_Period\ UTRA_TDD}$ = 480 ms is the period used for calculating the measurement period $T_{measurement_UTRA_TDD}$ for UTRA TDD P-CCPCH RSCP measurements.

$T_{basic_identify_UTRA_TDD}$ = 800 ms is the time period used in the inter RAT equation where the maximum allowed time for the UE to identify a new UTRA TDD cell is defined.

$T_{basic_measurement_UTRA_TDD}$ = 50 ms is the time period used in the equation for defining the measurement period for inter RAT P-CCPCH RSCP measurements.

$N_{freq}$ and $T_{inter1}$ are defined in TS 36.133 [4] section 8.1.2.1.1

Reported measurements in periodically triggered measurement reports shall meet the requirements in TS 36.133 [4] section 9.

Reported measurements in event triggered measurement reports shall meet the requirements in TS 36.133 [4] section 9.

The UE shall not send any event triggered measurement reports, as long as the reporting criteria is not fulfilled.

The measurement reporting delay is defined as the time between any event that will trigger a measurement report until the UE starts to transmit the measurement report over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify,\ UTRA_TDD}$ defined in TS 36.133 [4] Section 8.1.2.4.3.1.1 When L3 filtering is used an additional delay can be expected.

Reported measurements contained in event triggered periodic measurement reports shall meet the requirements in TS 36.133 [4] section 9.

The first report in event triggered periodic measurement reporting shall meet the requirements specified in TS 36.133 [4] section 8.1.2.4.3.1.4 Event Triggered Reporting.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.3 and A.8.7.1.

## 8.7.1.4 Test description

### 8.7.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.23.
2. The general test parameter settings are set up according to Table 8.7.1.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.7.1.4.3.
5. There is one E-UTRA TDD serving cell and one UTRA TDD cell specified in the test. Cell 1 (E-UTRA TDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.7.1.4.1-1: General test parameters for E-UTRA TDD to UTRA(1.28 Mcps TDD OPTION) cell search in fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Active cell | | Cell 1 | E-UTRA TDD cell |
| Neighbour cell | | Cell 2 | UTRA 1.28Mcps TDD Cell |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| Uplink-downlink configuration of cell 1 | | 1 | As specified in table 4.2.2 in TS 36.211 |
| Special subframe configuration of cell 1 | | 6 | As specified in table 4.2.1 in TS 36.211 |
| CP length of cell 1 | | normal | |
| Hysteresis | dB | 0 | |
| TimeToTrigger | dB | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| Time offset between cells | ms | 3 | Asynchronous cells<br>3ms or 92160*Ts |
| Ofn | dB | 0 | |
| Thresh | dBm | -87 | |
| T1 | s | 5 | |
| T2 | s | 10 | |

### 8.7.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event B1 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table's 8.7.1.5-1 and 8.7.1.5-2. Propagation conditions are set according to Annex B clause B.1.1 and B.2.2. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 8.7.1.5-1 and 8.7.1.5-2.
6. UE shall transmit a MeasurementReport message triggered by Event B1. If the overall delay measured from the beginning of time period T2 is less than 6402 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.
7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
8. SS shall change to set cell 2 cell parameter id =(current cell 2 cell parameter id +4) mod 16.
9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.7.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.7.1.4.3-1: Common Exception messages for E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-4<br>Table H.3.1-7 |

**Table 8.7.1.4.3-2: *ReportConfigInterRAT-B1-UTRA*: Additional E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.508 clause 4.6.6 Table 4.6.6-7B | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-UTRA(UTRA-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| b1-ThresholdUTRA CHOICE { | | | |
| utra-RSCP | 28 | UTRA-Thres is actual RSCP value in dBm<br>UTRA-Thres + 115 | UTRA-TDD |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0dB) | The actual value is IE value * 0.5 dB. | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.7.1.4.3-3: *MeasResults*: Additional E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 8.7.1.4.3-4: *MeasResultListUTRA*: Additional E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| tdd | PhysCellIdUTRA-TDD | INTEGER (0..127) | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-RSCP | | Set according to specific test INTEGER (-5..91) | |
| } | | | |
| } | | | |

## 8.7.1.5 Test requirement

Tables 8.7.1.4.1-1, 8.7.1.5-1 and 8.7.1.5-2 define the primary level settings including test tolerances for E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions in synchronous cells test.

**Table 8.7.1.5-1: Cell specific test parameters for cell search E-UTRA TDD to UTRA TDD test case (cell 1)**

| **Parameter** | **Unit** | **Cell 1** | |
|---|---|---|---|
| | | **T1** | **T2** |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | |
| PBCH_RA | dB | 0 | 0 |
| PBCH_RB | dB | | |
| PSS_RB | dB | | |
| SSS_RB | dB | | |
| PCFICH_PA | dB | | |
| PHICH_PA | dB | | |
| PHICH_PB | dB | | |
| PDCCH_PA | dB | | |
| PDCCH_PB | dB | | |
| PDSCH_PA | dB | | |
| PDSCH_PB | dB | | |
| OCNG_RA Note1 | dB | | |
| OCNG_RB Note1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 9 | 9 |
| $\hat{E}_s / N_{oc}$ | dB | 9 | 9 |
| $N_{oc}$ | dBm/15kHz | -98 | |
| RSRP | dBm/15kHz | -89 | -89 |
| SCH_RP | dBm/15kHz | -89 | -89 |
| Propagation Condition | | ETU70 | |

Note 1: OCNG shall be used such that cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

**Table 8.7.1.5-2: Cell specific test parameters for cell search E-UTRA TDD to UTRA TDD test case (cell 2)**

| Parameter | Unit | Cell 2 (UTRA) | | | |
|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPTS | |
| | | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number NOTE1 | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 |
| OCNS_Ec/Ior NOTE2 | dB | -3 | -3 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -inf | 5 | -inf | 5 |
| $I_{oc}$ | dBm/1.28 MHz | -80 | | | |
| PCCPCH RSCP | dBm | -inf | -78 | n.a. | n.a. |
| Propagation Condition | | Case 3 NOTE3 | | | |
| Note 1: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number.<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.<br>Note 3: Case 3 propagation conditions are defined in Annex B of 3GPP TS 25.102 | | | | | |

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event B1 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify,\ UTRA_TDD}$

$$T_{identify,\ UTRA_TDD} = Max\left\{5000, T_{basic\ identify\ UTRA_TDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms$$

$T_{basic_identify_UTRA_TDD}$ = 800 ms

$T_{Inter1}$ = 60 ms

$N_{Freq}$ = 1

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 6402 ms in this test case (note: this gives a total of 6400 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.7.2 E-UTRAN TDD - UTRAN TDD cell search when DRX is used under fading propagation conditions

Editor's note: It only includes 1.28 Mcps TDD option related requirements

### 8.7.2.1 Test purpose

The test cases are to verify that the UE makes correct reporting of an event in DRX. These tests will partly verify E-UTRA TDD to UTRA TDD cell search requirements when DRX is used in TS 36.133 [4] section 8.1.2.4.3.2.

### 8.7.2.2 Test applicability

This test applies to all types of release 8 E-UTRA TDD UEs that support release 9 and forward UTRA TDD. Applicability requires support for FGI bits 5, 15 and 22.

This test applies to all types of release 9 and forward E-UTRA TDD UEs that support release 9 and forward UTRA TDD and not supporting UTRA FDD. Applicability requires support for FGI bits 5, 15 and 22.

This test applies to all types of release 9 and forward E-UTRA TDD UEs that support release 9 and forward UTRA TDD. Applicability requires support for FGI bits 5, 15 and 39.

### 8.7.2.3 Minimum conformance requirements

When explicit neighbour list is provided and DRX is used the UE shall be able to identify a new detectable cell belonging to the neighbour cell list within $T_{identify_UTRA_TDD}$ as shown in table 8.7.2.3-1

**Table 8.7.2.3-1: Requirement to identify a newly detectable UTRA TDD cell**

| DRX cycle length (s) | $T_{identify_UTRA_TDD}$ (s) (DRX cycles) | |
|---|---|---|
| | Gap period = 40 ms | Gap period = 80 ms |
| ≤0.32 | Non DRX Requirements in TS 36.133 [4] section 8.1.2.4.3.1 are applicable | Non DRX Requirements in TS 36.133 [4] section 8.1.2.4.3.1 are applicable |
| 0.32<DRX-cycle≤0.512 | Note (20* Nfreq) | Note (25* Nfreq) |
| 0.512<DRX-cycle≤2.56 | Note (20* Nfreq) | Note (20* Nfreq) |
| Note: Time depends upon the DRX cycle in use | | |

A cell shall be considered detectable provided following conditions are fulfilled: A cell shall be considered detectable when

- P-CCPCH Ec/Io $\geq$ -8 dB,
- DwPCH_Ec/Io $\geq$ -5 dB.

When L3 filtering is used an additional delay can be expected.

The UE shall be capable of performing UTRA TDD P-CCPCH RSCP measurements of at least 6 UTRA cells per UTRA TDD carrier and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period defined in table 8.7.2.3-2.

**Table 8.7.2.3-2: Requirement to measure UTRA TDD cells**

| DRX cycle length (s) | $T_{measure_UTRA_TDD}$ (s) (DRX cycles) | |
|---|---|---|
| | Gap period = 40 ms | Gap period = 80 ms |
| ≤0.04 | Non DRX Requirements in TS 36.133 [4] section 8.1.2.4.3.1 are applicable | Non DRX Requirements in TS 36.133 [4] section 8.1.2.4.3.1 are applicable |
| 0.064 | 0.48*$N_{freq}$ (7.5*$N_{freq}$) | 0.8*$N_{freq}$ (12.5*$N_{freq}$) |
| 0.08 | 0.48*$N_{freq}$ (6*$N_{freq}$) | 0.8*$N_{freq}$ (10*$N_{freq}$) |
| 0.128 | 0.64*$N_{freq}$ (5*$N_{freq}$) | 0.8*$N_{freq}$ (6.25*$N_{freq}$) |
| 0.128<DRX-cycle≤2.56 | Note (5*$N_{freq}$) | Note (5*$N_{freq}$) |
| Note: Time depends upon the DRX cycle in use | | |

The measurement accuracy for all measured cells shall be as specified in the TS 36.133 [4] sub-clause 9.1.

Reported measurements in periodically triggered measurement reports shall meet the requirements in TS 36.133 [4] section 9.

Reported measurements in event triggered measurement reports shall meet the requirements in TS 36.133 [4] section 9.

The UE shall not send any event triggered measurement reports, as long as the reporting criteria is not fulfilled.

The measurement reporting delay is defined as the time between any event that will trigger a measurement report until the UE starts to transmit the measurement report over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify, UTRA_TDD}$ defined in TS 36.133 [4] Section 8.1.2.4.3.2 When L3 filtering is used an additional delay can be expected.

Reported measurements contained in event triggered periodic measurement reports shall meet the requirements TS 36.133 [4] in section 9.

The first report in event triggered periodic measurement reporting shall meet the requirements specified in TS 36.133 [4] section 8.1.2.4.3.2.2 Event Triggered Reporting.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.3.2 and A.8.7.2.

## 8.7.2.4 Test description

### 8.7.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.23.
2. The general test parameter settings are set up according to Table 8.7.2.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.7.2.4.3.
5. There is one E-UTRA TDD serving cell and one UTRA TDD cell specified in the test. Cell 1 (E-UTRA TDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 (UTRAN TDD cell) is to be searched. Gap pattern configuration # 0 as defined in TS 36.133 [4] table 8.1.2.1-1 is provided.

**Table 8.7.2.4.1-1: General test parameters for E-UTRAN TDD to UTRAN 1.28Mcps TDD cell search when DRX is used in fading propagation conditions**

| Parameter | Unit | Test 1 | Test 2 | Comment |
|---|---|---|---|---|
| | | Value | | |
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | | As specified in section A.1.2. Note that UE may only be allocated at *On Duration* |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | | As specified in section A.2.2. |
| Active cell | | Cell 1 | | E-UTRAN TDD cell |
| Neighbour cell | | Cell 2 | | UTRAN 1.28Mcps TDD cell |
| Gap Pattern Id | | 0 | | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| Uplink-downlink configuration | | 1 | | As specified in 3GPP TS 36.211 section 4.2 Table 4.2-2 |
| Special subframe configuration | | 6 | | As specified in table 4.2-1 in TS 36.211. The same configuration in both cells |
| PRACH configuration | | 4 | | As specified in table 5.7.1-3 in 3GPP TS 36.211 |
| CP length of cell 1 | | Normal | | |
| Ofn | dB | 0 | | |
| Thresh | dBm | -83 | | Absolute P-CCPCH RSCP threshold for event B1 |
| Hysteresis | dB | 0 | | |
| TimeToTrigger | s | 0 | | |
| Filter coefficient | | 0 | | L3 filtering is not used |
| Access Barring Information | - | Not Sent | | No additional delays in random access procedure. |
| DRX | | ON | | DRX related parameters are defined in Table 8.7.2.5-3 |
| Time offset between cells | ms | 3 | | Asynchronous cells<br>3ms or 92160*Ts |
| T1 | s | 5 | | |
| T2 | s | 8 | 30 | |

## 8.7.2.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event B1 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

In Test 1 UE needs to be provided at least once every 500ms with new Timing Advance Command MAC control element to restart the Time alignment timer to keep UE uplink time alignment. Furthermore UE is allocated with PUSCH resource at every DRX cycle. In Test 2 the uplink time alignment is not maintained and UE needs to use RACH to obtain UL allocation for measurement reporting.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table's 8.7.2.5-1 and 8.7.2.5-2. Propagation conditions are set according to Annex B clause B.1.1 and B.2.2. T1 starts.
3. The SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 8.7.2.5-1 and 8.7.2.5-2. T2 starts.
6. UE shall transmit a MeasurementReport message triggered by Event B1. If the overall delay measured from the beginning of time period T2 is less than 6442 ms for Test1 or less than 26882 ms for Test2 then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. SS shall set Cell 2 cell parameter id = (current Cell 2 cell parameter id+4) mod 16 for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

11. Repeat step 1-10 for each sub-test in Table 8.7.2.4.1-1 as appropriate.

### 8.7.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.7.2.4.3-1: Common Exception messages for E-UTRAN TDD to UTRAN 1.28Mcps TDD cell search when DRX is used in fading propagation**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-4<br>Table H.3.1-7<br>Table H.3.7-1<br>Table H.3.7-2<br>Table H.3.7-3 |

**Table 8.7.2.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN TDD to UTRAN 1.28Mcps TDD cell search when DRX is used in fading propagation**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.7.2.4.3-3: *ReportConfigInterRAT-B1-UTRA*: Additional E-UTRAN TDD to UTRAN 1.28Mcps TDD cell search when DRX is used in fading propagation**

| Derivation Path: 36.508 clause 4.6.6 Table 4.6.6-7B | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-UTRA(UTRA-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| b1-ThresholdUTRA CHOICE { | | | |
| thresholdUTRA-RSCP | 32 | UTRA-Thres + 115 UTRA-Thres is actual RSCP value in dBm | UTRA-TDD |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 | The actual value is IE value * 0.5 dB INTEGER (0..30) | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.7.2.4.3-4: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN TDD to UTRAN 1.28Mcps TDD cell search when DRX is used in fading propagation**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20: SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | Channel-bandwidth-dependent parameter | |
| sr-ConfigIndex | 2 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.7.2.4.3-5: *MeasResults*: Additional E-UTRAN TDD to UTRAN 1.28Mcps TDD cell search when DRX is used in fading propagation**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 8.7.2.4.3-6: *MeasResultListUTRA*: Additional E-UTRAN TDD to UTRAN 1.28Mcps TDD cell search when DRX is used in fading propagation**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| tdd | PhysCellIdUTRA-TDD | PhysCellIdUTRA-TDD INTEGER (0..127) | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-RSCP | | Set according to specific test INTEGER (-5..91) | |
| } | | | |
| } | | | |

**Table 8.7.2.4.3-7: *PRACH-Config-DEFAULT*: Additional E-UTRAN TDD to UTRAN 1.28Mcps TDD cell search when DRX is used in fading propagation**

| Derivation Path: TS 36.508 [7] clause 7.3.2 Table 7.3.2-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PRACH-Config-DEFAULT ::= SEQUENCE { | | | |
| prach-ConfigIndex | 4 | | |
| } | | | |

## 8.7.2.5 Test requirement

The common test parameters are given in Tables 8.7.2.4.1-1, 8.7.2.5-1 and 8.7.2.5-2. DRX configuration for Test1 and Test2 are given in Table 8.7.2.5-3 and time alignment timer and scheduling request related parameters in Table 8.7.2.5-4.

**Table 8.7.2.5-1: Cell specific test parameters for E-UTRAN TDD to UTRAN 1.28Mcps TDD cell search when DRX is used in fading propagation conditions (cell 1)**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | |
| PBCH_RA | dB | 0 | 0 |
| PBCH_RB | dB | | |
| PSS_RB | dB | | |
| SSS_RB | dB | | |
| PCFICH_PA | dB | | |
| PHICH_PA | dB | | |
| PHICH_PB | dB | | |
| PDCCH_PA | dB | | |
| PDCCH_PB | dB | | |
| PDSCH_PA | dB | | |
| PDSCH_PB | dB | | |
| OCNG_RA[Note1] | dB | | |
| OCNG_RB[Note1] | dB | | |

| | | | |
|---|---|---|---|
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| $N_{oc}$ Note 2 | dBm/15kHz | -98 | |
| RSRP Note 3 | dBm/15kHz | -94 | -94 |
| SCH_RP Note 3 | dBm/15kHz | -94 | -94 |
| Propagation Condition | | ETU70 | |
| Note 1: OCNG shall be used such that cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.<br>Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | |

**Table 8.7.2.5-2: Cell specific test parameters for E-UTRAN TDD to UTRAN 1.28Mcps TDD cell search when DRX is used in fading propagation conditions (cell 2)**

| Parameter | Unit | Cell 2 (UTRA) | | | |
|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPTS | |
| | | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number NOTE1 | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 |
| OCNS_Ec/Ior NOTE2 | dB | -3 | -3 | | |
| $\hat{I}_{or} / I_{oc}$ | dB | -inf | 9 | -inf | 9 |
| $I_{oc}$ | dBm/1.28 MHz | -80.40 | | | |
| PCCPCH RSCP | dBm | -inf | -74.40 | n.a. | n.a. |
| Propagation Condition | | Case 3 NOTE3 | | | |
| Note 1: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number.<br>Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to Ior.<br>Note 3: Case 3 propagation conditions are defined in Annex B of 3GPP TS 25.102 | | | | | |

**Table 8.7.2.5-3: drx-Configuration to be used in E-UTRAN TDD to UTRAN 1.28Mcps TDD cell search when DRX is used in fading propagation conditions**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf40 | sf1280 | |
| shortDRX | disable | disable | |

**Table 8.7.2.5-4: *TimeAlignmentTimer* and *sr-ConfigIndex* -Configuration to be used in E-UTRAN TDD to UTRAN 1.28Mcps TDD cell search when DRX is used in fading propagation conditions**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | For further information see section 6.3.2 in 3GPP TS 36.331. |
| sr-ConfigIndex | 2 | 2 | For further information see section 6.3.2 in 3GPP TS 36.331 and 10.1 in 3GPP TS 36.213. |

In Test 1 when DRX cycle length = 40 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment when the UE send one Event B1 triggered the measurement report on PUSCH.

In Test 2 when DRX cycle length = 1280 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment when the UE starts to send preambles on the PRACH for scheduling request (SR) to obtain allocation to send the measurement report on PUSCH.

For both tests:

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event B1 measurement report

The overall delay measured when DRX cycle length is 40 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify,\ UTRA_TDD}$=

$$T_{identify,\ UTRA_TDD} = Max\left\{5000, T_{basic\ identify\ UTRA_TDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms$$

Where:

$T_{basic_identify_UTRA_TDD}$ = 800 ms. It is the time period used in the inter RAT equation where the maximum allowed time for the UE to identify a new UTRA TDD cell is defined.

$T_{Inter1}$ = 60 ms.

$N_{freq}$ = 1. It is defined in TS 36.133 [4] clause 8.1.2.1.1.

TTI insertion uncertainty = 2 ms

DRX cycle length = 40 ms

The overall delay measured when DRX cycle length is 40 ms shall be less than a total of 6442 ms.

The overall delay measured when DRX cycle length is 1280 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_UTRA_TDD}$

$T_{identify_UTRA_TDD}$ = 25600 ms. When DRX cycle length is 1280 ms the $T_{identify_UTRA_TDD}$ is 20 x 1280 ms.

TTI insertion uncertainty = 2 ms

DRX cycle length = 1280 ms

The overall delay measured when DRX cycle length is 1280 ms shall be less than a total of 26882 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.7.3 E-UTRAN TDD - UTRAN TDD SON ANR cell search reporting under AWGN propagation conditions

### 8.7.3.1 Test purpose

The purpose of this test is to verify that the UE makes correct reporting of the strongest UTRAN TDD cell for SON automatic neighbour relations. This test will partly verify the E-UTRAN TDD - UTRAN TDD cell search requirements for identification of a new UTRA TDD cell for SON given in TS 36.133 [4] section 8.1.2.4.13.1.

### 8.7.3.2 Test applicability

This test applies to all types of release 8 E-UTRA TDD UEs that support release 9 and forward UTRA TDD. Applicability requires support for FGI bits 19 and 22.

This test applies to all types of release 9 and forward E-UTRA TDD UEs that support release 9 and forward UTRA TDD and not supporting UTRA FDD. Applicability requires support for FGI bits 22 and 37.

This test applies to all types of release 9 and forward E-UTRA TDD UEs that support release 9 and forward UTRA TDD. Applicability requires support for FGI bits 37 and 39.

### 8.7.3.3 Minimum conformance requirements

When no DRX is used the UE shall be able to identify a new cell within:

$$T_{identify,\ UTRA_TDD} = T_{basic_identify_UTRA_TDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq} \quad ms$$

$T_{basic_identify_UTRA_TDD}$ = 800 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new UTRA TDD cell is defined.

A cell shall be considered identifiable following conditions are fulfilled:

- P-CCPCH Ec/Io $\geq$ -8 dB,
- DwPCH_Ec/Io $\geq$ -5 dB.

When L3 filtering is used an additional delay can be expected.

If the UE is unable to identify the UTRA TDD cell for SON within $8*T_{identify,\ UTRA_TDD}$ ms, the UE may stop searching UTRA TDD cells for SON.

The UE shall not report the physical cell identity of an identifiable cell for SON as long as the reporting criteria are not fulfilled.

The reporting delay is defined as the time between the identification of the strongest cell for SON until the UE starts to transmit its physical cell identity over the Uu interface. This requirement assumes that the reporting of the physical cell identity is not delayed by other RRC signalling on the DCCH. This reporting delay excludes a delay uncertainty resulted when inserting the physical cell identity of the strongest cell for SON to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. This reporting delay excludes any delay caused by unavailability of UL resources for UE sending the physical cell identity of the strongest cell for SON.

The reporting delay of the physical cell identity of the strongest cell for SON without L3 filtering shall be less than $T_{identify,\ UTRA_TDD}$ defined in section 8.1.2.4.13.1.1 in TS 36.133 [4] and in section 8.1.2.4.13.1.2 in TS 36.133 [4] for non DRX and DRX cases respectively. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.13 and A.8.7.3.

## 8.7.3.4 Test description

### 8.7.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.

2. The general test parameter settings are set up according to Table 8.7.3.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.7.3.4.3.

5. There is one E-UTRA TDD serving cell and one UTRA TDD cell specified in the test. Cell 1 (E-UTRA TDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.7.3.4.1-1: General test parameters for E-UTRAN TDD-UTRAN TDD cell search reporting for SON ANR in AWGN propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Gap Pattern Id | | 1 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on UTRA RF channel number 1. |
| CP length | | Normal | Applicable to cell 1 |
| E-UTRA RF Channel Number | | 1 | One E-UTRA TDD carrier frequency is used. |
| UTRA RF Channel Number | | 1 | One UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Uplink-downlink configuration of cell 1 | | 1 | As specified in table 4.2.2 in TS 36.211 [9] |
| Special subframe configuration of cell 1 | | 6 | As specified in table 4.2.1 in TS 36.211 [9] |
| Inter-RAT (UTRA TDD) measurement quantity | | P-CCPCH RSCP | |
| Filter coefficient | | 0 | L3 filtering is not used. |
| DRX | | OFF | |
| Monitored UTRA TDD cell list size | | None | No explicit neighbour list is provided to the UE |
| T1 | s | >5 | During T1, cell 2 shall be powered off, and during the off time the scrambling code shall be changed, The intention is to ensure that cell 2 has not been detected by the UE prior to the start of period T2. |
| T2 | s | 14 | |

### 8.7.3.4.2 Test procedure

The test consists of one active E-UTRAN cell and one neighbour UTRAN cell. In the measurement control information it is indicated to the UE that periodical reporting with the purpose 'reportStrongestCellsForSON' is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Tables 8.7.3.5-1 and 8.7.3.5-2. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Tables 8.7.3.5-1 and 8.7.3.5-2.

6. The UE shall transmit a MeasurementReport message containing the cell parameter id of cell 2. If the overall delays measured from the beginning of time period T2 is less than 12802 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 7) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. The SS shall set Cell 2 cell parameter id = (current Cell 2 cell parameter id+4) mod 16 for the next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.7.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.7.3.4.3-1: Common Exception messages for E-UTRAN TDD-UTRAN TDD cell search reporting for SON ANR in AWGN propagation conditions**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7 |

**Table 8.7.3.4.3-2: *MeasConfig-DEFAULT*: Additional E-UTRAN TDD-UTRAN TDD cell search reporting for SON ANR in AWGN propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | f1 is the frequency of the serving cell (E-UTRA Cell) | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | | |
| measObject CHOICE { | | | |
| MeasObjectUTRA | MeasObjectUTRA-GENERIC(f2) | f2 is the frequency of the neighbouring cell (UTRA Cell) | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| ReportConfigToAddMod SEQUENCE { | | | |
| reportConfigId | idReportConfig | | |
| reportConfig CHOICE { | | | |
| reportConfigInterRAT | ReportConfigInterRAT-SON-UTRA | | |
| } | | | |
| } | | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | UTRAN |
| measGapConfig | MeasGapConfig-GP2 | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 8.7.3.4.3-3: *ReportConfigInterRAT-SON-UTRA*: Additional E-UTRAN TDD-UTRAN TDD cell search reporting for SON ANR in AWGN propagation conditions**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-SON-UTRA::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose ENUMERATED { | reportStrongestCellsForSON | | |
| } | | | |
| } | | | |
| } | | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

**Table 8.7.3.4.3-4: *MeasResults*: Additional E-UTRAN TDD-UTRAN TDD cell search reporting for SON ANR in AWGN propagation conditions**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 8.7.3.4.3-5: *MeasResultListUTRA*: Additional E-UTRAN TDD-UTRAN TDD cell search reporting for SON ANR in AWGN propagation conditions**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultUTRA | | | |
| MeasResultUTRA ::= SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| tdd | PhysCellIdUTRA-TDD | The cell parameter id, as defined in TS 25.331 | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-RSCP | | Set according to specific test | |
| } | | | |
| } | | | |

## 8.7.3.5 Test requirement

Tables 8.7.3.4.1-1, 8.7.3.5-1 and 8.7.3.5-2 define the primary level settings including test tolerances for UTRAN TDD cell search for SON ANR under AWGN propagation conditions test.

**Table 8.7.3.5-1: Cell specific test parameters for E-UTRAN TDD (cell # 1) for UTRAN TDD cell search for SON ANR under AWGN propagation conditions**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.7.3.5-2: Cell specific test parameters for UTRAN TDD (cell # 2) for UTRAN TDD cell search for SON ANR under AWGN propagation conditions**

| Parameter | Unit | Cell 2 | | | |
|---|---|---|---|---|---|
| | | T1 | | T2 | |
| UTRA RF Channel number Note2 | | Channel 2 | | | |
| DL timeslot number | | 0 | DwPTS | 0 | DwPTS |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | |
| DwPCH_Ec/Ior | dB | | 0 | | 0 |
| OCNS_Ec/Ior | dB | -3 | | -3 | |
| Îor/Ioc | dB | -Infinity | | 5 | |
| PCCPCH RSCP Note1 | dBm | -Infinity | n.a. | -73 | n.a. |
| Io Note1 | dBm/1.28MHz | -Infinity | | -70.88 | |
| Ioc | dBm/1.28MHz | -75 | | | |
| Propagation condition | | AWGN | | | |

Note 1: PCCPCH RSCP and Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves.

Note 2: In the case of multi-frequency network of 1.28 Mcps TDD, the UTRA RF Channel Number can be set for the primary frequency in this test.

The UE shall send the first measurement report containing the physical cell identity of cell 2, with a measurement reporting delay less than 12800 ms from the beginning of time period T2.

When no DRX is used the UE shall be able to identify a new cell within:

$$T_{identify,\ UTRA_TDD} = T_{basic_identify_UTRA_TDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq} \quad ms$$

$T_{basic_identify_UTRA_TDD}$ = 800 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new UTRA TDD cell is defined.

$T_{Inter1}$ = 30 ms. TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 12802 ms in this test case (note: this gives a total of 12800 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The rate of correct measurement reports observed with this delay during repeated tests shall be at least 90% with a confidence level of 95%.

## 8.7.4 E-UTRAN TDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions

### 8.7.4.1 Test purpose

The purpose of this test is to verify that the UE makes correct enhanced reporting of UTRAN TDD cells. This test will partly verify the Enhanced UTRA TDD cell identification requirements in TS 36.133 [4] section 8.1.2.4.3.1.1a under AWGN propagation conditions.

### 8.7.4.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward that support UTRA TDD and not supporting UTRA FDD. Applicability requires support for FGI bits 15 and 22.

This test applies to all types of E-UTRA TDD UE release 9 and forward that support UTRA TDD. Applicability requires support for FGI bits 15 and 39.

### 8.7.4.3 Minimum conformance requirements

When explicit neighbour list is provided and no DRX is used or when DRX cycle length ≤ 40 ms the UE shall be able to identify a new detectable cell belonging to the monitored set within $T_{identify,\ enhanced_UTRA_TDD}$:

$$T_{identify,\ enhanced_UTRA_TDD} = (T_{basic_identify_enhanced_UTRA_TDD} \cdot \frac{480}{T_{inter1}} + 480) \cdot N_{Freq} \quad ms$$

If the UE does not require transmit gap to perform inter-RAT UTRA TDD measurements, the UE shall be able to identify a new detectable inter-RAT UTRA TDD cell belonging to the monitored set within 500 ms.

A cell shall be considered detectable when:

- P-CCPCH_Ec/Io ≥ -6 dB,
- DwPCH_Ec/Io ≥ -1 dB

When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.3.1.1a and A.8.7.4.

### 8.7.4.4 Test description

#### 8.7.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.

2. The general test parameter settings are set up according to Table 8.7.4.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.7.4.4.3.

5. There is one E-UTRA TDD serving cell and one UTRA TDD cell specified in the test. Cell 1 (E-UTRA TDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 (UTRAN TDD cell) is to be searched. Gap pattern configuration # 0 as defined in TS 36.133 [4] table 8.1.2.1-1 is provided.

**Table 8.7.4.4.1-1: General test parameters for E-UTRAN TDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2. |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on UTRA RF channel number 1. |
| Time offset between cells | ms | 3 | Asynchronous cells |
| E-UTRA RF Channel Number | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| CP length | | Normal | Applicable to cell 1 |
| Uplink-downlink configuration of cell 1 | | 1 | As specified in table 4.2.2 in TS 36.211 [9] |
| Special subframe configuration of cell 1 | | 6 | As specified in table 4.2.1 in TS 36.211 [9] |
| UTRA RF Channel Number | | 1 | One UTRA TDD carrier frequency is used. |
| Inter-RAT (UTRA TDD) measurement quantity | | P-CCPCH RSCP | |
| Thresh | dBm | -83 | Absolute P-CCPCH RSCP threshold for event B1 |
| Hysteresis | dB | 0 | |
| Time To Trigger | ms | 0 | |
| Monitored UTRA TDD cell list size | | 12 | UTRA cells on UTRA RF channel 1 provided in the cell list |
| T1 | s | 5 | |
| T2 | s | 2 | |

### 8.7.4.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event B1 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Tables 8.7.4.5-1 and 8.7.4.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. The SS shall transmit an *RRCConnectionReconfiguration* with event B1 configured message.

4. The UE shall transmit *RRCConnectionReconfigurationComplete* message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Tables 8.7.4.5-1 and 8.7.4.5-2. T2 starts.

6. UE shall transmit a *MeasurementReport* message triggered by Event B1. If the overall delay measured from the beginning of time period T2 is less than 1122 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receives the *MeasurementReport* message in step 6) or when T2 expires, the SS shall transmit *RRCConnectionRelease* message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. SS shall set Cell 2 cell parameter id = (current Cell 2 cell parameter id+4) mod 16 for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.7.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.7.4.4.3-1: Common Exception messages for E-UTRAN TDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-4<br>Table H.3.1-7 |

**Table 8.7.4.4.3-2: *ReportConfigInterRAT-B1-UTRA*: Additional E-UTRAN TDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6 Table 4.6.6-7B | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-UTRA(UTRA-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| b1-ThresholdUTRA CHOICE { | | | |
| utra-RSCP | 32 | UTRA-Thres is actual RSCP value in dBm<br>UTRA-Thres + 115 | UTRA-TDD |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0dB) | The actual value is IE value * 0.5 dB. | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.7.4.4.3-3: *MeasResults*: Additional E-UTRAN TDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 8.7.4.4.3-4: *MeasResultListUTRA*: E-UTRAN TDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| tdd | PhysCellIdUTRA-TDD | INTEGER (0..127) | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-RSCP | | Set according to specific test<br>INTEGER (-5..91) | |
| } | | | |
| } | | | |

## 8.7.4.5 Test requirement

Tables 8.7.4.4.1-1, 8.7.4.5-1 and 8.7.4.5-2 define the primary level settings including test tolerances for E-UTRAN TDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions test.

**Table 8.7.4.5-1: Cell specific test parameters for cell 1 in E-UTRAN TDD - UTRAN TDD enhanced cell identification test under AWGN propagation conditions**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.7.4.5-2: Cell specific test parameters for cell 2 in E-UTRAN TDD - UTRAN TDD enhanced cell identification test under AWGN propagation conditions**

| Parameter | Unit | Cell 2 (UTRA TDD) | | | |
|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPTS | |
| | | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number[Note1] | | 1 | | | |
| P-CCPCH_Ec/Ior | dB | -4.77 | -4.77 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 |
| OCNS_Ec/Ior[Note2] | dB | -1.76 | -1.76 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | - infinity | 8 | - infinity | 8 |
| $I_{oc}$ | dBm/1.28 MHz | -79.4 | | | |
| P-CCPCH RSCP [Note3] | dBm | - infinity | -76.17 | n.a. | n.a. |
| P-CCPCH_Ec/Io [Note3] | dB | - infinity | -5.41 | n.a. | n.a. |
| DwPCH_Ec/Io [Note3] | dB | n.a. | n.a. | - infinity | -0.64 |
| Propagation Condition | | AWGN | | | |

Note 1: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number.
Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.
Note 3: P-CCPCH RSRP, PCCPCH_Ec/Io and DwPCH_Ec/Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event B1 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify,\ enhanced_UTRA_TDD}$

$$T_{identify,\ enhanced_UTRA_TDD} = (T_{basic_identify_enhanced_UTRA_TDD} \cdot \frac{480}{T_{inter1}} + 480) \cdot N_{Freq} \quad ms$$

$T_{basic_identify_enhanced_UTRA_TDD}$ = 80 ms. It is the time period used in the inter RAT equation where the maximum allowed time for the UE to identify a new UTRA TDD cell is defined.

$T_{Inter1}$ = 60 ms

$N_{Freq}$ = 1. It is defined in TS 36.133 [4] clause 8.1.2.1.1.

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 1122 ms in this test case (note: this gives a total of 1120 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 8.8 E-UTRAN FDD - GSM measurements

## 8.8.1 E-UTRAN FDD - GSM event triggered reporting in AWGN

### 8.8.1.1 Test purpose

To verify the UE's ability to make a correct reporting of an event when doing inter-RAT (GSM) measurements under AWGN condition within the E-UTRA FDD - GSM cell search requirements.

### 8.8.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support GSM. Applicability requires support for FGI bits 15 and 23.

### 8.8.1.3 Minimum conformance requirements

Measurements on GSM cells can be requested with BSIC verified.

In RRC_CONNECTED state when a supported measurement gap pattern sequence according to Table 8.1.2.1-1 in TS 36.133 [4] is configured by E-UTRAN the UE shall continuously measure GSM cells, search for new GSM cells given in the monitored set and re-confirm the BSIC for already detected cells. During DRX periods the UE may use other periods of time outside the specified measurement gap patterns.

This measurement shall be based on measurement gaps allocated for GSM carrier RSSI measurement as described in section 8.1.2.1 in TS 36.133 [4] A UE supporting GSM measurements shall measure minimum number of 10 GSM carrier RSSI measurement samples ($N_{GSM\ carrier\ RSSI}$) per measurement gap. In RRC_CONNECTED state the measurement period, $T_{Measurement\ Period,\ GSM}$, for the GSM carrier RSSI measurement is $N_{freq}$ * 480 ms. The parameter $N_{freq}$ is defined in clause 8.1.2.1.1 [4] as $N_{freq} = N_{freq,\ E\text{-}UTRA} + N_{freq,\ UTRA} + M_{gsm}$

Where:

$N_{freq,\ E\text{-}UTRA}$ is the number of E-UTRA carriers being monitored

$N_{freq,\ UTRA}$ is the number of UTRA carriers being monitored

$M_{GSM}$ is an integer which is a function of the number of GSM carriers on which measurements are being performed. $M_{GSM}$ is equal to 0 if no GSM carrier is being monitored. For a Measurement Gap Repetition Period (MGRP) of 40 ms, $M_{GSM}$ is equal to 1 if cells on up to 32 GSM carriers are being measured. For a MGRP of 80 ms, $M_{GSM}$ is equal to ceil ($N_{carriers,\ GSM}$ /20) where $N_{carriers,\ GSM}$ is the number of GSM carriers on which cells are being measured

The UE shall meet the measurement accuracy requirements stated for RXLEV in TS 45.008 [15], when the given measurement time allows the UE to take at least 3 GSM carrier RSSI samples per GSM carrier in the monitored set during the measurement period.

In case the UE is not able to acquire the required number of samples per GSM carrier during one measurement period, the UE shall measure as many GSM carriers as possible during that measurement period using at least 3 samples per GSM carrier. The GSM carriers that were not measured during that measurement period shall be measured in the following measurement periods.

Measurements on a GSM cell can be requested with BSIC verified. The UE shall be able to report the GSM cells with BSIC verified for those cells where the verification of BSIC has been successful.

The procedure for BSIC verification on a GSM cell can be divided into the following two tasks:

- **Initial BSIC identification:** Includes searching for the BSIC and decoding the BSIC for the first time when there is no knowledge about the relative timing between the E-UTRAN FDD and GSM cells. The UE shall trigger the initial BSIC identification within the available measurement gap pattern sequence. The requirements for BSIC re-confirmation can be found in section 8.1.2.4.5.1.2.1 of TS 36.133 [4].

- **BSIC re-confirmation:** Tracking and decoding the BSIC of a GSM cell after initial BSIC identification is performed. The UE shall trigger the BSIC re-confirmation within the available measurement gap pattern. The requirements for BSIC re-confirmation can be found in section 8.1.2.4.5.1.2.2 of TS 36.133 [4].

If the network requests measurements on a GSM cell, the UE shall behave as follows:

- The UE shall perform GSM carrier RSSI measurements according to section 8.1.2.4.5.1 in TS 36.133 [4] when a measurement gap pattern sequence is activated.
- The UE shall perform measurement reporting as defined in TS 36.331 [5].
- The UE shall perform BSIC identification. The UE shall use the most recently available GSM carrier RSSI measurement results for arranging GSM cells in signal strength order for performing BSIC identification.
- The UE shall perform BSIC re-confirmation on all the GSM cells that have been successfully identified.
- The UE shall perform all configured event evaluation for event-triggered reporting after the BSIC has been verified for a GSM cell. The UE shall use the most recently available GSM carrier RSSI measurement results in event evaluation and event-triggered reporting.
- Event-triggered and periodic reports shall be triggered according to TS 36.331 [5].

The BSIC of a GSM cell is considered to be “verified” if the UE has decoded the SCH of the BCCH carrier and identified the BSIC at least one time (initial BSIC identification). Once a GSM cell has been identified the BSIC shall be re-confirmed at least once every $8*T_{re\text{-}confirm,GSM}$ seconds. Otherwise the BSIC of the GSM cell is considered as "non-verified". If a measurement gap pattern sequence is deactivated by the network after BSIC has been identified or verified, the UE shall consider the BSIC as non-verified.

$T_{identify,GSM}$ indicates the maximum time allowed for the UE to decode the unknown BSIC of the GSM cell in one GSM BCCH carrier in the initial BSIC identification procedure.

$T_{re\text{-}confirm,GSM}$ indicates the maximum time allowed for the re-confirmation of the BSIC of one GSM cell in the BSIC re-confirmation procedure.

The UE shall be able to decode a BSIC within a measurement gap when the time difference between the middle of the received GSM synchronisation burst at the UE and the middle of the effective measurement gap is within the limits specified in table 8.1.2.4.5.1.2-1 of TS 36.133 [4].

The UE shall be able to perform BSIC verification at levels down to the reference sensitivity level or reference interference levels as specified in TS 45.005 [16].

This measurement shall be based on the measurement gaps used for Initial BSIC identification as described in section 8.1.2.4.5.1.2 of TS 36.133 [4].

The UE shall continuously attempt to decode the BSIC of SCH on the BCCH carrier of the 8 strongest BCCH carriers of the GSM cells indicated in the Inter-RAT cell info list. The UE shall give priority for BSIC decoding attempts in decreasing signal strength order to BCCH carriers with unknown BSIC. The strongest BCCH carrier is defined as the BCCH carrier having the highest measured GSM carrier RSSI value.

If the BSIC of the GSM BCCH carrier has been successfully decoded the UE shall immediately continue BSIC identification with the next GSM BCCH carrier, in signal strength order, with unknown BSIC. The GSM cell for which the BSIC has been successfully identified shall be moved to the BSIC re-confirmation procedure.

If the UE has not successfully decoded the BSIC of the GSM BCCH carrier within $T_{identify,GSM}$ ms, the UE shall abort the BSIC identification attempts for that GSM BCCH carrier. The UE shall continue to try to perform BSIC identification of the next GSM BCCH carrier in signal strength order. The GSM BCCH carrier for which the BSIC identification failed shall not be re-considered for BSIC identification until BSIC identification attempts have been made for all the rest of the 8 strongest GSM BCCH carriers in the monitored set with unknown BSIC.

$T_{identify,GSM}$ values are given for a set of reference gap patterns in table 8.1.2.4.5.1.2.1-1 of TS 36.133 [4]. The requirements in the table represent the time required to guarantee at least two attempts to decode the BSIC for one GSM BCCH carrier.

Reported measurements in event triggered measurement reports shall meet the requirements in section TS 36.331 [5].

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH.

The event triggered reporting delay requirement is valid when the UE for each GSM carrier in the monitored set can take the required number of samples during the measurement period $T_{Measurement\ Period,\ GSM}$ (see section 8.1.2.4.5.1 of TS 36.133 [4]).

The event triggered measurement reporting delay for a GSM cell with verified BSIC , measured without L3 filtering shall be less than $2*T_{Measurement\ Period,\ GSM}$, where $T_{Measurement\ Period,\ GSM}$ is defined in section 8.1.2.4.5.1. of TS 36.133 [4] When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.5 and A.8.8.1

## 8.8.1.4 Test description

### 8.8.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.14.

2. The general test parameter settings are set up according to Table 8.8.1.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.8.1.4.3.

5. There is one E-UTRA FDD carrier and one GSM cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.8.1.4.1-1: General test parameters for E-UTRAN FDD-GSM event triggered reporting in AWGN**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1. |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on Absolute RF Channel Number 1 (GSM cell) |
| CP length | | Normal | Applicable to cell 1 |
| E-UTRA RF Channel Number | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Inter-RAT (GSM) measurement quantity | | GSM Carrier RSSI | |
| b1-Threshold-GERAN | dBm | -80 | GSM Carrier RSSI threshold for event B1. |
| Hysteresis | dB | 0 | |
| Time To Trigger | ms | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used. |
| DRX | | OFF | |
| Monitored GSM cell list size | | 6 GSM neighbours including ARFCN 1 | List of GSM cells provided before T2 starts. |
| T1 | s | 5 | |
| T2 | s | 5 | |

#### 8.8.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event B1 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.8.1.5-1. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 8.8.1.5-1 and 8.8.1.5-2.

6. UE shall transmit a MeasurementReport message triggered by Event B1. If the measurement reporting delay measured from the beginning of time period T2 is less than 3122 ms then the number of successful tests is increased by one. If the UE fails to report the event within the measurement reporting delay requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. The SS shall set a different BSIC on Cell 2, as the previous timing information of Cell 2 is invalid in the UE for the next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 8.8.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.8.1.4.3-1: Common Exception messages for E-UTRAN FDD - GSM Event triggered reporting in AWGN**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br><br>Table H.3.1-7<br>Table H.3.1-10 |

**Table 8.8.1.4.3-2: *SystemInformationBlockType7*: Additional E-UTRAN FDD - GSM Event triggered reporting in AWGN**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-3 SystemInformationBlockType7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| commonInfo SEQUENCE { | | | |
| p-MaxGERAN | 33 (33 dBm) | | GSM 400 & GSM 900 & GSM 850 & GSM 700 |
| | 30 (30 dBm) | | DCS 1800 & PCS 1900 |
| } | | | |

**Table 8.8.1.4.3-3: *ReportConfigInterRAT-B1-GERAN*: Additional E-UTRAN FDD - GSM Event triggered reporting in AWGN**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7A ReportConfigInterRAT-B1-GERAN(GERAN-Thres) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-GERAN(GERAN-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventIdeventide CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| B1-ThresholdGERAN CHOICE { | | | |
| thresholdGERAN | 30 (-80 dBm) | -80 is actual value in dBm (30 - 110 dBm) | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.8.1.4.3-4: *MeasResults*: Additional E-UTRAN FDD - GSM Event triggered reporting in AWGN**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListGERAN | MeasResultListGERAN | | |
| } | | | |
| } | | | |

**Table 8.6.1.4.3-5: *MeasResultListGERAN*: Additional E-UTRAN FDD - GSM Event triggered reporting in AWGN**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListGERAN ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| MeasResultGERAN SEQUENCE { | | | |
| carrierFreq | CarrierFreqGERAN | Contains the carrier frequency of the target GERAN cell | |
| physCellId | PhysCellIdGERAN | Contains the Base Station Identity Code (BSIC) | |
| } | | | |
| measResult SEQUENCE { | | | |
| rssi | | Set according to specific test | |
| } | | | |
| } | | | |

## 8.8.1.5 Test requirement

Tables 8.8.1.4.1-1, 8.8.1.5-1 and 8.8.1.5-2 define the primary level settings including test tolerances for E-UTRAN FDD-GSM event triggered reporting under AWGN conditions.

**Table 8.8.1.5-1: Cell specific test parameters for E-UTRAN FDD (Cell # 1) for event triggered reporting of GSM cell in AWGN**

| **Parameter** | **Unit** | **Cell 1** | |
|---|---|---|---|
| | | **T1** | **T2** |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $\hat{E}_s / N_{oc}$ | | 4 | 4 |
| $N_{oc}$ | dBm/15 kHz | -98 | |
| RSRP | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2. | | | |

**Table 8.8.1.5-2: Cell specific test parameters for GSM (Cell # 2) for event triggered reporting of GSM cell in AWGN**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| Absolute RF Channel Number | | ARFNC 1 | |
| RXLEV | dBm | -Infinity | -75 |
| GSM BSIC | | N/A | Valid |

The overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE sends one Event B1 triggered measurement report including BSIC of Cell 2.

The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delay measured shall be less than a total of 3122 ms in this test case. (The delay for GSM cell identification with BSIC verified is equal to 3120 ms, which is the sum of the event triggered measurement reporting delay and the initial BSIC identification delay plus the TTI insertion uncertainty of 2ms).

The event triggered measurement reporting delay = $2*T_{Measurement\ Period,\ GSM}$ = 2* 480 ms = 960 ms.

Initial BSIC identification delay = 2160 ms.

TTI insertion uncertainty = 2 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.8.2 E-UTRAN FDD - GSM event triggered reporting when DRX is used in AWGN

### 8.8.2.1 Test purpose

To verify the UE's ability to make a correct reporting of an event in DRX when doing inter-RAT (GSM) measurements under AWGN condition within the E-UTRA FDD - GSM cell search requirements.

### 8.8.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support GSM. Applicability requires support for FGI bits 5, 15 and 23.

### 8.8.2.3 Minimum conformance requirements

Measurements on GSM cells can be requested with BSIC verified.

In RRC_CONNECTED state when a supported measurement gap pattern sequence according to Table 8.1.2.1-1 in TS 36.133 [4] is configured by E-UTRAN the UE shall continuously measure GSM cells, search for new GSM cells given in the monitored set and re-confirm the BSIC for already detected cells. During DRX periods the UE may use other periods of time outside the specified measurement gap patterns. The UE is not required to make measurements of GSM cells during DRX periods if a measurement gap pattern has not been configured.

This measurement shall be based on measurement gaps allocated for GSM carrier RSSI measurement as described in section 8.1.2.1 in TS 36.133 [4]. A UE supporting GSM measurements shall measure minimum number of 10 GSM carrier RSSI measurement samples ($N_{GSM\ carrier\ RSSI}$) per DRX cycle. In RRC_CONNECTED state the measurement period, $T_{Measurement\ Period,\ GSM}$, for the GSM carrier RSSI measurement is shown in Table 8.1.2.4.5.2.1-1 in TS 36.133 [4]. The parameter $N_{freq}$ is defined in clause 8.1.2.1.1 [4] as $N_{freq} = N_{freq,\ E\text{-}UTRA} + N_{freq,\ UTRA} + M_{gsm}$

Where:

$N_{freq,\ E\text{-}UTRA}$ is the number of E-UTRA carriers being monitored

$N_{freq, UTRA}$ is the number of UTRA carriers being monitored

$M_{GSM}$ is an integer which is a function of the number of GSM carriers on which measurements are being performed. $M_{GSM}$ is equal to 0 if no GSM carrier is being monitored. For a Measurement Gap Repetition Period (MGRP) of 40 ms, $M_{GSM}$ is equal to 1 if cells on up to 32 GSM carriers are being measured. For a MGRP of 80 ms, $M_{GSM}$ is equal to ceil ($N_{carriers, GSM}$ /20) where $N_{carriers, GSM}$ is the number of GSM carriers on which cells are being measured

The UE shall meet the measurement accuracy requirements stated for RXLEV in TS 45.008 [15], when the given measurement time allows the UE to take at least 3 GSM carrier RSSI samples per GSM carrier in the monitored set during the measurement period.

In case the UE is not able to acquire the required number of samples per GSM carrier during one measurement period, the UE shall measure as many GSM carriers as possible during that measurement period using at least 3 samples per GSM carrier. The GSM carriers that were not measured during that measurement period shall be measured in the following measurement periods.

Measurements on a GSM cell can be requested with BSIC verified. The UE shall be able to report the GSM cells with BSIC verified for those cells where the verification of BSIC has been successful.

The procedure for BSIC verification on a GSM cell can be divided into the following two tasks:

- **Initial BSIC identification:** Includes searching for the BSIC and decoding the BSIC for the first time when there is no knowledge about the relative timing between the E-UTRAN FDD and GSM cells.
- **BSIC re-confirmation:** Tracking and decoding the BSIC of a GSM cell after initial BSIC identification is performed. The UE shall trigger the BSIC re-confirmation within the available measurement gap pattern. The requirements for BSIC re-confirmation can be found in section 8.1.2.4.5.1.2.2 of TS 36.133 [4].

If the network requests measurements on a GSM cell, the UE shall behave as follows:

- The UE shall perform GSM carrier RSSI measurements according to section 8.1.2.4.5.2.1 in TS 36.133 [4] when a measurement gap pattern sequence is activated.
- The UE shall perform measurement reporting as defined in TS 36.331 [5].
- The UE shall perform BSIC identification. The UE shall use the most recently available GSM carrier RSSI measurement results for arranging GSM cells in signal strength order for performing BSIC identification.
- The UE shall perform BSIC re-confirmation on all the GSM cells that have been successfully identified.
- The UE shall perform all configured event evaluation for event-triggered reporting after the BSIC has been verified for a GSM cell. The UE shall use the most recently available GSM carrier RSSI measurement results in event evaluation and event-triggered reporting.
- Event-triggered and periodic reports shall be triggered according to TS 36.331 [5].

The BSIC of a GSM cell is considered to be “verified” if the UE has decoded the SCH of the BCCH carrier and identified the BSIC at least one time (initial BSIC identification). Once a GSM cell has been identified the BSIC shall be re-confirmed at least once every $8*T_{re\text{-}confirm,GSM}$ seconds. Otherwise the BSIC of the GSM cell is considered as "non-verified". If a measurement gap pattern sequence is deactivated by the network after BSIC has been identified or verified, the UE shall consider the BSIC as non-verified.

The UE shall be able to perform BSIC verification at levels down to the reference sensitivity level or reference interference levels as specified in TS 45.005 [16].

This measurement shall be made on GSM cells that are required with BSIC verified.

For DRX cycle length $\leq$ 40 ms, the initial GSM BSIC identification requirements corresponding to the non DRX requirements as specified in section 8.1.2.4.5.1.2.1 in TS 36.133 [4] shall apply.

For DRX cycle length > 40 ms, the UE shall make at least one attempt every $N_{freq}$*30s to decode the BSIC of SCH on the BCCH carrier of the 8 strongest BCCH carriers of the GSM cells indicated in the Inter-RAT cell info list. If the UE has not successfully decoded the BSIC of the GSM BCCH carrier within $N_{freq}$*60 s, the UE shall abort the BSIC identification attempts for that GSM BCCH carrier. The UE shall give priority for BSIC decoding attempts in decreasing signal strength order to BCCH carriers with unknown BSIC. The strongest BCCH carrier is defined as the

BCCH carrier having the highest measured GSM carrier RSSI value. The parameter $N_{freq}$ is defined in section 8.1.2.1.1 in TS 36.133 [4].

If the BSIC of the GSM BCCH carrier has been successfully decoded the UE shall continue BSIC identification with the next GSM BCCH carrier, in signal strength order, with unknown BSIC. The GSM cell for which the BSIC has been successfully identified shall be moved to the BSIC re-confirmation procedure.

The UE shall maintain the timing information of up to 8 identified GSM cells. Initial timing information is obtained from the initial BSIC identification. The timing information shall be updated every time the BSIC is decoded.

For DRX cycle length ≤ 40 ms, GSM BSIC re-confirmation requirements corresponding to the non DRX requirements as specified in section 8.1.2.4.5.1.2.2 in TS 36.133 [4] shall apply.

For DRX cycle length > 40 ms, at least every $N_{freq}$*30 seconds, the UE shall attempt to decode the BSIC of each identified GSM cell. If the UE fails to decode the BSIC after two successive attempts or if the UE has not been able to re-confirm the BSIC for a GSM cell within $N_{freq}$*60 seconds, the UE shall abort the BSIC re-confirmation attempts for that GSM cell. The GSM cell shall be treated as a new GSM cell with unidentified BSIC and the GSM cell shall be moved to the initial BSIC identification procedure, see section 8.1.2.4.5.2.2.1 in TS 36.133 [4]. The parameter $N_{freq}$ is defined in section 8.1.2.1.1 in TS 36.133 [4].

Reported measurements in event triggered measurement reports shall meet the requirements in section TS 36.331 [5].

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH.

The event triggered reporting delay requirement is valid when the UE for each GSM carrier in the monitored set can take the required number of samples during the measurement period $T_{Measurement\ Period,\ GSM}$ (see section 8.1.2.4.5.1 of TS 36.133 [4]).

The event triggered measurement reporting delay for a GSM cell with verified BSIC, measured without L3 filtering shall be less than 2*$T_{Measurement\ Period,\ GSM}$, where $T_{Measurement\ Period,\ GSM}$ is defined in section 8.1.2.4.5.1 of TS 36.133 [4]. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.5.2 and A.8.8.2.

## 8.8.2.4 Test description

### 8.8.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.14.

2. The general test parameter settings are set up according to Table 8.8.2.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.8.2.4.3.

5. There is one E-UTRA FDD carrier and one GSM cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.8.2.4.1-1: General test parameters for E-UTRAN FDD-GSM event triggered reporting when DRX is used in AWGN**

| Parameter | Unit | Test 1 | Test 2 | Comment |
|---|---|---|---|---|
| | | Value | | |
| PDSCH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.0 FDD | | As specified in section A.1.1. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.6 FDD | | As specified in section A.1.2. |
| Gap Pattern Id | | 0 | | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| Active cell | | Cell 1 | | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | | Cell 2 is on Absolute RF Channel Number 1 (GSM cell) |
| CP length | | Normal | | Applicable to cell 1 |
| E-UTRA RF Channel Number | | 1 | | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| Inter-RAT (GSM) measurement quantity | | GSM Carrier RSSI | | |
| B1-Threshold-GERAN | dBm | -80 | | GSM Carrier RSSI threshold for event B1. |
| Hysteresis | dB | 0 | | |
| TimeToTrigger | s | 0 | | |
| Filter coefficient | | 0 | | L3 filtering is not used |
| PRACH configuration | | 4 | | As specified in table 5.7.1-2 in TS 36.211 [9] |
| Access Barring Information | - | Not Sent | | No additional delays in random access procedure. |
| DRX | | ON | | DRX related parameters are defined in Table 8.8.2.5-2 |
| Monitored GSM cell list size | | 6 GSM neighbours including ARFCN 1 | | List of GSM cells provided before T2 starts. |
| T1 | s | 5 | | |
| T2 | s | 5 | 45 | |

### 8.8.2.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event B1 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

In Test 1 when DRX = 40 ms is used, UE needs to be provided at least once every 500 ms with new Timing Advance Command MAC control element to restart the Timer Alignment Timer to keep the UE uplink time alignment. Furthermore, the UE is allocated with PUSCH resource at every DRX cycle. In Test 2 when DRX = 1280 ms is used, the uplink time alignment is not maintained and the UE needs to use RACH to obtain uplink allocation for measurement reporting.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.8.2.5-1, 8.8.2.5-2, 8.8.2.5-3 and 8.8.2.5-4. propagation conditions are set according to Annex B clause B.1.1.T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 8.8.2.5-1 and 8.8.1.5-4. T2 starts.
6. UE shall transmit a MeasurementReport message triggered by Event B1. If the overall delays measured from the beginning of time period T2 is less than 3162 ms for Test 1 or less than 44082 ms for Test 2 then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. The SS shall set a different BSIC on Cell 2, as the previous timing information of Cell 2 is invalid in the UE for the next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

11. Repeat step 1-10 for each sub-test in Table 8.8.2.4.1-1 as appropriate.

### 8.8.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.8.2.4.3-1: Common Exception messages for E-UTRAN FDD - GSM Event triggered reporting when DRX is used in AWGN**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-7<br>Table H.3.1-10<br>Table H.3.7-1<br>Table H.3.7-2<br>Table H.3.7-3 |

**Table 8.8.2.4.3-2: *SystemInformationBlockType7*: Additional E-UTRAN FDD - GSM Event triggered reporting in AWGN**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-3 SystemInformationBlockType7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| commonInfo SEQUENCE { | | | |
| p-MaxGERAN | 33 (33 dBm) | | GSM 400 & GSM 900 & GSM 850 & GSM 700 |
| | 30 (30 dBm) | | DCS 1800 & PCS 1900 |
| } | | | |

**Table 8.8.2.4.3-3: *RRCConnectionReconfiguration*: Additional E-UTRAN FDD - GSM Event triggered reporting in AWGN**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.8.2.4.3-4: *ReportConfigInterRAT-B1-GERAN*: Additional E-UTRAN FDD - GSM Event triggered reporting in AWGN**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7A ReportConfigInterRAT-B1-GERAN(GERAN-Thres) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-GERAN(GERAN-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventIdeventide CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| B1-ThresholdGERAN CHOICE { | | | |
| thresholdGERAN | 30 (-80 dBm) | -80 is actual value in dBm (30 - 110 dBm) | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.8.2.4.3-5: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD-GSM Event triggered reporting when DRX is used in AWGN**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.8.2.4.3-6: *MeasResults*: Additional E-UTRAN FDD - GSM Event triggered reporting when DRX is used in AWGN**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListGERAN | MeasResultListGERAN | | |
| } | | | |
| } | | | |

**Table 8.8.2.4.3-7: *MeasResultListGERAN:* Additional E-UTRAN FDD - GSM Event triggered reporting when DRX is used in AWGN**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListGERAN ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| MeasResultGERAN SEQUENCE { | | | |
| carrierFreq | CarrierFreqGERAN | Contains the carrier frequency of the target GERAN cell | |
| physCellId | PhysCellIdGERAN | Contains the Base Station Identity Code (BSIC) | |
| measResult SEQUENCE { | | | |
| rssi | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 8.8.2.4.3-8: *PRACH-Config-DEFAULT*: Additional E-UTRAN FDD - GSM Event triggered reporting when DRX is used in AWGN**

| Derivation Path: TS 36.508 [7] clause 7.3.2 Table 7.3.2-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PRACH-Config-DEFAULT ::= SEQUENCE { | | | |
| prach-ConfigIndex | 4 | | |
| } | | | |

## 8.8.2.5 Test requirement

Tables 8.8.2.4.1-1, 8.8.2.5-1 and 8.8.2.5-4 define the primary level settings including test tolerances for E-UTRAN FDD-GSM event triggered reporting when DRX is used under AWGN conditions.

**Table 8.8.2.5-1: Cell specific test parameters for E-UTRAN FDD (Cell # 1) for event triggered reporting of GSM cell when DRX is used in AWGN**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | |
| RSRP Note 3 | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.8.2.5-2: DRX-Configuration to be used in E-UTRAN FDD - GSM Event triggered reporting when DRX is used in AWGN**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf40 | sf1280 | |
| shortDRX | disable | disable | |
| Note: For further information see section 6.3.2 in 3GPP TS 36.331 [5]. | | | |

**Table 8.8.2.5-3: *TimeAlignmentTimer* and *sr-ConfigIndex*-Configuration to be used in E-UTRAN FDD-GSM Event triggered reporting when DRX is used in AWGN**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | For further information see section 6.3.2 in 3GPP TS 36.331 [5]. |
| sr-ConfigIndex | 0 | 0 | For further information see section 6.3.2 in 3GPP TS 36.331 and section10.1 in 3GPP TS 36.213 [8]. |

**Table 8.8.2.5-4: Cell specific test parameters for GSM (Cell # 2) for event triggered reporting of GSM cell when DRX is used in AWGN**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| Absolute RF Channel Number | | ARFNC 1 | |
| RXLEV | dBm | -Infinity | -75 |
| GSM BSIC | | N/A | Valid |

In Test 1 when DRX cycle length = 40 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event B1 triggered measurement report to Cell 2 on PUSCH.

In Test 2 when DRX cycle length = 1280 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE starts to send preambles on the PRACH for Scheduling Request (SR) to obtain allocation to send the measurement report to Cell 2 on PUSCH.

For both tests:

The overall delay may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event B1 measurement report.

The overall delay measured when DRX cycle length is 40 ms test requirement is expressed as:

Overall delay measured = $2*T_{Measurement\ Period,\ GSM} + T_{identify,\ GSM}$ + TTI insertion uncertainty + DRX cycle length

$T_{Measurement\ Period,\ GSM}$ = 480 ms (as specified in table 8.1.2.4.5.2.1-1 of TS 36.133 [4] clause 8.1.2.4.5.2.1)

$T_{identify,\ GSM}$ = 2160 ms (as specified in table 8.1.2.4.5.1.2.1-1 of TS 36.133 [4] clause 8.1.2.4.5.1.2.1)

TTI insertion uncertainty = 2 ms

DRX cycle length = 40 ms

The overall delay measured when DRX cycle length is 40 ms shall be less than a total of 3162 ms.

The overall delay measured when DRX cycle length is 1280 ms test requirement is expressed as:

Overall delay measured = $2*T_{Measurement\ Period,\ GSM} + N_{freq} * 30s$ + TTI insertion uncertainty + DRX cycle length

$T_{Measurement\ Period,\ GSM}$ = 6400 ms (as specified in table 8.1.2.4.5.2.1-1 of TS 36.133 [4] clause 8.1.2.4.5.2.1)

$N_{freq}$ = 1 (as specified in TS 36.133 clause 8.1.2.1.1)

$N_{freq} * 30$ s = 30000 ms (as specified in TS 36.133 [4] clause 8.1.2.4.5.2.2.2 for when DRX cycle length > 40 ms)

TTI insertion uncertainty = 2 ms

DRX cycle length = 1280 ms

The overall delay measured when DRX cycle length is 1280 ms shall be less than a total of 44082 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 8.9 E-UTRAN FDD - UTRAN TDD measurements

## 8.9.1 E-UTRAN FDD - UTRAN TDD event triggered reporting under fading propagation conditions

### 8.9.1.1 Test purpose

To verify the UE's ability to make a correct reporting of an event under fading propagation conditions within the E-UTRA FDD - UTRA TDD cell search requirements.

### 8.9.1.2 Test applicability

This test applies to all types of release 8 E-UTRA FDD UEs that support release 9 and forward UTRA TDD. Applicability requires support for FGI bits 15 and 22.

This test applies to all types of release 9 and forward E-UTRA FDD UEs that support release 9 and forward UTRA TDD and not supporting UTRA FDD. Applicability requires support for FGI bits 15 and 22.

This test applies to all types of release 9 and forward E-UTRA FDD UEs that support release 9 and forward UTRA TDD. Applicability requires support for FGI bits 15 and 39.

### 8.9.1.3 Minimum requirement

The measurement reporting delay shall be less than $T_{identify,\ UTRA_TDD}$ in RRC_CONNECTED state.

When explicit neighbour list is provided and no DRX is used the UE shall be able to identify a new detectable cell belonging to the monitored set within

$$T_{identify,\ UTRA_TDD} = Max\left\{5000, T_{basic\ identify\ UTRA_TDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms$$

where

$T_{basic_identify_UTRA_TDD}$ = 800 ms is the time period used in the inter RAT equation where the maximum allowed time for the UE to identify a new UTRA TDD cell is defined.

$T_{Inter1}$ = 30 ms. This is the minimum available time for inter-RAT measurement during 480 ms period

$N_{Freq}$: This is the number of UTRA carriers being monitored

If the UE does not require transmit gap to perform inter-RAT UTRA TDD measurements, the UE shall be able to identify a new detectable inter-RAT UTRA TDD cell belonging to the monitored set within 5000 ms.

A cell shall be considered detectable when

- P-CCPCH Ec/Io $\geq$ -8 dB,
- DwPCH_Ec/Io $\geq$ -5 dB.

When L3 filtering is used an additional delay can be expected.

When measurement gaps are scheduled for UTRA TDD inter RAT measurements the UE physical layer shall be capable of reporting measurements to higher layers with measurement accuracy as specified in Section 9.3 with measurement period given by

$$T_{measurement\ UTRA_TDD} = Max\left\{T_{Measurement_Period\ UTRA_TDD}, T_{basic\ measurement\ UTRA_TDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms$$

Where:

$T_{Measurement_Period\ UTRA_TDD}$ = 480 ms is the period used for calculating the measurement period $T_{measurement_UTRA_TDD}$ for UTRA TDD P-CCPCH RSCP measurements.

$T_{basic_identify_UTRA_TDD}$ = 800 ms is the time period used in the inter RAT equation where the maximum allowed time for the UE to identify a new UTRA TDD cell is defined.

$T_{basic_measurement_UTRA_TDD}$ = 50 ms is the time period used in the equation for defining the measurement period for inter RAT P-CCPCH RSCP measurements.

$N_{freq}$ and $T_{inter1}$ are defined in section 8.1.2.1.1

If the UE does not need measurement gaps to perform UTRA TDD measurements, the measurement period for UTRA TDD measurements is 480 ms.

The UE shall be capable of performing UTRA TDD P-CCPCH RSCP measurements for $X_{basic\ measurementUTRA_TDD}$ inter-frequency cells per TDD frequency of the monitored set and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{Measurement_UTRA_TDD.}$ Where $X_{basic\ measurementUTRA_TDD}$= 6.

Reported measurements in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as the reporting criteria is not fulfilled.

The measurement reporting delay is defined as the time between any event that will trigger a measurement report until the UE starts to transmit the measurement report over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify,\ UTRA_TDD}$ defined in TS 36.133 [4] Section 8.1.2.4.3.1.1 When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.3 and A.8.9.

## 8.9.1.4 Test description

### 8.9.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.23.
2. The general test parameter settings are set up according to Table 8.9.1.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.9.1.4.3.
5. There is one E-UTRA FDD serving cell and one UTRA TDD cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.9.1.4.1-1: General test parameters for Event triggered reporting in fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Active cell | | Cell 1 | E-UTRA FDD Cell 1 |
| Neighbour cell | | Cell 2 | UTRA TDD Cell 2 is to be identified. |
| Gap Pattern Id | | 1 | As specified in TS 36.133 section8.1.2.1. Measurement Gap Repetition Period = 80ms |
| Inter-RAT measurement quantity | | UTRA TDD PCCPCH RSCP | |
| Threshold other system | dBm | -75 | UTRA TDD PCCPCH RSCP threshold for event B1. |
| Hysteresis | dB | 0 | |
| CP length | | Normal | |
| TimeToTrigger | dB | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | | OFF |
| T1 | s | 5 | |
| T2 | s | 15 | |

### 8.9.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table's 8.9.1.5-1 and 8.9.1.5-2. Propagation conditions are set according to Annex B clause B.1.1 and B.2.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 8.9.1.5-1 and 8.9.1.5-2.

6. UE shall transmit a MeasurementReport message triggered by Event B1. If the overall delay measured from the beginning of time period T2 is less than 12880 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 4) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. SS shall change set cell 2 cell parameter id =(current cell 2 cell parameter id +4) mod 16.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.9.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.9.1.4.3-1: Common Exception messages for E-UTRAN FDD - UTRAN TDD event triggered reporting under fading propagation conditions**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-4<br>Table H.3.1-7 |

**Table 8.9.1.4.3-2: *ReportConfigInterRAT-B1-UTRA*: Additional E-UTRAN FDD - UTRAN TDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-UTRA(UTRA-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| b1-ThresholdUTRA CHOICE { | | | |
| utra-RSCP | 40 | UTRA-Thres is actual RSCP value in dBm<br>UTRA-Thres + 115 | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.9.1.4.3-3: *MeasuredResults*: Additional E-UTRAN FDD - UTRAN TDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasuredResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCellsCHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 8.9.1.4.3-4: *MeasResultListUTRA*: Additional E-UTRAN FDD - UTRAN TDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellIdphysicallCellIdentity CHOICE { | | | |
| tdd | UTRA-TDD-CellIdentity | | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-RSCP | Set according to specific test | | |
| } | | | |
| } | | | |

**Table 8.9.1.4.3-5: CellGlobalId-UTRA: Additional E-UTRAN FDD - UTRAN TDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGlobalIdUTRA ::= SEQUENCE { | | | |
| plmn-Identity | PLMN-Identity | | |
| cellIdentity | | BIT STRING (SIZE (28)) | |
| } | | | |

## 8.9.1.5 Test requirement

Tables 8.9.1.4.1-1, 8.9.1.5-1 and 8.9.1.5-2 define the primary level settings including test tolerances for E-UTRAN FDD - UTRAN TDD event triggered reporting under fading propagation conditions in synchronous cells test.

**Table 8.9.1.5-1: Cell specific test parameters for Event triggered reporting of UTRA TDD neighbours in fading propagation conditions (cell1)**

| **Parameter** | **Unit** | **Cell 1** | |
|---|---|---|---|
| | | **T1** | **T2** |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |

| | | | |
|---|---|---|---|
| $N_{oc}$ | dBm/15KHz | -98 | |
| RSRP | dBm | -94 | -94 |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| P-SCH_RP | dBm | -94 | |
| S-SCH_RP | dBm | -94 | |
| Propagation Condition | | ETU70 | |
| Note 1: OCNG shall be used such that cell 1 is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2. | | | |

**Table 8.9.1.5-2: Cell specific test parameters for Event triggered reporting of UTRA TDD neighbours in fading propagation conditions (cell2)**

| Parameter | Unit | Cell 2 | | | |
|---|---|---|---|---|---|
| | | T1 | | T2 | |
| Timeslot Number | | 0 | DwPTS | 0 | DwPTS |
| UTRA RF Channel Number (NOTE1) | | Channel1 | | | |
| PCCPCH_Ec/Ior | dB | -Infinity | | -3 | |
| DwPCH_Ec/Ior | dB | -Infinity | | | 0 |
| OCNS_Ec/Ior | | -Infinity | | -3 | |
| $\hat{I}_{or} / I_{oc}$ | dB | -Infinity | | 9 | |
| $I_{oc}$ | dBm/1.28 MHz | -70 | | | |
| PCCPCH_RSCP | dB | -Infinity | | -64 | |
| PropagationCondition | | Case 3 (NOTE2) | | | |
| NOTE 1: The DPCH of the cell is located in a timeslot other than 0.<br>NOTE 2: Case 3 propagation conditions are specified in TS 25.102 Annex B. | | | | | |

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event B1 triggered measurement report to Cell 2.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify, UTRA_TDD}$

$$T_{identify,\ UTRA_TDD} = Max\left\{5000, T_{basic\ identify\ UTRA_TDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms \backslash$$

$T_{basic_identify_UTRA_TDD}$ = 800 ms

$T_{Inter1}$ = 30 ms

$N_{Freq}$ = 1

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 12802 ms in this test case (note: this gives a total of 12800 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.9.2 E-UTRAN FDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions

### 8.9.2.1 Test purpose

The purpose of this test is to verify that the UE makes correct enhanced reporting of UTRAN TDD cells. This test will partly verify the Enhanced UTRA TDD cell identification requirements in TS 36.133 [4] section 8.1.2.4.4 under AWGN propagation conditions.

### 8.9.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward that support UTRA TDD and not supporting UTRA FDD. Applicability requires support for FGI bit 15 and 22.

This test applies to all types of E-UTRA FDD UE release 9 and forward that support UTRA TDD. Applicability requires support for FGI bit 15 and 39.

### 8.9.2.3 Minimum conformance requirements

When explicit neighbour list is provided and no DRX is used or when DRX cycle length ≤ 40 ms the UE shall be able to identify a new detectable cell belonging to the monitored set within $T_{identify,\ enhanced_UTRA_TDD}$:

$$T_{identify,\ enhanced_UTRA_TDD} = (T_{basic_identify_enhanced_UTRA_TDD} \cdot \frac{480}{T_{inter1}} + 480) \cdot N_{Freq} \quad ms$$

If the UE does not require transmit gap to perform inter-RAT UTRA TDD measurements, the UE shall be able to identify a new detectable inter-RAT UTRA TDD cell belonging to the monitored set within 500 ms.

A cell shall be considered detectable when:

- P-CCPCH_Ec/Io ≥ -6 dB,
- DwPCH_Ec/Io ≥ -1 dB

When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.4 and A.8.9.2.

### 8.9.2.4 Test description

#### 8.9.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.
2. The general test parameter settings are set up according to Table 8.9.2.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.9.2.4.3.
5. There is one E-UTRA TDD serving cell and one UTRA TDD cell specified in the test. Cell 1 (E-UTRA TDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 (UTRAN TDD cell) is to be searched. Gap pattern configuration # 0 as defined in TS 36.133 [4] table 8.1.2.1-1 is provided.

**Table 8.9.2.4.1-1: General test parameters for E-UTRAN FDD- UTRAN TDD enhanced cell identification under AWGN propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1. |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on UTRA RF channel number 1. |
| CP length | | Normal | Applicable to cell 1 |
| E-UTRA RF Channel Number | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| UTRA RF Channel Number | | 1 | One UTRA TDD carrier frequency is used. |
| Inter-RAT (UTRA TDD) measurement quantity | | P-CCPCH RSCP | |
| Thresh | dBm | -83 | Absolute P-CCPCH RSCP threshold for event B1 |
| Hysteresis | dB | 0 | |
| Time To Trigger | ms | 0 | |
| Monitored UTRA TDD cell list size | | 12 | UTRA cells on UTRA RF channel 1 provided in the cell list |
| Time offset between cells | ms | 3 | |
| T1 | s | 5 | |
| T2 | s | 2 | |

### 8.9.2.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event B1 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Tables 8.9.2.5-1 and 8.9.2.5-2. Propagation conditions are set according to Annex B clause B.1.1. T1 starts.

3. The SS shall transmit an *RRCConnectionReconfiguration* with Event B1 configured message.

4. The UE shall transmit *RRCConnectionReconfigurationComplete* message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Tables 8.9.2.5-1 and 8.9.2.5-2. T2 starts.

6. UE shall transmit a *MeasurementReport* message triggered by Event B1. If the overall delay measured from the beginning of time period T2 is less than 1122 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. SS shall set Cell 2 cell parameter id = (current Cell 2 cell parameter id+4) mod 16 for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),

or

- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 8.9.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.9.2.4.3-1: Common Exception messages for E-UTRAN FDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-4<br>Table H.3.1-7 |

**Table 8.9.2.4.3-2: *ReportConfigInterRAT-B1-UTRA*: Additional E-UTRAN FDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6 Table 4.6.6-7B | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-UTRA(UTRA-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| b1-ThresholdUTRA CHOICE { | | | |
| utra-RSCP | 32 | UTRA-Thres is actual RSCP value in dBm<br>UTRA-Thres + 115 | UTRA-TDD |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0dB) | The actual value is IE value * 0.5 dB. | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.9.2.4.3-3: *MeasResults*: Additional E-UTRAN FDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 8.9.2.4.3-4: *MeasResultListUTRA*: E-UTRAN FDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| tdd | PhysCellIdUTRA-TDD | INTEGER (0..127) | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-RSCP | | Set according to specific test INTEGER (-5..91) | |
| } | | | |
| } | | | |

## 8.9.2.5 Test requirement

Tables 8.9.2.4.1-1, 8.9.2.5-1 and 8.9.2.5-2 define the primary level settings including test tolerances for E-UTRAN FDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions test.

**Table 8.9.2.5-1: Cell specific test parameters for cell #1 E-UTRAN FDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.9.2.5-2: Cell specific test parameters for cell #2 E-UTRAN FDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions**

| Parameter | Unit | Cell 2 (UTRA TDD) | | | |
|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPTS | |
| | | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number[Note1] | | 1 | | | |
| P-CCPCH_Ec/Ior | dB | -4.77 | -4.77 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 |
| OCNS_Ec/Ior[Note2] | dB | -1.76 | -1.76 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | - infinity | 8 | - infinity | 8 |
| $I_{oc}$ | dBm/1.28 MHz | -79.4 | | | |
| P-CCPCH RSCP [Note3] | dBm | - infinity | -76.17 | n.a. | n.a. |
| P-CCPCH_Ec/Io [Note3] | dB | - infinity | -5.41 | n.a. | n.a. |
| DwPCH_Ec/Io [Note3] | dB | n.a. | n.a. | - infinity | -0.64 |
| Propagation Condition | | AWGN | | | |

Note 1: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number.
Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.
Note 3: P-CCPCH RSRP, PCCPCH_Ec/Io and DwPCH_Ec/Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event B1 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify,\ enhanced_UTRA_TDD}$

$$T_{identify,\ enhanced_UTRA_TDD} = (T_{basic_identify_enhanced_UTRA_TDD} \cdot \frac{480}{T_{inter1}} + 480) \cdot N_{Freq} \quad ms$$

$T_{basic_identify_enhanced_UTRA_TDD}$ = 80 ms. It is the time period used in the inter RAT equation where the maximum allowed time for the UE to identify a new UTRA TDD cell is defined.

$T_{Inter1}$ = 60 ms

$N_{Freq}$ = 1. It is defined in TS 36.133 [4] clause 8.1.2.1.1.

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 1122 ms in this test case (note: this gives a total of 1120 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 8.10 E-UTRAN TDD - GSM measurements

## 8.10.1 E-UTRAN TDD - GSM event triggered reporting in AWGN

### 8.10.1.1 Test purpose

The purpose of this test is to verify that the UE makes correct reporting of an event when doing inter-RAT (GSM) measurements. This test will partly verify the E-UTRAN TDD - GSM cell search requirements in TS 36.133 [4] section 8.1.2.4.6.

### 8.10.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward that support GSM. Applicability requires support for FGI bits 15 and 23.

### 8.10.1.3 Minimum conformance requirements

The requirements in this section apply only to UE supporting E-UTRAN TDD and GSM.

Measurements on GSM cells can be requested with BSIC verified.

In RRC_CONNECTED state when a supported measurement gap pattern sequence according to TS 36.133 [4] Table 8.1.2.1-1 is configured by E-UTRAN the UE shall continuously measure GSM cells, search for new GSM cells given in the monitored set and re-confirm the BSIC for already detected cells.

This measurement shall be based on measurement gaps allocated for GSM carrier RSSI measurement as described in TS 36.133 [4] section 8.1.2.1. A UE supporting GSM measurements shall measure minimum number of 10 GSM carrier RSSI measurement samples ($N_{GSM\ carrier\ RSSI}$) per measurement gap. In RRC_CONNECTED state the measurement period, $T_{Measurement\ Period,\ GSM}$, for the GSM carrier RSSI measurement is $N_{freq}$*480 ms. The parameter $N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1.

The UE shall meet the measurement accuracy requirements stated for RXLEV in TS 45.008 [15], when the given measurement time allows the UE to take at least 3 GSM carrier RSSI samples per GSM carrier in the monitored set during the measurement period.

In case the UE is not able to acquire the required number of samples per GSM carrier during one measurement period, the UE shall measure as many GSM carriers as possible during that measurement period using at least 3 samples per GSM carrier. The GSM carriers that were not measured during that measurement period shall be measured in the following measurement periods.

Measurements on a GSM cell can be requested with BSIC verified. The UE shall be able to report the GSM cells with BSIC verified for those cells where the verification of BSIC has been successful.

The procedure for BSIC verification on a GSM cell can be divided into the following two tasks:

- **Initial BSIC identification:** Includes searching for the BSIC and decoding the BSIC for the first time when there is no knowledge about the relative timing between the E-UTRAN TDD and GSM cells. The UE shall trigger the initial BSIC identification within the available measurement gap pattern sequence. The requirements for BSIC re-confirmation can be found in TS 36.133 [4] section 8.1.2.4.5.1.2.1.

- **BSIC re-confirmation:** Tracking and decoding the BSIC of a GSM cell after initial BSIC identification is performed. The UE shall trigger the BSIC re-confirmation within the available measurement gap pattern. The requirements for BSIC re-confirmation can be found in TS 36.133 [4] section 8.1.2.4.5.1.2.2.

If the network requests measurements on a GSM cell, the UE shall behave as follows:

- The UE shall perform GSM carrier RSSI measurements according to TS 36.133 [4] section 8.1.2.4.5.1 when a measurement gap pattern sequence is activated.

- The UE shall perform measurement reporting as defined in TS 36.331 [5].

- The UE shall perform BSIC identification. The UE shall use the most recently available GSM carrier RSSI measurement results for arranging GSM cells in signal strength order for performing BSIC identification.
- The UE shall perform BSIC re-confirmation on all the GSM cells that have been successfully identified.
- The UE shall perform all configured event evaluation for event-triggered reporting after the BSIC has been verified for a GSM cell. The UE shall use the most recently available GSM carrier RSSI measurement results in event evaluation and event-triggered reporting.
- Event-triggered and periodic reports shall be triggered according to TS 36.331 [5].

The BSIC of a GSM cell is considered to be “verified” if the UE has decoded the SCH of the BCCH carrier and identified the BSIC at least one time (initial BSIC identification). Once a GSM cell has been identified the BSIC shall be re-confirmed at least once every $8*T_{re\text{-}confirm,GSM}$ seconds. Otherwise the BSIC of the GSM cell is considered as "non-verified". If a measurement gap pattern sequence is deactivated by the network after BSIC has been identified or verified, the UE shall consider the BSIC as non-verified.

$T_{identify,GSM}$ indicates the maximum time allowed for the UE to decode the unknown BSIC of the GSM cell in one GSM BCCH carrier in the initial BSIC identification procedure.

$T_{re\text{-}confirm,GSM}$ indicates the maximum time allowed for the re-confirmation of the BSIC of one GSM cell in the BSIC re-confirmation procedure.

The UE shall be able to decode a BSIC within a measurement gap when the time difference between the middle of the received GSM synchronisation burst at the UE and the middle of the effective measurement gap is within the limits specified in table 8.10.1.3-1.

**Table 8.10.1.3-1: The gap length and maximum time difference for BSIC verification**

| Gap length [ms] | Maximum time difference [µs] |
|---|---|
| 6 | ± 2350 µs |

The UE shall be able to perform BSIC verification at levels down to the reference sensitivity level or reference interference levels as specified in TS 45.005 [16].

This measurement shall be based on the measurement gaps used for Initial BSIC identification as described in TS 36.133 [4] section 8.1.2.4.5.1.2

The UE shall continuously attempt to decode the BSIC of SCH on the BCCH carrier of the 8 strongest BCCH carriers of the GSM cells indicated in the Inter-RAT cell info list. The UE shall give priority for BSIC decoding attempts in decreasing signal strength order to BCCH carriers with unknown BSIC. The strongest BCCH carrier is defined as the BCCH carrier having the highest measured GSM carrier RSSI value.

If the BSIC of the GSM BCCH carrier has been successfully decoded the UE shall immediately continue BSIC identification with the next GSM BCCH carrier, in signal strength order, with unknown BSIC. The GSM cell for which the BSIC has been successfully identified shall be moved to the BSIC re-confirmation procedure.

If the UE has not successfully decoded the BSIC of the GSM BCCH carrier within $T_{identify,GSM}$ ms, the UE shall abort the BSIC identification attempts for that GSM BCCH carrier. The UE shall continue to try to perform BSIC identification of the next GSM BCCH carrier in signal strength order. The GSM BCCH carrier for which the BSIC identification failed shall not be re-considered for BSIC identification until BSIC identification attempts have been made for all the rest of the 8 strongest GSM BCCH carriers in the monitored set with unknown BSIC.

$T_{identify,GSM}$ values are given for a set of reference gap patterns in table 8.10.1.3-2. The requirements in the table represent the time required to guarantee at least two attempts to decode the BSIC for one GSM BCCH carrier.

**Table 8.10.1.3-2**

| Number of carriers other than GSM | $T_{identify,gsm}$(ms) | | $T_{reconfirm,gsm}$(ms) | |
|---|---|---|---|---|
| | 40ms gap configuration (ID 0) | 80ms gap configuration (ID 1) | 40ms gap configuration (ID 0) | 80ms gap configuration (ID 1) |
| 0 | 2160 | 5280 | 1920 | 5040 |
| 1 | 5280 | 21760 | 5040 | 17280 |
| 2 | 5280 | 31680 | 5040 | 29280 |
| 3 | 19440 | No requirement | 13320 | No requirement |
| 4 | 31680 | No requirement | 29280 | No requirement |
| 5 | 31680 | No requirement | 29280 | No requirement |

The UE shall maintain the timing information of up to 8 identified GSM cells. Initial timing information is obtained from the initial BSIC identification. The timing information shall be updated every time the BSIC is decoded.

For each measurement gap used for GSM BSIC reconfirmation as described in TS 36.133 [4] section 8.1.2.4.5.1.2, the UE shall attempt to decode the BSIC falling within the measurement gap according to table 8.10.1.3 - 2. If more than one BSIC can be decoded within the same measurement gap, priority shall be given to the least recently decoded BSIC.

If the UE fails to decode the BSIC after two successive attempts or if the UE has not been able to re-confirm the BSIC for a GSM cell within $T_{re\text{-}confirm,GSM}$ seconds, the UE shall abort the BSIC re-confirmation attempts for that GSM cell. The GSM cell shall be treated as a new GSM cell with unidentified BSIC and the GSM cell shall be moved to the initial BSIC identification procedure, see TS 36.133 [4] section 8.1.2.4.5.1.2.1.

Reported measurements in periodically triggered measurement reports shall meet the requirements in TS 36.133 [4] section 9.

Reported measurements in event triggered measurement reports shall meet the requirements in TS 36.133 [4] section 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH.

The event triggered reporting delay requirement is valid when the UE for each GSM carrier in the monitored set can take the required number of samples during the measurement period $T_{Measurement\ Period,\ GSM}$ (see TS 36.133 [4] section 8.1.2.4.5.1).

The event triggered measurement reporting delay for a GSM cell with verified BSIC , measured without L3 filtering shall be less than $2*T_{Measurement\ Period,\ GSM}$, where $T_{Measurement\ Period,\ GSM}$ is defined in TS 36.133 [4] section 8.1.2.4.5.1. When L3 filtering is used an additional delay can be expected.

Reported measurements contained in event triggered periodic measurement reports shall meet the requirements in TS 36.133 [4] section 9.

The first report in event triggered periodic measurement reporting shall meet the requirements specified in TS 36.133 [4] section 8.1.2.4.5.1.4 Event Triggered Reporting.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.5.1 and A8.10.1

## 8.10.1.4 Test description

### 8.10.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14 for UEs that support receive diversity.
2. The general test parameter settings are set up according to Table 8.10.1.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.10.1.4.3.
5. There is one E-UTRA TDD serving cell and one GSM cell specified in the test. Cell 1 (E-UTRA TDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.10.1.4.1-1: General test parameters for E-UTRAN TDD-GSM event triggered reporting in AWGN**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2. |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on Absolute RF Channel Number 1 (GSM cell) |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in 3GPP TS 36.211 |
| Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in 3GPP TS 36.211 |
| CP length | | Normal | Applicable to cell 1 |
| E-UTRA RF Channel Number | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Inter-RAT (GSM) measurement quantity | | GSM Carrier RSSI | |
| b1-Threshold-GERAN | dBm | -80 | GSM Carrier RSSI threshold for event B1. |
| Hysteresis | dB | 0 | |
| Time To Trigger | ms | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used. |
| DRX | | OFF | |
| Monitored GSM cell list size | | 6 GSM neighbours including ARFCN 1 | List of GSM cells provided before T2 starts. |
| T1 | s | 5 | |
| T2 | s | 5 | |

### 8.10.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table's 8.10.1.5-1 and 8.10.1.5-2. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 8.10.1.5-1 and 8.10.1.5-2.

6. UE shall transmit a MeasurementReport message triggered by Event B1. If the overall delay measured from the beginning of time period T2 is less than 3122 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. SS shall set a different BSIC on Cell 2, as the previous timing information of Cell 2 is invalid in the UE for the next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.10.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.10.1.4.3-1: Common Exception messages for E-UTRAN TDD-GSM event triggered reporting in AWGN**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.3.1-10 |

**Table 8.10.1.4.3-2: *SystemInformationBlockType7*: Additional E-UTRAN TDD - GSM event triggered reporting in AWGN**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-3 SystemInformationBlockType7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| commonInfo SEQUENCE { | | | |
| p-MaxGERAN | 33 (33 dBm) | | GSM 400 & GSM 900 & GSM 850 & GSM 700 |
| | 30 (30 dBm) | | DCS 1800 & PCS 1900 |
| } | | | |

**Table 8.10.1.4.3-3: *ReportConfigInterRAT-B1-GERAN*: Additional E-UTRAN TDD - GSM Event triggered reporting in AWGN**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7A ReportConfigInterRAT-B1-GERAN(GERAN-Thres) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-GERAN(GERAN-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventIdeventide CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| B1-ThresholdGERAN CHOICE { | | | |
| b1-ThresholdGERAN | 30 (-80 dBm) | -80 is actual value in dBm (30 - 110 dBm) | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0dB) | The actual value is IE value * 0.5 dB INTEGER(0..30) | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |

**Table 8.10.1.4.3-4: *MeasResults*: Additional E-UTRAN TDD - GSM event triggered reporting in AWGN**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListGERAN | MeasResultListGERAN | | |
| } | | | |
| } | | | |

**Table 8.10.1.4.3-5: *MeasResultListGERAN*: Additional E-UTRAN TDD - GSM event triggered in AWGN**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListGERAN ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| carrierFreq | CarrierFreqGERAN | | |
| physCellId SEQUENCE { | | | |
| networkColourCode | | BIT STRING(SIZE(3)) | |
| baseStationColourCode | | BIT STRING(SIZE(3)) | |
| } | | | |
| measResult SEQUENCE { | | | |
| rssi | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 8.10.1.4.3-6: *CarrierFreqGERAN*: Additional E-UTRAN TDD - GSM event triggered reporting in AWGN**

| Derivation Path: 36.331 clause 6.3.4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CarrierFreqGERAN ::= SEQUENCE { | | | |
| arfcn | | INTEGER (0..1023) | |
| bandIndicator | | ENUMERATED {dcs1800, pcs1900} | |
| } | | | |

## 8.10.1.5 Test requirement

The test parameters are given in Tables 8.10.1.4.1-1, 8.10.1.5-1 and 8.10.1.5-2 below. In the measurement control information it is indicated to the UE that event-triggered reporting with Event B1 is used. The test consists of two successive time periods, with time duration of T1, and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2.

**Table 8.10.1.5-1: Cell specific test parameters for E-UTRAN TDD (cell # 1) for event triggered reporting of GSM cell in AWGN**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.10.1.5-2: Cell specific test parameters for GSM (cell # 2) for event triggered reporting of GSM cell in AWGN**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| Absolute RF Channel Number | | ARFCN 1 | |
| RXLEV | dBm | -Infinity | -75 |
| GSM BSIC | | N/A | Valid |

The UE shall send one Event B1 triggered measurement report including the valid BSIC of cell # 2, with a measurement reporting delay less than 3120 ms from the beginning of time period T2.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

NOTE 1: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 2: The delay for GSM cell identification with BSIC verified is equal to 3120 ms, which is the sum of the event triggered measurement reporting delay and the initial BSIC identification delay.

The event triggered measurement reporting delay = $2*T_{Measurement\ Period,\ GSM}$ = 2* 480ms =960ms.

Initial BSIC identification delay = 2160 ms.

The overall delays measured shall be less than a total of 3122 ms in this test case (note: this gives 960 ms for measurement reporting delay plus 2160 for BSIC identification and plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.10.2 E-UTRAN TDD-GSM event triggered reporting when DRX is used in AWGN

### 8.10.2.1 Test purpose

The purpose of these tests is to verify that the UE makes correct reporting of an event in DRX. These tests will partly verify the E-UTRAN TDD-GSM cell search requirements when DRX is used in TS 36.133 [4] section 8.1.2.4.6.

### 8.10.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward that support GSM. Applicability requires support for FGI bits 5, 15 and 23.

### 8.10.2.3 Minimum conformance requirements

The requirements in this section apply only to UE supporting E-UTRAN TDD and GSM.

Measurements on GSM cells can be requested with BSIC verified.

In RRC_CONNECTED state when a supported measurement gap pattern sequence according to TS 36.133 [4] Table 8.1.2.1-1 is configured by E-UTRAN the UE shall continuously measure GSM cells, search for new GSM cells given in the monitored set and re-confirm the BSIC for already detected cells. During DRX periods the UE may use other periods of time outside the specified measurement gap patterns. The UE is not required to make measurements of GSM cells during DRX periods if a measurement gap pattern has not been configured.

This measurement shall be based on measurement gaps allocated for GSM carrier RSSI measurement as described in TS 36.133 [4] section 8.1.2.1. A UE supporting GSM measurements shall measure minimum number of 10 GSM carrier RSSI measurement samples ($N_{GSM\ carrier\ RSSI}$) per DRX cycle. In RRC_CONNECTED state the measurement period, $T_{Measurement\ Period,\ GSM}$, for the GSM carrier RSSI measurement is shown in table 8.10.2.3-1. The parameter $N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1.

**Table 8.10.2.3-1: GSM measurement period for large DRX**

| DRX cycle length (s) | $T_{measure,GSM}$ (s) (DRX cycles) |
|---|---|
| ≤0.04 | Non DRX Requirements are applicable |
| 0.04<DRX-cycle≤ 0.08 | Note ($6*N_{freq}$) |
| 0.08<DRX-cycle≤ 2.56 | Note ($5*N_{freq}$) |
| Note: Time depends upon the DRX cycle in use | |

The UE shall meet the measurement accuracy requirements stated for RXLEV in TS 45.008 [15], when the given measurement time allows the UE to take at least 3 GSM carrier RSSI samples per GSM carrier in the monitored set during the measurement period.

In case the UE is not able to acquire the required number of samples per GSM carrier during one measurement period, the UE shall measure as many GSM carriers as possible during that measurement period using at least 3 samples per GSM carrier. The GSM carriers that were not measured during that measurement period shall be measured in the following measurement periods.

Measurements on a GSM cell can be requested with BSIC verified. The UE shall be able to report the GSM cells with BSIC verified for those cells where the verification of BSIC has been successful.

The procedure for BSIC verification on a GSM cell can be divided into the following two tasks:

- **Initial BSIC identification:** Includes searching for the BSIC and decoding the BSIC for the first time when there is no knowledge about the relative timing between the E-UTRAN TDD and GSM cells.
- **BSIC re-confirmation:** Tracking and decoding the BSIC of a GSM cell after initial BSIC identification is performed. The UE shall trigger the BSIC re-confirmation within the available measurement gap pattern

If the network requests measurements on a GSM cell, the UE shall behave as follows:

- The UE shall perform GSM carrier RSSI measurements according to TS 36.133 [4] section 8.1.2.4.5.2.1 when a measurement gap pattern sequence is activated.
- The UE shall perform measurement reporting as defined in TS 36.331 [5].
- The UE shall perform BSIC identification. The UE shall use the most recently available GSM carrier RSSI measurement results for arranging GSM cells in signal strength order for performing BSIC identification.
- The UE shall perform BSIC re-confirmation on all the GSM cells that have been successfully identified.
- The UE shall perform all configured event evaluation for event-triggered reporting after the BSIC has been verified for a GSM cell. The UE shall use the most recently available GSM carrier RSSI measurement results in event evaluation and event-triggered reporting.
- Event-triggered and periodic reports shall be triggered according to TS 36.331 [5].

The BSIC of a GSM cell is considered to be “verified” if the UE has decoded the SCH of the BCCH carrier and identified the BSIC at least one time (initial BSIC identification). Once a GSM cell has been identified the BSIC shall be re-confirmed at least once every 30 seconds. Otherwise the BSIC of the GSM cell is considered as "non-verified".

The UE shall be able to perform BSIC verification at levels down to the reference sensitivity level or reference interference levels as specified in TS 45.005 [16].

This measurement shall be made on GSM cells that are requested with BSIC verified.

For DRX cycle length ≤ 40 ms, the initial GSM BSIC identification requirements corresponding to the non DRX requirements as specified in section TS 36.133 [4] 8.1.2.4.5.1.2.1 shall apply.

For DRX cycle length > 40 ms, the UE shall make at least one attempt every $N_{freq}$*30s to decode the BSIC of SCH on the BCCH carrier of the 8 strongest BCCH carriers of the GSM cells indicated in the Inter-RAT cell info list. If the UE has not successfully decoded the BSIC of the GSM BCCH carrier within $N_{freq}$*60 s, the UE shall abort the BSIC identification attempts for that GSM BCCH carrier. The UE shall give priority for BSIC decoding attempts in decreasing signal strength order to BCCH carriers with unknown BSIC. The strongest BCCH carrier is defined as the BCCH carrier having the highest measured GSM carrier RSSI value. The parameter $N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1.

If the BSIC of the GSM BCCH carrier has been successfully decoded the UE shall continue BSIC identification with the next GSM BCCH carrier, in signal strength order, with unknown BSIC. The GSM cell for which the BSIC has been successfully identified shall be moved to the BSIC re-confirmation procedure.

The UE shall maintain the timing information of up to 8 identified GSM cells. Initial timing information is obtained from the initial BSIC identification. The timing information shall be updated every time the BSIC is decoded.

For DRX cycle length ≤ 40 ms, the GSM BSIC re-conformation requirements corresponding to the non DRX requirements as specified in TS 36.133 [4] section 8.1.2.4.5.1.2.2 shall apply.

For DRX cycle length > 40 ms, at least every $N_{freq}$*30 seconds, the UE shall attempt to decode the BSIC of each identified GSM cell.

If the UE fails to decode the BSIC after two successive attempts or if the UE has not been able to re-confirm the BSIC for a GSM cell within $N_{freq}$*60 seconds, the UE shall abort the BSIC re-confirmation attempts for that GSM cell. The GSM cell shall be treated as a new GSM cell with unidentified BSIC and the GSM cell shall be moved to the initial BSIC identification procedure, see TS 36.133 [4] section 8.1.2.4.5.2.2.1. The parameter $N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1.

Reported measurements in periodically triggered measurement reports shall meet the requirements in TS 36.133 [4] section 9.

Reported measurements in event triggered measurement reports shall meet the requirements in TS 36.133 [4] section 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH.

The event triggered reporting delay requirement is valid when the UE for each GSM carrier in the monitored set can take the required number of samples during the measurement period $T_{Measurement\ Period,\ GSM}$ (see TS 36.133 [4] section 8.1.2.4.5.2.1).

The event triggered measurement reporting delay for a GSM cell with verified BSIC , measured without L3 filtering shall be less than 2*$T_{Measurement\ Period,\ GSM}$, where $T_{Measurement\ Period,\ GSM}$ is defined in TS 36.133 [4] section 8.1.2.4.5.2.1. When L3 filtering is used an additional delay can be expected.

Reported measurements contained in event triggered periodic measurement reports shall meet the requirements in TS 36.133 [4] section 9.

The first report in event triggered periodic measurement reporting shall meet the requirements specified in TS 36.133 [4] section 8.1.2.4.5.2.4 Event Triggered Reporting.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.5.2 and A8.10.2

## 8.10.2.4 Test description

### 8.10.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14 for UEs that support receive diversity.
2. The general test parameter settings are set up according to Table 8.10.2.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.10.2.4.3.
5. In these tests, there are two cells, one E-UTRAN cell and one GSM cell, and gap pattern configuration # 0 as defined in TS 36.133 [4] Table 8.1.2.1-1 is provided. Cell 1 (E-UTRA TDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.10.2.4.1-1: General test parameters for E-UTRAN TDD-GSM event triggered reporting when DRX is used in AWGN**

| Parameter | Unit | Test 1 | Test 2 | Comment |
|---|---|---|---|---|
| | | Value | | |
| PDSCH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.0 TDD | | As specified in section A.1.2. Note that UE may only be allocated at *On Duration* |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.6 TDD | | As specified in section A.2.2. |
| Gap Pattern Id | | 0 | | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| Active cell | | Cell 1 | | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | | Cell 2 is on Absolute RF Channel Number 1 (GSM cell) |
| Special subframe configuration | | 6 | | As specified in table 4.2-1 in TS 36.211. |
| Uplink-downlink configuration | | 1 | | As specified in 3GPP TS 36.211 section 4.2 Table 4.2-2 |
| CP length | | Normal | | Applicable to cell 1 |
| E-UTRA RF Channel Number | | 1 | | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| Inter-RAT (GSM) measurement quantity | | GSM Carrier RSSI | | |
| B1-Threshold-GERAN | dBm | -80 | | GSM Carrier RSSI threshold for event B1. |
| Hysteresis | dB | 0 | | |
| TimeToTrigger | s | 0 | | |
| Filter coefficient | | 0 | | L3 filtering is not used |
| PRACH configuration | | 4 | | As specified in table 5.7.1-2 in TS 36.211 |
| Access Barring Information | - | Not Sent | | No additional delays in random access procedure. |
| DRX | | ON | | DRX related parameters are defined in Table 8.10.2.5-2 |
| Monitored GSM cell list size | | 6 GSM neighbours including ARFCN 1 | | List of GSM cells provided before T2 starts. |
| T1 | s | 5 | | |
| T2 | s | 5 | 45 | |

## 8.10.2.4.2 Test procedure

In Test 1 UE needs to be provided at least once every 500ms with new Timing Advance Command MAC control element to restart the Time alignment timer to keep UE uplink time alignment. Furthermore UE is allocated with PUSCH resource at every DRX cycle. In Test 2 the uplink time alignment is not maintained and UE needs to use RACH to obtain UL allocation for measurement reporting.

In the measurement control information, it is indicated to the UE that event-triggered reporting with Event B1 is used. The tests consist of two successive time periods, with time duration of T1, and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table's 8.10.2.5-1 and 8.10.2.5-4. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 8.10.2.5-1 and 8.10.2.5-4. T2 starts.
6. UE shall transmit a MeasurementReport message triggered by Event B1. If the overall delay measured from the beginning of time period T2 is less than 3162 ms for Test1 or less than 44082 ms for Test2 then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.
7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
8. SS shall set a different BSIC on Cell 2, as the previous timing information of Cell 2 is invalid in the UE for next iteration of the test procedure loop.
9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.
11. Repeat step 1-10 for each sub-test in Table 8.10.2.4.1-1 as appropriate.

### 8.10.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.10.2.4.3-1: Common Exception messages for E-UTRAN TDD-GSM event triggered reporting when DRX is used in AWGN**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-7<br>Table H.3.1-10<br>Table H.3.7-1<br>Table H.3.7-2<br>Table H.3.7-3 |

**Table 8.10.2.4.3-2: *SystemInformationBlockType7*: Additional E-UTRAN TDD-GSM event triggered reporting when DRX is used in AWGN**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-3 SystemInformationBlockType7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| commonInfo SEQUENCE { | | | |
| p-MaxGERAN | 33 (33 dBm) | | GSM 400 & GSM 900 & GSM 850 & GSM 700 |
| | 30 (30 dBm) | | DCS 1800 & PCS 1900 |
| } | | | |

**Table 8.10.2.4.3-3: *PRACH-ConfigSIB-DEFAULT*: Additional E-UTRAN TDD-GSM event triggered reporting when DRX is used in AWGN**

| Derivation Path: 36.508 clause 4.6.3 Table 4.6.3-7a PRACH-ConfigSIB-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PRACH-ConfigSIB-DEFAULT ::= SEQUENCE { | | | |
| prach-ConfigInfo SEQUENCE { | | | |
| prach-ConfigIndex | 4 | | TDD |
| } | | | |
| } | | | |

**Table 8.10.2.4.3-4: *RRCConnectionReconfiguration*: Additional E-UTRAN TDD - GSM Event triggered reporting when DRX is used in AWGN**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.10.2.4.3-5: *ReportConfigInterRAT-B1-GERAN*: Additional E-UTRAN TDD-GSM event triggered reporting when DRX is used in AWGN**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-GERAN(GERAN-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| B1-ThresholdGERAN CHOICE { | | | |
| b1-ThresholdGERAN | 30 | GERAN-Thres is actual value in dBm | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 | The actual value is IE value * 0.5 dB INTEGER(0..30) | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.10.2.4.3-6: SchedulingRequest-Config-DEFAULT: Additional E-UTRAN TDD-GSM event triggered reporting when DRX is used in AWGN**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20: SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | Channel-bandwidth-dependent parameter | |
| sr-ConfigIndex | 2 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.10.2.4.3-7: *MeasResults*: Additional E-UTRAN TDD-GSM event triggered reporting when DRX is used in AWGN**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListGERAN | MeasResultListGERAN | | |
| } | | | |
| } | | | |

**Table 8.10.2.4.3-8: *MeasResultListGERAN*: Additional E-UTRAN TDD-GSM event triggered reporting when DRX is used in AWGN**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListGERAN ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| carrierFreq | CarrierFreqGERAN | | |
| physCellId SEQUENCE { | | | |
| networkColourCode | | BIT STRING(SIZE(3)) | |
| baseStationColourCode | | BIT STRING(SIZE(3)) | |
| } | | | |
| measResult SEQUENCE { | | | |
| rssi | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 8.10.2.4.3-9: *CarrierFreqGERAN*: Additional E-UTRAN TDD-GSM event triggered reporting when DRX is used in AWGN**

| Derivation Path: 36.331 clause 6.3.4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CarrierFreqGERAN ::= SEQUENCE { | | | |
| arfcn | | INTEGER (0..1023) | |
| bandIndicator | | ENUMERATED {dcs1800, pcs1900} | |
| } | | | |

## 8.10.2.5 Test requirement

Cell specific test parameters are given in Table 8.10.2.5-1 for E-UTRAN and in Table A.8.10.2.5-4 for GSM. DRX configuration for Test1 and Test2 are given in Table 8.10.2.5-2 and time alignment timer and scheduling request related parameters in Table 8.10.2.5-3.

**Table 8.10.2.5-1: Cell specific test parameters for E-UTRAN TDD (cell #1) event triggered reporting of GSM cell when DRX is used in AWGN**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |

| | | | |
|---|---|---|---|
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | |
| RSRP Note 3 | dBm/15 kHz | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| Propagation Condition | | AWGN | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.<br>Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | |

**Table 8.10.2.5-2: drx-Configuration to be used in E-UTRAN TDD-GSM event triggered reporting when DRX is used in AWGN**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf40 | sf1280 | |
| shortDRX | Disable | Disable | |
| Note: For further information see section 6.3.2 in 3GPP TS 36.331. | | | |

**Table 8.10.2.5-3: *TimeAlignmentTimer* and *sr-ConfigIndex* -Configuration to be used in E-UTRAN TDD-GSM event triggered reporting when DRX is used in AWGN**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | For further information see section 6.3.2 in 3GPP TS 36.331. |
| sr-ConfigIndex | 2 | 2 | For further information see section 6.3.2 in 3GPP TS 36.331 and section10.1 in 3GPP TS 36.213. |

**Table 8.10.2.5-4: Cell specific test parameters for GSM (cell # 2) for event triggered reporting of GSM cell when DRX is used in AWGN**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| Absolute RF Channel Number | | ARFCN 1 | |
| RXLEV | dBm | -Infinity | -75 |
| GSM BSIC | | N/A | Valid |

In Test 1 when DRX cycle length = 40 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event B1 triggered measurement report to Cell 2 on PUSCH. In Test 2 when DRX cycle length = 1280 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE starts to send preambles on the PRACH for Scheduling Request (SR) to obtain allocation to send the measurement report to Cell 2 on PUSCH.

For both tests:

The overall delay may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event B1 measurement report

The overall delay measured when DRX cycle length is 40 ms test requirement is expressed as:

Overall delay measured = $2*T_{Measurement\ Period,\ GSM} + T_{identify,\ GSM}$ + TTI insertion uncertainty + DRX cycle length

$T_{Measurement\ Period,\ GSM}$ = 480 ms (as specified in table 8.1.2.4.5.2.1-1 of TS 36.133 [4] clause 8.1.2.4.5.2.1)

$T_{identify,\ GSM}$ = 2160 ms (as specified in table 8.1.2.4.5.1.2.1-1 of TS 36.133 [4] clause 8.1.2.4.5.1.2.1)

TTI insertion uncertainty = 2 ms

DRX cycle length = 40 ms

The overall delay measured when DRX cycle length is 40 ms shall be less than a total of 3162 ms.

The overall delay measured when DRX cycle length is 1280 ms test requirement is expressed as:

Overall delay measured = $2*T_{Measurement\ Period,\ GSM} + N_{freq} * 30s$ + TTI insertion uncertainty + DRX cycle length

$T_{Measurement\ Period,\ GSM}$ = 6400 ms (as specified in table 8.1.2.4.5.2.1-1 of TS 36.133 [4] clause 8.1.2.4.5.2.1)

$N_{freq}$ = 1 (as specified in TS 36.133 clause 8.1.2.1.1)

$N_{freq}$ * 30 s = 30000 ms (as specified in TS 36.133 [4] clause 8.1.2.4.5.2.2.2 for when DRX cycle length > 40 ms)

TTI insertion uncertainty = 2 ms

DRX cycle length = 1280 ms

The overall delay measured when DRX cycle length is 1280 ms shall be less than a total of 44082 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 8.11 Monitoring of Multiple Layers

## 8.11.1 Multiple E-UTRAN FDD-FDD Inter-frequency event triggered reporting under fading propagation conditions

### 8.11.1.1 Test purpose

To verify that the UE makes correct reporting of multiple events under fading propagation conditions within the E-UTRA FDD inter-frequency cell search requirements.

### 8.11.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward. Applicability requires support for FGI bit 25.

### 8.11.1.3 Minimum conformance requirements

When measurement gaps are scheduled the UE shall be able to identify a new FDD inter-frequency within $T_{Identify_Inter}$ according to the following expression:

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new FDD inter-frequency cell is defined.

$N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1 and $T_{inter1}$ is defined in TS 36.133 [4] section 8.1.2.1

A cell shall be considered detectable provided following conditions are fulfilled:

- RSRP and RSRP Ês/Iot according to Annex I.2.3 for a corresponding Band,
- other RSRP related side conditions given in TS 36.133 [4] Section 9.1 are fulfilled,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band.

When measurement gaps are scheduled for FDD inter frequency measurements the UE physical layer shall be capable of reporting measurements to higher layers with measurement accuracy as specified in TS 36.133 [4] sub-clause 9.1.3 with measurement period given by Table 8.11.1.3-1.

**Table 8.11.1.3-1: RSRP measurement period and measurement bandwidth**

| Configuration | Physical Layer Measurement period: $T_{Measurement_Period_Inter_FDD}$ [ms] | Measurement bandwidth [RB] |
|---|---|---|
| 0 | 480 x $N_{freq}$ | 6 |
| 1 (Note) | 240 x $N_{freq}$ | 50 |
| Note: This configuration is optional | | |

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per FDD inter-frequency for up to 3 FDD inter-frequencies and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period defined in Table 8.11.1.3-1.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_inter}$ defined in TS 36.133 [4] clause 8.1.2.3.1.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period Tidentify_intra defined in TS 36.133 [4] section 8.1.2.2.1.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than TMeasurement_Period Intra provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.1 and A.8.11.1.

## 8.11.1.4 Test description

### 8.11.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.19.

2. The general test parameter settings are set up according to Table 8.11.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are as defined in clause 8.11.1.4.3.

5. In this test, there are three E-UTRA cells, cell1, cell2 and cell3, all on different frequencies. Cell 1 (EUTRA FDD cell on RF channel number 1) is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.11.1.4.1-1: General test parameters for Inter-frequency E-UTRA FDD – E-UTRA FDD and E-UTRA FDD cell search under fading**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| E-UTRA RF Channel Number | | 1, 2, 3 | Three FDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active cell | | Cell 1 | Cell 1 is on RF channel number 1 |
| Neighbour cell | | Cell 2 and cell 3 | Cell 2 is on RF channel number 2 and cell 3 is on RF channel number 3 |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| A3-Offset | dB | -6 | |
| Hysteresis | dB | 0 | |
| CP length | | Normal | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | OFF |
| Time offset between E-UTRAN FDD cells | | 3 ms | Asynchronous cells |
| T1 | s | 5 | |
| T2 | s | 10 | |

### 8.11.1.4.2 Test procedure

This test scenario comprised of 3 E-UTRA FDD cells operating on different frequencies. In the measurement control information, it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time duration of T1, and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2 or cell 3. At T1 the UE is camped on to Cell 1.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.11.1.5-1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.11.1.5-1.

6. UE shall transmit MeasurementReport messages triggered by event A3 for cell 2 and cell 3, respectively.

6a. If the measurement reporting delay for cell 2 from the beginning of the time period T2 is less than 7682 ms the number of “cell 2 successes” is increased by one.

6b. If the measurement reporting delay for cell 3 from the beginning of time period T2 is less than 7682 ms the number of “cell 3 successes” is increased by one.

7. After the SS receives the MeasurementReport messages in step 6 or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. Set Cell 3 physical cell identity = ((current cell 3 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. The pass/ fail decisions are done separated for procedure steps 6a) and 6b). In general the number of repetitions up to the decision are different for 6a) and 6b). If one of 6a) or 6b) is passed, steps 1-8 are repeated, however the counter for the passed event is not anymore served. If one of 6a) or 6b) is failed, the test can be stopped.

### 8.11.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.11.1.4.3-1: Common Exception messages**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7 |

**Table 8.11.1.4.3-2: *MeasConfig-DEFAULT*: Additional Multiple E-UTRAN FDD-FDD inter frequency event triggered reporting under fading propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 *MeasConfig-DEFAULT*: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 3 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | serving frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the neighbouring cell(inter frequency cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | inter frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f3 | f3 is the frequency of the neighbouring cell(inter frequency cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f3) | inter frequency | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-A3 | | |
| reportConfig | ReportConfigEUTRA-A3 | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 2 entry | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-A3 | | |
| } | | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f3 | | |
| reportConfigId | idReportConfig-A3 | | |
| } | | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | MeasGapConfig-GP1 | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |

**Table 8.11.1.4.3-3: *ReportConfigEUTRA-A3*: Additional Multiple E-UTRAN FDD-FDD inter frequency event triggered reporting under fading propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-6 * 0.5 dB) | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.11.1.4.3-4: *MeasResults*: Additional Multiple E-UTRAN FDD-FDD inter frequency event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.11.1.4.3-5: *MeasResults*: Additional Multiple E-UTRAN FDD-FDD inter frequency event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.11.1.4.3-6: *MeasResultListEUTRA*: Additional Multiple E-UTRAN FDD-FDD inter frequency event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| MeasResultEUTRA ::= SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

## 8.11.1.5 Test requirement

Table 8.11.1.4.1-1 and 8.11.1.5-1 define the primary level settings including test tolerances for three E-UTRAN FDD cells.

**Table 8.11.1.5-1: Cell specific test parameters for Inter-frequency E-UTRA FDD – E-UTRA FDD and E-UTRA FDD cell search under fading conditions**

| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | | **Cell 3** | |
|---|---|---|---|---|---|---|---|
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| E-UTRA RF Channel Number | | 1 | | 2 | | 3 | |
| $BW_{channel}$ | MHz | 10 | | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | | 0 | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA[Note 1] | dB | | | | | | |
| OCNG_RB[Note 1] | dB | | | | | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | | | |
| RSRP Note 4 | dBm/15 kHz | -98.00 | -98.00 | -Infinity | -94.40 | -Infinity | -94.40 |
| $\hat{E}_s / I_{ot}$ | dB | 0 | 0 | -Infinity | 3.60 | -Infinity | 3.60 |
| SCH_RP Note 4 | dBm/15 kHz | -98.00 | -98.00 | -Infinity | -94.40 | -Infinity | -94.40 |
| $\hat{E}_s / N_{oc}$ | dB | 0 | 0 | -Infinity | 3.60 | -Infinity | 3.60 |
| Propagation Condition | | AWGN | | ETU70 | | ETU70 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall send one Event A3 triggered measurement report for cell 2 with a measurement reporting delay less than 7682 ms from the beginning of time period T2.

The UE shall send one Event A3 triggered measurement report for cell 3 with a measurement reporting delay less than 7682 ms from the beginning of time period T2.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

The statistical pass/ fail decisions are done separated for cell 2 and cell 3.

Decide the test pass, if events for cell 2 **and** cell 3 are passed, otherwise fail the UE.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured test requirement for Event A3 is expressed as:

Overall delays measured = Measurement reporting delay + TTI insertion uncertainty

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

Measurement reporting delay = $T_{identify_inter}$

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new FDD inter-frequency cell is defined.

$T_{inter1}$=60ms

$N_{freq}$=2.

TTI insertion uncertainty = 2 ms

Therefore, the overall delays measured shall be less than a total of 7682ms in this test case (note: this gives a total of 7680 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

## 8.11.2 E-UTRAN TDD - E-UTRAN TDD and E-UTRAN TDD Inter-frequency event triggered reporting under fading propagation conditions

### 8.11.2.1 Test purpose

To verify that the UE makes correct reporting of two event when doing inter frequency measurements.

### 8.11.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward. Applicability requires support for FGI bit 25.

### 8.11.2.3 Minimum conformance requirements

When measurement gaps are scheduled the UE shall be able to identify a new TDD inter-frequency within $T_{Identify_Inter}$ according to the following expression:

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new TDD inter-frequency cell is defined.

$T_{inter1}$ is defined in TS 36.133 [4] section 8.1.2.1

$N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1.

A cell shall be considered detectable provided following conditions are fulfilled:

- RSRP|$_{dBm}$ and RSRP Ês/Iot according to Annex I.2.3 for a corresponding Band,
- other RSRP related side conditions given in TS 36.133 [4] Section 9.1 are fulfilled,
- SCH_RP|dBm and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band.

When measurement gaps are scheduled for TDD inter frequency measurements the UE physical layer shall be capable of reporting measurements to higher layers with measurement accuracy as specified in TS 36.133 [4] sub-clause 9.1.3 with measurement period ($T_{Measurement_Period_TDD_Inter}$) given by table 8.11.2.3-1.

**Table 8.11.2.3-1: $T_{Measurement_Period_TDD_Inter}$ for different configurations**

| Configuration | Measurement bandwidth [RB] | Number of UL/DL sub-frames per half frame (5 ms) | | DwPTS | | $T_{Measurement_Period_TDD_Inter}$ [ms] |
|---|---|---|---|---|---|---|
| | | DL | UL | Normal CP | Extended CP | |
| 0 | 6 | 2 | 2 | $19760 \cdot T_s$ | $20480 \cdot T_s$ | 480 x $N_{freq}$ |
| 1 (Note 1) | 50 | 2 | 2 | $19760 \cdot T_s$ | $20480 \cdot T_s$ | 240 x $N_{freq}$ |
| Note 1: This configuration is optional.<br>Note 2: $T_s$ is defined in 3GPP TS 36.211 [9]. | | | | | | |

Where:

$T_{inter1}$ is defined in TS 36.133 [4] section 8.1.2.1

$N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1.

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per TDD inter-frequency and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period $T_{Measurement_Period_TDD_Inter}$.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{Identify_Inter}$ defined in TS 36.133 [4] clause 8.1.2.3.2.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra}$ defined in TS 36.133 [4] section 8.1.2.2.2.1 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.2 and A.8.11.2.

## 8.11.2.4 Test description

### 8.11.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.19.

2. The general test parameter settings are set up according to Table 8.11.2.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are as defined in clause 8.11.2.4.3.

5. There are three E-UTRA cells, cell1, cell2 and cell3, all on different frequencies in the test. Cell 1 (EUTRA TDD cell on RF channel number 1) is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.11.2.4.1-1: General test parameters for E-UTRAN TDD - E-UTRAN TDD and E-UTRAN TDD Inter-frequency event triggered reporting under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211. The same configuration in both cells |
| Uplink-downlink configuration | | 1 | As specified in 3GPP TS 36.211 section 4.2 Table 4.2-2 |
| E-UTRA RF Channel Number | | 1, 2, 3 | Three TDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active cell | | Cell 1 | Cell 1 is on RF channel number 1 |
| Neighbour cells | | Cell 2 and Cell 3 | Cell 2 and 3 are on RF channel numbers 2 and 3 respectively |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| A3-Offset | dB | -6 | |
| Hysteresis | dB | 0 | |
| CP length | | Normal | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | OFF |
| Time offset between cells | | 3 μs | Synchronous cells<br>3μs or 92*Ts |
| T1 | s | 5 | |
| T2 | s | 10 | |

### 8.11.2.4.2 Test procedure

This test scenario comprised of 3 E-UTRA TDD cells operating on different frequency. The test consists of two successive time periods, with time duration T1 and T2. In the measurement control information, it is indicated to the UE that event-triggered reporting with Event A3 is used. At T1 the UE is camped on to Cell 1.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.11.2.5-1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.11.2.5-1.

6. UE shall transmit two MeasurementReport message triggered by two events A3 for cell 2 and cell 3, respectively.

6a. If the overall delay measured from the beginning of the time period T2 is less than 7682 ms for event A3 for cell 2 report then the number of “cell 2 successes” is increased by one.

6b. If the overall delay measured from the beginning of time period T2 is less than 7682ms for event A3 for cell 3 report then the number of “cell 3 successes” is increased by one.

7. After the SS receives the MeasurementReport messages in step 6 or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. Set Cell 3 physical cell identity = ((current cell 3 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. The pass/ fail decisions are done separated for procedure steps 6a) and 6b). In general the number of repetitions up to the decision are different for 6a) and 6b). If one of 6a) or 6b) is passed, steps 1-9 are repeated, however the counter for the passed event is not anymore served. If one of 6a) or 6b) is failed, the test can be stopped.

#### 8.11.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.11.2.4.3-1: Common Exception messages**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7 |

**Table 8.11.2.4.3-2: *MeasConfig-DEFAULT*: Additional E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 *MeasConfig-DEFAULT*: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 3 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | serving frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the neighbouring cell(inter frequency cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | inter frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f3 | f3 is the frequency of the neighbouring cell(inter frequency cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f3) | inter frequency | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-A3 | | |
| reportConfig | ReportConfigEUTRA-A3 | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 2 entry | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-A3 | | |
| } | | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f3 | | |
| reportConfigId | idReportConfig-A3 | | |
| } | | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | MeasGapConfig-GP1 | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |

| | | | |
|---|---|---|---|
| } | | | |

**Table 8.11.2.4.3-3: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-6 * 0.5 dB) | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.11.2.4.3-4: *MeasResults*: Additional E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.11.2.4.3-5: *MeasResults*: Additional E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.11.2.4.3-6: *MeasResultListEUTRA*: Additional E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| MeasResultEUTRA ::= SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

## 8.11.2.5 Test requirement

Tables 8.11.2.4.1-1 and 8.11.2.5-1 define the primary level settings including test tolerances for three E-UTRAN TDD cells.

**Table 8.11.2.5-1: Cell specific test parameters for E-UTRAN TDD - E-UTRAN TDD and E-UTRAN TDD Inter-frequency event triggered reporting under fading propagation conditions cells**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | | 3 | |
| $BW_{channel}$ | MHz | 10 | | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | | OP.2 TDD | | OP.2 TDD | |
| PBCH_RA | dB | 0 | | 0 | | 0 | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | | | |
| RSRP Note 4 | dBm/15 kHz | -98.00 | -98.00 | -infinity | -94.40 | -infinity | -94.40 |
| $\hat{E}_s / I_{ot}$ | dB | 0 | 0 | -infinity | 3.60 | -infinity | 3.60 |
| SCH_RP Note 4 | dBm/15 kHz | -98.00 | -98.00 | -infinity | -94.40 | -infinity | -94.40 |
| $\hat{E}_s / N_{oc}$ | dB | 0 | 0 | -infinity | 3.60 | -infinity | 3.60 |
| Propagation Condition | | AWGN | | ETU70 | | ETU70 | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement for Event A3 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_inter}$

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new TDD inter-frequency cell is defined.

$T_{inter1}$=60ms

$N_{freq}$=2.

TTI insertion uncertainty = 2 ms

The UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 7682ms from the beginning of time period T2 (note: this gives a total of 7680 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

The statistical pass/ fail decisions are done separated for cell 2 and cell 3.

Decide the test pass, if the events for cell 2 and cell 3 are passed, otherwise fail the UE.

## 8.11.3 E-UTRAN FDD-FDD Inter-frequency and UTRAN FDD event triggered reporting under fading propagation conditions

### 8.11.3.1 Test purpose

To verify that the UE makes correct reporting of an event when doing inter frequency and UTRAN FDD measurements under fading propagation conditions.

### 8.11.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward. Applicability requires support for FGI bits 22, and 25.

### 8.11.3.3 Minimum conformance requirements

When measurement gaps are scheduled the UE shall be able to identify a new detectable FDD inter-frequency cell within $T_{Identify_Inter}$ according to the following expression:

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new FDD inter-frequency cell is defined.

$N_{freq}$ is defined in section 8.1.2.1.1 and $T_{inter1}$ is defined in section 8.1.2.1

A cell shall be considered detectable provided following conditions are fulfilled:

- RSRP and RSRP Ês/Iot according to Annex I.2.3 for a corresponding Band,
- other RSRP related side conditions given in Section 9.1 are fulfilled,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band.

When measurement gaps are scheduled for FDD inter frequency measurements the UE physical layer shall be capable of reporting measurements to higher layers with measurement accuracy as specified in sub-clause 9.1.3 with measurement period given by Table 8.11.3.3-1.

**Table 8.11.3.3-1: RSRP measurement period and measurement bandwidth**

| Configuration | Physical Layer Measurement period: $T_{Measurement_Period_Inter_FDD}$ [ms] | Measurement bandwidth [RB] |
|---|---|---|
| 0 | 480 x $N_{freq}$ | 6 |
| 1 (Note) | 240 x $N_{freq}$ | 50 |
| Note: This configuration is optional | | |

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per FDD inter-frequency for up to 3 FDD inter-frequencies and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period defined in Table 8.11.3.3-1.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_inter}$ defined in TS 36.133 [4] clause 8.1.2.3.1.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period Tidentify_intra defined in TS 36.133 [4] section 8.1.2.2.1.1 becomes undetectable for a period ≤ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than TMeasurement_Period Intra provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.1 and A.8.11.3.

When explicit neighbour list is provided and no DRX is used the UE shall be able to identify a new detectable FDD UTRA cell belonging to the monitored set within.

$$T_{identify,\ UTRA_FDD} = T_{basic_identify_UTRA_FDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq} \quad ms$$

A cell shall be considered detectable when

- CPICH Ec/Io ≥ -20 dB,
- SCH_Ec/Io ≥ -17 dB for at least one channel tap and SCH_Ec/Ior is equally divided between primary synchronisation code and secondary synchronisation code. When L3 filtering is used an additional delay can be expected.

When measurement gaps are scheduled for UTRA FDD inter RAT measurements the UE physical layer shall be capable of reporting measurements to higher layers with measurement accuracy as specified in TS 36.133 [4] section 9.2 with measurement period given by

$$T_{measurement_UTRA_FDD} = Max\left\{T_{Measurement_Period\ UTRA_FDD}, T_{basic_measurement_UTRA_FDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms$$

If the UE does not need measurement gaps to perform UTRA FDD measurements, the measurement period for UTRA FDD measurements is 480 ms.

The UE shall be capable of performing UTRA FDD CPICH measurements for $X_{basic\ measurementUTRA_FDD}$ inter-frequency cells per FDD frequency and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{Measurement_UTRA_FDD}$.

$X_{basic\ measurement\ UTRA_FDD} = 6$

$T_{Measurement_Period\ UTRA_FDD}$ = 480 ms. The period used for calculating the measurement period $T_{measurement_UTRA_FDD}$ for UTRA FDD CPICH measurements.

$T_{basic_identify_UTRA_FDD}$ = 300 ms. This is the time period used in the inter RAT equation in TS 36.133 [4] section 8.1.2.4.1.1.1 where the maximum allowed time for the UE to identify a new UTRA FDD cell is defined.

$T_{basic_identify_enhanced_UTRA_FDD}$ = 60 ms. This is the time period used in the inter RAT equation in section 8.1.2.4.1.1.1a where the maximum allowed time for the UE to identify a new UTRA FDD cell is defined.

$T_{basic_measurement_UTRA_FDD}$ = 50 ms. This is the time period used in the equation for defining the measurement period for inter RAT CPICH measurements.

$N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1 and $T_{inter1}$ is defined in TS 36.133 [4] section 8.1.2.1

Reported measurements in periodically triggered measurement reports shall meet the requirements in TS 36.133 [4] section 9.

Reported measurements in event triggered measurement reports shall meet the requirements in TS 36.133 [4] section 9.

The UE shall not send any event triggered measurement reports, as long as the reporting criteria is not fulfilled.

The measurement reporting delay is defined as the time between any event that will trigger a measurement report until the UE starts to transmit the measurement report over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify,\ UTRA_FDD}$ defined in TS 36.133 [4]section 8.1.2.4.1.1.1 for the minimum requirements or $T_{identify,\ enhanced_UTRA_FDD}$ defined in TS 36.133 [4] section 8.1.2.4.1.1.1a for the enhanced requirements When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify,\ UTRA_FDD}$ defined in TS 36.133 [4] section 8.1.2.4.1.1.1 for the minimum requirements or $T_{identify,\ enhanced_UTRA_FDD}$ defined in TS 36.133 [4] section 8.1.2.4.1.1.1a for the enhanced requirements and then enters or leaves the reporting range, the event triggered measurement reporting delay shall be less than $T_{measurement_UTRA_FDD}$ defined in TS 36.133 [4] section 8.1.2.4.1.1.2 provided the timing to that cell has not changed more than ± 32 chips while measurement gap has not been available and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

Reported measurements contained in event triggered periodic measurement reports shall meet the requirements in TS 36.133 [4] section 9.

The first report in event triggered periodic measurement reporting shall meet the requirements specified in TS 36.133 [4] section 8.1.2.4.1.1.4 Event Triggered Reporting.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.1 and A.8.11.3.

## 8.11.3.4 Test description

### 8.11.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.24.

2. The general test parameter settings are set up according to Table 8.11.3.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are as defined in clause 8.11.3.4.3.

5. In this test, there are two cells on different carrier frequencies and one cell on UTRAN carrier frequency. Cell 1 (EUTRA FDD cell on RF channel number 1) is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.11.3.4.1-1: General test parameters for Combined inter-frequency and UTRAN event triggered reporting in fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cells | | Cell 2, 3 | Cell 2 is on E-UTRA RF channel number 2. Cell 3 is on UTRA RF channel number 1. |
| CP length | | Normal | Applicable to cell 1 |
| E-UTRA RF Channel Number | | 1, 2 | Two FDD carrier frequencies are used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| UTRA RF Channel Number | | 1 | One UTRA FDD carrier frequency is used. |
| E-UTRAN FDD measurement quantity | | RSRP | |
| Inter-RAT (UTRA FDD) measurement quantity | | CPICH Ec/N0 | |
| A3-Offset | dB | -6 | |
| b2-Threshold-E-UTRA | dB | -86 | RSRP threshold for event B2. |
| b2-Threshold-UTRA | dB | -18 | CPICH Ec/N0 threshold for event B2. |
| Hysteresis | dB | 0 | |
| Time To Trigger | ms | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used. |
| DRX | | OFF | |
| Monitored UTRA FDD cell list size | | 12 | UTRA cells on UTRA RF channel 1 provided in the cell list. |
| Time offset between cells | | 3 ms | Asynchronous cells |
| T1 | s | 5 | |
| T2 | s | 8 | |

### 8.11.3.4.2 Test procedure

In the measurement control information, it is indicated to the UE that event-triggered reporting with Event A3 and B2 is used. The test consists of two successive time periods, with time duration of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2 and cell 3. At T1 the UE is camped on to Cell 1.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.11.3.5-1. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.11.3.5-1.

6. UE shall transmit MeasurementReport messages triggered by event A3 and B2.

6a. If the measurement reporting delay for event A3 from the beginning of the time period T2 is less than 7682 ms the number of “A3 successes” is increased by one.

6b. If the measurement reporting delay for event B2 from the beginning of time period T2 is less than 4802 ms the number of “B2 successes” is increased by one.

7. After the SS receives the MeasurementReport messages in step 6 or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. Set Cell 3 primary scrambling code = ((current cell 3 primary scrambling code - 50) mod 200 + 100) for the next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. The pass/ fail decisions are done separated for procedure steps 6a) and 6b). In general the number of repetitions up to the decision are different for 6a) and 6b). If one of 6a) or 6b) is passed, steps 1-8 are repeated, however the counter for the passed event is not anymore served. If one of 6a) or 6b) is failed, the test can be stopped.

### 8.11.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.11.3.4.3-1: Common Exception messages**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7 |

**Table 8.11.3.4.3-2: *MeasConfig-DEFAULT*: Additional E-UTRAN FDD-FDD Inter-frequency and UTRAN FDD event triggered reporting under fading propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 *MeasConfig-DEFAULT*: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 3 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell (E-UTRA serving cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | serving frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the neighbouring cell(E-UTRA inter frequency cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | inter frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f3 | f3 is the frequency of the neighbouring cell(UTRA cell) | |
| measObject CHOICE { | | | |
| MeasObjectUTRA | MeasObjectUTRA-GENERIC(f3) | inter frequency | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 2 entry | | |
| reportConfigToAddMod ::= SEQUENCE { | | | |
| reportConfigId | idReportConfig-A3 | | |
| reportConfig | ReportConfigEUTRA-A3 | | |
| } | | | |
| eportConfigToAddMod ::= SEQUENCE { | | | |
| reportConfigId | idReportConfig-B2 | | |
| reportConfig | ReportConfigInterRAT-B2-UTRA | | |
| } | | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 2 entry | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-A3 | | |
| } | | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f3 | | |
| reportConfigId | idReportConfig-B2 | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | MeasGapConfig-GP1 | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 8.11.3.4.3-3: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD Inter-frequency and UTRAN FDD event triggered reporting under fading propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-6 * 0.5 dB) | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.11.3.4.3-4: *ReportConfigInterRAT-B2-UTRA*: Additional E-UTRAN FDD-FDD Inter-frequency and UTRAN FDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.508 clause 4.6.6 Table 4.6.6-7B | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B2-UTRA(UTRA-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB2 SEQUENCE { | | | |
| b2-Threshold1 CHOICE { | | | |
| threshold-RSRP | 55(-86dBm) | INTEGER(0..97), the mapping table is Table 9.1.4-1 in [36.133] | |
| b2-Threshold2 CHOICE { | | | |
| B2-Threshold-UTRA CHOICE { | | | |
| thresholdUTRA- EcN0 | 13 (-18dBm) | UTRA-Thres is actual CPICH Ec/N0 value in dBm | UTRA-FDD |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | ms0 | Value range FFS | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.11.3.4.3-5 *MeasResults*: Additional E-UTRAN FDD-FDD Inter-frequency and UTRAN FDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | [1] | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.11.3.4.3-6: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD Inter-frequency and UTRAN FDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| MeasResultEUTRA ::= SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 8.11.3.4.3-7: *MeasuredResults*: Additional E-UTRAN FDD-FDD Inter-frequency and UTRAN FDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | [2] | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 8.11.3.4.3-8: *MeasResultListUTRA*: Additional E-UTRAN FDD-FDD Inter-frequency and UTRAN FDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| fdd | PhysCellIdUTRA-FDD | INTEGER (0..127) | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra- EcN0 | | Set according to specific test INTEGER (-5..91) | |
| } | | | |
| } | | | |

## 8.11.3.5 Test requirement

Table 8.11.3.5-1 and 8.11.3.5-2 define the primary level settings including test tolerances for two E-UTRAN FDD cells and one UTRAN FDD cell.

**Table 8.11.3.5-1: Cell specific test parameters for Combined inter-frequency and UTRAN event triggered reporting in fading propagation conditions**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 | | 1x2 Low | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |

| | | | | | |
|---|---|---|---|---|---|
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | |
| RSRP Note 4 | dBm/15 kHz | -95.05 | -95.05 | -Infinity | -91.0 |
| $\hat{E}_s / I_{ot}$ | dB | 2.95 | 2.95 | -Infinity | 7.0 |
| SCH_RP Note 4 | dBm/15 kHz | -95.05 | -95.05 | -Infinity | -91.0 |
| $\hat{E}_s / N_{oc}$ | dB | 2.95 | 2.95 | -Infinity | 7.0 |
| Propagation Condition | | AWGN | | ETU70 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.11.3.5-2: Cell specific test parameters for UTRAN FDD (cell # 3) for event triggered reporting of UTRAN FDD cell under fading propagation conditions**

| Parameter | Unit | Cell 3 | |
|---|---|---|---|
| | | T1 | T2 |
| UTRA RF Channel Number | | 1 | |
| CPICH_Ec/Ior | dB | -10 | |
| PCCPCH_Ec/Ior | dB | -12 | |
| SCH_Ec/Ior | dB | -12 | |
| PICH_Ec/Ior | dB | -15 | |
| DPCH_Ec/Ior | dB | N/A | |
| OCNS | | -0.941 | |
| $\hat{I}_{or} / I_{oc}$ | dB | -Infinity | -1.8 |
| $I_{oc}$ | dBm/3.84 MHz | -70 | |
| CPICH_Ec/Io | dB | -Infinity | -14 |
| Propagation Condition | | Case 5 (Note 3) | |

Note 1: The DPCH level is controlled by the power control loop.
Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.
Note 3: Case 5 propagation conditions are defined in Annex A of 3GPP TS 25.101.

The UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 7682ms from the beginning of time period T2.

The UE shall send one Event B2 triggered measurement report, with a measurement reporting delay less than 4802 ms from the beginning of time period T2.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

The statistical pass/ fail decisions are done separated for event A3 and event B2.

Decide the test pass, if events A3 and B2 are passed, otherwise fail the UE.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement for Event A3 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_inter}$

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new FDD inter-frequency cell is defined.

$T_{inter1}$=60ms

$N_{freq}$=2.

TTI insertion uncertainty = 2 ms

Therefore, the overall delays of Event A3 triggered measurement report measured shall be less than a total of 7682ms in this test case (note: this gives a total of 7680 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

Similarly, the overall delays measured test requirement for Event B2 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify,\ UTRA_FDD}$

$$T_{identify,\ UTRA_FDD} = T_{basic_identify_UTRA_FDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq} \quad ms$$

$T_{basic_identify_UTRA_FDD}$ = 300 ms

$T_{Inter1}$ = 30 ms

$N_{Freq}$ = 1

TTI insertion uncertainty = 2 ms

Therefore, the overall delays of Event B2 triggered measurement report measured shall be less than a total of 4802 ms in this test case (note: this gives a total of 4800 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

## 8.11.4 InterRAT E-UTRA TDD to E-UTRA TDD and UTRA TDD cell search

### 8.11.4.1 Test purpose

To verify that the UE makes correct reporting of an event when doing inter frequency measurements and UTRA TDD measurements.

### 8.11.4.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 that support UTRA TDD. Applicability requires support for FGI bits 22, and 25.

This test applies to all types of E-UTRA TDD UE release 9 and forward that support UTRA TDD and not supporting UTRA FDD. Applicability requires support for FGI bits 22, and 25.

This test applies to all types of E-UTRA TDD UE release 9 and forward that support UTRA TDD. Applicability requires support for FGI bits 39, and 25.

### 8.11.4.3 Minimum conformance requirements

When measurement gaps are scheduled the UE shall be able to identify a new TDD inter-frequency within $T_{Identify_Inter}$ according to the following expression:

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new TDD inter-frequency cell is defined.

$T_{inter1}$ is defined in TS 36.133 [4] section 8.1.2.1

$N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1.

A cell shall be considered detectable provided following conditions are fulfilled:

- $RSRP|_{dBm}$ and RSRP Ês/Iot according to Annex I.2.3 for a corresponding Band,
- other RSRP related side conditions given in TS 36.133 [4] Section 9.1 are fulfilled
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band.

When measurement gaps are scheduled for TDD inter frequency measurements the UE physical layer shall be capable of reporting measurements to higher layers with measurement accuracy as specified in TS 36.133 sub-clause 9.1.3 with measurement period ($T_{Measurement_Period_TDD_Inter}$) given by table 8.11.4.3-1.

**Table 8.11.4.3-1: $T_{Measurement_Period_TDD_Inter}$ for different configurations**

| Configuration | Measurement bandwidth [RB] | Number of UL/DL sub-frames per half frame (5 ms) | | DwPTS | | $T_{Measurement_Period_TDD_Inter}$ [ms] |
|---|---|---|---|---|---|---|
| | | DL | UL | Normal CP | Extended CP | |
| 0 | 6 | 2 | 2 | $19760 \cdot T_s$ | $20480 \cdot T_s$ | 480 x $N_{freq}$ |
| 1 (Note 1) | 50 | 2 | 2 | $19760 \cdot T_s$ | $20480 \cdot T_s$ | 240 x $N_{freq}$ |
| Note 1: This configuration is optional.<br>Note 2: $T_s$ is defined in 3GPP TS 36.211 [9]. | | | | | | |

Where:

$T_{inter1}$ is defined in TS 36.133 [4] section 8.1.2.1

$N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1.

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per TDD inter-frequency and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period $T_{Measurement_Period_TDD_Inter}$.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: [2] x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{Identify_Inter}$ defined in TS 36.133 [4] clause 8.1.2.3.2.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra}$ defined in TS 36.133 [4] section 8.1.2.2.1.1 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ Intra}$ provided the timing to that cell has

not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.2.

The measurement reporting delay shall be less than $T_{identify,\ UTRA_TDD}$ in RRC_CONNECTED state.

When explicit neighbour list is provided and no DRX is used the UE shall be able to identify a new detectable cell belonging to the monitored set within

$$T_{identify,\ UTRA_TDD} = Max\left\{5000, T_{basic\ identify\ UTRA_TDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms$$

If the UE does not require transmit gap to perform inter-RAT UTRA TDD measurements, the UE shall be able to identify a new detectable inter-RAT UTRA TDD cell belonging to the monitored set within 5000 ms.

A cell shall be considered detectable when

- P-CCPCH Ec/Io $\geq$ -8 dB,
- DwPCH_Ec/Io $\geq$ -5 dB.

When L3 filtering is used an additional delay can be expected.

When measurement gaps are scheduled for UTRA TDD inter RAT measurements the UE physical layer shall be capable of reporting measurements to higher layers with measurement accuracy as specified in TS 36.133 [4] clause 9.3 with measurement period given by

$$T_{measurement\ UTRA_TDD} = Max\left\{T_{Measurement_Period\ UTRA_TDD}, T_{basic\ measurement\ UTRA_TDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms$$

If the UE does not need measurement gaps to perform UTRA TDD measurements, the measurement period for UTRA TDD measurements is 480 ms.

The UE shall be capable of performing UTRA TDD P-CCPCH RSCP measurements for $X_{basic\ measurementUTRA_TDD}$ inter-frequency cells per TDD frequency of the monitored set and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{Measurement_UTRA_TDD.}$

$X_{basic\ measurementUTRA_TDD} = 6$

$T_{Measurement_Period\ UTRA_TDD}$ = 480 ms is the period used for calculating the measurement period $T_{measurement_UTRA_TDD}$ for UTRA TDD P-CCPCH RSCP measurements.

$T_{basic_identify_UTRA_TDD}$ = 800 ms is the time period used in the inter RAT equation where the maximum allowed time for the UE to identify a new UTRA TDD cell is defined.

$T_{basic_measurement_UTRA_TDD}$ = 50 ms is the time period used in the equation for defining the measurement period for inter RAT P-CCPCH RSCP measurements.

$N_{freq}$ and $T_{inter1}$ are defined in TS 36.133 [4] section 8.1.2.1.1

Reported measurements in periodically triggered measurement reports shall meet the requirements in TS 36.133 [4] section 9.

Reported measurements in event triggered measurement reports shall meet the requirements in TS 36.133 [4] section 9.

The UE shall not send any event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The measurement reporting delay is defined as the time between any events that will trigger a measurement report until the UE starts to transmit the measurement report over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify, UTRA_TDD}$ defined in TS 36.133 [4] Section 8.1.2.4.3.1.1 When L3 filtering is used an additional delay can be expected.

Reported measurements contained in event triggered periodic measurement reports shall meet the requirements in TS 36.133 [4] section 9.

The first report in event triggered periodic measurement reporting shall meet the requirements specified in TS 36.133 [4] section 8.1.2.4.3.1.4 Event Triggered Reporting.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.3.

## 8.11.4.4 Test description

### 8.11.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.24.

2. The general test parameter settings are set up according to Table 8.11.4.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are as defined in clause 8.11.4.4.3.

5. There are two E-UTRA TDD cells operating on different frequency and one UTRA TDD cell specified in the test. Cell 1 (EUTRA TDD cell on RF channel number 1) is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.11.4.4.1-1: General test parameters for combined E-UTRA TDD inter-frequency and UTRA TDD cells search under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Active cell | | Cell 1 | E-UTRA TDD cell is on RF channel number 1 |
| Neighbour cell | | Cell 2 | E-UTRA TDD cell is on RF channel number 2 |
| | | Cell 3 | 1.28Mcps TDD cell |
| CP length of cell1 and cell2 | | Normal | |
| Uplink-downlink configuration of cell1 and cell2 | | 1 | As specified in Table 4.2-2 in TS 36.211. The same configuration in both cells |
| Special subframe configuration of cell1 and cell2 | | 6 | As specified in table 4.2-1 in TS 36.211. The same configuration in both cells |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| E-UTRAN TDD measurement quantity | | RSRP | |
| UTRAN TDD measurement quantity | | RSCP | |
| DRX | | OFF | |
| Ofn | dB | 0 | Parameter for A3 and B2 event |
| Ocn | dB | 0 | Parameter for A3 event |
| Hys | dB | 0 | Parameter for A3 and B2 event |
| Ofs | dB | 0 | Parameter for A3 event |
| Ocs | dB | 0 | Parameter for A3 event |
| A3-Offset | dB | -6 | Parameter for A3 event |
| Thresh1 | dBm | -86 | Absolute E-UTRAN RSRP threshold for event B2 |
| Thresh2 | dBm | -84 | Absolute UTRAN RSCP threshold for event B2 |
| Hysteresis | dB | 0 | |
| TimeToTrigger | dB | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| Time offset between E-UTRAN TDD cells | μs | 3 | Synchronous cells |
| T1 | s | >5 | During T1, cell 2 and cell 3 shall be powered off. During the off time the physical layer cell identity of cell 2 shall be changed, and the scrambling code of cell 3 shall be changed. |
| T2 | s | 15 | |

### 8.11.4.4.2 Test procedure

This test scenario comprised of 2 E-UTRA TDD cells operating on different frequency, and 1 UTRA TDD cell. The test consists of 2 successive time periods, with time duration T1 and T2. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 and B2 shall be used. At T1 the UE is camped on to Cell 1.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.11.4.5-1and Table 8.11.4.5-2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.11.4.5-1and Table 8.11.4.5-2.

6. UE shall transmit a MeasurementReport message triggered by event A3 and B2.

6a. If the overall delays measured from the beginning of time period T2 is less than 7760 ms for event A3 report then the number of "A3 successes" is increased by one.

6b. If the overall delay measured from the beginning of time period T2 is less than 12.88s for event B2 report then the number of "B2 successes" is increased by one.

7. After the SS receives the MeasurementReport messages in step 6 or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. Set Cell 3 cell parameter id = (current Cell 3 cell parameter id+4) mod 16 for the next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. The pass/ fail decisions are done separated for procedure steps 6a) and 6b). In general the number of repetitions up to the decision are different for 6a) and 6b). If one of 6a) or 6b) is passed, steps 1-9 are repeated, however the counter for the passed event is not anymore served. If one of 6a) or 6b) is failed, the test can be stopped.

#### 8.11.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.11.4.4.3-1: Common Exception messages**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7 |

**Table 8.11.4.4.3-2: *MeasConfig-DEFAULT*: Additional E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 *MeasConfig-DEFAULT*: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 3 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell (E-UTRA serving cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | serving frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the neighbouring cell(E-UTRA inter frequency cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | inter frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f3 | f3 is the frequency of the neighbouring cell(UTRA cell) | |
| measObject CHOICE { | | | |
| MeasObjectUTRA | MeasObjectUTRA-GENERIC(f3) | inter frequency | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 2 entry | | |
| reportConfigToAddMod ::= SEQUENCE { | | | |
| reportConfigId | idReportConfig-A3 | | |
| reportConfig | ReportConfigEUTRA-A3 | | |
| } | | | |
| reportConfigToAddMod ::= SEQUENCE { | | | |
| reportConfigId | idReportConfig-B2 | | |
| reportConfig | ReportConfigInterRAT-B2-UTRA | | |
| } | | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 2 entry | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-A3 | | |
| } | | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f3 | | |
| reportConfigId | idReportConfig-B2 | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | MeasGapConfig-GP1 | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 8.11.4.4.3-3: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-6 * 0.5 dB) | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.11.4.4.3-4: *ReportConfigInterRAT-B2-UTRA*: Additional E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.508 clause 4.6.6 Table 4.6.6-8 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B2-UTRA(UTRA-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB2 SEQUENCE { | | | |
| b2-Threshold1 CHOICE { | | | |
| threshold-RSRP | 54(-86dBm) | INTEGER(0..97), the mapping table is Table 9.1.4-1 in [36.133] | |
| b2-Threshold2 CHOICE { | | | |
| B2-Threshold-UTRA CHOICE { | | | |
| thresholdUTRA-RSCP | 31(UTRA-Thres + 115) | UTRA-Thres is actual RSCP value in dBm | UTRA-TDD |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | ms0 | Value range FFS | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.11.4.4.3-5: *MeasResults*: Additional E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.11.4.4.3-6: *MeasResultListEUTRA*: Additional E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| MeasResultEUTRA ::= SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.11.4.4.3-7: *MeasuredResults*: Additional E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 8.11.4.4.3-8: *MeasResultListUTRA*: Additional E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| tdd | PhysCellIdUTRA-TDD | INTEGER (0..127) | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-RSRP | | Set according to specific test INTEGER (-5..91) | |
| } | | | |
| } | | | |

## 8.11.4.5 Test requirement

Tables 8.11.4.5-1 and 8.11.4.5-2 define the primary level settings including test tolerances for two E-UTRAN TDD cells and one UTRAN TDD cell.

**Table 8.11.4.5-1: Cell specific test parameters for combined E-UTRAN TDD inter-frequency and UTRA TDD cell search under fading propagation conditions(cell1 and cell2)**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| BWchannel | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 | | 1x2 Low | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | | OP.2 TDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RANote 1 | dB | | | | |
| OCNG_RBNote 1 | dB | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 2.95 | 2.95 | -Infinity | 7.0 |
| $\hat{E}_s / N_{oc}$ | dB | 2.95 | 2.95 | -Infinity | 7.0 |
| $N_{oc}$ | dBm/15 kHz | -98 | | | |
| RSRP | dBm/15 kHz | -95.05 | -95.05 | -Infinity | -91.0 |
| SCH_RP | dBm/15 kHz | -95.05 | -95.05 | -infinity | -91.0 |
| Propagation Condition | | AWGN | | ETU70 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE priori to the start of time period T2.
Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.11.4.5-2: Cell specific test parameters for combined E-UTRA TDD inter-frequency and UTRA TDD cell search under fading propagation conditions (cell3)**

| Parameter | Unit | Cell 3 (UTRA) | | | |
|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPTS | |
| | | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number* | | Channel 3 | | | |
| PCCPCH_Ec/Ior | dB | -3 | | | |
| DwPCH_Ec/Ior | dB | | | 0 | |
| OCNS_Ec/Ior | dB | -3 | | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -Infinity | 9.0 | -Infinity | 9.0 |
| $I_{oc}$ | dBm/1.28 MHz | -80.4 | | | |
| PCCPCH RSCP | dBm | -Infinity | -74.4 | n.a. | |
| Propagation Condition | | Case 3 | | | |
| Note1: The DPCH of all cells are located in a timeslot other than 0.<br>Note2: In the case of multi-frequency network, the UTRA RF Channel Number can be set for the primary frequency in this test.<br>Note3: P-CCPCH RSCP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | | | |

The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement for Event A3 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_inter}$

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new TDD inter-frequency cell is defined.

$T_{inter1}$=60ms

$N_{freq}$=2.

TTI insertion uncertainty = 2 ms

The UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 7682ms from the beginning of time period T2 (note: this gives a total of 7680 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

Similarly, the overall delays measured test requirement for Event B2 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify,UTRA_TDD}$

$$T_{identify,\ UTRA_TDD} = Max\left\{5000, T_{basic\ identify\ UTRA_TDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms$$

Where:

$T_{basic_identify_UTRA_TDD}$ = 800 ms is the time period used in the inter RAT equation where the maximum allowed time for the UE to identify a new UTRA TDD cell is defined.

$T_{basic_measurement_UTRA_TDD}$ = 50 ms is the time period used in the equation for defining the measurement period for inter RAT P-CCPCH RSCP measurements.

$T_{inter1}$=60ms

$N_{freq}$=2.

TTI insertion uncertainty = 2 ms

The UE shall send one Event B2 triggered measurement report, with a measurement reporting delay less than 1202m8s from the beginning of time period T2 (note: this gives a total of 12.8 s for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

The statistical pass/ fail decisions are done separated for event A3 and event B2.

Decide the test pass, if events A3 **and** B2 are passed, otherwise fail the UE.

## 8.11.5 Combined E-UTRAN FDD - E-UTRA FDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions

### 8.11.5.1 Test purpose

The purpose of this test is to verify that the UE makes correct reporting of multiple events when doing inter frequency and GSM measurements.

### 8.11.5.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward that support GSM. Applicability requires support for FGI bit 23, and 25.

### 8.11.5.3 Minimum conformance requirements

E-UTRAN part:

When measurement gaps are scheduled the UE shall be able to identify a new FDD inter-frequency within $T_{Identify_Inter}$ according to the following expression:

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new FDD inter-frequency cell is defined.

$N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1 and $T_{inter1}$ is defined in TS 36.133 [4] section 8.1.2.1

A cell shall be considered detectable provided following conditions are fulfilled:

- $RSRP|_{dBm}$ and RSRP Ês/Iot according to Annex I.2.3 for a corresponding Band,
- other RSRP related side conditions given in TS 36.133 [4] Section 9.1 are fulfilled,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band.

When measurement gaps are scheduled for FDD inter frequency measurements the UE physical layer shall be capable of reporting measurements to higher layers with measurement accuracy as specified in TS 36.133 [4] sub-clause 9.1.3 with measurement period given by table 8.11.5.3-1.

**Table 8.11.5.3-1: RSRP measurement period and measurement bandwidth**

| Configuration | Physical Layer Measurement period: $T_{Measurement_Period_Inter_FDD}$ [ms] | Measurement bandwidth [RB] |
|---|---|---|
| 0 | 480 x $N_{freq}$ | 6 |
| 1 (Note) | 240 x $N_{freq}$ | 50 |
| Note: This configuration is optional | | |

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per FDD inter-frequency for up to 3 FDD inter-frequencies and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period defined in Table 8.11.5.3-1.

Reported measurements contained in event triggered periodic measurement reports shall meet the requirements in TS 36.133 [4] section 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ -inter}$ defined in TS 36.133 [4] section 8.1.2.3.1.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_inter}$ defined in TS 36.133 [4] section 8.1.2.3.1.1 and then triggers the measurement report as per TS 36.331 [5], the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_Inter_FDD}$ defined in TS 36.133 [4] section 8.1.2.3.1.1 provided the timing to that cell has not changed more than ± 50 Ts while measurement gap has not been available and the L3 filter has not been used.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.1.

GSM part:

Measurements on GSM cells can be requested with BSIC verified.

In RRC_CONNECTED state when a supported measurement gap pattern sequence according to Table 8.1.2.1-1 in TS 36.133 [4] is configured by E-UTRAN the UE shall continuously measure GSM cells, search for new GSM cells given in the monitored set and re-confirm the BSIC for already detected cells.

This measurement shall be based on measurement gaps allocated for GSM carrier RSSI measurement as described in section 8.1.2.1 in TS 36.133 [4] A UE supporting GSM measurements shall measure minimum number of 10 GSM carrier RSSI measurement samples ($N_{GSM\ carrier\ RSSI}$) per measurement gap. In RRC_CONNECTED state the measurement period, $T_{Measurement\ Period,\ GSM}$, for the GSM carrier RSSI measurement is $N_{freq}$ * 480 ms. The parameter $N_{freq}$ is defined in clause 8.1.2.1.1 of TS 36.133 [4] as:

$$N_{freq} = N_{freq,\ E\text{-}UTRA} + N_{freq,\ UTRA} + M_{gsm} + N_{freq,\ cdma2000} + N_{freq,\ HRPD}$$

Where:

$N_{freq,\ E\text{-}UTRA}$ is the number of E-UTRA carriers being monitored

$N_{freq,\ UTRA}$ is the number of UTRA carriers being monitored

$N_{freq,\ cdma2000}$ is the number of cdma2000 carriers being monitored

$N_{freq,\ HRPD}$ is the number of HRPD carriers being monitored

$M_{GSM}$ is an integer which is a function of the number of GSM carriers on which measurements are being performed. $M_{GSM}$ is equal to 0 if no GSM carrier is being monitored. For a Measurement Gap Repetition Period (MGRP) of 40 ms, $M_{GSM}$ is equal to 1 if cells on up to 32 GSM carriers are being measured. For a MGRP of 80 ms, $M_{GSM}$ is equal to ceil ($N_{carriers,\ GSM}$ /20) where $N_{carriers,\ GSM}$ is the number of GSM carriers on which cells are being measured

The UE shall meet the measurement accuracy requirements stated for RXLEV in TS 45.008 [15], when the given measurement time allows the UE to take at least 3 GSM carrier RSSI samples per GSM carrier in the monitored set during the measurement period.

In case the UE is not able to acquire the required number of samples per GSM carrier during one measurement period, the UE shall measure as many GSM carriers as possible during that measurement period using at least 3 samples per GSM carrier. The GSM carriers that were not measured during that measurement period shall be measured in the following measurement periods.

BSIC verification:

Measurements on a GSM cell can be requested with BSIC verified. The UE shall be able to report the GSM cells with BSIC verified for those cells where the verification of BSIC has been successful.

The procedure for BSIC verification on a GSM cell can be divided into the following two tasks:

- **Initial BSIC identification:** Includes searching for the BSIC and decoding the BSIC for the first time when there is no knowledge about the relative timing between the E-UTRAN FDD and GSM cells. The UE shall trigger the initial BSIC identification within the available measurement gap pattern sequence. The requirements for BSIC re-confirmation can be found in section 8.1.2.4.5.1.2.1 of TS 36.133 [4].

- **BSIC re-confirmation:** Tracking and decoding the BSIC of a GSM cell after initial BSIC identification is performed. The UE shall trigger the BSIC re-confirmation within the available measurement gap pattern. The requirements for BSIC re-confirmation can be found in section 8.1.2.4.5.1.2.2 of TS 36.133 [4].

If the network requests measurements on a GSM cell, the UE shall behave as follows:

- The UE shall perform GSM carrier RSSI measurements according to section 8.1.2.4.5.1 in TS 36.133 [4] when a measurement gap pattern sequence is activated.

- The UE shall perform measurement reporting as defined in TS 36.331 [5].

- The UE shall perform BSIC identification if BSIC verified measurements are activated by RRC. The UE shall use the most recently available GSM carrier RSSI measurement results for arranging GSM cells in signal strength order for performing BSIC identification.

- The UE shall perform BSIC re-confirmation on all the GSM cells that have been successfully identified.

- The UE shall perform all configured event evaluation for event-triggered reporting after the BSIC has been verified for a GSM cell. The UE shall use the most recently available GSM carrier RSSI measurement results in event evaluation and event-triggered reporting.

- Event-triggered and periodic reports shall be triggered according to TS 36.331 [5].

The BSIC of a GSM cell is considered to be “verified” if the UE has decoded the SCH of the BCCH carrier and identified the BSIC at least one time (initial BSIC identification). Once a GSM cell has been identified the BSIC shall be re-confirmed at least once every $8*T_{re\text{-}confirm,GSM}$ seconds. Otherwise the BSIC of the GSM cell is considered as "non-verified". If a measurement gap pattern sequence is deactivated by the network after BSIC has been identified or verified, the UE shall consider the BSIC as non-verified.

$T_{identify,GSM}$ indicates the maximum time allowed for the UE to decode the unknown BSIC of the GSM cell in one GSM BCCH carrier in the initial BSIC identification procedure.

$T_{re\text{-}confirm,GSM}$ indicates the maximum time allowed for the re-confirmation of the BSIC of one GSM cell in the BSIC re-confirmation procedure.

The UE shall be able to decode a BSIC within a measurement gap when the time difference between the middle of the received GSM synchronisation burst at the UE and the middle of the effective measurement gap is within the limits specified in table 8.1.2.4.5.1.2-1 of TS 36.133 [4].

The UE shall be able to perform BSIC verification at levels down to the reference sensitivity level or reference interference levels as specified in TS 45.005 [16].

This measurement shall be based on the measurement gaps used for Initial BSIC identification as described in section 8.1.2.4.5.1.2 of TS 36.133 [4].

The UE shall continuously attempt to decode the BSIC of SCH on the BCCH carrier of the 8 strongest BCCH carriers of the GSM cells indicated in the Inter-RAT cell info list. The UE shall give priority for BSIC decoding attempts in decreasing signal strength order to BCCH carriers with unknown BSIC. The strongest BCCH carrier is defined as the BCCH carrier having the highest measured GSM carrier RSSI value.

If the BSIC of the GSM BCCH carrier has been successfully decoded the UE shall immediately continue BSIC identification with the next GSM BCCH carrier, in signal strength order, with unknown BSIC. The GSM cell for which the BSIC has been successfully identified shall be moved to the BSIC re-confirmation procedure.

If the UE has not successfully decoded the BSIC of the GSM BCCH carrier within $T_{identify,GSM}$ ms, the UE shall abort the BSIC identification attempts for that GSM BCCH carrier. The UE shall continue to try to perform BSIC identification of the next GSM BCCH carrier in signal strength order. The GSM BCCH carrier for which the BSIC identification failed shall not be re-considered for BSIC identification until BSIC identification attempts have been made for all the rest of the 8 strongest GSM BCCH carriers in the monitored set with unknown BSIC.

$T_{identify,GSM}$ values are given for a set of reference gap patterns in table 8.1.2.4.5.1.2.1-1 of TS 36.133 [4]. The requirements in the table represent the time required to guarantee at least two attempts to decode the BSIC for one GSM BCCH carrier.

Reported measurements in event triggered measurement reports shall meet the requirements in section TS 36.331 [5].

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH.

The event triggered reporting delay requirement is valid when the UE for each GSM carrier in the monitored set can take the required number of samples during the measurement period $T_{Measurement\ Period,\ GSM}$ (see section 8.1.2.4.5.1 of TS 36.133 [4]).

The event triggered measurement reporting delay for a GSM cell with verified BSIC , measured without L3 filtering shall be less than $2*T_{Measurement\ Period,\ GSM}$, where $T_{Measurement\ Period,\ GSM}$ is defined in section 8.1.2.4.5.1. of TS 36.133 [4] When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.1, 8.1.2.4.5 and A.8.11.5

## 8.11.5.4 Test description

### 8.11.5.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), AWGN noise source and Fader to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.25.

2. The general test parameter settings are set up according to Table 8.11.5.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are as defined in clause 8.11.5.4.3.

5. There are two E-UTRA FDD cells operating on different frequency and one GSM cell specified in the test. Cell 1 (EUTRA FDD cell on RF channel number 1) is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.11.5.4.1-1: General test parameters for Combined E-UTRAN FDD - E-UTRA FDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cells | | Cell 2, 3 | Cell 2 is on E-UTRA RF channel number 2. Cell 3 is on Absolute RF Channel Number 3 (GSM cell). |
| CP length | | Normal | Applicable to cell 1 and cell 2 |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| E-UTRAN FDD measurement quantity | | RSRP | |
| Hysteresis | dB | 0 | Parameter for A3 and B2 event |
| A3-Offset | dB | -6 | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | OFF |
| Time offset between E-UTRAN FDD cells | ms | 3 ms | Asynchronous cells |
| Inter-RAT (GSM) measurement quantity | | GSM Carrier RSSI | |
| b2-Threshold-E-UTRA | dBm | -83 | RSRP threshold for event B2. This is the threshold for E-UTRA in the B2 configuration. E-UTRA Pcell RSRP is below this throughout the test to account for measurement accuracy and fading |
| b2-Threshold-GERAN | dBm | -80 | GSM Carrier RSSI threshold for event B2. |
| Monitored GSM cell list size | | 6 GSM neighbours including ARFCN 3 | List of GSM cells provided before T2 starts. |
| T1 | s | 5 | |
| T2 | s | 10 | |

### 8.11.5.4.2 Test procedure

This test scenario comprised of 2 E-UTRA FDD cells operating on different frequency, and 1 GSM cell. The test consists of 2 successive time periods, with time duration T1 and T2. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 and B2 shall be used. At T1 the UE is camped on to Cell 1.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.11.5.5-1and Table 8.11.5.5-2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.11.5.5-1and Table 8.11.5.5-2.

6. UE shall transmit a MeasurementReport message triggered by event A3 and B2.

6a. If the overall delays measured from the beginning of time period T2 is less than 7682 ms for event A3 report then the number of “A3 successes” is increased by one.

6b. If the overall delays measured from the beginning of time period T2 is less than 7202 ms for event B2 report then the number of “B2 successes” is increased by one.

7. After the SS receives the MeasurementReport messages in step 6 or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. The SS shall set a different BSIC on Cell 3, as the previous timing information of Cell 3 is invalid in the UE for the next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. The pass/ fail decisions are done separated for procedure steps 6a) and 6b). In general the number of repetitions up to the decision are different for 6a) and 6b). If one of 6a) or 6b) is passed, steps 1-9 are repeated, however the counter for the passed event is not anymore served. If one of 6a) or 6b) is failed, the test can be stopped.

### 8.11.5.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.11.5.4.3-1: Common Exception messages for Combined E-UTRAN FDD – E-UTRA FDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7 |

**Table 8.11.5.4.3-2: *MeasConfig-DEFAULT*: Additional Combined E-UTRAN FDD – E-UTRA FDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 8 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell (E-UTRA serving cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | serving frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the neighbouring cell(E-UTRA inter frequency cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | inter frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f3 | f3 is the frequency of the neighbouring cell(GERAN cell) | |
| measObject CHOICE { | | | |
| MeasObjectGERAN | MeasObjectGERAN-GENERIC(f3) | inter frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f4 | f4 is the frequency of the neighbouring cell(GERAN cell) | |
| measObject CHOICE { | | | |
| MeasObjectGERAN | MeasObjectGERAN-GENERIC(f4) | inter frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f5 | f5 is the frequency of the neighbouring cell(GERAN cell) | |
| measObject CHOICE { | | | |
| MeasObjectGERAN | MeasObjectGERAN-GENERIC(f5) | inter frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f6 | f6 is the frequency of the neighbouring cell(GERAN cell) | |
| measObject CHOICE { | | | |
| MeasObjectGERAN | MeasObjectGERAN- | inter frequency | |

| | | | |
|---|---|---|---|
| | GENERIC(f6) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f7 | f7 is the frequency of the neighbouring cell(GERAN cell) | |
| measObject CHOICE { | | | |
| MeasObjectGERAN | MeasObjectGERAN-GENERIC(f7) | inter frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f8 | f8 is the frequency of the neighbouring cell(GERAN cell) | |
| measObject CHOICE { | | | |
| MeasObjectGERAN | MeasObjectGERAN-GENERIC(f8) | inter frequency | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 2 entry | | |
| reportConfigToAddMod ::= SEQUENCE { | | | |
| reportConfigId | idReportConfig-A3 | | |
| reportConfig | ReportConfigEUTRA-A3 | | |
| } | | | |
| reportConfigToAddMod ::= SEQUENCE { | | | |
| reportConfigId | idReportConfig-B2 | | |
| reportConfig | ReportConfigInterRAT-B2-GERAN | | |
| } | | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 2 entry | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-A3 | | |
| } | | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f3 | | |
| reportConfigId | idReportConfig-B2 | | |
| } | | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | MeasGapConfig-GP1 | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 8.11.5.4.3-3: *ReportConfigEUTRA-A3*: Additional Combined E-UTRAN FDD – E-UTRA FDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.11.5.4.3-4: *ReportConfigInterRAT-B2-GERAN*: Additional Combined E-UTRAN FDD – E-UTRA FDD Inter-frequency and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6 Table 4.6.6-7E | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B2-GERAN(EUTRA-Thres, GERAN-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB2 SEQUENCE { | | | |
| b2-Threshold1 CHOICE { | | | |
| threshold-RSRP | 57(-83dBm) | INTEGER(0..97), the mapping table is Table 9.1.4-1 in TS 36.133 [4] | |
| } | | | |
| b2-Threshold2 CHOICE { | | | |
| b2-Threshold2-GERAN | 30 (-80dBm) | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | INTEGER(0..30), 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.11.5.4.3-5 *MeasResults*: Additional Combined E-UTRAN FDD – E-UTRA FDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.11.5.4.3-6: *MeasResultListEUTRA*: Additional Combined E-UTRAN FDD – E-UTRA FDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| MeasResultEUTRA ::= SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.11.5.4.3-7: *MeasuredResults*: Additional Combined E-UTRAN FDD – E-UTRA FDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | RSRP-Range | Set according to specific test | |
| rsrqResult | RSRQ-Range | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListGERAN | MeasResultListGERAN | | |
| } | | | |
| } | | | |

**Table 8.11.5.4.3-8: *MeasResultListGERAN*: Additional Combined E-UTRAN FDD – E-UTRA FDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListGERAN ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| carrierFreq | CarrierFreqGERAN | | |
| physCellId | physCellId GERAN | | |
| measResult SEQUENCE { | | | |
| rssi | INTEGER (0..63) | Set according to specific test | |
| } | | | |
| } | | | |

## 8.11.5.5 Test requirement

Tables 8.11.5.5-1 and 8.11.5.5-2 define the primary level settings including test tolerances for two E-UTRAN FDD cells and one GSM cell.

**Table 8.11.5.5-1: Cell specific test parameters for Combined E-UTRAN FDD - E-UTRA FDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions (cell1 and cell2)**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | -98 | |
| RSRP Note 4 | dBm/15 kHz | -94.20 | -94.20 | -Infinity | -90.50 |
| $\hat{E}_s / I_{ot}$ | dB | 3.80 | 3.80 | -Infinity | 7.50 |
| SCH_RP Note 4 | dBm/15 kHz | -94.20 | -94.20 | -Infinity | -90.50 |
| $\hat{E}_s / N_{oc}$ | dB | 3.80 | 3.80 | -Infinity | 7.50 |
| Propagation Condition | | ETU70 | | ETU70 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.11.5.5-2: Cell specific test parameters for Combined E-UTRAN FDD - E-UTRA FDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions (cell3)**

| Parameter | Unit | Cell 3 | |
|---|---|---|---|
| | | T1 | T2 |
| Absolute RF Channel Number | | ARFCN3 | |
| RXLEV | dBm | -Infinity | -75 |
| GSM BSIC | | N/A | Valid |

The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement for Event A3 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_inter}$

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new FDD inter-frequency cell is defined.

$T_{inter1}$=60ms

$N_{freq}$=2.

TTI insertion uncertainty = 2 ms

The UE shall send one Event A3 triggered measurement report for cell 2, with a measurement reporting delay less than 7682ms from the beginning of time period T2 (note: this gives a total of 7680 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The overall delays measured test requirement for Event B2 is:

The delay for GSM cell identification with BSIC verified is the sum of the event triggered measurement reporting delay and the initial BSIC identification delay.

The event triggered measurement reporting delay = $2*T_{Measurement\ Period,\ GSM} = 2* N_{freq}*480ms = 1920ms$.

Initial BSIC identification delay = 5280 ms, when one carrier frequency other than GSM is monitored in the gaps (table 8.1.2.4.5.1.2.1-1 of TS 36.133 [4]).

The UE shall send one Event B2 triggered measurement report including BSIC of cell 3, with a measurement reporting delay less than 7202ms from the beginning of time period T2 (note: this gives a total of 7200ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

The statistical pass/ fail decisions are done separated for event A3 and event B2.

Decide the test pass, if events A3 **and** B2 are passed, otherwise fail the UE.

# 8.11.6 Combined E-UTRAN TDD - E-UTRA TDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions

## 8.11.6.1 Test purpose

The purpose of this test is to verify that the UE makes correct reporting of multiple events when doing inter frequency and GSM measurements.

### 8.11.6.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward that support GSM. Applicability requires support for FGI bit 23, and 25.

### 8.11.6.3 Minimum conformance requirements

E-UTRAN part:

When measurement gaps are scheduled the UE shall be able to identify a new TDD inter-frequency within $T_{Identify_Inter}$ according to the following expression:

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new TDD inter-frequency cell is defined.

$N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1 and $T_{inter1}$ is defined in TS 36.133 [4] section 8.1.2.1

A cell shall be considered detectable provided following conditions are fulfilled:

- $RSRP|_{dBm}$ and RSRP Ês/Iot according to Annex I.2.3 for a corresponding Band,
- other RSRP related side conditions given in TS 36.133 [4] Section 9.1 are fulfilled,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band.

When measurement gaps are scheduled for TDD inter frequency measurements the UE physical layer shall be capable of reporting measurements to higher layers with measurement accuracy as specified in TS 36.133 [4] sub-clause 9.1.3 with measurement period given by table 8.11.6.3-1.

**Table 8.11.6.3-1: $T_{Measurement_Period_TDD_Inter}$ for different configurations**

| Configuration | Measurement bandwidth [RB] | Number of UL/DL sub-frames per half frame (5 ms) | | DwPTS | | $T_{Measurement_Period_TDD_Inter}$ [ms] |
|---|---|---|---|---|---|---|
| | | DL | UL | Normal CP | Extended CP | |
| 0 | 6 | 2 | 2 | $19760 \cdot T_s$ | $20480 \cdot T_s$ | 480 x $N_{freq}$ |
| 1 (Note 1) | 50 | 2 | 2 | $19760 \cdot T_s$ | $20480 \cdot T_s$ | 240 x $N_{freq}$ |
| Note 1: This configuration is optional<br>Note 2: $T_s$ is defined in 3GPP TS 36.211 [9] | | | | | | |

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per TDD inter-frequency and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period $T_{Measurement_Period_TDD_Inter}$.

Reported measurements contained in event triggered periodic measurement reports shall meet the requirements in TS 36.133 [4] section 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\text{-}inter}$ defined in TS 36.133 [4] section 8.1.2.3.2.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_inter}$ defined in TS 36.133 [4] section 8.1.2.3.2.1 and then triggers the measurement report as per TS 36.331 [5], the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_TDD\ Inter}$ defined in TS 36.133 [4] section 8.1.2.3.2.1 provided the timing to that cell has not changed more than ± 50 Ts while measurement gap has not been available and the L3 filter has not been used.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.2.

GSM part:

Measurements on GSM cells can be requested with BSIC verified.

In RRC_CONNECTED state when a supported measurement gap pattern sequence according to Table 8.1.2.1-1 in TS 36.133 [4] is configured by E-UTRAN the UE shall continuously measure GSM cells, search for new GSM cells given in the monitored set and re-confirm the BSIC for already detected cells.

This measurement shall be based on measurement gaps allocated for GSM carrier RSSI measurement as described in section 8.1.2.1 in TS 36.133 [4] A UE supporting GSM measurements shall measure minimum number of 10 GSM carrier RSSI measurement samples ($N_{GSM\ carrier\ RSSI}$) per measurement gap. In RRC_CONNECTED state the measurement period, $T_{Measurement\ Period,\ GSM}$, for the GSM carrier RSSI measurement is $N_{freq}$ * 480 ms. The parameter $N_{freq}$ is defined in clause 8.1.2.1.1 of TS 36.133 [4] as:

$N_{freq} = N_{freq,\ E\text{-}UTRA} + N_{freq,\ UTRA} + M_{gsm} + N_{freq,\ cdma2000} + N_{freq,\ HRPD}$

Where:

$N_{freq,\ E\text{-}UTRA}$ is the number of E-UTRA carriers being monitored

$N_{freq,\ UTRA}$ is the number of UTRA carriers being monitored

$N_{freq,\ cdma2000}$ is the number of cdma2000 carriers being monitored

$N_{freq,\ HRPD}$ is the number of HRPD carriers being monitored

$M_{GSM}$ is an integer which is a function of the number of GSM carriers on which measurements are being performed. $M_{GSM}$ is equal to 0 if no GSM carrier is being monitored. For a Measurement Gap Repetition Period (MGRP) of 40 ms, $M_{GSM}$ is equal to 1 if cells on up to 32 GSM carriers are being measured. For a MGRP of 80 ms, $M_{GSM}$ is equal to ceil ($N_{carriers,\ GSM}$ /20) where $N_{carriers,\ GSM}$ is the number of GSM carriers on which cells are being measured

The UE shall meet the measurement accuracy requirements stated for RXLEV in TS 45.008 [15], when the given measurement time allows the UE to take at least 3 GSM carrier RSSI samples per GSM carrier in the monitored set during the measurement period.

In case the UE is not able to acquire the required number of samples per GSM carrier during one measurement period, the UE shall measure as many GSM carriers as possible during that measurement period using at least 3 samples per GSM carrier. The GSM carriers that were not measured during that measurement period shall be measured in the following measurement periods.

BSIC verification:

Measurements on a GSM cell can be requested with BSIC verified. The UE shall be able to report the GSM cells with BSIC verified for those cells where the verification of BSIC has been successful.

The procedure for BSIC verification on a GSM cell can be divided into the following two tasks:

- **Initial BSIC identification:** Includes searching for the BSIC and decoding the BSIC for the first time when there is no knowledge about the relative timing between the E-UTRAN TDD and GSM cells. The UE shall trigger the initial BSIC identification within the available measurement gap pattern sequence. The requirements for BSIC re-confirmation can be found in section 8.1.2.4.5.1.2.1 of TS 36.133 [4].

- **BSIC re-confirmation:** Tracking and decoding the BSIC of a GSM cell after initial BSIC identification is performed. The UE shall trigger the BSIC re-confirmation within the available measurement gap pattern. The requirements for BSIC re-confirmation can be found in section 8.1.2.4.5.1.2.2 of TS 36.133 [4].

If the network requests measurements on a GSM cell, the UE shall behave as follows:

- The UE shall perform GSM carrier RSSI measurements according to section 8.1.2.4.5.1 in TS 36.133 [4] when a measurement gap pattern sequence is activated.

- The UE shall perform measurement reporting as defined in TS 36.331 [5].

- The UE shall perform BSIC identification if BSIC verified measurements are activated by RRC. The UE shall use the most recently available GSM carrier RSSI measurement results for arranging GSM cells in signal strength order for performing BSIC identification.

- The UE shall perform BSIC re-confirmation on all the GSM cells that have been successfully identified.

- The UE shall perform all configured event evaluation for event-triggered reporting after the BSIC has been verified for a GSM cell. The UE shall use the most recently available GSM carrier RSSI measurement results in event evaluation and event-triggered reporting.

- Event-triggered and periodic reports shall be triggered according to TS 36.331 [5].

The BSIC of a GSM cell is considered to be “verified” if the UE has decoded the SCH of the BCCH carrier and identified the BSIC at least one time (initial BSIC identification). Once a GSM cell has been identified the BSIC shall be re-confirmed at least once every $8*T_{re\text{-}confirm,GSM}$ seconds. Otherwise the BSIC of the GSM cell is considered as "non-verified". If a measurement gap pattern sequence is deactivated by the network after BSIC has been identified or verified, the UE shall consider the BSIC as non-verified.

$T_{identify,GSM}$ indicates the maximum time allowed for the UE to decode the unknown BSIC of the GSM cell in one GSM BCCH carrier in the initial BSIC identification procedure.

$T_{re\text{-}confirm,GSM}$ indicates the maximum time allowed for the re-confirmation of the BSIC of one GSM cell in the BSIC re-confirmation procedure.

The UE shall be able to decode a BSIC within a measurement gap when the time difference between the middle of the received GSM synchronisation burst at the UE and the middle of the effective measurement gap is within the limits specified in table 8.1.2.4.5.1.2-1 of TS 36.133 [4].

The UE shall be able to perform BSIC verification at levels down to the reference sensitivity level or reference interference levels as specified in TS 45.005 [16].

This measurement shall be based on the measurement gaps used for Initial BSIC identification as described in section 8.1.2.4.5.1.2 of TS 36.133 [4].

The UE shall continuously attempt to decode the BSIC of SCH on the BCCH carrier of the 8 strongest BCCH carriers of the GSM cells indicated in the Inter-RAT cell info list. The UE shall give priority for BSIC decoding attempts in decreasing signal strength order to BCCH carriers with unknown BSIC. The strongest BCCH carrier is defined as the BCCH carrier having the highest measured GSM carrier RSSI value.

If the BSIC of the GSM BCCH carrier has been successfully decoded the UE shall immediately continue BSIC identification with the next GSM BCCH carrier, in signal strength order, with unknown BSIC. The GSM cell for which the BSIC has been successfully identified shall be moved to the BSIC re-confirmation procedure.

If the UE has not successfully decoded the BSIC of the GSM BCCH carrier within $T_{identify,GSM}$ ms, the UE shall abort the BSIC identification attempts for that GSM BCCH carrier. The UE shall continue to try to perform BSIC identification of the next GSM BCCH carrier in signal strength order. The GSM BCCH carrier for which the BSIC identification failed shall not be re-considered for BSIC identification until BSIC identification attempts have been made for all the rest of the 8 strongest GSM BCCH carriers in the monitored set with unknown BSIC.

$T_{identify,GSM}$ values are given for a set of reference gap patterns in table 8.1.2.4.5.1.2.1-1 of TS 36.133 [4]. The requirements in the table represent the time required to guarantee at least two attempts to decode the BSIC for one GSM BCCH carrier.

Reported measurements in event triggered measurement reports shall meet the requirements in section TS 36.331 [5].

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH.

The event triggered reporting delay requirement is valid when the UE for each GSM carrier in the monitored set can take the required number of samples during the measurement period $T_{Measurement\ Period,\ GSM}$ (see section 8.1.2.4.5.1 of TS 36.133 [4]).

The event triggered measurement reporting delay for a GSM cell with verified BSIC , measured without L3 filtering shall be less than $2*T_{Measurement\ Period,\ GSM}$, where $T_{Measurement\ Period,\ GSM}$ is defined in section 8.1.2.4.5.1 of TS 36.133 [4] When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clauses 8.1.2.3.2, 8.1.2.4.6 and A.8.11.6.

## 8.11.6.4 Test description

### 8.11.6.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator), AWGN noise source and Fader to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.25.

2. The general test parameter settings are set up according to Table 8.11.6.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are as defined in clause 8.11.6.4.3.

5. There are two E-UTRA TDD cells operating on different frequency and one GSM cell specified in the test. Cell 1 (EUTRA TDD cell on RF channel number 1) is the cell used for registration with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.11.6.4.1-1: General test parameters for Combined E-UTRAN TDD - E-UTRA TDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2. |
| Special subframe configuration of cell1 and cell2 | | 6 | As specified in table 4.2-1 in TS 36.211 [9]. The same configuration in both cells |
| Uplink-downlink configuration of cell1 and cell2 | | 1 | As specified in 3GPP TS 36.211 [9] section 4.2 Table 4.2-2 |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cells | | Cell 2, 3 | Cell 2 is on E-UTRA RF channel number 2. Cell 3 is on Absolute RF Channel Number 3 (GSM cell). |
| CP length | | Normal | Applicable to cell 1 and cell 2 |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| E-UTRAN TDD measurement quantity | | RSRP | |
| Hysteresis | dB | 0 | Parameter for A3 and B2 event |
| A3-Offset | dB | -6 | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | OFF |
| Time offset between E-UTRAN TDD cells | µs | 3 | Synchronous cells |
| Inter-RAT (GSM) measurement quantity | | GSM Carrier RSSI | |
| b2-Threshold-E-UTRA | dBm | -83 | RSRP threshold for event B2. This is the threshold for E-UTRA in the B2 configuration. E-UTRA Pcell RSRP is below this throughout the test to account for measurement accuracy and fading |
| b2-Threshold-GERAN | dBm | -80 | GSM Carrier RSSI threshold for event B2. |
| Monitored GSM cell list size | | 6 GSM neighbours including ARFCN 3 | List of GSM cells provided before T2 starts. |
| T1 | s | 5 | |
| T2 | s | 10 | |

### 8.11.6.4.2 Test procedure

This test scenario comprised of 2 E-UTRA TDD cells operating on different frequency, and 1 GSM cell. The test consists of 2 successive time periods, with time duration T1 and T2. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 and B2 shall be used. At T1 the UE is camped on to Cell 1.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.11.6.5-1and Table 8.11.6.5-2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.11.6.5-1and Table 8.11.6.5-2.

6. UE shall transmit a MeasurementReport message triggered by event A3 and B2.

6a. If the overall delays measured from the beginning of time period T2 is less than 7682 ms for event A3 report then the number of "A3 successes" is increased by one.

6b. If the overall delays measured from the beginning of time period T2 is less than 7202 ms for event B2 report then the number of "B2 successes" is increased by one.

7. After the SS receives the MeasurementReport messages in step 6 or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. The SS shall set a different BSIC on Cell 3, as the previous timing information of Cell 3 is invalid in the UE for the next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Table G.2.3-1 in Annex G clause G.2 is achieved. The pass/ fail decisions are done separated for procedure steps 6a) and 6b). In general the number of repetitions up to the decision are different for 6a) and 6b). If one of 6a) or 6b) is passed, steps 1-9 are repeated, however the counter for the passed event is not anymore served. If one of 6a) or 6b) is failed, the test can be stopped.

### 8.11.6.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.11.6.4.3-1: Common Exception messages for Combined E-UTRAN TDD - E-UTRA TDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7 |

**Table 8.11.6.4.3-2: *MeasConfig-DEFAULT*: Additional Combined E-UTRAN TDD - E-UTRA TDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 8 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell (E-UTRA serving cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | serving frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the neighbouring cell(E-UTRA inter frequency cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | inter frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f3 | f3 is the frequency of the neighbouring cell(GERAN cell) | |
| measObject CHOICE { | | | |
| MeasObjectGERAN | MeasObjectGERAN-GENERIC(f3) | inter frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f4 | f4 is the frequency of the neighbouring cell(GERAN cell) | |
| measObject CHOICE { | | | |
| MeasObjectGERAN | MeasObjectGERAN-GENERIC(f4) | inter frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f5 | f5 is the frequency of the neighbouring cell(GERAN cell) | |
| measObject CHOICE { | | | |
| MeasObjectGERAN | MeasObjectGERAN-GENERIC(f5) | inter frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f6 | f6 is the frequency of the neighbouring cell(GERAN cell) | |
| measObject CHOICE { | | | |
| MeasObjectGERAN | MeasObjectGERAN- | inter frequency | |

| | | | |
|---|---|---|---|
| | GENERIC(f6) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f7 | f7 is the frequency of the neighbouring cell(GERAN cell) | |
| measObject CHOICE { | | | |
| MeasObjectGERAN | MeasObjectGERAN-GENERIC(f7) | inter frequency | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f8 | f8 is the frequency of the neighbouring cell(GERAN cell) | |
| measObject CHOICE { | | | |
| MeasObjectGERAN | MeasObjectGERAN-GENERIC(f8) | inter frequency | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 2 entry | | |
| reportConfigToAddMod ::= SEQUENCE { | | | |
| reportConfigId | idReportConfig-A3 | | |
| reportConfig | ReportConfigEUTRA-A3 | | |
| } | | | |
| reportConfigToAddMod ::= SEQUENCE { | | | |
| reportConfigId | idReportConfig-B2 | | |
| reportConfig | ReportConfigInterRAT-B2-GERAN | | |
| } | | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 2 entry | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-A3 | | |
| } | | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f3 | | |
| reportConfigId | idReportConfig-B2 | | |
| } | | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | MeasGapConfig-GP1 | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 8.11.6.4.3-3: *ReportConfigEUTRA-A3*: Additional Combined E-UTRAN TDD - E-UTRA TDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.11.6.4.3-4: *ReportConfigInterRAT-B2-GERAN*: Additional Combined E-UTRAN TDD - E-UTRA TDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6 Table 4.6.6-7E | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B2-GERAN(EUTRA-Thres, GERAN-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB2 SEQUENCE { | | | |
| b2-Threshold1 CHOICE { | | | |
| threshold-RSRP | 57(-83dBm) | INTEGER(0..97), the mapping table is Table 9.1.4-1 in TS 36.133 [4] | |
| } | | | |
| b2-Threshold2 CHOICE { | | | |
| b2-Threshold2-GERAN | 30 (-80dBm) | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | INTEGER(0..30), 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.11.6.4.3-5 *MeasResults*: Additional Combined E-UTRAN TDD - E-UTRA TDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.11.6.4.3-6: *MeasResultListEUTRA*: Additional Combined E-UTRAN TDD - E-UTRA TDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| MeasResultEUTRA ::= SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.11.6.4.3-7: *MeasuredResults*: Additional Combined E-UTRAN TDD - E-UTRA TDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | RSRP-Range | Set according to specific test | |
| rsrqResult | RSRQ-Range | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListGERAN | MeasResultListGERAN | | |
| } | | | |
| } | | | |

**Table 8.11.6.4.3-8: *MeasResultListGERAN*: Additional Combined E-UTRAN TDD - E-UTRA TDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListGERAN ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| carrierFreq | CarrierFreqGERAN | | |
| physCellId | physCellId GERAN | | |
| measResult SEQUENCE { | | | |
| rssi | INTEGER (0..63) | Set according to specific test | |
| } | | | |
| } | | | |

## 8.11.6.5 Test requirement

Tables 8.11.6.5-1 and 8.11.6.5-2 define the primary level settings including test tolerances for two E-UTRAN TDD cells and one GSM cell.

**Table 8.11.6.5-1: Cell specific test parameters for Combined E-UTRAN TDD - E-UTRA TDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions (cell1 and cell2)**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | | OP.2 TDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | -98 | |
| RSRP Note 4 | dBm/15 kHz | -94.20 | -94.20 | -Infinity | -90.50 |
| $\hat{E}_s / I_{ot}$ | dB | 3.80 | 3.80 | -Infinity | 7.50 |
| SCH_RP Note 4 | dBm/15 kHz | -94.20 | -94.20 | -Infinity | -90.50 |
| $\hat{E}_s / N_{oc}$ | dB | 3.80 | 3.80 | -Infinity | 7.50 |
| Propagation Condition | | ETU70 | | ETU70 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.11.6.5-2: Cell specific test parameters for Combined E-UTRAN TDD - E-UTRA TDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions (cell3)**

| Parameter | Unit | Cell 3 | |
|---|---|---|---|
| | | T1 | T2 |
| Absolute RF Channel Number | | ARFCN3 | |
| RXLEV | dBm | -Infinity | -75 |
| GSM BSIC | | N/A | Valid |

The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement for Event A3 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify_inter}$

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new TDD inter-frequency cell is defined.

$T_{inter1}$=60ms

$N_{freq}$=2.

TTI insertion uncertainty = 2 ms

The UE shall send one Event A3 triggered measurement report for cell 2, with a measurement reporting delay less than 7682 ms from the beginning of time period T2 (note: this gives a total of 7680 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The overall delays measured test requirement for Event B2 is:

The delay for GSM cell identification with BSIC verified is the sum of the event triggered measurement reporting delay and the initial BSIC identification delay.

The event triggered measurement reporting delay = $2*T_{Measurement\ Period,\ GSM}$ = $2*\ N_{freq}*480ms$ = 1920ms.

Initial BSIC identification delay = 5280 ms, when one carrier frequency other than GSM is monitored in the gaps (table 8.1.2.4.5.1.2.1-1 of TS 36.133 [4]).

The UE shall send one Event B2 triggered measurement report including BSIC of cell 3, with a measurement reporting delay less than 7202 ms from the beginning of time period T2 (note: this gives a total of 7200 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

The statistical pass/ fail decisions are done separated for event A3 and event B2.

Decide the test pass, if events A3 **and** B2 are passed, otherwise fail the UE.

# 8.12 Void

# 8.13 Void

# 8.14 E-UTRAN TDD - FDD Inter-frequency Measurements

## 8.14.1 E-UTRAN TDD-FDD Inter-frequency event triggered reporting under fading propagation conditions in asynchronous cells

### 8.14.1.1 Test purpose

To verify the UE's ability to make a correct reporting of an event under fading propagation conditions in asynchronous cells within the E-UTRA TDD-FDD inter-frequency cell search requirements. This test will partly verify the TDD-FDD inter-frequency cell search requirements in section 8.1.2.3.3.

### 8.14.1.2 Test applicability

This test applies to all types of E-UTRA UE supporting FDD and TDD release 9 and forward. Applicability requires support for FGI bit 25.

### 8.14.1.3 Minimum conformance requirements

When measurement gaps are scheduled the UE shall be able to identify a new FDD inter-frequency within $T_{Identify_Inter}$ according to the following expression:

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new FDD inter-frequency cell is defined.

$N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1 and $T_{Inter1}$ is defined in TS36.133 [4] section 8.1.2.1.

A cell shall be considered detectable provided following conditions are fulfilled:

- RSRP and RSRP Ês/Iot according to Annex I.2.3 for a corresponding Band,
- other RSRP related side conditions given in Section 9.1 are fulfilled,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band.

When measurement gaps are scheduled for FDD inter frequency measurements the UE physical layer shall be capable of reporting measurements to higher layers with measurement accuracy as specified in TS 36.133 sub-clause 9.1.3 with measurement period given by table 8.14.1.3-1.

**Table 8.14.1.3-1: RSRP measurement period and measurement bandwidth**

| Configuration | Physical Layer Measurement period: $T_{Measurement_Period_Inter_FDD}$ [ms] | Measurement bandwidth [RB] |
|---|---|---|
| 0 | 480 x $N_{freq}$ | 6 |
| 1 (Note) | 240 x $N_{freq}$ | 50 |
| Note: This configuration is optional. | | |

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per FDD inter-frequency and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period defined in Table 8.14.1.3-1.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_inter}$ defined in TS 36.133 [4] clause 8.1.2.3.1.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra}$ defined in TS 36.133 [4] section 8.1.2.2.1.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.1 and A.8.14.1.

## 8.14.1.4 Test description

### 8.14.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15.

2. The general test parameter settings are set up according to Table 8.14.1.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.14.1.4.3.

5. There are two E-UTRA carriers and one E-UTRA TDD Cell 1 and one E-UTRA FDD Cell 2 on each carrier specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.14.1.4.1-1: General test parameters for E-UTRAN TDD-FDD inter-frequency event triggered reporting in fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Cell 1 PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| Cell 1 PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Cell1 Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211. |
| Cell1 Uplink-downlink configuration | | 1 | As specified in TS 36.211 section 4.2 Table 4.2-2. |
| Cell 2 PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| Cell 2 PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Cell 1 E-UTRA TDD RF Channel Number | | 1 | One TDD carrier frequency is used. |
| Cell 2 E-UTRA FDD RF Channel Number | | 2 | One FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active cell | | Cell 1 | Cell 1 is on RF channel number 1 |
| Neighbour cell | | Cell 2 | Cell 2 is on RF channel number 2 |
| Gap Pattern Id | | 0 | As specified in TS 36.133[4] section 8.1.2.1. |
| A3-Offset | dB | -6 | |
| Hysteresis | dB | 0 | |
| CP length | | Normal | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | OFF |
| Time offset between cells | | 3 ms | Asynchronous cells |
| T1 | s | 5 | |
| T2 | s | 5 | |

### 8.14.1.4.2 Test procedure

The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2. Gap pattern configuration #0 is configured before T2 begins to enable inter-frequency monitoring.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.14.1.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.14.1.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the measurement reporting delay from the beginning of time period T2 is less than 3842ms the number of successful tests is increased by one. If the UE fails to report the event within the measurement reporting delay requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.14.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.14.1.4.3-1: Common Exception messages for E-UTRAN TDD-FDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.3.1-9 |

**Table 8.14.1.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD-FDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.14.1.4.3-3: *MeasResults*: Additional E-UTRAN TDD-FDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.14.1.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN TDD-FDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | Cell 2 | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

## 8.14.1.5 Test requirement

Tables 8.14.1.4.1-1 and 8.14.1.5-1 define the primary level settings including test tolerances for event triggered reporting under fading propagation conditions in asynchronous TDD-FDD inter frequency cells test.

**Table 8.14.1.5-1: Cell specific test parameters for E-UTRAN TDD-FDD inter-frequency event triggered reporting under fading propagation conditions in asynchronous cells**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.1.2 (OP.2 FDD) | | OP.1 TDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | |
| RSRP Note 4 | dBm/15 kHz | -94.00 | -94.00 | -Infinity | -90.30 |
| $\hat{E}_s/I_{ot}$ | dB | 4.00 | 4.00 | -Infinity | 7.70 |
| SCH_RP Note 4 | dBm/15 kHz | -94.00 | -94.00 | -Infinity | -90.30 |
| $\hat{E}_s/N_{oc}$ | dB | 4.00 | 4.00 | -Infinity | 7.70 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 3842 ms from the beginning of time period T2.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = 3840 ms

TTI insertion uncertainty = $TTI_{DCCH}$ = 1 ms; $2xTTI_{DCCH}$ = 2 ms

The overall delays measured shall be less than a total of 3842 ms in this test case (note: this gives a total of 3840 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 8.14.2 E-UTRAN TDD-FDD Inter-frequency event triggered reporting when DRX is used under fading propagation conditions in asynchronous cells

### 8.14.2.1 Test purpose

To verify the UE's ability to make a correct reporting of an event in DRX under fading propagation conditions in asynchronous cells within the E-UTRA TDD-FDD inter-frequency cell search requirements.

### 8.14.2.2 Test applicability

This test applies to all types of E-UTRA UE supporting FDD and TDD release 9 and forward. Applicability requires support for FGI bits 5 and 25.

### 8.14.2.3 Minimum conformance requirements

When measurement gaps are scheduled the UE shall be able to identify a new FDD inter-frequency within $T_{Identify_Inter}$ as shown in table 8.14.2.3-1:

**Table 8.14.2.3-1: Requirement to identify a newly detectable FDD inter-frequency cell**

| DRX cycle length (s) | $T_{identify_inter}$ (s) (DRX cycles) | |
|---|---|---|
| | Gap period = 40 ms | Gap period = 80 ms |
| ≤0.16 | Non DRX Requirements in TS 36.133 [4] clause 8.1.2.3.1.1 are applicable | Non DRX Requirements in TS 36.133 [4] clause 8.1.2.3.1.1 are applicable |
| 0.256 | 5.12*$N_{freq}$ (20*$N_{freq}$) | 7.68*$N_{freq}$ (30*$N_{freq}$) |
| 0.32 | 6.4*$N_{freq}$ (20*$N_{freq}$) | 7.68*$N_{freq}$ (24*$N_{freq}$) |
| 0.32 < DRX-cycle ≤ 2.56 | Note (20*$N_{freq}$) | Note (20*$N_{freq}$) |
| Note: Time depends upon the DRX cycle in use | | |

The non DRX requirements in TS 36.133 [4] clause 8.1.2.3.1.1 states that when measurement gaps are scheduled the UE shall be able to identify a new FDD inter-frequency within $T_{Identify_Inter}$ according to the following expression:

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new FDD inter-frequency cell is defined.

$N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1 and $T_{Inter1}$ is defined in TS36.133 [4] section 8.1.2.1.

A cell shall be considered detectable provided following conditions are fulfilled:

- RSRP and RSRP Ês/Iot according to Annex I.2.3 for a corresponding Band,
- other RSRP related side conditions given in TS 36.133 [4] Section 9.1 are fulfilled,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band.

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per FDD inter-frequency and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period defined in Table 8.14.2.3-2.

**Table 8.14.2.3-2: Requirement to measure FDD inter-frequency cells**

| DRX cycle length (s) | $T_{measure_inter}$ (s) (DRX cycles) |
|---|---|
| ≤0.08 | Non DRX Requirements in TS 36.133 [4] clause 8.1.2.3.1.1 are applicable |
| 0.08 < DRX-cycle ≤ 2.56 | Note ($5*N_{freq}$) |
| Note: Time depends upon the DRX cycle in use | |

The measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clause 9.1.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x TTI DCCH. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_inter}$ defined in TS 36.133 [4] clause 8.1.2.3.1.2. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra}$ defined in TS 36.133 [4] section 8.1.2.3.1.2 becomes undetectable for a period ≤ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.3.2 and A.8.14.2.

## 8.14.2.4 Test description

### 8.14.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15.

2. The general test parameter settings are set up according to Table 8.14.2.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.14.2.4.3.

5. There are two E-UTRA carriers and one E-UTRA TDD Cell 1 and one E-UTRA FDD Cell 2 on each carrier specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.14.2.4.1-1: General test parameters for E-UTRAN TDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions**

| Parameter | Unit | Test 1 | Test 2 | Comment |
|---|---|---|---|---|
| | | Value | | |
| Cell1 PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | | As specified in section A.1.2. Note that UE may only be allocated at *On Duration* |
| Cell1PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | | As specified in section A.2.2. |
| Cell2 PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | | As specified in section A.1.1. Note that UE may only be allocated at *On Duration* |
| Cell2PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | | As specified in section A.2.1. |
| E-UTRA RF Channel Number | | 1 | | one TDD carrier frequencies is used. |
| E-UTRA RF Channel Number | | 2 | | one FDD carrier frequencies is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| Active cell | | Cell 1 | | Cell 1 is on RF channel number 1 |
| Neighbour cell | | Cell 2 | | Cell 2 is on RF channel number 2 |
| Gap Pattern Id | | 0 | | As specified in 3GPP TS 36.133[4] section 8.1.2.1. |
| Cell1 Uplink-downlink configuration | | 1 | | As specified in 3GPP TS 36.211 section 4.2 Table 4.2-2 |
| Cell1 Special subframe configuration | | 6 | | As specified in table 4.2-1 in TS 36.211. The same configuration in both cells |
| A3-Offset | dB | -6 | | |
| Hysteresis | dB | 0 | | |
| CP length | | Normal | | |
| TimeToTrigger | s | 0 | | |
| Filter coefficient | | 0 | | L3 filtering is not used |
| PRACH configuration | | 4 | | As specified in table 5.7.1-3 in TS 36.211 |
| Access Barring Information | - | Not Sent | | No additional delays in random access procedure. |
| DRX | | ON | | DRX related parameters are defined in Table 8.14.2.5-2 |
| Time offset between cells | | 3 ms | | Asynchronous cells |
| T1 | s | 5 | | |
| T2 | s | 5 | 30 | |

## 8.14.2.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

In Test 1 when DRX = 40 ms is used, UE needs to be provided at least once every 500 ms with new Timing Advance Command MAC control element to restart the Timer Alignment Timer to keep the UE uplink time alignment. Furthermore, the UE is allocated with PUSCH resource at every DRX cycle. In Test 2 when DRX = 1280 ms is used, the uplink time alignment is not maintained and the UE needs to use RACH to obtain uplink allocation for measurement reporting.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.14.2.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.14.2.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 3882 ms for Test 1 or less than 26882 ms for Test 2 then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

11. Repeat step 1-10 for each sub-test in Table 8.14.2.4.1-1 as appropriate.

### 8.14.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.14.2.4.3-1: Common Exception messages for E-UTRAN TDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test 1 and 2 requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-7<br>Table H.3.1-9<br>Table H.3.7-1<br>Table H.3.7-2<br>Table H.3.7-3 |

**Table 8.14.2.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN TDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test 1 and 2 requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.14.2.4.3-3: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test 1 and 2 requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventid CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.14.2.4.3-4: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN TDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test 1 and 2 requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 2 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.14.2.4.3-5: *MeasResults*: Additional E-UTRAN TDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test 1 and 2 requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.14.2.4.3-6: *MeasResultListEUTRA*: Additional E-UTRAN TDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test 1 and 2 requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| physCellId | PhysCellId | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.14.2.4.3-7: *PRACH-Config-DEFAULT*: Additional E-UTRAN TDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test 1 and 2 requirements**

| Derivation Path: TS 36.508 [7] clause 7.3.2 Table 7.3.2-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PRACH-Config-DEFAULT ::= SEQUENCE { | | | |
| prach-ConfigIndex | 4 | | |
| } | | | |

## 8.14.2.5 Test requirement

Tables 8.14.2.4.1-1, 8.14.2.5-1, 8.14.2.5-2 and 8.14.2.5-3 define the primary level settings including test tolerances for E-UTRAN TDD-FDD inter-frequency event triggered reporting when DRX is used under fading propagation conditions in asynchronous cells test.

**Table 8.14.2.5-1: Cell specific test parameters for E-UTRAN TDD-FDD inter-frequency event triggered reporting under fading propagation conditions in asynchronous cells**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.1.2 (OP.2 FDD) | | OP.1 TDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | | |
| RSRP Note 3 | dBm/15 kHz | -94 | -94 | -Infinity | -90.30 |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | -Infinity | 7.70 |
| SCH_RP Note 3 | dBm/15 kHz | -94 | -94 | -Infinity | -90.30 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | -Infinity | 7.70 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.14.2.5-2: DRX-Configuration to be used in E-UTRAN TDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf40 | sf1280 | |
| shortDRX | disable | disable | |
| Note: For further information see section 6.3.2 in 3GPP TS 36.331 [5]. | | | |

**Table 8.14.2.5-3: *TimeAlignmentTimer* and *sr-ConfigIndex*-Configuration to be used in E-UTRAN TDD-FDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | For further information see section 6.3.2 in 3GPP TS 36.331 [5]. |
| sr-ConfigIndex | 2 | 2 | For further information see section 6.3.2 in 3GPP TS 36.331 and section10.1 in 3GPP TS 36.213 [8]. |

In Test 1 when DRX cycle length = 40 ms is used, the overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2 on PUSCH.

In Test 2 when DRX cycle length = 1280 ms is used, the overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE starts to send preambles on the PRACH for Scheduling Request (SR) to obtain allocation to send the measurement report to Cell 2 on PUSCH.

For both tests:

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event A3 measurement report.

The overall delay measured when DRX cycle length is 40 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty+ DRX cycle length

Measurement reporting delay = $T_{Identify_Inter}$

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new FDD inter-frequency cell is defined.

$T_{Inter1}$ = 60 ms. It is defined in table 8.1.2.1-1 of TS36.133 [4] clause 8.1.2.1.

$N_{freq}$ = 1. It is defined in TS 36.133 [4] clause 8.1.2.1.1.

TTI insertion uncertainty = 2 ms

DRX cycle length = 40 ms

The overall delays measured when DRX cycle length is 40 ms shall be less than a total of 3882 ms.

The overall delay measured when DRX cycle length is 1280 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_inter}$

$T_{identify_inter}$ = 25600 ms. When DRX cycle length is 1280 ms then the $T_{identify_inter}$ is 20 x 1280 ms, as defined in Table 8.14.2.3-1.

TTI insertion uncertainty = 2 ms

DRX cycle length = 1280 ms

The overall delays measured when DRX cycle length is 1280 ms shall be less than a total of 26882 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.14.3 E-UTRAN TDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps

### 8.14.3.1 Test purpose

To verify the requirement for identification of a new CGI of E-UTRA cell with autonomous gaps.

### 8.14.3.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward with support of inter-frequency SI acquisition for HO. Applicability requires support for FGI bit 25.

### 8.14.3.3 Minimum conformance requirements

No explicit neighbour list is provided to the UE for identifying a new CGI of E-UTRA cell. The UE shall identify and report the CGI when requested by the network for the purpose of 'reportCGI'. The UE may make autonomous gaps in both downlink reception and uplink transmission for receiving MIB and SIB1 message according to section 5.5.3.1 of TS 36.331 [5]. Note that a UE is not required to use autonomous gap if si-RequestForHO is set to false. If autonomous gaps are used for measurement with the purpose of 'reportCGI', regardless of whether DRX is used or not, the UE shall be able to identify a new CGI of E-UTRA cell within:

$$T_{identify_CGI,\,inter} = T_{basic_identify_CGI,\,inter} \quad ms$$

Where

$T_{basic_identify_CGI,\,inter}$ = 150 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new CGI of E-UTRA cell is defined, provided that the E-UTRA cell has been already identified by the UE.

A cell shall be considered identifiable following conditions are fulfilled:

- RSRP related side conditions given in TS36.133 [4] Section 9.1 are fulfilled for a corresponding Band,
- SCH_RP|$_{dBm}$ and SCH Ês/Iot according to Annex I.2.4 for a corresponding Band

The MIB of an E-UTRA cell whose CGI is identified shall be considered decodable by the UE provided the PBCH demodulation requirements are met according to TS 36.331[5].

The requirement for identifying a new CGI of an E-UTRA cell within $T_{basic_identify_CGI,inter}$ is applicable when no DRX is used as well as when all the DRX cycles specified in 3GPP TS 36.331 [5] are used.

Given that continuous DL data allocation and no DRX is used, no measurement gaps are configured, and TDD configuration as in TS36.133 [4] Table 8.1.2.3.2.1-1 is used, the UE shall have more than 30 ACK/NACK transmitted during the identification of a new CGI of E-UTRA cell.

The ECGI reporting delay occurs due to the delay uncertainty when inserting the ECGI measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. In case DRX is used, the ECGI reporting may be delayed until the next DRX cycle.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.6 and A.8.14.3.

## 8.14.3.4 Test description

### 8.14.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. The general test parameter settings are set up according to Table 8.14.3.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.14.3.4.3.

5. There is one E-UTRA TDD carrier and one E-UTRA FDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.14.3.4.1-1: General test parameters for E-UTRAN TDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Cell1PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| Cell1 PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Cell2 PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| Cell2 PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Cell1 E-UTRA RF channel number | | 1 | One TDD carrier is used |
| Cell2 E-UTRA RF channel number | | 2 | One FDD carrier is used |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active cell | | Cell 1 | Cell 1 is on RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 1 is on RF channel number 2. |
| CP length | | Normal | |
| Cell1 special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211[9]. |
| Cell1 Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in TS 36.211[9]. |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 section 8.1.2.1 [5]. |
| A3-Offset | dB | -6 | |
| Hysteresis | dB | 0 | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| si-RequestForHO | | TRUE | As specified in section 5.5.3.1 in TS 36.331[5]. |
| Time offset between cells | ms | 3 | Asynchronous cells |
| T1 | s | 5 | |
| T2 | s | ≤10 | |
| T3 | s | 5 | |

### 8.14.3.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. PDCCHs indicating new transmissions should be sent continuously to ensure that the UE would have ACK/NACK sending during identifying a new CGI of E-UTRAN cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE does not have any timing information of cell 2. Gap pattern configuration is configured before T2

begins to enable inter-frequency monitoring. Starting T2, cell 2 becomes detectable and the UE is expected to detect and send a measurement report.

1. Ensure the UE is in 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.14.3.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. The SS shall transmit an RRCConnectionReconfiguration with event A3 configured message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.14.3.5-1.
6. The UE shall transmit a MeasurementReport message triggered by Event A3.
7. The SS shall transmit an RRCConnectionReconfiguration message during period T2. The RRC message shall release MeasGapConfig according to Table 8.14.3.4.3-6.
8. The UE shall transmit RRCConnectionReconfigurationComplete message.
9. The SS shall transmit an RRCConnectionReconfiguration message during period T2, The RRC message shall create a measurement report configuration with purpose *reportCGI* and *si-RequestForHO* set to TRUE.
10. The SS shall start T3 timer when the last TTI containing the RRC message implying SI reading is sent to the UE.
11. The UE shall transmit RRCConnectionReconfigurationComplete message.
12. The UE shall transmit a MeasurementReport message containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3. The UE shall be scheduled continuously throughout the test, and from the start of T3 until 170 ms at least 42 ACK/NACK shall be detected as being transmitted by the UE. If the overall delays measured from the beginning of time period T3 is less than 170 ms, and the UE have more than 42 ACK/NACKs transmitted from the start of T3 until 170 ms, then the number of successful tests is increased by one. If the UE fails to report the cell global identifier within the overall delays measured requirement, or the UE have less than 42 ACK/NACKs transmitted from the start of T3 until 170 ms, then the number of failure tests is increased by one.
13. After the SS receive the MeasurementReport message (in step 12) or when T3 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
14. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
15. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
16. Repeat step 2-15 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.14.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.14.3.4.3-1: Common Exception messages for E-UTRAN TDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.3.1-9 |

**Table 8.14.3.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.14.3.4.3-3: *MeasResults*: Additional E-UTRAN TDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.14.3.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN TDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step6)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.14.3.4.3-5: *ReportConfigEUTRA-PERIODICAL*: Additional E-UTRAN TDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step9)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportCGI | | |
| } | | | |
| } | | | |
| reportAmount | 1 | | |
| si-RequestForHO-r9 | setup | | |
| } | | | |

**Table 8.14.3.4.3-6: MeasGapConfig- Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step7)**

| Derivation Path: 36.331, clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasGapConfig ::= CHOICE { | | | |
| release | NULL | | |
| } | | | |

**Table 8.14.3.4.3-7: *MeasResultListEUTRA*: Additional E-UTRAN TDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step12)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| cgi-Info SEQUENCE{ | | | |
| cellGlobalId SEQUENCE{ | | | |
| plmn-Identity | plmn-Identity | | |
| cellIdentity | cellIdentity of cell 2 | | |
| } | | | |
| trackingAreaCode | | | |
| plmn-IdentityList | | | |
| } | | | |
| } | | | |

**Table 8.14.3.4.3-8: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN TDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step9)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(f2) ::= SEQUENCE { | | f2 is the frequency of the neighbouring cell | |
| cellForWhichToReportCGI | Physical Cell ID of Cell 2 | | |
| } | | | |

**Table 8.14.3.4.3-9: *MeasConfig-DEFAULT*: Additional E-UTRAN TDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| } | | | |

**Table 8.14.3.4.3-10: *MeasConfig-DEFAULT*: Additional E-UTRAN TDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step9)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | IdReportConfig-P | | |
| } | | | |
| } | | | |

**Table 8.14.3.4.3-11: *MeasResults*: Additional E-UTRAN TDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step12)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

## 8.14.3.5 Test requirement

Tables 8.14.3.4.1-1 and 8.14.3.5-1 define the primary level settings including test tolerances for E-UTRAN TDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test.

**Table 8.14.3.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.1.2 (OP.2 FDD) | | OP.1 TDD | OP.1 TDD | OP.1 TDD | OP.2 FDD | OP.2 FDD | OP.2 FDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | 4 | -Infinity | 7 | 7 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | 4 | -Infinity | 7 | 7 |
| RSRP Note 3 | dBm/15 KHz | -94 | -94 | -94 | -Infinity | -91 | -91 |
| SCH_RP Note3 | dBm/15 KHz | -94 | -94 | -94 | -Infinity | -91 | -91 |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall transmit a measurement report containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Test requirement = RRC Procedure delay + $T_{identify_CGI,\ inter}$ + TTI insertion uncertainty

RRC procedure delay=15ms

$$T_{identify_CGI,\ inter} = T_{basic_identify_CGI,\ inter} \quad ms$$

$T_{basic_identify_CGI,\ intra}$ = 150 ms

TTI insertion uncertainty = 2 ms

The overall delays measured is 167 ms, allow 170 ms. The UE shall be scheduled continuously throughout the test, and from the start of T3 until 170 ms at least 42 ACK/NACK shall be detected as being transmitted by the UE.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

NOTE: The overall 42 ACK/NACK number is caused by two parts. Firstly, at least 30 ACK/NACK shall be sent during identifying the cell global identifier of cell 2 according to the requirement in TS 36.133 [4] Section 8.1.2.3.7.1. Secondly, given that continuous DL data allocation, and the measurement gaps have been deconfigured before the start of T3, additional 12 ACK/NACK shall be sent from the start of T3 until 170 ms excludes 150 ms for identifying the cell global identifier of cell 2.

# 8.15 E-UTRAN FDD - TDD Inter-frequency Measurements

## 8.15.1 E-UTRAN FDD-TDD Inter-frequency event triggered reporting under fading propagation conditions in asynchronous cells

### 8.15.1.1 Test purpose

To verify the UE's ability to make a correct reporting of an event under fading propagation conditions in asynchronous cells within the E-UTRA FDD - TDD inter-frequency cell search requirements. This test will verify the FDD-TDD inter-frequency cell search requirements in section 8.1.2.3.4.

### 8.15.1.2 Test applicability

This test applies to all types of E-UTRA UE supporting FDD and TDD release 9 and forward. Applicability requires support for FGI bit 25.

### 8.15.1.3 Minimum conformance requirements

When measurement gaps are scheduled the UE shall be able to identify a new TDD inter-frequency within $T_{Identify_Inter}$ according to the following expression:

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new TDD inter-frequency cell is defined.

$T_{inter1}$ is defined in TS 36.133 [4] section 8.1.2.1

$N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1.

A cell shall be considered detectable provided following conditions are fulfilled:

- RSRP and RSRP Ês/Iot according to Annex I.2.3 for a corresponding Band,
- other RSRP related side conditions given in TS 36.133 [4] Section 9.1 are fulfilled,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band.

When measurement gaps are scheduled for TDD inter frequency measurements the UE physical layer shall be capable of reporting measurements to higher layers with measurement accuracy as specified in TS 36.133 sub-clause 9.1.3 with measurement period ($T_{Measurement_Period_TDD_Inter}$) given by table 8.4.4.1.3-1.

**Table 8.15.1.3-1: $T_{Measurement_Period_TDD_Inter}$ for different configurations**

| Configuration | Measurement bandwidth [RB] | Number of UL/DL sub-frames per half frame (5 ms) | | DwPTS | | $T_{Measurement_Period_TDD_Inter}$ [ms] |
|---|---|---|---|---|---|---|
| | | DL | UL | Normal CP | Extended CP | |

| | | | | | | |
|---|---|---|---|---|---|---|
| 0 | 6 | 2 | 2 | $19760 \cdot T_s$ | $20480 \cdot T_s$ | 480 x $N_{freq}$ |
| 1 (Note 1) | 50 | 2 | 2 | $19760 \cdot T_s$ | $20480 \cdot T_s$ | 240 x $N_{freq}$ |
| Note 1: This configuration is optional.<br>Note 2: $T_s$ is defined in 3GPP TS 36.211 [9]. | | | | | | |

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per TDD inter-frequency and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period $T_{Measurement_Period_TDD_Inter}$.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{Identify_Inter}$ defined in TS 36.133 [4] clause 8.1.2.3.2.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra}$ defined in TS 36.133 [4] section 8.1.2.2.1.1 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.4 and A.8.15.1.

## 8.15.1.4 Test description

### 8.15.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15.

2. The general test parameter settings are set up according to Table 8.15.1.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.15.1.4.3.

5. There are two E-UTRA carriers and one E-UTRA FDD Cell 1 and one E-UTRA TDD Cell 2 on each carrier specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.15.1.4.1-1: General test parameters for E-UTRAN FDD-TDD inter frequency event triggered reporting in fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Cell 1 PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| Cell 1 PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Cell 2 PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| Cell 2 PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Gap Pattern Id | | 1 | As specified in TS 36.133 section 8.1.2.1. |
| Cell2 Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211. Applicable to Cell 2. |
| Cell2 Uplink-downlink configuration | | 1 | As specified in TS 36.211 section 4.2 Table 4.2-2. Applicable to Cell 2. |
| CP length | | Normal | |
| Cell 1 E-UTRA FDD RF Channel Number | | 1 | One TDD carrier frequency is used. |
| Cell 2 E-UTRA TDD RF Channel Number | | 2 | One FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active cell | | Cell 1 | Cell 1 is on RF channel number 1 |
| Neighbour cell | | Cell 2 | Cell 2 is on RF channel number 2 |
| A3-Offset | dB | -6 | |
| Hysteresis | dB | 0 | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| Time offset between cells | | 3 ms | Asynchronous cells |
| T1 | s | 5 | |
| T2 | s | 10 | |

### 8.15.1.4.2 Test procedure

The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2. Gap pattern configuration #1is configured before T2 begins to enable inter-frequency monitoring.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.15.1.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.15.1.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the measurement reporting delay from the beginning of time period T2 is less than 7682 ms the number of successful tests is increased by one. If the UE fails to report the event within the measurement reporting delay requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit an RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.15.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.15.1.4.3-1: Common Exception messages for Additional E-UTRAN FDD-TDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H3.1-1<br>Table H3.1-3<br>Table H.3.1-7 |

**Table 8.15.1.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-TDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.15.1.4.3-3: *MeasResults*: Additional E-UTRAN FDD-TDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.15.1.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN FDD-TDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| MeasResultEUTRA ::= SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | Cell 2 | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.15.1.5 Test requirement

Tables 8.15.1.4.1-1 and 8.15.1.5-1 define the primary level settings including test tolerances for event triggered reporting under fading propagation conditions in asynchronous FDD-TDD inter frequency cells test.

**Table 8.15.1.5-1: Cell specific test parameters for E-UTRAN FDD-TDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells**

| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | |
|---|---|---|---|---|---|
| | | **T1** | **T2** | **T1** | **T2** |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Pattern defined in D.1.1 (OP.1 FDD) and in D.2.2 (OP.2 TDD) | | OP.1 FDD | | OP.2 TDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA[Note 1] | dB | | | | |
| OCNG_RB[Note 1] | dB | | | | |

| | | | | | |
|---|---|---|---|---|---|
| $\hat{E}_s/I_{ot}$ | dB | 4.00 | 4.00 | -Infinity | 7.70 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | |
| RSRP Note 4 | dBm/15 kHz | -94.00 | -94.00 | -Infinity | -90.30 |
| SCH_RP Note 4 | dBm/15 kHz | -94.00 | -94.00 | -infinity | -90.30 |
| $\hat{E}_s/N_{oc}$ | dB | 4.00 | 4.00 | -Infinity | 7.70 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE priori to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 7682 ms from the beginning of time period T2.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = 7680 ms

TTI insertion uncertainty = $TTI_{DCCH}$ = 1 ms; $2xTTI_{DCCH}$ = 2 ms

The overall delays measured shall be less than a total of 7682 ms in this test case (note: a total of 7680 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

## 8.15.2 E-UTRAN FDD-TDD Inter-frequency event triggered reporting when DRX is used under fading propagation conditions in asynchronous cells

### 8.15.2.1 Test purpose

To verify the UE's ability to make a correct reporting of an event in DRX under fading propagation conditions in asynchronous cells within the E-UTRA FDD-TDD inter-frequency cell search requirements.

### 8.15.2.2 Test applicability

This test applies to all types of E-UTRA UE supporting FDD and TDD release 9 and forward. Applicability requires support for FGI bits 5 and 25.

### 8.15.2.3 Minimum conformance requirements

When DRX is in use the UE shall be able to identify a new detectable E-UTRAN TDD inter frequency cell within $T_{identify_inter}$ as shown in table 8.15.2.3-1.

**Table 8.15.2.3-1: Requirement to identify a newly detectable TDD inter frequency cell**

| DRX cycle length (s) | $T_{identify_inter}$ (s) (DRX cycles) | |
|---|---|---|
| | Gap period = 40 ms | Gap period = 80 ms |
| ≤0.16 | Non DRX Requirements in TS 36.133 [4] clause 8.1.2.3.4.1 are applicable | Non DRX Requirements in TS 36.133 [4] clause 8.1.2.3.4.1 are applicable |
| 0.256 | 5.12*Nfreq (20*Nfreq) | 7.68*Nfreq (30*Nfreq) |
| 0.32 | 6.4*Nfreq (20*Nfreq) | 7.68*Nfreq (24*Nfreq) |
| 0.32<DRX-cycle≤2.56 | Note (20*Nfreq) | Note (20*Nfreq) |
| Note: Time depends upon the DRX cycle in use | | |

A cell shall be considered detectable provided following conditions are fulfilled:

- RSRP and RSRP Ês/Iot according to Annex I.2.3 for a corresponding Band,
- other RSRP related side conditions given in TS 36.133 [4] Section 9.1 are fulfilled,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band.

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per TDD inter-frequency and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period defined in Table 8.15.2.3-2.

**Table 8.15.2.3-2: Requirement to measure TDD inter frequency cells**

| DRX cycle length (s) | $T_{measure_inter}$ (s) (DRX cycles) |
|---|---|
| ≤0.84 | Non DRX Requirements in TS 36.133 [4] clause 8.1.2.3.4.1 are applicable |
| 0.08<DRX-cycle≤2.56 | Note ($5*N_{freq}$) |
| Note: Time depends upon the DRX cycle in use | |

The measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] sub-clause 9.1.

Reported measurements contained in periodically triggered measurement reports shall meet the requirements in TS 36.133 [4] section 9.

Reported measurements contained in event triggered periodic measurement reports shall meet the requirements in TS 36.133 [4] section 9.

The first report in event triggered periodic measurement reporting shall meet the requirements specified in TS 36.133 [4] section 8.1.2.3.2.2.1.3 Event Triggered Reporting.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] section 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a

delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{Identify_Inter}$ defined in TS 36.133 [4] Section 8.1.2.3.2.2 When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra}$ defined in TS 36.133 [4] section 8.1.2.2.1.1 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.4.2 and A.8.15.2.

## 8.15.2.4 Test description

### 8.15.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.15.

2. The general test parameter settings are set up according to Table 8.15.2.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.15.2.4.3.

5. There are two E-UTRA carriers and one E-UTRA FDD Cell 1 and one E-UTRA TDD Cell 2 on each carrier specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.15.2.4.1-1: General test parameters for E-UTRAN FDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions**

| Parameter | Unit | Test 1 | Test 2 | Comment |
|---|---|---|---|---|
| | | Value | | |
| Cell 1 PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | | As specified in section A.1.1 Note that UE may only be allocated at *On Duration* |
| Cell 1 PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | | As specified in section A.2.1. |
| Cell 2 PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | | As specified in section A.1.2 Note that UE may only be allocated at *On Duration* |
| Cell 2 PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | | As specified in section A.2.2 |
| Cell 1 E-UTRA FDD RF Channel Number | | 1 | | One FDD carrier frequency is used. |
| Cell 2 E-UTRA TDD RF Channel Number | | 2 | | One TDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | | |
| Active cell | | Cell 1 | | Cell 1 is on RF channel number 1 |
| Neighbour cell | | Cell 2 | | Cell 2 is on RF channel number 2 |
| Gap Pattern Id | | 0 | | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| A3-Offset | dB | -6 | | |
| Hysteresis | dB | 0 | | |
| CP length | | Normal | | |
| TimeToTrigger | s | 0 | | |
| Filter coefficient | | 0 | | L3 filtering is not used |
| Cell 2 Special subframe configuration | | 6 | | As specified in table 4.2-1 in TS 36.211 |
| Cell 2 Uplink-downlink configuration | | 1 | | As specified in table 4.2-2 in TS 36.211 |
| E-UTRA TDD PRACH configuration | | 4 | | As specified in table 5.7.1-2 in TS 36.211 |
| E-UTRA TDD Access Barring Information | - | Not Sent | | No additional delays in random access procedure. |
| DRX | | ON | | DRX related parameters are defined in Table 8.15.2.5-2 |
| Time offset between cells | ms | 3 | | Asynchronous cells |
| T1 | s | 5 | | |
| T2 | s | 5 | 30 | |

## 8.15.2.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A3 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

In Test 1 UE needs to be provided at least once every 500ms with new Timing Advance Command MAC control element to restart the Time alignment timer to keep UE uplink time alignment. Furthermore, the UE is allocated with PUSCH resource at every DRX cycle. In Test 2 the uplink time alignment is not maintained and UE needs to use RACH to obtain UL allocation for measurement reporting.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.15.2.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

3. The SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.15.2.5-1.

6. The UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 3882 ms for Test 1 and 26882 ms for Test 2 then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receives the MeasurementReport message in step 6 or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

11. Repeat step 1-10 for each sub-test in Table 8.15.2.4.1-1 as appropriate.

### 8.15.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.15.2.4.3-1: Common Exception messages for Additional E-UTRAN FDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-7<br>Table H.3.1-9<br>Table H.3.7-1<br>Table H.3.7-2<br>Table H.3.7-3 |

**Table 8.15.2.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN FDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig-DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.15.2.4.3-3: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.15.2.4.3-4: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20: SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | Channel-bandwidth-dependent parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.15.2.4.3-5: *MeasResults:* Additional E-UTRAN FDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | Set according to specific test | |
| rsrqResult | INTEGER(0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.15.2.4.3-6: *MeasResultListEUTRA*: Additional E-UTRAN FDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 8.15.2.4.3-7: *PRACH-Config-DEFAULT*: Additional E-UTRAN FDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions test requirement**

| Derivation Path: TS 36.508 [7] clause 7.3.2 Table 7.3.2-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PRACH-Config-DEFAULT ::= SEQUENCE { | | | |
| prach-ConfigIndex | 4 | | |
| } | | | |

## 8.15.2.5 Test requirement

Tables 8.15.2.5-1, 8.15.2.5-2 and 8.15.2.5-3 define the primary level settings including test tolerances for E-UTRAN FDD-TDD inter-frequency event triggered reporting when DRX is used under fading propagation conditions in asynchronous cells test.

**Table 8.15.2.5-1: Cell specific test parameters for E-UTRAN FDD-TDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Pattern defined in D.1.1 (OP.1 FDD) and in D.2.2 (OP.2 TDD) | | OP.1 FDD | | OP.2 TDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |

| | | | | | |
|---|---|---|---|---|---|
| $\hat{E}_s/I_{ot}$ | dB | 4.00 | 4.00 | -Infinity | 7.70 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | |
| RSRP Note 4 | dBm/15 kHz | -94.00 | -94.00 | -Infinity | -90.30 |
| SCH_RP Note 4 | dBm/15 kHz | -94.00 | -94.00 | -infinity | -90.30 |
| $\hat{E}_s/N_{oc}$ | dB | 4.00 | 4.00 | -Infinity | 7.70 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE priori to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.15.2.5-2: DRX-Configuration to be used in E-UTRAN FDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf40 | sf1280 | |
| shortDRX | disable | disable | |

**Table 8.15.2.5-3: *TimeAlignmentTimer* and *sr-ConfigIndex*-Configuration to be used in E-UTRAN FDD-TDD inter-frequency event triggered reporting when DRX is used in fading propagation conditions**

| Field | Test1 | Test2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | For further information see section 6.3.2 in 3GPP TS 36.331 [5]. |
| sr-ConfigIndex | 0 | 0 | For further information see section 6.3.2 in 3GPP TS 36.331 and 10.1 in 3GPP TS 36.213 [8]. |

In Test1 when DRX cycle length = 40 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment when the UE send one Event A3 triggered measurement report on PUSCH.

In Test 2 when DRX cycle length = 1280 ms is used, the overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE starts to send preambles on the PRACH for Scheduling Request (SR) to obtain allocation to send the measurement report to Cell 2 on PUSCH.TTI insertion uncertainty = 2 ms.

For both tests:

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event A3 measurement report.

The overall delay measured when DRX cycle length is 40 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{Identify_Inter}$

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new FDD inter-frequency cell is defined.

$T_{Inter1}$ = 60 ms. It is defined in table 8.1.2.1-1 of TS36.133 [4] clause 8.1.2.1.

$N_{freq}$ = 1. It is defined in TS 36.133 [4] clause 8.1.2.1.1.

TTI insertion uncertainty = 2 ms

DRX cycle length = 40 ms

The overall delay measured when DRX cycle length is 40 ms shall be less than a total of 3882 ms.

The overall delay measured when DRX cycle length is 1280 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_inter}$

$T_{identify_inter}$ = 25600 ms. When DRX cycle length is 1280 ms then the $T_{identify_inter}$ is 20 x 1280 ms, as defined in Table 8.15.2.3-1.

TTI insertion uncertainty = 2 ms

DRX cycle length = 1280 ms

The overall delay measured when DRX cycle length is 1280 ms shall be less than a total of 26882 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.15.3 E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps

### 8.15.3.1 Test purpose

To verify the requirement for identification of a new CGI of E-UTRA cell with autonomous gaps.

### 8.15.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward with support of inter-frequency SI acquisition for HO. Applicability requires support for FGI bit 25.

### 8.15.3.3 Minimum conformance requirements

No explicit neighbour list is provided to the UE for identifying a new CGI of E-UTRA cell. The UE shall identify and report the CGI when requested by the network for the purpose of 'reportCGI'. The UE may make autonomous gaps in both downlink reception and uplink transmission for receiving MIB and SIB1 message according to section 5.5.3.1 of TS 36.331 [5]. Note that a UE is not required to use autonomous gap if si-RequestForHO is set to false. If autonomous gaps are used for measurement with the purpose of 'reportCGI', regardless of whether DRX is used or not, the UE shall be able to identify a new CGI of E-UTRA cell within:

$$T_{identify_CGI,inter} = T_{basic_identify_CGI,inter} \quad ms$$

Where

$T_{basic_identify_CGI,\ inter}$ = 150 ms. This is the time period used in the above equation where the maximum allowed time for the UE to identify a new CGI of E-UTRA cell is defined, provided that the E-UTRA cell has been already identified by the UE.

A cell shall be considered identifiable following conditions are fulfilled:

- RSRP related side conditions given in TS 36.133[4] Section 9.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.4 for a corresponding Band.

The MIB of an E-UTRA cell whose CGI is identified shall be considered decodable by the UE provided the PBCH demodulation requirements are met according to TS 36.331[5].

The requirement for identifying a new CGI of an E-UTRA cell within $T_{basic_identify_CGI,inter}$ is applicable when no DRX is used as well as when all the DRX cycles specified in 3GPP TS 36.331 [5] are used.

Given that continuous DL data allocation and no DRX is used, and no measurement gaps are configured, the UE shall have more than 60 ACK/NACKs transmitted during the identification of a new CGI of E-UTRA cell.

The ECGI reporting delay occurs due to the delay uncertainty when inserting the ECGI measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. In case DRX is used, the ECGI reporting may be delayed until the next DRX cycle.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.8 and A.8.15.3.

## 8.15.3.4 Test description

### 8.15.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. The general test parameter settings are set up according to Table 8.15.3.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.15.3.4.3.

5. There is one E-UTRA FDD carrier and one E-UTRA TDD carrier and two cells specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.15.3.4.1-1: General test parameters for E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Cell1 PDSCH parameters | | DL Reference Measurement Channel R.3 FDD | As specified in section A.1.1 |
| Cell1 PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Cell2 PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| E-UTRA RF channel number | | 1, 2 | One FDD and one TDD carrier frequency are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active cell | | Cell 1 | Cell 1 is on RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 1 is on RF channel number 2. |
| CP length | | Normal | |
| Cell 2 Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211[9]. |
| Cell 2 Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in TS 36.211[9]. |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| A3-Offset | dB | -6 | |
| Hysteresis | dB | 0 | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| si-RequestForHO | | TRUE | As specified in section 5.5.3.1 in TS 36.331[5]. |
| Time offset between cells | ms | 3 | Asynchronous cells |
| T1 | s | 5 | |
| T2 | s | ≤10 | |
| T3 | s | 5 | |

### 8.15.3.4.2 Test procedure

The test comprises of one active cell and one neighbour cell. PDCCHs indicating new transmissions should be sent continuously to ensure that the UE would have ACK/NACK sending during identifying a new CGI of E-UTRAN cell. The test consists of three successive time periods, with time durations of T1, T2 and T3 respectively. At the start of time duration T1, the UE does not have any timing information of cell 2. Gap pattern configuration is configured before T2 begins to enable inter-frequency monitoring. Starting T2, cell 2 becomes detectable and the UE is expected to detect and send a measurement report.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.15.3.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
3. The SS shall transmit an RRCConnectionReconfiguration with event A3 configured message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.15.3.5-1.
6. The UE shall transmit a MeasurementReport message triggered by Event A3.
7. The SS shall transmit an RRCConnectionReconfiguration message during period T2. The RRC message shall release MeasGapConfig according to Table 8.15.3.4.3-6.
8. The UE shall transmit RRCConnectionReconfigurationComplete message.
9. The SS shall transmit an RRCConnectionReconfiguration message during period T2, The RRC message shall create a measurement report configuration with purpose *reportCGI* and *si-RequestForHO* set to TRUE.
10. The SS shall start T3 timer when the last TTI containing the RRC message implying SI reading is sent to the UE.

11. The UE shall transmit RRCConnectionReconfigurationComplete message.

12. The UE shall transmit a MeasurementReport message containing the cell global identifier of cell 2 within 170 milliseconds from the start of T3. The UE shall be scheduled continuously throughout the test, and from the start of T3 until 170 ms at least 80 ACK/NACK shall be detected as being transmitted by the UE. If the overall delays measured from the beginning of time period T3 is less than 170 ms, and the UE have more than 80 ACK/NACKs transmitted from the start of T3 until 170 ms, then the number of successful tests is increased by one. If the UE fails to report the cell global identifier within the overall delays measured requirement, or the UE have less than 80 ACK/NACKs transmitted from the start of T3 until 170 ms, then the number of failure tests is increased by one.

13. After the SS receives the MeasurementReport message (in step 12) or when T3 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

14. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

15. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

16. Repeat step 2-15 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.15.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.15.3.4.3-1: Common Exception messages for E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.3.1-9 |

**Table 8.15.3.4.3-2: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.15.3.4.3-3: *MeasResults*: Additional E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.15.3.4.3-4: *MeasResultListEUTRA*: Additional E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step6)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.15.3.4.3-5: *ReportConfigEUTRA-PERIODICAL*: Additional E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step9)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportCGI | | |
| } | | | |
| } | | | |
| reportAmount | 1 | | |
| si-RequestForHO-r9 | setup | | |
| } | | | |

**Table 8.15.3.4.3-6: MeasGapConfig- Additional E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step7)**

| Derivation Path: 36.331, clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasGapConfig ::= CHOICE { | | | |
| Release | NULL | | |
| } | | | |

**Table 8.15.3.4.3-7: *MeasResultListEUTRA*: Additional E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step12)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| cgi-Info SEQUENCE{ | | | |
| cellGlobalId SEQUENCE{ | | | |
| plmn-Identity | plmn-Identity | | |
| cellIdentity | cellIdentity | | |
| } | | | |
| trackingAreaCode | | | |
| plmn-IdentityList | | | |
| } | | | |
| } | | | |

**Table 8.15.3.4.3-8: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step9)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(f1) ::= SEQUENCE { | | f1 is the frequency of the serving cell | |
| neighCellConfig | '01'B (No MBSFN subframes are present in all neighbour cells) | | |
| } | | | |
| MeasObjectEUTRA-GENERIC(f2) ::= SEQUENCE { | | f2 is the frequency of the neighbouring cell | |
| cellForWhichToReportCGI | Physical Cell ID of Cell 2 | | |
| } | | | |

**Table 8.15.3.4.3-9: *SystemInformationBlockType2*: Additional E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-1 SystemInformationBlockType2 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfig | Not present | | Cell 1 |
| } | | | |

**Table 8.15.3.4.3-10: *SystemInformationBlockType5*: Additional E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-3 SystemInformationBlockType5 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType5 ::= SEQUENCE { | | | |
| neighCellConfig[*n*] | 01'B (No MBSFN subframes are present in all neighbour cells) | | Cell 1 |
| } | | | |

**Table 8.15.3.4.3-11: *MeasConfig-DEFAULT*: Additional E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| } | | | |

**Table 8.15.3.4.3-12: *MeasConfig-DEFAULT*: Additional E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step9)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | IdReportConfig-P | | |
| } | | | |
| } | | | |

**Table 8.15.3.4.3-13: *MeasResults*: Additional E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step12)**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.15.3.4.3-14: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test requirement (step3)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(f1) ::= SEQUENCE { | | f1 is the frequency of the serving cell | |
| neighCellConfig | '01'B (No MBSFN subframes are present in all neighbour cells) | | |
| } | | | |

## 8.15.3.5 Test requirement

Tables 8.15.3.4.1-1 and 8.15.3.5-1 define the primary level settings including test tolerances for E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps test.

**Table 8.15.3.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.10 (OP.10 FDD) and in D.2.2 (OP.2 TDD) | | OP.10 FDD | OP.10 FDD | OP.10 FDD | OP.2 TDD | OP.2 TDD | OP.2 TDD |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA[Note 1] | dB | | | | | | |
| OCNG_RB[Note 1] | dB | | | | | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | 4 | -Infinity | 7 | 7 |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | 4 | -Infinity | 7 | 7 |
| RSRP Note 3 | dBm/15 KHz | -94 | -94 | -94 | -Infinity | -91 | -91 |
| SCH_RP Note3 | dBm/15 KHz | -94 | -94 | -94 | -Infinity | -91 | -91 |
| Propagation Condition | | AWGN | | | | | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | | | | | |
| Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. | | | | | | | |
| Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | | | | | |

The UE shall transmit a measurement report containing the cell global identifier of cell 2 within [170] milliseconds from the start of T3.

Test requirement = RRC Procedure delay + $T_{identify_CGI,inter}$ + reporting delay

= 15 + 150 + 2ms from the start of T3

= 167 ms, allow 170 ms.

The UE shall be scheduled continuously throughout the test, and from the start of T3 until 170 ms at least 60 ACK/NACK shall be detected as being transmitted by the UE.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

NOTE: The overall 80 ACK/NACK number is caused by two parts. Firstly, at least 60 ACK/NACK shall be sent during identifying the cell global identifier of cell 2 according to the requirement in TS 36.133[4] Section 8.1.2.3.5.1. Secondly, given that continuous DL data allocation, and the measurement gaps have been deconfigured before the start of T3, additional 20 ACK/NACK shall be sent from the start of T3 until 170 ms excludes 150 ms for identifying the cell global identifier of cell 2.

# 8.16 E-UTRAN Carrier Aggregation Measurements

## 8.16.1 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX

### 8.16.1.1 Test purpose

To verify the UE's ability to make correct reportings of Event A2 and A6 under deactivated SCell in non-DRX within the requirements in TS 36.133[4] section 8.3.3.2.1.

### 8.16.1.2 Test applicability

This test applies to all types of E-UTRA UE FDD release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA. Applicability requires support for FGI bit 111.

This test applies to all types of E-UTRA UE FDD release 11 and forward that support CA with Intra-band non-contiguous CA. Applicability requires support for FGI bit 111.

### 8.16.1.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD or TDD cell on the secondary component carrier within $T_{identify_scc}$, according to the parameter *measCycleSCell* where $T_{identify_scc}$ = 20 *measCycleSCell*

A cell shall be considered detectable when

- RSRP related side condition given in Section 9.1 are fulfilled for a corresponding Band,
- SCH_RP|$_{dBm}$ and SCH Ês/Iot according to Annex I.2.7 for a corresponding Band

The measurement period for deactivated SCell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell* where $T_{measure_scc}$ = 5 *measCycleSCell*. The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified cells on the secondary component carrier, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_scc}$.

The measurement accuracy for all measured cells shall be as specified in TS 36.133[4] sub-clause 9.1.11 (Carrier aggregation measurement accuracy)

A UE may reconfigure the receiver bandwidth or turn on/off one of the RF chains when performing measurements on an SCC with deactivated SCell.This may cause interruptions on PCell that are specified in TS 36.133 [4] sub-clause 7.8.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc}$ defined in TS 36.133 [4] clause 8.3.3.2.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_scc}$ defined in TS 36.133 [4] section 8.3.3.2.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_scc}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.3.3.2.1 and A.8.16.1.

## 8.16.1.4 Test description

### 8.16.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.42 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.1.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.1.4.3.

5. Cell1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier, and Cell3 is the neighbouring cell of Cell 2 on the secondary component carrier, Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test, Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.1.4.1-1: General test parameters for E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| E-UTRA RF Channel Number | | | 1, 2 | Two radio channels are used for this test |
| Active PCell | | | Cell 1 | Primary cell on RF channel number 1. |
| Configured deactivated SCell | | | Cell 2 | Configured deactivated secondary cell on RF channel number 2. |
| Neighbour cell | | | Cell 3 | Neighbour cell to be identified on RF channel number 2. |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | Channel bandwidth for cells on primary and secondary component carriers |
| CP length | | | Normal | |
| DRX | | | OFF | Continuous monitoring of primary cell |
| A2 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A2. |
| | Threshold RSRP | dBm | -93 | Actual RSRP threshold for event A2. Needs to take absolute accuracy tolerance in TS 36.133 [4] section 9.1.11.1 into account plus margin. |
| | Time To Trigger | s | 0 | |
| A6 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A6. |
| | Offset | dB | -6 | Offset parameter for evaluation of event A6. Needs to take relative accuracy tolerance in TS 36.133 [4] section 9.1.11.2 into account plus margin. |
| | Report on leave | | False | |
| | Time To Trigger | s | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| SCell measurement cycle (measCycleSCell) | | ms | 320 | |
| Cell2 timing offset to cell1 | | µs | 0 | |
| Time alignment error between cell2 and cell1 | | µs | ≤ Time alignment error as specified in TS 36.104 [29] clause 6.5.3.1. | The value of time alignment error depends upon the type of carrier aggregation. |
| Cell3 timing offset to cell1 | | µs | 3 | Synchronous cells<br>3µs or 92*Ts |
| T1 | | s | 5 | During this time the UE shall be aware of cells 1 and 2 but not cell 3. |
| T2 | | s | ≤12 | UE shall report Event A6 within 6.4s (20×measCycleSCell) |
| T3 | | s | 5 | UE shall report Event A2 within 200 ms and 1.6s for cells 1 and 2, respectively. |
| NOTE: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | | |

## 8.16.1.4.2 Test procedure

The test consists of three successive time periods, with duration of T1, T2 and T3, respectively. During T1 the UE shall not have any information on cell 3. Immediately at beginning of T2 the transmission power of cell 3 is increased to same level as for cell 2, and due to usage of an offset this shall result in reporting of Event A6. At beginning of T3 the transmission powers of cells 1, 2 and 3 are reduced below a threshold value and this shall result in reporting of Event A2 for PCell and SCell, respectively.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0, C.1 for all downlink physical channels.

3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.1.4.3.

4. Set the parameters according to T1 in Table 8.16.1.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

5. After 2s from start of T1 SS shall transmit an RRCConnectionReconfiguration message with events A6 and A2 configured.

6. The UE shall transmit an RRCConnectionReconfigurationComplete message.

7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.16.1.5-1.

8. The UE shall transmit a MeasurementReport message triggered by Event A6. If the measurement reporting delay from the beginning of time period T2 is less than 6402ms, then count a success for the event “A6”. Otherwise count a fail for the event “A6”.

9. After the SS receives the MeasurementReport message in step 8) or when T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 8.16.1.5-1.

10. The UE shall transmit MeasurementReport messages triggered by Event A2 for Cell 1 and Cell 2, respectively.

10a. If the measurement reporting delay for Cell 1 from the beginning of time period T3 is less than 202ms, then count a success for the event “Cell 1 A2”. If the UE sends event-triggered measurement reports before the beginning of time period T3 or fails to report the event within the overall delays measured requirement, then the number of failure for the event “Cell 1 A2” is increased by one.

10b. If the measurement reporting delay for Cell 2 from the beginning of time period T3 is less than 1602ms, then count a success for the event “Cell 2 A2”. If the UE sends event-triggered measurement reports before the beginning of time period T3 or fails to report the event within the overall delays measured requirement, then the number of failure for the event “Cell 2 A2 is increased by one.

11. After the SS receives the MeasurementReport message in step 8) and 10) or when T3 expires, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

12. Set Cell 3 physical cell identity = ((current cell 3 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. If physical cell identity of Cell 3 = physical cell identity of Cell 2 then skip this physical cell identity value for Cell 3.

13. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 3-13 until a test verdict has been achieved.

Each of the events “A6”, “Cell 1 A2” and “Cell 2 A2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.16.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.1.4.3-1: Common Exception messages for E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.4.1-5<br>Table H.4.1-6 |

**Table 8.16.1.4.3-2: *SCellToAddMod-r10:* Additional E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation Path: 36.508, Table 4.6.3-19D | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SCellToAddMod-r10 ::= SEQUENCE { | | | |
| sCellIndex-r10 | 1 | | |
| cellIdentification-r10 SEQUENCE { | | | |
| physCellId-r10 | PhysicalCellIdentity of Cell 2 | | |
| dl-CarrierFreq-r10 | Same downlink EARFCN as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 8.16.1.4.3-3: *RadioResourceConfigCommonSCell-r10:* Additional E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation Path: 36.508, Table 4.6.3-13A | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigCommonSCell-r10 ::= SEQUENCE { | | | |
| nonUL-Configuration-r10 SEQUENCE { | | | |
| dl-Bandwidth-r10 | Same downlink system bandwidth as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 8.16.1.4.3-4: *MeasConfig:* Additional E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation path: 36.508 clause 4.6.6 table 4.6.6-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasConfig ::= SEQUENCE { | | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entries | | |
| measObjectId[1] | IdMeasObject-f1 | | |
| measObject[1] | MeasObjectEUTRA-GENERIC(f1) | Cell 1 | |
| measObjectId[2] | IdMeasObject-f2 | | |
| measObject[2] | MeasObjectEUTRA-GENERIC(f2) | Cell 2, 3 | |
| } | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId)) OF SEQUENCE { | 2 entries | | |
| reportConfigId[1] | IdReportConfig-A6 | | |
| reportConfig[1] | ReportConfig-A6 | | |
| reportConfigId[2] | IdReportConfig-A2 | | |
| reportConfig[2] | ReportConfig-A2-H | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) OF SEQUENCE { | 3 entries | | |
| measId[1] | 1 | | |
| measObjectId[1] | IdMeasObject-f2 | | |
| reportConfigId[1] | IdReportConfig-A6 | | |
| measId[2] | 2 | | |
| measObjectId[2] | IdMeasObject-f1 | | |
| reportConfigId[2] | IdReportConfig-A2 | | |
| measId[3] | 3 | | |
| measObjectId[3] | IdMeasObject-f2 | | |
| reportConfigId[3] | IdReportConfig-A2 | | |
| } | | | |
| } | | | |

**Table 8.16.1.4.3-5: MeasObjectEUTRA-GENERIC(f2): Additional E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation Path: 36.508 clause 7.3.3 Table 7.3.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | |
| } | | | |

**Table 8.16.1.4.3-6: *ReportConfig-A2-H*: Additional E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation Path: 36.508 clause 4.6.6 table 4.6.6-5 ReportConfigEUTRA-A2(-93) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A2(Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA2 SEQUENCE { | | | |
| a2-Threshold CHOICE { | | | |
| threshold-RSRP | 47 | The actual value is IE value – 140 dBm | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

**Table 8.16.1.4.3-7: *MeasurementReport*: Additional E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 2 | | |
| measResultPCell ::= SEQUENCE { | | Report Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | Not present | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | | |
| rsrqResultSCell-r10 | (0..34) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.1.4.3-8: *MeasurementReport*: Additional E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 3 | | |
| measResultPCell ::= SEQUENCE { | | Report Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | Not present | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | | |
| rsrqResultSCell-r10 | (0..34) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 8.16.1.5 Test requirement

Table 8.16.1.4-1 and Table 8.16.1.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX test.

**Table 8.16.1.5-1: Cell specific test parameters for E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | | 10 | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | 1x2 Low | | | 1x2 Low | | |
| Timing offset to Cell 1 | μs | - | | | 0 | | | 3 | | |
| OCNG Patterns defined inD.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | | | OP.2 FDD | | | OP.2 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -101.00 | | | -101.00 | | | | | |
| RSRP Note 3 | dBm/15 kHz | -81.80 | -81.80 | -104.00 | -81.80 | -81.80 | -104.20 | -infinity | -82.00 | -104.00 |
| $\hat{E}_s/I_{ot}$ | dB | 19.20 | 19.20 | -3.00 | 19.20 | 0.15 | -4.96 | -infinity | -0.25 | -4.70 |
| SCH_RP Note 3 | dBm/15 kHz | -81.80 | -81.80 | -104.00 | -81.80 | -81.80 | -104.20 | -infinity | -82.00 | -104.00 |
| $\hat{E}_s/N_{oc}$ | dB | 19.20 | 19.20 | -3.00 | 19.20 | 19.20 | -3.20 | -infinity | 19.00 | -3.00 |
| Propagation Condition | | ETU70 | | | | | | | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface

The actual overall delays measured in the tests may be up to $2\times TTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc}$

The overall delays measured test requirement for Event A6 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Where:

measurement reporting delay $= T_{identify_scc}$

$T_{identify_scc} = 20$ *measCycleSCell*

*measCycleSCell* = 320 ms

The UE shall send one Event A6 triggered measurement report with a measurement reporting delay of less than 6402 ms from the beginning of time T2 (note: this gives a total of 6400 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

In the RRC_CONNECTED state the measurement period for intra frequency RSRP and RSRQ measurements is 200 ms as per TS 36.133[4] section 8.1.2.2.1

The UE shall send one Event A2 triggered measurement report for Cell 1 with a measurement reporting delay of less than 202 ms (including 2 ms for TTI insertion uncertainty) from beginning of time T3.

The measurement period for deactivated scell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell*

The overall delays measured test requirement for Event A2 for Cell 2 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Where:

measurement reporting delay = $T_{measure_scc}$

where $T_{measure_scc}$ = 5 *measCycleSCell*.

The UE shall send one Event A2 triggered measurement report for Cell 2 with a measurement reporting delay of less than 1602ms from beginning of time T3 (note: this gives a total of 1600 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall not send event triggered measurement reports as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95% for each of the events.

The statistical pass/ fail decisions are done separately for Event A6 and Event A2.

Decide the test pass, if events A6 **and** A2 are passed, otherwise fail the UE.

## 8.16.2 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX

### 8.16.2.1 Test purpose

To verify the UE's ability to make correct reportings of Event A2 and A6 under deactivated SCell in non-DRX within the requirements in TS 36.133[4] section 8.3.3.2.1.

### 8.16.2.2 Test applicability

This test applies to all types of E-UTRA UE TDD release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA. Applicability requires support for FGI bit 111.

This test applies to all types of E-UTRA UE TDD release 11 and forward that support CA with Intra-band non-contiguous CA. Applicability requires support for FGI bit 111.

### 8.16.2.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD or TDD cell on the secondary component carrier within $T_{identify_scc}$, according to the parameter *measCycleSCell* where $T_{identify_scc}$ = 20 *measCycleSCell*

A cell shall be considered detectable when

- RSRP related side condition given in Section 9.1 are fulfilled for a corresponding Band,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.7 for a corresponding Band

The measurement period for deactivated scell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell* where $T_{measure_scc}$ = 5 *measCycleSCell*. The UE shall be capable of performing RSRP and RSRQ measurements for 8

identified cells on the secondary component carrier, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_scc}$.

The measurement accuracy for all measured cells shall be as specified in TS 36.133[4] sub-clause 9.1.11 (Carrier aggregation measurement accuracy)

A UE may reconfigure the receiver bandwidth or turn on/off one of the RF chains when performing measurements on an SCC with deactivated SCell.This may cause interruptions on PCell that are specified in TS 36.133 [4] sub-clause 7.8.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc}$ defined in TS 36.133 [4] clause 8.3.3.2.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_scc}$ defined in TS 36.133 [4] section 8.3.3.2.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_scc}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.3.3.2.1 and A.8.16.2.

## 8.16.2.4 Test description

### 8.16.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.42 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.2.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.2.4.3.

5. Cell1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier, and Cell3 is the neighbouring cell of Cell 2 on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test, Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.2.4.1-1: General test parameters for E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| E-UTRA RF Channel Number | | | 1, 2 | Two radio channels are used for this test |
| Active PCell | | | Cell 1 | Primary cell on RF channel number 1. |
| Configured deactivated SCell | | | Cell 2 | Configured deactivated secondary cell on RF channel number 2. |
| Neighbour cell | | | Cell 3 | Neighbour cell to be identified on RF channel number 2. |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | Channel bandwidth for cells on primary and secondary component carriers |
| CP length | | | Normal | |
| Special subframe configuration | | | 6 | As specified in table 4.2.1 in TS 36.211. The same configuration applies to all cells. |
| Uplink-downlink configuration | | | 1 | |
| DRX | | | OFF | Continuous monitoring of primary cell |
| A2 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A2. |
| | Threshold RSRP | dBm | -93 | Actual RSRP threshold for event A2. Needs to take absolute accuracy tolerance in TS 36.133 [4] section 9.1.11.1 into account plus margin. |
| | Time To Trigger | s | 0 | |
| A6 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A6. |
| | Offset | dB | -6 | Offset parameter for evaluation of event A6. Needs to take relative accuracy tolerance in TS 36.133 [4] section 9.1.11.2 into account plus margin. |
| | Report on leave | | False | |
| | Time To Trigger | s | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| SCell measurement cycle (measCycleSCell) | | ms | 320 | |
| Cell2 timing offset to cell1 | | µs | 0 | |
| Time alignment error between cell2 and cell1 | | µs | ≤ Time alignment error as specified in TS 36.104 [29] clause 6.5.3.1. | The value of time alignment error depends upon the type of carrier aggregation. |
| Cell3 timing offset to cell1 | | µs | 3 | Synchronous cells<br>3µs or 92*Ts |
| T1 | | s | 5 | During this time the UE shall be aware of cells 1 and 2 but not cell 3. |
| T2 | | s | ≤12 | UE shall report Event A6 within 6.4s (20×scellMeasCycle) |
| T3 | | s | 5 | UE shall report Event A2 within 200 ms and 1.6s for cells 1 and 2, respectively. |
| NOTE: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | | |

## 8.16.2.4.2 Test procedure

The test consists of three successive time periods, with duration of T1, T2 and T3, respectively. During T1 the UE shall not have any information on cell 3. Immediately at beginning of T2 the transmission power of cell 3 is increased to same level as for cell 2, and due to usage of an offset this shall result in reporting of Event A6. At beginning of T3 the transmission powers of cells 1, 2 and 3 are reduced below a threshold value and this shall result in reporting of Event A2 for PCell and SCell, respectively.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0, C.1 for all downlink physical channels.

3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.2.4.3.

4. Set the parameters according to T1 in Table 8.16.2.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

5. After 2s from start of T1, SS shall transmit an RRCConnectionReconfiguration message with events A6 and A2 configured.

6. The UE shall transmit an RRCConnectionReconfigurationComplete message.

7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.16.2.5-1.

8. The UE shall transmit a MeasurementReport message triggered by Event A6. If the measurement reporting delay from the beginning of time period T2 is less than 6402ms, then count a success for the event “A6”. Otherwise count a fail for the event “A6”.

9. After the SS receives the MeasurementReport message in step 8) or when T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 8.16.2.5-1.

10. The UE shall transmit MeasurementReport messages triggered by Event A2 for Cell 1 and Cell 2, respectively.

10a. If the measurement reporting delay for Cell 1 from the beginning of time period T3 is less than 202ms, then count a success for the event “Cell 1 A2”. If the UE sends event-triggered measurement reports before the beginning of time period T3 or fails to report the event within the overall delays measured requirement, then the number of failure for the event “Cell 1 A2” is increased by one.

10b. If the measurement reporting delay for Cell 2 from the beginning of time period T3 is less than 1602ms, then count a success for the event “Cell 2 A2”. If the UE sends event-triggered measurement reports before the beginning of time period T3 or fails to report the event within the overall delays measured requirement, then the number of failure for the event “Cell 2 A2” is increased by one.

11. After the SS receives the MeasurementReport message in step 8) and 10) or when T3 expires, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

12. Set Cell 3 physical cell identity = ((current cell 3 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. If physical cell identity of Cell 3 = physical cell identity of Cell 2 then skip this physical cell identity value for Cell 3.

13. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 3-13 until a test verdict has been achieved.

Each of the events “A6”, “Cell 1 A2” and “Cell 2 A2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.16.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.2.4.3-1: Common Exception messages for E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.4.1-5<br>Table H.4.1-6 |

**Table 8.16.2.4.3-2: *SCellToAddMod-r10:* Additional E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation Path: 36.508, Table 4.6.3-19D | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SCellToAddMod-r10 ::= SEQUENCE { | | | |
| sCellIndex-r10 | 1 | | |
| cellIdentification-r10 SEQUENCE { | | | |
| physCellId-r10 | PhysicalCellIdentity of Cell 2 | | |
| dl-CarrierFreq-r10 | Same downlink EARFCN as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 8.16.2.4.3-3: *RadioResourceConfigCommonSCell-r10:* Additional E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation Path: 36.508, Table 4.6.3-13A | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigCommonSCell-r10 ::= SEQUENCE { | | | |
| nonUL-Configuration-r10 SEQUENCE { | | | |
| dl-Bandwidth-r10 | Same downlink system bandwidth as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 8.16.2.4.3-4: *MeasConfig:* Additional E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation path: 36.508 clause 4.6.6 table 4.6.6-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasConfig ::= SEQUENCE { | | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entries | | |
| measObjectId[1] | IdMeasObject-f1 | | |
| measObject[1] | MeasObjectEUTRA-GENERIC(f1) | Cell 1 | |
| measObjectId[2] | IdMeasObject-f2 | | |
| measObject[2] | MeasObjectEUTRA-GENERIC(f2) | Cell 2, 3 | |
| } | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId)) OF SEQUENCE { | 2 entries | | |
| reportConfigId[1] | IdReportConfig-A6 | | |
| reportConfig[1] | ReportConfig-A6 | | |
| reportConfigId[2] | IdReportConfig-A2 | | |
| reportConfig[2] | ReportConfig-A2-H | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) OF SEQUENCE { | 3 entries | | |
| measId[1] | 1 | | |
| measObjectId[1] | IdMeasObject-f2 | | |
| reportConfigId[1] | IdReportConfig-A6 | | |
| measId[2] | 2 | | |
| measObjectId[2] | IdMeasObject-f1 | | |
| reportConfigId[2] | IdReportConfig-A2 | | |
| measId[3] | 3 | | |
| measObjectId[3] | IdMeasObject-f2 | | |
| reportConfigId[3] | IdReportConfig-A2 | | |
| } | | | |
| } | | | |

**Table 8.16.2.4.3-5: MeasObjectEUTRA-GENERIC(f2): Additional E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation Path: 36.508 clause 4.6.6-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | |
| } | | | |

**Table 8.16.2.4.3-6: *ReportConfig-A2-H*: Additional E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation Path: 36.508 clause 4.6.6 table 4.6.6-5 ReportConfigEUTRA-A2(-90) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A2(Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA2 SEQUENCE { | | | |
| a2-Threshold CHOICE { | | | |
| threshold-RSRP | 47 | The actual value is IE value – 140 dBm | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

**Table 8.16.2.4.3-7: *MeasurementReport*: Additional E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 2 | | |
| measResultPCell::= SEQUENCE { | | Report Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | Not present | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | | |
| rsrqResultSCell-r10 | (0..34) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.2.4.3-8: *MeasurementReport*: Additional E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 3 | | |
| measResultPCell::= SEQUENCE { | | Report Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | Not present | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | | |
| rsrqResultSCell-r10 | (0..34) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 8.16.2.5 Test requirement

Table 8.16.2.4-1 and Table 8.16.2.5-1 define the primary level settings including test tolerances for E-UTRAN TDD-event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX test.

**Table 8.16.2.5-1: Cell specific test parameters for E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | | 10 | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | 1x2 Low | | | 1x2 Low | | |
| Timing offset to Cell 1 | µs | - | | | 0 | | | 3 | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | | | OP.2 TDD | | | OP.2 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -101.00 | | | -101.00 | | | | | |
| RSRP Note 3 | dBm/15 kHz | -81.80 | -81.80 | -104.00 | -81.80 | -81.80 | -104.20 | -infinity | -82.00 | -104.00 |
| $\hat{E}_s/I_{ot}$ | dB | 19.20 | 19.20 | -3.00 | 19.20 | 0.15 | -4.96 | -infinity | -0.25 | -4.70 |
| SCH_RP Note 3 | dBm/15 kHz | -81.80 | -81.80 | -104.00 | -81.80 | -81.80 | -104.20 | -infinity | -82.00 | -104.00 |
| $\hat{E}_s/N_{oc}$ | dB | 19.20 | 19.20 | -3.00 | 19.20 | 19.20 | -3.20 | -infinity | 19.00 | -3.00 |
| Propagation Condition | | ETU70 | | | | | | | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface

The actual overall delays measured in the tests may be up to $2\times TTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc}$

The overall delays measured test requirement for Event A6 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Where:

measurement reporting delay $= T_{identify_scc}$

$T_{identify_scc} = 20$ *measCycleSCell*

*measCycleSCell* = 320 ms

The UE shall send one Event A6 triggered measurement report with a measurement reporting delay of less than 6402 ms from the beginning of time T2 (note: this gives a total of 6400 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

In the RRC_CONNECTED state the measurement period for intra frequency RSRP and RSRQ measurements is 200 ms as per TS 36.133[4] section 8.1.2.2.1

The UE shall send one Event A2 triggered measurement report for Cell 1 with a measurement reporting delay of less than 202 ms (including 2 ms for TTI insertion uncertainty) from beginning of time T3.

The measurement period for deactivated scell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell*

The overall delays measured test requirement for Event A2 for Cell 2 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Where:

measurement reporting delay = $T_{measure_scc}$

where $T_{measure_scc}$ = 5 *measCycleSCell*.

The UE shall send one Event A2 triggered measurement report for Cell 2 with a measurement reporting delay of less than 1602ms from beginning of time T3 (note: this gives a total of 1600 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall not send event triggered measurement reports as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95% for each of the events.

The statistical pass/ fail decisions are done separately for Event A6 and Event A2.

Decide the test pass, if events A6 **and** A2 are passed, otherwise fail the UE.

## 8.16.3 E-UTRAN FDD-FDD Event triggered reporting on deactivated SCell with PCell interruption in non-DRX

### 8.16.3.1 Test purpose

To verify the UE's ability to make a correct reporting of an event on deactivated SCell with PCell interruption in non-DRX within the E-UTRA FDD-FDD measurements of the secondary component carrier cell search requirements in section 8.3.3.2.1 while at the same time fulfilling the requirements on interruption rate.

### 8.16.3.2 Test applicability

This test applies to all types of E-UTRA UE FDD release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA. Applicability requires support for FGI bit 111.

This test applies to all types of E-UTRA UE FDD release 11 and forward that support CA with Intra-band non-contiguous CA. Applicability requires support for FGI bit 111.

### 8.16.3.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD or TDD cell on the secondary component carrier within $T_{identify_scc}$, according to the parameter *measCycleSCell* where $T_{identify_scc}$ = 20 *measCycleSCell*

A cell shall be considered detectable when

- RSRP related side condition given in Section 9.1 are fulfilled for a corresponding Band,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.7 for a corresponding Band

The measurement period for deactivated scell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell* where $T_{measure_scc}$ = 5 *measCycleSCell*. The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified cells on the secondary component carrier, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_scc}$.

The measurement accuracy for all measured cells shall be as specified in TS 36.133[4] sub-clause 9.1.11 (Carrier aggregation measurement accuracy)

A UE may reconfigure the receiver bandwidth or turn on/off one of the RF chains when performing measurements on an SCC with deactivated SCell.This may cause interruptions on PCell that are specified in TS 36.133 [4] sub-clause 7.8.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc}$ defined in TS 36.133 [4] clause 8.3.3.2.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_scc}$ defined in TS 36.133 [4] section 8.3.3.2.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_scc}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.3.3.2.1 and A.8.16.3.

## 8.16.3.4 Test description

### 8.16.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1 [10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.3.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.3.4.3.

5. There are two E-UTRA FDD carriers with one cell on one E-UTRA FDD carrier and two cells on the other E-UTRA FDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.3.4.1-1: General test parameters for E-UTRAN FDD-FDD Event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.3 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| E-UTRA RF Channel Number | | | 1, 2 | Two radio channels are used for this test |
| Active PCell | | | Cell 1 | Primary cell on RF channel number 1 |
| Configured deactivated SCell | | | Cell 2 | Configured deactivated secondary cell on RF channel number 2 |
| Neighbour cell | | | Cell 3 | Neighbour cell to be identified on RF channel number 2 |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | Channel bandwidth for cells on primary and secondary component carriers |
| CP length | | | Normal | |
| DRX | | | OFF | Continuous monitoring of primary cell |
| A6 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A6 |
| | Offset | dB | -3 | Offset parameter for evaluation of event A6. Needs to take relative accuracy tolerance in section 9.1.11.2 in TS 36.133 [4] into account plus margin |
| | Report on leave | | False | |
| | Time To Trigger | s | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on secondary component carrier |
| Filter coefficient | | | 0 | L3 filtering is not used |
| SCell measurement cycle | | ms | 1280 | |
| Cell2 timing offset to cell1 | | µs | 0 | |
| Time alignment error between cell2 and cell1 | | µs | ≤ Time alignment error as specified in TS 36.104 [29] clause 6.5.3.1. | The value of time alignment error depends upon the type of carrier aggregation. |
| Cell3 timing offset to cell1 | | µs | 3 | Synchronous cells<br>3µs or 92*Ts |
| T1 | | s | 5 | During this time the UE shall be aware of cells 1 and 2 but not cell 3 |
| T2 | | s | ≤30 | UE shall report Event A6 within 25.6s (20×scellMeasCycle) |
| NOTE: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | | |

## 8.16.3.4.2 Test procedure

The test consists of Cell 1 the PCell, Cell 2 the SCell on the Secondary Component Carrier (SCC), and Cell 3 the neighbouring cell on the SCC. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A6 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 3. Starting T2, Cell 3 becomes detectable and the UE is expected to detect and send a measurement report. The UE shall be scheduled on PCell continuously throughout the test.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. Set the parameters according to T1 in Table 8.16.3.5-1 as appropriate. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.

5. The SS shall transmit an RRCConnectionReconfiguration message with Event A6 configured.

6. The UE shall transmit RRCConnectionReconfigurationComplete message.

7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.16.3.5-1.

8. UE shall transmit a MeasurementReport message triggered by Event A6. If the overall delays measured from the beginning of time period T2 is less than 25602 ms and at least 99.5% of all expected ACK/NACKS are detected by the system simulator from the start of T1 until the measurement report is received during T2 then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement or less than 99.5% of all expected ACK/NACKS are detected by the system simulator then the number of failure tests is increased by one.

9. After the SS receives the MeasurementReport message in step 8) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

10. Set Cell 3 physical cell identity = ((current cell 3 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. If physical cell identity of Cell 3 = physical cell identity of Cell 2 then skip this physical cell identity value for Cell 3.

11. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

12. Repeat step 3-11 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.16.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.3.4.3-1: Common Exception messages for E-UTRAN FDD-FDD Event triggered reporting on deactivated SCell with PCell interruption in non-DRX**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.4.1-2<br>Table H.4.1-3<br>Table H.4.1-5<br>Table H.4.1-6 |

**Table 8.16.3.4.3-2: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD-FDD Event triggered reporting on deactivated SCell with PCell interruption in non-DRX test requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1, Cell 2, Cell 3 |
| } | | | |

**Table 8.16.3.4.3-3: *SystemInformationBlockType2*: Additional E-UTRAN FDD-FDD Event triggered reporting on deactivated SCell with PCell interruption in non-DRX test requirements**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-1 SystemInformationBlockType2 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfig | Not present | | Cell 1 |
| } | | | |

**Table 8.16.3.4.3-4: *SystemInformationBlockType3*: Additional E-UTRAN FDD-FDD Event triggered reporting on deactivated SCell with PCell interruption in non-DRX test requirements**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-2 SystemInformationBlockType3 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType3 ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1, Cell 2, Cell 3 |
| } | | | |

## 8.16.3.5 Test requirement

Table 8.16.3.4.1-1 and Table 8.16.3.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-FDD Event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX test.

**Table 8.16.3.5-1: Cell specific test parameters for E-UTRAN FDD-FDD Event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | | 10 | |
| OCNG Pattern defined in D.1.10 (OP.10 FDD) and in D.1.2 (OP.2 FDD) | | OP.10 FDD | | OP.2 FDD | | OP.2 FDD | |
| Timing offset to Cell1 | µs | - | | 0 | | 3 | |
| PBCH_RA | dB | 0 | | 0 | | 0 | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | | | |
| RSRP Note 4 | dBm/15 kHz | -82 | -82 | -82 | -82 | -Infinity | -82 |
| $\hat{E}_s / I_{ot}$ | dB | 16 | 16 | 16 | -0.11 | -Infinity | -0.11 |
| SCH_RP Note 4 | dBm/15 kHz | -82 | -82 | -82 | -82 | -Infinity | -82 |
| $\hat{E}_s / N_{oc}$ | dB | 16 | 16 | 16 | 16 | -Infinity | 16 |
| Propagation Condition | | AWGN | | | | | |

NOTE 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
NOTE 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall send one Event A6 triggered measurement report, with a measurement reporting delay less than 25602 ms from the beginning of time period T2.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A6 triggered measurement report to Cell 3.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = 25.6s

TTI insertion uncertainty = $TTI_{DCCH}$ = 1 ms; 2x$TTI_{DCCH}$ = 2 ms

The overall delays measured shall be less than a total of 25602 ms in this test case (note: this gives a total of 25.6s for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The UE shall be scheduled on PCell continuously throughout the test. From the start of T1 until the measurement report is received during T2, at least 99.5% of all expected ACK/NACKs shall be transmitted by the UE.

For a test to be considered successful requirements on both event detection and percentage of transmitted ACK/NACKs have to be fulfilled simultaneously.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

## 8.16.4 E-UTRAN TDD-TDD Event triggered reporting on deactivated SCell with PCell interruption in non-DRX

### 8.16.4.1 Test purpose

To verify the UE's ability to make a correct reporting of an event on deactivated SCell with PCell interruption in non-DRX within the E-UTRA TDD-TDD measurements of the secondary component carrier cell search requirements in section 8.3.3.2.1 while at the same time fulfilling the requirements on interruption rate.

### 8.16.4.2 Test applicability

This test applies to all types of E-UTRA UE TDD release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA. Applicability requires support for FGI bit 111.

This test applies to all types of E-UTRA UE TDD release 11 and forward that support CA with Intra-band non-contiguous CA. Applicability requires support for FGI bit 111.

### 8.16.4.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD or TDD cell on the secondary component carrier within $T_{identify_scc}$, according to the parameter *measCycleSCell* where $T_{identify_scc}$ = 20 *measCycleSCell.*

A cell shall be considered detectable when

- RSRP related side condition given in Section 9.1 are fulfilled for a corresponding Band,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.7 for a corresponding Band

The measurement period for deactivated scell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell* where $T_{measure_scc}$ = 5 *measCycleSCell*. The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified cells on the secondary component carrier, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_scc}$.

The measurement accuracy for all measured cells shall be as specified in TS 36.133[4] sub-clause 9.1.11 (Carrier aggregation measurement accuracy)

A UE may reconfigure the receiver bandwidth or turn on/off one of the RF chains when performing measurements on an SCC with deactivated SCell.This may cause interruptions on PCell that are specified in TS 36.133 [4] sub-clause 7.8.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc}$ defined in TS 36.133 [4] clause 8.3.3.2.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_scc}$ defined in TS 36.133 [4] section 8.3.3.2.1 becomes undetectable for a period ≤ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_scc}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.3.3.2.1 and A.8.16.4.

## 8.16.4.4 Test description

### 8.16.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1 [10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.4.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.4.4.3.

5. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.4.4.1-1: General test parameters for E-UTRAN TDD-TDD Event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| E-UTRA RF Channel Number | | | 1, 2 | Two radio channels are used for this test |
| Active PCell | | | Cell 1 | Primary cell on RF channel number 1 |
| Configured deactivated SCell | | | Cell 2 | Configured deactivated secondary cell on RF channel number 2 |
| Neighbour cell | | | Cell 3 | Neighbour cell to be identified on RF channel number 2 |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | Channel bandwidth for cells on primary and secondary component carriers |
| CP length | | | Normal | |
| Special subframe configuration | | | 6 | As specified in table 4.2.1 in TS 36.211 [8] The same configuration applies to all cells |
| Uplink-downlink configuration | | | 1 | |
| DRX | | | OFF | Continuous monitoring of primary cell |
| A6 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A6. |
| | Offset | dB | -3 | Offset parameter for evaluation of event A6. Needs to take relative accuracy tolerance in section 9.1.11.2 in TS 36.133 [4] into account plus margin |
| | Report on leave | | False | |
| | Time To Trigger | s | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on secondary component carrier |
| Filter coefficient | | | 0 | L3 filtering is not used |
| SCell measurement cycle | | ms | 1280 | |
| Cell2 timing offset to cell1 | | µs | 0 | |
| Time alignment error between cell2 and cell1 | | µs | ≤ Time alignment error as specified in TS 36.104 [29] clause 6.5.3.1. | The value of time alignment error depends upon the type of carrier aggregation. |
| Cell3 timing offset to cell1 | | µs | 3 | Synchronous cells<br>3µs or 92*Ts |
| T1 | | s | 5 | During this time the UE shall be aware of cells 1 and 2 but not cell 3 |
| T2 | | s | ≤30 | UE shall report Event A6 within 25.6s (20×scellMeasCycle) |
| NOTE: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | | |

## 8.16.4.4.2 Test procedure

The test consists of Cell 1 the PCell, Cell 2 the SCell on the Secondary Component Carrier (SCC), and Cell 3 the neighbouring cell on the SCC. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A6 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 3. Starting T2, Cell 3 becomes detectable and the UE is expected to detect and send a measurement report. The UE shall be scheduled on PCell continuously throughout the test.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. Set the parameters according to T1 in Table 8.16.4.5-1 as appropriate. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.

5. SS shall transmit an RRCConnectionReconfiguration message with Event A6 configured.

6. The UE shall transmit RRCConnectionReconfigurationComplete message.

7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.16.4.5-1.

8. UE shall transmit a MeasurementReport message triggered by Event A6. If the overall delays measured from the beginning of time period T2 is less than 25602 ms and at least 99.5% of all expected ACK/NACKS are detected by the system simulator from the start of T1 until the measurement report is received during T2 then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement or less than 99.5% of all expected ACK/NACKS are detected by the system simulator then the number of failure tests is increased by one.

9. After the SS receive the MeasurementReport message in step 8) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

10. Set Cell 3 physical cell identity = ((current cell 3 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. If physical cell identity of Cell 3 = physical cell identity of Cell 2 then skip this physical cell identity value for Cell 3.

11. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

12. Repeat step 3-11 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.16.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.4.4.3-1: Common Exception messages for E-UTRAN TDD-TDD Event triggered reporting on deactivated SCell with PCell interruption in non-DRX**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.4.1-2<br>Table H.4.1-3<br>Table H.4.1-5<br>Table H.4.1-6 |

## 8.16.4.5 Test requirement

Table 8.16.4.4.1-1 and Table 8.16.4.5-1 define the primary level settings including test tolerances for E-UTRAN TDD-TDD Event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX test.

**Table 8.16.4.5-1: Cell specific test parameters for E-UTRAN TDD-TDD Event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | | 10 | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | | OP.2 TDD | | OP.2 TDD | |
| Timing offset to Cell1 | µs | - | | 0 | | 3 | |
| PBCH_RA | dB | 0 | | 0 | | 0 | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | | | |
| RSRP Note 4 | dBm/15 kHz | -82 | -82 | -82 | -82 | -Infinity | -82 |
| $\hat{E}_s / I_{ot}$ | dB | 16 | 16 | 16 | -0.11 | -Infinity | -0.11 |
| SCH_RP Note 4 | dBm/15 kHz | -82 | -82 | -82 | -82 | -Infinity | -82 |
| $\hat{E}_s / N_{oc}$ | dB | 16 | 16 | 16 | 16 | -Infinity | 16 |
| Propagation Condition | | AWGN | | | | | |

NOTE 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
NOTE 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall send one Event A6 triggered measurement report, with a measurement reporting delay less than 25602 ms from the beginning of time period T2.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A6 triggered measurement report to Cell 3.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = 25.6s

TTI insertion uncertainty = $TTI_{DCCH}$ = 1 ms; 2x$TTI_{DCCH}$ = 2 ms

The overall delays measured shall be less than a total of 25602 ms in this test case (note: this gives a total of 25.6s for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The UE shall be scheduled on PCell continuously throughout the test. From the start of T1 until the measurement report is received during T2, at least 99.5% of all expected ACK/NACKs shall be transmitted by the UE.

For a test to be considered successful requirements on both event detection and percentage of transmitted ACK/NACKs have to be fulfilled simultaneously.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

## 8.16.5 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX for 20 MHz bandwidth

### 8.16.5.1 Test purpose

To verify the UE's ability to make correct reportings of Event A2 and A6 under deactivated SCell in non-DRX within the requirements in TS 36.133[4] section 8.3.3.2.1.

### 8.16.5.2 Test applicability

This test applies to all types of E-UTRA UE FDD release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA with 20 MHz channel bandwidth for both PCell and SCell. Applicability requires support for FGI bit 111.

This test applies to all types of E-UTRA UE FDD release 11 and forward that support CA with Intra-band non-contiguous CA with 20 MHz channel bandwidth for both PCell and SCell. Applicability requires support for FGI bit 111.

### 8.16.5.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD or TDD cell on the secondary component carrier within $T_{identify_scc}$, according to the parameter *measCycleSCell* where $T_{identify_scc}$ = 20 *measCycleSCell.*

A cell shall be considered detectable when

- RSRP related side condition given in Section 9.1 are fulfilled for a corresponding Band,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.7 for a corresponding Band.

The measurement period for deactivated scell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell* where $T_{measure_scc}$ = 5 *measCycleSCell*. The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified cells on the secondary component carrier, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_scc}$.

The measurement accuracy for all measured cells shall be as specified in TS 36.133[4] sub-clause 9.1.11 (Carrier aggregation measurement accuracy).

A UE may reconfigure the receiver bandwidth or turn on/off one of the RF chains when performing measurements on an SCC with deactivated SCell.This may cause interruptions on PCell that are specified in TS 36.133 [4] sub-clause 7.8.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$.This measurement reporting delay excludes a delay which is caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc}$ defined in TS 36.133 [4] section 8.3.3.2.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_scc}$ defined in TS 36.133 [4] section 8.3.3.2.1 becomes undetectable for a period ≤ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_scc}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.3.3.2.1 and A.8.16.5.

## 8.16.5.4 Test description

### 8.16.5.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.42 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.5.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.5.4.3.

5. Cell1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier, and Cell3 is the neighbouring cell of Cell 2 on the secondary component carrier, Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test, Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.5.4.1-1: General test parameters for E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions, 20 MHz**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.4 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.10 FDD | As specified in section A.2.1 |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 20 | Channel bandwidth for cells on primary and secondary component carriers |
| A2 | Threshold RSRP | dBm | -96 | Actual RSRP threshold for event A2. Needs to take absolute accuracy tolerance in clause 9.1.11.1 into account plus margin. |
| Note 1: See Table 8.16.1.4.1-1 for other general test parameters.<br>Note 2: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | | |

### 8.16.5.4.2 Test procedure

The test consists of three successive time periods, with duration of T1, T2 and T3, respectively. During T1 the UE shall not have any information on cell 3. Immediately at beginning of T2 the transmission power of cell 3 is increased to same level as for cell 2, and due to usage of an offset this shall result in reporting of Event A6. At beginning of T3 the transmission powers of cells 1, 2 and 3 are reduced below a threshold value and this shall result in reporting of Event A2 for PCell and SCell, respectively.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0, C.1 for all downlink physical channels.

3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.5.4.3.

4. Set the parameters according to T1 in Table 8.16.5.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

5. After 2s from start of T1 SS shall transmit an RRCConnectionReconfiguration message with events A6 and A2 configured.

6. The UE shall transmit an RRCConnectionReconfigurationComplete message.

7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.16.5.5-1.

8. The UE shall transmit a MeasurementReport message triggered by Event A6. If the measurement reporting delay from the beginning of time period T2 is less than 6402ms, then count a success for the event “A6”. Otherwise count a fail for the event “A6”.

9. After the SS receives the MeasurementReport message in step 8) or when T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 8.16.5.5-1.

10. The UE shall transmit MeasurementReport messages triggered by Event A2 for Cell 1 and Cell 2, respectively.

10a. If the measurement reporting delay for Cell 1 from the beginning of time period T3 is less than 202ms, then count a success for the event “Cell 1 A2”. If the UE sends event-triggered measurement reports before the beginning of time period T3 or fails to report the event within the overall delays measured requirement, then the number of failure for the event “Cell 1 A2” is increased by one.

10b. If the measurement reporting delay for Cell 2 from the beginning of time period T3 is less than 1602ms, then count a success for the event “Cell 2 A2”. If the UE sends event-triggered measurement reports before the beginning of time period T3 or fails to report the event within the overall delays measured requirement, then the number of failure for the event "Cell 2 A2" is increased by one.

11. After the SS receives the MeasurementReport message in step 8) and 10) or when T3 expires, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

12. Set Cell 3 physical cell identity = ((current cell 3 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. If physical cell identity of Cell 3 = physical cell identity of Cell 2 then skip this physical cell identity value for Cell 3.

13. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 3-13 until a test verdict has been achieved.

Each of the events “A6”, “Cell 1 A2” and “Cell 2 A2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.16.5.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.5.4.3-1: Common Exception messages for E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions, 20 MHz requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.4.1-5<br>Table H.4.1-6 |

**Table 8.16.5.4.3-2: *SCellToAddMod-r10:* Additional E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions, 20 MHz requirements**

| Derivation Path: 36.508, Table 4.6.3-19D | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SCellToAddMod-r10 ::= SEQUENCE { | | | |
| sCellIndex-r10 | 1 | | |
| cellIdentification-r10 SEQUENCE { | | | |
| physCellId-r10 | PhysicalCellIdentity of Cell 2 | | |
| dl-CarrierFreq-r10 | Same downlink EARFCN as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 8.16.5.4.3-3: *RadioResourceConfigCommonSCell-r10:* Additional E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions, 20 MHz requirements**

| Derivation Path: 36.508, Table 4.6.3-13A | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigCommonSCell-r10 ::= SEQUENCE { | | | |
| nonUL-Configuration-r10 SEQUENCE { | | | |
| dl-Bandwidth-r10 | Same downlink system bandwidth as used for Cell 2 | n100 | |
| } | | | |
| } | | | |

**Table 8.16.5.4.3-4: *MeasConfig:* Additional E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions, 20 MHz requirements**

| Derivation path: 36.508 clause 4.6.6 table 4.6.6-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasConfig ::= SEQUENCE { | | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entries | | |
| measObjectId[1] | IdMeasObject-f1 | | |
| measObject[1] | MeasObjectEUTRA-GENERIC(f1) | Cell 1 | |
| measObjectId[2] | IdMeasObject-f2 | | |
| measObject[2] | MeasObjectEUTRA-GENERIC(f2) | Cell 2, 3 | |
| } | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId)) OF SEQUENCE { | 2 entries | | |
| reportConfigId[1] | IdReportConfig-A6 | | |
| reportConfig[1] | ReportConfig-A6 | | |
| reportConfigId[2] | IdReportConfig-A2 | | |
| reportConfig[2] | ReportConfig-A2-H | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) OF SEQUENCE { | 3 entries | | |
| measId[1] | 1 | | |
| measObjectId[1] | IdMeasObject-f2 | | |
| reportConfigId[1] | IdReportConfig-A6 | | |
| measId[2] | 2 | | |
| measObjectId[2] | IdMeasObject-f1 | | |
| reportConfigId[2] | IdReportConfig-A2 | | |
| measId[3] | 3 | | |
| measObjectId[3] | IdMeasObject-f2 | | |
| reportConfigId[3] | IdReportConfig-A2 | | |
| } | | | |
| } | | | |

**Table 8.16.5.4.3-5: MeasObjectEUTRA-GENERIC(f2): Additional E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions, 20 MHz requirements**

| Derivation Path: 36.508 clause 7.3.3 Table 7.3.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | |
| } | | | |

**Table 8.16.5.4.3-6: *ReportConfig-A2-H*: Additional E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions, 20 MHz requirements**

| Derivation Path: 36.508 clause 4.6.6 table 4.6.6-5 ReportConfigEUTRA-A2(-90) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A2(Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA2 SEQUENCE { | | | |
| a2-Threshold CHOICE { | | | |
| threshold-RSRP | 44 | The actual value is IE value – 140 dBm | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

**Table 8.16.5.4.3-7: *MeasurementReport*: Additional E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions, 20 MHz requirements**

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 2 | | |
| measResultPCell ::= SEQUENCE { | | Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | Not present | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | | |
| rsrqResultSCell-r10 | (0..34) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.5.4.3-8: *MeasurementReport*: Additional E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions, 20 MHz requirements**

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 3 | | |
| measResultPCell ::= SEQUENCE { | | Report Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | Not present | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | | |
| rsrqResultSCell-r10 | (0..34) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 8.16.5.5 Test requirement

Table 8.16.5.4-1 and Table 8.16.5.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions for 20 MHz channel BW.

**Table 8.16.5.5-1: Cell specific test parameters for E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions, 20 MHz**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| $BW_{channel}$ | MHz | 20 | | | 20 | | | 20 | | |
| OCNG Patterns defined in A.3.2.1.11 (OP.11 FDD) and in A.3.2.1.12 (OP.12 FDD) | | OP.11 FDD | | | OP.12 FDD | | | OP.12 FDD | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -104 | | | -104.1 | | | | | |
| RSRP Note 3 | dBm/15 kHz | -84.8 | -84.8 | -107 | -84.8 | -84.8 | -107.2 | -infinity | -85.1 | -107.1 |
| $\hat{E}_s/I_{ot}$ | dB | 19.2 | 19.2 | -3 | 19.3 | 0.25 | -4.86 | -infinity | -0.35 | -4.73 |
| SCH_RP Note 3 | dBm/15 kHz | -84.8 | -84.8 | -107 | -84.8 | -84.8 | -107.2 | -infinity | -85.1 | -107.1 |
| $\hat{E}_s/N_{oc}$ | dB | 19.2 | 19.2 | -3 | 19.3 | 19.3 | -3.1 | -infinity | 19 | -3 |
| Note: See Table 8.16.1.5.1-1 for other cell-specific test parameters. | | | | | | | | | | |

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface.

The actual overall delays measured in the tests may be up to $2\times TTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc.}$

The overall delays measured test requirement for Event A6 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty.

Where:

measurement reporting delay $= T_{identify_scc.}$

$T_{identify_scc} = 20$ *measCycleSCell*

*measCycleSCell* = 320 ms

The UE shall send one Event A6 triggered measurement report with a measurement reporting delay of less than 6402 ms from the beginning of time T2 (note: this gives a total of 6400 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

In the RRC_CONNECTED state the measurement period for intra frequency RSRP and RSRQ measurements is 200 ms as per TS 36.133[4] section 8.1.2.2.1.

The UE shall send one Event A2 triggered measurement report for Cell 1 with a measurement reporting delay of less than 202 ms (including 2 ms for TTI insertion uncertainty) from beginning of time T3.

The measurement period for deactivated scell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell*

The overall delays measured test requirement for Event A2 for Cell 2 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty.

Where:

measurement reporting delay $= T_{measure_scc}$

where $T_{measure_scc} = 5$ *measCycleSCell*.

The UE shall send one Event A2 triggered measurement report for Cell 2 with a measurement reporting delay of less than 1602ms from beginning of time T3 (note: this gives a total of 1600 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall not send event triggered measurement reports as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95% for each of the events.

The statistical pass/ fail decisions are done separately for Event A6 and Event A2.

Decide the test pass, if events A6 **and** A2 are passed, otherwise fail the UE.

## 8.16.6 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 20 MHz bandwidth

### 8.16.6.1 Test purpose

To verify the UE's ability to make correct reportings of Event A2 and A6 under deactivated SCell in non-DRX within the requirements in TS 36.133[4] section 8.3.3.2.1.

### 8.16.6.2 Test applicability

This test applies to all types of E-UTRA UE TDD release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA with 20 MHz channel bandwidth for both PCell and SCell. Applicability requires support for FGI bit 111.

This test applies to all types of E-UTRA UE TDD release 11 and forward that support CA with Intra-band non-contiguous CA with 20 MHz channel bandwidth for both PCell and SCell. Applicability requires support for FGI bit 111.

### 8.16.6.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD or TDD cell on the secondary component carrier within $T_{identify_scc}$, according to the parameter *measCycleSCell* where $T_{identify_scc}$ = 20 *measCycleSCell.*

A cell shall be considered detectable when

- RSRP related side condition given in Section 9.1 are fulfilled for a corresponding Band,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.7 for a corresponding Band

The measurement period for deactivated scell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell* where $T_{measure_scc}$ = 5 *measCycleSCell*. The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified cells on the secondary component carrier, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_scc}$.

The measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] sub-clause 9.1.11 (Carrier aggregation measurement accuracy).

A UE may reconfigure the receiver bandwidth or turn on/off one of the RF chains when performing measurements on an SCC with deactivated SCell.This may cause interruptions on PCell that are specified in TS 36.133 [4] sub-clause 7.8.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$.This measurement reporting delay excludes a delay which is caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc}$ defined in TS 36.133 [4] section 8.3.3.2.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_scc}$ defined in TS 36.133 [4] section 8.3.3.2.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_scc}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.3.3.2.1 and A.8.16.6.

### 8.16.6.4 Test description

#### 8.16.6.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.42 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.6.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.6.4.3.

5. Cell1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier, and Cell3 is the neighbouring cell of Cell 2 on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test, Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.6.4.1-1: General test parameters for E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions, 20 MHz**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PDSCH parameters | | | DL Reference Measurement Channel R.3 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | | DL Reference Measurement Channel R.10 TDD | As specified in section A.2.2 |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 20 | Channel bandwidth for cells on primary and secondary component carriers |
| A2 | Threshold RSRP | dBm | -96 | Actual RSRP threshold for event A2. Needs to take absolute accuracy tolerance in clause 9.1.11.1 into account plus margin. |
| Note 1: See Table 8.16.2.4.1-1 for other general test parameters.<br>Note 2: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | | |

## 8.16.6.4.2 Test procedure

The test consists of three successive time periods, with duration of T1, T2 and T3, respectively. During T1 the UE shall not have any information on cell 3. Immediately at beginning of T2 the transmission power of cell 3 is increased to same level as for cell 2, and due to usage of an offset this shall result in reporting of Event A6. At beginning of T3 the transmission powers of cells 1, 2 and 3 are reduced below a threshold value and this shall result in reporting of Event A2 for PCell and SCell, respectively.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0, C.1 for all downlink physical channels.

3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.6.4.3.

4. Set the parameters according to T1 in Table 8.16.6.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

5. After 2s from start of T1 SS shall transmit an RRCConnectionReconfiguration message with events A6 and A2 configured.

6. The UE shall transmit an RRCConnectionReconfigurationComplete message.

7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.16.6.5-1.

8. The UE shall transmit a MeasurementReport message triggered by Event A6. If the measurement reporting delay from the beginning of time period T2 is less than 6402ms, then count a success for the event “A6”. Otherwise count a fail for the event “A6”.

9. After the SS receives the MeasurementReport message in step 8) or when T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 8.16.6.5-1.

10. The UE shall transmit MeasurementReport messages triggered by Event A2 for Cell 1 and Cell 2, respectively.

10a. If the measurement reporting delay for Cell 1 from the beginning of time period T3 is less than 202ms, then count a success for the event “Cell 1 A2”. If the UE sends event-triggered measurement reports before the beginning of time period T3 or fails to report the event within the overall delays measured requirement, then the number of failure for the event “Cell 1 A2” is increased by one.

10b. If the measurement reporting delay for Cell 2 from the beginning of time period T3 is less than 1602ms, then count a success for the event “Cell 2 A2”. If the UE sends event-triggered measurement reports before the beginning of time period T3 or fails to report the event within the overall delays measured requirement, then the number of failure for the event “Cell 2 A2” is increased by one.

11. After the SS receives the MeasurementReport message in step 8) and 10) or when T3 expires, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

12. Set Cell 3 physical cell identity = ((current cell 3 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. If physical cell identity of Cell 3 = physical cell identity of Cell 2 then skip this physical cell identity value for Cell 3.

13. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 3-13 until a test verdict has been achieved.

Each of the events “A6”, “Cell 1 A2” and “Cell 2 A2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

#### 8.16.6.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.6.4.3-1: Common Exception messages for E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements, 20 MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.4.1-5<br>Table H.4.1-6 |

**Table 8.16.6.4.3-2: *SCellToAddMod-r10:* Additional E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements, 20 MHz**

| Derivation Path: 36.508, Table 4.6.3-19D | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SCellToAddMod-r10 ::= SEQUENCE { | | | |
| sCellIndex-r10 | 1 | | |
| cellIdentification-r10 SEQUENCE { | | | |
| physCellId-r10 | PhysicalCellIdentity of Cell 2 | | |
| dl-CarrierFreq-r10 | Same downlink EARFCN as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 8.16.6.4.3-3: *RadioResourceConfigCommonSCell-r10:* Additional E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements, 20 MHz**

| Derivation Path: 36.508, Table 4.6.3-13A | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigCommonSCell-r10 ::= SEQUENCE { | | | |
| nonUL-Configuration-r10 SEQUENCE { | | | |
| dl-Bandwidth-r10 | Same downlink system bandwidth as used for Cell 2 | n100 | |
| } | | | |
| } | | | |

**Table 8.16.6.4.3-4: *MeasConfig:* Additional E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements, 20 MHz**

| Derivation path: 36.508 clause 4.6.6 table 4.6.6-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasConfig ::= SEQUENCE { | | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entries | | |
| measObjectId[1] | IdMeasObject-f1 | | |
| measObject[1] | MeasObjectEUTRA-GENERIC(f1) | Cell 1 | |
| measObjectId[2] | IdMeasObject-f2 | | |
| measObject[2] | MeasObjectEUTRA-GENERIC(f2) | Cell 2, 3 | |
| } | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId)) OF SEQUENCE { | 2 entries | | |
| reportConfigId[1] | IdReportConfig-A6 | | |
| reportConfig[1] | ReportConfig-A6 | | |
| reportConfigId[2] | IdReportConfig-A2 | | |
| reportConfig[2] | ReportConfig-A2-H | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) OF SEQUENCE { | 3 entries | | |
| measId[1] | 1 | | |
| measObjectId[1] | IdMeasObject-f2 | | |
| reportConfigId[1] | IdReportConfig-A6 | | |
| measId[2] | 2 | | |
| measObjectId[2] | IdMeasObject-f1 | | |
| reportConfigId[2] | IdReportConfig-A2 | | |
| measId[3] | 3 | | |
| measObjectId[3] | IdMeasObject-f2 | | |
| reportConfigId[3] | IdReportConfig-A2 | | |
| } | | | |
| } | | | |

**Table 8.16.6.4.3-5: MeasObjectEUTRA-GENERIC(f2): Additional E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions, 20 MHz requirements**

| Derivation Path: 36.508 clause 7.3.3 Table 7.3.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | |
| } | | | |

**Table 8.16.6.4.3-6: *ReportConfig-A2-H*: Additional E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions, 20 MHz requirements**

| Derivation Path: 36.508 clause 4.6.6 table 4.6.6-5 ReportConfigEUTRA-A2(-90) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A2(Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA2 SEQUENCE { | | | |
| a2-Threshold CHOICE { | | | |
| threshold-RSRP | 44 | The actual value is IE value – 140 dBm | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

**Table 8.16.6.4.3-7: *MeasurementReport*: Additional E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions, 20 MHz requirements**

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 2 | | |
| measResultPCell::= SEQUENCE { | | Report Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | Not present | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | | |
| rsrqResultSCell-r10 | (0..34) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.6.4.3-8: *MeasurementReport*: Additional E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions, 20 MHz requirements**

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 3 | | |
| measResultPCell::= SEQUENCE { | | Report Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | Not present | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | | |
| rsrqResultSCell-r10 | (0..34) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 8.16.6.5 Test requirement

Table 8.16.6.4-1 and Table 8.16.6.5-1 define the primary level settings including test tolerances for E-UTRAN TDD-e vent triggered reporting on configured but deactivated SCell in non-DRX test under fading propagation conditions for 20 MHz channel BW.

**Table 8.16.6.5-1: Cell specific test parameters for E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions, 20 MHz**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| $BW_{channel}$ | MHz | 20 | | | 20 | | | 20 | | |
| OCNG Patterns defined in D.2.7 (OP.7 TDD) and in D.2.8 (OP.8 TDD) | | OP.7 TDD | | | OP.8 TDD | | | OP.8 TDD | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -104 | | | -104.1 | | | | | |
| RSRP Note 3 | dBm/15 kHz | -84.8 | -84.8 | -107 | -84.8 | -84.8 | -107.2 | -infinity | -85.1 | -107.1 |
| $\hat{E}_s/I_{ot}$ | dB | 19.2 | 19.2 | -3 | 19.3 | 0.25 | -4.86 | -infinity | -0.35 | -4.73 |
| SCH_RP Note 3 | dBm/15 kHz | -84.8 | -84.8 | -107 | -84.8 | -84.8 | -107.2 | -infinity | -85.1 | -107.1 |
| $\hat{E}_s/N_{oc}$ | dB | 19.2 | 19.2 | -3 | 19.3 | 19.3 | -3.1 | -infinity | 19 | -3 |
| Note: See Table 8.16.2.5-1 for other cell-specific test parameters. | | | | | | | | | | |

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface.

The actual overall delays measured in the tests may be up to $2\times TTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc.}$

The overall delays measured test requirement for Event A6 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Where:

measurement reporting delay = $T_{identify_scc}$

$T_{identify_scc}$ = 20 *measCycleSCell*

*measCycleSCell* = 320 ms

The UE shall send one Event A6 triggered measurement report with a measurement reporting delay of less than 6402 ms from the beginning of time T2 (note: this gives a total of 6400 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

In the RRC_CONNECTED state the measurement period for intra frequency RSRP and RSRQ measurements is 200 ms as per TS 36.133[4] section 8.1.2.2.1.

The UE shall send one Event A2 triggered measurement report for Cell 1 with a measurement reporting delay of less than 202 ms (including 2 ms for TTI insertion uncertainty) from beginning of time T3.

The measurement period for deactivated scell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell*.

The overall delays measured test requirement for Event A2 for Cell 2 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Where:

measurement reporting delay = $T_{measure_scc}$

where $T_{measure_scc}$ = 5 *measCycleSCell*.

The UE shall send one Event A2 triggered measurement report for Cell 2 with a measurement reporting delay of less than 1602ms from beginning of time T3 (note: this gives a total of 1600 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall not send event triggered measurement reports as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95% for each of the events.

The statistical pass/ fail decisions are done separately for Event A6 and Event A2.

Decide the test pass, if events A6 **and** A2 are passed, otherwise fail the UE.

## 8.16.7 E-UTRA FDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 20 MHz bandwidth

### 8.16.7.1 Test purpose

To verify the UE's ability to make a correct reporting of an event on deactivated SCell with PCell interruption in non-DRX within the E-UTRA FDD measurements of the secondary component carrier cell search requirements in TS 36.133[4] section 8.3.3.2.1 while at the same time fulfilling the requirement on interruption rate.

### 8.16.7.2 Test applicability

This test applies to all types of E-UTRA UE FDD release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA with 20 MHz channel bandwidth for both PCell and SCell. Applicability requires support for FGI bit 111.

This test applies to all types of E-UTRA UE FDD release 11 and forward that support CA with Intra-band non-contiguous CA with 20 MHz channel bandwidth for both PCell and SCell. Applicability requires support for FGI bit 111.

### 8.16.7.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD or TDD cell on the secondary component carrier within $T_{identify_scc}$, according to the parameter *measCycleSCell* where $T_{identify_scc}$ = 20 *measCycleSCell.*

A cell shall be considered detectable when

- RSRP related side condition given in Section 9.1 are fulfilled for a corresponding Band,

- SCH_RP|$_{dBm}$ and SCH Ês/Iot according to Annex I.2.7 for a corresponding Band

The measurement period for deactivated scell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell* where $T_{measure_scc}$ = 5 *measCycleSCell*. The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified cells on the secondary component carrier, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_scc}$.

The measurement accuracy for all measured cells shall be as specified in TS 36.133[4] sub-clause 9.1.11 (Carrier aggregation measurement accuracy).

A UE may reconfigure the receiver bandwidth or turn on/off one of the RF chains when performing measurements on an SCC with deactivated SCell. This may cause interruptions on PCell that are specified in TS 36.133 [4] sub-clause 7.8.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc}$ defined in TS 36.133 [4] clause 8.3.3.2.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_scc}$ defined in TS 36.133 [4] section 8.3.3.2.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_scc}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.3.3.2.1 and A.8.16.7.

### 8.16.7.4 Test description

#### 8.16.7.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1 [10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.7.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.7.4.3.

5. There are two E-UTRA FDD carriers with one cell on one E-UTRA FDD carrier and two cells on the other E-UTRA FDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.7.4.1-1: General test parameters for E-UTRAN FDD event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX, 20 MHz**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.10 FDD | As specified in section A.2.1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 20 | Channel bandwidth for cells on primary and secondary component carriers |
| Note 1: See Table 8.16.3.4.1-1 for other general test parameters.<br>Note 2: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | |

## 8.16.7.4.2 Test procedure

The test consists of Cell 1 the PCell, Cell 2 the SCell on the Secondary Component Carrier (SCC), and Cell 3 the neighbouring cell on the SCC. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A6 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 3. Starting T2, Cell 3 becomes detectable and the UE is expected to detect and send a measurement report. The UE shall be scheduled on PCell continuously throughout the test.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. Set the parameters according to T1 in Table 8.16.7.5-1 as appropriate. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.

5. The SS shall transmit an RRCConnectionReconfiguration message with Event A6 configured.

6. The UE shall transmit RRCConnectionReconfigurationComplete message.

7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.16.7.5-1.

8. UE shall transmit a MeasurementReport message triggered by Event A6. If the overall delays measured from the beginning of time period T2 is less than 25602 ms and at least 99.5% of all expected ACK/NACKS are detected by the system simulator from the start of T1 until the measurement report is received during T2 then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement or less than 99.5% of all expected ACK/NACKS are detected by the system simulator then the number of failure tests is increased by one.

9. After the SS receives the MeasurementReport message in step 8) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

10. Set Cell 3 physical cell identity = ((current cell 3 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. If physical cell identity of Cell 3 = physical cell identity of Cell 2 then skip this physical cell identity value for Cell 3.

11. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

12. Repeat step 3-11 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.16.7.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.7.4.3-1: Common Exception messages for E-UTRAN FDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX, 20 MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.4.1-2<br>Table H.4.1-3<br>Table H.4.1-5<br>Table H.4.1-6 |

**Table 8.16.7.4.3-2: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX test requirements, 20 MHz**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1, Cell 2, Cell 3 |
| } | | | |

**Table 8.16.7.4.3-3: *SystemInformationBlockType2*: Additional E-UTRAN FDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX test requirements, 20 MHz**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-1 SystemInformationBlockType2 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfig | Not present | | Cell 1 |
| } | | | |

**Table 8.16.7.4.3-4: *SystemInformationBlockType3*: Additional E-UTRAN FDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX test requirements, 20 MHz**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-2 SystemInformationBlockType3 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType3 ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1, Cell 2, Cell 3 |
| } | | | |

## 8.16.7.5 Test requirement

Table 8.16.7.4.1-1 and Table 8.16.7.5-1 define the primary level settings including test tolerances for E-UTRAN TDD-event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX test for 20 MHz channel bandwidth.

**Table 8.16.7.5-1: Cell specific test parameters for E-UTRAN FDD event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX, 20 MHz**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| $BW_{channel}$ | MHz | 20 | | 20 | | 20 | |
| OCNG Patterns defined in D.1.17 (OP.17 FDD) and in D.1.12 (OP.12 FDD) | | OP.17 FDD | | OP.12 FDD | | OP.12 FDD | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -101 | | -101 | | | |
| RSRP Note 4 | dBm/15 kHz | -85 | -85 | -85 | -85 | -Infinity | -85 |
| $\hat{E}_s / I_{ot}$ | dB | 16 | 16 | 16 | -0.11 | -Infinity | -0.11 |
| SCH_RP Note 4 | dBm/15 kHz | -85 | -85 | -85 | -85 | -Infinity | -85 |
| $\hat{E}_s / N_{oc}$ | dB | 16 | 16 | 16 | 16 | -Infinity | 16 |
| Note: See Table 8.16.3.5-1 for other cell-specific test parameters. | | | | | | | |

The UE shall send one Event A6 triggered measurement report, with a measurement reporting delay less than 25602 ms from the beginning of time period T2.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A6 triggered measurement report to Cell 3.

The overall delays measured test requirement is expressed as:

- Overall delays measured = measurement reporting delay + TTI insertion uncertainty

- Measurement reporting delay = 25.6s

  - TTI insertion uncertainty = $TTI_{DCCH}$ = 1 ms; $2xTTI_{DCCH}$ = 2 ms

The overall delays measured shall be less than a total of 25602 ms in this test case (note: this gives a total of 25.6s for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The UE shall be scheduled on PCell continuously throughout the test. From the start of T1 until the measurement report is received during T2, at least 99.5% of all expected ACK/NACKs shall be transmitted by the UE.

For a test to be considered successful requirements on both event detection and percentage of transmitted ACK/NACKs have to be fulfilled simultaneously.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

## 8.16.8 E-UTRAN TDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 20 MHz bandwidth

### 8.16.8.1 Test purpose

To verify the UE's ability to make a correct reporting of an event on deactivated SCell with PCell interruption in non-DRX within the E-UTRA TDD measurements of the secondary component carrier cell search requirements in TS 36.133[4] section 8.3.3.2.1 while at the same time fulfilling the requirements on interruption rate.

### 8.16.8.2 Test applicability

This test applies to all types of E-UTRA UE TDD release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA with 20 MHz channel bandwidth for both PCell and SCell. Applicability requires support for FGI bit 111.

This test applies to all types of E-UTRA UE TDD release 11 and forward that support CA with Intra-band non-contiguous CA with 20 MHz channel bandwidth for both PCell and SCell. Applicability requires support for FGI bit 111.

### 8.16.8.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD or TDD cell on the secondary component carrier within $T_{identify_scc}$, according to the parameter *measCycleSCell* where $T_{identify_scc}$ = 20 *measCycleSCell*.

A cell shall be considered detectable when

- RSRP related side condition given in Section 9.1 are fulfilled for a corresponding Band,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.7 for a corresponding Band

The measurement period for deactivated scell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell* where $T_{measure_scc}$ = 5 *measCycleSCell*. The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified cells on the secondary component carrier, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_scc}$.

The measurement accuracy for all measured cells shall be as specified in TS 36.133[4] sub-clause 9.1.11 (Carrier aggregation measurement accuracy).

A UE may reconfigure the receiver bandwidth or turn on/off one of the RF chains when performing measurements on an SCC with deactivated SCell. This may cause interruptions on PCell that are specified in TS 36.133 [4] sub-clause 7.8.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc}$ defined in TS 36.133 [4] clause 8.3.3.2.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_scc}$ defined in TS 36.133 [4] section 8.3.3.2.1 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_scc}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.3.3.2.1 and A.8.16.8.

## 8.16.8.4 Test description

### 8.16.8.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1 [10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.8.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.8.4.3.

5. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.8.4.1-1: General test parameters for E-UTRAN TDD event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX, 20 MHz**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.3 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.10 TDD | As specified in section A.2.2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 20 | Channel bandwidth for cells on primary and secondary component carriers |
| Note 1: See Table 8.16.4.4.1-1 for other general test parameters.<br>Note 2: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | |

### 8.16.8.4.2 Test procedure

The test consists of Cell 1 the PCell, Cell 2 the SCell on the Secondary Component Carrier (SCC), and Cell 3 the neighbouring cell on the SCC. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A6 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 3. Starting T2, Cell 3 becomes detectable and the UE is expected to detect and send a measurement report. The UE shall be scheduled on PCell continuously throughout the test.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. Set the parameters according to T1 in Table 8.16.8.5-1 as appropriate. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.

5. The SS shall transmit an RRCConnectionReconfiguration message with Event A6 configured.

6. The UE shall transmit RRCConnectionReconfigurationComplete message.

7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.16.8.5-1.

8. The UE shall transmit a MeasurementReport message triggered by Event A6. If the overall delays measured from the beginning of time period T2 is less than 25602 ms and at least 99.5% of all expected ACK/NACKS are detected by the system simulator from the start of T1 until the measurement report is received during T2 then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement or less than 99.5% of all expected ACK/NACKS are detected by the system simulator then the number of failure tests is increased by one.

9. After the SS receives the MeasurementReport message in step 8) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

10. Set Cell 3 physical cell identity = ((current cell 3 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. If physical cell identity of Cell 3 = physical cell identity of Cell 2 then skip this physical cell identity value for Cell 3.

11. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

12. Repeat step 3-11 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.16.8.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.8.4.3-1: Common Exception messages for E-UTRAN TDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX, 20 MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.4.1-2<br>Table H.4.1-3<br>Table H.4.1-5<br>Table H.4.1-6 |

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## 8.16.8.5 Test requirement

Table 8.16.8.4.1-1 and Table 8.16.8.5-1 define the primary level settings including test tolerances for E-UTRAN TDD-event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX test for 20 MHz channel bandwidth.

**Table 8.16.8.5-1: Cell specific test parameters for E-UTRAN TDD event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX, 20 MHz**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| $BW_{channel}$ | MHz | 20 | | 20 | | 20 | |
| OCNG Patterns defined in D.2.7 (OP.7 TDD) and in D.2.8 (OP.8 TDD) | | OP.7 TDD | | OP.8 TDD | | OP.8 TDD | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -101 | | -101 | | | |
| RSRP Note 4 | dBm/15 kHz | -85 | -85 | -85 | -85 | -Infinity | -85 |
| $\hat{E}_s / I_{ot}$ | dB | 16 | 16 | 16 | -0.11 | -Infinity | -0.11 |
| SCH_RP Note 4 | dBm/15 kHz | -85 | -85 | -85 | -85 | -Infinity | -85 |
| $\hat{E}_s / N_{oc}$ | dB | 16 | 16 | 16 | 16 | -Infinity | 16 |
| Note: See Table 8.16.4.5-1 for other cell-specific test parameters. | | | | | | | |

The UE shall send one Event A6 triggered measurement report, with a measurement reporting delay less than 25602 ms from the beginning of time period T2.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A6 triggered measurement report to Cell 3.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = 25.6s

TTI insertion uncertainty = $TTI_{DCCH}$ = 1 ms; $2xTTI_{DCCH}$ = 2 ms

The overall delays measured shall be less than a total of 25602 ms in this test case (note: this gives a total of 25.6s for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The UE shall be scheduled on PCell continuously throughout the test. From the start of T1 until the measurement report is received during T2, at least 99.5% of all expected ACK/NACKs shall be transmitted by the UE.

For a test to be considered successful requirements on both event detection and percentage of transmitted ACK/NACKs have to be fulfilled simultaneously.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

## 8.16.9 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX for 10MHz+5MHz

### 8.16.9.1 Test purpose

Same test purpose as in clause 8.16.1.1.

### 8.16.9.2 Test applicability

This test applies to all types of E-UTRA UE FDD release 11 and forward that support CA. Applicability requires support for FGI bit 111.

## 8.16.9.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.16.1.3.

## 8.16.9.4 Test description

### 8.16.9.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz for cells on the PCC and 5 MHz for cells on the SCC as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.42 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.9.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.9.4.3.

5. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier, and Cell3 is the neighbouring cell of Cell 2 on the secondary component carrier, Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test, Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.9.4.1-1: General test parameters for E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions for 10MHz+5MHz bandwidth**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Channel bandwidth for cells on primary carriers ($BW_{channel}$) | MHz | 10 | Channel bandwidth for cells on primary carriers |
| Channel bandwidth for cells on secondary carriers ($BW_{channel}$) | MHz | 5 | Channel bandwidth for cells on secondary carriers |
| Note 1: See Table 8.16.1.4.1-1 for the other general parameters.<br>Note 2: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | |

### 8.16.9.4.2 Test procedure

Same test procedure as in clause 8.16.1.4.2 with the following exceptions:

- Instead of Table 8.16.1.4.3 → use Table 8.16.9.4.3.
- Instead of Table 8.16.1.5-1 → use Table 8.16.9.5-1.

### 8.16.9.4.3 Message contents

Same message contents as in clause 8.16.1.4.3.

## 8.16.9.5 Test requirement

Same test requirement as in clause 8.16.1.5 with the following exceptions:

- Instead of Table 8.16.1.4.1-1 → use Table 8.16.9.4.1-1.
- Instead of Table 8.16.1.5-1 → use Table 8.16.9.5-1.

**Table 8.16.9.5-1: Cell specific test parameters for E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions for 10MHz+5MHz bandwidth**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 5 | | | 5 | | |
| PDSCH Reference measurement channel defined in A.1.1 | | R.0 FDD | | | N/A | | | N/A | | |
| PDSCH allocation | $n_{PRB}$ | 13—36 | | | N/A | | | N/A | | |
| PCFICH/PDCCH/PHICH parameters defined in A.2.1 | | R.6 FDD | | | R.11 FDD | | | R.11 FDD | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | 1x2 Low | | | 1x2 Low | | |
| Timing offset to Cell 1 | μs | - | | | 0 | | | 3 | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.16 (OP.16 FDD) | | OP.1 FDD | | | OP.16 FDD | | | OP.16 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| $OCNG_RA^{Note\ 1}$ | dB | | | | | | | | | |
| $OCNG_RB^{Note\ 1}$ | dB | | | | | | | | | |
| $N_{oc}^{Note\ 2}$ | dBm/15 kHz | -101.00 | | | -101.00 | | | | | |
| RSRP Note 3 | dBm/15 kHz | -81.80 | -81.80 | -104.00 | -81.80 | -81.80 | -104.00 | -infinity | -82.00 | -104.00 |
| $\hat{E}_s/I_{ot}$ | dB | 19.20 | 19.20 | -3.00 | 19.20 | 0.15 | -4.76 | -infinity | -0.25 | -4.70 |
| SCH_RP Note 3 | dBm/15 kHz | -81.80 | -81.80 | -104.00 | -81.80 | -81.80 | -104.00 | -infinity | -82.00 | -104.00 |
| $\hat{E}_s/N_{oc}$ | dB | 19.20 | 19.20 | -3.00 | 19.20 | 19.20 | -3.00 | -infinity | 19.00 | -3.00 |
| Propagation Condition | | ETU70 | | | | | | | | |

NOTE 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

## 8.16.10 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 10MHz+5MHz

### 8.16.10.1 Test purpose

Same test purpose as in clause 8.16.2.1.

### 8.16.10.2 Test applicability

This test applies to all types of E-UTRA UE TDD release 11 and forward that support CA. Applicability requires support for FGI bit 111.

### 8.16.10.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.16.2.3.

### 8.16.10.4 Test description

#### 8.16.10.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz for cells on PCC, 5 MHz for cells on SCC as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.42 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.10.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.10.4.3.

5. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier, and Cell 3 is the neighbouring cell of Cell 2 on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test, Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.10.4.1-1: General test parameters for E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions for 10MHz+5MHz bandwidth**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Channel bandwidth for cells on primary carriers ($BW_{channel}$) | MHz | 10 | Channel bandwidth for cells on primary carriers |
| Channel bandwidth for cells on secondary carriers ($BW_{channel}$) | MHz | 5 | Channel bandwidth for cells on secondary carriers |
| Note 1: See Table 8.16.2.4.1-1 for the other general parameters.<br>Note 2: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | |

#### 8.16.10.4.2 Test procedure

Same test procedure as in clause 8.16.1.4.2 with the following exceptions:

- Instead of Table 8.16.2.4.3 → use Table 8.16.10.4.3.
- Instead of Table 8.16.2.5-1 → use Table 8.16.10.5-1.

#### 8.16.10.4.3 Message contents

Same message contents as in clause 8.16.2.4.3.

## 8.16.10.5 Test requirement

Same test requirement as in clause 8.16.2.5 with the following exceptions:

- Instead of Table 8.16.2.4.1-1 → use Table 8.16.10.4.1-1.
- Instead of Table 8.16.2.5-1 → use Table 8.16.10.5-1.

**Table 8.16.10.5-1: Cell specific test parameters for E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions for 10MHz+5MHz bandwidth**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 5 | | | 5 | | |
| PDSCH Reference measurement channel defined in A.1.2 | | R.0 TDD | | | N/A | | | N/A | | |
| PDSCH allocation | $n_{PRB}$ | 13—36 | | | N/A | | | N/A | | |
| PCFICH/PDCCH/PHICH parameters defined in A.2.2 | | R.6 TDD | | | R.12 TDD | | | R.12 TDD | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | 1x2 Low | | | 1x2 Low | | |
| Timing offset to Cell 1 | µs | - | | | 0 | | | 3 | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.10 (OP.10 TDD) | | OP.1 TDD | | | OP.10 TDD | | | OP.10 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -101.00 | | | -101.00 | | | | | |
| RSRP Note 3 | dBm/15 kHz | -81.80 | -81.80 | -104.00 | -81.80 | -81.80 | -104.00 | -infinity | -82.00 | -104.00 |
| $\hat{E}_s/I_{ot}$ | dB | 19.20 | 19.20 | -3.00 | 19.20 | 0.15 | -4.76 | -infinity | -0.25 | -4.70 |
| SCH_RP Note 3 | dBm/15 kHz | -81.80 | -81.80 | -104.00 | -81.80 | -81.80 | -104.00 | -infinity | -82.00 | -104.00 |
| $\hat{E}_s/N_{oc}$ | dB | 19.20 | 19.20 | -3.00 | 19.20 | 19.20 | -3.00 | -infinity | 19.00 | -3.00 |
| Propagation Condition | | ETU70 | | | | | | | | |

NOTE 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

# 8.16.11 E-UTRAN FDD event triggered reporting on deactivating SCell with PCell interruption in non-DRX for 10MHz+5MHz

## 8.16.11.1 Test purpose

Same test purpose as in clause 8.16.3.1.

## 8.16.11.2 Test applicability

This test applies to all types of E-UTRA UE FDD release 11 and forward that support CA. Applicability requires support for FGI bit 111.

## 8.16.11.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.16.3.3.

## 8.16.11.4 Test description

### 8.16.11.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz for cells on PCC, 5 MHz for cells on SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1 [10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.11.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.11.4.3.

5. There are two E-UTRA FDD carriers with one cell on one E-UTRA FDD carrier and two cells on the other E-UTRA FDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.11.4.1-1: General test parameters for E-UTRAN FDD event triggered reporting on deactivating SCell with PCell interruption in non-DRX for 10MHz+5MHz**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Channel bandwidth for cells on primary carrier ($BW_{channel}$) | MHz | 10 | Channel bandwidth for cells on primary component carrier |
| PDSCH parameters for cells on primary carriers | | DL Reference Measurement Channel R.3 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters for cells on primary carriers | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Channel bandwidth for cells on secondary carriers ($BW_{channel}$) | MHz | 5 | Channel bandwidth for cells on secondary component carrier |
| PCFICH/PDCCH/PHICH parameters for cells on secondary carrier | | DL Reference Measurement Channel R.11 FDD | As specified in section A.2.1 |
| Note 1: See Table 8.16.3.4.1-1 for other general test parameters.<br>Note 2: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | |

### 8.16.11.4.2 Test procedure

Same test procedure as in clause 8.16.3.4.2 with the following exceptions:

- Instead of Table 8.16.3.4.3 → use Table 8.16.11.4.3.
- Instead of Table 8.16.3.5-1 → use Table 8.16.11.5-1.

### 8.16.11.4.3 Message contents

Same message contents as in clause 8.16.3.4.3.

## 8.16.11.5 Test requirement

Same test requirement as in clause 8.16.3.5 with the following exceptions:

- Instead of Table 8.16.3.4.1-1 → use Table 8.16.11.4.1-1.
- Instead of Table 8.16.3.5-1 → use Table 8.16.11.5-1.

**Table 8.16.11.5-1: Cell specific test parameters for E-UTRAN FDD event triggered reporting on deactivating SCell with PCell interruption in non-DRX for 10MHz+5MHz**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| $BW_{channel}$ | MHz | 10 | | 5 | | 5 | |
| OCNG Patterns defined in D.1.10 (OP.10 FDD) and D.1.16 (OP.16 FDD) | | OP.10 FDD | | OP.16 FDD | | OP.16 FDD | |
| Note: See Table 8.16.3.5-1 for other cell-specific test parameters. | | | | | | | |

# 8.16.12 E-UTRAN TDD event triggered reporting on deactivating SCell with PCell interruption in non-DRX for 10MHz+5MHz

## 8.16.12.1 Test purpose

Same test purpose as in clause 8.16.4.1.

### 8.16.12.2 Test applicability

This test applies to all types of E-UTRA UE TDD release 11 and forward that support CA. Applicability requires support for FGI bit 111.

### 8.16.12.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.16.4.3.

### 8.16.12.4 Test description

#### 8.16.12.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz for cells on PCC, 5 MHz for cells on SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1 [10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.12.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.12.4.3.

5. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.12.4.1-1: General test parameters for E-UTRAN TDD event triggered reporting on deactivating SCell with PCell interruption in non-DRX for 10MHz+5MHz**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Channel bandwidth for cells on primary carrier ($BW_{channel}$) | MHz | 10 | Channel bandwidth for cells on primary component carrier |
| PDSCH parameters for cells on primary carriers | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters for cells on primary carriers | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Channel bandwidth for cells on secondary carriers ($BW_{channel}$) | MHz | 5 | Channel bandwidth for cells on secondary component carrier |
| PCFICH/PDCCH/PHICH parameters for cells on secondary carrier | | DL Reference Measurement Channel R.11 TDD | As specified in section A.2.2 |
| Note 1: See Table 8.16.4.4.1-1 for other general test parameters.<br>Note 2: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | |

#### 8.16.12.4.2 Test procedure

Same test procedure as in clause 8.16.4.4.2 with the following exceptions:

- Instead of Table 8.16.4.4.3 → use Table 8.16.12.4.3.
- Instead of Table 8.16.4.5-1 → use Table 8.16.12.5-1.

#### 8.16.12.4.3 Message contents

Same message contents as in clause 8.16.4.4.3.

### 8.16.12.5 Test requirement

Same test requirement as in clause 8.16.4.5 with the following exceptions:

- Instead of Table 8.16.4.4.1-1 → use Table 8.16.12.4.1-1.
- Instead of Table 8.16.4.5-1 → use Table 8.16.12.5-1.

**Table 8.16.12.5-1: Cell specific test parameters for E-UTRAN TDD event triggered reporting on deactivating SCell with PCell interruption in non-DRX for 10MHz+5MHz**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| $BW_{channel}$ | MHz | 10 | | 5 | | 5 | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.10 (OP.10 TDD) | | OP.1 TDD | | OP.10 TDD | | OP.10 TDD | |
| Note: See Table 8.16.4.5-1 for other cell-specific test parameters. | | | | | | | |

## 8.16.13 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX for 5MHz+5MHz

### 8.16.13.1 Test purpose

Same test purpose as in clause 8.16.1.1.

### 8.16.13.2 Test applicability

This test applies to all types of E-UTRA UE FDD release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA. Applicability requires support for FGI bit 111.

This test applies to all types of E-UTRA UE FDD release 11 and forward that support CA with Intra-band non-contiguous CA. Applicability requires support for FGI bit 111.

### 8.16.13.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.16.1.3.

### 8.16.13.4 Test description

#### 8.16.13.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz for cells on both the PCC and the SCC as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.42 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.13.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.13.4.3.

5. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier, and Cell3 is the neighbouring cell of Cell 2 on the secondary component carrier, Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test, Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.13.4.1-1: General test parameters for E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions for 5MHz+5MHz bandwidth**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.5 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.11 FDD | As specified in section A.2.1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 | Channel bandwidth for cells on primary component carrier |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 | Channel bandwidth for cells on secondary component carrier |
| Note 1: See Table 8.16.1.4.1-1 for other general test parameters.<br>Note 2: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | |

### 8.16.13.4.2 Test procedure

Same test procedure as in clause 8.16.1.4.2 with the following exceptions:

- Instead of Table 8.16.1.4.3 → use Table 8.16.13.4.3.

- Instead of Table 8.16.1.5-1 → use Table 8.16.13.5-1.

### 8.16.13.4.3 Message contents

Same message contents as in clause 8.16.1.4.3.

## 8.16.13.5 Test requirement

Same test requirement as in clause 8.16.1.5 with the following exceptions:

- Instead of Table 8.16.1.4.1-1 → use Table 8.16.13.4.1-1.

- Instead of Table 8.16.1.5-1 → use Table 8.16.13.5-1.

**Table 8.16.13.5-1: Cell specific test parameters for E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions for 5MHz+5MHz bandwidth**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | | | 2 | | | 2 | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | 1x2 Low | | | 1x2 Low | |
| Timing offset to Cell 1 | µs | - | | | 0 | | | 3 | |
| OCNG Patterns defined inD.1.15 (OP.15 FDD) and in D.1.16 (OP.16 FDD) | | OP.15 FDD | | | OP.16 FDD | | | OP.16 FDD | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | |
| PBCH_RB | dB | | | | | | | | |
| PSS_RA | dB | | | | | | | | |
| SSS_RA | dB | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | |
| PHICH_RA | dB | | | | | | | | |
| PHICH_RB | dB | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -101.00 | | | -101.00 | | | | |
| RSRP Note 3 | dBm/15 kHz | -81.80 | -81.80 | -104.00 | -81.80 | -81.80 | -104.00 | -infinity | -82.00 | -104.00 |
| $\hat{E}_s/I_{ot}$ | dB | 19.20 | 19.20 | -3.00 | 19.20 | 0.15 | -4.76 | -infinity | -0.25 | -4.70 |
| SCH_RP Note 3 | dBm/15 kHz | -81.80 | -81.80 | -104.00 | -81.80 | -81.80 | -104.00 | -infinity | -82.00 | -104.00 |
| $\hat{E}_s/N_{oc}$ | dB | 19.20 | 19.20 | -3.00 | 19.20 | 19.20 | -3.00 | -infinity | 19.00 | -3.00 |
| Propagation Condition | | ETU70 | | | | | | | | |

NOTE 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

## 8.16.14 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 5MHz+5MHz bandwidth

### 8.16.14.1 Test purpose

Same test purpose as in clause 8.16.2.1.

### 8.16.14.2 Test applicability

This test applies to all types of E-UTRA UE TDD release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA. Applicability requires support for FGI bit 111.

This test applies to all types of E-UTRA UE TDD release 11 and forward that support CA with Intra-band non-contiguous CA. Applicability requires support for FGI bit 111.

### 8.16.14.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.16.2.3.

## 8.16.14.4 Test description

### 8.16.14.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz for cells on both the PCC and the SCC as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.42 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.14.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.14.4.3.

5. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier, and Cell 3 is the neighbouring cell of Cell 2 on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test, Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.14.4.1-1: General test parameters for E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions for 5MHz+5MHz bandwidth**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.4.TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.11 TDD | As specified in section A.2.2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 | Channel bandwidth for cells on primary component carrier |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 | Channel bandwidth for cells on secondary component carrier |
| Note 1: See Table 8.16.2.4.1-1 for other general test parameters.<br>Note 2: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | |

### 8.16.14.4.2 Test procedure

Same test procedure as in clause 8.16.2.4.2 with the following exceptions:

- Instead of Table 8.16.2.4.3 → use Table 8.16.14.4.3.
- Instead of Table 8.16.2.5-1 → use Table 8.16.14.5-1.

### 8.16.14.4.3 Message contents

Same message contents as in clause 8.16.2.4.3.

## 8.16.14.5 Test requirement

Same test requirement as in clause 8.16.2.5 with the following exceptions:

- Instead of Table 8.16.2.4.1-1 → use Table 8.16.14.4.1-1.
- Instead of Table 8.16.2.5-1 → use Table 8.16.14.5-1.

**Table 8.16.14.5-1: Cell specific test parameters for E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions for 5MHz+5MHz bandwidth**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | | 10 | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | 1x2 Low | | | 1x2 Low | | |
| Timing offset to Cell 1 | µs | - | | | 0 | | | 3 | | |
| OCNG Patterns defined in D.2.9 (OP.9 TDD) and in D.2.10 (OP.10 TDD) | | OP.9 TDD | | | OP.10 TDD | | | OP.10 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -101.00 | | | -101.00 | | | | | |
| RSRP Note 3 | dBm/15 kHz | -81.80 | -81.80 | -104.00 | -81.80 | -81.80 | -104.00 | -infinity | -82.00 | -104.00 |
| $\hat{E}_s/I_{ot}$ | dB | 19.20 | 19.20 | -3.00 | 19.20 | 0.15 | -4.76 | -infinity | -0.25 | -4.70 |
| SCH_RP Note 3 | dBm/15 kHz | -81.80 | -81.80 | -104.00 | -81.80 | -81.80 | -104.00 | -infinity | -82.00 | -104.00 |
| $\hat{E}_s/N_{oc}$ | dB | 19.20 | 19.20 | -3.00 | 19.20 | 19.20 | -3.00 | -infinity | 19.00 | -3.00 |
| Propagation Condition | | ETU70 | | | | | | | | |

NOTE 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

## 8.16.15 E-UTRA FDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 5MHz+5MHz bandwidth

### 8.16.15.1 Test purpose

Same test purpose as in clause 8.16.3.1.

### 8.16.15.2 Test applicability

This test applies to all types of E-UTRA UE FDD release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA. Applicability requires support for FGI bit 111.

This test applies to all types of E-UTRA UE FDD release 11 and forward that support CA with Intra-band non-contiguous CA. Applicability requires support for FGI bit 111.

## 8.16.15.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.16.3.3.

## 8.16.15.4 Test description

### 8.16.15.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 5MHz for cells on both the PCC and the SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1 [10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.15.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.15.4.3.

5. There are two E-UTRA FDD carriers with one cell on one E-UTRA FDD carrier and two cells on the other E-UTRA FDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.15.4.1-1: General test parameters for E-UTRA FDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 5MHz+5MHz bandwidth**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.5 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.11 FDD | As specified in section A.2.1 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 | Channel bandwidth for cells on primary and secondary component carriers |
| Note 1: See Table 8.16.3.4.1-1 for other general test parameters.<br>Note 2: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | |

### 8.16.15.4.2 Test procedure

Same test procedure as in clause 8.16.3.4.2 with the following exceptions:

- Instead of Table 8.16.3.4.3 → use Table 8.16.15.4.3.
- Instead of Table 8.16.3.5-1 → use Table 8.16.15.5-1.

### 8.16.15.4.3 Message contents

Same message contents as in clause 8.16.3.4.3.

## 8.16.15.5 Test requirement

Same test requirement as in clause 8.16.3.5 with the following exceptions:

- Instead of Table 8.16.3.4.1-1 → use Table 8.16.15.4.1-1.
- Instead of Table 8.16.3.5-1 → use Table 8.16.15.5-1.

**Table 8.16.15.5-1: Cell specific test parameters for E-UTRA FDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 5MHz+5MHz bandwidth**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| $BW_{channel}$ | MHz | 5 | | 5 | | 5 | |
| OCNG Patterns defined in D.1.20 (OP.20 FDD) and in D.1.16 (OP.16 FDD) | | OP.20 FDD | | OP.16 FDD | | OP.16 FDD | |
| Note: See Table 8.16.3.5-1 for other cell-specific test parameters. | | | | | | | |

## 8.16.16 E-UTRA TDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 5MHz+5MHz bandwidth

### 8.16.16.1 Test purpose

Same test purpose as in clause 8.16.4.1.

### 8.16.16.2 Test applicability

This test applies to all types of E-UTRA UE TDD release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA. Applicability requires support for FGI bit 111.

This test applies to all types of E-UTRA UE TDD release 11 and forward that support CA with Intra-band non-contiguous CA. Applicability requires support for FGI bit 111.

### 8.16.16.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.16.4.3.

### 8.16.16.4 Test description

#### 8.16.16.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 5MHz for cells on both the PCC and the SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1 [10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.16.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.16.4.3.

5. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.16.4.1-1: General test parameters for E-UTRA TDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 5MHz+5 MHz bandwidth**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.4 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.11 TDD | As specified in section A.2.2 |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 | Channel bandwidth for cells on primary and secondary component carriers |
| Note 1: See Table 8.16.4.4.1-1 for other general test parameters.<br>Note 2: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C3.3.1. | | | |

### 8.16.16.4.2 Test procedure

Same test procedure as in clause 8.16.4.4.2 with the following exceptions:

- Instead of Table 8.16.4.4.3 → use Table 8.16.16.4.3.
- Instead of Table 8.16.4.5-1 → use Table 8.16.16.5-1.

### 8.16.16.4.3 Message contents

Same message contents as in clause 8.16.4.4.3.

## 8.16.16.5 Test requirement

Same test requirement as in clause 8.16.4.5 with the following exceptions:

- Instead of Table 8.16.4.4.1-1 → use Table 8.16.16.4.1-1.
- Instead of Table 8.16.4.5-1 → use Table 8.16.16.5-1.

**Table 8.16.16.5-1: Cell specific test parameters for E-UTRA TDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 5MHz+5 MHz bandwidth**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| $BW_{channel}$ | MHz | 5 | | 5 | | 5 | |
| OCNG Patterns defined in D.2.9 (OP.9 TDD) and in D.2.10 (OP.10 TDD) | | OP.9 TDD | | OP.10 TDD | | OP.10 TDD | |
| Note: See Table 8.16.4.5-1 for other cell-specific test parameters. | | | | | | | |

# 8.16.17 E-UTRAN FDD activation and deactivation of known SCell in non-DRX

## 8.16.17.1 Test purpose

To verify that the SCell activation and deactivation times are within the requirements when the SCell is known by the UE at the time of activation.

## 8.16.17.2 Test applicability

This test applies to all types of E-UTRA UE FDD release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA. Applicability requires support for FGI bit 25.

This test applies to all types of E-UTRA UE FDD release 11 and forward that support CA with Intra-band non-contiguous CA. Applicability requires support for FGI bit 25.

### 8.16.17.3 Minimum conformance requirements

The delay within which the UE shall be able to activate the deactivated SCell depends upon the specified conditions.

Upon receiving SCell activation command in subframe *n*, the UE shall be capable to transmit valid CSI report and apply actions related to the activation command as specified in [26] for the SCell being activated no later than in subframe *n*+24 provided the following conditions are met for the SCell:

- During the period equal to max(5 measCycleSCell, 5 DRX cycles) before the reception of the SCell activation command:
  - the UE has sent a valid measurement report for the SCell being activated and
  - the SCell being activated remains detectable according to the cell identification conditions specified in TS 36.133 [4] section 8.3.3.2,
- SCell being activated also remains detectable during the SCell activation delay according to the cell identification conditions specified in section TS 36.133 [4] 8.3.3.2.

Otherwise upon receiving the SCell activation command in subframe *n*, the UE shall be capable to transmit valid CSI report and apply actions related to the activation command as specified in [26] for the SCell being activated no later than in subframe *n*+34 provided the SCell can be successfully detected on the first attempt.

If there is no reference signal received for the CSI measurement over the delay corresponding to the minimum requirements specified above, then the UE shall report corresponding valid CSI for the activated SCell on the next available uplink reporting resource after receiving the reference signal.

If there are no uplink resources for reporting the valid CSI in subframe *n*+24 or *n*+34 then the UE shall use the next available uplink resource for reporting the corresponding valid CSI.

The valid CSI is based on the UE measurement and corresponds to any CQI value specified in [8] with the exception of CQI index = 0 (out of range) provided:

- the conditions in section 7.7 are met over the entire SCell activation delay and
- the conditions for CQI reporting defined in Section 7.2.3 of [8] are met.

In addition to CSI reporting defined above, UE shall also apply other actions related to the activation command specified in [26] for an SCell at the first opportunities for the corresponding actions once the SCell is activated.

The PCell interruption specified in TS 36.133 [4] section 8.3.3 shall not occur before subframe n+5 and not occur after subframe *n+9* when PCell belongs to E-UTRA FDD.

Starting from the subframe specified in TS 36.213 [8] section 4.3 and until the UE has completed the SCell activation, the UE shall report CQI index = 0 (out of range) if the UE has available uplink resources to report CQI for the SCell.

Upon receiving SCell deactivation command or upon expiry of the *sCellDeactivationTimer* in subframe *n*, the UE shall accomplish the deactivation actions specified in [26] for the SCell being deactivated no later than in subframe *n+8*.

The PCell interruption specified in TS 36.133 [4] section 8.3.3 shall not occur before subframe n+5 and not occur after subframe *n*+9 when PCell belongs to E-UTRA FDD.

The normative reference for this requirement is TS 36.133 [4] clause 7.7 and A.8.16.17.

### 8.16.17.4 Test description

#### 8.16.17.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.42 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.17.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.17.4.3.

5. Cell1 is PCell on the primary component carrier and Cell 2 is SCell on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test, Cell 2 shall be powered OFF.

**Table 8.16.17.4.1-1: General test parameters for known SCell activation case**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.3 FDD (Cell 1) | As specified in section A.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2 |
| E-UTRA RF Channel Number | | 1, 2 | Two radio channels are used for this test |
| Active PCell | | Cell 1 | Primary cell on RF channel number 1. |
| Configured deactivated SCell | | Cell 2 | Configured deactivated secondary cell on RF channel number 2. |
| CP length | | Normal | |
| DRX | | OFF | Continuous monitoring of primary cell |
| CQI/PMI periodicity and offset configuration index | | 0 | CQI reporting for SCell every second subframe |
| Cell-individual offset for cells on RF channel number 1 | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | dB | 0 | Individual offset for cells on secondary component carrier. |
| SCell measurement cycle (measCycleSCell) | ms | 320 | |
| Cell2 timing offset to cell1 | µs | 0 | |
| Time alignment error between cell2 and cell1 | µs | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1. | The value of time alignment error depends upon the type of carrier aggregation. |
| T1 | s | 7 | During this time the PCell shall be known and the SCell configured and detected. |
| T2 | s | 1 | During this time the UE shall activate the SCell. |
| T3 | s | 1 | During this time the UE shall deactivate the SCell. |
| Note: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | |

## 8.16.17.4.2 Test procedure

The test consists of three successive time periods, with duration of T1, T2 and T3, respectively. There are two carriers, each with one cell. Both cells have constant signal levels throughout the test. Before the test starts the UE is connected to Cell 1 (PCell) on radio channel 1 (PCC) but is not aware of Cell 2 on radio channel 2. The UE is only monitoring the PCC. The UE shall be continuously scheduled in the PCell throughout the whole test.

During T2 the Test procedure requires the UE to send the first CSI report for SCell1 in a subframe (m+8), but also allows a subframe (m+10) if the subframe (m+8) was subject to interruption. The SS determines whether the CSI report in subframe (m+8) was interrupted or not by monitoring ACK/NACK sent in PCell in subframe (m+8).

NOTE: The requirements are dependant on the CA configuration (e.g. intra band, inter band; see TS 36.133 [4], clause 7.8) hence the test procedure has been divided into two variants, found in steps 6a and 6b and in steps 8a and 8b. Depending on the CA configuration under test only one of those shall be run respectively.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.16.17.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.
3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.17.4.3.
4. The SS shall configure transmission of PDSCH with a maximum number of 1 HARQ transmission.
5. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe # denoted m which is an even number. If the SS receives ACK for MAC-CE sent by the UE, T2 starts in subframe m, and the test proceeds to step 6, otherwise go to step 9.
6. The UE shall start sending CSI reports for SCell and the SS shall monitor CSI reports for SCell sent from the UE and ACK/NACK sent in PCell during SCell activation.

   6a. For intra-band CA UE,

   - If the first CSI report for SCell is received by the SS in a subframe (m+8),
     - or (m+10) if the subframe (m+8) was subject to interruption,
   - and CSI report with non-zero CQI index is received by the SS earlier than or equal to subframe (m+24),
     - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+24)
   - and DTX is not observed by the SS outside the subframes (m+5) to (m+13) up to the end of T2,
   - Then the number of successes for the event “Activation” is increased by one. Otherwise, count a fail for the event “Activation” and go to step 9.

   6b. For inter-band CA UE,

   - If the first CSI report is received by the SS in a subframe (m+8),
     - or (m+10) if the subframe (m+8) was subject to interruption,
   - and CSI report with non-zero CQI index is received by the SS earlier than or equal to subframe (m+24),
     - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+24)
   - and $\leq$ 2 non-consecutive DTX subframes are observed by the SS during the subframes (m+5) to (m+13) up to the end of T2,
   - Then the number of successes for the event “Activation” is increased by one. Otherwise, count a fail for the event “Activation” and go to step 9.
7. When T2 expires, the SS deactivate SCC by sending the deactivation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe # denoted n. If the SS receives ACK for MAC-CE sent by the UE, T3 starts in subframe n, and the test proceeds to step 8, otherwise go to step 9.
8. The UE shall stop sending CSI reports for SCell and the SS shall monitor CSI reports for SCell sent from the UE and ACK/NACK sent in PCell during SCell deactivation.

   8a. For intra-band CA UE,

   - If the last CSI report is received by the SS earlier than or equal to subframe (n+8)
   - and DTX is not observed by the SS outside the subframes time span (n+5) to (n+13) up to the end of T3,ms from the beginning of time period T3 by the SS
   - Then the number of successes for the event “Deactivation” is increased by one. Otherwise, count a fail for the event “Deactivation”.

8b. For inter-band CA UE,

- If the last CSI report is received by the SS earlier than or equal to subframe (n+8),
- and ≤2 non-consecutive DTX subframes are observed by the SS during the subframes (n+5) to (n+13) up to the end of T3,
- Then the number of successes for the event "Deactivation" is increased by one. Otherwise, count a fail for the event "Deactivation".

9. When T3 expires, or Activation in step 5 was not acknowledged, or a fail was counted for the event "Activation" in step 6a or 6b, or Deactivation in step 7 was not acknowledged, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
10. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
11. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
12. Repeat steps 2-11 until a test verdict has been achieved.

Each of the events "Activation" and "Deactivation" is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.16.17.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.17.4.3-1: Common Exception messages for E-UTRAN FDD known SCell activation**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | |

**Table 8.16.17.4.3-2: *PhysicalConfigDedicated-DEFAULT: Additional E-UTRAN FDD known SCell activation***

| Derivation Path: 36.508 clause 4.8.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | CQI-ReportConfig-DEFAULT using condition CQI_PERIODIC | | RBC |
| cqi-ReportConfig-r10 | Not present | | SCell_AddMod |
| } | | | |

**Table 8.16.17.4.3-3: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 158 | | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | NULL | | |
| simultaneousAckNackAndCQI | TRUE | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.17.4.3-4: *PhysicalConfigDedicatedSCell-r10-DEFAULT: Additional E-UTRAN FDD known SCell activation***

| Derivation Path: 36.508 clause 7.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicatedSCell-r10 ::= SEQUENCE { | | | |
| ul-Configuration-r10 SEQUENCE { | | | |
| antennaInfoUL-r10 | Not present | | |
| pusch-ConfigDedicatedSCell-r10 | Not present | | |
| uplinkPowerControlDedicatedSCell-r10 | Not present | | |
| cqi-ReportConfigSCell-r10 | CQI-ReportConfigSCell-r10-DEFAULT | | |
| soundingRS-UL-ConfigDedicated-r10 | Not present | | |
| soundingRS-UL-ConfigDedicated-v1020 | Not present | | |
| soundingRS-UL-ConfigDedicatedAperiodic-r10 | Not present | | |
| } | | | |
| } | | | |

**Table 8.16.17.4.3-5: *CQI-ReportConfigSCell-r10-DEFAULT: Additional E-UTRAN FDD known SCell activation***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfigSCell-r10 ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic-r10 | Not present | | |
| nomPDSCH-RS-EPRE-Offset-r10 | 0 | | |
| cqi-ReportPeriodicSCell-r10 | | | |
| | *CQI-ReportPeriodic-r10-DEFAULT* | | CQI_PERIODIC |
| pmi-RI-Report-r10 | Not present | | |
| } | | | |

**Table 8.16.17.4.3-6: *CQI-ReportPeriodic-r10-DEFAULT: Additional E-UTRAN FDD known SCell activation***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportPeriodic-r10 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex-r10 | 0 | | |
| cqi-PUCCH-ResourceIndexP1-r10 | Not present | | |
| cqi-pmi-ConfigIndex | 0 | | |
| cqi-FormatIndicatorPeriodic-r10 CHOICE { | | | |
| widebandCQI-r10 SEQUENCE { | | | |
| csi-ReportMode-r10 | Not present | | |
| } | | | |
| ri-ConfigIndex | NULL | | |
| simultaneousAckNackAndCQI | TRUE | | |
| cqi-Mask-r9 | Not present | | |
| csi-ConfigIndex-r10 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.17.4.3-7: *MeasObjectEUTRA-GENERIC:* Additional E-UTRAN FDD known SCell activation**

| Derivation Path: 36.508 clause 4.6.6 Table 4.6.6-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | Cell 2 |
| } | | | |

## 8.16.17.5 Test requirement

Table 8.16.17.4.1-1 and Table 8.16.17.5-1 define the primary level settings including test tolerances for E-UTRAN FDD activation and deactivation of known SCell in non-DRX test.

**Table 8.16.17.5-1: Cell specific test parameters for E-UTRAN FDD known SCell activation**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.1.10 (OP.10 FDD) and in D.1.2 (OP.2 FDD) | | OP.10 FDD | | | OP.2 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -104 | | | -104 | | |
| RSRP Note 3 | dBm/15 kHz | -87 | | | -87 | | |
| $\hat{E}_s/I_{ot}$ | dB | 17 | | | 17 | | |
| SCH_RP Note 3 | dBm/15 kHz | -87 | | | -87 | | |
| $\hat{E}_s/N_{oc}$ | dB | 17 | | | 17 | | |
| Propagation Condition | | AWGN | | | | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: Es/Iot, RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: The uplink resources for CSI reporting are assigned to the UE prior to the start of time period T2.

During T2 the UE shall send the first CSI report for SCell in a subframe (m+8), or in a subframe (m+10) if the suframe (m+8) was subject to interruption. Whether CSI report in subframe (m+8) was interrupted or not is checked by monitoring ACK/NACK sent in PCell in subframe (m+8).

During T2 the UE shall start sending CSI reports for SCell with non-zero CQI index at latest in a subframe (m+24).

During T3 the UE shall stop sending CSI reports for SCell at latest in a subframe (n+8).

During T2 interruption of PCell during SCell activation shall not happen outside the subframes (m+5) to (m+9).

Figures 8.16.17.5-1 and 8.16.17.5-2 show the deriviation of the Test procedure requreiment for DTX during T2, based on the core requirements for interruption. The applicable figure is selected according to the CA configuration.

NOTE: The differences between activation and deactivation are as follows:

- Activation ➔ Deactivation
- First CSI ➔ Last CSI
- Latest valid CSI report ➔ Not exist (no need to check since CSI report was already stopped)

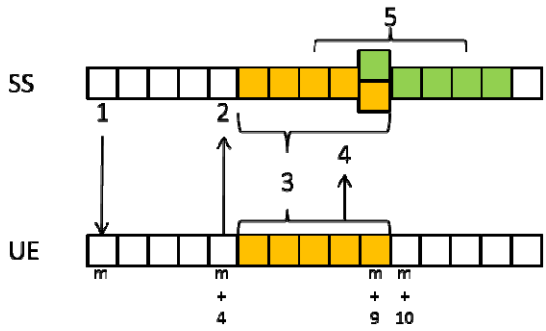


**Figure 8.16.17.5-1: Procedure derivation for Activation (Intra-band case)**

1) Activation command for SCell
2) ACK for MAC-CE for SCell1 activation
3) Possible interruption period by SCell1 activation
4) First CSI report timing (could be invalid CQI)
5) Possible DTX reception period on SS due to interruption by SCell1 activation

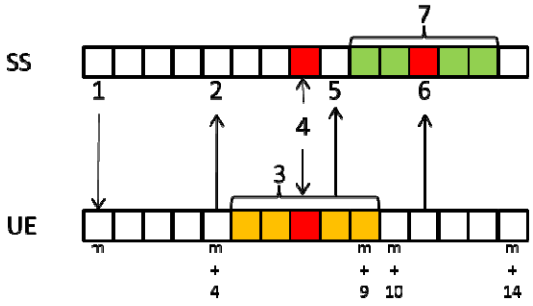


**Figure 8.16.17.5-2: Procedure derivation for Activation (Inter-band case)**

1) Activation command for SCell
2) ACK for MAC-CE for SCell activation
3) Possible interruption period by SCell activation
4) Allowed interruption timing(1 subframe in m+5~m+9) on UE by SCell activation,

   that is, possible DTX timing on SS
5) First CSI report timing(could be invalid CQI) at m+8 (might be m+10 due to interruption)
6) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell activation
7) Possible DTX reception period on SS due to interruption by SCell1 activation

During T3 interruption of PCell during SCell deactivation shall not happen outside the subframes (n+5) to (n+ 9).

The interruption of PCell shall not be more than 5 subframes for intra-band CA and for 1 subframe for inter-band CA specified in TS 36.133 [4] Section 7.8.2.

All of the above test requirements shall be fulfilled in order for the observed SCell activation delay and SCell deactivation delay to be counted as correct. The rate of correct observed SCell activation delay and SCell deactivation delay during repeated tests shall be at least 90% with a confidence level of 95% for each of the events.

The statistical pass/ fail decisions are done separately for activation and deactivation.

Decide the test pass, if activation and deactivation are passed, otherwise fail the UE.

# 8.16.17AE-UTRAN FDD activation and deactivation of known SCell in non-DRX for 20MHz +20MHz bandwidth

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- The test procedure and requirement for interruption of PCell needs further investigation

## 8.16.17A.1 Test purpose

Same test purpose as defined in clause 8.16.17.1.

## 8.16.17A.2 Test applicability

This test applies to all types of E-UTRA UE FDD release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA. Applicability requires support for FGI bit 25.

This test applies to all types of E-UTRA UE FDD release 11 and forward that support CA with Intra-band non-contiguous CA. Applicability requires support for FGI bit 25.

## 8.16.17A.3 Minimum conformance requirements

Same minimum conformance requirements as defined in clause 8.16.17.3 with the following exception:

- The normative reference for this requirement is TS 36.133 [4] clause 7.7 and A.8.16.17A.

## 8.16.17A.4 Test description

### 8.16.17A.4.1 Initial conditions

Same initial conditions as in clause 8.16.17.4.1 with the following exceptions:

- Channel Bandwidth to be tested: 20 MHz for both PCell and SCell
- The listed parameter values in Table 8.16.17A.4.1-1 will replace the values of corresponding parameters in Table 8.16.17.4.1-1 other parameters keep the same.

**Table 8.16.17A.4.1-1: General test parameters for known SCell activation case for 20MHz +20MHz**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.10 FDD | As specified in section A.2 |

### 8.16.17A.4.2 Test procedure

Same test procedure as in clause 8.16.17.4.2.

### 8.16.17A.4.3 Message contents

Same as Message contents as in clause 8.16.17.4.3.

### 8.16.17A.5 Test requirement

Table 8.16.17A.4.1-1 and Table 8.16.17A.5-1 define the primary level settings including test tolerances for E-UTRAN FDD activation and deactivation of known SCell in non-DRX test. The listed parameter values in Table 8.16.17A.5-1 will replace the values of corresponding parameters in Table 8.16.17.5-1 other parameters keep the same.

**Table 8.16.17A.5-1: Cell specific test parameters for E-UTRAN FDD known SCell activation**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| $BW_{channel}$ | MHz | 20 | | | 20 | | |
| OCNG Patterns defined in D.1.17 (OP.17 FDD) and in D.1.12 (OP.12 FDD) | | OP.17 FDD | | | OP.12 FDD | | |

The UE shall send the first CSI report for SCell at latest 9ms into T2.

The UE shall start sending CSI reports for SCell with non-zero CQI index at latest 24ms into T2.

The UE shall stop sending CSI reports for SCell in at latest 8ms into T3.

Interruption of PCell during SCell activation shall not happen outside the time span 5 to 9ms into T2.

Interruption of PCell during SCell deactivation shall not happen outside the time span 5 to 9ms into T3.

The interruption of PCell shall not be more than 5 subframes for intra-band CA and for 1 subframe for inter-band CA specified in TS 36.133 [4] Section 7.8.2.

All of the above test requirements shall be fulfilled in order for the observed SCell activation delay and SCell deactivation delay to be counted as correct. The rate of correct observed SCell activation delay and SCell deactivation delay during repeated tests shall be at least 90% with a confidence level of 95% for each of the events.

The statistical pass/ fail decisions are done separately for activation and deactivation.

Decide the test pass, if activation and deactivation are passed, otherwise fail the UE.

## 8.16.18 E-UTRAN TDD activation and deactivation of known SCell in non-DRX

### 8.16.18.1 Test purpose

To verify that the SCell activation and deactivation times are within the requirements when the SCell is known by the UE at the time of activation.

### 8.16.18.2 Test applicability

This test applies to all types of E-UTRA UE TDD release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA. Applicability requires support for FGI bit 25.

This test applies to all types of E-UTRA UE TDD release 11 and forward that support CA with Intra-band non-contiguous CA. Applicability requires support for FGI bit 25.

### 8.16.18.3 Minimum conformance requirements

The delay within which the UE shall be able to activate the deactivated SCell depends upon the specified conditions.

Upon receiving SCell activation command in subframe *n*, the UE shall be capable to transmit valid CSI report and apply actions related to the activation command as specified in [26] for the SCell being activated no later than in subframe *n*+24 provided the following conditions are met for the SCell:

- During the period equal to max(5 measCycleSCell, 5 DRX cycles) before the reception of the SCell activation command:

  - the UE has sent a valid measurement report for the SCell being activated and

  - the SCell being activated remains detectable according to the cell identification conditions specified in TS 36.133 [4] section 8.3.3.2,

- SCell being activated also remains detectable during the SCell activation delay according to the cell identification conditions specified in TS 36.133 [4] section 8.3.3.2.

Otherwise upon receiving the SCell activation command in subframe *n*, the UE shall be capable to transmit valid CSI report and apply actions related to the activation command as specified in [26] for the SCell being activated no later than in subframe *n*+34 provided the SCell can be successfully detected on the first attempt.

If there is no reference signal received for the CSI measurement over the delay corresponding to the minimum requirements specified above, then the UE shall report corresponding valid CSI for the activated SCell on the next available uplink reporting resource after receiving the reference signal.

If there are no uplink resources for reporting the valid CSI in subframe *n*+24 or *n*+34 then the UE shall use the next available uplink resource for reporting the corresponding valid CSI.

The valid CSI is based on the UE measurement and corresponds to any CQI value specified in [8] with the exception of CQI index = 0 (out of range) provided:

- the conditions in TS 36.133 [4] section 7.7 are met over the entire SCell activation delay and

- the conditions for CQI reporting defined in Section 7.2.3 of [8] are met.

In addition to CSI reporting defined above, UE shall also apply other actions related to the activation command specified in [26] for an SCell at the first opportunities for the corresponding actions once the SCell is activated.

The PCell interruption specified in TS 36.133 [4] section 8.3.3 shall not occur before subframe n+5 and not occur after subframe *n+11* when PCell belongs to E-UTRA TDD.

Starting from subframe n+11 when PCell belongs to E-UTRA TDD and until the UE has completed the SCell activation, the UE shall send CSI with CQI index = 0 (out of range) if the UE is configured to report the CQI in SCell.

Upon receiving SCell deactivation command or upon expiry of the *sCellDeactivationTimer* in subframe *n*, the UE shall accomplish the deactivation actions specified in [26] for the SCell being deactivated no later than in subframe *n+8*.

The PCell interruption specified in TS 36.133 [4] section 8.3.3 shall not occur before subframe n+5 and not occur after subframe *n+11* when PCell belongs to E-UTRA TDD.

The normative reference for this requirement is TS 36.133 [4] clause 7.7 and A.8.16.18.

## 8.16.18.4 Test description

### 8.16.18.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in 3GPP TS 36.508 [7] Annex A, Figure group A.45 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.18.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.18.4.3.

5. Cell1 is PCell on the primary component carrier and Cell 2 is SCell on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test, Cell 2 shall be powered OFF.

**Table 8.16.18.4.1-1: General test parameters for known SCell activation case**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2 |
| E-UTRA RF Channel Number | | 1, 2 | Two radio channels are used for this test |
| Active PCell | | Cell 1 | Primary cell on RF channel number 1. |
| Configured deactivated SCell | | Cell 2 | Configured deactivated secondary cell on RF channel number 2. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | Channel bandwidth for cells on primary and secondary component carriers |
| CP length | | Normal | |
| Special subframe configuration | | 6 | As specified in table 4.2.1 in TS 36.211 [9]. The same configuration applies to all cells. |
| Uplink-downlink configuration | | 1 | |
| DRX | | OFF | Continuous monitoring of primary cell |
| CQI/PMI periodicity and offset configuration index | | 0 | CQI reporting for SCell every UL subframe |
| Cell-individual offset for cells on RF channel number 1 | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | dB | 0 | Individual offset for cells on secondary component carrier. |
| SCell measurement cycle (measCycleSCell) | ms | 320 | |
| Cell2 timing offset to cell1 | µs | 0 | |
| Time alignment error between cell2 and cell1 | µs | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1. | The value of time alignment error depends upon the type of carrier aggregation. |
| Cell3 timing offset to cell1 | µs | 3 | Synchronous cells |
| T1 | s | 7 | During this time the PCell shall be known and the SCell configured, detected and reported. |
| T2 | s | 1 | During this time the UE shall activate the SCell. |
| T3 | s | 1 | During this time the UE shall deactivate the SCell. |
| Note: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | |

## 8.16.18.4.2 Test procedure

The test consists of three successive time periods, with duration of T1, T2 and T3, respectively. There are two carriers, each with one cell. Both cells have constant signal levels throughout the test. Before the test starts the UE is connected to Cell 1 (PCell) on radio channel 1 (PCC) but is not aware of Cell 2 on radio channel 2. The UE is only monitoring the PCC. The UE shall be continuously scheduled in the PCell throughout the whole test.

NOTE: The requirements are dependant on the CA configuration (e.g. intra band, inter band; see 3GPP TS 36.133 [4], clause 7.8) hence the test procedure has been divided into two variants, found in steps 6a and 6b and in steps 8a and 8b. Depending on the CA configuration under test only one of those shall be run respectively.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.16.18.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.18.4.3.

4. The SS shall configure transmission of PDSCH with a maximum number of 1 HARQ transmission.

5. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8). T2 starts.

6. The UE shall start sending CSI reports for SCell and the SS shall monitor CSI reports for SCell sent from the UE and ACK/NACK sent in PCell during SCell activation.

6a. For intra-band CA UE

- If the first CSI report for SCell is received by the SS in a subframe (m+8),
  - or (m+9) if the subframe (m+8) was subject to interruption
  - or (m+13) if the subframe (m+8) and subframe (m+9) were subject to interruption
- and CSI report with non-zero CQI index is received by the SS earlier than or equal to subframe (m+24) ,
  - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+24)
- and DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframes (m+5) to (m+9) and (m+13) up to the end of T2,
  - or DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframes (m+6) to (m+10) and (m+13) to (m+14) up to the end of T2,
  - or DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframes (m+7) to (m+11) and (m+13) to (m+14) and (m+18) up to the end of T2,
- Then the number of successes for the event “Activation” is increased by one. Otherwise, count a fail for the event “Activation” and go to step 9.

6b. For inter-band CA UE

- If the first CSI report for SCell1 is received by the SS in a subframe (m+8),
  - or (m+9) if the subframe (m+8) was subject to interruption,
- and CSI report with non-zero CQI index is received by the SS earlier than or equal to subframe (m+24),
  - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+24)
- and DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframes (m+5), (m+9) up to the end of T2,
  - or DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframes (m+6), (m+13) up to the end of T2,
  - or DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframes (m+7), (m+13) up to the end of T2,
  - or DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframe (m+8) up to the end of T2,
  - or DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframe (m+9) up to the end of T2,
  - or DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframes (m+10), (m+14) up to the end of T2,
  - or DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframes (m+11), (m+18) up to the end of T2,

- Then the number of successes for the event “Activation” is increased by one. Otherwise, count a fail for the event “Activation” and go to step 9.

7. When T2 expires, the SS deactivate SCC by sending the deactivation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe # denoted n, where n is 4 or 9 . If the SS receives ACK for MAC-CE sent by the UE, T3 starts in subframe n, and the test proceeds to step 8, otherwise go to step 9.

8. The UE shall stop sending CSI reports for SCell and the SS shall monitor CSI reports for SCell sent from the UE and ACK/NACK sent in PCell during SCell deactivation. If the last CSI report is received less than or equal to 8ms from the beginning of time period T3 by the SS

8a. For intra-band CA UE

- If the last CSI report is received by the SS earlier than or equal to subframe (n+8),
- and either DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframes (n+5) to (n+9) and (n+13) up to the end of T3,
  - or either DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframes (n+6) to (n+10) and (n+13) to (n+14) up to the end of T3,
  - or either DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframes (n+7) to (n+11) and (n+13) to (n+14) and (n+18) up to the end of T3,
- Then the number of successes for the event “Deactivation” is increased by one. Otherwise, count a fail for the event “Deactivation”.

8b. For inter-band CA UE

- If the last CSI report is received by the SS earlier than or equal to subframe (n+8),
- and DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframes (n+5), (n+9) up to the end of T3,
  - or either DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframes (n+6), (n+13) up to the end of T3,
  - or either DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframes (n+7), (n+13) up to the end of T3,
  - or either DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframes (n+8) up to the end of T3,
  - or either DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframes (n+9) up to the end of T3,
  - or either DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframes (n+10), (n+14) up to the end of T3,
  - or either DTX or “NACK/DTX” (defined in TS36.213 clause 10.1) are not observed by the SS outside the subframes (n+11), (n+18) up to the end of T3,
- Then the number of successes for the event “Deactivation” is increased by one. Otherwise, count a fail for the event “Deactivation”.

9. When T3 expires, or Activation in step 5 was not acknowledged, or a fail was counted for the event “Activation” in step 6a or 6b, or Deactivation in step 7 was not acknowledged, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

10. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

11. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),

or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

12. Repeat steps 2-11 until a test verdict has been achieved.

Each of the events “Activation” and “Deactivation” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.16.18.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.18.4.3-1: Common Exception messages for E-UTRAN TDD known SCell activation**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | |

**Table 8.16.18.4.3-2: *PhysicalConfigDedicated-DEFAULT: Additional E-UTRAN TDD known SCell activation***

| Derivation Path: 36.508 clause 4.8.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | CQI-ReportConfig-DEFAULT using condition CQI_PERIODIC | | RBC |
| cqi-ReportConfig-r10 | Not present | | SCell_AddMod |
| } | | | |

**Table 8.16.18.4.3-3: *CQI-ReportConfig-DEFAULT: Additional E-UTRAN TDD known SCell activation***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 159 | | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | NULL | | |
| simultaneousAckNackAndCQI | TRUE | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.18.4.3-4: *PhysicalConfigDedicatedSCell-r10-DEFAULT: Additional E-UTRAN TDD known SCell activation***

| Derivation Path: 36.508 clause 7.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicatedSCell-r10 ::= SEQUENCE { | | | |
| ul-Configuration-r10 SEQUENCE { | | | |
| antennaInfoUL-r10 | Not present | | |
| pusch-ConfigDedicatedSCell-r10 | Not present | | |
| uplinkPowerControlDedicatedSCell-r10 | Not present | | |
| cqi-ReportConfigSCell-r10 | CQI-ReportConfigSCell-r10-DEFAULT | | |
| soundingRS-UL-ConfigDedicated-r10 | Not present | | |
| soundingRS-UL-ConfigDedicated-v1020 | Not present | | |
| soundingRS-UL-ConfigDedicatedAperiodic-r10 | Not present | | |
| } | | | |
| } | | | |

**Table 8.16.18.4.3-5: *CQI-ReportConfigSCell-r10-DEFAULT: Additional E-UTRAN TDD known SCell activation***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfigSCell-r10 ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic-r10 | Not present | | |
| nomPDSCH-RS-EPRE-Offset-r10 | 0 | | |
| cqi-ReportPeriodicSCell-r10 | | | |
| | *CQI-ReportPeriodic-r10-DEFAULT* | | CQI_PERIODIC |
| pmi-RI-Report-r10 | Not present | | |
| } | | | |

**Table 8.16.18.4.3-6: *CQI-ReportPeriodic-r10-DEFAULT: Additional E-UTRAN TDD known SCell activation***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportPeriodic-r10 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex-r10 | 0 | | |
| cqi-PUCCH-ResourceIndexP1-r10 | Not present | | |
| cqi-pmi-ConfigIndex | 0 | | |
| cqi-FormatIndicatorPeriodic-r10 CHOICE { | | | |
| widebandCQI-r10 SEQUENCE { | | | |
| csi-ReportMode-r10 | Not present | | |
| } | | | |
| ri-ConfigIndex | NULL | | |
| simultaneousAckNackAndCQI | FALSE | | |
| cqi-Mask-r9 | Not present | | |
| csi-ConfigIndex-r10 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.18.4.3-7: *MeasObjectEUTRA-GENERIC: Additional E-UTRAN TDD known SCell activation***

| Derivation Path: 36.508 clause 4.6.6 Table 4.6.6-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | Cell 2 |
| } | | | |

## 8.16.18.5 Test requirement

Table 8.16.18.4.1-1 and Table 8.16.18.5-1 define the primary level settings including test tolerances for E-UTRAN TDD activation and deactivation of known SCell in non-DRX test.

**Table 8.16.18.5-1: Cell specific test parameters for E-UTRAN TDD known SCell activation**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | | | OP.2 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -104 | | | -104 | | |
| RSRP Note 3 | dBm/15 kHz | -87 | | | -87 | | |
| $\hat{E}_s/I_{ot}$ | dB | 17 | | | 17 | | |
| SCH_RP Note 3 | dBm/15 kHz | -87 | | | -87 | | |
| $\hat{E}_s/N_{oc}$ | dB | 17 | | | 17 | | |
| Propagation Condition | | AWGN | | | AWGN | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The uplink resources for CSI reporting are assigned to the UE prior to the start of time period T2.

The UE shall send the first CSI report for SCell in a subframe (m+8), or in a subframe (m+9) if the subframe (m+8) was subject to interruption, or in a subframe (m+13) if the subframes (m+8) and (m+9) were subject to interruption when an intra-band SCell is activated. Whether CSI report in subframe (m+8) and/or (m+9) was interrupted or not is checked by monitoring ACK/NACK sent in PCell in subframe (m+8) and/or (m+9).

The UE shall start sending CSI reports for SCell with non-zero CQI index at latest 24ms into T2.

The UE shall stop sending CSI reports for SCell in at latest 8ms into T3.

Interruption of PCell during SCell activation shall not happen outside the time span 5 to 11ms into T2.

Interruption of PCell during SCell deactivation shall not happen outside the time span 5 to 11ms into T3.

[The interruption of PCell shall not be more than 5 subframes for intra-band CA and for 1 subframe for inter-band CA specified in TS 36.133 [4] Section 7.8.2.]

All of the above test requirements shall be fulfilled in order for the observed SCell activation delay and SCell deactivation delay to be counted as correct. The rate of correct observed SCell activation delay and SCell deactivation delay during repeated tests shall be at least 90% with a confidence level of 95% for each of the events.

The statistical pass/ fail decisions are done separately for activation and deactivation.

Decide the test pass, if activation and deactivation are passed, otherwise fail the UE.

# 8.16.18AE-UTRAN TDD activation and deactivation of known SCell in non-DRX for 20MHz +20MHz bandwidth

## 8.16.18A.1 Test purpose

Same test purpose as defined in clause 8.16.18.1.

## 8.16.18A.2 Test applicability

This test applies to all types of E-UTRA UE TDD release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA. Applicability requires support for FGI bit 25.

This test applies to all types of E-UTRA UE TDD release 11 and forward that support CA with Intra-band non-contiguous CA. Applicability requires support for FGI bit 25.

## 8.16.18A.3 Minimum conformance requirements

Same minimum conformance requirements as defined in clause 8.16.18.3 with the following exception:

- The normative reference for this requirement is TS 36.133 [4] clause 7.7 and A.8.16.18A.

## 8.16.18A.4 Test description

### 8.16.18A.4.1 Initial conditions

Same initial conditions as in clause 8.16.18.4.1 with the following exceptions:

- Channel Bandwidth to be tested: 20 MHz for both PCell and SCell
- The listed parameter values in Table 8.16.18A.4.1-1 will replace the values of corresponding parameters in Table 8.16.18.4.1-1 other parameters keep the same.

**Table 8.16.18A.4.1-1: General test parameters for known SCell activation case for 20MHz +20MHz**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.3 TDD | As specified in section A.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.10 TDD | As specified in section A.2 |

### 8.16.18A.4.2 Test procedure

Same test procedure as in clause 8.16.18.4.2.

### 8.16.18A.4.3 Message contents

Same message contents as in clause 8.16.18.4.3.

### 8.16.18A.5 Test requirement

Table 8.16.18A.4.1-1 and Table 8.16.18A.5-1 define the primary level settings including test tolerances for E-UTRAN TDD activation and deactivation of known SCell in non-DRX test. The listed parameter values in Table 8.16.18A.5-1 will replace the values of corresponding parameters in Table 8.16.18.5-1 other parameters keep the same.

**Table 8.16.18A.5-1: Cell specific test parameters for E-UTRAN TDD known SCell activation**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| $BW_{channel}$ | MHz | 20 | | | 20 | | |
| OCNG Patterns defined in D.2.7 (OP.7 TDD) and in D.2.8 (OP.8 TDD) | | OP.7 TDD | | | OP.8.TDD | | |

The UE shall send the first CSI report for SCell in a subframe (m+8), or in a subframe (m+9) if the subframe (m+8) was subject to interruption, or in a subframe (m+13) if the subframes (m+8) and (m+9) were subject to interruption when an intra-band SCell is activated. Whether CSI report in subframe (m+8) and/or (m+9) was interrupted or not is checked by monitoring ACK/NACK sent in PCell in subframe (m+8) and/or (m+9).

The UE shall start sending CSI reports for SCell with non-zero CQI index at latest 24ms into T2.

The UE shall stop sending CSI reports for SCell in at latest 8ms into T3.

Interruption of PCell during SCell activation shall not happen outside the time span 5 to 11ms into T2.

Interruption of PCell during SCell deactivation shall not happen outside the time span 5 to 11ms into T3.

[The interruption of PCell shall not be more than 5 subframes for intra-band CA and for 1 subframe for inter-band CA specified in TS 36.133 [4] Section 7.8.2.]

The UE shall not send event triggered measurement reports as long as the reporting criteria are not fulfilled.

All of the above test requirements shall be fulfilled in order for the observed SCell activation delay and SCell deactivation delay to be counted as correct. The rate of correct observed SCell activation delay and SCell deactivation delay during repeated tests shall be at least 90% with a confidence level of 95% for each of the events.

The statistical pass/ fail decisions are done separately for activation and deactivation.

Decide the test pass, if activation and deactivation are passed, otherwise fail the UE.

## 8.16.19

## 8.16.20

## 8.16.21 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 20MHz+10MHz bandwidth

### 8.16.21.1 Test purpose

Same test purpose as in clause 8.16.2.1.

### 8.16.21.2 Test applicability

This test applies to all types of E-UTRA UE TDD release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA. Applicability requires support for FGI bit 111.

This test applies to all types of E-UTRA UE TDD release 11 and forward that support CA with Intra-band non-contiguous CA. Applicability requires support for FGI bit 111.

### 8.16.21.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.16.2.3.

### 8.16.21.4 Test description

#### 8.16.21.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 20 MHz for cell on PCC and 10MHz for cell on SCC as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.42 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.21.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.21.4.3.

5. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier, and Cell 3 is the neighbouring cell of Cell 2 on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test, Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.21.4.1-1: General test parameters for E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions for 20MHz+10MHz bandwidth**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Channel bandwidth for cells on primary carrier ($BW_{channel}$) | | MHz | 20 | Channel bandwidth for cells on primary component carrier |
| PDSCH parameters for cells on primary carriers | | | DL Reference Measurement Channel R.3 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters for cells on primary carriers | | | DL Reference Measurement Channel R.10 TDD | As specified in section A.2.2 |
| Channel bandwidth for cells on secondary carriers ($BW_{channel}$) | | MHz | 10 | Channel bandwidth for cells on secondary component carrier |
| PCFICH/PDCCH/PHICH parameters for cells on secondary carrier | | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| A2 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A2. |
| | Threshold RSRP | dBm | -96 | Actual RSRP threshold for event A2. Needs to take absolute accuracy tolerance in clause 9.1.11.1 into account plus margin. |
| | Time To Trigger | s | 0 | |
| NOTE 1: See Table 8.16.2.4.1-1 for other general test parameters.<br>NOTE 2: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | | |

### 8.16.21.4.2 Test procedure

Same test procedure as in clause 8.16.2.4.2 with the following exceptions:

- Instead of Table 8.16.2.4.3 → use Table 8.16.21.4.3.
- Instead of Table 8.16.2.5-1 → use Table 8.16.21.5-1.

### 8.16.21.4.3 Message contents

Same message contents as in clause 8.16.2.4.3 with the following exceptions:

**Table 8.16.21.4.3-6: *ReportConfig-A2-H*: Additional E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions for 20MHz+10MHz bandwidth**

| Derivation Path: 36.508 clause 4.6.6 table 4.6.6-5 ReportConfigEUTRA-A2(-90) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A2(Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA2 SEQUENCE { | | | |
| a2-Threshold CHOICE { | | | |
| threshold-RSRP | 44 | The actual value is IE value – 140 dBm | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

## 8.16.21.5 Test requirement

Same test requirement as in clause 8.16.2.5 with the following exceptions:

- Instead of Table 8.16.2.4.1-1 → use Table 8.16.21.4.1-1.
- Instead of Table 8.16.2.5-1 → use Table 8.16.21.5-1.

**Table 8.16.21.5-1: Cell specific test parameters for E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions for 20MHz+10MHz bandwidth**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | 2 | | |
| $BW_{channel}$ | MHz | 20 | | | 10 | | | 10 | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | 1x2 Low | | | 1x2 Low | | |
| OCNG Patterns defined in D.2.7 for OP.7 TDD and D.2.2 for OP.2 TDD | | OP.7 TDD | | | OP.2 TDD | | | OP.2 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -104 | | | -104 | | | | | |
| RSRP Note 3 | dBm/15 kHz | -84.80 | -84.80 | -107 | -84.80 | -84.80 | -107 | -infinity | -85 | -107 |
| $\hat{E}_s/I_{ot}$ | dB | 19.20 | 19.20 | -3 | 19.20 | 0.15 | -4.76 | -infinity | -0.25 | -4.76 |
| SCH_RP Note 3 | dBm/15 kHz | -84.80 | -84.80 | -107 | -84.80 | -84.80 | -107 | -infinity | -85 | -107 |
| $\hat{E}_s/N_{oc}$ | dB | 19.20 | 19.20 | -3 | 19.20 | 19.20 | -3 | -infinity | 19 | -3 |
| Propagation Condition | | ETU70 | | | | | | | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

## 8.16.22 E-UTRAN TDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 20MHz+10MHz bandwidth

### 8.16.22.1 Test purpose

Same test purpose as in clause 8.16.4.1.

### 8.16.22.2 Test applicability

This test applies to all types of E-UTRA UE TDD release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA. Applicability requires support for FGI bit 111.

This test applies to all types of E-UTRA UE TDD release 11 and forward that support CA with Intra-band non-contiguous CA. Applicability requires support for FGI bit 111.

### 8.16.22.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.16.4.3.

## 8.16.22.4 Test description

### 8.16.22.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 20MHz for cell on PCC and 10MHz for cell on SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1 [10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.22.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.22.4.3.

5. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.22.4.1-1: General test parameters for E-UTRA TDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 20MHz+10MHz bandwidth**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Channel bandwidth for cells on primary carrier ($BW_{channel}$) | MHz | 20 | Channel bandwidth for cells on primary component carrier |
| PDSCH parameters for cells on primary carriers | | DL Reference Measurement Channel R.3 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters for cells on primary carriers | | DL Reference Measurement Channel R.10 TDD | As specified in section A.2.2 |
| Channel bandwidth for cells on secondary carriers ($BW_{channel}$) | MHz | 10 | Channel bandwidth for cells on secondary component carrier |
| PCFICH/PDCCH/PHICH parameters for cells on secondary carrier | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| NOTE 1: See Table 8.16.4.4.1-1 for other general test parameters.<br>NOTE 2: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C3.3.1. | | | |

### 8.16.22.4.2 Test procedure

Same test procedure as in clause 8.16.4.4.2 with the following exceptions:

- Instead of Table 8.16.4.4.3 → use Table 8.16.22.4.3.
- Instead of Table 8.16.4.5-1 → use Table 8.16.22.5-1.

### 8.16.22.4.3 Message contents

Same message contents as in clause 8.16.4.4.3.

## 8.16.22.5 Test requirement

Same test requirement as in clause 8.16.4.5 with the following exceptions:

- Instead of Table 8.16.4.4.1-1 → use Table 8.16.22.4.1-1.

- Instead of Table 8.16.4.5-1 → use Table 8.16.22.5-1.

**Table 8.16.22.5-1: Cell specific test parameters for E-UTRA TDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 20MHz+10MHz bandwidth**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| $BW_{channel}$ | MHz | 20 | | 10 | | 10 | |
| OCNG Patterns defined in D.2.7 (OP.7 TDD) and in D.2.2 (OP.2 TDD) | | OP.7 TDD | | OP.2 TDD | | OP.2 TDD | |
| NOTE: See Table 8.16.4.5-1 for other cell-specific test parameters. | | | | | | | |

# 8.16.23 E-UTRAN TDD-FDD CA Event Triggered Reporting Under Deactivated SCell in Non-DRX with PCell in FDD

## 8.16.23.1 Test purpose

To verify that in TDD-FDD CA with PCell in FDD the UE's ability to make correct reportings of Event A2 and A6 under deactivated SCell in non-DRX within the requirements for SCell stated in TS 36.133[4] section 8.3.3.2.1 and the requirements for PCell stated in clause 8.3.2.

## 8.16.23.2 Test applicability

This test applies to all types of E-UTRA UE release 12 and forward that support E-UTRA FDD and TDD and 2DL TDD-FDD CA with FDD as PCell. Applicability requires support for FGI bit 111.

## 8.16.23.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD or TDD cell on a secondary component carrier within $T_{identify_scc}$, according to the parameter *measCycleSCell* where $T_{identify_scc}$ = 20 *measCycleSCell*

A cell shall be considered detectable when

- RSRP related side condition given in TS 36.133 [4] Clause 9.1 are fulfilled for a corresponding Band,

- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.7 for a corresponding Band

The measurement period for deactivated scell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell* where $T_{measure_scc}$ = 5 *measCycleSCell*. The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified cells on a secondary component carrier, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_scc}$.

The measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] sub-clause 9.1.11 (Carrier aggregation measurement accuracy).

A UE may reconfigure the receiver bandwidth or turn on/off one of the RF chains when performing measurements on one or two SCCs with deactivated SCell. This may cause interruptions on PCell or an activated SCell or both that are specified in TS 36.133 [4] Section 7.8.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty

is: 2 x $TTI_{DCCH}$.This measurement reporting delay excludes a delay which is caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc}$ defined in TS 36.133 [4] Clause 8.3.3.2.1. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_scc}$ defined in TS 36.133 [4] clause 8.3.3.2.1 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_scc}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

Measurements of cells on the primary component carrier shall meet all applicable requirements (FDD or TDD) in TS 36.133 [4] clause 8.1.2.2 (E-UTRAN intra frequency measurements and E-UTRAN intra frequency measurements with autonomous gaps).

The normative reference for this requirement is TS 36.133 [4] clause 8.3.3.2.1, 8.3.2 and A.8.16.23.

## 8.16.23.4 Test description

### 8.16.23.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: Any one of the supported TDD-FDD CA configurations with PCell in FDD with the largest aggregated CA bandwidth combination supported by UE from Table 8.16.23.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.42 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.23.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.23.4.3.

5. Cell 1 is the PCell on the FDD primary component carrier, Cell 2 is the configured and deactivated SCell on the TDD secondary component carrier, and Cell 3 is the neighbour cell on the TDD secondary component carrier, Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test, Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.23.4.1-1: General test parameters for E-UTRAN TDD-FDD CA event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions with PCell in FDD**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2 | Two radio channels are used for this test |
| Active PCell | | | Cell 1 | Primary cell on RF channel number 1. |
| Configured deactivated SCell | | | Cell 2 | Configured deactivated secondary cell on RF channel number 2. |
| Neighbour cell | | | Cell 3 | Neighbor cell to be identified on RF channel number 2. |
| CP length | | | Normal | |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211[9]. The same configuration in TDD cells |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211[9]. The same configuration in TDD cells |
| DRX | | | OFF | Continuous monitoring of primary cell |
| A2 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A2. |
| | Threshold RSRP | dBm | -98 | Actual RSRP threshold for event A2. Needs to take absolute accuracy tolerance in TS 36.133 [4] clause 9.1.11.1 into account plus margin. |
| | Time To Trigger | s | 0 | |
| A6 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A6. |
| | Offset | dB | -6 | Offset parameter for evaluation of event A6. Needs to take relative accuracy tolerance in TS 36.133 [4] clause 9.1.11.2 into account plus margin. |
| | Report on leave | | False | |
| | Time To Trigger | s | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| SCell measurement cycle (measCycleSCell) | | ms | 320 | |
| T1 | | s | 5 | During this time the UE shall be aware of cells 1 and 2 but not cell 3. |
| T2 | | s | ≤12 | UE shall report Event A6 within 6.4s (20×scellMeasCycle) |
| T3 | | s | 5 | UE shall report Event A2 within 200 ms and 1.6s for cells 1 and 2, respectively. |

## 8.16.23.4.2 Test procedure

Same test procedure as in clause 8.16.1.4.2 with the following exceptions:

- Instead of clause 8.16.1.4.3 → use clause 8.16.23.4.3.
- Instead of Table 8.16.1.5-1 → use Table 8.16.23.5-1.

## 8.16.23.4.3 Message contents

Same message contents as in clause 8.16.1.4.3 with the following exceptions:

**Table 8.16.23.4.3-1: *ReportConfig-A2-H*: Additional E-UTRAN TDD-FDD CA event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions with PCell in FDD requirements**

| Derivation Path: 36.508 clause 4.6.6 table 4.6.6-5 ReportConfigEUTRA-A2(-93) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A2(Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA2 SEQUENCE { | | | |
| a2-Threshold CHOICE { | | | |
| threshold-RSRP | 42 | The actual value is IE value – 140 dBm | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

## 8.16.23.5 Test requirement

Same test requirement as in clause 8.16.1.5 with the following exceptions:

- Instead of Table 8.16.1.4.1-1 → use Table 8.16.23.4.1-1.
- Instead of Table 8.16.1.5-1 → use Table 8.16.23.5-1.

**Table 8.16.23.5-1: Cell specific test parameters for E-UTRAN TDD-FDD CA event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions with PCell in FDD**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | | | |
| $BW_{channel}$ | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | | | |
| PDSCH parameters: DL Reference Measurement Channel | | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | | | - | | | - | | |
| PCFICH/PDCCH/ PHICH parameters: DL Reference Measurement Channel | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | |
| OCNG Patterns defined | | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA[Note 1] | dB | | | | | | | | | |
| OCNG_RB[Note 1] | dB | | | | | | | | | |
| $N_{oc}$[Note 2] | dBm/15 kHz | -104 | | | -104.5 | | | | | |
| $\hat{E}_s/N_{oc}$ | dB | 17.20 | 17.20 | -3 | 17.70 | 17.70 | -3.70 | -infinity | 17 | -3.7 |
| $\hat{E}_s/I_{ot}$ | dB | 17.20 | 17.20 | -3 | 17.70 | 0.61 | -5.24 | -infinity | -0.77 | -5.24 |
| RSRP [Note 3] | dBm/15 kHz | -86.80 | -86.80 | -107 | -86.80 | -86.80 | -108.2 | -infinity | -87.5 | -108.2 |
| SCH_RP [Note 3] | dBm/15 kHz | -86.80 | -86.80 | -107 | -86.80 | -86.80 | -108.2 | -infinity | -87.5 | -108.2 |
| Io [Note 3] | dBm/Ch BW | -58.94 +10log ($N_{RB,c}$ /50) | -58.94 +10log ($N_{RB,c}$ /50) | -74.45 +10log ($N_{RB,c}$ /50) | -58.94 +10log ($N_{RB,c}$ /50) | -56.30 +10log ($N_{RB,c}$ /50) | -74.03 +10log ($N_{RB,c}$ /50) | Specified in columns for Cell 2 | | |
| Propagation Condition | | AWGN | | | ETU70 | | | ETU70 | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | 1x2 Low | | | 1x2 Low | | |
| Timing offset to Cell 1 | μs | - | | | 0 | | | 3 | | |
| Time alignment error relative to cell 1 [Note 5] | μs | - | | | ≤ TAE | | | N/A | | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | Es/Iot, RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 4: | The resources for uplink transmission are assigned to the UE prior to the start of time period T2. |
| Note 5: | Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation. |

## 8.16.24 E-UTRAN TDD-FDD CA Event Triggered Reporting Under Deactivated SCell in Non-DRX with PCell in TDD

### 8.16.24.1 Test purpose

To verify that in TDD-FDD CA with PCell in TDD the UE's ability to make correct reportings of Event A2 and A6 under deactivated SCell in non-DRX within the requirements for SCell stated in TS 36.133[4] section 8.3.3.2.1 and the requirements for PCell stated in clause 8.3.2.

### 8.16.24.2 Test applicability

This test applies to all types of E-UTRA UE release 12 and forward that support E-UTRA FDD and TDD and 2DL TDD-FDD CA with TDD as PCell. Applicability requires support for FGI bit 111.

### 8.16.24.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.16.23.3.

### 8.16.24.4 Test description

#### 8.16.24.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: Any one of the supported TDD-FDD CA configurations with PCell in TDD with the largest aggregated CA bandwidth combination supported by UE from Table 8.16.24.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.42 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.24.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.24.4.3.

5. Cell 1 is the PCell on the TDD primary component carrier, Cell 2 is the configured and deactivated SCell on the FDD secondary component carrier, and Cell 3 is the neighbour cell on the FDD secondary component carrier, Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test, Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.24.4.1-1: General test parameters for E-UTRAN TDD-FDD CA event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions with PCell in TDD**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2 | Two radio channels are used for this test |
| Active PCell | | | Cell 1 | Primary cell on RF channel number 1. |
| Configured deactivated SCell | | | Cell 2 | Configured deactivated secondary cell on RF channel number 2. |
| Neighbour cell | | | Cell 3 | Neighbour cell to be identified on RF channel number 2. |
| CP length | | | Normal | |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211[9]. |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211[9]. |
| DRX | | | OFF | Continuous monitoring of primary cell |
| A2 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A2. |
| | Threshold RSRP | dBm | -98 | Actual RSRP threshold for event A2. Needs to take absolute accuracy tolerance in TS 36.133 [4] clause 9.1.11.1 into account plus margin. |
| | Time To Trigger | s | 0 | |
| A6 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A6. |
| | Offset | dB | -6 | Offset parameter for evaluation of event A6. Needs to take relative accuracy tolerance in TS 36.133 [4] clause 9.1.11.2 into account plus margin. |
| | Report on leave | | False | |
| | Time To Trigger | s | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| SCell measurement cycle (measCycleSCell) | | ms | 320 | |
| T1 | | s | 5 | During this time the UE shall be aware of cells 1 and 2 but not cell 3. |
| T2 | | s | ≤12 | UE shall report Event A6 within 6.4s (20×scellMeasCycle) |
| T3 | | s | 5 | UE shall report Event A2 within 200 ms and 1.6s for cells 1 and 2, respectively. |

## 8.16.24.4.2 Test procedure

Same test procedure as in clause 8.16.2.4.2 with the following exceptions:

- Instead of clause 8.16.2.4.3 → use clause 8.16.24.4.3.
- Instead of Table 8.16.2.5-1 → use Table 8.16.24.5-1.

## 8.16.24.4.3 Message contents

Same message contents as in clause 8.16.2.4.3 with the following exceptions:

**Table 8.16.24.4.3-1: *ReportConfig-A2-H*: Additional E-UTRAN TDD-FDD CA event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions with PCell in TDD requirements**

| Derivation Path: 36.508 clause 4.6.6 table 4.6.6-5 ReportConfigEUTRA-A2(-90) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A2(Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA2 SEQUENCE { | | | |
| a2-Threshold CHOICE { | | | |
| threshold-RSRP | 42 | The actual value is IE value – 140 dBm | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

## 8.16.24.5 Test requirement

Same test requirement as in clause 8.16.2.5 with the following exceptions:

- Instead of Table 8.16.2.4.1-1 → use Table 8.16.24.4.1-1.
- Instead of Table 8.16.2.5-1 → use Table 8.16.24.5-1.

**Table 8.16.24.5-1: Cell specific test parameters for E-UTRAN TDD-FDD CA event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions with PCell in TDD**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | | | |
| $BW_{channel}$ | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | | | |
| PDSCH parameters: DL Reference Measurement Channel | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | | | - | | | - | | |
| PCFICH/PDCCH/ PHICH parameters: DL Reference Measurement Channel | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | |
| OCNG Patterns defined | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -104 | | | -104.5 | | | | | |
| $\hat{E}_s/N_{oc}$ | dB | 17.20 | 17.20 | -3 | 17.70 | 17.70 | -3.70 | -infinity | 17 | -3.7 |
| $\hat{E}_s/I_{ot}$ | dB | 17.20 | 17.20 | -3 | 17.70 | 0.61 | -5.24 | -infinity | -0.77 | -5.24 |
| RSRP Note 3 | dBm/15 kHz | -86.80 | -86.80 | -107 | -86.80 | -86.80 | -108.2 | -infinity | -87.5 | -108.2 |
| SCH_RP Note 3 | dBm/15 kHz | -86.80 | -86.80 | -107 | -86.80 | -86.80 | -108.2 | -infinity | -87.5 | -108.2 |
| Io Note 3 | dBm/Ch BW | -58.94 +10log ($N_{RB,c}$ /50) | -58.94 +10log ($N_{RB,c}$ /50) | -74.45 +10log ($N_{RB,c}$ /50) | -58.94 +10log ($N_{RB,c}$ /50) | -56.30 +10log ($N_{RB,c}$ /50) | -74.03 +10log ($N_{RB,c}$ /50) | Specified in columns for Cell 2 | | |
| Propagation Condition | | AWGN | | | ETU70 | | | ETU70 | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | 1x2 Low | | | 1x2 Low | | |
| Timing offset to Cell 1 | µs | - | | | 0 | | | 3 | | |
| Time alignment error relative to cell 1 Note 5 | µs | - | | | ≤ TAE | | | N/A | | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | Es/Iot, RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 4: | The resources for uplink transmission are assigned to the UE prior to the start of time period T2. |
| Note 5: | Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation. |

## 8.16.25 E-UTRAN TDD-FDD CA Event triggered reporting on deactivated SCell with PCell interruption in non-DRX with PCell in FDD

### 8.16.25.1 Test purpose

To verify that in TDD-FDD CA with PCell in FDD the UE's ability to make a correct reporting of Event A6 on deactivated SCell with PCell interruption in non-DRX within the requirements for SCell stated in TS 36.133[4] section 8.3.3.2.1 and the requirements for PCell stated in TS 36.133[4] clause 8.3.2 while at the same time fulfilling the requirements on interruption rate.

### 8.16.25.2 Test applicability

This test applies to all types of E-UTRA UE release 12 and forward that support E-UTRA FDD and TDD and 2DL TDD-FDD CA with FDD as PCell. Applicability requires support for FGI bit 111.

### 8.16.25.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.16.23.3.

### 8.16.25.4 Test description

#### 8.16.25.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: Any one of the supported TDD-FDD CA configurations with PCell in FDD with the largest aggregated CA bandwidth combination supported by UE from Table 8.16.25.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.25.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.25.4.3.

5. Cell 1 is the PCell on the FDD primary component carrier, Cell 2 is the configured and deactivated SCell on the TDD secondary component carrier, and Cell 3 is the neighbour cell on the TDD secondary component carrier, Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test, Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.25.4.1-1: General test parameters for E-UTRAN TDD-FDD CA Event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX with PCell in FDD**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2 | Two radio channels are used for this test |
| Active PCell | | | Cell 1 | Primary cell on RF channel number 1. |
| Configured deactivated SCell | | | Cell 2 | Configured deactivated secondary cell on RF channel number 2. |
| Neighbour cell | | | Cell 3 | Neighbour cell to be identified on RF channel number 2. |
| CP length | | | Normal | |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211. The same configuration in TDD cells |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211. The same configuration in TDD cells |
| DRX | | | OFF | Continuous monitoring of primary cell |
| A6 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A6. |
| | Offset | dB | -3 | Offset parameter for evaluation of event A6. Needs to take relative accuracy tolerance in clause 9.1.11.2 into account plus margin. |
| | Report on leave | | False | |
| | Time To Trigger | s | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| SCell measurement cycle (measCycleSCell) | | ms | 1280 | |
| T1 | | s | 5 | During this time the UE shall be aware of cells 1 and 2 but not cell 3. |
| T2 | | s | ≤30 | UE shall report Event A6 within 25.6s (20×scellMeasCycle) |

### 8.16.25.4.2 Test procedure

Same test procedure as in clause 8.16.3.4.2 with the following exceptions:

- Instead of clause 8.16.3.4.3 → use clause 8.16.25.4.3.
- Instead of Table 8.16.3.5-1 → use Table 8.16.25.5-1.

### 8.16.25.4.3 Message contents

Same message contents as in clause 8.16.3.4.3.

## 8.16.25.5 Test requirement

Same test requirement as in clause 8.16.3.5 with the following exceptions:

- Instead of Table 8.16.3.4.1-1 → use Table 8.16.25.4.1-1.
- Instead of Table 8.16.3.5-1 → use Table 8.16.25.5-1.

**Table 8.16.25.5-1: Cell specific test parameters for E-UTRAN TDD-FDD CA Event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX with PCell in FDD**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | | | |
| $BW_{channel}$ | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | |
| PDSCH parameters: DL Reference Measurement Channel | | 5MHz: R.7 FDD<br>10MHz: R.3 FDD<br>20MHz: R.6 FDD | | - | | - | |
| PCFICH/PDCCH/ PHICH parameters: DL Reference Measurement Channel | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | |
| OCNG Patterns defined | | 5MHz: OP.20 FDD<br>10MHz: OP.10 FDD<br>20MHz: OP.17 FDD | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | |
| PBCH_RA | dB | 0 | | 0 | | 0 | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA[Note 1] | dB | | | | | | |
| OCNG_RB[Note 1] | dB | | | | | | |
| $N_{oc}$[Note 2] | dBm/15 kHz | -101 | | -101 | | | |
| $\hat{E}_s/N_{oc}$ | dB | 16 | 16 | 16 | 16 | -infinity | 16 |
| $\hat{E}_s/I_{ot}$ | dB | 16 | 16 | 16 | -0.11 | -infinity | -0.11 |
| RSRP [Note 3] | dBm/15 kHz | -85 | -85 | -85 | -85 | -infinity | -85 |
| SCH_RP [Note 3] | dBm/15 kHz | -85 | -85 | -85 | -85 | -infinity | -85 |
| Io [Note 3] | dBm/Ch BW | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -54.15 +10log ($N_{RB,c}$ /50) | Specified in columns for Cell 2 | |
| Propagation Condition | | AWGN | | AWGN | | AWGN | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| Timing offset to Cell 1 | µs | - | | 0 | | 3 | |
| Time alignment error relative to cell 1 [Note 5] | µs | - | | ≤ TAE | | N/A | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: Es/Iot, RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 5: Time alignment error (TAE) as specified in TS 36.104 [30] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

## 8.16.26 E-UTRAN TDD-FDD CA Event triggered reporting on deactivated SCell with PCell interruption in non-DRX with PCell in TDD

### 8.16.26.1 Test purpose

To verify that in TDD-FDD CA with PCell in TDD the UE's ability to make a correct reporting of Event A6 on deactivated SCell with PCell interruption in non-DRX within the requirements for SCell stated in TS 36.133[4] section 8.3.3.2.1 and the requirements for PCell stated in TS 36.133[4] clause 8.3.2 while at the same time fulfilling the requirements on interruption rate.

### 8.16.26.2 Test applicability

This test applies to all types of E-UTRA UE release 12 and forward that support E-UTRA FDD and TDD and 2DL TDD-FDD CA with TDD as PCell. Applicability requires support for FGI bit 111.

### 8.16.26.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.16.23.3.

### 8.16.26.4 Test description

#### 8.16.26.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: Any one of the supported TDD-FDD CA configurations with PCell in TDD with the largest aggregated CA bandwidth combination supported by UE from Table 8.16.26.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.26.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.26.4.3.

5. Cell 1 is the PCell on the TDD primary component carrier, Cell 2 is the configured and deactivated SCell on the FDD secondary component carrier, and Cell 3 is the neighbour cell on the FDD secondary component carrier, Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test, Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.26.4.1-1: General test parameters for E-UTRAN TDD-FDD CA Event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX with PCell in TDD**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2 | Two radio channels are used for this test |
| Active PCell | | | Cell 1 | Primary cell on RF channel number 1. |
| Configured deactivated SCell | | | Cell 2 | Configured deactivated secondary cell on RF channel number 2. |
| Neighbour cell | | | Cell 3 | Neighbour cell to be identified on RF channel number 2. |
| CP length | | | Normal | |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211. The same configuration in TDD cells |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211. The same configuration in TDD cells |
| DRX | | | OFF | Continuous monitoring of primary cell |
| A6 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A6. |
| | Offset | dB | -3 | Offset parameter for evaluation of event A6. Needs to take relative accuracy tolerance in clause 9.1.11.2 into account plus margin. |
| | Report on leave | | False | |
| | Time To Trigger | s | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| SCell measurement cycle (measCycleSCell) | | ms | 1280 | |
| T1 | | s | 5 | During this time the UE shall be aware of cells 1 and 2 but not cell 3. |
| T2 | | s | ≤30 | UE shall report Event A6 within 25.6s (20×scellMeasCycle) |

### 8.16.26.4.2 Test procedure

Same test procedure as in clause 8.16.4.4.2 with the following exceptions:

- Instead of clause 8.16.4.4.3 → use clause 8.16.26.4.3.
- Instead of Table 8.16.4.5-1 → use Table 8.16.26.5-1.

### 8.16.26.4.3 Message contents

Same message contents as in clause 8.16.4.4.3.

## 8.16.26.5 Test requirement

Same test requirement as in clause 8.16.4.5 with the following exceptions:

- Instead of Table 8.16.4.4.1-1 → use Table 8.16.26.4.1-1.
- Instead of Table 8.16.4.5-1 → use Table 8.16.26.5-1.

**Table 8.16.26.5-1: Cell specific test parameters for E-UTRAN TDD-FDD CA Event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX with PCell in TDD**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | | | |
| $BW_{channel}$ | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | |
| PDSCH parameters: DL Reference Measurement Channel | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | | - | | - | |
| PCFICH/PDCCH/ PHICH parameters: DL Reference Measurement Channel | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | |
| OCNG Patterns defined | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | |
| PBCH_RA | dB | 0 | | 0 | | 0 | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA[Note 1] | dB | | | | | | |
| OCNG_RB[Note 1] | dB | | | | | | |
| $N_{oc}$[Note 2] | dBm/15 kHz | -101 | | -101 | | | |
| $\hat{E}_s/N_{oc}$ | dB | 16 | 16 | 16 | 16 | -infinity | 16 |
| $\hat{E}_s/I_{ot}$ | dB | 16 | 16 | 16 | -0.11 | -infinity | -0.11 |
| RSRP [Note 3] | dBm/15 kHz | -85 | -85 | -85 | -85 | -infinity | -85 |
| SCH_RP [Note 3] | dBm/15 kHz | -85 | -85 | -85 | -85 | -infinity | -85 |
| Io [Note 3] | dBm/Ch BW | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -54.15 +10log ($N_{RB,c}$ /50) | Specified in columns for Cell 2 | |
| Propagation Condition | | AWGN | | AWGN | | AWGN | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| Timing offset to Cell 1 | µs | - | | 0 | | 3 | |
| Time alignment error relative to cell 1 [Note 5] | µs | - | | ≤ TAE | | N/A | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: Es/Iot, RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 5: Time alignment error (TAE) as specified in TS 36.104 [30] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

# 8.16.27 E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting under Deactivated SCells in Non-DRX with PCell in FDD

## 8.16.27.1 Test purpose

To verify the UE's ability to make correct reportings of Events A1, A2 and A6 under deactivated SCells in non-DRX within the requirements in TS 36.133 [4] section 8.3.3.2.1.

## 8.16.27.2 Test applicability

This test applies to all types of E-UTRA UE release 12 and forward that support E-UTRA FDD and TDD and 3DL CA with FDD as PCell.

## 8.16.27.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD or TDD cell on the secondary component carrier within $T_{identify_scc}$, according to the parameter *measCycleSCell* where $T_{identify_scc}$ = 20 *measCycleSCell*

A cell shall be considered detectable when

- RSRP related side condition given in TS 36.133 [4] sub-clause 9.1 are fulfilled for a corresponding Band,
- SCH_RP|$_{dBm}$ and SCH Ês/Iot according to Annex I.2.7 for a corresponding Band

The measurement period for deactivated SCell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell* where $T_{measure_scc}$ = 5 *measCycleSCell*. The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified cells on the secondary component carrier, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_scc}$.

The measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] sub-clause 9.1.11 (Carrier aggregation measurement accuracy).

A UE may reconfigure the receiver bandwidth or turn on/off one of the RF chains when performing measurements on one or two SCCs with deactivated SCell. This may cause interruptions on PCell or an activated SCell or both that are specified in TS 36.133 [4] sub-clause 7.8.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc}$ defined in TS 36.133 [4] clause 8.3.3.2.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_scc}$ defined in TS 36.133 [4] clause 8.3.3.2.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_scc}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.3.3.2.1 and A.8.16.27.

## 8.16.27.4 Test description

### 8.16.27.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.16.27.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.64 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.27.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.27.4.3.

5. Cell 1 is the PCell on the primary component carrier, Cell 2 is a SCell on RF channel number 2, Cell 3 is a SCell on RF channel number 3, and Cell 4 is a neighbour cell on RF channel number 3. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2, Cell 3 and Cell 4 shall be powered OFF.

**Table 8.16.27.4.1-1: General test parameters for E-UTRAN TDD-FDD 3 DL CA event triggered reporting under Deactivated SCells in Non-DRX with PCell in FDD**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2, 3 | Three radio channels are used for this test |
| Active PCell | | | Cell 1 | Primary cell on RF channel number 1. |
| Configured deactivated SCell | | | Cell 2 | Configured deactivated secondary cell on RF channel number 2. |
| Configured deactivated SCell | | | Cell 3 | Configured deactivated secondary cell on RF channel number 3. |
| Neighbour cell | | | Cell 4 | Neighbour cell to be identified on RF channel number 3. |
| CP length | | | Normal | |
| DRX | | | OFF | Continuous monitoring of primary cell |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211. The same configuration applies to all TDD cells (cell2, cell3 and cell4). |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211. The same configuration applies to all TDD cells (cell2, cell3 and cell4). |
| A1 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A1. |
| | Threshold RSRP | dBm | -98 | Actual RSRP threshold for event A1. Needs to take absolute accuracy tolerance in section 9.1.11.1 into account plus margin. |
| | Time To Trigger | s | 0 | |
| A2 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A2. |
| | Threshold RSRP | dBm | -98 | Actual RSRP threshold for event A2. Needs to take absolute accuracy tolerance in section 9.1.11.1 into account plus margin. |
| | Time To Trigger | s | 0 | |
| A6 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A6. |
| | Offset | dB | -6 | Offset parameter for evaluation of event A6. Needs to take relative accuracy tolerance in section 9.1.11.2 into account plus margin. |
| | Report on leave | | False | |
| | Time To Trigger | s | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Cell-individual offset for cells on RF channel number 3 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| SCell measurement cycle (measCycleSCell) | | ms | 320 | |
| T1 | | s | 5 | During this time cell 1 and cell 3 shall be known to the UE; but cell 2 and cell 4 shall be unknown to the UE. |
| T2 | | s | ≤12 | UE should report Event A1 for cell 2 and event A6 for cell 4 |
| T3 | | s | 5 | UE should report Event A2 for cells 1, 2 and 3. |

## 8.16.27.4.2 Test procedure

The test consists of three successive time periods, with duration of T1, T2 and T3, respectively. During T1 the UE shall not have any information on cell 4. At the beginning of T2 the transmission power of cell 4 is increased to the same level as cell 3, and due to usage of an offset this shall result in reporting of Event A6. Also, at the beginning of T2 the transmission power of cell 2 is increased to the same level as cell 1, which shall result in reporting of Event A1. At the beginning of T3 the transmission powers of cells 1, 2, 3 and 4 are reduced below a threshold value and this shall result in reporting of Event A2 for Cell 1, for Cell 2 and for Cell 3.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCCs according to Annex C.0, C.1 for all downlink physical channels.

3. The SS shall configure SCells (cell 2 and cell 3) on the SCCs as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.27.4.3.

4. Set the parameters according to T1 in Table 8.16.27.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

5. After 2s from start of T1 SS shall transmit an RRCConnectionReconfiguration message with events A1, A2 and A6 configured.

6. The UE shall transmit an RRCConnectionReconfigurationComplete message.

7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.16.27.5-1.

8. The UE shall transmit a MeasurementReport message triggered by Event A1. If the measurement reporting delay from the beginning of time period T2 is less than 6402ms, then count a success for the event “A1”. Otherwise count a fail for the event “A1”.

9. The UE shall transmit a MeasurementReport message triggered by Event A6. If the measurement reporting delay from the beginning of time period T2 is less than 6402ms, then count a success for the event “A6”. Otherwise count a fail for the event “A6”.

10. After the SS receives the MeasurementReport messages in steps 8 and 9, or when T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 8.16.27.5-1.

11. The UE shall transmit MeasurementReport messages triggered by Event A2 for cell 1, cell 2 and cell 3, respectively.

12. If the measurement reporting delay for cell 1 from the beginning of time period T3 is less than 202ms, then count a success for the event “Cell 1 A2”. If the UE sends event-triggered measurement reports during time period T2 or fails to report the event within the overall delays measured requirement, then the number of failure for the event “Cell 1 A2” is increased by one.

13. If the measurement reporting delay for cell 2 from the beginning of time period T3 is less than 1602ms, then count a success for the event “Cell 2 A2”. If the UE sends event-triggered measurement reports during time period T2 or fails to report the event within the overall delays measured requirement, then the number of failure for the event “Cell 2 A2 is increased by one.

14. If the measurement reporting delay for cell 3 from the beginning of time period T3 is less than 1602ms, then count a success for the event “Cell 3 A2”. If the UE sends event-triggered measurement reports during time period T2 or fails to report the event within the overall delays measured requirement, then the number of failure for the event “Cell 3 A2 is increased by one.

15. After the SS receives the MeasurementReport message in steps 12, 13 and 14 or when T3 expires, the SS shall transmit an RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

16. Set cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

17. Set cell 4 physical cell identity = ((current cell 4 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. If physical cell identity of cell 4 = physical cell identity of cell 3 then skip this physical cell identity value for cell 4.

18. After the RRC connection release, the SS:
    - transmits in cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

19. Repeat step 3-18 until a test verdict has been achieved.

Each of the events “A1”, “A6”, “Cell 1 A2”, “Cell 2 A2” and “Cell 3 A2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table

G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.16.27.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.27.4.3-1: Common Exception messages for E-UTRAN TDD-FDD 3DL CA event triggered reporting under Deactivated SCells in Non-DRX with PCell in FDD**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7Table H.4.1-6 |

**Table 8.16.27.4.3-2: *MeasConfig-DEFAULT*: Additional E-UTRAN TDD-FDD 3DL CA event triggered reporting under Deactivated SCells in Non-DRX with PCell in FDD**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 3 entries | | |
| MeasObjectToAddMod[1] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the PCell (Cell 1) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[2] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the SCell on the SCC (Cell 2) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[3] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f3 | f3 is the frequency of the SCell on the SCC (Cell 3)and neighbouring cell on the SCC (Cell 4) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f3) | | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 3 entries | | |
| reportConfigId[1] | idReportConfig-A1 | | |
| reportConfig[1] | ReportConfigEUTRA-A1 | | |
| reportConfigId[2] | idReportConfig-A2 | | |
| reportConfig[2] | ReportConfigEUTRA-A2 | | |
| reportConfigId[3] | IdReportConfig-A6 | | |
| reportConfig[3] | ReportConfig-A6 | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 5 entries | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-A1 | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f1 | | |
| reportConfigId | idReportConfig-A2 | | |
| measId | 3 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-A2 | | |
| measId | 4 | | |
| measObjectId | IdMeasObject-f3 | | |
| reportConfigId | idReportConfig-A2 | | |
| measId | 5 | | |
| measObjectId | IdMeasObject-f3 | | |
| reportConfigId | idReportConfig-A6 | | |

| | | | |
|---|---|---|---|
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 8.16.27.4.3-3: MeasObjectEUTRA-GENERIC: Additional E-UTRAN TDD-FDD 3DL CA event triggered reporting under Deactivated SCells in Non-DRX with PCell in FDD**

| Derivation Path: 36.508 clause 7.3.3 Table 7.3.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| offsetFreq | 0 (dB 0) | | |
| measCycleSCell-r10 | sf320 | | Cell 2, Cell 3 |
| } | | | |

**Table 8.16.27.4.3-4: *ReportConfig-A1*: Additional E-UTRAN TDD-FDD 3DL CA event triggered reporting under Deactivated SCells in Non-DRX with PCell in FDD**

| Derivation Path: 36.508 clause 4.6.6 table 4.6.6-4 ReportConfigEUTRA-A1(-98) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A1(Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA1 SEQUENCE { | | | |
| a1-Threshold CHOICE { | | | |
| threshold-RSRP | 42 | The actual value is IE value – 140 dBm | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

**Table 8.16.27.4.3-5: *ReportConfig-A2*: Additional E-UTRAN TDD-FDD 3DL CA event triggered reporting under Deactivated SCells in Non-DRX with PCell in FDD**

| Derivation Path: 36.508 clause 4.6.6 table 4.6.6-5 ReportConfigEUTRA-A2(-98) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A2(Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA2 SEQUENCE { | | | |
| a2-Threshold CHOICE { | | | |
| threshold-RSRP | 42 | The actual value is IE value – 140 dBm | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

**Table 8.16.27.4.3-6: *MeasurementReport*: Additional E-UTRAN TDD-FDD 3DL CA event triggered reporting under Deactivated SCells in Non-DRX with PCell in FDD**

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell ::= SEQUENCE { | | Report Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | Not present | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| MeasResultServFreq-r10[1] { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| MeasResultServFreq-r10[2] { | | | |
| servFreqId-r10 | 2 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 3 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.27.4.3-7: *MeasurementReport*: Additional E-UTRAN TDD-FDD 3DL CA event triggered reporting under Deactivated SCells in Non-DRX with PCell in FDD**

| Derivation path: TS 36.508 [7] clause 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 2 | | |
| measResultPCell::= SEQUENCE { | | PCell | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| MeasResultEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | Neighbour Cell 4 | |
| physCellId | physCellId of Neighbour Cell | | |
| cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | 2 entries | | |
| MeasResultServFreq-r10[1] { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| MeasResultServFreq-r10[2] { | | | |
| servFreqId-r10 | 2 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 3 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.27.4.3-8: *MeasurementReport*: Additional E-UTRAN TDD-FDD 3DL CA event triggered reporting under Deactivated SCells in Non-DRX with PCell in FDD**

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 3 | | |
| measResultPCell ::= SEQUENCE { | | Report Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | Not present | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| MeasResultServFreq-r10[1] { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| MeasResultServFreq-r10[2] { | | | |
| servFreqId-r10 | 2 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 3 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.27.4.3-9: *MeasurementReport*: Additional E-UTRAN TDD-FDD 3DL CA event triggered reporting under Deactivated SCells in Non-DRX with PCell in FDD**

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 4 | | |
| measResultPCell ::= SEQUENCE { | | Report Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | Not present | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| MeasResultServFreq-r10[1] { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| MeasResultServFreq-r10[2] { | | | |
| servFreqId-r10 | 2 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 3 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.27.4.3-10: *MeasurementReport*: Additional E-UTRAN TDD-FDD 3DL CA event triggered reporting under Deactivated SCells in Non-DRX with PCell in FDD**

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 5 | | |
| measResultPCell ::= SEQUENCE { | | Report Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | | | |
| MeasResultEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | Neighbour Cell 4 | |
| physCellId | physCellId of Neighbour Cell | | |
| cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| MeasResultServFreq-r10[1] { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| MeasResultServFreq-r10[2] { | | | |
| servFreqId-r10 | 2 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 3 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 8.16.27.5 Test requirement

Table 8.16.27.5-1 defines the primary level settings including test tolerances for E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting under Deactivated SCells in Non-DRX with PCell in FDD test.

**Table 8.16.27.5-1: Cell specific test parameters for E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting under Deactivated SCells in Non-DRX with PCell in FDD**

| ameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | | Cell 4 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 |
| RF Channel | | 1 | | | 2 | | | 3 | | | | |
| | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | | |
| arameters:<br>ence<br>nent | | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | | | - | | | - | | | - | |
| PDCCH/PHI<br>neters:<br>ence<br>nent | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | 5MHz: R.11<br>10MHz: R.6<br>20MHz: R.10 | |
| atterns | | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | | 5MHz: OP.10<br>10MHz: OP.2<br>20MHz: OP.8 | |
| A | dB | 0 | | | 0 | | | 0 | | | 0 | |
| B | dB | | | | | | | | | | | |
| | dB | | | | | | | | | | | |
| | dB | | | | | | | | | | | |
| RB | dB | | | | | | | | | | | |
| A | dB | | | | | | | | | | | |
| B | dB | | | | | | | | | | | |
| RA | dB | | | | | | | | | | | |
| RB | dB | | | | | | | | | | | |
| RA | dB | | | | | | | | | | | |
| RB | dB | | | | | | | | | | | |
| A Note 1 | dB | | | | | | | | | | | |
| B Note 1 | dB | | | | | | | | | | | |
| | dBm/15 KHz | -104 | | | -104 | | | -104.5 | | | | |
| | dB | 17 | 17 | -3 | -infinity | 17.2 | -3.2 | 17.7 | 17.7 | -3.7 | -infinity | 17 |
| | dB | 17 | 17 | -3 | -infinity | 17.2 | -3.2 | 17.7 | 0.61 | -5.24 | -infinity | -0.77 |
| e 3 | dBm/15 kHz | -87 | -87 | -107 | -infinity | -86.8 | -107.2 | -86.8 | -86.8 | -108.2 | -infinity | -87.5 |
| Note 3 | dBm/15 kHz | -87 | -87 | -107 | -infinity | -86.8 | -107.2 | -86.8 | -86.8 | -108.2 | -infinity | -87.5 |
| | dBm/Ch BW | -59.13 +10log ($N_{RB,c}$ /50) | -59.13 +10log ($N_{RB,c}$ /50) | -74.45 +10log ($N_{RB,c}$ /50) | -76.22 +10log ($N_{RB,c}$ /50) | -58.94 +10log ($N_{RB,c}$ /50) | -74.52 +10log ($N_{RB,c}$ /50) | -58.94 +10log ($N_{RB,c}$ /50) | -56.30 +10log ($N_{RB,c}$ /50) | -74.03 +10log ($N_{RB,c}$ /50) | Specified in colu Cell 3 | |
| ion Condition | | AWGN | | | ETU70 | | | ETU70 | | | ETU70 | |
| n Matrix and Configuration | | 1x2 | | | 1x2 Low | | | 1x2 Low | | | 1x2 Low | |
| fset to Cell 1 | µs | - | | | 0 | | | 0 | | | 3 | |
| nment error cell 1 Note 5 | µs | - | | | ≤ TAE | | | ≤ TAE | | | N/A | |
| ment error cell 2 Note 5 | µs | - | | | - | | | ≤ TAE | | | N/A | |

OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM syn
Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be mod
AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Es/Iot, RSRP, SCH_RP and Io have been derived from other parameters for information purposes. They are not settable parameters themselv
The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface

The actual overall delays measured in the tests may be up to $2\times TTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc}$

The overall delay measured test requirement for Event A1 is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty

Where:

- measurement reporting delay = $T_{identify_scc}$
- $T_{identify_scc}$ = 20 *measCycleSCell*
- *measCycleSCell* = 320 ms

The UE shall send one Event A1 triggered measurement report with a measurement reporting delay of less than 6402 ms from the beginning of time T2 (note: this gives a total of 6400 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The overall delay measured test requirement for Event A6 is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty

Where:

- measurement reporting delay = $T_{identify_scc}$
- $T_{identify_scc}$ = 20 *measCycleSCell*
- *measCycleSCell* = 320 ms

The UE shall send one Event A6 triggered measurement report with a measurement reporting delay of less than 6402 ms from the beginning of time T2 (note: this gives a total of 6400 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

In the RRC_CONNECTED state the measurement period for intra frequency RSRP and RSRQ measurements is 200 ms as per TS 36.133 [4] section 8.1.2.2.1

The UE shall send one Event A2 triggered measurement report for Cell 1 with a measurement reporting delay of less than 202 ms (including 2 ms for TTI insertion uncertainty) from beginning of time T3.

The measurement period for deactivated scell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell*.

The overall delays measured test requirement for Event A2 for cell 2 and cell 3 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty.

Where:

- measurement reporting delay = $T_{measure_scc}$
- where $T_{measure_scc}$ = 5 *measCycleSCell*.

The UE shall send one Event A2 triggered measurement report for cell 2 and one Event A2 triggered measurement report for cell 3 with a measurement reporting delay of less than 1602ms from beginning of time T3 (note: this gives a total of 1600 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall not send event triggered measurement reports as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95% for each of the events.

The statistical pass/ fail decisions are done separately for Event A1, Event A6, Cell 1 Event A2, Cell 2 Event A2, and Cell 3 Event A2.

Decide the test pass, if events A1 **and** A6 **and all** the A2 events are passed, otherwise fail the UE.

# 8.16.28 E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting under Deactivated SCells in Non-DRX with PCell in TDD

## 8.16.28.1 Test purpose

To verify the UE's ability to make correct reportings of Events A1, A2 and A6 under deactivated SCells in non-DRX within the requirements in TS 36.133 [4] section 8.3.3.2.1.

## 8.16.28.2 Test applicability

This test applies to all types of E-UTRA UE FDD release 12 and forward that support E-UTRA FDD and TDD and 3DL CA with TDD as PCell.

## 8.16.28.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.16.27.3.

The normative reference for this requirement is TS 36.133 [4] clause 8.3.3.2.1 and A.8.16.28.

## 8.16.28.4 Test description

### 8.16.28.4.1 Initial conditions

Same initial conditions as in clause 8.16.27.4.1 with the following exceptions:

- The general test parameter settings are set up according to Table 8.16.28.4.1-1.
- Message contents are defined in clause 8.16.28.4.3.

**Table 8.16.28.4.1-1: General test parameters for E-UTRAN TDD-FDD 3 DL CA event triggered reporting under Deactivated SCells in Non-DRX with PCell in TDD**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2, 3 | Three radio channels are used for this test |
| Active PCell | | | Cell 1 | Primary cell on RF channel number 1. |
| Configured deactivated SCell | | | Cell 2 | Configured deactivated secondary cell on RF channel number 2. |
| Configured deactivated SCell | | | Cell 3 | Configured deactivated secondary cell on RF channel number 3. |
| Neighbour cell | | | Cell 4 | Neighbour cell to be identified on RF channel number 3. |
| CP length | | | Normal | |
| DRX | | | OFF | Continuous monitoring of primary cell |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211. The same configuration applies to TDD cell (cell1). |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211. The same configuration applies to TDD cell (cell1). |
| A1 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A1. |
| | Threshold RSRP | dBm | -98 | Actual RSRP threshold for event A1. Needs to take absolute accuracy tolerance in section 9.1.11.1 into account plus margin. |
| | Time To Trigger | s | 0 | |
| A2 | Hysteresis | dB | 0 | Hysteresis for evaluation of events A2. |
| | Threshold RSRP | dBm | -98 | Actual RSRP threshold for events A2. Needs to take absolute accuracy tolerance in section 9.1.11.1 into account plus margin. |
| | Time To Trigger | s | 0 | |
| A6 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A6. |
| | Offset | dB | -6 | Offset parameter for evaluation of event A6. Needs to take relative accuracy tolerance in section 9.1.11.2 into account plus margin. |
| | Report on leave | | False | |
| | Time To Trigger | s | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Cell-individual offset for cells on RF channel number 3 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| SCell measurement cycle (measCycleSCell) | | ms | 320 | |
| T1 | | s | 5 | During this time cell 1 and cell 3 shall be known to the UE; but cell 2 and cell 4 shall be unknown to the UE. |
| T2 | | s | ≤12 | UE should report Event A1 for cell 2 and event A6 for cell 4 |
| T3 | | s | 5 | UE should report Event A2 for cells 1, 2 and 3. |

## 8.16.28.4.2 Test procedure

Same test procedure as in clause 8.16.27.4.2 with the following exceptions:

- Instead of Table 8.16.27.5-1 → use Table 8.16.28.5-1.
- Message content exceptions are defined in clause 8.16.28.4.3.

#### 8.16.28.4.3 Message contents

Same message contents as in clause 8.16.27.4.3.

### 8.16.28.5 Test requirement

Table 8.16.28.5-1 defines the primary level settings including test tolerances for E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting under Deactivated SCells in Non-DRX with PCell in TDD test.

**Table 8.16.28.5-1: Cell specific test parameters for E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting under Deactivated SCells in Non-DRX with PCell in TDD**

| ameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | | Cell 4 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 |
| RF Channel | | 1 | | | 2 | | | 3 | | | | |
| | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | | |
| arameters:<br>ence<br>nent | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | | | - | | | - | | | - | |
| PDCCH/PHI<br>neters:<br>ence<br>nent | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | 5MHz: R.11<br>10MHz: R.6<br>20MHz: R.10 | |
| atterns | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | | 5MHz: OP.16<br>10MHz: OP.2<br>20MHz: OP.12 | |
| A | dB | 0 | | | 0 | | | 0 | | | 0 | |
| B | dB | | | | | | | | | | | |
| | dB | | | | | | | | | | | |
| | dB | | | | | | | | | | | |
| RB | dB | | | | | | | | | | | |
| A | dB | | | | | | | | | | | |
| B | dB | | | | | | | | | | | |
| RA | dB | | | | | | | | | | | |
| RB | dB | | | | | | | | | | | |
| RA | dB | | | | | | | | | | | |
| RB | dB | | | | | | | | | | | |
| A Note 1 | dB | | | | | | | | | | | |
| B Note 1 | dB | | | | | | | | | | | |
| | dBm/15 KHz | -104 | | | -104 | | | -104.5 | | | | |
| | dB | 17 | 17 | -3 | -infinity | 17.2 | -3.2 | 17.7 | 17.7 | -3.7 | -infinity | 17 |
| | dB | 17 | 17 | -3 | -infinity | 17.2 | -3.2 | 17.7 | 0.61 | -5.24 | -infinity | -0.77 |
| e 3 | dBm/15 kHz | -87 | -87 | -107 | -infinity | -86.8 | -107.2 | -86.8 | -86.8 | -108.2 | -infinity | -87.5 |
| Note 3 | dBm/15 kHz | -87 | -87 | -107 | -infinity | -86.8 | -107.2 | -86.8 | -86.8 | -108.2 | -infinity | -87.5 |
| | dBm/Ch BW | -59.13 +10log ($N_{RB,c}$ /50) | -59.13 +10log ($N_{RB,c}$ /50) | -74.45 +10log ($N_{RB,c}$ /50) | -76.22 +10log ($N_{RB,c}$ /50) | -58.94 +10log ($N_{RB,c}$ /50) | -74.52 +10log ($N_{RB,c}$ /50) | -58.94 +10log ($N_{RB,c}$ /50) | -56.30 +10log ($N_{RB,c}$ /50) | -74.03 +10log ($N_{RB,c}$ /50) | Specified in colu Cell 3 | |
| ion Condition | | AWGN | | | ETU70 | | | ETU70 | | | ETU70 | |
| n Matrix and Configuration | | 1x2 | | | 1x2 Low | | | 1x2 Low | | | 1x2 Low | |
| fset to Cell 1 | µs | - | | | 0 | | | 0 | | | 3 | |
| nment error cell 1 Note 5 | µs | - | | | ≤ TAE | | | ≤ TAE | | | N/A | |
| nment error cell 2 Note 5 | µs | - | | | - | | | ≤ TAE | | | N/A | |

OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM sym
Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be mod
AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Es/Iot, RSRP, SCH_RP and Io have been derived from other parameters for information purposes. They are not settable parameters themsel
The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

Same Event reporting requirements for A1, A6 and A2 as in clause 8.16.27.5.

The UE shall not send event triggered measurement reports as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95% for each of the events.

The statistical pass/ fail decisions are done separately for Event A1, Event A6, Cell 1 Event A2, Cell 2 Event A2, and Cell 3 Event A2.

Decide the test pass, if events A1 **and** A6 **and all** the A2 events are passed, otherwise fail the UE.

## 8.16.29 3 DL FDD CA Event Triggered Reporting under Deactivated SCells in Non-DRX

### 8.16.29.1 Test purpose

To verify that the UE correctly detects Events A1 (Serving cell becomes better than threshold), A2 (Serving cell becomes worse than threshold) and A6 (Neighbour becomes better than SCell) defined in TS 36.331 [5] within the requirements stated in TS 36.133[4] clause 8.3.3.2.1.

### 8.16.29.2 Test applicability

This test case applies to all types of E-UTRA FDD UE release 10 and forward that support 3DL with Intra-band contiguous CA, or 3DL with Inter-band CA, or 3DL with Intra-band contiguous and Inter-band CA.

This test case also applies to all types of E-UTRA FDD UE release 11 and forward that support 3DL with Intra-band non-contiguous and Inter-band CA, or 3DL with Intra-band non-contiguous and Intra-band contiguous.

### 8.16.29.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD or TDD cell on a secondary component carrier within $T_{identify_scc}$, according to the parameter *measCycleSCell* where $T_{identify_scc}$ = 20 *measCycleSCell.*

A cell shall be considered detectable when

- RSRP related side condition given in TS 36.133[4] Clause 9.1 are fulfilled for a corresponding Band,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.7 for a corresponding Band

The measurement period for deactivated scell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell* where $T_{measure_scc}$ = 5 *measCycleSCell*. The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified cells on a secondary component carrier, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_scc}$.

The measurement accuracy for all measured cells shall be as specified in TS 36.133[4] the sub-clause 9.1.11 (Carrier aggregation measurement accuracy).

A UE may reconfigure the receiver bandwidth or turn on/off one of the RF chains when performing measurements on one or two SCCs with deactivated SCell. This may cause interruptions on PCell or an activated SCell or both that are specified in TS 36.133[4] Section 7.8.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which is caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc}$ defined in TS 36.133[4] Clause 8.3.3.2.1. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_scc}$ defined in TS 36.133[4] clause 8.3.3.2.1 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_scc}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.3.3.2.1 and A.8.16.29.

## 8.16.29.4 Test description

### 8.16.29.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.16.29.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.64 as appropriate.
2. The general test parameter settings are set up according to Table 8.16.29.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.16.29.4.3.
5. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on SCC1, Cell 3 is SCell on SCC2 and Cell 4 is the neighbouring cell on SCC2. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2, Cell 3 and Cell 4 shall be powered OFF.

**Table 8.16.29.4.1-1: General test parameters for E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2, 3 | Three radio channels are used for this test |
| Active PCell | | | Cell 1 | Primary cell on RF channel number 1. |
| Configured deactivated SCell | | | Cell 2 (SCell 1) | Configured deactivated secondary cell 1 on RF channel number 2. |
| | | | Cell 3 (SCell 2) | Configured deactivated secondary cell 2 on RF channel number 3. |
| Neighbour cell | | | Cell 4 | Neighbour cell to be identified on RF channel number 3. |
| CP length | | | Normal | |
| DRX | | | OFF | Continuous monitoring of primary cell |
| A1 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A1. |
| | Threshold RSRP | dBm | -98 | Actual RSRP threshold for event A1. Needs to take absolute accuracy tolerance in TS 36.133 [4] clause 9.1.11.1 into account plus margin. No event A1 reporting is configured for cell 1 and 3. |
| | Time To Trigger | s | 0 | |
| A2 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A2. |
| | Threshold RSRP | dBm | -98 | Actual RSRP threshold for event A2. Needs to take absolute accuracy tolerance in TS 36.133 [4] clause 9.1.11.1 into account plus margin. |
| | Time To Trigger | s | 0 | |
| A6 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A6. |
| | Offset | dB | -6 | Offset parameter for evaluation of event A6. Needs to take relative accuracy tolerance in TS 36.133 [4] clause 9.1.11.2 into account plus margin. |
| | Report on leave | | False | |
| | Time To Trigger | s | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on secondary component carrier 1. |
| Cell-individual offset for cells on RF channel number 3 | | dB | 0 | Individual offset for cells on secondary component carrier 2. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| SCell measurement cycle (measCycleSCell) for SCell 1 and 2 | | ms | 320 | |
| T1 | | s | 5 | During this time the UE shall be aware of cells 1, 2 and 3 but not cell 4. |
| T2 | | s | ≤12 | UE should report Event A1 within 1.6s (5×scellMeasCycle)<br>UE should report Event A6 within 6.4s (20×scellMeasCycle) |
| T3 | | s | 5 | UE should report Event A2 within 200 ms and 1.6s for cells 1, 2 and 3, respectively. |

## 8.16.29.4.2 Test procedure

It is indicated to the UE in the measurement control information that event-triggered reporting with Events A1 (SCell 1), Events A2 (PCell and SCell 1/2) and A6 is used.

The test consists of three successive time periods, with duration of T1, T2 and T3, respectively.
During T1 the UE shall not have any information on cell 4. Immediately at beginning of T2 the transmission power of cell 2 is increased above a threshold value such that this shall result in reporting of Event A1 for SCell 1, and cell 4 is increased to the same level as for Cell 3. Due to usage of an offset this shall result in reporting of Event A6. At beginning of T3 the transmission powers of cells 1, 2, 3 and 4 are reduced below a threshold value and this shall result in reporting of Event A2 for PCell and SCell 1/2, respectively.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure the SCCs according to Annex C.0, C.1 for all downlink physical channels.

3. The SS shall configure SCells (cell 2 and cell 3) on the SCCs as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.29.4.3.

4. Set the parameters according to T1 in Table 8.16.29.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

5. After 2s from start of T1 SS shall transmit an RRCConnectionReconfiguration message configuring events A1, A2 and A6 according to Table 8.16.29.4.1-1.

6. The UE shall transmit an RRCConnectionReconfigurationComplete message.

7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.16.1.5-1.

8. The UE shall transmit a MeasurementReport message triggered by Event A1 for Cell 2. If the measurement reporting delay from the beginning of time period T2 is less than 6402ms, then count a success for the event "A1". Otherwise count a fail for the event "A1".
   The UE shall also transmit a MeasurementReport message triggered by Event A6. If the measurement reporting delay from the beginning of time period T2 is less than 6402ms, then count a success for the event "A6". Otherwise count a fail for the event "A6".

9. After the SS receives the MeasurementReport message(s) in step 8) or when T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 8.16.29.5-1.

10. The UE shall transmit MeasurementReport messages triggered by Event A2 for Cell 1, Cell 2 and Cell 3, respectively.

10a. If the measurement reporting delay for Cell 1 from the beginning of time period T3 is less than 202ms, then count a success for the event "Cell 1 A2". If the UE sends event-triggered measurement reports during time period T2 or fails to report the event within the overall delays measured requirement, then the number of failure for the event "Cell 1 A2" is increased by one.

10b. If the measurement reporting delay for Cell 2 from the beginning of time period T3 is less than 1602ms, then count a success for the event "Cell 2 A2". If the UE sends event-triggered measurement reports during time period T2 or fails to report the event within the overall delays measured requirement, then the number of failure for the event "Cell 2 A2 is increased by one.

10c. If the measurement reporting delay for Cell 3 from the beginning of time period T3 is less than 1602ms, then count a success for the event "Cell 3 A2". If the UE sends event-triggered measurement reports during time period T2 or fails to report the event within the overall delays measured requirement, then the number of failure for the event "Cell 3 A2 is increased by one.

11. After the SS receives the MeasurementReport messages in steps 10a, 10b and 10c or when T3 expires, the SS shall transmit an RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

12. Set Cell 4 physical cell identity = ((current cell 4 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. If physical cell identity of Cell 4 = physical cell identity of Cell 3 then skip this physical cell identity value for Cell 4.

13. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 3-13 until a test verdict has been achieved.

Each of the events "A1", "A6", "Cell 1 A2", "Cell 2 A2" and Cell3 A2" is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table

G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.16.29.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.29.4.3-1: Common Exception messages for E-UTRAN FDD 3DL CA Event triggered reporting under deactivated SCell in non-DRX under fading propagation conditions**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.4.1-6 |

**Table 8.16.29.4.3-2: *MeasConfig-DEFAULT*: E-UTRAN FDD 3 DL CA Event triggered Reporting under Deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 3 entries | | |
| MeasObjectToAddMod[1] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the PCell (Cell 1) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[2] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the SCell on the SCC1 (Cell 2) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[3] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f3 | f3 is the frequency of the SCell on the SCC2 (Cell 3)and neighbouring cell on the SCC2 (Cell 4) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f3) | | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 3 entry | | |
| reportConfigId[1] | idReportConfig-A1 | | |
| reportConfig[1] | ReportConfigEUTRA-A1 | | |
| reportConfigId[2] | idReportConfig-A2 | | |
| reportConfig[2] | ReportConfigEUTRA-A2 | | |
| reportConfigId[3] | idReportConfig-A6 | | |
| reportConfig[3] | ReportConfigEUTRA-A6 | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 5 entry | | |
| measId[1] | 1 | | |
| measObjectId[1] | IdMeasObject-f2 | | |
| reportConfigId[1] | idReportConfig-A1 | | |
| measId[2] | 2 | | |
| measObjectId[2] | IdMeasObject-f1 | | |
| reportConfigId[2] | idReportConfig-A2 | | |
| measId[3] | 3 | | |
| measObjectId[3] | IdMeasObject-f2 | | |
| reportConfigId[3] | idReportConfig-A2 | | |
| measId[4] | 4 | | |
| measObjectId[4] | IdMeasObject-f3 | | |
| reportConfigId[4] | idReportConfig-A2 | | |
| measId[5] | 5 | | |
| measObjectId[5] | IdMeasObject-f3 | | |

| | | | |
|---|---|---|---|
| reportConfigId[5] | idReportConfig-A6 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 8.16.29.4.3-3: MeasObjectEUTRA-GENERIC: Additional E-UTRAN FDD 3 DL CA Event triggered Reporting under Deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation Path: 36.508 clause 7.3.3 Table 7.3.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | Cell 2, Cell 3 |
| } | | | |

**Table 8.16.29.4.3-4: *ReportConfig-A1-H*: Additional E-UTRAN FDD 3 DL CA Event triggered Reporting under Deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation Path: 36.508 clause 4.6.6 table 4.6.6-4 ReportConfigEUTRA-A1(-98) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A1(Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA1 SEQUENCE { | | | |
| A1-Threshold CHOICE { | | | |
| threshold-RSRP | 42 | The actual value is IE value – 140 dBm | |
| } | | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

**Table 8.16.29.4.3-5: *ReportConfig-A2-H*: Additional E-UTRAN FDD 3 DL CA Event triggered Reporting under Deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation Path: 36.508 clause 4.6.6 table 4.6.6-5 ReportConfigEUTRA-A2(-98) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A2(Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA2 SEQUENCE { | | | |
| a2-Threshold CHOICE { | | | |
| threshold-RSRP | 42 | The actual value is IE value – 140 dBm | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

**Table 8.16.29.4.3-6: MeasurementReport: Additional E-UTRAN FDD 3 DL CA Event triggered Reporting under Deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation path: TS 36.508 [7] clause 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell::= SEQUENCE { | | PCell | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| MeasResultEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | Neighbour Cell 4 | |
| physCellId | physCellId of Neighbour Cell | | |
| cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | 2 entries | | |
| MeasResultServFreq-r10[1] { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| MeasResultServFreq-r10[2] { | | | |
| servFreqId-r10 | 2 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 3 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.29.4.3-7: MeasurementReport: Additional E-UTRAN FDD 3 DL CA Event triggered Reporting under Deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation path: TS 36.508 [7] clause 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 2 | | |
| measResultPCell::= SEQUENCE { | | PCell | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| MeasResultEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | Neighbour Cell 4 | |
| physCellId | physCellId of Neighbour Cell | | |
| cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | 2 entries | | |
| MeasResultServFreq-r10[1] { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| MeasResultServFreq-r10[2] { | | | |
| servFreqId-r10 | 2 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 3 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.29.4.3-8: MeasurementReport: Additional E-UTRAN FDD 3 DL CA Event triggered Reporting under Deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation path: TS 36.508 [7] clause 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 3 | | |
| measResultPCell::= SEQUENCE { | | PCell | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| MeasResultEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | Neighbour Cell 4 | |
| physCellId | physCellId of Neighbour Cell | | |
| cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | 2 entries | | |
| MeasResultServFreq-r10[1] { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| MeasResultServFreq-r10[2] { | | | |
| servFreqId-r10 | 2 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 3 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.29.4.3-9: MeasurementReport: Additional E-UTRAN FDD 3 DL CA Event triggered Reporting under Deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation path: TS 36.508 [7] clause 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 4 | | |
| measResultPCell::= SEQUENCE { | | PCell | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| MeasResultEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | Neighbour Cell 4 | |
| physCellId | physCellId of Neighbour Cell | | |
| cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | 2 entries | | |
| MeasResultServFreq-r10[1] { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| MeasResultServFreq-r10[2] { | | | |
| servFreqId-r10 | 2 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 3 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.29.4.3-10: MeasurementReport: Additional E-UTRAN FDD 3 DL CA Event triggered Reporting under Deactivated SCell in non-DRX under fading propagation conditions requirements**

| Derivation path: TS 36.508 [7] clause 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 5 | | |
| measResultPCell::= SEQUENCE { | | PCell | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| MeasResultEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | Neighbour Cell 4 | |
| physCellId | physCellId of Neighbour Cell | | |
| cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | 2 entries | | |
| MeasResultServFreq-r10[1] { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| MeasResultServFreq-r10[2] { | | | |
| servFreqId-r10 | 2 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 3 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 8.16.29.5 Test requirement

Table 8.16.29.5-1 defines the primary level settings including test tolerances for 3 DL FDD CA Event Triggered Reporting under Deactivated SCells in Non-DRX.

**Table 8.16.29.5-1: Cell specific test parameters for E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | | Cell 4 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | 3 | | | | | |
| $BW_{channel}$ | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | | | |
| PDSCH parameters: DL Reference Measurement Channel | | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | | | - | | | - | | | - | | |
| PCFICH/PDCCH/PHICH parameters: DL Reference Measurement Channel | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | |
| OCNG Patterns | | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | | | | |
| $N_{oc}$ Note 2 | | -104 | | | -104 | | | -104.5 | | | | | |
| $\hat{E}_s/N_{oc}$ | dB | 17 | 17 | -3 | -infinity | 17.2 | -3.2 | 17.7 | 17.7 | -3.7 | -infinity | 17 | -3.7 |
| $\hat{E}_s/I_{ot}$ Note 3 | dB | 17 | 17 | -3 | -infinity | 17.2 | -3.2 | 17.7 | 0.61 | -5.24 | -infinity | -0.77 | -5.24 |
| RSRP Note 3 | dBm/15 kHz | -87 | -87 | -107 | -infinity | -86.8 | -107.2 | -86.8 | -86.8 | -108.2 | -infinity | -87.5 | -108.2 |
| SCH_RP Note 3 | dBm/15 kHz | -87 | -87 | -107 | -infinity | -86.8 | -107.2 | -86.8 | -86.8 | -108.2 | -infinity | -87.5 | -108.2 |
| Io Note 3 | dBm/Ch BW | -59.13 +10log ($N_{RB,c}$ /50) | -59.13 +10log ($N_{RB,c}$ /50) | -74.45 +10log ($N_{RB,c}$ /50) | -76.22 +10log ($N_{RB,c}$ /50) | -58.94 +10log ($N_{RB,c}$ /50) | -74.52 +10log ($N_{RB,c}$ /50) | -58.94 +10log ($N_{RB,c}$ /50) | -56.30 +10log ($N_{RB,c}$ /50) | -74.03 +10log ($N_{RB,c}$ /50) | Specified in columns for Cell 3 | | |

| | | | | | |
|---|---|---|---|---|---|
| Propagation Condition | | AWGN | ETU70 | ETU70 | ETU70 |
| Correlation Matrix and Antenna Configuration | | 1x2 | 1x2 Low | 1x2 Low | 1x2 Low |
| Timing offset to Cell 1 | μs | - | 0 | 0 | 3 |
| Time alignment error relative to cell 1 Note 5 | μs | - | ≤ TAE | ≤ TAE | N/A |
| Time alignment error relative to cell 2 Note 5 | μs | - | - | ≤ TAE | N/A |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: Es/Iot, RSRP, SCH_RP and Io have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 5: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

The overall delays measured test requirement for Events A1, A2 and A6 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Where:

measurement reporting delay = $T_{identify_scc}$

TTI insertion uncertainty = $2 \times TTI_{DCCH}$ = 2ms

In the test measCycleSCell for SCell1 and SCell2 = 320ms

This gives, for the different Events:

The UE shall send one Event A6 triggered measurement report for cell 4 with a measurement reporting delay of less than 6402s (($T_{identify_scc}$ = 20×measCycleSCell) + TTI insertion uncertainty) from the beginning of time T2.

The UE shall send one Event A1 triggered measurement report for Cell 2 with a measurement reporting delay of less than 6402s (($T_{identify_scc}$ = 20×measCycleSCell) + TTI insertion uncertainty) from beginning of time T2.

The UE shall send one Event A2 triggered measurement report for Cell 2 with a measurement reporting delay of less than 1602s (($T_{identify_scc}$ = 5× measCycleSCell) + TTI insertion uncertainty) from beginning of time T3.

The UE shall send one Event A2 triggered measurement report for Cell 3 with a measurement reporting delay of less than 1602s (($T_{identify_scc}$ = 5× measCycleSCell) + TTI insertion uncertainty) from beginning of time T3.

In the RRC_CONNECTED state the measurement period for intra frequency RSRP and RSRQ measurements is 200 ms as per TS 36.133[4] section 8.1.2.2.1 giving for Cell 1:

The UE shall send one Event A2 triggered measurement report for Cell 1 with a measurement reporting delay of less than 202 ms (including 2 ms for TTI insertion uncertainty) from beginning of time T3.

The UE shall not send event triggered measurement reports as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95% for each of the events.

The statistical pass/fail decisions are done separately for Event A1, Event A2 and Event A6.

Decide the test pass, if events A1 **and** A2 **and** A6 are passed, otherwise fail the UE.

## 8.16.30 3 DL TDD CA Event Triggered Reporting under Deactivated SCells in Non-DRX

### 8.16.30.1 Test purpose

Same test purpose as in clause 8.16.29.1.

### 8.16.30.2 Test applicability

This test case applies to all types of E-UTRA TDD UE release 10 and forward that support 3DL with Intra-band contiguous CA, or 3DL with Inter-band CA, or 3DL with Intra-band contiguous and Inter-band CA.

This test case also applies to all types of E-UTRA TDD UE release 11 and forward that support 3DL with Intra-band non-contiguous and Inter-band CA, or 3DL with Intra-band non-contiguous and Intra-band contiguous

### 8.16.30.3 Minimum conformance requirements

Same minimum requirements as in clause 8.16.29.3.

The normative reference for this requirement is TS 36.133 [4] clause 8.3.3.2.1 and A.8.16.30.

## 8.16.30.4 Test description

### 8.16.30.4.1 Initial conditions

Same initial conditions as 8.16.29.4.1 with the following exceptions:

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.16.30.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

### 8.16.30.4.2 Test procedure

Same test procedure as 8.16.29.4.2 with the following exceptions:

- Instead of Table 8.16.29.5-1 → use Table 8.16.30.5-1.
- Instead of referring to clause 8.16.29.4.3 → refer to clause 8.16.30.4.3.

### 8.16.30.4.3 Message contents

Same message content as in 8.16.29.4.3.

## 8.16.30.5 Test requirement

Same test requirements as in clause 8.16.29.5 with the exception of table 8.16.30.5.-1.

Table 8.16.30.5-1 defines the primary level settings including test tolerances for 3 DL TDD CA Event Triggered Reporting under Deactivated SCells in Non-DRX.

**Table 8.16.30.5-1: Cell specific test parameters for E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | | | |
| $BW_{channel}$ | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | | | 5MHz: N<br>10MHz:<br>20MHz: N |
| PDSCH parameters: DL Reference Measurement Channel | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | | | - | | | - | | |
| PCFICH/PDCCH/PHICH parameters: DL Reference Measurement Channel | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | |
| OCNG Patterns | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | |
| $N_{oc}$ Note 2 | | -104 | | | -104 | | | | | -10 |
| $\hat{E}_s/N_{oc}$ | dB | 17 | 17 | -3 | -infinity | 17.2 | -3.2 | 17.7 | 17.7 | -3.7 |
| $\hat{E}_s/I_{ot}$ Note 3 | dB | 17 | 17 | -3 | -infinity | 17.2 | -3.2 | 17.7 | 0.61 | -5.24 |
| RSRP Note 3 | dBm/15 kHz | -87 | -87 | -107 | -infinity | -86.8 | -107.2 | -86.8 | -86.8 | -108.2 |
| SCH_RP Note 3 | dBm/15 kHz | -87 | -87 | -107 | -infinity | -86.8 | -107.2 | -86.8 | -86.8 | -108.2 |
| Io Note 3 | dBm/Ch BW | -59.13 +10log ($N_{RB,c}$ /50) | -59.13 +10log ($N_{RB,c}$ /50) | -74.45 +10log ($N_{RB,c}$ /50) | -76.22 +10log ($N_{RB,c}$ /50) | -58.94 +10log ($N_{RB,c}$ /50) | -74.52 +10log ($N_{RB,c}$ /50) | -58.94 +10log ($N_{RB,c}$ /50) | -56.30 +10log ($N_{RB,c}$ /50) | -74.03 +10log ($N_{RB,c}$ /50) |
| Propagation Condition | | AWGN | | | ETU70 | | | ETU70 | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | 1x2 Low | | | 1x2 Low | | |
| Timing offset to Cell 1 | µs | - | | | 0 | | | 0 | | |
| Time alignment error relative to cell 1 Note 5 | µs | - | | | ≤ TAE | | | ≤ TAE | | |
| Time alignment error relative to cell 2 Note 5 | µs | - | | | - | | | ≤ TAE | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achiev
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and ti AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: Es/Iot, RSRP, SCH_RP and Io have been derived from other parameters for information purposes. They are not settable
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 5: Time alignment error (TAE) as specified in TS 36.104 [30] clause 6.5.3.1. The TAE value depends upon the type of carrie

# 8.16.31 E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in FDD

## 8.16.31.1 Test purpose

To verify the UE's ability to make a correct reporting of event A6 on deactivated SCell with PCell and SCell interruptions in non-DRX and with PCell in FDD within the requirements in section 8.3.3.2.1 while at the same time fulfilling the requirements on interruption rate.

## 8.16.31.2 Test applicability

This test case applies to all types of UE release 12 and forward supporting E-UTRA FDD and TDD and 3DL CA with FDD as PCell.

## 8.16.31.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD or TDD cell on a secondary component carrier within $T_{identify_scc}$, according to the parameter *measCycleSCell* where $T_{identify_scc}$ = 20 *measCycleSCell*

A cell shall be considered detectable when

- RSRP related side condition given in Clause 9.1 are fulfilled for a corresponding Band,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex B.2.7 for a corresponding Band

The measurement period for deactivated scell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell* where $T_{measure_scc}$ = 5 *measCycleSCell*. The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified cells on a secondary component carrier, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_scc}$.

The measurement accuracy for all measured cells shall be as specified in the TS 36.133[4] sub-clause 9.1.11 (Carrier aggregation measurement accuracy)

A UE may reconfigure the receiver bandwidth or turn on/off one of the RF chains when performing measurements on one or two SCCs with deactivated SCell. This may cause interruptions on PCell or an activated SCell or both that are specified in Section 7.8.

Reported measurements contained in event triggered measurement reports shall meet the requirements in clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which is caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc}$ defined in TS 36.133[4] Clause 8.3.3.2.1. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_scc}$ defined in TS 36.133[4] clause 8.3.3.2.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_scc}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected

The normative reference for this requirement is TS 36.133 [4] clause 8.3.3.2.1 and A.8.16.31.

## 8.16.31.4 Test description

### 8.16.31.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.16.31.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1 [10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.65 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.31.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.31.4.3.

5. There are three E-UTRA carriers and four cells: Cell1, Cell2, Cell3 and Cell4. Cell 1 is PCell on the FDD primary component (RF Channel 1), Cell 2 is the SCell on the TDD Secondary Component (RF Channel 2), and Cell 3 is SCell on the TDD secondary component (RF Channel 3) and Cell 4 is the neighbour cell on the TDD secondary component (RF Channel 3). PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2, Cell 3 and Cell 4 shall be powered OFF.

**Table 8.16.31.4.1-1: General test parameters for E-UTRAN TDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2, 3 | three radio channels are used for this test |
| Active PCell | | | Cell 1 | Primary cell on RF channel number 1. |
| Configured deactivated SCell | | | Cell 2 | Configured deactivated secondary cell on RF channel number 2. |
| Configured deactivated SCell | | | Cell 3 | Configured deactivated secondary cell on RF channel number 3. |
| Neighbour cell | | | Cell 4 | Neighbour cell to be identified on RF channel number 3. |
| CP length | | | Normal | |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211[9]. The same configuration in TDD cells |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211[9]. The same configuration in TDD cells |
| DRX | | | OFF | Continuous monitoring of primary cell |
| A6 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A6. |
| | Offset | dB | -3 | Offset parameter for evaluation of event A6. Needs to take relative accuracy tolerance in TS 36.133[4] clause 9.1.11.2 into account plus margin. |
| | Report on leave | | False | |
| | Time To Trigger | s | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Cell-individual offset for cells on RF channel number 3 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| SCell measurement cycle | | ms | 640 | |
| T1 | | s | 4 | During this time the UE shall be aware of cells 1, 2 and 3 but not cell 4. |
| T2 | | s | ≤15 | UE should report Event A6 within 12.8s (20×scellMeasCycle) |
| T3 | | s | 4 | During this time the UE shall activate cell 2 |
| T4 | | s | ≤4 | UE should report Event A6 within 3.2s (5×scellMeasCycle) |

## 8.16.31.4.2 Test procedure

The test consists of FDD Cell 1 the PCell on the FDD primary component (RF Channel 1), Cell 2 the SCell on the TDD Secondary Component Carrier (RF Channel 2), and Cell 3 SCell on the TDD secondary component (RF Channel 3) and Cell 4 the neighbour cell on the TDD secondary component (RF Channel 3). In the measurement control information it is indicated to the UE that event-triggered reporting with Event A6 is used. The test consists of four successive time periods, with duration of T1, T2, T3 and T4, respectively. During T1 and T2, both Cell2 and Cell3 are deactivated. During T1 the UE shall not have any information of cell 4. Immediately at beginning of T2 the transmission power of cell 4 is increased to same level as for cell 3, and due to usage of an offset this shall result in reporting of Event A6. At the beginning of T3, the transmission power of Cell 4 is turned off and the test equipment sends a MAC message for activation of Cell 2 to UE but Cell3 remains deactivated. Immediately at beginning of T4 the transmission power of Cell 4 is increased to same level as for Cell 3, and due to usage of an offset this shall result in reporting of Event A6. The UE shall be continuously scheduled in the PCell throughout the whole test and continuously scheduled in Cell2 during T4.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCCs according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCell Cell 2 and SCell Cell 3 on the SCCs (RF Channel 2 and RF Channel 3) as per TS 36.508 [7] clause 5.2A.4.

4. Set the parameters according to T1 in Table 8.16.31.5-1 as appropriate. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.

5. The SS shall transmit an RRCConnectionReconfiguration message with Event A6 configured.

6. The UE shall transmit RRCConnectionReconfigurationComplete message.

7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.16.31.5-1.

8. The UE shall transmit a MeasurementReport message triggered by Event A6. If the overall delays measured from the beginning of time period T2 is less than 12802 ms and at least 99.5% of all expected ACK/NACKS are detected by the system simulator from the start of T1 until the measurement report is received during T2 for PCell then the number of successful tests for event “T2 A6” is increased by one. If the UE fails to report the event 6 within the overall delays measured requirement or less than 99.5% of all expected ACK/NACKS are detected by the system simulator for PCell then the number of failure tests for event “T2 A6” is increased by one.

9. After the SS receives the MeasurementReport message in step 8) or when T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 8.16.31.5-1. The SS activates SCell (Cell 2) by sending the activation MAC-CE (Refer TS 36.321 [13], clauses 5.13, 6.1.3.8).

10. When T3 expires, the SS shall switch the power setting from T3 to T4 as specified in Table 8.16.31.5-1.

11. The UE shall transmit a MeasurementReport message triggered by Event A6. If the overall delays measured from the beginning of time period T4 is less than 3202 ms and at least 99.5% of all expected ACK/NACKS are detected by the system simulator from the start of T4 until the measurement report is received during T4 for both PCell and SCell2, then the number of successful tests for event “T4 A6” is increased by one. If the UE fails to report the event within the overall delays measured requirement or less than 99.5% of all expected ACK/NACKS are detected by the system simulator for either PCell or SCell then the number of failure tests for event “T4 A6” is increased by one.

12. After the SS receives the MeasurementReport message in step 11) or when T4 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

13. Set Cell 4 physical cell identity = ((current cell 4 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. If physical cell identity of Cell 4 = physical cell identity of Cell 3 then skip this physical cell identity value for Cell 4.

14. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

15. Repeat step 3-14 until a test verdict has been achieved.

Each of the events “T2 A6”and “T4 A6” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.16.31.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.31.4.3-1: Common Exception messages for E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in FDD**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.4.1-3<br>Table H.4.1-6 |

**Table 8.16.31.4.3-2: *MeasConfig-DEFAULT*: E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in FDD test requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 3 entries | | |
| MeasObjectToAddMod[1] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the PCell (Cell 1) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[2] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the SCell on the SCC1 (Cell 2) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[3] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f3 | f3 is the frequency of the SCell on the SCC2 (Cell 3)and neighbouring cell on the SCC2 (Cell 4) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f3) | | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-A6 | | |
| reportConfig | ReportConfigEUTRA-A6 | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f3 | | |
| reportConfigId | idReportConfig-A6 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 8.16.31.4.3-3: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in FDD test requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1, Cell 2, Cell 3, Cell 4 |
| } | | | |

**Table 8.16.31.4.3-4: MeasurementReport: E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in FDD test requirements**

| Derivation path: TS 36.508 [7] clause 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell::= SEQUENCE { | | PCell | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| MeasResultEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | Neighbour Cell 4 | |
| physCellId | physCellId of Neighbour Cell | | |
| cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | 2 entries | | |
| MeasResultServFreq-r10[1] { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| MeasResultServFreq-r10[2] { | | | |
| servFreqId-r10 | 2 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 3 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.31.4.3-5: *SystemInformationBlockType2*: Additional E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in FDD test requirements**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-1 SystemInformationBlockType2 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfig | Not present | | Cell 1 |
| } | | | |

**Table 8.16.31.4.3-6: *SystemInformationBlockType3*: Additional E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in FDD test requirements**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-2 SystemInformationBlockType3 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType3 ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1, Cell 2, Cell 3. Cell 4 |
| } | | | |

## 8.16.31.5 Test requirement

Table 8.16.31.4.1-1 and Table 8.16.31.5-1 define the primary level settings including test tolerances for E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in FDD test.

**Table 8.16.31.5-1: Cell specific test parameters for E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in FDD**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | | Cell 4 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T1 | T2 | T3 | T4 | T1 | T2 | T3 | T4 | T1 | T2 | T3 | T4 |
| E-UTRA RF Channel Number | | 1 | | | | 2 | | | | 3 | | | | 3 | | | |
| $BW_{channel}$ | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | |
| PDSCH parameters: DL Reference Measurement Channel | | 5MHz: R.7 FDD<br>10MHz: R.3 FDD<br>20MHz: R.6 FDD | | | | N/A | N/A | N/A | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | N/A | | | | N/A | | | |
| PCFICH/PDCCH/ PHICH parameters: DL Reference Measurement Channel | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | |
| OCNG Pattern defined in D.1 and D.2 | | 5MHz: OP.20 FDD<br>10MHz: OP.10 FDD<br>20MHz: OP.17 FDD | | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | | |
| PBCH_RA | dB | 0 | | | | 0 | | | | 0 | | | | 0 | | | |
| PBCH_RB | dB | | | | | | | | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | | | | | | | | |

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| OCNG_RA Note 1 | dB | | | | | | | | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | | | | | | | | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -101 | | | | -101 | | | | -101 | | | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | -Infinity | 16 | -Infinity | 16 |
| $\hat{E}_s / I_{ot}$ Note 4 | dB | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | -0.11 | 16 | -0.11 | -Infinity | -0.11 | -Infinity | -0.11 |
| RSRP Note 4 | dBm/15 kHz | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -Infinity | -85 | -Infinity | -85 |
| SCH_RP Note 4 | dBm/15 kHz | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -Infinity | -85 | -Infinity | -85 |
| Io Note 4 | dBm/Ch BW | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -54.15 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -54.15 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -54.15 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -54.15 +10log ($N_{RB,c}$ /50) |
| Propagation Condition | | AWGN | | | | AWGN | | | | AWGN | | | | AWGN | | | |
| Antenna Configuration | | 1x2 | | | | 1x2 | | | | 1x2 | | | | 1x2 | | | |
| Timing offset to Cell 1 | μs | - | | | | 0 | | | | 0 | | | | 3 | | | |
| Time alignment error relative to cell 1 Note 5 | μs | - | | | | ≤ TAE | | | | ≤ TAE | | | | N/A | | | |
| Time alignment error relative to cell 2 Note 5 | μs | - | | | | - | | | | ≤ TAE | | | | N/A | | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE on cell1 prior to the start of time period T2 and on cell2 prior to the start of time period T4.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Es/Iot, RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

The UE shall send one Event A6 triggered measurement report, with a measurement reporting delay less than 12802 ms from the beginning of time period T2.

The overall delays measured for Event A6 in T2 is defined as the time from the beginning of time period T2, to the moment the UE send one Event A6 triggered measurement report to Cell 4.

The overall delays measured test requirement is expressed as:

- Overall delays measured = measurement reporting delay + TTI insertion uncertainty

  Measurement reporting delay = $T_{identify_scc}$ = 20 *measCycleSCell* = 12800ms

- TTI insertion uncertainty = $TTI_{DCCH}$ = 1 ms; $2xTTI_{DCCH}$ = 2 ms

The overall delays measured shall be less than a total of 12802 ms in this test case in T2 (note: this gives a total of 12.8s for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall send one Event A6 triggered measurement report, with a measurement reporting delay less than 3202mss from the beginning of time period T4.

The overall delays measured for Event A6 in T4 is defined as the time from the beginning of time period T4, to the moment the UE send one Event A6 triggered measurement report to Cell 4.

The overall delays measured test requirement is expressed as:

- Overall delays measured = measurement reporting delay + TTI insertion uncertainty
- Measurement reporting delay = $T_{measure_scc}$ = 5 *measCycleSCell* = 3200ms
- TTI insertion uncertainty = $TTI_{DCCH}$ = 1 ms; $2xTTI_{DCCH}$ = 2 ms

The overall delays measured shall be less than a total of 3202 ms in this test case in T4 (note: this gives a total of 3.2s for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The UE shall be scheduled on PCell continuously throughout the test. From the start of T1 until the measurement report is received during T2, at least 99.5% of all expected ACK/NACKs shall be transmitted by the UE.

The UE shall be scheduled on Cell2 continuously from the beginning of T4 to the end. From the start of T4 until the measurement report is received during T4, at least 99.5% of all expected ACK/NACKs shall be transmitted by the UE.

For a test to be considered successful requirements on both event detection and percentage of transmitted ACK/NACKs have to be fulfilled simultaneously.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

## 8.16.32 E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in TDD

### 8.16.32.1 Test purpose

To verify the UE's ability to make a correct reporting of event A6 on deactivated SCell with PCell and SCell interruptions in non-DRX and with PCell in FDD within the requirements in section 8.3.3.2.1 while at the same time fulfilling the requirements on interruption rate.

### 8.16.32.2 Test applicability

This test case applies to all types of UE release 12 and forward supporting E-UTRA FDD and TDD and 3DL CA with TDD as PCell.

### 8.16.32.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD or TDD cell on a secondary component carrier within $T_{identify_scc}$, according to the parameter *measCycleSCell* where $T_{identify_scc}$ = 20 *measCycleSCell*

A cell shall be considered detectable when

- RSRP related side condition given in Clause 9.1 are fulfilled for a corresponding Band,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex B.2.7 for a corresponding Band

The measurement period for deactivated scell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell* where $T_{measure_scc}$ = 5 *measCycleSCell*. The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified cells on a secondary component carrier, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_scc}$.

The measurement accuracy for all measured cells shall be as specified in the TS 36.133[4] sub-clause 9.1.11 (Carrier aggregation measurement accuracy).

A UE may reconfigure the receiver bandwidth or turn on/off one of the RF chains when performing measurements on one or two SCCs with deactivated SCell. This may cause interruptions on PCell or an activated SCell or both that are specified in Section 7.8.

Reported measurements contained in event triggered measurement reports shall meet the requirements in clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which is caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc}$ defined in TS 36.133[4] Clause 8.3.3.2.1. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_scc}$ defined in TS 36.133[4] clause 8.3.3.2.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_scc}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.3.3.2.1 and A.8.16.32.

### 8.16.32.4 Test description

#### 8.16.32.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.16.32.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1 [10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.65 as appropriate.
2. The general test parameter settings are set up according to Table 8.16.32.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.32.4.3.

5. There are three E-UTRA carriers and four cells: Cell1, Cell2, Cell3 and Cell4. Cell 1 is PCell on the TDD primary component (RF Channel 1), Cell 2 is the SCell on the FDD Secondary Component (RF Channel 2), and Cell 3 is SCell on the FDD secondary component (RF Channel 3) and Cell 4 is the neighbour cell on the FDD secondary component (RF Channel 3). PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2, Cell 3 and Cell 4 shall be powered OFF.

**Table 8.16.32.4.1-1: General test parameters for E-UTRAN TDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2, 3 | Three radio channels are used for this test |
| Active PCell | | | Cell 1 | Primary cell on RF channel number 1. |
| Configured SCell | | | Cell 2 | Configured secondary cell on RF channel number 2. |
| Configured SCell | | | Cell 3 | Configured secondary cell on RF channel number 3. |
| Neighbour cell | | | Cell 4 | Neighbour cell to be identified on RF channel number 3. |
| CP length | | | Normal | |
| Special subframe configuration on PCell | | | 6 | As specified in table 4.2.1 in TS 36.211[9]. The same configuration applies to all cells. |
| Uplink-downlink configuration on PCell | | | 1 | |
| DRX | | | OFF | Continuous monitoring of primary cell |
| A6 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A6. |
| | Offset | dB | -3 | Offset parameter for evaluation of event A6. Needs to take relative accuracy tolerance in TS 36.133[4] clause 9.1.11.2 into account plus margin. |
| | Report on leave | | False | |
| | Time To Trigger | s | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Cell-individual offset for cells on RF channel number 3 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| SCell measurement cycle | | ms | 640 | |
| T1 | | s | 4 | During this time the UE shall be aware of cells 1, 2 and 3 but not cell 4. |
| T2 | | s | ≤15 | UE should report Event A6 within 12.8s (20×scellMeasCycle) |
| T3 | | s | 4 | During this time the UE shall activate cell 2 |
| T4 | | s | ≤4 | UE should report Event A6 within 3.2s (5×scellMeasCycle) |

## 8.16.32.4.2 Test procedure

The test consists of TDD Cell 1 the PCell on the TDD primary component (RF Channel 1), Cell 2 the SCell on the FDD Secondary Component Carrier (RF Channel 2), and Cell 3 SCell on the FDD secondary component (RF Channel 3) and Cell 4 the neighbour cell on the FDD secondary component (RF Channel 3). In the measurement control information it is indicated to the UE that event-triggered reporting with Event A6 is used. The test consists of four successive time periods, with duration of T1, T2, T3 and T4, respectively. During T1 and T2, both Cell2 and Cell3 are deactivated. During T1 the UE shall not have any information of cell 4. Immediately at beginning of T2 the transmission power of cell 4 is increased to same level as for cell 3, and due to usage of an offset this shall result in reporting of Event A6. At the beginning of T3, the transmission power of Cell 4 is turned off and the test equipment sends a MAC message for activation of Cell 2 to UE but Cell3 remains deactivated. Immediately at beginning of T4 the transmission power of Cell 4 is increased to same level as for Cell 3, and due to usage of an offset this shall result in reporting of Event A6. The UE shall be continuously scheduled in the PCell throughout the whole test and continuously scheduled in Cell2 during T4.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCCs according to Annex C.0 and C.1 for all downlink physical channels.
3. The SS shall configure the SCell Cell 2 and SCell Cell 3 on the SCCs (RF Channel 2 and RF Channel 3) as per TS 36.508 [7] clause 5.2A.4.
4. Set the parameters according to T1 in Table 8.16.32.5-1 as appropriate. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
5. The SS shall transmit an RRCConnectionReconfiguration message with Event A6 configured.
6. The UE shall transmit RRCConnectionReconfigurationComplete message.
7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.16.32.5-1.
8. The UE shall transmit a MeasurementReport message triggered by Event A6. If the overall delays measured from the beginning of time period T2 is less than 12802 ms and at least 99.5% of all expected ACK/NACKS are detected by the system simulator from the start of T1 until the measurement report is received during T2 for PCell then the number of successful tests for event “T2 A6” is increased by one. If the UE fails to report the event 6 within the overall delays measured requirement or less than 99.5% of all expected ACK/NACKS are detected by the system simulator for PCell then the number of failure tests for event “T2 A6” is increased by one.
9. After the SS receives the MeasurementReport message in step 8) or when T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 8.16.32.5-1. The SS activates SCell (Cell 2) by sending the activation MAC-CE (Refer TS 36.321 [13], clauses 5.13, 6.1.3.8).
10. When T3 expires, the SS shall switch the power setting from T3 to T4 as specified in Table 8.16.32.5-1.
11. The UE shall transmit a MeasurementReport message triggered by Event A6. If the overall delays measured from the beginning of time period T4 is less than 3202 ms and at least 99.5% of all expected ACK/NACKS are detected by the system simulator from the start of T4 until the measurement report is received during T4 for both PCell and Cell 2, then the number of successful tests for event “T4 A6” is increased by one. If the UE fails to report the event within the overall delays measured requirement or less than 99.5% of all expected ACK/NACKS are detected by the system simulator for either PCell or SCell, then the number of failure tests for event “T4 A6” is increased by one.
12. After the SS receives the MeasurementReport message in step 11) or when T4 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
13. Set Cell 4 physical cell identity = ((current cell 4 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. If physical cell identity of Cell 4 = physical cell identity of Cell 3 then skip this physical cell identity value for Cell 4.
14. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
15. Repeat step 3-14 until a test verdict has been achieved.

Each of the events “T2 A6”and “T4 A6” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.16.32.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.32.4.3-1: Common Exception messages for E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in TDD**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.4.1-3<br>Table H.4.1-6 |

**Table 8.16.32.4.3-2: *MeasConfig-DEFAULT*: E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in TDD test requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 3 entries | | |
| MeasObjectToAddMod[1] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the PCell (Cell 1) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[2] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the SCell on the SCC1 (Cell 2) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[3] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f3 | f3 is the frequency of the SCell on the SCC2 (Cell 3)and neighbouring cell on the SCC2 (Cell 4) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f3) | | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-A6 | | |
| reportConfig | ReportConfigEUTRA-A6 | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f3 | | |
| reportConfigId | idReportConfig-A6 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 8.16.32.4.3-3: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in TDD test requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1, Cell 2, Cell 3, Cell 4 |
| } | | | |

**Table 8.16.32.4.3-4: MeasurementReport: E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in TDD test requirements**

| Derivation path: TS 36.508 [7] clause 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell::= SEQUENCE { | | PCell | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| MeasResultEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | Neighbour Cell 4 | |
| physCellId | physCellId of Neighbour Cell | | |
| cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | 2 entries | | |
| MeasResultServFreq-r10[1] { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| MeasResultServFreq-r10[2] { | | | |
| servFreqId-r10 | 2 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 3 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.32.4.3-5: *SystemInformationBlockType2*: Additional E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in TDD test requirements**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-1 SystemInformationBlockType2 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfig | Not present | | Cell 2 |
| } | | | |

**Table 8.16.32.4.3-6: *SystemInformationBlockType3*: Additional E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in TDD test requirements**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-2 SystemInformationBlockType3 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType3 ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1, Cell 2, Cell 3. Cell 4 |
| } | | | |

**Table 8.16.32.4.3-7: *RadioResourceConfigCommonSCell-r10*: Additional E-UTRAN FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX test requirements**

| Derivation Path:TS 36.508 [7] clause 7.3.2 Table 7.3.2-2 RadioResourceConfigCommonSCell-r10-DEFAULT exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigCommonSCell-r10 ::= SEQUENCE { | | | |
| nonUL-Configuration-r10 SEQUENCE { | | | |
| mbsfn-SubframeConfigList-r10 | Not present | | Cell2 |
| } | | | |
| } | | | |

## 8.16.32.5 Test requirement

Table 8.16.32.4.1-1 and Table 8.16.32.5-1 define the primary level settings including test tolerances for E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in TDD test.

**Table 8.16.32.5-1: Cell specific test parameters for E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in TDD**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | | Cell 4 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T1 | T2 | T3 | T4 | T1 | T2 | T3 | T4 | T1 | T2 | T3 | T4 |
| E-UTRA RF Channel Number | | 1 | | | | 2 | | | | 3 | | | | 3 | | | |
| $BW_{channel}$ | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | |
| PDSCH parameters: DL Reference Measurement Channel | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | | | | N/A | N/A | N/A | 5MHz: R.7 FDD<br>10MHz: R.3 FDD<br>20MHz: R.6 FDD | N/A | | | | N/A | | | |
| PCFICH/PDCCH/PHICH parameters | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | |
| OCNG Pattern defined in D.1 and D.2 | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | | | | 5MHz: OP.19 FDD; 10MHz :OP.6 FDD; 20MHz : OP.14 FDD | 5MHz: OP.19 FDD; 10MHz :OP.6 FDD; 20MHz : OP.14 FDD | 5MHz: OP.19 FDD; 10MHz :OP.6 FDD; 20MHz : OP.14 FDD | 5MHz: OP.20 FDD; 10MHz :OP.10 FDD; 20MHz : OP.17 FDD | 5MHz: OP.16FDD;<br>10MHz:OP.2 FDD;<br>20MHz: OP.12FDD | | | | 5MHz: OP.16FDD;<br>10MHz:OP.2 FDD;<br>20MHz: OP.12FDD | | | |
| PBCH_RA | dB | 0 | | | | 0 | | | | 0 | | | | 0 | | | |
| PBCH_RB | dB | | | | | | | | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | | | | | | | | |
| OCNG_RA[Note 1] | dB | | | | | | | | | | | | | | | | |
| OCNG_RB[Note 1] | dB | | | | | | | | | | | | | | | | |

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $N_{oc}$ Note 3 | dBm/15 kHz | -101 | | | | -101 | | | | -101 | | | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | -Infinity | 16 | -Infinity | 16 |
| $\hat{E}_s / I_{ot}$ Note 4 | dB | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | -0.11 | 16 | -0.11 | -Infinity | -0.11 | -Infinity | -0.11 |
| RSRP Note 4 | dBm/15 kHz | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -Infinity | -85 | -Infinity | -85 |
| SCH_RP Note 4 | dBm/15 kHz | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -Infinity | -85 | -Infinity | -85 |
| Io Note 4 | dBm/Ch BW | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -54.15 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -54.15 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -54.15 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -54.15 +10log ($N_{RB,c}$ /50) |
| Propagation Condition | | AWGN | | | | AWGN | | | | AWGN | | | | AWGN | | | |
| Antenna Configuration | | 1x2 | | | | 1x2 | | | | 1x2 | | | | 1x2 | | | |
| Timing offset to Cell 1 | μs | - | | | | 0 | | | | 0 | | | | 3 | | | |
| Time alignment error relative to cell 1 Note5 | μs | - | | | | ≤ TAE | | | | ≤ TAE | | | | N/A | | | |
| Time alignment error relative to cell 2 Note5 | μs | - | | | | - | | | | ≤ TAE | | | | N/A | | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE on cell1 prior to the start of time period T2 and on cell2 prior to the start of time period T4.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Es/Iot, RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

The UE shall send one Event A6 triggered measurement report, with a measurement reporting delay less than 12802 ms from the beginning of time period T2.

The overall delays measured for Event A6 in T2 is defined as the time from the beginning of time period T2, to the moment the UE send one Event A6 triggered measurement report to Cell 4.

The overall delays measured test requirement is expressed as:

- Overall delays measured = measurement reporting delay + TTI insertion uncertainty
- Measurement reporting delay = $T_{identify_scc}$ = 20 *measCycleSCell* = 12800ms
- TTI insertion uncertainty = $TTI_{DCCH}$ = 1 ms; $2xTTI_{DCCH}$ = 2 ms

The overall delays measured shall be less than a total of 12802 ms in this test case in T2 (note: this gives a total of 12.8s for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall send one Event A6 triggered measurement report, with a measurement reporting delay less than 3202ms from the beginning of time period T4.

The overall delays measured for Event A6 in T4 is defined as the time from the beginning of time period T4, to the moment the UE send one Event A6 triggered measurement report to Cell 4.

The overall delays measured test requirement is expressed as:

- Overall delays measured = measurement reporting delay + TTI insertion uncertainty
- Measurement reporting delay = $T_{measure_scc}$ = 5 *measCycleSCell* = 3200ms
- TTI insertion uncertainty = $TTI_{DCCH}$ = 1 ms; $2xTTI_{DCCH}$ = 2 ms

The overall delays measured shall be less than a total of 3202 ms in this test case in T4 (note: this gives a total of 3,2s for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The UE shall be scheduled on PCell continuously throughout the test. From the start of T1 until the measurement report is received during T2, at least 99.5% of all expected ACK/NACKs shall be transmitted by the UE.

The UE shall be scheduled on Cell2 continuously from the beginning of T4 to the end. From the start of T4 until the measurement report is received during T4, at least 99.5% of all expected ACK/NACKs shall be transmitted by the UE.

For a test to be considered successful requirements on both event detection and percentage of transmitted ACK/NACKs have to be fulfilled simultaneously.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

## 8.16.33 E-UTRAN FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX

### 8.16.33.1 Test purpose

To verify the UE's ability to make a correct reporting of event A6 on deactivated SCell with PCell and SCell interruptions in non-DRX within the requirements in section 8.3.3.2.1 while at the same time fulfilling the requirements on interruption rate.

### 8.16.33.2 Test applicability

This test case applies to all types of E-UTRA FDD UE release 10 and forward that support 3DL Intra-band contiguous CA or 3DL Inter-band CA, or 3DL with Intra-band contiguous and Inter-band CA.

This test case also applies to all types of E-UTRA FDD UE release 11 and forward that support 3DL with Intra-band non-contiguous and Inter-band CA, or 3DL with Intra-band non-contiguous and Intra-band contiguous CA.

### 8.16.33.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD or TDD cell on a secondary component carrier within $T_{identify_scc}$, according to the parameter *measCycleSCell* where $T_{identify_scc}$ = 20 *measCycleSCell*

A cell shall be considered detectable when

- RSRP related side condition given in Clause 9.1 are fulfilled for a corresponding Band,
- SCH_RP|$_{dBm}$ and SCH Ês/Iot according to Annex B.2.7 for a corresponding Band

The measurement period for deactivated scell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell* where $T_{measure_scc}$ = 5 *measCycleSCell*. The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified cells on a secondary component carrier, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_scc}$.

The measurement accuracy for all measured cells shall be as specified in the TS 36.133[4] sub-clause 9.1.11 (Carrier aggregation measurement accuracy)

A UE may reconfigure the receiver bandwidth or turn on/off one of the RF chains when performing measurements on one or two SCCs with deactivated SCell. This may cause interruptions on PCell or an activated SCell or both that are specified in Section 7.8.

Reported measurements contained in event triggered measurement reports shall meet the requirements in clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which is caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc}$ defined in TS 36.133[4] Clause 8.3.3.2.1. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_scc}$ defined in TS 36.133[4] clause 8.3.3.2.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_scc}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.3.3.2.1 and A.8.16.33.

### 8.16.33.4 Test description

#### 8.16.33.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.16.33.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1 [10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.65 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.33.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.33.4.3.

5. There are three E-UTRA FDD carriers and four cells: Cell1, Cell2, Cell3 and Cell4. Cell 1 is PCell on the FDD primary component (RF Channel 1), Cell 2 is the SCell on the FDD Secondary Component (RF Channel 2), and Cell 3 is SCell on the FDD secondary component (RF Channel 3) and Cell 4 is the neighbour cell on the FDD secondary component (RF Channel 3). PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2, Cell 3 and Cell 4 shall be powered OFF.

**Table 8.16.33.4.1-1: General test parameters for E-UTRAN FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2, 3 | Three radio channels are used for this test |
| Active PCell | | | Cell 1 | Primary cell on RF channel number 1. |
| Configured deactivated SCell | | | Cell 2 | Configured deactivated secondary cell on RF channel number 2. |
| Configured deactivated SCell | | | Cell 3 | Configured deactivated secondary cell on RF channel number 3. |
| Neighbour cell | | | Cell 4 | Neighbour cell to be identified on RF channel number 3. |
| CP length | | | Normal | |
| DRX | | | OFF | Continuous monitoring of primary cell |
| A6 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A6. |
| | Offset | dB | -3 | Offset parameter for evaluation of event A6. Needs to take relative accuracy tolerance in TS 36.133[4] clause 9.1.11.2 into account plus margin. |
| | Report on leave | | False | |
| | Time To Trigger | s | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Cell-individual offset for cells on RF channel number 3 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| SCell measurement cycle | | ms | 640 | |
| T1 | | s | 4 | During this time the UE shall be aware of cells 1, 2 and 3 but not cell 4. |
| T2 | | s | ≤15 | UE should report Event A6 within 12.8s (20×scellMeasCycle) |
| T3 | | s | 4 | During this time the UE shall activate cell 2 |
| T4 | | s | ≤4 | UE should report Event A6 within 63.2s (5×scellMeasCycle) |

## 8.16.33.4.2 Test procedure

The test consists of Cell 1 the PCell, Cell 2 the SCell on the Secondary Component Carrier (RF Channel 2), and Cell 3 SCell on the FDD secondary component (RF Channel 3) and Cell 4 the neighbour cell on the FDD secondary component (RF Channel 3). In the measurement control information it is indicated to the UE that event-triggered reporting with Event A6 is used. The test consists of four successive time periods, with duration of T1, T2, T3 and T4, respectively. During T1 and T2, both Cell2 and Cell3 are deactivated. During T1 the UE shall not have any information of cell 4. Immediately at beginning of T2 the transmission power of cell 4 is increased to same level as for cell 3, and due to usage of an offset this shall result in reporting of Event A6. At the beginning of T3, the transmission power of Cell 4 is turned off and the test equipment sends a MAC message for activation of Cell 2 to UE but Cell3 remains deactivated. Immediately at beginning of T4 the transmission power of Cell 4 is increased to same level as for Cell 3, and due to usage of an offset this shall result in reporting of Event A6. The UE shall be continuously scheduled in the PCell throughout the whole test and continuously scheduled in Cell2 during T4.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCCs according to Annex C.0 and C.1 for all downlink physical channels.
3. The SS shall configure the SCell Cell 2 and SCell Cell 3 on the SCCs (RF Channel 2 and RF Channel 3) as per TS 36.508 [7] clause 5.2A.4.
4. Set the parameters according to T1 in Table 8.16.33.5-1 as appropriate. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
5. The SS shall transmit an RRCConnectionReconfiguration message with Event A6 configured.
6. The UE shall transmit RRCConnectionReconfigurationComplete message.
7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.16.33.5-1.
8. The UE shall transmit a MeasurementReport message triggered by Event A6. If the overall delays measured from the beginning of time period T2 is less than 12802 ms and at least 99.5% of all expected ACK/NACKS are detected by the system simulator from the start of T1 until the measurement report is received during T2 for PCell then the number of successful tests for event “T2 A6” is increased by one. If the UE fails to report the event 6 within the overall delays measured requirement or less than 99.5% of all expected ACK/NACKS are detected by the system simulator for PCell then the number of failure tests for event “T2 A6” is increased by one.
9. After the SS receives the MeasurementReport message in step 8) or when T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 8.16.33.5-1. The SS activates SCell (Cell 2) by sending the activation MAC-CE (Refer TS 36.321 [13], clauses 5.13, 6.1.3.8).
10. When T3 expires, the SS shall switch the power setting from T3 to T4 as specified in Table 8.16.33.5-1.
11. The UE shall transmit a MeasurementReport message triggered by Event A6. If the overall delays measured from the beginning of time period T4 is less than 3202 ms and at least 99.5% of all expected ACK/NACKS are detected by the system simulator from the start of T4 until the measurement report is received during T4 for both PCell and SCell, then the number of successful tests for event “T4 A6” is increased by one. If the UE fails to report the event within the overall delays measured requirement or less than 99.5% of all expected ACK/NACKS are detected by the system simulator for either PCell or SCell then the number of failure tests for event “T4 A6” is increased by one.
12. After the SS receives the MeasurementReport message in step 11) or when T4 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
13. Set Cell 4 physical cell identity = ((current cell 4 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. If physical cell identity of Cell 4 = physical cell identity of Cell 3 then skip this physical cell identity value for Cell 4.
14. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
15. Repeat step 3-14 until a test verdict has been achieved.

Each of the events “T2 A6”and “T4 A6” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.16.33.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.33.4.3-1: Common Exception messages for E-UTRAN FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.4.1-3<br>Table H.4.1-6 |

**Table 8.16.33.4.3-2: *MeasConfig-DEFAULT*: E-UTRAN FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 3 entries | | |
| MeasObjectToAddMod[1] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the PCell (Cell 1) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[2] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the SCell on the SCC1 (Cell 2) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[3] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f3 | f3 is the frequency of the SCell on the SCC2 (Cell 3)and neighbouring cell on the SCC2 (Cell 4) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f3) | | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-A6 | | |
| reportConfig | ReportConfigEUTRA-A6 | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f3 | | |
| reportConfigId | idReportConfig-A6 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 8.16.33.4.3-3: *MeasObjectEUTRA-GENERIC*: Additional E-UTRAN FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX test requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1, Cell 2, Cell 3, Cell 4 |
| } | | | |

**Table 8.16.33.4.3-4: MeasurementReport: E-UTRAN FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX**

| Derivation path: TS 36.508 [7] clause 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell::= SEQUENCE { | | PCell | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| MeasResultEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | Neighbour Cell 4 | |
| physCellId | physCellId of Neighbour Cell | | |
| cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | 2 entries | | |
| MeasResultServFreq-r10[1] { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| MeasResultServFreq-r10[2] { | | | |
| servFreqId-r10 | 2 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 3 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.33.4.3-5: *SystemInformationBlockType2*: Additional E-UTRAN FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX test requirements**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-1 SystemInformationBlockType2 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfig | Not present | | Cell 1, Cell 2 |
| } | | | |

**Table 8.16.33.4.3-6: *SystemInformationBlockType3*: Additional E-UTRAN FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX test requirements**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-2 SystemInformationBlockType3 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType3 ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1, Cell 2, Cell 3. Cell 4 |
| } | | | |

**Table 8.16.33.4.3-7: *SystemInformationBlockType5*: Additional E-UTRAN FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX test requirements**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-3 SystemInformationBlockType5 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType5 ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1, Cell 2, Cell 3, Cell 4 |
| } | | | |

**Table 8.16.33.4.3-8: *RadioResourceConfigCommonSCell-r10*: Additional E-UTRAN FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX test requirements**

| Derivation Path:TS 36.508 [7] clause 7.3.2 Table 7.3.2-2 RadioResourceConfigCommonSCell-r10-DEFAULT exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigCommonSCell-r10 ::= SEQUENCE { | | | |
| nonUL-Configuration-r10 SEQUENCE { | | | |
| mbsfn-SubframeConfigList-r10 | Not present | | Cell2 |
| } | | | |
| } | | | |

## 8.16.33.5 Test requirement

Table 8.16.33.4.1-1 and Table 8.16.33.5-1 define the primary level settings including test tolerances for E-UTRAN FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX test.

**Table 8.16.33.5-1: Cell specific test parameters for E-UTRAN FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | | Cell 4 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T1 | T2 | T3 | T4 | T1 | T2 | T3 | T4 | T1 | T2 | T3 | T4 |
| E-UTRA RF Channel Number | | 1 | | | | 2 | | | | 3 | | | | 3 | | | |
| $BW_{channel}$ | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | |
| PDSCH parameters: DL Reference Measurement Channel | | 5MHz: R.7 FDD<br>10MHz: R.3 FDD<br>20MHz: R.6 FDD | | | | N/A | N/A | N/A | 5MHz: R.7 FDD<br>10MHz: R.3 FDD<br>20MHz: R.6 FDD | N/A | | | | N/A | | | |
| PCFICH/PDCCH/PHICH parameters | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | |
| OCNG Pattern defined in D.1 | | 5MHz: OP.20 FDD<br>10MHz: OP.10 FDD<br>20MHz: OP.17 FDD | | | | 5MHz: OP.19 FDD<br>10MHz: OP.6 FDD<br>20MHz: OP.14 FDD | 5MHz: OP.19 FDD<br>10MHz: OP.6 FDD<br>20MHz: OP.14 FDD | 5MHz: OP.19 FDD<br>10MHz: OP.6 FDD<br>20MHz: OP.14 FDD | 5MHz: OP.20 FDD<br>10MHz: OP.10 FDD<br>20MHz: OP.17 FDD | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | | |
| PBCH_RA | dB | 0 | | | | 0 | | | | 0 | | | | 0 | | | |
| PBCH_RB | dB | | | | | | | | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | | | | | | | | |
| OCNG_RA[Note 1] | dB | | | | | | | | | | | | | | | | |
| OCNG_RB[Note 1] | dB | | | | | | | | | | | | | | | | |

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $N_{oc}$ Note 3 | dBm/15 kHz | -101 | | | | -101 | | | | -101 | | | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | - Infinity | 16 | - Infinity | 16 |
| $\hat{E}_s / I_{ot}$ Note 4 | dB | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | -0.11 | 16 | -0.11 | - Infinity | -0.11 | - Infinity | -0.11 |
| RSRP Note 4 | dBm/15 kHz | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | - Infinity | -85 | - Infinity | -85 |
| SCH_RP Note 4 | dBm/15 kHz | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | - Infinity | -85 | - Infinity | -85 |
| Io Note 4 | dBm/Ch BW | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -54.15 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -54.15 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -54.15 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -54.15 +10log ($N_{RB,c}$ /50) |
| Propagation Condition | | AWGN | | | | AWGN | | | | AWGN | | | | AWGN | | | |
| Antenna Configuration | | 1x2 | | | | 1x2 | | | | 1x2 | | | | 1x2 | | | |
| Timing offset to Cell 1 | μs | - | | | | 0 | | | | 0 | | | | 3 | | | |
| Time alignment error relative to cell 1 Note 5 | μs | - | | | | ≤ TAE | | | | ≤ TAE | | | | N/A | | | |
| Time alignment error relative to cell 2 Note 5 | μs | - | | | | - | | | | ≤ TAE | | | | N/A | | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE on cell1 from the start of time period T2 and on cell2 from the start of time period T4.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Es/Iot, RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

The UE shall send one Event A6 triggered measurement report, with a measurement reporting delay less than 12802 ms from the beginning of time period T2.

The overall delays measured for Event A6 in T2 is defined as the time from the beginning of time period T2, to the moment the UE send one Event A6 triggered measurement report to Cell 4.

The overall delays measured test requirement is expressed as:

- Overall delays measured = measurement reporting delay + TTI insertion uncertainty
- Measurement reporting delay = $T_{identify_scc}$ = 20 *measCycleSCell* = 12800ms
- TTI insertion uncertainty = $TTI_{DCCH}$ = 1 ms; $2xTTI_{DCCH}$ = 2 ms

The overall delays measured shall be less than a total of 12802 ms in this test case (note: this gives a total of 12.8s for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall send one Event A6 triggered measurement report, with a measurement reporting delay less than 3202ms from the beginning of time period T4.

The overall delays measured for Event A6 in T4 is defined as the time from the beginning of time period T4, to the moment the UE send one Event A6 triggered measurement report to Cell 4.

The overall delays measured test requirement is expressed as:

- Overall delays measured = measurement reporting delay + TTI insertion uncertainty
- Measurement reporting delay = $T_{measure_scc}$ = 5 *measCycleSCell* = 3200ms
- TTI insertion uncertainty = $TTI_{DCCH}$ = 1 ms; $2xTTI_{DCCH}$ = 2 ms

The overall delays measured shall be less than a total of 3202 ms in this test case (note: this gives a total of 3.2s for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The UE shall be scheduled on PCell continuously throughout the test. From the start of T1 until the measurement report is received during T2, at least 99.5% of all expected ACK/NACKs shall be transmitted by the UE.

The UE shall be scheduled on Cell2 continuously from the beginning of T4 to the end. From the start of T4 until the measurement report is received during T4, at least 99.5% of all expected ACK/NACKs shall be transmitted by the UE.

For a test to be considered successful requirements on both event detection and percentage of transmitted ACK/NACKs have to be fulfilled simultaneously.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

## 8.16.34 E-UTRAN TDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX

### 8.16.34.1 Test purpose

To verify the UE's ability to make a correct reporting of event A6 on deactivated SCell with PCell and SCell interruptions in non-DRX within the requirements in section 8.3.3.2.1 while at the same time fulfilling the requirements on interruption rate.

### 8.16.34.2 Test applicability

This test case applies to all types of E-UTRA TDD UE release 10 and forward that support 3DL Intra-band contiguous CA or 3DL Inter-band CA, or 3DL with Intra-band contiguous and Inter-band CA.

This test case also applies to all types of E-UTRA TDD UE release 11 and forward that support 3DL with Intra-band non-contiguous and Inter-band CA, or 3DL with Intra-band non-contiguous and Intra-band contiguous CA.

### 8.16.34.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD or TDD cell on a secondary component carrier within $T_{identify_scc}$, according to the parameter *measCycleSCell* where $T_{identify_scc}$ = 20 *measCycleSCell*

A cell shall be considered detectable when

- RSRP related side condition given in Clause 9.1 are fulfilled for a corresponding Band,
- SCH_RP|$_{dBm}$ and SCH Ês/Iot according to Annex B.2.7 for a corresponding Band

The measurement period for deactivated scell measurements is $T_{measure_scc}$ according to the parameter *measCycleSCell* where $T_{measure_scc}$ = 5 *measCycleSCell*. The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified cells on a secondary component carrier, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_scc}$.

The measurement accuracy for all measured cells shall be as specified in the TS 36.133[4] sub-clause 9.1.11 (Carrier aggregation measurement accuracy)

A UE may reconfigure the receiver bandwidth or turn on/off one of the RF chains when performing measurements on one or two SCCs with deactivated SCell. This may cause interruptions on PCell or an activated SCell or both that are specified in Section 7.8.

Reported measurements contained in event triggered measurement reports shall meet the requirements in clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which is caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc}$ defined in TS 36.133[4] Clause 8.3.3.2.1. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_scc}$ defined in TS 36.133[4] clause 8.3.3.2.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_scc}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.3.3.2.1 and A.8.16.34.

### 8.16.34.4 Test description

#### 8.16.34.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.16.34.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1 [10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.65 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.34.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.34.4.3.

5. There are three E-UTRA TDD carriers and four cells: Cell1, Cell2, Cell3 and Cell4. Cell 1 is PCell on the TDD primary component (RF Channel 1), Cell 2 is the SCell on the TDD Secondary Component (RF Channel 2), and Cell 3 is SCell on the TDD secondary component (RF Channel 3) and Cell 4 is the neighbour cell on the TDD secondary component (RF Channel 3). PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2, Cell 3 and Cell 4 shall be powered OFF.

**Table 8.16.34.4.1-1: General test parameters for E-UTRAN TDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2, 3 | three radio channels are used for this test |
| Active PCell | | | Cell 1 | Primary cell on RF channel number 1. |
| Configured deactivated SCell | | | Cell 2 | Configured deactivated secondary cell on RF channel number 2. |
| Configured deactivated SCell | | | Cell 3 | Configured deactivated secondary cell on RF channel number 3. |
| Neighbour cell | | | Cell 4 | Neighbour cell to be identified on RF channel number 3. |
| CP length | | | Normal | |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211[9]. The same configuration in both cells |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211[9]. The same configuration in both cells |
| DRX | | | OFF | Continuous monitoring of primary cell |
| A6 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A6. |
| | Offset | dB | -3 | Offset parameter for evaluation of event A6. Needs to take relative accuracy tolerance in TS 36.133[4] clause 9.1.11.2 into account plus margin. |
| | Report on leave | | False | |
| | Time To Trigger | s | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Cell-individual offset for cells on RF channel number 3 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| SCell measurement cycle | | ms | 640 | |
| T1 | | s | 4 | During this time the UE shall be aware of cells 1, 2 and 3 but not cell 4. |
| T2 | | s | ≤15 | UE should report Event A6 within 12.8s (20×scellMeasCycle) |
| T3 | | s | 4 | During this time the UE shall activate cell 2 |
| T4 | | s | ≤4 | UE should report Event A6 within 3.2s (5×scellMeasCycle) |

## 8.16.34.4.2 Test procedure

The test consists of Cell 1 the PCell, Cell 2 the SCell on the Secondary Component Carrier (RF Channel 2), and Cell 3 SCell on the TDD secondary component (RF Channel 3) and Cell 4 the neighbour cell on the TDD secondary component (RF Channel 3). In the measurement control information it is indicated to the UE that event-triggered reporting with Event A6 is used. The test consists of four successive time periods, with duration of T1, T2, T3 and T4, respectively. During T1 and T2, both Cell2 and Cell3 are deactivated. During T1 the UE shall not have any information of cell 4. Immediately at beginning of T2 the transmission power of cell 4 is increased to same level as for cell 3, and due to usage of an offset this shall result in reporting of Event A6. At the beginning of T3, the transmission power of Cell 4 is turned off and the test equipment sends a MAC message for activation of Cell 2 to UE but Cell3 remains

deactivated. Immediately at beginning of T4 the transmission power of Cell 4 is increased to same level as for Cell 3, and due to usage of an offset this shall result in reporting of Event A6. The UE shall be continuously scheduled in the PCell throughout the whole test and continuously scheduled in Cell2 during T4.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCCs according to Annex C.0 and C.1 for all downlink physical channels.
3. The SS shall configure the SCell Cell 2 and SCell Cell 3 on the SCCs (RF Channel 2 and RF Channel 3) as per TS 36.508 [7] clause 5.2A.4.
4. Set the parameters according to T1 in Table 8.16.34.5-1 as appropriate. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts.
5. The SS shall transmit an RRCConnectionReconfiguration message with Event A6 configured.
6. The UE shall transmit RRCConnectionReconfigurationComplete message.
7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.16.34.5-1.
8. The UE shall transmit a MeasurementReport message triggered by Event A6. If the overall delays measured from the beginning of time period T2 is less than 12802 ms and at least 99.5% of all expected ACK/NACKS are detected by the system simulator from the start of T1 until the measurement report is received during T2 for PCell then the number of successful tests for event “T2 A6” is increased by one. If the UE fails to report the event 6 within the overall delays measured requirement or less than 99.5% of all expected ACK/NACKS are detected by the system simulator for PCell then the number of failure tests for event “T2 A6” is increased by one.
9. After the SS receives the MeasurementReport message in step 8) or when T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 8.16.34.5-1. The SS activates SCell (Cell 2) by sending the activation MAC-CE (Refer TS 36.321 [13], clauses 5.13, 6.1.3.8)..
10. When T3 expires, the SS shall switch the power setting from T3 to T4 as specified in Table 8.16.34.5-1.
11. The UE shall transmit a MeasurementReport message triggered by Event A6. If the overall delays measured from the beginning of time period T4 is less than 3202 ms and at least 99.5% of all expected ACK/NACKS are detected by the system simulator from the start of T4 until the measurement report is received during T4 for both PCell and SCell, then the number of successful tests for event “T4 A6” is increased by one. If the UE fails to report the event within the overall delays measured requirement or less than 99.5% of all expected ACK/NACKS are detected by the system simulator for either PCell or SCell then the number of failure tests for event “T4 A6” is increased by one.
12. After the SS receives the MeasurementReport message in step 11) or when T4 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
13. Set Cell 4 physical cell identity = ((current cell 4 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. If physical cell identity of Cell 4 = physical cell identity of Cell 3 then skip this physical cell identity value for Cell 4.
14. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
15. Repeat step 3-14 until a test verdict has been achieved.

Each of the events “T2 A6”and “T4 A6” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.16.34.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.34.4.3-1: Common Exception messages for E-UTRAN TDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.4.1-3<br>Table H.4.1-6 |

**Table 8.16.34.4.3-2: *MeasConfig-DEFAULT*: E-UTRAN TDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 3 entries | | |
| MeasObjectToAddMod[1] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the PCell (Cell 1) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[2] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the SCell on the SCC1 (Cell 2) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[3] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f3 | f3 is the frequency of the SCell on the SCC2 (Cell 3)and neighbouring cell on the SCC2 (Cell 4) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f3) | | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-A6 | | |
| reportConfig | ReportConfigEUTRA-A6 | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f3 | | |
| reportConfigId | idReportConfig-A6 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 8.16.34.4.3-3: MeasurementReport: E-UTRAN TDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX**

| Derivation path: TS 36.508 [7] clause 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell::= SEQUENCE { | | PCell | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| MeasResultEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | Neighbour Cell 4 | |
| physCellId | physCellId of Neighbour Cell | | |
| cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | 2 entries | | |
| MeasResultServFreq-r10[1] { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| MeasResultServFreq-r10[2] { | | | |
| servFreqId-r10 | 2 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 3 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 8.16.34.5 Test requirement

Table 8.16.34.4.1-1 and Table 8.16.34.5-1 define the primary level settings including test tolerances for E-UTRAN TDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX test.

**Table 8.16.34.5-1: Cell specific test parameters for E-UTRAN TDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell interruptions in Non-DRX**

| Parameter | Unit | Cell 1 | | | | Cell 2 | | | | Cell 3 | | | | Cell 4 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T4 | T1 | T2 | T3 | T4 | T1 | T2 | T3 | T4 | T1 | T2 | T3 | T4 |
| E-UTRA RF Channel Number | | 1 | | | | 2 | | | | 3 | | | | 3 | | | |
| $BW_{channel}$ | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | |
| PDSCH parameters: DL Reference Measurement Channel | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | | | | N/A | N/A | N/A | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | N/A | | | | N/A | | | |
| PCFICH/PDCCH/PHICH parameters: DL Reference Measurement Channel | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | |
| OCNG Pattern defined in D.2 | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | | |
| PBCH_RA | dB | 0 | | | | 0 | | | | 0 | | | | 0 | | | |
| PBCH_RB | dB | | | | | | | | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | | | | | | | | |
| OCNG_RA[Note 1] | dB | | | | | | | | | | | | | | | | |

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| OCNG_RB Note 1 | dB | | | | | | | | | | | | | | | | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -101 | | | | -101 | | | | -101 | | | | | | | |
| $\hat{E}_s / N_{oc}$ | dB | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | -Infinity | 16 | -Infinity | 16 |
| $\hat{E}_s / I_{ot}$ Note 4 | dB | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | -0.11 | 16 | -0.11 | -Infinity | -0.11 | -Infinity | -0.11 |
| RSRP Note 4 | dBm/15 kHz | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -Infinity | -85 | -Infinity | -85 |
| SCH_RP Note 4 | dBm/15 kHz | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -Infinity | -85 | -Infinity | -85 |
| Io Note 4 | dBm/Ch BW | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -54.15 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -54.15 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -54.15 +10log ($N_{RB,c}$ /50) | -57.11 +10log ($N_{RB,c}$ /50) | -54.15 +10log ($N_{RB,c}$ /50) |
| Propagation Condition | | AWGN | | | | AWGN | | | | AWGN | | | | AWGN | | | |
| Antenna Configuration | | 1x2 | | | | 1x2 | | | | 1x2 | | | | 1x2 | | | |
| Timing offset to Cell 1 | μs | - | | | | 0 | | | | 0 | | | | 3 | | | |
| Time alignment error relative to cell 1 Note 5 | μs | - | | | | ≤ TAE | | | | ≤ TAE | | | | N/A | | | |
| Time alignment error relative to cell 2 Note 5 | μs | - | | | | - | | | | ≤ TAE | | | | N/A | | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE on cell1 prior to the start of time period T2 and on cell2 prior to the start of time period T4.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Es/Iot, RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

The UE shall send one Event A6 triggered measurement report, with a measurement reporting delay less than 12802 ms from the beginning of time period T2.

The overall delays measured for Event A6 in T2 is defined as the time from the beginning of time period T2, to the moment the UE send one Event A6 triggered measurement report to Cell 4.

The overall delays measured test requirement is expressed as:

- Overall delays measured = measurement reporting delay + TTI insertion uncertainty
- Measurement reporting delay = $T_{identify_scc}$ = 20 *measCycleSCell* = 12800ms
- TTI insertion uncertainty = $TTI_{DCCH}$ = 1 ms; 2x$TTI_{DCCH}$ = 2 ms

The overall delays measured shall be less than a total of 12802 ms in this test case in T2 (note: this gives a total of 12.8s for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall send one Event A6 triggered measurement report, with a measurement reporting delay less than 3202ms from the beginning of time period T4.

The overall delays measured for Event A6 in T4 is defined as the time from the beginning of time period T4, to the moment the UE send one Event A6 triggered measurement report to Cell 4.

The overall delays measured test requirement is expressed as:

- Overall delays measured = measurement reporting delay + TTI insertion uncertainty
- Measurement reporting delay = $T_{measure_scc}$ = 5 *measCycleSCell* = 3200ms
- TTI insertion uncertainty = $TTI_{DCCH}$ = 1 ms; 2x$TTI_{DCCH}$ = 2 ms

The overall delays measured shall be less than a total of 3202 ms in this test case in T4 (note: this gives a total of 3.2s for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The UE shall be scheduled on PCell continuously throughout the test. From the start of T1 until the measurement report is received during T2, at least 99.5% of all expected ACK/NACKs shall be transmitted by the UE.

The UE shall be scheduled on Cell2 continuously from the beginning of T4 to the end. From the start of T4 until the measurement report is received during T4, at least 99.5% of all expected ACK/NACKs shall be transmitted by the UE.

For a test to be considered successful requirements on both event detection and percentage of transmitted ACK/NACKs have to be fulfilled simultaneously.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

NOTE: The actual overall delays measured in the test may be up to 2x$TTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

## 8.16.35 3 DL PCell in FDD CA Activation and Deactivation of Known SCell in Non-DRX

### 8.16.35.1 Test purpose

To verify that SCell activation and deactivation times are within the requirements in TS 36.133 [4] section 7.7 for UE configured with two downlink SCells, when the SCells are known by the UE at the time of activation and PCell is in FDD.

### 8.16.35.2 Test applicability

This test applies to all types of E-UTRA UE release 12 and forward that support E-UTRA FDD and TDD and 3DL CA with FDD as PCell. Applicability requires support for FGI bit 25.

### 8.16.35.3 Minimum conformance requirements

While activating a SCell if the UE does not receive any command to activate, deactivate, configure or deconfigure any other SCell during the SCell activation delay then the UE shall meet the SCell activation delay requirements specified in TS 36.133 [4] section 7.7.2.

While activating a SCell if any other SCell is activated, deactivated, configured or deconfigured by the UE then the UE shall meet the SCell activation delay requirements ($T_{activate_total}$) according to the following expression:

$$T_{activate_total} = T_{activate_basic} + 5 \times \sum_{i=1}^{N-1} K_i$$

Where:

$T_{activate_total}$ is the total time to activate a SCell and is expressed in subframes.

$T_{activate_basic}$ is the SCell activation delay specified in section 7.7.2;

$K_i$ ($0 \leq K_i \leq [3]$) is the number of times the other $i^{th}$ SCell is activated, deactivated, configured or deconfigured while the SCell is being activated.

N ($2 \leq N \leq 3$) is the maximum number of SCells supported by the UE.

While activating a SCell:

- The interruption on the PCell and/or on the activated SCell due to the SCell activation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+11 if:
  - the PCell and/or the activated SCell being interrupted and the SCell being activated belong to E-UTRA TDD, or
  - the activated SCell being interrupted and the SCell being activated belong to E-UTRA FDD and the PCell belongs to E-UTRA TDD.
- Otherwise, the interruption on PCell and/or on the activated SCell due to the SCell activation specified in TS36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+9.

Starting from the subframe specified in section 4.3 of TS 36.213 [8] and until the UE has completed the SCell activation, the UE shall report CQI index = 0 (out of range) if the UE has available uplink resources to report CQI for an SCell.

While deactivating a SCell:

- The interruption on the PCell and/or on the activated SCell due to the SCell deactivation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+11 if:
  - The PCell and/or the activated SCell being interrupted and the SCell being deactivated belong to E-UTRA TDD or
  - The activated SCell being interrupted and the SCell being deactivated belong to E-UTRA FDD and the PCell belongs to E-UTRA TDD.
- Otherwise, the interruption on PCell and/or the activated SCell due to the SCell deactivation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+9.

The normative reference for this requirement is TS 36.133 [4] clause 7.7 and A.8.16.35.

### 8.16.35.4 Test description

#### 8.16.35.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.16.35.5-1 as defined in TS 36.508 [7] clause 4.3.1, for different CA configurations as defined in TS 36.521-1 [10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.63 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.35.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.35.4.3.

5. Cell 1 is the PCell on the primary component carrier, Cell 2 is the SCell on RF channel number 2, and Cell 3 is the deactivated SCell on RF channel number 3. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.35.4.1-1: General test parameters for 3 DL PCell in FDD CA activation and deactivation of known SCell in non-DRX**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1, 2, 3 | Three radio channels are used for this test |
| Active PCell | | Cell 1 | Primary cell on RF channel number 1. |
| Deconfigured deactivated SCell1 | | Cell 2 | Deconfigured deactivated secondary cell on RF channel number 2. |
| Configured deactivated SCell2 | | Cell 3 | Configured deactivated secondary cell on RF channel number 3. |
| CP length | | Normal | |
| DRX | | OFF | Continuous monitoring of primary cell |
| CQI/PMI periodicity and offset configuration index | | 0 | CQI reporting for SCell every second subframe |
| Cell-individual offset for cells on RF channel number 1 | dB | 0 | Individual offset for cells on PCC. |
| Cell-individual offset for cells on RF channel number 2 | dB | 0 | Individual offset for cells on SCC1. |
| Cell-individual offset for cells on RF channel number 3 | dB | 0 | Individual offset for cells on SCC2. |
| SCell measurement cycle (measCycleSCell) | ms | 320 | |
| T1 | s | 7 | During this time the PCell and SCell2 shall be known and the SCell1 configured, detected and reported |
| T2 | s | 1 | During this time the UE shall activate the SCell1. |
| T3 | s | 1 | During this time the UE shall deactivate the SCell1. |

### 8.16.35.4.2 Test procedure

The test consists of three successive time periods, with duration of T1, T2 and T3, respectively. There are three carriers, each with one cell. All cells have constant signal levels throughout the test.

Before the test starts the UE is connected to Cell 1 (PCell) on radio channel 1 (PCC) and Cell 3 (deactivated SCell2) on radio channel 3 (SCC2), but is not aware of Cell 2 (SCell1) on radio channel 2 (SCC1). The UE is only monitoring the PCC and SCC2. The UE shall be continuously scheduled in the PCell throughout the whole test.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC2 according to Annex C.0, C.1 for all downlink physical channels.

3. The SS shall configure SCell2 (Cell 3) on the SCC2 as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.35.4.3.

4. The SS shall configure transmission of PDSCH with a maximum number of 1 HARQ transmission.

5. Set the parameters according to T1 in Table 8.16.35.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts. The SS shall configure SCell1 (Cell 2) on the SCC1 as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.35.4.3.

6. The SS activates SCell1 by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe # denoted m which is an even number. If the SS receives ACK for MAC-CE sent by the UE, T2 starts in subframe m, and the test proceeds to step 7, otherwise go to step 12.

7. The SS activates SCell2 by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in subframe (m+10).

8. The UE shall start sending CSI reports for SCell1 and the SS shall monitor CSI reports for SCell1 sent from the UE and ACK/NACK sent in PCell during SCell activation.

    - If the first CSI report for SCell1 is received by the SS in a subframe (m+8),

        - or (m+10) if the subframe (m+8) was subject to interruption,

    - and CSI report with non-zero CQI index is received by the SS earlier than or equal to subframe (m+29),

        - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+29)

    - and DTX is not observed by the SS anywhere outside the subframes (m+5) to (m+13) and the subframes (m+15) to (m+23) up to the end of T2,

    - and ≤2 non-consecutive DTX subframes are observed by the SS anywhere during the subframes (m+5) to (m+13) and the subframes (m+15) to (m+23) up to the end of T2,

    - Then the number of successes for the event “Activation” is increased by one. Otherwise, count a fail for the event “Activation” and go to step 12.

9. When T2 expires, the SS deactivate SCell1 by sending the deactivation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe # denoted n which is an even number. If the SS receives ACK for MAC-CE sent by the UE, T3 starts in subframe n, and the test proceeds to step 10, otherwise go to step 12.

10. The SS deactivates SCell2 by sending the deactivation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe (n+10).

11. The UE shall stop sending CSI reports for SCell1 and the SS shall monitor CSI reports for SCell1 sent from the UE and ACK/NACK sent in PCell during SCell1 deactivation.

    - If the last CSI report is received by the SS earlier than or equal to subframe (n+8),

    - and DTX is not observed by the SS anywhere outside the subframes (n+5) to (n+13) and the subframes (n+15) to (n+23) up to the end of T3,

    - and ≤2 inconsecutive DTX subframes are observed by the SS anywhere during the subframes (n+5) to (n+13) and the subframes (n+15) to (n+23) up to the end of T3

    - Then the number of successes for the event “Deactivation” is increased by one. Otherwise, count a fail for the event “Deactivation”.

12. When T3 expires, or Activation in step 6 was not acknowledged, or a fail was counted for the event “Activation” in step 8, or Deactivation in step 9 was not acknowledged, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

13. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

14. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

15. Repeat steps 2-14 until a test verdict has been achieved.

Each of the events "Activation" and "Deactivation" is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.16.35.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.35.4.3-1: Common Exception messages for 3 DL PCell in FDD CA Activation and Deactivation of Known SCell in Non-DRX**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | |

**Table 8.16.35.4.3-2: *PhysicalConfigDedicated-DEFAULT:* Additional 3 DL PCell in FDD CA Activation and Deactivation of Known SCell in Non-DRX condition requirements**

| Derivation Path: TS 36.508 [7] clause 4.8.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | CQI-ReportConfig-DEFAULT using condition CQI_PERIODIC | | RBC |
| cqi-ReportConfig-r10 | Not present | | SCell_Add Mod |
| } | | | |

**Table 8.16.35.4.3-3: *CQI-ReportConfig-DEFAULT:* Additional 3 DL PCell in FDD activation and deactivation of known SCell in non-DRX condition requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 158 | | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | NULL | | |
| simultaneousAckNackAndCQI | TRUE | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.35.4.3-4: *PhysicalConfigDedicatedSCell-r10-DEFAULT:* Additional 3 DL PCell in FDD CA Activation and Deactivation of Known SCell in Non-DRX condition requirements**

| Derivation Path: TS 36.508 [4] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicatedSCell-r10-DEFAULT ::= SEQUENCE { | | | |
| ul-Configuration-r10 SEQUENCE { | | | |
| antennaInfoUL-r10 | Not present | | |
| pusch-ConfigDedicatedSCell-r10 | Not present | | |
| uplinkPowerControlDedicatedSCell-r10 | Not present | | |
| cqi-ReportConfigSCell-r10 | CQI-ReportConfigSCell-r10-DEFAULT | | SCell1 |
| soundingRS-UL-ConfigDedicated-r10 | Not present | | |
| soundingRS-UL-ConfigDedicated-v1020 | Not present | | |
| soundingRS-UL-ConfigDedicatedAperiodic-r10 | Not present | | |
| } | | | |
| } | | | |

**Table 8.16.35.4.3-5: *CQI-ReportConfigSCell-r10-DEFAULT:* Additional 3 DL PCell in FDD CA Activation and Deactivation of Known SCell in Non-DRX condition requirements**

| Derivation Path:TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfigSCell-r10 ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic-r10 | Not present | | |
| nomPDSCH-RS-EPRE-Offset-r10 | 0 | | |
| cqi-ReportPeriodicSCell-r10 | | | |
| | *CQI-ReportPeriodic-r10-DEFAULT* | | CQI_PERIODIC |
| pmi-RI-Report-r10 | Not present | | |
| } | | | |

**Table 8.16.35.4.3-6: *CQI-ReportPeriodic-r10-DEFAULT:* Additional 3 DL PCell in FDD CA Activation and Deactivation of Known SCell in Non-DRX condition requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportPeriodic-r10 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex-r10 | 0 | | |
| cqi-PUCCH-ResourceIndexP1-r10 | Not present | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1C in TS 36.213) | Cell2 |
| cqi-FormatIndicatorPeriodic-r10 CHOICE { | | | |
| widebandCQI-r10 SEQUENCE { | | | |
| csi-ReportMode-r10 | Not present | | |
| } | | | |
| ri-ConfigIndex | 484 | (see Table 7.2.2-1B in TS 36.213) | Cell2 |
| simultaneousAckNackAndCQI | TRUE | | |
| cqi-Mask-r9 | Not present | | |
| csi-ConfigIndex-r10 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.35.4.3-7: *MeasObjectEUTRA-GENERIC:* Additional 3 DL PCell in FDD CA Activation and Deactivation of Known SCell in Non-DRX condition requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6 Table 4.6.6-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | Cell 2, Cell 3 |
| } | | | |

**Table 8.16.35.4.3-8: *SystemInformationBlockType2:* Additional 3 DL PCell in FDD CA Activation and Deactivation of Known SCell in Non-DRX condition requirements**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-1 SystemInformationBlockType2 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfig | Not present | | Cell 1 |
| } | | | |

## 8.16.35.5 Test requirement

Table 8.16.35.5-1 defines the primary level settings including test tolerances for 3 DL PCell in FDD CA activation and deactivation of known SCell in non-DRX test.

**Table 8.16.35.5-1: Cell specific test parameters for 3 DL PCell in FDD CA activation and deactivation of known SCell in non-DRX**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | 3 | | |
| $BW_{channel}$ | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | |
| PDSCH parameters: DL Reference Measurement Channel | | 5MHz: R.7 FDD<br>10MHz: R.3 FDD<br>20MHz: R.6 FDD | | | - | | | - | | |
| PCFICH/PDCCH/PHICH parameters: DL Reference Measurement Channel | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | |
| OCNG Patterns | | 5MHz: OP.20 FDD<br>10MHz: OP.10 FDD<br>20MHz: OP.17 FDD | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -104 | | | -104 | | | -104 | | |
| $\hat{E}_s/N_{oc}$ | dB | 17 | | | 17 | | | 17 | | |
| $\hat{E}_s/I_{ot}$ | dB | 17 | | | 17 | | | 17 | | |
| RSRP Note 3 | dBm/15 kHz | -87 | | | -87 | | | -87 | | |
| SCH_RP Note 3 | dBm/15 kHz | -87 | | | -87 | | | -87 | | |
| Io Note 3 | dBm/Ch BW | -59.13 +10log ($N_{RB,c}$ /50) | | | -59.13 +10log ($N_{RB,c}$ /50) | | | -59.13 +10log ($N_{RB,c}$ /50) | | |
| Propagation Condition | | AWGN | | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x2 | | | 1x2 | | | 1x2 | | |
| Timing offset to Cell 1 | µs | - | | | 0 | | | 0 | | |
| Time alignment error relative to cell 1 Note 5 | µs | - | | | ≤ TAE | | | ≤ TAE | | |
| Time alignment error relative to cell 2 Note 5 | µs | - | | | - | | | ≤ TAE | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: Es/Iot, RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The uplink resources for CSI reporting are assigned to the UE prior to the start of time period T2.
Note 5: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

During T2 interruption of PCell during SCells activation shall not happen outside the subframes (m+5) to (m+9) and outside the subframes (m+15) to (m+19).

Figure 8.16.35.5-1 shows the derivation of the Test procedure requirements for DTX during T2, based on the core requirements for interruption.

NOTE: The differences between activation and deactivation are as follows:

- Activation ➔ Deactivation
- First CSI ➔ Last CSI
- Latest valid CSI report ➔ Not exist (no need to check since CSI report was already stopped)

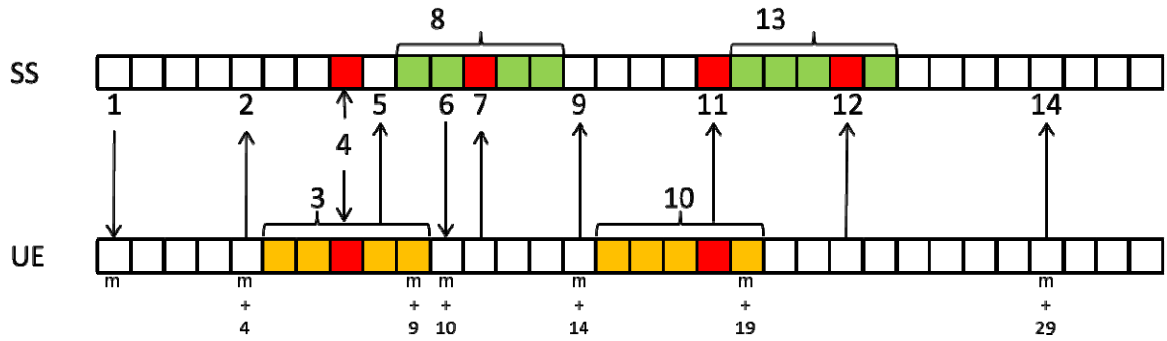


**Figure: 8.16.35.5-1 Procedure derivation for Activation (the case where the band relation between PCC and SCells are inter-band)**

1) Activation command for SCell1 (T2 starts)
2) ACK for MAC-CE for SCell1 activation
3) Possible interruption period by SCell1 activation
4) Allowed interruption timing (1 subframe in m+5~m+9) on UE by SCell1 activation, that is, possible DTX timing on SS
5) First CSI report timing (could be an invalid CQI) at m+8 (might be m+10 due to interruption)
6) Activation command for SCell2
7) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell1 activation
8) Possible DTX reception period on SS due to interruption by SCell1 activation
9) ACK for MAC-CE for SCell2 activation
10) Possible interruption period by SCell2 activation
11) Allowed interruption timing (1 subframe in m+15~m+19) on UE by SCell2 activation, that is, possible DTX timing on SS
12) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell2 activation
13) Possible DTX reception period on SS due to interruption by SCell2 activation
14) Latest valid CSI report timing (valid CQI)

During T3 interruption of PCell during SCells deactivation shall not happen outside the subframes (n+5) to (n+9) and outside the subframes (n+15) to (n+19).

The derivation of the Test procedure requirements for DTX during T3 is based on the core requirements for interruption, using the applicable figure according to the CA configuration, in a similar way to T2.

All of the above test requirements shall be fulfilled in order for the observed SCell1 activation delay and SCell1 deactivation delay to be counted as correct. The rate of correct observed SCell1 activation delay and SCell1 deactivation delay during repeated tests shall be at least 90% with a confidence level of 95% for each of the events.

The statistical pass/ fail decisions are done separately for activation and deactivation.

Decide the test pass, if activation and deactivation are passed, otherwise fail the UE.

## 8.16.36 3 DL PCell in TDD CA Activation and Deactivation of Known SCell in Non-DRX

### 8.16.36.1 Test purpose

To verify that SCell activation and deactivation times are within the requirements in TS 36.133 [4] section 7.7 for UE configured with two downlink SCells, when the SCells are known by the UE at the time of activation and PCell is in TDD.

### 8.16.36.2 Test applicability

This test applies to all types of E-UTRA UE release 12 and forward that support E-UTRA FDD and TDD and 3DL CA with TDD as PCell. Applicability requires support for FGI bit 25.

### 8.16.36.3 Minimum conformance requirements

While activating a SCell if the UE does not receive any command to activate, deactivate, configure or deconfigure any other SCell during the SCell activation delay then the UE shall meet the SCell activation delay requirements specified in TS 36.133 [4] section 7.7.2.

While activating a SCell if any other SCell is activated, deactivated, configured or deconfigured by the UE then the UE shall meet the SCell activation delay requirements ($T_{activate_total}$) according to the following expression:

$$T_{activate_total} = T_{activate_basic} + 5 \times \sum_{i=1}^{N-1} K_i$$

Where:

$T_{activate_total}$ is the total time to activate a SCell and is expressed in subframes.

$T_{activate_basic}$ is the SCell activation delay specified in section 7.7.2;

$K_i$ ($0 \leq K_i \leq [3]$) is the number of times the other $i^{th}$ SCell is activated, deactivated, configured or deconfigured while the SCell is being activated.

N ($2 \leq N \leq 3$) is the maximum number of SCells supported by the UE.

While activating a SCell:

- The interruption on the PCell and/or on the activated SCell due to the SCell activation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+11 if:
  - the PCell and/or the activated SCell being interrupted and the SCell being activated belong to E-UTRA TDD, or
  - the activated SCell being interrupted and the SCell being activated belong to E-UTRA FDD and the PCell belongs to E-UTRA TDD.
- Otherwise, the interruption on PCell and/or on the activated SCell due to the SCell activation specified in TS36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+9.

Starting from the subframe specified in section 4.3 of TS 36.213 [8] and until the UE has completed the SCell activation, the UE shall report CQI index = 0 (out of range) if the UE has available uplink resources to report CQI for an SCell.

While deactivating a SCell:

- The interruption on the PCell and/or on the activated SCell due to the SCell deactivation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+11 if:
  - The PCell and/or the activated SCell being interrupted and the SCell being deactivated belong to E-UTRA TDD or
  - The activated SCell being interrupted and the SCell being deactivated belong to E-UTRA FDD and the PCell belongs to E-UTRA TDD.
- Otherwise, the interruption on PCell and/or the activated SCell due to the SCell deactivation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+9.

The normative reference for this requirement is TS 36.133 [4] clause 7.7 and A.8.16.36.

## 8.16.36.4 Test description

### 8.16.36.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.16.36.5-1 as defined in TS 36.508 [7] clause 4.3.1, for different CA configurations as defined in TS 36.521-1 [10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.63 as appropriate.
2. The general test parameter settings are set up according to Table 8.16.36.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.16.36.4.3.
5. Cell 1 is the PCell on the primary component carrier, Cell 2 is the SCell on RF channel number 2, and Cell 3 is the deactivated SCell on RF channel number 3. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.36.4.1-1: General test parameters for 3 DL PCell in TDD CA activation and deactivation of known SCell in non-DRX**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1, 2, 3 | Three radio channels are used for this test |
| Active PCell | | Cell 1 | Primary cell on RF channel number 1. |
| Deconfigured deactivated SCell1 | | Cell 2 | Deconfigured deactivated secondary cell on RF channel number 2. |
| Configured deactivated SCell2 | | Cell 3 | Configured deactivated secondary cell on RF channel number 3. |
| CP length | | Normal | |
| Special subframe configuration | | 6 | As specified in table 4.2.1 in TS 36.211. The same configuration applies to all cells. |
| Uplink-downlink configuration | | 1 | |
| DRX | | OFF | Continuous monitoring of primary cell |
| CQI/PMI periodicity and offset configuration index | | 0 | CQI reporting for SCell every second subframe |
| Cell-individual offset for cells on RF channel number 1 | dB | 0 | Individual offset for cells on PCC. |
| Cell-individual offset for cells on RF channel number 2 | dB | 0 | Individual offset for cells on SCC1. |
| Cell-individual offset for cells on RF channel number 3 | dB | 0 | Individual offset for cells on SCC2. |
| SCell measurement cycle (measCycleSCell) | ms | 320 | |
| T1 | s | 7 | During this time the PCell and SCell2 shall be known and the SCell1 configured, detected and reported |
| T2 | s | 1 | During this time the UE shall activate the SCell1. |
| T3 | s | 1 | During this time the UE shall deactivate the SCell1. |

### 8.16.36.4.2 Test procedure

The test consists of three successive time periods, with duration of T1, T2 and T3, respectively. There are three carriers, each with one cell. All cells have constant signal levels throughout the test.

Before the test starts the UE is connected to Cell 1 (PCell) on radio channel 1 (PCC) and Cell 3 (deactivated SCell2) on radio channel 3 (SCC2), but is not aware of Cell 2 (SCell1) on radio channel 2 (SCC1). The UE is only monitoring the PCC and SCC2. The UE shall be continuously scheduled in the PCell throughout the whole test.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCC2 according to Annex C.0, C.1 for all downlink physical channels.
3. The SS shall configure SCell (Cell 3) on the SCC2 as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause Tables 8.16.36.4.3-1 to 8.16.36.4.3-6.
4. The SS shall configure transmission of PDSCH with a maximum number of 1 HARQ transmission.
5. Set the parameters according to T1 in Table 8.16.36.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts. The SS shall configure SCell1 (Cell 2) on the SCC1 as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.36.4.3.
6. The SS activates SCell1 by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe # denoted m, where m is 4 or 9. If the SS receives ACK for MAC-CE sent by the UE, T2 starts in subframe m, and the test proceeds to step 7, otherwise go to step 12.
7. The SS activates SCell2 by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in subframe (m+15).

8. The UE shall start sending CSI reports for SCell1 and the SS shall monitor CSI reports for SCell1 sent from the UE and ACK/NACK sent in PCell during SCell activation.

   - If the first CSI report for SCell1 is received by the SS in a subframe (m+8),
     - or (m+9) if the subframe (m+8) was subject to interruption,
     - or in a subframe (m+13) if the subframes (m+8) and (m+9) were subject to interruption when an intra-band SCell is activated.
   - and CSI report with non-zero CQI index is received by the SS earlier than or equal to subframe (m+29),
     - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+29)
   - and DTX is not observed by the SS outside the subframes (m+5), (m+9), (m+20) and (m+24), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+6), (m+13), (m+21) and (m+28), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+7), (m+13), (m+22) and (m+28), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+8) and (m+23), up to the end of T2,
     - Or DTX is not observed by the SS outside the subframes (m+9) and (m+24), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+10), (m+14), (m+25) and (m+29), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+11), (m+18), (m+26) and (m+33), up to the end of T2,
   - then the number of successes for the event “Activation” is increased by one. Otherwise, count a fail for the event “Activation” and go to step 12.

9. When T2 expires, the SS deactivate SCell1 by sending the deactivation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe # denoted n, where n is 4 or 9. If the SS receives ACK for MAC-CE sent by the UE, T3 starts in subframe n, and the test proceeds to step 11, otherwise go to step 12.

10. The SS deactivates SCell2 by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe (n+15).

11. The UE shall stop sending CSI reports for SCell1 and the SS shall monitor CSI reports for SCell1 sent from the UE and ACK/NACK sent in PCell during SCell1 deactivation.

    - If the last CSI report is received by the SS earlier than or equal to subframe (n+8),
    - and DTX is not observed by the SS outside the subframes (n+5), (n+9), (n+20) and (n+24), up to the end of T3,
      - or DTX is not observed by the SS outside the subframes (n+6), (n+13), (n+21) and (n+28), up to the end of T3,
      - or DTX is not observed by the SS outside the subframes (n+7), (n+13), (n+22) and (n+28), up to the end of T3,
      - or DTX is not observed by the SS outside the subframes (n+8) and (n+23), up to the end of T3,
      - or DTX is not observed by the SS outside the subframes (n+9) and (n+24), up to the end of T3,
      - or DTX is not observed by the SS outside the subframes (n+10), (n+14), (n+25) and (n+29), up to the end of T3,
      - or DTX is not observed by the SS outside the subframes (n+11), (n+18), (n+26) and (n+33), up to the end of T3,

- Then the number of successes for the event “Deactivation” is increased by one. Otherwise, count a fail for the event “Deactivation”.

12. When T3 expires, or Activation in step 6 was not acknowledged, or a fail was counted for the event “Activation” in step 8, or Deactivation in step 9 was not acknowledged, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

13. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

14. After the RRC connection release, the SS:

    - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),

    or

    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

15. Repeat steps 2-14 until a test verdict has been achieved.

Each of the events “Activation” and “Deactivation” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.16.36.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.36.4.3-1: Common Exception messages for 3 DL PCell in TDD CA Activation and Deactivation of Known SCell in Non-DRX**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | |

**Table 8.16.36.4.3-2: *PhysicalConfigDedicated-DEFAULT:* Additional 3 DL PCell in TDD CA Activation and Deactivation of Known SCell in Non-DRX condition requirements**

| Derivation Path: TS 36.508 [7] clause 4.8.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | CQI-ReportConfig-DEFAULT using condition CQI_PERIODIC | | RBC |
| cqi-ReportConfig-r10 | Not present | | SCell_AddMod |
| } | | | |

**Table 8.16.38.4.3-3: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 159 | | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | NULL | | |
| simultaneousAckNackAndCQI | TRUE | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.36.4.3-4: *PhysicalConfigDedicatedSCell-r10-DEFAULT:* Additional 3 DL PCell in TDD CA Activation and Deactivation of Known SCell in Non-DRX condition requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicatedSCell-r10-DEFAULT ::= SEQUENCE { | | | |
| ul-Configuration-r10 SEQUENCE { | | | |
| antennaInfoUL-r10 | Not present | | |
| pusch-ConfigDedicatedSCell-r10 | Not present | | |
| uplinkPowerControlDedicatedSCell-r10 | Not present | | |
| cqi-ReportConfigSCell-r10 | CQI-ReportConfigSCell-r10-DEFAULT | | SCell1 |
| soundingRS-UL-ConfigDedicated-r10 | Not present | | |
| soundingRS-UL-ConfigDedicated-v1020 | Not present | | |
| soundingRS-UL-ConfigDedicatedAperiodic-r10 | Not present | | |
| } | | | |
| } | | | |

**Table 8.16.36.4.3-5: *CQI-ReportConfigSCell-r10-DEFAULT:* Additional 3 DL PCell in TDD CA Activation and Deactivation of Known SCell in Non-DRX condition requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfigSCell-r10 ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic-r10 | Not present | | |
| nomPDSCH-RS-EPRE-Offset-r10 | 0 | | |
| cqi-ReportPeriodicSCell-r10 | | | |
| | *CQI-ReportPeriodic-r10-DEFAULT* | | CQI_PERIODIC |
| pmi-RI-Report-r10 | Not present | | |
| } | | | |

**Table 8.16.36.4.3-6: *CQI-ReportPeriodic-r10-DEFAULT:* Additional 3 DL PCell in TDD CA Activation and Deactivation of Known SCell in Non-DRX condition requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportPeriodic-r10 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex-r10 | 0 | | |
| cqi-PUCCH-ResourceIndexP1-r10 | Not present | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | Cell2 |
| cqi-FormatIndicatorPeriodic-r10 CHOICE { | | | |
| widebandCQI-r10 SEQUENCE { | | | |
| csi-ReportMode-r10 | Not present | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | Cell2 |
| simultaneousAckNackAndCQI | TRUE | | |
| cqi-Mask-r9 | Not present | | |
| csi-ConfigIndex-r10 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.36.4.3-7: *MeasObjectEUTRA-GENERIC:* Additional 3 DL PCell in TDD CA Activation and Deactivation of Known SCell in Non-DRX condition requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6 Table 4.6.6-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | Cell 2, Cell 3 |
| } | | | |

## 8.16.36.5 Test requirement

Table 8.16.36.5-1 defines the primary level settings including test tolerances for 3 DL PCell in TDD CA activation and deactivation of known SCell in non-DRX test.

**Table 8.16.36.5-1: Cell specific test parameters for 3 DL PCell in TDD CA activation and deactivation of known SCell in non-DRX**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | 3 | | |
| $BW_{channel}$ | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | |
| PDSCH parameters: DL Reference Measurement Channel | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | | | - | | | - | | |
| PCFICH/PDCCH/PHICH parameters: DL Reference Measurement Channel | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | |
| OCNG Patterns | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -104 | | | -104 | | | -104 | | |
| $\hat{E}_s/N_{oc}$ | dB | 17 | | | 17 | | | 17 | | |
| $\hat{E}_s/I_{ot}$ | dB | 17 | | | 17 | | | 17 | | |
| RSRP Note 3 | dBm/15 kHz | -87 | | | -87 | | | -87 | | |
| SCH_RP Note 3 | dBm/15 kHz | -87 | | | -87 | | | -87 | | |
| Io Note 3 | dBm/Ch BW | -59.13 +10log ($N_{RB,c}$ /50) | | | -59.13 +10log ($N_{RB,c}$ /50) | | | -59.13 +10log ($N_{RB,c}$ /50) | | |
| Propagation Condition | | AWGN | | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x2 | | | 1x2 | | | 1x2 | | |
| Timing offset to Cell 1 | µs | - | | | 0 | | | 0 | | |
| Time alignment error relative to cell 1 Note 5 | µs | - | | | ≤ TAE | | | ≤ TAE | | |
| Time alignment error relative to cell 2 Note 5 | µs | - | | | - | | | ≤ TAE | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: Es/Iot, RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The uplink resources for CSI reporting are assigned to the UE prior to the start of time period T2.
Note 5: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

During T2 interruption of PCell during SCells activation shall not happen outside the subframes (m+5) to (m+11) and outside the subframes (m+20) to (m+26).

Figure 8.16.36.5-1 shows the derivation of the Test procedure requirements for DTX during T2, based on the core requirements for interruption.

NOTE: The differences between activation and deactivation are as follows:

- Activation ➔ Deactivation
- First CSI ➔ Last CSI
- Latest valid CSI report ➔ Not exist (no need to check since CSI report was already stopped)

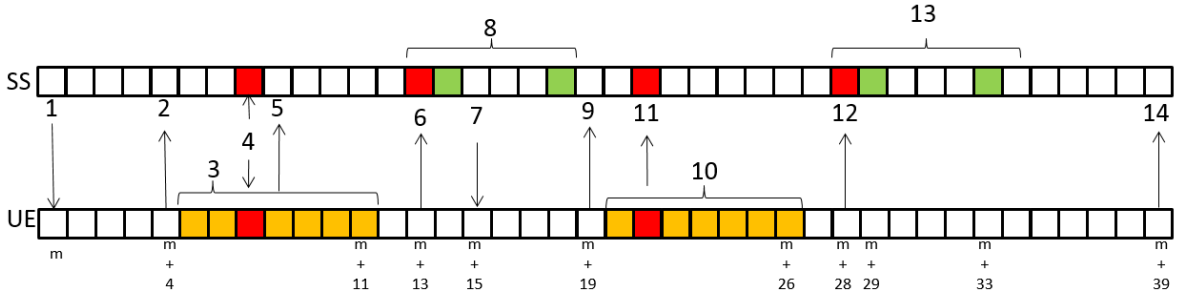


**Figure 8.16.36.5-1: Procedure derivation for Activation (the case where the band relation between PCC and SCells are inter-band)**

1) Activation command for SCell1(T2 starts)
2) ACK for MAC-CE for SCell1 activation
3) Possible interruption period by SCell1 activation
4) Allowed interruption timing (1 subframe in m+5~m+11) on UE by SCell1 activation, that is, possible DTX timing on SS
5) First CSI report timing(invalid CQI) at m+8 (might be m+9 or m+13 due to interruption)
6) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell1 activation
7) Activation command for SCell2
8) Possible DTX reception period on SS due to interruption by SCell1 activation (only m+13, m+14 or m+18 are allowed)
9) ACK for MAC-CE for SCell2 activation
10) Possible interruption period by SCell2 activation
11) Allowed interruption timing(1 subframe in m+20~m+26) on UE by SCell2 activation, that is, possible DTX timing on SS
12) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell2 activation (only m+28, m+29 or m+33 are allowed)
13) Possible DTX reception period on SS due to interruption by SCell2 activation
14) Latest valid CSI report timing(valid CQI)

During T3 interruption of PCell during SCells deactivation shall not happen outside the subframes (n+5) to (n+11) and outside the subframes (n+20) to (n+26).

The derivation of the Test procedure requirements for DTX during T3 is based on the core requirements for interruption, in a similar way to T2.

All of the above test requirements shall be fulfilled in order for the observed SCell1 activation delay and SCell1 deactivation delay to be counted as correct. The rate of correct observed SCell1 activation delay and SCell1 deactivation delay during repeated tests shall be at least 90% with a confidence level of 95% for each of the events.

NOTE: If there are no uplink resources for reporting the valid CSI 29ms into T2 then the UE shall use the next available uplink resource for reporting the corresponding valid CSI.

The statistical pass/ fail decisions are done separately for activation and deactivation.

Decide the test pass, if activation and deactivation are passed, otherwise fail the UE.

# 8.16.37 3DL FDD CA activation and deactivation of known SCell in non-DRX

## 8.16.37.1 Test purpose

To verify that SCell activation and deactivation times are within the requirements in TS 36.133 [4] section 7.7 for 3DL FDD carrier aggregation, when the SCells are known by the UE at the time of activation.

## 8.16.37.2 Test applicability

This test applies to all types of E-UTRA UE FDD release 10 and forward that support 3DL with intra-band contiguous CA or 3DL with inter-band CA, or 3DL with intra-band contiguous and inter-band CA. Applicability requires support for FGI bit 25.

This test case also applies to all types of E-UTRA FDD UE release 11 and forward that support 3DL with intra-band non-contiguous and inter-band CA, or 3DLwith intra-band non-contiguous and intra-band contiguous CA. Applicability requires support for FGI bit 25.8.16.37.3 Minimum conformance requirements

While activating a SCell if the UE does not receive any command to activate, deactivate, configure or deconfigure the other SCell during the SCell activation delay then the UE shall meet the SCell activation delay requirements specified in TS 36.133 [4] section 7.7.2.

While activating a SCell if the other SCell is activated, deactivated, configured or deconfigured by the UE then the UE shall meet the SCell activation delay requirements ($T_{activate_total}$) according to the following expression:

$$T_{activate_total} = T_{activate_basic} + K*5$$

Where:

$T_{activate_total}$ is the total time to activate a SCell and is expressed in subframes.

$T_{activate_basic}$ is the SCell activation delay specified in TS 36.133 [4] section 7.7.2;

K ($1 \leq K \leq [3]$) is the number of times the other SCell is activated, deactivated, configured or deconfigured while the SCell is being activated;

While activating a SCell:

- The interruption on the PCell and/or on the activated SCell due to the SCell activation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+11 if:
  - the PCell and/or the activated SCell being interrupted and the SCell being activated belong to E-UTRA TDD, or
  - the activated SCell being interrupted and the SCell being activated belong to E-UTRA FDD and the PCell belongs to E-UTRA TDD.
- Otherwise, the interruption on PCell and/or on the activated SCell due to the SCell activation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+9.

Starting from the subframe specified in section 4.3 of TS 36.213 [8] and until the UE has completed the SCell activation, the UE shall report CQI index = 0 (out of range) if the UE has available uplink resources to report CQI for an SCell.

The UE shall deactivate a SCell and meet the SCell deactivation delay requirements specified in TS 36.133 [4] section 7.7.3 regardless of whether the other SCell is activated, deactivated, configured or deconfigured or not by the UE during the SCell deactivation delay.

While deactivating a SCell:

- The interruption on the PCell and/or on the activated SCell due to the SCell deactivation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+11 if:
  - the PCell and/or the activated SCell being interrupted and the SCell being deactivated belong to E-UTRA TDD or
  - the activated SCell being interrupted and the SCell being deactivated belong to E-UTRA FDD and the PCell belongs to E-UTRA TDD.
- Otherwise, the interruption on PCell and/or the activated SCell due to the SCell deactivation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+9.

The normative reference for this requirement is TS 36.133 [4] clause 7.7 and A.8.16.37.

## 8.16.37.4 Test description

### 8.16.37.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated CA bandwidth combination supported by UE from Table 8.16.37.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.63 as appropriate.
2. The general test parameter settings are set up according to Table 8.16.37.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.16.37.4.3.
5. Cell 1 is the PCell on the primary component carrier, Cell 2 is the deconfigured SCell1 on RF channel number 2, and Cell 3 is the deactivated SCell2 on RF channel number 3. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.37.4-1: General test parameters for E-UTRAN FDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1, 2, 3 | Three radio channels are used for this test |
| Active PCell | | Cell 1 | Primary cell on RF channel number 1. |
| Deconfigured SCell | | Cell 2 | Deconfigured secondary cell on RF channel number 2. |
| Configured deactivated SCell | | Cell 3 | Configured deactivated secondary cell on RF channel number 3. |
| CP length | | Normal | |
| DRX | | OFF | Continuous monitoring of primary cell |
| CQI/PMI periodicity and offset configuration index | | 0 | CQI reporting for SCells every second subframe |
| Cell-individual offset for cells on RF channel number 1 | dB | 0 | Individual offset for cells on PCC. |
| Cell-individual offset for cells on RF channel number 2 | dB | 0 | Individual offset for cells on SCC1. |
| Cell-individual offset for cells on RF channel number 3 | dB | 0 | Individual offset for cells on SCC2. |
| Filter coefficient | | 0 | L3 filtering is not used |
| SCell measurement cycle (measCycleSCell) | ms | 320 | |
| T1 | s | 7 | During this time the PCell and SCell2 shall be known and the SCell1 configured and detected. |
| T2 | s | 1 | During this time the UE shall activate the SCell1 and SCell2. |
| T3 | s | 1 | During this time the UE shall deactivate the SCell1 and SCell2. |

### 8.16.37.4.2 Test procedure

The test consists of three successive time periods, with duration of T1, T2 and T3, respectively. There are three carriers, each with one cell. All cells have constant signal levels throughout the test.

Before the test starts the UE is connected to Cell 1 (PCell) on radio channel 1 (PCC) and Cell 3 (deactivated SCell2) on radio channel 3 (SCC2), but is not aware of Cell 2 (SCell1) on radio channel 2 (SCC1). The UE is only monitoring the PCC and SCC2. The UE shall be continuously scheduled in the PCell throughout the whole test.

During T2 the Test procedure requires the UE to send the first CSI report for SCell1 in a subframe (m+8), but also allows a subframe (m+10) if the subframe (m+8) was subject to interruption. The SS determines whether the CSI report in subframe (m+8) was interrupted or not by monitoring ACK/NACK sent in PCell in subframe (m+8).

NOTE: The requirements are dependant on the CA configuration (e.g. intra band, inter band; see TS 36.133 [4], clause 7.8) hence the test procedure has been divided into three variants, found in steps 8a, 8b and 8c and in steps 11a, 11b and 11c. Depending on the CA configuration under test only one of those shall be run respectively.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCC2 according to Annex C.0, C.1 for all downlink physical channels.
3. The SS shall configure SCell2 (Cell 3) on the SCC2 as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.37.4.3.
4. The SS shall configure transmission of PDSCH with a maximum number of 1 HARQ transmission.
5. Set the parameters according to T1 in Table 8.16.37.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts. The SS shall configure SCell1 (Cell 2) on the SCC1 as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.37.4.3.

6. The SS activates SCell1 by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe # denoted m which is an even number. If the SS receives ACK for MAC-CE sent by the UE, T2 starts in subframe m, and the test proceeds to step 7, otherwise go to step 12.

7. The SS activates SCell2 by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe (m+10).

8. The UE shall start sending CSI reports for SCell1 and the SS shall monitor CSI reports for SCell1 sent from the UE and ACK/NACK sent in PCell during SCells activation.

   8a. For all intra-band case

   - If the first CSI report for SCell1 is received by the SS in a subframe (m+8),
     - or (m+10) if the subframe (m+8) was subject to interruption
   - and CSI report with non-zero CQI index is received by the SS earlier than or equal to subframe (m+29),
     - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+29)
   - and DTX is not observed by the SS anywhere outside the subframes (m+5) to (m+13) and the subframes (m+15) to (m+23) up to the end of T2
   - Then the number of successes for the event “Activation” is increased by one. Otherwise, count a fail for the event “Activation” and go to step 12.

   8b. For the case where the band relation between PCC and SCells are inter-band

   - If the first CSI report for SCell1 is received by the SS in a subframe (m+8),
     - or (m+10) if the subframe (m+8) was subject to interruption,
   - and CSI report with non-zero CQI index is received by the SS earlier than or equal to subframe (m+29),
     - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+29)
   - and DTX is not observed by the SS anywhere outside the subframes (m+5) to (m+13) and the subframes (m+15) to (m+23) up to the end of T2,
   - and $\leq$2 non-consecutive DTX subframes are observed by the SS anywhere during the subframes (m+5) to (m+13) and the subframes (m+15) to (m+23) up to the end of T2,
   - Then the number of successes for the event “Activation” is increased by one. Otherwise, count a fail for the event “Activation” and go to step 12.

   8c. For the case where the band relation between PCC and SCell1 is intra-band and SCell2 is inter-band

   - If the first CSI report for SCell1 is received by the SS in a subframe (m+8),
     - or (m+10) if the subframe (m+8) was subject to interruption
   - and CSI report with non-zero CQI index is received by the SS earlier than or equal to subframe (m+29),
     - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+29)
   - and DTX is not observed by the SS anywhere outside the subframes (m+5) to (m+13) and the subframes (m+15) to (m+23) up to the end of T2,
   - and $\leq$2 non-consecutive DTX subframes are observed by the SS during the subframes (m+15) to (m+23) up to the end of T2,
   - Then the number of successes for the event “Activation” is increased by one. Otherwise, count a fail for the event “Activation” and go to step 12.

9. When T2 expires, the SS deactivate SCell1 by sending the deactivation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe # denoted n. If the SS receives ACK for MAC-CE sent by the UE, T3 starts in subframe n, and the test proceeds to step 10, otherwise go to step 12.

10. The SS deactivates SCell2 by sending the deactivation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe (n+10).

11. The UE shall stop sending CSI reports for SCell1 and the SS shall monitor CSI reports for SCell1 sent from the UE and ACK/NACK sent in PCell during SCell1 deactivation.

    11a. For all intra-band case

    - If the last CSI report is received by the SS earlier than or equal to subframe (n+8),
    - and DTX is not observed by the SS anywhere outside the subframes (n+5) to (n+13) and the subframes (n+15) to (n+23) up to the end of T3
    - Then the number of successes for the event “Deactivation” is increased by one. Otherwise, count a fail for the event “Deactivation”.

    11b. For the case where the band relation between PCC and SCells are inter-band

    - If the last CSI report is received by the SS earlier than or equal to subframe (n+8),
    - and DTX is not observed by the SS anywhere outside the subframes (n+5) to (n+13) and the subframes (n+15) to (n+23) up to the end of T3,
    - and ≤2 non-consecutive DTX subframes are observed by the SS anywhere during the subframes (n+5) to (n+13) and the subframes (n+15) to (n+23) up to the end of T3
    - Then the number of successes for the event “Deactivation” is increased by one. Otherwise, count a fail for the event “Deactivation”.

    11c. For the case where the band relation between PCC and SCell1 is intra-band and SCell2 is inter-band,

    - If the last CSI report is received by the SS earlier than or equal to subframe (n+8),
    - and DTX is not observed by the SS outside the subframes (n+5) to (n+13) and the subframes (n+15) to (n+23) up to the end of T3,
    - and ≤2 non-consecutive DTX subframes are observed by the SS the subframes (n+15) to (n+23) up to the end of T3,
    - Then the number of successes for the event “Deactivation” is increased by one. Otherwise, count a fail for the event “Deactivation”.

12. When T3 expires, or Activation in step 6 was not acknowledged, or a fail was counted for the event “Activation” in step 8a, 8b or 8c, or Deactivation in step 9 was not acknowledged, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

13. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

14. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

15. Repeat steps 2-14 until a test verdict has been achieved.

Each of the events “Activation” and “Deactivation” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is

achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.16.37.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.37.4.3-1: Common Exception messages for E-UTRAN FDD 3DL CA activation and deactivation of known SCell in non-DRX**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | |

**Table 8.16.37.4.3-2: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FDD 3DL CA activation and deactivation of known SCell in non-DRX**

| Derivation Path: 36.508 clause 4.8.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | CQI-ReportConfig-DEFAULT using condition CQI_PERIODIC | | RBC |
| cqi-ReportConfig-r10 | Not present | | SCell_AddMod |
| } | | | |

**Table 8.16.37.4.3-3: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 158 | | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | NULL | | |
| simultaneousAckNackAndCQI | TRUE | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.37.4.3-4: *PhysicalConfigDedicatedSCell-r10-DEFAULT:* Additional E-UTRAN FDD 3DL CA activation and deactivation of known SCell in non-DRX**

| Derivation Path: 36.508 clause 7.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicatedSCell-r10 ::= SEQUENCE { | | | |
| ul-Configuration-r10 SEQUENCE { | | | |
| antennaInfoUL-r10 | Not present | | |
| pusch-ConfigDedicatedSCell-r10 | Not present | | |
| uplinkPowerControlDedicatedSCell-r10 | Not present | | |
| cqi-ReportConfigSCell-r10 | CQI-ReportConfigSCell-r10-DEFAULT | | SCell1 |
| soundingRS-UL-ConfigDedicated-r10 | Not present | | |
| soundingRS-UL-ConfigDedicated-v1020 | Not present | | |
| soundingRS-UL-ConfigDedicatedAperiodic-r10 | Not present | | |
| } | | | |
| } | | | |

**Table 8.16.37.4.3-5: *CQI-ReportConfigSCell-r10-DEFAULT:* Additional E-UTRAN FDD 3DL CA activation and deactivation of known SCell in non-DRX**

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfigSCell-r10 ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic-r10 | Not present | | |
| nomPDSCH-RS-EPRE-Offset-r10 | 0 | | |
| cqi-ReportPeriodicSCell-r10 | | | |
| | *CQI-ReportPeriodic-r10-DEFAULT* | | CQI_PERIODIC |
| pmi-RI-Report-r10 | Not present | | |
| } | | | |

**Table 8.16.37.4.3-6: *CQI-ReportPeriodic-r10-DEFAULT:* Additional E-UTRAN FDD 3DL CA activation and deactivation of known SCell in non-DRX**

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportPeriodic-r10 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex-r10 | 0 | | |
| cqi-PUCCH-ResourceIndexP1-r10 | Not present | | |
| cqi-pmi-ConfigIndex | 0 | | |
| cqi-FormatIndicatorPeriodic-r10 CHOICE { | | | |
| widebandCQI-r10 SEQUENCE { | | | |
| csi-ReportMode-r10 | Not present | | |
| } | | | |
| ri-ConfigIndex | NULL | | |
| simultaneousAckNackAndCQI | TRUE | | |
| cqi-Mask-r9 | Not present | | |
| csi-ConfigIndex-r10 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.37.4.3-7: *MeasObjectEUTRA-GENERIC:* Additional E-UTRAN FDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Derivation Path: 36.508 clause 4.6.6 Table 4.6.6-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | Cell 2, Cell 3 |
| } | | | |

**Table 8.16.37.4.3-8: *MeasObjectEUTRA-GENERIC:* Additional E-UTRAN FDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1, Cell 2, Cell 3 |
| } | | | |

**Table 8.16.37.4.3-9: *QuantityConfig-DEFAULT:* Additional E-UTRAN FDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Derivation Path: TS 36.508 clause 4.6.6, Table 4.6.6-3A: QuantityConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| QuantityConfig-DEFAULT ::= SEQUENCE { | | | |
| quantityConfigEUTRA SEQUENCE { | | | |
| filterCoefficientRSRP | fc0 | | |
| filterCoefficientRSRQ | fc0 | | |
| } | | | |
| } | | | |

**Table 8.16.37.4.3-10: *SystemInformationBlockType2:* Additional E-UTRAN FDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-1 SystemInformationBlockType2 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfig | Not present | | Cell 1 |
| } | | | |

**Table 8.16.37.4.3-11: *SystemInformationBlockType3:* Additional E-UTRAN FDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-2 SystemInformationBlockType3 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType3 ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1, Cell 2, Cell 3 |
| } | | | |

**Table 8.16.37.4.3-12: *SystemInformationBlockType5:* Additional E-UTRAN FDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-3 SystemInformationBlockType5 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType5 ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1, Cell 2, Cell 3 |
| } | | | |

## 8.16.37.5 Test requirement

Table 8.16.37.5-1 defines the primary level settings including test tolerances for E-UTRAN FDD 3DL CA activation and deactivation of known SCell in non-DRX test.

**Table 8.16.37.5-1: Cell specific test parameters for E-UTRAN FDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |

| E-UTRA RF Channel Number | | 1 | 2 | 3 |
|---|---|---|---|---|
| $BW_{channel}$ | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 |
| PDSCH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.7 FDD<br>10MHz: R.3 FDD<br>20MHz: R.6 FDD | - | - |
| PCFICH/PDCCH/PHICH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD |
| OCNG Patterns | | 5MHz: OP.20 FDD<br>10MHz: OP.10 FDD<br>20MHz: OP.17 FDD | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD |
| PBCH_RA | dB | 0 | 0 | 0 |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -104 | -104 | -104 |
| $\hat{E}_s/N_{oc}$ | dB | 17 | 17 | 17 |
| $\hat{E}_s/I_{ot}$ Note 3 | dB | 17 | 17 | 17 |
| RSRP Note 3 | dBm/15 kHz | -87 | -87 | -87 |
| SCH_RP Note 3 | dBm/15 kHz | -87 | -87 | -87 |
| Io Note 3 | dBm/Ch BW | -59.13+10log ($N_{RB,c}$ /50) | -59.13+10log ($N_{RB,c}$ /50) | -59.13+10log ($N_{RB,c}$ /50) |
| Propagation Condition | | AWGN | AWGN | AWGN |
| Antenna Configuration | | 1x2 | 1x2 | 1x2 |
| Timing offset to Cell 1 | µs | - | 0 | 0 |
| Time alignment error relative to cell 1 Note 5 | µs | - | ≤ TAE | ≤ TAE |
| Time alignment error relative to cell 2 Note 5 | µs | - | - | ≤ TAE |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: $E_s/I_{ot}$, RSRP, SCH_RP and Io have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: The uplink resources for CSI reporting are assigned to the UE prior to the start of time period T2.

Note 5: Time alignment error (TAE) as specified in TS 36.104 [30] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

During T2 interruption of PCell during SCells activation shall not happen outside the subframes (m+5) to (m+9) and outside the subframes (m+15) to (m+19).

Figures 8.16.37.5-1 to 8.16.37.5-3 show the derivation of the Test procedure requirements for DTX during T2, based on the core requirements for interruption. The applicable figure is selected according to the CA configuration.

NOTE: The differences between activation and deactivation are as follows:

- Activation → Deactivation
- First CSI → Last CSI
- Latest valid CSI report → Not exist (no need to check since CSI report was already stopped)

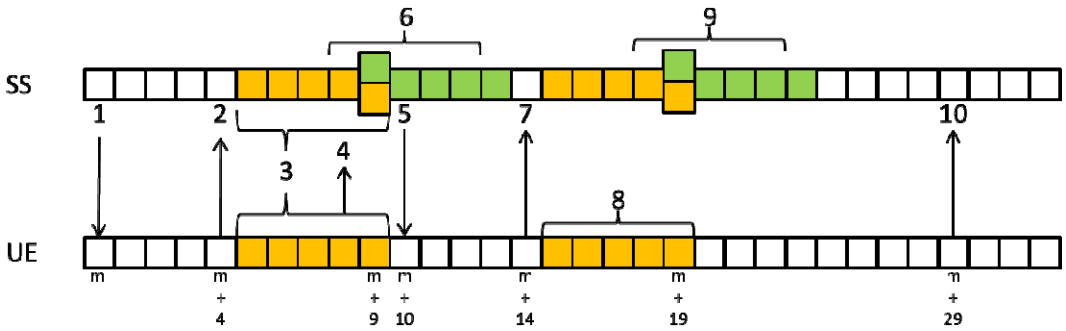


**Figure 8.16.37.5-1: Procedure derivation for Activation (All intra-band case)**

1) Activation command for SCell1 (T2 starts)
2) ACK for MAC-CE for SCell1 activation
3) Possible interruption period by SCell1 activation
4) First CSI report timing (could be invalid CQI)
5) Activation command for SCell2
6) Possible DTX reception period on SS due to interruption by SCell1 activation
7) ACK for MAC-CE for SCell2 activation
8) Possible interruption period by SCell2 activation
9) Possible DTX reception period on SS due to interruption by SCell2 activation
10 Latest valid CSI report timing (valid CQI)

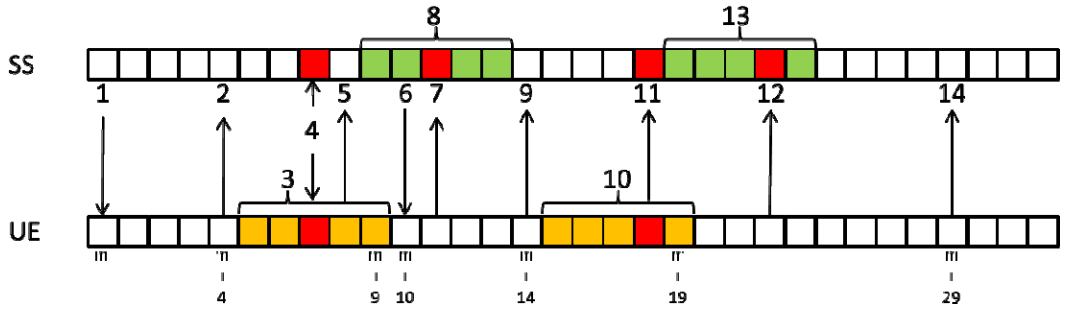


**Figure: 8.16.37.5-2 Procedure derivation for Activation**
**(the case where the band relation between PCC and SCells are inter-band)**

1) Activation command for SCell1 (T2 starts)
2) ACK for MAC-CE for SCell1 activation
3) Possible interruption period by SCell1 activation
4) Allowed interruption timing (1 subframe in m+5~m+9) on UE by SCell1 activation,
   that is, possible DTX timing on SS
5) First CSI report timing (could be invalid CQI) at m+8 (might be m+10 due to interruption)
6) Activation command for SCell2

7) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell1 activation

8) Possible DTX reception period on SS due to interruption by SCell1 activation

9) ACK for MAC-CE for SCell2 activation

10) Possible interruption period by SCell2 activation

11) Allowed interruption timing (1 subframe in m+15~m+19) on UE by SCell2 activation, that is, possible DTX timing on SS

12) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell2 activation

13) Possible DTX reception period on SS due to interruption by SCell2 activation

14) Latest valid CSI report timing (valid CQI)

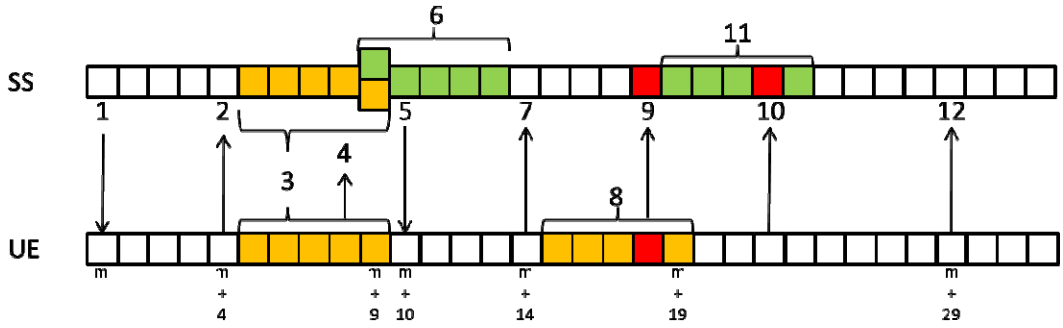


**Figure 8.16.37.5-3: Procedure derivation for Activation (The case where the band relation between PCC and SCell1 is intra-band and SCell2 is inter-band)**

1) Activation command for SCell1 (T2 starts)

2) ACK for MAC-CE for SCell1 activation

3) Possible interruption period by SCell1 activation

4) First CSI report timing (could be invalid CQI)

5) Activation command for SCell2

6) Possible DTX reception period on SS due to interruption by SCell1 activation

7) ACK for MAC-CE for SCell2 activation

8) Possible interruption period by SCell2 activation

9) Allowed interruption timing (1 subframe in m+15~m+19) on UE by SCell2 activation, that is, possible DTX timing on SS

10) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell2 activation

11) Possible DTX reception period on SS due to interruption by SCell2 activation

12) Latest valid CSI report timing (valid CQI)

During T3 interruption of PCell during SCells deactivation shall not happen outside the subframes (n+5) to (n+9) and outside the subframes (n+15) to (n+19).

The derivation of the Test procedure requirements for DTX during T3 is based on the core requirements for interruption, using the applicable figure according to the CA configuration, in a similar way to T2.

All of the requirements defined in the Test Procedure shall be fulfilled in order for the observed SCell1 activation delay and SCell1 deactivation delay to be counted as correct. The rate of correct observed SCell1 activation delay and SCell1 deactivation delay during repeated tests shall be at least 90% with a confidence level of 95% for each of the events.

The statistical pass/ fail decisions are done separately for activation and deactivation.

Decide the test pass, if activation and deactivation are passed, otherwise fail the UE.

## 8.16.38 3 DL TDD CA Activation and Deactivation of known SCell in non-DRX

### 8.16.38.1 Test purpose

To verify that SCell activation and deactivation times are within the requirements in TS 36.133 [4] section 7.7 for 3DL TDD carrier aggregation, when the SCells are known by the UE at the time of activation.

### 8.16.38.2 Test applicability

This test applies to all types of E-UTRA UE TDD release 10 and forward that support 3DL with intra-band contiguous CA or 3DL with inter-band CA, or 3DL with intra-band contiguous and inter-band CA. Applicability requires support for FGI bit 25.

This test case also applies to all types of E-UTRA TDD UE release 11 and forward that support 3DL with intra-band non-contiguous and inter-band CA, or 3DLwith intra-band non-contiguous and intra-band contiguous CA. Applicability requires support for FGI bit 25.

### 8.16.38.3 Minimum conformance requirements

While activating a SCell if the UE does not receive any command to activate, deactivate, configure or deconfigure the other SCell during the SCell activation delay then the UE shall meet the SCell activation delay requirements specified in TS 36.133 [4] section 7.7.2.

While activating a SCell if the other SCell is activated, deactivated, configured or deconfigured by the UE then the UE shall meet the SCell activation delay requirements ($T_{activate_total}$) according to the following expression:

$$T_{activate_total} = T_{activate_basic} + K*5$$

Where:

$T_{activate_total}$ is the total time to activate a SCell and is expressed in subframes.

$T_{activate_basic}$ is the SCell activation delay specified in TS 36.133 [4] section 7.7.2;

K ($1 \leq K \leq [3]$) is the number of times the other SCell is activated, deactivated, configured or deconfigured while the SCell is being activated;

While activating a SCell:

- The interruption on the PCell and/or on the activated SCell due to the SCell activation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+11 if:
  - The PCell and/or the activated SCell being interrupted and the SCell being activated belong to E-UTRA TDD, or
  - The activated SCell being interrupted and the SCell being activated belong to E-UTRA FDD and the PCell belongs to E-UTRA TDD.
- Otherwise, the interruption on PCell and/or on the activated SCell due to the SCell activation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+9.

Starting from the subframe specified in section 4.3 of TS 36.213 [8] and until the UE has completed the SCell activation, the UE shall report CQI index = 0 (out of range) if the UE has available uplink resources to report CQI for an SCell.

The UE shall deactivate a SCell and meet the SCell deactivation delay requirements specified in TS 36.133 [4] section 7.7.3 regardless of whether the other SCell is activated, deactivated, configured or deconfigured or not by the UE during the SCell deactivation delay.

While deactivating a SCell:

- The interruption on the PCell and/or on the activated SCell due to the SCell deactivation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+11 if:
  - The PCell and/or the activated SCell being interrupted and the SCell being deactivated belong to E-UTRA TDD or
  - The activated SCell being interrupted and the SCell being deactivated belong to E-UTRA FDD and the PCell belongs to E-UTRA TDD.
- Otherwise, the interruption on PCell and/or the activated SCell due to the SCell deactivation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+9.

The normative reference for this requirement is TS 36.133 [4] clause 7.7 and A.8.16.38.

## 8.16.38.4 Test description

### 8.16.38.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.16.38.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.63 as appropriate.
2. The general test parameter settings are set up according to Table 8.16.38.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.16.38.4.3.
5. Cell 1 is the PCell on the primary component carrier, Cell 2 is the SCell on RF channel number 2, and Cell 3 is the deactivated SCell on RF channel number 3. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.38.4.1-1: General test parameters for E-UTRAN TDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1, 2, 3 | Three radio channels are used for this test |
| Active PCell | | Cell 1 | Primary cell on RF channel number 1. |
| Deconfigured SCell | | Cell 2 | Deconfigured secondary cell on RF channel number 2. |
| Configured deactivated SCell | | Cell 3 | Configured deactivated secondary cell on RF channel number 3. |
| CP length | | Normal | |
| Special subframe configuration | | 6 | As specified in Table 4.2-1 in TS 36.211 [9]. The same configuration applies to all cells. |
| Uplink-downlink configuration | | 1 | As specified in Table 4.2-2 in TS 36.211 [9]. The same configuration applies to all cells |
| DRX | | OFF | Continuous monitoring of primary cell |
| CQI/PMI periodicity and offset configuration index | | 0 | CQI reporting for SCells every UL subframe |
| Cell-individual offset for cells on RF channel number 1 | dB | 0 | Individual offset for cells on PCC. |
| Cell-individual offset for cells on RF channel number 2 | dB | 0 | Individual offset for cells on SCC1. |
| Cell-individual offset for cells on RF channel number 3 | dB | 0 | Individual offset for cells on SCC2. |
| Filter coefficient | | 0 | L3 filtering is not used |
| SCell measurement cycle (measCycleSCell) | ms | 320 | |
| T1 | s | 7 | During this time the PCell and SCell2 shall be known and the SCell1 configured and detected. |
| T2 | s | 1 | During this time the UE shall activate the SCell1 and SCell2. |
| T3 | s | 1 | During this time the UE shall deactivate the SCell1 and SCell2 |

### 8.16.38.4.2 Test procedure

The test consists of three successive time periods, with duration of T1, T2 and T3, respectively. There are three carriers, each with one cell. All cells have constant signal levels throughout the test.

Before the test starts the UE is connected to Cell 1 (PCell) on radio channel 1 (PCC) and Cell 3 (deactivated SCell2) on radio channel 3 (SCC2), but is not aware of Cell 2 (SCell1) on radio channel 2 (SCC1). The UE is only monitoring the PCC and SCC2. The UE shall be continuously scheduled in the PCell throughout the whole test.

NOTE: The requirements are dependant on the CA configuration (e.g. intra band, inter band; see TS 36.133 [4], clause 7.8) hence the test procedure has been divided into three variants, found in steps 8a, 8b and 8c and in steps 11a, 11b and 11c. Depending on the CA configuration under test only one of those shall be run respectively.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCC2 according to Annex C.0, C.1 for all downlink physical channels.
3. The SS shall configure SCell2 (Cell 3) on the SCC2 as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.38.4.3.
4. The SS shall configure transmission of PDSCH with a maximum number of 1 HARQ transmission.
5. Set the parameters according to T1 in Table 8.16.38.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts. The SS shall configure SCell1 (Cell 2) on the SCC1 as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.38.4.3.

6. The SS activates SCell1 by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe # denoted m, where m is 4 or 9. If the SS receives ACK for MAC-CE sent by the UE, T2 starts in subframe m, and the test proceeds to step 7, otherwise go to step 12.

7. The SS activates SCell2 by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe (m+15).

8. The UE shall start sending CSI reports for SCell1 and the SS shall monitor CSI reports for SCell1 sent from the UE and ACK/NACK sent in PCell during SCells activation.

   8a. For all intra-band case

   - If the first CSI report for SCell1 is received by the SS in a subframe (m+8),
     - or (m+9) if the subframe (m+8) was subject to interruption,
     - or in a subframe (m+13) if the subframes (m+8) and (m+9) were subject to interruption when an intra-band SCell is activated,
   - and CSI report with non-zero CQI index is received by the SS earlier than or equal to subframe (m+29),
     - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+29).
   - and DTX is not observed by the SS outside the subframes (m+5) to (m+9) and (m+13) and the subframes (m+20) to (m+24) and (m+28), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+6) to (m+10) and (m+13) to (m+14) and the subframes (m+21) to (m+25) and (m+28) to (m+29), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+7) to (m+11) and (m+13) to (m+14) and (m+18) and the subframes (m+22) to (m+26) and (m+28) to (m+29) and (m+33), up to the end of T2,
   - Then the number of successes for the event “Activation” is increased by one. Otherwise, count a fail for the event “Activation” and go to step 12.

   8b. For the case where the band relation between PCC and SCells are inter-band

   - If the first CSI report for SCell1 is received by the SS in a subframe (m+8),
     - or (m+9) if the subframe (m+8) was subject to interruption,
     - or in a subframe (m+13) if the subframes (m+8) and (m+9) were subject to interruption when an intra-band SCell is activated,
   - and CSI report with non-zero CQI index is received by the SS earlier than or equal to (m+29),
     - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+29),
   - and DTX is not observed by the SS outside the subframes (m+5), (m+9), (m+20) and (m+24), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+6), (m+13), (m+21) and (m+28), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+7), (m+13), (m+22) and (m+28), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+8) and (m+23), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+9) and (m+24), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+10), (m+14), (m+25) and (m+29), up to the end of T2,

- or DTX is not observed by the SS outside the subframes (m+11), (m+18), (m+26) and (m+33), up to the end of T2,

- Then the number of successes for the event “Activation” is increased by one. Otherwise, count a fail for the event “Activation” and go to step 12.

8c. For the case where the band relation between PCC and SCell1 is intra-band and SCell2 is inter-band

- If the first CSI report for SCell1 is received by the SS in a subframes (m+8),

  - or (m+9) if the subframe (m+8) was subject to interruption,

  - or in a subframe (m+13) if the subframes (m+8) and (m+9) were subject to interruption when an intra-band SCell is activated,

- and CSI report with non-zero CQI index is received by the SS earlier than or equal to subframe (m+29),

  - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+29),

- and DTX is not observed by the SS outside the subframes (m+5) to (m+9) and (m+13),

  - or DTX is not observed by the SS outside the subframes (m+6) to (m+10) and (m+13) to (m+14),

  - or DTX is not observed by the SS outside the subframes (m+7) to (m+11) and (m+13) to (m+14) and (m+18),

- and DTX is not observed by the SS outside the subframes (m+20) and (m+24), up to the end of T2,

  - or DTX is not observed by the SS outside the subframes (m+21) and (m+28), up to the end of T2,

  - or DTX is not observed by the SS outside the subframes (m+22) and (m+28), up to the end of T2,

  - or DTX is not observed by the SS outside the subframe (m+23), up to the end of T2,

  - or DTX is not observed by the SS outside the subframe (m+24), up to the end of T2,

  - or DTX is not observed by the SS outside the subframes (m+25) and (m+29), up to the end of T2,

  - or DTX is not observed by the SS outside the subframes (m+26) and (m+33), up to the end of T2,

- Then the number of successes for the event “Activation” is increased by one. Otherwise, count a fail for the event “Activation” and go to step 12.

9. When T2 expires, the SS deactivate SCell1 by sending the deactivation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe # denoted n, where n is 4 or 9. If the SS receives ACK for MAC-CE sent by the UE, T3 starts in subframe n, and the test proceeds to step 10, otherwise go to step 12.

10. The SS deactivates SCell2 by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13.6.1.3.8) in a subframe (n+15).

11. The UE shall stop sending CSI reports for SCell1 and the SS shall monitor CSI reports for SCell1 sent from the UE and ACK/NACK sent in PCell during SCell1 deactivation.

    11a. For all intra-band case

    - If the last CSI report is received by the SS earlier than or equal to subframe (n+8),

    - and DTX is not observed by the SS outside the subframes (n+5) to (n+9) and (n+13) and the subframes (n+20) to (n+24) and (n+28), up to the end of T3,

      - or DTX is not observed by the SS outside the subframes (n+6) to (n+10) and (n+13) to (n+14) and the subframes (n+21) to (n+25) and (n+28) to (n+29), up to the end of T3,

      - or DTX is not observed by the SS outside the subframes (n+7) to (n+11) and (n+13) to (n+14) and (n+18) and the subframes (n+22) to (n+26) and (n+28) to (n+29) and (n+33), up to the end of T3,

- Then the number of successes for the event “Deactivation” is increased by one. Otherwise, count a fail for the event “Deactivation”.

11b. For the case where the band relation between PCC and SCells are inter-band

- If the last CSI report is received by the SS earlier than or equal to subframe (n+8),
- and DTX is not observed by the SS outside the subframes (n+5), (n+9), (n+20) and (n+24), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+6), (n+13), (n+21) and (n+28), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+7), (n+13), (n+22) and (n+28), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+8) and (n+23), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+9) and (n+24), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+10), (n+14), (n+25) and (n+29), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+11), (n+18), (n+26) and (n+33), up to the end of T3,
- Then the number of successes for the event “Deactivation” is increased by one. Otherwise, count a fail for the event “Deactivation”.

11c. For the case where the band relation between PCC and SCell1 is intra-band and SCell2 is inter-band

- If the last CSI report is received by the SS earlier than or equal to subframe (n+8),
- and DTX is not observed by the SS outside the subframes (n+5) to (n+9) and (n+13)
  - or DTX is not observed by the SS outside the subframes (n+6) to (n+10) and (n+13) to (n+14),
  - or DTX is not observed by the SS outside the subframes (n+7) to (n+11) and (n+13) to (n+14) and (n+18),
- and DTX is not observed by the SS outside the subframes (n+20) and (n+24), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+21) and (n+28), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+22) and (n+28), up to the end of T3,
  - or DTX is not observed by the SS outside the subframe (n+23), up to the end of T3,
  - or DTX is not observed by the SS outside the subframe (n+24), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+25) and (n+29), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+26) and (n+33), up to the end of T3,
- Then the number of successes for the event “Deactivation” is increased by one. Otherwise, count a fail for the event “Deactivation”.

12. When T3 expires, or Activation in step 6 was not acknowledged, or a fail was counted for the event “Activation” in step 8a, 8b or 8c, or Deactivation in step 9 was not acknowledged, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

13. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

14. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the

UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

15. Repeat steps 2-14 until a test verdict has been achieved.

Each of the events "Activation" and "Deactivation" is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.16.38.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.38.4.3-1: Common Exception messages for E-UTRAN TDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | |

**Table 8.16.38.4.3-2: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.8.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | CQI-ReportConfig-DEFAULT using condition CQI_PERIODIC | | RBC |
| cqi-ReportConfig-r10 | Not present | | SCell_AddMod |
| } | | | |

**Table 8.16.38.4.3-3: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 159 | | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | NULL | | |
| simultaneousAckNackAndCQI | TRUE | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.38.4.3-4: *PhysicalConfigDedicatedSCell-r10-DEFAULT:* Additional E-UTRAN TDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicatedSCell-r10-DEFAULT ::= SEQUENCE { | | | |
| ul-Configuration-r10 SEQUENCE { | | | |
| antennaInfoUL-r10 | Not present | | |
| pusch-ConfigDedicatedSCell-r10 | Not present | | |
| uplinkPowerControlDedicatedSCell-r10 | Not present | | |
| cqi-ReportConfigSCell-r10 | CQI-ReportConfigSCell-r10-DEFAULT | | SCell1 |
| soundingRS-UL-ConfigDedicated-r10 | Not present | | |
| soundingRS-UL-ConfigDedicated-v1020 | Not present | | |
| soundingRS-UL-ConfigDedicatedAperiodic-r10 | Not present | | |
| } | | | |
| } | | | |

**Table 8.16.38.4.3-5: *CQI-ReportConfigSCell-r10-DEFAULT:* Additional E-UTRAN TDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfigSCell-r10 ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic-r10 | Not present | | |
| nomPDSCH-RS-EPRE-Offset-r10 | 0 | | |
| cqi-ReportPeriodicSCell-r10 | | | |
| | *CQI-ReportPeriodic-r10-DEFAULT* | | CQI_PERIODIC |
| pmi-RI-Report-r10 | Not present | | |
| } | | | |

**Table 8.16.38.4.3-6: *CQI-ReportPeriodic-r10-DEFAULT:* Additional E-UTRAN TDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportPeriodic-r10 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex-r10 | 0 | | |
| cqi-PUCCH-ResourceIndexP1-r10 | Not present | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | Cell2 |
| cqi-FormatIndicatorPeriodic-r10 CHOICE { | | | |
| widebandCQI-r10 SEQUENCE { | | | |
| csi-ReportMode-r10 | Not present | | |
| } | | | |
| ri-ConfigIndex | 484 | (see Table 7.2.2-1B in TS 36.213) | Cell2 |
| simultaneousAckNackAndCQI | TRUE | | |
| cqi-Mask-r9 | Not present | | |
| csi-ConfigIndex-r10 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.38.4.3-7: *MeasObjectEUTRA-GENERIC:* Additional E-UTRAN TDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.6 Table 4.6.6-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | Cell 2, Cell 3 |
| } | | | |

## 8.16.38.5 Test requirement

Table 8.16.38.5-1 defines the primary level settings including test tolerances for E-UTRAN TDD 3 DL CA activation and deactivation of known SCell in non-DRX test.

**Table 8.16.38.5-1: Cell specific test parameters for E-UTRAN TDD 3 DL CA activation and deactivation of known SCell in non-DRX**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | 3 | | |
| $BW_{channel}$ | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | |
| PDSCH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | | | - | | | - | | |
| PCFICH/PDCCH/PHICH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | |
| OCNG Patterns | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -104 | | | -104 | | | -104 | | |
| $\hat{E}_s/N_{oc}$ | dB | 17 | | | 17 | | | 17 | | |
| $\hat{E}_s/I_{ot}$ Note 3 | dB | 17 | | | 17 | | | 17 | | |
| RSRP Note 3 | dBm/15 kHz | -87 | | | -87 | | | -87 | | |
| SCH_RP Note 3 | dBm/15 kHz | -87 | | | -87 | | | -87 | | |
| Io Note 3 | dBm/Ch BW | -59.13+10log ($N_{RB,c}$ /50) | | | -59.13+10log ($N_{RB,c}$ /50) | | | -59.13+10log ($N_{RB,c}$ /50) | | |
| Propagation Condition | | AWGN | | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x2 | | | 1x2 | | | 1x2 | | |
| Timing offset to Cell 1 | µs | - | | | 0 | | | 0 | | |
| Time alignment error relative to cell 1 Note 5 | µs | - | | | ≤ TAE | | | ≤ TAE | | |
| Time alignment error relative to cell 2 Note 5 | µs | - | | | - | | | ≤ TAE | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: $E_s/I_{ot}$, RSRP, SCH_RP and Io have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The uplink resources for CSI reporting are assigned to the UE prior to the start of time period T2.
Note 5: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

During T2 interruption of PCell during SCells activation shall not happen outside the subframes (m+5) to (m+11) and outside the subframes (m+20) to (m+26).

Figures 8.16.38.5-1 to 8.16.38.5-3 shows the derivation of the Test procedure requirements for DTX during T2, based on the core requirements for interruption. The applicable figure is selected according to the CA configuration.

NOTE: The differences between activation and deactivation are as follows:

- Activation ➔ Deactivation

- First CSI ➔ Last CSI
- Latest valid CSI report  Not exist (no need to check since CSI report was already stopped)

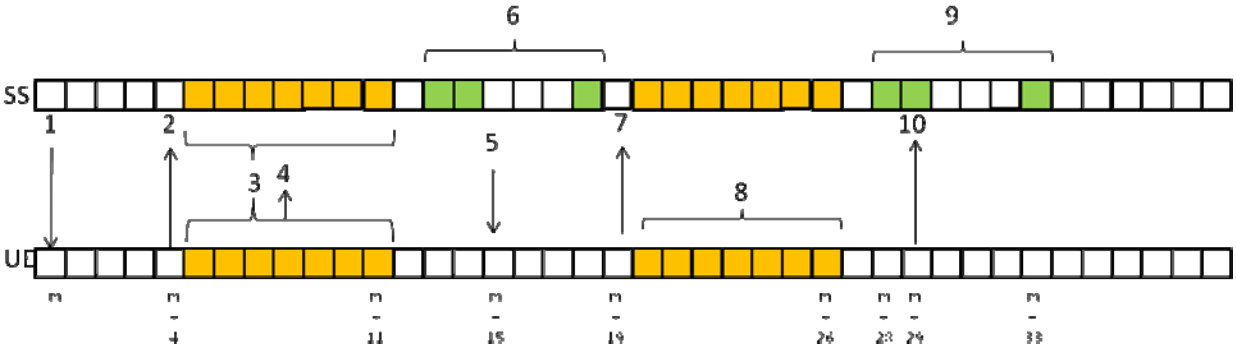


**Figure 8.16.38.5-1: Procedure derivation for Activation (All intra-band case)**

1) Activation command for SCell1(T2 starts)
2) ACK for MAC-CE for SCell1 activation
3) Possible interruption period by SCell1 activation
4) First CSI report timing(invalid CQI)
5) Activation command for SCell2
6) Possible DTX reception period on SS due to interruption by SCell1 activation

   (Only green subframes are allowed)
7) ACK for MAC-CE for SCell2 activation
8) Possible interruption period by SCell2 activation
9) Possible DTX reception period on SS due to interruption by SCell2 activation

   (Only green subframes are allowed)
10) Latest valid CSI report timing(valid CQI)

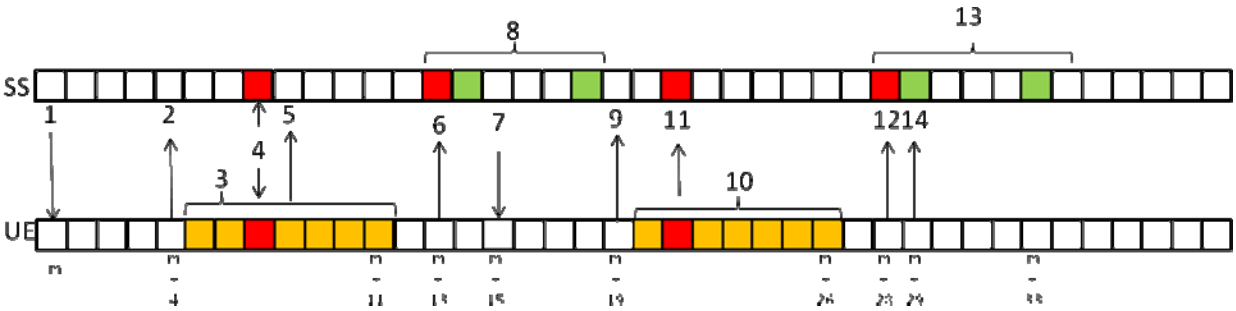


**Figure 8.16.38.5-2: Procedure derivation for Activation**
**(the case where the band relation between PCC and SCells are inter-band)**

1) Activation command for SCell1(T2 starts)
2) ACK for MAC-CE for SCell1 activation

3) Possible interruption period by SCell1 activation
4) Allowed interruption timing(1 subframe in m+5~m+11) on UE by SCell1 activation,
   that is, possible DTX timing on SS
5) First CSI report timing(invalid CQI) at m+8 (might be m+12 due to interruption)
6) Activation command for SCell2
7) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell1 activation
8) Possible DTX reception period on SS due to interruption by SCell1 activation
   (Only m+13, m+14 or m+18 are allowed)
9) ACK for MAC-CE for SCell2 activation
10) Possible interruption period by SCell2 activation
11) Allowed interruption timing(1 subframe in m+20~m+26) on UE by SCell2 activation,
    that is, possible DTX timing on SS
12) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell2 activation
    (Only m+28, m+29 or m+33 are allowed)
13) Possible DTX reception period on SS due to interruption by SCell2 activation
14) Latest valid CSI report timing(valid CQI)

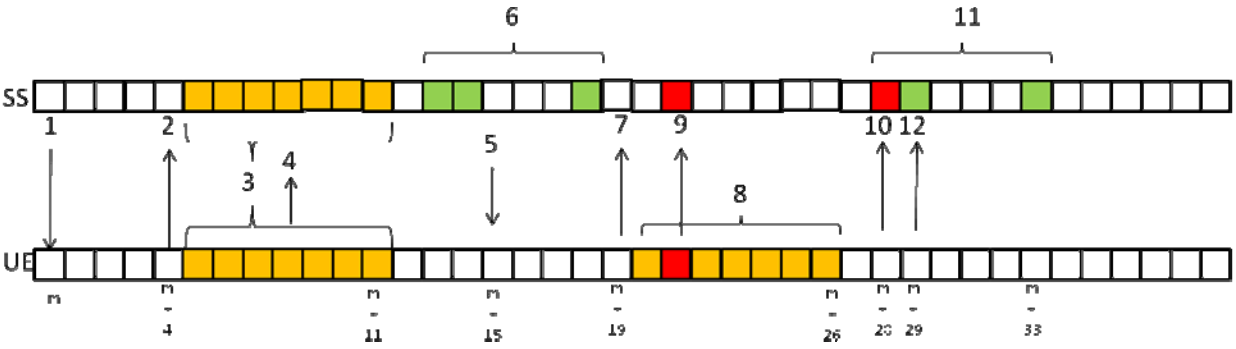


**Figure 8.16.38.5-3: Procedure derivation for Activation**
**(the case where the band relation between PCC and SCell1 is intra-band and SCell2 is inter-band)**

1) Activation command for SCell1(T2 starts)
2) ACK for MAC-CE for SCell1 activation
3) Possible interruption period by SCell1 activation
4) First CSI report timing(invalid CQI)
5) Activation command for SCell2
6) Possible DTX reception period on SS due to interruption by SCell1 activation
   (Only green subframes are allowed)
7) ACK for MAC-CE for SCell2 activation
8) Possible interruption period by SCell2 activation

9) Allowed interruption timing(1 subframe in m+20~m+26) on UE by SCell2 activation,

that is, possible DTX timing on SS

10) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell2 activation

(Only m+28, m+29 or m+33 are allowed)

11) Possible DTX reception period on SS due to interruption by SCell2 activation

12) Latest valid CSI report timing(valid CQI)

During T3 interruption of PCell during SCells deactivation shall not happen outside the subframes (n+5) to (n+11) and outside the subframes (n+20) to (n+26).

The derivation of the Test procedure requirements for DTX during T3 is based on the core requirements for interruption, using the applicable figure according to the CA configuration, in a similar way to T2.

All of the above test requirements shall be fulfilled in order for the observed SCell1 activation delay and SCell1 deactivation delay to be counted as correct. The rate of correct observed SCell1 activation delay and SCell1 deactivation delay during repeated tests shall be at least 90% with a confidence level of 95% for each of the events.

The statistical pass/ fail decisions are done separately for activation and deactivation.

Decide the test pass, if activation and deactivation are passed, otherwise fail the UE.

## 8.16.39 E-UTRA TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in FDD

### 8.16.39.1 Test purpose

To verify that SCell activation and deactivation times are within the requirements in TS 36.133 [4] section 7.7 for UE configured with two downlink SCells, when the SCell is unknown by the UE at the time of activation.

### 8.16.39.2 Test applicability

This test applies to all types of E-UTRA FDD and TDD Release 12 and forward UE that support E-UTRA FDD and TDD and 3DL CA with FDD as PCell. Applicability requires support for FGI bit 25.

### 8.16.39.3 Minimum conformance requirements

While activating a SCell if the UE does not receive any command to activate, deactivate, configure or deconfigure the other SCell during the SCell activation delay then the UE shall meet the SCell activation delay requirements specified in TS 36.133 [4] section 7.7.2.

While activating a SCell if the other SCell is activated, deactivated, configured or deconfigured by the UE then the UE shall meet the SCell activation delay requirements ($T_{activate_total}$) according to the following expression:

$$T_{activate_total} = T_{activate_basic} + K*5$$

Where:

$T_{activate_total}$ is the total time to activate a SCell and is expressed in subframes.

$T_{activate_basic}$ is the SCell activation delay specified in TS 36.133 [4] section 7.7.2;

K ($1 \leq K \leq [3]$) is the number of times the other SCell is activated, deactivated, configured or deconfigured while the SCell is being activated;

While activating a SCell:

- The interruption on the PCell and/or on the activated SCell due to the SCell activation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+11 if:

- The PCell and/or the activated SCell being interrupted and the SCell being activated belong to E-UTRA TDD, or

- The activated SCell being interrupted and the SCell being activated belong to E-UTRA FDD and the PCell belongs to E-UTRA TDD.

- Otherwise, the interruption on PCell and/or on the activated SCell due to the SCell activation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+9.

Starting from the subframe specified in section 4.3 of TS 36.213 [8] and until the UE has completed the SCell activation, the UE shall send CSI with CQI index = 0 (out of range) if the UE is configured to report the CQI for the SCell being activated.

The UE shall deactivate a SCell and meet the SCell deactivation delay requirements specified in TS 36.133 [4] section 7.7.3 regardless of whether the other SCell is activated, deactivated, configured or deconfigured or not by the UE during the SCell deactivation delay.

While deactivating a SCell:

- The interruption on the PCell and/or on the activated SCell due to the SCell deactivation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+11 if:

  - The PCell and/or the activated SCell being interrupted and the SCell being deactivated belong to E-UTRA TDD or

  - The activated SCell being interrupted and the SCell being deactivated belong to E-UTRA FDD and the PCell belongs to E-UTRA TDD.

- Otherwise, the interruption on PCell and/or the activated SCell due to the SCell deactivation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+9.

The normative reference for this requirement is TS 36.133 [4] clause 7.7 and A.8.16.39.

## 8.16.39.4 Test description

### 8.16.39.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.16.39.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.63 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.39.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.39.4.3.

5. Cell 1 is the PCell on the primary component carrier, Cell 2 is the Deconfigured deactivated SCell on RF channel number 2, and Cell 3 is Configured deactivated SCell on RF channel number 3. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.39.4.1-1: General test parameters for unknown SCell1 activation case**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1, 2, 3 | Three radio channels are used for this test |
| Active PCell | | Cell 1 | Primary cell on RF channel number 1. |
| Deconfigured deactivated SCell1 | | Cell 2 | Deconfigured deactivated secondary cell on RF channel number 2. |
| Configured deactivated SCell2 | | Cell 3 | Configured deactivated secondary cell on RF channel number 3. |
| CP length | | Normal | |
| DRX | | OFF | Continuous monitoring of primary cell |
| CQI/PMI periodicity and offset configuration index | | 0 | CQI reporting for SCell every second subframe |
| Cell-individual offset for cells on RF channel number 1 | dB | 0 | Individual offset for cells on PCC. |
| Cell-individual offset for cells on RF channel number 2 | dB | 0 | Individual offset for cells on SCC1. |
| Cell-individual offset for cells on RF channel number 3 | dB | 0 | Individual offset for cells on SCC2. |
| SCell measurement cycle (measCycleSCell) | ms | 320 | |
| T1 | ms | 100 | During this time the PCell and SCell2 shall be known and the SCell1 configured. |
| T2 | s | 1 | During this time the UE shall activate the SCell1 and SCell2. |
| T3 | s | 1 | During this time the UE shall deactivate the SCell1 and SCell2. |

### 8.16.39.4.2 Test procedure

The test consists of three successive time periods, with duration of T1, T2 and T3, respectively. There are three carriers, each with one cell. Cell 1 is FDD cell, and Cell 2 and Cell 3 are TDD cells. Cell 1 and Cell 3 have constant signal levels throughout the test.

Before the test starts the UE is connected to Cell 1 (PCell) on radio channel 1 (PCC) and Cell 3 (deactivated SCell2) on radio channel 3 (SCC2), but is not aware of Cell 2 (SCell1) on radio channel 2 (SCC1). The UE is only monitoring the PCC and SCC2. The UE shall be continuously scheduled in the PCell throughout the whole test.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCC2 according to Annex C.0, C.1 for all downlink physical channels.
3. The SS shall configure SCell (Cell 3) on the SCC2 as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.39.4.3.
4. The SS shall configure transmission of PDSCH with a maximum number of 1 HARQ transmission.
5. Set the parameters according to T1 in Table 8.16.39.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts. Configure SCC1 according to Annex C.0, C.1 for all downlink physical channels. The SS shall configure SCell1 (Cell 2) on the SCC1 as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.39.4.3.
6. The SS activates SCell1 by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe # denoted m which is an even number. If the SS receives ACK for MAC-CE sent by the UE, T2 starts in subframe m, and the test proceeds to step 7, otherwise go to step 12.
7. The SS activates SCell2 by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe (m+10).
8. The UE shall start sending CSI reports for SCell1 and the SS shall monitor CSI reports for SCell1 sent from the UE and ACK/NACK sent in PCell during SCells activation.
   - If the first CSI report for SCell1 is received by the SS in a subframe (m+8),
     - or (m+10) if the subframe (m+8) was subject to interruption,

- and CSI report with non-zero CQI index is received by the SS earlier than or equal to subframe (m+39),
  - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+39)
- and DTX is not observed by the SS anywhere outside the subframes (m+5) to (m+13) and the subframes (m+15) to (m+23) up to the end of T2,
- and ≤2 non-consecutive DTX subframes are observed by the SS anywhere during the subframes (m+5) to (m+13) and ≤2 non-consecutive DTX subframes are observed by the SS during the subframes (m+15) to (m+23) up to the end of T2,
- Then the number of successes for the event "Activation" is increased by one. Otherwise, count a fail for the event "Activation" and go to step 12.

9. When T2 expires, the SS deactivate SCC1 and SCC2 by sending the deactivation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe # denoted n which is an even number. If the SS receives ACK for MAC-CE sent by the UE, T3 starts in subframe n, and the test proceeds to step 10, otherwise go to step 12.
10. The SS deactivates SCell2 by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe (n+10).
11. The UE shall stop sending CSI reports for SCell1 and the SS shall monitor CSI reports for SCell1 sent from the UE and ACK/NACK sent in PCell during SCell1 deactivation.
    - If the last CSI report is received by the SS earlier than or equal to subframe (n+8),
    - and DTX is not observed by the SS anywhere outside the subframes (m+5) to (m+13) and the subframes (m+15) to (m+23) up to the end of T2,
    - and ≤2 non-consecutive DTX subframes are observed by the SS anywhere during the subframes (n+5) to (n+13) and the subframes (n+15) to (n+23) up to the end of T3.
    - Then the number of successes for the event "Deactivation" is increased by one. Otherwise, count a fail for the event "Deactivation".
12. When T3 expires or Activation in step 6 was not acknowledged, or a fail was counted for the event "Activation" in step 8, or Deactivation in step 9 was not acknowledged, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
13. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
14. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3,
15. Repeat steps 2-14 until a test verdict has been achieved,

Each of the events "Activation" and "Deactivation" is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.16.39.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.39.4.3-1: Common Exception messages for E-UTRAN TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in FDD**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | |

**Table 8.16.39.4.3-2: *PhysicalConfigDedicated-DEFAULT: Additional E-UTRAN* TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in FDD**

| Derivation Path: TS 36.508 [7] clause 4.8.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | CQI-ReportConfig-DEFAULT using condition CQI_PERIODIC | | RBC |
| cqi-ReportConfig-r10 | Not present | | SCell_AddMod |
| } | | | |

**Table 8.16.39.4.3-3: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in FDD**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 158 | | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | NULL | | |
| simultaneousAckNackAndCQI | TRUE | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.39.4.3-4: *PhysicalConfigDedicatedSCell-r10-DEFAULT: Additional E-UTRAN* TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in FDD**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicatedSCell-r10-DEFAULT ::= SEQUENCE { | | | |
| ul-Configuration-r10 SEQUENCE { | | | |
| antennaInfoUL-r10 | Not present | | |
| pusch-ConfigDedicatedSCell-r10 | Not present | | |
| uplinkPowerControlDedicatedSCell-r10 | Not present | | |
| cqi-ReportConfigSCell-r10 | CQI-ReportConfigSCell-r10-DEFAULT | | SCell1 |
| soundingRS-UL-ConfigDedicated-r10 | Not present | | |
| soundingRS-UL-ConfigDedicated-v1020 | Not present | | |
| soundingRS-UL-ConfigDedicatedAperiodic-r10 | Not present | | |
| } | | | |
| } | | | |

**Table 8.16.39.4.3-5: *CQI-ReportConfigSCell-r10-DEFAULT: Additional E-UTRAN* TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in FDD**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfigSCell-r10 ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic-r10 | Not present | | |
| nomPDSCH-RS-EPRE-Offset-r10 | 0 | | |
| cqi-ReportPeriodicSCell-r10 | | | |
| | *CQI-ReportPeriodic-r10-DEFAULT* | | CQI_PERIODIC |
| pmi-RI-Report-r10 | Not present | | |
| } | | | |

**Table 8.16.39.4.3-6: *CQI-ReportPeriodic-r10-DEFAULT: Additional E-UTRAN* TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in FDD**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportPeriodic-r10 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex-r10 | 0 | | |
| cqi-PUCCH-ResourceIndexP1-r10 | Not present | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | Cell2 |
| cqi-FormatIndicatorPeriodic-r10 CHOICE { | | | |
| widebandCQI-r10 SEQUENCE { | | | |
| csi-ReportMode-r10 | Not present | | |
| } | | | |
| ri-ConfigIndex | 484 | (see Table 7.2.2-1B in TS 36.213) | Cell2 |
| simultaneousAckNackAndCQI | TRUE | | |
| cqi-Mask-r9 | Not present | | |
| csi-ConfigIndex-r10 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.39.4.3-7: *MeasObjectEUTRA-GENERIC:* Additional E-UTRAN TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in FDD**

| Derivation Path: TS 36.508 [7] clause 4.6.6 Table 4.6.6-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | Cell 2, Cell 3 |
| } | | | |

**Table 8.16.39.4.3-8: *SystemInformationBlockType2:* Additional E-UTRAN TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in FDD**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-1 SystemInformationBlockType2 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfig | Not present | | Cell 1 |
| } | | | |

## 8.16.39.5 Test requirement

Table 8.16.39.5-1 defines the primary level settings including [test tolerances for E-UTRAN TDD-FDD 3DL CA activation and deactivation of Unknown SCell in non-DRX with PCell in FDD test.

**Table 8.16.39.5-1: Cell specific test parameters for E-UTRAN TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in FDD**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |

| E-UTRA RF Channel Number | | 1 | 2 | | 3 |
|---|---|---|---|---|---|
| TDD special subframe configuration | | - | 6 | | 6 |
| TDD uplink-downlink configuration | | - | 1 | | 1 |
| $BW_{channel}$ | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 |
| PDSCH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.7 FDD<br>10MHz: R.3 FDD<br>20MHz: R.6 FDD | - | | - |
| PCFICH/PDCCH/PHICH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD |
| OCNG Patterns | | 5MHz: OP.20 FDD<br>10MHz: OP.10 FDD<br>20MHz: OP.17 FDD | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD |
| PBCH_RA | dB | 0 | 0 | | 0 |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -104 | -104 | | -104 |
| $\hat{E}_s/N_{oc}$ | dB | 17 | -infinity | 17 | 17 |
| $\hat{E}_s/I_{ot}$ | dB | 17 | -infinity | 17 | 17 |
| RSRP Note 3 | dBm/15 kHz | -87 | -infinity | -87 | -87 |
| SCH_RP Note 3 | dBm/15 kHz | -87 | -infinity | -87 | -87 |
| Io Note 3 | dBm/Ch BW | -59.13 +10log ($N_{RB,c}$ /50) | -76.22 +10log ($N_{RB,c}$ /50) | -59.13 +10log ($N_{RB,c}$ /50) | -59.13 +10log ($N_{RB,c}$ /50) |
| Propagation Condition | | AWGN | AWGN | | AWGN |
| Antenna Configuration | | 1x2 | 1x2 | | 1x2 |
| Timing offset to Cell 1 | µs | - | 0 | | 0 |
| Time alignment error relative to cell 1 Note 5 | µs | - | ≤ TAE | | ≤ TAE |
| Time alignment error relative to cell 2 Note 5 | µs | - | - | | ≤ TAE |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: Es/Iot, RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The uplink resources for CSI reporting are assigned to the UE prior to the start of time period T2.
Note 5: Time alignment error (TAE) as specified in TS 36.104 [30] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.
Note 6: TDD special subframe configuration and uplink-downlink configuration are as specified in Table 4.2-1 and 4.2-2 in TS36.211 [16]. The same configuration applies to all TDD cells.

During T2 interruption of PCell during SCells activation shall not happen outside the subframes (m+5) to (m+9) and outside the subframes (m+15) to (m+19).

Figure 8.16.39.5-1 shows the derivation of the Test procedure requirements for DTX during T2, based on the core requirements for interruption.

NOTE: The differences between activation and deactivation are as follows:

- Activation ➔ Deactivation
- First CSI ➔ Last CSI
- Latest valid CSI report ➔ Not exist (no need to check since CSI report was already stopped)

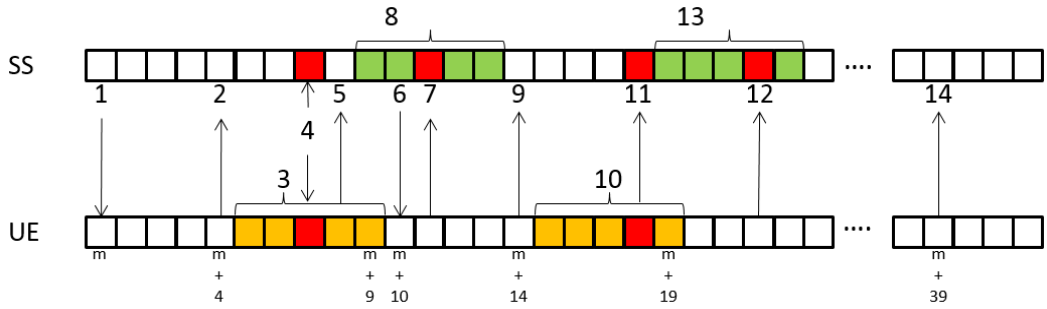


**Figure 8.16.39.5-1: Procedure derivation for Activation (the case where the band relation between PCC and SCells are inter-band)**

1) Activation command for SCell1 (T2 starts)
2) ACK for MAC-CE for SCell1 activation
3) Possible interruption period by SCell1 activation
4) Allowed interruption timing (1 subframe in m+5~m+9) on UE by SCell1 activation, that is, possible DTX timing on SS
5) First CSI report timing (could be invalid CQI) at m+8 (might be m+10 due to interruption)
6) Activation command for SCell2
7) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell1 activation
8) Possible DTX reception period on SS due to interruption by SCell1 activation
9) ACK for MAC-CE for SCell2 activation
10) Possible interruption period by SCell2 activation
11) Allowed interruption timing (1 subframe in m+15~m+19) on UE by SCell2 activation, that is, possible DTX timing on SS
12) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell2 activation
13) Possible DTX reception period on SS due to interruption by SCell2 activation
14) Latest valid CSI report timing (valid CQI)

During T3 interruption of PCell during SCells deactivation shall not happen outside the subframes (n+5) to (n+9) and outside the subframes (n+15) to (n+19).

The derivation of the Test procedure requirements for DTX during T3 is based on the core requirements for interruption, in a similar way to T2.

All of the above test requirements shall be fulfilled in order for the observed SCell1 activation delay and SCell1 deactivation delay to be counted as correct. The rate of correct observed SCell1 activation delay and SCell1 deactivation delay during repeated tests shall be at least 90%.

The statistical pass/ fail decisions are done separately for activation and deactivation.

Decide the test pass, if activation and deactivation are passed, otherwise fail the UE.

## 8.16.40 E-UTRA TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in TDD

### 8.16.40.1 Test purpose

To verify that SCell activation and deactivation times are within the requirements in TS 36.133 [4] section 7.7 for UE configured with two downlink SCells, when the SCell is unknown by the UE at the time of activation.

### 8.16.40.2 Test applicability

This test applies to all types of E-UTRA FDD and TDD Release 12 and forward UE that support E-UTRA FDD and TDD and 3DL CA with FDD as PCell. Applicability requires support for FGI bit 25.

### 8.16.40.3 Minimum conformance requirements

While activating a SCell if the UE does not receive any command to activate, deactivate, configure or deconfigure the other SCell during the SCell activation delay then the UE shall meet the SCell activation delay requirements specified in TS 36.133 [4] section 7.7.2.

While activating a SCell if the other SCell is activated, deactivated, configured or deconfigured by the UE then the UE shall meet the SCell activation delay requirements ($T_{activate_total}$) according to the following expression:

$$T_{activate_total} = T_{activate_basic} + K*5$$

Where:

$T_{activate_total}$ is the total time to activate a SCell and is expressed in subframes.

$T_{activate_basic}$ is the SCell activation delay specified in TS 36.133 [4] section 7.7.2;

K ($1 \leq K \leq [3]$) is the number of times the other SCell is activated, deactivated, configured or deconfigured while the SCell is being activated;

While activating a SCell:

- The interruption on the PCell and/or on the activated SCell due to the SCell activation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+11 if:
  - The PCell and/or the activated SCell being interrupted and the SCell being activated belong to E-UTRA TDD, or
  - The activated SCell being interrupted and the SCell being activated belong to E-UTRA FDD and the PCell belongs to E-UTRA TDD.
- Otherwise, the interruption on PCell and/or on the activated SCell due to the SCell activation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+9.

Starting from the subframe specified in section 4.3 of TS 36.213 [8] and until the UE has completed the SCell activation, the UE shall send CSI with CQI index = 0 (out of range) if the UE is configured to report the CQI for the SCell being activated.

The UE shall deactivate a SCell and meet the SCell deactivation delay requirements specified in TS 36.133 [4] section 7.7.3 regardless of whether the other SCell is activated, deactivated, configured or deconfigured or not by the UE during the SCell deactivation delay.

While deactivating a SCell:

- The interruption on the PCell and/or on the activated SCell due to the SCell deactivation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+11 if:

- The PCell and/or the activated SCell being interrupted and the SCell being deactivated belong to E-UTRA TDD or

- The activated SCell being interrupted and the SCell being deactivated belong to E-UTRA FDD and the PCell belongs to E-UTRA TDD.

- Otherwise, the interruption on PCell and/or the activated SCell due to the SCell deactivation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+9.

The normative reference for this requirement is TS 36.133 [4] clause 7.7 and A.8.16.40.

## 8.16.40.4 Test description

### 8.16.40.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.16.40.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.63 as appropriate.

2. The general test parameter settings are set up according to Table 8.16.40.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.16.40.4.3.

5. Cell 1 is the PCell on the primary component carrier, Cell 2 is the Deconfigured deactivated SCell on RF channel number 2, and Cell 3 is Configured deactivated SCell on RF channel number 3. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.40.4.1-1: General test parameters for unknown SCell1 activation case**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1, 2, 3 | Three radio channels are used for this test |
| Active PCell | | Cell 1 | Primary cell on RF channel number 1. |
| Deconfigured deactivated SCell1 | | Cell 2 | Deconfigured deactivated secondary cell on RF channel number 2. |
| Configured deactivated SCell2 | | Cell 3 | Configured deactivated secondary cell on RF channel number 3. |
| CP length | | Normal | |
| DRX | | OFF | Continuous monitoring of primary cell |
| CQI/PMI periodicity and offset configuration index | | 0 | CQI reporting for SCell every second subframe |
| Cell-individual offset for cells on RF channel number 1 | dB | 0 | Individual offset for cells on PCC. |
| Cell-individual offset for cells on RF channel number 2 | dB | 0 | Individual offset for cells on SCC1. |
| Cell-individual offset for cells on RF channel number 3 | dB | 0 | Individual offset for cells on SCC2. |
| SCell measurement cycle (measCycleSCell) | ms | 320 | |
| T1 | ms | 100 | During this time the PCell and SCell2 shall be known and the SCell1 configured. |
| T2 | s | 1 | During this time the UE shall activate the SCell1 and SCell2. |
| T3 | s | 1 | During this time the UE shall deactivate the SCell1 and SCell2. |

### 8.16.40.4.2 Test procedure

The test consists of three successive time periods, with duration of T1, T2 and T3, respectively. There are three carriers, each with one cell. Cell 1 is FDD cell, and Cell 2 and Cell 3 are TDD cells. Cell 1 and Cell 3 have constant signal levels throughout the test.

Before the test starts the UE is connected to Cell 1 (PCell) on radio channel 1 (PCC) and Cell 3 (deactivated SCell2) on radio channel 3 (SCC2), but is not aware of Cell 2 (SCell1) on radio channel 2 (SCC1). The UE is only monitoring the PCC and SCC2. The UE shall be continuously scheduled in the PCell throughout the whole test.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC2 according to Annex C.0, C.1 for all downlink physical channels.

3. The SS shall configure SCell (Cell 3) on the SCC2 as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.40.4.3.

4. The SS shall configure transmission of PDSCH with a maximum number of 1 HARQ transmission.

5. Set the parameters according to T1 in Table 8.16.40.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts. Configure SCC1 according to Annex C.0, C.1 for all downlink physical channels. The SS shall configure SCell1 (Cell 2) on the SCC1 as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.40.4.3.

6. The SS activates SCell1 by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe # denoted m is 4 or 9.If the SS receives ACK for MAC-CE sent by the UE, T2 starts in subframe m, and the test proceeds to step 7, otherwise go to step12.

7. The SS activates SCell2 by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe (m+15).

8. The UE shall start sending CSI reports for SCell1 and the SS shall monitor CSI reports for SCell1 sent from the UE and ACK/NACK sent in PCell during SCell1 activation.

   - If the first CSI report for SCell1 is received by the SS in a subframe (m+8),

- or (m+9) if the subframe (m+8) was subject to interruption,

- and CSI report with non-zero CQI index is received by the SS earlier than or equal to subframe (m+39),
  - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+39)

- and DTX is not observed by the SS outside the subframes (m+5), (m+9), (m+20) and (m+24), up to the end of T2,
  - or DTX is not observed by the SS outside the subframes (m+6), (m+13), (m+21) and (m+28), up to the end of T2,
  - or DTX is not observed by the SS outside the subframes (m+7), (m+13), (m+22) and (m+28), up to the end of T2,
  - or DTX is not observed by the SS outside the subframes (m+8) and (m+23), up to the end of T2,
  - or DTX is not observed by the SS outside the subframes (m+9) and (m+24), up to the end of T2,
  - or DTX is not observed by the SS outside the subframes (m+10), (m+14), (m+25) and (m+29), up to the end of T2,
  - or DTX is not observed by the SS outside the subframes (m+11), (m+18), (m+26) and (m+33), up to the end of T2.

- Then the number of successes for the event “Activation” is increased by one, Otherwise, count a fail for the event “Activation” and go to step 12.

9. When T2 expires, the SS deactivates SCell1 by sending the deactivation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8), in a subframe # denoted n, where n is 4 or 9. If the SS receives ACK for MAC-CE sent by the UE, T3 starts in subframe n, and the test proceeds to step 10, otherwise go to step 12.

10. The SS deactivates SCell2 by sending the deactivation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8), in subframe (n+10).

11. The UE shall stop sending CSI reports for SCell1 and the SS shall monitor CSI reports for SCell1 sent from the UE and ACK/NACK sent in PCell during SCell deactivation.

    - If the last CSI report is received by the SS earlier than or equal to subframe (n+8),

    - and DTX is not observed by the SS outside the subframes (n+5), (n+9), (n+20) and (n+24), up to the end of T3,
      - or DTX is not observed by the SS outside the subframes (n+6), (n+13), (n+21) and (n+28), up to the end of T3,
      - or DTX is not observed by the SS outside the subframes (n+7), (n+13), (n+22) and (n+28), up to the end of T3,
      - or DTX is not observed by the SS outside the subframes (n+8) and (n+23), up to the end of T3,
      - or DTX is not observed by the SS outside the subframes (n+9) and (n+24), up to the end of T3,
      - or DTX is not observed by the SS outside the subframes (n+10), (n+14), (n+25) and (n+29), up to the end of T3,
      - or DTX is not observed by the SS outside the subframes (n+11), (n+18), (n+26) and (n+33), up to the end of T3,

    - Then the number of successes for the event “Deactivation” is increased by one. Otherwise, count a fail for the event “Deactivation”.

12. When T3 expires or Activation in step 6 was not acknowledged, or a fail was counted for the event “Activation” in step 8, or Deactivation in step 9 was not acknowledged, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

13. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

14. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3,

15. Repeat steps 2-14 until a test verdict has been achieved,

Each of the events “Activation” and “Deactivation” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.16.40.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.40.4.3-1: Common Exception messages for E-UTRAN TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in TDD**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | |

**Table 8.16.40.4.3-2: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in TDD**

| Derivation Path: TS 36.508 [7] clause 4.8.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | CQI-ReportConfig-DEFAULT using condition CQI_PERIODIC | | RBC |
| cqi-ReportConfig-r10 | Not present | | SCell_Add Mod |
| } | | | |

**Table 8.16.40.4.3-3: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in TDD**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 159 | | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | NULL | | |
| simultaneousAckNackAndCQI | TRUE | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.40.4.3-4: *PhysicalConfigDedicatedSCell-r10-DEFAULT:* Additional E-UTRAN TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in TDD**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicatedSCell-r10-DEFAULT ::= SEQUENCE { | | | |
| ul-Configuration-r10 SEQUENCE { | | | |
| antennaInfoUL-r10 | Not present | | |
| pusch-ConfigDedicatedSCell-r10 | Not present | | |
| uplinkPowerControlDedicatedSCell-r10 | Not present | | |
| cqi-ReportConfigSCell-r10 | CQI-ReportConfigSCell-r10-DEFAULT | | SCell1 |
| soundingRS-UL-ConfigDedicated-r10 | Not present | | |
| soundingRS-UL-ConfigDedicated-v1020 | Not present | | |
| soundingRS-UL-ConfigDedicatedAperiodic-r10 | Not present | | |
| } | | | |
| } | | | |

**Table 8.16.40.4.3-5: *CQI-ReportConfigSCell-r10-DEFAULT:* Additional E-UTRAN TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in TDD**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfigSCell-r10 ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic-r10 | Not present | | |
| nomPDSCH-RS-EPRE-Offset-r10 | 0 | | |
| cqi-ReportPeriodicSCell-r10 | | | |
| | *CQI-ReportPeriodic-r10-DEFAULT* | | CQI_PERIODIC |
| pmi-RI-Report-r10 | Not present | | |
| } | | | |

**Table 8.16.40.4.3-6: *CQI-ReportPeriodic-r10-DEFAULT:* Additional E-UTRAN TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in TDD**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportPeriodic-r10 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex-r10 | 0 | | |
| cqi-PUCCH-ResourceIndexP1-r10 | Not present | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | Cell2 |
| cqi-FormatIndicatorPeriodic-r10 CHOICE { | | | |
| widebandCQI-r10 SEQUENCE { | | | |
| csi-ReportMode-r10 | Not present | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | Cell2 |
| simultaneousAckNackAndCQI | TRUE | | |
| cqi-Mask-r9 | Not present | | |
| csi-ConfigIndex-r10 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.40.4.3-7: *MeasObjectEUTRA-GENERIC:* Additional E-UTRAN TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in TDD**

| Derivation Path: TS 36.508 [7] clause 4.6.6 Table 4.6.6-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | Cell 2, Cell 3 |
| } | | | |

## 8.16.40.5 Test requirement

Table 8.16.40.5-1 defines the primary level settings including [test tolerances for E-UTRAN TDD-FDD 3DL CA activation and deactivation of Unknown SCell in non-DRX with PCell in TDD test.

**Table 8.16.40.5-1: Cell specific test parameters for E-UTRAN TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in TDD**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |

| E-UTRA RF Channel Number | | 1 | 2 | | 3 |
|---|---|---|---|---|---|
| TDD special subframe configuration | | 6 | - | | - |
| TDD uplink-downlink configuration | | 1 | - | | - |
| $BW_{channel}$ | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 |
| PDSCH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | - | | - |
| PCFICH/PDCCH/PHICH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD |
| OCNG Patterns | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD |
| PBCH_RA | dB | 0 | 0 | | 0 |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -104 | -104 | | -104 |
| $\hat{E}_s/N_{oc}$ | dB | 17 | -infinity | 17 | 17 |
| $\hat{E}_s/I_{ot}$ | dB | 17 | -infinity | 17 | 17 |
| RSRP Note 3 | dBm/15 kHz | -87 | -infinity | -87 | -87 |
| SCH_RP Note 3 | dBm/15 kHz | -87 | -infinity | -87 | -87 |
| Io Note 3 | dBm/Ch BW | -59.13 +10log ($N_{RB,c}$ /50) | -76.22 +10log ($N_{RB,c}$ /50) | -59.13 +10log ($N_{RB,c}$ /50) | -59.13 +10log ($N_{RB,c}$ /50) |
| Propagation Condition | | AWGN | AWGN | | AWGN |
| Antenna Configuration | | 1x2 | 1x2 | | 1x2 |
| Timing offset to Cell 1 | µs | - | 0 | | 0 |
| Time alignment error relative to cell 1 Note 5 | µs | - | ≤ TAE | | ≤ TAE |
| Time alignment error relative to cell 2 Note 5 | µs | - | - | | ≤ TAE |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: Es/Iot, RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The uplink resources for CSI reporting are assigned to the UE prior to the start of time period T2.
Note 5: Time alignment error (TAE) as specified in TS 36.104 [30] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.
Note 6: TDD special subframe configuration and uplink-downlink configuration are as specified in Table 4.2-1 and 4.2-2 in TS36.211 [16]. The same configuration applies to all TDD cells.

During T2 interruption of PCell during SCells activation shall not happen outside the subframes (m+5) to (m+11) and outside the subframes (m+20) to (m+26).

Figure 8.16.40.5-1 shows the derivation of the Test procedure requirements for DTX during T2, based on the core requirements for interruption.

NOTE: The differences between activation and deactivation are as follows:

- Activation ➔ Deactivation
- First CSI ➔ Last CSI
- Latest valid CSI report ➔ Not exist (no need to check since CSI report was already stopped).

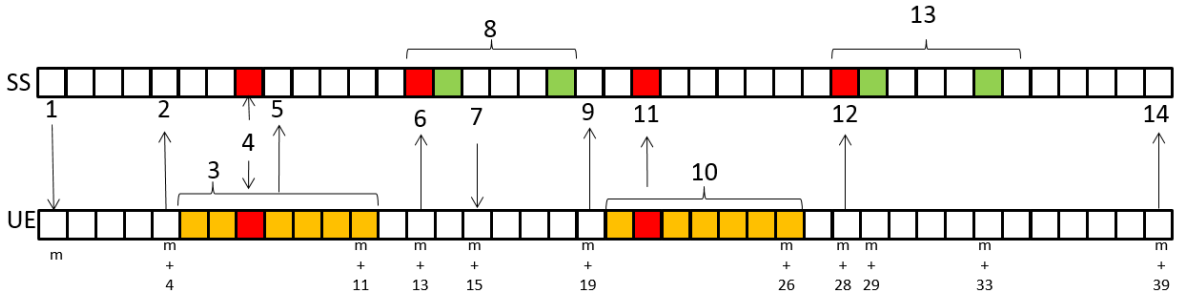


**Figure 8.16.40.5-1: Procedure derivation for Activation (the case where the band relation between PCC and SCells are inter-band)**

1) Activation command for SCell1(T2 starts)
2) ACK for MAC-CE for SCell1 activation
3) Possible interruption period by SCell1 activation
4) Allowed interruption timing(1 subframe in m+5~m+11) on UE by SCell1 activation, that is, possible DTX timing on SS
5) First CSI report timing(invalid CQI) at m+8 (might be m+9 or m+13 due to interruption)
6) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell1 activation
7) Activation command for SCell2
8) Possible DTX reception period on SS due to interruption by SCell1 activation (only m+13, m+14 or m+18 are allowed)
9) ACK for MAC-CE for SCell2 activation
10) Possible interruption period by SCell2 activation
11) Allowed interruption timing(1 subframe in m+20~m+26) on UE by SCell2 activation, that is, possible DTX timing on SS
12) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell2 activation (only m+28, m+29 or m+33 are allowed)
13) Possible DTX reception period on SS due to interruption by SCell2 activation
14) Latest valid CSI report timing(valid CQI)

During T3 interruption of PCell during SCells deactivation shall not happen outside the subframes (n+5) to (n+11) and outside the subframes (n+20) to (n+26).

The derivation of the Test procedure requirements for DTX during T3 is based on the core requirements for interruption, in a similar way to T2.

All of the above test requirements shall be fulfilled in order for the observed SCell1 activation delay and SCell1 deactivation delay to be counted as correct. The rate of correct observed SCell1 activation delay and SCell1 deactivation delay during repeated tests shall be at least 90%.

The statistical pass/ fail decisions are done separately for activation and deactivation.

Decide the test pass, if activation and deactivation are passed, otherwise fail the UE.

## 8.16.41 3 DL FDD CA activation and deactivation of unknown SCell in non-DRX

### 8.16.41.1 Test purpose

To verify that the SCell activation and deactivation times are within the requirements stated in TS 36.133 [4] clause 7.7 for 3DL FDD carrier aggregation, when the SCell is unknown by the UE at the time of activation.

### 8.16.41.2 Test applicability

This test applies to all types of E-UTRA UE FDD release 10 and forward that support 3DL intra-band contiguous CA or 3DL inter-band CA, or 3DL intra-band contiguous with inter-band CA. Applicability requires support for FGI bit 25.

This test case also applies to all types of E-UTRA FDD UE release 11 and forward that support 3DL intra-band non-contiguous and inter-band CA, or 3DL intra-band non-contiguous with intra-band contiguous CA. Applicability requires support for FGI bit 25.

### 8.16.41.3 Minimum conformance requirements

While activating a SCell if the UE does not receive any command to activate, deactivate, configure or deconfigure the other SCell during the SCell activation delay then the UE shall meet the SCell activation delay requirements specified in TS 36.133 [4] section 7.7.2.

While activating a SCell if the other SCell is activated, deactivated, configured or deconfigured by the UE then the UE shall meet the SCell activation delay requirements ($T_{activate_total}$) according to the following expression:

$$T_{activate_total} = T_{activate_basic} + K*5$$

Where:

$T_{activate_total}$ is the total time to activate a SCell and is expressed in subframes.

$T_{activate_basic}$ is the SCell activation delay specified in TS 36.133 [4] section 7.7.2;

K ($1 \leq K \leq [3]$) is the number of times the other SCell is activated, deactivated, configured or deconfigured while the SCell is being activated;

While activating a SCell:

- The interruption on the PCell and/or on the activated SCell due to the SCell activation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+11 if:
  - The PCell and/or the activated SCell being interrupted and the SCell being activated belong to E-UTRA TDD, or
  - The activated SCell being interrupted and the SCell being activated belong to E-UTRA FDD and the PCell belongs to E-UTRA TDD.
- Otherwise, the interruption on PCell and/or on the activated SCell due to the SCell activation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+9.

Starting from the subframe specified in section 4.3 of TS 36.213 [8] and until the UE has completed the SCell activation, the UE shall send CSI with CQI index = 0 (out of range) if the UE is configured to report the CQI for the SCell being activated.

The UE shall deactivate a SCell and meet the SCell deactivation delay requirements specified in TS 36.133 [4] section 7.7.3 regardless of whether the other SCell is activated, deactivated, configured or deconfigured or not by the UE during the SCell deactivation delay.

While deactivating a SCell:

- The interruption on the PCell and/or on the activated SCell due to the SCell deactivation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+11 if:
  - The PCell and/or the activated SCell being interrupted and the SCell being deactivated belong to E-UTRA TDD or
  - The activated SCell being interrupted and the SCell being deactivated belong to E-UTRA FDD and the PCell belongs to E-UTRA TDD.
- Otherwise, the interruption on PCell and/or the activated SCell due to the SCell deactivation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+9.

The normative reference for this requirement is TS 36.133 [4] clause 7.7 and A.8.16.41.

## 8.16.41.4 Test description

### 8.16.41.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.16.41.5-1 as defined in TS 36.508 [7] clause 4.3.1, for different CA configurations as defined in TS 36.521-1 [10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.63 as appropriate.
2. The general test parameter settings are set up according to Table 8.16.41.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.16.41.4.3.
5. Cell 1 is the PCell on the primary component carrier, Cell 2 is the SCell on RF channel number 2, and Cell 3 is the deactivated SCell on RF channel number 3. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.41.4.1-1: General test parameters for E-UTRAN FDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1, 2, 3 | Three radio channels are used for this test |
| Active PCell | | Cell 1 | Primary cell on RF channel number 1. |
| Deconfigured SCell1 | | Cell 2 | Deconfigured secondary cell on RF channel number 2. |
| Configured deactivated SCell2 | | Cell 3 | Configured deactivated secondary cell on RF channel number 3. |
| CP length | | Normal | |
| DRX | | OFF | Continuous monitoring of primary cell |
| CQI/PMI periodicity and offset configuration index | | 0 | CQI reporting for SCells every second subframe |
| Cell-individual offset for cells on RF channel number 1 | dB | 0 | Individual offset for cells on PCC. |
| Cell-individual offset for cells on RF channel number 2 | dB | 0 | Individual offset for cells on SCC1. |
| Cell-individual offset for cells on RF channel number 3 | dB | 0 | Individual offset for cells on SCC2. |
| SCell measurement cycle (measCycleSCell) | ms | 320 | |
| T1 | ms | 100 | During this time the PCell and SCell2 shall be known and the SCell1 configured. |
| T2 | s | 1 | During this time the UE shall activate the SCell1 and SCell2. |
| T3 | s | 1 | During this time the UE shall deactivate the SCell1 and SCell2. |

### 8.16.41.4.2 Test procedure

The test consists of three successive time periods, with duration of T1, T2 and T3, respectively. There are three carriers, each with one cell. Cell 1 and Cell 3 have constant signal levels throughout the test.

Before the test starts the UE is connected to Cell 1 (PCell) on radio channel 1 (PCC) and Cell 3 (deactivated SCell2) on radio channel 3 (SCC2), but is not aware of Cell 2 (SCell1) on radio channel 2 (SCC1). The UE is only monitoring the PCC and SCC2. The UE shall be continuously scheduled in the PCell throughout the whole test.

During T2 the Test procedure requires the UE to send the first CSI report for SCell1 in a subframe (m+8), but also allows a subframe (m+10) if the subframe (m+8) was subject to interruption. The SS determines whether the CSI report in subframe (m+8) was interrupted or not by monitoring ACK/NACK sent in PCell in subframe (m+8).

NOTE: The requirements are dependant on the CA configuration (e.g. intra band, inter band; see TS 36.133 [4], clause 7.8) hence the test procedure has been divided into three variants, found in steps 8a, 8b and 8c and in steps 11a, 11b and 11c. Depending on the CA configuration under test only one of those shall be run respectively.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC2 according to Annex C.0, C.1 for all downlink physical channels.

3. The SS shall configure SCell2 (Cell 3) on the SCC2 as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.41.4.3.

4. The SS shall configure transmission of PDSCH with a maximum number of 1 HARQ transmission.

5. Set the parameters according to T1 in Table 8.16.41.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts. Configure SCC1 according to Annex C.0, C.1 for all downlink physical channels. The SS shall configure SCell1 (Cell 2) on the SCC1 as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.41.4.3.

6. The SS activates SCell1 by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe # denoted m, which is an even number. If the SS receives ACK for MAC-CE sent by the UE, T2 starts in subframe m, and the test proceeds to step 7, otherwise go to step 12.

7. The SS activates SCell2 by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe (m+10).

8. The UE shall start sending CSI reports for SCell and the SS shall monitor CSI reports for SCell sent from the UE and ACK/NACK sent in PCell during SCell activation..

   8a. For all intra-band case

   - If the first CSI report for SCell1 is received by the SS in a subframe (m+8),
     - or (m+10) if the subframe (m+8) was subject to interruption
     - and CSI report with non-zero CQI index is received by the SS earlier than or equal to subframe (m+39),
     - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+39)
   - and DTX is not observed by the SS anywhere outside the subframes (m+5) to (m+13) and the subframes (m+15) to (m+23) up to the end of T2.
   - Then the number of successes for the event “Activation” is increased by one, Otherwise, count a fail for the event “Activation” and go to step 12.

   8b. For the case where the band relation between PCC and SCells are inter-band

   - If the first CSI report for SCell1 is received by the SS in a subframe (m+8)
     - or (m+10) if the subframe (m+8) was subject to interruption,
   - and CSI report with non-zero CQI index is received by the SS earlier than or equal to subframe (m+39),
     - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+39)
   - and DTX is not observed by the SS anywhere outside the subframes (m+5) to (m+13) and the subframes (m+15) to (m+23) up to the end of T2,
   - and $\leq$2 non-consecutive DTX subframes are observed by the SS during the subframes (m+5) to (m+13) and $\leq$2 non-consecutive DTX subframes are observed by the SS during the subframes (m+15) to (m+23),
   - Then the number of successes for the event “Activation” is increased by one, Otherwise, count a fail for the event “Activation” and go to step 12.

   8c. For the case where the band relation between PCC and SCell1 is intra-band and SCell2 is inter-band

   - If the first CSI report for SCell1 is received by the SS in a subframe (m+8),
     - or (m+10) if the subframe (m+8) was subject to interruption
   - and CSI report with non-zero CQI index is received by the SS earlier than or equal to subframe (m+39),
     - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+39)
   - and DTX is not observed by the SS anywhere outside the subframes (m+5) to (m+13) and the subframes (m+15) to (m+23) up to the end of T2,
   - and $\leq$2 non-consecutive DTX subframes are observed by the SS during the subframes (m+15) to (m+23),
   - Then the number of successes for the event “Activation” is increased by one, Otherwise, count a fail for the event “Activation” and go to step 12.

9. When T2 expires, the SS deactivates SCell1 by sending the deactivation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe # denoted n which is an even number. If the SS receives ACK for MAC-CE sent by the UE, T3 starts in subframe n, and the test proceeds to step 10, otherwise go to step 12.

10. The SS deactivates SCell2 by sending the deactivation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8), in subframe (n+10).

11. The UE shall stop sending CSI reports for SCell1 and the SS shall monitor CSI reports for SCell1 sent from the UE and ACK/NACK sent in PCell during SCell1 deactivation.

    11a. For all intra-band case

    - If the last CSI report is received by the SS earlier than or equal to subframe (n+8),
    - and DTX is not observed by the SS anywhere outside the subframes (n+5) to (n+13) and the subframes (n+15) to (n+23) up to the end of T3,
    - Then the number of successes for the event “Deactivation” is increased by one. Otherwise, count a fail for the event “Deactivation”.

    11b. For the case where the band relation between PCC and SCells are inter-band

    - If the last CSI report is received by the SS earlier than or equal to subframe (n+8),
    - and DTX is not observed by the SS anywhere outside the subframes (n+5) to (n+13) and the subframes (n+15) to (n+23) up to the end of T3,
    - and ≤2 non-consecutive DTX subframes are observed by the SS anywhere during the subframes (n+5) to (n+13) and ≤2 non-consecutive DTX subframes are observed by the SS during the subframes (n+15) to (n+23)
    - Then the number of successes for the event “Deactivation” is increased by one. Otherwise, count a fail for the event “Deactivation”.

    11c. For the case where the band relation between PCC and SCell1 is intra-band and SCell2 is inter-band,

    - If the last CSI report is received by the SS earlier than or equal to subframe (n+8),
    - and DTX is not observed by the SS anywhere outside the subframes (n+5) to (n+13) and the subframes (n+15) to (n+23) up to the end of T3,
    - and ≤2 non-consecutive DTX subframes are observed by the SS the subframes (n+15) to (n+23),
    - Then the number of successes for the event “Deactivation” is increased by one. Otherwise, count a fail for the event “Deactivation”.

12. When T3 expires, or Activation in step 6 was not acknowledged, or a fail was counted for the event “Activation” in step 8a, 8b or 8c, or Deactivation in step 9 was not acknowledged, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

13. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

14. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3,

15. Repeat steps 2-14 until a test verdict has been achieved,

Each of the events “Activation” and “Deactivation” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.16.41.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.41.4.3-1: Common Exception messages for E-UTRAN FDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | |

**Table 8.16.41.4.3-2: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN FDD unknown SCell activation**

| Derivation Path: TS 36.508 [7] clause 4.8.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | CQI-ReportConfig-DEFAULT using condition CQI_PERIODIC | | RBC |
| cqi-ReportConfig-r10 | Not present | | SCell_AddMod |
| } | | | |

**Table 8.16.41.4.3-3: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN FDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 158 | | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | NULL | | |
| simultaneousAckNackAndCQI | TRUE | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.41.4.3-4: *PhysicalConfigDedicatedSCell-r10-DEFAULT:* Additional E-UTRAN FDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicatedSCell-r10-DEFAULT ::= SEQUENCE { | | | |
| ul-Configuration-r10 SEQUENCE { | | | |
| antennaInfoUL-r10 | Not present | | |
| pusch-ConfigDedicatedSCell-r10 | Not present | | |
| uplinkPowerControlDedicatedSCell-r10 | Not present | | |
| cqi-ReportConfigSCell-r10 | CQI-ReportConfigSCell-r10-DEFAULT | | SCell1 |
| soundingRS-UL-ConfigDedicated-r10 | Not present | | |
| soundingRS-UL-ConfigDedicated-v1020 | Not present | | |
| soundingRS-UL-ConfigDedicatedAperiodic-r10 | Not present | | |
| } | | | |
| } | | | |

**Table 8.16.41.4.3-5: *CQI-ReportConfigSCell-r10-DEFAULT:* Additional E-UTRAN FDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfigSCell-r10 ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic-r10 | Not present | | |
| nomPDSCH-RS-EPRE-Offset-r10 | 0 | | |
| cqi-ReportPeriodicSCell-r10 | | | |
| | *CQI-ReportPeriodic-r10-DEFAULT* | | CQI_PERIODIC |
| pmi-RI-Report-r10 | Not present | | |
| } | | | |

**Table 8.16.41.4.3-6: *CQI-ReportPeriodic-r10-DEFAULT:* Additional E-UTRAN FDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportPeriodic-r10 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex-r10 | 0 | | |
| cqi-PUCCH-ResourceIndexP1-r10 | Not present | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | Cell2 |
| cqi-FormatIndicatorPeriodic-r10 CHOICE { | | | |
| widebandCQI-r10 SEQUENCE { | | | |
| csi-ReportMode-r10 | Not present | | |
| } | | | |
| ri-ConfigIndex | 484 | (see Table 7.2.2-1B in TS 36.213) | Cell2 |
| simultaneousAckNackAndCQI | TRUE | | |
| cqi-Mask-r9 | Not present | | |
| csi-ConfigIndex-r10 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.41.4.3-7: *MeasObjectEUTRA-GENERIC:* Additional E-UTRAN FDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.6 Table 4.6.6-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | Cell 2, Cell 3 |
| } | | | |

**Table 8.16.41.4.3-8: *MeasObjectEUTRA-GENERIC:* Additional E-UTRAN FDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1, Cell 2, Cell 3 |
| } | | | |

**Table 8.16.41.4.3-9: *SystemInformationBlockType2:* Additional E-UTRAN FDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-1 SystemInformationBlockType2 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfig | Not present | | Cell 1 |
| } | | | |

**Table 8.16.41.4.3-10: *SystemInformationBlockType3:* Additional E-UTRAN FDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-2 SystemInformationBlockType3 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType3 ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1, Cell 2, Cell 3 |
| } | | | |

**Table 8.16.41.4.3-11: *SystemInformationBlockType5:* Additional E-UTRAN FDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 7.2.1.3, Table 7.2.1.3-3 SystemInformationBlockType5 exceptions | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType5 ::= SEQUENCE { | | | |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 1, Cell 2, Cell 3 |
| } | | | |

## 8.16.41.5 Test requirement

Table 8.16.41.5-1 defines the primary level settings including test tolerances for E-UTRAN FDD 3 DL CA activation and deactivation of unknown SCell in non-DRX.

**Table 8.16.41.5-1: Cell specific test parameters for E-UTRAN FDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | 3 | | |
| $BW_{channel}$ | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | |
| PDSCH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.7 FDD<br>10MHz: R.3 FDD<br>20MHz: R.6 FDD | | | - | | | - | | |
| PCFICH/PDCCH/PHICH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | |
| OCNG Patterns | | 5MHz: OP.20 FDD<br>10MHz: OP.10 FDD<br>20MHz: OP.17 FDD | | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA[Note 1] | dB | | | | | | | | | |
| OCNG_RB[Note 1] | dB | | | | | | | | | |
| $N_{oc}$[Note 2] | dBm/ 15 kHz | -104 | | | -104 | | | -104 | | |
| $\hat{E}_s/N_{oc}$ | dB | 17 | | | -infinity | 17 | | 17 | | |
| $\hat{E}_s/I_{ot}$ [Note 3] | dB | 17 | | | -infinity | 17 | | 17 | | |
| RSRP [Note 3] | dBm/ 15 kHz | -87 | | | -infinity | -87 | | -87 | | |
| SCH_RP [Note 3] | dBm/ 15 kHz | -87 | | | -infinity | -87 | | -87 | | |
| Io [Note 3] | dBm/ Ch BW | -59.13+10log ($N_{RB,c}$ /50) | | | -76.22 +10log ($N_{RB,c}$ /50) | -59.13+10log ($N_{RB,c}$ /50) | | -59.13+10log ($N_{RB,c}$ /50) | | |
| Propagation Condition | | AWGN | | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x2 | | | 1x2 | | | 1x2 | | |
| Timing offset to Cell 1 | µs | - | | | 0 | | | 0 | | |
| Time alignment error relative to cell 1 [Note 5] | µs | - | | | ≤ TAE | | | ≤ TAE | | |
| Time alignment error relative to Cell 2 [Note 5] | µs | - | | | - | | | ≤ TAE | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: $E_s/I_{ot}$, RSRP, SCH_RP and Io have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The uplink resources for CSI reporting are assigned to the UE prior to the start of time period T2.
Note 5: The time alignment error (TAE) between two cells specified in TS36.104 [30] clause 6.5.3.1 (value depends upon the type of carrier aggregation).

During T2 interruption of PCell during SCells activation shall not happen outside the subframes (m+5) to (m+9) and outside the subframes (m+15) to (m+19).

Figures 8.16.41.5-1 to 8.16.41.5-3 show the derivation of the Test procedure requirements for DTX during T2, based on the core requirements for interruption. The applicable figure is selected according to the CA configuration.

NOTE: The differences between activation and deactivation are as follows:

- Activation ➔ Deactivation
- First CSI ➔ Last CSI
- Latest valid CSI report ➔ Not exist (no need to check since CSI report was already stopped)

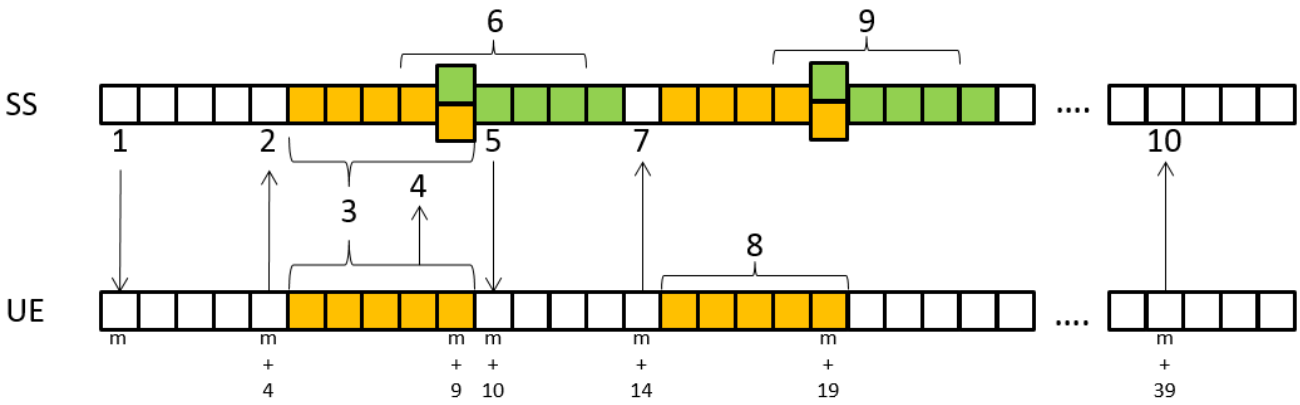


**Figure 8.16.41.5-1: Procedure derivation for Activation (all intra-band case)**

1) Activation command for SCell1 (T2 starts)
2) ACK for MAC-CE for SCell1 activation
3) Possible interruption period by SCell1 activation
4) First CSI report timing (could be invalid CQI)
5) Activation command for SCell2
6) Possible DTX reception period on SS due to interruption by SCell1 activation
7) ACK for MAC-CE for SCell2 activation
8) Possible interruption period by SCell2 activation
9) Possible DTX reception period on SS due to interruption by SCell2 activation
10) Latest valid CSI report timing (valid CQI)

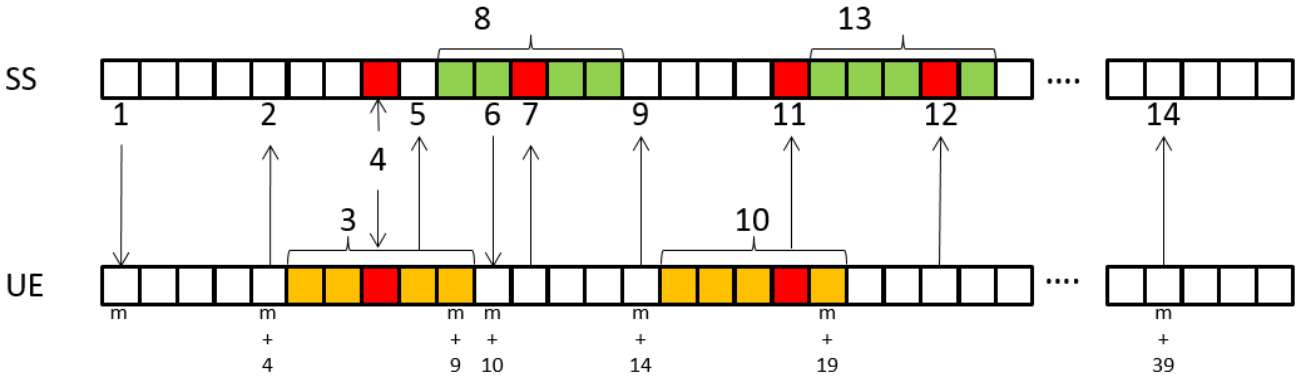


**Figure 8.16.41.5-2: Procedure derivation for Activation (the case where the band relation between PCC and SCells are inter-band)**

1) Activation command for SCell1 (T2 starts)
2) ACK for MAC-CE for SCell1 activation
3) Possible interruption period by SCell1 activation
4) Allowed interruption timing (1 subframe in m+5~m+9) on UE by SCell1 activation,

that is, possible DTX timing on SS

5) First CSI report timing (invalid CQI) at m+8 (might be m+10 due to interruption)
6) Activation command for SCell2
7) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell1 activation
8) Possible DTX reception period on SS due to interruption by SCell1 activation
9) ACK for MAC-CE for SCell2 activation
10) Possible interruption period by SCell2 activation
11) Allowed interruption timing (1 subframe in m+15~m+19) on UE by SCell2 activation,

that is, possible DTX timing on SS

12) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell2 activation
13) Possible DTX reception period on SS due to interruption by SCell2 activation
14) Latest valid CSI report timing (valid CQI)

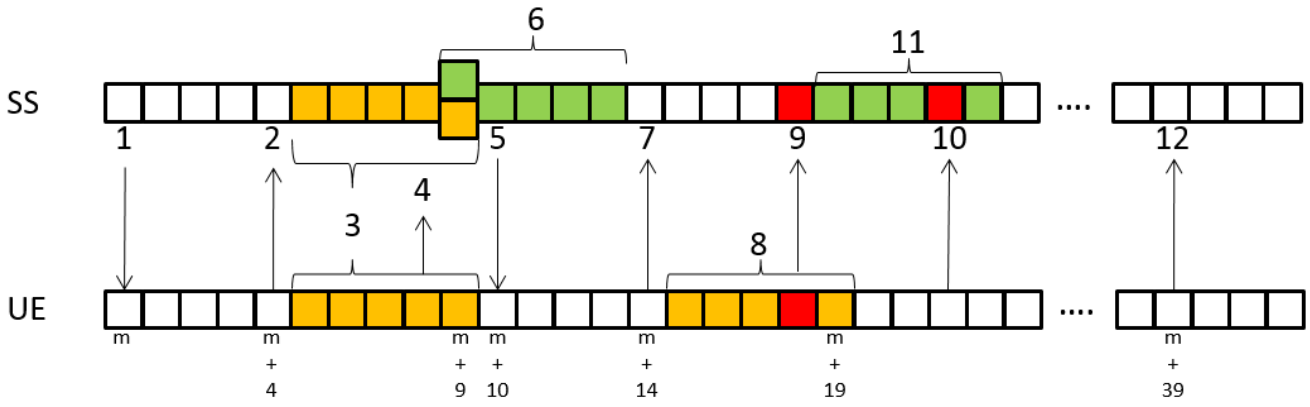


**Figure 8.16.41.5-3: Procedure derivation for Activation (the case where the band relation between PCC and SCell1 is intra-band and SCell2 is inter-band)**

1) Activation command for SCell1 (T2 starts)
2) ACK for MAC-CE for SCell1 activation
3) Possible interruption period by SCell1 activation
4) First CSI report timing (could be invalid CQI)
5) Activation command for SCell2
6) Possible DTX reception period on SS due to interruption by SCell1 activation
7) ACK for MAC-CE for SCell2 activation
8) Possible interruption period by SCell2 activation

9) Allowed interruption timing (1 subframe in m+15~m+19) on UE by SCell2 activation,

that is, possible DTX timing on SS

10) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell2 activation

11) Possible DTX reception period on SS due to interruption by SCell2 activation

12) Latest valid CSI report timing (valid CQI)

During T3 interruption of PCell during SCells deactivation shall not happen outside the subframes (n+5) to (n+9) and outside the subframes (n+15) to (n+19).

The derivation of the Test procedure requirements for DTX during T3 is based on the core requirements for interruption, using the applicable figure according to the CA configuration, in a similar way to T2.

All of the requirements defined in the Test Procedure shall be fulfilled in order for the observed SCell1 activation delay and SCell1 deactivation delay to be counted as correct. The rate of correct observed SCell1 activation delay and SCell1 deactivation delay during repeated tests shall be at least 90%.

The statistical pass/ fail decisions are done separately for activation and deactivation.

Decide the test pass, if activation and deactivation are passed, otherwise fail the UE.

## 8.16.42 3 DL TDD CA activation and deactivation of unknown SCell in non-DRX

### 8.16.42.1 Test purpose

To verify that the SCell activation and deactivation times are within the requirements stated in TS 36.133 [4] clause 7.7 for 3DL TDD carrier aggregation, when the SCell is unknown by the UE at the time of activation.

### 8.16.42.2 Test applicability

This test applies to all types of E-UTRA UE TDD release 10 and forward that support 3DL with intra-band contiguous CA or 3DL with inter-band CA, or 3DL with intra-band contiguous and inter-band CA. Applicability requires support for FGI bit 25.

This test case also applies to all types of E-UTRA TDD UE release 11 and forward that support 3DL with intra-band non-contiguous and inter-band CA, or 3DLwith intra-band non-contiguous and intra-band contiguous CA. Applicability requires support for FGI bit 25.

### 8.16.42.3 Minimum conformance requirements

While activating a SCell if the UE does not receive any command to activate, deactivate, configure or deconfigure the other SCell during the SCell activation delay then the UE shall meet the SCell activation delay requirements specified in TS 36.133 [4] section 7.7.2.

While activating a SCell if the other SCell is activated, deactivated, configured or deconfigured by the UE then the UE shall meet the SCell activation delay requirements ($T_{activate_total}$) according to the following expression:

$$T_{activate_total} = T_{activate_basic} + K*5$$

Where:

$T_{activate_total}$ is the total time to activate a SCell and is expressed in subframes.

$T_{activate_basic}$ is the SCell activation delay specified in TS 36.133 [4] section 7.7.2;

K ($1 \leq K \leq [3]$) is the number of times the other SCell is activated, deactivated, configured or deconfigured while the SCell is being activated;

While activating a SCell:

- The interruption on the PCell and/or on the activated SCell due to the SCell activation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+11 if:
  - The PCell and/or the activated SCell being interrupted and the SCell being activated belong to E-UTRA TDD, or
  - The activated SCell being interrupted and the SCell being activated belong to E-UTRA FDD and the PCell belongs to E-UTRA TDD.
- Otherwise, the interruption on PCell and/or on the activated SCell due to the SCell activation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+9.

Starting from the subframe specified in section 4.3 of TS 36.213 [8] and until the UE has completed the SCell activation, the UE shall send CSI with CQI index = 0 (out of range) if the UE is configured to report the CQI for the SCell being activated.

The UE shall deactivate a SCell and meet the SCell deactivation delay requirements specified in TS 36.133 [4] section 7.7.3 regardless of whether the other SCell is activated, deactivated, configured or deconfigured or not by the UE during the SCell deactivation delay.

While deactivating a SCell:

- The interruption on the PCell and/or on the activated SCell due to the SCell deactivation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+11 if:
  - The PCell and/or the activated SCell being interrupted and the SCell being deactivated belong to E-UTRA TDD or
  - The activated SCell being interrupted and the SCell being deactivated belong to E-UTRA FDD and the PCell belongs to E-UTRA TDD.
- Otherwise, the interruption on PCell and/or the activated SCell due to the SCell deactivation specified in TS 36.133 [4] section 7.8.2 shall not occur before subframe *n*+5 and not occur after subframe *n*+9.

The normative reference for this requirement is TS 36.133 [4] clause 7.7 and A.8.16.42.

## 8.16.42.4 Test description

### 8.16.42.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.16.42.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.63 as appropriate.
2. The general test parameter settings are set up according to Table 8.16.42.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.16.42.4.3.
5. Cell 1 is the PCell on the primary component carrier, Cell 2 is the SCell on RF channel number 2, and Cell 3 is the deactivated SCell on RF channel number 3. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.16.42.4.1-1: General test parameters for E-UTRAN TDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1, 2, 3 | Three radio channels are used for this test |
| Active PCell | | Cell 1 | Primary cell on RF channel number 1. |
| Deconfigured SCell1 | | Cell 2 | Deconfigured secondary cell on RF channel number 2. |
| Configured deactivated SCell2 | | Cell 3 | Configured deactivated secondary cell on RF channel number 3. |
| CP length | | Normal | |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211 [16]. The same configuration applies to all cells. |
| Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in TS 36.211 [16]. The same configuration applies to all cells |
| DRX | | OFF | Continuous monitoring of primary cell |
| CQI/PMI periodicity and offset configuration index | | 0 | CQI reporting for SCells every UL subframe |
| Cell-individual offset for cells on RF channel number 1 | dB | 0 | Individual offset for cells on PCC. |
| Cell-individual offset for cells on RF channel number 2 | dB | 0 | Individual offset for cells on SCC1. |
| Cell-individual offset for cells on RF channel number 3 | dB | 0 | Individual offset for cells on SCC2. |
| Filter coefficient | | 0 | L3 filtering is not used |
| SCell measurement cycle (measCycleSCell) | Ms | 320 | |
| T1 | Ms | 100 | During this time the PCell and SCell2 shall be known and the SCell1 configured, |
| T2 | S | 1 | During this time the UE shall activate the SCell1 and SCell2. |
| T3 | S | 1 | During this time the UE shall deactivate the SCell1 and SCell2 |

### 8.16.42.4.2 Test procedure

The test consists of three successive time periods, with duration of T1, T2 and T3, respectively. There are three carriers, each with one cell. Cell 1 and Cell 3 have constant signal levels throughout the test.

Before the test starts the UE is connected to Cell 1 (PCell) on radio channel 1 (PCC) and Cell 3 (deactivated SCell2) on radio channel 3 (SCC2), but is not aware of Cell 2 (SCell1) on radio channel 2 (SCC1). The UE is only monitoring the PCC and SCC2. The UE shall be continuously scheduled in the PCell throughout the whole test.

During T2 the Test procedure requires the UE to send the first CSI report for SCell1 in a subframe (m+8), but also allows a subframe (m+10) if the subframe (m+8) was subject to interruption. The SS determines whether the CSI report in subframe (m+8) was interrupted or not by monitoring ACK/NACK sent in PCell in subframe (m+8).

NOTE: The requirements are dependant on the CA configuration (e.g. intra band, inter band; see [3], clause 7.8) hence the test procedure has been divided into three variants, found in steps 8a, 8b and 8c and in steps 11a, 11b and 11c. Depending on the CA configuration under test only one of those shall be run respectively.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC2 according to Annex C.0, C.1 for all downlink physical channels.

3. The SS shall configure SCell2 (Cell 3) on the SCC2 as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.42.4.3.

4. The SS shall configure transmission of PDSCH with a maximum number of 1 HARQ transmission.

5. Set the parameters according to T1 in Table 8.16.42.5-1. Propagation conditions are set according to Annex B clauses B.1.1. T1 starts. Configure SCC1 according to Annex C.0, C.1 for all downlink physical channels. The SS shall configure SCell1 (Cell 2) on the SCC1 as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.16.42.4.3.

6. The SS activates SCell1 by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe # denoted m where m is 4 or 9. If the SS receives ACK for MAC-CE sent by the UE, T2 starts in subframe m, and the test proceeds to step 7, otherwise go to step 12.

7. The SS activates SCell2 by sending the activation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8) in a subframe (m+15).

8. The UE shall start sending CSI reports for SCell and the SS shall monitor CSI reports for SCell sent from the UE and ACK/NACK sent in PCell during SCell activation.

   8a. For all intra-band case

   - If the first CSI report for SCell1 is received by the SS in a subframe (m+8),
     - or (m+9) if the subframe (m+8) was subject to interruption
     - or (m+13) if the subframes (m+8) and (m+9) were subject to interruption
   - and CSI report with non-zero CQI index is received by the SS earlier than or equal to subframe (m+39),
     - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+39)
   - and DTX is not observed by the SS outside the subframes (m+5) to (m+9) and (m+13) and the subframes (m+20) to (m+24) and (m+28), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+6) to (m+10) and (m+13) to (m+14) and the subframes (m+21) to (m+25) and (m+28) to (m+29), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+7) to (m+11) and (m+13) to (m+14) and (m+18) and the subframes (m+22) to (m+26) and (m+28) to (m+29) and (m+33), up to the end of T2,
   - Then the number of successes for the event “Activation” is increased by one, Otherwise, count a fail for the event “Activation” and go to step 12.

   8b. For the case where the band relation between PCC and SCells are inter-band

   - If the first CSI report for SCell1 is received by the SS in a subframe (m+8),
     - or (m+9) if the subframe (m+8) was subject to interruption,
   - and CSI report with non-zero CQI index is received by the SS earlier than or equal to subframe (m+39),
     - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+39)
   - and DTX is not observed by the SS outside the subframes (m+5), (m+9), (m+20) and (m+24), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+6), (m+13), (m+21) and (m+28), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+7), (m+13), (m+22) and (m+28), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+8) and (m+23), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+9) and (m+24), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+10), (m+14), (m+25) and (m+29), up to the end of T2,
     - or DTX is not observed by the SS outside the subframes (m+11), (m+18), (m+26) and (m+33), up to the end of T2,
   - Then the number of successes for the event “Activation” is increased by one, Otherwise, count a fail for the event “Activation” and go to step 12.

8c. For the case where the band relation between PCC and SCell1 is intra-band and SCell2 is inter-band

- If the first CSI report for SCell1 is received by the SS in a subframe (m+8),
  - or (m+9) if the subframe (m+8) was subject to interruption
  - or (m+13) if the subframes (m+8) and (m+9) were subject to interruption
- and CSI report with non-zero CQI index is received by the SS earlier than or equal to subframe (m+39),
  - or the next available uplink resource if there are no uplink resources for reporting the valid CSI in a subframe (m+39)
- and DTX is not observed by the SS outside the subframes (m+5) to (m+9) and (m+13),
  - or DTX is not observed by the SS outside the subframes (m+6) to (m+10) and (m+13) to (m+14),
  - or DTX is not observed by the SS outside the subframes (m+7) to (m+11) and (m+13) to (m+14) and (m+18),
- and DTX is not observed by the SS outside the subframes (m+20) and (m+24), up to the end of T2,
  - or DTX is not observed by the SS outside the subframes (m+21) and (m+28), up to the end of T2,
  - or DTX is not observed by the SS outside the subframes (m+22) and (m+28), up to the end of T2,
  - or DTX is not observed by the SS outside the subframe (m+23), up to the end of T2,
  - or DTX is not observed by the SS outside the subframe (m+24), up to the end of T2,
  - or DTX is not observed by the SS outside the subframes (m+25) and (m+29), up to the end of T2,
  - or DTX is not observed by the SS outside the subframes (m+26) and (m+33), up to the end of T2,
- Then the number of successes for the event “Activation” is increased by one, Otherwise, count a fail for the event “Activation” and go to step 12.

9. When T2 expires, the SS deactivates SCell1 by sending the deactivation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8), in a subframe # denoted n, where n is 4 or 9. If the SS receives ACK for MAC-CE sent by the UE, T3 starts in subframe n, and the test proceeds to step 10, otherwise go to step 12.

10. The SS deactivates SCell2 by sending the deactivation MAC-CE (Refer TS 36.321 [26], clauses 5.13, 6.1.3.8), in subframe (n+10).

11. The UE shall stop sending CSI reports for SCell1 and the SS shall monitor CSI reports for SCell1 sent from the UE and ACK/NACK sent in PCell during SCell deactivation.

11a. For all intra-band case

- If the last CSI report is received by the SS earlier than or equal to subframe (n+8),
- and DTX is not observed by the SS outside the subframes (n+5) to (n+9) and (n+13) and the subframes (n+20) to (n+24) and (n+28), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+6) to (n+10) and (n+13) to (n+14) and the subframes (n+21) to (n+25) and (n+28) to (n+29), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+7) to (n+11) and (n+13) to (n+14) and (n+18) and the subframes (n+22) to (n+26) and (n+28) to (n+29) and (n+33), up to the end of T3,
- Then the number of successes for the event “Deactivation” is increased by one. Otherwise, count a fail for the event “Deactivation”.

11b. For the case where the band relation between PCC and SCells are inter-band

- If the last CSI report is received by the SS earlier than or equal to subframe (n+8),

- and DTX is not observed by the SS outside the subframes (n+5), (n+9), (n+20) and (n+24), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+6), (n+13), (n+21) and (n+28), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+7), (n+13), (n+22) and (n+28), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+8) and (n+23), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+9) and (n+24), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+10), (n+14), (n+25) and (n+29), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+11), (n+18), (n+26) and (n+33), up to the end of T3,
- Then the number of successes for the event “Deactivation” is increased by one. Otherwise, count a fail for the event “Deactivation” and go to step 14.

11c. For the case where the band relation between PCC and SCell1 is intra-band and SCell2 is inter-band,

- If the last CSI report is received by the SS earlier than or equal to subframe (n+8),
- and DTX is not observed by the SS outside the subframes (n+5) to (n+9) and (n+13)
  - or DTX is not observed by the SS outside the subframes (n+6) to (n+10) and (n+13) to (n+14),
  - or DTX is not observed by the SS outside the subframes (n+7) to (n+11) and (n+13) to (n+14) and (n+18),
- and DTX is not observed by the SS outside the subframes (n+20) and (n+24), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+21) and (n+28), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+22) and (n+28), up to the end of T3,
  - or DTX is not observed by the SS outside the subframe (n+23), up to the end of T3,
  - or DTX is not observed by the SS outside the subframe (n+24), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+25) and (n+29), up to the end of T3,
  - or DTX is not observed by the SS outside the subframes (n+26) and (n+33), up to the end of T3,
- Then the number of successes for the event “Deactivation” is increased by one. Otherwise, count a fail for the event “Deactivation”.

12. When T3 expires, or Activation in step 6 was not acknowledged, or a fail was counted for the event “Activation” in step 8a, 8b or 8c, or Deactivation in step 9 was not acknowledged, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

13. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

14. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3,

15. Repeat steps 2-14 until a test verdict has been achieved,

Each of the events "Activation" and "Deactivation" is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.16.42.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.16.42.4.3-1: Common Exception messages for E-UTRAN TDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | |

**Table 8.16.42.4.3-2: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRAN TDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.8.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | CQI-ReportConfig-DEFAULT using condition CQI_PERIODIC | | RBC |
| cqi-ReportConfig-r10 | Not present | | SCell_AddMod |
| } | | | |

**Table 8.16.42.4.3-3: *CQI-ReportConfig-DEFAULT:* Additional E-UTRAN TDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | CQI_PERIODIC |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 159 | | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | NULL | | |
| simultaneousAckNackAndCQI | TRUE | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.42.4.3-4: *PhysicalConfigDedicatedSCell-r10-DEFAULT:* Additional E-UTRAN TDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicatedSCell-r10-DEFAULT ::= SEQUENCE { | | | |
| ul-Configuration-r10 SEQUENCE { | | | |
| antennaInfoUL-r10 | Not present | | |
| pusch-ConfigDedicatedSCell-r10 | Not present | | |
| uplinkPowerControlDedicatedSCell-r10 | Not present | | |
| cqi-ReportConfigSCell-r10 | CQI-ReportConfigSCell-r10-DEFAULT | | SCell1 |
| soundingRS-UL-ConfigDedicated-r10 | Not present | | |
| soundingRS-UL-ConfigDedicated-v1020 | Not present | | |
| soundingRS-UL-ConfigDedicatedAperiodic-r10 | Not present | | |
| } | | | |
| } | | | |

**Table 8.16.42.4.3-5: *CQI-ReportConfigSCell-r10-DEFAULT:* Additional E-UTRAN TDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfigSCell-r10 ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic-r10 | Not present | | |
| nomPDSCH-RS-EPRE-Offset-r10 | 0 | | |
| cqi-ReportPeriodicSCell-r10 | | | |
| | *CQI-ReportPeriodic-r10-DEFAULT* | | CQI_PERIODIC |
| pmi-RI-Report-r10 | Not present | | |
| } | | | |

**Table 8.16.42.4.3-6: *CQI-ReportPeriodic-r10-DEFAULT:* Additional E-UTRAN TDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportPeriodic-r10 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex-r10 | 0 | | |
| cqi-PUCCH-ResourceIndexP1-r10 | Not present | | |
| cqi-pmi-ConfigIndex | 0 | (see Table 7.2.2-1A in TS 36.213) | Cell2 |
| cqi-FormatIndicatorPeriodic-r10 CHOICE { | | | |
| widebandCQI-r10 SEQUENCE { | | | |
| csi-ReportMode-r10 | Not present | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | Cell2 |
| simultaneousAckNackAndCQI | TRUE | | |
| cqi-Mask-r9 | Not present | | |
| csi-ConfigIndex-r10 | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.42.4.3-7: *MeasObjectEUTRA-GENERIC:* Additional E-UTRAN TDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Derivation Path: TS 36.508 [7] clause 4.6.6 Table 4.6.6-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | Cell 2, Cell 3 |
| } | | | |

## 8.16.42.5 Test requirement

Table 8.16.42.5-1 defines the primary level settings including test tolerances for E-UTRAN TDD 3 DL CA activation and deactivation of unknown SCell in non-DRX.

**Table 8.16.42.5-1: Cell specific test parameters for E-UTRAN TDD 3 DL CA activation and deactivation of unknown SCell in non-DRX**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | 3 | | |
| $BW_{channel}$ | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | |
| PDSCH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | | | - | | | - | | |
| PCFICH/PDCCH/PHICH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | |
| OCNG Patterns | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | |
| $N_{oc}$ Note 2 | dBm/ 15 kHz | -104 | | | -104 | | | -104 | | |
| $\hat{E}_s/N_{oc}$ | dB | 17 | | | -infinite | 17 | | 17 | | |
| $\hat{E}_s/I_{ot}$ Note 3 | dB | 17 | | | -infinite | 17 | | 17 | | |
| RSRP Note 3 | dBm/ 15 kHz | -87 | | | -infinite | -87 | | -87 | | |
| SCH_RP Note 3 | dBm/ 15 kHz | -87 | | | -infinite | -87 | | -87 | | |
| Io Note 3 | dBm/ Ch BW | -59.13+10log ($N_{RB,c}$ /50) | | | -76.22 +10log ($N_{RB,c}$ /50) | -59.13+10log ($N_{RB,c}$ /50) | | -59.13+10log ($N_{RB,c}$ /50) | | |
| Propagation Condition | | AWGN | | | AWGN | | | AWGN | | |
| Antenna Configuration | | 1x2 | | | 1x2 | | | 1x2 | | |
| Timing offset to Cell 1 | µs | - | | | 0 | | | 0 | | |
| Time alignment error relative to cell 1 Note 5 | µs | - | | | ≤ TAE | | | ≤ TAE | | |
| Time alignment error relative to cell 2 Note 5 | µs | - | | | - | | | ≤ TAE | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: $E_s/I_{ot}$, RSRP, SCH_RP and Io have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The uplink resources for CSI reporting are assigned to the UE prior to the start of time period T2.
Note 5: The time alignment error (TAE) between two cells specified in TS 36.104 [30] clause 6.5.3.1 (value depends upon the type of carrier aggregation).

During T2 interruption of PCell during SCells activation shall not happen outside the subframes (m+5) to (m+11) and outside the subframes (m+20) to (m+26).

Figures 8.16.42.5-1 to 8.16.42.5-3 shows the derivation of the Test procedure requirements for DTX during T2, based on the core requirements for interruption. The applicable figure is selected according to the CA configuration.

NOTE: The differences between activation and deactivation are as follows:

- Activation ➔ Deactivation
- First CSI ➔ Last CSI
- Latest valid CSI report ➔ Not exist (no need to check since CSI report was already stopped)

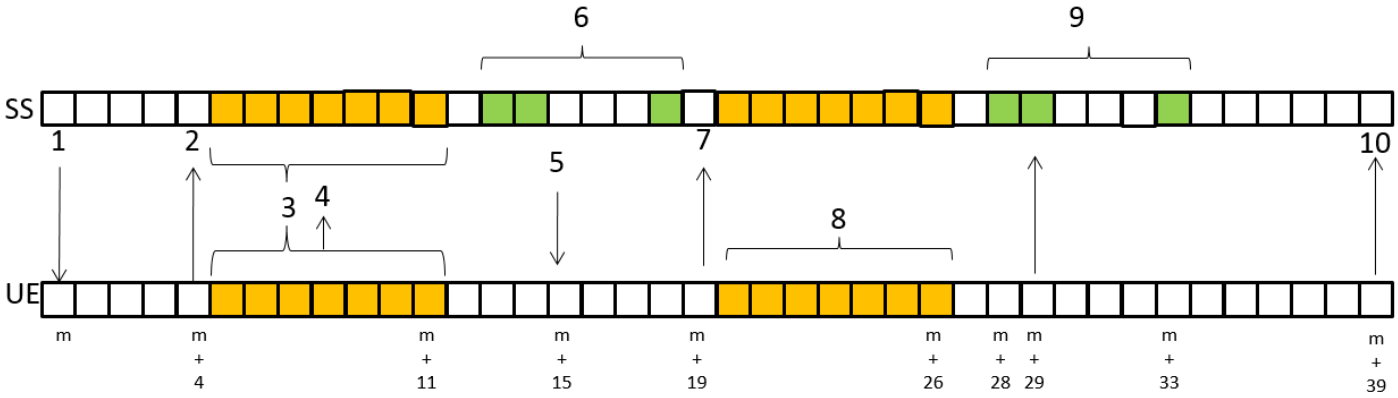


**Figure 8.16.42.5-1: Procedure derivation for Activation (all intra-band case)**

1) Activation command for SCell1(T2 starts)
2) ACK for MAC-CE for SCell1 activation
3) Possible interruption period by SCell1 activation
4) First CSI report timing(invalid CQI)
5) Activation command for SCell2
6) Possible DTX reception period on SS due to interruption by SCell1 activation
   (only green subframes are allowed)
7) ACK for MAC-CE for SCell2 activation
8) Possible interruption period by SCell2 activation
9) Possible DTX reception period on SS due to interruption by SCell2 activation
   (only green subframes are allowed)
10) Latest valid CSI report timing(valid CQI)

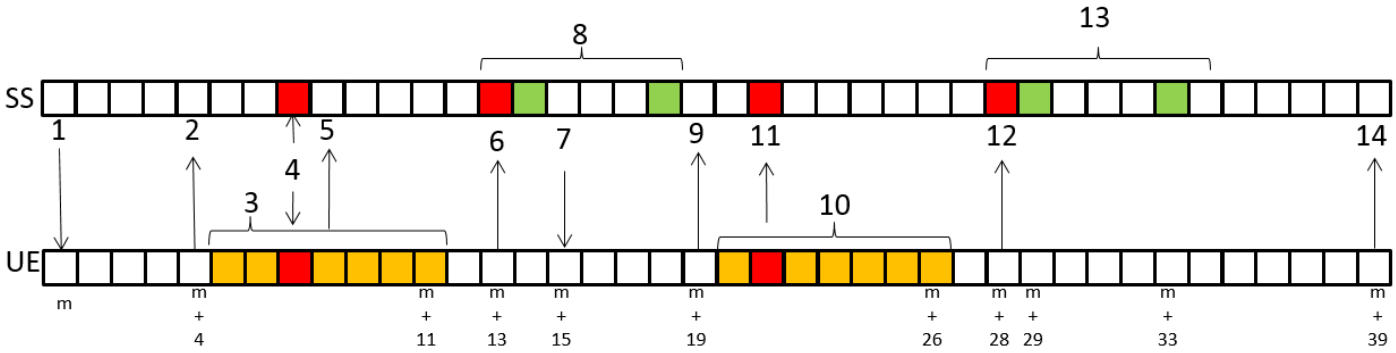


**Figure 8.16.42.5-2: Procedure derivation for Activation (the case where the band relation between PCC and SCells are inter-band)**

1) Activation command for SCell1(T2 starts)
2) ACK for MAC-CE for SCell1 activation
3) Possible interruption period by SCell1 activation
4) Allowed interruption timing(1 subframe in m+5~m+11) on UE by SCell1 activation, that is, possible DTX timing on SS
5) First CSI report timing(invalid CQI) at m+8 (might be m+9 or m+13 due to interruption)
6) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell1 activation
7) Activation command for SCell2
8) Possible DTX reception period on SS due to interruption by SCell1 activation (only m+13, m+14 or m+18 are allowed)
9) ACK for MAC-CE for SCell2 activation
10) Possible interruption period by SCell2 activation
11) Allowed interruption timing(1 subframe in m+20~m+26) on UE by SCell2 activation, that is, possible DTX timing on SS
12) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell2 activation (only m+28, m+29 or m+33 are allowed)
13) Possible DTX reception period on SS due to interruption by SCell2 activation
14) Latest valid CSI report timing(valid CQI)

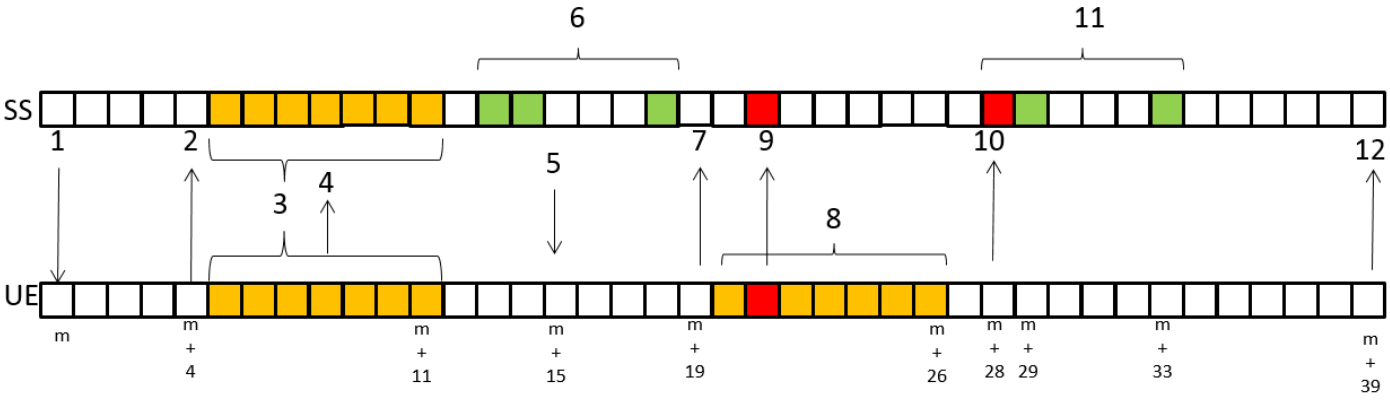


**Figure 8.16.42.5-3: Procedure derivation for Activation**
**(the case where the band relation between PCC and SCell1 is intra-band and SCell2 is inter-band)**

1) Activation command for SCell1(T2 starts)
2) ACK for MAC-CE for SCell1 activation
3) Possible interruption period by SCell1 activation
4) First CSI report timing(invalid CQI)
5) Activation command for SCell2
6) Possible DTX reception period on SS due to interruption by SCell1 activation
   (only green subframes are allowed)
7) ACK for MAC-CE for SCell2 activation
8) Possible interruption period by SCell2 activation
9) Allowed interruption timing(1 subframe in m+20~m+26) on UE by SCell2 activation,
   that is, possible DTX timing on SS
10) Corresponding DTX timing on PUCCH by allowed interruption timing by SCell2 activation
    (only m+28, m+29 or m+33 are allowed)
11) Possible DTX reception period on SS due to interruption by SCell2 activation
12) Latest valid CSI report timing(valid CQI)

During T3 interruption of PCell during SCells deactivation shall not happen outside the subframes (n+5) to (n+11) and outside the subframes (n+20) to (n+26).

The derivation of the Test procedure requirements for DTX during T3 is based on the core requirements for interruption, using the applicable figure according to the CA configuration, in a similar way to T2.

All of the requirements defined in the Test Procedure shall be fulfilled in order for the observed SCell1 activation delay and SCell1 deactivation delay to be counted as correct. The rate of correct observed SCell1 activation delay and SCell1 deactivation delay during repeated tests shall be at least 90%.

The statistical pass/ fail decisions are done separately for activation and deactivation.

Decide the test pass, if activation and deactivation are passed, otherwise fail the UE.

# 8.17 RSTD Measurements for E-UTRAN Carrier Aggregation

The RSTD measurements for EUTRAN Carrier Aggregation test cases can be found in TS 37.571-1 [27].

# 8.18 E-UTRAN TDD – HRPD Measurements

## 8.18.1 E-UTRAN TDD-HRPD event triggered reporting under fading propagation conditions

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

- The Test system uncertainties applicable to this test are undefined
- The Test tolerances applicable to this test are undefined

### 8.18.1.1 Test purpose

To verify the UE's ability to make a correct reporting of an event under fading propagation conditions within the E-UTRA TDD - HRPD cell search requirements.

### 8.18.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward that support HRPD. Applicability requires support for FGI bit 15.

### 8.18.1.3 Minimum requirement

UE shall perform HRPD measurements according to the procedure defined in 3GPP2 C.S0024-B on the HRPD neighbour cells indicated by the serving eNode B. If measurement gaps are required, the UE shall perform HRPD measurements only during the measurement gaps configured by the serving eNode B.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.12 and A.8.18.1.

### 8.18.1.4 Test description

#### 8.18.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.23.

2. The general test parameter settings are set up according to Table 8.18.1.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.18.1.4.3.

5. There is one E-UTRA TDD serving cell and one HRPD cell specified in the test. Cell 1 (E-UTRA TDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.18.1.4.1-1: General test parameters for E-UTRAN TDD to HRPD event triggered reporting under fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Active cell | | Cell 1 | E-UTRAN TDD cell |
| Neighbouring cell | | Cell 2 | HRPD cell |
| Gap Pattern Id | | 1 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| E-UTRAN TDD measurement quantity | | RSRP | |
| Inter-RAT (HRPD) measurement quantity | | CDMA2000 HRPD Pilot Strength | |
| b1-ThresholdCDMA2000 | dB | -7 | Absolute 'CDMA2000 HRPD Pilot Strength' threshold for event B1 |
| Hysteresis | dB | 0 | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | Non-DRX test |
| Access Barring Information | - | Not sent | No additional delays in random access procedure |
| E-UTRA RF Channel Number | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Uplink-downlink configuration of cell 1 | | 1 | As specified in table 4.2.2 in TS 36.211 |
| Special subframe configuration of cell 1 | | 6 | As specified in table 4.2.1 in TS 36.211 |
| HRPD RF Channel Number | | 1 | One HRPD carrier frequency is used. |
| HRPD neighbour cell list size | | 8 | HRPD cells on HRPD RF channel 1 provided in the cell list before T2. |
| cdma2000-SearchWindowSize | | 8 (60 PN chips) | Search window size as defined in section 6.3.5 in 3GPP TS 36.331 |
| T1 | s | 5 | |
| T2 | s | 3 | |

### 8.18.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Tables 8.18.1.5-1 and 8.18.1.5-2. Propagation conditions are set according to Annex B clause B.1.1 and B.2.2. T1 starts.

3. The neighbour cell shall broadcast its own PN offset and the measurement cell list of Cell 1 shall contain the PN offset of Cell 2. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Tables 8.19.1.5-1 and 8.19.1.5-2.

6. UE shall transmit a MeasurementReport message triggered by Event B1. If the measurement reporting delay measured from the beginning of time period T2 is less than 2136 ms then the number of successful tests is increased by one. If the UE fails to report the event within the measurement reporting delay requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. The SS shall set a different PN Offset on Cell 2, as the previous timing information of Cell 2 is invalid in the UE for the next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 8.18.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.18.1.4.3-1: Common Exception messages for E-UTRAN TDD - HRPD event triggered reporting under fading propagation conditions**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7 |

**Table 8.18.1.4.3-2: *MeasConfig-DEFAULT*: Additional E-UTRAN TDD - HRPD event triggered reporting under fading propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell(E-UTRA Cell) | |
| measObject CHOICE { | | | |
| measObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | E-UTRA Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f14 | f14 is the frequency of the neighbouring cell(CDMA2000 Cell) | |
| measObject CHOICE { | | | |
| measObjectCDMA2000 | MeasObjectCDMA2000 | CDMA2000 Cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId)) OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-B1 | | |
| reportConfig | ReportConfigInterRAT-B1-CDMA2000 | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) OF SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f14 | | |
| reportConfigId | IdReportConfig-B1 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | CDMA2000 |
| measGapConfig | MeasGapConfig-GP2 | | Gap Pattern Id = 1 |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 8.18.1.4.3-3: *ReportConfigInterRAT-B1-CDMA2000*: Additional E-UTRAN TDD - HRPD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-CDMA2000(CDMA2000-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| b1-ThresholdCDMA2000 CHOICE { | | | |
| ThresholdCDMA2000 | 14(-7) | -2*10*log10(Ec/Io) in units of 0.5dB, see C.S0005-A for details | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.18.1.4.3-4: *MeasResults*: Additional E-UTRAN TDD - HRPD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| MeasResultNeighCells CHOICE { | | | |
| measResultsCDMA2000 | MeasResultsCDMA2000 | | |
| } | | | |
| } | | | |

**Table 8.18.1.4.3-5: *MeasResultsCDMA2000*: Additional E-UTRAN TDD - HRPD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsCDMA2000::= SEQUENCE { | | | |
| preRegistrationStatusHRPD | true | | |
| measResultListCDMA2000 | MeasResultListCDMA2000 | | |
| } | | | |

**Table 8.18.1.4.3-6: *MeasResultListCDMA2000*: Additional E-UTRAN TDD - HRPD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListCDMA2000::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellIdCDMA2000 | | |
| measResult SEQUENCE { | | | |
| pilotStrength | 6(-3) | -2*10*log10(Ec/Io) in units of 0.5dB, see C.S0005-A for details | |
| } | | | |
| } | | | |

## 8.18.1.5 Test requirement

Tables 8.18.1.4.1-1, 8.18.1.5-1 and 8.18.1.5-2 define the primary level settings including test tolerances for E-UTRAN TDD - HRPD event triggered reporting under fading propagation conditions test.

**Table 8.18.1.5-1: Cell specific test parameters for E-UTRAN TDD cell#1 for event triggered reporting under fading propagation conditions**

| Parameter | Unit | Cell 1 (E-UTRA) | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | |
| RSRP Note 3 | dBm/15 KHz | -98 +TT | -98+TT |
| $\hat{E}_s / N_{oc}$ | dB | 0 | 0 |
| $\hat{E}_s / I_{ot}$ | dB | 0+TT | 0+TT |
| Propagation Condition | | ETU70 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.18.1.5-2: Cell specific test parameters for HRPD (cell # 2) for event triggered reporting under fading propagation conditions**

| Parameter | Unit | Cell 2 (HRPD) | |
|---|---|---|---|
| | | T1 | T2 |
| $\frac{\text{Control } E_b}{N_t}$ (38.4 kbps) | dB | 21 | |
| $\frac{\text{Control } E_b}{N_t}$ (76.8 kbps) | dB | 18 | |
| $\hat{I}_{or}/I_{oc}$ | dB | -infinity | 0 |
| $I_{oc}$ | dBm/1.2288 MHz | -55 | |
| CDMA2000 HRPD Pilot Strength | dB | -infinity | -3+TT |
| Propagation Condition | | ETU70 | |

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event B1 triggered measurement report to Cell 2.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{measurement_HRPD} = T_{measurement_CDMA2000_1x}$

$$T_{measurement_CDMA2000_1x} = T_{basic_measurement_CDMA2000_1x} \cdot N_{Freq} \cdot S_{gap}$$

$T_{basic_identify_UTRA_TDD}$ = 100 ms

$N_{Freq}$ = 1

$S_{gap}$ = 64/3

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 2136 ms in this test case (note: this gives a total of 2134 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 8.19 E-UTRAN TDD – CDMA2000 1X Measurements

## 8.19.1 E-UTRAN TDD-CDMA2000 1X event triggered reporting under fading propagation conditions

Editor's note: This section is incomplete. The following aspects are either missing or not yet determined:

- The Test system uncertainties applicable to this test are undefined
- The Test tolerances applicable to this test are undefined

### 8.19.1.1 Test purpose

To verify the UE's ability to make a correct reporting of an event under fading propagation conditions within the E-UTRA TDD-CDMA2000 1X inter-frequency cell search requirements.

### 8.19.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward that support cdma2000 1xRTT. Applicability requires support for FGI bit 15.

### 8.19.1.3 Minimum conformance requirements

When measurement gaps are scheduled for CDMA2000 1xRTT inter RAT measurements, or the UE supports capability of conducting such measurements without gaps, the UE physical layer shall be capable of reporting CDMA2000 1xRTT Pilot Strength measurements to higher layers with measurement accuracy as specified in TS36.133 [4] Section 9.5, corresponding to a 90% measurement success rate, with measurement period given by

$$T_{measurement_CDMA2000_1x} = T_{basic_measurement_CDMA2000_1x} \cdot N_{Freq} \cdot S_{gap}$$

where $T_{basic_measurement_CDMA2000_1x}$ = 100 ms and the measurement gap specific scale factor $S_{gap}$ is based on the measurement gap pattern in use as defined in Table 8.19.1.3-1. If inter-frequency RSTD measurements are configured as a part of the measurement configuration, $S_{gap}$ shall be based to the Gap Pattern Id 1.

**Table 8.19.1.3-1: Gap Pattern Specific Scale Factor**

| Gap Pattern Id | $S_{gap}$ |
|---|---|
| 0 | 32/3 |
| 1 | 64/3 |

Reported measurements in periodically triggered measurement reports shall meet the requirements in TS 36.133 [4] section 9.

The measurement reporting delay of each periodic report is defined as the time between the end of the last measurement period and the moment when the UE starts to transmit the measurement report over the Uu interface. This delay shall be less than $T_{71m}$ defined in 3GPP2 C.S0005-D for each periodic report. This measurement reporting delay excludes a delay which is caused by the unavailability of the uplink resources for the UE to send the measurement report.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.10 and A.8.19.1.

### 8.19.1.4 Test description

#### 8.19.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.23.

2. The general test parameter settings are set up according to Table 8.19.1.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.19.1.4.3.

5. There is one E-UTRA FDD serving cell and one cdma2000 1xRTT cell specified in the test. Cell 1 (E-UTRA FDD cell) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.19.1.4.1-1: General test parameters for E-UTRAN TDD-CDMA2000 1X event triggered reporting in fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Gap Pattern Id | | 1 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on CDMA2000 1X RF channel number 1. |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in 3GPP TS 36.211. Applicable to cell 1. |
| Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in 3GPP TS 36.211. Applicable to cell 1. |
| CP length | | Normal | Applicable to cell 1. |
| E-UTRA RF Channel Number | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| CDMA2000 1X Channel Number | | 1 | One CDMA2000 1X carrier frequency is used. |
| Inter-RAT (CDMA2000 1X) measurement quantity | | CDMA2000 1xRTT Pilot Strength | |
| B1-Threshold-CDMA2000 | dB | -14 | Absolute 'CDMA2000 1xRTT Pilot Strength' threshold for event B1 |
| Hysteresis | dB | 0 | |
| Time To Trigger | ms | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used. |
| DRX | | OFF | |
| cdma2000 1X neighbour cell list size | | 8 | cdma2000 1X cells on cdma2000 1X RF channel 1 provided in the cell list before T2. |
| cdma2000-SearchWindowSize | | 8 (60 PN chips) | Search window size as defined in section 6.3.5 in 3GPP TS 36.331 |
| T1 | s | 5 | |
| T2 | s | 3 | |

### 8.19.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event B1 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3. Cell 1 is the active cell.

2. Set the parameters according to T1 in Tables 8.19.1.5-1 and 8.19.1.5-2. Propagation conditions are set according to Annex B clause B.1.1 and B.2.2. T1 starts.

3. The neighbour cell shall broadcast its own PN offset and the measurement cell list of Cell 1 shall contain the PN offset of Cell 2. The SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Tables 8.19.1.5-1 and 8.19.1.5-2.

6. The UE shall transmit a MeasurementReport message triggered by Event B1. If the measurement reporting delay measured from the beginning of time period T2 is less than 2136 ms then the number of successful tests is increased by one. If the UE fails to report the event within the measurement reporting delay requirement then the number of failure tests is increased by one.

7. After the SS receives the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. The SS shall set a different PN Offset on Cell 2, as the previous timing information of Cell 2 is invalid in the UE for the next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 8.19.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.19.1.4.3-1: Common Exception messages for E-UTRAN TDD – CDMA 2000 1X event triggered reporting under fading propagation conditions**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7 |

**Table 8.18.1.4.3-2: *MeasConfig-DEFAULT*: Additional E-UTRAN TDD – CDMA2000 1X event triggered reporting under fading propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell(E-UTRA Cell) | |
| measObject CHOICE { | | | |
| measObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | E-UTRA Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f17 | f17 is the frequency of the neighbouring cell(CDMA2000 Cell) | |
| measObject CHOICE { | | | |
| measObjectCDMA2000 | MeasObjectCDMA2000 | CDMA2000 Cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId)) OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-B1 | | |
| reportConfig | ReportConfigInterRAT-B1-CDMA2000 | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) OF SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f17 | | |
| reportConfigId | IdReportConfig-B1 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | CDMA2000 |
| measGapConfig | MeasGapConfig-GP2 | | Gap Pattern Id = 1 |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 8.19.1.4.3-3: *ReportConfigInterRAT-B1-CDMA2000*: Additional E-UTRAN TDD – CDMA2000 1X event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-CDMA2000 (CDMA2000-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| b1-Threshold CDMA2000 | 28(-14) | INTEGER (0..63) | CDMA2000 |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0dB) | The actual value is IE value * 0.5 dB. | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.19.1.4.3-4: *MeasResults*: Additional E-UTRAN TDD – CDMA2000 1X event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultsCDMA2000 | MeasResultsCDMA2000 | | |
| } | | | |
| } | | | |

**Table 8.19.1.4.3-5: MeasResultsCDMA2000: Additional E-UTRAN TDD – CDMA 2000 1X event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsCDMA2000::= SEQUENCE ( | | | |
| preRegistrationStatusHRPD | FALSE | | |
| measResultListCDMA2000 | MeasResultListCDMA2000 | | |
| } | | | |

**Table 8.19.1.4.3-6: MeasResultListCDMA2000: Additional E-UTRAN TDD – CDMA2000 1X event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListCDMA2000::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultCDMA2000 { | | | |
| MeasResultCDMA2000 ::= SEQUENCE { | | | |
| physCellId | PhysCellIdCDMA2000 | | |
| measResult ::= SEQUENCE { | | | |
| pilotStrength | 20 (-10) | INTEGER (0..63) | |
| } | | | |
| } | | | |
| } | | | |

## 8.19.1.5 Test requirement

Tables 8.19.1.4.1-1, 8.19.1.5-1 and 8.19.1.5-2 define the primary level settings including test tolerances for E-UTRAN TDD-CDMA2000 1X event triggered reporting under fading propagation conditions.

**Table 8.19.1.5-1: Cell specific test parameters for E-UTRAN TDD (cell # 1) for event triggered reporting of CDMA2000 1X cell under fading propagation conditions**

| **Parameter** | **Unit** | **Cell 1** | |
|---|---|---|---|
| | | **T1** | **T2** |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 1 | dB | | |
| OCNG_RB Note 1 | dB | | |
| $\hat{E}_s / I_{ot}$ | dB | 4+TT | 4+TT |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 |
| $N_{oc}$ | dBm/15 kHz | -98 | |
| RSRP | dBm/15 kHz | -94+TT | -94+TT |
| SCH_RP | dBm/15 kHz | -94+TT | -94+TT |
| Propagation Condition | | ETU70 | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2. | | | |

**Table 8.19.1.5-2: Cell specific test parameters for CDMA2000 1X (cell # 2) for event triggered reporting of CDMA2000 1X cell under fading propagation conditions**

| Parameter | Unit | Cell 2 (cdma2000 1X) | |
|---|---|---|---|
| | | T1 | T2 |
| $\frac{\text{Pilot } E_c}{I_{or}}$ | dB | -7 | |
| $\frac{\text{Sync } E_c}{I_{or}}$ | dB | -16 | |
| $\frac{\text{Paging } E_c}{I_{or}}$ (4.8 kbps) | dB | -12 | |
| $\hat{I}_{or} / I_{oc}$ | dB | -infinity | 0 |
| $I_{oc}$ | dBm/1.2288 MHz | -55 | |
| CDMA2000 1xRTT Pilot Strength | dB | -infinity | -10+TT |
| Propagation Condition | | ETU70 | |

The overall delay measured is defined as the time from the beginning of time period T2, to the moment the UE sends one Event B1 triggered measurement report including PN offset of Cell 2.

The actual overall delays measured in the test may be up to 2xTTIDCCH higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{measurement_CDMA2000_1x}$

$$T_{measurement_CDMA2000_1x} = T_{basic_measurement_CDMA2000_1x} \cdot N_{Freq} \cdot S_{gap}$$

$T_{basic_measurement_CDMA2000_1x}$ = 100 ms.

$N_{Freq}$ = 1. It is defined in TS 36.133 [4] clause 8.1.2.1.1.

$S_{gap}$=64/3. It is based on the measurement gap pattern in use as defined in Table 8.19.1.3-1.

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 2136 ms in this test case (note: this gives a total of 2134 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 8.20 Inter-frequency/RAT Measurements in CA mode

## 8.20.1 E-UTRAN FDD-FDD Inter-frequency event triggered reporting under fading propagation conditions in asynchronous cells

### 8.20.1.1 Test purpose

To verify the UE's ability to make a correct reporting of an event under fading propagation conditions in asynchronous cells within the E-UTRA FDD inter-frequency cell search requirements in CA mode. This test will partly verify the FDD-FDD inter-frequency cell search requirements.

### 8.20.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA.

This test applies to all types of E-UTRA FDD UE release 11 and forward that support CA with Intra-band non-contiguous CA.

### 8.20.1.3 Minimum conformance requirements

When measurement gaps are scheduled the UE shall be able to identify a new FDD inter-frequency within $T_{Identify_Inter}$ according to the following expression:

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new FDD inter-frequency cell is defined.

$N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1 and $T_{Inter1}$ is defined in TS36.133 [4] section 8.1.2.1.

A cell shall be considered detectable provided following conditions are fulfilled:

- RSRP and RSRP Ês/Iot according to Annex I.2.3 for a corresponding Band,

- other RSRP related side conditions given in TS 36.133 [4] Sections 9.1.3.1 and 9.1.3.2 and RSRQ related side conditions given in TS 36.133 [4] Sections 9.1.6.1 and 9.1.6.2 are fulfilled,

- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band.

When measurement gaps are scheduled for FDD inter frequency measurements the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with measurement accuracy as specified in TS 36.133 [4] sub-clauses 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively, with measurement period given by table 8.20.1.3-1.

**Table 8.20.1.3-1: Measurement period and measurement bandwidth**

| Configuration | Physical Layer Measurement period: $T_{Measurement_Period_Inter_FDD}$ [ms] | Measurement bandwidth [RB] |
|---|---|---|
| 0 | 480 x $N_{freq}$ | 6 |
| 1 (Note) | 240 x $N_{freq}$ | 50 |
| Note: This configuration is optional. | | |

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per FDD inter-frequency and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period defined in Table 8.20.1.3-1.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clauses 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_inter}$ defined in TS 36.133 [4] clause 8.1.2.3.1.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra}$ defined in TS 36.133 [4] section 8.1.2.2.1.1 becomes undetectable for a period ≤ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.1 and A.8.20.1.

## 8.20.1.4 Test description

### 8.20.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.43 as appropriate.

2. The general test parameter settings are set up according to Table 8.20.1.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.20.1.4.3.

5. There are three E-UTRA FDD carriers and three cells specified in the test. Cell 1 is the PCell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 is the inter-frequency neighbour cell, Cell 3 is the SCell. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.20.1.4.1-1: General test parameters for E-UTRAN FDD-FDD inter-frequency event triggered reporting in fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| E-UTRA RF Channel Number | | 1, 2 | Two FDD carrier frequencies are used. |
| E-UTRA RF Channel Number for SCell | | 3 | One FDD carrier frequencies is used |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active cell | | Cell 1 | Cell 1 is on RF channel number 1 |
| Neighbour cell | | Cell 2 | Cell 2 is on RF channel number 2 |
| Configured active SCell | | Cell 3 | Cell 3 is on RF channel number 3 |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| A3-Offset | dB | -6 | |
| Hysteresis | dB | 0 | |
| CP length | | Normal | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | OFF |
| Cell2 timing offset to cell1 | ms | 3 | Asynchronous cells |
| Cell3 timing offset to cell1 | µs | 0 | Synchronous cells |
| Time alignment error between cell3 and cell1 | µs | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1. | The value of time alignment error depends upon the type of carrier aggregation. |
| T1 | s | 5 | |
| T2 | s | 5 | |

### 8.20.1.4.2 Test procedure

The test consists of two successive time periods, with time duration of T1, and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0, C.1 for all downlink physical channels.

3 The SS shall configure the SCell (Cell 3) on the SCC as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.20.1.4.3.

4. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [13], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).

5. Set the parameters according to T1 in Table 8.20.1.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

6. The SS shall transmit an RRCConnectionReconfiguration message with event A3 configured.

7. The UE shall transmit an RRCConnectionReconfigurationComplete message.

8. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.20.1.5-1.

9. UE shall transmit a MeasurementReport message triggered by Event A3. If the measurement reporting delay from the beginning of time period T2 is less than 3842ms the number of successful tests is increased by one. If the UE fails to report the event within the measurement reporting delay requirement then the number of failure tests is increased by one.

10. After the SS receives the MeasurementReport message in step 9) or when T2 expires, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

11. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

12. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

13. Repeat steps 2-12 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.20.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.20.1.4.3-1: Common Exception messages for E-UTRAN FDD-FDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.4.1-5 |

**Table 8.20.1.4.3-2: *MeasConfig:* Additional E-UTRAN FDD-FDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation path: TS 36.508 [7] clause 4.6.6 table 4.6.6-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasConfig ::= SEQUENCE { | | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 3 entries | | |
| measObjectId[1] | IdMeasObject-f1 | | |
| measObject[1] | MeasObjectEUTRA-GENERIC(f1) | Cell 1 | |
| measObjectId[2] | IdMeasObject-f2 | | |
| measObject[2] | MeasObjectEUTRA-GENERIC(f2) | Cell 2 | |
| measObjectId[3] | IdMeasObject-f3 | | |
| measObject[3] | MeasObjectEUTRA-GENERIC(f3) | Cell 3 | |
| } | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId)) OF SEQUENCE { | 1 entry | | |
| reportConfigId[1] | IdReportConfig-A3 | | |
| reportConfig[1] | ReportConfigEUTRA-A3 | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) OF SEQUENCE { | 1 entry | | |
| measId[1] | 1 | | |
| measObjectId[1] | IdMeasObject-f2 | | |
| reportConfigId[1] | IdReportConfig-A3 | | |
| } | | | |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| } | | | |

**Table 8.20.1.4.3-3: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

## 8.20.1.5 Test requirement

Tables 8.20.1.4.1-1 and 8.20.1.5-1 define the primary level settings including test tolerances for event triggered reporting under fading propagation conditions in asynchronous inter frequency cells test.

**Table 8.20.1.5-1: Cell specific test parameters for E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in asynchronous cells**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | | 3 | |
| $BW_{channel}$ | MHz | 10 | | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | | OP.1 FDD | |
| PBCH_RA | dB | 0 | | 0 | | 0 | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | | | |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 | -Infinity | 90.3 | -94 | -94 |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | -Infinity | 7.7 | 4 | 4 |
| SCH_RP Note 4 | dBm/15 kHz | -94 | -94 | -Infinity | -90.3 | -94 | -94 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | -Infinity | 7.7 | 4 | 4 |
| Propagation Condition | | ETU70 | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 3842 ms from the beginning of time period T2.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event A3 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = 3840 ms

TTI insertion uncertainty = $TTI_{DCCH}$ = 1 ms; $2xTTI_{DCCH}$ = 2 ms

The overall delays measured shall be less than a total of 3842 ms in this test case (note: this gives a total of 3840 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

## 8.20.2 E-UTRAN TDD-TDD Inter-frequency event triggered reporting under fading propagation conditions in synchronous cells

### 8.20.2.1 Test purpose

To verify the UE's ability to make a correct reporting of an event under fading propagation conditions in synchronous cells within the E-UTRA TDD inter-frequency cell search requirements in CA mode. This test will partly verify the TDD-TDD inter-frequency cell search requirements.

### 8.20.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA.

This test applies to all types of E-UTRA TDD UE release 11 and forward that support CA with Intra-band non-contiguous CA.

### 8.20.2.3 Minimum conformance requirements

When measurement gaps are scheduled the UE shall be able to identify a new TDD inter-frequency within $T_{Identify_Inter}$ according to the following expression:

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq} \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new TDD inter-frequency cell is defined.

$N_{freq}$ is defined in TS 36.133 [4] section 8.1.2.1.1 and $T_{Inter1}$ is defined in TS36.133 [4] section 8.1.2.1.

A cell shall be considered detectable provided following conditions are fulfilled:

- RSRP and RSRP Ês/Iot according to Annex I.2.3 for a corresponding Band,

- other RSRP related side conditions given in TS 36.133 [4] Sections 9.1.3.1 and 9.1.3.2 and RSRQ related side conditions given in TS 36.133 [4] Sections 9.1.6.1 and 9.1.6.2 are fulfilled,

- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band.

When measurement gaps are scheduled for TDD inter frequency measurements the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with measurement accuracy as specified in TS 36.133 [4] sub-clauses 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively, with measurement period ($T_{Measurement_Period_TDD_Inter}$) given by table 8.20.2.3-1.

**Table 8.20.2.3-1: $T_{Measurement_Period_TDD_Inter}$ for different configurations**

| Configuration | Measurement bandwidth [RB] | Number of UL/DL sub-frames per half frame (5 ms) | | DwPTS | | $T_{Measurement_Period_TDD_Inter}$ [ms] |
|---|---|---|---|---|---|---|
| | | DL | UL | Normal CP | Extended CP | |
| 0 | 6 | 2 | 2 | $19760 \cdot T_s$ | $20480 \cdot T_s$ | 480 x $N_{freq}$ |
| 1 (Note 1) | 50 | 2 | 2 | $19760 \cdot T_s$ | $20480 \cdot T_s$ | 240 x $N_{freq}$ |
| Note 1: This configuration is optional.<br>Note 2: $T_s$ is defined in 3GPP TS 36.211 [9]. | | | | | | |

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per TDD inter-frequency and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period $T_{Measurement_Period_TDD_Inter}$.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clauses 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_inter}$ defined in TS 36.133 [4] clause 8.1.2.3.2.1. When L3 filtering is used an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra}$ defined in TS 36.133 [4] section 8.1.2.2.1.1 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ Intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.3.2 and A.8.20.2.

## 8.20.2.4 Test description

### 8.20.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.43 as appropriate.

2. The general test parameter settings are set up according to Table 8.20.2.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.20.2.4.3.

5. There are three E-UTRA TDD carriers and three cells specified in the test. Cell 1 is the PCell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 is the inter-frequency neighbour cell, Cell 3 is the SCell. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.20.2.4.1-1: General test parameters for E-UTRAN TDD-TDD inter-frequency event triggered reporting in fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| Gap Pattern Id | | 1 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211 [9]. The same configuration in both cells |
| Uplink-downlink configuration | | 1 | As specified in 3GPP TS 36.211 [9] section 4.2 Table 4.2-2 |
| CP length | | Normal | |
| E-UTRA RF Channel Number | | 1, 2 | Two TDD carrier frequencies are used. |
| E-UTRA RF Channel Number for SCell | | 3 | One TDD carrier frequencies is used |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active cell | | Cell 1 | Cell 1 is on RF channel number 1 |
| Neighbour cell | | Cell 2 | Cell 2 is on RF channel number 2 |
| Configured active SCell | | Cell 3 | Cell 3 is on RF channel number 3 |
| A3-Offset | dB | -6 | |
| Hysteresis | dB | 0 | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| Cell2 timing offset to cell1 | μs | 3 | Synchronous cells |
| Cell3 timing offset to cell1 | μs | 0 | Synchronous cells |
| Time alignment error between cell3 and cell1 | μs | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1. | The value of time alignment error depends upon the type of carrier aggregation. |
| T1 | s | 5 | |
| T2 | s | 10 | |

### 8.20.2.4.2 Test procedure

The test consists of two successive time periods, with time duration of T1, and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 2

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0, C.1 for all downlink physical channels.

3 The SS shall configure the SCell (Cell 3) on the SCC as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.20.2.4.3.

4. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [13], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).

5. Set the parameters according to T1 in Table 8.20.2.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

6. The SS shall transmit an RRCConnectionReconfiguration message with event A3 configured.

7. The UE shall transmit an RRCConnectionReconfigurationComplete message.

8. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.20.2.5-1.

9. UE shall transmit a MeasurementReport message triggered by Event A3. If the measurement reporting delay from the beginning of time period T2 is less than 7682ms the number of successful tests is increased by one. If the UE fails to report the event within the measurement reporting delay requirement then the number of failure tests is increased by one.

10. After the SS receives the MeasurementReport message in step 9) or when T2 expires, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

11. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

12. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

13. Repeat steps 2-12 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.20.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.20.2.4.3-1: Common Exception messages for E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.4.1-5 |

**Table 8.20.2.4.3-2: *MeasConfig:* Additional E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation path: TS 36.508 [7] clause 4.6.6 table 4.6.6-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasConfig ::= SEQUENCE { | | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 3 entries | | |
| measObjectId[1] | IdMeasObject-f1 | | |
| measObject[1] | MeasObjectEUTRA-GENERIC(f1) | Cell 1 | |
| measObjectId[2] | IdMeasObject-f2 | | |
| measObject[2] | MeasObjectEUTRA-GENERIC(f2) | Cell 2 | |
| measObjectId[3] | IdMeasObject-f3 | | |
| measObject[3] | MeasObjectEUTRA-GENERIC(f3) | Cell 3 | |
| } | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId)) OF SEQUENCE { | 1 entry | | |
| reportConfigId[1] | IdReportConfig-A3 | | |
| reportConfig[1] | ReportConfigEUTRA-A3 | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) OF SEQUENCE { | 1 entry | | |
| measId[1] | 1 | | |
| measObjectId[1] | IdMeasObject-f2 | | |
| reportConfigId[1] | IdReportConfig-A3 | | |
| } | | | |
| measGapConfig | MeasGapConfig-GP2 | | Gap Pattern Id = 1 |
| } | | | |

**Table 8.20.2.4.3-3: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in synchronous cells test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

## 8.20.2.5 Test requirement

Tables 8.20.2.4.1-1 and 8.20.2.5-1 define the primary level settings including test tolerances for event triggered reporting under fading propagation conditions in synchronous inter frequency cells test.

**Table 8.20.2.5-1: Cell specific test parameters for E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in synchronous cells**

| **Parameter** | **Unit** | **Cell 1** | | **Cell 2** | | **Cell 3** | |
|---|---|---|---|---|---|---|---|
| | | **T1** | **T2** | **T1** | **T2** | **T1** | **T2** |
| E-UTRA RF Channel Number | | 1 | | 2 | | 3 | |
| $BW_{channel}$ | MHz | 10 | | 10 | | 10 | |
| **Correlation Matrix and Antenna Configuration** | | **1x2 Low** | | **1x2 Low** | | **1x2 Low** | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2) | | OP.1 TDD | | OP.2 TDD | | OP.1 TDD | |
| PBCH_RA | dB | 0 | | 0 | | 0 | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA[Note 1] | dB | | | | | | |
| OCNG_RB[Note 1] | dB | | | | | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $\hat{E}_s/I_{ot}$ | dB | 4 | 4 | -Infinity | 7.7 | 4 | 4 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | | | |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 | -Infinity | -90.3 | -94 | -94 |
| SCH_RP Note 4 | dBm/15 kHz | -94 | -94 | -infinity | -90.3 | -94 | -94 |
| $\hat{E}_s/N_{oc}$ | dB | 4 | 4 | -Infinity | 7.7 | 4 | 4 |
| Propagation Condition | | ETU70 | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 7682 ms from the beginning of time period T2.

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 7682 ms in this test case (note: a total of 7680 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

## 8.20.2A E-UTRAN TDD-TDD Inter-frequency event triggered reporting under fading propagation conditions in synchronous cells for 20 MHz +20 MHz bandwidth

### 8.20.2A.1 Test purpose

Same test purpose as in clause 8.20.2.1.

### 8.20.2A.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA.

This test applies to all types of E-UTRA TDD UE release 11 and forward that support CA with Intra-band non-contiguous CA.

### 8.20.2A.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.20.2.3.

### 8.20.2A.4 Test description

#### 8.20.2A.4.1 Initial conditions

Same initial conditions as in clause 8.20.2.4.1 with the following exceptions:

- Channel Bandwidth to be tested: 20 MHz.
  - Instead of Table 8.20.2.4.1-1 → use Table 8.20.2A.4.1-1.

**Table 8.20.2A.4.1-1: General test parameters for E-UTRAN TDD-TDD inter-frequency event triggered reporting in fading propagation conditions for 20 MHz +20 MHz bandwidth**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.3 TDD | As specified in clause A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.10 TDD | As specified in clause A.2.2 |
| Gap Pattern Id | | 1 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211 [9]. The same configuration in both cells |
| Uplink-downlink configuration | | 1 | As specified in 3GPP TS 36.211 [9] section 4.2 Table 4.2-2 |
| CP length | | Normal | |
| E-UTRA RF Channel Number | | 1, 2 | Two TDD carrier frequencies are used. |
| E-UTRA RF Channel Number for SCell | | 3 | One TDD carrier frequencies is used |
| Channel Bandwidth ($BW_{channel}$) | MHz | 20 | |
| Active cell | | Cell 1 | Cell 1 is on RF channel number 1 |
| Neighbour cell | | Cell 2 | Cell 2 is on RF channel number 2 |
| Configured active SCell | | Cell 3 | Cell 3 is on RF channel number 3 |
| A3-Offset | dB | -6 | |
| Hysteresis | dB | 0 | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| Cell2 timing offset to cell1 | μs | 3 | Synchronous cells |
| Cell3 timing offset to cell1 | μs | 0 | Synchronous cells |
| Time alignment error between cell3 and cell1 | μs | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1. | The value of time alignment error depends upon the type of carrier aggregation. |
| T1 | s | 5 | |
| T2 | s | 10 | |
| Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3 | | | |

### 8.20.2A.4.2 Test procedure

Same test procedure as in clause 8.20.2.4.2 with the following exceptions:

- Instead of Table 8.20.2.5-1 → use Table 8.20.2A.5-1.

### 8.20.2A.4.3 Message contents

Same message contents as in clause 8.20.2.4.3.

## 8.20.2A.5 Test requirement

Tables 8.20.2A.4.1-1 and 8.20.2A.5-1 define the primary level settings including test tolerances for event triggered reporting under fading propagation conditions in synchronous inter frequency cells test.

**Table 8.20.2A.5-1: Cell specific test parameters for E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in synchronous cells for 20 MHz + 20 MHz bandwidth**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | | 3 | |
| $BW_{channel}$ | MHz | 20 | | 20 | | 20 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| OCNG Pattern defined in D.2.7 (OP.7 TDD) and in D.2.8 (OP.8) | | OP.7 TDD | | OP.8 TDD | | OP.7 TDD | |
| PBCH_RA | dB | 0 | | 0 | | 0 | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | -Infinity | 7.7 | 4 | 4 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | | | |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 | -Infinity | -90.3 | -94 | -94 |
| SCH_RP Note 4 | dBm/15 kHz | -94 | -94 | -infinity | -90.3 | -94 | -94 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | -Infinity | 7.7 | 4 | 4 |
| Propagation Condition | | ETU70 | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 7682 ms from the beginning of time period T2.

TTI insertion uncertainty = 2 ms.

The overall delays measured shall be less than a total of 7682 ms in this test case (note: a total of 7680 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

# 8.20.2B E-UTRAN TDD-TDD Inter-frequency event triggered reporting under fading propagation conditions in synchronous cells for 20 MHz +10 MHz bandwidth

## 8.20.2B.1 Test purpose

Same test purpose as in clause 8.20.2.1.

## 8.20.2B.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA.

This test applies to all types of E-UTRA TDD UE release 11 and forward that support CA with Intra-band non-contiguous CA.

## 8.20.2B.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.20.2.3.

## 8.20.2B.4 Test description

### 8.20.2B.4.1 Initial conditions

Same initial conditions as in clause 8.20.2.4.1 with the following exceptions:

- Frequencies to be tested: According to Annex E table E-3 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.
- Channel Bandwidth to be tested:The bandwidth combination supported by UE from Table 8.20.2B.5-1 and Annex E table E-3 as defined in 3GPP TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in 3GPP TS 36.521-1 [10] clause 5.4.2A.

- Instead of Table 8.20.2.4.1-1 → use Table 8.20.2B.4.1-1.

**Table 8.20.2B.4.1-1: General test parameters for E-UTRAN TDD-TDD inter-frequency event triggered reporting in fading propagation conditions for 20 MHz +10 MHz bandwidth**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Gap Pattern Id | | 1 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211 [9]. The same configuration in both cells |
| Uplink-downlink configuration | | 1 | As specified in 3GPP TS 36.211 [9] section 4.2 Table 4.2-2 |
| CP length | | Normal | |
| E-UTRA RF Channel Number | | 1, 2 | Two TDD carrier frequencies are used. |
| E-UTRA RF Channel Number for SCell | | 3 | One TDD carrier frequencies is used |
| Channel Bandwidth ($BW_{channel}$) | MHz | 20 | |
| Active cell | | Cell 1 | Cell 1 is on RF channel number 1 |
| Neighbour cell | | Cell 2 | Cell 2 is on RF channel number 2 |
| Configured active SCell | | Cell 3 | Cell 3 is on RF channel number 3 |
| A3-Offset | dB | -6 | |
| Hysteresis | dB | 0 | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| Cell2 timing offset to cell1 | μs | 3 | Synchronous cells |
| Cell3 timing offset to cell1 | μs | 0 | Synchronous cells |
| Time alignment error between cell3 and cell1 | μs | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1. | The value of time alignment error depends upon the type of carrier aggregation. |
| T1 | s | 5 | |
| T2 | s | 10 | |
| Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3. | | | |

### 8.20.2B.4.2 Test procedure

Same test procedure as in clause 8.20.2.4.2 with the following exceptions:

- Instead of Table 8.20.2.5-1 → use Table 8.20.2B.5-1.

### 8.20.2B.4.3 Message contents

Same message contents as in clause 8.20.2.4.3.

## 8.20.2B.5 Test requirement

Tables 8.20.2B.4.1-1 and 8.20.2B.5-1 define the primary level settings including test tolerances for event triggered reporting under fading propagation conditions in synchronous inter frequency cells test.

**Table 8.20.2B.5-1: Cell specific test parameters for E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in synchronous cells for 20 MHz +10 MHz bandwidth**

| Parameter | Unit | Cell 1 | | Cell 2 | | Cell 3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | | 3 | |
| $BW_{channel}$ | MHz | 20 | | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| OCNG Pattern defined in D.2.1 (OP.1) and in D.2.2 (OP.2 TDD) and in D.2.7 (OP.7) | | OP.7 TDD | | OP.2 TDD | | OP.1 TDD | |
| PBCH_RA | dB | 0 | | 0 | | 0 | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_RB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_RB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | -Infinity | 7.7 | 4 | 4 |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | | | | |
| RSRP Note 4 | dBm/15 kHz | -94 | -94 | -Infinity | -90.3 | -94 | -94 |
| SCH_RP Note 4 | dBm/15 kHz | -94 | -94 | -infinity | -90.3 | -94 | -94 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | -Infinity | 7.7 | 4 | 4 |
| Propagation Condition | | ETU70 | | | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

The UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 7682 ms from the beginning of time period T2.

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 7682 ms in this test case (note: a total of 7680 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95%.

NOTE: The actual overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

## 8.20.3 E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions

### 8.20.3.1 Test purpose

To verify the UE's ability to make a correct reporting of an event under fading propagation conditions within the E-UTRA FDD - UTRA FDD cell search requirements in CA mode.

### 8.20.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA and UTRA FDD. Applicability requires support for FGI bit 15.

This test applies to all types of E-UTRA FDD UE release 11 and forward that support CA with Intra-band non-contiguous CA and UTRA FDD. Applicability requires support for FGI bit 15.

### 8.20.3.3 Minimum conformance requirements

The measurement reporting delay shall be less than $T_{identify,\ UTRA_FDD}$ in RRC_CONNECTED state.

When explicit neighbour list is provided and no DRX is used the UE shall be able to identify a new detectable cell within

$$T_{identify,\ UTRA_FDD} = T_{basic_identify_UTRA_FDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq} \quad ms$$

Where:

$T_{basic_identify_UTRA_FDD}$ = 300 ms. This is the time period used in the inter-RAT equation where the maximum allowed time for the UE to identify a new UTRA FDD cell is defined.

$T_{Inter1}$ = 30 ms. This is the minimum available time for inter-RAT measurement during 480 ms period

$N_{Freq}$: This is the number of UTRA carriers being monitored

A cell shall be considered detectable when

- CPICH Ec/Io $\geqq$ -20 dB,

- SCH_Ec/Io ≥ -17 dB for at least one channel tap and SCH_Ec/Ior is equally divided between primary synchronisation code and secondary synchronisation code. When L3 filtering is used an additional delay can be expected.

When measurement gaps are scheduled for UTRA FDD inter RAT measurement the UE physical layer shall be capable of reporting measurements to higher layers with measurement accuracy as specified in TS 36.133 [4] clause 9.2 of with measurement period given by

$$T_{measurement_UTRA_FDD} = Max\left\{T_{Measurement_Period\ UTRA_FDD}, T_{basic_measurement_UTRA_FDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms$$

Where:

$X_{basic\ measurement\ UTRA_FDD}$ = 6 (cells)

$T_{Measurement_Period\ UTRA_FDD}$ = 480 ms. The period used for calculating the measurement period.

$T_{basic_measurement_UTRA_FDD}$ = 50 ms. This is the time period used in the equation where the maximum allowed time for the UE to identify a new UTRA FDD cell is defined.

$T_{Inter1}$ = 30 ms. This is the minimum available time for inter-RAT measurement during 480 ms period

$N_{Freq}$: This is the number of UTRA carriers being monitored

If the UE does not need measurement gaps to perform UTRA FDD measurements, the measurement period for UTRA FDD measurements is 480 ms.

The UE shall be capable of performing UTRA FDD CPICH measurements for $X_{basic\ measurement\ UTRA_FDD}$ inter-frequency cells per FDD frequency and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{Measurement_UTRA_FDD}$.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between any events that will trigger a measurement report until the UE starts to transmit the measurement report over the Uu interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: twice the TTI of the uplink DCCH. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify\ UTRA_FDD}$ defined in TS 36.133 [4] clause 8.1.2.4.1.1.1. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.1 and A.8.20.3.

## 8.20.3.4 Test description

### 8.20.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.44 as appropriate.

2. The general test parameter settings are set up according to Table 8.20.3.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.20.3.4.3.

5. There is one E-UTRA FDD Cell 1 for PCC, one E-UTRA FDD Cell 3 for SCC and one UTRA FDD Cell 2 specified in the test. Cell 1 (E-UTRA FDD cell for PCC) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.20.3.4.1-1: General test parameters for E-UTRAN FDD-UTRAN FDD event triggered reporting in fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1. |
| Gap Pattern Id | | 1 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| Active cell | | Cell 1 | Cell 1 is on E-UTRA RF channel number 1. |
| Neighbour cell | | Cell 2 | Cell 2 is on UTRA RF channel number 1. |
| Configured active SCell | | Cell 3 | Cell 3 is on E-UTRA RF channel number 2. |
| CP length | | Normal | Applicable to cell 1 |
| E-UTRA RF Channel Number | | 1 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA RF Channel Number for SCell | | 2 | One E-UTRA FDD carrier frequency is used. |
| E-UTRA Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| UTRA RF Channel Number | | 1 | One UTRA FDD carrier frequency is used. |
| Inter-RAT (UTRA FDD) measurement quantity | | CPICH Ec/Io | |
| b1-Threshold-UTRA | dB | -18 | CPICH Ec/Io threshold for event B1. |
| Hysteresis | dB | 0 | |
| Time To Trigger | ms | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used. |
| DRX | | OFF | |
| Monitored UTRA FDD cell list size | | 12 | UTRA cells on UTRA RF channel 1 provided in the cell list. |
| T1 | s | 5 | |
| T2 | s | 6 | |

### 8.20.3.4.2 Test procedure

The test consists of one active E-UTRA Cell 1 and one UTRA neighbour cell 2 and one configured active SCell Cell 3. In the measurement control information it is indicated to the UE that event-triggered reporting with Event B1 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0, C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 3) on the SCC as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.20.3.4.3.

4. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [13], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133, clauses 8.3.3.2).

5. Set the parameters according to T1 in Table's 8.20.3.5-1 and 8.20.3.5-2. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

6. SS shall transmit an RRCConnectionReconfiguration message with Event B1 configured.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 8.20.3.5-1 and 8.20.3.5-2.

9. The UE shall transmit a MeasurementReport message triggered by Event B1. If the overall delay measured from the beginning of time period T2 is less than 4802 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

10. After the SS receives the MeasurementReport message in step 9) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

11. The SS shall set Cell 2 primary scrambling code = ((current cell 2 primary scrambling code - 50) mod 200 + 100) for the next iteration of the test procedure loop.

12. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

13. Repeat steps 2-12 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.20.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.20.3.4.3-1: Common Exception messages for E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7 |

**Table 8.20.3.4.3-2: *MeasConfig-DEFAULT*: E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 3 entries | | |
| measObjectId[1] | IdMeasObject-f1 | f1 is the frequency of the serving cell (E-UTRA Cell) | |
| measObject[1] | MeasObjectEUTRA-GENERIC(f1) | E-UTRA Cell | |
| measObjectId[2] | IdMeasObject-f2 | f2 is the frequency of the configured SCell (E-UTRA Cell) | |
| measObject[2] | MeasObjectEUTRA-GENERIC(f2) | E-UTRA Cell | |
| measObjectId[3] | IdMeasObject-f10 | f10 is the frequency of the neighbouring cell(UTRA Cell) | |
| measObject[3] | MeasObjectUTRA-GENERIC(f10) | UTRA Cell | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-B1 | | |
| reportConfig | ReportConfigInterRAT-B1-UTRA | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f10 | | |
| reportConfigId | idReportConfig-B1 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | UTRAN |
| measGapConfig | MeasGapConfig-GP2 | | Gap Pattern Id = 1 |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 8.20.3.4.3-3: *ReportConfigInterRAT-B1-UTRA*: Additional E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7B ReportConfigInterRAT-B1(EUTRA-Thres) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-UTRA(UTRA-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| b1-ThresholdUTRA CHOICE { | | | |
| utra-EcN0 | 13 (-18 dB) | -18 dB is actual EcNO value in dB ((13 - 49)/2 dB) | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.20.3.4.3-4: *MeasResults*: Additional E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | Cell 3 | |
| rsrpResultSCell-r10 | (0..97) | | |
| rsrqResultSCell-r10 | (0..34) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.20.3.4.3-5: *MeasResultListUTRA*: Additional E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| fdd | PhysCellIdUTRA-FDD | PhysCellIdUTRA-FDD INTEGER (0..511) | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-EcN0 | | Set according to specific test | |
| } | | | |
| } | | | |

## 8.20.3.5 Test requirement

Tables 8.20.3.4.1-1, 8.20.3.5-1 and 8.20.3.5-2 define the primary level settings including test tolerances for event triggered reporting under fading propagation conditions in asynchronous inter frequency cells test in CA mode.

**Table 8.20.3.5-1: Cell specific test parameters for E-UTRAN FDD (cell # 1, cell # 3) for event triggered reporting of UTRAN FDD cell under fading propagation conditions**

| Parameter | Unit | Cell 1 | | Cell 3 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| **Correlation Matrix and Antenna Configuration** | | **1x2 Low** | | **1x2 Low** | |
| OCNG Pattern defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | | OP.1 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA[Note 1] | dB | | | | |
| OCNG_RB[Note 1] | dB | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | 4 | 4 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | 4 | 4 |
| $N_{oc}$ | dBm/15 kHz | -98 | | -98 | |
| RSRP | dBm/15 kHz | -94 | -94 | -94 | -94 |
| SCH_RP | dBm/15 kHz | -94 | -94 | -94 | -94 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

**Table 8.20.3.5-2: Cell specific test parameters for UTRAN FDD (cell # 2) for event triggered reporting of UTRAN FDD cell under fading propagation conditions**

| Parameter | Unit | Cell 2 | |
|---|---|---|---|
| | | T1 | T2 |
| UTRA RF Channel Number | | 1 | |
| CPICH_Ec/Ior | dB | -10 | |
| PCCPCH_Ec/Ior | dB | -12 | |
| SCH_Ec/Ior | dB | -12 | |
| PICH_Ec/Ior | dB | -15 | |
| DPCH_Ec/Ior | dB | N/A | |
| OCNS | | -0.941 | |
| $\hat{I}_{or} / I_{oc}$ | dB | -Infinity | -1.8 |
| $I_{oc}$ | dBm/3.84 MHz | -70 | |
| CPICH_Ec/Io | dB | -Infinity | -14 |
| Propagation Condition | | AWGN | |

Note 1: The DPCH level is controlled by the power control loop.
Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event B1 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify,\ UTRA_FDD}$

$$T_{identify,\ UTRA_FDD} = T_{basic_identify_UTRA_FDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq} \quad ms$$

$T_{basic_identify_UTRA_FDD}$ = 300 ms

$T_{Inter1}$ = 30 ms

$N_{Freq}$ = 1

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 4802 ms in this test case (note: this gives a total of 4800 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.20.4 E-UTRAN TDD to UTRAN TDD cell search under fading propagation conditions

### 8.20.4.1 Test purpose

To verify the UE's ability to make a correct reporting of an event under fading propagation conditions within the E-UTRA TDD - UTRA TDD cell search requirements in CA mode.

### 8.20.4.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA and UTRA TDD and not supporting UTRA FDD. Applicability requires support for FGI bit 15 and 22.

This test applies to all types of E-UTRA TDD UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA and UTRA TDD. Applicability requires support for FGI bits 15 and 39.

This test applies to all types of E-UTRA TDD UE release 11 and forward that support CA with Intra-band non-contiguous CA and UTRA TDD and not supporting UTRA FDD. Applicability requires support for FGI bit 15 and 22.

This test applies to all types of E-UTRA TDD UE release 11 and forward that support CA with Intra-band non-contiguous CA and UTRA TDD. Applicability requires support for FGI bits 15 and 39.

### 8.20.4.3 Minimum conformance requirements

The measurement reporting delay shall be less than $T_{identify,\ UTRA_TDD}$ in RRC_CONNECTED state.

When explicit neighbour list is provided and no DRX is used the UE shall be able to identify a new detectable cell belonging to the monitored set within

$$T_{identify,\ UTRA_TDD} = Max\left\{5000, T_{basic\ identify\ UTRA_TDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms$$

If the UE does not require transmit gap to perform inter-RAT UTRA TDD measurements, the UE shall be able to identify a new detectable inter-RAT UTRA TDD cell belonging to the monitored set within 5000 ms.

A cell shall be considered detectable when

- P-CCPCH Ec/Io $\geq$ -8 dB,
- DwPCH_Ec/Io $\geq$ -5 dB.

When L3 filtering is used an additional delay can be expected.

When measurement gaps are scheduled for UTRA TDD inter RAT measurements the UE physical layer shall be capable of reporting measurements to higher layers with measurement accuracy as specified in TS 36.133 [4] clause 9.3 with measurement period given by

$$T_{measurement\ UTRA_TDD} = Max\left\{T_{Measurement_Period\ UTRA_TDD}, T_{basic\ measurement\ UTRA_TDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms$$

If the UE does not need measurement gaps to perform UTRA TDD measurements, the measurement period for UTRA TDD measurements is 480 ms.

The UE shall be capable of performing UTRA TDD P-CCPCH RSCP measurements for $X_{basic\ measurementUTRA_TDD}$ inter-frequency cells per TDD frequency of the monitored set and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{Measurement_UTRA_TDD.}$

$X_{basic\ measurementUTRA_TDD}$ = 6

$T_{Measurement_Period\ UTRA_TDD}$ = 480 ms is the period used for calculating the measurement period $T_{measurement_UTRA_TDD}$ for UTRA TDD P-CCPCH RSCP measurements.

$T_{basic_identify_UTRA_TDD}$ = 800 ms is the time period used in the inter RAT equation where the maximum allowed time for the UE to identify a new UTRA TDD cell is defined.

$T_{basic_measurement_UTRA_TDD}$ = 50 ms is the time period used in the equation for defining the measurement period for inter RAT P-CCPCH RSCP measurements.

$N_{freq}$ and $T_{inter1}$ are defined in TS 36.133 [4] section 8.1.2.1.1

Reported measurements in event triggered measurement reports shall meet the requirements in TS 36.133 [4] section 9.

The UE shall not send any event triggered measurement reports, as long as the reporting criteria is not fulfilled.

The measurement reporting delay is defined as the time between any event that will trigger a measurement report until the UE starts to transmit the measurement report over the Uu interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is twice the TTI of the uplink DCCH. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify,\ UTRA_TDD}$ defined in TS 36.133 [4] Section 8.1.2.4.3.1.1 When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.1.2.4.3 and A.8.20.4.

## 8.20.4.4 Test description

### 8.20.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7], Figure group A.44 as appropriate.

2. The general test parameter settings are set up according to Table 8.20.4.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.20.4.4.3.

5. There is one E-UTRA TDD Cell 1 for PCC, one E-UTRA TDD Cell 3 for SCC and one UTRA TDD Cell 2 specified in the test. Cell 1 (E-UTRA TDD cell for PCC) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

**Table 8.20.4.4.1-1: General test parameters for E-UTRA TDD to UTRA(1.28 Mcps TDD OPTION) cell search in fading propagation conditions**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| E-UTRA RF Channel Number | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA RF Channel Number for SCell | | 2 | One E-UTRA TDD carrier frequency is used. |
| Active cell | | Cell 1 | E-UTRA TDD cell |
| Neighbour cell | | Cell 2 | UTRA 1.28Mcps TDD Cell |
| Configured active SCell | | Cell 3 | E-UTRA TDD cell |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| Uplink-downlink configuration of cell 1 | | 1 | As specified in table 4.2.2 in TS 36.211 |
| Special subframe configuration of cell 1 | | 6 | As specified in table 4.2.1 in TS 36.211 |
| CP length of cell 1 | | normal | |
| Hysteresis | dB | 0 | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| Time offset between cells | | 3 ms | Asynchronous cells |
| Ofn | dB | 0 | |
| Thresh | dBm | -87 | |
| T1 | s | 5 | |
| T2 | s | 10 | |

### 8.20.4.4.2 Test procedure

This test scenario comprised of 1 E-UTRA TDD PCell, 1 E-UTRA TDD SCell and 1 UTRA TDD cell to be searched. In the measurement control information it is indicated to the UE that event-triggered reporting with Event B1 is used. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0, C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 3) on the SCC as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.20.4.4.3.

4. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [13], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133, clauses 8.3.3.2).

5. Set the parameters according to T1 in Table's 8.20.4.5-1 and 8.20.4.5-2. Propagation conditions are set according to Annex B clause B.1.1 and B.2.2. T1 starts.

6. SS shall transmit an RRCConnectionReconfiguration message with Event B1 configured.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table's 8.20.4.5-1 and 8.20.4.5-2.

9. The UE shall transmit a MeasurementReport message triggered by Event B1. If the overall delay measured from the beginning of time period T2 is less than 6402 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

10. After the SS receives the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

11. SS shall change to set cell 2 cell parameter id = (current cell 2 cell parameter id +4) mod 16.

12. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

13. Repeat step 2-12 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.20.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.20.4.4.3-1: Common Exception messages for E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7 |

**Table 8.20.4.4.3-2: *MeasConfig-DEFAULT*: E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 3 entries | | |
| measObjectId[1] | IdMeasObject-f1 | f1 is the frequency of the serving cell (E-UTRA Cell) | |
| measObject[1] | MeasObjectEUTRA-GENERIC(f1) | E-UTRA Cell | |
| measObjectId[2] | IdMeasObject-f2 | f2 is the frequency of the configured SCell (E-UTRA Cell) | |
| measObject[2] | MeasObjectEUTRA-GENERIC(f2) | E-UTRA Cell | |
| measObjectId[3] | IdMeasObject-f9 | f9 is the frequency of the neighbouring cell(UTRA Cell) | |
| measObject[3] | MeasObjectUTRA-GENERIC(f9) | UTRA Cell | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-B1 | | |
| reportConfig | ReportConfigInterRAT-B1-UTRA | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f9 | | |
| reportConfigId | idReportConfig-B1 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | UTRAN |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 8.20.4.4.3-3: *ReportConfigInterRAT-B1-UTRA*: Additional E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.508 clause 4.6.6 Table 4.6.6-7B | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-B1-UTRA(UTRA-Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventB1 SEQUENCE { | | | |
| b1-Threshold CHOICE { | | | |
| b1-ThresholdUTRA CHOICE { | | | |
| utra-RSCP | 28 | UTRA-Thres is actual RSCP value in dBm<br>UTRA-Thres + 115 | UTRA-TDD |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0dB) | The actual value is IE value * 0.5 dB. | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.20.4.4.3-4: *MeasResults*: Additional E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | Cell 3 | |
| rsrpResultSCell-r10 | (0..97) | | |
| rsrqResultSCell-r10 | (0..34) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.20.4.4.3-4: *MeasResultListUTRA*: Additional E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| tdd | PhysCellIdUTRA-TDD | INTEGER (0..127) | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-RSCP | | Set according to specific test INTEGER (-5..91) | |
| } | | | |
| } | | | |

## 8.20.4.5 Test requirement

Tables 8.20.4.4.1-1, 8.20.4.5-1 and 8.20.4.5-2 define the primary level settings including test tolerances for E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions in synchronous cells test in CA mode.

**Table 8.20.4.5-1: Cell specific test parameters for cell search E-UTRA TDD to UTRA TDD test case (cell 1, cell3)**

| Parameter | Unit | Cell 1 | | Cell 3 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | | OP.1 TDD | |
| PBCH_RA | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note1 | dB | | | | |
| OCNG_RB Note1 | dB | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 9 | 9 | 9 | 9 |
| $\hat{E}_s / N_{oc}$ | dB | 9 | 9 | 9 | 9 |
| $N_{oc}$ | dBm/15kHz | -98 | | | |
| RSRP | dBm/15kHz | -89 | -89 | -89 | -89 |
| SCH_RP | dBm/15kHz | -89 | -89 | -89 | -89 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

**Table 8.20.4.5-2: Cell specific test parameters for cell search E-UTRA TDD to UTRA TDD test case (cell 2)**

| Parameter | Unit | Cell 2 (UTRA) | | | |
|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPTS | |
| | | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number NOTE1 | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 |
| OCNS_Ec/Ior NOTE2 | dB | -3 | -3 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -inf | 5 | -inf | 5 + |
| $I_{oc}$ | dBm/1.28 MHz | -80 | | | |
| PCCPCH RSCP | dBm | -inf | -78 | n.a. | n.a. |
| Propagation Condition | | Case 3 NOTE3 | | | |

Note 1: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number.
Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.
Note 3: Case 3 propagation conditions are defined in Annex B of 3GPP TS 25.102

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event B1 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to 2x$TTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify, UTRA_TDD}$

$$T_{identify,\ UTRA_TDD} = Max\left\{5000, T_{basic\ identify\ UTRA_TDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms$$

$T_{basic_identify_UTRA_TDD}$ = 800 ms

$T_{Inter1}$ = 60 ms

$N_{Freq}$ = 1

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 6402 ms in this test case (note: this gives a total of 6400 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.20.4A E-UTRAN TDD to UTRAN TDD cell search under fading propagation conditions for 20 MHz + 20 MHz bandwidth

### 8.20.4A.1 Test purpose

Same test purpose as in clause 8.20.4.1.

## 8.20.4A.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA and UTRA TDD and not supporting UTRA FDD. Applicability requires support for FGI bit 15 and 22.

This test applies to all types of E-UTRA TDD UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA and UTRA TDD. Applicability requires support for FGI bits 15 and 39.

This test applies to all types of E-UTRA TDD UE release 11 and forward that support CA with Intra-band non-contiguous CA and UTRA TDD and not supporting UTRA FDD. Applicability requires support for FGI bit 15 and 22.

This test applies to all types of E-UTRA TDD UE release 11 and forward that support CA with Intra-band non-contiguous CA and UTRA TDD. Applicability requires support for FGI bits 15 and 39.

## 8.20.4A.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.20.4.3.

## 8.20.4A.4 Test description

### 8.20.4A.4.1 Initial conditions

Same initial conditions as in clause 8.20.4.4.1 with the following exceptions:

- Channel Bandwidth to be tested: 20 MHz
- Instead of Table 8.20.4.4.1-1 → use Table 8.20.4A.4.1-1.

**Table 8.20.4A.4.1-1: General test parameters for E-UTRA TDD to UTRA (1.28 Mcps TDD OPTION) cell search in fading propagation conditions for 20 MHz + 20 MHz bandwidth**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.3 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.10 TDD | As specified in section A.2.2 |
| E-UTRA RF Channel Number | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA RF Channel Number for SCell | | 2 | One E-UTRA TDD carrier frequency is used. |
| Active cell | | Cell 1 | E-UTRA TDD cell |
| Neighbour cell | | Cell 2 | UTRA 1.28Mcps TDD Cell |
| Configured active SCell | | Cell 3 | E-UTRA TDD cell |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| Uplink-downlink configuration of cell 1 | | 1 | As specified in table 4.2.2 in TS 36.211 |
| Special subframe configuration of cell 1 | | 6 | As specified in table 4.2.1 in TS 36.211 |
| CP length of cell 1 | | normal | |
| Hysteresis | dB | 0 | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| Time offset between cells | | 3 ms | Asynchronous cells |
| Ofn | dB | 0 | |
| Thresh | dBm | -87 | |
| T1 | s | 5 | |
| T2 | s | 10 | |
| Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is .performed according to the principle defined in section C.3.3 | | | |

### 8.20.4A.4.2 Test procedure

Same test procedure as in clause 8.20.4.4.2 with the following exceptions:

- Instead of Table 8.20.4.5-1 → use Table 8.20.4A.5-1.
- Instead of Table 8.20.4.5-2 → use Table 8.20.4A.5-2.

### 8.20.4A.4.3 Message contents

Same message contents as in clause 8.20.4.4.3.

## 8.20.4A.5 Test requirement

Table 8.20.4A.4.1-1, 8.20.4A.5-1 and 8.20.4A.5-2 define the primary level settings including test tolerances for E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions in synchronous cells test in CA mode.

**Table 8.20.4A.5-1: Cell specific test parameters for cell search E-UTRA TDD to UTRA TDD test case for 20 MHz + 20 MHz bandwidth (cell 1, cell3)**

| Parameter | Unit | Cell 1 | | Cell 3 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 20 | | 20 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Pattern defined in D.2.7 (OP.7 TDD) | | OP.7 TDD | | OP.7 TDD | |
| PBCH_RA | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA[Note1] | dB | | | | |
| OCNG_RB[Note1] | dB | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 9 | 9 | 9 | 9 |
| $\hat{E}_s / N_{oc}$ | dB | 9 | 9 | 9 | 9 |
| $N_{oc}$ | dBm/15kHz | -98 | | | |
| RSRP | dBm/15kHz | -89 | -89 | -89 | -89 |
| SCH_RP | dBm/15kHz | -89 | -89 | -89 | -89 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

**Table 8.20.4A.5-2: Cell specific test parameters for cell search E-UTRA TDD to UTRA TDD test case (cell 2)**

| Parameter | Unit | Cell 2 (UTRA) | | | |
|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPTS | |
| | | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number NOTE1 | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 |
| OCNS_Ec/Ior NOTE2 | dB | -3 | -3 | | |
| $\hat{I}_{or}/I_{oc}$ | dB | -inf | 5 | -inf | 5 |
| $I_{oc}$ | dBm/1.28 MHz | -80 | | | |
| PCCPCH RSCP | dBm | -inf | -78 | n.a. | n.a. |
| Propagation Condition | | Case 3 NOTE3 | | | |

Note 1: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number.
Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.
Note 3: Case 3 propagation conditions are defined in Annex B of 3GPP TS 25.102.

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event B1 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to 2x$TTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify, UTRA_TDD}$

$$T_{identify,\ UTRA_TDD} = Max\left\{5000, T_{basic\ identify\ UTRA_TDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms$$

$T_{basic_identify_UTRA_TDD}$ = 800 ms

$T_{Inter1}$ = 60 ms

$N_{Freq}$ = 1

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 6402 ms in this test case (note: this gives a total of 6400 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.20.4B E-UTRAN TDD to UTRAN TDD cell search under fading propagation conditions for 20 MHz + 10 MHz bandwidth

### 8.20.4B.1 Test purpose

Same test purpose as in clause 8.20.4.1

## 8.20.4B.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA and UTRA TDD and not supporting UTRA FDD. Applicability requires support for FGI bit 15 and 22.

This test applies to all types of E-UTRA TDD UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA and UTRA TDD. Applicability requires support for FGI bits 15 and 39.

This test applies to all types of E-UTRA TDD UE release 11 and forward that support CA with Intra-band non-contiguous CA and UTRA TDD and not supporting UTRA FDD. Applicability requires support for FGI bit 15 and 22.

This test applies to all types of E-UTRA TDD UE release 11 and forward that support CA with Intra-band non-contiguous CA and UTRA TDD. Applicability requires support for FGI bits 15 and 39.

## 8.20.4B.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.20.4.3

## 8.20.4B.4 Test description

### 8.20.4B.4.1 Initial conditions

Same initial conditions as in clause 8.20.4.4.1 with the following exceptions:

- Channel Bandwidth to be tested: 20 MHz for Cell 1 on the PCC, 10MHz for Cell 3 on the SCC
- Instead of Table 8.20.4.4.1-1 → use Table 8.20.4B.4.1-1.

**Table 8.20.4B.4.1-1: General test parameters for E-UTRA TDD to UTRA (1.28 Mcps TDD OPTION) cell search in fading propagation conditions for 20 MHz + 10 MHz bandwidth**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Channel bandwidth for Cell 1 ($BW_{channel}$) | MHz | 20 | |
| PDSCH parameters for Cell 1 | | DL Reference Measurement Channel R.3 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters for Cell 1 | | DL Reference Measurement Channel R.10 TDD | As specified in section A.2.2 |
| Channel bandwidth for Cell 3 ($BW_{channel}$) | MHz | 10 | |
| PDSCH parameters for Cell 3 | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters for Cell 3 | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| E-UTRA RF Channel Number | | 1 | One E-UTRA TDD carrier frequency is used. |
| E-UTRA RF Channel Number for SCell | | 2 | One E-UTRA TDD carrier frequency is used. |
| Active cell | | Cell 1 | E-UTRA TDD cell |
| Neighbour cell | | Cell 2 | UTRA 1.28Mcps TDD Cell |
| Configured active SCell | | Cell 3 | E-UTRA TDD cell |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| Uplink-downlink configuration of cell 1 | | 1 | As specified in table 4.2.2 in TS 36.211 |
| Special subframe configuration of cell 1 | | 6 | As specified in table 4.2.1 in TS 36.211 |
| CP length of cell 1 | | normal | |
| Hysteresis | dB | 0 | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| Time offset between cells | | 3 ms | Asynchronous cells |
| Ofn | dB | 0 | |
| Thresh | dBm | -87 | |
| T1 | s | 5 | |
| T2 | s | 10 | |
| Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3. | | | |

### 8.20.4B.4.2 Test procedure

Same test procedure as in clause 8.20.4.4.2 with the following exceptions:

- Instead of Table 8.20.4.5-1 → use Table 8.20.4B.5-1.

### 8.20.4B.4.3 Message contents

Same message contents as in clause 8.20.4.4.3.

## 8.20.4B.5 Test requirement

Table 8.20.4B.4.1-1, 8.20.4B.5-1 and 8.20.4B.5-2 define the primary level settings including test tolerances for E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions in synchronous cells test in CA mode.

**Table 8.20.4B.5-1: Cell specific test parameters for cell search E-UTRA TDD to UTRA TDD test case for 20 MHz + 10 MHz bandwidth (cell 1, cell3)**

| Parameter | Unit | Cell 1 | | Cell 3 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 20 | | 10 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Pattern defined in D.2.1 (OP.1 TDD) and D.2.7 (OP.7 TDD) | | OP.7 TDD | | OP.1 TDD | |
| PBCH_RA | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA[Note1] | dB | | | | |
| OCNG_RB[Note1] | dB | | | | |
| $\hat{E}_s / I_{ot}$ | dB | 9 | 9 | 9 | 9 |
| $\hat{E}_s / N_{oc}$ | dB | 9 | 9 | 9 | 9 |
| $N_{oc}$ | dBm/15kHz | -98 | | | |
| RSRP | dBm/15kHz | -89 | -89 | -89 | -89 |
| SCH_RP | dBm/15kHz | -89 | -89 | -89 | -89 |
| Propagation Condition | | ETU70 | | | |

Note 1: OCNG shall be used such that cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

**Table 8.20.4B.5-2: Cell specific test parameters for cell search E-UTRA TDD to UTRA TDD test case (cell 2)**

| Parameter | Unit | Cell 2 (UTRA) | | | |
|---|---|---|---|---|---|
| Timeslot Number | | 0 | | DwPTS | |
| | | T1 | T2 | T1 | T2 |
| UTRA RF Channel Number [NOTE1] | | Channel 2 | | | |
| PCCPCH_Ec/Ior | dB | -3 | -3 | | |
| DwPCH_Ec/Ior | dB | | | 0 | 0 |
| OCNS_Ec/Ior[NOTE2] | dB | -3 | -3 | | |
| $\hat{I}_{or} / I_{oc}$ | dB | -inf | 5 | -inf | 5 |
| $I_{oc}$ | dBm/1.28 MHz | -80 | | | |
| PCCPCH RSCP | dBm | -inf | -78 | n.a. | n.a. |
| Propagation Condition | | Case 3[NOTE3] | | | |

Note 1: In the case of multi-frequency cell, the UTRA RF Channel Number is the primary frequency's channel number.
Note 2: The power of the OCNS channel that is added shall make the total power from the cell to be equal to $I_{or}$.
Note 3: Case 3 propagation conditions are defined in Annex B of 3GPP TS 25.102

The overall delays measured is defined as the time from the beginning of time period T2, to the moment the UE send one Event B1 triggered measurement report to Cell 2.

The overall delays measured in the test may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The overall delays measured test requirement is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Measurement reporting delay = $T_{identify, UTRA_TDD}$

$$T_{identify,\ UTRA_TDD} = Max\left\{5000, T_{basic\ identify\ UTRA_TDD} \cdot \frac{480}{T_{inter1}} \cdot N_{Freq}\right\} ms$$

$T_{basic_identify_UTRA_TDD}$ = 800 ms

$T_{Inter1}$ = 60 ms

$N_{Freq}$ = 1

TTI insertion uncertainty = 2 ms

The overall delays measured shall be less than a total of 6402 ms in this test case (note: this gives a total of 6400 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 8.21

# 8.22 E-UTRAN Discovery Signal Measurements

## 8.22.1 E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal

### 8.22.1.1 Test purpose

To verify the UE's ability to make a correct reporting of an event when discovery signal is configured in DRX under fading propagation conditions in synchronous cells within the E-UTRA FDD-FDD intra frequency cell search requirements.

### 8.22.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward and supports CRS based discovery signals measurement. Applicability requires support for FGI bit 5.

### 8.22.1.3 Minimum conformance requirements

When DRX is in use the UE shall be able to identify a new detectable FDD intra frequency cell within $T_{identify_intra_SCE_DRX}$.

$$T_{identify_intra_SCE_DRX} = 16 * Max\ \{ T_{DMTC_periodicity}, \text{DRX cycle length}\} + T_{Measurement_Period_intra_FDD_CRS_DRX}$$

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] Sections 9.1.14.2 and RSRQ related side conditions given in TS 36.133 [4] Clause 9.1.14.4 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.1 for a corresponding Band

$T_{DMTC_periodicity}$ is the discovery signal measurement timing configuration periodicity of higher layer.

$T_{Measurement_Period_intra_FDD_CRS_DRX}$ is the intra-frequency period for measurements as shown in table 8.22.1.3-1

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{Measurement_Period\ intra_FDD_CRS_DRX}$ when DRX is used. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is $T_{Measurement_Period_intra_FDD_CRS_DRX}$ as shown in table 8.22.1.3-1, when DRX is in use. The UE shall be capable of performing RSRP and RSRQ measurements for 3 identified intra-frequency cells, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{Measurement_Period_intra_FDD_CRS_DRX}$.

**Table 8.22.1.3-1: Requirement to measure FDD intra frequency cell**

| Measurement bandwidth [RB] | Discovery signal occasion duration (*ds-OccasionDuration*) [ms] | $T_{Measurement_Period\ intra_FDD_CRS_DRX}$ [ms] |
|---|---|---|
| ≥ 6 | ≥1 | 5 * *Max*{ $T_{DMTC_periodicity}$, DRX cycle length } |
| ≥ 25 | ≥1 | 3 * *Max*{ $T_{DMTC_periodicity}$, DRX cycle length } |

The RSRP measurement accuracy for all measured cells shall be as specified in the TS 36.133 [4] sub-clauses 9.1.14.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in the sub-clause 9.1.14.4.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.14.2 and 9.1.14.4 respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$.This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra_SCE_DRX}$ defined in TS 36.133 [4] Clause 8.6.2.1.1.2. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_SCE_DRX}$ defined in TS 36.133 [4] clause 8.6.2.1.1.2 becomes undetectable for a period ≤ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_intra_FDD_CRS_DRX}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.6.2.1.1.2 and A.8.22.1.

## 8.22.1.4 Test description

### 8.22.1.4.1 Initial conditions

Same Initial conditions as defined in clause 8.1.3.4.1 with the following exceptions:

- Instead of Table 8.1.3.4.1-1 -> use Table 8.22.1.4.1-1.
- Instead of clause 8.1.3.4.3 -> use clause 8.22.1.4.3.

**Table 8.22.1.4.1-1: General test parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | One FDD carrier frequency is used. |
| DMTC period | ms | 160 | As specified in IE MeasDS-Config in TS 36.331[5] |
| dmtc-PeriodOffset | ms | 10 | As specified in IE MeasDS-Config in TS 36.331[5] |
| Discovery signal occasion duration | ms | 1 | As specified in IE MeasDS-Config in TS 36.331[5] |
| A3-Offset | dB | -6 | |
| CP length | | Normal | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | ON | DRX related parameters are defined in Table 8.22.1.5-2 |
| T1 | s | 5 | |
| T2 | s | 10 | |

### 8.22.1.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information, it is indicated to the UE performing CRS based discovery signals measurement and event-triggered reporting with Event A3 is configured. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

UE needs to be provided at least once every 500 ms with new Timing Advance Command MAC control element to restart the Timer Alignment Timer to keep the UE uplink time alignment. Furthermore, the UE is allocated with PUSCH resource at every DRX cycle.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.22.1.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.
3. The SS shall transmit an RRCConnectionReconfiguration message and the SS shall provide the UE with new Timing Advance Command MAC control element at least once every 500 ms to restart the Timer Alignment Timer in order to keep the UE uplink time alignment.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.22.1.5-1 and 8.22.1.5-2.
6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 5122 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.
7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

#### 8.22.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.22.1.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-2<br>Table H.3.1-7<br>Table H.3.7-3 |

**Table 8.22.1.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.22.1.4.3-3: MeasObjectEUTRA-GENERIC(f1): Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(f1) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| carrierFreq | Downlink EARFCN for Freq | | |
| allowedmeasBandwidth | 25 | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationFDD-r12 | 1 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.1.4.3-4: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventide CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.22.1.4.3-5: *MAC-MainConfig-RBC*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal requirements**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf256 | 9 | To avoid overlapping with measurement gap. | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | sf500 | | |
| } | | | |

**Table 8.22.1.4.3-6: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.22.1.4.3-7: *MeasResults*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.22.1.4.3-8: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| physCellId | PhysCellId | Cell 2 | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.22.1.5 Test requirement

Tables 8.22.1.4.1-1, 8.22.1.5-1, 8.22.1.5-2 and 8.22.1.5-3 define the primary level settings including test tolerances for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal test.

**Table 8.22.1.5-1: Cell specific test parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Measurement bandwidth | $n_{PRB}$ | 13-37 | | 13-37 | |
| PDSCH parameters: DL Reference Measurement Channel | | R.0 FDD | | - | |
| PCFICH/PDCCH/PHICH parameters: DL Reference Measurement Channel | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | 6.10 | 6.10 | -Infinity | 6.10 |
| $\hat{E}_s / I_{ot}$ | dB | 6.10 | -0.95 | -Infinity | -0.95 |
| RSRP Note 3 | dBm/15 KHz | -91.90 | --91.90 | -Infinity | -91.90 |
| SCH_RP Note 3 | dBm/15 KHz | -91.90 | -91.90 | -Infinity | -91.90 |
| Io Note 3 | dBm/9MHz | -63.17 | -60.61 | Specified in columns for Cell 1 | |
| Propagation Condition | | ETU30 | | ETU30 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| Timing offset to Cell 1 | μs | - | | 2.3 (CP/2) | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: Es/Iot, RSRP and SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.22.1.5-2: DRX-Configuration for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf256 | |
| shortDRX | disable | |

**Table 8.22.1.5-3: *TimeAlignmentTimer* and sr-Configuration for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | sf500 | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213. |

The UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 4864 ms from the beginning of time period T2. The measurement reporting delay is defined as the time from the beginning of time period T2 to the moment when the UE send the measurement report on PUSCH.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event A3 measurement report.

The overall delay measured test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra_SCE_DRX}$

$T_{identify_intra_SCE_DRX}$ = 16* *Max* { $T_{DMTC_periodicity}$, DRX cycle length}+ $T_{Measurement_Period_intra_FDD_CRS_DRX}$

Where $T_{DMTC_periodicity}$ = 160ms,

DRX cycle length = 256ms.

$T_{Measurement_Period_intra_FDD_CRS_DRX}$ = 3 * *Max*{ $T_{DMTC_periodicity}$, DRX cycle length } = 768ms

$T_{identify_intra_SCE_DRX}$ = 4864ms.

TTI insertion uncertainty = 2 ms

DRX cycle length = 256 ms

The overall delay measured shall be less than a total of 5122 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.22.2 E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal

### 8.22.2.1 Test purpose

To verify the UE's ability to make a correct reporting of an event when discovery signal is configured in DRX under fading propagation conditions in synchronous cells within the E-UTRA TDD-TDD intra frequency cell search in DRX requirements.

### 8.22.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward and supports CRS based discovery signals measurement. Applicability requires support for FGI bit 5.

### 8.22.2.3 Minimum conformance requirements

When DRX is in use the UE shall be able to identify a new detectable TDD intra frequency cell within $T_{identify_intra_SCE_DRX}$.

$$T_{identify_intra_SCE_DRX} = 16* \max \{ T_{DMTC_periodicity}, \text{DRX cycle length}\} + T_{Measurement_Period_intra_TDD_CRS_DRX}$$

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] Sections 9.1.14.2 and RSRQ related side conditions given in TS 36.133 [4] Clause 9.1.14.4 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.1 for a corresponding Band

$T_{DMTC_periodicity}$ is the discovery signal measurement timing configuration periodicity of higher layer.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{Measurement_Period\ intra_TDD_CRS_DRX}$ when DRX is used. If higher layer filtering is used, an additional cell identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is $T_{Measurement_Period_intra_TDD_CRS_DRX}$ as shown in table 8.22.2.3-1, when DRX is in use. The UE shall be capable of performing RSRP and RSRQ measurements for 3 identified intra-frequency cells, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{Measurement_Period_intra_TDD_CRS_DRX}$

**Table 8.22.2.3-1: Requirement to measure TDD intra frequency cell**

| Measurement bandwidth[RB] | Discovery signal occasion duration (*ds-OccasionDuration*) [ms] | $T_{Measurement_Period\ intra_TDD_CRS_DRX}$ [ms] |
|---|---|---|
| ≥ 6 | ≥2 | 5 * *Max*{ $T_{DMTC_periodicity}$, DRX cycle length } |
| ≥ 25 | ≥2 | 3 * *Max*{ $T_{DMTC_periodicity}$, DRX cycle length } |

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] sub-clauses 9.1.14.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] sub-clause 9.1.14.4.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.14.2 and 9.1.14.4 respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay

uncertainty is: 2 x $TTI_{DCCH}$.This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra_SCE_DRX}$ defined in TS 36.133 [4] Clause 8.6.2.1.2.2. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra_SCE_DRX}$ defined in TS 36.133 [4] clause 8.6.2.1.2.2 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_intra_TDD_CRS_DRX}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC.

The normative reference for this requirement is TS 36.133 [4] clause 8.6.2.1.2 and A.8.22.2.

## 8.22.2.4 Test description

### 8.22.2.4.1 Initial conditions

Same Initial conditions as defined in clause 8.2.2.4.1 with the following exceptions:

- Instead of Table 8.2.2.4.1-1 -> use Table 8.22.2.4.1-1.
- Instead of clause 8.2.2.4.3 -> use Table 8.22.2.4.3.

**Table 8.22.2.4.1-1: General test parameters for E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | One TDD carrier frequency is used. |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211[9]. The same configuration in both cells |
| Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in TS 36.211[9]. The same configuration in both cells |
| DMTC period | ms | 160 | As specified in IE MeasDS-Config in TS 36.331[5] |
| dmtc-PeriodOffset | ms | 10 | As specified in IE MeasDS-Config in TS 36.331[5] |
| Discovery signal occasion duration | ms | 2 | As specified in IE MeasDS-Config in TS 36.331[5] |
| A3-Offset | dB | -6 | |
| CP length | | Normal | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | ON | DRX related parameters are defined in Table 8.22.2.5-2 |
| T1 | s | 5 | |
| T2 | s | 10 | |

### 8.22.2.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information, it is indicated to the UE performing CRS based discovery signals measurement and event-triggered reporting with Event A3 is configured.

The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

UE needs to be provided at least once every 500ms with new Timing Advance Command MAC control element to restart the Time alignment timer to keep UE uplink time alignment. Furthermore, the UE is allocated with PUSCH resource at every DRX cycle.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.22.2.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message and the SS shall provide the UE with new Timing Advance Command MAC control element at least once every 500 ms to restart the Timer Alignment Timer in order to keep the UE uplink time alignment.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.22.2.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 5122 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.
7. After the SS receives the MeasurementReport message in step 6 or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.22.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.22.2.4.3-1: Common Exception messages for E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-2<br>Table H.3.1-7 |

**Table 8.22.2.4.3-2: RRCConnectionReconfiguration: Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig-DEFAULT | | |
| radioResourceConfigDedicated SEQUENCE { | | | |
| MAC-MainConfig-RBC SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf256 | 9 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | sf500 | | |
| } | | | |
| PhysicalConfigDedicated SEQUENCE { | | | |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.2.4.3-3: MeasObjectEUTRA-GENERIC(f1): Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(f1) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| carrierFreq | Downlink EARFCN for Freq | | |
| allowedmeasBandwidth | 25 | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationTDD-r12 | 2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.2.4.3-4: ReportConfigEUTRA-A3: Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.2.4.3-5: SchedulingRequest-Config-DEFAULT: Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20: SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | Channel-bandwidth-dependent parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.22.2.4.3-6: MeasResults: Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | Set according to specific test | |
| rsrqResult | INTEGER(0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.22.2.4.3-6: MeasResultListEUTRA: Additional E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| 1 entry physCellId | PhysCellId | INTEGER (0..503) of Cell 2 | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

## 8.22.2.5 Test requirement

Tables 8.22.2.5-1, 8.22.2.5-2 and 8.22.2.5-3 defines the primary level settings including test tolerances for E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal test.

**Table 8.22.2.5-1: Cell specific test parameters for E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Measurement bandwidth | $n_{PRB}$ | 13-37 | | 13-37 | |
| PDSCH parameters: DL Reference Measurement Channel | | R.0 TDD | | - | |
| PCFICH/PDCCH/PHICH parameters: DL Reference Measurement Channel | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | | OP.2 TDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 2 | dBm/15 KHz | -98+TT | | | |
| $\hat{E}_s / N_{oc}$ | dB | 6.60 | 6.60 | -Infinity | 6.60 |
| $\hat{E}_s / I_{ot}$ | dB | 6.60 | -0.86 | -Infinity | -0.86 |
| RSRP Note 3 | dBm/15 KHz | -91.40 | -91.40 | -Infinity | -91.40 |
| SCH_RP Note 3 | dBm/15 KHz | -91.40 | -91.40 | -Infinity | -91.40 |
| Io Note 3 | dBm/9MHz | -62.76 | -60.16 | Specified in columns for Cell 1 | |
| Propagation Condition | | ETU30 | | ETU30 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| Timing offset to Cell 1 | μs | - | | 2.3 (CP/2) | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: Es/Iot, RSRP and SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.22.2.5-2: DRX-Configuration for E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in clause 6.3.2 in TS 36.331[5] |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf256 | |
| shortDRX | disable | |

**Table 8.22.2.5-3: *TimeAlignmentTimer* and sr-Configuration for E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | sf500 | As specified in clause 6.3.2 in TS 36.331[5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331[5] and section10.1 in TS 36.213[8]. |

The UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 4864ms from the beginning of time period T2. The measurement reporting delay is defined as the time from the beginning of time period T2 to the moment when the UE send the measurement report on PUSCH.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event A3 measurement report.

The overall delay measured test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra_SCE_DRX}$

$T_{identify_intra_SCE_DRX} = 16 * Max \{ T_{DMTC_periodicity}, \text{DRX cycle length}\} + T_{Measurement_Period_intra_TDD_CRS_DRX}$

Where $T_{DMTC_periodicity}$ = 160ms,

DRX cycle length = 256ms

$T_{Measurement_Period_intra_TDD_CRS_DRX} = 3 * Max\{ T_{DMTC_periodicity}, \text{DRX cycle length} \} = 768ms$

$T_{identify_intra_SCE_DRX}$ = 4864ms.

TTI insertion uncertainty = 2 ms

DRX cycle length = 256 ms

The overall delay measured shall be less than a total of 5122 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.22.3 E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal

### 8.22.3.1 Test purpose

To verify the UE's ability to make a correct reporting of an event when discovery signal is configured in DRX under fading propagation conditions within the E-UTRA FDD-FDD inter-frequency cell search requirements.

### 8.22.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward and forward and supports CRS based discovery signals measurement. Applicability requires support for FGIs bits 5 and 25.

### 8.22.3.3 Minimum conformance requirements

The UE shall be able to identify new inter-frequency cells and perform RSRP and RSRQ measurements of identified inter-frequency cells if carrier frequency information is provided by the PCell, even if no explicit neighbour list with physical layer cell identities is provided. The discovery signal occasion and the measurement gap should be aligned, provided that also the following additional conditions are fulfilled:

- Entire discovery signal occasion should be contained in the measurement gap.
- The subframe contained discovery signal for the measurement is not overlapped with the first 0.5ms period and the last 0.5ms period in every gap.

When DRX is in use, either measurement gaps are scheduled or the UE supports capability of conducting such measurements without gaps, the UE shall be able to identify a new detectable E-UTRAN FDD inter frequency cell within $T_{identify_inter_SCE_DRX}$.

$$T_{identify_inter_SCE\ DRX} = 17 * Max\ \{ T_{DMTC_periodicity}, \text{DRX cycle length, MGRP}\} * N_{freq} + T_{Measurement_Period_inter_FDD_CRS_DRX}$$

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] Sections 9.1.14.2 and RSRQ related side conditions given in TS 36.133 [4] Clause 9.1.14.4 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band

$T_{DMTC_periodicity}$ is the discovery signal measurement timing configuration periodicity of higher layer.

$T_{Measurement_Period_inter_FDD_CRS_DRX}$ is the inter-frequency period for measurements as shown in Table 8.22.3.3-1. $N_{freq}$ is defined in clause TS 36.133 [4] 8.1.2.1.1.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{Measurement_Period_inter_FDD_CRS_DRX}$ when DRX is used. If higher layer filtering is used, an additional cell identification delay can be expected.

The UE shall be capable of performing RSRP and RSRQ measurements of at least 3 identified inter-frequency cells per FDD inter-frequency for up to 3 FDD inter-frequencies and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period defined in table 8.22.3.3-1 when DRX is in use, either measurement gaps are scheduled or the UE supports capability of conducting such measurements without gaps.

**Table 8.22.3.3-1: Requirement to measure FDD inter frequency cell**

| Measurement bandwidth[RB] | Discovery signal occasion duration (*ds-OccasionDuration*) [ms] | $T_{Measurement_Period\ inter_FDD_CRS_DRX}$ [ms] |
|---|---|---|
| ≥ 6 | ≥1 | 5 * *Max*{ $T_{DMTC_periodicity}$, DRX cycle length,MGRP}*$N_{freq}$ |
| ≥ 25 | ≥1 | 3 * *Max*{ $T_{DMTC_periodicity}$, DRX cycle length,MGRP}*$N_{freq}$ |

The RSRP measurement accuracy for all measured cells shall be as specified in the TS 36.133 [4] sub-clauses 9.1.14.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in the TS 36.133 [4] sub-clause 9.1.14.4.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.14.2, and 9.1.14.4 respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_inter_SCE\ DRX}$ defined in TS 36.133 [4] Clause 8.6.2.2.1.2. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_inter_SCE\ DRX}$ defined in TS 36.133 [4] clause 8.6.2.2.1.2 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_inter_FDD_CRS_DRX}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.6.2.2.1 and A.8.22.3.

## 8.22.3.4 Test description

### 8.22.3.4.1 Initial conditions

Same Initial conditions as defined in clause 8.3.2.4.1 with the following exceptions:

- Instead of Table 8.3.2.4.1-1 -> use Table 8.22.3.4.1-1.
- Instead of clause 8.3.2.4.3 -> use Table 8.22.3.4.3.

**Table 8.22.3.4.1-1: General test parameters for E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1, 2 | Two FDD carrier frequencies are used. |
| Active cell | | Cell 1 | Cell 1 is on RF channel number 1 |
| Neighbour cell | | Cell 2 | Cell 1 is on RF channel number 2 |
| Gap Pattern Id | | 0 | As specified in TS 36.133 [4] clause 8.1.2.1. |
| Gap Offset | | 9 | As specified in TS 36.331 [5] clause 6.3.5 |
| DMTC period | ms | 160 | As specified in IE MeasDS-Config in TS 36.331 [5] |
| dmtc-PeriodOffset | ms | 10 | As specified in IE MeasDS-Config in TS 36.331 [5] |
| Discovery signal occasion duration | ms | 1 | As specified in IE MeasDS-Config in TS 36.331 [5] |
| A3-Offset | dB | -6 | |
| Hysteresis | dB | 0 | |
| CP length | | Normal | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| PRACH configuration | | 4 | As specified in table 5.7.1-2 in TS 36.211[9] |
| Access Barring Information | - | Not Sent | No additional delays in random access procedure. |
| DRX | | ON | DRX related parameters are defined in Table 8.22.3.5-2 |
| T1 | s | 5 | |
| T2 | s | 10 | |

### 8.22.3.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information, it is indicated to the UE performing CRS based discovery signals measurement and event-triggered reporting with Event A3 is configured. Entire discovery signal occasion should be contained in the measurement gap. The subframe contained discovery signal for the measurement is not overlapped with the first 0.5ms period and the last 0.5ms period in every gap. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

UE needs to be provided at least once every 500 ms with new Timing Advance Command MAC control element to restart the Timer Alignment Timer to keep the UE uplink time alignment. Furthermore the uplink time alignment is not maintained and UE needs to use RACH to obtain UL allocation for measurement reporting.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.22.3.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message and the SS shall provide the UE with new Timing Advance Command MAC control element at least once every 500 ms to restart the Timer Alignment Timer in order to keep the UE uplink time alignment.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.22.3.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 5378 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.22.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.22.3.4.3-1: Common Exception messages for E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-7<br>Table H.3.1-9 with Gap Pattern Id = 0<br>Table H.3.7-3 |

**Table 8.22.3.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.22.3.4.3-3: MeasObjectEUTRA-GENERIC(Freq): Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(Freq) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| carrierFreq | Downlink EARFCN for Freq | | |
| allowedmeasBandwidth | 25 | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationFDD-r12 | 1 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.3.4.3-4: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventide CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.22.3.4.3-5: *MAC-MainConfig-RBC*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test requirements**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf256 | 9 | To avoid overlapping with measurement gap. | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | sf500 | | |
| } | | | |

**Table 8.22.3.4.3-6: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.22.3.4.3-7: *MeasResults*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.22.3.4.3-8: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| physCellId | PhysCellId | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.22.3.4.3-9: *PRACH-Config-DEFAULT*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test requirements**

| Derivation Path: TS 36.508 [7] clause 7.3.2 Table 7.3.2-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PRACH-Config-DEFAULT ::= SEQUENCE { | | | |
| prach-ConfigIndex | 4 | | |
| } | | | |

## 8.22.3.5 Test requirement

Tables 8.22.3.4.1-1, 8.22.3.5-1, 8.22.3.5-2 and 8.22.3.5-3 define the primary level settings including test tolerances for E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test.

**Table 8.22.3.5-1: Cell specific test parameters for E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |

| | | | | | |
|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Measurement bandwidth | $n_{PRB}$ | 13-37 | | 13-37 | |
| PDSCH parameters: DL Reference Measurement Channel | | R.0 FDD | | - | |
| PCFICH/PDCCH/PHICH parameters: DL Reference Measurement Channel | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | -Infinity | 7 |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | -Infinity | 7 |
| RSRP Note 3 | dBm/15 kHz | -94 | -94 | -Infinity | -91 |
| SCH_RP Note 3 | dBm/15 kHz | -94 | -94 | -Infinity | -91 |
| Io Note 3 | dBm/9MHz | -64.76 | -64.76 | -70.22 | -62.43 |
| Propagation Condition | | ETU30 | | ETU30 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| Timing offset to Cell 1 | µs | - | | 3 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: Es/Iot, RSRP and SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.22.3.5-2: DRX-Configuration for E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in clause 6.3.2 in TS 36.331[5] |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf256 | |
| shortDRX | disable | |

**Table 8.22.3.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | sf500 | As specified in clause 6.3.2 in TS 36.331[5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331[5] and section10.1 in TS 36.213[8]. |

UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 5120 ms from the beginning of time period T2. The measurement reporting delay is defined as the time from the beginning of time period T2, to the moment when the UE starts to send preambles on the PRACH for scheduling request (SR) to obtain allocation to send the measurement report on PUSCH.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event A3 measurement report.

The overall delay measured test requirement is expressed as:

- Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length
- Measurement reporting delay = $T_{identify_inter_SCE_DRX.}$

$T_{identify_inter_SCE\ DRX}$ = 17 * *Max* { $T_{DMTC_periodicity}$, DRX cycle length, MGRP} *$N_{freq}$ +$T_{Measurement_Period_inter_FDD_CRS_DRX}$

Where $T_{DMTC_periodicity}$ = 160ms,

- DRX cycle length = 256ms.
- $T_{Measurement_Period_inter_FDD_CRS_DRX}$ = 3 * *Max*{ $T_{DMTC_periodicity}$, DRX cycle length,MGRP}*$N_{freq}$ = 768ms

  Where $N_{freq}$ = 1. It is defined in TS 36.133 [4] clause 8.1.2.1.1.

  Gap Pattern Id = 0, MGRP = 40ms

$T_{identify_inter_SCE\ DRX}$ = 5120 ms.

TTI insertion uncertainty = 2 ms

DRX cycle length = 256 ms

The overall delay measured shall be less than a total of 5378 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.22.4 E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal

### 8.22.4.1 Test purpose

To verify the UE's ability to make a correct reporting of an event when discovery signal is configured in DRX under fading propagation conditions in synchronous cells within the E-UTRA TDD-TDD inter frequency cell search in DRX requirements.

### 8.22.4.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and12 and forward and supports CRS based discovery signals measurement. Applicability requires support for FGI bits 5, and 25.

### 8.22.4.3 Minimum conformance requirements

When DRX is in use, either measurement gaps are scheduled or the UE supports capability of conducting such measurements without gaps, the UE shall be able to identify a new detectable E-UTRAN TDD inter frequency cell within $T_{identify_inter_SCE_DRX}$.

$T_{identify_inter_SCE_DRX} = 17 * Max \{ T_{DMTC_periodicity}$, DRX cycle length, MGRP$\} * N_{freq} + T_{Measurement_Period_inter_TDD_CRS_DRX}$

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] Clause 9.1.14.2 and RSRQ related side conditions given in TS 36.133 [4] Sections 9.1.14.4 are fulfilled,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to Annex I.2.3 for a corresponding Band

$T_{DMTC_periodicity}$ is the discovery signal measurement timing configuration periodicity of higher layer.

$T_{Measurement_Period_inter_TDD_CRS_DRX}$ is the inter-frequency period for measurements as shown in Table 8.22.4.3-1. $N_{freq}$ is defined in TS 36.133 [4] clause 8.1.2.1.1.

Identification of a cell shall include detection of the cell and additionally performing a single measurement with measurement period of $T_{Measurement_Period\ intra_TDD_CRS_DRX}$ when DRX is used. If higher layer filtering is used, an additional cell identification delay can be expected.

The UE shall be capable of performing RSRP and RSRQ measurements of at least 3 identified inter-frequency cells per TDD inter-frequency for up to 3 TDD inter-frequencies and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period defined in table 8.22.4.3-1 when DRX is in use, either measurement gaps are scheduled or the UE supports capability of conducting such measurements without gaps.

**Table 8.22.4.3-1: Requirement to measure TDD inter frequency cell**

| Measurement bandwidth[RB] | Discovery signal occasion duration (*ds-OccasionDuration*) [ms] | $T_{Measurement_Period\ inter_TDD_CRS_DRX}$ [ms] |
|---|---|---|
| ≥ 6 | ≥2 | 5 * *Max*{ $T_{DMTC_periodicity}$, DRX cycle length,MGRP}*$N_{freq}$ |
| ≥ 25 | ≥2 | 3 * *Max*{ $T_{DMTC_periodicity}$, DRX cycle length,MGRP}*$N_{freq}$ |

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] sub-clauses 9.1.14.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] sub-clause 9.1.14.4.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.14.2, and 9.1.14.4, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{Identify_Inter}$ defined in TS 36.133 [4] clause 8.1.2.3.2.1. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_inter_SCE_DRX}$ defined in TS 36.133 [4] clause 8.6.2.2.2.2 and then triggers the measurement report as per TS 36.331 [5], the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_inter_TDD_CRS_DRX}$ defined in TS 36.133 [4] clause 8.6.2.2.2.2 provided the timing to that cell has not changed more than ± 50 Ts while measurementgap has not been available and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.6.2.2.2 and A.8.22.4.

## 8.22.4.4 Test description

### 8.22.4.4.1 Initial conditions

Same Initial conditions as defined in clause 8.4.2.4.1 with the following exceptions:

- Instead of Table 8.4.2.4.1-1 -> use Table 8.22.4.4.1-1.
- Instead of clause 8.4.2.4.3 -> use Table 8.22.4.4.3.

**Table 8.22.4.4.1-1: General test parameters for E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1, 2 | Two TDD carrier frequencies are used. |
| Active cell | | Cell 1 | Cell 1 is on RF channel number 1 |
| Neighbour cell | | Cell 2 | Cell 2 is on RF channel number 2 |
| Gap Pattern Id | | 0 | As specified in TS 36.133 clause 8.1.2.1. |
| Gap Offset | | 9 | As specified in TS 36.331 clause 6.3.5 |
| Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in TS 36.211. The same configuration in both cells |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211. The same configuration in both cells |
| DMTC period | ms | 160 | As specified in IE MeasDS-Config in TS 36.331 |
| dmtc-PeriodOffset | ms | 10 | As specified in IE MeasDS-Config in TS 36.331 |
| Discovery signal occasion duration | ms | 2 | As specified in IE MeasDS-Config in TS 36.331 |
| A3-Offset | dB | -6 | |
| Hysteresis | dB | 0 | |
| CP length | | Normal | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| PRACH configuration | | 4 | As specified in table 5.7.1-2 in TS 36.211 |
| Access Barring Information | - | Not Sent | No additional delays in random access procedure. |
| DRX | | ON | DRX related parameters are defined in Table A.8.22.4.1-3 |
| T1 | s | 5 | |
| T2 | s | 10 | |

### 8.22.4.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information, it is indicated to the UE performing CRS based discovery signals measurement and event-triggered reporting with Event A3 is configured. Entire discovery signal occasion should be contained in the measurement gap. The subframe contained discovery signal for the measurement is not overlapped with the first 0.5ms period and the last 0.5ms period in every gap. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

UE needs to be provided at least once every 500ms with new Timing Advance Command MAC control element to restart the Time alignment timer to keep UE uplink time alignment. Furthermore the uplink time alignment is not maintained and UE needs to use RACH to obtain UL allocation for measurement reporting.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.22.4.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message and The SS shall provide the UE with new Timing Advance Command MAC control element at least once every 500 ms to restart the Timer Alignment Timer in order to keep the UE uplink time alignment.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.22.4.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 5378 ms then the number of successful tests is increased by one. If the

UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receives the MeasurementReport message in step 6 or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
- transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
or
- switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.22.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.22.4.4.3-1: Common Exception messages for Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-7<br>Table H.3.1-9 with the Gap Pattern Id = 0<br>Table H.3.7-3 |

**Table 8.22.4.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | | | |
| | MeasConfig-DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated- HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.4.4.3-3: MeasObjectEUTRA-GENERIC(Freq): Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(Freq) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| carrierFreq | Downlink EARFCN for Freq | | |
| allowedmeasBandwidth | 25 | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationFDD-r12 | 2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.4.4.3-4: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |

**Table 8.22.4.4.3-5: *MAC-MainConfig-RBC*: Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test requirements**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf256 | 9 | To avoid overlapping with measurement gap. | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | sf500 | | |
| } | | | |

**Table 8.22.4.4.3-6: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20: SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | Channel-bandwidth-dependent parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.22.4.4.3-7: *MeasResults:* Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | Set according to specific test | |
| rsrqResult | INTEGER(0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 8.22.4.4.3-8: *MeasResultListEUTRA*: Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 8.22.4.4.3-9: *PRACH-Config-DEFAULT*: Additional E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test requirement**

| Derivation Path: TS 36.508 [7] clause 7.3.2 Table 7.3.2-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PRACH-Config-DEFAULT ::= SEQUENCE { | | | |
| prach-ConfigIndex | 4 | | |
| } | | | |

## 8.22.4.5 Test requirement

Tables 8.22.4.5-1, 8.22.4.5-2 and 8.22.4.5-3 define the primary level settings including test tolerances for E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal test.

**Table 8.22.4.5-1: Cell specific test parameters for E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |

| | | | | | |
|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Measurement bandwidth | $n_{PRB}$ | 13-37 | | 13-37 | |
| PDSCH parameters: DL Reference Measurement Channel | | R.0 TDD | | - | |
| PCFICH/PDCCH/PHICH parameters: DL Reference Measurement Channel | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | | OP.2 TDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_RB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_RB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | -Infinity | 7 |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | -Infinity | 7 |
| RSRP Note 3 | dBm/15 kHz | -94 | -94 | -Infinity | -91 |
| SCH_RP Note 3 | dBm/15 kHz | -94 | -94 | -Infinity | -91 |
| Io Note 3 | dBm/9MHz | -64.76 | -64.76 | -70.22 | -62.43 |
| Propagation Condition | | ETU30 | | ETU30 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| Timing offset to Cell 1 | µs | - | | 3 (Synchronous cells) | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: Es/Iot, RSRP and SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.22.4.5-2: DRX-Configuration for E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in clause 6.3.2 in TS 36.331[5] |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf256 | |
| shortDRX | disable | |

**Table 8.22.4.5-3: *TimeAlignmentTimer* and sr-Configuration for E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | sf500 | As specified in clause 6.3.2 in TS 36.331[5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331[5] and section10.1 in TS 36.213[8]. |

UE shall send one Event A3 triggered measurement report, with a measurement reporting delay less than 5120 ms from the beginning of time period T2. The measurement reporting delay is defined as the time from the beginning of time period T2, to the moment when the UE starts to send preambles on the PRACH for scheduling request (SR) to obtain allocation to send the measurement report on PUSCH.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event A3 measurement report.

The overall delay measured test requirement is expressed as:

- Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length
- Measurement reporting delay = $T_{identify_inter_SCE_DRX.}$

$T_{identify_inter_SCE\ DRX} = 17 * Max\ \{ T_{DMTC_periodicity}$, DRX cycle length, MGRP$\} * N_{freq} + T_{Measurement_Period_inter_TDD_CRS_DRX}$

Where:

- $T_{DMTC_periodicity}$ = 160ms,
- DRX cycle length = 256ms.
- $T_{Measurement_Period_inter_TDD_CRS_DRX} = 3 * Max\{ T_{DMTC_periodicity}$, DRX cycle length,MGRP$\} * N_{freq}$ = 768ms
  - Where $N_{freq}$ = 1. It is defined in TS 36.133 [4] clause 8.1.2.1.1.
  - Gap Pattern Id = 0, MGRP = 40ms
- $T_{identify_inter_SCE\ DRX}$ = 5120 ms.
- TTI insertion uncertainty = 2 ms
- DRX cycle length = 256 ms

The overall delay measured shall be less than a total of 5378 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.22.5 E-UTRAN FDD-FDD intra-frequency event triggered reporting in DRX based on CSI-RS based discovery signal

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- The Test system uncertainties applicable to this test are undefined
- The Test tolerances applicable to this test are undefined

### 8.22.5.1 Test purpose

The purpose of the test is to verify that the UE makes correct reporting of an event in DRX. The test will partly verify the FDD-FDD intra-frequency cell search in DRX requirements in TS 36.133 [4] clause 8.6.3.1.1.2.

### 8.22.5.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward and supports CSI-RS based discovery signals measurement. Applicability requires support for FGI bit 5.

### 8.22.5.3 Minimum conformance requirements

When DRX is in use the UE shall be able to identify a new detectable FDD intra frequency TP within $T_{identify_intra_TP_SCE_DRX}$.

$T_{identify_intra_TP_SCE_DRX} = T_{identify_intra_SCE_DRX} + T_{Measurement_Period\ _intra_FDD_CSI\text{-}RS_DRX}$

A TP shall be considered detectable when

- CSI-RSRP related side conditions given in TS 36.133 [4] Sections 9.1.14.3 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.10 for a corresponding Band.

$T_{identify_intra_SCE_DRX}$ is the intra-frequency period for cell identification in TS 36.133 [4] section 8.6.2.1.1.2. $T_{Measurement_Period\ _intra_FDD_CSI\text{-}RS_DRX}$ is the intra-frequency period for TP measurement as shown in table 8.22.5.3-1.

Identification of a TP shall include identification of the cell and additionally performing a single measurement with measurement period of $T_{Measurement_Period_intra_FDD_CSI\text{-}RS_DRX}$ when DRX is used. If higher layer filtering is used, an additional TP identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is $T_{Measurement_Period\ _intra_FDD_CSI\text{-}RS_DRX}$ as shown in table 8.22.5.3-1, when DRX is in use. The UE shall be capable of performing CSI-RSRP measurements for 3 identified intra-frequency TPs, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{Measurement_Period\ _intra_FDD_\ CSI\text{-}RS_DRX}$.

**Table 8.22.5.3-1: Requirement to measure FDD intra frequency TP**

| Measurement bandwidth [RB] | Discovery signal occasion duration (*ds-OccasionDuration*) [ms] | $T_{Measurement_Period\ intra_FDD_\ CSI\text{-}RS_DRX}$ [ms] |
|---|---|---|
| ≥ 6 | ≥1 | 5**Max*{$T_{DMTC_periodicity}$, DRX cycle length} |
| ≥ 25 | ≥1 | 3**Max*{$T_{DMTC_periodicity}$, DRX cycle length} |

$T_{DMTC_periodicity}$ is the discovery signal measurement timing configuration periodicity of higher layer.

The CSI-RSRP measurement accuracy for all measured TPs shall be as specified in the TS 36.133 [4] sub-clauses 9.1.14.3.

Reported CSI-RSRP measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.14.3.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra_TP_SCE_DRX}$ defined in TS 36.133 [4] Clause 8.6.3.1.1.2. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a TP which has been detectable at least for the time period $T_{identify_intra_TP_SCE_DRX}$ defined in TS 36.133 [4] clause 8.6.3.1.1.2 becomes undetectable for a period $\leq 5$ seconds and then the TP becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_intra_FDD_CSI-RS_DRX}$ provided the timing to that TP has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.6.3.1.1.2 and A.8.22.5.

## 8.22.5.4 Test description

### 8.22.5.4.1 Initial conditions

Same initial conditions as defined in clause 8.1.3.4.1 with the following exceptions:

- Instead of Table 8.1.3.4.1-1 -> use Table 8.22.5.4.1-1.
- Instead of clause 8.1.3.4.3 -> use clause 8.22.5.4.3.

**Table 8.22.5.4.1-1: General test parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| | | Test 1 | |
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in clause A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in clause A.2.1 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | One FDD carrier frequency is used. |
| DMTC period [2] | ms | 160 | |
| DMTC period offset [2] | ms | 10 | |
| Discovery signal occasion duration | ms | 1 | |
| c2-Offset | dB | -6 | |
| CP length | | Normal | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | ON | DRX related parameters are defined in Table 8.22.5.5-2 |
| Time offset between cells | | 2.3 μs | CP/2 or Synchronous cells |
| T1 | s | 5 | |
| T2 | s | 10 | |

### 8.22.5.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information, it is indicated to the UE performing CSI-RS based discovery signals measurement and event-triggered reporting with Event C2 is configured. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

UE needs to be provided at least once every 500 ms with new Timing Advance Command MAC control element to restart the Timer Alignment Timer to keep the UE uplink time alignment. Furthermore, the UE is allocated with PUSCH resource at every DRX cycle.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.22.5.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

3. The SS shall transmit an RRCConnectionReconfiguration message and the SS shall provide the UE with new Timing Advance Command MAC control element at least once every 500 ms to restart the Timer Alignment Timer in order to keep the UE uplink time alignment.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.22.5.5-1 and 8.22.5.5-2.

6. UE shall transmit a MeasurementReport message triggered by Event C2. If the overall delays measured from the beginning of time period T2 is less than 5890ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receives the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.22.5.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.22.5.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in DRX based on CSI-RS based discovery signal requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-2<br>Table H.3.7-3 |

**Table 8.22.5.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in DRX based on CSI-RS based discovery signal requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.22.5.4.3-3: *MeasObjectEUTRA-GENERIC(f1)*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in DRX based on CSI-RS based discovery signal requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(f1) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| carrierFreq | Downlink EARFCN for Freq | | |
| allowedmeasBandwidth | mbw50 | | |
| measDS-Config-r12 CHOICE { | | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationFDD-r12 | 1 | | |
| } | | | |
| measCSI-RS-ToAddModList-r12 SEQUENCE (SIZE (1..maxCSI-RS-Meas-r12)) OF SEQUENCE { | | | |
| MeasCSI-RS-Config-r12[0] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| physCellId-r12 | PhysCellId | Cell1 | |
| scramblingIdentity-r12 | 0 | | |
| resourceConfig-r12 | 2 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | 0 | | |
| } | | | |
| MeasCSI-RS-Config-r12[1] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 2 | | |
| physCellId-r12 | PhysCellId | Cell2 | |
| scramblingIdentity-r12 | 1 | | |
| resourceConfig-r12 | 4 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.5.4.3-4: *QuantityConfig-DEFAULT*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in DRX based on CSI-RS based discovery signal requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-3A: QuantityConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| QuantityConfig-DEFAULT ::= SEQUENCE { | | | |
| quantityConfigEUTRA SEQUENCE { | | | |
| filterCoefficientRSRP | fc0 | | |
| filterCoefficientRSRQ | fc0 | | |
| } | | | |
| quantityConfigEUTRA-v1250 SEQUENCE { | | | |
| filterCoefficientCSI-RSRP-r12 | fc0 | | |
| } | | | |
| } | | | |

**Table 8.22.5.4.3-5: *ReportConfigEUTRA-C2*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in DRX based on CSI-RS based discovery signal requirements**

| Derivation Path: 36.508 clause 4.6.6, Table 4.6.6-11: ReportConfigEUTRA-C2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-C2 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventC2-r12 SEQUENCE { | | | |
| c2-RefCSI-RS-r12 | 1 | | |
| c2-Offset-r12 | -12 (-6dB) | -6 is actual value in dB (-12*0.5 dB) | |
| c2-ReportOnLeave-r12 | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| reportStrongestCSI-RSs-r12 | false | | |
| reportCRS-Meas-r12 | false | | |
| triggerQuantityCSI-RS-r12 | true | | |
| } | | | |

**Table 8.22.5.4.3-5: *MAC-MainConfig-RBC*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in DRX based on CSI-RS based discovery signal requirements**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf256 | 9 | To avoid overlapping with measurement gap. | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | sf500 | | |
| } | | | |

**Table 8.22.5.4.3-6: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in DRX based on CSI-RS based discovery signal requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.22.5.4.3-7: *MeasResults*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in DRX based on CSI-RS based discovery signal requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| measResultCSI-RS-List-r12 SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| MeasResultCSI-RS-r12[0] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| csi-RSRP-Result-r12 | INTEGER(0..97) | Set according to specific test | |
| } | | | |
| MeasResultCSI-RS-r12[1] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 2 | | |
| csi-RSRP-Result-r12 | INTEGER(0..97) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.5.4.3-8: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in DRX based on CSI-RS based discovery signal requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| physCellId | PhysCellId | Cell 2 | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.22.5.5 Test requirement

Tables 8.22.5.4.1-1, 8.22.5.5-1, 8.22.5.5-2 and 8.22.5.5-3 define the primary level settings including test tolerances for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in DRX based on CSI-RS based discovery signal test.

**Table 8.22.5.5-1: Cell specific test parameters for E-UTRAN FDD-FDD intra-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Measurement bandwidth | $n_{PRB}$ | 13-37 | | 13-37 | |
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD as in A.1.1 | | DL Reference Measurement Channel R.0 FDD as in A.1.1 | |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD as in A.2.1 | | DL Reference Measurement Channel R.6 FDD as in A.2.1 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_PB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_PB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| p-C-r10 [5] | dB | 6 | 6 | 6 | 6 |
| $N_{oc}$ Note 3 | dBm/15 KHz | -98+TT | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4+TT | 4+TT | -Infinity | 4+TT |
| CRS $\hat{E}_s / I_{ot}$ | dB | 4+TT | -1.46+TT | -Infinity | -1.46+TT |
| CSI-RS $\hat{E}_s / I_{ot}$ | dB | 10+TT | 4.54+TT | -Infinity | 4.54+TT |
| RSRP Note 4 | dBm/15 KHz | -94+TT | -94+TT | -Infinity | -94+TT |
| CSI-RSRP Note 4 | dBm/15 KHz | -88+TT | -88+TT | -Infinity | -88+TT |
| SCH_RP Note 4 | dBm/15 KHz | -94+TT | -94+TT | -Infinity | -94+TT |
| Io | dBm/9 MHz | -60+TT | -60+TT | Specified in columns for cell1 | |
| CSI reference signal configurations [9] | | 2 | | 4 | |
| CSI-RS subframe offset | | 0 | | 0 | |
| CSI-RS individual offset [5] | dB | 0 | | 0 | |
| CSI-RS muting | | Enable | | Enable | |
| Propagation Condition | | ETU30 | | ETU30 | |
| Timing offset to cell 1 | us | - | | 2.3 (CP/2) | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.22.5.5-2: DRX-Configuration for E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in DRX based on CSI-RS based discovery signal**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in clause 6.3.2 in TS 36.331[5] |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf256 | |
| shortDRX | disable | |

**Table 8.22.5.5-3: *TimeAlignmentTimer* and sr-Configuration for E-UTRAN FDD-FDD intra-frequency event triggered reporting in DRX based on CSI-RS based discovery signal**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | sf500 | As specified in clause 6.3.2 in TS 36.331[5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331[5] and section10.1 in TS 36.213[8]. |

The UE shall send one Event C2 triggered measurement report, with a measurement reporting delay less than 5890ms from the beginning of time period T2. The measurement reporting delay is defined as the time from the beginning of time period T2 to the moment when the UE send the measurement report on PUSCH.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event C2 measurement report.

The overall delay measured test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra_TP_SCE_DRX}$

$T_{identify_intra_TP_SCE_DRX} = T_{identify_intra_SCE_DRX} + T_{Measurement_Period\ _intra_FDD_CSI\text{-}RS_DRX}$

Where

$T_{identify_intra_SCE_DRX} = 16 * Max \{ T_{DMTC_periodicity}, \text{DRX cycle length}\} + T_{Measurement_Period\ _intra_FDD_CRS_DRX,}$

$T_{Measurement_Period\ _intra_FDD_CRS_DRX} = 3 * Max\{ T_{DMTC_periodicity}, \text{DRX cycle length } \},$

$T_{Measurement_Period\ _intra_FDD_CSI\text{-}RS_DRX} = 3 * Max\{ T_{DMTC_periodicity}, \text{DRX cycle length } \} = 768ms,$

$T_{DMTC_periodicity} = 160ms,$

DRX cycle length = 256ms.

$T_{identify_intra_TP_SCE_DRX} = 22 * Max \{ T_{DMTC_periodicity}, \text{DRX cycle length}\} = 5632ms$

TTI insertion uncertainty = 2 ms

DRX cycle length = 256 ms

The overall delay measured shall be less than a total of 5890ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.22.6 E-UTRAN TDD-TDD intra-frequency event triggered reporting in DRX based on CSI-RS based discovery signal

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- The Test system uncertainties applicable to this test are undefined
- The Test tolerances applicable to this test are undefined

### 8.22.6.1 Test purpose

To verify the UE's ability to make a correct reporting of an event when discovery signal is configured in DRX based on CSI-RS based discovery signal under fading propagation conditions within the E-UTRA TDD-TDD intra frequency cell search requirements.

### 8.22.6.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward and supports CSI-RS based discovery signals measurement. Applicability requires support for FGI bit 5.

### 8.22.6.3 Minimum conformance requirements

When DRX is in use the UE shall be able to identify a new detectable TDD intra frequency TP within $T_{identify_intra_TP_SCE_DRX}$.

$T_{identify_intra_TP_SCE_DRX} = T_{identify_intra_SCE_DRX} + T_{Measurement_Period\ _intra_TDD_CSI\text{-}RS_DRX}$

A TP shall be considered detectable when

- CSI-RSRP related side conditions given in TS 36.133 [4] Sections 9.1.14.3 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.10 for a corresponding Band

$T_{identify_intra_SCE_DRX}$ is the intra-frequency period for cell identification as shown in section 8.22.2.3.
$T_{Measurement_Period_intra_TDD_CSI\text{-}RS_DRX}$ is the intra-frequency period for TP measurement as shown in Table 8.22.6.3-1.

Identification of a TP shall include identification of the cell and additionally performing a single measurement with measurement period of $T_{Measurement_Period_intra_TDD_CSI\text{-}RS_DRX}$ when DRX is used. If higher layer filtering is used, an additional TP identification delay can be expected.

In the RRC_CONNECTED state the measurement period for intra frequency measurements is $T_{Measurement_Period\ _intra_TDD_CSI\text{-}RS_DRX}$ as shown in table 8.22.6.3-1, when DRX is in use. The UE shall be capable of performing CSI-RSRP measurements for 3 identified intra-frequency TPs, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{Measurement_Period\ _intra_TDD_\ CSI\text{-}RS_DRX}$.

**Table 8.22.6.3-1: Requirement to measure TDD intrafrequency TP**

| Measurement bandwidth [RB] | Discovery signal occasion duration (*ds-OccasionDuration*) [ms] | $T_{Measurement_Period\ intra_TDD_\ CSI\text{-}RS_DRX}$ [ms] |
|---|---|---|
| ≥ 6 | ≥2 | 5**Max*{$T_{DMTC_periodicity}$, DRX cycle length} |
| ≥ 25 | ≥2 | 3**Max*{$T_{DMTC_periodicity}$, DRX cycle length} |

$T_{DMTC_periodicity}$ is the discovery signal measurement timing configuration periodicity of higher layer.

The CSI-RSRP measurement accuracy for all measured TPs shall be as specified in the TS 36.133 [4] sub-clauses 9.1.14.3.

Reported CSI-RSRP measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.14.3.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra_TP_SCE_DRX}$ defined in TS 36.133 [4] Clause 8.6.3.1.2.2. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a TP which has been detectable at least for the time period $T_{identify_intra_TP_SCE_DRX}$ defined in clause TS 36.133 [4] 8.6.3.1.2.2 becomes undetectable for a period $\leq 5$ seconds and then the TP becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_intra_TDD_CSI-RS_DRX}$ provided the timing to that TP has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.6.3.1.2.2 and A.8.22.6.

## 8.22.6.4 Test description

### 8.22.6.4.1 Initial conditions

Same Initial conditions as defined in clause 8.2.2.4.1 with the following exceptions:

- Instead of Table 8.2.2.4.1-1 -> use Table 8.22.6.4.1-1.
- Instead of clause 8.2.2.4.3 -> use clause 8.22.6.4.3.

**Table 8.22.6.4.1-1: General test parameters for E-UTRAN TDD-TDD intra-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| | | Test 1 | |
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in clause A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in clause A.2.2 |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1 | One TDD carrier frequency is used. |
| DMTC period [5] | ms | 160 | |
| DMTC period offset [5] | ms | 10 | |
| Discovery signal occasion duration | ms | 2 | |
| c2-Offset | dB | -6 | |
| CP length | | Normal | |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211 [9]. The same configuration in both cells |
| Uplink-downlink configuration | | 1 | As specified in table 4.2-2 in TS 36.211 [9]. The same configuration in both cells |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | ON | DRX related parameters are defined in Table 8.22.6.5-2 |
| T1 | s | 5 | |
| T2 | s | 10 | |

### 8.22.6.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information, it is indicated to the UE performing CSI-RS based discovery signals measurement and event-triggered reporting with Event C2 is configured. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

UE needs to be provided at least once every 500 ms with new Timing Advance Command MAC control element to restart the Timer Alignment Timer to keep the UE uplink time alignment. Furthermore, the UE is allocated with PUSCH resource at every DRX cycle.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.22.6.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.
3. The SS shall transmit an RRCConnectionReconfiguration message and the SS shall provide the UE with new Timing Advance Command MAC control element at least once every 500 ms to restart the Timer Alignment Timer in order to keep the UE uplink time alignment.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.22.6.5-1 and 8.22.6.5-2.
6. UE shall transmit a MeasurementReport message triggered by Event C2. If the overall delays measured from the beginning of time period T2 is less than 5890ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.
7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.22.6.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.22.6.4.3-1: Common Exception messages for E-UTRAN TDD-TDD intra frequency event triggered reporting under fading propagation conditions for DRS measurement based on CSI-RS with DRX**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-2<br>Table H.3.7-3 |

**Table 8.22.6.4.3-2: *RRCConnectionReconfiguration***

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.22.6.4.3-3: MeasObjectEUTRA-GENERIC(f1)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(f1) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| carrierFreq | Downlink EARFCN for Freq | | |
| allowedmeasBandwidth | mbw50 | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationTDD-r12 | 2 | | |
| } | | | |
| measCSI-RS-ToAddModList-r12 SEQUENCE (SIZE (1..maxCSI-RS-Meas-r12)) OF SEQUENCE { | | | |
| MeasCSI-RS-Config-r12[0] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| physCellId-r12 | PhysCellId | Cell1 | |
| scramblingIdentity-r12 | 0 | | |
| resourceConfig-r12 | 2 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | 0 | | |
| } | | | |
| MeasCSI-RS-Config-r12[1] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 2 | | |
| physCellId-r12 | PhysCellId | Cell2 | |
| scramblingIdentity-r12 | 1 | | |
| resourceConfig-r12 | 4 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.6.4.3-4: *QuantityConfig-DEFAULT***

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-3A: QuantityConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| QuantityConfig-DEFAULT ::= SEQUENCE { | | | |
| quantityConfigEUTRA SEQUENCE { | | | |
| filterCoefficientRSRP | fc0 | | |
| filterCoefficientRSRQ | fc0 | | |
| } | | | |
| quantityConfigEUTRA-v1250 SEQUENCE { | | | |
| filterCoefficientCSI-RSRP-r12 | fc0 | | |
| } | | | |
| } | | | |

**Table 8.22.6.4.3-5: *ReportConfigEUTRA-C2***

| Derivation Path: 36.508 clause 4.6.6, Table 4.6.6-11: ReportConfigEUTRA-C2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-C2 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventC2-r12 SEQUENCE { | | | |
| c2-RefCSI-RS-r12 | 1 | | |
| c2-Offset-r12 | -12 (-6dB) | -6 is actual value in dB (-12*0.5 dB) | |
| c2-ReportOnLeave-r12 | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| reportStrongestCSI-RSs-r12 | false | | |
| reportCRS-Meas-r12 | false | | |
| triggerQuantityCSI-RS-r12 | true | | |
| } | | | |

**Table 8.22.6.4.3-6: *MAC-MainConfig-RBCs***

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf256 | 9 | To avoid overlapping with measurement gap. | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | sf500 | | |
| } | | | |

**Table 8.22.6.4.3-6: *SchedulingRequest-Config-DEFAULT***

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.22.6.4.3-7: *MeasResults***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| measResultCSI-RS-List-r12 | MeasResultCSI-RS-List-r12 | | |
| } | | | |
| } | | | |

**Table 8.22.6.4.3-8: *MeasResultListEUTRA***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| physCellId | PhysCellId | Cell 2 | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.22.6.4.3-9: *MeasResultCSI-RS-List-r12***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultCSI-RS-List-r12 ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| MeasResultCSI-RS-r12[0] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| csi-RSRP-Result-r12 | INTEGER(0..97) | Set according to specific test | |
| } | | | |
| MeasResultCSI-RS-r12[1] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 2 | | |
| csi-RSRP-Result-r12 | INTEGER(0..97) | Set according to specific test | |
| } | | | |
| } | | | |

## 8.22.6.5 Test requirement

Tables 8.22.6.4.1-1, 8.22.6.5-1, 8.22.6.5-2 and 8.22.6.5-3 define the primary level settings including test tolerances for E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CSI-RS based discovery signal test.

**Table 8.22.6.5-1: Cell specific test parameters for E-UTRAN TDD-TDD intra-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |

| | | | | | |
|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Measurement bandwidth | $n_{PRB}$ | 13-37 | | 13-37 | |
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD as in A.1.2 | | DL Reference Measurement Channel R.0 TDD as in A.1.2 | |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD as in A.2.2 | | DL Reference Measurement Channel R.6 TDD as in A.2.2 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | | OP.2 TDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_PB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_PB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| p-C-r10 [5] | dB | 6+TT | 6+TT | 6+TT | 6+TT |
| $N_{oc}$ Note 3 | dBm/15 KHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4+TT | 4+TT | -Infinity | 4+TT |
| CRS $\hat{E}_s / I_{ot}$ | dB | 4+TT | -1.46+TT | -Infinity | -1.46+TT |
| CSI-RS $\hat{E}_s / I_{ot}$ | dB | 10+TT | 4.54+TT | -Infinity | 4.54+TT |
| RSRP Note 4 | dBm/15 KHz | -94+TT | -94+TT | -Infinity | -94+TT |
| CSI-RSRP Note 4 | dBm/15 KHz | -88+TT | -88+TT | -Infinity | -88+TT |
| Io | dBm/9 MHz | -60+TT | -57+TT | Specified in columns for cell1 | |
| SCH_RP Note 4 | dBm/15 KHz | -94+TT | -94+TT | -Infinity | -94+TT |
| Propagation Condition | | ETU30 | | | |
| CSI reference signal configurations [9] | | 2 | | 4 | |
| CSI-RS subframe offset | | 0 | | 0 | |
| CSI-RS individual offset [5] | dB | 0 | | 0 | |
| CSI-RS muting | | Enable | | Enable | |
| Timing offset to cell 1 | us | 0 | | 2.3 (CP/2) | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.22.6.5-2: DRX-Configuration for E-UTRAN TDD-TDD intra-frequency event triggered reporting in DRX under fading propagation conditions in synchronous cells**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in clause 6.3.2 in TS 36.331 [5] |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | Sf256 | |
| shortDRX | disable | |

**Table 8.22.6.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN TDD-TDD intra-frequency event triggered reporting in DRX under fading propagation conditions in synchronous cells**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | sf500 | As specified in clause 6.3.2 in TS 36.331 [5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331 [5] and section10.1 in TS 36.213 [8]. |

The UE shall send one Event C2 triggered measurement report, with a measurement reporting delay less than 5632 ms from the beginning of time period T2. The measurement reporting delay is defined as the time from the beginning of time period T2 to the moment when the UE send the measurement report on PUSCH.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The rate of correct events observed during repeated tests shall be at least 90%.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event C2 measurement report.

The overall delay measured test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_intra_TP_SCE_DRX}$

$T_{identify_intra_TP_SCE_DRX} = T_{identify_intra_SCE_DRX} + T_{Measurement_Period\ _intra_TDD_CSI\text{-}RS_DRX}$

Where $T_{identify_intra_SCE_DRX} = 16 * Max\ \{T_{DMTC_periodicity}, \text{DRX cycle length}\} + T_{Measurement_Period\ _intra_TDD_CRS_DRX}$

$T_{DMTC_periodicity}$ = 160ms,

DRX cycle length = 256ms.

$T_{Measurement_Period\ _intra_TDD_CRS_DRX} = 3 * Max\{ T_{DMTC_periodicity}, \text{DRX cycle length} \} = 768ms$

$T_{Measurement_Period\ _intra_TDD_CSI\text{-}RS_DRX} = 3 * Max\{ T_{DMTC_periodicity}, \text{DRX cycle length} \} = 768ms$

$T_{identify_intra_SCE_DRX} = 19 * Max\{ T_{DMTC_periodicity}, \text{DRX cycle length} \} = 4864$ ms.

$T_{identify_intra_TP_SCE_DRX} = 22 * Max\{ T_{DMTC_periodicity}, \text{DRX cycle length} \} = 5632$ ms.

TTI insertion uncertainty = 2 ms

DRX cycle length = 256 ms

The overall delay measured shall be less than a total of 5890 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.22.7 E-UTRAN FDD-FDD Inter-frequency event triggered reporting in DRX based on CSI-RS based discovery signal

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- The Test system uncertainties applicable to this test are undefined
- The Test tolerances applicable to this test are undefined

### 8.22.7.1 Test purpose

To verify the UE's ability to make a correct reporting of an event when discovery signal is configured in DRX under fading propagation conditions within the E-UTRA FDD-FDD inter-frequency cell search requirements.

### 8.22.7.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward and supports CSI-RS based discovery signals measurement. Applicability requires support for FGIs bits 5 and 25.

### 8.22.7.3 Minimum conformance requirements

When DRX is in use, either measurement gaps are scheduled or the UE supports capability of conducting such measurements without gaps, the UE shall be able to identify a new detectable E-UTRAN FDD inter frequency TP within $T_{identify_inter_TP_SCE_DRX}$ according to the following expression:

$T_{identify_inter_TP_SCE_DRX} = T_{identify_inter_SCE_DRX} + T_{Measurement_Period_inter_FDD_CSI-RS_DRX}$

A TP shall be considered detectable when

- CSI-RSRP related side conditions given in TS 36.133 [4] Sections 9.1.14.3 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.11 for a corresponding Band.

$T_{identify_inter_SCE_DRX}$ is the inter-frequency period for cell identification as shown in TS 36.133 [4] section 8.6.2.2.1.2. $N_{freq}$ is defined in TS 36.133 [4] clause 8.1.2.1.1. $T_{Measurement_Period_inter_FDD_CSI-RS_DRX}$ is the inter-frequency period for TP measurement as shown in table 8.22.7.3-1.

When measurement gaps are scheduled for FDD inter frequency measurements, or the UE supports capability of conducting such measurements without gaps, the UE physical layer shall be capable of reporting CSI-RSRP measurements to higher layers with measurement accuracy as specified in TS 36.133 [4] sub-clauses 9.1.14.3, with measurement period given by table 8.22.7.3-1.

The UE shall be capable of performing CSI-RSRP measurements of at least 3 identified inter-frequency TPs per FDD inter-frequency for up to 3 FDD inter-frequencies and the UE physical layer shall be capable of reporting CSI-RSRP measurements to higher layers with the measurement period defined in table 8.22.7.3-1 when DRX is in use, either measurement gaps are scheduled or the UE supports capability of conducting such measurements without gaps.

**Table 8.22.7.3-1: Requirement to measure FDD inter frequency TP**

| Measurement bandwidth [RB] | Discovery signal occasion duration (*ds-OccasionDuration*) [ms] | $T_{Measurement_Period\ inter_FDD_\ CSI-RS_DRX}$ [ms] |
|---|---|---|
| ≥ 6 | ≥1 | 5**Max*{$T_{DMTC_periodicity}$, DRX cycle length, MGRP}*$N_{freq}$ |
| ≥ 25 | ≥1 | 3**Max*{$T_{DMTC_periodicity}$, DRX cycle length, MGRP}*$N_{freq}$ |

$T_{DMTC_periodicity}$ is the discovery signal measurement timing configuration periodicity of higher layer.

The CSI-RSRP measurement accuracy for all measured TPs shall be as specified in the TS 36.133 [4] sub-clauses 9.1.14.3.

Reported CSI-RSRP measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.14.3.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_inter_TP_SCE_DRX}$ defined in TS 36.133 [4] clause 8.6.3.2.1.2. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a TP which has been detectable at least for the time period $T_{identify_inter_TP_SCE_DRX}$ defined in TS 36.133 [4] clause 8.6.3.2.1.2 and then triggers the measurement report as per TS 36.331 [5], the event triggered measurement reporting delay shall be less than $T_{Measurement_Period_inter_FDD_CSI-RS_DRX}$ provided the timing to that TP has not changed more than ± 50 Ts while measurement gap has not been available and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.6.3.2.1.2 and A.8.22.7.

## 8.22.7.4 Test description

### 8.22.7.4.1 Initial conditions

Same Initial conditions as defined in clause 8.3.2.4.1 with the following exceptions:

- Instead of Table 8.3.2.4.1-1 -> use Table 8.22.7.4.1-1.
- Instead of clause 8.3.2.4.3 -> use Table 8.22.7.4.3.

**Table 8.22.7.4.1-1: General test parameters for E-UTRAN FDD-FDD inter-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| | | Test 1 | |
| Active cell | | Cell 1 | |
| Neighbour cell | | Cell 2 | Cell to be identified. |
| E-UTRA RF Channel Number | | 1,2 | Two FDD carrier frequency is used. |
| DMTC period [2] | ms | 160 | |
| DMTC period offset [2] | ms | 10 | |
| Discovery signal occasion duration | ms | 1 | |
| C1-Threshold | dB | -93 | |
| CP length | | Normal | |
| Hysteresis | dB | 0 | |
| Time To Trigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | ON | DRX related parameters are defined in Table 8.22.7.5-2 |
| T1 | s | 5 | |
| T2 | s | 10 | |

### 8.22.7.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information, it is indicated to the UE performing CSI-RS based discovery signals measurement and event-triggered reporting with Event C1 is configured. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

UE needs to be provided at least once every 500 ms with new Timing Advance Command MAC control element to restart the Time Alignment Timer to keep the UE uplink time alignment. Furthermore UE is allocated with PUSCH resource at every DRX cycle.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to T1 in Table 8.22.7.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.
3. SS shall transmit an RRCConnectionReconfiguration message and the SS shall provide the UE with new Timing Advance Command MAC control element at least once every 500 ms to restart the Time Alignment Timer in order to keep the UE uplink time alignment.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.22.7.5-1.
6. UE shall transmit a MeasurementReport message triggered by Event A3. If the overall delays measured from the beginning of time period T2 is less than 6146 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.
7. After the SS receive the MeasurementReport message in step 6) or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.
8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.
9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.22.7.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.22.7.4.3-1: Common Exception messages for E-UTRAN FDD-FDD inter-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX test requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-9 with Gap Pattern Id = 0<br>Table H.3.7-3 |

**Table 8.22.7.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.22.7.4.3-3: MeasObjectEUTRA-GENERIC(Freq): Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX test requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(f1) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| carrierFreq | Downlink EARFCN for Freq | | |
| allowedmeasBandwidth | mbw50 | | |
| measDS-Config-r12 CHOICE { | | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationFDD-r12 | 1 | | |
| } | | | |
| measCSI-RS-ToAddModList-r12 SEQUENCE (SIZE (1..maxCSI-RS-Meas-r12)) OF SEQUENCE { | | | |
| MeasCSI-RS-Config-r12[0] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| physCellId-r12 | PhysCellId | Cell1 | |
| scramblingIdentity-r12 | 0 | | |
| resourceConfig-r12 | 2 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | 0 | | |
| } | | | |
| MeasCSI-RS-Config-r12[1] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 2 | | |
| physCellId-r12 | PhysCellId | Cell2 | |
| scramblingIdentity-r12 | 1 | | |
| resourceConfig-r12 | 4 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.5.4.3-4: *QuantityConfig-DEFAULT*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX test requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-3A: QuantityConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| QuantityConfig-DEFAULT ::= SEQUENCE { | | | |
| quantityConfigEUTRA SEQUENCE { | | | |
| filterCoefficientRSRP | fc0 | | |
| filterCoefficientRSRQ | fc0 | | |
| } | | | |
| quantityConfigEUTRA-v1250 SEQUENCE { | | | |
| filterCoefficientCSI-RSRP-r12 | fc0 | | |
| } | | | |
| } | | | |

**Table 8.22.7.4.3-5: *ReportConfigEUTRA-C1*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX test requirement**

| Derivation Path: 36.508 clause 4.6.6, Table 4.6.6-10: ReportConfigEUTRA-C1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-C1 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventC1-r12 SEQUENCE { | | | |
| c1-Threshold-r12 | 47 (Thres+140) | Thres is actual threshold value in dBm | |
| c1-ReportOnLeave-r12 | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| reportStrongestCSI-RSs-r12 | false | | |
| reportCRS-Meas-r12 | false | | |
| triggerQuantityCSI-RS-r12 | true | | |
| } | | | |

**Table 8.22.7.4.3-6: *MAC-MainConfig-RBC*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX test requirements**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf256 | 9 | To avoid overlapping with measurement gap. | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | sf500 | | |
| } | | | |

**Table 8.22.7.4.3-7: *SchedulingRequest-Config-DEFAULT*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX test requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.22.7.4.3-8: *MeasResults*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX test requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| measResultCSI-RS-List-r12 SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| MeasResultCSI-RS-r12[0] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| csi-RSRP-Result-r12 | INTEGER(0..97) | Set according to specific test | |
| } | | | |
| MeasResultCSI-RS-r12[1] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 2 | | |
| csi-RSRP-Result-r12 | INTEGER(0..97) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.7.4.3-9: *MeasResultListEUTRA*: Additional E-UTRAN FDD-FDD inter-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX test requirements**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| physCellId | PhysCellId | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

## 8.22.7.5 Test requirement

Tables 8.22.7.4.1-1, 8.22.7.5-1, 8.22.7.5-2 and 8.22.7.5-3 define the primary level settings including test tolerances for E-UTRAN FDD-FDD inter-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX test.

**Table 8.22.7.5-1: Cell specific test parameters for E-UTRAN FDD-FDD inter-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |

| | | | | | |
|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Measurement bandwidth | $n_{PRB}$ | 13-37 | | 13-37 | |
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD as in A.1.1 | | DL Reference Measurement Channel R.0 FDD as in A.1.1 | |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD as in A.2.1 | | DL Reference Measurement Channel R.6 FDD as in A.2.1 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.2 FDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_PB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_PB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| p-C-r10 [5] | dB | 6 | 6 | 6 | 6 |
| $N_{oc}$ Note 3 | dBm/15 KHz | -98+TT | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4+TT | 4+TT | -Infinity | 4+TT |
| CRS $\hat{E}_s / I_{ot}$ | dB | 4+TT | -1.46+TT | -Infinity | -1.46+TT |
| CSI-RS $\hat{E}_s / I_{ot}$ | dB | 10+TT | 4.54+TT | -Infinity | 4.54+TT |
| RSRP Note 4 | dBm/15 KHz | -94+TT | -94+TT | -Infinity | -94+TT |
| CSI-RSRP Note 4 | dBm/15 KHz | -88+TT | -88+TT | -Infinity | -88+TT |
| SCH_RP Note 4 | dBm/15 KHz | -94+TT | -94+TT | -Infinity | -94+TT |
| Io | dBm/9 MHz | -60+TT | -60+TT | As specified for cell 1 | |
| Propagation Condition | | ETU30 | | | |
| CSI reference signal configurations [9] | | 2 | | 4 | |
| CSI-RS subframe offset | | 0 | | 0 | |
| CSI-RS individual offset [5] | dB | 0 | | 0 | |
| CSI-RS muting | | Enable | | Enable | |
| Timing offset to cell 1 | us | - | | 3us | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.22.7.5-2: DRX-Configuration for E-UTRAN FDD-FDD inter-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in clause 6.3.2 in TS 36.331[5] |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | sf256 | |
| shortDRX | disable | |

**Table 8.22.7.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD-FDD inter-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | sf500 | As specified in clause 6.3.2 in TS 36.331[5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331[5] and section10.1 in TS 36.213[8]. |

The UE shall send one Event C1 triggered measurement report, with a measurement reporting delay less than 6146ms from the beginning of time period T2. The measurement reporting delay is defined as the time from the beginning of time period T2, to the moment when the UE send the measurement report on PUSCH.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event C1 measurement report.

The overall delay measured test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_inter_TP_SCE_DRX} = T_{identify_inter_SCE_DRX} + T_{Measurement_Period_inter_FDD_CSI\text{-}RS_DRX}$

$T_{identify_inter_SCE\ DRX} = 17 * Max\ \{ T_{DMTC_periodicity}$, DRX cycle length, MGRP$\} * N_{freq} + T_{Measurement_Period_inter_FDD_CRS_DRX}$

$T_{Measurement_Period_inter_FDD_CSI\text{-}RS_DRX} = 3*Max\{T_{DMTC_periodicity}$, DRX cycle length, MGRP$\}*N_{freq}$

$T_{identify_inter_TP_SCE_DRX} = 23*Max\{T_{DMTC_periodicity}$, DRX cycle length, MGRP$\}*N_{freq}$

Where:

$T_{DMTC_periodicity}$ = 160ms,

DRX cycle length = 256ms.

Where $N_{freq}$ = 1. It is defined in TS 36.133 [4] clause 8.1.2.1.1.

Gap Pattern Id = 0, MGRP = 40ms

$T_{identify_inter_TP_SCE_DRX}$ = 5888 ms.

TTI insertion uncertainty = 2 ms

DRX cycle length = 256 ms

The overall delay measured shall be less than a total of 6146 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.22.8 E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation condition in DRX based on CSI-RS based discovery signal

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- The Test system uncertainties applicable to this test are undefined
- The Test tolerances applicable to this test are undefined

### 8.22.8.1 Test purpose

To verify the UE's ability to make a correct reporting of an event when discovery signal is configured in DRX based on CSI-RS based discovery signal under fading propagation conditions within the E-UTRA TDD-TDD inter-frequency cell search requirements.

### 8.22.8.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward and supports CSI-RS based discovery signals measurement. Applicability requires support for FGI bit 5 and 25.

### 8.22.8.3 Minimum conformance requirements

When DRX is in use, either measurement gaps are scheduled or the UE supports capability of conducting such measurements without gaps, the UE shall be able to identify a new detectable E-UTRAN TDD inter frequency TP within $T_{identify_inter_TP_SCE_DRX}$ according to the following expression:

$T_{identify_inter_TP_SCE_DRX} = T_{identify_inter_SCE_DRX} + T_{Measurement_Period\ _inter_TDD_CSI\text{-}RS_DRX}$

A TP shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] Sections 9.1.14.3 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to Annex I.2.11 for a corresponding Band.

$T_{identify_inter_SCE_DRX}$ is the inter-frequency period for cell identification as shown in TS 36.133 [4] section 8.6.2.2.2.2. $N_{freq}$ is defined in TS 36.133 [4] clause 8.1.2.1.1. $T_{Measurement_Period\ _inter_TDD_CSI\text{-}RS_DRX}$ is the inter-frequency period for TP measurement as shown in table 8.22.8.3-1.

When measurement gaps are scheduled for TDD inter frequency measurements, or the UE supports capability of conducting such measurements without gaps, the UE physical layer shall be capable of reporting CSI-RSRP measurements to higher layers with measurement accuracy as specified in TS 36.133 [4] sub-clauses 9.1.14.3, with measurement period given by table 8.22.8.3-1.

The UE shall be capable of performing CSI-RSRP measurements of at least 3 identified inter-frequency TPs per TDD inter-frequency for up to 3 TDD inter-frequencies and the UE physical layer shall be capable of reporting CSI-RSRP measurements to higher layers with the measurement period defined in table 8.22.8.3-1 when DRX is in use, either measurement gaps are scheduled or the UE supports capability of conducting such measurements without gaps.

**Table 8.22.8.3-1: Requirement to measure TDD inter frequency TP**

| Measurement bandwidth [RB] | Discovery signal occasion duration (*ds-OccasionDuration*) [ms] | $T_{Measurement_Period\ inter_TDD_\ CSI\text{-}RS_DRX}$ [ms] |
|---|---|---|
| ≥ 6 | ≥2 | 5**Max*{$T_{DMTC_periodicity}$, DRX cycle length, MGRP}*$N_{freq}$ |
| ≥ 25 | ≥2 | 3**Max*{$T_{DMTC_periodicity}$, DRX cycle length, MGRP}*$N_{freq}$ |

$T_{DMTC_periodicity}$ is the discovery signal measurement timing configuration periodicity of higher layer.

The CSI-RSRP measurement accuracy for all measured TPs shall be as specified in the TS 36.133 [4] sub-clauses 9.1.14.3.

Reported CSI-RSRP measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] sections 9.1.14.3.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_inter_TP_SCE_DRX}$ defined in TS 36.133 [4] Clause 8.6.3.2.2.2. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a TP which has been detectable at least for the time period $T_{identify_inter_TP_SCE_DRX}$ in TS 36.133 [4] clause 8.6.3.2.2.2 and then triggers the measurement report as per TS 36.331 [5], the event triggered measurement reporting delay shall be less than $T_{Measurement_Period\ _inter_TDD_CSI\text{-}RS_DRX}$ provided the timing to that TP has not changed more than ± 50 Ts while measurement gap has not been available and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.6.3.2.2.2 and A.8.22.8.

## 8.22.8.4 Test description

### 8.22.8.4.1 Initial conditions

Same Initial conditions as defined in clause 8.4.2.4.1 with the following exceptions:

- Instead of Table 8.4.2.4.1-1 -> use Table 8.22.8.4.1-1.
- Instead of clause 8.4.2.4.3 -> use clause 8.22.8.4.3.

**Table 8.22.8.4.1-1: General test parameters for E-UTRAN TDD-TDD inter-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX**

| Parameter | Unit | Test 1 | Comment |
|---|---|---|---|
| | | Value | |

| | | | |
|---|---|---|---|
| | | | |
| E-UTRA RF Channel Number | | 1, 2 | Two TDD carrier frequencies are used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active cell | | Cell 1 | Cell 1 is on RF channel number 1 |
| Neighbour cell | | Cell 2 | Cell 2 is on RF channel number 2 |
| Gap Pattern Id | | 0 | As specified in TS 36.133[4] clause 8.1.2.1. |
| Uplink-downlink configuration | | 1 | As specified in TS 36.211[9] clause 4.2 Table 4.2-2 |
| Special subframe configuration | | 6 | As specified in table 4.2-1 in TS 36.211[9]. The same configuration in both cells |
| DMTC period [5] | ms | 160 | |
| DMTC period offset [5] | ms | 10 | |
| Discovery signal occasion duration | ms | 2 | |
| C1-Threshold | dB | -93 | |
| Hysteresis | dB | 0 | |
| CP length | | Normal | |
| TimeToTrigger | s | 0 | |
| Filter coefficient | | 0 | L3 filtering is not used |
| PRACH configuration | | 4 | As specified in table 5.7.1-3 in TS 36.211[9] |
| Access Barring Information | - | Not Sent | No additional delays in random access procedure. |
| DRX | | ON | DRX related parameters are defined in Table 8.22.8.5-2 |
| T1 | s | 5 | |
| T2 | s | 10 | |

### 8.22.8.4.2 Test procedure

The test consists of one active cell and one neighbour cell. In the measurement control information, it is indicated to the UE performing CSI-RS based discovery signals measurement and event-triggered reporting with Event C1 is configured. Entire discovery signal occasion should be contained in the measurement gap. The subframe contained discovery signal for the measurement is not overlapped with the first 0.5ms period and the last 0.5ms period in every gap. The test consists of two successive time periods, with time durations of T1 and T2 respectively. During time duration T1, the UE shall not have any timing information of Cell 2.

UE needs to be provided at least once every 500ms with new Timing Advance Command MAC control element to restart the Time alignment timer to keep UE uplink time alignment. Furthermore, the UE is allocated with PUSCH resource at every DRX cycle.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to T1 in Table 8.22.8.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

3. SS shall transmit an RRCConnectionReconfiguration message and The SS shall provide the UE with new Timing Advance Command MAC control element at least once every 500 ms to restart the Timer Alignment Timer in order to keep the UE uplink time alignment.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.22.8.5-1.

6. UE shall transmit a MeasurementReport message triggered by Event C1. If the overall delays measured from the beginning of time period T2 is less than 6146 ms then the number of successful tests is increased by one. If the UE fails to report the event within the overall delays measured requirement then the number of failure tests is increased by one.

7. After the SS receives the MeasurementReport message in step 6 or when T2 expires, the SS shall transmit RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

8. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

9. After the RRC connection release, the SS:
   - transmits in Cell 1 a Paging message (including PagingRecord with ue-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
   or
   - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.22.8.4.3 Message contents

**Table 8.22.8.4.3-1: Common Exception messages for E-UTRAN TDD-TDD intra frequency event triggered reporting under fading propagation conditions for DRS measurement based on CSI-RS with DRX**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-9 with the Gap Pattern Id = 0<br>Table H.3.7-3 |

**Table 8.22.8.4.3-2: *RRCConnectionReconfiguration***

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-HO | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.22.8.4.3-3: MeasObjectEUTRA-GENERIC(f1)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(f1) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| carrierFreq | Downlink EARFCN for Freq | | |
| allowedmeasBandwidth | mbw50 | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationTDD-r12 | 2 | | |
| } | | | |
| measCSI-RS-ToAddModList-r12 SEQUENCE (SIZE (1..maxCSI-RS-Meas-r12)) OF SEQUENCE { | | | |
| MeasCSI-RS-Config-r12[0] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| physCellId-r12 | PhysCellId | Cell1 | |
| scramblingIdentity-r12 | 0 | | |
| resourceConfig-r12 | 2 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | 0 | | |
| } | | | |
| MeasCSI-RS-Config-r12[1] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 2 | | |
| physCellId-r12 | PhysCellId | Cell2 | |
| scramblingIdentity-r12 | 1 | | |
| resourceConfig-r12 | 4 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | 0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.8.4.3-4: *QuantityConfig-DEFAULT***

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-3A: QuantityConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| QuantityConfig-DEFAULT ::= SEQUENCE { | | | |
| quantityConfigEUTRA SEQUENCE { | | | |
| filterCoefficientRSRP | fc0 | | |
| filterCoefficientRSRQ | fc0 | | |
| } | | | |
| quantityConfigEUTRA-v1250 SEQUENCE { | | | |
| filterCoefficientCSI-RSRP-r12 | fc0 | | |
| } | | | |
| } | | | |

**Table 8.22.8.4.3-5: *ReportConfigEUTRA-C1***

| Derivation Path: 36.508 clause 4.6.6, Table 4.6.6-10: ReportConfigEUTRA-C1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-C1 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventC1-r12 SEQUENCE { | | | |
| c1-Threshold-r12 | 47 (Thres+140) | Thres is actual threshold value in dBm | |
| c1-ReportOnLeave-r12 | FALSE | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| reportStrongestCSI-RSs-r12 | false | | |
| reportCRS-Meas-r12 | false | | |
| triggerQuantityCSI-RS-r12 | true | | |
| } | | | |

**Table 8.22.8.4.3-6: *MAC-MainConfig-RBC*s**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | | |
| sf256 | 9 | To avoid overlapping with measurement gap. | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | sf500 | | |
| } | | | |

**Table 8.22.8.4.3-6: *SchedulingRequest-Config-DEFAULT***

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-20 SchedulingRequest-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequest-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| sr-PUCCH-ResourceIndex | 41 | 10 MHz channel bandwidth parameter | |
| sr-ConfigIndex | 0 | | |
| dsr-TransMax | n4 | | |
| } | | | |
| } | | | |

**Table 8.22.8.4.3-7: *MeasResults***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| measResultCSI-RS-List-r12 | MeasResultCSI-RS-List-r12 | | |
| } | | | |
| } | | | |

**Table 8.22.8.4.3-8: *MeasResultListEUTRA***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF MeasResultEUTRA { | | | |
| physCellId | PhysCellId | Cell 2 | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test INTEGER(0..97) | |
| rsrqResult | | Set according to specific test INTEGER(0..34) | |
| } | | | |
| } | | | |

**Table 8.22.8.4.3-9: *MeasResultCSI-RS-List-r12***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultCSI-RS-List-r12 ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| MeasResultCSI-RS-r12[0] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| csi-RSRP-Result-r12 | INTEGER(0..97) | Set according to specific test | |
| } | | | |
| MeasResultCSI-RS-r12[1] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 2 | | |
| csi-RSRP-Result-r12 | INTEGER(0..97) | Set according to specific test | |
| } | | | |
| } | | | |

## 8.22.8.5 Test requirement

Tables 8.22.8.4.1-1, 8.22.8.5-1, 8.22.8.5-2 and 8.22.8.5-3 define the primary level settings including test tolerances for E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CSI-RS based discovery signal test.

**Table 8.22.8.5-1: Cell specific test parameters for E-UTRAN TDD-TDD inter-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX**

| Parameter | Unit | Cell 1 | | Cell 2 | |
|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 |

| | | | | | |
|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | | 2 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Measurement bandwidth | $n_{PRB}$ | 13-37 | | 13-37 | |
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD as in A.1.2 | | DL Reference Measurement Channel R.0 TDD as in A.1.2 | |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD as in A.2.2 | | DL Reference Measurement Channel R.6 TDD as in A.2.2 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | | OP.2 TDD | |
| PBCH_RA | dB | 0 | | 0 | |
| PBCH_RB | dB | | | | |
| PSS_RA | dB | | | | |
| SSS_RA | dB | | | | |
| PCFICH_RB | dB | | | | |
| PHICH_RA | dB | | | | |
| PHICH_PB | dB | | | | |
| PDCCH_RA | dB | | | | |
| PDCCH_PB | dB | | | | |
| PDSCH_RA | dB | | | | |
| PDSCH_RB | dB | | | | |
| OCNG_RA Note 1 | dB | | | | |
| OCNG_RB Note 1 | dB | | | | |
| p-C-r10 [5] | dB | 6+TT | 6+TT | 6+TT | 6+TT |
| $N_{oc}$ Note 3 | dBm/15 KHz | -98 | | | |
| $\hat{E}_s / N_{oc}$ | dB | 4+TT | 4+TT | -Infinity | 4+TT |
| CRS $\hat{E}_s / I_{ot}$ | dB | 4+TT | -1.46+TT | -Infinity | -1.46+TT |
| CSI-RS $\hat{E}_s / I_{ot}$ | dB | 10+TT | 4.54+TT | -Infinity | 4.54+TT |
| RSRP Note 4 | dBm/15 KHz | -94+TT | -94+TT | -Infinity | -94+TT |
| CSI-RSRP Note 4 | dBm/15 KHz | -88+TT | -88+TT | -Infinity | -88+TT |
| SCH_RP Note 4 | dBm/15 KHz | -94+TT | -94+TT | -Infinity | -94+TT |
| Io | dBm/9 MHz | -60+TT | -60+TT | As specified for cell1 | |
| Propagation Condition | | ETU30 | | | |
| CSI reference signal configurations [9] | | 2 | | 4 | |
| CSI-RS subframe offset | | 0 | | 0 | |
| CSI-RS individual offset [5] | dB | 0 | | 0 | |
| CSI-RS muting | | Enable | | Enable | |
| Timing offset to cell 1 | us | - | | 3 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 8.22.8.5-2: DRX-Configuration for E-UTRAN TDD-TDD inter-frequency event triggered reporting in DRX under fading propagation conditions in synchronous cells**

| Field | Value | Comment |
|---|---|---|
| onDurationTimer | psf1 | As specified in clause 6.3.2 in TS 36.331 [5] |
| drx-InactivityTimer | psf1 | |
| drx-RetransmissionTimer | psf1 | |
| longDRX-CycleStartOffset | Sf256 | |
| shortDRX | disable | |

**Table 8.22.8.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN TDD-TDD inter-frequency event triggered reporting in DRX under fading propagation conditions in synchronous cells**

| Field | Value | Comment |
|---|---|---|
| TimeAlignmentTimer | sf500 | As specified in clause 6.3.2 in TS 36.331[5] |
| sr-ConfigIndex | 0 | For further information see clause 6.3.2 in TS 36.331 [5]and section10.1 in TS 36.213 [8]. |

The UE shall send one Event C1 triggered measurement report, with a measurement reporting delay less than 5888 ms from the beginning of time period T2. The measurement reporting delay is defined as the time from the beginning of time period T2 to the moment when the UE send the measurement report on PUSCH.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The overall delay measured may be up to $2xTTI_{DCCH}$ higher than the measurement reporting delays because of TTI insertion uncertainty of the measurement report in DCCH.

The rate of correct events observed during repeated tests shall be at least 90%.

NOTE 1: The actual overall delays measured in the test may be up to one DRX cycle higher than the measurement reporting delays above because UE is allowed to delay the initiation of the measurement reporting procedure to the next until the Active Time.

NOTE 2: In order to calculate the rate of correct events the system simulator shall verify that it has received correct Event C1 measurement report.

The overall delay measured test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{identify_inter_TP_SCE_DRX}$

$T_{identify_inter_TP_SCE_DRX} = T_{identify_inter_SCE_DRX} + T_{Measurement_Period\ _inter_TDD_CSI\text{-}RS_DRX}$

Where $T_{identify_inter_SCE_DRX} = 17 * Max \{ T_{DMTC_periodicity}, \text{DRX cycle length, MGRP} \} * N_{freq} + T_{Measurement_Period_inter_TDD_CRS_DRX}$

$T_{DMTC_periodicity}$ = 160ms,

DRX cycle length = 256ms.

MGRP = 40ms,

$N_{freq}$ = 1,

$T_{Measurement_Period_inter_TDD_CRS_DRX} = 3 * Max \{ T_{DMTC_periodicity}, \text{DRX cycle length, MGRP} \}$

$T_{Measurement_Period\ _inter_TDD_CSI\text{-}RS_DRX} = 3 * Max \{ T_{DMTC_periodicity}, \text{DRX cycle length, MGRP} \}$

$T_{identify_inter_SCE_DRX} = 20 * Max \{ T_{DMTC_periodicity}, \text{DRX cycle length, MGRP} \}$

$T_{identify_intra_TP_SCE_DRX}$ = 23 * *Max* { $T_{DMTC_periodicity}$, DRX cycle length, MGRP} = 5888 ms.

TTI insertion uncertainty = 2 ms

DRX cycle length = 256 ms

The overall delay measured shall be less than a total of 6146 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.22.9 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX based on CRS based discovery signal

### 8.22.9.1 Test purpose

The purpose of this test is to verify that the UE correctly detects events A2 (Serving cell becomes worse than threshold) and A6 (Neighbour becomes better than SCell) defined in TS 36.331 [6] when CRS based discovery signal is configured within the requirements stated in TS36.133 [4] clause 8.7.2.4.1.

### 8.22.9.2 Test applicability

This test applies to all types of E-UTRA UE FDD release 12 and forward that support [CA and CRS based discovery signals measurement]. Applicability requires support for FGI bit 111.

### 8.22.9.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD or TDD cell on a secondary component carrier within $T_{identify_scc_SCE}$, according to the parameter *measCycleSCell* where $T_{identify_scc_CRS}$ = 13 **measCycleSCell*+ $T_{measure_scc_CRS}$

A cell shall be considered detectable when

- RSRP related side condition given in TS 36.133 [4] Clause 9.1.15 are fulfilled for a corresponding Band,
- SCH_RP|$_{dBm}$ and SCH Ês/Iot according to Annex I.2.10 for a corresponding Band.

The measurement period for deactivated scell measurements is $T_{measure_scc_CRS}$ according to the parameter *measCycleSCell* shown in Tables 8.22.9.3-1 and 8.22.9.3-2.

**Table 8.22.9.3-1: Requirement to measure intra frequency cell on FDD SCC with deactivated SCell**

| Measurement bandwidth[RB] | Discovery signal occasion duration (***ds-OccasionDuration***) [ms] | $T_{measure_scc_CRS}$ [ms] |
|---|---|---|
| ≥ 6 | ≥1 | 5* *measCycleSCell* |
| ≥ 25 | ≥1 | 3 * *measCycleSCell* |

**Table 8.22.9.3-2: Requirement to measure intra frequency cell on TDD SCC with deactivated SCell**

| Measurement bandwidth [RB] | Discovery signal occasion duration (***ds-OccasionDuration***) [ms] | $T_{measure_scc_CRS}$ [ms] |
|---|---|---|
| ≥ 6 | ≥2 | 5* *measCycleSCell* |
| ≥ 25 | ≥2 | 3 * *measCycleSCell* |

The UE shall be capable of performing RSRP and RSRQ measurements for 3 identified cells on a secondary component carrier, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_scc_CRS}$.

The measurement accuracy for all measured cells shall be as specified in the TS 36.133 [4] sub-clause 9.1.15.

A UE may reconfigure the receiver bandwidth or turn on/off one of the RF chains when performing measurements on one or two SCCs with deactivated SCell. This may cause interruptions on PCell or an activated SCell or both that are specified in TS 36.133 [4] Section 7.8.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which is caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc_CRS}$ defined in TS 36.133 [4] Clause 8.7.2.4.1. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_scc_SCE}$ defined in TS 36.133 [4] clause 8.7.2.4.1 becomes undetectable for a period $\leq$ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_scc_CRS}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.7.2.4.1 and A.8.22.9.

## 8.22.9.4 Test description

### 8.22.9.4.1 Initial conditions

Same Initial conditions as defined in clause 8.16.2.4.1 with the following exceptions:

- Instead of Table 8.16.1.4.1-1 -> use Table 8.22.9.4.1-1.
- Instead of clause 8.16.1.4.3 -> use clause 8.22.9.4.3.

**Table 8.22.9.4.1-1: General test parameters for E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX based on CRS based discovery signal**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2 | Two radio channels are used for this test |
| Active PCell | | | Cell 1 | Primary cell on RF channel number 1. |
| Configured deactivated SCell | | | Cell 2 | Configured deactivated secondary cell on RF channel number 2. |
| Neighbour cell | | | Cell 3 | Neighbour cell to be identified on RF channel number 2. |
| CP length | | | Normal | |
| DRX | | | OFF | Continuous monitoring of primary cell |
| DMTC period | | ms | 160 | As specified in IE MeasDS-Config in TS 36.331[5] |
| dmtc-PeriodOffset for cells 2 and 3 | | ms | 10 | As specified in IE MeasDS-Config in TS 36.331[5] |
| Discovery signal occasion duration | | ms | 1 | As specified in IE MeasDS-Config in TS 36.331[5] |
| A2 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A2. |
| | Threshold RSRP | dBm | -93 | Actual RSRP threshold for event A2. Needs to take absolute accuracy tolerance in clause 9.1.14.2 into account plus margin. |
| | Time To Trigger | s | 0 | |
| A6 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A6. |
| | Offset | dB | -6 | Offset parameter for evaluation of event A6. Needs to take relative accuracy tolerance in TS 36.133 [4] clause 9.1.14.2 into account plus margin. |
| | Report on leave | | False | |
| | Time To Trigger | s | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| SCell measurement cycle (measCycleSCell) | | ms | 320 | |
| T1 | | s | 10 | |
| T2 | | s | 10 | |
| T3 | | s | 5 | |
| NOTE: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | | |

## 8.22.9.4.2 Test procedure

It is indicated to the UE in the measurement control information that event-triggered reporting with Events A2 (PCell and SCell) and A6 is used. The test consists of three successive time periods, with duration of T1, T2 and T3, respectively. During T1 the UE shall not have any information on cell 3. Immediately at beginning of T2 the transmission power of cell 3 is increased to same level as for cell 2, and due to usage of an offset this shall result in reporting of Event A6. At beginning of T3 the transmission powers of cells 1, 2 and 3 are reduced below a threshold value and this shall result in reporting of Event A2 for PCell and SCell, respectively.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCC according to Annex C.0, C.1 for all downlink physical channels.
3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.22.9.4.3.
4. Set the parameters according to T1 in Table 8.22.9.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

5. After 2s from start of T1, SS shall transmit an RRCConnectionReconfiguration message with events A6 and A2 configured.

6. The UE shall transmit an RRCConnectionReconfigurationComplete message.

7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.22.9.5-1.

8. The UE shall transmit a MeasurementReport message triggered by Event A6. If the measurement reporting delay from the beginning of time period T2 is less than 5122ms, then count a success for the event "A6". Otherwise count a fail for the event "A6".

9. After the SS receives the MeasurementReport message in step 8) or when T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 8.22.9.5-1.

10. The UE shall transmit MeasurementReport messages triggered by Event A2 for Cell 1 and Cell 2, respectively.

10a. If the measurement reporting delay for Cell 1 from the beginning of time period T3 is less than 482ms, then count a success for the event "Cell 1 A2". If the UE sends event-triggered measurement reports before the beginning of time period T3 or fails to report the event within the overall delays measured requirement, then the number of failure for the event "Cell 1 A2" is increased by one.

10b. If the measurement reporting delay for Cell 2 from the beginning of time period T3 is less than 962ms, then count a success for the event "Cell 2 A2". If the UE sends event-triggered measurement reports before the beginning of time period T3 or fails to report the event within the overall delays measured requirement, then the number of failure for the event "Cell 2 A2" is increased by one.

11. After the SS receives the MeasurementReport message in step 8) and 10) or when T3 expires, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

12. Set Cell 3 physical cell identity = ((current cell 3 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. If physical cell identity of Cell 3 = physical cell identity of Cell 2 then skip this physical cell identity value for Cell 3.

13. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 3-13 until a test verdict has been achieved.

Each of the events "A6", "Cell 1 A2" and "Cell 2 A2" is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.22.9.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.22.9.4.3-1: Common Exception messages for E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.4.1-5<br>Table H.4.1-6 with A6-Offset = - 6dB |

**Table 8.22.9.4.3-2: *SCellToAddMod-r10***

| Derivation Path: 36.508, Table 4.6.3-19D | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SCellToAddMod-r10 ::= SEQUENCE { | | | |
| sCellIndex-r10 | 1 | | |
| cellIdentification-r10 SEQUENCE { | | | |
| physCellId-r10 | PhysicalCellIdentity of Cell 2 | | |
| dl-CarrierFreq-r10 | Same downlink EARFCN as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 8.22.9.4.3-3: RadioResourceConfigCommonSCell-r10**

| Derivation Path: 36.508, Table 4.6.3-13A | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigCommonSCell-r10 ::= SEQUENCE { | | | |
| nonUL-Configuration-r10 SEQUENCE { | | | |
| dl-Bandwidth-r10 | Same downlink system bandwidth as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 8.22.9.4.3-4: *MeasConfig***

| Derivation path: 36.508 clause 4.6.6 table 4.6.6-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasConfig ::= SEQUENCE { | | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entries | | |
| measObjectId[1] | IdMeasObject-f1 | | |
| measObject[1] | MeasObjectEUTRA-GENERIC(f1) | Cell 1 | |
| measObjectId[2] | IdMeasObject-f2 | | |
| measObject[2] | MeasObjectEUTRA-GENERIC(f2) | Cell 2, 3 | |
| } | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId)) OF SEQUENCE { | 2 entries | | |
| reportConfigId[1] | IdReportConfig-A6 | | |
| reportConfig[1] | ReportConfig-A6 | | |
| reportConfigId[2] | IdReportConfig-A2 | | |
| reportConfig[2] | ReportConfig-A2-H | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) OF SEQUENCE { | 3 entries | | |
| measId[1] | 1 | | |
| measObjectId[1] | IdMeasObject-f2 | | |
| reportConfigId[1] | IdReportConfig-A6 | | |
| measId[2] | 2 | | |
| measObjectId[2] | IdMeasObject-f1 | | |
| reportConfigId[2] | IdReportConfig-A2 | | |
| measId[3] | 3 | | |
| measObjectId[3] | IdMeasObject-f2 | | |
| reportConfigId[3] | IdReportConfig-A2 | | |
| } | | | |
| } | | | |

**Table 8.22.1.4.3-5: MeasObjectEUTRA-GENERIC(Freq)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(Freq) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| carrierFreq | Downlink EARFCN for Freq | | |
| allowedmeasBandwidth | mbw50 | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationFDD-r12 | 1 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.2.4.3-6: MeasObjectEUTRA-GENERIC(f2)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | |
| } | | | |

**Table 8.22.9.4.3-7: *ReportConfig-A2-H***

| Derivation Path: 36.508 clause 4.6.6 table 4.6.6-5 ReportConfigEUTRA-A2(-90) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A2(Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA2 SEQUENCE { | | | |
| a2-Threshold CHOICE { | | | |
| threshold-RSRP | 47 | The actual value is IE value – 140 dBm | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

**Table 8.22.9.4.3-8: *MeasurementReport***

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 2 | | |
| measResultPCell::= SEQUENCE { | | Report Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | Not present | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | | |
| rsrqResultSCell-r10 | (0..34) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.9.4.3-9: *MeasurementReport***

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 3 | | |
| measResultPCell::= SEQUENCE { | | Report Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | Not present | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | | |
| rsrqResultSCell-r10 | (0..34) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 8.22.9.5 Test requirement

Table 8.22.9.4-1 and Table 8.22.9.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX test based on CRS based discovery signal.

**Table 8.22.9.5-1: Cell specific test parameters for E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX based on CRS based discovery signal**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | | 10 | | |
| Measurement bandwidth | $n_{PRB}$ | 13-37 | | | 13-37 | | | 13-37 | | |
| PDSCH parameters: DL Reference Measurement Channel | | R.0 FDD | | | - | | | - | | |
| PCFICH/PDCCH/PHICH parameters: DL Reference Measurement Channel | | R.6 FDD | | | R.6 FDD | | | R.6 FDD | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | | | OP.2 FDD | | | OP.2 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -101.00 | | | -101.00 | | | | | |
| $\hat{E}_s/N_{oc}$ | dB | 19.20 | 19.20 | -3.00 | 19.20 | 19.20 | -3.20 | -infinity | 19.00 | -3.00 |
| $\hat{E}_s/I_{ot}$ | dB | 19.20 | 19.20 | -3.00 | 19.20 | -0.15 | -4.96 | -infinity | -0.25 | -4.70 |
| RSRP Note 3 | dBm/15 kHz | -81.80 | -81.80 | -104.00 | -81.80 | -81.80 | -104.20 | -infinity | -82.00 | -104.00 |
| SCH_RP Note 3 | dBm/15 kHz | -81.80 | -81.80 | -104.00 | -81.80 | -81.80 | -104.20 | -infinity | -82.00 | -104.00 |
| Io Note 3 | dBm/9MHz | -53.97 | -53.97 | -71.45 | -53.97 | -51.08 | -70.25 | Specified in columns for Cell 2 | | |
| Propagation Condition | | ETU30 | | | ETU30 | | | ETU30 | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | 1x2 Low | | | 1x2 Low | | |
| Timing offset to Cell 1 | μs | - | | | 0 | | | - | | |
| Time alignment error relative to cell 1 Note 5 | μs | - | | | ≤ TAE | | | N/A | | |
| Timing offset to Cell 2 | μs | - | | | - | | | 2.3 (CP/2) | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: Es/Iot, RSRP and SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 5: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface

The actual overall delays measured in the tests may be up to $2\times TTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc_SCE}$

The overall delays measured test requirement for Event A6 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Where:

measurement reporting delay = $T_{identify_scc_SCE}$

$T_{identify_scc_CRS}$ = 13 **measCycleSCell*+ $T_{measure_scc_CRS}$

$T_{measure_scc_CRS}$ = 3 * *measCycleSCell*

*measCycleSCell* = 320 ms

$T_{identify_scc_CRS}$ = 16 **measCycleSCell* = 5120 ms

The UE shall send one Event A6 triggered measurement report with a measurement reporting delay of less than 5122 ms from the beginning of time T2 (note: this gives a total of 5120 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

In the RRC_CONNECTED state, the measurements of the primary component carrier shall meet all applicable requirements (FDD or TDD) in TS 36.133 [4] clause 8.6.2.1 in which $T_{Measurement_Period\ intra_FDD_CRS} = 3 * T_{DMTC_periodicity}$ = 480ms.

The UE shall send one Event A2 triggered measurement report for Cell 1 with a measurement reporting delay of less than 482 ms (including 2 ms for TTI insertion uncertainty) from beginning of time T3.

The measurement period for deactivated scell measurements is $T_{measure_scc_CRS}$ according to the parameter *measCycleSCell*

The overall delays measured test requirement for Event A2 for Cell 2 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Where:

measurement reporting delay = $T_{measure_scc_CRS}$

where $T_{measure_scc_CRS}$ = 3 * *measCycleSCell*.

The UE shall send one Event A2 triggered measurement report for Cell 2 with a measurement reporting delay of less than 962ms from beginning of time T3 (note: this gives a total of 960 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall not send event triggered measurement reports as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95% for each of the events.

The statistical pass/ fail decisions are done separately for Event A6 and Event A2.

Decide the test pass, if events A6 and A2 are passed, otherwise fail the UE.

## 8.22.10 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX based on CRS based discovery signal

### 8.22.10.1 Test purpose

Same test purpose as in clause 8.22.9.1.

### 8.22.10.2 Test applicability

This test applies to all types of E-UTRA UE TDD release 12 and forward that support [CA and CRS based discovery signals measurement]. Applicability requires support for FGI bit 111.

### 8.22.10.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.22.9.3.

The normative reference for this requirement is TS 36.133 [4] clause 8.7.2.4.1 and A.8.22.10.

### 8.22.10.4 Test description

#### 8.22.10.4.1 Initial conditions

Same Initial conditions as defined in clause 8.16.2.4.1 with the following exceptions:

- Instead of Table 8.16.2.4.1-1 -> use Table 8.22.10.4.1-1.
- Instead of clause 8.16.2.4.3 -> use clause 8.22.10.4.3.

**Table 8.22.10.4.1-1: General test parameters for E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX based on CRS based discovery signal**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2 | Two radio channels are used for this test |
| Active PCell | | | Cell 1 | Primary cell on RF channel number 1. |
| Configured deactivated SCell | | | Cell 2 | Configured deactivated secondary cell on RF channel number 2. |
| Neighbour cell | | | Cell 3 | Neighbour cell to be identified on RF channel number 2. |
| CP length | | | Normal | |
| Special subframe configuration | | | 6 | As specified in table 4.2.1 in TS 36.211[9]. The same configuration applies to all cells. |
| Uplink-downlink configuration | | | 1 | |
| DRX | | | OFF | Continuous monitoring of primary cell |
| DMTC period | | ms | 160 | As specified in IE MeasDS-Config in TS 36.331 [5] |
| dmtc-PeriodOffset for cells 2 and 3 | | ms | 10 | As specified in IE MeasDS-Config in TS 36.331 [5] |
| Discovery signal occasion duration | | ms | 2 | As specified in IE MeasDS-Config in TS 36.331 [5] |
| A2 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A2. |
| | Threshold RSRP | dBm | -93 | Actual RSRP threshold for event A2. Needs to take absolute accuracy tolerance in clause 9.1.14.2 into account plus margin. |
| | Time To Trigger | s | 0 | |
| A6 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A6. |
| | Offset | dB | -6 | Offset parameter for evaluation of event A6. Needs to take relative accuracy tolerance in TS 36.133 [4] clause 9.1.14.2 into account plus margin. |
| | Report on leave | | False | |
| | Time To Trigger | s | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| SCell measurement cycle (measCycleSCell) | | ms | 320 | |
| T1 | | s | 10 | |
| T2 | | s | 10 | |
| T3 | | s | 5 | |
| NOTE: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | | |

## 8.22.10.4.2 Test procedure

Same test procedure as in clause 8.22.9.4.2 with the following exceptions:

- Instead of clause 8.22.9.4.3 for message content exceptions → use clause 8.22.10.4.3.
- Instead of Table 8.22.9.5-1 → use Table 8.22.10.5-1.

## 8.22.10.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.22.10.4.3-1: Common Exception messages for E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.4.1-5<br>Table H.4.1-6 with A6-Offset = - 6dB |

**Table 8.22.10.4.3-2: *SCellToAddMod-r10***

| Derivation Path: 36.508, Table 4.6.3-19D | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SCellToAddMod-r10 ::= SEQUENCE { | | | |
| sCellIndex-r10 | 1 | | |
| cellIdentification-r10 SEQUENCE { | | | |
| physCellId-r10 | PhysicalCellIdentity of Cell 2 | | |
| dl-CarrierFreq-r10 | Same downlink EARFCN as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 8.22.10.4.3-3: *RadioResourceConfigCommonSCell-r10***

| Derivation Path: 36.508, Table 4.6.3-13A | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigCommonSCell-r10 ::= SEQUENCE { | | | |
| nonUL-Configuration-r10 SEQUENCE { | | | |
| dl-Bandwidth-r10 | Same downlink system bandwidth as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 8.22.10.4.3-4: *MeasConfig***

| Derivation path: 36.508 clause 4.6.6 table 4.6.6-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasConfig ::= SEQUENCE { | | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entries | | |
| measObjectId[1] | IdMeasObject-f1 | | |
| measObject[1] | MeasObjectEUTRA-GENERIC(f1) | Cell 1 | |
| measObjectId[2] | IdMeasObject-f2 | | |
| measObject[2] | MeasObjectEUTRA-GENERIC(f2) | Cell 2, 3 | |
| } | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId)) OF SEQUENCE { | 2 entries | | |
| reportConfigId[1] | IdReportConfig-A6 | | |
| reportConfig[1] | ReportConfig-A6 | | |
| reportConfigId[2] | IdReportConfig-A2 | | |
| reportConfig[2] | ReportConfig-A2-H | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) OF SEQUENCE { | 3 entries | | |
| measId[1] | 1 | | |
| measObjectId[1] | IdMeasObject-f2 | | |
| reportConfigId[1] | IdReportConfig-A6 | | |
| measId[2] | 2 | | |
| measObjectId[2] | IdMeasObject-f1 | | |
| reportConfigId[2] | IdReportConfig-A2 | | |
| measId[3] | 3 | | |
| measObjectId[3] | IdMeasObject-f2 | | |
| reportConfigId[3] | IdReportConfig-A2 | | |
| } | | | |
| } | | | |

**Table 8.22.1.4.3-5: MeasObjectEUTRA-GENERIC(Freq)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(Freq) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| carrierFreq | Downlink EARFCN for Freq | | |
| allowedmeasBandwidth | mbw50 | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationTDD-r12 | 2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.2.4.3-6: MeasObjectEUTRA-GENERIC(f2)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | |
| } | | | |

**Table 8.22.10.4.3-7: *ReportConfig-A2-H***

| Derivation Path: 36.508 clause 4.6.6 table 4.6.6-5 ReportConfigEUTRA-A2(-90) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A2(Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA2 SEQUENCE { | | | |
| a2-Threshold CHOICE { | | | |
| threshold-RSRP | 47 | The actual value is IE value – 140 dBm | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

**Table 8.22.10.4.3-8: *MeasurementReport***

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 2 | | |
| measResultPCell::= SEQUENCE { | | Report Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | Not present | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | | |
| rsrqResultSCell-r10 | (0..34) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.10.4.3-9: *MeasurementReport***

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 3 | | |
| measResultPCell::= SEQUENCE { | | Report Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | Not present | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | | |
| rsrqResultSCell-r10 | (0..34) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 8.22.10.5 Test requirement

Table 8.22.10.4-1 and Table 8.22.10.5-1 define the primary level settings including test tolerances for E-UTRAN TDD-event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX test based on CRS based discovery signal.

**Table 8.22.10.5-1: Cell specific test parameters for E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX based on CRS based discovery signal**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | | 10 | | |
| Measurement bandwidth | $n_{PRB}$ | 13-37 | | | 13-37 | | | 13-37 | | |
| PDSCH parameters: DL Reference Measurement Channel | | R.0 TDD | | | - | | | - | | |
| PCFICH/PDCCH/PHICH parameters: DL Reference Measurement Channel | | R.6 TDD | | | R.6 TDD | | | R.6 TDD | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | | | OP.2 TDD | | | OP.2 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -101.00 | | | -101.00 | | | | | |
| $\hat{E}_s/N_{oc}$ | dB | 19.20 | 19.20 | -3.00 | 19.20 | 19.20 | -3.20 | -infinity | 19.00 | -3.00 |
| $\hat{E}_s/I_{ot}$ | dB | 19.20 | 19.20 | -3.00 | 19.20 | -0.15 | -4.96 | -infinity | -0.25 | -4.70 |
| RSRP Note 3 | dBm/15 kHz | -81.80 | -81.80 | -104.00 | -81.80 | -81.80 | -104.20 | -infinity | -82.00 | -104.00 |
| SCH_RP Note 3 | dBm/15 kHz | -81.80 | -81.80 | -104.00 | -81.80 | -81.80 | -104.20 | -infinity | -82.00 | -104.00 |
| Io Note 3 | dBm/9MHz | -53.97 | -53.97+TT | -71.45 | -53.97 | -51.08 | -70.25 | Specified in columns for Cell 2 | | |
| Propagation Condition | | ETU30 | | | ETU30 | | | ETU30 | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | 1x2 Low | | | 1x2 Low | | |
| Timing offset to Cell 1 | μs | - | | | 0 | | | - | | |
| Time alignment error relative to cell 1 Note 5 | μs | - | | | ≤ TAE | | | N/A | | |
| Timing offset to Cell 2 | μs | - | | | - | | | 2.3 (CP/2) | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: Es/Iot, RSRP and SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 5: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface

The actual overall delays measured in the tests may be up to $2\times TTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc_SCE}$

The overall delays measured test requirement for Event A6 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Where:

measurement reporting delay = $T_{identify_scc_SCE}$

$T_{identify_scc_CRS}$ = 13 **measCycleSCell*+ $T_{measure_scc_CRS}$

$T_{measure_scc_CRS}$ = 3 * *measCycleSCell*

*measCycleSCell* = 320 ms

$T_{identify_scc_CRS}$ = 16 **measCycleSCell* = 5120 ms

The UE shall send one Event A6 triggered measurement report with a measurement reporting delay of less than 5122 ms from the beginning of time T2 (note: this gives a total of 5120 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

In the RRC_CONNECTED state, the measurements of the primary component carrier shall meet all applicable requirements (FDD or TDD) in TS 36.133 [4] clause 8.6.2.1 in which $T_{Measurement_Period\ intra_TDD_CRS} = 3 * T_{DMTC_periodicity}$ = 480ms.

The UE shall send one Event A2 triggered measurement report for Cell 1 with a measurement reporting delay of less than 482 ms (including 2 ms for TTI insertion uncertainty) from beginning of time T3.

The measurement period for deactivated scell measurements is $T_{measure_scc_CRS}$ according to the parameter *measCycleSCell*

The overall delays measured test requirement for Event A2 for Cell 2 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Where:

measurement reporting delay = $T_{measure_scc_CRS}$

where $T_{measure_scc_CRS}$ = 3 * *measCycleSCell*.

The UE shall send one Event A2 triggered measurement report for Cell 2 with a measurement reporting delay of less than 962ms from beginning of time T3 (note: this gives a total of 960 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall not send event triggered measurement reports as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95% for each of the events.

The statistical pass/ fail decisions are done separately for Event A6 and Event A2.

Decide the test pass, if events A6 and A2 are passed, otherwise fail the UE.

## 8.22.11 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX based on CSI-RS based discovery signal

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- The Test system uncertainties applicable to this test are undefined
- The Test tolerances applicable to this test are undefined

- Test procedure, message contents part are not updated according to latest change to C1 event.
- Test requirement part stll need to be updated except the table part

### 8.22.11.1 Test purpose

The purpose of this test is to verify that the UE correctly detects events C1 (CSI-RS resource becomes better than threshold) defined in TS 36.331 [6] when CSI-RS based discovery signal is configured within the requirements stated in TS36.133 [4] clause 8.7.3.4.1.

### 8.22.11.2 Test applicability

This test applies to all types of E-UTRA UE FDD release 12 and forward that support [CA and CSI-RS based discovery signals measurement]. Applicability requires support for FGI bit 111.

### 8.22.11.3 Minimum conformance requirements

When no DRX is in use the UE shall be able to identify a new detectable FDD or TDD TP on a secondary component carrier within $T_{identify_scc_TP_SCE}$, according to the parameter *measCycleSCell*, where $T_{identify_scc_TP_SCE} = T_{identify_scc_SCE} + T_{measure_scc_CSI-RS}$,

A cell shall be considered detectable when

- CSI-RSRP related side condition given in TS 36.133 [4] Clause 9.1.15 are fulfilled for a corresponding Band,
- SCH_RP$|_{dBm}$ and SCH Ês/Iot according to Annex I.2.11 for a corresponding Band

$T_{identify_scc_SCE}$ is the intra-frequency period for cell identification in TS 36.133 [4]section 8.7.2.4.1. $T_{measure_scc_CSI-RS}$ is the intra-frequency period for TP measurement in table 8.22.11.3-1.

The measurement period for deactivated scell measurements is $T_{measure_scc_CSI-RS}$ according to the parameter *measCycleSCell* as shown in tables 8.22.11.3-1 and 8.22.11.3-1.

**Table 8.22.11.3-1: Requirement to measure intra frequency TP on FDD SCC with deactivated SCell**

| Measurement bandwidth [RB] | Discovery signal occasion duration (*ds-OccasionDuration*) [ms] | $T_{measure_scc_CSI-RS}$ [ms] |
|---|---|---|
| ≥ 6 | ≥1 | 5* *measCycleSCell* |
| ≥ 25 | ≥1 | 3* *measCycleSCell* |

**Table 8.22.11.3-2: Requirement to measure intra frequency TP on TDD SCC with deactivated SCell**

| Measurement bandwidth [RB] | Discovery signal occasion duration (*ds-OccasionDuration*) [ms] | $T_{measure_scc_CSI-RS}$ [ms] |
|---|---|---|
| ≥ 6 | ≥2 | 5* *measCycleSCell* |
| ≥ 25 | ≥2 | 3* *measCycleSCell* |

The UE shall be capable of performing RSRP measurements for 3 identified TPs on a secondary component carrier, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_scc_CSI-RS}$.

The measurement accuracy for all measured TPs shall be as specified in the TS 36.133 [4] sub-clause 9.1.15.

A UE may reconfigure the receiver bandwidth or turn on/off one of the RF chains when performing measurements on one or two SCCs with deactivated SCell. This may cause interruptions on PCell or an activated SCell or both that are specified in TS 36.133 [4] Section 7.8.

Reported measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clause 9.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which is caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc_TP_SCE}$ defined in Clause 8.7.3.4.1. When L3 filtering is used or IDC autonomous denial is configured an additional delay can be expected.

If a TP which has been detectable at least for the time period $T_{identify_scc_TP_SCE}$ defined in TS 36.133 [4] clause 8.7.3.4.1 becomes undetectable for a period $\leq 5$ seconds and then the TP becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_scc_CSI\text{-}RS}$ provided the timing to that TP has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.7.3.4.1 and A.8.22.11.

## 8.22.11.4 Test description

### 8.22.11.4.1 Initial conditions

Same Initial conditions as defined in clause 8.16.2.4.1 with the following exceptions:

- Instead of Table 8.16.1.4.1-1 > use Table 8.22.11.4.1-1.
- Instead of clause 8.16.1.4.3 > use clause 8.22.11.4.3.

**Table 8.22.11.4.1-1: General test parameters for E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions for CSI-RS based measurements under discovery signal**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2 | Two radio channels are used for this test |
| Active PCell | | | Cell 1 | Primary cell on RF channel number 1. |
| Configured deactivated SCell | | | Cell 2 | Configured deactivated secondary cell on RF channel number 2. |
| Neighbour cell | | | Cell 3 | Neighbour cell to be identified on RF channel number 2. |
| DMTC period | | ms | 160 | As specified in IE MeasDS-Config in TS 36.331[5] |
| dmtc-PeriodOffset for cells 2 and 3 | | ms | 10 | As specified in IE MeasDS-Config in TS 36.331[5] |
| Discovery signal occasion duration | | ms | 1 | As specified in IE MeasDS-Config in TS 36.331[5] |
| CP length | | | Normal | |
| DRX | | | OFF | Continuous monitoring of primary cell |
| C1 | Hysteresis | dB | 0 | Hysteresis for evaluation of event C1. |
| | Threshold CSI-RSRP | dBm | -93 | Actual RSRP threshold for event C1. Needs to take absolute accuracy tolerance in TS 36.133 [4] clause 9.1.14.3 into account plus margin. |
| | Time To Trigger | s | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| SCell measurement cycle (measCycleSCell) | | ms | 320 | |
| Cell3 timing offset to cell1 | | μs | 3 | Synchronous cells |
| T1 | | s | 5 | |
| T2 | | s | 5 | |
| T3 | | s | 10 | |
| NOTE: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | | |

## 8.22.11.4.2 Test procedure

It is indicated to the UE in the measurement control information that event-triggered reporting with Events A2 (PCell and SCell) and A6 is used. The test consists of three successive time periods, with duration of T1, T2 and T3, respectively. During T1 the UE shall not have any information on cell 3. Immediately at beginning of T2 the transmission power of cell 3 is increased to same level as for cell 2, and due to usage of an offset this shall result in reporting of Event A6. At beginning of T3 the transmission powers of cells 1, 2 and 3 are reduced below a threshold value and this shall result in reporting of Event A2 for PCell and SCell, respectively.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCC according to Annex C.0, C.1 for all downlink physical channels.
3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4 with the message content exceptions defined in clause 8.22.11.4.3.
4. Set the parameters according to T1 in Table 8.22.11.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.
5. After 2s from start of T1, SS shall transmit an RRCConnectionReconfiguration message with events A6 and A2 configured.
6. The UE shall transmit an RRCConnectionReconfigurationComplete message.
7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.22.11.5-1.

8. The UE shall transmit a MeasurementReport message triggered by Event A6. If the measurement reporting delay from the beginning of time period T2 is less than 6082ms, then count a success for the event "A6". Otherwise count a fail for the event "A6".

9. After the SS receives the MeasurementReport message in step 8) or when T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 8.22.11.5-1.

10. The UE shall transmit MeasurementReport messages triggered by Event A2 for Cell 1 and Cell 2, respectively.

10a. If the measurement reporting delay for Cell 1 from the beginning of time period T3 is less than 482ms, then count a success for the event "Cell 1 A2". If the UE sends event-triggered measurement reports before the beginning of time period T3 or fails to report the event within the overall delays measured requirement, then the number of failure for the event "Cell 1 A2" is increased by one.

10b. If the measurement reporting delay for Cell 2 from the beginning of time period T3 is less than 962ms, then count a success for the event "Cell 2 A2". If the UE sends event-triggered measurement reports before the beginning of time period T3 or fails to report the event within the overall delays measured requirement, then the number of failure for the event "Cell 2 A2" is increased by one.

11. After the SS receives the MeasurementReport message in step 8) and 10) or when T3 expires, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

12. Set Cell 3 physical cell identity = ((current cell 3 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. If physical cell identity of Cell 3 = physical cell identity of Cell 2 then skip this physical cell identity value for Cell 3.

13. After the RRC connection release, the SS:
    - transmits in Cell 1 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3),
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

14. Repeat step 3-13 until a test verdict has been achieved.

Each of the events "A6", "Cell 1 A2" and "Cell 2 A2" is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.22.11.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.22.10.4.3-1: Common Exception messages for E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.1-5<br>Table H.4.1-6 with A6-Offset = - 6dB |

**Table 8.22.10.4.3-2: *SCellToAddMod-r10***

| Derivation Path: 36.508, Table 4.6.3-19D | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SCellToAddMod-r10 ::= SEQUENCE { | | | |
| sCellIndex-r10 | 1 | | |
| cellIdentification-r10 SEQUENCE { | | | |
| physCellId-r10 | PhysicalCellIdentity of Cell 2 | | |
| dl-CarrierFreq-r10 | Same downlink EARFCN as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 8.22.10.4.3-3: RadioResourceConfigCommonSCell-r10**

| Derivation Path: 36.508, Table 4.6.3-13A | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigCommonSCell-r10 ::= SEQUENCE { | | | |
| nonUL-Configuration-r10 SEQUENCE { | | | |
| dl-Bandwidth-r10 | Same downlink system bandwidth as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 8.22.10.4.3-4: *MeasConfig***

| Derivation path: 36.508 clause 4.6.6 table 4.6.6-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasConfig ::= SEQUENCE { | | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entries | | |
| measObjectId[1] | IdMeasObject-f1 | | |
| measObject[1] | MeasObjectEUTRA-GENERIC(f1) | Cell 1 | |
| measObjectId[2] | IdMeasObject-f2 | | |
| measObject[2] | MeasObjectEUTRA-GENERIC(f2) | Cell 2, 3 | |
| } | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId)) OF SEQUENCE { | 2 entries | | |
| reportConfigId[1] | IdReportConfig-A6 | | |
| reportConfig[1] | ReportConfig-A6 | | |
| reportConfigId[2] | IdReportConfig-A2 | | |
| reportConfig[2] | ReportConfig-A2-H | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) OF SEQUENCE { | 3 entries | | |
| measId[1] | 1 | | |
| measObjectId[1] | IdMeasObject-f2 | | |
| reportConfigId[1] | IdReportConfig-A6 | | |
| measId[2] | 2 | | |
| measObjectId[2] | IdMeasObject-f1 | | |
| reportConfigId[2] | IdReportConfig-A2 | | |
| measId[3] | 3 | | |
| measObjectId[3] | IdMeasObject-f2 | | |
| reportConfigId[3] | IdReportConfig-A2 | | |
| } | | | |
| } | | | |

**Table 8.22.1.4.3-5: MeasObjectEUTRA-GENERIC(Freq)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(Freq) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| carrierFreq | Downlink EARFCN for Freq | | |
| allowedmeasBandwidth | mbw50 | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationFDD-r12 | 1 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.6.4.3-6: MeasObjectEUTRA-GENERIC(f1)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(f1) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| carrierFreq | Downlink EARFCN for Freq | | |
| allowedmeasBandwidth | mbw50 | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationFDD-r12 | 1 | | |
| } | | | |
| measCSI-RS-ToAddModList-r12 SEQUENCE (SIZE (1..maxCSI-RS-Meas-r12)) OF SEQUENCE { | | | |
| MeasCSI-RS-Config-r12[0] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| physCellId-r12 | PhysCellId | Cell1 | |
| scramblingIdentity-r12 | 0 | | |
| resourceConfig-r12 | 2 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | 0 | | |
| } | | | |
| MeasCSI-RS-Config-r12[1] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 2 | | |
| physCellId-r12 | PhysCellId | Cell2 | |
| scramblingIdentity-r12 | 1 | | |
| resourceConfig-r12 | 4 | | |
| subframeOffset-r12 | 1 | | |
| csi-RS-IndividualOffset-r12 | 1 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.16.2.4.3-7: MeasObjectEUTRA-GENERIC(f2)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | |
| } | | | |

**Table 8.22.10.4.3-8: *ReportConfig-A2-H***

| Derivation Path: 36.508 clause 4.6.6 table 4.6.6-5 ReportConfigEUTRA-A2(-90) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A2(Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA2 SEQUENCE { | | | |
| a2-Threshold CHOICE { | | | |
| threshold-RSRP | 47 | The actual value is IE value – 140 dBm | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| reportCRS-Meas-r12 | false | | |
| triggerQuantityCSI-RS-r12 | false | | |
| } | | | |

**Table 8.22.10.4.3-9: *MeasurementReport***

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 2 | | |
| measResultPCell::= SEQUENCE { | | Report Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | Not present | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | | |
| rsrqResultSCell-r10 | (0..34) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.10.4.3-10: *MeasurementReport***

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 3 | | |
| measResultPCell::= SEQUENCE { | | Report Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | Not present | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | | |
| rsrqResultSCell-r10 | (0..34) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.6.4.3-11: *QuantityConfig-DEFAULT***

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-3A: QuantityConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| QuantityConfig-DEFAULT ::= SEQUENCE { | | | |
| quantityConfigEUTRA SEQUENCE { | | | |
| filterCoefficientRSRP | fc0 | | |
| filterCoefficientRSRQ | fc0 | | |
| } | | | |
| quantityConfigEUTRA-v1250 SEQUENCE { | | | |
| filterCoefficientCSI-RSRP-r12 | fc0 | | |
| } | | | |
| } | | | |

## 8.22.11.5 Test requirement

Table 8.22.11.4-1 and Table 8.22.11.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX test based on CSI-RS based discovery signal.

**Table 8.22.11.5-1: Cell specific test parameters for E-UTRAN FDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions for CSI-RS based measurements under discovery signal**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | | 10 | | |
| Measurement bandwidth | $n_{PRB}$ | 13-37 | | | 13-37 | | | 13-37 | | |
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD as in A.1.1 | | | DL Reference Measurement Channel R.0 FDD as in A.1.1 | | | DL Reference Measurement Channel R.0 FDD as in A.1.1 | | |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD as in A.2.1 | | | DL Reference Measurement Channel R.6 FDD as in A.2.1 | | | DL Reference Measurement Channel R.6 FDD as in A.2.1 | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | 1x2 Low | | | 1x2 Low | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and in D.1.2 (OP.2 FDD) | | OP.1 FDD | | | OP.2 FDD | | | OP.2 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA[Note 1] | dB | | | | | | | | | |
| OCNG_RB[Note 1] | dB | | | | | | | | | |
| $N_{oc}$[Note 2] | dBm/15 kHz | -101+TT | | | -101+TT | | | | | |
| RSRP[Note 3] | dBm/15 kHz | -104+TT | -82+TT | -82+TT | -104+TT | -82+TT | -82+TT | -infinity | -infinity | -82+TT |
| CSI-RSRP[Note 3] | dBm/15 kHz | -98+TT | -76+TT | -76+TT | -98+TT | -76+TT | -76+TT | -infinity | -infinity | -76+TT |
| SCH_RP[Note 3] | dBm/15 kHz | -104+TT | -82+TT | -82+TT | -104+TT | -82+TT | -82+TT | -infinity | -infinity | -82+TT |
| $\hat{E}_s/N_{oc}$ | dB | -3+TT | 19+TT | 19+TT | -3+TT | 19+TT | 19+TT | -infinity | -infinity | 19+TT |
| CRS $\hat{E}_s/I_{ot}$ | dB | -3+TT | 19+TT | 19+TT | -3+TT | -0.05+TT | -0.05+TT | -infinity | -infinity | -0.05+TT |
| CSI-RS $\hat{E}_s/I_{ot}$ | dB | 3+TT | 25+TT | 25+TT | -3+TT | 5.95+TT | 5.95+TT | -infinity | -infinity | 5.95+TT |
| CSI-RS resource configurations [9] | | 2 | | | 4 | | | 6 | | |
| p-C-r10 [5] | dB | -6 | | | -6 | | | -6 | | |
| CSI-RS subframe offset | | 0 | | | 0 | | | 0 | | |
| CSI-RS individual offset [5] | [dB] | 0 | | | 0 | | | 0 | | |
| CSI-RS muting | | Enable | | | Enable | | | Enable | | |
| Propagation Condition | | ETU30 | | | ETU30 | | | ETU30 | | |
| Time offset to cell 1 | us | 0 | | | 0 | | | 2.3 (CP/2) | | |

| | | | | |
|---|---|---|---|---|
| Time alignment error relative to cell1 Note 5 | us | - | ≤ TAE | N/A |
| Timing offset to Cell 2 | µs | - | - | 2.3 (CP/2) |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP, CSI-RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 5: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface

The actual overall delays measured in the tests may be up to $2\times TTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc_TP_SCE}$

The overall delays measured test requirement for Event A6 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Where:

measurement reporting delay $= T_{identify_scc_TP_SCE} = T_{identify_scc_SCE} + T_{measure_scc_CSI\text{-}RS}$

$T_{identify_scc_SCE} = T_{identify_scc_CRS} = 13 * measCycleSCell + T_{measure_scc_CRS}$

$T_{measure_scc_CRS} = 3 * measCycleSCell$

$T_{measure_scc_CSI\text{-}RS} = 3 * measCycleSCell$

*measCycleSCell* = 320 ms

$T_{identify_scc_CRS} = 19 * measCycleSCell = 6080$ ms

The UE shall send one Event A6 triggered measurement report with a measurement reporting delay of less than 6082 ms from the beginning of time T2 (note: this gives a total of 6080 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

In the RRC_CONNECTED state, the measurements of the primary component carrier shall meet all applicable requirements (FDD or FDD) in TS 36.133 [4] clause 8.6.3.1 in which $T_{Measurement_Period_intra_FDD_CSI\text{-}RS} = 3 * T_{DMTC_periodicity}$ = 480ms.

The UE shall send one Event A2 triggered measurement report for Cell 1 with a measurement reporting delay of less than 482 ms (including 2 ms for TTI insertion uncertainty) from beginning of time T3.

The measurement period for deactivated scell measurements is $T_{measure_scc_CSI\text{-}RS}$ according to the parameter *measCycleSCell*

The overall delays measured test requirement for Event A2 for Cell 2 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Where:

measurement reporting delay $= T_{measure_scc_CSI\text{-}RS}$

where $T_{measure_scc_CSI\text{-}RS} = 3 * measCycleSCell$.

The UE shall send one Event A2 triggered measurement report for Cell 2 with a measurement reporting delay of less than 962ms from beginning of time T3 (note: this gives a total of 960 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall not send event triggered measurement reports as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95% for each of the events.

The statistical pass/ fail decisions are done separately for Event A6 and Event A2.

Decide the test pass, if events A6 and A2 are passed, otherwise fail the UE.

## 8.22.12 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX based on CSI-RS based discovery signal

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- The Test system uncertainties applicable to this test are undefined
- The Test tolerances applicable to this test are undefined
- Test procedure, message contents part are not updated according to latest change to C1 event.
- Test requirement part stll need to be updated except the table part

### 8.22.12.1 Test purpose

Same test purpose as in clause 8.22.11.1.

### 8.22.12.2 Test applicability

This test applies to all types of E-UTRA UE TDD release 12 and forward that support [CA and CSI-RS based discovery signals measurement]. Applicability requires support for FGI bit 111.

### 8.22.12.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.22.11.3.

The normative reference for this requirement is TS 36.133 [4] clause 8.7.3.4.1 and A.8.22.12.

### 8.22.12.4 Test description

#### 8.22.12.4.1 Initial conditions

Same Initial conditions as defined in clause 8.16.2.4.1 with the following exceptions:

- Instead of Table 8.16.2.4.1-1 -> use Table 8.22.12.4.1-1.
- Instead of clause 8.16.2.4.3 -> use clause 8.22.12.4.3.

**Table 8.22.12.4.1-1: General test parameters for E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions for CSI-RS based measurements under discovery signal**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2 | Two radio channels are used for this test |
| Active PCell | | | Cell 1 | Primary cell on RF channel number 1. |
| Configured deactivated SCell | | | Cell 2 | Configured deactivated secondary cell on RF channel number 2. |
| Neighbour cell | | | Cell 3 | Neighbour cell to be identified on RF channel number 2. |
| DMTC period | | ms | 160 | As specified in IE MeasDS-Config in TS 36.331[5] |
| dmtc-PeriodOffset for cells 2 and 3 | | ms | 10 | As specified in IE MeasDS-Config in TS 36.331[5] |
| Discovery signal occasion duration | | ms | 1 | As specified in IE MeasDS-Config in TS 36.331[5] |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 10 | Channel bandwidth for cells on primary and secondary component carriers |
| CP length | | | Normal | |
| Special subframe configuration | | | 6 | As specified in table 4.2.1 in TS 36.211[9]. The same configuration applies to all cells. |
| Uplink-downlink configuration | | | 1 | |
| DRX | | | OFF | Continuous monitoring of primary cell |
| A2 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A2. |
| | Threshold RSRP | dBm | -93 | Actual RSRP threshold for event A2. Needs to take absolute accuracy tolerance in TS 36.133 [4] clause 9.1.11.1 into account plus margin. |
| | Time To Trigger | s | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on secondary component carrier. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| SCell measurement cycle (measCycleSCell) | | ms | 320 | |
| T1 | | s | 5 | During this time the UE shall be aware of cells 1 and 2 but not cell 3. |
| T2 | | s | 5 | UE should report Event A6 within 6.08s (19×scellMeasCycle) |
| T3 | | s | 10 | UE should report Event A2 within 200 ms and 960s for cells 1 and 2, respectively. |
| NOTE: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | | |

## 8.22.12.4.2 Test procedure

Same test procedure as in clause 8.22.11.4.2 with the following exceptions:

- Instead of clause 8.22.11.4.3 for message content exceptions → use clause 8.22.10.4.3.
- Instead of Table 8.22.11.5-1 → use Table 8.22.12.5-1.

## 8.22.12.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.22.12.4.3-1: Common Exception messages for E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.1-5<br>Table H.4.1-6 with A6-Offset = - 6dB |

**Table 8.22.12.4.3-2: *SCellToAddMod-r10***

| Derivation Path: 36.508, Table 4.6.3-19D | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SCellToAddMod-r10 ::= SEQUENCE { | | | |
| sCellIndex-r10 | 1 | | |
| cellIdentification-r10 SEQUENCE { | | | |
| physCellId-r10 | PhysicalCellIdentity of Cell 2 | | |
| dl-CarrierFreq-r10 | Same downlink EARFCN as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 8.22.12.4.3-3: RadioResourceConfigCommonSCell-r10**

| Derivation Path: 36.508, Table 4.6.3-13A | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigCommonSCell-r10 ::= SEQUENCE { | | | |
| nonUL-Configuration-r10 SEQUENCE { | | | |
| dl-Bandwidth-r10 | Same downlink system bandwidth as used for Cell 2 | | |
| } | | | |
| } | | | |

**Table 8.22.12.4.3-4: *MeasConfig***

| Derivation path: 36.508 clause 4.6.6 table 4.6.6-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasConfig ::= SEQUENCE { | | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entries | | |
| measObjectId[1] | IdMeasObject-f1 | | |
| measObject[1] | MeasObjectEUTRA-GENERIC(f1) | Cell 1 | |
| measObjectId[2] | IdMeasObject-f2 | | |
| measObject[2] | MeasObjectEUTRA-GENERIC(f2) | Cell 2, 3 | |
| } | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId)) OF SEQUENCE { | 2 entries | | |
| reportConfigId[1] | IdReportConfig-A6 | | |
| reportConfig[1] | ReportConfig-A6 | | |
| reportConfigId[2] | IdReportConfig-A2 | | |
| reportConfig[2] | ReportConfig-A2-H | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) OF SEQUENCE { | 3 entries | | |
| measId[1] | 1 | | |
| measObjectId[1] | IdMeasObject-f2 | | |
| reportConfigId[1] | IdReportConfig-A6 | | |
| measId[2] | 2 | | |
| measObjectId[2] | IdMeasObject-f1 | | |
| reportConfigId[2] | IdReportConfig-A2 | | |
| measId[3] | 3 | | |
| measObjectId[3] | IdMeasObject-f2 | | |
| reportConfigId[3] | IdReportConfig-A2 | | |
| } | | | |
| } | | | |

**Table 8.22.12.4.3-5: MeasObjectEUTRA-GENERIC(Freq)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(Freq) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| carrierFreq | Downlink EARFCN for Freq | | |
| allowedmeasBandwidth | mbw50 | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationTDD-r12 | 2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.12.4.3-6: MeasObjectEUTRA-GENERIC(f1)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(f1) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| carrierFreq | Downlink EARFCN for Freq | | |
| allowedmeasBandwidth | mbw50 | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationTDD-r12 | 2 | | |
| } | | | |
| measCSI-RS-ToAddModList-r12 SEQUENCE (SIZE (1..maxCSI-RS-Meas-r12)) OF SEQUENCE { | | | |
| MeasCSI-RS-Config-r12[0] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| physCellId-r12 | PhysCellId | Cell1 | |
| scramblingIdentity-r12 | 0 | | |
| resourceConfig-r12 | 2 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | 0 | | |
| } | | | |
| MeasCSI-RS-Config-r12[1] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 2 | | |
| physCellId-r12 | PhysCellId | Cell2 | |
| scramblingIdentity-r12 | 1 | | |
| resourceConfig-r12 | 4 | | |
| subframeOffset-r12 | 1 | | |
| csi-RS-IndividualOffset-r12 | 1 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.12.4.3-7: MeasObjectEUTRA-GENERIC(f2)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | |
| } | | | |

**Table 8.22.12.4.3-8: *ReportConfig-A2-H***

| Derivation Path: 36.508 clause 4.6.6 table 4.6.6-5 ReportConfigEUTRA-A2(-90) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A2(Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA2 SEQUENCE { | | | |
| a2-Threshold CHOICE { | | | |
| threshold-RSRP | 47 | The actual value is IE value – 140 dBm | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| reportCRS-Meas-r12 | **true** | | |
| triggerQuantityCSI-RS-r12 | **false** | | |
| } | | | |

**Table 8.22.12.4.3-9: *MeasurementReport***

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 2 | | |
| measResultPCell::= SEQUENCE { | | Report Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | Not present | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | | |
| rsrqResultSCell-r10 | (0..34) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.12.4.3-10: *MeasurementReport***

| Derivation path: 36.508 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 3 | | |
| measResultPCell::= SEQUENCE { | | Report Cell 1 | |
| rsrpResult | (0..97) | | |
| rsrqResult | (0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { } | Not present | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | | |
| rsrqResultSCell-r10 | (0..34) | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.22.12.4.3-11: *QuantityConfig-DEFAULT***

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-3A: QuantityConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| QuantityConfig-DEFAULT ::= SEQUENCE { | | | |
| quantityConfigEUTRA SEQUENCE { | | | |
| filterCoefficientRSRP | fc0 | | |
| filterCoefficientRSRQ | fc0 | | |
| } | | | |
| quantityConfigEUTRA-v1250 SEQUENCE { | | | |
| filterCoefficientCSI-RSRP-r12 | fc0 | | |
| } | | | |
| } | | | |

## 8.22.12.5 Test requirement

Table 8.22.12.4-1 and Table 8.22.12.5-1 define the primary level settings including test tolerances for E-UTRAN TDD-event triggered reporting on configured but deactivated SCell with PCell interruption in non-DRX test based on CSI-RS based discovery signal.

**Table 8.22.12.5-1: Cell specific test parameters for E-UTRAN TDD event triggered reporting on configured but deactivated SCell in non-DRX under fading propagation conditions for CSI-RS based measurements under discovery signal**

| Parameter | Unit | Cell 1 | | | Cell 2 | | | Cell 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | | 2 | | |
| $BW_{channel}$ | MHz | 10 | | | 10 | | | 10 | | |
| Measurement bandwidth | $n_{PRB}$ | 13-37 | | | 13-37 | | | 13-37 | | |
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD As in A.1.2 | | | DL Reference Measurement Channel R.0 TDD As in A.1.2 | | | DL Reference Measurement Channel R.0 TDD As in A.1.2 | | |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD as in A.2.2 | | | DL Reference Measurement Channel R.6 TDD as in A.2.2 | | | DL Reference Measurement Channel R.6 TDD as in A.2.2 | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | 1x2 Low | | | 1x2 Low | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and in D.2.2 (OP.2 TDD) | | OP.1 TDD | | | OP.2 TDD | | | OP.2 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | | | | |
| PSS_RA | dB | | | | | | | | | |
| SSS_RA | dB | | | | | | | | | |
| PCFICH_RB | dB | | | | | | | | | |
| PHICH_RA | dB | | | | | | | | | |
| PHICH_RB | dB | | | | | | | | | |
| PDCCH_RA | dB | | | | | | | | | |
| PDCCH_RB | dB | | | | | | | | | |
| PDSCH_RA | dB | | | | | | | | | |
| PDSCH_RB | dB | | | | | | | | | |
| OCNG_RA[Note 1] | dB | | | | | | | | | |
| OCNG_RB[Note 1] | dB | | | | | | | | | |
| $N_{oc}$[Note 2] | dBm/15 kHz | -101+TT | | | -101+TT | | | | | |
| RSRP[Note 3] | dBm/15 kHz | dBm/15 | -82+TT | -82+TT | -82+TT | -82+TT | -82+TT | -infinity | -infinity | -82+TT |
| CSI-RSRP[Note 3] | dBm/15 kHz | dBm/15 | -76+TT | -76+TT | -98+TT | -76+TT | -76+TT | -infinity | -infinity | -76+TT |
| SCH_RP[Note 3] | dBm/15 kHz | dBm/15 | -82+TT | -82+TT | -82+TT | -82+TT | -82+TT | -infinity | -infinity | -82+TT |
| $\hat{E}_s/N_{oc}$ | dB | dBm/15 | 19+TT | 19+TT | 19+TT | 19+TT | 19+TT | -infinity | -infinity | 19+TT |
| CRS $\hat{E}_s/I_{ot}$ | dB | dBm/15 | 19+TT | 19+TT | 19+TT | -0.05+TT | -0.05+TT | -infinity | -infinity | -0.05+TT |
| CSI-RS $\hat{E}_s/I_{ot}$ | dB | dBm/15 | 25+TT | 25+TT | 25+TT | 5.95+TT | 5.95+TT | -infinity | -infinity | 5.95+TT |
| Propagation Condition | | ETU30 | | | | | | | | |
| CSI-RS resource configurations [9] | | 0 | | | 2 | | | 4 | | |
| CSI-RS subframe offset | | 0 | | | 0 | | | 0 | | |
| CSI-RS individual offset [5] | [dB] | 0 | | | 0 | | | 0 | | |
| CSI-RS muting | | Enable | | | Enable | | | Enable | | |
| p-C-r10 [5] | dB | -6 | | | -6 | | | -6 | | |
| Time offset to cell 1 | us | 0 | | | 0 | | | 2.3 (CP/2) | | |

| | | | | |
|---|---|---|---|---|
| Time alignment error relative to cell1 Note 5 | us | - | ≤ TAE | N/A |
| Timing offset to Cell 2 | μs | - | - | 2.3 (CP/2) |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission are assigned to the UE prior to the start of time period T2.
Note 5: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface

The actual overall delays measured in the tests may be up to $2\times TTI_{DCCH}$ higher than the measurement reporting delays above because of TTI insertion uncertainty of the measurement report in DCCH.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_scc_TP_SCE}$

The overall delays measured test requirement for Event A6 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Where:

measurement reporting delay $= T_{identify_scc_TP_SCE} = T_{identify_scc_SCE} + T_{measure_scc_CSI\text{-}RS}$

$T_{identify_scc_SCE} = T_{identify_scc_CRS} = 13 * measCycleSCell + T_{measure_scc_CRS}$

$T_{measure_scc_CRS} = 3 * measCycleSCell$

$T_{measure_scc_CSI\text{-}RS} = 3 * measCycleSCell$

*measCycleSCell* = 320 ms

$T_{identify_scc_CRS} = 19 * measCycleSCell = 6080$ ms

The UE shall send one Event A6 triggered measurement report with a measurement reporting delay of less than 6082 ms from the beginning of time T2 (note: this gives a total of 6080 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

In the RRC_CONNECTED state, the measurements of the primary component carrier shall meet all applicable requirements (FDD or TDD) in TS 36.133 [4] clause 8.6.3.1 in which $T_{Measurement_Period_intra_TDD_CSI\text{-}RS} = 3 * T_{DMTC_periodicity} = 480$ms.

The UE shall send one Event A2 triggered measurement report for Cell 1 with a measurement reporting delay of less than 482 ms (including 2 ms for TTI insertion uncertainty) from beginning of time T3.

The measurement period for deactivated scell measurements is $T_{measure_scc_CSI\text{-}RS}$ according to the parameter *measCycleSCell.*

The overall delays measured test requirement for Event A2 for Cell 2 is expressed as:

Overall delays measured = measurement reporting delay + TTI insertion uncertainty

Where:

measurement reporting delay $= T_{measure_scc_CSI\text{-}RS}$

where $T_{measure_scc_CSI\text{-}RS} = 3 * measCycleSCell$.

The UE shall send one Event A2 triggered measurement report for Cell 2 with a measurement reporting delay of less than 962ms from beginning of time T3 (note: this gives a total of 960 ms for measurement reporting delay plus 2 ms for TTI insertion uncertainty).

The UE shall not send event triggered measurement reports as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90% with a confidence level of 95% for each of the events.

The statistical pass/ fail decisions are done separately for Event A6 and Event A2.

Decide the test pass, if events A6 and A2 are passed, otherwise fail the UE.

# 8.23 E-UTRAN Dual Connectivity Measurements

## 8.23.1 E-UTRAN FDD-FDD DC intra-frequency event triggered reporting with DRX in synchronous DC

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- *The Test system uncertainties applicable to this test are undefined.*
- *The Test tolerances applicable to this test are undefined.*

### 8.23.1.1 Test purpose

To verify the Dual Connectivity UE's ability to make correct reporting of an event in DRX under fading propagation conditions in synchronous cells within the E-UTRA FDD-FDD intra-frequency cell search requirements.

### 8.23.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward of synchronous Dual Connectivity. Applicability requires support for FGI bit 5.

### 8.23.1.3 Minimum conformance requirements

The UE shall be able to identify new intra-frequency cells and perform RSRP measurements of identified intra-frequency cells without an explicit intra-frequency neighbour cell list containing physical layer cell identities. During the RRC_CONNECTED state the UE shall continuously measure identified intra frequency cells and additionally search for and identify new intra frequency cells.

When DRX is in use the UE shall be able to identify a new detectable FDD intra frequency cell within $T_{identify_intra}$ as shown in Table 8.23.1.3-1.

**Table 8.23.1.3-1: Requirement to identify a newly detectable FDD intrafrequency cell**

| DRX cycle length (s) | $T_{identify_intra}$ (s) (DRX cycles) |
|---|---|
| ≤0.04 | 0.8 (NOTE1) |
| 0.04<DRX-cycle≤0.08 | NOTE2 (40) |
| 0.128 | 3.2 (25) |
| 0.128<DRX-cycle≤2.56 | NOTE2(20) |
| NOTE1: Number of DRX cycle depends upon the DRX cycle in use<br>NOTE2: Time depends upon the DRX cycle in use | |

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] clauses 9.1.2.1 and 9.1.2.2 and RSRQ related side conditions given in TS 36.133 [4] clause 9.1.5.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to TS 36.133 [4] Annex B.2 for a corresponding Band

In the RRC_CONNECTED state the measurement period for intra frequency measurements is $T_{measure_intra}$ as shown in Table 8.23.1.3-2 The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified-intra-

frequency cells, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_intra}$.

**Table 8.23.1.3-2: Requirement to measure FDD intrafrequency cells**

| DRX cycle length (s) | $T_{measure_intra}$ (s) (DRX cycles) |
|---|---|
| ≤0.04 | 0.2 (NOTE1) |
| 0.04<DRX-cycle≤2.56 | NOTE2 (5) |
| NOTE1: Number of DRX cycle depends upon the DRX cycle in use | |
| NOTE2: Time depends upon the DRX cycle in use | |

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clauses 9.1.2.1 and 9.1.2.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clause 9.1.5.1.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clauses 9.1.2.1, 9.1.2.2, and 9.1.5.1, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra}$ defined in TS 36.133 [4] clause 8.1.2.2.1.2. When L3 filtering is used or IDC autonomous denial is configured or the UE is performing reception and/or transmission for ProSe Direct Discovery and/or ProSe Direct Communication, an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra}$ defined in TS 36.133 [4] clause 8.1.2.2.1.2 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured or the UE is performing reception and/or transmission for ProSe Direct Discovery and/or ProSe Direct Communication, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.8.2, 8.8.3, 8.1.2.2 and A.8.23.1.

## 8.23.1.4 Test description

### 8.23.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.23.1.5-1 as defined in TS 36.508 [7] clause [4.3.1] for different DC configurations (per band) as defined in TS 36.521-1 [10] clause 5.4.2..

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.42 as appropriate.
2. The general test parameter settings are set up according to Table 8.23.1.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.23.1.4.3.

5. There are two E-UTRA FDD carriers and two cells in the test. Cell 1 is PCell on the primary component carrier, Cell 2 is PSCell on the secondary component carrier, Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test, Cell 2 shall be powered OFF.

**Table 8.23.1.4.1-1: General test parameters for E-UTRAN FDD-FDD DC intra-frequency event triggered reporting with DRX in synchronous DC**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2 | Two radio channels are used for this test. |
| Active PCell | | | Cell1 | PCell on RF channel number 1. |
| Configured PSCell | | | Cell2 | PSCell on RF channel number 2. |
| A1 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A1. |
| | Threshold RSRP | dBm | -95 | Actual RSRP threshold for event A1. Needs to take absolute accuracy tolerance in TS 36.133 [4] clause 9.1.11.1 and 9.1.11.2 into account plus margin. |
| | Time To Trigger | s | 0 | |
| A2 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A2. |
| | Threshold RSRP | dBm | -99 | Actual RSRP threshold for event A2. Needs to take absolute accuracy tolerance in TS 36.133 [4] clause 9.1.11.1 and 9.1.11.2 into account plus margin. |
| | Time To Trigger | s | 0 | |
| CP length | | | Normal | |
| DRX | | | ON | DRX related parameters are defined in Table 8.23.1.5.1-2 |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on carrier frequency of Cell2. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| T1 | | s | 2 | |
| T2 | | s | 10 | |
| T3 | | s | 1 | |
| Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in TS 36.133[4] section A.3.6.1. | | | | |
| Note 2: A UE capable of both synchronous and asynchronous DC operations is not required to pass this test case in accordance with the principle defined in TS 36.133[4] section A.3.11. | | | | |

### 8.23.1.4.2 Test procedure

The test consists of two cells: Cell1 and Cell2. Cell1 is PCell and Cell2 is PSCell. In the measurement control information it is indicated to the UE that event-triggered reporting with Event A1 (PCell and PSCell) and A2 (PCell and PSCell) is used. The test consists of three successive time periods, with time durations of T1, T2 and T3, respectively. During time duration T1, the UE shall not have any timing information of Cell 2. Prior to the start of the time duration T1, the UE shall be fully synchronized to Cell1 and Cell2. Prior to the start of the time duration T1, DRX configurations on MCG and SCG are enabled and DRX inactivity timers for MCG and SCG have already been expired. Immediately at beginning of T2 the transmission powers of Cell1 and Cell2 are reduced below a threshold value of event A2 and this shall result in reporting of Event A2 for PCell and PSCell, respectively. Immediately after receiving the reporting of event A2 for both PCell and PSCell, PDCCH indicating a new transmission on Cell1 shall be sent continuously to ensure that the UE would not enter the DRX state on MCG throughout T3. At beginning of T3 the transmission powers of Cell1 and Cell2 are increased above a threshold value of event A1 and this shall result in reporting of Event A1 for PCell and PSCell, respectively.

1. Ensure the UE is in [State 3C-RF] according to TS 36.508 [7] clause 7.2A.4A.

2. Configure MCG and SCG according to Annex C.0, C.1 for all downlink physical channels.

3. The SS shall configure PCell (Cell1) and PSCell (Cell2) on the MCG and SCG as per TS 36.508 [7] clause 5.2A.2A with the message content exceptions defined in clause 8.23.1.4.3. 4. Set the parameters according to T1 in Table 8.23.1.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.

5. After 2s from start of T1 SS shall transmit an RRCConnectionReconfiguration message with events A2 and A1 configured.

6. The UE shall transmit an RRCConnectionReconfigurationComplete message.

7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.23.1.5-1.

8. The UE shall transmit a MeasurementReport message triggered by Event A2 for Cell1 and Cell2. If the measurement reporting delay from the beginning of time period T1 is less than 6402ms for Cell1 or less than 402ms for Cell2, then count a success for the event “A2”. Otherwise count a fail for the event “A2”.

9. After the SS receives the MeasurementReport message in step 8) or when T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 8.23.1.5-1.

10. The UE shall transmit MeasurementReport messages triggered by Event A2 for Cell 1 and Cell 2, respectively.

10a. If the measurement reporting delay for Cell 1 from the beginning of time period T3 is less than 202ms for Cell1, then count a success for the event “Cell 1 A1”. If the UE sends event-triggered measurement reports before the beginning of time period T3 or fails to report the event within the overall delays measured requirement, then the number of failure for the event “Cell 1 A1” is increased by one.

10b. If the measurement reporting delay for Cell 2 from the beginning of time period T3 is less than 402ms, then count a success for the event “Cell 2 A1”. If the UE sends event-triggered measurement reports before the beginning of time period T3 or fails to report the event within the overall delays measured requirement, then the number of failure for the event “Cell 2 A1 is increased by one.

11. After the SS receives the MeasurementReport message in step 8) and 10) or when T3 expires, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

12. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop. If physical cell identity of Cell 2 = physical cell identity of Cell 1 then skip this physical cell identity value for Cell 2.

13. After the RRC connection release, the SS:
- transmits in Cell 1 and Cell 2 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in State 3A according to TS 36.508 [7] clause 4.5.3A (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF-DC1 according to TS 36.508 [7] clause 7.2A.3A),
or
- switches off and on the UE and ensures the UE is in State 3A-RF-DC1 according to TS 36.508 [7] clause 7.2A.3A.

14. Repeat step 3-13 until a test verdict has been achieved.

Each of the events “A2”, “Cell 1 A1” and “Cell 2 A1” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 8.23.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.23.1.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra frequency event triggered reporting with DC in DRX under fading propagation conditions requirements**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-7<br>Table H.4.1-5 |

**Table 8.23.1.4.3-2: *RRCConnectionReconfiguration*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting with DC in DRX under fading propagation conditions requirements**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | MEAS |
| dedicatedInfoNASList | Set according to specific message content | | AM-DRB-ADD(1) |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-AM-DRB-ADD(1) | | AM-DRB-ADD(1) |
| } | | | |
| nonCriticalExtension SEQUENCE { | | | |
| scg-Configuration-r12 | SCG-Configuration-r12-DEFAULT | | PSCell_AddMod DC_Setup_SCG_DRB |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.23.1.4.3-3: MeasConfig: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting with DC in DRX under fading propagation conditions requirements**

| Derivation path: 36.508 clause 4.6.6 table 4.6.6-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasConfig ::= SEQUENCE { | | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entries | | |
| measObjectId[1] | IdMeasObject-f1 | | |
| measObject[1] | MeasObjectEUTRA-GENERIC(f1) | Cell 1 | |
| measObjectId[2] | IdMeasObject-f2 | | |
| measObject[2] | MeasObjectEUTRA-GENERIC(f2) | Cell 2 | |
| } | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId)) OF SEQUENCE { | 2 entries | | |
| reportConfigId[1] | IdReportConfig-A2 | | |
| reportConfig[1] | ReportConfig-A2 | | |
| reportConfigId[2] | IdReportConfig-A1 | | |
| reportConfig[2] | ReportConfig-A1-H | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) OF SEQUENCE { | 2 entries | | |
| measId[1] | 1 | | |
| measObjectId[1] | IdMeasObject-f2 | | |
| reportConfigId[1] | IdReportConfig-A2 | | |
| measId[2] | 2 | | |
| measObjectId[2] | IdMeasObject-f1 | | |
| reportConfigId[2] | IdReportConfig-A1 | | |
| } | | | |
| } | | | |

**Table 8.23.1.4.3-4: MeasObjectEUTRA-GENERIC(f2): Additional E-UTRAN FDD-FDD intra frequency event triggered reporting with DC in DRX under fading propagation conditions requirements**

| Derivation Path: 36.508 clause 7.3.3 Table 7.3.4-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| cellsToAddModList SEQUENCE (SIZE (1..maxCellMeas)) { } | Not present | | |
| measCycleSCell-r10 | sf320 | | |
| } | | | |

**Table 8.23.1.4.3-5: *ReportConfig-A1*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting with DC in DRX under fading propagation conditions requirements**

| Derivation Path: 36.508 clause 4.6.6 table 4.6.6-4 ReportConfigEUTRA-A1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A1(Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA1 SEQUENCE { | | | |
| a1-Threshold CHOICE { | | | |
| threshold-RSRP | 45 | The actual value is IE value – 140 dBm | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | 0 | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

**Table 8.23.1.4.3-6: *ReportConfig-A2*: Additional E-UTRAN FDD-FDD intra frequency event triggered reporting with DC in DRX under fading propagation conditions requirements**

| Derivation Path: 36.508 clause 4.6.6 table 4.6.6-5 ReportConfigEUTRA-A2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A1(Thres) ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA2 SEQUENCE { | | | |
| a2-Threshold CHOICE { | | | |
| threshold-RSRP | 41 | The actual value is IE value – 140 dBm | |
| } | | | |
| } | | | |
| } | | | |
| hysteresis | 0 (0 dB) | | |
| timeToTrigger | 0 | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 | | |
| reportAmount | r1 | | |
| } | | | |

## 8.23.1.5 Test requirement

Tables 8.23.1.4.1-1 and 8.23.1.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-FDD event triggered reporting under fading propagation conditions in synchronous cells test.

**Table 8.23.1.5-1: Cell specific test parameters for E-UTRAN FDD-FDD DC intra-frequency event triggered reporting with DRX in synchronous DC**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | |
| PDSCH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | | | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | | |
| PCFICH/PDCCH/PHICH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | |
| OCNG Patterns | | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | | | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_PB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_PB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 KHz | -104 | | | -104 | | |
| $\hat{E}_s / N_{oc}$ | dB | 16+TT | -2.5+TT | 20+TT | 16+TT | -2.5+TT | 20+TT |
| $\hat{E}_s / I_{ot}$ | dB | 16+TT | -2.5+TT | 20+TT | 16+TT | -2.5+TT | 20+TT |
| RSRP Note 3 | dBm/15 KHz | -88+TT | -106.5+TT | -84+TT | -88+TT | -106.5+TT | -84+TT |
| SCH_RP Note 3 | dBm/15 KHz | -88+TT | -106.5+TT | -84+TT | -88+TT | -106.5+TT | -84+TT |
| Io Note 3 | dBm/Ch BW | -60.11 +10log ($N_{RB,c}$ /50) | -74.28 +10log ($N_{RB,c}$ /50) | -56.18 +10log ($N_{RB,c}$ /50) | -60.11 +10log ($N_{RB,c}$ /50) | -74.28 +10log ($N_{RB,c}$ /50) | -56.18 +10log ($N_{RB,c}$ /50) |
| Propagation Condition | | ETU70 | | | ETU70 | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | 1x2 Low | | |
| Receive Time offset to cell1 Note 5 | μs | - | | | 33 | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for Noc to be fulfilled.
Note 3: RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: The resources for uplink transmission in Cell1 are assigned to the UE prior to the start of time period T3.
Note 5: Receive time difference between subframe boundaries of signals received from the two cells at the UE antenna connector including time alignment error between the two cells.

**Table 8.23.1.5-2: DRX-Configuration for E-UTRAN FDD-FDD DC intra-frequency event triggered reporting with DRX in synchronous DC**

| Field | Cell1 | Cell2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf1280 | sf80 | |
| shortDRX | disable | disable | |

**Table 8.23.1.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD-FDD DC intra-frequency event triggered reporting with DRX in synchronous DC**

| Field | Cell1 | Cell2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | 0 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213. |

The UE shall send one Event A2 triggered report for PCell on PCell with a measurement reporting delay less than 6.4s (5*MCG_DRX cycle) from the beginning of time period T2. The measurement reporting delay is defined as the time from the beginning of time period T2 to the moment when the UE starts to send preambles on the PRACH to obtain allocation to send the measurement report on PUSCH.

The UE shall send one Event A2 triggered report for PSCell on PCell with a measurement reporting delay less than 400ms (5*SCG_DRX cycle) from the beginning of time period T2. The measurement reporting delay is defined as the time from the beginning of time period T2 to the moment when the UE starts to send preambles on the PRACH to obtain allocation to send the measurement report on PUSCH.

The UE shall send one Event A1 triggered report for PCell on PCell with a measurement reporting delay less than 200ms from the beginning of time period T3. The measurement reporting delay is defined as the time from the beginning of time period T3 to the moment when the UE sends the measurement report on PUSCH.

The UE shall send one Event A1 triggered report for PSCell on PCell with a measurement reporting delay less than 400ms (5*SCG_DRX cycle) from the beginning of time period T3. The measurement reporting delay is defined as the time from the beginning of time period T3 to the moment when the UE sends the measurement report on PUSCH.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90%.

## 8.23.2 E-UTRAN FDD-FDD DC intra-frequency event triggered reporting with DRX in asynchronous DC

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- *The Test system uncertainties applicable to this test are undefined.*
- *The Test tolerances applicable to this test are undefined.*

### 8.23.2.1 Test purpose

To verify the Dual Connectivity UE's ability to make correct reporting of an event in DRX under fading propagation conditions in asynchronous cells within the E-UTRA FDD-FDD intra-frequency cell search requirements.

### 8.23.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward of asynchronous Dual Connectivity. Applicability requires support for FGI bit 5.

### 8.23.2.3 Minimum conformance requirements

The UE shall be able to identify new intra-frequency cells and perform RSRP measurements of identified intra-frequency cells without an explicit intra-frequency neighbour cell list containing physical layer cell identities. During the RRC_CONNECTED state the UE shall continuously measure identified intra frequency cells and additionally search for and identify new intra frequency cells.

When DRX is in use the UE shall be able to identify a new detectable FDD intra frequency cell within $T_{identify_intra}$ as shown in Table 8.23.2.3-1.

**Table 8.23.2.3-1: Requirement to identify a newly detectable FDD intrafrequency cell**

| DRX cycle length (s) | $T_{identify_intra}$ (s) (DRX cycles) |
|---|---|
| ≤0.04 | 0.8 (NOTE1) |
| 0.04<DRX-cycle≤0.08 | NOTE2 (40) |
| 0.128 | 3.2 (25) |
| 0.128<DRX-cycle≤2.56 | NOTE2(20) |
| NOTE1: Number of DRX cycle depends upon the DRX cycle in use<br>NOTE2: Time depends upon the DRX cycle in use | |

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133 [4] clauses 9.1.2.1 and 9.1.2.2 and RSRQ related side conditions given in TS 36.133 [4] clause 9.1.5.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to TS 36.133 [4] Annex B.2 for a corresponding Band

In the RRC_CONNECTED state the measurement period for intra frequency measurements is $T_{measure_intra}$ as shown in Table 8.23.2.3-2 The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified-intra-frequency cells, and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_intra}$.

**Table 8.23.2.3-2: Requirement to measure FDD intrafrequency cells**

| DRX cycle length (s) | $T_{measure_intra}$ (s) (DRX cycles) |
|---|---|
| ≤0.04 | 0.2 (NOTE1) |
| 0.04<DRX-cycle≤2.56 | NOTE2 (5) |
| NOTE1: Number of DRX cycle depends upon the DRX cycle in use<br>NOTE2: Time depends upon the DRX cycle in use | |

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clauses 9.1.2.1 and 9.1.2.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in TS 36.133 [4] clause 9.1.5.1.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133 [4] clauses 9.1.2.1, 9.1.2.2, and 9.1.5.1, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_intra}$ defined in TS 36.133 [4] clause 8.1.2.2.1.2. When L3 filtering is used or IDC autonomous denial is configured or the UE is performing reception and/or transmission for ProSe Direct Discovery and/or ProSe Direct Communication, an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_intra}$ defined in TS 36.133 [4] clause 8.1.2.2.1.2 becomes undetectable for a period $\leq 5$ seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than $T_{measure_intra}$ provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured or the UE is performing reception and/or transmission for ProSe Direct Discovery and/or ProSe Direct Communication, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.8.2, 8.8.3, 8.1.2.2 and A.8.23.2.

## 8.23.2.4 Test description

### 8.23.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.23.2.5-1 as defined in TS 36.508 [7] clause [4.3.1] for different DC configurations (per band) as defined in TS 36.521-1 [10] clause 5.4.2.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.42 as appropriate.
2. The general test parameter settings are set up according to Table 8.23.2.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.23.2.4.3.

**5. There are two E-UTRA FDD carriers and two cells in the test. Cell 1 is PCell on the primary component carrier. Cell 2 is PSCell on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 shall be powered OFF.Table 8.23.2.4.1-1: General test parameters for E-UTRAN FDD-FDD DC intra-frequency event triggered reporting with DRX in asynchronous DC**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2 | Two radio channels are used for this test. |
| Active PCell | | | Cell1 | PCell on RF channel number 1. |
| Configured PSCell | | | Cell2 | PSCell on RF channel number 2. |
| A1 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A1. |
| | Threshold RSRP | dBm | -95 | Actual RSRP threshold for event A1. Needs to take absolute accuracy tolerance in TS 36.133[4] clause 9.1.11.1 and 9.1.11.2 into account plus margin. |
| | Time To Trigger | s | 0 | |
| A2 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A2. |
| | Threshold RSRP | dBm | -99 | Actual RSRP threshold for event A2. Needs to take absolute accuracy tolerance in TS 36.133[4] clause 9.1.11.1 and 9.1.11.2 into account plus margin. |
| | Time To Trigger | s | 0 | |
| CP length | | | Normal | |
| DRX | | | ON | DRX related parameters are defined in Table 8.23.2.5-2 |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on carrier frequency of Cell2. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| T1 | | s | 2 | |
| T2 | | s | 10 | |
| T3 | | s | 1 | |
| Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in TS 36.133[4] section A.3.6.1.<br>Note 2: Even a UE capable of both synchronous and asynchronous DC operations is required to pass this test case in accordance with the principle defined in TS 36.133[4] section A.3.11. | | | | |

### 8.23.2.4.2 Test procedure

Same test procedure as in clause 8.23.1.4.2 with the following exceptions:

- Instead of message content exceptions defined in clause 8.23.1.4.3 → use message content exception defined in clause 8.23.2.4.3.
- Instead of Table 8.23.1.5-1 → use Table 8.23.2.5-1

### 8.23.2.4.3 Message contents

Same message contents as in clause 8.23.1.4.3.

## 8.23.2.5 Test requirement

Tables 8.23.2.4.1-1 and 8.23.2.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-FDD event triggered reporting under fading propagation conditions in asynchronous cells test.

**Table 8.23.2.5-1: Cell specific test parameters for E-UTRAN FDD-FDD DC intra-frequency event triggered reporting with DRX in asynchronous DC**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | |
| PDSCH parameters: DL Reference Measurement Channel | | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | | | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | | |
| PCFICH/PDCCH/PHICH parameters: DL Reference Measurement Channel | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | |
| OCNG Patterns | | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | | | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_PB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_PB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 KHz | -104 | | | -104 | | |
| $\hat{E}_s / N_{oc}$ | dB | 16+TT | -2.5+TT | 20+TT | 16+TT | -2.5+TT | 20+TT |
| $\hat{E}_s / I_{ot}$ | dB | 16+TT | -2.5+TT | 20+TT | 16+TT | -2.5+TT | 20+TT |
| RSRP Note 3 | dBm/15 KHz | -88+TT | -106.5+TT | -84+TT | -88+TT | -106.5+TT | -84+TT |
| SCH_RP Note 3 | dBm/15 KHz | -88+TT | -106.5+TT | -84+TT | -88+TT | -106.5+TT | -84+TT |
| Io Note 3 | dBm/Ch BW | -60.11 +10log ($N_{RB,c}$ /50) | -74.28 +10log ($N_{RB,c}$ /50) | -56.18 +10log ($N_{RB,c}$ /50) | -60.11 +10log ($N_{RB,c}$ /50) | -74.28 +10log ($N_{RB,c}$ /50) | -56.18 +10log ($N_{RB,c}$ /50) |
| Propagation Condition | | ETU70 | | | ETU70 | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | 1x2 Low | | |
| Receive Time offset to cell1 Note 5 | μs | - | | | 500 | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: The resources for uplink transmission in Cell1 are assigned to the UE prior to the start of time period T3.

Note 5: Receive time difference between subframe boundaries of signals received from the two cells at the UE antenna connector including time alignment error between the two cells.

**Table 8.23.2.5-2: DRX-Configuration for E-UTRAN FDD-FDD DC intra-frequency event triggered reporting with DRX in asynchronous DC**

| Field | Cell1 | Cell2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf1280 | sf80 | |
| shortDRX | disable | disable | |

**Table 8.23.2.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD-FDD DC intra-frequency event triggered reporting with DRX in asynchronous DC**

| Field | Cell1 | Cell2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | 0 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213. |

The UE shall send one Event A2 triggered report for PCell on PCell with a measurement reporting delay less than 6.4s (5*MCG_DRX cycle) from the beginning of time period T2. The measurement reporting delay is defined as the time from the beginning of time period T2 to the moment when the UE starts to send preambles on the PRACH to obtain allocation to send the measurement report on PUSCH.

The UE shall send one Event A2 triggered report for PSCell on PCell with a measurement reporting delay less than 400ms (5*SCG_DRX cycle) from the beginning of time period T2. The measurement reporting delay is defined as the time from the beginning of time period T2 to the moment when the UE starts to send preambles on the PRACH to obtain allocation to send the measurement report on PUSCH.

The UE shall send one Event A1 triggered report for PCell on PCell with a measurement reporting delay less than 200ms from the beginning of time period T3. The measurement reporting delay is defined as the time from the beginning of time period T3 to the moment when the UE sends the measurement report on PUSCH.

The UE shall send one Event A1 triggered report for PSCell on PCell with a measurement reporting delay less than 400ms (5*SCG_DRX cycle) from the beginning of time period T3. The measurement reporting delay is defined as the time from the beginning of time period T3 to the moment when the UE sends the measurement report on PUSCH.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90%.

## 8.23.3 E-UTRAN TDD-TDD DC intra-frequency event triggered reporting with DRX in synchronous DC

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- *The Test system uncertainties applicable to this test are undefined.*
- *The Test tolerances applicable to this test are undefined.*

### 8.23.3.1 Test purpose

To verify the Dual Connectivity UE's ability to make correct reporting of an event in DRX under fading propagation conditions in synchronous cells within the E-UTRA TDD-TDD intra-frequency cell search requirements.

### 8.23.3.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward of synchronous Dual Connectivity. Applicability requires support for FGI bit 5.

### 8.23.3.3 Minimum conformance requirements

The UE shall be able to identify new intra-frequency cells and perform RSRP measurements of identified intra-frequency cells without an explicit intra-frequency neighbour cell list containing physical layer cell identities. During the RRC_CONNECTED state the UE shall continuously measure identified intra frequency cells and additionally search for and identify new intra frequency cells.

When DRX is in use the UE shall be able to identify a new detectable TDD intra frequency cell within $T_{identify_intra}$ as shown in Table 8.23.3.3-1.

**Table 8.23.3.3-1: Requirement to identify a newly detectable TDD intrafrequency cell**

| DRX cycle length (s) | $T_{identify_intra}$ (s) (DRX cycles) |
|---|---|
| ≤0.04 | 0.8 (NOTE1) |
| 0.04<DRX-cycle≤0.08 | NOTE2 (40) |
| 0.128 | 3.2 (25) |
| 0.128<DRX-cycle≤2.56 | NOTE2(20) |
| NOTE1: Number of DRX cycle depends upon the DRX cycle in use.<br>NOTE2: Time depends upon the DRX cycle in use. | |

A cell shall be considered detectable when

- RSRP related side conditions given in TS 36.133[4] Clause 9.1.2.1 and 9.1.2.2 and RSRQ related side conditions given in TS 36.133[4] Clause 9.1.5.1 are fulfilled for a corresponding Band,
- SCH_RP and SCH Ês/Iot according to TS 36.133[4] Annex B.2.1 for a corresponding Band

In the RRC_CONNECTED state the measurement period for intra frequency measurements is $T_{measure_intra}$ as shown in Table 8.23.3.3-2. The UE shall be capable of performing RSRP and RSRQ measurements for 8 identified intra-frequency cells and the UE physical layer shall be capable of reporting measurements to higher layers with the measurement period of $T_{measure_intra}$.

**Table 8.23.3.3-2: Requirement to measure TDD intra frequency cells**

| DRX cycle length (s) | $T_{measure_intra}$ (s) (DRX cycles) |
|---|---|
| ≤0.04 | 0.2 (NOTE1) |
| 0.04<DRX-cycle≤2.56 | NOTE2 (5) |
| NOTE1: Number of DRX cycle depends upon the DRX cycle in use.<br>NOTE2: Time depends upon the DRX cycle in use. | |

The RSRP measurement accuracy for all measured cells shall be as specified in the TS 36.133[4] sub-clauses 9.1.2.1 and 9.1.2.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in the TS 36.133[4] sub-clause 9.1.5.1.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133[4] sections 9.1.2.1, 9.1.2.2, and 9.1.5.1, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x TTIDCCH. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than T identify_intra defined in TS 36.133[4] Clause 8.1.2.2.2.2. When L3 filtering is used or IDC autonomous denial is configured or the UE is performing reception and/or transmission for ProSe Direct Discovery and/or ProSe Direct Communication, an additional delay can be expected.

If a cell which has been detectable at least for the time period Tidentify_intra defined in TS 36.133[4] clause 8.1.2.2.2.2 becomes undetectable for a period ≤ 5 seconds and then the cell becomes detectable again and triggers an event, the event triggered measurement reporting delay shall be less than Tmeasure_intra provided the timing to that cell has not changed more than ± 50 Ts and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured or the UE is performing reception and/or transmission for ProSe Direct Discovery and/or ProSe Direct Communication, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.8.2, 8.8.4, 8.1.2.2 and A.8.23.3.

## 8.23.3.4 Test description

### 8.23.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.23.3.5-1 as defined in TS 36.508 [7] clause [4.3.1] for different DC configurations (per band) as defined in TS 36.521-1 [10] clause 5.4.2.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.42 as appropriate .
2. The general test parameter settings are set up according to Table 8.23.3.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.23.3.4.3.
5. There are two E-UTRA TDD carriers and two cells in the test. Cell 1 is PCell on the primary component carrier, Cell 2 is PSCell on the secondary component carrier, Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test, Cell 2 shall be powered OFF.

**Table 8.23.3.4.1-1: General test parameters for E-UTRAN TDD-TDD DC intra-frequency event triggered reporting with DRX in synchronous DC**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2 | Two radio channels are used for this test. |
| Active PCell | | | Cell1 | PCell on RF channel number 1. |
| Configured PSCell | | | Cell2 | PSCell on RF channel number 2. |
| A1 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A1. |
| | Threshold RSRP | dBm | -95 | Actual RSRP threshold for event A1. Needs to take absolute accuracy tolerance in TS 36.133[4] clause 9.1.11.1 and 9.1.11.2 into account plus margin. |
| | Time To Trigger | s | 0 | |
| A2 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A2. |
| | Threshold RSRP | dBm | -99 | Actual RSRP threshold for event A2. Needs to take absolute accuracy tolerance in TS 36.133[4] clause 9.1.11.1 and 9.1.11.2 into account plus margin. |
| | Time To Trigger | s | 0 | |
| CP length | | | Normal | |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211. The same configuration in both cells |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211. The same configuration in both cells |
| DRX | | | ON | DRX related parameters are defined in Table 8.23.3.5-2 |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on carrier frequency of Cell2. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| T1 | | s | 5 | |
| T2 | | s | 10 | |
| T3 | | s | 1 | |
| NOTE: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in TS 36.133[4] section A.3.6.1. | | | | |

### 8.23.3.4.2 Test procedure

Same test procedure as in clause 8.23.1.4.2 with the following exceptions:

- Instead of message content exceptions defined in clause 8.23.1.4.2 → use message content exception defined in clause 8.23.3.4.3.
- Instead of Table 8.23.1.5-1 → use Table 8.23.3.5-1

### 8.23.3.4.3 Message contents

Same message contents as in clause 8.23.1.4.3.

## 8.23.3.5 Test requirement

Tables 8.23.3.4.1-1 and 8.23.3.5-1 define the primary level settings including test tolerances for E-UTRAN TDD-TDD event triggered reporting under fading propagation conditions in synchronous cells test.

**Table 8.23.3.5-1: Cell specific test parameters for E-UTRAN TDD-TDD DC intra-frequency event triggered reporting with DRX in synchronous DC**

| Parameter | Unit | Cell 1 | | | Cell 2 | | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T3 | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | | 2 | | |
| $BW_{channel}$ | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | |
| PDSCH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | | |
| PCFICH/PDCCH/PHICH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | |
| OCNG Patterns | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | | |
| PBCH_RA | dB | 0 | | | 0 | | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_PB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_PB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 KHz | -104 | | | -104 | | |
| $\hat{E}_s / N_{oc}$ | dB | 16+TT | -2.5+TT | 20+TT | 16+TT | -2.5+TT | 20+TT |
| $\hat{E}_s / I_{ot}$ | dB | 16+TT | -2.5+TT | 20+TT | 16+TT | -2.5+TT | 20+TT |
| RSRP Note 3 | dBm/15 KHz | -88+TT | -106.5+TT | -84+TT | -88+TT | -106.5+TT | -84+TT |
| SCH_RP Note 3 | dBm/15 KHz | -88+TT | -106.5+TT | -84+TT | -88+TT | -106.5+TT | -84+TT |
| Io Note 3 | dBm/Ch BW | -60.11 +10log ($N_{RB,c}$ /50) | -74.28 +10log ($N_{RB,c}$ /50) | -56.18 +10log ($N_{RB,c}$ /50) | -60.11 +10log ($N_{RB,c}$ /50) | -74.28 +10log ($N_{RB,c}$ /50) | -56.18 +10log ($N_{RB,c}$ /50) |
| Propagation Condition | | ETU70 | | | ETU70 | | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | | 1x2 Low | | |
| Receive Time offset to cell1 Note 5 | µs | - | | | 33 | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: The resources for uplink transmission in Cell1 are assigned to the UE prior to the start of time period T3.

Note 5: Receive time difference between subframe boundaries of signals received from the two cells at the UE antenna connector including time alignment error between the two cells.

**Table 8.23.3.5-2: DRX-Configuration for E-UTRAN TDD-TDD DC intra-frequency event triggered reporting with DRX in synchronous DC**

| Field | Cell1 | Cell2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf1280 | sf80 | |
| shortDRX | disable | disable | |

**Table 8.23.3.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN TDD-TDD DC intra-frequency event triggered reporting with DRX in synchronous DC**

| Field | Cell1 | Cell2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | 0 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213. |

The UE shall send one Event A2 triggered report for PCell on PCell with a measurement reporting delay less than 6.4s (5*MCG_DRX cycle) from the beginning of time period T2. The measurement reporting delay is defined as the time from the beginning of time period T2 to the moment when the UE starts to send preambles on the PRACH to obtain allocation to send the measurement report on PUSCH.

The UE shall send one Event A2 triggered report for PSCell on PCell with a measurement reporting delay less than 400ms (5*SCG_DRX cycle) from the beginning of time period T2. The measurement reporting delay is defined as the time from the beginning of time period T2 to the moment when the UE starts to send preambles on the PRACH to obtain allocation to send the measurement report on PUSCH.

The UE shall send one Event A1 triggered report for PCell on PCell with a measurement reporting delay less than 200ms from the beginning of time period T3. The measurement reporting delay is defined as the time from the beginning of time period T3 to the moment when the UE sends the measurement report on PUSCH.

The UE shall send one Event A1 triggered report for PSCell on PCell with a measurement reporting delay less than 400ms (5*SCG_DRX cycle) from the beginning of time period T3. The measurement reporting delay is defined as the time from the beginning of time period T3 to the moment when the UE sends the measurement report on PUSCH.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The rate of correct events observed during repeated tests shall be at least 90%.

## 8.23.4 E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in synchronous DC

### 8.23.4.1 Test purpose

To verify the Dual Connectivity UE's ability to make correct reporting of an event in DRX under fading propagation conditions in synchronous cells within the E-UTRA FDD-FDD inter-frequency cell search requirements.

### 8.23.4.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward of Dual Connectivity. Applicability requires support for FGI bit 5.

### 8.23.4.3 Minimum conformance requirements

The UE shall be able to identify new inter-frequency cells and perform RSRP and RSRQ measurements of identified inter-frequency cells if carrier frequency information is provided by the PCell, even if no explicit neighbour list with physical layer cell identities is provided.

When DRX is in use, either measurement gaps are scheduled or the UE supports capability of conducting such measurements without gaps, the UE shall be able to identify a new detectable E-UTRAN FDD inter frequency cell within $T_{identify_inter}$ as shown in Table 8.23.4.3-1

**Table 8.23.4.3-1: Requirement to identify a newly detectable FDD interfrequency cell**

| DRX cycle length (s) | $T_{identify_inter}$ (s) (DRX cycles), normal performance | | $T_{identify_inter}$ (s) (DRX cycles), reduced performance | |
|---|---|---|---|---|
| | Gap period = 40 ms | Gap period = 80 ms | Gap period = 40 ms | Gap period = 80 ms |
| ≤0.16 | Non DRX Requirements in TS 36.133[4] clause 8.1.2.3.1.1 are applicable | Non DRX Requirements in TS 36.133[4] clause 8.1.2.3.1.1 are applicable | Non DRX Requirements in TS 36.133[4] clause 8.1.2.3.1.1 are applicable | Non DRX Requirements in TS 36.133[4] clause 8.1.2.3.1.1 are applicable |
| 0.256 | $5.12*K_n*N_{freq,n}$ ($20*K_n*N_{freq,n}$) | $7.68*K_n*N_{freq,n}$ ($30*K_n*N_{freq,n}$) | $5.12*K_r*N_{freq,r}$ ($20*K_r*N_{freq,r}$) | $7.68*K_r*N_{freq,r}$ ($30*K_r*N_{freq,r}$) |
| 0.32 | $6.4*K_n*N_{freq,n}$ ($20*K_n*N_{freq,n}$) | $7.68*K_n*N_{freq,nl}$ ($24*K_n*N_{freq,n}$) | $6.4*K_r*N_{freq,r}$ ($20*K_r*N_{freq,r}$) | $7.68*K_r*N_{freq,r}$ ($24*K_r*N_{freq,r}$) |
| 0.32< DRX-cycle≤2.56 | Note ($20*K_n*N_{freq,n}$) | Note ($20*K_n*N_{freq,n}$) | Note ($20*K_r*N_{freq,r}$) | Note ($20*K_r*N_{freq,r}$) |
| Note: Time depends upon the DRX cycle in use | | | | |

A cell shall be considered detectable provided following conditions are fulfilled:

- $RSRP|_{dBm}$ RSRP Ês/Iot according to TS 36.133[4] Annex B.2.3 for a corresponding Band
- other RSRP related side conditions given in TS 36.133[4] Clause 9.1.3.1 and 9.1.3.2 and RSRQ related side conditions given in TS 36.133[4] Sections 9.1.6.1 and 9.1.6.2 are fulfilled,
- $SCH_RP|_{dBm}$ SCH Ês/Iot according to TS 36.133[4] Annex B.2.3 for a corresponding Band,

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per FDD inter-frequency and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period defined in Table 8.23.4.3-2 when DRX is in use, either measurement gaps are scheduled or the UE supports capability of conducting such measurements without gaps.

**Table 8.23.4.3-2: Requirement to measure FDD interfrequency cells**

| DRX cycle length (s) | $T_{measure_inter}$ (s) (DRX cycles) (normal performance) | $T_{measure_inter}$ (s) (DRX cycles) (reduced performance) |
|---|---|---|
| ≤0.08 | Non DRX Requirements in TS 36.133[4] clause 8.1.2.3.1.1 are applicable | Non DRX Requirements in TS 36.133[4] clause 8.1.2.3.1.1 are applicable |
| 0.08<DRX-cycle≤2.56 | Note ($5*K_n*N_{freq,n}$) | Note ($5*K_r*N_{freq,r}$) |
| Note: Time depends upon the DRX cycle in use | | |

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133[4] sub-clauses 9.1.3.1 and 9.1.3.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in TS 36.133[4] sub-clauses 9.1.6.1 and 9.1.6.2.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133[4] sections 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_inter}$ defined in TS 36.133[4] clause 8.1.2.3.1.2. When L3 filtering is used or IDC autonomous denial is configured or the UE is performing reception and/or transmission for ProSe Direct Discovery and/or ProSe Direct Communication, an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_inter}$ defined in TS 36.133[4] clause 8.1.2.3.1.2 and then triggers the measurement report as per TS 36.331 [5], the event triggered measurement reporting delay shall be less than $T_{measure_inter}$ defined in TS 36.133[4] clause 8.1.2.3.1.2 provided the timing to that cell has not changed more than ± 50 $T_s$ while measurement gap has not been available and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured or the UE is performing reception and/or transmission for ProSe Direct Discovery and/or ProSe Direct Communication, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.8.4, 8.1.2.3 and A.8.23.4.

## 8.23.4.4 Test description

### 8.23.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.23.4.5-1 as defined in TS 36.508 [7] clause [4.3.1] for different DC configurations (per band) as defined in TS 36.521-1 [10] clause 5.4.2.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in 3GPP TS 36.508 [7] Annex A figure group A.43 as appropriate.

2. The general test parameter settings are set up according to Table 8.23.4.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.23.4.4.3.

5. There are three E-UTRA FDD cells in the test. Cell 1 is PCell, Cell 2 is PSCell and Cell 3 is a inter-frequency neighbour cell. Cell 1 and Cell 2 are the cells used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 3 shall be powered OFF.

**Table A.8.23.4.4.1-1: General test parameters for E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in synchronous DC**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2, 3 | Three radio channels are used for this test. |
| Active PCell | | | Cell1 | PCell on RF channel number 1. |
| Configured PSCell | | | Cell2 | PSCell on RF channel number 2. |
| Neighbour cell | | | Cell3 | Neighbour cell on RF channel number 3. |
| A3 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A3. |
| | A3-offset | dB | -6 | |
| | Time To Trigger | s | 0 | |
| CP length | | | Normal | |
| DRX | | | ON | DRX related parameters are defined in Table 8.23.4.5-2 |
| Measurement gap pattern Id | | | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on carrier frequency of Cell2. |
| Cell-individual offset for cells on RF channel number 3 | | dB | 0 | Individual offset for cells on carrier frequency of Cell3. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| T1 | | s | 5 | |
| T2 | | s | 5 | |
| Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in TS 36.133[4] section A.3.6.1.<br>Note 2: A UE capable of both synchronous and asynchronous DC operations is not required to pass this test case in accordance with the principle defined in TS 36.133[4] section A.3.6.11. | | | | |

### 8.23.4.4.2 Test procedure

The test consists of two successive time periods, with time duration of T1, and T2 respectively. During time duration T1, the UE shall not have any timing information of cell 3.

1. Ensure the UE is in State 3A-RF-DC1 according to TS 36.508 [7] clause 7.2A.3A.
2. Configure MCG and SCG according to Annex C.0, C.1 for all downlink physical channels.
3. The SS shall configure the PCell (Cell 1) and PSCell (Cell 2) on the MCG and SCG as per TS 36.508 [7] clause 7.2A.3A with the message content exceptions defined in clause 8.23.4.4.3.
4. Set the parameters according to T1 in Table 8.23.4.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.
5. The SS shall transmit an RRCConnectionReconfiguration message with event A3 configured.
6. The UE shall transmit an RRCConnectionReconfigurationComplete message.
7. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.23.4.5-1.
8. UE shall transmit a MeasurementReport message triggered by Event A3 for Cell 3 on PCell (Cell 1). If the measurement reporting delay from the beginning of time period T2 is less than 3922ms the number of successful tests is increased by one. If the UE fails to report the event within the measurement reporting delay requirement then the number of failure tests is increased by one.

9.. After the SS receives the MeasurementReport message in step 9) or when T2 expires, the SS shall transmit a RRCConnectionRelease message to release the RRC connection which includes the release of the established radio bearers as well as all radio resources.

10. Set Cell 3 physical cell identity = ((current cell 3 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

11. After the RRC connection release, the SS:
    - transmits in Cell 1 and Cell 2 a Paging message (including PagingRecord with UE-Identity) for the UE and ensures the UE is in [State 3A] according to TS 36.508 [7] clause [4.5.3A] (if the paging fails, switches off and on the UE and ensures the UE is in State 3A-RF-DC1 according to TS 36.508 [7] clause 7.2A.3A,
    or
    - switches off and on the UE and ensures the UE is in State 3A-RF-DC1 according to TS 36.508 [7] clause 7.2A.3A.

12. Repeat steps 2-12 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.23.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.23.4.4.3-1: Common Exception messages for E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in synchronous DC test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-7<br>Table H.4.1-5 |

**Table 8.23.4.4.3-2: *RRCConnectionReconfiguration:* Additional E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in synchronous DC test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | MEAS |
| dedicatedInfoNASList | Set according to specific message content | | AM-DRB-ADD(1) |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-AM-DRB-ADD(1) | | AM-DRB-ADD(1) |
| } | | | |
| nonCriticalExtension SEQUENCE { | | | |
| scg-Configuration-r12 | SCG-Configuration-r12-DEFAULT | | PSCell_AddMod DC_Setup_SCG_DRB |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.23.4.4.3-3: *MeasConfig:* Additional E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in synchronous DC test requirement**

| Derivation path: TS 36.508 [7] clause 4.6.6 table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasConfig ::= SEQUENCE { | | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 3 entries | | |
| measObjectId[1] | IdMeasObject-f1 | | |
| measObject[1] | MeasObjectEUTRA-GENERIC(f1) | Cell 1 | |
| measObjectId[2] | IdMeasObject-f2 | | |
| measObject[2] | MeasObjectEUTRA-GENERIC(f2) | Cell 2 | |
| measObjectId[3] | IdMeasObject-f3 | | |
| measObject[3] | MeasObjectEUTRA-GENERIC(f3) | Cell 3 | |
| } | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId)) OF SEQUENCE { | 1 entry | | |
| reportConfigId[1] | IdReportConfig-A3 | | |
| reportConfig[1] | ReportConfigEUTRA-A3 | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) OF SEQUENCE { | 1 entry | | |
| measId[1] | 1 | | |
| measObjectId[1] | IdMeasObject-f3 | | |
| reportConfigId[1] | IdReportConfig-A3 | | |
| } | | | |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| } | | | |

**Table 8.23.4.4.3-4: *ReportConfigEUTRA-A3*: Additional E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in synchronous DC test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

## 8.23.4.5 Test requirement

Tables 8.23.4.4.1-1 and 8.23.4.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-FDD event triggered reporting under fading propagation conditions in synchronous cells test.

**Table 8.23.4.5-1: Cell specific test parameters for E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in synchronous DC**

| Parameter | Unit | Cell1 | | Cell2 | | Cell3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | | 3 | |
| $BW_{channel}$ | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | |
| PDSCH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | | - | |
| PCFICH/PDCCH/PHICH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | |
| OCNG Patterns | | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | |
| PBCH_RA | dB | 0 | | 0 | | 0 | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_PB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_PB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 KHz | -101 | | -101 | | N/A | -101 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | 4 | 4 | -infinity | 7 |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | 4 | 4 | -infinity | 7 |
| RSRP Note 3 | dBm/15 KHz | -97 | -97 | -97 | -97 | -infinity | -94 |
| SCH_RP Note 3 | dBm/15 KHz | -97 | -97 | -97 | -97 | -infinity | -94 |
| Io Note 3 | dBm/Ch BW | -67.76 +10log ($N_{RB,c}$ /50) | -67.76 +10log ($N_{RB,c}$ /50) | -67.76 +10log ($N_{RB,c}$ /50) | -67.76 +10log ($N_{RB,c}$ /50) | N/A | -65.43 +10log ($N_{RB,c}$ /50) |
| Propagation Condition | | ETU70 | | ETU70 | | ETU70 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| Receive Time offset to cell1 Note 4 | μs | - | | 33 | | - | |
| Time offset to cell1 | μs | - | | - | | 3 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: Receive time difference between subframe boundaries of signals received from the two cells at the UE antenna connector including time alignment error between the two cells.

**Table 8.23.4.5-2: DRX-Configuration for E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in synchronous DC**

| Field | Cell1 | Cell2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf80 | sf1280 | |
| shortDRX | disable | disable | |

**Table 8.23.4.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in synchronous DC**

| Field | Cell1 | Cell2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | 0 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213. |

The UE shall send one Event A3 triggered measurement report for Cell 3 on PCell, with a measurement reporting delay less than 3922 ms from the beginning of time period T2. The measurement reporting delay is defined as the time from the beginning of time period T2 to the moment when the UE sends the measurement report on PUSCH.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The overall delay measured when DRX cycle length is 80 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{Identify_Inter}$

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq,n} \cdot K_n \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new FDD inter-frequency cell is defined.

$T_{Inter1}$ = 60 ms. It is defined in table 8.1.2.1-1 of TS 36.133 [4] clause 8.1.2.1.

$N_{freq,n}$ = 1. It is defined in TS 36.133 [4] clause 8.1.2.1.1.

$K_n$ = 1. It is defined in TS 36.133 [4] clause 8.1.2.1.1.

TTI insertion uncertainty = 2 ms

DRX cycle length = 80 ms

The overall delay measured when DRX cycle length is 80 ms shall be less than a total of 3922 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.23.5 E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in asynchronous DC

### 8.23.5.1 Test purpose

To verify the Dual Connectivity UE's ability to make correct reporting of an event in DRX under fading propagation conditions in asynchronous cells within the E-UTRA FDD-FDD inter-frequency cell search requirements.

### 8.23.5.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward of Dual Connectivity. Applicability requires support for FGI bit 5.

### 8.23.5.3 Minimum conformance requirements

The UE shall be able to identify new inter-frequency cells and perform RSRP and RSRQ measurements of identified inter-frequency cells if carrier frequency information is provided by the PCell, even if no explicit neighbour list with physical layer cell identities is provided.

When DRX is in use, either measurement gaps are scheduled or the UE supports capability of conducting such measurements without gaps, the UE shall be able to identify a new detectable E-UTRAN FDD inter frequency cell within $T_{identify_inter}$ as shown in Table 8.23.5.3-1

**Table 8.23.5.3-1: Requirement to identify a newly detectable FDD interfrequency cell**

| DRX cycle length (s) | $T_{identify_inter}$ (s) (DRX cycles), normal performance | | $T_{identify_inter}$ (s) (DRX cycles), reduced performance | |
|---|---|---|---|---|
| | Gap period = 40 ms | Gap period = 80 ms | Gap period = 40 ms | Gap period = 80 ms |
| ≤0.16 | Non DRX Requirements in TS 36.133[4] clause 8.1.2.3.1.1 are applicable | Non DRX Requirements in TS 36.133[4] clause 8.1.2.3.1.1 are applicable | Non DRX Requirements in TS 36.133[4] clause 8.1.2.3.1.1 are applicable | Non DRX Requirements in TS 36.133[4] clause 8.1.2.3.1.1 are applicable |
| 0.256 | $5.12*K_n*N_{freq,n}$ ($20*K_n*N_{freq,n}$) | $7.68*K_n*N_{freq,n}$ ($30*K_n*N_{freq,n}$) | $5.12*K_r*N_{freq,r}$ ($20*K_r*N_{freq,r}$) | $7.68*K_r*N_{freq,r}$ ($30*K_r*N_{freq,r}$) |
| 0.32 | $6.4*K_n*N_{freq,n}$ ($20*K_n*N_{freq,n}$) | $7.68*K_n*N_{freq,nl}$ ($24*K_n*N_{freq,n}$) | $6.4*K_r*N_{freq,r}$ ($20*K_r*N_{freq,r}$) | $7.68*K_r*N_{freq,r}$ ($24*K_r*N_{freq,r}$) |
| 0.32< DRX-cycle≤2.56 | Note ($20*K_n*N_{freq,n}$) | Note ($20*K_n*N_{freq,n}$) | Note ($20*K_r*N_{freq,r}$) | Note ($20*K_r*N_{freq,r}$) |
| Note: Time depends upon the DRX cycle in use | | | | |

A cell shall be considered detectable provided following conditions are fulfilled:

- $RSRP|_{dBm}$ RSRP Ês/Iot according to TS 36.133[4] Annex B.2.3 for a corresponding Band,
- other RSRP related side conditions given in TS 36.133[4] Clause 9.1.3.1 and 9.1.3.2 and RSRQ related side conditions given in TS 36.133[4] Sections 9.1.6.1 and 9.1.6.2 are fulfilled,
- $SCH_RP|_{dBm}$ SCH Ês/Iot according to TS 36.133[4] Annex B.2.3 for a corresponding Band,

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per FDD inter-frequency and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period defined in Table 8.23.5.3-2 when DRX is in use, either measurement gaps are scheduled or the UE supports capability of conducting such measurements without gaps.

**Table 8.23.5.3-2: Requirement to measure FDD interfrequency cells**

| DRX cycle length (s) | $T_{measure_inter}$ (s) (DRX cycles) (normal performance) | $T_{measure_inter}$ (s) (DRX cycles) (reduced performance) |
|---|---|---|
| ≤0.08 | Non DRX Requirements in TS 36.133[4] clause 8.1.2.3.1.1 are applicable | Non DRX Requirements in TS 36.133[4] clause 8.1.2.3.1.1 are applicable |
| 0.08<DRX-cycle≤2.56 | Note ($5*K_n*N_{freq,n}$) | Note ($5*K_r*N_{freq,r}$) |
| Note: Time depends upon the DRX cycle in use | | |

The RSRP measurement accuracy for all measured cells shall be as specified in TS 36.133[4] sub-clauses 9.1.3.1 and 9.1.3.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in TS 36.133[4] sub-clauses 9.1.6.1 and 9.1.6.2.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133[4] sections 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{identify_inter}$ defined in TS 36.133[4] clause 8.1.2.3.1.2. When L3 filtering is used or IDC autonomous denial is configured or the UE is performing reception and/or transmission for ProSe Direct Discovery and/or ProSe Direct Communication, an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{identify_inter}$ defined in TS 36.133[4] clause 8.1.2.3.1.2 and then triggers the measurement report as per TS 36.331 [5], the event triggered measurement reporting delay shall be less than $T_{measure_inter}$ defined in TS 36.133[4] clause 8.1.2.3.1.2 provided the timing to that cell has not changed more than ± 50 Ts while measurement gap has not been available and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured or the UE is performing reception and/or transmission for ProSe Direct Discovery and/or ProSe Direct Communication, an additional delay can be expected.

The normative reference for this requirement is TS 36.133 [4] clause 8.8.4, 8.1.2.3 and A.8.23.5.

## 8.23.5.4 Test description

### 8.23.5.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.23.5.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different DC configurations (per band) as defined in TS 36.521-1 [10] clause 5.4.2.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure group A.43 as appropriate..
2. The general test parameter settings are set up according to Table 8.23.5.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.23.5.4.3.

5. There are three E-UTRA FDD cells in the test. Cell1 is PCell, Cell2 is PSCell and Cell3 is a inter-frequency neighbour cell. Cell1 and Cell2 are the cells used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 3 shall be powered OFF.

**Table 8.23.5.4.1-1: General test parameters for E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in asynchronous DC**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2, 3 | Three radio channels are used for this test. |
| Active PCell | | | Cell1 | PCell on RF channel number 1. |
| Configured PSCell | | | Cell2 | PSCell on RF channel number 2. |
| Neighbour cell | | | Cell3 | Neighbour cell on RF channel number 3. |
| A3 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A3. |
| | A3-offset | dB | -6 | |
| | Time To Trigger | s | 0 | |
| CP length | | | Normal | |
| DRX | | | ON | DRX related parameters are defined in Table 8.23.5.5-2 |
| Measurement gap pattern Id | | | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on carrier frequency of Cell2. |
| Cell-individual offset for cells on RF channel number 3 | | dB | 0 | Individual offset for cells on carrier frequency of Cell3. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| T1 | | s | 5 | |
| T2 | | s | 5 | |
| Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in TS 36.133[4] section A.3.6.1. | | | | |
| Note 2: Even a UE capable of both synchronous and asynchronous DC operations is required to pass this test case in accordance with the principle defined in TS 36.133[4] section A.3.11. | | | | |

### 8.23.5.4.2 Test procedure

Same Test Procedure as in clause 8.23.4.4.2 with the following exceptions:

- Instead of clause 8.23.4.4.3 → use clause 8.23.5.4.3.
- Instead of Table 8.23.4.5-1 → use Table 8.23.5.5-1.

### 8.23.5.4.3 Message contents

Same Message contents as in clause 8.23.4.4.3.

## 8.23.5.5 Test requirement

Tables 8.23.5.4.1-1 and 8.23.5.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-FDD event triggered reporting under fading propagation conditions in asynchronous cells test.

**Table 8.23.5.5-1: Cell specific test parameters for E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in asynchronous DC**

| Parameter | Unit | Cell1 | | Cell2 | | Cell3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | | 3 | |
| $BW_{channel}$ | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | |
| PDSCH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | | - | |
| PCFICH/PDCCH/PHICH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | |
| OCNG Patterns | | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | |
| PBCH_RA | dB | 0 | | 0 | | 0 | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_PB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_PB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 KHz | -101 | | -101 | | N/A | -101 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | 4 | 4 | -infinity | 7 |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | 4 | 4 | -infinity | 7 |
| RSRP Note 3 | dBm/15 KHz | -97 | -97 | -97 | -97 | -infinity | -94 |
| SCH_RP Note 3 | dBm/15 KHz | -97 | -97 | -97 | -97 | -infinity | -94 |
| Io Note 3 | dBm/Ch BW | -67.76 +10log ($N_{RB,c}$ /50) | -67.76 +10log ($N_{RB,c}$ /50) | -67.76 +10log ($N_{RB,c}$ /50) | -67.76 +10log ($N_{RB,c}$ /50) | N/A | -65.43 +10log ($N_{RB,c}$ /50) |
| Propagation Condition | | ETU70 | | ETU70 | | ETU70 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| Receive time offset to cell1 Note 4 | μs | - | | 500 | | - | |
| Time offset to cell1 | μs | - | | - | | 400 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: Receive time difference between subframe boundaries of signals received from the two cells at the UE antenna connector including time alignment error between the two cells.

**Table 8.23.5.5-2: DRX-Configuration for E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in asynchronous DC**

| Field | Cell1 | Cell2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf80 | sf1280 | |
| shortDRX | disable | disable | |

**Table 8.23.5.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in asynchronous DC**

| Field | Cell1 | Cell2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | 0 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213. |

The UE shall send one Event A3 triggered measurement report for Cell 3 on PCell, with a measurement reporting delay less than 3922 ms from the beginning of time period T2. The measurement reporting delay is defined as the time from the beginning of time period T2 to the moment when the UE sends the measurement report on PUSCH.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The overall delay measured when DRX cycle length is 80 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{Identify_Inter}$

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq,n} \cdot K_n \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new FDD inter-frequency cell is defined.

$T_{Inter1}$ = 60 ms. It is defined in table 8.1.2.1-1 of TS 36.133 [4] clause 8.1.2.1.

$N_{freq,n}$ = 1. It is defined in TS 36.133 [4] clause 8.1.2.1.1.

$K_n$ = 1. It is defined in TS 36.133 [4] clause 8.1.2.1.1.

TTI insertion uncertainty = 2 ms

DRX cycle length = 80 ms

The overall delay measured when DRX cycle length is 80 ms shall be less than a total of 3922 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

## 8.23.6 E-UTRAN TDD-TDD DC inter-frequency event triggered reporting with DRX in synchronous DC

### 8.23.6.1 Test purpose

To verify the Dual Connectivity UE's ability to make correct reporting of an event in DRX under fading propagation conditions in synchronous cells within the E-UTRA TDD-TDD inter-frequency cell search requirements.

### 8.23.6.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward of Dual Connectivity. Applicability requires support for FGI bit 5.

### 8.23.6.3 Minimum conformance requirements

The UE shall be able to identify new inter-frequency cells and perform RSRP and RSRQ measurements of identified inter-frequency cells if carrier frequency information is provided by the PCell, even if no explicit neighbour list with physical layer cell identities is provided.

When DRX is in use, either measurement gaps are scheduled or the UE supports capability of conducting such measurements without gaps, the UE shall be able to identify a new detectable E-UTRAN TDD inter frequency cell within $T_{identify_inter}$ as shown in Table 8.23.6.3-1.

**Table 8.23.6.3-1: Requirement to identify a newly detectable TDD interfrequency cell**

| DRX cycle length (s) | $T_{identify_inter}$ (s) (DRX cycles) (normal performance) | | $T_{identify_inter}$ (s) (DRX cycles) (reduced performance) | |
|---|---|---|---|---|
| | Gap period = 40 ms | Gap period = 80 ms | Gap period = 40 ms | Gap period = 80 ms |
| ≤0.16 | Non DRX Requirements in TS 36.133[4] clause 8.1.2.3.2.1 are applicable | Non DRX Requirements in TS 36.133[4] clause 8.1.2.3.2.1 are applicable | Non DRX Requirements in TS 36.133[4] clause 8.1.2.3.2.1 are applicable | Non DRX Requirements in TS 36.133[4] clause 8.1.2.3.2.1 are applicable |
| 0.256 | $5.12*K_n*N_{freq,n}$ ($20*K_n*N_{freq,n}$) | $7.68*K_n*N_{freq,n}$ ($30*K_n*N_{freq,n}$) | $5.12*K_r*N_{freq,r}$ ($20*K_r*N_{freq,r}$) | $7.68*K_r*N_{freq,r}$ ($30*K_r*N_{freq,r}$) |
| 0.32 | $6.4*K_n*N_{freq,n}$ ($20*K_n*N_{freq,n}$) | $7.68*K_n*N_{freq,n}$ ($24*K_n*N_{freq,n}$) | $6.4*K_r*N_{freq,r}$ ($20*K_r*N_{freq,r}$) | $7.68*K_r*N_{freq,r}$ ($24*K_r*N_{freq,r}$) |
| 0.32<DRX-cycle≤2.56 | Note ($20*K_n*N_{freq,n}$) | Note ($20*K_n*N_{freq,n}$) | Note ($20*K_r*N_{freq,r}$) | Note ($20*K_r*N_{freq,r}$) |
| Note: Time depends upon the DRX cycle in use | | | | |

A cell shall be considered detectable provided following conditions are fulfilled:

- $RSRP|_{dBm}$ and RSRP Ês/Iot according to TS 36.133[4] Annex B.2.3 for a corresponding Band
- RSRP related side conditions given in TS 36.133[4] Clause 9.1.3.1 and 9.1.3.2 and RSRQ related side conditions given in TS 36.133[4] Sections 9.1.6.1 and 9.1.6.2 are fulfilled,
- $SCH_RP|_{dBm}$ and SCH Ês/Iot according to TS 36.133[4] Annex B.2.3 for a corresponding Band,

The UE shall be capable of performing RSRP and RSRQ measurements of at least 4 inter-frequency cells per TDD inter-frequency for up to 3 TDD inter-frequencies and the UE physical layer shall be capable of reporting RSRP and RSRQ measurements to higher layers with the measurement period defined in Table 8.23.6.3-2 when DRX is in use, either measurement gaps are scheduled or the UE supports capability of conducting such measurements without gaps.

**Table 8.23.6.3-2: Requirement to measure TDD interfrequency cells**

| DRX cycle length (s) | $T_{measure_inter}$ (s) (DRX cycles) (normal requirement) | $T_{measure_inter}$ (s) (DRX cycles) (reduced requirement) |
|---|---|---|
| ≤0.08 | Non DRX Requirements in TS 36.133[4] clause 8.1.2.3.2.1 are applicable | Non DRX Requirements in TS 36.133[4] clause 8.1.2.3.2.1 are applicable |
| 0.128 | When configuration 2 non DRX Requirements in TS 36.133[4] clause 8.1.2.3.2.1 are applicable, Otherwise Note ($5*K_n*N_{freq,n}$) | When configuration 2 non DRX Requirements in TS 36.133[4] clause 8.1.2.3.2.1 are applicable, Otherwise Note ($5*K_r*N_{freq,r}$) |
| 0.128<DRX-cycle≤2.56 | Note ($5*K_n*N_{freq,n}$) | Note ($5*K_r*N_{freq,r}$) |
| Note: Time depends upon the DRX cycle in use | | |

The RSRP measurement accuracy for all measured cells shall be as specified in the TS 36.133[4] sub-clauses 9.1.3.1 and 9.1.3.2, and the RSRQ measurement accuracy for all measured cells shall be as specified in the TS 36.133[4] sub-clauses 9.1.6.1 and 9.1.6.2.

Reported RSRP and RSRQ measurements contained in event triggered measurement reports shall meet the requirements in TS 36.133[4] clause 9.1.3.1, 9.1.3.2, 9.1.6.1, and 9.1.6.2, respectively.

The UE shall not send any event triggered measurement reports, as long as no reporting criteria are fulfilled.

The measurement reporting delay is defined as the time between an event that will trigger a measurement report and the point when the UE starts to transmit the measurement report over the air interface. This requirement assumes that that the measurement report is not delayed by other RRC signalling on the DCCH. This measurement reporting delay excludes a delay uncertainty resulted when inserting the measurement report to the TTI of the uplink DCCH. The delay uncertainty is: 2 x $TTI_{DCCH}$. This measurement reporting delay excludes a delay which caused by no UL resources for UE to send the measurement report.

The event triggered measurement reporting delay, measured without L3 filtering shall be less than $T_{Identify_Inter}$ defined in TS 36.133[4] Clause 8.1.2.3.2.2. When L3 filtering is used or IDC autonomous denial is configured or the UE is performing reception and/or transmission for ProSe Direct Discovery and/or ProSe Direct Communication, an additional delay can be expected.

If a cell which has been detectable at least for the time period $T_{Identify_Inter}$ in TS 36.133[4] clause 8.1.2.3.2.2 and then triggers the measurement report as per TS 36.331[5], the event triggered measurement reporting delay shall be less than $T_{measure_inter}$ in TS 36.133[4] clause 8.1.2.3.2.2 provided the timing to that cell has not changed more than ± 50 Ts while measurement gap has not been available and the L3 filter has not been used. When L3 filtering is used or IDC autonomous denial is configured or the UE is performing reception and/or transmission for ProSe Direct Discovery and/or ProSe Direct Communication, an additional delay can be expected.

The normative reference for this requirement is TS 36.133[4] clause 8.8.4, 8.1.2.3 and A.8.23.6.

## 8.23.6.4 Test description

### 8.23.6.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 8.23.6.5-1 as defined in TS 36.508 [7] clause 4.3.1- for different DC configurations (per band) as defined in TS 36.521-1 [10] clause 5.4.2.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure group A.43 as appropriate.

2. The general test parameter settings are set up according to Table 8.23.6.4.1-1.

3. Propagation conditions are set according to Annex B clauses B.0.

4. Message contents are defined in clause 8.23.6.4.3.

5. There are three E-UTRA TDD cells in the test. Cell1 is PCell, Cell2 is PSCell and Cell3 is a inter-frequency neighbour cell. Cell1 and Cell2 are the cells used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 3 shall be powered OFF.

**Table 8.23.6.4.1-1: General test parameters for E-UTRAN TDD-TDD DC inter-frequency event triggered reporting with DRX in synchronous DC**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1, 2, 3 | Three radio channels are used for this test. |
| Active PCell | | | Cell1 | PCell on RF channel number 1. |
| Configured PSCell | | | Cell2 | PSCell on RF channel number 2. |
| Neighbour cell | | | Cell3 | Neighbour cell on RF channel number 3. |
| A3 | Hysteresis | dB | 0 | Hysteresis for evaluation of event A3. |
| | A3-offset | dB | -6 | |
| | Time To Trigger | s | 0 | |
| CP length | | | Normal | |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211. The same configuration in both cells |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211. The same configuration in both cells |
| DRX | | | ON | DRX related parameters are defined in Table 8.23.6.5-2 |
| Measurement gap pattern Id | | | 0 | |
| Cell-individual offset for cells on RF channel number 1 | | dB | 0 | Individual offset for cells on primary component carrier. |
| Cell-individual offset for cells on RF channel number 2 | | dB | 0 | Individual offset for cells on carrier frequency of Cell2. |
| Cell-individual offset for cells on RF channel number 3 | | dB | 0 | Individual offset for cells on carrier frequency of Cell3. |
| Filter coefficient | | | 0 | L3 filtering is not used |
| T1 | | s | 5 | |
| T2 | | s | 5 | |
| Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in TS 36.133[4] section A.3.6.1. | | | | |

### 8.23.6.4.2 Test procedure

Same Test Procedure as in clause 8.23.4.4.2 with the following exceptions:

- Instead of clause 8.23.4.4.3 → use clause 8.23.6.4.3.
- Instead of Table 8.23.4.5-1 → use Table 8.23.6.5-1.

### 8.23.6.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.23.6.4.3-1: Common Exception messages for E-UTRAN TDD-TDD DC inter-frequency event triggered reporting with DRX in synchronous DC test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-7<br>Table H.4.1-5 |

**Table 8.23.6.4.3-2: *RRCConnectionReconfiguration:* Additional E-UTRAN TDD-TDD DC inter-frequency event triggered reporting with DRX in synchronous DC test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | MEAS |
| dedicatedInfoNASList | Set according to specific message content | | AM-DRB-ADD(1) |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-AM-DRB-ADD(1) | | AM-DRB-ADD(1) |
| } | | | |
| nonCriticalExtension SEQUENCE { | | | |
| scg-Configuration-r12 | SCG-Configuration-r12-DEFAULT | | PSCell_AddMod DC_Setup_SCG_DRB |
| } | | | |
| } | | | |
| } | | | |
| | | | |

**Table 8.23.6.4.3-3: *MeasConfig:* Additional E-UTRAN TDD-TDD DC inter-frequency event triggered reporting with DRX in synchronous DC test requirement**

| Derivation path: TS 36.508 [7] clause 4.6.6 table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasConfig ::= SEQUENCE { | | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 3 entries | | |
| measObjectId[1] | IdMeasObject-f1 | | |
| measObject[1] | MeasObjectEUTRA-GENERIC(f1) | Cell 1 | |
| measObjectId[2] | IdMeasObject-f2 | | |
| measObject[2] | MeasObjectEUTRA-GENERIC(f2) | Cell 2 | |
| measObjectId[3] | IdMeasObject-f3 | | |
| measObject[3] | MeasObjectEUTRA-GENERIC(f3) | Cell 3 | |
| } | | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId)) OF SEQUENCE { | 1 entry | | |
| reportConfigId[1] | IdReportConfig-A3 | | |
| reportConfig[1] | ReportConfigEUTRA-A3 | | |
| } | | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) OF SEQUENCE { | 1 entry | | |
| measId[1] | 1 | | |
| measObjectId[1] | IdMeasObject-f3 | | |
| reportConfigId[1] | IdReportConfig-A3 | | |
| } | | | |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| } | | | |

**Table 8.23.6.4.3-4: *ReportConfigEUTRA-A3*: Additional E-UTRAN TDD-TDD DC inter-frequency event triggered reporting with DRX in synchronous DC test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-6 ReportConfigEUTRA-A3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A3 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA3 SEQUENCE { | | | |
| a3-Offset | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | |
| reportOnLeave | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | 0 (0 ms) | | |
| } | | | |
| } | | | |
| } | | | |

## 8.23.6.5 Test requirement

Tables 8.23.6.4.1-1 and 8.23.6.5-1 define the primary level settings including test tolerances for E-UTRAN TDD-TDD event triggered reporting under fading propagation conditions in synchronous cells test.

**Table 8.23.6.5-1: Cell specific test parameters for E-UTRAN TDD-TDD DC inter-frequency event triggered reporting with DRX in synchronous DC**

| Parameter | Unit | Cell1 | | Cell2 | | Cell3 | |
|---|---|---|---|---|---|---|---|
| | | T1 | T2 | T1 | T2 | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | | 2 | | 3 | |
| $BW_{channel}$ | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | |
| PDSCH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | | - | |
| PCFICH/PDCCH/PHICH parameters:<br>DL Reference Measurement Channel | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | |
| OCNG Patterns | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | |
| PBCH_RA | dB | 0 | | 0 | | 0 | |
| PBCH_RB | dB | | | | | | |
| PSS_RA | dB | | | | | | |
| SSS_RA | dB | | | | | | |
| PCFICH_RB | dB | | | | | | |
| PHICH_RA | dB | | | | | | |
| PHICH_PB | dB | | | | | | |
| PDCCH_RA | dB | | | | | | |
| PDCCH_PB | dB | | | | | | |
| PDSCH_RA | dB | | | | | | |
| PDSCH_RB | dB | | | | | | |
| OCNG_RA Note 1 | dB | | | | | | |
| OCNG_RB Note 1 | dB | | | | | | |
| $N_{oc}$ Note 2 | dBm/15 KHz | -101 | | -101 | | N/A | -101 |
| $\hat{E}_s / N_{oc}$ | dB | 4 | 4 | 4 | 4 | -infinity | 7 |
| $\hat{E}_s / I_{ot}$ | dB | 4 | 4 | 4 | 4 | -infinity | 7 |
| RSRP Note 3 | dBm/15 KHz | -97 | -97 | -97 | -97 | -infinity | -94 |
| SCH_RP Note 3 | dBm/15 KHz | -97 | -97 | -97 | -97 | -infinity | -94 |
| Io Note 3 | dBm/Ch BW | -67.76 +10log ($N_{RB,c}$ /50) | -67.76 +10log ($N_{RB,c}$ /50) | -67.76 +10log ($N_{RB,c}$ /50) | -67.76 +10log ($N_{RB,c}$ /50) | N/A | -65.43 +10log ($N_{RB,c}$ /50) |
| Propagation Condition | | ETU70 | | ETU70 | | ETU70 | |
| Correlation Matrix and Antenna Configuration | | 1x2 Low | | 1x2 Low | | 1x2 Low | |
| Receive Time offset to cell1 Note 4 | μs | - | | 33 | | - | |
| Time offset to cell1 | μs | - | | - | | 3 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: Receive time difference between subframe boundaries of signals received from the two cells at the UE antenna connector including time alignment error between the two cells.

**Table 8.23.6.5-2: DRX-Configuration for E-UTRAN TDD-TDD DC inter-frequency event triggered reporting with DRX in synchronous DC**

| Field | Cell1 | Cell2 | Comment |
|---|---|---|---|
| | Value | Value | |
| onDurationTimer | psf1 | psf1 | As specified in clause 6.3.2 in TS 36.331 |
| drx-InactivityTimer | psf1 | psf1 | |
| drx-RetransmissionTimer | psf1 | psf1 | |
| longDRX-CycleStartOffset | sf80 | sf1280 | |
| shortDRX | disable | disable | |

**Table 8.23.6.5-3: *TimeAlignmentTimer* -Configuration for E-UTRAN TDD-TDD DC inter-frequency event triggered reporting with DRX in synchronous DC**

| Field | Cell1 | Cell2 | Comment |
|---|---|---|---|
| | Value | Value | |
| TimeAlignmentTimer | sf500 | sf500 | As specified in clause 6.3.2 in TS 36.331 |
| sr-ConfigIndex | 0 | 0 | For further information see clause 6.3.2 in TS 36.331 and section10.1 in TS 36.213. |

The UE shall send one Event A3 triggered measurement report for Cell 3 on PCell, with a measurement reporting delay less than 3922 ms from the beginning of time period T2. The measurement reporting delay is defined as the time from the beginning of time period T2 to the moment when the UE sends the measurement report on PUSCH.

The UE shall not send event triggered measurement reports, as long as the reporting criteria are not fulfilled.

The overall delay measured when DRX cycle length is 80 ms test requirement is expressed as:

Overall delay measured = measurement reporting delay + TTI insertion uncertainty + DRX cycle length

Measurement reporting delay = $T_{Identify_Inter}$

$$T_{Identify_Inter} = T_{Basic_Identify_Inter} \cdot \frac{480}{T_{Inter1}} \cdot N_{freq,n} \cdot K_n \quad ms$$

Where:

$T_{Basic_Identify_Inter}$ = 480 ms. It is the time period used in the inter frequency equation where the maximum allowed time for the UE to identify a new TDD inter-frequency cell is defined.

$T_{Inter1}$ = 60 ms. It is defined in table 8.1.2.1-1 of TS 36.133 [4] clause 8.1.2.1.

$N_{freq,n}$ = 1. It is defined in TS 36.133 [4] clause 8.1.2.1.1.

$K_n$ = 1. It is defined in TS 36.133 [4] clause 8.1.2.1.1.

TTI insertion uncertainty = 2 ms

DRX cycle length = 80 ms

The overall delay measured when DRX cycle length is 80 ms shall be less than a total of 3922 ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 8.24 Proximity-based Services

## 8.24.1 FFS

## 8.24.2 FFS

## 8.24.3 E-UTRAN FDD - Initiation/Cease of SLSS Transmission with ProSe Direct Discovery

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

- The Test system uncertainties applicable to this test are undefined

- The Test tolerances applicable to this test are undefined

- Message contents are TBD.

- Connection diagrams are TBD

- Annex E updates are pending.

### 8.24.3.1 Test purpose

To verify the ProSe UE's ability to initiate and cease SLSS transmissions and if the UE meets the maximum evaluation time.

### 8.24.3.2 Test applicability

This test case applies to all types of E-UTRA UE release 12 and forward which support ProSe Direct Communication.

### 8.24.3.3 Minimum conformance requirements

The SLSS transmission initiation delay is defined as the time from the beginning of time period T2 up to the moment when the UE initiates the SLSS transmission.

The SLSS transmission initiation delay shall be less than 1.96 s.

The SLSS transmission cease delay is defined as the time from the beginning of time period T3 up to the moment when the UE ceases the SLSS transmission.

The SLSS transmission cease delay shall be less than 1.96 s.

The rate of correct initiation/cease delay of SLSS transmissions observed during repeated tests shall be at least 90%.

NOTE: The initiation/cease delay of SLSS transmissions can be expressed as: $T_{evaluate,SLSS}$ + SLSS period,

Where:

$T_{evaluate,SLSS}$ is the evaluation time for initiate/cease of SLSS, and is 1.92 sec (see Table 8.24.3.4.1-1 in clause 8.24.3.4.1) for the parameters in this test;

SLSS period is set to 40ms.

### 8.24.3.4 Test description

#### 8.24.3.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz or 10 Mhz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure FFS.
2. The general test parameter settings are set up according to Table 8.24.3.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 8.24.3.4.3.
5. There is one E-UTRA FDD carrier specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 8.24.3.4.1-1: Test parameters for initiation/cease of SLSS transmissions test for E-UTRAN FDD**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | Only one FDD carrier frequency is used. |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 or 10 | According to principle defined in clause A.3.12.3 |
| Active cell | | Cell 1 | E-UTRA FDD Cell1 on RF channel number 1 |
| CP length of Cell 1 | | Normal | |
| Layer 3 filtering | | Disabled | L3 filtering is not used |
| drx-Configuration | | DRX_P1 | As specified in Table A.3.12.2-1 |
| T1 | s | 3 | |
| T2 | s | 5.24 | |
| T3 | s | 5.24 | |

**Table 8.24.3.4.1-2: ProSe Direct Communication configuration for initiation/cease of SLSS transmissions test for E-UTRAN FDD**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | UL carrier frequency |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 or 10 | According to principle defined in clause A.3.12.3 |
| ProSe Direct Communication configuration | | As specified in Table A.3.12.5-1 (Configuration #1) | IE values unless specified otherwise in this test. |
| networkControlledSyncTx | | Not configured | |
| syncTxThreshIC | dBm/15 kHz | -95 | In SIB18 |

**Table 8.24.3.4.1-3: General Test Parameters for initiation/cease of SLSS transmissions test for E-UTRAN FDD**

| Parameter | Unit | Cell 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | |
| $BW_{channel}$ Note 4 | MHz | 5 or 10 | | |
| OCNG Patterns defined in A.3.2.1.2 Note 4 | | 5MHz: OP.16 FDD<br>10 MHz: OP.2 FDD | | |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA Note 1 | | | | |
| OCNG_RB Note 1 | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -95 | | |
| $\hat{E}_s / N_{oc}$ | dB | 4.5 | -4.5 | 4.5 |
| RSRP Note3 | dBm/15 kHz | -90.5 | -99.5 | -90.5 |
| SCH_RP Note 3 | dBm/15 kHz | -90.5 | -99.5 | -90.5 |
| Propagation Condition | | AWGN | | |

Note 1: OCNG shall be used such that cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: This test is according to the principle defined in section A.3.12.3.

### 8.24.3.4.2 Test procedure

There is one active cell (PCell) in this test. The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. During T1, the RSRP of the PCell is above *syncTxThreshIC* and the UE is not expected to be transmitting SLSS. During T2, the RSRP of the PCell is lowered below *syncTxThreshIC* and the UE is expected to initiate SLSS transmissions. During T3, the RSRP of the PCell is increased back to be above *syncTxThreshIC* and the UE is expected to cease SLSS transmissions.

1. Ensure the UE is in State 4 according to TS 36.508 [7] clause 4.5.4 and UE test loop Mode D is activated.
2. Set the parameters according to T1 in Table 8.24.3.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.
3. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 8.24.3.5-1.
4. UE is expected to initiate a SLSS transmissions inside 1.96 s from the start of T2.
5. After the SS is able to measure the SLSS in step 6) or when T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 8.24.3.5-1.
6. UE is expected to cease SLSS transmission inside 1.96s from the start of T3.
7. After the RRC connection release, the SS switches off and on the UE and ensures the UE is in State 4 with test loop mode 4 activated according to TS 36.508 [7] clause 4.5.4.
8. Repeat step 2-7 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 8.24.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with FFS:

**Table 8.24.3.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | FFS |

## 8.24.3.5 Test requirement

Tables 8.24.3.4.1-1 and 8.24.3.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-Initiation/Cease of SLSS Transmission with ProSe Direct Discovery.

**Table 8.24.3.5-1: Cell Specific Test requirement Parameters for initiation/cease of SLSS transmissions test for E-UTRAN FDD**

| Parameter | Unit | Cell 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | |
| $BW_{channel}$ Note 4 | MHz | 5 or 10 | | |
| OCNG Patterns defined in A.3.2.1.2 Note 4 | | 5MHz: OP.16 FDD<br>10 MHz: OP.2 FDD | | |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA Note 1 | | | | |
| OCNG_RB Note 1 | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -95 | | |
| $\hat{E}_s / N_{oc}$ | dB | 4.5 | -4.5 | 4.5 |
| RSRP Note3 | dBm/15 kHz | -90.5 | -99.5 | -90.5 |
| SCH_RP Note 3 | dBm/15 kHz | -90.5 | -99.5 | -90.5 |
| Propagation Condition | | AWGN | | |

Note 1: OCNG shall be used such that cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: This test is according to the principle defined in section A.3.12.3.

The SLSS transmission initiation delay is defined as the time from the beginning of time period T2 up to the moment when the UE initiates the SLSS transmission.

The SLSS transmission initiation delay shall be less than 1.96 s.

The SLSS transmission cease delay is defined as the time from the beginning of time period T3 up to the moment when the UE ceases the SLSS transmission.

The SLSS transmission cease delay shall be less than 1.96 s.

The rate of correct initiation/cease delay of SLSS transmissions observed during repeated tests shall be at least 90%.

NOTE: The initiation/cease delay of SLSS transmissions can be expressed as: $T_{evaluate,SLSS}$ + SLSS period,

Where:

$T_{evaluate,SLSS}$ is the evaluation time for initiate/cease of SLSS, and is 1.92 sec (see Table 8.24.3.4.1-1 in clause 8.24.3.4.1) for the parameters in this test;

SLSS period is set to 40ms.

For the test to pass, the total number of successful tests shall be more than 90% of the cases with a confidence level of 95%.

# 9 Measurement Performance Requirements

When the UE is in RRC_CONNECTED state on a cell, physical layer measurements as defined in TS 36.214 [12] clause 5 are initiated and reported to higher layers. To initiate a specific measurement, the System Simulator sends a 'RRC Connection Reconfiguration message' to the UE including a measurement ID and type, a command (setup, modify, release), the measurement objects, the measurement quantity, the reporting quantities and the reporting criteria (periodical/event-triggered), the physical layer measurement process takes place. In this process when the reporting criteria are fulfilled the UE sends a 'Measurement Report message' to the System Simulator including the measurement ID and the results. The reporting criteria that trigger the UE to send a 'Measurement Report message' to the System Simulator is periodical as defined in TS 36.331 [5] clause 5.5.4. The physical layer measurements succeed only if the performance results in terms of accuracy are within the specified limits.

Since the UE reference sensitivity requirements are different depending on supported band, this is noted in each case with definition of the range Io for each frequency band.

The accuracy requirements in this clause are applicable for AWGN radio propagation conditions and assume independent interference (noise) at each receiver antenna port.

The reported measurement results after layer 1 filtering shall be an estimate of the average value of the measured quantity over the measurement period.

The accuracy requirements are valid for the reported measurement results after layer 1 filtering.

Unless explicitly stated:

- In state RRC_CONNECTED
- Reported measurements shall be within defined range in 90 % of the cases.
- Measurement channel is as defined in Annex A. This measurement channel is used both in active cell and cells to be measured.
- The reference channels assume transmission of PDSCH with a maximum number of 5 HARQ transmissions unless otherwise specified.
- SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Annex A1. The SS sends downlink MAC padding bits on the DL RMC.
- Uplink is configured according to Annex A.3.
- Propagation condition is AWGN as defined in Annex B.
- Physical channels used as defined in Annex C.
- Cell 1 is the active cell.
- Single task reporting.
- Power control is active.

NOTE: For Band 26, the tests shall be performed with the assigned E-UTRA channel bandwidth within 865-894 MHz.

# 9.1 RSRP

## 9.1.1 FDD Intra frequency RSRP Accuracy

### 9.1.1.1 FDD Intra Frequency Absolute RSRP Accuracy

#### 9.1.1.1.1 Test purpose

To verify that the FDD intra-frequency absolute RSRP measurement accuracy is within the specified limit for all bands.

#### 9.1.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE from release 8 to release 11. Applicability requires support for FGI bit 16.

#### 9.1.1.1.3 Minimum conformance requirements

The absolute accuracy of RSRP is defined as the RSRP measured from a cell on the same frequency as that of the serving cell.

The accuracy requirements in table 9.1.1.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.1.1.1.3-1: RSRP FDD Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 / dBm/$BW_{Channel}$ | dBm/$BW_{Channel}$ |
| ±6 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 / N/A | -70 |
| | | | FDD_B | -120.5 / N/A | -70 |
| | | | FDD_C, TDD_C | -120 / N/A | -70 |
| | | | FDD_D | -119.5 / N/A | -70 |
| | | | FDD_E, TDD_E | -119 / N/A | -70 |
| | | | FDD_F | -118.5 / N/A | -70 |
| | | | FDD_G | -118 / N/A | -70 |
| | | | FDD_H | -117.5 / N/A | -70 |
| | | | FDD_N | -114.5 / N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 / -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.1.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.1.1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.1, clause 9.1.4 and A.9.1.1.

### 9.1.1.1.4 Test description

#### 9.1.1.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.1.1.1.4.3.
4. All cells are in the same carrier frequency. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.1.1.1.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Table 9.1.1.1.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. UE shall transmit periodically MeasurementReport messages.
6. After 10s wait from Step 3, SS shall check RSRP reported value in periodical MeasurementReport messages. The RSRP value of Cell 2 reported by the UE is compared to the actual RSRP value according to Table 9.1.1.1.5-3. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.
7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.
8. Repeat step 1-7 for each sub-test in Table 9.1.1.1.5-2 as appropriate.

#### 9.1.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.1.1.4.3-1: Common Exception messages for RSRP FDD Intra frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3 |

**Table 9.1.1.1.4.3-2: *MeasResults*: Additional RSRP FDD Intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.1.1.4.3-3: *MeasResultListEUTRA*: Additional RSRP FDD Intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

### 9.1.1.1.5 Test requirement

Table 9.1.1.1.5-2 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD intra-frequency relative accuracy test shall meet the reported values test requirements in table 9.1.1.1.5-3.

**Table 9.1.1.1.5-1: Void**

**Table 9.1.1.1.5-2: RSRP FDD Intra frequency absolute accuracy test parameters**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - | R.0 FDD | - | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | |
| PSS_RA | | | | | | | | |
| SSS_RA | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA Note1 | | | | | | | | |
| OCNG_RB Note1 | | | | | | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -107.0 | | -88.0 | | -116.0 | |
| | 115.0 | | | | | | 115.5 | |
| | Bands FDD_C | | | | | | -115.0 | |
| | Bands FDD_D | | | | | | -114.5 | |
| | Bands FDD_E Note 5, FDD_F | | | | | | -114.0 | |
| | Bands FDD_G | | | | | | -113.0 | |
| | Bands FDD_H | | | | | | -112.5 | |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 | 1.88 | -4.97 | 0.09 | -4.96 |
| RSRP Note3 | Bands FDD_A | dBm/15 kHz | -101.0 | -105.0 | -82.0 | -86.0 | -113.0 | -116.2 |
| | Bands FDD_B | | | | | | -112.5 | -115.7 |
| | Bands FDD_C | | | | | | -112.0 | -115.2 |
| | Bands FDD_D | | | | | | -111.5 | -114.7 |
| | Bands FDD_E, FDD_F Note 5 | | | | | | -111.0 | -114.2 |
| | Bands FDD_G | | | | | | -110.0 | -113.2 |
| | Bands FDD_H | | | | | | -109.5 | -112.7 |
| Io Note3 | Bands FDD_A | dBm/9 MHz | -71.05 | | -52.05 | | -82.25 | |
| | Bands FDD_B | | | | | | -81.75 | |
| | Bands FDD_C | | | | | | -81.25 | |
| | Bands FDD_D | | | | | | -80.75 | |
| | Bands FDD_E, FDD_F Note 5 | | | | | | -80.25 | |
| | Bands FDD_G | | | | | | -79.25 | |
| | Bands FDD_H | | | | | | -78.75 | |
| $\hat{E}_s / N_{oc}$ | | dB | 6.0 | 2.0 | 6.0 | 2.0 | 3.0 | -0.2 |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 6: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.1.1.5-3: RSRP FDD Intra frequency absolute accuracy requirements for the reported values**

| Normal Conditions | Test 1 All bands | Test 2 All bands | Test 3 | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_28 | RSRP_45 | Bands FDD_A | RSRP_17 |
| | | | Bands FDD_B | RSRP_17 |
| | | | Bands FDD_C | RSRP_18 |
| | | | Bands FDD_D | RSRP_18 |
| | | | Bands FDD_E, FDD_F | RSRP_19 |
| | | | Bands FDD_G | RSRP_20 |
| | | | Bands FDD_H | RSRP_20 |
| Highest reported value (Cell 2) | RSRP_43 | RSRP_64 | Bands FDD_A | RSRP_32 |
| | | | Bands FDD_B | RSRP_32 |
| | | | Bands FDD_C | RSRP_33 |
| | | | Bands FDD_D | RSRP_33 |
| | | | Bands FDD_E, FDD_F | RSRP_34 |
| | | | Bands FDD_G | RSRP_35 |
| | | | Bands FDD_D | RSRP_35 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_25 | RSRP_42 | Bands FDD_A | RSRP_14 |
| | | | Bands FDD_B | RSRP_14 |
| | | | Bands FDD_C | RSRP_15 |
| | | | Bands FDD_D | RSRP_15 |
| | | | Bands FDD_E, FDD_F | RSRP_16 |
| | | | Bands FDD_G | RSRP_17 |
| | | | Bands FDD_H | RSRP_17 |
| Highest reported value (Cell 2) | RSRP_46 | RSRP_67 | Bands FDD_A | RSRP_35 |
| | | | Bands FDD_B | RSRP_35 |
| | | | Bands FDD_C | RSRP_36 |
| | | | Bands FDD_D | RSRP_36 |
| | | | Bands FDD_E, FDD_F | RSRP_37 |
| | | | Bands FDD_G | RSRP_38 |
| | | | Bands FDD_H | RSRP_38 |
| Note 1: E-UTRA operating band groups are as defined in Section 3.5. | | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.1.1_1 FDD Intra Frequency Absolute RSRP Accuracy (Rel-12 and forward)

### 9.1.1.1_1.1 Test purpose

Same test purpose as in clause 9.1.1.1.1.

### 9.1.1.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward. Applicability requires support for FGI bit 16.

### 9.1.1.1_1.3 Minimum conformance requirements

The absolute accuracy of RSRP is defined as the RSRP measured from a cell on the same frequency as that of the serving cell.

The accuracy requirements in table 9.1.1.1_1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.1.1.1_1.3-1: RSRP FDD Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 / dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 / N/A | -70 |
| | | | FDD_B | -120.5 / N/A | -70 |
| | | | FDD_C, TDD_C | -120 / N/A | -70 |
| | | | FDD_D | -119.5 / N/A | -70 |
| | | | FDD_E, TDD_E | -119 / N/A | -70 |
| | | | FDD_F | -118.5 / N/A | -70 |
| | | | FDD_G | -118 / N/A | -70 |
| | | | FDD_H | -117.5 / N/A | -70 |
| | | | FDD_N | -114.5 / N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.1.1_1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.1.1_1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.1, clause 9.1.4 and A.9.1.1.

### 9.1.1.1_1.4 Test description

#### 9.1.1.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.1.1.4.1 with the following exceptions:

- Instead of 9.1.1.1.4.3 → use 9.1.1.1_1.4.3.

#### 9.1.1.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.1.1.4.2 with the following exceptions:

- Instead of Table 9.1.1.1.5-2 → use Table 9.1.1.1_1.5-1.
- Instead of Table 9.1.1.1.5-3 → use Table 9.1.1.1_1.5-2.

#### 9.1.1.1_1.4.3 Message contents

Same message contents as in clause 9.1.1.1.4.3.

### 9.1.1.1_1.5 Test requirement

Table 9.1.1.1_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD intra-frequency relative accuracy test shall meet the reported values test requirements in table 9.1.1.1_1.5-2.

**Table 9.1.1.1_1.5-1: RSRP FDD Intra frequency absolute accuracy test parameters**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - | R.0 FDD | - | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | |
| PSS_RA | | | | | | | | |
| SSS_RA | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA[Note1] | | | | | | | | |
| OCNG_RB[Note1] | | | | | | | | |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -107.0 | | -88.0 | | -116.0 | |
| | Bands FDD_B | | | | | | 115.5 | |
| | Bands FDD_C | | | | | | -115.0 | |
| | Bands FDD_D | | | | | | -114.5 | |
| | Bands FDD_E FDD_F [Note 5] | | | | | | -114.0 | |
| | Bands FDD_G | | | | | | -113.0 | |
| | Bands FDD_H | | | | | | -112.5 | |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 | 1.88 | -4.97 | 0.09 | -4.96 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -101.0 | -105.0 | -82.0 | -86.0 | -113.0 | -116.2 |
| | Bands FDD_B | | | | | | -112.5 | -115.7 |
| | Bands FDD_C | | | | | | -112.0 | -115.2 |
| | Bands FDD_D | | | | | | -111.5 | -114.7 |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -111.0 | -114.2 |
| | Bands FDD_G | | | | | | -110.0 | -113.2 |
| | Bands FDD_H | | | | | | -109.5 | -112.7 |
| Io[Note3] | Bands FDD_A | dBm/9 MHz | -71.05 | | -52.05 | | -82.25 | |
| | Bands FDD_B | | | | | | -81.75 | |
| | Bands FDD_C | | | | | | -81.25 | |
| | Bands FDD_D | | | | | | -80.75 | |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -80.25 | |
| | Bands FDD_G | | | | | | -79.25 | |
| | Bands FDD_H | | | | | | -78.75 | |
| $\hat{E}_s / N_{oc}$ | | dB | 6.0 | 2.0 | 6.0 | 2.0 | 3.0 | -0.2 |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |

NOTE 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
NOTE 5: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
NOTE 6: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.1.1_1.5-2: RSRP FDD Intra frequency absolute accuracy requirements for the reported values**

| Normal Conditions | Test 1 All bands | Test 2 All bands | Test 3 | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_29 | RSRP_45 | Bands FDD_A | RSRP_18 |
| | | | Bands FDD_B | RSRP_19 |
| | | | Bands FDD_C | RSRP_19 |
| | | | Bands FDD_D | RSRP_20 |
| | | | Bands FDD_E, FDD_F | RSRP_20 |
| | | | Bands FDD_G | RSRP_21 |
| | | | Bands FDD_H | RSRP_22 |
| Highest reported value (Cell 2) | RSRP_42 | RSRP_64 | Bands FDD_A | RSRP_30 |
| | | | Bands FDD_B | RSRP_31 |
| | | | Bands FDD_C | RSRP_31 |
| | | | Bands FDD_D | RSRP_32 |
| | | | Bands FDD_E, FDD_F | RSRP_32 |
| | | | Bands FDD_G | RSRP_33 |
| | | | Bands FDD_D | RSRP_34 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_25 | RSRP_42 | Bands FDD_A | RSRP_14 |
| | | | Bands FDD_B | RSRP_14 |
| | | | Bands FDD_C | RSRP_15 |
| | | | Bands FDD_D | RSRP_15 |
| | | | Bands FDD_E, FDD_F | RSRP_16 |
| | | | Bands FDD_G | RSRP_17 |
| | | | Bands FDD_H | RSRP_17 |
| Highest reported value (Cell 2) | RSRP_46 | RSRP_67 | Bands FDD_A | RSRP_35 |
| | | | Bands FDD_B | RSRP_35 |
| | | | Bands FDD_C | RSRP_36 |
| | | | Bands FDD_D | RSRP_36 |
| | | | Bands FDD_E, FDD_F | RSRP_37 |
| | | | Bands FDD_G | RSRP_38 |
| | | | Bands FDD_H | RSRP_38 |
| NOTE 1: E-UTRA operating band groups are as defined in Section 3.5. | | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.1.2 FDD Intra Frequency Relative Accuracy of RSRP

### 9.1.1.2.1 Test purpose

To verify that the FDD intra-frequency relative accuracy measurement of RSRP is within the specified limit for all bands.

#### 9.1.1.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward. Applicability requires support for FGI bit 16.

#### 9.1.1.2.3 Minimum conformance requirements

The relative accuracy of RSRP is defined as the RSRP measured from one cell compared to the RSRP measured from another cell on the same frequency.

The accuracy requirements in table 9.1.1.2.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

RSRP1,2$|_{dBm}$ according to Annex I.3.8 for a corresponding Band.

**Table 9.1.1.2.3-1: RSRP FDD Intra frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/$BW_{Channel}$ |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.1.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.2, clause 9.1.4 and A.9.1.1.

#### 9.1.1.2.4 Test description

##### 9.1.1.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20 .

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.1.2.4.3.

4. All cells are in the same carrier frequency. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.1.1.2.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.1.1.2.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message on cell.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the RSRP value of Cell 1 and Cell 2 in periodical MeasurementReport messages. The reported RSRP value for Cell 2 is compared to the reported RSRP value for Cell 1 for each MeasurementReport message according to Table 9.1.1.2.5-3. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.1.2.5-2 as appropriate.

#### 9.1.1.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.1.2.4.3-1: Common Exception messages for RSRP FDD Intra frequency relative accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3 |

**Table 9.1.1.2.4.3-2: *MeasResults*: Additional RSRP FDD intra frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.1.2.4.3-3: *MeasResultListEUTRA*: Additional RSRP FDD intra frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

## 9.1.1.2.5 Test requirement

Table 9.1.1.2.5-2 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD intra-frequency relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.1.2.5-3.

**Table 9.1.1.2.5-1: Void**

**Table 9.1.1.2.5-2: RSRP FDD Intra frequency relative accuracy test parameters**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |

| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
|---|---|---|---|---|---|---|---|---|
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - | R.0 FDD | - | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.1.2.1 | | | R.6 FDD | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA[Note1]<br>OCNG_RB[Note1] | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -106.00 | -106.00 | -88.00 | -88.00 | -116.00 | |
| | Bands FDD_B | | | | | | -115.50 | |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -114.00 | |
| | Bands FDD_H | | | | | | -112.50 | |
| | Bands FDD_D | | | | | | -114.50 | |
| | Bands FDD_G | | | | | | -113.00 | |
| | Bands FDD_C | | | | | | -115.00 | |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 | 1.88 | -4.97 | -0.01 | -4.76 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -100.00 | -104.00 | -82.00 | -86.00 | -113.00 | -116.00 |
| | Bands FDD_B | | | | | | -112.50 | -115.50 |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -111.00 | -114.00 |
| | Bands FDD_H | | | | | | -109.50 | -112.50 |
| | Bands FDD_D | | | | | | -111.50 | -114.50 |
| | Bands FDD_G | | | | | | -110.00 | -113.00 |
| | Bands FDD_C | | | | | | -112.00 | -115.00 |
| Io[Note3] | Bands FDD_A | dBm/9 MHz | -70.05 | -70.05 | -52.05 | -52.05 | -82.20 | |
| | Bands FDD_B | | | | | | -81.70 | |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -80.20 | |
| | Bands FDD_H | | | | | | -78.70 | |
| | Bands FDD_D | | | | | | -80.70 | |
| | Bands FDD_G | | | | | | -79.20 | |
| | Bands FDD_C | | | | | | -81.20 | |
| $\hat{E}_s / N_{oc}$ | | dB | 6.00 | 2.00 | 6.00 | 2.00 | 3.00 | 0.00 |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 6: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.1.2.5-3: RSRP FDD Intra frequency relative accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| | All bands | All bands | All bands |
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRP_x - 9 | RSRP_x - 9 | RSRP_x - 8 |
| Highest reported value (Cell 2) | RSRP_x + 1 | RSRP_x + 1 | RSRP_x + 2 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRP_x - 9 | RSRP_x - 9 | RSRP_x - 8 |
| Highest reported value (Cell 2) | RSRP_x + 1 | RSRP_x + 1 | RSRP_x + 2 |
| RSRP_x is the reported value of Cell 1 | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.2 TDD Intra frequency RSRP Accuracy

### 9.1.2.1 TDD Intra Frequency Absolute RSRP Accuracy

#### 9.1.2.1.1 Test purpose

To verify that the TDD intra-frequency absolute RSRP measurement accuracy is within the specified limit for all bands.

#### 9.1.2.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE from release 8 to release 11. Applicability requires support for FGI bit 16.

#### 9.1.2.1.3 Minimum conformance requirements

The absolute accuracy of RSRP is defined as the RSRP measured from a cell on the same frequency as that of the serving cell.

The accuracy requirements in Table 9.1.2.1-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.1.2.1.3-1: RSRP TDD Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | | |
| | | | E-UTRA operating band groups [Note 3] | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 2] | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±6 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_B | -120.5 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.2.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.2.1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.1, clause 9.1.4 and A.9.1.2.

## 9.1.2.1.4 Test description

### 9.1.2.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.2.1.4.3.

4. All cells are in the same carrier frequency. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.1.2.1.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.1.2.1.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message on cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check RSRP reported value in periodical MeasurementReport messages. The RSRP value of Cell 2 reported by the UE is compared to actual RSRP value according to Table 9.1.2.1.5-3.

7 If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.. SS shall check the MeasurementReport message transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.2.1.5-2 as appropriate.

#### 9.1.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.1.2.1.4.3-1: Common Exception message for RSRP TDD intra frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3 |

**Table 9.1.2.1.4.3-2: *MeasResults*: Additional RSRP TDD intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.2.1.4.3-3: *MeasResultListEUTRA*: Additional RSRP TDD intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of Cell 2 | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

### 9.1.2.1.5 Test requirement

Table 9.1.2.1.5-2 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD intra-frequency absolute accuracy test shall meet the reported values test requirements in table 9.1.2.1.5-3.

**Table 9.1.2.1.5-1: Void**

**Table 9.1.2.1.5-2: RSRP TDD Intra frequency absolute accuracy test parameters**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Special subframe configuration[Note1] | | | 6 | | 6 | | 6 | |
| Uplink/downlink configuration[Note1] | | | 1 | | 1 | | 1 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | R.0 TDD | - | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | |
| PSS_RA | | | | | | | | |
| SSS_RA | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA[Note2] | | | | | | | | |
| OCNG_RB[Note2] | | | | | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -107.0 | | -88.0 | | -116.0 | |
| | Bands TDD_C | | | | | | -115.0 | |
| | Bands TDD_E | | | | | | -114.0 | |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 | 1.88 | -4.97 | 0.09 | -4.96 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -101.0 | -105.0 | -82.0 | -86.0 | -113 | -116.2 |
| | Bands TDD_C | | | | | | -112 | -115.2 |
| | Bands TDD_E | | | | | | -111 | -114.2 |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -71.05 | | -52.05 | | -82.25 | |
| | Bands TDD_C | | | | | | -81.25 | |
| | Bands TDD_E | | | | | | -80.25 | |
| $\hat{E}_s / N_{oc}$ | | dB | 6.0 | 2.0 | 6.0 | 2.0 | 3.0 | -0.20 |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in 3GPP TS 36.211.
Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 6: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.2.1.5-3: RSRP TDD Intra frequency absolute accuracy requirements for the reported values**

| Normal Conditions | Test 1 All bands | Test 2 All bands | Test 3 | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_28 | RSRP_45 | Bands TDD_A | RSRP_17 |
| | | | Bands TDD_C | RSRP_18 |
| | | | Bands TDD_E | RSRP_19 |
| Highest reported value (Cell 2) | RSRP_43 | RSRP_64 | Bands TDD_A | RSRP_32 |
| | | | Bands TDD_C | RSRP_33 |
| | | | Bands TDD_E | RSRP_34 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_25 | RSRP_42 | Bands TDD_A | RSRP_14 |
| | | | Bands TDD_C | RSRP_15 |
| | | | Bands TDD_E | RSRP_16 |
| Highest reported value (Cell 2) | RSRP_46 | RSRP_67 | Bands TDD_A | RSRP_35 |
| | | | Bands TDD_C | RSRP_36 |
| | | | Bands TDD_E | RSRP_37 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.2.1_1 TDD Intra Frequency Absolute RSRP Accuracy (Rel-12 and forward)

### 9.1.2.1_1.1 Test purpose

Same test purpose as in clause 9.1.2.1.1.

### 9.1.2.1_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward. Applicability requires support for FGI bit 16.

### 9.1.2.1_1.3 Minimum conformance requirements

The absolute accuracy of RSRP is defined as the RSRP measured from a cell on the same frequency as that of the serving cell.

The accuracy requirements in Table 9.1.2.1_1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.1.2.1_1.3-1: RSRP TDD Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 3] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 / dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 / N/A | -70 |
| | | | FDD_B | -120.5 / N/A | -70 |
| | | | FDD_C, TDD_C | -120 / N/A | -70 |
| | | | FDD_D | -119.5 / N/A | -70 |
| | | | FDD_E, TDD_E | -119 / N/A | -70 |
| | | | FDD_F | -118.5 / N/A | -70 |
| | | | FDD_G | -118 / N/A | -70 |
| | | | FDD_H | -117.5 / N/A | -70 |
| | | | FDD_N | -114.5 / N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, GDD_B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.2.1_1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.2.1_1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.1_1, clause 9.1.4 and A.9.1.2.

## 9.1.2.1_1.4 Test description

### 9.1.2.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.2.1.4.1 with the following exceptions:

- Instead of 9.1.2.1.4.3 → use 9.1.2.1_1.4.3.

### 9.1.2.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.2.1.4.2 with the following exceptions:

- Instead of Table 9.1.2.1.5-2 → use Table 9.1.2.1_1.5-1.
- Instead of Table 9.1.2.1.5-3 → use Table 9.1.2.1_1.5-2.

### 9.1.2.1_1.4.3 Message contents

Same message contents as in clause 9.1.2.1.4.3

### 9.1.2.1_1.5 Test requirement

Table 9.1.2.1_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD intra-frequency absolute accuracy test shall meet the reported values test requirements in table 9.1.2.1_1.5-2.

**Table 9.1.2.1_1.5-1: RSRP TDD Intra frequency absolute accuracy test parameters**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Special subframe configuration[Note1] | | | 6 | | 6 | | 6 | |
| Uplink/downlink configuration[Note1] | | | 1 | | 1 | | 1 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | R.0 TDD | - | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA[Note2]<br>OCNG_RB[Note2] | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -107.0 | | -88.0 | | -116.0 | |
| | Bands TDD_C | | | | | | -115.0 | |
| | Bands TDD_E | | | | | | -114.0 | |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 | 1.88 | -4.97 | 0.09 | -4.96 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -101.0 | -105.0 | -82.0 | -86.0 | -113 | -116.2 |
| | Bands TDD_C | | | | | | -112 | -115.2 |
| | Bands TDD_E | | | | | | -111 | -114.2 |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -71.05 | | -52.05 | | -82.25 | |
| | Bands TDD_C | | | | | | -81.25 | |
| | Bands TDD_E | | | | | | -80.25 | |
| $\hat{E}_s / N_{oc}$ | | dB | 6.0 | 2.0 | 6.0 | 2.0 | 3.0 | -0.20 |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |

NOTE 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in 3GPP TS 36.211.
NOTE 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
NOTE 6: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.2.1_1.5-2: RSRP TDD Intra frequency absolute accuracy requirements for the reported values**

| Normal Conditions | Test 1 All bands | Test 2 All bands | Test 3 | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_29 | RSRP_45 | Bands TDD_A | RSRP_18 |
| | | | Bands TDD_C | RSRP_19 |
| | | | Bands TDD_E | RSRP_20 |
| Highest reported value (Cell 2) | RSRP_42 | RSRP_64 | Bands TDD_A | RSRP_30 |
| | | | Bands TDD_C | RSRP_31 |
| | | | Bands TDD_E | RSRP_32 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_25 | RSRP_42 | Bands TDD_A | RSRP_14 |
| | | | Bands TDD_C | RSRP_15 |
| | | | Bands TDD_E | RSRP_16 |
| Highest reported value (Cell 2) | RSRP_46 | RSRP_67 | Bands TDD_A | RSRP_35 |
| | | | Bands TDD_C | RSRP_36 |
| | | | Bands TDD_E | RSRP_37 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.2.2 TDD Intra Frequency Relative Accuracy of RSRP

### 9.1.2.2.1 Test purpose

To verify that the TDD intra-frequency relative accuracy measurement of RSRP is within the specified limit for all bands.

### 9.1.2.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward. Applicability requires support for FGI bit 16.

### 9.1.2.2.3 Minimum conformance requirements

The relative accuracy of RSRP is defined as the RSRP measured from one cell compared to the RSRP measured from another cell on the same frequency.

The accuracy requirements in Table 9.1.2.2.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP1,2|_{dBm}$ according to Annex I.3.8 for a corresponding Band.

**Table 9.1.2.2.3-1: RSRP TDD Intra frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/$BW_{Channel}$ |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.2.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.2, clause 9.1.4 and A.9.1.2.

## 9.1.2.2.4 Test description

### 9.1.2.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.20.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.1.2.2.4.3.
4. All cells are in the same carrier frequency. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

### 9.1.2.2.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Table 9.1.2.2.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the reported RSRP value of Cell 1 and Cell 2 in periodical MeasurementReport messages. The reported RSRP measurement value for Cell 2 is compared to the reported RSRP measurement value for Cell 1 for each MeasurementReport message according to Table 9.1.2.2.5-3. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.2.2.5-2 as appropriate.

#### 9.1.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.1.2.2.4.3-1: Common Exception messages for RSRP TDD intra frequency relative accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3 |

**Table 9.1.2.2.4.3-2: *MeasResults*: Additional RSRP TDD intra frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.2.2.4.3-3: *MeasResultListEUTRA*: Additional RSRP TDD intra frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | physCellId of Cell2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | According to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

### 9.1.2.2.5 Test requirement

Table 9.1.2.2.5-2 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD intra-frequency relative accuracy test shall meet the reported values test requirements in table 9.1.2.2.5-3. The mapping of measured quantity is defined in Table 9.1.2.2.5-3. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.2.2.5-1: Void**

**Table 9.1.2.2.5-2: RSRP TDD Intra frequency relative accuracy test parameters**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Special subframe configuration[Note1] | | | 6 | | 6 | | 6 | |
| Uplink/downlink configuration[Note1] | | | 1 | | 1 | | 1 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | R.0 TDD | - | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | |
| PSS_RA | | | | | | | | |
| SSS_RA | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA[Note2] | | | | | | | | |
| OCNG_RB[Note2] | | | | | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -106.0 | -106.0 | -88.0 | -88.0 | -116.0 | |
| | Bands TDD_C | | | | | | -115.00 | |
| | Bands TDD_E | | | | | | -114.00 | |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 | 1.88 | -4.97 | -0.01 | -4.76 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -100.0 | -104.0 | -82.0 | -86.0 | -113.0 | -116.0 |
| | Bands TDD_C | | | | | | -112.00 | -115.00 |
| | Bands TDD_E | | | | | | -111.00 | -114.00 |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -70.05 | -70.05 | -52.05 | -52.05 | -82.20 | |
| | Bands TDD_C | | | | | | -81.20 | |
| | Bands TDD_E | | | | | | -80.20 | |
| $\hat{E}_s / N_{oc}$ | | dB | 6.0 | 2.0 | 6.0 | 2.0 | 3.0 | 0.0 |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in 3GPP TS 36.211.
Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 6: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.2.2.5-3: RSRP TDD Intra frequency relative accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 | | |
|---|---|---|---|---|---|
| | All bands | All bands | Bands 33, 34, 35, 36, 37, 38, 39, 40 | Band 42, 43 | Band 41 |
| Normal Conditions | | | | | |
| Lowest reported value (Cell 2) | RSRP_x - 9 | RSRP_x - 9 | RSRP_x - 8 | RSRP_x - 8 | RSRP_x - 8 |
| Highest reported value (Cell 2) | RSRP_x + 1 | RSRP_x + 1 | RSRP_x + 2 | RSRP_x + 2 | RSRP_x + 2 |
| Extreme Conditions | | | | | |
| Lowest reported value (Cell 2) | RSRP_x - 9 | RSRP_x - 9 | RSRP_x - 8 | RSRP_x - 8 | RSRP_x - 8 |
| Highest reported value (Cell 2) | RSRP_x + 1 | RSRP_x + 1 | RSRP_x + 2 | RSRP_x + 2 | RSRP_x + 2 |
| RSRP_x is the reported value of Cell 1 | | | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.3 FDD Inter frequency RSRP Accuracy

### 9.1.3.1 FDD - FDD Inter Frequency Absolute RSRP Accuracy

#### 9.1.3.1.1 Test purpose

To verify that the FDD inter-frequency absolute RSRP measurement accuracy is within the specified limit for all bands.

#### 9.1.3.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE from release 8 to release 11. Applicability requires support for FGI bits 16 and 25.

#### 9.1.3.1.3 Minimum conformance requirements

The absolute accuracy of RSRP is defined as the RSRP measured from a cell that has different carrier frequency from the serving cell.

The accuracy requirements in table 9.1.3.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.3 for a corresponding Band.

**Table 9.1.3.1.3-1: RSRP FDD Inter frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 3] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 2] | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±6 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_B | -120.5 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.1.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.3.1, clause 9.1.4 and A.9.1.3.

### 9.1.3.1.4 Test description

#### 9.1.3.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.1.3.1.4.3.
4. Cell 1 and Cell 2 are on the different carrier frequencies. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.1.3.1.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Table 9.1.3.1.5-2 for the test interval as appropriate. Propagation conditions are set according to Annex B clause B.1.1
3. SS shall transmit an RRCConnectionReconfiguration message on cell.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check RSRP reported value in periodical MeasurementReport messages. The RSRP value of Cell 2 reported by the UE is compared to the actual RSRP according to Table 9.1.3.1.5-3.

If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.7. SS shall check the MeasurementReport message transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.3.1.5-2 as appropriate.

#### 9.1.3.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.3.1.4.3-1: Common Exception messages for RSRP FDD Inter frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-2<br>Table H.3.5-3 |

**Table 9.1.3.1.4.3-2: *MeasResults*: Additional RSRP FDD Inter frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.3.1.4.3-3: *MeasResultListEUTRA*: Additional RSRP FDD Inter frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

### 9.1.3.1.5 Test requirement

Table 9.1.3.1.5-2 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD inter-frequency absolute accuracy test shall meet the reported values test requirements in table 9.1.3.1.5-3.

**Table 9.1.3.1.5-1: Void**

**Table 9.1.3.1.5-2: RSRP FDD - FDD Inter frequency absolute accuracy test parameters**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 | 1 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 | 10 | 10 |
| Gap Pattern Id | | | 0 | - | 0 | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | |
| PSS_RA | | | | | | |
| SSS_RA | | | | | | |
| PCFICH_RB | | | | | | |
| PHICH_RA | | | | | | |
| PHICH_RB | | | | | | |
| PDCCH_RA | | | | | | |
| PDCCH_RB | | | | | | |
| PDSCH_RA | | | | | | |
| PDSCH_RB | | | | | | |
| OCNG_RA Note1 | | | | | | |
| OCNG_RB Note1 | | | | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -89.25 | -89.25 | ($N_{oc}$ for Channel 2 +8dB) | -117 |
| | Bands FDD_B | | | | | -116.5 |
| | Bands FDD_C | | | | | -116 |
| | Bands FDD_D | | | | | -115.5 |
| | Bands FDD_E, FDD_F Note 5 | | | | | -115 |
| | Bands FDD_G | | | | | -114 |
| | Bands FDD_H | | | | | -113.5 |
| $\hat{E}_s / I_{ot}$ | | dB | 10.00 | 10.00 | 13.00 | -3.20 |
| RSRP Note3 | Bands FDD_A | dBm/15 kHz | -79.25 | -79.25 | (RSRP for Cell 2 +24.2dB) | -120.20 |
| | Bands FDD_B | | | | | -119.70 |
| | Bands FDD_C | | | | | -119.20 |
| | Bands FDD_D | | | | | -118.70 |
| | Bands FDD_E, FDD_F Note 5 | | | | | -118.20 |
| | Bands FDD_G | | | | | -117.20 |
| | Bands FDD_H | | | | | -116.70 |
| Io Note3 | Bands FDD_A | dBm/9 MHz | -51.05 | -51.05 | (Io for Channel 2 +19.51dB) | -87.52 |
| | Bands FDD_B | | | | | -87.02 |
| | Bands FDD_C | | | | | -86.52 |
| | Bands FDD_D | | | | | -86.02 |
| | Bands FDD_E, FDD_F Note 5 | | | | | -85.52 |
| | Bands FDD_G | | | | | -84.52 |
| | Bands FDD_H | | | | | -84.02 |
| $\hat{E}_s / N_{oc}$ | | dB | 10 | 10 | 13 | -3.2 |
| Propagation condition | | - | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

| | |
|---|---|
| Note 4: | RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz. |
| Note 6: | E-UTRA operating band groups are as defined in Section 3.5. |

**Table 9.1.3.1.5-3: RSRP FDD Inter frequency absolute accuracy requirements for the reported values**

| **Normal Conditions** | **Test 1 All bands** | **Test 2** | |
|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_52 | Bands FDD_A | RSRP_13 |
| | | Bands FDD_B | RSRP_13 |
| | | Bands FDD_C | RSRP_14 |
| | | Bands FDD_D | RSRP_14 |
| | | Bands FDD_E, FDD_F | RSRP_15 |
| | | Bands FDD_G | RSRP_16 |
| | | Bands FDD_H | RSRP_16 |
| Highest reported value (Cell 2) | RSRP_71 | Bands FDD_A | RSRP_28 |
| | | Bands FDD_B | RSRP_28 |
| | | Bands FDD_C | RSRP_29 |
| | | Bands FDD_D | RSRP_29 |
| | | Bands FDD_E, FDD_F | RSRP_30 |
| | | Bands FDD_G | RSRP_31 |
| | | Bands FDD_H | RSRP_31 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2** | |
| Lowest reported value (Cell 2) | RSRP_49 | Bands FDD_A | RSRP_10 |
| | | Bands FDD_B | RSRP_10 |
| | | Bands FDD_C | RSRP_11 |
| | | Bands FDD_D | RSRP_11 |
| | | Bands FDD_E, FDD_F | RSRP_12 |
| | | Bands FDD_G | RSRP_13 |
| | | Bands FDD_H | RSRP_13 |
| Highest reported value (Cell 2) | RSRP_74 | Bands FDD_A | RSRP_31 |
| | | Bands FDD_B | RSRP_31 |
| | | Bands FDD_C | RSRP_32 |
| | | Bands FDD_D | RSRP_32 |
| | | Bands FDD_E, FDD_F | RSRP_33 |
| | | Bands FDD_G | RSRP_34 |
| | | Bands FDD_H | RSRP_34 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.3.1_1 FDD - FDD Inter Frequency Absolute RSRP Accuracy (Rel-12 and forward)

### 9.1.3.1_1.1 Test purpose

Same test purpose as in clause 9.1.3.1.1

### 9.1.3.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward. Applicability requires support for FGI bits 16 and 25.

### 9.1.3.1_1.3 Minimum conformance requirements

The absolute accuracy of RSRP is defined as the RSRP measured from a cell that has different carrier frequency from the serving cell.

The accuracy requirements in table 9.1.3.1_1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.3 for a corresponding Band.

**Table 9.1.3.1_1.3-1: RSRP FDD Inter frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 3 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_B | -120.5 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3..<br>NOTE 3: E-UTRA operating band groups are as defined in Section 3.5. | | | | | | |

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.1.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.3.1_1, clause 9.1.4 and A.9.1.3.

## 9.1.3.1_1.4 Test description

### 9.1.3.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.3.1.4.1 with the following exceptions:

- Instead of 9.1.3.1.4.3 → use 9.1.3.1_1.4.3.

### 9.1.3.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.3.1.4.2 with the following exceptions:

- Instead of Table 9.1.3.1.5-2 → use Table 9.1.3.1_1.5-1.
- Instead of Table 9.1.3.1.5-3 → use Table 9.1.3.1_1.5-2.

### 9.1.3.1_1.4.3 Message contents

Same message contents as in clause 9.1.3.1.4.3

## 9.1.3.1_1.5 Test requirement

Table 9.1.3.1_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD inter-frequency absolute accuracy test shall meet the reported values test requirements in table 9.1.3.1_1.5-2.

**Table 9.1.3.1_1.5-1: RSRP FDD - FDD Inter frequency absolute accuracy test parameters**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 | 1 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 | 10 | 10 |
| Gap Pattern Id | | | 0 | - | 0 | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | |
| PSS_RA | | | | | | |
| SSS_RA | | | | | | |
| PCFICH_RB | | | | | | |
| PHICH_RA | | | | | | |
| PHICH_RB | | | | | | |
| PDCCH_RA | | | | | | |
| PDCCH_RB | | | | | | |
| PDSCH_RA | | | | | | |
| PDSCH_RB | | | | | | |
| OCNG_RA Note1 | | | | | | |
| OCNG_RB Note1 | | | | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -89.25 | -89.25 | ($N_{oc}$ for Channel 2 +8dB) | -117 |
| | Bands FDD_B | | | | | -116.5 |
| | Bands FDD_C | | | | | -116 |
| | Bands FDD_D | | | | | -115.5 |
| | Bands FDD_E, FDD_F Note 5 | | | | | -115 |
| | Bands FDD_G | | | | | -114 |
| | Bands FDD_H | | | | | -113.5 |
| $\hat{E}_s/I_{ot}$ | | dB | 10.00 | 10.00 | 13.00 | -3.20 |
| RSRP Note3 | Bands FDD_A | dBm/15 kHz | -79.25 | -79.25 | (RSRP for Cell 2 +24.2dB) | -120.20 |
| | Bands FDD_B | | | | | -119.70 |
| | Bands FDD_C | | | | | -119.20 |
| | Bands FDD_D | | | | | -118.70 |
| | Bands FDD_E, FDD_F Note 5 | | | | | -118.20 |
| | Bands FDD_G | | | | | -117.20 |
| | Bands FDD_H | | | | | -116.70 |
| Io Note3 | Bands FDD_A | dBm/9 MHz | -51.05 | -51.05 | (Io for Channel 2 +19.51dB) | -87.52 |
| | Bands FDD_B | | | | | -87.02 |
| | Bands FDD_C | | | | | -86.52 |
| | Bands FDD_D | | | | | -86.02 |
| | Bands FDD_E, FDD_F Note 5 | | | | | -85.52 |
| | Bands FDD_G | | | | | -84.52 |
| | Bands FDD_H | | | | | -84.02 |

| | | | | | |
|---|---|---|---|---|---|
| $\hat{E}_s / N_{oc}$ | dB | 10 | 10 | 13 | -3.2 |
| Propagation condition | - | AWGN | | AWGN | |

NOTE 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
NOTE 5: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
NOTE 6: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.3.1_1.5-2: RSRP FDD Inter frequency absolute accuracy requirements for the reported values**

| **Normal Conditions** | **Test 1 All bands** | **Test 2** | |
|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_52 | Bands FDD_A | RSRP_14 |
| | | Bands FDD_B | RSRP_15 |
| | | Bands FDD_C | RSRP_15 |
| | | Bands FDD_D | RSRP_16 |
| | | Bands FDD_E, FDD_F | RSRP_16 |
| | | Bands FDD_G | RSRP_17 |
| | | Bands FDD_H | RSRP_18 |
| Highest reported value (Cell 2) | RSRP_71 | Bands FDD_A | RSRP_26 |
| | | Bands FDD_B | RSRP_27 |
| | | Bands FDD_C | RSRP_27 |
| | | Bands FDD_D | RSRP_28 |
| | | Bands FDD_E, FDD_F | RSRP_28 |
| | | Bands FDD_G | RSRP_29 |
| | | Bands FDD_H | RSRP_30 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2** | |
| Lowest reported value (Cell 2) | RSRP_49 | Bands FDD_A | RSRP_10 |
| | | Bands FDD_B | RSRP_10 |
| | | Bands FDD_C | RSRP_11 |
| | | Bands FDD_D | RSRP_11 |
| | | Bands FDD_E, FDD_F | RSRP_12 |
| | | Bands FDD_G | RSRP_13 |
| | | Bands FDD_H | RSRP_13 |
| Highest reported value (Cell 2) | RSRP_74 | Bands FDD_A | RSRP_31 |
| | | Bands FDD_B | RSRP_31 |
| | | Bands FDD_C | RSRP_32 |
| | | Bands FDD_D | RSRP_32 |
| | | Bands FDD_E, FDD_F | RSRP_33 |
| | | Bands FDD_G | RSRP_34 |
| | | Bands FDD_H | RSRP_34 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.3.2 FDD - FDD Inter Frequency Relative Accuracy of RSRP

### 9.1.3.2.1 Test purpose

To verify that the FDD inter-frequency relative accuracy measurement of RSRP is within the specified limit for all bands.

#### 9.1.3.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE from release 8 to release 11. Applicability requires support for FGI bits 16 and 25.

#### 9.1.3.2.3 Minimum conformance requirements

The relative accuracy of RSRP in inter frequency case is defined as the RSRP measured from one cell compared to the RSRP measured from another cell on a different frequency.

The accuracy requirements in table 9.1.3.2.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

RSRP1,2$|_{dBm}$ according to Annex I.3.4 for a corresponding Band.

$$\left| RSRP1|_{dBm} - RSRP2|_{dBm} \right| \leq 27dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.3.2.3-1: RSRP FDD Inter frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/BW$_{Channel}$ |
| ±6 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.1.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.3.2, clause 9.1.4 and A.9.1.3.

#### 9.1.3.2.4 Test description

##### 9.1.3.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.3.2.4.3.

4. Cell 1 and Cell 2 are on the different carrier frequencies. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.1.3.2.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.1.3.2.5-2 for the test interval as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message on cell.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the RSRP value of Cell 1 and Cell 2 in periodical MeasurementReport messages. The reported RSRP value for Cell 2 is compared to the reported RSRP value for Cell 1 for each MeasurementReport message according to Table 9.1.3.2.5-3. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.3.2.5-2 as appropriate.

#### 9.1.3.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.3.2.4.3-1: Common Exception messages for RSRP FDD Inter frequency relative accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-2<br>Table H.3.5-3 |

**Table 9.1.3.2.4.3-2: *MeasResults*: Additional RSRP FDD Inter frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.3.2.4.3-3: *MeasResultListEUTRA*: Additional RSRP FDD Inter frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

### 9.1.3.2.5 Test requirement

Table 9.1.3.2.5-2 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD inter-frequency relative accuracy test shall meet the reported values test requirements in table 9.1.3.2.5-3.

**Table 9.1.3.2.5-1: Void**

**Table 9.1.3.2.5-2: RSRP FDD - FDD Inter frequency relative accuracy test parameters**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 | 1 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 | 10 | 10 |
| Gap Pattern Id | | | 0 | - | 0 | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA Note1<br>OCNG_RB Note1 | | dB | 0 | 0 | 0 | 0 |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -89.25 | -89.25 | ($N_{oc}$ for Channel 2 +7dB) | -117 |
| | Bands FDD_B | | | | | -116.5 |
| | Bands FDD_C | | | | | -116 |
| | Bands FDD_D | | | | | -115.5 |
| | Bands FDD_E, FDD_F Note 5 | | | | | -115 |
| | Bands FDD_G | | | | | -114 |
| | Bands FDD_H | | | | | -113.5 |
| $\hat{E}_s / I_{ot}$ | | dB | 10.00 | 10.00 | 13.00 | -3.20 |
| RSRP Note3 | Bands FDD_A | dBm/15 kHz | -79.25 | -79.25 | (RSRP for Cell 2 +23.2dB) | -120.20 |
| | Bands FDD_B | | | | | -119.70 |
| | Bands FDD_C | | | | | -119.20 |
| | Bands FDD_D | | | | | -118.70 |
| | Bands FDD_E, FDD_F Note 5 | | | | | -118.20 |
| | Bands FDD_G | | | | | -117.20 |
| | Bands FDD_H | | | | | -116.70 |
| Io Note3 | Bands FDD_A | dBm/9 MHz | -51.05 | -51.05 | (Io for Channel 2 +18.51dB) | -87.52 |
| | Bands FDD_B | | | | | -87.02 |
| | Bands FDD_C | | | | | -86.52 |
| | Bands FDD_D | | | | | -86.02 |
| | Bands FDD_E, FDD_F Note 5 | | | | | -85.52 |
| | Bands FDD_G | | | | | -84.52 |
| | Bands FDD_H | | | | | -84.02 |
| $\hat{E}_s / N_{oc}$ | | dB | 10 | 10 | 13 | -3.2 |
| Propagation condition | | - | AWGN | | AWGN | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of | | | | | | |

| | |
|---|---|
| | appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 4: | RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz. |
| Note 6: | E-UTRA operating band groups are as defined in Section 3.5. |

**Table 9.1.3.2.5-3: RSRP FDD Inter frequency relative accuracy requirements for the reported values**

| | Test 1<br>All bands | Test 2<br>All bands |
|---|---|---|
| Normal Conditions | | |
| Lowest reported value (Cell 2) | RSRP_(x - 9) | RSRP_(x - 32) |
| Highest reported value (Cell 2) | RSRP_(x + 9) | RSRP_(x - 16) |
| Extreme Conditions | | |
| Lowest reported value (Cell 2) | RSRP_(x - 9) | RSRP_(x - 32) |
| Highest reported value (Cell 2) | RSRP_(x + 9) | RSRP_(x - 16) |
| RSRP_x is the reported value of Cell 1 | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.3.2_1 FDD - FDD Inter Frequency Relative Accuracy of RSRP (Rel-12 and forward)

### 9.1.3.2_1.1 Test purpose

Same test purpose as in clause 9.1.3.2.1.

### 9.1.3.2_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward. Applicability requires support for FGI bits 16 and 25.

### 9.1.3.2_1.3 Minimum conformance requirements

The relative accuracy of RSRP in inter frequency case is defined as the RSRP measured from one cell compared to the RSRP measured from another cell on a different frequency.

The accuracy requirements in table 9.1.3.2_1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

RSRP1,2$|_{dBm}$ according to Annex I.3.4 for a corresponding Band.

$$\left| RSRP1|_{dBm} - RSRP2|_{dBm} \right| \leq 27dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.3.2_1.3-1: RSRP FDD Inter frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/$BW_{Channel}$ |
| ±4.5 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.1.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.3.2, clause 9.1.4 and A.9.1.3.

## 9.1.3.2_1.4 Test description

### 9.1.3.2_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.3.2.4.1 with the following exceptions:

- Instead of 9.1.3.2.4.3 → use 9.1.3.2_1.4.3

### 9.1.3.2_1.4.2 Test procedure

Same test procedure as in clause 9.1.3.2.4.2 with the following exceptions:

- Instead of Table 9.1.3.2.5-2 → use Table 9.1.3.2_1.5-1.
- Instead of Table 9.1.3.2.5-3 → use Table 9.1.3.2_1.5-2.

### 9.1.3.2_1.4.3 Message contents

Same message contents as in clause 9.1.3.2.4.3.

## 9.1.3.2_1.5 Test requirement

Table 9.1.3.2_1.5-2 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD inter-frequency relative accuracy test shall meet the reported values test requirements in table 9.1.3.2_1.5-3.

**Table 9.1.3.2_1.5-1: RSRP FDD - FDD Inter frequency relative accuracy test parameters**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 | 1 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 | 10 | 10 |
| Gap Pattern Id | | | 0 | - | 0 | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | |
| PSS_RA | | | | | | |
| SSS_RA | | | | | | |
| PCFICH_RB | | | | | | |
| PHICH_RA | | | | | | |
| PHICH_RB | | | | | | |
| PDCCH_RA | | | | | | |
| PDCCH_RB | | | | | | |
| PDSCH_RA | | | | | | |
| PDSCH_RB | | | | | | |
| OCNG_RA Note1 | | | | | | |
| OCNG_RB Note1 | | | | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -89.25 | -89.25 | ($N_{oc}$ for Channel 2 +7dB) | -117 |
| | Bands FDD_B | | | | | -116.6 |
| | Bands FDD_C | | | | | -116 |
| | Bands FDD_D | | | | | -115.5 |
| | Bands FDD_E, FDD_F Note 5 | | | | | -115 |
| | Bands FDD_G | | | | | -114 |
| | Bands FDD_H | | | | | -113.5 |
| $\hat{E}_s/I_{ot}$ | | dB | 10.00 | 10.00 | 13.00 | -3.20 |
| RSRP Note3 | Bands FDD_A | dBm/15 kHz | -79.25 | -79.25 | (RSRP for Cell 2 +23.2dB) | -120.20 |
| | Bands FDD_B | | | | | -119.70 |
| | Bands FDD_C | | | | | -119.20 |
| | Bands FDD_D | | | | | -118.70 |
| | Bands FDD_E, FDD_F Note 5 | | | | | -118.20 |
| | Bands FDD_G | | | | | -117.20 |
| | Bands FDD_H | | | | | -116.70 |
| Io Note3 | Bands FDD_A | dBm/9 MHz | -51.05 | -51.05 | (Io for Channel 2 +18.51dB) | -87.52 |
| | Bands FDD_B | | | | | -87.02 |
| | Bands FDD_C | | | | | -86.52 |
| | Bands FDD_D | | | | | -86.02 |
| | Bands FDD_E, FDD_F Note 5 | | | | | -85.52 |
| | Bands FDD_G | | | | | -84.52 |
| | Bands FDD_H | | | | | -84.02 |
| $\hat{E}_s/N_{oc}$ | | dB | 10 | 10 | 13 | -3.2 |
| Propagation condition | | - | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

| | |
|---|---|
| Note 3: | RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 4: | RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz. |
| Note 6: | E-UTRA operating band groups are as defined in Section 3.5. |

**Table 9.1.3.2_1.5-2: RSRP FDD Inter frequency relative accuracy requirements for the reported values**

| | Test 1<br>All bands | Test 2<br>All bands |
|---|---|---|
| Normal Conditions | | |
| Lowest reported value (Cell 2) | RSRP_(x - 7) | RSRP_(x - 30) |
| Highest reported value (Cell 2) | RSRP_(x + 7) | RSRP_(x - 17) |
| Extreme Conditions | | |
| Lowest reported value (Cell 2) | RSRP_(x - 9) | RSRP_(x - 32) |
| Highest reported value (Cell 2) | RSRP_(x + 9) | RSRP_(x - 16) |
| RSRP_x is the reported value of Cell 1 | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.4 TDD Inter frequency RSRP Accuracy

### 9.1.4.1 TDD - TDD Inter Frequency Absolute RSRP Accuracy

#### 9.1.4.1.1 Test purpose

To verify that the TDD inter-frequency absolute RSRP measurement accuracy is within the specified limit for all bands.

#### 9.1.4.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE from release 8 to release 11. Applicability requires support for FGI bits 16 and 25.

#### 9.1.4.1.3 Minimum conformance requirements

The absolute accuracy of RSRP is defined as the RSRP measured from a cell that has different carrier frequency from the serving cell.

The accuracy requirements in Table 9.1.4.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.3 for a corresponding Band.

**Table 9.1.4.1.3-1: RSRP TDD-TDD Inter frequency absolute accuracy**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | | |
| | | | E-UTRA operating band groups [Note 3] | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 2] | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±6 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_B | -120.5 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table9.1.2.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.3.1 , clause 9.1.4 and A.9.1.4.

### 9.1.4.1.4 Test description

#### 9.1.4.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.1.4.1.4.3.
4. Cell 1 and Cell 2 are on the different carrier frequencies. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.1.4.1.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Table 9.1.4.1.5-2 for the test interval as appropriate. Propagation conditions are set according to Annex B clause B.1.1
3. SS shall transmit an RRCConnectionReconfiguration message on cell.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3 SS shall check the reported RSRP value in periodical MeasurementReport messages. The RSRP value of Cell 2 reported by the UE is compared to the actual RSRP value according to Table 9.1.4.1.5-3. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport message transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.4.1.5-2 as appropriate.

#### 9.1.4.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.1.4.1.4.3-1: Common Exception messages for RSRP TDD - TDD Inter frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-2<br>Table H.3.5-3 |

**Table 9.1.4.1.4.3-2: *MeasResults*: Additional RSRP TDD - TDD Inter frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.4.1.4.3-3: *MeasResultListEUTRA*: Additional RSRP TDD - TDD Inter frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | According to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

### 9.1.4.1.5 Test requirement

Table 9.1.4.1.5-2 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD inter-frequency absolute accuracy test shall meet the reported values test requirements in table 9.1.4.1.5-3.

**Table 9.1.4.1.5-1: Void**

**Table 9.1.4.1.5-2: RSRP TDD-TDD Inter frequency absolute accuracy test parameters**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 | 1 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 | 10 | 10 |
| Special subframe configuration[Note1] | | | 6 | | 6 | |
| Uplink-downlink configuration[Note1] | | | 1 | | 1 | |
| Gap Pattern Id | | | 0 | - | 0 | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | |
| PSS_RA | | | | | | |
| SSS_RA | | | | | | |
| PCFICH_RB | | | | | | |
| PHICH_RA | | | | | | |
| PHICH_RB | | | | | | |
| PDCCH_RA | | | | | | |
| PDCCH_RB | | | | | | |
| PDSCH_RA | | | | | | |
| PDSCH_RB | | | | | | |
| OCNG_RA [Note2] | | | | | | |
| OCNG_RB [Note2] | | | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -89.25 | -89.25 | ($N_{oc}$ for Channel 2 +8dB) | -117 |
| | Bands TDD_C | | | | | -116 |
| | Bands TDD_E | | | | | -115 |
| $\hat{E}_s / I_{ot}$ | | dB | 10.00 | 10.00 | 13.00 | -3.20 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -79.25 | -79.25 | (RSRP for Cell 2 +24.2dB) | -120.20 |
| | Bands TDD_C | | | | | -119.20 |
| | Bands TDD_E | | | | | -118.20 |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -51.05 | -51.05 | (Io for Channel 2 +19.51dB) | -87.52 |
| | Bands TDD_C | | | | | -86.52 |
| | Bands TDD_E | | | | | -85.52 |
| $\hat{E}_s / N_{oc}$ | | dB | 10 | 10 | 13 | -3.2 |
| Propagation condition | | - | AWGN | | AWGN | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.

Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 6: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.4.1.5-3: RSRP TDD-TDD Inter frequency absolute accuracy requirements for the reported values**

| **Normal Conditions** | **Test 1 All bands** | **Test 2** | |
|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_52 | Bands TDD_A | RSRP_13 |
| | | Bands TDD_C | RSRP_14 |
| | | Bands TDD_E | RSRP_15 |
| Highest reported value (Cell 2) | RSRP_71 | Bands TDD_A | RSRP_28 |
| | | Bands TDD_C | RSRP_29 |
| | | Bands TDD_E | RSRP_30 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2** | |
| Lowest reported value (Cell 2) | RSRP_49 | Bands TDD_A | RSRP_10 |
| | | Bands TDD_C | RSRP_11 |
| | | Bands TDD_E | RSRP_12 |
| Highest reported value (Cell 2) | RSRP_74 | Bands TDD_A | RSRP_31 |
| | | Bands TDD_C | RSRP_32 |
| | | Bands TDD_E | RSRP_33 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.4.1_1 TDD - TDD Inter Frequency Absolute RSRP Accuracy (Rel-12 and forward)

### 9.1.4.1_1.1 Test purpose

Same test purpose as in clause 9.1.4.1.1

### 9.1.4.1_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward. Applicability requires support for FGI bits 16 and 25.

### 9.1.4.1_1.3 Minimum conformance requirements

The absolute accuracy of RSRP is defined as the RSRP measured from a cell that has different carrier frequency from the serving cell.

The accuracy requirements in Table 9.1.4.1_1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.3 for a corresponding Band.

**Table 9.1.4.1_1.3-1: RSRP TDD-TDD Inter frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 3] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 2] | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_B | -120.5 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table9.1.2.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.3.1 , clause 9.1.4 and A.9.1.4.

## 9.1.4.1_1.4 Test description

### 9.1.4.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.4.1.4.1 with the following exceptions:

- Instead of 9.1.4.1.4.3 → use 9.1.4.1_1.4.3.

### 9.1.4.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.4.1.4.2 with the following exceptions:

- Instead of Table 9.1.4.1.5-2 → use Table 9.1.4.1_1.5-1.
- Instead of Table 9.1.4.1.5-3 → use Table 9.1.4.1_1.5-2.

### 9.1.4.1_1.4.3 Message contents

Same message contents as in clause 9.1.4.1.4.3

## 9.1.4.1_1.5 Test requirement

Table 9.1.4.1_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD inter-frequency absolute accuracy test shall meet the reported values test requirements in table 9.1.4.1_1.5-2.

**Table 9.1.4.1_1.5-1: RSRP TDD-TDD Inter frequency absolute accuracy test parameters**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 | 1 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 | 10 | 10 |
| Special subframe configuration Note1 | | | 6 | | 6 | |
| Uplink-downlink configuration Note1 | | | 1 | | 1 | |
| Gap Pattern Id | | | 0 | - | 0 | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA Note2<br>OCNG_RB Note2 | | dB | 0 | 0 | 0 | 0 |
| $N_{oc}$ Note3 | Bands TDD_A | dBm/15 kHz | -89.25 | -89.25 | ($N_{oc}$ for Channel 2 +8dB) | -117 |
| | Bands TDD_C | | | | | -116 |
| | Bands TDD_E | | | | | -115 |
| $\hat{E}_s / I_{ot}$ | | dB | 10.00 | 10.00 | 13.00 | -3.20 |
| RSRP Note4 | Bands TDD_A | dBm/15 kHz | -79.25 | -79.25 | (RSRP for Cell 2 +24.2dB) | -120.20 |
| | Bands TDD_C | | | | | -119.20 |
| | Bands TDD_E | | | | | -118.20 |
| Io Note4 | Bands TDD_A | dBm/9 MHz | -51.05 | -51.05 | (Io for Channel 2 +19.51 dB) | -87.52 |
| | Bands TDD_C | | | | | -86.52 |
| | Bands TDD_E | | | | | -85.52 |
| $\hat{E}_s / N_{oc}$ | | dB | 10 | 10 | 13 | -3.2 |
| Propagation condition | | - | AWGN | | AWGN | |

NOTE 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
NOTE 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

NOTE 6: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.4.1_1.5-2: RSRP TDD-TDD Inter frequency absolute accuracy requirements for the reported values**

| Normal Conditions | Test 1 All bands | Test 2 | |
|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_52 | Bands TDD_A | RSRP_14 |
| | | Bands TDD_C | RSRP_15 |
| | | Bands TDD_E | RSRP_16 |
| Highest reported value (Cell 2) | RSRP_71 | Bands TDD_A | RSRP_26 |
| | | Bands TDD_C | RSRP_27 |
| | | Bands TDD_E | RSRP_28 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2** | |
| Lowest reported value (Cell 2) | RSRP_49 | Bands TDD_A | RSRP_10 |
| | | Bands TDD_C | RSRP_11 |
| | | Bands TDD_E | RSRP_12 |
| Highest reported value (Cell 2) | RSRP_74 | Bands TDD_A | RSRP_31 |
| | | Bands TDD_C | RSRP_32 |
| | | Bands TDD_E | RSRP_33 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.4.2 TDD - TDD Inter Frequency Relative Accuracy of RSRP

### 9.1.4.2.1 Test purpose

To verify that the TDD inter-frequency relative accuracy measurement of RSRP is within the specified limit for all bands.

### 9.1.4.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE from release 8 to release 11. Applicability requires support for FGI bits 16 and 25.

### 9.1.4.2.3 Minimum conformance requirements

The relative accuracy of RSRP in inter frequency case is defined as the RSRP measured from one cell compared to the RSRP measured from another cell on a different frequency.

The accuracy requirements in Table 9.1.4.2.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP1,2|_{dBm}$ according to Annex I.3.4 for a corresponding Band.

$$\left| RSRP1|_{dBm} - RSRP2|_{dBm} \right| \leq 27dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.4.2.3-1: RSRP TDD-TDD Inter frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/$BW_{Channel}$ |
| ±6 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table9.1.2.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.3.2, clause 9.1.4 and A.9.1.4.

### 9.1.4.2.4 Test description

#### 9.1.4.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.4.2.4.3.

4. Cell 1 and Cell 2 are on the different carrier frequencies. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.1.4.2.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.1.4.2.5-2 for the test interval as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message on cell.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the reported RSRP value of Cell 1 and Cell 2 in MeasurementReport messages. The reported RSRP value for Cell 2 is compared to the reported RSRP value for Cell 1 for each

MeasurementReport message according to Table 9.1.4.2.5-3. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.4.2.5-2 as appropriate.

#### 9.1.4.2.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.1.4.2.4.3-1: Common Exception messages for RSRP TDD - TDD Inter frequency relative accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-2<br>Table H.3.5-3 |

**Table 9.1.4.2.4.3-2: *MeasResults*: Additional RSRP TDD - TDD Inter frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | According to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.4.2.4.3-3: *MeasResultListEUTRA*: Additional RSRP TDD - TDD Inter frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | According to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

### 9.1.4.2.5 Test requirement

Table 9.1.4.2.5-2 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD inter-frequency relative accuracy test shall meet the reported values test requirements in table 9.1.4.2.5-3. The mapping of measured quantity is defined in Table 9.1.4.2.5-3. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.4.2.5-1: Void**

**Table 9.1.4.2.5-2: RSRP TDD-TDD Inter frequency relative accuracy test parameters**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 | 1 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 | 10 | 10 |
| Special subframe configuration[Note1] | | | 6 | | 6 | |
| Uplink-downlink configuration[Note1] | | | 1 | | 1 | |
| Gap Pattern Id | | | 0 | - | 0 | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA [Note2]<br>OCNG_RB [Note2] | | dB | 0 | 0 | 0 | 0 |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -89.25 | -89.25 | ($N_{oc}$ for Channel 2 +7dB) | -117 |
| | Bands TDD_C | | | | | -116 |
| | Bands TDD_E | | | | | -115 |
| $\hat{E}_s / I_{ot}$ | | dB | 10.00 | 10.00 | 13.00 | -3.20 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -79.25 | -79.25 | (RSRP for Cell 2 +23.2dB) | -120.20 |
| | Bands TDD_C | | | | | -119.20 |
| | Bands TDD_E | | | | | -118.20 |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -51.05 | -51.05 | (Io for Channel 2 +18.51dB) | -87.52 |
| | Bands TDD_C | | | | | -86.52 |
| | Bands TDD_E | | | | | -85.52 |
| $\hat{E}_s / N_{oc}$ | | dB | 10 | 10 | 13 | -3.2 |
| Propagation condition | | - | AWGN | | AWGN | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.

Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 6: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.4.2.5-3: RSRP TDD-TDD Inter frequency relative accuracy requirements for the reported values**

| | Test 1 | Test 2 |
|---|---|---|
| | All bands | All bands |
| Normal Conditions | | |
| Lowest reported value (Cell 2) | RSRP_(x - 9) | RSRP_(x - 32) |
| Highest reported value (Cell 2) | RSRP_(x + 9) | RSRP_(x - 16) |
| Extreme Conditions | | |
| Lowest reported value (Cell 2) | RSRP_(x - 9) | RSRP_(x - 32) |
| Highest reported value (Cell 2) | RSRP_(x + 9) | RSRP_(x - 16) |
| RSRP_x is the reported value of Cell 1 | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.4.2_1 TDD - TDD Inter Frequency Relative Accuracy of RSRP (Rel-12 and forward)

### 9.1.4.2_1.1 Test purpose

Same test purpose as in clause 9.1.4.2.1.

### 9.1.4.2_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward. Applicability requires support for FGI bits 16 and 25.

### 9.1.4.2_1.3 Minimum conformance requirements

The relative accuracy of RSRP in inter frequency case is defined as the RSRP measured from one cell compared to the RSRP measured from another cell on a different frequency.

The accuracy requirements in Table 9.1.4.2_1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

RSRP1,2$|_{dBm}$ according to Annex I.3.4 for a corresponding Band.

$$\left| RSRP1|_{dBm} - RSRP2|_{dBm} \right| \leq 27dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.4.2_1.3-1: RSRP TDD-TDD Inter frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/$BW_{Channel}$ |
| ±4.5 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table9.1.2.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.3.2, clause 9.1.4 and A.9.1.4.

## 9.1.4.2_1.4 Test description

### 9.1.4.2_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.4.2.4.1 with the following exceptions:

- Instead of 9.1.4.2.4.3 → use 9.1.4.2_1.4.3.

### 9.1.4.2_1.4.2 Test procedure

Same test procedure as in clause 9.1.4.2.4.2 with the following exceptions:

- Instead of Table 9.1.4.2.5-2 → use Table 9.1.4.2_1.5-1.
- Instead of Table 9.1.4.2.5-3 → use Table 9.1.4.2_1.5-2.

### 9.1.4.2_1.4.3 Message contents

Same message contents as in clause 9.1.4.2.4.3.

## 9.1.4.2_1.5 Test requirement

Table 9.1.4.2_1.5-2 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD inter-frequency relative accuracy test shall meet the reported values test requirements in table 9.1.4.2_1.5-3. The mapping of measured quantity is defined in Table 9.1.4.2_1.5-3. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.4.2_1.5-1: RSRP TDD-TDD Inter frequency relative accuracy test parameters**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |

| | | | | | | |
|---|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | 2 | 1 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 | 10 | 10 |
| Special subframe configuration Note1 | | | 6 | | 6 | |
| Uplink-downlink configuration Note1 | | | 1 | | 1 | |
| Gap Pattern Id | | | 0 | - | 0 | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | |
| PSS_RA | | | | | | |
| SSS_RA | | | | | | |
| PCFICH_RB | | | | | | |
| PHICH_RA | | | | | | |
| PHICH_RB | | | | | | |
| PDCCH_RA | | | | | | |
| PDCCH_RB | | | | | | |
| PDSCH_RA | | | | | | |
| PDSCH_RB | | | | | | |
| OCNG_RA Note2 | | | | | | |
| OCNG_RB Note2 | | | | | | |
| $N_{oc}$ Note3 | Bands TDD_A | dBm/15 kHz | -89.25 | -89.25 | ($N_{oc}$ for Channel 2 +7dB) | -117 |
| | Bands TDD_C | | | | | -116 |
| | Bands TDD_E | | | | | -115 |
| $\hat{E}_s/I_{ot}$ | | dB | 10.00 | 10.00 | 13.00 | -3.20 |
| RSRP Note4 | Bands TDD_A | dBm/15 kHz | -79.25 | -79.25 | (RSRP for Cell 2 +23.2dB) | -120.20 |
| | Bands TDD_C | | | | | -119.20 |
| | Bands TDD_E | | | | | -118.20 |
| Io Note4 | Bands TDD_A | dBm/9 MHz | -51.05 | -51.05 | (Io for Channel 2 +18.51dB) | -87.52 |
| | Bands TDD_C | | | | | -86.52 |
| | Bands TDD_E | | | | | -85.52 |
| $\hat{E}_s/N_{oc}$ | | dB | 10 | 10 | 13 | -3.2 |
| Propagation condition | | - | AWGN | | AWGN | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.

Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 6: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.4.2_1.5-2: RSRP TDD-TDD Inter frequency relative accuracy requirements for the reported values**

| | Test 1 | Test 2 |
|---|---|---|
| | All bands | All bands |
| Normal Conditions | | |
| Lowest reported value (Cell 2) | RSRP_(x - 7) | RSRP_(x - 30) |
| Highest reported value (Cell 2) | RSRP_(x + 7) | RSRP_(x - 17) |
| Extreme Conditions | | |
| Lowest reported value (Cell 2) | RSRP_(x - 9) | RSRP_(x - 32) |
| Highest reported value (Cell 2) | RSRP_(x + 9) | RSRP_(x - 16) |
| RSRP_x is the reported value of Cell 1 | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.5 FDD - TDD Inter frequency RSRP Accuracy

### 9.1.5.1 FDD - TDD Inter Frequency Absolute RSRP Accuracy

#### 9.1.5.1.1 Test purpose

To verify that the FDD - TDD inter-frequency absolute RSRP measurement accuracy is within the specified limit for all bands.

#### 9.1.5.1.2 Test applicability

This test applies to all types of E-UTRA UE supporting FDD and TDD from release 9 to release 11. Applicability requires support for FGI bit 25.

#### 9.1.5.1.3 Minimum conformance requirements

The absolute accuracy of RSRP is defined as the RSRP measured from a cell that has different carrier frequency from the serving cell.

The accuracy requirements in table 9.1.5.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.3 for a corresponding Band.

**Table 9.1.5.1.3-1: RSRP FDD - TDD Inter frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 3] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 2] / dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±6 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 / N/A | -70 |
| | | | FDD_C, TDD_C | -120 / N/A | -70 |
| | | | FDD_D | -119.5 / N/A | -70 |
| | | | FDD_E, TDD_E | -119 / N/A | -70 |
| | | | FDD_F | -118.5 / N/A | -70 |
| | | | FDD_G | -118 / N/A | -70 |
| | | | FDD_H | -117.5 / N/A | -70 |
| | | | FDD_N | -114.5 / N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.5.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.5.1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.3.1, clause 9.1.4 and A.9.1.5.

### 9.1.5.1.4 Test description

#### 9.1.5.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.5.1.4.3.

4. Cell 1 and Cell 2 are on the different carrier frequencies. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is FDD cell and Cell 2 is TDD cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.1.5.1.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.1.5.1.5-1 and Table 9.1.5.1.5-2 for the test interval as appropriate. Propagation conditions are set according to Annex B clause B.1.1

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check RSRP reported value in periodical MeasurementReport messages. The RSRP value of Cell 2 reported by the UE is compared to the actual RSRP according to Table 9.1.5.1.5-3. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport message transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.5.1.5-1 and Table 9.1.5.1.5-2 as appropriate.

#### 9.1.5.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.5.1.4.3-1: Common Exception messages for RSRP FDD-TDD Inter frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-2<br>Table H.3.5-3 |

**Table 9.1.5.1.4.3-2: *MeasResults*: Additional RSRP FDD-TDD Inter frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.5.1.4.3-3: *MeasResultListEUTRA*: Additional RSRP FDD-TDD Inter frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | Cell 2 | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

### 9.1.5.1.5 Test requirement

Table 9.1.5.1.5-1 and Table 9.1.5.1.5-2 define the primary level settings including test tolerances for all tests.

Each RSRP FDD-TDD inter-frequency absolute accuracy test shall meet the reported values test requirements in table 9.1.5.1.5-3.

**Table 9.1.5.1.5-1: RSRP FDD—TDD Inter frequency test parameters (FDD Cell1)**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | | Cell 1 | |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Gap Pattern Id | | 0 | | 0 | |
| Measurement bandwidth | $n_{PRB}$ | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.3.1.1.1 | | R.0 FDD | | R.0 FDD | |
| PDSCH allocation | $n_{PRB}$ | 13—36 | | 13—36 | |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.1 | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1 FDD) and A.3.2.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.1 FDD | |
| PBCH_RA | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RANote1 | | | | | |
| OCNG_RBNote | | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -89.25 | | -104 | |
| $\hat{E}_s / I_{ot}$ | dB | 10 | | 13 | |
| RSRP Note3 | dBm/15 kHz | -79.25 | | -91 | |
| Io Note3 | dBm/9 MHz | -51.05 | | -63.01 | |
| $\hat{E}_s / N_{oc}$ | dB | 10 | | 13 | |
| Propagation condition | - | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 9.1.5.1.5-2: RSRP FDD—TDD Inter frequency test parameters (TDD cell2)**

| Parameter | Unit | Test 1 | Test 2 |
|---|---|---|---|
| | | Cell 2 | Cell 2 |
| E-UTRA RF Channel Number | | 2 | 2 |
| $BW_{channel}$ | MHz | 10 | 10 |
| Special subframe configuration[Note1] | | 6 | 6 |
| Uplink-downlink configuration[Note1] | | 1 | 1 |
| Gap Pattern Id | | - | - |
| Measurement bandwidth | $n_{PRB}$ | 22—27 | 22—27 |
| PDSCH Reference measurement channel defined in A.3.1.1.2 | | - | - |
| PDSCH allocation | $n_{PRB}$ | - | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.2 | | R.6 TDD | R.6 TDD |
| OCNG Patterns defined in A.3.2.2.1 (OP.1 TDD) and A.3.2.2.2 (OP.2 TDD) | | OP.2 TDD | OP.2 TDD |
| PBCH_RA | dB | 0 | 0 |
| PBCH_RB | | | |
| PSS_RA | | | |
| SSS_RA | | | |
| PCFICH_RB | | | |
| PHICH_RA | | | |
| PHICH_RB | | | |
| PDCCH_RA | | | |
| PDCCH_RB | | | |
| PDSCH_RA | | | |
| PDSCH_RB | | | |
| OCNG_RA[Note2] | | | |
| OCNG_RB[Note2] | | | |
| $N_{oc}$ [Note3] | dBm/15 kHz | -89.25 | -112 |
| $\hat{E}_s / I_{ot}$ | dB | 10 | -3.2 |
| RSRP[Note4] | dBm/15 kHz | -79.25 | -115.2 |
| Io[Note4] | dBm/9 MHz | -51.05 | -82.52 |
| $\hat{E}_s / N_{oc}$ | dB | 10 | -3.2 |
| Propagation condition | - | AWGN | AWGN |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in 3GPP TS 36.211.

Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 9.1.5.1.5-3: RSRP FDD-TDD Inter frequency absolute accuracy requirements for the reported values**

| | Test 1 | Test 2 |
|---|---|---|
| Normal Conditions | | |
| Lowest reported value (Cell 2) | RSRP_52 | RSRP_18 |
| Highest reported value (Cell 2) | RSRP_71 | RSRP_33 |
| Extreme Conditions | | |
| Lowest reported value (Cell 2) | RSRP_49 | RSRP_15 |
| Highest reported value (Cell 2) | RSRP_74 | RSRP_36 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.5.1_1 FDD - TDD Inter Frequency Absolute RSRP Accuracy (Rel-12 and forward)

Editor's note: This Test case is incomplete for frequencies above 3GHz

- The Test system uncertainties applicable above 3GHz are undefined
- The Test Tolerances and Test Requirements applicable above 3GHz are undefined
- Test requirements for Test2 in normal conditions are TBD

### 9.1.5.1_1.1 Test purpose

Same test purpose as in clause 9.1.5.1.1.

### 9.1.5.1_1.2 Test applicability

This test applies to all types of E-UTRA UE supporting FDD and TDD release 12 and forward. Applicability requires support for FGI bit 25.

### 9.1.5.1_1.3 Minimum conformance requirements

The absolute accuracy of RSRP is defined as the RSRP measured from a cell that has different carrier frequency from the serving cell.

The accuracy requirements in table 9.1.5.1_1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.3 for a corresponding Band.

**Table 9.1.5.1_1.3-1: RSRP FDD - TDD Inter frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |

| | | | | | | |
|---|---|---|---|---|---|---|
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 3: E-UTRA operating band groups are as defined in Section 3.5. | | | | | | |

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.5.1_1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.5.1_1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.3.1, clause 9.1.4 and A.9.1.5.

## 9.1.5.1_1.4 Test description

### 9.1.5.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.5.1.4.1 with the following exceptions:

- Instead of 9.1.5.1.4.3 → use 9.1.5.1_1.4.3.

### 9.1.5.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.5.1.4.2 with the following exceptions:

- Instead of Table 9.1.5.1.5-1 → use Table 9.1.5.1_1.5-1.
- Instead of Table 9.1.5.1.5-2 → use Table 9.1.5.1_1.5-2.
- Instead of Table 9.1.5.1.5-3 → use Table 9.1.5.1_1.5-3.

### 9.1.5.1_1.4.3 Message contents

Same message contents as in clause 9.1.5.1.4.3

## 9.1.5.1_1.5 Test requirement

Table 9.1.5.1_1.5-1 and Table 9.1.5.1_1.5-2 define the primary level settings including test tolerances for all tests.

Each RSRP FDD-TDD inter-frequency absolute accuracy test shall meet the reported values test requirements in table 9.1.5.1_1.5-3.

**Table 9.1.5.1_1.5-1: RSRP FDD—TDD Inter frequency test parameters (FDD Cell1)**

| Parameter | Unit | Test 1 | Test 2 |
|---|---|---|---|
| | | Cell 1 | Cell 1 |

| | | | | | |
|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Gap Pattern Id | | 0 | | 0 | |
| Measurement bandwidth | $n_{PRB}$ | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.3.1.1.1 | | R.0 FDD | | R.0 FDD | |
| PDSCH allocation | $n_{PRB}$ | 13—36 | | 13—36 | |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.1 | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1 FDD) and A.3.2.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.1 FDD | |
| PBCH_RA | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RANote1 | | | | | |
| OCNG_RBNote | | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -88.95 | | -104 | |
| $\hat{E}_s / I_{ot}$ | dB | 10 | | 13 | |
| RSRP Note3 | dBm/15 kHz | -78.95 | | -91 | |
| Io Note3 | dBm/9 MHz | -50.75 | | -63.01 | |
| $\hat{E}_s / N_{oc}$ | dB | 10 | | 13 | |
| Propagation condition | - | AWGN | | AWGN | |

NOTE 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 9.1.5.1_1.5-2: RSRP FDD—TDD Inter frequency test parameters (TDD cell2)**

| Parameter | Unit | Test 1 | Test 2 |
|---|---|---|---|
| | | Cell 2 | Cell 2 |

| | | | |
|---|---|---|---|
| E-UTRA RF Channel Number | | 2 | 2 |
| $BW_{channel}$ | MHz | 10 | 10 |
| Special subframe configuration[Note1] | | 6 | 6 |
| Uplink-downlink configuration[Note1] | | 1 | 1 |
| Gap Pattern Id | | - | - |
| Measurement bandwidth | $n_{PRB}$ | 22—27 | 22—27 |
| PDSCH Reference measurement channel defined in A.3.1.1.2 | | - | - |
| PDSCH allocation | $n_{PRB}$ | - | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.2 | | R.6 TDD | R.6 TDD |
| OCNG Patterns defined in A.3.2.2.1 (OP.1 TDD) and A.3.2.2.2 (OP.2 TDD) | | OP.2 TDD | OP.2 TDD |
| PBCH_RA | dB | 0 | 0 |
| PBCH_RB | | | |
| PSS_RA | | | |
| SSS_RA | | | |
| PCFICH_RB | | | |
| PHICH_RA | | | |
| PHICH_RB | | | |
| PDCCH_RA | | | |
| PDCCH_RB | | | |
| PDSCH_RA | | | |
| PDSCH_RB | | | |
| OCNG_RA[Note2] | | | |
| OCNG_RB[Note2] | | | |
| $N_{oc}$ [Note3] | dBm/15 kHz | -88.95 | -112 |
| $\hat{E}_s / I_{ot}$ | dB | 10 | -3.2 |
| RSRP[Note4] | dBm/15 kHz | -78.95 | -115.2 |
| Io[Note4] | dBm/9 MHz | -50.75 | -82.52 |
| $\hat{E}_s / N_{oc}$ | dB | 10 | -3.2 |
| Propagation condition | - | AWGN | AWGN |

NOTE 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in 3GPP TS 36.211.
NOTE 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 9.1.5.1_1.5-3: RSRP FDD-TDD Inter frequency absolute accuracy requirements for the reported values**

| | Test 1 | Test 2 |
|---|---|---|
| Normal Conditions | | |
| Lowest reported value (Cell 2) | RSRP_52 | TBD |
| Highest reported value (Cell 2) | RSRP_71 | TBD |
| Extreme Conditions | | |
| Lowest reported value (Cell 2) | RSRP_49 | RSRP_15 |
| Highest reported value (Cell 2) | RSRP_74 | RSRP_36 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.5.2 FDD - TDD Inter Frequency Relative Accuracy of RSRP

### 9.1.5.2.1 Test purpose

To verify that the FDD-TDD inter-frequency relative accuracy measurement of RSRP is within the specified limit for all bands.

### 9.1.5.2.2 Test applicability

This test applies to all types of E-UTRA UE supporting FDD and TDD from release 9 to release 11. Applicability requires support for FGI bit 25.

### 9.1.5.2.3 Minimum conformance requirements

The relative accuracy of RSRP in inter frequency case is defined as the RSRP measured from one cell compared to the RSRP measured from another cell on a different frequency.

The accuracy requirements in table 9.1.5.2.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP1,2|_{dBm}$ according to Annex I.3.4 for a corresponding Band.

$$\left| RSRP1 \right|_{dBm} - RSRP2 \left|_{dBm} \right| \leq 27dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.5.2.3-1: RSRP FDD Inter frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | $dBm/BW_{Channel}$ |
| ±6 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.1.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.3.2, clause 9.1.4 and A.9.1.5.

### 9.1.5.2.4 Test description

#### 9.1.5.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.5.2.4.3.

4. Cell 1 and Cell 2 are on the different carrier frequencies. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is FDD cell and Cell 2 is TDD cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.1.5.2.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.1.5.2.5-1 and Table 9.1.5.2.5-2 for the test interval as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the RSRP value of Cell 1 and Cell 2 in periodical MeasurementReport messages. The reported RSRP value for Cell 2 is compared to the reported RSRP value for Cell 1 for each MeasurementReport message according to Table 9.1.5.2.5-3. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.5.2.5-1 and Table 9.1.5.2.5-2 as appropriate.

#### 9.1.5.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.5.2.4.3-1: Common Exception messages for RSRP FDD-TDD Inter frequency relative accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-2<br>Table H.3.5-3 |

**Table 9.1.5.2.4.3-2: *MeasResults*: Additional RSRP FDD-TDD Inter frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.5.2.4.3-3: *MeasResultListEUTRA*: Additional RSRP FDD-TDD Inter frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | Cell 2 | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

### 9.1.5.2.5 Test requirement

Table 9.1.5.2.5-1 and Table 9.1.5.2.5-2 define the primary level settings including test tolerances for all tests.

Each RSRP FDD-TDD inter-frequency relative accuracy test shall meet the reported values test requirements in table 9.1.5.2.5-3.

**Table 9.1.5.2.5-1: RSRP FDD—TDD Inter frequency test parameters (FDD Cell1)**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | | Cell 1 | |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Gap Pattern Id | | 0 | | 0 | |
| Measurement bandwidth | $n_{PRB}$ | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.3.1.1.1 | | R.0 FDD | | R.0 FDD | |
| PDSCH allocation | $n_{PRB}$ | 13—36 | | 13—36 | |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.1 | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1 FDD) and A.3.2.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.1 FDD | |
| PBCH_RA | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RANote1 | | | | | |
| OCNG_RBNote | | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -89.25 | | -105.0 | |
| $\hat{E}_s / I_{ot}$ | dB | 10.00 | | 13.00 | |
| RSRP Note3 | dBm/15 kHz | -79.25 | | -92.00 | |
| Io Note3 | dBm/9 MHz | -51.05 | | -64.01 | |
| $\hat{E}_s / N_{oc}$ | dB | 10.00 | | 13.00 | |
| Propagation condition | - | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 9.1.5.2.5-2: RSRP FDD—TDD Inter frequency test parameters (TDD cell2)**

| Parameter | Unit | Test 1 | Test 2 |
|---|---|---|---|
| | | Cell 2 | Cell 2 |
| E-UTRA RF Channel Number | | 2 | 2 |
| $BW_{channel}$ | MHz | 10 | 10 |
| Special subframe configuration[Note1] | | 6 | 6 |
| Uplink-downlink configuration[Note1] | | 1 | 1 |
| Gap Pattern Id | | - | - |
| Measurement bandwidth | $n_{PRB}$ | 22—27 | 22—27 |
| PDSCH Reference measurement channel defined in A.3.1.1.2 | | - | - |
| PDSCH allocation | $n_{PRB}$ | - | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.2 | | R.6 TDD | R.6 TDD |
| OCNG Patterns defined in A.3.2.2.1 (OP.1 TDD) and A.3.2.2.2 (OP.2 TDD) | | OP.2 TDD | OP.2 TDD |
| PBCH_RA | dB | 0 | 0 |
| PBCH_RB | | | |
| PSS_RA | | | |
| SSS_RA | | | |
| PCFICH_RB | | | |
| PHICH_RA | | | |
| PHICH_RB | | | |
| PDCCH_RA | | | |
| PDCCH_RB | | | |
| PDSCH_RA | | | |
| PDSCH_RB | | | |
| OCNG_RA[Note2] | | | |
| OCNG_RB[Note2] | | | |
| $N_{oc}$ [Note3] | dBm/15 kHz | -89.25 | -112.00 |
| $\hat{E}_s / I_{ot}$ | dB | 10.00 | -3.20 |
| RSRP[Note4] | dBm/15 kHz | -79.25 | -115.20 |
| Io[Note4] | dBm/9 MHz | -51.05 | -82.52 |
| $\hat{E}_s / N_{oc}$ | dB | 10.00 | -3.20 |
| Propagation condition | - | AWGN | AWGN |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in 3GPP TS 36.211.

Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 9.1.5.2.5-3: RSRP FDD-TDD Inter frequency relative accuracy requirements for the reported values**

| | Test 1 | Test 2 |
|---|---|---|
| Normal Conditions | | |
| Lowest reported value (Cell 2) | RSRP_(x - 9) | RSRP_(x - 32) |
| Highest reported value (Cell 2) | RSRP_(x + 9) | RSRP_(x - 16) |
| Extreme Conditions | | |
| Lowest reported value (Cell 2) | RSRP_(x -9) | RSRP_(x - 32) |
| Highest reported value (Cell 2) | RSRP_(x + 9) | RSRP_(x - 16) |
| RSRP_x is the reported value of Cell 1 | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.5.2_1 FDD - TDD Inter Frequency Relative Accuracy of RSRP (Rel-12 and forward)

Editor's note: This test case is incomplete for frequencies above 3GHz

- The test system uncertainties applicable above 3GHz are undefined

- The test tolerances and test requirements applicable above 3GHz are undefined

- Test requirements for Test 1 and 2 in normal conditions are TBD

### 9.1.5.2_1.1 Test purpose

Same test purpose as in clause 9.1.5.2.1.

### 9.1.5.2_1.2 Test applicability

This test applies to all types of E-UTRA UE supporting FDD and TDD release 12 and forward. Applicability requires support for FGI bit 25.

### 9.1.5.2_1.3 Minimum conformance requirements

The relative accuracy of RSRP in inter frequency case is defined as the RSRP measured from one cell compared to the RSRP measured from another cell on a different frequency.

The accuracy requirements in table 9.1.5.2_1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 Section 7.3 for reference sensitivity are fulfilled.

RSRP1,2$|_{dBm}$ according to Annex I.3.4 for a corresponding Band.

$$\left| RSRP1|_{dBm} - RSRP2|_{dBm} \right| \leq 27dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.5.2_1.3-1: RSRP FDD Inter frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | $dBm/BW_{Channel}$ |
| ±4.5 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.1.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.3.2, clause 9.1.4 and A.9.1.5.

## 9.1.5.2_1.4 Test description

### 9.1.5.2_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.5.2.4.1 with the following exceptions:

- Instead of 9.1.5.2.4.3 → use 9.1.5.2_1.4.3.

### 9.1.5.2_1.4.2 Test procedure

Same test procedure as in clause 9.1.5.2.4.2 with the following exceptions:

- Instead of Table 9.1.5.2.5-1→ use Table 9.1.5.2_1.5-1.
- Instead of Table 9.1.5.2.5-2 → use Table 9.1.5.2_1.5-2.
- Instead of Table 9.1.5.2.5-3 → use Table 9.1.5.2_1.5-3.

### 9.1.5.2_1.4.3 Message contents

Same message contents as in clause 9.1.5.2.4.3.

## 9.1.5.2_1.5 Test requirement

Table 9.1.5.2_1.5-1 and Table 9.1.5.2_1.5-2 define the primary level settings including test tolerances for all tests.

Each RSRP FDD-TDD inter-frequency relative accuracy test shall meet the reported values test requirements in table 9.1.5.2_1.5-3.

**Table 9.1.5.2_1.5-1: RSRP FDD—TDD Inter frequency test parameters (FDD Cell1)**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | | Cell 1 | |
| E-UTRA RF Channel Number | | 1 | | 1 | |
| $BW_{channel}$ | MHz | 10 | | 10 | |
| Gap Pattern Id | | 0 | | 0 | |
| Measurement bandwidth | $n_{PRB}$ | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.3.1.1.1 | | R.0 FDD | | R.0 FDD | |
| PDSCH allocation | $n_{PRB}$ | 13—36 | | 13—36 | |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.1 | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1 FDD) and A.3.2.1.2 (OP.2 FDD) | | OP.1 FDD | | OP.1 FDD | |
| PBCH_RA | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RANote1 | | | | | |
| OCNG_RBNote | | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -88.95 | | -104.60 | |
| $\hat{E}_s / I_{ot}$ | dB | 10.00 | | 13.00 | |
| RSRP Note3 | dBm/15 kHz | -78.95 | | -91.60 | |
| Io Note3 | dBm/9 MHz | -50.75 | | -63.61 | |
| $\hat{E}_s / N_{oc}$ | dB | 10.00 | | 13.00 | |
| Propagation condition | - | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 9.1.5.2_1.5-2: RSRP FDD—TDD Inter frequency test parameters (TDD cell2)**

| Parameter | Unit | Test 1 | Test 2 |
|---|---|---|---|
| | | Cell 2 | Cell 2 |
| E-UTRA RF Channel Number | | 2 | 2 |
| $BW_{channel}$ | MHz | 10 | 10 |
| Special subframe configuration[Note1] | | 6 | 6 |
| Uplink-downlink configuration[Note1] | | 1 | 1 |
| Gap Pattern Id | | - | - |
| Measurement bandwidth | $n_{PRB}$ | 22—27 | 22—27 |
| PDSCH Reference measurement channel defined in A.3.1.1.2 | | - | - |
| PDSCH allocation | $n_{PRB}$ | - | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.2 | | R.6 TDD | R.6 TDD |
| OCNG Patterns defined in A.3.2.2.1 (OP.1 TDD) and A.3.2.2.2 (OP.2 TDD) | | OP.2 TDD | OP.2 TDD |
| PBCH_RA | dB | 0 | 0 |
| PBCH_RB | | | |
| PSS_RA | | | |
| SSS_RA | | | |
| PCFICH_RB | | | |
| PHICH_RA | | | |
| PHICH_RB | | | |
| PDCCH_RA | | | |
| PDCCH_RB | | | |
| PDSCH_RA | | | |
| PDSCH_RB | | | |
| OCNG_RA[Note2] | | | |
| OCNG_RB[Note2] | | | |
| $N_{oc}$ [Note3] | dBm/15 kHz | -88.95 | -112.00 |
| $\hat{E}_s / I_{ot}$ | dB | 10.00 | -3.20 |
| RSRP[Note4] | dBm/15 kHz | -78.95 | -115.20 |
| Io[Note4] | dBm/9 MHz | -50.75 | -82.52 |
| $\hat{E}_s / N_{oc}$ | dB | 10.00 | -3.20 |
| Propagation condition | - | AWGN | AWGN |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in 3GPP TS 36.211.

Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 9.1.5.2_1.5-3: RSRP FDD-TDD Inter frequency relative accuracy requirements for the reported values**

| | Test 1 | Test 2 |
|---|---|---|
| Normal Conditions | | |
| Lowest reported value (Cell 2) | TBD | TBD |
| Highest reported value (Cell 2) | TBD | TBD |
| Extreme Conditions | | |
| Lowest reported value (Cell 2) | RSRP_(x -8) | RSRP_(x - 32) |
| Highest reported value (Cell 2) | RSRP_(x + 8) | RSRP_(x - 16) |
| RSRP_x is the reported value of Cell 1 | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.1.6 FDD RSRP Accuracy E-UTRA for Carrier Aggregation

## 9.1.6.1 FDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation

### 9.1.6.1.1 Test purpose

To verify that FDD absolute RSRP measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the FDD absolute RSRP accuracy requirements of cells on the primary component carrier and the secondary component carrier.

### 9.1.6.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and release 11 supporting CA with Intra-band contiguous CA, or Inter-band CA.

This test applies to all types of E-UTRA FDD UE release 11 supporting CA with Intra-band non-contiguous CA.

### 9.1.6.1.3 Minimum conformance requirements

The FDD RSRP measurements of cells on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1. The FDD RSRP measurements of cells on the secondary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1.

The accuracy requirements in table 9.1.6.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.1.6.1.3-1: RSRP FDD Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 3] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 2] | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±6 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | -70 | FDD_B | -120.5 | N/A |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_B FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3.<br>NOTE 3: E-UTRA operating band groups are as defined in Section 3.5. | | | | | | |

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.6.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.6.1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1, clause 9.1.11.2, clause 9.1.4, and A.9.1.6

### 9.1.6.1.4 Test description

#### 9.1.6.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.6.1.4.3.

4. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier and Cell 3 is the neighbouring cell on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.1.6.1.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCell according to Annex C.0, C.1 [and C.2] for all downlink physical channels.

3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).

5. Set the parameters according to Table 9.1.6.1.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.

6. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP values for Cell 1 and Cell 2 are compared to the actual RSRP values according to Table 9.1.6.1.5-3. This counts respectively as a Pass or Fail for the events “Cell 1” and “Cell 2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations for the respective event ”Cell 2” is increased by one.

10. SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

11. If test configuration is Inter-band carrier aggregation, repeat steps 1-10 for switched PCell/SCell scenario according to the UE declared capability for UL support (within CA operation) in the individual bands.

Each of the events “Cell 1” and “Cell 2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If both events (“Cell 1” and “Cell 2”) pass for each configuration (without and with switched PCell/SCell scenario), the test passes. If one event fails, the test fails.

#### 9.1.6.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.6.1.4.3-1: Common Exception messages for FDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-3<br>Table H.4.1-5<br>Table H.4.2-1 |

### 9.1.6.1.5 Test requirement

Table 9.1.6.1.5-2 defines the primary level settings including test tolerances for all tests.

The RSRP FDD absolute accuracy test shall meet the reported values test requirements in table 9.1.6.1.5-3.

**Table 9.1.6.1.5-1: Void**

**Table 9.1.6.1.5-2: RSRP FDD absolute accuracy carrier aggregation test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell3 |
| E-UTRA RF Channel Number | | | 1 | 2 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 | 10 |
| Timing offset to cell1 | | µs | - | 0 | 3µs or 92*Ts |
| Time alignment error between cell 2 and cell 1 | | | - | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1. | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RANote1 | | | | | |
| OCNG_RBNote | | | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -117 | ( $N_{oc}$ for Channel 1 +1dB) | |
| | Bands FDD_B | | -116.5 | | |
| | Bands FDD_C | | -116 | | |
| | Bands FDD_D | | -115.5 | | |
| | Bands FDD_E, FDD_F Note 6 | | -115 | | |
| | Bands FDD_G | | -114 | | |
| | Bands FDD_H | | -113.5 | | |
| $\hat{E}_s / I_{ot}$ | | dB | -4.00 | 0.46 | -5.76 |
| RSRP Note3 | Bands FDD_A | dBm/15 kHz | -121 | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4dB) |
| | Bands FDD_B | | -120.5 | | |
| | Bands FDD_C | | -120 | | |
| | Bands FDD_D | | -119.5 | | |
| | Bands FDD_E, FDD_F Note 6 | | -119 | | |
| | Bands FDD_G | | -118 | | |
| | Bands FDD_H | | -117.5 | | |

| | | | | | |
|---|---|---|---|---|---|
| Io[Note3] | Bands FDD_A | dBm/9 MHz | -87.76 | (Io for Channel 1 +5.33dB) | |
| | Bands FDD_B | | -87.26 | | |
| | Bands FDD_C | | -86.76 | | |
| | Bands FDD_D | | -86.26 | | |
| | Bands FDD_E, FDD_F [Note 6] | | -85.76 | | |
| | Bands FDD_G | | -84.76 | | |
| | Bands FDD_H | | -84.26 | | |
| $\hat{E}_s / N_{oc}$ | | dB | -4 | 3 | -1 |
| Propagation condition | | - | AWGN | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: The selection of the bands for testing depends on the configuration of the carrier aggregations supported by the UEs.
Note 6: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 7: Band 29 is used only for E-UTRA carrier aggregation with other E-UTRA bands.
Note 8: If test configuration is Inter-band carrier aggregation, the frequencies of PCell and SCell shall be switched and tested for each configuration according to the UE declared capability for UL support (within CA operation) in the individual bands.
Note 9: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 10: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.6.1.5-3: RSRP FDD Carrier Aggregation absolute accuracy requirements for the reported values**

| | Test 1 | | | | | | |
|---|---|---|---|---|---|---|---|
| **Band of Cell 1 on Primary Component Carrier** | **Bands FDD_A** | **Bands FDD_B** | **Bands FDD_C** | **Bands FDD_D** | **Bands FDD_E, FDD_F** | **Bands FDD_G** | **Bands FDD_H** |
| Normal Conditions | | | | | | | |
| Lowest reported value (Cell 1) | RSRP_12 | RSRP_12 | RSRP_13 | RSRP_13 | RSRP_14 | RSRP_15 | RSRP_15 |
| Highest reported value (Cell 1) | RSRP_27 | RSRP_28 | RSRP_28 | RSRP_29 | RSRP_29 | RSRP_30 | RSRP_31 |
| Lowest reported value (Cell 2) | RSRP_20 | RSRP_20 | RSRP_21 | RSRP_21 | RSRP_22 | RSRP_23 | RSRP_23 |
| Highest reported value (Cell 2) | RSRP_35 | RSRP_36 | RSRP_36 | RSRP_37 | RSRP_37 | RSRP_38 | RSRP_39 |
| Extreme Conditions | | | | | | | |
| Lowest reported value (Cell 1) | RSRP_9 | RSRP_9 | RSRP_10 | RSRP_10 | RSRP_11 | RSRP_12 | RSRP_12 |
| Highest reported value (Cell 1) | RSRP_30 | RSRP_31 | RSRP_31 | RSRP_32 | RSRP_32 | RSRP_33 | RSRP_34 |
| Lowest reported value (Cell 2) | RSRP_17 | RSRP_17 | RSRP_18 | RSRP_18 | RSRP_19 | RSRP_20 | RSRP_20 |
| Highest reported value (Cell 2) | RSRP_38 | RSRP_39 | RSRP_39 | RSRP_40 | RSRP_40 | RSRP_41 | RSRP_42 |

Note: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2.

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.6.1_1 FDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation (Rel-12 and forward)

### 9.1.6.1_1.1 Test purpose

Same test purpose as in clause 9.1.6.1.1.

### 9.1.6.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward supporting CA.

### 9.1.6.1_1.3 Minimum conformance requirements

The FDD RSRP measurements of cells on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1. The FDD RSRP measurements of cells on the secondary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1.

The accuracy requirements in table 9.1.6.1_1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.1.6.1_1.3-1: RSRP Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 / dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 / N/A | -70 |
| | | | FDD_B | -120.5 / N/A | -70 |
| | | | FDD_C, TDD_C | -120 / N/A | -70 |
| | | | FDD_D | -119.5 / N/A | -70 |
| | | | FDD_E, TDD_E | -119 / N/A | -70 |
| | | | FDD_F | -118.5 / N/A | -70 |
| | | | FDD_G | -118 / N/A | -70 |
| | | | FDD_H | -117.5 / N/A | -70 |
| | | | FDD_N | -114.5 / N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.6.1_1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.6.1_1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1, clause 9.1.11.2, clause 9.1.4, and A.9.1.6.

## 9.1.6.1_1.4 Test description

### 9.1.6.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.6.1.4.1 with the following exceptions:

- Instead of 9.1.6.1.4.3 → use 9.1.6.1_1.4.3.

### 9.1.6.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.6.1.4.2 with the following exceptions:

- Instead of Table 9.1.6.1.5-2 → use Table 9.1.6.1_1.5-1.
- Instead of Table 9.1.6.1.5-3 → use Table 9.1.6.1_1.5-2.

### 9.1.6.1_1.4.3 Message contents

Same message contents as in clause 9.1.6.1.4.3.

## 9.1.6.1_1.5 Test requirement

Table 9.1.6.1_1.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP FDD absolute accuracy test shall meet the reported values test requirements in table 9.1.6.1_1.5-2.

**Table 9.1.6.1_1.5-1: RSRP FDD absolute accuracy carrier aggregation test parameters**

| Parameter | Unit | Test 1 | | |
|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell3 |

| | | | | | |
|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | 2 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 | 10 |
| Timing offset to cell1 | | µs | - | 0 | 3µs or 92*Ts |
| Time alignment error between cell 2 and cell 1 | | | - | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1. | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RANote1 | | | | | |
| OCNG_RBNote | | | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -117 | ( $N_{oc}$ for Channel 1 +1dB) | |
| | Bands FDD_B | | -116.5 | | |
| | Bands FDD_C | | -116 | | |
| | Bands FDD_D | | -115.5 | | |
| | Bands FDD_E, FDD_F Note 6 | | -115 | | |
| | Bands FDD_G | | -114 | | |
| | Bands FDD_H | | -113.5 | | |
| $\hat{E}_s / I_{ot}$ | | dB | -4.00 | 0.46 | -5.76 |
| RSRPNote3 | Bands FDD_A | dBm/15 kHz | -121 | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4dB) |
| | Bands FDD_B | | -120.5 | | |
| | Bands FDD_C | | -120 | | |
| | Bands FDD_D | | -119.5 | | |
| | Bands FDD_E, FDD_F Note 6 | | -119 | | |
| | Bands FDD_G | | -118 | | |
| | Bands FDD_H | | -117.5 | | |
| IoNote3 | Bands FDD_A | dBm/9 MHz | -87.76 | (Io for Channel 1 +5.33dB) | |
| | Bands FDD_B | | -87.26 | | |
| | Bands FDD_C | | -86.76 | | |
| | Bands FDD_D | | -86.26 | | |
| | Bands FDD_E, FDD_F Note 6 | | -85.76 | | |
| | Bands FDD_G | | -84.76 | | |
| | Bands FDD_H | | -84.26 | | |

| | | | | |
|---|---|---|---|---|
| $\hat{E}_s / N_{oc}$ | dB | -4 | 3 | -1 |
| Propagation condition | - | AWGN | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: The selection of the bands for testing depends on the configuration of the carrier aggregations supported by the UEs.
Note 6: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 7: Band 29 is used only for E-UTRA carrier aggregation with other E-UTRA bands.
Note 8: If test configuration is Inter-band carrier aggregation, the frequencies of PCell and SCell shall be switched and tested for each configuration according to the UE declared capability for UL support (within CA operation) in the individual bands.
Note 9: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 10: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.6.1_1.5-2: RSRP FDD Carrier Aggregation absolute accuracy requirements for the reported values**

| | **Test 1** | | | | | | |
|---|---|---|---|---|---|---|---|
| **Band of Cell 1 on Primary Component Carrier** | **Bands FDD_A** | **Bands FDD_B** | **Bands FDD_C** | **Bands FDD_D** | **Bands FDD_E, FDD_F** | **Bands FDD_G** | **Bands FDD_H** |
| Normal Conditions | | | | | | | |
| Lowest reported value (Cell 1) | RSRP_13 | RSRP_14 | RSRP_14 | RSRP_15 | RSRP_15 | RSRP_16 | RSRP_17 |
| Highest reported value (Cell 1) | RSRP_26 | RSRP_26 | RSRP_27 | RSRP_27 | RSRP_28 | RSRP_29 | RSRP_29 |
| Lowest reported value (Cell 2) | RSRP_21 | RSRP_22 | RSRP_22 | RSRP_23 | RSRP_23 | RSRP_24 | RSRP_25 |
| Highest reported value (Cell 2) | RSRP_34 | RSRP_34 | RSRP_35 | RSRP_35 | RSRP_36 | RSRP_37 | RSRP_37 |
| Extreme Conditions | | | | | | | |
| Lowest reported value (Cell 1) | RSRP_9 | RSRP_9 | RSRP_10 | RSRP_10 | RSRP_11 | RSRP_12 | RSRP_12 |
| Highest reported value (Cell 1) | RSRP_30 | RSRP_31 | RSRP_31 | RSRP_32 | RSRP_32 | RSRP_33 | RSRP_34 |
| Lowest reported value (Cell 2) | RSRP_17 | RSRP_17 | RSRP_18 | RSRP_18 | RSRP_19 | RSRP_20 | RSRP_20 |
| Highest reported value (Cell 2) | RSRP_38 | RSRP_39 | RSRP_39 | RSRP_40 | RSRP_40 | RSRP_41 | RSRP_42 |

Note: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2.

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.6.2 FDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation

### 9.1.6.2.1 Test purpose

To verify that FDD relative RSRP measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the FDD relative RSRP accuracy requirements of cells between the primary and secondary component carrier and FDD relative RSRP accuracy between the secondary component carriers.

### 9.1.6.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and release 11 supporting CA with Intra-band contiguous CA, or Inter-band CA.

This test applies to all types of E-UTRA FDD UE release 11 supporting CA with Intra-band non-contiguous CA.

### 9.1.6.2.3 Minimum conformance requirements

The FDD RSRP relative measurements of cells on PCC and SCC shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.3.2. The FDD RSRP relative measurements of cells on the SCC shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.2.2.

The accuracy requirements in Table 9.1.6.2.3-1 for PCC-SCC relative accuracy requirements are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

RSRP1,2$|_{dBm}$ according to Annex I.3.4 for a corresponding Band

$$\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.6.2.3-1: FDD RSRP relative accuracy for PCC and SCC**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot Note 2 | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/BW$_{Channel}$ |
| ±6 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The requirements in Table 9.1.6.2.3-2 for SCC relative accuracy are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

RSRP1,2$|_{dBm}$ according to Annex I.3.8 for a corresponding Band.

**Table 9.1.6.2.3-2: FDD RSRP relative accuracy for SCCs**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 6 | $dBm/BW_{Channel}$ |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 Note 3 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 4 | Note 4 | Note 4 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The condition has the minimum Io of -119 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz.
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.
NOTE 6: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.2, clause 9.1.11.3, clause 9.1.4, and A.9.1.6.

### 9.1.6.2.4 Test description

#### 9.1.6.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.6.2.4.3.

4. Cell1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier and Cell 3 is the neighbouring cell on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.1.6.2.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCell according to Annex C.0, C.1 [and C.2] for all downlink physical channels.

3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).

5. Set the parameters according to Table 9.1.6.2.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.

6. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP value for Cell 2 is compared to the reported RSRP value for Cell 1 for each MeasurementReport message according to Table 9.1.6.2.5-3. Also the reported RSRP value for Cell 3 is compared to the reported RSRP value for Cell 2 for each MeasurementReport message according to Table 9.1.6.2.5-3. This counts respectively as a Pass or Fail for the events “Cell 1-2” and “Cell 2-3”. If the UE fails to report the measurement value for Cell 2 / Cell 3, the number of failed iterations for the respective affected event “Cell 1-2” / ”Cell 2-3” is increased by one.

10. SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

11. If test configuration is Inter-band carrier aggregation, repeat steps 1-10 for switched PCell/SCell scenario according to the UE declared capability for UL support (within CA operation) in the individual bands.

Each of the events “Cell 1-2” and “Cell 2-3” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If both events (“Cell 1-2” and “Cell 2-3”) pass for each configuration (without and with switched PCell/SCell scenario), the test passes. If one event fails, the test fails.

#### 9.1.6.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.6.2.4.3-1: Common Exception messages for FDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-3<br>Table H.4.1-5<br>Table H.4.2-1 |

### 9.1.6.2.5 Test requirement

Table 9.1.6.2.5-2 defines the primary level settings including test tolerances for all tests.

The RSRP FDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.6.2.5-3.

**Table 9.1.6.2.5-1: Void**

**Table 9.1.6.2.5-2: RSRP FDD relative accuracy carrier aggregation test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell3 |
| E-UTRA RF Channel Number | | | 1 | 2 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 | 10 |
| Timing offset to cell1 | | µs | - | 0 | 3µs or 92*Ts |
| Time alignment error between cell 2 and cell 1 | | | - | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1. | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.1 | | | R.6 FDD | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RANote1 | | | | | |
| OCNG_RBNote | | | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -117 | ( $N_{oc}$ for Channel 1 +1dB) | |
| | Bands FDD_B | | -116.5 | | |
| | Bands FDD_C | | -116 | | |
| | Bands FDD_D | | -115.5 | | |
| | Bands FDD_E, FDD_F Note 6 | | -115 | | |
| | Bands FDD_G | | -114 | | |
| | Bands FDD_H | | -113.5 | | |
| $\hat{E}_s / I_{ot}$ | | dB | -4.00 | 0.09 | -4.96 |
| RSRP Note3 | Bands FDD_A | dBm/15 kHz | -121 | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4.8dB) |
| | Bands FDD_B | | -120.5 | | |
| | Bands FDD_C | | -120 | | |
| | Bands FDD_D | | -119.5 | | |
| | Bands FDD_E, FDD_F Note 6 | | -119 | | |
| | Bands FDD_G | | -118 | | |
| | Bands FDD_H | | -117.5 | | |
| Io Note3 | Bands FDD_A | dBm/9 MHz | -87.76 | (Io for Channel 1 +5.51dB) | |
| | Bands FDD_B | | -87.26 | | |
| | Bands FDD_C | | -86.76 | | |
| | Bands FDD_D | | -86.26 | | |
| | Bands FDD_E, FDD_F Note 6 | | -85.76 | | |
| | Bands FDD_G | | -84.76 | | |
| | Bands FDD_H | | -84.26 | | |
| $\hat{E}_s / N_{oc}$ | | dB | -4 | 3 | -0.2 |

| | | |
|---|---|---|
| Propagation condition | - | AWGN |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | |
| Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. | | |
| Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | |
| Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. | | |
| Note 5: The selection of the bands for testing depends on the configuration of the carrier aggregations supported by the UEs. | | |
| Note 6: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz | | |
| Note 7: This band is used only for E-UTRA carrier aggregation with other E-UTRA bands. | | |
| Note 8: If test configuration is Inter-band carrier aggregation, the frequencies of PCell and SCell shall be switched and tested for each configuration according to the UE declared capability for UL support (within CA operation) in the individual bands. | | |
| Note 9: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | |
| Note 10: E-UTRA operating band groups are as defined in Section 3.5. | | |

**Table 9.1.6.2.5-3: RSRP FDD Carrier Aggregation relative accuracy requirements for the reported values**

| | **Test 1** |
|---|---|
| | All bands |
| Normal and Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRP_x – 1 |
| Highest reported value (Cell 2) | RSRP_x + 17 |
| Lowest reported value (Cell 3) | RSRP_y – 8 |
| Highest reported value (Cell 3) | RSRP_y + 1 |
| RSRP_x is the reported value of Cell 1<br>RSRP_y is the reported value of Cell 2 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.6.2_1 FDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation (Rel-12 and forward)

### 9.1.6.2_1.1 Test purpose

Same test purpose as defined in clause 9.1.6.2.1.

### 9.1.6.2_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward supporting CA.

### 9.1.6.2_1.3 Minimum conformance requirements

The FDD RSRP relative measurements of cells on PCC and SCC shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.3.2. The FDD RSRP relative measurements of cells on the SCC shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.2.2.

The accuracy requirements in Table 9.1.6.2_1.3-1 for PCC-SCC relative accuracy requirements are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

- RSRP1,2|$_{dBm}$ according to Annex I.3.4 for a corresponding Band
- $\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$
  - | Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.6.2_1.3-1: FDD RSRP relative accuracy for PCC and SCC**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/BW$_{Channel}$ |
| ±4.5 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The requirements in Table 9.1.6.2_1.3-2 for SCC relative accuracy are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.
- RSRP1,2|$_{dBm}$ according to Annex I.3.8 for a corresponding Band.

**Table 9.1.6.2_1.3-2: FDD RSRP relative accuracy for SCCs**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/BW$_{Channel}$ |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 3 | Note 3 | Note 3 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.<br>NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3.<br>NOTE 5: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.2, clause 9.1.11.3, clause 9.1.4, and A.9.1.6.

## 9.1.6.2_1.4 Test description

### 9.1.6.2_1.4.1 Initial conditions

Same initial conditions as defined in clause 9.1.6.2.4.1.

### 9.1.6.2_1.4.2 Test procedure

Same test procedure as defined in clause 9.1.6.2.4.2 with the following exceptions:

- Instead of Table 9.1.6.2.5-2 → use Table 9.1.6.2_1.5-1.
- Instead of Table 9.1.6.2.5-3 → use Table 9.1.6.2_1.5-2.

### 9.1.6.2_1.4.3 Message contents

Same message contents as defined in clause 9.1.6.2.4.3.

## 9.1.6.2_1.5 Test requirement

Table 9.1.6.2_1.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP FDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.6.2_1.5-2.

**Table 9.1.6.2_1.5-1: RSRP FDD relative accuracy carrier aggregation test parameters**

| Parameter | Unit | Test 1 | | |
|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell3 |

| | | | | | |
|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | 2 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 | 10 |
| Timing offset to cell1 | | µs | - | 0 | 3µs or 92*Ts |
| Time alignment error between cell 2 and cell 1 | | | - | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1. | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RANote1 | | | | | |
| OCNG_RBNote | | | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -117 | ( $N_{oc}$ for Channel 1 +1dB) | |
| | Bands FDD_B | | -116.5 | | |
| | Bands FDD_C | | -116 | | |
| | Bands FDD_D | | -115.5 | | |
| | Bands FDD_E, FDD_F Note 6 | | -115 | | |
| | Bands FDD_G | | -114 | | |
| | Bands FDD_H | | -113.5 | | |
| $\hat{E}_s / I_{ot}$ | | dB | -4.00 | 0.09 | -4.96 |
| RSRPNote3 | Bands FDD_A | dBm/15 kHz | -121 | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4.8dB) |
| | Bands FDD_B | | -120.5 | | |
| | Bands FDD_C | | -120 | | |
| | Bands FDD_D | | -119.5 | | |
| | Bands FDD_E, FDD_F Note 6 | | -119 | | |
| | Bands FDD_G | | -118 | | |
| | Bands FDD_H | | -117.5 | | |
| IoNote3 | Bands FDD_A | dBm/9 MHz | -87.76 | (Io for Channel 1 +5.51dB) | |
| | Bands FDD_B | | -87.26 | | |
| | Bands FDD_C | | -86.76 | | |
| | Bands FDD_D | | -86.26 | | |
| | Bands FDD_E, FDD_F Note 6 | | -85.76 | | |
| | Bands FDD_G | | -84.76 | | |
| | Bands FDD_H | | -84.26 | | |

| | | | | |
|---|---|---|---|---|
| $\hat{E}_s / N_{oc}$ | dB | -4 | 3 | -0.2 |
| Propagation condition | - | AWGN | | |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | | | |
| Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. | | | | |
| Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | | |
| Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. | | | | |
| Note 5: The selection of the bands for testing depends on the configuration of the carrier aggregations supported by the UEs. | | | | |
| Note 6: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz | | | | |
| Note 7: This band is used only for E-UTRA carrier aggregation with other E-UTRA bands. | | | | |
| Note 8: If test configuration is Inter-band carrier aggregation, the frequencies of PCell and SCell shall be switched and tested for each configuration according to the UE declared capability for UL support (within CA operation) in the individual bands. | | | | |
| Note 9: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | | |
| Note 10: E-UTRA operating band groups are as defined in Section 3.5. | | | | |

**Table 9.1.6.2_1.5-2: RSRP FDD Carrier Aggregation relative accuracy requirements for the reported values**

| | **Test 1** |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 2) | RSRP_x + 2 |
| Highest reported value (Cell 2) | RSRP_x + 15 |
| Lowest reported value (Cell 3) | RSRP_y - 8 |
| Highest reported value (Cell 3) | RSRP_y + 1 |
| Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRP_x – 1 |
| Highest reported value (Cell 2) | RSRP_x + 17 |
| Lowest reported value (Cell 3) | RSRP_y – 8 |
| Highest reported value (Cell 3) | RSRP_y + 1 |
| RSRP_x is the reported value of Cell 1<br>RSRP_y is the reported value of Cell 2 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.7 TDD RSRP Accuracy E-UTRA for Carrier Aggregation

### 9.1.7.1 TDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation

#### 9.1.7.1.1 Test purpose

To verify that TDD absolute RSRP measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the TDD absolute RSRP accuracy requirements of cells on the primary component carrier and the secondary component carrier.

#### 9.1.7.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and release 11 supporting CA with Intra-band contiguous CA, or Inter-band CA.

This test applies to all types of E-UTRA TDD UE release 11 supporting CA with Intra-band non-contiguous CA.

#### 9.1.7.1.3 Minimum conformance requirements

The TDD RSRP measurements of cells on the primary component carrier shall meet the absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1. The TDD RSRP measurements of cells on the secondary component carrier shall meet the absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1.

The accuracy requirements in table 9.1.7.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.1.7.1.3-1: RSRP TDD absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | $dBm/BW_{Channel}$ | $dBm/BW_{Channel}$ |
| ±6 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_B | -120.5 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_B FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.7.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.7.1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1, clause 9.1.11.2, clause 9.1.4, and A.9.1.7.

#### 9.1.7.1.4 Test description

##### 9.1.7.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.7.1.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

##### 9.1.7.1.4.2 Test procedure

The test consists of Cell 1 the PCell, Cell 2 the SCell on the Secondary Component Carrier (SCC), and Cell 3 the neighbouring cell on the SCC. The SCC is configured and activated. The absolute accuracy of RSRP is defined as the RSRP measured from the primary component carrier (Cell 1) and the RSRP measured from the secondary component carrier (Cell 2 and Cell 3).

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [13], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.1.7.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. The SS shall transmit an RRCConnectionReconfiguration message.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. The UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, the SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP values for Cell 1 and Cell 2 are compared to the actual RSRP values according to Table 9.1.7.1.5-2. This counts respectively as a Pass or Fail for the events “Cell 1” and “Cell 2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations for the respective event ”Cell 2” is increased by one.

10. The SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

11. If test configuration is Inter-band carrier aggregation, repeat steps 1-10 for switched PCell/SCell scenario according to the UE declared capability for UL support (within CA operation) in the individual bands.

Each of the events “Cell 1” and “Cell 2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If both events (“Cell 1” and “Cell 2”) pass for each configuration (without and with switched PCell/SCell scenario), the test passes. If one event fails, the test fails.

#### 9.1.7.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.7.1.4.3-1: Common Exception messages for TDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-3<br>Table H.4.1-5<br>Table H.4.2-1 |

### 9.1.7.1.5 Test requirement

Table 9.1.7.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD absolute accuracy test shall meet the reported values test requirements in table 9.1.7.1.5-2.

**Table 9.1.7.1.5-1: RSRP TDD absolute accuracy carrier aggregation test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| E-UTRA RF Channel Number | | | 1 | 2 | |
| $BW_{channel}$ | | MHz | 10 | | |
| Special subframe configuration[Note1] | | | 6 | | |
| Uplink/downlink configuration[Note1] | | | 1 | | |
| Timing offset to Cell 1 | | µs | - | 0 | 3µs or 92*Ts |
| Time alignment error between cell 2 and cell 1 | | | - | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1. | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | |
| PDSCH Reference measurement channel defined in A.3.1.1.2 | | | R.0 TDD | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.2 | | | R.6 TDD | | |
| OCNG Patterns defined in A.3.2.2.1 (OP.1 TDD) and A.3.2.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA[Note2] | | | | | |
| OCNG_RB[Note2] | | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -117 | ($N_{oc}$ for Channel 1 +1dB) | |
| | Bands TDD_C | | -116 | | |
| | Bands TDD_E | | -115 | | |
| $\hat{E}_s / I_{ot}$ | | dB | -4.00 | 0.46 | -5.76 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -121 | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4dB) |
| | Bands TDD_C | | -120 | | |
| | Bands TDD_E | | -119 | | |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -87.76 | (Io for Channel 1 +5.33dB) | |
| | Bands TDD_C | | -86.76 | | |
| | Bands TDD_E | | -85.76 | | |
| $\hat{E}_s / N_{oc}$ | | dB | -4 | 3 | -1 |
| Propagation condition | | - | AWGN | | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in 3GPP TS 36.211.

Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 5: RSRP minimum requirements are specified assuming independent interference and noise

| | |
|---|---|
| | at each receiver antenna port. |
| Note 6: | The selection of the bands for testing depends on the configuration of the carrier aggregations supported by the UEs. |
| Note 7: | If test configuration is Inter-band carrier aggregation, the frequencies of PCell and SCell shall be switched and tested for each configuration according to the UE declared capability for UL support (within CA operation) in the individual bands. |
| Note 8: | This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. |
| Note 9: | E-UTRA operating band groups are as defined in Section 3.5. |

**Table 9.1.7.1.5-2: RSRP TDD Carrier Aggregation absolute accuracy requirements for the reported values**

| Parameter | Test 1 | | |
|---|---|---|---|
| **Band of Cell 1 on Primary Component Carrier** | **Bands TDD_A** | **Bands TDD_C** | **Bands TDD_E** |
| Normal Conditions | | | |
| Lowest reported value (Cell 1) | RSRP_12 | RSRP_13 | RSRP_14 |
| Highest reported value (Cell 1) | RSRP_27 | RSRP_28 | RSRP_29 |
| Lowest reported value (Cell 2) | RSRP_20 | RSRP_21 | RSRP_22 |
| Highest reported value (Cell 2) | RSRP_35 | RSRP_36 | RSRP_37 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 1) | RSRP_9 | RSRP_10 | RSRP_11 |
| Highest reported value (Cell 1) | RSRP_30 | RSRP_31 | RSRP_32 |
| Lowest reported value (Cell 2) | RSRP_17 | RSRP_18 | RSRP_19 |
| Highest reported value (Cell 2) | RSRP_38 | RSRP_39 | RSRP_40 |
| Note: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2. | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.7.1_1 TDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation (Rel-12 and forward)

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

-The Test tolerances applicable to this test are undefined

### 9.1.7.1_1.1 Test purpose

Same test purpose as in clause 9.1.7.1.1.

### 9.1.7.1_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward supporting CA.

### 9.1.7.1_1.3 Minimum conformance requirements

The TDD RSRP measurements of cells on the primary component carrier shall meet the absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1. The TDD RSRP measurements of cells on the secondary component carrier shall meet the absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1.

The accuracy requirements in table 9.1.7.1_1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.1.7.1_1.3-1: RSRP TDD absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | $dBm/BW_{Channel}$ | $dBm/BW_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 3: E-UTRA operating band groups are as defined in Section 3.5. | | | | | | |

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.7.1_1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.7.1_1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1, clause 9.1.11.2, clause 9.1.4, and A.9.1.7.

## 9.1.7.1_1.4 Test description

### 9.1.7.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.7.1.4.1 with the following exceptions:

- Instead of 9.1.7.1.4.3 → use 9.1.7.1_1.4.3.

### 9.1.7.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.7.1.4.2 with the following exceptions:

- Instead of Table 9.1.7.1.5-1 → use Table 9.1.7.1_1.5-1.
- Instead of Table 9.1.7.1.5-2 → use Table 9.1.7.1_1.5-2.

#### 9.1.7.1_1.4.3 Message contents

Same message contents as in clause 9.1.7.1.4.3

### 9.1.7.1_1.5 Test requirement

Table 9.1.7.1_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD absolute accuracy test shall meet the reported values test requirements in table 9.1.7.1_1.5-2.

**Table 9.1.7.1_1.5-1: RSRP TDD absolute accuracy carrier aggregation test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| E-UTRA RF Channel Number | | | 1 | 2 | |
| $BW_{channel}$ | | MHz | 10 | | |
| Special subframe configuration[Note1] | | | 6 | | |
| Uplink/downlink configuration[Note1] | | | 1 | | |
| Timing offset to Cell 1 | | µs | - | 0 | 3µs or 92*Ts |
| Time alignment error between cell 2 and cell 1 | | | - | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1. | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | |
| PDSCH Reference measurement channel defined in A.3.1.1.2 | | | R.0 TDD | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.2 | | | R.6 TDD | | |
| OCNG Patterns defined in A.3.2.2.1 (OP.1 TDD) and A.3.2.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA[Note2] | | | | | |
| OCNG_RB[Note2] | | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -117 | ($N_{oc}$ for Channel 1 +1dB) | |
| | Bands TDD_C | | -116 | | |
| | Bands TDD_E | | -115 | | |
| $\hat{E}_s / I_{ot}$ | | dB | -4.00 | 0.46 | -5.76 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -121 | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4dB) |
| | Bands TDD_C | | -120 | | |
| | Bands TDD_E | | -119 | | |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -87.76 | (Io for Channel 1 +5.33dB) | |
| | Bands TDD_C | | -86.76 | | |
| | Bands TDD_E | | -85.76 | | |
| $\hat{E}_s / N_{oc}$ | | dB | -4 | 3 | -1 |
| Propagation condition | | - | AWGN | | |

NOTE 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in 3GPP TS 36.211.
NOTE 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 5: RSRP minimum requirements are specified assuming independent interference and noise

at each receiver antenna port.
NOTE 6: The selection of the bands for testing depends on the configuration of the carrier aggregations supported by the UEs.
NOTE 7: If test configuration is Inter-band carrier aggregation, the frequencies of PCell and SCell shall be switched and tested for each configuration according to the UE declared capability for UL support (within CA operation) in the individual bands.
NOTE 8: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
NOTE 9: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.7.1_1.5-2: RSRP TDD Carrier Aggregation absolute accuracy requirements for the reported values**

| Parameter | Test 1 | | |
|---|---|---|---|
| **Band of Cell 1 on Primary Component Carrier** | **Bands TDD_A** | **Bands TDD_C** | **Bands TDD_E** |
| Normal Conditions | | | |
| Lowest reported value (Cell 1) | TBD | TBD | TBD |
| Highest reported value (Cell 1) | TBD | TBD | TBD |
| Lowest reported value (Cell 2) | TBD | TBD | TBD |
| Highest reported value (Cell 2) | TBD | TBD | TBD |
| Extreme Conditions | | | |
| Lowest reported value (Cell 1) | RSRP_9 | RSRP_10 | RSRP_11 |
| Highest reported value (Cell 1) | RSRP_30 | RSRP_31 | RSRP_32 |
| Lowest reported value (Cell 2) | RSRP_17 | RSRP_18 | RSRP_19 |
| Highest reported value (Cell 2) | RSRP_38 | RSRP_39 | RSRP_40 |
| NOTE: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2. | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.7.2 TDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation

### 9.1.7.2.1 Test purpose

To verify that TDD relative RSRP measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the TDD relative RSRP accuracy requirements of cells between the primary and secondary component carrier and TDD relative RSRP accuracy of cells on the secondary component carriers.

### 9.1.7.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and release 11 supporting CA with Intra-band contiguous CA, or Inter-band CA.

This test applies to all types of E-UTRA TDD UE release 11 supporting CA with Intra-band non-contiguous CA.

### 9.1.7.2.3 Minimum conformance requirements

The TDD RSRP relative measurements of cells on PCC and SCC shall meet relative accuracy requirements defined in TS 36.133 clause 9.1.3.2. The TDD RSRP relative measurements of cells on the SCC shall meet relative accuracy requirements defined in TS 36.133 clause 9.1.2.2.

The accuracy requirements in Table 9.1.7.2.3-1 for PCC-SCC relative accuracy requirements are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

RSRP1,2|$_{dBm}$ according to Annex I.3.8 for a corresponding Band

$$\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.7.2.3-1: TDD RSRP relative accuracy for PCC and SCC**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io [Note 1] range | | |
| Normal condition | Extreme condition | Ês/Iot [Note 2] | E-UTRA operating band groups [Note 4] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 3] | dBm/BW$_{Channel}$ |
| ±6 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.3, clause 9.1.4, and A.9.1.7.

The TDD RSRP relative measurements of cells on SCCs shall meet relative accuracy requirements defined in TS 36.133 clause 9.1.2.2.

The accuracy requirements in Table 9.1.7.2.3-2 for SCCs relative accuracy requirements are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

RSRP1,2|$_{dBm}$ according to Annex I.3.8 for a corresponding Band.

**Table 9.1.7.2.3-2: TDD RSRP relative accuracy for SCCs**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/$BW_{Channel}$ |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.2, clause 9.1.11.3, clause 9.1.4, and A.9.1.7.

### 9.1.7.2.4 Test description

#### 9.1.7.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.7.2.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.1.7.2.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [13], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.1.7.2.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. SS shall transmit an RRCConnectionReconfiguration message.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP value for Cell 2 is compared to the reported RSRP value for Cell 1 for each MeasurementReport message according to Table 9.1.7.2.5-2. Also the reported RSRP value for Cell 3 is compared to the reported RSRP value for Cell 2 for each MeasurementReport message according to Table 9.1.7.2.5-2. This counts respectively as a Pass or Fail for the events “Cell 1-2” and “Cell 2-3”. If the UE fails to report the measurement value for Cell 2 / Cell 3, the number of failed iterations for the respective affected event “Cell 1-2” / ”Cell 2-3” is increased by one.

10. SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

11. If test configuration is Inter-band carrier aggregation, repeat steps 1-10 for switched PCell/SCell scenario according to the UE declared capability for UL support (within CA operation) in the individual bands.

Each of the events “Cell 1-2” and “Cell 2-3” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If both events (“Cell 1-2” and “Cell 2-3”) pass for each configuration (without and with switched PCell/SCell scenario), the test passes. If one event fails, the test fails.

#### 9.1.7.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.7.2.4.3-1: Common Exception messages for TDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-3<br>Table H.4.1-5<br>Table H.4.2-1 |

### 9.1.7.2.5 Test requirement

Table 9.1.7.2.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.7.2.5-2.

**Table 9.1.7.2.5-1: RSRP TDD relative accuracy carrier aggregation test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| E-UTRA RF Channel Number | | | 1 | 2 | |
| $BW_{channel}$ | | MHz | 10 | | |
| Special subframe configuration[Note1] | | | 6 | | |
| Uplink/downlink configuration[Note1] | | | 1 | | |
| Time alignment error between cell 2 and cell 1 | | | - | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1. | - |
| Timing offset to Cell 1 | | µs | - | 0 | 3µs or 92*Ts |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | |
| PDSCH Reference measurement channel defined in A.3.1.1.2 | | | R.0 TDD | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.2 | | | R.6 TDD | | |
| OCNG Patterns defined in A.3.2.2.1 (OP.1 TDD) and A.3.2.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA[Note2] | | | | | |
| OCNG_RB[Note2] | | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -117 | ($N_{oc}$ for Channel 1 +1dB) | |
| | Bands TDD_C | | -116 | | |
| | Bands TDD_E | | -115 | | |
| $\hat{E}_s / I_{ot}$ | | dB | -4.00 | 0.09 | -4.96 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -121 | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4.8dB) |
| | Bands TDD_C | | -120 | | |
| | Bands TDD_E | | -119 | | |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -87.76 | (Io for Channel 1 +5.51dB) | |
| | Bands TDD_C | | -86.76 | | |
| | Bands TDD_E | | -85.76 | | |
| $\hat{E}_s / N_{oc}$ | | dB | -4 | 3 | -0.2 |
| Propagation condition | | - | AWGN | | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in 3GPP TS 36.211.

Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 5: RSRP minimum requirements are specified assuming independent interference and noise

| | |
|---|---|
| | at each receiver antenna port. |
| Note 6: | The selection of the bands for testing depends on the configuration of the carrier aggregations supported by the UEs. |
| Note 7: | If test configuration is Inter-band carrier aggregation, the frequencies of PCell and SCell shall be switched and tested for each configuration according to the UE declared capability for UL support (within CA operation) in the individual bands. |
| Note 8: | This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. |
| Note 9: | E-UTRA operating band groups are as defined in Section 3.5. |

**Table 9.1.7.2.5-2: RSRP TDD Carrier Aggregation relative accuracy requirements for the reported values**

| | **Test 1** |
|---|---|
| | All bands |
| Normal and Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRP_x – 1 |
| Highest reported value (Cell 2) | RSRP_x + 17 |
| Lowest reported value (Cell 3) | RSRP_y – 8 |
| Highest reported value (Cell 3) | RSRP_y + 1 |
| RSRP_x is the reported value of Cell 1<br>RSRP_y is the reported value of Cell 2 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.7.2_1 TDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation (Rel-12 and forward)

Editor's note: This Test case is incomplete. The following aspects are either missing or not yet determined:

- *The Test system uncertainties applicable to this test are undefined*
- *The Test tolerances applicable to this test are undefined*
- *Test requirements in normal conditions need to be defined*

### 9.1.7.2_1.1 Test purpose

Same test purpose as defined in clause 9.1.7.2.1.

### 9.1.7.2_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward supporting CA.

### 9.1.7.2_1.3 Minimum conformance requirements

The TDD RSRP relative measurements of cells on PCC and SCC shall meet relative accuracy requirements defined in TS 36.133 clause 9.1.3.2. The TDD RSRP relative measurements of cells on the SCC shall meet relative accuracy requirements defined in TS 36.133 clause 9.1.2.2.

The accuracy requirements in Table 9.1.7.2_1.3-1 for PCC-SCC relative accuracy requirements are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.
- $RSRP1,2|_{dBm}$ according to Annex I.3.8 for a corresponding Band

- $\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$

- | Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.7.2_1.3-1: TDD RSRP relative accuracy for PCC and SCC**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/BW$_{Channel}$ |
| ±4.5 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.3, clause 9.1.4, and A.9.1.7.

The TDD RSRP relative measurements of cells on SCCs shall meet relative accuracy requirements defined in TS 36.133 clause 9.1.2.2.

The accuracy requirements in Table 9.1.7.2_1.3-2 for SCCs relative accuracy requirements are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.

- Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

- RSRP1,2|$_{dBm}$ according to Annex I.3.8 for a corresponding Band.

**Table 9.1.7.2_1.3-2: TDD RSRP relative accuracy for SCCs**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/BW$_{Channel}$ |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 3 | Note 3 | Note 3 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.<br>NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 5: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.2, clause 9.1.11.3, clause 9.1.4, and A.9.1.7.

### 9.1.7.2_1.4 Test description

#### 9.1.7.2_1.4.1 Initial conditions

Same initial conditions as defined in clause 9.1.7.2.4.1.

#### 9.1.7.2_1.4.2 Test procedure

Same test procedure as defined in clause 9.1.7.2.4.2 with the following exceptions:

- Instead of Table 9.1.7.2.4.2-1 → use Table 9.1.7.2_1.5-1.
- Instead of Table 9.1.7.2.4.2-2 → use Table 9.1.7.2_1.5-2.

#### 9.1.7.2_1.4.3 Message contents

Same message contents as defined in clause 9.1.7.2.4.2.

### 9.1.7.2_1.5 Test requirement

Table 9.1.7.2_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.7.2_1.5-2.

**Table 9.1.7.2_1.5-1: RSRP TDD relative accuracy carrier aggregation test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| E-UTRA RF Channel Number | | | 1 | 2 | |
| $BW_{channel}$ | | MHz | 10 | | |
| Special subframe configuration[Note1] | | | 6 | | |
| Uplink/downlink configuration[Note1] | | | 1 | | |
| Time alignment error between cell 2 and cell 1 | | | - | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1. | - |
| Timing offset to Cell 1 | | µs | - | 0 | 3µs or 92*Ts |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA[Note2] | | | | | |
| OCNG_RB[Note2] | | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -117+TT | ($N_{oc}$ for Channel 1 +1dB) | |
| | Bands TDD_C | | -116+TT | | |
| | Bands TDD_E | | -115+TT | | |
| $\hat{E}_s / I_{ot}$ | | dB | -4.00+TT | 0.09+TT | -4.96+TT |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -121+TT | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4.8dB) |
| | Bands TDD_C | | -120+TT | | |
| | Bands TDD_E | | -119+TT | | |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -87.76+TT | (Io for Channel 1 +5.51dB) | |
| | Bands TDD_C | | -86.76+TT | | |
| | Bands TDD_E | | -85.76+TT | | |
| $\hat{E}_s / N_{oc}$ | | dB | -4+TT | 3+TT | -0.2+TT |
| Propagation condition | | - | AWGN | | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in 3GPP TS 36.211[9].

Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 5: RSRP minimum requirements are specified assuming independent interference and noise

| | |
|---|---|
| | at each receiver antenna port. |
| Note 6: | The selection of the bands for testing depends on the configuration of the carrier aggregations supported by the UEs. |
| Note 7: | If test configuration is Inter-band carrier aggregation, the frequencies of PCell and SCell shall be switched and tested for each configuration according to the UE declared capability for UL support (within CA operation) in the individual bands. |
| Note 8: | This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. |
| Note 9: | E-UTRA operating band groups are as defined in Section 3.5. |

**Table 9.1.7.2_1.5-2: RSRP TDD Carrier Aggregation relative accuracy requirements for the reported values**

| | **Test 1** |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 2) | TBD |
| Highest reported value (Cell 2) | TBD |
| Lowest reported value (Cell 3) | TBD |
| Highest reported value (Cell 3) | TBD |
| Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRP_x – 1 |
| Highest reported value (Cell 2) | RSRP_x + 17 |
| Lowest reported value (Cell 3) | RSRP_y – 8 |
| Highest reported value (Cell 3) | RSRP_y + 1 |
| RSRP_x is the reported value of Cell 1<br>RSRP_y is the reported value of Cell 2 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.8 FDD RSRP Accuracy E-UTRA under Time-Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC)

### 9.1.8.1 FDD Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC)

#### 9.1.8.1.1 Test purpose

To verify that FDD absolute RSRP measurement accuracy measurements under time-domain measurement resource restriction with Non-MBSFN ABS configured in the aggressor cell is within the specified limits under AWGN propagation conditions.

#### 9.1.8.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and release 11. Applicability requires support for FGI bit 115.

#### 9.1.8.1.3 Minimum conformance requirements

The FDD RSRP measurements of cells shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.3.

The accuracy requirements in table 9.1.8.1.3-1 are valid under the following conditions:

Cell specific reference signals in the measured cell are transmitted either from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.9 for a corresponding Band.

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.1.8.1.3-1: RSRP Intra frequency absolute accuracy under time domain measurement resource restriction**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 2 range | | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 3 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±6 | ±9 | ≥-4 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-4 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRP measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 3: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.8.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.8.1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.3, clause 9.1.4 and A.9.1.8

### 9.1.8.1.4 Test description

#### 9.1.8.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. The general test parameter settings are set up according to Table 9.1.8.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.1.8.1.4.3.

5. Cell 1 is the serving cell as well as aggressor cell to Cell 2, Cell 2 is the neighbouring cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

**Table 9.1.8.1.4.1-1: General test parameters for E-UTRAN FDD RSRP intra frequency test parameters under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Serving cell (PCell) | | Cell 1 | The aggressor cell to Cell 2 |
| Neighbour cell | | Cell 2 | Cell to be measured |
| PCell ABS configuration | | Non-MBSFN ABS | As defined in Table C.3.1.1.1-1 |
| CP length | | Normal | For both cells in the test |
| DRX | | | OFF |
| Time offset between cells | | 3 μs | Synchronous cells |
| Physical cell ID PCI | | $(PCI_{cell1} - PCI_{cell2})mod6 != 0$ | Cell PCIs for Cell 1 and Cell 2 are selected randomly so that the condition is met |
| ABS pattern | | '10000000100000001000 00001000000010000000' | Non-MBSFN ABS. FDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 1. The first/leftmost bit corresponds to the subframe #0 of a radio frame satisfying SFN mod x = 0, where x is the size of the bit string (40) divided by 10. No MBSFN subframes are configured in the ABS subframes in Cell 1. |
| Time-domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | '10000000100000001000 00001000000010000000' | Configured for Cell 2 measurements by measSubframePattern-Neigh IE in measSubframePatternConfig-Neigh, as defined in TS 36.331 [5], clause 6.3.5. measSubframeCellList contains Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | '01000000010000000100 00001000000001000000' | Configured for measurements on Cell 1. |

### 9.1.8.1.4.2 Test procedure

In this test case all cells are on the same carrier frequency. The absolute accuracy of RSRP intra frequency measurements under time domain measurement resource restriction is tested by using the parameters in Tables 9.1.8.1.4.1-1 and Table 9.1.8.1.5-1 for non-MBSFN ABS with non-colliding CRS. Cell 1 is the serving cell and also the aggressor cell to Cell 2. Cell 2 is the target cell to be measured for RSRP.

The UE is configured by higher layers via Cell 1 with a time-domain measurement resource restriction pattern for performing E-UTRAN FDD intra-frequency measurements on neighbour cells and provided with a neighbour cell list associated with the pattern, where the cell list includes Cell 2. The UE is also configured with a time-domain measurement resource restriction pattern for the serving cell measurements. The information for both patterns shall be provided to the UE before the measurements start.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Tables 9.1.8.1.4.1-1 and Table 9.1.8.1.5-1. Propagation conditions are set according to Annex B clause B.1.

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP values for Cell 2 are compared to the actual RSRP values according to Table 9.1.8.1.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.8.1.5-1 as appropriate.

### 9.1.8.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.8.1.4.3-1: Common Exception messages for RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.7-1<br>Table H.2.7-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3<br>Table H.3.5-5 |

**Table 9.1.8.1.4.3-2: *MeasResults*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.8.1.4.3-3: *MeasResultListEUTRA*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.1.8.1.4.3-4: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '01000000010000000100 00000100000001000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.8.1.4.3-5: *MeasObjectEUTRA-GENERIC(Freq)*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternFDD-r10 | '10000000100000001000 00001000000010000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 2 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.1.8.1.5 Test requirement

Table 9.1.8.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD absolute accuracy test shall meet the reported values test requirements in table 9.1.8.1.5-2.

**Table 9.1.8.1.5-1: Cell Specific test parameters for E-UTRAN RSRP FDD absolute accuracy under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - | R.0 FDD | - | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in D.1.5 (OP.5 FDD) and D.1.6 (OP.6 FDD) | | | OP.5 FDD | OP.6 FDD | OP.5 FDD | OP.6 FDD | OP.5 FDD | OP.6 FDD |
| PBCH_RA<br>PBCH_RB<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA Note1<br>OCNG_RB Note1 | | dB | Note 6 | 0 | Note 6 | 0 | Note 6 | 0 |
| PSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| SSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -106 | | -88 | | -116 | |
| | Bands FDD_C | | | | | | -115 | |
| | Bands FDD_D | | | | | | -114.5 | |
| | Bands FDD_E, FDD_F Note 7 | | | | | | -114 | |
| | Bands FDD_G | | | | | | -113 | |
| | Bands FDD_H | | | | | | -112.5 | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 5 | -2 | 5 | -3.05 | 5 | -3.05 |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{meas}$ Note 5 | | dB | 2.88 | -2.00 | 3.25 | -3.05 | 3.25 | -3.05 |
| SCH $\hat{E}_s / I_{ot}$ | | dB | -1.12 | -5.54 | -0.75 | -6.59 | -0.75 | -6.59 |
| RSRP Note3,4,5 | Bands FDD_A | dBm/15 kHz | -101 | -108 | -83 | -91.05 | -111 | -119.05 |
| | Bands FDD_C | | | | | | -110 | -118.05 |
| | Bands FDD_D | | | | | | -109.5 | -117.55 |
| | Bands FDD_E, FDD_F Note 7 | | | | | | -109 | -117.05 |
| | Bands FDD_G | | | | | | -108 | -116.05 |
| | Bands FDD_H | | | | | | -107.5 | -115.55 |
| $(\mathrm{Io})_{meas}$ Note 3 | Bands FDD_A | dBm/9 MHz | -71.41 | -74.88 | -53.54 | -57.16 | -81.54 | -85.16 |
| | Bands FDD_C | | | | | | -80.54 | -84.16 |
| | Bands FDD_D | | | | | | -80.04 | -83.66 |
| | Bands FDD_E, FDD_F Note 7 | | | | | | -79.54 | -83.16 |
| | Bands FDD_G | | | | | | -78.54 | -82.16 |
| | Bands FDD_H | | | | | | -78.04 | -81.66 |
| Propagation condition | | | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Applies to all subframes.
Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes.
Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Applies to restricted measurement subframes of the respective cell.
Note 6: Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1.
Note 7: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 8: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.8.1.5-2: E-UTRAN RSRP FDD absolute accuracy under time-domain measurement resource restriction with non-MBSFN ABS requirements for the reported values**

| **Normal Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_25 | RSRP_40 | Bands FDD_A | RSRP_14 |
| | | | Bands FDD_C | RSRP_15 |
| | | | Bands FDD_D | RSRP_15 |
| | | | Bands FDD_E, FDD_F | RSRP_16 |
| | | | Bands FDD_G | RSRP_17 |
| | | | Bands FDD_H | RSRP_17 |
| Highest reported value (Cell 2) | RSRP_40 | RSRP_59 | Bands FDD_A | RSRP_29 |
| | | | Bands FDD_C | RSRP_30 |
| | | | Bands FDD_D | RSRP_30 |
| | | | Bands FDD_E, FDD_F | RSRP_31 |
| | | | Bands FDD_G | RSRP_32 |
| | | | Bands FDD_H | RSRP_32 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_22 | RSRP_37 | Bands FDD_A | RSRP_11 |
| | | | Bands FDD_C | RSRP_12 |
| | | | Bands FDD_D | RSRP_12 |
| | | | Bands FDD_E, FDD_F | RSRP_13 |
| | | | Bands FDD_G | RSRP_14 |
| | | | Bands FDD_H | RSRP_14 |
| Highest reported value (Cell 2) | RSRP_43 | RSRP_62 | Bands FDD_A | RSRP_32 |
| | | | Bands FDD_C | RSRP_33 |
| | | | Bands FDD_D | RSRP_33 |
| | | | Bands FDD_E, FDD_F | RSRP_34 |
| | | | Bands FDD_G | RSRP_35 |
| | | | Bands FDD_H | RSRP_35 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.8.1_1 FDD Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC) (Rel-12 and forward)

Editor's note: This test case is incomplete. The following item is missing or incomplete:

The test tolerances and test requirements for test 1, 2 and 3 in normal conditions are TBD

### 9.1.8.1_1.1 Test purpose

Same test purpose as in clause 9.1.8.1.1.

### 9.1.8.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward. Applicability requires support for FGI bit 115.

### 9.1.8.1_1.3 Minimum conformance requirements

The FDD RSRP measurements of cells shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.3.

The accuracy requirements in table 9.1.8.1_1.3-1 are valid under the following conditions:

Cell specific reference signals in the measured cell are transmitted either from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.9 for a corresponding Band.

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.1.8.1_1.3-1: RSRP Intra frequency absolute accuracy under time domain measurement resource restriction**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 2 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 3 / dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-4 dB | FDD_A, TDD_A | -121 / N/A | -70 |
| | | | FDD_C, TDD_C | -120 / N/A | -70 |
| | | | FDD_D | -119.5 / N/A | -70 |
| | | | FDD_E, TDD_E | -119 / N/A | -70 |
| | | | FDD_F | -118.5 / N/A | -70 |
| | | | FDD_G | -118 / N/A | -70 |
| | | | FDD_H | -117.5 / N/A | -70 |
| | | | FDD_N | -114.5 / N/A | -70 |
| ±8 | ±11 | ≥-4 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRP measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 3: The condition level is increased by $\Delta>0$, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.8.1_1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.8.1_1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.3, clause 9.1.4 and A.9.1.8

## 9.1.8.1_1.4 Test description

### 9.1.8.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.8.1.4.1.

### 9.1.8.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.8.1.4.2 with the following exceptions:

- Instead of Table 9.1.8.1.5-1→ use Table 9.1.8.1_1.5-1.
- Instead of Table 9.1.8.1.5-2→ use Table 9.1.8.1_1.5-2.

### 9.1.8.1_1.4.3 Message contents

Same message contents as in clause 9.1.8.1.4.3.

## 9.1.8.1_1.5 Test requirement

Table 9.1.8.1_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD absolute accuracy test shall meet the reported values test requirements in table 9.1.8.1_1.5-2.

**Table 9.1.8.1_1.5-1: Cell Specific test parameters for E-UTRAN RSRP FDD absolute accuracy under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.3.1.1.1 | | | R.0 FDD | - | R.0 FDD | - | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.1 | | | R.6 FDD | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in A.3.2.1.5 (OP.5 FDD) and A.3.2.1.6 (OP.6 FDD) | | | OP.5 FDD | OP.6 FDD | OP.5 FDD | OP.6 FDD | OP.5 FDD | OP.6 FDD |
| PBCH_RA<br>PBCH_RB<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA Note1<br>OCNG_RB Note1 | | dB | Note 6 | 0 | Note 6 | 0 | Note 6 | 0 |
| PSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| SSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -106 | | -88 | | -116 | |
| | Bands FDD_C | | | | | | -115 | |
| | Bands FDD_D | | | | | | -114.5 | |
| | Bands FDD_E, FDD_F Note 7 | | | | | | -114 | |
| | Bands FDD_G Note 9 | | | | | | -113 | |
| | Bands FDD_H | | | | | | -112.5 | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 5+TT | -2+TT | 5+TT | -4+TT | 5+TT | -4+TT |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{meas}$ Note 5 | | dB | 2.88+TT | -2+TT | 3.54+TT | -4+TT | 3.54+TT | -4+TT |
| SCH $\hat{E}_s / I_{ot}$ | | dB | -1.12+TT | -5.54+TT | -0.46+TT | -7.54+TT | -0.46+TT | -7.54+TT |
| RSRP Note3,4,5 | Bands FDD_A | dBm/15 kHz | -101+TT | -108+TT | -83+TT | -92+TT | -111+TT | -120+TT |
| | Bands FDD_C | | | | | | -110+TT | -119+TT |
| | Bands FDD_D | | | | | | -109.5+TT | -118.5+TT |
| | Bands FDD_E, FDD_F Note 7 | | | | | | -109+TT | -118+TT |
| | Bands FDD_G Note 9 | | | | | | -108+TT | -117+TT |
| | Bands FDD_H | | | | | | -107.5+TT | -116.5+TT |
| $(\mathrm{Io})_{meas}$ Note 3 | Bands FDD_A | dBm/9 MHz | -71.41+TT | -74.88+TT | -53.63+TT | -57.37+TT | -81.63+TT | -85.37+TT |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Bands FDD_C | | | | | | -80.63+TT | -84.37+TT |
| | Bands FDD_D | | | | | | -80.13+TT | -83.87+TT |
| | Bands FDD_E, FDD_F Note 7 | | | | | | -79.63+TT | -83.37+TT |
| | Bands FDD_G Note 9 | | | | | | -78.63+TT | -82.37+TT |
| | Bands FDD_H | | | | | | -78.13+TT | -81.87+TT |
| Propagation condition | | | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.
Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes.
Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Applies to restricted measurement subframes of the respective cell.
Note 6: Non-ABS and ABS subframe channel powers defined in Table A.3.4.1.1-1.
Note 7: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 8: E-UTRA operating band groups are as defined in Section 3.5.
Note 9: Except Band 29 and Band 32.

**Table 9.1.8.1_1.5-2: E-UTRAN RSRP FDD absolute accuracy under time-domain measurement resource restriction with non-MBSFN ABS requirements for the reported values**

| **Normal Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | TBD | TBD | Bands FDD_A | TBD |
| | | | Bands FDD_C | TBD |
| | | | Bands FDD_D | TBD |
| | | | Bands FDD_E, FDD_F | TBD |
| | | | Bands FDD_G | TBD |
| | | | Bands FDD_H | TBD |
| Highest reported value (Cell 2) | TBD | TBD | Bands FDD_A | TBD |
| | | | Bands FDD_C | TBD |
| | | | Bands FDD_D | TBD |
| | | | Bands FDD_E, FDD_F | TBD |
| | | | Bands FDD_G | TBD |
| | | | Bands FDD_H | TBD |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_22 | RSRP_37 | Bands FDD_A | RSRP_11 |
| | | | Bands FDD_C | RSRP_12 |
| | | | Bands FDD_D | RSRP_12 |
| | | | Bands FDD_E, FDD_F | RSRP_13 |
| | | | Bands FDD_G | RSRP_14 |
| | | | Bands FDD_H | RSRP_14 |
| Highest reported value (Cell 2) | RSRP_43 | RSRP_62 | Bands FDD_A | RSRP_32 |
| | | | Bands FDD_C | RSRP_33 |
| | | | Bands FDD_D | RSRP_33 |
| | | | Bands FDD_E, FDD_F | RSRP_34 |
| | | | Bands FDD_G | RSRP_35 |
| | | | Bands FDD_H | RSRP_35 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

### 9.1.8.2 FDD Relative RSRP Accuracy under Time-Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC)

#### 9.1.8.2.1 Test purpose

To verify that FDD relative RSRP measurement accuracy measurements under time-domain measurement resource restriction with Non-MBSFN ABS configured in the aggressor cell is within the specified limits under AWGN propagation condition.

#### 9.1.8.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward. Applicability requires support for FGI bit 115.

#### 9.1.8.2.3 Minimum conformance requirements

The FDD RSRP measurements of cells shall meet the intra frequency relative accuracy requirements defined in TS 36.133 [4] clause 9.1.2.4.

The accuracy requirements in table 9.1.8.2.3-1 are valid under the following conditions:

Cell specific reference signals in the measured cell are transmitted either from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.10 for a corresponding Band.

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.1.8.2.3-1: RSRP Intra frequency relative accuracy under time domain measurement resource restriction**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 3 range | | |
| | | | E-UTRA operating band groups Note 6 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 5 | dBm/BW$_{Channel}$ |
| ±2 | ±3 | ≥-2 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-4 dB | Note 4 | Note 4 | Note 4 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRP measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 5: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 6: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.8.2.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.8.2.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.4 , clause 9.1.4 and A.9.1.8

## 9.1.8.2.4 Test description

### 9.1.8.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. The general test parameter settings are set up according to Table 9.1.8.2.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.1.8.2.4.3.

5. Cell 1 is the serving cell as well as aggressor cell to Cell 2, Cell 2 is the neighbouring cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

**Table 9.1.8.2.4.1-1: General test parameters for E-UTRAN FDD RSRP intra frequency test parameters under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Serving cell (PCell) | | Cell 1 | The aggressor cell to Cell 2 |
| Neighbour cell | | Cell 2 | Cell to be measured |
| PCell ABS configuration | | Non-MBSFN ABS | As defined in Table C.3.1.1.1-1 |
| CP length | | Normal | For both cells in the test |
| DRX | | | OFF |
| Time offset between cells | | 3 μs | Synchronous cells |
| Physical cell ID PCI | | $(PCI_{cell1} - PCI_{cell2})mod6 != 0$ | Cell PCIs for Cell 1 and Cell 2 are selected randomly so that the condition is met |
| ABS pattern | | '10000000100000001000000010000000010000000' | Non-MBSFN ABS. FDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 1. The first/leftmost bit corresponds to the subframe #0 of a radio frame satisfying SFN mod x = 0, where x is the size of the bit string (40) divided by 10. No MBSFN subframes are configured in the ABS subframes in Cell 1. |
| Time-domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | '10000000100000001000000010000000010000000' | Configured for Cell 2 measurements by measSubframePattern-Neigh IE in measSubframePatternConfig-Neigh, as defined in TS 36.331 [5], clause 6.3.5. measSubframeCellList contains Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | '01000000010000000100000001000000001000000' | Configured for measurements on Cell 1. |

### 9.1.8.2.4.2 Test procedure

In this test case all cells are on the same carrier frequency. The relative accuracy of RSRP intra frequency measurements under time domain measurement resource restriction is tested by using the parameters in Tables 9.1.8.2.4.1-1 and Table 9.1.8.2.5-1 for non-MBSFN ABS with non-colliding CRS. Cell 1 is the serving cell and also the aggressor cell to Cell 2. Cell 1 and Cell 2 are both measured.

The UE is configured by higher layers via Cell 1 with a time-domain measurement resource restriction pattern for performing E-UTRAN FDD intra-frequency measurements on neighbour cells and provided with a neighbour cell list associated with the pattern, where the cell list includes Cell 2. The UE is also configured with a time-domain measurement resource restriction pattern for the serving cell measurements. The information for both patterns shall be provided to the UE before the measurements start.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Tables 9.1.8.2.4.1-1 and Table 9.1.8.2.5-1. Propagation conditions are set according to Annex B clause B.1.

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP value for Cell 2 is compared to the reported RSRP value for Cell 1 for each MeasurementReport message according to Table 9.1.8.2.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.8.2.5-1 as appropriate.

### 9.1.8.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.8.2.4.3-1: Common Exception messages for RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS relative accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.7-1<br>Table H.2.7-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3<br>Table H.3.5-5 |

**Table 9.1.8.2.4.3-2: *MeasResults*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.8.2.4.3-3: *MeasResultListEUTRA*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.1.8.2.4.3-4: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '01000000010000000100 00000100000001000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.8.2.4.3-5: *MeasObjectEUTRA-GENERIC(Freq)*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternFDD-r10 | '10000000100000001000 00001000000010000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 2 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.1.8.2.5 Test requirement

Table 9.1.8.2.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD relative accuracy test shall meet the reported values test requirements in table 9.1.8.2.5-2.

**Table 9.1.8.2.5-1: Cell Specific test parameters for E-UTRAN RSRP FDD relative accuracy under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - | R.0 FDD | - | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in D.1.5 (OP.5 FDD) and D.1.6 (OP.6 FDD) | | | OP.5 FDD | OP.6 FDD | OP.5 FDD | OP.6 FDD | OP.5 FDD | OP.6 FDD |
| PBCH_RA<br>PBCH_RB<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA Note1<br>OCNG_RB Note1 | | dB | Note 6 | 0 | Note 6 | 0 | Note 6 | 0 |
| PSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| SSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -106 | | -88 | | -116 | |
| | Bands FDD_C | | | | | | -115 | |
| | Bands FDD_D | | | | | | -114.5 | |
| | Bands FDD_E, FDD_F Note 7 | | | | | | -114 | |
| | Bands FDD_G | | | | | | -113 | |
| | Bands FDD_H | | | | | | -112.5 | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 5 | -1.2 | 5 | -3.05 | 5 | -3.05 |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{meas}$ Note 5 | | dB | 2.55 | -1.20 | 3.25 | -3.05 | 3.25 | -3.05 |
| SCH $\hat{E}_s / I_{ot}$ | | dB | -1.45 | -4.74 | -0.75 | -6.59 | -0.75 | -6.59 |
| RSRP Note3,4,5 | Bands FDD_A | dBm/15 kHz | -101 | -107.2 | -83 | -91.05 | -111 | -119.05 |
| | Bands FDD_C | | | | | | -110 | -118.05 |
| | Bands FDD_D | | | | | | -109.5 | -117.55 |
| | Bands FDD_E, FDD_F Note 7 | | | | | | -109 | -117.05 |
| | Bands FDD_G | | | | | | -108 | -116.05 |
| | Bands FDD_H | | | | | | -107.5 | -115.55 |
| $(\mathrm{Io})_{meas}$ Note 3 | Bands FDD_A | dBm/9 MHz | -71.30 | -74.63 | -53.54 | -57.16 | -81.54 | -85.16 |
| | Bands FDD_C | | | | | | -80.54 | -84.16 |
| | Bands FDD_D | | | | | | -80.04 | -83.66 |
| | Bands FDD_E, FDD_F Note 7 | | | | | | -79.54 | -83.16 |
| | Bands FDD_G | | | | | | -78.54 | -82.16 |
| | Bands FDD_H | | | | | | -78.04 | -81.66 |
| Propagation condition | | | AWGN | | AWGN | | AWGN | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes. |
| Note 3: | RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes. |
| Note 4: | RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | Applies to restricted measurement subframes of the respective cell. |
| Note 6: | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1. |
| Note 7: | For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz. |
| Note 8: | E-UTRA operating band groups are as defined in Section 3.5. |

**Table 9.1.8.2.5-2: E-UTRAN RSRP FDD relative accuracy under time-domain measurement resource restriction with non-MBSFN ABS requirements for the reported values**

| **Normal Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3 All bands** |
|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_x-10 | RSRP_x-13 | RSRP_x-13 |
| Highest reported value (Cell 2) | RSRP_x-4 | RSRP_x-4 | RSRP_x-4 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3 All bands** |
| Lowest reported value (Cell 2) | RSRP_x-11 | RSRP_x-13 | RSRP_x-13 |
| Highest reported value (Cell 2) | RSRP_x-3 | RSRP_x-4 | RSRP_x-4 |
| RSRP_x is the reported value of Cell 1 | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.9 TDD RSRP Accuracy E-UTRA under Time-Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC)

### 9.1.9.1 TDD Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC)

#### 9.1.9.1.1 Test purpose

To verify that TDD absolute RSRP measurement accuracy measurements under time-domain measurement resource restriction with Non-MBSFN ABS configured in the aggressor cell is within the specified limits under AWGN propagation conditions.

#### 9.1.9.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and release 11. Applicability requires support for FGI bit 115.

#### 9.1.9.1.3 Minimum conformance requirements

The TDD RSRP measurements of cells shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.3.

The accuracy requirements in table 9.1.9.1.3-1 are valid under the following conditions:

Cell specific reference signals in the measured cell are transmitted either from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.9 for a corresponding Band.

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.1.9.1.3-1: RSRP Intra frequency absolute accuracy under time domain measurement resource restriction**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 2 range | | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 3 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±6 | ±9 | ≥-4 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-4 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRP measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 3: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.9.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.9.1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.3 , clause 9.1.4 and A.9.1.9

### 9.1.9.1.4 Test description

#### 9.1.9.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. The general test parameter settings are set up according to Table 9.1.9.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.1.9.1.4.3.

5. Cell 1 is the serving cell as well as aggressor cell to Cell 2, Cell 2 is the neighbouring cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

**Table 9.1.9.1.4.1-1: General test parameters for E-UTRAN TDD RSRP intra frequency test parameters under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Serving cell (PCell) | | Cell 1 | Also the aggressor cell. |
| Neighbour cell | | Cell 2 | Cell to be measured |
| PCell ABS configuration | | Non-MBSFN ABS | As defined in Table C.3.1.1.1-1 |
| Special subframe configuration | | 6 | For Cell 1 and Cell 2. For special subframe configurations see Table 4.2-1 in [16]. |
| Uplink/downlink subframe configuration | | 1 | For Cell 1 and Cell 2. For uplink-downlink subframe configurations see Table 4.2-2 in [16]. |
| CP length | | Normal | For both cells in the test |
| DRX | | | OFF |
| Time offset between cells | | 3 μs | Synchronous cells |
| Physical cell ID PCI | | $(PCI_{cell1} - PCI_{cell2})mod6 != 0$ | Cell PCIs for Cell 1 and Cell 2 are randomly selected so that the condition is met |
| ABS pattern | | '00000000010000000001' | Non-MBSFN ABS. TDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 1. The first/leftmost bit corresponds to the subframe #0 of a radio frame satisfying SFN mod x = 0, where x is the size of the bit string (20) divided by 10. No MBSFN subframes are configured in the ABS subframes in Cell 1. |
| Time-domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | '00000000010000000001' | Configured for Cell 2 measurements by measSubframePattern-Neigh IE in measSubframePatternConfig-Neigh, as defined in TS 36.331 [2], clause 6.3.5. measSubframeCellList contains Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | '10000000001000000000' | Configured for Cell 1 measurements. |

### 9.1.9.1.4.2 Test procedure

In this test case all cells are on the same carrier frequency. The absolute accuracy of RSRP intra frequency measurements under time domain measurement resource restriction is tested by using the parameters in Tables 9.1.9.1.4.1-1 and Table 9.1.9.1.5-1 for non-MBSFN ABS with non-colliding CRS. Cell 1 is the serving cell and also the aggressor cell to Cell 2. Cell 2 is the target cell to be measured for RSRP.

The UE is configured by higher layers via Cell 1 with a time-domain measurement resource restriction pattern for performing E-UTRAN TDD intra-frequency measurements on neighbour cells and provided with a neighbour cell list associated with the pattern, where the cell list includes Cell 2. The UE is also configured with a time-domain measurement resource restriction pattern for the serving cell measurements. The information for both patterns shall be provided to the UE before the measurements start.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Tables 9.1.9.1.4.1-1 and Table 9.1.9.1.5-1. Propagation conditions are set according to Annex B clause B.1.

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP values for Cell 2 are compared to the actual RSRP values according to Table 9.1.9.1.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.9.1.5-1 as appropriate.

### 9.1.9.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.9.1.4.3-1: Common Exception message for RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.7-1 |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3 |

**Table 9.1.9.1.4.3-2: *MeasResults*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.9.1.4.3-3: *MeasResultListEUTRA*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of Cell 2 | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.1.9.1.4.3-4: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 CHOICE { | | | |
| subframeConfig1-5-r10 | '10000000001000000000' | BIT STRING (SIZE (20)) | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.9.1.4.3-5: *MeasObjectEUTRA-GENERIC(Freq)*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternTDD-r10 CHOICE { | | | |
| subframeConfig1-5-r10 | '00000000010000000001' | BIT STRING (SIZE (20)) | |
| } | | | |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 2 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.1.9.1.5 Test requirement

Table 9.1.9.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD absolute accuracy test shall meet the reported values test requirements in table 9.1.9.1.5-2.

**Table 9.1.9.1.5-1: Cell Specific test parameters for E-UTRAN RSRP TDD absolute accuracy under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | R.0 TDD | - | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA<br>PBCH_RB<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA Note1<br>OCNG_RB Note1 | | dB | Note 6 | 0 | Note 6 | 0 | Note 6 | 0 |
| PSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| SSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| $N_{oc}$ Note2 | Bands TDD_A | dBm/15 kHz | -106 | | -88 | | -116 | |
| | Bands TDD_C | | | | | | -115 | |
| | Bands TDD_E | | | | | | -114 | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 5 | -2 | 5 | -3.05 | 5 | -3.05 |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{meas}$ Note 5 | | dB | 2.88 | -2.00 | 3.25 | -3.05 | 3.25 | -3.05 |
| SCH $\hat{E}_s / I_{ot}$ | | dB | -1.12 | -5.54 | -0.75 | -6.59 | -0.75 | -6.59 |
| RSRP Note3,4,5 | Bands TDD_A | dBm/15 kHz | -101 | -108 | -83 | -91.05 | -111 | -119.05 |
| | Bands TDD_C | | | | | | -110 | -118.05 |
| | Bands TDD_E | | | | | | -109 | -117.05 |
| $(\mathrm{Io})_{meas}$ Note 3 | Bands TDD_A | dBm/9 MHz | -71.41 | -74.88 | -53.54 | -57.16 | -81.54 | -85.16 |
| | Bands TDD_C | | | | | | -80.54 | -84.16 |
| | Bands TDD_E | | | | | | -79.54 | -83.16 |
| Propagation condition | | | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Applies to all subframes.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: Applies to restricted measurement subframes of the respective cell.

Note 6: Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1.

Note 7: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.9.1.5-2: E-UTRAN RSRP TDD absolute accuracy under time-domain measurement resource restriction with non-MBSFN ABS requirements for the reported values**

| **Normal Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_25 | RSRP_40 | Bands TDD_A | RSRP_14 |
| | | | Bands TDD_C | RSRP_15 |
| | | | Bands TDD_E | RSRP_16 |
| Highest reported value (Cell 2) | RSRP_40 | RSRP_59 | Bands TDD_A | RSRP_29 |
| | | | Bands TDD_C | RSRP_30 |
| | | | Bands TDD_E | RSRP_31 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_22 | RSRP_37 | Bands TDD_A | RSRP_11 |
| | | | Bands TDD_C | RSRP_12 |
| | | | Bands TDD_E | RSRP_13 |
| Highest reported value (Cell 2) | RSRP_43 | RSRP_62 | Bands TDD_A | RSRP_32 |
| | | | Bands TDD_C | RSRP_33 |
| | | | Bands TDD_E | RSRP_34 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.9.1_1 TDD Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC) (Rel-12 and forward)

Editor's note: This test case is incomplete. The following item is missing or incomplete:

The test tolerances and test requirements for test 1, 2 and 3 in normal conditions are TBD

### 9.1.9.1_1.1 Test purpose

Same test purpose as in clause 9.1.9.1.1.

### 9.1.9.1_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward. Applicability requires support for FGI bit 115.

### 9.1.9.1_1.3 Minimum conformance requirements

The TDD RSRP measurements of cells shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.3.

The accuracy requirements in table 9.1.9.1_1.3-1 are valid under the following conditions:

Cell specific reference signals in the measured cell are transmitted either from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.9 for a corresponding Band.

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.1.9.1_1.3-1: RSRP Intra frequency absolute accuracy under time domain measurement resource restriction**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 2 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 3 / dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-4 dB | FDD_A, TDD_A | -121 / N/A | -70 |
| | | | FDD_C, TDD_C | -120 / N/A | -70 |
| | | | FDD_D | -119.5 / N/A | -70 |
| | | | FDD_E, TDD_E | -119 / N/A | -70 |
| | | | FDD_F | -118.5 / N/A | -70 |
| | | | FDD_G | -118 / N/A | -70 |
| | | | FDD_H | -117.5 / N/A | -70 |
| | | | FDD_N | -114.5 / N/A | -70 |
| ±8 | ±11 | ≥-4 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 / -50 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRP measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 3: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.9.1_1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.9.1_1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.3, clause 9.1.4 and A.9.1.9

## 9.1.9.1_1.4 Test description

### 9.1.9.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.9.1.4.1.

### 9.1.9.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.8.1.4.2 with the following exceptions:

- Instead of Table 9.1.8.1.5-1→ use Table 9.1.8.1_1.5-1.
- Instead of Table 9.1.8.1.5-2→ use Table 9.1.8.1_1.5-2.

#### 9.1.9.1_1.4.3 Message contents

Same message contents as in clause 9.1.9.1.4.3.

### 9.1.9.1_1.5 Test requirement

Table 9.1.9.1_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD absolute accuracy test shall meet the reported values test requirements in table 9.1.9.1_1.5-2.

**Table 9.1.9.1_1.5-1: Cell Specific test parameters for E-UTRAN RSRP TDD absolute accuracy under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.3.1.1.2 | | | R.0 TDD | - | R.0 TDD | - | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.2 | | | R.6 TDD | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in A.3.2.2.1 (OP.1 TDD) and A.3.2.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | Note 6 | 0 | Note 6 | 0 | Note 6 | 0 |
| PBCH_RB | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA Note1 | | | | | | | | |
| OCNG_RB Note1 | | | | | | | | |
| PSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| SSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| $N_{oc}$ Note 2 | Bands TDD_A | dBm/15 kHz | -106 | | -88 | | -116 | |
| | Bands TDD_C | | | | | | -115 | |
| | Bands TDD_E | | | | | | -114 | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 5+TT | -2+TT | 5+TT | -4+TT | 5+TT | -4+TT |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{meas}$ Note 5 | | dB | 2.88+TT | -2+TT | 3.54+TT | -4+TT | 5+TT | -4+TT |
| SCH $\hat{E}_s / I_{ot}$ | | dB | -1.12+TT | -5.54+TT | -0.46+TT | -7.54+TT | -0.46+TT | -7.54+TT |
| RSRP Note3,4,5 | Bands TDD_A | dBm/15 kHz | -101+TT | -108+TT | -83+TT | -92+TT | -111+TT | -120+TT |
| | Bands TDD_C | | | | | | -110+TT | -119+TT |
| | Bands TDD_E | | | | | | -109+TT | -118+TT |
| $(\mathrm{Io})_{meas}$ Note 3 | Bands TDD_A | dBm/9 MHz | -71.41+TT | -74.88+TT | -53.63+TT | -57.37+TT | -81.6+TT | -85.4+TT |
| | Bands TDD_C | | | | | | -80.6+TT | -84.4+TT |
| | Bands TDD_E | | | | | | -79.6+TT | -83.4+TT |
| Propagation condition | | | AWGN | | AWGN | | AWGN | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes. |
| Note 3: | RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes. |
| Note 4: | RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | Applies to restricted measurement subframes of the respective cell. |
| Note 6: | Non-ABS and ABS subframe channel powers defined in Table A.3.4.1.1-1. |
| Note 7: | E-UTRA operating band groups are as defined in Section 3.5. |

**Table 9.1.9.1_1.5-2: E-UTRAN RSRP TDD absolute accuracy under time-domain measurement resource restriction with non-MBSFN ABS requirements for the reported values**

| **Normal Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | TBD | TBD | Bands TDD_A | TBD |
| | | | Bands TDD_C | TBD |
| | | | Bands TDD_E | TBD |
| Highest reported value (Cell 2) | TBD | TBD | Bands TDD_A | TBD |
| | | | Bands TDD_C | TBD |
| | | | Bands TDD_E | TBD |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_22 | RSRP_37 | Bands TDD_A | RSRP_11 |
| | | | Bands TDD_C | RSRP_12 |
| | | | Bands TDD_E | RSRP_13 |
| Highest reported value (Cell 2) | RSRP_43 | RSRP_62 | Bands TDD_A | RSRP_32 |
| | | | Bands TDD_C | RSRP_33 |
| | | | Bands TDD_E | RSRP_34 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.9.2 TDD Relative RSRP Accuracy under Time-Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC)

### 9.1.9.2.1 Test purpose

To verify that TDD relative RSRP measurement accuracy measurements under time-domain measurement resource restriction with Non-MBSFN ABS configured in the aggressor cell is within the specified limits under AWGN propagation condition.

### 9.1.9.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward. Applicability requires support for FGI bit 115.

### 9.1.9.2.3 Minimum conformance requirements

The TDD RSRP measurements of cells shall meet the intra frequency relative accuracy requirements defined in TS 36.133 [4] clause 9.1.2.4.

The accuracy requirements in table 9.1.9.2.3-1 are valid under the following conditions:

Cell specific reference signals in the measured cell are transmitted either from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.10 for a corresponding Band.

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.1.9.2.3-1: RSRP Intra frequency relative accuracy under time domain measurement resource restriction**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 3 range | | |
| | | | E-UTRA operating band groups Note 6 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 5 | $dBm/BW_{Channel}$ |
| ±2 | ±3 | ≥-2 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-4 dB | Note 4 | Note 4 | Note 4 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRP measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 5: The condition level is increased by $\Delta>0$, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 6: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.9.2.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.9.2.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.4 , clause 9.1.4 and A.9.1.9

## 9.1.9.2.4 Test description

### 9.1.9.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. The general test parameter settings are set up according to Table 9.1.9.2.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.1.9.2.4.3.

5. Cell 1 is the serving cell as well as aggressor cell to Cell 2, Cell 2 is the neighbouring cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

**Table 9.1.9.2.4.1-1: General test parameters for E-UTRAN TDD RSRP intra frequency test parameters under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Serving cell (PCell) | | Cell 1 | Also the aggressor cell. |
| Neighbour cell | | Cell 2 | Cell to be measured |
| PCell ABS configuration | | Non-MBSFN ABS | As defined in Table C.3.1.1.1-1 |
| Special subframe configuration | | 6 | For Cell 1 and Cell 2. For special subframe configurations see Table 4.2-1 in [16]. |
| Uplink/downlink subframe configuration | | 1 | For Cell 1 and Cell 2. For uplink-downlink subframe configurations see Table 4.2-2 in [16]. |
| CP length | | Normal | For both cells in the test |
| DRX | | | OFF |
| Time offset between cells | | 3 μs | Synchronous cells |
| Physical cell ID PCI | | $(PCI_{cell1} - PCI_{cell2})mod6 != 0$ | Cell PCIs for Cell 1 and Cell 2 are randomly selected so that the condition is met |
| ABS pattern | | '00000000010000000001' | Non-MBSFN ABS. TDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 1. The first/leftmost bit corresponds to the subframe #0 of a radio frame satisfying SFN mod x = 0 , where x is the size of the bit string (20) divided by 10. No MBSFN subframes are configured in the ABS subframes in Cell 1. |
| Time-domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | '00000000010000000001' | Configured for Cell 2 measurements by measSubframePattern-Neigh IE in measSubframePatternConfig-Neigh, as defined in TS 36.331 [2], clause 6.3.5. measSubframeCellList contains Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | '10000000001000000000' | Configured for Cell 1 measurements. |

### 9.1.9.2.4.2 Test procedure

In this test case all cells are on the same carrier frequency. The relative accuracy of RSRP intra frequency measurements under time domain measurement resource restriction is tested by using the parameters in Tables 9.1.9.2.4.1-1 and Table 9.1.9.2.5-1 for non-MBSFN ABS with non-colliding CRS. Cell 1 is the serving cell and also the aggressor cell to Cell 2. Cell 1 and Cell 2 are both measured.

The UE is configured by higher layers via Cell 1 with a time-domain measurement resource restriction pattern for performing E-UTRAN TDD intra-frequency measurements on neighbour cells and provided with a neighbour cell list associated with the pattern, where the cell list includes Cell 2. The UE is also configured with a time-domain measurement resource restriction pattern for the serving cell measurements. The information for both patterns shall be provided to the UE before the measurements start.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Tables 9.1.9.2.4.1-1 and Table 9.1.9.2.5-1. Propagation conditions are set according to Annex B clause B.1.

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP value for Cell 2 is compared to the reported RSRP value for Cell 1 for each MeasurementReport message according to Table 9.1.9.2.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.9.2.5-1 as appropriate.

### 9.1.9.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.9.2.4.3-1: Common Exception message for RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3 |

**Table 9.1.9.2.4.3-2: *MeasResults*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.9.2.4.3-3: *MeasResultListEUTRA*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of Cell 2 | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.1.9.2.4.3-4: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 CHOICE { | | | |
| subframeConfig1-5-r10 | '10000000001000000000' | BIT STRING (SIZE (20)) | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.9.2.4.3-5: *MeasObjectEUTRA-GENERIC(Freq)*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternTDD-r10 CHOICE { | | | |
| subframeConfig1-5-r10 | '00000000010000000001' | BIT STRING (SIZE (20)) | |
| } | | | |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 2 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.1.9.2.5 Test requirement

Table 9.1.9.2.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD relative accuracy test shall meet the reported values test requirements in table 9.1.9.2.5-2.

**Table 9.1.9.2.5-1: Cell Specific test parameters for E-UTRAN RSRP TDD relative accuracy under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | R.0 TDD | - | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA<br>PBCH_RB<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA Note1<br>OCNG_RB Note1 | | dB | Note 6 | 0 | Note 6 | 0 | Note 6 | 0 |
| PSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| SSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| $N_{oc}$ Note2 | Bands TDD_A | dBm/15 kHz | -106 | | -88 | | -116 | |
| | Bands TDD_C | | | | | | -115 | |
| | Bands TDD_E | | | | | | -114 | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 5 | -1.2 | 5 | -3.05 | 5 | -3.05 |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{meas}$ Note 5 | | dB | 2.55 | -1.2 | 3.25 | -3.05 | 3.25 | -3.05 |
| SCH $\hat{E}_s / I_{ot}$ | | dB | -1.45 | -4.74 | -0.75 | -6.59 | -0.75 | -6.59 |
| RSRP Note3,4,5 | Bands TDD_A | dBm/15 kHz | -101 | -107.2 | -83 | -91.05 | -111 | -119.05 |
| | Bands TDD_C | | | | | | -110 | -118.05 |
| | Bands TDD_E | | | | | | -109 | -117.05 |
| $(\mathrm{Io})_{meas}$ Note 3 | Bands TDD_A | dBm/9 MHz | -71.30 | -74.63 | -53.54 | -57.16 | -81.54 | -85.16 |
| | Bands TDD_C | | | | | | -80.54 | -84.16 |
| | Bands TDD_E | | | | | | -79.54 | -83.16 |
| Propagation condition | | | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Applies to all subframes.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: Applies to restricted measurement subframes of the respective cell.

Note 6: Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1.

Note 7: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.9.2.5-2: E-UTRAN RSRP TDD relative accuracy under time-domain measurement resource restriction with non-MBSFN ABS requirements for the reported values**

| **Normal Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3 All bands** |
|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_x-10 | RSRP_x-13 | RSRP_x-13 |
| Highest reported value (Cell 2) | RSRP_x-4 | RSRP_x-4 | RSRP_x-4 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3 All bands** |
| Lowest reported value (Cell 2) | RSRP_x-11 | RSRP_x-13 | RSRP_x-13 |
| Highest reported value (Cell 2) | RSRP_x-3 | RSRP_x-4 | RSRP_x-4 |
| RSRP_x is the reported value of Cell 1 | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.10 FDD RSRP Accuracy E-UTRA under Time-Domain Measurement Resource Restriction with MBSFN ABS (eICIC)

### 9.1.10.1 FDD Absolute RSRP under Time-Domain Measurement Resource Restriction with MBSFN ABS (eICIC)

#### 9.1.10.1.1 Test purpose

To verify that FDD absolute RSRP measurement accuracy measurements under time-domain measurement resource restriction with MBSFN ABS configured in the aggressor cell is within the specified limits under AWGN propagation conditions. This test will verify the FDD absolute RSRP accuracy.

#### 9.1.10.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and release 11 supporting eICIC. Applicability requires support for FGI bit 115.

#### 9.1.10.1.3 Minimum conformance requirements

The FDD RSRP measurements of cells shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.3.

The accuracy requirements in table 9.1.10.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.1.10.1.3-1: RSRP Intra frequency absolute accuracy under time domain measurement resource restriction**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 2] range | | | |
| | | | E-UTRA operating band groups [Note 4] | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz [Note 1, 3] | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±6 | ±9 | ≥-4 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-4 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRP measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 3: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.10.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.10.1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.3 , clause 9.1.4 and A.9.1.10

## 9.1.10.1.4 Test description

### 9.1.10.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.10.1.4.3.

4. Cell 1 is the serving cell as well as aggressor cell to Cell 2, Cell 2 is the neighbouring cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.1.10.1.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2.A.3.

2. Set the parameters according to Table 9.1.10.1.5-1 and 9.1.10.1.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP values for Cell 1 and Cell 2 are compared to the actual RSRP values according to Table 9.1.10.1.5-3. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.10.1.5-2 as appropriate.

#### 9.1.10.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.10.1.4.3-1: Common Exception messages for RSRP FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3<br>Table H.3.5-5 |

**Table 9.1.10.1.4.3-2: *MeasResults*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.10.1.4.3-3: *MeasResultListEUTRA*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.1.10.1.4.3-4: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '00010000000100000001 00000001000000010000' | BIT STRING (SIZE (40)) | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.10.1.4.3-5: *MeasObjectEUTRA-GENERIC(Freq)*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternFDD-r10 | '01000000100000001000 00000010000001000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 2 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.10.1.4.3-6: *SystemInformationBlockType2*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 SystemInformationBlockType2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfiguration | Not present | | Cell 2 |
| } | | | |

**Table 9.1.10.1.4.3-7: *SystemInformationBlockType3*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| neighCellConfig | '01'B (No MBSFN subframes are present in all neighbour cells) | | Cell 1 |
| | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 2 |

## 9.1.10.1.5 Test requirement

Table 9.1.10.1.5-2 defines the primary level settings including test tolerances for all tests.

The RSRP FDD absolute accuracy test shall meet the reported values test requirements in table 9.1.10.1.5-3.

**Table 9.1.10.1.5-1: General test parameters for E-UTRAN RSRP FDD absolute accuracy under time-domain measurement resource restriction with MBSFN ABS**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Serving cell (PCell) | | Cell 1 | The aggressor cell to Cell 2 |
| Neighbour cell | | Cell 2 | Cell to be measured |
| E-UTRA RF Channel Number | | 1 | One FDD carrier frequency is used |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | For both cells in the test |
| CP length | | Normal | For both cells in the test |
| DRX | | | OFF |
| Time offset between cells | | 3 μs | Synchronous cells |
| Physical cell ID PCI | | $(PCI_{cell1} - PCI_{cell2})mod6 = 0$, $PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell PCIs for Cell 1 and Cell 2 are selected randomly so that the condition is met |
| ABS pattern | | '01000000100000001000000000010000001000000' | MBSFN ABS pattern. FDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 1. The first/leftmost bit corresponds to the subframe #0 of a radio frame satisfying SFN mod x = 0 , where x is the size of the bit string (40) divided by 10. All ABS subframes are MBSFN subframes. |
| Time-domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | '01000000100000001000000000010000001000000' | Configured for Cell 2 measurements by measSubframePatternNeigh IE in measSubframePatternConfigNeigh, as defined in TS 36.331, clause 6.3.5. measSubframeCellList contains Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | '00010000000100000001000000001000000010000' | Configured for measurements on Cell 1. |

**Table 9.1.10.1.5-2: Cell Specific test parameters for E-UTRAN RSRP FDD absolute accuracy under time-domain measurement resource restriction with MBSFN ABS**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - | R.0 FDD | - | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.1.2 | | | R.6 FDD | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in D.1.8 (OP.8 FDD) and D.1.6 (OP.6 FDD) Note5 | | | OP.8 FDD | OP.6 FDD | OP.8 FDD | OP.6 FDD | OP.8 FDD | OP.6 FDD |
| PBCH_RA<br>PBCH_RB<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA Note1<br>OCNG_RB Note1 | | dB | Note 6 | 0 | Note 6 | 0 | Note 6 | 0 |
| PSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| SSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| $N_{oc}$ Note 2 | Bands FDD_A | dBm/15 kHz | -106 | | -88 | | -116 | |
| | Bands FDD_C | | | | | | -115 | |
| | Bands FDD_D | | | | | | -114.5 | |
| | Bands FDD_E, FDD_F Note 8 | | | | | | -114 | |
| | Bands FDD_G | | | | | | -113 | |
| | Bands FDD_H | | | | | | -112.5 | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 5 | -2 | 5 | -3.05 | 5 | -3.05 |
| CRS $\left(\hat{E}_s / Iot\right)_{meas}$ note 5, 7 in the 1st OFDM symbol | | dB | 2.88 | -8.19 | 3.54 | -10.19 | 3.54 | -10.19 |
| CRS $\left(\hat{E}_s / Iot\right)_{meas}$ note 5 in OFDM symbols 4,7,11 | | dB | 2.88 | -2 | 3.25 | -3.05 | 3.25 | -3.05 |
| SCH $\hat{E}_s / I_{ot}$ | | dB | -1.12 | -5.54 | -0.75 | -6.59 | -0.75 | -6.59 |
| RSRP Note 3,4 | Bands FDD_A | dBm/15 kHz | -101 | -108 | -83 | -91.05 | -111 | -119.05 |
| | Bands FDD_C | | | | | | -110 | -118.05 |
| | Bands FDD_D | | | | | | -109.5 | -117.55 |
| | Bands FDD_E, FDD_F Note 8 | | | | | | -109 | -117.95 |
| | Bands FDD_G | | | | | | -108 | -116.05 |
| | Bands FDD_H | | | | | | -107.5 | -115.55 |
| $(\mathrm{Io})_{meas}$ Note 3 in the 1st OFDM symbol | Bands FDD_A | dBm/9 MHz | -71.41 | -74.88 | -53.63 | -57.37 | -81.63 | -85.37 |
| | Bands FDD_C | | | | | | -80.63 | -84.37 |
| | Bands FDD_D | | | | | | -80.13 | -83.87 |
| | Bands FDD_E, FDD_F Note 8 | | | | | | -79.63 | -83.37 |
| | Bands FDD_G | | | | | | -78.63 | -82.37 |
| | Bands FDD_H | | | | | | -78.13 | -81.87 |
| $(\mathrm{Io})_{meas}$ Note 3 in OFDM symbols other | Bands FDD_A | dBm/9 MHz | -71.41 | -76.09 | -53.54 | -57.16 | -81.54 | -85.16 |
| | Bands FDD_C | | | | | | -80.54 | -84.16 |
| | Bands FDD_D | | | | | | -80.04 | -83.66 |
| | Bands FDD_E, | | | | | | -79.54 | -83.16 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| than the 1st one | FDD_F Note 8 | | | | | | | |
| | Bands FDD_G | | | | | | -78.54 | -82.16 |
| | Bands FDD_H | | | | | | -78.04 | -81.66 |
| Propagation condition | | | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Applies to all subframes.
Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes.
Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Applies to restricted measurement subframes of the respective cell.
Note 6: Non-ABS and ABS subframe channel powers defined in Table C.3.1.2.1-1.
Note 7: In the 1st OFDM symbol, Cell 2 is not expected to meet the Es/Iot side condition in TS36.133 [4] clause 9.1.2.3 and 9.1.2.4.
Note 8: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 9: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.10.1.5-3: E-UTRAN RSRP FDD absolute accuracy under time-domain measurement resource restriction with MBSFN ABS requirements for the reported values**

| **Normal Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_25 | RSRP_40 | Bands FDD_A | RSRP_14 |
| | | | Bands FDD_C | RSRP_15 |
| | | | Bands FDD_D | RSRP_15 |
| | | | Bands FDD_E, FDD_F | RSRP_16 |
| | | | Bands FDD_G | RSRP_17 |
| | | | Bands FDD_H | RSRP_17 |
| Highest reported value (Cell 2) | RSRP_40 | RSRP_59 | Bands FDD_A | RSRP_29 |
| | | | Bands FDD_C | RSRP_30 |
| | | | Bands FDD_D | RSRP_30 |
| | | | Bands FDD_E, FDD_F | RSRP_31 |
| | | | Bands FDD_G | RSRP_32 |
| | | | Bands FDD_H | RSRP_32 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_22 | RSRP_37 | Bands FDD_A | RSRP_11 |
| | | | Bands FDD_C | RSRP_12 |
| | | | Bands FDD_D | RSRP_12 |
| | | | Bands FDD_E, FDD_F | RSRP_13 |
| | | | Bands FDD_G | RSRP_14 |
| | | | Bands FDD_H | RSRP_14 |
| Highest reported value (Cell 2) | RSRP_43 | RSRP_62 | Bands FDD_A | RSRP_32 |
| | | | Bands FDD_C | RSRP_33 |
| | | | Bands FDD_D | RSRP_33 |
| | | | Bands FDD_E, FDD_F | RSRP_34 |
| | | | Bands FDD_G | RSRP_35 |
| | | | Bands FDD_H | RSRP_35 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.10.1_1 FDD Absolute RSRP under Time-Domain Measurement Resource Restriction with MBSFN ABS (eICIC) (Rel-12 and forward)

Editor's note: This test case is incomplete. The following item is missing or incomplete:

The test tolerances and test requirements for test 1, 2 and 3 in normal conditions are TBD

### 9.1.10.1_1.1 Test purpose

Same test purpose as in clause 9.1.10.1.1.

### 9.1.10.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward supporting eICIC. Applicability requires support for FGI bit 115.

### 9.1.10.1_1.3 Minimum conformance requirements

The FDD RSRP measurements of cells shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.3.

The accuracy requirements in table 9.1.10.1_1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.1.10.1_1.3-1: RSRP Intra frequency absolute accuracy under time domain measurement resource restriction**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 2 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 3 / dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-4 dB | FDD_A, TDD_A | -121 / N/A | -70 |
| | | | FDD_C, TDD_C | -120 / N/A | -70 |
| | | | FDD_D | -119.5 / N/A | -70 |
| | | | FDD_E, TDD_E | -119 / N/A | -70 |
| | | | FDD_F | -118.5 / N/A | -70 |
| | | | FDD_G | -118 / N/A | -70 |
| | | | FDD_H | -117.5 / N/A | -70 |
| | | | FDD_N | -114.5 / N/A | -70 |
| ±8 | ±11 | ≥-4 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRP measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 3: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.10.1_1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.10.1_1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.3, clause 9.1.4 and A.9.1.10.

## 9.1.10.1_1.4 Test description

### 9.1.10.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.10.1.4.1.

### 9.1.10.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.10.1.4.2 with the following exceptions:

- Instead of Table 9.1.10.1.5-1→ use Table 9.1.10.1_1.5-1.
- Instead of Table 9.1.10.1.5-2→ use Table 9.1.10.1_1.5-2.
- Instead of Table 9.1.10.1.5-3→ use Table 9.1.10.1_1.5-3.

### 9.1.10.1_1.4.3 Message contents

Same message contents as in clause 9.1.10.1.4.3.

## 9.1.10.1_1.5 Test requirement

Table 9.1.10.1_1.5-2 defines the primary level settings including test tolerances for all tests.

The RSRP FDD absolute accuracy test shall meet the reported values test requirements in table 9.1.10.1_1.5-3.

**Table 9.1.10.1_1.5-1: General test parameters for E-UTRAN RSRP FDD absolute accuracy under time-domain measurement resource restriction with MBSFN ABS**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Serving cell (PCell) | | Cell 1 | The aggressor cell to Cell 2 |
| Neighbour cell | | Cell 2 | Cell to be measured |
| E-UTRA RF Channel Number | | 1 | One FDD carrier frequency is used |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | For both cells in the test |
| CP length | | Normal | For both cells in the test |
| DRX | | | OFF |
| Time offset between cells | | 3 μs | Synchronous cells |
| Physical cell ID PCI | | ($PCI_{cell1}$ - $PCI_{cell2}$ )mod6 =0, $PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell PCIs for Cell 1 and Cell 2 are selected randomly so that the condition is met |
| ABS pattern | | '01000000100000001000 00000010000001000000' | MBSFN ABS pattern. FDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 1. The first/leftmost bit corresponds to the subframe #0 of a radio frame satisfying SFN mod x = 0 , where x is the size of the bit string (40) divided by 10. All ABS subframes are MBSFN subframes. |
| Time-domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | '01000000100000001000 00000010000001000000' | Configured for Cell 2 measurements by measSubframePatternNeigh IE in measSubframePatternConfigNeigh, as defined in TS 36.331, clause 6.3.5. measSubframeCellList contains Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | '00010000000100000001 00000001000000010000' | Configured for measurements on Cell 1. |

**Table 9.1.10.1_1.5-2: Cell Specific test parameters for E-UTRAN RSRP FDD absolute accuracy under time-domain measurement resource restriction with MBSFN ABS**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.3.1.1.1 | | | R.0 FDD | - | R.0 FDD | - | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.1 | | | R.6 FDD | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in A.3.2.1.8 (OP.8 FDD) and A.3.2.1.6 (OP.6 FDD) | | | OP.8 FDD | OP.6 FDD | OP.8 FDD | OP.6 FDD | OP.8 FDD | OP.6 FDD |
| PBCH_RA<br>PBCH_RB<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA Note1<br>OCNG_RB Note1 | | dB | Note 6 | 0 | Note 6 | 0 | Note 6 | 0 |
| PSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| SSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| $N_{oc}$ Note 2 | Bands FDD_A | dBm/15 kHz | -106 | | -88 | | -116 | |
| | Bands FDD_C | | | | | | -115 | |
| | Bands FDD_D | | | | | | -114.5 | |
| | Bands FDD_E, FDD_F Note 8 | | | | | | -114 | |
| | Bands FDD_G Note 10 | | | | | | -113 | |
| | Bands FDD_H | | | | | | -112.5 | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 5+TT | -2+TT | 5+TT | -4+TT | 5+TT | -4+TT |
| CRS $\left(\hat{E}_s / Iot\right)_{meas}$ Note 5, 7 in the 1st OFDM symbol | | dB | 2.88+TT | -8.19+TT | 3.54+TT | -10.19+TT | 3.54+TT | -10.19+TT |
| CRS $\left(\hat{E}_s / Iot\right)_{meas}$ Note 5 in OFDM symbols 4,7,11 | | dB | 2.88+TT | -2+TT | 3.54+TT | -4+TT | 3.54+TT | -4+TT |
| SCH $\hat{E}_s / I_{ot}$ | | dB | -1.12+TT | -5.54+TT | -0.46+TT | -7.54+TT | -0.46+TT | -7.54+TT |
| RSRP Note 3,4 | Bands FDD_A | dBm/15 kHz | -101+TT | -108+TT | -83+TT | -92+TT | -111+TT | -120+TT |
| | Bands FDD_C | | | | | | -110+TT | -119+TT |
| | Bands FDD_D | | | | | | -109.5+TT | -118.5+TT |
| | Bands FDD_E, FDD_F Note 8 | | | | | | -109+TT | -118+TT |
| | Bands FDD_G Note 10 | | | | | | -108+TT | -117+TT |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Bands FDD_H | | | | | | -107.5+TT | -116.5+TT |
| $(Io)_{meas}$ Note 3 in the 1st OFDM symbol | Bands FDD_A | dBm/9 MHz | -71.41+TT | -74.88+TT | -53.63+TT | -57.37+TT | -81.63+TT | -85.37+TT |
| | Bands FDD_C | | | | | | -80.63+TT | -84.37+TT |
| | Bands FDD_D | | | | | | -80.13+TT | -83.87+TT |
| | Bands FDD_E, FDD_F Note 8 | | | | | | -79.63+TT | -83.37+TT |
| | Bands FDD_G Note 10 | | | | | | -78.63+TT | -82.37+TT |
| | Bands FDD_H | | | | | | -78.13+TT | -81.87+TT |
| $(Io)_{meas}$ Note 3 in OFDM symbols other than the 1st one | Bands FDD_A | dBm/9 MHz | -71.41+TT | -76.09+TT | -53.63+TT | -58.76+TT | -81.63+TT | -86.76+TT |
| | Bands FDD_C | | | | | | -80.63+TT | -85.76+TT |
| | Bands FDD_D | | | | | | -80.13+TT | -85.26+TT |
| | Bands FDD_E, FDD_F Note 8 | | | | | | -79.63+TT | -84.76+TT |
| | Bands FDD_G Note 10 | | | | | | -78.63+TT | -83.76+TT |
| | Bands FDD_H | | | | | | -78.13+TT | -83.26+TT |
| Propagation condition | | | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Applies to all subframes.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: Applies to restricted measurement subframes of the respective cell.

Note 6: Non-ABS and ABS subframe channel powers defined in Table A.3.4.2.1-1.

Note 7: In the 1st OFDM symbol, Cell 2 is not expected to meet the Es/Iot side condition in 9.1.2.3 and 9.1.2.4.

Note 8: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 9: E-UTRA operating band groups are as defined in Section 3.5.

Note 10: Except Band 29 and Band 32.

**Table 9.1.10.1_1.5-3: E-UTRAN RSRP FDD absolute accuracy under time-domain measurement resource restriction with MBSFN ABS requirements for the reported values**

| Normal Conditions | Test 1 All bands | Test 2 All bands | Test 3 | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | TBD | TBD | Bands FDD_A | TBD |
| | | | Bands FDD_C | TBD |
| | | | Bands FDD_D | TBD |
| | | | Bands FDD_E, FDD_F | TBD |
| | | | Bands FDD_G | TBD |
| | | | Bands FDD_H | TBD |
| Highest reported value (Cell 2) | TBD | TBD | Bands FDD_A | TBD |
| | | | Bands FDD_C | TBD |
| | | | Bands FDD_D | TBD |
| | | | Bands FDD_E, FDD_F | TBD |
| | | | Bands FDD_G | TBD |
| | | | Bands FDD_H | TBD |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_22 | RSRP_37 | Bands FDD_A | RSRP_11 |
| | | | Bands FDD_C | RSRP_12 |
| | | | Bands FDD_D | RSRP_12 |
| | | | Bands FDD_E, FDD_F | RSRP_13 |
| | | | Bands FDD_G | RSRP_14 |
| | | | Bands FDD_H | RSRP_14 |
| Highest reported value (Cell 2) | RSRP_43 | RSRP_62 | Bands FDD_A | RSRP_32 |
| | | | Bands FDD_C | RSRP_33 |
| | | | Bands FDD_D | RSRP_33 |
| | | | Bands FDD_E, FDD_F | RSRP_34 |
| | | | Bands FDD_G | RSRP_35 |
| | | | Bands FDD_H | RSRP_35 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.10.2 FDD Relative RSRP under Time-Domain Measurement Resource Restriction with MBSFN ABS (eICIC)

### 9.1.10.2.1 Test purpose

To verify that FDD relative RSRP measurement accuracy under time-domain measurement resource restriction with MBSFN ABS configured in the aggressor cell is within the specified limits under AWGN propagation conditions. This test will verify the FDD relative RSRP accuracy requirements.

### 9.1.10.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward supporting eICIC. Applicability requires support for FGI bit 115.

### 9.1.10.2.3 Minimum conformance requirements

The FDD RSRP relative measurements of cells shall meet relative accuracy requirements defined in TS 36.133 clause 9.1.2.4.

The accuracy requirements in Table 9.1.10.2.3-1 for relative accuracy requirements are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

RSRP1,2|dBm according to Annex I.3.1 for a corresponding Band

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.1.10.2.3-1: RSRP Intra frequency relative accuracy under time domain measurement resource restriction**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 3 range | | |
| | | | E-UTRA operating band groups Note 6 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 5 | dBm/BW$_{Channel}$ |
| ±2 | ±3 | ≥-2 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-4 dB | Note 4 | Note 4 | Note 4 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRP measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 5: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 6: E-UTRA operating band groups are as defined in Section 3.5.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.4, clause 9.1.4 and A.9.1.10.

### 9.1.10.2.4 Test description

#### 9.1.10.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.10.2.4.3.

4. Cell 1 is the serving cell as well as aggressor cell to Cell 2, Cell 2 is the neighbouring cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.1.10.2.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2.A.3.

2. Set the parameters according to Table 9.1.10.2.5-1 and 9.1.10.2.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP value for Cell 2 is compared to the reported RSRP value for Cell 1 for each MeasurementReport message according to Table 9.1.10.2.5-3. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.10.2.5-2 as appropriate.

### 9.1.10.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.10.2.4.3-1: Common Exception messages for RSRP FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS relative accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3<br>Table H.3.5-5 |

**Table 9.1.10.2.4.3-2: *MeasResults*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.10.2.4.3-3: *MeasResultListEUTRA*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.1.10.2.4.3-4: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '00010000000100000001 00000001000000010000' | BIT STRING (SIZE (40)) | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.10.2.4.3-5: *MeasObjectEUTRA-GENERIC(Freq)*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternFDD-r10 | '01000000100000001000 00000010000001000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 2 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.10.2.4.3-6: *SystemInformationBlockType2*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 SystemInformationBlockType2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfiguration | Not present | | Cell 2 |
| } | | | |

**Table 9.1.10.2.4.3-7: *SystemInformationBlockType3*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| neighCellConfig | '01'B (No MBSFN subframes are present in all neighbour cells) | | Cell 1 |
| | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 2 |

## 9.1.10.2.5 Test requirement

Table 9.1.10.2.5-2 defines the primary level settings including test tolerances for all tests.

The RSRP FDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.10.2.5-3.

**Table 9.1.10.2.5-1: General test parameters for E-UTRAN RSRP FDD relative accuracy under time-domain measurement resource restriction with MBSFN ABS**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Serving cell (PCell) | | Cell 1 | The aggressor cell to Cell 2 |
| Neighbour cell | | Cell 2 | Cell to be measured |
| E-UTRA RF Channel Number | | 1 | One FDD carrier frequency is used |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | For both cells in the test |
| CP length | | Normal | For both cells in the test |
| DRX | | | OFF |
| Time offset between cells | | 3 μs | Synchronous cells |
| Physical cell ID PCI | | $(PCI_{cell1} - PCI_{cell2})mod6 =0$, $PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell PCIs for Cell 1 and Cell 2 are selected randomly so that the condition is met |
| ABS pattern | | '01000000100000001000 00000010000001000000' | MBSFN ABS pattern. FDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 1. The first/leftmost bit corresponds to the subframe #0 of a radio frame satisfying SFN mod x = 0 , where x is the size of the bit string (40) divided by 10. All ABS subframes are MBSFN subframes. |
| Time-domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | '01000000100000001000 00000010000001000000' | Configured for Cell 2 measurements by measSubframePatternNeigh IE in measSubframePatternConfigNeigh, as defined in TS 36.331, clause 6.3.5. measSubframeCellList contains Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | '00010000000100000001 00000001000000010000' | Configured for measurements on Cell 1. |

**Table 9.1.10.2.5-2: Cell Specific test parameters for E-UTRAN RSRP FDD relative accuracy under time-domain measurement resource restriction with MBSFN ABS**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - | R.0 FDD | - | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.1.2 | | | R.6 FDD | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in D.1.8 (OP.8 FDD) and D.1.6 (OP.6 FDD) Note5 | | | OP.8 FDD | OP.6 FDD | OP.8 FDD | OP.6 FDD | OP.8 FDD | OP.6 FDD |
| PBCH_RA | | dB | Note 6 | 0 | Note 6 | 0 | Note 6 | 0 |
| PBCH_RB | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA Note1 | | | | | | | | |
| OCNG_RB Note1 | | | | | | | | |
| PSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| SSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| $N_{oc}$ Note 2 | Bands FDD_A | dBm/15 kHz | -106 | | -88 | | -116 | |
| | Bands FDD_C | | | | | | -115 | |
| | Bands FDD_D | | | | | | -114.5 | |
| | Bands FDD_E, FDD_F Note 8 | | | | | | -114 | |
| | Bands FDD_G | | | | | | -113 | |
| | Bands FDD_H | | | | | | -112.5 | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 5 | -1.2 | 5 | -3.05 | 5 | -3.05 |
| CRS $\left(\hat{E}_s / Iot\right)_{meas}$ note 5, 7 in the 1st OFDM symbol | | dB | 2.88 | -8.19 | 3.54 | -10.19 | 3.54 | -10.19 |
| CRS $\left(\hat{E}_s / Iot\right)_{meas}$ note 5 in OFDM symbols 4,7,11 | | dB | 2.55 | -1.2 | 3.25 | -3.05 | 3.25 | -3.05 |
| SCH $\hat{E}_s / I_{ot}$ | | dB | -1.45 | -4.74 | -0.75 | -6.59 | -0.75 | -6.59 |
| RSRP Note 3,4 | Bands FDD_A | dBm/15 kHz | -101 | -107.2 | -83 | -91.05 | -111 | -119.05 |
| | Bands FDD_C | | | | | | -110 | -118.05 |
| | Bands FDD_D | | | | | | -109.5 | -117.55 |
| | Bands FDD_E, FDD_F Note 8 | | | | | | -109 | -117.05 |
| | Bands FDD_G | | | | | | -108 | -116.05 |
| | Bands FDD_H | | | | | | -107.5 | -115.55 |
| $(\mathrm{Io})_{meas}$ Note 3 in the 1st OFDM symbol | Bands FDD_A | dBm/9 MHz | -71.3 | -74.63 | -53.54 | -57.16 | -81.54 | -85.16 |
| | Bands FDD_C | | | | | | -80.54 | -84.16 |
| | Bands FDD_D | | | | | | -80.04 | -83.66 |
| | Bands FDD_E, FDD_F Note 8 | | | | | | -79.54 | -83.16 |
| | Bands FDD_G | | | | | | -78.54 | -82.16 |
| | Bands FDD_H | | | | | | -78.04 | -81.66 |
| $(\mathrm{Io})_{meas}$ Note 3 in OFDM symbols other than the 1st one | Bands FDD_A | dBm/9 MHz | -71.30 | -75.77 | -53.54 | -58.47 | -81.54 | -86.47 |
| | Bands FDD_C | | | | | | -80.54 | -85.47 |
| | Bands FDD_D | | | | | | -80.04 | -84.97 |
| | Bands FDD_E, FDD_F Note 8 | | | | | | -79.54 | -84.47 |
| | Bands FDD_G | | | | | | -78.54 | -83.47 |
| | Bands FDD_H | | | | | | -78.04 | -82.97 |

| Propagation condition | | AWGN | AWGN | AWGN |
|---|---|---|---|---|

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: Applies to restricted measurement subframes of the respective cell.

Note 6: Non-ABS and ABS subframe channel powers defined in Table C.3.1.2.1-1.

Note 7: In the 1st OFDM symbol, Cell 2 is not expected to meet the Es/Iot side condition in TS36.133 [4] clause 9.1.2.3 and 9.1.2.4.

Note 8: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 9: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.10.2.5-3: E-UTRAN RSRP FDD relative accuracy under time-domain measurement resource restriction with MBSFN ABS requirements for the reported values**

| **Normal Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3 All bands** |
|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_x-10 | RSRP_x-13 | RSRP_x-13 |
| Highest reported value (Cell 2) | RSRP_x-4 | RSRP_x-4 | RSRP_x-4 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3 All bands** |
| Lowest reported value (Cell 2) | RSRP_x-11 | RSRP_x-13 | RSRP_x-13 |
| Highest reported value (Cell 2) | RSRP_x-3 | RSRP_x-4 | RSRP_x-4 |
| RSRP_x is the reported value of Cell 1 | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.11 TDD RSRP Accuracy E-UTRA under Time-Domain Measurement Resource Restriction with MBSFN ABS (eICIC)

### 9.1.11.1 TDD Absolute RSRP under Time-Domain Measurement Resource Restriction with MBSFN ABS (eICIC)

#### 9.1.11.1.1 Test purpose

To verify that TDD absolute RSRP measurement accuracy measurements under time-domain measurement resource restriction with MBSFN ABS configured in the aggressor cell is within the specified limits under AWGN propagation conditions. This test will verify the TDD absolute RSRP accuracy.

#### 9.1.11.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and release 11 supporting eICIC. Applicability requires support for FGI bit 115.

#### 9.1.11.1.3 Minimum conformance requirements

The TDD RSRP measurements of cells shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.3.

The accuracy requirements in table 9.1.11.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.1.11.1.3-1: RSRP Intra frequency absolute accuracy under time domain measurement resource restriction**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 2 range | | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 3 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±6 | ±9 | ≥-4 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-4 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRP measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 3: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.11.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.11.1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.3 , clause 9.1.4 and A.9.1.11

## 9.1.11.1.4 Test description

### 9.1.11.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.11.1.4.3.

4. Cell 1 is the serving cell as well as aggressor cell to Cell 2, Cell 2 is the neighbouring cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

### 9.1.11.1.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3..

2. Set the parameters according to Table 9.1.11.1.5-1 and 9.1.11.1.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP values for Cell 1 and Cell 2 are compared to the actual RSRP values according to Table 9.1.11.1.5-3. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved.

### 9.1.11.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.11.1.4.3-1: Common Exception message for RSRP TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3 |

**Table 9.1.11.1.4.3-2: *MeasResults*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.11.1.4.3-3: *MeasResultListEUTRA*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of Cell 2 | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.1.11.1.4.3-4: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 CHOICE { | | | |
| subframeConfig1-5-r10 | '10000000001000000000' | BIT STRING (SIZE (20)) | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.11.1.4.3-5: *MeasObjectEUTRA-GENERIC(Freq)*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternTDD-r10 CHOICE { | | | |
| subframeConfig1-5-r10 | '00001000000000100000' | BIT STRING (SIZE (20)) | |
| } | | | |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 2 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.11.1.4.3-6: *SystemInformationBlockType2*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: Clause 4.4.3.3, Table 4.4.3.3-1 SystemInformationBlockType2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| mbsfn-SubframeConfigList ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF SEQUENCE { | | | Cell 1 |
| radioframeAllocationPeriod | n1 | Every radio frame is with MBSFN subframe | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| oneFrame | '010000'B | Subframe 4 is used for MBSFN | |
| } | | | |
| } | | | |

**Table 9.1.11.1.4.3-7: *SystemInformationBlockType3*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 2 |

### 9.1.11.1.5 Test requirement

Table 9.1.11.1.5-2 defines the primary level settings including test tolerances for all tests.

The RSRP TDD absolute accuracy test shall meet the reported values test requirements in table 9.1.11.1.5-3.

**Table 9.1.11.1.5-1: General test parameters for E-UTRAN TDD RSRP intra-frequency test parameters under time-domain measurement resource restriction with MBSFN ABS**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Serving cell (PCell) | | Cell 1 | The aggressor cell to Cell 2 |
| Neighbour cell | | Cell 2 | Cell to be measured |
| E-UTRA RF Channel Number | | 1 | One TDD carrier frequency is used |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | For both cells in the test |
| Special subframe configuration | | 6 | For Cell 1 and Cell 2. For special subframe configurations see Table 4.2-1 in [9]. |
| Uplink/downlink subframe configuration | | 1 | For Cell 1 and Cell 2. For uplink-downlink subframe configurations see Table 4.2-2 in [9]. |
| CP length | | Normal | For both cells in the test |
| DRX | | | OFF |
| Time offset between cells | | 3 µs | Synchronous cells |
| Physical cell ID PCI | | $(PCI_{cell1} - PCI_{cell2})mod6 =0$, $PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell PCIs for Cell 1 and Cell 2 are selected randomly so that the condition is met |
| ABS pattern | | '00001000000000100000' | MBSFN ABS pattern. TDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 1. The first/leftmost bit corresponds to the subframe #0 of a radio frame satisfying SFN mod x = 0, where x is the size of the bit string (20) divided by 10. All ABS subframes are MBSFN subframes. |
| Time-domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | '00001000000000100000' | Configured for Cell 2 measurements by measSubframePatternNeigh IE in measSubframePatternConfigNeigh, as defined in TS 36.331 clause 6.3.5. measSubframeCellList contains Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | '10000000001000000000' | Configured for measurements on Cell 1. |

**Table 9.1.11.1.5-2: Cell-specific test parameters for E-UTRAN TDD RSRP intra-frequency test parameters under time-domain measurement resource restriction with MBSFN ABS**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | R.0 TDD | - | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.5 (OP.5 TDD) and D.2.2 (OP.2 TDD) Note5 | | | OP.5 TDD | OP.2 TDD | OP.5 TDD | OP.2 TDD | OP.5 TDD | OP.2 TDD |
| PBCH_RA | | dB | Note 6 | 0 | Note 6 | 0 | Note 6 | 0 |
| PBCH_RB | | | | | | | | |
| PSS_RA | | | | | | | | |
| SSS_RA | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA Note1 | | | | | | | | |
| OCNG_RB Note1 | | | | | | | | |
| PSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| SSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| $N_{oc}$ Note 2 | Bands TDD_A | dBm/15 kHz | -106 | | -88 | | -116 | |
| | Bands TDD_C | | | | | | -115 | |
| | Bands TDD_E | | | | | | -114 | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 5 | -2 | 5 | -3.05 | 5 | -3.05 |
| CRS $\left(\hat{E}_s / Iot\right)_{meas}$ note 5, 7 in the 1st OFDM symbol | | dB | 2.88 | -8.19 | 3.54 | -10.19 | 3.54 | -10.19 |
| CRS $\left(\hat{E}_s / Iot\right)_{meas}$ note 5 in OFDM symbols 4,7,11 | | dB | 2.88 | -2 | 3.25 | -3.05 | 3.25 | -3.05 |
| SCH $\hat{E}_s / I_{ot}$ | | dB | -1.12 | -5.54 | -0.75 | -6.59 | -0.75 | -6.59 |
| RSRP Note 3,4 | Bands TDD_A | dBm/15 kHz | -101 | -108 | -83 | -91.05 | -111 | -119.05 |
| | Bands TDD_C | | | | | | -110 | -118.05 |
| | Bands TDD_E | | | | | | -109 | -117.05 |
| $(\mathrm{Io})_{meas}$ Note 3 in the 1st OFDM symbol | Bands TDD_A | dBm/9 MHz | -71.41 | -74.88 | -53.63 | -57.37 | -81.63 | -85.37 |
| | Bands TDD_C | | | | | | -80.63 | -84.37 |
| | Bands TDD_E | | | | | | -79.63 | -83.37 |
| $(\mathrm{Io})_{meas}$ Note 3 in OFDM symbols other than the 1st one | Bands TDD_A | dBm/9 MHz | -71.41 | -76.09 | -53.54 | -57.16 | -81.54 | -85.16 |
| | Bands TDD_C | | | | | | -80.54 | -84.16 |
| | Bands TDD_E | | | | | | -79.54 | -83.16 |
| Propagation condition | | | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.
Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes.
Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port
Note 5: Applies to restricted measurement subframes of the respective cell.
Note 6: Non-ABS and ABS subframe channel powers defined in Table C.3.1.2.1-1.
Note 7: In the 1st OFDM symbol, Cell 2 is not expected to meet the Es/Iot side condition in 9.1.2.3 and 9.1.2.4.
Note 8: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.11.1.5-3: E-UTRAN RSRP TDD absolute accuracy under time-domain measurement resource restriction with MBSFN ABS requirements for the reported values**

| Normal Conditions | Test 1 All bands | Test 2 All bands | Test 3 | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_25 | RSRP_40 | Bands TDD_A | RSRP_14 |
| | | | Bands TDD_C | RSRP_15 |
| | | | Bands TDD_E | RSRP_16 |
| Highest reported value (Cell 2) | RSRP_40 | RSRP_59 | Bands TDD_A | RSRP_29 |
| | | | Bands TDD_C | RSRP_30 |
| | | | Bands TDD_E | RSRP_31 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_22 | RSRP_37 | Bands TDD_A | RSRP_11 |
| | | | Bands TDD_C | RSRP_12 |
| | | | Bands TDD_E | RSRP_13 |
| Highest reported value (Cell 2) | RSRP_43 | RSRP_62 | Bands TDD_A | RSRP_32 |
| | | | Bands TDD_C | RSRP_33 |
| | | | Bands TDD_E | RSRP_34 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.11.1_1 TDD Absolute RSRP under Time-Domain Measurement Resource Restriction with MBSFN ABS (eICIC) (Rel-12 and forward)

Editor's note: This test case is incomplete. The following item is missing or incomplete:

The test tolerances and test requirements for test 1, 2 and 3 in normal conditions are TBD

### 9.1.11.1_1.1 Test purpose

Same test purpose as in clause 9.1.11.1.1.

### 9.1.11.1_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward supporting eICIC. Applicability requires support for FGI bit 115.

### 9.1.11.1_1.3 Minimum conformance requirements

The TDD RSRP measurements of cells shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.3.

The accuracy requirements in table 9.1.11.1_1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.1.11.1_1.3-1: RSRP Intra frequency absolute accuracy under time domain measurement resource restriction**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 2 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 3 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-4 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-4 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRP measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 3: The condition level is increased by $\Delta>0$, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.11.1_1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.11.1_1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.3, clause 9.1.4 and A.9.1.11.

## 9.1.11.1_1.4 Test description

### 9.1.11.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.11.1.4.1.

### 9.1.11.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.11.1.4.2 with the following exceptions:

- Instead of Table 9.1.11.1.5-1→ use Table 9.1.11.1_1.5-1.
- Instead of Table 9.1.11.1.5-2→ use Table 9.1.11.1_1.5-2.
- Instead of Table 9.1.11.1.5-3→ use Table 9.1.11.1_1.5-3.

### 9.1.11.1_1.4.3 Message contents

Same message contents as in clause 9.1.11.1.4.3.

## 9.1.11.1_1.5 Test requirement

Table 9.1.11.1_1.5-2 defines the primary level settings including test tolerances for all tests.

The RSRP TDD absolute accuracy test shall meet the reported values test requirements in table 9.1.11.1_1.5-3.

**Table 9.1.11.1_1.5-1: General test parameters for E-UTRAN TDD RSRP intra-frequency test parameters under time-domain measurement resource restriction with MBSFN ABS**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Serving cell (PCell) | | Cell 1 | The aggressor cell to Cell 2 |
| Neighbour cell | | Cell 2 | Cell to be measured |
| E-UTRA RF Channel Number | | 1 | One TDD carrier frequency is used |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | For both cells in the test |
| Special subframe configuration | | 6 | For Cell 1 and Cell 2. For special subframe configurations see Table 4.2-1 in [9]. |
| Uplink/downlink subframe configuration | | 1 | For Cell 1 and Cell 2. For uplink-downlink subframe configurations see Table 4.2-2 in [9]. |
| CP length | | Normal | For both cells in the test |
| DRX | | | OFF |
| Time offset between cells | | 3 μs | Synchronous cells |
| Physical cell ID PCI | | ($PCI_{cell1}$ - $PCI_{cell2}$ )mod6 =0, $PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell PCIs for Cell 1 and Cell 2 are selected randomly so that the condition is met |
| ABS pattern | | '00001000000000100000' | MBSFN ABS pattern. TDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 1. The first/leftmost bit corresponds to the subframe #0 of a radio frame satisfying SFN mod x = 0, where x is the size of the bit string (20) divided by 10. All ABS subframes are MBSFN subframes. |
| Time-domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | '00001000000000100000' | Configured for Cell 2 measurements by measSubframePatternNeigh IE in measSubframePatternConfigNeigh, as defined in TS 36.331 clause 6.3.5. measSubframeCellList contains Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | '10000000001000000000' | Configured for measurements on Cell 1. |

**Table 9.1.11.1_1.5-2: Cell-specific test parameters for E-UTRAN TDD RSRP intra-frequency test parameters under time-domain measurement resource restriction with MBSFN ABS**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | R.0 TDD | - | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.5 (OP.5 TDD) and D.2.2 (OP.2 TDD) Note5 | | | OP.5 TDD | OP.2 TDD | OP.5 TDD | OP.2 TDD | OP.5 TDD | OP.2 TDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA Note1<br>OCNG_RB Note1 | | dB | Note 6 | 0 | Note 6 | 0 | Note 6 | 0 |
| PSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| SSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| $N_{oc}$ Note 2 | Bands TDD_A | dBm/15 kHz | -106 | | -88 | | -116 | |
| | Bands TDD_C | | | | | | -115 | |
| | Bands TDD_E | | | | | | -114 | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 5+TT | -2+TT | 5+TT | -3.05+TT | 5+TT | -3.05+TT |
| CRS $\left(\hat{E}_s / Iot\right)_{meas}$ note 5, 7 in the 1st OFDM symbol | | dB | 2.88+TT | -8.19+TT | 3.54+TT | -10.19+TT | 3.54+TT | -10.19+TT |
| CRS $\left(\hat{E}_s / Iot\right)_{meas}$ note 5 in OFDM symbols 4,7,11 | | dB | 2.88+TT | -2+TT | 3.25+TT | -3.05+TT | 3.25+TT | -3.05+TT |
| SCH $\hat{E}_s / I_{ot}$ | | dB | -1.12+TT | -5.54+TT | -0.75+TT | -6.59+TT | -0.75+TT | -6.59+TT |
| RSRP Note 3,4 | Bands TDD_A | dBm/15 kHz | -101+TT | -108+TT | -83+TT | -91.05+TT | -111+TT | -119.05+TT |
| | Bands TDD_C | | | | | | -110+TT | -118.05+TT |
| | Bands TDD_E | | | | | | -109+TT | -117.05+TT |
| $(\mathrm{Io})_{meas}$ Note 3 in the 1st OFDM symbol | Bands TDD_A | dBm/9 MHz | -71.41+TT | -74.88+TT | -53.63+TT | -57.37+TT | -81.63+TT | -85.37+TT |
| | Bands TDD_C | | | | | | -80.63+TT | -84.37+TT |
| | Bands TDD_E | | | | | | -79.63+ | -83.37+T |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | TT | T |
| $(Io)_{meas}$ Note 3 in OFDM symbols other than the 1st one | Bands TDD_A | dBm/9 MHz | -71.41+TT | -76.09+TT | -53.54+TT | -57.16+TT | -81.54+TT | -85.16+T T |
| | Bands TDD_C | | | | | | -80.54+TT | -84.16+T T |
| | Bands TDD_E | | | | | | -79.54+TT | -83.16+T T |
| Propagation condition | | | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.
Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes.
Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port
Note 5: Applies to restricted measurement subframes of the respective cell.
Note 6: Non-ABS and ABS subframe channel powers defined in Table C.3.1.2.1-1.
Note 7: In the 1st OFDM symbol, Cell 2 is not expected to meet the Es/Iot side condition in 9.1.2.3 and 9.1.2.4.
Note 8: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.11.1_1.5-3: E-UTRAN RSRP TDD absolute accuracy under time-domain measurement resource restriction with MBSFN ABS requirements for the reported values**

| **Normal Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | TBD | TBD | Bands TDD_A | TBD |
| | | | Bands TDD_C | TBD |
| | | | Bands TDD_E | TBD |
| Highest reported value (Cell 2) | TBD | TBD | Bands TDD_A | TBD |
| | | | Bands TDD_C | TBD |
| | | | Bands TDD_E | TBD |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_22 | RSRP_37 | Bands TDD_A | RSRP_11 |
| | | | Bands TDD_C | RSRP_12 |
| | | | Bands TDD_E | RSRP_13 |
| Highest reported value (Cell 2) | RSRP_43 | RSRP_62 | Bands TDD_A | RSRP_32 |
| | | | Bands TDD_C | RSRP_33 |
| | | | Bands TDD_E | RSRP_34 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.11.2 TDD Relative RSRP under Time-Domain Measurement Resource Restriction with MBSFN ABS (eICIC)

### 9.1.11.2.1 Test purpose

To verify that TDD relative RSRP measurement accuracy under time-domain measurement resource restriction with MBSFN ABS configured in the aggressor cell is within the specified limits under AWGN propagation conditions. This test will verify the TDD relative RSRP accuracy requirements.

### 9.1.11.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward supporting eICIC. Applicability requires support for FGI bit 115.

### 9.1.11.2.3 Minimum conformance requirements

The TDD RSRP relative measurements of cells shall meet relative accuracy requirements defined in TS 36.133 clause 9.1.2.4.

The accuracy requirements in Table 9.1.11.2.3-1 for relative accuracy requirements are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

RSRP1,2|dBm according to Annex I.3.1 for a corresponding Band

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.1.11.2.3-1: RSRP Intra frequency relative accuracy under time domain measurement resource restriction**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 3 range | | |
| Normal condition | Extreme condition | Ês/Iot Note 2 | E-UTRA operating band groups Note 6 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 5 | dBm/BW$_{Channel}$ |
| ±2 | ±3 | ≥-2 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-4 dB | Note 4 | Note 4 | Note 4 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRP measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 5: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 6: E-UTRA operating band groups are as defined in Section 3.5.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.4, clause 9.1.4 and A.9.1.11.

### 9.1.11.2.4 Test description

#### 9.1.11.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.11.2.4.3.

4. Cell 1 is the serving cell as well as aggressor cell to Cell 2, Cell 2 is the neighbouring cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

### 9.1.11.2.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3..

2. Set the parameters according to Table 9.1.11.2.5-1 and 9.1.11.2.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP value for Cell 2 is compared to the reported RSRP value for Cell 1 for each MeasurementReport message according to Table 9.1.11.2.5-3.

If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved.

### 9.1.11.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.11.2.4.3-1: Common Exception message for RSRP TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3 |

**Table 9.1.11.2.4.3-2: *MeasResults*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.11.2.4.3-3: *MeasResultListEUTRA*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of Cell 2 | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.1.11.2.4.3-4: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 CHOICE { | | | |
| subframeConfig1-5-r10 | '10000000001000000000' | BIT STRING (SIZE (20)) | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.11.2.4.3-5: *MeasObjectEUTRA-GENERIC(Freq)*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternTDD-r10 CHOICE { | | | |
| subframeConfig1-5-r10 | '00001000000000100000' | BIT STRING (SIZE (20)) | |
| } | | | |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 2 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.11.2.4.3-6: *SystemInformationBlockType2*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS relative accuracy test requirement**

| Derivation Path: Clause 4.4.3.3, Table 4.4.3.3-1 SystemInformationBlockType2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| mbsfn-SubframeConfigList ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF SEQUENCE { | | | Cell 1 |
| radioframeAllocationPeriod | n1 | Every radio frame is with MBSFN subframe | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| oneFrame | '010000'B | Subframe 4 is used for MBSFN | |
| } | | | |
| } | | | |

**Table 9.1.11.2.4.3-7: *SystemInformationBlockType3*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 2 |

### 9.1.11.2.5 Test requirement

Table 9.1.11.2.5-2 defines the primary level settings including test tolerances for all tests.

The RSRP TDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.11.2.5-3.

**Table 9.1.11.2.5-1: General test parameters for E-UTRAN RSRP TDD relative accuracy under time-domain measurement resource restriction with MBSFN ABS**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Serving cell (PCell) | | Cell 1 | The aggressor cell to Cell 2 |
| Neighbour cell | | Cell 2 | Cell to be measured |
| E-UTRA RF Channel Number | | 1 | One TDD carrier frequency is used |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | For both cells in the test |
| Special subframe configuration | | 6 | For Cell 1 and Cell 2. For special subframe configurations see Table 4.2-1 in [16]. |
| Uplink/downlink subframe configuration | | 1 | For Cell 1 and Cell 2. For uplink-downlink subframe configurations see Table 4.2-2 in [9]. |
| CP length | | Normal | For both cells in the test |
| DRX | | | OFF |
| Time offset between cells | | 3 µs | Synchronous cells |
| Physical cell ID PCI | | $(PCI_{cell1} - PCI_{cell2})mod6 =0$, $PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell PCIs for Cell 1 and Cell 2 are selected randomly so that the condition is met |
| ABS pattern | | '00001000000000100000' | MBSFN ABS pattern. TDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 1. The first/leftmost bit corresponds to the subframe #0 of a radio frame satisfying SFN mod x = 0, where x is the size of the bit string (20) divided by 10. All ABS subframes are MBSFN subframes. |
| Time-domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | '00001000000000100000' | Configured for Cell 2 measurements by measSubframePatternNeigh IE in measSubframePatternConfigNeigh, as defined in TS 36.331 ,clause 6.3.5. measSubframeCellList contains Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | '10000000001000000000' | Configured for measurements on Cell 1. |

**Table 9.1.11.2.5-2: Cell-specific test parameters for E-UTRAN TDD RSRP intra-frequency test parameters under time-domain measurement resource restriction with MBSFN ABS**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | R.0 TDD | - | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.5 (OP.5 TDD) and D.2.2 (OP.2 TDD) Note5 | | | OP.5 TDD | OP.2 TDD | OP.5 TDD | OP.2 TDD | OP.5 TDD | OP.2 TDD |
| PBCH_RA<br>PBCH_RB<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA Note1<br>OCNG_RB Note1 | | dB | Note 6 | 0 | Note 6 | 0 | Note 6 | 0 |
| PSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| SSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| $N_{oc}$ Note 2 | Bands TDD_A | dBm/15 kHz | -106 | | -88 | | -116 | |
| | Bands TDD_C | | | | | | -115 | |
| | Bands TDD_E | | | | | | -114 | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 5 | -1.2 | 5 | -3.05 | 5 | -3.05 |
| CRS $\left(\hat{E}_s / Iot\right)_{meas}$ note 5, 7 in the 1st OFDM symbol | | dB | 2.88 | -8.19 | 3.54 | -10.19 | 3.54 | -10.19 |
| CRS $\left(\hat{E}_s / Iot\right)_{meas}$ note 5 in OFDM symbols 4,7,11 | | dB | 2.55 | -1.2 | 3.25 | -3.05 | 3.25 | -3.05 |
| SCH $\hat{E}_s / I_{ot}$ | | dB | -1.45 | -4.74 | -0.75 | -6.59 | -0.75 | -6.59 |
| RSRP Note 3,4 | Bands TDD_A | dBm/15 kHz | -101 | -107.2 | -83 | -91.05 | -111 | -119.05 |
| | Bands TDD_C | | | | | | -110 | -118.05 |
| | Bands TDD_E | | | | | | -109 | -117.05 |
| $(\mathrm{Io})_{meas}$ Note 3 in the 1st OFDM symbol | Bands TDD_A | dBm/9 MHz | -71.41 | -74.88 | -53.63 | -57.37 | -81.63 | -85.37 |
| | Bands TDD_C | | | | | | -80.63 | -84.37 |
| | Bands TDD_E | | | | | | -79.63 | -83.37 |
| $(\mathrm{Io})_{meas}$ Note 3 in OFDM symbols other than the 1st one | Bands TDD_A | dBm/9 MHz | -71.30 | -76.09 | -53.54 | -58.76 | -81.54 | -86.47 |
| | Bands TDD_C | | | | | | -80.54 | -85.47 |
| | Bands TDD_E | | | | | | -79.54 | -84.47 |
| Propagation condition | | | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.
Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes.
Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Applies to restricted measurement subframes of the respective cell.
Note 6: Non-ABS and ABS subframe channel powers defined in Table C.3.1.2.1-1.
Note 7: In the 1st OFDM symbol, Cell 2 is not expected to meet the Es/Iot side condition in 9.1.2.3 and 9.1.2.4.
Note 8: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.11.2.5-3: E-UTRAN RSRP TDD relative accuracy under time-domain measurement resource restriction with MBSFN ABS requirements for the reported values**

| **Normal Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3 All bands** |
|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_x-10 | RSRP_x-13 | RSRP_x-13 |
| Highest reported value (Cell 2) | RSRP_x-4 | RSRP_x-4 | RSRP_x-4 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3 All bands** |
| Lowest reported value (Cell 2) | RSRP_x-11 | RSRP_x-13 | RSRP_x-13 |
| Highest reported value (Cell 2) | RSRP_x-3 | RSRP_x-4 | RSRP_x-4 |
| RSRP_x is the reported value of Cell 1 | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.1.12 FDD RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz

## 9.1.12.1 FDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz

### 9.1.12.1.1 Test purpose

To verify that FDD absolute RSRP measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the FDD absolute RSRP accuracy requirements of cells on the primary component carrier and the secondary component carrier under bandwidth of 20MHz.

### 9.1.12.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and release 11 supporting CA with Intra-band contiguous CA, or Inter-band CA.

This test applies to all types of E-UTRA FDD UE release 11 supporting CA with Intra-band non-contiguous CA.

### 9.1.12.1.3 Minimum conformance requirements

The FDD RSRP measurements of cells on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1. The FDD RSRP measurements of cells on the secondary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1.

The accuracy requirements in table 9.1.12.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.1.6.1.3-1: RSRP FDD Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 1 range | | | |
| | | | E-UTRA operating band groups Note 3 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | $dBm/BW_{Channel}$ | $dBm/BW_{Channel}$ |
| ±6 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_B | -120.5 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.12.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.12.1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1, clause 9.1.11.2, clause 9.1.4, and A.9.1.12

### 9.1.12.1.4 Test description

#### 9.1.12.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.12.1.4.3.

4. There are two E-UTRA FDD carriers with one cell on one E-UTRA FDD carrier and two cells on the other E-UTRA FDD carrier specified in the test. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.1.12.1.4.2 Test procedure

The test consists of Cell 1 the PCell, Cell 2 the SCell on the Secondary Component Carrier (SCC), and Cell 3 the neighbouring cell on the SCC. The SCC is configured and activated. The absolute accuracy of RSRP is defined as the RSRP measured from the primary component carrier (Cell 1) and the RSRP measured from the secondary component carrier (Cell 2 and Cell 3).

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [13], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.1.12.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. The SS shall transmit an RRCConnectionReconfiguration message.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. The UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, the SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP values for Cell 1 and Cell 2 are compared to the actual RSRP values according to Table 9.1.12.1.5-2. This counts respectively as a Pass or Fail for the events “Cell 1” and “Cell 2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations for the respective event ”Cell 2” is increased by one.

10. The SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

11. If test configuration is Inter-band carrier aggregation, repeat steps 1-10 for switched PCell/SCell scenario according to the UE declared capability for UL support (within CA operation) in the individual bands.

Each of the events “Cell 1” and “Cell 2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If both events pass (“Cell 1” and “Cell 2”), the test passes ofr each configuration (without and with switched PCell/SCell scenario). If one event fails, the test fails.

#### 9.1.12.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.12.1.4.3-1: Common Exception messages for FDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-3<br>Table H.4.1-5<br>Table H.4.2-1 |

### 9.1.12.1.5 Test requirement

Table 9.1.12.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD absolute accuracy test shall meet the reported values test requirements in table 9.1.12.1.5-2.

**Table 9.1.12.1.5-1: RSRP FDD absolute accuracy carrier aggregation test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 20 | 20 | 20 |
| Measurement bandwidth | | $n_{PRB}$ | 47—52 | | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.4 FDD | R.4 FDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 38—61 | 38—61 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.10 FDD | | |
| OCNG Patterns defined in D.1.11 (OP.11 FDD) and D.1.12 (OP.12 FDD) | | | OP.11 FDD | OP.11 FDD | OP.12 FDD |
| Io Note2 | Bands FDD_A Note 5 | dBm/18 MHz | -84.75 | (Io for Channel 1 +5.33dB) | |
| | Bands FDD_B Note 5 | | -84.25 | | |
| | Bands FDD_C Note 5 | | -83.75 | | |
| | Bands FDD_D Note 5 | | -83.25 | | |
| | Bands FDD_E Note 5 | | -82.75 | | |
| | Bands FDD_G Note 5 | | -81.75 | | |
| | Bands FDD_H Note 5 | | -81.25 | | |

NOTE 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
NOTE 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 3: See Table 9.1.6.1.5-1 for the other parameters.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.
NOTE 5: The test applies for E-UTRA operating bands in this band group which are supporting 20 MHz channel bandwidth.

**Table 9.1.12.1.5-2: RSRP FDD Carrier Aggregation absolute accuracy requirements for the reported values**

| | Test 1 | | | | | | |
|---|---|---|---|---|---|---|---|
| **Band of Cell 1 on Primary Component Carrier** | **Bands FDD_A** | **Bands FDD_B** | **Bands FDD_C** | **Bands FDD_D** | **Bands FDD_E** | **Bands FDD_G** | **Bands FDD_H** |
| Normal Conditions | | | | | | | |
| Lowest reported value (Cell 1) | RSRP_12 | RSRP_12 | RSRP_13 | RSRP_13 | RSRP_14 | RSRP_15 | RSRP_15 |
| Highest reported value (Cell 1) | RSRP_27 | RSRP_28 | RSRP_28 | RSRP_29 | RSRP_29 | RSRP_30 | RSRP_31 |
| Lowest reported value (Cell 2) | RSRP_20 | RSRP_20 | RSRP_21 | RSRP_21 | RSRP_22 | RSRP_23 | RSRP_23 |
| Highest reported value (Cell 2) | RSRP_35 | RSRP_36 | RSRP_36 | RSRP_37 | RSRP_37 | RSRP_38 | RSRP_39 |
| Extreme Conditions | | | | | | | |
| Lowest reported value (Cell 1) | RSRP_9 | RSRP_9 | RSRP_10 | RSRP_10 | RSRP_11 | RSRP_12 | RSRP_12 |
| Highest reported value (Cell 1) | RSRP_30 | RSRP_31 | RSRP_31 | RSRP_32 | RSRP_32 | RSRP_33 | RSRP_34 |
| Lowest reported value (Cell 2) | RSRP_17 | RSRP_17 | RSRP_18 | RSRP_18 | RSRP_19 | RSRP_20 | RSRP_20 |
| Highest reported value (Cell 2) | RSRP_38 | RSRP_39 | RSRP_39 | RSRP_40 | RSRP_40 | RSRP_41 | RSRP_42 |
| NOTE: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2. | | | | | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.12.1_1 FDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz (Rel-12 and forward)

### 9.1.12.1_1.1 Test purpose

Same test purpose as in clause 9.1.12.1.1.

### 9.1.12.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward supporting CA.

### 9.1.12.1_1.3 Minimum conformance requirements

The FDD RSRP measurements of cells on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1. The FDD RSRP measurements of cells on the secondary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1.

The accuracy requirements in table 9.1.12.1_1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 3] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 2] / dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 / N/A | -70 |
| | | | FDD_B | -120.5 / N/A | -70 |
| | | | FDD_C, TDD_C | -120 / N/A | -70 |
| | | | FDD_D | -119.5 / N/A | -70 |
| | | | FDD_E, TDD_E | -119 / N/A | -70 |
| | | | FDD_F | -118.5 / N/A | -70 |
| | | | FDD_G | -118 / N/A | -70 |
| | | | FDD_H | -117.5 / N/A | -70 |
| | | | FDD_N | -114.5 / N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.12.1_1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.12.1_1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1, clause 9.1.11.2, clause 9.1.4, and A.9.1.12.

## 9.1.12.1_1.4 Test description

### 9.1.12.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.12.1.4.1 with the following exceptions:

- Instead of 9.1.12.1.4.3 → use 9.1.12.1_1.4.3.

### 9.1.12.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.12.1.4.2 with the following exceptions:

- Instead of Table 9.1.12.1.5-1 → use Table 9.1.12.1_1.5-1.
- Instead of Table 9.1.12.1.5-2 → use Table 9.1.12.1_1.5-2.

### 9.1.12.1_1.4.3 Message contents

Same message contents as in clause 9.1.12.1.4.3.

### 9.1.12.1_1.5 Test requirement

Table 9.1.12.1_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD absolute accuracy test shall meet the reported values test requirements in table 9.1.12.1_1.5-2.

**Table 9.1.12.1_1.5-1: RSRP FDD absolute accuracy carrier aggregation test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 20 | 20 | 20 |
| Measurement bandwidth | | $n_{PRB}$ | 47—52 | | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.4 FDD | R.4 FDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 38—61 | 38—61 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.10 FDD | | |
| OCNG Patterns defined in D.1.11 (OP.11 FDD) and D.1.12 (OP.12 FDD) | | | OP.11 FDD | OP.11 FDD | OP.12 FDD |
| Io Note2 | Bands FDD_A Note 5 | dBm/18 MHz | -84.75 | (Io for Channel 1 +5.33dB) | |
| | Bands FDD_B Note 5 | | -84.25 | | |
| | Bands FDD_C Note 5 | | -83.75 | | |
| | Bands FDD_D Note 5 | | -83.25 | | |
| | Bands FDD_E Note 5 | | -82.75 | | |
| | Bands FDD_G Note 5 | | -81.75 | | |
| | Bands FDD_H Note 5 | | -81.25 | | |

NOTE 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
NOTE 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 3: See Table 9.1.6.1.5-1 for the other parameters.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.
NOTE 5: The test applies for E-UTRA operating bands in this band group which are supporting 20 MHz channel bandwidth.

**Table 9.1.12.1_1.5-2: RSRP FDD Carrier Aggregation absolute accuracy requirements for the reported values**

| | Test 1 | | | | | | |
|---|---|---|---|---|---|---|---|
| **Band of Cell 1 on Primary Component Carrier** | **Bands FDD_A** | **Bands FDD_B** | **Bands FDD_C** | **Bands FDD_D** | **Bands FDD_E** | **Bands FDD_G** | **Bands FDD_H** |
| Normal Conditions | | | | | | | |
| Lowest reported value (Cell 1) | RSRP_13 | RSRP_14 | RSRP_14 | RSRP_15 | RSRP_15 | RSRP_16 | RSRP_17 |
| Highest reported value (Cell 1) | RSRP_26 | RSRP_26 | RSRP_27 | RSRP_27 | RSRP_28 | RSRP_29 | RSRP_29 |
| Lowest reported value (Cell 2) | RSRP_21 | RSRP_22 | RSRP_22 | RSRP_23 | RSRP_23 | RSRP_24 | RSRP_25 |
| Highest reported value (Cell 2) | RSRP_34 | RSRP_34 | RSRP_35 | RSRP_35 | RSRP_36 | RSRP_37 | RSRP_37 |
| Extreme Conditions | | | | | | | |
| Lowest reported value (Cell 1) | RSRP_9 | RSRP_9 | RSRP_10 | RSRP_10 | RSRP_11 | RSRP_12 | RSRP_12 |
| Highest reported value (Cell 1) | RSRP_30 | RSRP_31 | RSRP_31 | RSRP_32 | RSRP_32 | RSRP_33 | RSRP_34 |
| Lowest reported value (Cell 2) | RSRP_17 | RSRP_17 | RSRP_18 | RSRP_18 | RSRP_19 | RSRP_20 | RSRP_20 |
| Highest reported value (Cell 2) | RSRP_38 | RSRP_39 | RSRP_39 | RSRP_40 | RSRP_40 | RSRP_41 | RSRP_42 |
| NOTE: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2. | | | | | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.12.2 FDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz

### 9.1.12.2.1 Test purpose

To verify that FDD relative RSRP measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the FDD relative RSRP accuracy requirements of cells between the primary and secondary component carrier and FDD relative RSRP accuracy of cells on the secondary component carriers.

### 9.1.12.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and release 11 supporting CA with Intra-band contiguous CA, or Inter-band CA.

This test applies to all types of E-UTRA FDD UE release 11 supporting CA with Intra-band non-contiguous CA.

### 9.1.12.2.3 Minimum conformance requirements

The FDD RSRP relative measurements of cells on PCC and SCC shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.3.2. The FDD RSRP relative measurements of cells on the SCC shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.2.2.

The accuracy requirements in Table 9.1.12.2.3-1 for PCC-SCC relative accuracy requirements are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

RSRP1,2$|_{dBm}$ according to Annex I.3.4 for a corresponding Band

$$\left| RSRP1 \right|_{dBm} - RSRP2 \left|_{dBm} \right| \leq 27dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.12.2.3-1: FDD RSRP relative accuracy for PCC and SCC**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 6 | $dBm/BW_{Channel}$ |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 Note 3 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 4 | Note 4 | Note 4 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The condition has the minimum Io of -119 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz.
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.
NOTE 6: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.

The requirements in Table 9.1.12.2.3-2 for SCC relative accuracy are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

$RSRP1,2|_{dBm}$ according to Annex I.3.8 for a corresponding Band.

**Table 9.1.12.2.3-2: FDD RSRP relative accuracy for SCCs**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 6 | $dBm/BW_{Channel}$ |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 Note 3 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 4 | Note 4 | Note 4 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The condition has the minimum Io of -119 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz.
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.
NOTE 6: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.2, clause 9.1.11.3, clause 9.1.4, and A.9.1.12.

### 9.1.12.2.4 Test description

#### 9.1.12.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.12.2.4.3.

4. There are two E-UTRA FDD carriers with one cell on one E-UTRA FDD carrier and two cells on the other E-UTRA FDD carrier specified in the test. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.1.12.2.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [13], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.1.12.2.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. SS shall transmit an RRCConnectionReconfiguration message.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP value for Cell 2 is compared to the reported RSRP value for Cell 1 for each MeasurementReport message according to Table 9.1.12.2.5-2. Also the reported RSRP value for Cell 3 is compared to the reported RSRP value for Cell 2 for each MeasurementReport message according to Table 9.1.12.2.5-2. This counts respectively as a Pass or Fail for the events “Cell 1-2” and “Cell 2-3”. If the UE fails to report the measurement value for Cell 2 / Cell 3, the number of failed iterations for the respective affected event “Cell 1-2” / ”Cell 2-3” is increased by one.

10. SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

11. If test configuration is Inter-band carrier aggregation, repeat steps 1-10 for switched PCell/SCell scenario according to the UE declared capability for UL support (within CA operation) in the individual bands.

Each of the events “Cell 1-2” and “Cell 2-3” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If both events (“Cell 1-2” and “Cell 2-3”) pass for each configuration (without and with switched PCell/SCell scenario), the test passes. If one event fails, the test fails.

9.1.12.2.4.3 Message contents

**Table 9.1.12.2.4.3-1: Common Exception messages for FDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-3<br>Table H.4.1-5<br>Table H.4.2-1 |

## 9.1.12.2.5 Test requirement

Table 9.1.12.2.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.12.2.5-2.

**Table 9.1.12.2.5-1: RSRP TDD relative accuracy carrier aggregation test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 20 | 20 | 20 |
| Measurement bandwidth | | $n_{PRB}$ | 47—52 | | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.4 FDD | R.4 FDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 38—61 | 38—61 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.10 FDD | | |
| OCNG Patterns defined in D.1.11 (OP.11 FDD) and D.1.12 (OP.12 FDD) | | | OP.11 FDD | OP.11 FDD | OP.12 FDD |
| Io Note2 | Bands FDD_A Note 5 | dBm/18 MHz | -84.75 | (Io for Channel 1 +5.33dB) | |
| | Bands FDD_B Note 5 | | -84.25 | | |
| | Bands FDD_C Note 5 | | -83.75 | | |
| | Bands FDD_D Note 5 | | -83.25 | | |
| | Bands FDD_E Note 5 | | -82.75 | | |
| | Bands FDD_G Note 5 | | -81.75 | | |
| | Bands FDD_H Note 5 | | -81.25 | | |

NOTE 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
NOTE 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 3: See Table 9.1.6.2.5-1 for the other parameters.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.
NOTE 5: The test applies for E-UTRA operating bands in this band group which are supporting 20 MHz channel bandwidth.

**Table 9.1.12.2.5-2: RSRP TDD Carrier Aggregation relative accuracy requirements for the reported values**

| | Test 1 |
|---|---|
| | All bands |
| Normal and Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRP_x – 1+TT |
| Highest reported value (Cell 2) | RSRP_x + 17+TT |
| Lowest reported value (Cell 3) | RSRP_y – 8+TT |
| Highest reported value (Cell 3) | RSRP_y + 1+TT |
| RSRP_x is the reported value of Cell 1<br>RSRP_y is the reported value of Cell 2 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.12.2_1FDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz (Rel-12 and forward)

### 9.1.12.2_1.1 Test purpose

Same test purpose as defined in clause 9.1.12.2.1.

### 9.1.12.2_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward supporting CA.

### 9.1.12.2_1.3 Minimum conformance requirements

Same minimum conformance requirements as defined in clause 9.1.6.2_1.3.

### 9.1.12.2_1.4 Test description

#### 9.1.12.2_1.4.1 Initial conditions

Same initial conditions as defined in clause 9.1.12.2.4.1.

#### 9.1.12.2_1.4.2 Test procedure

Same test procedure as defined in clause 9.1.12.2.4.2 with the following exceptions:

- Instead of Table 9.1.12.2.5-1 → use Table 9.1.12.2_1.5-1.
- Instead of Table 9.1.12.2.5-2 → use Table 9.1.12.2_1.5-2.

#### 9.1.12.2_1.4.3 Message contents

Same message contents as defined in clause 9.1.12.2.4.3.

### 9.1.12.2_1.5 Test requirement

Table 9.1.12.2_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.12.2_1.5-2.

**Table 9.1.12.2_1.5-1: RSRP TDD relative accuracy carrier aggregation test parameters**

| Parameter | Unit | Test 1 | | |
|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 3 |

| | | | | | |
|---|---|---|---|---|---|
| $BW_{channel}$ Note 1 | | MHz | 20 | 20 | 20 |
| Measurement bandwidth | | $n_{PRB}$ | 47—52 | | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.4 FDD | R.4 FDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 38—61 | 38—61 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.10 FDD | | |
| OCNG Patterns defined in D.1.11 (OP.11 FDD) and D.1.12 (OP.12 FDD) | | | OP.11 FDD | OP.11 FDD | OP.12 FDD |
| Io Note2 | Bands FDD_A Note 5 | dBm/18 MHz | -84.75 | (Io for Channel 1 +5.33dB) | |
| | Bands FDD_B Note 5 | | -84.25 | | |
| | Bands FDD_C Note 5 | | -83.75 | | |
| | Bands FDD_D Note 5 | | -83.25 | | |
| | Bands FDD_E Note 5 | | -82.75 | | |
| | Bands FDD_G Note 5 | | -81.75 | | |
| | Bands FDD_H Note 5 | | -81.25 | | |

NOTE 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
NOTE 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 3: See Table 9.1.6.2.5-1 for the other parameters.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.
NOTE 5: The test applies for E-UTRA operating bands in this band group which are supporting 20 MHz channel bandwidth.

**Table 9.1.12.2_1.5-2: RSRP TDD Carrier Aggregation relative accuracy requirements for the reported values**

| | **Test 1** |
|---|---|
| | **All bands** |
| Normal Conditions | |
| Lowest reported value (Cell 2) | RSRP_x +2 |
| Highest reported value (Cell 2) | RSRP_x +15 |
| Lowest reported value (Cell 3) | RSRP_y - 8 |
| Highest reported value (Cell 3) | RSRP_y + 1 |
| Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRP_x – 1 |
| Highest reported value (Cell 2) | RSRP_x + 17 |
| Lowest reported value (Cell 3) | RSRP_y – 8 |
| Highest reported value (Cell 3) | RSRP_y + 1 |
| RSRP_x is the reported value of Cell 1<br>RSRP_y is the reported value of Cell 2 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.1.13 TDD RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz

## 9.1.13.1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz

### 9.1.13.1.1 Test purpose

To verify that TDD absolute RSRP measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the TDD absolute RSRP accuracy requirements of cells on the primary component carrier and the secondary component carrier under bandwidth of 20MHz.

### 9.1.13.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and release 11 supporting CA with Intra-band contiguous CA, or Inter-band CA.

This test applies to all types of E-UTRA TDD UE release 11 supporting CA with Intra-band non-contiguous CA.

### 9.1.13.1.3 Minimum conformance requirements

The TDD RSRP measurements of cells on the primary component carrier shall meet the absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1. The TDD RSRP measurements of cells on the secondary component carrier shall meet the absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1.

The accuracy requirements in table 9.1.13.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.1.13.1.3-1: RSRP TDD absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 / dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±6 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 / N/A | -70 |
| | | | FDD_B | -120.5 / N/A | -70 |
| | | | FDD_C, TDD_C | -120 / N/A | -70 |
| | | | FDD_D | -119.5 / N/A | -70 |
| | | | FDD_E, TDD_E | -119 / N/A | -70 |
| | | | FDD_F | -118.5 / N/A | -70 |
| | | | FDD_G | -118 / N/A | -70 |
| | | | FDD_H | -117.5 / N/A | -70 |
| | | | FDD_N | -114.5 / N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.13.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.13.1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1,clause 9.1.11.2 clause 9.1.4 and A.9.1.13.

### 9.1.13.1.4 Test description

#### 9.1.13.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.13.1.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.1.13.1.4.2 Test procedure

The test consists of Cell 1 the PCell, Cell 2 the SCell on the Secondary Component Carrier (SCC), and Cell 3 the neighbouring cell on the SCC. The SCC is configured and activated. The absolute accuracy of RSRP is defined as the RSRP measured from the primary component carrier (Cell 1) and the RSRP measured from the secondary component carrier (Cell 2 and Cell 3).

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [13], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.1.13.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. The SS shall transmit an RRCConnectionReconfiguration message.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. The UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, the SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP values for Cell 1 and Cell 2 are compared to the actual RSRP values according to Table

9.1.13.1.5-2. This counts respectively as a Pass or Fail for the events “Cell 1” and “Cell 2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations for the respective event ”Cell 2” is increased by one.

10. The SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events “Cell 1” and “Cell 2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If both events pass, the test passes. If one event fails, the test fails.

#### 9.1.13.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.13.1.4.3-1: Common Exception messages for TDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-3<br>Table H.4.1-5<br>Table H.4.2-1 |

### 9.1.13.1.5 Test requirement

Table 9.1.13.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD absolute accuracy test shall meet the reported values test requirements in table 9.1.13.1.5-2.

**Table 9.1.13.1.5-1: RSRP TDD absolute accuracy carrier aggregation test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 20 | | |
| Measurement bandwidth | | $n_{PRB}$ | 47—52 | | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.3 TDD | R.3 TDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 38—61 | 38—61 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.10 TDD | | |
| OCNG Patterns defined in D.2.7 (OP.7 TDD) and D.2.8 (OP.8 TDD) | | | OP.7 TDD | OP.7 TDD | OP.8 TDD |
| $Io^{Note2}$ | Bands TDD_A Note 5 | dBm/18 MHz | -84.75 | (Io for Channel 1 +5.33dB) | |
| | Bands TDD_C Note 5 | | -83.75 | | |
| | Bands TDD_E Note 5 | | -82.75 | | |

Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in TS 36.133 [4] section A.3.6.1.
Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: See Table 9.1.7.1.5-1 for the other parameters.
Note 4: E-UTRA operating band groups are as defined in Section 3.5.
Note 5: The test applies for E-UTRA operating bands in this band group which are supporting 20 MHz channel bandwidth.

**Table 9.1.13.1.5-2: RSRP TDD Carrier Aggregation absolute accuracy requirements for the reported values**

| Parameter | Test 1 | | |
|---|---|---|---|
| **Band of Cell 1 on Primary Component Carrier** | **Bands TDD_A** | **Bands TDD_C** | **Bands TDD_E** |
| Normal Conditions | | | |
| Lowest reported value (Cell 1) | RSRP_12 | RSRP_13 | RSRP_14 |
| Highest reported value (Cell 1) | RSRP_27 | RSRP_28 | RSRP_29 |
| Lowest reported value (Cell 2) | RSRP_20 | RSRP_21 | RSRP_22 |
| Highest reported value (Cell 2) | RSRP_35 | RSRP_36 | RSRP_37 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 1) | RSRP_9 | RSRP_10 | RSRP_11 |
| Highest reported value (Cell 1) | RSRP_30 | RSRP_31 | RSRP_32 |
| Lowest reported value (Cell 2) | RSRP_17 | RSRP_18 | RSRP_19 |
| Highest reported value (Cell 2) | RSRP_38 | RSRP_39 | RSRP_40 |
| Note: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2. | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.13.1_1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz (Rel-12 and forward)

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

-The Test tolerances applicable to this test are undefined

### 9.1.13.1_1.1 Test purpose

Same test purpose as in clause 9.1.13.1.1

### 9.1.13.1_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward supporting CA.

### 9.1.13.1_1.3 Minimum conformance requirements

The TDD RSRP measurements of cells on the primary component carrier shall meet the absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1. The TDD RSRP measurements of cells on the secondary component carrier shall meet the absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1.

The accuracy requirements in table 9.1.13.1_1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.1.13.1_1.3-1: RSRP TDD absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 3] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 2] / dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 / N/A | -70 |
| | | | FDD_C, TDD_C | -120 / N/A | -70 |
| | | | FDD_D | -119.5 / N/A | -70 |
| | | | FDD_E, TDD_E | -119 / N/A | -70 |
| | | | FDD_F | -118.5 / N/A | -70 |
| | | | FDD_G | -118 / N/A | -70 |
| | | | FDD_H | -117.5 / N/A | -70 |
| | | | FDD_N | -114.5 / N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 / -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.13.1_1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.13.1_1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1,clause 9.1.11.2 clause 9.1.4 and A.9.1.13.

## 9.1.13.1_1.4 Test description

### 9.1.13.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.13.1.4.1 with the following exceptions:

- Instead of 9.1.13.1.4.3 → use 9.1.13.1_1.4.3.

### 9.1.13.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.13.1.4.2 with the following exceptions:

- Instead of Table 9.1.13.1.5-1 → use Table 9.1.13.1_1.5-1.
- Instead of Table 9.1.13.1.5-2 → use Table 9.1.13.1_1.5-2.

### 9.1.13.1_1.4.3 Message contents

Same message contents as in clause 9.1.13.1.4.3

### 9.1.13.1_1.5 Test requirement

Table 9.1.13.1_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD absolute accuracy test shall meet the reported values test requirements in table 9.1.13.1_1.5-2.

**Table 9.1.13.1_1.5-1: RSRP TDD absolute accuracy carrier aggregation test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 20 | | |
| Measurement bandwidth | | $n_{PRB}$ | 47—52 | | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.3 TDD | R.3 TDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 38—61 | 38—61 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.10 TDD | | |
| OCNG Patterns defined in D.2.7 (OP.7 TDD) and D.2.8 (OP.8 TDD) | | | OP.7 TDD | OP.7 TDD | OP.8 TDD |
| Io Note2 | Bands TDD_A Note 5 | dBm/18 MHz | -84.75 | (Io for Channel 1 +5.33dB) | |
| | Bands TDD_C Note 5 | | -83.75 | | |
| | Bands TDD_E Note 5 | | -82.75 | | |
| [NOTE 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in TS 36.133 [4] section A.3.6.1.]<br>NOTE 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>NOTE 3: See Table 9.1.7.1.5-1 for the other parameters.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.<br>NOTE 5: The test applies for E-UTRA operating bands in this band group which are supporting 20 MHz channel bandwidth. | | | | | |

**Table 9.1.13.1_1.5-2: RSRP TDD Carrier Aggregation absolute accuracy requirements for the reported values**

| Parameter | Test 1 | | |
|---|---|---|---|
| **Band of Cell 1 on Primary Component Carrier** | **Bands TDD_A** | **Bands TDD_C** | **Bands TDD_E** |
| Normal Conditions | | | |
| Lowest reported value (Cell 1) | TBD | TBD | TBD |
| Highest reported value (Cell 1) | TBD | TBD | TBD |
| Lowest reported value (Cell 2) | TBD | TBD | TBD |
| Highest reported value (Cell 2) | TBD | TBD | TBD |
| Extreme Conditions | | | |
| Lowest reported value (Cell 1) | RSRP_9 | RSRP_10 | RSRP_11 |
| Highest reported value (Cell 1) | RSRP_30 | RSRP_31 | RSRP_32 |
| Lowest reported value (Cell 2) | RSRP_17 | RSRP_18 | RSRP_19 |
| Highest reported value (Cell 2) | RSRP_38 | RSRP_39 | RSRP_40 |
| NOTE: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2. | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.13.2 TDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz

### 9.1.13.2.1 Test purpose

To verify that TDD relative RSRP measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the TDD relative RSRP accuracy requirements of cells between the primary and secondary component carrier and TDD relative RSRP accuracy of cells on the secondary component carriers.

### 9.1.13.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and release 11 supporting CA with Intra-band contiguous CA, or Inter-band CA.

This test applies to all types of E-UTRA TDD UE release 11 supporting CA with Intra-band non-contiguous CA.

### 9.1.13.2.3 Minimum conformance requirements

The TDD RSRP relative measurements of cells on PCC and SCC shall meet relative accuracy requirements defined in TS 36.133 clause 9.1.3.2. The TDD RSRP relative measurements of cells on the SCC shall meet relative accuracy requirements defined in TS 36.133 clause 9.1.2.2.

The accuracy requirements in Table 9.1.13.2.3-1 for PCC-SCC relative accuracy requirements are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

RSRP1,2$|_{dBm}$ according to Annex I.3.8 for a corresponding Band

$$\left| RSRP1\big|_{dBm} - RSRP2\big|_{dBm} \right| \le 27dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.13.2.3-1: TDD RSRP relative accuracy for PCC and SCC**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io [Note 1] range | | |
| Normal condition | Extreme condition | Ês/Iot [Note 2] | E-UTRA operating band groups [Note 4] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 3] | dBm/BW$_{Channel}$ |
| ±6 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.3, clause 9.1.4 and A.9.1.13.

The TDD RSRP relative measurements of cells on SCCs shall meet relative accuracy requirements defined in TS 36.133 clause 9.1.2.2.

The accuracy requirements in Table 9.1.13.2.3-2 for SCCs relative accuracy requirements are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

RSRP1,2$|_{dBm}$ according to Annex I.3.8 for a corresponding Band.

**Table 9.1.13.2.3-2: TDD RSRP relative accuracy for SCCs**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/BW$_{Channel}$ |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.2, 9.1.11.3, clause 9.1.4 and A.9.1.13.

## 9.1.13.2.4 Test description

### 9.1.13.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.13.2.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

### 9.1.13.2.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [13], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.1.13.2.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. SS shall transmit an RRCConnectionReconfiguration message.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP value for Cell 2 is compared to the reported RSRP value for Cell 1 for each MeasurementReport message according to Table 9.1.13.2.5-2. Also the reported RSRP value for Cell 3 is compared to the reported RSRP value for Cell 2 for each MeasurementReport message according to Table 9.1.13.2.5-2. This counts respectively as a Pass or Fail for the events “Cell 1-2” and “Cell 2-3”. If the UE fails to report the measurement value for Cell 2 / Cell 3, the number of failed iterations for the respective affected event “Cell 1-2” / ”Cell 2-3” is increased by one.

10. SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events “Cell 1-2” and “Cell 2-3” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If both events pass, the test passes. If one event fails, the test fails.

#### 9.1.13.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.13.2.4.3-1: Common Exception messages for TDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-3<br>Table H.4.1-5<br>Table H.4.2-1 |

### 9.1.13.2.5 Test requirement

Table 9.1.13.2.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.13.2.5-2.

**Table 9.1.13.2.5-1: RSRP TDD relative accuracy carrier aggregation test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 20 | | |
| Measurement bandwidth | | $n_{PRB}$ | 47—52 | | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.3 TDD | R.3 TDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 38—61 | 38—61 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.10 TDD | | |
| OCNG Patterns defined in D.2.7 (OP.7 TDD) and D.2.8 (OP.8 TDD) | | | OP.7 TDD | OP.7 TDD | OP.8 TDD |
| Io Note2 | Bands TDD_A Note 5 | dBm/18 MHz | -84.75 | (Io for Channel 1 +5.33dB) | |
| | Bands TDD_C Note 5 | | -83.75 | | |
| | Bands TDD_E Note 5 | | -82.75 | | |

Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in TS 36.133 [4] section A.3.6.1.
Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: See Table 9.1.7.2.5-1 for the other parameters.
Note 4: E-UTRA operating band groups are as defined in Section 3.5.
Note 5: The test applies for E-UTRA operating bands in this band group which are supporting 20 MHz channel bandwidth.

**Table 9.1.13.2.5-2: RSRP TDD Carrier Aggregation relative accuracy requirements for the reported values**

| | Test 1 |
|---|---|
| | All bands |
| Normal and Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRP_x – 1 |
| Highest reported value (Cell 2) | RSRP_x + 17 |
| Lowest reported value (Cell 3) | RSRP_y – 8 |
| Highest reported value (Cell 3) | RSRP_y + 1 |
| RSRP_x is the reported value of Cell 1<br>RSRP_y is the reported value of Cell 2 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.13.2_1 TDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz (Rel-12 and forward)

Editor's note: This Test case is incomplete. The following aspects are either missing or not yet determined:

- *The Test system uncertainties applicable to this test are undefined*
- *The Test tolerances applicable to this test are undefined*
- *Test requirements in normal conditions need to be defined*

### 9.1.13.2_1.1 Test purpose

Same test purpose as defined in clause 9.1.13.2.1.

### 9.1.13.2_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward supporting CA.

### 9.1.13.2_1.3 Minimum conformance requirements

Same minimum conformance requirements as defined in clause 9.1.7.2_1.3.

### 9.1.13.2_1.4 Test description

#### 9.1.13.2_1.4.1 Initial conditions

Same initial conditions as defined in clause 9.1.13.2.4.1 with the following exceptions:

- Instead of Table 9.1.13.2.5-1 → use Table 9.1.13.2_1.5-1.
- Instead of Table 9.1.13.2.5-2 → use Table 9.1.13.2_1.5-2.

#### 9.1.13.2_1.4.2 Test procedure

Same test procedure as defined in clause 9.1.13.2.4.2.

#### 9.1.13.2_1.4.3 Message contents

Same message contents as defined in clause 9.1.13.2.4.3.

### 9.1.13.2_1.5 Test requirement

Table 9.1.13.2_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.13.2_1.5-2.

**Table 9.1.13.2_1.5-1: RSRP TDD relative accuracy carrier aggregation test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 20 | | |
| Measurement bandwidth | | $n_{PRB}$ | 47—52 | | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.3 TDD | R.3 TDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 38—61 | 38—61 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.10 TDD | | |
| OCNG Patterns defined in D.2.7 (OP.7 TDD) and D.2.8 (OP.8 TDD) | | | OP.7 TDD | OP.7 TDD | OP.8 TDD |
| Io Note2 | Bands TDD_A Note 5 | dBm/18 MHz | -84.75+TT | (Io for Channel 1 +5.33dB) | |
| | Bands TDD_C Note 5 | | -83.75+TT | | |
| | Bands TDD_E Note 5 | | -82.75+TT | | |

Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in TS 36.133 [4] section A.3.6.1.
Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: See Table 9.1.7.2.5-1 for the other parameters.
Note 4: E-UTRA operating band groups are as defined in Section 3.5.
Note 5: The test applies for E-UTRA operating bands in this band group which are supporting 20 MHz channel bandwidth.

**Table 9.1.13.2_1.5-2: RSRP TDD Carrier Aggregation relative accuracy requirements for the reported values**

| | **Test 1** |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 2) | TBD |
| Highest reported value (Cell 2) | TBD |
| Lowest reported value (Cell 3) | TBD |
| Highest reported value (Cell 3) | TBD |
| Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRP_x – 1 |
| Highest reported value (Cell 2) | RSRP_x + 17 |
| Lowest reported value (Cell 3) | RSRP_y – 8 |
| Highest reported value (Cell 3) | RSRP_y + 1 |
| RSRP_x is the reported value of Cell 1<br>RSRP_y is the reported value of Cell 2 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.14 FDD RSRP under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC)

### 9.1.14.1 FDD Intra Frequency Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC)

#### 9.1.14.1.1 Test purpose

To verify that the Absolute RSRP measurement accuracy under time-domain measurement resource restriction with CRS Assistance Information and non-MBSFN ABS configured in the aggressor cells is within the specified limits.

#### 9.1.14.1.2 Test applicability

This test applies to all types of E-UTRA FDD and CRS interference handling UE release 11. Applicability requires support of FGI bit 115.

#### 9.1.14.1.3 Minimum conformance requirements

The FDD RSRP measurements of cells shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.3.

The accuracy requirements in table 9.1.14.1.3-1 are valid under the following conditions:

Cell specific reference signals in the measured cell are transmitted either from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.9 for a corresponding Band.

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.1.14.1.3-1: RSRP Intra frequency absolute accuracy under time domain measurement resource restriction**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 2 range | | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 3 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±6 | ±9 | ≥-4 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-4 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRP measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 3: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.14.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.14.1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.3, clause 9.1.4 and A.9.1.14.

### 9.1.14.1.4 Test description

#### 9.1.14.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.54 (without faders).

2. The general test parameter settings are set up according to Table 9.1.14.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.1.14.1.4.3.

5. Cell 1 is the serving cell as well as aggressor cell to Cell 3, Cell 2 is the neighbour aggressor cell without CRS colliding to Cell 3. Cell 3 is the cell to be measured for RSRP absolute accuracy, whilst both Cell 1 and Cell 3 are measured for RSRP relative accuracy. Non-MBSFN ABS pattern is configured for Cell 1 and Cell 2 during the test. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

**Table 9.1.14.1.4.1-1: General test parameters for E-UTRAN FDD RSRP intra frequency test parameters under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Serving cell (PCell) | | | Cell 1 | The aggressor cell to Cell 3 |
| Neighbour cell | | | Cell 2 | The aggressor cell to Cell 3 |
| Neighbour cell | | | Cell 3 | Cell to be measured |
| PCell ABS configuration | | | Non-MBSFN ABS | As defined in Table C.3.1.1.1-1 |
| CP length | | | Normal | For three cells in the test |
| DRX | | | | OFF |
| Cell 2 time offset with respect to Cell 1 | | | 0μs | Synchronous cells |
| Cell 3 time offset with respect to Cell 1 | | | -2.5 μs | Synchronous cells |
| Physical cell ID PCI | | | Colliding CRS: ($PCI_{cell1}$ – $PCI_{cell3}$)mod6=0, $PCI_{cell1}$ not equal to $PCI_{cell3}$<br>Non-colliding CRS: ($PCI_{cell2}$ – $PCI_{cell3}$)mod6 !=0 | Cell PCIs for three cells are selected randomly so that all conditions are met |
| ABS pattern | | | '10000000100000001000000010000000010000000' | Non-MBSFN ABS. FDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 1.<br>The first/leftmost bit corresponds to the PCell subframe #0 of a radio frame satisfying SFN mod x = 0, where x is the size of the bit string (40) divided by 10. No MBSFN subframes are configured in the ABS subframes in Cell 1. |
| Time-domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | | '10000000100000001000000010000000010000000' | Time domain measurement resource restriction pattern for neighbour cell measurement signalled to the UE in measSubframePatternNeigh IE in measSubframePatternConfigNeigh, as defined in TS 36.331, clause 6.3.5.<br>Configured before the measurements start. The cell list in measSubframeCellList IE shall contain Cell 3 but not Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | | '01000000010000000100000001000000010000000' | Configured for measurements on Cell 1. |
| CRS assistance information | physCellId | | see PCI conditions above | The CRS assistance information is provided for Cell 2 only in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation *one Frame*='000000'. |
| | antennaPortsCount | | 1 | |
| | mbsfn-SubframeConfigList | | *oneFrame* = '000000' | |

### 9.1.14.1.4.2 Test procedure

In this test case all cells are on the same carrier frequency. The absolute accuracy of RSRP intra frequency measurements under time domain measurement resource restriction with CRS assistance information is tested by using the parameters in Table 9.1.14.1.4.1-1 and Table 9.1.14.1.5-1 for non-MBSFN ABS with colliding CRS between Cell 1 and Cell 3 and non-colliding CRS between Cell 1 and Cell 2. Cell 1 is the serving cell and also the aggressor cell to Cell 3. Cell 2 is the neighbour aggressor cell without CRS colliding to Cell 3. Cell 3 is the cell to be measured for RSRP absolute accuracy.

The UE is configured by higher layers via Cell 1 with a time-domain measurement resource restriction pattern for performing E-UTRAN FDD intra-frequency measurements on neighbour cells, namely Cell 3 measurements with a neighbour cell list, where the cell list includes Cell 3. The UE is also provided via higher layers with the CRS assistance information of Cell 2. The information for both measurement pattern and the CRS assistance information shall be provided to the UE before the measurements start.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3 with exceptions listed in 7.2A.6.

2. Set the parameters according to Tables 9.1.14.1.4.1-1 and Table 9.1.14.1.5-1. Propagation conditions are set according to Annex B clause B.1.

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP values for Cell 3 are compared to the actual RSRP values according to Table 9.1.14.1.5-2. If the UE fails to report the measurement value for Cell 3, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.14.1.5-1 as appropriate.

### 9.1.14.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 7.2A.6 with the following exceptions:

**Table 9.1.14.1.4.3-1: Common Exception messages for RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.7-1<br>Table H.2.7-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3a<br>Table H.3.5-5<br>Table H.5.1-1<br>Table H.5.1-2<br>Table H.5.1-3 |

**Table 9.1.14.1.4.3-2: *MeasResults*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.14.1.4.3-3: *MeasResultListEUTRA*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity of Cell 3 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.1.14.1.4.3-4: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 5.2A.5, Table 5.2A.5.1.1-2 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '01000000010000000100000001000000001000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.14.1.4.3-5: *MeasObjectEUTRA-GENERIC(Freq)*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternFDD-r10 | '10000000100000001000000001000000010000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 3 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.1.14.1.5 Test requirement

Table 9.1.14.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD absolute accuracy test shall meet the reported values test requirements in table 9.1.14.1.5-2.

**Table 9.1.14.1.5-1: Cell Specific test parameters for E-UTRAN RSRP FDD absolute accuracy under time-domain measurement resource restriction with CRS Assistance Information and non-MBSFN ABS**

| Parameter | | Unit | Test 1 | | | Test 2 | | | Test 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 | Cell 1 | Cell 2 | Cell 3 | Cell 1 | Cell 2 | Cell 3 |
| E-UTRA RF Channel Number | | | 1 | | | 1 | | | 1 | | |
| $BW_{channel}$ | | MHz | 10 | | | 10 | | | 10 | | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | | 22—27 | | | 22—27 | | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - | - | R.0 FDD | - | - | R.0 FDD | - | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | - | 13—36 | - | - | 13—36 | - | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | | R.6 FDD | | | R.6 FDD | | |
| OCNG Patterns defined in D.1.5 (OP.5 FDD) and D.1.6 (OP.6 FDD) | | | OP.5 FDD | OP.6 FDD | OP.6 FDD | OP.5 FDD | OP.6 FDD | OP.6 FDD | OP.5 FDD | OP.6 FDD | OP.6 FDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA Note1<br>OCNG_RB Note1 | | dB | Note 6 | Note 6 | 0 | Note 6 | Note 6 | 0 | Note 6 | Note 6 | 0 |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -106 | | | -88 | | | -116 | | |
| | Bands FDD_C | | | | | | | | -115 | | |
| | Bands FDD_D | | | | | | | | -114.5 | | |
| | Bands FDD_E, FDD_F Note 7 | | | | | | | | -114 | | |
| | Bands FDD_G | | | | | | | | -113 | | |
| | Bands FDD_H | | | | | | | | -112.5 | | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 4 | 2 | -1.5 | 4 | 2 | -3 | 4 | 2 | -3 |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{meas}$ Note 5 | | dB | -1.18 | -0.32 | -6.96 | -0.89 | 0.24 | -8.46 | -0.89 | 0.24 | -8.46 |
| RSRP Note3,4,5 | Bands FDD_A | dBm/15 kHz | -102 | -104 | -107.5 | -84 | -86 | -91 | -112 | -114 | -119 |
| | Bands FDD_C | | | | | | | | -111 | -113 | -118 |
| | Bands FDD_D | | | | | | | | -110.5 | -112.5 | -117.5 |
| | Bands FDD_E, FDD_F Note 7 | | | | | | | | -110 | -112 | -117 |
| | Bands FDD_G | | | | | | | | -109 | -111 | -116 |
| | Bands FDD_H | | | | | | | | -108.5 | -110.5 | -115.5 |
| $(\mathrm{Io})_{meas}$ Note 3,5 | Bands FDD_A | dBm/9 MHz | -70.58 | -74.43 | | -52.74 | -56.83 | | -80.74 | -84.83 | |
| | Bands FDD_C | | | | | | | | -79.74 | -83.83 | |
| | Bands FDD_D | | | | | | | | -79.24 | -83.33 | |
| | Bands FDD_E, FDD_F Note 7 | | | | | | | | -78.74 | -82.83 | |
| | Bands FDD_G | | | | | | | | -77.74 | -81.83 | |
| | Bands FDD_H | | | | | | | | -77.24 | -81.33 | |
| Propagation condition | | | AWGN | | | AWGN | | | AWGN | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.
Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes.
Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Applies to restricted measurement subframes for only Cell 2 and Cell 3. For Cell 1, the corresponding value is derived from the normal subframes other than the subframes indicated in the time domain measurement resource restriction pattern for intra-frequency measurements.
Note 6: Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1.
Note 7: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 8: E-UTRA operating band groups are as defined in TS 36.133Section 3.5.

**Table 9.1.14.1.5-2: E-UTRAN RSRP FDD absolute accuracy under time-domain measurement resource restriction with non-MBSFN ABS requirements for the reported values**

| Normal Conditions | Test 1 All bands | Test 2 All bands | Test 3 | |
|---|---|---|---|---|
| Lowest reported value (Cell 3) | RSRP_25 | RSRP_40 | Bands FDD_A | RSRP_14 |
| | | | Bands FDD_C | RSRP_15 |
| | | | Bands FDD_D | RSRP_15 |
| | | | Bands FDD_E, FDD_F | RSRP_16 |
| | | | Bands FDD_G | RSRP_17 |
| | | | Bands FDD_H | RSRP_17 |
| Highest reported value (Cell 3) | RSRP_41 | RSRP_59 | Bands FDDA | RSRP_29 |
| | | | Bands FDD_C | RSRP_30 |
| | | | Bands FDD_D | RSRP_31 |
| | | | Bands FDD_E, FDD_F | RSRP_31 |
| | | | Bands FDD_G | RSRP_32 |
| | | | Bands FDD_H | RSRP_33 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 3) | RSRP_22 | RSRP_37 | Bands FDD_A | RSRP_11 |
| | | | Bands FDD_C | RSRP_12 |
| | | | Bands FDD_D | RSRP_12 |
| | | | Bands FDD_E, FDD_F | RSRP_13 |
| | | | Bands FDD_G | RSRP_14 |
| | | | Bands FDD_H | RSRP_14 |
| Highest reported value (Cell 3) | RSRP_44 | RSRP_62 | Bands FDD_A | RSRP_32 |
| | | | Bands FDD_C | RSRP_33 |
| | | | Bands FDD_D | RSRP_34 |
| | | | Bands FDD_E, FDD_F | RSRP_34 |
| | | | Bands FDD_G | RSRP_35 |
| | | | Bands FDD_H | RSRP_36 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.14.1_1 FDD Intra Frequency Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) (Rel-12 and forward)

Editor's note: This test case is incomplete. The following item is missing or incomplete:

The test tolerances and test requirements for test 1, 2 and 3 in normal conditions are TBD

### 9.1.14.1_1.1 Test purpose

Same test purpose as in clause 9.1.14.1.1.

### 9.1.14.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD and CRS interference handling UE release 12 and forward. Applicability requires support of FGI bit 115.

### 9.1.14.1_1.3 Minimum conformance requirements

The FDD RSRP measurements of cells shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.3.

The accuracy requirements in table 9.1.14.1_1.3-1 are valid under the following conditions:

Cell specific reference signals in the measured cell are transmitted either from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.9 for a corresponding Band.

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.1.14.1_1.3-1: RSRP Intra frequency absolute accuracy under time domain measurement resource restriction**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 2 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 4 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 3 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-4 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-4 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRP measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 3: The condition level is increased by $\Delta>0$, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.14.1_1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.14.1_1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.3, clause 9.1.4 and A.9.1.14.

## 9.1.14.1_1.4 Test description

### 9.1.14.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.14.1.4.1.

### 9.1.14.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.14.1.4.2 with the following exceptions:

- Instead of Table 9.1.14.1.5-1→ use Table 9.1.14.1_1.5-1.
- Instead of Table 9.1.14.1.5-2→ use Table 9.1.14.1_1.5-2.

### 9.1.14.1_1.4.3 Message contents

Same message contents as in clause 9.1.14.1.4.3.

## 9.1.14.1_1.5 Test requirement

Table 9.1.14.1_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD absolute accuracy test shall meet the reported values test requirements in table 9.1.14.1_1.5-2.

**Table 9.1.14.1_1.5-1: Cell Specific test parameters for E-UTRAN RSRP FDD absolute accuracy under time-domain measurement resource restriction with CRS Assistance Information and non-MBSFN ABS**

| Parameter | Unit | Test 1 | | | Test 2 | | | Test 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 3 | Cell 1 | Cell 2 | Cell 3 | Cell 1 | Cell 2 | Cell 3 |

| E-UTRA RF Channel Number | | | 1 | | | 1 | | | 1 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $BW_{channel}$ | | MHz | 10 | | | 10 | | | 10 | | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | | 22—27 | | | 22—27 | | |
| PDSCH Reference measurement channel defined in A.3.1.1.1 | | | R.0 FDD | - | - | R.0 FDD | - | - | R.0 FDD | - | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | - | 13—36 | - | - | 13—36 | - | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.1 | | | R.6 FDD | | | R.6 FDD | | | R.6 FDD | | |
| OCNG Patterns defined in A.3.2.1.5 (OP.5 FDD) and A.3.2.1.6 (OP.6 FDD) | | | OP.5 FDD | OP.6 FDD | OP.6 FDD | OP.5 FDD | OP.6 FDD | OP.6 FDD | OP.5 FDD | OP.6 FDD | OP.6 FDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA Note1<br>OCNG_RB Note1 | | dB | Note 6 | Note 6 | 0 | Note 6 | Note 6 | 0 | Note 6 | Note 6 | 0 |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -106 | | | -88 | | | -116 | | |
| | Bands FDD_C | | | | | | | | -115 | | |
| | Bands FDD_D | | | | | | | | -114.5 | | |
| | Bands FDD_E, FDD_F Note 7 | | | | | | | | -114 | | |
| | Bands FDD_G Note 9 | | | | | | | | -113 | | |
| | Bands FDD_H | | | | | | | | -112.5 | | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 4+TT | 2+TT | -1.5+TT | 4+TT | 2+TT | -4+TT | 4+TT | 2+TT | -4+TT |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{meas}$ Note 5 | | dB | -1.18+TT | -0.32+TT | -6.96+TT | -0.75+TT | 0.54+TT | -9.46+TT | -0.75+TT | 0.54+TT | -9.46+TT |
| RSRP Note3,4,5 | Bands FDD_A | dBm/15 kHz | -102+TT | -104+TT | -107.5+TT | -84+TT | -86+TT | -92+TT | -112+TT | -114+TT | -120+TT |
| | Bands FDD_C | | | | | | | | -111+TT | -113+TT | -119+TT |
| | Bands FDD_D | | | | | | | | -110.5+TT | -112.5+TT | -118.5+TT |
| | Bands FDD_E, FDD_F Note 7 | | | | | | | | -110+TT | -112+TT | -118+TT |
| | Bands FDD_G Note 9 | | | | | | | | -109+TT | -111+TT | -117+TT |
| | Bands FDD_H | | | | | | | | -108.5+TT | -110.5+TT | -116.5+TT |
| $(\mathrm{Io})_{meas}$ Note 3,5 | Bands FDD_A | dBm/9 MHz | -70.58+TT | -74.43+TT | | -52.82+TT | -57.04+TT | | -80.82+TT | -85.04+TT | |
| | Bands FDD_C | | | | | | | | -79.82+TT | -84.04+TT | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Bands FDD_D | | | | | | -79.32+TT | -83.54+TT |
| | Bands FDD_E, FDD_F Note 7 | | | | | | -78.82+TT | -83.04+TT |
| | Bands FDD_G Note 9 | | | | | | -77.82+TT | -82.04+TT |
| | Bands FDD_H | | | | | | -77.32+TT | -81.54+TT |
| Propagation condition | | | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.
Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes.
Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Applies to restricted measurement subframes for only Cell 2 and Cell 3. For Cell 1, the corresponding value is derived from the normal subframes other than the subframes indicated in the time domain measurement resource restriction pattern for intra-frequency measurements.
Note 6: Non-ABS and ABS subframe channel powers defined in Table A.3.4.1.1-1.
Note 7: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 8: E-UTRA operating band groups are as defined in Section 3.5.
Note 9: Except Band 29 and Band 32.

**Table 9.1.14.1_1.5-2: E-UTRAN RSRP FDD absolute accuracy under time-domain measurement resource restriction with non-MBSFN ABS requirements for the reported values**

| **Normal Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
|---|---|---|---|---|
| Lowest reported value (Cell 3) | TBD | TBD | Bands FDD_A | TBD |
| | | | Bands FDD_C | TBD |
| | | | Bands FDD_D | TBD |
| | | | Bands FDD_E, FDD_F | TBD |
| | | | Bands FDD_G | TBD |
| | | | Bands FDD_H | TBD |
| Highest reported value (Cell 3) | TBD | TBD | Bands FDDA | TBD |
| | | | Bands FDD_C | TBD |
| | | | Bands FDD_D | TBD |
| | | | Bands FDD_E, FDD_F | TBD |
| | | | Bands FDD_G | TBD |
| | | | Bands FDD_H | TBD |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 3) | RSRP_22 | RSRP_37 | Bands FDD_A | RSRP_11 |
| | | | Bands FDD_C | RSRP_12 |
| | | | Bands FDD_D | RSRP_12 |
| | | | Bands FDD_E, FDD_F | RSRP_13 |
| | | | Bands FDD_G | RSRP_14 |
| | | | Bands FDD_H | RSRP_14 |
| Highest reported value (Cell 3) | RSRP_44 | RSRP_62 | Bands FDD_A | RSRP_32 |
| | | | Bands FDD_C | RSRP_33 |
| | | | Bands FDD_D | RSRP_34 |
| | | | Bands FDD_E, FDD_F | RSRP_34 |
| | | | Bands FDD_G | RSRP_35 |
| | | | Bands FDD_H | RSRP_36 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.14.2 FDD Intra Frequency Relative RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC)

### 9.1.14.2.1 Test purpose

To verify that the Relative RSRP measurement accuracy under time-domain measurement resource restriction with CRS Assistance Information and non-MBSFN ABS configured in the aggressor cells is within the specified limits.

### 9.1.14.2.2 Test applicability

This test applies to all types of E-UTRA FDD and CRS interference handling UE release 11 and forward. Applicability requires support of FGI bit 115.

### 9.1.14.2.3 Minimum conformance requirements

The FDD RSRP measurements of cells shall meet the intra frequency relative accuracy requirements defined in TS 36.133 [4] clause 9.1.2.4.

The accuracy requirements in table 9.1.14.2.3-1 are valid under the following conditions:

Cell specific reference signals in the measured cell are transmitted either from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.9 for a corresponding Band.

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.1.14.2.3-1: RSRP Intra frequency relative accuracy under time domain measurement resource restriction**

| **Accuracy** | | **Conditions** | | | |
|---|---|---|---|---|---|
| **Normal condition** | **Extreme condition** | **Ês/Iot** Note 2 | **Io** Note 3 **range** | | |
| | | | **E-UTRA operating band groups** Note 6 | **Minimum Io** | **Maximum Io** |
| **dB** | **dB** | **dB** | | dBm/ **15kHz** Note 1, 5 | **dBm/BW$_{Channel}$** |
| ±2 | ±3 | ≥-2 dB | FDD_A | -121 | -50 |
| | | | FDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-4 dB | Note 4 | Note 4 | Note 4 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRP measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 5: The condition level is increased by $\Delta>0$, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 6: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.14.2.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.14.2.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.4, clause 9.1.4 and A.9.1.14.

## 9.1.14.2.4 Test description

### 9.1.14.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.54 (without faders).

2. The general test parameter settings are set up according to Table 9.1.14.2.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.1.14.2.4.3.

5. Cell 1 is the serving cell as well as aggressor cell to Cell 3, Cell 2 is the neighbour aggressor cell without CRS colliding to Cell 3. Cell 3 is the cell to be measured for RSRP relative accuracy, whilst both Cell 1 and Cell 3 are measured for RSRP relative accuracy. Non-MBSFN ABS pattern is configured for Cell 1 and Cell 2 during the test. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

**Table 9.1.14.2.4.1-1: General test parameters for E-UTRAN FDD RSRP intra frequency test parameters under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Serving cell (PCell) | | | Cell 1 | The aggressor cell to Cell 3 |
| Neighbour cell | | | Cell 2 | The aggressor cell to Cell 3 |
| Neighbour cell | | | Cell 3 | Cell to be measured |
| PCell ABS configuration | | | Non-MBSFN ABS | As defined in Table C.3.1.1.1-1 |
| CP length | | | Normal | For three cells in the test |
| DRX | | | | OFF |
| Cell 2 time offset with respect to Cell 1 | | | 0μs | Synchronous cells |
| Cell 3 time offset with respect to Cell 1 | | | -2.5 μs | Synchronous cells |
| Physical cell ID PCI | | | Colliding CRS: ($PCI_{cell1}$ – $PCI_{cell3}$)mod6=0, $PCI_{cell1}$ not equal to $PCI_{cell3}$<br>Non-colliding CRS: ($PCI_{cell2}$ – $PCI_{cell3}$)mod6 !=0 | Cell PCIs for three cells are selected randomly so that all conditions are met |
| ABS pattern | | | '10000000100000001000000010000000010000000' | Non-MBSFN ABS. FDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 1.<br>The first/leftmost bit corresponds to the PCell subframe #0 of a radio frame satisfying SFN mod x = 0, where x is the size of the bit string (40) divided by 10. No MBSFN subframes are configured in the ABS subframes in Cell 1. |
| Time-domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | | '10000000100000001000000010000000010000000' | Time domain measurement resource restriction pattern for neighbour cell measurement signalled to the UE in measSubframePatternNeigh IE in measSubframePatternConfigNeigh, as defined in TS 36.331, clause 6.3.5.<br>Configured before the measurements start. The cell list in measSubframeCellList IE shall contain Cell 3 but not Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | | '01000000010000000100000001000000010000000' | Configured for measurements on Cell 1. |
| CRS assistance information | physCellId | | see PCI conditions above | The CRS assistance information is provided for Cell 2 only in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation *one Frame*='000000'. |
| | antennaPortsCount | | 1 | |
| | mbsfn-SubframeConfigList | | *oneFrame* = '000000' | |

9.1.14.2.4.2 Test procedure

In this test case all cells are on the same carrier frequency. The relative accuracy of RSRP intra frequency measurements under time domain measurement resource restriction with CRS assistance information is tested by using the parameters in Table 9.1.14.2.4.1-1 and Table 9.1.14.2.5-1 for non-MBSFN ABS with colliding CRS between Cell 1 and Cell 3 and non-colliding CRS between Cell 1 and Cell 2. Cell 1 is the serving cell and also the aggressor cell to Cell 3. Cell 2 is the neighbour aggressor cell without CRS colliding to Cell 3. Cell 3 is the cell to be measured for RSRP relative accuracy.

The UE is configured by higher layers via Cell 1 with a time-domain measurement resource restriction pattern for performing E-UTRAN FDD intra-frequency measurements on neighbour cells, namely Cell 3 measurements with a neighbour cell list, where the cell list includes Cell 3. The UE is also provided via higher layers with the CRS assistance information of Cell 2. The information for both measurement pattern and the CRS assistance information shall be provided to the UE before the measurements start.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3 with exceptions listed in 7.2A.6.

2. Set the parameters according to Tables 9.1.14.2.4.1-1 and Table 9.1.14.2.5-1. Propagation conditions are set according to Annex B clause B.1.

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP values for Cell 3 are compared to the actual RSRP values according to Table 9.1.14.2.5-2. If the UE fails to report the measurement value for Cell 3, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.14.2.5-1 as appropriate.

### 9.1.14.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 7.2A.6 with the following exceptions:

**Table 9.1.14.2.4.3-1: Common Exception messages for RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS relative accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.7-1<br>Table H.2.7-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3a<br>Table H.3.5-5<br>Table H.5.1-1<br>Table H.5.1-2<br>Table H.5.1-3 |

**Table 9.1.14.2.4.3-2: *MeasResults*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.14.2.4.3-3: *MeasResultListEUTRA*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity of Cell 3 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.1.14.2.4.3-4: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 5.2A.5, Table 5.2A.5.1.1-2 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '01000000010000000100000001000000001000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.14.2.4.3-5: *MeasObjectEUTRA-GENERIC(Freq)*: Additional RSRP FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternFDD-r10 | '10000000100000001000000001000000010000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 3 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.1.14.2.5 Test requirement

Table 9.1.14.2.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD relative accuracy test shall meet the reported values test requirements in table 9.1.14.2.5-2.

**Table 9.1.14.2.5-1: Cell Specific test parameters for E-UTRAN RSRP FDD relative accuracy under time-domain measurement resource restriction with CRS Assistance Information and non-MBSFN ABS**

| Parameter | | Unit | Test 1 | | | Test 2 | | | Test 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 | Cell 1 | Cell 2 | Cell 3 | Cell 1 | Cell 2 | Cell 3 |
| E-UTRA RF Channel Number | | | 1 | | | 1 | | | 1 | | |
| $BW_{channel}$ | | MHz | 10 | | | 10 | | | 10 | | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | | 22—27 | | | 22—27 | | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - | - | R.0 FDD | - | - | R.0 FDD | - | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | - | 13—36 | - | - | 13—36 | - | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | | R.6 FDD | | | R.6 FDD | | |
| OCNG Patterns defined in D.1.5 (OP.5 FDD) and D.1.6 (OP.6 FDD) | | | OP.5 FDD | OP.6 FDD | OP.6 FDD | OP.5 FDD | OP.6 FDD | OP.6 FDD | OP.5 FDD | OP.6 FDD | OP.6 FDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA Note1<br>OCNG_RB Note1 | | dB | Note 6 | Note 6 | 0 | Note 6 | Note 6 | 0 | Note 6 | Note 6 | 0 |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -106 | | | -88 | | | -116 | | |
| | Bands FDD_C | | | | | | | | -115 | | |
| | Bands FDD_D | | | | | | | | -114.5 | | |
| | Bands FDD_E, FDD_F Note 7 | | | | | | | | -114 | | |
| | Bands FDD_G | | | | | | | | -113 | | |
| | Bands FDD_H | | | | | | | | -112.5 | | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 4 | 2 | -0.5 | 4 | 2 | -3 | 4 | 2 | -3 |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{meas}$ Note 5 | | dB | -1.41 | -0.77 | -5.96 | -0.89 | 0.24 | -8.46 | -0.89 | 0.24 | -8.46 |
| RSRP Note3,4,5 | Bands FDD_A | dBm/15 kHz | -102 | -104 | -106.5 | -84 | -86 | -91 | -112 | -114 | -119 |
| | Bands FDD_C | | | | | | | | -111 | -113 | -118 |
| | Bands FDD_D | | | | | | | | -110.5 | -112.5 | -117.5 |
| | Bands FDD_E, FDD_F Note 7 | | | | | | | | -110 | -112 | -117 |
| | Bands FDD_G | | | | | | | | -109 | -111 | -116 |
| | Bands FDD_H | | | | | | | | -108.5 | -110.5 | -115.5 |
| $(\mathrm{Io})_{meas}$ Note 3,5 | Bands FDD_A | dBm/9 MHz | 70.45 | -74.11 | | 52.74 | -56.83 | | -80.74 | -84.83 | |
| | Bands FDD_C | | | | | | | | -79.74 | -83.83 | |
| | Bands FDD_D | | | | | | | | -79.24 | -83.33 | |
| | Bands FDD_E, FDD_F Note 7 | | | | | | | | -78.74 | -82.83 | |
| | Bands FDD_G | | | | | | | | -77.74 | -81.83 | |
| | Bands FDD_H | | | | | | | | -77.24 | -81.33 | |
| Propagation condition | | | AWGN | | | AWGN | | | AWGN | | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes. |
| Note 3: | RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes. |
| Note 4: | RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | Applies to restricted measurement subframes for only Cell 2 and Cell 3. For Cell 1, the corresponding value is derived from the normal subframes other than the subframes indicated in the time domain measurement resource restriction pattern for intra-frequency measurements. |
| Note 6: | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1. |
| Note 7: | For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz. |
| Note 8: | E-UTRA operating band groups are as defined in TS 36.133Section 3.5. |

**Table 9.1.14.2.5-2: E-UTRAN RSRP FDD relative accuracy under time-domain measurement resource restriction with non-MBSFN ABS requirements for the reported values**

| **Normal Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3 All bands** |
|---|---|---|---|
| Lowest reported value (Cell 3) | RSRP_x-8 | RSRP_x-12 | RSRP_x-12 |
| Highest reported value (Cell 3) | RSRP_x-2 | RSRP_x-3 | RSRP_x-3 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3 All bands** |
| Lowest reported value (Cell 3) | RSRP_x-9 | RSRP_x-12 | RSRP_x-12 |
| Highest reported value (Cell 3) | RSRP_x-1 | RSRP_x-3 | RSRP_x-3 |
| RSRP_x is the reported value of Cell 1 | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.15 TDD RSRP under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC)

### 9.1.15.1 TDD Intra Frequency Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC)

#### 9.1.15.1.1 Test purpose

To verify that the Absolute RSRP measurement accuracy under time-domain measurement resource restriction with CRS Assistance Information and non-MBSFN ABS configured in the aggressor cells is within the specified limits.

#### 9.1.15.1.2 Test applicability

This test applies to all types of E-UTRA TDD and CRS interference handling UE release 11. Applicability requires support of FGI bit 115.

#### 9.1.15.1.3 Minimum conformance requirements

The TDD RSRP measurements of cells shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.3.

The accuracy requirements in table 9.1.15.1.3-1 are valid under the following conditions:

Cell specific reference signals in the measured cell are transmitted either from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.9 for a corresponding Band.

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.1.15.1.3-1: RSRP Intra frequency absolute accuracy under time domain measurement resource restriction**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 2 range | | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 3 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±6 | ±9 | ≥-4 dB | TDD_A | -121 | N/A | -70 |
| | | | TDD_C | -120 | N/A | -70 |
| | | | | | | |
| | | | TDD_E | -119 | N/A | -70 |
| | | | | | | |
| | | | | | | |
| | | | | | | |
| | | | | | | |
| ±8 | ±11 | ≥-4 dB | TDD_A, TDD_C, TDD_E | N/A | -70 | -50 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRP measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.15.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.15.1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.3, clause 9.1.4 and A.9.1.15.

## 9.1.15.1.4 Test description

### 9.1.15.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.54 (without faders).

2. The general test parameter settings are set up according to Table 9.1.15.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.1.15.1.4.3.

5. Cell 1 is the serving cell as well as aggressor cell to Cell 3, Cell 2 is the neighbour aggressor cell without CRS colliding to Cell 3. Cell 3 is the cell to be measured for RSRP absolute accuracy, whilst both Cell 1 and Cell 3 are measured for RSRP relative accuracy. Non-MBSFN ABS pattern is configured for Cell 1 and Cell 2 during the test. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

**Table 9.1.15.1.4.1-1: General test parameters for E-UTRAN TDD RSRP intra frequency test parameters under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Serving cell (PCell) | | | Cell 1 | The aggressor cell to Cell 3 |
| Neighbour cell | | | Cell 2 | The aggressor cell to Cell 3 |
| Neighbour cell | | | Cell 3 | Cell to be measured |
| PCell ABS configuration | | | Non-MBSFN ABS | As defined in Table C.3.1.1.1-1 |
| CP length | | | Normal | For three cells in the test |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211. The same configuration in both cells |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211. The same configuration in both cells |
| DRX | | | | OFF |
| Cell 2 time offset with respect to Cell 1 | | | 0μs | Synchronous cells |
| Cell 3 time offset with respect to Cell 1 | | | -2.5 μs | Synchronous cells |
| Physical cell ID PCI | | | Colliding CRS: ($PCI_{cell1}$ – $PCI_{cell3}$)mod6=0, $PCI_{cell1}$ not equal to $PCI_{cell3}$<br>Non-colliding CRS: ($PCI_{cell2}$ – $PCI_{cell3}$)mod6 !=0 | Cell PCIs for three cells are selected randomly so that all conditions are met |
| ABS pattern | | | '00000000010000000001' | TDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54.<br>The first/leftmost bit corresponds to the PCell subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (20) divided by 10. No MBSFN subframes are configured in the ABS subframes.<br>Provided to the UE for Cell 1 and Cell 2 before the measurements start. |
| Time-domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | | '00000000010000000001' | Time domain measurement resource restriction pattern for neighbour cell measurement signalled to the UE in measSubframePatternNeigh IE in measSubframePatternConfigNeigh, as defined in TS 36.331, clause 6.3.5.<br>Configured before the measurements start. The cell list in measSubframeCellList IE shall contain Cell 3 but not Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | | '10000000001000000000' | Configured for Cell 1 measurements. |
| CRS assistance information | physCellId | | see PCI conditions above | The CRS assistance information is provided for Cell 2 only in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation *one Frame*='000000'. |
| | antennaPortsCount | | 1 | |
| | mbsfn-SubframeConfigList | | *oneFrame* = '000000' | |

#### 9.1.15.1.4.2 Test procedure

In this test case all cells are on the same carrier frequency. The absolute accuracy of RSRP intra frequency measurements under time domain measurement resource restriction with CRS assistance information is tested by using the parameters in Table 9.1.15.1.4.1-1 and Table 9.1.15.1.5-1 for non-MBSFN ABS with colliding CRS between Cell 1 and Cell 3 and non-colliding CRS between Cell 1 and Cell 2. Cell 1 is the serving cell and also the aggressor cell to Cell 3. Cell 2 is the neighbour aggressor cell without CRS colliding to Cell 3. Cell 3 is the cell to be measured for RSRP absolute accuracy.

The UE is configured by higher layers via Cell 1 with a time-domain measurement resource restriction pattern for performing E-UTRAN TDD intra-frequency measurements on neighbour cells, namely Cell 3 measurements with a neighbour cell list, where the cell list includes Cell 3. The UE is also provided via higher layers with the CRS assistance information of Cell 2. The information for both measurement pattern and the CRS assistance information shall be provided to the UE before the measurements start.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3 with exceptions listed in 7.2A.6.

2. Set the parameters according to Tables 9.1.15.1.4.1-1 and Table 9.1.15.1.5-1. Propagation conditions are set according to Annex B clause B.1.

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP values for Cell 3 are compared to the actual RSRP values according to Table 9.1.15.1.5-2. If the UE fails to report the measurement value for Cell 3, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.15.1.5-1 as appropriate.

#### 9.1.15.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 7.2A.6 with the following exceptions:

**Table 9.1.15.1.4.3-1: Common Exception messages for RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.7-1<br>Table H.2.7-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3a<br>Table H.5.1-1<br>Table H.5.1-2<br>Table H.5.1-3 |

**Table 9.1.15.1.4.3-2: *MeasResults*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.15.1.4.3-3: *MeasResultListEUTRA*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of Cell 3 | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.1.15.1.4.3-4: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 5.2A.5, Table 5.2A.5.1.1-2 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 CHOICE { | | | |
| subframeConfig1-5-r10 | '10000000001000000000' | BIT STRING (SIZE (20)) | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.15.1.4.3-5: *MeasObjectEUTRA-GENERIC(Freq)*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternTDD-r10 CHOICE { | | | |
| subframeConfig1-5-r10 | '00000000010000000001' | BIT STRING (SIZE (20)) | |
| } | | | |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 3 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.1.15.1.5 Test requirement

Table 9.1.15.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD absolute accuracy test shall meet the reported values test requirements in table 9.1.15.1.5-2.

**Table 9.1.15.1.5-1: Cell Specific test parameters for E-UTRAN RSRP TDD absolute accuracy under time-domain measurement resource restriction with CRS Assistance Information and non-MBSFN ABS**

| Parameter | | Unit | Test 1 | | | Test 2 | | | Test 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 | Cell 1 | Cell 2 | Cell 3 | Cell 1 | Cell 2 | Cell 3 |
| E-UTRA RF Channel Number | | | 1 | | | 1 | | | 1 | | |
| $BW_{channel}$ | | MHz | 10 | | | 10 | | | 10 | | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | | 22—27 | | | 22—27 | | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | - | R.0 TDD | - | - | R.0 TDD | - | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | - | 13—36 | - | - | 13—36 | - | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | | R.6 TDD | | | R.6 TDD | | |
| OCNG Patterns defined in D.2 | | | OP.1 TDD | OP.2 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD | OP.2 TDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA[Note1]<br>OCNG_RB[Note1] | | dB | Note 6 | | 0 | Note 6 | | 0 | Note 6 | | 0 |
| $N_{oc}$ [Note2] | Bands TDD_A | dBm/15 kHz | -106 | | | -88 | | | -116 | | |
| | Bands TDD_C | | | | | | | | -115 | | |
| | Bands TDD_E | | | | | | | | -114 | | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 4 | 2 | -1.5 | 4 | 2 | -3 | 4 | 2 | -3 |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{meas}$ [Note 5] | | dB | -1.18 | -0.32 | -6.96 | -0.89 | 0.24 | -8.46 | -0.89 | 0.24 | -8.46 |
| RSRP [Note3,4,5] | Bands TDD_A | dBm/15 kHz | -102 | -104 | -107.5 | -84 | -86 | -91 | -112 | -114 | -119 |
| | Bands TDD_C | | | | | | | | -111 | -113 | -118 |
| | Bands TDD_E | | | | | | | | -110 | -112 | -117 |
| $(\mathrm{Io})_{meas}$ [Note 3, 5] | Bands TDD_A | dBm/9 MHz | -70.58 | -74.43 | | -52.74 | -56.83 | | -80.74 | -84.83 | |
| | Bands TDD_C | | | | | | | | -79.74 | -83.83 | |
| | Bands TDD_E | | | | | | | | -78.74 | -82.83 | |
| Propagation condition | | | AWGN | | | AWGN | | | AWGN | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: Applies to restricted measurement subframes for only Cell 2 and Cell 3. For Cell 1, the corresponding value is derived from the normal subframes other than the subframes indicated in the time domain measurement resource restriction pattern for intra-frequency measurements.

Note 6: Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1.

Note 7: E-UTRA operating band groups are as defined in TS 36.133 Section 3.5.

**Table 9.1.15.1.5-2: E-UTRAN RSRP TDD absolute accuracy under time-domain measurement resource restriction with non-MBSFN ABS requirements for the reported values**

| Normal Conditions | Test 1 All bands | Test 2 All bands | Test 3 | |
|---|---|---|---|---|
| Lowest reported value (Cell 3) | RSRP_25 | RSRP_40 | Bands TDD_A | RSRP_14 |
| | | | Bands TDD_C | RSRP_15 |
| | | | | |
| | | | Bands TDD_E | RSRP_16 |
| | | | | |
| | | | | |
| Highest reported value (Cell 3) | RSRP_41 | RSRP_59 | Bands TDD_A | RSRP_29 |
| | | | Bands TDD_C | RSRP_30 |
| | | | | |
| | | | Bands TDD_E, FDD_F | RSRP_31 |
| | | | | |
| | | | | |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 3) | RSRP_22 | RSRP_37 | Bands TDD_A | RSRP_11 |
| | | | Bands TDD_C | RSRP_12 |
| | | | | |
| | | | Bands TDD_E | RSRP_13 |
| | | | | |
| | | | | |
| Highest reported value (Cell 3) | RSRP_44 | RSRP_62 | Bands TDD_A | RSRP_32 |
| | | | Bands TDD_C | RSRP_33 |
| | | | | |
| | | | Bands TDD_E | RSRP_34 |
| | | | | |
| | | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.15.1_1 TDD Intra Frequency Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) (Rel-12 and forward)

Editor's note: This test case is incomplete. The following item is missing or incomplete:

The test tolerances and test requirements for test 1, 2 and 3 in normal conditions are TBD

### 9.1.15.1_1.1 Test purpose

Same test purpose as in clause 9.1.15.1.1.

### 9.1.15.1_1.2 Test applicability

This test applies to all types of E-UTRA TDD and CRS interference handling UE release 12 and forward. Applicability requires support of FGI bit 115.

### 9.1.15.1_1.3 Minimum conformance requirements

The TDD RSRP measurements of cells shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.3.

The accuracy requirements in table 9.1.15.1_1.3-1 are valid under the following conditions:

Cell specific reference signals in the measured cell are transmitted either from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.9 for a corresponding Band.

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.1.15.1_1.3-1: RSRP Intra frequency absolute accuracy under time domain measurement resource restriction**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 2 range | | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 3 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-4 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-4 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRP measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 3: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.15.1_1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.15.1_1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.3, clause 9.1.4 and A.9.1.15.

## 9.1.15.1_1.4 Test description

### 9.1.15.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.15.1.4.1.

#### 9.1.15.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.15.1.4.2 with the following exceptions:

- Instead of Table 9.1.15.1.5-1→ use Table 9.1.15.1_1.5-1
- Instead of Table 9.1.15.1.5-2→ use Table 9.1.15.1_1.5-2

#### 9.1.15.1_1.4.3 Message contents

Same message contents as in clause 9.1.15.1.4.3.

### 9.1.15.1_1.5 Test requirement

Table 9.1.15.1_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD absolute accuracy test shall meet the reported values test requirements in table 9.1.15.1_1.5-2.

**Table 9.1.15.1_1.5-1: Cell Specific test parameters for E-UTRAN RSRP TDD absolute accuracy under time-domain measurement resource restriction with CRS Assistance Information and non-MBSFN ABS**

| Parameter | | Unit | Test 1 | | | Test 2 | | | Test 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 | Cell 1 | Cell 2 | Cell 3 | Cell 1 | Cell 2 | Cell 3 |
| E-UTRA RF Channel Number | | | 1 | | | 1 | | | 1 | | |
| $BW_{channel}$ | | MHz | 10 | | | 10 | | | 10 | | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | | 22—27 | | | 22—27 | | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | - | R.0 TDD | - | - | R.0 TDD | - | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | - | 13—36 | - | - | 13—36 | - | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | | R.6 TDD | | | R.6 TDD | | |
| OCNG Patterns defined in D.2.5 (OP.5 TDD) and D.2.6 (OP.6 TDD) | | | OP.5 TDD | OP.6 TDD | OP.6 TDD | OP.5 TDD | OP.6 TDD | OP.6 TDD | OP.5 TDD | OP.6 TDD | OP.6 TDD |
| PBCH_RA | | dB | Note 6 | | 0 | Note 6 | | 0 | Note 6 | | 0 |
| PBCH_RB | | | | | | | | | | | |
| PSS_RA | | | | | | | | | | | |
| SSS_RA | | | | | | | | | | | |
| PCFICH_RB | | | | | | | | | | | |
| PHICH_RA | | | | | | | | | | | |
| PHICH_RB | | | | | | | | | | | |
| PDCCH_RA | | | | | | | | | | | |
| PDCCH_RB | | | | | | | | | | | |
| PDSCH_RA | | | | | | | | | | | |
| PDSCH_RB | | | | | | | | | | | |
| OCNG_RA[Note1] | | | | | | | | | | | |
| OCNG_RB[Note1] | | | | | | | | | | | |
| $N_{oc}$ [Note2] | Bands TDD_A | dBm/15 kHz | -106 | | | -88 | | | -116 | | |
| | Bands TDD_C | | | | | | | | -115 | | |
| | Bands TDD_E | | | | | | | | -114 | | |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CRS $\hat{E}_s / N_{oc}$ | | dB | 4+TT | 2+TT | -1.5+TT | 4+TT | 2+TT | -3+TT | 4+TT | 2+TT | -3+TT |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{meas}$ Note 5 | | dB | -1.18+TT | -0.32+TT | -6.96+TT | -0.89+TT | 0.24+TT | -8.46+TT | -0.89+TT | 0.24+TT | -8.46+TT |
| RSRP Note3,4,5 | Bands TDD_A | dBm/15 kHz | -102+TT | -104+TT | -107.5+TT | -84+TT | -86+TT | -91+TT | -112+TT | -114+TT | -119+TT |
| | Bands TDD_C | | | | | | | | -111+TT | -113+TT | -118+TT |
| | Bands TDD_E | | | | | | | | -110+TT | -112+TT | -117+TT |
| $(\mathrm{Io})_{meas}$ Note 3, 5 | Bands TDD_A | dBm/9 MHz | -70.58 | -74.43+TT | | -52.74+TT | -56.83+TT | | -80.74+TT | -84.83+TT | |
| | Bands TDD_C | | | | | | | | -79.74+TT | -83.83+TT | |
| | Bands TDD_E | | | | | | | | -78.74+TT | -82.83+TT | |
| Propagation condition | | | AWGN | | | AWGN | | | AWGN | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: Applies to restricted measurement subframes for only Cell 2 and Cell 3. For Cell 1, the corresponding value is derived from the normal subframes other than the subframes indicated in the time domain measurement resource restriction pattern for intra-frequency measurements.

Note 6: Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1.

Note 7: E-UTRA operating band groups are as defined in TS 36.133 Section 3.5.

**Table 9.1.15.1_1.5-2: E-UTRAN RSRP TDD absolute accuracy under time-domain measurement resource restriction with non-MBSFN ABS requirements for the reported values**

| **Normal Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
|---|---|---|---|---|
| Lowest reported value (Cell 3) | TBD | TBD | Bands TDD_A | TBD |
| | | | Bands TDD_C | TBD |
| | | | | |
| | | | Bands TDD_E | TBD |
| | | | | |
| | | | | |
| Highest reported value (Cell 3) | TBD | TBD | Bands TDD_A | TBD |
| | | | Bands TDD_C | TBD |
| | | | | |
| | | | Bands TDD_E, FDD_F | TBD |
| | | | | |
| | | | | |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 3) | RSRP_22 | RSRP_37 | Bands TDD_A | RSRP_11 |
| | | | Bands TDD_C | RSRP_12 |
| | | | | |
| | | | Bands TDD_E | RSRP_13 |
| | | | | |
| | | | | |
| Highest reported value (Cell 3) | RSRP_44 | RSRP_62 | Bands TDD_A | RSRP_32 |
| | | | Bands TDD_C | RSRP_33 |

| | | | | |
|---|---|---|---|---|
| | | | Bands TDD_E | RSRP_34 |
| | | | | |
| | | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

### 9.1.15.2 TDD Intra Frequency Relative RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC)

#### 9.1.15.2.1 Test purpose

To verify that the Relative RSRP measurement accuracy under time-domain measurement resource restriction with CRS Assistance Information and non-MBSFN ABS configured in the aggressor cells is within the specified limits.

#### 9.1.15.2.2 Test applicability

This test applies to all types of E-UTRA TDD and CRS interference handling UE release 11 and forward. Applicability requires support of FGI bit 115.

#### 9.1.15.2.3 Minimum conformance requirements

The TDD RSRP measurements of cells shall meet the intra frequency relative accuracy requirements defined in TS 36.133 [4] clause 9.1.2.4.

The accuracy requirements in table 9.1.15.2.3-1 are valid under the following conditions:

Cell specific reference signals in the measured cell are transmitted either from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.9 for a corresponding Band.

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRP measurement,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.1.15.2.3-1: RSRP Intra frequency relative accuracy under time domain measurement resource restriction**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 3 range | | |
| | | | E-UTRA operating band groups Note 6 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 5 | dBm/BW$_{Channel}$ |
| ±2 | ±3 | ≥-2 dB | TDD_A | -121 | -50 |
| | | | TDD_C | -120 | -50 |
| | | | | | |
| | | | TDD_E | -119 | -50 |
| | | | | | |
| | | | | | |
| | | | | | |
| | | | | | |
| ±3 | ±3 | ≥-4 dB | Note 4 | Note 4 | Note 4 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRP measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 5: The condition level is increased by $\Delta$>0, when applicable, as described in TS 36.133 [4] Sections I.4.2 and I.4.3.
NOTE 6: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.15.2.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.15.2.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.4, clause 9.1.4 and A.9.1.15.

## 9.1.15.2.4 Test description

### 9.1.15.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.54 (without faders).

2. The general test parameter settings are set up according to Table 9.1.15.2.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.1.15.2.4.3.

5. Cell 1 is the serving cell as well as aggressor cell to Cell 3, Cell 2 is the neighbour aggressor cell without CRS colliding to Cell 3. Cell 3 is the cell to be measured for RSRP relative accuracy, whilst both Cell 1 and Cell 3 are measured for RSRP relative accuracy. Non-MBSFN ABS pattern is configured for Cell 1 and Cell 2 during the test. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

**Table 9.1.15.2.4.1-1: General test parameters for E-UTRAN TDD RSRP intra frequency test parameters under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Serving cell (PCell) | | | Cell 1 | The aggressor cell to Cell 3 |
| Neighbour cell | | | Cell 2 | The aggressor cell to Cell 3 |
| Neighbour cell | | | Cell 3 | Cell to be measured |
| PCell ABS configuration | | | Non-MBSFN ABS | As defined in Table C.3.1.1.1-1 |
| CP length | | | Normal | For three cells in the test |
| Special subframe configuration | | | 6 | As specified in table 4.2-1 in TS 36.211. The same configuration in both cells |
| Uplink-downlink configuration | | | 1 | As specified in table 4.2-2 in TS 36.211. The same configuration in both cells |
| DRX | | | | OFF |
| Cell 2 time offset with respect to Cell 1 | | | 0μs | Synchronous cells |
| Cell 3 time offset with respect to Cell 1 | | | -2.5 μs | Synchronous cells |
| Physical cell ID PCI | | | Colliding CRS: ($PCI_{cell1}$ – $PCI_{cell3}$)mod6=0, $PCI_{cell1}$ not equal to $PCI_{cell3}$<br>Non-colliding CRS: ($PCI_{cell2}$ – $PCI_{cell3}$)mod6 !=0 | Cell PCIs for three cells are selected randomly so that all conditions are met |
| ABS pattern | | | '00000000010000000001' | TDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54.<br>The first/leftmost bit corresponds to the PCell subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (20) divided by 10. No MBSFN subframes are configured in the ABS subframes.<br>Provided to the UE for Cell 1 and Cell 2 before the measurements start. |
| Time-domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | | '00000000010000000001' | Time domain measurement resource restriction pattern for neighbour cell measurement signalled to the UE in measSubframePatternNeigh IE in measSubframePatternConfigNeigh, as defined in TS 36.331, clause 6.3.5.<br>Configured before the measurements start. The cell list in measSubframeCellList IE shall contain Cell 3 but not Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | | '10000000001000000000' | Configured for Cell 1 measurements. |
| CRS assistance information | physCellId | | see PCI conditions above | The CRS assistance information is provided for Cell 2 only in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation *one Frame*='000000'. |
| | antennaPortsCount | | 1 | |
| | mbsfn-SubframeConfigList | | *oneFrame* = '000000' | |

#### 9.1.15.2.4.2 Test procedure

In this test case all cells are on the same carrier frequency. The relative accuracy of RSRP intra frequency measurements under time domain measurement resource restriction with CRS assistance information is tested by using the parameters in Table 9.1.15.2.4.1-1 and Table 9.1.15.2.5-1 for non-MBSFN ABS with colliding CRS between Cell 1 and Cell 3 and non-colliding CRS between Cell 1 and Cell 2. Cell 1 is the serving cell and also the aggressor cell to Cell 3. Cell 2 is the neighbour aggressor cell without CRS colliding to Cell 3. Cell 3 is the cell to be measured for RSRP relative accuracy.

The UE is configured by higher layers via Cell 1 with a time-domain measurement resource restriction pattern for performing E-UTRAN TDD intra-frequency measurements on neighbour cells, namely Cell 3 measurements with a neighbour cell list, where the cell list includes Cell 3. The UE is also provided via higher layers with the CRS assistance information of Cell 2. The information for both measurement pattern and the CRS assistance information shall be provided to the UE before the measurements start.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3 with exceptions listed in 7.2A.6.

2. Set the parameters according to Tables 9.1.15.2.4.1-1 and Table 9.1.15.2.5-1. Propagation conditions are set according to Annex B clause B.1.

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP values for Cell 3 are compared to the actual RSRP values according to Table 9.1.15.2.5-2. If the UE fails to report the measurement value for Cell 3, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.15.2.5-1 as appropriate.

#### 9.1.15.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 7.2A.6 with the following exceptions:

**Table 9.1.15.2.4.3-1: Common Exception messages for RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS relative accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.7-1<br>Table H.2.7-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3a<br>Table H.5.1-1<br>Table H.5.1-2<br>Table H.5.1-3 |

**Table 9.1.15.2.4.3-2: *MeasResults*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.15.2.4.3-3: *MeasResultListEUTRA*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of Cell 3 | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.1.15.2.4.3-4: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 5.2A.5, Table 5.2A.5.1.1-2 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 CHOICE { | | | |
| subframeConfig1-5-r10 | '10000000001000000000' | BIT STRING (SIZE (20)) | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.15.2.4.3-5: *MeasObjectEUTRA-GENERIC(Freq)*: Additional RSRP TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS relative accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternTDD-r10 CHOICE { | | | |
| subframeConfig1-5-r10 | '00000000010000000001' | BIT STRING (SIZE (20)) | |
| } | | | |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 3 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.1.15.2.5 Test requirement

Table 9.1.15.2.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD relative accuracy test shall meet the reported values test requirements in table 9.1.15.2.5-2.

**Table 9.1.15.2.5-1: Cell Specific test parameters for E-UTRAN RSRP TDD relative accuracy under time-domain measurement resource restriction with CRS Assistance Information and non-MBSFN ABS**

| Parameter | | Unit | Test 1 | | | Test 2 | | | Test 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 | Cell 1 | Cell 2 | Cell 3 | Cell 1 | Cell 2 | Cell 3 |
| E-UTRA RF Channel Number | | | 1 | | | 1 | | | 1 | | |
| $BW_{channel}$ | | MHz | 10 | | | 10 | | | 10 | | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | | 22—27 | | | 22—27 | | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | - | R.0 TDD | - | - | R.0 TDD | - | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | - | 13—36 | - | - | 13—36 | - | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | | R.6 TDD | | | R.6 TDD | | |
| OCNG Patterns defined in D.2 | | | OP.1 TDD | OP.2 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD | OP.2 TDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA[Note1]<br>OCNG_RB[Note1] | | dB | Note 6 | | 0 | Note 6 | | 0 | Note 6 | | 0 |
| $N_{oc}$ [Note2] | Bands TDD_A | dBm/15 kHz | -106 | | | -88 | | | -116 | | |
| | Bands TDD_C | | | | | | | | -115 | | |
| | Bands TDD_E | | | | | | | | -114 | | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 4 | 2 | -0.5 | 4 | 2 | -3 | 4 | 2 | -3 |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{meas}$ [Note 5] | | dB | -1.41 | -0.77 | -5.96 | -0.89 | 0.24 | -8.46 | -0.89 | 0.24 | -8.46 |
| RSRP [Note3,4,5] | Bands TDD_A | dBm/15 kHz | -102 | -104 | -106.5 | -84 | -86 | -91 | -112 | -114 | -119 |
| | Bands TDD_C | | | | | | | | -111 | -113 | -118 |
| | Bands TDD_E | | | | | | | | -110 | -112 | -117 |
| $(\mathrm{Io})_{meas}$ [Note 3, 5] | Bands TDD_A | dBm/9 MHz | -70.45 | -74.11 | | -52.74 | -56.83 | | -80.74 | -84.83 | |
| | Bands TDD_C | | | | | | | | -79.74 | -83.83 | |
| | Bands TDD_E | | | | | | | | -78.74 | -82.83 | |
| Propagation condition | | | AWGN | | | AWGN | | | AWGN | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: Applies to restricted measurement subframes for only Cell 2 and Cell 3. For Cell 1, the corresponding value is derived from the normal subframes other than the subframes indicated in the time domain measurement resource restriction pattern for intra-frequency measurements.

Note 6: Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1.

Note 7: E-UTRA operating band groups are as defined in TS 36.133 Section 3.5.

**Table 9.1.15.2.5-2: E-UTRAN RSRP TDD relative accuracy under time-domain measurement resource restriction with non-MBSFN ABS requirements for the reported values**

| Normal Conditions | Test 1 All bands | Test 2 All bands | Test 3 All bands |
|---|---|---|---|
| Lowest reported value (Cell 3) | RSRP_x-8 | RSRP_x-12 | RSRP_x-12 |
| Highest reported value (Cell 3) | RSRP_x-2 | RSRP_x-3 | RSRP_x-3 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3 All bands** |
| Lowest reported value (Cell 3) | RSRP_x-9 | RSRP_x-12 | RSRP_x-12 |
| Highest reported value (Cell 3) | RSRP_x-1 | RSRP_x-3 | RSRP_x-3 |
| RSRP_x is the reported value of Cell 1 | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.1.16 FDD Intra frequency RSRP Accuracy for 5MHz Bandwidth

## 9.1.16.1 FDD Intra Frequency Absolute RSRP Accuracy for 5MHz Bandwidth

### 9.1.16.1.1 Test purpose

To verify that the FDD intra-frequency absolute RSRP measurement accuracy is within the specified limit for Band 31.

### 9.1.16.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE from release 8 to release 11 that support Band 31. Applicability requires support for FGI bit 16.

### 9.1.16.1.3 Minimum conformance requirements

The requirements for absolute accuracy of RSRP in this clause apply to a cell on the same frequency as that of the serving cell.

The accuracy requirements in Table 9.1.16.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101[2] Clause 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.1 for a corresponding Band.

**Table 9.1.16.1.3-1: RSRP Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 3] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 2] / dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±6 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 / N/A | -70 |
| | | | FDD_B | -120.5 / N/A | -70 |
| | | | FDD_C, TDD_C | -120 / N/A | -70 |
| | | | FDD_D | -119.5 / N/A | -70 |
| | | | FDD_E, TDD_E | -119 / N/A | -70 |
| | | | FDD_F | -118.5 / N/A | -70 |
| | | | FDD_G | -118 / N/A | -70 |
| | | | FDD_H | -117.5 / N/A | -70 |
| | | | FDD_N | -114.5 / N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 / -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in section 3.5.1.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.16.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.16.1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.1, clause 9.1.4 and A.9.1.16.

## 9.1.16.1.4 Test description

### 9.1.16.1.4.1 Initial conditions

Same initial conditions as defined in clause 9.1.1.1.4.1 with the exception that the Channel Bandwidth to be tested is 5MHz.

### 9.1.16.1.4.2 Test procedure

Same test procedure as defined in clause 9.1.1.1.4.2 with the following exceptions:

- Instead of Table 9.1.1.1.5-2 → use Table 9.1.16.1.5-1.
- Instead of Table 9.1.1.1.5-3 → use Table 9.1.16.1.5-2.

### 9.1.16.1.4.3 Message contents

Same message contents as defined in clause 9.1.1.1.4.3.

### 9.1.16.1.5 Test requirement

Table 9.1.16.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD intra-frequency relative accuracy test shall meet the reported values test requirements in table 9.1.16.1.5-2.

**Table 9.1.16.1.5-1: RSRP FDD Intra frequency absolute accuracy test parameters**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 5 | | 5 | | 5 | |
| Measurement bandwidth | | $n_{PRB}$ | 10—15 | | 10—15 | | 10—15 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.5 FDD | - | R.5 FDD | - | R.5 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 7—17 | - | 7-17 | - | 7-17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.11 FDD | | R.11 FDD | | R.11 FDD | |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | | OP.15 FDD | OP.16 FDD | OP.15 FDD | OP.16 FDD | OP.15 FDD | OP.16 FDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA[Note1]<br>OCNG_RB[Note1] | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $N_{oc}$ [Note2] | Band 31 | dBm/15 kHz | -104 | | -84 | | -109.5 | |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 | 1.88 | -4.97 | 0.09 | -4.96 |
| RSRP[Note3] | Band 31 | dBm/15 kHz | -98.0 | -102.0 | -78.0 | -82.0 | -106.5 | -109.7 |
| Io[Note3] | Band 31 | dBm/4.5 MHz | -71.06 | | -51.06 | | -78.76 | |
| $\hat{E}_s / N_{oc}$ | | dB | 6 | 2 | 6 | 2 | 3 | -0.2 |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 9.1.16.1.5-2: RSRP FDD Intra frequency absolute accuracy requirements for the reported values for 5MHz bandwidth**

| Normal Conditions | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_31 | RSRP_49 | RSRP_23 |
| Highest reported value (Cell 2) | RSRP_46 | RSRP_68 | RSRP_38 |
| **Extreme Conditions** | **Test 1** | **Test 2** | **Test 3** |
| Lowest reported value (Cell 2) | RSRP_28 | RSRP_46 | RSRP_20 |
| Highest reported value (Cell 2) | RSRP_49 | RSRP_71 | RSRP_41 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.16.1_1 FDD Intra Frequency Absolute RSRP Accuracy for 5MHz Bandwidth (Rel-12 and forward)

### 9.1.16.1_1.1 Test purpose

Same test purpose as in clause 9.1.16.1.1.

### 9.1.16.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward that support Band 31. Applicability requires support for FGI bit 16.

### 9.1.16.1_1.3 Minimum conformance requirements

The requirements for absolute accuracy of RSRP in this clause apply to a cell on the same frequency as that of the serving cell.

The accuracy requirements in Table 9.1.16.1_1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101[2] Clause 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.1 for a corresponding Band.

**Table 9.1.16.1_1.3-1: RSRP Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 3] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 2] / dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 / N/A | -70 |
| | | | FDD_B | -120.5 / N/A | -70 |
| | | | FDD_C, TDD_C | -120 / N/A | -70 |
| | | | FDD_D | -119.5 / N/A | -70 |
| | | | FDD_E, TDD_E | -119 / N/A | -70 |
| | | | FDD_F | -118.5 / N/A | -70 |
| | | | FDD_G | -118 / N/A | -70 |
| | | | FDD_H | -117.5 / N/A | -70 |
| | | | FDD_N | -114.5 / N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 / -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in section 3.5.1.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.16.1_1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.16.1_1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.1, clause 9.1.4 and A.9.1.16.

## 9.1.16.1_1.4 Test description

### 9.1.16.1_1.4.1 Initial conditions

Same initial conditions as defined in clause 9.1.1.1.4.1 with the exception that the Channel Bandwidth to be tested is 5MHz.

### 9.1.16.1_1.4.2 Test procedure

Same test procedure as defined in clause 9.1.1.1.4.2 with the following exceptions:

- Instead of Table 9.1.1.1.5-2 → use Table 9.1.16.1_1.5-1.
- Instead of Table 9.1.1.1.5-3 → use Table 9.1.16.1_1.5-2.

### 9.1.16.1_1.4.3 Message contents

Same message contents as defined in clause 9.1.1.1.4.3.

### 9.1.16.1_1.5 Test requirement

Table 9.1.16.1_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD intra-frequency relative accuracy test shall meet the reported values test requirements in table 9.1.16.1_1.5-2.

**Table 9.1.16.1_1.5-1: RSRP FDD Intra frequency absolute accuracy test parameters**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 5 | | 5 | | 5 | |
| Measurement bandwidth | | $n_{PRB}$ | 10—15 | | 10—15 | | 10—15 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.5 FDD | - | R.5 FDD | - | R.5 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 7—17 | - | 7-17 | - | 7-17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.11 FDD | | R.11 FDD | | R.11 FDD | |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | | OP.15 FDD | OP.16 FDD | OP.15 FDD | OP.16 FDD | OP.15 FDD | OP.16 FDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA[Note1]<br>OCNG_RB[Note1] | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $N_{oc}$ [Note2] | Band 31 | dBm/15 kHz | -104 | | -84 | | -109.5 | |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 | 1.88 | -4.97 | 0.09 | -4.96 |
| RSRP[Note3] | Band 31 | dBm/15 kHz | -98.0 | -102.0 | -78.0 | -82.0 | -106.5 | -109.7 |
| Io[Note3] | Band 31 | dBm/4.5 MHz | -71.06 | | -51.06 | | -78.76 | |
| $\hat{E}_s / N_{oc}$ | | dB | 6 | 2 | 6 | 2 | 3 | -0.2 |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |

NOTE 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

NOTE 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

NOTE 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 9.1.16.1_1.5-2: RSRP FDD Intra frequency absolute accuracy requirements for the reported values for 5MHz bandwidth**

| Normal Conditions | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_32 | RSRP_49 | RSRP_25 |
| Highest reported value (Cell 2) | RSRP_45 | RSRP_68 | RSRP_37 |
| **Extreme Conditions** | **Test 1** | **Test 2** | **Test 3** |
| Lowest reported value (Cell 2) | RSRP_28 | RSRP_46 | RSRP_20 |
| Highest reported value (Cell 2) | RSRP_49 | RSRP_71 | RSRP_41 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.16.2 FDD Intra Frequency Relative Accuracy of RSRP for 5MHz Bandwidth

### 9.1.16.2.1 Test purpose

To verify that the FDD intra-frequency relative accuracy measurement of RSRP is within the specified limit for Band 31.

### 9.1.16.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support Band 31. Applicability requires support for FGI bit 16.

### 9.1.16.2.3 Minimum conformance requirements

The relative accuracy of RSRP is defined as the RSRP measured from one cell compared to the RSRP measured from another cell on the same frequency.

The accuracy requirements in Table 9.1.16.2.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101[2] Clause 7.3 for reference sensitivity are fulfilled.

$RSRP1,2|_{dBm}$ according to Annex I.3.8 for a corresponding Band.

**Table 9.1.16.2.3-1: RSRP Intra frequency relative accuracy**

| **Accuracy** | | **Conditions** | | | |
|---|---|---|---|---|---|
| **Normal condition** | **Extreme condition** | **Ês/Iot** Note 2 | **Io** Note 1 **range** | | |
| | | | **E-UTRA operating band groups** Note 5 | **Minimum Io** | **Maximum Io** |
| **dB** | **dB** | **dB** | | **dBm/15kHz** Note 4 | **dBm/BW$_{Channel}$** |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.16.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.2, clause 9.1.4 and A.9.1.16.

### 9.1.16.2.4 Test description

#### 9.1.16.2.4.1 Initial conditions

Same initial conditions as defined in clause 9.1.1.2.4.1 with the exception that the Channel Bandwidth to be tested is 5MHz.

#### 9.1.16.2.4.2 Test procedure

Same test procedure as defined in clause 9.1.1.2.4.2 with the following exceptions:

- Instead of Table 9.1.1.2.5-2 → use Table 9.1.16.2.5-1.
- Instead of Table 9.1.1.2.5-3 → use Table 9.1.16.2.5-2.

#### 9.1.16.2.4.3 Message contents

Same message contents as defined in clause 9.1.1.2.4.3.

### 9.1.16.2.5 Test requirement

Table 9.1.16.2.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD intra-frequency relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.16.2.5-2.

**Table 9.1.16.2.5-1: RSRP FDD Intra frequency relative accuracy test parameters**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 5 | | 5 | | 5 | |
| Measurement bandwidth | | $n_{PRB}$ | 10—15 | | 10—15 | | 10—15 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.5 FDD | - | R.5 FDD | - | R.5 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 7—17 | - | 7-17 | - | 7-17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.11 FDD | | R.11 FDD | | R.11 FDD | |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | | OP.15 FDD | OP.16 FDD | OP.15 FDD | OP.16 FDD | OP.15 FDD | OP.16 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | |
| PSS_RA | | | | | | | | |
| SSS_RA | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA[Note1] | | | | | | | | |
| OCNG_RB[Note1] | | | | | | | | |
| $N_{oc}$ [Note2] | Band 31 | dBm/15 kHz | -103 | | -84 | | -109.5 | |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 | 1.88 | -4.97 | -0.01 | -4.76 |
| RSRP[Note3] | Band 31 | dBm/15 kHz | -97.0 | -101.0 | -78.0 | -82.0 | -106.5 | -109.5 |
| Io[Note3] | Band 31 | dBm/4.5 MHz | -70.06 | | -51.06 | | -78.71 | |
| $\hat{E}_s / N_{oc}$ | | dB | 6 | 2 | 6 | 2 | 3 | 0 |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 9.1.16.2.5-2: RSRP FDD Intra frequency relative accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRP_x - 9 | RSRP_x - 9 | RSRP_x - 8 |
| Highest reported value (Cell 2) | RSRP_x + 1 | RSRP_x + 1 | RSRP_x + 2 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRP_x - 9 | RSRP_x - 9 | RSRP_x - 8 |
| Highest reported value (Cell 2) | RSRP_x + 1 | RSRP_x + 1 | RSRP_x + 2 |
| RSRP_x is the reported value of Cell 1 | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.1.17 FDD Inter frequency RSRP Accuracy for 5MHz Bandwidth

## 9.1.17.1 FDD - FDD Inter Frequency Absolute RSRP Accuracy for 5MHz Bandwidth

### 9.1.17.1.1 Test purpose

To verify that the FDD inter-frequency absolute RSRP measurement accuracy for 5MHz Bandwidth is within the specified limit.

### 9.1.17.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE from release 8 to release 11 that support Band 31. Applicability requires support for FGI bits 16 and 25.

### 9.1.17.1.3 Minimum conformance requirements

The absolute accuracy of RSRP is defined as the RSRP measured from a cell that has different carrier frequency from the serving cell.

The accuracy requirements in table 9.1.17.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.3 for a corresponding Band.

**Table 9.1.17.1-1: RSRP Inter frequency absolute accuracy**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | | |
| | | | E-UTRA operating band groups [Note 3] | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 2] | dBm/$BW_{Channel}$ | dBm/$BW_{Channel}$ |
| ±6 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_B | -120.5 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.1.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.3.1, clause 9.1.4 and A.9.1.17.

## 9.1.17.1.4 Test description

### 9.1.17.1.4.1 Initial conditions

Same initial conditions as defined in clause 9.1.3.1.4.1 with the exception that the Channel Bandwidth to be tested is 5MHz.

### 9.1.17.1.4.2 Test procedure

Same test procedure as defined in clause 9.1.3.1.4.2 with the following exceptions:

- Instead of Table 9.1.3.1.5-2 → use Table 9.1.17.1.5-1.
- Instead of Table 9.1.3.1.5-3 → use Table 9.1.17.1.5-2.

### 9.1.17.1.4.3 Message contents

Same message contents as defined in clause 9.1.3.1.4.3.

## 9.1.17.1.5 Test requirement

Table 9.1.17.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD inter-frequency absolute accuracy test shall meet the reported values test requirements in table 9.1.17.1.5-2.

**Table 9.1.17.1.5-1: RSRP FDD-FDD Inter frequency absolute accuracy test parameters for 5MHz Bandwidth**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 | 1 | 2 |
| $BW_{channel}$ | | MHz | 5 | 5 | 5 | 5 |
| Gap Pattern Id | | | 0 | - | 0 | - |
| Measurement bandwidth | | $n_{PRB}$ | 10-15 | | 10-15 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.5 FDD | - | R.5 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 7-17 | - | 7-17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.11 FDD | | R.11 FDD | |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | | OP.15 FDD | OP.16 FDD | OP.15 FDD | OP.16 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | |
| PSS_RA | | | | | | |
| SSS_RA | | | | | | |
| PCFICH_RB | | | | | | |
| PHICH_RA | | | | | | |
| PHICH_RB | | | | | | |
| PDCCH_RA | | | | | | |
| PDCCH_RB | | | | | | |
| PDSCH_RA | | | | | | |
| PDSCH_RB | | | | | | |
| OCNG_RANote1 | | | | | | |
| OCNG_RBNote | | | | | | |
| $N_{oc}$ Note2 | Cell 2: Band 31 | dBm/15 kHz | -86.25 | -86.25 | -102.5 | -110.5 |
| $\hat{E}_s / I_{ot}$ | | dB | 10 | 10 | 13 | -3.2 |
| RSRP Note3 | Cell 2: Band 31 | dBm/15 kHz | -76.25 | -76.25 | -89.5 | -113.7 |
| Io Note3 | Cell 2: Band 31 | dBm/4.5 MHz | -51.06 | -51.06 | -64.52 | -84.03 |
| $\hat{E}_s / N_{oc}$ | | dB | 10 | 10 | 13 | -3.2 |
| Propagation condition | | - | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: For Band 26, the tests shall be performed with the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 6: Band 29 is used only for E-UTRA carrier aggregation with other E-UTRA bands.

Note 7: This test is only applicable for testing inter-frequency requirements for Band 31.

**Table 9.1.17.1.5-2: RSRP FDD Inter frequency absolute accuracy requirements for the reported values, 5MHz Bandwidth**

| Normal Conditions | Test 1 All bands | Test 2 | |
|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_55 | Cell 2: Band 31 | RSRP_19 |
| Highest reported value (Cell 2) | RSRP_74 | Cell 2: Band 31 | RSRP_34 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2** | |
| Lowest reported value (Cell 2) | RSRP_52 | Cell 2: Band 31 | RSRP_16 |
| Highest reported value (Cell 2) | RSRP_77 | Cell 2: Band 31 | RSRP_37 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.17.1_1 FDD - FDD Inter Frequency Absolute RSRP Accuracy for 5MHz Bandwidth (Rel-12 and forward)

Editor's notes: This test case is incomplete. The following item is missing or incomplete:

- Test requirements for Test2 in normal conditions are TBD.

### 9.1.17.1_1.1 Test purpose

Same test purpose as in clause 9.1.17.1.1.

### 9.1.17.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward that support Band 31. Applicability requires support for FGI bits 16 and 25.

### 9.1.17.1_1.3 Minimum conformance requirements

The absolute accuracy of RSRP is defined as the RSRP measured from a cell that has different carrier frequency from the serving cell.

The accuracy requirements in table 9.1.17.1_1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.3 for a corresponding Band.

**Table 9.1.17.1_1.3-1: RSRP Inter frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 / dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 / N/A | -70 |
| | | | FDD_C, TDD_C | -120 / N/A | -70 |
| | | | FDD_D | -119.5 / N/A | -70 |
| | | | FDD_E, TDD_E | -119 / N/A | -70 |
| | | | FDD_F | -118.5 / N/A | -70 |
| | | | FDD_G | -118 / N/A | -70 |
| | | | FDD_H | -117.5 / N/A | -70 |
| | | | FDD_N | -114.5 / N/A | -70 |

| | | | | | | |
|---|---|---|---|---|---|---|
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 3: E-UTRA operating band groups are as defined in Section 3.5. | | | | | | |

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.1.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.3.1, clause 9.1.4 and A.9.1.17.

## 9.1.17.1_1.4 Test description

### 9.1.17.1_1.4.1 Initial conditions

Same initial conditions as defined in clause 9.1.3.1.4.1 with the exception that the Channel Bandwidth to be tested is 5MHz.

### 9.1.17.1_1.4.2 Test procedure

Same test procedure as defined in clause 9.1.3.1.4.2 with the following exceptions:

- Instead of Table 9.1.3.1.5-2 → use Table 9.1.17.1_1.5-1.
- Instead of Table 9.1.3.1.5-3 → use Table 9.1.17.1_1.5-2.

### 9.1.17.1_1.4.3 Message contents

Same message contents as defined in clause 9.1.3.1.4.3.

## 9.1.17.1_1.5 Test requirement

Table 9.1.17.1_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD inter-frequency absolute accuracy test shall meet the reported values test requirements in table 9.1.17.1_1.5-2.

**Table 9.1.17.1_1.5-1: RSRP FDD-FDD Inter frequency absolute accuracy test parameters for 5MHz Bandwidth**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |

| | | | | | | |
|---|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | 2 | 1 | 2 |
| $BW_{channel}$ | | MHz | 5 | 5 | 5 | 5 |
| Gap Pattern Id | | | 0 | - | 0 | - |
| Measurement bandwidth | | $n_{PRB}$ | 10-15 | | 10-15 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.5 FDD | - | R.5 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 7-17 | - | 7-17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.11 FDD | | R.11 FDD | |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | | OP.15 FDD | OP.16 FDD | OP.15 FDD | OP.16 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | |
| PSS_RA | | | | | | |
| SSS_RA | | | | | | |
| PCFICH_RB | | | | | | |
| PHICH_RA | | | | | | |
| PHICH_RB | | | | | | |
| PDCCH_RA | | | | | | |
| PDCCH_RB | | | | | | |
| PDSCH_RA | | | | | | |
| PDSCH_RB | | | | | | |
| OCNG_RANote1 | | | | | | |
| OCNG_RBNote | | | | | | |
| $N_{oc}$ Note2 | Cell 2: Band 31 | dBm/15 kHz | -86.25 | -86.25 | -102.5 | -110.5 |
| $\hat{E}_s / I_{ot}$ | | dB | 10 | 10 | 13 | -3.2 |
| RSRP Note3 | Cell 2: Band 31 | dBm/15 kHz | -76.25 | -76.25 | -89.5 | -113.7 |
| Io Note3 | Cell 2: Band 31 | dBm/4.5 MHz | -51.06 | -51.06 | -64.52 | -84.03 |
| $\hat{E}_s / N_{oc}$ | | dB | 10 | 10 | 13 | -3.2 |
| Propagation condition | | - | AWGN | | AWGN | |

NOTE 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
NOTE 5: For Band 26, the tests shall be performed with the assigned E-UTRA channel bandwidth within 865-894 MHz.
NOTE 6: Band 29 is used only for E-UTRA carrier aggregation with other E-UTRA bands.
NOTE 7: This test is only applicable for testing inter-frequency requirements for Band 31.

**Table 9.1.17.1_1.5-2: RSRP FDD Inter frequency absolute accuracy requirements for the reported values, 5MHz Bandwidth**

| **Normal Conditions** | **Test 1 All bands** | **Test 2** | |
|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_55 | Cell 2: Band 31 | TBD |
| Highest reported value (Cell 2) | RSRP_74 | Cell 2: Band 31 | TBD |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2** | |
| Lowest reported value (Cell 2) | RSRP_52 | Cell 2: Band 31 | RSRP_16 |
| Highest reported value (Cell 2) | RSRP_77 | Cell 2: Band 31 | RSRP_37 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.17.2 FDD - FDD Inter Frequency Relative Accuracy of RSRP for 5MHz Bandwidth

### 9.1.17.2.1 Test purpose

To verify that the FDD inter-frequency relative accuracy measurement of RSRP is within the specified limit for Band 31.

### 9.1.17.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE from release 8 to release 11 that support Band 31. Applicability requires support for FGI bits 16 and 25.

### 9.1.17.2.3 Minimum conformance requirements

The relative accuracy of RSRP in inter frequency case is defined as the RSRP measured from one cell compared to the RSRP measured from another cell on a different frequency.

The accuracy requirements in Table 9.1.17.2.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101[2] Clause 7.3 for reference sensitivity are fulfilled.

$RSRP1,2|_{dBm}$ according to Annex B.3.4 for a corresponding Band

$$\left| RSRP1 \right|_{dBm} - RSRP2 \left|_{dBm} \right| \leq 27dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.17.2.3-1: RSRP Inter frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | $dBm/BW_{Channel}$ |
| ±6 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in section 3.5.1. | | | | | |

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.16.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.3.2, clause 9.1.4 and A.9.1.17.

### 9.1.17.2.4 Test description

#### 9.1.17.2.4.1 Initial conditions

Same initial conditions as defined in clause 9.1.3.2.4.1 with the exception that the Channel Bandwidth to be tested is 5MHz.

#### 9.1.17.2.4.2 Test procedure

Same test procedure as defined in clause 9.1.3.2.4.2 with the following exceptions:

- Instead of Table 9.1.3.2.5-2 → use Table 9.1.17.2.5-1.
- Instead of Table 9.1.3.2.5-3 → use Table 9.1.17.2.5-2.

#### 9.1.17.2.4.3 Message contents

Same message contents as defined in clause 9.1.3.2.4.3.

### 9.1.17.2.5 Test requirement

Table 9.1.17.2.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD inter-frequency relative accuracy test shall meet the reported values test requirements in table 9.1.17.2.5-2.

**Table 9.1.17.2.5-1: RSRP FDD - FDD Inter frequency relative accuracy test parameters for 5MHz bandwidth**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 | 1 | 2 |
| $BW_{channel}$ | | MHz | 5 | 5 | 5 | 5 |
| Gap Pattern Id | | | 0 | - | 0 | - |
| Measurement bandwidth | | $n_{PRB}$ | 10—15 | | 10—15 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.5 FDD | - | R.5 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 7—17 | - | 7-17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.11 FDD | | R.11 FDD | |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | | OP.15 FDD | OP.16 FDD | OP.15 FDD | OP.16 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | |
| PSS_RA | | | | | | |
| SSS_RA | | | | | | |
| PCFICH_RB | | | | | | |
| PHICH_RA | | | | | | |
| PHICH_RB | | | | | | |
| PDCCH_RA | | | | | | |
| PDCCH_RB | | | | | | |
| PDSCH_RA | | | | | | |
| PDSCH_RB | | | | | | |
| OCNG_RANote1 | | | | | | |
| OCNG_RBNote | | | | | | |
| $N_{oc}$ Note2 | Cell 2: Band 31 | dBm/15 kHz | -86.25 | -86.25 | -103.50 | -110.50 |
| $\hat{E}_s / I_{ot}$ | | dB | 10 | 10 | 13 | -3.2 |
| RSRP Note3 | Cell 2: Band 31 | dBm/15 kHz | -76.25 | -76.25 | -90.50 | -113.70 |
| Io Note3 | Cell 2: Band 31 | dBm/4.5 MHz | -51.06 | -51.06 | -65.52 | -84.03 |
| $\hat{E}_s / N_{oc}$ | | dB | 10 | 10 | 13 | -3.2 |
| Propagation condition | | - | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: For Band 26, the tests shall be performed with the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 6: Band 29 is used only for E-UTRA carrier aggregation with other E-UTRA bands.

Note 7: This test is only applicable for testing inter-frequency requirements for Band 31. Cell 2 is on the Band under test, and Cell 1 is on another band supported by the UE.

**Table 9.1.17.2.5-2: RSRP FDD Inter frequency relative accuracy requirements for the 5MHz bandwidth for the reported values**

| | Test 1<br>All bands | Test 2<br>All bands |
|---|---|---|
| Normal Conditions | | |
| Lowest reported value (Cell 2) | RSRP_(x - 9) | RSRP_(x - 32) |
| Highest reported value (Cell 2) | RSRP_(x + 9) | RSRP_(x - 16) |
| Extreme Conditions | | |
| Lowest reported value (Cell 2) | RSRP_(x - 9) | RSRP_(x - 32) |
| Highest reported value (Cell 2) | RSRP_(x + 9) | RSRP_(x - 16) |
| RSRP_x is the reported value of Cell 1 | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.17.2_1 FDD - FDD Inter Frequency Relative Accuracy of RSRP for 5MHz Bandwidth (Rel-12 and forward)

Editor's notes: This test case is incomplete. The following item is missing or incomplete:

- Test requirements for Test 1 and 2 in normal conditions are TBD

### 9.1.17.2_1.1 Test purpose

Same test purpose as in clause 9.1.17.2.1.

### 9.1.17.2_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward that support Band 31. Applicability requires support for FGI bits 16 and 25.

### 9.1.17.2_1.3 Minimum conformance requirements

The relative accuracy of RSRP in inter frequency case is defined as the RSRP measured from one cell compared to the RSRP measured from another cell on a different frequency.

The accuracy requirements in Table 9.1.17.2_1.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101[2] Clause 7.3 for reference sensitivity are fulfilled.

RSRP1,2$|_{dBm}$ according to Annex B.3.4 for a corresponding Band

$$\left| RSRP1 \right|_{dBm} - RSRP2 \left|_{dBm} \right| \leq 27dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.17.2_1.3-1: RSRP Inter frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/$BW_{Channel}$ |
| ±4.5 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in section 3.5.1. | | | | | |

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.16.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.3.2, clause 9.1.4 and A.9.1.17.

## 9.1.17.2_1.4 Test description

### 9.1.17.2_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.17.2.4.1.

### 9.1.17.2_1.4.2 Test procedure

Same test procedure as defined in clause 9.1.17.2.4.2 with the following exceptions:

- Instead of Table 9.1.17.2.5-1 → use Table 9.1.17.2_1.5-1.
- Instead of Table 9.1.17.2.5-2 → use Table 9.1.17.2_1.5-2.

### 9.1.17.2_1.4.3 Message contents

Same message contents as defined in clause 9.1.17.2.4.3.

## 9.1.17.2_1.5 Test requirement

Table 9.1.17.2_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD inter-frequency relative accuracy test shall meet the reported values test requirements in table 9.1.17.2_1.5-2.

**Table 9.1.17.2_1.5-1: RSRP FDD - FDD Inter frequency relative accuracy test parameters for 5MHz bandwidth**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 | 1 | 2 |
| $BW_{channel}$ | | MHz | 5 | 5 | 5 | 5 |
| Gap Pattern Id | | | 0 | - | 0 | - |
| Measurement bandwidth | | $n_{PRB}$ | 10—15 | | 10—15 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.5 FDD | - | R.5 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 7—17 | - | 7-17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.11 FDD | | R.11 FDD | |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | | OP.15 FDD | OP.16 FDD | OP.15 FDD | OP.16 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | |
| PSS_RA | | | | | | |
| SSS_RA | | | | | | |
| PCFICH_RB | | | | | | |
| PHICH_RA | | | | | | |
| PHICH_RB | | | | | | |
| PDCCH_RA | | | | | | |
| PDCCH_RB | | | | | | |
| PDSCH_RA | | | | | | |
| PDSCH_RB | | | | | | |
| OCNG_RANote1 | | | | | | |
| OCNG_RBNote | | | | | | |
| $N_{oc}$ Note2 | Cell 2: Band 31 | dBm/15 kHz | -86.25 | -86.25 | -103.50 | -110.50 |
| $\hat{E}_s / I_{ot}$ | | dB | 10 | 10 | 13 | -3.2 |
| RSRP Note3 | Cell 2: Band 31 | dBm/15 kHz | -76.25 | -76.25 | -90.50 | -113.70 |
| Io Note3 | Cell 2: Band 31 | dBm/4.5 MHz | -51.06 | -51.06 | -65.52 | -84.03 |
| $\hat{E}_s / N_{oc}$ | | dB | 10 | 10 | 13 | -3.2 |
| Propagation condition | | - | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: For Band 26, the tests shall be performed with the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 6: Band 29 is used only for E-UTRA carrier aggregation with other E-UTRA bands.

Note 7: This test is only applicable for testing inter-frequency requirements for Band 31. Cell 2 is on the Band under test, and Cell 1 is on another band supported by the UE.

**Table 9.1.17.2_1.5-2: RSRP FDD Inter frequency relative accuracy requirements for the 5MHz bandwidth for the reported values**

| | Test 1<br>All bands | Test 2<br>All bands |
|---|---|---|
| Normal Conditions | | |
| Lowest reported value (Cell 2) | TBD | TBD |
| Highest reported value (Cell 2) | TBD | TBD |
| Extreme Conditions | | |
| Lowest reported value (Cell 2) | RSRP_(x - 9) | RSRP_(x - 32) |
| Highest reported value (Cell 2) | RSRP_(x + 9) | RSRP_(x - 16) |
| RSRP_x is the reported value of Cell 1 | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.18 FDD RSRP Accuracy for E-UTRAN Carrier Aggregation for 10MHz + 5MHz

### 9.1.18.1 FDD Absolute RSRP Accuracy for E-UTRA for Carrier Aggregation for 10MHz + 5MHz

#### 9.1.18.1.1 Test purpose

Same test purpose as in clause 9.1.6.1.1.

#### 9.1.18.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 11 supporting CA.

#### 9.1.18.1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.6.1.3.

#### 9.1.18.1.4 Test description

##### 9.1.18.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz for Cell 1 on the PCC, 5MHz for Cell 2 on the SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.18.1.4.3.

4. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier and Cell 3 is the neighbouring cell on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

##### 9.1.18.1.4.2 Test procedure

Same test procedure as in clause 9.1.6.1.4.2 with the following exceptions:

- Instead of Table 9.1.6.1.5-2 → use Table 9.1.18.1.5-1.
- Instead of Table 9.1.6.1.5-3 → use Table 9.1.18.1.5-2.

#### 9.1.18.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.18.1.4.3-1: Common Exception messages for FDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation test requirement for 10MHz + 5MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-3<br>Table H.4.1-5 |

### 9.1.18.1.5 Test requirement

Table 9.1.18.1.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP FDD absolute accuracy test shall meet the reported values test requirements in table 9.1.18.1.5-2.

**Table 9.1.18.1.5-1: RSRP FDD absolute accuracy carrier aggregation test parameters for 10MHz + 5MHz**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 10 | 5 | |
| Measurement bandwidth | | $n_{PRB}$ | 22-27 | 10-15 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | R.5 FDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 13-36 | 7-17 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | R.11 FDD | |
| OCNG Patterns defined in D.1 (FDD) | | | OP.1 FDD | OP.15 FDD | OP.16 FDD |
| Io Note2 | Bands FDD_A | dBm/9 MHz | -87.76 | N/A | |
| | Bands FDD_C | | -86.76 | | |
| | Bands FDD_D | | -86.26 | | |
| | Bands FDD_E, FDD_F | | -85.76 | | |
| | Bands FDD_G | | -84.76 | | |
| | Bands FDD_H | | -84.26 | | |
| | Bands FDD_A | dBm/4.5 MHz | N/A | (Io for Channel 1 +2.32dB) | |
| | Bands FDD_C | | | | |
| | Bands FDD_D | | | | |
| | Bands FDD_E, FDD_F | | | | |
| | Bands FDD_G | | | | |
| | Bands FDD_H | | | | |

Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: See Table 9.1.6.1.5-2 for the other parameters.

**Table 9.1.18.1.5-2: RSRP FDD Carrier Aggregation absolute accuracy requirements for the reported values**

| | Test 1 | | | | | |
|---|---|---|---|---|---|---|
| **Band of Cell 1 on Primary Component Carrier** | **Bands FDD_A** | **Bands FDD_C** | **Band FDD_D** | **Bands FDD_E, FDD_F** | **Bands FDD_G** | **Bands FDD_H** |
| Normal Conditions | | | | | | |
| Lowest reported value (Cell 1) | RSRP_12 | RSRP_13 | RSRP_13 | RSRP_14 | RSRP_15 | RSRP_15 |
| Highest reported value (Cell 1) | RSRP_27 | RSRP_28 | RSRP_29 | RSRP_29 | RSRP_30 | RSRP_31 |
| Lowest reported value (Cell 2) | RSRP_20 | RSRP_21 | RSRP_21 | RSRP_22 | RSRP_23 | RSRP_23 |
| Highest reported value (Cell 2) | RSRP_35 | RSRP_36 | RSRP_37 | RSRP_37 | RSRP_38 | RSRP_39 |
| Extreme Conditions | | | | | | |
| Lowest reported value (Cell 1) | RSRP_9 | RSRP_10 | RSRP_10 | RSRP_11 | RSRP_12 | RSRP_12 |
| Highest reported value (Cell 1) | RSRP_30 | RSRP_31 | RSRP_32 | RSRP_32 | RSRP_33 | RSRP_34 |
| Lowest reported value (Cell 2) | RSRP_17 | RSRP_18 | RSRP_18 | RSRP_19 | RSRP_20 | RSRP_20 |
| Highest reported value (Cell 2) | RSRP_38 | RSRP_39 | RSRP_40 | RSRP_40 | RSRP_41 | RSRP_42 |
| Note: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2. | | | | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.18.1_1 FDD Absolute RSRP Accuracy for E-UTRA for Carrier Aggregation for 10MHz + 5MHz (Rel-12 and forward)

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

-The test tolerances applicable to this test are undefined

### 9.1.18.1_1.1 Test purpose

Same test purpose as in clause 9.1.18.1.1.

### 9.1.18.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward supporting CA.

### 9.1.18.1_1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.6.1_1.3.

### 9.1.18.1_1.4 Test description

#### 9.1.18.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.18.1.4.1 with the following exceptions:

- Instead of 9.1.18.1.4.3 → use 9.1.18.1_1.4.3.

#### 9.1.18.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.18.1.4.2 with the following exceptions:

- Instead of Table 9.1.18.1.5-1 → use Table 9.1.18.1_1.5-1.
- Instead of Table 9.1.18.1.5-2 → use Table 9.1.18.1_1.5-2.

### 9.1.18.1_1.4.3 Message contents

Same message contents as in clause 9.1.18.1.4.3.

## 9.1.18.1_1.5 Test requirement

Table 9.1.18.1_1.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP FDD absolute accuracy test shall meet the reported values test requirements in table 9.1.18.1_1.5-2.

**Table 9.1.18.1_1.5-1: RSRP FDD absolute accuracy carrier aggregation test parameters for 10MHz + 5MHz**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 10 | 5 | |
| Measurement bandwidth | | $n_{PRB}$ | 22-27 | 10-15 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | R.5 FDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 13-36 | 7-17 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | R.11 FDD | |
| OCNG Patterns defined in D.1 (FDD) | | | OP.1 FDD | OP.15 FDD | OP.16 FDD |
| $Io^{Note2}$ | Bands FDD_A | dBm/9 MHz | -87.76 | N/A | |
| | Bands FDD_C | | -86.76 | | |
| | Bands FDD_D | | -86.26 | | |
| | Bands FDD_E, FDD_F | | -85.76 | | |
| | Bands FDD_G | | -84.76 | | |
| | Bands FDD_H | | -84.26 | | |
| | Bands FDD_A | dBm/4.5 MHz | N/A | (Io for Channel 1 +2.32dB) | |
| | Bands FDD_C | | | | |
| | Bands FDD_D | | | | |
| | Bands FDD_E, FDD_F | | | | |
| | Bands FDD_G | | | | |
| | Bands FDD_H | | | | |

Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: See Table 9.1.6.1.5-2 for the other parameters.

**Table 9.1.18.1_1.5-2: RSRP FDD Carrier Aggregation absolute accuracy requirements for the reported values**

| | Test 1 | | | | | |
|---|---|---|---|---|---|---|
| **Band of Cell 1 on Primary Component Carrier** | **Bands FDD_A** | **Bands FDD_C** | **Band FDD_D** | **Bands FDD_E, FDD_F** | **Bands FDD_G** | **Bands FDD_H** |
| Normal Conditions | | | | | | |
| Lowest reported value (Cell 1) | TBD | TBD | TBD | TBD | TBD | TBD |
| Highest reported value (Cell 1) | TBD | TBD | TBD | TBD | TBD | TBD |
| Lowest reported value (Cell 2) | TBD | TBD | TBD | TBD | TBD | TBD |
| Highest reported value (Cell 2) | TBD | TBD | TBD | TBD | TBD | TBD |
| Extreme Conditions | | | | | | |
| Lowest reported value (Cell 1) | RSRP_9 | RSRP_10 | RSRP_10 | RSRP_11 | RSRP_12 | RSRP_12 |
| Highest reported value (Cell 1) | RSRP_30 | RSRP_31 | RSRP_32 | RSRP_32 | RSRP_33 | RSRP_34 |
| Lowest reported value (Cell 2) | RSRP_17 | RSRP_18 | RSRP_18 | RSRP_19 | RSRP_20 | RSRP_20 |
| Highest reported value (Cell 2) | RSRP_38 | RSRP_39 | RSRP_40 | RSRP_40 | RSRP_41 | RSRP_42 |
| Note: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2. | | | | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.18.2 FDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation for 10MHz + 5MHz

### 9.1.18.2.1 Test purpose

Same test purpose as in clause 9.1.6.2.1.

### 9.1.18.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 11 supporting CA.

### 9.1.18.2.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.6.2.3.

### 9.1.18.2.4 Test description

#### 9.1.18.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz for Cell 1 on the PCC, 5MHz for Cell 2 on the SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.18.2.4.3.

4. Cell1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier and Cell 3 is the neighbouring cell on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.1.18.2.4.2 Test procedure

Same test procedure as in clause 9.1.6.2.4.2 with the following exceptions:

- Instead of Table 9.1.6.2.5-2 → use Table 9.1.18.2.5-1.
- Instead of Table 9.1.6.2.5-3 → use Table 9.1.18.2.5-2.

#### 9.1.18.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.18.2.4.3-1: Common Exception messages for FDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation test requirement for 10MHz + 5MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-3<br>Table H.4.1-5 |

### 9.1.18.2.5 Test requirement

Table 9.1.18.2.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP FDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.18.2.5-2.

**Table 9.1.18.2.5-1: RSRP FDD relative accuracy carrier aggregation test parameters for 10MHz + 5MHz**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 10 | 5 | |
| Measurement bandwidth | | $n_{PRB}$ | 22-27 | 10-15 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | R.5 FDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 13-36 | 7-17 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | R.11 FDD | |
| OCNG Patterns defined in D.1 (FDD) | | | OP.1 FDD | OP.15 FDD | OP.16 FDD |
| Io Note2 | Bands FDD_A | dBm/9 MHz | -87.76 | N/A | |
| | Bands FDD_C | | -86.76 | | |
| | Bands FDD_D | | -86.26 | | |
| | Bands FDD_E, FDD_F | | -85.76 | | |
| | Bands FDD_G | | -84.76 | | |
| | Bands FDD_H | | -84.26 | | |
| | Bands FDD_A | dBm/4.5 MHz | N/A | (Io for Channel 1 +2.50dB) | |
| | Bands FDD_C | | | | |
| | Bands FDD_D | | | | |
| | Bands FDD_E, FDD_F | | | | |
| | Bands FDD_G | | | | |
| | Bands FDD_H | | | | |
| Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.<br>Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 3: See Table 9.1.6.2.5-2 for the other parameters. | | | | | |

**Table 9.1.18.2.5-2: RSRP FDD Carrier Aggregation relative accuracy requirements for the reported values for 10MHz + 5MHz**

| | Test 1 |
|---|---|
| | All bands |
| Normal and Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRP_x – 1 |
| Highest reported value (Cell 2) | RSRP_x + 17 |
| Lowest reported value (Cell 3) | RSRP_y – 8 |
| Highest reported value (Cell 3) | RSRP_y + 1 |
| RSRP_x is the reported value of Cell 1<br>RSRP_y is the reported value of Cell 2 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.18.2_1 FDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation for 10MHz + 5MHz (Rel-12 and forward)

Editor's note: This Test case is incomplete. The following aspects are either missing or not yet determined:

- *The Test system uncertainties applicable to this test are undefined*
- *The Test tolerances applicable to this test are undefined*
- *Test requirements in normal conditions need to be defined*

### 9.1.18.2_1.1 Test purpose

Same test purpose as in clause 9.1.18.2.1.

### 9.1.18.2_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward supporting CA.

### 9.1.18.2_1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.6.2_1.3.

### 9.1.18.2_1.4 Test description

#### 9.1.18.2_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.18.2.4.1.

#### 9.1.18.2_1.4.2 Test procedure

Same test procedure as in clause 9.1.18.2.4.2 with the following exceptions:

- Instead of Table 9.1.18.2.5-1 → use Table 9.1.18.2_1.5-1.
- Instead of Table 9.1.18.2.5-2 → use Table 9.1.18.2_1.5-2.

#### 9.1.18.2_1.4.3 Message contents

Same message contents as in clause 9.1.18.2.4.3

### 9.1.18.2_1.5 Test requirement

Table 9.1.18.2_1.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP FDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.18.2_1.5-2.

**Table 9.1.18.2_1.5-1: RSRP FDD relative accuracy carrier aggregation test parameters for 10MHz + 5MHz**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 10 | 5 | |
| Measurement bandwidth | | $n_{PRB}$ | 22-27 | 10-15 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | R.5 FDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 13-36 | 7-17 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | R.11 FDD | |
| OCNG Patterns defined in D.1 (FDD) | | | OP.1 FDD | OP.15 FDD | OP.16 FDD |
| Io Note2 | Bands FDD_A | dBm/9 MHz | -87.76+TT | N/A | |
| | Bands FDD_C | | -86.76+TT | | |
| | Bands FDD_D | | -86.26+TT | | |
| | Bands FDD_E, FDD_F | | -85.76+TT | | |
| | Bands FDD_G | | -84.76+TT | | |
| | Bands FDD_H | | -84.26+TT | | |
| | Bands FDD_A | dBm/4.5 MHz | N/A | (Io for Channel 1 +2.50dB) | |
| | Bands FDD_C | | | | |
| | Bands FDD_D | | | | |

| | | | | |
|---|---|---|---|---|
| | Bands FDD_E, FDD_F | | | |
| | Bands FDD_G | | | |
| | Bands FDD_H | | | |
| Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.<br>Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 3: See Table 9.1.6.2.5-2 for the other parameters. | | | | |

**Table 9.1.18.2_1.5-2: RSRP FDD Carrier Aggregation relative accuracy requirements for the reported values for 10MHz + 5MHz**

| | **Test 1** |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 2) | TBD |
| Highest reported value (Cell 2) | TBD |
| Lowest reported value (Cell 3) | TBD |
| Highest reported value (Cell 3) | TBD |
| Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRP_x – 1 |
| Highest reported value (Cell 2) | RSRP_x + 17 |
| Lowest reported value (Cell 3) | RSRP_y – 8 |
| Highest reported value (Cell 3) | RSRP_y + 1 |
| RSRP_x is the reported value of Cell 1<br>RSRP_y is the reported value of Cell 2 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.19 TDD RSRP Accuracy for E-UTRAN Carrier Aggregation for 10MHz + 5MHz

### 9.1.19.1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 10MHz + 5MHz

#### 9.1.19.1.1 Test purpose

Same test purpose as in clause 9.1.7.1.1.

#### 9.1.19.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 11 supporting CA.

#### 9.1.19.1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.7.1.2.

#### 9.1.19.1.4 Test description

##### 9.1.19.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz for Cell 1 on the PCC, 5MHz for Cell 2 on the SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.19.1.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.1.19.1.4.2 Test procedure

Same test procedure as in clause 9.1.7.1.4.2 with the following exceptions:

- Instead of Table 9.1.7.1.5-1 → use Table 9.1.19.1.5-1.

- Instead of Table 9.1.7.1.5-2 → use Table 9.1.19.1.5-2.

#### 9.1.19.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.19.1.4.3-1: Common Exception messages for TDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation test requirement for 10MHz + 5MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-3<br>Table H.4.1-5 |

### 9.1.19.1.5 Test requirement

Table 9.1.19.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD absolute accuracy test shall meet the reported values test requirements in table 9.1.19.1.5-2.

**Table 9.1.19.1.5-1: RSRP TDD absolute accuracy carrier aggregation test parameters for 10MHz + 5MHz**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 10 | 5 | |
| Measurement bandwidth | | $n_{PRB}$ | 22-27 | 10-15 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | R.5 TDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 13-36 | 7-17 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | R.12 TDD | |
| OCNG Patterns defined in D.2 (TDD) | | | OP.1 TDD | OP.9 TDD | OP.10 TDD |
| Io Note2 | Bands TDD_A | dBm/9 MHz | -87.76 | N/A | |
| | Bands TDD_C | | -86.76 | | |
| | Bands TDD_E | | -85.76 | | |
| | Bands TDD_A | dBm/4.5MHz | N/A | (Io for Channel 1 +2.32dB) | |
| | Bands TDD_C | | | | |
| | Bands TDD_E | | | | |
| Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.<br>Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 3: See Table 9.1.7.1.5-1 for the other parameters. | | | | | |

**Table 9.1.19.1.5-2: RSRP TDD Carrier Aggregation absolute accuracy requirements for the reported values for 10MHz + 5MHz**

| Parameter | Test 1 | | |
|---|---|---|---|
| **Band of Cell 1 on Primary Component Carrier** | **Bands TDD_A** | **Bands TDD_C** | **Band TDD_E** |
| Normal Conditions | | | |
| Lowest reported value (Cell 1) | RSRP_12 | RSRP_13 | RSRP_14 |
| Highest reported value (Cell 1) | RSRP_27 | RSRP_28 | RSRP_29 |
| Lowest reported value (Cell 2) | RSRP_9 | RSRP_10 | RSRP_11 |
| Highest reported value (Cell 2) | RSRP_30 | RSRP_31 | RSRP_32 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 1) | RSRP_20 | RSRP_21 | RSRP_22 |
| Highest reported value (Cell 1) | RSRP_35 | RSRP_36 | RSRP_37 |
| Lowest reported value (Cell 2) | RSRP_17 | RSRP_18 | RSRP_19 |
| Highest reported value (Cell 2) | RSRP_38 | RSRP_39 | RSRP_40 |
| Note: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2. | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.19.1_1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 10MHz + 5MHz (Rel-12 and forward)

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

-The Test system uncertainties applicable to this test are undefined

-The Test tolerances applicable to this test are undefined

-The cell configuration mapping for this test case is undefined in Annex E

-Test case applicability is still being discussed.

### 9.1.19.1_1.1 Test purpose

Same test purpose as in clause 9.1.7.1_1.1.

### 9.1.19.1_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward supporting CA.

### 9.1.19.1_1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.7.1_1.3

### 9.1.19.1_1.4 Test description

#### 9.1.19.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.19.1.4.1 with the following exceptions:

- Instead of 9.1.19.1.4.3 → use 9.1.19.1_1.4.3.

#### 9.1.19.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.7.1_1.4.2 with the following exceptions:

- Instead of Table 9.1.7.1_1.5-1 → use Table 9.1.19.1_1.5-1.
- Instead of Table 9.1.7.1_1.5-2 → use Table 9.1.19.1_1.5-2.

#### 9.1.19.1_1.4.3 Message contents

Same message contents as in clause 9.1.19.1.4.3

### 9.1.19.1_1.5 Test requirement

Table 9.1.19.1_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD absolute accuracy test shall meet the reported values test requirements in table 9.1.19.1_1.5-2.

**Table 9.1.19.1_1.5-1: RSRP TDD absolute accuracy carrier aggregation test parameters for 10MHz + 5MHz**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 10 | 5 | |
| Measurement bandwidth | | $n_{PRB}$ | 22-27 | 10-15 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | R.5 TDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 13-36 | 7-17 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | R.12 TDD | |
| OCNG Patterns defined in D.2 (TDD) | | | OP.1 TDD | OP.9 TDD | OP.10 TDD |
| Io Note2 | Bands TDD_A | dBm/9 MHz | -87.76+TT | N/A | |
| | Bands TDD_C | | -86.76+TT | | |
| | Bands TDD_E | | -85.76+TT | | |
| | Bands TDD_A | dBm/4.5MHz | N/A | -85.44+TT | |
| | Bands TDD_C | | | -84.44+TT | |
| | Bands TDD_E | | | -83.44+TT | |
| NOTE 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.<br>NOTE 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>NOTE 3: See Table 9.1.7.1_1.5-1 for the other parameters. | | | | | |

**Table 9.1.19.1_1.5-2: RSRP TDD Carrier Aggregation absolute accuracy requirements for the reported values for 10MHz + 5MHz**

| Parameter | Test 1 | | |
|---|---|---|---|
| **Band of Cell 1 on Primary Component Carrier** | **Bands TDD_A** | **Bands TDD_C** | **Band TDD_E** |
| Normal Conditions | | | |
| Lowest reported value (Cell 1) | TBD | TBD | TBD |
| Highest reported value (Cell 1) | TBD | TBD | TBD |
| Lowest reported value (Cell 2) | TBD | TBD | TBD |
| Highest reported value (Cell 2) | TBD | TBD | TBD |
| Extreme Conditions | | | |
| Lowest reported value (Cell 1) | RSRP_xx | RSRP_xx | RSRP_xx |
| Highest reported value (Cell 1) | RSRP_xx | RSRP_xx | RSRP_xx |
| Lowest reported value (Cell 2) | RSRP_xx | RSRP_xx | RSRP_xx |
| Highest reported value (Cell 2) | RSRP_xx | RSRP_xx | RSRP_xx |
| NOTE: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2. | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.19.2 TDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 10MHz + 5MHz

### 9.1.19.2.1 Test purpose

Same test purpose as in clause 9.1.7.2.1.

### 9.1.19.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 11 supporting CA.

### 9.1.19.2.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.7.2.3.

### 9.1.19.2.4 Test description

#### 9.1.19.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz for Cell 1 on the PCC, 5MHz for Cell 2 on the SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.19.2.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.1.19.2.4.2 Test procedure

Same test procedure as in clause 9.1.7.2.4.2 with the following exceptions:

- Instead of Table 9.1.7.2.5-1 → use Table 9.1.19.2.5-1.

- Instead of Table 9.1.7.2.5-2 → use Table 9.1.19.2.5-2.

#### 9.1.19.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.19.2.4.3-1: Common Exception messages for TDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation test requirement for 10MHz + 5MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-3<br>Table H.4.1-5 |

### 9.1.19.2.5 Test requirement

Table 9.1.19.2.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.19.2.5-2.

**Table 9.1.19.2.5-1: RSRP TDD relative accuracy carrier aggregation test parameters for 10MHz + 5MHz**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 10 | 5 | |
| Measurement bandwidth | | $n_{PRB}$ | 22-27 | 10-15 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | R.5 TDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 13-36 | 7-17 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | R.12 TDD | |
| OCNG Patterns defined in D.2 (TDD) | | | OP.1 TDD | OP.9 TDD | OP.10 TDD |
| Io Note2 | Bands TDD_A | dBm/9 MHz | -87.76 | N/A | |
| | Bands TDD_C | | -86.76 | | |
| | Bands TDD_E | | -85.76 | | |
| | Bands TDD_A | dBm/4.5MHz | N/A | (Io for Channel 1 +2.50dB) | |
| | Bands TDD_C | | | | |
| | Bands TDD_E | | | | |
| Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.<br>Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 3: See Table 9.1.7.2.5-1 for the other parameters. | | | | | |

**Table 9.1.19.2.5-2: RSRP TDD Carrier Aggregation relative accuracy requirements for the reported values for 10MHz + 5MHz**

| | Test 1 |
|---|---|
| | All bands |
| Normal and Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRP_x – 1 |
| Highest reported value (Cell 2) | RSRP_x + 17 |
| Lowest reported value (Cell 3) | RSRP_y – 8 |
| Highest reported value (Cell 3) | RSRP_y + 1 |
| RSRP_x is the reported value of Cell 1<br>RSRP_y is the reported value of Cell 2 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.19.2_1 TDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 10MHz + 5MHz (Rel-12 and forward)

Editor's note: This Test case is incomplete. The following aspects are either missing or not yet determined:

- *The Test system uncertainties applicable to this test are undefined*
- *The Test tolerances applicable to this test are undefined*
- *Test requirements in normal conditions need to be defined*

### 9.1.19.2_1.1 Test purpose

Same test purpose as in clause 9.1.19.2.1.

### 9.1.19.2_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward supporting CA.

### 9.1.19.2_1.3 Minimum conformance requirements

Same minimum conformance requirements as defined in clause 9.1.7.2_1.3.

### 9.1.19.2_1.4 Test description

#### 9.1.19.2_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.19.2.4.1.

#### 9.1.19.2_1.4.2 Test procedure

Same test procedure as in clause 9.1.19.2.4.2 with the following exceptions:

- Instead of Table 9.1.19.2.5-1 → use Table 9.1.19.2_1.5-1.
- Instead of Table 9.1.19.2.5-2 → use Table 9.1.19.2_1.5-2.

#### 9.1.19.2_1.4.3 Message contents

Same message contents as in clause 9.1.19.2.4.3

### 9.1.19.2_1.5 Test requirement

Table 9.1.19.2_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.19.2_1.5-2.

**Table 9.1.19.2_1.5-1: RSRP TDD relative accuracy carrier aggregation test parameters for 10MHz + 5MHz**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 10 | 5 | |
| Measurement bandwidth | | $n_{PRB}$ | 22-27 | 10-15 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | R.5 TDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 13-36 | 7-17 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | R.12 TDD | |
| OCNG Patterns defined in D.2 (TDD) | | | OP.1 TDD | OP.9 TDD | OP.10 TDD |
| Io Note2 | Bands TDD_A | dBm/9 MHz | -87.76+TT | N/A | |
| | Bands TDD_C | | -86.76+TT | | |
| | Bands TDD_E | | -85.76+TT | | |
| | Bands TDD_A | dBm/4.5MHz | N/A | (Io for Channel 1 +2.50dB) | |
| | Bands TDD_C | | | | |
| | Bands TDD_E | | | | |

Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: See Table 9.1.7.2.5-1 for the other parameters.

**Table 9.1.19.2_1.5-2: RSRP TDD Carrier Aggregation relative accuracy requirements for the reported values for 10MHz + 5MHz**

| | **Test 1** |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 2) | TBD |
| Highest reported value (Cell 2) | TBD |
| Lowest reported value (Cell 3) | TBD |
| Highest reported value (Cell 3) | TBD |
| Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRP_x – 1 |
| Highest reported value (Cell 2) | RSRP_x + 17 |
| Lowest reported value (Cell 3) | RSRP_y – 8 |
| Highest reported value (Cell 3) | RSRP_y + 1 |
| RSRP_x is the reported value of Cell 1<br>RSRP_y is the reported value of Cell 2 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.20 FDD RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz bandwidth

### 9.1.20.1 FDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz bandwidth

#### 9.1.20.1.1 Test purpose

Same test purpose as in clause 9.1.6.1.1.

#### 9.1.20.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and release 11 supporting CA with Intra-band contiguous CA, or Inter-band CA.

This test applies to all types of E-UTRA FDD UE release 11 supporting CA with Intra-band non-contiguous CA.

#### 9.1.20.1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.6.1.3.

#### 9.1.20.1.4 Test description

##### 9.1.20.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 5 MHz for cells on both the PCC and the SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.20.1.4.3.

4. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier and Cell 3 is the neighbouring cell on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.1.20.1.4.2 Test procedure

Same test procedure as in clause 9.1.6.1.4.2 with the following exceptions:

- Instead of Table 9.1.6.1.5-2 → use Table 9.1.20.1.5-1.
- Instead of Table 9.1.6.1.5-3 → use Table 9.1.20.1.5-2.

#### 9.1.20.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.20.1.4.3-1: Common Exception messages for FDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation test requirement for 5MHz + 5MHz bandwidth**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-3<br>Table H.4.1-5 |

### 9.1.20.1.5 Test requirement

Table 9.1.20.1.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP FDD absolute accuracy test shall meet the reported values test requirements in table 9.1.20.1.5-2.

**Table 9.1.20.1.5-1: RSRP FDD absolute accuracy carrier aggregation test parameters for 5MHz + 5MHz bandwidth**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 5 | 5 | 5 |
| Measurement bandwidth | | $n_{PRB}$ | 10-15 | 10-15 | 10-15 |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | R.5 FDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 7-17 | 7-17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.11 FDD | R.11 FDD | R.11 FDD |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | | OP.15 FDD | OP.15 FDD | OP.16 FDD |
| Io Note2 | Bands FDD_A Note 5 | dBm/4.5 MHz | -90.76 | (Io for Channel 1 +5.33dB) | |
| | Bands FDD_C Note 5 | | -89.76 | | |
| | Bands FDD_D Note 5 | | -89.26 | | |
| | Bands FDD_E, FDD_F Note 5 | | -88.76 | | |
| | Bands FDD_G Note 5 | | -87.76 | | |
| | Bands FDD_H Note 5 | | -87.26 | | |

Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: See Table 9.1.6.1.5-1 for the other parameters.
Note 4: E-UTRA operating band groups are as defined in Section 3.5.1.
Note 5: The test applies for E-UTRA operating bands in this band group which are supporting 5MHz + 5MHz channel bandwidth.

**Table 9.1.20.1.5-2: RSRP FDD Carrier Aggregation absolute accuracy requirements for the reported values for 5MHz + 5MHz bandwidth**

| | Test 1 | | | | | |
|---|---|---|---|---|---|---|
| **Band of Cell 1 on Primary Component Carrier** | **Bands FDD_A** | **Bands FDD_C** | **Band FDD_D** | **Bands FDD_E, FDD_F** | **Bands FDD_G** | **Bands FDD_H** |
| Normal Conditions | | | | | | |
| Lowest reported value (Cell 1) | RSRP_12 | RSRP_13 | RSRP_13 | RSRP_14 | RSRP_15 | RSRP_15 |
| Highest reported value (Cell 1) | RSRP_27 | RSRP_28 | RSRP_29 | RSRP_29 | RSRP_30 | RSRP_31 |
| Lowest reported value (Cell 2) | RSRP_20 | RSRP_21 | RSRP_21 | RSRP_22 | RSRP_23 | RSRP_23 |
| Highest reported value (Cell 2) | RSRP_35 | RSRP_36 | RSRP_37 | RSRP_37 | RSRP_38 | RSRP_39 |
| Extreme Conditions | | | | | | |
| Lowest reported value (Cell 1) | RSRP_9 | RSRP_10 | RSRP_10 | RSRP_11 | RSRP_12 | RSRP_12 |
| Highest reported value (Cell 1) | RSRP_30 | RSRP_31 | RSRP_32 | RSRP_32 | RSRP_33 | RSRP_34 |
| Lowest reported value (Cell 2) | RSRP_17 | RSRP_18 | RSRP_18 | RSRP_19 | RSRP_20 | RSRP_20 |
| Highest reported value (Cell 2) | RSRP_38 | RSRP_39 | RSRP_40 | RSRP_40 | RSRP_41 | RSRP_42 |

Note: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2.

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.20.1_1 FDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz bandwidth (Rel-12 and forward)

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

-The test tolerances applicable to this test are undefined

### 9.1.20.1_1.1 Test purpose

Same test purpose as in clause 9.1.20.1.1.

### 9.1.20.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward supporting CA.

### 9.1.20.1_1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.6.1_1.3.

### 9.1.20.1_1.4 Test description

#### 9.1.20.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.20.1.4.1 with the following exceptions:

- Instead of 9.1.20.1.4.3 → use 9.1.20.1_1.4.3.

#### 9.1.20.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.20.1.4.2 with the following exceptions:

- Instead of Table 9.1.20.1.5-1 → use Table 9.1.20.1_1.5-1.
- Instead of Table 9.1.20.1.5-2 → use Table 9.1.20.1_1.5-2.

#### 9.1.20.1_1.4.3 Message contents

Same message contents as in clause 9.1.20.1.4.3.

### 9.1.20.1_1.5 Test requirement

Table 9.1.20.1_1.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP FDD absolute accuracy test shall meet the reported values test requirements in table 9.1.20.1_1.5-2.

**Table 9.1.20.1_1.5-1: RSRP FDD absolute accuracy carrier aggregation test parameters for 5MHz + 5MHz bandwidth**

| Parameter | Unit | Test 1 | | |
|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 3 |

| | | | | | |
|---|---|---|---|---|---|
| $BW_{channel}$ Note 1 | | MHz | 5 | 5 | 5 |
| Measurement bandwidth | | $n_{PRB}$ | 10-15 | 10-15 | 10-15 |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | R.5 FDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 7-17 | 7-17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.11 FDD | R.11 FDD | R.11 FDD |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | | OP.15 FDD | OP.15 FDD | OP.16 FDD |
| Io Note2 | Bands FDD_A Note 5 | dBm/4.5 MHz | -90.76 | (Io for Channel 1 +5.33dB) | |
| | Bands FDD_C Note 5 | | -89.76 | | |
| | Bands FDD_D Note 5 | | -89.26 | | |
| | Bands FDD_E, FDD_F Note 5 | | -88.76 | | |
| | Bands FDD_G Note 5 | | -87.76 | | |
| | Bands FDD_H Note 5 | | -87.26 | | |

Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: See Table 9.1.6.1.5-1 for the other parameters.
Note 4: E-UTRA operating band groups are as defined in Section 3.5.1.
Note 5: The test applies for E-UTRA operating bands in this band group which are supporting 5MHz + 5MHz channel bandwidth.

**Table 9.1.20.1_1.5-2: RSRP FDD Carrier Aggregation absolute accuracy requirements for the reported values for 5MHz + 5MHz bandwidth**

| | Test 1 | | | | | |
|---|---|---|---|---|---|---|
| **Band of Cell 1 on Primary Component Carrier** | **Bands FDD_A** | **Bands FDD_C** | **Band FDD_D** | **Bands FDD_E, FDD_F** | **Bands FDD_G** | **Bands FDD_H** |
| Normal Conditions | | | | | | |
| Lowest reported value (Cell 1) | TBD | TBD | TBD | TBD | TBD | TBD |
| Highest reported value (Cell 1) | TBD | TBD | TBD | TBD | TBD | TBD |
| Lowest reported value (Cell 2) | TBD | TBD | TBD | TBD | TBD | TBD |
| Highest reported value (Cell 2) | TBD | TBD | TBD | TBD | TBD | TBD |
| Extreme Conditions | | | | | | |
| Lowest reported value (Cell 1) | RSRP_9 | RSRP_10 | RSRP_10 | RSRP_11 | RSRP_12 | RSRP_12 |
| Highest reported value (Cell 1) | RSRP_30 | RSRP_31 | RSRP_32 | RSRP_32 | RSRP_33 | RSRP_34 |
| Lowest reported value (Cell 2) | RSRP_17 | RSRP_18 | RSRP_18 | RSRP_19 | RSRP_20 | RSRP_20 |
| Highest reported value (Cell 2) | RSRP_38 | RSRP_39 | RSRP_40 | RSRP_40 | RSRP_41 | RSRP_42 |

Note: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2.

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.20.2 FDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz bandwidth

### 9.1.20.2.1 Test purpose

Same test purpose as in clause 9.1.6.2.1.

### 9.1.20.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and release 11 supporting CA with Intra-band contiguous CA, or Inter-band CA.

This test applies to all types of E-UTRA FDD UE release 11 supporting CA with Intra-band non-contiguous CA.

### 9.1.20.2.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.6.2.3.

### 9.1.20.2.4 Test description

#### 9.1.20.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 5 MHz for cells on both the PCC and the SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.20.2.4.3.

4. Cell1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier and Cell 3 is the neighbouring cell on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.1.20.2.4.2 Test procedure

Same test procedure as in clause 9.1.6.2.4.2 with the following exceptions:

- Instead of Table 9.1.6.2.5-2 → use Table 9.1.20.2.5-1.
- Instead of Table 9.1.6.2.5-3 → use Table 9.1.20.2.5-2.

#### 9.1.20.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.20.2.4.3-1: Common Exception messages for FDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation test requirement for 5MHz + 5MHz bandwidth**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-3<br>Table H.4.1-5 |

## 9.1.20.2.5 Test requirement

Table 9.1.20.2.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP FDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.20.2.5-2.

**Table 9.1.20.2.5-1: RSRP FDD relative accuracy carrier aggregation test parameters for 5MHz + 5MHz bandwidth**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 5 | 5 | 5 |
| Measurement bandwidth | | $n_{PRB}$ | 10-15 | 10-15 | 10-15 |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | R.5 FDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 7-17 | 7-17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.11 FDD | R.11 FDD | R.11 FDD |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | | OP.15 FDD | OP.15 FDD | OP.16 FDD |
| Io Note2 | Bands FDD_A Note 5 | dBm/4.5 MHz | -90.76 | (Io for Channel 1 +5.51dB) | |
| | Bands FDD_C Note 5 | | -89.76 | | |
| | Bands FDD_D Note 5 | | -89.26 | | |
| | Bands FDD_E, FDD_F Note 5 | | -88.76 | | |
| | Bands FDD_G Note 5 | | -87.76 | | |
| | Bands FDD_H Note 5 | | -87.26 | | |

Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: See Table 9.1.6.2.5-1 for the other parameters.
Note 4: E-UTRA operating band groups are as defined in Section 3.5.1.
Note 5: The test applies for E-UTRA operating bands in this band group which are supporting 5MHz + 5MHz channel bandwidth.

**Table 9.1.20.2.5-2: RSRP FDD Carrier Aggregation relative accuracy requirements for the reported values for 5MHz + 5MHz bandwidth**

| | **Test 1** |
|---|---|
| | All bands |
| Normal and Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRP_x – 1 |
| Highest reported value (Cell 2) | RSRP_x + 17 |
| Lowest reported value (Cell 3) | RSRP_y – 8 |
| Highest reported value (Cell 3) | RSRP_y + 1 |
| RSRP_x is the reported value of Cell 1<br>RSRP_y is the reported value of Cell 2 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.20.2_1 FDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz bandwidth (Rel-12 and forward)

Editor's note: This Test case is incomplete. The following aspects are either missing or not yet determined:

- *The Test system uncertainties applicable to this test are undefined*
- *The Test tolerances applicable to this test are undefined*
- *Test requirements in normal conditions need to be defined*

### 9.1.20.2_1.1 Test purpose

Same test purpose as in clause 9.1.20.2.1.

### 9.1.20.2_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward supporting CA.

### 9.1.20.2_1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.6.2_1.3.

### 9.1.20.2_1.4 Test description

#### 9.1.20.2_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.20.2.4.1.

#### 9.1.20.2_1.4.2 Test procedure

Same test procedure as in clause 9.1.20.2.4.2 with the following exceptions:

- Instead of Table 9.1.20.2.5-1 → use Table 9.1.20.2_1.5-1.
- Instead of Table 9.1.20.2.5-2 → use Table 9.1.20.2_1.5-2.

#### 9.1.20.2_1.4.3 Message contents

Same message contents as in clause 9.1.20.2.4.2.

### 9.1.20.2_1.5 Test requirement

Table 9.1.20.2_1.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP FDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.20.2_1.5-2.

**Table 9.1.20.2_1.5-1: RSRP FDD relative accuracy carrier aggregation test parameters for 5MHz + 5MHz bandwidth**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 5 | 5 | 5 |
| Measurement bandwidth | | $n_{PRB}$ | 10-15 | 10-15 | 10-15 |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | R.5 FDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 7-17 | 7-17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.11 FDD | R.11 FDD | R.11 FDD |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | | OP.15 FDD | OP.15 FDD | OP.16 FDD |
| $Io^{Note2}$ | Bands FDD_A Note 5 | dBm/4.5 MHz | -90.76+TT | (Io for Channel 1 +5.51dB) | |
| | Bands FDD_C Note 5 | | -89.76+TT | | |
| | Bands FDD_D Note 5 | | -89.26+TT | | |
| | Bands FDD_E, FDD_F Note 5 | | -88.76+TT | | |
| | Bands FDD_G Note 5 | | -87.76+TT | | |
| | Bands FDD_H Note 5 | | -87.26+TT | | |

Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: See Table 9.1.6.2.5-1 for the other parameters.
Note 4: E-UTRA operating band groups are as defined in Section 3.5.1.
Note 5: The test applies for E-UTRA operating bands in this band group which are supporting 5MHz + 5MHz channel bandwidth.

**Table 9.1.20.2_1.5-2: RSRP FDD Carrier Aggregation relative accuracy requirements for the reported values for 5MHz + 5MHz bandwidth**

| | Test 1 |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 2) | TBD |
| Highest reported value (Cell 2) | TBD |
| Lowest reported value (Cell 3) | TBD |
| Highest reported value (Cell 3) | TBD |
| Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRP_x – 1 |
| Highest reported value (Cell 2) | RSRP_x + 17 |
| Lowest reported value (Cell 3) | RSRP_y – 8 |
| Highest reported value (Cell 3) | RSRP_y + 1 |
| RSRP_x is the reported value of Cell 1<br>RSRP_y is the reported value of Cell 2 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.1.21 TDD RSRP Accuracy for E-UTRAN Carrier Aggregation for 5MHz + 5MHz

## 9.1.21.1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz

### 9.1.21.1.1 Test purpose

Same test purpose as in clause 9.1.7.1.1.

### 9.1.21.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and release 11 supporting CA with Intra-band contiguous CA, or Inter-band CA.

This test applies to all types of E-UTRA TDD UE release 11 supporting CA with Intra-band non-contiguous CA.

### 9.1.21.1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.7.1.2.

### 9.1.21.1.4 Test description

#### 9.1.21.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 5 MHz for cells on both the PCC and the SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.21.1.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.1.21.1.4.2 Test procedure

Same test procedure as in clause 9.1.7.1.4.2 with the following exceptions:

- Instead of Table 9.1.7.1.5-1 → use Table 9.1.21.1.5-1.
- Instead of Table 9.1.7.1.5-2 → use Table 9.1.21.1.5-2.

#### 9.1.21.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.21.1.4.3-1: Common Exception messages for TDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation test requirement for 5MHz + 5MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-3<br>Table H.4.1-5 |

## 9.1.21.1.5 Test requirement

Table 9.1.21.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD absolute accuracy test shall meet the reported values test requirements in table 9.1.21.1.5-2.

**Table 9.1.21.1.5-1: RSRP TDD absolute accuracy carrier aggregation test parameters for 5MHz + 5MHz**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 5 | 5 | 5 |
| Measurement bandwidth | | $n_{PRB}$ | 10-15 | 10-15 | 10-15 |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.4 TDD | R.4 TDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 7-17 | 7-17 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.11 TDD | | |
| OCNG Patterns defined in D.2.9 (OP.9 TDD) and D.2.10 (OP.10 TDD) | | | OP.9 TDD | OP.9 TDD | OP.10 TDD |
| Io Note2 | Bands TDD_A Note 5 | dBm/4.5MHz | -90.76 | (Io for Channel 1 +5.33dB) | |
| | Bands TDD_C Note 5 | | -89.76 | | |
| | Bands TDD_E Note 5 | | -88.76 | | |

Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: See Table 9.1.7.1.5-1 for the other parameters.
Note 4: E-UTRA operating band groups are as defined in Section 3.5.1.
Note 5: The test applies for E-UTRA operating bands in this band group which are supporting 5MHz + 5MHz channel bandwidth.

**Table 9.1.21.1.5-2: RSRP TDD Carrier Aggregation absolute accuracy requirements for the reported values for 5MHz + 5MHz**

| Parameter | Test 1 | | |
|---|---|---|---|
| **Band of Cell 1 on Primary Component Carrier** | **Bands TDD_A** | **Bands TDD_C** | **Band TDD_E** |
| Normal Conditions | | | |
| Lowest reported value (Cell 1) | RSRP_12 | RSRP_13 | RSRP_14 |
| Highest reported value (Cell 1) | RSRP_27 | RSRP_28 | RSRP_29 |
| Lowest reported value (Cell 2) | RSRP_20 | RSRP_21 | RSRP_22 |
| Highest reported value (Cell 2) | RSRP_35 | RSRP_36 | RSRP_37 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 1) | RSRP_9 | RSRP_10 | RSRP_11 |
| Highest reported value (Cell 1) | RSRP_30 | RSRP_31 | RSRP_32 |
| Lowest reported value (Cell 2) | RSRP_17 | RSRP_18 | RSRP_19 |
| Highest reported value (Cell 2) | RSRP_38 | RSRP_39 | RSRP_40 |
| Note: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2. | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.21.1_1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz (Rel-12 and forward)

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

-The Test tolerances applicable to this test are undefined

### 9.1.21.1_1.1 Test purpose

Same test purpose as in clause 9.1.7.1_1.1.

### 9.1.21.1_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and release 11 supporting CA.

### 9.1.21.1_1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.7.1_1.2

### 9.1.21.1_1.4 Test description

#### 9.1.21.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.21.1.4.1 with the following exceptions:

- Instead of 9.1.21.1.4.3 → use 9.1.21.1_1.4.3.

#### 9.1.21.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.7.1_1.4.2 with the following exceptions:

- Instead of Table 9.1.7.1_1.5-1 → use Table 9.1.21.1_1.5-1.
- Instead of Table 9.1.7.1_1.5-2 → use Table 9.1.21.1_1.5-2.

9.1.21.1_1.4.3 Message contents

Same message contents as in clause 9.1.19.1.4.3

## 9.1.21.1_1.5 Test requirement

Table 9.1.21.1_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD absolute accuracy test shall meet the reported values test requirements in table 9.1.21.1_1.5-2.

**Table 9.1.21.1_1.5-1: RSRP TDD absolute accuracy carrier aggregation test parameters for 5MHz + 5MHz**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 5 | 5 | 5 |
| Measurement bandwidth | | $n_{PRB}$ | 10-15 | 10-15 | 10-15 |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.4 TDD | R.4 TDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 7-17 | 7-17 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.11 TDD | | |
| OCNG Patterns defined in D.2.9 (OP.9 TDD) and D.2.10 (OP.10 TDD) | | | OP.9 TDD | OP.9 TDD | OP.10 TDD |
| Io Note2 | Bands TDD_A Note 5 | dBm/4.5MHz | -90.76+TT | (Io for Channel 1 +5.33dB) | |
| | Bands TDD_C Note 5 | | -89.76+TT | | |
| | Bands TDD_E Note 5 | | -88.76+TT | | |

NOTE 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
NOTE 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 3: See Table 9.1.7.1.5-1 for the other parameters.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.1.
NOTE 5: The test applies for E-UTRA operating bands in this band group which are supporting 5MHz + 5MHz channel bandwidth.

**Table 9.1.21.1_1.5-2: RSRP TDD Carrier Aggregation absolute accuracy requirements for the reported values for 5MHz + 5MHz**

| Parameter | Test 1 | | |
|---|---|---|---|
| **Band of Cell 1 on Primary Component Carrier** | **Bands TDD_A** | **Bands TDD_C** | **Band TDD_E** |
| Normal Conditions | | | |
| Lowest reported value (Cell 1) | RSRP_xx | RSRP_xx | RSRP_xx |
| Highest reported value (Cell 1) | RSRP_xx | RSRP_xx | RSRP_xx |
| Lowest reported value (Cell 2) | RSRP_xx | RSRP_xx | RSRP_xx |
| Highest reported value (Cell 2) | RSRP_xx | RSRP_xx | RSRP_xx |
| Extreme Conditions | | | |
| Lowest reported value (Cell 1) | RSRP_xx | RSRP_xx | RSRP_xx |
| Highest reported value (Cell 1) | RSRP_xx | RSRP_xx | RSRP_xx |
| Lowest reported value (Cell 2) | RSRP_xx | RSRP_xx | RSRP_xx |
| Highest reported value (Cell 2) | RSRP_xx | RSRP_xx | RSRP_xx |

NOTE: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2.

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.21.2 TDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz

### 9.1.21.2.1 Test purpose

Same test purpose as in clause 9.1.7.2.1.

### 9.1.21.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward supporting CA with Intra-band contiguous CA, or Inter-band CA.

This test applies to all types of E-UTRA TDD UE release 11 supporting CA with Intra-band non-contiguous CA.

### 9.1.21.2.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.7.2.3.

### 9.1.21.2.4 Test description

#### 9.1.21.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 5 MHz for cells on both the PCC and the SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.21.2.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.1.21.2.4.2 Test procedure

Same test procedure as in clause 9.1.7.2.4.2 with the following exceptions:

- Instead of Table 9.1.7.2.5-1 → use Table 9.1.21.2.5-1.
- Instead of Table 9.1.7.2.5-2 → use Table 9.1.21.2.5-2.

#### 9.1.21.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.21.2.4.3-1: Common Exception messages for TDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation test requirement for 5MHz + 5MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-3<br>Table H.4.1-5 |

## 9.1.21.2.5 Test requirement

Table 9.1.21.2.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.21.2.5-2.

**Table 9.1.21.2.5-1: RSRP TDD relative accuracy carrier aggregation test parameters for 5MHz + 5MHz**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 5 | 5 | 5 |
| Measurement bandwidth | | $n_{PRB}$ | 10-15 | 10-15 | 10-15 |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.4 TDD | R.4 TDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 7-17 | 7-17 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.11 TDD | | |
| OCNG Patterns defined in D.2.9 (OP.9 TDD) and D.2.10 (OP.10 TDD) | | | OP.9 TDD | OP.9 TDD | OP.10 TDD |
| Io Note2 | Bands TDD_A Note 5 | dBm/4.5MHz | -90.76 | (Io for Channel 1 +5.51dB) | |
| | Bands TDD_C Note 5 | | -89.76 | | |
| | Bands TDD_E Note 5 | | -88.76 | | |

Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: See Table 9.1.7.1.5-1 for the other parameters.
Note 4: E-UTRA operating band groups are as defined in Section 3.5.1.
Note 5: The test applies for E-UTRA operating bands in this band group which are supporting 5MHz + 5MHz channel bandwidth.

**Table 9.1.21.2.5-2: RSRP TDD Carrier Aggregation relative accuracy requirements for the reported values for 5MHz + 5MHz**

| | Test 1 |
|---|---|
| | All bands |
| Normal and Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRP_x – 1 |
| Highest reported value (Cell 2) | RSRP_x + 17 |
| Lowest reported value (Cell 3) | RSRP_y – 8 |
| Highest reported value (Cell 3) | RSRP_y + 1 |
| RSRP_x is the reported value of Cell 1<br>RSRP_y is the reported value of Cell 2 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.21.2_1 TDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz (Rel-12 and forward)

Editor's note: This Test case is incomplete. The following aspects are either missing or not yet determined:

- *The Test system uncertainties applicable to this test are undefined*
- *The Test tolerances applicable to this test are undefined*
- *Test requirements in normal conditions need to be defined*

### 9.1.21.2_1.1 Test purpose

Same test purpose as in clause 9.1.21.2.1.

### 9.1.21.2_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward supporting CA.

### 9.1.21.2_1.3 Minimum conformance requirements

Same minimum conformance requirements as defined in clause 9.1.7.2_1.3.

### 9.1.21.2_1.4 Test description

#### 9.1.21.2_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.21.2.4.1.

#### 9.1.21.2_1.4.2 Test procedure

Same test procedure as in clause 9.1.21.2.4.2 with the following exceptions:

- Instead of Table 9.1.21.2.5-1 → use Table 9.1.21.2_1.5-1.
- Instead of Table 9.1.21.2.5-2 → use Table 9.1.21.2_1.5-2.

#### 9.1.21.2_1.4.3 Message contents

Same message contents as in clause 9.1.21.2.4.3.

### 9.1.21.2_1.5 Test requirement

Table 9.1.21.2_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.21.2_1.5-2.

**Table 9.1.21.2_1.5-1: RSRP TDD relative accuracy carrier aggregation test parameters for 5MHz + 5MHz**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 5 | 5 | 5 |
| Measurement bandwidth | | $n_{PRB}$ | 10-15 | 10-15 | 10-15 |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.4 TDD | R.4 TDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 7-17 | 7-17 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.11 TDD | | |
| OCNG Patterns defined in D.2.9 (OP.9 TDD) and D.2.10 (OP.10 TDD) | | | OP.9 TDD | OP.9 TDD | OP.10 TDD |
| Io Note2 | Bands TDD_A Note 5 | dBm/4.5MHz | -90.76+TT | (Io for Channel 1 +5.51dB) | |
| | Bands TDD_C Note 5 | | -89.76+TT | | |
| | Bands TDD_E Note 5 | | -88.76+TT | | |

Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: See Table 9.1.7.1.5-1 for the other parameters.
Note 4: E-UTRA operating band groups are as defined in Section 3.5.1.
Note 5: The test applies for E-UTRA operating bands in this band group which are supporting 5MHz + 5MHz channel bandwidth.

**Table 9.1.21.2_1.5-2: RSRP TDD Carrier Aggregation relative accuracy requirements for the reported values for 5MHz + 5MHz**

| | Test 1 |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 2) | TBD |
| Highest reported value (Cell 2) | TBD |
| Lowest reported value (Cell 3) | TBD |
| Highest reported value (Cell 3) | TBD |
| Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRP_x – 1 |
| Highest reported value (Cell 2) | RSRP_x + 17 |
| Lowest reported value (Cell 3) | RSRP_y – 8 |
| Highest reported value (Cell 3) | RSRP_y + 1 |
| RSRP_x is the reported value of Cell 1<br>RSRP_y is the reported value of Cell 2 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.22 FDD-TDD RSRP Accuracy E-UTRA for Carrier Aggregation with PCell in FDD

*Editor's note:*

*- TS36.521-2 need to be updated to be consistent with the test applicability.*

### 9.1.22.1 Test purpose

To verify that FDD and TDD absolute and relative RSRP measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the FDD and TDD absolute RSRP accuracy requirements of cells on the primary component carrier and the secondary component carrier, and the relative RSRP

accuracy requirements of the secondary component carriers.This test will also verify the relative RSRP accuracy requirements of primary and secondary component carriers.

### 9.1.22.2 Test applicability

This test applies to all types of E-UTRA FDD and TDD Release 12 and forward UE that support E-UTRA FDD and TDD and 2DL TDD-FDD CA with FDD as PCell.

### 9.1.22.3 Minimum conformance requirements

The FDD RSRP measurements of cells on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.11.1. The TDD RSRP measurements of cells on the secondary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1.

The accuracy requirements in table 9.1.22.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.1.22.3-1: RSRP FDD Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_B | -120.5 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.22.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.22.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The FDD-TDD RSRP relative measurements of cells on PCC and SCC shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.3.2. The TDD RSRP relative measurements of cells on the SCC shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.2.2.

The accuracy requirements in Table 9.1.22.3-3 for PCC-SCC relative accuracy requirements are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

RSRP1,2$|_{dBm}$ according to Annex I.3.4 for a corresponding Band

$$\left| RSRP1|_{dBm} - RSRP2|_{dBm} \right| \leq 27dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.22.3-3: FDD RSRP relative accuracy for PCC and SCC**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot Note 2 | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/BW$_{Channel}$ |
| ±4.5 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The requirements in Table 9.1.22.3-4 for SCC relative accuracy are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

RSRP1,2$|_{dBm}$ according to Annex I.3.8 for a corresponding Band.

**Table 9.1.22.3-4: FDD RSRP relative accuracy for SCCs**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/$BW_{Channel}$ |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1, clause 9.1.11.2, clause 9.1.11.3, clause 9.1.4, and A.9.1.22.

## 9.1.22.4 Test description

### 9.1.22.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: Any one of the supported TDD-FDD CA configurations with PCell in FDD with the largest aggregated CA bandwidth combination as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.1.22.4.3.
4. Cell 1 is PCell on the FDD primary component carrier, Cell 2 is SCell on the TDD secondary component carrier and Cell 3 is the neighbouring cell on the TDD secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

### 9.1.22.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCell according to Annex C.0, C.1 [and C.2] for all downlink physical channels.
3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.
4. SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).

5. Set the parameters according to Table 9.1.22.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.

6. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10 s wait from step 6, SS shall check the reported RSRP values in periodical measurement report messages. The reported RSRP values for Cell 1 and Cell 2 are compared to the actual RSRP values according to Table 9.1.22.5-3. This counts respectively as a Pass or Fail for the events “Cell 1” and “Cell 2”.

   The reported RSRP value for Cell 2 is compared to the reported RSRP value for Cell 1 for each MeasurementReport message according to Table 9.1.22.5-3. Also the reported RSRP value for Cell 3 is compared to the reported RSRP value for Cell 2 for each MeasurementReport message according to Table 9.1.22.5-3. This counts respectively as a Pass or Fail for the events “Cell 1-2” and “Cell 2-3”.

10. SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events “Cell 1”, “Cell 2”, “Cell 1-2”, “Cell 2-3” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all the events (“Cell 1”, “Cell 2”, “Cell 1-2”, “Cell 2-3”) pass ,the test passes. If one event fails, the test fails.

### 9.1.22.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.22.4.3-1: Common Exception messages for FDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-3<br>Table H.4.1-5<br>Table H.4.2-1 |

## 9.1.22.5 Test requirement

Table 9.1.22.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP FDD absolute accuracy test shall meet the reported values test requirements in table 9.1.22.5-2.

**Table 9.1.22.5-1: RSRP TDD-FDD absolute accuracy carrier aggregation test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell3 |
| E-UTRA RF Channel Number | | | 1 | 2 | 2 |
| $BW_{channel}$ | | | 5MHz: $N_{RB,c}$ = 25 10MHz: $N_{RB,c}$ = 50 20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25 10MHz: $N_{RB,c}$ = 50 20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25 10MHz: $N_{RB,c}$ = 50 20MHz: $N_{RB,c}$ = 100 |
| Special subframe configuration Note9 | | | - | 6 | 6 |
| Uplink-downlink configuration Note9 | | | - | 1 | 1 |
| Measurement bandwidth | | $n_{PRB}$ | 5MHz: 10-15 10MHz: 22-27 20MHz: 47-52 | 5MHz: 10-15 10MHz: 22-27 20MHz: 47-52 | 5MHz: 10-15 10MHz: 22-27 20MHz: 47-52 |
| PDSCH Reference measurement channel defined in A.1.1 and A.1.2 | | | 5MHz: R.5 FDD 10MHz: R.0 FDD 20MHz: R.4 FDD | 5MHz: R.4 TDD 10MHz: R.0 TDD 20MHz: R.3 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 5MHz: 7-17 10MHz: 13-36 20MHz: 38-61 | 5MHz: 7-17 10MHz: 13-36 20MHz: 38-61 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 and A.2.2 | | | 5MHz: R.11 FDD 10MHz: R.6 FDD 20MHz: R.10 FDD | 5MHz: R.11 TDD 10MHz: R.6 TDD 20MHz: R.10 TDD | 5MHz: R.11 TDD 10MHz: R.6 TDD 20MHz: R.10 TDD |
| OCNG Patterns defined in D.1.1 (OP.1 FDD), D.2.1 (OP.1 TDD), and D.2.2 (OP.2 TDD) | | | 5MHz: OP.15 FDD 10MHz: OP.1 FDD 20MHz: OP.11 FDD | 5MHz: OP.9 TDD 10MHz: OP.1 TDD 20MHz: OP.7 TDD | 5MHz: OP.10 TDD 10MHz: OP.2 TDD 20MHz: OP.8 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA Note 2 | | | | | |
| OCNG_RB Note 2 | | | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -117 | - | |
| | Bands FDD_B | | -116.5 | - | |
| | Bands FDD_C | | -116 | - | |
| | Bands FDD_D | | -115.5 | - | |
| | Bands FDD_E, FDD_F Note 6 | | -115 | - | |
| | Bands FDD_G | | -114 | - | |

| | | | | | |
|---|---|---|---|---|---|
| | Bands FDD_H | | -113.5 | - | |
| | Bands TDD_A | | - | ( $N_{oc}$ for Channel 1 +1dB) | |
| | Bands TDD_C | | - | | |
| | Bands TDD_E | | - | | |
| $\hat{E}_s / N_{oc}$ | | dB | -4 | 3 | -1 |
| $\hat{E}_s / I_{ot}$ | | dB | -4 | 0.46 | -5.76 |
| RSRP Note3 | Bands FDD_A | dBm/15 kHz | -121 | - | - |
| | Bands FDD_B | | -120.5 | - | - |
| | Bands FDD_C | | -120 | - | - |
| | Bands FDD_D | | -119.5 | - | - |
| | Bands FDD_E, FDD_F Note 6 | | -119 | - | - |
| | Bands FDD_G | | -118 | - | - |
| | Bands FDD_H | | -117.5 | - | - |
| | Bands TDD_A | | - | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4dB) |
| | Bands TDD_C | | - | | |
| | Bands TDD_E | | - | | |
| Io Note3 | Bands FDD_A | dBm/ $BW_{channel}$ | -87.76 + 10log($N_{RB, c}$/50) | - | |
| | Bands FDD_B | | -87.26 + 10log($N_{RB, c}$/50) | - | |
| | Bands FDD_C | | -86.76+ 10log($N_{RB, c}$/50) | - | |
| | Bands FDD_D | | -86.26+ 10log($N_{RB, c}$/50) | - | |
| | Bands FDD_E, FDD_F Note 6 | | -85.76+ 10log($N_{RB, c}$/50) | - | |
| | Bands FDD_G | | -84.76+ 10log($N_{RB, c}$/50) | - | |
| | Bands FDD_H | | -84.26+ 10log($N_{RB, c}$/50) | - | |
| | Bands TDD_A | | - | (Io for Channel 1 +5.33dB +10log ($N_{RB\ channel2}$ / $N_{RB\ channel\ 1}$)) | |
| | Bands TDD_C | | - | | |
| | Bands TDD_E | | - | | |
| Propagation condition | | - | AWGN | AWGN | AWGN |
| Antenna Configuration | | - | 1x2 | 1x2 | 1x2 |
| Timing offset to cell 1 | | µs | - | 0 | 3 |
| Time alignment error relative to cell 1 Note 8 | | - | - | ≤ TAE | - |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for to be fulfilled.
Note 4: Es/Iot, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 6: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.
Note 7: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 8: Time alignment error (TAE) as specified in TS 36.104 [30] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.
Note 9: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.22.5-2: 2DL PCELL in FDD RSRP for EUTRA Carrier Aggregation absolute accuracy requirements for the reported values**

| Band of Cell 1 on Primary Component Carrier | Event | Bands FDD_A | Bands FDD_B | Bands FDD_C | Bands FDD_D | Bands FDD_E, FDD_F | Bands FDD_G | Bands FDD_H |
|---|---|---|---|---|---|---|---|---|
| Normal Conditions | | | | | | | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRP_13 | RSRP_14 | RSRP_14 | RSRP_15 | RSRP_15 | RSRP_16 | RSRP_17 |
| Highest reported value (Cell 1) | | RSRP_26 | RSRP_26 | RSRP_27 | RSRP_27 | RSRP_28 | RSRP_29 | RSRP_29 |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_21 | RSRP_22 | RSRP_22 | RSRP_23 | RSRP_23 | RSRP_24 | RSRP_25 |
| Highest reported value (Cell 2) | | RSRP_34 | RSRP_34 | RSRP_35 | RSRP_35 | RSRP_36 | RSRP_37 | RSRP_37 |
| Extreme Conditions | | | | | | | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRP_9 | RSRP_9 | RSRP_10 | RSRP_10 | RSRP_11 | RSRP_12 | RSRP_12 |
| Highest reported value (Cell 1) | | RSRP_30 | RSRP_31 | RSRP_31 | RSRP_32 | RSRP_32 | RSRP_33 | RSRP_34 |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_17 | RSRP_17 | RSRP_18 | RSRP_18 | RSRP_19 | RSRP_20 | RSRP_20 |
| Highest reported value (Cell 2) | | RSRP_38 | RSRP_39 | RSRP_39 | RSRP_40 | RSRP_40 | RSRP_41 | RSRP_42 |
| Note: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2. | | | | | | | | |

**Table 9.1.22.5-3: 2DL PCELL in FDD RSRP for EUTRA FDD Carrier Aggregation relative accuracy requirements for the reported values**

| | Event | |
|---|---|---|
| Normal Conditions, all bands | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 + 2 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 15 |
| Extreme Conditions, all bands | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 – 1 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 17 |
| Normal and Extreme Conditions, all bands | | |
| Lowest reported value (Cell 3) | Cell 2-3 | RSRP_Cell 2 – 8 |
| Highest reported value (Cell 3) | | RSRP_Cell 2 + 1 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.23 FDD-TDD RSRP Accuracy E-UTRA for Carrier Aggregation with PCell in TDD

*Editor's note:*

*- TS36.521-2 need to be updated to be consistent with the test applicability.*

### 9.1.23.1 Test purpose

To verify that FDD and TDD absolute RSRP measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the TDD and FDD absolute RSRP accuracy requirements of cells on the primary component carrier and the secondary component carrier, and the relative RSRP accuracy requirements of the secondary component carriers. This test will also verify the relative RSRP accuracy requirements of primary and secondary component carriers.

### 9.1.23.2 Test applicability

This test applies to all types of E-UTRA FDD and TDD Release 12 and forward UE that support E-UTRA FDD and TDD and 2DL TDD-FDD CA with TDD as PCell.

### 9.1.23.3 Minimum conformance requirements

The TDD RSRP measurements of cells on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.11.1. The FDD RSRP measurements of cells on the secondary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1.

The accuracy requirements in table 9.1.23.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.1.23.3-1: RSRP FDD Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | $dBm/BW_{Channel}$ | $dBm/BW_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_B | -120.5 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.23.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.23.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The FDD RSRP relative measurements of cells on PCC and SCC shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.3.2. The FDD RSRP relative measurements of cells on the SCC shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.2.2.

The accuracy requirements in Table 9.1.23.3-3 for PCC-SCC relative accuracy requirements are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

RSRP1,2|$_{dBm}$ according to Annex I.3.4 for a corresponding Band

$$\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.23.3-3: FDD RSRP relative accuracy for PCC and SCC**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/BW$_{Channel}$ |
| ±4.5 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The requirements in Table 9.1.23.3-4 for SCC relative accuracy are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

RSRP1,2|$_{dBm}$ according to Annex I.3.8 for a corresponding Band.

**Table 9.1.23.3-4: FDD RSRP relative accuracy for SCCs**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/BW$_{Channel}$ |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 3 | Note 3 | Note 3 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.<br>NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 5: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1, clause 9.1.11.2, clause 9.1.4, 9.1.11.3 and A.9.1.23.

## 9.1.23.4 Test description

### 9.1.23.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: Any one of the supported TDD-FDD CA configurations with PCell in FDD with the largest aggregated CA bandwidth combination as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.1.23.4.3.
4. Cell 1 is PCell on the TDD primary component carrier, Cell 2 is SCell on the FDD secondary component carrier and Cell 3 is the neighbouring cell on the FDD secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

### 9.1.23.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCell according to Annex C.0, C.1 [and C.2] for all downlink physical channels.
3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.
4. SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).
5. Set the parameters according to Table 9.1.23.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.
6. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
7. The UE shall transmit RRCConnectionReconfigurationComplete message.
8. UE shall transmit periodically MeasurementReport messages.
9. After 10 s wait from step 6, SS shall check the reported RSRP values in periodical measurement report messages.

   The reported RSRP values for Cell 1 and Cell 2 are compared to the actual RSRP values according to Table 9.1.23.5-3. This counts respectively as a Pass or Fail for the events “Cell 1” and “Cell 2”.

   The reported RSRP value for Cell 2 is compared to the reported RSRP value for Cell 1 for each MeasurementReport message according to Table 9.1.23.5-3. Also the reported RSRP value for Cell 3 is compared to the reported RSRP value for Cell 2 for each MeasurementReport message according to Table 9.1.23.5-3. This counts respectively as a Pass or Fail for the events “Cell 1-2” and “Cell 2-3”.
10. SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events “Cell 1”, “Cell 2”, “Cell 1-2” and “Cell 2-3” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in

Annex G.2 is achieved. Different events may require different times for a verdict.
If all events (“Cell 1”, “Cell 2”, “Cell 1-2” and “Cell 2-3”) pass, the test passes. If one event fails, the test fails.

#### 9.1.23.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.23.4.3-1: Common Exception messages for FDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-3<br>Table H.4.1-5<br>Table H.4.2-1 |

### 9.1.23.5 Test requirement

Table 9.1.23.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP FDD absolute accuracy test shall meet the reported values test requirements in table 9.1.23.5-2.

**Table 9.1.23.5-1: RSRP TDD-FDD absolute accuracy carrier aggregation test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell3 |
| E-UTRA RF Channel Number | | | 1 | 2 | 2 |
| $BW_{channel}$ | | | 5MHz: $N_{RB,c}$ = 25 10MHz: $N_{RB,c}$ = 50 20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25 10MHz: $N_{RB,c}$ = 50 20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25 10MHz: $N_{RB,c}$ = 50 20MHz: $N_{RB,c}$ = 100 |
| Special subframe configuration[Note1] | | | 6 | - | - |
| Uplink-downlink configuration[Note1] | | | 1 | - | - |
| Measurement bandwidth | | $n_{PRB}$ | 5MHz: 10-15 10MHz: 22-27 20MHz: 47-52 | 5MHz: 10-15 10MHz: 22-27 20MHz: 47-52 | 5MHz: 10-15 10MHz: 22-27 20MHz: 47-52 |
| PDSCH Reference measurement channel defined in A.1.1 and A.1.2 | | | 5MHz: R.4 TDD 10MHz: R.0 TDD 20MHz: R.3 TDD | 5MHz: R.5 FDD 10MHz: R.0 FDD 20MHz: R.4 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 5MHz: 7-17 10MHz: 13-36 20MHz: 38-61 | 5MHz: 7-17 10MHz: 13-36 20MHz: 38-61 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 and A.2.2 | | | 5MHz: R.11 TDD 10MHz: R.6 TDD 20MHz: R.10 TDD | 5MHz: R.11 FDD 10MHz: R.6 FDD 20MHz: R.10 FDD | 5MHz: R.11 FDD 10MHz: R.6 FDD 20MHz: R.10 FDD |
| OCNG Patterns defined in D.1.1 (OP.1 FDD), D.2.1 (OP.1 TDD), and D.2.2 (OP.2 FDD) | | | 5MHz: OP.9 TDD 10MHz: OP.1 TDD 20MHz: OP.7 TDD | 5MHz: OP.15 FDD 10MHz: OP.1 FDD 20MHz: OP.11 FDD | 5MHz: OP.16 FDD 10MHz: OP.2 FDD 20MHz: OP.12 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA [Note 2] | | | | | |
| OCNG_RB [Note 2] | | | | | |
| $N_{oc}$ [Note 3] | Bands FDD_A | dBm/15 kHz | - | ( $N_{oc}$ for Channel 1 +1dB) | |
| | Bands FDD_B | | - | | |
| | Bands FDD_C | | - | | |
| | Bands FDD_D | | - | | |
| | Bands FDD_E, FDD_F [Note 9] | | - | | |
| | Bands FDD_G | | - | | |

| | | | | | |
|---|---|---|---|---|---|
| | Bands FDD_H | | - | | |
| | Bands TDD_A | | -117 | - | |
| | Bands TDD_C | | -116 | - | |
| | Bands TDD_E | | -115 | - | |
| $\hat{E}_s / N_{oc}$ | | dB | -4 | 3 | -1 |
| $\hat{E}_s / I_{ot}$ | | dB | -4 | 0.46 | -5.76 |
| RSRP Note 4 | Bands FDD_A | dBm/15 kHz | - | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4dB) |
| | Bands FDD_B | | - | | |
| | Bands FDD_C | | - | | |
| | Bands FDD_D | | - | | |
| | Bands FDD_E, FDD_F Note 9 | | - | | |
| | Bands FDD_G | | - | | |
| | Bands FDD_H | | - | | |
| | Bands TDD_A | | -121 | - | - |
| | Bands TDD_C | | -120 | | |
| | Bands TDD_E | | -119 | | |
| Io Note 4 | Bands FDD_A | dBm/ $BW_{channel}$ | - | (Io for Channel 1 +5.33dB+10log ($N_{RB\ channel2}$ / $N_{RB\ channel\ 1}$)) | |
| | Bands FDD_B | | - | | |
| | Bands FDD_C | | - | | |
| | Bands FDD_D | | - | | |
| | Bands FDD_E, FDD_F Note 9 | | - | | |
| | Bands FDD_G | | - | | |
| | Bands FDD_H | | - | | |
| | Bands TDD_A | | -87.76 + 10log($N_{RB,c}$/50) | - | |
| | Bands TDD_C | | -86.76 + 10log($N_{RB,c}$/50) | - | |
| | Bands TDD_E | | -85.76 + 10log($N_{RB,c}$/50) | - | |
| Propagation condition | | - | AWGN | AWGN | AWGN |
| Antenna Configuration | | - | 1x2 | 1x2 | 1x2 |
| Timing offset to cell 1 | | µs | - | 0 | 3 |
| Time alignment error relative to cell 1 Note 8 | | - | - | ≤ TAE | - |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Es/Iot, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 6: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.
Note 7: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 8: Time alignment error (TAE) as specified in TS 36.104 [30] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.
Note 9: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.23.5-2: 2DL PCELL in TDD RSRP for EUTRAN FDD Carrier Aggregation absolute accuracy requirements for the reported values**

| Band of Cell 1 on Primary Component Carrier | Event | Bands TDD_A | Bands TDD_C | Bands TDD_E |
|---|---|---|---|---|
| Normal Conditions | | | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRP_13 | RSRP_14 | RSRP_15 |
| Highest reported value (Cell 1) | | RSRP_26 | RSRP_27 | RSRP_28 |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_21 | RSRP_22 | RSRP_23 |
| Highest reported value (Cell 2) | | RSRP_34 | RSRP_35 | RSRP_36 |
| Extreme Conditions | | | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRP_9 | RSRP_10 | RSRP_11 |
| Highest reported value (Cell 1) | | RSRP_30 | RSRP_31 | RSRP_32 |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_17 | RSRP_18 | RSRP_19 |
| Highest reported value (Cell 2) | | RSRP_38 | RSRP_39 | RSRP_40 |
| Note: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2. | | | | |

**Table 9.1.23.5-3: 2DL PCELL in TDD RSRP for EUTRAN FDD Carrier Aggregation relative accuracy requirements for the reported values**

| | Event | |
|---|---|---|
| Normal Conditions, all bands | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 + 2 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 15 |
| Extreme Conditions, all bands | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 – 1 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 17 |
| Normal and Extreme Conditions, all bands | | |
| Lowest reported value (Cell 3) | Cell 2-3 | RSRP_Cell 2 – 8 |
| Highest reported value (Cell 3) | | RSRP_Cell 2 + 1 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.24 TDD RSRP Accuracy for E-UTRAN Carrier Aggregation for 20MHz + 10MHz

### 9.1.24.1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20MHz + 10MHz

#### 9.1.24.1.1 Test purpose

Same test purpose as in clause 9.1.7.1.1.

#### 9.1.24.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and release 11 supporting CA with Intra-band contiguous CA, or Inter-band CA.

This test applies to all types of E-UTRA TDD UE release 11 supporting CA with Intra-band non-contiguous CA.

#### 9.1.24.1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.7.1.3

### 9.1.24.1.4 Test description

#### 9.1.24.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-3 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The bandwidth combination supported by UE from Table 9.1.24.1.5-1 and Annex E table E-3 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.24.1.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.1.24.1.4.2 Test procedure

Same test procedure as in clause 9.1.7.1.4.2 with the following exceptions:

- Instead of Table 9.1.7.1.5-1 → use Table 9.1.24.1.5-1.

- Instead of Table 9.1.7.1.5-2 → use Table 9.1.24.1.5-2.

#### 9.1.24.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.24.1.4.3-1: Common Exception messages for TDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation test requirement for 20MHz + 10MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-3<br>Table H.4.1-5 |

### 9.1.24.1.5 Test requirement

Table 9.1.24.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD absolute accuracy test shall meet the reported values test requirements in table 9.1.24.1.5-2.

**Table 9.1.24.1.5-1: RSRP TDD absolute accuracy carrier aggregation test parameters for 20MHz + 10MHz**

| Parameter | | Unit | Combination | Test 1 | | |
|---|---|---|---|---|---|---|
| | | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 20MHz+10MHz | 20MHz: $N_{RB,c}$ = 100 | 10MHz: $N_{RB,c}$ = 50 | |
| | | | 10MHz+20MHz | 10MHz: $N_{RB,c}$ = 50 | 20MHz: $N_{RB,c}$ = 100 | |
| Measurement bandwidth | | $n_{PRB}$ | 20MHz+10MHz | 47-52 | 22-27 | |
| | | | 10MHz+20MHz | 22-27 | 47-52 | |
| PDSCH Reference measurement channel defined in A.3.1.1.2 | | | 20MHz+10MHz | R.3 TDD | R.0 TDD | N/A |
| | | | 10MHz+20MHz | R.0 TDD | R3.TDD | |
| PDSCH allocation | | $n_{PRB}$ | 20MHz+10MHz | 38-61 | 13-36 | N/A |
| | | | 10MHz+20MHz | 13-36 | 38-61 | |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.2 | | | 20MHz+10MHz | R.10 TDD | R.6 TDD | |
| | | | 10MHz+20MHz | R.6 TDD | R.10 TDD | |
| OCNG Patterns defined in A.3.2.2 (TDD) | | | 20MHz+10MHz | OP.7 TDD | OP.1 TDD | OP.2 TDD |
| | | | 10MHz+20MHz | OP.1 TDD | OP.7 TDD | OP.8 TDD |
| Io Note2 | Bands TDD_A | dBm/$BW_{channel}$ | All | -87.76 + 10log($N_{RB,c}$/50) | N/A | |
| | Bands TDD_C | | | -86.76 + 10log($N_{RB,c}$/50) | | |
| | Bands TDD_E | | | -85.76 + 10log($N_{RB,c}$/50) | | |
| | Bands TDD_A | dBm/ $BW_{channel}$ | All | N/A | (Io for Channel 1 +5.33dB) +10log ($N_{RB\ channel2}$ / $N_{RB\ channel\ 1}$)) | |
| | Bands TDD_C | | | | | |
| | Bands TDD_E | | | | | |

Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: See Table 9.1.7.1.5-1 for the other parameters.
Note 4: For each parameter, the allowed combinations are shown in separate rows.

**Table 9.1.24.1.5-2: RSRP TDD Carrier Aggregation absolute accuracy requirements for the reported values for 20MHz + 10MHz**

| Parameter | Test 1 | | |
|---|---|---|---|
| **Band of Cell 1 on Primary Component Carrier** | **Bands TDD_A** | **Bands TDD_C** | **Band TDD_E** |
| Normal Conditions | | | |
| Lowest reported value (Cell 1) | RSRP_12 | RSRP_13 | RSRP_14 |
| Highest reported value (Cell 1) | RSRP_27 | RSRP_28 | RSRP_29 |
| Lowest reported value (Cell 2) | RSRP_20 | RSRP_21 | RSRP_22 |
| Highest reported value (Cell 2) | RSRP_35 | RSRP_36 | RSRP_37 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 1) | RSRP_9 | RSRP_10 | RSRP_11 |
| Highest reported value (Cell 1) | RSRP_30 | RSRP_31 | RSRP_32 |
| Lowest reported value (Cell 2) | RSRP_17 | RSRP_18 | RSRP_19 |
| Highest reported value (Cell 2) | RSRP_38 | RSRP_39 | RSRP_40 |
| NOTE: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2. | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.24.1_1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20MHz + 10MHz (Rel-12 and forward)

### 9.1.24.1_1.1 Test purpose

Same test purpose as in clause 9.1.7.1_1.1.

### 9.1.24.1_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward supporting CA.

### 9.1.24.1_1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.7.1_1.3

### 9.1.24.1_1.4 Test description

#### 9.1.24.1_1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 20 MHz for Cell 1 on the PCC, 10MHz for Cell 2 on the SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.24.1_1.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.1.24.1_1.4.2 Test procedure

Same test procedure as in clause 9.1.7.1_1.4.2 with the following exceptions:

- Instead of Table 9.1.7.1_1.5-1 → use Table 9.1.24.1_1.5-1.
- Instead of Table 9.1.7.1_1.5-2 → use Table 9.1.24.1_1.5-2.

#### 9.1.24.1_1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.24.1_1.4.3-1: Common Exception messages for TDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation test requirement for 20MHz + 10MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-3<br>Table H.4.1-5 |

### 9.1.24.1_1.5 Test requirement

Table 9.1.24.1_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD absolute accuracy test shall meet the reported values test requirements in table 9.1.24.1_1.5-2.

**Table 9.1.24.1_1.5-1: RSRP TDD absolute accuracy carrier aggregation test parameters for 20MHz + 10MHz**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 20 | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 47-52 | 22-27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.3 TDD | R.0 TDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 38-61 | 13-36 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.10 TDD | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD), D.2.2 (OP.2 TDD) and D.2.7 (OP.7 TDD) | | | OP.7 TDD | OP.1 TDD | OP.2 TDD |
| Io Note2 | Bands TDD_A | dBm/18 MHz | -84.76 | N/A | |
| | Bands TDD_C | | -83.76 | | |
| | Bands TDD_E | | -82.76 | | |
| | Bands TDD_A | dBm/9MHz | N/A | (Io for Channel 1 +2.33dB) | |
| | Bands TDD_C | | | | |
| | Bands TDD_E | | | | |
| NOTE 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.<br>NOTE 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>NOTE 3: See Table 9.1.7.1_1.5-1 for the other parameters. | | | | | |

**Table 9.1.24.1_1.5-2: RSRP TDD Carrier Aggregation absolute accuracy requirements for the reported values for 20MHz + 10MHz**

| Parameter | Test 1 | | |
|---|---|---|---|
| **Band of Cell 1 on Primary Component Carrier** | **Bands TDD_A** | **Bands TDD_C** | **Band TDD_E** |
| Normal Conditions | | | |
| Lowest reported value (Cell 1) | RSRP_13 | RSRP_14 | RSRP_15 |
| Highest reported value (Cell 1) | RSRP_26 | RSRP_27 | RSRP_28 |
| Lowest reported value (Cell 2) | RSRP_21 | RSRP_22 | RSRP_23 |
| Highest reported value (Cell 2) | RSRP_34 | RSRP_35 | RSRP_36 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 1) | RSRP_9 | RSRP_10 | RSRP_11 |
| Highest reported value (Cell 1) | RSRP_30 | RSRP_31 | RSRP_32 |
| Lowest reported value (Cell 2) | RSRP_17 | RSRP_18 | RSRP_19 |
| Highest reported value (Cell 2) | RSRP_38 | RSRP_39 | RSRP_40 |
| NOTE: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2. | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.24.2 TDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 20MHz + 10MHz

### 9.1.24.2.1 Test purpose

Same test purpose as in clause 9.1.7.2.1.

### 9.1.24.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and release 11 supporting CA with Intra-band contiguous CA, or Inter-band CA.

This test applies to all types of E-UTRA TDD UE release 11 supporting CA with Intra-band non-contiguous CA.

### 9.1.24.2.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.7.2.3.

### 9.1.24.2.4 Test description

#### 9.1.24.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-3 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The bandwidth combination supported by UE from Table 9.1.24.2.5-1 and Annex E table E-3 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.24.2.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.1.24.2.4.2 Test procedure

Same test procedure as in clause 9.1.7.2.4.2 with the following exceptions:

- Instead of Table 9.1.7.2.5-1 → use Table 9.1.24.2.5-1.

- Instead of Table 9.1.7.2.5-2 → use Table 9.1.24.2.5-2.

#### 9.1.24.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.24.2.4.3-1: Common Exception messages for TDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation test requirement for 20MHz + 10MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-3<br>Table H.4.1-5 |

## 9.1.24.2.5 Test requirement

Table 9.1.24.2.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.24.2.5-2.

**Table 9.1.24.2.5-1: RSRP TDD relative accuracy carrier aggregation test parameters for 20MHz + 10MHz**

| Parameter | | Unit | Combination | Test 1 | | |
|---|---|---|---|---|---|---|
| | | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 20MHz+10MHz | 20MHz: $N_{RB,c}$ = 100 | 10MHz: $N_{RB,c}$ = 50 | |
| | | | 10MHz+20MHz | 10MHz: $N_{RB,c}$ = 50 | 20MHz: $N_{RB,c}$ = 100 | |
| Measurement bandwidth | | $n_{PRB}$ | 20MHz+10MHz | 47-52 | 22-27 | |
| | | | 10MHz+20MHz | 22-27 | 47-52 | |
| PDSCH Reference measurement channel defined in A.3.1.1.2 | | | 20MHz+10MHz | R.3 TDD | R.0 TDD | N/A |
| | | | 10MHz+20MHz | R.0 TDD | R3.TDD | |
| PDSCH allocation | | $n_{PRB}$ | 20MHz+10MHz | 38-61 | 13-36 | N/A |
| | | | 10MHz+20MHz | 13-36 | 38-61 | |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.2 | | | 20MHz+10MHz | R.10 TDD | R.6 TDD | |
| | | | 10MHz+20MHz | R.6 TDD | R.10 TDD | |
| OCNG Patterns defined in A.3.2.2 (TDD) | | | 20MHz+10MHz | OP.7 TDD | OP.1 TDD | OP.2 TDD |
| | | | 10MHz+20MHz | OP.1 TDD | OP.7 TDD | OP.8 TDD |
| Io Note2 | Bands TDD_A | dBm/$BW_{channel}$ | All | -87.76 + 10log($N_{RB,c}$/50) | N/A | |
| | Bands TDD_C | | | -86.76 + 10log($N_{RB,c}$/50) | | |
| | Bands TDD_E | | | -85.76 + 10log($N_{RB,c}$/50) | | |
| | Bands TDD_A | dBm/ $BW_{channel}$ | All | N/A | (Io for Channel 1 +5.33dB) +10log ($N_{RB\ channel2}$ / $N_{RB\ channel\ 1}$)) | |
| | Bands TDD_C | | | | | |
| | Bands TDD_E | | | | | |

Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: See Table 9.1.7.2.5-1 for the other parameters.
Note 4: For each parameter, the allowed combinations are shown in separate rows.

**Table 9.1.24.2.5-2: RSRP TDD Carrier Aggregation relative accuracy requirements for the reported values for 20MHz + 10MHz**

| | Test 1 |
|---|---|
| | All bands |
| Normal and Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRP_x – 1 |
| Highest reported value (Cell 2) | RSRP_x + 17 |
| Lowest reported value (Cell 3) | RSRP_y – 8 |
| Highest reported value (Cell 3) | RSRP_y + 1 |
| RSRP_x is the reported value of Cell 1<br>RSRP_y is the reported value of Cell 2 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.24.2_1 TDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 20MHz + 10MHz (Rel-12 and forward)

Editor's note: This Test case is incomplete. The following aspects are either missing or not yet determined:

- *The Test system uncertainties applicable to this test are undefined*
- *The Test tolerances applicable to this test are undefined*
- *Test requirements in normal conditions need to be defined*

### 9.1.24.2_1.1 Test purpose

Same test purpose as in clause 9.1.24.2.1.

### 9.1.24.2_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward supporting CA.

### 9.1.24.2_1.3 Minimum conformance requirements

Same minimum conformance requirements as defined in clause 9.1.7.2_1.3.

### 9.1.24.2_1.4 Test description

#### 9.1.24.2_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.24.2.4.1.

#### 9.1.24.2_1.4.2 Test procedure

Same test procedure as in clause 9.1.24.2.4.2 with the following exceptions:

- Instead of Table 9.1.24.2.5-1 → use Table 9.1.24.2_1.5-1.
- Instead of Table 9.1.24.2.5-2 → use Table 9.1.24.2_1.5-2.

#### 9.1.24.2_1.4.3 Message contents

Same message contents as in clause 9.1.24.2.4.2

### 9.1.24.2_1.5 Test requirement

Table 9.1.24.2_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD relative accuracy test shall meet the RSRP accuracy requirements in table 9.1.24.2_1.5-2.

**Table 9.1.24.2_1.5-1: RSRP TDD relative accuracy carrier aggregation test parameters for 20MHz + 10MHz**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel}$ Note 1 | | MHz | 20 | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 47-52 | 22-27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.3 TDD | R.0 TDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 38-61 | 13-36 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.10 TDD | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD), D.2.2 (OP.2 TDD) and D.2.7 (OP.7 TDD) | | | OP.7 TDD | OP.1 TDD | OP.2 TDD |
| Io Note2 | Bands TDD_A | dBm/18 MHz | -84.76+TT | N/A | |
| | Bands TDD_C | | -83.76+TT | | |
| | Bands TDD_E | | -82.76+TT | | |
| | Bands TDD_A | dBm/9MHz | N/A | (Io for Channel 1 +2.50dB) | |
| | Bands TDD_C | | | | |
| | Bands TDD_E | | | | |
| NOTE 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.<br>NOTE 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>NOTE 3: See Table 9.1.7.1.5-1 for the other parameters. | | | | | |

**Table 9.1.24.2_1.5-2: RSRP TDD Carrier Aggregation relative accuracy requirements for the reported values for 20MHz + 10MHz**

| | Test 1 |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 2) | TBD |
| Highest reported value (Cell 2) | TBD |
| Lowest reported value (Cell 3) | TBD |
| Highest reported value (Cell 3) | TBD |
| Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRP_x – 1 |
| Highest reported value (Cell 2) | RSRP_x + 17 |
| Lowest reported value (Cell 3) | RSRP_y – 8 |
| Highest reported value (Cell 3) | RSRP_y + 1 |
| RSRP_x is the reported value of Cell 1<br>RSRP_y is the reported value of Cell 2 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.25 FDD intra-frequency absolute and relative RSRP accuracies in CRS based discovery signal

### 9.1.25.1 Test purpose

The purpose of this test is to verify that the RSRP absolute and relative measurement accuracies in CRS based discovery signal are within the specified limits.

### 9.1.25.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward and supports CRS based discovery signals measurement. Applicability requires support for FGI bit 16.

### 9.1.25.3 Minimum conformance requirements

Intra-frequency absolute RSRP measurement accuracy requirements in discovery signal occasions are the same as specified in TS36.133 [4] Section 9.1.2.1.Intra-frequency relative RSRP measurement accuracy requirements in discovery signal occasions are the same as specified in TS36.133 [4] Section 9.1.2.2.

The absolute accuracy of RSRP is defined as the RSRP measured from a cell on the same frequency as that of the serving cell.

The accuracy requirements in Table 9.1.25.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101[2] Clause 7.3 for reference sensitivity are fulfilled.
- RSRP|dBm according to Annex I.3.1 for a corresponding Band.

**Table 9.1.25.3-1: RSRP Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_B | -120.5 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The relative accuracy of RSRP is defined as the RSRP measured from one cell compared to the RSRP measured from another cell on the same frequency.

The accuracy requirements in Table 9.1.25.3-2 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101[2] Clause 7.3 for reference sensitivity are fulfilled.
- RSRP1,2|$_{dBm}$ according to Annex I.3.8 for a corresponding Band.

**Table 9.1.25.3-2: RSRP Intra frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/$BW_{Channel}$ |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.25.3-3. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.25.3-3: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.14.2, clause 9.1.4 and A.9.1.25.

## 9.1.25.4 Test description

### 9.1.25.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.1.25.4.3.
4. All cells are in the same carrier frequency. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

### 9.1.25.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Table 9.1.25.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
3. SS shall transmit an RRCConnectionReconfiguration message on cell 1.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. UE shall transmit periodically MeasurementReport messages.
6. After 10s wait from Step 3, SS shall check the reported RSRP values in periodical MeasurementReport messages.

   The reported RSRP value of Cell 2 reported by the UE is compared to actual RSRP value according to Table 9.1.25.5-2. This counts as a Pass or Fail for the events “Cell 2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

   The reported RSRP value for Cell 2 is compared to the reported RSRP value for Cell 1 for each MeasurementReport message according to Table 9.1.25.5-3. This counts as a Pass or Fail for the events “Cell 1-2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.
7. SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events “Cell 1” and “Cell 1-2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.

If all events pass for each scenario, the test passes. If one event fails, the test fails.

### 9.1.25.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.25.4.3-1: Common Exception message**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3 |

**Table 9.1.25.4.3-2: *MeasResults***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell EQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.25.4.3-3: *MeasResultListEUTRA***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of Cell 2 | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.1.25.4.3-4: MeasObjectEUTRA-GENERIC**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(f1) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationFDD-r12 | 1 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.1.25.5 Test requirement

Table 9.1.25.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP absolute accuracy shall meet the reported values test requirements in table 9.1.25.5-2.

The RSRP relative accuracy shall meet the reported values test requirements in table 9.1.25.5-3.

**Table 9.1.25.5-1: RSRP FDD Intra frequency test parameters**

| Parameter | | Unit | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | |
| $BW_{channel}$ | | MHz | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | |
| DTMC period | | ms | N/A | 160 |
| DTMC period offset | | | N/A | 10 |
| Discovery signal occasion duration | | ms | N/A | 1 |
| Time offset between cells | | µs | 2.3 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.2.2 (OP.2 FDD) | | | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA Note1 | | | | |
| OCNG_RB Note1 | | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -107 | -107 |
| | Bands FDD_B | | | |
| | Bands FDD_C | | | |
| | Bands FDD_D | | | |
| | Bands FDD_E, FDD_F Note 5 | | | |
| | Bands FDD_G Note 7 | | | |
| | Bands FDD_H | | | |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 |
| RSRP Note3 | Bands FDD_A | dBm/15 kHz | -101.0 | -105.0 |
| | Bands FDD_B | | | |
| | Bands FDD_C | | | |
| | Bands FDD_D | | | |
| | Bands FDD_E, FDD_F Note 5 | | | |
| | Bands FDD_G Note 7 | | | |
| | Bands FDD_H | | | |
| Io Note3 | Bands FDD_A | dBm/9 MHz | -71.05 | -71.05 |
| | Bands FDD_B | | | |
| | Bands FDD_C | | | |
| | Bands FDD_D | | | |
| | Bands FDD_E, FDD_F Note 5 | | | |
| | Bands FDD_G Note 7 | | | |
| | Bands FDD_H | | | |
| $\hat{E}_s / N_{oc}$ | | dB | 6.0 | 2.0 |
| Propagation condition | | - | AWGN | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 4: | RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz. |
| Note 6: | E-UTRA operating band groups are as defined in Section 3.5. |
| Note 7: | Except Band 29 and Band 32. |

**Table 9.1.25.5-2: RSRP FDD Intra frequency absolute accuracy requirements for the reported values**

| | Event | Test 1<br>All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_29 |
| Highest reported value (Cell 2) | | RSRP_42 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_25 |
| Highest reported value (Cell 2) | | RSRP_46 |

**Table 9.1.25.5-3: RSRP FDD Intra frequency relative accuracy requirements for the reported values**

| | Event | Test 1<br>All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 - 9 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 1 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 - 9 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 1 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.26 TDD intra-frequency absolute and relative RSRP accuracies in CRS based discovery signal

### 9.1.26.1 Test purpose

Same test purpose as in clause 9.1.25.1.

### 9.1.26.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward and supports CRS based discovery signals measurement. Applicability requires support for FGI bit 16.

### 9.1.26.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.25.3.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.14.2, clause 9.1.4 and A.9.1.26.

## 9.1.26.4 Test description

### 9.1.26.4.1 Initial conditions

Same initial conditions as in clause 9.1.25.4.1 with the following exceptions:

- Message contents are defined in clause 9.1.26.4.3.

### 9.1.26.4.2 Test procedure

Same test procedure as in clause 9.1.25.4.2 with the following exceptions:

- Instead of Table 9.1.25.5-1 → use Table 9.1.26.5-1.
- Instead of Table 9.1.25.5-2 → use Table 9.1.26.5-2.
- Instead of Table 9.1.25.5-3 → use Table 9.1.26.5-3.

### 9.1.26.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.26.4.3-1: Common Exception message**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3 |

**Table 9.1.26.4.3-2: *MeasResults***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell EQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.26.4.3-3: *MeasResultListEUTRA***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of Cell 2 | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.1.26.4.3-4: MeasObjectEUTRA-GENERIC**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(f1) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationTDD-r12 | 2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.1.26.5 Test requirement

Table 9.1.26.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP absolute accuracy shall meet the reported values test requirements in table 9.1.26.5-2.

The RSRP relative accuracy shall meet the reported values test requirements in table 9.1.26.5-3.

**Table 9.1.26.5-1: RSRP TDD Intra frequency test parameters**

| Parameter | | Unit | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | |
| $BW_{channel}$ | | MHz | 10 | |
| Special subframe configuration[Note1] | | | 6 | |
| Uplink/downlink configuration[Note1] | | | 1 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | |
| DTMC period | | ms | N/A | 160 |
| DTMC period offset | | | N/A | 10 |
| Discovery signal occasion duration | | ms | N/A | 2 |
| Time offset between cells | | µs | 2.3 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA[Note2] | | | | |
| OCNG_RB[Note2] | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -107 | -107 |
| | Bands TDD_C | | | |
| | Bands TDD_E | | | |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -101.0 | -105.0 |
| | Bands TDD_C | | | |
| | Bands TDD_E | | | |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -71.05 | -71.05 |
| | Bands TDD_C | | | |
| | Bands TDD_E | | | |
| $\hat{E}_s / N_{oc}$ | | dB | 6.0 | 2.0 |
| Propagation condition | | - | AWGN | |

| | |
|---|---|
| Note 1: | For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211. |
| Note 2: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 3: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 4: | RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 5: | RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 6: | E-UTRA operating band groups are as defined in Section 3.5. |

**Table 9.1.26.5-2: RSRP TDD Intra frequency absolute accuracy requirements for the reported values**

| | Event | Test 1<br>All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_29 |
| Highest reported value (Cell 2) | | RSRP_42 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_25 |
| Highest reported value (Cell 2) | | RSRP_46 |

**Table 9.1.26.5-3: RSRP TDD Intra frequency relative accuracy requirements for the reported values**

| | Event | Test 1<br>All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 - 9 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 1 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 - 9 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 1 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.27 FDD—FDD inter-frequency absolute and relative RSRP accuracies in CRS based discovery signal

### 9.1.27.1 Test purpose

The purpose of this test is to verify that the RSRP absolute and relative measurement accuracies in CRS based discovery signal are within the specified limits.

### 9.1.27.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward and supports CRS based discovery signals measurement. Applicability requires support for FGI bit 16 and 25.

### 9.1.27.3 Minimum conformance requirements

Inter-frequency absolute RSRP measurement accuracy requirements in discovery signal occasions are the same as specified in TS36.133 [4] Section 9.1.3.1. Inter-frequency relative RSRP measurement accuracy requirements in discovery signal occasions are the same as specified in TS36.133 [4] Section 9.1.3.2.

The requirements for absolute accuracy of RSRP apply to a cell that has different carrier frequency from the serving cell.

The accuracy requirements in Table 9.1.27.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101[2] Clause 7.3 for reference sensitivity are fulfilled.
- RSRP|dBm according to Annex I.3.3 for a corresponding Band

**Table 9.1.27.3-1: RSRP Inter frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_B | -120.5 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The relative accuracy of RSRP in inter frequency case is defined as the RSRP measured from one cell compared to the RSRP measured from another cell on a different frequency.

The accuracy requirements in Table 9.1.27.3-2 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101[2] Clause 7.3 for reference sensitivity are fulfilled.
- RSRP1,2$|_{dBm}$ according to Annex I.3.4 for a corresponding Band
- $\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$
- | Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.27.3-2: RSRP Inter frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/$BW_{Channel}$ |
| ±4.5 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.27.3-3. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.27.3-3: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.14.2, clause 9.1.4 and A.9.1.27.

## 9.1.27.4 Test description

### 9.1.27.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.1.27.4.3.
4. Cell 1 and Cell 2 are on the different carrier frequencies. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

### 9.1.27.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.1.27.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message on cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the reported RSRP values in periodical MeasurementReport messages.

   The reported RSRP value of Cell 2 reported by the UE is compared to actual RSRP value according to Table 9.1.27.5-2. This counts as a Pass or Fail for the events “Cell 2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

   The reported RSRP value for Cell 2 is compared to the reported RSRP value for Cell 1 for each MeasurementReport message according to Table 9.1.27.5-3. This counts as a Pass or Fail for the events “Cell 1-2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved..

Each of the events “Cell 1” and “Cell 1-2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.

If all events pass for each scenario, the test passes. If one event fails, the test fails.

### 9.1.27.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.27.4.3-1: Common Exception message**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-2<br>Table H.3.5-3 |

**Table 9.1.27.4.3-2: *MeasResults***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell EQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.27.4.3-3: *MeasResultListEUTRA***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of Cell 2 | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.1.27.4.3-4: MeasObjectEUTRA-GENERIC**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(f2) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationFDD-r12 | 1 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.1.27.5 Test requirement

Table 9.1.27.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP absolute accuracy shall meet the reported values test requirements in table 9.1.27.5-2.

The RSRP relative accuracy shall meet the reported values test requirements in table 9.1.27.5-3.

**Table 9.1.27.5-1: CRS RSRP FDD-FDD Inter frequency test parameters**

| Parameter | | Unit | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 |
| Gap Pattern Id | | | 0 | - |
| gapOffset | | ms | 9 | |
| DMTC period | | ms | - | 160 |
| DMTC period offset | | ms | - | 10 |
| Discovery signal occasion duration | | ms | - | 1 |
| Time offset between cells | | µs | - | 3 |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.2.2 (OP.2 FDD) | | | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA Note1 | | | | |
| OCNG_RB Note1 | | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | ($N_{oc}$ for Channel 2 +6dB) | -115 |
| | Bands FDD_B | | | -114.5 |
| | Bands FDD_C | | | -114 |
| | Bands FDD_D | | | -113.5 |
| | Bands FDD_E, FDD_F Note 5 | | | -113 |
| | Bands FDD_G Note 7 | | | -112 |
| | Bands FDD_H | | | -111.5 |
| $\hat{E}_s / I_{ot}$ | | dB | 13 | -5.2 |
| RSRP Note3 | Bands FDD_A | dBm/15 kHz | (RSRP for Cell 2 +24.2dB) | -120.2 |
| | Bands FDD_B | | | -119.7 |
| | Bands FDD_C | | | -119.2 |
| | Bands FDD_D | | | -118.7 |
| | Bands FDD_E, FDD_F Note 5 | | | -118.2 |
| | Bands FDD_G Note 7 | | | -117.2 |
| | Bands FDD_H | | | -116.7 |
| Io Note3 | Bands FDD_A | dBm/9 MHz | (Io for Channel 2 +18.07dB) | -86.07 |
| | Bands FDD_B | | | -85.57 |
| | Bands FDD_C | | | -85.07 |
| | Bands FDD_D | | | -84.57 |
| | Bands FDD_E, FDD_F Note 5 | | | -84.04 |
| | Bands FDD_G Note 7 | | | -83.07 |
| | Bands FDD_H | | | -82.57 |

| | | | |
|---|---|---|---|
| $\hat{E}_s / N_{oc}$ | dB | 13 | -5.2 |
| Propagation condition | - | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: For Band 26, the tests shall be performed with the carrier frequency of assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 6: E-UTRA operating band groups are as defined in Section 3.5.
Note 7: Except Band 29 and Band 32.
Note 8: DMTC is provided to the UE in the *measDS-Config* (in TS36.331) before the beginning of the test

**Table 9.1.27.5-2: RSRP FDD Inter frequency absolute accuracy requirements for the reported values**

| **Normal Conditions** | **Event** | **Test 1** | |
|---|---|---|---|
| Lowest reported value (Cell 2) | Cell 2 | Bands FDD_A | RSRP_14 |
| | | Bands FDD_B | RSRP_15 |
| | | Bands FDD_C | RSRP_15 |
| | | Bands FDD_D | RSRP_16 |
| | | Bands FDD_E, FDD_F | RSRP_16 |
| | | Bands FDD_G | RSRP_17 |
| | | Bands FDD_H | RSRP_18 |
| Highest reported value (Cell 2) | Cell 2 | Bands FDD_A | RSRP_26 |
| | | Bands FDD_B | RSRP_27 |
| | | Bands FDD_C | RSRP_27 |
| | | Bands FDD_D | RSRP_28 |
| | | Bands FDD_E, FDD_F | RSRP_28 |
| | | Bands FDD_G | RSRP_29 |
| | | Bands FDD_H | RSRP_30 |
| **Extreme Conditions** | **Event** | **Test 1** | |
| Lowest reported value (Cell 2) | Cell 2 | Bands FDD_A | RSRP_10 |
| | | Bands FDD_B | RSRP_10 |
| | | Bands FDD_C | RSRP_11 |
| | | Bands FDD_D | RSRP_11 |
| | | Bands FDD_E, FDD_F | RSRP_12 |
| | | Bands FDD_G | RSRP_13 |
| | | Bands FDD_H | RSRP_13 |
| Highest reported value (Cell 2) | Cell 2 | Bands FDD_A | RSRP_31 |
| | | Bands FDD_B | RSRP_31 |
| | | Bands FDD_C | RSRP_32 |
| | | Bands FDD_D | RSRP_32 |
| | | Bands FDD_E, FDD_F | RSRP_33 |
| | | Bands FDD_G | RSRP_34 |
| | | Bands FDD_H | RSRP_34 |

**Table 9.1.27.5-3: RSRP FDD Inter frequency relative accuracy requirements for the reported values**

| | Event | Test 1<br>All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 - 31 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 - 18 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 - 33 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 - 17 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.28 TDD—TDD inter-frequency absolute and relative RSRP accuracies in CRS based discovery signal

### 9.1.28.1 Test purpose

Same test purpose as in clause 9.1.27.1.

### 9.1.28.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward and supports CRS based discovery signals measurement. Applicability requires support for FGI bit 16 and 25.

### 9.1.28.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.27.3.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.14.2, clause 9.1.4 and A.9.1.28.

### 9.1.28.4 Test description

#### 9.1.28.4.1 Initial conditions

Same initial conditions as in clause 9.1.27.4.1 with the following exceptions:

- Message contents are defined in clause 9.1.28.4.3.

#### 9.1.28.4.2 Test procedure

Same test procedure as in clause 9.1.25.4.2 with the following exceptions:

- Instead of Table 9.1.27.5-1 → use Table 9.1.28.5-1.
- Instead of Table 9.1.27.5-2 → use Table 9.1.28.5-2.
- Instead of Table 9.1.27.5-3 → use Table 9.1.28.5-3.

#### 9.1.28.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.28.4.3-1: Common Exception message**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-2<br>Table H.3.5-3 |

**Table 9.1.28.4.3-2: *MeasResults***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell EQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.28.4.3-3: *MeasResultListEUTRA***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of Cell 2 | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.1.28.4.3-4: MeasObjectEUTRA-GENERIC**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(f2) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationTDD-r12 | 2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.1.28.5 Test requirement

Table 9.1.28.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP absolute accuracy shall meet the reported values test requirements in table 9.1.28.5-2.

The RSRP relative accuracy shall meet the reported values test requirements in table 9.1.28.5-3.

**Table 9.1.28.5-1: CRS RSRP TDD—TDD Inter frequency test parameters**

| Parameter | | Unit | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 |
| Special subframe configuration[Note1] | | | 6 | |
| Uplink-downlink configuration[Note1] | | | 1 | |
| Gap Pattern Id | | | 0 | - |
| gapOffset | | ms | 9 | |
| DMTC period | | ms | - | 160 |
| DMTC period offset | | ms | - | 10 |
| Discovery signal occasion duration | | ms | - | 2 |
| Time offset between cells | | µs | - | 3 |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA[Note2] | | | | |
| OCNG_RB[Note2] | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | ($N_{oc}$ for Channel 2 +6dB) | -115 |
| | Bands TDD_C | | | -114 |
| | Bands TDD_E | | | -113 |
| $\hat{E}_s / I_{ot}$ | | dB | 13 | -5.2 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | (RSRP for Cell 2 +24.2dB) | -120.2 |
| | Bands TDD_C | | | -119.2 |
| | Bands TDD_E | | | -118.2 |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | (Io for Channel 2 +18.07B) | -86.07 |
| | Bands TDD_C | | | -85.07 |
| | Bands TDD_E | | | -84.07 |
| $\hat{E}_s / N_{oc}$ | | dB | 13 | -5.2 |
| Propagation condition | | - | AWGN | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211[9].

Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

| | |
|---|---|
| Note 4: | RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 5: | RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 6: | E-UTRA operating band groups are as defined in Section 3.5. |
| Note 7: | DMTC is provided to the UE in the measDS-Config (in TS36.331 [5]) before the beginning of test |

**Table 9.1.28.5-2: RSRP TDD Inter frequency absolute accuracy requirements for the reported values**

| Normal Conditions | Event | Test 1 | |
|---|---|---|---|
| Lowest reported value (Cell 2) | Cell 2 | Bands TDD_A | RSRP_14 |
| | | Bands TDD_C | RSRP_15 |
| | | Bands TDD_E | RSRP_16 |
| Highest reported value (Cell 2) | Cell 2 | Bands TDD_A | RSRP_26 |
| | | Bands TDD_C | RSRP_27 |
| | | Bands TDD_E | RSRP_28 |
| **Extreme Conditions** | **Event** | **Test 1** | |
| Lowest reported value (Cell 2) | Cell 2 | Bands TDD_A | RSRP_10 |
| | | Bands TDD_C | RSRP_11 |
| | | Bands TDD_E | RSRP_12 |
| Highest reported value (Cell 2) | Cell 2 | Bands TDD_A | RSRP_31 |
| | | Bands TDD_C | RSRP_32 |
| | | Bands TDD_E | RSRP_33 |

**Table 9.1.28.5-3: RSRP TDD Inter frequency relative accuracy requirements for the reported values**

| | Event | Test 1<br>All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 - 31 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 - 18 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 - 33 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 - 17 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.29 FDD intra frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal

### 9.1.29.1 Test purpose

The purpose of this test is to verify that the CSI-RSRP absolute and relative measurement accuracies in CSI-RS based discovery signal are within the specified limits.

### 9.1.29.2 Test applicability

This test applies to all types to E-UTRA FDD UE release 12 and forward that supports CSI-RS based discovery signals measurement. Applicability requires support for FGI bit 16.

### 9.1.29.3 Minimum conformance requirements

Intra-frequency absolute CSI-RSRP measurement accuracy requirements in discovery signal occasions are the same as specified in TS36.133 [4] Section 9.1.14.3.1.1. Intra-frequency relative CSI-RSRP measurement accuracy requirements in discovery signal occasions are the same as specified in TS36.133 [4] Section 9.1.14.3.1.2.

The absolute CSI-RSRP measurement accuracy requirements in discovery signal occasions apply to a cell or TP on the same frequency as that of the serving cell.

The accuracy requirements in Table 9.1.29.3-1 are valid under the following conditions:

- CSI reference signals in discovery signal occasions are transmitted on one antenna port only from each TP.
- Conditions defined in 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.
- CSI-RSRP is specified in Annex I.3.14.2 for a corresponding Band.

**Table 9.1.29.3-1: Intra-frequency absolute CSI-RSRP measurement accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | CSI Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥ 0 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_B | -120.5 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥ 0 dB | FDD_A, TDD_A, FDD_B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The relative CSI-RSRP measurement is defined as the CSI-RSRP measured from one cell or TP compared to the CSI-RSRP measured on the same frequency from another cell or from another TP. If two TPs are compared, they may belong to the same or different cells.

The accuracy requirements in Table 9.1.29.3-2 are valid under the following conditions:

- CSI reference signals in discovery signal occasions are transmitted on one antenna port only from each TP.
- Conditions defined in 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.
- CSI-RSRP is specified in Annex I.3.15.2 for a corresponding Band.

**Table 9.1.29.3-2: Intra-frequency relative CSI-RSRP measurement accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | CSI Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/$BW_{Channel}$ |
| ±2 | ±3 | ≥ 0 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥ 0 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter CSI Ês/Iot is the minimum CSI Ês/Iot of the pair of cells or TPs to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of CSI-RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.29.3-3. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.29.3-3: CSI-RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| CSI_RSRP_00 | CSI_RSRP < -140 | dBm |
| CSI_RSRP_01 | -140 ≤ CSI_RSRP < -139 | dBm |
| CSI_RSRP_02 | -139 ≤ CSI_RSRP < -138 | dBm |
| … | … | … |
| CSI_RSRP_95 | -46 ≤ CSI_RSRP < -45 | dBm |
| CSI_RSRP_96 | -45 ≤ CSI_RSRP < -44 | dBm |
| CSI_RSRP_97 | -44 ≤ CSI_RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.14.3 and A.9.1.29.

## 9.1.29.4 Test description

### 9.1.29.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.20 as appropriate.

2. Propagation conditions are set according to Annex B.0.

3. Message contents are defined in clause 9.1.29.4.3.

4. All cells are in the same carrier frequency. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

### 9.1.29.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Table 9.1.29.5-1 as appropriate. Propagation conditions are set according to Annex B.1.
3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. UE shall transmit periodically MeasurementReport messages.
6. After 10s wait from Step 3, SS shall check the reported CSI-RSRP values in periodical MeasurementReport messages.

   The reported CSI-RSRP value of Cell 2 reported by the UE is compared to actual CSI-RSRP value according to Table 9.1.29.5-2. This counts as a Pass or Fail for the event “Cell 2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

   The reported CSI-RSRP value of Cell 2 is compared to the reported CSI-RSRP value of Cell 1 for each MeasurementReport message according to Table 9.1.29.5-3. This counts as a Pass or Fail for the event “Cell 1-2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.
7. SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events “Cell 2” and “Cell 1-2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.

If all events pass for each scenario, the test passes. If one event fails, the test fails.

### 9.1.29.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.29.4.3-1: Common Exception message for CSI-RSRP FDD intra frequency accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.8-1<br>Table H.3.8-4<br>Table H.3.8-5 |

**Table 9.1.29.4.3-2: *MeasDS-Config-DEFAULT*: Additional CSI-RSRP FDD intra frequency accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1B: MeasDS-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasDS-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationFDD-r12 | 1 | | |
| } | | | |
| measCSI-RS-ToAddModList-r12 SEQUENCE (SIZE (1..maxCSI-RS-Meas-r12)) OF SEQUENCE { | 2 entries | | |
| MeasCSI-RS-Config-r12[0] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| physCellId-r12 | PhysCellId of Cell 1 | INTEGER(0..503) | |
| scramblingIdentity-r12 | 0 | | |
| resourceConfig-r12 | 2 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | dB0 | | |
| } | | | |
| MeasCSI-RS-Config-r12[1] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 2 | | |
| physCellId-r12 | PhysCellId of Cell 2 | INTEGER(0..503) | |
| scramblingIdentity-r12 | 1 | | |
| resourceConfig-r12 | 4 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | dB0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.29.4.3-3: *MeasResults*: Additional CSI-RSRP FDD intra frequency accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultCSI-RS-List-r12 SEQUENCE (SIZE(1..maxCellReport)) OF SEQUENCE { | | | |
| MeasResultCSI-RS-r12[0] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | Cell 1 | |
| csi-RSRP-Result-r12 | INTEGER(0..97) | Set according to specific test | |
| } | | | |
| MeasResultCSI-RS-r12[1] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 2 | Cell 2 | |
| csi-RSRP-Result-r12 | INTEGER(0.97) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |

## 9.1.29.5 Test requirement

Table 9.1.29.5-1 defines the primary level settings including test tolerances for all tests.

The CSI-RSRP absolute accuracy shall meet the reported values test requirements in Table 9.1.29.5-2.

The CSI-RSRP relative accuracy shall meet the reported values test requirements in Table 9.1.29.5-3.

**Table 9.1.29.5-1: CSI-RSRP FDD Intra frequency test parameters**

| Parameter | | Unit | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | |
| $BW_{channel}$ | | MHz | 10 | |
| DMTC period | | ms | 160 | |
| DMTC period offset | | ms | 10 | |
| Discovery signal occasion duration | | ms | 1 | |
| CSI-RS resource configuration | | | 2 | 4 |
| CSI-RS periodicity | | ms | 10 | |
| CSI-RS subframe offset | | ms | 0 | |
| CSI-RS individual offset[2] | | dB | 0 | 0 |
| CSI-RS muting | | | Enable | Enable |
| Time offset between cells | | µs | - | 2.3 |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA Note1 | | | | |
| OCNG_RB Note1 | | | | |
| p-C-r10[2] | | dB | 6 | 6 |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -116 | |
| | Bands FDD_B | | -115.5 | |
| | Bands FDD_C | | -115 | |
| | Bands FDD_D | | -114.5 | |
| | Bands FDD_E, FDD_F Note 5 | | -114 | |
| | Bands FDD_G Note 7 | | -113 | |
| | Bands FDD_H | | -112.5 | |
| CRS $\hat{E}_s / I_{ot}$ | | dB | 0.46 | -5.76 |
| CSI-RS $\hat{E}_s / I_{ot}$ | | dB | 6.46 | 1.04 |
| RSRP Note3 | Bands FDD_A | dBm/15 kHz | -113 | -117 |
| | Bands FDD_B | | -112.5 | -116.5 |
| | Bands FDD_C | | -112 | -116 |
| | Bands FDD_D | | -111.5 | -115.5 |
| | Bands FDD_E, FDD_F Note 5 | | -111 | -115 |
| | Bands FDD_G Note 7 | | -110 | -114 |
| | Bands FDD_H | | -109.5 | -113.5 |
| CSI-RSRP Note3 | Bands FDD_A | dBm/15 kHz | (RSRP for Cell 1 +6dB) | (RSRP for Cell 2 +6.8dB) |
| | Bands FDD_B | | | |
| | Bands FDD_C | | | |
| | Bands FDD_D | | | |

| | | | | |
|---|---|---|---|---|
| | Bands FDD_E, FDD_F Note 5 | | | |
| | Bands FDD_G Note 7 | | | |
| | Bands FDD_H | | | |
| Io Note3 | Bands FDD_A | dBm/9 MHz | -82.43 | |
| | Bands FDD_B | | -81.93 | |
| | Bands FDD_C | | -81.43 | |
| | Bands FDD_D | | -80.93 | |
| | Bands FDD_E, FDD_F Note 5 | | -80.43 | |
| | Bands FDD_G Note 7 | | -79.43 | |
| | Bands FDD_H | | -78.93 | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 3 | -1 |
| CSI-RS $\hat{E}_s / N_{oc}$ | | dB | 9 | 5.8 |
| Propagation condition | | - | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement subframe.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: For Band 26, the tests shall be performed with the carrier frequency of assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 6: E-UTRA operating band groups are as defined in Section 3.5.

Note 7: Except Band 29 and Band 32.

Note 8: DMTC is provided to the UE in the *measDS-Config* (in TS36.331) before the beginning of the test

**Table 9.1.29.5-2: CSI-RSRP FDD Intra frequency absolute accuracy requirements for the reported values**

| | Event | Bands FDD_A | Bands FDD_B | Bands FDD_C | Bands FDD_D | Bands FDD_E, FDD_F | Bands FDD_G | Bands FDD_H |
|---|---|---|---|---|---|---|---|---|
| **Normal Conditions** | | | | | | | | |
| Lowest reported value (Cell 2) | Cell 2 | CSI_RSRP_24 | CSI_RSRP_25 | CSI_RSRP_25 | CSI_RSRP_26 | CSI_RSRP_26 | CSI_RSRP_27 | CSI_RSRP_28 |
| Highest reported value (Cell 2) | | CSI_RSRP_36 | CSI_RSRP_37 | CSI_RSRP_37 | CSI_RSRP_38 | CSI_RSRP_38 | CSI_RSRP_39 | CSI_RSRP_40 |
| **Extreme Conditions** | | | | | | | | |
| Lowest reported value (Cell 2) | Cell 2 | CSI_RSRP_20 | CSI_RSRP_20 | CSI_RSRP_21 | CSI_RSRP_21 | CSI_RSRP_22 | CSI_RSRP_23 | CSI_RSRP_23 |
| Highest reported value (Cell 2) | | CSI_RSRP_41 | CSI_RSRP_41 | CSI_RSRP_42 | CSI_RSRP_42 | CSI_RSRP_43 | CSI_RSRP_44 | CSI_RSRP_45 |

**Table 9.1.29.5-3: CSI-RSRP FDD Intra frequency relative accuracy requirements for the reported values**

| | Event | All bands |
|---|---|---|
| **Normal and Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | CSI_RSRP_Cell 1 – 8 |
| Highest reported value (Cell 2) | | CSI_RSRP_Cell 1 + 1 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.30 TDD intra-frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal

### 9.1.30.1 Test purpose

The purpose of this test is to verify that the CSI-RSRP absolute and relative measurement accuracies in CSI-RS based discovery signal are within the specified limits.

### 9.1.30.2 Test applicability

This test applies to all types to E-UTRA TDD UE release 12 and forward that supports CSI-RS based discovery signals measurement. Applicability requires support for FGI bit 16.

### 9.1.30.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.29.3

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.14.3 and A.9.1.30.

### 9.1.30.4 Test description

#### 9.1.30.4.1 Initial conditions

Same initial conditions as in clause 9.1.29.4.1 with the following exception:

- Message contents are defined in clause 9.1.30.4.3.

#### 9.1.30.4.2 Test procedure

Same test procedure as in clause 9.1.29.4.2 with the following exceptions:

- Instead of Table 9.1.29.5-1 → use Table 9.1.30.5-1.
- Instead of Table 9.1.29.5-2 → use Table 9.1.30.5-2.
- Instead of Table 9.1.29.5-3 → use Table 9.1.30.5-3.

#### 9.1.30.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.30.4.3-1: Common Exception message for CSI-RSRP TDD intra frequency accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.8-1<br>Table H.3.8-4<br>Table H.3.8-5 |

**Table 9.1.30.4.3-2: *MeasDS-Config-DEFAULT*: Additional CSI-RSRP TDD intra frequency accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1B: MeasDS-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasDS-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationTDD-r12 | 2 | | |
| } | | | |
| measCSI-RS-ToAddModList-r12 SEQUENCE (SIZE (1..maxCSI-RS-Meas-r12)) OF SEQUENCE { | 2 entries | | |
| MeasCSI-RS-Config-r12[0] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| physCellId-r12 | PhysCellId of Cell 1 | INTEGER(0..503) | |
| scramblingIdentity-r12 | 0 | | |
| resourceConfig-r12 | 2 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | dB0 | | |
| } | | | |
| MeasCSI-RS-Config-r12[1] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 2 | | |
| physCellId-r12 | PhysCellId of Cell 2 | INTEGER(0..503) | |
| scramblingIdentity-r12 | 1 | | |
| resourceConfig-r12 | 4 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | dB0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.30.4.3-3: *MeasResults*: Additional CSI-RSRP TDD intra frequency accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultCSI-RS-List-r12 SEQUENCE (SIZE(1..maxCellReport)) OF SEQUENCE { | | | |
| MeasResultCSI-RS-r12[0] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | Cell 1 | |
| csi-RSRP-Result-r12 | INTEGER(0..97) | Set according to specific test | |
| } | | | |
| MeasResultCSI-RS-r12[1] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 2 | Cell 2 | |
| csi-RSRP-Result-r12 | INTEGER(0.97) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |

## 9.1.30.5 Test requirement

Table 9.1.30.5-1 defines the primary level settings including test tolerances for all tests.

The CSI-RSRP absolute accuracy shall meet the reported values test requirements in Table 9.1.30.5-2.

The CSI-RSRP relative accuracy shall meet the reported values test requirements in Table 9.1.30.5-3.

**Table 9.1.30.5-1: CSI-RSRP TDD Intra frequency test parameters**

| Parameter | | Unit | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | |
| $BW_{channel}$ | | MHz | 10 | |
| Special subframe configuration[Note1] | | | 6 | |
| Uplink-downlink configuration[Note1] | | | 1 | |
| DMTC period | | ms | 160 | |
| DMTC period offset | | ms | 10 | |
| Discovery signal occasion duration | | ms | 2 | |
| CSI-RS resource configuration | | | 2 | 4 |
| CSI-RS periodicity | | ms | 10 | |
| CSI-RS subframe offset | | ms | 0 | |
| CSI-RS individual offset[2] | | dB | 0 | 0 |
| CSI-RS muting | | | Enable | Enable |
| Time offset between cells | | µs | - | 2.3 |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA[Note1] | | | | |
| OCNG_RB[Note1] | | | | |
| p-C-r10[2] | | dB | 6 | 6 |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -116 | |
| | Bands TDD_C | | -115 | |
| | Bands TDD_E | | -114 | |
| CRS $\hat{E}_s / I_{ot}$ | | dB | 0.46 | -5.76 |
| CSI-RS $\hat{E}_s / I_{ot}$ | | dB | 6.46 | 1.04 |
| RSRP[Note3] | Bands TDD_A | dBm/15 kHz | -113 | -117 |
| | Bands TDD_C | | -112 | -116 |
| | Bands TDD_E | | -111 | -115 |
| CSI-RSRP [Note3] | Bands TDD_A | dBm/15 kHz | (RSRP for Cell 1 +6dB) | (RSRP for Cell 2 +6.8dB) |
| | Bands TDD_C | | | |
| | Bands TDD_E | | | |
| Io[Note3] | Bands TDD_A | dBm/9 MHz | -82.43 | |
| | Bands TDD_C | | -81.43 | |
| | Bands TDD_E | | -80.43 | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 3 | -1 |
| CSI-RS $\hat{E}_s / N_{oc}$ | | dB | 9 | 5.8 |
| Propagation condition | | - | AWGN | |
| Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.<br>Note 2: OCNG shall be used such that both cells are fully allocated and a | | | | |

| | |
|---|---|
| | constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 3: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 4: | RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement subframe. |
| Note 5: | RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 6: | E-UTRA operating band groups are as defined in Section 3.5. |
| Note 7: | DMTC is provided to the UE in the *measDS-Config* (in TS36.331) before the beginning of the test. |

**Table 9.1.30.5-2: CSI-RSRP TDD Intra frequency absolute accuracy requirements for the reported values**

| | Event | Bands TDD_A | Bands TDD_C | Bands TDD_E |
|---|---|---|---|---|
| **Normal Conditions** | | | | |
| Lowest reported value (Cell 2) | Cell 2 | CSI_RSRP_24 | CSI_RSRP_25 | CSI_RSRP_26 |
| Highest reported value (Cell 2) | | CSI_RSRP_36 | CSI_RSRP_37 | CSI_RSRP_38 |
| **Extreme Conditions** | | | | |
| Lowest reported value (Cell 2) | Cell 2 | CSI_RSRP_20 | CSI_RSRP_21 | CSI_RSRP_22 |
| Highest reported value (Cell 2) | | CSI_RSRP_41 | CSI_RSRP_42 | CSI_RSRP_43 |

**Table 9.1.30.5-3: CSI-RSRP TDD Intra frequency relative accuracy requirements for the reported values**

| | Event | All bands |
|---|---|---|
| **Normal and Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | CSI_RSRP_Cell 1 – 8 |
| Highest reported value (Cell 2) | | CSI_RSRP_Cell 1 + 1 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.31 FDD-FDD inter-frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal

### 9.1.31.1 Test purpose

The purpose of this test is to verify that the CSI-RSRP absolute and relative measurement accuracies in CSI-RS based discovery signal are within the specified limits.

### 9.1.31.2 Test applicability

This test applies to all types to E-UTRA FDD UE release 12 and forward that supports CSI-RS based discovery signals measurement. Applicability requires support for FGI bits 16 and 25.

### 9.1.31.3 Minimum conformance requirements

Inter-frequency absolute CSI-RSRP measurement accuracy requirements in discovery signal occasions are the same as specified in TS36.133 [4] Section 9.1.14.3.2.1. Inter-frequency relative CSI-RSRP measurement accuracy requirements in discovery signal occasions are the same as specified in TS36.133 [4] Section 9.1.14.3.2.2.

The absolute CSI-RSRP measurement accuracy requirements for discovery signal measurements apply to a cell or TP on a different carrier frequency from that of the serving cell.

The accuracy requirements in Table 9.1.31.3-1 are valid under the following conditions:

- CSI reference signals in discovery signal occasions are transmitted on one antenna port only from each TP.
- Conditions defined in 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.
- CSI-RSRP is specified in Annex I.3.16.2 for a corresponding band.

**Table 9.1.31.3-1: Inter-frequency absolute CSI-RSRP measurement accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | CSI Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥0 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_B | -120.5 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥0 dB | FDD_A, TDD_A, FDD B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The relative CSI-RSRP measurement is defined as the CSI-RSRP measured from one cell or TP compared to the CSI-RSRP measured on a different frequency from another cell or from another TP.

The accuracy requirements in Table 9.1.31.3-2 are valid under the following conditions:

- CSI reference signals in discovery signal occasions are transmitted on one antenna port only from each TP.
- Conditions defined in 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.
- CSI-RSRP is specified in Annex I.3.17.2 for a corresponding band.

$$\left| CSI_RSRP1\big|_{dBm} - CSI_RSRP2\big|_{dBm} \right| \le 27dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.31.3-2: Inter-frequency relative CSI-RSRP measurement accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io [Note 1] range | | |
| Normal condition | Extreme condition | CSI Ês/Iot [Note 2] | E-UTRA operating band groups [Note 4] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 3] | dBm/$BW_{Channel}$ |
| ±4.5 | ±6 | ≥0 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter CSI Ês/Iot is the minimum CSI Ês/Iot of the pair of cells or TPs to which the requirement applies.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of CSI-RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.31.3-3. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.31.3-3: CSI-RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| CSI_RSRP_00 | CSI_RSRP < -140 | dBm |
| CSI_RSRP_01 | -140 ≤ CSI_RSRP < -139 | dBm |
| CSI_RSRP_02 | -139 ≤ CSI_RSRP < -138 | dBm |
| … | … | … |
| CSI_RSRP_95 | -46 ≤ CSI_RSRP < -45 | dBm |
| CSI_RSRP_96 | -45 ≤ CSI_RSRP < -44 | dBm |
| CSI_RSRP_97 | -44 ≤ CSI_RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.14.3 and A.9.1.31.

## 9.1.31.4 Test description

### 9.1.31.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.14 as appropriate.
2. Propagation conditions are set according to Annex B.0.
3. Message contents are defined in clause 9.1.31.4.3.
4. Cell 1 and Cell 2 are on the different carrier frequencies. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

### 9.1.31.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Table 9.1.31.5-1 as appropriate. Propagation conditions are set according to Annex B.1.
3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. UE shall transmit periodically MeasurementReport messages.
6. After 10s wait from Step 3, SS shall check the reported CSI-RSRP values in periodical MeasurementReport messages.

   The reported CSI-RSRP value of Cell 2 reported by the UE is compared to actual CSI-RSRP value according to Table 9.1.31.5-2. This counts as a Pass or Fail for the event “Cell 2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

   The reported CSI-RSRP value of Cell 2 is compared to the reported CSI-RSRP value of Cell 1 for each MeasurementReport message according to Table 9.1.31.5-3. This counts as a Pass or Fail for the event “Cell 1-2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.
7. SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events “Cell 2” and “Cell 1-2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.

If all events pass for each scenario, the test passes. If one event fails, the test fails.

### 9.1.31.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.31.4.3-1: Common Exception message for CSI-RSRP FDD-FDD inter frequency accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.8-2<br>Table H.3.8-4<br>Table H.3.8-5<br>Table H.3.8-6 |

**Table 9.1.31.4.3-2: *MeasDS-Config-DEFAULT*: Additional CSI-RSRP FDD-FDD inter frequency accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1B: MeasDS-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasDS-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 0 | | Cell 1 |
| | 10 | | Cell 2 |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationFDD-r12 | 1 | | |
| } | | | |
| measCSI-RS-ToAddModList-r12 SEQUENCE (SIZE (1..maxCSI-RS-Meas-r12)) OF SEQUENCE { | 1 entry | | |
| MeasCSI-RS-Config-r12 SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| physCellId-r12 | PhysCellId of Cell 1 | INTEGER(0..503) | Cell 1 |
| | PhysCellId of Cell 2 | INTEGER(0..503) | Cell 2 |
| scramblingIdentity-r12 | 0 | | |
| resourceConfig-r12 | 2 | | Cell 1 |
| | 4 | | Cell 2 |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | dB0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.31.4.3-3: *MeasResults*: Additional CSI-RSRP FDD-FDD inter frequency accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | Cell 1 |
| | 2 | | Cell 2 |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultCSI-RS-List-r12 SEQUENCE (SIZE(1..maxCellReport)) OF SEQUENCE { | | | |
| MeasResultCSI-RS-r12 SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| csi-RSRP-Result-r12 | INTEGER(0..97) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |

## 9.1.31.5 Test requirement

Table 9.1.31.5-1 defines the primary level settings including test tolerances for all tests.

The CS-RSRP absolute accuracy shall meet the reported values test requirements in Table 9.1.31.5-2.

The CSI-RSRP relative accuracy shall meet the reported values test requirements in Table 9.1.31.5-3.

**Table 9.1.31.5-1: CSI-RSRP FDD-FDD Inter frequency test parameters**

| Parameter | | Unit | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 |
| Gap Pattern Id | | | 0 | - |
| gapOffset | | ms | 9 | |
| DMTC period | | ms | 160 | 160 |
| DMTC period offset | | ms | 0 | 10 |
| Discovery signal occasion duration | | ms | 1 | 1 |
| CSI-RS resource configuration | | | 2 | 4 |
| CSI-RS periodicity | | ms | 10 | |
| CSI-RS subframe offset | | ms | 0 | |
| CSI-RS individual offset[2] | | dB | 0 | 0 |
| CSI-RS muting | | | Enable | Enable |
| Time offset between cells | | µs | - | 3 |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA Note1 | | | | |
| OCNG_RB Note1 | | | | |
| p-C-r10[2] | | dB | 0 | 6 |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | ($N_{oc}$ for Channel 2 +6dB) | -115 |
| | Bands FDD_B | | | -114.5 |
| | Bands FDD_C | | | -114 |
| | Bands FDD_D | | | -113.5 |
| | Bands FDD_E, FDD_F Note 5 | | | -113 |
| | Bands FDD_G Note 7 | | | -112 |
| | Bands FDD_H | | | -111.5 |
| CRS $\hat{E}_s/I_{ot}$ | | dB | 13 | -6 |
| CSI-RS $\hat{E}_s/I_{ot}$ | | dB | 13 | 0.8 |
| RSRP Note3 | Bands FDD_A | dBm/15 kHz | (RSRP for Cell 2 +25dB) | -121 |
| | Bands FDD_B | | | -120.5 |
| | Bands FDD_C | | | -120 |
| | Bands FDD_D | | | -119.5 |
| | Bands FDD_E, FDD_F Note 5 | | | -119 |
| | Bands FDD_G Note 7 | | | -118 |
| | Bands FDD_H | | | -117.5 |

| | | | | |
|---|---|---|---|---|
| CSI-RSRP Note3 | Bands FDD_A | dBm/15 kHz | (RSRP for Cell 1 +0dB) | (RSRP for Cell 2 +6.8dB) |
| | Bands FDD_B | | | |
| | Bands FDD_C | | | |
| | Bands FDD_D | | | |
| | Bands FDD_E, FDD_F Note 5 | | | |
| | Bands FDD_G Note 7 | | | |
| | Bands FDD_H | | | |
| Io Note3 | Bands FDD_A | dBm/9 MHz | (Io for Channel 2 +19.68dB) | -86.25 |
| | Bands FDD_B | | | -85.75 |
| | Bands FDD_C | | | -85.25 |
| | Bands FDD_D | | | -84.75 |
| | Bands FDD_E, FDD_F Note 5 | | | -84.25 |
| | Bands FDD_G Note 7 | | | -83.25 |
| | Bands FDD_H | | | -82.75 |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 13 | -6 |
| CSI-RS $\hat{E}_s / N_{oc}$ | | dB | 13 | 0.8 |
| Propagation condition | | - | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement subframe.
Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: For Band 26, the tests shall be performed with the carrier frequency of assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 6: E-UTRA operating band groups are as defined in Section 3.5.
Note 7: Except Band 29 and Band 32.
Note 8: DMTC is provided to the UE in the *measDS-Config* (in TS36.331) before the beginning of the test.

**Table 9.1.31.5-2: CSI-RSRP FDD-FDD Inter frequency absolute accuracy requirements for the reported values**

| | Event | Bands FDD_A | Bands FDD_B | Bands FDD_C | Bands FDD_D | Bands FDD_E, FDD_F | Bands FDD_G | Bands FDD_H |
|---|---|---|---|---|---|---|---|---|
| **Normal Conditions** | | | | | | | | |
| Lowest reported value (Cell 2) | Cell 2 | CSI_RSRP_20 | CSI_RSRP_21 | CSI_RSRP_22 | CSI_RSRP_23 | CSI_RSRP_24 | CSI_RSRP_25 | CSI_RSRP_26 |
| Highest reported value (Cell 2) | | CSI_RSRP_27 | CSI_RSRP_28 | CSI_RSRP_29 | CSI_RSRP_30 | CSI_RSRP_31 | CSI_RSRP_32 | CSI_RSRP_33 |
| **Extreme Conditions** | | | | | | | | |
| Lowest reported value (Cell 2) | Cell 2 | CSI_RSRP_16 | CSI_RSRP_16 | CSI_RSRP_17 | CSI_RSRP_17 | CSI_RSRP_18 | CSI_RSRP_19 | CSI_RSRP_19 |
| Highest reported value (Cell 2) | | CSI_RSRP_37 | CSI_RSRP_37 | CSI_RSRP_38 | CSI_RSRP_38 | CSI_RSRP_39 | CSI_RSRP_40 | CSI_RSRP_40 |

**Table 9.1.31.5-3: CSI-RSRP FDD-FDD Inter frequency relative accuracy requirements for the reported values**

| | Event | All bands |
|---|---|---|
| **Normal and Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | CSI_RSRP_Cell 1 – 25 |
| Highest reported value (Cell 2) | | CSI_RSRP_Cell 1 + 12 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | CSI_RSRP_Cell 1 – 27 |
| Highest reported value (Cell 2) | | CSI_RSRP_Cell 1 – 11 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.32 TDD-TDD inter-frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal

### 9.1.32.1 Test purpose

The purpose of this test is to verify that the CSI-RSRP absolute and relative measurement accuracies in CSI-RS based discovery signal are within the specified limits.

### 9.1.32.2 Test applicability

This test applies to all types to E-UTRA TDD UE release 12 and forward that supports CSI-RS based discovery signals measurement. Applicability requires support for FGI bits 16 and 25.

### 9.1.32.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.31.3.

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.14.3 and A.9.1.32.

### 9.1.32.4 Test description

#### 9.1.32.4.1 Initial conditions

Same initial conditions as in clause 9.1.31.4.1 with the following exception:

- Message contents are defined in clause 9.1.32.4.3.

#### 9.1.32.4.2 Test procedure

Same test procedure as in clause 9.1.31.4.2 with the following exceptions:

- Instead of Table 9.1.31.5-1 → use Table 9.1.32.5-1.
- Instead of Table 9.1.31.5-2 → use Table 9.1.32.5-2.
- Instead of Table 9.1.31.5-3 → use Table 9.1.32.5-3.

#### 9.1.32.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.32.4.3-1: Common Exception message for CSI-RSRP TDD-TDD inter frequency accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.8-2<br>Table H.3.8-4<br>Table H.3.8-5<br>Table H.3.8-6 |

**Table 9.1.32.4.3-2: *MeasDS-Config-DEFAULT*: Additional CSI-RSRP TDD-TDD inter frequency accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1B: MeasDS-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasDS-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 0 | | Cell 1 |
| | 10 | | Cell 2 |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationTDD-r12 | 2 | | |
| } | | | |
| measCSI-RS-ToAddModList-r12 SEQUENCE (SIZE (1..maxCSI-RS-Meas-r12)) OF SEQUENCE { | 1 entry | | |
| MeasCSI-RS-Config-r12 SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| physCellId-r12 | PhysCellId of Cell 1 | INTEGER(0..503) | Cell 1 |
| | PhysCellId of Cell 2 | INTEGER(0..503) | Cell 2 |
| scramblingIdentity-r12 | 0 | | |
| resourceConfig-r12 | 2 | | Cell 1 |
| | 4 | | Cell 2 |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | dB0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.32.4.3-3: *MeasResults*: Additional CSI-RSRP TDD-TDD inter frequency accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | Cell 1 |
| | 2 | | Cell 2 |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultCSI-RS-List-r12 SEQUENCE (SIZE(1..maxCellReport)) OF SEQUENCE { | | | |
| MeasResultCSI-RS-r12[0] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| csi-RSRP-Result-r12 | INTEGER(0..97) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |

## 9.1.32.5 Test requirement

Table 9.1.32.5-1 defines the primary level settings including test tolerances for all tests.

The CS-RSRP absolute accuracy shall meet the reported values test requirements in Table 9.1.32.5-2.

The CSI-RSRP relative accuracy shall meet the reported values test requirements in Table 9.1.32.5-3.

**Table 9.1.32.5-1: CSI-RSRP TDD-TDD Inter frequency test parameters**

| Parameter | | Unit | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 |
| Special subframe configuration[Note1] | | | 6 | |
| Uplink-downlink configuration[Note1] | | | 1 | |
| Gap Pattern Id | | | 0 | - |
| gapOffset | | ms | 9 | |
| DMTC period | | ms | 160 | 160 |
| DMTC period offset | | ms | 0 | 10 |
| Discovery signal occasion duration | | ms | 2 | 2 |
| CSI-RS resource configuration | | | 2 | 4 |
| CSI-RS periodicity | | ms | 10 | |
| CSI-RS subframe offset | | ms | 0 | |
| CSI-RS individual offset[2] | | dB | 0 | 0 |
| CSI-RS muting | | | Enable | Enable |
| Time offset between cells | | µs | - | 3 |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA[Note2] | | | | |
| OCNG_RB[Note2] | | | | |
| p-C-r10[2] | | dB | 0 | 6 |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | ( $N_{oc}$ for Channel 2 +6dB) | -115 |
| | Bands TDD_C | | | -114 |
| | Bands TDD_E | | | -113 |
| CRS $\hat{E}_s/I_{ot}$ | | dB | 13 | -6 |
| CSI-RS $\hat{E}_s/I_{ot}$ | | dB | 13 | 0.8 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | (RSRP for Cell 2 +25dB) | -121 |
| | Bands TDD_C | | | -120 |
| | Bands TDD_E | | | -119 |
| CSI-RSRP [Note3] | Bands TDD_A | dBm/15 kHz | (RSRP for Cell 1 +0dB) | (RSRP for Cell 2 +6.8dB) |
| | Bands TDD_C | | | |
| | Bands TDD_E | | | |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | (Io for Channel 2 +18.24dB) | -86.25 |
| | Bands TDD_C | | | -85.25 |
| | Bands TDD_E | | | -84.25 |

| | | | |
|---|---|---|---|
| CRS $\hat{E}_s / N_{oc}$ | dB | 13 | -6 |
| CSI-RS $\hat{E}_s / N_{oc}$ | dB | 13 | 0.8 |
| Propagation condition | - | AWGN | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement subframe.
Note 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 6: E-UTRA operating band groups are as defined in Section 3.5.
Note 7: DMTC is provided to the UE in the measDS-Config (in TS36.331) before the beginning of test.

**Table 9.1.32.5-2: CSI-RSRP TDD-TDD Inter frequency absolute accuracy requirements for the reported values**

| | Event | Bands TDD_A | Bands TDD_C | Bands TDD_E |
|---|---|---|---|---|
| **Normal Conditions** | | | | |
| Lowest reported value (Cell 2) | Cell 2 | CSI_RSRP_20 | CSI_RSRP_21 | CSI_RSRP_22 |
| Highest reported value (Cell 2) | | CSI_RSRP_[32 | CSI_RSRP_33 | CSI_RSRP_34 |
| **Extreme Conditions** | | | | |
| Lowest reported value (Cell 2) | Cell 2 | CSI_RSRP_16 | CSI_RSRP_17 | CSI_RSRP_18 |
| Highest reported value (Cell 2) | | CSI_RSRP_37 | CSI_RSRP_38 | CSI_RSRP_39 |

**Table 9.1.32.5-3: CSI-RSRP TDD-TDD Inter frequency relative accuracy requirements for the reported values**

| | Event | All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | CSI_RSRP_Cell 1 – 25 |
| Highest reported value (Cell 2) | | CSI_RSRP_Cell 1 + 12 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | CSI_RSRP_Cell 1 – 27 |
| Highest reported value (Cell 2) | | CSI_RSRP_Cell 1 – 11 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.33 FDD absolute and relative RSRP accuracies for E-UTRAN Carrier Aggregation in CRS based discovery signal

### 9.1.33.1 Test purpose

The purpose of this test is to verify that the RSRP absolute and relative measurement accuracies in carrier aggregation in CRS based discovery signal are within the specified limits.

### 9.1.33.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward and supports [CA and CRS based discovery signals measurement].

### 9.1.33.3 Minimum conformance requirements

RSRP measurement of cells on the primary component carrier or any of the secondary component carrier(s) in discovery signal occasions shall meet the intra frequency absolute accuracy requirements in TS36.133 [4] section 9.1.2.1.

The accuracy requirements in Table 9.1.33.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101[2] Clause 7.3 for reference sensitivity are fulfilled.
- RSRP|dBm according to Annex I.3.1 for a corresponding Band

**Table 9.1.33.3-1: RSRP Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | $dBm/BW_{Channel}$ | $dBm/BW_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_B | -120.5 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

Comparisons between RSRP of cells on the same secondary component carrier shall meet the intra-frequency relative accuracy requirements in TS36.133 [4] sections 9.1.2.2.

The accuracy requirements in Table 9.1.33.3-2 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101[2] Clause 7.3 for reference sensitivity are fulfilled.
- $RSRP1,2|_{dBm}$ according to Annex I.3.8 for a corresponding Band.

**Table 9.1.33.3-2: RSRP Intra frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | $dBm/BW_{Channel}$ |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

When measurements of cells on the primary component carrier are compared with measurements of cells on any of the secondary component carrier(s), the applicable relative accuracy requirements are the RSRP inter-frequency accuracy requirements in TS36.133 [4] sections 9.1.3.2.

The accuracy requirements in Table 9.1.33.3-3 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101[2] Clause 7.3 for reference sensitivity are fulfilled.
- $RSRP1,2|_{dBm}$ according to Annex I.3.4 for a corresponding Band
- $\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$
- | Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.33.3-3: RSRP Inter frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | $dBm/BW_{Channel}$ |
| ±4.5 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.33.3-4. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.33.3-4: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.15.1, clause 9.1.14.2, clause 9.1.4 and A.9.1.33.

## 9.1.33.4 Test description

### 9.1.33.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.1.33.4.3.
4. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier and Cell 3 is the neighbouring cell on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

### 9.1.33.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCell according to Annex C.0, C.1 and C.2 for all downlink physical channels.
3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.
4. SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).
5. Set the parameters according to Table 9.1.33.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.
6. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
7. The UE shall transmit RRCConnectionReconfigurationComplete message.
8. UE shall transmit periodically MeasurementReport messages.
9. After 10s wait from Step 6, SS shall check the reported RSRP values in periodical MeasurementReport messages.
   The reported RSRP value of Cell 3 reported by the UE is compared to actual RSRP value according to Table

9.1.33.5-2. This counts as a Pass or Fail for the events “Cell 3”. If the UE fails to report the measurement value for Cell 3, the number of failed iterations is increased by one.
The reported RSRP value for Cell 3 is compared to the reported RSRP value for Cell2 and Cell1 for each MeasurementReport message according to Table 9.1.33.5-3. This counts as a Pass or Fail for the events “Cell 2-3” and “Cell 1-3”. If the UE fails to report the measurement value for Cell 3, the number of failed iterations is increased by one.

10. SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

11. If test configuration is Inter-band carrier aggregation, repeat steps 1-10 for switched PCell/SCell scenario according to the UE declared capability for UL support (within CA operation) in the individual bands.

Each of the events “Cell 3”, “Cell 2-3” and “Cell 1-3” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass for each configuration (without and with switched PCell/SCell scenario), the test passes. If one event fails, the test fails.

### 9.1.33.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.33.4.3-1: Common Exception message**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-3<br>Table H.4.1-5<br>Table H.4.2-1 |

**Table 9.1.33.4.3-2: MeasObjectEUTRA-GENERIC**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(Freq) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | Freq is the frequency of the SCell on the SCC (Cell 2) and neighbouring cell on the SCC(Cell 3) |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationFDD-r12 | 1 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.1.33.5 Test requirement

Table 9.1.33.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP absolute accuracy shall meet the reported values test requirements in table 9.1.33.5-2.

The RSRP relative accuracy shall meet the reported values test requirements in table 9.1.33.5-3.

**Table 9.1.33.5-1: RSRP FDD carrier aggregation test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell3 |
| E-UTRA RF Channel Number | | | 1 | 2 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 | 10 |
| DMTC period | | ms | N/A | N/A | 160 |
| DMTC period offset | | | N/A | N/A | 10 |
| Discovery signal occasion duration | | ms | N/A | N/A | 1 |
| Timing offset to cell1 | | µs | - | 0 | 3 |
| Time alignment error between cell 2 and cell 1 | | | - | ≤ Time alignment error as specified in 3GPP TS 36.104 [30] clause 6.5.3.1. | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.2.2 (OP.2 FDD) | | | OP.1 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RANote1 | | | | | |
| OCNG_RBNote | | | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -117 | ( $N_{oc}$ for Channel 1 +1dB) | |
| | Bands FDD_B | | -116.5 | | |
| | Bands FDD_C | | -116 | | |
| | Bands FDD_D | | -115.5 | | |
| | Bands FDD_E, FDD_F Note 6 | | -115 | | |
| | Bands FDD_G | | -114 | | |
| | Bands FDD_H | | -113.5 | | |
| $\hat{E}_s / I_{ot}$ | | dB | -4 | 0.09 | -4.96 |
| RSRP Note3 | Bands FDD_A | dBm/15 kHz | -121 | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4.8dB) |
| | Bands FDD_C | | -120.5 | | |
| | Bands FDD_C | | -120 | | |
| | Bands FDD_D | | -119.5 | | |
| | Bands FDD_E, FDD_F Note 6 | | -119 | | |
| | Bands FDD_G | | -118 | | |
| | Bands FDD_H | | -117.5 | | |
| Io Note3 | Bands FDD_A | dBm/9 MHz | -87.76 | (Io for Channel 1 +5.51dB) | |
| | Bands FDD_C | | -87.26 | | |
| | Bands FDD_C | | -86.76 | | |
| | Bands FDD_D | | -86.26 | | |
| | Bands FDD_E, FDD_F Note 6 | | -85.76 | | |

| | | | | | |
|---|---|---|---|---|---|
| | Bands FDD_G | | -84.76 | | |
| | Bands FDD_H | | -84.26 | | |
| $\hat{E}_s / N_{oc}$ | | dB | -4 | 3 | -0.2 |
| Propagation condition | | - | AWGN | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.

Note 6: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 7: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.

Note 8: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.33.5-2: RSRP FDD Intra frequency absolute accuracy requirements for the reported values**

| **Normal Conditions** | **Event** | **Test 1** | |
|---|---|---|---|
| Lowest reported value (Cell 3) | Cell 3 | Bands FDD_A | RSRP_18 |
| | | Bands FDD_B | RSRP_19 |
| | | Bands FDD_C | RSRP_19 |
| | | Bands FDD_D | RSRP_20 |
| | | Bands FDD_E, FDD_F | RSRP_20 |
| | | Bands FDD_G | RSRP_21 |
| | | Bands FDD_H | RSRP_22 |
| Highest reported value (Cell 3) | Cell 3 | Bands FDD_A | RSRP_30 |
| | | Bands FDD_B | RSRP_31 |
| | | Bands FDD_C | RSRP_31 |
| | | Bands FDD_D | RSRP_32 |
| | | Bands FDD_E, FDD_F | RSRP_32 |
| | | Bands FDD_G | RSRP_33 |
| | | Bands FDD_H | RSRP_34 |
| **Extreme Conditions** | **Event** | **Test 1** | |
| Lowest reported value (Cell 3) | Cell 3 | Bands FDD_A | RSRP_14 |
| | | Bands FDD_B | RSRP_14 |
| | | Bands FDD_C | RSRP_15 |
| | | Bands FDD_D | RSRP_15 |
| | | Bands FDD_E, FDD_F | RSRP_16 |
| | | Bands FDD_G | RSRP_17 |
| | | Bands FDD_H | RSRP_17 |
| Highest reported value (Cell 3) | Cell 3 | Bands FDD_A | RSRP_35 |
| | | Bands FDD_B | RSRP_35 |
| | | Bands FDD_C | RSRP_36 |
| | | Bands FDD_D | RSRP_36 |
| | | Bands FDD_E, FDD_F | RSRP_37 |
| | | Bands FDD_G | RSRP_38 |
| | | Bands FDD_H | RSRP_38 |

**Table 9.1.33.5-3: RSRP FDD Intra frequency relative accuracy requirements for the reported values**

| | Event | Test 1<br>All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 3) | Cell 2-3 | RSRP_Cell 2 - 8 |
| Highest reported value (Cell 3) | | RSRP_Cell 2 + 1 |
| Lowest reported value (Cell 3) | Cell 1-3 | RSRP_Cell 1 - 2 |
| Highest reported value (Cell 3) | | RSRP_Cell 1 + 12 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 3) | Cell 2-3 | RSRP_Cell 2 - 8 |
| Highest reported value (Cell 3) | | RSRP_Cell 2 + 1 |
| Lowest reported value (Cell 3) | Cell 1-3 | RSRP_Cell 1 - 4 |
| Highest reported value (Cell 3) | | RSRP_Cell 1 + 13 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.34 TDD absolute and relative RSRP accuracies for E-UTRAN Carrier Aggregation in CRS based discovery signal

### 9.1.34.1 Test purpose

Same test purpose as in clause 9.1.33.1.

### 9.1.34.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward and supports [CA and CRS based discovery signals measurement].

### 9.1.34.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.33.3.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.15.1, clause 9.1.14.2, clause 9.1.4 and A.9.1.34.

### 9.1.34.4 Test description

#### 9.1.34.4.1 Initial conditions

Same initial conditions as in clause 9.1.33.4.1 with the following exceptions:

- Message contents are defined in clause 9.1.34.4.3.

#### 9.1.34.4.2 Test procedure

Same test procedure as in clause 9.1.33.4.2 with the following exceptions:

- Instead of Table 9.1.33.5-1 → use Table 9.1.34.5-1.
- Instead of Table 9.1.33.5-2 → use Table 9.1.34.5-2.
- Instead of Table 9.1.33.5-3 → use Table 9.1.34.5-3.

#### 9.1.34.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.34.4.3-1: Common Exception message**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-3<br>Table H.4.1-5<br>Table H.4.2-1 |

**Table 9.1.34.4.3-2: MeasObjectEUTRA-GENERIC**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(Freq) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | Freq is the frequency of the SCell on the SCC (Cell 2) and neighbouring cell on the SCC(Cell 3) |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationTDD-r12 | 2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.1.34.5 Test requirement

Table 9.1.34.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP absolute accuracy shall meet the reported values test requirements in table 9.1.34.5-2.

The RSRP relative accuracy shall meet the reported values test requirements in table 9.1.34.5-3.

**Table 9.1.34.5-1: Carrier aggregation RSRP test parameters for TDD**

| Parameter | Unit | Test 1 | | |
|---|---|---|---|---|
| | | Cell 1 | Cell 2 | Cell 3 |

| | | | | | |
|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | 2 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 | 10 |
| DMTC period | | ms | N/A | N/A | 160 |
| DMTC period offset | | | N/A | N/A | 10 |
| Discovery signal occasion duration | | ms | N/A | N/A | 2 |
| Special subframe configuration[Note1] | | | 6 | | |
| Uplink/downlink configuration[Note1] | | | 1 | | |
| Timing offset to Cell 1 | | μs | - | 0 | 3 |
| Time alignment error between cell 2 and cell 1 | | | - | ≤ Time alignment error as specified in 3GPP TS 36.104 [30] clause 6.5.3.1 | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA[Note2] | | | | | |
| OCNG_RB[Note2] | | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -117 | ($N_{oc}$ for Channel 1 +1dB) | |
| | Bands TDD_C | | -116 | | |
| | Bands TDD_E | | -115 | | |
| $\hat{E}_s / I_{ot}$ | | dB | -4 | 0.09 | -4.96 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -121 | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4.8dB) |
| | Bands TDD_C | | -120 | | |
| | Bands TDD_E | | -119 | | |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -87.76 | (Io for Channel 1 +5.51dB) | |
| | Bands TDD_C | | -86.76 | | |
| | Bands TDD_E | | -85.76 | | |

| | | | | |
|---|---|---|---|---|
| $\hat{E}_s / N_{oc}$ | dB | -4 | 3 | -0.2 |
| Propagation condition | - | AWGN | | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 6: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.
Note 7: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 8: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.34.5-2: RSRP TDD Intra frequency absolute accuracy requirements for the reported values**

| **Normal Conditions** | **Event** | **Test 1** | |
|---|---|---|---|
| Lowest reported value (Cell 3) | Cell 3 | Bands TDD_A | RSRP_18 |
| | | Bands TDD_C | RSRP_19 |
| | | Bands TDD_E | RSRP_20 |
| Highest reported value (Cell 3) | Cell 3 | Bands TDD_A | RSRP_30 |
| | | Bands TDD_C | RSRP_31 |
| | | Bands TDD_E | RSRP_32 |
| **Extreme Conditions** | **Event** | **Test 1** | |
| Lowest reported value (Cell 3) | Cell 3 | Bands TDD_A | RSRP_14 |
| | | Bands TDD_C | RSRP_15 |
| | | Bands TDD_E | RSRP_16 |
| Highest reported value (Cell 3) | Cell 3 | Bands TDD_A | RSRP_35 |
| | | Bands TDD_C | RSRP_36 |
| | | Bands TDD_E | RSRP_37 |

**Table 9.1.34.5-3: RSRP TDD Intra frequency relative accuracy requirements for the reported values**

| | **Event** | **Test 1 All bands** |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 3) | Cell 2-3 | RSRP_Cell 2 - 8 |
| Highest reported value (Cell 3) | | RSRP_Cell 2 + 1 |
| Lowest reported value (Cell 3) | Cell 1-3 | RSRP_Cell 1 - 2 |
| Highest reported value (Cell 3) | | RSRP_Cell 1 + 12 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 3) | Cell 2-3 | RSRP_Cell 2 - 8 |
| Highest reported value (Cell 3) | | RSRP_Cell 2 + 1 |
| Lowest reported value (Cell 3) | Cell 1-3 | RSRP_Cell 1 - 4 |
| Highest reported value (Cell 3) | | RSRP_Cell 1 + 13 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.35 FDD absolute and relative CSI-RSRP accuracies for E-UTRAN Carrier Aggregation in CSI-RS based discovery signal

### 9.1.35.1 Test purpose

The purpose of this test is to verify that the CSI-RSRP absolute and relative measurement accuracies for carrier aggregation in CSI-RS based discovery signal are within the specified limits. This test will verify the absolute CSI-RSRP accuracy requirements of the primary component carrier, the absolute CSI-RSRP accuracy requirements of the secondary component carrier, and the relative CSI-RSRP accuracy requirements of the secondary component carrier. The test will also verify the primary and secondary component carrier relative CSI-RSRP accuracy requirement.

### 9.1.35.2 Test applicability

This test applies to all types to E-UTRA FDD UE release 12 and forward that supports CA and CSI-RS based discovery signals measurement.

### 9.1.35.3 Minimum conformance requirements

The CSI-RSRP measurements of cells on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.14.3.1.1. The CSI-RSRP measurements of cells on the secondary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.14.3.1.2.

The absolute CSI-RSRP measurement accuracy requirements in discovery signal occasions apply to a cell or TP on the same frequency as that of the serving cell.

The accuracy requirements in Table 9.1.35.3-1 are valid under the following conditions:

- CSI reference signals in discovery signal occasions are transmitted on one antenna port only from each TP.
- Conditions defined in 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.
- CSI-RSRP is specified in Annex I.3.14.2 for a corresponding Band.

**Table 9.1.35.3-1: Intra-frequency absolute CSI-RSRP measurement accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | CSI $\hat{E}s/Iot$ | E-UTRA operating band groups Note 3 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | $dBm/BW_{Channel}$ | $dBm/BW_{Channel}$ |
| ±4.5 | ±9 | ≥ 0 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_B | -120.5 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥ 0 dB | FDD_A, TDD_A, FDD_B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

Comparisons between CSI-RSRP of cells on the primary component carrier shall also meet the intra-frequency relative accuracy requirements defined in TS 36.133 [4] clause 9.1.14.3.1.2. Comparisons between CSI-RSRP of cells on the

same secondary component carrier shall meet the intra-frequency relative accuracy requirements defined in TS 36.133 [4] clause 9.1.14.3.1.2.

The relative CSI-RSRP measurement is defined as the CSI-RSRP measured from one cell or TP compared to the CSI-RSRP measured on the same frequency from another cell or from another TP. If two TPs are compared, they may belong to the same or different cells.

The accuracy requirements in Table 9.1.35.3-2 are valid under the following conditions:

- CSI reference signals in discovery signal occasions are transmitted on one antenna port only from each TP.
- Conditions defined in 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.
- CSI-RSRP is specified in Annex I.3.15.2 for a corresponding Band.

**Table 9.1.35.3-2: Intra-frequency relative CSI-RSRP measurement accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | CSI Ês/Iot Note 2 | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/$BW_{Channel}$ |
| ±2 | ±3 | ≥ 0 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥ 0 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter CSI Ês/Iot is the minimum CSI Ês/Iot of the pair of cells or TPs to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

When measurements of cells on the primary component carrier are compared with measurements of cells on any of the secondary component carrier(s), the applicable relative accuracy requirements are the CSI-RSRP inter-frequency accuracy requirements defined in TS 36.133 [4] clause 9.1.14.3.2.2.

The relative CSI-RSRP measurement is defined as the CSI-RSRP measured from one cell or TP compared to the CSI-RSRP measured on a different frequency from another cell or from another TP.

The accuracy requirements in Table 9.1.35.3-3 are valid under the following conditions:

- CSI reference signals in discovery signal occasions are transmitted on one antenna port only from each TP.
- Conditions defined in 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.
- CSI-RSRP is specified in Annex I.3.17.2 for a corresponding band.

$$\left|CSI_RSRP1\right|_{dBm} - CSI_RSRP2\left|_{dBm}\right| \leq 27dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.35.3-3: Inter-frequency relative CSI-RSRP measurement accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | CSI Ês/Iot Note 2 | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/$BW_{Channel}$ |
| ±4.5 | ±6 | ≥0 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter CSI Ês/Iot is the minimum CSI Ês/Iot of the pair of cells or TPs to which the requirement applies.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of CSI-RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.35.3-4. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.35.3-4: CSI-RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| CSI_RSRP_00 | CSI_RSRP < -140 | dBm |
| CSI_RSRP_01 | -140 ≤ CSI_RSRP < -139 | dBm |
| CSI_RSRP_02 | -139 ≤ CSI_RSRP < -138 | dBm |
| … | … | … |
| CSI_RSRP_95 | -46 ≤ CSI_RSRP < -45 | dBm |
| CSI_RSRP_96 | -45 ≤ CSI_RSRP < -44 | dBm |
| CSI_RSRP_97 | -44 ≤ CSI_RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.14.3, 9.1.15.2, and A.9.1.35.

## 9.1.35.4 Test description

### 9.1.35.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1 [10] clause 5.4.2A.

1. Connect the SS and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.14 as appropriate.

2. Propagation conditions are set according to Annex B.0.

3. Message contents are defined in clause 9.1.35.4.3.

4. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on the secondary component carrier and Cell 3 is the neighbouring cell on the secondary component carrier. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

### 9.1.35.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCell according to Annex C.0 and C.1 for all downlink physical channels.
3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.
4. The SS activates the SCC by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13 and 6.1.3.8). Wait for at least 2 seconds (Refer to TS 36.133 [4] clauses 8.3.3.2).
5. Set the parameters according to Table 9.1.35.5-1 as appropriate. Propagation conditions are set according to Annex B.1.
6. The SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
7. The UE shall transmit RRCConnectionReconfigurationComplete message.
8. The UE shall transmit periodically MeasurementReport messages.
9. After 10s wait from Step 6, the SS shall check the reported CSI-RSRP values in periodical MeasurementReport messages.
   The reported CSI-RSRP values of Cell 1 and Cell 2 reported by the UE are compared to actual CSI-RSRP values according to Table 9.1.35.5-2. This counts respectively as a Pass or Fail for the events “Cell 1” and “Cell 2”. If the UE fails to report the measurement value for Cell 1 or Cell 2, the number of failed iterations is increased by one.
   The reported CSI-RSRP values of Cell 1 and Cell 2 reported by the UE are compared to the reported CSI-RSRP values of other cells for each MeasurementReport message according to Table 9.1.35.5-3. This counts as a Pass or Fail for the events “Cell 1-2” and “Cell 2-3”. If the UE fails to report the measurement value of Cell 1 or Cell 2, the number of failed iterations is increased by one.
10. The SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.
11. If the test configuration is inter-band carrier aggregation, repeat steps 1-10 for all relevant PCell/SCells scenarios defined in Table E-2 according to the UE declared capability for UL support (within CA operation) in the individual bands.

Each of the events “Cell 1”, “Cell 2”, “Cell 1-2”, and “Cell 2-3” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass for each configuration (without and with switched PCell/SCells scenario), the test passes. If one event fails, the test fails.

### 9.1.35.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.35.4.3-1: Common Exception message for CSI-RSRP FDD Carrier Aggregation accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.8-3<br>Table H.3.8-4<br>Table H.3.8-5 |

**Table 9.1.35.4.3-2: *MeasDS-Config-DEFAULT-PCell*: Additional CSI-RSRP FDD Carrier Aggregation accuracy test requirement for PCell**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1B: MeasDS-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasDS-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 0 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationFDD-r12 | 1 | | |
| } | | | |
| measCSI-RS-ToAddModList-r12 SEQUENCE (SIZE (1..maxCSI-RS-Meas-r12)) OF SEQUENCE { | 1 entry | | |
| MeasCSI-RS-Config-r12 SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| physCellId-r12 | PhysCellId of Cell 1 | INTEGER(0..503) | |
| scramblingIdentity-r12 | 0 | | |
| resourceConfig-r12 | 2 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | dB0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.35.4.3-3: *MeasDS-Config-DEFAULT-SCell*: Additional CSI-RSRP FDD Carrier Aggregation accuracy test requirement for SCell**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1B: MeasDS-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasDS-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationFDD-r12 | 1 | | |
| } | | | |
| measCSI-RS-ToAddModList-r12 SEQUENCE (SIZE (1..maxCSI-RS-Meas-r12)) OF SEQUENCE { | 2 entries | | |
| MeasCSI-RS-Config-r12[0] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| physCellId-r12 | PhysCellId of Cell 2 | INTEGER(0..503) | |
| scramblingIdentity-r12 | 0 | | |
| resourceConfig-r12 | 4 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | dB0 | | |
| } | | | |
| MeasCSI-RS-Config-r12[1] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 2 | | |
| physCellId-r12 | PhysCellId of Cell 3 | INTEGER(0..503) | |
| scramblingIdentity-r12 | 1 | | |
| resourceConfig-r12 | 6 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | dB0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.35.4.3-4: *MeasResults-PCell*: Additional CSI-RSRP FDD Carrier Aggregation accuracy test requirement for PCell**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell EQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultCSI-RS-List-r12 SEQUENCE (SIZE(1..maxCellReport)) OF SEQUENCE { | | | |
| MeasResultCSI-RS-r12 SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | Cell 1 | |
| csi-RSRP-Result-r12 | INTEGER(0..97) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.35.4.3-5: *MeasResults-SCell*: Additional CSI-RSRP FDD Carrier Aggregation accuracy test requirement for SCell**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | | |
| measResultPCell EQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultCSI-RS-List-r12 SEQUENCE (SIZE(1..maxCellReport)) OF SEQUENCE { | | | |
| MeasResultCSI-RS-r12[0] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | Cell 2 | |
| csi-RSRP-Result-r12 | INTEGER(0..97) | Set according to specific test | |
| } | | | |
| MeasResultCSI-RS-r12[1] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 2 | Cell 3 | |
| csi-RSRP-Result-r12 | INTEGER(0..97) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |

## 9.1.35.5 Test requirement

Table 9.1.35.5-1 defines the primary level settings including test tolerances for all tests.

The CSI-RSRP absolute accuracy shall meet the reported values test requirements in Table 9.1.35.5-2.

The CSI-RSRP relative accuracy shall meet the reported values test requirements in Table 9.1.35.5-3.

**Table 9.1.35.5-1: CSI-RSRP FDD Carrier Aggregation test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell3 |
| E-UTRA RF Channel Number | | | 1 | 2 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 | 10 |
| Timing offset to cell1 | | µs | - | 0 | 3 |
| Time alignment error between cell 2 and cell 1 | | | - | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1. | - |
| DMTC period | | ms | 160 | 160 | |
| DMTC period offset | | ms | 0 | 10 | |
| Discovery signal occasion duration | | ms | 1 | 1 | |
| CSI-RS resource configuration | | | 2 | 4 | 6 |
| CSI-RS periodicity | | ms | 10 | 10 | 10 |
| CSI-RS subframe offset | | ms | 0 | 0 | 0 |
| CSI-RS individual offset[2] | | dB | 0 | 0 | 0 |
| CSI-RS muting | | | Enable | Enable | Enable |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RANote1 | | | | | |
| OCNG_RBNote | | | | | |
| p-C-r10[2] | | dB | 6 | 6 | 6 |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -117 | ( $N_{oc}$ for Channel 1 +1dB) | |
| | Bands FDD_B | | -116.5 | | |
| | Bands FDD_C | | -116 | | |
| | Bands FDD_D | | -115.5 | | |
| | Bands FDD_E, FDD_F Note 6 | | -115 | | |
| | Bands FDD_G | | -114 | | |
| | Bands FDD_H | | -113.5 | | |
| CRS $\hat{E}_s / I_{ot}$ | | dB | -4 | 0.46 | -5.76 |
| CSI-RS $\hat{E}_s / I_{ot}$ | | dB | 2 | 6.46 | 0.24 |
| RSRPNote3 | Bands FDD_A | dBm/15 kHz | -121 | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4dB) |
| | Bands FDD_B | | -120.5 | | |
| | Bands FDD_C | | -120 | | |
| | Bands FDD_D | | -119.5 | | |
| | Bands FDD_E, FDD_F Note 6 | | -119 | | |

| | | | | | |
|---|---|---|---|---|---|
| | Bands FDD_G | | -118 | | |
| | Bands FDD_H | | -117.5 | | |
| CSI-RSRP Note3 | Bands FDD_A | dBm/15 kHz | -115 | (CSI-RSRP for Cell 1 +8dB) | (CSI-RSRP for Cell 1 +4dB) |
| | Bands FDD_B | | -114.5 | | |
| | Bands FDD_C | | -114 | | |
| | Bands FDD_D | | -113.5 | | |
| | Bands FDD_E, FDD_F Note 6 | | -113 | | |
| | Bands FDD_G | | -112 | | |
| | Bands FDD_H | | -111.5 | | |
| Io Note3 | Bands FDD_A | dBm/9 MHz | -87.76 | (Io for Channel 1 +5.33dB) | |
| | Bands FDD_B | | -87.26 | | |
| | Bands FDD_C | | -86.76 | | |
| | Bands FDD_D | | -86.26 | | |
| | Bands FDD_E, FDD_F Note 6 | | -85.76 | | |
| | Bands FDD_G | | -84.76 | | |
| | Bands FDD_H | | -84.26 | | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | -4 | 3 | -1 |
| CSI-RS $\hat{E}_s / N_{oc}$ | | dB | 2 | 9 | 5 |
| Propagation condition | | - | AWGN | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement subframe.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.

Note 6: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 7: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.

Note 8: E-UTRA operating band groups are as defined in Section 3.5.

Note 9: DMTC configurations are provided to the UE in the *measDS-Config* (in TS36.331) before the beginning of the test.

**Table 9.1.35.5-2: CSI-RSRP FDD Carrier Aggregation absolute accuracy requirements for the reported values**

| | Event | Bands FDD_A | Bands FDD_B | Bands FDD_C | Bands FDD_D | Bands FDD_E, FDD_F | Bands FDD_G | Bands FDD_H |
|---|---|---|---|---|---|---|---|---|
| **Normal Conditions** | | | | | | | | |
| Lowest reported value (Cell 1) | Cell 1 | CSI_RSRP_19 | CSI_RSRP_20 | CSI_RSRP_20 | CSI_RSRP_21 | CSI_RSRP_21 | CSI_RSRP_22 | CSI_RSRP_23 |
| Highest reported value (Cell 1) | | CSI_RSRP_32 | CSI_RSRP_32 | CSI_RSRP_33 | CSI_RSRP_33 | CSI_RSRP_34 | CSI_RSRP_35 | CSI_RSRP_35 |
| Lowest reported value (Cell 2) | Cell 2 | CSI_RSRP_27 | CSI_RSRP_28 | CSI_RSRP_28 | CSI_RSRP_29 | CSI_RSRP_29 | CSI_RSRP_30 | CSI_RSRP_31 |
| Highest reported value (Cell 2) | | CSI_RSRP_40 | CSI_RSRP_40 | CSI_RSRP_41 | CSI_RSRP_41 | CSI_RSRP_42 | CSI_RSRP_43 | CSI_RSRP_43 |
| **Extreme Conditions** | | | | | | | | |
| Lowest reported value (Cell 1) | Cell 1 | CSI_RSRP_15 | CSI_RSRP_15 | CSI_RSRP_16 | CSI_RSRP_16 | CSI_RSRP_17 | CSI_RSRP_18 | CSI_RSRP_18 |
| Highest reported value (Cell 1) | | CSI_RSRP_36 | CSI_RSRP_37 | CSI_RSRP_37 | CSI_RSRP_38 | CSI_RSRP_38 | CSI_RSRP_39 | CSI_RSRP_40 |
| Lowest reported value (Cell 2) | Cell 2 | CSI_RSRP_23 | CSI_RSRP_23 | CSI_RSRP_24 | CSI_RSRP_24 | CSI_RSRP_25 | CSI_RSRP_26 | CSI_RSRP_26 |
| Highest reported value (Cell 2) | | CSI_RSRP_44 | CSI_RSRP_45 | CSI_RSRP_45 | CSI_RSRP_46 | CSI_RSRP_46 | CSI_RSRP_47 | CSI_RSRP_48 |
| Note: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2. | | | | | | | | |

**Table 9.1.35.5-3: CSI-RSRP FDD Carrier Aggregation relative accuracy requirements for the reported values**

| | Event | All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | CSI_RSRP_Cell 1 – 8 |
| Highest reported value (Cell 2) | | CSI_RSRP_Cell 1 – 1 |
| Lowest reported value (Cell 3) | Cell 2-3 | CSI_RSRP_Cell 2 + 2 |
| Highest reported value (Cell 3) | | CSI_RSRP_Cell 2 + 15 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | CSI_RSRP_Cell 1 – 9 |
| Highest reported value (Cell 2) | | CSI_RSRP_Cell 1 + 1 |
| Lowest reported value (Cell 3) | Cell 2-3 | CSI_RSRP_Cell 2 – 1 |
| Highest reported value (Cell 3) | | CSI_RSRP_Cell 2 + 17 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.36 TDD absolute and relative CSI-RSRP accuracies for E-UTRAN Carrier Aggregation in CSI-RS based discovery signal

### 9.1.36.1 Test purpose

The purpose of this test is to verify that the TDD CSI-RSRP absolute and relative measurement accuracies for carrier aggregation in CSI-RS based discovery signal are within the specified limits. This test will verify the absolute CSI-RSRP accuracy requirements of the primary component carrier, the absolute CSI-RSRP accuracy requirements of the secondary component carrier, and the relative CSI-RSRP accuracy requirements of the secondary component carrier. The test will also verify the primary and secondary component carrier relative CSI-RSRP accuracy requirement.

### 9.1.36.2 Test applicability

This test applies to all types to E-UTRA TDD UE release 12 and forward that supports CA and CSI-RS based discovery signals measurement.

### 9.1.36.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.35.3

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.14.3, 9.1.15.2, and A.9.1.36.

### 9.1.36.4 Test description

#### 9.1.36.4.1 Initial conditions

Same initial conditions as in clause 9.1.35.4.1 with the following exception:

- Message contents are defined in clause 9.1.36.4.3.

#### 9.1.36.4.2 Test procedure

Same test procedure as in clause 9.1.35.4.2 with the following exceptions:

- Instead of Table 9.1.35.5-1 → use Table 9.1.36.5-1.
- Instead of Table 9.1.35.5-2 → use Table 9.1.36.5-2.
- Instead of Table 9.1.35.5-3 → use Table 9.1.36.5-3.

#### 9.1.36.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.36.4.3-1: Common Exception message for CSI-RSRP TDD Carrier Aggregation accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.8-3<br>Table H.3.8-4<br>Table H.3.8-5 |

**Table 9.1.36.4.3-2: *MeasDS-Config-DEFAULT-PCell*: Additional CSI-RSRP TDD Carrier Aggregation accuracy test requirement for PCell**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1B: MeasDS-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasDS-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 0 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationTDD-r12 | 2 | | |
| } | | | |
| measCSI-RS-ToAddModList-r12 SEQUENCE (SIZE (1..maxCSI-RS-Meas-r12)) OF SEQUENCE { | 1 entry | | |
| MeasCSI-RS-Config-r12 SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| physCellId-r12 | PhysCellId of Cell 1 | INTEGER(0..503) | |
| scramblingIdentity-r12 | 0 | | |
| resourceConfig-r12 | 2 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | dB0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.36.4.3-3: *MeasDS-Config-DEFAULT-SCell*: Additional CSI-RSRP TDD Carrier Aggregation accuracy test requirement for SCell**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1B: MeasDS-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasDS-Config-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationTDD-r12 | 2 | | |
| } | | | |
| measCSI-RS-ToAddModList-r12 SEQUENCE (SIZE (1..maxCSI-RS-Meas-r12)) OF SEQUENCE { | 2 entries | | |
| MeasCSI-RS-Config-r12[0] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | | |
| physCellId-r12 | PhysCellId of Cell 2 | INTEGER(0..503) | |
| scramblingIdentity-r12 | 0 | | |
| resourceConfig-r12 | 4 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | dB0 | | |
| } | | | |
| MeasCSI-RS-Config-r12[1] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 2 | | |
| physCellId-r12 | PhysCellId of Cell 3 | INTEGER(0..503) | |
| scramblingIdentity-r12 | 1 | | |
| resourceConfig-r12 | 6 | | |
| subframeOffset-r12 | 0 | | |
| csi-RS-IndividualOffset-r12 | dB0 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.36.4.3-4: *MeasResults-PCell*: Additional CSI-RSRP TDD Carrier Aggregation accuracy test requirement for PCell**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell EQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultCSI-RS-List-r12 SEQUENCE (SIZE(1..maxCellReport)) OF SEQUENCE { | | | |
| MeasResultCSI-RS-r12 SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | Cell 1 | |
| csi-RSRP-Result-r12 | INTEGER(0..97) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.1.36.4.3-5: *MeasResults-SCell*: Additional CSI-RSRP TDD Carrier Aggregation accuracy test requirement for SCell**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 2 | | |
| measResultPCell EQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultCSI-RS-List-r12 SEQUENCE (SIZE(1..maxCellReport)) OF SEQUENCE { | | | |
| MeasResultCSI-RS-r12[0] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 1 | Cell 2 | |
| csi-RSRP-Result-r12 | INTEGER(0..97) | Set according to specific test | |
| } | | | |
| MeasResultCSI-RS-r12[1] SEQUENCE { | | | |
| measCSI-RS-Id-r12 | 2 | Cell 3 | |
| csi-RSRP-Result-r12 | INTEGER(0..97) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |

## 9.1.36.5 Test requirement

Table 9.1.36.5-1 defines the primary level settings including test tolerances for all tests.

The CSI-RSRP absolute accuracy shall meet the reported values test requirements in Table 9.1.36.5-2.

The CSI-RSRP relative accuracy shall meet the reported values test requirements in Table 9.1.36.5-3.

**Table 9.1.36.5-1: CSI-RSRP TDD Carrier Aggregation test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| E-UTRA RF Channel Number | | | 1 | 2 | |
| $BW_{channel}$ | | MHz | 10 | | |
| Special subframe configuration[Note1] | | | 6 | | |
| Uplink/downlink configuration[Note1] | | | 1 | | |
| Timing offset to Cell 1 | | µs | - | 0 | 3 |
| Time alignment error between cell 2 and cell 1 | | | - | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1 | - |
| DMTC period | | ms | 160 | 160 | |
| DMTC period offset | | ms | 0 | 10 | |
| Discovery signal occasion duration | | ms | 2 | 2 | |
| CSI-RS resource configuration | | | 2 | 4 | 6 |
| CSI-RS periodicity | | ms | 10 | 10 | 10 |
| CSI-RS subframe offset | | ms | 0 | 0 | 0 |
| CSI-RS individual offset[2] | | dB | 0 | 0 | 0 |
| CSI-RS muting | | | Enable | Enable | Enable |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA[Note2] | | | | | |
| OCNG_RB[Note2] | | | | | |
| p-C-r10[2] | | dB | 6 | 6 | 6 |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -117 | ( $N_{oc}$ for Channel 1 +1dB) | |
| | Bands TDD_C | | -116 | | |
| | Bands TDD_E | | -115 | | |
| CRS $\hat{E}_s / I_{ot}$ | | dB | -4 | 0.46 | -5.76 |
| CSI-RS $\hat{E}_s / I_{ot}$ | | dB | 2 | 6.46 | 0.24 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -121 | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4dB) |
| | Bands TDD_C | | -120 | | |
| | Bands TDD_E | | -119 | | |
| CSI-RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -115 | (CSI-RSRP for Cell 1 +8dB) | (CSI-RSRP for Cell 1 +4dB) |
| | Bands TDD_C | | -114 | | |
| | Bands TDD_E | | -113 | | |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -87.76 | (Io for Channel 1 +5.33dB) | |
| | Bands TDD_C | | -86.76 | | |
| | Bands TDD_E | | -85.76 | | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | -4 | 3 | -1 |

| | | | | |
|---|---|---|---|---|
| CSI-RS $\hat{E}_s / N_{oc}$ | dB | 2 | 9 | 5 |
| Propagation condition | - | AWGN | | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement subframe.
Note 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 6: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.
Note 7: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 8: E-UTRA operating band groups are as defined in Section 3.5.
Note 9: DMTC configurations are provided to the UE in the *measDS-Config* (in TS36.331) before the beginning of the test.

**Table 9.1.36.5-2: CSI-RSRP TDD Carrier Aggregation absolute accuracy requirements for the reported values**

| | Event | Bands TDD_A | Bands TDD_C | Bands TDD_E |
|---|---|---|---|---|
| **Normal Conditions** | | | | |
| Lowest reported value (Cell 1) | Cell 1 | CSI_RSRP_19 | CSI_RSRP_20 | CSI_RSRP_21 |
| Highest reported value (Cell 1) | | CSI_RSRP_32 | CSI_RSRP_33 | CSI_RSRP_34 |
| Lowest reported value (Cell 2) | Cell 2 | CSI_RSRP_27 | CSI_RSRP_28 | CSI_RSRP_29 |
| Highest reported value (Cell 2) | | CSI_RSRP_40 | CSI_RSRP_41 | CSI_RSRP_42 |
| **Extreme Conditions** | | | | |
| Lowest reported value (Cell 1) | Cell 1 | CSI_RSRP_15 | CSI_RSRP_16 | CSI_RSRP_17 |
| Highest reported value (Cell 1) | | CSI_RSRP_36 | CSI_RSRP_37 | CSI_RSRP_38 |
| Lowest reported value (Cell 2) | Cell 2 | CSI_RSRP_23 | CSI_RSRP_24 | CSI_RSRP_25 |
| Highest reported value (Cell 2) | | CSI_RSRP_44 | CSI_RSRP_45 | CSI_RSRP_46 |

Note: The band of Cell 1 determines the levels for both Cell 1 and Cell 2, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2.

**Table 9.1.36.5-3: CSI-RSRP TDD Carrier Aggregation relative accuracy requirements for the reported values**

| | Event | All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | CSI_RSRP_Cell 1 – 8 |
| Highest reported value (Cell 2) | | CSI_RSRP_Cell 1 – 1 |
| Lowest reported value (Cell 3) | Cell 2-3 | CSI_RSRP_Cell 2 + 2 |
| Highest reported value (Cell 3) | | CSI_RSRP_Cell 2 + 15 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | CSI_RSRP_Cell 1 – 9 |
| Highest reported value (Cell 2) | | CSI_RSRP_Cell 1 + 1 |
| Lowest reported value (Cell 3) | Cell 2-3 | CSI_RSRP_Cell 2 – 1 |
| Highest reported value (Cell 3) | | CSI_RSRP_Cell 2 + 17 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.1.37 3DL PCell in FDD RSRP for E-UTRAN in Carrier Aggregation

## 9.1.37.1 Test purpose

To verify that TDD-FDD RSRP absolute and relative measurement accuracy in carrier aggregation with PCell in FDD are within the specified limits under AWGN propagation conditions. This test will verify the absolute RSRP accuracy requirements of the primary component carrier, the absolute RSRP accuracy requirements of the secondary component carriers, and the relative RSRP accuracy requirements of the secondary component carriers. The test will also verify the primary and secondary component carriers' relative RSRP accuracy requirement.

## 9.1.37.2 Test applicability

This test applies to all types of E-UTRA UE release 12 and forward that support E-UTRA FDD and TDD and 3DL CA with FDD as PCell.

## 9.1.37.3 Minimum conformance requirements

The RSRP measurements of cells on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1. The RSRP measurements of cells on the secondary component carriers shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1.

The absolute accuracy requirements in table 9.1.37.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.
- $RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.1.37.3-1: RSRP Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 / dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 / N/A | -70 |
| | | | FDD_B | -120.5 / N/A | -70 |
| | | | FDD_C, TDD_C | -120 / N/A | -70 |
| | | | FDD_D | -119.5 / N/A | -70 |
| | | | FDD_E, TDD_E | -119 / N/A | -70 |
| | | | FDD_F | -118.5 / N/A | -70 |
| | | | FDD_G | -118 / N/A | -70 |
| | | | FDD_H | -117.5 / N/A | -70 |
| | | | FDD_N | -114.5 / N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The RSRP relative measurements of cells on the PCC compared with measurements of cells on any of the SCCs shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.3.2. The RSRP relative measurements of cells on any SCC shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.2.2.

The accuracy requirements in Table 9.1.37.3-2 for PCC-SCC relative accuracy requirements are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.

- Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.
- RSRP1,2$|_{dBm}$ according to Annex I.3.4 for a corresponding Band
- $\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$
- | Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.37.3-2: RSRP PCC-SCC relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/BW$_{Channel}$ |
| ±4.5 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The requirements in Table 9.1.37.3-3 for SCC relative accuracy are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.
- RSRP1,2$|_{dBm}$ according to Annex I.3.8 for a corresponding Band.

**Table 9.1.37.3-3: RSRP relative accuracy for SCCs**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/BW$_{Channel}$ |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 3 | Note 3 | Note 3 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.<br>NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 5: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.37.3-4. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.37.3-4: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.11.1, 9.1.11.2, 9.1.11.3, 9.1.4, and A.9.1.37.

## 9.1.37.4 Test description

### 9.1.37.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 9.1.37.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.66 as appropriate.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.1.37.4.3.
4. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on SCC1, and Cell 3 is the neighbouring cell on SCC1. Cell 4 is SCell on SCC2, and Cell 5 is the neighbouring cell on SCC2. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2, Cell 3, Cell 4 and Cell 5 shall be powered OFF.

### 9.1.37.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCell according to Annex C.0, C.1 and C.2 for all downlink physical channels.
3. The SS shall configure SCells (cell 2 and cell 4) on the SCCs as per TS 36.508 [7] clause 5.2A.4 [with the message content exceptions defined in clause 9.1.37.4.3].
4. SS activates SCCs by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).
5. Set the parameters according to Table 9.1.37.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.
6. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
7. The UE shall transmit RRCConnectionReconfigurationComplete message.
8. UE shall transmit periodically MeasurementReport messages.
9. After 10s wait from Step 6, SS shall check the reported RSRP values in periodical MeasurementReport messages.

The reported RSRP values for Cell 1, Cell 2 and Cell 4 are compared to the actual RSRP values according to Table 9.1.37.5-2. This counts respectively as a Pass or Fail for the events “Cell 1”, “Cell 2” and “Cell 4”. If the UE fails to report the measurement value for Cell 2 or Cell 4, the number of failed iterations for the respective event is increased by one.

The reported RSRP value for Cell 2, Cell 4, Cell 3 and Cell 5 are compared to the reported RSRP value of other cells for each MeasurementReport message according to Table 9.1.37.5-3. These count respectively as a Pass or Fail for the events “Cell 1-2”, “Cell 1-4”, “Cell 2-3”, and “Cell 4-5”. If the UE fails to report the measurement value for Cell 2, Cell 3, Cell 4, or Cell 5, the number of failed iterations for the respective event is increased by one.

10. SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events “Cell 1”, “Cell 2”, “Cell 4”, “Cell 1-2”, “Cell 1-4”, “Cell 2-3”, and “Cell 4-5” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.

If all events pass for each scenario, the test passes. If one event fails, the test fails.

### 9.1.37.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.37.4.3-1: Common Exception messages for FDD RSRP Accuracy E-UTRA for Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-3<br>Table H.4.1-7<br>Table H.4.2-5 |

## 9.1.37.5 Test requirement

Table 9.1.37.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP absolute accuracy shall meet the reported values test requirements in table 9.1.37.5-2.

The RSRP relative accuracy shall meet the reported values test requirements in table 9.1.37.5-3.

**Table 9.1.37.5-1: 3DL PCell in FDD RSRP for E-UTRAN in Carrier Aggregation test parameters**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 | Cell 4 | Cell 5 |
|---|---|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | 2 | | 3 | |
| $BW_{channel}$ | | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | |
| Special subframe configuration[Note1] | | | - | 6 | | 6 | |
| Uplink/downlink configuration[Note1] | | | - | 1 | | 1 | |
| Measurement bandwidth | | $n_{PRB}$ | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | |
| PDSCH Reference measurement channel | | | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | - | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | - | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD |
| OCNG Patterns | | | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | |
| PSS_RA | | | | | | | |
| SSS_RA | | | | | | | |
| PCFICH_RB | | | | | | | |
| PHICH_RA | | | | | | | |
| PHICH_RB | | | | | | | |
| PDCCH_RA | | | | | | | |
| PDCCH_RB | | | | | | | |
| PDSCH_RA | | | | | | | |
| PDSCH_RB | | | | | | | |
| OCNG_RA[Note2] | | | | | | | |
| OCNG_RB[Note2] | | | | | | | |
| $N_{oc}$[Note3] | Bands TDD_A | dBm/15 kHz | - | ( $N_{oc}$ for Channel 1 +1dB) | | ( $N_{oc}$ for Channel 1 +1dB) | |
| | Bands TDD_C | | | | | | |
| | Bands TDD_E | | | | | | |
| | Bands FDD_A | | -117 | - | | - | |
| | Bands FDD_B | | -116.5 | | | | |
| | Bands FDD_C | | -116 | | | | |
| | Bands FDD_D | | -115.5 | | | | |
| | Bands FDD_E, FDD_F[Note 7] | | -115 | | | | |
| | Bands FDD_G | | -114 | | | | |
| | Bands FDD_H | | -113.5 | | | | |
| $\hat{E}_s / N_{oc}$ | | dB | -4 | 3 | -0.2 | 3 | -0.2 |
| $\hat{E}_s / I_{ot}$ | | dB | -4 | 0.09 | -4.96 | 0.09 | -4.96 |
| RSRP[Note4] | Bands TDD_A | dBm/15 | - | (RSRP for Cell | (RSRP for | (RSRP for | (RSRP for |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | Bands TDD_C | kHz | | 1 +8dB) | Cell 1 +4.8dB) | Cell 1 +8dB) | Cell 1 +4.8dB) |
| | Bands TDD_E | | | | | | |
| | Bands FDD_A | | -121 | - | - | - | - |
| | Bands FDD_B | | -120.5 | | | | |
| | Bands FDD_C | | -120 | | | | |
| | Bands FDD_D | | -119.5 | | | | |
| | Bands FDD_E, FDD_F Note 7 | | -119 | | | | |
| | Bands FDD_G | | -118 | | | | |
| | Bands FDD_H | | -117.5 | | | | |
| Io Note4 | Bands TDD_A | dBm/ $BW_{channel}$ | - | (Io for Channel 1 +5.51dB +10log ($N_{RB\ channel2}$ / $N_{RB\ channel\ 1}$)) | | (Io for Channel 1 +5.51dB +10log ($N_{RB\ channel3}$ / $N_{RB\ channel\ 1}$)) | |
| | Bands TDD_C | | | | | | |
| | Bands TDD_E | | | | | | |
| | Bands FDD_A | | -87.76+10log($N_{RB,c}$/50) | - | | - | |
| | Bands FDD_B | | -87.26+10log($N_{RB,c}$/50) | | | | |
| | Bands FDD_C | | -86.76+10log($N_{RB,c}$/50) | | | | |
| | Bands FDD_D | | -86.26+10log($N_{RB,c}$/50) | | | | |
| | Bands FDD_E, FDD_F Note 7 | | -85.76 +10log($N_{RB,c}$/50) | | | | |
| | Bands FDD_G | | -84.76 +10log($N_{RB,c}$/50) | | | | |
| | Bands FDD_H | | -84.26 +10log($N_{RB,c}$/50) | | | | |
| Propagation condition | | - | AWGN | AWGN | AWGN | AWGN | AWGN |
| Antenna Configuration | | - | 1x2 | 1x2 | 1x2 | 1x2 | 1x2 |
| Timing offset to cell 1 | | µs | - | 0 | 3 | 0 | 3 |
| Time alignment error relative to cell 1 Note 8 | | | - | ≤ TAE | - | ≤ TAE | - |
| Time alignment error relative to cell 2 Note 8 | | | - | - | - | ≤ TAE | - |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 6: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.
Note 7: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 8: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.
Note 9: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.37.5-2: 3DL PCell in FDD RSRP for E-UTRAN in Carrier Aggregation absolute accuracy requirements for the reported values**

| Band of Cell 1 on Primary Component Carrier | Event | Bands FDD_A | Bands FDD_B | Bands FDD_C | Bands FDD_D | Bands FDD_E, FDD_F | Bands FDD_G | Bands FDD_H |
|---|---|---|---|---|---|---|---|---|
| Normal Conditions | | | | | | | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRP_13 | RSRP_14 | RSRP_14 | RSRP_15 | RSRP_15 | RSRP_16 | RSRP_17 |
| Highest reported value (Cell 1) | | RSRP_26 | RSRP_26 | RSRP_27 | RSRP_27 | RSRP_28 | RSRP_29 | RSRP_29 |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_21 | RSRP_22 | RSRP_22 | RSRP_23 | RSRP_23 | RSRP_24 | RSRP_25 |
| Highest reported value (Cell 2) | | RSRP_34 | RSRP_34 | RSRP_35 | RSRP_35 | RSRP_36 | RSRP_37 | RSRP_37 |
| Lowest reported value (Cell 4) | Cell 4 | RSRP_21 | RSRP_22 | RSRP_22 | RSRP_23 | RSRP_23 | RSRP_24 | RSRP_25 |
| Highest reported value (Cell 4) | | RSRP_34 | RSRP_34 | RSRP_35 | RSRP_35 | RSRP_36 | RSRP_37 | RSRP_37 |
| Extreme Conditions | | | | | | | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRP_9 | RSRP_9 | RSRP_10 | RSRP_10 | RSRP_11 | RSRP_12 | RSRP_12 |
| Highest reported value (Cell 1) | | RSRP_30 | RSRP_31 | RSRP_31 | RSRP_32 | RSRP_32 | RSRP_33 | RSRP_34 |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_17 | RSRP_17 | RSRP_18 | RSRP_18 | RSRP_19 | RSRP_20 | RSRP_20 |
| Highest reported value (Cell 2) | | RSRP_38 | RSRP_39 | RSRP_39 | RSRP_40 | RSRP_40 | RSRP_41 | RSRP_42 |
| Lowest reported value (Cell 4) | Cell 4 | RSRP_17 | RSRP_17 | RSRP_18 | RSRP_18 | RSRP_19 | RSRP_20 | RSRP_20 |
| Highest reported value (Cell 4) | | RSRP_38 | RSRP_39 | RSRP_39 | RSRP_40 | RSRP_40 | RSRP_41 | RSRP_42 |
| Note: The band of Cell 1 determines the levels for Cell 1, Cell 2 and Cell 4, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2 or Cell 4. | | | | | | | | |

**Table 9.1.37.5-3: 3DL PCell in FDD RSRP for E-UTRAN in Carrier Aggregation relative accuracy requirements for the reported values**

| | Event | |
|---|---|---|
| Normal Conditions, all bands | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 + 2 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 15 |
| Lowest reported value (Cell 4) | Cell 1-4 | RSRP_Cell 1 + 2 |
| Highest reported value (Cell 4) | | RSRP_Cell 1 + 15 |
| Extreme Conditions, all bands | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 – 1 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 17 |
| Lowest reported value (Cell 4) | Cell 1-4 | RSRP_Cell 1 – 1 |
| Highest reported value (Cell 4) | | RSRP_Cell 1 + 17 |
| Normal and Extreme Conditions, all bands | | |
| Lowest reported value (Cell 3) | Cell 2-3 | RSRP_Cell 2 – 8 |
| Highest reported value (Cell 3) | | RSRP_Cell 2 + 1 |
| Lowest reported value (Cell 5) | Cell 4-5 | RSRP_Cell 4 – 8 |
| Highest reported value (Cell 5) | | RSRP_Cell 4 + 1 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.38 3DL PCell in TDD RSRP for E-UTRAN in Carrier Aggregation

### 9.1.38.1 Test purpose

To verify that TDD-FDD RSRP absolute and relative measurement accuracy in carrier aggregation with PCell in TDD are within the specified limits under AWGN propagation conditions. This test will verify the absolute RSRP accuracy requirements of the primary component carrier, the absolute RSRP accuracy requirements of the secondary component carriers, and the relative RSRP accuracy requirements of the secondary component carriers. The test will also verify the primary and secondary component carriers' relative RSRP accuracy requirement.

### 9.1.38.2 Test applicability

This test applies to all types of E-UTRA UE release 12 and forward that support E-UTRA FDD and TDD and 3DL CA with TDD as PCell.

### 9.1.38.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.37.3.

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.11.1, 9.1.11.2, 9.1.11.3, 9.1.4, and A.9.1.38.

### 9.1.38.4 Test description

#### 9.1.38.4.1 Initial conditions

Same initial conditions as in clause 9.1.37.4.1 with the following exceptions:

- Message contents are defined in clause 9.1.38.4.3.

#### 9.1.38.4.2 Test procedure

Same test procedure as in clause 9.1.37.4.2 with the following exceptions:

- Instead of Table 9.1.37.5-1 → use Table 9.1.38.5-1.
- Instead of Table 9.1.37.5-2 → use Table 9.1.38.5-2.
- Instead of Table 9.1.37.5-3 → use Table 9.1.38.5-3.

#### 9.1.38.4.3 Message contents

Same message contents as in clause 9.1.37.4.3.9.1.38.5 Test requirement

Table 9.1.38.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP absolute accuracy shall meet the reported values test requirements in table 9.1.38.5-2.

The RSRP relative accuracy shall meet the reported values test requirements in table 9.1.38.5-3.

**Table 9.1.38.5-1: 3DL PCell in TDD RSRP for E-UTRAN in Carrier Aggregation test parameters**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 | Cell 4 | Cell 5 |
|---|---|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | 2 | | 3 | |
| $BW_{channel}$ | | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | |
| Special subframe configuration Note1 | | | 6 | - | | - | |
| Uplink/downlink configuration Note1 | | | 1 | - | | - | |
| Measurement bandwidth | | $n_{PRB}$ | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | |
| PDSCH Reference measurement channel | | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | - | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | - | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD |
| OCNG Patterns | | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | |
| PSS_RA | | | | | | | |
| SSS_RA | | | | | | | |
| PCFICH_RB | | | | | | | |
| PHICH_RA | | | | | | | |
| PHICH_RB | | | | | | | |
| PDCCH_RA | | | | | | | |
| PDCCH_RB | | | | | | | |
| PDSCH_RA | | | | | | | |
| PDSCH_RB | | | | | | | |
| OCNG_RA Note1 | | | | | | | |
| OCNG_RB Note1 | | | | | | | |
| $N_{oc}$ Note3 | Bands FDD_A | dBm/15 kHz | - | ( $N_{oc}$ for Channel 1 +1dB) | | ( $N_{oc}$ for Channel 1 +1dB) | |
| | Bands FDD_B | | | | | | |
| | Bands FDD_C | | | | | | |
| | Bands FDD_D | | | | | | |
| | Bands FDD_E, FDD_F Note 7 | | | | | | |
| | Bands FDD_G | | | | | | |
| | Bands FDD_H | | | | | | |
| | Bands TDD_A | | -117 | - | | - | |
| | Bands TDD_C | | -116 | | | | |
| | Bands TDD_E | | -115 | | | | |
| $\hat{E}_s / N_{oc}$ | | dB | -4 | 3 | -0.2 | 3 | -0.2 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $\hat{E}_s/I_{ot}$ | | dB | -4 | 0.09 | -4.96 | 0.09 | -4.96 |
| RSRP[Note4] | Bands FDD_A | dBm/15 kHz | - | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4.8dB) | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4.8dB) |
| | Bands FDD_B | | | | | | |
| | Bands FDD_C | | | | | | |
| | Bands FDD_D | | | | | | |
| | Bands FDD_E, FDD_F [Note 7] | | | | | | |
| | Bands FDD_G | | | | | | |
| | Bands FDD_H | | | | | | |
| | Bands TDD_A | | -121 | - | - | - | - |
| | Bands TDD_C | | -120 | | | | |
| | Bands TDD_E | | -119 | | | | |
| Io[Note4] | Bands FDD_A | dBm/ $BW_{channel}$ | - | (Io for Channel 1 +5.51dB +10log ($N_{RB\ channel2}$ / $N_{RB\ channel\ 1}$)) | | (Io for Channel 1 +5.51dB +10log ($N_{RB\ channel3}$ / $N_{RB\ channel\ 1}$)) | |
| | Bands FDD_B | | | | | | |
| | Bands FDD_C | | | | | | |
| | Bands FDD_D | | | | | | |
| | Bands FDD_E, FDD_F [Note 7] | | | | | | |
| | Bands FDD_G | | | | | | |
| | Bands FDD_H | | | | | | |
| | Bands TDD_A | | -87.76+10log($N_{RB,c}$/50) | - | | - | |
| | Bands TDD_C | | -86.76+10log($N_{RB,c}$/50) | | | | |
| | Bands TDD_E | | -85.76+10log($N_{RB,c}$/50) | | | | |
| Propagation condition | | - | AWGN | AWGN | AWGN | AWGN | AWGN |
| Antenna Configuration | | - | 1x2 | 1x2 | 1x2 | 1x2 | 1x2 |
| Timing offset to cell 1 | | µs | - | 0 | 3 | 0 | 3 |
| Time alignment error relative to cell 1 [Note 8] | | | - | ≤ TAE | - | ≤ TAE | - |
| Time alignment error relative to cell 2[Note8] | | | - | - | - | ≤ TAE | - |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 6: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.
Note 7: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 8: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.
Note 9: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.38.5-2: 3DL PCell in TDD RSRP for E-UTRAN in Carrier Aggregation absolute accuracy requirements for the reported values**

| Band of Cell 1 on Primary Component Carrier | Event | Bands TDD_A | Bands TDD_C | Bands TDD_E |
|---|---|---|---|---|
| Normal Conditions | | | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRP_13 | RSRP_14 | RSRP_15 |
| Highest reported value (Cell 1) | | RSRP_26 | RSRP_27 | RSRP_28 |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_21 | RSRP_22 | RSRP_23 |
| Highest reported value (Cell 2) | | RSRP_34 | RSRP_35 | RSRP_36 |
| Lowest reported value (Cell 4) | Cell 4 | RSRP_21 | RSRP_22 | RSRP_23 |
| Highest reported value (Cell 4) | | RSRP_34 | RSRP_35 | RSRP_36 |
| Extreme Conditions | | | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRP_9 | RSRP_10 | RSRP_11 |
| Highest reported value (Cell 1) | | RSRP_30 | RSRP_31 | RSRP_32 |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_17 | RSRP_18 | RSRP_19 |
| Highest reported value (Cell 2) | | RSRP_38 | RSRP_39 | RSRP_40 |
| Lowest reported value (Cell 4) | Cell 4 | RSRP_17 | RSRP_18 | RSRP_19 |
| Highest reported value (Cell 4) | | RSRP_38 | RSRP_39 | RSRP_40 |
| Note: The band of Cell 1 determines the levels for Cell 1, Cell 2 and Cell 4, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2 or Cell 4. | | | | |

**Table 9.1.38.5-3: 3DL PCell in TDD RSRP for E-UTRAN in Carrier Aggregation relative accuracy requirements for the reported values**

| | Event | |
|---|---|---|
| Normal Conditions, all bands | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 + 2 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 15 |
| Lowest reported value (Cell 4) | Cell 1-4 | RSRP_Cell 1 + 2 |
| Highest reported value (Cell 4) | | RSRP_Cell 1 + 15 |
| Extreme Conditions, all bands | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 – 1 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 17 |
| Lowest reported value (Cell 4) | Cell 1-4 | RSRP_Cell 1 – 1 |
| Highest reported value (Cell 4) | | RSRP_Cell 1 + 17 |
| Normal and Extreme Conditions, all bands | | |
| Lowest reported value (Cell 3) | Cell 2-3 | RSRP_Cell 2 – 8 |
| Highest reported value (Cell 3) | | RSRP_Cell 2 + 1 |
| Lowest reported value (Cell 5) | Cell 4-5 | RSRP_Cell 4 – 8 |
| Highest reported value (Cell 5) | | RSRP_Cell 4 + 1 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.39 3DL FDD RSRP for E-UTRAN in Carrier Aggregation

### 9.1.39.1 Test purpose

To verify that FDD absolute and relative RSRP measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the absolute RSRP accuracy requirements of the primary component carrier, the absolute RSRP accuracy requirements of the secondary component carriers, and the relative RSRP accuracy requirements of the secondary component carriers. The test will also verify the primary and secondary component carriers' relative RSRP accuracy requirement.

### 9.1.39.2 Test applicability

This test case applies to all types of E-UTRA FDD UE release 10 and 11 that support 3DL with Intra-band contiguous CA, or 3DL with Inter-band CA, or 3DL with Intra-band contiguous and Inter-band CA.

This test case also applies to all types of E-UTRA FDD UE release 11 that support 3DL with Intra-band non-contiguous and Inter-band CA, or 3DL with Intra-band non-contiguous and Intra-band contiguous CA.

### 9.1.39.3 Minimum conformance requirements

The RSRP measurements of cells on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1. The FDD RSRP measurements of cells on the secondary component carriers shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1.

The accuracy requirements in table 9.1.39.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports
- Conditions defined in 36.101[2] Section 7.3 for reference sensitivity are fulfilled.
- $RSRP_{|dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.1.39.3-1: RSRP FDD Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | $dBm/BW_{Channel}$ | $dBm/BW_{Channel}$ |
| ±6 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The FDD RSRP relative measurements of cells on the PCC compared with measurements of cells on any of the SCCs shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.3.2. The RSRP relative measurements of cells on any SCC shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.2.2.

The accuracy requirements in Table 9.1.39.3-2 for PCC-SCC relative accuracy requirements are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.
- $RSRP1,2_{|dBm}$ according to Annex I.3.4 for a corresponding Band
- $\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$
- | Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.39.3-2: RSRP PCC-SCC relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/$BW_{Channel}$ |
| ±6 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The requirements in Table 9.1.39.3-3 for SCC relative accuracy are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.
- $RSRP1,2|_{dBm}$ according to Annex I.3.8 for a corresponding Band.

**Table 9.1.39.3-3: RSRP relative accuracy for SCCs**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/$BW_{Channel}$ |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.39.3-4. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.39.3-4: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.11.1, 9.1.11.2, 9.1.11.3, 9.1.4, and A.9.1.39.

## 9.1.39.4 Test description

### 9.1.39.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 9.1.39.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.66 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.39.4.3.

4. Cell 1 is PCell on channel 1, and cell 2 and cell 4 are activated SCells on secondary component carriers SCC1 and SCC2 respectively. Cell 3 and cell 5 are neighbouring cells on secondary component carriers SCC1 and SCC2 respectively. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2, Cell 3, Cell 4 and Cell 5 shall be powered OFF.

### 9.1.39.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCells according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure SCells (Cell 2 and Cell 4) on the SCC1 and SCC2 as per TS 36.508 [7] clause 5.2A.4.

4. SS activates SCCs by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).

5. Set the parameters according to Tables 9.1.39.5-1 and 9.1.39.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.

6. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, SS shall check the reported RSRP values in periodical MeasurementReport messages.

   The reported RSRP values for Cell 1, Cell 2 and Cell 4 are compared to the actual RSRP values according to Table 9.1.39.5-3. These count respectively as a Pass or Fail for the events “Cell 1”, “Cell 2” and “Cell 4”. If the UE fails to report the measurement value for Cell 1, Cell 2 or Cell 4, the number of failed iterations for the respective event is increased by one.

The reported RSRP values for Cell 2, Cell 4, Cell 3 and Cell 5 are compared to the reported RSRP value of other cells for each MeasurementReport message according to Table 9.1.39.5-4. These count respectively as a Pass or Fail for the events “Cell 1-2”, “Cell 1-4”, “Cell 2-3”, and “Cell 4-5”. If the UE fails to report the measurement value for Cell 2, Cell 3, Cell 4, or Cell 5, the number of failed iterations for the respective event is increased by one.

10. SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

11. If test configuration is Inter-band carrier aggregation, repeat steps 1-10 for all relevant PCell/SCells scenarios defined in Table E-2 according to the UE declared capability for UL support (within CA operation) in the individual bands.

Each of the events “Cell 1”, “Cell 2”, “Cell 4”, “Cell 1-2”, “Cell 1-4”, “Cell 2-3”, and “Cell 4-5” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass for each configuration (without and with switched PCell/SCells scenario), the test passes. If one event fails, the test fails.

### 9.1.39.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.39.4.3-1: Common Exception messages for FDD RSRP Accuracy E-UTRA for Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-3<br>Table H.4.1-7<br>Table H.4.2-5 |

## 9.1.39.5 Test requirement

Tables 9.1.39.5-1 and 9.1.39.5-2 define the primary level settings including test tolerances for all tests.

The RSRP FDD absolute accuracy test shall meet the reported values test requirements in table 9.1.39.5-3.

The RSRP FDD relative accuracy test shall meet the reported values test requirements in table 9.1.39.5-4.

**Table 9.1.39.5-1: 3 DL FDD RSRP test parameters for E-UTRAN Carrier aggregation (cell #1, cell #2 and cell #3)**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | 2 | |
| $BW_{channel}$ | | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | |
| Measurement bandwidth | | $n_{PRB}$ | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD |
| OCNG Patterns defined in D.1 | | | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA Note1 | | | | | |
| OCNG_RB Note1 | | | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/ 15kHz | -117 | ( $N_{oc}$ for Channel 1 +1dB) | |
| | Bands FDD_C | | -116 | | |
| | Bands FDD_D | | -115.5 | | |
| | Bands FDD_E, FDD_F Note 6 | | -115 | | |
| | Bands FDD_G | | -114 | | |
| | Bands FDD_H | | -113.5 | | |
| $\hat{E}_s / N_{oc}$ | | dB | -4 | 3 | -0.2 |
| $\hat{E}_s / I_{ot}$ | | dB | -4 | 0.09 | -4.96 |
| RSRP Note3 | Bands FDD_A | dBm/ 15kHz | -121 | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4.8dB) |
| | Bands FDD_C | | -120 | | |
| | Bands FDD_D | | -119.5 | | |
| | Bands FDD_E, FDD_F Note 6 | | -119 | | |
| | Bands FDD_G | | -118 | | |
| | Bands FDD_H | | -117.5 | | |
| Io Note3 | Bands FDD_A | dBm/ $BW_{channel}$ | -87.76 +10log($N_{RB,c}$/50) | (Io for Channel 1 +5.51dB +10log ($N_{RB\ channel2}$ / $N_{RB\ channel\ 1}$)) | |
| | Bands FDD_C | | -86.76 +10log($N_{RB,c}$/50) | | |
| | Bands FDD_D | | -86.26 +10log($N_{RB,c}$/50) | | |
| | Bands FDD_E, FDD_F Note 6 | | -85.76 +10log($N_{RB,c}$/50) | | |
| | Bands FDD_G | | -84.76 +10log($N_{RB,c}$/50) | | |
| | Bands FDD_H | | -84.26 | | |

| | | | | | |
|---|---|---|---|---|---|
| | | | +10log($N_{RB,c}$/50) | | |
| Propagation Condition | | | AWGN | AWGN | AWGN |
| Antenna Configuration | | | 1x2 | 1x2 | 1x2 |
| Timing offset to Cell 1 | | μs | - | 0 | 3 |
| Time alignment error relative to cell 1 Note 7 | | | - | ≤ TAE | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.

Note 6: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 7: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

Note 8: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.39.5-2: 3 DL FDD RSRP test parameters for E-UTRAN Carrier aggregation (cell #4 and cell #5)**

| Parameter | | Unit | Cell 4 | Cell 5 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 3 | |
| $BW_{channel}$ | | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | |
| Measurement bandwidth | | $n_{PRB}$ | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD |
| OCNG Patterns defined in D.1 | | | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | 5MHz: OP.16 FDD<br>10MHz: OP.2 FDD<br>20MHz: OP.12 FDD |
| PBCH_RA | | dB | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA Note1 | | | | |
| OCNG_RB Note1 | | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/ 15kHz | ( $N_{oc}$ for Channel 1 +1dB) | |
| | Bands FDD_C | | | |
| | Bands FDD_D | | | |
| | Bands FDD_E, FDD_F Note 6 | | | |
| | Bands FDD_G | | | |
| | Bands FDD_H | | | |
| $\hat{E}_s / N_{oc}$ | | dB | 3 | -0.2 |
| $\hat{E}_s / I_{ot}$ | | dB | 0.09 | -4.96 |
| RSRP Note3 | Bands FDD_A | dBm/ 15kHz | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4.8dB) |
| | Bands FDD_C | | | |
| | Bands FDD_D | | | |
| | Bands FDD_E, FDD_F Note 6 | | | |
| | Bands FDD_G | | | |
| | Bands FDD_H | | | |
| Io Note3 | Bands FDD_A | dBm/ $BW_{channel}$ | (Io for Channel 1 +5.51dB +10log ($N_{RB\ channel3}$ / $N_{RB\ channel\ 1}$)) | |
| | Bands FDD_C | | | |
| | Bands FDD_D | | | |
| | Bands FDD_E, FDD_F Note 6 | | | |
| | Bands FDD_G | | | |
| | Bands FDD_H | | | |
| Propagation Condition | | | AWGN | AWGN |
| Antenna Configuration | | | 1x2 | 1x2 |
| Timing offset to Cell 1 | | µs | 0 | 3 |

| | | | |
|---|---|---|---|
| Time alignment error relative to cell 1[Note 7] | | ≤ TAE | - |
| Time alignment error relative to cell 2[Note 7] | | ≤ TAE | - |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.
Note 6: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 7: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.
Note 8: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.39.5-3: RSRP FDD Carrier Aggregation absolute accuracy requirements for the reported values**

| Band of Cell 1 on Primary Component Carrier | Event | Bands FDD_A | Bands FDD_C | Bands FDD_D | Bands FDD_E, FDD_F | Bands FDD_G | Bands FDD_H |
|---|---|---|---|---|---|---|---|
| Normal Conditions | | | | | | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRP_12 | RSRP_13 | RSRP_13 | RSRP_14 | RSRP_15 | RSRP_15 |
| Highest reported value (Cell 1) | | RSRP_27 | RSRP_28 | RSRP_29 | RSRP_29 | RSRP_30 | RSRP_31 |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_20 | RSRP_21 | RSRP_21 | RSRP_22 | RSRP_23 | RSRP_23 |
| Highest reported value (Cell 2) | | RSRP_35 | RSRP_36 | RSRP_37 | RSRP_37 | RSRP_38 | RSRP_39 |
| Lowest reported value (Cell 4) | Cell 4 | RSRP_20 | RSRP_21 | RSRP_21 | RSRP_22 | RSRP_23 | RSRP_23 |
| Highest reported value (Cell 4) | | RSRP_35 | RSRP_36 | RSRP_37 | RSRP_37 | RSRP_38 | RSRP_39 |
| Extreme Conditions | | | | | | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRP_09 | RSRP_10 | RSRP_10 | RSRP_11 | RSRP_12 | RSRP_12 |
| Highest reported value (Cell 1) | | RSRP_30 | RSRP_31 | RSRP_32 | RSRP_32 | RSRP_33 | RSRP_34 |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_17 | RSRP_18 | RSRP_18 | RSRP_19 | RSRP_20 | RSRP_20 |
| Highest reported value (Cell 2) | | RSRP_38 | RSRP_39 | RSRP_40 | RSRP_40 | RSRP_41 | RSRP_42 |
| Lowest reported value (Cell 4) | Cell 4 | RSRP_17 | RSRP_18 | RSRP_18 | RSRP_19 | RSRP_20 | RSRP_20 |
| Highest reported value (Cell 4) | | RSRP_38 | RSRP_39 | RSRP_40 | RSRP_40 | RSRP_41 | RSRP_42 |

Note: The band of Cell 1 determines the levels for Cell 1, Cell 2 and Cell 4, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2 or Cell 4.

**Table 9.1.39.5-4: RSRP FDD Carrier Aggregation relative accuracy requirements for the reported values**

| | Event | |
|---|---|---|
| Normal and Extreme Conditions, all bands | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 - 1 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 17 |
| Lowest reported value (Cell 4) | Cell 1-4 | RSRP_Cell 1 - 1 |
| Highest reported value (Cell 4) | | RSRP_Cell 1 + 17 |
| Normal and Extreme Conditions, all bands | | |
| Lowest reported value (Cell 3) | Cell 2-3 | RSRP_Cell 2 - 8 |
| Highest reported value (Cell 3) | | RSRP_Cell 2 + 1 |
| Lowest reported value (Cell 5) | Cell 4-5 | RSRP_Cell 4 - 8 |
| Highest reported value (Cell 5) | | RSRP_Cell 4 + 1 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.1.39_1 3DL FDD RSRP for E-UTRAN in Carrier Aggregation(Rel-12 and forward)

## 9.1.39_1.1 Test purpose

Same test purpose as in clause 9.1.39.1.

## 9.1.39_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward supporting 3DL CA.

## 9.1.39_1.3 Minimum conformance requirements

The FDD RSRP measurements of cells on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1. The FDD RSRP measurements of cells on the secondary component carriers shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1.

The accuracy requirements in table 9.1.39_1.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.
- $RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.1.39_1.3-1: RSRP Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 / dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 / N/A | -70 |
| | | | FDD_B | -120.5 / N/A | -70 |
| | | | FDD_C, TDD_C | -120 / N/A | -70 |
| | | | FDD_D | -119.5 / N/A | -70 |
| | | | FDD_E, TDD_E | -119 / N/A | -70 |
| | | | FDD_F | -118.5 / N/A | -70 |
| | | | FDD_G | -118 / N/A | -70 |
| | | | FDD_H | -117.5 / N/A | -70 |
| | | | FDD_N | -114.5 / N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 / -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The RSRP relative measurements of cells on the PCC compared with measurements of cells on any of the SCCs shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.3.2. The RSRP relative measurements of cells on any SCC shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.2.2.

The accuracy requirements in Table 9.1.39_1.3-2 for PCC-SCC relative accuracy requirements are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.
- RSRP1,2$|_{dBm}$ according to Annex I.3.4 for a corresponding Band
- $\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$
- | Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.39_1.3-2: RSRP PCC-SCC relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/BW$_{Channel}$ |
| ±4.5 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The requirements in Table 9.1.39_1.3-3 for SCC relative accuracy are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.
- RSRP1,2$|_{dBm}$ according to Annex I.3.8 for a corresponding Band.

**Table 9.1.39_1.3-3: RSRP relative accuracy for SCCs**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/BW$_{Channel}$ |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 3 | Note 3 | Note 3 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.<br>NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 5: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.39.1_1.3-4. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.39_1.3-4: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.11.1, 9.1.11.2, 9.1.11.3, 9.1.4, and A.9.1.39.

## 9.1.39_1.4 Test description

### 9.1.39_1.4.1 Initial conditions

Same initial conditions as in clause 9.1.39.4.1 with the following exceptions:

- Instead of clause 9.1.39.4.3 → use clause 9.1.39_1.4.3.

### 9.1.39_1.4.2 Test procedure

Same test procedure as in clause 9.1.39.4.2 with the following exceptions:

- Instead of Table 9.1.39.5-3 → use Table 9.1.39_1.5-1.
- Instead of Table 9.1.39.5-4 → use Table 9.1.39_1.5-2.

### 9.1.39_1.4.3 Message contents

Same message contents as in clause 9.1.39.4.3.

## 9.1.39_1.5 Test requirement

Same test requirements as in clause 9.1.39.5 with the following exceptions:

- Instead of Table 9.1.39.5-3 → use Table 9.1.39_1.5-1.
- Instead of Table 9.1.39.5-4 → use Table 9.1.39_1.5-2.

**Table 9.1.39_1.5-1: RSRP FDD Carrier Aggregation absolute accuracy requirements for the reported values**

| Band of Cell 1 on Primary Component Carrier | Event | Bands FDD_A | Bands FDD_B | Bands FDD_C | Bands FDD_D | Bands FDD_E, FDD_F | Bands FDD_G | Bands FDD_H |
|---|---|---|---|---|---|---|---|---|
| Normal Conditions | | | | | | | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRP_13 | RSRP_14 | RSRP_14 | RSRP_15 | RSRP_15 | RSRP_16 | RSRP_17 |
| Highest reported value (Cell 1) | | RSRP_26 | RSRP_26 | RSRP_27 | RSRP_27 | RSRP_28 | RSRP_29 | RSRP_29 |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_21 | RSRP_22 | RSRP_22 | RSRP_23 | RSRP_23 | RSRP_24 | RSRP_25 |
| Highest reported value (Cell 2) | | RSRP_34 | RSRP_34 | RSRP_35 | RSRP_35 | RSRP_36 | RSRP_37 | RSRP_37 |
| Lowest reported value (Cell 4) | Cell 4 | RSRP_21 | RSRP_22 | RSRP_22 | RSRP_23 | RSRP_23 | RSRP_24 | RSRP_25 |
| Highest reported value (Cell 4) | | RSRP_34 | RSRP_34 | RSRP_35 | RSRP_35 | RSRP_36 | RSRP_37 | RSRP_37 |
| Extreme Conditions | | | | | | | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRP_9 | RSRP_9 | RSRP_10 | RSRP_10 | RSRP_11 | RSRP_12 | RSRP_12 |
| Highest reported value (Cell 1) | | RSRP_30 | RSRP_31 | RSRP_31 | RSRP_32 | RSRP_32 | RSRP_33 | RSRP_34 |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_17 | RSRP_17 | RSRP_18 | RSRP_18 | RSRP_19 | RSRP_20 | RSRP_20 |
| Highest reported value (Cell 2) | | RSRP_38 | RSRP_39 | RSRP_39 | RSRP_40 | RSRP_40 | RSRP_41 | RSRP_42 |
| Lowest reported value (Cell 4) | Cell 4 | RSRP_17 | RSRP_17 | RSRP_18 | RSRP_18 | RSRP_19 | RSRP_20 | RSRP_20 |
| Highest reported value (Cell 4) | | RSRP_38 | RSRP_39 | RSRP_39 | RSRP_40 | RSRP_40 | RSRP_41 | RSRP_42 |
| Note: The band of Cell 1 determines the levels for Cell 1, Cell 2 and Cell 4, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2 or Cell 4. | | | | | | | | |

**Table 9.1.39_1.5-2: RSRP FDD for E-UTRAN in Carrier Aggregation relative accuracy requirements for the reported values**

| | Event | |
|---|---|---|
| Normal Conditions, all bands | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 + 2 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 15 |
| Lowest reported value (Cell 4) | Cell 1-4 | RSRP_Cell 1 +2 |
| Highest reported value (Cell 4) | | RSRP_Cell 1 + 15 |
| Extreme Conditions, all bands | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 – 1 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 17 |
| Lowest reported value (Cell 4) | Cell 1-4 | RSRP_Cell 1 – 1 |
| Highest reported value (Cell 4) | | RSRP_Cell 1 + 17 |
| Normal and Extreme Conditions, all bands | | |
| Lowest reported value (Cell 3) | Cell 2-3 | RSRP_Cell 2 – 8 |
| Highest reported value (Cell 3) | | RSRP_Cell 2 + 1 |
| Lowest reported value (Cell 5) | Cell 4-5 | RSRP_Cell 4 – 8 |
| Highest reported value (Cell 5) | | RSRP_Cell 4 + 1 |

## 9.1.40 3DL TDD RSRP Accuracy for E-UTRA in Carrier Aggregation

### 9.1.40.1 Test purpose

To verify the TDD RSRP absolute and relative accuracy requirements in carrier aggregation are within the specified limits. This test will verify the absolute RSRP accuracy requirements of the primary component carrier, the absolute RSRP accuracy requirements of the secondary component carriers, and the relative RSRP accuracy requirements of the secondary component carriers. The test will also verify the primary and secondary component carriers' relative RSRP accuracy requirement.

### 9.1.40.2 Test applicability

This test case applies to all types of E-UTRA TDD UE release 10 and 11 that support 3DL Intra-band contiguous CA or 3DL Inter-band CA, or 3DL Intra-band contiguous with Inter-band CA.

This test case also applies to all types of E-UTRA TDD UE release 11 that support 3DL Intra-band non-contiguous with Inter-band CA, or 3DL Intra-band non-contiguous with Intra-band contiguous CA.

### 9.1.40.3 Minimum conformance requirements

The TDD RSRP measurements of cells on the primary component carrier shall meet the absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1. The TDD RSRP measurements of cells on the secondary component carrier 1 and 2 shall meet the absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1.

The accuracy requirements in table 9.1.40.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.
- $RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.1.40.3-1: RSRP TDD absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±6 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 3: E-UTRA operating band groups are as defined in Section 3.5. | | | | | | |

The TDD RSRP relative measurements of cells on PCC and SCC shall meet relative accuracy requirements defined in TS 36.133 clause 9.1.3.2. The TDD RSRP relative measurements of cells on the SCC shall meet relative accuracy requirements defined in TS 36.133 clause 9.1.2.2.

The accuracy requirements in Table 9.1.40.3-2 for PCC-SCC relative accuracy requirements are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

- Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.
- $RSRP1,2|_{dBm}$ according to Annex I.3.8 for a corresponding Band
- $\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$
- | Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.40.3-2: TDD RSRP relative accuracy for PCC and SCC**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 6 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 5 | $dBm/BW_{Channel}$ |
| ±6 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 Note 3 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The condition has the minimum Io of -119 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz.
NOTE 4: Band 29 is used only for E-UTRA carrier aggregation with other E-UTRA bands.
NOTE 5: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 6: E-UTRA operating band groups are as defined in Section 3.5.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.3, clause 9.1.4, and A.9.1.40.

The TDD RSRP relative measurements of cells on SCCs shall meet relative accuracy requirements defined in TS 36.133 clause 9.1.2.2.

The accuracy requirements in Table 9.1.40.3-3 for SCCs relative accuracy requirements are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.
- $RSRP1,2|_{dBm}$ according to Annex I.3.8 for a corresponding Band.

**Table 9.1.40.3-3: TDD RSRP relative accuracy for SCCs**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | $dBm/BW_{Channel}$ |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.2, clause 9.1.4, and A.9.1.40.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.40.3-4. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.40.3-4: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1, clause 9.1.11.2, clause 9.1.11.3, clause 9.1.4, and A.9.1.40.

## 9.1.40.4 Test description

### 9.1.40.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 9.1.40.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.66 as appropriate.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.1.40.4.3.
4. Cell 1 is PCell on the primary component carrier, Cell 2 is SCell on SCC1, and Cell 3 is the neighbouring cell on SCC1, Cell 4 is SCell on SCC2, and Cell 5 is the neighbouring cell on SCC2. Cell 1 is used for connection setup with the power level set according to Annex C.0 and C.1. Cell 2, Cell 3, Cell 4 and Cell 5 shall be powered OFF.

### 9.1.40.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCC 1 and SCC 2 according to Annex C.0 and C.1 for all downlink physical channels except PHICH.
3. The SS shall configure SCell 1 (Cell 2) and SCell 2 (Cell 4) on the SCCs as per TS 36.508 [7] clause 5.2A.4 [with the message content exceptions defined in clause 9.1.40.4.3].
4. The SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [13], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.
5. Set the parameters according to Table 9.1.40.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
6. The SS shall transmit an RRCConnectionReconfiguration message.
7. The UE shall transmit RRCConnectionReconfigurationComplete message.
8. The UE shall transmit periodically MeasurementReport messages.
9. After 10s wait from Step 6, the SS shall check the reported RSRP values in periodical MeasurementReport messages. The reported RSRP values for Cell 1, Cell 2 and Cell 4 are compared to the actual RSRP values

according to Table 9.1.40.5-2. This counts respectively as a Pass or Fail for the events “Cell 1”, “Cell 2” and “Cell 4”. If the UE fails to report the measurement value for Cell 1, Cell 2, Cell 4, the number of failed iterations for the respective event is increased by one.

The reported RSRP value for Cell 2, Cell 3, Cell 4 and Cell 5 are compared to the reported RSRP value of other cells for each MeasurementReport message according to Table 9.1.40.5-3. These count respectively as a Pass or Fail for the events “Cell 1-2”, “Cell 1-4”, “Cell 2-3”, and “Cell 4-5”. If the UE fails to report the measurement value for Cell 2, Cell 3, Cell 4, or Cell 5, the number of failed iterations for the respective event is increased by one.

10. The SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

11. If test configuration is Inter-band carrier aggregation, repeat steps 1-10 for all relevant PCell/SCells scenarios defined in Table E-2 according to the UE declared capability for UL support (within CA operation) in the individual bands.

Each of the events “Cell 1”, “Cell 2”, “Cell 4”, “Cell 1-2”, “Cell 1-4”, “Cell 2-3” and “Cell 4-5” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass for each configuration (without and with switched PCell/SCell scenario), the test passes. If one event fails, the test fails.

### 9.1.40.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.40.4.3-1: Common Exception messages for 3DL TDD RSRP Accuracy for E-UTRA Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-3<br>Table H.4.1-7<br>Table H.4.2-5 |

## 9.1.40.5 Test requirement

Table 9.1.40.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD absolute accuracy test shall meet the reported values test requirements in table 9.1.40.5-2.

**Table 9.1.40.5-1: 3 DL TDD RSRP absolute accuracy carrier aggregation test parameters**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 | Cell 4 | Cell 5 |
|---|---|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | 2 | | 3 | |
| $BW_{channel}$ | | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | |
| Special subframe configuration[Note1] | | | 6 | | | | |
| Uplink/downlink configuration[Note1] | | | 1 | | | | |
| Measurement bandwidth | | $n_{PRB}$ | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | - | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | - | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD |
| OCNG Patterns defined in A.3.2.2 | | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA[Note2]<br>OCNG_RB[Note2] | | dB | 0 | 0 | 0 | 0 | 0 |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/ 15kHz | -117 | ( $N_{oc}$ for Channel 1 +1dB) | | ( $N_{oc}$ for Channel 1 +1dB) | |
| | Bands TDD_C | | -116 | | | | |
| | Bands TDD_E | | -115 | | | | |
| $\hat{E}_s / N_{oc}$ | | dB | -4 | 3 | -0.2 | 3 | -0.2 |
| $\hat{E}_s / I_{ot}$ | | dB | -4 | 0.09 | -4.96 | 0.09 | -4.96 |
| RSRP[Note4] | Bands TDD_A | dBm/ 15kHz | -121 | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4.8dB) | (RSRP for Cell 1 +8dB) | (RSRP for Cell 1 +4.8dB) |
| | Bands TDD_C | | -120 | | | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | Bands TDD_E | | -119 | | | | |
| Io[Note4] | Bands TDD_A | dBm/ $BW_{channel}$ | -87.76 + 10log($N_{RB,c}$/50) | (Io for Channel 1 +5.51dB +10log ($N_{RB\ channel2}$ / $N_{RB\ channel\ 1}$)) | | (Io for Channel 1 +5.51dB +10log ($N_{RB\ channel3}$ / $N_{RB\ channel\ 1}$)) | |
| | Bands TDD_C | | -86.76 + 10log($N_{RB,c}$/50) | | | | |
| | Bands TDD_E | | -85.76 + 10log($N_{RB,c}$/50) | | | | |
| Propagation Condition | | | AWGN | AWGN | AWGN | AWGN | AWGN |
| Antenna Configuration | | | 1x2 | 1x2 | 1x2 | 1x2 | 1x2 |
| Timing offset to Cell 1 | | µs | - | 0 | 3 | 0 | 3 |
| Time alignment error relative to cell 1 [Note 7] | | | - | ≤ TAE | - | ≤ TAE | - |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 6: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.
Note 7: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.
Note 8: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.40.5-2: RSRP TDD Carrier Aggregation absolute accuracy requirements for the reported values**

| Band of Cell 1 on Primary Component Carrier | Event | Bands TDD_A | Bands TDD_C | Bands TDD_E |
|---|---|---|---|---|
| Normal Conditions | | | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRP_12 | RSRP_13 | RSRP_14 |
| Highest reported value (Cell 1) | | RSRP_27 | RSRP_28 | RSRP_29 |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_20 | RSRP_21 | RSRP_22 |
| Highest reported value (Cell 2) | | RSRP_35 | RSRP_36 | RSRP_37 |
| Lowest reported value (Cell 4) | Cell 4 | RSRP_20 | RSRP_21 | RSRP_22 |
| Highest reported value (Cell 4) | | RSRP_35 | RSRP_36 | RSRP_37 |
| Extreme Conditions | | | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRP_09 | RSRP_10 | RSRP_11 |
| Highest reported value (Cell 1) | | RSRP_30 | RSRP_31 | RSRP_32 |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_17 | RSRP_18 | RSRP_19 |
| Highest reported value (Cell 2) | | RSRP_38 | RSRP_39 | RSRP_40 |
| Lowest reported value (Cell 4) | Cell 4 | RSRP_17 | RSRP_18 | RSRP_19 |
| Highest reported value (Cell 4) | | RSRP_38 | RSRP_39 | RSRP_40 |
| Note: The band of Cell 1 determines the levels for Cell 1, Cell 2 and Cell 4, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2 or Cell 4. | | | | |

**Table 9.1.40.5-3: RSRP TDD Carrier Aggregation relative accuracy requirements for the reported values**

| | Event | |
|---|---|---|
| Normal and Extreme Conditions, all bands | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 - 1 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 17 |
| Lowest reported value (Cell 4) | Cell 1-4 | RSRP_Cell 1 - 1 |
| Highest reported value (Cell 4) | | RSRP_Cell 1 + 17 |
| Normal and Extreme Conditions, all bands | | |
| Lowest reported value (Cell 3) | Cell 2-3 | RSRP_Cell 2 - 8 |
| Highest reported value (Cell 3) | | RSRP_Cell 2 + 1 |
| Lowest reported value (Cell 5) | Cell 4-5 | RSRP_Cell 4 - 8 |
| Highest reported value (Cell 5) | | RSRP_Cell 4 + 1 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.1.40_1 3DL TDD RSRP Accuracy for E-UTRA in Carrier Aggregation (Rel-12 and forward)

## 9.1.40_1.1 Test purpose

Same test purpose as defined in clause 9.1.40.1.

## 9.1.40_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward supporting 3DL CA.

## 9.1.40_1.3 Minimum conformance requirements

The TDD RSRP measurements of cells on the primary component carrier shall meet the absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1. The TDD RSRP measurements of cells on the secondary component carrier 1 and 2 shall meet the absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1.

The accuracy requirements in table 9.1.40_1.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.
- $RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.1.40_1.3-1: RSRP absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 3 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_B | -120.5 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The TDD RSRP relative measurements of cells on PCC and SCC shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.3.2. The TDD RSRP relative measurements of cells on the SCC shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.2.2.

The accuracy requirements in Table 9.1.40_1.3-2 for PCC-SCC relative accuracy requirements are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.
- $RSRP1,2|_{dBm}$ according to Annex I.3.8 for a corresponding Band
- $\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$
- | Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.40_1.3-2: RSRP relative accuracy for PCC and SCC**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/BW$_{Channel}$ |
| ±4.5 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.3, clause 9.1.4, and A.9.1.40.

The TDD RSRP relative measurements of cells on SCCs shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.2.2.

The accuracy requirements in Table 9.1.40_1.3-3 for SCCs relative accuracy requirements are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.
- $RSRP1,2|_{dBm}$ according to Annex I.3.8 for a corresponding Band.

**Table 9.1.40_1.3-3: RSRP relative accuracy for SCCs**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | $dBm/BW_{Channel}$ |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.2, clause 9.1.4, and A.9.1.40.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.40_1.3-4. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.40_1.3-4: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1, clause 9.1.11.2, clause 9.1.11.3, clause 9.1.4, and A.9.1.40.

## 9.1.40_1.4 Test description

### 9.1.40_1.4.1 Initial conditions

Same initial conditions as defined in clause 9.1.40.4.1.

### 9.1.40_1.4.2 Test procedure

Same test procedure as defined in clause 9.1.40.4.2 with the following exceptions:

- Instead of Table 9.1.40.5-2 → use Table 9.1.40_1.5-1.
- Instead of Table 9.1.40.5-3 → use Table 9.1.40_1.5-2.

### 9.1.40_1.4.3 Message contents

Same message contents as defined in clause 9.1.40.4.3

## 9.1.40_1.5 Test requirement

Same test requirements as in clause 9.1.40.5 with the following exceptions:

- Instead of Table 9.1.40.5-2 → use Table 9.1.40_1.5-1.
- Instead of Table 9.1.40.5-3 → use Table 9.1.40_1.5-2.

**Table 9.1.40_1.5-1: RSRP TDD Carrier Aggregation absolute accuracy requirements for the reported values**

| Band of Cell 1 on Primary Component Carrier | Event | Bands TDD_A | Bands TDD_C | Bands TDD_E |
|---|---|---|---|---|
| Normal Conditions | | | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRP_13 | RSRP_14 | RSRP_15 |
| Highest reported value (Cell 1) | | RSRP_26 | RSRP_27 | RSRP_28 |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_21 | RSRP_22 | RSRP_23 |
| Highest reported value (Cell 2) | | RSRP_34 | RSRP_35 | RSRP_36 |
| Lowest reported value (Cell 4) | Cell 4 | RSRP_21 | RSRP_22 | RSRP_23 |
| Highest reported value (Cell 4) | | RSRP_34 | RSRP_35 | RSRP_36 |
| Extreme Conditions | | | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRP_9 | RSRP_10 | RSRP_11 |
| Highest reported value (Cell 1) | | RSRP_30 | RSRP_31 | RSRP_32 |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_17 | RSRP_18 | RSRP_19 |
| Highest reported value (Cell 2) | | RSRP_38 | RSRP_39 | RSRP_40 |
| Lowest reported value (Cell 4) | Cell 4 | RSRP_17 | RSRP_18 | RSRP_19 |
| Highest reported value (Cell 4) | | RSRP_38 | RSRP_39 | RSRP_40 |
| Note: The band of Cell 1 determines the levels for Cell 1, Cell 2 and Cell 4, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2 or Cell 4. | | | | |

**Table 9.1.40_1.5-2: RSRP TDD Carrier Aggregation relative accuracy requirements for the reported values**

| | Event | |
|---|---|---|
| Normal Conditions, all bands | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 + 2 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 15 |
| Lowest reported value (Cell 4) | Cell 1-4 | RSRP_Cell 1 + 2 |
| Highest reported value (Cell 4) | | RSRP_Cell 1 + 15 |
| Extreme Conditions, all bands | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 – 1 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 17 |
| Lowest reported value (Cell 4) | Cell 1-4 | RSRP_Cell 1 – 1 |
| Highest reported value (Cell 4) | | RSRP_Cell 1 + 17 |
| Normal and Extreme Conditions, all bands | | |
| Lowest reported value (Cell 3) | Cell 2-3 | RSRP_Cell 2 – 8 |
| Highest reported value (Cell 3) | | RSRP_Cell 2 + 1 |
| Lowest reported value (Cell 5) | Cell 4-5 | RSRP_Cell 4 – 8 |
| Highest reported value (Cell 5) | | RSRP_Cell 4 + 1 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.1.41 FD-FDD Intra frequency RSRP Accuracy for UE category 0

## 9.1.41.1 FD-FDD Intra Frequency Absolute RSRP Accuracy for UE category 0

### 9.1.41.1.1 Test purpose

To verify that the FDD intra-frequency absolute RSRP measurement accuracy is within the specified limit for all bands.

### 9.1.41.1.2 Test applicability

This test applies to E-UTRA FD-FDD Category 0 UE from release 12 and forward. Applicability requires support for FGI bit 16.

### 9.1.41.1.3 Minimum conformance requirements

The requirements for absolute accuracy of RSRP in this clause apply to a cell on the same frequency as that of the serving cell for UE category 0.

The accuracy requirements in table 9.1.41.1.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101 Clause 7.3 for reference sensitivity are fulfilled.
- RSRP|dBm according to Annex I.3.1 for a corresponding Band.
- At least 1 DL subframe per radio frame of measured cell is available at the UE for RSRP measurement assuming measured cell is identified cell.

**Table 9.1.41.1.3-1: RSRP Intra frequency absolute accuracy for UE category 0**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 3] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 2] / dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±7 | ±10 | ≥-6 dB | FDD_A, TDD_A | -121 / N/A | -70 |
| | | | FDD_C, TDD_C | -120 / N/A | -70 |
| | | | FDD_D | -119.5 / N/A | -70 |
| | | | FDD_E, TDD_E | -119 / N/A | -70 |
| | | | FDD_F | -118.5 / N/A | -70 |
| | | | FDD_G | -118 / N/A | -70 |
| | | | FDD_H | -117.5 / N/A | -70 |
| | | | FDD_N | -114.5 / N/A | -70 |
| ±9 | ±12 | ≥-6 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 / -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.41.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.41.1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.13.1, clause 9.1.4 and A.9.1.41.

## 9.1.41.1.4 Test description

### 9.1.41.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20 using only main UE Tx/Rx antenna.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.1.41.1.4.3.
4. All cells are in the same carrier frequency. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.1.41.1.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Table 9.1.41.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.
3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. UE shall transmit periodically MeasurementReport messages.
6. After 10s wait from Step 3, SS shall check RSRP reported value in periodical MeasurementReport messages. The RSRP value of Cell 2 reported by the UE is compared to the actual RSRP value according to Table 9.1.41.1.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.
7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.
8. Repeat step 1-7 for each sub-test in Table 9.1.41.1.5-1 as appropriate.

#### 9.1.41.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.41.1.4.3-1: Common Exception messages for RSRP FDD Intra frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3 |

**Table 9.1.41.1.4.3-2: *MeasResults*: Additional RSRP FDD Intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement identity for which the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | Set according to specific test | |
| rsrqResult | INTEGER(0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.41.1.4.3-3: *MeasResultListEUTRA*: Additional RSRP FDD Intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId from Cell2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

## 9.1.41.1.5 Test requirement

Table 9.1.41.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD intra-frequency absolute accuracy test shall meet the reported values test requirements in table 9.1.41.1.5-2.

**Table 9.1.41.1.5-1: FD-FDD RSRP Intra frequency test parameters for UE category 0**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.3 | | | R.13 FDD | - | R.13 FDD | - | R.13 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in D.1 | | | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA[Note1]<br>OCNG_RB[Note1] | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -107 | | -87 | | -116 | |
| | Bands FDD_B | | | | | | -115.5 | |
| | Bands FDD_C | | | | | | -115 | |
| | Bands FDD_D | | | | | | -114.5 | |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -114 | |
| | Bands FDD_G [Note 6] | | | | | | -113 | |
| | Bands FDD_H | | | | | | -112.5 | |
| $\hat{E}_s / N_{oc}$ | | dB | 6 | 2 | 6 | 2 | 3 | -0.2 |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 | 1.88 | -4.97 | 0.09 | -4.96 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -101 | -105 | -81 | -85 | -113 | -116.2 |
| | Bands FDD_B | | | | | | -112.5 | -115.7 |
| | Bands FDD_C | | | | | | -112 | -115.2 |
| | Bands FDD_D | | | | | | -111.5 | -114.7 |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -111 | -114.2 |
| | Bands FDD_G [Note 6] | | | | | | -110 | -113.2 |
| | Bands FDD_H | | | | | | -109.5 | -112.7 |
| Io[Note3] | Bands FDD_A | dBm/9 MHz | *-71.05* | | -51.05 | | -82.25 | |
| | Bands FDD_B | | | | | | -81.75 | |
| | Bands FDD_C | | | | | | -81.25 | |
| | Bands FDD_D | | | | | | -80.75 | |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -80.25 | |
| | Bands FDD_G [Note 6] | | | | | | -79.25 | |
| | Bands FDD_H | | | | | | -78.75 | |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |
| Correlation Matrix and Antenna Configuration | | | 1x1 | | 1x1 | | 1x1 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: Es/Iot, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

| | |
|---|---|
| Note 4: | For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz. |
| Note 5: | E-UTRA operating band groups are as defined in Section 3.5. |
| Note 6: | Except Band 29 and Band 32. |

**Table 9.1.41.1.5-2: RSRP FDD Intra frequency absolute accuracy requirements for the reported values**

| Normal Conditions | Test 1 All bands | Test 2 All bands | Test 3 | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_27 | RSRP_45 | Bands FDD_A | RSRP_16 |
| | | | Bands FDD_B | RSRP_16 |
| | | | Bands FDD_C | RSRP_17 |
| | | | Bands FDD_D | RSRP_17 |
| | | | Bands FDD_E, FDD_F | RSRP_18 |
| | | | Bands FDD_G | RSRP_19 |
| | | | Bands FDD_H | RSRP_19 |
| Highest reported value (Cell 2) | RSRP_44 | RSRP_66 | Bands FDD_A | RSRP_33 |
| | | | Bands FDD_B | RSRP_33 |
| | | | Bands FDD_C | RSRP_34 |
| | | | Bands FDD_D | RSRP_34 |
| | | | Bands FDD_E, FDD_F | RSRP_35 |
| | | | Bands FDD_G | RSRP_36 |
| | | | Bands FDD_D | RSRP_36 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_24 | RSRP_42 | Bands FDD_A | RSRP_13 |
| | | | Bands FDD_B | RSRP_13 |
| | | | Bands FDD_C | RSRP_14 |
| | | | Bands FDD_D | RSRP_14 |
| | | | Bands FDD_E, FDD_F | RSRP_15 |
| | | | Bands FDD_G | RSRP_16 |
| | | | Bands FDD_H | RSRP_16 |
| Highest reported value (Cell 2) | RSRP_47 | RSRP_69 | Bands FDD_A | RSRP_36 |
| | | | Bands FDD_B | RSRP_36 |
| | | | Bands FDD_C | RSRP_37 |
| | | | Bands FDD_D | RSRP_37 |
| | | | Bands FDD_E, FDD_F | RSRP_38 |
| | | | Bands FDD_G | RSRP_39 |
| | | | Bands FDD_H | RSRP_39 |
| Note 1: E-UTRA operating band groups are as defined in Section 3.5. | | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.41.2 FD-FDD Intra Frequency Relative RSRP Accuracy for UE category 0

### 9.1.41.2.1 Test purpose

To verify that the FD-FDD intra-frequency relative RSRP measurement accuracy is within the specified limit for all bands.

### 9.1.41.2.2 Test applicability

This test applies to all types of E-UTRA FD-FDD Category 0 UE from release 12 and forward. Applicability requires support for FGI bit 16.

### 9.1.41.2.3 Minimum conformance requirements

The relative accuracy of RSRP is defined as the RSRP measured from one cell compared to the RSRP measured from another cell on the same frequency for category 0 UE.

The accuracy requirements in Table 9.1.41.2.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

- Conditions defined in 36.101 Clause 7.3 for reference sensitivity are fulfilled.
- RSRP1,2|$_{dBm}$ according to Annex I.3.8 for a corresponding Band.
- At least 1 DL subframe per radio frame of measured cell is available at the UE for RSRP measurement assuming measured cell is identified cell.

**Table 9.1.41.2.3-1: RSRP Intra frequency relative accuracy for UE category 0**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/BW$_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.41.2.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.41.2.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.13.2, clause 9.1.4 and A.9.1.41.

## 9.1.41.2.4 Test description

### 9.1.41.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20 using only main UE Tx/Rx antenna.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.1.41.2.4.3.
4. All cells are in the same carrier frequency. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.1.41.2.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Table 9.1.41.2.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.
3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. UE shall transmit periodically MeasurementReport messages.
6. After 10s wait from Step 3, SS shall check the RSRP value of Cell 1 and Cell 2 in periodical MeasurementReport messages. The reported RSRP value for Cell 2 is compared to the reported RSRP value for Cell 1 for each MeasurementReport message according to Table 9.1.41.2.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.
7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.
8. Repeat step 1-7 for each sub-test in Table 9.1.41.2.5-1 as appropriate.

#### 9.1.41.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.41.2.4.3-1: Common Exception messages for RSRP FD-FDD Intra frequency relative accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3 |

**Table 9.1.41.2.4.3-2: *MeasResults*: Additional RSRP FD-FDD Intra frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement identity for which the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | Set according to specific test | |
| rsrqResult | INTEGER(0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.41.2.4.3-3: *MeasResultListEUTRA*: Additional RSRP FD-FDD Intra frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId from Cell2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

## 9.1.41.2.5 Test requirement

Table 9.1.41.2.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FD-FDD intra-frequency relative accuracy test shall meet the reported values test requirements in table 9.1.41.2.5-2.

**Table 9.1.41.2.5-1: FD-FDD RSRP Intra frequency test parameters for UE category 0**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.3 | | | R.13 FDD | - | R.13 FDD | - | R.13 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in D.1 | | | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA[Note1]<br>OCNG_RB[Note1] | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -106 | | -87 | | -116 | |
| | Bands FDD_B | | | | | | -115.5 | |
| | Bands FDD_C | | | | | | -115 | |
| | Bands FDD_D | | | | | | -114.5 | |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -114 | |
| | Bands FDD_G [Note 6] | | | | | | -113 | |
| | Bands FDD_H | | | | | | -112.5 | |
| $\hat{E}_s / N_{oc}$ | | dB | 6 | 2 | 6 | 2 | 3 | 0 |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 | 1.88 | -4.97 | -0.01 | -4.76 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -100 | -104 | -81 | -85 | -113 | -116 |
| | Bands FDD_B | | | | | | -112.5 | -115.5 |
| | Bands FDD_C | | | | | | -112 | -115 |
| | Bands FDD_D | | | | | | -111.5 | -114.5 |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -111 | -114 |
| | Bands FDD_G [Note 6] | | | | | | -110 | -113 |
| | Bands FDD_H | | | | | | -109.5 | -112.5 |
| Io[Note3] | Bands FDD_A | dBm/9 MHz | -70.05 | | -51.05 | | -82.20 | |
| | Bands FDD_B | | | | | | -81.70 | |
| | Bands FDD_C | | | | | | -81.20 | |
| | Bands FDD_D | | | | | | -80.70 | |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -80.20 | |
| | Bands FDD_G [Note 6] | | | | | | -79.20 | |
| | Bands FDD_H | | | | | | -78.70 | |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |
| Correlation Matrix and Antenna Configuration | | | 1x1 | | 1x1 | | 1x1 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: Es/Iot, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

| | |
|---|---|
| Note 4: | For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz. |
| Note 5: | E-UTRA operating band groups are as defined in Section 3.5. |
| Note 6: | Except Band 29 and Band 32. |

**Table 9.1.41.2.5-2: RSRP FD-FDD Intra frequency relative accuracy requirements for the reported values**

| Normal Conditions | Test 1 All bands | Test 2 All bands | Test 3 | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_x-10 | RSRP_x-10 | Bands FDD_A | RSRP_x-9 |
| | | | Bands FDD_B | RSRP_x-9 |
| | | | Bands FDD_C | RSRP_x-9 |
| | | | Bands FDD_D | RSRP_x-9 |
| | | | Bands FDD_E, FDD_F | RSRP_x-9 |
| | | | Bands FDD_G | RSRP_x-9 |
| | | | Bands FDD_H | RSRP_x-9 |
| Highest reported value (Cell 2) | RSRP_x+2 | RSRP_x+2 | Bands FDD_A | RSRP_x+3 |
| | | | Bands FDD_B | RSRP_x+3 |
| | | | Bands FDD_C | RSRP_x+3 |
| | | | Bands FDD_D | RSRP_x+3 |
| | | | Bands FDD_E, FDD_F | RSRP_x+3 |
| | | | Bands FDD_G | RSRP_x+3 |
| | | | Bands FDD_D | RSRP_x+3 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_x-10 | RSRP_x-10 | Bands FDD_A | RSRP_x-9 |
| | | | Bands FDD_B | RSRP_x-9 |
| | | | Bands FDD_C | RSRP_x-9 |
| | | | Bands FDD_D | RSRP_x-9 |
| | | | Bands FDD_E, FDD_F | RSRP_x-9 |
| | | | Bands FDD_G | RSRP_x-9 |
| | | | Bands FDD_H | RSRP_x-9 |
| Highest reported value (Cell 2) | RSRP_x+2 | RSRP_x+2 | Bands FDD_A | RSRP_x+3 |
| | | | Bands FDD_B | RSRP_x+3 |
| | | | Bands FDD_C | RSRP_x+3 |
| | | | Bands FDD_D | RSRP_x+3 |
| | | | Bands FDD_E, FDD_F | RSRP_x+3 |
| | | | Bands FDD_G | RSRP_x+3 |
| | | | Bands FDD_H | RSRP_x+3 |
| Note 1: E-UTRA operating band groups are as defined in Section 3.5. | | | | |
| Note 2: RSRP_x is the reported value of Cell 1 | | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.42 HD-FDD Intra frequency RSRP Accuracy for UE category 0

### 9.1.42.1 HD-FDD Intra Frequency Absolute RSRP Accuracy for UE category 0

#### 9.1.42.1.1 Test purpose

To verify that the FDD intra-frequency absolute RSRP measurement accuracy is within the specified limit for all bands.

#### 9.1.42.1.2 Test applicability

This test applies to all types of E-UTRA HD-FDD operation Category 0 UE from release 12 and forward. Applicability requires support for FGI bit 16.

### 9.1.42.1.3 Minimum conformance requirements

The requirements for absolute accuracy of RSRP in this clause apply to a cell on the same frequency as that of the serving cell for UE category 0.

The accuracy requirements in table 9.1.42.1.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101 Clause 7.3 for reference sensitivity are fulfilled.
- RSRP|dBm according to Annex I.3.1 for a corresponding Band.
- At least 1 DL subframe per radio frame of measured cell is available at the UE for RSRP measurement assuming measured cell is identified cell.

**Table 9.1.42.1.3-1: RSRP Intra frequency absolute accuracy for UE category 0**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±7 | ±10 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±9 | ±12 | ≥-6 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.42.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.42.1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.13.1, clause 9.1.4 and A.9.1.42.

### 9.1.42.1.4 Test description

#### 9.1.42.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20 using only main UE Tx/Rx antenna.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.1.42.1.4.3.
4. All cells are in the same carrier frequency. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.1.42.1.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Table 9.1.42.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.
3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. UE shall transmit periodically MeasurementReport messages.
6. After 10s wait from Step 3, SS shall check RSRP reported value in periodical MeasurementReport messages. The RSRP value of Cell 2 reported by the UE is compared to the actual RSRP value according to Table 9.1.42.1.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.
7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.
8. Repeat step 1-7 for each sub-test in Table 9.1.42.1.5-1 as appropriate.

#### 9.1.42.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.42.1.4.3-1: Common Exception messages for RSRP FDD Intra frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3 |

**Table 9.1.42.1.4.3-2: *MeasResults*: Additional RSRP FDD Intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement identity for which the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | Set according to specific test | |
| rsrqResult | INTEGER(0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.42.1.4.3-3: *MeasResultListEUTRA*: Additional RSRP FDD Intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId from Cell2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

## 9.1.42.1.5 Test requirement

Table 9.1.42.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD intra-frequency absolute accuracy test shall meet the reported values test requirements in table 9.1.42.1.5-2.

**Table 9.1.42.1.5-1: HD-FDD RSRP Intra frequency test parameters for UE category 0**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.4 | | | R.1 HD-FDD | - | R.1 HD-FDD | - | R.1 HD-FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.3 | | | R.3 HD-FDD | | R.3 HD-FDD | | R.3 HD-FDD | |
| OCNG Patterns defined in D.1 | | | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | |
| PSS_RA | | | | | | | | |
| SSS_RA | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA Note1 | | | | | | | | |
| OCNG_RB Note1 | | | | | | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -107 | | -87 | | -116 | |
| | Bands FDD_B | | | | | | -115.5 | |
| | Bands FDD_C | | | | | | -115 | |
| | Bands FDD_D | | | | | | -114.5 | |
| | Bands FDD_E, FDD_F Note 5 | | | | | | -114 | |
| | Bands FDD_G Note 7 | | | | | | -113 | |
| | Bands FDD_H | | | | | | -112.5 | |
| $\hat{E}_s / N_{oc}$ | | dB | 6 | 2 | 6 | 2 | 3 | -0.2 |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 | 1.88 | -4.97 | 0.09 | -4.96 |
| RSRP Note3 | Bands FDD_A | dBm/15 kHz | -101 | -105 | -81 | -85 | -113 | -116.20 |
| | Bands FDD_B | | | | | | -112.5 | -115.7 |
| | Bands FDD_C | | | | | | -112 | -115.20 |
| | Bands FDD_D | | | | | | -111.5 | -114.70 |
| | Bands FDD_E, FDD_F Note 4 | | | | | | -111 | -114.20 |
| | Bands FDD_G Note 6 | | | | | | -110 | -113.20 |
| | Bands FDD_H | | | | | | -109.5 | -112.70 |
| Io Note3 | Bands FDD_A | dBm/9 MHz | -71.05 | | -51.05 | | -82.25 | |
| | Bands FDD_B | | | | | | -81.75 | |
| | Bands FDD_C | | | | | | -81.25 | |
| | Bands FDD_D | | | | | | -80.75 | |
| | Bands FDD_E, FDD_F Note 4 | | | | | | -80.25 | |
| | Bands FDD_G Note 6 | | | | | | -79.25 | |
| | Bands FDD_H | | | | | | -78.75 | |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |
| Correlation Matrix and Antenna Configuration | | | 1x1 | | 1x1 | | 1x1 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

| | |
|---|---|
| Note 3: | Es/Iot, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 4: | For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz. |
| Note 5: | E-UTRA operating band groups are as defined in Section 3.5. |
| Note 6: | Except Band 29 and Band 32. |

**Table 9.1.42.1.5-2: RSRP FDD Intra frequency absolute accuracy requirements for the reported values**

| Normal Conditions | Test 1 All bands | Test 2 All bands | Test 3 | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_27 | RSRP_45 | Bands FDD_A | RSRP_16 |
| | | | Bands FDD_B | RSRP_16 |
| | | | Bands FDD_C | RSRP_17 |
| | | | Bands FDD_D | RSRP_17 |
| | | | Bands FDD_E, FDD_F | RSRP_18 |
| | | | Bands FDD_G | RSRP_19 |
| | | | Bands FDD_H | RSRP_19 |
| Highest reported value (Cell 2) | RSRP_44 | RSRP_66 | Bands FDD_A | RSRP_33 |
| | | | Bands FDD_B | RSRP_33 |
| | | | Bands FDD_C | RSRP_34 |
| | | | Bands FDD_D | RSRP_34 |
| | | | Bands FDD_E, FDD_F | RSRP_35 |
| | | | Bands FDD_G | RSRP_36 |
| | | | Bands FDD_D | RSRP_36 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_24 | RSRP_42 | Bands FDD_A | RSRP_13 |
| | | | Bands FDD_B | RSRP_13 |
| | | | Bands FDD_C | RSRP_14 |
| | | | Bands FDD_D | RSRP_14 |
| | | | Bands FDD_E, FDD_F | RSRP_15 |
| | | | Bands FDD_G | RSRP_16 |
| | | | Bands FDD_H | RSRP_16 |
| Highest reported value (Cell 2) | RSRP_47 | RSRP_69 | Bands FDD_A | RSRP_36 |
| | | | Bands FDD_B | RSRP_36 |
| | | | Bands FDD_C | RSRP_37 |
| | | | Bands FDD_D | RSRP_37 |
| | | | Bands FDD_E, FDD_F | RSRP_38 |
| | | | Bands FDD_G | RSRP_39 |
| | | | Bands FDD_H | RSRP_39 |
| Note 1: E-UTRA operating band groups are as defined in Section 3.5. | | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.42.2 HD-FDD Intra Frequency Relative RSRP Accuracy for UE category 0

### 9.1.42.2.1 Test purpose

To verify that the HD-FDD intra-frequency relative RSRP measurement accuracy is within the specified limit for all bands.

### 9.1.42.2.2 Test applicability

This test applies to all types of E-UTRA HD-FDD Category 0 UE from release 12 and forward. Applicability requires support for FGI bit 16.

### 9.1.42.2.3 Minimum conformance requirements

The relative accuracy of RSRP is defined as the RSRP measured from one cell compared to the RSRP measured from another cell on the same frequency for category 0 UE.

The accuracy requirements in Table 9.1.42.2.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101 Clause 7.3 for reference sensitivity are fulfilled.
- $RSRP1,2|_{dBm}$ according to Annex I.3.8 for a corresponding Band.
- At least 1 DL subframe per radio frame of measured cell is available at the UE for RSRP measurement assuming measured cell is identified cell.

**Table 9.1.42.2.3-1: RSRP Intra frequency relative accuracy for UE category 0**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | $dBm/BW_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.42.2.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.42.2.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.13.2, clause 9.1.4 and A.9.1.42.

### 9.1.42.2.4 Test description

#### 9.1.42.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20 using only main UE Tx/Rx antenna.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.1.42.2.4.3.
4. All cells are in the same carrier frequency. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.1.42.2.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Table 9.1.42.2.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.
3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. UE shall transmit periodically MeasurementReport messages.
6. After 10s wait from Step 3, SS shall check the RSRP value of Cell 1 and Cell 2 in periodical MeasurementReport messages. The reported RSRP value for Cell 2 is compared to the reported RSRP value for Cell 1 for each MeasurementReport message according to Table 9.1.42.2.5-3. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.
7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.
8. Repeat step 1-7 for each sub-test in Table 9.1.42.2.5-1 as appropriate.

#### 9.1.42.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.42.2.4.3-1: Common Exception messages for RSRP HD-FDD Intra frequency relative accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3 |

**Table 9.1.42.2.4.3-2: *MeasResults*: Additional RSRP HD-FDD Intra frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement identity for which the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | Set according to specific test | |
| rsrqResult | INTEGER(0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.42.2.4.3-3: *MeasResultListEUTRA*: Additional RSRP HD-FDD Intra frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId from Cell2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

## 9.1.42.2.5 Test requirement

Table 9.1.42.2.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD intra-frequency relative accuracy test shall meet the reported values test requirements in table 9.1.42.2.5-2.

**Table 9.1.42.2.5-1: HD-FDD RSRP Intra frequency test parameters for UE category 0**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.4 | | | R.1 HD-FDD | - | R.1 HD-FDD | - | R.1 HD-FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.3 | | | R.3 HD-FDD | | R.3 HD-FDD | | R.3 HD-FDD | |
| OCNG Patterns defined in D.1 | | | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA[Note1]<br>OCNG_RB[Note1] | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -106 | | -87 | | -116 | |
| | Bands FDD_B | | | | | | -115.5 | |
| | Bands FDD_C | | | | | | -115 | |
| | Bands FDD_D | | | | | | -114.5 | |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -114 | |
| | Bands FDD_G [Note 7] | | | | | | -113 | |
| | Bands FDD_H | | | | | | -112.5 | |
| $\hat{E}_s / N_{oc}$ | | dB | 6 | 2 | 6 | 2 | 3 | 0 |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 | 1.88 | -4.97 | -0.01 | -4.76 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -100 | -104 | -81 | -85 | -113 | -116 |
| | Bands FDD_B | | | | | | -112.5 | -115.5 |
| | Bands FDD_C | | | | | | -112 | -115 |
| | Bands FDD_D | | | | | | -111.5 | -114.5 |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -111 | -114 |
| | Bands FDD_G [Note 6] | | | | | | -110 | -113 |
| | Bands FDD_H | | | | | | -109.5 | -112.5 |
| Io[Note3] | Bands FDD_A | dBm/9 MHz | -70.05 | | -51.05 | | -82.20 | |
| | Bands FDD_B | | | | | | -81.70 | |
| | Bands FDD_C | | | | | | -81.20 | |
| | Bands FDD_D | | | | | | -80.70 | |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -80.20 | |
| | Bands FDD_G [Note 6] | | | | | | -79.20 | |
| | Bands FDD_H | | | | | | -78.70 | |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |
| Correlation Matrix and Antenna Configuration | | | 1x1 | | 1x1 | | 1x1 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: Es/Iot, RSRP and Io levels have been derived from other parameters for information purposes. They are

not settable parameters themselves.
Note 4: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 5: E-UTRA operating band groups are as defined in Section 3.5.
Note 6: Except Band 29 and Band 32.

**Table 9.1.42.2.5-2: RSRP HD-FDD Intra frequency relative accuracy requirements for the reported values**

| Normal Conditions | Test 1 All bands | Test 2 All bands | Test 3 | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_x-10 | RSRP_x-10 | Bands FDD_A | RSRP_x-9 |
| | | | Bands FDD_B | RSRP_x-9 |
| | | | Bands FDD_C | RSRP_x-9 |
| | | | Bands FDD_D | RSRP_x-9 |
| | | | Bands FDD_E, FDD_F | RSRP_x-9 |
| | | | Bands FDD_G | RSRP_x-9 |
| | | | Bands FDD_H | RSRP_x-9 |
| Highest reported value (Cell 2) | RSRP_x+2 | RSRP_x+2 | Bands FDD_A | RSRP_x+3 |
| | | | Bands FDD_B | RSRP_x+3 |
| | | | Bands FDD_C | RSRP_x+3 |
| | | | Bands FDD_D | RSRP_x+3 |
| | | | Bands FDD_E, FDD_F | RSRP_x+3 |
| | | | Bands FDD_G | RSRP_x+3 |
| | | | Bands FDD_D | RSRP_x+3 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_x-10 | RSRP_x-10 | Bands FDD_A | RSRP_x-9 |
| | | | Bands FDD_B | RSRP_x-9 |
| | | | Bands FDD_C | RSRP_x-9 |
| | | | Bands FDD_D | RSRP_x-9 |
| | | | Bands FDD_E, FDD_F | RSRP_x-9 |
| | | | Bands FDD_G | RSRP_x-9 |
| | | | Bands FDD_H | RSRP_x-9 |
| Highest reported value (Cell 2) | RSRP_x+2 | RSRP_x+2 | Bands FDD_A | RSRP_x+3 |
| | | | Bands FDD_B | RSRP_x+3 |
| | | | Bands FDD_C | RSRP_x+3 |
| | | | Bands FDD_D | RSRP_x+3 |
| | | | Bands FDD_E, FDD_F | RSRP_x+3 |
| | | | Bands FDD_G | RSRP_x+3 |
| | | | Bands FDD_H | RSRP_x+3 |
| Note 1: E-UTRA operating band groups are as defined in Section 3.5. | | | | |
| Note 2: RSRP_x is the reported value of Cell 1 | | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.1.43 TDD Intra frequency RSRP Accuracy for UE category 0

## 9.1.43.1 TDD Intra Frequency Absolute RSRP Accuracy for UE category 0

### 9.1.43.1.1 Test purpose

To verify that the TDD intra-frequency absolute RSRP measurement accuracy is within the specified limit for all bands.

### 9.1.43.1.2 Test applicability

This test applies to all types of E-UTRA TDD Category 0 UE from release 12 and forward. Applicability requires support for FGI bit 16.

### 9.1.43.1.3 Minimum conformance requirements

The requirements for absolute accuracy of RSRP in this clause apply to a cell on the same frequency as that of the serving cell for UE category 0.

The accuracy requirements in table 9.1.43.1.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101 Clause 7.3 for reference sensitivity are fulfilled.
- RSRP|dBm according to Annex I.3.1 for a corresponding Band.
- At least 1 DL subframe per radio frame of measured cell is available at the UE for RSRP measurement assuming measured cell is identified cell.

**Table 9.1.43.1.3-1: RSRP Intra frequency absolute accuracy for UE category 0**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 / dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±7 | ±10 | ≥-6 dB | FDD_A, TDD_A | -121 / N/A | -70 |
| | | | FDD_C, TDD_C | -120 / N/A | -70 |
| | | | FDD_D | -119.5 / N/A | -70 |
| | | | FDD_E, TDD_E | -119 / N/A | -70 |
| | | | FDD_F | -118.5 / N/A | -70 |
| | | | FDD_G | -118 / N/A | -70 |
| | | | FDD_H | -117.5 / N/A | -70 |
| | | | FDD_N | -114.5 / N/A | -70 |
| ±9 | ±12 | ≥-6 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 / -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.43.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.43.1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.13.1, clause 9.1.4 and A.9.1.43.

### 9.1.43.1.4 Test description

#### 9.1.43.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20 using only main UE Tx/Rx antenna.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.1.43.1.4.3.
4. All cells are in the same carrier frequency. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.1.43.1.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Table 9.1.43.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.
3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. UE shall transmit periodically MeasurementReport messages.
6. After 10s wait from Step 3, SS shall check RSRP reported value in periodical MeasurementReport messages. The RSRP value of Cell 2 reported by the UE is compared to the actual RSRP value according to Table 9.1.43.1.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.
7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.
8. Repeat step 1-7 for each sub-test in Table 9.1.43.1.5-1 as appropriate.

#### 9.1.43.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.43.1.4.3-1: Common Exception messages for RSRP TDD Intra frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3 |

**Table 9.1.43.1.4.3-2: *MeasResults*: Additional RSRP TDD Intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement identity for which the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | Set according to specific test | |
| rsrqResult | INTEGER(0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.43.1.4.3-3: *MeasResultListEUTRA*: Additional RSRP TDD Intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId from Cell2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

### 9.1.43.1.5 Test requirement

Table 9.1.43.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD intra-frequency absolute accuracy test shall meet the reported values test requirements in table 9.1.43.1.5-2.

**Table 9.1.43.1.5-1: TDD RSRP Intra frequency test parameters for UE category 0**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Special subframe configuration Note1 | | | 6 | | 6 | | 6 | |
| Uplink/downlink configuration Note1 | | | 1 | | 1 | | 1 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.5 | | | R.12 TDD | - | R.12 TDD | - | R.12 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | |
| PSS_RA | | | | | | | | |
| SSS_RA | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA Note2 | | | | | | | | |
| OCNG_RB Note2 | | | | | | | | |
| $N_{oc}$ Note3 | Bands TDD_A | dBm/15 kHz | -1076 | | -87 | | -116 | |
| | Bands TDD_C | | | | | | -115 | |
| | Bands TDD_E | | | | | | -114 | |
| $\hat{E}_s / N_{oc}$ | | dB | 6 | 2 | 6 | 2 | 3 | -0.2 |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 | 1.88 | -4.97 | 0.09 | -4.96 |
| RSRP Note4 | Bands TDD_A | dBm/15 kHz | -101 | -105 | -81 | -85 | -113 | -116.20 |
| | Bands TDD_C | | | | | | -112 | -115.20 |
| | Bands TDD_E | | | | | | -111 | -114.20 |
| Io Note4 | Bands TDD_A | dBm/9 MHz | -71.05 | | -51.05 | | -82.25 | |
| | Bands TDD_C | | | | | | -81.25 | |
| | Bands TDD_E | | | | | | -80.25 | |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |
| Correlation Matrix and Antenna Configuration | | | 1x1 | | 1x1 | | 1x1 | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Es/Iot, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.43.1.5-2: RSRP TDD Intra frequency absolute accuracy requirements for the reported values**

| Normal Conditions | Test 1 All bands | Test 2 All bands | Test 3 | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_27 | RSRP_45 | Bands TDD_A | RSRP_16 |
| | | | Bands TDD_C | RSRP_17 |
| | | | Bands TDD_E | RSRP_18 |
| Highest reported value (Cell 2) | RSRP_44 | RSRP_66 | Bands TDD_A | RSRP_33 |
| | | | Bands TDD_C | RSRP_34 |
| | | | Bands TDD_E | RSRP_35 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_24 | RSRP_42 | Bands TDD_A | RSRP_13 |
| | | | Bands TDD_C | RSRP_14 |
| | | | Bands TDD_E | RSRP_15 |
| Highest reported value (Cell 2) | RSRP_47 | RSRP_69 | Bands TDD_A | RSRP_36 |
| | | | Bands TDD_C | RSRP_37 |
| | | | Bands TDD_E | RSRP_38 |
| Note 1: E-UTRA operating band groups are as defined in Section 3.5. | | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.43.2 TDD Intra Frequency Relative RSRP Accuracy for UE category 0

### 9.1.43.2.1 Test purpose

To verify that the TDD intra-frequency relative RSRP measurement accuracy is within the specified limit for all bands.

### 9.1.43.2.2 Test applicability

This test applies to all types of E-UTRA TDD Category 0 UE from release 12 and forward. Applicability requires support for FGI bit 16.

### 9.1.43.2.3 Minimum conformance requirements

The relative accuracy of RSRP is defined as the RSRP measured from one cell compared to the RSRP measured from another cell on the same frequency for category 0 UE.

The accuracy requirements in Table 9.1.43.2.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101 Clause 7.3 for reference sensitivity are fulfilled.
- RSRP1,2|dBm according to Annex I.3.8 for a corresponding Band.
- At least 1 DL subframe per radio frame of measured cell is available at the UE for RSRP measurement assuming measured cell is identified cell.

**Table 9.1.43.2.3-1: RSRP Intra frequency relative accuracy for UE category 0**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/$BW_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.43.2.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.43.2.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.13.2, clause 9.1.4 and A.9.1.43.

## 9.1.43.2.4 Test description

### 9.1.43.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20 using only main UE Tx/Rx antenna.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.1.43.2.4.3.
4. All cells are in the same carrier frequency. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.1.43.2.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Table 9.1.43.2.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.
3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. UE shall transmit periodically MeasurementReport messages.
6. After 10s wait from Step 3, SS shall check the RSRP value of Cell 1 and Cell 2 in periodical MeasurementReport messages. The reported RSRP value for Cell 2 is compared to the reported RSRP value for Cell 1 for each MeasurementReport message according to Table 9.1.43.2.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.
7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.
8. Repeat step 1-7 for each sub-test in Table 9.1.43.2.5-1 as appropriate.

#### 9.1.43.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.43.2.4.3-1: Common Exception messages for RSRP TDD Intra frequency relative accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3 |

**Table 9.1.43.2.4.3-2: *MeasResults*: Additional RSRP TDD Intra frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement identity for which the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | Set according to specific test | |
| rsrqResult | INTEGER(0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.43.2.4.3-3: *MeasResultListEUTRA*: Additional RSRP TDD Intra frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId from Cell2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

## 9.1.43.2.5 Test requirement

Table 9.1.43.2.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD intra-frequency absolute accuracy test shall meet the reported values test requirements in table 9.1.43.2.5-2.

**Table 9.1.43.2.5-1: TDD RSRP Intra frequency test parameters for UE category 0**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Special subframe configuration Note1 | | | 6 | | 6 | | 6 | |
| Uplink/downlink configuration Note1 | | | 1 | | 1 | | 1 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.5 | | | R.12 TDD | - | R.12 TDD | - | R.12 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | |
| PSS_RA | | | | | | | | |
| SSS_RA | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA Note2 | | | | | | | | |
| OCNG_RB Note2 | | | | | | | | |
| $N_{oc}$ Note3 | Bands TDD_A | dBm/15 kHz | -106 | | -87 | | -116 | |
| | Bands TDD_C | | | | | | -115 | |
| | Bands TDD_E | | | | | | -114 | |
| $\hat{E}_s / N_{oc}$ | | dB | 6 | 2 | 6 | 2 | 3 | 0 |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 | 1.88 | -4.97 | -0.01 | -4.76 |
| RSRP Note4 | Bands TDD_A | dBm/15 kHz | -100 | -104 | -81 | -85 | -113 | -116 |
| | Bands TDD_C | | | | | | -112 | -115 |
| | Bands TDD_E | | | | | | -111 | -114 |
| Io Note4 | Bands TDD_A | dBm/9 MHz | -70.05 | | -51.05 | | -82.20 | |
| | Bands TDD_C | | | | | | -81.20 | |
| | Bands TDD_E | | | | | | -80.20 | |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |
| Correlation Matrix and Antenna Configuration | | | 1x1 | | 1x1 | | 1x1 | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Es/Iot, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.43.2.5-2: RSRP TDD Intra frequency relative accuracy requirements for the reported values**

| Normal Conditions | Test 1 All bands | Test 2 All bands | Test 3 | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_x-10 | RSRP_x-10 | Bands TDD_A | RSRP_x-9 |
| | | | Bands TDD_C | RSRP_x-9 |
| | | | Bands TDD_E | RSRP_x-9 |
| Highest reported value (Cell 2) | RSRP_x+2 | RSRP_x+2 | Bands TDD_A | RSRP_x+3 |
| | | | Bands TDD_C | RSRP_x+3 |
| | | | Bands TDD_E | RSRP_x+3 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_x-10 | RSRP_x-10 | Bands TDD_A | RSRP_x-9 |
| | | | Bands TDD_C | RSRP_x-9 |
| | | | Bands TDD_E | RSRP_x-9 |
| Highest reported value (Cell 2) | RSRP_x+2 | RSRP_x+2 | Bands TDD_A | RSRP_x+3 |
| | | | Bands TDD_C | RSRP_x+3 |
| | | | Bands TDD_E | RSRP_x+3 |
| Note 1: E-UTRA operating band groups are as defined in Section 3.5. | | | | |
| Note 2: RSRP_x is the reported value of Cell 1. | | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.44 4 DL CA PCell in FDD FDD-TDD RSRP for E-UTRAN in Carrier Aggregation

Editor's Note: This test case is incomplete. The following items are missing or are incomplete:

- Test tolerances are undefined

- Connection Diagram is undefined in TS 36.508

- Message contents are undefined

- The test applicability needs to be added to TS36.521-2

### 9.1.44.1 Test purpose

To verify that TDD-FDD RSRP absolute and relative measurement accuracy in carrier aggregation with PCell in FDD are within the specified limits under AWGN propagation conditions. This test will verify the absolute RSRP accuracy requirements of the primary component carrier, the absolute RSRP accuracy requirements of the secondary component carriers, and the relative RSRP accuracy requirements of the secondary component carriers. The test will also verify the primary and secondary component carriers' relative RSRP accuracy requirement.

### 9.1.44.2 Test applicability

This test applies to all types of E-UTRA UE release 12 and forward that support E-UTRA FDD and TDD and 4DL CA with FDD as PCell.

### 9.1.44.3 Minimum conformance requirements

The RSRP measurements of cells on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1. The RSRP measurements of cells on the secondary component carriers shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.2.1.

The absolute accuracy requirements in table 9.1.44.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.

- Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.

- RSRP$|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.1.44.3-1: RSRP Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_B | -120.5 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD B, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The RSRP relative measurements of cells on the PCC compared with measurements of cells on any of the SCCs shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.3.2. The RSRP relative measurements of cells on any SCC shall meet relative accuracy requirements defined in TS 36.133 [4] clause 9.1.2.2.

The accuracy requirements in Table 9.1.44.3-2 for PCC-SCC relative accuracy requirements are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.
- RSRP1,2$|_{dBm}$ according to Annex I.3.4 for a corresponding Band
- $\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$
- | Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.1.44.3-2: RSRP PCC-SCC relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/$BW_{Channel}$ |
| ±4.5 | ±6 | ≥-6 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The requirements in Table 9.1.44.3-3 for SCC relative accuracy are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled.
- $RSRP1,2|_{dBm}$ according to Annex I.3.8 for a corresponding Band.

**Table 9.1.37.3-3: RSRP relative accuracy for SCCs**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/$BW_{Channel}$ |
| ±2 | ±3 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3 | ±3 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.44.3-4. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.44.3-4: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.11.1, 9.1.11.2, 9.1.11.3, 9.1.4, and A.9.1.44.

## 9.1.44.4 Test description

### 9.1.44.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 9.1.44.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure TBD as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.1.44.4.3.

4. Cell 1 is PCell on the primary component carrier, and Cell 2, Cell 4 and Cell 6 are activated SCells on secondary component carriers SCC1, SCC2 and SCC3 respectively. Cell 3, Cell 5 and Cell 7 are neighbouring cells on secondary component carriers SCC1, SCC2 and SCC3 respectively. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2, Cell 3, Cell 4, Cell 5, Cell 6, and Cell 7 shall be powered OFF.

### 9.1.44.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCell according to Annex C.0, C.1 and C.2 for all downlink physical channels.

3. The SS shall configure SCells (cell 2, cell 4, and cell 6) on the SCCs as per TS 36.508 [7] clause 5.2A.4 [with the message content exceptions defined in clause 9.1.44.4.3].

4. SS activates SCCs by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).

5. Set the parameters according to Table 9.1.44.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.

6. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, SS shall check the reported RSRP values in periodical MeasurementReport messages.

   The reported RSRP values for Cell 1, Cell 2, Cell 4 and Cell 6 are compared to the actual RSRP values according to Table 9.1.44.5-2. This counts respectively as a Pass or Fail for the events “Cell 1”, “Cell 2”, “Cell

4", and "Cell 6". If the UE fails to report the measurement value for Cell 2 or Cell 4 or Cell 6, the number of failed iterations for the respective event is increased by one.

The reported RSRP value for Cell 2, Cell 3, Cell 4, Cell5, Cell 6, and Cell 7 are compared to the reported RSRP value of other cells for each MeasurementReport message according to Table 9.1.44.5-3. These count respectively as a Pass or Fail for the events "Cell 1-2", "Cell 1-4", "Cell 1-6", "Cell 2-3", "Cell 4-5", and "Cell 6-7". If the UE fails to report the measurement value for Cell 2, Cell 3, Cell 4, Cell 5, Cell6, or Cell 7, the number of failed iterations for the respective event is increased by one.

10. SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events "Cell 1", "Cell 2", "Cell 4", "Cell 6", "Cell 1-2", "Cell 1-4", "Cell 1-6", "Cell 2-3", "Cell 4-5", and "Cell 6-7" is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.

If all events pass for each scenario, the test passes. If one event fails, the test fails.

### 9.1.44.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.1.44.4.3-1: Common Exception messages for FDD RSRP Accuracy E-UTRA for Carrier Aggregation test requirement**

**TBD**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-3<br>Table H.4.1-7<br>Table H.4.2-5 |

## 9.1.44.5 Test requirement

Table 9.1.44.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP absolute accuracy shall meet the reported values test requirements in table 9.1.44.5-2.

The RSRP relative accuracy shall meet the reported values test requirements in table 9.1.44.5-3.

**Table 9.1.44.5-1: 4DL PCell in FDD RSRP for E-UTRAN in Carrier Aggregation test parameters**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 | Cell 4 | Cell 5 | Cell 6 | Cell 7 |
|---|---|---|---|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | 2 | | 3 | | 4 | |
| $BW_{channel}$ | | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | |
| Special subframe configuration[Note1] | | | - | 6 | | 6 | | 6 | |
| Uplink/downlink configuration[Note1] | | | - | 1 | | 1 | | 1 | |
| Measurement bandwidth | | $n_{PRB}$ | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | |
| PDSCH Reference measurement channel | | | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | - | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | - | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | - | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | - | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD |
| OCNG Patterns | | | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.10 TDD<br>10MHz: OP.2 TDD<br>20MHz: OP.8 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | | |
| PSS_RA | | | | | | | | | |
| SSS_RA | | | | | | | | | |
| PCFICH_RB | | | | | | | | | |
| PHICH_RA | | | | | | | | | |
| PHICH_RB | | | | | | | | | |
| PDCCH_RA | | | | | | | | | |
| PDCCH_RB | | | | | | | | | |
| PDSCH_RA | | | | | | | | | |
| PDSCH_RB | | | | | | | | | |
| OCNG_RA[Note2] | | | | | | | | | |
| OCNG_RB[Note2] | | | | | | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | - | ($N_{oc}$ for Channel 1 +1dB) +TT | | ($N_{oc}$ for Channel 1 +1dB) +TT | | ($N_{oc}$ for Channel 1 +1dB) +TT | |
| | Bands TDD_C | | | | | | | | |
| | Bands TDD_E | | | | | | | | |
| | Bands FDD_A | | -117+TT | - | | - | | - | |
| | Bands FDD_B | | -116.5+TT | | | | | | |
| | Bands FDD_C | | -116+TT | | | | | | |
| | Bands | | -115.5+TT | | | | | | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | FDD_D | | | | | | | | |
| | Bands FDD_E, FDD_F Note 7 | | -115+TT | | | | | | |
| | Bands FDD_G | | -114+TT | | | | | | |
| | Bands FDD_H | | -113.5+TT | | | | | | |
| $\hat{E}_s / N_{oc}$ | | dB | -4+TT | 3+TT | -1+TT | 3+TT | -1+TT | 3+TT | -1+TT |
| $\hat{E}_s / I_{ot}$ | | dB | -4+TT | 0.46+TT | -5.76+TT | 0.46+TT | -5.76+TT | 0.46+TT | -5.76+TT |
| RSRP Note4 | Bands TDD_A | dBm/15 kHz | - | (RSRP for Cell 1 +8dB+TT) | (RSRP for Cell 1 +4dB+TT) | (RSRP for Cell 1 +8dB+TT) | (RSRP for Cell 1 +4dB+TT) | (RSRP for Cell 1 +8dB+TT) | (RSRP for Cell 1 +4dB+TT) |
| | Bands TDD_C | | | | | | | | |
| | Bands TDD_E | | | | | | | | |
| | Bands FDD_A | | -121+TT | - | - | - | - | - | - |
| | Bands FDD_B | | -120.5+TT | | | | | | |
| | Bands FDD_C | | -120+TT | | | | | | |
| | Bands FDD_D | | -119.5+TT | | | | | | |
| | Bands FDD_E, FDD_F Note 7 | | -119+TT | | | | | | |
| | Bands FDD_G | | -118+TT | | | | | | |
| | Bands FDD_H | | -117.5+TT | | | | | | |
| Io Note4 | Bands TDD_A | dBm/ $BW_{channel}$ | - | (Io for Channel 1 +5.33dB +10log ($N_{RB\ channel2}$ / $N_{RB\ channel\ 1}$)) +TT | | (Io for Channel 1 +5.33dB +10log ($N_{RB\ channel3}$ / $N_{RB\ channel\ 1}$)) +TT | | (Io for Channel 1 +5.33dB +10log ($N_{RB\ channel4}$ / $N_{RB\ channel\ 1}$)) +TT | |
| | Bands TDD_C | | | | | | | | |
| | Bands TDD_E | | | | | | | | |
| | Bands FDD_A | | -87.76+10log($N_{RB,c}$/50) +TT | - | | - | | - | |
| | Bands FDD_B | | -87.26+10log($N_{RB,c}$/50) +TT | | | | | | |
| | Bands FDD_C | | -86.76+10log($N_{RB,c}$/50) +TT | | | | | | |
| | Bands FDD_D | | -86.26+10log($N_{RB,c}$/50) +TT | | | | | | |
| | Bands FDD_E, FDD_F Note 7 | | -85.76 +10log($N_{RB,c}$/50) +TT | | | | | | |
| | Bands FDD_G | | -84.76 +10log($N_{RB,c}$/50) +TT | | | | | | |
| | Bands FDD_H | | -84.26 +10log($N_{RB,c}$/50) +TT | | | | | | |
| Propagation condition | | - | AWGN | AWGN | AWGN | AWGN | AWGN | AWGN | AWGN |
| Antenna Configuration | | - | 1x2 | 1x2 | 1x2 | 1x2 | 1x2 | 1x2 | 1x2 |
| Timing offset to cell 1 | | µs | - | 0 | 3 | 0 | 3 | 0 | 3 |
| Time alignment error relative to cell 1 Note 8 | | | - | ≤ TAE | - | ≤ TAE | - | ≤ TAE | - |
| Time alignment error relative to cell 2 Note 8 | | | - | - | - | ≤ TAE | - | ≤ TAE | - |
| Time alignment error relative to cell 4 Note 8 | | | - | - | - | - | - | ≤ TAE | - |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over

subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 6: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.

Note 7: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 8: Time alignment error (TAE) as specified in TS 36.104 [30] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

Note 9: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.44.5-2: 4DL PCell in FDD RSRP for E-UTRAN in Carrier Aggregation absolute accuracy requirements for the reported values**

| Band of Cell 1 on Primary Component Carrier | Event | Bands FDD_A | Bands FDD_B | Bands FDD_C | Bands FDD_D | Bands FDD_E, FDD_F | Bands FDD_G | Bands FDD_H |
|---|---|---|---|---|---|---|---|---|
| Normal Conditions | | | | | | | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] |
| Highest reported value (Cell 1) | | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] |
| Highest reported value (Cell 2) | | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] |
| Lowest reported value (Cell 4) | Cell 4 | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] |
| Highest reported value (Cell 4) | | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] |
| Lowest reported value (Cell 6) | Cell 6 | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] |
| Highest reported value (Cell 6) | | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] |
| Extreme Conditions | | | | | | | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] |
| Highest reported value (Cell 1) | | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] |
| Lowest reported value (Cell 2) | Cell 2 | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] |
| Highest reported value (Cell 2) | | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] |
| Lowest reported value (Cell 4) | Cell 4 | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] |
| Highest reported value (Cell 4) | | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] |
| Lowest reported value (Cell 6) | Cell 6 | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] |
| Highest reported value (Cell 6) | | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] | RSRP_[TBD] |

Note: The band of Cell 1 determines the levels for Cell 1, Cell 2 and Cell 4, so the reported RSRP values depend only on the band of Cell 1, and are not dependent on the band of Cell 2 or Cell 4.

**Table 9.1.44.5-3: 4DL PCell in FDD RSRP for E-UTRAN in Carrier Aggregation relative accuracy requirements for the reported values**

| | Event | |
|---|---|---|
| Normal Conditions, all bands | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 + [TBD] |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + [TBD] |
| Lowest reported value (Cell 4) | Cell 1-4 | RSRP_Cell 1 + [TBD] |
| Highest reported value (Cell 4) | | RSRP_Cell 1 + [TBD] |
| Lowest reported value (Cell 6) | Cell 1-6 | RSRP_Cell 1 + [TBD] |
| Highest reported value (Cell 6) | | RSRP_Cell 1 + [TBD] |
| Extreme Conditions, all bands | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 + [TBD] |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + [TBD] |
| Lowest reported value (Cell 4) | Cell 1-4 | RSRP_Cell 1 + [TBD] |
| Highest reported value (Cell 4) | | RSRP_Cell 1 + [TBD] |
| Lowest reported value (Cell 6) | Cell 1-6 | RSRP_Cell 1 + [TBD] |
| Highest reported value (Cell 6) | | RSRP_Cell 1 + [TBD] |
| Normal and Extreme Conditions, all bands | | |
| Lowest reported value (Cell 3) | Cell 2-3 | RSRP_Cell 2 + [TBD] |
| Highest reported value (Cell 3) | | RSRP_Cell 2 + [TBD] |
| Lowest reported value (Cell 5) | Cell 4-5 | RSRP_Cell 4 + [TBD] |
| Highest reported value (Cell 5) | | RSRP_Cell 4 + [TBD] |
| Lowest reported value (Cell 7) | Cell 6-7 | RSRP_Cell 6 + [TBD] |
| Highest reported value (Cell 7) | | RSRP_Cell 6 + [TBD] |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.52 FD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeA

### 9.1.52.1 Test purpose

The purpose of this test is to verify that the RSRP measurement accuracy is within the specified limits. This test will verify the requirements in the intra frequency absolute and relative RSRP accuracies for FD-FDD intra frequency RSRP measurements for Cat-M1 UE in CEModeA.

### 9.1.52.2 Test applicability

This test applies to all types of E-UTRA FD-FDD UE release 13 and forward of UE category M1 that support CE Mode A. Applicability requires support for FGI bit 16.

### 9.1.52.3 Minimum conformance requirements

The requirements for absolute accuracy of RSRP in this clause apply to a cell on the same frequency as that of the serving cell for UE category M1.

The accuracy requirements in Table 9.1.52.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band

At least 1 DL subframe per radio frame of measured cell is available at the UE for RSRP measurement assuming measured cell is identified cell.

**Table 9.1.52.3-1: RSRP Intra frequency absolute accuracy for UE category M1 with CE mode A**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 3] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 2] | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±7 | ±10 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±9 | ±12 | ≥-6 dB | FDD_A, TDD_A, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_N | N/A | -70 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 3: E-UTRA operating band groups are as defined in Section 3.5. | | | | | | |

The relative accuracy of RSRP is defined as the RSRP measured from one cell compared to the RSRP measured from another cell on the same frequency for category M1 UE.

The accuracy requirements in Table 9.1.52.3-2 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Clause 7.3 for reference sensitivity are fulfilled.

RSRP1,2$|_{dBm}$ according to Annex I.3.8 for a corresponding Band.

At least 1 DL subframe per radio frame of measured cell is available at the UE for RSRP measurement assuming measured cell is identified cell.

**Table 9.1.52.3-2: RSRP Intra frequency relative accuracy for UE category M1 with CE mode A**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot [Note 2] | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 5] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 4] | dBm/BW$_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.<br>NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.<br>NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 5: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.52.3-3. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.52.3-3: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.21.1, 9.1.21.2, 9.1.4 and A.9.1.52.

## 9.1.52.4 Test description

### 9.1.52.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E Table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A Figure A.20 using only main UE Tx/Rx antenna.
2. Propagation conditions are set according to Annex B.0.
3. Message contents are defined in clause 9.1.52.4.3.
4. All cells are in the same carrier frequency. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

### 9.1.52.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF-CE with the condition CEModeA according to TS 36.508 [7] clause 7.2A.3AA.
2. Set the parameters according to Table 9.1.52.5-1 as appropriate. Propagation conditions are set according to Annex B.1.
3. The SS shall transmit the RRCConnectionReconfiguration message on Cell 1.
4. The UE shall transmit the RRCConnectionReconfigurationComplete message.
5. The UE shall transmit periodically MeasurementReport messages.
6. After 10s wait from Step 3, the SS shall check the reported RSRP values in periodical MeasurementReport messages.
   The reported RSRP value of Cell 2 reported by the UE is compared to the actual RSRP value according to Table 9.1.52.5-2. This counts as a Pass or Fail for the event “Cell 2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.
   The reported RSRP value of Cell 2 is compared to the reported RSRP value of Cell 1 for each MeasurementReport message according to Table 9.1.52.5-3. This counts as a Pass or Fail for the event “Cell 1-2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.
7. The SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.
8. Repeat step 1-7 for each sub-test in Table 9.1.52.5-1 as appropriate.

Each of the events “Cell 2” and “Cell 1-2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.

If all events pass for each scenario, the test passes. If one event fails, the test fails.

### 9.1.52.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the condition CEModeA and the following exceptions:

**Table 9.1.52.4.3-1: Common Exception messages for FD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeA test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1a with Gap Pattern Id = 0<br>Table H.3.5-3a |

**Table 9.1.52.4.3-2: *MeasResults*: Additional FD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeA test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement identity for which the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | Set according to specific test | |
| rsrqResult | INTEGER(0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.52.4.3-3: *MeasResultListEUTRA*: Additional FD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeA test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId from Cell2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

## 9.1.52.5 Test requirement

Table 9.1.52.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP absolute accuracy shall meet the reported values test requirements in Table 9.1.52.5-2.

The RSRP relative accuracy shall meet the reported values test requirements in Table 9.1.52.5-3.

**Table 9.1.52.5-1: FD-FDD RSRP Intra frequency test parameters for Cat-M1 UE in CEModeA**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel | | | R.20 FDD | - | R.20 FDD | - | R.20 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| MPDCCH Reference measurement channel | | | R.16 FDD | | R.16 FDD | | R.16 FDD | |
| OCNG Patterns | | | OP.21 FDD | OP.2 FDD | OP.21 FDD | OP.2 FDD | OP.21 FDD | OP.2 FDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>MPDCCH_RA<br>MPDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA[Note1]<br>OCNG_RB[Note1] | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -107 | | -87 | | -116 | |
| | Bands FDD_B | | | | | | -115.5 | |
| | Bands FDD_C | | | | | | -115 | |
| | Bands FDD_D | | | | | | -114.5 | |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -114 | |
| | Bands FDD_G [Note 6] | | | | | | -113 | |
| | Bands FDD_H | | | | | | -112.5 | |
| $\hat{E}_s / N_{oc}$ | | dB | 6 | 2 | 6 | 2 | 3 | -0.2 |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 | 1.88 | -4.97 | 0.09 | -4.96 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -101 | -105 | -81 | -85 | -113 | -116.2 |
| | Bands FDD_B | | | | | | -112.5 | -115.7 |
| | Bands FDD_C | | | | | | -112 | -115.2 |
| | Bands FDD_D | | | | | | -111.5 | -114.7 |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -111 | -114.2 |
| | Bands FDD_G [Note 6] | | | | | | -110 | -113.2 |
| | Bands FDD_H | | | | | | -109.5 | -112.7 |
| Io[Note3] | Bands FDD_A | dBm/9 MHz | -71.05 | | -51.05 | | -82.25 | |
| | Bands FDD_B | | | | | | -81.75 | |
| | Bands FDD_C | | | | | | -81.25 | |
| | Bands FDD_D | | | | | | -80.75 | |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -80.25 | |
| | Bands FDD_G [Note 6] | | | | | | -79.25 | |
| | Bands FDD_H | | | | | | -78.75 | |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |
| Antenna Configuration | | | 1x1 | | 1x1 | | 1x1 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: Es/Iot, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

| |
|---|
| Note 5: E-UTRA operating band groups are as defined in Section 3.5.<br>Note 6: Except Band 29 and Band 32. |

**Table 9.1.52.5-2: FD-FDD RSRP Intra frequency absolute accuracy requirements for the reported values for Cat-M1 UE in CEModeA**

| Normal Conditions | Test 1<br>All bands | Test 2<br>All bands | Test 3 | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_27 | RSRP_45 | Bands FDD_A | RSRP_16 |
| | | | Bands FDD_B | RSRP_16 |
| | | | Bands FDD_C | RSRP_17 |
| | | | Bands FDD_D | RSRP_17 |
| | | | Bands FDD_E, FDD_F | RSRP_18 |
| | | | Bands FDD_G | RSRP_19 |
| | | | Bands FDD_H | RSRP_19 |
| Highest reported value (Cell 2) | RSRP_44 | RSRP_66 | Bands FDD_A | RSRP_33 |
| | | | Bands FDD_B | RSRP_33 |
| | | | Bands FDD_C | RSRP_34 |
| | | | Bands FDD_D | RSRP_34 |
| | | | Bands FDD_E, FDD_F | RSRP_35 |
| | | | Bands FDD_G | RSRP_36 |
| | | | Bands FDD_D | RSRP_36 |
| **Extreme Conditions** | **Test 1<br>All bands** | **Test 2<br>All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_24 | RSRP_42 | Bands FDD_A | RSRP_13 |
| | | | Bands FDD_B | RSRP_13 |
| | | | Bands FDD_C | RSRP_14 |
| | | | Bands FDD_D | RSRP_14 |
| | | | Bands FDD_E, FDD_F | RSRP_15 |
| | | | Bands FDD_G | RSRP_16 |
| | | | Bands FDD_H | RSRP_16 |
| Highest reported value (Cell 2) | RSRP_47 | RSRP_69 | Bands FDD_A | RSRP_36 |
| | | | Bands FDD_B | RSRP_36 |
| | | | Bands FDD_C | RSRP_37 |
| | | | Bands FDD_D | RSRP_37 |
| | | | Bands FDD_E, FDD_F | RSRP_38 |
| | | | Bands FDD_G | RSRP_39 |
| | | | Bands FDD_H | RSRP_39 |
| NOTE 1: E-UTRA operating band groups are as defined in Section 3.5. | | | | |

**Table 9.1.52.5-3: FD-FDD RSRP Intra frequency relative accuracy requirements for the reported values for Cat-M1 UE in CEModeA**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| | All bands | All bands | All bands |
| Normal and Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRP_Cell 1 - 10 | RSRP_Cell 1 -10 | RSRP_Cell 1 - 9 |
| Highest reported value (Cell 2) | RSRP_Cell 1 + 2 | RSRP_Cell 1 + 2 | RSRP_Cell 1 + 2 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.53 HD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeA

### 9.1.53.1 Test purpose

The purpose of this test is to verify that the RSRP measurement accuracy is within the specified limits. This test will verify the requirements in the intra frequency absolute and relative RSRP accuracies for HD-FDD intra frequency RSRP measurements for Cat-M1 UE in CEModeA.

### 9.1.53.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 13 and forward of UE category M1 that support CE Mode A. Applicability requires support for FGI bit 16.

### 9.1.53.3 Minimum conformance requirements

Same minimum conformance requirements as in Clause 9.1.52.3.

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.21.1, 9.1.21.2, 9.1.4 and A.9.1.53.

### 9.1.53.4 Test description

Same test description as in clause 9.1.52.4 with the following exceptions:

- Instead of Table 9.1.52.5-1 → use Table 9.1.53.5-1.
- Instead of Table 9.1.52.5-2 → use Table 9.1.53.5-2.
- Instead of Table 9.1.52.5-3 → use Table 9.1.53.5-3.

### 9.1.53.5 Test requirement

Table 9.1.53.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP absolute accuracy shall meet the reported values test requirements in Table 9.1.53.5-2.

The RSRP relative accuracy shall meet the reported values test requirements in Table 9.1.53.5-3.

**Table 9.1.53.5-1: HD-FDD RSRP Intra frequency test parameters for Cat-M1 UE in CEModeA**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel | | | R.10 HD-FDD | - | R.10 HD-FDD | - | R.10 HD-FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| MPDCCH Reference measurement channel | | | R.6 HD-FDD | | R.6 HD-FDD | | R.6 HD-FDD | |
| OCNG Patterns | | | OP.21 FDD | OP.2 FDD | OP.21 FDD | OP.2 FDD | OP.21 FDD | OP.2 FDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>MPDCCH_RA<br>MPDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA[Note1]<br>OCNG_RB[Note1] | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -107 | | -87 | | -116 | |
| | Bands FDD_B | | | | | | -115.5 | |
| | Bands FDD_C | | | | | | -115 | |
| | Bands FDD_D | | | | | | -114.5 | |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -114 | |
| | Bands FDD_G [Note 7] | | | | | | -113 | |
| | Bands FDD_H | | | | | | -112.5 | |
| $\hat{E}_s / N_{oc}$ | | dB | 6 | 2 | 6 | 2 | 3 | -0.2 |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 | 1.88 | -4.97 | 0.09 | -4.96 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -101 | -105 | -81 | -85 | -113 | -116.2 |
| | Bands FDD_B | | | | | | -112.5 | -115.7 |
| | Bands FDD_C | | | | | | -112 | -115.2 |
| | Bands FDD_D | | | | | | -111.5 | -114.7 |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -111 | -114.2 |
| | Bands FDD_G [Note 6] | | | | | | -110 | -113.2 |
| | Bands FDD_H | | | | | | -109.5 | -112.7 |
| Io[Note3] | Bands FDD_A | dBm/9 MHz | -71.05 | | -51.05 | | -82.25 | |
| | Bands FDD_B | | | | | | -81.75 | |
| | Bands FDD_C | | | | | | -81.25 | |
| | Bands FDD_D | | | | | | -80.75 | |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -80.25 | |
| | Bands FDD_G [Note 6] | | | | | | -79.25 | |
| | Bands FDD_H | | | | | | -78.75 | |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |
| Antenna Configuration | | | 1x1 | | 1x1 | | 1x1 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: Es/Iot, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel

| | |
|---|---|
| | bandwidth within 865-894 MHz. |
| Note 5: | E-UTRA operating band groups are as defined in Section 3.5. |
| Note 6: | Except Band 29 and Band 32. |

**Table 9.1.53.5-2: HD-FDD RSRP Intra frequency absolute accuracy requirements for the reported values for Cat-M1 UE in CEModeA**

| Normal Conditions | Test 1 All bands | Test 2 All bands | Test 3 | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_27 | RSRP_45 | Bands FDD_A | RSRP_16 |
| | | | Bands FDD_B | RSRP_16 |
| | | | Bands FDD_C | RSRP_17 |
| | | | Bands FDD_D | RSRP_17 |
| | | | Bands FDD_E, FDD_F | RSRP_18 |
| | | | Bands FDD_G | RSRP_19 |
| | | | Bands FDD_H | RSRP_19 |
| Highest reported value (Cell 2) | RSRP_44 | RSRP_66 | Bands FDD_A | RSRP_33 |
| | | | Bands FDD_B | RSRP_33 |
| | | | Bands FDD_C | RSRP_34 |
| | | | Bands FDD_D | RSRP_34 |
| | | | Bands FDD_E, FDD_F | RSRP_35 |
| | | | Bands FDD_G | RSRP_36 |
| | | | Bands FDD_D | RSRP_36 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_24 | RSRP_42 | Bands FDD_A | RSRP_13 |
| | | | Bands FDD_B | RSRP_13 |
| | | | Bands FDD_C | RSRP_14 |
| | | | Bands FDD_D | RSRP_14 |
| | | | Bands FDD_E, FDD_F | RSRP_15 |
| | | | Bands FDD_G | RSRP_16 |
| | | | Bands FDD_H | RSRP_16 |
| Highest reported value (Cell 2) | RSRP_47 | RSRP_69 | Bands FDD_A | RSRP_36 |
| | | | Bands FDD_B | RSRP_36 |
| | | | Bands FDD_C | RSRP_37 |
| | | | Bands FDD_D | RSRP_37 |
| | | | Bands FDD_E, FDD_F | RSRP_38 |
| | | | Bands FDD_G | RSRP_39 |
| | | | Bands FDD_H | RSRP_39 |
| NOTE 1: E-UTRA operating band groups are as defined in Section 3.5. | | | | |

**Table 9.1.53.5-3: HD-FDD RSRP Intra frequency relative accuracy requirements for the reported values for Cat-M1 UE in CEModeA**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| | All bands | All bands | All bands |
| Normal and Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRP_Cell 1 - 10 | RSRP_Cell 1 -10 | RSRP_Cell 1 - 9 |
| Highest reported value (Cell 2) | RSRP_Cell 1 + 2 | RSRP_Cell 1 + 2 | RSRP_Cell 1 + 2 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.54 TDD RSRP Intra frequency case for Cat-M1 UE in CEModeA

### 9.1.54.1 Test purpose

The purpose of this test is to verify that the RSRP measurement accuracy is within the specified limits. This test will verify the requirements in the intra frequency absolute and relative RSRP accuracies for TDD intra frequency RSRP measurements for Cat-M1 UE in CEModeA.

### 9.1.54.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 13 and forward of UE category M1 that support CE Mode A. Applicability requires support for FGI bit 16.

### 9.1.54.3 Minimum conformance requirements

Same minimum conformance requirements as in Clause 9.1.52.3.

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.21.1, 9.1.21.2, 9.1.4 and A.9.1.54.

### 9.1.54.4 Test description

Same test description as in clause 9.1.52.4 with the following exceptions:

- Instead of Table 9.1.52.5-1 → use Table 9.1.54.5-1.
- Instead of Table 9.1.52.5-2 → use Table 9.1.54.5-2.
- Instead of Table 9.1.52.5-3 → use Table 9.1.54.5-3.

### 9.1.54.5 Test requirement

Table 9.1.54.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP absolute accuracy shall meet the reported values test requirements in Table 9.1.54.5-2.

The RSRP relative accuracy shall meet the reported values test requirements in Table 9.1.54.5-3.

**Table 9.1.54.5-1: TDD RSRP Intra frequency test parameters for Cat-M1 UE in CEModeA**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Special subframe configuration[Note1] | | | 6 | | 6 | | 6 | |
| Uplink/downlink configuration[Note1] | | | 1 | | 1 | | 1 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel | | | R.16 TDD | - | R.16 TDD | - | R.16 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| MPDCCH Reference measurement channel | | | R.14 TDD | | R.14 TDD | | R.14 TDD | |
| OCNG Patterns | | | OP.11 TDD | OP.2 TDD | OP.11 TDD | OP.2 TDD | OP.11 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | |
| PSS_RA | | | | | | | | |
| SSS_RA | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| MPDCCH_RA | | | | | | | | |
| MPDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA[Note2] | | | | | | | | |
| OCNG_RB[Note2] | | | | | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -107 | | -87 | | -116 | |
| | Bands TDD_C | | | | | | -115 | |
| | Bands TDD_E | | | | | | -114 | |
| $\hat{E}_s / N_{oc}$ | | dB | 6 | 2 | 6 | 2 | 3 | -0.2 |
| $\hat{E}_s / I_{ot}$ | | dB | 1.88 | -4.97 | 1.88 | -4.97 | 0.09 | -4.96 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -101 | -105 | -81 | -85 | -113 | -116.2 |
| | Bands TDD_C | | | | | | -112 | -115.2 |
| | Bands TDD_E | | | | | | -111 | -114.2 |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -71.05 | | -51.05 | | -82.25 | |
| | Bands TDD_C | | | | | | -81.25 | |
| | Bands TDD_E | | | | | | -80.25 | |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |
| Correlation Matrix and Antenna Configuration | | | 1x1 | | 1x1 | | 1x1 | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Es/Iot, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.54.5-2: TDD RSRP Intra frequency absolute accuracy requirements for the reported values for Cat-M1 UE in CEModeA**

| Normal Conditions | Test 1 All bands | Test 2 All bands | Test 3 | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_27 | RSRP_45 | Bands TDD_A | RSRP_16 |
| | | | Bands TDD_C | RSRP_17 |
| | | | Bands TDD_E | RSRP_18 |
| Highest reported value (Cell 2) | RSRP_44 | RSRP_66 | Bands TDD_A | RSRP_33 |
| | | | Bands TDD_C | RSRP_34 |
| | | | Bands TDD_E | RSRP_35 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_24 | RSRP_42 | Bands TDD_A | RSRP_13 |
| | | | Bands TDD_C | RSRP_14 |
| | | | Bands TDD_E | RSRP_15 |
| Highest reported value (Cell 2) | RSRP_47 | RSRP_69 | Bands TDD_A | RSRP_36 |
| | | | Bands TDD_C | RSRP_37 |
| | | | Bands TDD_E | RSRP_38 |
| NOTE 1: E-UTRA operating band groups are as defined in Section 3.5. | | | | |

**Table 9.1.54.5-3: TDD RSRP Intra frequency relative accuracy requirements for the reported values for Cat-M1 UE in CEModeA**

| | Event | All bands |
|---|---|---|
| **Normal and Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 – [FFS] |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + [FFS] |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.57 FD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeB

### 9.1.57.1 Test purpose

The purpose of this test is to verify that the RSRP measurement accuracy is within the specified limits. This test will verify the requirements for FD-FDD intra-frequency RSRP measurements for Cat-M1 UE in CEModeB.

### 9.1.57.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 13 and forward of UE category M1. Applicability requires support for CE Mode B and FGI bit 16.

### 9.1.57.3 Minimum conformance requirements

The Intra-frequency absolute RSRP measurements for UE category M1 shall meet the absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.21.3. The Intra-frequency relative RSRP measurements for UE category M1 shall meet the relative accuracy requirements defined in TS 36.133 [4] clause 9.1.21.4.

The requirements for absolute accuracy of RSRP in this clause apply to a cell on the same frequency as that of the serving cell for UE category M1.

The accuracy requirements in Table 9.1.57.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Clause 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.1 for a corresponding Band

At least 1 DL subframe per radio frame of measured cell is available at the UE for RSRP measurement assuming measured cell is identified cell.

**Table 9.1.57.3-1: RSRP Intra-frequency absolute accuracy for UE category M1 with CE mode B for FDD and TDD**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 1 range | | | |
| | | | E-UTRA operating band groups Note 3 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±8 | ±11 | -15≤Ês/Iot≤-12 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| ±7 | ±10 | ≥-12 dB | FDD_G | -118 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±10 | ±13 | -15≤Ês/Iot≤-12 dB | FDD_A, TDD_A, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_N | N/A | -70 | -50 |
| ±9 | ±12 | ≥-12 dB | | | | |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The relative accuracy of RSRP is defined as the RSRP measured from one cell compared to the RSRP measured from another cell on the same frequency for category M1 UE.

The accuracy requirements in Table 9.1.57.3-2 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Clause 7.3 for reference sensitivity are fulfilled.

RSRP1,2$|_{dBm}$ according to Annex I.3.8 for a corresponding Band.

At least 1 DL subframe per radio frame of measured cell is available at the UE for RSRP measurement assuming measured cell is identified cell.

**Table 9.1.57.3-2: SRP Intra-frequency relative accuracy for UE category M1 with CE mode B for FDD and TDD**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/$BW_{Channel}$ |
| ±4 | ±4 | ≥-12 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±5 | ±5 | -15≤Ês/Iot≤-12 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.1.57.3-3. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.1.57.3-3: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.21.3, 9.1.21.4, 9.1.21.5, 9.1.4 and A.9.1.57.

## 9.1.57.4 Test description

### 9.1.57.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20 using only main UE Tx/Rx antenna.

2. Propagation conditions are set according to Annex B clauses B.0.

3. Message contents are defined in clause 9.1.57.4.3.

4. All cells are in the same carrier frequency. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

### 9.1.57.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF-CE with the condition CEModeB according to TS 36.508 [7] clause 7.2A.3AA.

2. Set the parameters according to Table 9.1.57.5-1 as appropriate. Propagation conditions are set according to Annex B.1.

3. The SS shall transmit the RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit the RRCConnectionReconfigurationComplete message.

5. The UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, the SS shall check the reported RSRP values in periodical MeasurementReport messages.
   The reported RSRP value of Cell 2 reported by the UE is compared to the actual RSRP value according to Table 9.1.57.5-2. This counts as a Pass or Fail for the event “Cell 2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.
   The reported RSRP value of Cell 2 is compared to the reported RSRP value of Cell 1 for each MeasurementReport message according to Table 9.1.57.5-3. This counts as a Pass or Fail for the event “Cell 1-2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. The SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

8. Repeat step 1-7 for each sub-test in Table 9.1.57.5-1 as appropriate.

Each of the events “Cell 2” and “Cell 1-2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.

If all events pass for each scenario, the test passes. If one event fails, the test fails.

### 9.1.57.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the condition CEModeB and the following exceptions:

**Table 9.1.57.4.3-1: Common Exception messages for FD-FDD RSRP Intra-frequency case for Cat-M1 UE in CEModeB test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1a with Gap Pattern Id = 0<br>Table H.3.5-3a |

**Table 9.1.57.4.3-2: *MeasResults*: Additional FD-FDD RSRP Intra-frequency case for Cat-M1 UE in CEModeB test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement identity for which the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | Set according to specific test | |
| rsrqResult | INTEGER(0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.1.57.4.3-3: *MeasResultListEUTRA*: Additional FD-FDD RSRP Intra-frequency case for Cat-M1 UE in CEModeB test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId from Cell2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

## 9.1.57.5 Test requirement

Table 9.1.57.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP absolute accuracy shall meet the reported values test requirements in table 9.1.57.7-2.

The RSRP relative accuracy shall meet the reported values test requirements in table 9.1.57.7-3.

**Table 9.1.57.5-1: FD-FDD RSRP Intra-frequency test parameters for Cat-M1 UE in CEModeB**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel | | | R.22 FDD | - | R.22 FDD | - | R.22 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| MPDCCH Reference measurement channel | | | R.18 FDD | | R.18 FDD | | R.18 FDD | |
| OCNG Patterns | | | OP.21 FDD | OP.2 FDD | OP.21 FDD | OP.2 FDD | OP.21 FDD | OP.2 FDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>MPDCCH_RA<br>MPDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA[Note1]<br>OCNG_RB[Note1] | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -99.3 | | -79.3 | | -107 | |
| | Bands FDD_B | | | | | | -106.5 | |
| | Bands FDD_C | | | | | | -106 | |
| | Bands FDD_D | | | | | | -105.5 | |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -105 | |
| | Bands FDD_G | | | | | | -104 | |
| | Bands FDD_H | | | | | | -103.5 | |
| $\hat{E}_s / N_{oc}$ | | dB | -12 | -13.9 | -12 | -13.9 | -12 | -13.9 |
| $\hat{E}_s / I_{ot}$ | | dB | -12.17 | -14.17 | -12.17 | -14.17 | -12.17 | -14.17 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -111.3 | -113.2 | -91.3 | -93.2 | -119 | -120.9 |
| | Bands FDD_B | | | | | | -118.5 | -120.4 |
| | Bands FDD_C | | | | | | -118 | -119.9 |
| | Bands FDD_D | | | | | | -117.5 | -119.4 |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -117 | -118.9 |
| | Bands FDD_G | | | | | | -116 | -117.9 |
| | Bands FDD_H | | | | | | -115.5 | -117.4 |
| Io[Note3] | Bands FDD_A | dBm/9 MHz | -71.09 | | -51.09 | | -78.79 | |
| | Bands FDD_B | | | | | | -78.29 | |
| | Bands FDD_C | | | | | | -77.79 | |
| | Bands FDD_D | | | | | | -77.29 | |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -76.79 | |
| | Bands FDD_G | | | | | | -75.79 | |
| | Bands FDD_H | | | | | | -75.29 | |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |
| Antenna Configuration | | | 1x1 | | 1x1 | | 1x1 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: Es/Iot, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 5: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.57.5-2: FD-FDD RSRP Intra-frequency absolute accuracy requirements for the reported values for Cat-M1 UE in CEModeB**

| Normal Conditions | Test 1 All bands | Test 2 All bands | Test 3 | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_18 | RSRP_36 | Bands FDD_A | RSRP_10 |
| | | | Bands FDD_B | RSRP_11 |
| | | | Bands FDD_C | RSRP_11 |
| | | | Bands FDD_D | RSRP_12 |
| | | | Bands FDD_E, FDD_F | RSRP_12 |
| | | | Bands FDD_G | RSRP_13 |
| | | | Bands FDD_H | RSRP_14 |
| Highest reported value (Cell 2) | RSRP_37 | RSRP_59 | Bands FDD_A | RSRP_29 |
| | | | Bands FDD_B | RSRP_30 |
| | | | Bands FDD_C | RSRP_30 |
| | | | Bands FDD_D | RSRP_31 |
| | | | Bands FDD_E, FDD_F | RSRP_31 |
| | | | Bands FDD_G | RSRP_32 |
| | | | Bands FDD_H | RSRP_34 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_15 | RSRP_33 | Bands FDD_A | RSRP_7 |
| | | | Bands FDD_B | RSRP_8 |
| | | | Bands FDD_C | RSRP_8 |
| | | | Bands FDD_D | RSRP_9 |
| | | | Bands FDD_E, FDD_F | RSRP_9 |
| | | | Bands FDD_G | RSRP_10 |
| | | | Bands FDD_H | RSRP_11 |
| Highest reported value (Cell 2) | RSRP_40 | RSRP_62 | Bands FDD_A | RSRP_32 |
| | | | Bands FDD_B | RSRP_33 |
| | | | Bands FDD_C | RSRP_33 |
| | | | Bands FDD_D | RSRP_34 |
| | | | Bands FDD_E, FDD_F | RSRP_34 |
| | | | Bands FDD_G | RSRP_35 |
| | | | Bands FDD_H | RSRP_36 |
| Note 1: E-UTRA operating band groups are as defined in Section 3.5. | | | | |

**Table 9.1.57.5-3: FD-FDD RSRP Intra-frequency relative accuracy requirements for the reported values for Cat-M1 UE in CEModeB**

| | Event | Test 1 All bands | Test 2 All bands | Test 3 All bands |
|---|---|---|---|---|
| **Normal Conditions** | | | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 – 9 | RSRP_Cell 1 – 9 | RSRP_Cell 1 – 9 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 5 | RSRP_Cell 1 + 5 | RSRP_Cell 1 + 5 |
| **Extreme Conditions** | | | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 – 9 | RSRP_Cell 1 – 9 | RSRP_Cell 1 – 9 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 5 | RSRP_Cell 1 + 5 | RSRP_Cell 1 + 5 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.1.58 HD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeB

## 9.1.58.1 Test purpose

The purpose of this test is to verify that the RSRP measurement accuracy is within the specified limits. This test will verify the requirements in the intra frequency absolute and relative RSRP accuracies for HD-FDD intra frequency RSRP measurements for Cat-M1 UE in CEModeB.

## 9.1.58.2 Test applicability

This test applies to all types of E-UTRA HD-FDD UE release 13 and forward of UE category M1 that support CE Mode B. Applicability requires support for FGI bit 16.

## 9.1.58.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.57.3 with the following exceptions:

- Instead of Table 9.1.57.3-1 → use Table 9.1.58.3-1.
- Instead of Table 9.1.57.3-2 → use Table 9.1.58.3-2.

**Table 9.1.58.3-1: RSRP Intra frequency absolute accuracy for UE category M1 with CE mode B for HD-FDD**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| | | | Io Note 1 range | | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±8 | ±11 | -15≤Ês/Iot≤-12 dB | FDD_A | -121 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| ±7 | ±10 | ≥-12 dB | FDD_G | -118 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±10 | ±13 | -15≤Ês/Iot≤-12 dB | FDD_A, FDD_D, FDD_E, FDD_F, FDD_G, FDD_N | N/A | -70 | -50 |
| ±9 | ±12 | ≥-12 dB | | | | |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.58.3-2: RSRP Intra frequency relative accuracy for UE category M1 with CE mode B for HD-FDD**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/BW$_{Channel}$ |
| ±4 | ±4 | ≥-12 dB | FDD_A | -121 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±5 | ±5 | -15≤Ês/Iot≤-12 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.21.3, 9.1.21.4, 9.1.4, and A.9.1.58.

## 9.1.58.4 Test description

Same test description as in clause 9.1.57.4 with the following exceptions:

- Instead of Table 9.1.57.5-1 → use Table 9.1.58.5-1.
- Instead of Table 9.1.57.5-2 → use Table 9.1.58.5-2.
- Instead of Table 9.1.57.5-3 → use Table 9.1.58.5-3.

## 9.1.58.5 Test requirement

Table 9.1.58.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP absolute accuracy shall meet the reported values test requirements in Table 9.1.58.5-2.

The RSRP relative accuracy shall meet the reported values test requirements in Table 9.1.58.5-3.

**Table 9.1.58.5-1: HD-FDD RSRP Intra frequency test parameters for Cat-M1 UE in CEModeB**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel | | | R.12 HD-FDD | - | R.12 HD-FDD | - | R.12 HD-FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| MPDCCH Reference measurement channel | | | R.8 HD-FDD | | R.8 HD-FDD | | R.8 HD-FDD | |
| OCNG Patterns | | | OP.21 FDD | OP.2 FDD | OP.21 FDD | OP.2 FDD | OP.21 FDD | OP.2 FDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>MPDCCH_RA<br>MPDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA[Note1]<br>OCNG_RB[Note1] | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -99.3 | | -79.3 | | -107 | |
| | Bands FDD_B | | | | | | -106.5 | |
| | Bands FDD_C | | | | | | -106 | |
| | Bands FDD_D | | | | | | -105.5 | |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -105 | |
| | Bands FDD_G [Note 7] | | | | | | -104 | |
| | Bands FDD_H | | | | | | -103.5 | |
| $\hat{E}_s / N_{oc}$ | | dB | -12 | -13.9 | -12 | -13.9 | -12 | -13.9 |
| $\hat{E}_s / I_{ot}$ | | dB | -12.17 | -14.17 | -12.17 | -14.17 | -12.17 | -14.17 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -111.3 | -113.2 | -91.3 | -93.2 | -119 | -120.9 |
| | Bands FDD_B | | | | | | -118.5 | -120.4 |
| | Bands FDD_C | | | | | | -118 | -119.9 |
| | Bands FDD_D | | | | | | -117.5 | -119.4 |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -117 | -118.9 |
| | Bands FDD_G [Note 6] | | | | | | -116 | -117.9 |
| | Bands FDD_H | | | | | | -115.5 | -117.4 |
| Io[Note3] | Bands FDD_A | dBm/9 MHz | -71.79 | | -51.79 | | -78.79 | |
| | Bands FDD_B | | | | | | -78.29 | |
| | Bands FDD_C | | | | | | -77.79 | |
| | Bands FDD_D | | | | | | -77.29 | |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -76.79 | |
| | Bands FDD_G [Note 6] | | | | | | -75.79 | |
| | Bands FDD_H | | | | | | -75.29 | |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |
| Antenna Configuration | | | 1x1 | | 1x1 | | 1x1 | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: Es/Iot, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel

| |
|---|
| bandwidth within 865-894 MHz.<br>Note 5: E-UTRA operating band groups are as defined in Section 3.5.<br>Note 6: Except Band 29 and Band 32. |

**Table 9.1.58.5-2: HD-FDD RSRP Intra frequency absolute accuracy requirements for the reported values for Cat-M1 UE in CEModeB**

| Normal Conditions | Test 1 All bands | Test 2 All bands | Test 3 | |
|---|---|---|---|---|
| Lowest reported value (Cell 2) | RSRP_18 | RSRP_36 | Bands FDD_A | RSRP_10 |
| | | | Bands FDD_B | RSRP_11 |
| | | | Bands FDD_C | RSRP_11 |
| | | | Bands FDD_D | RSRP_12 |
| | | | Bands FDD_E, FDD_F | RSRP_12 |
| | | | Bands FDD_G | RSRP_13 |
| | | | Bands FDD_H | RSRP_14 |
| Highest reported value (Cell 2) | RSRP_37 | RSRP_59 | Bands FDD_A | RSRP_29 |
| | | | Bands FDD_B | RSRP_30 |
| | | | Bands FDD_C | RSRP_30 |
| | | | Bands FDD_D | RSRP_31 |
| | | | Bands FDD_E, FDD_F | RSRP_31 |
| | | | Bands FDD_G | RSRP_32 |
| | | | Bands FDD_H | RSRP_33 |
| **Extreme Conditions** | **Test 1 All bands** | **Test 2 All bands** | **Test 3** | |
| Lowest reported value (Cell 2) | RSRP_15 | RSRP_33 | Bands FDD_A | RSRP_7 |
| | | | Bands FDD_B | RSRP_8 |
| | | | Bands FDD_C | RSRP_8 |
| | | | Bands FDD_D | RSRP_9 |
| | | | Bands FDD_E, FDD_F | RSRP_9 |
| | | | Bands FDD_G | RSRP_10 |
| | | | Bands FDD_H | RSRP_11 |
| Highest reported value (Cell 2) | RSRP_40 | RSRP_62 | Bands FDD_A | RSRP_32 |
| | | | Bands FDD_B | RSRP_33 |
| | | | Bands FDD_C | RSRP_33 |
| | | | Bands FDD_D | RSRP_34 |
| | | | Bands FDD_E, FDD_F | RSRP_34 |
| | | | Bands FDD_G | RSRP_35 |
| | | | Bands FDD_H | RSRP_36 |
| Note 1: E-UTRA operating band groups are as defined in Section 3.5. | | | | |

**Table 9.1.58.5-3: HD-FDD RSRP Intra frequency relative accuracy requirements for the reported values for Cat-M1 UE in CEModeB**

| | Event | Test 1 All bands | Test 2 All bands | Test 3 All bands |
|---|---|---|---|---|
| **Normal and Extreme Conditions** | | | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 – 9 | RSRP_Cell 1 – 9 | RSRP_Cell 1 – 9 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 5 | RSRP_Cell 1 + 5 | RSRP_Cell 1 + 5 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.1.59 TDD RSRP Intra frequency case for Cat-M1 UE in CEModeB

### 9.1.59.1 Test purpose

The purpose of this test is to verify that the RSRP measurement accuracy is within the specified limits. This test will verify the requirements for TDD Intra-frequency RSRP measurements for Cat-M1 UE in CEModeB.

### 9.1.59.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 13 and forward of UE category M1. Applicability requires support for CE Mode B and FGI bit 16.

### 9.1.59.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.1.57.3.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.21.3, clause 9.1.21.4, clause 9.1.21.5 and A.9.1.59.

### 9.1.59.4 Test description

#### 9.1.59.4.1 Initial conditions

Same initial conditions as in clause 9.1.57.4.1.

#### 9.1.59.4.2 Test procedure

Same test procedure as in clause 9.1.57.4.2 with following exceptions:

- Instead of Table 9.1.57.5-1 → use Table 9.1.59.5-1.
- Instead of Table 9.1.57.5-2 → use Table 9.1.59.5-2.
- Instead of Table 9.1.57.5-2 → use Table 9.1.59.5-2.

#### 9.1.59.4.3 Message contents

Same message content as in clause 9.1.57.4.3.

### 9.1.59.5 Test requirement

Table 9.1.59.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP absolute accuracy shall meet the reported values test requirements in table 9.1.59.5-2.

**The RSRP relative accuracy shall meet the reported values test requirements in table 9.1.59.5-3.Table 9.1.59.5-1: TDD RSRP Intra-frequency test parameters for Cat-M1 UE in CEModeB**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Special subframe configuration[Note1] | | | 6 | | 6 | | 6 | |
| Uplink/downlink configuration[Note1] | | | 1 | | 1 | | 1 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in clause A.8.6 | | | R.18 TDD | - | R.18 TDD | - | R.18 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| MPDCCH Reference measurement channel defined in clause A.7.6 | | | R.16 TDD | | R.16 TDD | | R.16 TDD | |
| OCNG Patterns defined in clause D.2 | | | OP.11 TDD | OP.2 TDD | OP.11 TDD | OP.2 TDD | OP.11 TDD | OP.2 TDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>MPDCCH_RA<br>MPDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA[Note2]<br>OCNG_RB[Note2] | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -99.3 | | -79.3 | | -107 | |
| | Bands TDD_C | | | | | | -106 | |
| | Bands TDD_E | | | | | | -105 | |
| $\hat{E}_s / N_{oc}$ | | dB | -12 | -13.9 | -12 | -13.9 | -12 | -13.9 |
| $\hat{E}_s / I_{ot}$ | | dB | -12.17 | -14.17 | -12.17 | -14.17 | -12.17 | -14.17 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -111.3 | -113.2 | -91.3 | -93.2 | -119 | -120.9 |
| | Bands TDD_C | | | | | | -118 | -119.9 |
| | Bands TDD_E | | | | | | -117 | -118.9 |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -71.09 | | -51.09 | | -78.79 | |
| | Bands TDD_C | | | | | | -77.79 | |
| | Bands TDD_E | | | | | | -76.79 | |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |
| Correlation Matrix and Antenna Configuration | | | 1x1 | | 1x1 | | 1x1 | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Es/Iot, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.1.59.5-2: TDD RSRP Intra-frequency absolute accuracy requirements for the reported values for Cat-M1 UE in CEModeB**

| Normal Conditions | Event | | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|---|
| Lowest reported value (Cell 2) | Cell 2 | Bands TDD_A | RSRP_18 | RSRP_36 | RSRP_10 |
| | | Bands TDD_C | | | RSRP_11 |
| | | Bands TDD_E | | | RSRP_12 |
| Highest reported value (Cell 2) | Cell 2 | Bands TDD_A | RSRP_37 | RSRP_59 | RSRP_29 |
| | | Bands TDD_C | | | RSRP_30 |
| | | Bands TDD_E | | | RSRP_31 |
| **Extreme Conditions** | **Event** | | **Test 1** | **Test 2** | **Test 3** |
| Lowest reported value (Cell 2) | Cell 2 | Bands TDD_A | RSRP_15 | RSRP_33 | RSRP_7 |
| | | Bands TDD_C | | | RSRP_8 |
| | | Bands TDD_E | | | RSRP_9 |
| Highest reported value (Cell 2) | Cell 2 | Bands TDD_A | RSRP_40 | RSRP_62 | RSRP_32 |
| | | Bands TDD_C | | | RSRP_33 |
| | | Bands TDD_E | | | RSRP_34 |

**Table 9.1.59.5-3: TDD RSRP Intra-frequency relative accuracy requirements for the reported values for Cat-M1 UE in CEModeB**

| | Event | Test 1 All bands | Test 2 All bands | Test 3 All bands |
|---|---|---|---|---|
| **Normal Conditions** | | | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 – 9 | RSRP_Cell 1 – 9 | RSRP_Cell 1 – 9 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 5 | RSRP_Cell 1 + 5 | RSRP_Cell 1 + 5 |
| **Extreme Conditions** | | | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRP_Cell 1 – 9 | RSRP_Cell 1 – 9 | RSRP_Cell 1 – 9 |
| Highest reported value (Cell 2) | | RSRP_Cell 1 + 5 | RSRP_Cell 1 + 5 | RSRP_Cell 1 + 5 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.2 RSRQ

## 9.2.1 FDD Intra frequency RSRQ Accuracy

### 9.2.1.1 FDD Intra Frequency Absolute RSRQ Accuracy

#### 9.2.1.1.1 Test purpose

To verify that the FDD intra frequency absolute RSRQ measurement accuracy is within the specified limit for all bands.

#### 9.2.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward. Applicability requires support for FGI bit 16.

#### 9.2.1.1.3 Minimum conformance requirements

The absolute accuracy of RSRQ is defined as the RSRQ measured from a cell on the same frequency as that of the serving cell.

The accuracy requirements in table 9.2.1.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.1 for a corresponding Band.

**Table 9.2.1.1.3-1: RSRQ FDD intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/BW$_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.1.1.3-2.

**Table 9.2.1.1.3-2: RSRQ FDD Intra frequency absolute accuracy measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.5.1 and A.9.2.1.

### 9.2.1.1.4 Test description

#### 9.2.1.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.1.1.4.3.

4. There is one E-UTRA FDD carrier and two cells specified in each test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

#### 9.2.1.1.4.2 Test procedure

The three tests consist of Cell 1, serving cell and Cell 2, target cell. The absolute accuracy of RSRQ is defined as the RSRQ measured from a cell on the same frequency as that of the serving cell, Cell 1.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.2.1.1.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the RSRQ value in periodical MeasurementReport messages. The RSRQ value of Cell 2 reported by the UE is compared to the actual RSRQ value according to Table 9.2.1.1.5-3. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.2.1.1.5-2 as appropriate.

#### 9.2.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.1.1.4.3-1: Common Exception messages for RSRQ FDD intra frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-4 |

**Table 9.2.1.1.4.3-2: *MeasResults*: Additional RSRQ FDD intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.1.1.4.3-3: *MeasResultListEUTRA*: Additional RSRQ FDD intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | Not present | | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

### 9.2.1.1.5 Test requirement

Table 9.2.1.1.5-2 defines the primary level settings including test tolerances for all tests.

Each RSRQ FDD intra frequency absolute accuracy test shall meet the reported values test requirements in Table 9.2.1.1.5-3.

**Table 9.2.1.1.5-1: Void**

**Table 9.2.1.1.5-2: Cell Specific Test requirement Parameters for RSRQ FDD intra frequency absolute accuracy**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - | R.0 FDD | - | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | |
| PSS_RA | | | | | | | | |
| SSS_RA | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA Note1 | | | | | | | | |
| OCNG_RB Note1 | | | | | | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -85.51 | | -103.85 | | -116 | |
| | Bands FDD_C | | | | | | -115 | |
| | Bands FDD_D | | | | | | -114.5 | |
| | Bands FDD_E, FDD_F Note 5 | | | | | | -114 | |
| | Bands FDD_G | | | | | | -113 | |
| | Bands FDD_H | | | | | | -112.5 | |
| $\hat{E}_s / I_{ot}$ | | dB | -1.76 | -1.76 | -4.7 | -4.7 | -5.17 | -5.17 |
| RSRP Note3 | Bands FDD_A | dBm/15 kHz | -82.51 | -82.51 | -106.75 | -106.75 | -119.60 | -119.60 |
| | Bands FDD_C | | | | | | -118.60 | -118.60 |
| | Bands FDD_D | | | | | | -118.10 | -118.10 |
| | Bands FDD_E, FDD_F Note 5 | | | | | | -117.60 | -117.60 |
| | Bands FDD_G | | | | | | -116.60 | -116.60 |
| | Bands FDD_H | | | | | | -116.10 | -116.10 |
| RSRQ Note3 | Bands FDD_A | dB | -14.77 | -14.77 | -16.76 | -16.76 | -17.12 | -17.12 |
| | Bands FDD_C | | | | | | | |
| | Bands FDD_D | | | | | | | |
| | Bands FDD_E, FDD_F Note 5 | | | | | | | |
| | Bands FDD_G | | | | | | | |
| | Bands FDD_H | | | | | | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Io[Note3] | Bands FDD_A | dBm/9 MHz | -50.75 | | -73.00 | | -85.49 | |
| | Bands FDD_C | | | | | | -84.49 | |
| | Bands FDD_D | | | | | | -83.99 | |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -83.49 | |
| | Bands FDD_G | | | | | | -82.49 | |
| | Bands FDD_H | | | | | | -81.99 | |
| $\hat{E}_s / N_{oc}$ | | dB | 3 | 3 | -2.9 | -2.9 | -3.6 | -3.6 |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 6: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.2.1.1.5-3: RSRQ FDD Intra frequency absolute accuracy requirements for the reported values**

| | **Test 1** | **Test 2** | **Test 3** |
|---|---|---|---|
| | All bands | All bands | All bands |
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_04 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_16 | RSRQ_14 | RSRQ_14 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_01 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_19 | RSRQ_15 | RSRQ_15 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.2 TDD Intra frequency RSRQ Accuracy

### 9.2.2.1 TDD Intra Frequency Absolute RSRQ Accuracy

#### 9.2.2.1.1 Test purpose

To verify that the TDD intra frequency absolute RSRQ measurement accuracy is within the specified limit for all TDD bands.

#### 9.2.2.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward. Applicability requires support for FGI bit 16.

#### 9.2.2.1.3 Minimum conformance requirements

The absolute accuracy of RSRQ is defined as the RSRQ measured from a cell on the same frequency as that of the serving cell.

The accuracy requirements in table 9.2.2.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.1 for a corresponding Band.

**Table 9.2.2.1.3-1: RSRQ TDD intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 4] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 3] | dBm/$BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution.

The mapping of measured quantity is defined in Table 9.2.2.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.2.2.1.3-2: RSRQ TDD Intra frequency absolute accuracy measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.5.1, 9.1.7 and A.9.2.2.

## 9.2.2.1.4 Test description

### 9.2.2.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.2.1.4.3.

4. There is one E-UTRA TDD carrier and two cells specified in each test. Cell 1 is the cell used for call setup with the power level set according to clause C.0 and C.1 for this test.

#### 9.2.2.1.4.2 Test procedure

The three tests consist of Cell 1, serving cell and Cell 2, target cell. The absolute accuracy of RSRQ is defined as the RSRQ measured from a cell on the same frequency as that of the serving cell, Cell 1.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.2.2.1.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the RSRQ value in periodical MeasurementReport messages. The RSRQ value of Cell 2 reported by the UE is compared to the actual RSRQ according to Table 9.2.2.1.5-3. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.2.2.1.5-2 as appropriate.

#### 9.2.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.2.1.4.3-1: Common Exception messages for RSRQ TDD intra frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-4 |

**Table 9.2.2.1.4.3-2: *MeasResults*: Additional RSRQ TDD intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| meaResuCellItsServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.2.1.4.3-3: *MeasResultListEUTRA*: Additional RSRQ TDD intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| measResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of Cell 2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | Not present | | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

## 9.2.2.1.5 Test requirement

Table 9.2.2.1.5-2 defines the primary level settings including test tolerances for all tests.

Each RSRQ TDD intra frequency absolute accuracy test shall meet the reported values test requirements in Table 9.2.2.1.5-3.

**Table 9.2.2.1.5-1: Void**

**Table 9.2.2.1.5-2: Cell Specific Test requirement Parameters for RSRQ TDD intra frequency absolute accuracy**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Special subframe configuration[Note1] | | | 6 | | 6 | | 6 | |
| Uplink-downlink configuration[Note1] | | | 1 | | 1 | | 1 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | R.0 TDD | - | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | |
| PSS_RA | | | | | | | | |
| SSS_RA | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA[Note2] | | | | | | | | |
| OCNG_RB[Note2] | | | | | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -85.51 | | -103.85 | | -116 | |
| | Bands TDD_C | | | | | | -115 | |
| | Bands TDD_E | | | | | | -114 | |
| $\hat{E}_s / I_{ot}$ | | dB | -1.76 | -1.76 | -4.7 | -4.7 | -5.17 | -5.17 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -82.51 | -82.51 | -106.75 | -106.75 | -119.60 | -119.60 |
| | Bands TDD_C | | | | | | -118.60 | -118.60 |
| | Bands TDD_E | | | | | | -117.60 | -117.60 |
| RSRQ[Note4] | Bands TDD_A, TDD_C, TDD_E | dB | -14.77 | -14.77 | -16.76 | -16.76 | -17.12 | -17.12 |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -50.75 | | -73.00 | | -85.49 | |
| | Bands TDD_C | | | | | | -84.49 | |
| | Bands TDD_E | | | | | | -83.49 | |
| $\hat{E}_s / N_{oc}$ | | dB | 3 | 3 | -2.9 | -2.9 | -3.6 | -3.6 |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in 3GPP TS 36.211.
Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 6: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.2.2.1.5-3: RSRQ TDD Intra frequency absolute accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| | All bands | All bands | All bands |
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_04 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_16 | RSRQ_14 | RSRQ_14 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_01 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_19 | RSRQ_15 | RSRQ_15 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.3 FDD - FDD Inter frequency RSRQ Accuracy

### 9.2.3.1 FDD - FDD Inter Frequency Absolute RSRQ Accuracy

#### 9.2.3.1.1 Test purpose

To verify that the FDD - FDD inter frequency absolute RSRQ measurement accuracy is within the specified limit for all bands.

#### 9.2.3.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward. Applicability requires support for FGI bits 16 and 25.

#### 9.2.3.1.3 Minimum conformance requirements

The absolute accuracy of RSRQ is defined as the RSRQ measured from a cell that has different carrier frequency from the serving cell.

The accuracy requirements in table 9.2.3.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.3 for a corresponding Band.

**Table 9.2.3.1.3-1: RSRQ FDD - FDD inter frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 4] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 3] | dBm/$BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.3.1.3-2.

**Table 9.2.3.1.3-2: RSRQ FDD - FDD Inter frequency absolute accuracy measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.6.1 and A.9.2.3.

### 9.2.3.1.4 Test description

#### 9.2.3.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.3.1.4.3.

4. There are two E-UTRA FDD carriers and one cell on each carrier specified in each test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

#### 9.2.3.1.4.2 Test procedure

The three tests consist of Cell 1, serving cell and Cell 2, target cell. The absolute accuracy of RSRQ is defined as the RSRQ measured from a cell that has different carrier frequency from the serving cell, Cell 1.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.2.3.1.5-2 for the test interval as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the RSRQ value in periodical MeasurementReport messages. The RSRQ value of Cell 2 reported by the UE is compared to the actual RSRQ value according to Table 9.2.3.1.5-3. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to table G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each test interval in Table 9.2.3.1.5-2 as appropriate.

#### 9.2.3.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.3.1.4.3-1: Common Exception messages for RSRQ FDD - FDD inter frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-2<br>Table H.3.5-4 |

**Table 9.2.3.1.4.3-2: *MeasResults*: Additional RSRQ FDD - FDD inter frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.3.1.4.3-3: *MeasResultListEUTRA*: Additional RSRQ FDD - FDD inter frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId | | |
| measResult SEQUENCE { | | | |
| rsrpResult | Not present | | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

### 9.2.3.1.5 Test requirement

Table 9.2.3.1.5-2 defines the primary level settings including test tolerances for all tests.

Each RSRQ FDD - FDD inter frequency absolute accuracy test shall meet the reported values test requirements in Table 9.2.3.1.5-3.

**Table 9.2.3.1.5-1: Void**

**Table 9.2.3.1.5-2: Cell Specific Test requirement Parameters for RSRQ FDD - FDD inter frequency absolute accuracy**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 | 1 | 2 | 1 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 | 10 | 10 | 10 | 10 |
| Gap Pattern Id | | | 0 | - | 0 | - | 0 | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - | R.0 FDD | - | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | |
| PSS_RA | | | | | | | | |
| SSS_RA | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA[Note1] | | | | | | | | |
| OCNG_RB[Note1] | | | | | | | | |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -80.0 | -81.1 | -104.7 | -104.7 | -119.5 | -119.2 |
| | Bands FDD_C | | | | | | -118.5 | -118.2 |
| | Bands FDD_D | | | | | | -118 | -117.7 |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -117.5 | -117.2 |
| | Bands FDD_G | | | | | | -116.5 | -116.2 |
| | Bands FDD_H | | | | | | -116 | -115.7 |
| $\hat{E}_s / I_{ot}$ | | dB | -1.75 | -1.75 | -4.0 | -3.2 | -4.0 | -3.2 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -81.75 | -82.85 | -108.70 | -107.90 | -123.5 | -122.4 |
| | Bands FDD_C | | | | | | -122.5 | -121.4 |
| | Bands FDD_D | | | | | | -122.0 | -120.9 |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -121.5 | -120.4 |
| | Bands FDD_G | | | | | | -120.5 | -119.4 |
| | Bands FDD_H | | | | | | -120.0 | -118.9 |
| RSRQ[Note3] | Bands FDD_A | dB | -14.76 | -14.76 | -16.25 | -15.69 | -16.25 | -15.69 |
| | Bands FDD_C | | | | | | | |
| | Bands FDD_D | | | | | | | |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | | |
| | Bands FDD_G | | | | | | | |
| | Bands FDD_H | | | | | | | |
| Io[Note3] | Bands FDD_A | dBm/9 MHz | -50.00 | -51.10 | -75.46 | -75.22 | -90.26 | -89.72 |
| | Bands FDD_C | | | | | | -89.26 | -88.72 |
| | Bands FDD_D | | | | | | -88.76 | -88.22 |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -88.26 | -87.72 |
| | Bands FDD_G | | | | | | -87.26 | -86.72 |
| | Bands FDD_H | | | | | | -86.76 | -86.22 |
| $\hat{E}_s / N_{oc}$ | | dB | -1.75 | -1.75 | -4.0 | -3.2 | -4.0 | -3.2 |

| | | | | |
|---|---|---|---|---|
| Propagation condition | - | AWGN | AWGN | AWGN |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 6: E-UTRA operating band groups are as defined in TS 36.133 [4] section 3.5.

**Table 9.2.3.1.5-3: RSRQ FDD - FDD inter frequency absolute accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_04 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_16 | RSRQ_16 | RSRQ_16 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_01 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_19 | RSRQ_17 | RSRQ_17 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.3.2 FDD - FDD Inter Frequency Relative Accuracy of RSRQ

### 9.2.3.2.1 Test purpose

To verify that the FDD - FDD inter frequency relative accuracy measurement of RSRQ is within the specified limit for all bands.

### 9.2.3.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward. Applicability requires support for FGI bits 16 and 25.

### 9.2.3.2.3 Minimum conformance requirements

The relative accuracy of RSRQ in inter frequency case is defined as the RSRQ measured from one cell compared to the RSRQ measured from another cell on a different frequency.

The accuracy requirements in table 9.2.3.2.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

RSRP1,2$|_{dBm}$ according to Annex I.3.4 for a corresponding Band

$$\left| RSRP1|_{dBm} - RSRP2|_{dBm} \right| \leq [27] dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.2.3.2.3-1: RSRQ FDD - FDD inter frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/$BW_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.3.2.3-2.

**Table 9.2.3.2.3-2: RSRQ FDD - FDD Inter frequency relative accuracy measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.6.2 and A.9.2.3.

## 9.2.3.2.4 Test description

### 9.2.3.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.3.2.4.3.

4. There are two E-UTRA FDD carriers and one cell on each carrier specified in the test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

#### 9.2.3.2.4.2 Test procedure

The three tests consist of Cell 1, serving cell and Cell 2, target cell. The relative accuracy of RSRQ is defined as the RSRQ measured from one cell compared to the RSRQ measured from another cell on a different frequency.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.2.3.2.5-2 for the test interval as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the RSRQ value of Cell 1 and Cell 2 in periodical MeasurementReport messages. The reported RSRQ value for Cell 2 is compared to the reported RSRQ value for Cell 1 for each MeasurementReport message. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each test interval in Table 9.2.3.2.5-2 as appropriate.

#### 9.2.3.2.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.3.2.4.3-1: Common Exception messages for RSRQ FDD - FDD inter frequency relative accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-2<br>Table H.3.5-4 |

**Table 9.2.3.2.4.3-2: *MeasResults*: Additional RSRQ FDD - FDD inter frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.3.2.4.3-3: *MeasResultListEUTRA*: Additional RSRQ FDD - FDD inter frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | Not present | | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

## 9.2.3.2.5 Test requirement

Table 9.2.3.2.5-2 defines the primary level settings including test tolerances for all tests.

Each RSRQ FDD - FDD inter frequency relative accuracy test shall meet the reported values test requirements in Table 9.2.3.2.5-3.

**Table 9.2.3.2.5-1: Void**

**Table 9.2.3.2.5-2: Cell Specific Test requirement Parameters for RSRQ FDD - FDD inter frequency relative accuracy**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 | 1 | 2 | 1 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 | 10 | 10 | 10 | 10 |
| Gap Pattern Id | | | 0 | - | 0 | - | 0 | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - | R.0 FDD | - | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | |
| PSS_RA | | | | | | | | |
| SSS_RA | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA[Note1] | | | | | | | | |
| OCNG_RB[Note1] | | | | | | | | |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -81.1 | -81.1 | -104.7 | -104.7 | -119.5 | -119.2 |
| | Bands FDD_C | | | | | | -118.5 | -118.2 |
| | Bands FDD_D | | | | | | -118 | -117.7 |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -117.5 | -117.2 |
| | Bands FDD_G | | | | | | -116.5 | -116.2 |
| | Bands FDD_H | | | | | | -116 | -115.7 |
| $\hat{E}_s / I_{ot}$ | | dB | -1.75 | -1.75 | -4.0 | -3.2 | -4.0 | -3.2 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -82.85 | -82.85 | -108.70 | -107.90 | -123.5 | -122.4 |
| | Bands FDD_C | | | | | | -122.5 | -121.4 |
| | Bands FDD_D | | | | | | -122.0 | -120.9 |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -121.5 | -120.4 |
| | Bands FDD_G | | | | | | -120.5 | -119.4 |
| | Bands FDD_H | | | | | | -120.0 | -118.9 |
| RSRQ[Note3] | Bands FDD_A | dB | -14.76 | -14.76 | -16.25 | -15.69 | -16.25 | -15.69 |
| | Bands FDD_C | | | | | | | |
| | Bands FDD_D | | | | | | | |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | | |
| | Bands FDD_G | | | | | | | |
| | Bands FDD_H | | | | | | | |
| Io[Note3] | Bands FDD_A | dBm/9 MHz | -51.10 | -51.10 | -75.46 | -75.22 | -90.26 | -89.72 |
| | Bands FDD_C | | | | | | -89.26 | -88.72 |
| | Bands FDD_D | | | | | | -88.76 | -88.22 |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -88.26 | -87.72 |
| | Bands FDD_G | | | | | | -87.26 | -86.72 |
| | Bands FDD_H | | | | | | -86.76 | -86.22 |
| $\hat{E}_s / N_{oc}$ | | dB | -1.75 | -1.75 | -4.0 | -3.2 | -4.0 | -3.2 |

| | | | | |
|---|---|---|---|---|
| Propagation condition | - | AWGN | AWGN | AWGN |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 6: E-UTRA operating band groups are as defined in section 3.5.

**Table 9.2.3.2.5-3: RSRQ FDD inter frequency relative accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_x - 8 | RSRQ_x - 9 | RSRQ_x - 9 |
| Highest reported value (Cell 2) | RSRQ_x +8 | RSRQ_x +11 | RSRQ_x +11 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_x - 10 | RSRQ_x - 9 | RSRQ_x - 9 |
| Highest reported value (Cell 2) | RSRQ_x + 10 | RSRQ_x + 11 | RSRQ_x + 11 |
| RSRQ_x is the reported value of Cell 1 | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.4 TDD - TDD Inter frequency RSRQ Accuracy

### 9.2.4.1 TDD - TDD Inter Frequency Absolute RSRQ Accuracy

#### 9.2.4.1.1 Test purpose

To verify that the TDD - TDD inter frequency absolute RSRQ measurement accuracy is within the specified limit for all bands.

#### 9.2.4.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward. Applicability requires support for FGI bits 16 and 25.

#### 9.2.4.1.3 Minimum conformance requirements

The absolute accuracy of RSRQ is defined as the RSRQ measured from a cell that has different carrier frequency from the serving cell.

The accuracy requirements in table 9.2.4.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or RSRQ value of Cell 2 reported by the UE four antenna ports.

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.3 for a corresponding Band.

**Table 9.2.4.1.3-1: RSRQ TDD - TDD inter frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 4] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 3] | dBm/$BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.2.1.3-2.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.6.1 and A.9.2.4.

### 9.2.4.1.4 Test description

#### 9.2.4.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.4.1.4.3.

4. There are two E-UTRA TDD carriers and one cell on each carrier specified in each test. Cell 1 is the cell used for call setup with the power level set according to clause C.0 and C.1 for this test.

#### 9.2.4.1.4.2 Test procedure

The three tests consist of Cell 1, serving cell and Cell 2, target cell. The absolute accuracy of RSRQ is defined as the RSRQ measured from a cell that has different carrier frequency from the serving cell, Cell 1. The Gap pattern configuration is with id#0 as defined in TS 36.133 clause 8.1.2.1.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.2.4.1.5-2 for the test interval as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the RSRQ value in periodical MeasurementReport messages. The RSRQ value of Cell 2 reported by the UE is compared to the actual RSRQ value according to Table 9.2.4.1.5-3. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each test interval in Table 9.2.4.1.5-2 as appropriate.

#### 9.2.4.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.4.1.4.3-1: Common Exception messages for RSRQ TDD - TDD inter frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-2<br>Table H.3.5-4 |

**Table 9.2.4.1.4.3-2: *MeasResults*: Additional RSRQ TDD - TDD inter frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell | SEQUENCE { | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.4.1.4.3-3: *MeasResultListEUTRA*: Additional RSRQ TDD - TDD inter frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | Not present | | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

### 9.2.4.1.5 Test requirement

Table 9.2.4.1.5-2 defines the primary level settings including test tolerances for all tests.

Each RSRQ TDD inter frequency absolute accuracy test shall meet the reported values test requirements in Table 9.2.4.2.5-3.

**Table 9.2.4.1.5-1: Void**

**Table 9.2.4.1.5-2: Cell Specific Test requirement Parameters for RSRQ TDD - TDD inter frequency absolute accuracy**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 | 1 | 2 | 1 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 | 10 | 10 | 10 | 10 |
| Gap Pattern Id | | | 0 | - | 0 | - | 0 | - |
| Special subframe configuration Note1 | | | 6 | | 6 | | 6 | |
| Uplink-downlink configuration Note1 | | | 1 | | 1 | | 1 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | R.0 TDD | - | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | |
| PSS_RA | | | | | | | | |
| SSS_RA | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA Note2 | | | | | | | | |
| OCNG_RB Note2 | | | | | | | | |
| $N_{oc}$ Note3 | Bands TDD_A | dBm/15 kHz | -80.0 | -81.1 | -104.70 | -104.70 | -119.50 | -119.20 |
| | Bands TDD_C | | | | | | -118.50 | -118.20 |
| | Bands TDD_E | | | | | | -117.50 | -117.20 |
| $\hat{E}_s / I_{ot}$ | | dB | -1.75 | -1.75 | -4.0 | -3.2 | -4.0 | -3.2 |
| RSRP Note4 | Bands TDD_A | dBm/15 kHz | -81.75 | -82.85 | -108.70 | -107.90 | -123.50 | -122.40 |
| | Bands TDD_C | | | | | | -122.50 | -121.40 |
| | Bands TDD_E | | | | | | -121.50 | -120.40 |
| RSRQ Note4 | Bands TDD_A, TDD_C, TDD_E | dB | -14.76 | -14.76 | -16.25 | -15.69 | -16.25 | -15.69 |
| Io Note4 | Bands TDD_A | dBm/9 MHz | -50.00 | -51.10 | -75.46 | -75.22 | -90.26 | -89.72 |
| | Bands TDD_C | | | | | | -89.26 | -88.72 |
| | Bands TDD_E | | | | | | -88.26 | -87.72 |
| $\hat{E}_s / N_{oc}$ | | dB | -1.75 | -1.75 | -4.0 | -3.2 | -4.0 | -3.2 |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |

| | |
|---|---|
| Note 1: | For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211. |
| Note 2: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 3: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 4: | RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 5: | RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 6: | E-UTRA operating band groups are as defined in TS 36.133 [4] section 3.5. |

**Table 9.2.4.1.5-3: RSRQ TDD - TDD inter frequency absolute accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_04 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_16 | RSRQ_16 | RSRQ_16 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_01 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_19 | RSRQ_17 | RSRQ_17 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.4.2 TDD - TDD Inter Frequency Relative Accuracy of RSRQ

### 9.2.4.2.1 Test purpose

To verify that the TDD - TDDinter frequency relative accuracy measurement of RSRQ is within the specified limit for all bands.

### 9.2.4.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward. Applicability requires support for FGI bits 16 and 25.

### 9.2.4.2.3 Minimum conformance requirements

The relative accuracy of RSRQ in inter frequency case is defined as the RSRQ measured from one cell compared to the RSRQ measured from another cell on a different frequency.

The accuracy requirements in table 9.2.4.2.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

RSRP1,2$|_{\text{dBm}}$ according to Annex I.3.4 for a corresponding Band.

$$\left| RSRP1|_{dBm} - RSRP2|_{dBm} \right| \le [27] dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.2.4.2.3-1: RSRQ TDD - TDD Inter frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/$BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.2.1.3-2.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.6.2 and A.9.2.4.

### 9.2.4.2.4 Test description

#### 9.2.4.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.4.2.4.3.

4. There are two E-UTRA TDD carriers and one cell on each carrier specified in the test. Cell 1 is the cell used for call setup with the power level set according to clause C.0 and C.1 for this test.

#### 9.2.4.2.4.2 Test procedure

The three tests consist of Cell 1, serving cell and Cell 2, target cell. The relative accuracy of RSRQ is defined as the RSRQ measured from one cell compared to the RSRQ measured from another cell on a different frequency. The Gap pattern configuration is with id#0 as defined in TS 36.133 clause 8.1.2.1.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.2.4.2.5-2 for the test interval as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the RSRQ value of Cell 1 and Cell 2 in periodical MeasurementReport messages. The reported RSRQ value for Cell 2 is compared to the reported RSRQ value for Cell 1 for each MeasurementReport message according to Table 9.2.4.2.5-3. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each test interval in Table 9.2.4.1.5-2 as appropriate.

#### 9.2.4.2.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.4.2.4.3-1: Common Exception messages for RSRQ TDD - TDD inter frequency relative accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-2<br>Table H.3.5-4 |

**Table 9.2.4.1.4.3-2: *MeasResults*: Additional RSRQ TDD - TDD inter frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.4.1.4.3-3: *MeasResultListEUTRA*: Additional RSRQ TDD - TDD inter frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | Not present | | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

### 9.2.4.2.5 Test requirement

Table 9.2.4.2.5-2 defines the primary level settings including test tolerances for all tests.

Each RSRQ TDD inter frequency relative accuracy test shall meet the reported values test requirements in Table 9.2.4.2.5-3.

**Table 9.2.4.2.5-1: Void**

**Table 9.2.4.2.5-2: Cell Specific Test requirement Parameters for RSRQ TDD - TDD inter frequency relative accuracy**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 | 1 | 2 | 1 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 | 10 | 10 | 10 | 10 |
| Gap Pattern Id | | | 0 | - | 0 | - | 0 | - |
| Special subframe configuration Note1 | | | 6 | | 6 | | 6 | |
| Uplink-downlink configuration Note1 | | | 1 | | 1 | | 1 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | R.0 TDD | - | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA Note2<br>OCNG_RB Note2 | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $N_{oc}$ Note3 | Bands TDD_A | dBm/15 kHz | -81.1 | -81.1 | -104.70 | -104.70 | -119.50 | -119.20 |
| | Bands TDD_C | | | | | | -118.50 | -118.20 |
| | Bands TDD_E | | | | | | -117.50 | -117.20 |
| $\hat{E}_s / I_{ot}$ | | dB | -1.75 | -1.75 | -4.0 | -3.2 | -4.0 | -3.2 |
| RSRP Note4 | Bands TDD_A | dBm/15 kHz | -82.85 | -82.85 | -108.70 | -107.90 | -123.50 | -122.40 |
| | Bands TDD_C | | | | | | -122.50 | -121.40 |
| | Bands TDD_E | | | | | | -121.50 | -120.40 |
| RSRQ Note4 | Bands TDD_A, TDD_C, TDD_E | dB | -14.76 | -14.76 | -16.25 | -15.69 | -16.25 | -15.69 |
| Io Note4 | Bands TDD_A | dBm/9 MHz | -51.10 | -51.10 | -75.46 | -75.22 | -90.26 | -89.72 |
| | Bands TDD_C | | | | | | -89.26 | -88.72 |
| | Bands TDD_E | | | | | | -88.26 | -87.72 |
| $\hat{E}_s / N_{oc}$ | | dB | -1.75 | -1.75 | -4.0 | -3.2 | -4.0 | -3.2 |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |

| | |
|---|---|
| Note 1: | For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211. |
| Note 2: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 3: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 4: | RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 5: | RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 6: | E-UTRA operating band groups are as defined in section 3.5. |

**Table 9.2.4.2.5-3: RSRQ TDD - TDD inter frequency relative accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_x - 8 | RSRQ_x -9 | RSRQ_x -9 |
| Highest reported value (Cell 2) | RSRQ_x + 8 | RSRQ_x + 11 | RSRQ_x + 11 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_x - 10 | RSRQ_x - 9 | RSRQ_x - 9 |
| Highest reported value (Cell 2) | RSRQ_x + 10 | RSRQ_x + 11 | RSRQ_x + 11 |
| RSRQ_x is the reported value of Cell 1 | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.2.4A FDD - TDD Inter frequency RSRQ Accuracy

## 9.2.4A.1 FDD - TDD Inter Frequency Absolute RSRQ Accuracy

### 9.2.4A.1.1 Test purpose

To verify that the FDD - TDD inter frequency absolute RSRQ measurement accuracy is within the specified limit for all bands.

### 9.2.4A.1.2 Test applicability

This test applies to all types of E-UTRA UE supporting FDD and TDD release 9 and forward. Applicability requires support for FGI bits 16 and 25.

### 9.2.4A.1.3 Minimum conformance requirements

The absolute accuracy of RSRQ is defined as the RSRQ measured from a cell that has different carrier frequency from the serving cell.

The accuracy requirements in table 9.2.4A.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.3 for a corresponding Band.

**Table 9.2.4A.1.3-1: RSRQ FDD - TDD inter frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 4] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 3] | dBm/$BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.4A.1.3-2.

**Table 9.2.4A.1.3-2: RSRQ FDD - TDD Inter frequency absolute accuracy measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.6.1 and A.9.2.4A.

## 9.2.4A.1.4 Test description

### 9.2.4A.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.4A.1.4.3.

4. Cell 1 and Cell 2 are on the different carrier frequencies. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is FDD cell and Cell 2 is TDD cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.2.4A.1.4.2 Test procedure

The three tests consist of Cell 1, serving cell and Cell 2, target cell. The absolute accuracy of RSRQ is defined as the RSRQ measured from a cell that has different carrier frequency from the serving cell, Cell 1.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.2.4A.1.5-1 and 9.2.4A.1.5-2 for the test interval as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. The SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. The UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, the SS shall check the RSRQ value in periodical MeasurementReport messages. The RSRQ value of Cell 2 reported by the UE is compared to the actual RSRQ value according to Table 9.2.4A.1.5-3. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to table G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.2.4A.1.5-1 and 9.2.4A.1.5-2 as appropriate.

#### 9.2.4A.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.4A.1.4.3-1: Common Exception messages for RSRQ FDD - TDD inter frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-2<br>Table H.3.5-4 |

**Table 9.2.4A.1.4.3-2: *MeasResults*: Additional RSRQ FDD - TDD inter frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.4A.1.4.3-3: *MeasResultListEUTRA*: Additional RSRQ FDD - TDD inter frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId | | |
| measResult SEQUENCE { | | | |
| rsrpResult | Not present | | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

## 9.2.4A.1.5 Test requirement

Table 9.2.4A.1.5-1 and 9.2.4A.1.5-2 define the primary level settings including test tolerances for all tests.

Each RSRQ FDD - TDD inter frequency absolute accuracy test shall meet the reported values test requirements in Table 9.2.4A.1.5-3.

**Table 9.2.4A.1.5-1: Cell Specific Test requirement Parameters for RSRQ FDD - TDD inter frequency absolute accuracy (FDD Cell1)**

| Parameter | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|
| | | Cell 1 | Cell 1 | Cell 1 |
| E-UTRA RF Channel Number | | 1 | 1 | 1 |
| $BW_{channel}$ | MHz | 10 | 10 | 10 |
| Gap Pattern Id | | 0 | 0 | 0 |
| Measurement bandwidth | $n_{PRB}$ | 22—27 | 22—27 | 22—27 |
| PDSCH Reference measurement channel defined in A.1.1 | | R.0 FDD | R.0 FDD | R.0 FDD |
| PDSCH allocation | $n_{PRB}$ | 13—36 | 13—36 | 13—36 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | R.6 FDD | R.6 FDD | R.6 FDD |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.1 FDD | OP.1 FDD |
| PBCH_RA | dB | 0 | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA Note1 | | | | |
| OCNG_RB Note1 | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -80 | -104.70 | -114.5 |
| $\hat{E}_s / I_{ot}$ | dB | -1.75 | -4.0 | -4.0 |
| RSRP Note3 | dBm/15 kHz | -81.75 | -108.70 | -118.5 |
| RSRQ Note3 | dB | -14.76 | -16.25 | -16.25 |
| Io Note3 | dBm/9 MHz | -50 | -75.46 | -85.26 |
| $\hat{E}_s / N_{oc}$ | dB | -1.75 | -4.0 | -4.0 |
| Propagation condition | - | AWGN | AWGN | AWGN |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 9.2.4A.1.5-2: Cell Specific Test requirement Parameters for RSRQ FDD - TDD inter frequency absolute accuracy (TDD Cell2)**

| Parameter | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|
| | | Cell 2 | Cell 2 | Cell 2 |
| E-UTRA RF Channel Number | | 2 | 2 | 2 |
| $BW_{channel}$ | MHz | 10 | 10 | 10 |
| Gap Pattern Id | | - | - | - |
| Special subframe configuration Note1 | | 6 | 6 | 6 |
| Uplink-downlink configuration Note1 | | 1 | 1 | 1 |
| Measurement bandwidth | $n_{PRB}$ | 22—27 | 22—27 | 22—27 |
| PDSCH Reference measurement channel defined in A.1.2 | | - | - | - |
| PDSCH allocation | $n_{PRB}$ | - | - | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | R.6 TDD | R.6 TDD | R.6 TDD |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | OP.2 TDD | OP.2 TDD | OP.2 TDD |
| PBCH_RA | dB | 0 | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA Note2 | | | | |
| OCNG_RB Note2 | | | | |
| $N_{oc}$ Note3 | dBm/15 kHz | -81.10 | -104.70 | -114.50 |
| $\hat{E}_s / I_{ot}$ | dB | -1.75 | -3.20 | -3.20 |
| RSRP Note4 | dBm/15 kHz | -82.85 | -107.9 | -117.70 |
| RSRQ Note4 | dB | -14.76 | -15.69 | -15.69 |
| Io Note4 | dBm/9 MHz | -51.10 | -75.22 | -85.02 |
| $\hat{E}_s / N_{oc}$ | dB | -1.75 | -3.20 | -3.20 |
| Propagation condition | - | AWGN | AWGN | AWGN |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in 3GPP TS 36.211.

Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 5: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 9.2.4A.1.5-3: RSRQ FDD - TDD inter frequency absolute accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_04 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_16 | RSRQ_16 | RSRQ_16 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_01 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_19 | RSRQ_17 | RSRQ_17 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.4A.2 FDD - TDD Inter Frequency Relative Accuracy of RSRQ

### 9.2.4A.2.1 Test purpose

To verify that the FDD - TDD inter frequency relative accuracy measurement of RSRQ is within the specified limit for all bands.

### 9.2.4A.2.2 Test applicability

This test applies to all types of E-UTRA UE supporting FDD and TDD release 9 and forward. Applicability requires support for FGI bits 16 and 25.

### 9.2.4A.2.3 Minimum conformance requirements

The relative accuracy of RSRQ in inter frequency case is defined as the RSRQ measured from one cell compared to the RSRQ measured from another cell on a different frequency.

The accuracy requirements in table 9.2.4A.2.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

$RSRP1,2|_{dBm}$ according to Annex I.3.4 for a corresponding Band

$$\left| RSRP1|_{dBm} - RSRP2|_{dBm} \right| \le [27] dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.2.4A.2.3-1: RSRQ FDD - TDD inter frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/BW$_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.4A.2.3-2.

**Table 9.2.4A.2.3-2: RSRQ FDD - TDD Inter frequency relative accuracy measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.6.2 and A.9.2.4A.

## 9.2.4A.2.4 Test description

### 9.2.4A.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.4A.2.4.3.

4. Cell 1 and Cell 2 are on the different carrier frequencies. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is FDD cell and Cell 2 is TDD cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

### 9.2.4A.2.4.2 Test procedure

The three tests consist of Cell 1, serving cell and Cell 2, target cell. The relative accuracy of RSRQ is defined as the RSRQ measured from one cell compared to the RSRQ measured from another cell on a different frequency.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.2.4A.2.5-1 and 9.2.4A.2.5-2 for the test interval as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. The SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. The UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, the SS shall check the RSRQ value of Cell 1 and Cell 2 in periodical MeasurementReport messages. The reported RSRQ value for Cell 2 is compared to the reported RSRQ value for Cell 1 for each MeasurementReport message. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.2.4A.2.5-1 and 9.2.4A.2.5-2 as appropriate.

### 9.2.4A.2.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.4A.2.4.3-1: Common Exception messages for RSRQ FDD - TDD inter frequency relative accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-2<br>Table H.3.5-4 |

**Table 9.2.4A.2.4.3-2: *MeasResults*: Additional RSRQ FDD - TDD inter frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.4A.2.4.3-3: *MeasResultListEUTRA*: Additional RSRQ FDD - TDD inter frequency relative accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | Not present | | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

## 9.2.4A.2.5 Test requirement

Table 9.2.4A.2.5-1 and 9.2.4A.2.5-2 defines the primary level settings including test tolerances for all tests.

Each RSRQ FDD - TDD inter frequency relative accuracy test shall meet the reported values test requirements in Table 9.2.4A.2.5-3.

**Table 9.2.4A.2.5-1: Cell Specific Test requirement Parameters for RSRQ FDD - TDD inter frequency relative accuracy (FDD Cell1)**

| Parameter | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|
| | | Cell 1 | Cell 1 | Cell 1 |
| E-UTRA RF Channel Number | | 1 | 1 | 1 |
| $BW_{channel}$ | MHz | 10 | 10 | 10 |
| Gap Pattern Id | | 0 | 0 | 0 |
| Measurement bandwidth | $n_{PRB}$ | 22—27 | 22—27 | 22—27 |
| PDSCH Reference measurement channel defined in A.1.1 | | R.0 FDD | R.0 FDD | R.0 FDD |
| PDSCH allocation | $n_{PRB}$ | 13—36 | 13—36 | 13—36 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | R.6 FDD | R.6 FDD | R.6 FDD |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | OP.1 FDD | OP.1 FDD | OP.1 FDD |
| PBCH_RA | dB | 0 | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA Note1 | | | | |
| OCNG_RB Note1 | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -81.10 | -104.70 | -114.5 |
| $\hat{E}_s / I_{ot}$ | dB | -1.75 | -4.0 | -4.0 |
| RSRP Note3 | dBm/15 kHz | -82.85 | -108.70 | -118.5 |
| RSRQ Note3 | dB | -14.76 | -16.25 | -16.25 |
| Io Note3 | dBm/9 MHz | -51.10 | -75.46 | -85.26 |
| $\hat{E}_s / N_{oc}$ | dB | -1.75 | -4.0 | -4.0 |
| Propagation condition | - | AWGN | AWGN | AWGN |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 9.2.4A.2.5-2: Cell Specific Test requirement Parameters for RSRQ FDD - TDD inter frequency relative accuracy (TDD cell2)**

| Parameter | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|
| | | Cell 2 | Cell 2 | Cell 2 |
| E-UTRA RF Channel Number | | 2 | 2 | 2 |
| $BW_{channel}$ | MHz | 10 | 10 | 10 |
| Gap Pattern Id | | - | - | - |
| Special subframe configuration Note1 | | 6 | 6 | 6 |
| Uplink-downlink configuration Note1 | | 1 | 1 | 1 |
| Measurement bandwidth | $n_{PRB}$ | 22—27 | 22—27 | 22—27 |
| PDSCH Reference measurement channel defined in A.1.2 | | - | - | - |
| PDSCH allocation | $n_{PRB}$ | - | - | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | R.6 TDD | R.6 TDD | R.6 TDD |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | OP.2 TDD | OP.2 TDD | OP.2 TDD |
| PBCH_RA | dB | 0 | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA Note2 | | | | |
| OCNG_RB Note2 | | | | |
| $N_{oc}$ Note3 | dBm/15 kHz | -81.10 | -104.70 | -114.50 |
| $\hat{E}_s / I_{ot}$ | dB | -1.75 | -3.20 | -3.20 |
| RSRP Note4 | dBm/15 kHz | -82.85 | -107.90 | -117.70 |
| RSRQ Note4 | dB | -14.76 | -15.69 | -15.69 |
| Io Note4 | dBm/9 MHz | -51.10 | -75.22 | -85.02 |
| $\hat{E}_s / N_{oc}$ | dB | -1.75 | -3.20 | -3.20 |
| Propagation condition | - | AWGN | AWGN | AWGN |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in 3GPP TS 36.211.

Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 5: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 9.2.4A.2.5-3: RSRQ FDD-TDD inter frequency relative accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_x-8 | RSRQ_x-10 | RSRQ_x-10 |
| Highest reported value (Cell 2) | RSRQ_x+8 | RSRQ_x+10 | RSRQ_x+10 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_x-10 | RSRQ_x-10 | RSRQ_x-10 |
| Highest reported value (Cell 2) | RSRQ_x+10 | RSRQ_x+10 | RSRQ_x+10 |
| RSRQ_x is the reported value of Cell 1 | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.5 FDD RSRQ for E-UTRA Carrier Aggregation

### 9.2.5.1 FDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation

#### 9.2.5.1.1 Test purpose

To verify that FDD absolute RSRQ measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the FDD absolute RSRQ accuracy requirements of the primary component carrier and the secondary component carrier.

#### 9.2.5.1.2 Test applicability

This test case applies to all types of E-UTRA UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA.

This test case applies to all types of E-UTRA UE release 11 and forward that support CA with Intra-band non-contiguous CA.

#### 9.2.5.1.3 Minimum conformance requirements

The FDD RSRQ measurements of cells on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1. The FDD RSRQ measurements of cells on the secondary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1.

The accuracy requirements in Table 9.2.5.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.1 for a corresponding Band.

**Table 9.2.5.1.3-1: FDD RSRQ absolute accuracy for Carrier Aggregation**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | $dBm/BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.5.1.3-2.

**Table 9.2.5.1.3-2: FDD RSRQ absolute accuracy measurement report mapping for Carrier Aggregation**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1, clause 9.1.11.2, clause 9.1.5.1 and A.9.2.5.

### 9.2.5.1.4 Test description

#### 9.2.5.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.5.1.4.3.

4. There are two E-UTRA FDD carriers with one cell on one E-UTRA FDD carrier and two cells on the other E-UTRA FDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.2.5.1.4.2 Test procedure

The test consists of Cell 1 the PCell, Cell 2 the SCell on the Secondary Component Carrier (SCC), and Cell 3 the neighbouring cell on the SCC. The SCell (Cell 2) and neighbouring cell (Cell 3) on the SCC are configured. The SCell (Cell 2) on the SCC is activated. The absolute accuracy of RSRQ is defined as the RSRQ measured from the primary component carrier (Cell 1) and the RSRQ measured from the secondary component carrier (Cell 2).

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. SS activates SCC by sending the MAC-CE according to TS 36.321 [11] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.2.5.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. SS shall transmit an RRCConnectionReconfiguration message.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, SS shall check the reported RSRQ values in periodical MeasurementReport messages. The reported RSRQ values for Cell 1 and Cell 2 by the UE are compared to the actual RSRQ values according to Table 9.2.5.1.5-2. This counts respectively as a Pass or Fail for the events “Cell 1” and “Cell 2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations for the respective event ”Cell 2” is increased by one.

10. The SS shall check MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events “Cell 1” and “Cell 2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If both events pass, the test passes. If one event fails, the test fails.

#### 9.2.5.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.5.1.4.3-1: Common Exception messages for FDD RSRQ absolute accuracy for Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-4<br>Table H.4.1-5<br>Table H.4.2-1 |

### 9.2.5.1.5 Test requirement

Table 9.2.5.1.5-1 defines the primary level settings including test tolerances for all tests.

The FDD RSRQ absolute accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.5.1.5-2.

**Table 9.2.5.1.5-1: Cell Specific Test requirement Parameters for FDD RSRQ absolute accuracy for Carrier Aggregation**

| Parameters | | Test 1 | | | |
|---|---|---|---|---|---|
| | | Units | Cell 1 | Cell 2 | Cell 3 |
| E-UTRA RF Channel Number | | | 1 | 2 | 2 |
| $BW_{channel_CA}$ | | MHz | 10 | 10 | 10 |
| Time offset to Cell 1 | | µs | - | 0 | 3µs or 92*Ts |
| Time alignment error between cell 2 and cell 1 | | | - | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1. | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | 22—27 | 22—27 |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | R.6 FDD | R.6 FDD |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA[Note1] | | | | | |
| OCNG_RB[Note1] | | | | | |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -119.5 | -116 | |
| | Bands FDD_C | | -118.5 | -115 | |
| | Bands FDD_D | | -118 | -114.5 | |
| | Bands FDD_E, FDD_F [Note 6] | | -117.5 | -114 | |
| | Bands FDD_G | | -116.5 | -113 | |
| | Bands FDD_H | | -116 | -112.5 | |
| $\hat{E}_s / I_{ot}$ | | dB | -4.0 | -5.16 | -5.54 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -123.5 | -119.7 | -120 |
| | Bands FDD_C | | -122.5 | -118.7 | -119 |
| | Bands FDD_D | | -122 | -118.2 | -118.5 |
| | Bands FDD_E, FDD_F [Note 6] | | -121.5 | -117.7 | -118 |
| | Bands FDD_G | | -120.5 | -116.7 | -117 |

| | | | | | |
|---|---|---|---|---|---|
| | Bands FDD_H | | -120 | -116.2 | -116.5 |
| RSRQ[Note3] | Bands FDD_A | dB | -16.25 | -17.10 | -17.40 |
| | Bands FDD_C | | | | |
| | Bands FDD_D | | | | |
| | Bands FDD_E, FDD_F Note 6 | | | | |
| | Bands FDD_G | | | | |
| | Bands FDD_H | | | | |
| Io[Note3] | Bands FDD_A | dBm/9 MHz | -90.26 | -85.61 | |
| | Bands FDD_C | | -89.26 | -84.61 | |
| | Bands FDD_D | | -88.76 | -84.11 | |
| | Bands FDD_E, FDD_F Note 6 | | -88.26 | -83.67 | |
| | Bands FDD_G | | -87.26 | -82.61 | |
| | Bands FDD_H | | -86.76 | -82.11 | |
| $\hat{E}_s / N_{oc}$ | | dB | -4.0 | -3.7 | -4.0 |
| Propagation condition | | - | AWGN | | |

NOTE 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 3: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 4: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
NOTE 5: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs
NOTE 6: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz
NOTE 7: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 8: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.2.5.1.5-2: FDD RSRQ absolute accuracy requirements for the reported values for Carrier Aggregation**

| | **Test 1** |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_15 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_14 |
| Extreme Conditions | |

| | |
|---|---|
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_16 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_15 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.5.2 FDD Relative RSRQ Accuracy E-UTRA for Carrier Aggregation

### 9.2.5.2.1 Test purpose

To verify that FDD relative RSRQ measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the FDD relative RSRQ accuracy requirements between the primary and secondary component carriers.

### 9.2.5.2.2 Test applicability

This test case applies to all types of E-UTRA UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA.

This test case applies to all types of E-UTRA UE release 11 and forward that support CA with Intra-band non-contiguous CA.

### 9.2.5.2.3 Minimum conformance requirements

The FDD RSRQ measurements of cells on the primary component carrier are compared with measurements of cells on the secondary component carrier defined in TS 36.133 [4] clause 9.1.6.2. The applicable relative accuracy requirements are the RSRQ inter-frequency accuracy requirements as defined in TS 36.133 [4] clause 9.1.6.2.

The accuracy requirements in Table 9.2.5.2.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

RSRP1,2$|_{dBm}$ according to Annex I.3.4 for a corresponding Band.

$$\left| RSRP1|_{dBm} - RSRP2|_{dBm} \right| \leq [27] dB$$

|Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.2.5.2.3-1: FDD RSRQ relative accuracy for Carrier Aggregation**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | $dBm/BW_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by $\Delta>0$, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.5.2.3-2.

**Table 9.2.5.2.3-2: FDD RSRQ relative accuracy measurement report mapping for Carrier Aggregation**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.3, clause 9.1.6.2 and A.9.2.5.

### 9.2.5.2.4 Test description

#### 9.2.5.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.5.2.4.3.

4. There are two E-UTRA FDD carriers with one cell on one E-UTRA FDD carrier and two cells on the other E-UTRA FDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.2.5.2.4.2 Test procedure

The test consists of Cell 1 the PCell, Cell 2 the SCell on the Secondary Component Carrier (SCC), and Cell 3 the neighbouring cell on the SCC. The SCell (Cell 2) and neighbouring cell (Cell 3) on the SCC are configured. The SCell (Cell 2) on the SCC is activated. The relative accuracy of RSRQ is defined as the RSRQ measured cells on the primary component carrier compared with measurements of cells on the secondary component carrier,

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. SS activates SCC by sending the MAC-CE according to TS 36.321 [11] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.2.5.2.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. SS shall transmit an RRCConnectionReconfiguration message.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, SS shall check the reported RSRQ values in periodical MeasurementReport messages. The reported RSRQ value for Cell 2 is compared to the reported RSRQ value for Cell 1 for each MeasurementReport message according to Table 9.2.5.2.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

10. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to table G.2.3-1 in Annex G.2 is achieved.

#### 9.2.5.2.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.5.2.4.3-1: Common Exception messages for FDD RSRQ relative accuracy for Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-4<br>Table H.4.1-5<br>Table H.4.2-1 |

### 9.2.5.2.5 Test requirement

Table 9.2.5.2.5-1 defines the primary level settings including test tolerances for all tests.

The FDD RSRQ relative accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.5.2.5-2.

**Table 9.2.5.2.5-1: Cell Specific Test requirement Parameters for FDD RSRQ relative accuracy for Carrier Aggregation**

| Parameters | | Test 1 | | | |
|---|---|---|---|---|---|
| | | Units | Cell 1 | Cell 2 | Cell 3 |
| E-UTRA RF Channel Number | | | 1 | 2 | 2 |
| $BW_{channel_CA}$ | | MHz | 10 | 10 | 10 |
| Time offset to Cell 1 | | µs | - | 0 | 3µs or 92*Ts |
| Time alignment error between cell 2 and cell 1 | | | - | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1. | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | 22—27 | 22—27 |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | R.6 FDD | R.6 FDD |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA[Note1] | | | | | |
| OCNG_RB[Note1] | | | | | |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -119.5 | -116 | |
| | Bands FDD_C | | -118.5 | -115 | |
| | Bands FDD_D | | -118 | -114.5 | |
| | Bands FDD_E, FDD_F Note 6 | | -117.5 | -114 | |
| | Bands FDD_G | | -116.5 | -113 | |
| | Bands FDD_H | | -116 | -112.5 | |
| $\hat{E}_s / I_{ot}$ | | dB | -4.0 | -5.16 | -5.54 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -123.5 | -119.7 | -120 |
| | Bands FDD_C | | -122.5 | -118.7 | -119 |
| | Bands FDD_D | | -122 | -118.2 | -118.5 |
| | Bands FDD_E, FDD_F Note 6 | | -121.5 | -117.7 | -118 |
| | Bands FDD_G | | -120.5 | -116.7 | -117 |

| | | | | | |
|---|---|---|---|---|---|
| | Bands FDD_H | | -120 | -116.2 | -116.5 |
| RSRQ[Note3] | Bands FDD_A | dB | -16.25 | -17.10 | -17.40 |
| | Bands FDD_C | | | | |
| | Bands FDD_D | | | | |
| | Bands FDD_E, FDD_F Note 6 | | | | |
| | Bands FDD_G | | | | |
| | Bands FDD_H | | | | |
| Io[Note3] | Bands FDD_A | dBm/9 MHz | -90.26 | -85.61 | |
| | Bands FDD_C | | -89.26 | -84.61 | |
| | Bands FDD_D | | -88.76 | -84.11 | |
| | Bands FDD_E, FDD_F Note 6 | | -88.26 | -83.61 | |
| | Bands FDD_G | | -87.26 | -82.61 | |
| | Bands FDD_H | | -86.76 | -82.11 | |
| $\hat{E}_s / N_{oc}$ | | dB | -4.0 | -3.7 | -4.0 |
| Propagation condition | | - | AWGN | | |

NOTE 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

NOTE 3: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

NOTE 4: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

NOTE 5: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs

NOTE 6: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz

NOTE 7: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.

NOTE 8: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.2.5.2.5-2: FDD RSRQ relative accuracy requirements for the reported values for Carrier Aggregation**

| | **Test 1** |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 2) | RSRQ_x - 12 |
| Highest reported value (Cell 2) | RSRQ_x + 9 |
| Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRQ_x - 12 |

| | |
|---|---|
| Highest reported value (Cell 2) | RSRQ_x + 9 |
| RSRQ_x is the reported value of Cell 1 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.6 TDD RSRQ for E-UTRA Carrier Aggregation

### 9.2.6.1 TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation

#### 9.2.6.1.1 Test purpose

To verify that TDD absolute RSRQ measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the TDD absolute RSRQ accuracy requirements of the primary component carrier and the secondary component carrier.

#### 9.2.6.1.2 Test applicability

This test case applies to all types of E-UTRA UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA.

This test case applies to all types of E-UTRA UE release 11 and forward that support CA with Intra-band non-contiguous CA.

#### 9.2.6.1.3 Minimum conformance requirements

The TDD RSRQ measurements of cells on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1. The TDD RSRQ measurements of cells on the secondary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1.

The accuracy requirements in Table 9.2.6.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.1 for a corresponding Band.

**Table 9.2.6.1.3-1: TDD RSRQ absolute accuracy for Carrier Aggregation**

| **Accuracy** | | **Conditions** | | | |
|---|---|---|---|---|---|
| | | | **Io Note 1 range** | | |
| **Normal condition** | **Extreme condition** | **Ês/Iot** | **E-UTRA operating band groups Note 4** | **Minimum Io** | **Maximum Io** |
| **dB** | **dB** | **dB** | | **dBm/15kHz Note 3** | **dBm/BW$_{Channel}$** |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.6.1.3-2.

**Table 9.2.6.1.3-2: TDD RSRQ absolute accuracy measurement report mapping for Carrier Aggregation**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1, clause 9.1.11.2, clause 9.1.5.1 and A.9.2.6.

### 9.2.6.1.4 Test description

#### 9.2.6.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.6.1.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.2.6.1.4.2 Test procedure

The test consists of Cell 1 the PCell, Cell 2 the SCell on the Secondary Component Carrier (SCC), and Cell 3 the neighbouring cell on the SCC. The SCell (Cell 2) and neighbouring cell (Cell 3) on the SCC are configured. The SCell (Cell 2) on the SCC is activated. The absolute accuracy of RSRQ is defined as the RSRQ measured from the primary component carrier (Cell 1) and the RSRQ measured from the secondary component carrier (Cell 2) [and (Cell 3)].

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. SS activates SCC by sending the MAC-CE according to TS 36.321 [11] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.2.6.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. SS shall transmit an RRCConnectionReconfiguration message.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, SS shall check the reported RSRQ values in periodical MeasurementReport messages. The reported RSRQ values for Cell 1 and Cell 2 by the UE are compared to the actual RSRQ values according to Table 9.2.6.1.5-2. This counts respectively as a Pass or Fail for the events “Cell 1” and “Cell 2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations for the respective event ”Cell 2” is increased by one.

10. The SS shall check MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events “Cell 1” and “Cell 2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If both events pass, the test passes. If one event fails, the test fails.

#### 9.2.6.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.6.1.4.3-1: Common Exception messages for TDD RSRQ absolute accuracy for Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-4<br>Table H.4.1-5<br>Table H.4.2-1 |

### 9.2.6.1.5 Test requirement

Table 9.2.6.1.5-1 defines the primary level settings including test tolerances for all tests.

The TDD RSRQ absolute accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.6.1.5-2.

**Table 9.2.6.1.5-1: Cell Specific Test requirement Parameters for TDD RSRQ absolute accuracy for Carrier Aggregation**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| E-UTRA RF Channel Number | | | 1 | 2 | 2 |
| $BW_{channel}$ | | MHz | 10 | | |
| Timing offset to cell1 | | µs | - | 0 | 3µs or 92*Ts |
| Time alignment error between cell 2 and cell 1 | | | - | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1. | - |
| Special subframe configuration[Note1] | | | 6 | | |
| Uplink-downlink configuration[Note1] | | | 1 | | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | R.6 TDD | R.6 TDD |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA[Note2] | | | | | |
| OCNG_RB[Note2] | | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -119.5 | -116 | |
| | Bands TDD_C | | -118.5 | -115 | |
| | Bands TDD_E | | -117.5 | -114 | |
| $\hat{E}_s / I_{ot}$ | | dB | -4.0 | -5.16 | -5.54 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -123.50 | -119.7 | -120 |
| | Bands TDD_C | | -122.50 | -118.7 | -119 |
| | Bands TDD_E | | -121.50 | -117.7 | -118 |
| RSRQ[Note4] | Bands TDD_A | dB | -16.25 | -17.10 | -17.40 |
| Io[Note4] | Bands TDD_C | dBm/9 MHz | -90.26 | -85.61 | |
| | Bands TDD_E | | -89.26 | -84.61 | |
| | Bands TDD_A | | -88.26 | -83.61 | |
| $\hat{E}_s / N_{oc}$ | | dB | -4.0 | -3.70 | -4.0 |
| Propagation condition | | - | AWGN | | |

| | |
|---|---|
| NOTE 1: | For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in 3GPP TS 36.211. |
| NOTE 2: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| NOTE 3: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| NOTE 4: | RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| NOTE 5: | RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| NOTE 6: | The selection of the bands for testing depends on the configuration of the carrier aggregations supported by the UEs. |
| NOTE 7: | This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. |
| NOTE 8: | E-UTRA operating band groups are as defined in Section 3.5. |

**Table 9.2.6.1.5-2: TDD RSRQ absolute accuracy requirements for the reported values for Carrier Aggregation**

| | **Test 1** |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_15 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value Cell 2) | RSRQ_14 |
| Extreme Conditions | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_16 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value Cell 2) | RSRQ_15 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.6.2 TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation

### 9.2.6.2.1 Test purpose

To verify that TDD relative RSRQ measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the TDD relative RSRQ accuracy requirements between the primary and secondary component carriers.

### 9.2.6.2.2 Test applicability

This test case applies to all types of E-UTRA UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA.

This test case applies to all types of E-UTRA UE release 11 and forward that support CA with Intra-band non-contiguous CA.

### 9.2.6.2.3 Minimum conformance requirements

The TDD RSRQ measurements of cells on the primary component carrier are compared with measurements of cells on the secondary component carrier defined in TS 36.133 [4] clause 9.1.6.2. The applicable relative accuracy requirements are the RSRQ inter-frequency accuracy requirements as defined in TS 36.133 [4] clause 9.1.6.2.

The accuracy requirements in Table 9.2.6.2.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

RSRP1,2|$_{dBm}$ according to Annex I.3.4 for a corresponding Band.

$$\left|RSRP1|_{dBm} - RSRP2|_{dBm}\right| \leq [27]dB$$

|Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.2.6.2.3-1: TDD RSRQ relative accuracy for Carrier Aggregation**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/BW$_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.6.2.3-2.

**Table 9.2.6.2.3-2: TDD RSRQ relative accuracy measurement report mapping for Carrier Aggregation**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.3, clause 9.1.6.2 and A.9.2.6.

## 9.2.6.2.4 Test description

### 9.2.6.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.6.2.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.2.6.2.4.2 Test procedure

The test consists of Cell 1 the PCell, Cell 2 the SCell on the Secondary Component Carrier (SCC), and Cell 3 the neighbouring cell on the SCC. The SCell (Cell 2) and neighbouring cell (Cell 3) on the SCC are configured. The SCell (Cell 2) on the SCC is activated. The relative accuracy of RSRQ is defined as the RSRQ measured cells on the primary component carrier compared with measurements of cells on the secondary component carrier,

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. SS activates SCC by sending the MAC-CE according to TS 36.321 [11] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.2.6.2.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. SS shall transmit an RRCConnectionReconfiguration message.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, SS shall check the reported RSRQ values in periodical MeasurementReport messages. The reported RSRQ value for Cell 2 is compared to the reported RSRQ value for Cell 1 for each MeasurementReport message according to Table 9.2.6.2.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

10. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to table G.2.3-1 in Annex G.2 is achieved.

#### 9.2.6.2.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.6.2.4.3-1: Common Exception messages for TDD RSRQ relative accuracy for Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-4<br>Table H.4.1-5<br>Table H.4.2-1 |

### 9.2.6.2.5 Test requirement

Table 9.2.6.2.5-1 defines the primary level settings including test tolerances for all tests.

The TDD RSRQ relative accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.5.2.5-2.

**Table 9.2.6.2.5-1: Cell Specific Test requirement Parameters for TDD RSRQ relative accuracy for Carrier Aggregation**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| E-UTRA RF Channel Number | | | 1 | 2 | 2 |
| $BW_{channel}$ | | MHz | 10 | | |
| Timing offset to cell1 | | µs | - | 0 | 3µs or 92*Ts |
| Time alignment error between cell 2 and cell 1 | | | - | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1. | - |
| Special subframe configuration[Note1] | | | 6 | | |
| Uplink-downlink configuration[Note1] | | | 1 | | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | R.6 TDD | R.6 TDD |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA[Note2] | | | | | |
| OCNG_RB[Note2] | | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -119.5 | -116 | |
| | Bands TDD_C | | -118.5 | -115 | |
| | Bands TDD_E | | -117.5 | -114 | |
| $\hat{E}_s / I_{ot}$ | | dB | -4.0 | -5.16 | -5.54 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -123.50 | -119.7 | -120 |
| | Bands TDD_C | | -122.50 | -118.7 | -119 |
| | Bands TDD_E | | -121.5 | -117.7 | -118 |
| RSRQ[Note4] | Bands TDD_A, TDD_C, TDD_E | dB | -16.25 | -17.10 | -17.40 |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -90.26 | -85.61 | |
| | Bands TDD_C | | -89.26 | -84.61 | |
| | Bands TDD_E | | -88.26 | -83.61 | |
| $\hat{E}_s / N_{oc}$ | | dB | -4.0 | -3.7 | -4.0 |
| Propagation condition | | - | AWGN | | |

NOTE 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in 3GPP TS 36.211.

NOTE 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

NOTE 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 4: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 5: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
NOTE 6: The selection of the bands for testing depends on the configuration of the carrier aggregations supported by the UEs.
NOTE 7: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
NOTE 8: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.2.6.2.5-2: TDD RSRQ relative accuracy requirements for the reported values for Carrier Aggregation**

| | **Test 1** |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 2) | RSRQ_x - 12 |
| Highest reported value (Cell 2) | RSRQ_x + 9 |
| Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRQ_x - 12 |
| Highest reported value (Cell 2) | RSRQ_x + 9 |
| RSRQ_x is the reported value of Cell 1 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.7 FDD RSRQ Accuracy under Time Domain Measurement Resource Restriction with Non-MBSFN ABS

### 9.2.7.1 FDD RSRQ under Time Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC)

#### 9.2.7.1.1 Test purpose

The purpose of this test is to verify that the RSRQ measurement accuracy under time domain measurement resource restriction is within the specified limits for all bands.

#### 9.2.7.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward. Applicability requires support for FGI bit 115.

#### 9.2.7.1.3 Minimum conformance requirements

The requirements for absolute accuracy of RSRQ in this section shall apply to a cell on the same frequency as that of the serving cell when a time domain measurement resource restriction pattern for performing RSRQ measurements of this cell is configured by higher layers according to TS 36.331 [5].

The accuracy requirements in Table 9.2.7.1.3-1 are valid under the following conditions:

Cell specific reference signals in the measured cell are transmitted either from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled,

$RSRP|_{dBm}$ according to Annex I.3.9 for a corresponding Band,

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRQ measurement,

The RSRQ measurement is not performed in any subframe other than those indicated by the time domain measurement resource restriction pattern configured for the measured cell,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.2.7.1.3-1: RSRQ Intra frequency absolute accuracy under time domain measurement resource restriction**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 2 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 4 | dBm/BW$_{Channel}$ |
| ±2.5 | ±4 | ≥-2 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-4 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRQ measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.7.1.3-2.

**Table 9.2.7.1.3-2: RSRQ FDD Intra frequency absolute accuracy measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.5.2 and A.9.2.7.

## 9.2.7.1.4 Test description

### 9.2.7.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. The general test parameter settings are set up according to Table 9.2.7.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.2.7.1.4.3.

5. There is one E-UTRA FDD carrier and two cells specified in each test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 9.2.7.1.4.1-1: General test parameters for E-UTRAN FDD RSRQ intra frequency test parameters under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Serving cell (PCell) | | Cell 1 | The aggressor cell to Cell 2 |
| Neighbour cell | | Cell 2 | Cell to be measured |
| PCell ABS configuration | | Non-MBSFN ABS | As defined in Table C.3.1.1.1-1 |
| CP length | | Normal | For both cells in the test |
| DRX | | | OFF |
| Time offset between cells | | 3 µs | Synchronous cells<br>3µs or 92*Ts |
| Physical cell ID PCI | | $(PCI_{cell1} - PCI_{cell2})mod6 != 0$ | Cell PCIs for Cell 1 and Cell 2 are selected randomly so that the condition is met. |
| ABS pattern | | '10000000100000001000000010000000010000000' | Non-MBSFN ABS. FDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 1.<br>The first/leftmost bit corresponds to the subframe #0 of a radio frame satisfying SFN mod x = 0, where x is the size of the bit string (40) divided by 10. No MBSFN subframes are configured in the ABS subframes in Cell 1. |
| Time-domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | '10000000100000001000000010000000010000000' | Configured for Cell 2 measurements by measSubframePattern-Neigh IE in measSubframePatternConfig-Neigh, as defined in TS 36.331 [5], clause 6.3.5. measSubframeCellList contains Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | '01000000010000000100000001000000001000000' | Configured for measurements on Cell 1. |

### 9.2.7.1.4.2 Test procedure

In this test case all cells are on the same carrier frequency. The absolute accuracy of RSRQ intra frequency measurements under time domain measurement resource restriction is tested by using the parameters in Table 9.2.7.1.4.1-1 and Table 9.2.7.1.5-1 for non-MBSFN ABS with non-colliding CRS. In all test cases, Cell 1 is the serving cell and also the aggressor cell to Cell 2. Cell 2 is the target cell to be measured for RSRQ.

The UE is configured by higher layers via Cell 1 with a time-domain measurement resource restriction pattern for performing E-UTRAN FDD intra-frequency measurements on neighbour cells and provided with a neighbour cell list associated with the pattern, where the cell list includes Cell 2. The UE is also configured with a time-domain measurement resource restriction pattern for the serving cell measurements. The information for both patterns shall be provided to the UE before the measurements start.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.2.7.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the RSRQ value in periodical MeasurementReport messages. The RSRQ value of Cell 2 reported by the UE is compared to the actual RSRQ value according to Table 9.2.7.1.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.2.7.1.5-1 as appropriate.

### 9.2.7.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.2.7.1.4.3-1: Common Exception messages for RSRQ FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.7-1<br>Table H.2.7-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-4<br>Table H.3.5-5 |

**Table 9.2.7.1.4.3-2: *MeasResults*: Additional RSRQ FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.7.1.4.3-3: *MeasResultListEUTRA*: Additional RSRQ FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | Not present | | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 9.2.7.1.4.3-4: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional RSRQ FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '01000000010000000100 00000100000001000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.7.1.4.3-5: *MeasObjectEUTRA-GENERIC(Freq)*: Additional RSRQ FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternFDD-r10 | '10000000100000001000 00001000000010000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 2 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.2.7.1.5 Test requirement

Table 9.2.7.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRQ FDD intra frequency absolute accuracy test shall meet the reported values test requirements in Table 9.2.7.1.5-2.

**Table 9.2.7.1.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD RSRQ intra frequency test parameters under time domain measurement resource restriction with non-MBSFN ABS**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - | R.0 FDD | - | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in D.1.5 (OP.5 FDD) and D.1.6 (OP.6 FDD) | | | OP.5 FDD | OP.6 FDD | OP.5 FDD | OP.6 FDD | OP.5 FDD | OP.6 FDD |
| PBCH_RA | | dB | Note 6 | 0 | Note 6 | 0 | Note 6 | 0 |
| PBCH_RB | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA Note1 | | | | | | | | |
| OCNG_RB Note1 | | | | | | | | |
| PSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| SSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -85.76 | | -103.85 | | -116 | |
| | Bands FDD_C | | | | | | -115 | |
| | Bands FDD_D | | | | | | -114.5 | |
| | Bands FDD_E, FDD_F Note 7 | | | | | | -114 | |
| | Bands FDD_G | | | | | | -113 | |
| | Bands FDD_H | | | | | | -112.5 | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 5 | -1.2 | 5 | -1.2 | 5 | -3.05 |
| CRS $(\hat{E}_s / I_{ot})_{meas}$ Note 5 | | dB | 2.55 | -1.20 | 2.55 | -1.20 | 3.25 | -3.05 |
| SCH $\hat{E}_s / I_{ot}$ | | dB | -1.45 | -4.74 | -1.45 | -4.74 | -0.75 | -6.59 |
| RSRP Note3,4,5 | Bands FDD_A | dBm/15 kHz | -80.76 | -86.96 | -98.85 | -105.05 | -111 | -119.05 |
| | Bands FDD_C | | | | | | -110 | -118.05 |
| | Bands FDD_D | | | | | | -109.5 | -117.55 |
| | Bands FDD_E, FDD_F Note 7 | | | | | | -109 | -117.05 |
| | Bands FDD_G | | | | | | -108 | -116.05 |
| | Bands FDD_H | | | | | | -107.5 | -115.55 |
| $(\mathrm{RSRQ})_{meas}$ Note3,4,5 | Bands FDD_A, FDD_C, FDD_E, FDD_F Note 7, FDD_G, FDD_H | dB | -12.71 | -14.80 | -12.71 | -14.80 | -12.47 | -16.01 |
| $(\mathrm{Io})_{meas}$ Note3 | Bands FDD_A | dBm/9 MHz | -51.06 | -54.39 | -69.15 | -72.48 | -81.54 | -85.16 |
| | Bands FDD_C | | | | | | -80.54 | -84.16 |
| | Bands FDD_D | | | | | | -80.04 | -83.66 |
| | Bands FDD_E, FDD_F Note 7 | | | | | | -79.54 | -83.16 |
| | Bands FDD_G | | | | | | -78.54 | -82.16 |
| | Bands FDD_H | | | | | | -78.04 | -81.66 |

| Propagation condition | - | AWGN | AWGN | AWGN |
|---|---|---|---|---|

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.

Note 3: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes.

Note 4: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: Applies to restricted measurement subframes of the respective cell.

Note 6: Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1

Note 7: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 8: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.2.7.1.5-2: RSRQ FDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy requirements for the reported values**

| | **Test 1** | **Test 2** | **Test 3** |
|---|---|---|---|
| | All bands | All bands | All bands |
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_04 | RSRQ_04 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_16 | RSRQ_16 | RSRQ_16 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_01 | RSRQ_01 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_19 | RSRQ_19 | RSRQ_19 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.2.8 TDD RSRQ Accuracy under Time Domain Measurement Resource Restriction with Non-MBSFN ABS

## 9.2.8.1 TDD RSRQ under Time Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC)

### 9.2.8.1.1 Test purpose

The purpose of this test is to verify that the RSRQ measurement accuracy under time domain measurement resource restriction is within the specified limits for all bands.

### 9.2.8.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward. Applicability requires support for FGI bit 115.

### 9.2.8.1.3 Minimum conformance requirements

The requirements for absolute accuracy of RSRQ in this section shall apply to a cell on the same frequency as that of the serving cell when a time domain measurement resource restriction pattern for performing RSRQ measurements of this cell is configured by higher layers according to TS 36.331 [5].

The accuracy requirements in Table 9.2.8.1.3-1 are valid under the following conditions:

Cell specific reference signals in the measured cell are transmitted either from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled,

$RSRP|_{dBm}$ according to Annex I.3.9 for a corresponding Band,

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRQ measurement,

The RSRQ measurement is not performed in any subframe other than those indicated by the time domain measurement resource restriction pattern configured for the measured cell,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.2.8.1.3-1: RSRQ Intra frequency absolute accuracy under time domain measurement resource restriction**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 2 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 4 | $dBm/BW_{Channel}$ |
| ±2.5 | ±4 | ≥-2 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-4 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRQ measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.8.1.3-2.

**Table 9.2.8.1.3-2: RSRQ TDD Intra frequency absolute accuracy measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.5.2 and A.9.2.8.

## 9.2.8.1.4 Test description

### 9.2.8.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. The general test parameter settings are set up according to Table 9.2.8.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.2.8.1.4.3.

5. There is one E-UTRA TDD carrier and two cells specified in each test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 9.2.8.1.4.1-1: General test parameters for E-UTRAN TDD RSRQ intra frequency test parameters under time-domain measurement resource restriction with non-MBSFN ABS**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Serving cell (PCell) | | Cell 1 | Also the aggressor cell. |
| Neighbour cell | | Cell 2 | Cell to be measured |
| PCell ABS configuration | | Non-MBSFN ABS | As defined in Table C.3.1.1.1-1 |
| | | | |
| | | | |
| Special subframe configuration | | 6 | For Cell 1 and Cell 2. For special subframe configurations see Table 4.2-1 in [9]. |
| Uplink/downlink subframe configuration | | 1 | For Cell 1 and Cell 2. For uplink-downlink subframe configurations see Table 4.2-2 in [9]. |
| CP length | | Normal | For both cells in the test |
| DRX | | | OFF |
| Time offset between cells | | 3 μs | Synchronous cells<br>3μs or 92*Ts |
| Physical cell ID PCI | | $(PCI_{cell1} - PCI_{cell2})mod6 != 0$ | Cell PCIs for Cell 1 and Cell 2 are randomly selected so that the condition is met |
| ABS pattern | | '00000000010000000001' | Non-MBSFN ABS. TDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 1.<br>The first/leftmost bit corresponds to the subframe #0 of a radio frame satisfying SFN mod x = 0, where x is the size of the bit string (20) divided by 10. No MBSFN subframes are configured in the ABS subframes in Cell 1. |
| Time-domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | '00000000010000000001' | Configured for Cell 2 measurements by measSubframePattern-Neigh IE in measSubframePatternConfig-Neigh, as defined in TS 36.331 [5], clause 6.3.5. measSubframeCellList contains Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | '10000000001000000000' | Configured for Cell 1 measurements. |

### 9.2.8.1.4.2 Test procedure

In this test case all cells are on the same carrier frequency. The absolute accuracy of RSRQ intra frequency measurements under time domain measurement resource restriction is tested by using the parameters in Table 9.2.8.1.4.1-1 and Table 9.2.8.1.5-1 for non-MBSFN ABS with non-colliding CRS. In all test cases, Cell 1 is the serving cell and also the aggressor cell to Cell 2. Cell 2 is the target cell to be measured for RSRQ.

The UE is configured by higher layers via Cell 1 with a time-domain measurement resource restriction pattern for performing E-UTRAN TDD intra-frequency measurements on neighbour cells and provided with a neighbour cell list associated with the pattern, where the cell list includes Cell 2. The UE is also configured with a time-domain measurement resource restriction pattern for the serving cell measurements. The information for both patterns shall be provided to the UE before the measurements start.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.2.8.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the RSRQ value in periodical MeasurementReport messages. The RSRQ value of Cell 2 reported by the UE is compared to the actual RSRQ value according to Table 9.2.8.1.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.2.8.1.5-1 as appropriate.

### 9.2.8.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.2.8.1.4.3-1: Common Exception messages for RSRQ TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.7-1 |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-4 |

**Table 9.2.8.1.4.3-2: *MeasResults*: Additional RSRQ TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.8.1.4.3-3: *MeasResultListEUTRA*: Additional RSRQ TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | Not present | | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 9.2.8.1.4.3-4: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional RSRQ TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 CHOICE { | | | |
| subframeConfig1-5-r10 | '10000000001000000000' | BIT STRING (SIZE (20)) | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.8.1.4.3-5: *MeasObjectEUTRA-GENERIC(Freq)*: Additional RSRQ TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternTDD-r10 CHOICE { | | | |
| subframeConfig1-5-r10 | '00000000010000000001' | BIT STRING (SIZE (20)) | |
| } | | | |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 2 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.2.8.1.5 Test requirement

Table 9.2.8.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRQ TDD intra frequency absolute accuracy test shall meet the reported values test requirements in Table 9.2.8.1.5-2.

**Table 9.2.8.1.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD RSRQ intra frequency test parameters under time domain measurement resource restriction with non-MBSFN ABS**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | R.0 TDD | - | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | Note 6 | 0 | Note 6 | 0 | Note 6 | 0 |
| PBCH_RB | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA Note1 | | | | | | | | |
| OCNG_RB Note1 | | | | | | | | |
| PSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| SSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| $N_{oc}$ Note2 | Bands TDD_A | dBm/15 kHz | -85.76 | | -103.85 | | -116 | |
| | Bands TDD_C | | | | | | -115 | |
| | Bands TDD_E | | | | | | -114 | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 5 | -1.2 | 5 | -1.2 | 5 | -3.05 |
| CRS $(\hat{E}_s / I_{ot})_{meas}$ Note 5 | | dB | 2.55 | -1.20 | 2.55 | -1.20 | 3.25 | -3.05 |
| SCH $\hat{E}_s / I_{ot}$ | | dB | -1.45 | -4.74 | -1.45 | -4.74 | -0.75 | -6.59 |
| RSRP Note3,4,5 | Bands TDD_A | dBm/15 kHz | -80.76 | -86.96 | -98.85 | -105.05 | -111 | -119.05 |
| | Bands TDD_C | | | | | | -110 | -118.05 |
| | Bands TDD_E | | | | | | -109 | -117.05 |
| $(\mathrm{RSRQ})_{meas}$ Note3,4,5 | Bands TDD_A, TDD_C, TDD_E | dB | -12.71 | -14.80 | -12.71 | -14.80 | -12.47 | -16.01 |
| $(\mathrm{Io})_{meas}$ Note3 | Bands TDD_A | dBm/9 MHz | -51.06 | -54.39 | -69.15 | -72.48 | -81.54 | -85.16 |
| | Bands TDD_C | | | | | | -80.54 | -84.16 |
| | Bands TDD_E | | | | | | -79.54 | -83.16 |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.

Note 3: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes.

Note 4: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: Applies to restricted measurement subframes of the respective cell.

Note 6: Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1.

**Table 9.2.8.1.5-2: RSRQ TDD Intra frequency under time domain measurement resource restriction with non-MBSFN ABS absolute accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| | All bands | All bands | All bands |
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_04 | RSRQ_04 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_16 | RSRQ_16 | RSRQ_16 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_01 | RSRQ_01 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_19 | RSRQ_19 | RSRQ_19 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.9 FDD RSRQ Accuracy under Time Domain Measurement Resource Restriction with MBSFN ABS

### 9.2.9.1 FDD Absolute RSRQ under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)

#### 9.2.9.1.1 Test purpose

The purpose of this test is to verify that the RSRQ measurement accuracy under time domain measurement resource restriction with MBSFN ABS is within the specified limits for all bands.

#### 9.2.9.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward. Applicability requires support of FGI bit 115.

#### 9.2.9.1.3 Minimum conformance requirements

The requirements for absolute accuracy of RSRQ in this section shall apply to a cell on the same frequency as that of the serving cell when a time domain measurement resource restriction pattern for performing RSRQ measurements of this cell is configured by higher layers according to TS 36.331 [5].

The accuracy requirements in Table 9.2.9.1.3-1 are valid under the following conditions:

Cell specific reference signals in the measured cell are transmitted either from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled,

$RSRP|_{dBm}$ according to Annex I.3.9 for a corresponding Band,

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRQ measurement,

The RSRQ measurement is not performed in any subframe other than those indicated by the time domain measurement resource restriction pattern configured for the measured cell,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.2.9.1.3-1: RSRQ Intra frequency absolute accuracy under time domain measurement resource restriction with MBSFN ABS**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 2 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 4 | dBm/BW$_{Channel}$ |
| ±2.5 | ±4 | ≥-2 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-4 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRQ measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.9.1.3-2.

**Table 9.2.9.1.3-2: RSRQ FDD Intra frequency absolute accuracy measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.5.2 and A.9.2.9.

### 9.2.9.1.4 Test description

#### 9.2.9.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. The general test parameter settings are set up according to Table 9.2.9.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.2.9.1.4.3.

5. Cell 1 is the serving cell as well as aggressor cell to Cell 2, Cell 2 is the neighbouring cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

**Table 9.2.9.1.4.1-1: General test parameters for E-UTRAN FDD RSRQ intra frequency test parameters under time-domain measurement resource restriction with MBSFN ABS**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters Note 1 | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters Note 1 | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| Serving cell (PCell) | | Cell 1 | Also the aggressor cell on E-UTRA RF channel number 1 |
| Neighbour cell | | Cell 2 | Cell to be identified on E-UTRA RF channel number 1 |
| E-UTRA RF Channel Number | | 1 | One E-UTRA FDD carrier frequency is used |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | For all cells in the test |
| PCell ABS configuration | | MBSFN ABS | As defined in Table C.3.1.2.1-1 |
| CP length | | Normal | |
| DRX | | OFF | |
| Time offset between cells | | 3 µs | Synchronous cells |
| Physical cell ID PCI | | ($PCI_{cell1}$ - $PCI_{cell2}$ ) mod 6 = 0, $PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell PCIs are selected so that the condition is met (colliding CRS) |
| Cell 1 MBSFN ABS pattern | | '01000000100000001000 00000010000001000000' | ABS subframe is only MBSFN subframe. FDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. The first/leftmost bit corresponds to the subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (40) divided by 10. Configured in Cell 1. |
| Time domain measurement resource restriction pattern for PCell (Cell 1) measurements on RF Channel 1 | | '00010000000100000001 00000001000000010000' | Time domain measurement resource restriction pattern for PCell measurement signalled to the UE in measSubframePatternPCell. The IE MeasSubframePattern is used to specify the time domain measurement resource restriction as defined in TS 36.331 [5], clause 6.3.6. Configured for Cell 1 measurements. |
| Time domain measurement resource restriction pattern for neighbour cell (Cell 2) measurements on RF Channel 1 | | '01000000100000001000 00000010000001000000' | Time domain measurement resource restriction pattern for neighbour cell measurement signalled to the UE in measSubframePatternNeigh The IE MeasSubframePattern is used to specify the time domain measurement resource restriction as defined in TS 36.331 [5], clause 6.3.6. Configured for Cell 2 measurements. |
| Note 1: Applies to restricted measurement subframes of the respective cell. | | | |

### 9.2.9.1.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2.A.3.

2. Set the parameters according to Table 9.2.9.1.5-1 and 9.2.9.1.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the reported RSRQ values in periodical MeasurementReport messages. The RSRQ value of Cell 2 reported by the UE is compared to the actual RSRQ value according to Table 9.2.9.1.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.2.9.1.5-1 as appropriate.

9.2.9.1.4.3 Message contents

**Table 9.2.9.1.4.3-1: Common Exception messages for RSRQ FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-4<br>Table H.3.5-5 |

**Table 9.2.9.1.4.3-2: *MeasResults*: Additional RSRQ FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.9.1.4.3-3: *MeasResultListEUTRA*: Additional RSRQ FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | Not present | | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 9.2.9.1.4.3-4: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional RSRQ FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '00010000000100000001 00000001000000010000' | BIT STRING (SIZE (40)) | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.9.1.4.3-5: *MeasObjectEUTRA-GENERIC(Freq)*: Additional RSRQ FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternFDD-r10 | '01000000100000001000 00000010000001000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 2 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.9.1.4.3-6: *SystemInformationBlockType2*: Additional RSRQ FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 SystemInformationBlockType2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfiguration | Not present | | Cell 2 |
| } | | | |

**Table 9.2.9.1.4.3-7: *SystemInformationBlockType3*: Additional RSRQ FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| neighCellConfig | '01'B (No MBSFN subframes are present in all neighbour cells) | | Cell 1 |
| | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 2 |

#### 9.2.9.1.5 Test requirement

Table 9.2.9.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRQ FDD intra frequency absolute accuracy test shall meet the reported values test requirements in Table 9.2.9.1.5-2.

**Table 9.2.9.1.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD RSRQ intra frequency test parameters under time domain measurement resource restriction with MBSFN ABS**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| OCNG Patterns defined in D.1.8 (OP.8 FDD) and D.1.6 (OP.6 FDD) Note5 | | | OP.8 FDD | OP.6 FDD | OP.8 FDD | OP.6 FDD | OP.8 FDD | OP.6 FDD |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PBCH_RA<br>PBCH_RB<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA Note1<br>OCNG_RB Note1 | | dB | Note 6 | 0 | Note 6 | 0 | Note 6 | 0 |
| PSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| SSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -85.76 | | -103.85 | | -116 | |
| | Bands FDD_C | | | | | | -115 | |
| | Bands FDD_D | | | | | | -114.5 | |
| | Bands FDD_E, FDD_F Note 8 | | | | | | -114 | |
| | Bands FDD_G | | | | | | -113 | |
| | Bands FDD_H | | | | | | -112.5 | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 5 | -1.2 | 5 | -1.2 | 5 | -3.05 |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{meas}$ Note5, Note 7 in the 1st OFDM symbol | | dB | 2.55 | -7.39 | 2.55 | -7.39 | 3.25 | -9.39 |
| CRS $\left(\hat{E}_s / Iot\right)_{meas}$ note 5 in OFDM symbols 4,7,11 | | dB | 2.55 | -1.2 | 2.55 | -1.2 | 3.25 | -3.05 |
| SCH $\hat{E}_s / I_{ot}$ | | dB | -1.45 | -4.74 | -1.45 | -4.74 | -0.75 | -6.59 |
| RSRP Note3,4,5 | Bands FDD_A | dBm/15 kHz | -80.76 | -86.96 | -98.85 | -105.05 | -111 | -119.05 |
| | Bands FDD_C | | | | | | -110 | -118.05 |
| | Bands FDD_D | | | | | | -109.5 | -117.55 |
| | Bands FDD_E, FDD_F Note 8 | | | | | | -109 | -117.05 |
| | Bands FDD_G | | | | | | -108 | -116.05 |
| | Bands FDD_H | | | | | | -107.5 | -115.55 |
| (RSRQ) *meas* Note3,4,5 | Bands FDD_A | dB | -12.71 | -14.54 | -12.71 | -14.54 | -12.47 | -15.70 |
| | Bands FDD_C | | | | | | | |
| | Bands FDD_D | | | | | | | |
| | Bands FDD_E, FDD_F Note 8 | | | | | | | |
| | Bands FDD_G | | | | | | | |
| | Bands FDD_H | | | | | | | |
| (Io) *meas* Note3 1st OFDM symbol | Bands FDD_A | dBm/9 MHz | -51.06 | -54.39 | -69.15 | -72.48 | -81.55 | -85.20 |
| | Bands FDD_C | | | | | | -80.55 | -84.20 |
| | Bands FDD_D | | | | | | -80.05 | -83.70 |
| | Bands FDD_E, FDD_F Note 8 | | | | | | -79.55 | -83.20 |
| | Bands FDD_G | | | | | | -78.55 | -82.20 |
| | Bands FDD_H | | | | | | -78.05 | -81.70 |
| (Io) *meas* Note3 OFDM | Bands FDD_A | dBm/9 MHz | -51.06 | -55.53 | -69.15 | -73.62 | -81.54 | -86.47 |
| | Bands FDD_C | | | | | | -80.54 | -85.47 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| symbols other than the 1st one | Bands FDD_D | | | | | | -80.04 | -84.97 |
| | Bands FDD_E, FDD_F Note 8 | | | | | | -79.54 | -84.47 |
| | Bands FDD_G | | | | | | -78.54 | -83.47 |
| | Bands FDD_H | | | | | | -78.04 | -82.97 |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Applies to all subframes.

Note 3: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes

Note 4: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: Applies to restricted measurement subframes of the respective cell.

Note 6: Non-ABS and ABS subframe channel powers defined in Table C.3.1.2.1-1.

Note 7: In the 1st OFDM symbol, Cell 2 is not expected to meet the Es/Iot side condition in TS 36.133 [4] 9.1.5.2.

Note 8: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 9: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.2.9.1.5-2: RSRQ FDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| | All bands | All bands | All bands |
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_04 | RSRQ_04 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_16 | RSRQ_16 | RSRQ_16 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_01 | RSRQ_01 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_19 | RSRQ_19 | RSRQ_17 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.10 TDD RSRQ Accuracy under Time Domain Measurement Resource Restriction with MBSFN ABS

### 9.2.10.1 TDD RSRQ under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC)

#### 9.2.10.1.1 Test purpose

The purpose of this test is to verify that the RSRQ measurement accuracy under time domain measurement resource restriction with MBSFN ABS is within the specified limits for all bands.

#### 9.2.10.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward. Applicability requires support of FGI bit 115

#### 9.2.10.1.3 Minimum conformance requirements

The requirements for absolute accuracy of RSRQ in this section shall apply to a cell on the same frequency as that of the serving cell when a time domain measurement resource restriction pattern for performing RSRQ measurements of this cell is configured by higher layers according to TS 36.331 [5].

The accuracy requirements in Table 9.2.10.1.3-1 are valid under the following conditions:

Cell specific reference signals in the measured cell are transmitted either from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled,

$RSRP|_{dBm}$ according to Annex I.3.9 for a corresponding Band,

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRQ measurement,

The RSRQ measurement is not performed in any subframe other than those indicated by the time domain measurement resource restriction pattern configured for the measured cell,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern.

**Table 9.2.10.1.3-1: RSRQ Intra frequency absolute accuracy under time domain measurement resource restriction with MBSFN ABS**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 2 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/ 15kHz Note 1, 4 | dBm/BW$_{Channel}$ |
| ±2.5 | ±4 | ≥-2 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-4 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: This minimum Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRQ measurements of this cell. The Io range defined by the minimum and the maximum Io levels applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by $\Delta$>0, when applicable, as described in Sections B.4.2 and B.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.10.1.3-2.

**Table 9.2.10.1.3-2: RSRQ TDD Intra frequency absolute accuracy measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.5.2 and A.9.2.10.

### 9.2.10.1.4 Test description

#### 9.2.10.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.

2. The general test parameter settings are set up according to Table 9.2.10.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.2.9.1.4.3.

5. Cell 1 is the serving cell as well as aggressor cell to Cell 2, Cell 2 is the neighbouring cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

**Table 9.2.10.1.4.1-1: General test parameters for E-UTRAN TDD RSRQ intra frequency test parameters under time-domain measurement resource restriction with MBSFN ABS**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| Serving cell (PCell) | | Cell 1 | The aggressor cell to Cell 2 |
| Neighbour cell | | Cell 2 | Cell to be measured |
| E-UTRA RF Channel Number | | 1 | One TDD carrier frequency is used |
| Channel Bandwidth ($BW_{channel}$) | MHz | 10 | For both cells in the test |
| PCell ABS configuration | | MBSFN ABS | As defined in Table C.3.1.2.1-1 |
| Special subframe configuration | | 6 | For Cell 1 and Cell 2. For special subframe configurations see Table 4.2-1 in [16]. |
| Uplink/downlink subframe configuration | | 1 | For Cell 1 and Cell 2. For uplink-downlink subframe configurations see Table 4.2-2 in [16]. |
| CP length | | Normal | For both cells in the test |
| DRX | | | OFF |
| Time offset between cells | | 3 µs | Synchronous cells |
| Physical cell ID PCI | | ($PCI_{cell1}$ - $PCI_{cell2}$ )mod6 =0 $PCI_{cell1}$ not equal to $PCI_{cell2}$ | Cell PCIs for Cell 1 and Cell 2 are selected randomly so that the condition is met |
| ABS pattern | | '00001000000000100000' | MBSFN ABS pattern. TDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54. Configured in Cell 1. The first/leftmost bit corresponds to the subframe #0 of a radio frame satisfying SFN mod x = 0, where x is the size of the bit string (20) divided by 10. All ABS subframes are MBSFN subframes. |
| Time-domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | '00001000000000100000' | Configured for Cell 2 measurements by measSubframePattern-Neigh IE in measSubframePatternConfig-Neigh, as defined in TS 36.331 [5], clause 6.3.5. measSubframeCellList contains Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | '10000000001000000000' | Configured for measurements on Cell 1. |

#### 9.2.10.1.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3..

2. Set the parameters according to Table 9.2.9.1.5-1 and 9.2.9.1.5-2 as appropriate. Propagation conditions are set according to Annex B clause B.1.

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the reported RSRQ values in periodical MeasurementReport messages. The RSRQ value of Cell 2 reported by the UE is compared to the actual RSRQ values according to Table 9.2.10.1.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved.

### 9.2.10.1.4.3 Message contents

**Table 9.2.10.1.4.3-1: Common Exception messages for RSRQ TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-4 |

**Table 9.2.10.1.4.3-2: *MeasResults*: Additional RSRQ TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.10.1.4.3-3: *MeasResultListEUTRA*: Additional RSRQ TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity | | |
| measResult SEQUENCE { | | | |
| rsrpResult | Not present | | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 9.2.10.1.4.3-4: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional RSRQ TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 CHOICE { | | | |
| subframeConfig1-5-r10 | '10000000001000000000' | BIT STRING (SIZE (20)) | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.10.1.4.3-5: *MeasObjectEUTRA-GENERIC(Freq)*: Additional RSRQ TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measSubframePatternConfigNeigh-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| measSubframePatternNeigh-r10 CHOICE { | | | |
| subframePatternTDD-r10 CHOICE { | | | |
| subframeConfig1-5-r10 | '00001000000000100000' | BIT STRING (SIZE (20)) | |
| } | | | |
| } | | | |
| measSubframeCellList-r10 SEQUENCE { | | | |
| start | Physical Cell ID of Cell 2 | | |
| range | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.10.1.4.3-6: *SystemInformationBlockType2*: Additional RSRQ TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: Clause 4.4.3.3, Table 4.4.3.3-1 SystemInformationBlockType2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| mbsfn-SubframeConfigList ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF SEQUENCE { | | | Cell 1 |
| radioframeAllocationPeriod | n1 | Every radio frame is with MBSFN subframe | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| oneFrame | '010000'B | Subframe 4 is used for MBSFN | |
| } | | | |
| } | | | |

**Table 9.2.10.1.4.3-7: *SystemInformationBlockType3*: Additional RSRQ TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| neighCellConfig | '00'B (Not all neighbour cells have the same MBSFN subframe allocation as serving cell) | | Cell 2 |

## 9.2.10.1.5 Test requirement

Table 9.2.10.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRQ TDD intra frequency absolute accuracy test shall meet the reported values test requirements in Table 9.2.10.1.5-2.

**Table 9.2.10.1.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD RSRQ intra frequency test parameters under time domain measurement resource restriction with MBSFN ABS**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | R.0 TDD | - | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in D.2.5 (OP.5 TDD) and D.2.2 (OP.2 TDD) | | | OP.5 TDD | OP.2 TDD | OP.5 TDD | OP.2 TDD | OP.5 TDD | OP.2 TDD |
| PBCH_RA<br>PBCH_RB<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA Note1<br>OCNG_RB Note1 | | dB | Note 6 | 0 | Note 6 | 0 | Note 6 | 0 |
| PSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| SSS_RA | | dB | -4 | 0 | -4 | 0 | -4 | 0 |
| $N_{oc}$ Note2 | Bands TDD_A | dBm/15 kHz | -85.76 | | -103.85 | | -116 | |
| | Bands TDD_C | | | | | | -115 | |
| | Bands TDD_E | | | | | | -114 | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 5 | -1.2 | 5 | -1.2 | 5 | -3.05 |
| CRS $\left(\hat{E}_s / Iot\right)_{meas}$ Note 5 In the 1st OFDM symbol | | dB | 2.55 | -7.39 | 2.55 | -7.39 | 3.25 | -9.39 |
| CRS $\left(\hat{E}_s / Iot\right)_{meas}$ note 5 in OFDM symbols 4,7,11 | | dB | 2.55 | -1.2 | 2.55 | -1.2 | 3.25 | -3.05 |
| SCH $\hat{E}_s / I_{ot}$ | | dB | -1.45 | -4.74 | -1.45 | -4.74 | -0.75 | -6.59 |
| RSRP Note 3,4,5 | Bands TDD_A | dBm/15 kHz | -80.76 | -86.96 | -98.85 | -105.05 | -111 | -119.05 |
| | Bands TDD_C | | | | | | -110 | -118.05 |
| | Bands TDD_E | | | | | | -109 | -117.05 |
| $(\mathrm{RSRQ})_{meas}$ Note3,4,5 | Bands TDD_A, TDD_C, TDD_E | dB | -12.71 | -14.54 | -12.71 | -14.54 | -12.47 | -15.70 |
| $(\mathrm{Io})_{meas}$ Note3 in the 1st OFDM symbol | Bands TDD_A | dBm/9 MHz | -51.06 | -54.39 | -69.15 | -72.48 | -81.55 | -85.20 |
| | Bands TDD_C | | | | | | -80.55 | -84.20 |
| | Bands TDD_E | | | | | | -79.55 | -83.20 |
| $(\mathrm{Io})_{meas}$ Note3 in OFDM symbols other than the 1st one | Bands TDD_A | dBm/9 MHz | -51.06 | -55.53 | -69.15 | -73.62 | -81.54 | -86.47 |
| | Bands TDD_C | | | | | | -80.54 | -85.47 |
| | Bands TDD_E | | | | | | -79.54 | -84.47 |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.
Note 3: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes.
Note 4: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Applies to restricted measurement subframes of the respective cell.
Note 6: Non-ABS and ABS subframe channel powers defined in Table C.3.1.2.1-1.
Note 7: In the 1st OFDM symbol, Cell 2 is not expected to meet the Es/Iot side condition in TS 36.133 [4] 9.1.5.2.

**Table 9.2.10.1.5-2: RSRQ TDD Intra frequency under time domain measurement resource restriction with MBSFN ABS absolute accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| | All bands | All bands | All bands |
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_04 | RSRQ_04 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_16 | RSRQ_16 | RSRQ_16 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_01 | RSRQ_01 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_19 | RSRQ_19 | RSRQ_17 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%

## 9.2.11 FDD RSRQ for E-UTRA Carrier Aggregation for 20MHz

### 9.2.11.1 FDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz

#### 9.2.11.1.1 Test purpose

To verify that FDD absolute RSRQ measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions for 20MHz. This test will verify the FDD absolute RSRQ accuracy requirements of the primary component carrier and the secondary component carrier.

#### 9.2.11.1.2 Test applicability

This test case applies to all types of E-UTRA FDD UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA.

This test case applies to all types of E-UTRA FDD UE release 11 and forward that support CA with Intra-band non-contiguous CA.

#### 9.2.11.1.3 Minimum conformance requirements

The FDD RSRQ measurements of cells on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1. The FDD RSRQ measurements of cells on the secondary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1.

The accuracy requirements in Table 9.2.11.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.1 for a corresponding Band.

**Table 9.2.11.1.3-1: FDD RSRQ absolute accuracy for Carrier Aggregation**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/BW$_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.11.1.3-2.

**Table 9.2.11.1.3-2: FDD RSRQ absolute accuracy measurement report mapping for Carrier Aggregation**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1, clause 9.1.11.2, clause 9.1.5.1 and A.9.2.11.

## 9.2.11.1.4 Test description

### 9.2.11.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.11.1.4.3.

4. There are two E-UTRA FDD carriers with one cell on one E-UTRA FDD carrier and two cells on the other E-UTRA FDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.2.11.1.4.2 Test procedure

The test consists of Cell 1 the PCell, Cell 2 the SCell on the Secondary Component Carrier (SCC), and Cell 3 the neighbouring cell on the SCC. The SCell (Cell 2) and neighbouring cell (Cell 3) on the SCC are configured. The SCell (Cell 2) on the SCC is activated. The absolute accuracy of RSRQ is defined as the RSRQ measured from the primary component carrier (Cell 1) and the RSRQ measured from the secondary component carrier (Cell 2).

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. The SS activates SCC by sending the MAC-CE according to TS 36.321 [11] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.2.11.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. The SS shall transmit an RRCConnectionReconfiguration message.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, the SS shall check the reported RSRQ values in periodical MeasurementReport messages. The reported RSRQ values for Cell 1 and Cell 2 by the UE are compared to the actual RSRQ values according to Table 9.2.11.1.5-2. This counts respectively as a Pass or Fail for the events “Cell 1” and “Cell 2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations for the respective event ”Cell 2” is increased by one.

10. The SS shall check MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events “Cell 1” and “Cell 2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If both events pass, the test passes. If one event fails, the test fails.

#### 9.2.11.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.11.1.4.3-1: Common Exception messages for FDD RSRQ absolute accuracy for Carrier Aggregation for 20MHz test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-4<br>Table H.4.1-5<br>Table H.4.2-1 |

### 9.2.11.1.5 Test requirement

Table 9.2.11.1.5-1 defines the primary level settings including test tolerances for all tests.

The FDD RSRQ absolute accuracy for carrier aggregation for 20MHz test shall meet the reported values test requirements in Table 9.2.11.1.5-2.

**Table 9.2.11.1.5-1: Cell Specific Test requirement Parameters for FDD RSRQ absolute accuracy for Carrier Aggregation for 20MHz**

| Parameters | | Test 1 | | | |
|---|---|---|---|---|---|
| | | Units | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel_CA}$ Note 1 | | MHz | 20 | 20 | 20 |
| Measurement bandwidth | | $n_{PRB}$ | 47-52 | 47-52 | 47-52 |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.4 FDD | R.4 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 38-61 | 38-61 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.10 FDD | R.10 FDD | R.10 FDD |
| OCNG Patterns defined in D.1.11 (OP.11 FDD) and D.1.12 (OP.12 FDD) | | | OP.11 FDD | OP.11 FDD | OP.12 FDD |
| Io Note2 | Bands FDD_A Note 5 | dBm/18 MHz | -87.26 | -82.67 | |
| | Bands FDD_C Note 5 | | -86.26 | -81.67 | |
| | Bands FDD_D Note 5 | | -85.76 | -81.17 | |
| | Bands FDD_E Note 5 | | -85.26 | -80.67 | |
| | Bands FDD_G Note 5 | | -84.26 | -79.67 | |
| | Bands FDD_H Note 5 | | -83.76 | -79.17 | |
| NOTE 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.<br>NOTE 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>NOTE 3: See Table 9.2.5.2.5-1 for the other parameters.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.<br>NOTE 5: The test applies for E-UTRA operating bands in this band group which are supporting 20 MHz channel bandwidth. | | | | | |

**Table 9.2.11.1.5-2: FDD RSRQ absolute accuracy requirements for the reported values for Carrier Aggregation for 20MHz**

| | Test 1 |
|---|---|
| | All bands |
| **Normal Conditions** | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_15 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_14 |
| **Extreme Conditions** | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_16 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_15 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.11.2 FDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz

### 9.2.11.2.1 Test purpose

To verify that FDD relative RSRQ measurement accuracy in carrier aggregation for 20MHz is within the specified limits under AWGN propagation conditions. This test will verify the FDD relative RSRQ accuracy requirements between the primary and secondary component carriers.

### 9.2.11.2.2 Test applicability

This test case applies to all types of E-UTRA FDD UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA.

This test case applies to all types of E-UTRA FDD UE release 11 and forward that support CA with Intra-band non-contiguous CA.

### 9.2.11.2.3 Minimum conformance requirements

The FDD RSRQ measurements of cells on the primary component carrier are compared with measurements of cells on the secondary component carrier defined in TS 36.133 [4] clause 9.1.6.2. The applicable relative accuracy requirements are the RSRQ inter-frequency accuracy requirements as defined in TS 36.133 [4] clause 9.1.6.2.

The accuracy requirements in Table 9.2.11.2.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

$RSRP1,2|_{dBm}$ according to Annex I.3.4 for a corresponding Band.

$$\left| RSRP1|_{dBm} - RSRP2|_{dBm} \right| \le [27] dB$$

|Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.2.11.2.3-1: FDD RSRQ relative accuracy for Carrier Aggregation**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot Note 2 | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/$BW_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.11.2.3-2.

**Table 9.2.11.2.3-2: FDD RSRQ relative accuracy measurement report mapping for Carrier Aggregation**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.3, clause 9.1.6.2 and A.9.2.11.

### 9.2.11.2.4 Test description

#### 9.2.11.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.11.2.4.3.

4. There are two E-UTRA FDD carriers with one cell on one E-UTRA FDD carrier and two cells on the other E-UTRA FDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.2.11.2.4.2 Test procedure

The test consists of Cell 1 the PCell, Cell 2 the SCell on the Secondary Component Carrier (SCC), and Cell 3 the neighbouring cell on the SCC. The SCell (Cell 2) and neighbouring cell (Cell 3) on the SCC are configured. The SCell (Cell 2) on the SCC is activated. The relative accuracy of RSRQ is defined as the RSRQ measured cells on the primary component carrier compared with measurements of cells on the secondary component carrier,

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. The SS activates SCC by sending the MAC-CE according to TS 36.321 [11] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.2.11.2.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. The SS shall transmit an RRCConnectionReconfiguration message.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. The UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, the SS shall check the reported RSRQ values in periodical MeasurementReport messages. The reported RSRQ value for Cell 2 is compared to the reported RSRQ value for Cell 1 for each

MeasurementReport message according to Table 9.2.11.2.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

10. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to table G.2.3-1 in Annex G.2 is achieved.

#### 9.2.11.2.4.3 Message contents

**Table 9.2.11.2.4.3-1: Common Exception messages for FDD RSRQ relative accuracy for Carrier Aggregation for 20MHz test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-4<br>Table H.4.1-5<br>Table H.4.2-1 |

### 9.2.11.2.5 Test requirement

Table 9.2.11.2.5-1 defines the primary level settings including test tolerances for all tests.

The FDD RSRQ relative accuracy for carrier aggregation for 20MHz test shall meet the reported values test requirements in Table 9.2.11.2.5-2.

**Table 9.2.11.2.5-1: Cell Specific Test requirement Parameters for FDD RSRQ relative accuracy for Carrier Aggregation**

| Parameters | Test 1 | | | |
|---|---|---|---|---|
| | Units | Cell 1 | Cell 2 | Cell 3 |

| | | | | | |
|---|---|---|---|---|---|
| $BW_{channel_CA}$ Note 1 | | MHz | 20 | 20 | 20 |
| Measurement bandwidth | | $n_{PRB}$ | 47-52 | 47-52 | 47-52 |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.4 FDD | R.4 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 38-61 | 38-61 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.10 FDD | R.10 FDD | R.10 FDD |
| OCNG Patterns defined in D.1.11 (OP.11 FDD) and D.1.12 (OP.12 FDD) | | | OP.11 FDD | OP.11 FDD | OP.12 FDD |
| $Io^{Note2}$ | Bands FDD_A Note 5 | dBm/18 MHz | -87.26 | -82.67 | |
| | Bands FDD_C Note 5 | | -86.26 | -81.67 | |
| | Bands FDD_D Note 5 | | -85.76 | -81.17 | |
| | Bands FDD_E Note 5 | | -85.26 | -80.67 | |
| | Bands FDD_G Note 5 | | -84.26 | -79.67 | |
| | Bands FDD_H Note 5 | | -83.76 | -79.17 | |

NOTE 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
NOTE 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 3: See Table 9.2.5.2.5-1 for the other parameters.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.
NOTE 5: The test applies for E-UTRA operating bands in this band group which are supporting 20 MHz channel bandwidth.

**Table 9.2.12.2.5-2: FDD RSRQ relative accuracy requirements for the reported values for Carrier Aggregation for 20MHz**

| | **Test 1** |
|---|---|
| | **All bands** |
| **Normal Conditions** | |
| Lowest reported value (Cell 2) | RSRQ_x – 12 |
| Highest reported value (Cell 2) | RSRQ_x + 9 |
| **Extreme Conditions** | |
| Lowest reported value (Cell 2) | RSRQ_x – 12 |
| Highest reported value (Cell 2) | RSRQ_x + 9 |
| RSRQ_x is the reported value of Cell 1 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.2.12 TDD RSRQ for E-UTRA Carrier Aggregation for 20MHz

## 9.2.12.1 TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz

### 9.2.12.1.1 Test purpose

To verify that TDD absolute RSRQ measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions for 20MHz. This test will verify the TDD absolute RSRQ accuracy requirements of the primary component carrier and the secondary component carrier.

### 9.2.12.1.2 Test applicability

This test case applies to all types of E-UTRA TDD UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA.

This test case applies to all types of E-UTRA TDD UE release 11 and forward that support CA with Intra-band non-contiguous CA.

### 9.2.12.1.3 Minimum conformance requirements

The TDD RSRQ measurements of cells on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1. The TDD RSRQ measurements of cells on the secondary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1.

The accuracy requirements in Table 9.2.12.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.1 for a corresponding Band.

**Table 9.2.12.1.3-1: TDD RSRQ absolute accuracy for Carrier Aggregation**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/BW$_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.12.1.3-2.

**Table 9.2.12.1.3-2: TDD RSRQ absolute accuracy measurement report mapping for Carrier Aggregation**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1, clause 9.1.11.2, clause 9.1.5.1 and A.9.2.12.

### 9.2.12.1.4 Test description

#### 9.2.12.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.12.1.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.2.12.1.4.2 Test procedure

The test consists of Cell 1 the PCell, Cell 2 the SCell on the Secondary Component Carrier (SCC), and Cell 3 the neighbouring cell on the SCC. The SCell (Cell 2) and neighbouring cell (Cell 3) on the SCC are configured. The SCell (Cell 2) on the SCC is activated. The absolute accuracy of RSRQ is defined as the RSRQ measured from the primary component carrier (Cell 1) and the RSRQ measured from the secondary component carrier (Cell 2).

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. The SS activates SCC by sending the MAC-CE according to TS 36.321 [11] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.2.12.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. The SS shall transmit an RRCConnectionReconfiguration message.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, The SS shall check the reported RSRQ values in periodical MeasurementReport messages. The reported RSRQ values for Cell 1 and Cell 2 by the UE are compared to the actual RSRQ values according to Table 9.2.12.1.5-2. This counts respectively as a Pass or Fail for the events “Cell 1” and “Cell 2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations for the respective event ”Cell 2” is increased by one.

10. The SS shall check MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events “Cell 1” and “Cell 2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If both events pass, the test passes. If one event fails, the test fails.

#### 9.2.12.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.12.1.4.3-1: Common Exception messages for TDD RSRQ absolute accuracy for Carrier Aggregation for 20MHz test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-4<br>Table H.4.1-5<br>Table H.4.2-1 |

### 9.2.12.1.5 Test requirement

Table 9.2.12.1.5-1 defines the primary level settings including test tolerances for all tests.

The TDD RSRQ absolute accuracy for carrier aggregation for 20MHz test shall meet the reported values test requirements in Table 9.2.12.1.5-2.

**Table 9.2.12.1.5-1: Cell Specific Test requirement Parameters for TDD RSRQ absolute accuracy for Carrier Aggregation for 20MHz**

| Parameters | | Test 1 | | | |
|---|---|---|---|---|---|
| | | Units | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel_CA}$ Note1 | | MHz | 20 | 20 | 20 |
| Measurement bandwidth | | $n_{PRB}$ | 47-52 | 47-52 | 47-52 |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.3 TDD | R.3 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 38-61 | 38-61 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.10 TDD | R.10 TDD | R.10 TDD |
| OCNG Patterns defined in D.2.7 (OP.7 TDD) and D.2.8 (OP.8 TDD) | | | OP.7 TDD | OP.7 TDD | OP.8 TDD |
| Io Note2 | Bands TDD_A Note 5 | dBm/18 MHz | -87.26 | -82.60 | |
| | Bands TDD_C Note 5 | | -86.26 | -81.60 | |
| | Bands TDD_E Note 5 | | -85.26 | -80.60 | |
| Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in TS 36.133 [4] section A.3.6.1.<br>Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 3: See Table 9.2.6.1.5-1 for the other parameters.<br>Note 4: E-UTRA operating band groups are as defined in Section 3.5.<br>Note 5: The test applies for E-UTRA operating bands in this band group which are supporting 20 MHz channel bandwidth. | | | | | |

**Table 9.2.12.1.5-2: TDD RSRQ absolute accuracy requirements for the reported values for Carrier Aggregation for 20MHz**

| | Test 1 |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_15 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value Cell 2) | RSRQ_14 |
| Extreme Conditions | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_16 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value Cell 2) | RSRQ_15 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.12.2 TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz

### 9.2.12.2.1 Test purpose

To verify that TDD relative RSRQ measurement accuracy in carrier aggregation for 20MHz is within the specified limits under AWGN propagation conditions. This test will verify the TDD relative RSRQ accuracy requirements between the primary and secondary component carriers.

### 9.2.12.2.2 Test applicability

This test case applies to all types of E-UTRA TDD UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA.

This test case applies to all types of E-UTRA TDD UE release 11 and forward that support CA with Intra-band non-contiguous CA.

### 9.2.12.2.3 Minimum conformance requirements

The TDD RSRQ measurements of cells on the primary component carrier are compared with measurements of cells on the secondary component carrier defined in TS 36.133 [4] clause 9.1.6.2. The applicable relative accuracy requirements are the RSRQ inter-frequency accuracy requirements as defined in TS 36.133 [4] clause 9.1.6.2.

The accuracy requirements in Table 9.2.12.2.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

RSRP1,2$|_{dBm}$ according to Annex I.3.4 for a corresponding Band.

$$\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \le [27]dB$$

|Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.2.12.2.3-1: TDD RSRQ relative accuracy for Carrier Aggregation**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/$BW_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.12.2.3-2.

**Table 9.2.12.2.3-2: TDD RSRQ relative accuracy measurement report mapping for Carrier Aggregation**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.3, clause 9.1.6.2 and A.9.2.12.

### 9.2.12.2.4 Test description

#### 9.2.12.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 20 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.12.2.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.2.12.2.4.2 Test procedure

The test consists of Cell 1 the PCell, Cell 2 the SCell on the Secondary Component Carrier (SCC), and Cell 3 the neighbouring cell on the SCC. The SCell (Cell 2) and neighbouring cell (Cell 3) on the SCC are configured. The SCell (Cell 2) on the SCC is activated. The relative accuracy of RSRQ is defined as the RSRQ measured cells on the primary component carrier compared with measurements of cells on the secondary component carrier,

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. The SS activates SCC by sending the MAC-CE according to TS 36.321 [11] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.2.12.2.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. The SS shall transmit an RRCConnectionReconfiguration message.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. The UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, the SS shall check the reported RSRQ values in periodical MeasurementReport messages. The reported RSRQ value for Cell 2 is compared to the reported RSRQ value for Cell 1 for each MeasurementReport message according to Table 9.2.12.2.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

10. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to table G.2.3-1 in Annex G.2 is achieved.

#### 9.2.12.2.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.12.2.4.3-1: Common Exception messages for TDD RSRQ relative accuracy for Carrier Aggregation for 20MHz test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-4<br>Table H.4.1-5<br>Table H.4.2-1 |

## 9.2.12.2.5 Test requirement

Table 9.2.12.2.5-1 defines the primary level settings including test tolerances for all tests.

The TDD RSRQ relative accuracy for carrier aggregation for 20MHz test shall meet the reported values test requirements in Table 9.2.12.2.5-2.

**Table 9.2.12.2.5-1: Cell Specific Test requirement Parameters for TDD RSRQ relative accuracy for Carrier Aggregation**

| Parameters | | Test 1 | | | |
|---|---|---|---|---|---|
| | | Units | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel_CA}$ Note1 | | MHz | 20 | 20 | 20 |
| Measurement bandwidth | | $n_{PRB}$ | 47-52 | 47-52 | 47-52 |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.3 TDD | R.3 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 38-61 | 38-61 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.10 TDD | R.10 TDD | R.10 TDD |
| OCNG Patterns defined in D.2.7 (OP.7 TDD) and D.2.8 (OP.8 TDD) | | | OP.7 TDD | OP.7 TDD | OP.8 TDD |
| Io Note2 | Bands TDD_A Note 5 | dBm/18 MHz | -87.26 | -82.60 | |
| | Bands TDD_C Note 5 | | -86.26 | -81.60 | |
| | Bands TDD_E Note 5 | | -85.26 | -80.60 | |

Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in TS 36.133 [4] section A.3.6.1.
Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: See Table 9.2.6.2.5-1 for the other parameters.
Note 4: E-UTRA operating band groups are as defined in Section 3.5.
Note 5: The test applies for E-UTRA operating bands in this band group which are supporting 20 MHz channel bandwidth.

**Table 9.2.12.2.5-2: TDD RSRQ relative accuracy requirements for the reported values for Carrier Aggregation for 20MHz**

| | Test 1 |
|---|---|

| | All bands |
|---|---|
| Normal Conditions | |
| Lowest reported value (Cell 2) | RSRQ_x - 12 |
| Highest reported value (Cell 2) | RSRQ_x + 9 |
| Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRQ_x - 12 |
| Highest reported value (Cell 2) | RSRQ_x + 9 |
| RSRQ_x is the reported value of Cell 1 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.13 Void

## 9.2.14 Void

## 9.2.15 FDD RSRQ Accuracy under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS

### 9.2.15.1 FDD RSRQ Accuracy under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC)

#### 9.2.15.1.1 Test purpose

The purpose of this test is to verify that the RSRQ measurement accuracy under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS within the specified limits.

#### 9.2.15.1.2 Test applicability

This test applies to all types of E-UTRA FDD and CRS interference handling UE release 11 and forward. Applicability requires support of FGI bit 115.

#### 9.2.15.1.3 Minimum conformance requirements

The requirements for absolute accuracy of RSRQ in this section shall apply to a cell on the same frequency as that of the serving cell when a time domain measurement resource restriction pattern for performing RSRQ measurements of this cell is configured by higher layers according to TS 36.331 [5].

The accuracy requirements in Table 9.2.15.1.3-1 are valid under the following conditions:

Cell specific reference signals in the measured cell are transmitted either from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled,

$RSRP|_{dBm}$ according to Annex I.3.9 for a corresponding Band,

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRQ measurement,

The RSRQ measurement is not performed in any subframe other than those indicated by the time domain measurement resource restriction pattern configured for the measured cell,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern,

The UE is provided with the CRS assistance information (TS 36.331 [5]) and the CRS assistance information is valid throughout the entire evaluation period.

**Table 9.2.15.1.3-1: RSRQ Intra frequency absolute accuracy under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | $\hat{E}s/Iot$ Note 5 | Io Note 2 range | | |
| | | | E-UTRA operating band groups Note 6 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 1, 4 | $dBm/BW_{Channel}$ |
| ±2.5 | ±4 | ≥-6.96 | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-9.46 | Note 3 | Note 3 | Note 3 |

NOTE 1: This Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRQ measurements of this cell. The specified Io range applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: The gap between the Es/Iot level in TS36.133 [4] table 9.1.5.3-1 and TS36.133 [4] 9.1.5.2-1 is due to the interference from either PCell or at least one neighbour cell indicated within the CRS assistance information.
NOTE 6: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.15.1.3-2.

**Table 9.2.15.1.3-2: RSRQ FDD Intra frequency absolute accuracy measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.5.3 and A.9.2.15.

### 9.2.15.1.4 Test description

#### 9.2.15.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.54 (without faders).

2. The general test parameter settings are set up according to Table 9.2.15.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.2.15.1.4.3.

5. There is one E-UTRA FDD carrier and three cells specified in each test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 9.2.15.1.4.1-1: General test parameters for E-UTRAN FDD RSRQ intra frequency under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PCell | | | Cell 1 | Serving/aggressor cell |
| Neighbour cells | | | Cell 2 | Neighbour/aggressor cell |
| | | | Cell 3 | Cell to be measured |
| ABS transmission configuration | | | Non-MBSFN ABS | As defined in Table C.3.1.1.1-1 |
| CP length | | | Normal | For all cells in the test |
| DRX | | | | OFF |
| Time offset between cells | | µs | Cell 2 offset with respect to Cell 1: 0<br>Cell 3 offset with respect to Cell 1: -2.5 | Three synchronous cells |
| Physical cell IDs | | | $(PCI_{cell1} - PCI_{cell3})mod6 = 0$<br>$(PCI_{cell2} - PCI_{cell3})mod6 != 0$<br><br>$PCI_{cell1}$ not equal to $PCI_{cell3}$ | Cell PCIs are selected so that all conditions are met |
| ABS pattern | | | '10000000100000001000000010000000010000000' | FDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54.<br>The first/leftmost bit corresponds to the Pcell subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (40) divided by 10. No MBSFN subframes are configured in the ABS subframes.<br>Configured in Cell 1 and Cell 2. |
| Time domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | | '10000000100000001000000010000000010000000' | Time domain measurement resource restriction pattern for neighbour cell measurement signalled to the UE in measSubframePatternNeigh IE in measSubframePatternConfigNeigh, as defined in TS 36.331, clause 6.3.5.<br>Configured for Cell 3 measurements. The cell list in measSubframeCellList IE shall contain Cell 3 but not Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | | '01000000010000000100000001000000001000000' | Configured for measurements on Cell 1. |
| CRS assistance information | physCellId | | see PCI conditions above | Only the CRS information of Cell 2 is provided in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation *one Frame*='000000'. |
| | antennaPortsCount | | 1 | |
| | mbsfn-SubframeConfigList | | *oneFrame* = '000000' | |

### 9.2.15.1.4.2 Test procedure

In this test case all cells are on the same carrier frequency. The absolute accuracy of RSRQ intra frequency measurements under time domain measurement resource restriction with CRS assistance information is tested by using the parameters in Table 9.2.15.1.4.1-1 and Table .9.2.15.1.5-1 for non-MBSFN ABS with colliding CRS between Cell 1 and Cell 3 and non-colliding CRS between Cell1 and Cell 2. In all test cases, Cell 1 is the serving/aggressor cell, Cell 2 is the neighbour/aggressor cell and Cell 3 is the target cell to be measured for RSRQ.

The UE is configured by higher layers via Cell 1 with a time domain measurement resource restriction pattern for performing E-UTRAN FDD intra-frequency measurements on neighbour cells, namely Cell 3 measurements with a neighbour cell list, where the cell list includes Cell 3. The UE is also provided via higher layers with the CRS assistance information of Cell 2. The information for both measurement pattern and the CRS assistance information shall be provided to the UE before the measurements start.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3 with exceptions listed in 7.2A.6.

2. Set the parameters according to Table 9.2.15.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the RSRQ value in periodical MeasurementReport messages. The RSRQ value of Cell 3 reported by the UE is compared to the actual RSRQ value according to Table 9.2.15.1.5-2. If the UE fails to report the measurement value for Cell 3, the number of failed iterations is increased by one.

7. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.2.15.1.5-1 as appropriate.

### 9.2.15.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 7.2A.6 with the following exceptions:

**Table 9.2.15.1.4.3-1: Common Exception messages for E-UTRAN FDD RSRQ intra frequency under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.7-1<br>Table H.2.7-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-4a<br>Table H.3.5-5<br>Table H.5.1-1<br>Table H.5.1-2<br>Table H.5.1-3 |

**Table 9.2.15.1.4.3-2: *MeasResults*: Additional E-UTRAN FDD RSRQ intra frequency under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.15.1.4.3-3: *MeasResultListEUTRA*: Additional E-UTRAN FDD RSRQ intra frequency under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity of Cell 3 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | Not present | | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 9.2.15.1.4.3-4: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional E-UTRAN FDD RSRQ intra frequency under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternFDD-r10 | '01000000010000000100000001000000001000000' | BIT STRING (SIZE (40)) | |
| } | | | |
| } | | | |
| } | | | |

## 9.2.15.1.5 Test requirement

Table 9.2.15.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRQ FDD intra frequency absolute accuracy test shall meet the reported values test requirements in Table 9.2.15.1.5-2.

**Table 9.2.15.1.5-1: Cell Specific Test requirement Parameters for E-UTRAN FDD RSRQ intra frequency under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Parameter | | Unit | Test 1 | | | Test 2 | | | Test 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 | Cell 1 | Cell 2 | Cell 3 | Cell 1 | Cell 2 | Cell 3 |
| E-UTRA RF Channel Number | | | 1 | | | 1 | | | 1 | | |
| $BW_{channel}$ | | MHz | 10 | | | 10 | | | 10 | | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | | 22—27 | | | 22—27 | | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - | | R.0 FDD | - | | R.0 FDD | - | |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | | 13—36 | - | | 13—36 | - | |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | | | R.6 FDD | | | R.6 FDD | | |
| OCNG Patterns defined in D.1 | | | OP.5 FDD | OP.6 FDD | OP.6 FDD | OP.5 FDD | OP.6 FDD | OP.6 FDD | OP.5 FDD | OP.6FDD | OP.6 FDD |
| PBCH_RA | | dB | Note 6 | Note 6 | 0 | Note 6 | Note 6 | 0 | Note 6 | Note 6 | 0 |
| PBCH_RB | | | | | | | | | | | |
| PSS_RA | | | | | | | | | | | |
| SSS_RA | | | | | | | | | | | |
| PCFICH_RB | | | | | | | | | | | |
| PHICH_RA | | | | | | | | | | | |
| PHICH_RB | | | | | | | | | | | |
| PDCCH_RA | | | | | | | | | | | |
| PDCCH_RB | | | | | | | | | | | |
| PDSCH_RA | | | | | | | | | | | |
| PDSCH_RB | | | | | | | | | | | |
| OCNG_RA Note 1 | | | | | | | | | | | |
| OCNG_RB Note 1 | | | | | | | | | | | |
| $N_{oc}$ Note 2 | Bands FDD_A | dBm/15 kHz | -84.76 | | | -103.85 | | | -116 | | |
| | Bands FDD_C | | | | | | | | -115 | | |
| | Bands FDD_D | | | | | | | | -114.5 | | |
| | Bands FDD_E, FDD_F Note 7 | | | | | | | | -114 | | |
| | Bands FDD_G | | | | | | | | -113 | | |
| | Bands FDD_H | | | | | | | | -112.5 | | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 4 | 2 | -0.5 | 4 | 2 | -0.5 | 4 | 2 | -3 |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{meas}$ Note 5 | | dB | -1.41 | -0.77 | -5.96 | -1.41 | -0.77 | -5.96 | -0.89 | 0.24 | -8.46 |
| RSRP Note 3,4,5 | Bands FDD_A | dBm/15 kHz | -80.76 | -82.76 | -85.26 | -99.85 | -101.85 | -104.35 | -112 | -114 | -119 |
| | Bands FDD_C | | | | | | | | -111 | -113 | -118 |
| | Bands FDD_D | | | | | | | | -110.5 | -112.5 | -117.5 |
| | Bands FDD_E, FDD_F Note 7 | | | | | | | | -110 | -112 | -117 |
| | Bands FDD_G | | | | | | | | -109 | -111 | -116 |
| | Bands FDD_H | | | | | | | | -108.5 | -110.5 | -115.5 |
| $(\mathrm{RSRQ})_{meas}$ Note 3,4,5 | Bands FDD_A, FDD_C, FDD_D, FDD_E, FDD_F Note 7, FDD_G, FDD_H | dB | -14.56 | -11.99 | -14.49 | -14.56 | -11.99 | -14.49 | -14.27 | -11.09 | -16.09 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $(\mathrm{Io})_{meas}$ Note 3 | Bands FDD_A | dBm/9 MHz | -49.21 | -52.87 | -68.30 | -71.96 | -80.74 | -84.83 |
| | Bands FDD_C | | | | | | -79.74 | -83.83 |
| | Bands FDD_D | | | | | | -79.24 | -83.33 |
| | Bands FDD_E, FDD_F Note 7 | | | | | | -78.74 | -82.83 |
| | Bands FDD_G Note 9 | | | | | | -77.74 | -81.83 |
| | Bands FDD_H | | | | | | -77.24 | -81.33 |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes.
Note 3: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes.
Note 4: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: Applies to restricted measurement subframes for only Cell 2 and Cell 3. For Cell 1, the corresponding value is derived from the normal subframes other than the subframes indicated in the time domain measurement resource restriction pattern for intra-frequency measurements.
Note 6: Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1.
Note 7: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 8: E-UTRA operating band groups are as defined in Section 3.5.
Note 9: Except Band 29 and Band 32.

**Table 9.2.15.1.5-2: RSRQ FDD Intra frequency under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS absolute accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| | All bands | All bands | All bands |
| Normal Conditions | | | |
| Lowest reported value (Cell 3) | RSRQ_05 | RSRQ_05 | RSRQ_00 |
| Highest reported value (Cell 3) | RSRQ_17 | RSRQ_17 | RSRQ_16 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 3) | RSRQ_02 | RSRQ_02 | RSRQ_00 |
| Highest reported value (Cell 3) | RSRQ_20 | RSRQ_20 | RSRQ_17 |

## 9.2.16 TDD RSRQ Accuracy under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS

### 9.2.16.1 TDD RSRQ Accuracy under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC)

#### 9.2.16.1.1 Test purpose

The purpose of this test is to verify that the RSRQ measurement accuracy under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS within the specified limits.

### 9.2.16.1.2 Test applicability

This test applies to all types of E-UTRA TDD and CRS interference handling UE release 11 and forward. Applicability requires support of FGI bit 115.

### 9.2.16.1.3 Minimum conformance requirements

The requirements for absolute accuracy of RSRQ in this section shall apply to a cell on the same frequency as that of the serving cell when a time domain measurement resource restriction pattern for performing RSRQ measurements of this cell is configured by higher layers according to TS 36.331 [5].

The accuracy requirements in Table 9.2.16.1.3-1 are valid under the following conditions:

Cell specific reference signals in the measured cell are transmitted either from one, two or four antenna ports,

Conditions defined in TS 36.101 [2] Section 7.3 for reference sensitivity are fulfilled,

$RSRP|_{dBm}$ according to Annex I.3.9 for a corresponding Band,

The time domain measurement resource restriction pattern configured for the measured cell indicates at least one subframe per radio frame for performing the RSRQ measurement,

The RSRQ measurement is not performed in any subframe other than those indicated by the time domain measurement resource restriction pattern configured for the measured cell,

Four symbols containing CRS are available in all subframes indicated by the time domain measurement resource restriction pattern,

The UE is provided with the CRS assistance information (TS 36.331 [5]) and the CRS assistance information is valid throughout the entire evaluation period.

**Table 9.2.16.1.3-1: RSRQ Intra frequency absolute accuracy under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 2 range | | |
| Normal condition | Extreme condition | Ês/Iot Note 5 | E-UTRA operating band groups Note 6 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 1, 4 | $dBm/BW_{Channel}$ |
| ±2.5 | ±4 | ≥-6.96 | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-9.46 | Note 3 | Note 3 | Note 3 |

NOTE 1: This Io condition is expressed as the average Io per RE over all REs in that symbol.
NOTE 2: Io is defined in subframes indicated by the time domain measurement resource restriction pattern configured for performing RSRQ measurements of this cell. The specified Io range applies to CRS and non-CRS symbols. Io may be different in different symbols within a subframe.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: The gap between the Es/Iot level in TS36.133[4] table 9.1.5.3-1 and TS36.133[4] 9.1.5.2-1 is due to the interference from either PCell or at least one neighbour cell indicated within the CRS assistance information.
NOTE 6: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.16.1.3-2.

**Table 9.2.16.1.3-2: RSRQ FDD Intra frequency absolute accuracy measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.5.3 and A.9.2.16.

## 9.2.16.1.4 Test description

### 9.2.16.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.54 (without faders).

2. The general test parameter settings are set up according to Table 9.2.16.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.2.16.1.4.3.

5. There is one E-UTRA TDD carrier and three cells specified in each test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 9.2.16.1.4.1-1: General test parameters for E-UTRAN TDD RSRQ intra frequency under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| PCell | | | Cell 1 | Serving/aggressor cell |
| Neighbour cells | | | Cell 2 | Neighbour/aggressor cell |
| | | | Cell3 | Cell to be measured |
| Special subframe configuration | | | 6 | For Cell 1, Cell 2 and Cell 3. For special subframe configurations see Table 4.2-1 in 36.211 [9]. |
| Uplink/downlink subframe configuration | | | 1 | For Cell 1, Cell 2 and Cell 2. For uplink-downlink subframe configurations see Table 4.2-2 in TS 36.211 [9]. |
| ABS transmission configuration | | | Non-MBSFN ABS | As defined in Table C.3.1.1.1-1 |
| CP length | | | Normal | For all cells in the test |
| DRX | | | | OFF |
| Time offset between cells | | µs | Cell 2 offset with respect to Cell 1: 0<br>Cell 3 offset with respect to Cell 1: -2.5 | Three synchronous cells |
| Physical cell IDs | | | $(PCI_{cell1} - PCI_{cell3})mod6 = 0$<br>$(PCI_{cell2} - PCI_{cell3})mod6 != 0$<br><br>$PCI_{cell1}$ not equal to $PCI_{cell3}$ | Cell PCIs are selected so that all conditions are met |
| ABS pattern | | | '00000000010000000001' | TDD ABS Pattern Info IE, as defined in TS 36.423 [28], clause 9.2.54.<br>The first/leftmost bit corresponds to the Pcell subframe #0 of the radio frame satisfying SFN mod x = 0, where x is the size of the bit string (20) divided by 10. No MBSFN subframes are configured in the ABS subframes.<br>Provided to the UE for Cell 1 and Cell 2. |
| Time domain measurement resource restriction pattern for neighbour cell measurements on RF Channel 1 | | | '00000000010000000001' | Time domain measurement resource restriction pattern for neighbour cell measurement signalled to the UE in measSubframePatternNeigh IE in measSubframePatternConfigNeigh, as defined in TS 36.331, clause 6.3.5.<br>Provided to the UE for Cell 3 measurements.<br>The cell list in measSubframeCellList IE shall contain Cell 3 but not Cell 2. |
| Time-domain measurement resource restriction pattern for serving cell measurements | | | '10000000001000000000' | Configured for Cell 1 measurements. |
| CRS assistance information | physCellId | | see PCI conditions above | Only the CRS assistance information of cell 2 is provided for Cell 2 only in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation *one Frame*='000000'. |
| | antennaPortsCount | | 1 | |
| | mbsfn-SubframeConfigList | | *oneFrame* = '000000' | |

### 9.2.16.1.4.2 Test procedure

In this test case all cells are on the same carrier frequency. The absolute accuracy of RSRQ intra frequency measurements under time domain measurement resource restriction with CRS assistance information is tested by using the parameters in Table 9.2.16.1.4.1-1 and Table.9.2.16.1.5-1 for non-MBSFN ABS with colliding CRS between Cell 1 and Cell3 and non-colliding CRS between Cell 1 and Cell 2. In all test cases, Cell 1 is the serving/aggressor cell, Cell 2 is the neighbour/aggressor cell and Cell 3 is the target cell to be measured for RSRQ.

The UE is configured by higher layers via Cell 1 with a time domain measurement resource restriction pattern for performing E-UTRAN TDD intra-frequency measurements on neighbour cells, namely Cell 3 measurements with a neighbour cell list, where the cell list includes Cell 3. The UE is also provided via higher layers with the CRS assistance information of Cell 2. The information for both measurement pattern and the CRS assistance information shall be provided to the UE before the measurements start.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3 with exceptions listed in 7.2A.6.

2. Set the parameters according to Table 9.2.16.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the RSRQ value in periodical MeasurementReport messages. The RSRQ value of Cell 3 reported by the UE is compared to the actual RSRQ value according to Table 9.2.16.1.5-2. If the UE fails to report the measurement value for Cell 3, the number of failed iterations is increased by one.

7. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.2.16.1.5-1 as appropriate.

### 9.2.16.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 7.2A.6 with the following exceptions:

**Table 9.2.16.1.4.3-1: Common Exception messages for E-UTRAN TDD RSRQ intra frequency under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | Table H.2.7-1<br>Table H.2.7-2 |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-4a<br>Table H.3.5-5<br>Table H.5.1-1<br>Table H.5.1-2<br>Table H.5.1-3 |

**Table 9.2.16.1.4.3-2: *MeasResults*: Additional E-UTRAN TDD RSRQ intra frequency under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.16.1.4.3-3: *MeasResultListEUTRA*: Additional E-UTRAN TDD RSRQ intra frequency under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: TS 36.331 [5] clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysicalCellIdentity of Cell 3 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | Not present | | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

**Table 9.2.16.1.4.3-4: *RadioResourceConfigDedicated-SRB2-DRB(n,m)*: Additional E-UTRAN TDD RSRQ intra frequency under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 RadioResourceConfigDedicated-SRB2-DRB(n,m) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| MeasSubframePatternPCell-r10 CHOICE { | | | |
| setup SEQUENCE { | | | |
| subframePatternTDD-r10 CHOICE { | | | |
| subframeConfig1-5-r10 | '10000000001000000000' | BIT STRING (SIZE (20)) | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.2.16.1.5 Test requirement

Table 9.2.16.1.5-1 defines the primary level settings including test tolerances for all tests.

**Table 9.2.16.1.5-1: Cell Specific Test requirement Parameters for E-UTRAN TDD RSRQ intra frequency under time-domain measurement resource restriction with CRS assistance information and non-MBSFN ABS**

| Parameter | | Unit | Test 1 | | | Test 2 | | | Test 3 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 | Cell 1 | Cell 2 | Cell 3 | Cell 1 | Cell 2 | Cell 3 |
| E-UTRA RF Channel Number | | | 1 | | | 1 | | | 1 | | |
| $BW_{channel}$ | | MHz | 10 | | | 10 | | | 10 | | |
| Measurement bandwidth | | $n_{PRB}$ | 22-27 | | | 22-27 | | | 22-27 | | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - | | R.0 TDD | - | | R.0 TDD | - | |
| PDSCH allocation | | $n_{PRB}$ | 13-36 | - | | 13—36 | - | | 13-36 | - | |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | | | R.6 TDD | | | R.6 TDD | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD | OP.2 TDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA[Note1]<br>OCNG_RB[Note1] | | dB | Note 6 | Note 6 | 0 | Note 6 | Note 6 | 0 | Note 6 | Note 6 | 0 |
| $N_{oc}$ [Note2] | Bands TDD_A | dBm/15 kHz | -84.76 | | | -103.85 | | | -116 | | |
| | Bands TDD_C | | | | | | | | -115 | | |
| | Bands TDD_E | | | | | | | | -114 | | |
| CRS $\hat{E}_s / N_{oc}$ | | dB | 4 | 2 | -0.5 | 4 | 2 | -0.5 | 4 | 2 | -3 |
| CRS $\left(\hat{E}_s / I_{ot}\right)_{meas}$ [Note 5] | | dB | -1.41 | -0.77 | -5.96 | -1.41 | -0.77 | -5.96 | -0.89 | 0.24 | -8.46 |
| RSRP[Note3,4,5] | Bands TDD_A | dBm/15 kHz | -80.76 | -82.76 | -85.26 | -99.85 | -101.85 | -104.35 | -112 | -114 | -119 |
| | Bands TDD_C | | | | | | | | -111 | -113 | -118 |
| | Bands TDD_E | | | | | | | | -110 | -112 | -117 |
| $(\mathrm{RSRQ})_{meas}$ Note3,4,5 | Bands TDD_A, TDD_C, TDD_E | dB | -14.56 | -11.96 | -14.49 | -14.56 | -11.99 | -14.49 | -14.27 | -11.09 | -16.09 |
| $(\mathrm{Io})_{meas}$ [Note3] | Bands TDD_A | dBm/9 MHz | -49.21 | -52.87 | | -68.30 | -71.96 | | -80.74 | -84.83 | |
| | Bands TDD_C | | | | | | | | -79.74 | -83.83 | |
| | Bands TDD_E | | | | | | | | -78.74 | -82.83 | |
| Propagation condition | | - | AWGN | | | AWGN | | | AWGN | | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. Applies to all subframes. |
| Note 3: | RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. Io levels are calculated in CRS symbols of measurement restricted subframes. |
| Note 4: | RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | Applies to restricted measurement subframes for only Cell 2 and Cell 3. For Cell 1, the corresponding value is derived from the normal subframes other than the subframes indicated in the time domain measurement resource restriction pattern for intra-frequency measurements. |
| Note 6: | Non-ABS and ABS subframe channel powers defined in Table C.3.1.1.1-1. |
| Note 7: | E-UTRA operating band groups are as defined in Section 3.5. |

**Table 9.2.16.1.5-2: RSRQ TDD Intra frequency under time domain measurement resource restriction with CRS assistance information and non-MBSFN ABS absolute accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| | All bands | All bands | All bands |
| Normal Conditions | | | |
| Lowest reported value (Cell 3) | RSRQ_05 | RSRQ_05 | RSRQ_00 |
| Highest reported value (Cell 3) | RSRQ_17 | RSRQ_17 | RSRQ_16 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 3) | RSRQ_02 | RSRQ_02 | RSRQ_00 |
| Highest reported value (Cell 3) | RSRQ_20 | RSRQ_20 | RSRQ_17 |

## 9.2.17 FDD Intra frequency RSRQ Accuracy for 5MHz Bandwidth

### 9.2.17.1 FDD Intra Frequency Absolute RSRQ Accuracy for 5MHz Bandwidth

#### 9.2.17.1.1 Test purpose

To verify that the FDD intra frequency absolute RSRQ measurement accuracy for 5MHz Bandwidth is within the specified limit.

#### 9.2.17.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support Band 31. Applicability requires support for FGI bit 16.

#### 9.2.17.1.3 Minimum conformance requirements

The absolute accuracy of RSRQ is defined as the RSRQ measured from a cell on the same frequency as that of the serving cell.

The accuracy requirements in table 9.2.17.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.1 for a corresponding Band.

**Table 9.2.17.1.3-1: RSRQ Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 4] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 3] | dBm/$BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.17.1.3-2.

**Table 9.2.17.1.3-2: RSRQ FDD Intra frequency absolute accuracy measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.5.1 and A.9.2.17.

## 9.2.17.1.4 Test description

### 9.2.17.1.4.1 Initial conditions

Same initial conditions as defined in clause 9.2.1.1.4.1 with the exception that the Channel Bandwidth to be tested is 5MHz.

### 9.2.17.1.4.2 Test procedure

Same test procedure as defined in clause 9.2.1.1.4.2 with the following exceptions:

- Instead of Table 9.2.1.1.5-2 → use Table 9.2.17.1.5-1.
- Instead of Table 9.2.2.1.5-3 → use Table 9.2.17.1.5-2.

### 9.2.17.1.4.3 Message contents

Same message contents as defined in clause 9.2.1.1.4.3.

## 9.2.17.1.5 Test requirement

Table 9.2.17.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRQ FDD intra frequency absolute accuracy test shall meet the reported values test requirements in Table 9.2.17.1.5-2.

**Table 9.2.17.1.5-1: RSRQ FDD Intra frequency absolute accuracy test parameters for 5MHz Bandwidth**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 5 | | 5 | | 5 | |
| Measurement bandwidth | | $n_{PRB}$ | 10-15 | | 10-15 | | 10-15 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.5 FDD | - | R.5 FDD | - | R.5 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 7-17 | - | 7-17 | - | 7-17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.11 FDD | | R.11 FDD | | R.11 FDD | |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | | OP.15 FDD | OP.16 FDD | OP.15 FDD | OP.16 FDD | OP.15 FDD | OP.16 FDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA[Note1]<br>OCNG_RB[Note1] | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $N_{oc}$ [Note2] | Band 31 | dBm/15 kHz | -82.81 | | -100.85 | | -109.5 | |
| $\hat{E}_s / I_{ot}$ | | dB | -1.76 | -1.76 | -4.70 | -4.70 | -5.17 | -5.17 |
| RSRP[Note3] | Band 31 | dBm/15 kHz | -79.81 | -79.81 | -103.75 | -103.75 | -113.10 | -113.10 |
| RSRQ[Note3] | Band 31 | dB | -14.77 | -14.77 | -16.76 | -16.76 | -17.12 | -17.12 |
| Io[Note3] | Band 31 | dBm/4.5 MHz | -51.06 | | -73.01 | | -82.00 | |
| $\hat{E}_s / N_{oc}$ | | dB | 3 | 3 | -2.9 | -2.9 | -3.6 | -3.6 |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 9.2.17.1.5-2: RSRQ FDD Intra frequency absolute accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| | All bands | All bands | All bands |
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_04 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_16 | RSRQ_14 | RSRQ_14 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_01 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_19 | RSRQ_15 | RSRQ_15 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.2.18 FDD - FDD Inter frequency RSRQ Accuracy for 5MHz Bandwidth

## 9.2.18.1 FDD - FDD Inter Frequency Absolute RSRQ Accuracy for 5MHz Bandwidth

### 9.2.18.1.1 Test purpose

To verify that the FDD - FDD inter frequency absolute RSRQ measurement accuracy is within the specified limit for band 31.

### 9.2.18.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support Band 31. Applicability requires support for FGI bits 16 and 25.

### 9.2.18.1.3 Minimum conformance requirements

The requirements for absolute accuracy of RSRQ in this clause apply to a cell that has different carrier frequency from the serving cell.

The accuracy requirements in Table 9.2.18.1.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101[2] Clause 7.3 for reference sensitivity are fulfilled.
- RSRP|dBm according to Annex B.3.3 for a corresponding Band.

**Table 9.2.18.1.3-1: RSRQ Inter frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 4] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 3] | dBm/$BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.13.1.3-2.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.6.1, clause 9.1.7 and A.9.2.18.

## 9.2.18.1.4 Test description

### 9.2.18.1.4.1 Initial conditions

Same initial conditions as defined in clause 9.2.3.1.4.1 with the exception that the Channel Bandwidth to be tested is 5MHz.

### 9.2.18.1.4.2 Test procedure

Same test procedure as defined in clause 9.2.3.1.4.2 with the following exceptions:

- Instead of Table 9.2.3.1.5-2 → use Table 9.2.18.1.5-1.
- Instead of Table 9.2.3.1.5-3 → use Table 9.2.18.1.5-2.

### 9.2.18.1.4.3 Message contents

Same message contents as defined in clause 9.2.3.1.4.3.

## 9.2.18.1.5 Test requirement

Table 9.2.18.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRQ FDD - FDD inter frequency absolute accuracy test shall meet the reported values test requirements in Table 9.2.18.1.5-2.

**Table 9.2.18.1.5-1: Cell Specific Test requirement Parameters for RSRQ FDD - FDD inter frequency absolute accuracy**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 | 1 | 2 | 1 | 2 |
| $BW_{channel}$ | | MHz | 5 | 5 | 5 | 5 | 5 | 5 |
| Gap Pattern Id | | | 0 | - | 0 | - | 0 | - |
| Measurement bandwidth | | $n_{PRB}$ | 10—15 | | 10—15 | | 10—15 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.5 FDD | - | R.5 FDD | - | R.5 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 7—17 | - | 7—17 | - | 7—17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.11 FDD | | R.11 FDD | | R.11 FDD | |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | | OP.15 FDD | OP.16 FDD | OP.15 FDD | OP.16 FDD | OP.15 FDD | OP.16 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | |
| PSS_RA | | | | | | | | |
| SSS_RA | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA[Note1] | | | | | | | | |
| OCNG_RB[Note1] | | | | | | | | |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -77.0 | -78.1 | -101.70 | -101.70 | -119.5 | N/A |
| | Bands FDD_C | | | | | | -118.5 | N/A |
| | Bands FDD_D | | | | | | -118.0 | N/A |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -117.5 | N/A |
| | Bands FDD_G | | | | | | -116.5 | N/A |
| | Bands FDD_H | | | | | | -116.0 | N/A |
| | Bands FDD_N | | | | | | - N/A | -112.7 |
| $\hat{E}_s / I_{ot}$ | | dB | -1.75 | -1.75 | -4.00 | -3.20 | -4.00 | -3.20 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -78.75 | -79.85 | -105.70 | -104.90 | -123.5 | N/A |
| | Bands FDD_C | | | | | | -122.5 | N/A |
| | Bands FDD_D | | | | | | -122.0 | N/A |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -121.5 | N/A |
| | Bands FDD_G | | | | | | -120.5 | N/A |
| | Bands FDD_H | | | | | | -120.0 | N/A |
| | Bands FDD_N | | | | | | N/A | -115.9 |
| RSRQ[Note3] | Bands FDD_A | dB | -14.76 | -14.76 | -16.25 | -15.69 | -16.25 | -15.69 |
| | Bands FDD_C | | | | | | | |
| | Bands FDD_D | | | | | | | |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | | |
| | Bands FDD_G | | | | | | | |
| | Bands FDD_H | | | | | | | |
| | Bands FDD_N | | | | | | | |
| Io[Note3] | Bands FDD_A | dBm/4.5 MHz | -50.01 | -51.11 | -75.47 | -75.23 | -93.27 | N/A |
| | Bands FDD_C | | | | | | -92.27 | N/A |
| | Bands FDD_D | | | | | | -91.77 | N/A |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -91.27 | N/A |
| | Bands FDD_G | | | | | | -90.27 | N/A |
| | Bands FDD_H | | | | | | -89.77 | N/A |
| | Bands FDD_N | | | | | | N/A | -86.23 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $\hat{E}_s / N_{oc}$ | dB | -1.75 | -1.75 | -4.0 | -3.2 | -4.0 | -3.2 |
| Propagation condition | - | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: For Band 26, the tests shall be performed with the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 6: This test is only applicable for testing inter-frequency requirements for Band 31. Cell 2 is on the Band under test, and Cell 1 is on another band supported by the UE.
Note 7: E-UTRA operating band groups are as defined in section 3.5.

**Table 9.2.18.1.5-2: RSRQ FDD - FDD inter frequency absolute accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_04 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_16 | RSRQ_16 | RSRQ_16 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_01 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_19 | RSRQ_17 | RSRQ_17 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.18.2 FDD - FDD Inter Frequency Relative Accuracy of RSRQ for 5MHz Bandwidth

### 9.2.18.2.1 Test purpose

To verify that the FDD - FDD inter frequency relative accuracy measurement of RSRQ is within the specified limit for band 31.

### 9.2.18.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward that support Band 31. Applicability requires support for FGI bits 16 and 25.

### 9.2.18.2.3 Minimum conformance requirements

The relative accuracy of RSRQ in inter frequency case is defined as the RSRQ measured from one cell compared to the RSRQ measured from another cell on a different frequency.

The accuracy requirements in Table 9.2.18.2.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101[2] Clause 7.3 for reference sensitivity are fulfilled.
- $RSRP1,2|_{dBm}$ according to Annex B.3.4 for a corresponding Band.

$$\left| RSRP1|_{dBm} - RSRP2|_{dBm} \right| \leq 27dB$$

| Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.2.18.2.3-1: RSRQ Inter frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | $dBm/BW_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.13.1.3-2.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.6.2, clause 9.1.7 and A.9.2.18.

## 9.2.18.2.4 Test description

### 9.2.18.2.4.1 Initial conditions

Same initial conditions as defined in clause 9.2.3.2.4.1 with the exception that the Channel Bandwidth to be tested is 5MHz.

### 9.2.18.2.4.2 Test procedure

Same test procedure as defined in clause 9.2.3.2.4.2 with the following exceptions:

- Instead of Table 9.2.3.2.5-2 → use Table 9.2.18.2.5-1.
- Instead of Table 9.2.3.2.5-3 → use Table 9.2.18.2.5-2.

### 9.2.18.2.4.3 Message contents

Same message contents as defined in clause 9.2.3.2.4.3.

## 9.2.18.2.5 Test requirement

Table 9.2.18.2.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRQ FDD - FDD inter frequency relative accuracy test shall meet the reported values test requirements in Table 9.2.18.2.5-2.

**Table 9.2.18.2.5-1: Cell Specific Test requirement Parameters for RSRQ FDD - FDD inter frequency relative accuracy for 5MHz bandwidth**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 | 1 | 2 | 1 | 2 |
| $BW_{channel}$ | | MHz | 5 | 5 | 5 | 5 | 5 | 5 |
| Gap Pattern Id | | | 0 | - | 0 | - | 0 | - |
| Measurement bandwidth | | $n_{PRB}$ | 10—15 | | 10—15 | | 10—15 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.5 FDD | - | R.5 FDD | - | R.5 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 7—17 | - | 7—17 | - | 7—17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.11 FDD | | R.11 FDD | | R.11 FDD | |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | | OP.15 FDD | OP.16 FDD | OP.15 FDD | OP.16 FDD | OP.15 FDD | OP.16 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | |
| PSS_RA | | | | | | | | |
| SSS_RA | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA[Note1] | | | | | | | | |
| OCNG_RB[Note1] | | | | | | | | |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -78.1 | -78.1 | -101.70 | -101.70 | -119.5 | N/A |
| | Bands FDD_C | | | | | | -118.5 | N/A |
| | Bands FDD_D | | | | | | -118.0 | N/A |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -117.5 | N/A |
| | Bands FDD_G | | | | | | -116.5 | N/A |
| | Bands FDD_H | | | | | | -116.0 | N/A |
| | Bands FDD_N | | | | | | - N/A | -112.7 |
| $\hat{E}_s / I_{ot}$ | | dB | -1.75 | -1.75 | -4.00 | -3.20 | -4.00 | -3.20 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -79.85 | -79.85 | -105.70 | -104.90 | -123.5 | N/A |
| | Bands FDD_C | | | | | | -122.5 | N/A |
| | Bands FDD_D | | | | | | -122.0 | N/A |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -121.5 | N/A |
| | Bands FDD_G | | | | | | -120.5 | N/A |
| | Bands FDD_H | | | | | | -120.0 | N/A |
| | Bands FDD_N | | | | | | N/A | -115.9 |
| RSRQ[Note3] | Bands FDD_A | dB | -14.76 | -14.76 | -16.25 | -15.69 | -16.25 | -15.69 |
| | Bands FDD_C | | | | | | | |
| | Bands FDD_D | | | | | | | |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | | |
| | Bands FDD_G | | | | | | | |
| | Bands FDD_H | | | | | | | |
| | Bands FDD_N | | | | | | | |
| Io[Note3] | Bands FDD_A | dBm/4.5 MHz | -51.11 | -51.11 | -75.47 | -75.23 | -93.27 | N/A |
| | Bands FDD_C | | | | | | -92.27 | N/A |
| | Bands FDD_D | | | | | | -91.77 | N/A |
| | Bands FDD_E, FDD_F [Note 5] | | | | | | -91.27 | N/A |
| | Bands FDD_G | | | | | | -90.27 | N/A |
| | Bands FDD_H | | | | | | -89.77 | N/A |
| | Bands FDD_N | | | | | | N/A | -86.23 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $\hat{E}_s / N_{oc}$ | dB | -1.75 | -1.75 | -4.0 | -3.2 | -4.0 | -3.2 |
| Propagation condition | - | AWGN | | AWGN | | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: For Band 26, the tests shall be performed with the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 6: This test is only applicable for testing inter-frequency requirements for Band 31. Cell 2 is on the Band under test, and Cell 1 is on another band supported by the UE.
Note 7: E-UTRA operating band groups are as defined in section 3.5.

**Table 9.2.18.2.5-2: RSRQ FDD inter frequency relative accuracy requirements for 5MHz bandwidth for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_x - 8 | RSRQ_x - 9 | RSRQ_x - 9 |
| Highest reported value (Cell 2) | RSRQ_x + 8 | RSRQ_x + 11 | RSRQ_x + 11 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_x - 10 | RSRQ_x - 9 | RSRQ_x - 9 |
| Highest reported value (Cell 2) | RSRQ_x + 10 | RSRQ_x + 11 | RSRQ_x + 11 |
| RSRQ_x is the reported value of Cell 1 | | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.19 FDD-FDD Inter Frequency WB-RSRQ

### 9.2.19.1 FDD-FDD Inter Frequency absolute WB-RSRQ accuracy

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

Exceptions for Message contents are FFS

The Test Tolerances and test requirements are undefined

#### 9.2.19.1.1 Test purpose

To verify that the RSRQ measurement accuracy is within the specified limits when the measurement configuration message received by the UE contains *widebandRSRQ-Meas* parameter in TS 36.331 [5]. In the test the UE shall also be configured with the *AllowedMeasBandwidth* parameter defined in TS 36.331 [5]. The test shall verify the WB-RSRQ inter frequency absolute accuracy requirements.

#### 9.2.19.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 11 and forward that support WB-RSRQ measurement. Applicability requires support for FGI bits 16 and 25.

#### 9.2.19.1.3 Minimum conformance requirements

The requirements in this section shall apply when the measurement configuration message received by the UE contains *widebandRSRQ-Meas* parameter in TS 36.331 [5]. The WB-RSRQ accuracy figures in Table 9.2.19.1.3-1 are relative to the value that would be obtained by using the *AllowedMeasBandwidth* in TS 36.331 [5].

The accuracy requirements in Table 9.2.19.1.3-1 are valid under the following conditions:

The value of the parameter, *AllowedMeasBandwidth* in TS 36.331 [5], is 50 resource blocks or larger

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex B.3.1 for a corresponding Band.

**Table 9.2.19.1.3-1: WB-RSRQ Inter frequency absolute accuracy**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 3 | Io1-Io2 Note 2 | Io range Note 1 | | |
| | | | | E-UTRA operating band groups Note 6 | Minimum Io Note 5 | Maximum Io |
| dB | dB | dB | dB | | dBm/15kHz | dBm/$BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | 0 ≤Io1-Io2 | FDD_A, TDD_A | -121 | -50 |
| | | | | FDD_C, TDD_C | -120 | -50 |
| | | | | FDD_D | -119.5 | -50 |
| | | | | FDD_E, TDD_E | -119 | -50 |
| | | | | FDD_F | -118.5 | -50 |
| | | | | FDD_G | -118 | -50 |
| | | | | FDD_H | -117.5 | -50 |
| | | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | | Note 4 | Note 4 | Note 4 |

NOTE 1: Io is the average across all the resource blocks within the *AllowedMeasBandwidth* in TS 36.331 [5].
NOTE 2: Io1 is the Io level in the resource blocks other than central 6 resource blocks within the *AllowedMeasBandwidth* in TS 36.331 [5] and Io2 is the Io level in central 6 resource blocks. The Io1 and Io2 have the same range as defined for Io.
NOTE 3: Iot is the received power spectrum density of total interference and noise for all the resource blocks, other than central 6 resource blocks.
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 5: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 6: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.19.1.3-2.

**Table 9.2.19.1.3-2: FDD WB-RSRQ absolute accuracy measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.6.3 and A.9.2.19.

## 9.2.19.1.4 Test description

### 9.2.19.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.19.1.4.3.

4. There are two E-UTRA FDD carriers and one cell on each carrier specified in each test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

#### 9.2.19.1.4.2 Test procedure

The test consists of Cell 1, serving cell and Cell 2, target cell. The absolute accuracy of WB-RSRQ is defined as the RSRQ measured from a cell that has different carrier frequency from the serving cell, Cell 1.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.2.19.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the RSRQ value in periodical MeasurementReport messages. The RSRQ value of Cell 2 reported by the UE is compared to the actual RSRQ value according to Table 9.2.19.1.5-1. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to table G.2.3-1 in Annex G.2 is achieved.

#### 9.2.19.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

FFS

### 9.2.19.1.5 Test requirement

Table 9.2.19.1.5-1 defines the primary level settings including test tolerances for all tests.

The FDD WB-RSRQ absolute accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.19.1.5-2.

**Table 9.2.19.1.5-1: WB-RSRQ FDD-FDD Inter frequency test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | |
| E-UTRA RF Channel Number | | | 1 | 2 | |
| $BW_{channel}$ | | MHz | 10 | 10 | |
| Antenna Configuration | | | 1x2 | 1x2 | |
| Gap Pattern Id | | | 0 | - | |
| PBCH_RA | | dB | 0 | 0 | |
| PBCH_RB | | | | 0 | |
| PSS_RA | | | | 0 | |
| SSS_RA | | | | 0 | |
| PCFICH_RB | | | | -∞ | |
| PHICH_RA | | | | -∞ | |
| PHICH_RB | | | | -∞ | |
| PDCCH_RA | | | | -∞ | |
| PDCCH_RB | | | | -∞ | |
| PDSCH_RA | | | | -∞ | |
| PDSCH_RB | | | | -∞ | |
| OCNG_RA[Note1] | | | | -∞ | |
| OCNG_RB[Note1] | | | | -∞ | |
| *AllowedMeasBandwidth* in TS 36.331 [2] | | RB | 6 | 50 | |
| PDSCH Reference measurement channel defined in A.3.1.1.1 | | | R.0 FDD | - | |
| PDSCH allocation | | $n_{PRB}$ | 13-36 | - | |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.1 | | | R.6 FDD | - | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1 FDD) and A.3.2.1.2 (OP.2 FDD) | | | OP.1 FDD | - | |
| $I_{ot}$ [Note2] | bandwidth | $n_{PRB}$ | 0-49 | 0-21<br>28-49 | 22-27 |
| | | dBm/15 kHz | -94 + TT | -87 | -110 + TT |
| $\hat{E}_s / I_{ot}$ | bandwidth | $n_{PRB}$ | 0-49 | 0-21<br>28-49 | 22-27 |
| | | dB | -4 + TT | -3 + TT | 20 + TT |
| RSRP[Note3] | | dBm/15 kHz | -98 | -90 | |
| RSRQ[Note3] | | dB | -16.25 | - | |
| $WB\text{-}RSRQ_0$[Note3] in subframe 0 | | dB | - | -13.68 | |
| $WB\text{-}RSRQ_1$[Note3] in subframe ≠ 0 | | dB | - | -13.63 | |
| Io[Note3] | | dBm/ 9 MHz | -64.76 | - | |
| Io[Note3] in symbol 0, 4, 11 of subframe 0 | | dBm/ 9 MHz | - | -82.38 | |
| Io[Note3] in symbol 7 of subframe 0 | | dBm/ 9 MHz | - | -82.20 | |
| Io[Note3] in symbol 0, 4, 7, 11 of subframes ≠ 0 | | dBm/ 9 MHz | - | -82.38 | |
| Propagation condition | | - | AWGN | AWGN | |

Note 1: OCNG shall be used such that Cell 1 is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells not specified in the test, assumed to be constant over time and modelled as noise.

Note 3: RSRQ, RSRP, WB-RSRQ and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. The stated values represent the weighted average over the allowed measurement bandwidth, and the WB-RSRQ values assume averaging over symbols 0, 4, 7 and 11 of the subframe.

Note 4: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: This test case is applicable to all FDD frequency bands except band 31.

**Table 9.2.19.1.5-2: FDD WB-RSRQ accuracy for the reported values**

| | Test 1 |
|---|---|
| | **All bands** |
| Normal Conditions | |
| Lowest reported value (Cell 1) | FFS |
| Highest reported value Cell 1) | FFS |
| Lowest reported value (Cell 2) | FFS |
| Highest reported value (Cell 2) | FFS |
| Extreme Conditions | |
| Lowest reported value (Cell 1) | FFS |
| Highest reported value Cell 1) | FFS |
| Lowest reported value (Cell 2) | FFS |
| Highest reported value (Cell 2) | FFS |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.20 TDD-TDD Inter Frequency WB-RSRQ

### 9.2.20.1 TDD-TDD Inter Frequency absolute WB-RSRQ accuracy

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

Exceptions for Message contents are FFS

The Test Tolerances and test requirements are undefined

#### 9.2.20.1.1 Test purpose

To verify that the RSRQ measurement accuracy is within the specified limits when the measurement configuration message received by the UE contains *widebandRSRQ-Meas* parameter in TS 36.331 [5]. In the test the UE shall also be configured with the *AllowedMeasBandwidth* parameter defined in TS 36.331 [5]. The test shall verify the WB-RSRQ inter frequency absolute accuracy requirements.

#### 9.2.20.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 11 and forward that support wideband RSRQ measurement. Applicability requires support for FGI bits 16 and 25.

#### 9.2.20.1.3 Minimum conformance requirements

The requirements in this section shall apply when the measurement configuration message received by the UE contains *widebandRSRQ-Meas* parameter in TS 36.331 [5]. The WB-RSRQ accuracy figures in Table 9.2.20.1.3-1 are relative to the value that would be obtained by using the *AllowedMeasBandwidth* in TS 36.331 [5].

The accuracy requirements in Table 9.2.20.1.3-1 are valid under the following conditions:

The value of the parameter, *AllowedMeasBandwidth* in TS 36.331 [5], is 50 resource blocks or larger

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex B.3.1 for a corresponding Band.

**Table 9.2.20.1.3-1: WB-RSRQ Inter frequency absolute accuracy**

| Accuracy | | Conditions | | | | |
|---|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 3 | Io1-Io2 Note 2 | Io range Note 1 | | |
| | | | | E-UTRA operating band groups Note 6 | Minimum Io Note 5 | Maximum Io |
| dB | dB | dB | dB | | dBm/15kHz | dBm/$BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | 0 ≤Io1-Io2 | FDD_A, TDD_A | -121 | -50 |
| | | | | FDD_C, TDD_C | -120 | -50 |
| | | | | FDD_D | -119.5 | -50 |
| | | | | FDD_E, TDD_E | -119 | -50 |
| | | | | FDD_F | -118.5 | -50 |
| | | | | FDD_G | -118 | -50 |
| | | | | FDD_H | -117.5 | -50 |
| | | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | | Note 4 | Note 4 | Note 4 |

NOTE 1: Io is the average across all the resource blocks within the *AllowedMeasBandwidth* in TS 36.331 [5].
NOTE 2: Io1 is the Io level in the resource blocks other than central 6 resource blocks within the *AllowedMeasBandwidth* in TS 36.331 [5] and Io2 is the Io level in central 6 resource blocks. The Io1 and Io2 have the same range as defined for Io.
NOTE 3: Iot is the received power spectrum density of total interference and noise for all the resource blocks, other than central 6 resource blocks.
NOTE 4: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 5: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 6: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.20.1.3-2.

**Table 9.2.20.1.3-2: TDD WB-RSRQ absolute accuracy measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.6.3 and A.9.2.20.

## 9.2.20.1.4 Test description

### 9.2.20.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.20.1.4.3.

4. There are two E-UTRA FDD carriers and one cell on each carrier specified in each test. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

#### 9.2.20.1.4.2 Test procedure

The test consists of Cell 1, serving cell and Cell 2, target cell. The absolute accuracy of WB-RSRQ is defined as the RSRQ measured from a cell that has different carrier frequency from the serving cell, Cell 1.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.2.20.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the RSRQ value in periodical MeasurementReport messages. The RSRQ value of Cell 2 reported by the UE is compared to the actual RSRQ value according to Table 9.2.20.1.5-1. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to table G.2.3-1 in Annex G.2 is achieved.

#### 9.2.20.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

FFS

### 9.2.20.1.5 Test requirement

Table 9.2.10.1.5-1 defines the primary level settings including test tolerances for all tests.

The TDD WB-RSRQ absolute accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.20.1.5-2.

**Table 9.2.20.1.5-1: WB-RSRQ TDD-TDD Inter frequency test parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | |
| E-UTRA RF Channel Number | | | 1 | 2 | |
| $BW_{channel}$ | | MHz | 10 | 10 | |
| Special subframe configuration Note1 | | | 6 | 6 | |
| Uplink-downlink configuration Note1 | | | 1 | 1 | |
| Antenna Configuration | | | 1x2 | 1x2 | |
| Gap Pattern Id | | | 0 | - | |
| PBCH_RA | | dB | 0 | 0 | |
| PBCH_RB | | | | 0 | |
| PSS_RA | | | | 0 | |
| SSS_RA | | | | 0 | |
| PCFICH_RB | | | | -∞ | |
| PHICH_RA | | | | -∞ | |
| PHICH_RB | | | | -∞ | |
| PDCCH_RA | | | | -∞ | |
| PDCCH_RB | | | | -∞ | |
| PDSCH_RA | | | | -∞ | |
| PDSCH_RB | | | | -∞ | |
| OCNG_RA Note2 | | | | -∞ | |
| OCNG_RB Note2 | | | | -∞ | |
| *AllowedMeasBandwidth* in TS 36.331 [2] | | RB | 6 | 50 | |
| PDSCH Reference measurement channel defined in A.3.1.1.2 | | | R.0 TDD | - | |
| PDSCH allocation | | $n_{PRB}$ | 13-36 | - | |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.2 | | | R.6 TDD | - | |
| OCNG Patterns defined in A.3.2.2.1 (OP.1 TDD) | | | OP.1 TDD | - | |
| $I_{ot}$ Note3 | bandwidth | $n_{PRB}$ | 0-49 | 0-21<br>28-49 | 22-27 |
| | | dBm/15 kHz | -94 + TT | -87 + TT | -110 + TT |
| $\hat{E}_s / I_{ot}$ | bandwidth | $n_{PRB}$ | 0—49 | 0-21<br>28-49 | 22-27 |
| | | dB | -4 + TT | -3 + TT | 20 + TT |
| RSRP Note4 | | dBm/15 kHz | -98 | -90 | |
| RSRQ Note4 | | dB | -16.25 | - | |
| WB-RSRQ$_0$ Note4 in subframe 0 | | dB | - | -13.68 | |
| WB-RSRQ$_1$ Note4 in subframe ≠ 0 | | dB | - | -13.63 | |
| Io Note4 | | dBm/ 9 MHz | -64.76 | - | |
| Io Note4 in symbol 0, 4, 11 of subframe 0 | | dBm/ 9 MHz | - | -82.38 | |
| Io Note4 in symbol 7 of subframe 0 | | dBm/ 9 MHz | - | -82.20 | |
| Io Note4 in symbol 0, 4, 7, 11 of subframes ≠ 0 | | dBm/ 9 MHz | - | -82.38 | |
| Propagation condition | | - | AWGN | AWGN | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211 [9].
Note 2: OCNG shall be used such that Cell 1 is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells not specified in the test, assumed to be constant over time and modelled as noise.
Note 4: RSRQ, RSRP, WB-RSRQ and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. The stated values represent the weighted average over the allowed measurement bandwidth, and the WB-RSRQ values assume averaging over symbols 0, 4, 7

| | |
|---|---|
| | and 11 of the subframe. |
| Note 5: | RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |

**Table 9.2.20.1.5--2: TDD WB-RSRQ accuracy for the reported values**

| | **Test 1** |
|---|---|
| | All Bands |
| Normal Conditions | |
| Lowest reported value (Cell 1) | FFS |
| Highest reported value Cell 1) | FFS |
| Lowest reported value (Cell 2) | FFS |
| Highest reported value (Cell 2) | FFS |
| Extreme Conditions | |
| Lowest reported value (Cell 1) | FFS |
| Highest reported value Cell 1) | FFS |
| Lowest reported value (Cell 2) | FFS |
| Highest reported value (Cell 2) | FFS |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.21 FDD RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz

### 9.2.21.1 FDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz

#### 9.2.21.1.1 Test purpose

Same test purpose as in clause 9.2.5.1.

#### 9.2.21.1.2 Test applicability

This test case applies to all types of E-UTRA UE release 11 and forward that support CA.

#### 9.2.21.1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.2.5.1.3.

#### 9.2.21.1.4 Test description

##### 9.2.21.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz for Cell 1 on the PCC, 5MHz for Cell 2 on the SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.21.1.4.3.

4. There are two E-UTRA FDD carriers with one cell on one E-UTRA FDD carrier and two cells on the other E-UTRA FDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.2.21.1.4.2 Test procedure

Same test procedure as in clause 9.2.5.1.4.2 with the following exceptions:

- Instead of Table 9.2.5.1.5-1 → use Table 9.2.21.1.5-1.
- Instead of Table 9.2.5.1.5-2 → use Table 9.2.21.1.5-2.

#### 9.2.21.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.21.1.4.3-1: Common Exception messages for FDD RSRQ absolute accuracy for Carrier Aggregation test requirement for 10MHz+5MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-4<br>Table H.4.1-5 |

### 9.2.21.1.5 Test requirement

Table 9.2.21.1.5-1 defines the primary level settings including test tolerances for all tests.

The FDD RSRQ absolute accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.21.1.5-2.

**Table 9.2.21.1.5-1: Cell Specific Test requirement Parameters for FDD RSRQ absolute accuracy for Carrier Aggregation for 10MHz+5MHz**

| Parameters | | Test 1 | | | |
|---|---|---|---|---|---|
| | | Units | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel_CA}$ Note 1 | | MHz | 10 | 5 | |
| Measurement bandwidth | | $n_{PRB}$ | 22-27 | 10-15 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | R.5 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13-36 | 7-17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | R.11 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD), D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | | OP.1 FDD | OP.15 FDD | OP.16 FDD |
| $Io^{Note2}$ | Bands FDD_A | dBm/9MHz | -90.26 | N/A | |
| | Bands FDD_C | | -89.26 | | |
| | Bands FDD_D | | -88.76 | | |
| | Bands FDD_E, FDD_F | | -88.26 | | |
| | Bands FDD_G | | -87.26 | | |
| | Bands FDD_H | | -86.76 | | |
| | Bands FDD_A | dBm/4.5MHz | N/A | -88.62 | |
| | Bands FDD_C | | | -87.62 | |
| | Bands FDD_D | | | -87.12 | |
| | Bands FDD_E, FDD_F | | | -86.62 | |
| | Bands FDD_G | | | -85.62 | |
| | Bands FDD_H | | | -85.12 | |
| Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.<br>Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 3: See Table 9.2.5.1.5-1 for the other parameters. | | | | | |

**Table 9.2.21.1.5-2: FDD RSRQ absolute accuracy requirements for the reported values for Carrier Aggregation for 10MHz+5MHz**

| | Test 1 |
|---|---|
| | All bands |
| **Normal Conditions** | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_15 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_14 |
| **Extreme Conditions** | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_16 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_15 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.21.2 FDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz

### 9.2.21.2.1 Test purpose

Same test purpose as in clause 9.2.5.2.1.

### 9.2.21.2.2 Test applicability

This test case applies to all types of E-UTRA UE release 11 and forward that support CA.

### 9.2.21.2.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.2.5.2.3.

### 9.2.21.2.4 Test description

#### 9.2.21.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz for Cell 1 on the PCC, 5MHz for Cell 2 on the SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.21.2.4.3.

4. There are two E-UTRA FDD carriers with one cell on one E-UTRA FDD carrier and two cells on the other E-UTRA FDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.2.21.2.4.2 Test procedure

Same test procedure as in clause 9.2.5.2.4.2 with the following exceptions:

- Instead of Table 9.2.5.2.5-1 → use Table 9.2.21.2.5-1.
- Instead of Table 9.2.5.2.5-2 → use Table 9.2.21.2.5-2.

#### 9.2.21.2.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.21.2.4.3-1: Common Exception messages for FDD RSRQ relative accuracy for Carrier Aggregation test requirement for 10MHz+5MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-4<br>Table H.4.1-5 |

### 9.2.21.2.5 Test requirement

Table 9.2.21.2.5-1 defines the primary level settings including test tolerances for all tests.

The FDD RSRQ relative accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.21.2.5-2.

**Table 9.2.21.2.5-1: Cell Specific Test requirement Parameters for FDD RSRQ relative accuracy for Carrier Aggregation for 10MHz+5MHz**

| Parameters | | Test 1 | | | |
|---|---|---|---|---|---|
| | | Units | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel_CA}$ Note 1 | | MHz | 10 | 5 | |
| Measurement bandwidth | | $n_{PRB}$ | 22-27 | 10-15 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | R.5 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13-36 | 7-17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | R.11 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD), D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | | OP.1 FDD | OP.15 FDD | OP.16 FDD |
| $Io^{Note2}$ | Bands FDD_A | dBm/9MHz | -90.26 | N/A | |
| | Bands FDD_C | | -89.26 | | |
| | Bands FDD_D | | -88.76 | | |
| | Bands FDD_E, FDD_F | | -88.26 | | |
| | Bands FDD_G | | -87.26 | | |
| | Bands FDD_H | | -86.76 | | |
| | Bands FDD_A | dBm/4.5MHz | N/A | -88.62 | |
| | Bands FDD_C | | | -87.62 | |
| | Bands FDD_D | | | -87.12 | |
| | Bands FDD_E, FDD_F | | | -86.62 | |
| | Bands FDD_G | | | -85.62 | |
| | Bands FDD_H | | | -85.12 | |
| Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.<br>Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 3: See Table 9.2.5.2.5-1 for the other parameters | | | | | |

**Table 9.2.21.2.5-2: FDD RSRQ relative accuracy requirements for the reported values for Carrier Aggregation for 10MHz+5MHz**

| | Test 1 |
|---|---|
| | All bands |
| **Normal Conditions** | |
| Lowest reported value (Cell 2) | RSRQ_x - 12 |
| Highest reported value (Cell 2) | RSRQ_x +9 |
| **Extreme Conditions** | |
| Lowest reported value (Cell 2) | RSRQ_x - 12 |
| Highest reported value (Cell 2) | RSRQ_x + 9 |
| RSRQ_x is the reported value of Cell 1 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.22 TDD RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz

### 9.2.22.1 TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz

#### 9.2.22.1.1 Test purpose

Same test purpose as in clause 9.2.6.1.1.

#### 9.2.22.1.2 Test applicability

This test case applies to all types of E-UTRA UE release 11 and forward that support CA.

#### 9.2.22.1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.2.6.2.13.

#### 9.2.22.1.4 Test description

##### 9.2.22.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 5 MHz for Cells on both the PCC and the SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.22.1.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

##### 9.2.22.1.4.2 Test procedure

Same test procedure as in clause 9.2.6.1.4.2 with the following exceptions:

- Instead of Table 9.2.6.1.5-1 → use Table 9.2.22.1.5-1.
- Instead of Table 9.2.6.1.5-2 → use Table 9.2.22.1.5-2.

##### 9.2.22.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.22.1.4.3-1: Common Exception messages for TDD RSRQ absolute accuracy for Carrier Aggregation test requirement for 10MHz+5MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-4<br>Table H.4.1-5 |

## 9.2.22.1.5 Test requirement

Table 9.2.22.1.5-1 defines the primary level settings including test tolerances for all tests.

The TDD RSRQ absolute accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.22.1.5-2.

**Table 9.2.22.1.5-1: Cell Specific Test requirement Parameters for TDD RSRQ absolute accuracy for Carrier Aggregation for 10MHz+5MHz**

| Parameters | | Test 1 | | | |
|---|---|---|---|---|---|
| | | Units | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel_CA}$ Note1 | | MHz | 10 | 5 | |
| Measurement bandwidth | | $n_{PRB}$ | 22-27 | 10-15 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | R.4TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13-36 | 7-17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | R.11 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD), D.2.9 (OP.9 TDD) and D.2.10 (OP.10 TDD) | | | OP.1 TDD | OP.9 TDD | OP.10 TDD |
| Io Note2 | Bands TDD_A | dBm/9MHz | -90.26 | N/A | |
| | Bands TDD_C | | -89.26 | | |
| | Bands TDD_E | | -88.26 | | |
| | Bands TDD_A | dBm/4.5MHz | N/A | -88.62 | |
| | Bands TDD_C | | | -87.62 | |
| | Bands TDD_E | | | -86.62 | |
| Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.<br>Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves<br>Note 3: See Table 9.2.6.1.5-1 for the other parameters | | | | | |

**Table 9.2.22.1.5-2: TDD RSRQ absolute accuracy requirements for the reported values for Carrier Aggregation for 10MHz+5MHz**

| | Test 1 |
|---|---|
| | All bands |
| **Normal Conditions** | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_15 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value Cell 2) | RSRQ_14 |
| **Extreme Conditions** | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_16 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value Cell 2) | RSRQ_15 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.22.2 TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz

### 9.2.22.2.1 Test purpose

Same test purpose as in clause 9.2.6.2.1.

### 9.2.22.2.2 Test applicability

This test case applies to all types of E-UTRA UE release 11 and forward that support CA.

### 9.2.22.2.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.2.6.2.3.

### 9.2.22.2.4 Test description

#### 9.2.22.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz for Cell 1 on the PCC, 5MHz for Cell 2 on the SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.22.2.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.2.22.2.4.2 Test procedure

Same test procedure as in clause 9.2.6.2.4.2 with the following exceptions:

- Instead of Table 9.2.6.2.5-1 → use Table 9.2.22.2.5-1.
- Instead of Table 9.2.6.2.5-2 → use Table 9.2.22.2.5-2.

#### 9.2.22.2.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.22.2.4.3-1: Common Exception messages for TDD RSRQ relative accuracy for Carrier Aggregation test requirement for 10MHz+5MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-4<br>Table H.4.1-5 |

### 9.2.22.2.5 Test requirement

Table 9.2.22.2.5-1 defines the primary level settings including test tolerances for all tests.

The TDD RSRQ relative accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.5.2.5-2.

**Table 9.2.22.2.5-1: Cell Specific Test requirement Parameters for TDD RSRQ relative accuracy for Carrier Aggregation for 10MHz+5MHz**

| Parameters | | Test 1 | | | |
|---|---|---|---|---|---|
| | | Units | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel_CA}$ Note1 | | MHz | 10 | 5 | |
| Measurement bandwidth | | $n_{PRB}$ | 22-27 | 10-15 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | R.4TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13-36 | 7-17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | R.11 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD), D.2.9 (OP.9 TDD) and D.2.10 (OP.10 TDD) | | | OP.1 TDD | OP.9 TDD | OP.10 TDD |
| Io Note2 | Bands TDD_A | dBm/9MHz | -90.26 | N/A | |
| | Bands TDD_C | | -89.26 | | |
| | Bands TDD_E | | -88.26 | | |
| | Bands TDD_A | dBm/4.5MHz | N/A | -88.62 | |
| | Bands TDD_C | | | -87.62 | |
| | Bands TDD_E | | | -86.62 | |
| Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.<br>Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 3: See Table 9.2.6.2.5-1 for the other parameters. | | | | | |

**Table 9.2.22.2.5-2: TDD RSRQ relative accuracy requirements for the reported values for Carrier Aggregation for 10MHz+5MHz**

| | Test 1 |
|---|---|
| | All bands |
| **Normal Conditions** | |
| Lowest reported value (Cell 2) | RSRQ_x - 12 |
| Highest reported value (Cell 2) | RSRQ_x + 9 |
| **Extreme Conditions** | |
| Lowest reported value (Cell 2) | RSRQ_x - 12 |
| Highest reported value (Cell 2) | RSRQ_x + 9 |
| RSRQ_x is the reported value of Cell 1 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.2.23 FDD RSRQ Accuracy for E-UTRA Carrier Aggregation for 5MHz+5MHz

## 9.2.23.1 FDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 5MHz+5MHz

### 9.2.23.1.1 Test purpose

Same test purpose as in clause 9.2.5.1.

### 9.2.23.1.2 Test applicability

This test case applies to all types of E-UTRA UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA.

This test case applies to all types of E-UTRA UE release 11 and forward that support CA with Intra-band non-contiguous CA.

### 9.2.23.1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.2.5.1.3.

### 9.2.23.1.4 Test description

#### 9.2.23.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 5 MHz for cells on both the PCC and the SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.23.1.4.3.

4. There are two E-UTRA FDD carriers with one cell on one E-UTRA FDD carrier and two cells on the other E-UTRA FDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.2.23.1.4.2 Test procedure

Same test procedure as in clause 9.2.5.1.4.2 with the following exceptions:

- Instead of Table 9.2.5.1.5-1 → use Table 9.2.23.1.5-1.

- Instead of Table 9.2.5.1.5-2 → use Table 9.2.23.1.5-2.

#### 9.2.23.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.23.1.4.3-1: Common Exception messages for FDD RSRQ absolute accuracy for Carrier Aggregation test requirement for 5MHz+5MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-4<br>Table H.4.1-5 |

## 9.2.23.1.5 Test requirement

Table 9.2.23.1.5-1 defines the primary level settings including test tolerances for all tests.

The FDD RSRQ absolute accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.23.1.5-2.

**Table 9.2.23.1.5-1: Cell Specific Test requirement Parameters for FDD RSRQ absolute accuracy for Carrier Aggregation for 5MHz+5MHz**

| Parameters | | Test 1 | | | |
|---|---|---|---|---|---|
| | | Units | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel_CA}$ Note 1 | | MHz | 5 | 5 | 5 |
| Measurement bandwidth | | $n_{PRB}$ | 10-15 | 10-15 | 10-15 |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.5 FDD | R.5 FDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 7-17 | 7-17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.11 FDD | R.11 FDD | R.11 FDD |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | | OP.15 FDD | OP.15 FDD | OP.16FDD |
| $Io^{Note2}$ | Bands FDD_A Note 5 | dBm/4.5MHz | -93.26 | -88.62 | |
| | Bands FDD_C Note 5 | | -92.26 | -87.62 | |
| | Bands FDD_D | | -91.76 | -87.12 | |
| | Bands FDD_E, FDD_F Note 5 | | -91.26 | -86.62 | |
| | Bands FDD_G Note 5 | | -90.26 | -85.62 | |
| | Bands FDD_H Note 5 | | -89.76 | -85.12 | |

Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: See Table 9.2.5.1.5-1 for the other parameters.
Note 4: E-UTRA operating band groups are as defined in Section 3.5.1.
Note 5: The test applies for E-UTRA operating bands in this band group which are supporting 5MHz + 5MHz channel bandwidth.

**Table 9.2.23.1.5-2: FDD RSRQ absolute accuracy requirements for the reported values for Carrier Aggregation for 5MHz+5MHz**

| | Test 1 |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_15 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_14 |
| Extreme Conditions | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_16 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_15 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.23.2 FDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 5MHz+5MHz

### 9.2.23.2.1 Test purpose

Same test purpose as in clause 9.2.5.2.1.

### 9.2.23.2.2 Test applicability

This test case applies to all types of E-UTRA UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA.

This test case applies to all types of E-UTRA UE release 11 and forward that support CA with Intra-band non-contiguous CA.

### 9.2.23.2.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.2.5.2.3.

### 9.2.23.2.4 Test description

#### 9.2.23.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 5 MHz for cells on both the PCC and the SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.23.2.4.3.

4. There are two E-UTRA FDD carriers with one cell on one E-UTRA FDD carrier and two cells on the other E-UTRA FDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.2.23.2.4.2 Test procedure

Same test procedure as in clause 9.2.5.2.4.2 with the following exceptions:

- Instead of Table 9.2.5.2.5-1 → use Table 9.2.23.2.5-1.
- Instead of Table 9.2.5.2.5-2 → use Table 9.2.23.2.5-2.

#### 9.2.23.2.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.23.2.4.3-1: Common Exception messages for FDD RSRQ relative accuracy for Carrier Aggregation test requirement for 5MHz+5MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-4<br>Table H.4.1-5 |

### 9.2.23.2.5 Test requirement

Table 9.2.23.2.5-1 defines the primary level settings including test tolerances for all tests.

The FDD RSRQ relative accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.23.2.5-2.

**Table 9.2.23.2.5-1: Cell Specific Test requirement Parameters for FDD RSRQ relative accuracy for Carrier Aggregation for 5MHz+5MHz**

| Parameters | | Test 1 | | | |
|---|---|---|---|---|---|
| | | Units | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel_CA}$ Note 1 | | MHz | 5 | 5 | 5 |
| Measurement bandwidth | | $n_{PRB}$ | 10-15 | 10-15 | 10-15 |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.5 FDD | R.5 FDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 7-17 | 7-17 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.11 FDD | R.11 FDD | R.11 FDD |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) and D.1.16 (OP.16 FDD) | | | OP.15 FDD | OP.15 FDD | OP.16FDD |
| Io Note2 | Bands FDD_A Note 5 | dBm/4.5MHz | -93.26 | -88.62 | |
| | Bands FDD_C Note 5 | | -92.26 | -87.62 | |
| | Bands FDD_D | | -91.76 | -87.12 | |
| | Bands FDD_E, FDD_F Note 5 | | -91.26 | -86.62 | |
| | Bands FDD_G Note 5 | | -90.26 | -85.62 | |
| | Bands FDD_H Note 5 | | -89.76 | -85.12 | |

Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: See Table 9.2.5.1.5-1 for the other parameters.
Note 4: E-UTRA operating band groups are as defined in Section 3.5.1.
Note 5: The test applies for E-UTRA operating bands in this band group which are supporting 5MHz + 5MHz channel bandwidth.

**Table 9.2.23.2.5-2: FDD RSRQ relative accuracy requirements for the reported values for Carrier Aggregation for 5MHz+5MHz**

| | Test 1 |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 2) | RSRQ_x - 12 |
| Highest reported value (Cell 2) | RSRQ_x + 9 |
| Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRQ_x - 12 |
| Highest reported value (Cell 2) | RSRQ_x + 9 |
| RSRQ_x is the reported value of Cell 1 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.2.24 TDD RSRQ Accuracy for E-UTRA Carrier Aggregation for 5MHz+5MHz

## 9.2.24.1 TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 5MHz+5MHz

### 9.2.24.1.1 Test purpose

Same test purpose as in clause 9.2.6.1.1.

### 9.2.24.1.2 Test applicability

This test case applies to all types of E-UTRA UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA.

This test case applies to all types of E-UTRA UE release 11 and forward that support CA with Intra-band non-contiguous CA.

### 9.2.24.1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.2.6.2.13.

### 9.2.24.1.4 Test description

#### 9.2.24.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 5 MHz for Cells on both the PCC and SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.24.1.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.2.24.1.4.2 Test procedure

Same test procedure as in clause 9.2.6.1.4.2 with the following exceptions:

- Instead of Table 9.2.6.1.5-1 → use Table 9.2.24.1.5-1.

- Instead of Table 9.2.6.1.5-2 → use Table 9.2.24.1.5-2.

#### 9.2.24.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.24.1.4.3-1: Common Exception messages for TDD RSRQ absolute accuracy for Carrier Aggregation test requirement for 5MHz+5MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-4<br>Table H.4.1-5 |

## 9.2.24.1.5 Test requirement

Table 9.2.24.1.5-1 defines the primary level settings including test tolerances for all tests.

The TDD RSRQ absolute accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.24.1.5-2.

**Table 9.2.24.1.5-1: Cell Specific Test requirement Parameters for TDD RSRQ absolute accuracy for Carrier Aggregation for 5MHz+5MHz**

| Parameters | | Test 1 | | | |
|---|---|---|---|---|---|
| | | Units | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel_CA}$ Note1 | | MHz | 10 | 5 | 5 |
| Measurement bandwidth | | $n_{PRB}$ | 10-15 | 10-15 | 10-15 |
| PDSCH Reference measurement channel defined in A.2.1 | | | R.4 TDD | R.4 TDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 7-17 | 7-17 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.11 TDD | R.11 TDD | R.11 TDD |
| OCNG Patterns defined in D.2.9 (OP.9 TDD) and D.2.10 (OP.10 TDD) | | | OP.9 TDD | OP.9 TDD | OP.10 TDD |
| Io Note2 | Bands TDD_A Note 5 | dBm4.5MHz | -93.26 | -88.62 | |
| | Bands TDD_C Note 5 | | -92.26 | -87.62 | |
| | Bands TDD_E Note 5 | | -91.26 | -86.62 | |

Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves
Note 3: See Table 9.2.6.1.5-1 for the other parameters
Note 4: E-UTRA operating band groups are as defined in Section 3.5.1.
Note 5: The test applies for E-UTRA operating bands in this band group which are supporting 5MHz + 5MHz channel bandwidth.

**Table 9.2.24.1.5-2: TDD RSRQ absolute accuracy requirements for the reported values for Carrier Aggregation for 5MHz+5MHz**

| | **Test 1** |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_15 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value Cell 2) | RSRQ_14 |
| Extreme Conditions | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_16 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value Cell 2) | RSRQ_15 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.24.2 TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 5MHz+5MHz

### 9.2.24.2.1 Test purpose

Same test purpose as in clause 9.2.6.2.1.

### 9.2.24.2.2 Test applicability

This test case applies to all types of E-UTRA UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA.

This test case applies to all types of E-UTRA UE release 11 and forward that support CA with Intra-band non-contiguous CA.

### 9.2.24.2.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.2.6.2.3.

### 9.2.24.2.4 Test description

#### 9.2.24.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 5 MHz for Cells on both the PCC and SCC as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.24.2.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.2.24.2.4.2 Test procedure

Same test procedure as in clause 9.2.6.2.4.2 with the following exceptions:

- Instead of Table 9.2.6.2.5-1 → use Table 9.2.24.2.5-1.
- Instead of Table 9.2.6.2.5-2 → use Table 9.2.24.2.5-2.

#### 9.2.24.2.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.24.2.4.3-1: Common Exception messages for TDD RSRQ relative accuracy for Carrier Aggregation test requirement for 5MHz+5MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-4<br>Table H.4.1-5 |

### 9.2.24.2.5 Test requirement

Table 9.2.24.2.5-1 defines the primary level settings including test tolerances for all tests.

The TDD RSRQ relative accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.5.2.5-2.

**Table 9.2.24.2.5-1: Cell Specific Test requirement Parameters for TDD RSRQ relative accuracy for Carrier Aggregation for 5MHz+5MHz**

| Parameters | | Test 1 | | | |
|---|---|---|---|---|---|
| | | Units | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel_CA}$ Note1 | | MHz | 10 | 5 | 5 |
| Measurement bandwidth | | $n_{PRB}$ | 10-15 | 10-15 | 10-15 |
| PDSCH Reference measurement channel defined in A.2.1 | | | R.4 TDD | R.4 TDD | N/A |
| PDSCH allocation | | $n_{PRB}$ | 7-17 | 7-17 | N/A |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.11 TDD | R.11 TDD | R.11 TDD |
| OCNG Patterns defined in D.2.9 (OP.9 TDD) and D.2.10 (OP.10 TDD) | | | OP.9 TDD | OP.9 TDD | OP.10 TDD |
| Io Note2 | Bands TDD_A Note 5 | dBm4.5MHz | -93.26 | -88.62 | |
| | Bands TDD_C Note 5 | | -92.26 | -87.62 | |
| | Bands TDD_E Note 5 | | -91.26 | -86.62 | |
| Note 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.<br>Note 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 3: See Table 9.2.6.1.5-1 for the other parameters.<br>Note 4: E-UTRA operating band groups are as defined in Section 3.5.1.<br>Note 5: The test applies for E-UTRA operating bands in this band group which are supporting 5MHz + 5MHz channel bandwidth. | | | | | |

**Table 9.2.24.2.5-2: TDD RSRQ relative accuracy requirements for the reported values for Carrier Aggregation for 5MHz+5MHz**

| | Test 1 |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 2) | RSRQ_x - 12 |
| Highest reported value (Cell 2) | RSRQ_x + 9 |
| Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRQ_x - 12 |
| Highest reported value (Cell 2) | RSRQ_x + 9 |
| RSRQ_x is the reported value of Cell 1 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.25 RSRQ for E-UTRAN TDD-FDD Carrier Aggregation with PCell in FDD

*Editor's note:*

*- TS36.521-2 need to be updated to be consistent with the test applicability.*

## 9.2.25.1 Absolute RSRQ Accuracy for E-UTRAN TDD-FDD Carrier Aggregation with PCell in FDD

*Editor's note:*

- TS36.521-2 need to be updated to be consistent with the test applicability.

### 9.2.25.1.1 Test purpose

To verify that FDD and TDD absolute RSRQ measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the FDD absolute RSRQ accuracy requirements of the primary component carrier and the TDD absolute RSRQ accuracy requirements of the secondary component carrier.

### 9.2.25.1.2 Test applicability

This test case applies to all types of E-UTRA UE release 12 and forward that support E-UTRA FDD and TDD and 2DL TDD-FDD CA with FDD as PCell.

### 9.2.25.1.3 Minimum conformance requirements

The RSRQ measurement of an FDD cell on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1. The RSRQ measurement of a TDD cell on the secondary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1.

The accuracy requirements in Table 9.2.25.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.1 for a corresponding Band.

**Table 9.2.25.1.3-1: RSRQ absolute accuracy for Carrier Aggregation**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io [Note 1] range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups [Note 4] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 3] | dBm/$BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.25.1.3-2.

**Table 9.2.25.1.3-2: RSRQ absolute accuracy measurement report mapping for Carrier Aggregation**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1, clause 9.1.11.2, clause 9.1.5.1 and A.9.2.25.

## 9.2.25.1.4 Test description

### 9.2.25.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and [4.3.1] for different CA bandwidth classes.

Channel Bandwidth to be tested: Any one of the supported TDD-FDD CA configurations with PCell in FDD with the largest aggregated CA bandwidth combination supported by UE from Table 9.2.25.1.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.45 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.25.1.4.3.

4. There are two E-UTRA carriers with one cell on one E-UTRA FDD carrier and one cell on the other E-UTRA TDD carrier specified in the test. Cell 1 is PCell on the PCC and Cell 2 is the SCell on the Secondary Component Carrier (SCC). PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 shall be powered OFF.

### 9.2.25.1.4.2 Test procedure

The test consists of Cell 1 the PCell, and Cell 2 the SCell on the Secondary Component Carrier (SCC). The SCell (Cell 2) is configured. The SCell (Cell 2) on the SCC is activated. The absolute accuracy of RSRQ is defined as the RSRQ measured from the primary component carrier (Cell 1) and the RSRQ measured from the secondary component carrier (Cell 2).

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. SS activates SCC by sending the MAC-CE according to TS 36.321 [11] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.2.25.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. SS shall transmit an RRCConnectionReconfiguration message.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, SS shall check the reported RSRQ values in periodical MeasurementReport messages. The reported RSRQ values for Cell 1 and Cell 2 by the UE are compared to the actual RSRQ values according to Table 9.2.25.1.5-2. This counts respectively as a Pass or Fail for the events “Cell 1” and “Cell 2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations for the respective event ”Cell 2” is increased by one.

10. The SS shall check MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events “Cell 1” and “Cell 2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If both events pass, the test passes. If one event fails, the test fails.

#### 9.2.25.1.4.3 Message contents

Message contents are according to TS 36.508 [7] with the following exceptions:

**Table 9.2.25.1.4.3-1: Common Exception messages for RSRQ for E-UTRAN TDD-FDD Carrier Aggregation with PCell in FDD test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-4<br>Table H.4.1-5a |

**Table 9.2.25.1.4.3-2: *MeasConfig-DEFAULT*: E-UTRAN RSRP and RSRQ Accuracy for CA without NeighCell**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entries | | |
| MeasObjectToAddMod[1] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the PCell (Cell 1) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the PCell (Cell 1) | Switch PCell/SCell scenario |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[2] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the SCell on the SCC (Cell 2) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | | |
| } | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the SCell on the SCC (Cell 2) | Switch PCell/SCell scenario |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the SCell on the SCC (Cell 2) | |
| | IdMeasObject-f1 | f1 is the frequency of the SCell on the SCC (Cell 2) | Switch PCell/SCell scenario |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

#### 9.2.25.1.5 Test requirement

Table 9.2.25.1.5-1 defines the primary level settings including test tolerances for all tests.

The RSRQ absolute accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.25.1.5-2.

**Table 9.2.25.1.5-1: Cell Specific Test requirement Parameters for TDD-FDD Carrier Aggregation with PCell in FDD test parameters**

| Parameter | | Unit | Cell 1 | Cell 2 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | 2 |
| $BW_{channel}$ | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 |
| Special subframe configuration[Note1] | | | - | 6 |
| Uplink-downlink configuration[Note1] | | | - | 1 |
| Measurement bandwidth | | $n_{PRB}$ | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 |
| PDSCH Reference measurement channel defined in A.3.1.1 | | | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD |
| PDSCH allocation | | $n_{PRB}$ | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2 | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD |
| OCNG Patterns defined in A.3.2 | | | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD |
| PBCH_RA | | dB | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA[Note2] | | | | |
| OCNG_RB[Note2] | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | - | -116 |
| | Bands TDD_C | | - | -115 |
| | Bands TDD_E | | - | -114 |
| | Bands FDD_A | | -119.2 | - |
| | Bands FDD_B | | -118.7 | - |
| | Bands FDD_C | | -118.2 | - |
| | Bands FDD_D | | -117.7 | - |
| | Bands FDD_E, Bands FDD_F [Note 6] | | -117.2 | - |
| | Bands FDD_G | | -116.2 | - |
| | Bands FDD_H | | -115.7 | - |
| $\hat{E}_s / N_{oc}$ | | dB | -5.2 | -5.2 |
| $\hat{E}_s / I_{ot}$ | | dB | -5.2 | -5.2 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | - | -121.2 |
| | Bands TDD_C | | - | -120.2 |
| | Bands TDD_E | | - | -119.2 |
| | Bands FDD_A | | -124.4 | - |
| | Bands FDD_B | | -123.9 | - |
| | Bands FDD_C | | -123.4 | - |
| | Bands FDD_D | | -122.9 | - |

| | | | | |
|---|---|---|---|---|
| | Bands FDD_E, Bands FDD_F Note 6 | | -122.4 | - |
| | Bands FDD_G | | -121.4 | - |
| | Bands FDD_H | | -120.9 | - |
| RSRQ Note4 | Bands TDD_A | dB | - | -17.14 |
| | Bands TDD_C | | | |
| | Bands TDD_E | | | |
| | Bands FDD_A | | -17.14 | - |
| | Bands FDD_B | | | |
| | Bands FDD_C | | | |
| | Bands FDD_D | | | |
| | Bands FDD_E, Bands FDD_F Note 6 | | | |
| | Bands FDD_G | | | |
| | Bands FDD_H | | | |
| Io Note4 | Bands TDD_A | dBm/BW$_{channel}$ | - | -87.07 + 10log($N_{RB,c}$/50) |
| | Bands TDD_C | | - | -86.07 + 10log($N_{RB,c}$/50) |
| | Bands TDD_E | | - | -85.07 + 10log($N_{RB,c}$/50) |
| | Bands FDD_A | | -90.27 + 10log($N_{RB,c}$/50) | - |
| | Bands FDD_B | | -89.77 + 10log($N_{RB,c}$/50) | - |
| | Bands FDD_C | | -89.27 + 10log($N_{RB,c}$/50) | - |
| | Bands FDD_D | | -88.77 + 10log($N_{RB,c}$/50) | - |
| | Bands FDD_E, Bands FDD_F Note 6 | | -88.27 + 10log($N_{RB,c}$/50) | - |
| | Bands FDD_G | | -87.27 + 10log($N_{RB,c}$/50) | - |
| | Bands FDD_H | | -86.77 + 10log($N_{RB,c}$/50) | - |
| Propagation Condition | | | AWGN | AWGN |
| Antenna Configuration | | | 1x2 | 1x2 |
| Timing offset to Cell 1 | | µs | - | 0 |
| Time alignment error relative to cell 1 Note 10 | | | - | ≤ TAE |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.

Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: Es/Iot, RSRP, RSRQ and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 5: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 6: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 7: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.

Note 8: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.

Note 9: E-UTRA operating band groups are as defined in Section 3.5.

Note 10: Time alignment error (TAE) as specified in TS 36.104 [30] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

**Table 9.2.25.1.5-2: RSRQ absolute accuracy requirements for the reported values for TDD-FDD Carrier Aggregation with PCell in FDD**

| | Test 1 |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_14 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_14 |
| Extreme Conditions | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_15 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_15 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.25.2 Relative RSRQ Accuracy for E-UTRAN TDD-FDD Carrier Aggregation with PCell in FDD

*Editor's note:*

- TS36.521-2 need to be updated to be consistent with the test applicability.

### 9.2.25.2.1 Test purpose

To verify that TDD-FDD relative RSRQ measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the TDD-FDD relative RSRQ accuracy requirements between the primary and secondary component carriers.

### 9.2.25.2.2 Test applicability

This test case applies to all types of E-UTRA UE release 12 and forward that support E-UTRA FDD and TDD and 2DL TDD-FDD CA with FDD as PCell.

### 9.2.25.2.3 Minimum conformance requirements

The RSRQ measurement of a TDD cell on the secondary component carrier is compared with the RSRQ measurement of an FDD cell on the primary component carrier, and shall meet the applicable relative accuracy requirements which are the RSRQ inter-frequency accuracy requirements defined in TS 36.133 [4] clause 9.1.6.2.

The accuracy requirements in Table 9.2.25.2.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.4 for a corresponding Band.

$$\left| RSRP1 \right|_{dBm} - RSRP2 \left|_{dBm} \right| \le [27] dB$$

|Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.2.25.2.3-1: RSRQ Inter-frequency relative accuracy for Carrier Aggregation**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/$BW_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.25.2.3-2.

**Table 9.2.25.2.3-2: RSRQ relative accuracy measurement report mapping for Carrier Aggregation**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.3, clause 9.1.6.2 and A.9.2.25.

## 9.2.25.2.4 Test description

### 9.2.25.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and [4.3.1] for different CA bandwidth classes.

Channel Bandwidth to be tested: Any one of the supported TDD-FDD CA configurations with PCell in FDD with the largest aggregated CA bandwidth combination supported by UE from Table 9.2.25.2.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.45 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.25.2.4.3.

4. There are two E-UTRA carriers with one cell on one E-UTRA FDD carrier and one cell on the other E-UTRA TDD carrier specified in the test. Cell 1 is PCell on the PCC and Cell 2 is the SCell on the Secondary Component Carrier

(SCC). PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 shall be powered OFF.

#### 9.2.25.2.4.2 Test procedure

The test consists of Cell 1 the PCell, and Cell 2 the SCell on the Secondary Component Carrier (SCC). The SCell (Cell 2)is configured. The SCell (Cell 2) on the SCC is activated. The relative accuracy of RSRQ is defined as the RSRQ measured from the secondary component carrier (Cell 2) compared with the RSRQ measured from the primary component carrier (Cell 1).

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. SS activates SCC by sending the MAC-CE according to TS 36.321 [11] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.2.25.2.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. SS shall transmit an RRCConnectionReconfiguration message.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, SS shall check the reported RSRQ values in periodical MeasurementReport messages. The reported RSRQ value for Cell 2 is compared to the reported RSRQ value for Cell 1 for each MeasurementReport message according to Table 9.2.25.2.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

10. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to table G.2.3-1 in Annex G.2 is achieved.

#### 9.2.25.2.4.3 Message contents

Message contents are according to TS 36.508 [7] with the following exceptions:

**Table 9.2.25.2.4.3-1: Common Exception messages for RSRQ for E-UTRAN TDD-FDD Carrier Aggregation with PCell in FDD test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-4<br>Table H.4.1-5a |

**Table 9.2.25.2.4.3-2: *MeasConfig-DEFAULT*: E-UTRAN RSRP and RSRQ Accuracy for CA without NeighCell**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entries | | |
| MeasObjectToAddMod[1] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the PCell (Cell 1) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the PCell (Cell 1) | Switch PCell/SCell scenario |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[2] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the SCell on the SCC (Cell 2) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | | |
| } | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the SCell on the SCC (Cell 2) | Switch PCell/SCell scenario |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the SCell on the SCC (Cell 2) | |
| | IdMeasObject-f1 | f1 is the frequency of the SCell on the SCC (Cell 2) | Switch PCell/SCell scenario |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

#### 9.2.25.2.5 Test requirement

Table 9.2.25.2.5-1 defines the primary level settings including test tolerances for all tests.

The RSRQ relative accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.25.2.5-2.

**Table 9.2.25.2.5-1: Cell Specific Test requirement Parameters for TDD-FDD Carrier Aggregation with PCell in FDD test parameters**

| Parameter | | Unit | Cell 1 | Cell 2 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | 2 |
| $BW_{channel}$ | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 |
| Special subframe configuration[Note1] | | | - | 6 |
| Uplink-downlink configuration[Note1] | | | - | 1 |
| Measurement bandwidth | | $n_{PRB}$ | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 |
| PDSCH Reference measurement channel defined in A.3.1.1 | | | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD |
| PDSCH allocation | | $n_{PRB}$ | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2 | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD |
| OCNG Patterns defined in A.3.2 | | | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD |
| PBCH_RA | | dB | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA[Note2] | | | | |
| OCNG_RB[Note2] | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | - | -116 |
| | Bands TDD_C | | - | -115 |
| | Bands TDD_E | | - | -114 |
| | Bands FDD_A | | -119.2 | - |
| | Bands FDD_B | | -118.7 | - |
| | Bands FDD_C | | -118.2 | - |
| | Bands FDD_D | | -117.7 | - |
| | Bands FDD_E, Bands FDD_F [Note 6] | | -117.2 | - |
| | Bands FDD_G | | -116.2 | - |
| | Bands FDD_H | | -115.7 | - |
| $\hat{E}_s / N_{oc}$ | | dB | -5.2 | -5.2 |
| $\hat{E}_s / I_{ot}$ | | dB | -5.2 | -5.2 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | - | -121.2 |
| | Bands TDD_C | | - | -120.2 |
| | Bands TDD_E | | - | -119.2 |
| | Bands FDD_A | | -124.4 | - |
| | Bands FDD_B | | -123.9 | - |
| | Bands FDD_C | | -123.4 | - |
| | Bands FDD_D | | -122.9 | - |

| | | | | |
|---|---|---|---|---|
| | Bands FDD_E, Bands FDD_F Note 6 | | -122.4 | - |
| | Bands FDD_G | | -121.4 | - |
| | Bands FDD_H | | -120.9 | - |
| RSRQ Note4 | Bands TDD_A | dB | - | -17.14 |
| | Bands TDD_C | | | |
| | Bands TDD_E | | | |
| | Bands FDD_A | | -17.14 | - |
| | Bands FDD_B | | | |
| | Bands FDD_C | | | |
| | Bands FDD_D | | | |
| | Bands FDD_E, Bands FDD_F Note 6 | | | |
| | Bands FDD_G | | | |
| | Bands FDD_H | | | |
| Io Note4 | Bands TDD_A | dBm/BW$_{channel}$ | - | -87.07 + 10log($N_{RB,c}$/50) |
| | Bands TDD_C | | - | -86.07 + 10log($N_{RB,c}$/50) |
| | Bands TDD_E | | - | -85.07 + 10log($N_{RB,c}$/50) |
| | Bands FDD_A | | -90.27 + 10log($N_{RB,c}$/50) | - |
| | Bands FDD_B | | -89.77 + 10log($N_{RB,c}$/50) | - |
| | Bands FDD_C | | -89.27 + 10log($N_{RB,c}$/50) | - |
| | Bands FDD_D | | -88.77 + 10log($N_{RB,c}$/50) | - |
| | Bands FDD_E, Bands FDD_F Note 6 | | -88.27 + 10log($N_{RB,c}$/50) | - |
| | Bands FDD_G | | -87.27 + 10log($N_{RB,c}$/50) | - |
| | Bands FDD_H | | -86.77 + 10log($N_{RB,c}$/50) | - |
| Propagation Condition | | | AWGN | AWGN |
| Antenna Configuration | | | 1x2 | 1x2 |
| Timing offset to Cell 1 | | µs | - | 0 |
| Time alignment error relative to cell 1 Note 10 | | | - | ≤ TAE |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.

Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: Es/Iot, RSRP, RSRQ and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 5: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 6: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 7: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.

Note 8: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.

Note 9: E-UTRA operating band groups are as defined in Section 3.5.

Note 10: Time alignment error (TAE) as specified in TS 36.104 [30] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

**Table 9.2.25.2.5-2: RSRQ relative accuracy requirements for the reported values for TDD-FDD Carrier Aggregation with PCell in FDD**

| | Test 1 |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 2) | RSRQ_x – 10 |
| Highest reported value (Cell 2) | RSRQ_x + 10 |
| Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRQ_x - 10 |
| Highest reported value (Cell 2) | RSRQ_x + 10 |
| RSRQ_x is the reported value of Cell 1 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.26 RSRQ for E-UTRAN TDD-FDD Carrier Aggregation with PCell in TDD

*Editor's note:*

- TS36.521-2 need to be updated to be consistent with the test applicability.

### 9.2.26.1 Absolute RSRQ Accuracy for E-UTRAN TDD-FDD Carrier Aggregation with PCell in TDD

*Editor's note:*

- TS36.521-2 need to be updated to be consistent with the test applicability.

#### 9.2.26.1.1 Test purpose

To verify that FDD and TDD absolute RSRQ measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the TDD absolute RSRQ accuracy requirements of the primary component carrier and the FDD absolute RSRQ accuracy requirements of the secondary component carrier.

#### 9.2.26.1.2 Test applicability

This test case applies to all types of E-UTRA UE release 12 and forward that support E-UTRA FDD and TDD and 2DL TDD-FDD CA with TDD as PCell.

#### 9.2.26.1.3 Minimum conformance requirements

The RSRQ measurement of a TDD cell on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1. The RSRQ measurement of an FDD cell on the secondary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1.

The accuracy requirements in Table 9.2.26.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.1 for a corresponding Band.

**Table 9.2.26.1.3-1: RSRQ absolute accuracy for Carrier Aggregation**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 4] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 3] | dBm/$BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.26.1.3-2.

**Table 9.2.26.1.3-2: RSRQ absolute accuracy measurement report mapping for Carrier Aggregation**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1, clause 9.1.11.2, clause 9.1.5.1 and A.9.2.26.

## 9.2.26.1.4 Test description

### 9.2.26.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and [4.3.1] for different CA bandwidth classes.

Channel Bandwidth to be tested: Any one of the supported TDD-FDD CA configurations with PCell in TDD with the largest aggregated CA bandwidth combination supported by UE from Table 9.2.26.1.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.45 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.26.1.4.3.

4. There are two E-UTRA carriers with one cell on one E-UTRA TDD carrier and one cell on the other E-UTRA FDD carrier specified in the test. Cell 1 is PCell on the PCCand Cell 2 is the SCell on the Secondary Component Carrier

(SCC). PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 shall be powered OFF.

#### 9.2.26.1.4.2 Test procedure

The test consists of Cell 1 the PCell, and Cell 2 the SCell on the Secondary Component Carrier (SCC). The SCell (Cell 2) is configured. The SCell (Cell 2) on the SCC is activated. The absolute accuracy of RSRQ is defined as the RSRQ measured from the primary component carrier (Cell 1) and the RSRQ measured from the secondary component carrier (Cell 2).

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. SS activates SCC by sending the MAC-CE according to TS 36.321 [11] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.2.26.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. SS shall transmit an RRCConnectionReconfiguration message.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, SS shall check the reported RSRQ values in periodical MeasurementReport messages. The reported RSRQ values for Cell 1 and Cell 2 by the UE are compared to the actual RSRQ values according to Table 9.2.26.1.5-2. This counts respectively as a Pass or Fail for the events “Cell 1” and “Cell 2”. If the UE fails to report the measurement value for Cell 2, the number of failed iterations for the respective event ”Cell 2” is increased by one.

10. The SS shall check MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events “Cell 1” and “Cell 2” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If both events pass, the test passes. If one event fails, the test fails.

#### 9.2.26.1.4.3 Message contents

Message contents are according to TS 36.508 [7] with the following exceptions:

**Table 9.2.26.1.4.3-1: Common Exception messages for RSRQ for E-UTRAN TDD-FDD Carrier Aggregation with PCell in FDD test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-4<br>Table H.4.1-5a |

**Table 9.2.26.1.4.3-2: *MeasConfig-DEFAULT*: E-UTRAN RSRP and RSRQ Accuracy for CA without NeighCell**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entries | | |
| MeasObjectToAddMod[1] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the PCell (Cell 1) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the PCell (Cell 1) | Switch PCell/SCell scenario |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[2] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the SCell on the SCC (Cell 2) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | | |
| } | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the SCell on the SCC (Cell 2) | Switch PCell/SCell scenario |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the SCell on the SCC (Cell 2) | |
| | IdMeasObject-f1 | f1 is the frequency of the SCell on the SCC (Cell 2) | Switch PCell/SCell scenario |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

#### 9.2.26.1.5 Test requirement

Table 9.2.26.1.5-1 defines the primary level settings including test tolerances for all tests.

The RSRQ absolute accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.26.1.5-2.

**Table 9.2.26.1.5-1: Cell Specific Test requirement Parameters for TDD-FDD Carrier Aggregation with PCell in TDD test parameters**

| Parameter | | Unit | Cell 1 | Cell 2 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | 2 |
| $BW_{channel}$ | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 |
| Special subframe configuration Note1 | | | 6 | - |
| Uplink-downlink configuration Note1 | | | 1 | - |
| Measurement bandwidth | | $n_{PRB}$ | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 |
| PDSCH Reference measurement channel defined in A.3.1.1 | | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD |
| PDSCH allocation | | $n_{PRB}$ | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2 | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD |
| OCNG Patterns defined in A.3.2 | | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD |
| PBCH_RA | | dB | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA Note2 | | | | |
| OCNG_RB Note2 | | | | |
| $N_{oc}$ Note3 | Bands TDD_A | dBm/15 kHz | -119.2 | - |
| | Bands TDD_C | | -118.2 | - |
| | Bands TDD_E | | -117.2 | - |
| | Bands FDD_A | | - | -116 |
| | Bands FDD_B | | - | -115.5 |
| | Bands FDD_C | | - | -115 |
| | Bands FDD_D | | - | -114.5 |
| | Bands FDD_E, Bands FDD_F Note 6 | | - | -114 |
| | Bands FDD_G | | - | -113 |
| | Bands FDD_H | | - | -112.5 |
| $\hat{E}_s / N_{oc}$ | | dB | -5.2 | -5.2 |
| $\hat{E}_s / I_{ot}$ | | dB | -5.2 | -5.2 |
| RSRP Note4 | Bands TDD_A | dBm/15 kHz | -124.4 | - |
| | Bands TDD_C | | -123.4 | - |
| | Bands TDD_E | | -122.4 | - |
| | Bands FDD_A | | - | -121.2 |
| | Bands FDD_B | | - | -120.7 |
| | Bands FDD_C | | - | -120.2 |
| | Bands FDD_D | | - | -119.7 |

| | | | | |
|---|---|---|---|---|
| | Bands FDD_E, Bands FDD_F Note 6 | | - | -119.2 |
| | Bands FDD_G | | - | -118.2 |
| | Bands FDD_H | | - | -117.7 |
| RSRQ Note4 | Bands TDD_A | dB | -17.14 | - |
| | Bands TDD_C | | | |
| | Bands TDD_E | | | |
| | Bands FDD_A | | - | -17.14 |
| | Bands FDD_B | | | |
| | Bands FDD_C | | | |
| | Bands FDD_D | | | |
| | Bands FDD_E, Bands FDD_F Note 6 | | | |
| | Bands FDD_G | | | |
| | Bands FDD_H | | | |
| Io Note4 | Bands TDD_A | dBm/BW$_{channel}$ | -90.27 + 10log($N_{RB,c}$/50) | - |
| | Bands TDD_C | | -89.27 + 10log($N_{RB,c}$/50) | - |
| | Bands TDD_E | | -88.27 + 10log($N_{RB,c}$/50) | - |
| | Bands FDD_A | | - | -87.07 + 10log($N_{RB,c}$/50) |
| | Bands FDD_B | | | -86.57 + 10log($N_{RB,c}$/50) |
| | Bands FDD_C | | - | -86.07 + 10log($N_{RB,c}$/50) |
| | Bands FDD_D | | - | -85.57 + 10log($N_{RB,c}$/50) |
| | Bands FDD_E, Bands FDD_F Note 6 | | - | -85.07 + 10log($N_{RB,c}$/50) |
| | Bands FDD_G | | - | -84.07 + 10log($N_{RB,c}$/50) |
| | Bands FDD_H | | - | -83.57 + 10log($N_{RB,c}$/50) |
| Propagation Condition | | | AWGN | AWGN |
| Antenna Configuration | | | 1x2 | 1x2 |
| Timing offset to Cell 1 | | µs | - | 0 |
| Time alignment error relative to cell 1 Note 10 | | | - | ≤ TAE |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.

Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: Es/Iot, RSRP, RSRQ and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 5: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 6: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 7: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.

Note 8: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.

Note 9: E-UTRA operating band groups are as defined in Section 3.5.

Note 10: Time alignment error (TAE) as specified in TS 36.104 [30] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

**Table 9.2.26.1.5-2: RSRQ absolute accuracy requirements for the reported values for TDD-FDD Carrier Aggregation with PCell in TDD**

| | Test 1 |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_14 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_14 |
| Extreme Conditions | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_15 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_15 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.26.2 Relative RSRQ Accuracy for E-UTRAN TDD-FDD Carrier Aggregation with PCell in TDD

*Editor's note:*

- TS36.521-2 need to be updated to be consistent with the test applicability.

### 9.2.26.2.1 Test purpose

To verify that TDD-FDD relative RSRQ measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the TDD-FDD relative RSRQ accuracy requirements between the primary and secondary component carriers.

### 9.2.26.2.2 Test applicability

This test case applies to all types of E-UTRA UE release 12 and forward that support E-UTRA FDD and TDD and 2DL TDD-FDD CA with TDD as PCell.

### 9.2.26.2.3 Minimum conformance requirements

The RSRQ measurement of an FDD cell on the secondary component carrier is compared with the RSRQ measurement of a TDD cell on the primary component carrier, and shall meet the applicable relative accuracy requirements which are the RSRQ inter-frequency accuracy requirements defined in TS 36.133 [4] clause 9.1.6.2.

The accuracy requirements in Table 9.2.26.2.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.4 for a corresponding Band.

$$\left| RSRP1 \right|_{dBm} - RSRP2 \left|_{dBm} \right| \le [27] dB$$

|Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.2.26.2.3-1: RSRQ Inter-frequency relative accuracy for Carrier Aggregation**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/$BW_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.26.2.3-2.

**Table 9.2.26.2.3-2: RSRQ relative accuracy measurement report mapping for Carrier Aggregation**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.3, clause 9.1.6.2 and A.9.2.26.

## 9.2.26.2.4 Test description

### 9.2.26.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and [4.3.1] for different CA bandwidth classes.

Channel Bandwidth to be tested: Any one of the supported TDD-FDD CA configurations with PCell in TDD with the largest aggregated CA bandwidth combination supported by UE from Table 9.2.26.2.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.45 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.26.2.4.3.

4. There are two E-UTRA carriers with one cell on one E-UTRA TDD carrier and one cell on the other E-UTRA FDD carrier specified in the test. Cell 1 is PCell on the PCC and Cell 2 is the SCell on the Secondary Component

Carrier (SCC). PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 shall be powered OFF.

#### 9.2.26.2.4.2 Test procedure

The test consists of Cell 1 the PCell, and Cell 2 the SCell on the Secondary Component Carrier (SCC). The SCell (Cell 2) is configured. The SCell (Cell 2) on the SCC is activated. The relative accuracy of RSRQ is defined as the RSRQ measured from the secondary component carrier (Cell 2) compared with the RSRQ measured from the primary component carrier (Cell 1).

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. SS activates SCC by sending the MAC-CE according to TS 36.321 [11] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.2.26.2.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. SS shall transmit an RRCConnectionReconfiguration message.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, SS shall check the reported RSRQ values in periodical MeasurementReport messages. The reported RSRQ value for Cell 2 is compared to the reported RSRQ value for Cell 1 for each MeasurementReport message according to Table 9.2.26.2.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

10. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to table G.2.3-1 in Annex G.2 is achieved.

#### 9.2.26.2.4.3 Message contents

Message contents are according to TS 36.508 [7] with the following exceptions:

**Table 9.2.26.2.4.3-1: Common Exception messages for RSRQ for E-UTRAN TDD-FDD Carrier Aggregation with PCell in TDD test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-4<br>Table H.4.1-5a |

**Table 9.2.26.2.4.3-2: *MeasConfig-DEFAULT*: E-UTRAN RSRP and RSRQ Accuracy for CA without NeighCell**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entries | | |
| MeasObjectToAddMod[1] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the PCell (Cell 1) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the PCell (Cell 1) | Switch PCell/SCell scenario |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[2] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the SCell on the SCC (Cell 2) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | | |
| } | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the SCell on the SCC (Cell 2) | Switch PCell/SCell scenario |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the SCell on the SCC (Cell 2) | |
| | IdMeasObject-f1 | f1 is the frequency of the SCell on the SCC (Cell 2) | Switch PCell/SCell scenario |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

#### 9.2.26.2.5 Test requirement

Table 9.2.26.2.5-1 defines the primary level settings including test tolerances for all tests.

The RSRQ relative accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.26.2.5-2.

**Table 9.2.26.2.5-1: Cell Specific Test requirement Parameters for TDD-FDD Carrier Aggregation with PCell in TDD test parameters**

| Parameter | | Unit | Cell 1 | Cell 2 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | 2 |
| $BW_{channel}$ | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 |
| Special subframe configuration Note1 | | | 6 | - |
| Uplink-downlink configuration Note1 | | | 1 | - |
| Measurement bandwidth | | $n_{PRB}$ | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 |
| PDSCH Reference measurement channel defined in A.3.1.1 | | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD |
| PDSCH allocation | | $n_{PRB}$ | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2 | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD |
| OCNG Patterns defined in A.3.2 | | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD |
| PBCH_RA | | dB | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA Note2 | | | | |
| OCNG_RB Note2 | | | | |
| $N_{oc}$ Note3 | Bands TDD_A | dBm/15 kHz | -119.2 | - |
| | Bands TDD_C | | -118.2 | - |
| | Bands TDD_E | | -117.2 | - |
| | Bands FDD_A | | - | -116 |
| | Bands FDD_B | | - | -115.5 |
| | Bands FDD_C | | - | -115 |
| | Bands FDD_D | | - | -114.5 |
| | Bands FDD_E, Bands FDD_F Note 6 | | - | -114 |
| | Bands FDD_G | | - | -113 |
| | Bands FDD_H | | - | -112.5 |
| $\hat{E}_s / N_{oc}$ | | dB | -5.2 | -5.2 |
| $\hat{E}_s / I_{ot}$ | | dB | -5.2 | -5.2 |
| RSRP Note4 | Bands TDD_A | dBm/15 kHz | -124.4 | - |
| | Bands TDD_C | | -123.4 | - |
| | Bands TDD_E | | -122.4 | - |
| | Bands FDD_A | | - | -121.2 |
| | Bands FDD_B | | - | -120.7 |
| | Bands FDD_C | | - | -120.2 |
| | Bands FDD_D | | - | -119.7 |

| | | | | |
|---|---|---|---|---|
| | Bands FDD_E, Bands FDD_F Note 6 | | - | -119.2 |
| | Bands FDD_G | | - | -118.2 |
| | Bands FDD_H | | - | -117.7 |
| RSRQ Note4 | Bands TDD_A | dB | -17.14 | - |
| | Bands TDD_C | | | |
| | Bands TDD_E | | | |
| | Bands FDD_A | | - | -17.14 |
| | Bands FDD_B | | | |
| | Bands FDD_C | | | |
| | Bands FDD_D | | | |
| | Bands FDD_E, Bands FDD_F Note 6 | | | |
| | Bands FDD_G | | | |
| | Bands FDD_H | | | |
| Io Note4 | Bands TDD_A | dBm/BW$_{channel}$ | -90.27 + 10log($N_{RB,c}$/50) | - |
| | Bands TDD_C | | -89.27 + 10log($N_{RB,c}$/50) | - |
| | Bands TDD_E | | -88.27 + 10log($N_{RB,c}$/50) | - |
| | Bands FDD_A | | - | -87.07 + 10log($N_{RB,c}$/50) |
| | Bands FDD_B | | | -86.57 + 10log($N_{RB,c}$/50) |
| | Bands FDD_C | | - | -86.07 + 10log($N_{RB,c}$/50) |
| | Bands FDD_D | | - | -85.57 + 10log($N_{RB,c}$/50) |
| | Bands FDD_E, Bands FDD_F Note 6 | | - | -85.07 + 10log($N_{RB,c}$/50) |
| | Bands FDD_G | | - | -84.07 + 10log($N_{RB,c}$/50) |
| | Bands FDD_H | | - | -83.57 + 10log($N_{RB,c}$/50) |
| Propagation Condition | | | AWGN | AWGN |
| Antenna Configuration | | | 1x2 | 1x2 |
| Timing offset to Cell 1 | | µs | - | 0 |
| Time alignment error relative to cell 1 Note 10 | | | - | ≤ TAE |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.

Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: Es/Iot, RSRP, RSRQ and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 5: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 6: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 7: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.

Note 8: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.

Note 9: E-UTRA operating band groups are as defined in Section 3.5.

Note 10: Time alignment error (TAE) as specified in TS 36.104 [30] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

**Table 9.2.26.2.5-2: RSRQ relative accuracy requirements for the reported values for TDD-FDD Carrier Aggregation with PCell in TDD**

| | Test 1 |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 2) | RSRQ_x – 10 |
| Highest reported value (Cell 2) | RSRQ_x + 10 |
| Extreme Conditions | |
| Lowest reported value (Cell 2) | RSRQ_x - 10 |
| Highest reported value (Cell 2) | RSRQ_x + 10 |
| RSRQ_x is the reported value of Cell 1 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.2.27 TDD RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz+10MHz

## 9.2.27.1 TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz+10MHz

### 9.2.27.1.1 Test purpose

Same test purpose as in clause 9.2.6.1.1.

### 9.2.27.1.2 Test applicability

This test case applies to all types of E-UTRA UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA.

This test case applies to all types of E-UTRA UE release 11 and forward that support CA with Intra-band non-contiguous CA.

### 9.2.27.1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.2.6.2.13.

### 9.2.27.1.4 Test description

#### 9.2.27.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-3 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The bandwidth combination supported by UE from Table 9.2.27.1.5-1 and Annex E table E-3 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.27.1.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

#### 9.2.27.1.4.2 Test procedure

Same test procedure as in clause 9.2.6.1.4.2 with the following exceptions:

- Instead of Table 9.2.6.1.5-1 → use Table 9.2.27.1.5-1.
- Instead of Table 9.2.6.1.5-2 → use Table 9.2.27.1.5-2.

#### 9.2.27.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.27.1.4.3-1: Common Exception messages for TDD RSRQ absolute accuracy for Carrier Aggregation test requirement for 20MHz+10MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-4<br>Table H.4.1-5 |

### 9.2.27.1.5 Test requirement

Table 9.2.27.1.5-1 defines the primary level settings including test tolerances for all tests.

The TDD RSRQ absolute accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.27.1.5-2.

**Table 9.2.27.1.5-1: Cell Specific Test requirement Parameters for TDD RSRQ absolute accuracy for Carrier Aggregation for 20MHz+10MHz**

| Parameters | | Units | Combination | Test 1 | | |
|---|---|---|---|---|---|---|
| | | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel_CA}$ Note1 | | MHz | 20MHz+10MHz | 20MHz: $N_{RB,c}$ = 100 | 10MHz: $N_{RB,c}$ = 50 | |
| | | | 10MHz+20MHz | 10MHz: $N_{RB,c}$ = 50 | 20MHz: $N_{RB,c}$ = 100 | |
| Measurement bandwidth | | $n_{PRB}$ | 20MHz+10MHz | 47-52 | 22-27 | |
| | | | 10MHz+20MHz | 22-27 | 47-52 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | 20MHz+10MHz | R.3 TDD | R.0 TDD | N/A |
| | | | 10MHz+20MHz | R.0 TDD | R.3 TDD | |
| PDSCH allocation | | $n_{PRB}$ | 20MHz+10MHz | 38-61 | 13-36 | N/A |
| | | | 10MHz+20MHz | 13-36 | 38-61 | |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | 20MHz+10MHz | R.10 TDD | R.6 TDD | R.6 TDD |
| | | | 10MHz+20MHz | R.6 TDD | R.10 TDD | R.10 TDD |
| OCNG Patterns defined in D.2 (TDD) | | | 20MHz+10MHz | OP.7 TDD | OP.1 TDD | OP.2 TDD |
| | | | 10MHz+20MHz | OP.1 TDD | OP.7 TDD | OP.8 TDD |
| $Io^{Note2}$ | Bands TDD_A | $dBm/BW_{channel}$ | All | -90.26 + 10log($N_{RB,c}$/50) | N/A | |
| | Bands TDD_C | | | -89.26 + 10log($N_{RB,c}$/50) | | |

| | | | | | |
|---|---|---|---|---|---|
| | Bands TDD_E | | | -88.26 + 10log($N_{RB,c}$/50) | |
| | Bands TDD_A | dBm/$BW_{channel}$ | All | N/A | -85.61 + 10log($N_{RB,c}$/50) |
| | Bands TDD_C | | | | -84.61 + 10log($N_{RB,c}$/50) |
| | Bands TDD_E | | | | -83.61 + 10log($N_{RB,c}$/50) |
| NOTE 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. | | | | | |
| NOTE 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | | | | |
| NOTE 3: See Table 9.2.6.1.5-1 for the other parameters. | | | | | |

**Table 9.2.27.1.5-2: TDD RSRQ absolute accuracy requirements for the reported values for Carrier Aggregation for 20MHz+10MHz**

| | **Test 1** |
|---|---|
| | All bands |
| **Normal Conditions** | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_15 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value Cell 2) | RSRQ_14 |
| **Extreme Conditions** | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_16 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value Cell 2) | RSRQ_15 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.27.2 TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz+10MHz

### 9.2.27.2.1 Test purpose

Same test purpose as in clause 9.2.6.2.1.

### 9.2.27.2.2 Test applicability

This test case applies to all types of E-UTRA UE release 10 and forward that support CA with Intra-band contiguous CA, or Inter-band CA.

This test case applies to all types of E-UTRA UE release 11 and forward that support CA with Intra-band non-contiguous CA.

### 9.2.27.2.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.2.6.2.3.

### 9.2.27.2.4 Test description

#### 9.2.27.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-3 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The bandwidth combination supported by UE from Table 9.2.27.2.5-1 and Annex E table E-3 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.27.2.4.3.

4. There are two E-UTRA TDD carriers with one cell on one E-UTRA TDD carrier and two cells on the other E-UTRA TDD carrier specified in the test. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

### 9.2.27.2.4.2 Test procedure

Same test procedure as in clause 9.2.6.2.4.2 with the following exceptions:

- Instead of Table 9.2.6.2.5-1 → use Table 9.2.27.2.5-1.

- Instead of Table 9.2.6.2.5-2 → use Table 9.2.27.2.5-2.

### 9.2.27.2.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.27.2.4.3-1: Common Exception messages for TDD RSRQ relative accuracy for Carrier Aggregation test requirement for 20MHz+10MHz**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.4.2-1<br>Table H.3.5-4<br>Table H.4.1-5 |

## 9.2.27.2.5 Test requirement

Table 9.2.27.2.5-1 defines the primary level settings including test tolerances for all tests.

The TDD RSRQ relative accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.5.2.5-2.

**Table 9.2.27.2.5-1: Cell Specific Test requirement Parameters for TDD RSRQ relative accuracy for Carrier Aggregation for 20MHz+10MHz**

**Table 9.2.27.2.5-1: Cell Specific Test requirement Parameters for TDD RSRQ relative accuracy for Carrier Aggregation for 20MHz+10MHz**

| Parameters | Units | Combination | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| $BW_{channel_CA}$ Note1 | MHz | 20MHz+10MHz | 20MHz: $N_{RB,c}$ = 100 | 10MHz: $N_{RB,c}$ = 50 | |
| | | 10MHz+20MHz | 10MHz: $N_{RB,c}$ = 50 | 20MHz: $N_{RB,c}$ = 100 | |
| Measurement bandwidth | $n_{PRB}$ | 20MHz+10MHz | 47-52 | 22-27 | |
| | | 10MHz+20MHz | 22-27 | 47-52 | |
| PDSCH Reference measurement channel defined in A.1.2 | | 20MHz+10MHz | R.3 TDD | R.0 TDD | N/A |
| | | 10MHz+20MHz | R.0 TDD | R.3 TDD | |
| PDSCH allocation | $n_{PRB}$ | 20MHz+10MHz | 38-61 | 13-36 | N/A |
| | | 10MHz+20MHz | 13-36 | 38-61 | |

| | | | | | | |
|---|---|---|---|---|---|---|
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | 20MHz+10MHz | R.10 TDD | R.6 TDD | R.6 TDD |
| | | | 10MHz+20MHz | R.6 TDD | R.10 TDD | R.10 TDD |
| OCNG Patterns defined in D.2 (TDD) | | | 20MHz+10MHz | OP.7 TDD | OP.1 TDD | OP.2 TDD |
| | | | 10MHz+20MHz | OP.1 TDD | OP.7 TDD | OP.8 TDD |
| $Io^{Note2}$ | Bands TDD_A | $dBm/BW_{channel}$ | All | -90.26 + $10log(N_{RB,c}/50)$ | N/A | |
| | Bands TDD_C | | | -89.26 + $10log(N_{RB,c}/50)$ | | |
| | Bands TDD_E | | | -88.26 + $10log(N_{RB,c}/50)$ | | |
| | Bands TDD_A | $dBm/BW_{channel}$ | All | N/A | -85.61 + $10log(N_{RB,c}/50)$ | |
| | Bands TDD_C | | | | -84.61 + $10log(N_{RB,c}/50)$ | |
| | Bands TDD_E | | | | -83.61 + $10log(N_{RB,c}/50)$ | |

NOTE 1: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
NOTE 2: Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 3: See Table 9.2.6.1.5-1 for the other parameters.

**Table 9.2.27.2.5-2: TDD RSRQ relative accuracy requirements for the reported values for Carrier Aggregation for 20MHz+10MHz**

| | **Test 1** |
|---|---|
| | All bands |
| **Normal Conditions** | |
| Lowest reported value (Cell 2) | RSRQ_x - 12 |
| Highest reported value (Cell 2) | RSRQ_x + 9 |
| **Extreme Conditions** | |
| Lowest reported value (Cell 2) | RSRQ_x - 12 |
| Highest reported value (Cell 2) | RSRQ_x + 9 |
| RSRQ_x is the reported value of Cell 1 | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.28 FDD intra-frequency absolute RSRQ accuracy with CRS based discovery signal

### 9.2.28.1 Test purpose

The purpose of this test is to verify that the RSRQ measurement accuracy is within the specified limits.

### 9.2.28.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward and supports CRS based discovery signals measurement. Applicability requires support for FGI bit 16.

### 9.2.28.3 Minimum conformance requirements

Intra-frequency absolute RSRQ measurement accuracy requirements in discovery signal occasions are the same as specified in TS36.133 [4] Section 9.1.5.1.

The requirements for absolute accuracy of RSRQ in this clause apply to a cell on the same frequency as that of the serving cell.

The accuracy requirements in Table 9.2.28.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101[2] Clause 7.3 for reference sensitivity are fulfilled.
- RSRP|dBm according to Annex I.3.1 for a corresponding Band

**Table 9.2.28.3-1: RSRQ Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/$BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -34 dB to 2.5 dB with 0.5 dB resolution.

The mapping of measured quantity is defined in table 9.2.28.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.2.28.3-2: RSRQ measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_-30 | RSRQ < -34 | dB |
| RSRQ_-29 | -34 ≤ RSRQ < -33.5 | dB |
| … | … | … |
| RSRQ_-02 | -20.5 ≤ RSRQ < -20 | dB |
| RSRQ_-01 | -20 ≤ RSRQ < -19.5 | dB |
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |
| RSRQ_35 | -3 ≤ RSRQ < -2.5 | dB |
| RSRQ_36 | -2.5 ≤ RSRQ < -2 | dB |
| … | … | … |
| RSRQ_45 | 2 ≤ RSRQ < 2.5 | dB |
| RSRQ_46 | 2.5 ≤ RSRQ | dB |

Note: The ranges from RSRQ_-30 to RSRQ_-01 and from RSRQ_35 to RSRQ_46 apply for the UE who can support extended RSRQ range in TS 36.306 [14].

The normative reference for this requirement is TS 36.133 [4] clause 9.1.14.4, clause 9.1.7 and A.9.2.28.

## 9.2.28.4 Test description

### 9.2.28.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20.
2. propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.2.28.4.3.
4. All cells are in the same carrier frequency. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

### 9.2.28.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Table 9.2.28.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
3. SS shall transmit an RRCConnectionReconfiguration message.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. UE shall transmit periodically MeasurementReport messages.
6. After 10s wait from Step 3, SS shall check the RSRQ value in periodical MeasurementReport messages. The RSRQ value of Cell 2 reported by the UE is compared to the actual RSRQ value according to Table 9.2.28.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.
7. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to G.2.3-1 in Annex G.2 is achieved.

### 9.2.28.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.2.28.4.3-1: Common Exception message**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-4 |

**Table 9.2.28.4.3-2: *MeasResults***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell EQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.28.4.3-3: *MeasResultListEUTRA***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of Cell 2 | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.2.28.4.3-4: MeasObjectEUTRA-GENERIC**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(f1) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationFDD-r12 | 1 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.2.28.5 Test requirement

Table 9.2.28.5-1 defines the primary level settings including test tolerances for all tests.

The RSRQ absolute accuracy shall meet the reported values test requirements in table 9.2.28.5-2.

**Table 9.2.28.5-1: RSRQ FDD Intra frequency test parameters**

| Parameter | | Unit | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | |
| $BW_{channel}$ | | MHz | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | |
| DMTC period | | ms | N/A | 160 |
| DMTC period offset | | ms | N/A | 10 |
| Discovery signal occasion duration | | ms | N/A | 1 |
| Time offset between cell 1 and cell 2 | | µs | 0 | 2.3 |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - |
| *PDSCH allocation* | | $n_{PRB}$ | *13—36* | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.2.2 (OP.2 FDD) | | | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA[Note1] | | | | |
| OCNG_RB[Note1] | | | | |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -116 | |
| | Bands FDD_B | | -115.5 | |
| | Bands FDD_C | | -115 | |
| | Bands FDD_D | | -114.5 | |
| | Bands FDD_E, FDD_F [Note 5] | | -114 | |
| | Bands FDD_G [Note 7] | | -113 | |
| | Bands FDD_H | | -112.5 | |
| $\hat{E}_s / I_{ot}$ | | dB | -5.17 | -5.17 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -119.60 | -119.60 |
| | Bands FDD_B | | -119.10 | -119.10 |
| | Bands FDD_C | | -118.60 | -118.60 |
| | Bands FDD_D | | -118.10 | -118.10 |
| | Bands FDD_E, FDD_F [Note 5] | | -117.60 | -117.60 |
| | Bands FDD_G [Note 7] | | -116.60 | -116.60 |
| | Bands FDD_H | | -116.10 | -116.10 |
| RSRQ[Note3] | Bands FDD_A | dB | -17.12 | -17.12 |
| | Bands FDD_B | | | |
| | Bands FDD_C | | | |
| | Bands FDD_D | | | |
| | Bands FDD_E, FDD_F [Note 5] | | | |
| | Bands FDD_G [Note 7] | | | |
| | Bands FDD_H | | | |
| Io[Note3] | Bands FDD_A | dBm/9 MHz | -85.49 | |
| | Bands FDD_B | | -84.99 | |
| | Bands FDD_C | | -84.49 | |
| | Bands FDD_D | | -83.99 | |
| | Bands FDD_E, FDD_F [Note 5] | | -83.49 | |
| | Bands FDD_G [Note 7] | | -82.49 | |
| | Bands FDD_H | | -81.99 | |
| $\hat{E}_s / N_{oc}$ | | dB | -3.60 | -3.60 |
| Propagation condition | | - | AWGN | |

| | |
|---|---|
| Note 1: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 2: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 3: | RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 4: | RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 5: | For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz. |
| Note 6: | E-UTRA operating band groups are as defined in Section 3.5. |
| Note 7: | Except Band 29 and Band 32. |

**Table 9.2.28.5-2: RSRQ FDD Intra frequency absolute accuracy requirements for the reported values**

| | Test 1<br>All bands |
|---|---|
| **Normal Conditions** | |
| Lowest reported value (Cell 2) | RSRQ_0 |
| Highest reported value (Cell 2) | RSRQ_14 |
| **Extreme Conditions** | |
| Lowest reported value (Cell 2) | RSRQ_0 |
| Highest reported value (Cell 2) | RSRQ_15 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.29 TDD intra-frequency absolute RSRQ accuracy with CRS based discovery signal

### 9.2.29.1 Test purpose

Same test purpose as in clause 9.2.28.1.

### 9.2.29.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward and supports CRS based discovery signals measurement. Applicability requires support for FGI bit 16.

### 9.2.29.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.2.28.3.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.14.4, clause 9.1.7 and A.9.2.29.

### 9.2.29.4 Test description

#### 9.2.29.4.1 Initial conditions

Same initial conditions as in clause 9.2.28.4.1 with the following exceptions:

- Message contents are defined in clause 9.2.29.4.3.

#### 9.2.29.4.2 Test procedure

Same test procedure as in clause 9.2.28.4.2 with the following exceptions:

- Instead of Table 9.2.28.5-1 → use Table 9.2.29.5-1.

- Instead of Table 9.2.28.5-2 → use Table 9.2.29.5-2.

### 9.2.29.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.2.29.4.3-1: Common Exception message**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-4 |

**Table 9.2.29.4.3-2: *MeasResults***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell EQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.29.4.3-3: *MeasResultListEUTRA***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of Cell 2 | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.2.29.4.3-4: MeasObjectEUTRA-GENERIC**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(f1) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationTDD-r12 | 2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.2.29.5 Test requirement

Table 9.2.29.5-1 defines the primary level settings including test tolerances for all tests.

The RSRQ absolute accuracy shall meet the reported values test requirements in table 9.2.29.5-2.

**Table 9.2.29.5-1: RSRQ TDD Intra frequency test parameters**

| Parameter | | Unit | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | |
| $BW_{channel}$ | | MHz | 10 | |
| Special subframe configuration[Note1] | | | 6 | |
| Uplink-downlink configuration[Note1] | | | 1 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | |
| DMTC period | | ms | N/A | 160 |
| DMTC period offset | | ms | N/A | 10 |
| Discovery signal occasion duration | | ms | N/A | 2 |
| Time offset between cell 1 and cell 2 | | µs | 0 | 2.3 |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - |
| *PDSCH allocation* | | $n_{PRB}$ | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA[Note1] | | | | |
| OCNG_RB[Note1] | | | | |
| $N_{oc}$ [Note2] | Bands TDD_A | dBm/15 kHz | -116 | |
| | Bands TDD_C | | -115 | |
| | Bands TDD_E | | -114 | |
| $\hat{E}_s / I_{ot}$ | | dB | -5.17 | -5.17 |
| RSRP[Note3] | Bands TDD_A | dBm/15 kHz | -119.60 | -119.60 |
| | Bands TDD_C | | -118.60 | -118.60 |
| | Bands TDD_E | | -117.60 | -117.60 |
| RSRQ[Note3] | Bands TDD_A | dB | -17.12 | -17.12 |
| | Bands TDD_C | | | |
| | Bands TDD_E | | | |
| Io[Note3] | Bands TDD_A | dBm/9 MHz | -85.49 | |
| | Bands TDD_C | | -84.49 | |
| | Bands TDD_E | | -83.49 | |
| $\hat{E}_s / N_{oc}$ | | dB | -3.60 | -3.60 |
| Propagation condition | | - | AWGN | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211[9].

Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 5: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver

| | |
|---|---|
| | antenna port. |
| Note 6: | E-UTRA operating band groups are as defined in Section 3.5. |

**Table 9.2.29.5-2: RSRQ TDD Intra frequency absolute accuracy requirements for the reported values**

| | Test 1<br>All bands |
|---|---|
| **Normal Conditions** | |
| Lowest reported value (Cell 2) | RSRQ_0 |
| Highest reported value (Cell 2) | RSRQ_14 |
| **Extreme Conditions** | |
| Lowest reported value (Cell 2) | RSRQ_0 |
| Highest reported value (Cell 2) | RSRQ_15 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.30 FDD-FDD inter-frequency absolute and relative RSRQ accuracies with CRS based discovery signal

### 9.2.30.1 Test purpose

The purpose of this test is to verify that the RSRQ absolute and relative measurement accuracies in CRS based discovery signal are within the specified limits.

### 9.2.30.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward and supports CRS based discovery signals measurement. Applicability requires support for FGI bit 16 and 25.

### 9.2.30.3 Minimum conformance requirements

Inter-frequency absolute RSRQ measurement accuracy requirements in discovery signal occasions are the same as specified in TS36.133 [4] Section 9.1.6.1. Inter-frequency relative RSRQ measurement accuracy requirements in discovery signal occasions are the same as specified in TS36.133 [4] Section 9.1.6.2.

The requirements for absolute accuracy of RSRQ in this clause apply to a cell that has different carrier frequency from the serving cell.

The accuracy requirements in Table 9.2.30.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.
- RSRP|dBm according to Annex I.3.3 for a corresponding Band

**Table 9.2.30.3-1: RSRQ Inter frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 4] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 3] | dBm/BW$_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The relative accuracy of RSRQ in inter frequency case is defined as the RSRQ measured from one cell compared to the RSRQ measured from another cell on a different frequency.

The accuracy requirements in Table 9.2.30.3-2 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.
- RSRP1,2$|_{dBm}$ according to Annex I.3.4 for a corresponding Band
- $\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$
- | Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.2.30.3-2: RSRQ Inter frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot [Note 2] | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 5] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 4] | dBm/BW$_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -34 dB to 2.5 dB with 0.5 dB resolution.

The mapping of measured quantity is defined in table I.4.2 and I.4.3. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.2.30.3-3: RSRQ measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_-30 | RSRQ < -34 | dB |
| RSRQ_-29 | -34 ≤ RSRQ < -33.5 | dB |
| … | … | … |
| RSRQ_-02 | -20.5 ≤ RSRQ < -20 | dB |
| RSRQ_-01 | -20 ≤ RSRQ < -19.5 | dB |
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |
| RSRQ_35 | -3 ≤ RSRQ < -2.5 | dB |
| RSRQ_36 | -2.5 ≤ RSRQ < -2 | dB |
| … | … | … |
| RSRQ_45 | 2 ≤ RSRQ < 2.5 | dB |
| RSRQ_46 | 2.5 ≤ RSRQ | dB |

NOTE: The ranges from RSRQ_-30 to RSRQ_-01 and from RSRQ_35 to RSRQ_46 apply for the UE who can support extended RSRQ range in TS 36.306 [14].

The normative reference for this requirement is TS 36.133 [4] clause 9.1.14.2, clause 9.1.6 and A.9.2.30.

## 9.2.30.4 Test description

### 9.2.30.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.2.30.4.3.
4. Cell 1 and Cell 2 are on the different carrier frequencies. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

### 9.2.30.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Table 9.2.30.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
3. SS shall transmit an RRCConnectionReconfiguration message on cell 1.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the reported RSRQ values in periodical MeasurementReport messages.
   The reported RSRQ value of Cell 2 reported by the UE is compared to actual RSRQ value according to Table 9.2.30.5-2. This counts as a Pass or Fail for the events "Cell 2". If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.
   The reported RSRQ value for Cell 2 is compared to the reported RSRQ value for Cell 1 for each MeasurementReport message according to Table 9.2.30.5-3. This counts as a Pass or Fail for the events "Cell 1-2". If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved..

Each of the events "Cell 1" and "Cell 1-2" is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.

If all events pass for each scenario, the test passes. If one event fails, the test fails.

### 9.2.30.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.2.30.4.3-1: Common Exception message**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-2<br>Table H.3.5-4 |

**Table 9.2.30.4.3-2: *MeasResults***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell EQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.30.4.3-3: *MeasResultListEUTRA***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of Cell 2 | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.2.30.4.3-4: MeasObjectEUTRA-GENERIC**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(f2) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationFDD-r12 | 1 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.2.30.5 Test requirement

Table 9.2.30.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP absolute accuracy shall meet the reported values test requirements in table 9.2.30.5-2.

The RSRP relative accuracy shall meet the reported values test requirements in table 9.2.30.5-3.

**Table 9.2.30.5-1: RSRQ in discovery signal occasions FDD—FDD Inter frequency test parameters**

| Parameter | | Unit | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 |
| Gap Pattern Id | | | 0 | - |
| Gap Offset | | ms | 9 | - |
| DMTC period | | ms | - | 160 |
| DMTC period offset | | ms | - | 10 |
| Discovery signal occasion duration | | ms | - | 1 |
| Time offset between cell 2 and cell 1 | | µs | 3 | |
| Measurement bandwidth | | $n_{PRB}$ | 22-27 | |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13-36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.2.2 (OP.2 FDD) | | | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA[Note1] | | | | |
| OCNG_RB[Note1] | | | | |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -117.5 | -117.2 |
| | Bands FDD_B | | -117 | -116.7 |
| | Bands FDD_C | | -116.5 | -116.2 |
| | Bands FDD_D | | -116 | -115.7 |
| | Bands FDD_E, FDD_F [Note 5] | | -115.5 | -115.2 |
| | Bands FDD_G [Note 7] | | -114.5 | -114.2 |
| | Bands FDD_H | | -114 | -113.7 |
| $\hat{E}_s / I_{ot}$ | | dB | -5.20 | -5.20 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -122.70 | -122.40 |
| | Bands FDD_B | | -122.20 | -121.90 |
| | Bands FDD_C | | -121.70 | -121.40 |
| | Bands FDD_D | | -121.20 | -120.90 |
| | Bands FDD_E, FDD_F [Note 5] | | -120.70 | -120.40 |
| | Bands FDD_G [Note 7] | | -119.70 | -119.40 |
| | Bands FDD_H | | -119.20 | -118.90 |
| RSRQ[Note3] | Bands FDD_A | dB | -17.14 | -17.14 |
| | Bands FDD_B | | | |
| | Bands FDD_C | | | |
| | Bands FDD_D | | | |
| | Bands FDD_E, FDD_F [Note 5] | | | |
| | Bands FDD_G [Note 7] | | | |
| | Bands FDD_H | | | |
| Io[Note3] | Bands FDD_A | dBm/ 9 MHz | -88.57 | -88.27 |
| | Bands FDD_B | | -88.07 | -87.77 |
| | Bands FDD_C | | -87.57 | -87.27 |
| | Bands FDD_D | | -87.07 | -86.77 |
| | Bands FDD_E, FDD_F [Note 5] | | -86.57 | -86.27 |

| | | | | |
|---|---|---|---|---|
| | Bands FDD_G Note 7 | | -85.57 | -85.27 |
| | Bands FDD_H | | -85.07 | -84.77 |
| $\hat{E}_s / N_{oc}$ | | dB | -5.20 | -5.20 |
| Propagation condition | | - | AWGN | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 6: E-UTRA operating band groups are as defined in Section 3.5.
Note 7: Except Band 29 and Band 32.

**Table 9.2.30.5-2: RSRQ FDD Inter frequency absolute accuracy requirements for the reported values**

| | Event | Test 1<br>All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 2) | Cell 2 | RSRQ_0 |
| Highest reported value (Cell 2) | | RSRQ_14 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 2 | RSRQ_0 |
| Highest reported value (Cell 2) | | RSRQ_15 |

**Table 9.2.30.5-3: RSRQ FDD Inter frequency relative accuracy requirements for the reported values**

| | Event | Test 1<br>All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRQ_Cell 1 - 10 |
| Highest reported value (Cell 2) | | RSRQ_Cell 1 + 10 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRQ_Cell 1 - 10 |
| Highest reported value (Cell 2) | | RSRQ_Cell 1 + 10 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.31 TDD-TDD inter-frequency absolute and relative RSRQ accuracies with CRS based discovery signal

### 9.2.31.1 Test purpose

Same test purpose as in clause 9.2.30.1.

### 9.2.31.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward and supports CRS based discovery signals measurement. Applicability requires support for FGI bit 16 and 25.

### 9.2.31.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.2.30.3.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.14.2, clause 9.1.7 and A.9.2.31.

## 9.2.31.4 Test description

### 9.2.31.4.1 Initial conditions

Same initial conditions as in clause 9.2.30.4.1 with the following exceptions:

- Message contents are defined in clause 9.2.31.4.3.

### 9.2.31.4.2 Test procedure

Same test procedure as in clause 9.2.30.4.2 with the following exceptions:

- Instead of Table 9.2.30.5-1 → use Table 9.2.31.5-1.
- Instead of Table 9.2.30.5-2 → use Table 9.2.31.5-2.
- Instead of Table 9.2.30.5-3 → use Table 9.2.31.5-3.

### 9.2.31.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.2.31.4.3-1: Common Exception message**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-2<br>Table H.3.5-4 |

**Table 9.2.31.4.3-2: *MeasResults***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell EQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.31.4.3-3: *MeasResultListEUTRA***

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId of Cell 2 | INTEGER (0..503) | |
| measResult SEQUENCE { | | | |
| rsrpResult | | | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

**Table 9.2.31.4.3-4: MeasObjectEUTRA-GENERIC**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(f2) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationTDD-r12 | 2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.2.31.5 Test requirement

Table 9.2.31.5-1 defines the primary level settings including test tolerances for all tests.

The RSRP absolute accuracy shall meet the reported values test requirements in table 9.2.31.5-2.

The RSRP relative accuracy shall meet the reported values test requirements in table 9.2.31.5-3.

**Table 9.2.31.5-1: RSRQ TDD—TDD Inter frequency test parameters for TDD configuration 1**

| Parameter | | Unit | Test 1 | |
|---|---|---|---|---|
| | | | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | 2 |
| $BW_{channel}$ | | MHz | 10 | 10 |
| Gap Pattern Id | | | 0 | - |
| Gap Offset | | | 9 | - |
| DMTC period | | ms | - | 160 |
| DMTC period offset | | ms | - | 10 |
| Discovery signal occasion duration | | ms | - | 2 |
| Time offset between cells | | µs | 0 | 3 |
| Special subframe configuration Note1 | | | 6 | |
| Uplink-downlink configuration Note1 | | | 1 | |
| Measurement bandwidth | | $n_{PRB}$ | 22-27 | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA Note2 | | | | |
| OCNG_RB Note2 | | | | |
| $N_{oc}$ Note3 | Bands TDD_A | dBm/15 kHz | -117.50 | -117.20 |
| | Bands TDD_C | | -116.50 | -116.20 |
| | Bands TDD_E | | -115.50 | -115.20 |
| $\hat{E}_s / I_{ot}$ | | dB | -5.20 | -5.20 |
| RSRP Note4 | Bands TDD_A | dBm/15 kHz | -122.70 | -122.40 |
| | Bands TDD_C | | -121.70 | -121.40 |
| | Bands TDD_E | | -120.70 | -120.40 |
| RSRQ Note4 | Bands TDD_A, TDD_C, TDD_E | dB | -17.14 | -17.14 |
| Io Note4 | Bands TDD_A | dBm/9 MHz | -88.57 | -88.27 |
| | Bands TDD_C | | -87.57 | -87.27 |
| | Bands TDD_E | | -86.57 | -86.27 |
| $\hat{E}_s / N_{oc}$ | | dB | -5.20 | -5.20 |
| Propagation condition | | - | AWGN | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.

Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 5: RSRP minimum requirements are specified assuming independent

| | |
|---|---|
| | interference and noise at each receiver antenna port. |
| Note 6: | E-UTRA operating band groups are as defined in Section 3.5. |

**Table 9.2.31.5-2: RSRQ TDD Inter frequency absolute accuracy requirements for the reported values**

| | Event | Test 1<br>All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 2) | Cell 2 | RSRQ_0 |
| Highest reported value (Cell 2) | | RSRQ_14 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 2 | RSRQ_0 |
| Highest reported value (Cell 2) | | RSRQ_15 |

**Table 9.2.31.5-3: RSRQ TDD Inter frequency relative accuracy requirements for the reported values**

| | Event | Test 1<br>All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRQ_Cell 1 - 10 |
| Highest reported value (Cell 2) | | RSRQ_Cell 1 + 10 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRQ_Cell 1 - 10 |
| Highest reported value (Cell 2) | | RSRQ_Cell 1 + 10 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.32 FDD absolute and relative RSRQ accuracy for E-UTRAN Carrier Aggregation in CRS based discovery signal

### 9.2.32.1 Test purpose

The purpose of this test is to verify that the RSRQ measurement accuracy for carrier aggregation in CRS based discovery signal is within the specified limits under AWGN propagation conditions.

### 9.2.32.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward and supports [CA and CRS based discovery signals measurement].

### 9.2.32.3 Minimum conformance requirements

RSRQ measurement of cells on the primary component carrier or any of the secondary component carrier(s) in discovery signal occasions shall meet the intrafrequency absolute accuracy requirements in TS36.133 [4] section 9.1.5.1.

The accuracy requirements in Table 9.2.32.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101[2] Clause 7.3 for reference sensitivity are fulfilled.
- RSRP|dBm according to Annex I.3.1 for a corresponding Band

**Table 9.2.32.3-1: RSRQ Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 4] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 3] | dBm/$BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

When measurements of cells on the primary component carrier are compared with measurements of cells on any of the secondary component carrier(s), the applicable relative accuracy requirements are the RSRQ inter-frequency accuracy requirements in TS36.133 [4] sections 9.1.6.2.

The accuracy requirements in Table 9.2.32.3-2 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.
- $RSRP1,2|_{dBm}$ according to Annex I.3.4 for a corresponding Band
- $\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$
- | Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.2.32.3-2: RSRQ Inter frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot [Note 2] | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 5] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 4] | dBm/$BW_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -34 dB to 2.5 dB with 0.5 dB resolution.

The mapping of measured quantity is defined in table I.4.2 and I.4.3. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.2.30.3-3: RSRQ measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_-30 | RSRQ < -34 | dB |
| RSRQ_-29 | -34 ≤ RSRQ < -33.5 | dB |
| … | … | … |
| RSRQ_-02 | -20.5 ≤ RSRQ < -20 | dB |
| RSRQ_-01 | -20 ≤ RSRQ < -19.5 | dB |
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |
| RSRQ_35 | -3 ≤ RSRQ < -2.5 | dB |
| RSRQ_36 | -2.5 ≤ RSRQ < -2 | dB |
| … | … | … |
| RSRQ_45 | 2 ≤ RSRQ < 2.5 | dB |
| RSRQ_46 | 2.5 ≤ RSRQ | dB |

NOTE: The ranges from RSRQ_-30 to RSRQ_-01 and from RSRQ_35 to RSRQ_46 apply for the UE who can support extended RSRQ range in TS 36.306 [14].

The normative reference for this requirement is TS 36.133 [4] clause 9.1.15.1, clause 9.1.14.4, clause 9.1.7 and A.9.2.32.

## 9.2.32.4 Test description

### 9.2.32.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure group A.41 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.32.4.3.

4. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the Secondary Component Carrier (SCC), and Cell 3 is a neighbour cell on the SCC. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

### 9.2.32.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCell according to Annex C.0, C.1 and C.2 for all downlink physical channels.

3. The SS shall configure SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.

4. SS activates SCC by sending the activation MAC-CE (Refer TS 36.321 [11], clauses 5.13, 6.1.3.8). Wait for at least 2 seconds (Refer TS 36.133 [4], clauses 8.3.3.2).

5. Set the parameters according to Table 9.2.32.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.

6. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, SS shall check the reported RSRP values in periodical MeasurementReport messages.

   The reported RSRQ value of Cell 3 reported by the UE is compared to actual RSRP value according to Table 9.2.32.5-2. This counts as a Pass or Fail for the events “Cell 3”. If the UE fails to report the measurement value for Cell 3, the number of failed iterations is increased by one.

   The reported RSRQ value for Cell 3 is compared to the reported RSRP value for Cell1 for each MeasurementReport message according to Table 9.2.32.5-3. This counts as a Pass or Fail for the events “Cell 1-3”. If the UE fails to report the measurement value for Cell 3, the number of failed iterations is increased by one.

10. SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

11. If test configuration is Inter-band carrier aggregation, repeat steps 1-10 for switched PCell/SCell scenario according to the UE declared capability for UL support (within CA operation) in the individual bands.

### 9.2.32.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.2.32.4.3-1: Common Exception message**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-4<br>Table H.4.1-5<br>Table H.4.2-1 |

**Table 9.2.32.4.3-2: MeasObjectEUTRA-GENERIC**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(Freq) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | Freq is the frequency of the SCell on the SCC (Cell 2) and neighbouring cell on the SCC(Cell 3) |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationFDD-r12 | 1 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 9.2.32.5 Test requirement

Table 9.2.32.5-1 defines the primary level settings including test tolerances for all tests.

The RSRQ absolute accuracy shall meet the reported values test requirements in table 9.2.32.5-2.

The RSRQ relative accuracy shall meet the reported values test requirements in table 9.2.32.5-3.

**Table 9.2.32.5-1: FDD RSRQ Carrier Aggregation Test Parameters**

| Parameters | | Test 1 | | | |
|---|---|---|---|---|---|
| | | Units | Cell 1 | Cell 2 | Cell 3 |
| E-UTRA RF Channel Number | | | 1 | 2 | 2 |
| $BW_{channel_CA}$ | | MHz | 10 | 10 | 10 |
| DMTC period | | | N/A | N/A | 160 |
| DMTC period offset | | | N/A | N/A | 10 |
| Discovery signal occasion duration | | | N/A | N/A | 1 |
| Timing offset to Cell 1 | | µs | - | 0 | 3 |
| Time alignment error between cell 2 and cell 1 | | | - | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1 | - |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | 22—27 | 22—27 |
| PDSCH Reference measurement channel defined in A.1.1 | | | R.0 FDD | R.0 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | R.6 FDD | R.6FDD | R.6 FDD |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) and D.2.2 (OP.2 FDD) | | | OP.1 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA[Note1] | | | | | |
| OCNG_RB[Note1] | | | | | |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -119.5 | -116 | -116 |
| | Bands FDD_C | | -118.5 | -115 | -115 |
| | Bands FDD_D | | -118 | -114.5 | -114.5 |
| | Bands FDD_E, FDD_F [Note 6] | | -117.5 | -114 | -114 |
| | Bands FDD_G | | -116.5 | -113 | -113 |
| | Bands FDD_H | | -116 | -112.5 | -112.5+ |
| $\hat{E}_s / I_{ot}$ | | dB | -4.0 | -5.54 | -5.16 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -123.5 | -120 | -119.70 |
| | Bands FDD_C | | -122.5 | -119 | -118.70 |
| | Bands FDD_D | | -122 | -118.5 | -118.20 |
| | Bands FDD_E, FDD_F [Note 6] | | -121.5 | -118 | -117.70 |
| | Bands FDD_G | | -120.5 | -117 | -116.70 |
| | Bands FDD_H | | -120 | -116.5 | -116.20 |
| RSRQ[Note3] | Bands FDD_A | dB | -16.25 | -17.40 | -17.10 |
| | Bands FDD_C | | | | |
| | Bands FDD_D | | | | |
| | Bands FDD_E, FDD_F [Note 6] | | | | |
| | Bands FDD_G | | | | |
| | Bands FDD_H | | | | |
| Io[Note3] | Bands FDD_A | dBm/9 | -90.26 | -85.61 | -85.61 |

| | | | | | |
|---|---|---|---|---|---|
| | Bands FDD_C | MHz | -89.26 | -84.61 | -84.61 |
| | Bands FDD_D | | -88.76 | -84.11 | -84.11 |
| | Bands FDD_E, FDD_F Note 6 | | -88.26 | -83.61 | -83.61 |
| | Bands FDD_G | | -87.26 | -82.61 | -82.61 |
| | Bands FDD_H | | -86.76 | -82.17+TT | -82.17+TT |
| $\hat{E}_s / N_{oc}$ | | dB | -4.0 | -4.0 | -3.70 |
| Propagation condition | | - | AWGN | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 5: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs
Note 6: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 7: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1..
Note 8: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.2.32.5-2: RSRQ FDD Intra frequency absolute accuracy requirements for the reported values**

| | Event | Test 1<br>All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 3) | Cell 3 | RSRQ_0 |
| Highest reported value (Cell 3) | | RSRQ _14 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 3) | Cell 3 | RSRQ _0 |
| Highest reported value (Cell 3) | | RSRQ _15 |

**Table 9.2.32.5-3: RSRQ FDD Intra frequency relative accuracy requirements for the reported values**

| | Event | Test 1<br>All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 3) | Cell 1-3 | RSRQ _Cell 1 - 13 |
| Highest reported value (Cell 3) | | RSRQ _Cell 1 + 8 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 3) | Cell 1-3 | RSRQ _Cell 1 - 13 |
| Highest reported value (Cell 3) | | RSRQ _Cell 1 + 8 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.33 TDD absolute and relative RSRQ accuracy for E-UTRAN Carrier Aggregation in CRS based discovery signal

### 9.2.33.1 Test purpose

The purpose of this test is to verify that the RSRQ measurement accuracy for carrier aggregation in CRS based discovery signal is within the specified limits under AWGN propagation conditions.

### 9.2.33.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward and supports [CA and CRS based discovery signals measurement].

### 9.2.33.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 9.2.32.3.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.15.1, clause 9.1.14.4, clause 9.1.7 and A.9.2.33.

### 9.2.33.4 Test description

#### 9.2.33.4.1 Initial conditions

Same initial conditions as in clause 9.2.32.4.1 with the following exceptions:

- Message contents are defined in clause 9.2.33.4.3.

#### 9.2.33.4.2 Test procedure

Same test procedure as in clause 9.2.32.4.2 with the following exceptions:

- Instead of Table 9.2.32.5-1 → use Table 9.2.33.5-1.
- Instead of Table 9.2.32.5-2 → use Table 9.2.33.5-2.
- Instead of Table 9.2.32.5-3 → use Table 9.2.33.5-3

#### 9.2.33.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.2.33.4.3-1: Common Exception message**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-4<br>Table H.4.1-5<br>Table H.4.2-1 |

**Table 9.2.33.4.3-2: MeasObjectEUTRA-GENERIC**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2: MeasObjectEUTRA-GENERIC(Freq) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | Freq is the frequency of the SCell on the SCC (Cell 2) and neighbouring cell on the SCC(Cell 3) |
| measDS-Config-r12 CHOICE { | Not present | | |
| setup SEQUENCE { | | | |
| dmtc-PeriodOffset-r12 CHOICE { | | | |
| ms160-r12 | 10 | | |
| } | | | |
| ds-OccasionDuration-r12 CHOICE { | | | |
| durationTDD-r12 | 2 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 9.2.33.5 Test requirement

Table 9.2.33.5-1 defines the primary level settings including test tolerances for all tests.

The RSRQ absolute accuracy shall meet the reported values test requirements in table 9.2.33.5-2.

The RSRQ relative accuracy shall meet the reported values test requirements in table 9.2.33.5-3.

**Table 9.2.33.5-1: TDD RSRQ Carrier Aggregation Test Parameters**

| Parameter | | Unit | Test 1 | | |
|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 3 |
| E-UTRA RF Channel Number | | | 1 | 2 | 2 |
| $BW_{channel}$ | | MHz | 10 | | |
| DMTC period | | | N/A | N/A | 160 |
| DMTC period offset | | | N/A | N/A | 10 |
| Discovery signal occasion duration | | | N/A | N/A | 1 |
| Timing offset to cell 1 | | µs | - | 0 | 3 |
| Time alignment error between cell 2 and cell 1 | | | - | ≤ Time alignment error as specified in 3GPP TS 36.104 [29] clause 6.5.3.1 | - |
| Special subframe configuration[Note1] | | | 6 | | |
| Uplink-downlink configuration[Note1] | | | 1 | | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | |
| PDSCH Reference measurement channel defined in A.1.2 | | | R.0 TDD | R.0 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | R.6 TDD | R.6 TDD | R.6 TDD |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) and D.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA[Note2] | | | | | |
| OCNG_RB[Note2] | | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -119.5 | -116 | |
| | Bands TDD_C | | -118.5 | -115 | |
| | Bands TDD_E | | -117.5 | -114 | |
| $\hat{E}_s / I_{ot}$ | | dB | -4.0 | -5.54 | -5.16 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -123.5 | -120 | -119.70 |
| | Bands TDD_C | | -122.5 | -119 | -118.70 |
| | Bands TDD_E | | -121.5 | -118 | -117.70 |
| RSRQ[Note4] | Bands TDD_A, TDD_C, TDD_E | dB | -16.25 | -17.40 | -17.10 |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -90.26 | -85.61 | |
| | Bands TDD_C | | -89.26 | -84.61 | |
| | Bands TDD_E | | -88.26 | -83.61 | |
| $\hat{E}_s / N_{oc}$ | | dB | -4.0 | -4.0 | -3.70 |
| Propagation condition | | - | AWGN | | |

| | |
|---|---|
| Note 1: | For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211 [9]. |
| Note 2: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 3: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 4: | RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 5: | RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 6: | The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs. |
| Note 7: | This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1. |
| Note 8: | E-UTRA operating band groups are as defined in Section 3.5. |

**Table 9.2.33.5-2: RSRQ TDD Intra frequency absolute accuracy requirements for the reported values**

| | Event | Test 1<br>All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 3) | Cell 3 | RSRQ_0 |
| Highest reported value (Cell 3) | | RSRQ_14 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 3) | Cell 3 | RSRQ_0 |
| Highest reported value (Cell 3) | | RSRQ_15 |

**Table 9.2.33.5-3: RSRQ TDD Intra frequency relative accuracy requirements for the reported values**

| | Event | Test 1<br>All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 3) | Cell 1-3 | RSRQ _Cell 1 - 13 |
| Highest reported value (Cell 3) | | RSRQ _Cell 1 + 8 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 3) | Cell 1-3 | RSRQ _Cell 1 - 13 |
| Highest reported value (Cell 3) | | RSRQ _Cell 1 + 8 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.34 to 9.2.37

## 9.2.38 3DL PCell in FDD RSRQ for E-UTRAN in Carrier Aggregation

### 9.2.38.1 Test purpose

To verify that TDD-FDD RSRQ absolute and relative measurement accuracy in carrier aggregation with PCell in FDD are within the specified limits under AWGN propagation conditions. This test will verify the absolute accuracy of intra-frequency RSRQ measurements for the primary component carrier, the absolute accuracy of intra-frequency RSRQ measurements for the secondary component carrier and also the relative inter-frequency RSRQ accuracy requirement between primary and secondary component carriers.

### 9.2.38.2 Test applicability

This test applies to all types of E-UTRA FDD and TDD UE Release 12 and forward that support 3DL TDD-FDD CA with FDD as PCell.

### 9.2.38.3 Minimum conformance requirements

The RSRQ measurements of cells on the primary component carrier shall meet the intra-frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1. The RSRQ measurements of cells on any of the secondary component carrier(s) shall meet the intra-frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1.

The accuracy requirements in Table 9.2.38.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.
- RSRP|dBm according to Annex I.3.1 for a corresponding Band

**Table 9.2.38.3-1: RSRQ Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/$BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

When measurements of cells on the primary component carrier are compared with measurements of cells on any of the secondary component carrier(s), the applicable relative accuracy requirements are the RSRQ inter-frequency accuracy requirements defined in TS 36.133 [4] clause 9.1.6.2.

The accuracy requirements in Table 9.2.38.3-2 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
  - Conditions defined in 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.
  - RSRP1,2|dBm according to Annex I.3.4 for a corresponding Band
  - $\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$
- | Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.2.38.3-2: RSRQ Inter frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | $dBm/BW_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.38.3-3.

**Table 9.2.38.3-3: RSRQ measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.5.1, 9.1.6.2, 9.1.7, 9.1.11, and A.9.2.38.

## 9.2.38.4 Test description

### 9.2.38.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 9.2.38.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1 [10] clause 5.4.2A.

1 Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.63 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.38.4.3.

4. Cell 1 is PCell in FDD on the primary component carrier, Cell 2 is SCell on SCC1, and Cell 3 is SCell on SCC2. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

### 9.2.38.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCells according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure SCells (Cell 2 and Cell 3) on the SCCs as per TS 36.508 [7] clause 5.2A.4.

4. The SS activates SCCs by sending the MAC-CE according to TS 36.321 [11] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.2.38.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.

6. The SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

7. The UE shall transmit an RRCConnectionReconfigurationComplete message.

8. The UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, the SS shall check the reported RSRQ values in periodical MeasurementReport messages.

   The reported RSRQ values for Cell 1, Cell 2, and Cell 3 are compared to the actual RSRQ values according to Table 9.2.38.5-2. This counts respectively as a Pass or Fail for the events “Cell 1”, “Cell 2”, and “Cell 3”. If the UE fails to report the measurement value for Cell 1, Cell 2, or Cell 3, the number of failed iterations for the respective event is increased by one.

   The reported RSRQ values for Cell 2 and Cell 3 are compared to the reported RSRQ value for Cell 1 for each MeasurementReport message according to Table 9.2.38.5-3. These count respectively as a Pass or Fail for the events “Cell 1-2” and “Cell 1-3”. If the UE fails to report the measurement value for Cell 2 or Cell 3, the number of failed iterations for the respective event is increased by one.

10. The SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events “Cell 1”, “Cell 2”, “Cell 3”, “Cell 1-2” and “Cell 1-3” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.

If all events pass for each scenario, the test passes. If one event fails, the test fails.

### 9.2.38.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.2.38.4.3-1: Common Exception messages for 3DL PCell in FDD RSRQ for E-UTRAN Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-4<br>Table H.4.1-8<br>Table H.4.2-6 |

## 9.2.38.5 Test requirement

Table 9.2.38.5-1 defines the primary level settings including test tolerances for all tests.

The RSRQ absolute accuracy shall meet the reported values test requirements in Table 9.2.38.5-2.

The RSRQ relative accuracy shall meet the reported values test requirements in Table 9.2.38.5-3.

**Table 9.2.38.5-1: 3 DL PCell in FDD RSRQ for E-UTRAN in Carrier Aggregation test parameters (cell #1, cell #2 and cell #3)**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | 2 | 3 |
| $BW_{channel}$ | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 |
| Special subframe configuration[Note1] | | | - | 6 | 6 |
| Uplink/downlink configuration[Note1] | | | - | 1 | 1 |
| Measurement bandwidth | | $n_{PRB}$ | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 |
| PDSCH Reference measurement channel defined in A.3.1.1 | | | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD |
| PDSCH allocation | | $n_{PRB}$ | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2 | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD |
| OCNG Patterns defined in A.3.2 | | | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA[Note2] | | | | | |
| OCNG_RB[Note2] | | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/ 15kHz | - | -116 | -116 |
| | Bands TDD_C | | - | -115 | -115 |
| | Bands TDD_E | | - | -114 | -114 |
| | Bands FDD_A | | -119.2 | - | - |
| | Bands FDD_C | | -118.2 | - | - |
| | Bands FDD_D | | -117.7 | - | - |
| | Bands FDD_E, Bands FDD_F [Note 6] | | -117.2 | - | - |
| | Bands FDD_G | | -116.2 | - | - |
| | Bands FDD_H | | -115.7 | - | - |
| $\hat{E}_s / N_{oc}$ | | dB | -5.2 | -5.2 | -5.2 |
| $\hat{E}_s / I_{ot}$ | | dB | -5.2 | -5.2 | -5.2 |
| RSRP[Note4] | Bands TDD_A | dBm/ 15kHz | - | -121.2 | -121.2 |
| | Bands TDD_C | | - | -120.2 | -120.2 |
| | Bands TDD_E | | - | -119.2 | -119.2 |
| | Bands FDD_A | | -124.4 | - | - |
| | Bands FDD_C | | -123.4 | - | - |
| | Bands FDD_D | | -122.9 | - | - |
| | Bands FDD_E, | | -122.4 | - | - |

| | | | | | |
|---|---|---|---|---|---|
| | Bands FDD_F Note 6 | | | | |
| | Bands FDD_G | | -121.4 | - | - |
| | Bands FDD_H | | -120.9 | - | - |
| RSRQ Note4 | Bands TDD_A | dB | - | -17.14 | -17.14 |
| | Bands TDD_C | | | | |
| | Bands TDD_E | | | | |
| | Bands FDD_A | | -17.14 | - | - |
| | Bands FDD_C | | | | |
| | Bands FDD_D | | | | |
| | Bands FDD_E, Bands FDD_F Note 6 | | | | |
| | Bands FDD_G | | | | |
| | Bands FDD_H | | | | |
| Io Note4 | Bands TDD_A | dBm/ $BW_{channel}$ | - | -87.07 + 10log($N_{RB,c}$/50) | -87.07 + 10log($N_{RB,c}$/50) |
| | Bands TDD_C | | - | -86.07 + 10log($N_{RB,c}$/50) | -86.07 + 10log($N_{RB,c}$/50) |
| | Bands TDD_E | | - | -85.07 + 10log($N_{RB,c}$/50) | -85.07 + 10log($N_{RB,c}$/50) |
| | Bands FDD_A | | -90.27 + 10log($N_{RB,c}$/50) | - | - |
| | Bands FDD_C | | -89.27 + 10log($N_{RB,c}$/50) | - | - |
| | Bands FDD_D | | -88.77 + 10log($N_{RB,c}$/50) | - | - |
| | Bands FDD_E, Bands FDD_F Note 6 | | -88.27 + 10log($N_{RB,c}$/50) | - | - |
| | Bands FDD_G | | -87.27 + 10log($N_{RB,c}$/50) | - | - |
| | Bands FDD_H | | -86.77 + 10log($N_{RB,c}$/50) | - | - |
| Propagation Condition | | | AWGN | AWGN | AWGN |
| Antenna Configuration | | | 1x2 | 1x2 | 1x2 |
| Timing offset to Cell 1 | | µs | - | 0 | 0 |
| Time alignment error relative to cell 1 Note 10 | | | - | ≤ TAE | ≤ TAE |
| Time alignment error relative to cell 2 Note 10 | | | ≤ TAE | - | ≤ TAE |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP, RSRQ and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 6: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 7: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.
Note 8: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section A.3.6.1.
Note 9: E-UTRA operating band groups are as defined in Section 3.5.
Note 10: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

**Table 9.2.38.5-2: RSRQ absolute accuracy requirements for the reported values for 3 DL PCell in FDD for E-UTRAN in Carrier Aggregation**

| | Event | All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRQ_00 |
| Highest reported value Cell 1) | | RSRQ_14 |
| Lowest reported value (Cell 2) | Cell 2 | RSRQ_00 |
| Highest reported value (Cell 2) | | RSRQ_14 |
| Lowest reported value (Cell 3) | Cell 3 | RSRQ_00 |
| Highest reported value (Cell 3) | | RSRQ_14 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRQ_00 |
| Highest reported value Cell 1) | | RSRQ_15 |
| Lowest reported value (Cell 2) | Cell 2 | RSRQ_00 |
| Highest reported value (Cell 2) | | RSRQ_15 |
| Lowest reported value (Cell 3) | Cell 3 | RSRQ_00 |
| Highest reported value (Cell 3) | | RSRQ_15 |

**Table 9.2.38.5-3: RSRQ relative accuracy requirements for the reported values for 3 DL PCell in FDD for E-UTRAN in Carrier Aggregation**

| | Event | All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRQ_x – 10 |
| Highest reported value (Cell 2) | | RSRQ_x + 10 |
| Lowest reported value (Cell 3) | Cell 1-3 | RSRQ_x – 10 |
| Highest reported value (Cell 3) | | RSRQ_x + 10 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRQ_x – 10 |
| Highest reported value (Cell 2) | | RSRQ_x + 10 |
| Lowest reported value (Cell 3) | Cell 1-3 | RSRQ_x – 10 |
| Highest reported value (Cell 3) | | RSRQ_x + 10 |
| RSRQ_x is the reported value of Cell 1 | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.39 3DL PCell in TDD RSRQ for E-UTRAN in Carrier Aggregation

### 9.2.39.1 Test purpose

The purpose of this test is to verify that the TDD-FDD RSRQ measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the absolute accuracy of intra-frequency RSRQ measurements for the primary component carrier, the absolute accuracy of intra-frequency RSRQ measurements for the secondary component carrier, and also the relative inter-frequency RSRQ accuracy requirement between primary and secondary component carriers.

### 9.2.39.2 Test applicability

This test case applies to all types of UE Release 12 and forward supporting E-UTRA FDD and TDD and 3DL CA with TDD as PCell.

### 9.2.39.3 Minimum conformance requirements

The TDD RSRQ measurements of cells on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1. The TDD RSRQ measurements of cells on the secondary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1.

The accuracy requirements in Table 9.2.39.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.
- RSRP|dBm according to Annex I.3.1 for a corresponding Band.

**Table 9.2.39.3-1: TDD RSRQ absolute accuracy for Carrier Aggregation**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/BW$_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.39.3-3.

The TDD RSRQ measurements of cells on the primary component carrier are compared with measurements of cells on the secondary component carrier defined in TS 36.133 [4] clause 9.1.6.2. The applicable relative accuracy requirements are the RSRQ inter-frequency accuracy requirements as defined in TS 36.133 [4] clause 9.1.6.2.

The accuracy requirements in Table 9.2.39.3-2 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.
- RSRP1,2|$_{dBm}$ according to Annex I.3.4 for a corresponding Band.
- $\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$
- |Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.2.39.3-2: TDD RSRQ relative accuracy for Carrier Aggregation**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/$BW_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.39.3-3.

**Table 9.2.39.3-3: TDD RSRQ absolute accuracy measurement report mapping for Carrier Aggregation**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1, clause 9.1.11.2, clause 9.1.11.3, clause 9.1.5.1, clause 9.1.6.2 and A.9.2.6.

## 9.2.39.4 Test description

### 9.2.39.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 9.2.39.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.63 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.39.4.3.

4. There are three E-UTRA TDD carriers. Cell 1 is PCell on the PCC, Cell 2 is the SCell on the SCC1, and Cell 3 is the SCell on the SCC2. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

### 9.2.39.4.2 Test procedure

The absolute accuracy of RSRQ is defined as the RSRQ measured from the PCC (Cell 1) and the RSRQ measured from the SCC1 (Cell 2) and SCC2 (Cell 3). The relative accuracy of RSRQ is defined as the RSRQ measured on PCC compared with RSRQ measured on SCC1 and SCC2,

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels except PHICH.
3. The SS shall configure the SCells (Cell 2 and Cell 3) on the SCC1 and SCC2 as per TS 36.508 [7] clause 5.2A.4.
4. SS activates SCC1 and SCC2 by sending the MAC-CE according to TS 36.321 [11] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.
5. Set the parameters according to Table 9.2.39.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
6. SS shall transmit an RRCConnectionReconfiguration message.
7. The UE shall transmit RRCConnectionReconfigurationComplete message.
8. UE shall transmit periodically MeasurementReport messages.
9. After 10s wait from Step 6, SS shall check the reported RSRQ values in periodical MeasurementReport messages.

   The reported RSRQ values for Cell 1, Cell 2 and Cell 3 by the UE are compared to the actual RSRQ values according to Table 9.2.39.5-2. This counts respectively as a Pass or Fail for the events “Cell 1”, “Cell 2” and “Cell 3”. If the UE fails to report the measurement value for Cell 1, Cell 2, or Cell 3, the number of failed iterations for the respective event is increased by one.

   The reported RSRQ values for Cell 2 and Cell 3 are compared to the reported RSRQ value for Cell 1 for each MeasurementReport message according to Table 9.2.39.5-3. If the UE fails to report the measurement value for Cell 2 or Cell 3, the number of failed iterations for the respective event is increased by one.
10. The SS shall check MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events “Cell 1”, “Cell 2”, “Cell 3”, “Cell 1-2” and “Cell 1-3” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.

If all events pass, the test passes. If one event fails, the test fails.

### 9.2.39.4.3 Message contents

**Table 9.1.40.4.3-1: Common Exception messages for 3DL TDD RSRP Accuracy for E-UTRA Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-4<br>Table H.4.1-8<br>Table H.4.2-6 |

## 9.2.39.5 Test requirement

Table 9.2.39.5-1 defines the primary level settings including test tolerances for all tests.

The TDD RSRQ absolute accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.39.5-2 and Table 9.2.39.5-3.

**Table 9.2.39.5-1: Cell Specific Test requirement Parameters for TDD RSRQ absolute and relative accuracy for Carrier Aggregation**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | 2 | 3 |
| $BW_{channel}$ | | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 |
| Special subframe configuration Note1 | | | 6 | - | - |
| Uplink/downlink configuration Note1 | | | 1 | - | - |
| Measurement bandwidth | | $n_{PRB}$ | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 |
| PDSCH Reference measurement channel defined in A.1.1 and A.1.2 | | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD |
| PDSCH allocation | | $n_{PRB}$ | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 and A.2.2 | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD |
| OCNG Patterns defined in D.1 and D.2 | | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA Note2 | | | | | |
| OCNG_RB Note2 | | | | | |
| $N_{oc}$ Note3 | Bands TDD_A | dBm/ 15kHz | -119.2 | - | - |
| | Bands TDD_C | | -118.2 | - | - |
| | Bands TDD_E | | -117.2 | - | - |
| | Bands FDD_A | | - | -116 | -116 |
| | Bands FDD_C | | - | -115 | -115 |
| | Bands FDD_D | | - | -114.5 | -114.5 |
| | Bands FDD_E, Bands FDD_F Note 6 | | - | -114 | -114 |
| | Bands FDD_G | | - | -113 | -113 |
| | Bands FDD_H | | - | -112.5 | -112.5 |
| $\hat{E}_s / N_{oc}$ | | dB | -5.20 | -5.20 | -5.20 |
| $\hat{E}_s / I_{ot}$ | | dB | -5.20 | -5.20 | -5.20 |
| RSRP Note4 | Bands TDD_A | dBm/ 15kHz | -124.4 | - | - |
| | Bands TDD_C | | -123.4 | - | - |
| | Bands TDD_E | | -122.4 | - | - |
| | Bands FDD_A | | - | -121.2 | -121.2 |
| | Bands FDD_C | | - | -120.2 | -120.2 |
| | Bands FDD_D | | - | -119.7 | -119.7 |
| | Bands FDD_E, Bands FDD_F Note 6 | | - | -119.2 | -119.2 |

| | | | | | |
|---|---|---|---|---|---|
| | Bands FDD_G | | - | -118.2 | -118.2 |
| | Bands FDD_H | | - | -117.7 | -117.7 |
| RSRQ Note4 | Bands TDD_A | dB | -17.14 | - | - |
| | Bands TDD_C | | | | |
| | Bands TDD_E | | | | |
| | Bands FDD_A | | - | -17.14 | -17.14 |
| | Bands FDD_C | | | | |
| | Bands FDD_D | | | | |
| | Bands FDD_E, Bands FDD_F Note 6 | | | | |
| | Bands FDD_G | | | | |
| | Bands FDD_H | | | | |
| Io Note4 | Bands TDD_A | dBm/ $BW_{channel}$ | $-90.27 + 10\log(N_{RB,c}/50)$ | - | - |
| | Bands TDD_C | | $-89.27 + 10\log(N_{RB,c}/50)$ | - | - |
| | Bands TDD_E | | $-88.27 + 10\log(N_{RB,c}/50)$ | - | - |
| | Bands FDD_A | | - | $-87.07 + 10\log(N_{RB,c}/50)$ | $-87.07 + 10\log(N_{RB,c}/50)$ |
| | Bands FDD_C | | - | $-86.07 + 10\log(N_{RB,c}/50)$ | $-86.07 + 10\log(N_{RB,c}/50)$ |
| | Bands FDD_D | | - | $-85.57 + 10\log(N_{RB,c}/50)$ | $-85.57 + 10\log(N_{RB,c}/50)$ |
| | Bands FDD_E, Bands FDD_F Note 6 | | - | $-85.07 + 10\log(N_{RB,c}/50)$ | $-85.07 + 10\log(N_{RB,c}/50)$ |
| | Bands FDD_G | | - | $-84.07 + 10\log(N_{RB,c}/50)$ | $-84.07 + 10\log(N_{RB,c}/50)$ |
| | Bands FDD_H | | - | $-83.57 + 10\log(N_{RB,c}/50)$ | $-83.57 + 10\log(N_{RB,c}/50)$ |
| Propagation Condition | | | AWGN | AWGN | AWGN |
| Antenna Configuration | | | 1x2 | 1x2 | 1x2 |
| Timing offset to Cell 1 | | μs | - | 0 | 0 |
| Time alignment error relative to cell 1 Note 10 | | | - | ≤ TAE | ≤ TAE |
| Time alignment error relative to cell 2 Note 10 | | | - | - | ≤ TAE |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP, RSRQ and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 6: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 7: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.
Note 8: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section C.3.3.1.
Note 9: E-UTRA operating band groups are as defined in Section 3.5.
Note 10: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

**Table 9.2.39.5-2: TDD RSRQ absolute accuracy requirements for the reported values for Carrier Aggregation**

| | All bands |
|---|---|
| **Normal Conditions** | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_14 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value Cell 2) | RSRQ_14 |
| Lowest reported value (Cell 3) | RSRQ_00 |
| Highest reported value Cell 3) | RSRQ_14 |
| **Extreme Conditions** | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_15 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value Cell 2) | RSRQ_15 |
| Lowest reported value (Cell 3) | RSRQ_00 |
| Highest reported value Cell 3) | RSRQ_15 |

**Table 9.2.39.5-3: TDD RSRQ relative accuracy requirements for the reported values for Carrier Aggregation**

| | Event | All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRQ_x – 10 |
| Highest reported value (Cell 2) | | RSRQ_x + 10 |
| Lowest reported value (Cell 3) | Cell 1-3 | RSRQ_x – 10 |
| Highest reported value (Cell 3) | | RSRQ_x + 10 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRQ_x – 10 |
| Highest reported value (Cell 2) | | RSRQ_x + 10 |
| Lowest reported value (Cell 3) | Cell 1-3 | RSRQ_x – 10 |
| Highest reported value (Cell 3) | | RSRQ_x + 10 |
| RSRQ_x is the reported value of Cell 1 | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.40 3DL FDD RSRQ for E-UTRAN in Carrier Aggregation

### 9.2.40.1 Test purpose

To verify that FDD absolute RSRQ measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the FDD absolute RSRQ accuracy requirements of the primary component carrier and the secondary component carriers, the test will also verify the primary and secondary component carriers' relative RSRQ accuracy requirement.

### 9.2.40.2 Test applicability

This test case applies to all types of E-UTRA FDD UE release 10 and forward that support 3DL with Intra-band contiguous CA, or 3DL with Inter-band CA, or 3DL with Intra-band contiguous and Inter-band CA.

This test case also applies to all types of E-UTRA FDD UE release 11 and forward that support 3DL with Intra-band non-contiguous and Inter-band CA, or 3DL with Intra-band non-contiguous and Intra-band contiguous CA.

### 9.2.40.3 Minimum conformance requirements

The FDD RSRQ measurements of cells on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1. The FDD RSRQ measurements of cells on the

secondary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1.

The accuracy requirements in Table 9.2.40.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

- Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.
- RSRP|dBm according to Annex I.3.1 for a corresponding Band.

**Table 9.2.40.3-1: FDD RSRQ absolute accuracy for Carrier Aggregation**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/$BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.<br>NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 4: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The FDD RSRQ relative measurements of cells on the primary component carrier are compared with measurements of cells on the secondary component carrier defined in TS 36.133 [4] clause 9.1.6.2. The applicable relative accuracy requirements are the RSRQ inter-frequency accuracy requirements as defined in TS 36.133 [4] clause 9.1.6.2.

The accuracy requirements in Table 9.2.40.3-2 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.
- $RSRP1,2|_{dBm}$ according to Annex I.3.4 for a corresponding Band.
- $\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$
- |Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.2.40.3-2: FDD RSRQ relative accuracy for Carrier Aggregation**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/$BW_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.40.3-3.

**Table 9.2.40.3-3: FDD RSRQ absolute accuracy measurement report mapping for Carrier Aggregation**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.11.1, 9.1.11.2, 9.1.11.3, 9.1.7, and A.9.2.40.

## 9.2.40.4 Test description

### 9.2.40.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 9.2.40.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.63 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.40.4.3.

4. There are three E-UTRA FDD carriers with one cell on each E-UTRA FDD carrier. Cell 1 is PCell on the PCC, Cell 2 and Cell 3 are the SCells on the Secondary Component Carriers SCC1 and SCC2 respectively. PCell (Cell 1) is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

### 9.2.40.4.2 Test procedure

The test consists of Cell 1 the PCell, Cell 2 and Cell 3 the SCells on the Secondary Component Carrier SCC1 and SCC2. The SCell1 (Cell 2) and Scell2 (Cell 3) are configured and activated. The absolute accuracy of RSRQ is defined as the RSRQ measured from the primary component carrier (Cell 1) and the RSRQ measured from the secondary component carriers (Cell 2 and Cell 3).

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Configure SCC according to Annex C.0 and C.1 for all downlink physical channels.
3. The SS shall configure the SCell (Cell 2) on the SCC as per TS 36.508 [7] clause 5.2A.4.
4. SS activates SCC by sending the MAC-CE according to TS 36.321 [11] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.
5. Set the parameters according to Table 9.2.40.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.
6. SS shall transmit an RRCConnectionReconfiguration message.
7. The UE shall transmit RRCConnectionReconfigurationComplete message.
8. UE shall transmit periodically MeasurementReport messages.
9. After 10s wait from Step 6, SS shall check the reported RSRQ values in periodical MeasurementReport messages.

   The reported RSRQ values for Cell 1, Cell 2 and Cell 3 by the UE are compared to the actual RSRQ values according to Table 9.2.40.5-2. These count respectively as a Pass or Fail for the events “Cell 1”, “Cell 2” and “Cell 3”. If the UE fails to report the measurement value for Cell 1, Cell 2 or Cell 3, the number of failed iterations for the respective event is increased by one.

   The reported RSRQ values for Cell 2 and Cell3 are compared to the reported RSRQ value for other cells for each MeasurementReport message according to Table 9.2.40.5-2. These count respectively as a Pass or Fail for the events “Cell 1-2” and “Cell 1-3”. If the UE fails to report the measurement value for Cell 2 or Cell 3, the number of failed iterations is increased by one.
10. The SS shall check MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events “Cell 1”, “Cell 2”, “Cell 3”, “Cell 1-2” and “Cell 1-3” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.
If all events pass, the test passes. If one event fails, the test fails.

### 9.2.40.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.2.40.4.3-1: Common Exception messages for FDD RSRQ Accuracy E-UTRA for Carrier Aggregation test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-4<br>Table H.4.1-8<br>Table H.4.2-6 |

## 9.2.40.5 Test requirement

Table 9.2.40.5-1 defines the primary level settings including test tolerances for all tests.

The FDD RSRQ absolute accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.40.5-2.

The FDD RSRQ relative accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.40.5-3.

**Table 9.2.40.5-1: 3 DL FDD RSRQ carrier aggregation test parameters**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 |
|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | 2 | 3 |
| $BW_{channel}$ | | MHz | 5MHz:$N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz:$N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 |
| Measurement bandwidth | | $n_{PRB}$ | 5MHz:10-15<br>10MHz:22-27<br>20MHz:47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 |
| PDSCH Reference measurement channel defined in A.1.1 | | | 5MHz:R.5 FDD<br>10MHz:R.0 FDD<br>20MHz:R.4 FDD | 5MHz:R.5 FDD<br>10MHz:R.0 FDD<br>20MHz:R.4 FDD | 5MHz:R.5 FDD<br>10MHz:R.0 FDD<br>20MHz:R.4 FDD |
| PDSCH allocation | | $n_{PRB}$ | 5MHz:7-17<br>10MHz:13-36<br>20MHz:38-61 | 5MHz:7-17<br>10MHz:13-36<br>20MHz:38-61 | 5MHz:7-17<br>10MHz:13-36<br>20MHz:38-61 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1 | | | 5MHz:R.11 FDD<br>10MHz:R.6 FDD<br>20MHz:R.10 FDD | 5MHz:R.11 FDD<br>10MHz:R.6 FDD<br>20MHz:R.10 FDD | 5MHz:R.11 FDD<br>10MHz:R.6 FDD<br>20MHz:R.10 FDD |
| OCNG Patterns defined in D.1 | | | 5MHz:OP.15 FDD<br>10MHz:OP.1 FDD<br>20MHz:OP.11 FDD | 5MHz:OP.15 FDD<br>10MHz:OP.1 FDD<br>20MHz:OP.11 FDD | 5MHz:OP.15 FDD<br>10MHz:OP.1 FDD<br>20MHz:OP.11 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA[Note1] | | | | | |
| OCNG_RB[Note1] | | | | | |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -119.2 | -116 | -116 |
| | Bands FDD_C | | -118.2 | -115 | -115 |
| | Bands FDD_D | | -117.7 | -114.5 | -114.5 |
| | Bands FDD_E, FDD_F [Note 6] | | -117.2 | -114 | -114 |
| | Bands FDD_G | | -116.2 | -113 | -113 |
| | Bands FDD_H | | -115.7 | -112.5 | -112.5 |
| $\hat{E}_s / N_{oc}$ | | dB | -5.20 | -5.20 | -5.20 |
| $\hat{E}_s / I_{ot}$ [Note3] | | dB | -5.20 | -5.20 | -5.20 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -124.4 | -121.2 | -121.2 |
| | Bands FDD_C | | -123.4 | -120.2 | -120.2 |
| | Bands FDD_D | | -122.9 | -119.7 | -119.7 |
| | Bands FDD_E, FDD_F [Note 6] | | -122.4 | -119.2 | -119.2 |
| | Bands FDD_G | | -121.4 | -118.2 | -118.2 |
| | Bands FDD_H | | -120.9 | -117.7 | -117.7 |
| RSRQ[Note3] | Bands FDD_A | dB | -17.14 | -17.14 | -17.14 |
| | Bands FDD_C | | | | |
| | Bands FDD_D | | | | |
| | Bands FDD_E, FDD_F [Note 6] | | | | |
| | Bands FDD_G | | | | |
| | Bands FDD_H | | | | |
| Io[Note3] | Bands FDD_A | dBm/ $BW_{channel}$ | -90.27+ 10log($N_{RB,c}$ /50) | -87.07+ 10log($N_{RB,c}$ /50) | -87.07+ 10log($N_{RB,c}$ /50) |
| | Bands FDD_C | | -89.27+ 10log($N_{RB,c}$ /50) | -86.07+ 10log($N_{RB,c}$ /50) | -86.07+ 10log($N_{RB,c}$ /50) |

| | | | | | |
|---|---|---|---|---|---|
| | Bands FDD_D | | -88.77+ 10log($N_{RB,c}$ /50) | -85.57+ 10log($N_{RB,c}$ /50) | -85.57+ 10log($N_{RB,c}$ /50) |
| | Bands FDD_E, FDD_F Note 6 | | -88.27+ 10log($N_{RB,c}$ /50) | -85.07+ 10log($N_{RB,c}$ /50) | -85.07+ 10log($N_{RB,c}$ /50) |
| | Bands FDD_G | | -87.27+ 10log($N_{RB,c}$ /50) | -84.07+ 10log($N_{RB,c}$ /50) | -84.07+ 10log($N_{RB,c}$ /50) |
| | Bands FDD_H | | -86.77+ 10log($N_{RB,c}$ /50) | -83.57+ 10log($N_{RB,c}$ /50) | -83.57+ 10log($N_{RB,c}$ /50) |
| Propagation condition | | - | AWGN | AWGN | AWGN |
| Antenna Configuration | | - | 1x2 | 1x2 | 1x2 |
| Timing offset to Cell 1 | | μs | - | 0 | 0 |
| Time alignment error relative to cell 1 Note 7 | | | - | ≤ TAE | ≤ TAE |
| Time alignment error relative to cell 2 Note 7 | | | - | - | ≤ TAE |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: Es/Iot, RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.

Note 6: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 7: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

Note 8: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.2.40.5-2: FDD RSRQ absolute accuracy requirements for the reported values for Carrier Aggregation**

| | **Test 1** |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_14 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_14 |
| Lowest reported value (Cell 3) | RSRQ_00 |
| Highest reported value (Cell 3) | RSRQ_14 |
| Extreme Conditions | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value Cell 1) | RSRQ_15 |
| Lowest reported value (Cell 2) | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_15 |
| Lowest reported value (Cell 3) | RSRQ_00 |
| Highest reported value (Cell 3) | RSRQ_15 |

**Table 9.2.40.5-3: 3DL PCell in FDD RSRP for E-UTRAN in Carrier Aggregation relative accuracy requirements for the reported values**

| | **Event** | |
|---|---|---|
| Normal and Extreme Conditions, all bands | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRQ_x – 10 |
| Highest reported value (Cell 2) | | RSRQ_x + 10 |
| Lowest reported value (Cell 3) | Cell 1-3 | RSRQ_x – 10 |
| Highest reported value (Cell 3) | | RSRQ_x + 10 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.2.41 3DL TDD RSRQ for E-UTRAN in Carrier Aggregation

### 9.2.41.1 Test purpose

To verify that TDD absolute and relative RSRQ measurement accuracy in carrier aggregation is within the specified limits under AWGN propagation conditions. This test will verify the absolute RSRQ accuracy requirements of the primary component carrier, the absolute RSRQ accuracy requirements of the secondary component carriers, and also the relative RSRQ accuracy requirement between the primary and secondary component carriers.

### 9.2.41.2 Test applicability

This test applies to all types of E-UTRA UE TDD release 10 and forward that support 3DL with intra-band contiguous CA or 3DL with inter-band CA, or 3DL with intra-band contiguous and inter-band CA.

This test case also applies to all types of E-UTRA TDD UE release 11 and forward that support 3DL with intra-band non-contiguous and inter-band CA, or 3DLwith intra-band non-contiguous and intra-band contiguous CA.

### 9.2.41.3 Minimum conformance requirements

The TDD RSRQ measurements of cells on the primary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1. The TDD RSRQ measurements of cells on the secondary component carrier shall meet the intra frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1.

The accuracy requirements in Table 9.2.41.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.
- RSRP|dBm according to Annex I.3.1 for a corresponding Band.

**Table 9.2.41.3-1: 3DL TDD RSRQ absolute for Carrier Aggregation**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/$BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The TDD RSRQ measurements of cells on the primary component carrier are compared with measurements of cells on the secondary component carrier defined in TS 36.133 [4] clause 9.1.6.2. The applicable relative accuracy requirements are the RSRQ inter-frequency accuracy requirements as defined in TS 36.133 [4] clause 9.1.6.2.

The accuracy requirements in Table 9.2.41.3-2 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in TS 36.101 [2] clause 7.3 for reference sensitivity are fulfilled.
- RSRP1,2$|_{dBm}$ according to Annex I.3.4 for a corresponding Band.
- $\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$
- |Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.2.41.3-2: 3DL TDD RSRQ relative for Carrier Aggregation**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot Note 2 | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 5 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 4 | dBm/BW$_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution. The mapping of measured quantity is defined in Table 9.2.41.3-3.

**Table 9.2.41.3-3: 3DL TDD RSRQ measurement report mapping for Carrier Aggregation**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.11.1, clause 9.1.11.2, clause 9.1.11.3, clause 9.1.5.1 and A.9.2.41.

## 9.2.41.4 Test description

### 9.2.41.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 9.2.41.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1[10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.63 as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.41.4.3.

4. There are three E-UTRA TDD carriers and three cells specified in the test. Cell 1 is PCell on the primary component carrier (PCC), Cell 2 is the SCell on the Secondary Component Carrier (SCC1), and Cell 3 is the SCell on the Secondary Component Carrier (SCC2). Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 and Cell 3 shall be powered OFF.

### 9.2.41.4.2 Test procedure

The test consists of Cell 1 the primary component carrier, Cell 2 and Cell 3 are the SCells on the secondary component carrier SCC1 and SCC2, respectively. The SCell1 (Cell 2) and SCell2 (Cell 3) on the SCC1 and SCC2 are configured. The SCell1 (Cell 2) on the SCC1 is activated and SCell2 (Cell 3) on the SCC2 is activated. The absolute accuracy of RSRQ is defined as the RSRQ measured from the primary component carrier (Cell 1) and the RSRQ measured from the secondary component carrier SCC1 (Cell 2) and secondary component carrier SCC2 (Cell 3). The relative accuracy of RSRQ is defined as the RSRQ measured cells on the primary component carrier (Cell 1) compared with measurements of cells on the secondary component carrier SCC1 (Cell 2) and secondary component carrier SCC2 (Cell 3),

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCCs according to Annex C.0, C.1 and C.2 for all downlink physical channels.

3. The SS shall configure the SCell1 (Cell 2) on the SCC1 and SCell2 (Cell 3) on the SCC2 as per TS 36.508 [7] clause 5.2A.4.

4. SS activates SCC1 and SCC2 by sending the MAC-CE according to TS 36.321 [11] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.2.41.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

6. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

7. The UE shall transmit RRCConnectionReconfigurationComplete message.

8. UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6; SS shall check the reported RSRQ values in periodical MeasurementReport messages.

   The reported RSRQ values for Cell 1, Cell 2 and Cell 3 by the UE are compared to the actual RSRQ values according to Table 9.2.41.5-2. These counts respectively as a Pass or Fail for the events “Cell 1”, “Cell 2” and “Cell 3”. If the UE fails to report the measurement value for Cell 2 or Cell 3, the number of failed iterations for the respective event is increased by one.

   The reported RSRQ value for Cell 2 and Cell 3 are compared to the reported RSRQ value for Cell 1 for each MeasurementReport message according to Table 9.2.41.5-3. These count respectively as a Pass or Fail for the events “Cell 1-2” and “Cell 1-3”. If the UE fails to report the measurement value for Cell 2 or Cell 3, the number of failed iterations for the respective affected event is increased by one.

10. The SS shall check MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events “Cell 1”, “Cell 2”, “Cell 3”, “Cell 1-2”, and “Cell 1-3” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.

If all events pass for each scenario, the test passes. If one event fails, the test fails.

### 9.2.41.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.2.41.4.3-1: Common Exception messages for 3DL TDD RSRQ absolute for Carrier Aggregation**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-4<br>Table H.4.1-8<br>Table H.4.2-6 |

## 9.2.41.5 Test requirement

Table 9.2.41.5-1 defines the primary level settings including test tolerances for all tests.

The TDD RSRQ absolute accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.41.5-2.

The TDD RSRQ relative accuracy for carrier aggregation test shall meet the reported values test requirements in Table 9.2.41.5-3.

**Table 9.2.41.5-1: Cell Specific Test requirement Parameters for 3DL TDD RSRQ absolute for Carrier Aggregation**

| Parameter | | Unit | Cell 1 | Cell2 | Cell3 |
|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | 2 | 3 |
| $BW_{channel}$ | | MHz | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 |
| Special subframe configuration Note1 | | | 6 | | |
| Uplink-downlink configuration Note1 | | | 1 | | |
| Measurement bandwidth | | $n_{PRB}$ | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 |
| PDSCH Reference measurement channel defined in A.1.2 | | | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD |
| PDSCH allocation | | $n_{PRB}$ | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2 | | | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz:R.11 TDD<br>10MHz:R.6 TDD<br>20MHz:R.10 TDD | 5MHz:R.11 TDD<br>10MHz:R.6 TDD<br>20MHz:R.10 TDD |
| OCNG Patterns defined in D.2.1, D.2.7, D.2.9 | | | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 |
| PBCH_RB | | | | | |
| PSS_RA | | | | | |
| SSS_RA | | | | | |
| PCFICH_RB | | | | | |
| PHICH_RA | | | | | |
| PHICH_RB | | | | | |
| PDCCH_RA | | | | | |
| PDCCH_RB | | | | | |
| PDSCH_RA | | | | | |
| PDSCH_RB | | | | | |
| OCNG_RA Note2 | | | | | |
| OCNG_RB Note2 | | | | | |
| $N_{oc}$ Note3 | Bands TDD_A | dBm/15 kHz | -119.2 | -116 | -116 |
| | Bands TDD_C | | -118.2 | -115 | -115 |
| | Bands TDD_E | | -117.2 | -114 | -114 |
| $\hat{E}_s / N_{oc}$ | | dB | -5.20 | -5.20 | -5.20 |
| $\hat{E}_s / I_{ot}$ Note4 | | dB | -5.20 | -5.20 | -5.20 |
| RSRP Note4 | Bands TDD_A | dBm/15 kHz | -124.4 | -121.2 | -121.2 |
| | Bands TDD_C | | -123.4 | -120.2 | -120.2 |
| | Bands TDD_E | | -122.4 | -119.2 | -119.2 |
| RSRQ Note4 | Bands TDD_A, TDD_C, TDD_E | dB | -17.14 | -17.14 | -17.14 |
| Io Note4 | Bands TDD_A | dBm/ $BW_{channel}$ | -90.27+ 10log($N_{RB,c}$ /50) | -87.07+ 10log($N_{RB,c}$ /50) | -87.07+ 10log($N_{RB,c}$ /50) |
| | Bands TDD_C | | -89.27+ 10log($N_{RB,c}$ /50) | -86.07+ 10log($N_{RB,c}$ /50) | -86.07+ 10log($N_{RB,c}$ /50) |
| | Bands TDD_E | | -88.27+ 10log($N_{RB,c}$ /50) | -85.07+ 10log($N_{RB,c}$ /50) | -85.07+ 10log($N_{RB,c}$ /50) |
| Propagation condition | | - | AWGN | AWGN | AWGN |

| | | | | |
|---|---|---|---|---|
| Antenna Configuration | - | 1x2 | 1x2 | 1x2 |
| Timing offset to Cell 1 | μs | - | 0 | 0 |
| Time alignment error relative to cell 1 Note 7 | | - | ≤ TAE | ≤ TAE |
| Time alignment error relative to cell 2 Note 7 | | - | - | ≤ TAE |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211 [9].
Note 2: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: Es/Iot, RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 6: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.
Note 7: Time alignment error (TAE) as specified in TS 36.104 [29] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.
Note 8: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.2.41.5-2: TDD RSRQ absolute requirements for the reported values for Carrier Aggregation**

| | Event | All bands |
|---|---|---|
| Normal Conditions | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRQ_00 |
| Highest reported value (Cell 1) | | RSRQ_14 |
| Lowest reported value (Cell 2) | Cell 2 | RSRQ_00 |
| Highest reported value (Cell 2) | | RSRQ_14 |
| Lowest reported value (Cell3) | Cell 3 | RSRQ_00 |
| Highest reported value (Cell 3) | | RSRQ_14 |
| Extreme Conditions | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRQ_00 |
| Highest reported value (Cell 1) | | RSRQ_15 |
| Lowest reported value (Cell 2) | Cell 2 | RSRQ_00 |
| Highest reported value (Cell 2) | | RSRQ_15 |
| Lowest reported value (Cell 3) | Cell 3 | RSRQ_00 |
| Highest reported value (Cell 3) | | RSRQ_15 |

**Table 9.2.41.5-3: 3DL TDD RSRQ relative requirements for the reported values for Carrier Aggregation**

| | Event | All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRQ_x – 10 |
| Highest reported value (Cell 2) | | RSRQ_x + 10 |
| Lowest reported value (Cell 3) | Cell 1-3 | RSRQ_x – 10 |
| Highest reported value (Cell 3) | | RSRQ_x + 10 |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRQ_x – 10 |
| Highest reported value (Cell 2) | | RSRQ_x + 10 |
| Lowest reported value (Cell 3) | Cell 1-3 | RSRQ_x – 10 |
| Highest reported value (Cell 3) | | RSRQ_x + 10 |
| RSRQ_x is the reported value of Cell 1 | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.2.42 FD-FDD RSRQ Intra frequency case for UE category 0

## 9.2.42.1 FD-FDD Intra Frequency Absolute RSRQ Accuracy for UE category 0

### 9.2.42.1.1 Test purpose

To verify the FD-FDD intra frequency absolute RSRQ measurement accuracy is within the specified limit for all bands.

### 9.2.42.1.2 Test applicability

This test applies to all types of E-UTRA FD-FDD Category 0 UE from release 12 and forward. Applicability requires support for FGI bit 16.

### 9.2.42.1.3 Minimum conformance requirements

The requirements for absolute accuracy of RSRQ in this clause apply to a cell on the same frequency as that of the serving cell for category 0 UE.

The accuracy requirements in Table 9.2.42.1.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101 Clause 7.3 for reference sensitivity are fulfilled.
- RSRP|dBm according to Annex B.3.1 for a corresponding Band.
- At least 1 DL subframe per radio frame of measured cell is available at the UE for RSRQ measurement assuming measured cell is identified cell.

**Table 9.2.42.1.3-1: RSRQ Intra frequency absolute accuracy for UE category 0**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/BW$_{Channel}$ |
| ±3.5 | ±5 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4.5 | ±5 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.13.3 and A.9.2.42.1.

### 9.2.42.1.4 Test description

#### 9.2.42.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20 using only main UE Tx/Rx antenna.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.2.42.1.4.3.
4. All cells are in the same carrier frequency. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.2.42.1.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Table 9.2.42.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.
3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. UE shall transmit periodically MeasurementReport messages.
6. After 10s wait from Step 3, SS shall check RSRQ reported value in periodical MeasurementReport messages. The RSRQ value of Cell 2 reported by the UE is compared to the actual RSRQ value according to Table 9.2.42.1.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.
7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.
8. Repeat step 1-7 for each sub-test in Table 9.2.42.1.5-1 as appropriate.

#### 9.2.42.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.2.42.1.4.3-1: Common Exception messages for RSRQ FD-FDD Intra frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-2<br>Table H.3.5-4 |

**Table 9.2.42.1.4.3-2: *MeasResults*: Additional RSRQ FD-FDD Intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement identity for which the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | Set according to specific test | |
| rsrqResult | INTEGER(0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.42.1.4.3-3: *MeasResultListEUTRA*: Additional RSRQ FD-FDD Intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId from Cell2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

## 9.2.42.1.5 Test requirement

Table 9.2.42.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRQ FDD intra-frequency absolute accuracy test shall meet the reported values test requirements in table 9.2.42.1.5-2.

**Table 9.2.42.1.5-1: FD-FDD RSRQ Intra frequency test parameters for UE category 0**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.3.1.1.3 | | | R.13 FDD | - | R.13 FDD | - | R.13 FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.1 | | | R.6 FDD | | R.6 FDD | | R.6 FDD | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1 FDD) and A.3.2.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | |
| PSS_RA | | | | | | | | |
| SSS_RA | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA Note1 | | | | | | | | |
| OCNG_RB Note1 | | | | | | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -85.81 | | -103.85 | | -116 | |
| | Bands FDD_B | | | | | | -115.5 | |
| | Bands FDD_C | | | | | | -115 | |
| | Bands FDD_D | | | | | | -114.5 | |
| | Bands FDD_E, FDD_F Note 5 | | | | | | -114 | |
| | Bands FDD_G Note 7 | | | | | | -113 | |
| | Bands FDD_H | | | | | | -112.5 | |
| $\hat{E}_s / N_{oc}$ | | dB | 3 | 3 | -2.9 | -2.9 | -3.6 | -3.6 |
| $\hat{E}_s / I_{ot}$ | | dB | - 1.76 | -1.76 | -4.7 | -4.7 | -5.17 | -5.17 |
| RSRP Note3 | Bands FDD_A | dBm/15 kHz | -82.81 | -82.81 | -106.75 | -106.75 | -119.60 | -119.60 |
| | Bands FDD_B | | | | | | -119.10 | -119.10 |
| | Bands FDD_C | | | | | | -118.60 | -118.60 |
| | Bands FDD_D | | | | | | -118.10 | -118.10 |
| | Bands FDD_E, FDD_F Note 4 | | | | | | -117.60 | -117.60 |
| | Bands FDD_G Note 6 | | | | | | -116.60 | -116.60T |
| | Bands FDD_H | | | | | | -116.10 | -116.10 |
| RSRQ Note3 | Bands FDD_A | dB | -14.77 | -14.77 | -16.76 | -16.76 | -17.12 | -17.12 |
| | Bands FDD_B | | | | | | | |
| | Bands FDD_C | | | | | | | |
| | Bands FDD_D | | | | | | | |
| | Bands FDD_E, FDD_F Note 45 | | | | | | | |
| | Bands FDD_G Note 6 | | | | | | | |
| | Bands FDD_H | | | | | | | |
| Io Note3 | Bands FDD_A | dBm/9 MHz | -51.05 | | -73 | | -85.49 | |
| | Bands FDD_B | | | | | | -84.99 | |
| | Bands FDD_C | | | | | | -84.49 | |
| | Bands FDD_D | | | | | | -83.99 | |
| | Bands FDD_E, | | | | | | -83.49 | |

| | | | | | |
|---|---|---|---|---|---|
| | FDD_F Note 4 | | | | |
| | Bands FDD_G Note 6 | | | | -82.49 |
| | Bands FDD_H | | | | -81.99 |
| Propagation condition | | - | AWGN | AWGN | AWGN |
| Correlation Matrix and Antenna Configuration | | | 1x1 | 1x1 | 1x1 |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: Es/Iot, RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 5: E-UTRA operating band groups are as defined in Section 3.5.
Note 6: Except Band 29 and Band 32.

**Table 9.2.42.1.5-2: RSRQ FD-FDD Intra frequency absolute accuracy requirements for the reported values for UE category 0**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| | All bands | All bands | All bands |
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_02 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_18 | RSRQ_16 | RSRQ_16 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_00 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_21 | RSRQ_17 | RSRQ_17 |

# 9.2.43 HD-FDD RSRQ Intra frequency case for UE category 0

## 9.2.43.1 HD-FDD Intra Frequency Absolute RSRQ Accuracy for UE category 0

### 9.2.43.1.1 Test purpose

To verify the HD-FDD intra frequency absolute RSRQ measurement accuracy is within the specified limit for all bands.

### 9.2.43.1.2 Test applicability

This test applies to all types of E-UTRA HD-FDD Category 0 UE from release 12 and forward. Applicability requires support for FGI bit 16.

### 9.2.43.1.3 Minimum conformance requirements

The requirements for absolute accuracy of RSRQ in this clause apply to a cell on the same frequency as that of the serving cell for category 0 UE.

The accuracy requirements in Table 9.2.43.1.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101 Clause 7.3 for reference sensitivity are fulfilled.
- RSRP|dBm according to Annex B.3.1 for a corresponding Band.
- At least 1 DL subframe per radio frame of measured cell is available at the UE for RSRQ measurement assuming measured cell is identified cell.

**Table 9.2.43.1.3-1: RSRQ Intra frequency absolute accuracy for UE category 0**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/$BW_{Channel}$ |
| ±3.5 | ±5 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4.5 | ±5 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.13.3 and A.9.2.43.1.

## 9.2.43.1.4 Test description

### 9.2.43.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20 using only main UE Tx/Rx antenna.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.2.43.1.4.3.
4. All cells are in the same carrier frequency. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

### 9.2.43.1.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Table 9.2.43.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.
3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. UE shall transmit periodically MeasurementReport messages.
6. After 10s wait from Step 3, SS shall check RSRQ reported value in periodical MeasurementReport messages. The RSRQ value of Cell 2 reported by the UE is compared to the actual RSRQ value according to Table 9.2.43.1.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.2.43.1.5-1 as appropriate.

#### 9.2.43.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.2.43.1.4.3-1: Common Exception messages for RSRQ HD-FDD Intra frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-2<br>Table H.3.5-4 |

**Table 9.2.43.1.4.3-2: *MeasResults*: Additional RSRQ HD-FDD Intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement identity for which the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | Set according to specific test | |
| rsrqResult | INTEGER(0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.43.1.4.3-3: *MeasResultListEUTRA*: Additional RSRQ HD-FDD Intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId from Cell2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

### 9.2.43.1.5 Test requirement

Table 9.2.43.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRQ FDD intra-frequency absolute accuracy test shall meet the reported values test requirements in table 9.2.43.1.5-2.

**Table 9.2.43.1.5-1: HD-FDD RSRQ Intra frequency test parameters for UE category 0**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.3.1.1.4 | | | R.1 HD-FDD | - | R.1 HD-FDD | - | R.1 HD-FDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.3 | | | R.3 HD-FDD | | R.3 HD-FDD | | R.3 HD-FDD | |
| OCNG Patterns defined in A.3.2.1.1 (OP.1 FDD) and A.3.2.1.2 (OP.2 FDD) | | | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD | OP.1 FDD | OP.2 FDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | | | |
| PSS_RA | | | | | | | | |
| SSS_RA | | | | | | | | |
| PCFICH_RB | | | | | | | | |
| PHICH_RA | | | | | | | | |
| PHICH_RB | | | | | | | | |
| PDCCH_RA | | | | | | | | |
| PDCCH_RB | | | | | | | | |
| PDSCH_RA | | | | | | | | |
| PDSCH_RB | | | | | | | | |
| OCNG_RA[Note1] | | | | | | | | |
| OCNG_RB[Note1] | | | | | | | | |
| $N_{oc}$ [Note2] | Bands FDD_A | dBm/15 kHz | -85.81 | | -103.85 | | -116 | |
| | Bands FDD_B | | | | | | -115.5 | |
| | Bands FDD_C | | | | | | -115 | |
| | Bands FDD_D | | | | | | -114.5 | |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -114 | |
| | Bands FDD_G [Note 7] | | | | | | -113 | |
| | Bands FDD_H | | | | | | -112.5 | |
| $\hat{E}_s / N_{oc}$ | | dB | 3 | 3 | -2.9 | -2.9 | -3.6 | -3.6 |
| $\hat{E}_s / I_{ot}$ | | dB | -1.76 | -1.76 | -4.7 | -4.7 | -5.17 | -5.17 |
| RSRP[Note3] | Bands FDD_A | dBm/15 kHz | -82.81 | -82.81 | -106.75 | -106.75 | -119.60 | -119.60 |
| | Bands FDD_B | | | | | | -119.10 | -119.10 |
| | Bands FDD_C | | | | | | -118.60 | -118.60 |
| | Bands FDD_D | | | | | | -118.10 | -118.10 |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | -117.60 | -117.60 |
| | Bands FDD_G [Note 6] | | | | | | -116.60 | -116.60 |
| | Bands FDD_H | | | | | | -116.10 | -116.10 |
| RSRQ[Note3] | Bands FDD_A | dB | -14.77 | -14.77 | -16.76 | -16.76 | -17.12 | -17.12 |
| | Bands FDD_B | | | | | | | |
| | Bands FDD_C | | | | | | | |
| | Bands FDD_D | | | | | | | |
| | Bands FDD_E, FDD_F [Note 4] | | | | | | | |
| | Bands FDD_G [Note 6] | | | | | | | |
| | Bands FDD_H | | | | | | | |
| Io[Note3] | Bands FDD_A | dBm/9 MHz | -51.05 | | -73 | | -85.49 | |
| | Bands FDD_B | | | | | | -84.99 | |
| | Bands FDD_C | | | | | | -84.49 | |
| | Bands FDD_D | | | | | | -83.99 | |
| | Bands FDD_E, | | | | | | -83.49 | |

| | | | | | |
|---|---|---|---|---|---|
| | FDD_F Note 4 | | | | |
| | Bands FDD_G Note 6 | | | | -82.49 |
| | Bands FDD_H | | | | -81.99 |
| Propagation condition | | - | AWGN | AWGN | AWGN |
| Correlation Matrix and Antenna Configuration | | | 1x1 | 1x1 | 1x1 |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: Es/Iot, RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 5: E-UTRA operating band groups are as defined in Section 3.5.
Note 6: Except Band 29 and Band 32.

**Table 9.2.43.1.5-2: RSRQ FD-FDD Intra frequency absolute accuracy requirements for the reported values for UE category 0**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| | All bands | All bands | All bands |
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_02 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_18 | RSRQ_16 | RSRQ_16 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_00 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_21 | RSRQ_17 | RSRQ_17 |

# 9.2.44 TDD RSRQ Intra frequency case for UE category 0

## 9.2.44.1 TDD Intra Frequency Absolute RSRQ Accuracy for UE category 0

*Editor's notes: This test is incomplete. The following aspects are either missing or not yet determined:*

### 9.2.44.1.1 Test purpose

To verify the TDD intra frequency absolute RSRQ measurement accuracy is within the specified limit for all bands.

### 9.2.44.1.2 Test applicability

This test applies to all types of E-UTRA TDD Category 0 UE from release 12 and forward. Applicability requires support for FGI bit 16.

### 9.2.44.1.3 Minimum conformance requirements

The requirements for absolute accuracy of RSRQ in this clause apply to a cell on the same frequency as that of the serving cell for category 0 UE.

The accuracy requirements in Table 9.2.44.1.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
- Conditions defined in 36.101 Clause 7.3 for reference sensitivity are fulfilled.
- RSRP|dBm according to Annex B.3.1 for a corresponding Band.
- At least 1 DL subframe per radio frame of measured cell is available at the UE for RSRQ measurement assuming measured cell is identified cell.

**Table 9.2.44.1.3-1: RSRQ Intra frequency absolute accuracy for UE category 0**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 4] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 3] | dBm/$BW_{Channel}$ |
| ±3.5 | ±5 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_B | -120.5 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4.5 | ±5 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

The normative reference for this requirement is TS 36.133 [4] clause 9.1.13.3 and A.9.2.44.1.

## 9.2.44.1.4 Test description

### 9.2.44.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.20 using only main UE Tx/Rx antenna.
2. Propagation conditions are set according to Annex B clause B.0.
3. Message contents are defined in clause 9.2.44.1.4.3.
4. All cells are in the same carrier frequency. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

### 9.2.44.1.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.
2. Set the parameters according to Table 9.2.44.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.
3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.
4. The UE shall transmit RRCConnectionReconfigurationComplete message.
5. UE shall transmit periodically MeasurementReport messages.
6. After 10s wait from Step 3, SS shall check RSRQ reported value in periodical MeasurementReport messages. The RSRQ value of Cell 2 reported by the UE is compared to the actual RSRQ value according to Table 9.2.44.1.5-2. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.2.44.1.5-1 as appropriate.

### 9.2.44.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.2.44.1.4.3-1: Common Exception messages for RSRQ TDD Intra frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-2<br>Table H.3.5-4 |

**Table 9.2.44.1.4.3-2: *MeasResults*: Additional RSRQ TDD Intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement identity for which the reporting being performed | |
| measResultPCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | Set according to specific test | |
| rsrqResult | INTEGER(0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListEUTRA | MeasResultListEUTRA | | |
| } | | | |
| } | | | |

**Table 9.2.44.1.4.3-3: *MeasResultListEUTRA*: Additional RSRQ TDD Intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListEUTRA::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId | PhysCellId from Cell2 | | |
| measResult SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | Not present | | |
| } | | | |
| } | | | |

## 9.2.44.1.5 Test requirement

Table 9.2.44.1.5-1 defines the primary level settings including test tolerances for all tests.

**Each RSRQ TDD intra-frequency absolute accuracy test shall meet the reported values test requirements in table 9.2.44.1.5-2.Table 9.2.44.1.5-1: TDD RSRQ Intra frequency test parameters for UE category 0**

| Parameter | | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|---|
| | | | Cell 1 | Cell 2 | Cell 1 | Cell 2 | Cell 1 | Cell 2 |
| E-UTRA RF Channel Number | | | 1 | | 1 | | 1 | |
| $BW_{channel}$ | | MHz | 10 | | 10 | | 10 | |
| Special subframe configuration[Note1] | | | 6 | | 6 | | 6 | |
| Uplink-downlink configuration[Note1] | | | 1 | | 1 | | 1 | |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 | | 22—27 | | 22—27 | |
| PDSCH Reference measurement channel defined in A.3.1.1.5 | | | R.12 TDD | - | R.12 TDD | - | R.12 TDD | - |
| PDSCH allocation | | $n_{PRB}$ | 13—36 | - | 13—36 | - | 13—36 | - |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.2 | | | R.6 TDD | | R.6 TDD | | R.6 TDD | |
| OCNG Patterns defined in A.3.2.2.1 (OP.1 TDD) and A.3.2.2.2 (OP.2 TDD) | | | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD | OP.1 TDD | OP.2 TDD |
| PBCH_RA<br>PBCH_RB<br>PSS_RA<br>SSS_RA<br>PCFICH_RB<br>PHICH_RA<br>PHICH_RB<br>PDCCH_RA<br>PDCCH_RB<br>PDSCH_RA<br>PDSCH_RB<br>OCNG_RA[Note2]<br>OCNG_RB[Note2] | | dB | 0 | 0 | 0 | 0 | 0 | 0 |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -85.81 | | -103.85 | | -116 | |
| | Bands TDD_C | | | | | | -115 | |
| | Bands TDD_E | | | | | | -114 | |
| $\hat{E}_s / N_{oc}$ | | dB | 3 | 3 | -2.9 | -2.9 | -3.6 | -3.6 |
| $\hat{E}_s / I_{ot}$ | | dB | -1.76 | -1.76 | -4.7 | -4.7 | -5.17 | -5.17 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -82.81 | -82.81 | -106.75 | -106.75 | -119.60 | -119.60 |
| | Bands TDD_C | | | | | | -118.60 | -118.60 |
| | Bands TDD_E | | | | | | -117.60 | -117.60 |
| RSRQ[Note4] | Bands TDD_A, TDD_C, TDD_E | dB | -14.77 | -14.77 | -16.76 | -16.76 | -17.12 | -17.12 |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -51.05 | | -73 | | -85.49 | |
| | Bands TDD_C | | | | | | -84.49 | |
| | Bands TDD_E | | | | | | -83.49 | |
| Propagation condition | | - | AWGN | | AWGN | | AWGN | |
| Correlation Matrix and Antenna Configuration | | | 1x1 | | 1x1 | | 1x1 | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRQ, RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 6: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.2.44.1.5-2: RSRQ FD-FDD Intra frequency absolute accuracy requirements for the reported values for UE category 0**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| | All bands | All bands | All bands |
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_02 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_18 | RSRQ_16 | RSRQ_16 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | RSRQ_00 | RSRQ_00 | RSRQ_00 |
| Highest reported value (Cell 2) | RSRQ_21 | RSRQ_17 | RSRQ_17 |

## 9.2.45 4 DL CA PCell in FDD FDD-TDD RSRQ for E-UTRAN in Carrier Aggregation

Editor's Note: This test case is incomplete. The following items are missing or are incomplete:

- Test tolerances are undefined

- Connection Diagram is undefined in TS 36.508

- Message contents are undefined

- The test applicability needs to be added to TS36.521-2

### 9.2.45.1 Test purpose

To verify that TDD-FDD RSRQ absolute and relative measurement accuracy in carrier aggregation with PCell in FDD are within the specified limits under AWGN propagation conditions. This test will verify the absolute accuracy of intra-frequency RSRQ measurements for the primary component carrier, the absolute accuracy of intra-frequency RSRQ measurements for the secondary component carrier and also the relative inter-frequency RSRQ accuracy requirement between primary and secondary component carriers.

### 9.2.45.2 Test applicability

This test applies to all types of E-UTRA FDD and TDD UE Release 12 and forward that support 4DL TDD-FDD CA with FDD as PCell.

### 9.2.45.3 Minimum conformance requirements

The RSRQ measurements of cells on the primary component carrier shall meet the intra-frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1. The RSRQ measurements of cells on any of the secondary component carrier(s) shall meet the intra-frequency absolute accuracy requirements defined in TS 36.133 [4] clause 9.1.5.1.

The accuracy requirements in Table 9.2.45.3-1 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.

- Conditions defined in 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.

- RSRP|dBm according to Annex I.3.1 for a corresponding Band

**Table 9.2.45.3-1: RSRQ Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 4] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 3] | dBm/$BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 3: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 4: E-UTRA operating band groups are as defined in Section 3.5.

When measurements of cells on the primary component carrier are compared with measurements of cells on any of the secondary component carrier(s), the applicable relative accuracy requirements are the RSRQ inter-frequency accuracy requirements defined in TS 36.133 [4] clause 9.1.6.2.

The accuracy requirements in Table 9.2.45.3-2 are valid under the following conditions:

- Cell specific reference signals are transmitted either from one, two or four antenna ports.
  - Conditions defined in 36.101 [2] Clause 7.3 for reference sensitivity are fulfilled.
  - $RSRP1,2|_{dBm}$ according to Annex I.3.4 for a corresponding Band
  - $\left|RSRP1\right|_{dBm} - RSRP2\left|_{dBm}\right| \leq 27dB$
  - | Channel 1_Io -Channel 2_Io | ≤ 20 dB

**Table 9.2.45.3-2: RSRQ Inter frequency relative accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot [Note 2] | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 5] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 4] | dBm/$BW_{Channel}$ |
| ±3 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±4 | ±4 | ≥-6 dB | Note 3 | Note 3 | Note 3 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The parameter Ês/Iot is the minimum Ês/Iot of the pair of cells to which the requirement applies.
NOTE 3: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
NOTE 4: The condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 5: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -34 dB to 2.5 dB with 0.5 dB resolution. The mapping of measured quantity is defined in table 9.1.7-1. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.2.45.3-3: RSRQ measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_-30 | RSRQ < -34 | dB |
| RSRQ_-29 | -34 ≤ RSRQ < -33.5 | dB |
| … | … | … |
| RSRQ_-02 | -20.5 ≤ RSRQ < -20 | dB |
| RSRQ_-01 | -20 ≤ RSRQ < -19.5 | dB |
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |
| RSRQ_35 | -3 ≤ RSRQ < -2.5 | dB |
| RSRQ_36 | -2.5 ≤ RSRQ < -2 | dB |
| … | … | … |
| RSRQ_45 | 2 ≤ RSRQ < 2.5 | dB |
| RSRQ_46 | 2.5 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clauses 9.1.5.1, 9.1.6.2, 9.1.7, 9.1.11, and A.9.2.45.

## 9.2.45.4 Test description

### 9.2.45.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-2 and TS 36.508 [7] clauses 4.4.2 and 4.3.1 for different CA bandwidth classes.

Channel Bandwidth to be tested: The largest aggregated bandwidth combination supported by UE from Table 9.2.45.5-1 as defined in TS 36.508 [7] clause 4.3.1 for different CA configurations as defined in TS 36.521-1 [10] clause 5.4.2A.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure TBD as appropriate.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.2.45.4.3.

4. Cell 1 is PCell in FDD on the primary component carrier, Cell 2 is SCell on SCC1, Cell 3 is SCell on SCC2, and Cell 4 is SCell on SCC3. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test. Cell 2, Cell 3, Cell 4 shall be powered OFF.

### 9.2.45.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Configure SCells according to Annex C.0 and C.1 for all downlink physical channels.

3. The SS shall configure SCells (Cell 2 and Cell 3) on the SCCs as per TS 36.508 [7] clause 5.2A.4.

4. The SS activates SCCs by sending the MAC-CE according to TS 36.321 [11] clauses 5.13 and 6.1.3.8. Wait for at least 2 seconds as per TS 36.133 [4] clause 8.3.3.2.

5. Set the parameters according to Table 9.2.45.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.

6. The SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

7. The UE shall transmit an RRCConnectionReconfigurationComplete message.

8. The UE shall transmit periodically MeasurementReport messages.

9. After 10s wait from Step 6, the SS shall check the reported RSRQ values in periodical MeasurementReport messages.

The reported RSRQ values for Cell 1, Cell 2, Cell 3, and Cell 4 are compared to the actual RSRQ values according to Table 9.2.45.5-2. This counts respectively as a Pass or Fail for the events “Cell 1”, “Cell 2”, “Cell 3”, and “Cell 4”. If the UE fails to report the measurement value for Cell 1, Cell 2, Cell 3, or Cell 4, the number of failed iterations for the respective event is increased by one.

The reported RSRQ values for Cell 2, Cell 3 and Cell 4 are compared to the reported RSRQ value for Cell 1 for each MeasurementReport message according to Table 9.2.45.5-3. These count respectively as a Pass or Fail for the events “Cell 1-2”, “Cell 1-3”, and “Cell 1-4”. If the UE fails to report the measurement value for Cell 2, Cell 3, or Cell 4, the number of failed iterations for the respective event is increased by one.

10. The SS shall check the MeasurementReport messages transmitted by the UE until a test verdict has been achieved.

Each of the events “Cell 1”, “Cell 2”, “Cell 3”, “Cell 4”, “Cell 1-2”, “Cell 1-3”, and “Cell 1-4” is evaluated independently for the statistic, resulting in an event verdict: pass or fail. Each event is evaluated only until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved. Different events may require different times for a verdict.

If all events pass for each scenario, the test passes. If one event fails, the test fails.

### 9.2.45.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.2.45.4.3-1: Common Exception messages for 4DL PCell in FDD RSRQ for E-UTRAN Carrier Aggregation test requirement**

**TBD**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-4<br>Table H.4.1-8<br>Table H.4.2-6 |

## 9.2.45.5 Test requirement

Table 9.2.45.5-1 defines the primary level settings including test tolerances for all tests.

The RSRQ absolute accuracy shall meet the reported values test requirements in Table 9.2.45.5-2.

The RSRQ relative accuracy shall meet the reported values test requirements in Table 9.2.45.5-3.

**Table 9.2.45.5-1: 4 DL PCell in FDD RSRQ for E-UTRAN in Carrier Aggregation test parameters (Cell #1, Cell #2, Cell #3 and Cell #4)**

| Parameter | | Unit | Cell 1 | Cell 2 | Cell 3 | Cell 4 |
|---|---|---|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 | 2 | 3 | 4 |
| $BW_{channel}$ | | | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 | 5MHz: $N_{RB,c}$ = 25<br>10MHz: $N_{RB,c}$ = 50<br>20MHz: $N_{RB,c}$ = 100 |
| Special subframe configuration[Note1] | | | - | 6 | 6 | 6 |
| Uplink/downlink configuration[Note1] | | | - | 1 | 1 | 1 |
| Measurement bandwidth | | $n_{PRB}$ | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 | 5MHz: 10-15<br>10MHz: 22-27<br>20MHz: 47-52 |
| PDSCH Reference measurement channel defined in A.3.1.1 | | | 5MHz: R.5 FDD<br>10MHz: R.0 FDD<br>20MHz: R.4 FDD | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD | 5MHz: R.4 TDD<br>10MHz: R.0 TDD<br>20MHz: R.3 TDD |
| PDSCH allocation | | $n_{PRB}$ | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 | 5MHz: 7-17<br>10MHz: 13-36<br>20MHz: 38-61 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2 | | | 5MHz: R.11 FDD<br>10MHz: R.6 FDD<br>20MHz: R.10 FDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD | 5MHz: R.11 TDD<br>10MHz: R.6 TDD<br>20MHz: R.10 TDD |
| OCNG Patterns defined in A.3.2 | | | 5MHz: OP.15 FDD<br>10MHz: OP.1 FDD<br>20MHz: OP.11 FDD | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD | 5MHz: OP.9 TDD<br>10MHz: OP.1 TDD<br>20MHz: OP.7 TDD |
| PBCH_RA | | dB | 0 | 0 | 0 | 0 |
| PBCH_RB | | | | | | |
| PSS_RA | | | | | | |
| SSS_RA | | | | | | |
| PCFICH_RB | | | | | | |
| PHICH_RA | | | | | | |
| PHICH_RB | | | | | | |
| PDCCH_RA | | | | | | |
| PDCCH_RB | | | | | | |
| PDSCH_RA | | | | | | |
| PDSCH_RB | | | | | | |
| OCNG_RA[Note2] | | | | | | |
| OCNG_RB[Note2] | | | | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/ 15kHz | - | -116+TT | -116+TT | -116+TT |
| | Bands TDD_C | | - | -115+TT | -115+TT | -115+TT |
| | Bands TDD_E | | - | -114+TT | -114+TT | -114+TT |
| | Bands FDD_A | | -119.5+TT | - | - | - |
| | Bands FDD_B | | -119+TT | - | - | - |
| | Bands FDD_C | | -118.5+TT | - | - | - |
| | Bands FDD_D | | -118+TT | - | - | - |
| | Bands FDD_E, Bands FDD_F [Note 6] | | -117.5+TT | - | - | - |
| | Bands | | -116.5+TT | - | - | - |

| | | | | | | |
|---|---|---|---|---|---|---|
| | FDD_G | | | | | |
| | Bands FDD_H | | -116+TT | - | - | - |
| $\hat{E}_s / N_{oc}$ | | dB | -6.0+TT | -6.0+TT | -6.0+TT | -6.0+TT |
| $\hat{E}_s / I_{ot}$ | | dB | -6.0+TT | -6.0+TT | -6.0+TT | -6.0+TT |
| RSRP Note4 | Bands TDD_A | dBm/ 15kHz | - | -122+TT | -122+TT | -122+TT |
| | Bands TDD_C | | - | -121+TT | -121+TT | -121+TT |
| | Bands TDD_E | | - | -120+TT | -120+TT | -120+TT |
| | Bands FDD_A | | -125.5+TT | - | - | - |
| | Bands FDD_B | | -125+TT | - | - | - |
| | Bands FDD_C | | -124.5+TT | - | - | - |
| | Bands FDD_D | | -124+TT | - | - | - |
| | Bands FDD_E, Bands FDD_F Note 6 | | -123.5+TT | - | - | - |
| | Bands FDD_G | | -122.5+TT | - | - | - |
| | Bands FDD_H | | -122+TT | - | - | - |
| RSRQ Note4 | Bands TDD_A | dB | - | -17.77+TT | -17.77+TT | -17.77+TT |
| | Bands TDD_C | | | | | |
| | Bands TDD_E | | | | | |
| | Bands FDD_A | | -17.77+TT | - | - | - |
| | Bands FDD_B | | | | | |
| | Bands FDD_C | | | | | |
| | Bands FDD_D | | | | | |
| | Bands FDD_E, Bands FDD_F Note 6 | | | | | |
| | Bands FDD_G | | | | | |
| | Bands FDD_H | | | | | |
| Io Note4 | Bands TDD_A | dBm/ $BW_{channel}$ | - | -87.25 + 10log($N_{RB,c}$/50) +TT | -87.25 + 10log($N_{RB,c}$/50) +TT | -87.25 + 10log($N_{RB,c}$/50) +TT |
| | Bands TDD_C | | - | -86.25 + 10log($N_{RB,c}$/50) +TT | -86.25 + 10log($N_{RB,c}$/50) +TT | -86.25 + 10log($N_{RB,c}$/50) +TT |
| | Bands TDD_E | | - | -85.25 + 10log($N_{RB,c}$/50) +TT | -85.25 + 10log($N_{RB,c}$/50) +TT | -85.25 + 10log($N_{RB,c}$/50) +TT |
| | Bands FDD_A | | -90.75 + 10log($N_{RB,c}$/50) +TT | - | - | - |
| | Bands FDD_B | | -90.25 + 10log($N_{RB,c}$/50) +TT | - | - | - |
| | Bands | | -89.75 + | - | - | - |

| | | | | | | |
|---|---|---|---|---|---|---|
| | FDD_C | | $10\log(N_{RB,c}/50)$ +TT | | | |
| | Bands FDD_D | | -89.25 + $10\log(N_{RB,c}/50)$ +TT | - | - | - |
| | Bands FDD_E, Bands FDD_F Note 6 | | -88.75 + $10\log(N_{RB,c}/50)$ +TT | - | - | - |
| | Bands FDD_G | | -87.75 + $10\log(N_{RB,c}/50)$ +TT | - | - | - |
| | Bands FDD_H | | -87.25 + $10\log(N_{RB,c}/50)$ +TT | - | - | - |
| Propagation Condition | | | AWGN | AWGN | AWGN | AWGN |
| Antenna Configuration | | | 1x2 | 1x2 | 1x2 | 1x2 |
| Timing offset to Cell 1 | | μs | - | 0 | 0 | 0 |
| Time alignment error relative to cell 1 Note 10 | | | - | ≤ TAE | ≤ TAE | ≤ TAE |
| Time alignment error relative to cell 2 Note 10 | | | - | - | ≤ TAE | ≤ TAE |
| Time alignment error relative to cell 3 Note 10 | | | - | - | - | ≤ TAE |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
Note 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 4: RSRP, RSRQ and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 5: RSRP and RSRQ minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
Note 6: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.
Note 7: The selection of the bands for testing depends on the configuration of the carrier aggregation supported by the UEs.
Note 8: This test verifies the RRM requirement which is independent of channel bandwidth and is performed according to the principle defined in section A.3.6.1.
Note 9: E-UTRA operating band groups are as defined in Section 3.5.
Note 10: Time alignment error (TAE) as specified in TS 36.104 [30] clause 6.5.3.1. The TAE value depends upon the type of carrier aggregation.

**Table 9.2.45.5-2: RSRQ absolute accuracy requirements for the reported values for 4 DL PCell in FDD for E-UTRAN in Carrier Aggregation**

| | Event | All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRQ_[TBD] |
| Highest reported value Cell 1) | | RSRQ_[TBD] |
| Lowest reported value (Cell 2) | Cell 2 | RSRQ_[TBD] |
| Highest reported value (Cell 2) | | RSRQ_[TBD] |
| Lowest reported value (Cell 3) | Cell 3 | RSRQ_[TBD] |
| Highest reported value (Cell 3) | | RSRQ_[TBD] |
| Lowest reported value (Cell 4) | Cell 4 | RSRQ_[TBD] |
| Highest reported value (Cell 4) | | RSRQ_[TBD] |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 1) | Cell 1 | RSRQ_[TBD] |
| Highest reported value Cell 1) | | RSRQ_[TBD] |
| Lowest reported value (Cell 2) | Cell 2 | RSRQ_[TBD] |
| Highest reported value (Cell 2) | | RSRQ_[TBD] |
| Lowest reported value (Cell 3) | Cell 3 | RSRQ_[TBD] |
| Highest reported value (Cell 3) | | RSRQ_[TBD] |
| Lowest reported value (Cell 4) | Cell 4 | RSRQ_[TBD] |
| Highest reported value (Cell 4) | | RSRQ_[TBD] |

**Table 9.2.45.5-3: RSRQ relative accuracy requirements for the reported values for 4 DL PCell in FDD for E-UTRAN in Carrier Aggregation**

| | Event | All bands |
|---|---|---|
| **Normal Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRQ_x + [TBD] |
| Highest reported value (Cell 2) | | RSRQ_x + [TBD] |
| Lowest reported value (Cell 3) | Cell 1-3 | RSRQ_x + [TBD] |
| Highest reported value (Cell 3) | | RSRQ_x + [TBD] |
| Lowest reported value (Cell 4) | Cell 1-4 | RSRQ_x + [TBD] |
| Highest reported value (Cell 4) | | RSRQ_x + [TBD] |
| **Extreme Conditions** | | |
| Lowest reported value (Cell 2) | Cell 1-2 | RSRQ_x + [TBD] |
| Highest reported value (Cell 2) | | RSRQ_x + [TBD] |
| Lowest reported value (Cell 3) | Cell 1-3 | RSRQ_x + [TBD] |
| Highest reported value (Cell 3) | | RSRQ_x + [TBD] |
| Lowest reported value (Cell 4) | Cell 1-4 | RSRQ_x + [TBD] |
| Highest reported value (Cell 4) | | RSRQ_x + [TBD] |
| RSRQ_x is the reported value of Cell 1 | | |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.3 UTRA FDD CPICH RSCP

## 9.3.1 E-UTRAN FDD - UTRA FDD CPICH RSCP absolute accuracy

### 9.3.1.1 Test purpose

To verify that the CPICH RSCP absolute measurement accuracy is within the specified limits.

### 9.3.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward that support UTRA FDD.

### 9.3.1.3 Minimum conformance requirements

The accuracy requirements shall be the same as the inter-frequency measurement accuracy requirements for FDD CPICH RSCP.

In RRC_CONNECTED state the accuracy requirements shall meet the absolute accuracy requirements in table 9.3.1.3-1.

**Table 9.3.1.3-1: UTRAN FDD CPICH_RSCP absolute accuracy**

| Accuracy | | Conditions | | |
|---|---|---|---|---|
| Normal condition | Extreme condition | Io range | | |
| | | UTRA operating bands | Minimum Io | Maximum Io |
| dB | dB | | dBm/3.84 MHz | dBm/3.84 MHz |
| ±6 | ±9 | Band I, IV, VI, X XI, XIX and XXI | -94 | -70 |
| | | Band IX | -93 | -70 |
| | | Band II, V and VII | -92 | -70 |
| | | Band III, VIII, XII, XIII, XIV , XX and XXII | -91 | -70 |
| | | Band XXV, XXVI Note 1 | -90.5 | -70 |
| ±8 | ±11 | Note 2 | -70 | -50 |

NOTE 1: For Band XXVI, the condition has the minimum Io of -92 dBm/3.84 MHz when the carrier frequency of the assigned UTRA channel is within 869-894 MHz for the UE which supports both Band V and Band XXVI operating frequencies.
NOTE 2: The same bands apply for this requirement as for the corresponding highest accuracy requirement.

If the UE, in RRC_CONNECTED state, needs measurement gaps to perform UTRAN FDD measurements, the UTRAN FDD measurement procedure and measurement gap pattern stated in TS 36.133 [4] clause 8.1.2.4.1 shall apply.

The reporting range and mapping specified for FDD CPICH RSCP is defined in Table 9.3.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.3.1.3-2: CPICH RSCP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| CPICH_RSCP_LEV_-05 | CPICH RSCP <-120 | dBm |
| CPICH_RSCP_LEV_-04 | -120 ≤ CPICH RSCP < -119 | dBm |
| CPICH_RSCP_LEV_-03 | -119 ≤ CPICH RSCP < -118 | dBm |
| … | … | … |
| CPICH_RSCP_LEV_89 | -27 ≤ CPICH RSCP < -26 | dBm |
| CPICH_RSCP_LEV_90 | -26 ≤ CPICH RSCP < -25 | dBm |
| CPICH_RSCP_LEV_91 | -25 ≤ CPICH RSCP | dBm |

The normative reference for this requirement is TS 25.133 [21] clauses 9.1.1.2 and 9.1.1.3 and TS 36.133 [4] clause 9.2.1 and A.9.3.1.

### 9.3.1.4 Test description

#### 9.3.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.

2. The general test parameter settings are set up according to Table 9.3.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.3.1.4.3.

5. Cell 1 is the serving E-UTRAN FDD cell and Cell 2 is the target UTRAN FDD cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

**Table 9.3.1.4.1-1: General test parameters for UTRAN FDD CPICH RSCP absolute measurement accuracy test in E-UTRAN FDD**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| E-UTRAN RF Channel Number | | 1 | One E-UTRAN FDD carrier frequency is used. |
| UTRAN RF Channel Number | | 1 | One UTRAN FDD carrier frequency is used. |
| E-UTRAN Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active cell | | Cell 1 | E-UTRAN cell 1 on RF channel number 1 |
| Neighbour cells | | Cell 2 | UTRAN cell 2 on RF channel number 1 |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| Inter-RAT (UTRAN FDD) measurement quantity | | CPICH RSCP | |
| Monitored UTRA FDD cell list size | | 12 | UTRA cells on UTRA RF channel 1 provided in the cell list. |
| CP length | | Normal | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | OFF |

### 9.3.1.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Tables 9.3.1.5-1 and 9.3.1.5-2 for Test 1. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit a RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit a RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check CPICH_RSCP reported values of Cell 2 in periodical MeasurementReport messages according to Table 9.3.1.5-3 as appropriate for Test 1. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport message transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. Set the parameters according to Tables 9.3.1.5-1 and 9.3.1.5-2 for Test 2. While RF parameters are being changed any MeasurementReport messages send from the UE shall be ignored by the SS. SS shall wait for an additional 1s and still ignore any MeasurementReport messages send from the UE. Then, step 6 and 7 above are repeated as appropriate for Test 2.

### 9.3.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.3.1.4.3-1: *MeasConfig- DEFAULT:* Additional E-UTRAN FDD - UTRA FDD CPICH RSCP absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | | |
| measObject CHOICE { | | | |
| measObject EUTRA | MeasObjectEUTRA-GENERIC(f1) | E-UTRA Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | | |
| measObject CHOICE { | | | |
| measObjectUTRA | MeasObjectUTRA-GENERIC(f2) | UTRA Cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId)) OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigInterRAT-PERIODICAL | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | MeasGapConfig-GP1 | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 9.3.1.4.3-2: *MeasResults*: Additional E-UTRAN FDD - UTRA FDD CPICH RSCP absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 9.3.1.4.3-3: *MeasResultListUTRA*: Additional E-UTRAN FDD - UTRA FDD CPICH RSCP absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| fdd | PhysCellIdUTRA-FDD | | |
| tdd | Not Present | | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-RSCP | Set according to specific test INTEGER (-5..91) | | |
| } | | | |
| } | | | |

**Table 9.3.1.4.3-4: *QuantityConfig-DEFAULT*: Additional E-UTRAN FDD - UTRA FDD CPICH RSCP absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-3A: QuantityConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| QuantityConfig-DEFAULT ::= SEQUENCE { | | | |
| quantityConfigEUTRA SEQUENCE { | | | |
| filterCoefficientRSRP | fc0 | | |
| filterCoefficientRSRQ | fc0 | | |
| } | | | |
| quantityConfigUTRA SEQUENCE { | | | |
| measQuantityUTRA-FDD | cpich-RSCP | | |
| filterCoefficient | fc0 | | |
| } | | | |

## 9.3.1.5 Test requirement

The test parameters are given in Tables 9.3.1.4.1-1, 9.3.1.5-1 and 9.3.1.5-2 as below. Table 9.3.1.5-2 and 9.3.1.5-3 define the primary level settings including test tolerances for all tests.

**Table 9.3.1.5-1: E-UTRAN FDD cell specific test parameters for UTRAN FDD CPICH RSCP absolute measurement accuracy test in E-UTRAN FDD**

| Parameter | Unit | Test 1 | Test 2 |
|---|---|---|---|
| E-UTRAN RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | |

| | | |
|---|---|---|
| PBCH_RA | dB | 0 |
| PBCH_RB | dB | |
| PSS_RA | dB | |
| SSS_RA | dB | |
| PCFICH_RB | dB | |
| PHICH_RA | dB | |
| PHICH_RB | dB | |
| PDCCH_RA | dB | |
| PDCCH_RB | dB | |
| PDSCH_RA | dB | |
| PDSCH_RB | dB | |
| OCNG_RA Note 1 | dB | |
| OCNG_RB Note 1 | dB | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 |
| RSRP Note 3 | dBm/15 kHz | -94 |
| $\hat{E}_s / I_{ot}$ | dB | 4 |
| SCH_RP Note 3 | dBm/15 kHz | -94 |
| $\hat{E}_s / N_{oc}$ | dB | 4 |
| Propagation Condition | | AWGN |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 9.3.1.5-2: UTRAN FDD cell specific test parameters for UTRAN FDD CPICH RSCP absolute measurement accuracy test in E-UTRAN FDD**

| Parameter | | Unit | Test 1 | Test 2 |
|---|---|---|---|---|
| | | | Cell 2 | Cell 2 |
| CPICH_Ec/Ior | | dB | -10 | -10 |
| PCCPCH_Ec/Ior | | dB | -12 | -12 |
| SCH_Ec/Ior | | dB | -12 | -12 |
| PICH_Ec/Ior | | dB | -15 | -15 |
| DPCH_Ec/Ior | | dB | - | - |
| OCNS_Ec/Ior | | dB | -0.94 | -0.94 |
| Ioc | Band I, IV, VI, X, XI, XIX, XXI | dBm/3.84 MHz | -60.75 | -93.76 |
| | Band II, V, VII | | | -91.76 |
| | Band XXV, XXVI | | | -90.26 (Note 3) |
| | Band III, VIII, XII, XIII, XIV, XX, XXII | | | -90.76 |
| | Band IX (Note 2) | | | -92.76 |
| Îor/Ioc | | dB | 9.54 | -9.19 |
| CPICH RSCP, Note 1 | Band I, IV, VI, X, XI, XIX, XXI | dBm | -61.21 | -112.95 |
| | Band II, V, VII | | | -110.95 |
| | Band XXV, XXVI | | | -109.45 (Note 3) |
| | Band III, VIII, XII, XIII, XIV, XX, XXII | | | -109.95 |
| | Band IX (Note 2) | | | -111.95 |
| Io, Note 1 | Band I, IV, VI, X, XI, XIX, XXI | dBm/3.84 MHz | -50.75 | -93.27 |
| | Band II, V, VII | | | -91.27 |
| | Band XXV, XXVI | | | -89.77 (Note 3) |
| | Band III, VIII, XII, XIII, XIV, XX, XXII | | | -90.27 |
| | Band IX (Note 2) | | | -92.27 |
| Propagation condition | | - | AWGN | AWGN |
| NOTE 1: CPICH RSCP and Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves. | | | | |
| NOTE 2: For the UE which supports both Band III and Band IX operating frequencies, the measurement performance requirements for Band III shall apply to the multi-band UE. | | | | |
| NOTE 3: The test parameter is modified by -1.5 dB when the carrier frequency of the assigned UTRA channel is within 869-894 MHz for the UE which supports both Band V and Band XXVI operating frequencies. | | | | |
| Tests shall be done sequentially. Test 1 shall be done first. After test 1 has been executed test parameters for test 2 shall be set within 5 seconds so that UE does not loose the Cell 2 in between the tests. | | | | |

Each UTRAN FDD CPICH RSCP absolute measurement accuracy test shall meet the reported values test requirements in table 9.3.1.5-3.

**Table 9.3.1.5-3: UTRAN FDD CPICH RSCP absolute measurement accuracy requirements for the reported values**

| | Test 1 | Test 2 | | | | |
|---|---|---|---|---|---|---|
| | All bands | Band I, IV, VI, X, XI, XIX, XXI | Band II, V, VII | Band XXV, XXVI (Note 2) | Band III, VIII, XII, XIII, XIV, XX, XXII | Band IX (Note 1) |
| Normal Conditions | | | | | | |
| Lowest reported value (Cell 2) | CPICH_RSCP_46 | CPICH_RSCP_-04 | CPICH_RSCP_-02 | CPICH_RSCP_-01 | CPICH_RSCP_-01 | CPICH_RSCP_-03 |
| Highest reported value (Cell 2) | CPICH_RSCP_63 | CPICH_RSCP_9 | CPICH_RSCP_11 | CPICH_RSCP_13 | CPICH_RSCP_12 | CPICH_RSCP_10 |
| Extreme Conditions | | | | | | |
| Lowest reported value (Cell 2) | CPICH_RSCP_43 | CPICH_RSCP_-05 | CPICH_RSCP_-05 | CPICH_RSCP_-04 | CPICH_RSCP_-04 | CPICH_RSCP_-05 |
| Highest reported value (Cell 2) | CPICH_RSCP_66 | CPICH_RSCP_12 | CPICH_RSCP_14 | CPICH_RSCP_16 | CPICH_RSCP_15 | CPICH_RSCP_13 |

NOTE 1: For a multiband UE supporting both Band III and Band IX, for Band IX apply the test requirements of Band III. (Reference Table 9.3.1.5-2, Note 2).
NOTE 2: For a multiband UE supporting both Band V and Band XXVI, for Band XXVI when the carrier frequency of the assigned UTRA channel in is within 869-894 MHz apply the test requirements of Band V. (Reference Table 9.3.1.5-2, Note 3).

## 9.3.2 E-UTRAN TDD - UTRA FDD CPICH RSCP absolute accuracy

### 9.3.2.1 Test purpose

To verify that the E-UTRAN TDD - UTRA FDD CPICH RSCP absolute measurement accuracy is within the specified limits.

### 9.3.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward that support UTRA FDD.

### 9.3.2.3 Minimum conformance requirements

The accuracy requirements shall be the same as the inter-frequency measurement accuracy requirements for FDD CPICH RSCP.

In RRC_CONNECTED state the accuracy requirements shall meet the absolute accuracy requirements in table 9.3.2.3-1.

**Table 9.3.2.3-1: UTRAN FDD CPICH_RSCP absolute accuracy**

| Accuracy | | Conditions | | |
|---|---|---|---|---|
| Normal condition | Extreme condition | Io range | | |
| | | UTRA operating bands | Minimum Io | Maximum Io |
| dB | dB | | dBm/3.84 MHz | dBm/3.84 MHz |
| ±6 | ±9 | Band I, IV, VI, X XI, XIX and XXI | -94 | -70 |
| | | Band IX | -93 | -70 |
| | | Band II, V and VII | -92 | -70 |
| | | Band III, VIII, XII, XIII, XIV , XX and XXII | -91 | -70 |
| | | Band XXV, XXVI Note 1 | -90.5 | -70 |
| ±8 | ±11 | Note 2 | -70 | -50 |

NOTE 1: For Band XXVI, the condition has the minimum Io of -92 dBm/3.84 MHz when the carrier frequency of the assigned UTRA channel is within 869-894 MHz for the UE which supports both Band V and Band XXVI operating frequencies.
NOTE 2: The same bands apply for this requirement as for the corresponding highest accuracy requirement.

If the UE, in RRC_CONNECTED state, needs measurement gaps to perform UTRAN FDD measurements, the UTRAN FDD measurement procedure and measurement gap pattern stated in TS 36.133 [4] clause 8.1.2.4.1 shall apply.

The reporting range and mapping specified for FDD CPICH RSCP is defined in Table 9.3.2.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.3.2.3-2: CPICH RSCP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| CPICH_RSCP_LEV_-05 | CPICH RSCP <-120 | dBm |
| CPICH_RSCP_LEV_-04 | -120 ≤ CPICH RSCP < -119 | dBm |
| CPICH_RSCP_LEV_-03 | -119 ≤ CPICH RSCP < -118 | dBm |
| … | … | … |
| CPICH_RSCP_LEV_89 | -27 ≤ CPICH RSCP < -26 | dBm |
| CPICH_RSCP_LEV_90 | -26 ≤ CPICH RSCP < -25 | dBm |
| CPICH_RSCP_LEV_91 | -25 ≤ CPICH RSCP | dBm |

The normative reference for this requirement is TS 25.133 [21] clauses 9.1.1.2 and 9.1.1.3 and TS 36.133 [4] clause 9.2.1 and A.9.3.2.

## 9.3.2.4 Test description

### 9.3.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.

2. The general test parameter settings are set up according to Table 9.3.2.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.3.2.4.3.

5. Cell 1 is the serving E-UTRAN TDD cell and Cell 2 is the target UTRAN FDD cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

**Table 9.3.2.4.1-1: General test parameters for UTRAN FDD CPICH RSCP absolute measurement accuracy test in E-UTRAN TDD**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.1 |
| E-UTRAN RF Channel Number | | 1 | One E-UTRAN TDD carrier frequency is used. |
| UTRAN RF Channel Number | | 1 | One UTRAN FDD carrier frequency is used. |
| E-UTRAN Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active cell | | Cell 1 | E-UTRAN cell 1 on RF channel number 1 |
| Neighbour cells | | Cell 2 | UTRAN cell 2 on RF channel number 1 |
| Gap Pattern Id | | 1 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| Inter-RAT (UTRAN FDD) measurement quantity | | CPICH RSCP | |
| Monitored UTRA FDD cell list size | | 12 | UTRA cells on UTRA RF channel 1 provided in the cell list. |
| CP length | | Normal | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | OFF |

### 9.3.2.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.3.2.5-1 and 9.3.2.5-2 for Test 1. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check CPICH_RSCP reported values of Cell 2 in periodical MeasurementReport messages according to Table 9.3.2.5-3 as appropriate for Test 1. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport message transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. Set the parameters according to Tables 9.3.2.5-1and 9.3.2.5-2 for Test 2. While RF parameters are being changed any MeasurementReport messages send from the UE shall be ignored by the SS. SS shall wait for an additional 1s and still ignore any MeasurementReport messages send from the UE. Then, step 6 and 7 above are repeated as appropriate for Test 2.

### 9.3.2.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.3.2.4.3-1: *MeasConfig- DEFAULT:* Additional E-UTRAN TDD - UTRA FDD CPICH RSCP absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | | |
| measObject CHOICE { | | | |
| measObject EUTRA | MeasObjectEUTRA-GENERIC(f1) | E-UTRA Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f8 | | |
| measObject CHOICE { | | | |
| measObjectUTRA | MeasObjectUTRA-GENERIC(f8) | UTRA Cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigInterRAT-PERIODICAL | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f8 | | |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | MeasGapConfig-GP2 | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 9.3.2.4.3-2: *MeasResults*: Additional E-UTRAN TDD - UTRA FDD CPICH RSCP absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 9.3.2.4.3-3: MeasResultListUTRA: Additional E-UTRAN TDD - UTRA FDD CPICH RSCP absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| fdd | PhysCellIdUTRA-FDD | | |
| tdd | Not Present | | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-RSCP | According to specific test | | |
| } | | | |
| } | | | |

## 9.3.2.5 Test requirement

The test parameters are given in Tables 9.3.2.4.1-1, 9.3.2.5-1 and 9.3.2.5-2 as below. Table 9.3.2.5-2 and 9.3.2.5-3 define the primary level settings including test tolerances for all tests.

Each UTRAN FDD CPICH RSCP absolute measurement accuracy test shall meet the reported values test requirements in table 9.3.2.5-3.

**Table 9.3.2.5-1: E-UTRAN TDD cell specific test parameters for UTRAN FDD CPICH RSCP absolute measurement accuracy test in E-UTRAN TDD**

| Parameter | Unit | Test 1 | Test 2 |
|---|---|---|---|
| E-UTRAN RF Channel Number | | 1 | |
| $BW_{channel}$ | MHz | 10 | |
| Special subframe configuration Note1 | | 6 | |
| Uplink-downlink configuration Note1 | | 1 | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | dB | | |
| PSS_RA | dB | | |
| SSS_RA | dB | | |
| PCFICH_RB | dB | | |
| PHICH_RA | dB | | |
| PHICH_RB | dB | | |
| PDCCH_RA | dB | | |
| PDCCH_RB | dB | | |
| PDSCH_RA | dB | | |
| PDSCH_RB | dB | | |
| OCNG_RA Note 2 | dB | | |
| OCNG_RB Note 2 | dB | | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | |
| RSRP Note 4 | dBm/15 kHz | -94 | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | |
| SCH_RP Note 4 | dBm/15 kHz | -94 | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | |
| Propagation Condition | | AWGN | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in 3GPP TS 36.211.

Note 2: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 9.3.2.5-2: UTRAN FDD cell specific test parameters for UTRAN FDD CPICH RSCP absolute measurement accuracy test in E-UTRAN TDD**

| Parameter | | Unit | Test 1 | Test 2 |
|---|---|---|---|---|
| | | | Cell 2 | Cell 2 |
| CPICH_Ec/Ior | | dB | -10 | -10 |
| PCCPCH_Ec/Ior | | dB | -12 | -12 |
| SCH_Ec/Ior | | dB | -12 | -12 |
| PICH_Ec/Ior | | dB | -15 | -15 |
| DPCH_Ec/Ior | | dB | - | - |
| OCNS_Ec/Ior | | dB | -0.94 | -0.94 |
| Ioc | Band I, IV, VI, X, XI, XIX, XXI | dBm/3.84 MHz | -60.75 | -93.76 |
| | Band II, V, VII | | | -91.76 |
| | Band XXV, XXVI | | | -90.26 (Note 3) |
| | Band III, VIII, XII, XIII, XIV, XXII | | | -90.76 |
| | Band IX (Note 2) | | | -92.76 |
| Îor/Ioc | | dB | 9.54 | -9.19 |
| CPICH RSCP, Note 1 | Band I, IV, VI, X, XI, XIX, XXI | dBm | -61.21 | -112.95 |
| | Band II, V, VII | | | -110.95 |
| | Band XXV, XXVI | | | -109.45 (Note 3) |
| | Band III, VIII, XII, XIII, XIV, XXII | | | -109.95 |
| | Band IX (Note 2) | | | -111.95 |
| Io, Note 1 | Band I, IV, VI, X, XI, XIX, XXI | dBm/3.84 MHz | -50.75 | -93.27 |
| | Band II, V, VII | | | -91.27 |
| | Band XXV, XXVI | | | -89.77 (Note 3) |
| | Band III, VIII, XII, XIII, XIV, XXII | | | -90.27 |
| | Band IX (Note 2) | | | -92.27 |
| Propagation condition | | - | AWGN | AWGN |
| NOTE 1: CPICH RSCP and Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves.<br>NOTE 2: For the UE which supports both Band III and Band IX operating frequencies, the measurement performance requirements for Band III shall apply to the multi-band UE.<br>NOTE 3: The test parameter is modified by -1.5 dB when the carrier frequency of the assigned UTRA channel is within 869-894 MHz for the UE which supports both Band V and Band XXVI operating frequencies. | | | | |
| Tests shall be done sequentially. Test 1 shall be done first. After test 1 has been executed test parameters for test 2 shall be set within 5 seconds so that UE does not loose the Cell 2 in between the tests. | | | | |

**Table 9.3.2.5-3: UTRAN FDD CPICH RSCP absolute measurement accuracy requirements for the reported values**

| | Test 1 | Test 2 | | | | |
|---|---|---|---|---|---|---|
| | All bands | Band I, IV, VI, X, XI, XIX, XXI | Band II, V, VII | Band XXV, XXVI | Band III, VIII, XII, XIII, XIV, XX, XXII | Band IX |
| Normal Conditions | | | | | | |
| Lowest reported value (Cell 2) | CPICH_RSCP_46 | CPICH_RSCP_-04 | CPICH_RSCP_-02 | CPICH_RSCP_-01 | CPICH_RSCP_-01 | CPICH_RSCP_-03 |
| Highest reported value (Cell 2) | CPICH_RSCP_63 | CPICH_RSCP_9 | CPICH_RSCP_11 | CPICH_RSCP_13 | CPICH_RSCP_12 | CPICH_RSCP_10 |
| Extreme Conditions | | | | | | |
| Lowest reported value (Cell 2) | CPICH_RSCP_43 | CPICH_RSCP_-05 | CPICH_RSCP_-05 | CPICH_RSCP_-04 | CPICH_RSCP_-04 | CPICH_RSCP_-05 |
| Highest reported value (Cell 2) | CPICH_RSCP_66 | CPICH_RSCP_12 | CPICH_RSCP_14 | CPICH_RSCP_16 | CPICH_RSCP_15 | CPICH_RSCP_13 |

## 9.3.3 E-UTRAN FDD - UTRA FDD CPICH RSCP absolute accuracy for 5MHz bandwidth

### 9.3.3.1 Test purpose

To verify that the CPICH RSCP absolute measurement accuracy is within the specified limits for Band 31.

### 9.3.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward that support E-UTRA Band 31 and UTRA FDD.

### 9.3.3.3 Minimum conformance requirements

The requirements in this clause are valid for terminals supporting this capability.

The measurement period for RRC_CONNECTED state is specified in TS 36.133[4] clauses 8.1.2.4.1 and 8.1.2.4.2.

In RRC_CONNECTED state the accuracy requirements shall meet the absolute accuracy requirements in TS 36.133[4] table 9.3.3.3-1, under the following conditions:

- CPICH Ec/Io condition for a detectable cell is as specified in TS 36.133[4] clauses 8.1.2.4.1, 8.1.2.4.2, 8.1.2.4.7, 8.1.2.4.8;

- SCH_Ec/Io condition for a detectable cell is as specified in TS 36.133[4] clauses 8.1.2.4.1, 8.1.2.4.2, 8.1.2.4.7, 8.1.2.4.8.

**Table 9.3.3.3-1: UTRAN FDD CPICH_RSCP absolute accuracy**

| Accuracy | | Conditions | | |
|---|---|---|---|---|
| Normal condition | Extreme condition | Io range | | |
| | | UTRA operating bands | Minimum Io | Maximum Io |
| dB | dB | | dBm/3.84 MHz | dBm/3.84 MHz |
| ±6 | ±9 | Band I, IV, VI, X XI, XIX and XXI | -94 | -70 |
| | | Band IX | -93 | -70 |
| | | Band II, V and VII | -92 | -70 |
| | | Band III, VIII, XII, XIII, XIV , XX and XXII | -91 | -70 |
| | | Band XXV, XXVI Note 1 | -90.5 | -70 |
| ±8 | ±11 | Note 2 | -70 | -50 |

NOTE 1: For Band XXVI, the condition has the minimum Io of -92 dBm/3.84 MHz when the carrier frequency of the assigned UTRA channel is within 869-894 MHz for the UE which supports both Band V and Band XXVI operating frequencies.
NOTE 2: The same bands apply for this requirement as for the corresponding highest accuracy requirement.

If the UE, in RRC_CONNECTED state, needs measurement gaps to perform UTRAN FDD measurements, the relevant UTRAN FDD measurement procedure and measurement gap pattern stated in TS 36.133[4] clause 8.1.2.4 shall apply.

The reporting range and mapping specified for FDD CPICH RSCP is defined in Table 9.3.3.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.3.3.3-2: CPICH RSCP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| CPICH_RSCP_LEV_-05 | CPICH RSCP <-120 | dBm |
| CPICH_RSCP_LEV_-04 | -120 ≤ CPICH RSCP < -119 | dBm |
| CPICH_RSCP_LEV_-03 | -119 ≤ CPICH RSCP < -118 | dBm |
| … | … | … |
| CPICH_RSCP_LEV_89 | -27 ≤ CPICH RSCP < -26 | dBm |
| CPICH_RSCP_LEV_90 | -26 ≤ CPICH RSCP < -25 | dBm |
| CPICH_RSCP_LEV_91 | -25 ≤ CPICH RSCP | dBm |

The normative reference for this requirement is TS 25.133 [21] clause 9.1.1.3 and TS 36.133 [4] clause 9.2.1 and A.9.3.3.

## 9.3.3.4 Test description

### 9.3.3.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.

2. The general test parameter settings are set up according to Table 9.3.3.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.3.3.4.3.

5. Cell 1 is the serving E-UTRAN FDD cell and Cell 2 is the target UTRAN FDD cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

**Table 9.3.3.4.1-1: General test parameters for UTRAN FDD CPICH RSCP absolute measurement accuracy test in E-UTRAN FDD for 5MHz bandwidth**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.5 FDD | As specified in clause A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.11 FDD | As specified in clause A.2.1 |
| E-UTRAN Channel Bandwidth ($BW_{channel}$) | MHz | 5 | |
| Note 1: See Table 9.3.1.4.1-1 for other general test parameters. | | | |

### 9.3.3.4.2 Test procedure

Same test procedure as defined in clause 9.3.1.4.2 with the following exceptions:

- Instead of Table 9.3.1.5-1 → use Table 9.3.3.5-1.

### 9.3.3.4.3 Message contents

Same message contents as defined in clause 9.3.1.4.3.

### 9.3.3.5 Test requirement

The test parameters are given in Tables 9.3.3.4.1-1, 9.3.3.5-1 and 9.3.1.5-2 as below. Table 9.3.1.5-2 and 9.3.1.5-3 define the primary level settings including test tolerances for all tests.

**Table 9.3.3.5-1: E-UTRAN FDD cell specific test parameters for UTRAN FDD CPICH RSCP absolute measurement accuracy test in E-UTRAN FDD for 5MHz bandwidth**

| Parameter | | Unit | Test 1 | Test 2 |
|---|---|---|---|---|
| E-UTRAN RF Channel Number | | | 1 | |
| $BW_{channel}$ | | MHz | 5 | |
| OCNG Patterns defined in D.1.15 (OP.15 FDD) | | | OP.15 FDD | |
| PBCH_RA | | dB | 0 | |
| PBCH_RB | | dB | | |
| PSS_RA | | dB | | |
| SSS_RA | | dB | | |
| PCFICH_RB | | dB | | |
| PHICH_RA | | dB | | |
| PHICH_RB | | dB | | |
| PDCCH_RA | | dB | | |
| PDCCH_RB | | dB | | |
| PDSCH_RA | | dB | | |
| PDSCH_RB | | dB | | |
| OCNG_RA Note 1 | | dB | | |
| OCNG_RB Note 1 | | dB | | |
| $N_{oc}$ Note 2 | Band 31 | dBm/15 kHz | -98 | |
| RSRP Note 3 | Band 31 | dBm/15 kHz | -94 | |
| $\hat{E}_s / I_{ot}$ | | dB | 4 | |
| SCH_RP Note 3 | Band 31 | dBm/15 kHz | -94 | |
| $\hat{E}_s / N_{oc}$ | | dB | 4 | |
| Io Note3 | Band 31 | dBm/4.5 MHz | -67.8 | |
| Propagation Condition | | | AWGN | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP, SCH_RP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

# 9.4 UTRAN FDD CPICH Ec/No

## 9.4.1 E-UTRAN FDD - UTRA FDD CPICH Ec/No absolute accuracy

### 9.4.1.1 Test purpose

To verify that the E-UTRAN FDD - UTRA FDD CPICH Ec/No absolute measurement accuracy is within the specified limits.

### 9.4.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward that support UTRA FDD.

### 9.4.1.3 Minimum conformance requirements

The accuracy requirements shall be the same as the inter-frequency measurement accuracy requirements for FDD CPICH Ec/No

The accuracy requirements in table 9.4.1.3-1 are valid under the following conditions:

CPICH_RSCP$|_{dBm}$ ≥ -114 dBm for Bands I, IV, VI, X, XI, XIX and XXI

CPICH_RSCP$|_{dBm}$ ≥ -113 dBm for Band IX,

CPICH_RSCP$|_{dBm}$ ≥ -112 dBm for Bands II, V and VII,

CPICH_RSCP$|_{dBm}$ ≥ -111 dBm for Band III, VIII, XII, XIII, XIV, XX and XXII

CPICH_RSCP$|_{dBm}$ ≥ -110.5 dBm for Band XXV.

$$\left.\frac{I_o}{\left(\hat{I}_{or}\right)}\right|_{in\ dB} - \left.\left(\frac{CPICH_E_c}{I_{or}}\right)\right|_{in\ dB} \leq 20dB$$

**Table 9.4.1.3-1: UTRA FDD CPICH_Ec/Io absolute accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions | | | | |
|---|---|---|---|---|---|---|---|---|
| | | Normal condition | Extreme condition | Band I, IV VI, X, XI, XIX and XXI | Band II, V and VII | Band XXV and XXVI | Band III, VIII, XII, XIII, XIV, XX and XXII | Band IX |
| | | | | Io [dBm/3,84 MHz] | Io [dBm/3,84 MHz] | Io [dBm/3,84 MHz] | Io [dBm/3,84 MHz] | Io [dBm/3,84 MHz] |
| CPICH_Ec/Io | dB | ± 1.5 for -14 ≤ CPICH Ec/Io<br>± 2 for -16 ≤ CPICH Ec/Io < -14<br>± 3 for -20 ≤ CPICH Ec/Io < -16 | ± 3 | -94...-50 | -92…-50 | -90.5…-50 (Note 1) | -91…-50 | -93...-50 |
| NOTE 1: The condition is -92…-50 dBm/3.84 MHz when the carrier frequency of the assigned UTRA channel is within 869-894 MHz for the UE which supports both Band V and Band XXVI operating frequencies. | | | | | | | | |

If the UE, in RRC_CONNECTED state, needs measurement gaps to perform UTRAN FDD measurements, the UTRAN FDD measurement procedure and measurement gap pattern stated in TS 36.133 [4] clause 8.1.2.4.1 shall apply.

The reporting range is for *CPICH Ec/Io* is from -24 ...0 dB.

In table 9.4.1.3-2 the mapping of measured quantity is defined.

The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.4.1.3-2: UTRA FDD CPICH_Ec/Io measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| CPICH_Ec/No_00 | CPICH Ec/Io < -24 | dB |
| CPICH_Ec/No_01 | -24 ≤ CPICH Ec/Io < -23.5 | dB |
| CPICH_Ec/No_02 | -23.5 ≤ CPICH Ec/Io < -23 | dB |
| … | … | … |
| CPICH_Ec/No_47 | -1 ≤ CPICH Ec/Io < -0.5 | dB |
| CPICH_Ec/No_48 | -0.5 ≤ CPICH Ec/Io < 0 | dB |
| CPICH_Ec/No_49 | 0 ≤ CPICH Ec/Io | dB |

The normative reference for this requirement is TS 25.133 [21] clauses 9.1.2.2.1 and 9.1.2.3 and TS 36.133 [4] clause 9.2.3 and A.9.4.1.

### 9.4.1.4 Test description

#### 9.4.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.4.1.4.3.

4. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.4.1.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Tables 9.4.1.5-2 and 9.4.1.5-3 for Test 1. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit a RRCConnectionReconfiguration message on cell1.

4. The UE shall transmit a RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check UTRA FDD CPICH Ec/Io reported values of Cell 2 in periodical MeasurementReport messages according to Table 9.4.1.5-4 as appropriate for Test 1. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport message transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. If Test 2 not done, set the parameters according to Tables 9.4.1.5-2 and 9.4.1.5-3 for Test 2. While RF parameters are being changed any MeasurementReport messages send from the UE shall be ignored by the SS. SS shall wait for an additional 1s and still ignore any MeasurementReport messages send from the UE. Then, step 6 and 7 above are repeated as appropriate for Test 2.

9. If Test 3 not done, set the parameters according to Tables 9.4.1.5-2 and 9.4.1.5-3 for Test 3. While RF parameters are being changed any MeasurementReport messages send from the UE shall be ignored by the SS. SS shall wait for an additional 1s and still ignore any MeasurementReport messages send from the UE. Then, step 6 and 7 above are repeated as appropriate for Test 3.

#### 9.4.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.4.1.4.3-1: CPICH_Ec/Io measurement configuration**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-7 |

**Table 9.4.1.4.3-2: *MeasConfig- DEFAULT:* CPICH_Ec/Io measurement configuration**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | | |
| measObject CHOICE { | | | |
| measObject EUTRA | MeasObjectEUTRA-GENERIC(f1) | E-UTRA Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | | |
| measObject CHOICE { | | | |
| measObjectUTRA | MeasObjectUTRA-GENERIC(f2) | UTRA Cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigInterRAT-PERIODICAL | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | MeasGapConfig-GP1 | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 9.4.1.4.3-3: *MeasResults*: CPICH_Ec/Io measurement configuration**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 9.4.1.4.3-4: MeasResultListUTRA: CPICH_Ec/Io measurement configuration**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| fdd | PhysCellIdUTRA-FDD | | |
| tdd | Not Present | | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-EcN0 | According to specific test | | |
| } | | | |
| } | | | |

## 9.4.1.5 Test requirement

The test parameters are given in Tables 9.4.1.5-1, 9.4.1.5-2 and 9.4.1.5-3 as below. Table, 9.4.1.5-2 and 9.4.1.5-3 define the primary level settings including test tolerances for all tests.

**Table 9.4.1.5-1: General test parameters for UTRAN FDD CPICH Ec/No absolute measurement accuracy test in E-UTRAN FDD**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| E-UTRAN RF Channel Number | | 1 | One E-UTRAN FDD carrier frequency is used. |
| UTRAN RF Channel Number | | 1 | One UTRAN FDD carrier frequency is used. |
| E-UTRAN Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active cell | | Cell 1 | E-UTRAN cell 1 on RF channel number 1 |
| Neighbour cells | | Cell 2 | UTRAN cell 2 on RF channel number 1 |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133[4] section 8.1.2.1. |
| Inter-RAT (UTRAN FDD) measurement quantity | | CPICH Ec/N0 | |
| Monitored UTRA FDD cell list size | | 12 | UTRA cells on UTRA RF channel 1 provided in the cell list. |
| CP length | | Normal | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | OFF |

**Table 9.4.1.5-2: E-UTRAN FDD cell specific test parameters for UTRAN FDD CPICH Ec/No absolute measurement accuracy test in E-UTRAN FDD**

| Parameter | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|
| E-UTRAN RF Channel Number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | | |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 1 | dB | | | |
| OCNG_RB Note 1 | dB | | | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 | | |
| RSRP Note 3 | dBm/15 kHz | -94 | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | | |
| SCH_RP Note 3 | dBm/15 kHz | -94 | | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | | |
| Propagation Condition | | AWGN | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 9.4.1.5-3: UTRAN FDD cell specific test parameters for UTRAN FDD CPICH Ec/No absolute measurement accuracy test in E-UTRAN FDD**

| Parameter | | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|---|
| | | | Cell 2 | Cell 2 | Cell 2 |
| CPICH_Ec/Ior | | dB | -10 | -10 | -10 |
| PCCPCH_Ec/Ior | | dB | -12 | -12 | -12 |
| SCH_Ec/Ior | | dB | -12 | -12 | -12 |
| PICH_Ec/Ior | | dB | -15 | -15 | -15 |
| DPCH_Ec/Ior | | dB | - | - | - |
| OCNS_Ec/Ior | | dB | -0.94 | -0.94 | -0.94 |
| Ioc | Band I, IV, VI, X, XI, XIX, XXI | dBm/ 3.84 MHz | -53.12 | -87.27 | -93.76 |
| | Band II, V, VII | | | | -91.76 |
| | Band XXV, XXVI | | | | -90.26 (Note 3) |
| | Band III, VIII, XII, XIII, XIV, XX, XXII | | | | -90.76 |
| | Band IX (Note 2) | | | | -92.76 |
| Îor/Ioc | | dB | -1.45 | -4.4 | -9.14 |
| CPICH Ec/Io, Note 1 | | dBm | -13.8 | -15.75 | -19.64 |
| Io, Note 1 | Band I, IV, VI, X, XI, XIX, XXI | dBm/ 3.84 MHz | -50.77 | -85.92 | -93.26 |
| | Band II, V, VII | | | | -91.26 |
| | Band XXV, XXVI | | | | -89.76 (Note 3) |
| | Band III, VIII, XII, XIII, XIV, XX, XXII | | | | -90.26 |
| | Band IX (Note 2) | | | | -92.26 |
| Propagation condition | | - | AWGN | AWGN | AWGN |
| NOTE 1: CPICH Ec/Io and Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves.<br>NOTE 2: For the UE which supports both Band III and Band IX operating frequencies, the measurement performance requirements for Band III shall apply to the multi-band UE.<br>NOTE 3: The test parameter is modified by -1.5 dB when the carrier frequency of the assigned UTRA channel is within 869-894 MHz for the UE which supports both Band V and Band XXVI operating frequencies. | | | | | |
| Tests shall be done sequentially. Test 1 shall be done first. After test 1 has been executed test parameters for tests 2 and 3 shall be set within 5 seconds so that UE does not lose the Cell 2 in between the tests. | | | | | |

Each UTRAN FDD CPICH Ec/No absolute measurement accuracy test shall meet the reported values test requirements in table 9.4.1.5-4.

**Table 9.4.1.5-4: UTRAN FDD CPICH Ec/No absolute measurement accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| Normal Conditions | | | |
| Lowest reported value | CPICH_Ec/No_17 | CPICH_Ec/No_13 | CPICH_Ec/No_0 |
| Highest reported value | CPICH_Ec/No_24 | CPICH_Ec/No_22 | CPICH_Ec/No_16 |
| Extreme Conditions | | | |
| Lowest reported value | CPICH_Ec/No_14 | CPICH_Ec/No_11 | CPICH_Ec/No_0 |
| Highest reported value | CPICH_Ec/No_27 | CPICH_Ec/No_24 | CPICH_Ec/No_16 |

## 9.4.2 E-UTRAN TDD - UTRA FDD CPICH Ec/No absolute accuracy

### 9.4.2.1 Test purpose

To verify that the E-UTRAN TDD - UTRA FDD CPICH Ec/No absolute measurement accuracy is within the specified limits.

### 9.4.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward that support UTRA FDD.

### 9.4.2.3 Minimum conformance requirements

The accuracy requirements shall be the same as the inter-frequency measurement accuracy requirements for FDD CPICH Ec/No.

The accuracy requirements in table 9.4.2.3-1 are valid under the following conditions:

CPICH_RSCP$|_{dBm}$ ≥ -114 dBm for Bands I, IV, VI, X, XI, XIX and XXI,

CPICH_RSCP$|_{dBm}$ ≥ -113 dBm for Band IX

CPICH_RSCP$|_{dBm}$ ≥ -112 dBm for Bands II, V and VII,

CPICH_RSCP$|_{dBm}$ ≥ -111 dBm for Band III, VIII, XII, XIII, XIV and XX,

CPICH_RSCP$|_{dBm}$ ≥ -110.5 dBm for Band XXV.

$$\left.\frac{I_o}{\left(\hat{I}_{or}\right)}\right|_{in\ dB} - \left.\left(\frac{CPICH_E_c}{I_{or}}\right)\right|_{in\ dB} \leq 20dB$$

**Table 9.4.2.3-1: UTRAN FDD CPICH_Ec/Io absolute accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions | | | | |
|---|---|---|---|---|---|---|---|---|
| | | Normal condition | Extreme condition | Band I, IV, VI, X, XI, XIX and XXI | Band II, V and VII | Band XXV and XXVI | Band III, VIII, XII, XIII, XIV, XX and XXII | Band IX |
| | | | | Io [dBm/3,84 MHz] | Io [dBm/3,84 MHz] | Io [dBm/3,84 MHz] | Io [dBm/3,84 MHz] | Io [dBm/3,84 MHz] |
| CPICH_Ec/Io | dB | ± 1.5 for -14 ≤ CPICH Ec/Io ± 2 for -16 ≤ CPICH Ec/Io < -14 ± 3 for -20 ≤ CPICH Ec/Io < -16 | ± 3 | -94...-50 | -92…-50 | -90.5...-50 (Note 1) | -91…-50 | -93...-50 |
| NOTE 1: The condition is -92…-50 dBm/3.84 MHz when the carrier frequency of the assigned UTRA channel is within 869-894 MHz for the UE which supports both Band V and Band XXVI operating frequencies. | | | | | | | | |

If the UE, in RRC_CONNECTED state, needs measurement gaps to perform UTRAN FDD measurements, the UTRAN FDD measurement procedure and measurement gap pattern stated in TS 36.133 [4] clause 8.1.2.4.1 shall apply.

The reporting range is for *CPICH Ec/Io* is from -24 ...0 dB.

In table 9.4.2.3-2 the mapping of measured quantity is defined. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.4.2.3-2: UTRAN FDD CPICH_Ec/Io measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| CPICH_Ec/No_00 | CPICH Ec/Io < -24 | dB |
| CPICH_Ec/No_01 | -24 ≤ CPICH Ec/Io < -23.5 | dB |
| CPICH_Ec/No_02 | -23.5 ≤ CPICH Ec/Io < -23 | dB |
| … | … | … |
| CPICH_Ec/No_47 | -1 ≤ CPICH Ec/Io < -0.5 | dB |
| CPICH_Ec/No_48 | -0.5 ≤ CPICH Ec/Io < 0 | dB |
| CPICH_Ec/No_49 | 0 ≤ CPICH Ec/Io | dB |

The normative reference for this requirement is TS 25.133 [21] clauses 9.1.2.2.1 and 9.1.2.3 and TS 36.133 [4] clause 9.2.3 and A.9.4.2.

## 9.4.2.4 Test description

### 9.4.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.

2. The general test parameter settings are set up according to Table 9.4.2.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.4.2.4.3.

5. Cell 1 is the serving E-UTRAN TDD cell and Cell 2 is the target UTRAN FDD cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

**Table 9.4.2.4.1-1: General test parameters for UTRAN FDD CPICH Ec/No absolute measurement accuracy test in E-UTRAN TDD**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| E-UTRAN RF Channel Number | | 1 | One E-UTRAN TDD carrier frequency is used. |
| UTRAN RF Channel Number | | 1 | One UTRAN FDD carrier frequency is used. |
| E-UTRAN Channel Bandwidth ($BW_{channel}$) | MHz | 10 | |
| Active cell | | Cell 1 | E-UTRAN cell 1 on RF channel number 1 |
| Neighbour cells | | Cell 2 | UTRAN cell 2 on RF channel number 1 |
| Gap Pattern Id | | 0 | As specified in 3GPP TS 36.133 section 8.1.2.1. |
| Inter-RAT (UTRAN FDD) measurement quantity | | CPICH Ec/N0 | |
| Monitored UTRA FDD cell list size | | 12 | UTRA cells on UTRA RF channel 1 provided in the cell list. |
| CP length | | Normal | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | OFF |

### 9.4.2.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Tables 9.4.2.5-1 and 9.4.2.5-2 as appropriate for Test 1. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check CPICH_Ec/Io reported values of Cell 2 in periodical MeasurementReport messages according to Table 9.4.2.5-3 as appropriate for Test 1. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport message transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. If Test 2 not done, set the parameters according to Tables 9.4.2.5-1 and 9.4.2.5-2 for Test 2. While RF parameters are being changed any MeasurementReport messages send from the UE shall be ignored by the SS. SS shall wait for an additional 1s and still ignore any MeasurementReport messages send from the UE. Then, step 6 and 7 above are repeated as appropriate for Test 2.

9. If Test 3 not done, set the parameters according to Tables 9.4.2.5-1 and 9.4.2.5-2 for Test 3. While RF parameters are being changed any MeasurementReport messages send from the UE shall be ignored by the SS. SS shall wait for an additional 1s and still ignore any MeasurementReport messages send from the UE. Then, step 6 and 7 above are repeated as appropriate for Test 3.

#### 9.4.2.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.4.2.4.3-1: *MeasConfig- DEFAULT:* Additional E-UTRAN TDD - UTRA FDD CPICH Ec/No absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | | |
| measObject CHOICE { | | | |
| measObject EUTRA | MeasObjectEUTRA-GENERIC(f1) | E-UTRA Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f8 | | |
| measObject CHOICE { | | | |
| measObjectUTRA | MeasObjectUTRA-GENERIC(f8) | UTRA Cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigInterRAT-PERIODICAL | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f8 | | |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | MeasGapConfig-GP1 | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 9.4.2.4.3-2: *MeasResults*: Additional E-UTRAN TDD - UTRA FDD CPICH Ec/No absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 9.4.2.4.3-3: MeasResultListUTRA: Additional E-UTRAN TDD - UTRA FDD CPICH Ec/No absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsLIstEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| fdd | PhysCellIdUTRA-FDD | | |
| tdd | Not Present | | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-EcN0 | According to specific test | | |
| } | | | |
| } | | | |

## 9.4.2.5 Test requirement

The test parameters are given in Tables 9.4.2.4.1-1, 9.4.2.5-1 and 9.4.2.5-2 as below. Table 9.4.2.5-1 and 9.4.2.5-2 define the primary level settings including test tolerances for all tests.

Each UTRAN FDD CPICH Ec/No absolute measurement accuracy test shall meet the reported values test requirements in table 9.4.2.5-3.

**Table 9.4.2.5-1: E-UTRAN TDD cell specific test parameters for UTRAN FDD CPICH Ec/No absolute measurement accuracy test in E-UTRAN TDD**

| Parameter | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|
| E-UTRAN RF Channel Number | | 1 | | |
| $BW_{channel}$ | MHz | 10 | | |
| Special subframe configuration Note1 | | 6 | | |
| Uplink-downlink configuration Note1 | | 1 | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | | |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | dB | | | |
| PSS_RA | dB | | | |
| SSS_RA | dB | | | |
| PCFICH_RB | dB | | | |
| PHICH_RA | dB | | | |
| PHICH_RB | dB | | | |
| PDCCH_RA | dB | | | |
| PDCCH_RB | dB | | | |
| PDSCH_RA | dB | | | |
| PDSCH_RB | dB | | | |
| OCNG_RA Note 2 | dB | | | |
| OCNG_RB Note 2 | dB | | | |
| $N_{oc}$ Note 3 | dBm/15 kHz | -98 | | |
| RSRP Note 4 | dBm/15 kHz | -94 | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | | |
| SCH_RP Note 4 | dBm/15 kHz | -94 | | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | | |
| Propagation Condition | | AWGN | | |

Note 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in 3GPP TS 36.211.

Note 2: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 4: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 9.4.2.5-2: UTRAN FDD cell specific test parameters for UTRAN FDD CPICH Ec/No absolute measurement accuracy test in E-UTRAN TDD**

| Parameter | | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|---|
| | | | Cell 2 | Cell 2 | Cell 2 |
| CPICH_Ec/Ior | | dB | -10 | -10 | -10 |
| PCCPCH_Ec/Ior | | dB | -12 | -12 | -12 |
| SCH_Ec/Ior | | dB | -12 | -12 | -12 |
| PICH_Ec/Ior | | dB | -15 | -15 | -15 |
| DPCH_Ec/Ior | | dB | - | - | - |
| OCNS_Ec/Ior | | dB | -0.94 | -0.94 | -0.94 |
| Ioc | Band I, IV, VI, X, XIX | dBm/ 3.84 MHz | -53.12 | -87.27 | -93.76 |
| | Band II, V, VII, XI | | | | -91.76 |
| | Band XXV, XXVI | | | | -90.26 (Note 3) |
| | Band III, VIII, XII, XIII, XIV, XXII | | | | -90.76 |
| | Band IX (Note 2) | | | | -92.76 |
| Îor/Ioc | | dB | -1.45 | -4.4 | -9.14 |
| CPICH Ec/Io, Note 1 | | dBm | -13.8 | -15.75 | -19.64 |
| Io, Note 1 | Band I, IV, VI, X, XIX | dBm/ 3.84 MHz | -50.77 | -85.92 | -93.26 |
| | Band II, V, VII, XI | | | | -91.26 |
| | Band XXV, XXVI | | | | -89.76 (Note 3) |
| | Band III, VIII, XII, XIII, XIV, XXII | | | | -90.26 |
| | Band IX (Note 2) | | | | -92.26 |
| Propagation condition | | - | AWGN | AWGN | AWGN |

NOTE 1: CPICH Ec/Io and Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves.
NOTE 2: For the UE which supports both Band III and Band IX operating frequencies, the measurement performance requirements for Band III shall apply to the multi-band UE.
NOTE 3: The test parameter is modified by -1.5 dB when the carrier frequency of the assigned UTRA channel is within 869-894 MHz for the UE which supports both Band V and Band XXVI operating frequencies.
Tests shall be done sequentially. Test 1 shall be done first. After test 1 has been executed test parameters for tests 2 and 3 shall be set within 5 seconds so that UE does not loose the Cell 2 in between the tests.

**Table 9.4.2.5-3: UTRAN FDD CPICH Ec/No absolute measurement accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| Normal Conditions | | | |
| Lowest reported value | CPICH_Ec/No_17 | CPICH_Ec/No_13 | CPICH_Ec/No_0 |
| Highest reported value | CPICH_Ec/No_24 | CPICH_Ec/No_22 | CPICH_Ec/No_16 |
| Extreme Conditions | | | |
| Lowest reported value | CPICH_Ec/No_14 | CPICH_Ec/No_11 | CPICH_Ec/No_0 |
| Highest reported value | CPICH_Ec/No_27 | CPICH_Ec/No_24 | CPICH_Ec/No_16 |

## 9.4.3 E-UTRAN FDD - UTRA FDD CPICH Ec/No absolute accuracy for 5MHz bandwidth

### 9.4.3.1 Test purpose

To verify that the E-UTRAN FDD - UTRA FDD CPICH Ec/No absolute measurement accuracy is within the specified limits for Band 31.

### 9.4.3.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward that support E-UTRA Band 31 and UTRA FDD.

### 9.4.3.3 Minimum conformance requirements

The measurement period for RRC_CONNECTED state is specified in TS 36.133[4] clauses 8.1.2.4.1 and 8.1.2.4.2.

In RRC_CONNECTED state the accuracy requirements shall be the same as the inter-frequency measurement accuracy requirements for FDD CPICH Ec/No in TS 25.133 [18].

If the UE, in RRC_CONNECTED state, needs measurement gaps to perform UTRAN FDD measurements, the UTRAN FDD measurement procedure and measurement gap pattern stated in TS 36.133[4] clause 8.1.2.4.1 shall apply.

The reporting range is for *CPICH Ec/Io* is from -24 ...0 dB.

In table 9.4.3.3-2 the mapping of measured quantity is defined.

The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.4.3.3-2: UTRA FDD CPICH_Ec/Io measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| CPICH_Ec/No_00 | CPICH Ec/Io < -24 | dB |
| CPICH_Ec/No_01 | -24 ≤ CPICH Ec/Io < -23.5 | dB |
| CPICH_Ec/No_02 | -23.5 ≤ CPICH Ec/Io < -23 | dB |
| … | … | … |
| CPICH_Ec/No_47 | -1 ≤ CPICH Ec/Io < -0.5 | dB |
| CPICH_Ec/No_48 | -0.5 ≤ CPICH Ec/Io < 0 | dB |
| CPICH_Ec/No_49 | 0 ≤ CPICH Ec/Io | dB |

The normative reference for this requirement is TS 25.133 [21] clauses 9.1.2.2.1 and 9.1.2.3 and TS 36.133 [4] clause 9.2.3 and A.9.4.3.

### 9.4.3.4 Test description

#### 9.4.3.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.

2. The general test parameter settings are set up according to Table 9.4.3.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.4.3.4.3.

5. Cell 1 is the serving cell and Cell 2 is the target cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

**Table 9.4.3.4.1-1: General test parameters for UTRAN FDD CPICH Ec/No absolute measurement accuracy test in E-UTRAN FDD for 5MHz bandwidth**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.5 FDD | As specified in clause A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.11 FDD | As specified in clause A.2.1 |
| E-UTRAN Channel Bandwidth ($BW_{channel}$) | MHz | 5 | |
| Note 1: See Table 9.4.1.5-1 for other general test parameters. | | | |

#### 9.4.3.4.2 Test procedure

Same test procedure as defined in clause 9.4.1.4.2 with the following exceptions:

- Instead of Table 9.4.1.5-2 → use Table 9.4.3.5-1.

#### 9.4.3.4.3 Message contents

Same message contents as defined in clause 9.4.1.4.3.

### 9.4.3.5 Test requirement

The test parameters are given in Tables 9.4.1.4.1-1, 9.4.3.5-1 and 9.4.1.5-3. Tables 9.4.3.5-1 and 9.4.1.5-3 define the primary level settings including test tolerances for all tests.

Table 9.4.1.5-4 defines the UTRAN FDD CPICH Ec/No absolute measurement accuracy requirements for the reported values.

**Table 9.4.3.5-1: E-UTRAN FDD cell specific test parameters for UTRAN FDD CPICH Ec/No absolute measurement accuracy test in E-UTRAN FDD for 5MHz bandwidth**

| Parameter | | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|---|
| E-UTRAN RF Channel Number | | | 1 | | |
| $BW_{channel}$ | | MHz | 5 | | |
| OCNG Patterns defined in A.3.2.1.15 (OP.15 FDD) | | | OP.15 FDD | | |
| PBCH_RA | | dB | 0 | | |
| PBCH_RB | | dB | | | |
| PSS_RA | | dB | | | |
| SSS_RA | | dB | | | |
| PCFICH_RB | | dB | | | |
| PHICH_RA | | dB | | | |
| PHICH_RB | | dB | | | |
| PDCCH_RA | | dB | | | |
| PDCCH_RB | | dB | | | |
| PDSCH_RA | | dB | | | |
| PDSCH_RB | | dB | | | |
| OCNG_RA Note 1 | | dB | | | |
| OCNG_RB Note 1 | | dB | | | |
| $N_{oc}$ Note 2 | Band 31 | dBm/15 kHz | -98 | | |
| RSRP Note 3 | Band 31 | dBm/15 kHz | -94 | | |
| $\hat{E}_s / I_{ot}$ | | dB | 4 | | |
| SCH_RP Note 3 | Band 31 | dBm/15 kHz | -94 | | |
| $\hat{E}_s / N_{oc}$ | | dB | 4 | | |
| Io Note3 | Band 31 | dBm/4.5 MHz | -67.8 | | |
| Propagation Condition | | | AWGN | | |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

# 9.5 UTRAN TDD P-CCPCH RSCP

## 9.5.1 E-UTRAN FDD – UTRA TDD P-CCPCH RSCP absolute accuracy

### 9.5.1.1 Test purpose

To verify that the UTRAN TDD P-CCPCH RSCP absolute measurement accuracy is within the specified limits.

### 9.5.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward that support UTRA TDD. Applicability requires support for FGI bit 39.

This test applies to all types of E-UTRA TDD UE release 9 and forward that support UTRA TDD and not supporting UTRA FDD. Applicability requires support for FGI bit 22.

### 9.5.1.3 Minimum conformance requirements

In RRC_CONNECTED state the accuracy requirements shall be the same as the inter-frequency measurement accuracy requirements for URAN TDD P-CCPCH RSCP in 3GPP TS 25.123 [22].

The accuracy requirements in table 9.5.1.3-1 are valid under the following conditions:

P-CCPCH RSCP ≥ -102 dBm

P-CCPCH Ec/Io ≥ -8 dB

DwPCH_Ec/Io ≥ -5 dB

**Table 9.5.1.3-1: UTRAN TDD P-CCPCH absolute accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/ 1.28 MHz] |
| P-CCPCH_RSCP | dBm | ± 6 | ± 9 | -94...-70 |
| | dBm | ± 8 | ± 11 | -70...-50 |

If the UE, in RRC_CONNECTED state, needs measurement gaps to perform UTRAN TDD measurements, the UTRAN TDD measurement procedure and measurement gap pattern stated in TS 36.133 [4] clause 8.1.2.4.3 shall apply.

The reporting range is for URAN TDD P-CCPCH RSCP is from -115 …-25 dBm.

In table 9.5.1.3-2 the mapping of measured quantity is defined.

The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.5.1.3-2: UTRAN TDD P-CCPCH absolute accuracy measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| P-CCPCH RSCP_LEV_-05 | P-CCPCH RSCP <-120 | dBm |
| P-CCPCH RSCP_LEV_-04 | -120 ≤ P-CCPCH RSCP < -119 | dBm |
| P-CCPCH RSCP_LEV_-03 | -119 ≤ P-CCPCH RSCP < -118 | dBm |
| … | … | … |
| PCCPCH_RSCP_LEV_89 | -27 ≤ PCCPCH RSCP< -26 | dBm |
| PCCPCH_RSCP_LEV_90 | -26 ≤ PCCPCH RSCP< -25 | dBm |
| PCCPCH_RSCP_LEV_91 | -25 ≤ PCCPCH RSCP | dBm |

The normative reference for this requirement is TS 25.123 [22] clause 9.1.1.1.1.2, clause 9.1.1.1.3 and TS 36.133 [4] clause 9.3.1 and A.9.5.1.

### 9.5.1.4 Test description

#### 9.5.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.

2. The general test parameter settings are set up according to Table 9.5.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.5.1.4.3.

5. Cell 1 is the serving E-UTRAN FDD cell and Cell 2 is the target UTRAN TDD cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

**Table 9.5.1.4.1-1: General test parameters for UTRAN TDD P-CCPCH RSCP absolute measurement accuracy test in E-UTRAN FDD**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1 |
| E-UTRAN RF Channel Number | | 1 | One E-UTRAN FDD carrier frequency is used. |
| UTRAN RF Channel Number | | 2 | One UTRAN TDD carrier frequency is used. |
| E-UTRAN Channel Bandwidth (BWchannel) | MHz | 10 | |
| Active cell | | Cell 1 | E-UTRA FDD cell1 on RF channel number 1 |
| Neighbour cell | | Cell 2 | 1.28Mcps UTRA TDD Cell2 on RF channel number 2 |
| Gap Pattern Id | | 1 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| CP length of cell 1 | | Normal | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| Inter-RAT (UTRAN TDD) measurement quantity | | P-CCPCH RSCP | |

#### 9.5.1.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.5.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check P-CCPCH RSCP reported values of Cell 2 in periodical MeasurementReport messages according to Table 9.5.1.5-3. If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one.

7. SS shall check the MeasurementReport message transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.5.1.5-1 as appropriate.

### 9.5.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.5.1.4.3-1: Common Exception messages for E-UTRAN FDD - UTRA TDD P-CCPCH RSCP absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7 |

**Table 9.5.1.4.3-2: *MeasConfig- DEFAULT:* Additional E-UTRAN FDD - UTRA TDD P-CCPCH RSCP absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | | |
| measObject CHOICE { | | | |
| measObject EUTRA | MeasObjectEUTRA-GENERIC(f1) | E-UTRA Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | | |
| measObject CHOICE { | | | |
| measObjectUTRA | MeasObjectUTRA-GENERIC(f2) | UTRA Cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigInterRAT-PERIODICAL | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| quantityConfig SEQUENCE { | | | |
| quantityConfigUTRA SEQUENCE { | | | UTRAN |
| measQuantityUTRA-TDD | pccpch-RSCP | | |
| } | | | |
| } | | | |
| measGapConfig | MeasGapConfig-GP2 | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 9.5.1.4.3-3: *MeasResults*: Additional E-UTRAN FDD - UTRA TDD P-CCPCH RSCP absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 9.5.1.4.3-4: MeasResultListUTRA: Additional E-UTRAN FDD - UTRA TDD P-CCPCH RSCP absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| fdd | Not Present | | |
| tdd | PhysCellIdUTRA-TDD | | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-RSCP | Set according to specific test INTEGER (-5..91) | | |
| } | | | |
| } | | | |

## 9.5.1.5 Test requirement

The test parameters are given in Tables 9.5.1.4.1-1, 9.5.1.5-1 and 9.5.1.5-2 as below. Table 9.5.1.5-2 and 9.5.1.5-3 define the primary level settings including test tolerances for all tests.

**Table 9.5.1.5-1: E-UTRAN FDD cell specific test parameters for UTRAN TDD P-CCPCH RSCP absolute measurement accuracy test in E-UTRAN TDD**

| Parameter | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | | |
| BWchannel | MHz | 10 | | |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) | | OP.1 FDD | | |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA Note1 | | | | |
| OCNG_RB Note1 | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | | |
| RSRP Note3 | dBm/15 kHz | -94 | | |
| Io Note3 | dBm/9 MHz | -64.76 | | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | | |
| Propagation condition | - | AWGN | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 9.5.1.5-2: UTRAN TDD cell specific test parameters for UTRAN TDD P-CCPCH RSCP absolute measurement accuracy test in E-UTRAN FDD**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|
| DL timeslot number | | 0 | DwPTS | 0 | DwPTS | 0 | DwPTS |
| UTRA RF Channel number Note2 | | Channel 2 | | Channel 2 | | Channel 2 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | | -3 | |
| DwPCH_Ec/Ior | dB | | 0 | | 0 | | 0 |
| OCNS_Ec/Ior | dB | -3 | | -3 | | -3 | |
| Ioc | dBm/1.28MHz | -54.9 | | -75.2 | | -96.8 | |
| Îor/Ioc | dB | 2 | | 5 | | 0 | |
| PCCPCH RSCP Note1 | dBm | -55.9 | | -73.2 | | -99.2 | |
| Io Note1 | dBm/1.28MHz | -50.78 | | -69.01 | | -93.19 | |
| Propagation condition | | AWGN | | | | | |

Note 1: PCCPCH RSCP and Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves.
Note 2: In the case of multi-frequency network of 1.28 Mcps TDD, the UTRA RF Channel Number can be set for the primary frequency in this test.

Each UTRAN TDD P-CCPCH RSCP absolute measurement accuracy test shall meet the reported values test requirements in table 9.5.1.5-3.

**Table 9.5.1.5-3: UTRAN TDD P-CCPCH RSCP absolute measurement accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | P-CCPCH RSCP_LEV_51 | P-CCPCH RSCP_LEV 34 | P-CCPCH RSCP_LEV 10 |
| Highest reported value (Cell 2) | P-CCPCH RSCP_LEV_68 | P-CCPCH RSCP_LEV 51 | P-CCPCH RSCP_LEV 23 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | P-CCPCH RSCP_LEV_48 | P-CCPCH RSCP_LEV 31 | P-CCPCH RSCP_LEV 07 |
| Highest reported value (Cell 2) | P-CCPCH RSCP_LEV_71 | P-CCPCH RSCP_LEV 54 | P-CCPCH RSCP_LEV 26 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.5.2 E-UTRAN TDD – UTRA TDD P-CCPCH RSCP absolute accuracy

### 9.5.2.1 Test purpose

To verify that the UTRAN TDD P-CCPCH RSCP absolute measurement accuracy is within the specified limits.

### 9.5.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward that support UTRA TDD. Applicability requires support for FGI bit 39.

This test applies to all types of E-UTRA TDD UE release 9 and forward that support UTRA TDD and not supporting UTRA FDD. Applicability requires support for FGI bit 22.

### 9.5.2.3 Minimum conformance requirements

In RRC_CONNECTED state the accuracy requirements shall be the same as the inter-frequency measurement accuracy requirements for URAN TDD P-CCPCH RSCP in 3GPP TS 25.123 [22].

The accuracy requirements in table 9.5.2.3-1 are valid under the following conditions:

P-CCPCH RSCP ≥ -102 dBm

P-CCPCH Ec/Io ≥ -8 dB

DwPCH_Ec/Io ≥ -5 dB

**Table 9.5.2.3-1: UTRAN TDD P-CCPCH absolute accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Io [dBm/ 1.28 MHz] |
| P-CCPCH_RSCP | dBm | ± 6 | ± 9 | -94...-70 |
| | dBm | ± 8 | ± 11 | -70...-50 |

If the UE, in RRC_CONNECTED state, needs measurement gaps to perform UTRAN TDD measurements, the UTRAN TDD measurement procedure and measurement gap pattern stated in TS 36.133 [4] clause 8.1.2.4.3 shall apply.

The reporting range is for URAN TDD P-CCPCH RSCP is from -115 …-25 dBm.

In table 9.5.2.3-2 the mapping of measured quantity is defined.

The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.5.2.3-2: UTRAN TDD P-CCPCH absolute accuracy measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| PCCPCH_RSCP_LEV -05 | P-CCPCH RSCP <-120 | dBm |
| PCCPCH_RSCP_LEV -04 | -120 ≤ P-CCPCH RSCP < -119 | dBm |
| PCCPCH_RSCP_LEV -03 | -119 ≤ P-CCPCH RSCP < -118 | dBm |
| … | … | … |
| PCCPCH_RSCP_LEV_89 | -27 ≤ PCCPCH RSCP< -26 | dBm |
| PCCPCH_RSCP_LEV_90 | -26 ≤ PCCPCH RSCP< -25 | dBm |
| PCCPCH_RSCP_LEV_91 | -25 ≤ PCCPCH RSCP | dBm |

The normative reference for this requirement is TS 25.123 [22] clause 9.1.1.1.1.2, clause 9.1.1.1.3 and TS 36.133 [4] clause 9.3.1 and A.9.5.2.

## 9.5.2.4 Test description

### 9.5.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.22.

2. The general test parameter settings are set up according to Table 9.5.2.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.5.2.4.3.

5. Cell 1 is the serving E-UTRAN TDD cell and Cell 2 is the target UTRAN TDD cell. Cell 1 is the cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

**Table 9.5.2.4.1-1: General test parameters for UTRAN TDD P-CCPCH RSCP absolute measurement accuracy test in E-UTRAN TDD**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2 |
| PCFICH/PDCCH/PHICH parameters | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2 |
| E-UTRAN RF Channel Number | | 1 | One E-UTRAN TDD carrier frequency is used. |
| UTRAN RF Channel Number | | 2 | One UTRAN TDD carrier frequency is used. |
| E-UTRAN Channel Bandwidth (BWchannel) | MHz | 10 | |
| Active cell | | Cell 1 | E-UTRA TDD cell1 on RF channel number 1 |
| Neighbour cell | | Cell 2 | 1.28Mcps UTRA TDD Cell2 on RF channel number 2 |
| Gap Pattern Id | | 1 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| Uplink-downlink configuration of cell 1 | | 1 | As specified in table 4.2.2 in TS 36.211 [9] |
| Special subframe configuration of cell 1 | | 6 | As specified in table 4.2.1 in TS 36.211 [9] |
| CP length of cell 1 | | Normal | |
| Filter coefficient | | 0 | L3 filtering is not used |
| DRX | | OFF | |
| Time offset between cells | ms | 3 | Asynchronous cells |
| Inter-RAT (UTRAN TDD) measurement quantity | | P-CCPCH RSCP | |

### 9.5.2.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.5.2.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message on Cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check P-CCPCH RSCP reported values of Cell 2 in periodical MeasurementReport messages according to Table 9.5.2.5-3 If the UE fails to report the measurement value for Cell 2, the number of failed iterations is increased by one..

7. SS shall check the MeasurementReport message transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

8. Repeat step 1-7 for each sub-test in Table 9.5.2.5-1 as appropriate.

### 9.5.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.5.2.4.3-1: Common Exception messages for E-UTRAN TDD - UTRA TDD P-CCPCH RSCP absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7 |

**Table 9.5.2.4.3-2: *MeasConfig- DEFAULT:* Additional E-UTRAN TDD - UTRA TDD P-CCPCH RSCP absolute accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | | |
| measObject CHOICE { | | | |
| measObject EUTRA | MeasObjectEUTRA-GENERIC(f1) | E-UTRA Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | | |
| measObject CHOICE { | | | |
| measObjectUTRA | MeasObjectUTRA-GENERIC(f2) | UTRA Cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigInterRAT-PERIODICAL | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| quantityConfig SEQUENCE { | | | |
| quantityConfigUTRA SEQUENCE { | | | UTRAN |
| measQuantityUTRA-TDD | pccpch-RSCP | | |
| } | | | |
| } | | | |
| measGapConfig | MeasGapConfig-GP2 | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table 9.5.2.4.3-3: *MeasResults*: Additional E-UTRAN TDD - UTRA TDD P-CCPCH RSCP absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| measResultListUTRA | MeasResultListUTRA | | |
| } | | | |
| } | | | |

**Table 9.5.2.4.3-4: MeasResultListUTRA: Additional E-UTRAN TDD - UTRA TDD P-CCPCH RSCP absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultsListUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| physCellId CHOICE { | | | |
| fdd | Not Present | | |
| tdd | PhysCellIdUTRA-TDD | | |
| } | | | |
| measResult SEQUENCE { | | | |
| utra-RSCP | Set according to specific test INTEGER (-5..91) | | |
| } | | | |
| } | | | |

## 9.5.2.5 Test requirement

The test parameters are given in Tables 9.5.2.4.1-1, 9.5.2.5-1 and 9.5.2.5-2 as below. Table 9.5.2.5-2 and 9.5.2.5-3 define the primary level settings including test tolerances for all tests.

**Table 9.5.2.5-1: E-UTRAN TDD cell specific test parameters for UTRAN TDD P-CCPCH RSCP absolute measurement accuracy test in E-UTRAN TDD**

| Parameter | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | | |
| BWchannel | MHz | 10 | | |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) | | OP.1 TDD | | |
| PBCH_RA | dB | 0 | | |
| PBCH_RB | | | | |
| PSS_RA | | | | |
| SSS_RA | | | | |
| PCFICH_RB | | | | |
| PHICH_RA | | | | |
| PHICH_RB | | | | |
| PDCCH_RA | | | | |
| PDCCH_RB | | | | |
| PDSCH_RA | | | | |
| PDSCH_RB | | | | |
| OCNG_RA Note1 | | | | |
| OCNG_RB Note1 | | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | | |
| $\hat{E}_s / I_{ot}$ | dB | 4 | | |
| RSRP Note3 | dBm/15 kHz | -94 | | |
| Io Note3 | dBm/9 MHz | -64.76 | | |
| $\hat{E}_s / N_{oc}$ | dB | 4 | | |
| Propagation condition | - | AWGN | | |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

**Table 9.5.2.5-2: UTRAN TDD cell specific test parameters for UTRAN TDD P-CCPCH RSCP absolute measurement accuracy test in E-UTRAN TDD**

| Parameter | Unit | Test 1 | | Test 2 | | Test 3 | |
|---|---|---|---|---|---|---|---|
| DL timeslot number | | 0 | DwPTS | 0 | DwPTS | 0 | DwPTS |
| UTRA RF Channel number Note2 | | Channel 2 | | Channel 2 | | Channel 2 | |
| PCCPCH_Ec/Ior | dB | -3 | | -3 | | -3 | |
| DwPCH_Ec/Ior | dB | | 0 | | 0 | | 0 |
| OCNS_Ec/Ior | dB | -3 | | -3 | | -3 | |
| Ioc | dBm/1.28MHz | -54.9 | | -75.2 | | -96.2 | |
| Îor/Ioc | dB | 2 | | 5 | | 0 | |
| PCCPCH RSCP Note1 | dBm | -55.9 | | -73.2 | | -99.2 | |
| Io Note1 | dBm/1.28MHz | -50.78 | | -69.01 | | -93.19 | |
| Propagation condition | | AWGN | | | | | |

Note 1: PCCPCH RSCP and Io levels have been calculated from other parameters for information purposes. They are not settable parameters themselves.
Note 2: In the case of multi-frequency network of 1.28 Mcps TDD, the UTRA RF Channel Number can be set for the primary frequency in this test.

Each UTRAN TDD P-CCPCH RSCP absolute measurement accuracy test shall meet the reported values test requirements in table 9.5.2.5-3.

**Table 9.5.2.5-3: UTRAN TDD P-CCPCH RSCP absolute measurement accuracy requirements for the reported values**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| Normal Conditions | | | |
| Lowest reported value (Cell 2) | P-CCPCH RSCP_LEV_51 | P-CCPCH RSCP_LEV 34 | P-CCPCH RSCP_LEV 10 |
| Highest reported value (Cell 2) | P-CCPCH RSCP_LEV_68 | P-CCPCH RSCP_LEV 51 | P-CCPCH RSCP_LEV 23 |
| Extreme Conditions | | | |
| Lowest reported value (Cell 2) | P-CCPCH RSCP_LEV_48 | P-CCPCH RSCP_LEV 31 | P-CCPCH RSCP_LEV 07 |
| Highest reported value (Cell 2) | P-CCPCH RSCP_LEV_71 | P-CCPCH RSCP_LEV 54 | P-CCPCH RSCP_LEV 26 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 9.6 GSM carrier RSSI

## 9.6.1 GSM RSSI accuracy for E-UTRAN FDD

### 9.6.1.1 Test purpose

To verify that the GSM RSSI measurement accuracy is within the specified limits.

### 9.6.1.2 Test applicability

This test applies all the types of E-UTRA FDD UE release 9 and forward that support GSM. Applicability requires support for FGI bits 16 and 23.

### 9.6.1.3 Minimum conformance requirements

**Absolute accuracy**

The R.M.S received signal level at the receiver input shall be measured by the MS and the BSS over the full range of -110 dBm to -48 dBm with an absolute accuracy of ±4 dB from -110 dBm to -70 dBm under normal conditions and ±6 dB over the full range under both normal and extreme conditions. The R.M.S received signal level at the receiver input shall be measured by the MS above -48 dBm up to -38 dBm with an absolute accuracy of ± 9 dB under both normal and extreme conditions.

This requirement is summarized in Table 9.6.1.3-1.

**Table 9.6.1.3-1: GSM RXLEV absolute accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Input level dBm |
| RXLEV | dBm | ± 4 | ± 6 | -110…-70 |
| | dBm | ± 6 | ± 6 | -70…-48 |
| | dBm | ± 9 | ± 9 | -48..-38 |

The reporting range and mapping for RXLEV is summarized in Table 9.6.1.3-2.

**Table 9.6.1.3-2: GSM RSSI measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RXLEV_00 | RXLEV < -110 | dBm |
| RXLEV_01 | -110 ≤ RXLEV < -109 | dBm |
| RXLEV_02 | -109 ≤ RXLEV < -108 | dBm |
| … | … | … |
| RXLEV_61 | -50≤ RXLEV < -49 | dBm |
| RXLEV_62 | -49 ≤ RXLEV < -48 | dBm |
| RXLEV_63 | -48 ≤ RXLEV | dBm |

**Relative accuracy**

The relative accuracy shall be as follows:

If signals of level x1 and x2 dBm are received (where x1 ≤ x2) and levels y1 and y2 dBm respectively are measured, if x2 - x1 < 20 dB and x1 is not below the reference sensitivity level, then y1 and y2 shall be such that:

(x2 - x1) - a ≤ y2 - y1 ≤ (x2 - x1 + b) if the measurements are on the same or on different RF channel within the same frequency band;

and

(x2 - x1 ) - c ≤ y2 - y1 ≤( x2 - x1 + d) if the measurements are on different frequency bands:

a, b, c and d are in dB and depend on the value of x1 as follows:

| | **a** | **b** | **c** | **d** |
|---|---|---|---|---|
| x1 ≥ s+14, x2< -48 dBm | 2 | 2 | 4 | 4 |
| s+14 > x1 ≥ s+1 | 3 | 2 | 5 | 4 |
| s+1 > x1 | 4 | 2 | 6 | 4 |
| | | | | |

For single band MS and measurements between ARFCN in the same band for a multiband MS:

s = reference sensitivity level as specified in Table 9.6.1.3-3 (normative reference 3GPP TS 45.005 [16] Table 6.2-1a)

For measurements between ARFCN in different bands:

s = the reference sensitivity level as specified in Table 9.6.1.3-3 (normative reference 3GPP TS 45.005 [16] Table 6.2-1a) for the band including x1.

**Table 9.6.1.3-3: Reference sensitivity level for MS**

| | | |
|---|---|---|
| GSM 400 MS | for GSM 400 small MS | -102 dBm |
| | for other GSM 400 MS | -104 dBm |
| GSM 900 MS | for GSM 900 small MS | -102 dBm |
| | for other GSM 900 MS | -104 dBm |
| GSM 850 MS | for GSM 850 small MS | -102 dBm |
| | for other GSM 850 MS | -104 dBm |
| GSM 700 MS | for GSM 700 small MS | -102 dBm |
| | for other GSM 700 MS | -104 dBm |
| DCS 1 800 MS | for DCS 1 800 class 1 or class 2 MS | -100 / -102 dBm * |
| | for DCS 1 800 class 3 MS | -102 dBm |
| PCS 1 900 MS | for PCS 1 900 class 1 or class 2 MS | -102 dBm |
| | for other PCS 1 900 MS | -104 dBm |
| Note: For DCS 1 800 class 1 and class 2 MS, the 102 dBm level shall apply for the reference sensitivity performance as specified in table 1 for the normal conditions defined in TS 45.005 [16] Annex D and 100 dBm level shall be used to determine all other MS performances. | | |

The normative reference for this requirement is:

For E-UTRA: TS 36.133 [4] clause 9.4.1 and A.9.6.1

For GSM: TS 45.008 [15] clause 8.1.2 and 8.1.4 and TS 45.005 [16].

## 9.6.1.4 Test description

### 9.6.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. The general test parameter settings are set up according to Table 9.6.1.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.6.1.4.3.

5. There is one E-UTRA FDD cell (Cell 1) and two GSM cells (Cell 2 and Cell 3) specified in each test. Cell 1 is the cell used for call setup with the power level set according to Annex C.0 and C.1 for this test. Cell 2 (BCCH1) and Cell 3 (BCCH other than BCCH1 according to sub-test) are measured and reported by the UE.

**Table 9.6.1.4.1-1: General GSM Carrier RSSI test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.0 FDD | As specified in section A.1.1. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN FDD) | | DL Reference Measurement Channel R.6 FDD | As specified in section A.2.1. |
| Active cell | - | Cell 1 | |
| DRX | - | OFF | |
| Gap pattern Id | | 1 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| Filtering coefficient | - | 0 | L3 filtering is not used. |
| Inter-RAT measurement quantity | | GSM Carrier RSSI | |
| Monitored cell list size | | 6 GSM neighbours including ARFCN 1 | Included in the Measurement control information |

### 9.6.1.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.6.1.5-1, 9.6 1.5-2 and 9.6 1.5-3 as appropriate. Propagation conditions for the E-UTRA cell are set according to Annex B clause B.1. For sub-test 4 and sub-test 12 GERAN cell is initialized to -95dBm.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. Only for sub-tests 4 and 12, SS waits for 10s for initial measurement report by UE and ignores measurement reports received at this stage. If SS receives the measurement reports, then adjust the power levels of Cell 2 according to Table

9.6.1.5-2. SS ignores measurement reports for 1 reportInterval cycle (1024ms) and continues with step 7, else fail the UE for the relevant subtest and skip to Step.9

7. SS shall check the reported GSM RSSI values in periodical MeasurementReport messages. The reported RSSI value for Cell 2 is compared to the actual RSSI value according to Table 9.6.1.5-4. This counts as a Pass or Fail for the event "Absolute". Also the reported RSSI value for Cell 3 is compared to the reported RSSI value for Cell 2 for each MeasurementReport message according to Table 9.6.1.5-5. This counts as a Pass or Fail for the event "Relative".
If the UE fails to report the measurement value for Cell 2, the number of failed iterations for the event "Absolute" is increased by one.
If the UE fails to report the measurement value for any of Cell 2 or Cell 3, the number of failed iterations for the event "Relative" is increased by one.

8. The SS shall check the periodical MeasurementReport messages transmitted by the UE until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved for each of the events "Absolute" and "Relative". Each event is evaluated only until the confidence level is achieved. Different events may require different times for a verdict.

9. Repeat step 1-9 for each sub-test in Table 9.6.1.5-2 as appropriate.

### 9.6.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.6.1.4.3-1: Common Exception messages for GSM RSSI measurement accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.3.1-11 |

**Table 9.6.1.4.3-2: MeasuredResults: Additional GSM RSSI measurement accuracy test requirement**

| Derivation Path: TS 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | RSRP-Range | Set according to specific test | |
| rsrqResult | RSRQ-Range | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| MeasResultListGERAN | MeasResultListGERAN | | |
| } | | | |
| } | | | |

**Table 9.6.1.4.3-3: MeasResultListGERAN: Additional GSM measurement accuracy test requirement**

| Derivation Path: TS 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListGERAN::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| carrierFreq | CarrierFreqGERAN | | |
| physCellId | physCellId GERAN | | |
| Cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| Rssi | INTEGER (0..63) | Set according to specific test | |
| } | | | |
| } | | | |

**Table 9.6.1.4.3-4: ReportConfigInterRAT-PERIODICAL: Additional GSM measurement accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-9 ReportConfigInterRAT-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-PERIODICAL ::= SEQUENCE { | | | |
| maxReportCells | 6 | | |
| } | | | |

## 9.6.1.5 Test requirement

Tables 9.6.1.5-1, 9.6.1.5-2 and 9.6.1.5-3 define the primary level settings including test tolerances for all tests.

The GSM RSSI measurement accuracy test for the reported values shall meet the requirements in Table 9.6.1.5-4 and Table 9.6.1.5-5.

**Table 9.6.1.5-1: E-UTRAN FDD Cell specific test parameters for GSM Carrier RSSI accuracy test in E-UTRAN FDD**

| Parameter | Unit | Tests 1-12 |
|---|---|---|
| E-UTRAN RF Channel Number | | 1 |
| $BW_{channel}$ | MHz | 10 |
| OCNG Patterns defined in D.1.1 (OP.1 FDD) | | OP.1 FDD |

| | | |
|---|---|---|
| PBCH_RA | dB | 0 |
| PBCH_RB | dB | |
| PSS_RA | dB | |
| SSS_RA | dB | |
| PCFICH_RB | dB | |
| PHICH_RA | dB | |
| PHICH_RB | dB | |
| PDCCH_RA | dB | |
| PDCCH_RB | dB | |
| PDSCH_RA | dB | |
| PDSCH_RB | dB | |
| OCNG_RA Note 1 | dB | |
| OCNG_RB Note 1 | dB | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 |
| RSRP Note 3 | dBm/15 kHz | -94 |
| $\hat{E}_s / I_{ot}$ | dB | 4 |
| SCH_RP Note 3 | dBm/15 kHz | -94 |
| $\hat{E}_s / N_{oc}$ | dB | 4 |
| Propagation Condition | | AWGN |

Note 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 9.6.1.5-2: BCCH signal levels at receiver input in dBm**

| Sub-test | BCCH1 | BCCH2 | BCCH3 | BCCH4 | BCCH5 | BCCH6 |
|---|---|---|---|---|---|---|
| 1 | -38.7 | -38.5 | NA | NA | NA | NA |
| 2 | -48.7 | -50.0 | NA | NA | NA | NA |
| 3 | -70.7 | -70.5 | NA | NA | NA | NA |
| 4 | -109.3 | -109.5 | NA | NA | NA | NA |
| 5 | -57.5 | NA | -54.5 | NA | NA | NA |
| 6 | -64.5 | NA | -59.5 | NA | NA | NA |
| 7 | -71.5 | NA | NA | -64.5 | NA | NA |
| 8 | -78.5 | NA | NA | -69.5 | NA | NA |
| 9 | -85.5 | NA | NA | NA | -74.5 | NA |
| 10 | -92.5 | NA | NA | NA | -79.5 | NA |
| 11 | -99.5 | NA | NA | NA | NA | -84.5 |
| 12 | -106.5 | NA | NA | NA | NA | -89.5 |

**Table 9.6.1.5-3: ARFCN numbers for GSM cells**

| GSM band | BCCH1 | BCCH2 | BCCH3 | BCCH4 | BCCH5 | BCCH6 |
|---|---|---|---|---|---|---|
| GSM 450 | 276 | 293 | 264 | 269 | 281 | 288 |
| GSM 480 | 323 | 340 | 311 | 316 | 328 | 335 |
| GSM 750 | 475 | 511 | 440 | 455 | 485 | 500 |
| GSM 850 | 189 | 251 | 150 | 170 | 210 | 230 |
| GSM 900 | 62 | 124 | 20 | 40 | 80 | 100 |
| DCS 1800 | 700 | 885 | 585 | 660 | 790 | 835 |
| PCS 1900 | 700 | 805 | 585 | 660 | 790 | 550 |

Note: As defined in clause 3A.1, the test shall run without frequency overlapping between E-UTRA and GSM cells. The ARFCN numbers defined here, can be updated accordingly (even E-UTRA band specific) to avoid possible overlapping.

**Table 9.6.1.5-4: GSM Carrier RSSI absolute accuracy requirements for the reported values**

| Sub-test | Normal Condition | | Extreme condition | |
|---|---|---|---|---|
| | Lowest reported value for BCCH1 | Highest reported value for BCCH1 | Lowest reported value for BCCH1 | Highest reported value for BCCH1 |
| 1 | RXLEV_62 | RXLEV_63 | RXLEV_62 | RXLEV_63 |
| 2 | RXLEV_55 | RXLEV_63 | RXLEV_55 | RXLEV_63 |
| 3 | RXLEV_35 | RXLEV_45 | RXLEV_33 | RXLEV_47 |
| 4 | RXLEV_00 | RXLEV_06 | RXLEV_00 | RXLEV_08 |
| 5 | RXLEV_46 | RXLEV_60 | RXLEV_46 | RXLEV_60 |
| 6 | RXLEV_39 | RXLEV_53 | RXLEV_39 | RXLEV_53 |
| 7 | RXLEV_34 | RXLEV_44 | RXLEV_32 | RXLEV_46 |
| 8 | RXLEV_27 | RXLEV_37 | RXLEV_25 | RXLEV_39 |
| 9 | RXLEV_20 | RXLEV_30 | RXLEV_18 | RXLEV_32 |
| 10 | RXLEV_13 | RXLEV_23 | RXLEV_11 | RXLEV_25 |
| 11 | RXLEV_06 | RXLEV_16 | RXLEV_04 | RXLEV_18 |
| 12 | RXLEV_00 | RXLEV_09 | RXLEV_00 | RXLEV_11 |
| Note: It is not mandatory for the UE to report BCCH1 in sub-test 12. In case of no BCCH1 report in sub-test 12, the absolute accuracy for sub-test 12 is not tested. | | | | |

**Table 9.6.1.5-5: GSM Carrier RSSI Relative accuracy requirements for the reported values**

| Sub-test | Normal & Extreme condition | |
|---|---|---|
| | Lowest reported value for BCCH2 | Highest reported value for BCCH2 |
| 1 | N/A (Note3) | N/A(Note3) |
| 2 | RXLEV = x-6 | RXLEV = x+3 |
| 3 | RXLEV = x-4 | RXLEV = x+5 |
| 4 | N/A (Note3) | N/A (Note3) |
| | Lowest reported value for BCCH3 | Highest reported value for BCCH3 |
| 5 | RXLEV = x-2 | RXLEV = x+8 |
| 6 | RXLEV = x+1 | RXLEV = x+10 |
| | Lowest reported value for BCCH4 | Highest reported value for BCCH4 |
| 7 | RXLEV = x+3 | RXLEV = x+12 |
| 8 | RXLEV = x+5 | RXLEV = x+14 |
| | Lowest reported value for BCCH5 | Highest reported value for BCCH5 |
| 9 | RXLEV = x+7 | RXLEV = x+16 |
| 10 | RXLEV = x+8 | RXLEV = x+18 |
| | Lowest reported value for BCCH6 | Highest reported value for BCCH6 |
| 11 | RXLEV = x+10 | RXLEV = x+20 |
| 12 | N/A (Note3) | N/A (Note3) |
| Note 1: x is the reported value RXLEV for BCCH1.<br>Note 2: It is not mandatory for the UE to report BCCH1 in sub-test 12. In case of no BCCH1 report in sub-test 12, the relative accuracy for sub-test 12 is not tested.<br>Note 3: Sub-tests 1, 4 and 12 are not applicable for relative accuracy as they would be testing the UE outside the side conditions. | | |

For the test to pass, the ratio of successful reported values in each sub-test for absolute and relative accuracy shall be more than 90% with a confidence level of 95%.

## 9.6.2 GSM RSSI accuracy for E-UTRAN TDD

### 9.6.2.1 Test purpose

To verify that the GSM RSSI measurement accuracy is within the specified limits.

### 9.6.2.2 Test applicability

This test applies all the types of E-UTRA TDD UE release 9 and forward that support GSM. Applicability requires support for FGI bit 16 and 23.

### 9.6.2.3 Minimum conformance requirements

**Absolute accuracy**

The R.M.S received signal level at the receiver input shall be measured by the MS and the BSS over the full range of -110 dBm to -48 dBm with an absolute accuracy of ±4 dB from -110 dBm to -70 dBm under normal conditions and ±6 dB over the full range under both normal and extreme conditions. The R.M.S received signal level at the receiver input shall be measured by the MS above -48 dBm up to -38 dBm with an absolute accuracy of ± 9 dB under both normal and extreme conditions.

This requirement is summarized in Table 9.6.2.3-1.

**Table 9.6.2.3-1: GSM RXLEV absolute accuracy**

| Parameter | Unit | Accuracy [dB] | | Conditions |
|---|---|---|---|---|
| | | Normal condition | Extreme condition | Input level dBm |
| RXLEV | dBm | ± 4 | ± 6 | -110…-70 |
| | dBm | ± 6 | ± 6 | -70…-48 |
| | dBm | ± 9 | ± 9 | -48..-38 |

The reporting range and mapping for RXLEV is summarized in Table 9.6.2.3-2.

**Table 9.6.2.3-2: GSM RSSI measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RXLEV_00 | RXLEV < -110 | dBm |
| RXLEV_01 | -110 ≤ RXLEV < -109 | dBm |
| RXLEV_02 | -109 ≤ RXLEV < -108 | dBm |
| … | … | … |
| RXLEV_61 | -50 ≤ RXLEV < -49 | dBm |
| RXLEV_62 | -49 ≤ RXLEV < -48 | dBm |
| RXLEV_63 | -48 ≤ RXLEV | dBm |

**Relative accuracy**

The relative accuracy shall be as follows:

If signals of level x1 and x2 dBm are received (where x1 ≤ x2) and levels y1 and y2 dBm respectively are measured, if x2 - x1 < 20 dB and x1 is not below the reference sensitivity level, then y1 and y2 shall be such that:

(x2 - x1) - a ≤ y2 - y1 ≤ (x2 - x1 + b) if the measurements are on the same or on different RF channel within the same frequency band;

and

(x2 - x1 ) - c ≤ y2 - y1 ≤( x2 - x1 + d) if the measurements are on different frequency bands:

a, b, c and d are in dB and depend on the value of x1 as follows:

| | a | b | c | d |
|---|---|---|---|---|
| x1 ≥ s+14, x2< -48 dBm | 2 | 2 | 4 | 4 |
| s+14 > x1 ≥ s+1 | 3 | 2 | 5 | 4 |
| s+1 > x1 | 4 | 2 | 6 | 4 |
| | | | | |

For single band MS and measurements between ARFCN in the same band for a multiband MS:

s = reference sensitivity level as specified in Table 9.6.2.3-3 (normative reference 3GPP TS 45.005 [16] Table 6.2-1a)

For measurements between ARFCN in different bands:

s = the reference sensitivity level as specified in Table 9.6.2.3-3 (normative reference 3GPP TS 45.005 [16] Table 6.2-1a) for the band including x1.

**Table 9.6.2.3-3: Reference sensitivity level for MS**

| | | |
|---|---|---|
| GSM 400 MS | for GSM 400 small MS | -102 dBm |
| | for other GSM 400 MS | -104 dBm |
| GSM 900 MS | for GSM 900 small MS | -102 dBm |
| | for other GSM 900 MS | -104 dBm |
| GSM 850 MS | for GSM 850 small MS | -102 dBm |
| | for other GSM 850 MS | -104 dBm |
| GSM 700 MS | for GSM 700 small MS | -102 dBm |
| | for other GSM 700 MS | -104 dBm |
| DCS 1 800 MS | for DCS 1 800 class 1 or class 2 MS | -100 / -102 dBm * |
| | for DCS 1 800 class 3 MS | -102 dBm |
| PCS 1 900 MS | for PCS 1 900 class 1 or class 2 MS | -102 dBm |
| | for other PCS 1 900 MS | -104 dBm |
| Note: For DCS 1 800 class 1 and class 2 MS, the 102 dBm level shall apply for the reference sensitivity performance as specified in table 1 for the normal conditions defined in TS 45.005 [16] Annex D and 100 dBm level shall be used to determine all other MS performances. | | |

The normative reference for this requirement is:

For E-UTRA: TS 36.133 [4] clause 9.4.1 and A.9.6.2

For GSM: TS 45.008 [15] clause 8.1.2 and 8.1.4 and TS 45.005 [16].

## 9.6.2.4 Test description

### 9.6.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.14.

2. The general test parameter settings are set up according to Table 9.6.2.4.1-1.

3. Propagation conditions are set according to Annex B clause B.0.

4. Message contents are defined in clause 9.6.2.4.3.

5. There is one E-UTRA TDD cell (Cell 1) and two GSM cells (Cell 2 and Cell 3) specified in each test. Cell 1 is the cell used for call setup with the power level set according to Annexes C.0 and C.1 for this test. Cell 2 (BCCH1) and Cell 3 (BCCH other than BCCH1 according to sub-test) are measured and reported by the UE.

**Table 9.6.2.4.1-1: General GSM Carrier RSSI test parameters**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| PDSCH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.0 TDD | As specified in section A.1.2. |
| PCFICH/PDCCH/PHICH parameters (E-UTRAN TDD) | | DL Reference Measurement Channel R.6 TDD | As specified in section A.2.2. |
| Active cell | - | Cell 1 | |
| DRX | - | OFF | |
| Uplink-downlink configuration of cell 1 | | 1 | As specified in table 4.2.2 in TS 36.211 [9] |
| Special subframe configuration of cell 1 | | 6 | As specified in table 4.2.1 in TS 36.211 [9] |
| Gap pattern Id | | 1 | As specified in 3GPP TS 36.133 [4] section 8.1.2.1. |
| Filtering coefficient | - | 0 | L3 filtering is not used. |
| Inter-RAT measurement quantity | | GSM Carrier RSSI | |
| Monitored cell list size | | 6 GSM neighbours including ARFCN 1 | Included in the Measurement control information |

### 9.6.2.4.2 Test procedure

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.6.2.5-1, 9.6 2.5-2 and 9.6 2.5-3 as appropriate. Propagation conditions for the E-UTRA cell are set according to Annex B clause B.1. For sub-test 4 and sub-test 12 GERAN cell is initialized to -95dBm.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. Only for sub-tests 4 and 12, SS waits for 10s for initial measurement report by UE and ignores measurement reports received at this stage. If SS receives the measurement reports, then adjust the power levels of Cell 2 according to Table 9.6.2.5-2. SS ignores measurement reports for 1 reportInterval cycle (1024ms) and continues with step 7, else fails the UE for the relevant sub-test and skip to Step.9.

7. SS shall check the reported GSM RSSI value in periodical MeasurementReport messages. The reported RSSI value for Cell 2 is compared to the actual RSSI value according to Table 9.6.2.5-4. This counts as a Pass or Fail for the event “Absolute”. Also the reported RSSI value for Cell 3 is compared to the reported RSSI value for Cell 2 for each MeasurementReport message according to Table 9.6.2.5-5. This counts as a Pass or Fail for the event “Relative”.
If the UE fails to report the measurement value for Cell 2, the number of failed iterations for the event “Absolute” is increased by one.
If the UE fails to report the measurement value for any of Cell 2 or Cell 3, the number of failed iterations for the event ”Relative” is increased by one.

8. The SS shall check the periodical MeasurementReport messages transmitted by the UE until the confidence level according to Table G.2.3-1 in Annex G.2 is achieved for each of the events "Absolute” and “Relative”. Each event is evaluated only until the confidence level is achieved. Different events may require different times for a verdict.

9. Repeat step 1-9 for each sub-test in Table 9.6.2.5-2 as appropriate.

### 9.6.2.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.6.2.4.3-1: Common Exception messages for GSM RSSI measurement accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.1-7<br>Table H.3.1-11 |

**Table 9.6.2.4.3-2: MeasuredResults: Additional GSM RSSI measurement accuracy test requirement**

| Derivation Path: TS 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | RSRP-Range | Set according to specific test | |
| rsrqResult | RSRQ-Range | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| MeasResultListGERAN | MeasResultListGERAN | | |
| } | | | |
| } | | | |

**Table 9.6.2.4.3-3: MeasResultListGERAN: Additional GSM measurement accuracy test requirement**

| Derivation Path: TS 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResultListGERAN::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| carrierFreq | CarrierFreqGERAN | | |
| physCellId | physCellId GERAN | | |
| Cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| Rssi | INTEGER (0..63) | Set according to specific test | |
| } | | | |
| } | | | |

**Table 9.6.2.4.3-4: ReportConfigInterRAT-PERIODICAL: Additional GSM measurement accuracy test requirement**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-9 ReportConfigInterRAT-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigInterRAT-PERIODICAL ::= SEQUENCE { | | | |
| maxReportCells | 6 | | |
| } | | | |

## 9.6.2.5 Test requirement

Tables 9.6.2.5-1, 9.6.2.5-2 and 9.6.2.5-3 define the primary level settings including test tolerances for all tests.

The GSM RSSI measurement accuracy test for the reported values shall meet the requirements in Table 9.6.2.5-4 and Table 9.6.2.5-5.

**Table 9.6.2.5-1: E-UTRAN TDD Cell specific test parameters for GSM Carrier RSSI accuracy test in E-UTRAN TDD**

| Parameter | Unit | Tests 1 - 12 |
|---|---|---|
| E-UTRAN RF Channel Number | | 1 |
| $BW_{channel}$ | MHz | 10 |
| OCNG Patterns defined in D.2.1 (OP.1 TDD) | | OP.1 TDD |
| PBCH_RA | dB | 0 |
| PBCH_RB | dB | |
| PSS_RA | dB | |
| SSS_RA | dB | |
| PCFICH_RB | dB | |
| PHICH_RA | dB | |
| PHICH_RB | dB | |
| PDCCH_RA | dB | |
| PDCCH_RB | dB | |
| PDSCH_RA | dB | |
| PDSCH_RB | dB | |
| OCNG_RA Note 1 | dB | |
| OCNG_RB Note 1 | dB | |
| $N_{oc}$ Note 2 | dBm/15 kHz | -98 |
| RSRP Note 3 | dBm/15 kHz | -94 |
| $\hat{E}_s / I_{ot}$ | dB | 4 |
| SCH_RP Note 3 | dBm/15 kHz | -94 |
| $\hat{E}_s / N_{oc}$ | dB | 4 |
| **Propagation Condition** | | AWGN |

NOTE 1: OCNG shall be used such that all cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

NOTE 3: RSRP and SCH_RP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

**Table 9.6.2.5-2: BCCH signal levels at receiver input in dBm**

| Sub-test | BCCH1 | BCCH2 | BCCH3 | BCCH4 | BCCH5 | BCCH6 |
|---|---|---|---|---|---|---|
| 1 | -38.7 | -38.5 | NA | NA | NA | NA |
| 2 | -48.7 | -50.0 | NA | NA | NA | NA |
| 3 | -70.7 | -70.5 | NA | NA | NA | NA |
| 4 | -109.3 | -109.5 | NA | NA | NA | NA |
| 5 | -57.5 | NA | -54.5 | NA | NA | NA |
| 6 | -64.5 | NA | -59.5 | NA | NA | NA |
| 7 | -71.5 | NA | NA | -64.5 | NA | NA |
| 8 | -78.5 | NA | NA | -69.5 | NA | NA |
| 9 | -85.5 | NA | NA | NA | -74.5 | NA |
| 10 | -92.5 | NA | NA | NA | -79.5 | NA |
| 11 | -99.5 | NA | NA | NA | NA | -84.5 |
| 12 | -106.5 | NA | NA | NA | NA | -89.5 |

**Table 9.6.2.5-3: ARFCN numbers for GSM cells**

| GSM band | BCCH1 | BCCH2 | BCCH3 | BCCH4 | BCCH5 | BCCH6 |
|---|---|---|---|---|---|---|
| GSM 450 | 276 | 293 | 264 | 269 | 281 | 288 |
| GSM 480 | 323 | 340 | 311 | 316 | 328 | 335 |
| GSM 750 | 475 | 511 | 440 | 455 | 485 | 500 |
| GSM 850 | 189 | 251 | 150 | 170 | 210 | 230 |
| GSM 900 | 62 | 124 | 20 | 40 | 80 | 100 |
| DCS 1800 | 700 | 885 | 585 | 660 | 790 | 835 |
| PCS 1900 | 700 | 805 | 585 | 660 | 790 | 550 |
| Note: As defined in clause 3A.1, the test shall run without frequency overlapping between E-UTRA and GSM cells. The ARFCN numbers defined here, can be updated accordingly (even E-UTRA band specific) to avoid possible overlapping. | | | | | | |

**Table 9.6.2.5-4: GSM Carrier RSSI absolute accuracy requirements for the reported values**

| Sub-test | Normal condition | | Extreme condition | |
|---|---|---|---|---|
| | Lowest reported value for BCCH1 | Highest reported value for BCCH1 | Lowest reported value for BCCH1 | Highest reported value for BCCH1 |
| 1 | RXLEV_62 | RXLEV_63 | RXLEV_62 | RXLEV_63 |
| 2 | RXLEV_55 | RXLEV_63 | RXLEV_55 | RXLEV_63 |
| 3 | RXLEV_35 | RXLEV_45 | RXLEV_33 | RXLEV_47 |
| 4 | RXLEV_00 | RXLEV_06 | RXLEV_00 | RXLEV_08 |
| 5 | RXLEV_46 | RXLEV_60 | RXLEV_46 | RXLEV_60 |
| 6 | RXLEV_39 | RXLEV_53 | RXLEV_39 | RXLEV_53 |
| 7 | RXLEV_34 | RXLEV_44 | RXLEV_32 | RXLEV_46 |
| 8 | RXLEV_27 | RXLEV_37 | RXLEV_25 | RXLEV_39 |
| 9 | RXLEV_20 | RXLEV_30 | RXLEV_18 | RXLEV_32 |
| 10 | RXLEV_13 | RXLEV_23 | RXLEV_11 | RXLEV_25 |
| 11 | RXLEV_06 | RXLEV_16 | RXLEV_04 | RXLEV_18 |
| 12 | RXLEV_00 | RXLEV_09 | RXLEV_00 | RXLEV_11 |
| Note: It is not mandatory for the UE to report BCCH1 in sub-test 12. In case of no BCCH1 report in sub-test 12, the absolute accuracy for sub-test 12 is not tested. | | | | |

**Table 9.6.2.5-5: GSM Carrier RSSI Relative accuracy requirements for the reported values**

| Sub-test | Normal & Extreme condition | |
|---|---|---|
| | Lowest reported value for BCCH2 | Highest reported value for BCCH2 |
| 1 | N/A (Note3) | N/A (Note3) |
| 2 | RXLEV = x-6 | RXLEV = x+3 |
| 3 | RXLEV = x-4 | RXLEV = x+5 |
| 4 | N/A (Note3) | N/A (Note3) |
| | Lowest reported value for BCCH3 | Highest reported value for BCCH3 |
| 5 | RXLEV = x-2 | RXLEV = x+8 |
| 6 | RXLEV = x+1 | RXLEV = x+10 |
| | Lowest reported value for BCCH4 | Highest reported value for BCCH4 |
| 7 | RXLEV = x+3 | RXLEV = x+12 |
| 8 | RXLEV = x+5 | RXLEV = x+14 |
| | Lowest reported value for BCCH5 | Highest reported value for BCCH5 |
| 9 | RXLEV = x+7 | RXLEV = x+16 |
| 10 | RXLEV = x+8 | RXLEV = x+18 |
| | Lowest reported value for BCCH6 | Highest reported value for BCCH6 |
| 11 | RXLEV = x+10 | RXLEV = x+20 |
| 12 | N/A (Note3) | N/A (Note3) |
| Note 1: x is the reported value RXLEV for BCCH1.<br>Note 2: It is not mandatory for the UE to report BCCH1 in sub-test 12. In case of no BCCH1 report in sub-test 12, the relative accuracy for sub-test 12 is not tested.<br>Note 3: Sub-tests 1, 4 and 12 are not applicable for relative accuracy as they would be testing the UE outside the side conditions. | | |

For the test to pass, the ratio of successful reported values in each sub-test for absolute and relative accuracy shall be more than 90% with a confidence level of 95%.

# 9.7 UE Rx – Tx Time Difference

The UE Rx – Tx Time difference test cases can be found in TS 37.571-1 [27].

# 9.8 RSTD

The RSTD test cases can be found in TS 37.571-1 [27].

# 9.9 Serving Cell RSRP and RSRQ

## 9.9.1 FDD Serving Cell RSRP and RSRQ Accuracy

### 9.9.1.1 FDD Intra Frequency Serving Cell Absolute RSRP Accuracy

#### 9.9.1.1.1 Test purpose

To verify the FDD intra-frequency serving cell absolute RSRP measurement accuracy is within the specified limit.

#### 9.9.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and release 11. Applicability requires support for FGI bit 16.

#### 9.9.1.1.3 Minimum conformance requirements

The serving cell absolute accuracy of RSRP is defined as the RSRP measured of the serving cell.

The accuracy requirements in table 9.9.1.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.9.1.1.3-1: RSRP Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | dBm/BW$_{Channel}$ | dBm/BW$_{Channel}$ |
| ±6 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, FDD_C, FDD_D, FDD_E, FDD_F, FDD_G, FDD_H, TDD_A, TDD_C, TDD_E | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.9.1.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.9.1.1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.1, clause 9.1.4 and A.9.9.1.

### 9.9.1.1.4 Test description

#### 9.9.1.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.18.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.9.1.1.4.3.

4. Cell 1 is the serving cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.9.1.1.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.9.1.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message on cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check RSRP reported value in periodical MeasurementReport messages. The RSRP value of Cell 1 reported by the UE is compared to actual RSRP value according to Table 9.9.1.1.5-2.

7. SS shall check the MeasurementReport message transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

#### 9.9.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.9.1.1.4.3-1: Common Exception messages for RSRP FDD Intra frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3 |

**Table 9.9.1.1.4.3-2: *MeasResults*: Additional RSRP FDD Intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | Identifies the measurement id for the reporting being performed | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

## 9.9.1.1.5 Test requirement

Table 9.9.1.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD intra-frequency absolute accuracy test shall meet the reported values test requirements in table 9.9.1.1.5-2.

**Table 9.9.1.1.5-1: RSRP FDD Intra frequency absolute accuracy test parameters**

| Parameter | Unit | Test |
|---|---|---|
| | | Cell 1 |

| | | | |
|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 |
| $BW_{channel}$ | | MHz | 10 |
| Antenna Configuration | | | 1x2 |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 |
| PDSCH Reference measurement channel defined in A.1.1-1 | | | R.0 FDD |
| PDSCH allocation | | $n_{PRB}$ | 13—36 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1-1 | | | R.6 FDD |
| OCNG Patterns defined in D.1.1-1 (OP.1 FDD) | | | OP.1 FDD |
| PBCH_RA | | dB | 0 |
| PBCH_RB | | | |
| PSS_RA | | | |
| SSS_RA | | | |
| PCFICH_RB | | | |
| PHICH_RA | | | |
| PHICH_RB | | | |
| PDCCH_RA | | | |
| PDCCH_RB | | | |
| PDSCH_RA | | | |
| PDSCH_RB | | | |
| OCNG_RA Note1 | | | |
| OCNG_RB Note1 | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -120.4 |
| | Bands FDD_C | | -119.4 |
| | Bands FDD_D | | -118.9 |
| | Bands FDD_E, FDD_F Note 5 | | -118.4 |
| | Bands FDD_G Note 7 | | -117.4 |
| | Bands FDD_H | | -116.9 |
| $\hat{E}_s / I_{ot}$ | | dB | -4 |
| RSRP Note3 | Bands FDD_A | dBm/15 kHz | -124.4 |
| | Bands FDD_C | | -123.4 |
| | Bands FDD_D | | -122.9 |
| | Bands FDD_E, FDD_F Note 5 | | -122.4 |
| | Bands FDD_G Note 7 | | -121.4 |
| | Bands FDD_H | | -120.9 |
| RSRQ Note3 | Bands FDD_A | dB | -14.93 |
| | Bands FDD_C | | |
| | Bands FDD_D | | |
| | Bands FDD_E, FDD_F Note 5 | | |
| | Bands FDD_G Note 7 | | |
| | Bands FDD_H | | |
| Io Note3 | Bands FDD_A | dBm/9 MHz | -91.16 |
| | Bands FDD_C | | -90.16 |
| | Bands FDD_D | | -89.66 |
| | Bands FDD_E, FDD_F Note 5 | | -89.16 |
| | Bands FDD_G Note 7 | | -88.16 |
| | Bands FDD_H | | -87.66 |

| | | |
|---|---|---|
| $\hat{E}_s / N_{oc}$ | dB | -4 |
| Propagation condition | - | AWGN |
| Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.<br>Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.<br>Note 3: RSRP, RSRQ and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.<br>Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.<br>Note 5: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.<br>Note 6: E-UTRA operating band groups are as defined in Section 3.5.<br>Note 7: Except Band 29 and Band 32. | | |

**Table 9.9.1.1.5-2: RSRP FDD Intra frequency absolute accuracy requirements for the reported values**

| **Normal Conditions** | **Test 1** | |
|---|---|---|
| Lowest reported value (Cell 1) | Bands FDD_A | RSRP_9 |
| | Bands FDD_C | RSRP_10 |
| | Bands FDD_D | RSRP_10 |
| | Bands FDD_E<br>Bands FDD_F | RSRP_11 |
| | Bands FDD_G | RSRP_12 |
| | Bands FDD_H | RSRP_12 |
| Highest reported value (Cell 1) | Bands FDD_A | RSRP_24 |
| | Bands FDD_C | RSRP_25 |
| | Bands FDD_D | RSRP_25 |
| | Bands FDD_E<br>Bands FDD_F | RSRP_26 |
| | Bands FDD_G | RSRP_27 |
| | Bands FDD_H | RSRP_27 |
| Extreme Conditions | **Test 1** | |
| Lowest reported value (Cell 1) | Bands FDD_A | RSRP_6 |
| | Bands FDD_C | RSRP_7 |
| | Bands FDD_D | RSRP_7 |
| | Bands FDD_E<br>Bands FDD_F | RSRP_8 |
| | Bands FDD_G | RSRP_9 |
| | Bands FDD_H | RSRP_9 |
| Highest reported value (Cell 1) | Bands FDD_A | RSRP_27 |
| | Bands FDD_C | RSRP_28 |
| | Bands FDD_D | RSRP_28 |
| | Bands FDD_E<br>Bands FDD_F | RSRP_29 |
| | Bands FDD_G | RSRP_30 |
| | Bands FDD_H | RSRP_30 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.9.1.1_1 FDD Intra Frequency Serving Cell Absolute RSRP Accuracy (Rel-12 and forward)

Editor's notes: This test case is incomplete. The following item is missing or incomplete:

- Test tolerances are TBD

- Test requirements for normal conditions are TBD.

### 9.9.1.1_1.1 Test purpose

Same test purpose as in clause 9.9.1.1.1.

### 9.9.1.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 12 and forward. Applicability requires support for FGI bit 16.

### 9.9.1.1_1.3 Minimum conformance requirements

The serving cell absolute accuracy of RSRP is defined as the RSRP measured of the serving cell.

The accuracy requirements in table 9.9.1.1_1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.9.1.1_1.3-1: RSRP FDD Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | $dBm/BW_{Channel}$ | $dBm/BW_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.9.1.1_1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.9.1.1_1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.1, clause 9.1.4 and A.9.9.1.

## 9.9.1.1_1.4 Test description

### 9.9.1.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.9.1.1.4.1 with the following exceptions:

- Instead of 9.9.1.1.4.3 → use 9.9.1.1_1.4.3.

### 9.9.1.1_1.4.2 Test procedure

Same test procedure as in clause 9.9.1.1.4.2 with the following exceptions:

- Instead of Table 9.9.1.1.5-1 → use Table 9.9.1.1_1.5-1.
- Instead of Table 9.9.1.1.5-2 → use Table 9.9.1.1_1.5-2.

### 9.9.1.1_1.4.3 Message contents

Same message contents as in clause 9.9.1.1.4.3.

## 9.9.1.1_1.5 Test requirement

Table 9.9.1.1_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP FDD intra-frequency absolute accuracy test shall meet the reported values test requirements in table 9.9.1.1_1.5-2.

**Table 9.9.1.1_1.5-1: RSRP FDD Intra frequency absolute accuracy test parameters**

| Parameter | Unit | Test |
|---|---|---|
| | | Cell 1 |

| | | | |
|---|---|---|---|
| E-UTRA RF Channel Number | | | 1 |
| $BW_{channel}$ | | MHz | 10 |
| Antenna Configuration | | | 1x2 |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 |
| PDSCH Reference measurement channel defined in A.3.1.1.1 | | | R.0 FDD |
| PDSCH allocation | | $n_{PRB}$ | 13—36 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.1 | | | R.6 FDD |
| OCNG Patterns defined in A.3.2.1.1 (OP.1 FDD) | | | OP.1 FDD |
| PBCH_RA | | dB | 0 |
| PBCH_RB | | | |
| PSS_RA | | | |
| SSS_RA | | | |
| PCFICH_RB | | | |
| PHICH_RA | | | |
| PHICH_RB | | | |
| PDCCH_RA | | | |
| PDCCH_RB | | | |
| PDSCH_RA | | | |
| PDSCH_RB | | | |
| OCNG_RA Note1 | | | |
| OCNG_RB Note1 | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -122 |
| | Bands FDD_C | | -121 |
| | Bands FDD_D | | -120.5 |
| | Bands FDD_E, FDD_F Note 5 | | -120 |
| | Bands FDD_G Note 7 | | -119 |
| | Bands FDD_H | | -118.5 |
| $\hat{E}_s / I_{ot}$ | | dB | -4 |
| RSRP Note3 | Bands FDD_A | dBm/15 kHz | -126 |
| | Bands FDD_C | | -125 |
| | Bands FDD_D | | -124.5 |
| | Bands FDD_E, FDD_F Note 5 | | -124 |
| | Bands FDD_G Note 7 | | -123 |
| | Bands FDD_H | | -122.5 |
| RSRQ Note3 | Bands FDD_A | dB | -14.93 |
| | Bands FDD_C | | |
| | Bands FDD_D | | |
| | Bands FDD_E, FDD_F Note 5 | | |
| | Bands FDD_G Note 7 | | |
| | Bands FDD_H | | |
| Io Note3 | Bands FDD_A | dBm/9 MHz | -92.76+TT |
| | Bands FDD_C | | -91.76+TT |
| | Bands FDD_D | | -91.26+TT |
| | Bands FDD_E, FDD_F Note 5 | | -90.76+TT |
| | Bands FDD_G Note 7 | | -89.76+TT |
| | Bands FDD_H | | -89.26+TT |

| | | |
|---|---|---|
| $\hat{E}_s / N_{oc}$ | dB | -4 |
| Propagation condition | - | AWGN |
| NOTE 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. | | |
| NOTE 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. | | |
| NOTE 3: RSRP, RSRQ and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. | | |
| NOTE 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. | | |
| NOTE 5: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz. | | |
| NOTE 6: E-UTRA operating band groups are as defined in Section 3.5. | | |
| NOTE 7: Except Band 29. | | |

**Table 9.9.1.1_1.5-2: RSRP FDD Intra frequency absolute accuracy requirements for the reported values**

| **Normal Conditions** | **Test 1** | |
|---|---|---|
| Lowest reported value (Cell 1) | Bands FDD_A | TBD |
| | Bands FDD_C | TBD |
| | Bands FDD_D | TBD |
| | Bands FDD_E<br>Bands FDD_F | TBD |
| | Bands FDD_G | TBD |
| | Bands FDD_H | TBD |
| Highest reported value (Cell 1) | Bands FDD_A | TBD |
| | Bands FDD_C | TBD |
| | Bands FDD_D | TBD |
| | Bands FDD_E<br>Bands FDD_F | TBD |
| | Bands FDD_G | TBD |
| | Bands FDD_H | TBD |
| Extreme Conditions | **Test 1** | |
| Lowest reported value (Cell 1) | Bands FDD_A | RSRP_6+TT |
| | Bands FDD_C | RSRP_7+TT |
| | Bands FDD_D | RSRP_7+TT |
| | Bands FDD_E<br>Bands FDD_F | RSRP_8+TT |
| | Bands FDD_G | RSRP_9+TT |
| | Bands FDD_H | RSRP_9+TT |
| Highest reported value (Cell 1) | Bands FDD_A | RSRP_21+TT |
| | Bands FDD_C | RSRP_22+TT |
| | Bands FDD_D | RSRP_22+TT |
| | Bands FDD_E<br>Bands FDD_F | RSRP_23+TT |
| | Bands FDD_G | RSRP_24+TT |
| | Bands FDD_H | RSRP_24+TT |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.9.1.2 FDD Intra Frequency Serving Cell Absolute RSRQ Accuracy

### 9.9.1.2.1 Test purpose

To verify the FDD intra-frequency serving cell absolute RSRQ measurement accuracy is within the specified limit.

### 9.9.1.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward. Applicability requires support for FGI bit 16.

### 9.9.1.2.3 Minimum conformance requirements

The serving cell absolute accuracy of RSRQ is defined as the RSRQ measured of the serving cell.

The accuracy requirements in table 9.9.1.2.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.1 for a corresponding Band.

**Table 9.9.1.2.3-1: RSRQ FDD intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 4 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 3 | dBm/$BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

Note 1: Io is assumed to have constant EPRE across the bandwidth.
Note 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
Note 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
Note 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution.

The mapping of measured quantity is defined in Table 9.9.1.2.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.9.1.2.3-2: RSRQ FDD Intra frequency absolute accuracy measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.5.1, 9.1.7 and A.9.9.1.

#### 9.9.1.2.4 Test description

##### 9.9.1.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.18.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.9.1.2.4.3.

4. Cell 1 is the serving cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

##### 9.9.1.2.4.2 Test procedure

The serving cell absolute accuracy of RSRQ is defined as the RSRQ measured from the serving cell 1.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.9.1.2.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the RSRQ value in periodical MeasurementReport messages. The RSRQ value of serving Cell 1 reported by the UE is compared to the actual RSRQ according to Table 9.9.1.2.5-2.

7. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

##### 9.9.1.2.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.9.1.2.4.3-1: Common Exception messages for RSRQ FDD intra frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-4 |

**Table 9.9.1.2.4.3-2: *MeasResults*: Additional RSRQ FDD intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| meaResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

## 9.9.1.2.5 Test requirement

Table 9.9.1.2.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRQ FDD intra frequency absolute accuracy test shall meet the reported values test requirements in Table 9.9.1.2.5-2.

**Table 9.9.1.2.5-1: Cell Specific Test requirement Parameters for RSRQ FDD intra frequency absolute accuracy**

| **Parameter** | | **Unit** | **Test** |
|---|---|---|---|
| | | | **Cell 1** |
| E-UTRA RF Channel Number | | | 1 |
| $BW_{channel}$ | | MHz | 10 |
| Antenna Configuration | | | 1x2 |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 |
| PDSCH Reference measurement channel defined in A.1.1-1 | | | R.0 FDD |
| PDSCH allocation | | $n_{PRB}$ | 13—36 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.1-1 | | | R.6 FDD |
| OCNG Patterns defined in D.1.1-1 (OP.1 FDD) | | | OP.1 FDD |
| PBCH_RA | | dB | 0 |
| PBCH_RB | | | |
| PSS_RA | | | |
| SSS_RA | | | |
| PCFICH_RB | | | |
| PHICH_RA | | | |
| PHICH_RB | | | |
| PDCCH_RA | | | |
| PDCCH_RB | | | |
| PDSCH_RA | | | |
| PDSCH_RB | | | |
| OCNG_RA Note1 | | | |
| OCNG_RB Note1 | | | |
| $N_{oc}$ Note2 | Bands FDD_A | dBm/15 kHz | -120.4 |
| | Bands FDD_C | | -119.4 |
| | Bands FDD_D | | -118.9 |
| | Bands FDD_E, FDD_F Note 5 | | -118.4 |
| | Bands FDD_G Note 7 | | -117.4 |
| | Bands FDD_H | | -116.9 |
| $\hat{E}_s / I_{ot}$ | | dB | -4 |
| RSRP Note3 | Bands FDD_A | dBm/15 kHz | -124.4 |
| | Bands FDD_C | | -123.4 |

| | | | |
|---|---|---|---|
| | Bands FDD_D | | -122.9 |
| | Bands FDD_E, FDD_F Note 5 | | -122.4 |
| | Bands FDD_G Note 7 | | -121.4 |
| | Bands FDD_H | | -120.9 |
| RSRQ Note3 | Bands FDD_A | dB | -16.25 |
| | Bands FDD_C | | |
| | Bands FDD_D | | |
| | Bands FDD_E, FDD_F Note 5 | | |
| | Bands FDD_G Note 7 | | |
| | Bands FDD_H | | |
| Io Note3 | Bands FDD_A | dBm/9 MHz | -91.16 |
| | Bands FDD_C | | -90.16 |
| | Bands FDD_D | | -89.66 |
| | Bands FDD_E, FDD_F Note 5 | | -89.16 |
| | Bands FDD_G Note 7 | | -88.16 |
| | Bands FDD_H | | -87.66 |
| $\hat{E}_s / N_{oc}$ | | dB | -4 |
| Propagation condition | | - | AWGN |

Note 1: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.

Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 3: RSRP, RSRQ and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 4: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.

Note 5: For Band 26, the tests shall be performed with the carrier frequency of the assigned E-UTRA channel bandwidth within 865-894 MHz.

Note 6: E-UTRA operating band groups are as defined in Section 3.5.

Note 7: Except Band 29 and Band 32.

**Table 9.9.1.2.5-2: RSRQ FDD Intra frequency absolute accuracy requirements for the reported values**

| | **Test 1** |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value (Cell 1) | RSRQ_15 |
| Extreme Conditions | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value (Cell 1) | RSRQ_16 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.9.2 TDD Serving Cell RSRP and RSRQ Accuracy

### 9.9.2.1 TDD Intra Frequency Serving Cell Absolute RSRP Accuracy

#### 9.9.2.1.1 Test purpose

To verify the TDD intra-frequency serving cell absolute RSRP measurement accuracy is within the specified limit.

#### 9.9.2.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and release 11. Applicability requires support for FGI bit 16.

#### 9.9.2.1.3 Minimum conformance requirements

The serving cell absolute accuracy of RSRP is defined as the RSRP measured of the serving cell.

The accuracy requirements in Table 9.9.2.1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.9.2.1.3-1: RSRP Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| | | | Io Note 1 range | | |
| Normal condition | Extreme condition | Ês/Iot | E-UTRA operating band groups Note 3 | Minimum Io | | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 | $dBm/BW_{Channel}$ | $dBm/BW_{Channel}$ |
| ±6 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, FDD_C, FDD_D, FDD_E, FDD_F, FDD_G, FDD_H, TDD_A, TDD_C, TDD_E | N/A | -70 | -50 |

NOTE 1: Io is assumed to have constant EPRE across the bandwidth.
NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.9.2.1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.9.2.1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.1, clause 9.1.4 and A.9.9.2.

### 9.9.2.1.4 Test description

#### 9.9.2.1.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.18.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.9.2.1.4.3.

4. Cell 1 is the serving cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.9.2.1.4.2 Test procedure

1. Ensure that the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.9.2.1.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message on cell 1.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check RSRP reported value in periodical MeasurementReport messages. The RSRP value of Cell 1 reported by the UE is compared to actual RSRP value according to Table 9.9.2.1.5-2.

7. SS shall check the MeasurementReport message transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

#### 9.9.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.9.2.1.4.3-1: Common Exception message for RSRP TDD intra frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-3 |

**Table 9.9.2.1.4.3-2: *MeasResults*: Additional RSRP TDD intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultServCell SEQUENCE { | | | |
| rsrpResult | INTEGER(0..97) | | |
| rsrqResult | INTEGER(0..34) | | |
| } | | | |
| } | | | |

## 9.9.2.1.5 Test requirement

Table 9.9.2.1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD intra-frequency absolute accuracy test shall meet the reported values test requirements in table 9.9.2.1.5-2.

**Table 9.9.2.1.5-1: RSRP TDD Intra frequency absolute accuracy test parameters**

| Parameter | | Unit | Test |
|---|---|---|---|
| | | | Cell 1 |
| E-UTRA RF Channel Number | | | 1 |
| $BW_{channel}$ | | MHz | 10 |
| Antenna Configuration | | | 1x2 |
| Special subframe configuration[Note1] | | | 6 |
| Uplink/downlink configuration[Note1] | | | 1 |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 |
| PDSCH Reference measurement channel defined in A.1.2-1 | | | R.0 TDD |
| PDSCH allocation | | $n_{PRB}$ | 13—36 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2-1 | | | R.6 TDD |
| OCNG Patterns defined in D.2.1-1 (OP.1 TDD) | | | OP.1 TDD |
| PBCH_RA | | dB | 0 |
| PBCH_RB | | | |
| PSS_RA | | | |
| SSS_RA | | | |
| PCFICH_RB | | | |
| PHICH_RA | | | |
| PHICH_RB | | | |
| PDCCH_RA | | | |
| PDCCH_RB | | | |
| PDSCH_RA | | | |
| PDSCH_RB | | | |
| OCNG_RA[Note2] | | | |
| OCNG_RB[Note2] | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -120.4 |
| | Bands TDD_C | | -119.4 |
| | Bands TDD_E | | -118.4 |
| $\hat{E}_s / I_{ot}$ | | dB | -4 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -124.4 |
| | Bands TDD_C | | -123.4 |
| | Bands TDD_E | | -122.4 |
| RSRQ[Note4] | Bands TDD_A | dB | -14.93 |
| | Bands TDD_C | | |
| | Bands TDD_E | | |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -91.16 |
| | Bands TDD_C | | -90.16 |
| | Bands TDD_E | | -89.16 |
| $\hat{E}_s / N_{oc}$ | | dB | -4 |
| Propagation condition | | - | AWGN |

| | |
|---|---|
| Note 1: | For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211. |
| Note 2: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 3: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 4: | RSRP, RSRQ and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 5: | RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 6: | E-UTRA operating band groups are as defined in Section 3.5. |

**Table 9.9.2.1.5-2: RSRP TDD Intra frequency absolute accuracy requirements for the reported values**

| **Normal Conditions** | **Test 1** | |
|---|---|---|
| Lowest reported value (Cell 1) | Bands TDD_A | RSRP_9 |
| | Bands TDD_C | RSRP_10 |
| | Bands TDD_E | RSRP_11 |
| Highest reported value (Cell 1) | Bands TDD_A | RSRP_24 |
| | Bands TDD_C | RSRP_25 |
| | Bands TDD_E | RSRP_26 |
| **Extreme Conditions** | **Test 1** | |
| Lowest reported value (Cell 1) | Bands TDD_A | RSRP_6 |
| | Bands TDD_C | RSRP_7 |
| | Bands TDD_E | RSRP_8 |
| Highest reported value (Cell 1) | Bands TDD_A | RSRP_27 |
| | Bands TDD_C | RSRP_28 |
| | Bands TDD_E | RSRP_29 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.9.2.1_1 TDD Intra Frequency Serving Cell Absolute RSRP Accuracy (Rel-12 and forward)

Editor's notes: This test case is incomplete. The following item is missing or incomplete:

- Test tolerances are TBD

- Test requirements for normal conditions are TBD

### 9.9.2.1_1.1 Test purpose

Same test purpose as in clause 9.9.2.1.1.

### 9.9.2.1_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 12 and forward. Applicability requires support for FGI bit 16.

### 9.9.2.1_1.3 Minimum conformance requirements

The serving cell absolute accuracy of RSRP is defined as the RSRP measured of the serving cell.

The accuracy requirements in Table 9.9.2.1_1.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

$RSRP|_{dBm}$ according to Annex I.3.1 for a corresponding Band.

**Table 9.9.2.1_1.3-1: RSRP TDD Intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io Note 1 range | | |
| | | | E-UTRA operating band groups Note 3 | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz Note 2 / dBm/$BW_{Channel}$ | dBm/$BW_{Channel}$ |
| ±4.5 | ±9 | ≥-6 dB | FDD_A, TDD_A | -121 | N/A | -70 |
| | | | FDD_C, TDD_C | -120 | N/A | -70 |
| | | | FDD_D | -119.5 | N/A | -70 |
| | | | FDD_E, TDD_E | -119 | N/A | -70 |
| | | | FDD_F | -118.5 | N/A | -70 |
| | | | FDD_G | -118 | N/A | -70 |
| | | | FDD_H | -117.5 | N/A | -70 |
| | | | FDD_N | -114.5 | N/A | -70 |
| ±8 | ±11 | ≥-6 dB | FDD_A, TDD_A, FDD_C, TDD_C, FDD_D, FDD_E, TDD_E, FDD_F, FDD_G, FDD_H, FDD_N | N/A | -70 | -50 |
| NOTE 1: Io is assumed to have constant EPRE across the bandwidth.<br>NOTE 2: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 3: E-UTRA operating band groups are as defined in Section 3.5. | | | | | |

The reporting range of RSRP is defined from -140 dBm to -44 dBm with 1 dB resolution.

The mapping of measured quantity is defined in Table 9.9.2.1_1.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.9.2.1_1.3-2: RSRP measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRP_00 | RSRP < -140 | dBm |
| RSRP_01 | -140 ≤ RSRP < -139 | dBm |
| RSRP_02 | -139 ≤ RSRP < -138 | dBm |
| … | … | … |
| RSRP_95 | -46 ≤ RSRP < -45 | dBm |
| RSRP_96 | -45 ≤ RSRP < -44 | dBm |
| RSRP_97 | -44 ≤ RSRP | dBm |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.2.1, clause 9.1.4 and A.9.9.2.

## 9.9.2.1_1.4 Test description

### 9.9.2.1_1.4.1 Initial conditions

Same initial conditions as in clause 9.9.2.1.4.1 with the following exceptions:

- Instead of 9.9.2.1.4.3 → use 9.9.2.1_1.4.3.

### 9.9.2.1_1.4.2 Test procedure

Same test procedure as in clause 9.9.1.1.4.2 with the following exceptions:

- Instead of Table 9.9.2.1.5-1 → use Table 9.9.2.1_1.5-1.
- Instead of Table 9.9.2.1.5-2 → use Table 9.9.2.1_1.5-2.

#### 9.9.2.1_1.4.3 Message contents

Same message contents as in clause 9.9.2.1.4.3.

### 9.9.2.1_1.5 Test requirement

Table 9.9.2.1_1.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRP TDD intra-frequency absolute accuracy test shall meet the reported values test requirements in table 9.9.2.1_1.5-2.

**Table 9.9.2.1_1.5-1: RSRP TDD Intra frequency absolute accuracy test parameters**

| Parameter | | Unit | Test |
|---|---|---|---|
| | | | Cell 1 |
| E-UTRA RF Channel Number | | | 1 |
| $BW_{channel}$ | | MHz | 10 |
| Antenna Configuration | | | 1x2 |
| Special subframe configuration Note1 | | | 6 |
| Uplink/downlink configuration Note1 | | | 1 |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 |
| PDSCH Reference measurement channel defined in A.3.1.1.2 | | | R.0 TDD |
| PDSCH allocation | | $n_{PRB}$ | 13—36 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.3.1.2.2 | | | R.6 TDD |
| OCNG Patterns defined in A.3.2.2.1 (OP.1 TDD) | | | OP.1 TDD |
| PBCH_RA | | dB | 0 |
| PBCH_RB | | | |
| PSS_RA | | | |
| SSS_RA | | | |
| PCFICH_RB | | | |
| PHICH_RA | | | |
| PHICH_RB | | | |
| PDCCH_RA | | | |
| PDCCH_RB | | | |
| PDSCH_RA | | | |
| PDSCH_RB | | | |
| OCNG_RA Note2 | | | |
| OCNG_RB Note2 | | | |
| $N_{oc}$ Note3 | Bands TDD_A | dBm/15 kHz | -122 |
| | Bands TDD_C | | -121 |
| | Bands TDD_E | | -120 |
| $\hat{E}_s / I_{ot}$ | | dB | -4 |
| RSRP Note4 | Bands TDD_A | dBm/15 kHz | -126 |
| | Bands TDD_C | | -125 |
| | Bands TDD_E | | -124 |
| RSRQ Note4 | Bands TDD_A | dB | -14.93 |
| | Bands TDD_C | | |
| | Bands TDD_E | | |
| Io Note4 | Bands TDD_A | dBm/9 MHz | -92.76+TT |
| | Bands TDD_C | | -91.76+TT |
| | Bands TDD_E | | -90.76+TT |
| $\hat{E}_s / N_{oc}$ | | dB | -4 |
| Propagation condition | | - | AWGN |

NOTE 1: For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211.
NOTE 2: OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
NOTE 3: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
NOTE 4: RSRP, RSRQ and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
NOTE 5: RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port.
NOTE 6: E-UTRA operating band groups are as defined in Section 3.5.

**Table 9.9.2.1_1.5-2: RSRP TDD Intra frequency absolute accuracy requirements for the reported values**

| **Normal Conditions** | **Test 1** | |
|---|---|---|
| Lowest reported value (Cell 1) | Bands TDD_A | TBD |
| | Bands TDD_C | TBD |
| | Bands TDD_E | TBD |
| Highest reported value (Cell 1) | Bands TDD_A | TBD |
| | Bands TDD_C | TBD |
| | Bands TDD_E | TBD |
| **Extreme Conditions** | **Test 1** | |
| Lowest reported value (Cell 1) | Bands TDD_A | RSRP_6+TT |
| | Bands TDD_C | RSRP_7+TT |
| | Bands TDD_E | RSRP_8+TT |
| Highest reported value (Cell 1) | Bands TDD_A | RSRP_24+TT |
| | Bands TDD_C | RSRP_25+TT |
| | Bands TDD_E | RSRP_26+TT |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

## 9.9.2.2 TDD Intra Frequency Serving Cell Absolute RSRQ Accuracy

### 9.9.2.2.1 Test purpose

To verify the TDD intra-frequency serving cell absolute RSRQ measurement accuracy is within the specified limit.

### 9.9.2.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 10 and forward. Applicability requires support for FGI bit 16.

### 9.9.2.2.3 Minimum conformance requirements

The serving cell absolute accuracy of RSRQ is defined as the RSRQ measured of the serving cell.

The accuracy requirements in table 9.9.2.2.3-1 are valid under the following conditions:

Cell specific reference signals are transmitted either from one, two or four antenna ports.

Conditions defined in 36.101 Section 7.3 for reference sensitivity are fulfilled.

RSRP|dBm according to Annex I.3.1 for a corresponding Band.

**Table 9.9.2.2.3-1: RSRQ TDD intra frequency absolute accuracy**

| Accuracy | | Conditions | | | |
|---|---|---|---|---|---|
| Normal condition | Extreme condition | Ês/Iot | Io [Note 1] range | | |
| | | | E-UTRA operating band groups [Note 4] | Minimum Io | Maximum Io |
| dB | dB | dB | | dBm/15kHz [Note 3] | dBm/$BW_{Channel}$ |
| ±2.5 | ±4 | ≥-3 dB | FDD_A, TDD_A | -121 | -50 |
| | | | FDD_C, TDD_C | -120 | -50 |
| | | | FDD_D | -119.5 | -50 |
| | | | FDD_E, TDD_E | -119 | -50 |
| | | | FDD_F | -118.5 | -50 |
| | | | FDD_G | -118 | -50 |
| | | | FDD_H | -117.5 | -50 |
| | | | FDD_N | -114.5 | -50 |
| ±3.5 | ±4 | ≥-6 dB | Note 2 | Note 2 | Note 2 |

Note 1: Io is assumed to have constant EPRE across the bandwidth.
Note 2: The same bands and the same Io conditions for each band apply for this requirement as for the corresponding highest accuracy requirement.
Note 3: The condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
Note 4: E-UTRA operating band groups are as defined in Section 3.5.

The reporting range of RSRQ is defined from -19.5 dB to -3 with 0.5 dB resolution.

The mapping of measured quantity is defined in Table 9.9.2.2.3-2. The range in the signalling may be larger than the guaranteed accuracy range.

**Table 9.9.2.2.3-2: RSRQ TDD Intra frequency absolute accuracy measurement report mapping**

| Reported value | Measured quantity value | Unit |
|---|---|---|
| RSRQ_00 | RSRQ < -19.5 | dB |
| RSRQ_01 | -19.5 ≤ RSRQ < -19 | dB |
| RSRQ_02 | -19 ≤ RSRQ < -18.5 | dB |
| … | … | … |
| RSRQ_32 | -4 ≤ RSRQ < -3.5 | dB |
| RSRQ_33 | -3.5 ≤ RSRQ < -3 | dB |
| RSRQ_34 | -3 ≤ RSRQ | dB |

The normative reference for this requirement is TS 36.133 [4] clause 9.1.5.1, 9.1.7 and A.9.9.2.

## 9.9.2.2.4 Test description

### 9.9.2.2.4.1 Initial conditions

Test Environment: Normal, TL/VL, TL/VH, TH/VL, TH/VH; as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure A.18.

2. Propagation conditions are set according to Annex B clause B.0.

3. Message contents are defined in clause 9.9.2.2.4.3.

4. Cell 1 is the serving cell used for connection setup with the power levels set according to Annex C.0 and C.1 for this test.

#### 9.9.2.2.4.2 Test procedure

The serving cell absolute accuracy of RSRQ is defined as the RSRQ measured from the serving cell 1.

1. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 7.2A.3.

2. Set the parameters according to Table 9.9.2.2.5-1 as appropriate. Propagation conditions are set according to Annex B clause B.1.1.

3. SS shall transmit an RRCConnectionReconfiguration message.

4. The UE shall transmit RRCConnectionReconfigurationComplete message.

5. UE shall transmit periodically MeasurementReport messages.

6. After 10s wait from Step 3, SS shall check the RSRQ value in periodical MeasurementReport messages. The RSRQ value of serving Cell 1 reported by the UE is compared to the actual RSRQ according to Table 9.9.2.2.5-2.

7. The SS shall check MeasurementReport messages transmitted by the UE until the confidence level according to Tables G.2.3-1 in Annex G.2 is achieved.

#### 9.9.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] values 4.6 with the following exceptions:

**Table 9.9.2.2.4.3-1: Common Exception messages for RSRQ TDD intra frequency absolute accuracy test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | Table H.3.1-1<br>Table H.3.5-1<br>Table H.3.5-4 |

**Table 9.9.2.2.4.3-2: *MeasResults*: Additional RSRQ TDD intra frequency absolute accuracy test requirement**

| Derivation Path: 36.331 clause 6.3.5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| meaResultServCell SEQUENCE { | | | |
| rsrpResult | | Set according to specific test | |
| rsrqResult | | Set according to specific test | |
| } | | | |
| } | | | |

### 9.9.2.2.5 Test requirement

Table 9.9.2.2.5-1 defines the primary level settings including test tolerances for all tests.

Each RSRQ TDD intra frequency absolute accuracy test shall meet the reported values test requirements in Table 9.9.2.2.5-2.

**Table 9.9.2.2.5-1: Cell Specific Test requirement Parameters for RSRQ TDD intra frequency absolute accuracy**

| Parameter | | Unit | Test |
|---|---|---|---|
| | | | Cell 1 |
| E-UTRA RF Channel Number | | | 1 |
| $BW_{channel}$ | | MHz | 10 |
| Antenna Configuration | | | 1x2 |
| Special subframe configuration[Note1] | | | 6 |
| Uplink/downlink configuration[Note1] | | | 1 |
| Measurement bandwidth | | $n_{PRB}$ | 22—27 |
| PDSCH Reference measurement channel defined in A.1.2-1 | | | R.0 TDD |
| PDSCH allocation | | $n_{PRB}$ | 13—36 |
| PDCCH/PCFICH/PHICH Reference measurement channel defined in A.2.2-1 | | | R.6 TDD |
| OCNG Patterns defined in D.2.1-1 (OP.1 TDD) | | | OP.1 TDD |
| PBCH_RA | | dB | 0 |
| PBCH_RB | | | |
| PSS_RA | | | |
| SSS_RA | | | |
| PCFICH_RB | | | |
| PHICH_RA | | | |
| PHICH_RB | | | |
| PDCCH_RA | | | |
| PDCCH_RB | | | |
| PDSCH_RA | | | |
| PDSCH_RB | | | |
| OCNG_RA[Note2] | | | |
| OCNG_RB[Note2] | | | |
| $N_{oc}$ [Note3] | Bands TDD_A | dBm/15 kHz | -120.4 |
| | Bands TDD_C | | -119.4 |
| | Bands TDD_E | | -118.4 |
| $\hat{E}_s / I_{ot}$ | | dB | -4 |
| RSRP[Note4] | Bands TDD_A | dBm/15 kHz | -124.4 |
| | Bands TDD_C | | -123.4 |
| | Bands TDD_E | | -122.4 |
| RSRQ[Note4] | Bands TDD_A | dB | -16.25 |
| | Bands TDD_C | | |
| | Bands TDD_E | | |
| Io[Note4] | Bands TDD_A | dBm/9 MHz | -91.16 |
| | Bands TDD_C | | -90.16 |
| | Bands TDD_E | | -89.16 |
| $\hat{E}_s / N_{oc}$ | | dB | -4 |
| Propagation condition | | - | AWGN |

| | |
|---|---|
| Note 1: | For special subframe and uplink-downlink configurations see Tables 4.2-1 and 4.2-2 in TS 36.211. |
| Note 2: | OCNG shall be used such that both cells are fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols. |
| Note 3: | Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled. |
| Note 4: | RSRP, RSRQ and Io levels have been derived from other parameters for information purposes. They are not settable parameters themselves. |
| Note 5: | RSRP minimum requirements are specified assuming independent interference and noise at each receiver antenna port. |
| Note 6: | E-UTRA operating band groups are as defined in Section 3.5. |

**Table 9.9.2.2.5-2: RSRQ TDD Intra frequency absolute accuracy requirements for the reported values**

| | **Test 1** |
|---|---|
| | All bands |
| Normal Conditions | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value (Cell 1) | RSRQ_15 |
| Extreme Conditions | |
| Lowest reported value (Cell 1) | RSRQ_00 |
| Highest reported value (Cell 1) | RSRQ_16 |

For the test to pass, the ratio of successful reported values in each test shall be more than 90% with a confidence level of 95%.

# 10 Proximity-based Services in Any Cell Selection

## 10.1 FFS

## 10.2 E-UTRAN FDD – Initiation/Cease of SLSS Transmission with ProSe Direct Communication

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

- Message contents are TBD.

- Connection diagrams are TBD

- Annex E updates are pending.

### 10.2.1 Test purpose

To verify the ProSe UE's ability to initiate and cease SLSS transmissions during cell selection state and if the UE meets the maximum evaluation time.

### 10.2.2 Test applicability

This test case applies to all types of E-UTRA UE release 12 and forward which support ProSe Direct Communication.

## 10.2.3 Minimum conformance requirements

The SLSS transmission initiation delay is defined as the time from the beginning of time period T2 up to the moment when the UE initiates the SLSS transmission.

The SLSS transmission initiation delay shall be less than 0.84 s.

The SLSS transmission cease delay is defined as the time from the beginning of time period T3 up to the moment when the UE ceases the SLSS transmission.

The SLSS transmission cease delay shall be less than 0.84 s.

The rate of correct initiation/cease delay of SLSS transmissions observed during repeated tests shall be at least 90%.

NOTE: The initiation/cease delay of SLSS transmissions can be expressed as: $T_{evaluate,SLSS}$ + SLSS period,

Where:

$T_{evaluate,SLSS}$ is the evaluation time for initiate/cease of SLSS, and is 0.8 sec (clause 11.3.2) for the parameters in this test;

SLSS period is set as 40ms in this test.

The normative reference for this requirement is TS 36.133 [4] clause 11.3, and A.10.2.

## 10.2.4 Test description

### 10.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz or 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure FFS.
2. The general test parameter settings are set up according to Table 10.2.4.1-1.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 10.2.4.3.
5. There no active cells in this test case during the actual test loop, and SS is expected to simulate a reference UE, SyncRef UE 1. Cell 1 is the cell used for connection setup with the power level set according to Annex C.0 and C.1 for this test.

**Table 10.2.4.1-1: Test parameters for initiation/cease of SLSS transmissions test for E-UTRAN FDD**

| Parameter | Unit | Value | Comment |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | |
| Channel Bandwidth ($BW_{channel}$) | MHz | 5 or 10 | According to principle defined in clause A.3.12.3 |
| Active cell | | None | |
| Active SyncRef UE | | SyncRef UE 1 | Transmitting SLSS+MIB-SL on uplink of RF channel number 1 |
| ProSe Direct Communication preconfiguration | | As specified in Table A.3.12.5-2 (Configuration #2) | IE values unless specified otherwise in this test. |
| syncTxThreshOoC | dBm/15 kHz | -95 | |
| T1 | s | 3 | |
| T2 | s | 5.24 | |
| T3 | s | 5.24 | |

**Table 10.2.4.1-2: SyncRef UE specific test parameters for initiation/cease of SLSS transmissions test for E-UTRAN FDD**

| Parameter | Unit | SyncRef UE 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | |
| $BW_{channel}$ Note 4 | MHz | 5 or 10 | | |
| ProSe Direct Communication resource pool configuration | | As specified in Table A.3.12.5-1 (Configuration #1) Note resource pool is same as Configuration #2 used by ProSe UE. | | |
| syncOffsetIndicator | | Set same as *syncOffsetIndicator1* in ProSe Direct Communication preconfiguration | | |
| slssid | | 30 | | |
| inCoverage | | TRUE | | |
| networkControlledSyncTx | | ON | | |
| $N_{oc}$ Note1 | dBm/15 kHz | -96 | | |
| $\hat{E}_s / N_{oc}$ | dB | 5.5 | -3.5 | 5.5 |
| S-RSRP Note2, Note3 | dBm/15 kHz | -90.5 | -99.5 | -90.5 |
| Propagation Condition | | AWGN | | |

Note 1: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.

Note 2: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.

Note 3: SSSS Es/Iot is set the same as PSSS/PSBCH Es/Iot.

Note 4: This test is according to the principle defined in section A.3.12.3.

## 10.2.4.2 Test procedure

There are no active cells in this test. There is one active SyncRef UE (SyncRef UE 1) in this test. The test system shall emulate SyncRef UE 1 to transmit SLSS and MIB-SL every synchronization period Prior to start of test, test system is required to ensure that the ProSe UE is synchronized to the SyncRef UE 1 and is transmitting SLSS + MIB-SL as derived from the SLSS + MIB-SL of SyncRef. The test consists of three successive time periods, with time duration of T1, T2 and T3 respectively. During T1, the S-RSRP of SyncRef UE 1 is above *syncTxThreshOOC* and the UE is not expected to be transmitting SLSS. During T2, the S-RSRP of SyncRef UE 1 is lowered below *syncTxThreshOOC* and the UE is expected to initiate SLSS transmissions. During T3, the S-RSRP of SyncRef UE 1 is increased back to be above *syncTxThreshOOC* and the UE is expected to cease SLSS transmissions.

1. Ensure the UE is in State 4 according to TS 36.508 [7] clause 4.5.4 and UE test loop Mode E is activated on a EUTRA cell.

2. EUTRA cell is powered down for the remaining duration of the test case. SS would ensure that it enables a SyncRefUE1 at this stage and the ProSe UE under test is synchronized to SyncRefUE1.
3. Set the parameters according to T1 in Table 10.2.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2. T1 starts.
4. During T1, SS checks if the UE is not transmitting SLSS.
5. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 10.2.5-1.
6. UE is expected to initiate a SLSS transmissions inside 0.84 s from the start of T2. If the UE initiates SLSS transmission consider the loop to be pass, else the loop is considered as fail.
7. After the SS is able to measure the SLSS in step 6) or when T2 expires, the SS shall switch the power setting from T2 to T3 as specified in Table 10.2.5-1
8. UE is expected to cease SLSS transmission inside 0.84s from the start of T3. If the UE ceases SLSS transmission consider the loop to be pass, else the loop is considered as fail.
9. After the RRC connection release, the SS switches off and on the UE and ensures the UE is in State 4 with test loop mode E activated according to TS 36.508 [7] clause 4.5.4.
10. Repeat step 2-9 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved for both the events.

### 10.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with FFS:

**Table 10.2.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | FFS |

## 10.2.5 Test requirement

Tables 10.2.4.1-2 and 10.2.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-Initiation/Cease of SLSS Transmission with ProSe Direct Discovery.

**Table 10.2.5-1: SyncRef UE specific test parameters for initiation/cease of SLSS transmissions test for E-UTRAN FDD**

| Parameter | Unit | SyncRef UE 1 | | |
|---|---|---|---|---|
| | | T1 | T2 | T3 |
| E-UTRA RF Channel Number | | 1 | | |
| $BW_{channel}$ Note 4 | MHz | 5 or 10 | | |
| ProSe Direct Communication resource pool configuration | | As specified in Table A.3.12.5-1 (Configuration #1) Note resource pool is same as Configuration #2 used by ProSe UE. | | |
| syncOffsetIndicator | | Set same as *syncOffsetIndicator1* in ProSe Direct Communication preconfiguration | | |
| slssid | | 30 | | |
| inCoverage | | TRUE | | |
| networkControlledSyncTx | | ON | | |

| | | | | |
|---|---|---|---|---|
| $N_{oc}$ Note1 | dBm/15 kHz | -96 | | |
| $\hat{E}_s / N_{oc}$ | dB | 6.1 | -4.1 | 6.1 |
| S-RSRP Note2, Note3 | dBm/15 kHz | -89.9 | -100.1 | -89.9 |
| Propagation Condition | | AWGN | | |

Note 1: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 2: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: SSSS Es/Iot is set the same as PSSS/PSBCH Es/Iot.
Note 4: This test is according to the principle defined in section A.3.12.3.

The SLSS transmission initiation delay is defined as the time from the beginning of time period T2 up to the moment when the UE initiates the SLSS transmission.

The SLSS transmission initiation delay shall be less than 0.84 s.

The SLSS transmission cease delay is defined as the time from the beginning of time period T3 up to the moment when the UE ceases the SLSS transmission.

The SLSS transmission cease delay shall be less than 0.84 s.

The rate of correct initiation/cease delay of SLSS transmissions observed during repeated tests shall be at least 90%.

NOTE: The initiation/cease delay of SLSS transmissions can be expressed as: $T_{evaluate,SLSS}$ + SLSS period,

Where:

$T_{evaluate,SLSS}$ is the evaluation time for initiate/cease of SLSS, and is 0.8 sec (clause 11.3.2) for the parameters in this test;

SLSS period is set as 40ms in this test.

# 10.3 FFS

# 10.4 E-UTRAN FDD – Initiation/Cease of SLSS Transmission with ProSe Direct Communication

Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:

- Message contents are TBD.

- Connection diagrams are TBD

- Annex E updates are pending.

## 10.4.1 Test purpose

The purpose of this test is to verify cell identification delay requirement for a newly detectable cell on the downlink frequency associated with the pre-configured ProSe carrier frequency in Any Cell Selection.

## 10.4.2 Test applicability

This test case applies to all types of E-UTRA UE release 12 and forward which support ProSe Direct Communication.

## 10.4.3 Minimum conformance requirements

The cell selection delay to a newly detectable cell on the downlink associated with the preconfigured ProSe carrier is defined as the time from the beginning of T2 to the time UE camps on the cell and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST.

The cell selection delay to a newly detectable cell on the downlink associated with the preconfigured ProSe carrier shall be less than 7.68 s.

The cell selection delay can be expressed as $T_{basic_identify_OoC_ProSe\ Tx_ON} + T_{SI}$, where

- $T_{basic_identify_OoC_ProSe\ Tx_ON}$ = 6.4sec as specified in sub-clause 11.4.2.2
- $T_{SI}$ = Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell; 1280 ms is assumed in this test case

This gives a total of 7.68 sec.

The normative reference for this requirement is TS 36.133 [4] clause 11.4, and A.10.4.

## 10.4.4 Test description

### 10.4.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: According to Annex E table E-1 and TS 36.508 [7] clauses 4.4.2 and 4.3.1.

Channel Bandwidth to be tested: 5 MHz or 10 MHz as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS (node B emulator) and faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A figure FFS.
2. The general test parameter settings are set up according to Table 10.4.4.1-2.
3. Propagation conditions are set according to Annex B clauses B.0.
4. Message contents are defined in clause 10.4.4.3.
5. There is one active cell (Cell 1) and active SyncRef UE (SyncRef UE 1) in this test. The test system shall emulate SyncRef UE 1 to transmit SLSS and MIB-SL every SLSS period (40ms).

**Table 10.4.4.1-1: Test parameters for cell identification test on downlink frequency associated with ProSe frequency for E-UTRAN FDD (when UE is transmitting for ProSe)**

| Parameter | | Unit | Value | Comment |
|---|---|---|---|---|
| Initial condition | Active synchronization source | | Sync Ref UE 1 | |
| Final condition | Active synchronization source | | Cell1 | |
| E-UTRA RF Channel Number | | | 1 | |
| Channel Bandwidth ($BW_{channel}$) | | MHz | 5 or 10 | According to principle defined in clause A.3.12.3 |
| Active cell | | | Cell1 | |
| Active SyncRef UEs | | | SyncRef UE 1 | Transmitting SLSS+MIB-SL on uplink of RF channel number 1 |
| ProSe Direct Communication preconfiguration | | | As specified in Table A.3.12.5-2 (Configuration #2) | IE values unless specified otherwise in this test. |
| syncTxThreshOoC | | | 11 (+infinity) | |
| T1 | | s | 2 | |
| T2 | | s | 30 | |

**Table 10.4.4.1-2: Cell specific test parameters for cell identification test on downlink frequency associated with ProSe frequency for E-UTRAN FDD (when UE is transmitting for ProSe)**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ Note 4 | MHz | 5 or 10 | |
| OCNG Patterns defined in A.3.2.1.2 Note 4 | | 5 MHz: OP.16 FDD<br>10 MHz: OP.2 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | | | |
| PSS_RA | | | |
| SSS_RA | | | |
| PCFICH_RB | | | |
| PHICH_RA | | | |
| PHICH_RB | | | |
| PDCCH_RA | | | |
| PDCCH_RB | | | |
| PDSCH_RA | | | |
| PDSCH_RB | | | |
| OCNG_RA Note 1 | | | |
| OCNG_RB Note 1 | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | -infinity | -3 |
| RSRP Note3 | dBm/15 kHz | -infinity | -101 |
| SCH_RP Note3 | dBm/15 kHz | -infinity | -101 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: This test is according to the principle defined in section A.3.12.3.

**Table 10.4.4.1-3: SyncRef UE specific test parameters for cell identification test on downlink frequency associated with ProSe frequency for E-UTRAN FDD**

| Parameter | Unit | SyncRef UE 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 (Uplink) | |
| $BW_{channel}$ Note 4 | MHz | 5 or 10 | |
| ProSe Direct Communication resource pool configuration | | As specified in Table A.3.12.5-1 (Configuration #1) | |
| networkControlledSyncTx | | ON | |
| slssid | | 30 | |
| inCoverage (in MIB-SL) | | TRUE | |
| syncOffsetIndicator | | syncOffsetIndicator1 | |
| $N_{oc}$ Note1 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 13 | |
| S-RSRP Note2, Note3 | dBm/15 kHz | -85 | |
| Propagation Condition | | AWGN | |

Note 1: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 2: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: SSSS Es/Iot is set the same as PSSS/PSBCH Es/Iot.
Note 4: This test is according to the principle defined in section A.3.12.3.

### 10.4.4.2 Test procedure

There is one active cell (Cell 1) and active SyncRef UE (SyncRef UE 1) in this test. The test system shall emulate SyncRef UE 1 to transmit SLSS and MIB-SL every SLSS period (40ms). The test consists of two successive time periods, with time duration of T1 and T2 respectively. During T1, the cell is powered OFF and the ProSe UE is synchronized to SyncRef UE 1. During T2, the cell is powered ON and the ProSe UE will detect the cell and attempt to camp on the cell.

1 Ensure the UE is in State 4 according to TS 36.508 [7] clause 4.5.4 and UE test loop Mode D is activated.

2. Set the parameters according to T1 in Table 10.4.5-1. Propagation conditions are set according to Annex B clauses B.1.1 and B.2.2.

3. At the start of T1, cell 1 is powered off and UE is expected to synchronize with SyncRefUE1.Also TE is expected to verify that UE is able to transmit SLSS + MIB-SL before the end of T1.

3. When T1 expires, the SS shall switch the power setting from T1 to T2 as specified in Table 10.4.5-1.

4. UE is expected to initiate PRACH preambles inside 7.68 s from the start of T2. If the UE sends PRACH inside 7.68s consider the loop to be pass, else UE fails the iteration.

5. Set Cell 2 physical cell identity = ((current cell 2 physical cell identity + 1) mod 14 + 2) for next iteration of the test procedure loop.

6. After the RRC connection release, the SS switches off and on the UE and ensures the UE is in State 4 with test loop mode 4 activated according to TS 36.508 [7] clause 4.5.4.

7. Repeat step 2-6 until the confidence level according to Tables G.2.3-1 in Annex G clause G.2 is achieved.

### 10.4.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with FFS:

**Table 10.4.4.3-1: Common Exception messages for E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells test requirement**

| Default Message Contents | |
|---|---|
| Common contents of system information blocks exceptions | |
| Default RRC messages and information elements contents exceptions | FFS |

## 10.4.5 Test requirement

Tables 10.4.4.1-1 and 10.4.5-1 define the primary level settings including test tolerances for E-UTRAN FDD-Initiation/Cease of SLSS Transmission with ProSe Direct Discovery.

**Table 10.4.5-1: Cell specific test parameters for cell identification test on downlink frequency associated with ProSe frequency for E-UTRAN FDD (when UE is transmitting for ProSe)**

| Parameter | Unit | Cell 1 | |
|---|---|---|---|
| | | T1 | T2 |

| | | | |
|---|---|---|---|
| E-UTRA RF Channel Number | | 1 | |
| $BW_{channel}$ Note 4 | MHz | 5 or 10 | |
| OCNG Patterns defined in A.3.2.1.2 Note 4 | | 5 MHz: OP.16 FDD<br>10 MHz: OP.2 FDD | |
| PBCH_RA | dB | 0 | |
| PBCH_RB | | | |
| PSS_RA | | | |
| SSS_RA | | | |
| PCFICH_RB | | | |
| PHICH_RA | | | |
| PHICH_RB | | | |
| PDCCH_RA | | | |
| PDCCH_RB | | | |
| PDSCH_RA | | | |
| PDSCH_RB | | | |
| OCNG_RA Note 1 | | | |
| OCNG_RB Note 1 | | | |
| $N_{oc}$ Note2 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | -infinity | -2.4 |
| RSRP Note3 | dBm/15 kHz | -infinity | -100.4 |
| SCH_RP Note3 | dBm/15 kHz | -infinity | -100.4 |
| Propagation Condition | | AWGN | |

Note 1: OCNG shall be used such that cell is fully allocated and a constant total transmitted power spectral density is achieved for all OFDM symbols.
Note 2: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 3: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 4: This test is according to the principle defined in section A.3.12.3.

**Table 10.4.5-2: SyncRef UE specific test parameters for cell identification test on downlink frequency associated with ProSe frequency for E-UTRAN FDD**

| Parameter | Unit | SyncRef UE 1 | |
|---|---|---|---|
| | | T1 | T2 |
| E-UTRA RF Channel Number | | 1 (Uplink) | |
| $BW_{channel}$ Note 4 | MHz | 5 or 10 | |
| ProSe Direct Communication resource pool configuration | | As specified in Table A.3.12.5-1 (Configuration #1) | |
| networkControlledSyncTx | | ON | |
| slssid | | 30 | |
| inCoverage (in MIB-SL) | | TRUE | |
| syncOffsetIndicator | | syncOffsetIndicator1 | |
| $N_{oc}$ Note1 | dBm/15 kHz | -98 | |
| $\hat{E}_s / N_{oc}$ | dB | 13 | |
| S-RSRP Note2, Note3 | dBm/15 kHz | -85 | |
| Propagation Condition | | AWGN | |

Note 1: Interference from other cells and noise sources not specified in the test is assumed to be constant over subcarriers and time and shall be modelled as AWGN of appropriate power for $N_{oc}$ to be fulfilled.
Note 2: RSRP levels have been derived from other parameters for information purposes. They are not settable parameters themselves.
Note 3: SSSS Es/Iot is set the same as PSSS/PSBCH Es/Iot.
Note 4: This test is according to the principle defined in section A.3.12.3.

The cell selection delay to a newly detectable cell on the downlink associated with the preconfigured ProSe carrier is defined as the time from the beginning of T2 to the time UE camps on the cell and starts to send preambles on the PRACH for sending the RRC CONNECTION REQUEST.

The cell selection delay to a newly detectable cell on the downlink associated with the preconfigured ProSe carrier shall be less than 7.68 s.

The cell selection delay can be expressed as $T_{basic_identify_OoC_ProSe\ Tx_ON} + T_{SI}$, where

- $T_{basic_identify_OoC_ProSe\ Tx_ON}$ = 6.4sec as specified in sub-clause 11.4.2.2
- $T_{SI}$ = Maximum repetition period of relevant system info blocks that needs to be received by the UE to camp on a cell; 1280 ms is assumed in this test case

This gives a total of 7.68 sec.

# Annex A (normative): Reference Measurement Channels

## A.1 PDSCH

### A.1.1 FDD

**Table A.1.1-1: PDSCH Reference Measurement Channels for FDD**

| Parameter | Unit | Value | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Reference channel | | R.2 FDD | | R.5 FDD | R.7 FDD | R.0 FDD | R.1 FDD | R.3 FDD | R.4 FDD | R.6 FDD |
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 5 | 10 | 10 | 10 | 20 | 20 |
| Number of transmitter antennas | | 1 | | 1 | 1 | 1 | 2 | 1 | 1 | 1 |
| Allocated resource blocks (Note 4) | | 2 | | 11 | 11 | 24 | 24 | 24 | 24 | 24 |
| Allocated subframes per Radio Frame | | 10 | | 10 | 10 | 10 | 10 | 10 | 10 | 10 |
| Modulation | | QPSK | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding Rate | | 1/3 | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Information Bit Payload | | | | | | | | | | |
| For Sub-Frames 4, 9 | Bits | 120 | | 968 | 968 | 2088 | 2088 | 2088 | 2088 | 2088 |
| For Sub-Frame 5 | Bits | 104 | | 776 | 776 | 2088 | 1736 | 2088 | 2088 | 2088 |
| For Sub-Frame 0 | Bits | 32 | | 616 | 616 | 1736 | 1736 | 1736 | 1736 | 1736 |
| For Sub-Frame 1, 2, 3, 6, 7, 8 | Bits | 0 | | 0 | 968 | 0 | 0 | 2088 | 0 | 2088 |
| Number of Code Blocks per Sub-Frame (Note 5) | | | | | | | | | | |
| For Sub-Frames 4, 9 | | 1 | | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| For Sub-Frame 5 | | 1 | | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| For Sub-Frame 0 | | 1 | | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| For Sub-Frame 1, 2, 3, 6, 7, 8 | | 0 | | 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| Binary Channel Bits Per Sub-Frame | | | | | | | | | | |
| For Sub-Frames 4, 9 | Bits | 456 | | 2772 | 2772 | 6624 | 6336 | 6624 | 6624 | 6624 |
| For Sub-Frame 5 | Bits | 360 | | 2484 | 2484 | 6336 | 6048 | 6336 | 6336 | 6336 |
| For Sub-Frame 0 | Bits | 176 | | 1932 | 1932 | 5784 | 5520 | 5784 | 5784 | 5784 |
| For Sub-Frame 1, 2, 3, 6, 7, 8 | Bits | 0 | | 0 | 2772 | 0 | 0 | 6624 | 0 | 6624 |
| Max. Throughput averaged over 1 frame | kbps | 37.6 | | 332.8 | 913.6 | 800 | 765 | 2053 | 800 | 2053 |

Note 1: 2 symbols allocated to PDCCH for 10 MHz channel BW. 4 symbols allocated to PDCCH for 1.4 MHz channel BW. 3 OFDM symbols allocated to PDCCH for 5 MHz channel BW.
Note 2: Reference signal, synchronization signals and PBCH allocated as defined in 3GPP TS 36.211 [9].
Note 3: If necessary the information bit payload size can be adjusted to facilitate the test implementation. The payload sizes are defined in 3GPP TS 36.213 [8].
Note 4: Allocation is located in the middle of bandwidth.
Note 5: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)
Note 6: PDSCH allocation applies only to subframes not configured as PRS subframes.

## A.1.2 TDD

**Table A.1.2-1: PDSCH Reference Measurement Channels for TDD UL/DL configuration1**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | R.2 TDD | | R.4 TDD | R.0 TDD | R.1 TDD | R.3 TDD |
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 10 | 20 |
| Number of transmitter antennas | | 1 | | 1 | 1 | 2 | 1 |
| Allocated resource blocks (Note 4) | | 2 | | 11 | 24 | 24 | 24 |
| Uplink-Downlink Configuration (Note 5) | | 1 | | 1 | 1 | 1 | 1 |
| Special Subframe Configuration (Note 6) | | 6 | | 6 | 6 | 6 | 6 |
| Allocated subframes per Radio Frame | | 6 | | 6 | 6 | 6 | 6 |
| Modulation | | QPSK | | QPSK | QPSK | QPSK | QPSK |
| Target Coding Rate | | 1/3 | | 1/3 | 1/3 | 1/3 | 1/3 |
| Information Bit Payload | | | | | | | |
| For Sub-Frames 4,9 | Bits | 120 | | 968 | 2088 | 2088 | 2088 |
| For Sub-Frame 5 | Bits | 104 | | 968 | 2088 | 2088 | 2088 |
| For Sub-Frame 0 | Bits | 56 | | 616 | 2088 | 1736 | 2088 |
| For Sub-Frame 1, 6 (DwPTS) | Bits | 56 | | 552 | 1032 | 1032 | 1032 |
| Number of Code Blocks per Sub-Frame (Note 7) | | | | 1 | | | 1 |
| For Sub-Frames 4,9 | | 1 | | 1 | 1 | 1 | 1 |
| For Sub-Frame 5 | | 1 | | 1 | 1 | 1 | 1 |
| For Sub-Frame 0 | | 1 | | 1 | 1 | 1 | 1 |
| For Sub-Frame 1, 6 (DwPTS) | | 1 | | 1 | 1 | 1 | 1 |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 4,9 | Bits | 456 | | 2772 | 6624 | 6336 | 6624 |
| For Sub-Frame 5 | Bits | 408 | | 2628 | 6480 | 6192 | 6480 |
| For Sub-Frame 0 | Bits | 224 | | 2076 | 5928 | 5664 | 5928 |
| For Sub-Frame 1, 6 (DwPTS) | Bits | 272 | | 1616 | 3696 | 3504 | 3696 |
| Max. Throughput averaged over 1 frame | Mbps | 0.05612 | | 0.4624 | 1.0416 | 1.0064 | 1.0416 |

Note 1: 2 symbols allocated to PDCCH for 10 MHz channel BW. 3 OFDM symbols allocated to PDCCH for 5MHz channel BW. 4 symbols allocated to PDCCH for 1.4 MHz channel BW. For special subframe (1 & 6) only 2 OFDM symbols are allocated to PDCCH for all bandwidths.
Note 2: Reference signal, synchronization signals and PBCH allocated as defined in 3GPP TS 36.211 [9].4 symbols allocated to PDCCH for 1.4 MHz channel BW.
Note 3: If necessary the information bit payload size can be adjusted to facilitate the test implementation. The payload sizes are defined in 3GPP TS 36.213 [8].Reference signal, synchronization signals and PBCH allocated as defined in 3GPP TS 36.211 [9].
Note 4: Allocation is located in the middle of bandwidth. If necessary the information bit payload size can be adjusted to facilitate the test implementation. The payload sizes are defined in 3GPP TS 36.213 [8].
Note 5: As per Table 4.2-2 in TS 36.211 [9].
Note 6: As per Table 4.2-1 in TS 36.211 [9].
Note 7: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 8: PDSCH allocation applies only to subframes not configured as PRS subframes.

**Table A.1.2-2: PDSCH Reference Measurement Channels for TDD UL/DL configuration0**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | | | | R.5 TDD | | |
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 10 | 20 |
| Number of transmitter antennas | | | | | 1 | | |
| Allocated resource blocks (Note 4) | | | | | 24 | | |
| Uplink-Downlink Configuration (Note 5) | | | | | 0 | | |
| Special Subframe Configuration (Note 6) | | | | | 6 | | |
| Allocated subframes per Radio Frame | | | | | 4 | | |
| Modulation | | | | | QPSK | | |
| Target Coding Rate | | | | | 1/3 | | |
| Information Bit Payload | | | | | | | |
| For Sub-Frames 4,9 | Bits | | | | N/A | | |
| For Sub-Frame 5 | Bits | | | | 2088 | | |
| For Sub-Frame 0 | Bits | | | | 2088 | | |
| For Sub-Frame 1, 6 (DwPTS) | Bits | | | | 1032 | | |
| Number of Code Blocks per Sub-Frame (Note 7) | | | | | 1 | | |
| For Sub-Frames 4,9 | | | | | N/A | | |
| For Sub-Frame 5 | | | | | 1 | | |
| For Sub-Frame 0 | | | | | 1 | | |
| For Sub-Frame 1, 6 (DwPTS) | | | | | 1 | | |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 4,9 | Bits | | | | N/A | | |
| For Sub-Frame 5 | Bits | | | | 6480 | | |
| For Sub-Frame 0 | Bits | | | | 5928 | | |
| For Sub-Frame 1, 6 (DwPTS) | Bits | | | | 3696 | | |
| Max. Throughput averaged over 1 frame | Mbps | | | | 0.624 | | |

Note 1: 2 symbols allocated to PDCCH for 10 MHz channel BW. 3 OFDM symbols allocated to PDCCH for 5MHz channel BW. 4 symbols allocated to PDCCH for 1.4 MHz channel BW. For special subframe (1 & 6) only 2 OFDM symbols are allocated to PDCCH for all bandwidths.
Note 2: Reference signal, synchronization signals and PBCH allocated as defined in TS 36.211 [9].
Note 3: If necessary the information bit payload size can be adjusted to facilitate the test implementation. The payload sizes are defined in TS 36.213 [8].
Note 4: Allocation is located in the middle of bandwidth.
Note 5: As per Table 4.2-2 in TS 36.211 [9].
Note 6: As per Table 4.2-1 in TS 36.211 [9].
Note 7: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 8: PDSCH allocation applies only to subframes not configured as PRS subframes.

## A.1.3 FDD for UE Category 0

**Table A.1.3-1: PDSCH Reference Measurement Channels for FDD**

| Parameter | Unit | Value | | |
|---|---|---|---|---|
| Reference channel | | R.13 FDD | R.14 FDD | R.15 FDD |
| Channel bandwidth | MHz | 10 | 10 | 10 |
| Number of transmitter antennas | | 1 | 2 | 2 |
| Allocated resource blocks (Note 4) | | 24 | 24 | 24 |
| Allocated subframes per Radio Frame | | 10 | 10 | 10 |
| Modulation | | QPSK | QPSK | QPSK |
| Target Coding Rate | | 1/10 | 1/10 | 1/10 |
| Information Bit Payload | | | | |
| For Sub-Frames 4, 9 | Bits | 648 | 648 | 648 |
| For Sub-Frame 5 | Bits | 648 | 648 | 648 |
| For Sub-Frame 0 | Bits | 648 | 648 | 648 |
| For Sub-Frame 1, 2, 3, 6, 7, 8 | Bits | 0 | 0 | 648 |
| Number of Code Blocks per Sub-Frame (Note 5) | | | | |
| For Sub-Frames 4, 9 | | 1 | 1 | 1 |
| For Sub-Frame 5 | | 1 | 1 | 1 |
| For Sub-Frame 0 | | 1 | 1 | 1 |
| For Sub-Frame 1, 2, 3, 6, 7, 8 | | 1 | 1 | 1 |
| Binary Channel Bits Per Sub-Frame | | | | |
| For Sub-Frames 4, 9 | Bits | 6624 | 6336 | 6636 |
| For Sub-Frame 5 | Bits | 6336 | 6048 | 6408 |
| For Sub-Frame 0 | Bits | 5784 | 5520 | 5520 |
| For Sub-Frame 1, 2, 3, 6, 7, 8 | Bits | 0 | 0 | 6636 |
| Max. Throughput averaged over 1 frame | kbps | 259.2 | 259.2 | 648 |

Note 1: 2 symbols allocated to PDCCH for 10 MHz channel BW. 4 symbols allocated to PDCCH for 1.4 MHz channel BW. 3 OFDM symbols allocated to PDCCH for 5 MHz channel BW.
Note 2: Reference signal, synchronization signals and PBCH allocated as defined in 3GPP TS 36.211 [9].
Note 3: If necessary the information bit payload size can be adjusted to facilitate the test implementation. The payload sizes are defined in 3GPP TS 36.213 [8].
Note 4: Allocation is located in the middle of bandwidth.
Note 5: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 6: PDSCH allocation applies only to subframes not configured as PRS subframes.

## A.1.4 HD-FDD for UE Category 0

**Table A.1.4-1: PDSCH Reference Measurement Channels for HD-FDD**

| Parameter | Unit | Value | |
|---|---|---|---|
| Reference channel | | R.1 HD-FDD | R.2 HD-FDD |
| Channel bandwidth | MHz | 10 | 10 |
| Number of transmitter antennas | | 1 | 2 |
| Allocated resource blocks (Note 4) | | 24 | 24 |
| Allocated subframes per Radio Frame | | 10 | 10 |
| Modulation | | QPSK | QPSK |
| Target Coding Rate | | 1/10 | 1/10 |
| Information Bit Payload | | | |
| For Sub-Frames 4, 9, | Bits | 0 | 0 |
| For Sub-Frame 5 (Note 7) | Bits | 424 | 424 |
| For Sub-Frame 0 (Note 7) | | 648 | 648 |
| For Sub-Frame 1, 2, 3, 6, 7, 8 | Bits | 0 | 0 |
| For Sub-Frame 1, 6 | Bits | | |
| Number of Code Blocks per Sub-Frame (Note 5) | | | |
| For Sub-Frames 4, 9 | | 1 | 1 |
| For Sub-Frame 5 | | 1 | 1 |
| For Sub-Frame 0 | | 1 | 1 |
| For Sub-Frame 1, 2, 3, 6, 7, 8 | | 1 | 1 |
| Binary Channel Bits Per Sub-Frame | | | |
| For Sub-Frames 4, 9 | Bits | 0 | 0 |
| For Sub-Frame 5 | Bits | 6336 | 6048 |
| For Sub-Frame 0 | Bits | 5784 | 5520 |
| For Sub-Frame 1, 2, 3, 6, 7, 8 | Bits | 0 | 0 |
| Max. Throughput averaged over 1 frame | kbps | | |

Note 1: 2 symbols allocated to PDCCH for 10 MHz channel BW. 4 symbols allocated to PDCCH for 1.4 MHz channel BW. 3 OFDM symbols allocated to PDCCH for 5 MHz channel BW.
Note 2: Reference signal, synchronization signals and PBCH allocated as defined in 3GPP TS 36.211 [9].
Note 3: If necessary the information bit payload size can be adjusted to facilitate the test implementation. The payload sizes are defined in 3GPP TS 36.213 [8].
Note 4: Allocation is located in the middle of bandwidth.
Note 5: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 6: PDSCH allocation applies only to subframes not configured as PRS subframes.
Note 7: Sub-frame 0 or sub-frame 5 is randomly scheduled in downlink.

## A.1.5 TDD for UE Category 0

**Table A.1.5-1: PDSCH Reference Measurement Channels for TDD UL/DL configuration1**

| Parameter | Unit | Value | |
|---|---|---|---|
| Reference channel | | R.12 TDD | R.13 TDD |
| Channel bandwidth | MHz | 10 | 10 |
| Number of transmitter antennas | | 1 | 2 |
| Allocated resource blocks (Note 4) | | 24 | 24 |
| Uplink-Downlink Configuration (Note 5) | | 1 | 1 |
| Special Subframe Configuration (Note 6) | | 6 | 6 |
| Allocated subframes per Radio Frame | | 6 | 6 |
| Modulation | | QPSK | QPSK |
| Target Coding Rate | | 1/10 | 1/10 |
| Information Bit Payload | | | |
| For Sub-Frames 4,9 | Bits | 648 | 648 |
| For Sub-Frame 5 | Bits | 648 | 648 |
| For Sub-Frame 0 | Bits | 648 | 648 |
| For Sub-Frame 1, 6 (DwPTS) | Bits | 488 | 488 |
| Number of Code Blocks per Sub-Frame (Note 7) | | 1 | 1 |
| For Sub-Frames 4,9 | | 1 | 1 |
| For Sub-Frame 5 | | 1 | 1 |
| For Sub-Frame 0 | | 1 | 1 |
| For Sub-Frame 1, 6 (DwPTS) | | 1 | 1 |
| Binary Channel Bits Per Sub-Frame | | | |
| For Sub-Frames 4,9 | Bits | 6624 | 6336 |
| For Sub-Frame 5 | Bits | 6580 | 6192 |
| For Sub-Frame 0 | Bits | 5928 | 5664 |
| For Sub-Frame 1, 6 (DwPTS) | Bits | 3696 | 3408 |
| Max. Throughput averaged over 1 frame | Mbps | 0.3552 | 0.3552 |

Note 1: 2 symbols allocated to PDCCH for 10 MHz channel BW. 3 OFDM symbols allocated to PDCCH for 5MHz channel BW. 4 symbols allocated to PDCCH for 1.4 MHz channel BW. For special subframe (1 & 6) only 2 OFDM symbols are allocated to PDCCH for all bandwidths.
Note 2: Reference signal, synchronization signals and PBCH allocated as defined in TS 36.211 [9].
Note 3: If necessary the information bit payload size can be adjusted to facilitate the test implementation. The payload sizes are defined in TS 36.213 [8].
Note 4: Allocation is located in the middle of bandwidth.
Note 5: As per Table 4.2-2 in TS 36.211 [9]
Note 6: As per Table 4.2-1 in TS 36.211 [9]
Note 7: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)
Note 8: PDSCH allocation applies only to subframes not configured as PRS subframes.

# A.2 PCFICH/PDCCH/PHICH

## A.2.1 FDD

**Table A.2.1-1: PCFICH/PDCCH/PHICH Reference Channel for FDD**

| Parameter | Unit | Value | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Reference channel | | R.8 FDD | R.14 FDD | R.11 FDD | R.12 FDD | R.10 FDD | R.6 FDD | R.7 FDD | R.9 FDD |
| Channel bandwidth | MHz | 1.4 | 1.4 | 5 | 5 | 20 | 10 | 10 | 10 |
| Number of transmitter antennas | | 1 | 1 | 1 | 2 | 1 | 1 | 2 | 2 |
| Control region OFDM symbols[Note1] | symbols | 4 | 3 | 3 | 3 | 2 | 2 | 2 | 3 |
| Aggregation level | CCE | 2 (Note 6) | 2 (Note 6) | 8 | 8 | 8 | 8 | 8 | 8 |
| DCI Format | | Note 3 | Note 3 | Note 3 | Note 3 | Note 3 | Note 3 | Note 3 | Note 3 |
| Cell ID | | Note 4 | Note 4 | Note 4 | Note 4 | Note 4 | Note 4 | Note 4 | Note 4 |
| Payload (without CRC) | Bits | Note 5 | Note 5 | Note 5 | Note 5 | Note 5 | Note 5 | Note 5 | Note 5 |
| Note 1: The control region consists of PCFICH, PHICH and PDCCH.<br>Note 2: DCI formats are defined in TS 36.212.<br>Note 3: DCI format shall depend upon the test configuration.<br>Note 4: Cell ID shall depend upon the test configuration.<br>Note 5: Payload size shall depend upon the test configuration.<br>Note 6: For PDCCH using SI/RA/P-RNTI, Aggregation level 4 is used. | | | | | | | | | |

## A.2.2 TDD

**Table A.2.2-1: PCFICH/PDCCH/PHICH Reference Channel for TDD**

| Parameter | Unit | Value | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Reference channel | | R.8 TDD | R.14 TDD | R11 TDD | R.10 TDD | R.6 TDD | R.7 TDD | R.9 TDD |
| Channel bandwidth | MHz | 1.4 | 1.4 | 5 | 20 | 10 | 10 | 10 |
| Number of transmitter antennas | | 1 | 1 | 1 | 1 | 1 | 2 | 2 |
| Control region OFDM symbols[Note1] | symbols | 4 (Note 6) | 3 (Note 6) | 3 | 2 | 2 | 2 | 3 |
| Aggregation level | CCE | 2 (Note 7) | 2 (Note 7) | 8 | 8 | 8 | 8 | 8 |
| DCI Format | | Note 3 | Note 3 | Note 3 | Note 3 | Note 3 | Note 3 | Note 3 |
| Cell ID | | Note 4 | Note 4 | Note 4 | Note 4 | Note 4 | Note 4 | Note 4 |
| Payload (without CRC) | Bits | Note 5 | Note 5 | Note 5 | Note 5 | Note 5 | Note 5 | Note 5 |
| Note 1: The control region consists of PCFICH, PHICH and PDCCH.<br>Note 2: DCI formats are defined in 3GPP TS 36.212.<br>Note 3: DCI format shall depend upon the test configuration.<br>Note 4: Cell ID shall depend upon the test configuration.<br>Note 5: Payload size shall depend upon the test configuration.<br>Note 6: Only 2 OFDM symbols for special subframes 1 and 6.<br>Note 7: For PDCCH using SI/RA/P-RNTI, Aggregation level 4 is used. | | | | | | | | |

### A.2.3 HD-FDD for UE category 0

**Table A.2.3-1: PCFICH/PDCCH/PHICH Reference Channel for HD-FDD**

| Parameter | Unit | Value | | |
|---|---|---|---|---|
| Reference channel | | R.3 HD-FDD | R.4 HD-FDD | R.5 HD-FDD |
| Channel bandwidth | MHz | 10 | 10 | 10 |
| Number of transmitter antennas | | 1 | 2 | 2 |
| Control region OFDM symbols[Note1] | symbols | 2 | 2 | 3 |
| Aggregation level | CCE | 8 | 8 | 8 |
| DCI Format | | Note 3 | Note 3 | Note 3 |
| Cell ID | | Note 4 | Note 4 | Note 4 |
| Payload (without CRC) | Bits | Note 5 | Note 5 | Note 5 |
| Note 1: The control region consists of PCFICH, PHICH and PDCCH.<br>Note 2: DCI formats are defined in TS 36.212.<br>Note 3: DCI format shall depend upon the test configuration.<br>Note 4: Cell ID shall depend upon the test configuration.<br>Note 5: Payload size shall depend upon the test configuration.<br>Note 6: For PDCCH using SI/RA/P-RNTI, Aggregation level 4 is used.<br>Note 7: Sub-frame 0 or sub-frame 5 is randomly scheduled in downlink. | | | | |

## A.3 PUSCH

This rule applies to E-UTRA cell(s), which the UE is connected to. The UE is in RRC-CONNECTED mode.

When signalling or data payloads are expected to be sent on the PUSCH, the UE may be provided in advance with PUSCH resources by the SS. For sake of simplicity the PUSCH scheduling may also occur continuously over many consecutive subframes. These options shall not be used if:
1) stated otherwise in the test description, or
2) the transmission of PUSCH and UL scheduling information affects the test purpose (e.g. DRX, PUCCH reception etc.)

For handover test cases, after RRC Connection reconfiguration message implying handover is sent, the UE shall be provided continuously with PUSCH resources by the SS in the source cell. This is done in order to make the requirement UE implementation agnostic, w.r.t. different delays caused by different handling of positive RLC acknowledgements, which are not mandatory and of lower priority than the handover procedure progress (subclause 5.3.5.4 [5]).

If a PUSCH scheduling occurs, the SS sends uplink scheduling information via PDCCH DCI format 0 for C-RNTI to the UE. The UE sends uplink MAC padding bits on the PUSCH.

# A.4 Reference Measurement Channels for ProSe Direct Discovery

## A.4.1 FDD

**Table A.4.1-1: PSDCH Reference Measurement Channels for FDD**

| Parameter | Unit | Value |
|---|---|---|
| Reference channel | | D.1 FDD |
| Channel bandwidth | MHz | 5 |
| Allocated resource blocks | | 2 |
| Subcarriers per resource block | | 12 |
| Allocated subframes per Discovery period | | 1 |
| DFT-OFDM Symbols per subframe (see note) | | 11 |
| Modulation | | QPSK |
| Information Bit Payload | | 232 |
| Transport block CRC | Bits | 24 |
| Maximum number of HARQ transmissions | | 1 |
| Binary Channel Bits (see note) | Bits | 528 |
| NOTE1: PSDCH transmissions are rate-matched for 12 DFT-OFDM symbols per subframe, and the last symbol shall be punctured as per TS 36.211. | | |

# A.5 Reference measurement channels for ProSe Direct Communication

## A.5.1 FDD

**Table A.5.1-1: PSCCH Reference Measurement Channels for FDD**

| Parameter | | Unit | Value | |
|---|---|---|---|---|
| Reference channel | | | CC.1 FDD | CC.2 FDD |
| Channel bandwidth | | MHz | 5 | 10 |
| Allocated resource blocks | | | 1 | 1 |
| Subcarriers per resource block | | | 12 | 12 |
| DFT-OFDM Symbols per subframe (see Note 1) | | | 11 | 11 |
| Modulation | | | QPSK | QPSK |
| Information Bit Payload | | Bits | 41 | 43 |
| Information bits | Frequency hopping flag | | 0 | |
| | RB assignment | | Set as per PSSCH RB allocation specific in the test | |
| | Time resource pattern ($I_{TRP}$) | | 0 (Note 2) | |
| | Modulation and coding scheme | | Set as the PSSCH MCS specified in the test | |
| | Timing advance indication | | 0 | |
| | Group destination ID | | As set by higher layers | |
| Transport block CRC | | Bits | 16 | 16 |
| Maximum number of HARQ transmissions | | | 2 | 2 |
| Binary Channel Bits (see Note 1) | | Bits | 264 | 264 |
| NOTE1: PSCCH transmissions are rate-matched for 12 DFT-OFDM symbols per subframe, and the last symbol shall be punctured as per TS 36.211.<br>NOTE 2: For $N_{TRP}$ = 8 (FDD) and *trpt-Subset* = 001, $I_{TRP}$ = 0 corresponds to a time repetition pattern of (1,0,0,0,0,0,0,0) as per TS 36.213. | | | | |

**Table A.5.1-2: PSSCH Reference Measurement Channels for FDD**

| Parameter | Unit | Value | |
|---|---|---|---|
| Reference channel | | CD.1 FDD | CD.2 FDD |
| Channel bandwidth | MHz | 5 | 10 |
| Allocated resource blocks | | 2 | 3 |
| Subcarriers per resource block | | 12 | 12 |
| DFT-OFDM Symbols per subframe (see Note 1) | | 11 | 11 |
| Modulation | | QPSK | QPSK |
| Target Code Rate | | 1/3 | 1/3 |
| Information Bit Payload | | 176 | 256 |
| Transport block CRC | Bits | 24 | 24 |
| Maximum number of HARQ transmissions | | 3 | 3 |
| Binary Channel Bits (see note) | Bits | 528 | 1056 |
| NOTE1: PSDCH transmissions are rate-matched for 12 DFT-OFDM symbols per subframe, and the last symbol shall be punctured as per TS 36.211. | | | |

# A.6 ProSe Receive Traffic Generator

This clause defines the configuration for active Sidelink UEs used to generate receive traffic in ProSe RRM tests.

## A.6.1 ProSe Direct Communication Receive Traffic Generator for FDD

**Table A.6.1-1: Active Sidelink UE configuration for ProSe Direct Communication**

| Configuration | | | PCP.1.FDD |
|---|---|---|---|
| Channel BW | | MHz | 5 or 10 |
| Number of Active Sidelink UEs per sc-period | | | 5 MHz: 12<br>10 MHz: 16 |
| Active Sidelink UEs | PSCCH RMC (defined in A.5) | | 5 MHz: CC.1 FDD<br>10 MHz: CC.2 FDD |
| | PSCCH resource allocation | | 5MHz: [2i:2i], for Sidelink UE i=0,..,11<br>10MHz:[3i:3i], for Sidelink UE i = 0, .., 15 |
| | PSSCH RMC (defined in A.5) | | 5 MHz: CD.1 FDD<br>10 MHz: CD.2 FDD |
| | PSSCH resource allocation | | Non-overlapping RBs<br>5MHz: [2i:2i+1], for Sidelink UE i = 0, .., 11<br>10MHz:[3i:3i+2], for Sidelink UE i = 0, .., 15 |
| | RSRP | dBm/15kHz | -98 |

## A.6.2 ProSe Direct Discovery Receive Traffic Generator for FDD

**Table A.6.2-1: Active Sidelink UE configuration for ProSe Direct Discovery**

| Configuration | | | PDP.1.FDD | PDP.2.FDD |
|---|---|---|---|---|
| Channel BW | | MHz | 5 | |
| Number of Active Sidelink UEs per Discovery subframe | | | 12 | |
| Active Sidelink UEs | Sidelink UE Transmissions | | PSDCH (RMC D.1 FDD) | PSDCH (RMC D.1 FDD) + SLSS on synchronization subframe |
| | Resource allocation | | Non overlapping RBs in a subframe | |
| | RSRP | dBm/15kHz | -95 | |

# A.7 MPDCCH Reference Channels for Cat-M1 UEs

## A.7.1 FDD in CEModeA

**Table A.7.1-1: MPDCCH Reference Channel for Cat-M1 FDD UEs in CEModeA**

| Parameter | Unit | Value | |
|---|---|---|---|
| MPDCCH Reference channel | - | R.16 FDD | R.17 FDD |
| Carrier bandwidth | MHz | 10 | 10 |
| Number of transmitter antennas | - | 1 | 2 |
| DCI Format | - | 6-1A | 6-1A |
| Transmission Type | - | Distributed | Distributed |
| Number of PRB pairs per M-PDCCH set | - | 4 | 4 |
| Aggregation level | ECCE | 16 | 16 |
| Maximum number of repetitions | - | 8 | 8 |
| Frequency hopping | - | ON | ON |
| Number of narrowbands | - | 2 | 2 |
| MPDCCH Narrowband | - | 1st | 1st |
| Frequency hopping HoppingOffset | narrowbands | 7 | 7 |
| MPDCCH start subframe | subframes | 1 | 1 |
| MPDCCH start symbol | symbols | 2 | 2 |
| Frequency hopping interval | subframes | 4 | 4 |
| Payload (without CRC) | Bits | Note 1 | Note 1 |
| Cell ID | - | Note 2 | Note 2 |
| Note 1: Payload size shall depend upon the test configuration.<br>Note 2: Cell ID shall depend upon the test configuration. | | | |

## A.7.2 HD-FDD in CEModeA

**Table A.7.2-1: MPDCCH Reference Channel for Cat-M1 HD-FDD UEs in CEModeA**

| Parameter | Unit | Value | |
|---|---|---|---|
| MPDCCH Reference channel | - | R.6 HD-FDD | R.7 HD-FDD |
| Carrier bandwidth | MHz | 10 | 10 |
| Number of transmitter antennas | - | 1 | 2 |
| DCI Format | - | 6-1A | 6-1A |
| Transmission Type | - | Distributed | Distributed |
| Number of PRB pairs per M-PDCCH set | - | 4 | 4 |
| Aggregation level | ECCE | 16 | 16 |
| Maximum number of repetitions | - | 8 | 8 |
| Frequency hopping | - | ON | ON |
| Number of narrowbands | - | 2 | 2 |
| MPDCCH Narrowband | - | 1st | 1st |
| Frequency hopping HoppingOffset | narrowbands | 7 | 7 |
| MPDCCH start subframe | subframes | 1 | 1 |
| MPDCCH start symbol | symbols | 2 | 2 |
| Frequency hopping interval | subframes | 4 | 4 |
| Payload (without CRC) | Bits | Note 1 | Note 1 |
| Cell ID | - | Note 2 | Note 2 |
| Note 1: Payload size shall depend upon the test configuration.<br>Note 2: Cell ID shall depend upon the test configuration. | | | |

## A.7.3 TDD in CEModeA

**Table A.7.3-1: MPDCCH Reference Channel for Cat-M1 TDD UEs in CEModeA**

| Parameter | Unit | Value | |
|---|---|---|---|
| MPDCCH Reference channel | - | R.14 TDD | R.15 TDD |
| Carrier bandwidth | MHz | 10 | 10 |
| Number of transmitter antennas | - | 1 | 2 |
| DCI Format | - | 6-1A | 6-1A |
| Transmission Type | - | Distributed | Distributed |
| Number of PRB pairs per M-PDCCH set | - | 4 | 4 |
| Aggregation level | ECCE | 16 | 16 |
| Maximum number of repetitions | - | 8 | 8 |
| Frequency hopping | - | ON | ON |
| Number of narrowbands | - | 2 | 2 |
| MPDCCH Narrowband | - | 1st | 1st |
| Frequency hopping HoppingOffset | narrowbands | 7 | 7 |
| MPDCCH start subframe | subframes | 1 | 1 |
| MPDCCH start symbol | symbols | 2 | 2 |
| Frequency hopping interval | subframes | 10 | 10 |
| Payload (without CRC) | Bits | Note 1 | Note 1 |
| Cell ID | - | Note 2 | Note 2 |
| Note 1: Payload size shall depend upon the test configuration.<br>Note 2: Cell ID shall depend upon the test configuration. | | | |

## A.7.4 FDD in CEModeB

**Table A.7.4-1: MPDCCH Reference Channel for Cat-M1 FDD UEs in CEModeB**

| Parameter | Unit | Value | |
|---|---|---|---|
| MPDCCH Reference channel | - | R.18 FDD | R.19 FDD |
| Carrier bandwidth | MHz | 10 | 10 |
| Number of transmitter antennas | - | 1 | 2 |
| DCI Format | - | 6-1B | 6-1B |
| Transmission Type | - | Distributed | Distributed |
| Number of PRB pairs per M-PDCCH set | - | 6 | 6 |
| Aggregation level | ECCE | 24 | 24 |
| Maximum number of repetitions | - | 128 | 128 |
| Frequency hopping | - | ON | ON |
| Number of narrowbands | - | 2 | 2 |
| MPDCCH Narrowband | - | 1st | 1st |
| Frequency hopping HoppingOffset | narrowbands | 7 | 7 |
| MPDCCH start subframe | subframes | 1 | 1 |
| MPDCCH start symbol | symbols | 2 | 2 |
| Frequency hopping interval | subframes | 8 | 8 |
| Payload (without CRC) | Bits | 17 | 17 |
| Cell ID | - | Note 1 | Note 1 |
| Note 1: Cell ID shall depend upon the test configuration. | | | |

## A.7.5 HD-FDD in CEModeB

**Table A.7.5-1: MPDCCH Reference Channel for Cat-M1 HD-FDD UEs in CEModeB**

| Parameter | Unit | Value | |
|---|---|---|---|
| MPDCCH Reference channel | - | R.8 HD-FDD | R.9 HD-FDD |
| Carrier bandwidth | MHz | 10 | 10 |
| Number of transmitter antennas | - | 1 | 2 |
| DCI Format | - | 6-1B | 6-1B |
| Transmission Type | - | Distributed | Distributed |
| Number of PRB pairs per M-PDCCH set | - | 6 | 6 |
| Aggregation level | ECCE | 24 | 24 |
| Maximum number of repetitions | - | 128 | 128 |
| Frequency hopping | - | ON | ON |
| Number of narrowbands | - | 2 | 2 |
| MPDCCH Narrowband | - | 1st | 1st |
| Frequency hopping HoppingOffset | narrowbands | 7 | 7 |
| MPDCCH start subframe | subframes | 1 | 1 |
| MPDCCH start symbol | symbols | 2 | 2 |
| Frequency hopping interval | subframes | 8 | 8 |
| Payload (without CRC) | Bits | 17 | 17 |
| Cell ID | - | Note 1 | Note 1 |
| Note 1: Cell ID shall depend upon the test configuration. | | | |

## A.7.6 TDD in CEModeB

**Table A.7.6-1: MPDCCH Reference Channel for Cat-M1 TDD UEs in CEModeB**

| Parameter | Unit | Value | |
|---|---|---|---|
| MPDCCH Reference channel | - | R.15 TDD | R.16 TDD |
| Carrier bandwidth | MHz | 10 | 10 |
| Number of transmitter antennas | - | 1 | 2 |
| DCI Format | - | 6-1B | 6-1B |
| Transmission Type | - | Distributed | Distributed |
| Number of PRB pairs per M-PDCCH set | - | 6 | 6 |
| Aggregation level | ECCE | 24 | 24 |
| Maximum number of repetitions | - | 128 | 128 |
| Frequency hopping | - | ON | ON |
| Number of narrowbands | - | 2 | 2 |
| MPDCCH Narrowband | - | 1st | 1st |
| Frequency hopping HoppingOffset | narrowbands | 7 | 7 |
| MPDCCH start subframe | subframes | 1 | 1 |
| MPDCCH start symbol | symbols | 2 | 2 |
| Frequency hopping interval | subframes | 20 | 20 |
| Payload (without CRC) | Bits | 17 | 17 |
| Cell ID | - | Note 1 | Note 1 |
| Note 1: Cell ID shall depend upon the test configuration. | | | |

# A.8 PDSCH Reference Channels for Cat-M1 UEs

## A.8.1 FDD in CEModeA

**Table A.8.1-1: PDSCH Reference Channel for Cat-M1 FDD in CEModeA**

| Parameter | Unit | Value | |
|---|---|---|---|
| PDSCH Reference channel | | R.20 FDD | R.21 FDD |
| Carrier bandwidth | MHz | 10 | 10 |
| Number of transmitter antennas | | 1 | 2 |
| Allocated resource blocks[Note1] | PRBs | 2 | 2 |
| Allocated subframes per Radio Frame | subframes | 10 | 10 |
| Modulation | | QPSK | QPSK |
| Target coding rate | | 1/3 | 1/3 |
| Information Bit Payload | | | |
| Sub-Frames 0 ~ 9 | Bits | 32 | 32 |
| Number of Code Blocks per Sub-Frame | | | |
| Sub-Frames 0 ~ 9 | | 1 | 1 |
| Maximum number of repetitions | | 16 | 16 |
| Frequency hopping | | ON | ON |
| Number of narrowbands for frequency hopping | narrowbands | 2 | 2 |
| PDSCH Narrowband | | 2nd | 2nd |
| Frequency HoppingOffset | narrowbands | 5 | 5 |
| PDSCH start symbol | symbols | 2 | 2 |
| Frequency hopping interval | subframes | 4 | 4 |
| Max. Throughput averaged over 1 frame | kbps | TBD | TBD |
| Cell ID | | Note 2 | Note 2 |
| Note 1: Allocation is located in the middle of narrowband.<br>Note 2: Cell ID shall depend upon the test configuration. | | | |

## A.8.2 HD-FDD in CEModeA

**Table A.8.2-1: PDSCH Reference Channel for Cat-M1 HD-FDD in CEModeA**

| Parameter | Unit | Value | |
|---|---|---|---|
| PDSCH Reference channel | | R.10 HD-FDD | R.11 HD-FDD |
| Carrier bandwidth | MHz | 10 | 10 |
| Number of transmitter antennas | | 1 | 2 |
| Allocated resource blocks[Note1] | PRBs | 2 | 2 |
| Allocated subframes per Radio Frame | subframes | 10 | 10 |
| Modulation | | QPSK | QPSK |
| Target coding rate | | 1/3 | 1/3 |
| Information Bit Payload | | | |
| Sub-Frames 0 ~ 9 | Bits | 32 | 32 |
| Number of Code Blocks per Sub-Frame | | | |
| Sub-Frames 0 ~ 9 | | 1 | 1 |
| Maximum number of repetitions | | 16 | 16 |
| Frequency hopping | | ON | ON |
| Number of narrowbands for frequency hopping | narrowbands | 2 | 2 |
| PDSCH Narrowband | | 1st | 1st |
| Frequency HoppingOffset | narrowbands | 5 | 5 |
| PDSCH start symbol | symbols | 2 | 2 |
| Frequency hopping interval | subframes | 4 | 4 |
| Max. Throughput averaged over 1 frame | kbps | TBD | TBD |
| Cell ID | | Note 2 | Note 2 |
| Note 1: Allocation is located in the middle of narrowband.<br>Note 2: Cell ID shall depend upon the test configuration. | | | |

## A.8.3 TDD in CEModeA

**Table A.8.3-1: PDSCH Reference Channel for Cat-M1 TDD in CEModeA**

| Parameter | Unit | Value | |
|---|---|---|---|
| PDSCH Reference channel | | R.16 TDD | R.17 TDD |
| Carrier bandwidth | MHz | 10 | 10 |
| Number of transmitter antennas | | 1 | 2 |
| Allocated resource blocks[Note1] | PRBs | 2 | 2 |
| Allocated subframes per Radio Frame | | 4 | 4 |
| TDD Uplink-Downlink Configuration | | 0 | 0 |
| TDD Special Subframe Configuration | | 6 | 6 |
| Modulation | | QPSK | QPSK |
| Target coding rate | | 1/3 | 1/3 |
| Information Bit Payload | | | |
| All Sub-Frames except 1, 6 (TDD) | Bits | 32 | 32 |
| All Sub-Frames except 1, 6 (TDD DwPTS) | - | 0 | 0 |
| Number of Code Blocks per Sub-Frame | | 1 | 1 |
| All Sub-Frames except 1, 6 (TDD) | | 1 | 1 |
| All Sub-Frames except 1, 6 (TDD DwPTS) | | 0 | 0 |
| Maximum number of repetitions | | 16 | 16 |
| Frequency hopping | | ON | ON |
| Number of narrowbands for frequency hopping | | 2 | 2 |
| PDSCH Narrowband | | 2nd | 2nd |
| Frequency HoppingOffset | narrowbands | 5 | 5 |
| PDSCH start symbol | symbols | 2 | 2 |
| Frequency hopping interval | subframes | 10 | 10 |
| Max. Throughput averaged over 1 frame | kbps | TBD | TBD |
| Cell ID | | Note 2 | Note 2 |
| Note 1: Allocation is located in the middle of narrowband.<br>Note 2: Cell ID shall depend upon the test configuration. | | | |

## A.8.4 FDD in CEModeB

**Table A.8.4-1: PDSCH Reference Channel for Cat-M1 FDD in CEModeB**

| Parameter | Unit | Value | |
|---|---|---|---|
| PDSCH Reference channel | | R.22 FDD | R.23 FDD |
| Carrier bandwidth | MHz | 10 | 10 |
| Number of transmitter antennas | | 1 | 2 |
| Allocated resource blocks[Note1] | | 2 | 2 |
| Allocated subframes per Radio Frame | - | 10 | 10 |
| Modulation | - | QPSK | QPSK |
| Target coding rate | - | 1/3 | 1/3 |
| Information Bit Payload | | | |
| Sub-Frames 0 ~ 9 | Bits | 32 | 32 |
| Number of Code Blocks per Sub-Frame | | | |
| Sub-Frames 0 ~ 9 | - | 1 | 1 |
| Maximum number of repetitions | - | 192 | 192 |
| Frequency hopping | - | ON | ON |
| Number of narrowbands for frequency hopping | narrowbands | 2 | 2 |
| PDSCH Narrowband | - | 2nd | 2nd |
| Frequency HoppingOffset | narrowbands | 5 | 5 |
| PDSCH start symbol | symbols | 2 | 2 |
| Frequency hopping interval | subframes | 8 | 8 |
| Max. Throughput averaged over 1 frame | kbps | TBD | TBD |
| Cell ID | | Note 2 | Note 2 |
| Note 1: Allocation is located in the middle of narrowband.<br>Note 2: Cell ID shall depend upon the test configuration. | | | |

## A.8.5 HD-FDD in CEModeB

**Table A.8.5-1: PDSCH Reference Channel for Cat-M1 HD-FDD in CEModeB**

| Parameter | Unit | Value | |
|---|---|---|---|
| PDSCH Reference channel | | R.12 HD-FDD | R.13 HD-FDD |
| Carrier bandwidth | MHz | 10 | 10 |
| Number of transmitter antennas | | 1 | 2 |
| Allocated resource blocks[Note1] | PRBs | 2 | 2 |
| Allocated subframes per Radio Frame | subframes | 10 | 10 |
| Modulation | | QPSK | QPSK |
| Target coding rate | | 1/3 | 1/3 |
| Information Bit Payload | | | |
| Sub-Frames 0 ~ 9 | Bits | 32 | 32 |
| Number of Code Blocks per Sub-Frame | | | |
| Sub-Frames 0 ~ 9 | | 1 | 1 |
| Maximum number of repetitions | | 192 | 192 |
| Frequency hopping | | ON | ON |
| Number of narrowbands for frequency hopping | narrowbands | 2 | 2 |
| PDSCH Narrowband | | 1st | 1st |
| Frequency HoppingOffset | narrowbands | 5 | 5 |
| PDSCH start symbol | symbols | 2 | 2 |
| Frequency hopping interval | subframes | 8 | 8 |
| Max. Throughput averaged over 1 frame | kbps | TBD | TBD |
| Cell ID | | Note 2 | Note 2 |
| Note 1: Allocation is located in the middle of narrowband.<br>Note 2: Cell ID shall depend upon the test configuration. | | | |

## A.8.6 TDD in CEModeA

**Table A.8.6-1: PDSCH Reference Channel for Cat-M1 TDD in CEModeB**

| Parameter | Unit | Value | |
|---|---|---|---|
| PDSCH Reference channel | | R.18 TDD | R.19 TDD |
| Carrier bandwidth | MHz | 10 | 10 |
| Number of transmitter antennas | | 1 | 2 |
| Allocated resource blocks[Note1] | PRBs | 2 | 2 |
| Allocated subframes per Radio Frame | | 4 | 4 |
| TDD Uplink-Downlink Configuration | | 0 | 0 |
| TDD Special Subframe Configuration | | 6 | 6 |
| Modulation | | QPSK | QPSK |
| Target coding rate | | 1/3 | 1/3 |
| Information Bit Payload | | | |
| All Sub-Frames except 1, 6 (TDD) | Bits | 32 | 32 |
| All Sub-Frames except 1, 6 (TDD DwPTS) | - | 0 | 0 |
| Number of Code Blocks per Sub-Frame | | 1 | 1 |
| All Sub-Frames except 1, 6 (TDD) | | 1 | 1 |
| All Sub-Frames except 1, 6 (TDD DwPTS) | | 0 | 0 |
| Maximum number of repetitions | | 192 | 192 |
| Frequency hopping | | ON | ON |
| Number of narrowbands for frequency hopping | | 2 | 2 |
| PDSCH Narrowband | | 2nd | 2nd |
| Frequency HoppingOffset | narrowbands | 5 | 5 |
| PDSCH start symbol | symbols | 2 | 2 |
| Frequency hopping interval | subframes | 20 | 20 |
| Max. Throughput averaged over 1 frame | kbps | TBD | TBD |
| Cell ID | | Note 2 | Note 2 |
| Note 1: Allocation is located in the middle of narrowband.<br>Note 2: Cell ID shall depend upon the test configuration. | | | |

# A.9 Reference PRACH Configurations

**Table A.9-1: PRACH configuration parameters**

| Parameter | Value | | | Comments |
|---|---|---|---|---|
| Reference configuration | PRACH_2CE | PRACH_3CE | PRACH_4CE | |
| RsrpThresholdsPRACH | {-99} dBm | {-107, -99} dBm | {-107, -99, -92} dBm | As defined in TS36.331[5] |
| CE level0: | Configured | Configured | Configured | Up to 4 CE levels, each corrsesonding to a PRACH configuration |
| Prach Configation Index: | FDD : 4<br>TDD : 53 | FDD : 4<br>TDD : 53 | FDD : 4<br>TDD : 53 | See TS 36.211[9] section 5.7.1 |
| numRepetitionPerPreambleAttempt: | 1 | 1 | 1 | Number of PRACH repetitions per attempt for each CE level, See TS 36.211[9] |
| prach-HoppingConfig: | off | off | off | Coverage level specific frequency hopping configuration for PRACH |
| CE level1: | Configured | Configured | Configured | Up to 4 CE levels, each corrsesonding to a PRACH configuration |
| Prach Configation Index: | FDD : 4<br>TDD : 53 | FDD : 4<br>TDD : 53 | FDD : 4<br>TDD : 53 | see TS 36.211[9] section 5.7.1 |
| numRepetitionPerPreambleAttempt: | 128 | 64 | 32 | Number of PRACH repetitions per attempt for each CE level, See TS 36.211[9] |
| prach-HoppingConfig: | off | off | off | Coverage level specific frequency hopping configuration for PRACH |
| CE level2: | Not Configured | Configured | Configured | Up to 4 CE levels, each corrsesonding to a PRACH configuration |
| Prach Configation Index: | - | FDD : 4<br>TDD : 53 | FDD : 4<br>TDD : 53 | see TS 36.211[9] section 5.7.1 |
| numRepetitionPerPreambleAttempt: | - | 128 | 64 | Number of PRACH repetitions per attempt for each CE level, See TS 36.211[9] |
| prach-HoppingConfig: | - | off | off | Coverage level specific frequency hopping configuration for PRACH |
| CE level3: | Not Configured | Not Configured | Configured | Up to 4 CE levels, each corrsesonding to a PRACH configuration |
| Prach Configation Index: | - | - | FDD : 4<br>TDD : 53 | see TS 36.211[9] section 5.7.1 |
| numRepetitionPerPreambleAttempt: | - | - | 128 | Number of PRACH repetitions per attempt for each CE level, See TS 36.211[9] |
| prach-HoppingConfig: | - | - | off | Coverage level specific frequency hopping configuration for PRACH |

# A.10 General configuration for NB-IoT

This clause defines the configuration for category NB1 UEs used in NB-IoT RRM tests.

# A.10.1 NPDCCH Reference Channel for UE category NB1

## A.10.1.1 HD-FDD in-band operation

**Table A.10.1.1-1: NPDCCH Reference Channel for UE category NB1 for in-band operation**

| Parameter | Unit | Value | |
|---|---|---|---|
| NPDCCH Reference channel | - | R.26 HD-FDD | R.27 HD-FDD |
| LTE Carrier bandwidth | MHz | 10 | 10 |
| Allocated resource blocks Note1 | PRB | 1 | 1 |
| Number of transmitter antennas | - | 1 | 2 |
| DCI Format | - | N1 | N1 |
| Aggregation level | NCCE | 2 | 2 |
| Maximum number of repetitions | - | Note 2 | Note 2 |
| NPDCCH start symbol | symbols | 3 | 3 |
| Payload (without CRC) | Bits | Note 3 | Note 3 |
| Cell ID | - | Note 4 | Note 4 |
| Note 1: Allocation is located in the middle of narrowband.<br>Note 2: Maximum number of repetitions shall depend upon the test configuration.<br>Note 3: Payload size shall depend upon the test configuration.<br>Note 4: Cell ID shall depend upon the test configuration. | | | |

## A.10.1.2 HD-FDD standalone operation

**Table A.10.1.2-1: NPDCCH Reference Channel for UE category NB1 for standalone operation**

| Parameter | Unit | Value | |
|---|---|---|---|
| NPDCCH Reference channel | - | R.30 HD-FDD | R.31 HD-FDD |
| Channel bandwidth | KHz | 200 | 200 |
| Number of transmitter antennas | - | 1 | 2 |
| DCI Format | - | N1 | N1 |
| Aggregation level | NCCE | 2 | 2 |
| Maximum number of repetitions | - | Note 2 | Note 2 |
| Payload (without CRC) | Bits | Note 3 | Note 3 |
| Cell ID | - | Note 4 | Note 4 |
| Note 1: Allocation is located in the middle of narrowband.<br>Note 2: Maximum number of repetitions shall depend upon the test configuration.<br>Note 3: Payload size shall depend upon the test configuration.<br>Note 4: Cell ID shall depend upon the test configuration. | | | |

## A.10.1.3 HD-FDD guard band operation

**Table A.10.1.3-1: NPDCCH Reference Channel for UE category NB1 for guard band operation**

| Parameter | Unit | Value | |
|---|---|---|---|
| NPDCCH Reference channel | - | R.34 HD-FDD | R.35 HD-FDD |
| LTE Carrier bandwidth | MHz | 10 | 10 |
| Allocated resource blocks for NB-IoT Note1 | PRB | 1 | 1 |
| Number of transmitter antennas | - | 1 | 2 |
| DCI Format | - | N1 | N1 |
| Aggregation level | NCCE | 2 | 2 |
| Maximum number of repetitions | - | Note 2 | Note 2 |
| Payload (without CRC) | Bits | Note 3 | Note 3 |
| Cell ID | - | Note 4 | Note 4 |
| Note 1: Allocation is located in the middle of narrowband.<br>Note 2: Maximum number of repetitions shall depend upon the test configuration.<br>Note 3: Payload size shall depend upon the test configuration.<br>Note 4: Cell ID shall depend upon the test configuration. | | | |

# A.10.2 NPDSCH Reference Channel for UE category NB1

*Editor's note: The following aspects are either missing or not yet determined:*

- *Note 2 of all NPDSCH tables are still FFS and need clarification by RAN4.*
- *Note 1 of all NPDSCH tables not applicable for NB-IoT and need clarification by RAN4.*

## A.10.2.1 HD-FDD in-band operation

**Table A.10.2.1-1: NPDSCH Reference Channel for UE category NB1 for in-band operation**

| Parameter | Unit | Value | |
|---|---|---|---|
| NPDSCH Reference channel | - | R.14 HD-FDD | R.15 HD-FDD |
| LTE Carrier bandwidth | MHz | 10 | 10 |
| Allocated resource blocks Note1 | PRB | 1 | 1 |
| Number of transmitter antennas | - | 1 | 2 |
| Allocated subframes per Radio Frame | subframes | Note 2 | Note 2 |
| Modulation | | QPSK | QPSK |
| Target coding rate | | 1/3 | 1/3 |
| Information Bit Payload | | | |
| For Sub-Frame 1, 2, 3, 6, 7, 8 | Bits | 40 | 40 |
| For Sub-Frame 4, 9 | Bits | Note 3 | Note 3 |
| For Sub-Frame 0, 5 | Bits | 0 | 0 |
| Number of Code Blocks per Sub-Frame | | | |
| For Sub-Frame 1, 2, 3, 6, 7, 8 | | 1 | 1 |
| For Sub-Frame 4, 9 | | Note 4 | Note 4 |
| For Sub-Frame 0, 5 | | 0 | 0 |
| Maximum number of repetitions | - | Note 5 | Note 5 |
| NPDCCH start symbol | symbols | 3 | 3 |
| Cell ID | - | Note 6 | Note 6 |

Note 1: Allocation is located in the middle of narrowband.
Note 2: 6 when $n_f$ mod 2 = 0, otherwise 8. $n_f$ is specified in TS36.211 [16].
Note 3: 0 when $n_f$ mod 2 = 0, otherwise 40.
Note 4: 0 when $n_f$ mod 2 = 0, otherwise 1.
Note 5: Maximum number of repetitions shall depend upon the test configuration.
Note 6: Cell ID shall depend upon the test configuration.

## A.10.2.2 HD-FDD standalone operation

**Table A.10.2.2-1: NPDSCH Reference Channel for UE category NB1 for standalone operation**

| Parameter | Unit | Value | |
|---|---|---|---|
| NPDSCH Reference channel | - | R.18 HD-FDD | R.19 HD-FDD |
| Channel bandwidth | KHz | 200 | 200 |
| Number of transmitter antennas | - | 1 | 2 |
| Allocated subframes per Radio Frame | subframes | Note 2 | Note 2 |
| Modulation | | QPSK | QPSK |
| Target coding rate | | 1/3 | 1/3 |
| Information Bit Payload | | | |
| For Sub-Frame 1, 2, 3, 6, 7, 8 | Bits | 72 | 72 |
| For Sub-Frame 4, 9 | Bits | Note 3 | Note 3 |
| For Sub-Frame 0, 5 | Bits | 0 | 0 |
| Number of Code Blocks per Sub-Frame | | | |
| For Sub-Frame 1, 2, 3, 6, 7, 8 | | 1 | 1 |
| For Sub-Frame 4, 9 | | Note 4 | Note 4 |
| For Sub-Frame 0, 5 | | 0 | 0 |
| Maximum number of repetitions | - | Note 5 | Note 5 |
| Cell ID | - | Note 6 | Note 6 |
| Note 1: Allocation is located in the middle of narrowband. | | | |
| Note 2: 6 when $n_f$ mod 2 = 0, otherwise 8. $n_f$ is specified in TS36.211 [16]. | | | |
| Note 3: 0 when $n_f$ mod 2 = 0, otherwise 72. | | | |
| Note 4: 0 when $n_f$ mod 2 = 0, otherwise 1. | | | |
| Note 5: Maximum number of repetitions shall depend upon the test configuration. | | | |
| Note 6: Cell ID shall depend upon the test configuration. | | | |

### A.10.2.3 HD-FDD guard band operation

**Table A.10.2.3-1: NPDSCH Reference Channel for UE category NB1 for guard band operation**

| Parameter | Unit | Value | |
|---|---|---|---|
| NPDSCH Reference channel | - | R.22 HD-FDD | R.23 HD-FDD |
| LTE Carrier bandwidth | MHz | 10 | 10 |
| Allocated resource blocks for NB-IoT Note1 | PRB | 1 | 1 |
| Number of transmitter antennas | - | 1 | 2 |
| Allocated subframes per Radio Frame | subframes | Note 2 | Note 2 |
| Modulation | | QPSK | QPSK |
| Target coding rate | | 1/3 | 1/3 |
| Information Bit Payload | | | |
| For Sub-Frame 1, 2, 3, 6, 7, 8 | Bits | 72 | 72 |
| For Sub-Frame 4, 9 | Bits | Note 3 | Note 3 |
| For Sub-Frame 0, 5 | Bits | 0 | 0 |
| Number of Code Blocks per Sub-Frame | | | |
| For Sub-Frame 1, 2, 3, 6, 7, 8 | | 1 | 1 |
| For Sub-Frame 4, 9 | | Note 4 | Note 4 |
| For Sub-Frame 0, 5 | | 0 | 0 |
| Maximum number of repetitions | - | Note 5 | Note 5 |
| Cell ID | - | Note 6 | Note 6 |
| Note 1: Allocation is located in the middle of narrowband. | | | |
| Note 2: 6 when $n_f$ mod 2 = 0, otherwise 8. $n_f$ is specified in TS36.211 [16]. | | | |
| Note 3: 0 when $n_f$ mod 2 = 0, otherwise 72. | | | |
| Note 4: 0 when $n_f$ mod 2 = 0, otherwise 1. | | | |
| Note 5: Maximum number of repetitions shall depend upon the test configuration. | | | |
| Note 6: Cell ID shall depend upon the test configuration. | | | |

## A.10.3 Reference NPRACH Configurations

Table A.10.3-1 defines the reference NB-IoT PRACH configurations for a NB-IoT RRM test case where the UE is required to transmit NPRACH during the testing procedure, but the testing purpose of the RRM test case does not include testing NPRACH performance.

**Table A.10.3-1: NPRACH.R-1: Reference NPRACH Configuration**

| Field | Value | | | Comment |
|---|---|---|---|---|
| **Parameters not per NPRACH coverage level** | | | | |
| rsrp-ThresholdsPrach | {rsrp1, rsrp2} | | | The values of NPRACH RSRP thresholds for will be set according the requirement of individual test cases |
| nprach-CP-Length | us66dot7 | | | NPRACH format 0 |
| **Parameters per NPRACH coverage Level** | | | | |
| ***CE Level*** | ***Level 0*** | ***Level 1*** | ***Level 2*** | **Valid values as defined in TS 36.331** |
| nprach-Periodicity | ms40 | ms40 | ms40 | {ms40, ms80, ms160, ms240, ms320, ms640, ms1280, ms2560} |
| nprach-StartTime | ms8 | ms8 | ms8 | {ms8, ms16, ms32, ms64, ms128, ms256, ms512, ms1024} |
| nprach-SubcarrierOffset | n0 | n0 | n0 | {n0, n12, n24, n36, n2, n18, n34} |
| nprach-NumSubcarriers | n12 | n12 | n12 | {n12, n24, n36, n48} |
| nprach-SubcarrierMSG3-RangeStart | {one} | {one} | {one} | {zero, oneThird, twoThird, one} |
| maxNumPreambleAttemptCE | n3 | n5 | n7 | {n3, n4, n5, n6, n7, n8, n10} |
| numRepetitionsPerPreambleAttempt | n1 | n8 | n32 | {n1, n2, n4, n8, n16, n32, n64, n128} |
| npdcch-NumRepetitions-RA | r1 | r8 | n32 | {r1, r2, r4, r8, r16, r32, r64, r128, r256, r512, r1024, r2048} |
| npdcch-StartSF-CSS-RA | v2 | v2 | v2 | {v1dot5, v2, v4, v8, v16, v32, v48, v64} |
| npdcch-Offset-RA | zero | zero | zero | {zero, oneEighth, oneFourth, threeEighth} |
| Note 1: See Clause 6.3.2 in TS 36.331 for further information on the parameters in this table. | | | | |

# Annex B (normative): Propagation Conditions

## B.0 No interference

See TS 36.521-1[10] Annex B.0.

## B.1 Static propagation condition

See TS 36.521-1[10] Annex B.1 and B.1.1

## B.2 Multi-path fading Propagation Conditions

See TS 36.521-1[10] Annex B.2,B.2.1 and B.2.2

# Annex C (normative): Downlink Physical Channels

## C.0 Downlink signal

See TS 36.521-1[10] Annex C.0.

## C.1 General

See TS 36.521-1[10] Annex C.1.

For CA tests with 1 UL, no PHICH resources are assigned to the DUT in the SCell. In such case PHICH of SCell is entirely part of OCNG pattern used.

## C.2 Set-up

See TS 36.521-1[10] Annex C.2.

## C.3 Test specific scenarios

### C.3.1 ABS Transmission Configurations

#### C.3.1.1 Non-MBSFN ABS Transmission Configurations

##### C.3.1.1.1 Non-MBSFN ABS Transmission, 1x2 antenna with PBCH

**Table C.3.1.1.1-1: Transmission configuration with non-MBSFN ABS, 1x2 with PBCH**

| Physical Channels and Signals | Parameters | EPRE, [dB] | |
|---|---|---|---|
| | | Non-ABS | ABS |
| PBCH | PBCH_RA | 0 | 0 |
| | PBCH_RB | 0 | 0 |
| PSS | PSS_RA | 0 | 0 |
| SSS | SSS_RA | 0 | 0 |
| PCFICH | PCFICH_RB | 0 | 0 (Note 1) |
| PHICH | PHICH_RA | 0 | -Inf |
| | PHICH_RB | 0 | -Inf |
| PDCCH | PDCCH_RA | 0 | 0 (Note 1) |
| | PDCCH_RB | 0 | 0 (Note 1) |
| PDSCH | PDSCH_RA | 0 | 0 (Note 1) |
| | PDSCH_RB | 0 | 0 (Note 1) |
| OCNG | OCNG_RA | 0 | -Inf |
| | OCNG_RB | 0 | -Inf |
| NOTE 1: Only used for SIB1, otherwise EPRE is –Inf<br>NOTE 2: 1x2 antenna configuration is assumed | | | |

### C.3.1.1.2 Non-MBSFN ABS Transmission, 2x2 antenna without PBCH

**Table C.3.1.1.2-1: Transmission configuration #1 with non-MBSFN ABS, 2x2 without PBCH**

| Physical Channels and Signals | Parameters | EPRE [dB] | |
|---|---|---|---|
| | | Non-ABS | ABS |
| PBCH | PBCH_RA | -3 | -Inf |
| | PBCH_RB | -3 | -Inf |
| PSS | PSS_RA | -3 | -3 |
| SSS | SSS_RA | -3 | -3 |
| PCFICH | PCFICH_RB | 1 | -Inf |
| PHICH | PHICH_RA | -3 | -Inf |
| | PHICH_RB | -3 | -Inf |
| PDCCH | PDCCH_RA | 1 | -Inf |
| | PDCCH_RB | 1 | -Inf |
| PDSCH | PDSCH_RA | -3 | -Inf |
| | PDSCH_RB | -3 | -Inf |
| OCNG | OCNG_RA | -3 | -Inf |
| | OCNG_RB | -3 | -Inf |
| NOTE: 2x2 antenna configuration is assumed | | | |

**Table C.3.1.1.2-2: Transmission configuration #2 with non-MBSFN ABS, 2x2 without PBCH**

| Physical Channels and Signals | Parameters | EPRE [dB] | |
|---|---|---|---|
| | | Non-ABS | ABS |
| PBCH | PBCH_RA | -3 | -Inf |
| | PBCH_RB | -3 | -Inf |
| PSS | PSS_RA | -3 | -3 |
| SSS | SSS_RA | -3 | -3 |
| PCFICH | PCFICH_RB | 1 | -Inf |
| PHICH | PHICH_RA | -3 | -Inf |
| | PHICH_RB | -3 | -Inf |
| PDCCH | PDCCH_RA | -3 | -Inf |
| | PDCCH_RB | -3 | -Inf |
| PDSCH | PDSCH_RA | -3 | -Inf |
| | PDSCH_RB | -3 | -Inf |
| OCNG | OCNG_RA | -3 | -Inf |
| | OCNG_RB | -3 | -Inf |
| NOTE: 2x2 antenna configuration is assumed | | | |

## C.3.1.2 MBSFN ABS Transmission Configurations

### C.3.1.2.1 MBSFN ABS Transmission, 1x2 antenna

**Table C.3.1.2.1-1: Transmission configuration with MBSFN ABS, 1x2**

| Physical Channels and Signals | Parameters | EPRE, [dB] | |
|---|---|---|---|
| | | Non-ABS | ABS |
| PBCH | PBCH_RA | 0 | N/A |
| | PBCH_RB | 0 | N/A |
| PSS | PSS_RA | 0 | N/A |
| SSS | SSS_RA | 0 | N/A |
| PCFICH | PCFICH_RB | 0 | -Inf |
| PHICH | PHICH_RA | 0 | -Inf |
| | PHICH_RB | 0 | -Inf |
| PDCCH | PDCCH_RA | 0 | -Inf |
| | PDCCH_RB | 0 | -Inf |
| PDSCH | PDSCH_RA | 0 | -Inf |
| | PDSCH_RB | 0 | -Inf |
| PMCH | PMCH_RA | 0 | -Inf |
| | PMCH_RB | 0 | -Inf |
| OCNG | OCNG_RA | 0 | -Inf |
| | OCNG_RB | 0 | -Inf |
| NOTE: 1x2 antenna configuration is assumed | | | |

### C.3.1.2.2 MBSFN ABS Transmission, 2x2 antenna

**Table C.3.1.2.2-1: Transmission configuration #1 with MBSFN ABS, 2x2**

| Physical Channels and Signals | Parameters | EPRE, [dB] | |
|---|---|---|---|
| | | Non-ABS | ABS |
| PBCH | PBCH_RA | -3 | N/A |
| | PBCH_RB | -3 | N/A |
| PSS | PSS_RA | -3 | N/A |
| SSS | SSS_RA | -3 | N/A |
| PCFICH | PCFICH_RB | 1 | -Inf |
| PHICH | PHICH_RA | -3 | -Inf |
| | PHICH_RB | -3 | -Inf |
| PDCCH | PDCCH_RA | 1 | -Inf |
| | PDCCH_RB | 1 | -Inf |
| PDSCH | PDSCH_RA | -3 | -Inf |
| | PDSCH_RB | -3 | -Inf |
| PMCH | PMCH_RA | -3 | -Inf |
| | PMCH_RB | -3 | -Inf |
| OCNG | OCNG_RA | -3 | -Inf |
| | OCNG_RB | -3 | -Inf |
| NOTE: 2x2 antenna configuration is assumed | | | |

**Table C.3.1.2.2-2: Transmission configuration # 2 with MBSFN ABS, 2x2**

| Physical Channels and Signals | Parameters | EPRE, [dB] | |
|---|---|---|---|
| | | Non-ABS | ABS |
| PBCH | PBCH_RA | -3 | N/A |
| | PBCH_RB | -3 | N/A |
| PSS | PSS_RA | -3 | N/A |
| SSS | SSS_RA | -3 | N/A |
| PCFICH | PCFICH_RB | 1 | -Inf |
| PHICH | PHICH_RA | -3 | -Inf |
| | PHICH_RB | -3 | -Inf |
| PDCCH | PDCCH_RA | -3 | -Inf |
| | PDCCH_RB | -3 | -Inf |
| PDSCH | PDSCH_RA | -3 | -Inf |
| | PDSCH_RB | -3 | -Inf |
| PMCH | PMCH_RA | -3 | -Inf |
| | PMCH_RB | -3 | -Inf |
| OCNG | OCNG_RA | -3 | -Inf |
| | OCNG_RB | -3 | -Inf |
| NOTE: 2x2 antenna configuration is assumed | | | |

# C.3.2 Impact of Reference Sensitivity Degradation with Carrier Aggregation on Test Cases

## C.3.2.1 Impact of Reference Sensitivity Degradation due to Insertion Loss

For a UE supporting inter-band carrier aggregation configuration with uplink in one E-UTRA band, if there is a relaxation of receiver sensitivity $\Delta R_{IB,c}$>0 dB as defined in TS 36.101 [2], 7.3.1-1A, there is no adjustment of test parameters in the tests specified in TS 36.133 when $\Delta R_{IB,c} \leq 1$ dB.

# C.3.3 Carrier Aggregation Test Cases with Different Channel Bandwidth Combinations

## C.3.3.1 Introduction

In clauses 8 and 9 carrier aggregation test cases may be defined with different channel bandwidth combinations to verify the same RRM requirement.

If multiple carrier aggregation test cases with different channel bandwidth combinations are defined to verify the same RRM requirement that is channel bandwidth independent, then the UE needs to be tested only with one bandwidth combination out of the bandwidth combination sets supported by that UE.

# C.3.4 Proximity-based Services

## C.3.4.1 Introduction

This clause also defines the principle and the reference configurations that are applicable to test cases verifying RRM core requirements for ProSe Direct Discovery and ProSe Direct Communication.

## C.3.4.2 Reference DRX configurations for ProSe tests

**Table C.3.4.2-1: Reference DRX Configurations**

| Parameter | Value |
|---|---|
| Reference configuration | DRX_P1 |
| onDurationTimer | psf1 |
| drx-InactivityTimer | psf1 |
| drx-RetransmissionTimer | psf1 |
| longDRX-CycleStartOffset | sf320, 0 |
| shortDRX | Disabled |
| Note: For further information see clause 6.3.2 in TS 36.331. | |

## C.3.4.3 Test Cases with Different Channel Bandwidths

### C.3.4.3.1 Introduction

This clause defines a principle which is applicable to test cases verifying ProSe RRM requirements with different channel bandwidths.

### C.3.4.3.2 Principle of testing

Some ProSe test cases are defined for different channel bandwidths to verify the same RRM requirement.

If test cases with different channel bandwidth are defined to verify the same RRM requirement then the UE is required to pass the test cases only with one of the channel bandwidths.

# Annex D (normative): OFDMA Channel Noise Generator (OCNG)

## D.1 OCNG Patterns for FDD

The following OCNG patterns are used for modelling allocations to virtual UEs (which are not under test) and/or allocations used for MBSFN. The OCNG pattern for each sub frame specifies the allocations that shall be filled with OCNG, and furthermore, the relative power level of each such allocation.

In each test case the OCNG is expressed by parameters OCNG_RA and OCNG_RB which together with a relative power level ( $\gamma$ ) specifies the PDSCH EPRE-to-RS EPRE ratios in OFDM symbols with and without reference symbols, respectively. The relative power, which is used for modelling boosting per virtual UE allocation, is expressed by:

$$\gamma_i = PDSCH_i_RA / OCNG_RA = PDSCH_i_RB / OCNG_RB,$$

where $\gamma_i$ denotes the relative power level of the *i:th* virtual UE. The parameter settings of OCNG_RA, OCNG_RB, and the set of relative power levels $\gamma$ are chosen such that when also taking allocations to the UE under test into account, as given by a PDSCH reference channel, a constant transmitted power spectral density that is constant on an OFDM symbol basis is targeted.

Moreover the OCNG pattern is accompanied by a PCFICH/PDCCH/PHICH reference channel which specifies the control region. The number of PDCCH OFDM symbols in the non-MBSFN subframes is the same as specified in the RMC used in the test. The number of PDCCH OFDM symbols in the MBSFN subframes is the maximal allowed according to 3GPP TS 36.213 [8]. For any aggregation and PHICH allocation, the PDCCH and any unused PHICH groups are padded with resource element groups with a power level given respectively by PDCCH_RA/RB and PHICH_RA/RB as specified in the test case such that a total power spectral density in the control region that is constant on an OFDM symbol basis is targeted.

For subframes configured as PRS subframes the PDSCH allocation defined in the OCNG pattern does not apply.

For subframes configured as ABS subframes the PDSCH and PMCH allocation defined in the OCNG pattern does not apply.

The system information is scheduled in the allocations reserved for the OCNG patterns, in the subframes not configured for MBSFN. For this purpose the number of the RB-s allocated with PDSCH defined in the OCNG pattern can be reduced as necessary.

## D.1.1 OCNG FDD pattern 1: outer resource blocks allocation in 10 MHz

**Table D.1.1-1: OP.1 FDD: OCNG FDD Pattern 1**

| **Allocation** $n_{PRB}$ | **Relative power level** $\gamma_{PRB}$ **[dB]** | | | | **PDSCH Data** | **PMCH Data** |
|---|---|---|---|---|---|---|
| | **Subframe** | | | | | |
| | 0 | 5 | 4,9 | 1-3, 6-8 | | |
| 0 - 12 | 0 | 0 | 0 | N/A | Note 1 | N/A |
| 37 - 49 | 0 | 0 | 0 | N/A | | |
| 0-49 | N/A | N/A | N/A | Note 4 | N/A | Note 2 |

Note 1: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 2: Each physical resource block (PRB) is assigned to MBSFN transmission. The data in each PRB shall be uncorrelated with data in other PRBs over the period of any measurement. The MBSFN data shall be QPSK modulated. PMCH symbols shall not contain cell-specific Reference Signals. PMCH subframes shall contain cell-specific Reference Signals only in the first symbol of the first time slot. The parameter $\gamma_{PRB}$ is used to scale the power of PMCH.

Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

Note 4: 0 dB for 1 transmit antenna with CRS, +3 dB for 2 transmit antennas with CRS.

N/A: Not Applicable

## D.1.2 OCNG FDD pattern 2: full bandwidth allocation in 10 MHz

**Table D.1.2-1: OP.2 FDD: OCNG FDD Pattern 2**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data | PMCH Data |
|---|---|---|---|---|---|---|
| | Subframe | | | | | |
| | 0 | 5 | 4, 9 | 1 - 3, 6 - 8 | | |
| 0 - 49 | 0 | 0 | 0 | N/A | Note 1 | N/A |
| 0 - 49 | N/A | N/A | N/A | Note 4 | N/A | Note 2 |

Note 1: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 2: Each physical resource block (PRB) is assigned to MBSFN transmission. The data in each PRB shall be uncorrelated with data in other PRBs over the period of any measurement. The MBSFN data shall be QPSK modulated. PMCH symbols shall not contain cell-specific Reference Signals. PMCH subframes shall contain cell-specific Reference Signals only in the first symbol of the first time slot. The parameter $\gamma_{PRB}$ is used to scale the power of PMCH.

Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

Note 4: 0 dB for 1 transmit antenna with CRS, +3 dB for 2 transmit antennas with CRS.

N/A: Not Applicable

## D.1.3 OCNG FDD pattern 3: outer resource blocks allocation in 1.4 MHz

**Table A.3.2.1.3-1: OP.3 FDD: OCNG FDD Pattern 3**

| **Allocation** $n_{PRB}$ | **Relative power level** $\gamma_{PRB}$ **[dB]** | | | | **PDSCH Data** | **PMCH Data** |
|---|---|---|---|---|---|---|
| | **Subframe** | | | | | |
| | 0 | 5 | 4,9 | 1-3, 6-8 | | |
| 0 - 1 | 0 | 0 | 0 | N/A | Note 1 | N/A |
| 4 - 5 | 0 | 0 | 0 | N/A | | |
| 0 - 5 | N/A | N/A | N/A | Note 4 | N/A | Note 2 |

Note 1: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 2: Each physical resource block (PRB) is assigned to MBSFN transmission. The data in each PRB shall be uncorrelated with data in other PRBs over the period of any measurement. The MBSFN data shall be QPSK modulated. PMCH symbols shall not contain cell-specific Reference Signals. PMCH subframes shall contain cell-specific Reference Signals only in the first symbol of the first time slot. The parameter $\gamma_{PRB}$ is used to scale the power of PMCH.

Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

Note 4: 0 dB for 1 transmit antenna with CRS, +3 dB for 2 transmit antennas with CRS.

N/A: Not Applicable

## D.1.4 OCNG FDD pattern 4: full bandwidth allocation in 1.4 MHz

**Table A.3.2.1.4-1: OP.4 FDD: OCNG FDD Pattern 4**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data | PMCH Data |
|---|---|---|---|---|---|---|
| | Subframe | | | | | |
| | 0 | 5 | 4, 9 | 1 - 3, 6 - 8 | | |
| 0 - 5 | 0 | 0 | 0 | N/A | Note 1 | N/A |
| 0 - 5 | N/A | N/A | N/A | Note 4 | N/A | Note 2 |

Note 1: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 2: Each physical resource block (PRB) is assigned to MBSFN transmission. The data in each PRB shall be uncorrelated with data in other PRBs over the period of any measurement. The MBSFN data shall be QPSK modulated. PMCH subframes shall contain cell-specific Reference Signals only in the first symbol of the first time slot. The parameter $\gamma_{PRB}$ is used to scale the power of PMCH.

Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

Note 4: 0 dB for 1 transmit antenna with CRS, +3 dB for 2 transmit antennas with CRS.

N/A: Not Applicable

## D.1.5 OCNG FDD pattern 5: outer resource blocks allocation in 10 MHz (without MBSFN)

**Table D.1.5-1: OP.5 FDD: OCNG FDD Pattern 5**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data |
|---|---|---|---|---|---|
| | Subframe (Note 1) | | | | |
| | 0 | 5 | 4, 9 | 1 - 3, 6 - 8 | |
| 0 - 12 | 0 | 0 | 0 | N/A | Note 2 |
| 37 - 49 | 0 | 0 | 0 | N/A | |
| 0 - 49 | N/A | N/A | N/A | 0 | |

Note 1: The allocation of any PDSCH with or without SIB1 applies only to the subframes not configured as PRS subframes.

Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

N/A: Not Applicable

## D.1.6 OCNG FDD pattern 6: full bandwidth allocation in 10 MHz (without MBSFN)

**Table D.1.6-1: OP.6 FDD: OCNG FDD Pattern 6**

| **Allocation** $n_{PRB}$ | **Relative power level** $\gamma_{PRB}$ **[dB]** | | | | **PDSCH Data** |
|---|---|---|---|---|---|
| | **Subframe** (Note 1) | | | | |
| | 0 | 5 | 4, 9 | 1 - 3, 6 - 8 | |
| 0 - 49 | 0 | 0 | 0 | 0 | Note 2 |

Note 1: The allocation of any PDSCH with or without SIB1 applies only to the subframes not configured as PRS subframes.

Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated.

The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

N/A: Not Applicable

## D.1.7 OCNG FDD pattern 7: full bandwidth allocation in 1.4 MHz (without MBSFN)

**Table D.1.7-1: OP.7 FDD: OCNG FDD Pattern 7**

| **Allocation** $n_{PRB}$ | **Relative power level** $\gamma_{PRB}$ **[dB]** | | | | **PDSCH Data** |
|---|---|---|---|---|---|
| | **Subframe** (Note 1) | | | | |
| | 0 | 5 | 4, 9 | 1 - 3, 6 - 8 | |
| 0 - 5 | 0 | 0 | 0 | 0 | Note 2 |

Note 1: The allocation of any PDSCH with or without SIB1 applies only to the subframes not configured as PRS subframes.

Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated.

The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

N/A: Not Applicable

## D.1.8 OCNG FDD pattern 8: outer resource blocks allocation in 10 MHz for MBSFN ABS

**Table D.1.8-1: OP.8 FDD: OCNG FDD Pattern 8**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data |
|---|---|---|---|---|---|
| | Subframe (Note 1) | | | | |
| | 0 | 5 | 4,9 | (1-3, 6-8)Note4 | |
| 0 – 12 | 0 | 0 | 0 | N/A | Note 2 |
| 37 – 49 | 0 | 0 | 0 | N/A | |
| 0 – 49 | N/A | N/A | N/A | 0 | |

Note 1: The allocation of any PDSCH with or without SIB1 applies only to the subframes not configured as PRS subframes.
Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.
Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in clause 7.1 in TS 36.213.
Note 4: The subframe(s) configured as MBSFN ABS in a test shall not contain any PMCH data and shall contain CRS only in the first symbol of the first time slot. The subframe(s) configured as MBSFN ABS depend upon the MBSFN ABS pattern used in the test.
N/A: Not Applicable

## D.1.9 OCNG FDD pattern 9: full bandwidth allocation in 10 MHz for MBSFN ABS

**Table D.1.9-1: OP.9 FDD: OCNG FDD Pattern 9**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data |
|---|---|---|---|---|---|
| | Subframe (Note 1) | | | | |
| | 0 | 5 | 4, 9 | (1-3, 6-8)Note4 | |
| 0 – 49 | 0 | 0 | 0 | 0 | Note 2 |

Note 1: The allocation of any PDSCH with or without SIB1 applies only to the subframes not configured as PRS subframes.
Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.
Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in clause 7.1 in TS 36.213.
Note 4: The subframe(s) configured as MBSFN ABS in a test shall not contain any PMCH data and shall contain CRS only in the first symbol of the first time slot. The subframe(s) configured as MBSFN ABS depend upon the MBSFN ABS pattern used in the test.
N/A: Not Applicable

## D.1.10 OCNG FDD pattern 10: outer resource blocks allocation in 10 MHz with user data in every SF (without MBSFN)

**Table D.1.10-1: OP.10 FDD: OCNG FDD Pattern 10**

| **Allocation** $n_{PRB}$ | **Relative power level** $\gamma_{PRB}$ **[dB]** | | | | **PDSCH Data** |
|---|---|---|---|---|---|
| | **Subframe** (Note 1) | | | | |
| | 0 | 5 | 4, 9 | 1 - 3, 6 - 8 | |
| 0 - 12 | 0 | 0 | 0 | 0 | Note 2 |
| 37 - 49 | 0 | 0 | 0 | 0 | |

Note 1: The allocation of any PDSCH with or without SIB1 applies only to the subframes not configured as PRS subframes.

Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated.

The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

N/A: Not Applicable

## D.1.11 OCNG FDD pattern 11: outer resource blocks allocation in 20 MHz

**Table D.1.11-1: OP.11 FDD: OCNG FDD Pattern 11**

| **Allocation** $n_{PRB}$ | **Relative power level** $\gamma_{PRB}$ **[dB]** | | | | **PDSCH Data** | **PMCH Data** |
|---|---|---|---|---|---|---|
| | **Subframe** | | | | | |
| | 0 | 5 | 4,9 | 1-3, 6-8 | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| 0 – 37 | 0 | 0 | 0 | N/A | Note 1 | N/A |
| 62 – 99 | 0 | 0 | 0 | N/A | | |
| 0-99 | N/A | N/A | N/A | Note 4 | N/A | Note 2 |

Note 1: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 2: Each physical resource block (PRB) is assigned to MBSFN transmission. The data in each PRB shall be uncorrelated with data in other PRBs over the period of any measurement. The MBSFN data shall be QPSK modulated. PMCH symbols shall not contain cell-specific Reference Signals. PMCH subframes shall contain cell-specific Reference Signals only in the first symbol of the first time slot. The parameter $\gamma_{PRB}$ is used to scale the power of PMCH.

Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

Note 4: 0dB for 1 transmit antenna with CRS, +3dB for 2 transmit antennas with CRS.

N/A: Not Applicable

# D.1.12 OCNG FDD pattern 12: full bandwidth allocation in 20 MHz

**Table D.1.12-1: OP.12 FDD: OCNG FDD Pattern 12**

| **Allocation** $n_{PRB}$ | **Relative power level** $\gamma_{PRB}$ **[dB]** | | | | **PDSCH Data** | **PMCH Data** |
|---|---|---|---|---|---|---|
| | **Subframe** | | | | | |
| | 0 | 5 | 4, 9 | 1 – 3, 6 – 8 | | |
| 0 – 99 | 0 | 0 | 0 | N/A | Note 1 | N/A |
| 0 – 99 | N/A | N/A | N/A | Note 4 | N/A | Note 2 |

Note 1: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 2: Each physical resource block (PRB) is assigned to MBSFN transmission. The data in each PRB shall be uncorrelated with data in other PRBs over the period of any measurement. The MBSFN data shall be QPSK modulated. PMCH subframes shall contain cell-specific Reference Signals only in the first symbol of the first time slot. The parameter $\gamma_{PRB}$ is used to scale the power of PMCH.

Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

Note 4: 0dB for 1 transmit antenna with CRS, +3dB for 2 transmit antennas with CRS.

N/A: Not Applicable

# D.1.13 OCNG FDD pattern 13: outer resource blocks allocation in 20 MHz (without MBSFN)

**Table D.1.13-1: OP.13 FDD: OCNG FDD Pattern 13**

| **Allocation** $n_{PRB}$ | **Relative power level** $\gamma_{PRB}$ **[dB]** | | | | **PDSCH Data** |
|---|---|---|---|---|---|
| | **Subframe (Note 1)** | | | | |
| | 0 | 5 | 4,9 | 1-3, 6-8 | |
| 0 – 37 | 0 | 0 | 0 | N/A | Note 2 |
| 62 – 99 | 0 | 0 | 0 | N/A | |
| 0 – 99 | N/A | N/A | N/A | 0 | |

Note 1: The allocation of any PDSCH with or without SIB1 applies only to the subframes not configured as PRS subframes.

Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

N/A: Not Applicable

## D.1.14 OCNG FDD pattern 14: full bandwidth allocation in 20 MHz (without MBSFN)

**Table D.1.14-1: OP.14 FDD: OCNG FDD Pattern 14**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data |
|---|---|---|---|---|---|
| | Subframe (Note 1) | | | | |
| | 0 | 5 | 4, 9 | 1 – 3, 6 – 8 | |
| 0 – 99 | 0 | 0 | 0 | 0 | Note 2 |

Note 1: The allocation of any PDSCH with or without SIB1 applies only to the subframes not configured as PRS subframes.

Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

N/A: Not Applicable

## D.1.15 OCNG FDD pattern 15: outer resource blocks allocation in 5 MHz

**Table D.1.15-1: OP.15 FDD: OCNG FDD Pattern 15**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data | PMCH Data |
|---|---|---|---|---|---|---|
| | Subframe | | | | | |
| | 0 | 5 | 4,9 | 1-3, 6-8 | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| 0 – 6 | 0 | 0 | 0 | N/A | Note 1 | N/A |
| 18 – 24 | 0 | 0 | 0 | N/A | | |
| 0-24 | N/A | N/A | N/A | Note 4 | N/A | Note 2 |

Note 1: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 2: Each physical resource block (PRB) is assigned to MBSFN transmission. The data in each PRB shall be uncorrelated with data in other PRBs over the period of any measurement. The MBSFN data shall be QPSK modulated. PMCH symbols shall not contain cell-specific Reference Signals. PMCH subframes shall contain cell-specific Reference Signals only in the first symbol of the first time slot. The parameter $\gamma_{PRB}$ is used to scale the power of PMCH.

Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

Note 4: 0dB for 1 transmit antenna with CRS, +3dB for 2 transmit antennas with CRS

N/A: Not Applicable

# D.1.16 OCNG FDD pattern 16: full bandwidth allocation in 5 MHz

**Table D.1.16-1: OP.16 FDD: OCNG FDD Pattern 16**

| **Allocation** $n_{PRB}$ | **Relative power level** $\gamma_{PRB}$ **[dB]** | | | | **PDSCH Data** | **PMCH Data** |
|---|---|---|---|---|---|---|
| | **Subframe** | | | | | |
| | 0 | 5 | 4, 9 | 1 – 3, 6 – 8 | | |
| 0 – 24 | 0 | 0 | 0 | N/A | Note 1 | N/A |
| 0 – 24 | N/A | N/A | N/A | Note 4 | N/A | Note 2 |

Note 1: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 2: Each physical resource block (PRB) is assigned to MBSFN transmission. The data in each PRB shall be uncorrelated with data in other PRBs over the period of any measurement. The MBSFN data shall be QPSK modulated. PMCH subframes shall contain cell-specific Reference Signals only in the first symbol of the first time slot. The parameter $\gamma_{PRB}$ is used to scale the power of PMCH.

Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

Note 4: 0dB for 1 transmit antenna with CRS, +3dB for 2 transmit antennas with CRS

N/A: Not Applicable

## D.1.17 OCNG FDD pattern 17: outer resource blocks allocation in 20 MHz with user data in every subframe (without MBSFN)

**Table D.1.17-1: OP.17 FDD: OCNG FDD Pattern 17**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data |
|---|---|---|---|---|---|
| | Subframe (Note 1) | | | | |
| | 0 | 5 | 4, 9 | 1 - 3, 6 - 8 | |
| 0 - 37 | 0 | 0 | 0 | 0 | Note 2 |
| 62 - 99 | 0 | 0 | 0 | 0 | |

Note 1: The allocation of any PDSCH with or without SIB1 applies only to the subframes not configured as PRS subframes.
Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.
Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.
N/A: Not Applicable.

## D.1.18 OCNG FDD pattern 18: outer resource blocks allocation in 5 MHz (without MBSFN)

**Table D.1.18-1: OP.18 FDD: OCNG FDD Pattern 18**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data |
|---|---|---|---|---|---|
| | Subframe (Note 1) | | | | |
| | 0 | 5 | 4,9 | 1-3, 6-8 | |
| 0 – 6 | 0 | 0 | 0 | N/A | Note 2 |
| 18 – 24 | 0 | 0 | 0 | N/A | |
| 0 – 24 | N/A | N/A | N/A | 0 | |

Note 1: PDSCH allocation applies only to subframes not configured as PRS subframes.
Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.
Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.
N/A: Not Applicable

## D.1.19 OCNG FDD pattern 19: full bandwidth allocation in 5 MHz (without MBSFN)

**Table D.1.19-1: OP.19 FDD: OCNG FDD Pattern 19**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data |
|---|---|---|---|---|---|
| | Subframe (Note 1) | | | | |
| | 0 | 5 | 4, 9 | 1 – 3, 6 – 8 | |
| 0 – 24 | 0 | 0 | 0 | 0 | Note 2 |

Note 1: PDSCH allocation applies only to subframes not configured as PRS subframes.
Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.
Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.
N/A: Not Applicable

## D.1.20 OCNG FDD pattern 20: outer resource blocks allocation in 5 MHz with user data in every subframe (without MBSFN)

**Table D.1.20-1: OP.20 FDD: OCNG FDD Pattern 20**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data |
|---|---|---|---|---|---|
| | Subframe (Note 1) | | | | |
| | 0 | 5 | 4, 9 | 1 - 3, 6 - 8 | |
| 0 - 6 | 0 | 0 | 0 | 0 | Note 2 |
| 18 - 24 | 0 | 0 | 0 | 0 | |

Note 1: PDSCH allocation applies only to subframes not configured as PRS subframes.
Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.
Note 3: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.
N/A: Not Applicable.

## D.1.21 OCNG FDD pattern 21: Generic resource blocks allocation (without MBSFN)

**Table D.1.21-1: OP.21 FDD: OCNG FDD Pattern 21**

| OCNG Pattern Name | Bandwidth (MHz) | Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data |
|---|---|---|---|---|---|---|---|
| | | | Subframe | | | | |
| | | | 0 | 5 | 4, 9 | 1 – 3, 6 – 8 | |
| OP.21 FDD | 10 | 0 – 49 Note 1,2 | 0 | 0 | 0 | 0 | Note 3 |

Note 1: The OCNG pattern is used only for a serving cell of the UE under test.

Note 2: The OCNG allocation applied to all downlink physical resource blocks (PRBs) except the REs that are allocated for the PDSCH for the UE under test.

Note 3: OCNG PRBs are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 4: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in clause 7.1 in TS 36.213.

# D.2 OCNG Patterns for TDD

The following OCNG patterns are used for modelling allocations to virtual UEs (which are not under test). The OCNG pattern for each sub frame specifies the allocations that shall be filled with OCNG, and furthermore, the relative power level of each such allocation.

In each test case the OCNG is expressed by parameters OCNG_RA and OCNG_RB which together with a relative power level ( $\gamma$ ) specifies the PDSCH EPRE-to-RS EPRE ratios in OFDM symbols with and without reference symbols, respectively. The relative power, which is used for modelling boosting per virtual UE allocation, is expressed by:

$$\gamma_i = PDSCH_i_RA / OCNG_RA = PDSCH_i_RB / OCNG_RB,$$

where $\gamma_i$ denotes the relative power level of the *i:th* virtual UE. The parameter settings of OCNG_RA, OCNG_RB, and the set of relative power levels $\gamma$ are chosen such that when also taking allocations to the UE under test into account, as given by a PDSCH reference channel, a transmitted power spectral density that is constant on an OFDM symbol basis is targeted.

Moreover the OCNG pattern is accompanied by a PCFICH/PDCCH/PHICH reference channel which specifies the control region. The number of PDCCH OFDM symbols in the non-MBSFN subframes is the same as specified in the RMC used in the test. The number of PDCCH OFDM symbols in the MBSFN subframes is the maximal allowed according to 3GPP TS 36.213 [8]. For any aggregation and PHICH allocation, the PDCCH and any unused PHICH groups are padded with resource element groups with a power level given respectively by PDCCH_RA/RB and PHICH_RA/RB as specified in the test case such that a total power spectral density in the control region that is constant on an OFDM symbol basis is targeted.

For subframes configured as PRS subframes the PDSCH allocation defined in the OCNG pattern does not apply.

For subframes configured as ABS subframes the PDSCH and PMCH allocation defined in the OCNG pattern does not apply.

The system information is scheduled in the allocations reserved for the OCNG patterns, in the subframes not configured for MBSFN. For this purpose the number of the RB-s allocated with PDSCH defined in the OCNG pattern can be reduced as necessary.

## D.2.1 OCNG TDD pattern 1: outer resource blocks allocation in 10 MHz

**Table D.2.1-1: OP.1 TDD: OCNG TDD Pattern 1 for 5ms downlink-to-uplink switch-point periodicity**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data |
|---|---|---|---|---|---|
| | **Subframe** (Note 1) | | | | |
| | 0 | 5 | 3, 4, 8, 9 and 6 (as normal subframe) Note 3 | 1 and 6 (as special subframe) Note 3 | |
| 0 - 12 | 0 | 0 | 0 | 0 | Note 2 |
| 37 - 49 | 0 | 0 | 0 | 0 | |

Note 1: The allocation of any PDSCH with or without SIB1 applies only to the subframes not configured as PRS subframes.

Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 3: Subframes available for DL transmission depends on the Uplink-Downlink configuration defined in Table 4.2-2 in 3GPP TS 36.211 [9]. The control region consists of PCFICH, PHICH and PDCCH. Number of OFDM symbols belonging to the control region may vary between subframes.

Note 4: If two or more transmit antennas with CRS are used in the test, the OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

## D.2.2 OCNG TDD pattern 2: full bandwidth allocation in 10 MHz

**Table D.2.2-1: OP.2 TDD: OCNG TDD Pattern 2 for 5ms downlink-to-uplink switch-point periodicity**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data |
|---|---|---|---|---|---|
| | **Subframe** (Note 1) | | | | |
| | 0 | 5 | 3 , 4, 8, 9 and 6 (as normal subframe) Note 3 | 1 and 6 (as special subframe) Note 3 | |
| 0 - 49 | 0 | 0 | 0 | 0 | Note 2 |

Note 1: The allocation of any PDSCH with or without SIB1 applies only to the subframes not configured as PRS subframes.

Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 3: Subframes available for DL transmission depends on the Uplink-Downlink configuration in Table 4.2-2 in 3GPP TS 36.211[9]. The control region consists of PCFICH, PHICH and PDCCH. Number of OFDM symbols belonging to the control region may vary between subframes.

Note 4: If two or more transmit antennas with CRS are used in the test, the OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

## D.2.3 OCNG TDD pattern 3: outer resource blocks allocation in 1.4 MHz

**Table D.2.3-1: OP.3 TDD: OCNG TDD Pattern 3 for 5 ms downlink-to-uplink switch-point periodicity**

| **Allocation** $n_{PRB}$ | **Relative power level** $\gamma_{PRB}$ **[dB]** | | | | **PDSCH Data** |
|---|---|---|---|---|---|
| | **Subframe** (Note 1) | | | | |
| | 0 | 5 | 3 , 4, 8, 9 and 6 (as normal subframe) Note 3 | 1 and 6 (as special subframe) Note 3 | |
| 0 - 1 | 0 | 0 | 0 | 0 | Note 2 |
| 4 - 5 | 0 | 0 | 0 | 0 | |

Note 1: The allocation of any PDSCH with or without SIB1 applies only to the subframes not configured as PRS subframes.

Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 3: Subframes available for DL transmission depends on the Uplink-Downlink configuration defined in Table 4.2-2 in 3GPP TS 36.211 [9].

Note 4: If two or more transmit antennas with CRS are used in the test, the OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

## D.2.4 OCNG TDD pattern 4: full bandwidth allocation in 1.4 MHz

**Table D.2.4-1: OP.4 TDD: OCNG TDD Pattern 4 for 5 ms downlink-to-uplink switch-point periodicity**

| **Allocation** $n_{PRB}$ | **Relative power level** $\gamma_{PRB}$ **[dB]** | | | | **PDSCH Data** |
|---|---|---|---|---|---|
| | **Subframe** (Note 1) | | | | |
| | 0 | 5 | 3 , 4, 8, 9 and 6 (as normal subframe) Note 3 | 1 and 6 (as special subframe) Note 3 | |
| 0 – 5 | 0 | 0 | 0 | 0 | Note 2 |

Note 1: The allocation of any PDSCH with or without SIB1 applies only to the subframes not configured as PRS subframes.

Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 3: Subframes available for DL transmission depends on the Uplink-Downlink configuration in Table 4.2-2 in 3GPP TS 36.211 [9].

Note 4: If two or more transmit antennas with CRS are used in the test, the OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

# D.2.5 OCNG TDD pattern 5: outer resource blocks allocation in 10 MHz for MBSFN ABS

**Table D.2.5-1: OP.5 TDD: OCNG TDD Pattern 5 for 5ms downlink-to-uplink switch-point periodicity**

| **Allocation** $n_{PRB}$ | **Relative power level** $\gamma_{PRB}$ **[dB]** | | | | **PDSCH Data** |
|---|---|---|---|---|---|
| | **Subframe (Note 1)** | | | | |
| | 0 | 5 | 3 , 4, 8, 9 and 6 (as normal subframe) Note 3 | 1 and 6 (as special subframe) Note 3 | |
| 0 – 12 | 0 | 0 | 0 | Table D.2.5-2 | Note 2 |
| 37 – 49 | 0 | 0 | 0 | | |

Note 1: The allocation of any PDSCH with or without SIB1 applies only to the subframes not configured as PRS subframes.

Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 3: Subframes available for DL transmission depends on the Uplink-Downlink configuration defined in Table 4.2-2 in TS 36.211 [16]. Any of the subframes 3, 4, 8 and 9 configured as MBSFN ABS in a test shall not contain any PMCH data and shall contain CRS only in the first symbol of the first time slot. The subframe(s) configured as MBSFN ABS depend upon the MBSFN ABS pattern used in the test.

Note 4: If two or more transmit antennas with CRS are used in the test, the OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in clause 7.1 in TS 36.213.

**Table D.2.5-2: OP.5 TDD: OCNG TDD Pattern 5 for special subframe configuration with 5ms downlink-to-uplink switch-point periodicity**

| Allocation $n_{PRB}$ | CP length | Relative power level $\gamma_{PRB}$ [dB] | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Special subframe configuration | | | | | | | | | | | | | | | | | |
| | | 0 | | 1 | | 2 | | 3 | | 4 | | 5 | | 6 | | 7 | | 8 | |
| | | Control region OFDM symbols | | | | | | | | | | | | | | | | | |
| | | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 |
| 0 – 12 | N | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | |
| | | | | | | | | | | | | | | | | | | | |
| 37 – 49 | N | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | |
| | | | | | | | | | | | | | | | | | | | |
| Note 1: Special subframe configurations are defined in Table 4.2-1 in TS 36.211 [16]. | | | | | | | | | | | | | | | | | | | |

# D.2.6 OCNG TDD pattern 6: full bandwidth allocation in 10 MHz for MBSFN ABS

**Table D.2.6-1: OP.6 TDD: OCNG TDD Pattern 6 for 5ms downlink-to-uplink switch-point periodicity**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data |
|---|---|---|---|---|---|
| | Subframe (Note 1) | | | | |
| | 0 | 5 | 3 , 4, 8, 9 and 6 (as normal subframe)Note 3 | 1 and 6 (as special subframe) Note 3 | |
| 0 – 49 | 0 | 0 | 0 | 0 | Note 2 |

Note 1: The allocation of any PDSCH with or without SIB1 applies only to the subframes not configured as PRS subframes.

Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 3: Subframes available for DL transmission depends on the Uplink-Downlink configuration in Table 4.2-2 in TS 36.211 [16]. Any of the subframes 3, 4, 8 and 9 configured as MBSFN ABS in a test shall not contain any PMCH data and shall contain CRS only in the first symbol of the first time slot. The subframe(s) configured as MBSFN ABS depend upon the MBSFN ABS pattern used in the test.

Note 4: If two or more transmit antennas with CRS are used in the test, the OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in clause 7.1 in TS 36.213.

# D.2.7 OCNG TDD pattern 7: outer resource blocks allocation in 20 MHz

**Table D.2.7-1: OP.7 TDD: OCNG TDD Pattern 7 for 5ms downlink-to-uplink switch-point periodicity**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data |
|---|---|---|---|---|---|
| | **Subframe** (Note 1) | | | | |
| | 0 | 5 | 3 , 4, 8, 9 and 6 (as normal subframe) Note 3 | 1 and 6 (as special subframe) Note 3 | |

| | | | | | |
|---|---|---|---|---|---|
| 0 – 37 | 0 | 0 | 0 | Table D.2.7-2 | Note 2 |
| 62 – 99 | 0 | 0 | 0 | | |

Note 1: The allocation of any PDSCH with or without SIB1 applies only to the subframes not configured as PRS subframes.

Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 3: Subframes available for DL transmission depends on the Uplink-Downlink configuration defined in Table 4.2-2 in 3GPP TS 36.211 [16].

Note 4: If two or more transmit antennas with CRS are used in the test, the OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

**Table D.2.7-2: OP.7 TDD: OCNG TDD Pattern 7 for special subframe configuration with 5ms downlink-to-uplink switch-point periodicity**

| **Allocation** $n_{PRB}$ | **CP length** | **Relative power level $\gamma_{PRB}$ [dB]** | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | **Special subframe configuration** | | | | | | | | | | | | | | | | | |
| | | 0 | | 1 | | 2 | | 3 | | 4 | | 5 | | 6 | | 7 | | 8 | |
| | | **Control region OFDM symbols** | | | | | | | | | | | | | | | | | |
| | | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 |
| 0 – 37 | N | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | |
| 62 – 99 | N | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | |

Note 1: Special subframe configurations are defined in Table 4.2-1 in TS 36.211 [16].

# D.2.8 OCNG TDD pattern 8: full bandwidth allocation in 20 MHz

**Table D.2.8-1: OP.8 TDD: OCNG TDD Pattern 8 for 5ms downlink-to-uplink switch-point periodicity**

| **Allocation** $n_{PRB}$ | **Relative power level $\gamma_{PRB}$ [dB]** | | | | **PDSCH Data** |
|---|---|---|---|---|---|
| | **Subframe (Note 1)** | | | | |
| | 0 | 5 | 3 , 4, 8, 9 and 6 (as normal subframe)Note 3 | 1 and 6 (as special subframe) Note 3 | |
| 0 – 99 | 0 | 0 | 0 | 0 | Note 2 |

Note 1: The allocation of any PDSCH with or without SIB1 applies only to the subframes not configured as PRS subframes.

Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 3: Subframes available for DL transmission depends on the Uplink-Downlink configuration in Table 4.2-2 in 3GPP TS 36.211 [16].

Note 4: If two or more transmit antennas with CRS are used in the test, the OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

# D.2.9 OCNG TDD pattern 9: outer resource blocks allocation in 5 MHz

**Table D.2.9-1: OP.9 TDD: OCNG TDD Pattern 9 for 5ms downlink-to-uplink switch-point periodicity**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data |
|---|---|---|---|---|---|
| | Subframe (Note 1) | | | | |
| | 0 | 5 | 3 , 4, 8, 9 and 6 (as normal subframe) Note 3 | 1 and 6 (as special subframe) Note 3 | |
| 0 – 6 | 0 | 0 | 0 | Table A.3.2.1.7-2 | Note 2 |
| 18 – 24 | 0 | 0 | 0 | | |

Note 1: PDSCH allocation applies only to subframes not configured as PRS subframes.
Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.
Note 3: Subframes available for DL transmission depends on the Uplink-Downlink configuration defined in Table 4.2-2 in 3GPP TS 36.211 [16].
Note 4: If two or more transmit antennas with CRS are used in the test, the OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

**Table D.2.9-2: OP.9 TDD: OCNG TDD Pattern 9 for special subframe configuration with 5ms downlink-to-uplink switch-point periodicity**

| Allocation $n_{PRB}$ | CP length | Relative power level $\gamma_{PRB}$ [dB] | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Special subframe configuration | | | | | | | | | | | | | | | | | |
| | | 0 | | 1 | | 2 | | 3 | | 4 | | 5 | | 6 | | 7 | | 8 | |
| | | Control region OFDM symbols | | | | | | | | | | | | | | | | | |
| | | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 2 |
| 0 – 6 | N | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | |
| | | | | | | | | | | | | | | | | | | | |
| 18 – 24 | N | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | | 0 | |
| | | | | | | | | | | | | | | | | | | | |

Note 1: Special subframe configurations are defined in Table 4.2-1 in TS 36.211 [16].

# D.2.10 OCNG TDD pattern 10: full bandwidth allocation in 5 MHz

**Table D.2.10-1: OP.10 TDD: OCNG TDD Pattern 10 for 5ms downlink-to-uplink switch-point periodicity**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data |
|---|---|---|---|---|---|
| | Subframe (Note 1) | | | | |
| | 0 | 5 | 3 , 4, 8, 9 and 6 (as normal subframe) Note 3 | 1 and 6 (as special subframe) Note 3 | |
| 0 – 24 | 0 | 0 | 0 | 0 | Note 2 |

Note 1: PDSCH allocation applies only to subframes not configured as PRS subframes.
Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.
Note 3: Subframes available for DL transmission depends on the Uplink-Downlink configuration in Table 4.2-2 in 3GPP TS 36.211 [16].
Note 4: If two or more transmit antennas with CRS are used in the test, the OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

# D.2.11 OCNG TDD pattern 11: Generic resource blocks allocation (without MBSFN)

**Table D.2.11-1: OP.11 TDD: OCNG TDD Pattern 11**

| OCNG Pattern Name | Bandwidth (MHz) | Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data |
|---|---|---|---|---|---|---|---|
| | | | Subframe | | | | |
| | | | 0 | 5 | 3 , 4, 8, 9 and 6 (as normal subframe) Note 5 | 1 and 6 (as special subframe) Note 5 | |
| OP.11 TDD | 10 | 0 – 49 Note 1,2 | 0 | 0 | 0 | 0 | Note 3 |
| | | | | | | | |
| | | | | | | | |

Note 1: The OCNG pattern is used only for a serving cell of the UE under test.
Note 2: The OCNG allocation applied to all downlink physical resource blocks (PRBs) except the REs that are allocated for the PDSCH for the UE under test.
Note 3: OCNG PRBs are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.
Note 4: If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in clause 7.1 in TS 36.213.
Note 5: Subframes available for DL transmission depends on the Uplink-Downlink configuration in Table 4.2-2 in TS 36.211 [16]

# D.3 OCNG Patterns for Narrowband IoT

The following Narrowband OCNG patterns (NOCNG) are used for modelling allocations to UEs not under test in a Narrowband IoT cell. Depending on scenario, allocations may be for UEs of category NB1only, or for UEs of category NB1 as well as of other categories. The former is applicable to guard-band and stand-alone deployments of Narrowband IoT, whereas the latter is applicable to in-band deployment. In order to allow different power levels for the LTE cell and the Narrowband IoT cell, a distinction is made between OCNG and NOCNG where the latter is used for category NB1 UEs and the former is used for other UE categories.

OCNG in the LTE cell is expressed by parameters OCNG_RA and OCNG_RB which together with a relative power level ( $\gamma$ ) specifies the PDSCH-to-RS EPRE ratio in OFDM symbols with and without LTE cell-specific reference symbols, respectively. The relative power, which is used for modelling boosting per virtual LTE UE allocation, is expressed by:

$$\gamma_i = PDSCH_i_RA / OCNG_RA = PDSCH_i_RB / OCNG_RB,$$

where $\gamma_i$ denotes the relative power level of the *i:th* virtual LTE UE.

Moreover in each test case NOCNG is expressed by parameters NOCNG_RA and NOCNG_RB which together with a relative power level ( $\gamma$ ) specifies the <channel>-to-RS EPRE ratio in OFDM symbols with and without Narrowband reference symbols (NB-RS), respectively. The relative power, which is used for modelling boosting per virtual UE category NB1 allocation, is expressed by:

$$\gamma_k = <channel>_k_RA / NOCNG_RA = <channel>_k_RB / NOCNG_RB,$$

where $\gamma_k$ denotes the relative power level of the *k:th* virtual NB-IoT UE, and channel may be either of NPDCCH and NPDSCH.

The parameter settings of OCNG_RA, OCNG_RB, NOCNG_RA, NOCNG_RB and the set of relative power levels $\gamma$ are chosen such that when also taking allocations to the UE category NB1 under test into account, as given by a NPDCCH and NPDSCH reference channels, a transmitted power spectral density that is constant on an OFDM symbol basis is targeted.

## D.3.1 Narrowband IoT OCNG FDD pattern 1: In-band NB-IoT in 10 MHz EUTRAN cell

**Table D.3.1-1: NOP.1 FDD: OCNG FDD Pattern 1**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | PDSCH Data | NPDCCH and NPDSCH Data |
|---|---|---|---|---|---|
| | Subframe | | | | |
| | 0, 4 | 5, 9 | 1-3, 6-8 | | |
| 0 – 29, 35 – 49 | 0 | 0 | 0 | Note 1 | N/A |
| 30 – 34 | 0 (Note 3) | 0 (Note 3) | N/A | N/A | Note 2 |
| 30 – 34 | N/A | N/A | 0 | Note 1 | N/A |

Note 1: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated QPSK modulated pseudo random data. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.
If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in clause 7.1 in TS 36.213.

Note 2: This physical resource block is assigned to a virtual UE for transmission of NPDCCH or NPDSCH; the data transmitted over the NOCNG NPDCCH or NPDSCH shall be uncorrelated QPSK modulated pseudo random data. The parameter $\gamma_{PRB}$ is used to scale the power of NPDCCH and NPDSCH.
If two transmit antennas with NBRS are used in the test, the NPDCCH and NPDSCH parts of NOCNG shall be transmitted to the virtual users by both transmit antennas with NBRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the NPDCCH and NPDSCH parts of NOCNG is equal between both the transmit antennas with NBRS used in the test. The antenna transmission modes are specified in clause 7.1 in TS 36.213.

Note 3: Value of $\gamma_{PRB}$ is applicable to PRBs not used for transmission of NPSS, NSSS, and NPBCH in anchor cell ( $n_{PRB}$ = 30).

Note 4: SI transmissions, and NPDCCH and NPDSCH for the UE under test, have precedence over OCNG and NOCNG.

N/A: Not Applicable

## D.3.2 Narrowband IoT OCNG FDD pattern 2: guard band NB-IoT in 10 MHz EUTRAN cell

**Table D.3.2-1: NOP.2 FDD: OCNG FDD Pattern 2**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | PDSCH Data | NPDCCH and NPDSCH Data |
|---|---|---|---|---|---|
| | Subframe | | | | |
| | 0, 4 | 5, 9 | 1-3, 6-8 | | |

| | | | | | |
|---|---|---|---|---|---|
| 0-49 | 0 | 0 | 0 | Note 1 | N/A |
| 50-52 | 0 | 0 (Note 3) | 0 | N/A | Note 2 |

Note 1: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated QPSK modulated pseudo random data. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.
If two or more transmit antennas with CRS are used in the test, the PDSCH part of OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the PDSCH part of OCNG is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in clause 7.1 in TS 36.213.

Note 2: This physical resource block is assigned to a virtual UE for transmission of NPDCCH or NPDSCH; the data transmitted over the NOCNG NPDCCH or NPDSCH shall be uncorrelated QPSK modulated pseudo random data. The parameter $\gamma_{PRB}$ is used to scale the power of NPDCCH and NPDSCH.
If two transmit antennas with NBRS are used in the test, the NPDCCH and NPDSCH parts of NOCNG shall be transmitted to the virtual users by both transmit antennas with NBRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the NPDCCH and NPDSCH parts of NOCNG is equal between both the transmit antennas with NBRS used in the test. The antenna transmission modes are specified in clause 7.1 in TS 36.213.

Note 3: Value of $\gamma_{PRB}$ is applicable to PRBs not used for transmission of NPSS, NSSS and PBCH in anchor cell ($n_{PRB}$ = 50).

Note 4: SI transmissions, and NPDCCH and NPDSCH for the UE under test, have precedence over OCNG and NOCNG.

N/A: Not Applicable

## D.3.3 Narrowband IoT OCNG FDD pattern 3: standalone NB-IoT

**Table D.3.3-1: NOP.3 FDD: OCNG FDD Pattern 3**

| **Allocation** $n_{PRB}$ | **Relative power level** $\gamma_{PRB}$ **[dB]** | | | **NPDCCH and NPDSCH Data** |
|---|---|---|---|---|
| | **Subframe** | | | |
| | 0, 4 | 5, 9 | 1-3, 6-8 | |
| 0 – 5 | 0 (Note 1) | 0 (Note 1) | 0 | Note 2 |

Note 1: This physical resource block is assigned to a virtual UE for transmission of NPDCCH or NPDSCH; the data transmitted over the NOCNG NPDCCH or NPDSCH shall be uncorrelated QPSK modulated pseudo random data. The parameter $\gamma_{PRB}$ is used to scale the power of NPDCCH and NPDSCH.
If two transmit antennas with NBRS are used in the test, the NPDCCH and NPDSCH parts of NOCNG shall be transmitted to the virtual users by both transmit antennas with NBRS and according to the antenna transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power of the NPDCCH and NPDSCH parts of NOCNG is equal between both the transmit antennas with NBRS used in the test. The antenna transmission modes are specified in clause 7.1 in TS 36.213.

Note 2: Value of $\gamma_{PRB}$ is applicable to PRBs not used for transmission of NPSS, NSSS and PBCH in anchor cell ($n_{PRB}$ = 0).

Note 3: SI transmissions, and NPDCCH and NPDSCH for the UE under test, have precedence over NOCNG.

# Annex E (normative): Cell configuration mapping

The cells used in TS 36.521-3 do not correspond to the cells defined in TS 36.508 [7] section 4.4.2. Tables E-1, E-2 and E-3 describes the mapping between cells described in TS 36.521-3 and those defined in TS 36.508 [7]. Table E-4 describe the mapping between Ncells described in TS 36.521-3 and those defined in TS 36.508 [7] For each test case the cells as defined in TS 36.508 [7] section 4.4.2 and section 8.1.4.2 are listed in one row. The test case shall apply the RF parameters as defined in TS 36.521-3 according to the column heading.

NOTE: For example if the second cell in a test case is an inter-frequency cell then Cell3 from TS 36.508 [7] section 4.4.2 is used with the radio parameters as defined for Cell2 in TS 36.521-3.

**Table E-1: Cell configuration mapping for RRM testing**

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| **4.2.1** | RRC IDLE / E-UTRAN Cell Reselection / FDD - FDD cell re-selection intra frequency case | Cell1 | Cell11 | | |
| **4.2.2** | RRC IDLE / E-UTRAN Cell Reselection / TDD - TDD cell re-selection intra frequency case | Cell1 | Cell11 | | |
| **4.2.3** | RRC IDLE / E-UTRAN Cell Reselection / FDD - FDD cell re-selection inter frequency case | Cell6 | Cell23 | | |
| **4.2.4** | RRC IDLE / E-UTRAN Cell Reselection / FDD - TDD cell re-selection inter frequency case | Cell1 | Cell31 | Dual mode in single PLMN | |
| **4.2.5** | RRC IDLE / E-UTRAN Cell Reselection / TDD - FDD cell re-selection inter frequency case | Cell31 | Cell1 | | |
| **4.2.6** | RRC IDLE / E-UTRAN Cell Reselection / TDD - TDD cell re-selection inter frequency case | Cell6 | Cell23 | | |
| **4.2.7** | RRC IDLE / E-UTRAN Cell Reselection / FDD – FDD Inter frequency case in the existence of non-allowed CSG cell | Cell23 | Cell6 | Cell3 | |
| **4.2.8** | RRC IDLE / E-UTRAN Cell Reselection / TDD – TDD Inter frequency case in the existence of non-allowed CSG cell | Cell23 | Cell6 | Cell3 | |
| **4.2.9** | RRC IDLE / E-UTRAN Cell Reselection / E-UTRAN FDD-FDD intra-frequency Cell Re-selection case for 5MHz Bandwidth | Cell1 | Cell11 | | |
| **4.2.14** | RRC IDLE / E-UTRAN Cell Reselection / TDD - TDD intra frequency case for Cat-M1 UE in normal coverage | Cell1 | Cell11 | | |
| **4.3.1.1** | RRC IDLE / E-UTRAN to UTRAN Cell re-selection / E-UTRAN FDD - UTRAN FDD cell re-selection: UTRA is of higher priority | Cell3 | Cell9 | | |
| **4.3.1.2** | RRC IDLE / E-UTRAN to UTRAN Cell re-selection / E-UTRAN FDD - UTRAN FDD cell re-selection: UTRA is of lower priority | Cell3 | Cell9 | | |
| **4.3.1.3** | RRC IDLE / E-UTRAN to UTRAN Cell re-selection / E-UTRAN FDD - UTRAN FDD cell re-selection in fading propagation conditions: UTRA FDD is of lower priority | Cell3 | Cell9 | | |
| **4.3.1.4** | RRC IDLE / E-UTRAN to UTRAN Cell re-selection / E-UTRAN FDD - UTRAN FDD cell re-selection: UTRA FDD is of lower priority for 5MHz Bandwidth | Cell3 | Cell9 | | |
| **4.3.2** | RRC IDLE / E-UTRAN to UTRAN Cell re-selection / E-UTRAN FDD - UTRAN TDD cell re-selection | Cell6 | Cell8 | | |
| **4.3.3** | RRC IDLE / E-UTRAN to UTRAN Cell re-selection / E-UTRAN TDD - UTRAN FDD cell re-selection | Cell6 | Cell8 | | |
| **4.3.4.1** | RRC IDLE / E-UTRAN to UTRAN Cell re-selection / E-UTRAN TDD - UTRAN TDD cell re-selection: UTRA is of higher priority | Cell6 | Cell8 | | |
| **4.3.4.2** | RRC IDLE / E-UTRAN to UTRAN Cell re-selection / E-UTRAN TDD - UTRAN TDD cell re-selection: UTRA is of lower priority | Cell6 | Cell8 | | |
| **4.3.4.3** | RRC IDLE / E-UTRAN to UTRAN Cell re-selection /EUTRA TDD-UTRA TDD cell reselection in fading propagation conditions: UTRA TDD is of lower priority | Cell3 | Cell9 | | |
| **4.4.1** | RRC IDLE / E-UTRAN to GSM Cell re-selection / E-UTRAN FDD - GSM cell re-selection | Cell1 | Cell26 | | |
| **4.4.2** | RRC IDLE / E-UTRAN to GSM Cell re-selection / E-UTRAN TDD - GSM cell re-selection | Cell1 | Cell26 | | |
| **4.5.1.1** | RRC IDLE / E-UTRAN to HRPD Cell re-selection / E-UTRAN | Cell1 | Cell15 | | |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| | FDD - HRPD cell re-selection: HRPD is of lower priority | | | | |
| **4.5.2.1** | RRC IDLE / E-UTRAN to HRPD Cell re-selection / E-UTRAN TDD - HRPD Cell Reselection: HRPD is of Lower Priority | Cell1 | Cell15 | | |
| **4.6.1.1** | RRC IDLE / E-UTRAN to cdma2000 1xRTT Cell re-selection / E-UTRAN FDD - cdma2000 1xRTT cell re-selection: cdma2000 1x is of lower priority | Cell1 | Cell19 | | |
| **4.6.2.1** | RRC IDLE / E-UTRAN to cdma2000 1xRTT Cell re-selection / E-UTRAN TDD - cdma2000 1X Cell Reselection: cdma2000 1X is of Lower Priority | Cell1 | Cell19 | | |
| **5.1.1** | RRC CONNECTED / E-UTRAN Handover / FDD - FDD / Intra frequency case | Cell1 | Cell2 | | |
| **5.1.2** | RRC CONNECTED / E-UTRAN Handover / TDD - TDD / Intra frequency case | Cell1 | Cell2 | | |
| **5.1.3** | RRC CONNECTED / E-UTRAN Handover / FDD - FDD / Inter frequency case | Cell6 | Cell3 | | |
| **5.1.4** | RRC CONNECTED / E-UTRAN Handover / TDD - TDD / Inter frequency case | Cell6 | Cell3 | | |
| **5.1.5** | RRC CONNECTED / E-UTRAN Handover / FDD - FDD / Inter frequency case: unknown target cell | Cell6 | Cell3 | | |
| **5.1.6** | RRC CONNECTED / E-UTRAN Handover / TDD - TDD / Inter frequency case: unknown target cell | Cell6 | Cell3 | | |
| **5.1.7** | RRC CONNECTED / E-UTRAN Handover / FDD – TDD / Inter frequency case | Cell1 | Cell10 | Dual mode in multiple PLMN | |
| **5.1.8** | RRC CONNECTED / E-UTRAN Handover / TDD – FDD / Inter frequency case | Cell10 | Cell1 | | |
| **5.1.9** | RRC CONNECTED / E-UTRAN Handover / FDD - FDD / Intra frequency handover for 5MHz Bandwidth | Cell1 | Cell2 | | |
| **5.1.10** | E-UTRAN FDD-FDD Handover intra frequency handover for UE category 0 | Cell1 | Cell2 | | |
| **5.1.11** | E-UTRAN HD-FDD Handover intra frequency handover for UE category 0 | Cell1 | Cell2 | | |
| **5.1.12** | E-UTRAN TDD-TDD Handover intra frequency handover for UE category 0 | Cell1 | Cell2 | | |
| **5.1.13** | E-UTRAN FDD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA | Cell1 | Cell2 | | |
| **5.1.14** | E-UTRAN HD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA | Cell1 | Cell2 | | |
| **5.1.15** | E-UTRAN TDD Intra frequency handover for Cat-M1 UEs in CEModeA | Cell1 | Cell2 | | |
| **5.2.1** | RRC CONNECTED / Handover from E-UTRAN to other RATs / From E-UTRAN to UTRAN / E-UTRAN FDD - UTRAN FDD handover | Cell3 | Cell9 | | |
| **5.2.2** | RRC CONNECTED / Handover from E-UTRAN to other RATs / From E-UTRAN to UTRAN / E-UTRAN TDD - UTRAN FDD handover | Cell6 | Cell8 | | |
| **5.2.3** | RRC CONNECTED / Handover from E-UTRAN to other RATs / From E-UTRAN to GSM / E-UTRAN FDD - GSM handover | Cell1 | Cell26 | | |
| **5.2.4** | RRC CONNECTED / Handover from E-UTRAN to other RATs / From E-UTRAN to UTRAN / E-UTRAN TDD - UTRAN TDD handover | Cell3 | Cell9 | | |
| **5.2.5** | RRC CONNECTED / Handover from E-UTRAN to other RATs / From E-UTRAN to UTRAN / E-UTRAN FDD - UTRAN TDD handover | Cell6 | Cell8 | | |
| **5.2.6** | RRC CONNECTED / Handover from E-UTRAN to other RATs / From E-UTRAN to GSM / E-UTRA TDD - GSM handover | Cell1 | Cell26 | | |
| **5.2.7** | RRC CONNECTED / Handover from E-UTRAN to other RATs / E-UTRAN FDD - UTRAN FDD handover: unknown target cell | Cell3 | Cell9 | | |
| **5.2.8** | RRC CONNECTED / Handover from E-UTRAN to other RATs / E-UTRAN FDD - GSM handover: unknown target cell | Cell1 | Cell26 | | |
| **5.2.9** | RRC CONNECTED / Handover from E-UTRAN to other RATs / E-UTRAN TDD - GSM Handover: unknown target cell | Cell1 | Cell26 | | |
| **5.2.10** | RRC CONNECTED / Handover from E-UTRAN to other RATs / E-UTRAN TDD - UTRAN TDD HO test: unknown target cell | Cell6 | Cell8 | | |
| **5.2.11** | RRC CONNECTED / Handover from E-UTRAN to other RATs | Cell3 | Cell9 | | |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| | / E-UTRAN FDD - UTRAN FDD handover for 5MHz Bandwidth | | | | |
| **5.3.1** | RRC CONNECTED / Handover from E-UTRAN to non-3GPP RATs / E-UTRAN FDD – HRPD handover | Cell1 | Cell15 | | |
| **5.3.2** | RRC CONNECTED / Handover from E-UTRAN to non-3GPP RATs / E-UTRAN FDD – cdma2000 1xRTT handover | Cell1 | Cell19 | | |
| **5.3.3** | RRC CONNECTED / Handover from E-UTRAN to non-3GPP RATs / E-UTRAN FDD – HRPD Handover: unknown target cell | Cell1 | Cell15 | | |
| **5.3.4** | RRC CONNECTED / Handover from E-UTRAN to non-3GPP RATs / E-UTRAN FDD - cdma2000 1xRTT Handover: unknown target cell | Cell1 | Cell19 | | |
| **5.3.5** | RRC CONNECTED / Handover from E-UTRAN to non-3GPP RATs / E-UTRAN TDD-HRPD Handover | Cell1 | Cell15 | | |
| **5.3.6** | RRC CONNECTED / Handover from E-UTRAN to non-3GPP RATs / E-UTRAN TDD-cdma2000 1X Handover | Cell1 | Cell19 | | |
| **6.1.1** | RRC Connection Mobility Control / E-UTRAN FDD Intra-frequency RRC Re-establishment | Cell1 | Cell2 | | |
| **6.1.2** | RRC Connection Mobility Control / E-UTRAN FDD Inter-frequency RRC Re-establishment | Cell6 | Cell3 | | |
| **6.1.3** | RRC Connection Mobility Control / E-UTRAN TDD Intra-frequency RRC Re-establishment | Cell1 | Cell2 | | |
| **6.1.4** | RRC Connection Mobility Control / E-UTRAN TDD Inter-frequency RRC Re-establishment | Cell6 | Cell3 | | |
| **6.1.5** | RRC Connection Mobility Control / E-UTRAN FDD Intra-frequency RRC Re-establishment for 5MHz Bandwidth | Cell1 | Cell2 | | |
| **6.1.8** | E-UTRAN TDD Intra-frequency RRC Re-establishment for UE category 0 | Cell1 | Cell2 | | |
| **6.1.9** | E-UTRAN FD-FDD Intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA | Cell1 | Cell2 | | |
| **6.1.10** | E-UTRAN HD-FDD Intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA | Cell1 | Cell2 | | |
| **6.1.11** | E-UTRAN TDD Intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA | Cell1 | Cell2 | | |
| **6.2.1** | RRC Connection Mobility Control / Random Access / E-UTRAN FDD - Contention Based Random Access | Cell1 | | | |
| **6.2.2** | RRC Connection Mobility Control / Random Access / E-UTRAN FDD - Non-Contention Based Random Access | Cell1 | | | |
| **6.2.3** | RRC Connection Mobility Control / Random Access / E-UTRAN TDD - Contention Based Random Access | Cell1 | | | |
| **6.2.4** | RRC Connection Mobility Control / Random Access / E-UTRAN TDD - Non-Contention Based Random Access | Cell1 | | | |
| **6.2.5** | RRC Connection Mobility Control / Random Access / E-UTRAN FDD - Contention Based Random Access Test for 5MHz Bandwidth | Cell1 | | | |
| **6.2.6** | RRC Connection Mobility Control / Random Access / E-UTRAN FDD - Non-Contention Based Random Access Test for 5MHz Bandwidth | Cell1 | | | |
| **6.2.7** | E-UTRAN FDD – Non-Contention Based Random Access Test For SCell in sTAG | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **6.2.8** | E-UTRAN TDD – Non-Contention Based Random Access Test For SCell in sTAG | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **6.2.10** | E-UTRAN FDD Contention Based Random Access Test for Cat-M1 UEs in Normal Coverage | Cell1 | | | |
| **6.2.11** | E-UTRAN HD-FDD Contention Based Random Access Test for Cat-M1 UEs in Normal Coverage | Cell1 | | | |
| **6.2.12** | RRC Connection Mobility Control / Random Access / E-UTRAN TDD Contention Based Random Access Test for Cat-M1 UEs in Normal Coverage | Cell1 | | | |
| **6.3.1** | RRC Connection Mobility Control / RRC Connection Release with Redirection / Redirection from E-UTRAN FDD to UTRAN FDD | Cell3 | Cell9 | | |
| **6.3.2** | RRC Connection Mobility Control / RRC Connection Release | Cell6 | Cell8 | | |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| | with Redirection / Redirection from E-UTRAN TDD to UTRAN FDD | | | | |
| **6.3.3** | RRC Connection Mobility Control / RRC Connection Release with Redirection / Redirection from E-UTRAN FDD to GERAN when System Information is provided | Cell1 | Cell26 | | |
| **6.3.4** | RRC Connection Mobility Control / RRC Connection Release with Redirection / Redirection from E-UTRAN TDD to GERAN when System Information is provided | Cell1 | Cell26 | | |
| **6.3.5** | RRC Connection Mobility Control / RRC Connection Release with Redirection / E-UTRA TDD RRC connection release redirection to UTRA TDD | Cell3 | Cell9 | | |
| **6.3.6** | RRC Connection Mobility Control / RRC Connection Release with Redirection / E-UTRA FDD RRC connection release redirection to UTRA TDD | Cell6 | Cell8 | | |
| **6.3.7** | RRC Connection Mobility Control / RRC Connection Release with Redirection / E-UTRA TDD RRC connection release redirection to UTRA TDD without SI provided | Cell3 | Cell9 | | |
| **6.3.8** | RRC Connection Mobility Control / RRC Connection Release with Redirection / E-UTRA FDD RRC connection release redirection to UTRA TDD without SI provided | Cell6 | Cell8 | | |
| **6.3.9** | RRC Connection Mobility Control / RRC Connection Release with Redirection / Redirection from E-UTRAN FDD to UTRAN FDD without System Information | Cell3 | Cell9 | | |
| **6.3.10** | RRC Connection Mobility Control / RRC Connection Release with Redirection / Redirection from E-UTRAN FDD to GERAN when System Information is not provided | Cell1 | Cell26 | | |
| **6.3.11** | RRC Connection Mobility Control / RRC Connection Release with Redirection / Redirection from E-UTRAN TDD to GERAN when System Information is not provided | Cell1 | Cell26 | | |
| **6.3.12** | RRC Connection Mobility Control / RRC Connection Release with Redirection / E-UTRAN TDD RRC connection release redirection to UTRAN FDD without SI provided | Cell6 | Cell8 | | |
| **7.1.1** | E-UTRAN FDD-UE Transmit Timing Accuracy | Cell1 | | | |
| **7.1.2** | E-UTRAN TDD-UE Transmit Timing Accuracy | Cell1 | | | |
| **7.1.3** | E-UTRAN FDD – UE Transmit Timing Accuracy Tests for SCell | Cell1 | Cell2 | | Intra-band |
| | | Cell1 | Cell10 | | Inter-band |
| **7.1.3_1** | E-UTRAN FDD – UE Transmit Timing Accuracy Tests for SCell (Release 12 and forward) | Cell1 | Cell2 | | Intra-band |
| | | Cell1 | Cell10 | | Inter-band |
| **7.1.4** | E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell | Cell1 | Cell2 | | Intra-band |
| | | Cell1 | Cell10 | | Inter-band |
| **7.1.4A** | E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell for 20 MHz +10 MHz bandwidth | Cell1 | Cell2 | | Intra-band |
| | | Cell1 | Cell10 | | Inter-band |
| **7.1.4_1** | E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell (Release 12 and forward) | Cell1 | Cell2 | | Intra-band |
| | | Cell1 | Cell10 | | Inter-band |
| **7.1.5** | E-UTRAN FDD - UE Transmit Timing Accuracy Tests for 5MHz Bandwidth | Cell1 | | | |
| **7.1.6** | E-UTRAN FDD – UE Transmit Timing Accuracy Tests for SCell in sTAG | Cell1 | Cell2 | | Intra-band |
| | | Cell1 | Cell10 | | Inter-band |
| **7.1.7** | E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell in sTAG | Cell1 | Cell2 | | Intra-band |
| | | Cell1 | Cell10 | | Inter-band |
| **7.1.7A** | E-UTRAN TDD – UE Transmit Timing Accuracy Tests for | Cell1 | Cell2 | | Intra- |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| | SCell in sTAG for 20MHz +20MHz bandwidth | | | | band |
| | | Cell1 | Cell10 | | Inter-band |
| **7.1.7B** | E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell in sTAG for 20MHz +10MHz bandwidth | Cell1 | Cell2 | | Intra-band |
| | | Cell1 | Cell10 | | Inter-band |
| **7.1.8** | Void | | | | |
| **7.1.9** | Void | | | | |
| **7.1.10** | E-UTRAN FDD – UE Transmit Timing Accuracy Tests for Cat-M1 UE in CEModeA | Cell1 | | | |
| **7.1.11** | E-UTRAN HD-FDD – UE Transmit Timing Accuracy Tests for Cat-M1 UE in CEModeA | Cell1 | | | |
| **7.1.12** | E-UTRAN TDD - UE Transmit Timing Accuracy Tests for Cat-M1 UE in CEModeA | Cell1 | | | |
| **7.2.1** | E-UTRAN FDD-UE Timing Advance Adjustment Accuracy | Cell1 | | | |
| **7.2.2** | E-UTRAN TDD-UE Timing Advance Adjustment Accuracy | Cell1 | | | |
| **7.2.3** | E-UTRAN FDD - UE Timing Advance Adjustment Accuracy Test for 5MHz Bandwidth | Cell1 | | | |
| **7.2.4** | E-UTRAN FDD – UE Timing Advance Adjustment Accuracy Test For SCell in sTAG | Cell1 | Cell2 | | Intra-band |
| | | Cell1 | Cell10 | | Inter-band |
| **7.2.5** | E-UTRAN TDD – UE Timing Advance Adjustment Accuracy Test For SCell in sTAG | Cell1 | Cell2 | | Intra-band |
| | | Cell1 | Cell10 | | Inter-band |
| **7.2.5A** | E-UTRAN TDD - UE Timing Advance Adjustment Accuracy Test for SCell in sTAG for 20MHz +20MHz bandwidth | Cell1 | Cell2 | | Intra-band |
| | | Cell1 | Cell10 | | Inter-band |
| **7.2.5B** | E-UTRAN TDD - UE Timing Advance Adjustment Accuracy Test for SCell in sTAG for 20MHz +10MHz bandwidth | Cell1 | Cell2 | | Intra-band |
| | | Cell1 | Cell10 | | Inter-band |
| **7.3.1** | E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync | Cell1 | | | |
| **7.3.1_1** | E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync with 4 Rx antenna ports | Cell1 | | | |
| **7.3.2** | E-UTRAN FDD Radio Link Monitoring Test for In-sync | Cell1 | | | |
| **7.3.2_1** | E-UTRAN FDD Radio Link Monitoring Test for In-sync with 4 Rx antenna ports | Cell1 | | | |
| **7.3.3** | E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync | Cell1 | | | |
| **7.3.4** | E-UTRAN TDD Radio Link Monitoring Test for In-sync | Cell1 | | | |
| **7.3.5** | E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync in DRX | Cell1 | | | |
| **7.3.6** | E-UTRAN FDD Radio Link Monitoring Test for In-sync in DRX | Cell1 | | | |
| **7.3.7** | E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync in DRX | Cell1 | | | |
| **7.3.8** | E-UTRAN TDD Radio Link Monitoring Test for In-sync in DRX | Cell1 | | | |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| **7.3.9** | E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with Non MBSFN ABS (eICIC) | Cell1 | Cell2 | | |
| **7.3.10** | E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with Non MBSFN ABS (eICIC) | Cell1 | Cell2 | | |
| **7.3.11** | E-UTRAN FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with Non MBSFN ABS (eICIC) | Cell1 | Cell2 | | |
| **7.3.12** | E-UTRAN TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with Non MBSFN ABS (eICIC) | Cell1 | Cell2 | | |
| **7.3.13** | E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) (Note 3) | Cell1 | Cell3 | | |
| **7.3.14** | E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) (Note 3) | Cell1 | Cell3 | | |
| **7.3.15** | E-UTRAN FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) (Note 3) | Cell1 | Cell3 | | |
| **7.3.16** | E-UTRAN TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) (Note 3) | Cell1 | Cell3 | | |
| **7.3.17** | E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non MBSFN ABS (feICIC) (Note 3) | Cell1 | Cell3 | Cell4 | |
| **7.3.18** | E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non MBSFN ABS (feICIC) (Note 3) | Cell1 | Cell3 | Cell4 | |
| **7.3.19** | E-UTRAN FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non MBSFN ABS (feICIC) (Note 3) | Cell1 | Cell3 | Cell4 | |
| **7.3.20** | E-UTRAN TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non MBSFN ABS (feICIC) (Note 3) | Cell1 | Cell3 | Cell4 | |
| **7.3.21** | E-UTRAN FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and MBSFN ABS (feICIC) (Note 3) | Cell1 | Cell3 | Cell4 | |
| **7.3.22** | E-UTRAN TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and MBSFN ABS (feICIC) (Note 3) | Cell1 | Cell3 | Cell4 | |
| **7.3.23** | E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync for 5MHz Bandwidth | Cell1 | | | |
| **7.3.24** | E-UTRAN FDD Radio Link Monitoring Test for In-sync for 5MHz Bandwidth | Cell1 | | | |
| **7.3.25** | E-UTRAN FDD Radio Link Monitoring Test for In-sync in DRX for 5MHz Bandwidth | Cell1 | | | |
| **7.3.26** | E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync for UE category 0 | Cell1 | | | |
| **7.3.27** | E-UTRAN FD-FDD Radio Link Monitoring Test for In-sync for UE category 0 | Cell1 | | | |
| **7.3.28** | E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category 0 | Cell1 | | | |
| **7.3.29** | E-UTRAN FD-FDD Radio Link Monitoring Test for In-sync in | Cell1 | | | |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| | DRX for UE category 0 | | | | |
| **7.3.30** | E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync for UE category 0 | Cell1 | | | |
| **7.3.31** | E-UTRAN HD-FDD Radio Link Monitoring Test for In-sync for UE category 0 | Cell1 | | | |
| **7.3.32** | E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category 0 | Cell1 | | | |
| **7.3.33** | E-UTRAN HD-FDD Radio Link Monitoring Test for In-sync in DRX for UE category 0 | Cell1 | | | |
| **7.3.34** | E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync for UE category 0 | Cell1 | | | |
| **7.3.35** | E-UTRAN TDD Radio Link Monitoring Test for In-sync for UE category 0 | Cell1 | | | |
| **7.3.36** | E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category 0 | Cell1 | | | |
| **7.3.37** | E-UTRAN TDD Radio Link Monitoring Test for In-sync in DRX for UE category 0 | Cell1 | | | |
| **7.3.38** | E-UTRAN FDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC | Cell1 | Cell10 | | |
| **7.3.39** | E-UTRAN FDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in asynchronous DC | Cell1 | Cell10 | | |
| **7.3.40** | E-UTRAN TDD-TDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC | Cell1 | Cell10 | | |
| **7.3.41** | E-UTRAN FDD-FDD Radio Link Monitoring Test for In-sync in DRX in synchronous dual connectivity | Cell1 | Cell10 | | |
| **7.3.42** | E-UTRAN FDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in asynchronous DC | Cell1 | Cell10 | | |
| **7.3.43** | E-UTRAN TDD-TDD Radio Link Monitoring Test for In-sync in DRX in synchronous dual connectivity | Cell1 | Cell10 | | |
| **7.3.44** | E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in FDD | Cell1 | Cell10 | | |
| **7.3.45** | E-UTRAN TDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC with PCell in TDD | Cell1 | Cell10 | | |
| **7.3.46** | E-UTRAN TDD-FDD Radio Link Monitoring Test for In-sync in DRX for PSCell in synchronous DC with PCell in FDD | Cell1 | Cell10 | | |
| **7.3.47** | E-UTRAN TDD-FDD Radio Link Monitoring Test for In-sync in DRX for PSCell in synchronous DC with PCell in TDD | Cell1 | Cell10 | | |
| **7.3.48** | E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync for Cat-M1 UE in CEMode A | Cell1 | | | |
| **7.3.49** | E-UTRAN FD-FDD Radio Link Monitoring Test for In-Sync for Cat-M1 UE in CEMode A | Cell1 | | | |
| **7.3.52** | E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync for Cat-M1 UE in CEMode A | Cell1 | | | |
| **7.3.53** | E-UTRAN HD-FDD Radio Link Monitoring Test for In-Sync for Cat-M1 UE in CEMode A | Cell1 | | | |
| **7.5.1** | E-UTRAN FDD – UE ProSe Direct Discovery Transmission Timing Accuracy Test | Cell 1 | | | |
| **7.5.4** | E-UTRAN FDD – UE ProSe Direct Communication Transmission Timing Accuracy Test | Cell 1 | | | |
| **8.1.1** | UE Measurement Procedures / E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells | Cell1 | Cell2 | | |
| **8.1.2** | UE Measurement Procedures / E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells | Cell1 | Cell2 | | |
| **8.1.3** | UE Measurement Procedures / E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX | Cell1 | Cell2 | | |
| **8.1.4** | Void | | | | |
| **8.1.5** | UE Measurement Procedures / E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | Cell1 | Cell2 | | |
| **8.1.6** | UE Measurement Procedures / E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX | Cell1 | Cell2 | | |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| **8.1.7** | UE Measurement Procedures / E-UTRAN FDD-FDD Intra-frequency event-triggered reporting under time domain measurement resource restriction with non-MBSFN ABS (eICIC) | Cell1 | Cell2 | | |
| **8.1.8** | E-UTRAN FDD-FDD Intra-Frequency Event-Triggered Reporting under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) (Note 3) | Cell1 | Cell2 | Cell6 | |
| **8.1.9** | E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells for 5MHz bandwidth | Cell1 | Cell2 | | |
| **8.1.10** | E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX for 5MHz bandwidth | Cell1 | Cell2 | | |
| **8.1.11** | E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for UE category 0 | Cell1 | Cell2 | | |
| **8.1.12** | E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for UE category 0 | Cell1 | Cell2 | | |
| **8.1.13** | E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX for UE category 0 | Cell1 | Cell2 | | |
| **8.1.17** | E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells | Cell1 | Cell2 | | |
| **8.1.18** | E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX | Cell1 | Cell2 | | |
| **8.1.29** | E-UTRAN TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA | Cell1 | Cell2 | | |
| **8.1.30** | E-UTRAN TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA in DRX | Cell1 | Cell2 | | |
| **8.2.1** | UE Measurement Procedures / E-UTRAN TDD-TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells | Cell1 | Cell2 | | |
| **8.2.2** | UE Measurement Procedures / E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX | Cell1 | Cell2 | | |
| **8.2.3** | UE Measurement Procedures / E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | Cell1 | Cell2 | | |
| **8.2.4** | UE Measurement Procedures / E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX | Cell1 | Cell2 | | |
| **8.2.5** | UE Measurement Procedures / E-UTRAN TDD-TDD intra-frequency event-triggered Reporting under time domain measurement resource restriction with non-MBSFN ABS (eICIC) | Cell1 | Cell2 | | |
| **8.2.6** | E-UTRAN TDD-TDD Intra-Frequency Event-Triggered Reporting under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) (Note 3) | Cell1 | Cell2 | Cell6 | |
| **8.3.1** | UE Measurement Procedures / E-UTRAN FDD-FDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells | Cell6 | Cell3 | | |
| **8.3.2** | UE Measurement Procedures / E-UTRAN FDD-FDD inter frequency event triggered reporting when DRX is used under fading propagation conditions in asynchronous cells | Cell6 | Cell3 | | |
| **8.3.3** | UE Measurement Procedures / E-UTRAN FDD-FDD inter frequency event triggered reporting under AWGN propagation conditions in asynchronous cells with DRX when L3 filtering is | Cell6 | Cell3 | | |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| | used | | | | |
| **8.3.4** | UE Measurement Procedures / E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | Cell6 | Cell3 | | |
| **8.3.5** | UE Measurement Procedures / E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX | Cell6 | Cell3 | | |
| **8.3.6** | UE Measurement Procedures / E-UTRAN FDD-FDD inter frequency event triggered reporting without measurement gaps under AWGN propagation conditions in asynchronous cells | Cell6 | Cell3 | | |
| **8.4.1** | UE Measurement Procedures / E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in synchronous cells | Cell6 | Cell3 | | |
| **8.4.2** | UE Measurement Procedures / E-UTRAN TDD-TDD Inter-frequency event triggered reporting when DRX is used under fading propagation conditions in synchronous cells | Cell6 | Cell3 | | |
| **8.4.3** | UE Measurement Procedures / E-UTRAN TDD-TDD inter-frequency event triggered reporting under AWGN propagation conditions in synchronous cells with DRX when L3 filtering is used | Cell6 | Cell3 | | |
| **8.4.4** | UE Measurement Procedures / E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | Cell6 | Cell3 | | |
| **8.4.5** | UE Measurement Procedures / E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX | Cell6 | Cell3 | | |
| **8.5.1** | UE Measurement Procedures / E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions | Cell3 | Cell9 | | |
| **8.5.2** | UE Measurement Procedures / E-UTRAN FDD - UTRAN FDD SON ANR cell search reporting under AWGN propagation conditions | Cell3 | Cell9 | | |
| **8.5.3** | UE Measurement Procedures / E-UTRAN FDD - UTRAN FDD event triggered reporting when DRX is used under fading propagation conditions | Cell3 | Cell9 | | |
| **8.5.4** | UE Measurement Procedures / E-UTRAN FDD - UTRAN FDD enhanced cell identification under AWGN propagation conditions | Cell3 | Cell9 | | |
| **8.5.6** | UE Measurement Procedures / E-UTRAN FDD - UTRAN FDD event triggered reporting without measurement gaps under AWGN propagation conditions | Cell3 | Cell9 | | |
| **8.5.7** | UE Measurement Procedures / E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions for 5MHz bandwidth | Cell3 | Cell9 | | |
| **8.6.1** | UE Measurement Procedures / E-UTRAN TDD - UTRAN FDD event triggered reporting under fading propagation conditions | Cell1 | Cell8 | | |
| **8.7.1** | UE Measurement Procedures / E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions | Cell6 | Cell8 | | |
| **8.7.2** | UE Measurement Procedures / E-UTRAN TDD - UTRAN TDD cell search when DRX is used under fading propagation conditions | Cell6 | Cell8 | | |
| **8.7.3** | UE Measurement Procedures / E-UTRAN TDD - UTRAN TDD SON ANR cell search reporting under AWGN propagation conditions | Cell3 | Cell9 | | |
| **8.7.4** | UE Measurement Procedures / E-UTRAN TDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions | Cell3 | Cell9 | | |
| **8.8.1** | UE Measurement Procedures / E-UTRAN FDD - GSM event triggered reporting in AWGN | Cell6 | Cell26 | | |
| **8.8.2** | UE Measurement Procedures / E-UTRAN FDD - GSM event triggered reporting when DRX is used in AWGN | Cell6 | Cell26 | | |
| **8.9.1** | UE Measurement Procedures / E-UTRAN FDD - UTRAN TDD event triggered reporting under fading propagation conditions | Cell6 | Cell8 | | |
| **8.9.2** | UE Measurement Procedures / E-UTRAN FDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions | Cell6 | Cell8 | | |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| **8.10.1** | UE Measurement Procedures / E-UTRAN TDD - GSM event triggered reporting in AWGN | Cell6 | Cell26 | | |
| **8.10.2** | UE Measurement Procedures / E-UTRAN TDD - GSM event triggered reporting when DRX is used in AWGN | Cell6 | Cell26 | | |
| **8.11.1** | UE Measurement Procedures / Monitoring of multiple layers / E-UTRAN FDD - E-UTRAN FDD and E-UTRAN FDD Inter-frequency event triggered reporting under fading propagation conditions | Cell1 | Cell3 | Cell6 | |
| **8.11.2** | UE Measurement Procedures / Monitoring of multiple layers / E-UTRAN TDD - E-UTRAN TDD and E-UTRAN TDD Inter-frequency event triggered reporting under fading propagation conditions | Cell1 | Cell3 | Cell6 | |
| **8.11.3** | UE Measurement Procedures / Monitoring of multiple layers / InterRAT E-UTRA FDD to E-UTRA FDD and UTRA FDD cell search | Cell1 | Cell6 | Cell8 | |
| **8.11.4** | UE Measurement Procedures / Monitoring of multiple layers / InterRAT E-UTRA TDD to E-UTRA TDD and UTRA TDD cell search | Cell1 | Cell6 | Cell8 | |
| **8.11.5** | UE Measurement Procedures / Combined E-UTRAN FDD - E-UTRA FDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions | Cell6 | Cell3 | Cell24 | |
| **8.11.6** | UE Measurement Procedures / Combined E-UTRAN TDD - E-UTRA TDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions | Cell6 | Cell3 | Cell24 | |
| **8.14.1** | UE Measurement Procedures / E-UTRAN TDD-FDD Inter-frequency event triggered reporting under fading propagation conditions in asynchronous cells | Cell10 | Cell1 | | |
| **8.14.2** | UE Measurement Procedures / E-UTRAN TDD-FDD Inter-frequency event triggered reporting when DRX is used under fading propagation conditions in synchronous cells | Cell10 | Cell1 | | |
| **8.14.3** | UE Measurement Procedures / E-UTRAN TDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | Cell10 | Cell1 | | |
| **8.15.1** | UE Measurement Procedures / E-UTRAN FDD-TDD Inter-frequency event triggered reporting under fading propagation conditions in asynchronous cells | Cell1 | Cell10 | | |
| **8.15.2** | UE Measurement Procedures / E-UTRAN FDD-TDD Inter-frequency event triggered reporting when DRX is used under fading propagation conditions in asynchronous cells | Cell1 | Cell10 | | |
| **8.15.3** | UE Measurement Procedures / E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | Cell1 | Cell10 | | |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| **8.16.1** | UE Measurement Procedures / E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.16.2** | UE Measurement Procedures / E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.16.3** | UE Measurement Procedures / E-UTRAN FDD-FDD Event triggered reporting on deactivated SCell with PCell interruption in non-DRX | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.16.4** | UE Measurement Procedures / E-UTRAN TDD-TDD Event triggered reporting on deactivated SCell with PCell interruption in non-DRX | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.16.5** | UE Measurement Procedures / E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX for 20 MHz bandwidth | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.16.6** | UE Measurement Procedures / E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 20 MHz bandwidth | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.16.7** | UE Measurement Procedures / E-UTRAN FDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 20 MHz bandwidth | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.16.8** | UE Measurement Procedures / E-UTRAN TDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 20 MHz bandwidth | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.16.9** | E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX for 10MHz+5MHz | Cell6 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.16.10** | E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 10MHz+5MHz | Cell6 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.16.11** | E-UTRAN FDD event triggered reporting on deactivating SCell with PCell interruption in non-DRX for 10MHz+5MHz | Cell6 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.16.12** | E-UTRAN TDD event triggered reporting on deactivating SCell with PCell interruption in non-DRX for 10MHz+5MHz | Cell6 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.16.13** | E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX for 5MHz+5MHz | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.16.14** | E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 5MHz+5MHz | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.16.15** | E-UTRA FDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 5MHz+5MHz bandwidth | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| **8.16.16** | E-UTRA TDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 5MHz+5MHz bandwidth | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.16.17** | E-UTRAN FDD activation and deactivation of known SCell in non-DRX | Cell1 | Cell2 | | Intra-band |
| | | Cell1 | Cell10 | | Inter-band |
| **8.16.17A** | E-UTRAN FDD activation and deactivation of known SCell in non-DRX for 20MHz +20MHz bandwidth | Cell1 | Cell2 | | Intra-band |
| | | Cell1 | Cell10 | | Inter-band |
| **8.16.18** | E-UTRAN TDD activation and deactivation of known SCell in non-DRX | Cell1 | Cell2 | | Intra-band |
| | | Cell1 | Cell10 | | Inter-band |
| **8.16.18A** | E-UTRAN TDD activation and deactivation of known SCell in non-DRX for 20MHz +20MHz bandwidth | Cell1 | Cell2 | | Intra-band |
| | | Cell1 | Cell10 | | Inter-band |
| **8.16.21** | E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 20MHz+10MHz bandwidth | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.16.22** | E-UTRAN TDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 20MHz+10MHz bandwidth | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.16.23** | UE Measurement Procedures / E-UTRAN TDD-FDD CA Event Triggered Reporting Under Deactivated SCell in Non-DRX with PCell in FDD | Cell1 | Cell10 | Cell30 | |
| **8.16.24** | UE Measurement Procedures / E-UTRAN TDD-FDD CA Event Triggered Reporting Under Deactivated SCell in Non-DRX with PCell in TDD | Cell1 | Cell10 | Cell30 | |
| **8.16.25** | UE Measurement Procedures / E-UTRAN TDD-FDD CA Event triggered reporting on deactivated SCell with PCell interruption in non-DRX with PCell in FDD | Cell1 | Cell10 | Cell30 | |
| **8.16.26** | UE Measurement Procedures / E-UTRAN TDD-FDD CA Event triggered reporting on deactivated SCell with PCell interruption in non-DRX with PCell in TDD | Cell1 | Cell10 | Cell30 | |
| **8.18.1** | UE Measurement Procedures / E-UTRAN TDD-HRPD event triggered reporting under fading propagation conditions | Cell1 | Cell15 | | |
| **8.19.1** | UE Measurement Procedures / E-UTRAN TDD-CDMA2000 1X event triggered reporting under fading propagation conditions | Cell1 | Cell19 | | |
| **8.20.1** | UE Measurement Procedures / RAT Measurements in CA mode / E-UTRAN FDD-FDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells | Cell6 | Cell3 Note6 | Cell13 | Intra-band |
| | | Cell6 | Cell3 | Cell10 Note4 | Inter-band |
| **8.20.2** | UE Measurement Procedures / RAT Measurements in CA mode / E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells | Cell6 | Cell3 Note6 | Cell13 | Intra-band |
| | | Cell6 | Cell3 | Cell10 Note4 | Inter-band |
| **8.20.2A** | E-UTRAN TDD-TDD Inter-frequency event triggered reporting under fading propagation conditions in synchronous cells for 20 MHz +20 MHz bandwidth | Cell6 | Cell3 Note6 | Cell13 | Intra-band |
| | | Cell6 | Cell3 | Cell10 Note4 | Inter-band |
| **8.20.3** | UE Measurement Procedures / RAT Measurements in CA mode / E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions | Cell1 | Cell8 | Cell4 | Intra-band |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| | | Cell1 | Cell8 | Cell10 | Inter-band |
| **8.20.4** | UE Measurement Procedures / RAT Measurements in CA mode / E-UTRAN TDD to UTRAN TDD cell search under fading propagation conditions | Cell1 | Cell8 | Cell4 | Intra-band |
| | | Cell1 | Cell8 | Cell10 | Inter-band |
| **8.20.4A** | E-UTRAN TDD to UTRAN TDD cell search under fading propagation conditions for 20 MHz + 20 MHz bandwidth | Cell1 | Cell8 | Cell4 | Intra-band |
| | | Cell1 | Cell8 | Cell10 | Inter-band |
| **8.20.4B** | E-UTRAN TDD to UTRAN TDD cell search under fading propagation conditions for 20 MHz + 10 MHz bandwidth | Cell1 | Cell8 | Cell4 | Intra-band |
| | | Cell1 | Cell8 | Cell10 | Inter-band |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| **8.22.1** | UE Measurement Procedures / E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal | Cell1 | Cell2 | | |
| **8.22.2** | UE Measurement Procedures / E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal | Cell1 | Cell2 | | |
| **8.22.3** | UE Measurement Procedures / E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal | Cell6 | Cell3 | | |
| **8.22.4** | UE Measurement Procedures / E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal | Cell6 | Cell3 | | |
| **8.22.5** | UE Measurement Procedures / E-UTRAN FDD-FDD intra-frequency event triggered reporting in DRX based on CSI-RS based discovery signal | Cell1 | Cell2 | | |
| **8.22.6** | UE Measurement Procedures / E-UTRAN TDD-TDD intra-frequency event triggered reporting in DRX based on CSI-RS based discovery signal | Cell1 | Cell2 | | |
| **8.22.7** | UE Measurement Procedures / E-UTRAN FDD-FDD Inter-frequency event triggered reporting in DRX based on CSI-RS based discovery signal | Cell6 | Cell3 | | |
| **8.22.8** | UE Measurement Procedures / E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation condition in DRX based on CSI-RS based discovery signal | Cell6 | Cell3 | | |
| **8.22.9** | UE Measurement Procedures / E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX based on CRS based discovery signal | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.22.10** | UE Measurement Procedures / E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX based on CRS based discovery signal | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.22.11** | UE Measurement Procedures / E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX based on CSI-RS based discovery signal | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.22.12** | UE Measurement Procedures / E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX based on CSI-RS based discovery signal | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **8.23.1** | E-UTRAN FDD-FDD DC intra-frequency event triggered reporting with DRX in synchronous DC | Cell1 | Cell10 | | |
| **8.23.2** | E-UTRAN FDD-FDD DC intra-frequency event triggered reporting with DRX in asynchronous DC | Cell1 | Cell10 | | |
| **8.23.3** | E-UTRAN TDD-TDD DC intra-frequency event triggered reporting with DRX in synchronous DC | Cell1 | Cell10 | | |
| **8.23.4** | E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in synchronous DC | Cell6 | Cell3 | Cell10 Note4 | |
| **8.23.5** | E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in asynchronous DC | Cell6 | Cell3 | Cell10 Note4 | |
| **8.23.6** | E-UTRAN TDD-TDD DC inter-frequency event triggered reporting with DRX in synchronous DC | Cell6 | Cell3 | Cell10 Note4 | |
| **9.1.1.1** | Measurement Performance Requirements / E-UTRAN / FDD Intra frequency RSRP Accuracy / Absolute | Cell1 | Cell2 | | |
| **9.1.1.1_1** | Measurement Performance Requirements / E-UTRAN / FDD Intra Frequency Absolute RSRP Accuracy (Rel-12 and forward) | Cell1 | Cell2 | | |
| **9.1.1.2** | Measurement Performance Requirements / E-UTRAN / FDD Intra frequency RSRP Accuracy / Relative | Cell1 | Cell2 | | |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| **9.1.2.1** | Measurement Performance Requirements / E-UTRAN / TDD Intra Frequency RSRP Accuracy / Absolute | Cell1 | Cell2 | | |
| **9.1.2.1_1** | Measurement Performance Requirements / E-UTRAN / TDD Intra Frequency Absolute RSRP Accuracy (Rel-12 and forward) | Cell1 | Cell2 | | |
| **9.1.2.2** | Measurement Performance Requirements / E-UTRAN / TDD Intra Frequency RSRP Accuracy / Relative | Cell1 | Cell2 | | |
| **9.1.3.1** | Measurement Performance Requirements / E-UTRAN / FDD Inter frequency RSRP Accuracy / Absolute | Cell6 | Cell3 | | |
| **9.1.3.1_1** | Measurement Performance Requirements / E-UTRAN / FDD - FDD Inter Frequency Absolute RSRP Accuracy (Rel-12 and forward) | Cell6 | Cell3 | | |
| **9.1.3.2** | Measurement Performance Requirements / E-UTRAN / FDD Inter frequency RSRP Accuracy / Relative | Cell6 | Cell3 | | |
| **9.1.3.2_1** | Measurement Performance Requirements / E-UTRAN / FDD Inter frequency RSRP Accuracy / Relative (Rel-12 and forward) | Cell6 | Cell3 | | |
| **9.1.4.1** | Measurement Performance Requirements / E-UTRAN / TDD Inter Frequency RSRP Accuracy / Absolute | Cell6 | Cell3 | | |
| **9.1.4.1_1** | Measurement Performance Requirements / E-UTRAN / TDD - TDD Inter Frequency Absolute RSRP Accuracy (Rel-12 and forward) | Cell6 | Cell3 | | |
| **9.1.4.2** | Measurement Performance Requirements / E-UTRAN / TDD Inter Frequency RSRP Accuracy / Relative | Cell6 | Cell3 | | |
| **9.1.4.2_1** | Measurement Performance Requirements / E-UTRAN / TDD Inter Frequency RSRP Accuracy / Relative (Rel-12 and forward) | Cell6 | Cell3 | | |
| **9.1.5.1** | Measurement Performance Requirements / E-UTRAN / FDD – TDD Inter frequency RSRP Accuracy / Absolute | Cell1 | Cell10 | | |
| **9.1.5.1_1** | Measurement Performance Requirements / E-UTRAN / FDD - TDD Inter Frequency Absolute RSRP Accuracy (Rel-12 and forward) | Cell1 | Cell10 | | |
| **9.1.5.2** | Measurement Performance Requirements / E-UTRAN / FDD- TDD Inter frequency RSRP Accuracy / Relative | Cell1 | Cell10 | | |
| **9.1.5.2_1** | Measurement Performance Requirements / E-UTRAN / FDD- TDD Inter frequency RSRP Accuracy / Relative (Rel-12 and forward) | Cell1 | Cell10 | | |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| **9.1.6.1** | Measurement Performance Requirements / FDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.6.1_1** | Measurement Performance Requirements / FDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation (Rel 12 and forward) | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.6.2** | Measurement Performance Requirements / FDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.6.2_1** | Measurement Performance Requirements / FDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation (Rel-12 and forward) | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.7.1** | Measurement Performance Requirements / TDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.7.1_1** | Measurement Performance Requirements / TDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation (Rel-12 and forward) | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.7.2** | Measurement Performance Requirements / TDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.7.2_1** | Measurement Performance Requirements / TDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation (Rel-12 and forward) | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.8.1** | Measurement Performance Requirements / FDD Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC) | Cell1 | Cell2 | | |
| **9.1.8.1_1** | Measurement Performance Requirements / FDD Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC) (Rel-12 and forward) | Cell1 | Cell2 | | |
| **9.1.8.2** | Measurement Performance Requirements / FDD Relative RSRP Accuracy under Time-Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC) | Cell1 | Cell2 | | |
| **9.1.9.1** | Measurement Performance Requirements / TDD Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC) | Cell1 | Cell2 | | |
| **9.1.9.1_1** | Measurement Performance Requirements / TDD Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC) (Rel-12 and forward) | Cell1 | Cell2 | | |
| **9.1.9.2** | Measurement Performance Requirements / TDD Relative RSRP Accuracy under Time-Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC) | Cell1 | Cell2 | | |
| **9.1.10.1** | Measurement Performance Requirements / FDD Absolute RSRP under Time-Domain Measurement Resource Restriction with MBSFN ABS (eICIC) (Note 3) | Cell1 | Cell6 | | |
| **9.1.10.1_1** | Measurement Performance Requirements / FDD Absolute RSRP under Time-Domain Measurement Resource Restriction with MBSFN ABS (eICIC) (Rel-12 and forward) (Note 3) | Cell1 | Cell6 | | |
| **9.1.10.2** | Measurement Performance Requirements / FDD Relative | Cell1 | Cell6 | | |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| | RSRP under Time-Domain Measurement Resource Restriction with MBSFN ABS (eICIC) (Note 3) | | | | |
| **9.1.11.1** | Measurement Performance Requirements / TDD Absolute RSRP under Time-Domain Measurement Resource Restriction with MBSFN ABS (eICIC) (Note 3) | Cell1 | Cell6 | | |
| **9.1.11.1_1** | Measurement Performance Requirements / TDD Absolute RSRP under Time-Domain Measurement Resource Restriction with MBSFN ABS (eICIC) (Rel-12 and forward) (Note 3) | Cell1 | Cell6 | | |
| **9.1.11.2** | Measurement Performance Requirements / TDD Relative RSRP under Time-Domain Measurement Resource Restriction with MBSFN ABS (eICIC) (Note 3) | Cell1 | Cell6 | | |
| **9.1.12.1** | Measurement Performance Requirements / FDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.12.1_1** | Measurement Performance Requirements / FDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz (Rel 12 and forward) | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.12.2** | Measurement Performance Requirements / FDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.13.1** | Measurement Performance Requirements / TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.13.1_1** | Measurement Performance Requirements / TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz (Rel-12 and forward) | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.13.2** | Measurement Performance Requirements / TDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.14.1** | FDD Intra Frequency Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) (Note 3) | Cell1 | Cell2 | Cell6 | |
| **9.1.14.1_1** | FDD Intra Frequency Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) (Rel-12 and forward) (Note 3) | Cell1 | Cell2 | Cell6 | |
| **9.1.14.2** | FDD Intra Frequency Relative RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) (Note 3) | Cell1 | Cell2 | Cell6 | |
| **9.1.15.1** | TDD Intra Frequency Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) (Note 3) | Cell1 | Cell2 | Cell6 | |
| **9.1.15.1_1** | TDD Intra Frequency Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) (Note 3) | Cell1 | Cell2 | Cell6 | |
| **9.1.15.2** | TDD Intra Frequency Relative RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | Cell1 | Cell2 | Cell6 | |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| | (Note 3) | | | | |
| **9.1.16.1** | Measurement Performance Requirements / E-UTRAN / FDD Intra Frequency Absolute RSRP Accuracy for 5MHz Bandwidth | Cell1 | Cell2 | | |
| **9.1.16.1_1** | Measurement Performance Requirements / E-UTRAN / FDD Intra Frequency Absolute RSRP Accuracy for 5MHz Bandwidth (Rel-12 and forward) | Cell1 | Cell2 | | |
| **9.1.16.2** | Measurement Performance Requirements / E-UTRAN / FDD Intra Frequency Relative Accuracy of RSRP for 5MHz Bandwidth | Cell1 | Cell2 | | |
| **9.1.17.1** | Measurement Performance Requirements / E-UTRAN / FDD Inter frequency RSRP Accuracy / Absolute for 5 MHz | Cell1 | Cell10 | | |
| **9.1.17.1_1** | Measurement Performance Requirements / E-UTRAN / FDD - FDD Inter Frequency Absolute RSRP Accuracy for 5MHz Bandwidth (Rel-12 and forward) | Cell1 | Cell10 | | |
| **9.1.17.2** | Measurement Performance Requirements / E-UTRAN / FDD - FDD Inter Frequency Relative Accuracy of RSRP for 5MHz Bandwidth | Cell1 | Cell10 | | |
| **9.1.17.2_1** | Measurement Performance Requirements / E-UTRAN / FDD - FDD Inter Frequency Relative Accuracy of RSRP for 5MHz Bandwidth (Rel-12 and forward) | Cell1 | Cell10 | | |
| **9.1.18.1** | Measurement Performance Requirements / FDD Absolute RSRP Accuracy for E-UTRA for Carrier Aggregation for 10MHz + 5MHz | Cell6 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.18.1_1** | Measurement Performance Requirements / FDD Absolute RSRP Accuracy for E-UTRA for Carrier Aggregation for 10MHz + 5MHz (Rel 12 and forward) | Cell6 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.18.2** | Measurement Performance Requirements / FDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation for 10MHz + 5MHz | Cell6 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.19.1** | Measurement Performance Requirements / TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 10MHz + 5MHz | Cell6 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.19.1_1** | TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 10MHz + 5MHz (Rel 12 and forward) | Cell6 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.19.2** | Measurement Performance Requirements / TDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 10MHz + 5MHz | Cell6 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.20.1** | Measurement Performance Requirements / FDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz bandwidth | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.20.1_1** | Measurement Performance Requirements / FDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz bandwidth (Rel 12 and forward) | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.20.2** | Measurement Performance Requirements / FDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz bandwidth | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| **9.1.21.1** | Measurement Performance Requirements / TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.21.1_1** | Measurement Performance Requirements / TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz (Rel 12 and forward) | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.21.2** | Measurement Performance Requirements / TDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.22** | FDD-TDD RSRP Accuracy E-UTRA for Carrier Aggregation with PCell in FDD | Cell1 | Cell10 | Cell30 | |
| **9.1.23** | FDD-TDD RSRP Accuracy E-UTRA for Carrier Aggregation with PCell in TDD | Cell1 | Cell10 | Cell30 | |
| **9.1.24.1_1** | TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20MHz + 10MHz (Rel 12 and forward) | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.25** | Measurement Performance Requirements / FDD intra-frequency absolute and relative RSRP accuracies in CRS based discovery signal | Cell1 | Cell2 | | |
| **9.1.26** | Measurement Performance Requirements / TDD intra-frequency absolute and relative RSRP accuracies in CRS based discovery signal | Cell1 | Cell2 | | |
| **9.1.27** | Measurement Performance Requirements / FDD—FDD inter-frequency absolute and relative RSRP accuracies in CRS based discovery signal | Cell6 | Cell3 | | |
| **9.1.28** | Measurement Performance Requirements / TDD—TDD inter-frequency absolute and relative RSRP accuracies in CRS based discovery signal | Cell6 | Cell3 | | |
| **9.1.29** | FDD intra frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal | Cell1 | Cell2 | | |
| **9.1.30** | TDD intra frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal | Cell1 | Cell2 | | |
| **9.1.31** | Measurement Performance Requirements / FDD-FDD inter frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal | Cell6 | Cell3 | | |
| **9.1.32** | Measurement Performance Requirements / TDD-TDD inter frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal | Cell6 | Cell3 | | |
| **9.1.33** | Measurement Performance Requirements / FDD absolute and relative RSRP accuracies for E-UTRAN Carrier Aggregation in CRS based discovery signal | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.34** | Measurement Performance Requirements / TDD absolute and relative RSRP accuracies for E-UTRAN Carrier Aggregation in CRS based discovery signal | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.35** | Measurement Performance Requirements / FDD absolute and relative CSI-RSRP accuracies for E-UTRAN Carrier Aggregation in CSI-RS based discovery signal | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.36** | Measurement Performance Requirements / TDD absolute and relative CSI-RSRP accuracies for E-UTRAN Carrier Aggregation in CSI-RS based discovery signal | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.1.41.1** | FD-FDD Intra Frequency Absolute RSRP Accuracy for UE category 0 | Cell1 | Cell2 | | |
| **9.1.41.2** | FD-FDD Intra Frequency Relative RSRP Accuracy for UE category 0 | Cell1 | Cell2 | | |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| **9.1.42.1** | HD-FDD Intra Frequency Absolute RSRP Accuracy for UE category 0 | Cell1 | Cell2 | | |
| **9.1.42.2** | HD-FDD Intra Frequency Relative RSRP Accuracy for UE category 0 | Cell1 | Cell2 | | |
| **9.1.43.1** | TDD Intra Frequency Absolute RSRP Accuracy for UE category 0 | Cell1 | Cell2 | | |
| **9.1.43.2** | TDD Intra Frequency Relative RSRP Accuracy for UE category 0 | Cell1 | Cell2 | | |
| **9.1.52** | FD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeA | Cell1 | Cell2 | | |
| **9.1.53** | HD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeA | Cell1 | Cell2 | | |
| **9.1.54** | TDD RSRP Intra frequency case for Cat-M1 UE in CEModeA | Cell1 | Cell2 | | |
| **9.1.57** | FD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeB | Cell1 | Cell2 | | |
| **9.1.58** | HD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeB | Cell1 | Cell2 | | |
| **9.1.59** | TDD RSRP Intra frequency case for Cat-M1 UE in CEModeB | Cell1 | Cell2 | | |
| **9.2.1.1** | Measurement Performance Requirements / E-UTRAN / FDD Intra frequency RSRQ Accuracy / Absolute | Cell1 | Cell2 | | |
| **9.2.2.1** | Measurement Performance Requirements / E-UTRAN / TDD Intra Frequency RSRQ Accuracy / Absolute | Cell1 | Cell2 | | |
| **9.2.3.1** | Measurement Performance Requirements / E-UTRAN / FDD Inter frequency RSRQ Accuracy / Absolute | Cell6 | Cell3 | | |
| **9.2.3.2** | Measurement Performance Requirements / E-UTRAN / FDD Inter frequency RSRQ Accuracy / Relative | Cell6 | Cell3 | | |
| **9.2.4.1** | Measurement Performance Requirements / E-UTRAN / TDD - TDD Inter Frequency RSRQ Accuracy / Absolute | Cell6 | Cell3 | | |
| **9.2.4.2** | Measurement Performance Requirements / E-UTRAN / TDD - TDD Inter Frequency RSRQ Accuracy / Relative | Cell6 | Cell3 | | |
| **9.2.4A.1** | Measurement Performance Requirements / FDD - TDD Inter Frequency Absolute RSRQ Accuracy | Cell1 | Cell10 | | |
| **9.2.4A.2** | Measurement Performance Requirements / FDD - TDD Inter Frequency Relative Accuracy of RSRQ | Cell1 | Cell10 | | |
| **9.2.5.1** | Measurement Performance Requirements / FDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.2.5.2** | Measurement Performance Requirements / FDD Relative RSRQ Accuracy E-UTRA for Carrier Aggregation | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.2.6.1** | Measurement Performance Requirements / TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.2.6.2** | Measurement Performance Requirements / TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.2.7.1** | Measurement Performance Requirements / FDD RSRQ under Time Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC) | Cell1 | Cell2 | | |
| **9.2.8.1** | Measurement Performance Requirements / TDD RSRQ under Time Domain Measurement Resource Restriction with Non-MBSFN ABS (eICIC) | Cell1 | Cell2 | | |
| **9.2.9.1** | Measurement Performance Requirements / FDD Absolute RSRQ under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) (Note 3) | Cell1 | Cell6 | | |
| **9.2.10.1** | Measurement Performance Requirements / TDD RSRQ under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) (Note 3) | Cell1 | Cell6 | | |
| **9.2.11.1** | Measurement Performance Requirements / FDD Absolute | Cell1 | Cell2 | Cell4 | Intra- |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| | RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz | | | Note1 | band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.2.11.2** | Measurement Performance Requirements / FDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.2.12.1** | Measurement Performance Requirements / TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.2.12.2** | Measurement Performance Requirements / TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.2.15.1** | FDD RSRQ Accuracy under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) (Note 3) | Cell1 | Cell2 | Cell6 | |
| **9.2.16.1** | TDD RSRQ Accuracy under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) (Note 3) | Cell1 | Cell2 | Cell6 | |
| **9.2.17.1** | Measurement Performance Requirements / E-UTRAN / FDD Intra frequency RSRQ Accuracy / Absolute for 5MHz | Cell1 | Cell2 | | |
| **9.2.18.1** | Measurement Performance Requirements / FDD - FDD Inter Frequency Absolute RSRQ Accuracy for 5MHz Bandwidth | Cell1 | Cell10 | | |
| **9.2.18.2** | Measurement Performance Requirements / FDD - FDD Inter Frequency Relative Accuracy of RSRQ for 5MHz Bandwidth | Cell1 | Cell10 | | |
| **9.2.19.1** | Measurement Performance Requirements / FDD-FDD Inter Frequency absolute WB-RSRQ accuracy | Cell6 | Cell3 | | |
| **9.2.20.1** | Measurement Performance Requirements / TDD-TDD Inter Frequency absolute WB-RSRQ accuracy | Cell6 | Cell3 | | |
| **9.2.21.1** | Measurement Performance Requirements / FDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.2.21.2** | Measurement Performance Requirements / FDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.2.22.1** | Measurement Performance Requirements / TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.2.22.2** | Measurement Performance Requirements / TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.2.23.1** | Measurement Performance Requirements / FDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 5MHz+5MHz | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.2.23.2** | Measurement Performance Requirements / FDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 5MHz+5MHz | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.2.24.1** | Measurement Performance Requirements / TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for | Cell1 | Cell2 | Cell4 Note1 | Intra-band |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| | 5MHz+5MHz | | | | |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.2.24.2** | Measurement Performance Requirements / TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 5MHz+5MHz | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.2.25.1** | Absolute RSRQ Accuracy for E-UTRAN TDD-FDD Carrier Aggregation with PCell in FDD | Cell1 | Cell10 | | |
| **9.2.25.2** | Relative RSRQ Accuracy for E-UTRAN TDD-FDD Carrier Aggregation with PCell in FDD | Cell1 | Cell10 | | |
| **9.2.26.1** | Absolute RSRQ Accuracy for E-UTRAN TDD-FDD Carrier Aggregation with PCell in TDD | Cell1 | Cell10 | | |
| **9.2.26.2** | Relative RSRQ Accuracy for E-UTRAN TDD-FDD Carrier Aggregation with PCell in TDD | Cell1 | Cell10 | | |
| **9.2.28** | Measurement Performance Requirements / FDD intra-frequency absolute RSRQ accuracy with CRS based discovery signal | Cell1 | Cell2 | | |
| **9.2.29** | Measurement Performance Requirements / TDD intra-frequency absolute RSRQ accuracy with CRS based discovery signal | Cell1 | Cell2 | | |
| **9.2.30** | Measurement Performance Requirements / FDD-FDD inter-frequency absolute and relative RSRQ accuracies with CRS based discovery signal | Cell6 | Cell3 | | |
| **9.2.31** | Measurement Performance Requirements / TDD-TDD inter-frequency absolute and relative RSRQ accuracies with CRS based discovery signal | Cell6 | Cell3 | | |
| **9.2.32** | Measurement Performance Requirements / FDD absolute and relative RSRQ accuracy for E-UTRAN Carrier Aggregation in CRS based discovery signal | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.2.33** | Measurement Performance Requirements / TDD absolute and relative RSRQ accuracy for E-UTRAN Carrier Aggregation in CRS based discovery signal | Cell1 | Cell2 | Cell4 Note1 | Intra-band |
| | | Cell1 | Cell10 | Cell30 | Inter-band |
| **9.3.1** | Measurement Performance Requirements / E-UTRAN FDD - UTRA FDD CPICH RSCP absolute accuracy | Cell3 | Cell9 | | |
| **9.3.2** | Measurement Performance Requirements / E-UTRAN TDD- UTRA FDD CPICH RSCP absolute accuracy | Cell3 | Cell9 | | |
| **9.3.3** | Measurement Performance Requirements / E-UTRAN FDD - UTRA FDD CPICH RSCP absolute accuracy for 5MHz bandwidth | Cell3 | Cell9 | | |
| **9.4.1** | Measurement Performance Requirements / E- UTRAN FDD - UTRA FDD CPICH Ec/No absolute accuracy | Cell3 | Cell9 | | |
| **9.4.2** | Measurement Performance Requirements / E- UTRAN TDD - UTRA FDD CPICH Ec/No absolute accuracy | Cell3 | Cell9 | | |
| **9.4.3** | Measurement Performance Requirements / E-UTRAN FDD - UTRA FDD CPICH Ec/No absolute accuracy for 5MHz bandwidth | Cell3 | Cell9 | | |
| **9.5.1** | Measurement Performance Requirements / E- UTRAN FDD - UTRA TDD P-CCPCH RSCP absolute accuracy | Cell3 | Cell9 | | |
| **9.5.2** | Measurement Performance Requirements / E- UTRAN TDD - UTRA TDD P-CCPCH RSCP absolute accuracy | Cell3 | Cell9 | | |
| **9.6.1** | Measurement Performance Requirements / GSM RSSI accuracy for E-UTRAN FDD | Cell1 | Cell26 | Cell24 | |
| **9.6.2** | Measurement Performance Requirements / GSM RSSI accuracy for E-UTRAN TDD | Cell1 | Cell26 | Cell24 | |
| **9.9.1.1** | Measurement Performance Requirements / FDD Intra Frequency Serving Cell Absolute RSRP Accuracy | Cell1 | | | |
| **9.9.1.1_1** | Measurement Performance Requirements / FDD Intra Frequency Serving Cell Absolute RSRP Accuracy (Rel-12 and forward) | Cell1 | | | |

| Test Case | Description | 36.521-3 Cell1 | 36.521-3 Cell2 | 36.521-3 Cell3 | CA Type (Note 2) |
|---|---|---|---|---|---|
| **9.9.1.2** | Measurement Performance Requirements / FDD Intra Frequency Serving Cell Absolute RSRQ Accuracy | Cell1 | | | |
| **9.9.2.1** | Measurement Performance Requirements / TDD Intra Frequency Serving Cell Absolute RSRP Accuracy | Cell1 | | | |
| **9.9.2.1_1** | Measurement Performance Requirements / TDD Intra Frequency Serving Cell Absolute RSRP Accuracy (Rel-12 and forward) | Cell1 | | | |
| **9.9.2.2** | Measurement Performance Requirements / TDD Intra Frequency Serving Cell Absolute RSRQ Accuracy | Cell1 | | | |

Note 1: Neighbour cell uses same frequency as SCC.
Note 2: “Intra-band” notation in the table refers to both intra-band contiguous and intra-band non-contiguous CA configuration as applicable.
Note 3: All cells are intra frequency and shall use the same range (mid) independently of the range information given in TS 36.508 Table 4.4.2-1.
Note 4: Low range shall be used independently on the range information given in TS 36.508 Table 4.4.2-1.
Note 5: The frequency range remains unchanged after any change of Cell IDs in the test procedure.
Note 6: Mid range shall be used for intra-band non-contiguous CA independently on the range information given in TS 36.508 Table 4.4.2-1.
Note 7: For operating band 66 “High Range” simulated cells for all test cases shall be configured according to “High Range” as defined in TS36.508 Table 4.3.1.1.66 when UL is not configured on the cell. Otherwise “Paired High Range” shall be configured.

**Table E-2: Cell configuration mapping for 3DL CA RRM testing**

| Test Case | 36.521-3 Cell1 | | 36.521-3 Cell2 | | 36.521-3 Cell3 | | 36.521-3 Cell4 | | 36.521-3 Cell5 | |
|---|---|---|---|---|---|---|---|---|---|---|
| 8.16.27<br>8.16.31 | Default Test Settings for a CA_XA-YC Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | Neigh | | | |
| | Band | | Band | | Band | | Band | | | |
| | X | Cell1 | Y | Cell2 | Y | Cell3 | Y | Cell4 | | |
| | Default Test Settings for a XA-YA-ZA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | Neigh | | | |
| | Band | | Band | | Band | | Band | | | |
| | X | Cell1 | Y | Cell2 | Z | Cell3 | Z | Cell4 | | |
| | Default Test Settings for a XA-YA-YA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | Neigh | | | |
| | Band | | Band | | Band | | Band | | | |
| | X | Cell1 | Y | Cell2 | Y | Cell3 | Y | Cell4 | | |
| | Default Test Settings for a CA_XC-YA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | Neigh | | | |
| | Band | | Band | | Band | | Band | | | |
| | Y | Cell1 | X | Cell2 | X | Cell3 | X | Cell4 | | |
| 8.16.28<br>8.16.32 | Default Test Settings for a XA-YA-ZA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | Neigh | | | |
| | Band | | Band | | Band | | Band | | | |
| | Z | Cell1 | X | Cell2 | Y | Cell3 | Y | Cell4 | | |
| | Default Test Settings for a XA-XA-YA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | Neigh | | | |
| | Band | | Band | | Band | | Band | | | |
| | Y | Cell1 | X | Cell2 | X | Cell3 | X | Cell4 | | |
| 8.16.29<br>8.16.30<br>8.16.33<br>8.16.34 | Default Test Settings for a CA_XA-YC Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | Neigh | | | |
| | Band | | Band | | Band | | Band | | | |
| | X | Cell1 | Y | Cell2 | Y | Cell3 | Y | Cell4 | | |
| | Default Test Settings for a CA_XC-YA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | Neigh | | | |
| | Band | | Band | | Band | | Band | | | |
| | X | Cell1 | X | Cell2 | Y | Cell3 | Y | Cell4 | | |
| | Default Test Settings for a CA_XA-YA-ZA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | Neigh | | | |
| | Band | | Band | | Band | | Band | | | |
| | X | Cell1 | Y | Cell2 | Z | Cell3 | Z | Cell4 | | |
| | Default Test Settings for a CA_XA-XA-YA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | Neigh | | | |
| | Band | | Band | | Band | | Band | | | |
| | X | Cell1 | X | Cell2 | Y | Cell3 | Y | Cell4 | | |
| | Default Test Settings for a CA_XA-YA-YA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | Neigh | | | |
| | Band | | Band | | Band | | Band | | | |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | X | Cell1 | Y | Cell2 | Y | Cell3 | Y | Cell4 | | |
| | Default Test Settings for a CA_XD/XA-XC/XC-XA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | Neigh | | | |
| | Band | | Band | | Band | | Band | | | |
| | X | Cell1 | X | Cell2 | X | Cell3 | X | Cell4 | | |
| | Default Test Settings for a CA_XA-YC Configuration | | | | | | | | | |
| | PCC | | SCC1 | | Neigh | | SCC2 | | Neigh | |
| | Band | | Band | | Band | | Band | | Band | |
| 9.1.37 | X | Cell1 | Y | Cell2 | Y | Cell3 | Y | Cell4 | Y | Cell6 |
| | Default Test Settings for a XA-YA-ZA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | Neigh | | SCC2 | | Neigh | |
| | Band | | Band | | Band | | Band | | Band | |
| | X | Cell1 | Y | Cell2 | Y | Cell3 | Z | Cell4 | Z | Cell6 |
| | Default Test Settings for a XA-YA-YA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | Neigh | | SCC2 | | Neigh | |
| | Band | | Band | | Band | | Band | | Band | |
| | X | Cell1 | Y | Cell2 | Y | Cell3 | Y | Cell4 | Y | Cell6 |
| | Default Test Settings for a CA_XC-YA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | Neigh | | SCC2 | | Neigh | |
| | Band | | Band | | Band | | Band | | | |
| | Y | Cell1 | X | Cell2 | X | Cell3 | X | Cell4 | X | Cell6 |
| | Default Test Settings for a XA-YA-ZA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | Neigh | | SCC2 | | Neigh | |
| | Band | | Band | | Band | | Band | | Band | |
| 9.1.38 | Z | Cell1 | X | Cell2 | X | Cell3 | Y | Cell4 | Y | Cell6 |
| | Default Test Settings for a XA-XA-YA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | Neigh | | SCC2 | | Neigh | |
| | Band | | Band | | Band | | Band | | | |
| | Y | Cell1 | X | Cell2 | X | Cell3 | X | Cell4 | X | Cell6 |
| | Default Test Settings for a CA_XA-YC Configuration | | | | | | | | | |
| | PCC | | SCC1 | | Neigh | | SCC2 | | Neigh | |
| | Band | | Band | | Band | | Band | | Band | |
| | X | Cell1 | Y | Cell2 | Y | Cell3 | Y | Cell4 | Y | Cell6 |
| | Y | Cell1 | Y | Cell2 | Y | Cell3 | X | Cell4 | X | Cell6 |
| | Default Test Settings for a CA_XC-YA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | Neigh | | SCC2 | | Neigh | |
| 9.1.39<br>9.1.39_1<br>9.1.40<br>9.1.40_1 | Band | | Band | | Band | | Band | | Band | |
| | X | Cell1 | X | Cell2 | X | Cell3 | Y | Cell4 | Y | Cell6 |
| | Y | Cell1 | X | Cell2 | X | Cell3 | X | Cell4 | X | Cell6 |
| | Default Test Settings for a CA_XA-YA-ZA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | Neigh | | SCC2 | | Neigh | |
| | Band | | Band | | Band | | Band | | Band | |
| | X | Cell1 | Y | Cell2 | Y | Cell3 | Z | Cell4 | Z | Cell6 |
| | Y | Cell1 | X | Cell2 | X | Cell3 | Z | Cell4 | Z | Cell6 |
| | Z | Cell1 | X | Cell2 | X | Cell3 | Y | Cell4 | Y | Cell6 |
| | Default Test Settings for a CA_XA-XA-YA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | Neigh | | SCC2 | | Neigh | |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Band | | Band | | Band | | Band | | Band | |
| | X | Cell1 | X | Cell2 | X | Cell3 | Y | Cell4 | Y | Cell6 |
| | Y | Cell1 | X | Cell2 | X | Cell3 | X | Cell4 | X | Cell6 |
| | Default Test Settings for a CA_XA-YA-YA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | Neigh | | SCC2 | | Neigh | |
| | Band | | Band | | Band | | Band | | Band | |
| | X | Cell1 | Y | Cell2 | Y | Cell3 | Y | Cell4 | Y | Cell6 |
| | Y | Cell1 | Y | Cell2 | Y | Cell3 | X | Cell4 | X | Cell6 |
| | Default Test Settings for a CA_XD/CA_XA-XC/CA_XC-XA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | Neigh | | SCC2 | | Neigh | |
| | Band | | Band | | Band | | Band | | Band | |
| | X | Cell1 | X | Cell2 | X | Cell3 | X | Cell4 | X | Cell6 |
| 8.16.35<br>8.16.39<br>9.2.38 | Default Test Settings for a CA_XA-YC Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | | | | |
| | Band | | Band | | Band | | | | | |
| | X | Cell1 | Y | Cell2 | Y | Cell3 | | | | |
| | Default Test Settings for a CA_XA-YA-ZA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | | | | |
| | Band | | Band | | Band | | | | | |
| | X | Cell1 | Y | Cell2 | Z | Cell3 | | | | |
| | Default Test Settings for a XA-YA-YA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | | | | |
| | Band | | Band | | Band | | | | | |
| | X | Cell1 | Y | Cell2 | Y | Cell3 | | | | |
| | Default Test Settings for a CA_XC-YA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | | | | |
| | Band | | Band | | Band | | | | | |
| | Y | Cell1 | X | Cell2 | X | Cell3 | | | | |
| 8.16.36<br>8.16.40<br>9.2.39 | Default Test Settings for a CA_XA-YA-ZA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | | | | |
| | Band | | Band | | Band | | | | | |
| | Z | Cell1 | X | Cell2 | Y | Cell3 | | | | |
| | Default Test Settings for a XA-XA-YA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | | | | |
| | Band | | Band | | Band | | | | | |
| | Y | Cell1 | X | Cell2 | X | Cell3 | | | | |
| 8.16.37<br>8.16.38<br>8.16.41<br>8.16.42<br>9.2.40<br>9.2.41 | Default Test Settings for a CA_XA-YC Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | | | | |
| | Band | | Band | | Band | | | | | |
| | X | Cell1 | Y | Cell2 | Y | Cell3 | | | | |
| | Default Test Settings for a CA_XC-YA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | | | | |
| | Band | | Band | | Band | | | | | |
| | X | Cell1 | X | Cell2 | Y | Cell3 | | | | |
| | Default Test Settings for a CA_XA-YA-ZA Configuration | | | | | | | | | |
| | PCC | | SCC1 | | SCC2 | | | | | |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Band | | Band | | Band | | | | | | |
| X | Cell1 | Y | Cell2 | Z | Cell3 | | | | | |
| Default Test Settings for a CA_XA-XA-YA Configuration | | | | | | | | | | |
| PCC | | SCC1 | | SCC2 | | | | | | |
| Band | | Band | | Band | | | | | | |
| X | Cell1 | X | Cell2 | Y | Cell3 | | | | | |
| Default Test Settings for a CA_XA-YA-YA Configuration | | | | | | | | | | |
| PCC | | SCC1 | | SCC2 | | | | | | |
| Band | | Band | | Band | | | | | | |
| X | Cell1 | Y | Cell2 | Y | Cell3 | | | | | |
| Default Test Settings for a CA_XD / CA_XA-XC / CA_XC-XA Configuration | | | | | | | | | | |
| PCC | | SCC1 | | SCC2 | | | | | | |
| Band | | Band | | Band | | | | | | |
| X | Cell1 | X | Cell2 | X | Cell3 | | | | | |

Note1: All cells shall use the same range (mid) independently of the range information given in Table 4.4.2-1 of TS 36.508 [7].

Note2: X, Y, Z correspond to the different bands in the CA Configuration is specified in 5.4.2A.1 of TS 36.521-1[10]. E.g. for CA_1A-3A-19A, X=1, Y=3, Z=19.

**Table E-3: Cell configuration mapping for 2DL CA 20MHz+10MHz RRM testing**

| Test Case | 36.521-3 Cell1 | | 36.521-3 Cell2 | | 36.521-3 Cell3 | |
|---|---|---|---|---|---|---|
| 9.1.24.1<br>9.1.24.2<br>9.2.27.1<br>9.2.27.2 | Default Test Settings for a CA_XA-YA Configuration | | | | | |
| | PCC | | SCC1 | | Neigh | |
| | Band | | Band | | Band | |
| | X | Cell1 | Y | Cell10 | Y | Cell30 |
| | Y | Cell1 | X | Cell10 | X | Cell30 |
| | Default Test Settings for a CA_XC Configuration | | | | | |
| | PCC | | SCC1 | | Neigh | |
| | Band | | Band | | Band | |
| | X | Cell1 | X | Cell2 | X | Cell4 Note1 |
| | Default Test Settings for a CA_XA-XA Configuration | | | | | |
| | PCC | | SCC1 | | Neigh | |
| | Band | | Band | | Band | |
| | X | Cell1 | X | Cell2 | X | Cell4 Note1 |
| 8.20.2B | Default Test Settings for a CA_XA-YA Configuration | | | | | |
| | PCC | | Neigh | | SCC1 | |
| | Band | | Band | | Band | |
| | X | Cell6 | Y | Cell3 | Y | Cell10 Note5 |
| | Y | Cell6 | X | Cell3 | X | Cell10 Note5 |
| | Default Test Settings for a CA_XC Configuration | | | | | |
| | PCC | | Neigh | | SCC1 | |
| | Band | | Band | | Band | |
| | X | Cell6 | X | Cell3 | X | Cell13 |
| | Default Test Settings for a CA_XA-XA Configuration | | | | | |
| | PCC | | Neigh | | SCC1 | |
| | Band | | Band | | Band | |
| | X | Cell6 | X | Cell3 Note6 | X | Cell13 |

Note 1: Neighbour cell uses same frequency as SCC.
Note 2: X, Y correspond to the different bands in the CA Configuration is specified in 5.4.2A.1 of TS 36.521-1[10]. E.g. for CA_1A-3A, X=1, Y=3.
Note 3: For intra-band contiguous and intra-band non-contiguous CA, 20 MHz for PCC, 10MHz for SCC are tested.
Note 4: For inter-band CA configuration, 20 MHz for band X and 10 MHz for band Y are tested if X supports 20 MHz and Y supports 10 MHz. Otherwise, 10 MHz for band X and 20 MHz for band Y are tested.
Note 5: Low range shall be used independently on the range information given in TS 36.508 Table 4.4.2-1.
Note 6: Mid range shall be used independently on the range information given in TS 36.508 Table 4.4.2-1.

**Table E-4: Ncell and Cell configuration mapping for RRM testing of Category NB1**

| Test Case | Description | 36.521-3 Ncell1 | 36.521-3 Ncell2 | 36.521-3 Cell1 | 36.521-3 Cell2 |
|---|---|---|---|---|---|
| **6.1.15** | HD-FDD Intra-frequency RRC Re-establishment for UE category NB1 under normal coverage | Ncell1 | Ncell2 | Cell1 | |
| **6.1.16** | HD-FDD Inter-frequency RRC Re-establishment for UE | Ncell1 | Ncell3 | Cell1 | |

| Test Case | Description | 36.521-3 Ncell1 | 36.521-3 Ncell2 | 36.521-3 Cell1 | 36.521-3 Cell2 |
|---|---|---|---|---|---|
| | category NB1 under enhanced coverage | | | | |
| **7.2.9** | HD-FDD UE Timing Advance Adjustment Accuracy Test for UE Category NB1 in Standalone Mode under Enhance Coverage | Ncell1 | | | |

# Annex F (normative): Measurement uncertainties and Test Tolerances

The requirements of this clause apply to all applicable tests in the present document.

## F.1 Acceptable uncertainty of Test System (normative)

See TS 36.521-1[10] Annex F1.

### F.1.1 Measurement of test environments

See TS 36.521-1[10] Annex F1.1.

## F.1.2 Measurement of RRM requirements

**Table F.1.2-1: Maximum Test System Uncertainty for RRM Requirements**

| Subclause | Maximum Test System Uncertainty[1] | Derivation of Test System Uncertainty |
|---|---|---|
| 4.2.1 E-UTRA FDD - FDD cell re-selection intra frequency | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$ | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 4.2.2 E-UTRA TDD - TDD cell re-selection intra frequency | Same as 4.2.1 | |
| 4.2.3 E-UTRA FDD - FDD cell re-selection inter frequency | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$ | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br>$N_{oc2}$ is the AWGN on cell 2 frequency<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN |
| 4.2.4 E-UTRA FDD - TDD cell re-selection inter frequency | Same as 4.2.3 | |
| 4.2.5 E-UTRA TDD - FDD cell re-selection inter frequency | Same as 4.2.3 | |
| 4.2.6 E-UTRA TDD - TDD cell re-selection inter frequency | Same as 4.2.3 | |
| 4.2.7 E-UTRA FDD Inter frequency re-selection in the existence of non-allowed CSG cell | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_3 / N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br><br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$ | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_3 / N_{oc1}$ is the ratio of cell 3 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN |
| 4.2.8 E-UTRA TDD Inter frequency re-selection in the existence of non-allowed CSG cell | Same as 4.2.7 | |
| 4.2.9 E-UTRAN FDD - FDD intra frequency cell re-selection case for 5MHz bandwidth | Same as 4.2.1 | |
| 4.2.12 E-UTRAN FDD – FDD Intra frequency case for Cat-M1 UE in normal coverage | Same as 4.2.1 | |
| 4.2.13 E-UTRAN HD – FDD Intra frequency case for Cat-M1 UE in normal coverage | Same as 4.2.1 | |
| 4.2.14 E-UTRA TDD - TDD intra frequency case for Cat-M1 UE in normal coverage | Same as 4.2.1 | |
| 4.3.1.1 E-UTRA FDD - UTRAN FDD cell reselection: UTRA FDD is of higher priority | E-UTRA cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$I_{or} / I_{oc}$ ±0.3 dB<br>CPICH $E_c / I_{or}$ ±0.1 dB | Notes:<br><br>$N_{oc}$ is the AWGN on cell 1 (E-UTRA )frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>$I_{oc}$ is the AWGN on cell 2 (UTRA) frequency<br>$I_{or} / I_{oc}$ is the ratio of cell 2 signal / AWGN<br>CPICH $E_c / I_{or}$ is the fraction of cell 2 power assigned to the CPICH Physical channel |

| | | |
|---|---|---|
| 4.3.1.2 E-UTRAN FDD - UTRAN FDD cell re-selection: UTRA FDD is of lower priority | E-UTRA cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$I_{or} / I_{oc}$ ±0.3 dB<br>CPICH $E_c / I_{or}$ ±0.1 dB | Notes:<br><br>$N_{oc}$ is the AWGN on cell 1 (E-UTRA )frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>$I_{oc}$ is the AWGN on cell 2 (UTRA) frequency<br>$I_{or} / I_{oc}$ is the ratio of cell 2 signal / AWGN<br>CPICH $E_c / I_{or}$ is the fraction of cell 2 power assigned to the CPICH Physical channel |
| 4.3.1.3 EUTRA FDD-UTRA FDD cell reselection in fading propagation conditions: UTRA FDD is of lower priority | E-UTRA cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.6 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$I_{or} / I_{oc}$ ±0.3 dB<br>CPICH $E_c / I_{or}$ ±0.1 dB | Notes:<br><br>$N_{oc}$ is the AWGN on cell 1 (E-UTRA )frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>$\hat{E}s / N_{oc}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty$^2$ + Fading profile power uncertainty$^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB<br><br>$I_{oc}$ is the AWGN on cell 2 (UTRA) frequency<br>$I_{or} / I_{oc}$ is the ratio of cell 2 signal / AWGN<br>CPICH $E_c / I_{or}$ is the fraction of cell 2 power assigned to the CPICH Physical channel |
| 4.3.1.4 E-UTRAN FDD - UTRAN FDD cell re-selection: UTRA FDD is of lower priority for 5MHz bandwidth | Same as 4.3.1.2 | |
| 4.3.2 E-UTRA FDD - UTRAN TDD cell re-selection | E-UTRA cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$\hat{I}_{or} / I_{oc}$ ±0.3 dB<br>PCCPCH Ec/Ior ±0.1 dB<br>DwPCH_Ec/Ior ±0.1 dB | Note:<br>$N_{oc}$ is the AWGN on cell 1 frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>$I_{oc}$ is the AWGN on cell 2 frequency<br>$\hat{I}_{or} / I_{oc}$ is the ratio of cell 2 signal / AWGN<br>PCCPCH $E_c / I_{or}$ is the fraction of cell 2 power assigned to the PCCPCH Physical channel<br>DwPCH_Ec/Ior is the fraction of cell 2 power assigned to the DwPCH channel |
| 4.3.3 E-UTRAN TDD - UTRAN FDD cell re-selection: UTRA FDD is of lower priority | Same as 4.3.1.2 | |
| 4.3.4.1 E-UTRA TDD - UTRAN TDD cell re-selection : UTRA is of higher priority | E-UTRA cell<br>Noc ±1.0 dB averaged over BWConfig<br>Ês / Noc ±0.3 dB averaged over BWConfig<br><br>UTRA cell<br>Ioc ±0.7 dB<br>Îor / Ioc ±0.3 dB<br>PCCPCH Ec/Ior ±0.1 dB<br>DwPCH_Ec/Ior ±0.1 dB | Note:<br>Noc is the AWGN on cell 1 frequency<br>Ês / Noc is the ratio of cell 1 signal / AWGN<br><br>Ioc is the AWGN on cell 2 frequency<br>Îor / Ioc is the ratio of cell 2 signal / AWGN<br>PCCPCH Ec / Ior is the fraction of cell 2 power assigned to the PCCPCH Physical channel<br>DwPCH_Ec/Ior is the fraction of cell 2 power assigned to the DwPCH channel |
| 4.3.4.2 E-UTRA TDD - UTRAN TDD cell re-selection : UTRA is of lower priority | Same as 4.3.2 | |

| | | |
|---|---|---|
| 4.3.4.3 EUTRA TDD-UTRA TDD cell reselection in fading propagation conditions: UTRA TDD is of lower priority | E-UTRA cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.6 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$\hat{I}_{or} / I_{oc}$ ±0.3 dB<br>PCCPCH Ec/Ior ±0.1 dB<br>DwPCH_Ec/Ior ±0.1 dB | Note:<br>$N_{oc}$ is the AWGN on cell 1 (E-UTRA) frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>Each $\hat{E}s / N_{oc}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB<br><br>$I_{oc}$ is the AWGN on cell 2 (UTRA) frequency<br>$\hat{I}_{or} / I_{oc}$ is the ratio of cell 2 signal / AWGN<br>PCCPCH $E_c / I_{or}$ is the fraction of cell 2 power assigned to the PCCPCH Physical channel<br>DwPCH_Ec/Ior is the fraction of cell 2 power assigned to the DwPCH channel |
| 4.4.1 E-UTRAN FDD - GSM cell re-selection | E-UTRA cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>GSM cell<br>Signal level ±0.7 dB | Notes:<br>$N_{oc}$ is the AWGN on cell 1 (E-UTRA )frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>Cell 2 (GSM) has only the wanted signal, without AWGN |
| 4.4.2 E-UTRAN TDD - GSM cell re-selection | Same as 4.4.1 | |
| 4.5.1.1 RRC IDLE / E-UTRAN to HRPD Cell re-selection / E-UTRAN FDD - HRPD cell re-selection: HRPD is of lower priority | E-UTRA cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>HRPD cell<br>$I_{oc}$ ±2.0 dB<br>$\hat{I}_{or} / I_{oc}$ ±0.7 dB | Note:<br>$N_{oc}$ is the AWGN on cell 1 frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>$I_{oc}$ is the AWGN on cell 2 (HRPD) frequency<br>$I_{or} / I_{oc}$ is the ratio of cell 2 signal / AWGN |
| 4.5.2.1 E-UTRAN TDD - HRPD Cell Reselection: HRPD is of Lower Priority | Same as 4.5.1.1 | Same as 4.5.1.1 |
| 5.1.1 E-UTRAN FDD-FDD Handover intra frequency case | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$ | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 5.1.2 E-UTRAN TDD-TDD Handover intra frequency case | Same as 5.1.1 | Same as 5.1.1 |
| 5.1.3 E-UTRAN FDD-FDD Handover inter frequency case | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$ | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br>$N_{oc2}$ is the AWGN on cell 2 frequency<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN |
| 5.1.4 E-UTRAN TDD-TDD Handover inter frequency case | Same as 5.1.3 | Same as 5.1.3 |

| | | |
|---|---|---|
| 5.1.5 E-UTRAN FDD-FDD inter-frequency Handover with unknown target cell | Same as 5.1.1 | Same as 5.1.1 |
| 5.1.6 E-UTRAN TDD-TDD inter-frequency Handover with unknown target cell | Same as 5.1.1 | Same as 5.1.1 |
| 5.1.7 E-UTRAN FDD-TDD Handover inter frequency case | Same as 5.1.3 | Same as 5.1.3 |
| 5.1.8 E-UTRAN TDD-FDD Handover inter frequency case | Same as 5.1.3 | Same as 5.1.3 |
| 5.1.9 E-UTRAN FDD-FDD Intra frequency handover for 5MHz bandwidth | Same as 5.1.1 | Same as 5.1.1 |
| 5.1.10 E-UTRAN FDD-FDD Handover intra frequency handover for UE category 0 | Same as 5.1.1 | Same as 5.1.1 |
| 5.1.11 E-UTRAN HD-FDD Handover intra frequency handover for UE category 0 | Same as 5.1.1 | Same as 5.1.1 |
| 5.1.12 E-UTRAN TDD-TDD Handover intra frequency handover for UE category 0 | Same as 5.1.1 | Same as 5.1.1 |
| 5.1.13 E-UTRAN FDD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA | Same as 5.1.1 | Same as 5.1.1 |
| 5.1.14 E-UTRAN HD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA | Same as 5.1.1 | Same as 5.1.1 |
| 5.1.15 E-UTRAN TDD Intra frequency handover for Cat-M1 UEs in CEModeA | Same as 5.1.1 | Same as 5.1.1 |
| 5.2.1 E-UTRAN FDD - UTRAN FDD handover | E-UTRA cell<br>$N_{oc}$ ±1.0dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$I_{or} / I_{oc}$ ±0.3 dB<br>CPICH $Ec/I_{or}$ ±0.1 dB | Notes:<br><br>$N_{oc}$ is the AWGN on cell 1 (E-UTRA )frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br><br>$I_{oc}$ is the AWGN on cell 2 (UTRA) frequency<br>$I_{or} / I_{oc}$ is the ratio of cell 2 signal / AWGN<br>CPICH Ec / Ior is the fraction of cell 2 power assigned to the CPICH Physical channel |
| 5.2.2 E-UTRAN TDD - UTRAN FDD handover | Same as 5.2.1 | Same as 5.2.1 |
| 5.2.3 E-UTRAN FDD - GSM handover | E-UTRA Cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>GSM cell<br>Signal level ±0.7 dB | Note:<br>$N_{oc}$ is the AWGN on cell 1 frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br><br>Cell 2 (GSM) has only the wanted signal, without AWGN |
| 5.2.4 E-UTRA TDD – UTRA TDD handover | E-UTRA Cell:<br>$N_{oc}$ ± 1.0 dB averaged over $BW_{Config}$<br>$\hat{E}_s / N_{oc}$ ± 0.3 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ± 0.7 dB<br>$\hat{I}_{or} / I_{oc}$ ± 0.3 dB<br>P-CCPCH_$E_c$ / $I_{or}$ ± 0.1 dB<br>DwPCH_$E_c$ / $I_{or}$ ± 0.1 dB | Note:<br>$N_{oc}$ is the AWGN on cell 1 frequency<br>$\hat{E}_s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>$I_{oc}$ is the AWGN on cell 2 frequency<br>$\hat{I}_{or} / I_{oc}$ is the ratio of cell 2 signal / AWGN<br>P-CCPCH_$E_c$ / $I_{or}$ is the fraction of cell 2 power assigned to the P-CCPCH physical channel.<br>DwPCH_$E_c$ / $I_{or}$ is the fraction of cell 2 power assigned to the DwPCH channel |
| 5.2.5 E-UTRA FDD – UTRA TDD handover | Same as 5.2.4 | Same as 5.2.4 |
| 5.2.6 E-UTRA TDD - GSM handover | Same as 5.2.3 | Same as 5.2.3 |

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| 5.2.7 E-UTRAN FDD - UTRAN FDD handover: unknown target cell | E-UTRA cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$I_{or}/I_{oc}$ ±0.3 dB<br>CPICH Ec/Ior ±0.1 dB | Note:<br><br>$N_{oc}$ is the AWGN on cell 1 frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>Ioc is the AWGN on Cell 2 (UTRA) frequency<br>Ior/Ioc is the ratio of Cell 2 signal/AWGN<br>CPICH Ec/Ior is the fraction on Cell 2 power assigned to the CPICH physical channel |
| 5.2.8 E-UTRAN FDD - GSM handover: unknown target cell | Same as 5.2.3 | Same as 5.2.3 |
| 5.2.9 E-UTRAN TDD – GSM handover: unknown target cell | Same as 5.2.3 | Same as 5.2.3 |
| 5.2.10 E-UTRAN TDD - UTRAN TDD HO test: unknown target cell | E-UTRA cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>UTRA TDD cell<br>$I_{oc}$ ±0.7 dB<br>Ior/Ioc ±0.3 dB<br>PCCPCH_Ec /Ior ±0.1 dB<br>DwPCH_Ec /Ior ±0.1 dB | Note:<br><br>$N_{oc}$ is the AWGN on cell 1 frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>Ioc is the AWGN on Cell 2 (UTRA TDD) frequency<br>Ior/Ioc is the ratio of Cell 2 signal/AWGN<br>PCCPCH_Ec /Ior is the fraction of Cell 2 power assigned to the PCCPCH physical channel<br>DwPCH_Ec /Ior is the fraction of Cell 2 power assigned to the DwPCH physical channel |
| 5.2.11 E-UTRAN FDD – UTRAN FDD Handover for 5MHz Bandwidth | Same as 5.2.1 | Same as 5.2.1 |
| 5.3.1 RRC CONNECTED / Handover from E-UTRAN to non-3GPP RATs / E-UTRAN FDD – HRPD handover | E-UTRA cell<br>$N_{oc}$ ±0.7 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>HRPD cell<br>$I_{oc}$ ±2.0 dB<br>$\hat{I}_{or} / I_{oc}$ ±0.7 dB | Note:<br>$N_{oc}$ is the AWGN on cell 1 frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>$I_{oc}$ is the AWGN on cell 2 (HRPD) frequency<br>$I_{or} / I_{oc}$ is the ratio of cell 2 signal / AWGN |
| 5.3.5 E-UTRAN TDD - HRPD handover | Same as 5.3.1 | Same as 5.3.1 |
| 6.1.1 E-UTRAN FDD Intra-frequency RRC Re-establishment | Same as 5.1.1 | Same as 5.1.1 |
| 6.1.2 E-UTRAN FDD Inter-frequency RRC Re-establishment | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$ | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br>$N_{oc2}$ is the AWGN on cell 2 frequency<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN |
| 6.1.3 E-UTRAN TDD Intra-frequency RRC Re-establishment | Same as 6.1.1 | Same as 6.1.1 |
| 6.1.4 E-UTRAN TDD Inter-frequency RRC Re-establishment | Same as 6.1.3 | Same as 6.1.3 |
| 6.1.5 E-UTRAN FDD Intra-frequency RRC Re-establishment for 5MHz Bandwidth | Same as 6.1.1 | Same as 6.1.1 |
| 6.1.6 E-UTRAN FD-FDD Intra-frequency RRC Re-establishment for UE category 0 | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$ | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN |

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| 6.1.7 E-UTRAN HD-FDD Intra-frequency RRC Re-establishment for UE category 0 | Same as 6.1.6 | Same as 6.1.6 |
| 6.1.8 E-UTRAN TDD Intra-frequency RRC Re-establishment for UE category 0 | Same as 6.1.6 | Same as 6.1.6 |
| 6.1.9 E-UTRAN FD-FDD Intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA | Same as 6.1.1 | Same as 6.1.1 |
| 6.1.10 E-UTRAN HD-FDD Intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA | Same as 6.1.1 | Same as 6.1.1 |
| 6.1.11 E-UTRAN TDD Intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA | Same as 6.1.1 | Same as 6.1.1 |
| 6.2.1 E-UTRAN FDD - Contention Based Random Access Test | Test 1 and Test 2:<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>Uplink absolute power measurement ±1.0 dB<br><br>Uplink relative power measurement ±0.7 dB<br><br>±3Ts Uplink signal transmit timing relative to downlink | Note:<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 6.2.2 E-UTRAN FDD - Non Contention Based Random Access Test | Same as 6.2.1 | Same as 6.2.1 |
| 6.2.3 E-UTRAN TDD - Contention Based Random Access Test | Same as 6.2.1 | Same as 6.2.1 |
| 6.2.4 E-UTRAN TDD - Non Contention Based Random Access Test | Same as 6.2.1 | Same as 6.2.1 |
| 6.2.5 E-UTRAN FDD – Contention Based Random Access Test for 5MHz bandwidth | Same as 6.2.1 | Same as 6.2.1 |
| 6.2.6 E-UTRAN FDD – Non-contention Based Random Access Test for 5MHz bandwidth | Same as 6.2.1 | Same as 6.2.1 |
| 6.2.7 E-UTRAN FDD – Non-Contention Based Random Access Test for SCell in sTAG | Test 1, Test 2 and Test 3<br>$N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$<br><br>Uplink absolute power measurement ±1.0 dB<br><br>Uplink relative power measurement ±0.7 dB<br><br>±3Ts Uplink signal transmit timing relative to downlink | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br>$N_{oc2}$ is the AWGN on cell 2 frequency<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 6.2.8 E-UTRAN TDD – Non-Contention Based Random Access Test for SCell in sTAG | Same as 6.2.7 | Same as 6.2.7 |
| 6.2.12 E-UTRAN TDD Contention Based Random Access Test for Cat-M1 UEs in Normal Coverage | Same as 6.2.1 | Same as 6.2.1 |

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| 6.3.1 Redirection from E-UTRAN FDD to UTRAN FDD | E-UTRA cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$I_{or}/I_{oc}$ ±0.3 dB<br>CPICH Ec/Ior ±0.1 dB<br>SCH Ec/Ior ±0.1 dB | Note:<br><br>$N_{oc}$ is the AWGN on cell 1 frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br><br><br>Ioc is the AWGN on Cell 2 (UTRA) frequency<br>Ior/Ioc is the ratio of Cell 2 signal/AWGN<br>CPICH Ec/Ior and SCH_Ec/Ior are the fractions of Cell 2 power assigned to the CPICH and SCH physical channels |
| 6.3.2 Redirection from E-UTRAN TDD to UTRAN FDD | Same as 6.3.1 | Same as 6.3.1 |
| 6.3.3 Redirection from E-UTRAN FDD to GERAN when System Information is provided | E-UTRA Cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>GSM cell<br>Signal level ±0.7 dB | Note:<br>$N_{oc}$ is the AWGN on cell 1 frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br><br>Cell 2 (GSM) has only the wanted signal, without AWGN |
| 6.3.4 Redirection from E-UTRAN TDD to GERAN when System Information is provided | Same as 6.3.3 | Same as 6.3.3 |
| 6.3.5 E-UTRA TDD RRC connection release redirection to UTRA TDD | E-UTRA cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$I_{or}/I_{oc}$ ±0.3 dB<br>PCCPCH_Ec/Ior ±0.1 dB<br>DwPCH_Ec/Ior ±0.1 dB | Note:<br>$N_{oc}$ is the AWGN on cell 1 frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br><br>$I_{oc}$ is the AWGN on cell 2 frequency<br>$\hat{I}_{or} / I_{oc}$ is the ratio of cell 2 signal / AWGN<br>PCCPCH $E_c / I_{or}$ is the fraction of cell 2 power assigned to the PCCPCH Physical channel<br>DwPCH_Ec/Ior is the fraction of cell 2 power assigned to the DwPCH channel |
| 6.3.6 E-UTRA FDD RRC connection release redirection to UTRA TDD | Same as 6.3.5 | Same as 6.3.5 |
| 6.3.7 E-UTRA TDD RRC connection release redirection to UTRA TDD without SI provided | Same as 6.3.5 | Same as 6.3.5 |
| 6.3.8 E-UTRA FDD RRC connection release redirection to UTRA TDD without SI provided | Same as 6.3.5 | Same as 6.3.5 |
| 6.3.9 Redirection from E-UTRAN FDD to UTRAN FDD without System Information | Same as 6.3.1 | Same as 6.3.1 |
| 6.3.10 Redirection from E-UTRAN FDD to GERAN when System Information is not provided | Same as 6.3.3 | Same as 6.3.3 |
| 6.3.11 Redirection from E-UTRAN TDD to GERAN when System Information is not provided | Same as 6.3.3 | Same as 6.3.3 |
| 6.3.12 E-UTRAN TDD RRC connection release redirection to UTRAN FDD without SI provided | Same as 6.3.1 | Same as 6.3.1 |
| 7.1.1 E-UTRAN FDD - UE Transmit Timing Accuracy | $N_{oc}$ ±3.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB<br><br>±3Ts Uplink signal transmit timing relative to downlink<br><br>±0.5Ts relative during UE timing adjustment | Note:<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |

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| 7.1.2 E-UTRAN TDD - UE Transmit Timing Accuracy | Same as 7.1.1 | Same as 7.1.1 |
| 7.1.3 E-UTRAN FDD – UE Transmit Timing Accuracy Tests for SCell | $N_{oc1}$ ±3.0 dB averaged over $BW_{Config}$<br>$\hat{E}_{s1}$ / $N_{oc1}$ ±0.3 dB<br>$N_{oc2}$ ±3.0 dB averaged over $BW_{Config}$<br>$\hat{E}_{s2}$ / $N_{oc2}$ ±0.3 dB<br><br>±3$T_S$ Uplink signal transmit timing relative to downlink<br><br>±0.5$T_S$ relative during UE timing adjustment | Note:<br>$\hat{E}s$ / $N_{oc}$ is the ratio of cell signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 7.1.3_1 E-UTRAN FDD – UE Transmit Timing Accuracy Tests for SCell (Release 12 and forward) | Same as 7.1.3 | Same as 7.1.3 |
| 7.1.4 E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell | Same as 7.1.3 | Same as 7.1.3 |
| 7.1.4_1 E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell (Release 12 and forward) | Same as 7.1.3 | Same as 7.1.3 |
| 7.1.4A E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell for 20 MHz +10 MHz bandwidth | Same as 7.1.3 | Same as 7.1.3 |
| 7.1.5 E-UTRAN FDD - UE Transmit Timing Accuracy Tests for 5MHz Bandwidth | Same as 7.1.1 | Same as 7.1.1 |
| 7.1.6 E-UTRAN FDD - UE Transmit Timing Accuracy Tests for SCell in sTAG | $N_{oc1}$ ±3.0 dB averaged over $BW_{Config}$<br>$\hat{E}_{s1}$ / $N_{oc1}$ ±0.3 dB<br>$N_{oc2}$ ±3.0 dB averaged over $BW_{Config}$<br>$\hat{E}_{s2}$ / $N_{oc2}$ ±0.3 dB<br><br>±3Ts Uplink signal transmit timing relative to downlink<br><br>±0.5Ts relative during UE timing adjustment | Note:<br>$\hat{E}s$ / $N_{oc}$ is the ratio of cell signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 7.1.7 E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell in sTAG | Same as 7.1.6 | Same as 7.1.6 |
| 7.1.7A E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell in sTAG for 20MHz +20MHz bandwidth | Same as 7.1.6 | Same as 7.1.6 |
| 7.1.7B E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell in sTAG for 20MHz +10MHz bandwidth | Same as 7.1.6 | Same as 7.1.6 |
| 7.1.10 E-UTRAN FDD – UE Transmit Timing Accuracy Tests for Cat-M1 UE in CEModeA | Same as 7.1.1 | Same as 7.1.1 |
| 7.1.11 E-UTRAN HD-FDD – UE Transmit Timing Accuracy Tests for Cat-M1 UE in CEModeA | Same as 7.1.1 | Same as 7.1.1 |
| 7.2.1 E-UTRAN FDD - UE Timing Advance Adjustment Accuracy | $N_{oc1}$ ±3.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc1}$ ±0.3 dB<br><br>Timing Advance Adjustment: ±0.5$T_s$ | Note:<br>$\hat{E}s_1$ / $N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>The timing unit $T_S$ = 1/(15000 * 2048 ) seconds, as defined in TS.36.211 [9] |
| 7.2.2 E-UTRAN TDD - UE Timing Advance Adjustment Accuracy | Same as 7.2.1 | Same as 7.2.1 |
| 7.2.3 E-UTRAN FDD - UE Timing Advance Adjustment Accuracy Test for 5MHz Bandwidth | Same as 7.2.1 | Same as 7.2.1 |

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| 7.2.4 E-UTRAN FDD - UE Timing Advance Adjustment Accuracy Test for SCell in sTAG | $N_{oc1}$ ±3.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc1}$ ±0.3 dB<br>$N_{oc2}$ ±3.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc2}$ ±0.3 dB<br><br>Timing Advance Adjustment: ±0.5$T_s$ | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1$ / $N_{oc1}$ is the ratio of cell 1 signal / AWGN<br>$N_{oc2}$ is the AWGN on cell 2 frequency<br>$\hat{E}s_2$ / $N_{oc2}$ is the ratio of cell 2 signal / AWGN<br><br>The timing unit $T_S$ = 1/(15000 * 2048 ) seconds, as defined in TS.36.211 [9] |
| 7.2.5 E-UTRAN TDD - UE Timing Advance Adjustment Accuracy Test for SCell in sTAG | Same as 7.2.4 | Same as 7.2.4 |
| 7.2.5A E-UTRAN TDD - UE Timing Advance Adjustment Accuracy Test for SCell in sTAG for 20MHz +20MHz bandwidth | Same as 7.2.1 | Same as 7.2.1 |
| 7.2.5B E-UTRAN TDD - UE Timing Advance Adjustment Accuracy Test for SCell in sTAG for 20MHz +10MHz bandwidth | Same as 7.2.1 | Same as 7.2.1 |
| 7.2.6 E-UTRAN FDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA | Same as 7.2.1 | Same as 7.2.1 |
| 7.2.7 E-UTRAN HD-FDD UE Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA | Same as 7.2.1 | Same as 7.2.1 |
| 7.2.8 E-UTRAN TDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA | Same as 7.2.1 | Same as 7.2.1 |

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| 7.3.1 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync | ± 0.6dB (Subtest 1&2, AWGN conditions)<br><br>± 0.8dB (Subtest 3, Fading conditions, single antenna transmission)<br><br>± 0.9dB (Subtest 4, Fading conditions, two antenna transmission) | Subtests 1 & 2:<br>Overall system uncertainty for AWGN condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Effect of AWGN flatness and signal flatness<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>AWGN flatness and signal flatness ±2.0 dB<br><br>Subtests 3:<br>Overall system uncertainty for fading condition comprises three quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br><br>Items 1, 2 and 3 are assumed to be uncorrelated so can be root sum squared:<br>Test System uncertainty = SQRT (Average signal-to-noise ratio uncertainty $^2$ + Signal-to-noise ratio variation$^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Signal-to-noise ratio variation ±0.5 dB<br>Fading profile power uncertainty ±0.5 dB for single Tx<br><br>Subtest 4:<br>Same calculations as for subtest 3 but with Fading profile uncertainty of ±0.7 for two Tx. |
| 7.3.2 E-UTRAN FDD Radio Link Monitoring Test for In-sync | ± 0.8dB (Subtest 1, Fading conditions, single antenna transmission)<br><br>± 0.9dB (Subtest 2, Fading conditions, two antenna transmission) | Subtest 1:<br>See 7.3.1 subtest 3<br><br>Subtest 2:<br>See 7.3.1 subtest 4 |
| 7.3.3 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync | Same as 7.3.1 | Same as 7.3.1 |
| 7.3.4 E-UTRAN TDD Radio Link Monitoring Test for In-sync | Same as 7.3.2 | Same as 7.3.2 |
| 7.3.5 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync in DRX | ±0.9dB (Subtest 1, Fading conditions, two antenna transmission)<br><br>± 0.6dB (Subtest 2, AWGN conditions) | Subtest 1:<br>See 7.3.1, subtest 4<br><br>Subtest 2:<br>See 7.3.1, subtest 1 |
| 7.3.6 E-UTRAN FDD Radio Link Monitoring Test for In-sync in DRX | ± 0.6dB (AWGN conditions) | See 7.3.1, subtest 1 |
| 7.3.7 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync in DRX | ±0.9dB (Subtest 1, Fading conditions, two antenna transmission)<br><br>± 0.6dB (Subtest 2, AWGN conditions) | Subtest 1:<br>See 7.3.1, subtest 4<br><br>Subtest 2:<br>See 7.3.1, subtest 1 |
| 7.3.8 E-UTRAN TDD Radio Link Monitoring Test for In-sync in DRX | ± 0.6dB (AWGN conditions) | See 7.3.1, subtest 1 |

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| 7.3.9 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with Non MBSFN ABS (eICIC) | Cell 1: $\hat{E}s_1/N_{oc}$ ± 0.9 dB<br>Cell 2: $\hat{E}s_2/N_{oc}$ ± 0.9 dB<br>$\hat{E}s_2/\hat{E}s_1$ ± 1.1 dB | System uncertainties comprise a number of quantities which are selected according to their applicability in the equations below.<br><br>Uncertainties are assumed to be uncorrelated, so can be root sum squared.<br><br>For $\hat{E}s/N_{oc}$ uncertainty AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br>For $\hat{E}s_2/\hat{E}s_1$ uncertainty AWGN flatness effect has been excluded, because it would cancel out for the ratio $Es_2/Es_1$.<br><br>$\hat{E}s_1/N_{oc}$ or $\hat{E}s_2/N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$)<br><br>$\hat{E}s_2/\hat{E}s_1$ uncertainty = SQRT (Cell 1 SNR uncertainty $^2$ + Cell 1 Fading profile power uncertainty $^2$ + Cell 2 SNR uncertainty $^2$ + Cell 2 Fading profile power uncertainty $^2$)<br><br><u>Component uncertainties:</u><br>Cell 1 SNR uncertainty ($E_{s1}$ / $N_{oc}$ ratio before fading) ±0.3 dB<br>Cell 2 SNR uncertainty ($E_{s2}$ / $N_{oc}$ ratio before fading) ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for 2 Tx<br>AWGN flatness and signal flatness ±2.0 dB |
| 7.3.10 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with Non MBSFN ABS (eICIC) | Same as 7.3.9 | Same as 7.3.9 |

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| 7.3.11 E-UTRAN FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with Non MBSFN ABS (eICIC) | Cell 1: $\hat{E}s_1/N_{oc}$ ± 0.9 dB<br>Cell 2: $\hat{E}s_2/N_{oc}$ ± 0.9 dB<br>$\hat{E}s_2/\hat{E}s_1$ ± 1.1 dB | System uncertainties comprise a number of quantities which are selected according to their applicability in the equations below.<br><br>Uncertainties are assumed to be uncorrelated, so can be root sum squared.<br><br>For $\hat{E}s/N_{oc}$ uncertainty AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br>For $\hat{E}s_2/\hat{E}s_1$ uncertainty AWGN flatness effect has been excluded, because it would cancel out for the ratio $Es_2/Es_1$.<br><br>$\hat{E}s_1/N_{oc}$ or $\hat{E}s_2/N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$)<br><br>$\hat{E}s_2/\hat{E}s_1$ uncertainty = SQRT (Cell 1 SNR uncertainty $^2$ + Cell 1 Fading profile power uncertainty $^2$ + Cell 2 SNR uncertainty $^2$ + Cell 2 Fading profile power uncertainty $^2$)<br><br><u>Component uncertainties:</u><br>Cell 1 SNR uncertainty ($E_{s1}$ / $N_{oc}$ ratio before fading) ±0.3 dB<br>Cell 2 SNR uncertainty ($E_{s2}$ / $N_{oc}$ ratio before fading) ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for 2 Tx<br>AWGN flatness and signal flatness ±2.0 dB |
| 7.3.12 E-UTRAN TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with Non MBSFN ABS (eICIC) | Same as 7.3.11 | Same as 7.3.11 |
| 7.3.13 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) | Same as 7.3.9 | Same as 7.3.9 |
| 7.3.14 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) | Same as 7.3.9 | Same as 7.3.9 |
| 7.3.15 E-UTRAN FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) | Same as 7.3.11 | Same as 7.3.11 |
| 7.3.16 E-UTRAN TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) | Same as 7.3.11 | Same as 7.3.11 |

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| 7.3.17 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non MBSFN ABS (feICIC) | Cell 1: $\hat{E}s_1/N_{oc}$ ± 0.9 dB<br>Cell 2: $\hat{E}s_2/N_{oc}$ ± 0.9 dB<br>Cell 3: $\hat{E}s_3/N_{oc}$ ± 0.9 dB<br>$\hat{E}s_2/\hat{E}s_1$ ± 1.1 dB<br>$\hat{E}s_3/\hat{E}s_1$ ± 1.1 dB | System uncertainties comprise a number of quantities which are selected according to their applicability in the equations below.<br><br>Uncertainties are assumed to be uncorrelated, so can be root sum squared.<br><br>For $\hat{E}s/N_{oc}$ uncertainty AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br>For $\hat{E}s_2/\hat{E}s_1$ and $\hat{E}s_3/\hat{E}s_1$ uncertainty AWGN flatness effect has been excluded, because it would cancel out for the ratio $Es_x/Es_1$.<br><br>$\hat{E}s_1/N_{oc}$, $\hat{E}s_2/N_{oc}$ or $\hat{E}s_3/N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$)<br><br>$\hat{E}s_2/\hat{E}s_1$ or $\hat{E}s_3/\hat{E}s_1$ uncertainty = SQRT (Cell 1 SNR uncertainty $^2$ + Cell 1 Fading profile power uncertainty $^2$ + Cell 2 SNR uncertainty $^2$ + Cell 2 Fading profile power uncertainty $^2$)<br><br><u>Component uncertainties:</u><br>Cell 1 SNR uncertainty ($E_{s1}$ / $N_{oc}$ ratio before fading) ±0.3 dB<br>Cell 2 SNR uncertainty ($E_{s2}$ / $N_{oc}$ ratio before fading) ±0.3 dB<br>Cell 3 SNR uncertainty ($E_{s3}$ / $N_{oc}$ ratio before fading) ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for 2 Tx<br>AWGN flatness and signal flatness ±2.0 dB |
| 7.3.18 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non MBSFN ABS (feICIC) | Same as 7.3.17 | Same as 7.3.17 |
| 7.3.19 E-UTRAN FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non MBSFN ABS (feICIC) | Same as 7.3.17 | Same as 7.3.17 |
| 7.3.20 E-UTRAN TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non MBSFN ABS (feICIC) | Same as 7.3.17 | Same as 7.3.17 |
| 7.3.21 E-UTRAN FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and MBSFN ABS (feICIC) | Same as 7.3.17 | Same as 7.3.17 |
| 7.3.22 E-UTRAN TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and MBSFN ABS (feICIC) | Same as 7.3.17 | Same as 7.3.17 |
| 7.3.23 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync for 5MHz Bandwidth | ± 0.9dB (Subtest 4, Fading conditions, two antenna transmission) | See 7.3.1, subtest 4 |

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| 7.3.24 E-UTRAN FDD Radio Link Monitoring Test for In-sync for 5MHz Bandwidth | ± 0.9dB (Subtest 2, Fading conditions, two antenna transmission) | See 7.3.2, subtest 2 |
| 7.3.25 E-UTRAN FDD Radio Link Monitoring Test for In-sync in DRX for 5MHz Bandwidth | ±0.6dB (AWGN conditions) | See 7.3.1, subtest 1 |
| 7.3.26 E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync for UE category 0 | ± 0.9dB (Fading conditions, two antenna transmission) | Overall system uncertainty for fading condition comprises three quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br><br>Items 1, 2 and 3 are assumed to be uncorrelated so can be root sum squared:<br>Test System uncertainty = SQRT (Average signal-to-noise ratio uncertainty $^2$ + Signal-to-noise ratio variation$^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Signal-to-noise ratio variation ±0.5 dB<br>Fading profile power uncertainty ±0.7 dB for two Tx |
| 7.3.27 E-UTRAN FD-FDD Radio Link Monitoring Test for In-sync for UE category 0 | Same as 7.3.26 | Same as 7.3.26 |
| 7.3.28 E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category 0 | ± 0.6dB (AWGN conditions) | Overall system uncertainty for AWGN condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Effect of AWGN flatness and signal flatness<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>AWGN flatness and signal flatness ±2.0 dB |
| 7.3.29 E-UTRAN FD-FDD Radio Link Monitoring Test for In-sync in DRX for UE category 0 | Same as 7.3.28 | Same as 7.3.28 |
| 7.3.30 E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync for UE category 0 | Same as 7.3.26 | Same as 7.3.26 |
| 7.3.31 E-UTRAN HD-FDD Radio Link Monitoring Test for In-sync for UE category 0 | Same as 7.3.26 | [Same as 7.3.26 |
| 7.3.32 E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category 0 | Same as 7.3.28 | Same as 7.3.28 |
| 7.3.33 E-UTRAN HD-FDD Radio Link Monitoring Test for In-sync in DRX for UE category 0 | Same as 7.3.28 | Same as 7.3.28 |
| 7.3.34 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync for UE category 0 | Same as 7.3.26 | Same as 7.3.26 |
| 7.3.35 E-UTRAN TDD Radio Link Monitoring Test for In-sync for UE category 0 | Same as 7.3.26 | Same as 7.3.26 |
| 7.3.36 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category 0 | Same as 7.3.28 | Same as 7.3.28 |

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| 7.3.37 E-UTRAN TDD Radio Link Monitoring Test for In-sync in DRX for UE category 0 | Same as 7.3.28 | Same as 7.3.28 |
| 7.3.38 E-UTRAN FDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC | ± 0.9dB (Fading conditions, two antenna transmission) | Overall system uncertainty for fading condition comprises three quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br><br>Items 1, 2 and 3 are assumed to be uncorrelated so can be root sum squared:<br>Test System uncertainty = SQRT (Average signal-to-noise ratio uncertainty $^2$ + Signal-to-noise ratio variation$^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Signal-to-noise ratio variation ±0.5 dB<br>Fading profile power uncertainty ±0.7 dB for two Tx |
| 7.3.39 E-UTRAN FDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in asynchronous DC | Same as 7.3.38 | Same as 7.3.38 |
| 7.3.40 E-UTRAN TDD-TDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC | Same as 7.3.38 | Same as 7.3.38 |
| 7.3.41 E-UTRAN FDD-FDD Radio Link Monitoring Test for In-sync in DRX in synchronous dual connectivity | ± 0.6dB (AWGN conditions) | Overall system uncertainty for AWGN condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Effect of AWGN flatness and signal flatness<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>AWGN flatness and signal flatness ±2.0 dB |
| 7.3.42 E-UTRAN FDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in asynchronous DC | Same as 7.3.41 | Same as 7.3.41 |
| 7.3.43 E-UTRAN TDD-TDD Radio Link Monitoring Test for In-sync in DRX in synchronous dual connectivity | Same as 7.3.41 | Same as 7.3.41 |
| 7.3.48 E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync for Cat-M1 UE in CEMode A | ± 0.9dB (Fading conditions, two antenna transmission) | Overall system uncertainty for fading condition comprises three quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br><br>Items 1, 2 and 3 are assumed to be uncorrelated so can be root sum squared:<br>Test System uncertainty = SQRT (Average signal-to-noise ratio uncertainty $^2$ + Signal-to-noise ratio variation$^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Signal-to-noise ratio variation ±0.5 dB<br>Fading profile power uncertainty ±0.7 dB for one Tx |

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| 7.3.49 E-UTRAN FD-FDD Radio Link Monitoring Test for In-Sync for Cat-M1 UE in CEMode A | ± 0.9dB (Fading conditions, two antenna transmission) | Overall system uncertainty for fading condition comprises three quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br><br>Items 1, 2 and 3 are assumed to be uncorrelated so can be root sum squared:<br>Test System uncertainty = SQRT (Average signal-to-noise ratio uncertainty $^2$ + Signal-to-noise ratio variation$^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Signal-to-noise ratio variation ±0.5 dB<br>Fading profile power uncertainty ±0.7 dB for one Tx |
| 7.3.50: E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-Sync in DRX for UE Category M1 Configured in CEMode A | Same as 7.3.28 | Same as 7.3.28 |
| 7.3.51 E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE Category M1 Configured in CEMode A | Same as 7.3.28 | Same as 7.3.28 |
| 7.3.52 E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync for Cat-M1 UE in CEMode A | ± 0.9dB (Fading conditions, two antenna transmission) | Overall system uncertainty for fading condition comprises three quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br><br>Items 1, 2 and 3 are assumed to be uncorrelated so can be root sum squared:<br>Test System uncertainty = SQRT (Average signal-to-noise ratio uncertainty $^2$ + Signal-to-noise ratio variation$^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Signal-to-noise ratio variation ±0.5 dB<br>Fading profile power uncertainty ±0.7 dB for one Tx |
| 7.3.53 E-UTRAN HD-FDD Radio Link Monitoring Test for In-Sync for Cat-M1 UE in CEMode A | ± 0.9dB (Fading conditions, two antenna transmission) | Overall system uncertainty for fading condition comprises three quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br><br>Items 1, 2 and 3 are assumed to be uncorrelated so can be root sum squared:<br>Test System uncertainty = SQRT (Average signal-to-noise ratio uncertainty $^2$ + Signal-to-noise ratio variation$^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Signal-to-noise ratio variation ±0.5 dB<br>Fading profile power uncertainty ±0.7 dB for one Tx |
| 7.3.54: E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE Category M1 Configured in CEMode A | Same as 7.3.28 | Same as 7.3.28 |
| 7.3.55: E-UTRAN HD-FDD Radio Link Monitoring Test for In-sync in DRX for UE Category M1 configured in CEMode A | Same as 7.3.28 | Same as 7.3.28 |

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| 7.5.1 E-UTRAN FDD – UE ProSe Direct Discovery Transmission Timing Accuracy Test | $N_{oc}$ ±3.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB<br><br>±3Ts Uplink signal transmit timing relative to downlink | Note:<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 7.5.4 E-UTRAN FDD – UE ProSe Direct Communication Transmission Timing Accuracy Test | $N_{oc}$ ±3.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB<br><br>±3Ts Uplink signal transmit timing relative to downlink | Note:<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 8.1.1 E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc}$ ±0.6 dB averaged over $BW_{Config}$ | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN<br><br>$\hat{E}s / N_{oc}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty$^2$ + Fading profile power uncertainty$^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB |
| 8.1.2 E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.3 E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.5 E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$ | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 8.1.6 E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX | Same as 8.1.5 | Same as 8.1.5 |
| 8.1.7 UE Measurement Procedures / E-UTRAN FDD-FDD Intra-frequency event-triggered reporting under time domain measurement resource restriction with non-MBSFN ABS (eICIC) | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc}$ ±0.6 dB averaged over $BW_{Config}$ | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN<br><br>$\hat{E}s / N_{oc}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty$^2$ + Fading profile power uncertainty$^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB |

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| 8.1.8 E-UTRAN FDD-FDD Intra-Frequency Event-Triggered Reporting under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_3 / N_{oc}$ ±0.6 dB averaged over $BW_{Config}$ | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of Cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of Cell 2 signal / AWGN<br>$\hat{E}s_3 / N_{oc}$ is the ratio of Cell 3 signal / AWGN<br><br>$\hat{E}s / N_{oc}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty$^2$ + Fading profile power uncertainty$^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB |
| 8.1.9 E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells for 5MHz Bandwidth | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.10 E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX for 5MHz Bandwidth | Same as 8.1.3 | Same as 8.1.3 |
| 8.1.11 E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for UE category 0 | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc}$ ±0.6 dB averaged over $BW_{Config}$ | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN<br><br>$\hat{E}s / N_{oc}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty$^2$ + Fading profile power uncertainty$^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB |
| 8.1.12 E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for UE category 0 | Same as 8.1.11 | Same as 8.1.11 |
| 8.1.13 E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX for UE category 0 | Same as 8.1.11 | Same as 8.1.11 |
| 8.1.17 E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for UE category 0 | Same as 8.1.11 | Same as 8.1.11 |
| 8.1.18 E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX for UE category 0 | Same as 8.1.11 | Same as 8.1.11 |
| 8.1.19 E-UTRAN FD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps for UE category 0 | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$ | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 8.1.20 E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0 | Same as 8.1.19 | Same as 8.1.19 |

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| 8.1.21 E-UTRAN HD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps for UE category 0 | Same as 8.1.19 | Same as 8.1.19 |
| 8.1.22 E-UTRAN HD- FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0 | Same as 8.1.19 | Same as 8.1.19 |
| 8.1.33 E-UTRAN HD-FDD Intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeB | TBD | TBD |
| 8.1.34 E-UTRAN HD-FDD Intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB | TBD | TBD |
| 8.1.35 E-UTRAN TDD Intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB | TBD | TBD |
| 8.2.1 E-UTRAN TDD-TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells | Same as 8.1.1 | Same as 8.1.1 |
| 8.2.2 E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX | Same as 8.1.1 | Same as 8.1.1 |
| 8.2.3 E-UTRAN TDD-TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$ | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 8.2.4 E-UTRAN TDD-TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX | Same as 8.2.3 | Same as 8.2.3 |
| 8.2.5 UE Measurement Procedures / E-UTRAN TDD-TDD Intra-frequency event-triggered reporting under time domain measurement resource restriction with non-MBSFN ABS (eICIC) | Same as 8.1.7 | Same as 8.1.7 |
| 8.2.6 E-UTRAN TDD-TDD Intra-Frequency Event-Triggered Reporting under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | Same as 8.1.8 | Same as 8.1.8 |
| 8.2.7 E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps for UE category 0 | Same as 8.1.19 | Same as 8.1.19 |
| 8.2.8 E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0 | Same as 8.1.19 | Same as 8.1.19 |

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| 8.3.1 E-UTRAN FDD-FDD Inter-frequency event triggered reporting under fading propagation conditions in asynchronous cells | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.6 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$ | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on cell 2 frequency<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br><br>Each $\hat{E}s / N_{oc}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB |
| 8.3.2 E-UTRAN FDD-FDD Inter-frequency event triggered reporting when DRX is used under fading propagation conditions in asynchronous cells | Same as 8.3.1 | Same as 8.3.1 |
| 8.3.3 E-UTRAN FDD-FDD Inter frequency event triggered reporting under AWGN propagation conditions in asynchronous cells with DRX when L3 filtering is used | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$ | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br>$N_{oc2}$ is the AWGN on cell 2 frequency<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN |
| 8.3.4 E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$ | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br>$N_{oc2}$ is the AWGN on cell 2 frequency<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN |
| 8.3.5 E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX | Same as 8.3.4 | Same as 8.3.4 |
| 8.3.6 E-UTRAN FDD-FDD Inter-frequency event triggered reporting without measurement gaps under AWGN propagation conditions in asynchronous cells | Same as 8.3.4 | Same as 8.3.4 |

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| 8.4.1 E-UTRAN TDD-TDD Inter-frequency event triggered reporting under fading propagation conditions in synchronous cells | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.6 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$ | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on cell 2 frequency<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br><br>Each $\hat{E}s / N_{oc}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB |
| 8.4.2 E-UTRAN TDD-TDD Inter-frequency event triggered reporting when DRX is used under fading propagation conditions in synchronous cells | Same as 8.4.1 | Same as 8.4.1 |
| 8.4.3 E-UTRAN TDD-TDD inter-frequency event triggered reporting under AWGN propagation conditions in synchronous cells with DRX when L3 filtering is used | Same as 8.3.3 | Same as 8.3.3 |
| 8.4.4 E-UTRAN TDD-TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$ | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br>$N_{oc2}$ is the AWGN on cell 2 frequency<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN |
| 8.4.5 E-UTRAN TDD-TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX | Same as 8.4.4 | Same as 8.4.4 |
| 8.5.1 E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions | E-UTRAN cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.6 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$I_{or}/I_{oc}$ ±0.6 dB<br>CPICH Ec/Ior ±0.1 dB | Note:<br>$N_{oc}$ is the AWGN on Cell 1 frequency<br>$\hat{E}s / N_{oc}$ is the ratio of Cell 1 signal / AWGN<br><br>$\hat{E}s / N_{oc}$ uncertainty or $I_{or}/I_{oc}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB<br><br>Ioc is the AWGN on Cell 2 (UTRA) frequency<br>Ior/Ioc is the ratio of Cell 2 signal/AWGN<br>CPICH Ec/Ior is the fraction of Cell 2 power assigned to the CPICH physical channel |

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| 8.5.2 E-UTRAN FDD - UTRAN FDD SON ANR cell search reporting under AWGN propagation conditions | E-UTRA cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$I_{or}/I_{oc}$ ±0.3 dB<br>CPICH Ec/Ior ±0.1 dB<br>SCH Ec/Ior ±0.1 dB | Note:<br>$N_{oc}$ is the AWGN on cell 1 frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>Ioc is the AWGN on Cell 2 (UTRA) frequency<br>Ior/Ioc is the ratio of Cell 2 signal/AWGN<br>CPICH Ec/Ior is the fraction of Cell 2 power assigned to the CPICH physical channel<br>SCH Ec/Ior is the fraction of Cell 2 power assigned to the SCH physical channel |
| 8.5.3 E-UTRAN FDD - UTRAN FDD event triggered reporting when DRX is used under fading propagation conditions | Same as 8.5.1 | Same as 8.5.1 |
| 8.5.4 E-UTRAN FDD - UTRAN FDD enhanced cell identification under AWGN propagation conditions | E-UTRA cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}_s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$I_{or}/I_{oc}$ ±0.3 dB<br>CPICH $E_c/I_{or}$ ±0.1 dB<br>SCH $E_c/I_{or}$ ±0.1 dB | Note:<br>$N_{oc}$ is the AWGN on cell 1 frequency<br>$\hat{E}_s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>$I_{oc}$ is the AWGN on Cell 2 (UTRA) frequency<br>$I_{or}/I_{oc}$ is the ratio of Cell 2 signal/AWGN<br>CPICH $E_c/I_{or}$ is the fraction of Cell 2 power assigned to the CPICH physical channel<br>SCH $E_c/I_{or}$ is the fraction of Cell 2 power assigned to the SCH physical channel |
| 8.5.6 E-UTRAN FDD - UTRAN FDD event triggered reporting without measurement gaps under AWGN propagation conditions | Same as 8.5.4 | Same as 8.5.4 |
| 8.5.7 E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions for 5MHz Bandwidth | Same as 8.5.1 | Same as 8.5.1 |
| 8.6.1 E-UTRAN TDD - UTRAN FDD event triggered reporting under fading propagation conditions | Same as 8.5.1 | Same as 8.5.1 |
| 8.7.1 E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions | E-UTRA cell<br>$N_{oc}$ ±0.7 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.6 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$I_{or}/I_{oc}$ ±0.6 dB<br>PCCPCH Ec/Ior ±0.1 dB<br>DwPCH_Ec/Ior ±0.1 dB | Notes:<br><br>$N_{oc}$ is the AWGN on cell 1 (E-UTRA )frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>$\hat{E}s / N_{oc}$ uncertainty or $I_{or} / I_{oc}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ uncertainty or $I_{or} / I_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB<br><br>$I_{oc}$ is the AWGN on cell 2 (UTRA) frequency<br>$I_{or} / I_{oc}$ is the ratio of cell 2 signal / AWGN<br>PCCPCH $E_c / I_{or}$ is the fraction of cell 2 power assigned to the PCCPCH Physical channel<br>DwPCH_Ec/Ior is the fraction of cell 2 power assigned to the DwPCH channel |
| 8.7.2 E-UTRAN TDD - UTRAN TDD cell search when DRX is used under fading propagation conditions | Same as 8.7.1 | Same as 8.7.1 |

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| 8.7.3 E-UTRAN TDD - UTRAN TDD SON ANR cell search reporting under AWGN propagation conditions | E-UTRA cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$I_{or}/I_{oc}$ ±0.3 dB<br>PCCPCH Ec/Ior ±0.1 dB<br>DwPCH_Ec/Ior ±0.1 dB | Note:<br>$N_{oc}$ is the AWGN on cell 1 frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>$I_{oc}$ is the AWGN on cell 2 (UTRA) frequency<br>$I_{or} / I_{oc}$ is the ratio of cell 2 signal / AWGN<br>PCCPCH $E_c / I_{or}$ is the fraction of cell 2 power assigned to the PCCPCH Physical channel<br>DwPCH_Ec/Ior is the fraction of cell 2 power assigned to the DwPCH channel |
| 8.7.4 E-UTRAN TDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions | E-UTRA cell<br>Noc ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s$ / Noc ±0.3 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>Ioc ±0.7 dB<br>Ior/Ioc ±0.3 dB<br>PCCPCH Ec/Ior ±0.1 dB<br>DwPCH_Ec/Ior ±0.1 dB | Note:<br>Noc is the AWGN on cell 1 frequency<br>$\hat{E}s$ / Noc is the ratio of cell 1 signal / AWGN<br><br>Ioc is the AWGN on cell 2 (UTRA) frequency<br>Ior / Ioc is the ratio of cell 2 signal / AWGN<br>PCCPCH Ec/Ior is the fraction of cell 2 power assigned to the PCCPCH Physical channel<br>DwPCH_Ec/Ior is the fraction of cell 2 power assigned to the DwPCH channel |
| 8.8.1 E-UTRAN FDD - GSM event triggered reporting in AWGN | E-UTRA Cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>GSM cell<br>Signal level ±0.7 dB | Note:<br>$N_{oc}$ is the AWGN on cell 1 frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>Cell 2 (GSM) has only the wanted signal, without AWGN |
| 8.8.2 E-UTRAN FDD- GSM event triggered reporting when DRX is used in AWGN | Same as 8.8.1 | Same as 8.8.1 |
| 8.9.1 E-UTRAN FDD - UTRAN TDD event triggered reporting under fading propagation conditions | E-UTRA cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.6 dB averaged over $BW_{Config}$<br><br>UTRA TDD cell<br>$I_{oc}$ ±0.7 dB<br>Ior/Ioc ±0.6 dB<br>PCCPCH_Ec /Ior ±0.1 dB<br>DwPCH_Ec /Ior ±0.1 dB | $N_{oc}$ is the AWGN on cell 1 frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>Ioc is the AWGN on Cell 2 (UTRA TDD) frequency<br>Ior/Ioc is the ratio of Cell 2 signal/AWGN<br>PCCPCH_Ec /Ior is the fraction on Cell 2 power assigned to the CPCCPCH physical channel<br>DwPCH_Ec /Ior is the fraction on Cell 2 power assigned to the DwPCH physical channel<br><br>$\hat{E}s / N_{oc}$ and Ior/Ioc uncertainty for fading condition comprise two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty |
| 8.9.2 E-UTRAN FDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions | Same as 8.7.4 | Same as 8.7.4 |
| 8.10.1 E-UTRAN TDD - GSM event triggered reporting in AWGN | Same as 8.8.1 | Same as 8.8.1 |
| 8.10.2 E-UTRAN TDD- GSM event triggered reporting when DRX is used in AWGN | Same as 8.8.2 | Same as 8.8.2 |

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| 8.11.1 Multiple E-UTRAN FDD-FDD Inter-frequency event triggered reporting under fading propagation conditions | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br><br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$<br><br>$N_{oc3}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_3 / N_{oc3}$ ±0.6 dB averaged over $BW_{Config}$ | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on cell 2 frequency<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br><br>$N_{oc3}$ is the AWGN on cell 3 frequency<br>$\hat{E}s3 / N_{oc3}$ is the ratio of cell 3 signal / AWGN<br><br>$\hat{E}s_2 / N_{oc2}$ uncertainty or $\hat{E}s_3 / N_{oc3}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB |
| 8.11.2 E-UTRAN TDD - E-UTRAN TDD and E-UTRAN TDD Inter-frequency event triggered reporting under fading propagation conditions | Same as 8.11.1 | Same as 8.11.1 |
| 8.11.3 E-UTRAN FDD-FDD Inter-frequency and UTRAN FDD event triggered reporting under fading propagation conditions | E-UTRA cells<br>$N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br><br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$I_{or}/I_{oc}$ ±0.6 dB<br>CPICH Ec/Ior ±0.1 dB | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on cell 2 frequency<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br><br>Ioc is the AWGN on Cell 3 (UTRA) frequency<br>Ior/Ioc is the ratio of Cell 3 signal/AWGN<br>CPICH Ec/Ior is the fraction of Cell 3 power assigned to the CPICH physical channel<br><br>$\hat{E}s_2 / N_{oc2}$ uncertainty or $I_{or}/I_{oc}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB |

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| 8.11.4 InterRAT E-UTRA TDD to E-UTRA TDD and UTRA TDD cell search | E-UTRA cells<br>$N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br><br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$I_{or}/I_{oc}$ ±0.6 dB<br>PCCPCH Ec/Ior ±0.1 dB<br>DwPCH_Ec/Ior ±0.1 dB | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on cell 2 frequency<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br><br>$I_{oc}$ is the AWGN on cell 3 (UTRA) frequency<br>$I_{or} / I_{oc}$ is the ratio of cell 3 signal / AWGN<br>PCCPCH $E_c / I_{or}$ is the fraction of cell 3 power assigned to the PCCPCH Physical channel<br>DwPCH_Ec/Ior is the fraction of cell 3 power assigned to the DwPCH channel<br><br>$\hat{E}s_2 / N_{oc2}$ uncertainty or $I_{or}/I_{oc}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ or $I_{or}/I_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB |
| 8.11.5 Combined E-UTRAN FDD - E-UTRA FDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions | E-UTRA cells<br>$N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.6 dB averaged over $BW_{Config}$<br><br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$<br><br>GSM cell<br>Signal level ±0.7 dB | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on cell 2 frequency<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br><br>Cell 3 (GSM) has only the wanted signal, without AWGN<br><br>$\hat{E}s_1 / N_{oc1}$ uncertainty or $\hat{E}s_2 / N_{oc2}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB |
| 8.11.6 Combined E-UTRAN TDD - E-UTRA TDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions | Same as 8.11.5 | Same as 8.11.5 |
| 8.14.1 E-UTRAN TDD-FDD Inter-frequency event triggered reporting under fading propagation conditions in asynchronous cells | Same as 8.3.1 | Same as 8.3.1 |

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| 8.14.2 E-UTRAN TDD-FDD Inter-frequency event triggered reporting when DRX is used under fading propagation conditions in asynchronous cells | Same as 8.3.1 | Same as 8.3.1 |
| 8.14.3 E-UTRAN TDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | Same as 8.4.4 | Same as 8.4.4 |
| 8.15.1 E-UTRAN FDD-TDD Inter-frequency event triggered reporting under fading propagation conditions in asynchronous cells | Same as 8.4.1 | Same as 8.4.1 |
| 8.15.2 E-UTRAN FDD-TDD Inter-frequency event triggered reporting when DRX is used under fading propagation conditions in asynchronous cells | Same as 8.4.1 | Same as 8.4.1 |
| 8.15.3 E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | Same as 8.4.4 | Same as 8.4.4 |
| 8.16.1 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.6 dB averaged over $BW_{Config}$<br><br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_3 / N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$<br><br>Time alignment error cell 2 rel. to cell 1: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br><br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br><br>$\hat{E}s_3 / N_{oc2}$ is the ratio of cell 3 signal / AWGN<br><br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ uncertainty and $\hat{E}s_3 / N_{oc2}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty$^2$ + Fading profile power uncertainty$^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 8.16.2 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX | Same as 8.16.1 | Same as 8.16.1 |

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| 8.16.3 E-UTRAN FDD-FDD Event triggered reporting on deactivated SCell with PCell interruption in non-DRX | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br><br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2$ / $N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_3$ / $N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$<br><br>Time alignment error cell 2 rel. to cell 1: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1$ / $N_{oc1}$ is the ratio of Cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br><br>$\hat{E}s_2$ / $N_{oc2}$ is the ratio of Cell 2 signal / AWGN<br><br>$\hat{E}s_3$ / $N_{oc2}$ is the ratio of Cell 3 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 8.16.4 E-UTRAN TDD-TDD Event triggered reporting on deactivated SCell with PCell interruption in non-DRX | Same as 8.16.3 | Same as 8.16.3 |
| 8.16.5 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX for 20 MHz bandwidth | Same as 8.16.1 | Same as 8.16.1 |
| 8.16.6 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 20 MHz bandwidth | Same as 8.16.1 | Same as 8.16.1 |
| 8.16.7 E-UTRA FDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 20 MHz bandwidth | Same as 8.16.3 | Same as 8.16.3 |
| 8.16.8 E-UTRAN TDD Event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 20 MHz bandwidth | Same as 8.16.3 | Same as 8.16.3 |
| 8.16.9 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX for 10MHz+5MHz | Same as 8.16.1 | Same as 8.16.1 |
| 8.16.10 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 10MHz+5MHz | Same as 8.16.1 | Same as 8.16.1 |
| 8.16.11 E-UTRAN FDD event triggered reporting on deactivating SCell with PCell interruption in non-DRX for 10MHz+5MHz | Same as 8.16.3 | Same as 8.16.3 |
| 8.16.12 E-UTRAN TDD event triggered reporting on deactivating SCell with PCell interruption in non-DRX for 10MHz+5MHz | Same as 8.16.3 | Same as 8.16.3 |
| 8.16.13 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX for 5MHz+5MHz | Same as 8.16.1 | Same as 8.16.1 |
| 8.16.14 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 5MHz+5MHz | Same as 8.16.1 | Same as 8.16.1 |
| 8.16.15 E-UTRAN FDD event triggered reporting on deactivating SCell with PCell interruption in non-DRX for 5MHz+5MHz | Same as 8.16.3 | Same as 8.16.3 |
| 8.16.16 E-UTRAN TDD event triggered reporting on deactivating SCell with PCell interruption in non-DRX for 5MHz+5MHz | Same as 8.16.3 | Same as 8.16.3 |
| 8.16.17 E-UTRAN FDD activation and deactivation of known SCell in non-DRX | Same as 8.16.37 | Same as 8.16.37 |

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| 8.16.17A E-UTRAN FDD activation and deactivation of known SCell in non-DRX for 20MHz | Same as 8.16.17 | Same as 8.16.17 |
| 8.16.18 E-UTRAN TDD activation and deactivation of known SCell in non-DRX | Same as 8.16.17 | Same as 8.16.17 |
| 8.16.18 E-UTRAN TDD activation and deactivation of known SCell in non-DRX for 20MHz | Same as 8.16.17 | Same as 8.16.17 |
| 8.16.21 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 20MHz+10MHz bandwidth | Same as 8.16.1 | Same as 8.16.1 |
| 8.16.22 E-UTRAN TDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 20MHz+10MHz bandwidth | Same as 8.16.3 | Same as 8.16.3 |
| 8.16.23 E-UTRAN TDD-FDD CA Event Triggered Reporting Under Deactivated SCell in Non-DRX with PCell in FDD | Same as 8.16.1 | Same as 8.16.1 |
| 8.16.24 E-UTRAN TDD-FDD CA Event Triggered Reporting Under Deactivated SCell in Non-DRX with PCell in TDD | Same as 8.16.1 | Same as 8.16.1 |
| 8.16.25 E-UTRAN TDD-FDD CA Event triggered reporting on deactivated SCell with PCell interruption in non-DRX with PCell in FDD | Same as 8.16.3 | Same as 8.16.3 |
| 8.16.26 E-UTRAN TDD-FDD CA Event triggered reporting on deactivated SCell with PCell interruption in non-DRX with PCell in TDD | Same as 8.16.3 | Same as 8.16.3 |
| 8.16.27 E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting under Deactivated SCells in Non-DRX with PCell in FDD | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br><br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$<br><br>$N_{oc3}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_3 / N_{oc3}$ ±0.6 dB averaged over $BW_{Config}$<br>$\hat{E}s_4 / N_{oc3}$ ±0.6 dB averaged over $BW_{Config}$<br><br>Time alignment errors: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br><br>$N_{oc3}$ is the AWGN on frequency 3<br>$\hat{E}s_3 / N_{oc3}$ is the ratio of cell 3 signal / AWGN<br>$\hat{E}s_4 / N_{oc3}$ is the ratio of cell 4 signal / AWGN<br><br>$\hat{E}s_2 / N_{oc2}$, $\hat{E}s_3 / N_{oc3}$ uncertainty and $\hat{E}s_4 / N_{oc3}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty$^2$ + Fading profile power uncertainty$^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 8.16.28 E-UTRAN TDD-FDD 3DL CA Event Triggered Reporting under Deactivated SCells in Non-DRX with PCell in TDD | Same as 8.16.27 | Same as 8.16.27 |

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| 8.16.29, 3DL FDD CA Event Triggered Reporting under Deactivated SCells in Non-DRX | Same as 8.16.27 | Same as 8.16.27 |
| 8.16.30, 3DL TDD CA Event Triggered Reporting under Deactivated SCells in Non-DRX | Same as 8.16.27 | Same as 8.16.27 |
| 8.16.31 E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in FDD | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br><br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$<br><br>$N_{oc3}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_3 / N_{oc3}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_4 / N_{oc3}$ ±0.3 dB averaged over $BW_{Config}$<br><br>Time alignment errors: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br><br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br><br>$N_{oc3}$ is the AWGN on frequency 3<br><br>$\hat{E}s_3 / N_{oc3}$ is the ratio of cell 3 signal / AWGN<br><br>$\hat{E}s_4 / N_{oc3}$ is the ratio of cell 4 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 8.16.32 E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in TDD | Same as 8.16.31 | Same as 8.16.31 |
| 8.16.33 E-UTRAN FDD 3DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX | Same as 8.16.31 | Same as 8.16.31 |
| 8.16.34 E-UTRAN TDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX | Same as 8.16.31 | Same as 8.16.31 |
| 8.16.35 3 DL PCell in FDD CA Activation and Deactivation of Known SCell in Non-DRX | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.3 dB averaged over $BW_{Config}$<br><br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.3 dB averaged over $BW_{Config}$<br><br>$N_{oc3}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_3 / N_{oc3}$ ±0.3 dB averaged over $BW_{Config}$<br><br>Time alignment errors: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br><br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br><br>$N_{oc3}$ is the AWGN on frequency 3<br><br>$\hat{E}s_3 / N_{oc3}$ is the ratio of cell 3 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |

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| 8.16.36 3 DL PCell in TDD CA Activation and Deactivation of Known SCell in Non-DRX | Same as 8.16.35 | Same as 8.16.35 |
| 8.16.37 3DL FDD CA Activation and Deactivation of known SCell in non-DRX | Same as 8.16.35 | Same as 8.16.35 |
| 8.16.38 3DL TDD CA Activation and Deactivation of known SCell in non-DRX | Same as 8.16.35 | Same as 8.16.35 |
| 8.16.39 E-UTRA TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in FDD | Same as 8.16.35 | Same as 8.16.35 |
| 8.16.40 E-UTRA TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in TDD | Same as 8.16.35 | Same as 8.16.35 |
| 8.16.41 3 DL FDD CA activation and deactivation of unknown SCell in non-DRX | Same as 8.16.35 | Same as 8.16.35 |
| 8.16.42 3 DL TDD CA activation and deactivation of unknown SCell in non-DRX | Same as 8.16.35 | Same as 8.16.35 |
| 8.20.1 UE Measurement Procedures / RAT Measurements in CA mode / E-UTRAN FDD-FDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.6 dB averaged over $BW_{Config}$<br><br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$<br><br>$N_{oc3}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_3 / N_{oc3}$ ±0.6 dB averaged over $BW_{Config}$ | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on cell 3 frequency<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 3 signal / AWGN<br><br>$N_{oc3}$ is the AWGN on cell 3 frequency<br>$\hat{E}s_3 / N_{oc3}$ is the ratio of cell 3 signal / AWGN<br><br>$\hat{E}s_1 / N_{oc1}$ uncertainty, $\hat{E}s_2 / N_{oc2}$ uncertainty, $\hat{E}s_3 / N_{oc3}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty$^2$ + Fading profile power uncertainty$^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB |
| 8.20.2 UE Measurement Procedures / RAT Measurements in CA mode / E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells | Same as 8.20.1 | Same as 8.20.1 |
| 8.20.2A UE Measurement Procedures / RAT Measurements in CA mode / E-UTRAN TDD-TDD Inter-frequency event triggered reporting under fading propagation conditions in synchronous cells for 20 MHz +20 MHz bandwidth | Same as 8.20.1 | Same as 8.20.1 |
| 8.20.2B UE Measurement Procedures / RAT Measurements in CA mode / E-UTRAN TDD-TDD Inter-frequency event triggered reporting under fading propagation conditions in synchronous cells for 20 MHz +10 MHz bandwidth | Same as 8.20.1 | Same as 8.20.1 |

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| 8.20.3 E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions | E-UTRA cells<br>$N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.6 dB averaged over $BW_{Config}$<br><br>$N_{oc3}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_3 / N_{oc3}$ ±0.6 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$I_{or}/I_{oc}$ ±0.6 dB<br>CPICH Ec/Ior ±0.1 dB | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc3}$ is the AWGN on cell 3 frequency<br>$\hat{E}s_3 / N_{oc3}$ is the ratio of cell 3 signal / AWGN<br><br>Ioc is the AWGN on Cell 2 (UTRA) frequency<br>Ior/Ioc is the ratio of Cell 2 signal/AWGN<br>CPICH Ec/Ior is the fraction of Cell 2 power assigned to the CPICH physical channel<br><br>$\hat{E}s_1 / N_{oc1}$ uncertainty, $\hat{E}s_3 / N_{oc3}$ uncertainty or $I_{or}/I_{oc}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB |
| 8.20.4 E-UTRAN TDD to UTRAN TDD cell search under fading propagation conditions | E-UTRA cells<br>$N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.6 dB averaged over $BW_{Config}$<br><br>$N_{oc3}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_3 / N_{oc3}$ ±0.6 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$I_{or}/I_{oc}$ ±0.6 dB<br>PCCPCH Ec/Ior ±0.1 dB<br>DwPCH_Ec/Ior ±0.1 dB | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc3}$ is the AWGN on cell 3 frequency<br>$\hat{E}s_3 / N_{oc3}$ is the ratio of cell 3 signal / AWGN<br><br>$I_{oc}$ is the AWGN on cell 2 (UTRA) frequency<br>$I_{or} / I_{oc}$ is the ratio of cell 2 signal / AWGN<br>PCCPCH $E_c / I_{or}$ is the fraction of cell 2 power assigned to the PCCPCH Physical channel<br>DwPCH_Ec/Ior is the fraction of cell 2 power assigned to the DwPCH channel<br><br>$\hat{E}s_1 / N_{oc1}$ uncertainty, $\hat{E}s_3 / N_{oc3}$ uncertainty or $I_{or}/I_{oc}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ or $I_{or}/I_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB |
| 8.20.4A E-UTRAN TDD to UTRAN TDD cell search under fading propagation conditions for 20 MHz + 20 MHz bandwidth | Same as 8.20.4 | Same as 8.20.4 |

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| 8.20.4B E-UTRAN TDD to UTRAN TDD cell search under fading propagation conditions for 20 MHz + 10 MHz bandwidth | Same as 8.20.4 | Same as 8.20.4 |
| 8.22.1 E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal | Same as 8.1.1 | Same as 8.1.1 |
| 8.22.2 E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal | Same as 8.1.1 | Same as 8.1.1 |
| 8.22.3 E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal | Same as 8.3.1 | Same as 8.3.1 |
| 8.22.4 E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal | Same as 8.3.1 | Same as 8.3.1 |
| 8.22.9 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX based on CRS based discovery signal | Same as 8.16.1 | Same as 8.16.1 |
| 8.22.10 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX based on CRS based discovery signal | Same as 8.16.1 | Same as 8.16.1 |
| 8.23.4 E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in synchronous DC | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.6 dB averaged over $BW_{Config}$<br><br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_2 / N_{oc2}$ ±0.6 dB averaged over $BW_{Config}$<br><br>$N_{oc3}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s_3 / N_{oc3}$ ±0.6 dB averaged over $BW_{Config}$ | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on cell 3 frequency<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 3 signal / AWGN<br><br>$N_{oc3}$ is the AWGN on cell 3 frequency<br>$\hat{E}s_3 / N_{oc3}$ is the ratio of cell 3 signal / AWGN<br><br>$\hat{E}s_1 / N_{oc1}$ uncertainty, $\hat{E}s_2 / N_{oc2}$ uncertainty, $\hat{E}s_3 / N_{oc3}$ uncertainty for fading condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>$\hat{E}s / N_{oc}$ uncertainty = SQRT (Signal-to-noise ratio uncertainty$^2$ + Fading profile power uncertainty$^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB |
| 8.23.5 E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in asynchronous DC | Same as 8.23.4 | Same as 8.23.4 |
| 8.23.6 E-UTRAN TDD-TDD DC inter-frequency event triggered reporting with DRX in synchronous DC | Same as 8.23.4 | Same as 8.23.4 |

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| 9.1.1.1 FDD Intra Frequency Absolute RSRP Accuracy | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 9.1.1.1_1 FDD Intra Frequency Absolute RSRP Accuracy (Rel 12 and forward) | Same as 9.1.1.1 | Same as 9.1.1.1 |
| 9.1.1.2 FDD Intra Frequency Relative Accuracy of RSRP | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 9.1.2.1 TDD Intra Frequency Absolute RSRP Accuracy | Same as 9.1.1.1 | Same as 9.1.1.1 |
| 9.1.2.1_1 TDD Intra Frequency Absolute RSRP Accuracy (Rel 12 and forward) | Same as 9.1.1.1 | Same as 9.1.1.1 |
| 9.1.2.2 TDD Intra Frequency Relative RSRP Accuracy | Same as 9.1.1.2 | Same as 9.1.1.2 |
| 9.1.3.1 FDD Inter Frequency Absolute RSRP Accuracy | $N_{oc1}$ and $N_{oc2}$ each ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ and $N_{oc2}$ each ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc1}$ and $\hat{E}s_2 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ and $\hat{E}s_2 / N_{oc2}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br>$N_{oc2}$ is the AWGN on cell 2 frequency<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN |
| 9.1.3.1_1 FDD Inter Frequency Absolute RSRP Accuracy (Rel 12 and forward) | Same as 9.1.3.1 | Same as 9.1.3.1 |
| 9.1.3.2 FDD Inter Frequency Relative RSRP Accuracy | Same as 9.1.3.1 | Same as 9.1.3.1 |
| 9.1.3.2_1 FDD Inter Frequency Relative RSRP Accuracy (Rel 12 and forward) | Same as 9.1.3.1 | Same as 9.1.3.1 |
| 9.1.4.1 TDD Inter Frequency Absolute RSRP Accuracy | Same as 9.1.3.1 | Same as 9.1.3.1 |
| 9.1.4.1_1 TDD Inter Frequency Absolute RSRP Accuracy (Rel 12 and forward) | Same as 9.1.3.1 | Same as 9.1.3.1 |
| 9.1.4.2 TDD Inter Frequency Relative RSRP Accuracy | Same as 9.1.3.1 | Same as 9.1.3.1 |
| 9.1.4.2_1 TDD Inter Frequency Relative RSRP Accuracy (Rel 12 and forward) | Same as 9.1.3.1 | Same as 9.1.3.1 |
| 9.1.5.1 FDD - TDD Inter Frequency Absolute RSRP Accuracy | $N_{oc1}$ and $N_{oc2}$ each ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ and $N_{oc2}$ each ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc1}$ and $\hat{E}s_2 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ and $\hat{E}s_2 / N_{oc2}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br>$N_{oc2}$ is the AWGN on cell 2 frequency<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN |

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| 9.1.5.2 FDD - TDD Inter Frequency Relative Accuracy of RSRP | $N_{oc1}$ and $N_{oc2}$ each ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ and $N_{oc2}$ each ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc1}$ and $\hat{E}s_2 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ and $\hat{E}s_2 / N_{oc2}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br>$N_{oc2}$ is the AWGN on cell 2 frequency<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN |
| 9.1.6.1 FDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #22-27<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.8 dB for PRBs #22-27<br><br>Time alignment error cell 2 rel. to cell 1: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3 / N_{oc2}$ is the ratio of cell 3 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 9.1.6.1_1 Measurement Performance Requirements / FDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation (Rel 12 and forward) | Same as 9.1.6.1 | Same as 9.1.6.1 |
| 9.1.6.2 FDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation | Same as 9.1.6.1 | |
| 9.1.6.2_1 FDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation (Rel 12 and forward) | Same as 9.1.6.1 | Same as 9.1.6.1 |
| 9.1.7.1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation | Same as 9.1.6.1 | |
| 9.1.7.2 TDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation | Same as 9.1.6.1 | |
| 9.1.8.1 FDD Absolute RSRP Accuracy under Time Domain Measurement Resource Restriction with Non-MBSFN ABS | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 9.1.8.2 FDD Relative RSRP Accuracy under Time Domain Measurement Resource Restriction with Non-MBSFN ABS | Same as 9.1.8.1 | |
| 9.1.9.1 TDD Absolute RSRP Accuracy under Time Domain Measurement Resource Restriction with Non-MBSFN ABS | Same as 9.1.8.1 | |
| 9.1.9.2 TDD Relative RSRP Accuracy under Time Domain Measurement Resource Restriction with Non-MBSFN ABS | Same as 9.1.8.1 | |
| 9.1.10.1 FDD Absolute RSRP Accuracy under Time Domain Measurement Resource Restriction with MBSFN ABS | Same as 9.1.8.1 | |
| 9.1.10.2 FDD Relative RSRP Accuracy under Time Domain Measurement Resource Restriction with MBSFN ABS | Same as 9.1.8.1 | |

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| 9.1.11.1 TDD Absolute RSRP Accuracy under Time Domain Measurement Resource Restriction with MBSFN ABS | Same as 9.1.8.1 | |
| 9.1.11.2 TDD Relative RSRP Accuracy under Time Domain Measurement Resource Restriction with MBSFN ABS | Same as 9.1.8.1 | |
| 9.1.12.1 FDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20MHz | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #47-52<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #47-52<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.8 dB for PRBs #47-52<br><br>Time alignment error cell 2 rel. to cell 1: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3 / N_{oc2}$ is the ratio of cell 3 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 9.1.12.1_1 Measurement Performance Requirements / FDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz (Rel 12 and forward) | Same as 9.1.12.1 | Same as 9.1.12.1 |
| 9.1.12.2 FDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 20MHz | Same as 9.1.12.1 | |
| 9.1.12.2_1 FDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz (Rel 12 and forward) | Same as 9.1.12.1 | Same as 9.1.12.1 |
| 9.1.13.1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20MHz | Same as 9.1.12.1 | |
| 9.1.13.2 TDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 20MHz | Same as 9.1.12.1 | |
| 9.1.14.1 FDD Intra Frequency Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc}$, $\hat{E}s_2 / N_{oc}$ and $\hat{E}s_3 / N_{oc}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$, $\hat{E}s_2 / N_{oc}$ and $\hat{E}s_3 / N_{oc}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3 / N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 9.1.14.2 FDD Intra Frequency Relative RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc}$, $\hat{E}s_2 / N_{oc}$ and $\hat{E}s_3 / N_{oc}$ each ±0.3 dB averaged over $BW_{Config}\hat{E}s_1 / N_{oc}$, $\hat{E}s_2 / N_{oc}$ and $\hat{E}s_3 / N_{oc}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3 / N_{oc}$ is the ratio of cell 3 signal / AWGN |
| 9.1.15.1 TDD Intra Frequency Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | Same as 9.1.14.1 | Same as 9.1.14.1 |

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| 9.1.15.2 TDD Intra Frequency Relative RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | Same as 9.1.14.2 | |
| 9.1.16.1 FDD Intra Frequency Absolute RSRP Accuracy for 5MHz Bandwidth | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #10-15<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.8 dB for PRBs #10-15 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 9.1.16.1_1 FDD Intra Frequency Absolute RSRP Accuracy for 5MHz Bandwidth (Rel 12 and forward) | Same as 9.1.16.1 | Same as 9.1.16.1 |
| 9.1.16.2 FDD Intra Frequency Relative Accuracy of RSRP for 5MHz Bandwidth | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #10-15<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.8 dB for PRBs #10-15 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 9.1.17.1 FDD Inter Frequency Absolute RSRP Accuracy for 5MHz Bandwidth | $N_{oc1}$ and $N_{oc2}$ each ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ and $N_{oc2}$ each ±1.3 dB for PRBs #10-15<br>$\hat{E}s_1 / N_{oc1}$ and $\hat{E}s_2 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ and $\hat{E}s_2 / N_{oc2}$ each ±0.8 dB for PRBs #10-15 | Note:<br>$N_{oc1}$ is the AWGN on cell 1 frequency<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br>$N_{oc2}$ is the AWGN on cell 2 frequency<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN |
| 9.1.17.2 FDD - FDD Inter Frequency Relative Accuracy of RSRP for 5MHz Bandwidth | $N_{oc1}$ and $N_{oc2}$ each ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ and $N_{oc2}$ each ±1.3 dB for PRBs #10-15<br>$\hat{E}s_1 / N_{oc1}$ and $\hat{E}s_2 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ and $\hat{E}s_2 / N_{oc2}$ each ±0.8 dB for PRBs #10-15 | Note:<br>Noc1 is the AWGN on cell 1 frequency<br>Ês1 / Noc1 is the ratio of cell 1 signal / AWGN<br>Noc2 is the AWGN on cell 2 frequency<br>Ês2 / Noc2 is the ratio of cell 2 signal / AWGN |
| 9.1.18.1 FDD Absolute RSRP Accuracy for E-UTRA for Carrier Aggregation for 10MHz + 5MHz | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #22-27<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #10-15<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.8 dB for PRBs #22-27<br>$\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.8 dB for PRBs #10-15<br><br>Time alignment error cell 2 rel. to cell 1: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3 / N_{oc2}$ is the ratio of cell 3 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |

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| 9.1.18.2 FDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation for 10MHz + 5MHz | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #22-27<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #10-15<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.8 dB for PRBs #22-27<br>$\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.8 dB for PRBs #10-15<br><br>Time alignment error cell 2 rel. to cell 1: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3 / N_{oc2}$ is the ratio of cell 3 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 9.1.19.1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 10MHz + 5MHz | Same as 9.1.18.1 | Same as 9.1.18.1 |
| 9.1.19.2 TDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation for 10MHz + 5MHz | Same as 9.1.18.2 | Same as 9.1.18.2 |
| 9.1.20.1 FDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz bandwidth | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #10-15<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #10-15<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.8 dB for PRBs #10-15<br><br>Time alignment error cell 2 rel. to cell 1: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3 / N_{oc2}$ is the ratio of cell 3 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 9.1.20.2 FDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz bandwidth | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #10-15<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #10-15<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.8 dB for PRBs #10-15<br><br>Time alignment error cell 2 rel. to cell 1: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3 / N_{oc2}$ is the ratio of cell 3 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |

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| 9.1.21.1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz bandwidth | Same as 9.1.20.1 | Same as 9.1.20.1 |
| 9.1.21.2 TDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz bandwidth | Same as 9.1.20.2 | Same as 9.1.20.2 |
| 9.1.22 FDD-TDD RSRP Accuracy E-UTRA for Carrier Aggregation with PCell in FDD | Noc1 ±1.0 dB averaged over BWConfig<br>Noc1 ±1.3 dB over UE Measurement bandwidth<br>Noc2 ±1.0 dB averaged over BWConfig<br>Noc2 ±1.3 dB over UE Measurement bandwidth<br>Ês1 / Noc1, Ês2 / Noc2, Ês3 / Noc2 each ±0.3 dB averaged over BWConfig<br>Ês1 / Noc1, Ês2 / Noc2, Ês3 / Noc2 each ±0.8 dB over UE Measurement bandwidth<br><br>Time alignment errors: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Notes:<br>Noc1 is the AWGN on frequency 1<br>Ês1 / Noc1 is the ratio of cell 1 signal / AWGN<br><br>Noc2 is the AWGN on frequency 2<br>Ês2 / Noc2 is the ratio of cell 2 signal / AWGN<br>Ês3 / Noc2 is the ratio of cell 3 signal / AWGN<br><br>UE Measurement bandwidth:<br>5MHz Ch BW: PRBs 10-15<br>10MHz Ch BW: PRBs 22-27<br>20MHz Ch BW: PRBs 47-52<br><br>TS = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 9.1.23 FDD-TDD RSRP Accuracy E-UTRA for Carrier Aggregation with PCell in TDD | Same as 9.1.22 | Same as 9.1.22 |
| 9.1.24.1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20MHz + 10MHz | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #47-52<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1$ / $N_{oc1}$, $\hat{E}s_2$ / $N_{oc2}$ and $\hat{E}s_3$ / $N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc1}$ ±0.8 dB for PRBs #47-52<br>$\hat{E}s_2$ / $N_{oc2}$ and $\hat{E}s_3$ / $N_{oc2}$ each ±0.8 dB for PRBs #22-27<br><br>Time alignment error cell 2 rel. to cell 1: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1$ / $N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2$ / $N_{oc2}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3$ / $N_{oc2}$ is the ratio of cell 3 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 9.1.24.1_1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20MHz + 10MHz (Rel-12 and forward) | Same as 9.1.24.1 | Same as 9.1.24.1 |

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| 9.1.24.2 TDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 20MHz + 10MHz | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #47-52<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.8 dB for PRBs #47-52<br>$\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.8 dB for PRBs #22-27<br><br>Time alignment error cell 2 rel. to cell 1: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3 / N_{oc2}$ is the ratio of cell 3 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 9.1.25 FDD intra-frequency absolute and relative RSRP accuracies in CRS based discovery signal | Same as 9.1.1.1 | Same as 9.1.1.1 |
| 9.1.26 TDD intra-frequency absolute and relative RSRP accuracies in CRS based discovery signal | Same as 9.1.1.1 | Same as 9.1.1.1 |
| 9.1.27 FDD—FDD inter-frequency absolute and relative RSRP accuracies in CRS based discovery signal | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #22-27<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN |
| 9.1.28 TDD—TDD inter-frequency absolute and relative RSRP accuracies in CRS based discovery signal | Same as 9.1.27 | Same as 9.1.27 |
| 9.1.29 FDD intra frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #22-27<br>CRS $\hat{E}s_1 / N_{oc}$ and CRS $\hat{E}s_2 / N_{oc}$ each ±0.3 dB averaged over $BW_{Config}$<br>CSI-RS $\hat{E}s_1 / N_{oc}$ and CSI-RS $\hat{E}s_2 / N_{oc}$ each ±0.3 dB averaged over $BW_{Config}$<br>CRS $\hat{E}s_1 / N_{oc}$ and CRS $\hat{E}s_2 / N_{oc}$ each ±0.8 dB for PRBs #22-27<br>CSI-RS $\hat{E}s_1 / N_{oc}$ and CSI-RS $\hat{E}s_2 / N_{oc}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 9.1.30 TDD intra-frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal | Same as 9.1.29 | Same as 9.1.29 |
| 9.1.31 FDD-FDD inter-frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #22-27<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN |

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| 9.1.32 TDD-TDD inter-frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal | Same as 9.1.32 | Same as 9.1.32 |
| 9.1.33 FDD absolute and relative RSRP accuracies for E-UTRAN Carrier Aggregation in CRS based discovery signal | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #22-27<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1$ / $N_{oc1}$, $\hat{E}s_2$ / $N_{oc2}$ and $\hat{E}s_3$ / $N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc1}$, $\hat{E}s_2$ / $N_{oc2}$ and $\hat{E}s_3$ / $N_{oc2}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1$ / $N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2$ / $N_{oc2}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3$ / $N_{oc2}$ is the ratio of cell 3 signal / AWGN |
| 9.1.34 TDD absolute and relative RSRP accuracies for E-UTRAN Carrier Aggregation in CRS based discovery signal | Same as 9.1.33 | Same as 9.1.33 |
| 9.1.35 FDD absolute and relative CSI-RSRP accuracies for E-UTRAN Carrier Aggregation in CSI-RS based discovery signal | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #22-27<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #22-27<br>CSI-RS $\hat{E}s_1$ / $N_{oc1}$, CSI-RS $\hat{E}s_2$ / $N_{oc2}$, and CSI-RS $\hat{E}s_3$ / $N_{oc3}$ each ±0.3 dB averaged over $BW_{Config}$<br>CSI-RS $\hat{E}s_1$ / $N_{oc1}$, CSI-RS $\hat{E}s_2$ / $N_{oc2}$, and CSI-RS $\hat{E}s_3$ / $N_{oc3}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>CSI-RS $\hat{E}s_1$ / $N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>CSI-RS $\hat{E}s_2$ / $N_{oc2}$ is the ratio of cell 2 signal / AWGN<br>CSI-RS $\hat{E}s_3$ / $N_{oc2}$ is the ratio of cell 3 signal / AWGN |
| 9.1.36 TDD absolute and relative CSI-RSRP accuracies for E-UTRAN Carrier Aggregation in CSI-RS based discovery signal | Same as 9.1.35 | Same as 9.1.35 |
| 9.1.37 3DL PCell in FDD RSRP for E-UTRAN in Carrier Aggregation | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB over UE Measurement bandwidth<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB over UE Measurement bandwidth<br>$N_{oc3}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc3}$ ±1.3 dB over UE Measurement bandwidth<br>$\hat{E}s_1$ / $N_{oc1}$, $\hat{E}s_2$ / $N_{oc2}$, $\hat{E}s_3$ / $N_{oc2}$, $\hat{E}s_4$ / $N_{oc3}$, $\hat{E}s_5$ / $N_{oc3}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc1}$, $\hat{E}s_2$ / $N_{oc2}$, $\hat{E}s_3$ / $N_{oc2}$, $\hat{E}s_4$ / $N_{oc3}$, $\hat{E}s_5$ / $N_{oc3}$ each ±0.8 dB over UE Measurement bandwidth<br><br>Time alignment errors: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Notes:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1$ / $N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2$ / $N_{oc2}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3$ / $N_{oc2}$ is the ratio of cell 3 signal / AWGN<br><br>$N_{oc3}$ is the AWGN on frequency 3<br>$\hat{E}s_4$ / $N_{oc3}$ is the ratio of cell 4 signal / AWGN<br>$\hat{E}s_5$ / $N_{oc3}$ is the ratio of cell 5 signal / AWGN<br><br>UE Measurement bandwidth:<br>5MHz Ch BW: PRBs 10-15<br>10MHz Ch BW: PRBs 22-27<br>20MHz Ch BW: PRBs 47-52<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 9.1.38 3DL PCell in TDD RSRP for E-UTRAN in Carrier Aggregation | Same as 9.1.37 | Same as 9.1.37 |
| 9.1.39 3DL FDD RSRP for E-UTRAN in Carrier Aggregation | Same as 9.1.37 | Same as 9.1.37 |

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| 9.1.39_1 3DL FDD RSRP for E-UTRAN in Carrier Aggregation (Rel-12 and forward) | Same as 9.1.37 | Same as 9.1.37 |
| 9.1.40 3DL TDD RSRP for E-UTRAN in Carrier Aggregation | Same as 9.1.37 | Same as 9.1.37 |
| 9.1.40_1 3DL TDD RSRP for E-UTRAN in Carrier Aggregation (Rel-12 and forward) | Same as 9.1.37 | Same as 9.1.37 |
| 9.1.41.1 FD-FDD Intra Frequency Absolute RSRP Accuracy for UE category 0 | Same as 9.1.1.1 | Same as 9.1.1.1 |
| 9.1.41.2 FD-FDD Intra Frequency Relative RSRP Accuracy for UE category 0 | Same as 9.1.1.2 | Same as 9.1.1.2 |
| 9.1.42.1 HD-FDD Intra Frequency Absolute RSRP Accuracy for UE category 0 | Same as 9.1.1.1 | Same as 9.1.1.1 |
| 9.1.42.2 HD-FDD Intra Frequency Relative RSRP Accuracy for UE category 0 | Same as 9.1.1.2 | Same as 9.1.1.2 |
| 9.1.43.1 TDD Intra Frequency Absolute RSRP Accuracy for UE category 0 | Same as 9.1.1.1 | Same as 9.1.1.1 |
| 9.1.43.2 TDD Intra Frequency Relative RSRP Accuracy for UE category 0 | Same as 9.1.1.2 | Same as 9.1.1.2 |
| 9.1.52 FD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeA | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 9.1.53 HD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeA | Same as 9.1.52 | Same as 9.1.52 |
| 9.1.54 TDD RSRP Intra frequency case for Cat-M1 UE in CEModeA | Same as 9.1.52 | Same as 9.1.52 |
| 9.1.57 FD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeB | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc}$, $\hat{E}s_2 / N_{oc}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$, $\hat{E}s_2 / N_{oc}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 9.1.58 HD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeB | Same as 9.1.57 | Same as 9.1.57 |
| 9.1.59 TDD RSRP Intra frequency case for Cat-M1 UE in CEModeB | Same as 9.1.57 | Same as 9.1.57 |
| 9.2.1.1 FDD Intra Frequency Absolute RSRQ Accuracy | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 9.2.2.1 TDD Intra Frequency Absolute RSRQ Accuracy | Same as 9.2.1.1 | Same as 9.2.1.1 |
| 9.2.3.1 FDD - FDD Inter Frequency Absolute RSRQ Accuracy | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #22-27<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc1}$ and $\hat{E}s_2 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ and $\hat{E}s_2 / N_{oc2}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN on frequency 1<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN on frequency 2 |

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| 9.2.3.2 FDD - FDD Inter Frequency Relative RSRQ Accuracy | Same as 9.2.3.1 | |
| 9.2.4.1 TDD - TDD Inter Frequency Absolute RSRQ Accuracy | Same as 9.2.3.1 | |
| 9.2.4.2 TDD - TDD Inter Frequency Relative RSRQ Accuracy | Same as 9.2.3.1 | |
| 9.2.4A.1 FDD - TDD Inter Frequency Absolute RSRQ Accuracy | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #22-27<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc1}$ and $\hat{E}s_2 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ and $\hat{E}s_2 / N_{oc2}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN on frequency 1<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN on frequency 2 |
| 9.2.4A.2 FDD - TDD Inter Frequency Relative Accuracy of RSRQ | Same as 9.2.4A.1 | |
| 9.2.5.1 FDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #22-27<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.8 dB for PRBs #22-27<br><br>Time alignment error cell 2 rel. to cell 1: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3 / N_{oc2}$ is the ratio of cell 3 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 9.2.5.2 FDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation | Same as 9.2.5.1 | |
| 9.2.6.1 TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation | Same as 9.2.5.1 | |
| 9.2.6.2 TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation | Same as 9.2.5.1 | |
| 9.2.7.1 FDD RSRQ under Time Domain Measurement Resource Restriction with Non-MBSFN ABS | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 9.2.8.1 TDD RSRQ under Time Domain Measurement Resource Restriction with Non-MBSFN ABS | Same as 9.2.7.1 | Same as 9.2.7.1 |
| 9.2.9.1 FDD Absolute RSRQ under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 9.2.10.1 TDD Absolute RSRQ under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) | Same as 9.2.9.1 | |

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| 9.2.11.1 FDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #47-52<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #47-52<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.8 dB for PRBs #47-52<br><br>Time alignment error cell 2 rel. to cell 1: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3 / N_{oc2}$ is the ratio of cell 3 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 9.2.11.2 FDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz | Same as 9.2.11.1 | |
| 9.2.12.1 TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz | Same as 9.2.11.1 | |
| 9.2.12.2 TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz | Same as 9.2.11.1 | |
| 9.2.15.1 FDD RSRQ Accuracy under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc}$, $\hat{E}s_2 / N_{oc}$ and $\hat{E}s_3 / N_{oc}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$, $\hat{E}s_2 / N_{oc}$ and $\hat{E}s_3 / N_{oc}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3 / N_{oc}$ is the ratio of cell 3 signal / AWGN |
| 9.2.16.1 TDD RSRQ Accuracy under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | Same as 9.2.15.1 | Same as 9.2.15.1 |
| 9.2.17.1 FDD Intra Frequency Absolute RSRQ Accuracy for 5MHz Bandwidth | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #10-15<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.8 dB for PRBs #10-15 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 9.2.18.1 FDD - FDD Inter Frequency Absolute RSRQ Accuracy for 5MHz Bandwidth | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #10-15<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #10-15<br>$\hat{E}s_1 / N_{oc1}$ and $\hat{E}s_2 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ and $\hat{E}s_2 / N_{oc2}$ each ±0.8 dB for PRBs #10-15 | Note:<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN on frequency 1<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN on frequency 2 |
| 9.2.18.2 FDD - FDD Inter Frequency Relative RSRQ Accuracy for 5MHz Bandwidth | Same as 9.2.18.1 | |
| 9.2.19.1 FDD-FDD Inter Frequency absolute WB-RSRQ accuracy | FFS | FFS |

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| 9.2.20.1 TDD-TDD Inter Frequency absolute WB-RSRQ accuracy | FFS | FFS |
| 9.2.21.1 FDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #22-27<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #10-15<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ ±0.8 dB for PRBs #22-27<br>$\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.8 dB for PRBs #10-15<br><br>Time alignment error cell 2 rel. to cell 1: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3 / N_{oc2}$ is the ratio of cell 3 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 9.2.21.2 FDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz | Same as 9.2.21.1 | |
| 9.2.22.1 TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz | Same as 9.2.21.1 | |
| 9.2.22.2 TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz | Same as 9.2.21.1 | |
| 9.2.23.1 FDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 5MHz+5MHz | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #10-15<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #10-15<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.8 dB for PRBs #10-15<br><br>Time alignment error cell 2 rel. to cell 1: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3 / N_{oc2}$ is the ratio of cell 3 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 9.2.23.2 FDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 5MHz+5MHz | Same as 9.2.23.1 | Same as 9.2.23.1 |
| 9.2.24.1 TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 5MHz+5MHz | Same as 9.2.23.1 | Same as 9.2.23.1 |
| 9.2.24.2 TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 5MHz+5MHz | Same as 9.2.23.1 | Same as 9.2.23.1 |

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| 9.2.25.1 Absolute RSRQ Accuracy for E-UTRAN TDD-FDD Carrier Aggregation with PCell in FDD | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #22-27<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1$ / $N_{oc1}$ and $\hat{E}s_2$ / $N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc1}$ and $\hat{E}s_2$ / $N_{oc2}$ each ±0.8 dB for PRBs #22-27<br><br>Time alignment error cell 2 rel. to cell 1:<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1$ / $N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2$ / $N_{oc2}$ is the ratio of cell 2 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 9.2.25.2 Relative RSRQ Accuracy for E-UTRAN TDD-FDD Carrier Aggregation with PCell in FDD | Same as 9.2.25.1 | |
| 9.2.26.1 Absolute RSRQ Accuracy for E-UTRAN TDD-FDD Carrier Aggregation with PCell in TDD | Same as 9.2.25.1 | |
| 9.2.26.2 Relative RSRQ Accuracy for E-UTRAN TDD-FDD Carrier Aggregation with PCell in TDD | Same as 9.2.25.1 | |
| 9.2.27.1 TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz+10MHz | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #47-52<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1$ / $N_{oc1}$, $\hat{E}s_2$ / $N_{oc2}$ and $\hat{E}s_3$ / $N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc1}$ ±0.8 dB for PRBs #47-52<br>$\hat{E}s_2$ / $N_{oc2}$ and $\hat{E}s_3$ / $N_{oc2}$ each ±0.8 dB for PRBs #22-27<br><br>Time alignment error cell 2 rel. to cell 1: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1$ / $N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2$ / $N_{oc2}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3$ / $N_{oc2}$ is the ratio of cell 3 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 9.2.27.2 TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz+10MHz | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #47-52<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1$ / $N_{oc1}$, $\hat{E}s_2$ / $N_{oc2}$ and $\hat{E}s_3$ / $N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1$ / $N_{oc1}$ ±0.8 dB for PRBs #47-52<br>$\hat{E}s_2$ / $N_{oc2}$ and $\hat{E}s_3$ / $N_{oc2}$ each ±0.8 dB for PRBs #22-27<br><br>Time alignment error cell 2 rel. to cell 1: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1$ / $N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2$ / $N_{oc2}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3$ / $N_{oc2}$ is the ratio of cell 3 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |

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| 9.2.28 FDD intra-frequency absolute RSRQ accuracy with CRS based discovery signal | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ and $\hat{E}s_2 / N_{oc}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}s_2 / N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 9.2.29 TDD intra-frequency absolute RSRQ accuracy with CRS based discovery signal | Same as 9.2.28 | Same as 9.2.28 |
| 9.2.30 FDD-FDD inter-frequency absolute and relative RSRQ accuracies with CRS based discovery signal | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #22-27<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc1}$ and $\hat{E}s_2 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$ and $\hat{E}s_2 / N_{oc2}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN on frequency 1<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN on frequency 2 |
| 9.2.31 TDD-TDD inter-frequency absolute and relative RSRQ accuracies with CRS based discovery signal | Same as 9.2.30 | Same as 9.2.30 |
| 9.2.32 FDD absolute and relative RSRQ accuracy for E-UTRAN Carrier Aggregation in CRS based discovery signal | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB for PRBs #22-27<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc1}$, $\hat{E}s_2 / N_{oc2}$ and $\hat{E}s_3 / N_{oc2}$ each ±0.8 dB for PRBs #22-27<br><br>Time alignment error cell 2 rel. to cell 1: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Note:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}s_1 / N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}s_2 / N_{oc2}$ is the ratio of cell 2 signal / AWGN<br>$\hat{E}s_3 / N_{oc2}$ is the ratio of cell 3 signal / AWGN<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 9.2.33 TDD absolute and relative RSRQ accuracy for E-UTRAN Carrier Aggregation in CRS based discovery signal | Same as 9.2.32 | Same as 9.2.32 |

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| 9.2.38 3DL PCell in FDD RSRQ for E-UTRAN in Carrier Aggregation | $N_{oc1}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc1}$ ±1.3 dB over UE Measurement bandwidth<br>$N_{oc2}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc2}$ ±1.3 dB over UE Measurement bandwidth<br>$N_{oc3}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc3}$ ±1.3 dB over UE Measurement bandwidth<br>$\hat{E}_{s1}$ / $N_{oc1}$, $\hat{E}_{s2}$ / $N_{oc2}$, $\hat{E}_{s3}$ / $N_{oc3}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}_{s1}$ / $N_{oc1}$, $\hat{E}_{s2}$ / $N_{oc2}$, $\hat{E}_{s3}$ / $N_{oc3}$ each ±0.8 dB over UE Measurement bandwidth<br><br>Time alignment errors: Intra-band contiguous CA: ±130 ns (±4Ts)<br>Intra-band non-contiguous CA: ±260 ns (±8Ts)<br>Inter-band CA: ±260 ns (±8Ts) | Notes:<br>$N_{oc1}$ is the AWGN on frequency 1<br>$\hat{E}_{s1}$ / $N_{oc1}$ is the ratio of cell 1 signal / AWGN<br><br>$N_{oc2}$ is the AWGN on frequency 2<br>$\hat{E}_{s2}$ / $N_{oc2}$ is the ratio of cell 2 signal / AWGN<br><br>$N_{oc3}$ is the AWGN on frequency 3<br>$\hat{E}_{s3}$ / $N_{oc3}$ is the ratio of cell 3 signal / AWGN<br><br>UE Measurement bandwidth:<br>5MHz Ch BW: PRBs 10-15<br>10MHz Ch BW: PRBs 22-27<br>20MHz Ch BW: PRBs 47-52<br><br>$T_S$ = 1/(15000 x 2048) seconds, the basic timing unit defined in TS 36.211 [9] |
| 9.2.39 3DL PCell in TDD RSRQ for E-UTRAN in Carrier Aggregation | Same as 9.2.38 | Same as 9.2.38 |
| 9.2.40 3DL FDD RSRQ for E-UTRA Carrier Aggregation | Same as 9.2.38 | Same as 9.2.38 |
| 9.2.41 3DL TDD RSRQ for E-UTRA Carrier Aggregation | Same as 9.2.38 | Same as 9.2.38 |
| 9.2.42.1 FD-FDD Intra Frequency Absolute RSRQ Accuracy for UE category 0 | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}_{s1}$ / $N_{oc}$ and $\hat{E}_{s2}$ / $N_{oc}$ each ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}_{s1}$ / $N_{oc}$ and $\hat{E}_{s2}$ / $N_{oc}$ each ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}_{s1}$ / $N_{oc}$ is the ratio of cell 1 signal / AWGN<br>$\hat{E}_{s2}$ / $N_{oc}$ is the ratio of cell 2 signal / AWGN |
| 9.2.43.1 HD-FDD Intra Frequency Absolute RSRQ Accuracy for UE category 0 | Same as 9.2.42.1 | Same as 9.2.42.1 |
| 9.2.44.1 TDD Intra Frequency Absolute RSRQ Accuracy for UE category 0 | Same as 9.2.42.1 | Same as 9.2.42.1 |
| 9.3.1 E-UTRAN FDD - UTRA FDD CPICH<br>RSCP absolute accuracy | E-UTRA cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}_s$ / $N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$I_{or}$ / $I_{oc}$ ±0.3 dB<br>CPICH $E_c$ / $I_{or}$ ±0.1 dB | Notes:<br><br>$N_{oc}$ is the AWGN on cell 1 (E-UTRA )frequency<br>$\hat{E}_s$ / $N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>$I_{oc}$ is the AWGN on cell 2 (UTRA) frequency<br>$I_{or}$ / $I_{oc}$ is the ratio of cell 2 signal / AWGN<br>CPICH $E_c$ / $I_{or}$ is the fraction of cell 2 power assigned to the CPICH Physical channel |
| 9.3.2 E-UTRAN TDD - UTRA FDD CPICH<br>RSCP absolute accuracy | Same as 9.3.1 | |
| 9.3.3 E-UTRAN FDD - UTRA FDD CPICH RSCP absolute accuracy for 5MHz bandwidth | Same as 9.3.1 | |

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| 9.4.1 E-UTRAN FDD – UTRA FDD CPICH Ec/No absolute accuracy | E-UTRA cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$I_{or} / I_{oc}$ ±0.3 dB<br>CPICH $E_c / I_{or}$ ±0.1 dB | Notes:<br><br>$N_{oc}$ is the AWGN on cell 1 (E-UTRA )frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>$I_{oc}$ is the AWGN on cell 2 (UTRA) frequency<br>$I_{or} / I_{oc}$ is the ratio of cell 2 signal / AWGN<br>CPICH $E_c / I_{or}$ is the fraction of cell 2 power assigned to the CPICH Physical channel |
| 9.4.2 E-UTRAN TDD – UTRA FDD CPICH Ec/No absolute accuracy | Same as 9.4.1 | Same as 9.4.1 |
| 9.4.3 E-UTRAN FDD - UTRA FDD CPICH Ec/No absolute accuracy for 5MHz bandwidth | Same as 9.4.1 | Same as 9.4.1 |
| 9.5.1 E-UTRAN FDD – UTRA TDD P-CCPCH RSCP absolute accuracy | E-UTRA cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>UTRA cell<br>$I_{oc}$ ±0.7 dB<br>$I_{or} / I_{oc}$ ±0.3 dB<br>PCCPCH $E_C/I_{or}$ ±0.1 dB<br>DwPCH_$E_C/I_{or}$ ±0.1 dB | Notes:<br>$N_{oc}$ is the AWGN on cell 1 (E-UTRA )frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>$I_{oc}$ is the AWGN on cell 2 (UTRA) frequency<br>$I_{or} / I_{oc}$ is the ratio of cell 2 signal / AWGN<br>PCCPCH $E_c / I_{or}$ is the fraction of cell 2 power assigned to the PCCPCH Physical channel<br>DwPCH_$E_c / I_{or}$ is the fraction of cell 2 power assigned to the DwPCH channel |
| 9.5.2 E-UTRAN TDD – UTRA TDD P-CCPCH RSCP absolute accuracy | Same as 9.5.1 | |
| 9.6.1 GSM RSSI accuracy for E-UTRAN FDD | E-UTRA Cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>GSM cell BCCH1<br>Signal level ±0.7 dB<br><br>GSM cell BCCH 2 to 6<br>Signal level ±2.0 dB | Note:<br>$N_{oc}$ is the AWGN on cell 1 frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>GSM cells BCCH 1 to 6 have only the wanted signal, without AWGN |
| 9.6.2 GSM RSSI accuracy for E-UTRAN TDD | E-UTRA Cell<br>$N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$\hat{E}s / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br><br>GSM cell BCCH1<br>Signal level ±0.7 dB<br><br>GSM cell BCCH 2 to 6<br>Signal level ±2.0 dB | Note:<br>$N_{oc}$ is the AWGN on cell 1 frequency<br>$\hat{E}s / N_{oc}$ is the ratio of cell 1 signal / AWGN<br><br>GSM cells BCCH 1 to 6 have only the wanted signal, without AWGN |
| 9.9.1.1 FDD Intra Frequency Serving Cell Absolute RSRP Accuracy | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN |

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| 9.9.1.2 FDD Intra Frequency Serving Cell Absolute RSRQ Accuracy | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN |
| 9.9.2.1 TDD Intra Frequency Serving Cell Absolute RSRP Accuracy | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN |
| 9.9.2.2 TDD Intra Frequency Serving Cell Absolute RSRQ Accuracy | $N_{oc}$ ±1.0 dB averaged over $BW_{Config}$<br>$N_{oc}$ ±1.3 dB for PRBs #22-27<br>$\hat{E}s_1 / N_{oc}$ ±0.3 dB averaged over $BW_{Config}$<br>$\hat{E}s_1 / N_{oc}$ ±0.8 dB for PRBs #22-27 | Note:<br>$\hat{E}s_1 / N_{oc}$ is the ratio of cell 1 signal / AWGN |
| 10.2 E-UTRAN FDD – Initiation/Cease of SLSS Transmission with ProSe Direct Communication | ± 0.6dB | Overall system uncertainty for AWGN condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Effect of AWGN flatness and signal flatness<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared: AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>AWGN flatness and signal flatness ±2.0 dB |
| 10.4 E-UTRAN FDD – Initiation/Cease of SLSS Transmission with ProSe Direct Communication | ± 0.6dB | Overall system uncertainty for AWGN condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Effect of AWGN flatness and signal flatness<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared: AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>AWGN flatness and signal flatness ±2.0 dB |
| In addition, the following Test System uncertainties and related constraints apply.<br>Any additional constraints are defined in the specific tests. | | |
| AWGN Bandwidth | | ≥ 1.08MHz, 2.7MHz, 4.5MHz, 9MHz, 13.5MHz, 18MHz;<br>$N_{RB}$ x 180kHz according to $BW_{Config}$ |
| AWGN absolute power uncertainty | | Test-specific |
| AWGN flatness and signal flatness, max deviation for any Resource Block, relative to average over $BW_{Config}$ | | ±2 dB |
| AWGN peak to average ratio | | ≥10 dB @0.001% |
| Signal-to noise ratio uncertainty | | Test-specific |
| Fading profile power uncertainty | | ±0.5 dB |

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| Fading profile delay uncertainty, relative to frame timing | ±5 ns (excludes absolute errors related to baseband timing) |

# F.2 Interpretation of measurement results (normative)

See TS 36.521-1[10] Annex F2.

# F.3 Test Tolerance and Derivation of Test Requirements (informative)

See TS 36.521-1[10] Annex F3.

## F.3.1 Measurement of test environments

See TS 36.521-1[10] Annex F3.1.

## F.3.2 Measurement of RRM requirements

Because the relationships between the Test system uncertainties and the Test Tolerances are often complex, it is not always possible to give a simple derivation of the Test Requirement in this document. The analysis is recorded in 3GPP TR 36 903 [20].

**Table F.3.2-1: Derivation of Test Requirements (RRM tests)**

| Test | Minimum Requirement in TS 36.133 | Test Tolerance (TT) | Test Requirement in TS 36.521-3 |
|---|---|---|---|
| 4.2.1 E-UTRA FDD – FDD cell re-selection intra frequency | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +16.00dB<br>$\hat{E}s_2$ / $N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +13.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +16.00dB<br><br>During T3:<br>$N_{oc}$: -98dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +16.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +13.00dB | During T1:<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>0dB<br>0dB<br>+0.45dB<br><br>During T3:<br>0dB<br>+0.45dB<br>0dB | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +16.00dB<br>$\hat{E}s_2$ / $N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +13.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +16.45dB<br><br>During T3:<br>$N_{oc}$: -98dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +16.45dB<br>$\hat{E}s_2$ / $N_{oc}$: +13.00dB |
| 4.2.2 E-UTRA TDD – TDD cell re-selection intra frequency | Same as 4.2.1 | Same as 4.2.1 | Same as 4.2.1 |
| 4.2.3 E-UTRA FDD - FDD cell re-selection inter frequency | During T0:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: +14.00dB<br><br>During T1:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +14.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -4.00dB<br><br>During T2:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +14.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -infinity<br><br>During T3:<br>$N_{oc1}$: -98dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +14.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: +12.00dB | During T0:<br>-1.35dB<br>+0.3dB<br>-1.35dB<br>+0.4dB<br><br>During T1:<br>-1.35dB<br>+2.4dB<br>-1.35dB<br>+0.3dB<br><br>During T2:<br>-1.35dB<br>+2.4dB<br>-1.35dB<br>0dB<br><br>During T3:<br>-1.35dB<br>+2.4dB<br>-1.35dB<br>+2.4dB | During T0:<br>$N_{oc1}$: -99.35dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -3.70dB<br>$N_{oc2}$: -99.35dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: +14.40dB<br><br>During T1:<br>$N_{oc1}$: -99.35dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +16.40dB<br>$N_{oc2}$: -99.35dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -3.70dB<br><br>During T2:<br>$N_{oc1}$: -99.35dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +16.40dB<br>$N_{oc2}$: -99.35dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -infinity<br><br>During T3:<br>$N_{oc1}$: -99.35dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +16.40dB<br>$N_{oc2}$: -99.35dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: +14.40dB |
| 4.2.4 E-UTRA FDD - TDD cell re-selection inter frequency | Same as 4.2.3 | Same as 4.2.3 | Same as 4.2.3 |
| 4.2.5 E-UTRA TDD - FDD cell re-selection inter frequency | Same as 4.2.3 | Same as 4.2.3 | Same as 4.2.3 |
| 4.2.6 E-UTRA TDD - TDD cell re-selection inter frequency | Same as 4.2.3 | Same as 4.2.3 | Same as 4.2.3 |

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| 4.2.7 E-UTRA FDD Inter frequency re-selection in the existence of non-allowed CSG cell | During T0:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +13.00dB<br>$\hat{E}s_3$ / $N_{oc1}$: -infinity<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -3.00dB<br><br>During T1:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +8.00dB<br>$\hat{E}s_3$ / $N_{oc1}$: +8.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -infinity<br><br>During T2:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +8.00dB<br>$\hat{E}s_3$ / $N_{oc1}$: +13.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: +13.00dB<br><br>During T3:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +13.00dB<br>$\hat{E}s_3$ / $N_{oc1}$: +38.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: +8.00dB | During T0:<br>0dB<br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T1:<br>0dB<br>-0.2dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>0dB<br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T3:<br>0dB<br>0dB<br>0dB<br>0dB<br>0dB | During T0:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +13.00dB<br>$\hat{E}s_3$ / $N_{oc1}$: -infinitydB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -3.00dB<br><br>During T1:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +7.80dB<br>$\hat{E}s_3$ / $N_{oc1}$: +8.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -infinity<br><br>During T2:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +8.00dB<br>$\hat{E}s_3$ / $N_{oc1}$: +13.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: +13.00dB<br><br>During T3:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +13.00dB<br>$\hat{E}s_3$ / $N_{oc1}$: +38.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: +8.00dB |
| 4.2.8 E-UTRA TDD Inter frequency re-selection in the existence of non-allowed CSG cell | Same as 4.2.7 | Same as 4.2.7 | Same as 4.2.7 |
| 4.2.9 E-UTRAN FDD - FDD intra frequency cell re-selection case for 5MHz bandwidth | Same as 4.2.1 | Same as 4.2.1 | Same as 4.2.1 |
| 4.2.12 E-UTRAN FDD – FDD Intra frequency case for Cat-M1 UE in normal coverage | Same as 4.2.1 | Same as 4.2.1 | Same as 4.2.1 |
| 4.2.13 E-UTRAN HD – FDD Intra frequency case for Cat-M1 UE in normal coverage | Same as 4.2.1 | Same as 4.2.1 | Same as 4.2.1 |
| 4.2.14 E-UTRA TDD - TDD intra frequency case for Cat-M1 UE in normal coverage | Same as 4.2.1 | Same as 4.2.1 | Same as 4.2.1 |

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| 4.3.1.1 E-UTRA FDD - UTRAN FDD cell reselection: UTRA FDD is of higher priority | During T1:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.00dBm/15kHz<br>Ês / $N_{oc}$: +14.00dB<br>UTRA Cell 2<br>$I_{oc}$: -70.00dBm/3.84MHz<br>$I_{or}$ / $I_{oc}$: -∞dB<br>CPICH_$E_c$/$I_{or}$: -10.00dB<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.00dBm/15kHz<br>Ês / $N_{oc}$: +14.00dB<br>UTRA Cell 2<br>$I_{oc}$: -70.00dBm/3.84MHz<br>$I_{or}$ / $I_{oc}$: +11.00dB<br>CPICH_$E_c$/$I_{or}$: -10.00dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.00dBm/15kHz<br>Ês / $N_{oc}$: +14.00dB<br>UTRA Cell 2<br>$I_{oc}$: -70.00dBm/3.84MHz<br>$I_{or}$ / $I_{oc}$: -5.00dB<br>CPICH_$E_c$/$I_{or}$: -10.00dB | During T1:<br><br>0dB<br>+1.05dB<br><br>-0.1dB<br>0dB<br>0dB<br><br>During T2:<br><br>0dB<br>+1.05dB<br><br>-0.1dB<br>+0.9dB<br>0dB<br><br>During T3:<br><br>0dB<br>+1.05dB<br><br>-0.1dB<br>-0.7dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.00dBm/15kHz<br>Ês / $N_{oc}$: +1.05dB<br>UTRA Cell 2<br>$I_{oc}$: -70.10dBm/3.84MHz<br>$I_{or}$ / $I_{oc}$: -∞dB<br>CPICH_$E_c$/$I_{or}$: -10.00dB<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.00dBm/15kHz<br>Ês / $N_{oc}$: +1.05dB<br>UTRA Cell 2<br>$I_{oc}$: -70.10dBm/3.84MHz<br>$I_{or}$ / $I_{oc}$: +11.90dB<br>CPICH_$E_c$/$I_{or}$: -10.00dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.00dBm/15kHz<br>Ês / $N_{oc}$: +1.05dB<br>UTRA Cell 2<br>$I_{oc}$: -70.10dBm/3.84MHz<br>$I_{or}$ / $I_{oc}$: -5.70dB<br>CPICH_$E_c$/$I_{or}$: -10.00dB |
| 4.3.1.2 E-UTRAN FDD - UTRAN FDD cell re-selection: UTRA FDD is of lower priority | During T1:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.00dBm/15kHz<br>Ês / $N_{oc}$: +12.00dB<br>UTRA Cell 2<br>$I_{oc}$: -70.00dBm/3.84MHz<br>$I_{or}$ / $I_{oc}$: +13.00dB<br>CPICH_$E_c$/$I_{or}$: -10.00dB<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.00dBm/15kHz<br>Ês / $N_{oc}$: -4.00dB<br>UTRA Cell 2<br>$I_{oc}$: -70.00dBm/3.84MHz<br>$I_{or}$ / $I_{oc}$: +13.00dB<br>CPICH_$E_c$/$I_{or}$: -10.00dB | During T1:<br><br>-1.35dB<br>+2.40dB<br><br>0dB<br>+0.80dB<br>0dB<br><br>During T2:<br><br>-1.35dB<br>+0.30dB<br><br>0dB<br>+0.80dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>$N_{oc}$: -99.35dBm/15kHz<br>Ês / $N_{oc}$: +14.40dB<br>UTRA Cell 2<br>$I_{oc}$: -70.00dBm/3.84MHz<br>$I_{or}$ / $I_{oc}$: +13.80dB<br>CPICH_$E_c$/$I_{or}$: -10.00dB<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -99.35dBm/15kHz<br>Ês / $N_{oc}$: -3.70dB<br>UTRA Cell 2<br>$I_{oc}$: -70.00dBm/3.84MHz<br>$I_{or}$ / $I_{oc}$: +13.80dB<br>CPICH_$E_c$/$I_{or}$: -10.00dB |
| 4.3.1.3 EUTRA FDD-UTRA FDD cell reselection in fading propagation conditions: UTRA FDD is of lower priority | During T1, T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -104.00dBm/15kHz<br>Ês / $N_{oc}$: +22.00dB<br>UTRA Cell 2<br>$I_{oc}$: -70.00dBm/3.84MHz<br>$I_{or}$ / $I_{oc}$: +13.00dB<br>CPICH_$E_c$/$I_{or}$: -10.00dB<br><br>During T3, T4:<br>E-UTRA Cell 1<br>$N_{oc}$: -104.00dBm/15kHz<br>Ês / $N_{oc}$: -3.00dB<br>UTRA Cell 2<br>$I_{oc}$: -70.00dBm/3.84MHz<br>$I_{or}$ / $I_{oc}$: +13.00dB<br>CPICH_$E_c$/$I_{or}$: -10.00dB | During T1,T2<br><br>0dB<br>0dB<br><br>0dB<br>+0.80dB<br>0dB<br><br>During T3,T4<br><br>0dB<br>0dB<br><br>0dB<br>+0.80dB<br>0dB | During T1, T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -104.00dBm/15kHz<br>Ês / $N_{oc}$: +22.00dB<br>UTRA Cell 2<br>$I_{oc}$: -70.00dBm/3.84MHz<br>$I_{or}$ / $I_{oc}$: +13.80dB<br>CPICH_$E_c$/$I_{or}$: -10.00dB<br><br>During T3, T4:<br>E-UTRA Cell 1<br>$N_{oc}$: -104.00dBm/15kHz<br>Ês / $N_{oc}$: -3.00dB<br>UTRA Cell 2<br>$I_{oc}$: -70.00dBm/3.84MHz<br>$I_{or}$ / $I_{oc}$: +13.80dB<br>CPICH_$E_c$/$I_{or}$: -10.00dB |
| 4.3.1.4 E-UTRAN FDD - UTRAN FDD cell re-selection: UTRA FDD is of lower priority for 5MHz bandwidth | Same as 4.3.1.2 | | |

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| 4.3.2 E-UTRA FDD - UTRAN TDD cell re-selection | During T1:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.0dBm/15kHz<br>$\hat{E}s / N_{oc}$: +11.00dB<br>UTRA Cell 2<br>$I_{oc}$: -80.0dBm/1.28MHz<br>$\hat{I}_{or} / I_{oc}$: +11.0dB<br>PCCPCH_$E_c/I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.0dBm/15kHz<br>$\hat{E}s / N_{oc}$: -3.0dB<br>UTRA Cell 2<br>$I_{oc}$: -80.0dBm/1.28MHz<br>$\hat{I}_{or} / I_{oc}$: +11.0dB<br>PCCPCH_$E_c/I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB | During T1:<br><br>0dB<br>0.1dB<br><br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br><br>0dB<br>0.1dB<br><br>0dB<br>0dB<br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.0dBm/15kHz<br>$\hat{E}s / N_{oc}$: +11.10dB<br>UTRA Cell 2<br>$I_{oc}$: -80.0dBm/1.28MHz<br>$\hat{I}_{or} / I_{oc}$: +11.0dB<br>PCCPCH_$E_c/I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.0dBm/15kHz<br>$\hat{E}s / N_{oc}$: -3.1dB<br>UTRA Cell 2<br>$I_{oc}$: -80.0dBm/1.28MHz<br>$\hat{I}_{or} / I_{oc}$: +11.0dB<br>PCCPCH_$E_c/I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB |
| 4.3.3 E-UTRAN TDD - UTRAN FDD cell re-selection: UTRA FDD is of lower priority | Same as 4.3.1.2 | Same as 4.3.1.2 | Same as 4.3.1.2 |
| 4.3.4.1 E-UTRA TDD - UTRAN TDD cell re-selection : UTRA is of higher priority | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +11.00dB<br>UTRA Cell 2<br>Ioc: -80dBm/1.28MHz<br>Îor / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm /15kHz<br>Ês / Noc: +11.00dB<br>UTRA Cell 2<br>Ioc: -80dBm/1.28MHz<br>Îor / Ioc: +11.00dB<br>PCCPCH_Ec/Ior: -3dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T3:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +11.00dB<br>UTRA Cell 2<br>Ioc: -80dBm/1.28MHz<br>Îor / Ioc: -3.00dB<br>PCCPCH_Ec/Ior: -3dB<br>DwPCH_Ec/Ior: 0dB | During T1:<br><br>0dB<br>+0.1dB<br><br>0dB<br>0dB<br><br>During T2:<br><br>0dB<br>+0.1dB<br><br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T3:<br><br>0dB<br>+0.1dB<br><br>0dB<br>0dB<br>0dB<br>0dB | During T1:<br><br>$N_{oc}$: -98.0dBm/15kHz<br>$\hat{E}s / N_{oc}$: +11.1dB<br><br>$I_{oc}$: -80.0dBm/1.28MHz<br>$\hat{I}_{or} / I_{oc}$: -infinity<br><br>During T2:<br><br>$N_{oc}$: -98.0dBm /15kHz<br>$\hat{E}s / N_{oc}$: +11.1dB<br><br>$I_{oc}$: -80.0dBm/1.28MHz<br>$\hat{I}_{or} / I_{oc}$: +11.0dB<br>PCCPCH_$E_c/I_{or}$: -3dB<br>DwPCH_$E_c/I_{or}$: 0dB<br><br>During T3:<br><br>$N_{oc}$: -98.0dBm/15kHz<br>$\hat{E}s / N_{oc}$: +11.1dB<br><br>$I_{oc}$: -80.0dBm/1.28MHz<br>$\hat{I}_{or} / I_{oc}$: -3.0dB<br>PCCPCH_$E_c/I_{or}$: -3dB<br>DwPCH_$E_c/I_{or}$: 0dB |
| 4.3.4.2 E-UTRA TDD - UTRAN TDD cell re-selection : UTRA is of lower priority | Same as 4.3.2 | Same as 4.3.2 | Same as 4.3.2 |

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| 4.3.4.3 EUTRA TDD-UTRA TDD cell reselection in fading propagation conditions: UTRA TDD is of lower priority | During T1, T2:<br>E-UTRA Cell 1<br><br>$N_{oc}$: -104.0dBm/15kHz<br>$\hat{E}s / N_{oc}$: +22.00dB<br>UTRA Cell 2<br>$I_{oc}$: -80.0dBm/1.28MHz<br>$\hat{I}_{or} / I_{oc}$: +13.0dB<br>PCCPCH_$E_c/I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T3, T4:<br><br>E-UTRA Cell 1<br>$N_{oc}$: -104.0dBm/15kHz<br>$\hat{E}s / N_{oc}$: -3.0dB<br>UTRA Cell 2<br>$I_{oc}$: -80.0dBm/1.28MHz<br>$\hat{I}_{or} / I_{oc}$: +13.0dB<br>PCCPCH_$E_c/I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB | During T1, T2:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T3, T4:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br>0dB<br>0dB | During T1, T2:<br>E-UTRA Cell 1<br><br>$N_{oc}$: -104.0dBm/15kHz<br>$\hat{E}s / N_{oc}$: +22.00dB<br>UTRA Cell 2<br>$I_{oc}$: -80.0dBm/1.28MHz<br>$\hat{I}_{or} / I_{oc}$: +13.0dB<br>PCCPCH_$E_c/I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T3, T4:<br><br>E-UTRA Cell 1<br>$N_{oc}$: -104.0dBm/15kHz<br>$\hat{E}s / N_{oc}$: -3.0dB<br>UTRA Cell 2<br>$I_{oc}$: -80.0dBm/1.28MHz<br>$\hat{I}_{or} / I_{oc}$: +13.0dB<br>PCCPCH_$E_c/I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB |
| 4.4.1 E-UTRAN FDD - GSM cell re-selection | During T1:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.00dBm/15kHz<br>$\hat{E}s / N_{oc}$: +9.00dB<br>GSM Cell 2<br>Signal level: -90.00dBm<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.00dBm/15kHz<br>$\hat{E}s / N_{oc}$: -4.00dB<br>GSM Cell 2<br>Signal level: -75.0dBm | During T1:<br><br>-1.4dB<br>+1.5dB<br><br>0dB<br><br>During T2:<br><br>-1.4dB<br>+0.3dB<br><br>0dB | During T1:<br>E-UTRA Cell 1<br>$N_{oc}$: -99.40dBm/15kHz<br>$\hat{E}s / N_{oc}$: +10.5dB<br>GSM Cell 2<br>Signal level: -90.00dBm<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -99.40dBm/15kHz<br>$\hat{E}s / N_{oc}$: -3.70dB<br>GSM Cell 2<br>Signal level: -75.00dBm |
| 4.4.2 E-UTRAN TDD - GSM cell re-selection | Same as 4.4.1 | Same as 4.4.1 | Same as 4.4.1 |
| 4.5.1.1 RRC IDLE / E-UTRAN to HRPD Cell re-selection / E-UTRAN FDD - HRPD cell re-selection: HRPD is of lower priority | During T1:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.00dBm/15kHz<br>$\hat{E}s / N_{oc}$: +9.00dB<br>HRPD Cell 2<br>$I_{oc}$: -55.00dBm/1.2288MHz<br>$I_{or} / I_{oc}$: 0 dB<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.00dBm/15kHz<br>$\hat{E}s / N_{oc}$: -4.00dB<br>HRPD Cell 2<br>$I_{oc}$: -55.00dBm/1.2288MHz<br>$I_{or} / I_{oc}$: 0 dB | During T1<br><br>-1.4dB<br>1.5dB<br><br>0dB<br>0dB<br><br>During T2<br><br>-1.4dB<br>0.3dB<br><br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>$N_{oc}$: -99.4dBm/15kHz<br>$\hat{E}s / N_{oc}$: +10.5dB<br>HRPD Cell 2<br>$I_{oc}$: -55.00dBm/1.2288MHz<br>$I_{or} / I_{oc}$: 0 dB<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -99.4dBm/15kHz<br>$\hat{E}s / N_{oc}$: -3.70dB<br>HRPD Cell 2<br>$I_{oc}$: -55.00dBm/1.2288MHz<br>$I_{or} / I_{oc}$: 0 dB |
| 4.5.2.1 E-UTRAN TDD - HRPD Cell Reselection: HRPD is of Lower Priority | Same as 4.5.1.1 | Same as 4.5.1.1 | Same as 4.5.1.1 |

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| 5.1.1 E-UTRAN FDD-FDD Handover intra frequency case | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +8.00dB<br>$\hat{E}s_2$ / $N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +8.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +11.00dB<br><br>During T3:<br>$N_{oc}$: -98dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +8.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +11.00dB | During T1:<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>0dB<br>0dB<br>+0.5dB<br><br>During T3:<br>0dB<br>0dB<br>+0.5dB | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +8.00dB<br>$\hat{E}s_2$ / $N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +8.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +11.50dB<br><br>During T3:<br>$N_{oc}$: -98dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +8.00<br>$\hat{E}s_2$ / $N_{oc}$: +11.50dB |
| 5.1.2 E-UTRAN TDD-TDD Handover intra frequency case | Same as 5.1.1 | Same as 5.1.1 | Same as 5.1.1 |
| 5.1.3 E-UTRAN FDD-FDD Handover inter frequency case | During T1:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +4dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -infinity<br><br>During T2:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +4dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: +7.0dB<br><br>During T3:<br>$N_{oc1}$: -98dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +4dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: +7.0dB | During T1:<br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>0dB<br>0dB<br>0dB<br>+0.5dB<br><br>During T3:<br>0dB<br>0dB<br>0dB<br>+0.5dB | During T1:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +4dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -infinity<br><br>During T2:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +4dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: +7.50dB<br><br>During T3:<br>$N_{oc1}$: -98dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +4dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: +7.50dB |
| 5.1.4 E-UTRAN TDD-TDD Handover inter frequency case | Same as 5.1.3 | Same as 5.1.3 | Same as 5.1.3 |
| 5.1.5 E-UTRAN FDD-FDD inter-frequency Handover with unknown target cell | During T1:<br>Noc1: -98dBm/15kHz<br>Ês1 / Noc1: +4dB<br>Noc2: -98dBm/15kHz<br>Ês2 / Noc2: -infinity<br><br>During T2:<br>Noc1: -98dBm/15kHz<br>Ês1 / Noc1: +4dB<br>Noc2: -98dBm/15kHz<br>Ês2 / Noc2: +7.0dB | During T1:<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>0dB<br>0dB<br>0dB<br>0dB | During T1:<br>Noc1: -98dBm/15kHz<br>Ês1 / Noc1: +4dB<br>Noc2: -98dBm/15kHz<br>Ês2 / Noc2: -infinity<br><br>During T2:<br>Noc1: -98dBm/15kHz<br>Ês1 / Noc1: +4dB<br>Noc2: -98dBm/15kHz<br>Ês2 / Noc2: +7.0dB |
| 5.1.6 E-UTRAN TDD-TDD inter-frequency Handover with unknown target cell | During T1:<br>Noc1: -98dBm/15kHz<br>Ês1 / Noc1: +4dB<br>Noc2: -98dBm/15kHz<br>Ês2 / Noc2: -infinity<br><br>During T2:<br>Noc1: -98dBm/15kHz<br>Ês1 / Noc1: +4dB<br>Noc2: -98dBm/15kHz<br>Ês2 / Noc2: +5.0dB | During T1:<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>0dB<br>0dB<br>0dB<br>0dB | During T1:<br>Noc1: -98dBm/15kHz<br>Ês1 / Noc1: +4dB<br>Noc2: -98dBm/15kHz<br>Ês2 / Noc2: -infinity<br><br>During T2:<br>Noc1: -98dBm/15kHz<br>Ês1 / Noc1: +4dB<br>Noc2: -98dBm/15kHz<br>Ês2 / Noc2: +5.0dB |
| 5.1.7 E-UTRAN FDD-TDD Handover inter frequency case | Same as 5.1.3 | Same as 5.1.3 | Same as 5.1.3 |
| 5.1.8 E-UTRAN TDD-FDD Handover inter frequency case | Same as 5.1.3 | Same as 5.1.3 | Same as 5.1.3 |

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| 5.1.9 E-UTRAN FDD-FDD Intra frequency handover for 5MHz bandwidth | Same as 5.1.1 | Same as 5.1.1 | Same as 5.1.1 |
| 5.1.10 E-UTRAN FDD-FDD Handover intra frequency handover for UE category 0 | Same as 5.1.1 | Same as 5.1.1 | Same as 5.1.1 |
| 5.1.11 E-UTRAN HD-FDD Handover intra frequency handover for UE category 0 | Same as 5.1.1 | Same as 5.1.1 | Same as 5.1.1 |
| 5.1.12 E-UTRAN TDD-TDD Handover intra frequency handover for UE category 0 | Same as 5.1.1 | Same as 5.1.1 | Same as 5.1.1 |
| 5.1.13 E-UTRAN FDD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA | Same as 5.1.1 | Same as 5.1.1 | Same as 5.1.1 |
| 5.1.14 E-UTRAN HD-FDD Intra frequency handover for Cat-M1 UEs in CEModeA | Same as 5.1.1 | Same as 5.1.1 | Same as 5.1.1 |
| 5.1.15 E-UTRAN TDD Intra frequency handover for Cat-M1 UEs in CEModeA | Same as 5.1.1 | Same as 5.1.1 | Same as 5.1.1 |
| 5.2.1 E-UTRAN FDD - UTRAN FDD handover | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 0.00dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 0.00dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc:-1.80dB<br><br>During T3:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 0.00dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc:-1.80dB | During T1:<br><br>0dB<br>-1.10dB<br><br>0dB<br>-<br><br>During T2:<br><br>0dB<br>-1.10dB<br><br>0dB<br>0dB<br><br>During T3:<br><br>0dB<br>-1.10dB<br><br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: -1.10dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: -1.10dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc:-1.80dB<br><br>During T3:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: -1.10dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc:-1.80dB |
| 5.2.2 E-UTRAN TDD - UTRAN FDD handover | Same as 5.2.1 | Same as 5.2.1 | Same as 5.2.1 |

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| 5.2.3 E-UTRAN FDD - GSM handover | During T1:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -85dBm<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -75dBm<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm /15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -75dBm | During T1:<br><br>0dB<br>0dB<br><br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br><br>During T3:<br><br>0dB<br>0dB<br><br>0dB | During T1:<br>E-UTRAN Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -85dBm<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -75dBm<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm /15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -75dBm |
| 5.2.4 E-UTRA TDD – UTRA TDD handover | During T1:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}_s$ / $N_{oc}$: +13.00dB<br>UTRA Cell 2<br>$I_{oc}$: -80dBm/1.28MHz<br>$\hat{I}_{or}$ / $I_{oc}$: -3.00dB<br>PCCPCH_Ec/Ior: -3dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: -3.00dB<br>UTRA Cell 2<br>$I_{oc}$: -80dBm/1.28MHz<br>$\hat{I}_{or}$ / $I_{oc}$: 11.00dB<br>PCCPCH_Ec/Ior: -3dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: -3.00dB<br>UTRA Cell 2<br>$I_{oc}$: -80dBm/1.28MHz<br>$\hat{I}_{or}$ / $I_{oc}$: 11.00dB<br>PCCPCH_Ec/Ior: -3dB<br>DwPCH_Ec/Ior: 0dB | During T1:<br><br>-1.05dB<br>2.1dB<br><br>-0.8dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br><br>-1.05dB<br>0dB<br><br>-0.8dB<br>1.6dB<br>0dB<br>0dB<br><br>During T3:<br><br>-1.05dB<br>0dB<br><br>-0.8dB<br>+1.6dB<br>0dB<br>0dB | During T1:<br><br>$N_{oc}$: -99.05dBm/15kHz<br>Ês / $N_{oc}$: +15.1dB<br><br>$I_{oc}$: -80.8dBm/1.28MHz<br>$\hat{I}_{or}$ / $I_{oc}$: -3.0dB<br>PCCPCH_Ec/Ior: -3dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T2:<br><br>$N_{oc}$: -99.05dBm/15kHz<br>Ês / $N_{oc}$: -3.0dB<br><br>$I_{oc}$: -80.8dBm/1.28MHz<br>$\hat{I}_{or}$ / $I_{oc}$: 12.6dB<br>PCCPCH_Ec/Ior: -3dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T3:<br><br>$N_{oc}$: -99.05dBm/15kHz<br>Ês / $N_{oc}$: -3.0dB<br><br>$I_{oc}$: -80.8dBm/1.28MHz<br>$\hat{I}_{or}$ / $I_{oc}$: 12.6dB<br>PCCPCH_Ec/Ior: -3dB<br>DwPCH_Ec/Ior: 0dB |
| 5.2.5 E-UTRA FDD – UTRA TDD handover | Same as 5.2.4 | Same as 5.2.4 | Same as 5.2.4 |
| 5.2.6 E-UTRA TDD - GSM handover | Same as 5.2.3 | Same as 5.2.3 | Same as 5.2.3 |
| 5.2.7 E-UTRAN FDD - UTRAN FDD handover: unknown target cell | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 0dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 0dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -1.8 dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 0dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 0dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -1.8dB |

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| 5.2.8 E-UTRAN FDD - GSM handover: unknown target cell | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>GSM Cell 2<br>Signal level: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>GSM Cell 2<br>Signal level: -75 dBm | During T1:<br><br>0dB<br>0dB<br><br><br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>GSM Cell 2<br>Signal level: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>GSM Cell 2<br>Signal level: -75 dBm |
| 5.2.9 E-UTRAN TDD – GSM handover: unknown target cell | Same as 5.2.8 | Same as 5.2.8 | Same as 5.2.8 |
| 5.2.10 E-UTRAN TDD - UTRAN TDD HO test: unknown target cell | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 3dB<br>UTRA Cell 2<br>Ioc: -80dBm/1.28MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 3dB<br>UTRA Cell 2<br>Ioc: -80dBm/1.28MHz<br>Ior / Ioc: 13 dB<br>PCCPCH_$E_c/I_{or}$: -3.00dB<br>DwPCH_$E_c/I_{or}$: 0dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br><br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 3dB<br>UTRA Cell 2<br>Ioc: -80dBm/1.28MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 3dB<br>UTRA Cell 2<br>Ioc: -80dBm/1.28MHz<br>Ior / Ioc: 13 dB<br>PCCPCH_$E_c/I_{or}$: -3.00dB<br>DwPCH_$E_c/I_{or}$: 0dB |
| 5.2.11 E-UTRAN FDD – UTRAN FDD Handover for 5MHz Bandwidth | Same as 5.2.1 | Same as 5.2.1 | Same as 5.2.1 |
| 5.3.1 RRC CONNECTED / Handover from E-UTRAN to non-3GPP RATs / E-UTRAN FDD – HRPD handover | During T1:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}_s / N_{oc}$: 0dB<br>HRPD Cell 2<br>$I_{oc}$: -55.00dBm/1.2288MHz<br>$I_{or} / I_{oc}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}_s / N_{oc}$: 0dB<br>HRPD Cell 2<br>$I_{oc}$: -55.00dBm/1.2288MHz<br>$I_{or} / I_{oc}$: 0 dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}_s / N_{oc}$: 0dB<br>HRPD Cell 2<br>$I_{oc}$: -55.00dBm/1.2288MHz<br>$I_{or} / I_{oc}$: 0 dB | During T1:<br><br>0dB<br>-0.8dB<br><br>0dB<br><br><br>During T2:<br><br>0dB<br>-0.8dB<br><br>0dB<br>0dB<br><br>During T3:<br><br>0dB<br>-0.8dB<br><br>0dB<br>0dB | During T1:<br><br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}_s / N_{oc}$: -0.8dB<br><br>$I_{oc}$: -55.00dBm/1.2288MHz<br>$I_{or} / I_{oc}$: -infinity<br><br>During T2:<br><br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}_s / N_{oc}$: -0.8dB<br><br>$I_{oc}$: -55.00dBm/1.2288MHz<br>$I_{or} / I_{oc}$: 0 dB<br><br>During T3:<br><br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}_s / N_{oc}$: -0.8dB<br><br>$I_{oc}$: -55.00dBm/1.2288MHz<br>$I_{or} / I_{oc}$: 0 dB |
| 5.3.5 E-UTRAN TDD - HRPD handover | Same as 5.3.1 | Same as 5.3.1 | Same as 5.3.1 |

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| 6.1.1 E-UTRAN FDD Intra-frequency RRC Re-establishment | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +7.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +4.00dB<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -infinity<br>$\hat{E}s_2$ / $N_{oc}$: +4.00dB<br><br>During T3:<br>$N_{oc}$: -98dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -infinity<br>$\hat{E}s_2$ / $N_{oc}$: +4.00dB | During T1:<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>0dB<br>0dB<br>0dB<br><br>During T3:<br>0dB<br>0dB<br>0dB | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +7.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +4.00dB<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -infinity<br>$\hat{E}s_2$ / $N_{oc}$: +4.00dB<br><br>During T3:<br>$N_{oc}$: -98dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -infinity<br>$\hat{E}s_2$ / $N_{oc}$: +4.00dB |
| 6.1.2 E-UTRAN FDD Inter-frequency RRC Re-establishment | During T1:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -infinity<br><br>During T2:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -infinity<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -infinity<br><br>During T3:<br>$N_{oc1}$: -98dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -infinity<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: +7.00dB | During T1:<br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T3:<br>0dB<br>0dB<br>0dB<br>0dB | During T1:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -infinity<br><br>During T2:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -infinity<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -infinity<br><br>During T3:<br>$N_{oc1}$: -98dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -infinity<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: +7.00dB |
| 6.1.3 E-UTRAN TDD Intra-frequency RRC Re-establishment | Same as 6.1.1 | Same as 6.1.1 | Same as 6.1.1 |
| 6.1.4 E-UTRAN TDD Inter-frequency RRC Re-establishment | Same as 6.1.3 | Same as 6.1.3 | Same as 6.1.3 |
| 6.1.5 E-UTRAN FDD Intra-frequency RRC Re-establishment for 5MHz Bandwidth | Same as 6.1.1 | Same as 6.1.1 | Same as 6.1.1 |
| 6.1.6 E-UTRAN FD-FDD Intra-frequency RRC Re-establishment for UE category 0 | Same as 6.1.1 | Same as 6.1.1 | Same as 6.1.1 |
| 6.1.7 E-UTRAN HD-FDD Intra-frequency RRC Re-establishment for UE category 0 | Same as 6.1.1 | Same as 6.1.1 | Same as 6.1.1 |
| 6.1.8 E-UTRAN TDD Intra-frequency RRC Re-establishment for UE category 0 | Same as 6.1.1 | Same as 6.1.1 | Same as 6.1.1 |
| 6.1.9 E-UTRAN FD-FDD Intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA | Same as 6.1.1 | Same as 6.1.1 | Same as 6.1.1 |
| 6.1.10 E-UTRAN HD-FDD Intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA | Same as 6.1.1 | Same as 6.1.1 | Same as 6.1.1 |
| 6.1.11 E-UTRAN TDD Intra-frequency RRC Re-establishment for Cat-M1 UE in CEModeA | Same as 6.1.1 | Same as 6.1.1 | Same as 6.1.1 |

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| 6.2.1 E-UTRAN FDD - Contention Based Random Access Test | Test 1 and Test 2<br>Absolute uplink power:<br>Normal conditions ±9dB<br>Extreme conditions ±12dB<br><br>Relative uplink power step:<br>Normal conditions ±3dB<br>Extreme conditions ±5dB<br><br>Uplink timing $T_e$: ±12$T_s$ | <br><br>1.5dB<br>1.5dB<br><br><br>0.7dB<br>0.7dB<br><br>3$T_s$ | Test 1 and Test 2<br>Absolute uplink power:<br>Normal conditions ±10.5dB<br>Extreme conditions ±13.5dB<br><br>Relative uplink power step:<br>Normal conditions ±3.7dB<br>Extreme conditions ±5.7dB<br><br>Uplink timing $T_e$: ±15$T_s$ |
| 6.2.2 E-UTRAN FDD - Non Contention Based Random Access Test | Same as 6.2.1 | Same as 6.2.1 | Same as 6.2.1 |
| 6.2.3 E-UTRAN TDD - Contention Based Random Access Test | Same as 6.2.1 | Same as 6.2.1 | Same as 6.2.1 |
| 6.2.4 E-UTRAN TDD - Non Contention Based Random Access Test | Same as 6.2.1 | Same as 6.2.1 | Same as 6.2.1 |
| 6.2.5 E-UTRAN FDD – Contention Based Random Access Test for 5MHz bandwidth | Same as 6.2.1 | Same as 6.2.1 | Same as 6.2.1 |
| 6.2.6 E-UTRAN FDD – Non-contention Based Random Access Test for 5MHz bandwidth | Same as 6.2.1 | Same as 6.2.1 | Same as 6.2.1 |
| 6.2.7 E-UTRAN FDD – Non-Contention Based Random Access Test for SCell in sTAG | Test 1, Test 2 and Test 3<br>Absolute uplink power:<br>Normal conditions ±9dB<br>Extreme conditions ±12dB<br><br>Relative uplink power step:<br>Normal conditions ±3dB<br>Extreme conditions ±5dB<br><br>Uplink timing $T_e$: ±12$T_s$ | <br><br>1.5dB<br>1.5dB<br><br><br>0.7dB<br>0.7dB<br><br>3$T_s$ | Test 1, Test 2 and Test 3<br>Absolute uplink power:<br>Normal conditions ±10.5dB<br>Extreme conditions ±13.5dB<br><br>Relative uplink power step:<br>Normal conditions ±3.7dB<br>Extreme conditions ±5.7dB<br><br>Uplink timing $T_e$: ±15$T_s$ |
| 6.2.8 E-UTRAN TDD – Non-Contention Based Random Access Test for SCell in sTAG | Same as 6.2.7 | Same as 6.2.7 | Same as 6.2.7 |
| 6.2.12 E-UTRAN TDD Contention Based Random Access Test for Cat-M1 UEs in Normal Coverage | Same as 6.2.1 | Same as 6.2.1 | Same as 6.2.1 |
| 6.3.1 Redirection from E-UTRAN FDD to UTRAN FDD | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: 0.02 dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br><br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>0.4dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: 0.42dB |
| 6.3.2 Redirection from E-UTRAN TDD to UTRAN FDD | Same as 6.3.1 | Same as 6.3.1 | Same as 6.3.1 |

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| 6.3.3 Redirection from E-UTRAN FDD to GERAN when System Information is provided | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>GSM Cell 2<br>Signal level: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>GSM Cell 2<br>Signal level: -75 dBm | During T1:<br><br>0dB<br>0dB<br><br><br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>GSM Cell 2<br>Signal level: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>GSM Cell 2<br>Signal level: -75 dBm |
| 6.3.4 Redirection from E-UTRAN TDD to GERAN when System Information is provided | Same as 6.3.3 | Same as 6.3.3 | Same as 6.3.3 |
| 6.3.5 E-UTRA TDD RRC connection release redirection to UTRA TDD | During T1:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.0dBm/15kHz<br>Ês / $N_{oc}$: +4.00dB<br>UTRA Cell 2<br>$I_{oc}$: -80.0dBm/1.28MHz<br>$\hat{I}_{or}$ / $I_{oc}$: -infinity<br>PCCPCH_$E_c$/$I_{or}$: -4.77dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.0dBm/15kHz<br>Ês / $N_{oc}$: +4.0dB<br>UTRA Cell 2<br>$I_{oc}$: -80.0dBm/1.28MHz<br>$\hat{I}_{or}$ / $I_{oc}$: +8.0dB<br>PCCPCH_$E_c$/$I_{or}$: -4.77dB<br>DwPCH_Ec/Ior: 0dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.0dBm/15kHz<br>Ês / $N_{oc}$: +4.00dB<br>UTRA Cell 2<br>$I_{oc}$: -80.0dBm/1.28MHz<br>$\hat{I}_{or}$ / $I_{oc}$: -infinity<br>PCCPCH_$E_c$/$I_{or}$: -4.77dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -98.0dBm/15kHz<br>Ês / $N_{oc}$: +4.0dB<br>UTRA Cell 2<br>$I_{oc}$: -80.0dBm/1.28MHz<br>$\hat{I}_{or}$ / $I_{oc}$: +8.0dB<br>PCCPCH_$E_c$/$I_{or}$: -4.77dB<br>DwPCH_Ec/Ior: 0dB |
| 6.3.6 E-UTRA FDD RRC connection release redirection to UTRA TDD | Same as 6.3.5 | Same as 6.3.5 | Same as 6.3.5 |
| 6.3.7 E-UTRA TDD RRC connection release redirection to UTRA TDD without SI provided | Same as 6.3.5 | Same as 6.3.5 | Same as 6.3.5 |
| 6.3.8 E-UTRA FDD RRC connection release redirection to UTRA TDD without SI provided | Same as 6.3.5 | Same as 6.3.5 | Same as 6.3.5 |
| 6.3.9 Redirection from E-UTRAN FDD to UTRAN FDD without System Information | Same as 6.3.1 | Same as 6.3.1 | Same as 6.3.1 |
| 6.3.10 Redirection from E-UTRAN FDD to GERAN when System Information is not provided | Same as 6.3.3 | Same as 6.3.3 | Same as 6.3.3 |
| 6.3.11 Redirection from E-UTRAN TDD to GERAN when System Information is not provided | Same as 6.3.3 | Same as 6.3.3 | Same as 6.3.3 |
| 6.3.12 E-UTRAN TDD RRC connection release redirection to UTRAN FDD without SI provided | Same as 6.3.1 | Same as 6.3.1 | Same as 6.3.1 |

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| 7.1.1 E-UTRAN FDD - UE Transmit Timing Accuracy | Test 1 (10MHz Ch BW):<br>Uplink timing: ±12$T_s$<br>Max step size $T_q$: 3.5$T_s$<br>Min adjust rate: 7$T_s$<br>Max adjust rate: 3.5$T_s$<br>Ês / $N_{oc}$: +3.00dB<br><br>Test 2 (10MHz Ch BW):<br>Uplink timing: ±12$T_s$<br>Ês / $N_{oc}$: +3.00dB<br><br>Test 3: (1.4MHz Ch BW)<br>Uplink timing: ±24$T_s$<br>Max step size $T_q$: 17.5$T_s$<br>Max adjust rate: 17.5$T_s$<br>Ês / $N_{oc}$: +3.00dB | <br>±3$T_s$<br>+0.5$T_s$<br>-3.6$T_s$<br>+1.1$T_s$<br>+0.3dB<br><br><br>±3$T_s$<br>+0.3dB<br><br><br>±3$T_s$<br>+0.5$T_s$<br>+1.1$T_s$<br>+0.3dB | Test 1 (10MHz Ch BW):<br>Uplink timing: ±15$T_s$<br>Max step size $T_q$: 4.0T<br>Min adjust rate: 3.4$T_s$<br>Max adjust rate: 4.6$T_s$<br>Ês / $N_{oc}$: +3.30dB<br><br>Test 2 (10MHz Ch BW):<br>Uplink timing: ±15$T_s$<br>Ês / $N_{oc}$: +3.30dB<br><br>Test 3: (1.4MHz Ch BW)<br>Uplink timing: ±27$T_s$<br>Max step size $T_q$: 18$T_s$<br>Max adjust rate: 18.6$T_s$<br>Ês / $N_{oc}$: +3.30dB |
| 7.1.1_1 E-UTRAN FDD - UE Transmit Timing Accuracy (Non DRx UE) | Test 1 (10MHz Ch BW):<br>Uplink timing: ±12$T_s$<br>Max step size $T_q$: 3.5$T_s$<br>Min adjust rate: 7$T_s$<br>Max adjust rate: 3.5$T_s$<br>Ês / $N_{oc}$: +3.00dB<br><br>Test 2 not applicable<br><br>Test 3: (1.4MHz Ch BW)<br>Uplink timing: ±24$T_s$<br>Max step size $T_q$: 17.5$T_s$<br>Max adjust rate: 17.5$T_s$<br>Ês / $N_{oc}$: +3.00dB | <br>±3$T_s$<br>+0.5$T_s$<br>-3.6$T_s$<br>+1.1$T_s$<br>+0.3dB<br><br><br><br>±3$T_s$<br>+0.5$T_s$<br>+1.1$T_s$<br>+0.3dB | Test 1 (10MHz Ch BW):<br>Uplink timing: ±15$T_s$<br>Max step size $T_q$: 4$T_s$<br>Min adjust rate: 3.4$T_s$<br>Max adjust rate: 4.6$T_s$<br>Ês / $N_{oc}$: +3.30dB<br><br>Test 2 not applicable<br><br>Test 3: (1.4MHz Ch BW)<br>Uplink timing: ±27$T_s$<br>Max step size $T_q$: 18$T_s$<br>Max adjust rate: 18.6$T_s$<br>Ês / $N_{oc}$: +3.30dB |
| 7.1.2 E-UTRAN TDD - UE Transmit Timing Accuracy | Test 1 (10MHz Ch BW):<br>Uplink timing: (624 ±12) x $T_s$<br>Max step size $T_q$: 3.5$T_s$<br>Min adjust rate: 7$T_s$<br>Max adjust rate: 3.5$T_s$<br>Ês / $N_{oc}$: +3.00dB<br><br>Test 2 (10MHz Ch BW):<br>Uplink timing: (624 ±12) x $T_s$<br>Ês / $N_{oc}$: +3.00dB<br><br>Test 3: (1.4MHz Ch BW)<br>Uplink timing: (624 ±24) x $T_s$<br>Max step size $T_q$: 17.5$T_s$<br>Max adjust rate: 17.5$T_s$<br>Ês / $N_{oc}$: +3.00dB | <br>±3$T_s$<br>+0.5$T_s$<br>-3.6$T_s$<br>+1.1$T_s$<br>+0.3dB<br><br><br>±3$T_s$<br>+0.3dB<br><br><br>±3$T_s$<br>+0.5$T_s$<br>+1.1$T_s$<br>+0.3dB | Test 1 (10MHz Ch BW):<br>Uplink timing: (624 ±15) x $T_s$<br>Max step size $T_q$: 4$T_s$<br>Min adjust rate: 3.4$T_s$<br>Max adjust rate: 4.6$T_s$<br>Ês / $N_{oc}$: +3.30dB<br><br>Test 2 (10MHz Ch BW):<br>Uplink timing: (624 ±15) x $T_s$<br>Ês / $N_{oc}$: +3.30dB<br><br>Test 3: (1.4MHz Ch BW)<br>Uplink timing: (624 ±27) x $T_s$<br>Max step size $T_q$: 18$T_s$<br>Max adjust rate: 18.6$T_s$<br>Ês / $N_{oc}$: +3.30dB |
| 7.1.2_1 E-UTRAN TDD - UE Transmit Timing Accuracy (Non DRx UE) | Test 1 (10MHz Ch BW):<br>Uplink timing: (624 ±12) x $T_s$<br>Max step size $T_q$: 3.5$T_s$<br>Min adjust rate: 7$T_s$<br>Max adjust rate: 3.5$T_s$<br>Ês / $N_{oc}$: +3.00dB<br><br>Test 2 not applicable<br><br>Test 3: (1.4MHz Ch BW)<br>Uplink timing: (624 ±24) x $T_s$<br>Max step size $T_q$: 17.5$T_s$<br>Max adjust rate: 17.5$T_s$<br>Ês / $N_{oc}$: +3.00dB | <br>±3$T_s$<br>+0.5$T_s$<br>-3.6$T_s$<br>+1.1$T_s$<br>+0.3dB<br><br><br><br>±3$T_s$<br>+0.5$T_s$<br>+1.1$T_s$<br>+0.3dB | Test 1 (10MHz Ch BW):<br>Uplink timing: (624 ±15) x $T_s$<br>Max step size $T_q$: 4$T_s$<br>Min adjust rate: 3.4$T_s$<br>Max adjust rate: 4.6$T_s$<br>Ês / $N_{oc}$: +3.30dB<br><br>Test 2 not applicable<br><br>Test 3: (1.4MHz Ch BW)<br>Uplink timing: (624 ±27) x $T_s$<br>Max step size $T_q$: 18$T_s$<br>Max adjust rate: 18.6$T_s$<br>Ês / $N_{oc}$: +3.30dB |

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| 7.1.3 E-UTRAN FDD – UE Transmit Timing Accuracy Tests for SCell | Test 1:<br>Uplink timing: ±12$T_s$<br>Max step size $T_q$: 3.5$T_s$<br>Min adjust rate: 7$T_s$<br>Max adjust rate: 3.5$T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.00dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.00dB<br><br>Test 2:<br>Uplink timing: ±12$T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.00dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.00dB | ±3$T_s$<br>+0.5$T_s$<br>-3.6$T_s$<br>+1.1$T_s$<br>+0.3dB<br>+0.3dB<br><br>±3$T_s$<br>+0.3dB<br>+0.3dB | Test 1:<br>Uplink timing: ±15$T_s$<br>Max step size $T_q$: 4$T_s$<br>Min adjust rate: 3.4$T_s$<br>Max adjust rate: 4.6$T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.30dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.30dB<br><br>Test 2:<br>Uplink timing: ±15$T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.30dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.30dB |
| 7.1.3_1 E-UTRAN FDD – UE Transmit Timing Accuracy Tests for SCell (Release 12 and forward) | Test 1:<br>Uplink timing: ±12$T_s$<br>Max step size $T_q$: 3.5$T_s$<br>Min adjust rate: 7$T_s$<br>Max adjust rate: 3.5$T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.00dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.00dB<br><br>Test 2 and Test 3:<br>Uplink timing: ±12$T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.00dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.00dB | ±3$T_s$<br>+0.5$T_s$<br>-3.6$T_s$<br>+1.1$T_s$<br>+0.3dB<br>+0.3dB<br><br>±3$T_s$<br>+0.3dB<br>+0.3dB | Test 1:<br>Uplink timing: ±15$T_s$<br>Max step size $T_q$: 4$T_s$<br>Min adjust rate: 3.4$T_s$<br>Max adjust rate: 4.6$T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.30dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.30dB<br><br>Test 2 and Test 3:<br>Uplink timing: ±15$T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.30dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.30dB |
| 7.1.4 E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell | Test 1:<br>Uplink timing: (624 ±12) x $T_s$<br>Max step size $T_q$: 3.5$T_s$<br>Min adjust rate: 7$T_s$<br>Max adjust rate: 3.5$T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.00dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.00dB<br><br>Test 2:<br>Uplink timing: (624 ±12) x $T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.00dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.00dB | ±3$T_s$<br>+0.5$T_s$<br>-3.6$T_s$<br>+1.1$T_s$<br>+0.3dB<br>+0.3dB<br><br>±3$T_s$<br>+0.3dB<br>+0.3dB | Test 1:<br>Uplink timing: (624 ±15) x $T_s$<br>Max step size $T_q$: 4$T_s$<br>Min adjust rate: 3.4$T_s$<br>Max adjust rate: 4.6$T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.30dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.30dB<br><br>Test 2:<br>Uplink timing: (624 ±15) x $T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.30dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.30dB |
| 7.1.4_1 E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell (Release 12 and forward) | Test 1:<br>Uplink timing: (624 ±12) x $T_s$<br>Max step size $T_q$: 3.5$T_s$<br>Min adjust rate: 7$T_s$<br>Max adjust rate: 3.5$T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.00dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.00dB<br><br>Test 2 and Test:<br>Uplink timing: (624 ±12) x $T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.00dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.00dB | ±3$T_s$<br>+0.5$T_s$<br>-3.6$T_s$<br>+1.1$T_s$<br>+0.3dB<br>+0.3dB<br><br>±3$T_s$<br>+0.3dB<br>+0.3dB | Test 1:<br>Uplink timing: (624 ±15) x $T_s$<br>Max step size $T_q$: 4$T_s$<br>Min adjust rate: 3.4$T_s$<br>Max adjust rate: 4.6$T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.30dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.00dB<br><br>Test 2 and Test 3:<br>Uplink timing: (624 ±15) x $T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.30dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.30dB |
| 7.1.4A E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell for 20 MHz +10 MHz bandwidth | Same as 7.1.4 | Same as 7.1.4 | Same as 7.1.4 |
| 7.1.5 E-UTRAN FDD - UE Transmit Timing Accuracy Tests for 5MHz Bandwidth | Test 1 (5MHz Ch BW):<br>Uplink timing: ±12$T_s$<br>Max step size $T_q$: 5.5$T_s$<br>Min adjust rate: 11$T_s$ in 1.57s<br>Max adjust rate: 5.5$T_s$<br>$\hat{E}s$ / $N_{oc}$: +3.00dB | ±3$T_s$<br>+0.5$T_s$<br>-5.3$T_s$<br>+1.1$T_s$<br>+0.3dB | Test 1 (5MHz Ch BW):<br>Uplink timing: ±15$T_s$<br>Max step size $T_q$: 6.0$T_s$<br>Min adjust rate: 5.7$T_s$ in 1.57s<br>Max adjust rate: 6.6$T_s$<br>$\hat{E}s$ / $N_{oc}$: +3.30dB |

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| 7.1.6 E-UTRAN FDD - UE Transmit Timing Accuracy Tests for SCell in sTAG | Test 1:<br>Uplink timing: ±12$T_s$<br>Max step size $T_q$: 3.5$T_s$<br>Min adjust rate: 7$T_s$<br>Max adjust rate: 3.5$T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.00dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.00dB<br><br>Test 2:<br>Uplink timing: ±12$T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.00dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.00dB | ±3$T_s$<br>+0.5$T_s$<br>-3.6$T_s$<br>+1.1$T_s$<br>+0.3dB<br>+0.3dB<br><br>±3$T_s$<br>+0.3dB<br>+0.3dB | Test 1:<br>Uplink timing: ±15$T_s$<br>Max step size $T_q$: 4.0$T_s$<br>Min adjust rate: 3.4$T_s$<br>Max adjust rate: 4.6$T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.30dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.30dB<br><br>Test 2:<br>Uplink timing: ±15$T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.30dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.30dB |
| 7.1.7 E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell in sTAG | Test 1:<br>Uplink timing: (624 ±12) x $T_s$<br>Max step size $T_q$: 3.5$T_s$<br>Min adjust rate: 7$T_s$<br>Max adjust rate: 3.5$T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.00dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.00dB<br><br>Test 2:<br>Uplink timing: (624 ±12) x $T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.00dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.00dB | ±3$T_s$<br>+0.5$T_s$<br>-3.6$T_s$<br>+1.1$T_s$<br>+0.3dB<br>+0.3dB<br><br>±3$T_s$<br>+0.3dB<br>+0.3dB | Test 1:<br>Uplink timing: (624 ±15) x $T_s$<br>Max step size $T_q$: 4.0$T_s$<br>Min adjust rate: 3.4$T_s$<br>Max adjust rate: 4.6$T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.30dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.30dB<br><br>Test 2:<br>Uplink timing: (624 ±15) x $T_s$<br>$\hat{E}_{s1}$ / $N_{oc1}$: +3.30dB<br>$\hat{E}_{s2}$ / $N_{oc2}$: +3.30dB |
| 7.1.7A E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell in sTAG for 20MHz +20MHz bandwidth | Sams as 7.1.7 | Sams as 7.1.7 | Sams as 7.1.7 |
| 7.1.7B E-UTRAN TDD – UE Transmit Timing Accuracy Tests for SCell in sTAG for 20MHz +10MHz bandwidth | Sams as 7.1.7 | Sams as 7.1.7 | Sams as 7.1.7 |
| 7.1.10 E-UTRAN FDD – UE Transmit Timing Accuracy Tests for Cat-M1 UE in CEModeA | Test 1:<br>Same as 7.1.1 Test 1<br><br>Test 2 & 3:<br>Same as 7.1.1 Test 2 | Test 1:<br>Same as 7.1.1 Test 1<br><br>Test 2 & 3:<br>Same as 7.1.1 Test 2 | Test 1:<br>Same as 7.1.1 Test 1<br><br>Test 2 & 3:<br>Same as 7.1.1 Test 2 |
| 7.1.11 E-UTRAN HD-FDD – UE Transmit Timing Accuracy Tests for Cat-M1 UE in CEModeA | Test 1:<br>Same as 7.1.1 Test 1<br><br>Test 2 & 3:<br>Same as 7.1.1 Test 2 | Test 1:<br>Same as 7.1.1 Test 1<br><br>Test 2 & 3:<br>Same as 7.1.1 Test 2 | Test 1:<br>Same as 7.1.1 Test 1<br><br>Test 2 & 3:<br>Same as 7.1.1 Test 2 |
| 7.2.1 E-UTRAN FDD - UE Timing Advanced Adjustment Accuracy | Timing Advance Adjustment:<br>±4$T_s$ | 0.5$T_S$ | Timing Advance Adjustment:<br>±4.5$T_s$ |
| 7.2.2 E-UTRAN TDD - UE Timing Advance Adjustment Accuracy | Same as 7.2.1 | Same as 7.2.1 | Same as 7.2.1 |
| 7.2.3 E-UTRAN FDD - UE Timing Advance Adjustment Accuracy Test for 5MHz Bandwidth | Same as 7.2.1 | Same as 7.2.1 | Same as 7.2.1 |
| 7.2.4 E-UTRAN FDD - UE Timing Advance Adjustment Accuracy Test for SCell in sTAG | Same as 7.2.1 | Same as 7.2.1 | Same as 7.2.1 |

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| 7.2.5 E-UTRAN TDD - UE Timing Advance Adjustment Accuracy Test for SCell in sTAG | Same as 7.2.1 | Same as 7.2.1 | Same as 7.2.1 |
| 7.2.5A E-UTRAN TDD - UE Timing Advance Adjustment Accuracy Test for SCell in sTAG for 20MHz +20MHz bandwidth | Same as 7.2.1 | Same as 7.2.1 | Same as 7.2.1 |
| 7.2.5B E-UTRAN TDD - UE Timing Advance Adjustment Accuracy Test for SCell in sTAG for 20MHz +10MHz bandwidth | Same as 7.2.1 | Same as 7.2.1 | Same as 7.2.1 |
| 7.2.6 E-UTRAN FDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA | Same as 7.2.1 | Same as 7.2.1 | Same as 7.2.1 |
| 7.2.7 E-UTRAN HD-FDD UE Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA | Same as 7.2.1 | Same as 7.2.1 | Same as 7.2.1 |
| 7.2.8 E-UTRAN TDD Timing Advance Adjustment Accuracy Test for Cat-M1 UE in CEModeA | Same as 7.2.1 | Same as 7.2.1 | Same as 7.2.1 |
| 7.3.1 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync | SNRs as specified | 0.6dB (Subtests 1&2)<br><br>0.8dB (Subtest 3)<br><br>0.9dB (Subtest 4) | During T1:<br>Formula: SNR + TT<br><br>During T2:<br>Formula: SNR + TT<br><br>During T3:<br>Formula: SNR - TT |
| 7.3.2 E-UTRAN FDD Radio Link Monitoring Test for In-sync | SNRs as specified | 0.8dB (Subtest 1)<br><br>0.9dB (Subtest 2) | During T1:<br>Formula: SNR + TT<br><br>During T2:<br>Formula: SNR + TT<br><br>During T3:<br>Formula: SNR - TT<br><br>During T4:<br>Formula: SNR - TT<br><br>During T5:<br>Formula: SNR + TT |
| 7.3.3 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync | SNRs as specified | Same as 7.3.1 | Same as 7.3.1 |
| 7.3.4 E-UTRAN TDD Radio Link Monitoring Test for In-sync | SNRs as specified | Same as 7.3.2 | Same as 7.3.2 |
| 7.3.5 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync in DRX | SNRs as specified | 0.9dB<br>(Subtest 1)<br><br>0.6dB<br>(Subtest 2) | Same as 7.3.1 |
| 7.3.6 E-UTRAN FDD Radio Link Monitoring Test for In-sync in DRX | SNRs as specified | 0.6dB | Same as 7.3.2 |

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| 7.3.7 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync in DRX | SNRs as specified | 0.9dB<br>(Subtest 1)<br><br>0.6dB<br>(Subtest 2) | Same as 7.3.1 |
| 7.3.8 E-UTRAN TDD Radio Link Monitoring Test for In-sync in DRX | SNRs as specified | 0.6dB | Same as 7.3.2 |
| 7.3.9 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with Non MBSFN ABS (eICIC) | SNRs as specified | 0.9dB for cell 1 (TT1)<br>0.2dB for cell 2 (TT2) | During T1:<br>Cell 1 Formula: SNR + TT1<br>Cell 2 Formula: SNR - TT2<br><br>During T2:<br>Cell 1 Formula: SNR + TT1<br>Cell 2 Formula: SNR - TT2<br><br>During T3:<br>Cell 1 Formula: SNR - TT1<br>Cell 2 Formula: SNR + TT2<br><br>Formuale for Test Tolerance values:<br>TT1: $\hat{E}s_1/N_{oc}$ uncertainty<br>TT2: ($\hat{E}s_2/\hat{E}s_1$ uncertainty - $\hat{E}s_1/N_{oc}$ uncertainty) |
| 7.3.10 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with Non MBSFN ABS (eICIC) | Same as 7.3.9 | Same as 7.3.9 | Same as 7.3.9 |
| 7.3.11 E-UTRAN FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with Non MBSFN ABS (eICIC) | SNRs as specified | 0.9dB for cell 1 (TT1)<br>0.2dB for cell 2(TT2) | During T1:<br>Cell 1 Formula: SNR + TT1<br>Cell 2 Formula: SNR - TT2<br><br>During T2:<br>Cell 1 Formula: SNR + TT1<br>Cell 2 Formula: SNR - TT2<br><br>During T3:<br>Cell 1 Formula: SNR - TT1<br>Cell 2 Formula: SNR + TT2<br><br>During T4:<br>Cell 1 Formula: SNR - TT1<br>Cell 2 Formula: SNR + TT2<br><br>During T5:<br>Cell 1 Formula: SNR + TT1<br>Cell 2 Formula: SNR - TT2<br><br>Formuale for Test Tolerance values:<br>TT1: $\hat{E}s_1/N_{oc}$ uncertainty<br>TT2: ($\hat{E}s_2/\hat{E}s_1$ uncertainty - $\hat{E}s_1/N_{oc}$ uncertainty) |
| 7.3.12 E-UTRAN TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with Non MBSFN ABS (eICIC) | Same as 7.3.11 | Same as 7.3.11 | Same as 7.3.11 |

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| 7.3.13 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) | Same as 7.3.9 | Same as 7.3.9 | Same as 7.3.9 |
| 7.3.14 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) | Same as 7.3.9 | Same as 7.3.9 | Same as 7.3.9 |
| 7.3.15 E-UTRAN FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) | Same as 7.3.11 | Same as 7.3.11 | Same as 7.3.11 |
| 7.3.16 E-UTRAN TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) | Same as 7.3.11 | Same as 7.3.11 | Same as 7.3.11 |
| 7.3.17 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non MBSFN ABS (feICIC) | SNRs as specified | 0.9dB for cell 1 (TT1)<br>0.2dB for cell 2 (TT2)<br>0.2dB for cell 3 (TT3) | During T1:<br>Cell 1 Formula: SNR + TT1<br>Cell 2 Formula: SNR - TT2<br>Cell 3 Formula: SNR - TT3<br><br>During T2:<br>Cell 1 Formula: SNR + TT1<br>Cell 2 Formula: SNR - TT2<br>Cell 3 Formula: SNR - TT3<br><br>During T3:<br>Cell 1 Formula: SNR - TT1<br>Cell 2 Formula: SNR + TT2<br>Cell 3 Formula: SNR + TT3<br><br>Formuale for Test Tolerance values:<br>TT1: $\hat{E}s_1/N_{oc}$ uncertainty<br>TT2: ($\hat{E}s_2/\hat{E}s_1$ uncertainty - $\hat{E}s_1/N_{oc}$ uncertainty)<br>TT3: ($\hat{E}s_3/\hat{E}s_1$ uncertainty - $\hat{E}s_1/N_{oc}$ uncertainty) |
| 7.3.18 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non MBSFN ABS (feICIC) | Same as 7.3.17 | Same as 7.3.17 | Same as 7.3.17 |

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| 7.3.19 E-UTRAN FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non MBSFN ABS (feICIC) | SNRs as specified | 0.9dB for cell 1 (TT1)<br>0.2dB for cell 2(TT2)<br>0.2dB for cell 3 (TT3) | During T1:<br>Cell 1 Formula: SNR + TT1<br>Cell 2 Formula: SNR - TT2<br>Cell 3 Formula: SNR - TT3<br><br>During T2:<br>Cell 1 Formula: SNR + TT1<br>Cell 2 Formula: SNR - TT2<br>Cell 3 Formula: SNR - TT3<br><br>During T3:<br>Cell 1 Formula: SNR - TT1<br>Cell 2 Formula: SNR + TT2<br>Cell 3 Formula: SNR + TT3<br><br>During T4:<br>Cell 1 Formula: SNR - TT1<br>Cell 2 Formula: SNR + TT2<br>Cell 3 Formula: SNR + TT3<br><br>During T5:<br>Cell 1 Formula: SNR + TT1<br>Cell 2 Formula: SNR - TT2<br>Cell 3 Formula: SNR - TT3<br><br>Formuale for Test Tolerance values:<br>TT1: $\hat{E}s_1/N_{oc}$ uncertainty<br>TT2: ($\hat{E}s_2/\hat{E}s_1$ uncertainty - $\hat{E}s_1/N_{oc}$ uncertainty)<br>TT3: ($\hat{E}s_3/\hat{E}s_1$ uncertainty - $\hat{E}s_1/N_{oc}$ uncertainty) |
| 7.3.20 E-UTRAN TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non MBSFN ABS (feICIC) | Same as 7.3.19 | Same as 7.3.19 | Same as 7.3.19 |
| 7.3.21 E-UTRAN FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and MBSFN ABS (feICIC) | Same as 7.3.19 | Same as 7.3.19 | Same as 7.3.19 |
| 7.3.22 E-UTRAN TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and MBSFN ABS (feICIC) | Same as 7.3.19 | Same as 7.3.19 | Same as 7.3.19 |
| 7.3.23 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync for 5MHz Bandwidth | SNRs as specified | Same as 7.3.1 | Same as 7.3.1 |
| 7.3.24 E-UTRAN FDD Radio Link Monitoring Test for In-sync for 5MHz Bandwidth | SNRs as specified | 0.9dB | Same as 7.3.2 |

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| 7.3.25 E-UTRAN FDD Radio Link Monitoring Test for In-sync in DRX for 5MHz Bandwidth | SNRs as specified | 0.6dB | Same as 7.3.2 |
| 7.3.26 E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync for UE category 0 | SNRs as specified | 0.9dB | Same as 7.3.1 |
| 7.3.27 E-UTRAN FD-FDD Radio Link Monitoring Test for In-sync for UE category 0 | SNRs as specified | Same as 7.3.26 | Same as 7.3.2 |
| 7.3.28 E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category 0 | SNRs as specified | 0.6dB | Same as 7.3.1 |
| 7.3.29 E-UTRAN FD-FDD Radio Link Monitoring Test for In-sync in DRX for UE category 0 | SNRs as specified | Same as 7.3.28 | Same as 7.3.2 |
| 7.3.30 E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync for UE category 0 | SNRs as specified | Same as 7.3.26 | Same as 7.3.1 |
| 7.3.31 E-UTRAN HD-FDD Radio Link Monitoring Test for In-sync for UE category 0 | SNRs as specified | Same as 7.3.26 | Same as 7.3.2 |
| 7.3.32 E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category 0 | SNRs as specified | Same as 7.3.28 | Same as 7.3.1 |
| 7.3.33 E-UTRAN HD-FDD Radio Link Monitoring Test for In-sync in DRX for UE category 0 | SNRs as specified | Same as 7.3.28 | Same as 7.3.2 |
| 7.3.34 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync for UE category 0 | SNRs as specified | Same as 7.3.26 | Same as 7.3.1 |
| 7.3.35 E-UTRAN TDD Radio Link Monitoring Test for In-sync for UE category 0 | SNRs as specified | Same as 7.3.26 | Same as 7.3.2 |
| 7.3.36 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync in DRX for UE category 0 | SNRs as specified | Same as 7.3.28 | Same as 7.3.1 |
| 7.3.37 E-UTRAN TDD Radio Link Monitoring Test for In-sync in DRX for UE category 0 | SNRs as specified | Same as 7.3.28 | Same as 7.3.2 |
| 7.3.38 E-UTRAN FDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC | SNRs as specified | 0.9dB | Same as 7.3.1 |
| 7.3.39 E-UTRAN FDD-FDD DC Radio Link Monitoring Test for Out-of-sync in DRX in asynchronous DC | SNRs as specified | 0.9dB | Same as 7.3.1 |

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| 7.3.40 E-UTRAN TDD-TDD DC Radio Link Monitoring Test for Out-of-sync in DRX in synchronous DC | SNRs as specified | 0.9dB | Same as 7.3.1 |
| 7.3.41 E-UTRAN FDD-FDD Radio Link Monitoring Test for In-sync in DRX in synchronous dual connectivity | SNRs as specified | 0.6dB | Same as 7.3.2 |
| 7.3.42 E-UTRAN FDD-FDD DC Radio Link Monitoring Test for In-sync in DRX in asynchronous DC | SNRs as specified | 0.6dB | Same as 7.3.2 |
| 7.3.43 E-UTRAN TDD-TDD Radio Link Monitoring Test for In-sync in DRX in synchronous dual connectivity | SNRs as specified | 0.6dB | Same as 7.3.2 |
| 7.3.48 E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync for Cat-M1 UE in CEMode A | SNRs as specified | 0.9dB | Same as 7.3.1 |
| 7.3.49 E-UTRAN FD-FDD Radio Link Monitoring Test for In-Sync for Cat-M1 UE in CEMode A | SNRs as specified | 0.9dB | Same as 7.3.2 |
| 7.3.50: E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-Sync in DRX for UE Category M1 Configured in CEMode A | SNRs as specified | 0.6dB | Same as 7.3.1 |
| 7.3.51 E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE Category M1 Configured in CEMode A | SNRs as specified | 0.6dB | Same as 7.3.2 |
| 7.3.52 E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync for Cat-M1 UE in CEMode A | SNRs as specified | 0.9dB | Same as 7.3.1 |
| 7.3.53 E-UTRAN HD-FDD Radio Link Monitoring Test for In-Sync for Cat-M1 UE in CEMode A | SNRs as specified | 0.9dB | Same as 7.3.2 |
| 7.3.54: E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE Category M1 Configured in CEMode A | SNRs as specified | 0.6dB | Same as 7.3.1 |
| 7.3.55: E-UTRAN HD-FDD Radio Link Monitoring Test for In-sync in DRX for UE Category M1 configured in CEMode A | SNRs as specified | 0.6dB | Same as 7.3.2 |

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| 7.5.1 E-UTRAN FDD – UE ProSe Direct Discovery Transmission Timing Accuracy Test | Uplink timing: ±12$T_s$<br>$\hat{E}s$ / $N_{oc}$: +3.00dB | ±3$T_s$<br>+0.3dB | Uplink timing: ±15$T_s$<br>$\hat{E}s$ / $N_{oc}$: +3.30dB |
| 7.5.4 E-UTRAN FDD – UE ProSe Direct Communication Transmission Timing Accuracy Test | Uplink timing: ±12$T_s$<br>$\hat{E}s$ / $N_{oc}$: +3.00dB | ±3$T_s$<br>+0.3dB | Uplink timing: ±15$T_s$<br>$\hat{E}s$ / $N_{oc}$: +3.30dB |
| 8.1.1 E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +4.00dB<br>$\hat{E}s_2$ / $N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +4.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +4.00dB | During T1:<br>0dB<br>2.10dB<br>0dB<br><br>During T2:<br>0dB<br>2.10dB<br>2.10dB | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.10dB<br>$\hat{E}s_2$ / $N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.10dB<br>$\hat{E}s_2$ / $N_{oc}$: +6.10dB |
| 8.1.2 E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells | Same as 8.1.1 | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.3 E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX | Same as 8.1.1 | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.5 E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +8.00dB<br>$\hat{E}s_2$ / $N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +8.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +11.00dB<br><br>During T3:<br>$N_{oc}$: -98dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +8.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +11.00dB | During T1:<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>0dB<br>0dB<br>0dB<br><br>During T3:<br>0dB<br>0dB<br>0dB | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +8.00dB<br>$\hat{E}s_2$ / $N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +8.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +11.00dB<br><br>During T3:<br>$N_{oc}$: -98dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +8.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +11.00dB |
| 8.1.6 E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX | Same as 8.1.5 | Same as 8.1.5 | Same as 8.1.5 |
| 8.1.7 UE Measurement Procedures / E-UTRAN FDD-FDD Intra-frequency event-triggered reporting under time domain measurement resource restriction with non-MBSFN ABS (eICIC) | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +1.00dB<br>$\hat{E}s_2$ / $N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +1.00dB<br>$\hat{E}s_2$ / $N_{oc}$: -4.00dB | During T1:<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>0dB<br>0dB<br>0.8dB | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +1.00dB<br>$\hat{E}s_2$ / $N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +1.00dB<br>$\hat{E}s_2$ / $N_{oc}$: -3.20dB |

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| 8.1.8 E-UTRAN FDD-FDD Intra-Frequency Event-Triggered Reporting under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | During T1:<br>Noc: -98dBm/15kHz<br>Ês1 / Noc: +4.00dB<br>Ês2 / Noc: +2.00dB<br>Ês3 / Noc: -infinity<br><br>During T2:<br>Noc: -98dBm/15kHz<br>Ês1 / Noc: +4.00dB<br>Ês2 / Noc: +2.00dB<br>Ês3 / Noc: -4.00dB | During T1:<br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>0dB<br>0dB<br>0dB<br>0.8dB | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}_{s1}$ / $N_{oc}$: +4.00dB<br>$\hat{E}_{s2}$ / $N_{oc}$: +2.00dB<br>$\hat{E}_{s3}$ / $N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}_{s1}$ / $N_{oc}$: +4.00dB<br>$\hat{E}_{s1}$ / $N_{oc}$: +2.00dB<br>$\hat{E}_{s2}$ / $N_{oc}$: -3.20dB |
| 8.1.9 E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells for 5MHz Bandwidth | Same as 8.1.1 | Same as 8.1.1 | Same as 8.1.1 |
| 8.1.10 E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX for 5MHz Bandwidth | Same as 8.1.3 | Same as 8.1.3 | Same as 8.1.3 |
| 8.1.11 E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for UE category 0 | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}_{s1}$ / $N_{oc}$: +4.00dB<br>$\hat{E}_{s2}$ / $N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}_{s1}$ / $N_{oc}$: +4.00dB<br>$\hat{E}_{s2}$ / $N_{oc}$: +4.00dB | During T1:<br>0dB<br>2.10dB<br>0dB<br><br>During T2:<br>0dB<br>2.10dB<br>2.10dB | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}_{s1}$ / $N_{oc}$: +6.10dB<br>$\hat{E}_{s2}$ / $N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}_{s1}$ / $N_{oc}$: +6.10dB<br>$\hat{E}_{s2}$ / $N_{oc}$: +6.10dB |
| 8.1.12 E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for UE category 0 | Same as 8.1.11 | Same as 8.1.11 | Same as 8.1.11 |
| 8.1.13 E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX for UE category 0 | Same as 8.1.11 | Same as 8.1.11 | Same as 8.1.11 |
| 8.1.17 E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for UE category 0 | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}_{s1}$ / $N_{oc}$: +4.00dB<br>$\hat{E}_{s2}$ / $N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}_{s1}$ / $N_{oc}$: +4.00dB<br>$\hat{E}_{s2}$ / $N_{oc}$: +4.00dB | During T1:<br>0dB<br>2.60dB<br>0dB<br><br>During T2:<br>0dB<br>2.60dB<br>2.60dB | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}_{s1}$ / $N_{oc}$: +6.60dB<br>$\hat{E}_{s2}$ / $N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}_{s1}$ / $N_{oc}$: +6.60dB<br>$\hat{E}_{s2}$ / $N_{oc}$: +6.60dB |
| 8.1.18 E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX for UE category 0 | Same as 8.1.17 | Same as 8.1.17 | Same as 8.1.17 |

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| 8.1.19 E-UTRAN FD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps for UE category 0 | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +8.00dB<br>$\hat{E}s_2$ / $N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +8.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +11.00dB<br><br>During T3:<br>$N_{oc}$: -98dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +8.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +11.00dB | During T1:<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>0dB<br>0dB<br>0dB<br><br>During T3:<br>0dB<br>0dB<br>0dB | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +8.00dB<br>$\hat{E}s_2$ / $N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +8.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +11.00dB<br><br>During T3:<br>$N_{oc}$: -98dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +8.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +11.00dB |
| 8.1.20 E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0 | Same as 8.1.19 | Same as 8.1.19 | Same as 8.1.19 |
| 8.1.21 E-UTRAN HD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps for UE category 0 | Same as 8.1.19 | Same as 8.1.19 | Same as 8.1.19 |
| 8.1.22 E-UTRAN HD-FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0 | Same as 8.1.19 | Same as 8.1.19 | Same as 8.1.19 |
| 8.1.33 E-UTRAN HD-FDD Intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeB | TBD | TBD | TBD |
| 8.1.34 E-UTRAN HD-FDD Intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB | TBD | TBD | TBD |
| 8.1.35 E-UTRAN TDD Intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB | TBD | TBD | TBD |
| 8.2.1 E-UTRAN TDD-TDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +4.00dB<br>$\hat{E}s_2$ / $N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +4.00dB<br>$\hat{E}s_2$ / $N_{oc}$: +4.00dB | During T1:<br>0dB<br>2.10dB<br>0dB<br><br>During T2:<br>0dB<br>2.60dB<br>2.60dB | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.10dB<br>$\hat{E}s_2$ / $N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.60dB<br>$\hat{E}s_2$ / $N_{oc}$: +6.60dB |

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| 8.2.2 E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX | Same as 8.2.1 | Same as 8.2.1 | Same as 8.2.1 |
| 8.2.3 E-UTRAN TDD-TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +8.00dB<br>$\hat{E}s_2 / N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +8.00dB<br>$\hat{E}s_2 / N_{oc}$: +11.00dB<br><br>During T3:<br>$N_{oc}$: -98dBm /15kHz<br>$\hat{E}s_1 / N_{oc}$: +8.00dB<br>$\hat{E}s_2 / N_{oc}$: +11.00dB | During T1:<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>0dB<br>0dB<br>0dB<br><br>During T3:<br>0dB<br>0dB<br>0dB | During T1:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +8.00dB<br>$\hat{E}s_2 / N_{oc}$: -infinity<br><br>During T2:<br>$N_{oc}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +8.00dB<br>$\hat{E}s_2 / N_{oc}$: +11.00dB<br><br>During T3:<br>$N_{oc}$: -98dBm /15kHz<br>$\hat{E}s_1 / N_{oc}$: +8.00dB<br>$\hat{E}s_2 / N_{oc}$: +11.00dB |
| 8.2.4 E-UTRAN TDD-TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX | Same as 8.2.3 | Same as 8.2.3 | Same as 8.2.3 |
| 8.2.5 UE Measurement Procedures / E-UTRAN TDD-TDD Intra-frequency event-triggered reporting under time domain measurement resource restriction with non-MBSFN ABS (eICIC) | Same as 8.1.7 | Same as 8.1.7 | Same as 8.1.7 |
| 8.2.6 E-UTRAN TDD-TDD Intra-Frequency Event-Triggered Reporting under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | Same as 8.1.8 | Same as 8.1.8 | Same as 8.1.8 |
| 8.2.7 E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps for UE category 0 | Same as 8.1.19 | Same as 8.1.19 | Same as 8.1.19 |
| 8.2.8 E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0 | Same as 8.1.19 | Same as 8.1.19 | Same as 8.1.19 |

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| 8.3.1 E-UTRAN FDD-FDD Inter-frequency event triggered reporting under fading propagation conditions in asynchronous cells | During T1:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: -infinity<br><br>During T2:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +7.00dB | During T1:<br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>0dB<br>0dB<br>0dB<br>0.7dB | During T1:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: -infinity<br><br>During T2:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +7.70dB |
| 8.3.2 E-UTRAN FDD-FDD Inter-frequency event triggered reporting when DRX is used under fading propagation conditions in asynchronous cells | Same as 8.3.1 | Same as 8.3.1 | Same as 8.3.1 |
| 8.3.3 E-UTRAN FDD-FDD Inter frequency event triggered reporting under AWGN propagation conditions in asynchronous cells with DRX when L3 filtering is used | During T1:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +4.00dB<br><br>During T2:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +24.00dB | During T1:<br>+2.5dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>+2.5dB<br>-4.0dB<br>0dB<br>0dB | During T1:<br>$N_{oc1}$: -95.50dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +4.00dB<br><br>During T2:<br>$N_{oc1}$: -95.50dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +0.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +24.00dB |
| 8.3.4 E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | During T1:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: -infinity<br><br>During T2:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +7.00dB<br><br>During T3:<br>$N_{oc1}$: -98dBm /15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +7.00dB | During T1:<br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T3:<br>0dB<br>0dB<br>0dB<br>0dB | During T1:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: -infinity<br><br>During T2:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +7.00dB<br><br>During T3:<br>$N_{oc1}$: -98dBm /15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +7.00dB |
| 8.3.5 E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX | Same as 8.3.4 | Same as 8.3.4 | Same as 8.3.4 |
| 8.3.6 E-UTRAN FDD-FDD Inter-frequency event triggered reporting without measurement gaps under AWGN propagation conditions in asynchronous cells | During T1:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: -infinity<br><br>During T2:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +7.00dB | During T1:<br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>0dB<br>0dB<br>0dB<br>0dB | During T1:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: -infinity<br><br>During T2:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +7.00dB |

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| 8.4.1 E-UTRAN TDD-TDD Inter-frequency event triggered reporting under fading propagation conditions in synchronous cells | During T1:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: -infinity<br><br>During T2:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +7.00dB | During T1:<br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>0dB<br>0dB<br>0dB<br>0.7dB | During T1:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: -infinity<br><br>During T2:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +7.70dB |
| 8.4.2 E-UTRAN TDD-TDD Inter-frequency event triggered reporting when DRX is used under fading propagation conditions in synchronous cells | Same as 8.4.1 | Same as 8.4.1 | Same as 8.4.1 |
| 8.4.3 E-UTRAN TDD-TDD inter-frequency event triggered reporting under AWGN propagation conditions in synchronous cells with DRX when L3 filtering is used | Same as 8.3.3 | Same as 8.3.3 | Same as 8.3.3 |
| 8.4.4 E-UTRAN TDD-TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | During T1:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: -infinity<br><br>During T2:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +7.00dB<br><br>During T3:<br>$N_{oc1}$: -98dBm /15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +7.00dB | During T1:<br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T3:<br>0dB<br>0dB<br>0dB<br>0dB | During T1:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: -infinity<br><br>During T2:<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +7.00dB<br><br>During T3:<br>$N_{oc1}$: -98dBm /15kHz<br>$\hat{E}s_1 / N_{oc1}$: +4.00dB<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +7.00dB |
| 8.4.5 E-UTRAN TDD-TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX | Same as 8.4.4 | Same as 8.4.4 | Same as 8.4.4 |
| 8.5.1 E-UTRAN FDD – UTRAN FDD event triggered reporting under fading propagation conditions | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -1.8 dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br><br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -1.8dB |

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| 8.5.2 E-UTRAN FDD – UTRAN FDD SON ANR cell search reporting under AWGN propagation conditions | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -3.35 dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br><br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>0.4dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -2.95dB |
| 8.5.3 E-UTRAN FDD – UTRAN FDD event triggered reporting when DRX is used under fading propagation conditions | Same as 8.5.1 | Same as 8.5.1 | Same as 8.5.1 |
| 8.5.4 E-UTRAN FDD – UTRAN FDD enhanced cell identification under AWGN propagation conditions | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: 0.02 dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br><br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: 0.02dB |
| 8.5.6 E-UTRAN FDD - UTRAN FDD event triggered reporting without measurement gaps under AWGN propagation conditions | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -1.8 dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br><br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -1.8dB |
| 8.5.7 E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions for 5MHz Bandwidth | Same as 8.5.1 | Same as 8.5.1 | Same as 8.5.1 |
| 8.6.1 E-UTRAN TDD - UTRAN FDD event triggered reporting under fading propagation conditions | Same as 8.5.1 | Same as 8.5.1 | Same as 8.5.1 |

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| 8.7.1 E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +9dB<br>UTRA Cell 2<br>$I_{oc}$: -80dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: -inf<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +9dB<br>UTRA Cell 2<br>$I_{oc}$: -80dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: +5dB<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +9dB<br>UTRA Cell 2<br>$I_{oc}$: -80dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: -inf<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +9dB<br>UTRA Cell 2<br>$I_{oc}$: -80dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: +5dB<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB |
| 8.7.2 E-UTRAN TDD - UTRAN TDD cell search when DRX is used under fading propagation conditions | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>UTRA Cell 2<br>$I_{oc}$: -80dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: -inf<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>UTRA Cell 2<br>$I_{oc}$: -80dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: +9dB<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB | During T1:<br><br>0dB<br>0dB<br><br>-0.40dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>-0.40dB<br>0dB<br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>UTRA Cell 2<br>$I_{oc}$: -80.40dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: -inf<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>UTRA Cell 2<br>$I_{oc}$: -80.40dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: +9dB<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB |
| 8.7.3 E-UTRAN TDD - UTRAN TDD SON ANR cell search reporting under AWGN propagation conditions | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>UTRA Cell 2<br>$I_{oc}$: -75dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: -inf<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>UTRA Cell 2<br>$I_{oc}$: -75dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: +5dB<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>UTRA Cell 2<br>$I_{oc}$: -75dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: -inf<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>UTRA Cell 2<br>$I_{oc}$: -75dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: +5dB<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB |

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| 8.7.4 E-UTRAN TDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>UTRA Cell 2<br>Ioc: -80dBm/1.28MHz<br>Ior / Ioc: - infinity<br>PCCPCH_Ec/Ior: -4.77dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>UTRA Cell 2<br>Ioc: -80dBm/1.28MHz<br>Ior / Ioc: +8dB<br>PCCPCH_Ec/Ior: -4.77dB<br>DwPCH_Ec/Ior: 0dB | During T1:<br><br>0dB<br>0dB<br><br>+0.6dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>+0.6dB<br>0dB<br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>UTRA Cell 2<br>Ioc: -79.4dBm/1.28MHz<br>Ior / Ioc: - infinity<br>PCCPCH_Ec/Ior: -4.77dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>UTRA Cell 2<br>Ioc: -79.4dBm/1.28MHz<br>Ior / Ioc: +8dB<br>PCCPCH_Ec/Ior: -4.77dB<br>DwPCH_Ec/Ior: 0dB |
| 8.8.1 E-UTRAN FDD - GSM event triggered reporting in AWGN | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>GSM Cell 2<br>Signal level: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>GSM Cell 2<br>Signal level: -75 dBm | During T1:<br><br>0dB<br>0dB<br><br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>GSM Cell 2<br>Signal level: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4dB<br>GSM Cell 2<br>Signal level: -75 dBm |
| 8.8.2 E-UTRAN FDD- GSM event triggered reporting when DRX is used in AWGN | During T1:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4.00dB<br>GSM Cell 2<br>Signal level: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4.00dB<br>GSM Cell 2<br>Signal level: -75dBm | During T1:<br><br>0dB<br>0dB<br><br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB | During T1:<br>E-UTRAN Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4.00dB<br>GSM Cell 2<br>Signal level: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4.00dB<br>GSM Cell 2<br>Signal level: -75dBm |
| 8.9.1 E-UTRAN FDD - UTRAN TDD event triggered reporting under fading propagation conditions | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Iot: 4dB<br>UTRA Cell 2<br>Ioc: -70dBm/1.28MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Iot: 4dB<br>UTRA Cell 2<br>Ioc: -70dBm/1.28MHz<br>Ior / Ioc: 9 dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>0 dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Iot: 4dB<br>UTRA Cell 2<br>Ioc: -70dBm/1.28MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Iot: 4dB<br>UTRA Cell 2<br>Ioc: -70dBm/1.28MHz<br>Ior / Ioc: 9 dB |
| 8.9.2 E-UTRAN FDD - UTRAN TDD enhanced cell identification under AWGN propagation conditions | Same as 8.7.4 | Same as 8.7.4 | Same as 8.7.4 |
| 8.10.1 E-UTRAN TDD - GSM event triggered reporting in AWGN | Same as 8.8.1 | Same as 8.8.1 | Same as 8.8.1 |

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| 8.10.2 E-UTRAN TDD-GSM event triggered reporting when DRX is used in AWGN | Same as 8.8.2 | Same as 8.8.2 | Same as 8.8.2 |
| 8.11.1 Multiple E-UTRAN FDD-FDD Inter-frequency event triggered reporting under fading propagation conditions | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 0dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: -infinity<br>E-UTRA Cell 3<br>Noc: -98dBm/15kHz<br>Ês / Noc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 0dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: 3dB<br>E-UTRA Cell 3<br>Noc: -98dBm/15kHz<br>Ês / Noc: 3dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>+0.6dB<br><br>0dB<br>+0.6dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 0dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: -infinity<br>E-UTRA Cell 3<br>Noc: -98dBm/15kHz<br>Ês / Noc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 0dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: 3.6dB<br>E-UTRA Cell 3<br>Noc: -98dBm/15kHz<br>Ês / Noc: 3.6dB |
| 8.11.2 E-UTRAN TDD - E-UTRAN TDD and E-UTRAN TDD Inter-frequency event triggered reporting under fading propagation conditions | Same as 8.11.1 | Same as 8.11.1 | Same as 8.11.1 |
| 8.11.3 E-UTRAN FDD-FDD Inter-frequency and UTRAN FDD event triggered reporting under fading propagation conditions | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4.00dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: -infinity<br>UTRA Cell 3<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4.00dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: +7.00dB<br>UTRA Cell 3<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -1.8 dB | During T1:<br><br>0dB<br>-1.05dB<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>During T2:<br><br>0dB<br>-1.05dB<br><br>0dB<br>0dB<br><br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +2.95dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: -infinity<br>UTRA Cell 3<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +2.95dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: +7.00dB<br>UTRA Cell 3<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -1.8 dB |

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| 8.11.4 InterRAT E-UTRA TDD to E-UTRA TDD and UTRA TDD cell search | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4.00dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: -infinity<br>UTRA Cell 3<br>$I_{oc}$: -80dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: -infinity<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4.00dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: +7.00dB<br>UTRA Cell 3<br>$I_{oc}$: -80dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: 9.00dB<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB | During T1:<br><br>0dB<br>-1.05dB<br><br>0dB<br>0dB<br><br>-0.4dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br><br>0dB<br>-1.05dB<br><br>0dB<br>0dB<br><br>-0.4dB<br>0dB<br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +2.95dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: -infinity<br>UTRA Cell 3<br>$I_{oc}$: -80.4dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: -infinity<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +2.95dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: +7.00dB<br>UTRA Cell 3<br>$I_{oc}$: -80.4dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: 9.00dB<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB |
| 8.11.5 Combined E-UTRAN FDD - E-UTRA FDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4.00dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: -infinity<br>GSM Cell 3<br>Signal level: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4.00dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: +7.00dB<br>GSM Cell 3<br>Signal level: -75dBm | During T1:<br><br>0dB<br>-0.2dB<br><br>0dB<br>0dB<br><br>0dB<br><br>During T2:<br><br>0dB<br>-0.2dB<br><br>0dB<br>+0.5dB<br><br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +3.80dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: -infinity<br>GSM Cell 3<br>Signal level: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +3.80dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: +7.50dB<br>GSM Cell 3<br>Signal level: -75dBm |
| 8.11.6 Combined E-UTRAN TDD - E-UTRA TDD and GSM cell search. E-UTRA cells in fading; GSM cell in static propagation conditions | Same as 8.11.5 | Same as 8.11.5 | Same as 8.11.5 |
| 8.14.1 E-UTRAN TDD-FDD Inter-frequency event triggered reporting under fading propagation conditions in asynchronous cells | Same as 8.3.1 | Same as 8.3.1 | Same as 8.3.1 |
| 8.14.2 E-UTRAN TDD-FDD Inter-frequency event triggered reporting when DRX is used under fading propagation conditions in asynchronous cells | Same as 8.3.1 | Same as 8.3.1 | Same as 8.3.1 |
| 8.14.3 E-UTRAN TDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | Same as 8.4.4 | Same as 8.4.4 | Same as 8.4.4 |

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| 8.15.1 E-UTRAN FDD-TDD Inter-frequency event triggered reporting under fading propagation conditions in asynchronous cells | Same as 8.4.1 | Same as 8.4.1 | Same as 8.4.1 |
| 8.15.2 E-UTRAN FDD-TDD Inter-frequency event triggered reporting when DRX is used under fading propagation conditions in asynchronous cells | Same as 8.4.1 | Same as 8.4.1 | Same as 8.4.1 |
| 8.15.3 E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | Same as 8.4.4 | Same as 8.4.4 | Same as 8.4.4 |
| 8.16.1 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: 19dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: 19dB<br>E-UTRA Cell 3<br>$\hat{E}s_3 / N_{oc2}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: 19dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: 19dB<br>E-UTRA Cell 3<br>$\hat{E}s_3 / N_{oc2}$: 19dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: -3dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: -3dB<br>E-UTRA Cell 3<br>$\hat{E}s_3 / N_{oc2}$: -3dB | During T1:<br><br>0dB<br>+0.2dB<br><br>0dB<br>+0.2dB<br><br>0dB<br><br>During T2:<br><br>0dB<br>+0.2dB<br><br>0dB<br>+0.2dB<br><br>0dB<br><br>During T3:<br><br>0dB<br>0dB<br><br>0dB<br>-0.2dB<br><br>0dB | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: 19.2dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: 19.2dB<br>E-UTRA Cell 3<br>$\hat{E}s_3 / N_{oc2}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: 19.2dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: 19.2dB<br>E-UTRA Cell 3<br>$\hat{E}s_3 / N_{oc2}$: 19.0dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: -3.0dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: -3.2dB<br>E-UTRA Cell 3<br>$\hat{E}s_3 / N_{oc2}$: -3.0dB |
| 8.16.2 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX | Same as 8.16.1 | Same as 8.16.1 | Same as 8.16.1 |

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| 8.16.3 E-UTRAN FDD-FDD Event triggered reporting on deactivated SCell with PCell interruption in non-DRX | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: 16dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -98dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -98dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: 16dB |
| 8.16.4 E-UTRAN TDD-TDD Event triggered reporting on deactivated SCell with PCell interruption in non-DRX | Same as 8.16.3 | Same as 8.16.3 | Same as 8.16.3 |
| 8.16.5 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX for 20 MHz bandwidth | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 19dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 19dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 19dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 19dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: 19dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -3dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -3dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -3dB | During T1:<br><br>0dB<br>+0.2dB<br><br>-0.1dB<br>+0.3dB<br><br>0dB<br><br>During T2:<br><br>0dB<br>+0.2dB<br><br>-0.1dB<br>+0.3dB<br><br>0dB<br><br>During T3:<br><br>0dB<br>0dB<br><br>-0.1dB<br>-0.1dB<br><br>0dB | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 19.2dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104.1dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 19.3dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 19.2dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104.1dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 19.3dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: 19.0dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -3.0dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104.1dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -3.1dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -3.0dB |
| 8.16.6 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 20 MHz bandwidth | Same as 8.16.5 | Same as 8.16.5 | Same as 8.16.5 |

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| 8.16.7 E-UTRA FDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 20 MHz bandwidth | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: 16dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: 16dB |
| 8.16.8 E-UTRAN TDD Event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 20 MHz bandwidth | Same as 8.16.7 | Same as 8.16.7 | Same as 8.16.7 |
| 8.16.9 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX for 10MHz+5MHz | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 19dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 19dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 19dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 19dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: 19dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -3dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -3dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -3dB | During T1:<br><br>0dB<br>+0.2dB<br><br>0dB<br>+0.2dB<br><br>0dB<br><br>During T2:<br><br>0dB<br>+0.2dB<br><br>0dB<br>+0.2dB<br><br>0dB<br><br>During T3:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 19.2dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 19.2dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 19.2dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 19.2dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: 19.0dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -3.0dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -3.0dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -3.0dB |
| 8.16.10 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 10MHz+5MHz | Same as 8.16.9 | Same as 8.16.9 | Same as 8.16.9 |
| 8.16.11 E-UTRAN FDD event triggered reporting on deactivating SCell with PCell interruption in non-DRX for 10MHz+5MHz | Same as 8.16.3 | Same as 8.16.3 | Same as 8.16.3 |

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| 8.16.12 E-UTRAN FDD event triggered reporting on deactivating SCell with PCell interruption in non-DRX for 10MHz+5MHz | Same as 8.16.3 | Same as 8.16.3 | Same as 8.16.3 |
| 8.16.13 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX for 5MHz+5MHz | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 19dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 19dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 19dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 19dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: 19dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -3dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -3dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -3dB | During T1:<br><br>0dB<br>+0.2dB<br><br>0dB<br>+0.2dB<br><br>0dB<br><br>During T2:<br><br>0dB<br>+0.2dB<br><br>0dB<br>+0.2dB<br><br>0dB<br><br>During T3:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 19.2dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 19.2dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 19.2dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 19.2dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: 19.0dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -3.0dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -3.0dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -3.0dB |
| 8.16.14 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 5MHz+5MHz | Same as 8.16.13 | Same as 8.16.13 | Same as 8.16.13 |
| 8.16.15 E-UTRAN FDD event triggered reporting on deactivating SCell with PCell interruption in non-DRX for 5MHz+5MHz | Same as 8.16.3 | Same as 8.16.3 | Same as 8.16.3 |
| 8.16.16 E-UTRAN FDD event triggered reporting on deactivating SCell with PCell interruption in non-DRX for 5MHz+5MHz | Same as 8.16.3 | Same as 8.16.3 | Same as 8.16.3 |
| 8.16.17 E-UTRAN FDD activation and deactivation of known SCell in non-DRX | Same as 8.16.37 | Same as 8.16.37 | Same as 8.16.37 |
| 8.16.17A E-UTRAN FDD activation and deactivation of known SCell in non-DRX for 20MHz | Same as 8.16.17 | Same as 8.16.17 | Same as 8.16.17 |
| 8.16.18 E-UTRAN TDD activation and deactivation of known SCell in non-DRX | Same as 8.16.17 | Same as 8.16.17 | Same as 8.16.17 |

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| 8.16.18A E-UTRAN TDD activation and deactivation of known SCell in non-DRX for 20MHz | Same as 8.16.17 | Same as 8.16.17 | Same as 8.16.17 |
| 8.16.21 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 20MHz+10MHz bandwidth | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 19dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 19dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 19dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 19dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: 19dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -3dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -3dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -3dB | During T1:<br><br>0dB<br>+0.2dB<br><br>0dB<br>+0.2dB<br><br>0dB<br><br>During T2:<br><br>0dB<br>+0.2dB<br><br>0dB<br>+0.2dB<br><br>0dB<br><br>During T3:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 19.2dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 19.2dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 19.2dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 19.2dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: 19.0dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -3.0dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -3.0dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -3.0dB |
| 8.16.22 E-UTRAN TDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 20MHz+10MHz bandwidth | Same as 8.16.3 | Same as 8.16.3 | Same as 8.16.3 |

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| 8.16.23 E-UTRAN TDD-FDD CA Event Triggered Reporting Under Deactivated SCell in Non-DRX with PCell in FDD | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 17dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 17dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 17dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 17dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: 17dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -3dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -3dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -3dB | During T1:<br><br>0dB<br>+0.2dB<br><br>-0.5dB<br>+0.7dB<br><br>0dB<br><br>During T2:<br><br>0dB<br>+0.2dB<br><br>-0.5dB<br>+0.7dB<br><br>0dB<br><br>During T3:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 17.2dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104.5dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 17.7dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 17.2dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104.5dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 17.7dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: 17.0dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -3.0dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -3.0dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -3.0dB |
| 8.16.24 E-UTRAN TDD-FDD CA Event Triggered Reporting Under Deactivated SCell in Non-DRX with PCell in TDD | Same as 8.16.23 | Same as 8.16.23 | Same as 8.16.23 |
| 8.16.25 E-UTRAN TDD-FDD CA Event triggered reporting on deactivated SCell with PCell interruption in non-DRX with PCell in FDD | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: 16dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16dB<br>E-UTRA Cell 3<br>$\hat{E}s_3$ / $N_{oc2}$: 16dB |
| 8.16.26 E-UTRAN TDD-FDD CA Event triggered reporting on deactivated SCell with PCell interruption in non-DRX with PCell in TDD | Same as 8.16.25 | Same as 8.16.25 | Same as 8.16.25 |

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| 8.16.27 E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting under Deactivated SCells in Non-DRX with PCell in FDD | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 17dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -infinity<br>E-UTRA Cell 3<br>$N_{oc3}$: -104dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$: 17dB<br>E-UTRA Cell 4<br>$\hat{E}s_4$ / $N_{oc3}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 17dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 17dB<br>E-UTRA Cell 3<br>$N_{oc3}$: -104dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$: 17dB<br>E-UTRA Cell 4<br>$\hat{E}s_4$ / $N_{oc3}$: 17dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -3dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -3dB<br>E-UTRA Cell 3<br>$N_{oc3}$: -104dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$: -3dB<br>E-UTRA Cell 4<br>$\hat{E}s_4$ / $N_{oc3}$: -3dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>-0.5dB<br>+0.7dB<br><br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>+0.2dB<br><br>-0.5dB<br>+0.7dB<br><br>0dB<br><br>During T3:<br><br>0dB<br>0dB<br><br>0dB<br>-0.2dB<br><br>-0.5dB<br>-0.7dB<br><br>-0.7dB | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 17dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -infinity<br>E-UTRA Cell 3<br>$N_{oc3}$: -104.5dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$: 17.7dB<br>E-UTRA Cell 4<br>$\hat{E}s_4$ / $N_{oc3}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 17dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 17.2dB<br>E-UTRA Cell 3<br>$N_{oc3}$: -104.5dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$: 17.7dB<br>E-UTRA Cell 4<br>$\hat{E}s_4$ / $N_{oc3}$: 17dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -3dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -3.2dB<br>E-UTRA Cell 3<br>$N_{oc3}$: -104.5dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$: -3.7dB<br>E-UTRA Cell 4<br>$\hat{E}s_4$ / $N_{oc3}$: -3.7dB |
| 8.16.28 E-UTRAN TDD-FDD 3DL CA Event Triggered Reporting under Deactivated SCells in Non-DRX with PCell in TDD | Same as 8.16.27 | Same as 8.16.27 | Same as 8.16.27 |
| 8.16.29 3DL FDD CA Event Triggered Reporting under Deactivated SCells in Non-DRX | Same as 8.16.27 | Same as 8.16.27 | Same as 8.16.27 |
| 8.16.30 3DL TDD CA Event Triggered Reporting under Deactivated SCells in Non-DRX | Same as 8.16.27 | Same as 8.16.27 | Same as 8.16.27 |

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| 8.16.31 E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in FDD | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16dB<br>E-UTRA Cell 3<br>$N_{oc3}$: -101dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$: 16dB<br>E-UTRA Cell 4<br>$\hat{E}s_4$ / $N_{oc3}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16dB<br>E-UTRA Cell 3<br>$N_{oc3}$: -101dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$: 16dB<br>E-UTRA Cell 4<br>$\hat{E}s_4$ / $N_{oc3}$: 16dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16dB<br>E-UTRA Cell 3<br>$N_{oc3}$: -101dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$:16dB<br>E-UTRA Cell 4<br>$\hat{E}s_4$ / $N_{oc3}$: -infinity<br><br>During T4:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16dB<br>E-UTRA Cell 3<br>$N_{oc3}$: -101dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$: 16dB<br>E-UTRA Cell 4<br>$\hat{E}s_4$ / $N_{oc3}$: 16dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br><br>During T3:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br><br><br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16<br>E-UTRA Cell 3<br>$N_{oc3}$: -101dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$: 16dB<br>E-UTRA Cell 4<br>$\hat{E}s_4$ / $N_{oc3}$: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16dB<br>E-UTRA Cell 3<br>$N_{oc3}$: -101dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$: 16dB<br>E-UTRA Cell 4<br>$\hat{E}s_4$ / $N_{oc3}$: 16dB<br><br>During T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16dB<br>E-UTRA Cell 3<br>$N_{oc3}$: -101dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$: 16dB<br>E-UTRA Cell 4<br>$\hat{E}s_4$ / $N_{oc3}$: -infinity<br><br><br>E-UTRA Cell 1<br>$N_{oc1}$: -101dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 16dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -101dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 16dB<br>E-UTRA Cell 3<br>$N_{oc3}$: -101dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$: 16dB<br>E-UTRA Cell 4<br>$\hat{E}s_4$ / $N_{oc3}$: 16dB |
| 8.16.32 E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in TDD | Same as 8.16.31 | Same as 8.16.31 | Same as 8.16.31 |
| 8.16.33 E-UTRAN FDD 3DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX | Same as 8.16.31 | Same as 8.16.31 | Same as 8.16.31 |

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| 8.16.34 E-UTRAN TDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX | Same as 8.16.31 | Same as 8.16.31 | Same as 8.16.31 |
| 8.16.35 3 DL PCell in FDD CA Activation and Deactivation of Known SCell in Non-DRX | During T1, T2, T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 17dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 17dB<br>E-UTRA Cell 3<br>$N_{oc3}$: -104dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$: 17dB | During T1,T2,T3:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br>0dB | During T1, T2, T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 17dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 17dB<br>E-UTRA Cell 3<br>$N_{oc3}$: -104dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$: 17dB |
| 8.16.36 3 DL PCell in TDD CA Activation and Deactivation of Known SCell in Non-DRX | Same as 8.16.35 | Same as 8.16.35 | Same as 8.16.35 |
| 8.16.37 3DL FDD CA Activation and Deactivation of known SCell in non-DRX | Same as 8.16.35 | Same as 8.16.35 | Same as 8.16.35 |
| 8.16.38 3DL TDD CA Activation and Deactivation of known SCell in non-DRX | Same as 8.16.35 | Same as 8.16.35 | Same as 8.16.35 |
| 8.16.39 E-UTRA TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in FDD | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 17dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -infinity<br>E-UTRA Cell 3<br>$N_{oc3}$: -104dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$: 17dB<br><br>During T2, T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 17dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 17dB<br>E-UTRA Cell 3<br>$N_{oc3}$: -104dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$: 17dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>During T2, T3:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 17dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -infinity<br>E-UTRA Cell 3<br>$N_{oc3}$: -104dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$: 17dB<br><br>During T2, T3:<br>E-UTRA Cell 1<br>$N_{oc1}$: -104dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: 17dB<br>E-UTRA Cell 2<br>$N_{oc2}$: -104dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: 17dB<br>E-UTRA Cell 3<br>$N_{oc3}$: -104dBm/15kHz<br>$\hat{E}s_3$ / $N_{oc3}$: 17dB |
| 8.16.40 E-UTRA TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in TDD | Same as 8.16.39 | Same as 8.16.39 | Same as 8.16.39 |
| 8.16.41 3 DL FDD CA activation and deactivation of unknown SCell in non-DRX | Same as 8.16.39 | Same as 8.16.39 | Same as 8.16.39 |
| 8.16.42 3 DL TDD CA activation and deactivation of unknown SCell in non-DRX | Same as 8.16.39 | Same as 8.16.39 | Same as 8.16.39 |

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| 8.20.1 UE Measurement Procedures / RAT Measurements in CA mode / E-UTRAN FDD-FDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: -infinity<br>E-UTRA Cell 3<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: 7dB<br>E-UTRA Cell 3<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>0.7dB<br><br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: -infinity<br>E-UTRA Cell 3<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: 7.7dB<br>E-UTRA Cell 3<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB |
| 8.20.2 UE Measurement Procedures / RAT Measurements in CA mode / E-UTRAN TDD-TDD inter frequency event triggered reporting under fading propagation conditions in asynchronous cells | Same as 8.20.1 | Same as 8.20.1 | Same as 8.20.1 |
| 8.20.2A UE Measurement Procedures / RAT Measurements in CA mode / E-UTRAN TDD-TDD Inter-frequency event triggered reporting under fading propagation conditions in synchronous cells for 20 MHz +20 MHz bandwidth | Same as 8.20.1 | Same as 8.20.1 | Same as 8.20.1 |
| 8.20.2B UE Measurement Procedures / RAT Measurements in CA mode / E-UTRAN TDD-TDD Inter-frequency event triggered reporting under fading propagation conditions in synchronous cells for 20 MHz +10 MHz bandwidth | Same as 8.20.1 | Same as 8.20.1 | Same as 8.20.1 |

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| 8.20.3 E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4.00dB<br>E-UTRA Cell 3<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4.00dB<br>E-UTRA Cell 3<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -1.8 dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4.00dB<br>E-UTRA Cell 3<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4.00dB<br>E-UTRA Cell 3<br>Noc: -98dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -70dBm/3.84MHz<br>Ior / Ioc: -1.8 dB |
| 8.20.4 E-UTRAN TDD to UTRAN TDD cell search under fading propagation conditions | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +9.00dB<br>E-UTRA Cell 3<br>Noc: -98dBm/15kHz<br>Ês / Noc: +9.00dB<br>UTRA Cell 2<br>$I_{oc}$: -80dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: -infinity<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +9.00dB<br>E-UTRA Cell 3<br>Noc: -98dBm/15kHz<br>Ês / Noc: +9.00dB<br>UTRA Cell 2<br>$I_{oc}$: -80dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: 5.00dB<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +9.00dB<br>E-UTRA Cell 3<br>Noc: -98dBm/15kHz<br>Ês / Noc: +9.00dB<br>UTRA Cell 2<br>$I_{oc}$: -80dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: -infinity<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: +9.00dB<br>E-UTRA Cell 3<br>Noc: -98dBm/15kHz<br>Ês / Noc: +9.00dB<br>UTRA Cell 2<br>$I_{oc}$: -80dBm/1.28MHz<br>$I_{or}$ / $I_{oc}$: 5.00dB<br>PCCPCH_$E_c$/$I_{or}$: -3dB<br>DwPCH_Ec/Ior: 0dB |
| 8.20.4A E-UTRAN TDD to UTRAN TDD cell search under fading propagation conditions for 20 MHz + 20 MHz bandwidth | Same as 8.20.4 | Same as 8.20.4 | Same as 8.20.4 |
| 8.20.4B E-UTRAN TDD to UTRAN TDD cell search under fading propagation conditions for 20 MHz + 10 MHz bandwidth | Same as 8.20.4 | Same as 8.20.4 | Same as 8.20.4 |
| 8.22.1 E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal | Same as 8.1.1 | Same as 8.1.1 | Same as 8.1.1 |

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| 8.22.2 E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal | Same as 8.1.1 | Same as 8.1.1 | Same as 8.1.1 |
| 8.22.3 E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal | During T1:<br>Noc1: -98dBm/15kHz<br>Ês1 / Noc1: +4.00dB<br>Noc2: -98dBm/15kHz<br>Ês2 / Noc2: -infinity<br><br>During T2:<br>Noc1: -98dBm/15kHz<br>Ês1 / Noc1: +4.00dB<br>Noc2: -98dBm/15kHz<br>Ês2 / Noc2: +7.00dB | During T1:<br>0dB<br>0dB<br>0dB<br>0dB<br><br>During T2:<br>0dB<br>0dB<br>0dB<br>0dB | During T1:<br>Noc1: -98dBm/15kHz<br>Ês1 / Noc1: +4.00dB<br>Noc2: -98dBm/15kHz<br>Ês2 / Noc2: -infinity<br><br>During T2:<br>Noc1: -98dBm/15kHz<br>Ês1 / Noc1: +4.00dB<br>Noc2: -98dBm/15kHz<br>Ês2 / Noc2: +7.00dB |
| 8.22.4 E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal | Same as 8.22.3 | Same as 8.22.3 | Same as 8.22.3 |
| 8.22.9 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX based on CRS based discovery signal | Same as 8.16.1 | Same as 8.16.1 | Same as 8.16.1 |
| 8.22.10 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX based on CRS based discovery signal | Same as 8.16.1 | Same as 8.16.1 | Same as 8.16.1 |
| 8.23.4 E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in synchronous DC | During T1:<br>E-UTRA Cell 1<br>Noc: -101dBm/15kHz<br>Ês / Noc: 4dB<br>E-UTRA Cell 2<br>Noc: -101dBm/15kHz<br>Ês / Noc: 4dB<br>E-UTRA Cell 3<br>Noc: -101dBm/15kHz<br>Ês / Noc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -101dBm/15kHz<br>Ês / Noc: 4dB<br>E-UTRA Cell 2<br>Noc: -101dBm/15kHz<br>Ês / Noc: 4dB<br>E-UTRA Cell 3<br>Noc: -101dBm/15kHz<br>Ês / Noc: 7dB | During T1:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>During T2:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br><br>0dB<br>0dB | During T1:<br>E-UTRA Cell 1<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>E-UTRA Cell 2<br>Noc: -98dBm/15kHz<br>Ês / Noc: 4dB<br>E-UTRA Cell 3<br>Noc: -98dBm/15kHz<br>Ês / Noc: -infinity<br><br>During T2:<br>E-UTRA Cell 1<br>Noc: -101dBm/15kHz<br>Ês / Noc: 4dB<br>E-UTRA Cell 2<br>Noc: -101dBm/15kHz<br>Ês / Noc: 4dB<br>E-UTRA Cell 3<br>Noc: -101dBm/15kHz<br>Ês / Noc: 7dB |
| 8.23.5 E-UTRAN FDD-FDD DC inter-frequency event triggered reporting with DRX in asynchronous DC | Same as 8.23.4 | Same as 8.23.4 | Same as 8.23.4 |

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| 8.23.6 E-UTRAN TDD-TDD DC inter-frequency event triggered reporting with DRX in synchronous DC | Same as 8.23.4 | Same as 8.23.4 | Same as 8.23.4 |
| 9.1.1.1 FDD Intra Frequency Absolute RSRP Accuracy | Test 1:<br>$N_{oc}$: -106dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +1.0dB<br>Reported RSRP values: ±6dB<br><br>Test 2:<br>$N_{oc}$: -88dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +1.0dB<br>Reported RSRP values: ±8dB<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: +3.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -1.0dB<br>Reported RSRP values: ±6dB | Test 1:<br>-1.0dB<br>0dB<br>+1.0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>+1.0dB<br>Via mapping<br><br>Test 3:<br>0dB<br><br>0dB<br>+0.8dB<br>Via mapping | Test 1:<br>$N_{oc}$: -107.0dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>RSRP_29 to RSRP_43<br><br>Test 2:<br>$N_{oc}$: -88dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>RSRP_45 to RSRP_64<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: +3.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -0.2dB<br>RSRP_17 to RSRP_32<br>RSRP_18 to RSRP_33<br>RSRP_19 to RSRP_34<br>RSRP_20 to RSRP_35<br>depending on operating band |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 2 RSRP from $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for normal conditions. In all cases the RSRP values are 3dB wider at each end for extreme conditions. | | |
| 9.1.1.1_1 FDD Intra Frequency Absolute RSRP Accuracy (Rel 12 and forward) | Test 1:<br>$N_{oc}$: -106dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +1.0dB<br>Reported RSRP values: ±4.5dB<br><br>Test 2:<br>$N_{oc}$: -88dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +1.0dB<br>Reported RSRP values: ±8dB<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: +3.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -1.0dB<br>Reported RSRP values: ±4.5dB | Test 1:<br>-1.0dB<br>0dB<br>+1.0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>+1.0dB<br>Via mapping<br><br>Test 3:<br>0dB<br><br>0dB<br>+0.8dB<br>Via mapping | Test 1:<br>$N_{oc}$: -107.0dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>RSRP_29 to RSRP_42<br><br>Test 2:<br>$N_{oc}$: -88dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>RSRP_45 to RSRP_64<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: +3.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -0.2dB<br>RSRP_18 to RSRP_30<br>RSRP_19 to RSRP_31<br>RSRP_20 to RSRP_32<br>RSRP_21 to RSRP_33<br>RSRP_22 to RSRP_34<br>depending on operating band |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 2 RSRP from $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for normal conditions. The RSRP values are recalculated for extreme conditions. | | |

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| 9.1.1.2 FDD Intra Frequency Relative RSRP Accuracy | Test 1:<br>$N_{oc}$: -106dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +1.0dB<br>Reported relative RSRP values: ±3dB<br><br>Test 2:<br>$N_{oc}$: -88dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +1.0dB<br>Reported relative RSRP values: ±3dB<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br><br>$\hat{E}s_1$ / $N_{oc}$: +3.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -1.0dB<br>Reported relative RSRP values: ±3dB | Test 1:<br>0 dB<br>0 dB<br>+1.0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>+1.0dB<br>Via mapping<br><br>Test 3:<br>0dB<br><br><br><br>0dB<br>+1.0dB<br>Via mapping | Test 1:<br>$N_{oc}$: -106 dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>RSRP_x-9 to RSRP_x+1<br><br>Test 2:<br>$N_{oc}$: -88dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>RSRP_x-9 to RSRP_x+1<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: +3.0dB<br>$\hat{E}s_2$ / $N_{oc}$: 0dB<br>RSRP_x-8 to RSRP_x+2 |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 1 and Cell 2 RSRP from $N_{oc}$, $\hat{E}s_1$ / $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function. The RSRP values given above are for both normal and extreme conditions. | | |
| 9.1.2.1 TDD Intra Frequency Absolute RSRP Accuracy | Same as 9.1.1.1 | Same as 9.1.1.1 | Same as 9.1.1.1 |
| 9.1.2.1_1 TDD Intra Frequency Absolute RSRP Accuracy (Rel 12 and forward) | Same as 9.1.1.1_1 | Same as 9.1.1.1_1 | Same as 9.1.1.1_1 |
| 9.1.2.2 TDD Intra Frequency Relative RSRP Accuracy | Same as 9.1.1.2 | Same as 9.1.1.2 | Same as 9.1.1.2 |
| 9.1.3.1 FDD Inter Frequency Absolute RSRP Accuracy | Test 1:<br>$N_{oc1}$: -88.65dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +10.00dB<br>$N_{oc2}$: -88.65dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: +10.00dB<br>Reported RSRP values: ±8dB<br><br>Test 2:<br>$N_{oc1}$: ($N_{oc2}$ +8dB)<br>$\hat{E}s_1$ / $N_{oc1}$: +13.00dB<br>$N_{oc2}$: -117dBm to -113.5dBm /15kHz depending on operating band<br>$\hat{E}s_2$ / $N_{oc2}$: -4.00dB<br>Reported RSRP values: ±6dB | Test 1:<br>-0.6dB<br>0dB<br>-0.6dB<br>0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>0dB<br><br><br>0.8dB<br>Via mapping | Test 1:<br>$N_{oc1}$: -89.25dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +10.00dB<br>$N_{oc2}$: -89.25dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: +10.00dB<br>RSRP_52 to RSRP_71<br><br>Test 2:<br>$N_{oc1}$: ($N_{oc2}$ +8dB)<br>$\hat{E}s_1$ / $N_{oc1}$: +13.00dB<br>$N_{oc2}$: -117dBm to -113.5dBm /15kHz depending on operating band<br>$\hat{E}s_2$ / $N_{oc}$: -3.20dB<br>RSRP_13 to RSRP_28<br>RSRP_14 to RSRP_29<br>RSRP_15 to RSRP_30<br>RSRP_16 to RSRP_31<br>depending on operating band |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 2 RSRP from $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for normal conditions. In all cases the RSRP values are 3dB wider at each end for extreme conditions. | | |

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| 9.1.3.1_1 FDD Inter Frequency Absolute RSRP Accuracy (Rel 12 and forward) | Test 1:<br>$N_{oc1}$: -88.65dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +10.00dB<br>$N_{oc2}$: -88.65dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +10.00dB<br>Reported RSRP values: ±8dB<br><br>Test 2:<br>$N_{oc1}$: ($N_{oc2}$ +8dB)<br>$\hat{E}s_1 / N_{oc1}$: +13.00dB<br>$N_{oc2}$: -117dBm to -113.5dBm /15kHz depending on operating band<br>$\hat{E}s_2 / N_{oc2}$: -4.00dB<br>Reported RSRP values: ±6dB | Test 1:<br>-0.6dB<br>0dB<br>-0.6dB<br>0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>0dB<br><br>0.8dB<br>Via mapping | Test 1:<br>$N_{oc1}$: -89.25dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +10.00dB<br>$N_{oc2}$: -89.25dBm/15kHz<br>$\hat{E}s_2 / N_{oc2}$: +10.00dB<br>RSRP_52 to RSRP_71<br><br>Test 2:<br>$N_{oc1}$: ($N_{oc2}$ +8dB)<br>$\hat{E}s_1 / N_{oc1}$: +13.00dB<br>$N_{oc2}$: -117dBm to -113.5dBm /15kHz depending on operating band<br>$\hat{E}s_2 / N_{oc}$: -3.20dB<br>Normal condition:<br>RSRP_14 to RSRP_26<br>RSRP_15 to RSRP_27<br>RSRP_16 to RSRP_28<br>RSRP_17 to RSRP_29<br>RSRP_18 to RSRP_30<br>Extreme condition:<br>RSRP_10 to RSRP_31<br>RSRP_11 to RSRP_32<br>RSRP_12 to RSRP_33<br>RSRP_13 to RSRP_34<br>depending on operating band |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 2 RSRP from $N_{oc2}$ and $\hat{E}s_2 / N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function. | | |
| 9.1.3.2 FDD Inter Frequency Relative RSRP Accuracy | Test 1:<br>$N_{oc1}$: -88.65dBm/15kHz<br>$N_{oc2}$: -88.65dBm/15kHz<br>$\hat{E}s_1 / N_{oc1}$: +10dB<br>$\hat{E}s_2 / N_{oc2}$: +10dB<br><br>Reported relative RSRP values: ±6dB<br><br>Test 2:<br>$N_{oc1}$: ($N_{oc2}$ +8dB)<br>$N_{oc2}$: -117dBm to -113.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc1}$: +13dB<br>$\hat{E}s_2 / N_{oc2}$: -4.0dB<br><br>Reported relative RSRP values: ±6dB | Test 1:<br>-0.6dB<br>-0.6dB<br>0dB<br>0dB<br><br>Via mapping<br><br>Test 2:<br>-1.0dB<br>0dB<br><br>0dB<br>0.8dB<br><br>Via mapping | Test 1:<br>$N_{oc1}$: -89.25dBm/15kHz<br>$N_{oc2}$: -89.25dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +10dB<br>$\hat{E}s_2 / N_{oc}$: +10dB<br><br>RSRP_(x-9) to RSRP_(x+9)<br><br>Test 2:<br>$N_{oc1}$: ($N_{oc2}$ +7dB)<br>$N_{oc2}$: -117dBm to -113.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc1}$: +13dB<br>$\hat{E}s_2 / N_{oc2}$: -3.2dB<br><br>RSRP_(x-32) to RSRP_(x-16) |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 1 and Cell 2 RSRP from $N_{oc1}$ and $\hat{E}s_1 / N_{oc1}$ and $N_{oc2}$ and $\hat{E}s_2 / N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for both normal and extreme conditions. | | |

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| 9.1.3.2_1 FDD Inter Frequency Relative RSRP Accuracy (Rel 12 and forward) | Test 1:<br>$N_{oc1}$: -88.65dBm/15kHz<br>$N_{oc2}$: -88.65dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +10dB<br>$\hat{E}s_2$ / $N_{oc2}$: +10dB<br><br>Reported relative RSRP values normal: ±4.5dB<br>Reported relative RSRP values extreme : ±6dB<br><br>Test 2:<br>$N_{oc1}$: ($N_{oc2}$ +8dB)<br>$N_{oc2}$: -117dBm to -113.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc1}$: +13dB<br>$\hat{E}s_2$ / $N_{oc2}$: -4.0dB<br><br>Reported relative RSRP values normal: ±4.5dB<br>Reported relative RSRP values extreme : ±6dB | Test 1:<br>-0.6dB<br>-0.6dB<br>0dB<br>0dB<br><br>Via mapping<br><br><br><br>Test 2:<br>-1.0dB<br>0dB<br><br>0dB<br>0.8dB<br><br>Via mapping | Test 1:<br>$N_{oc1}$: -89.25dBm/15kHz<br>$N_{oc2}$: -89.25dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +10dB<br>$\hat{E}s_2$ / $N_{oc}$: +10dB<br><br>Normal condition:<br>RSRP_(x-7) to RSRP_(x+7)<br>Extreme condition:<br>RSRP_(x-9) to RSRP_(x+9)<br><br>Test 2:<br>$N_{oc1}$: ($N_{oc2}$ +7dB)<br>$N_{oc2}$: -117dBm to -113.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc1}$: +13dB<br>$\hat{E}s_2$ / $N_{oc2}$: -3.2dB<br><br>Normal condition:<br>RSRP_(x-30) to RSRP_(x-17)<br>Extreme condition:<br>RSRP_(x-32) to RSRP_(x-16) |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 1 and Cell 2 RSRP from $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$ and $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function. | | |
| 9.1.4.1 TDD Inter Frequency Absolute RSRP Accuracy | Same as 9.1.3.1 | Same as 9.1.3.1 | Same as 9.1.3.1 |
| 9.1.4.1_1 TDD Inter Frequency Absolute RSRP Accuracy (R12 and forward) | Same as 9.1.3.1 | Same as 9.1.3.1 | Same as 9.1.3.1 |
| 9.1.4.2 TDD Inter Frequency Relative RSRP Accuracy | Same as 9.1.3.2 | Same as 9.1.3.2 | Same as 9.1.3.2 |
| 9.1.4.2_1 TDD Inter Frequency Relative RSRP Accuracy (Rel 12 and forward) | Same as 9.1.3.2 | Same as 9.1.3.2 | Same as 9.1.3.2 |
| 9.1.5.1 FDD-TDD inter frequency absolute RSRP Accuracy | Test 1:<br>$N_{oc1}$: -88.65dBm/15kHz<br>$N_{oc2}$: -88.65dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +10dB<br>$\hat{E}s_2$ / $N_{oc2}$: +10dB<br><br>Reported absolute RSRP values: ±8dB<br><br>Test 2:<br>$N_{oc1}$: -104dBm /15kHz<br>$N_{oc2}$: -112dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +13dB<br>$\hat{E}s_2$ / $N_{oc2}$: -4.0dB<br><br>Reported absolute RSRP values: ±6dB | Test 1:<br>-0.6dB<br>-0.6dB<br>0dB<br>0dB<br><br>Via mapping<br><br><br>Test 2:<br>0dB<br>0dB<br>0dB<br>0.8dB<br><br>Via mapping | Test 1:<br>$N_{oc1}$: -89.25dBm/15kHz<br>$N_{oc2}$: -89.25dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +10dB<br>$\hat{E}s_2$ / $N_{oc}$: +10dB<br><br>RSRP_52 to RSRP_71<br><br><br>Test 2:<br>$N_{oc1}$: -104Bm /15kHz<br>$N_{oc2}$: -112dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +13dB<br>$\hat{E}s_2$ / $N_{oc2}$: -3.2dB<br><br>RSRP_18 to RSRP_33 |

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| 9.1.5.2 FDD-TDD Inter Frequency Relative RSRP Accuracy | Test 1:<br>$N_{oc1}$: -88.65dBm/15kHz<br>$N_{oc2}$: -88.65dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +10dB<br>$\hat{E}s_2$ / $N_{oc2}$: +10dB<br><br>Reported absolute RSRP values: ±6dB<br><br>Test 2:<br>$N_{oc1}$: -104dBm /15kHz<br>$N_{oc2}$: -112dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +13dB<br>$\hat{E}s_2$ / $N_{oc2}$: -4.0dB<br><br>Reported absolute RSRP values: ±6dB | Test 1:<br>-0.6dB<br>-0.6dB<br>0dB<br>0dB<br><br>Via mapping<br><br><br>Test 2:<br>-1.0dB<br>0dB<br>0dB<br>0.8dB<br><br>Via mapping | Test 1:<br>$N_{oc1}$: -89.25dBm/15kHz<br>$N_{oc2}$: -89.25dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +10dB<br>$\hat{E}s_2$ / $N_{oc}$: +10dB<br><br>RSRP_(x-9) to RSRP_(x+9)<br><br><br>Test 2:<br>$N_{oc1}$: -105.0Bm /15kHz<br>$N_{oc2}$: -112dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +13dB<br>$\hat{E}s_2$ / $N_{oc2}$: -3.2dB<br><br>RSRP_(x-32) to RSRP_(x-16) |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 1 and Cell 2 RSRP from $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$ and $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for both normal and extreme conditions. | | |
| 9.1.6.1 FDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation | $N_{oc1}$: -117dBm or -116dBm or -115.5dBm or -115 dBm or -114dBm or -113.5dBm /15kHz depending on operating band<br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: +3dB<br>$\hat{E}s_3$ / $N_{o2c}$: -1dB<br><br><u>Reported RSRP values:</u><br>±6dB for normal conditions and ±9dB for extreme conditions | 0dB<br><br><br><br>0dB<br>0dB<br>0dB<br>0dB<br><br><br>Via mapping<br>Via mapping | $N_{oc1}$: -117dBm or -116dBm or -115.5dBm or -115 dBm or -114dBm or -113.5dBm /15kHz depending on operating band<br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: +3dB<br>$\hat{E}s_3$ / $N_{o2c}$: -1dB<br><br><u>Cell 1:</u><br>RSRP_12 to RSRP_27<br>RSRP_13 to RSRP_28<br>RSRP_13 to RSRP_29<br>RSRP_14 to RSRP_29<br>RSRP_15 to RSRP_30<br>RSRP_15 to RSRP_31<br>depending on operating band<br><u>Cell 2:</u><br>RSRP_20 to RSRP_35<br>RSRP_21 to RSRP_36<br>RSRP_21 to RSRP_37<br>RSRP_22 to RSRP_37<br>RSRP_23 to RSRP_38<br>RSRP_23 to RSRP_39<br>depending on Cell 1 band |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 1 RSRP from $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$, the uncertainty in Cell 2 RSRP from $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for normal conditions. In all cases the RSRP values are 3dB wider at each end for extreme conditions. | | |

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| 9.1.6.1_1 Measurement Performance Requirements / FDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation (Rel 12 and forward) | Noc1: -117dBm or -116dBm or -115.5dBm or -115 dBm or -114dBm or -113.5dBm /15kHz depending on operating band<br>Noc2: = Noc1 +1dB<br>Es1 / Noc1: -4dB<br>Es2 / No2c: +3dB<br>Es3 / No2c: -1dB<br>Reported RSRP values:<br>4.5dB for normal conditions and 9dB for extreme conditions | 0dB<br><br><br><br>0dB<br>0dB<br>0dB<br>0dB<br><br><br>Via mapping<br>Via mapping | Noc1: -117dBm or -116dBm or -115.5dBm or -115 dBm or -114dBm or -113.5dBm /15kHz depending on operating band<br>Noc2: = Noc1 +1dB<br>Es1 / Noc1: -4dB<br>Es2 / No2c: +3dB<br>Es3 / No2c: -1dB<br><br><br>Normal condition<br>Cell 1:<br>RSRP_13 to RSRP_26<br>RSRP_14 to RSRP_26<br>RSRP_14 to RSRP_27<br>RSRP_15 to RSRP_27<br>RSRP_15 to RSRP_28<br>RSRP_16 to RSRP_29<br>RSRP_17 to RSRP_29<br>depending on operating band<br>Cell 2:<br>RSRP_21 to RSRP_34<br>RSRP_22 to RSRP_34<br>RSRP_22 to RSRP_35<br>RSRP_23 to RSRP_35<br>RSRP_23 to RSRP_36<br>RSRP_24 to RSRP_37<br>RSRP_25 to RSRP_37<br>depending on Cell 1 band<br><br>Exreme condition<br>Cell 1:<br>RSRP_9 to RSRP_30<br>RSRP_9 to RSRP_31<br>RSRP_10 to RSRP_31<br>RSRP_10 to RSRP_31<br>RSRP_11 to RSRP_32<br>RSRP_12 to RSRP_33<br>RSRP_12 to RSRP_34<br>depending on operating band<br>Cell 2:<br>RSRP_17 to RSRP_38<br>RSRP_17 to RSRP_39<br>RSRP_18 to RSRP_39<br>RSRP_18 to RSRP_40<br>RSRP_19 to RSRP_40<br>RSRP_20 to RSRP_41<br>RSRP_20 to RSRP_42<br>depending on Cell 1 band |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 1 RSRP from Noc1 and Es1 / Noc1, the uncertainty in Cell 2 RSRP from Noc2 and Es2 / Noc2, the allowed UE reporting accuracy, and the UE mapping function. | | |
| 9.1.6.2 FDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation | $N_{oc1}$: -117dBm or -116dBm or -115.5dBm or -115 dBm or -114dBm or -113.5dBm /15kHz depending on operating band<br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: +3dB<br>$\hat{E}s_3$ / $N_{o2c}$: -1dB<br><br>Reported relative RSRP values:<br>(Cell 2 – Cell 1):<br>±6dB<br>(Cell 3 – Cell 2):<br>±3dB | 0dB<br><br><br><br>0dB<br>0dB<br>0dB<br>+0.8dB<br><br><br><br>Via mapping<br><br>Via mapping | $N_{oc1}$: -117dBm or -116dBm or -115.5dBm or -115 dBm or -114dBm or -113.5dBm /15kHz depending on operating band<br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: +3dB<br>$\hat{E}s_3$ / $N_{o2c}$: -0.2dB<br><br><br>(Cell 2 – Cell 1):<br>RSRP_(x-1) to RSRP_(x+17)<br>(Cell 3 - Cell 2):<br>RSRP_(x-8) to RSRP_(x+1) |

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| | The derivation of the RSRP values takes into account the uncertainty in Cell 1 RSRP from $N_{oc1}$ and $\hat{E}s_1 / N_{oc1}$, the uncertainty in Cell 2 RSRP from $N_{oc2}$ and $\hat{E}s_2 / N_{oc2}$, the uncertainty in Cell 3 RSRP from $\hat{E}s_3 / N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for normal and extreme conditions. | | |
| 9.1.6.2_1 FDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation (Rel 12 and forward) | Noc1: -117dBm or -116dBm or -115.5dBm or -115 dBm or -114dBm or -113.5dBm /15kHz depending on operating band<br>Noc2: = Noc1 +1dB<br>Es1 / Noc1: -4dB<br>Es2 / No2c: +3dB<br>Es3 / No2c: -1dB<br><br>Reported relative RSRP values:<br>(Cell 2 - Cell 1):<br>±4.5dB for normal conditions and ±6dB for extreme conditions<br>(Cell 3 - Cell 2):<br>±3dB | 0dB<br><br><br><br>0dB<br>0dB<br>0dB<br>+0.8dB<br><br><br><br>Via mapping<br><br>Via mapping | Noc1: -117dBm or -116dBm or -115.5dBm or -115 dBm or -114dBm or -113.5dBm /15kHz depending on operating band<br>Noc2: = Noc1 +1dB<br>Es1 / Noc1: -4dB<br>Es2 / No2c: +3dB<br>Es3 / No2c: -0.2dB<br><br><br>Normal condition:<br>(Cell 2 - Cell 1):<br>RSRP_(x+2) to RSRP_(x+15)<br>(Cell 3 - Cell 2):<br>RSRP_(x-8) to RSRP_(x+1)<br><br>Extreme condition:<br>(Cell 2 - Cell 1):<br>RSRP_(x-1) to RSRP_(x+17)<br>(Cell 3 - Cell 2):<br>RSRP_(x-8) to RSRP_(x+1) |
| 9.1.7.1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation | Same as 9.1.6.1 | Same as 9.1.6.1 | Same as 9.1.6.1 |
| 9.1.7.2 TDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation | Same as 9.1.6.2 | Same as 9.1.6.2 | Same as 9.1.6.2 |
| 9.1.8.1 FDD Absolute RSRP Accuracy under Time Domain Measurement Resource Restriction with Non-MBSFN ABS | Test 1:<br>$N_{oc}$: -106.0dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +5.0dB<br>$\hat{E}s_2 / N_{oc}$: -2.0dB<br>Reported RSRP values: ±6dB<br><br>Test 2:<br>$N_{oc}$: -88.0dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +5.0dB<br>$\hat{E}s_2 / N_{oc}$: -4.0dB<br>Reported RSRP values: ±8dB<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc}$: +5.0dB<br>$\hat{E}s_2 / N_{oc}$: -4.0dB<br>Reported RSRP values: ±6dB | Test 1:<br>0dB<br>0dB<br>0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>+0.95dB<br>Via mapping<br><br>Test 3:<br>0dB<br><br><br>0dB<br>+0.95dB<br>Via mapping | Test 1:<br>$N_{oc}$: -106.0Bm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +5.0dB<br>$\hat{E}s_2 / N_{oc}$: -2.0dB<br>RSRP_25 to RSRP_40<br><br>Test 2:<br>$N_{oc}$: -88.0dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +5.0dB<br>$\hat{E}s_2 / N_{oc}$: -3.05dB<br>RSRP_40 to RSRP_59<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc}$: +5.0dB<br>$\hat{E}s_2 / N_{oc}$: -3.05dB<br>RSRP_14 to RSRP_29<br>RSRP_15 to RSRP_30<br>RSRP_16 to RSRP_31<br>RSRP_17 to RSRP_32<br>depending on operating band |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 2 RSRP from $N_{oc}$ and $\hat{E}s_2 / N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for normal conditions. In all cases the RSRP values are 3dB wider at each end for extreme conditions. | | |

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| 9.1.8.2 FDD Relative RSRP Accuracy under Time Domain Measurement Resource Restriction with Non-MBSFN ABS | Test 1:<br>$N_{oc}$: -106.0dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -2.0dB<br>Reported relative RSRP values: ±2dB<br><br>Test 2:<br>$N_{oc}$: -88.0dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -4.0dB<br>Reported relative RSRP values: ±3dB<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -4.0dB<br>Reported relative RSRP values: ±3dB | Test 1:<br>0dB<br>0dB<br>+0.8dB<br><br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>+0.95dB<br><br>Via mapping<br><br>Test 3:<br>0dB<br><br><br>0dB<br>+0.95dB<br><br>Via mapping | Test 1:<br>$N_{oc}$: -106.0Bm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -1.2dB<br><br>RSRP_x-10 to RSRP_x-4<br><br>Test 2:<br>$N_{oc}$: -88.0dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -3.05dB<br><br>RSRP_x-13 to RSRP_x-4<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -3.05dB<br><br>RSRP_x-13 to RSRP_x-4 |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 1 and Cell 2 RSRP from $N_{oc}$, $\hat{E}s_1$ / $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function. The RSRP values given above are for normal conditions. For Test 1 extreme conditions allow 1dB wider at each end. Test 2, Test 3 RSRP values for extreme conditions are the same for normal conditions. | | |
| 9.1.9.1 TDD Absolute RSRP Accuracy under Time Domain Measurement Resource Restriction with Non-MBSFN ABS | Same as 9.1.8.1 | Same as 9.1.8.1 | Same as 9.1.8.1 |
| 9.1.9.2 TDD Relative RSRP Accuracy under Time Domain Measurement Resource Restriction with Non-MBSFN ABS | Same as 9.1.8.2 | Same as 9.1.8.2 | Same as 9.1.8.2 |
| 9.1.10.1 FDD Absolute RSRP Accuracy under Time Domain Measurement Resource Restriction with MBSFN ABS | Test 1:<br>$N_{oc}$: -106.0dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -2.0dB<br>Reported RSRP values: ±6dB<br><br>Test 2:<br>$N_{oc}$: -88.0dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -4.0dB<br>Reported RSRP values: ±8dB<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -4.0dB<br>Reported RSRP values: ±6dB | Test 1:<br>0dB<br>0dB<br>0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>+0.95dB<br>Via mapping<br><br>Test 3:<br>0dB<br><br><br>0dB<br>+0.95dB<br>Via mapping | Test 1:<br>$N_{oc}$: -106.0Bm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -2.0dB<br>RSRP_25 to RSRP_40<br><br>Test 2:<br>$N_{oc}$: -88.0dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -3.05dB<br>RSRP_40 to RSRP_59<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -3.05dB<br>RSRP_14 to RSRP_29<br>RSRP_15 to RSRP_30<br>RSRP_16 to RSRP_31<br>RSRP_17 to RSRP_32<br>depending on operating band |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 2 RSRP from $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for normal conditions. In all cases the RSRP values are 3dB wider at each end for extreme conditions. | | |

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| 9.1.10.2 FDD Relative RSRP Accuracy under Time Domain Measurement Resource Restriction with MBSFN ABS | Test 1:<br>$N_{oc}$: -106.0dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -2.0dB<br>Reported relative RSRP values: ±2dB<br><br>Test 2:<br>$N_{oc}$: -88.0dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -4.0dB<br>Reported relative RSRP values: ±3dB<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -4.0dB<br>Reported relative RSRP values: ±3dB | Test 1:<br>0dB<br>0dB<br>+0.8dB<br><br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>+0.95dB<br><br>Via mapping<br><br>Test 3:<br>0dB<br><br><br>0dB<br>+0.95dB<br><br>Via mapping | Test 1:<br>$N_{oc}$: -106.0Bm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -1.2dB<br><br>RSRP_x-10 to RSRP_x-4<br><br>Test 2:<br>$N_{oc}$: -88.0dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -3.05dB<br><br>RSRP_x-13 to RSRP_x-4<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -3.05dB<br><br>RSRP_x-13 to RSRP_x-4 |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 1 and Cell 2 RSRP from $N_{oc}$, $\hat{E}s_1$ / $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function. The RSRP values given above are for normal conditions. For Test 1 extreme conditions allow 1dB wider at each end. Test 2, Test 3 RSRP values for extreme conditions are the same for normal conditions. | | |
| 9.1.11.1 TDD Absolute RSRP Accuracy under Time Domain Measurement Resource Restriction with MBSFN ABS | Same as 9.1.10.1 | Same as 9.1.10.1 | Same as 9.1.10.1 |
| 9.1.11.2 TDD Relative RSRP Accuracy under Time Domain Measurement Resource Restriction with MBSFN ABS | Same as 9.1.10.2 | Same as 9.1.10.2 | Same as 9.1.10.2 |
| 9.1.12.1 FDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20MHz | Same as 9.1.6.1 | Same as 9.1.6.1 | Same as 9.1.6.1 |
| 9.1.12.1_1 Measurement Performance Requirements / FDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz (Rel 12 and forward) | Same as 9.1.6.1_1 | Same as 9.1.6.1_1 | Same as 9.1.6.1_1 |
| 9.1.12.2 FDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 20MHz | Same as 9.1.6.2 | Same as 9.1.6.2 | Same as 9.1.6.2 |
| 9.1.12.2_1 FDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz (Rel 12 and forward) | Same as 9.1.6.2_1 | Same as 9.1.6.2_1 | Same as 9.1.6.2_1 |
| 9.1.13.1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20MHz | Same as 9.1.6.1 | Same as 9.1.6.1 | Same as 9.1.6.1 |

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| 9.1.13.2 TDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 20MHz | Same as 9.1.6.2 | Same as 9.1.6.2 | Same as 9.1.6.2 |
| 9.1.14.1 FDD Intra Frequency Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | Test 1:<br>$N_{oc}$: -106dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +4.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>$\hat{E}s_3$ / $N_{oc}$: -1.5dB<br>Reported RSRP values: ±6dB<br><br>Test 2:<br>$N_{oc}$: -88dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +4.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>$\hat{E}s_3$ / $N_{oc}$: -4.0dB<br>Reported RSRP values: ±8dB<br><br>Test 3:<br>$N_{oc}$: -116 to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: +4.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>$\hat{E}s_3$ / $N_{oc}$: -4.0dB<br>Reported RSRP values: ±6dB | Test 1:<br>0dB<br>0dB<br>0dB<br>0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>0dB<br>+1.0dB<br>Via mapping<br><br>Test 3:<br>0dB<br><br>0dB<br>0dB<br>+1.0dB<br>Via mapping | Test 1:<br>$N_{oc}$: -106.0dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +4.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>$\hat{E}s_3$ / $N_{oc}$: -1.5dB<br>RSRP_25 to RSRP_41<br><br>Test 2:<br>$N_{oc}$: -88dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +4.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>$\hat{E}s_3$ / $N_{oc}$: -3.0dB<br>RSRP_40 to RSRP_59<br><br>Test 3:<br>$N_{oc}$: -116 to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: +4.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>$\hat{E}s_3$ / $N_{oc}$: -3.0dB<br>RSRP_14 to RSRP_29<br>RSRP_15 to RSRP_30<br>RSRP_15 to RSRP_31<br>RSRP_16 to RSRP_31<br>RSRP_17 to RSRP_32<br>RSRP_17 to RSRP_33 depending on operating band |
| 9.1.14.2 FDD Intra Frequency Relative RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | Test 1:<br>$N_{oc}$: -106dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +4.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>$\hat{E}s_3$ / $N_{oc}$: -1.5dB<br>Reported relative RSRP values: ±2dB<br><br>Test 2:<br>$N_{oc}$: -88dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +4.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>$\hat{E}s_3$ / $N_{oc}$: -4.0dB<br>Reported relative RSRP values: ±3dB<br><br>Test 3:<br>$N_{oc}$: -116 to -112.5 dBm/15kHz depending on frequency band<br>$\hat{E}s_1$ / $N_{oc}$: +4.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -3.0dB<br>$\hat{E}s_3$ / $N_{oc}$: -4.0dB<br>Reported relative RSRP values: ±3dB | Test 1:<br>0dB<br>0dB<br>0dB<br>+1dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>0dB<br>+1dB<br>Via mapping<br><br>Test 3:<br>0dB<br><br>0dB<br>0dB<br>+1dB<br>Via mapping | Test 1:<br>$N_{oc}$: -106.0dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +4.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>$\hat{E}s_3$ / $N_{oc}$: -0.5dB<br><br>RSRP_x-8 to RSRP_x-2<br><br>Test 2:<br>$N_{oc}$: -88dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +4.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>$\hat{E}s_3$ / $N_{oc}$: -3.0dB<br>RSRP_x-12 to RSRP_x-3<br><br>Test 3:<br>$N_{oc}$: -116 to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: +4.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -3.0dB<br>$\hat{E}s_3$ / $N_{oc}$: -3.0dB<br>RSRP_x-12 to RSRP_x-3 |
| 9.1.15.1 TDD Intra Frequency Absolute RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | Same as 9.1.14.1 | Same as 9.1.14.1 | Same as 9.1.14.1 |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 2 RSRP from $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$, the uncertainty in Cell 3 RSRP from $N_{oc}$ and $\hat{E}s_3$ / $N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for normal conditions. For extreme conditions, the RSRP values are 3dB wider at each end in all cases. | | |

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| 9.1.15.2 TDD Intra Frequency Relative RSRP Accuracy under Time-Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | Same as 9.1.14.2 | Same as 9.1.14.2 | Same as 9.1.14.2 |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 3 RSRP from $N_{oc}$ and $\hat{E}s_3 / N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for normal conditions. For extreme conditions, the RSRP values are 1dB wider at each end for Test 1, the same than for normal conditions for Test 2, Test 3. | | |
| 9.1.16.1 FDD Intra Frequency Absolute RSRP Accuracy for 5MHz Bandwidth | Test 1:<br>$N_{oc}$: -103dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +6.0dB<br>$\hat{E}s_2 / N_{oc}$: +1.0dB<br>Reported relative RSRP values: ±3dB<br><br>Test 2:<br>$N_{oc}$: -83dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +6.0dB<br>$\hat{E}s_2 / N_{oc}$: +1.0dB<br>Reported relative RSRP values: ±3dB<br><br>Test 3:<br>$N_{oc}$: -109.5dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +3.0dB<br>$\hat{E}s_2 / N_{oc}$: -1.0dB<br>Reported relative RSRP values: ±3dB | Test 1:<br>-1.0dB<br>0 dB<br>+1.0dB<br>Via mapping<br><br>Test 2:<br>-1.0dB<br>0dB<br>+1.0dB<br>Via mapping<br><br>Test 3:<br>0dB<br>0dB<br>+0.8dB<br>Via mapping | Test 1:<br>$N_{oc}$: -104.0dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +6.0dB<br>$\hat{E}s_2 / N_{oc}$: +2.0dB<br>RSRP_31 to RSRP_46<br><br>Test 2:<br>$N_{oc}$: -84dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +6.0dB<br>$\hat{E}s_2 / N_{oc}$: +2.0dB<br>RSRP_49 to RSRP_68<br><br>Test 3:<br>$N_{oc}$: -109.5dBm /15kHz<br>$\hat{E}s_1 / N_{oc}$: +3.0dB<br>$\hat{E}s_2 / N_{oc}$: -0.2dB<br>RSRP_23 to RSRP_38 |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 2 RSRP from $N_{oc}$ and $\hat{E}s_2 / N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for normal conditions. In all cases the RSRP values are 3dB wider at each end for extreme conditions. | | |
| 9.1.16.1_1 FDD Intra Frequency Absolute RSRP Accuracy for 5MHz Bandwidth (Rel 12 and forward) | Test 1:<br>$N_{oc}$: -103dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +6.0dB<br>$\hat{E}s_2 / N_{oc}$: +1.0dB<br>Reported RSRP values: ±4.5dB<br><br>Test 2:<br>$N_{oc}$: -83dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +6.0dB<br>$\hat{E}s_2 / N_{oc}$: +1.0dB<br>Reported RSRP values: ±8dB<br><br>Test 3:<br>$N_{oc}$: -109.5dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +3.0dB<br>$\hat{E}s_2 / N_{oc}$: -1.0dB<br>Reported RSRP values: ±4.5dB | Test 1:<br>-1.0dB<br>0 dB<br>+1.0dB<br>Via mapping<br><br>Test 2:<br>-1.0dB<br>0dB<br>+1.0dB<br>Via mapping<br><br>Test 3:<br>0dB<br>0dB<br>+0.8dB<br>Via mapping | Test 1:<br>$N_{oc}$: -104.0dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +6.0dB<br>$\hat{E}s_2 / N_{oc}$: +2.0dB<br>RSRP_32 to RSRP_45<br><br>Test 2:<br>$N_{oc}$: -84dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +6.0dB<br>$\hat{E}s_2 / N_{oc}$: +2.0dB<br>RSRP_49 to RSRP_68<br><br>Test 3:<br>$N_{oc}$: -109.5dBm /15kHz<br>$\hat{E}s_1 / N_{oc}$: +3.0dB<br>$\hat{E}s_2 / N_{oc}$: -0.2dB<br>RSRP_25 to RSRP_37 |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 2 RSRP from $N_{oc}$ and $\hat{E}s_2 / N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for normal conditions. The RSRP values are recalculated for extreme conditions. | | |

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| 9.1.16.2 FDD Intra Frequency Relative Accuracy of RSRP for 5MHz Bandwidth | Test 1:<br>$N_{oc}$: -103dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +1.0dB<br>Reported relative RSRP values: ±3dB<br><br>Test 2:<br>$N_{oc}$: -83dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +1.0dB<br>Reported relative RSRP values: ±3dB<br><br>Test 3:<br>$N_{oc}$: -109.5.5dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +3.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -1.0dB<br>Reported relative RSRP values: ±3dB | Test 1:<br>0 dB<br>0 dB<br>+1.0dB<br>Via mapping<br><br>Test 2:<br>-1.0dB<br>0dB<br>+1.0dB<br>Via mapping<br><br>Test 3:<br>0dB<br>0dB<br>+1.0dB<br>Via mapping | Test 1:<br>$N_{oc}$: -103 dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>RSRP_x-9 to RSRP_x+1<br><br>Test 2:<br>$N_{oc}$: -84dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br>RSRP_x-9 to RSRP_x+1<br><br>Test 3:<br>$N_{oc}$: -109.5dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +3.0dB<br>$\hat{E}s_2$ / $N_{oc}$: 0dB<br>RSRP_x-8 to RSRP_x+2 |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 1 and Cell 2 RSRP from $N_{oc}$, $\hat{E}s_1$ / $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function. The RSRP values given above are for both normal and extreme conditions. | | |
| 9.1.17.1 FDD Inter Frequency Absolute RSRP Accuracy for 5MHz Bandwidth | Test 1:<br>$N_{oc1}$: -85.65dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +10.00dB<br>$N_{oc2}$: -85.65dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: +10.00dB<br>Reported RSRP values: ±8dB<br><br>Test 2:<br>$N_{oc1}$: ($N_{oc2}$ +8dB)<br>$\hat{E}s_1$ / $N_{oc1}$: +13.00dB<br>$N_{oc2}$: -110.5dBm /15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: -4.00dB<br>Reported RSRP values: ±6dB | Test 1:<br>-0.6dB<br>0dB<br>-0.6dB<br>0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>0dB<br>0.8dB<br>Via mapping | Test 1:<br>$N_{oc1}$: -86.25dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +10.00dB<br>$N_{oc2}$: -86.25dBm/15kHz<br>$\hat{E}s_2$ / $N_{oc2}$: +10.00dB<br>RSRP_55 to RSRP_74<br><br>Test 2:<br>$N_{oc1}$: ($N_{oc2}$ +8dB)<br>$\hat{E}s_1$ / $N_{oc1}$: +13.00dB<br>$N_{oc2}$: -110.5dBm /15kHz<br>$\hat{E}s_2$ / $N_{oc}$: -3.20dB<br>RSRP_19 to RSRP_34 |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 2 RSRP from $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for normal conditions. In all cases the RSRP values are 3dB wider at each end for extreme conditions. | | |
| 9.1.17.2 FDD - FDD Inter Frequency Relative Accuracy of RSRP for 5MHz Bandwidth | Test 1:<br>$N_{oc1}$: -85.65dBm/15kHz<br>$N_{oc2}$: -85.65dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +10dB<br>$\hat{E}s_2$ / $N_{oc2}$: +10dB<br><br>Reported relative RSRP values: ±6dB<br><br>Test 2:<br>$N_{oc1}$: -102.5dBm/15kHz<br>$N_{oc2}$: -110.5dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +13dB<br>$\hat{E}s_2$ / $N_{oc2}$: -4.0dB<br><br>Reported relative RSRP values: ±6dB | Test 1:<br>-0.6dB<br>-0.6dB<br>0dB<br>0dB<br><br>Via mapping<br><br>Test 2:<br>-1.0dB<br>0dB<br>0dB<br>0.8dB<br><br>Via mapping | Test 1:<br>$N_{oc1}$: -86.25dBm/15kHz<br>$N_{oc2}$: -86.25dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +10dB<br>$\hat{E}s_2$ / $N_{oc}$: +10dB<br><br>RSRP_(x-9) to RSRP_(x+9)<br><br>Test 2:<br>$N_{oc1}$: -103.5dBm/15kHz<br>$N_{oc2}$: -100.5dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: +13dB<br>$\hat{E}s_2$ / $N_{oc2}$: -3.2dB<br><br>RSRP_(x-32) to RSRP_(x-16) |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 1 and Cell 2 RSRP from $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$ and $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for both normal and extreme conditions. | | |
| 9.1.18.1 FDD Absolute RSRP Accuracy for E-UTRA for Carrier Aggregation for 10MHz + 5MHz | Same as 9.1.6.1 | Same as 9.1.6.1 | Same as 9.1.6.1 |

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| 9.1.18.2 FDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation for 10MHz + 5MHz | $N_{oc1}$: -117dBm or -116dBm or -115.5dBm or -115 dBm or -114dBm or -113.5dBm /15kHz depending on operating band<br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{oc2}$: +3dB<br>$\hat{E}s_3$ / $N_{oc2}$: -1dB<br><br>Reported relative RSRP values:<br>(Cell 2 – Cell 1):<br>±6dB<br>(Cell 3 – Cell 2):<br>±3dB | 0dB<br><br><br><br>0dB<br>0dB<br>0dB<br>+0.8dB<br><br><br><br>Via mapping<br><br>Via mapping | $N_{oc1}$: -117dBm or -116dBm or -115.5dBm or -115 dBm or -114dBm or -113.5dBm /15kHz depending on operating band<br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{oc2}$: +3dB<br>$\hat{E}s_3$ / $N_{oc2}$: -0.2dB<br><br><br>(Cell 2 – Cell 1):<br>RSRP_(x-1) to RSRP_(x+17)<br>(Cell 3 - Cell 2):<br>RSRP_(x-8) to RSRP_(x+1) |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 1 RSRP from $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$, the uncertainty in Cell 2 RSRP from $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the uncertainty in Cell 3 RSRP from $\hat{E}s_3$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for normal and extreme conditions. | | |
| 9.1.19.1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 10MHz + 5MHz | Same as 9.1.6.1 | Same as 9.1.6.1 | Same as 9.1.6.1 |
| 9.1.19.2 TDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation for 10MHz + 5MHz | Same as 9.1.18.2 | Same as 9.1.18.2 | Same as 9.1.18.2 |
| 9.1.20.1 FDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz bandwidth | $N_{oc1}$: -117dBm or -116dBm or -115.5dBm or -115 dBm or -114dBm or -113.5dBm /15kHz depending on operating band<br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{oc2}$: +3dB<br>$\hat{E}s_3$ / $N_{oc2}$: -1dB<br><br>Reported RSRP values:<br>±6dB for normal conditions and ±9dB for extreme conditions | 0dB<br><br><br><br>0dB<br>0dB<br>0dB<br>0dB<br><br><br>Via mapping<br>Via mapping | $N_{oc1}$: -117dBm or -116dBm or -115.5dBm or -115 dBm or -114dBm or -113.5dBm /15kHz depending on operating band<br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{oc2}$: +3dB<br>$\hat{E}s_3$ / $N_{oc2}$: -1dB<br><br><br>Cell 1:<br>RSRP_12 to RSRP_27<br>RSRP_13 to RSRP_28<br>RSRP_13 to RSRP_29<br>RSRP_14 to RSRP_29<br>RSRP_15 to RSRP_30<br>RSRP_15 to RSRP_31<br>depending on operating band<br>Cell 2:<br>RSRP_20 to RSRP_35<br>RSRP_21 to RSRP_36<br>RSRP_21 to RSRP_37<br>RSRP_22 to RSRP_37<br>RSRP_23 to RSRP_38<br>RSRP_23 to RSRP_39<br>depending on Cell 1 band |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 1 RSRP from $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$, the uncertainty in Cell 2 RSRP from $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for normal conditions. In all cases the RSRP values are 3dB wider at each end for extreme conditions. | | |

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| 9.1.20.2 FDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation for 5MHz + 5MHz | $N_{oc1}$: -117dBm or -116dBm or -115.5dBm or -115 dBm or -114dBm or -113.5dBm /15kHz depending on operating band<br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{oc2}$: +3dB<br>$\hat{E}s_3$ / $N_{oc2}$: -1dB<br><br>Reported relative RSRP values:<br>(Cell 2 – Cell 1):<br>±6dB<br>(Cell 3 – Cell 2):<br>±3dB | 0dB<br><br><br><br>0dB<br>0dB<br>0dB<br>+0.8dB<br><br><br><br>Via mapping<br><br>Via mapping | $N_{oc1}$: -117dBm or -116dBm or -115.5dBm or -115 dBm or -114dBm or -113.5dBm /15kHz depending on operating band<br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{oc2}$: +3dB<br>$\hat{E}s_3$ / $N_{oc2}$: -0.2dB<br><br><br>(Cell 2 – Cell 1):<br>RSRP_(x-1) to RSRP_(x+17)<br>(Cell 3 - Cell 2):<br>RSRP_(x-8) to RSRP_(x+1) |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 1 RSRP from $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$, the uncertainty in Cell 2 RSRP from $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the uncertainty in Cell 3 RSRP from $\hat{E}s_3$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for normal and extreme conditions. | | |
| 9.1.21.1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 5MHz + 5MHz bandwidth | Same as 9.1.20.1 | Same as 9.1.20.1 | Same as 9.1.20.1 |
| 9.1.21.2 TDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation for 5MHz + 5MHz | Same as 9.1.20.2 | Same as 9.1.20.2 | Same as 9.1.20.2 |
| 9.1.22 FDD-TDD RSRP Accuracy E-UTRA for Carrier Aggregation with PCell in FDD | Same as 9.1.37 Cell 1 and Cell 2 absolute RSRP; and Cell 2 –Cell 1 relative RSRP and Cell 3- Cell 2 relative RSRP | Same as 9.1.37 Cell 1, Cell 2 and Cell 3 | Same as 9.1.37 Cell 1 and Cell 2 absolute RSRP; and Cell 2 –Cell 1 relative RSRP and Cell 3- Cell 2 relative RSRP |
| 9.1.23 FDD-TDD RSRP Accuracy E-UTRA for Carrier Aggregation with PCell in TDD | Same as 9.1.37 Cell 1 and Cell 2 absolute RSRP; and Cell 2 –Cell 1 relative RSRP and Cell 3- Cell 2 relative RSRP | Same as 9.1.37 Cell 1, Cell 2 and Cell 3 | Same as 9.1.37 Cell 1 and Cell 2 absolute RSRP; and Cell 2 –Cell 1 relative RSRP and Cell 3- Cell 2 relative RSRP |
| 9.1.24.1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20MHz + 10MHz | $N_{oc1}$: -117dBm or -116dBm or -115dBm /15kHz depending on operating band<br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: +3dB<br>$\hat{E}s_3$ / $N_{o2c}$: -1dB<br><br>Reported RSRP values:<br>±6dB for normal conditions and ±9dB for extreme conditions | 0dB<br><br><br>0dB<br>0dB<br>0dB<br>0dB<br><br><br>Via mapping<br>Via mapping | $N_{oc1}$: -117dBm or -116dBm or -115dBm /15kHz depending on operating band<br><br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: +3dB<br>$\hat{E}s_3$ / $N_{o2c}$: -1dB<br><br>Cell 1:<br>RSRP_12 to RSRP_27<br>RSRP_13 to RSRP_28<br>RSRP_14 to RSRP_29<br>depending on operating band<br>Cell 2:<br>RSRP_20 to RSRP_35<br>RSRP_21 to RSRP_36<br>RSRP_22 to RSRP_37<br>depending on Cell 1 band |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 1 RSRP from $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$, the uncertainty in Cell 2 RSRP from $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for normal conditions. In all cases the RSRP values are 3dB wider at each end for extreme conditions. | | |

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| 9.1.24.1_1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20MHz + 10MHz (Rel-12 and forward) | $N_{oc1}$: -117dBm or -116dBm or -115dBm /15kHz depending on operating band<br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: +3dB<br>$\hat{E}s_3$ / $N_{o2c}$: -1dB<br><br>Reported RSRP values:<br>±4.5dB for normal conditions and ±9dB for extreme conditions | 0dB<br><br>0dB<br>0dB<br>0dB<br>0dB<br><br>Via mapping<br>Via mapping | $N_{oc1}$: -117dBm or -116dBm or -115dBm /15kHz depending on operating band<br><br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: +3dB<br>$\hat{E}s_3$ / $N_{o2c}$: -1dB<br><br>Cell 1:<br>RSRP_13 to RSRP_26<br>RSRP_14 to RSRP_27<br>RSRP_15 to RSRP_28<br>depending on operating band<br>Cell 2:<br>RSRP_21 to RSRP_34<br>RSRP_22 to RSRP_35<br>RSRP_23 to RSRP_36<br>depending on Cell 1 band |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 1 RSRP from $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$, the uncertainty in Cell 2 RSRP from $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for normal conditions. By using separate calculation for normal and extreme condition, all cases the RSRP values are 4dB wider at each end for extreme conditions. | | |
| 9.1.24.2 TDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 20MHz + 10MHz | $N_{oc1}$: -117dBm or -116dBm or -115dBm /15kHz depending on operating band<br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{oc2}$: +3dB<br>$\hat{E}s_3$ / $N_{oc2}$: -1dB<br><br>Reported relative RSRP values:<br>(Cell 2 – Cell 1):<br>±6dB<br>(Cell 3 – Cell 2):<br>±3dB | 0dB<br><br>0dB<br>0dB<br>0dB<br>+0.8dB<br><br><br><br>Via mapping<br><br>Via mapping | $N_{oc1}$: -117dBm or -116dBm or -115dBm /15kHz depending on operating band<br><br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{oc2}$: +3dB<br>$\hat{E}s_3$ / $N_{oc2}$: -0.2dB<br><br><br>(Cell 2 – Cell 1):<br>RSRP_(x-1) to RSRP_(x+17)<br>(Cell 3 - Cell 2):<br>RSRP_(x-8) to RSRP_(x+1) |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 1 RSRP from $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$, the uncertainty in Cell 2 RSRP from $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the uncertainty in Cell 3 RSRP from $\hat{E}s_3$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRP values given above are for normal and extreme conditions. | | |
| 9.1.25 FDD intra-frequency absolute and relative RSRP accuracies in CRS based discovery signal | Test 1:<br>$N_{oc}$: -106dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +1.0dB<br>Reported absolute RSRP values:<br>Normal condtion: ±4.5dB<br>Extreme condtion: ±9dB<br>Reported relative RSRP values: ±3dB | Test 1:<br>-1.0dB<br>0dB<br>+1.0dB<br>Via mapping | Test 1:<br>$N_{oc}$: -107.0dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br><br>RSRP_29 to RSRP_42<br>RSRP_25 to RSRP_46<br>RSRP_Cell 1 - 9 to RSRP_Cell 1 + 1 |
| | The derivation of the RSRP values takes into account the uncertainty in Cell1 and Cell 2 RSRP from $N_{oc}$, $\hat{E}s_1$ / $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function. | | |
| 9.1.26 TDD intra-frequency absolute and relative RSRP accuracies in CRS based discovery signal | Same as 9.1.25 | Same as 9.1.25 | Same as 9.1.25 |
| | Same as 9.1.25 | | |

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| 9.1.27 FDD—FDD inter-frequency absolute and relative RSRP accuracies in CRS based discovery signal | Test 1:<br>$N_{oc1}$: = $N_{oc2}$ +6dB<br>$N_{oc2}$: -115dBm to -111.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc1}$: 13dB<br>$\hat{E}s_2$ / $N_{oc2}$: -6dB<br><br>Reported absolute RSRP values:<br>Normal condtion: ±4.5dB<br><br><br><br><br><br><br><br><br>Extreme condtion: ±9dB<br><br><br><br><br><br><br><br><br>Reported relative RSRP values:<br>Normal condtion: ±4.5dB<br>Extreme condtion: ±6dB | Test 1:<br>0dB<br>0dB<br><br>0dB<br>+0.8dB<br><br>Via mapping<br><br><br><br><br><br><br><br><br><br>Via mapping<br><br><br><br><br><br><br><br><br><br>Via mapping | Test 1:<br>$N_{oc1}$: = $N_{oc2}$ +6dB<br>$N_{oc2}$: -115dBm to -111.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc1}$: 13dB<br>$\hat{E}s_2$ / $N_{oc2}$: -5.2dB<br><br>Cell 2, absolute:<br>RSRP_14 to RSRP_26<br>RSRP_15 to RSRP_27<br>RSRP_15 to RSRP_27<br>RSRP_16 to RSRP_28<br>RSRP_16 to RSRP_28<br>RSRP_17 to RSRP_29<br>RSRP_18 to RSRP_30<br>depending on operating band<br><br>Cell 2, absolute:<br>RSRP_10 to RSRP_31<br>RSRP_10 to RSRP_31<br>RSRP_11 to RSRP_32<br>RSRP_11 to RSRP_32<br>RSRP_12 to RSRP_33<br>RSRP_13 to RSRP_34<br>RSRP_13 to RSRP_34<br>depending on operating band<br><br>Relative:<br>RSRP_(x-31) to RSRP_(x-18)<br>RSRP_(x-33) to RSRP_(x-17) |
| | The derivation of the RSRP values takes into account the uncertainty from all applicable $N_{oc}$ and $\hat{E}s$ / $N_{oc}$ values, the allowed UE reporting accuracy, and the UE mapping function. Where extreme condition UE accuracies differ from normal condition accuracies, reported values are recalculated. | | |
| 9.1.28 TDD—TDD inter-frequency absolute and relative RSRP accuracies in CRS based discovery signal | Same as 9.1.27 | Same as 9.1.27 | Same as 9.1.27 |
| | Same as 9.1.27 | | |
| 9.1.29 FDD intra frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal | Test 1:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>CRS $\hat{E}s_1$ / $N_{oc}$: 3dB<br>CSI-RS $\hat{E}s_1$ / $N_{oc}$: 9dB<br>CRS $\hat{E}s_2$ / $N_{oc}$: -1dB<br>CSI-RS $\hat{E}s_2$ / $N_{oc}$: 5dB<br>Reported absolute CSI-RSRP values:<br>Normal condtion: ±4.5dB<br>Extreme condtion: ±9dB<br>Reported relative CSI-RSRP values: ±3dB | Test 1:<br>0 dB<br><br>0 dB<br>0 dB<br>0 dB<br>+0.8 dB<br>Via mapping | Test 1:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>CRS $\hat{E}s_1$ / $N_{oc}$: 3dB<br>CSI-RS $\hat{E}s_1$ / $N_{oc}$: 9dB<br>CRS $\hat{E}s_2$ / $N_{oc}$: -1dB<br>CSI-RS $\hat{E}s_2$ / $N_{oc}$: 5.8dB<br><br>CSI-RSRP_24 to CSI-RSRP_43<br>CSI-RSRP_20 to CSI-RSRP_47<br>CSI-RSRP_Cell 1 – 8 to CSI-RSRP_Cell 1 + 1 |
| | The derivation of the CSI-RSRP values takes into account the uncertainty in Cell 1 and Cell 2 CSI-RSRP from $N_{oc}$, CSI-RS $\hat{E}s_1$ / $N_{oc}$ and CSI-RS $\hat{E}s_2$ / $N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function. | | |
| 9.1.30 TDD intra-frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal | Same as 9.1.29 | Same as 9.1.29 | Same as 9.1.29 |

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| 9.1.31 FDD-FDD inter-frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal | $N_{oc1}$: = $N_{oc2}$ +6dB<br>$N_{oc2}$: -115dBm to -111.5dBm /15kHz depending on operating band<br>CSI-RS $\hat{E}s_1$ / $N_{oc1}$: 13dB<br>CSI-RS $\hat{E}s_2$ / $N_{oc2}$: 0dB<br><br>Reported absolute CSI-RSRP values:<br>Normal condtion: ±4.5dB<br><br><br><br><br><br><br><br>Extreme condtion: ±9dB<br><br><br><br><br><br><br><br>Reported relative CSI-RSRP values:<br>Normal condtion: ±4.5dB<br><br><br>Extreme condtion: ±6dB | 0dB<br>0dB<br><br>0dB<br>+0.8dB<br><br><br>Via mapping<br><br><br><br><br><br><br><br>Via mapping<br><br><br><br><br><br><br><br><br>Via mapping<br><br><br>Via mapping | $N_{oc1}$: = $N_{oc2}$ +6dB<br>$N_{oc2}$: -115dBm to -111.5dBm /15kHz depending on operating band<br>CSI-RS $\hat{E}s_1$ / $N_{oc1}$: 13dB<br>CSI-RS $\hat{E}s_2$ / $N_{oc2}$: 0.8dB<br><br>Cell 2:<br>CSI_RSRP_20 to CSI_RSRP_32<br>CSI_RSRP_21 to CSI_RSRP_33<br>CSI_RSRP_21 to CSI_RSRP_33<br>CSI_RSRP_22 to CSI_RSRP_34<br>CSI_RSRP_22 to CSI_RSRP_34<br>CSI_RSRP_23 to CSI_RSRP_35<br>CSI_RSRP_24 to CSI_RSRP_36<br><br>CSI_RSRP_16 to CSI_RSRP_37<br>CSI_RSRP_16 to CSI_RSRP_37<br>CSI_RSRP_17 to CSI_RSRP_38<br>CSI_RSRP_17 to CSI_RSRP_38<br>CSI_RSRP_18 to CSI_RSRP_39<br>CSI_RSRP_19 to CSI_RSRP_40<br>CSI_RSRP_19 to CSI_RSRP_40<br><br>Cell 2 – Cell 1<br>CSI_RSRP_Cell 1 – 25 to CSI_RSRP_Cell 1 – 12<br><br>CSI_RSRP_Cell 1 – 27 to CSI_RSRP_Cell 1 – 11 |
| | The derivation of the CSI-RSRP values takes into account the uncertainty in Cell 1 and Cell 2 CSI-RSRP from $N_{oc}$, CSI-RS $\hat{E}s_1$ / $N_{oc}$ and CSI-RS $\hat{E}s_2$ / $N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function. | | |

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| 9.1.33 FDD absolute and relative RSRP accuracies for E-UTRAN Carrier Aggregation in CRS based discovery signal | Test 1:<br>$N_{oc1}$: -117dBm to -113.5dBm /15kHz depending on operating band<br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{oc2}$: 3dB<br>$\hat{E}s_3$ / $N_{oc2}$: -1dB<br><br>Reported absolute RSRP values:<br>Normal condtion: ±4.5dB<br><br><br><br><br><br><br><br><br><br>Extreme condtion: ±9dB<br><br><br><br><br><br><br><br><br><br>Reported relative RSRP values:<br>(Cell 3 – Cell 2): ±3dB<br>(Cell 3 – Cell 1):<br>Normal condtion: ±4.5dB<br>Extreme condtion: ±6dB | Test 1:<br>0dB<br><br>0dB<br>0dB<br>0dB<br>+0.8dB<br><br>Via mapping<br><br><br><br><br><br><br><br><br><br>Via mapping<br><br><br><br><br><br><br><br><br><br>Via mapping | Test 1:<br>$N_{oc1}$: -117dBm to -113.5dBm /15kHz depending on operating band<br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{oc2}$: 3dB<br>$\hat{E}s_3$ / $N_{oc2}$: -0.2dB<br><br>Cell 3, absolute:<br>RSRP_18 to RSRP_30<br>RSRP_19 to RSRP_31<br>RSRP_19 to RSRP_31<br>RSRP_20 to RSRP_32<br>RSRP_20 to RSRP_32<br>RSRP_21 to RSRP_33<br>RSRP_22 to RSRP_34<br>depending on operating band<br><br>Cell 3, absolute:<br>RSRP_14 to RSRP_35<br>RSRP_14 to RSRP_35<br>RSRP_15 to RSRP_36<br>RSRP_15 to RSRP_36<br>RSRP_16 to RSRP_37<br>RSRP_17 to RSRP_38<br>RSRP_17 to RSRP_38<br>depending on operating band<br><br>Relative:<br>RSRP_(x-8) to RSRP_(x+1)<br><br>RSRP_(x-2) to RSRP_(x+12)<br>RSRP_(x-4) to RSRP_(x+13) |
| | The derivation of the RSRP values takes into account the uncertainty from all applicable $N_{oc}$ and $\hat{E}s$ / $N_{oc}$ values, the allowed UE reporting accuracy, and the UE mapping function. Where extreme condition UE accuracies differ from normal condition accuracies, reported values are recalculated. | | |
| 9.1.34 TDD absolute and relative RSRP accuracies for E-UTRAN Carrier Aggregation in CRS based discovery signal | Same as 9.1.33 | Same as 9.1.33 | Same as 9.1.33 |
| | Same as 9.1.33 | | |

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| 9.1.35 FDD absolute and relative CSI-RSRP accuracies for E-UTRAN Carrier Aggregation in CSI-RS based discovery signal | $N_{oc1}$: -117dBm to -113.5dBm /15kHz depending on operating band<br>$N_{oc2}$: = $N_{oc2}$ +1dB<br>CSI-RS $\hat{E}s_1$ / $N_{oc1}$: 2dB<br>CSI-RS $\hat{E}s_2$ / $N_{oc2}$: 9dB<br>CSI-RS $\hat{E}s_3$ / $N_{oc3}$: 5dB<br><br>Reported absolute CSI-RSRP values:<br>Normal condtion: ±4.5dB<br><br>Extreme condtion: ±9dB<br><br>Reported intra-freq relative CSI-RSRP values:<br>Normal condtion: ±2dB<br><br>Extreme condtion: ±3dB<br><br>Reported inter-freq relative CSI-RSRP values:<br>Normal condtion: ±4.5dB<br><br>Extreme condtion: ±6dB |
| | The derivation of the CSI-RSRP values takes into account the uncertainty in Cell 1 and Cell 2 CSI-RSRP from $N_{oc}$, CSI-RS $\hat{E}s$ / $N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function. |
| 9.1.36 TDD absolute and relative CSI-RSRP accuracies for E-UTRAN Carrier Aggregation in CSI-RS based discovery signal | Same as 9.1.35 |

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| 9.1.37 3DL PCell in FDD RSRP for E-UTRAN in Carrier Aggregation | $N_{oc1}$: -117dBm or -116.5dBm or -116dBm or -115.5dBm or -115 dBm or -114dBm or -113.5dBm /15kHz depending on operating band<br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$N_{oc3}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{oc2}$: +3dB<br>$\hat{E}s_3$ / $N_{oc2}$: -1dB<br>$\hat{E}s_4$ / $N_{oc3}$: +3dB<br>$\hat{E}s_5$ / $N_{oc3}$: -1dB<br><br><u>Reported absolute RSRP values:</u><br>±4.5dB for normal conditions and ±9dB for extreme conditions<br><br><br>

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| 9.1.39 3DL FDD RSRP for E-UTRAN in Carrier Aggregation | $N_{oc1}$: -117dBm or -116dBm or -115.5dBm or -115 dBm or -114dBm or -113.5dBm /15kHz depending on operating band<br>$N_{oc2}$: = $N_{oc1}$ +1dB<br>$N_{oc3}$: = $N_{oc1}$ +1dB<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{oc2}$: +3dB<br>$\hat{E}s_3$ / $N_{oc2}$: -1dB<br>$\hat{E}s_4$ / $N_{oc3}$: +3dB<br>$\hat{E}s_5$ / $N_{oc3}$: -1dB<br><br><u>Reported absolute RSRP values:</u><br>±6dB for normal conditions and ±9dB for extreme conditions<br><br><br>

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| 9.1.41.1 FD-FDD Intra Frequency Absolute RSRP Accuracy for UE category 0 | Test 1:<br>Noc: -106dBm/15kHz<br>Ês1 / Noc: +6.0dB<br>Ês2 / Noc: +1.0dB<br>Reported RSRP values: ±7dB<br><br>Test 2:<br>Noc: -86dBm/15kHz<br>Ês1 / Noc: +6.0dB<br>Ês2 / Noc: +1.0dB<br>Reported RSRP values: ±9dB<br><br>Test 3:<br>Noc: -116dBm to -112.5dBm /15kHz depending on operating band<br>Ês1 / Noc: +3.0dB<br>Ês2 / Noc: -1.0dB<br>Reported RSRP values: ±7dB | Test 1:<br>-1.0dB<br>0dB<br>+1.0dB<br>Via mapping<br><br>Test 2:<br>-1.0dBdB<br>0dB<br>+1.0dB<br>Via mapping<br><br>Test 3:<br>0dB<br><br>0dB<br>+0.8dB<br>Via mapping | Test 1:<br>Noc: -107.0dBm/15kHz<br>Ês1 / Noc: +6.0dB<br>Ês2 / Noc: +2.0dB<br>RSRP_27 to RSRP_44<br><br>Test 2:<br>Noc: -87dBm/15kHz<br>Ês1 / Noc: +6.0dB<br>Ês2 / Noc: +2.0dB<br>RSRP_45 to RSRP_66<br><br>Test 3:<br>Noc: -116dBm to -112.5dBm /15kHz depending on operating band<br>Ês1 / Noc: +3.0dB<br>Ês2 / Noc: -0.2dB<br>RSRP_16 to RSRP_33<br>RSRP_17 to RSRP_34<br>RSRP_18 to RSRP_35<br>RSRP_19 to RSRP_36<br>depending on operating band |
| 9.1.41.2 FD-FDD Intra Frequency Relative RSRP Accuracy for UE category 0 | Test 1:<br>Noc: -106dBm/15kHz<br>Ês1 / Noc: +6.0dB<br>Ês2 / Noc: +1.0dB<br>Reported relative RSRP values: ±4dB<br><br>Test 2:<br>Noc: -86dBm/15kHz<br>Ês1 / Noc: +6.0dB<br>Ês2 / Noc: +1.0dB<br>Reported relative RSRP values: ±4dB<br><br>Test 3:<br>Noc: -116dBm to -112.5dBm /15kHz depending on operating band<br>Ês1 / Noc: +3.0dB<br>Ês2 / Noc: -1.0dB<br>Reported relative RSRP values: ±4dB | Test 1:<br>0 dB<br>0 dB<br>+1.0dB<br>Via mapping<br><br>Test 2:<br>-1.0dB<br>0dB<br>+1.0dB<br>Via mapping<br><br>Test 3:<br>0dB<br><br>0dB<br>+1.0dB<br>Via mapping | Test 1:<br>Noc: -106 dBm/15kHz<br>Ês1 / Noc: +6.0dB<br>Ês2 / Noc: +2.0dB<br>RSRP_x-10 to RSRP_x+2<br><br>Test 2:<br>Noc: -87dBm/15kHz<br>Ês1 / Noc: +6.0dB<br>Ês2 / Noc: +2.0dB<br>RSRP_x-10 to RSRP_x+2<br><br>Test 3:<br>Noc: -116dBm to -112.5dBm /15kHz depending on operating band<br>Ês1 / Noc: +3.0dB<br>Ês2 / Noc: 0dB<br>RSRP_x-9 to RSRP_x+3 |
| 9.1.42.1 HD-FDD Intra Frequency Absolute RSRP Accuracy for UE category 0 | Same as 9.1.41.1 | Same as 9.1.41.1 | Same as 9.1.41.1 |
| 9.1.42.2 HD-FDD Intra Frequency Relative RSRP Accuracy for UE category 0 | Same as 9.1.41.2 | Same as 9.1.41.2 | Same as 9.1.41.2 |
| 9.1.43.1 TDD Intra Frequency Absolute RSRP Accuracy for UE category 0 | Same as 9.1.41.1 | Same as 9.1.41.1 | Same as 9.1.41.1 |
| 9.1.43.2 TDD Intra Frequency Relative RSRP Accuracy for UE category 0 | Same as 9.1.41.2 | Same as 9.1.41.2 | Same as 9.1.41.2 |

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| 9.1.52 FD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeA | Test 1:<br>$N_{oc}$: -106dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +1.0dB<br>Reported absolute RSRP values:<br>Normal conditions: ±7dB Extreme conditions: ±10dB<br>Reported relative RSRP values:<br>Normal and extreme conditions: ±4dB<br><br>Test 2:<br>$N_{oc}$: -86dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +1.0dB<br>Reported absolute RSRP values:<br>Normal conditions: ±9dB<br>Extreme conditions: ±12dB<br>Reported relative RSRP values:<br>Normal and extreme conditions: ±4dB<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5 dBm/15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: +3.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -1.0dB<br>Reported absolute RSRP values:<br>Normal conditions: ±7dB<br><br>Extreme conditions: ±10dB<br><br>Reported relative RSRP values:<br>Normal and extreme conditions: ±4dB | Test 1:<br>-1.0dB<br>0dB<br>+1.0dB<br><br>Via mapping<br><br><br><br>Test 2:<br>-1.0dB<br>0dB<br>+1.0dB<br><br>Via mapping<br><br><br><br>Test 3:<br>0dB<br><br>0dB<br>+0.8dB<br>Via mapping | Test 1:<br>$N_{oc}$: -107dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br><br>RSRP_27 to RSRP_44<br>RSRP_24 to RSRP_47<br><br>RSRP_Cell 1 – 10 to RSRP_Cell 1 + 2<br><br>Test 2:<br>$N_{oc}$: -87dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +6.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +2.0dB<br><br>RSRP_45 to RSRP_66<br>RSRP_52 to RSRP_69<br><br>RSRP_Cell 1 – 10 to RSRP_Cell 1 + 2<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5 dBm/15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: +3.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -0.2dB<br>RSRP_16 to RSRP_33<br>RSRP_16 to RSRP_33<br>RSRP_17 to RSRP_34<br>RSRP_17 to RSRP_34<br>RSRP_18 to RSRP_35<br>RSRP_19 to RSRP_36<br>RSRP_19 to RSRP_36<br><br>RSRP_13 to RSRP_36<br>RSRP_13 to RSRP_36<br>RSRP_14 to RSRP_37<br>RSRP_14 to RSRP_37<br>RSRP_15 to RSRP_38<br>RSRP_16 to RSRP_39<br>RSRP_16 to RSRP_39<br><br>RSRP_Cell 1 – 9 to RSRP_Cell 1 + 2 |
| | The derivation of the RSRP values takes into account the uncertainty in Cell 1 and Cell 2 RSRP from $N_{oc}$, $\hat{E}s_1$ / $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The absolute RSRP values given above are different for normal and extreme conditions.<br>The relative RSRP values given above are for both normal and extreme conditions. | | |
| 9.1.53 HD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeA | Same as 9.1.52 | Same as 9.1.52 | Same as 9.1.52 |
| 9.1.54 TDD RSRP Intra frequency case for Cat-M1 UE in CEModeA | Same as 9.1.52 | Same as 9.1.52 | Same as 9.1.52 |

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| 9.1.57 FD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeB | Test 1:<br>$N_{oc}$ = -99dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -12dB<br>$\hat{E}s_2$ / $N_{oc}$: -14dB<br><br>Reported absolute RSRP values:<br>Normal condtion: ±8dB<br>Extreme condtion: ±11dB<br><br>Reported relative RSRP values:<br>Normal condtion: ±5dB<br>Extreme condtion: ±5dB<br><br>Test 2:<br>$N_{oc}$ = -79dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -12dB<br>$\hat{E}s_2$ / $N_{oc}$: -14dB<br><br>Reported absolute RSRP values:<br>Normal condtion: ±10dB<br>Extreme condtion: ±13dB<br><br>Reported relative RSRP values:<br>Normal condtion: ±5dB<br>Extreme condtion: ±5dB<br><br>Test 3:<br>$N_{oc}$ = -107dBm~-103.5dBm/15kHz<br>depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: -12dB<br>$\hat{E}s_2$ / $N_{oc}$: -14dB<br><br>Reported absolute RSRP values:<br>Normal condtion: ±8dB<br><br>Extreme condtion: ±11dB<br><br>Reported relative RSRP values:<br>Normal condtion: ±5dB<br>Extreme condtion: ±5dB | Test 1:<br>-0.3dB<br>0dB<br>+0.1dB<br><br>Via mapping<br><br>Via mapping<br><br>Test 2:<br>-0.3dB<br>0dB<br>+0.1dB<br><br>Via mapping<br><br>Via mapping<br><br>Test 3:<br>0dB<br><br>0dB<br>+0.1dB<br><br>Via mapping<br><br>Via mapping<br><br>Via mapping | Test 1:<br>$N_{oc}$ = -99.3dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -12dB<br>$\hat{E}s_2$ / $N_{oc}$: -13.9dB<br><br>Cell 2, absolute:<br>RSRP_18 to RSRP_37<br>RSRP_15 to RSRP_40<br><br>Relative:<br>RSRP_(x-9) to RSRP_(x+5)<br>RSRP_(x-9) to RSRP_(x+5)<br><br>Test 2:<br>$N_{oc}$ = -79.3dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -12dB<br>$\hat{E}s_2$ / $N_{oc}$: -13.9dB<br><br>Cell 2, absolute:<br>RSRP_36 to RSRP_59<br>RSRP_33 to RSRP_62<br><br>Relative:<br>RSRP_(x-9) to RSRP_(x+5)<br>RSRP_(x-9) to RSRP_(x+5)<br><br>Test 3:<br>$N_{oc}$ = -107dBm~-103.5dBm/15kHz<br>depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: -12dB<br>$\hat{E}s_2$ / $N_{oc}$: -13,9dB<br><br>Cell 2, absolute:<br>RSRP_10 to RSRP_29<br>RSRP_11 to RSRP_30<br>RSRP_11 to RSRP_30<br>RSRP_12 to RSRP_31<br>RSRP_12 to RSRP_31<br>RSRP_13 to RSRP_32<br>RSRP_14 to RSRP_33<br>depending on operating band<br><br>Cell 2, absolute:<br>RSRP_7 to RSRP_32<br>RSRP_8 to RSRP_33<br>RSRP_8 to RSRP_33<br>RSRP_9 to RSRP_34<br>RSRP_9 to RSRP_34<br>RSRP_10 to RSRP_35<br>RSRP_11 to RSRP_36<br>depending on operating band<br><br>Relative:<br>RSRP_(x-9) to RSRP_(x+5)<br>RSRP_(x-9) to RSRP_(x+5) |
| | The derivation of the RSRP values takes into account the uncertainty from all applicable $N_{oc}$ and $\hat{E}s$ / $N_{oc}$ values, the allowed UE reporting accuracy, and the UE mapping function. Where extreme condition UE accuracies differ from normal condition accuracies, reported values are recalculated. | | |
| 9.1.58 HD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeB | Same as 9.1.57 | Same as 9.1.57 | Same as 9.1.57 |
| | Same as 9.1.57 | | |
| 9.1.59 TDD RSRP Intra frequency case for Cat-M1 UE in CEModeB | Same as 9.1.57 | Same as 9.1.57 | Same as 9.1.57 |

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| | Same as 9.1.57 | | |
| 9.2.1.1 FDD Intra Frequency Absolute RSRQ Accuracy | Test 1:<br>$N_{oc}$: -84.76dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +3.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +3.0dB<br>Reported RSRQ values: ±2.5dB<br><br>Test 2:<br>$N_{oc}$: -103.85dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -2.9dB<br>$\hat{E}s_2$ / $N_{oc}$: -2.9dB<br>Reported RSRQ values: ±3.5dB<br><br>Test 3:<br>$N_{oc}$: -116dBm or -114dBm or -113dBm or -115dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: -4.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -4.0dB<br>Reported RSRQ values: ±3.5dB | Test 1:<br>-0.75dB<br>0dB<br>0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>0dB<br>Via mapping<br><br>Test 3:<br>0dB<br><br><br>+0.4dB<br>+0.4dB<br>Via mapping | Test 1:<br>$N_{oc}$: -85.51Bm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +3.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +3.0dB<br>RSRQ_04 to RSRQ_16<br><br>Test 2:<br>$N_{oc}$: -103.85dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -2.9dB<br>$\hat{E}s_2$ / $N_{oc}$: -2.9dB<br>RSRQ_00 to RSRP_14<br><br>Test 3:<br>$N_{oc}$: -116dBm or -114dBm or -113dBm or -115dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: -3.6dB<br>$\hat{E}s_2$ / $N_{oc}$: -3.6dB<br>RSRQ_00 to RSRQ_14 |
| | The derivation of the RSRQ values takes into account the uncertainty in Cell 2 RSRQ from $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRQ values given above are for normal conditions. For test 1 the RSRQ values are 1.5dB wider at each end for extreme conditions, and for tests 2 and 3 the RSRQ values are 0.5dB wider at each end for extreme conditions. | | |
| 9.2.2.1 TDD Intra Frequency Absolute RSRQ Accuracy | Test 1:<br>$N_{oc}$: -84.76dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +3.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +3.0dB<br>Reported RSRQ values: ±2.5dB<br><br>Test 2:<br>$N_{oc}$: -103.85dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -2.9dB<br>$\hat{E}s_2$ / $N_{oc}$: -2.9dB<br>Reported RSRQ values: ±3.5dB<br><br>Test 3:<br>$N_{oc}$: -116dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -4.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -4.0dB<br>Reported RSRQ values: ±3.5dB | Test 1:<br>-0.75dB<br>0dB<br>0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>0dB<br>Via mapping<br><br>Test 3:<br>0dB<br>+0.4dB<br>+0.4dB<br>Via mapping | Test 1:<br>$N_{oc}$: -85.51Bm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +3.0dB<br>$\hat{E}s_2$ / $N_{oc}$: +3.0dB<br>RSRQ_04 to RSRQ_16<br><br>Test 2:<br>$N_{oc}$: -103.85dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -2.9dB<br>$\hat{E}s_2$ / $N_{oc}$: -2.9dB<br>RSRQ_00 to RSRQ_14<br><br>Test 3:<br>$N_{oc}$: -116dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -3.6dB<br>$\hat{E}s_2$ / $N_{oc}$: -3.6dB<br>RSRQ_00 to RSRQ_14 |
| | The derivation of the RSRQ values takes into account the uncertainty in Cell 2 RSRQ from $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRQ values given above are for normal conditions. For test 1 the RSRQ values are 1.5dB wider at each end for extreme conditions, and for test 2 the RSRQ values are 0.5dB wider at each end for extreme conditions. | | |

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| 9.2.3.1 FDD - FDD Inter Frequency Absolute RSRQ Accuracy | Test 1:<br>$N_{oc1}$: -80dBm/15kHz<br>$N_{oc2}$: -80dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -1.75dB<br>$\hat{E}s_2$ / $N_{oc2}$: -1.75dB<br>Reported RSRQ values:<br>±2.5dB for normal conditions and ±4dB for extreme conditions<br><br>Test 2:<br>$N_{oc1}$: -104.7dBm/15kHz<br>$N_{oc2}$: -104.7dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -4dB<br>Reported RSRQ values:<br>±3.5dB for normal conditions and ±4dB for extreme conditions<br><br>Test 3:<br>$N_{oc1}$: -119.5dBm to -116dBm/15kHz depending on operating band<br>$N_{oc2}$: -119.5dBm to -116dBm/15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -4dB<br><br>Reported RSRQ values:<br>±3.5dB for normal conditions and ±4dB for extreme conditions | Test 1:<br>0dB<br>-1.1dB<br>0dB<br>0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>0dB<br>0.8dB<br>Via mapping<br><br>Test 3:<br>0dB<br><br>0.3dB<br><br>0dB<br>0.8dB<br><br>Via mapping | Test 1:<br>$N_{oc1}$: -80dBm/15kHz<br>$N_{oc2}$: -81.1dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -1.75dB<br>$\hat{E}s_2$ / $N_{oc2}$: -1.75dB<br><br>RSRQ_04 to RSRQ_16 (NTC)<br>RSRQ_01 to RSRQ_19 (ETC)<br><br>Test 2:<br>$N_{oc1}$: -104.7dBm/15kHz<br>$N_{oc2}$: -104.7dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -3.2dB<br><br>RSRQ_00 to RSRQ_16 (NTC)<br>RSRQ_00 to RSRQ_17 (ETC)<br><br>Test 3:<br>$N_{oc1}$: -119.5dBm to-116 dBm /15kHz depending on operating band<br>$N_{oc2}$: -119.2dBm -115.7dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -3.2dB<br><br>RSRQ_00 to RSRQ_16 (NTC)<br>RSRQ_00 to RSRQ_17 (ETC) |
| | The derivation of the RSRQ values takes into account the uncertainty in Cell 2 RSRQ from $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function. | | |
| 9.2.3.2 FDD - FDD Inter Frequency Relative RSRQ Accuracy | Test 1:<br>$N_{oc1}$: -80dBm/15kHz<br>$N_{oc2}$: -80dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -1.75dB<br>$\hat{E}s_2$ / $N_{oc2}$: -1.75dB<br>Reported Relative RSRQ values:<br>±3dB for normal conditions and ±4dB for extreme conditions<br><br>Test 2:<br>$N_{oc1}$: -104.7dBm/15kHz<br>$N_{oc2}$: -104.7dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -4dB<br>Reported Relative RSRQ values:<br>±4dB<br><br>Test 3:<br>$N_{oc1}$: -119.5dBm to -116dBm/15kHz depending on operating band<br>$N_{oc2}$: -119.5dBm to -116dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -4dB<br><br>Reported Relative RSRQ values:<br>±4dB | Test 1:<br>-1.1dB<br>-1.1dB<br>0dB<br>0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>0dB<br>0.8dB<br>Via mapping<br><br>Test 3:<br>0dB<br><br>+0.3dB<br><br>0dB<br>0.8dB<br><br>Via mapping | Test 1:<br>$N_{oc1}$: -81.1dBm/15kHz<br>$N_{oc2}$: -81.1dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -1.75dB<br>$\hat{E}s_2$ / $N_{oc2}$: -1.75dB<br><br>RSRQ_x - 8 to RSRQ_x + 8 (NTC)<br>RSRQ_x - 10 to RSRQ_x + 10 (ETC)<br><br>Test 2:<br>$N_{oc1}$: -104.7dBm/15kHz<br>$N_{oc2}$: -104.7dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -3.2dB<br><br>RSRQ_x - 9 to RSRQ_x + 11<br><br>Test 3:<br>$N_{oc1}$: -119.5dBm to -116dBm/15kHz depending on operating band<br>$N_{oc2}$: -119.2dBm to -115.7dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -3.2dB<br><br>RSRQ_x - 9 to RSRQ_x + 11 |
| | The derivation of the relative RSRQ values takes into account the uncertainty in Cell 1 RSRQ from $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$ and Cell 2 RSRQ from $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function. | | |
| 9.2.4.1 TDD - TDD Inter Frequency Absolute RSRQ Accuracy | Same as 9.2.3.1 | | |

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| 9.2.4.2 TDD - TDD Inter Frequency Relative RSRQ Accuracy | Same as 9.2.3.2 | | |
| 9.2.4A.1 FDD - TDD Inter Frequency Absolute RSRQ Accuracy | Test 1:<br>$N_{oc1}$: -80dBm/15kHz<br>$N_{oc2}$: -80dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -1.75dB<br>$\hat{E}s_2$ / $N_{oc2}$: -1.75dB<br>Reported RSRQ values:<br>±2.5dB for normal conditions and ±4dB for extreme conditions<br><br>Test 2:<br>$N_{oc1}$: -104.7dBm/15kHz<br>$N_{oc2}$: -104.7dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4.0dB<br>$\hat{E}s_2$ / $N_{o2c}$: -4.0dB<br>Reported RSRQ values:<br>±3.5dB for normal conditions and ±4dB for extreme conditions<br><br>Test 3:<br>$N_{oc1}$: -114.5dBm/15kHz<br>$N_{oc2}$: -114.5dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -4dB<br><br>Reported RSRQ values:<br>±3.5dB for normal conditions and ±4dB for extreme conditions | Test 1:<br>0dB<br>-1.10dB<br>0dB<br>0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>0dB<br>0.8dB<br>Via mapping<br><br>Test 3:<br>0dB<br>0dB<br>0dB<br>0.8dB<br><br>Via mapping | Test 1:<br>$N_{oc1}$: -80dBm/15kHz<br>$N_{oc2}$: -81.10dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -1.75dB<br>$\hat{E}s_2$ / $N_{oc2}$: -1.75dB<br><br>RSRQ_04 to RSRQ_16 (NTC)<br>RSRQ_01 to RSRQ_19 (ETC)<br><br>Test 2:<br>$N_{oc1}$: -104.7dBm/15kHz<br>$N_{oc2}$: -104.7dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -3.2dB<br><br>RSRQ_00 to RSRQ_16 (NTC)<br>RSRQ_00 to RSRQ_17 (ETC)<br><br>Test 3:<br>$N_{oc1}$: -114.5dBm/15kHz<br>$N_{oc2}$: -114.5dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -3.2dB<br><br>RSRQ_00 to RSRQ_16 (NTC)<br>RSRQ_00 to RSRQ_17 (ETC) |
| | The derivation of the RSRQ values takes into account the uncertainty in Cell 2 RSRQ from $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function. | | |
| 9.2.4A.2 FDD - TDD Inter Frequency Relative Accuracy of RSRQ | Test 1:<br>$N_{oc1}$: -80dBm/15kHz<br>$N_{oc2}$: -80dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -1.75dB<br>$\hat{E}s_2$ / $N_{oc2}$: -1.75dB<br>Reported Relative RSRQ values:<br>±3dB for normal conditions and ±4dB for extreme conditions<br><br>Test 2:<br>$N_{oc1}$: -104.7dBm/15kHz<br>$N_{oc2}$: -104.7dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -4dB<br>Reported Relative RSRQ values:<br>±4dB<br><br>Test 3:<br>$N_{oc1}$: -114.5dBm/15kHz<br>$N_{oc2}$: -114.5dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -4dB<br><br>Reported Relative RSRQ values:<br>±4dB | Test 1:<br>-1.10dB<br>-1.10dB<br>0dB<br>0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>0dB<br>0.8dB<br>Via mapping<br><br>Test 3:<br>0dB<br>0dB<br>0dB<br>0.8dB<br><br>Via mapping | Test 1:<br>$N_{oc1}$: -81.10dBm/15kHz<br>$N_{oc2}$: -81.10dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -1.75dB<br>$\hat{E}s_2$ / $N_{oc2}$: -1.75dB<br><br>RSRQ_x - 8 to RSRQ_x + 8 (NTC)<br>RSRQ_x - 10 to RSRQ_x + 10 (ETC)<br><br>Test 2:<br>$N_{oc1}$: -104.7dBm/15kHz<br>$N_{oc2}$: -104.7dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -3.2dB<br><br>RSRQ_x - 10 to RSRQ_x + 10<br><br>Test 3:<br>$N_{oc1}$: -114.5dBm/15kHz<br>$N_{oc2}$: -114.5dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -3.2dB<br><br>RSRQ_x - 10 to RSRQ_x + 10 |
| | The derivation of the relative RSRQ values takes into account the uncertainty in Cell 1 RSRQ from $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$ and Cell 2 RSRQ from $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function. | | |

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| 9.2.5.1 FDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation | $N_{oc1}$: -119.5dBm or -118.5dBm or -118dBm or -117.5 dBm or -116.5dBm or -116dBm /15kHz depending on operating band<br>$N_{oc2}$: -116dBm or -115dBm or -114.5dBm or -114 dBm or -113dBm or -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -4dB<br>$\hat{E}s_3$ / $N_{o2c}$: -4dB<br><br><u>Reported RSRQ values:</u><br>±3.5dB for normal conditions and ±4dB for extreme conditions | 0dB<br><br>0dB<br><br>0dB<br>+0.3dB<br>0dB<br><br>Via mapping<br><br>Via mapping | $N_{oc1}$: -119.5dBm or -118.5dBm or -118dBm or -117.5 dBm or -116.5dBm or -116dBm /15kHz depending on operating band<br>$N_{oc2}$: -116dBm or -115dBm or -114.5dBm or -114 dBm or -113dBm or -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -3.7dB<br>$\hat{E}s_3$ / $N_{o2c}$: -4dB<br><br><u>Cell 1:</u><br>RSRQ_00 to RSRQ_15 (NTC)<br>RSRQ_00 to RSRQ_16 (ETC)<br><u>Cell 2:</u><br>RSRQ_00 to RSRQ_14 (NTC)<br>RSRQ_00 to RSRQ_15 (ETC) |
| | The derivation of the RSRQ values takes into account the uncertainty in Cell 1 RSRQ from $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$, the uncertainty in Cell 2 RSRQ from $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function. | | |
| 9.2.5.2 FDD Relative RSRQ Accuracy E-UTRA for Carrier Aggregation | $N_{oc1}$: -119.5dBm or -118.5dBm or -118dBm or -117.5 dBm or -116.5dBm or -116dBm /15kHz depending on operating band<br>$N_{oc2}$: -116dBm or -115dBm or -114.5dBm or -114 dBm or -113dBm or -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -4dB<br>$\hat{E}s_3$ / $N_{o2c}$: -4dB<br><br><u>Reported RSRQ values:</u><br>±4dB | 0dB<br><br>0dB<br><br>0dB<br>+0.3dB<br>0dB<br><br>Via mapping | $N_{oc1}$: -119.5dBm or -118.5dBm or -118dBm or -117.5 dBm or -116.5dBm or -116dBm /15kHz depending on operating band<br>$N_{oc2}$: -116dBm or -115dBm or -114.5dBm or -114 dBm or -113dBm or -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -3.7dB<br>$\hat{E}s_3$ / $N_{o2c}$: -4dB<br><br>RSRQ_x - 12 to RSRQ_x + 9 (NTC and ETC) |
| 9.2.6.1 TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation | Same as 9.2.5.1 | Same as 9.2.5.1 | Same as 9.2.5.1 |
| 9.2.6.2 TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation | Same as 9.2.5.2 | Same as 9.2.5.2 | Same as 9.2.5.2 |
| 9.2.7.1 FDD RSRQ under Time Domain Measurement Resource Restriction with Non-MBSFN ABS | <u>Test 1:</u><br>$N_{oc}$: -84.76dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -2.0dB<br>Reported RSRQ values: ±2.5dB<br><br><u>Test 2:</u><br>$N_{oc}$: -103.85dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -2.0dB<br>Reported RSRQ values: ±2.5dB<br><br><u>Test 3:</u><br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -4.0dB<br>Reported RSRQ values: ±3.5dB | <u>Test 1:</u><br>-1.0dB<br>0dB<br>+0.8dB<br>Via mapping<br><br><u>Test 2:</u><br>0dB<br>0dB<br>+0.8dB<br>Via mapping<br><br><u>Test 3:</u><br>0dB<br><br>0dB<br>+0.95dB<br>Via mapping | <u>Test 1:</u><br>$N_{oc}$: -85.76Bm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -1.2dB<br>RSRQ_04 to RSRQ_16<br><br><u>Test 2:</u><br>$N_{oc}$: -103.85dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -1.2dB<br>RSRQ_04 to RSRQ_16<br><br><u>Test 3:</u><br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: +5.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -3.05dB<br>RSRQ_00 to RSRQ_16 |
| | The derivation of the RSRQ values takes into account the uncertainty in Cell 2 RSRQ from $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRQ values given above are for normal conditions. For tests 1 and 2 the RSRQ values are 1.5dB wider at each end for extreme conditions, and for test 3 the RSRQ values are 0.5dB wider at each end for extreme conditions. | | |

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| 9.2.8.1 TDD RSRQ under Time Domain Measurement Resource Restriction with Non-MBSFN ABS | Subset of 9.2.7.1 | Same as 9.2.7.1 | Subset of 9.2.7.1 |
| 9.2.9.1 FDD Absolute RSRQ under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) | Test 1:<br>$N_{oc}$: -84.76dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +5.0dB<br>$\hat{E}s_2 / N_{oc}$: -2.0dB<br>Reported RSRQ values: ±2.5dB<br><br>Test 2:<br>$N_{oc}$: -103.85dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +5.0dB<br>$\hat{E}s_2 / N_{oc}$: -2.0dB<br>Reported RSRQ values: ±2.5dB<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc}$: +5.0dB<br>$\hat{E}s_2 / N_{oc}$: -4.0dB<br>Reported RSRQ values: ±3.5dB | Test 1:<br>-1.0dB<br>0dB<br>+0.8dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>+0.8dB<br>Via mapping<br><br>Test 3:<br>0dB<br><br><br>0dB<br>+0.95dB<br>Via mapping | Test 1:<br>$N_{oc}$: -85.76Bm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +5.0dB<br>$\hat{E}s_2 / N_{oc}$: -1.2dB<br>RSRQ_04 to RSRQ_16<br><br>Test 2:<br>$N_{oc}$: -103.85dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +5.0dB<br>$\hat{E}s_2 / N_{oc}$: -1.2dB<br>RSRQ_04 to RSRQ_16<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc}$: +5.0dB<br>$\hat{E}s_2 / N_{oc}$: -3.05dB<br>RSRQ_00 to RSRQ_16 |
| | The derivation of the RSRQ values takes into account the uncertainty in Cell 2 RSRQ from $N_{oc}$ and $\hat{E}s_2 / N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRQ values given above are for normal conditions. For tests 1 and 2 the RSRQ values are 1.5dB wider at each end for extreme conditions, and for test 3 the RSRQ values are 0.5dB wider at each end for extreme conditions. | | |
| 9.2.10.1 TDD Absolute RSRQ under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) | Subset of 9.2.9.1 | Same as 9.2.9.1 | Subset of 9.2.9.1 |
| 9.2.11.1 TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz | Same as 9.2.5.1 | Same as 9.2.5.1 | Same as 9.2.5.1 |
| 9.2.11.2 TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz | Same as 9.2.5.2 | Same as 9.2.5.2 | Same as 9.2.5.2 |
| 9.2.12.1 TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz | Same as 9.2.5.1 | Same as 9.2.5.1 | Same as 9.2.5.1 |
| 9.2.12.2 TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz | Same as 9.2.5.2 | Same as 9.2.5.2 | Same as 9.2.5.2 |

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| 9.2.15.1 FDD RSRQ Accuracy under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | Test 1:<br>$N_{oc}$: -84.76dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +4.0dB<br>$\hat{E}s_2 / N_{oc}$: +2.0dB<br>$\hat{E}s_3 / N_{oc}$: -1.5dB<br>Reported RSRQ values: ±2.5dB<br><br>Test 2:<br>$N_{oc}$: -103.85dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +4.0dB<br>$\hat{E}s_2 / N_{oc}$: +2.0dB<br>$\hat{E}s_3 / N_{oc}$: -1.5dB<br>Reported RSRQ values: ±2.5dB<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc}$: +4.0dB<br>$\hat{E}s_2 / N_{oc}$: +2.0dB<br>$\hat{E}s_3 / N_{oc}$: -4.0dB<br>Reported RSRQ values: ±3.5dB | Test 1:<br>0dB<br>0dB<br>0dB<br>+1.0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>0dB<br>+1.0dB<br>Via mapping<br><br>Test 3:<br>0dB<br><br><br>0dB<br>0dB<br>+1.0dB<br>Via mapping | Test 1:<br>$N_{oc}$: -84.76Bm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +4.0dB<br>$\hat{E}s_2 / N_{oc}$: +2.0dB<br>$\hat{E}s_3 / N_{oc}$: -0.5dB<br>RSRQ_05 to RSRQ_17<br><br>Test 2:<br>$N_{oc}$: -103.85dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +4.0dB<br>$\hat{E}s_2 / N_{oc}$: +2.0dB<br>$\hat{E}s_3 / N_{oc}$: -0.5dB<br>RSRQ_05 to RSRQ_17<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc}$: +4.0dB<br>$\hat{E}s_2 / N_{oc}$: +2.0dB<br>$\hat{E}s_3 / N_{oc}$: -3.0dB<br>RSRQ_00 to RSRQ_16 |
| | The derivation of the RSRQ values takes into account the uncertainty in Cell 3 RSRQ from $\hat{E}s_1 / N_{oc}$, $\hat{E}s_2 / N_{oc}$, and $\hat{E}s_3 / N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRQ values given above are for normal conditions. For tests 1 and 2 the RSRQ values are 1.5dB wider at each end for extreme conditions, and for test 3 the RSRQ values are 0.5dB wider at each end for extreme conditions. | | |
| 9.2.16.1 TDD RSRQ Accuracy under Time Domain Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | Subset of 9.2.15.1 | Same as 9.2.15.1 | Subset of 9.2.15.1 |
| 9.2.17.1 FDD Intra Frequency Absolute RSRQ Accuracy for 5MHz Bandwidth | Test 1:<br>$N_{oc}$: -81.76dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +3.0dB<br>$\hat{E}s_2 / N_{oc}$: +3.0dB<br>Reported RSRQ values: ±2.5dB<br><br>Test 2:<br>$N_{oc}$: -100.85dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: -2.9dB<br>$\hat{E}s_2 / N_{oc}$: -2.9dB<br>Reported RSRQ values: ±3.5dB<br><br>Test 3:<br>$N_{oc}$: -109.5dBm /15kHz<br>$\hat{E}s_1 / N_{oc}$: -4.0dB<br>$\hat{E}s_2 / N_{oc}$: -4.0dB<br>Reported RSRQ values: ±3.5dB | Test 1:<br>-1.05dB<br>0dB<br>0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>0dB<br>Via mapping<br><br>Test 3:<br>0dB<br>+0.4dB<br>+0.4dB<br>Via mapping | Test 1:<br>$N_{oc}$: -82.81Bm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +3.0dB<br>$\hat{E}s_2 / N_{oc}$: +3.0dB<br>RSRQ_04 to RSRQ_16<br><br>Test 2:<br>$N_{oc}$: -100.85dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: -2.9dB<br>$\hat{E}s_2 / N_{oc}$: -2.9dB<br>RSRQ_00 to RSRP_14<br><br>Test 3:<br>$N_{oc}$: -109.5dBm /15kHz<br>$\hat{E}s_1 / N_{oc}$: -3.6dB<br>$\hat{E}s_2 / N_{oc}$: -3.6dB<br>RSRQ_00 to RSRQ_14 |
| | The derivation of the RSRQ values takes into account the uncertainty in Cell 2 RSRQ from $N_{oc}$ and $\hat{E}s_2 / N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRQ values given above are for normal conditions. For test 1 the RSRQ values are 1.5dB wider at each end for extreme conditions, and for tests 2 and 3 the RSRQ values are 0.5dB wider at each end for extreme conditions. | | |

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| 9.2.18.1 FDD - FDD Inter Frequency Absolute RSRQ Accuracy for 5MHz Bandwidth | Test 1:<br>$N_{oc1}$: -77dBm/15kHz<br>$N_{oc2}$: -77dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -1.75dB<br>$\hat{E}s_2$ / $N_{oc2}$: -1.75dB<br>Reported RSRQ values:<br>±2.5dB for normal conditions and ±4dB for extreme conditions<br><br>Test 2:<br>$N_{oc1}$: -101.7dBm/15kHz<br>$N_{oc2}$: -101.7dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -4dB<br>Reported RSRQ values:<br>±3.5dB for normal conditions and ±4dB for extreme conditions<br><br>Test 3:<br>$N_{oc1}$: -119.5dBm to -116dBm/15kHz depending on operating band<br>$N_{oc2}$: 113dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -4dB<br><br>Reported RSRQ values:<br>±3.5dB for normal conditions and ±4dB for extreme conditions | Test 1:<br>0dB<br>-1.1dB<br>0dB<br>0dB<br>Via mapping<br><br><br><br>Test 2:<br>0dB<br>0dB<br>0dB<br>0.8dB<br>Via mapping<br><br><br><br>Test 3:<br>0dB<br><br><br>0.3dB<br>0dB<br>0.8dB<br><br><br>Via mapping | Test 1:<br>$N_{oc1}$: -77dBm/15kHz<br>$N_{oc2}$: -78.1dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -1.75dB<br>$\hat{E}s_2$ / $N_{oc2}$: -1.75dB<br><br>RSRQ_04 to RSRQ_16 (NTC)<br>RSRQ_01 to RSRQ_19 (ETC)<br><br>Test 2:<br>$N_{oc1}$: -101.7dBm/15kHz<br>$N_{oc2}$: -101.7dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -3.2dB<br><br>RSRQ_00 to RSRQ_16 (NTC)<br>RSRQ_00 to RSRQ_17 (ETC)<br><br>Test 3:<br>$N_{oc1}$: -119.5dBm to -116dBm /15kHz depending on operating band<br>$N_{oc2}$: -112.7dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -3.2dB<br><br>RSRQ_00 to RSRQ_16 (NTC)<br>RSRQ_00 to RSRQ_17 (ETC) |
| | The derivation of the RSRQ values takes into account the uncertainty in Cell 2 RSRQ from $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function. | | |
| 9.2.18.2 FDD - FDD Inter Frequency Relative RSRQ Accuracy for 5MHz Bandwidth | Test 1:<br>$N_{oc1}$: -77dBm/15kHz<br>$N_{oc2}$: -77dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -1.75dB<br>$\hat{E}s_2$ / $N_{oc2}$: -1.75dB<br>Reported Relative RSRQ values:<br>±3dB for normal conditions and ±4dB for extreme conditions<br><br><br><br>Test 2:<br>$N_{oc1}$: -101.7dBm/15kHz<br>$N_{oc2}$: -101.7dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -4dB<br>Reported Relative RSRQ values:<br>±4dB<br><br>Test 3:<br>$N_{oc1}$: -119.5dBm to -116dBm/15kHz depending on operating band<br>$N_{oc2}$: 113dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -4dB<br>Reported Relative RSRQ values:<br>±4dB | Test 1:<br>-1.1dB<br>-1.1dB<br>0dB<br>0dB<br>Via mapping<br><br><br><br><br><br>Test 2:<br>0dB<br>0dB<br>0dB<br>0.8dB<br>Via mapping<br><br><br><br>Test 3:<br>0dB<br><br><br>+0.3dB<br>0dB<br>0.8dB<br>Via mapping | Test 1:<br>$N_{oc1}$: -78.1dBm/15kHz<br>$N_{oc2}$: -78.1dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -1.75dB<br>$\hat{E}s_2$ / $N_{oc2}$: -1.75dB<br><br>RSRQ_x - 8 to RSRQ_x + 8 (NTC)<br>RSRQ_x - 10 to RSRQ_x + 10 (ETC)<br><br>Test 2:<br>$N_{oc1}$: -101.7dBm/15kHz<br>$N_{oc2}$: -101.7dBm/15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -3.2dB<br><br>RSRQ_x - 9 to RSRQ_x + 11<br><br>Test 3:<br>$N_{oc1}$: -119.5dBm to -116dBm/15kHz depending on operating band<br>$N_{oc2}$: -112.7dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -3.2dB<br><br>RSRQ_x - 9 to RSRQ_x + 11 |
| | The derivation of the relative RSRQ values takes into account the uncertainty in Cell 1 RSRQ from $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$ and Cell 2 RSRQ from $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function. | | |
| 9.2.19.1 FDD-FDD Inter Frequency absolute WB-RSRQ accuracy | FFS | FFS | FFS |
| 9.2.20.1 TDD-TDD Inter Frequency absolute WB-RSRQ accuracy | FFS | FFS | FFS |

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| 9.2.21.1 FDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz | $N_{oc1}$: -119.5dBm or -118.5dBm or -118dBm or -117.5 dBm or -116.5dBm or -116dBm /15kHz depending on operating band<br>$N_{oc2}$: -116dBm or -115dBm or -114.5dBm or -114 dBm or -113dBm or -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc1}$: -4dB<br>$\hat{E}s_2 / N_{o2c}$: -4dB<br>$\hat{E}s_3 / N_{o2c}$: -4dB<br><br>Reported RSRQ values:<br>±3.5dB for normal conditions and ±4dB for extreme conditions | 0dB<br><br>0dB<br><br>0dB<br>+0.3dB<br>0dB<br><br>Via mapping<br><br><br>Via mapping | $N_{oc1}$: -119.5dBm or -118.5dBm or -118dBm or -117.5 dBm or -116.5dBm or -116dBm /15kHz depending on operating band<br>$N_{oc2}$: -116dBm or -115dBm or -114.5dBm or -114 dBm or -113dBm or -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc1}$: -4dB<br>$\hat{E}s_2 / N_{o2c}$: -3.7dB<br>$\hat{E}s_3 / N_{o2c}$: -4dB<br><br>Cell 1:<br>RSRQ_00 to RSRQ_15 (NTC)<br>RSRQ_00 to RSRQ_16 (ETC)<br>Cell 2:<br>RSRQ_00 to RSRQ_14 (NTC)<br>RSRQ_00 to RSRQ_15 (ETC) |
| | The derivation of the RSRQ values takes into account the uncertainty in Cell 1 RSRQ from $N_{oc1}$ and $\hat{E}s_1 / N_{oc1}$, the uncertainty in Cell 2 RSRQ from $N_{oc2}$ and $\hat{E}s_2 / N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function. | | |
| 9.2.21.2 FDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz | $N_{oc1}$: -119.5dBm or -118.5dBm or -118dBm or -117.5 dBm or -116.5dBm or -116dBm /15kHz depending on operating band<br>$N_{oc2}$: -116dBm or -115dBm or -114.5dBm or -114 dBm or -113dBm or -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc1}$: -4dB<br>$\hat{E}s_2 / N_{o2c}$: -4dB<br>$\hat{E}s_3 / N_{o2c}$: -4dB<br><br>Reported RSRQ values:<br>±4dB | 0dB<br><br>0dB<br><br>0dB<br>+0.3dB<br>0dB<br><br>Via mapping | $N_{oc1}$: -119.5dBm or -118.5dBm or -118dBm or -117.5 dBm or -116.5dBm or -116dBm /15kHz depending on operating band<br>$N_{oc2}$: -116dBm or -115dBm or -114.5dBm or -114 dBm or -113dBm or -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc1}$: -4dB<br>$\hat{E}s_2 / N_{o2c}$: -3.7dB<br>$\hat{E}s_3 / N_{o2c}$: -4dB<br><br>RSRQ_x - 12 to RSRQ_x + 9 (NTC and ETC) |
| | The derivation of the RSRQ values takes into account the uncertainty in Cell 1 RSRQ from $N_{oc1}$ and $\hat{E}s_1 / N_{oc1}$, the uncertainty in Cell 2 RSRQ from $N_{oc2}$ and $\hat{E}s_2 / N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function. | | |
| 9.2.22.1 TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz | Same as 9.2.21.1 | Same as 9.2.21.1 | Same as 9.2.21.1 |
| 9.2.22.2 TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz | Same as 9.2.21.2 | Same as 9.2.21.2 | Same as 9.2.21.2 |
| 9.2.23.1 FDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 5MHz+5MHz | $N_{oc1}$: -119.5dBm or -118.5dBm or -118dBm or -117.5 dBm or -116.5dBm or -116dBm /15kHz depending on operating band<br>$N_{oc2}$: -116dBm or -115dBm or -114.5dBm or -114 dBm or -113dBm or -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc1}$: -4dB<br>$\hat{E}s_2 / N_{oc2}$: -4dB<br>$\hat{E}s_3 / N_{oc2}$: -4dB<br><br>Reported RSRQ values:<br>±3.5dB for normal conditions and ±4dB for extreme conditions | 0dB<br><br>0dB<br><br>0dB<br>+0.3dB<br>0dB<br><br>Via mapping<br><br><br>Via mapping | $N_{oc1}$: -119.5dBm or -118.5dBm or -118dBm or -117.5 dBm or -116.5dBm or -116dBm /15kHz depending on operating band<br>$N_{oc2}$: -116dBm or -115dBm or -114.5dBm or -114 dBm or -113dBm or -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc1}$: -4dB<br>$\hat{E}s_2 / N_{oc2}$: -3.7dB<br>$\hat{E}s_3 / N_{oc2}$: -4dB<br><br>Cell 1:<br>RSRQ_00 to RSRQ_15 (NTC)<br>RSRQ_00 to RSRQ_16 (ETC)<br>Cell 2:<br>RSRQ_00 to RSRQ_14 (NTC)<br>RSRQ_00 to RSRQ_15 (ETC) |

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| | The derivation of the RSRQ values takes into account the uncertainty in Cell 1 RSRQ from $N_{oc1}$ and $\hat{E}s_1 / N_{oc1}$, the uncertainty in Cell 2 RSRQ from $N_{oc2}$ and $\hat{E}s_2 / N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function. | | |
| 9.2.23.2 FDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 5MHz+5MHz | $N_{oc1}$: -119.5dBm or -118.5dBm or -118dBm or -117.5 dBm or -116.5dBm or -116dBm /15kHz depending on operating band<br>$N_{oc2}$: -116dBm or -115dBm or -114.5dBm or -114 dBm or -113dBm or -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc1}$: -4dB<br>$\hat{E}s_2 / N_{oc2}$: -4dB<br>$\hat{E}s_3 / N_{oc2}$: -4dB<br><br><u>Reported RSRQ values:</u><br>±4dB | 0dB<br><br><br><br>0dB<br><br><br><br>0dB<br>+0.3dB<br>0dB<br><br>Via mapping | $N_{oc1}$: -119.5dBm or -118.5dBm or -118dBm or -117.5 dBm or -116.5dBm or -116dBm /15kHz depending on operating band<br>$N_{oc2}$: -116dBm or -115dBm or -114.5dBm or -114 dBm or -113dBm or -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc1}$: -4dB<br>$\hat{E}s_2 / N_{oc2}$: -3.7dB<br>$\hat{E}s_3 / N_{oc2}$: -4dB<br><br>RSRQ_x - 12 to RSRQ_x + 9 (NTC and ETC) |
| 9.2.24.1 TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 5MHz+5MHz | Same as 9.2.23.1 | Same as 9.2.23.1 | Same as 9.2.23.1 |
| | The derivation of the RSRQ values takes into account the uncertainty in Cell 1 RSRQ from $N_{oc1}$ and $\hat{E}s_1 / N_{oc1}$, the uncertainty in Cell 2 RSRQ from $N_{oc2}$ and $\hat{E}s_2 / N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function. | | |
| 9.2.24.2 TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 5MHz+5MHz | Same as 9.2.23.2 | Same as 9.2.23.2 | Same as 9.2.23.2 |
| 9.2.25.1 Absolute RSRQ Accuracy for E-UTRAN TDD-FDD Carrier Aggregation with PCell in FDD | <u>Same as 9.2.38 Cell 1 and Cell 2, absolute RSRQ part</u> | <u>Same as 9.2.38 Cell 1 and Cell 2</u> | <u>Same as 9.2.38 Cell 1 and Cell 2, absolute RSRQ part</u> |
| 9.2.25.2 Relative RSRQ Accuracy for E-UTRAN TDD-FDD Carrier Aggregation with PCell in FDD | <u>Same as 9.2.38 Cell 1 and Cell 2, absolute RSRQ part</u> | <u>Same as 9.2.38 Cell 1 and Cell 2</u> | <u>Same as 9.2.38 Cell 1 and Cell 2, absolute RSRQ part</u> |
| 9.2.26.1 Absolute RSRQ Accuracy for E-UTRAN TDD-FDD Carrier Aggregation with PCell in TDD | <u>Same as 9.2.39 Cell 1 and Cell 2, absolute RSRQ part</u> | <u>Same as 9.2.39 Cell 1 and Cell 2</u> | <u>Same as 9.2.39 Cell 1 and Cell 2, absolute RSRQ part</u> |
| 9.2.26.2 Relative RSRQ Accuracy for E-UTRAN TDD-FDD Carrier Aggregation with PCell in TDD | <u>Same as 9.2.39 Cell 1 and Cell 2, absolute RSRQ part</u> | <u>Same as 9.2.39 Cell 1 and Cell 2</u> | <u>Same as 9.2.39 Cell 1 and Cell 2, absolute RSRQ part</u> |
| 9.2.27.1 TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz+10MHz | $N_{oc1}$: -119.5dBm or -118.5dBm or -117.5 dBm /15kHz depending on operating band<br>$N_{oc2}$: -116dBm or -115dBm or -114 dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc1}$: -4dB<br>$\hat{E}s_2 / N_{o2c}$: -4dB<br>$\hat{E}s_3 / N_{o2c}$: -4dB<br><br><u>Reported RSRQ values:</u><br>±3.5dB for normal conditions and ±4dB for extreme conditions | 0dB<br><br><br>0dB<br><br><br>0dB<br>+0.3dB<br>0dB<br><br>Via mapping<br><br><br><br>Via mapping | $N_{oc1}$: -119.5dBm or -118.5dBm or -117.5 dBm /15kHz depending on operating band<br>$N_{oc2}$: -116dBm or -115dBm or -114 dBm /15kHz depending on operating band<br><br>$\hat{E}s_1 / N_{oc1}$: -4dB<br>$\hat{E}s_2 / N_{o2c}$: -3.7dB<br>$\hat{E}s_3 / N_{o2c}$: -4dB<br><br><u>Cell 1:</u><br>RSRQ_00 to RSRQ_15 (NTC)<br>RSRQ_00 to RSRQ_16 (ETC)<br><u>Cell 2:</u><br>RSRQ_00 to RSRQ_14 (NTC)<br>RSRQ_00 to RSRQ_15 (ETC) |

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| | The derivation of the RSRQ values takes into account the uncertainty in Cell 1 RSRQ from $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$, the uncertainty in Cell 2 RSRQ from $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function. | | |
| 9.2.27.2 TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz+10MHz | $N_{oc1}$: -119.5dBm or -118.5dBm or -117.5 dBm /15kHz depending on operating band<br>$N_{oc2}$: -116dBm or -115dBm or -114 dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -4dB<br>$\hat{E}s_3$ / $N_{o2c}$: -4dB<br><br>Reported RSRQ values:<br>±4dB | 0dB<br><br>0dB<br><br>0dB<br>+0.3dB<br>0dB<br><br>Via mapping | $N_{oc1}$: -119.5dBm or -118.5dBm or -117.5 dBm /15kHz depending on operating band<br>$N_{oc2}$: -116dBm or -115dBm or -114 dBm /15kHz depending on operating band<br><br>$\hat{E}s_1$ / $N_{oc1}$: -4dB<br>$\hat{E}s_2$ / $N_{o2c}$: -3.7dB<br>$\hat{E}s_3$ / $N_{o2c}$: -4dB<br><br>RSRQ_x - 12 to RSRQ_x + 9 (NTC and ETC) |
| | The derivation of the RSRQ values takes into account the uncertainty in Cell 1 RSRQ from $N_{oc1}$ and $\hat{E}s_1$ / $N_{oc1}$, the uncertainty in Cell 2 RSRQ from $N_{oc2}$ and $\hat{E}s_2$ / $N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function. | | |
| 9.2.28 FDD intra-frequency absolute RSRQ accuracy with CRS based discovery signal | Test 1:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: -4.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -4.0dB<br>Reported RSRQ values: ±3.5dB | Test 1:<br>0dB<br><br>+0.4dB<br>+0.4dB<br>Via mapping | Test 1:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc}$: -3.6dB<br>$\hat{E}s_2$ / $N_{oc}$: -3.6dB<br>RSRQ_00 to RSRQ_14 |
| | The derivation of the RSRQ values takes into account the uncertainty in Cell 2 RSRQ from $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRQ values given above are for normal conditions. For test 1 the RSRQ values are 0.5dB wider at each end for extreme conditions. | | |
| 9.2.29 TDD intra-frequency absolute RSRQ accuracy with CRS based discovery signal | Test 1:<br>$N_{oc}$: -116dBm to -114dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -4.0dB<br>$\hat{E}s_2$ / $N_{oc}$: -4.0dB<br>Reported RSRQ values: ±3.5dB | Test 1:<br>0dB<br>+0.4dB<br>+0.4dB<br>Via mapping | Test 1:<br>$N_{oc}$: -116dBm to -114dBm /15kHz<br>$\hat{E}s_1$ / $N_{oc}$: -3.6dB<br>$\hat{E}s_2$ / $N_{oc}$: -3.6dB<br>RSRQ_00 to RSRQ_14 |
| | The derivation of the RSRQ values takes into account the uncertainty in Cell 2 RSRQ from $N_{oc}$ and $\hat{E}s_2$ / $N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRQ values given above are for normal conditions. For test 1 the RSRQ values are 0.5dB wider at each end for extreme conditions. | | |
| 9.2.38 3DL PCell in FDD RSRQ for E-UTRAN in Carrier Aggregation | $N_{oc1}$: -119.5dBm to -116dBm /15kHz depending on operating band<br>$N_{oc2}$: -116dBm to -114dBm /15kHz depending on operating band<br>$N_{oc3}$: -116dBm to -114dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc1}$: -6dB<br>$\hat{E}s_2$ / $N_{oc2}$: -6dB<br>$\hat{E}s_3$ / $N_{oc3}$: -6dB<br><br>Reported absolute RSRQ values:<br>±3.5dB for normal conditions and ±4dB for extreme conditions<br><br>Reported relative RSRQ values:<br>(Cell 2 – Cell 1) and (Cell 3 – Cell 1):<br>±4dB for normal conditions<br>±4dB for extreme conditions | 0.3dB<br><br>0dB<br><br>0dB<br><br>+0.8dB<br>+0.8dB<br>+0.8dB<br><br>Via mapping<br><br>Via mapping | $N_{oc1}$: -119.2dBm to -115.7dBm /15kHz depending on operating band<br><br>$N_{oc2}$: -116dBm to -114dBm /15kHz depending on operating band<br>$N_{oc3}$: -116dBm to -114dBm /15kHz depending on operating band<br>$\hat{E}s_1$ / $N_{oc1}$: -5.20dB<br>$\hat{E}s_2$ / $N_{oc2}$: -5.20dB<br>$\hat{E}s_3$ / $N_{oc3}$: -5.20dB<br><br>Absolute:<br>Cell 1, Cell 2, and Cell 3 RSRQ_0 to RSRQ_14<br><br>Relative:<br>(Cell 2 – Cell 1) and (Cell 3 – Cell 1): RSRQ_(x-10) to RSRQ_(x+10) |
| | The derivation of the RSRQ values takes into account the uncertainty from all applicable $N_{oc}$ and $\hat{E}s$ / $N_{oc}$ values, the allowed UE reporting accuracy, and the UE mapping function. Where extreme condition UE accuracies differ from normal condition accuracies, reported values are recalculated. | | |

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| 9.2.30 FDD-FDD inter-frequency absolute and relative RSRQ accuracies with CRS based discovery signal | Test 1:<br>$N_{oc1}$: -117.5dBm to -114dBm/15kHz depending on operating band<br>$N_{oc2}$: -117.5dBm to -114dBm/15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc1}$: -6dB<br>$\hat{E}s_2 / N_{o2c}$: -6dB<br><br>Reported absolute RSRQ values:<br>±3.5dB for normal conditions and ±4dB for extreme conditions<br>Reported absolute RSRQ values:<br>±4dB for both normal conditions and extreme conditions | Test 1:<br>0dB<br><br>0.3dB<br><br>0.8dB<br>0.8dB<br><br>Via mapping | Test 1:<br>$N_{oc1}$: -117.5dBm to-114 dBm /15kHz depending on operating band<br>$N_{oc2}$: -117.2dBm -113.7dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc1}$: -5.2dB<br>$\hat{E}s_2 / N_{o2c}$: -5.2dB<br><br><br>RSRQ_00 to RSRQ_14 (NTC)<br>RSRQ_00 to RSRQ_15 (ETC)<br><br>RSRQ_x - 10 to RSRQ_x + 10 |
| | The derivation of the absolute RSRQ values takes into account the uncertainty in Cell 2 RSRQ from $N_{oc2}$ and $\hat{E}s_2 / N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The derivation of the relative RSRQ values takes into account the uncertainty in Cell 1 RSRQ from $N_{oc1}$ and $\hat{E}s_1 / N_{oc1}$ and Cell 2 RSRQ from $N_{oc2}$ and $\hat{E}s_2 / N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function. | | |
| 9.2.31 TDD-TDD inter-frequency absolute and relative RSRQ accuracies with CRS based discovery signal | Test 1:<br>$N_{oc1}$: -117.5dBm to -115.5dBm/15kHz depending on operating band<br>$N_{oc2}$: -117.5dBm to -115.5dBm/15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc1}$: -6dB<br>$\hat{E}s_2 / N_{o2c}$: -6dB<br><br>Reported absolute RSRQ values:<br>±3.5dB for normal conditions and ±4dB for extreme conditions<br>Reported absolute RSRQ values:<br>±4dB for both normal conditions and extreme conditions | Test 1:<br>0dB<br><br>0.3dB<br><br>0.8dB<br>0.8dB<br><br>Via mapping | Test 1:<br>$N_{oc1}$: -117.5dBm to-115.5 dBm /15kHz depending on operating band<br>$N_{oc2}$: -117.2dBm -115.2dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc1}$: -5.2dB<br>$\hat{E}s_2 / N_{o2c}$: -5.2dB<br><br><br>RSRQ_00 to RSRQ_14 (NTC)<br>RSRQ_00 to RSRQ_15 (ETC)<br><br>RSRQ_x - 10 to RSRQ_x + 10 |
| | The derivation of the absolute RSRQ values takes into account the uncertainty in Cell 2 RSRQ from $N_{oc2}$ and $\hat{E}s_2 / N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The derivation of the relative RSRQ values takes into account the uncertainty in Cell 1 RSRQ from $N_{oc1}$ and $\hat{E}s_1 / N_{oc1}$ and Cell 2 RSRQ from $N_{oc2}$ and $\hat{E}s_2 / N_{oc2}$, the allowed UE reporting accuracy, and the UE mapping function. | | |
| 9.2.39 3DL PCell in TDD RSRQ for E-UTRAN in Carrier Aggregation | Same as 9.2.38 | Same as 9.2.38 | Same as 9.2.38 |
| 9.2.40 3DL FDD RSRQ for E-UTRA Carrier Aggregation | Same as 9.2.38 | Same as 9.2.38 | Same as 9.2.38 |
| 9.2.41 3DL TDD RSRQ for E-UTRA Carrier Aggregation | Same as 9.2.38 | Same as 9.2.38 | Same as 9.2.38 |
| 9.2.42.1 FD-FDD Intra Frequency Absolute RSRQ Accuracy for UE category 0 | Test 1:<br>$N_{oc}$: -84.76dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +3.0dB<br>$\hat{E}s_2 / N_{oc}$: +3.0dB<br>Reported RSRQ values: ±3.5dB<br><br>Test 2:<br>$N_{oc}$: -103.85dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: -2.9dB<br>$\hat{E}s_2 / N_{oc}$: -2.9dB<br>Reported RSRQ values: ±4.5dB<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc}$: -4.0dB<br>$\hat{E}s_2 / N_{oc}$: -4.0dB<br>Reported RSRQ values: ±4.5dB | Test 1:<br>-1.05dB<br>0dB<br>0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>0dB<br>Via mapping<br><br>Test 3:<br>0dB<br><br><br>+0.4dB<br>+0.4dB<br>Via mapping | Test 1:<br>$N_{oc}$: -85.81Bm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +3.0dB<br>$\hat{E}s_2 / N_{oc}$: +3.0dB<br>RSRQ_02 to RSRQ_18<br><br>Test 2:<br>$N_{oc}$: -103.85dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: -2.9dB<br>$\hat{E}s_2 / N_{oc}$: -2.9dB<br>RSRQ_00 to RSRP_16<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc}$: -3.6dB<br>$\hat{E}s_2 / N_{oc}$: -3.6dB<br>RSRQ_00 to RSRQ_16 |

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| | The derivation of the RSRQ values takes into account the uncertainty in Cell 2 RSRQ from $N_{oc}$ and $\hat{E}s_2 / N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRQ values given above are for normal conditions. For test 1 the RSRQ values are 1.5dB wider at each end for extreme conditions, and for tests 2 and 3 the RSRQ values are 0.5dB wider at each end for extreme conditions. | | |
| 9.2.43.1 HD-FDD Intra Frequency Absolute RSRQ Accuracy for UE category 0 | Same as 9.2.42.1 | Same as 9.2.42.1 | Same as 9.2.42.1 |
| 9.2.44.1 TDD Intra Frequency Absolute RSRQ Accuracy for UE category 0 | Test 1:<br>$N_{oc}$: -84.76dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +3.0dB<br>$\hat{E}s_2 / N_{oc}$: +3.0dB<br>Reported RSRQ values: ±3.5dB<br><br>Test 2:<br>$N_{oc}$: -103.85dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: -2.9dB<br>$\hat{E}s_2 / N_{oc}$: -2.9dB<br>Reported RSRQ values: ±4.5dB<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc}$: -4.0dB<br>$\hat{E}s_2 / N_{oc}$: -4.0dB<br>Reported RSRQ values: ±4.5dB | Test 1:<br>-1.05dB<br>0dB<br>0dB<br>Via mapping<br><br>Test 2:<br>0dB<br>0dB<br>0dB<br>Via mapping<br><br>Test 3:<br>0dB<br><br><br>+0.4dB<br>+0.4dB<br>Via mapping | Test 1:<br>$N_{oc}$: -85.81Bm/15kHz<br>$\hat{E}s_1 / N_{oc}$: +3.0dB<br>$\hat{E}s_2 / N_{oc}$: +3.0dB<br>RSRQ_02 to RSRQ_18<br><br>Test 2:<br>$N_{oc}$: -103.85dBm/15kHz<br>$\hat{E}s_1 / N_{oc}$: -2.9dB<br>$\hat{E}s_2 / N_{oc}$: -2.9dB<br>RSRQ_00 to RSRP_16<br><br>Test 3:<br>$N_{oc}$: -116dBm to -112.5dBm /15kHz depending on operating band<br>$\hat{E}s_1 / N_{oc}$: -3.6dB<br>$\hat{E}s_2 / N_{oc}$: -3.6dB<br>RSRQ_00 to RSRQ_16 |
| | The derivation of the RSRQ values takes into account the uncertainty in Cell 2 RSRQ from $N_{oc}$ and $\hat{E}s_2 / N_{oc}$, the allowed UE reporting accuracy, and the UE mapping function.<br>The RSRQ values given above are for normal conditions. For test 1 the RSRQ values are 1.5dB wider at each end for extreme conditions, and for test 2 the RSRQ values are 0.5dB wider at each end for extreme conditions. | | |
| 9.3.1 E-UTRAN FDD - UTRA FDD CPICH RSCP absolute accuracy | Test 1:<br>E-UTRA Cell 1<br>Noc: -98.00dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -60.00dBm/3.84MHz<br>Ior / Ioc: +9.54dB<br>CPICH_Ec/Ior: -10.00dB<br>Reported CPICH_RSCP values: ±8dB<br><br>Test 2:<br>E-UTRA Cell 1<br>Noc: -98.00dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -94.46dBm or -92.46dBm or -90.96dBm or -91.46dBm or -93.46dBm /3.84MHz depending on operating band<br>Ior / Ioc: -9.54dB<br>CPICH_Ec/Ior: -10.00dB<br>Reported CPICH_RSCP values: ±6dB | Test 1:<br><br>0dB<br>0dB<br><br>-0.75dB<br>0dB<br>0dB<br>Via mapping<br><br><br>Test 2:<br><br>0dB<br>0dB<br><br><br><br><br>0.7dB<br><br>0.35dB<br>0dB<br>Via mapping | Test 1:<br>E-UTRA Cell 1<br>Noc: -98.00dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -60.75dBm/3.84MHz<br>Ior / Ioc: +9.54dB<br>CPICH_Ec/Ior: -10.00dB<br>CPICH_RSCP_46 to CPICH_RSCP_63<br><br>Test 2:<br>E-UTRA Cell 1<br>Noc: -98.00dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -93.76dBm or -91.76dBm or -90.26dBm or -90.76dBm or -92.76dBm /3.84MHz depending on operating band<br>Ior / Ioc: -9.19dB<br>CPICH_Ec/Ior: -10.00dB<br>CPICH_RSCP_-04 to CPICH_RSCP_9<br>CPICH_RSCP_-02 to CPICH_RSCP_11<br>CPICH_RSCP_-01 to CPICH_RSCP_12<br>CPICH_RSCP_-03 to CPICH_RSCP_10<br>depending on operating band |

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| | The derivation of the CPICH_RSCP values takes into account the uncertainty in Cell 2 CPICH_RSCP from Ioc, Ior / Ioc and CPICH_Ec/Ior, the allowed UE reporting accuracy, and the UE mapping function.<br>The CPICH_RSCP values given above are for normal conditions. In all cases the CPICH_RSCP values are 3dB wider at each end for extreme conditions. | | |
| 9.3.2 E-UTRAN TDD - UTRA FDD CPICH RSCP absolute accuracy | Same as 9.3.1 | Same as 9.3.1 | Same as 9.3.1 |
| 9.3.3 E UTRAN FDD UTRA FDD CPICH RSCP absolute accuracy for 5MHz bandwidth | Same as 9.3.1 | Same as 9.3.1 | Same as 9.3.1 |
| 9.4.1 E-UTRAN FDD – UTRA FDD CPICH Ec/No absolute accuracy | Test 1:<br>E-UTRA Cell 1<br>Noc: -98.00dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -52.22dBm/3.84MHz<br>Ior / Ioc: -1.75dB<br>CPICH_Ec/Ior: -10.00dB<br>Reported CPICH_Ec/Io accuracy values: ±1.5dB for normal conditions and ±3dB for extreme conditions<br><br>Test 2:<br>E-UTRA Cell 1<br>Noc: -98.00dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -87.27dBm/3.84MHz<br>Ior / Ioc: -4.7dB<br>CPICH_Ec/Ior: -10.00dB<br>Reported CPICH_Ec/Io accuracy values: ±2dB for normal conditions and ±3dB for extreme conditions<br><br>Test 3:<br>E-UTRA Cell 1<br>Noc: -98.00dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -94.46dBm or -92.46dBm or 90.96dBm or -91.46dBm or -93.46dBm/3.84MHz depending on operating band<br>Ior / Ioc: -9.54dB<br>CPICH_Ec/Ior: -10.00dB<br>Reported CPICH_Ec/Io accuracy values: -4.2 to +3dB for normal conditions and extreme conditions. The additional 1.2 dB relaxation to the lower limit for both normal and extreme condition is to take into account the effect of thermal noise and noise generated in the receiver as specified in A.9.4.1.3 of TS 36.133 [4] | Test 1:<br><br>0dB<br>-0dB<br><br>-0.9dB<br>0.3dB<br>0dB<br>Via mapping<br><br>Test 2:<br><br>0dB<br>0dB<br><br>0dB<br>0.3dB<br>0dB<br>Via mapping<br><br>Test 3:<br><br>0dB<br>0dB<br><br>0.7dB<br><br><br><br>0.4dB<br>0dB<br>Via mapping | Test 1:<br>E-UTRA Cell 1<br>Noc: -98.00dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -53.12dBm/3.84MHz<br>Ior / Ioc: -1.45dB<br>CPICH_Ec/Ior: -10.00dB<br>CPICH_Ec/Io_17 to CPICH_Ec/Io_24 for normal conditions.<br>CPICH_Ec/Io_14 to CPICH_Ec/Io_27 for extreme conditions<br><br>Test 2:<br>E-UTRA Cell 1<br>Noc: -98.00dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -87.27dBm/3.84MHz<br>Ior / Ioc: -4.4dB<br>CPICH_Ec/Ior: -10.00dB<br>CPICH_Ec/Io_13 to CPICH_Ec/Io_22 for normal conditions.<br>CPICH_Ec/Io_11 to CPICH_Ec/Io_24 for extreme conditions<br><br>Test 3:<br>E-UTRA Cell 1<br>Noc: -98.00dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -93.76dBm or -91.76dBm or 90.26dBm or -90.76dBm or -92.76dBm /3.84MHz depending on operating band<br>Ior / Ioc: -9.14dB<br>CPICH_Ec/Ior: -10.00dB<br>CPICH_Ec/Io_0 to CPICH_Ec/Io_16 for normal and extreme conditions. |
| 9.4.2 E-UTRAN TDD – UTRA FDD CPICH Ec/No absolute accuracy | Same as 9.4.1 | Same as 9.4.1 | Same as 9.4.1 |
| 9.4.3 E-UTRAN FDD - UTRA FDD CPICH Ec/No absolute accuracy for 5MHz bandwidth | Same as 9.4.1 | Same as 9.4.1 | Same as 9.4.1 |

| | | | |
|---|---|---|---|
| 9.5.1 E-UTRAN FDD – UTRA TDD P-CCPCH RSCP absolute accuracy | Test 1:<br>E-UTRA Cell 1<br>Noc: -98.00dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -54.1dBm/1.28MHz<br>Ior / Ioc: 2.0dB<br>PCCPCH_$E_C/I_{or}$: -3dB<br>DwPCH_$E_C/I_{or}$: 0dB<br>Reported PCCPCH RSCP accuracy values: ±8dB for normal conditions and ±11dB for extreme conditions<br><br>Test 2:<br>E-UTRA Cell 1<br>Noc: -98.00dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -75.2dBm/1.28MHz<br>Ior / Ioc: 5.0dB<br>PCCPCH_$E_C/I_{or}$: -3dB<br>DwPCH_$E_C/I_{or}$: 0dB<br>Reported PCCPCH RSCP accuracy values: ±8dB for normal conditions and ±11dB for extreme conditions<br><br>Test 3:<br>E-UTRA Cell 1<br>Noc: -98.00dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -97.0dBm/1.28MHz<br>Ior / Ioc: 0dB<br>PCCPCH_$E_C/I_{or}$: -3dB<br>DwPCH_$E_C/I_{or}$: 0dB<br>Reported PCCPCH RSCP accuracy values: ±6dB for normal conditions and ±9dB for extreme conditions | Test 1:<br><br>0dB<br>0dB<br><br>-0.8dB<br>0dB<br>0dB<br>0dB<br><br>Via mapping<br><br><br><br><br>Test 2:<br><br>0dB<br>0dB<br><br>0dB<br>0dB<br>0dB<br>0dB<br>Via mapping<br><br><br><br><br>Test 3:<br><br>0dB<br>0dB<br><br>0.8dB<br>0dB<br>0dB<br>0dB<br>Via mapping | Test 1:<br>E-UTRA Cell 1<br>Noc: -98.00dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -54.9dBm/1.28MHz<br>Ior / Ioc: 2.0dB<br>PCCPCH_$E_C/I_{or}$: -3dB<br>DwPCH_$E_C/I_{or}$: 0dB<br>PCCPCH RSCP 51 to PCCPCH RSCP 68 for normal conditions. PCCPCH RSCP 48 to PCCPCH RSCP 71 for extreme conditions<br><br>Test 2:<br>E-UTRA Cell 1<br>Noc: -98.00dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -75.2dBm/1.28MHz<br>Ior / Ioc: 5.0dB<br>PCCPCH_$E_C/I_{or}$: -3dB<br>DwPCH_$E_C/I_{or}$: 0dB<br>PCCPCH RSCP 34 to PCCPCH RSCP 51 for normal conditions. PCCPCH RSCP 31 to PCCPCH RSCP 54 for extreme conditions<br><br>Test 3:<br>E-UTRA Cell 1<br>Noc: -98.00dBm/15kHz<br>Ês / Noc: +4.00dB<br>UTRA Cell 2<br>Ioc: -96.2dBm/1.28MHz<br>Ior / Ioc: 0dB<br>PCCPCH_$E_C/I_{or}$: -3dB<br>DwPCH_$E_C/I_{or}$: 0dB<br>PCCPCH RSCP 10 to PCCPCH RSCP 23 for normal conditions. PCCPCH RSCP 07 to PCCPCH RSCP 26 for extreme conditions |
| 9.5.2 E-UTRAN TDD – UTRA TDD P-CCPCH RSCP absolute accuracy | Same as 9.5.1 | Same as 9.5.1 | Same as 9.5.1 |

| | | | | |
|---|---|---|---|---|
| 9.6.1 GSM RSSI accuracy for E-UTRAN FDD | Subtest 1:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -38.5dBm<br>GSM Cell 3<br>Signal level: -38.5dBm | | Subtest 1:<br>E-UTRA Cell 1<br>0 dB<br>0 dB<br>GSM Cell 2<br>-0.2 dB<br>GSM Cell 3<br>0 dB | Subtest 1:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -38.7dBm<br>GSM Cell 3<br>Signal level: -38.5dBm |
| | Subtest 2:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -48.5dBm<br>GSM Cell 3<br>Signal level: -48.5dBm | | Subtest 2:<br>E-UTRA Cell 1<br>0 dB<br>0 dB<br>GSM Cell 2<br>-0.2 dB<br>GSM Cell 3<br>-1.5 dB | Subtest 2:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -48.7dBm<br>GSM Cell 3<br>Signal level: -50.0dBm |
| | Subtest 3:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -70.5dBm<br>GSM Cell 3<br>Signal level: -70.5dBm | | Subtest 3:<br>E-UTRA Cell 1<br>0 dB<br>0 dB<br>GSM Cell 2<br>-0.2 dB<br>GSM Cell 3<br>0 dB | Subtest 3:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -70.7dBm<br>GSM Cell 3<br>Signal level: -70.5dBm |
| | Subtest 4:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -109.5dBm<br>GSM Cell 3<br>Signal level: -109.5dBm | | Subtest 4:<br>E-UTRA Cell 1<br>0 dB<br>0 dB<br>GSM Cell 2<br>+0.2 dB<br>GSM Cell 3<br>0 dB | Subtest 4:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -109.3dBm<br>GSM Cell 3<br>Signal level: -109.5dBm |
| | Subtest 5:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -57.5dBm<br>GSM Cell 3<br>Signal level: -54.5dBm | | Subtest 5:<br>E-UTRA Cell 1<br>0 dB<br>0 dB<br>GSM Cell 2<br>0.dB<br>GSM Cell 3<br>0 dB | Subtest 5:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -57.5dBm<br>GSM Cell 3<br>Signal level: -54.5dBm |
| | Subtest 6:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -64.5dBm<br>GSM Cell 3<br>Signal level: -59.5dBm | | Subtest 6:<br>E-UTRA Cell 1<br>0 dB<br>0 dB<br>GSM Cell 2<br>0dB<br>GSM Cell 3<br>0 dB | Subtest 6:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -64.5dBm<br>GSM Cell 3<br>Signal level: -59.5dBm |
| | Subtest 7:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -71.5dBm<br>GSM Cell 3<br>Signal level: -64.5dBm | | Subtest 7:<br>E-UTRA Cell 1<br>0 dB<br>0 dB<br>GSM Cell 2<br>0dB<br>GSM Cell 3<br>0 dB | Subtest 7:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -71.5dBm<br>GSM Cell 3<br>Signal level: -64.5dBm |
| | Subtest 8:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -78.5dBm<br>GSM Cell 3 | | Subtest 8:<br>E-UTRA Cell 1<br>0 dB<br>0 dB<br>GSM Cell 2<br>0dB<br>GSM Cell 3 | Subtest 8:<br>E-UTRA Cell 1<br>$N_{oc}$: -98dBm/15kHz<br>Ês / $N_{oc}$: +4dB<br>GSM Cell 2<br>Signal level: -78.5dBm<br>GSM Cell 3<br>Signal level: -69.5dBm |

| | | | |
|---|---|---|---|
| 9.6.2 GSM RSSI accuracy for E-UTRAN TDD | Same as 9.6.1 | Same as 9.6.1 | Same as 9.6.1 |
| 9.9.1.1 FDD Intra Frequency Serving Cell Absolute RSRP Accuracy | Noc: -122dBm to -118.5dBm /15kHz<br>depending on operating band<br>Ês1 / Noc: -4.0dB<br>Reported RSRP values: ±6dB | 1.6dB<br><br>0dB<br>Via mapping | Noc: -120.4dBm to -116.9dBm /15kHz<br>depending on operating band<br>Ês1 / Noc: -4.0dB<br>RSRP_9 to RSRP_24<br>RSRP_10 to RSRP_25<br>RSRP_10 to RSRP_25<br>RSRP_11 to RSRP_26<br>RSRP_12 to RSRP_27<br>RSRP_12 to RSRP_27<br>depending on operating band |
| 9.9.1.2 FDD Intra Frequency Serving Cell Absolute RSRQ Accuracy | $N_{oc}$: -122dBm to -118.5dBm /15kHz<br>depending on operating band<br>$Ês_1$ / $N_{oc}$: -4.0dB<br>Reported RSRQ values:<br>±3.5dB for normal conditions and ±4dB for extreme conditions | 1.6dB<br><br>0dB<br><br>Via mapping | $N_{oc}$: -120.4dBm to -116.9dBm /15kHz<br>depending on operating band<br>$Ês_1$ / $N_{oc}$: -4.0dB<br><br>RSRQ_00 to RSRQ_15 (NTC)<br>RSRQ_00 to RSRQ_16 (ETC) |
| 9.9.2.1 TDD Intra Frequency Serving Cell Absolute RSRP Accuracy | Noc: -122dBm to -120dBm /15kHz<br>depending on operating band<br>Ês1 / Noc: -4.0dB<br>Reported RSRP values: ±6dB | 1.6dB<br><br>0dB<br>Via mapping | Noc: -120.4dBm to -118.4dBm /15kHz<br>depending on operating band<br>Ês1 / Noc: -4.0dB<br>RSRP_9 to RSRP_24<br>RSRP_10 to RSRP_25<br>RSRP_12 to RSRP_27<br>depending on operating band |
| 9.9.2.2 TDD Intra Frequency Serving Cell Absolute RSRQ Accuracy | Noc: -122dBm to -120dBm /15kHz<br>depending on operating band<br>Ês1 / Noc: -4.0dB<br>Reported RSRQ values:<br>±3.5dB for normal conditions and ±4dB for extreme conditions | 1.6dB<br><br>0dB<br><br>Via mapping | Noc: -120.4dBm to -118.4dBm /15kHz<br>depending on operating band<br>Ês1 / Noc: -4.0dB<br><br>RSRQ_00 to RSRQ_15 (NTC)<br>RSRQ_00 to RSRQ_16 (ETC) |
| 10.2 E-UTRAN FDD – Initiation/Cease of SLSS Transmission with ProSe Direct Communication | SNRs as specified | 0.6dB | During T1:<br>Formula: SNR + TT<br><br>During T2:<br>Formula: SNR - TT<br><br>During T3:<br>Formula: SNR+ TT |
| 10.4 E-UTRAN FDD – Initiation/Cease of SLSS Transmission with ProSe Direct Communication | During T1:<br>E-UTRA Cell<br>Noc: -98dBm /15kHz<br>Ês / Noc: -infinity<br>SyncRef UE<br>Noc: -98dBm /15kHz<br>Ês / Noc: 13.0dB<br><br>During T2:<br>E-UTRA Cell<br>Noc: -98dBm /15kHz<br>Ês / Noc: -3.0dB<br>SyncRef UE<br>Noc: -98dBm /15kHz<br>Ês / Noc: 13.0dB | During T1:<br>EUTRA Cell:<br>0dB<br>0dB<br>SyncRef UE:<br>0dB<br>0dB<br><br>During T2:<br>EUTRA Cell<br>0dB<br>0.6 dB<br>SyncRef UE :<br>0dB<br>0dB | During T1:<br>E-UTRA Cell<br>Noc: -98dBm /15kHz<br>Ês / Noc: -infinity<br>SyncRef UE<br>Noc: -98dBm /15kHz<br>Ês / Noc: 13.0dB<br><br>During T2:<br>E-UTRA Cell<br>Noc: -98dBm /15kHz<br>Ês / Noc: -2.4dB<br>SyncRef UE<br>Noc: -98dBm /15kHz<br>Ês / Noc: 13.0 dB |

# Annex G (normative): Statistical Testing

## G.1 General

FSS

## G.2 Statistical testing of delay and UE measurement performance in RRM tests

### G.2.1 General

The RRM tests are either of deterministic or of statistical nature. The pass fail limits in tests of statistical nature are expressed as a limit ( e.g. delay limit) and a success ratio applicable for the limit. The success ratio is 90% uniform (the complement is the error ratio ER = 10%).

### G.2.2 Design of the test

The test is defined by the following design principles (see TS 36.521-1 clause G.X, Theory):

1) The early decision concept is applied.

2) A second limit is introduced: bad DUT factor M>1

To decide the test pass:

Supplier risk is applied based on the bad DUT quality

To decide the test fails

Customer risk is applied based on the specified DUT quality

The test is defined by the following parameters:

1) Limit ER = 0.1 (success ratio = 90%)

2) Bad DUT factor M=1.5 (selectivity)

3) Confidence level CL = 95% (for specified DUT and bad DUT-quality)

## G.2.3 Numerical definition of the pass fail limits

**Table G.2.3-1: pass fail limits**

| ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 33 | NA | 43 | 408 | 283 | 86 | 737 | 644 | 129 | 1056 | 1021 |
| 1 | 46 | NA | 44 | 416 | 291 | 87 | 745 | 653 | 130 | 1064 | 1030 |
| 2 | 58 | 2 | 45 | 424 | 299 | 88 | 752 | 661 | 131 | 1071 | 1039 |
| 3 | 69 | 5 | 46 | 432 | 307 | 89 | 760 | 670 | 132 | 1078 | 1048 |
| 4 | 79 | 8 | 47 | 440 | 315 | 90 | 767 | 679 | 133 | 1086 | 1057 |
| 5 | 89 | 12 | 48 | 447 | 324 | 91 | 775 | 687 | 134 | 1093 | 1066 |
| 6 | 99 | 17 | 49 | 455 | 332 | 92 | 782 | 696 | 135 | 1100 | 1074 |
| 7 | 109 | 22 | 50 | 463 | 340 | 93 | 790 | 705 | 136 | 1108 | 1083 |
| 8 | 118 | 27 | 51 | 471 | 348 | 94 | 797 | 713 | 137 | 1115 | 1092 |
| 9 | 127 | 33 | 52 | 478 | 356 | 95 | 804 | 722 | 138 | 1122 | 1101 |
| 10 | 136 | 39 | 53 | 486 | 365 | 96 | 812 | 731 | 139 | 1130 | 1110 |
| 11 | 145 | 45 | 54 | 494 | 373 | 97 | 819 | 739 | 140 | 1137 | 1119 |
| 12 | 154 | 51 | 55 | 502 | 381 | 98 | 827 | 748 | 141 | 1144 | 1128 |
| 13 | 163 | 58 | 56 | 509 | 389 | 99 | 834 | 757 | 142 | 1152 | 1137 |
| 14 | 172 | 64 | 57 | 517 | 398 | 100 | 842 | 766 | 143 | 1159 | 1147 |
| 15 | 180 | 71 | 58 | 525 | 406 | 101 | 849 | 774 | 144 | 1166 | 1155 |
| 16 | 189 | 78 | 59 | 532 | 414 | 102 | 857 | 783 | 145 | 1174 | 1164 |
| 17 | 197 | 85 | 60 | 540 | 423 | 103 | 864 | 792 | 146 | 1181 | 1173 |
| 18 | 206 | 92 | 61 | 548 | 431 | 104 | 871 | 801 | 147 | NA | 1182 |
| 19 | 214 | 99 | 62 | 555 | 440 | 105 | 879 | 809 | 148 | | |
| 20 | 223 | 106 | 63 | 563 | 448 | 106 | 886 | 818 | 149 | | |
| 21 | 231 | 113 | 64 | 571 | 456 | 107 | 894 | 827 | 150 | | |
| 22 | 239 | 120 | 65 | 578 | 465 | 108 | 901 | 836 | 151 | | |
| 23 | 248 | 128 | 66 | 586 | 473 | 109 | 909 | 844 | 152 | | |
| 24 | 256 | 135 | 67 | 594 | 482 | 110 | 916 | 853 | 153 | | |
| 25 | 264 | 142 | 68 | 601 | 490 | 111 | 923 | 862 | 154 | | |
| 26 | 272 | 150 | 69 | 609 | 499 | 112 | 931 | 871 | 155 | | |
| 27 | 281 | 157 | 70 | 616 | 507 | 113 | 938 | 880 | 156 | | |
| 28 | 289 | 165 | 71 | 624 | 516 | 114 | 946 | 888 | 157 | | |
| 29 | 297 | 173 | 72 | 632 | 524 | 115 | 953 | 897 | 158 | | |
| 30 | 305 | 180 | 73 | 639 | 533 | 116 | 960 | 906 | 159 | | |
| 31 | 313 | 188 | 74 | 647 | 541 | 117 | 968 | 915 | 160 | | |
| 32 | 321 | 196 | 75 | 654 | 550 | 118 | 975 | 924 | 161 | | |
| 33 | 329 | 204 | 76 | 662 | 558 | 119 | 983 | 933 | 162 | | |
| 34 | 337 | 211 | 77 | 669 | 567 | 120 | 990 | 941 | 163 | | |
| 35 | 345 | 219 | 78 | 677 | 575 | 121 | 997 | 950 | 164 | | |
| 36 | 353 | 227 | 79 | 684 | 584 | 122 | 1005 | 959 | 165 | | |
| 37 | 361 | 235 | 80 | 692 | 592 | 123 | 1012 | 968 | 166 | | |
| 38 | 369 | 243 | 81 | 700 | 601 | 124 | 1019 | 977 | 167 | | |
| 39 | 377 | 251 | 82 | 707 | 610 | 125 | 1027 | 986 | 168 | | |
| 40 | 385 | 259 | 83 | 715 | 618 | 126 | 1034 | 994 | 169 | | |
| 41 | 393 | 267 | 84 | 722 | 627 | 127 | 1042 | 1003 | | | |
| 42 | 400 | 275 | 85 | 730 | 635 | 128 | 1049 | 1012 | | | |

The first column is the number of errors (ne = number of exceeded delays or number of wrong reports)

The second column is the number of samples for the pass limit ($ns_p$ , ns=Number of samples= number of successes + number of exceedings or number of reports)

The third column is the number of samples for the fail limit ($ns_f$)

## G.2.4 Pass fail decision rules

The pass fail decision rules apply for a single test, comprising one component in the test vector. The over all Pass /Fail conditions are defined in clause G.2.6

Having observed 0 errors, pass the test at 33+ samples, otherwise continue

Having observed 1 error, pass the test at 46+ samples, otherwise continue

Having observed 2 errors, pass the test at 58+ samples, fail the test at 2 samples, otherwise continue

Having observed 146 errors, pass the test at 1181+ samples, fail the test at 1173- samples, otherwise continue

Having observed 147 errors, fail the test at 1182- samples,

Where x+ means: x or more, x- means x or less

NOTE 1: an ideal DUT passes after 33 samples. The maximum test time is 1181 samples.

## G.2.5 Void

## G.2.6 Test conditions for delay tests and UE measurement performance

**Table G.2.6-1: test conditions**

| **Test** | **Statistical independence** | **Number of components in the test vector, as specified in the test requirements and initial conditions of the applicable test** | **-** | **Over all Pass/Fail condition** |
|---|---|---|---|---|
| All tests in clauses 4, 5, 6.1, 7.2, 7.3 and 8 are delay tests of statistical nature while 6.2 and 7.1 are not applicable, since deterministic. | Test procedure in all statistical tests ensures independency | 1 per operating band (if tested, see 3A.3.3) | | Full set of environmental conditions (5) per operating band |
| All tests in clause 9 are UE level reports of statistical nature | Independency is assumed, although Layer 1 filtering is applied to the reported results | Full set of environmental conditions (5) per operating band | | Full set of environmental conditions (5) per operating band |

# G.X Theory to derive the numbers in Table G.2.3-1 (informative)

TS 36.521-1 Annex G.X applies

# Annex H (normative): Default Message Contents

This annex contains the default values of common messages specific to RRM, other than those described in TS 36.508 [7]. The message contents shall apply to test cases accordingly and unless indicated otherwise in specific test cases, shall be transmitted and checked by the system simulator. The default message contents can be defined for FDD Mode, or TDD Mode or both FDD/TDD Modes. All the messages are listed in alphabetical order based on conformance tests.

NOTE: For example, test case 8.1.1 has an exception for RRCConnectionReconfiguration message and therefore uses message contents according to TS 36.508 [7] with the exception of the RRCConnectionReconfiguration message specified in Annex H.

## H.1 Common contents of system information messages exceptions

This clause contains the default values of common system information messages, other than those described in TS 36.508 [7].

## H.2 Common contents of system information blocks exceptions

This clause contains the default values of common system information blocks, other than those described in TS 36.508 [7].

### H.2.1 System information blocks message contents exceptions for E-UTRAN intra frequency cell re-selection

SystemInformationBlockType3: (FDD/TDD) for E-UTRAN intra-frequency cell re-selection

**Table H.2.1-1: *SystemInformationBlockType3*: E-UTRAN intra frequency cell re-selection**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| cellReselectionInfoCommon SEQUENCE { | | | |
| q-Hyst | dB0 (0 dB) | 0 is actual value in dB (0 * 2 dB) | |
| } | | | |

SystemInformationBlockType4: (FDD/TDD) for E-UTRAN intra-frequency cell re-selection

For Cell 2

**Table H.2.1-2: *SystemInformationBlockType4*: E-UTRAN intra frequency cell re-selection**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-3 SystemInformationBlockType4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| intraFreqNeighCellList SEQUENCE (SIZE (1..maxCellIntra)) OF SEQUENCE { | | | |
| IntraFreqNeighCellInfo ::= SEQUENCE { | | | |
| physCellId | 0 (Cell 1 Id) | INTEGER (0..503) | |
| q-OffsetCell | dB0 (0 dB) | 0 is actual value in dB (0 * 2 dB) | |
| } | | | |
| } | | | |

## H.2.2 System information blocks message contents exceptions for E-UTRAN inter frequency cell re-selection

SystemInformationBlockType3: (FDD/TDD) for E-UTRAN inter-frequency cell re-selection

**Table H.2.2-1: *SystemInformationBlockType3*: E-UTRAN inter frequency cell re-selection**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| cellReselectionServingFreqInfo SEQUENCE { | | | |
| s-NonIntraSearch | 25 (50 dB) | 50 is actual value in dB (25 * 2 dB); for Cell 1 | Cell 1 |
| | Not Present | | Cell 2 |
| threshServingLow | 22 (44 dB) | 44 is actual value in dB (22 * 2 dB) | |
| cellReselectionPriority | 4 | | Cell 1 |
| | 5 | | Cell 2 |
| } | | | |

SystemInformationBlockType5: (FDD/TDD) for E-UTRAN inter-frequency cell re-selection case

For Cell 1

**Table H.2.2-2: *SystemInformationBlockType5*: E-UTRAN inter frequency cell re-selection**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-4 SystemInformationBlockType5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| interFreqCarrierFreqList SEQUENCE (SIZE (1..maxFreq)) OF SEQUENCE { | | | |
| q-Rxlevmin | -70 (-140 dBm) | -140 is actual value in dBm (-70 * 2 dBm) | |
| threshX-High | 24 (48 dB) | 48 is actual value in dB (24 * 2 dB) | |
| threshX-Low | 25 (50 dB) | 50 is actual value in dB (25 * 2 dB) | |
| cellReselectionPriority[*n*] | 5 for cell 1 | | |
| } | | | |

For Cell 2

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-4 SystemInformationBlockType5 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| interFreqCarrierFreqList SEQUENCE (SIZE (1..maxFreq)) OF SEQUENCE { | | | |
| q-Rxlevmin | -70 (-140 dBm) | -140 is actual value in dBm (-70 * 2 dBm) | |
| threshX-High | 24 (48 dB) | 48 is actual value in dB (24 * 2 dB) | |
| threshX-Low | 25 (50 dB) | 50 is actual value in dB (25 * 2 dB) | |
| cellReselectionPriority[*n*] | 4 for cell 2 | | |
| } | | | |
| interFreqNeighCellList[*n*] SEQUENCE (SIZE (1..maxCellInter)) OF SEQUENCE { | | | |
| physCellId | 0 (Cell 1 Id) | INTEGER (0..503) | |
| q-OffsetCell | dB0 (0 dB) | 0 is actual value in dB (0 * 2 dB) | |
| } | | | |

## H.2.3 System information blocks message contents exceptions for E-UTRAN inter-RAT cell re-selection

SystemInformationBlockType3: for inter-RAT EUTRAN FDD - UTRA FDD is of higher priority cell re-selection

**Table H.2.3-1: *SystemInformationBlockType3*: Inter-RAT E-UTRAN FDD - UTRA FDD is of higher priority cell re-selection**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| cellReselectionServingFreqInfo SEQUENCE { | | | |
| s-NonIntraSearch | 25 (50 dB) | 50 is actual value in dB (25 * 2 dB); for Cell 1 (E-UTRA) | |
| } | | | |

SystemInformationBlockType6: for inter-RAT EUTRAN FDD - UTRA FDD is of higher priority cell re-selection

**Table H.2.3-2: *SystemInformationBlockType6*: Inter-RAT E-UTRAN FDD - UTRA FDD is of higher priority cell re-selection**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-5 SystemInformationBlockType6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CarrierFreqListUTRA-FDD ::= SEQUENCE (SIZE (1..maxUTRA-FDD-Carrier)) OF SEQUENCE { | | | UTRA-FDD |
| threshX-High | 20 (40 dB) | 40 is actual value in dB (20 * 2 dB) | |
| q-RxLevMin | -58 (-115 dBm) | -115 is actual value in dBm (-58 * 2 + 1 dBm) | |
| p-MaxUTRA | 21 (21 dBm) | Default value in TS 36.508 | |
| q-QualMin | -20 (-20 dB) | | |
| cellReselectionPriority[*n*] | 5 | UTRA is of higher priority than E-UTRAN | |
| } | | | |

SystemInformationBlockType3: for inter-RAT EUTRAN TDD - UTRA TDD is of higher priority cell re-selection

**Table H.2.3-3: *SystemInformationBlockType3*: Inter-RAT E-UTRAN TDD - UTRA TDD is of higher priority cell re-selection**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| cellReselectionServingFreqInfo SEQUENCE { | | | |
| s-NonIntraSearch | 23 (46 dB) | 46 is actual value in dB (23 * 2 dB); for Cell 1 (E-UTRA) | |
| } | | | |

SystemInformationBlockType6: for inter-RAT EUTRAN TDD - UTRA TDD is of higher priority cell re-selection

**Table H.2.3-4: *SystemInformationBlockType6*: Inter-RAT E-UTRAN TDD - UTRA TDD is of higher priority cell re-selection**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-5 SystemInformationBlockType6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CarrierFreqListUTRA-TDD ::= SEQUENCE (SIZE (1..maxUTRA-TDD-Carrier)) OF SEQUENCE { | | | UTRA-TDD |
| | | | |
| threshX-High | 12 (24 dB) | 24 is actual value in dB (12 * 2 dB) | |
| q-RxLevMin | -52 (-103 dBm) | -103 is actual value in dBm (-52 * 2 + 1 dBm) | |
| p-MaxUTRA | 21 (21 dBm) | | |
| cellReselectionPriority[*n*] | 5 | UTRA is of higher priority than E-UTRAN | |
| } | | | |

SystemInformationBlockType3: for inter-RAT EUTRAN FDD/TDD - UTRA FDD is of lower priority cell re-selection

**Table H.2.3-5: *SystemInformationBlockType3*: Inter-RAT E-UTRAN FDD/TDD - UTRA FDD is of lower priority cell re-selection**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| cellReselectionServingFreqInfo SEQUENCE { | | | |
| threshServingLow | 22 (44 dB) | 44 is actual value in dB (22 * 2 dB) | |

SystemInformationBlockType6: for inter-RAT EUTRAN FDD/TDD - UTRA FDD is of lower priority cell re-selection

**Table H.2.3-6: *SystemInformationBlockType6*: Inter-RAT E-UTRAN FDD/TDD - UTRA FDD is of lower priority cell re-selection**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-5 SystemInformationBlockType6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CarrierFreqListUTRA-FDD ::= SEQUENCE (SIZE (1..maxUTRA-FDD-Carrier)) OF SEQUENCE { | | | UTRA-FDD |
| threshX-Low | 21 (42 dB) | 42 is actual value in dB (21 * 2 dB) | |
| q-RxLevMin | -58 (-115 dBm) | -115 is actual value in dBm (-58 * 2 + 1 dBm) | |
| p-MaxUTRA | 21 (21 dBm) | | |
| q-QualMin | -20 (-20 dB) | | |
| } | | | |

SystemInformationBlockType3: for inter-RAT EUTRAN FDD/TDD - UTRA TDD is of lower priority cell re-selection

**Table H.2.3-7: *SystemInformationBlockType3*: Inter-RAT E-UTRAN FDD/TDD - UTRA TDD is of lower priority cell re-selection**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| cellReselectionServingFreqInfo SEQUENCE { | | | |
| threshServingLow | 23 (46 dB) | 46 is actual value in dB (23 * 2 dB) | |

SystemInformationBlockType6: for inter-RAT EUTRAN FDD/TDD - UTRA TDD is of lower priority cell re-selection

**Table H.2.3-8: *SystemInformationBlockType6*: Inter-RAT E-UTRAN FDD/TDD - UTRA TDD is of lower priority cell re-selection**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-5 SystemInformationBlockType6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CarrierFreqListUTRA-TDD ::= SEQUENCE (SIZE (1..maxUTRA-TDD-Carrier)) OF SEQUENCE { | | | UTRA-TDD |
| threshX-Low | 12 (24 dB) | 24 is actual value in dB (12 * 2 dB) | |
| q-RxLevMin | -52 (-103 dBm) | -103 is actual value in dBm (-52 * 2 + 1 dBm) | |
| p-MaxUTRA | 21 (21 dBm) | | |
| } | | | |

SystemInformationBlockType3: (FDD/TDD) for inter-RAT E-UTRAN - GSM cell re-selection

**Table H.2.3-9: *SystemInformationBlockType3*: Inter-RAT E-UTRAN - GSM cell re-selection**

| **Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3** | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| cellReselectionServingFreqInfo SEQUENCE { | | | |
| s-NonIntraSearch | Not sent | | |
| threshServingLow | 22 (44 dB) | 44 is actual value in dB (22 * 2 dB) | |
| cellReselectionPriority | 4 | | |

SystemInformationBlockType7: (FDD/TDD) for inter-RAT E-UTRAN - GSM cell re-selection

**Table H.2.3-10: *SystemInformationBlockType7*: Inter-RAT E-UTRAN - GSM cell re-selection**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-6 SystemInformationBlockType7 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| commonInfo SEQUENCE { | | | |
| cellReselectionPriority | 0 | | |
| ncc-Permitted | '11111111'B | | |
| q-RxLevMin | 5 (-105 dBm) | -105 is actual value in dBm (5 * 2 - 115 dBm ) | |
| p-MaxGERAN | 23 (23 dBm) | | GSM 400 & GSM 900 & GSM 850 & GSM 700 |
| | 24 (24 dBm) | | DCS 1800 & PCS 1900 |
| threshX-High | 12 (24 dB) | 24 is actual value in dB (12 * 2 dB) | |
| threshX-Low | 12 (24 dB) | 24 is actual value in dB (12 * 2 dB) | |

SystemInformationBlockType3: Inter-RAT E-UTRAN FDD/TDD - HRPD is of lower priority cell re-selection

**Table H.2.3-11: *SystemInformationBlockType3*: Inter-RAT E-UTRAN FDD/TDD - HRPD is of lower priority cell re-selection**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| cellReselectionServingFreqInfo SEQUENCE { | | | |
| s-NonIntraSearch | Not sent | | |
| threshServingLow | 22 (44 dB) | 44 is actual value in dB (22 * 2 dB) | |
| cellReselectionPriority | 1 | | |
| } | | | |
| intraFreqCellReselectionInfo SEQUENCE { | | | |
| q-RxLevMin | -70 (-140 dBm) | For RF/RRM test cases | |
| s-IntraSearch | Not present | | |
| t-ReselectionEUTRA | 0 | | |

SystemInformationBlockType8: Inter-RAT E-UTRAN FDD/TDD - HRPD is of lower priority cell re-selection

**Table H.2.3-12: *SystemInformationBlockType8*: Inter-RAT E-UTRAN FDD/TDD - HRPD is of lower priority cell re-selection**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-7 SystemInformationBlockType8 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| cellReselectionParametersHRPD SEQUENCE { | | | |
| bandClassList SEQUENCE (SIZE (1..maxCDMA -BandClass)) OF SEQUENCE { | 1 entry | | |
| cellReselectionPriority | 0 | | |
| threshX-High | 60(-30) | INTEGER (0..63) | |
| threshX-Low | 28(-14) | INTEGER (0..63) | |
| } | | | |
| } | | | |
| t-ReselectionCDMA2000 | 0 | INTEGER (0..7) | |
| } | | | |

System Information Block Type 3: for E-UTRAN to UTRAN inter-RAT cell re-selection

**Table H.2.3-13: System Information Block type3: Inter-RAT E-UTRAN FDD/TDD - UTRAN FDD cell re-selection**

| Derivation Path: 34.108 clause 6.1.0b | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| - Cell selection and reselection info | | | |
| - Qqualmin | -20 | | |
| - Qrxlevlmin | -58 (-115dBm) | | |
| - Maximum allowed UL TX power | 21 | | |

**Table H.2.3-14: System Information Block type 3 (1.28 Mcps TDD): inter-RAT E-UTRAN FDD/TDD – UTRAN TDD cell re-selection**

| Derivation Path: 34.108 clause 6.1.0b | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| - SIB4 Indicator | TRUE | | |
| - Cell identity | 0000 0000 0000 0000 0000 0000 0001B | | |
| - Cell selection and re-selection info | | | |
| - Mapping info | Not present | | |
| - Cell selection and reselection quality measure | (no data) | | |
| - CHOICE mode | TDD | | |
| - Sintrasearch | 10 dB | | |
| - Sintersearch | 10 dB | | |
| - SsearchHCS | Not present | | |
| - RAT List | Not present | | |
| - Qrxlevmin | -103 dBm | | |
| - Qhyst1s | 0 dB | | |
| - Treselections | 0 seconds | | |
| - HCS Serving cell information | Not present | | |

SystemInformationBlockType3: Inter-RAT E-UTRAN FDD/TDD – cdma2000 1xRTT is of lower priority cell re-selection

**Table H.2.3-15: *SystemInformationBlockType3*: Inter-RAT E-UTRAN FDD/TDD – cdma2000 1xRTT is of lower priority cell re-selection**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| cellReselectionServingFreqInfo SEQUENCE { | | | |
| s-NonIntraSearch | Not sent | | |
| threshServingLow | 22 (44 dB) | 44 is actual value in dB (22 * 2 dB) | |
| cellReselectionPriority | 1 | | |
| } | | | |
| intraFreqCellReselectionInfo SEQUENCE { | | | |
| q-RxLevMin | -70 (-140 dBm) | For RF/RRM test cases | |
| s-IntraSearch | Not present | | |
| t-ReselectionEUTRA | 0 | | |
| } | | | |

SystemInformationBlockType8: Inter-RAT E-UTRAN FDD/TDD – cdma2000 1xRTT is of lower priority cell re-selection

**Table H.2.3-16: *SystemInformationBlockType8*: Inter-RAT E-UTRAN FDD/TDD – cdma2000 1xRTT is of lower priority cell re-selection**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-7 SystemInformationBlockType8 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| cellReselectionParameters1XRTT SEQUENCE { | | | 1XRTT |
| longCodeState1XRTT | Not Present | | |
| cellReselectionParameters1XRTT SEQUENCE { | | | |
| bandClassList SEQUENCE (SIZE (1..maxCDMA -BandClass)) OF SEQUENCE { | 1 entry | | |
| cellReselectionPriority | 0 | | |
| threshX-High | 60(-30) | INTEGER (0..63) | |
| threshX-Low | 56(-28) | INTEGER (0..63) | |
| } | | | |
| t-ReselectionCDMA2000 | 0 | INTEGER (0..7) | |
| t-ReselectionCDMA2000-SF | Not Present | | |
| } | | | |
| } | | | |
| } | | | |

# H.2.4 System information blocks message contents exceptions for E-UTRAN radio link monitoring (RLM)

SystemInformationBlockType2: (FDD/TDD) for E-UTRAN Radio Link Monitoring test for out-of-sync

**Table H.2.4-1: *SystemInformationBlockType2*: E-UTRAN Radio Link Monitoring test for out-of-sync**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 SystemInformationBlockType2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType 2 ::= SEQUENCE { | | | |
| radioResourceConfigCommon SEQUENCE { | | | |
| pdsch-ConfigCommon SEQUENCE { | | | |
| referenceSignalPower | Set to an arbitrarily selected value above -11dBm and within the IE allowed range described in 36.331[5] | The selected IE value depends on the test system implementation and should be declared in the test report. | |
| } | | | |
| soundingRS-UL-ConfigCommon CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| ue-TimersAndConstants { | | | |
| t300 | ms1000 | | |
| t301 | ms1000 | | |
| t310 | ms0 | | |
| n310 | n1 | | |
| t311 | ms1000 | | |
| n311 | n1 | | |
| } | | | |

SystemInformationBlockType2: (FDD/TDD) for E-UTRAN Radio Link Monitoring test for in-sync

**Table H.2.4-2: *SystemInformationBlockType2*: E-UTRAN Radio Link Monitoring test for in-sync**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType 2 ::= SEQUENCE { | | | |
| radioResourceConfigCommon SEQUENCE { | | | |
| pdsch-ConfigCommon SEQUENCE { | | | |
| referenceSignalPower | Set to an arbitrarily selected value above -11dBm and within the IE allowed range described in 36.331[5] | The selected IE value depends on the test system implementation and should be declared in the test report. | |
| } | | | |
| soundingRS-UL-ConfigCommon CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| ue-TimersAndConstants { | | | |
| t300 | ms1000 | | |
| t301 | ms1000 | | |
| t310 | ms2000 | | |
| n310 | n1 | | |
| t311 | ms1000 | | |
| n311 | n1 | | |
| } | | | |

## H.2.5 System information blocks message contents exceptions for RRC Re-establishment

SystemInformationBlockType2: (FDD/TDD) for E-UTRAN Intra-frequency RRC Re-establishment

**Table H.2.5-1: *SystemInformationBlockType2*: E-UTRAN FDD Intra-frequency RRC Re-establishment**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 SystemInformationBlockType2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType 2 ::= SEQUENCE { | | | |
| ue-TimersAndConstants { | | | |
| t310 | ms0 | | |
| t311 | ms3000 | | |
| n310 | n1 | | |
| n311 | n1 | | |

SystemInformationBlockType2: (FDD/TDD) for E-UTRAN Inter-frequency RRC Re-establishment

**Table H.2.5-2: *SystemInformationBlockType2*: E-UTRAN FDD Inter-frequency RRC Re-establishment**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 SystemInformationBlockType2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType 2 ::= SEQUENCE { | | | |
| ue-TimersAndConstants { | | | |
| t310 | ms0 | | |
| t311 | ms5000 | | |
| n310 | n1 | | |
| n311 | n1 | | |

## H.2.6 System information block messages and information elements contents exceptions for E-UTRAN Random Access

SystemInformationBlockType1: (FDD/TDD) for E-UTRAN random access

**Table H.2.6-1: *SystemInformationBlockType1*: E-UTRAN FDD/TDD - Contention and Non-Contention Based Random Access**

| Derivation Path: TS 36.508 [7] clause 4.4.3.2, Table 4.4.3.2-3 SystemInformationBlockType1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType1 ::= SEQUENCE { | | | |
| p-Max | 23 (dBm) | | |

RACH-ConfigCommon-DEFAULT: (FDD/TDD) for E-UTRAN random access

**Table H.2.6-2: *RACH-ConfigCommon-DEFAULT*: E-UTRAN FDD/TDD - Contention and Non-Contention Based Random Access**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-12 RACH-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RACH-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| preambleInfo SEQUENCE { | | | |
| numberOfRA-Preambles | n52 | | |
| preamblesGroupAConfig SEQUENCE {} | Not present | | |
| } | | | |
| powerRampingParameters SEQUENCE { | | | |
| powerRampingStep | dB2 | | |
| preambleInitialReceivedTargetPower | dBm-120 | | |
| } | | | |
| ra-SupervisionInfo SEQUENCE { | | | Cat-M1 UE |
| preambleTransMax | n6 | | |
| } | | | |
| maxHARQ-Msg3Tx | 4 | | |
| } | | | |
| RAR-HoppingConfigInfo-r13 [1] SEQUENCE { | | | |
| rar-HoppingConfig | off | | |
| } | | | |

PDSCH-ConfigCommon-DEFAULT: (FDD/TDD) for E-UTRAN random access

**Table H.2.6-3: *PDSCH-ConfigCommon-DEFAULT*: E-UTRAN FDD/TDD - Contention and Non-Contention Based Random Access**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-5 PDSCH-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| referenceSignalPower | -5 (dBm) | | 1TX |

## H.2.7 System information blocks message contents exceptions for eICIC/feICIC

SystemInformationBlockType2: (FDD/TDD) for eICIC/feICIC

**Table H.2.7-1: *SystemInformationBlockType2*: Time Domain Measurement Resource Restriction with Non MBSFN ABS (eICIC/feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 SystemInformationBlockType2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfiguration | Not present | | |
| } | | | |

SystemInformationBlockType3: (FDD) for eICIC/feICIC

**Table H.2.7-2: *SystemInformationBlockType3*: Time Domain Measurement Resource Restriction with Non MBSFN ABS (eICIC/feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-2 SystemInformationBlockType3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| neighCellConfig | '01'B (No MBSFN subframes are present in all neighbour cells) | | FDD with E-UTRA FDD neighbour cell |

# H.3 Default RRC messages and information elements contents exceptions

This clause contains the default values of common sRRC messages and information elements, other than those described in TS 36.508 [7].

## H.3.1 RRC messages and information elements contents exceptions for E-UTRAN measurement configuration

RRCConnectionReconfiguration: (FDD/TDD) to setup E-UTRAN Measurement Configuration

**Table H.3.1-1: *RRCConnectionReconfiguration*: E-UTRAN Measurement Configuration**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| measConfig | MeasConfig -DEFAULT | | MEAS |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| | | | |

MeasConfig-DEFAULT (FDD/TDD) E-UTRAN measurement configuration for intra frequency measurement

**Table H.3.1-2: *MeasConfig-DEFAULT*: E-UTRAN intra frequency measurement configuration**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 1 entry | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-A3 | | |
| reportConfig | ReportConfigEUTRA-A3 | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f1 | | |
| reportConfigId | idReportConfig-A3 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

MeasConfig-DEFAULT (FDD/TDD) E-UTRAN measurement configuration for inter frequency handover

**Table H.3.1-3: *MeasConfig-DEFAULT*: E-UTRAN inter frequency measurement configuration for inter frequency handover**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | f1 is the frequency of the serving cell | |
| measObjectId | IdMeasObject-f1 | | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the neighbouring cell(inter frequency cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | inter frequency cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-A3 | | |
| reportConfig | ReportConfigEUTRA-A3 | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-A3 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| | MeasGapConfig-GP2 | | Gap Pattern Id = 1 |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

MeasConfig-DEFAULT (FDD/TDD) E-UTRAN measurement configuration for E-UTRAN to UTRAN cell search

**Table H.3.1-4: *MeasConfig-DEFAULT*: interRAT UTRAN measurement configuration for E-UTRAN to UTRAN cell search**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell (E-UTRA Cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | E-UTRA Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the neighbouring cell(UTRA Cell) | |
| measObject CHOICE { | | | |
| measObjectUTRA | MeasObjectUTRA-GENERIC(f2) | UTRA Cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-B1 | | |
| reportConfig | ReportConfigInterRAT-B1-UTRA | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-B1 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | UTRAN |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| | MeasGapConfig-GP2 | | Gap Pattern Id = 1 |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

MeasConfig-DEFAULT (FDD/TDD) E-UTRAN measurement configuration for E-UTRAN to UTRAN handover

**Table H.3.1-5: *MeasConfig-DEFAULT*: interRAT UTRAN measurement configuration for E-UTRAN to UTRAN handover**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell(E-UTRA Cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | E-UTRA Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the neighbouring cell(UTRA Cell) | |
| measObject CHOICE { | | | |
| measObjectUTRA | MeasObjectUTRA-GENERIC(f2) | UTRA Cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId)) OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-B2 | | |
| reportConfig | ReportConfigInterRAT-B2-UTRA | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) OF SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | IdReportConfig-B2 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | UTRAN |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| | MeasGapConfig-GP2 | | Gap Pattern Id = 1 |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

MeasConfig-DEFAULT (FDD/TDD) E-UTRAN measurement configuration for E-UTRAN to GSM handover

**Table H.3.1-6: *MeasConfig-DEFAULT*: interRAT GSM measurement configuration for E-UTRAN to GSM handover**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell(E-UTRA Cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | E-UTRA Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the neighbouring cell(GERAN Cell) | |
| measObject CHOICE { | | | |
| measObjectGERAN | MeasObjectGERAN-GENERIC(f2) | GERAN Cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-B1 | | |
| reportConfig | ReportConfigInterRAT-B1-GERAN | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-B1 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | GERAN |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| | MeasGapConfig-GP2 | | Gap Pattern Id = 1 |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

QuantityConfig-DEFAULT (FDD/TDD) E-UTRAN measurement configuration for L3 filtering is not used

**Table H.3.1-7: QuantityConfig-DEFAULT: measurement configuration for L3 filtering is not used**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-3A: QuantityConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| QuantityConfig-DEFAULT ::= SEQUENCE { | | | |
| quantityConfigEUTRA SEQUENCE { | | | |
| filterCoefficientRSRP | fc0 | | |
| filterCoefficientRSRQ | fc0 | | |
| } | | | |
| quantityConfigUTRA SEQUENCE {} | Not present | | |
| quantityConfigUTRA SEQUENCE { | | | UTRAN |
| measQuantityUTRA-FDD | cpich-EcN0 | | |
| measQuantityUTRA-TDD | pccpch-RSCP | | |
| filterCoefficient | fc0 | | |
| } | | | |
| quantityConfigGERAN SEQUENCE {} | Not present | | |
| quantityConfigGERAN SEQUENCE { | | | GERAN |
| measQuantityGERAN | rssi | | |
| filterCoefficient | fc0 | | |
| } | | | |
| quantityConfigCDMA2000 SEQUENCE {} | Not present | | |
| quantityConfigCDMA2000 SEQUENCE { | | | CDMA2000 |
| measQuantityCDMA2000 | pilotStrength | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| UTRAN | For inter-RAT measurements with UTRAN |
| GERAN | For inter-RAT measurements with GERAN |
| CDMA2000 | For inter-RAT measurements with CDMA2000 |

**Table H.3.1-8: *MeasConfig-DEFAULT*: interRAT HRPD measurement configuration for E-UTRAN to HRPD handover**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell(E-UTRA Cell) | |
| measObject CHOICE { | | | |
| measObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | E-UTRA Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the neighbouring cell(CDMA2000 Cell) | |
| measObject CHOICE { | | | |
| measObjectCDMA2000 | MeasObjectCDMA2000 | CDMA2000 Cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId)) OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-B2 | | |
| reportConfig | ReportConfigInterRAT-B2-CDMA2000 | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) OF SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | IdReportConfig-B2 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | CDMA2000 |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

MeasConfig-DEFAULT (FDD/TDD) E-UTRAN measurement configuration for inter frequency measurement

**Table H.3.1-9: *MeasConfig-DEFAULT*: E-UTRAN inter frequency measurement configuration**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the neighbouring cell(inter frequency cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | inter frequency cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-A3 | | |
| reportConfig | ReportConfigEUTRA-A3 | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-A3 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| | MeasGapConfig-GP2 | | Gap Pattern Id = 1 |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

MeasConfig-DEFAULT (FDD/TDD) E-UTRAN measurement configuration for E-UTRAN to GSM cell search

**Table H.3.1-10: *MeasConfig-DEFAULT*: interRAT GSM measurement configuration for E-UTRAN to GSM cell search**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 7 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell(E-UTRA Cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | E-UTRA Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the neighbouring cell(GERAN Cell) | |
| measObject CHOICE { | | | |
| measObjectGERAN | MeasObjectGERAN-GENERIC(f2) | GERAN Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f3 | f3 is the frequency of the neighbouring cell(GERAN Cell) | |
| measObject CHOICE { | | | |
| measObjectGERAN | MeasObjectGERAN-GENERIC(f3) | GERAN Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f4 | f4 is the frequency of the neighbouring cell(GERAN Cell) | |
| measObject CHOICE { | | | |
| measObjectGERAN | MeasObjectGERAN-GENERIC(f4) | GERAN Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f5 | f5 is the frequency of the neighbouring cell(GERAN Cell) | |
| measObject CHOICE { | | | |
| measObjectGERAN | MeasObjectGERAN-GENERIC(f5) | GERAN Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f6 | f6 is the frequency of the neighbouring cell(GERAN Cell) | |
| measObject CHOICE { | | | |

| | | | |
|---|---|---|---|
| measObjectGERAN | MeasObjectGERAN-GENERIC(f6) | GERAN Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f7 | f7 is the frequency of the neighbouring cell(GERAN Cell) | |
| measObject CHOICE { | | | |
| measObjectGERAN | MeasObjectGERAN-GENERIC(f7) | GERAN Cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-B1 | | |
| reportConfig | ReportConfigInterRAT-B1-GERAN | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-B1 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | GERAN |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| | MeasGapConfig-GP2 | | Gap Pattern Id = 1 |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

MeasConfig-DEFAULT (FDD/TDD) E-UTRAN measurement periodical configuration for E-UTRAN to GSM cell search

**Table H.3.1-11: *MeasConfig-DEFAULT*: interRAT GSM measurement periodical configuration for E-UTRAN to GSM cell search**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT: | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 7 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell(E-UTRA Cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | E-UTRA Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the neighbouring cell(GERAN Cell) | |
| measObject CHOICE { | | | |
| measObjectGERAN | MeasObjectGERAN-GENERIC(f2) | GERAN Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f3 | f3 is the frequency of the neighbouring cell(GERAN Cell) | |
| measObject CHOICE { | | | |
| measObjectGERAN | MeasObjectGERAN-GENERIC(f3) | GERAN Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f4 | f4 is the frequency of the neighbouring cell(GERAN Cell) | |
| measObject CHOICE { | | | |
| measObjectGERAN | MeasObjectGERAN-GENERIC(f4) | GERAN Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f5 | f5 is the frequency of the neighbouring cell(GERAN Cell) | |
| measObject CHOICE { | | | |
| measObjectGERAN | MeasObjectGERAN-GENERIC(f5) | GERAN Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f6 | f6 is the frequency of the neighbouring cell(GERAN Cell) | |

| | | | |
|---|---|---|---|
| measObject CHOICE { | | | |
| measObjectGERAN | MeasObjectGERAN-GENERIC(f6) | GERAN Cell | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f7 | f7 is the frequency of the neighbouring cell(GERAN Cell) | |
| measObject CHOICE { | | | |
| measObjectGERAN | MeasObjectGERAN-GENERIC(f7) | GERAN Cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigInterRAT-PERIODICAL | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 2 entry | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f3 | | |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | GERAN |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| | MeasGapConfig-GP2 | | Gap Pattern Id = 1 |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

## H.3.2 RRC messages and information elements contents exceptions for E-UTRAN cell re-selection and handover

PRACH-Config-DEFAULT: (FDD) for cell re-selection and handover

**Table H.3.2-1: *PRACH-Config-DEFAULT*: E-UTRAN FDD cell re-selection and handover**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-7 PRACH-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PRACH-ConfigInfo SEQUENCE { | | | |
| prach-ConfigIndex | 4 | | |

PRACH-Config-DEFAULT: (TDD) for cell re-selection and intra frequency / inter frequency handover

**Table H.3.2-2: *PRACH-Config-DEFAULT*: E-UTRAN TDD cell re-selection and handover**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-7 PRACH-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PRACH-ConfigInfo SEQUENCE { | | | |
| prach-ConfigIndex | 53 | | |

RRCConnectionReconfiguration: (FDD/TDD) for intra-frequency / inter-frequency handover

**Table H.3.2-3: *RRCConnectionReconfiguration*: E-UTRAN handover Configuration**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| mobilityControlInfo | | | |
| } | MobilityControlInfo-HO | | HO |
| } | | | |
| } | | | |
| } | | | |
| | | | |

## H.3.3 RRC messages and information elements contents exceptions for E-UTRAN inter-RAT handover

Handover: (FDD/TDD) to perform Inter-RAT E-UTRAN - UTRAN handover

**Table H.3.3-1: *Handover*: Inter-RAT E-UTRAN - UTRAN handover**

| **Derivation Path: 36.331 clause 6.2.2** | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| Handover ::= SEQUENCE { | | | |
| targetRAT-Type | utra | ENUMERATED {utra, geran, cdma2000-1XTT, cdma2000-HRPD, spare4, spare3, spare2, spare1, …} | |
| targetRAT-MessageContainer | | OCTET STRING | |
| nas-SecurityParamFromEUTRA | | OCTET STRING(SIZE (1)) | UTRAGERAN |
| } | | | |

Handover: (FDD/TDD) to perform Inter-RAT E-UTRAN - GSM handover

**Table H.3.3-2: *Handover*: Inter-RAT E-UTRAN - GSM handover**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| Handover ::= SEQUENCE { | | | |
| targetRAT-Type | geran | ENUMERATED {utran, geran, cdma2000-1XTT, cdma2000-HRPD, spare4, spare3, spare2, spare1, …} | |
| targetRAT-MessageContainer | | OCTET STRING | |
| nas-SecurityParamFromEUTRA | | OCTET STRING(SIZE (1)) | UTRAGERAN |
| } | | | |

MobilityFromEUTRACommand: (FDD/TDD) to setup Inter-RAT E-UTRAN handover

**Table H.3.3-3: *MobilityFromEUTRACommand*: Inter-RAT E-UTRAN handover**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-6 MobilityFromEUTRACommand | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MobilityFromEUTRACommand ::= SEQUENCE { | | | |
| rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| mobilityFromEUTRACommand-r8 SEQUENCE { | | | |
| csFallbackIndicator | FALSE | | |
| purpose CHOICE { | | | |
| Handover | Handover | | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| nonCriticalExtension SEQUENCE { | | | GERAN |
| lateNonCriticalExtension | Not present | | |
| nonCriticalExtension SEQUENCE { | | | |
| bandIndicator | | ENUMERATED {dcs1800, pcs1900} | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| Condition | Explanation |
|---|---|
| GERAN | The field should be present if the *purpose* is set to "*handover*" and the *targetRAT-Type* is set to "*geran*"; otherwise the field is not present |

Handover: (FDD/TDD) to perform Inter-RAT E-UTRAN – HRPD handover

**Table H.3.3-4: *Handover*: Inter-RAT E-UTRAN – HRPD handover**

| Derivation Path: 36.331 clause 6.2.2 | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| Handover ::= SEQUENCE { | | | |
| targetRAT-Type | cdma2000-HRPD | ENUMERATED {utran, geran, cdma2000-1XTT, cdma2000-HRPD, spare4, spare3, spare2, spare1, …} | |

## H.3.4 RRC messages and information elements exceptions for E-UTRAN UE transmit timing accuracy and UE timing advance adjustment accuracy

RRCConnectionReconfiguration: (FDD/TDD) to establish E-UTRAN Radio Resource Configuration

**Table H.3.4-1: *RRCConnectionReconfiguration*: E-UTRAN Radio Resource Configuration**

| Derivation Path: TS 36.508 [7] clause 4.6.1, Table 4.6.1-8 RRCConnectionReconfiguration | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| Rrc-TransactionIdentifier | RRC-TransactionIdentifier-DL | | |
| criticalExtensions CHOICE { | | | |
| C1 CHOICE{ | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| radioResourceConfigDedicated | | | |
| | RadioResourceConfigDedicated-HO-TO-EUTRA(n,m) | | HO-TO-EUTRA(n,m) |

PhysicalConfigDedicated-DEFAULT: (FDD/TDD) for E-UTRAN Physical Configuration

**Table H.3.4-2: *PhysicalConfigDedicated-DEFAULT*: E-UTRAN Physical Configuration**

| Derivation Path: TS 36.508 [7] clause 4.8.21.6, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | | | SRB1 |
| | SoundingRS-UL-ConfigDedicated-DEFAULT | | RBC |
| antennaInformation CHOICE { | | | |
| defaultValue | NULL | | |
| } | | | |
| schedulingRequestConfig | Not present | | SRB1 |

## H.3.5 RRC messages and information elements contents exceptions for E-UTRAN RSRP and RSRQ Accuracy

MeasConfig: (FDD/TDD) to perform Measurement Configuration for E-UTRAN intra frequency RSRP and RSRQ accuracy

**Table H.3.5-1: *MeasConfig-DEFAULT*: E-UTRAN intra frequency RSRP and RSRQ Accuracy**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 1 entry | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f1 | | |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

**Table H.3.5-1a: *MeasConfig-DEFAULT*: E-UTRAN intra frequency RSRP and RSRQ Accuracy for Cat-M1 UE**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 1 entry | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f1 | | |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| | MeasGapConfig-GP2 | | Gap Pattern Id = 1 |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

MeasConfig: (FDD/TDD) perform Measurement Configuration for E-UTRAN inter frequency RSRP and RSRQ accuracy

**Table H.3.5-2: *MeasConfig-DEFAULT*: E-UTRAN inter frequency RSRP and RSRQ Accuracy**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entry | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the neighbouring cell (inter frequency cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | inter frequency cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | MeasGapConfig-GP1 | | Gap Pattern Id = 0 |
| | MeasGapConfig-GP2 | | Gap Pattern Id = 1 |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

ReportConfigEUTRA-PEROIDICAL: (FDD/TDD) for periodical configuration reporting of E-UTRAN RSRP accuracy

**Table H.3.5-3: *ReportConfigEUTRA-PERIODICAL*: E-UTRAN RSRP Accuracy**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportStrongestCells | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | sameAsTriggerQuantity | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 (1024 ms) | | |
| reportAmount | Infinity | | |
| } | | | |

**Table H.3.5-3a: *ReportConfigEUTRA-PERIODICAL*: E-UTRAN RSRP Accuracy**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportStrongestCells | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | sameAsTriggerQuantity | | |
| maxReportCells | 2 | | |
| reportInterval | ms1024 (1024 ms) | | |
| reportAmount | Infinity | | |
| } | | | |

ReportConfigEUTRA-PEROIDICAL: (FDD/TDD) for periodical configuration reporting of E-UTRAN RSRQ accuracy

**Table H.3.5-4: *ReportConfigEUTRA-PERIODICAL*: E-UTRAN RSRQ Accuracy**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportStrongestCells | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrq | | |
| reportQuantity | sameAsTriggerQuantity | | |
| maxReportCells | 1 | | |
| reportInterval | ms1024 (1024 ms) | | |
| reportAmount | Infinity | | |
| } | | | |

ReportConfigEUTRA-PEROIDICAL: (FDD/TDD) for periodical configuration reporting of E-UTRAN RSRQ accuracy

**Table H.3.5-4a: *ReportConfigEUTRA-PERIODICAL*: E-UTRAN RSRQ Accuracy**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-7 ReportConfigEUTRA-PERIODICAL | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportStrongestCells | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrq | | |
| reportQuantity | sameAsTriggerQuantity | | |
| maxReportCells | 2 | | |
| reportInterval | ms1024 (1024 ms) | | |
| reportAmount | Infinity | | |
| } | | | |

MeasObjectEUTRA-GENERIC(Freq): (FDD/TDD) for eICIC/feICIC (Non-MBSFN ABS)

**Table H.3.5-5: *MeasObjectEUTRA-GENERIC*: Time Domain Measurement Resource Restriction with Non MBSFN ABS (eICIC/feICIC)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| neighCellConfig | '01'B (No MBSFN subframes are present in all neighbour cells) | | FDD with E-UTRA FDD neighbour cell |
| } | | | |

## H.3.6 RRC messages and information elements contents exceptions for E-UTRAN inter frequency handover and E-UTRAN intra frequency cell search

MAC-MainConfig-RBC: (FDD/TDD) to perform DRX Configuration for E-UTRAN - inter-frequency handover and E-UTRAN intra-frequency cell search with DRX_S

**Table H.3.6-1: *MAC-MainConfig-RBC*: E-UTRAN inter frequency handover and E-UTRAN intra frequency cell search with DRX_S**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | DRX_S |
| setup SEQUENCE { | | | |
| onDurationTimer | psf2 | | |
| drx-InactivityTimer | psf100 | | |
| drx-RetransmissionTimer | psf16 | | |
| longDRX-CycleStartOffset CHOICE { | | sf40 typical value in real network for best-effort services. | |
| sf40 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| } | | | |

MAC-MainConfig-RBC: (FDD/TDD) to perform DRX Configuration for E-UTRAN - inter-frequency handover and E-UTRAN intra-frequency cell search with DRX_L

**Table H.3.6-2: *MAC-MainConfig-RBC*: E-UTRAN inter frequency handover and E-UTRAN intra frequency cell search with DRX_L**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | DRX_L |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf6 | | |
| drx-InactivityTimer | psf1920 | | |
| drx-RetransmissionTimer | psf16 | | |
| longDRX-CycleStartOffset CHOICE { | | sf1280 typical value in real network for best-effort services. | |
| sf1280 | 0 | | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| } | | | |

## H.3.7 RRC messages and information elements contents exceptions for E-UTRAN inter frequency and E-UTRAN inter-RAT cell search when DRX is used

MAC-MainConfig-RBC: (FDD/TDD) to perform DRX Configuration for E-UTRAN - inter-frequency and E-UTRAN inter-RAT cell search when DRX = 40 ms

**Table H.3.7-1: *MAC-MainConfig-RBC*: E-UTRAN inter frequency cell search and E-UTRAN intra frequency cell search when DRX = 40 ms**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf40 typical value in real network for real-time services. | |
| sf40 | 9 | To avoid overlapping with measurement gap. | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | sf500 | | |
| } | | | |

MAC-MainConfig-RBC: (FDD/TDD) to perform DRX Configuration for E-UTRAN - inter-frequency and E-UTRAN inter-RAT cell search when DRX = 1280 ms

**Table H.3.7-2: *MAC-MainConfig-RBC*: E-UTRAN inter frequency and E-UTRAN inter-RAT cell search when DRX = 1280 ms**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n5 | | |
| periodicBSR-Timer | sf20 | | |
| retxBSR-Timer | sf1280 | | |
| ttiBundling | FALSE | | |
| } | | | |
| drx-Config CHOICE { | | | |
| Setup SEQUENCE { | | | |
| onDurationTimer | psf1 | | |
| drx-InactivityTimer | psf1 | | |
| drx-RetransmissionTimer | psf1 | | |
| longDRX-CycleStartOffset CHOICE { | | sf1280 typical value in real network for best-effort services. | |
| sf1280 | 9 | To avoid overlapping with measurement gap. | |
| } | | | |
| shortDRX | Not present | | |
| } | | | |
| } | | | |
| timeAlignmentTimerDedicated | sf500 | | |
| } | | | |

PhysicalConfigDedicated-DEFAULT: (FDD/TDD) to perform DRX Configuration for E-UTRAN - inter-frequency and E-UTRAN inter-RAT cell search

**Table H.3.7-3: *PhysicalConfigDedicated-DEFAULT*: E-UTRAN inter frequency and E-UTRAN inter-RAT cell search**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.6, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| schedulingRequestConfig | SchedulingRequest-Config-DEFAULT | | |
| } | | | |

## H.3.8 RRC messages and information elements contents exceptions for E-UTRAN CSI-RSRP Accuracy

MeasConfig: (FDD/TDD) to perform Measurement Configuration for E-UTRAN intra frequency CSI-RSRP

**Table H.3.8-1: *MeasConfig-DEFAULT*: E-UTRAN intra frequency CSI-RSRP Accuracy**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 1 entry | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) using condition DS_Meas | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL-CSI-RS | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f1 | | |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

MeasConfig: (FDD/TDD) to perform Measurement Configuration for E-UTRAN inter frequency CSI-RSRP

**Table H.3.8-2: *MeasConfig-DEFAULT*: E-UTRAN inter frequency CSI-RSRP Accuracy**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entries | | |
| MeasObjectToAddMod[1] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the serving cell | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) using condition DS_Meas | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[2] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the neighbouring cell (inter frequency cell) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) using condition DS_Meas | Inter frequency cell | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL-CSI-RS | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 2 entries | | |
| measIdToAddMod[0] SEQUENCE { | | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f1 | | |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| measIdToAddMod[1] SEQUENCE { | | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | MeasGapConfig-GP1 | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

MeasConfig: (FDD/TDD) to perform Measurement Configuration for E-UTRAN Carrier Aggregation CSI-RSRP

**Table H.3.8-3: *MeasConfig-DEFAULT*: E-UTRAN Carrier Aggregation CSI-RSRP Accuracy**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entries | | |
| MeasObjectToAddMod[1] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the PCell | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) using condition DS_Meas | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[2] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the SCell | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) using condition DS_Meas | | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL-CSI-RS | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 2 entries | | |
| measIdToAddMod[0] SEQUENCE { | | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f1 | | |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| measIdToAddMod[1] SEQUENCE { | | | |
| measId | 2 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

ReportConfigEUTRA-PEROIDICAL-CSI-RS: (FDD/TDD) for periodical configuration reporting of E-UTRAN CSI-RSRP accuracy

**Table H.3.8-4: *ReportConfigEUTRA-PERIODICAL-CSI-RS*: E-UTRAN CSI-RSRP Accuracy**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-12 ReportConfigEUTRA-PERIODICAL-CSI-RS | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-PERIODICAL ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| periodical SEQUENCE { | | | |
| purpose | reportStrongestCells | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | sameAsTriggerQuantity | | |
| maxReportCells | 2 | | |
| reportInterval | ms1024 | | |
| reportAmount | infinity | | |
| si-RequestForHO-r9 | Not Present | | |
| ue-RxTxTimeDiffPeriodical-r9 | Not Present | | |
| includeLocationInfo-r10 | Not Present | | |
| reportAddNeighMeas-r10 | Not Present | | |
| alternativeTimeToTrigger-r12 | Not Present | | |
| useT312-r12 | Not Present | | |
| usePSCell-r12 | Not Present | | |
| aN-Threshold1-v1250 | Not Present | | |
| a5-Threshold2-v1250 | Not Present | | |
| reportStrongestCSI-RSs-r12 | true | | |
| reportCRS-Meas-r12 | false | | |
| triggerQuantityCSI-RS-r12 | true | | |
| } | | | |

QuantityConfig-DEFAULT (FDD/TDD) E-UTRAN measurement configuration for E-UTRAN CSI-RSRP accuracy

**Table H.3.8-5: QuantityConfig-DEFAULT: measurement configuration for E-UTRAN CSI-RSRP accuracy**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-3A: QuantityConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| QuantityConfig-DEFAULT ::= SEQUENCE { | | | |
| quantityConfigEUTRA SEQUENCE { | | | |
| filterCoefficientRSRP | fc0 | | |
| filterCoefficientRSRQ | fc0 | | |
| } | | | |
| quantityConfigEUTRA-v1250 SEQUENCE { | | | |
| filterCoefficientCSI-RSRP-r12 | fc0 | | |
| } | | | |
| } | | | |

MeasGapConfig-GP1 (FDD/TDD) E-UTRAN measurement gap configuration for E-UTRAN CSI-RSRP accuracy

**Table H.3.8-6: MeasGapConfig-GP1: measurement configuration for E-UTRAN CSI-RSRP accuracy**

| Derivation Path: 36.508 [7] clause 4.6.6, Table 4.6.6-1A: MeasGapConfig-GP1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasGapConfig-GP1 ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| gapOffset CHOICE { | | | |
| gp0 | 9 | TGRP = 40 ms | |
| } | | | |
| } | | | |
| } | | | |

# H.4 Default RRC messages and information elements contents exceptions for Carrier Aggregation

This clause contains the default values of common RRC messages and information elements for Carrier Aggregation, other than those described in TS 36.508 [7].

## H.4.1 RRC messages and information elements contents exceptions for E-UTRAN measurement configuration for CA

RRCConnectionReconfiguration: (FDD/TDD) to setup E-UTRAN Measurement Configuration for CA

**Table H.4.1-1: *Void***

MeasConfig-DEFAULT: (FDD/TDD) E-UTRAN Measurement Configuration for Event Triggered Reporting for CA

**Table H.4.1-2: *MeasConfig-DEFAULT*: E-UTRAN Measurement Configuration for CA**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entries | | |
| MeasObjectToAddMod[1] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the PCell (Cell 1) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[2] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the SCell on the SCC (Cell 2)and neighbouring cell on the SCC (Cell 3) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-A6 | | |
| reportConfig | ReportConfigEUTRA-A6 | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | | |
| reportConfigId | idReportConfig-A6 | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

MeasObjectEUTRA-GENERIC: (FDD/TDD) E-UTRAN Measurement Configuration for Event Triggered Reporting for CA

**Table H.4.1-3: *MeasObjectEUTRA-GENERIC(Freq)*: E-UTRAN Measurement Configuration for CA with SCell measurement cycle 1280 ms)**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-2 MeasObjectEUTRA-GENERIC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasObjectEUTRA-GENERIC(Freq) ::= SEQUENCE { | | | |
| carrierFreq | Downlink EARFCN for Freq | | |
| allowedmeasBandwidth | The number of the resource blocks for Freq | | |
| presenceAntennaPort1 | FALSE | | |
| neighbourCellConfig | '01'B (No MBSFN subframes are present in all neighbour cells) | MBSFN doesn't apply by default. | TDD |
| | 10'B (The MBSFN subframe allocations of all neighbour cells are identical to or subsets of that in the serving cell) | | FDD |
| offsetFreq | 0 (dB 0) | | |
| cellsToRemoveList | Not present | | |
| cellsToAddModList | Not present | | |
| blackCellsToRemoveList | Not present | | |
| blackCellsToAddModList | Not present | | |
| cellForWhichToReportCGI | Not present | | |
| measCycleSCell-r10 | sf1280 | | |
| measSubframePatternConfigNeigh-r10 | Not present | | |
| } | | | |

**Table H.4.1-4: Void**

MeasurementReport: (FDD/TDD) E-UTRAN Measurement Report for CA and DC **without NeighCells**

**Table H.4.1-5: MeasurementReport: E-UTRAN Measurement Report for CA and DC**

| Derivation path: TS 36.508 [7] clause 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell::= SEQUENCE { | | PCell | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| MeasResultEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | Neighbour cell | |
| physCellId | physCellId of Neighbour Cell | | |
| cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | SCell | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table H.4.1-5a: MeasurementReport: E-UTRAN Measurement Report for CA and DC without Neigh Cells**

| Derivation path: TS 36.508 [7] clause 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell::= SEQUENCE { | | PCell | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | SCell | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

ReportConfigEUTRA-A6: (FDD/TDD) E-UTRAN Measurement Report Configuration for Event A6 for CA

**Table H.4.1-6: ReportConfig-A6: E-UTRAN Report config for Event A6 for CA**

| Derivation Path: TS 36.508 [7] clause 4.6.6 table 4.6.6-6A | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ReportConfigEUTRA-A6 ::= SEQUENCE { | | | |
| triggerType CHOICE { | | | |
| event SEQUENCE { | | | |
| eventId CHOICE { | | | |
| eventA6-r10 SEQUENCE { | | | |
| a6-Offset-r10 | -6 (-3 dB) | -3 is actual value in dB (-6 * 0.5 dB) | A6-Offset = -3dB |
| | -12 (-6 dB) | -6 is actual value in dB (-12 * 0.5 dB) | A6-Offset = -6dB |
| a6-ReportOnLeave-r10 | FALSE | | |
| } | | | |
| } | | | |
| Hysteresis | 0 (0 dB) | 0 is actual value in dB (0 * 0.5 dB) | |
| timeToTrigger | ms0 | | |
| } | | | |
| } | | | |
| triggerQuantity | rsrp | | |
| reportQuantity | both | | |
| maxReportCells | 1 | | |
| reportInterval | ms5120 | | |
| reportAmount | r1 | | |
| } | | | |

MeasurementReport: (FDD/TDD) E-UTRAN RSRP Measurement Report for 3CA

**Table H.4.1-7: MeasurementReport: E-UTRAN RSRP Measurement Report for 3CA**

| Derivation path: TS 36.508 [7] clause 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell::= SEQUENCE { | | PCell | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| MeasResultListEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | Neighbour cell | |
| MeasResultEUTRA SEQUENCE { | | | |
| physCellId | physCellId of Neighbour Cell | | |
| cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | 2 entries | | |
| MeasResultServFreq-r10[1] { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| MeasResultServFreq-r10[2] { | | | |
| servFreqId-r10 | 2 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 4 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

MeasurementReport: (FDD/TDD) E-UTRAN RSRQ Measurement Report for 3CA

**Table H.4.1-8: MeasurementReport: E-UTRAN RSRQ Measurement Report for 3CA**

| Derivation path: TS 36.508 [7] clause 4.6.1 table 4.6.1-5 | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell::= SEQUENCE { | | PCell | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | 2 entries | | |
| MeasResultServFreq-r10[1] { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 2 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| MeasResultServFreq-r10[2] { | | | |
| servFreqId-r10 | 2 | | |
| measResultSCell-r10 SEQUENCE { | | SCell Cell 3 | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## H.4.2 RRC messages and information elements contents exceptions for E-UTRAN RSRP and RSRQ Accuracy for CA

MeasConfig: (FDD/TDD) to perform Measurement Configuration for E-UTRAN RSRP and RSRQ accuracy for CA

**Table H.4.2-1: *MeasConfig-DEFAULT*: E-UTRAN RSRP and RSRQ Accuracy for CA**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | 2 entries | | |
| MeasObjectToAddMod[1] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the PCell (Cell 1) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the PCell (Cell 1) | Switch PCell/SCell scenario |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | | |
| } | | | |
| } | | | |
| MeasObjectToAddMod[2] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency of the SCell on the SCC (Cell 2) and neighbouring cell on the SCC(Cell 3) | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f2) | | |
| } | | | |
| measObjectId | IdMeasObject-f1 | f1 is the frequency of the SCell on the SCC (Cell 2) and neighbouring cell on the SCC(Cell 3) | Switch PCell/SCell scenario |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f1) | | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entry | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f2 | f2 is the frequency | |

| | | | |
|---|---|---|---|
| | | of the SCell on the SCC (Cell 2) and neighbouring cell on the SCC (Cell 3) | |
| | IdMeasObject-f1 | f1 is the frequency of the SCell on the SCC (Cell 2) and neighbouring cell on the SCC (Cell 3) | Switch PCell/SCell scenario |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

ReportConfigEUTRA-PEROIDICAL: (FDD/TDD) for periodical configuration reporting of E-UTRAN RSRP accuracy for CA

**Table H.4.2-2: Void**

ReportConfigEUTRA-PEROIDICAL: (FDD/TDD) for periodical configuration reporting of E-UTRAN RSRQ accuracy for CA

**Table H.4.2-3: Void**

MeasurementReport: (FDD/TDD) for periodical configuration reporting of E-UTRAN RSRP and RSRQ accuracy for CA

**Table H.4.2-4: Void**

MeasurementReport: (FDD/TDD) for periodical configuration reporting of E-UTRAN RSRP accuracy for 3CA

**Table H.4.2-5: *MeasConfig-DEFAULT*: E-UTRAN RSRP Accuracy for 3CA**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | N entries | N is E-UTRA RF Channel Number used in the test | |
| MeasObjectToAddMod[n] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f(n) | f(n), n=1,2,…N is the frequency of the PCell and Scells according to Annex E | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f(n)) | | |
| } | | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | N entries | N is E-UTRA RF Channel Number used in the test | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | n | n denotes the index of the entry. n=1,2,…N | |
| measObjectId | IdMeasObject-f(n) | | |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

MeasurementReport: (FDD/TDD) for periodical configuration reporting of E-UTRAN RSRQ accuracy for 3CA

**Table H.4.2-6: *MeasConfig-DEFAULT*: E-UTRAN RSRQ Accuracy for 3CA**

| Derivation Path: TS 36.508 [7] clause 4.6.6, Table 4.6.6-1 MeasConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MeasConfig-DEFAULT ::= SEQUENCE { | | | |
| measObjectToRemoveList | Not present | | |
| measObjectToAddModList SEQUENCE (SIZE (1..maxObjectId)) OF SEQUENCE { | N entries | N is E-UTRA RF Channel Number used in the test | |
| MeasObjectToAddMod[n] SEQUENCE { | | | |
| measObjectId | IdMeasObject-f(n) | f(n) , n=1,2,…N is the frequency of the PCell and Scells according to Annex E | |
| measObject CHOICE { | | | |
| MeasObjectEUTRA | MeasObjectEUTRA-GENERIC(f(n)) | | |
| } | | | |
| } | | | |
| reportConfigToRemoveList | Not present | | |
| reportConfigToAddModList SEQUENCE (SIZE (1..maxReportConfigId) )OF SEQUENCE { | 1 entry | | |
| reportConfigId | idReportConfig-P | | |
| reportConfig | ReportConfigEUTRA-PERIODICAL | | |
| } | | | |
| measIdToRemoveList | Not present | | |
| measIdToAddModList SEQUENCE (SIZE (1..maxMeasId)) of SEQUENCE { | 1 entries | | |
| measIdToAddMod ::= SEQUENCE { | | | |
| measId | 1 | | |
| measObjectId | IdMeasObject-f(1) | | |
| reportConfigId | idReportConfig-P | | |
| } | | | |
| quantityConfig | QuantityConfig-DEFAULT | | |
| measGapConfig | Not present | | |
| s-Measure | Not present | | |
| preRegistrationInfoHRPD | Not present | | |
| speedStatePars | Not present | | |
| } | | | |

# H.5 Default RRC messages and information elements contents exceptions for FeICIC

This clause contains the default values of common RRC messages and information elements for FeICIC, other than those described in TS 36.508 [7].

## H.5.1 RRC messages and information elements contents exceptions for E-UTRAN measurement configuration for FeICIC

RadioResourceConfigDedicated-SRB2-DRB(n, m): (FDD/TDD) E-UTRAN measurement configuration for NeighCells-Info-r11

**Table H.5.1-1: *RadioResourceConfigDedicated-SRB2-DRB(n, m)*: NeighCellsCRS-Info**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| neighCellsCRS-Info-r11 ::= CHOICE { | | | |
| NeighCellsCRS-Info-r11 ::= CHOICE { | | | |
| setup | CRS-AssistanceInfoList-r11 | | |
| } | | | |
| } | | | |

RadioResourceConfigDedicated-SRB2-DRB(n, m): (FDD/TDD) E-UTRAN measurement configuration for CRS-AssistancecInfoList-r11

**Table H.5.1-2: *RadioResourceConfigDedicated-SRB2-DRB(n, m)*: CRS-AssistanceInfoList-r11**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CRS-AssistancedInfoList-r11 ::= SEQUENCE (SIZE (1..maxCellReport)) OF CRS-AssistanceInfo-r11 | | | |
| CRS-AssistancedInfo-r11 ::= SEQUENCE { | | | |
| physCellId-r11 | ($PCI_{cell1}$ - $PCI_{cell3}$ )mod6 = 0<br>($PCI_{cell2}$ - $PCI_{cell3}$ )mod6 != 0 | Cell PCIs are selected so that both conditions are met | |
| antennaPortsCount-r11 | an1 | | |
| mbsfn-SubframeConfigList-r11 | MBSFN-SubframeConfigList | | |
| } | | | |

RadioResourceConfigDedicated-SRB2-DRB(n, m): (FDD/TDD) E-UTRAN measurement configuration for MBSFN-SubframeConfigList

**Table H.5.1-3: *RadioResourceConfigDedicated-SRB2-DRB(n, m)*: MBSFN-SubframeConfigList**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-16 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MBSFN-SubframeConfigList ::= SEQUENCE (SIZE (1..maxMBSFN-Allocations)) OF MBSFN-SubframeConfig | | | |
| MBSFN-SubframeConfig:: = SEQUENCE { | | | |
| radioframeAllocationPeriod | n1 | | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| oneFrame | '000000' | Only the CRS information of Cell 2 is provided in CRS-AssistanceInfo. It includes a single MBSFN-SubframeConfig element with subframe allocation *oneFrame*='000000' BIT STRING (SIZE(6)) | |
| } | | | |
| } | | | |

## H.5.2 RRC messages and information elements contents exceptions for E-UTRAN RSRP and RSRQ Accuracy for FeICIC

MeasurementReport: (FDD/TDD) for periodical configuration reporting of E-UTRAN RSRP and RSRQ accuracy for FeICIC

**Table H.5.2-1: *MeasurementReport*: E-UTRAN RSRP and RSRQ Accuracy for FeICIC**

| Derivation path: TS 36.508 [7] clause 4.6.1 Table 4.6.1-5 MeasurementReport | | | |
|---|---|---|---|
| **Information Element** | **Value/Remark** | **Comment** | **Condition** |
| MeasurementReport ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| measurementReport-r8 SEQUENCE { | | | |
| measResults ::= SEQUENCE { | | | |
| measId | 1 | | |
| measResultPCell::= SEQUENCE { | | PCell | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| measResultNeighCells CHOICE { | | | |
| MeasResultEUTRA ::= SEQUENCE (SIZE (1..maxCellReport)) OF SEQUENCE { | | | |
| MeasResultEUTRA SEQUENCE { | | | |
| physCellId | physCellId of best Cell | | |
| cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| MeasResultEUTRA SEQUENCE { | | | |
| physCellId | physCellId of 2nd best Cell | | |
| cgi-Info | Not present | | |
| measResult SEQUENCE { | | | |
| rsrpResult | (0..97) | Set according to specific test | |
| rsrqResult | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| measResultForECID-r9 | Not present | | |
| locationInfo-r10 | Not present | | |
| measResultServFreqList-r10 SEQUENCE (SIZE (1..maxServCell-r10)) OF SEQUENCE { | | | |
| servFreqId-r10 | 1 | | |
| measResultSCell-r10 SEQUENCE { | | SCell | |
| rsrpResultSCell-r10 | (0..97) | Set according to specific test | |
| rsrqResultSCell-r10 | (0..34) | Set according to specific test | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |

# H.6 Default RRC messages and information elements contents exceptions for Proximity based services

This clause contains the default values of common RRC messages and information elements for Proximity based services, other than those described in TS 36.508 [7].

# H.6.1 RRC messages and information elements contents exceptions for E-UTRAN measurement configuration for Proximity based services

## H.6.1.1 Reference resource pool configurations for ProSe Direct Discovery

**Table H.6.1.1-1: ProSe Direct Discovery configuration for E-UTRA FDD (Configuration #1)**

| Information Element | | | | Value |
|---|---|---|---|---|
| discRxPool | cp-Len | | | Normal |
| | discPeriod | | | rf32 |
| | numRetx | | | 0 |
| | numRepetition | | | 1 |
| | tf-ResourceConfig | prb-Num | | 12 |
| | | prb-Start | | 0 |
| | | prb-End | | 23 |
| | | offsetIndicator | | 160 |
| | | subframeBitmap | | 11000000 00000000 00000000 00000000 00000000 |
| | txParameters | | | not present |
| | rxParameters | | | not present |
| discTxPoolCommon | cp-Len | | | Normal |
| | discPeriod | | | rf32 |
| | numRetx | | | 0 |
| | numRepetition | | | 1 |
| | tf-ResourceConfig | prb-Num | | 2 |
| | | prb-Start | | 3 |
| | | prb-End | | 5 |
| | | offsetIndicator | | 160 |
| | | subframeBitmap | | 10000000 00000000 00000000 00000000 00000000 |
| | txParameters | txParametersGeneral | alpha | al0 |
| | | | p0 | 31 |
| | | ue-SelectedResourceConfig | poolSelection | random |
| | | | txProbability | p100 |
| | rxParameters | | | not present |
| discTxPowerInfo | discMaxTxPower | | | 23 |
| SL-SyncConfig | syncCP-Len | | | Normal |
| | syncOffsetIndicator | | | 155 |
| | slssid | | | 30 |
| | txParameters | txParametersGeneral | alpha | al0 |
| | | | p0 | 31 |
| | | syncTxThreshIC | | 0 (-infinity) |
| | rxParamsNCell | | | not present |
| discInterFreqList | | | | not present |

**Table H.6.1.1-2: ProSe Direct Discovery configuration for E-UTRA FDD (Configuration #2)**

| Information Element | | | | Value |
|---|---|---|---|---|
| *discRxPool* | *cp-Len* | | | Normal |
| | *discPeriod* | | | rf32 |
| | *numRetx* | | | 0 |
| | *numRepetition* | | | 1 |
| | *tf-ResourceConfig* | *prb-Num* | | 12 |
| | | *prb-Start* | | 0 |
| | | *prb-End* | | 23 |
| | | *offsetIndicator* | | 160 |
| | | *subframeBitmap* | | 11000000<br>00000000<br>00000000<br>00000000<br>00000000 |
| | *txParameters* | | | not present |
| | *rxParameters* | *tdd-Config* | | not present |
| | | *syncConfigIndex* | | 0 |
| *discTxPoolCommon* | | | | not present |
| *discTxPowerInfo* | *discMaxTxPower* | | | 23 |
| *SL-SyncConfig* | *syncCP-Len* | | | Normal |
| | *syncOffsetIndicator* | | | 140 |
| | *slssid* | | | 30 |
| | *txParameters* | | | not present |
| | *rxParamsNCell* | *physCellId* | | 1 |
| | | *discSyncWindow* | | w1 |
| *discInterFreqLis* | | | | not present |

**Table H.6.1.1-3: ProSe Direct Discovery configuration for E-UTRA TDD Config 0 (Configuration #3)**

| Information Element | | | | Value |
|---|---|---|---|---|
| *discRxPool* | *cp-Len* | | | Normal |
| | *discPeriod* | | | rf32 |
| | *numRetx* | | | 0 |
| | *numRepetition* | | | 1 |
| | *tf-ResourceConfig* | *prb-Num* | | 12 |
| | | *prb-Start* | | 0 |
| | | *prb-End* | | 23 |
| | | *offsetIndicator* | | 163 |
| | | *subframeBitmap* | | 11000000 00000000 00000000 00000000 00000000 00 |
| | *txParameters* | | | not present |
| | *rxParameters* | | | not present |
| *discTxPoolCommon* | *cp-Len* | | | Normal |
| | *discPeriod* | | | rf32 |
| | *numRetx* | | | 0 |
| | *numRepetition* | | | 1 |
| | *tf-ResourceConfig* | *prb-Num* | | 2 |
| | | *prb-Start* | | 3 |
| | | *prb-End* | | 5 |
| | | *offsetIndicator* | | 163 |
| | | *subframeBitmap* | | 10000000 00000000 00000000 00000000 00000000 00 |
| | *txParameters* | *txParametersGeneral* | *alpha* | al0 |
| | | | *p0* | 31 |
| | | *ue-SelectedResourceConfig* | *poolSelection* | random |
| | | | *txProbability* | p100 |
| | *rxParameters* | | | not present |
| *discTxPowerInfo* | *discMaxTxPower* | | | 23 |
| *SL-SyncConfig* | *syncCP-Len* | | | Normal |
| | *syncOffsetIndicator* | | | 158 |
| | *slssid* | | | 30 |
| | *txParameters* | *txParametersGeneral* | *alpha* | al0 |
| | | | *p0* | 31 |
| | | *syncTxThreshIC* | | 0 (-infinity) |
| | *rxParamsNCell* | | | not present |
| *discInterFreqList* | | | | not present |

## H.6.1.2 Reference resource pool configurations for ProSe Direct Communication

**Table H.6.1.2-1: ProSe Direct Communication configuration for E-UTRA FDD (Configuration #1)**

| Information Element | | | | Value (5MHz) | Value (10MHz) |
|---|---|---|---|---|---|
| *commRxPool* | *sc-CP-Len* | | | Normal | |
| | *sc-Period* | | | sf40 | |
| | *sc-TF-ResourceConfig* | *prb-Num* | | 12 | 25 |
| | | *prb-Start* | | 0 | 0 |
| | | *prb-End* | | 23 | 49 |
| | | *offsetIndicator* | | 0 | |
| | | *subframeBitmap* | | 00011000 00000000 00000000 00000000 00000000 | |
| | *data-CP-Len* | | | Normal | |
| | *dataHoppingConfig* | *hoppingParameter* | | 0 | |
| | | *numSubbands* | | ns1 | |
| | | *rb-Offset* | | 0 | |
| | *ue-SelectedResourceConfig* | *data-TF-ResourceConfig* | *prb-Num* | 12 | 25 |
| | | | *prb-Start* | 0 | 0 |
| | | | *prb-End* | 23 | 49 |
| | | | *offsetIndicator* | 0 | |
| | | | *subframeBitmap* | 00000000 11111111 11111111 11111111 11111111 | |
| | | *trpt-Subset-r12* | | 001 | |
| | *rxParametersNCell* | | | not present | |
| | *txParameters* | | | not present | |
| *commTxPoolNormalCommon* | *sc-CP-Len* | | | Normal | |
| | *sc-Period* | | | sf40 | |
| | *sc-TF-ResourceConfig* | *prb-Num* | | 12 | 25 |
| | | *prb-Start* | | 0 | 0 |
| | | *prb-End* | | 24 | 49 |
| | | *offsetIndicator* | | 0 | |
| | | *subframeBitmap* | | 00011000 00000000 00000000 00000000 00000000 | |
| | *data-CP-Len* | | | Normal | |
| | *dataHoppingConfig* | *hoppingParameter* | | 0 | |
| | | *numSubbands* | | ns1 | |
| | | *rb-Offset* | | 0 | |
| | *ue-SelectedResourceConfig* | *data-TF-ResourceConfig* | *prb-Num* | 12 | 25 |
| | | | *prb-Start* | 0 | 0 |
| | | | *prb-End* | 23 | 49 |
| | | | *offsetIndicator* | 0 | |
| | | | *subframeBitmap* | 00000000 11111111 11111111 11111111 11111111 | |
| | | *trpt-Subset-r12* | | 001 | |
| | *rxParametersNCell* | | | not present | |
| | *txParameters* | *sc-TxParameters* | *alpha* | al0 | |
| | | | *p0* | 31 | |
| | | *dataTxParameters* | *alpha* | al0 | |

| | | | | |
|---|---|---|---|---|
| | | | *p0* | 31 |
| *SL-SyncConfig* | *syncCP-Len* | | | Normal |
| | *syncOffsetIndicator* | | | 2 |
| | *slssid* | | | 30 |
| | *txParameters* | *txParametersGeneral* | *alpha* | al0 |
| | | | *p0* | 31 |
| | | *syncTxThreshIC* | | 0 (-infinity) |
| | *rxParamsNCell* | | | not present |

**Table H.6.1.2-2: ProSe Direct Communication pre-configuration for E-UTRAN FDD for out-of-network coverage operation (Configuration #2)**

| **Information Element** | | | | **Value (5MHz)** | **Value (10MHz)** |
|---|---|---|---|---|---|
| *preconfigSync* | *syncCP-Len-r12* | | | Normal | |
| | *syncOffsetIndicator1* | | | 2 | |
| | *syncOffsetIndicator2* | | | 1 | |
| | *syncTxParameters* | | | 31 | |
| | *syncTxThreshOoC* | | | 0 (-110dBm / 15kHz) | |
| | *filterCoefficient* | | | fc0 | |
| | *syncRefMinHyst* | | | dB0 | |
| | *syncRefDiffHyst* | | | dB0 | |
| *preconfigComm* | *sc-CP-Len* | | | Normal | |
| | *sc-Period* | | | sf40 | |
| | *sc-TF-ResourceConfig* | *prb-Num* | | 12 | 25 |
| | | *prb-Start* | | 0 | 0 |
| | | *prb-End* | | 23 | 49 |
| | | *offsetIndicator* | | 0 | |
| | | *subframeBitmap* | | 00011000 00000000 00000000 00000000 00000000 | |
| | *data-CP-Len* | | | Normal | |
| | *dataHoppingConfig* | *hoppingParameter* | | 0 | |
| | | *numSubbands* | | ns1 | |
| | | *rb-Offset* | | 0 | |
| | *ue-SelectedResourceConfig* | *data-TF-ResourceConfig* | *prb-Num* | 12 | 25 |
| | | | *prb-Start* | 0 | 0 |
| | | | *prb-End* | 23 | 49 |
| | | | *offsetIndicator* | 0 | |
| | | | *subframeBitmap* | 00000000 11111111 11111111 11111111 11111111 | |
| | | *trpt-Subset-r12* | | 001 | |

# Annex I (normative): Conditions for RRM requirements applicability for operating bands

## I.1 Conditions for E-UTRAN RRC_IDLE state mobility

### I.1.1 Conditions for measurements of intra-frequency E-UTRAN cells for cell re-selection

This section defines the E-UTRAN intra-frequency RSRP, RSRP Ês/Iot, SCH_RP and SCH Ês/Iot applicable for a corresponding operating band.

The conditions for measurements of intra-frequency E-UTRAN cells for cell re-selection are defined in Table I.1.1-1

**Table I.1.1-1: Conditions for measurements of intra-frequency E-UTRAN cells for cell re-selection**

| Parameter | E-UTRA operating band groups Note 3 | Minimum RSRP Note 1 | Minimum SCH_RP Note 1 | RSRP Ês/Iot | SCH Ês/Iot |
|---|---|---|---|---|---|
| | | dBm/15kHz | dBm/15kHz | dB | dB |
| Conditions | FDD_A, TDD_A | -124 | -124 | ≥ -4 | ≥ -4 |
| | FDD_B | -123.5 | -123.5 | | |
| | FDD_C, TDD_C | -123 | -123 | | |
| | FDD_D | -122.5 | -122.5 | | |
| | FDD_E, TDD_E | -122 | -122 | | |
| | FDD_F | -121.5 Note 2 | -121.5 Note 2 | | |
| | FDD_G | -121 | -121 | | |
| | FDD_H | -120.5 | -120.5 | | |
| | FDD_N | -117.5 | -117.5 | | |

NOTE 1: This condition level is increased by Δ>0, when applicable, as described in Section I.4.2.
NOTE 2: The condition is -122 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

### I.1.2 Conditions for measurements of inter-frequency E-UTRAN cells for cell re-selection

This section defines the E-UTRAN inter-frequency RSRP, RSRP Ês/Iot, SCH_RP and SCH Ês/Iot applicable for a corresponding operating band.

The conditions for measurements of intra-frequency E-UTRAN cells for cell re-selection defined in Table I.1.1-1 also apply for inter-frequency E-UTRAN cells in this section.

## I.2 Conditions for UE Measurements Procedures in RRC_CONNECTED State

### I.2.1 Conditions for E-UTRAN intra-frequency measurements

This section defines the E-UTRAN intra-frequency SCH_RP and SCH Ês/Iot applicable for a corresponding operating band

The conditions for intra-frequency E-UTRAN measurements are defined in Table I.2.1-1

**Table I.2.1-1: E-UTRAN intra-frequency measurements**

| Parameter | E-UTRA operating band groups Note 3 | Minimum SCH_RP Note 1 | SCH Ês/Iot |
|---|---|---|---|
| | | dBm/15kHz | dB |
| Conditions | FDD_A, TDD_A | -127 | ≥ -6 |
| | FDD_B | -126.5 | |
| | FDD_C, TDD_C | -126 | |
| | FDD_D | -125.5 | |
| | FDD_E, TDD_E | -125 | |
| | FDD_F | -124.5 Note 2 | |
| | FDD_G | -124 | |
| | FDD_H | -123.5 | |
| | FDD_N | -120.5 | |

NOTE 1: This condition level is increased by Δ>0, when applicable, as described in Sections I.4.2.
NOTE 2: The condition is -125 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

## I.2.2 Conditions for E-UTRAN intra-frequency measurements with autonomous gaps

This section defines the E-UTRAN intra-frequency SCH_RP and SCH Ês/Iot applicable for a corresponding operating band

The conditions for intra-frequency E-UTRAN measurements with autonomous gap are defined in Table I.2.1-1

**Table I.2.2-1: Void**

## I.2.3 Conditions for E-UTRAN inter-frequency measurements

This section defines the E-UTRAN inter-frequency SCH_RP, SCH Ês/Iot, RSRP and RSRP Ês/Iot applicable for a corresponding operating band

The conditions for inter-frequency E-UTRAN measurements with autonomous gap are defined in Table I.2.3-1

**Table I.2.3-1: E-UTRAN inter-frequency measurements**

| Parameter | E-UTRA operating band groups Note 3 | Minimum RSRP Note 1 | Minimum SCH_RP Note 1 | RSRP Ês/Iot | SCH Ês/Iot |
|---|---|---|---|---|---|
| | | dBm/15kHz | dBm/15kHz | dB | dB |
| Conditions | FDD_A, TDD_A | -125 | -125 | ≥ -4 | ≥ -4 |
| | FDD_B | -124.5 | -124.5 | | |
| | FDD_C, TDD_C | -124 | -124 | | |
| | FDD_D | -123.5 | -123.5 | | |
| | FDD_E, TDD_E | -123 | -123 | | |
| | FDD_F | -122.5 Note 2 | -122.5 Note 2 | | |
| | FDD_G | -122 | -122 | | |
| | FDD_H | -121.5 | -121.5 | | |
| | FDD_N | -118.5 | -118.5 | | |

NOTE 1: This condition level is increased by Δ>0, when applicable, as described in Sections B.4.2 and B.4.3.
NOTE 2: The condition is -123 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

## I.2.4 Conditions for E-UTRAN inter-frequency measurements with autonomous gaps

This section defines the E-UTRAN inter-frequency SCH_RP and SCH Ês/Iot applicable for a corresponding operating band.

The conditions for inter-frequency E-UTRAN measurements with autonomous gap are defined in Table I.2.4-1.

**Table I.2.4-1: E-UTRAN inter-frequency measurements with autonomous gaps**

| Parameter | E-UTRA operating band groups Note 3 | Minimum SCH_RP Note 1 | SCH Ês/Iot |
|---|---|---|---|
| | | dBm/15kHz | dB |
| Conditions | FDD_A, TDD_A | -125 | ≥ -4 |
| | FDD_B | -124.5 | |
| | FDD_C, TDD_C | -124 | |
| | FDD_D | -123.5 | |
| | FDD_E, TDD_E | -123 | |
| | FDD_F | -122.5 Note 2 | |
| | FDD_G | -122 | |
| | FDD_H | -121.5 | |
| | FDD_N | -118.5 | |

NOTE 1: This condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 2: The condition is -123 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

## I.2.5 Void

## I.2.6 Void

## I.2.7 Conditions for Measurements of the secondary component carrier with deactivated SCell

This section defines the SCH_RP and SCH Ês/Iot for measurements in the secondary component carrier applicable for a corresponding operating band

The conditions for measurements of the secondary component carrier with deactivated SCell are defined in Table I.2.7-1.

**Table I.2.7-1: Measurements of the secondary component carrier with deactivated SCell**

| Parameter | E-UTRA operating band groups Note 3 | Minimum SCH_RP Note 1 | SCH Ês/Iot |
|---|---|---|---|
| | | dBm/15kHz | dB |
| Conditions | FDD_A, TDD_A | -127 | ≥ -6 |
| | FDD_B | -126.5 | |
| | FDD_C, TDD_C | -126 | |
| | FDD_D | -125.5 | |
| | FDD_E, TDD_E | -125 | |
| | FDD_F | -124.5 Note 2 | |
| | FDD_G | -124 | |
| | FDD_H | -123.5 | |
| | FDD_N | -120.5 | |

NOTE 1: This condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 2: The condition is -125 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

## I.2.8 Conditions for E-UTRAN Intra-Frequency Measurements under Time Domain Measurement Resource Restriction

This section defines the E-UTRAN intra-frequency SCH_RP and SCH Ês/Iot applicable for a corresponding operating band.

The conditions for intra-frequency E-UTRAN measurements under time domain measurement resource restriction are defined in Table I.2.8-1.

**Table I.2.8-1 E-UTRAN intra-frequency measurements under time domain measurement resource restriction**

| Parameter | E-UTRA operating band groups Note 3 | Minimum SCH_RP Note 1 | SCH Ês/Iot |
|---|---|---|---|
| | | dBm/15kHz | dB |
| Conditions | FDD_A, TDD_A | -127 | ≥ -7.5 |
| | FDD_B | -126.5 | |
| | FDD_C, TDD_C | -126 | |
| | FDD_D | -125.5 | |
| | FDD_E, TDD_E | -125 | |
| | FDD_F | -124.5 Note 2 | |
| | FDD_G | -124 | |
| | FDD_H | -123.5 | |
| | FDD_N | -120.5 | |

NOTE 1: This condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 2: The condition is -125 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

## I.2.9 Conditions for E-UTRAN Intra-Frequency Measurements under Time Domain Measurement Resource Restriction with CRS Assistance Information

This clause defines the E-UTRAN intra-frequency SCH_RP and SCH Ês/Iot applicable for a corresponding operating band.

The conditions for intra-frequency E-UTRAN measurements under time domain measurement resource restriction with CRS assistance information are defined in Table I.2.9-1.

**Table I.2.9-1: E-UTRAN intra-frequency measurements under time domain measurement resource**

| Parameter | E-UTRA operating band groups Note 3 | Minimum SCH_RP Note 1 dBm/15kHz | SCH Ês/Iot dB |
|---|---|---|---|
| Conditions | FDD_A, TDD_A | -127 | ≥ -11.07 |
| | FDD_B | -126.5 | |
| | FDD_C, TDD_C | -126 | |
| | FDD_D | -125.5 | |
| | FDD_E, TDD_E | -125 | |
| | FDD_F | -124.5 Note 2 | |
| | FDD_G | -124 | |
| | FDD_H | -123.5 | |
| | FDD_N | -120.5 | |

NOTE 1: This condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 2: The condition is -125 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

**restriction with CRS assistance information**

# I.2.10 Conditions for E-UTRAN intra-frequency discovery signal measurements

This clause defines the E-UTRAN intra-frequency SCH_RP, SCH Ês/Iot in discovery signal occasions [9], applicable for a corresponding operating band for discovery signal measurements

The conditions for E-UTRAN intra-frequency discovery signal measurements are as in Table I.2.1-1.

# I.2.11 Conditions for E-UTRAN inter-frequency discovery signal measurements

## I.2.11.1 Conditions for E-UTRAN inter-frequency CRS-based measurements

This clause defines the E-UTRAN inter-frequency SCH_RP, SCH Ês/Iot, RSRP, and Ês/Iot in discovery signal occasions [9], applicable for a corresponding operating band for CRS based discovery signal measurements.

The conditions for E-UTRAN inter-frequency CRS-based discovery signal measurements in discovery signal occasions are specified in Table I.2.11.1-1.

**Table I.2.11.1-1: E-UTRAN inter-frequency discovery signal measurements**

| Parameter | E-UTRA operating band groups Note 3 | Minimum RSRP Note 1 | Minimum SCH_RP Note 1 | RSRP Ês/Iot | SCH Ês/Iot |
|---|---|---|---|---|---|
| | | dBm/15kHz | dBm/15kHz | dB | dB |
| Conditions | FDD_A, TDD_A | -125 | -125 | ≥ -6 | ≥ -6 |
| | FDD_B | -124.5 | -124.5 | | |
| | FDD_C, TDD_C | -124 | -124 | | |
| | FDD_D | -123.5 | -123.5 | | |
| | FDD_E, TDD_E | -123 | -123 | | |
| | FDD_F | -122.5 Note 2 | -122.5 Note 2 | | |
| | FDD_G | -122 | -122 | | |
| | FDD_H | -121.5 | -121.5 | | |
| | FDD_N | -118.5 | -118.5 | | |

NOTE 1: This condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 2: The condition is -123 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

### I.2.11.2 Conditions for E-UTRAN inter-frequency CSI-RS based measurements

This clause defines the E-UTRAN inter-frequency SCH_RP, SCH Ês/Iot, CSI-RSRP, and CSI-RS Ês/Iot in discovery signal occasions [9], applicable for a corresponding operating band for CSI-RS based discovery signal measurements.

The conditions for E-UTRAN inter-frequency CRS-based discovery signal measurements in discovery signal occasions are specified in Table I.2.11.2-1.

**Table I.2.11.2-1: E-UTRAN inter-frequency discovery signal measurements**

| Parameter | E-UTRA operating band groups Note 3 | Minimum CSI-RSRP Note 1 | Minimum SCH_RP Note 1 | CSI-RS Ês/Iot | SCH Ês/Iot |
|---|---|---|---|---|---|
| | | dBm/15kHz | dBm/15kHz | dB | dB |
| Conditions | FDD_A, TDD_A | -125 | -125 | ≥ 0 | ≥ -6 |
| | FDD_B | -124.5 | -124.5 | | |
| | FDD_C, TDD_C | -124 | -124 | | |
| | FDD_D | -123.5 | -123.5 | | |
| | FDD_E, TDD_E | -123 | -123 | | |
| | FDD_F | -122.5 Note 2 | -122.5 Note 2 | | |
| | FDD_G | -122 | -122 | | |
| | FDD_H | -121.5 | -121.5 | | |
| | FDD_N | -118.5 | -118.5 | | |

NOTE 1: This condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 2: The condition is -123 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

## I.3 Conditions for measurements performance requirements for UE

### I.3.1 Conditions for intra-frequency RSRP and RSRQ Accuracy Requirements

This section defines the E-UTRAN intra-frequency RSRP applicable for a corresponding operating band.

The conditions for intra-frequency absolute RSRP and RSRQ accuracy requirements are defined in Table I.3.1-1.

**Table I.3.1-1: Intra-frequency absolute RSRP and RSRQ Accuracy Requirements**

| Parameter | E-UTRA operating band groups Note 3 | Minimum RSRP Note 1 |
|---|---|---|
| | | dBm/15kHz |
| Conditions | FDD_A, TDD_A | -127 |
| | FDD_B | -126.5 |
| | FDD_C, TDD_C | -126 |
| | FDD_D | -125.5 |
| | FDD_E, TDD_E | -125 |
| | FDD_F | -124.5 Note 2 |
| | FDD_G | -124 |
| | FDD_H | -123.5 |
| | FDD_N | -120.5 |
| NOTE 1: This condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3.<br>NOTE 2: The condition is -125 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz.<br>NOTE 3: E-UTRA operating band groups are as defined in Section 3.5. | | |

## I.3.2 Void

## I.3.3 Conditions for inter-frequency RSRP and RSRQ Accuracy Requirements

This section defines the E-UTRAN inter-frequency RSRP applicable for a corresponding operating band.

The conditions for inter-frequency absolute RSRP and RSRQ accuracy requirements are defined in Table I.3.1-1.

## I.3.4 Conditions for inter-frequency relative RSRP and RSRQ Accuracy Requirements

This section defines the E-UTRAN inter-frequency RSRP1,2 applicable for a corresponding operating band.

The conditions for inter-frequency relative RSRP and RSRQ accuracy requirements are defined in Table I.3.8-1.

## I.3.5 to I.3.7Void

## I.3.8 Conditions for Intra-Frequency Relative RSRP Accuracy Requirements

This section defines the E-UTRAN intra-frequency RSRP1,2 applicable for a corresponding operating band.

The conditions for intra-frequency relative RSRP accuracy requirements are specified in Table I.3.8-1.

**Table I.3.8-1 Intra-frequency relative RSRP accuracy requirements**

| Parameter | E-UTRA operating band groups Note 3 | Minimum RSRP1,2 Note 1 |
|---|---|---|
| | | dBm/15kHz |
| Conditions | FDD_A, TDD_A | -127 |
| | FDD_B | -126.5 |
| | FDD_C, TDD_C | -126 |
| | FDD_D | -125.5 |
| | FDD_E, TDD_E | -125 |
| | FDD_F | -124.5 Note 2 |
| | FDD_G | -124 |
| | FDD_H | -123.5 |
| | FDD_N | -120.5 |

NOTE 1: This condition level is increased by $\Delta$>0, when applicable, as described in Sections I.4.2 and I.4.3.
NOTE 2: The condition is -125 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz.
NOTE 3: E-UTRA operating band groups are as defined in Section 3.5.

## I.3.9 Conditions for Intra-Frequency Absolute RSRP and RSRQ Accuracy Requirements under Time Domain Measurement Resource Restriction

This section defines the E-UTRAN intra-frequency RSRP applicable for a corresponding operating band.

The conditions for intra-frequency absolute RSRP and RSRQ accuracy requirements under time domain measurement resource restriction are as specified in Table I.3.1-1.

## I.3.10 Conditions for Intra-Frequency Relative RSRP Accuracy Requirements under Time Domain Measurement Resource Restriction

This section defines the E-UTRAN intra-frequency RSRP1,2 applicable for a corresponding operating band.

The conditions for intra-frequency relative RSRP accuracy requirements under time domain measurement resource restriction are defined in Table I.3.8-1.

## I.3.11 Conditions for Intra-Frequency Absolute RSRP and RSRQ Accuracy Requirements under Time Domain Measurement Resource Restriction with CRS Assistance Information

This clause defines the E-UTRAN intra-frequency RSRP applicable for a corresponding operating band.

The conditions for intra-frequency absolute RSRP and RSRQ accuracy requirements under time domain measurement resource restriction with CRS assistance information are as specified in Table I.3.1-1.

## I.3.12 Conditions for Intra-Frequency Relative RSRP Accuracy Requirements under Time Domain Measurement Resource Restriction with CRS Assistance Information

This clause defines the E-UTRAN intra-frequency RSRP1,2 applicable for a corresponding operating band.

The conditions for intra-frequency relative RSRP accuracy requirements under time domain measurement resource restriction with CRS assistance information are as specified in Table I.3.8-1.

# I.3.13 Void

# I.3.14 Conditions for Intra-Frequency Absolute Discovery Signal Measurement Accuracy Requirements

## I.3.14.1 Conditions for Intra-frequency CRS-based measurements

This clause defines the intra-frequency RSRP in discovery signal occasions [9], applicable for a corresponding operating band for CRS based discovery signal measurements.

The conditions for intra-frequency absolute RSRP and RSRQ accuracy requirements for CRS-based discovery signal measurements in discovery signal occasions are as in Table I.3.1-1

## I.3.14.2 Conditions for Intra-frequency CSI-RS-based measurements

This clause defines the intra-frequency CSI-RSRP in discovery signal occasions [9], applicable for a corresponding operating band for CSI-RS based discovery signal measurements.

The conditions for intra-frequency absolute CSI-RSRP accuracy requirements for CSI-RS-based discovery signal measurements in discovery signal occasions are specified in Table I.3.14.2-1

**Table I.3.14.2-1: Intra-frequency Absolute CSI-RSRP Accuracy Requirements**

| Parameter | E-UTRA operating band groups Note 3 | Minimum CSI-RSRP Note 1 |
|---|---|---|
| | | dBm/15kHz |
| Conditions | FDD_A, TDD_A | -127 |
| | FDD_B | -126.5 |
| | FDD_C, TDD_C | -126 |
| | FDD_D | -125.5 |
| | FDD_E, TDD_E | -125 |
| | FDD_F | -124.5 Note 2 |
| | FDD_G | -124 |
| | FDD_H | -123.5 |
| | FDD_N | -120.5 |
| NOTE 1: This condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3. | | |
| NOTE 2: The condition is -125 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz. | | |
| NOTE 3: E-UTRA operating band groups are as defined in Section 3.5. | | |

# I.3.15 Conditions for Intra-Frequency Relative Discovery Signal Measurement Accuracy Requirements

## I.3.15.1 Conditions for Intra-frequency CRS-based measurements

This clause defines the intra-frequency RSRP in discovery signal occasions [9], applicable for a corresponding operating band for CRS based discovery signal measurements.

The conditions for intra-frequency relative RSRP accuracy requirements for CRS-based discovery signal measurements in discovery signal occasions are as in Table I.3.8-1.

## I.3.15.2 Conditions for Intra-frequency CSI-RS-based measurements

This clause defines the intra-frequency CSI-RSRP in discovery signal occasions [9], applicable for a corresponding operating band for CSI-RS based discovery signal measurements.

The conditions for intra-frequency relative CSI-RSRP accuracy requirements for CSI-RS-based discovery signal measurements in discovery signal occasions are specified in Table I.3.15.2-1.

**Table I.3.15.2-1: Intra-frequency Relative CSI-RSRP Accuracy Requirements**

| Parameter | E-UTRA operating band groups Note 3 | Minimum CSI-RSRP1,2 Note 1 |
|---|---|---|
| | | dBm/15kHz |
| Conditions | FDD_A, TDD_A | -127 |
| | FDD_B | -126.5 |
| | FDD_C, TDD_C | -126 |
| | FDD_D | -125.5 |
| | FDD_E, TDD_E | -125 |
| | FDD_F | -124.5 Note 2 |
| | FDD_G | -124 |
| | FDD_H | -123.5 |
| | FDD_N | -120.5 |
| NOTE 1: This condition level is increased by Δ>0, when applicable, as described in Sections I.4.2 and I.4.3. | | |
| NOTE 2: The condition is -125 dBm/15kHz when the carrier frequency of the assigned E-UTRA channel bandwidth is within 865-894 MHz. | | |
| NOTE 3: E-UTRA operating band groups are as defined in Section 3.5. | | |

# I.3.16 Conditions for Inter-Frequency Absolute Discovery Signal Measurement Accuracy Requirements

## I.3.16.1 Conditions for Inter-frequency CRS-based measurements

This clause defines the inter-frequency RSRP in discovery signal occasions [9], applicable for a corresponding operating band for CRS based discovery signal measurements.

The conditions for inter-frequency absolute RSRP and RSRQ accuracy requirements for CRS-based discovery signal measurements in discovery signal occasions are as in Table I.3.1-1.

## I.3.16.2 Conditions for Inter-frequency CSI-RS-based measurements

This clause defines the inter-frequency CSI-RSRP in discovery signal occasions [9], applicable for a corresponding operating band for CSI-RS based discovery signal measurements.

The conditions for inter-frequency absolute CSI-RSRP accuracy requirements for CSI-RS-based discovery signal measurements in discovery signal occasions are as in Table I.3.14.2-1.

# I.3.17 Conditions for Inter-Frequency Relative Discovery Signal Measurement Accuracy Requirements

## I.3.17.1 Conditions for Inter-frequency CRS-based measurements

This clause defines the inter-frequency RSRP in discovery signal occasions [9], applicable for a corresponding operating band for CRS based discovery signal measurements.

The conditions for inter-frequency relative RSRP and RSRQ accuracy requirements for CRS-based discovery signal measurements in discovery signal occasions are as in Table I.3.8-1.

## I.3.17.2 Conditions for Inter-frequency CSI-RS-based measurements

This clause defines the inter-frequency CSI-RSRP in discovery signal occasions [9], applicable for a corresponding operating band for CSI-RS based discovery signal measurements.

The conditions for inter-frequency relative CSI-RSRP accuracy requirements for CSI-RS-based discovery signal measurements in discovery signal occasions are as in Table I.3.15.2-1.

# I.4 RRM Requirements Exceptions

## I.4.1 General

## I.4.2 Receiver sensitivity relaxation for UE supporting CA

For a UE supporting inter-band carrier aggregation configuration with uplink in one E-UTRA band, if there is a relaxation of receiver sensitivity $\Delta R_{IB,c}$>0 dB as defined in TS 36.101 [2], Table 7.3.1-1A, the relevant side conditions specifying received power levels (E-UTRA RSRP, SCH_RP, PRP, and Io) shall be increased by the amount $\Delta=\Delta R_{IB,c}$ defined for each of the downlink E-UTRA bands.

NOTE: This side condition adjustment applies only for a UE supporting a single inter-band LTE CA band combination. For a UE supporting additional inter-band LTE CA band combinations, the $\Delta R_{IB,c}$ for all bands supported by the UE, need to be studied TS 36.101 [2].

## I.4.3 Receiver sensitivity relaxation for UE configured with CA

### I.4.3.1 Inter-band carrier aggregation

In this section, requirements exceptions are described for the UE configured with inter-band carrier aggregation with one uplink active in low operating band.

A relevant side condition (e.g., E-UTRA RSRP, SCH_RP, PRP, and Io) in a requirement shall be increased by the amount Δ=L2-L1, where L1 is the reference sensitivity level specified in TS 36.101[2], Table 7.3.1-1, and L2 is the reference sensitivity level specified in TS 36.101[2], Table 7.3.1A-0a, when the following conditions are fulfilled,

- both downlink component carriers on different bands are configured with CA and active,
- the single uplink is active in the low operating band,
- the exception requirements specified in TS 36.101[2], Table 7.3.1A-0a, apply.

If the relaxation Δ specified in this section applies, then the relaxation specified in Section I.4.2 should not be applied.

### I.4.3.2 Intra-band non-contiguous carrier aggregation

For a UE configured with intra-band non-contiguous carrier aggregation configuration with uplink in one E-UTRA band, if there is a relaxation of receiver sensitivity $\Delta R_{IBNC}$>0 as defined in TS 36.101 [2], Table 7.3.1A-3, the relevant side conditions specifying received power levels (E-UTRA RSRP, SCH_RP, PRP, and Io) shall be increased by the amount $\Delta=\Delta R_{IBNC}$ defined for the downlink SCC, when the following conditions are fulfilled,

- both downlink component carriers are configured with CA and active,
- one uplink carrier is active,
- the exception requirements specified in TS36.101[2], Table 7.3.1A-3, apply.

If the relaxation Δ specified in this section applies, then the relaxation specified in Section I.4.2 should not be applied.

### I.4.3.3 Inter-band carrier aggregation with operating bands without uplink band

In this section, requirements are described for the UE configured with inter-band carrier aggregation involving one operating band without uplink band.

There is no relaxation in relevant side condition (e.g., E-UTRA RSRP, SCH_RP, PRP, and Io) in a requirement, i.e., Δ=0, when the following conditions are fulfilled,

- both downlink component carriers on different bands are configured with CA and active,
- the single uplink is active in the high operating band,
- conditions specified in TS36.101[2], Table 7.3.1A-0d, apply.

If Δ specified in this section applies, then no other additional relaxation to REFSENS shall be applied.

# Annex J (informative): Handling requirements and tests for different releases and UE capabilities

This annex gives guidance on how minimum requirements in different releases of 3GPP TS 36.133 [4] and different UE capabilities are handled in the specification 3GPP TS 36.521-3.

## J.1 General considerations

Same as TS 36.521-1 [10] Annex I with the following exceptions:

- Instead of "*TS 36.101*" → use "*TS 36.133*"

- Instead of "*TS 36.521-1*" → use "*TS 36.521-3*"

- Instead of "*Annex I*" → use "*Annex J*"

## J.2 Concrete scenarios

### J.2.1 Tests for minimum requirements varying between releases, without introduction of new features

Same as TS 36.521-1 [10] Annex I with the following exceptions:

- Instead of "*TS 36.101*" → use "*TS 36.133*"

- Instead of "*TS 36.521-1*" → use "*TS 36.521-3*"

- Instead of "*Annex I*" → use "*Annex J*"

### J.2.2 Tests for CA (Carrier aggregation)

FFS

# Annex K (informative): Change history

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| Date | TSG # | TSG Doc. | CR | Rev | Subject/Comment | Old | New |
| 2008-06 | RAN5#39bis | R5-082129 | | | R5-082129: Restructure of TS 36.521-1 and RRM proposal (Split of RRM from 36.521-1 v0.2.0 in its own specification 36.521-3 v0.1.0) | | 0.1.0 |
| *2008-06* | *RAN5#39bis* | *R5-082174* | | | Following approved TPs have been included:<br>R5-082129: Restructure of TS 36.521-1 and RRM proposal (Split of RRM from 36.521-1 v0.2.0 in its own specification 36.521-3 v0.1.0)<br>R5-082160: Cover for LTE E-UTRAN RRC_IDLE State Mobility text proposal<br>R5-082161: Cover for LTE E-UTRAN RRC_CONNECTED State Mobility text proposal<br>R5-082162: Update of 36.521-1: Introduction of HRPD and CDMA2000 in RRM test cases<br>R5-082163: Cover for LTE UE Transmit Timing Requirements text proposal<br>Editorial changes for Annexes | *0.1.0* | *0.2.0* |
| 2008-08 | RAN5#40 | R5-083164 | | | Following approved TPs have been included:<br>R5-083051: LTE E-UTRAN RRC_IDLE State Mobility text proposal<br>R5-083052: LTE E-UTRAN RRC_CONNECTED State Mobility text proposal<br>R5-083053: LTE UE Transmit Timing Requirements text proposal<br>R5-083054: LTE UE Measurement Procedures text proposal<br>R5-083813: LTE UE Measurement Performance Requirements text proposal<br>R5-083138: Text proposal for LTE E-UTRAN Cell Re-selection to HRPD or to cdma2000 1xRTT in TS 36.521-3<br>R5-083056: RRC Connection Mobility Control text proposal<br>R5-083164: LTE-RF 36-521-3 after RAN5#40<br>Editorial restructuring to section 4 | 0.2.0 | 0.3.0 |
| 2008-10 | RAN5#40Bis | R5-084073 | | | Following approved TPs have been included:<br>R5-084073: TS 36.521-3 after RAN5#40Bis<br>R5-084079: LTE Cell Re-Selection text proposal<br>R5-084322: LTE FDD/FDD Handover for intra/inter frequency text proposal | 0.3.0 | 0.4.0 |
| 2008-11 | RAN5#41 | R5-085084 | | | Following approved TPs have been included:<br>R5-085084 LTE-RF: TS 36.521-3 after RAN5#41<br>R5-085718 LTE RRM Cell Re-Selection text proposal<br>R5-085719 LTE RRM FDD/FDD Handover for intra/inter frequency text proposal<br>R5-085720 E-UTRAN FDD intra-frequency measurements text proposal<br>R5-085740 RSRQ Accuracy Measurement Performance Requirements text proposal<br>R5-085722 Text Proposal for Cell Configuration mapping annex in 36.521-3<br>Editor's cleanup | 0.4.0 | 0.5.0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2009-01 | RAN5#41Bis | R5-086067 | | | Following approved TPs have been included:<br>R5-086067 LTE-RF: TS 36.521-3 after RAN5#41Bis<br>R5-086149 References to connection diagrams<br>R5-086418 LTE RRM Cell Re-Selection text proposal<br>R5-086095 Cell configuration reference correction for RRM tests in 36.521-3 section 3A.3<br>R5-086419 LTE RRM FDD/FDD Handover for intra/inter frequency text proposal<br>R5-086420 E-UTRAN FDD intra-frequency measurements text proposal<br>R5-086431 RSRQ Accuracy Measurement Performance Requirements text proposal<br>R5-086082 LTE UE Transmit Timing Requirements text proposal<br>R5-086422 Text proposal for RSRP measurement accuracy test cases<br>R5-086432 E-UTRAN FDD- FDD Inter-Frequency Measurements text proposal<br>R5-086142 Measurement Reference Channels and OCNG for RRM testing<br>R5-086150 Statistical testing in RRM tests<br>Editor's cleanup | 0.5.0 | 0.6.0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2009-03 | RAN5#42 | R5-090191 | | | Following approved TPs have been included:<br>R5-091026 TDD Intra frequency RSRQ Accuracy<br>R5-091085 TDD Inter frequency RSRQ Accuracy<br>R5-091035 LTE RRM FDD/FDD Handover for intra/inter frequency text proposal<br>R5-091047 E-UTRAN FDD intra-frequency measurements text proposal<br>R5-091029 RSTQ Accuracy Measurement Performance Requirements text proposal<br>R5-091041 LTE RRM E-UTRA FDD to GSM Cell Re-Selection text proposal<br>R5-091040 LTE RRM E-UTRA FDD to GSM Handover text proposal<br>R5-090182 LTE UE Measurement Procedures Structure text proposal<br>R5-091048 LTE RRM E-UTRA FDD to UTRA FDD Cell Search text proposal<br>R5-090184 LTE UE inter-RAT Handover Structure text proposal<br>R5-091039 LTE RRM E-UTRA FDD to UTRA FDD Handover text proposal<br>R5-091053 LTE UE Transmit Timing Requirements text proposal<br>R5-090191 LTE-RF: TS 36.521-3 after RAN5#42<br>R5-091091 Intra-frequency cell search TDD<br>R5-091088 Intra-frequency Absolute RSRP measurement accuracy TDD<br>R5-091090 Intra-frequency Relative RSRP measurement accuracy TDD<br>R5-091089 Inter-frequency RSRP absolute accuracy TDD<br>R5-091087 Inter-frequency RSRP relative accuracy TDD<br>R5-091028 Text Proposal for RSRP Measurement Accuracy test cases<br>R5-091076 Text Proposal for Annex C of TS 36.521-3<br>R5-091051 TP of E-UTRAN TDD & GSM cell re-selection test case<br>R5-091043 TP of E-UTRAN TDD & TDD inter frequency cell re-selection test case<br>R5-091036 TP of E-UTRAN TDD & TDD inter frequency handover test case<br>R5-091044 TP of E-UTRAN TDD - TDD intra frequency cell re-selection test case<br>R5-091038 TP of E-UTRAN TDD & TDD intra frequency handover test case<br>R5-091045 TP of E-UTRAN TDD & UTRAN TDD cell re-selection test case<br>R5-091049 E-UTRAN FDD- FDD Inter-Frequency Measurements text proposal<br>R5-091050 E-UTRAN TDD- TDD Inter-Frequency Measurements text proposal<br>R5-091052 LTE-RF: Update to 36.521-3 Annex E Cell Configuration mapping<br>R5-091064 Correction to frequencies to be tested in RRM test cases<br>R5-091042 LTE RRM Cell Re-Selection text proposal<br>Editor's cleanup | 0.6.0 | 1.0.0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2009-03 | RAN5#42Bis | | | | R5-091263 LTE-RRM Cell Re-Selection text proposal<br>R5-091922 LTE-RRM E-UTRA FDD to GSM Cell Re-Selection text proposal<br>R5-091923 LTE-RRM E-UTRA FDD - UTRA TDD Cell Re-Selection text proposal<br>R5-091924 TP of E-UTRA TDD - GSM cell reselection<br>R5-091945 TP of E-UTRA TDD-UTRAN TDD cell re-selection : UTRA is of higher priority<br>R5-091926 TP of E-UTRA TDD - UTRA TDD cell reselection: UTRA is of lower priority test case<br>R5-091264 LTE-RRM FDD/FDD Handover for intra/inter frequency text proposal<br>R5-091931 LTE-RRM E-UTRA FDD to GSM Handover text proposal<br>R5-091928 LTE-RRM E-UTRA FDD to UTRA FDD Handover text proposal<br>R5-091946 LTE-RRM: E-UTRA TDD to UTRA FDD Handover text proposal<br>R5-091947 LTE-RRM: Handover test proposal<br>R5-091930 TP of E-UTRA TDD to UTRA TDD handover test case<br>R5-091265 LTE-RRM E-UTRAN FDD intra-frequency measurements text proposal<br>R5-091266 LTE-RRM RSRQ Accuracy Measurement Performance Requirements text proposal<br>R5-091932 LTE-RRM E-UTRA FDD to UTRA FDD Cell Search (fading) text proposal<br>R5-091933 LTE-RRM E-UTRAN FDD & UE Transmit Timing Accuracy text proposal<br>R5-091276 LTE-RRM E-UTRAN FDD & UE Timing Advance Adjustment Accuracy text proposal<br>R5-091934 LTE-RRM: E-UTRA TDD to UTRA FDD Cell Search (fading) text proposal<br>R5-091935 LTE-RRM E-UTRAN TDD & UE Transmit Timing Accuracy text proposal<br>R5-091936 LTE-RRM E-UTRAN TDD & UE Timing Advance Adjustment Accuracy text proposal<br>R5-091937 LTE-RRM E-UTRA FDD to UTRA TDD Cell Search (fading) text proposal<br>R5-091381 EUTRAN TDD to UTRAN TDD cell search (fading)<br>R5-091386 LTE RRM TDD Inter Frequency RSRP Accuracy text proposal<br>R5-091398 Text Proposal for RSRP Measurement Accuracy test cases<br>R5-091948 LTE-RRM: Measurements test proposal<br>R5-091431 RRM-EUTRAN FDD RLM test for out-of-sync<br>R5-091434 RRM-EUTRAN TDD RLM test for out-of-sync<br>R5-091435 RRM-EUTRAN FDD RLM test for In-sync<br>R5-091436 RRM-EUTRAN TDD RLM test for In-sync<br>R5-091468 RRM E-UTRAN FDD-FDD Inter-frequency Measurements<br>R5-091469 RRM E-UTRAN TDD-TDD Inter-frequency Measurements<br>R5-091939 LTE-RRM cell configuration mapping updates<br>R5-091407 Update of statistical requirements to 36.521-3<br>Editor's cleanup | 1.0.0 | 1.1.0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2009-05 | RAN5#43 | R5-092156 | | | R5-092156 LTE-RF: TS 36.521-3 after RAN5#43<br>R5-092066 E-UTRAN FDD- FDD Inter-Frequency Measurements text proposal<br>R5-092617 RRM E-UTRAN TDD-TDD Inter-frequency Measurement<br>R5-092068 RRM-EUTRAN FDD RLM test for out-of-sync and in-synch<br>R5-092069 RRM-EUTRAN TDD RLM test for out-of-sync and in-synch<br>R5-092071 Reference measurement Channels for Radio Link Monitoring Tests with 2 Antennas<br>R5-092127 Update of statistical requirements to 36.521-3<br>R5-092630 LTE RRM: 1→2 RX antenna<br>R5-092618 Text Proposal for E-UTRAN FDD-UTRAN FDD cell re-selection test cases<br>R5-092651 Text Proposal for E-UTRAN FDD - GSM Measurements test case<br>R5-092620 LTE-RRM E-UTRA TDD to GSM event triggered reporting in AWGN text proposal<br>R5-092360 LTE RRM TDD Inter Frequency RSRP Accuracy text proposal<br>R5-092621 LTE-RRM Default Message Contents for support of RRM text proposal<br>R5-092384 LTE-RRM Update of Message Contents for E-UTRAN FDD RRM tests to align with support of RRM text proposal<br>R5-092622 LTE-RRM E-UTRAN FDD Cell Re-Selection text proposal<br>R5-092386 LTE-RRM FDD Inter Frequency RSRQ Accuracy Measurement Performance Requirements text proposal<br>R5-092387 LTE-RRM E-UTRA FDD to GSM Cell Re-Selection text proposal<br>R5-092623 LTE-RRM E-UTRAN FDD intra frequency cell search (fading) in long DRX text proposal<br>R5-092624 LTE-RRM E-UTRAN FDD intra frequency cell search (fading) in short DRX text proposal<br>R5-092390 LTE-RRM E-UTRAN FDD - UE Transmit Timing Accuracy text proposal<br>R5-092626 LTE-RRM E-UTRAN FDD random access: contention based scenario text proposal<br>R5-092627 LTE-RRM E-UTRAN FDD random access: non-contention based scenario text proposal<br>R5-092628 LTE-RRM E-UTRA FDD to HRPD Handover text proposal<br>R5-092629 LTE-RRM E-UTRA FDD to cdma2000 1xRTT Handover text proposal<br>R5-092443 Addition of band 18 and 19 to LTE RRM test cases<br>Editor's cleanup | 1.1.0 | 2.0.0 |
| 2009-05 | RAN#44 | - | - | | Updated to v8.0.0 after RAN#44 with no technical change. | 2.0.0 | 8.0.0 |
| 2009-06 | - | - | - | | Editorial clean up | 8.0.0 | 8.0.1 |
| 2009-09 | RAN#45 | R5-094036 | 0001 | - | Correction CR to 36.521-3: Update of Requirements for E-UTRAN (FDD) cell re-selection tests | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094037 | 0002 | - | Correction CR to 36.521-3: Update of Requirements for E-UTRAN FDD - FDD Inter Frequency Handover test | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094038 | 0003 | - | Correction CR to 36.521-3: Update of Requirements for E-UTRAN FDD - FDD Intra Frequency Cell Search test | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094039 | 0004 | - | Correction CR to 36.521-3: Update of Requirements for E-UTRAN FDD - UE transmit timing accuracy test | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094040 | 0005 | - | Correction CR to 36.521-3: Update of Requirements for E-UTRAN FDD - GSM cell re-selection test | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094041 | 0006 | - | Correction CR to 36.521-3: Update of Requirements conditions for E-UTRAN FDD - UE timing advance adjustment accuracy test | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094042 | 0007 | - | Correction CR to 36.521-3: Update of Requirements for E-UTRAN FDD - GSM Handover test | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094043 | 0008 | - | Correction CR to 36.521-3: Update of Requirements for E-UTRAN FDD - UTRAN FDD Handover test | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094045 | 0009 | - | Correction CR to 36.521-3: Update of Requirements for E-UTRAN FDD - GSM Cell Search test | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094047 | 0010 | - | Correction CR to 36.521-3: Update of Requirements for E-UTRAN FDD - Contention Based Random Access test | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094048 | 0011 | - | Correction CR to 36.521-3: Update of Requirements for E-UTRAN FDD - Non-Contention Based Random Access test | 8.0.1 | 8.1.0 |

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|---|---|---|---|---|---|---|---|
| 2009-09 | RAN#45 | R5-094049 | 0012 | - | Addition of test scenario CR to 36.521-3: E-UTRAN FDD-FDD Inter-frequency cell search when DRX is used under fading propagation conditions | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094050 | 0013 | - | Correction CR to 36.521-3: Update of E-UTRAN FDD-FDD Intra-frequency cell search when DRX is used under fading propagation conditions | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094051 | 0014 | - | Correction CR to 36.521-3: Update of Annex H Default Message Contents for support of RRM | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094217 | 0015 | - | Update for E-UTRA FDD - UTRA TDD cell reselection | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094218 | 0016 | - | Test proposal for E-UTRA FDD - UTRA TDD HO | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094219 | 0017 | - | Test proposal for E-UTRA TDD random access: contention based scenario | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094220 | 0018 | - | Test proposal for E-UTRA TDD random access: non-contention based scenario | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094221 | 0019 | - | Update for TDD Intra-frequency RSRQ measurement accuracy | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094222 | 0020 | - | Update for TDD Inter-frequency RSRQ measurement accuracy | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094223 | 0021 | - | Update for E-UTRAN TDD Transmit timing accuracy | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094225 | 0022 | - | Update for E-UTRA FDD - UTRA TDD cell search(fading) | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094253 | 0023 | - | CR to 36.521-3: Addition of E-UTRAN FDD Intra-frequency RRC Re-establishment | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094254 | 0024 | - | CR to 36.521-3: Addition of E-UTRAN FDD Inter-frequency RRC Re-establishment | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094285 | 0025 | - | LTE-RRM: Introduction of Common Exception messages table for E-UTRAN TDD-UTRAN FDD handover and E-UTRAN TDD-UTRAN FDD measurements | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094358 | 0026 | - | Correction to RSRP measurement accuracy test cases | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094442 | 0027 | - | CR to 36.521-3: Update of E-UTRA FDD to HRPD Handover and E-UTRA FDD to cdma2000 1xRTT Handover test cases | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094709 | 0028 | - | LTE RRM: Correction to test cases 4.4.1 and 4.4.2 | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094713 | 0029 | - | Resubmission - Update to E-UTRAN to HRPD Cell Re-Selection (HRPD is of lower priority) test case | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094720 | 0030 | - | Resubmission - Update to E-UTRAN to CDMA2000 1xRTT Cell Re-Selection (CDMA2000 1xRTT is of lower priority) test case | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094743 | 0031 | - | RRM TCs in test mode | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094927 | 0032 | - | Correction CR to 36.521-3: Update of inter-frequency E-UTRAN TDD-TDD cell re-selection 4.2.6 | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094928 | 0033 | - | Correction CR to 36.521-3: Update of E-UTRAN TDD - UTRAN TDD cell re-selection 4.3.4 | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094929 | 0034 | - | Correction CR to 36.521-3: Update of Requirements for E-UTRAN FDD - UTRAN FDD cell re-selection test | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094930 | 0035 | - | LTE-RRM: Addition of common messages to Annex H | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094931 | 0036 | - | Test Proposal for E-UTRAN TDD Intra-frequency RRC Re-establishment | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094932 | 0037 | - | Test Proposal for E-UTRAN TDD Inter-frequency RRC Re-establishment | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094933 | 0038 | - | Update for E-UTRAN TDD Timing advanced adjustment accuracy | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094934 | 0039 | - | Correction CR to 36.521-3: Update of Requirements for E-UTRAN FDD - UTRAN FDD Cell Search test | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094935 | 0040 | - | E-UTRA TDD - TDD Intra frequency cell search with DRX | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094936 | 0041 | - | TDD - TDD RSRP measurement | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094937 | 0042 | - | Update 8.10.1 E-UTRAN TDD-GSM event triggered reporting in AWGN | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094938 | 0043 | - | Add new tc 8.10.2 EUTRAN TDD-GSM cell search with DRX | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094939 | 0044 | - | Add new tc 8.7.2 EUTRAN TDD - UTRAN TDD cell search with DRX | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094940 | 0045 | - | E-UTRA TDD - TDD Inter frequency cell search with DRX | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094942 | 0046 | - | Update to Annex E Cell Configuration Mapping | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094967 | 0047 | - | RRM Radio Link Monitoring FDD update | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094968 | 0048 | - | RRM Radio Link Monitoring TDD update | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094969 | 0050 | - | RRM: E-UTRAN FDD - UTRAN FDD SON ANR cell search reporting | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094970 | 0051 | - | CR to 36.521-3:Message updates for RSRP and RSRQ Accuracy measurement | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094971 | 0052 | - | RRM OCNG and RMC update | 8.0.1 | 8.1.0 |
| 2009-09 | RAN#45 | R5-094972 | 0053 | - | RRM: Update of Annex E for SON | 8.0.1 | 8.1.0 |
| 2009-12 | RAN#46 | R5-095492 | 0054 | - | Removal of test state 4 in RRM test cases | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-095493 | 0055 | - | CR to 36.521-3 Annexes of E-UTRAN cell reselection test cases | 8.1.0 | 8.2.0 |

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|---|---|---|---|---|---|---|---|
| 2009-12 | RAN#46 | R5-095499 | 0056 | - | CR for E-UTRAN FDD - UTRAN TDD handover | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-095501 | 0057 | - | CR for E-UTRAN TDD - UE Transmit Timing Accuracy | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-095503 | 0058 | - | CR for E-UTRAN FDD - UTRAN TDD event triggered reporting under fading propagation conditions | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-095504 | 0059 | - | Correction to TDD RSRP and RSRQ measurement requirements | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-095527 | 0060 | - | Correction CR to 36.521-3: E-UTRAN FDD - FDD cell re-selection intra frequency case and inter frequency case conformance minimum requirements updates | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-095528 | 0061 | - | Correction CR to 36.521-3: E-UTRAN FDD - UTRAN FDD cell re-selection: UTRA FDD is of higher priority and UTRA FDD is of lower priority conformance minimum requirements | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-095529 | 0062 | - | Correction CR to 36.521-3: E-UTRAN FDD - FDD Inter Frequency event triggered reporting under fading propagation conditions in asynchronous cells case | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-095530 | 0063 | - | Correction CR to 36.521-3: E-UTRAN FDD - FDD Inter Frequency event triggered reporting when DRX is used under fading propagation conditions | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-095531 | 0064 | - | Correction CR to 36.521-3: E-UTRAN FDD - FDD Intra Frequency event triggered reporting under fading propagation conditions | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-095537 | 0065 | - | Correction CR to 36.521-3: E-UTRAN FDD - UE Transmit Timing Accuracy case | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-095538 | 0066 | - | Correction CR to 36.521-3: E-UTRAN FDD - FDD inter frequency event triggered reporting when DRX is used | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-095557 | 0067 | - | Correction CR to 36.521-3: General RRM Updates | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-095572 | 0068 | - | Update TC 8.7.2 E-UTRAN TDD - UTRAN TDD cell search when DRX is used under fading propagation conditions | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-095573 | 0069 | - | Update TC 8.10.2 E-UTRAN TDD - GSM event triggered reporting when DRX is used in AWGN | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-095576 | 0070 | - | Update TC 8.2.1 E-UTRAN TDD - TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-095591 | 0071 | - | update of Annex H.2.3 in 36.521-3 | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-095741 | 0072 | - | CR to the inconsistent expression in UE Measurements Procedures | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-095917 | 0073 | - | Update: Radio Link Monitoring test cases: no DRX | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096145 | 0074 | - | Correction CR to 36.521-3: E-UTRAN FDD - FDD Inter Frequency Absolute RSRP Accuracy case | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096243 | 0075 | - | Update to RRM TC:E-UTRAN FDD - UTRAN TDD cell re-selection | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096244 | 0104 | 1 | Addition of new TC to 36.521-3: E-UTRAN TDD - UTRAN FDD cell re-selection test | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096246 | 0105 | - | Modification of section 4.2.2 in 36.521-3 | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096247 | 0106 | - | Modification of section 4.2.2 in 36.521-3 | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096255 | 0076 | - | CR to the RA response window's name in Random Access conformance requirements | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096257 | 0077 | - | Addition of test scenario CR to 36.521-3: E-UTRAN FDD - UTRAN FDD cell re-selection in fading propagation conditions: UTRA FDD is of lower priority | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096258 | 0078 | 1 | Addition of new TC to 36.521-3:E-UTRAN TDD - GSM Handover: Unknown Target Cell | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096263 | 0079 | 1 | Add new TC 5.1.6 E-UTRAN TDD - TDD inter frequency handover: unknown target cell | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096265 | 0080 | - | CR to 36.521-3: Update of E-UTRA FDD to HRPD Handover and E-UTRA FDD to cdma2000 1xRTT Handover test cases | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096267 | 0081 | - | Addition of test scenario CR to 36.521-3: E-UTRAN FDD - GSM Cell Search when DRX is used | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096268 | 0082 | - | Addition of test scenario CR to 36.521-3: E-UTRAN FDD - UTRAN FDD Cell Search when DRX is used | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096269 | 0083 | - | RRM: Update of test case 8.4.1 TDD inter-frequency event triggered reporting | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096271 | 0084 | - | LTE-RF: Update to Annex E Cell Configuration Mapping | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096272 | 0085 | - | Correction CR to 36.521-3: Addition of message contents exceptions for E-UTRAN inter frequency and inter-RAT Cell Search for when DRX is used | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096273 | 0086 | - | Correction CR to 36.521-3: E-UTRAN FDD - UTRA FDD Handover case | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096274 | 0087 | - | CR to 36.521-3: Update to FDD Intra-frequency RRC Re-establishment test case | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096275 | 0088 | - | CR to 36.521-3: Update to FDD Inter-frequency RRC Re-establishment test case | 8.1.0 | 8.2.0 |

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|---|---|---|---|---|---|---|---|
| 2009-12 | RAN#46 | R5-096276 | 0107 | - | Test Case of E-UTRAN TDD to GSM Handover | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096296 | 0089 | - | Update TC 8.7.1 E-UTRAN TDD - UTRAN TDD event triggered reporting under fading propagation conditions | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096302 | 0090 | - | Correction CR to 36.521-3: E-UTRAN FDD - GSM event triggered reporting in AWGN case | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096303 | 0091 | - | Addition of new TC to 36.521-3: E-UTRAN TDD to E-UTRAN TDD and UTRA TDD cell search test case | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096310 | 0092 | - | Addition of test scenario CR to 36.521-3: E-UTRAN FDD - UTRAN FDD Handover: Unknown Target Cell | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096324 | 0093 | - | Addition of new TC to 36.521-3 E-UTRAN TDD - UTRAN TDD HO test: unknown target cell | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096325 | 0094 | - | Addition of test scenario CR to 36.521-3: E-UTRAN FDD - FDD inter frequency Handover test cases: Unknown Target Cell | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096326 | 0095 | - | Addition of test scenario CR to 36.521-3: E-UTRAN FDD - GSM Handover: Unknown Target Cell | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096327 | 0096 | - | Addition of new TC to 36.521-3: E-UTRAN TDD - E-UTRAN TDD and E-UTRAN TDD Inter-frequency event triggered reporting under fading propagation conditions test case | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096328 | 0097 | - | E-UTRAN FDD Radio Link Monitoring test for Out-of-sync in DRX | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096329 | 0098 | - | E-UTRAN FDD Radio Link Monitoring test for in-sync in DRX | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096330 | 0099 | - | E-UTRAN TDD Radio Link Monitoring test for Out-of-sync in DRX | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096331 | 0100 | - | E-UTRAN TDD Radio Link Monitoring test for in-sync in DRX | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096332 | 0101 | - | RRM: Update of test case 8.3.1 FDD inter-frequency event triggered reporting | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096337 | 0102 | - | Correction CR to 36.521-3: E-UTRAN FDD - FDD Handover intra frequency and inter frequency case | 8.1.0 | 8.2.0 |
| 2009-12 | RAN#46 | R5-096340 | 0103 | - | Introduction of uncertainties for RRM test cases 4.2.1 and 4.2.2 | 8.1.0 | 8.2.0 |
| 2010-03 | RAN#47 | R5-100130 | 0108 | - | Test Tolerances and alignment with 36.133 for cell re-selection intra frequency cases | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100132 | 0109 | - | Uncertainties and Test Tolerances for inter frequency cell re-selection | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100135 | 0110 | - | Clarification of Extreme conditions for RSRP test cases | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100362 | 0113 | - | CR about the Cell Search Requirements for LTE FDD-FDD/ TDD-TDD Handover to Unknown Target Cell | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100365 | 0114 | - | CR on updating the handover delay requirements for E-UTRAN TDD - TDD both intra-frequency and inter-frequency handovers | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100367 | 0115 | - | CR to correct the test requirements of reselection from E-UTRAN FDD/TDD to UTRAN TDD | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100394 | 0116 | - | Correction of Annex H about measurement performance messages | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100401 | 0117 | - | RRM Inter frequency cell search updates, TC 8.3.1 and 8.4.1 | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100438 | 0118 | - | Update TC 8.7.1, 8.9.1 and 8.11.4 | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100460 | 0119 | - | Misc update on 521-3 | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100486 | 0120 | - | CR to 36.521-3: Addition of E-UTRA FDD to HRPD Handover: Unknown Target Cell and E-UTRA FDD to cdma2000 1xRTT Handover: Unknown Target Cell test cases | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100519 | 0121 | - | Correction to RSRP Accuracy test cases | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100546 | 0122 | - | CR to 36.521-3: Update to E-UTRAN FDD RRC Re-establishment test cases | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100562 | 0123 | - | CR to 36.521-3: Update LTE RRM test cases with test requirements for extended LTE1500 | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100714 | 0124 | - | Addition of missing Es/Noc parameters in RRM test cases | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100715 | 0125 | - | Correction to GSM measurement configuration in Annex H.3.1 | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100716 | 0126 | - | Update on Annex C for 36.521-3 | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100849 | 0127 | - | Text on exclusion of extra delay due to RRC retransmission | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100850 | 0128 | - | Correction to test iteration procedure in cell re-selection TCs | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100852 | 0129 | - | DL Mac Padding for RRM TCs | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100853 | 0130 | - | Update TC 5.1.6 E-UTRAN TDD-TDD inter frequency handover unknown target cell | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100854 | 0131 | - | New RRM test case, 8.7.3 E-UTRAN TDD SON ANR | 8.2.0 | 8.3.0 |

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|---|---|---|---|---|---|---|---|
| 2010-03 | RAN#47 | R5-100859 | 0132 | - | Update TC 8.2.2 E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100860 | 0133 | - | Update TC 8.2.1 E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100861 | 0134 | - | Update TC 8.4.2 E-UTRAN TDD-TDD Inter-frequency event triggered reporting when DRX is used under fading propagation conditions in synchronous cells | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100862 | 0135 | - | Misc update on test applicability | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100865 | 0136 | - | CR about corrections of PDSCH Reference Measurement Channels | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100866 | 0137 | - | CR about OFDMA Channel Noise Generator (OCNG) | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100873 | 0138 | - | CR to 36.521-3 Rel-8 Introduction of E-UTRAN FDD - FDD Intra Frequency Cell Search with DRX when L3 filtering is used | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100890 | 0139 | - | Update to RRM TC: TDD Intra frequency RSRP Accuracy | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100896 | 0140 | - | Clarification on Time offset between cells | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100897 | 0141 | - | Update to RRM TC:E-UTRAN TDD-TDD cell re-selection | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100898 | 0142 | - | Update to RRM TC: TDD Inter frequency RSRP Accuracy | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100900 | 0143 | - | Uncertainties and Test Tolerances for FDD Intra Frequency Absolute RSRP Accuracy | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | R5-100901 | 0144 | - | RRM TTIdcch and cell timing change, update of chapter 8 | 8.2.0 | 8.3.0 |
| 2010-03 | RAN#47 | - | - | - | Moved to v9.0.0 with no change | 8.3.0 | 9.0.0 |
| 2010-06 | RAN#48 | R5-103105 | 0145 | - | CR to 36.521-3: Update RSRP test cases with band 11 and 21 | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103116 | 0146 | - | Correction of CR conflict for Intra Frequency TDD reselection test | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103117 | 0147 | - | Reference to TR 36.903 in TS 36.521-3 | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103312 | 0149 | - | Connection diagram for test 8.11.2 (3 cells) | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103315 | 0150 | - | Correction to connection diagram reference for test 8.10.1 and 8.10.2 | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103330 | 0151 | - | update on test applicability | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103358 | 0152 | - | Annex E update | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103496 | 0153 | - | LTE-RRM: Update of test procedure for measurement performance test cases | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103526 | 0154 | - | CR 36.521-3 on corrections to requirements in Idle Mode | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103528 | 0155 | - | CR 36.521-3 on correction to InterRAT handover minimum requirements | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103531 | 0156 | - | CR 36.521-3 on correction to measurement requirements | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103532 | 0157 | - | CR 36.521-3 on correction to E-UTRA inter frequency cell search requirements | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103534 | 0158 | - | CR 36.521-3 on correction to UE transmit timing minimum and test requirements | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103541 | 0159 | - | Addition of test tolerances and system uncertainties for E-UTRAN FDD-FDD HO inter-frequency case | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103546 | 0160 | - | Addition of test tolerances and system uncertainties for E-UTRAN FDD-FDD intra frequency cell search under fading in asynchronous cells | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103547 | 0161 | - | Addition of test tolerances and system uncertainties for E-UTRAN FDD-FDD intra frequency cell search under fading in synchronous cells | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103548 | 0162 | - | Addition of test tolerances and system uncertainties for E-UTRAN TDD-TDD intra frequency cell search under fading in synchronous cells | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103607 | 0163 | - | Correction to step of physical cell identity change in 4.2.3 | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103608 | 0164 | - | Correction of test mode reference to 36.508 | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103611 | 0165 | - | Correction to the references of exceptional message | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103612 | 0166 | - | Correction to b2-Threshold1 in the exceptional message | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103613 | 0194 | - | Correction to Radio Resource Configuration in UE transmit timing and UE timing advance TCs | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103614 | 0195 | - | Correction to Gap Pattern Id in the exceptional message | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103615 | 0196 | - | Correction to Measure object and ID in the exceptional messages | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103658 | 0197 | - | Iteration in cell reselection tests | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103709 | 0167 | - | Connection diagram reference for intra-freq measurement TCs | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103724 | 0168 | - | LTE-RRM:CR to E-UTRAN TDD RRC Re-establishment test cases | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103734 | 0169 | - | Test Tolerances and alignment for RLM FDD TC 7.3.1, 7.3.2 | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103736 | 0170 | - | Uncertainties and Test Tolerances for Inter Frequency Absolute RSRP Accuracy | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103737 | 0171 | - | Uncertainties and Test Tolerances for TC 8.1.3 and 8.2.2 | 9.0.0 | 9.1.0 |

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|---|---|---|---|---|---|---|---|
| 2010-06 | RAN#48 | R5-103738 | 0172 | - | Uncertainties and Test Tolerances for TC 8.4.1 and 8.4.2 | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103739 | 0173 | - | LTE-RRM: CR for Test Tolerances of intra-freq hand over test cases (5.1.1 & 5.1.2) | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103740 | 0174 | - | LTE-RRM:CR for Test Tolerances of inter-freq absolute RSRQ accuracy test cases (9.2.3.1 & 9.2.4.1) | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103741 | 0175 | - | LTE-RRM:CR for Test Tolerances of inter-freq relative RSRQ accuracy test cases (9.2.3.2 & 9.2.4.2) | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103742 | 0176 | - | Uncertainties and Test Tolerances for Inter Frequency Relative RSRP Accuracy | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103743 | 0177 | - | LTE-RRM: CR on Test Tolerances for TDD intra-freq absolute RSRP accuracy Test | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103744 | 0178 | - | LTE-RRM: CR on Test Tolerances for TDD intra-freq relative RSRP accuracy Test case | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103745 | 0179 | - | Addition of test tolerances and system uncertainties for E-UTRAN TDD-TDD HO inter-frequency case | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103746 | 0180 | - | Additions to measurement uncertainties and Test Tolerances for E-UTRAN FDD-FDD and TDD-TDD HO inter-frequency case in Annex F | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103747 | 0181 | - | Additions to measurement uncertainties and Test Tolerances for E-UTRAN FDD-FDD and TDD-TDD intra frequency cell search in Annex F | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103748 | 0182 | - | Addition of test tolerances and system uncertainties for FDD intra frequency absolute RSRQ accuracy | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103749 | 0183 | - | Addition of test tolerances and system uncertainties for TDD intra frequency absolute RSRQ accuracy | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103750 | 0184 | - | Additions to measurement uncertainties and Test Tolerances for FDD and TDD intra frequency absolute RSRQ accuracy in Annex F | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103758 | 0185 | - | CR on 36.521-3 for corrections of missing Es/Noc parameters in RRM test cases | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103759 | 0186 | - | Adding new test case 8.11.5 Combined E-UTRAN - EUTRAN FDD and GSM cell search | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103760 | 0187 | - | Adding new test case 8.11.6 Combined E-UTRAN - EUTRAN TDD and GSM cell search. | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103761 | 0188 | - | Adding test case 8.7.3, 8.11.5, 8.11.6 to Annex E Cell configuration mapping. | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103769 | 0189 | - | Adding band 20, 800MHz in EU to TS36.521-3 | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103773 | 0190 | - | Iteration in Handover and Re-establishment test cases | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103779 | 0191 | - | LTE-RRM: Addition of new TC E-UTRAN FDD -UTRAN FDD CPICH Ec/No absolute accuracy | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103783 | 0192 | - | Correction to q-RxLevMin for E-UTRAN - GSM cell re-selection | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103784 | 0145 | - | DL Mac Padding for RRM TCs | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103105 | 0146 | - | CR to 36.521-3: Update RSRP test cases with band 11 and 21 | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103116 | 0147 | - | Correction of CR conflict for Intra Frequency TDD reselection test | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103117 | 0201 | - | Reference to FTR 36.903 in TS 36.521-3 | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103312 | 0150 | - | Connection diagram for test 8.11.2 (3 cells) | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103315 | 0151 | - | Correction to connection diagram reference for test 8.10.1 and 8.10.2 | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103330 | 0152 | - | update on test applicability | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103358 | 0153 | - | Annex E update | 9.0.0 | 9.1.0 |
| 2010-09 | RAN#49 | R5-104098 | 0198 | - | PUSCH Scheduling for RRM tests | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104103 | 0199 | - | Delay exclusion for retransmissions in RRM test cases | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104108 | 0200 | - | Expiry of contention resolution timer in Contention based PRACH test | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104160 | 0201 | - | Uncertainties and Test Tolerances for FDD Intra Frequency Relative RSRP Accuracy section 9.1.1.2 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104230 | 0202 | - | Correction CR to 36.521-3: Update of E-UTRAN TDD-TDD cell re-selection intra frequency case 4.2.2 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104231 | 0203 | - | Correction CR to 36.521-3: Update of E-UTRAN TDD-TDD cell re-selection inter frequency case 4.2.6 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104232 | 0204 | - | Correction CR to 36.521-3: Update of E-UTRAN TDD-TDD Handover intra frequency case 5.1.2 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104233 | 0205 | - | Correction CR to 36.521-3: Update of E-UTRAN TDD-TDD Handover inter frequency case 5.1.4 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104247 | 0206 | - | Addition of Cell Configuration Mapping for Cell Search Test | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104248 | 0207 | - | CR to 36.521-3 on Correction to cell search | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104249 | 0208 | - | CR to 36.521-3 on Correction to UE Measurement Procedures | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104250 | 0209 | - | CR to 36.521-3 on Correction to RRM Cell Search | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104251 | 0210 | - | CR to 36.521-3 on Correction to RRM General | 9.1.0 | 9.2.0 |

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| 2010-09 | RAN#49 | R5-104260 | 0211 | - | Addition of test tolerances and system uncertainties for E-UTRAN FDD random access tests | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104261 | 0212 | - | Addition of test tolerances and system uncertainties for E-UTRAN TDD random access tests | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104262 | 0213 | - | LTE-RRM: Addition of new TC E-UTRAN TDD -UTRAN FDD CPICH Ec/No absolute accuracy | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104263 | 0214 | - | LTE-RRM: Addition of new TC E-UTRAN TDD -UTRAN FDD CPICH RSCP absolute accuracy | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104451 | 0215 | - | Test Tolerances and alignment for RLM FDD TC 7.3.3, 7.3.4 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104452 | 0216 | - | Test Tolerances and alignment for RLM FDD TC 7.3.5, 7.3.6 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104453 | 0217 | - | Test Tolerances and alignment for RLM TDD TC 7.3.7, 7.3.8 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104456 | 0218 | - | Uncertainties and Test Tolerances for E-UTRAN FDD Intra-frequency RRC Re-establishment | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104460 | 0219 | - | Uncertainties and Test Tolerances for E-UTRAN FDD-FDD Inter-frequency event triggered reporting under fading propagation conditions | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104497 | 0220 | - | Clarification on the neighbour cell info | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104498 | 0221 | - | Addition of the exceptional message to UE Transmit Timing | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104499 | 0222 | - | Maintenance on exceptional messages for annex info | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104500 | 0223 | - | Correction to 6.1.1 and 6.1.2 of RRC Re-establishment test case | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104501 | 0224 | - | Maintenance on exceptional messages for Mobility Control Info | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104521 | 0225 | - | 36521-3 General update of sections 00 to 07: missing Introduction references formatting | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104563 | 0226 | - | Update on exclusion of extra delay due to RRC retransmission | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104616 | 0227 | - | 36.521-3 Correction to test procedure in 8.11.5 and 8.11.6 test cases | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104625 | 0228 | - | E-UTRAN TDD inter-frequency reselection test | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104650 | 0229 | - | Clarifications of test requirements in measurement accuracy tests | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104663 | 0230 | - | 36.521-3: Annex B and Annex C update | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104825 | 0231 | - | Missing cell Identity change step for test cases with unknown cell 2 timing | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104826 | 0232 | - | Addition of test tolerances and system uncertainties for FDD timing characteristics tests | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104827 | 0233 | - | Addition of test tolerances and system uncertainties for TDD timing characteristics tests | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104828 | 0234 | - | Additions to measurement uncertainties and test tolerances for timing characteristics tests in annex F | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104829 | 0235 | - | Uncertainties and Test Tolerances for E-UTRAN FDD Inter-frequency RRC Re-establishment | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104830 | 0236 | - | Uncertainties and Test Tolerances for E-UTRAN TDD-TDD inter-frequency Handover with unknown target cell | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104839 | 0237 | - | 36521-3: Editorial update of sections 08 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104849 | 0238 | - | Maintenance on the exceptional messages in Ch8 - Annex | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104855 | 0239 | - | Uncertainties, Test Tolerances and Test Requirements for UE Transmit Timing | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104856 | 0240 | - | GSM carrier RSSI measurement accuracy in E-UTRAN TDD | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104859 | 0241 | - | E-UTRAN_to_UTRAN_FDD_reselection | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104864 | 0242 | - | Applicability of RRM inter-frequency test cases to (narrow) frequency bands | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104865 | 0243 | - | Maintenance on the exceptional messages in Ch5 - Ch6 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104866 | 0244 | - | 36.521-3: Annex E update | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104880 | 0245 | - | Correction to E-UTRAN to UTRAN Cell Re-Selection test case | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104881 | 0246 | - | Redundant information in RRM Random Access Test Requirements | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104883 | 0247 | - | E-UTRAN TDD to UTRAN FDD Handover | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104885 | 0248 | - | Cell ID change time and iteration procedure for RRM test cases 4.2.1, 4.2.2 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104886 | 0249 | - | Cell ID change time for RRM test cases 4.2.3, 4.2.6 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104887 | 0250 | - | Scrambling code change time for RRM test cases 4.3.1.1, 4.3.4.1, 8.5.2, 8.7.3, 8.11.4 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104889 | 0251 | - | Iteration procedure for handover and re-establishment test cases | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104890 | 0252 | - | Correction to cell re-selection inter frequency test case | 9.1.0 | 9.2.0 |

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|---|---|---|---|---|---|---|---|
| 2010-09 | RAN#49 | R5-105057 | 0253 | - | Clarification of Radio link monitoring test cases | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | RP-100941 | 0254 | - | Correction of status for RRM test cases and missing information in Annex | 9.1.0 | 9.2.0 |
| - | - | - | - | - | Re-insertion of the ambiguous step 11 of cl. 5.2.2.4.2 according to R5-104825 after email discussion | 9.2.0 | 9.2.1 |
| 2010-12 | RAN#50 | R5-106079 | 0255 | - | HARQ delay exclusion for HO test: Clarification for UE-DTX-case | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106080 | 0256 | - | Iteration procedure for inter RAT handover test cases | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106082 | 0257 | - | Corrections to event triggered measurement tests using DRX (Clause 8) | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106083 | 0258 | - | Missing titles in the RRM specification | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106085 | 0259 | - | Scheduling of System information for RRM tests | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106086 | 0260 | - | Update of PDCCH aggregation level for channel BW 1,4 MHz | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106119 | 0261 | - | CR to 36.521-3: Update LTE RRM test requirements for EUTRA TDD LTE band 41. | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106313 | 0262 | - | Uncertainties and Test Tolerances for Connected State Mobility test | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106314 | 0263 | - | Addition to Measurement Uncertainties and Test Tolerances for Connected State Mobility Test in Annex | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106318 | 0264 | - | Correction to inter-RAT Connected State Mobility test setup | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106320 | 0265 | - | Correction to Inter-RAT Connected State Mobility test requirements | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106321 | 0266 | - | Correction to Inter-RAT Connected State Mobility for Alignment | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106322 | 0267 | - | Correction to Inter-RAT Connected State Mobility test requirements | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106448 | 0268 | - | Addition of SIB7 exceptional messages | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106451 | 0269 | - | Correction to UE transmit timing TC | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106455 | 0270 | - | Correction to the exceptional messages in RSRQ tests | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106456 | 0271 | - | Correction to Min Test time for RRM fading tests | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106483 | 0272 | - | Annex E update | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106493 | 0273 | - | CR to 36.521-3: Update to G.2.6 Test Conditions for Delay Tests and UE Measurement Performance | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106805 | 0274 | - | Correction to test case 5.1.2 - Update of E-UTRAN TDD-TDD Handover intra frequency case | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106806 | 0275 | - | Correction to test case 5.1.4 - Update of E-UTRAN TDD-TDD Handover inter frequency case | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106807 | 0276 | - | Correction to Inter-RAT UE Measurements Procedures | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106808 | 0277 | - | Correction to Inter-RAT UE Measurements Procedures under fading | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106810 | 0278 | - | Correction to test case 8.2.1 | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106811 | 0279 | - | Correction to test case 8.2.2 | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106812 | 0295 | - | Update of RRM OCNG patterns | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106829 | 0280 | - | General Corrections to RRC_IDLE State Mobility | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106830 | 0281 | - | Correction to test case 6.2.3 | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106831 | 0282 | - | Correction to test case 6.2.4 | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106832 | 0283 | - | Correction to MeasConfig-DEFAULT in RRM TCs | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106833 | 0284 | - | Adding support of inter-band test configuration for RRM inter-frequency/inter-RAT test cases | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106834 | 0285 | - | CR on UEs RRM Band applicability | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106835 | 0286 | - | Correction to test case 7.1.2 | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106836 | 0287 | - | Correction to test case 10.1.2.1, 9.1.2.2 and 9.2.2.1 | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106840 | 0288 | - | Update to Radio Link Monitoring Test Cases | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106857 | 0291 | - | Correction to DL configuration on Non-Contention Based Random Access Test | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106859 | 0292 | - | Corrections to UE transmit timing tests (Subclause 7.3) | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106862 | 0293 | - | Correction to DL configuration on Contention Based Random Access Test | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106864 | 0294 | - | Update of Annex G for RLM test in DRX | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106870 | 0289 | - | Uncertainties and Test Tolerances for UE measurements procedures test | 9.2.1 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106871 | 0290 | - | Addition to Measurement Uncertainties and Test Tolerances for UE Measurement Procedures test in Annex | 9.2.1 | 9.3.0 |
| 2011-03 | RAN#51 | R5-110150 | 0296 | - | RRC Re-establishment tests: Corrections to Message contents | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110151 | 0297 | - | Radio link monitoring tests: Corrections to Message contents | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110155 | 0298 | - | UE Measurements Procedures tests: Test loop | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110167 | 0299 | - | Removal of [ ] from PDSCH and PCFICH/PDCCH/PHICH Measurement Channel references | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110348 | 0300 | - | Revision of 36.521-3 Annex G - Statistical testing | 9.3.0 | 9.4.0 |

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|---|---|---|---|---|---|---|---|
| 2011-03 | RAN#51 | R5-110418 | 0301 | - | Correction to TDD cell re-selection | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110419 | 0302 | - | Correction to exception messages in 4.5.1 HRPD Re selection test | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110424 | 0303 | - | Alignment of exception messages for TDD event triggered measurement tests | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110435 | 0304 | - | Modification of message content definition for TC 8.4.1 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110437 | 0305 | - | Update to TC 8.6.1: E-UTRAN TDD - UTRAN FDD event triggered reporting under fading propagation conditions | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110438 | 0306 | - | Correction to TC 8.7.1: E-UTRA TDD to UTRA(1.28 Mcps TDD OPTION) cell search in fading propagation conditions | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110443 | 0307 | - | Update to TC 8.8.1: E-UTRAN FDD - GSM event triggered reporting in AWGN | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110445 | 0308 | - | Corrections to TC 8.9.1: E-UTRAN FDD - UTRAN TDD event triggered reporting under fading propagation conditions | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110520 | 0309 | - | Correction to Inter-RAT Connected State Mobility for Alignment | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110546 | 0310 | - | Uncertainties and Test Tolerances for Connected State Mobility Inter-RAT to UTRAN test | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110549 | 0312 | - | Uncertainties and Test Tolerances for Connected State Mobility Inter-RAT to GSM unknown test | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110584 | 0314 | - | Correction to gap pattern ID in test case 5.1.4 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110586 | 0315 | - | Clarification to 1.4 MHz testing and applicability in test case 7.1.1 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110588 | 0316 | - | Test time limit correction for DRX=40ms in test case 8.1.3 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110863 | 0330 | - | Higher SNR on event triggered measurement tests | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110866 | 0311 | - | Addition to Measurement Uncertainties and Test Tolerances for Connected State Mobility Inter-RAT to UTRAN test in Annex | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110868 | 0313 | - | Addition to Measurement Uncertainties and Test Tolerances for Connected State Mobility Inter-RAT to GSM unknown test in Annex | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110902 | 0317 | - | CR to 36.521-3: Update LTE RRM test requirements for EUTRA TDD LTE band 41 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110903 | 0318 | - | Correction to exception messages in 5.3.1 HRPD HHO test | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110904 | 0319 | - | MIMO Correlation scenario for RLM test cases | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110905 | 0320 | - | Enabling HARQ for section 8 and 9 RRM Tests | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110907 | 0321 | - | Re-ordering of Time periods, definition of uncertainties, and addition of Test Tolerances for RRM test case 4.3.1.1 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110910 | 0322 | - | Updated Test Tolerances for RRM Test cases 7.1.1 + 7.1.2 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110911 | 0323 | - | Uncertainties and Test Tolerances for Connected State Mobility Inter-RAT to GSM tests | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110912 | 0324 | - | Addition to Measurement Uncertainties and Test Tolerances for Connected State Mobility Inter-RAT to GSM test in Annex | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110927 | 0325 | - | Corrections to RRM TC 8.1.1, 8.1.2 and 8.1.3 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110928 | 0326 | - | Corrections to test cases about E-UTRAN FDD-FDD Inter-frequency measurement 8.3.1, 8.3.2 and 8.3.3 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110929 | 0327 | - | Corrections to TCs related to E-UTRAN FDD - UTRAN measurements: 8.5.1, 8.5.2 and 8.5.3 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110930 | 0328 | - | UE Measurement procedures tests: Corrections to Message contents | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110931 | 0329 | - | DL-RMC-s and OCNG for RRM tests: Updates | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110946 | 0331 | - | Uncertainties and Test Tolerances for RRM test case 4.3.1.2 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110948 | 0332 | - | Uncertainties and Test Tolerances for RRM test cases 4.4.1 and 4.4.2 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110956 | 0333 | - | Modification of test case 5.1.6 - E-UTRAN TDD-TDD inter frequency handover: unknown target cell | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110957 | 0334 | - | LTE RRM: reference to state 3A in 36.521-3 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110958 | 0335 | - | Correction to RRM testes for Alignment | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110959 | 0336 | - | CR to 36.521-3: E-UTRAN FDD - UTRA FDD CPICH RSCP absolute accuracy test case | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110960 | 0337 | - | CR to 36.521-3: Addition of Multiple E-UTRAN FDD-FDD Inter-frequency event triggered reporting test case | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110961 | 0338 | - | CR to 36.521-3: E-UTRAN FDD-FDD Inter-frequency and UTRAN FDD event triggered reporting test case | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110962 | 0339 | - | Correction to exception messages in Radio Link Monitoring Test | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110963 | 0340 | - | Correction to TC 8.4.2: E-UTRAN TDD-TDD Inter-frequency event triggered reporting when DRX is used under fading propagation conditions in synchronous cells | 9.3.0 | 9.4.0 |

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|---|---|---|---|---|---|---|---|
| 2011-03 | RAN#51 | R5-110964 | 0341 | - | Corrections to TC 8.7.3: E-UTRAN TDD - UTRAN TDD SON ANR cell search reporting under AWGN propagation conditions | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110965 | 0342 | - | Correct the message definitions related to the RSRP and RSRQ performance testing | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110966 | 0343 | - | Update of RRM test 8.5.2 FDD SON | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110974 | 0344 | - | PUSCH scheduling: Correction for considering DRX | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110980 | 0345 | - | Correction to TC 8.8.2: E-UTRAN FDD - GSM Event triggered reporting when DRX is used in AWGN | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110981 | 0346 | - | Update to TC 8.10.1: E-UTRAN TDD - GSM event triggered reporting in AWGN | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110982 | 0347 | - | Corrections to TC 8.10.2: E-UTRAN TDD-GSM event triggered reporting when DRX is used in AWGN | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110983 | 0348 | - | Modification to TC 8.7.2: E-UTRAN TDD to UTRAN 1.28Mcps TDD cell search when DRX is used in fading propagation conditions | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110995 | 0352 | - | Radio link monitoring test 7.3.4: Minor correction to the test requirement | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110996 | 0353 | - | Radio link monitoring tests: Corrections to the test procedure | 9.3.0 | 9.4.0 |
| 2011-06 | RAN#52 | R5-112124 | 0354 | - | Uncertainties and Test Tolerances for RRM test case 8.7.1 | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112126 | 0355 | - | Uncertainties and Test Tolerances for RRM test case 8.7.2 | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112128 | 0356 | - | Uncertainties and Test Tolerances for RRM test cases 8.8.1+8.10.1 | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112152 | 0357 | - | RRM TC-s 4.2: Transition between time intervals | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112153 | 0358 | - | RRM TC 4.2.6: Introduction of time duration T0 | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112155 | 0359 | - | RRM TC-s clause 8: Reference to the state 3A / 3A-RF in the test loop | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112185 | 0360 | - | Addition of new RRM TC 4.3.4.3: EUTRA TDD-UTRA TDD cell reselection in fading propagation conditions: UTRA TDD is of lower priority | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112314 | 0365 | - | Correction to E-UTRAN FDD - UTRAN FDD cell re-selection when UTRA FDD is under lower priority | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112315 | 0366 | - | Correction to E-UTRA FDD-high UTRA FDD inter RAT cell re-selection test case | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112316 | 0367 | - | Correction to E-UTRAN TDD-UTRAN TDD cell search reporting for SON ANR in AWGN propagation conditions | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112317 | 0368 | - | Correction on test cases of E-UTRA to UTRA cell reselection in idle state | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112318 | 0369 | - | Correction to E-UTRAN TDD - UTRAN TDD test case in 36.521-3 | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112418 | 0370 | - | Update of 4.3.1.3 E-UTRA-UTRA reselection test case | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112421 | 0371 | - | Correction to 6.1 RRC Re-establishment test case | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112423 | 0372 | - | Maintenance on Message contents in 8.5.3 | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112424 | 0373 | - | Correction to Annex H.3.3 for Inter-RAT E-UTRAN - HRPD handover | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112454 | 0374 | - | Wrong references into statistical annex | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112457 | 0375 | - | References into connection diagrams in 36.508,Annex A | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112470 | 0376 | - | Misalignment in Meas Gap configuration | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112475 | 0377 | - | Band 20 missing in section 9 test cases | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112533 | 0378 | - | Uncertainties and Test Tolerances for connected state mobility Inter-RAT TDD to GSM test | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112536 | 0379 | - | Addition to Measurement Uncertainties and Test Tolerances for connected state mobility Inter-RAT TDD to GSM test in Annex | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112543 | 0380 | - | Uncertainties and Test Tolerances for connected state mobility Inter-RAT TDD to GSM unknown test | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112544 | 0381 | - | Addition to Measurement Uncertainties and Test Tolerances for connected state mobility Inter-RAT TDD to GSM unknown test in Annex | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112546 | 0382 | - | Uncertainties and Test Tolerances for Inter-RAT to UTRAN event triggered reporting under fading test | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112554 | 0383 | - | Uncertainties and Test Tolerances for Inter-RAT to GSM event triggered when DRX is used in AWGN tests | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112555 | 0384 | - | Addition to Measurement Uncertainties and Test Tolerances for Inter-RAT to GSM event triggered when DRX is used in AWGN tests in Annex | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112734 | 0385 | - | Addition of Band 24 to section 9.1 and 9.2 , RSRP and RSRQ measurement performance requirements | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112741 | 0363 | - | Uncertainties and Test Tolerances for RRM test case 10.3.1 | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112742 | 0364 | - | Uncertainties and Test Tolerances for RRM test case 9.3.2 | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112745 | 0394 | - | Completing for E-UTRAN TDD-UTRAN TDD cell reselection_UTRA is of lower priority | 9.4.0 | 9.5.0 |

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|---|---|---|---|---|---|---|---|
| 2011-06 | RAN#52 | R5-112746 | 0395 | - | Completing for E-UTRAN FDD-UTRAN FDD-UTRAN TDD cell reselection | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112803 | 0386 | - | Addition to Measurement Uncertainties and Test Tolerances for Inter-RAT to UTRAN event triggered reporting under fading test in Annex | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112815 | 0387 | - | Correction to test frequency references in RRM initial condition | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112817 | 0388 | - | RRM TC-s 8.1.1, 8.1.2, 8.1.3, 8.2.1, 8.2.2: Changing SNR for serving cell | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112818 | 0389 | - | RRM TC 9.6.2: Overall corrections | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112819 | 0390 | - | CR for 9.4 UTRA FDD measurement accuracy | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112820 | 0391 | - | Add test frequencies for bands 42, 43 (3500MHz) | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112849 | 0398 | - | Update of clause 3A.3 RRM test configuration | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112853 | 0399 | - | Correction to inconsistent test procedures in RRM | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112855 | 0400 | - | Uncertainties and Test Tolerances for RRM test case 5.2.10 | 9.4.0 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112858 | 0401 | - | Addition of new RRM TC 8.4.3: E-UTRAN TDD-TDD inter-freq event triggered reporting under AWGN in synchronous cells with DRX when L3 filtering is used | 9.4.0 | 9.5.0 |
| 2011-09 | RAN#53 | R5-113183 | 0402 | - | RRM TC 8: Adding missing PRACH Configuration for some tests | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113226 | 0403 | - | Uncertainties and Test Tolerances for RRM test case 4.3.1.3 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113249 | 0404 | - | Uncertainties and Test Tolerances for TC 5.2.1 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113250 | 0405 | - | Uncertainties and Test Tolerances for TC 5.2.2 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113395 | 0406 | - | Not tested minimum requirement in Clause 8 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113460 | 0407 | - | Correction to 4.2.3 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113461 | 0408 | - | Correction to the exceptional messages in HO TCs with unknown target cell | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113462 | 0409 | - | Maintenance on the exceptional messages for Mobility Control Info | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113463 | 0410 | - | Correction to 6.2.3 and 6.2.4 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113466 | 0411 | - | Correction to FDD RSRP and RSRQ test | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113467 | 0412 | - | Correction to TDD RSRP and RSRQ test for band 41 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113468 | 0413 | - | Correction to the exceptional messages in Annex H | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113597 | 0414 | - | Abbreviation update and Editorial corrections in TS36.521-3 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113843 | 0443 | - | Adding FGI Applicabilities into Chapters 4 - 7 in 36.521-3 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113844 | 0440 | - | RRM TCs 5.1: PRACH power configuration | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113845 | 0444 | - | RRM TCs 7.3: Update of the test procedure and requirements | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113846 | 0425 | - | Statistical clarification for TC 8.3.3 and 8.3.4 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114005 | 0415 | - | LTE-RRM: Corrections to test iteration for test case 4.3.4.1 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114007 | 0416 | - | Correction on the inter-RAT cell identification time in DRX | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114009 | 0417 | - | Completing for E-UTRAN TDD - UTRAN TDD handover test case | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114013 | 0418 | - | Uncertainties and Test Tolerance for FDD SON ANR test case 8.5.2 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114016 | 0419 | - | Uncertainties and Test Tolerances for TC 9.4.1 and 9.4.2 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114019 | 0420 | - | CR Uncertainties and TT for 8.4.3 in 36.521-3 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114021 | 0421 | - | CR Uncertainties and TT for 4.3.4.3 in 36.521-3 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114026 | 0422 | - | Deletion of editor note for discrepancy between TT and 36.903 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114050 | 0423 | - | RRM: Use of State 3A-RF | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114055 | 0424 | - | RRM TCs 7.2: Transition between iteration loops | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114057 | 0426 | - | Statistical clarification in 6 Test cases in clause 8.11. | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114059 | 0427 | - | Completing for E-UTRAN TDD-UTRAN TDD cell re-selection_UTRA is of higher priority | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114060 | 0428 | - | Uncertainties and Test Tolerances for TC 8.9.1 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114072 | 0429 | - | Update LTE RRM test requirements for FDD LTE Band 23 in 36.521-3 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114084 | 0430 | - | Simplification of frequency dependent minimum requirements in TS36.521-3 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114097 | 0431 | - | Adding FGI Applicabilities into Chapters 8 - 9 in 36.521-3 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114099 | 0432 | - | Addition of new RRM TC 8.1.5 ÜE-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114100 | 0433 | - | Addition of new RRM TC 8.1.6: E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114101 | 0434 | - | Addition of new RRM TC 8.2.3: E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | 9.5.0 | 9.6.0 |

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| 2011-09 | RAN#53 | R5-114102 | 0435 | - | Addition of new RRM TC 8.2.4: E-UTRAN TDD - TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114103 | 0436 | - | Addition of new RRM TC 8.3.4´ UE-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114104 | 0437 | - | Addition of new RRM TC 8.3.5:E-UTRAN FDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114105 | 0438 | - | Addition of new RRM TC 8.4.4:E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114106 | 0439 | - | Addition of new RRM TC 8.4.5:E-UTRAN TDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114111 | 0441 | - | Correction to RLM | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114115 | 0442 | - | LTE-RRM: Correction to test procedure for inter-RAT cell reselection test cases | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114119 | 0445 | - | Introduction of Expanded 1900MHz Band (Band 25) into section 9 of 36.521-3 | 9.5.0 | 9.6.0 |
| 2011-12 | RAN#54 | R5-115121 | 0446 | - | RRM TC-s 7, 8: Iteration loop and usage of the UE states 3A / 3A-RF | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115140 | 0447 | - | Modify the test requirement table in the TC 5.2.1 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115142 | 0449 | - | LTE-RRM :Update to Annex E | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115189 | 0452 | - | Uncertainties and Test Tolerances for RRM test case 8.11.3 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115199 | 0453 | - | Correction of references to Annex I in TS36.521-3 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115200 | 0454 | - | Test System uncertainties for frequencies between 3000MHz to 4200MHz in 36.521-3 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115202 | 0456 | - | Uncertainties and Test Tolerance for TDD SON ANR test case 8.7.3 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115327 | 0457 | - | Correction to RRM tests 7.1.2 and 7.2.2 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115379 | 0458 | - | Update of operating band configuration | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115381 | 0459 | - | Correction to FGI in test applicability for Cell reselection test case | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115385 | 0460 | - | Correction to 5.2.2 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115386 | 0461 | - | Alignment of the exceptional messages in 7.3.x RLM | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115387 | 0462 | - | Correction to 8.10.1 and 8.10.2 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115388 | 0463 | - | Correction to the exceptional message in 8.6.1.1 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115403 | 0465 | - | Uncertainties and Test Tolerances for RRM test case 4.3.3 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115433 | 0466 | - | Corrections to TC 5.1.5 and TC 5.1.6 inter-f HO: unknown target cell | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115435 | 0467 | - | Updates of TC 5.2.3: E-UTRAN FDD - GSM handover | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115479 | 0468 | - | Corrections to RSRQ in Intra-Frequency Measurement Minimum Requirements | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115482 | 0469 | - | Addition to measurement uncertainties and test tolerances E-UTRAN FDD - UTRAN FDD event triggered reporting w/DRX under fading test in Annex | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115787 | 0471 | - | Uncertainties and Test Tolerances for RRM test case 8.11.1 and 8.11.2 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115582 | 0472 | - | RRM TC 6.2: Corrections to power settings | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115814 | 0473 | - | Incomplete test case for 7.1.1 and 7.1.2 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115823 | 0474 | - | Uncertainties and TT for TC 6.1.3 and 6.1.4 in 36.521-3 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115827 | 0477 | - | Correction to Test Tolerances for RRM ch.9 test cases | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115833 | 0478 | - | Adding band 22 (3500MHz FDD) to 36.521-3 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115834 | 0479 | - | RRM: Phase rotation for intra frequency tests in static conditions | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115835 | 0480 | - | Addition of the exceptional message in 4.6.1.1 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115836 | 0481 | - | Addition of undefined UTRA system information for TC 4.3.1.1 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115838 | 0482 | - | Corrections to TC 5.2.1 and TC 5.2.2: E-UTRAN FDD/TDD - UTRAN FDD handover | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115839 | 0483 | - | Corrections to message content definition for TC 5.1.1 and TC 5.1.2: intra-f HO | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115840 | 0484 | - | Updates to TC 5.1.3 and TC 5.1.4: inter-f HO | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115841 | 0485 | - | Updates of TC 5.2.4 and TC 5.2.5: E-UTRAN FDD/TDD - UTRAN TDD handover | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115842 | 0486 | - | Removal of measurement related message definitions in TC 5.2.7, TC 5.2.9 and TC 5.2.10 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115843 | 0487 | - | Modification of message definitions in the Annex H | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115844 | 0488 | - | Modification of the test cases of Random Access | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115845 | 0489 | - | RRM TC-s 9: Missing bands in specification | 9.6.0 | 9.7.0 |

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| 2011-12 | RAN#54 | R5-115846 | 0490 | - | RRM TC-s 7.3: SRS configuration in radio link monitoring tests | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115847 | 0491 | - | Correction to test frequency in MeasConfig-DEFAULT | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115848 | 0492 | - | Corrections to TC 7.1.1 and TC 7.1.2: UE Transmit Timing Accuracy | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115850 | 0493 | - | Correction to 5.2.6 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115878 | 0494 | - | Correction to cell reselection delay in test procedure | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115882 | 0495 | - | Addition of undefined UTRA system information for TC 4.3.1.2 and TC 4.3.1.3 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115883 | 0496 | - | Addition of undefined UTRA system information for TC 4.3.2 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115884 | 0497 | - | Addition of undefined UTRA system information for TC 4.3.3 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115885 | 0498 | - | Addition of undefined UTRA system information for TC 4.3.4.1 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115886 | 0499 | - | Addition of UTRA system information definitions for TC 4.3.4.2 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115887 | 0500 | - | Updates of the message content definitions for TC 4.3.4.3 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115888 | 0501 | - | Updates of TC 4.4.1 and TC 4.4.2: E-UTRAN FDD/TDD - GSM cell re-selection | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115786 | 0502 | - | Corrections to test cases for E-UTRAN RRC Re-establishment | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115893 | 0503 | - | RRM TC-s 4: General review of the test procedures of cell re-selection test cases | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115481 | 0504 | - | Uncertainties and test tolerances E-UTRAN FDD - UTRAN FDD event triggered reporting w/DRX under fading test | 9.6.0 | 9.7.0 |
| 2012-03 | RAN#55 | R5-120107 | 0505 | - | Uncertainties and Test Tolerance for E-UTRAN TDD Intra-frequency new CGI test cases 8.2.3 and 8.2.4 | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120124 | 0506 | - | Uncertainties and Test Tolerance for E-UTRAN TDD Inter-frequency new CGI test cases 8.4.4 and 8.4.5. | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120141 | 0507 | - | RRM: Iteration loop in cdma2000 reselection tests | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120178 | 0508 | - | RF/RRM: Correction on TC 8.4.1 message content definition | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120183 | 0509 | - | RF/RRM: Addition of new TC 4.2.4 E-UTRAN FDD - TDD cell re-selection inter frequency case | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120184 | 0510 | - | RF/RRM: Addition of new TC 4.2.5 E-UTRAN TDD - FDD cell re-selection inter frequency case | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120185 | 0511 | - | RF/RRM: Addition of new TC 5.1.7 E-UTRAN FDD - TDD handover inter frequency case | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120186 | 0512 | - | RF/RRM: Addition of new TC 5.1.8 E-UTRAN TDD - FDD handover Inter frequency case | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120187 | 0513 | - | RF/RRM: Addition of new TC 8.12.1 E-UTRAN TDD - FDD Inter-frequency event triggered reporting under fading propagation conditions in asynchronous cells | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120189 | 0514 | - | RF/RRM: Addition of new TC 9.1.5.1 FDD - TDD Inter Frequency Absolute RSRP Accuracy | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120190 | 0515 | - | RF/RRM: Addition of new TC 9.1.5.2 FDD - TDD Inter Frequency Relative Accuracy of RSRP | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120191 | 0516 | - | RF/RRM: Addition of Cell configuration mapping for those new RRM test cases | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120245 | 0517 | - | Update of 36.521-3 Test Cases 9.1.4.1 and 9.1.4.2, Io difference band-independent | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120249 | 0518 | - | Uncertainties and Test Tolerances for RRM test case 8.11.4 | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120321 | 0519 | - | Update of 4.3.1 E-UTRAN FDD - UTRAN FDD cell re-selection | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120322 | 0520 | - | Update of 4.3.4 E-UTRAN TDD - UTRAN TDD cell re-selection | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120323 | 0521 | - | Adding E-UTRAN test parameter reference to messages exception | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120324 | 0522 | - | Correction to 4.6.1.1 E-UTRAN FDD c2k cell re-selection | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120325 | 0523 | - | Correction to 5.2.4 E-UTRAN TDD - UTRAN TDD handover | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120339 | 0524 | - | Addition of FGI bit 16 into test cases 9.1.x.x and 9.2.x.x | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120341 | 0525 | - | Correction to FGI bits in test case 8.5.2 | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120424 | 0526 | - | Correction to TIntra in Minimum Conformance Requirements | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120425 | 0527 | - | Correction to the identification time in DRX for UTRA TDD | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120515 | 0528 | - | Addition of FGI bit 15 into test cases configuring event 1B | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120808 | 0535 | - | RF/RRM: Corrections on RSRP and RSRQ accuracy related test cases | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120809 | 0536 | - | TS 36.521-3: 8.3.3 and 8.4.3 T2 value correction | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120810 | 0537 | - | TS 36.521-3: 6.2.3 Extreme conditions test tolerance correction | 9.7.0 | 9.8.0 |

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| 2012-03 | RAN#55 | R5-120827 | 0538 | - | Correction to CQI report configuration of 7.1.1 in TS36.521-3 | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120846 | 0539 | - | Test configuration for Inter RAT testcases, delete note | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120847 | 0540 | - | RF/RRM: Addition of new TC 8.13.1 E-UTRAN FDD - TDD Inter-frequency event triggered reporting under fading propagation conditions in asynchronous cells | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120848 | 0541 | - | Correction to 8.7.3 E-UTRAN TDD - UTRAN TDD SON ANR | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120892 | 0542 | - | Update of PRACH test case | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120894 | 0543 | - | Correction to 5.2.10 E-UTRAN TDD - UTRAN TDD handover | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120895 | 0544 | - | Correction to test frequency in MeasConfig-DEFAULT for E-UTRAN to GSM cell search test case | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120907 | 0547 | - | Uncertainties and Test Tolerances for E-UTRAN to HRPD Cell reselection TC 4.5.1.1 | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120917 | 0546 | - | Uncertainties and Test Tolerances for E-UTRAN to HRPD HO TC 5.3.1 | 9.7.0 | 9.8.0 |
| 2012-03 | RAN#55 | R5-120530 | 0529 | - | Introduction to FDD RSRQ for E-UTRA Carrier Aggregation | 9.8.0 | 10.0.0 |
| 2012-03 | RAN#55 | R5-120531 | 0530 | - | Introduction to TDD RSRQ for E-UTRA Carrier Aggregation | 9.8.0 | 10.0.0 |
| 2012-03 | RAN#55 | R5-120532 | 0531 | - | Introduction to FDD RSRQ for E-UTRA Carrier Aggregation in Annex | 9.8.0 | 10.0.0 |
| 2012-03 | RAN#55 | R5-120533 | 0532 | - | Introduction to TDD RSRQ for E-UTRA Carrier Aggregation in Annex | 9.8.0 | 10.0.0 |
| 2012-03 | RAN#55 | R5-120535 | 0533 | - | Introduction to Carrier Aggregation in Radio Resource Management | 9.8.0 | 10.0.0 |
| 2012-03 | RAN#55 | R5-120536 | 0534 | - | Introduction to Carrier Aggregation in Default Message Contents | 9.8.0 | 10.0.0 |
| 2012-06 | RAN#56 | R5-121229 | 0549 | - | Correction of test procedures for Autonomous gap test cases, section 8 | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121240 | 0550 | - | Revise test frequencies for FDD-TDD interworking Test cases | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121247 | 0551 | - | RRM: Removal of Editors note on connection diagram used in intra frequency tests in static conditions | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121527 | 0552 | - | Addition of Handover to UTRAN commands in 36.521-3 | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121528 | 0553 | - | Correction of drx-RetransmissionTimer parameters | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121529 | 0554 | - | Correction to Test2 in 7.1.1 and 7.1.2 of 36.521-3 | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121530 | 0555 | - | Correction to DRX offset in 7.1.2 | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121532 | 0557 | - | Correction to 8.11.1 and 8.11.2 | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121547 | 0558 | - | Addition of new TCs for UTRAN TDD P-CCPCH RSCP measurement | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121691 | 0559 | - | TS 36.521-3: 8.3.3 and 8.4.3 update | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121901 | 0560 | - | Introduction of E-UTRAN Inter Introduction of E-UTRAN Inter frequency case reselection in the existence of non-allowed CSG cell | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121902 | 0561 | - | Addition of new RRM TC 6.3.1: Redirection from E-UTRAN FDD to UTRAN FDD | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121922 | 0562 | - | Addition of new RRM TC 6.3.2: Redirection from E-UTRAN TDD to UTRAN FDD | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121923 | 0563 | - | Addition of new RRM TC 6.3.3: Redirection from E-UTRAN FDD to GERAN when System Information is provided | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121924 | 0564 | - | Addition of new RRM TC 6.3.4: Redirection from E-UTRAN TDD to GERAN when System Information is provided | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121927 | 0565 | - | Uncertainties and Test Tolerances for TC 9.6.2 | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121929 | 0566 | - | Uncertainties and Test Tolerances for RRM test cases 4.2.4 and 4.2.5 | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121930 | 0567 | - | Uncertainties and Test Tolerances for RRM test cases 5.1.7 and 5.1.8 | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121931 | 0568 | - | Uncertainties and Test Tolerances for RRM test cases 8.14.1 | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121932 | 0569 | - | Uncertainties and Test Tolerances for RRM test case 8.15.1 | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121933 | 0570 | - | Uncertainties and Test Tolerances for RRM test case 9.1.5.1 | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121934 | 0571 | - | Uncertainties and Test Tolerances for FDD - TDD Inter Frequency Relative Accuracy of RSRP test case 9.1.5.2 | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121940 | 0572 | - | Correction to the test mode references for RRM tests | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121941 | 0573 | - | Addition of new RRM TC 6.3.5: E-UTRA TDD RRC connection release redirection to UTRA TDD | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121942 | 0574 | - | RRM: Introduction of Annex for handling of different releases and UE capabilities | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121969 | 0575 | - | Adding operating band 26 to TS 36.521-3 | 10.0.0 | 10.1.0 |

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| 2012-06 | RAN#56 | R5-121977 | 0576 | - | Change in transmit timing tests based on DRX feature group indicator | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121980 | 0577 | - | Clarifications to FDD RSRQ for E-UTRA Carrier Aggregation | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121981 | 0578 | - | Clarifications to TDD RSRQ for E-UTRA Carrier Aggregation | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121991 | 0579 | - | Further Test Tolerance analysis for operating band 25 and 41 in 36.521-3 | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121998 | 0580 | - | Addition of new RRM TC 6.3.6: E-UTRA FDD RRC connection release redirection to UTRA TDD | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-121999 | 0581 | - | Addition of new RRM TC 8.5.4 E-UTRAN FDD - UTRAN FDD enhanced cell identification under AWGN propagation conditions | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-122000 | 0582 | - | Addition of new RRM TC 8.7.4 E-UTRA TDD-UTRA TDD enhanced cell identification under AWGN propagation conditions | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-122001 | 0583 | - | Addition of new RRM TC 8.9.2 E-UTRA FDD-UTRA TDD enhanced cell identification under AWGN propagation conditions | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-122002 | 0584 | - | RRM: Clarifications to the OCNG patterns | 10.0.0 | 10.1.0 |
| 2012-06 | RAN#56 | R5-122007 | 0556 | - | Correction to SR Config Index for TDD DRX test cases | 10.0.0 | 10.1.0 |
| 2012-09 | RAN#57 | R5-123065 | 0585 | - | Correction to References in Annex I | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123149 | 0587 | - | Corrections to E-UTRAN FDD intra frequency measurements requirements | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123151 | 0588 | - | Corrections to E-UTRAN FDD inter frequency measurements requirements | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123152 | 0589 | - | Corrections to E-UTRAN TDD intra frequency measurements requirements | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123153 | 0590 | - | Corrections to E-UTRAN TDD inter frequency measurements requirements | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123163 | 0591 | - | Introduction of E-UTRAN TDD-TDD Event Triggered reporting on deactivated SCell with PCell interruption in non-DRX for CA | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123166 | 0592 | - | Correction to FDD RSRQ for E-UTRA Carrier Aggregation tests | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123167 | 0593 | - | Correction to TDD RSRQ for E-UTRA Carrier Aggregation tests | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123168 | 0594 | - | Introduction of default RRC messages exceptions for Carrier Aggregation | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123281 | 0596 | - | Addition of new TC 4.6.2.1 E-UTRAN TDD-cdma2000 1X Cell Reselection: cdma2000 1X is of Lower Priority | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123290 | 0597 | - | Addition of new TC 8.18.1 E-UTRAN TDD-HRPD event triggered reporting under fading propagation conditions | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123291 | 0598 | - | Addition of new TC 8.19.1 E-UTRAN TDD-CDMA2000 1X event triggered reporting under fading propagation conditions | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123292 | 0599 | - | Annex H message content updates | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123301 | 0600 | - | Addition of Cell configuration mapping for new RRM test cases | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123335 | 0601 | - | Correction to transmit timing test cases | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123336 | 0602 | - | Addition of band indicator for GERAN | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123426 | 0603 | - | RRM: Update of Annex J | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123903 | 0606 | - | Addition of new RRM TC 4.5.2.1 E-UTRAN TDD-HRPD Cell Reselection: HRPD is of Lower Priority | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123904 | 0607 | - | Addition of new TC 5.3.5 E-UTRAN TDD-HRPD Handover | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123905 | 0608 | - | Addition of new TC 6.3.8 E-UTRA FDD RRC connection release redirection to UTRA TDD without SI provided | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123906 | 0609 | - | Addition of new TC 6.3.9 Redirection from E-UTRAN FDD to UTRAN FDD without System Information | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123907 | 0610 | - | Addition of new TC 6.3.10 Redirection from E-UTRAN FDD to GERAN when System Information is not provided | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123908 | 0611 | - | Addition of new TC 6.3.11 Redirection from E-UTRAN TDD to GERAN when System Information is not provided | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123916 | 0612 | - | Introduction of Chapter9 absolute and relative RSRP measurement test cases for carrier aggregation (TC 9.1.6.1 and 9.1.6.2) | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123920 | 0613 | - | Uncertainties and Test Tolerances for E-UTRAN FDD, TDD Inter frequency reselection in the existence of non-allowed CSG cell Test cases 4.2.7 and 4.2.8 | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123935 | 0614 | - | Addition of new TC 5.3.6 E-UTRAN TDD-cdma2000 1X Handover | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123936 | 0615 | - | Addition of new TC 6.3.7 E-UTRA TDD RRC connection release redirection to UTRA TDD without SI provided | 10.1.0 | 10.2.0 |

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|---|---|---|---|---|---|---|---|
| 2012-09 | RAN#57 | R5-123937 | 0616 | - | Addition of new TC 6.3.12 E-UTRAN TDD RRC connection release redirection to UTRAN FDD without SI provided | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123938 | 0617 | - | Addition of new TC 9.2.4A.1 FDD - TDD Inter Frequency Absolute RSRQ Accuracy | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123939 | 0618 | - | Addition of new TC 9.2.4A.2 FDD - TDD Inter Frequency Relative Accuracy of RSRQ | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123940 | 0619 | - | Addition of new TC 8.14.3 E-UTRAN TDD - FDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123941 | 0620 | - | Addition of new TC 8.15.3 E-UTRAN FDD - TDD Inter-frequency identification of a new CGI of E-UTRA cell using autonomous gaps | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123959 | 0621 | - | Introduction of E-UTRAN FDD-FDD Event Triggered reporting on deactivated SCell with PCell interruption in non-DRX for CA | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123960 | 0622 | - | Introduction of E-UTRAN Event Triggered reporting on deactivated SCell with PCell interruption in non-DRX for CA in Annex | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123995 | 0623 | - | Addition of new TC 8.14.2 E-UTRAN TDD-FDD Inter-frequency event triggered reporting when DRX is used under fading propagation conditions in synchronous cells | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123996 | 0624 | - | Addition of new TC 8.15.2 E-UTRAN FDD-TDD Inter-frequency event triggered reporting when DRX is used under fading propagation conditions in asynchronous cells | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123998 | 0625 | - | RRM: Further avoidance of frequency overlapping for inter-frequency and inter-RAT | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123999 | 0626 | - | Adding clauses in TS36.521-3 with references for positioning test cases | 10.1.0 | 10.2.0 |
| 2012-09 | RAN#57 | R5-123789 | 0605 | - | Implementation of only partly implemented CR: Clarification of the release of UTRAN-EUTRAN Inter-RAT RRM test cases in 36.521-3 | 10.2.0 | 10.2.1 |
| 2012-12 | RAN#58 | R5-124155 | 0667 | - | New TC(8.20) introduction of inter-frequency/RAT measurements in CA mode | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-124172 | 0668 | - | Correction to TC 8.16.3 E-UTRAN FDD-FDD Event Triggered reporting on deactivated SCell with PCell interruption in non-DRX for CA | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-124173 | 0669 | - | Corrections to TC 8.16.4 E-UTRAN TDD-TDD Event Triggered reporting on deactivated SCell with PCell interruption in non-DRX for CA | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-124174 | 0670 | - | Introduction of default RRC messages exceptions for Carrier Aggregation | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-124180 | 0671 | - | Addition of new cell configuration mapping for CA related test cases | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125209 | 0627 | - | RRM TC 9.1.6.1: General updates and corrections | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125210 | 0628 | - | RRM TC 9.1.6.2: General updates and corrections | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125328 | 0629 | - | RRM Annex C: Addition of physical channel settings for eICIC tests | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125335 | 0630 | - | RRM Annex A: Addition of measurement channels for eICIC tests | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125337 | 0631 | - | RRM New TC 7.3.9: Test skeleton for E-UTRAN FDD Radio Link Monitoring Out-of-sync with non-MBSFN ABS | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125338 | 0632 | - | RRM New TC 7.3.10: Test skeleton for E-UTRAN TDD Radio Link Monitoring Out-of-sync with non-MBSFN ABS | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125339 | 0633 | - | RRM New TC 7.3.11: Test skeleton for E-UTRAN FDD Radio Link Monitoring In-sync with non-MBSFN ABS | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125340 | 0634 | - | RRM New TC 7.3.12: Test skeleton for E-UTRAN TDD Radio Link Monitoring In-sync with non-MBSFN ABS | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125341 | 0635 | - | RRM New TC 8.1.7: Test skeleton for E-UTRAN FDD-FDD Intra-Frequency Event-Triggered Reporting with non-MBSFN ABS | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125342 | 0636 | - | RRM New TC 8.2.5: Test skeleton for E-UTRAN TDD-TDD Intra-Frequency Event-Triggered Reporting with non-MBSFN ABS | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125344 | 0637 | - | RRM TC 9.6.2: General corrections | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125363 | 0638 | - | Update Test cases 5.2.3+5.2.6 uncertainties for >3GHz | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125364 | 0639 | - | Fading margin for RRM Test cases 8.3.1, 8.3.2, 8.4.1, 8.4.2, 8.14.1 and 8.15.1 | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125366 | 0640 | - | Procedure and requirements for Test cases 4.2.7 and 4.2.8 | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125368 | 0641 | - | Correction of RSRP values in the Test Requirement for RRM Test case 5.3.1 | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125382 | 0642 | - | Clean up of TDD related RRM tests in 36.521-3 | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125427 | 0643 | - | Correction to TC 9.2.5.1 FDD Absolute RSRQ for E-UTRA Carrier Aggregation tests | 10.3.0 | 10.4.0 |

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|---|---|---|---|---|---|---|---|
| 2012-12 | RAN#58 | R5-125430 | 0644 | - | Correction to TC 9.2.6.1 TDD Absolute RSRQ for E-UTRA Carrier Aggregation tests | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125538 | 0645 | - | RRM TC 9.6.1: Introduction of GSM RSSI accuracy for E-UTRAN FDD | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125549 | 0646 | - | Uncertainties and Test Tolerances for TC 4.5.2.1 | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125551 | 0647 | - | Uncertainties and Test Tolerances for TC 5.3.5 | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125553 | 0648 | - | Uncertainties and Test Tolerances for TC 8.14.2 | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125555 | 0649 | - | Uncertainties and Test Tolerances for TC 8.15.2 | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125557 | 0650 | - | Uncertainties and Test Tolerances for TC 8.14.3 and 8.15.3 | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125559 | 0651 | - | Uncertainties and Test Tolerances for TC 9.2.4A.1 | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125561 | 0652 | - | Uncertainties and Test Tolerances for TC 9.2.4A.2 | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125576 | 0665 | - | Correction to Table 9.2.1.1.5-2 in subclause 9.2.1.1.5 | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125812 | 0653 | - | Correction to RRM 9.3.1 in 36.521-3 | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125813 | 0654 | - | Addition of new TC 8.16.1 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX including uncertainties and Test Tolerances | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125814 | 0655 | - | Addition of new TC 8.16.2 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX including uncertainties and Test Tolerances | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125865 | 0656 | - | Introduction of Band 27 to TS 36.521-3 | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125878 | 0657 | - | Correction to TC 9.2.5.2 FDD Relative RSRQ for E-UTRA Carrier Aggregation tests | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125879 | 0658 | - | Correction to TC 9.2.6.2 TDD Relative RSRQ for E-UTRA Carrier Aggregation tests | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125888 | 0659 | - | Correction to accuracy requirements in RSRP/RSRQ test for Band 26in 36.521-3 | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125889 | 0660 | - | Correction to RRM 9.4.1 in 36.521-3 | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-125922 | 0661 | - | Update Test Procedure and Test Tolerances for UE Transmit Timing Accuracy | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-126045 | 0663 | - | Addition of a new TC 9.1.7.1 TDD Absolute RSRP Accuracy E-UTRA for Carrier Aggregation | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-126046 | 0664 | - | Addition of a new TC 9.1.7.2 TDD Relative RSRP Accuracy E-UTRA for Carrier Aggregation | 10.3.0 | 10.4.0 |
| 2012-12 | RAN#58 | R5-126064 | 0662 | - | Correction to RRM 4.2.7 in 36.521-3 | 10.3.0 | 10.4.0 |
| 2013-03 | RAN#59 | R5-130058 | 0666 | - | Uncertainties and Test Tolerances for RRM test cases 9.1.6.1 and 9.1.7.1 | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130060 | 0667 | - | Uncertainties and Test Tolerances for RRM test cases 9.1.6.2 and 9.1.7.2 | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130062 | 0668 | - | Uncertainties and Test Tolerances for RRM test cases 9.2.5.1 and 9.2.6.1 | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130162 | 0705 | - | Modifying test requirements for handover test from E-UTRAN to UTRAN TDD | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130290 | 0669 | - | RRM:Corrections to TC 9.3.1 | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130300 | 0670 | - | RRM: Corrections to TC 9.6.1 | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130301 | 0671 | - | RRM: Corrections to TC 9.6.2 | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130302 | 0672 | - | Editors note for test cases where Test Requirement not valid above 3GHz | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130395 | 0673 | - | Uncertainties and Test Tolerances for TC 6.3.1 Redirection from E-UTRAN FDD to UTRAN FDD | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130396 | 0674 | - | Uncertainties and Test Tolerances for TC 6.3.3 Redirection from E-UTRAN FDD to GERAN when System Information is provided | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130397 | 0675 | - | Uncertainties and Test Tolerances for TC 6.3.9 Redirection from E-UTRAN FDD to UTRAN FDD without System Information | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130398 | 0676 | - | Uncertainties and Test Tolerances for TC 6.3.10 Redirection from E-UTRAN FDD to GERAN when System Information is not provided | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130430 | 0677 | - | Correction to RRM 4.2.7 and 4.2.8 | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130436 | 0678 | - | Update of cell configuration mapping in Annex E | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130468 | 0706 | - | Correction to RRM measurement accuracy tests | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130803 | 0680 | - | Updates to TC 8.16.3 and 8.16.4 | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130931 | 0683 | - | Update Test Procedure and Test Tolerances for UE Transmit Timing Accuracy | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130932 | 0684 | - | Uncertainties and Test Tolerances for RRM test cases 9.2.5.2 and 9.2.6.2 | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130934 | 0685 | - | Test Tolerances to TCs 8.16.3 and 8.16.4 Event Triggered reporting on deactivated SCell with PCell interruption in non-DRX | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130781 | 0686 | - | RRM: Updates and corrections to TC 8.1.5 | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130943 | 0687 | - | Correction to sr-ConfigIndex in RRM FDD-TDD dual mode tests | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130953 | 0688 | - | CA RRM: Corrections to tests with independent events | 10.4.0 | 10.5.0 |

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| 2013-03 | RAN#59 | R5-130962 | 0689 | - | Add structure for new eICIC test cases 9.2.7.1 and 9.2.8.1, FDD and TDD RSRQ | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130981 | 0690 | - | Introduction of Chapter 9 RRM test cases for RSRP accuracy for FDD EUTRA -eICIC | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130982 | 0691 | - | Introduction of Chapter 9 RRM test cases for RSRP accuracy for TDD EUTRA -eICIC | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130983 | 0692 | - | Additions to TC 8.16.3 E-UTRAN FDD-FDD Event triggered reporting on deactivated SCell with PCell interruption in non-DRX | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130984 | 0693 | - | Additions to TC 8.16.4 E-UTRAN TDD-TDD Event triggered reporting on deactivated SCell with PCell interruption in non-DRX | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130985 | 0694 | - | Additions to TC 9.2.5.1 FDD Absolute RSRQ for E-UTRA Carrier Aggregation tests | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130986 | 0695 | - | Additions to TC 9.2.5.2 FDD Relative RSRQ for E-UTRA Carrier Aggregation tests | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130987 | 0696 | - | Additions to TC 9.2.6.1 TDD Absolute RSRQ for E-UTRA Carrier Aggregation tests | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130988 | 0697 | - | Additions to TC 9.2.6.2 TDD Relative RSRQ for E-UTRA Carrier Aggregation tests | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130989 | 0698 | - | Updates to TC 8.16.1 and 8.16.2 | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130990 | 0699 | - | Updates to TCs in section 8.20 | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130991 | 0700 | - | Updates to TCs about FDD and TDD RSRP accuracy for E-UTRA CA | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130992 | 0701 | - | Updates to TCs about FDD and TDD RSRQ accuracy for E-UTRA CA | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130993 | 0702 | - | Updates to H.4 default RRC message content for CA | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130995 | 0703 | - | Correction to RRC Connection Release with Redirection tests | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130782 | 0704 | - | Correction to new CGI E-UTRA cell with autonomous gaps tests | 10.4.0 | 10.5.0 |
| 2013-03 | RAN#59 | R5-130452 | 0679 | - | Adding operating bands 28 and 44 to Annex I in TS36.521-3 | 10.5.0 | 11.0.0 |
| 2013-03 | RAN#59 | R5-130906 | 0681 | - | Introduction of Band 44 for APAC 700 MHz | 10.5.0 | 11.0.0 |
| 2013-03 | RAN#59 | R5-130907 | 0682 | - | Introduction of Band 28 for APAC 700 MHz | 10.5.0 | 11.0.0 |
| 2013-06 | RAN#60 | R5-131105 | 0707 | - | Correction to Test requirement for tests 7.1.X | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131109 | 0708 | - | eICIC RRM: Addition of MBSFN ABS configuration in annex | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131110 | 0709 | - | eICIC RRM: Further specification of TC 7.3.9 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131111 | 0710 | - | eICIC RRM: Further specification of TC 7.3.10 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131112 | 0711 | - | eICIC RRM: Further specification of TC 7.3.11 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131113 | 0712 | - | eICIC RRM: Further specification of TC 7.3.12 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131118 | 0713 | - | CA RRM: References to connection diagrams | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131148 | 0714 | - | Updates to RRM test case for CA band combo CA_2A-29A | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131177 | 0715 | - | RRM: FGI bit support in test applicability statement | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131178 | 0716 | - | RRM TC 9.6: Clarification on testing requirement when no BCCH1 report available | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131180 | 0718 | - | RRM: Uncertainties and test tolerances for TCs 8.1.5 and 8.1.6 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131182 | 0719 | - | RRM: Uncertainties and test tolerances for TCs 8.3.4 and 8.3.5 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131185 | 0720 | - | RRM TC 8.14.2: Minor correction to cross references | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131186 | 0721 | - | RRM: Several corrections to CGI related test cases | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131282 | 0722 | - | Uncertainties and Test Tools for eICIC RRM test cases 9.2.7.1, 9.2.8.1 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131286 | 0723 | - | Uncertainties and Test Tools for RRM test cases 8.11.5, 8.11.6 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131288 | 0724 | - | Uncertainties and Test Tools for RRM test cases 9.5.1, 9.5.2 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131291 | 0725 | - | Cleanup of Annex F Introductory Text | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131403 | 0726 | - | Uncertainties and Test Tolerances for TS 36.521-3 test cases 5.2.4 and 5.2.5 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131404 | 0727 | - | Uncertainties and Test Tolerances for TS 36.521-3 test case 8.5.4 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131405 | 0728 | - | Uncertainties and Test Tolerances for TS 36.521-3 test cases 8.7.4 and 8.9.2 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131445 | 0729 | - | Addition of inter-freq/RAT without measurement gaps TCs | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131463 | 0730 | - | Uncertainties and Test Tolerances for RRM test cases 6.3.2 and 6.3.12 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131468 | 0731 | - | Uncertainties and Test Tolerances for RRM test cases 6.3.5+6.3.6+6.3.7+6.3.8 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131469 | 0732 | - | Modification to test cases 8.7.1 and 8.7.2 | 11.0.0 | 11.1.0 |

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|---|---|---|---|---|---|---|---|
| 2013-06 | RAN#60 | R5-131517 | 0733 | - | Corrections to RRM requirements for interruption in single CA | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131529 | 0734 | - | Corrections for RSRQ E-UTRA CA | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131586 | 0735 | - | Correction to the test cases with the existence of non-allowed CSG cell | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131590 | 0736 | - | Correction to PRACH configuration | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131591 | 0737 | - | Correction to Time Alignment Timer | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131613 | 0738 | - | Editors note for test cases where Test Requirement not valid above 3GHz, clause 8 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131615 | 0739 | - | Editors note for test cases where Test Requirement not valid above 3GHz, clause 8 - 9 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131740 | 0740 | - | Corrections of band 26 notes in TS36.521-3 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131913 | 0741 | - | Change the IE value of System Information Block type 19 for TS36.521-3 TDD test case 4.3.4.1. | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131922 | 0742 | - | eICIC RRM: Further specification of TC 8.1.7 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131923 | 0743 | - | eICIC RRM: Further specification of TC 8.2.5 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131924 | 0744 | - | Structure for new TCs 9.1.8.1, 9.1.8.2, FDD Absolute and Relative eICIC RSRP, Non-MBSFN | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131925 | 0745 | - | Structure for new TCs 9.1.9.1, 9.1.9.2, TDD Absolute and Relative eICIC RSRP, Non-MBSFN | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131926 | 0746 | - | Procedure and messages for eICIC RRM test cases 9.2.7.1, 9.2.8.1 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131938 | 0747 | - | Uncertainties and Test Tolerances for RRM test cases 6.3.4 and 6.3.11 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131940 | 0748 | - | RRM: Uncertainties and test tolerances for TCs 9.6.1 and 9.6.2 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131965 | 0749 | - | Correction to RRC Connection Release with Redirection tests | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131966 | 0750 | - | Correction to Monitored UTRA cell list size | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131967 | 0751 | - | Correction to UE Transmit Timing Accuracy | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131986 | 0752 | - | Editors note for test cases where Test Requirement not valid above 3GHz, clauses 4 to 7 | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-131992 | 0753 | - | Addition of Chapter 9 RRM test cases for absolute RSRQ accuracy under Time-Domain Measurement Resource Restriction with MBSFN ABS | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-132075 | 0754 | - | Correction to new CGI E-UTRA cell with autonomous gaps tests | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-132107 | 0755 | - | Update of Annex E | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-132112 | 0756 | - | Correction to Combined E-UTRAN - E-UTRA and GSM cell search | 11.0.0 | 11.1.0 |
| 2013-06 | RAN#60 | R5-132114 | 0757 | - | Addition of Band 27 to overlooked sections of TS 36.521-3 | 11.0.0 | 11.1.0 |
| 2013-09 | RAN#61 | R5-133101 | 0758 | - | Uncertainties and Test Tolerances for eICIC Absolute RSRP test cases 9.1.8.1+9.1.9.1 | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133103 | 0759 | - | Uncertainties and Test Tolerances for eICIC Relative RSRP test cases 9.1.8.2+9.1.9.2 | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133105 | 0760 | - | Uncertainties and Test Tolerances update for Inter-freq RSRP Test cases 9.1.3.x and 9.1.4.x | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133142 | 0761 | - | Addition of test cases 7.3.13 and 7.3.15 | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133220 | 0762 | - | RRM: Uncertainties and test tolerances for TCs 9.6.1 and 9.6.2 | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133351 | 0764 | - | Uncertainties and Test Tolerances update for Intra-freq RSRP Test cases 9.1.1.1 and 9.1.2.1 | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133354 | 0765 | - | Band 28 update for Intra-freq relative RSRP Test case 9.1.1.2 | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133421 | 0766 | - | Correction to Common Exception messages for ReportConfig-A6 | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133430 | 0767 | - | Correction to 8.7.3 of 36.521-3 | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133432 | 0768 | - | Correction to 8.11.5 and 8.11.6 of 36.521-3 | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133433 | 0769 | - | Correction to 9.5.1 and 9.5.2 of 36.521-3 | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133435 | 0770 | - | Correction to Common Exception messages in RLM tests | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133440 | 0771 | - | Addition of TC 7.3.14 E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133441 | 0772 | - | Addition of TC 7.3.16 E-UTRAN TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133447 | 0793 | - | Correction of configurations in RSRP TDD absolute and relative accuracy for CA test cases | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133528 | 0773 | - | Corrections to Conditions for UE Measurements Procedures in RRC_CONNECTED State | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133722 | 0763 | - | eICIC RRM: Message contents for TC-s 7.3.9-12, 8.1.7 and 8.2.5 | 11.1.0 | 11.2.0 |

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|---|---|---|---|---|---|---|---|
| 2013-09 | RAN#61 | R5-133727 | 0788 | - | Corrections to RRM CA measurement accuracy CA test cases with PCell and SCell switching | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133728 | 0789 | - | Correction to E-UTRAN RSRP and RSRQ accuracy for CA test cases of 36.521-3 | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133729 | 0791 | - | Correction to CGI test cases | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133809 | 0774 | - | Correction to RRM CA test case 8.16.1 | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133810 | 0775 | - | Correction to RRM CA test case 8.16.3 | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133813 | 0776 | - | Correction to 8.11.4 InterRAT E-UTRA TDD to E-UTRA TDD and UTRA TDD cell search | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133814 | 0777 | - | Addition of Uplink-downlink configuration in 6.3.5 of 36.521-3 | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133836 | 0778 | - | Cell Timing offsets and Cell Timing uncertainties for CA 8.x and 9.x test cases | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133842 | 0779 | - | Correction to 9.6.1 and 9.6.2 of 36.521-3 | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133855 | 0780 | - | Addition of new CA TC 8.16.5 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX for 20 MHz bandwidth | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133856 | 0781 | - | Addition of new CA TC 8.16.6 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 20 MHz bandwidth | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133857 | 0782 | - | Addition of new CA TC 9.2.12.1 TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133858 | 0783 | - | Addition of new CA TC 9.2.12.2 TDD Relative RSRQ Accuracy for E-UTRA Carrier Aggregation for 20MHz | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133859 | 0784 | - | Correction to RRM CA test case 8.20.1 | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133860 | 0785 | - | New RRM TC 9.1.13.2 TDD Relative RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133861 | 0786 | - | New RRM TC 9.1.13.1 TDD Absolute RSRP Accuracy for E-UTRA Carrier Aggregation for 20 MHz | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133862 | 0787 | - | Cell configuration mapping for new CA TCs for 20MHz | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133896 | 0790 | - | Addition of RMC and OCNG pattern for CA 20 MHz | 11.1.0 | 11.2.0 |
| 2013-09 | RAN#61 | R5-133901 | 0792 | - | CA RRM: Clarification of applicability and completeness status of CA RRM TC-s | 11.1.0 | 11.2.0 |
| 2013-12 | RAN#62 | R5-134145 | 0794 | - | CA RRM: Uncertainties and test tolerances for TC 9.1.13.1 | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134147 | 0795 | - | CA RRM: Uncertainties and test tolerances for TC 9.1.13.2 | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134149 | 0796 | - | CA RRM: Uncertainties and test tolerances for TC 9.2.12.1 | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134151 | 0797 | - | CA RRM: Uncertainties and test tolerances for TC 9.2.12.2 | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134168 | 0798 | - | Clarification of CGI minimum conformance requirements | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134186 | 0799 | - | Introduction of TC 8.1.8 E-UTRAN FDD-FDD Intra-Frequency Event-Triggered Reporting (feICIC) | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134187 | 0800 | - | Introduction of TC 8.2.6 E-UTRAN TDD-TDD Intra-Frequency Event-Triggered Reporting (feICIC) | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134188 | 0801 | - | Introduction 9.2.13 FDD RSRQ under Time Domain Measurement Resource Restriction (feICIC) | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134189 | 0802 | - | Introduction 9.2.14 TDD RSRQ under Time Domain Measurement Resource Restriction (feICIC) | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134197 | 0803 | - | Receiver sensitivity relaxation of UE for High and Low operating band | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134221 | 0808 | - | Uncertainties and Test Tolerances for E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions test case 8.20.3 | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134223 | 0809 | - | Uncertainties and Test Tolerances for eICIC RSRQ test cases 9.2.9.1 and 9.2.10.1 | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134246 | 0810 | - | Reported value correction for RRM test case 9.1.7.1 | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134319 | 0811 | - | RRM: Correction to FGI bit support for TCs 9.1.5.1 and 9.1.5.2 | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134483 | 0813 | - | Update of cell configuration mapping for eICIC RRM in Annex E | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134491 | 0814 | - | Correction to InterRat HO test case with UTRAN TDD | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134492 | 0815 | - | Correction to Cell time offset in TDD Inter-RAT test cases | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134813 | 0820 | - | eICIC RRM: Editorial corrections to section 7 tests | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134814 | 0821 | - | Updates to Chapter 9 RSRP and RSRQ accuracy test cases under Time-Domain Measurement Resource Restriction with MBSFN ABS (eICIC) | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134815 | 0822 | - | Updates to Chapter 7 FDD RLM test cases under Time Domain Measurement Resource Restriction with MBSFN ABS (eICIC) | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134852 | 0829 | - | Uncertainties and Test Tolerances update for Intra-freq RSRQ Test cases 9.2.1.1 and 9.2.2.1 | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134862 | 0830 | - | RRM: Updates to the reference measurement channels and OCNG patterns | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134863 | 0831 | - | Introduction of TDD configuration definitions | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134864 | 0832 | - | Update of RRM Requirements Exceptions | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134872 | 0833 | - | Clarification of RRM tests for multiple bandwidths | 11.2.0 | 11.3.0 |

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|---|---|---|---|---|---|---|---|
| 2013-12 | RAN#62 | R5-134873 | 0834 | - | Addition of new TC E-UTRA TDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 20 MHz bandwidth | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134874 | 0835 | - | Update of cell configuration mapping for CA in Annex E | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134990 | 0837 | - | CA RRM: Corrections to CA event triggered tests on deactivated SCell with PCell interruption in non-DRX | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-135066 | 0853 | - | CA RRM: Corrections to the message contents exceptions for some CA 20 MHz test cases | 11.2.0 | 11.3.0 |
| 2013-12 | RAN#62 | R5-134216 | 0805 | - | Introduction of FDD Inter frequency absolute RSRP for 5MHz Bandwidth Test case 9.1.17.1 | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-134218 | 0807 | - | Introduction of FDD Intra frequency absolute RSRQ for 5MHz Bandwidth Test case 9.2.17.1 | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-134579 | 0816 | - | Introduction of Band 31 to Annex I conditions for RRM measurements | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-134583 | 0817 | - | Addition of new TC 5.2.11 E-UTRAN FDD - UTRAN FDD handover for 5MHz Bandwidth | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-134585 | 0818 | - | Addition of new TC 6.2.5 E-UTRAN FDD - Contention Based Random Access Test for 5MHz Bandwidth | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-134586 | 0819 | - | Addition of new TC 6.2.6 E-UTRAN FDD - Non-Contention Based Random Access Test for 5MHz Bandwidth | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-134819 | 0823 | - | Addition of new TC 9.1.16 FDD Intra frequency RSRP Accuracy for 5MHz Bandwidth | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-134820 | 0824 | - | Addition of new TC 9.1.17.2 FDD Inter frequency RSRP Accuracy for 5MHz Bandwidth | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-134822 | 0826 | - | Addition of new TC 9.2.18 FDD Inter frequency RSRQ Accuracy for 5MHz Bandwidth | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-134823 | 0827 | - | Addition of new TC 9.3.3 E-UTRAN FDD - UTRA FDD CPICH RSCP absolute accuracy for 5MHz bandwidth | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-134824 | 0828 | - | Addition of new TC 9.4.3 E-UTRAN FDD - UTRA FDD CPICH Ec/No absolute accuracy for 5MHz bandwidth | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-134898 | 0836 | - | Update of band configuration table for Inter Band | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-135033 | 0838 | - | Addition of new TC 4.2.9 E-UTRAN FDD-FDD intra-frequency Cell Re-selection case for 5MHz | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-135034 | 0839 | - | Addition of new TC 4.3.1.4 E-UTRAN FDD - UTRAN FDD cell re-selection: UTRA FDD is of lower priority for 5MHz | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-135035 | 0840 | - | Addition of new TC 5.1.9 E-UTRAN FDD-FDD Intra frequency handover for 5MHz | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-135036 | 0841 | - | Addition of new TC 6.1.5 E-UTRAN FDD Intra-frequency RRC Re-establishment for 5MHz Bandwidth | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-135037 | 0842 | - | Addition of new TC 7.1.5 E-UTRAN FDD - UE Transmit Timing Accuracy Tests for 5MHz Bandwidth | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-135038 | 0843 | - | Addition of new TC 7.2.3 E-UTRAN FDD - UE Timing Advance Adjustment Accuracy Test for 5MHz Bandwidth | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-135039 | 0844 | - | Addition of new TC 7.3.23 E-UTRAN FDD Radio Link Monitoring Test for Out-of-sync for 5MHz Bandwidth | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-135040 | 0845 | - | Addition of new TC 7.3.24 E-UTRAN FDD Radio Link Monitoring Test for In-sync for 5MHz Bandwidth | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-135041 | 0846 | - | Addition of new TC 7.3.25 E-UTRAN FDD Radio Link Monitoring Test for In-sync in DRX for 5MHz Bandwidth | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-135042 | 0847 | - | Addition of new TC 8.1.9 E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in asynchronous cells for 5MHz bandwidth | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-135043 | 0848 | - | Addition of new TC 8.1.10 E-UTRAN FDD-FDD intra frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX for 5MHz bandwidth | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-135044 | 0849 | - | Addition of new TC 8.5.7 E-UTRAN FDD - UTRAN FDD event triggered reporting under fading propagation conditions for 5MHz bandwidth | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-135047 | 0850 | - | Updates to TCs 4.3.1.2 and 4.3.3 | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-135048 | 0851 | - | Updates to TCs 5.2.1 and 5.2.2 | 11.3.0 | 12.0.0 |
| 2013-12 | RAN#62 | R5-135049 | 0852 | - | Updates the test system uncertainties and test tolerance for TC 6.2.1, 6.2.2, 6.2.3 and 6.2.4 | 11.3.0 | 12.0.0 |
| 2014-03 | RAN#63 | R5-140085 | 0854 | - | CA RRM: Uncertainties and test tolerances for TC 8.20.1 and 8.20.2 | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140088 | 0855 | - | Corrections to UE Transmit Timing minimum requirements | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140089 | 0856 | - | Corrections to measurements of the SCC with deactivated SCell | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140090 | 0857 | - | Corrections to TC 8.16.3 E-UTRAN FDD-FDD Event triggered reporting on deactivated SCell with PCell | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140234 | 0859 | - | Uncertainties and Test Tolerances for FDD Intra frequency relative RSRP for 5MHz Bandwidth Test case 9.1.16.2 and update 9.1.1.2, 9.1.2.2. | 12.0.0 | 12.1.0 |

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|---|---|---|---|---|---|---|---|
| 2014-03 | RAN#63 | R5-140236 | 0860 | - | Uncertainties and Test Tolerances for FDD Inter Frequency Absolute RSRQ Accuracy for 5MHz Bandwidth 9.2.18.1, and update Test cases 9.2.3.1, 9.2.4.1. | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140238 | 0861 | - | Uncertainties and Test Tolerances for FDD Inter Frequency Relative RSRQ Accuracy for 5MHz Bandwidth 9.2.18.2, and update Test cases 9.2.3.2, 9.2.4.2. | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140240 | 0862 | - | Uncertainties and Test Tolerances for E-UTRAN TDD to UTRAN TDD cell search under fading propagation conditions test case 8.20.4 | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140241 | 0863 | - | Noc uncertainty for Test Cases 8.5.1, 8.5.3, 8.5.7 and 8.6.1 | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140255 | 0864 | - | CA RRM: Corrections to event triggered reporting under deactivated SCell in non-DRX tests | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140259 | 0865 | - | CA RRM: Correction and clarification of messages exception on the maximum number of cells to be measured | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140289 | 0867 | - | Uncertainties and Test Tolerances for RRM test case 7.3.24 | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140295 | 0868 | - | Uncertainties and Test Tolerances for RRM test case 9.3.3 | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140297 | 0869 | - | Uncertainties and Test Tolerances for RRM test case 9.4.3 | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140301 | 0870 | - | Addition of cell configuration mapping for LTE 450 related test cases | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140401 | 0871 | - | Introduction of operating band groups to TS36.521-3, clause 3 | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140404 | 0872 | - | Introduction of operating band groups to TS36.521-3, Annex I | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140423 | 0873 | - | Correction to test applicability of measurement without gaps | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140483 | 0874 | - | Completion and alignment of 5MHz Bandwidth Test cases 9.1.17.1, 9.2.17.1 | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140501 | 0875 | - | Update of exceptions in eICIC RRM with Non-MBSFN | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140502 | 0876 | - | Update of eICIC RLM test cases | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140512 | 0877 | - | Correction of UTRA TDD Redirection tests | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140513 | 0878 | - | Correction to CA RRM Inter frequency tests | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140562 | 0879 | - | Correction to GSM RSSI absolute accuracy tests | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140597 | 0880 | - | TC 9.3.1, 9.4.1 Corrections to test procedures in TS36.521-3 | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140810 | 0881 | - | Correction of b2-threshold for EUTRA and GERAN cell | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140849 | 0883 | - | Additions to TC 8.1.8 E-UTRAN FDD-FDD Intra-Frequency Event-Triggered Reporting (feICIC) | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140850 | 0884 | - | Additions to TC 8.2.6 E-UTRAN TDD-TDD Intra-Frequency Event-Triggered Reporting (feICIC) | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140851 | 0885 | - | Additions to TC 9.2.13 FDD RSRQ under Time Domain Measurement Resource Restriction (feICIC) | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140852 | 0886 | - | Additions to TC 9.2.14 TDD RSRQ under Time Domain Measurement Resource Restriction (feICIC) | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140853 | 0887 | - | Addition to TC 7.3.21 FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and MBSFN ABS (feICIC) | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140854 | 0888 | - | Addition to TC 7.3.22 TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and MBSFN ABS (feICIC) | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140855 | 0889 | - | Addition to TC 9.1.15.2 TDD Intra Frequency Relative RSRP Accuracy under TD Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140863 | 0890 | - | Corrections to minimum requirements for time domain measurement | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140865 | 0891 | - | Uncertainties and Test Tolerances for RRM test case 9.1.16.1 | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140866 | 0892 | - | Uncertainties and Test Tolerances for RRM test case 9.1.17.2 | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140869 | 0893 | - | Corrections to test cases 8.16.3 and 8.16.4 | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140876 | 0895 | - | Introduction of feICIC to Annex H | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140882 | 0896 | - | Addition to TC 7.3.19 FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non-MBSFN ABS (feICIC) | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140883 | 0897 | - | Addition to TC 7.3.20 TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non-MBSFN ABS (feICIC) | 12.0.0 | 12.1.0 |

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| 2014-03 | RAN#63 | R5-140884 | 0898 | - | Addition to TC 9.1.14.1 FDD Intra Frequency Absolute RSRP Accuracy under TD Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140885 | 0899 | - | Addition to TC 9.1.14.2 FDD Intra Frequency Relative RSRP Accuracy under TD Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-140886 | 0900 | - | Addition to TC 9.1.15.1 TDD Intra Frequency Absolute RSRP Accuracy under TD Measurement Resource Restriction with CRS Assistance Information and Non-MBSFN ABS (feICIC) | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-141021 | 0901 | - | Introduction of operating band groups to TS36.521-3, test cases | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-141022 | 0902 | - | Addition of WB-RSRQ measurement test cases | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-141023 | 0903 | - | Update of intra-band CA cell mapping in Annex E | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-141024 | 0904 | - | Clarification on operating band configuration for inter-RAT RRM test cases | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-141031 | 0905 | - | update to TC 9.4.1 L2W CPICH Ec/No Absolute accuracy | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-141036 | 0906 | - | Addition of new CA RRM TC 7.1.3 E-UTRAN FDD - UE Transmit Timing Accuracy Tests for SCell | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-141037 | 0907 | - | Addition of new CA RRM TC 7.1.4 E-UTRAN TDD - UE Transmit Timing Accuracy Tests for SCell | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-141049 | 0908 | - | Correction to Test2 of UE Transmit Timing Accuracy test cases 7.1.1 and 7.1.2 | 12.0.0 | 12.1.0 |
| 2014-03 | RAN#63 | R5-141050 | 0866 | - | Updates Test cases about UE Transmit Timing Accuracy Tests | 12.0.0 | 12.1.0 |
| 2014-06 | RAN#64 | R5-142167 | 0909 | - | Correction of clause 9.2.7.1 in 36.521-3 | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-142180 | 0910 | - | Correction on PDSCH allocation in PRS subframe | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-142212 | 0911 | - | Additions to feICIC RRM test cases in Annex F | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-142222 | 0912 | - | Cleanup of operating band groups in TS36.521-3 | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-142547 | 0913 | - | Uncertainties and Test Tolerances for eICIC Intra-Frequency Event-Triggered Reporting test cases 8.1.7 and 8.2.5 | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-142550 | 0914 | - | Uncertainties and Test Tolerances for eICIC RLM out-of-sync Test cases 7.3.9,7.3.10,7.3.13 and 7.3.14 | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-142553 | 0915 | - | Uncertainties and Test Tolerances for eICIC RLM In-sync Test cases 7.3.11,7.3.12,7.3.15 and 7.3.16 | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-142600 | 0916 | - | Clean-up of eICIC RRM tests | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-142621 | 0918 | - | Clean-up for E-UTRAN RRM test cases | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-142623 | 0919 | - | Update of Annex E for CA RRM Inter frequency tests for Intra-band | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143018 | 0920 | - | Correction of clause 9.2.8.1 in 36.521-3 | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143019 | 0921 | - | Update eICIC Absolute RSRP Test cases 9.1.8.1+9.1.9.1 | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143020 | 0922 | - | Update eICIC Absolute RSRQ Test cases 9.2.7.1+9.2.8.1 | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143022 | 0923 | - | Uncertainties and test tolerance for RRM Test case 9.1.10.1, 9.1.11.1 | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143023 | 0924 | - | Uncertainties and test tolerance for RRM Test case 9.1.10.2, 9.1.11.2 | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143024 | 0925 | - | Update to eICIC Absolute RSRQ with MBSFN ABS for chapter 9 RRM test cases | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143057 | 0926 | - | Updates to CA RRM 8.16.x test cases | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143058 | 0927 | - | Correction to UL timing accuracy Tx test cases | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143059 | 0928 | - | Update of Annex E for InterRAT with TDD-TD-SCDMA | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143062 | 0929 | - | Additions to TC 8.2.6 E-UTRAN TDD-TDD Intra-Frequency Event-Triggered Reporting (feICIC) | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143063 | 0930 | - | Update eICIC Relative RSRP Test cases 9.1.8.2+9.1.9.2 | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143064 | 0931 | - | Additions to TC 9.2.13 FDD RSRQ under Time Domain Measurement Resource Restriction (feICIC) | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143065 | 0932 | - | Additions to TC 9.2.14 TDD RSRQ under Time Domain Measurement Resource Restriction (feICIC) | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143066 | 0933 | - | Addition of RLM test cases with non-MBSFN for fEICIC | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143067 | 0934 | - | Test Cases 9.1.14.1, 9.1.14.2, 9.1.15.1, 9.1.15.2 for feICIC | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143085 | 0935 | - | Addition of new TC 8.16.10 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 10MHz+5MHz | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143086 | 0936 | - | Addition of new TC 9.1.18 FDD RSRP Accuracy for E-UTRAN Carrier Aggregation for 10MHz + 5MHz | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143087 | 0937 | - | Addition of new TC 9.1.19 TDD RSRP Accuracy for E-UTRAN Carrier Aggregation for 10MHz + 5MHz | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143088 | 0938 | - | Addition of new TC 9.2.21 FDD RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143089 | 0939 | - | Addition of new TC 9.2.22 TDD RSRQ Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz | 12.1.0 | 12.2.0 |

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|---|---|---|---|---|---|---|---|
| 2014-06 | RAN#64 | R5-143090 | 0940 | - | Addition of new TC 8.16.14 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 5MHz+5MHz | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143091 | 0941 | - | Addition of new TC 9.1.20 FDD RSRP Accuracy for E-UTRAN Carrier Aggregation for 5MHz + 5MHz | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143092 | 0942 | - | Addition of new TC 9.1.21 TDD RSRP Accuracy for E-UTRAN Carrier Aggregation for 5MHz + 5MHz | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143093 | 0943 | - | Addition of new TC 9.2.23 FDD RSRQ Accuracy for E-UTRAN Carrier Aggregation for 5MHz + 5MHz | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143094 | 0944 | - | Addition of new TC 9.2.24 TDD RSRQ Accuracy for E-UTRAN Carrier Aggregation for 5MHz + 5MHz | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143102 | 0945 | - | Additions to TC 8.1.8 E-UTRAN FDD-FDD Intra-Frequency Event-Triggered Reporting (feICIC) | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143103 | 0946 | - | Additions to feICIC to Annex H | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143108 | 0947 | - | Correction to periodicity of ABS pattern in eICIC RRM | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143112 | 0948 | - | Additions to TDD interruption requirements for SCell | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143152 | 0949 | - | Correction to event triggered reporting tests for CA | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143153 | 0950 | - | Correction to Intra CA test cases | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143157 | 0951 | - | TC 7.3.19 FDD Radio Link Monitoring for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non-MBSFN ABS (feICIC) | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143158 | 0952 | - | TC 7.3.20 TDD Radio Link Monitoring for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and Non-MBSFN ABS (feICIC) | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143159 | 0953 | - | TC 7.3.21 FDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and MBSFN ABS (feICIC) | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143160 | 0954 | - | TC 7.3.22 TDD Radio Link Monitoring Test for In-sync under Time Domain Measurement Resource Restriction with CRS assistance information and MBSFN ABS (feICIC) | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143172 | 0955 | - | Addition of new TC 8.16.9 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX for 10MHz+5MHz | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143173 | 0956 | - | Addition of new TC 8.16.13 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX for 5MHz+5MHz | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143176 | 0957 | - | Addition of new TC 8.16.7 E-UTRA FDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 20 MHz bandwidth | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143191 | 0958 | - | Clarification of tracking area updating procedure | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | R5-143209 | 0917 | 1 | Correction to 9.6.2 GSM RSSI accuracy for E-UTRAN TDD | 12.1.0 | 12.2.0 |
| 2014-06 | RAN#64 | - | - | - | Correction of misplaced Table F.3.2-1 entry of R5-142547 | 12.2.0 | 12.2.1 |
| 2014-09 | RAN#65 | R5-144124 | 0959 | - | Corrections to RRM Physical Cell Id (PCI) Configuration Conditions in feICIC | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144169 | 0962 | - | RRM: Corrections to TCs 9.2.18.x (5MHz) | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144170 | 0963 | - | eICIC RRM: Corrections to TCs 7.3.14 and 7.3.16 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144171 | 0964 | - | RRM: Removal of redundant (time limited) editor's notes | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144172 | 0965 | - | CA RRM: Correction of wrong test applicability | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144197 | 0966 | - | Addition of new RRM TC 8.16.11 E-UTRAN FDD event triggered reporting on deactivating SCell with PCell interruption in non-DRX for 10MHz+5MHz | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144202 | 0967 | - | Addition of new RRM TC 8.16.16 E-UTRA TDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 5+5MHz BW | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144355 | 0973 | - | Correction to Inter-frequency event triggered reporting test in 8.3.6 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144356 | 0974 | - | Correction to InterRAT event triggered reporting test in 8.5.6 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144357 | 0975 | - | Update of Annex E | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144361 | 0976 | - | Correction to CQI reporting periodicity in RLM with eICIC | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144362 | 0977 | - | Addition of message exceptions for eICIC test cases with MBSFN | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144767 | 0979 | - | Uncertainties and Test Tolerances for RRM TCs 8.16.7+8.16.8 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144768 | 0980 | - | Uncertainties and Test Tolerances for RRM TCs 8.16.5+8.16.6 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144785 | 0988 | - | Uncertainties and test tolerance for 36.521-3 Test case 9.4.1, 9.4.2 and 9.4.3 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144788 | 0978 | - | Updates to Annex I of TS 36.521-3 | 12.2.1 | 12.3.0 |

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| 2014-09 | RAN#65 | R5-144810 | 0981 | - | Antenna correlation for RRM Test cases | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144811 | 0982 | - | Correction to RRM event triggered tests | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144812 | 0983 | - | Correction to E-UTRAN CA Measurements | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144830 | 0984 | - | Uncertainties and Test Tolerances for Test case 8.3.6 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144832 | 0985 | - | Uncertainties and Test Tolerances for Test case 8.5.6 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144833 | 0986 | - | Uncertainties and Test Tolerances update for Test cases 8.3.3+8.4.3 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144850 | 0989 | - | Addition of new RRM TC 8.16.12 E-UTRAN TDD Event triggered reporting on deactivating Scells and interruption probability (0.5%) without DRX for 10MHz+5MHz BW | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144851 | 0990 | - | Cell configuration mapping for 5MHz+5MHz and 10MHz+5MHz test cases | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144852 | 0991 | - | Addition of new RRM TC 8.16.15 E-UTRA FDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 5+5 MHz BW | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144858 | 0992 | - | Update to test procedure for GSM carrier measurement test cases | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144908 | 0993 | - | New TC: FDD Intra Frequency Serving Cell Absolute RSRP Accuracy | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144909 | 0994 | - | New TC: FDD Intra Frequency Serving Cell Absolute RSRQ Accuracy | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144910 | 0995 | - | New TC: TDD Intra Frequency Serving Cell Absolute RSRP Accuracy | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144912 | 0996 | - | Removable of editor's notes for maxReportCells | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144913 | 0997 | - | Correction to neighCellConfig of MeasObjectEUTRA in CGI tests | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144927 | 0998 | - | New TC: TDD Intra Frequency Serving Cell Absolute RSRQ Accuracy | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144124 | 0959 | - | Corrections to RRM Physical Cell Id (PCI) Configuration Conditions in feICIC | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144169 | 0962 | - | RRM: Corrections to TCs 9.2.18.x (5MHz) | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144170 | 0963 | - | eICIC RRM: Corrections to TCs 7.3.14 and 7.3.16 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144171 | 0964 | - | RRM: Removal of redundant (time limited) editor's notes | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144172 | 0965 | - | CA RRM: Correction of wrong test applicability | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144197 | 0966 | - | Addition of new RRM TC 8.16.11 E-UTRAN FDD event triggered reporting on deactivating SCell with PCell interruption in non-DRX for 10MHz+5MHz | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144202 | 0967 | - | Addition of new RRM TC 8.16.16 E-UTRA TDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 5+5MHz BW | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144355 | 0973 | - | Correction to Inter-frequency event triggered reporting test in 8.3.6 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144356 | 0974 | - | Correction to InterRAT event triggered reporting test in 8.5.6 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144357 | 0975 | - | Update of Annex E | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144361 | 0976 | - | Correction to CQI reporting periodicity in RLM with eICIC | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144362 | 0977 | - | Addition of message exceptions for eICIC test cases with MBSFN | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144767 | 0979 | - | Uncertainties and Test Tolerances for RRM TCs 8.16.7+8.16.8 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144768 | 0980 | - | Uncertainties and Test Tolerances for RRM TCs 8.16.5+8.16.6 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144785 | 0988 | - | Uncertainties and test tolerance for 36.521-3 Test case 9.4.1, 9.4.2 and 9.4.3 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144788 | 0978 | 1 | Updates to Annex I of TS 36.521-3 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144810 | 0981 | - | Antenna correlation for RRM Test cases | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144811 | 0982 | - | Correction to RRM event triggered tests | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144812 | 0983 | - | Correction to E-UTRAN CA Measurements | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144830 | 0984 | - | Uncertainties and Test Tolerances for Test case 8.3.6 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144832 | 0985 | - | Uncertainties and Test Tolerances for Test case 8.5.6 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144833 | 0986 | - | Uncertainties and Test Tolerances update for Test cases 8.3.3+8.4.3 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144850 | 0989 | - | Addition of new RRM TC 8.16.12 E-UTRAN TDD Event triggered reporting on deactivating Scells and interruption probability (0.5%) without DRX for 10MHz+5MHz BW | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144851 | 0990 | - | Cell configuration mapping for 5MHz+5MHz and 10MHz+5MHz test cases | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144852 | 0991 | - | Addition of new RRM TC 8.16.15 E-UTRA FDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 5+5 MHz BW | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144858 | 0992 | - | Update to test procedure for GSM carrier measurement test cases | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144908 | 0993 | - | New TC: FDD Intra Frequency Serving Cell Absolute RSRP Accuracy | 12.2.1 | 12.3.0 |

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| 2014-09 | RAN#65 | R5-144909 | 0994 | - | New TC: FDD Intra Frequency Serving Cell Absolute RSRQ Accuracy | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144910 | 0995 | - | New TC: TDD Intra Frequency Serving Cell Absolute RSRP Accuracy | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144912 | 0996 | - | Removable of editor's notes for maxReportCells | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144913 | 0997 | - | Correction to neighCellConfig of MeasObjectEUTRA in CGI tests | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144927 | 0998 | - | New TC: TDD Intra Frequency Serving Cell Absolute RSRQ Accuracy | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144124 | 0959 | - | Corrections to RRM Physical Cell Id (PCI) Configuration Conditions in feICIC | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144169 | 0962 | - | RRM: Corrections to TCs 9.2.18.x (5MHz) | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144170 | 0963 | - | eICIC RRM: Corrections to TCs 7.3.14 and 7.3.16 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144171 | 0964 | - | RRM: Removal of redundant (time limited) editor's notes | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144172 | 0965 | - | CA RRM: Correction of wrong test applicability | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144197 | 0966 | - | Addition of new RRM TC 8.16.11 E-UTRAN FDD event triggered reporting on deactivating SCell with PCell interruption in non-DRX for 10MHz+5MHz | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144202 | 0967 | - | Addition of new RRM TC 8.16.16 E-UTRA TDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 5+5MHz BW | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144355 | 0973 | - | Correction to Inter-frequency event triggered reporting test in 8.3.6 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144356 | 0974 | - | Correction to InterRAT event triggered reporting test in 8.5.6 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144357 | 0975 | - | Update of Annex E | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144361 | 0976 | - | Correction to CQI reporting periodicity in RLM with eICIC | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144362 | 0977 | - | Addition of message exceptions for eICIC test cases with MBSFN | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144767 | 0979 | - | Uncertainties and Test Tolerances for RRM TCs 8.16.7+8.16.8 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144768 | 0980 | - | Uncertainties and Test Tolerances for RRM TCs 8.16.5+8.16.6 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144785 | 0988 | - | Uncertainties and test tolerance for 36.521-3 Test case 9.4.1, 9.4.2 and 9.4.3 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144788 | 0978 | 1 | Updates to Annex I of TS 36.521-3 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144810 | 0981 | - | Antenna correlation for RRM Test cases | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144811 | 0982 | - | Correction to RRM event triggered tests | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144812 | 0983 | - | Correction to E-UTRAN CA Measurements | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144830 | 0984 | - | Uncertainties and Test Tolerances for Test case 8.3.6 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144832 | 0985 | - | Uncertainties and Test Tolerances for Test case 8.5.6 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144833 | 0986 | - | Uncertainties and Test Tolerances update for Test cases 8.3.3+8.4.3 | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144850 | 0989 | - | Addition of new RRM TC 8.16.12 E-UTRAN TDD Event triggered reporting on deactivating Scells and interruption probability (0.5%) without DRX for 10MHz+5MHz BW | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144851 | 0990 | - | Cell configuration mapping for 5MHz+5MHz and 10MHz+5MHz test cases | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144852 | 0991 | - | Addition of new RRM TC 8.16.15 E-UTRA FDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 5+5 MHz BW | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144858 | 0992 | - | Update to test procedure for GSM carrier measurement test cases | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144908 | 0993 | - | New TC: FDD Intra Frequency Serving Cell Absolute RSRP Accuracy | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144909 | 0994 | - | New TC: FDD Intra Frequency Serving Cell Absolute RSRQ Accuracy | 12.2.1 | 12.3.0 |
| 2014-09 | RAN#65 | R5-144910 | 0995 | - | New TC: TDD Intra Frequency Serving Cell Absolute RSRP Accuracy | 12.2.1 | 12.3.0 |
| 2014-12 | RAN#66 | R5-145129 | 1000 | - | CA-RRM: Corrections to TC 8.16.5 and 8.16.6 | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145130 | 1001 | - | RRM: Correction to MBSFN subframes configuration | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145245 | 1002 | - | eICIC-RRM: Missing uncertainties for TC 8.1.7 and 8.2.5 | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145299 | 1003 | - | Update Uncertainties for Test case 8.5.3 | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145301 | 1004 | - | Uncertainties and Test Tolerances for feICIC RSRQ accuracy Test cases | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145303 | 1005 | - | Uncertainties and Test Tolerances for Test cases 9.2.21.1 and 9.2.22.1 | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145305 | 1006 | - | Uncertainties and Test Tolerances for Test cases 9.2.21.2 and 9.2.22.2 | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145457 | 1007 | - | Addition of new RRM TC 7.1.7 E-UTRAN TDD - UE Transmit Timing Accuracy Tests for SCell in sTAG | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145464 | 1008 | - | Corrections to RSRP and RSRQ tests | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145477 | 1009 | - | Corrections to TC 7.3.20 E-UTRAN TDD RL Monitoring Test for In-sync under for feICIC | 12.3.0 | 12.4.0 |

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|---|---|---|---|---|---|---|---|
| 2014-12 | RAN#66 | R5-145488 | 1011 | - | Test Tolerances for TC 8.1.8 E-UTRAN FDD-FDD Intra-Frequency Event-Triggered Reporting (feICIC) | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145489 | 1012 | - | Test Tolerances for TC 8.2.6 E-UTRAN TDD-TDD Intra-Frequency Event-Triggered Reporting (feICIC) | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145493 | 1013 | - | Uncertainties and Test Tolerances to Annex F for feICIC RRM test cases | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145501 | 1014 | - | Correction to periodicity of ABS pattern in RRM for feICIC | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145512 | 1015 | - | Alignment of Annex I | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145528 | 1017 | - | Editorial removal of brackets from TC 7.3.15 | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145587 | 1018 | - | Update of feICIC Out-Of-Sync test cases with assistance data | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145639 | 1019 | - | Updates to RRM feICIC test cases in Clause 9 | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145801 | 1020 | - | RRM: Modification of test description to refer to State 2A-RF | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145802 | 1021 | - | Update of Annex E cell configuration mapping for feICIC | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145816 | 1022 | - | Updates to RRM TC 7.1.3 and 7.1.4 | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145817 | 1023 | - | Correction to the band group in 9.2.1.1 | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145819 | 1024 | - | Correction to 8.16.6 | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145820 | 1025 | - | Correction to Phys Cell Id of MeasResult in RRM test cases | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145831 | 1026 | - | Uncertainties and Test Tolerances for feICIC Test cases 7.3.17, 7.3.18, 7.3.19, 7.3.20, 7.3.21, 7.3.22 | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145832 | 1027 | - | Uncertainties and Test Tolerances for feICIC test cases 9.1.14.1 and 9.1.15.1 | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145837 | 1028 | - | Uncertainties and Test Tolerances for feICIC test cases 9.1.14.2 and 9.1.15.2 | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145877 | 1029 | - | Addition of new TCs for activation and deactivation of known SCell | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145878 | 1030 | - | Update to test procedure for LTE CA Event reporting test cases | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145906 | 1031 | - | Addition of new RRM TC 6.2.7 E-UTRAN FDD - Non-Contention Based Random Access Test For SCell in sTAG | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145907 | 1032 | - | Addition of new RRM TC 6.2.8 E-UTRAN TDD - Non-Contention Based Random Access Test For SCell in sTAG | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145908 | 1033 | - | Addition of new RRM TC 7.1.6 E-UTRAN FDD - UE Transmit Timing Accuracy Tests for SCell in sTAG | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145909 | 1034 | - | Addition of new RRM TC 7.2.4 E-UTRAN FDD - UE Timing Advance Adjustment Accuracy Test For SCell in sTAG | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145910 | 1035 | - | Addition of new RRM TC 7.2.5 E-UTRAN TDD - UE Timing Advance Adjustment Accuracy Test For SCell in sTAG | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145911 | 1036 | - | Updates to CA TCs about 10+5MHz and 5+5MHz BW | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145912 | 1037 | - | New TCs 9.1.12 for FDD RSRP Accuracy for CA 20MHz | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145913 | 1038 | - | New TCs 9.2.11 for FDD RSRQ Accuracy for CA 20MHz | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145915 | 1039 | - | Correction to measurement performance test cases 9.3.2 | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145918 | 1040 | - | Update of Annex H for feICIC test cases | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145919 | 1041 | - | Update of feICIC RLM In-sync test cases with assistance data | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145927 | 1042 | - | Correction to the test procedure in 9.6.1 and 9.6.2 | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145977 | 0999 | - | CA-RRM: Review of test frequencies in Annex E | 12.3.0 | 12.4.0 |
| 2014-12 | RAN#66 | R5-145982 | 1010 | - | Cell mapping for some RRM TCs | 12.3.0 | 12.4.0 |
| 2015-01 | - | - | - | - | Re-implementation of R5-145927 | 12.4.0 | 12.4.1 |
| 2015-01 | - | - | - | - | Re-implementation of R5-145911 | 12.4.0 | 12.4.1 |
| 2015-03 | RAN#67 | R5-150213 | 1043 | - | Corrections to multicarrier handover from E-UTRA to UTRA requirements | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150218 | 1044 | - | Test Tolerances for 4.3.1.3 E-UTRAN FDD - UTRAN FDD cell re-selection | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150220 | 1045 | - | Test Tolerance for 4.3.4.3 EUTRA TDD-UTRA TDD cell reselection | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150222 | 1046 | - | Test Tolerances for 8.5.2 E-UTRAN FDD - UTRAN FDD SON ANR cell search reporting | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150224 | 1047 | - | Test Tolerances for 8.7.3 E-UTRAN TDD - UTRAN TDD SON ANR cell search reporting | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150226 | 1048 | - | Test Tolerances for 8.9.1 E-UTRAN FDD - UTRAN TDD event triggered reporting | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150228 | 1049 | - | Test Tolerances for 8.8.1+8.10.1 E-UTRAN - GSM event triggered reporting | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150230 | 1050 | - | Test Tolerances for 8.8.2+8.10.2 E-UTRAN - GSM event triggered reporting when DRX is used | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150232 | 1051 | - | Uncertainties to Annex F for RRM test cases | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150242 | 1052 | - | Update Uncertainties for Test case 5.1.5 | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150244 | 1053 | - | Update Uncertainties for Test case 5.1.6 | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150246 | 1054 | - | Update Uncertainties for Test case 5.2.3, 5.2.6 | 12.4.1 | 12.5.0 |

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| 2015-03 | RAN#67 | R5-150256 | 1055 | - | Introduction of Test Cases for RSRQ Accuracy for TDD-FDD CA | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150306 | 1056 | - | Uncerts and Test Tools for TCs 8.16.11+8.16.12 36.521-3 CR | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150310 | 1057 | - | Uncerts and Test Tools for TCs 8.16.15+8.16.16 36.521-3 CR | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150314 | 1058 | - | Uncerts and Test Tools for TCs 9.1.18.2+9.1.19.2 36.521-3 CR | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150316 | 1059 | - | Uncerts and Test Tools for TCs 9.1.20.1+9.1.21.1 36.521-3 CR | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150318 | 1060 | - | Uncerts and Test Tools for TCs 9.1.20.2+9.1.21.2 36.521-3 CR | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150320 | 1061 | - | RRM: Corrections to event triggered tests without measurement gap | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150324 | 1062 | - | Uncerts and Test Tools for TCs 9.2.23.1+9.2.24.1 36.521-3 CR | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150327 | 1063 | - | Uncerts and Test Tools for TCs 9.2.23.2+9.2.24.2 36.521-3 CR | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150328 | 1064 | - | Correction to RRM TCs 9.1.6.2 and 9.1.7.2 | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150361 | 1065 | - | Addition of cell mapping for some test cases | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150381 | 1066 | - | Update of Annex E and F for TCs 9.1.12 and 9.2.11 | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150390 | 1067 | - | Correction to test procedure for LTE CA event reporting test case | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150421 | 1068 | - | Correction to message exceptions in eICIC MBSFN ABS | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150423 | 1069 | - | Correction to feICIC common exceptions in Annex H | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150424 | 1070 | - | Clean up on cl.7 and cl.8 feICIC RRM test cases | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150425 | 1071 | - | Clean up on cl.9 feICIC RRM test cases | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150517 | 1072 | - | Error correction to TDD Absolute RSRQ Accuracy for E-UTRA Carrier Aggregation | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150523 | 1073 | - | Cell configuration mapping for Rel 12 and forward absolute RSRP accuracy RRM cases | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150526 | 1074 | - | Update R8 intra frequency absolute RSRP accuracy requirements | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150527 | 1075 | - | Update R8 inter frequency absolute RSRP accuracy requirements | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150529 | 1076 | - | Update R9 inter frequency absolute RSRP accuracy requirements | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150530 | 1077 | - | Update R10 intra frequency absolute RSRP accuracy requirements | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150531 | 1078 | - | Update TDD RSRP Accuracy requirements for Carrier Aggregation | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150806 | 1079 | - | New TCs: intra frequency absolute RSRP accuracy requirements | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150807 | 1080 | - | New TCs: inter frequency absolute RSRP accuracy requirements | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150808 | 1081 | - | New TCs: RSRP absolute accuracy for E-UTRA CA | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150812 | 1082 | - | New TC: E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX for 20MHz+10MHz | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150813 | 1083 | - | New TC: E-UTRAN TDD event triggered reporting on deactivated SCell with PCell interruption in non-DRX for 20MHz+10MHz bandwidth | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150814 | 1084 | - | New TC: TDD RSRP Accuracy for E-UTRAN Carrier Aggregation for 20MHz + 10MHz | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150815 | 1085 | - | New TC: TDD RSRQ Accuracy for E-UTRAN Carrier Aggregation for 20MHz + 10MHz | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150841 | 1086 | - | Corrections to on time to identify the target UTRA TDD cell for blind redirection | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150842 | 1087 | - | Update of TCs 9.1.12 for FDD RSRP Accuracy for CA 20MHz | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150843 | 1088 | - | Update TCs 9.2.11 for FDD RSRQ Accuracy for CA 20MHz | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150844 | 1089 | - | Correction to eICIC Radio link monitoring test cases | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150845 | 1090 | - | Clean up on cl.7 and cl.9 eICIC RRM test cases | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150847 | 1092 | - | Update R11 RSRP Accuracy for E-UTRA Carrier Aggregation for 10MHz+5MHz | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150852 | 1093 | - | Uncerts and Test Tools for TCs 8.16.13+8.16.14 36.521-3 CR | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150856 | 1094 | - | Uncerts and Test Tools for TCs 8.16.9+8.16.10 36.521-3 CR | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150857 | 1104 | - | RRM: Inter-frequency testing for Band 30 | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150884 | 1095 | - | Cell configuration mapping for 20MHz +10MHz CA RRM new cases | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150885 | 1096 | - | Editorial correction to 8.14.2 and 8.15.2 | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150886 | 1097 | - | Corrections to RRM test cases in clauses 5 and 8 | 12.4.1 | 12.5.0 |

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| 2015-03 | RAN#67 | R5-150887 | 1098 | - | Uncerts and Test Tools for TCs 9.1.18.1+9.1.19.1 36.521-3 CR | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150901 | 1099 | - | CA RRM: Correction to test frequencies and Cell-IDs | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150902 | 1100 | - | Addition of CA capability in RRM tests | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150903 | 1101 | - | feICIC RRM: Corrections to Cell-IDs | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150930 | 1102 | - | RRM: Consideration of unsent measurement reports | 12.4.1 | 12.5.0 |
| 2015-03 | RAN#67 | R5-150908 | 1103 | - | Updates of test procedures in activation and deactivation test cases | 12.4.1 | 12.5.0 |
| 2015-06 | RAN#68 | R5-151126 | 1107 | - | RRM: Corrections to message configuration of TCs 9.3.2 and 9.4.2 | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151127 | 1108 | - | RRM: Corrections to test procedures of TCs 9.3.x and 9.4.x | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151128 | 1109 | - | CA RRM: Corrections to TC 9.1.24.2 | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151133 | 1111 | - | Uncertainties and Test tolerances for Test cases 9.9.1.1 and 9.9.2.1 | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151135 | 1112 | - | Uncertainties and Test tolerances for Test cases 9.9.1.2 and 9.9.2.2 | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151197 | 1122 | - | Corrections to CPICH Ec/No to CPICH Ec/Io in EUTRA FDD to UTRA FDD HO | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151277 | 1129 | - | Correction to RRM test case 8.2.4 | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151278 | 1130 | - | Correction to RRM test case 8.11.4 | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151330 | 1150 | - | Addition of band 32 to TS36.521-3 | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151464 | 1155 | - | Correction to applicability of TD-LTE to UTRA TDD periodic measurements | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151485 | 1158 | - | Update of eICIC meas pattern for PCell, TC 8.2.5 | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151542 | 1162 | - | Uncerts and Test Tools for TC 8.16.21 36.521-3 CR | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151544 | 1163 | - | Uncerts and Test Tools for TC 8.16.22 36.521-3 CR | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151546 | 1164 | - | Uncerts and Test Tools for TCs 9.1.24.1 36.521-3 CR | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151548 | 1165 | - | Uncerts and Test Tools for TCs 9.1.24.1_1 36.521-3 CR | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151550 | 1166 | - | Uncerts and Test Tools for TC 9.1.24.2 36.521-3 CR | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151552 | 1167 | - | Uncerts and Test Tools for TC 9.2.27.1 36.521-3 CR | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151554 | 1168 | - | Uncerts and Test Tools for TC 9.2.27.2 36.521-3 CR | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151830 | 1105 | 1 | RRM: Consideration of unsent measurement reports in recently added test cases | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151831 | 1113 | 1 | Test Tolerances for 8.3.1+8.3.2+8.14.1+8.14.2 inter-frequency event triggered reporting under fading propagation conditions | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151832 | 1115 | 1 | Test Tolerances for 8.4.1+8.4.2+8.15.1+8.15.2 inter-frequency event triggered reporting under fading propagation conditions | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151833 | 1117 | 1 | Test Tolerances for 9.2.4A.1 Absolute Accuracy RSRQ | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151834 | 1119 | 1 | Test Tolerances for 9.2.4A.2 Relative Accuracy RSRQ | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151835 | 1121 | 1 | Uncertainties to Annex F for above 3GHz RRM test cases | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151838 | 1127 | 1 | Update Test Tolerances and Test requirements for TS 36.521-3 Test cases 4.2.3, 4.2.4, 4.2.5, 4.2.6 | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151844 | 1128 | 1 | Uncertainties and Test Tolerances for Rel-12 Improved RSRP accuracy | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151846 | 1159 | 1 | Update Uncerts and Test Tools for TCs 8.11.1+8.11.2 for frequencies above 3GHz 36.521-3 CR | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151848 | 1160 | 1 | Update Uncerts and Test Tools for TC 8.11.3 for frequencies above 3GHz 36.521-3 CR | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151850 | 1161 | 1 | Update Uncerts and Test Tools for TC 8.11.4 for frequencies above 3GHz 36.521-3 CR | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151861 | 1171 | 1 | Addition of new RRM TC 8.16.25 E-UTRAN TDD-FDD CA Event Triggered Reporting on Deactivated SCell with PCell interruption in Non-DRX with PCell in FDD | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151862 | 1172 | 1 | Addition of new RRM TC 8.16.26 E-UTRAN TDD-FDD CA Event Triggered Reporting on Deactivated SCell with PCell interruption in Non-DRX with PCell in TDD | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151864 | 1123 | 1 | Corrections to RRM test cases for Core Alignment | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151866 | 1132 | 1 | New TC: E-UTRAN TDD-TDD Inter-frequency event triggered reporting under fading propagation conditions in synchronous cells for 20 MHz +10 MHz bandwidth | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151867 | 1133 | 1 | New TC: E-UTRAN TDD to UTRAN TDD cell search under fading propagation conditions for 20 MHz + 20 MHz bandwidth | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151868 | 1134 | 1 | New TC: E-UTRAN TDD to UTRAN TDD cell search under fading propagation conditions for 20 MHz + 10 MHz bandwidth | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151869 | 1135 | 1 | Cell configuration mapping for Rel 12 and forward RSRP relative accuracy RRM cases | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151870 | 1136 | 1 | Cell configuration mapping for Rel 12 and forward FDD absolute RSRP accuracy RRM cases | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151871 | 1137 | 1 | New TCs: R12 RSRP accuracy in eICIC | 12.5.0 | 12.6.0 |

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| 2015-06 | RAN#68 | R5-151872 | 1138 | 1 | New TCs: R12 RSRP accuracy in FeICIC | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151873 | 1139 | 1 | New TCs: RSRP absolute accuracy for FDD E-UTRA CA | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151874 | 1140 | 1 | New TCs: R12 inter frequency relative accuracy of RSRP | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151875 | 1141 | 1 | Update of R12 CA RSRP measurement accuracy test cases | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151876 | 1142 | 1 | Update of R12 RSRP inter frequency measurement accuracy test cases | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151877 | 1143 | 1 | Update of R12 RSRP intra frequency measurement accuracy test cases | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151878 | 1144 | 1 | Update test applicabilities for R8 inter frequency relative accuracy of RSRP | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151879 | 1145 | 1 | Update test applicabilities for R9 inter frequency relative accuracy of RSRP | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151880 | 1146 | 1 | Update test applicabilities of R10 RSRP measurement accuracy | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151882 | 1147 | 1 | Update test applicabilities of R11 RSRP measurement accuracy | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151883 | 1148 | 1 | Update FDD CA RSRP Accuracy requirements | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151904 | 1174 | 1 | Correction to the titles of 2UL CA RRM test cases | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151905 | 1149 | 1 | Correction of allowed CQI report delays in sCell Activation Tests | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151906 | 1154 | 1 | Correction to feICIC Radio link monitoring test cases | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151915 | 1153 | 1 | Correction to test case 9.2.10.1 of 36.521-3 | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151916 | 1175 | 1 | Correction to FeICIC measurement performance test cases and Annex H.3 | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151917 | 1177 | - | Update Test Tolerances and Test requirements for TS 36.521-3 Test case 9.1.5.1 | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151918 | 1178 | - | Update Test Tolerances and Test requirements for TS 36.521-3 Test case 9.1.5.2 | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151923 | 1169 | 1 | Addition of new RRM TC 8.16.23 E-UTRAN TDD-FDD CA Event Triggered Reporting Under Deactivated SCell in Non-DRX with PCell in FDD | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151924 | 1170 | 1 | Addition of new RRM TC 8.16.24 E-UTRAN TDD-FDD CA Event Triggered Reporting Under Deactivated SCell in Non-DRX with PCell in TDD | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151925 | 1173 | 1 | Applicability for newly added TDD-FDD CA RRM test cases | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-151939 | 1152 | 1 | Addition of FDD-TDD RSRP accuracy test cases for FDD-TDD CA. | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-152010 | 1151 | 1 | Correction to RLM test in feICIC | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-152015 | 1126 | 2 | Corrections to Derivation of Test Requirements for above 3GHz RRM test cases | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-152130 | 1110 | 1 | CA RRM: Corrections to cell configuration mapping table | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-152131 | 1156 | 1 | Correction to Event Triggered Reporting test in feICIC | 12.5.0 | 12.6.0 |
| 2015-06 | RAN#68 | R5-152134 | 1157 | 2 | Clarification of 2DL CA RRM inter-frequency tests with smaller channels bandwidth | 12.5.0 | 12.6.0 |
| 2015-09 | RAN#69 | R5-153094 | 1183 | - | Addition of cell configuration mapping for RRM test cases 8.16.17 and 8.16.18 | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153095 | 1184 | - | Correction of cell configuration mapping for RRM cases | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153096 | 1185 | - | Correction of RMC used in UE Timing Advance Adjustment Accuracy Test for SCell in sTAG | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153164 | 1193 | - | New TC: E-UTRAN TDD - UE Transmit Timing Accuracy Tests for SCell for 20MHz + 10MHz bandwidth | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153167 | 1194 | - | Cell configuration mapping for newly introduced 20+20MHz and 20+10MHz cases | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153176 | 1197 | - | Addition of missing titles to section 8 RRM test cases | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153177 | 1198 | - | Update Test Tolerances and Test requirements for Test case 4.3.1.1 | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153179 | 1199 | - | Update Test Tolerances and Test requirements for Test case 4.3.4.1 | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153181 | 1200 | - | Update Test Tolerances and Test requirements for Test cases 5.1.3+5.1.4+5.1.7+5.1.8 | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153249 | 1204 | - | CA RRM: Clarification of PHICH configuration | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153250 | 1205 | - | RRM: Minor correction to TC 8.7.2 | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153251 | 1206 | - | Uncertainties and Test tolerances for TS 36.521-3 Test case 8.20.2B | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153310 | 1209 | - | Correction of L2G PSHO applicability for TS 36.521-3 spec | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153357 | 1210 | - | Correction to RLM tests for eICIC | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153358 | 1211 | - | Correction to initial conditions in event triggered reporting tests | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153359 | 1212 | - | Correction to RSRP and RSRQ test cases | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153387 | 1215 | - | Correction to applicability of feICIC RRM test cases. | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153461 | 1222 | - | New TC: 8.20.2A for RRM | 12.6.0 | 12.7.0 |

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| 2015-09 | RAN#69 | R5-153471 | 1226 | - | Update of test applicabilites for Rel-10 CA RSRP relative accuracy tests | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153473 | 1227 | - | Update of test applicabilites for Rel-11 CA RSRP relative accuracy tests | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153475 | 1228 | - | Update of Annex E for RRM 9.1.6.2_1 and 9.1.7.2_1 | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153532 | 1233 | - | Update Uncertainties for Test case 5.2.7 | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153544 | 1234 | - | Update Uncertainties for Test cases 5.2.8 and 5.2.9 | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153547 | 1235 | - | Update Uncertainties for Test case 5.2.10 | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153553 | 1236 | - | Correction to 8.16.17 and 8.16.18 activation and deactivation of known Scell | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153687 | 1239 | - | Uncerts and Test Tools for TCs 7.1.3+7.1.4 36.521-3 CR | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153688 | 1240 | - | Cell mapping for new RRM TCs 7.1.3_1+7.1.4_1 | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153812 | 1190 | 1 | New TC: E-UTRAN TDD activation and deactivation of known SCell in non-DRX for 20MHz +20MHz bandwidth | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153814 | 1224 | 1 | New TCs: FDD Rel-12 CA RSRP relative accuracy tests | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153815 | 1225 | 1 | New TCs: TDD Rel-12 CA RSRP relative accuracy tests | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153816 | 1241 | 1 | Addition of new RRM TCs 7.1.3_1+7.1.4_1 | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153818 | 1186 | 1 | Correction of descriptions for test cases 8.16.18 | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153819 | 1187 | 1 | Update editor's note for RRM test case 8.16.17 and 8.16.18 | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153829 | 1201 | 1 | Introduction of 3DL CA Event triggered reporting Test cases 8.16.27 and 8.16.28 | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153830 | 1202 | 1 | Introduction of 3DL CA RSRP accuracy Test cases 9.1.37 and 9.1.38 | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153831 | 1203 | 1 | Introduction of 3DL Pcell in FDD RSRQ for E-UTRAN in Carrier Aggregation | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153832 | 1207 | 1 | Addition of new TC of 3DL FDD RSRP for E-UTRAN in Carrier Aggregation | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153833 | 1208 | 1 | Addition of new TC of 3 DL FDD RSRQ for E-UTRAN in Carrier Aggregation | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153834 | 1217 | 1 | Updates to RRM TCs 8.16.23, 8.16.24, 8.16.25 and 8.16.26 | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153871 | 1229 | 1 | New TCs: 9.1.40 and 9.1.40_1 TDD RSRP Accuracy for 3DL CA | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153872 | 1230 | 1 | New TCs: 9.2.39 TDD RSRQ Accuracy for 3DL CA | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153898 | 1218 | 1 | Addition of new TC 8.16.31 E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in FDD | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153899 | 1219 | 1 | Addition of new TC 8.16.32 E-UTRAN TDD-FDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX and with PCell in TDD | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153900 | 1220 | 1 | Addition of new TC 8.16.33 E-UTRAN FDD 3DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153901 | 1221 | 1 | Addition of new TC 8.16.34 E-UTRAN TDD 3 DL CA Event Triggered Reporting on Deactivated SCell with PCell and SCell Interruptions in Non-DRX | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153930 | 1188 | 1 | New TC: E-UTRAN TDD - UE Timing Advance Adjustment Accuracy Test for Scell in sTAG for 20 MHz +20 MHz bandwidth | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153931 | 1189 | 1 | New TC: E-UTRAN TDD - UE Timing Advance Adjustment Accuracy Test for Scell in sTAG for 20 MHz +10 MHz bandwidth | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153934 | 1195 | 1 | Correction to applicability of TD-LTE to UTRA TDD connected mode measurements | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153936 | 1191 | 1 | New TC: E-UTRAN TDD - UE Transmit Timing Accuracy Tests for SCell in sTAG for 20MHz + 20MHz bandwidth | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153937 | 1192 | 1 | New TC: E-UTRAN TDD - UE Transmit Timing Accuracy Tests for SCell in sTAG for 20MHz + 10MHz bandwidth | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153938 | 1196 | 1 | Correction to cell mapping of eICIC Chapter 9 test cases | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153939 | 1213 | 1 | Clarification of neighbour cell frequency for intra-band non-contiguous CA | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153940 | 1223 | 1 | Update of minimum requirements for relative RSRP Accuracy TC 9.1.6.2 and 9.1.7.2 | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-153943 | 1238 | 1 | Introduction of new 3DL CA Test cases 8.16.30 | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-154000 | 1179 | 1 | Introduction of TC 8.16.38 3DL TDD CA activation and deactivation of known SCell in non-DRX | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-154001 | 1182 | 1 | Introduction of 3 DL TDD Absolute/Relative RSRQ for E-UTRAN in Carrier Aggregation | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-154028 | 1214 | 1 | Update of Annex E for 3DL CA RRM | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-154029 | 1237 | 2 | Introduction of new 3DL CA Test cases 8.16.29 | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-154033 | 1242 | 1 | Correction to relative measurement test cases of feICIC RRM test cases | 12.6.0 | 12.7.0 |
| 2015-09 | RAN#69 | R5-154049 | 1243 | - | CA RRM: Additions to cell mapping Table E-1 | 12.6.0 | 12.7.0 |

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|---|---|---|---|---|---|---|---|
| 2015-09 | RAN#69 | R5-154058 | 1244 | - | eICIC RRM: Corrections to cell mapping Table E-1 | 12.6.0 | 12.7.0 |
| 2015-12 | RAN#70 | R5-155108 | 1258 | - | Correction to test procedure of inter-frequency autonomous gaps test cases | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155134 | 1262 | - | Updating Annex E and F with newly introduced test cases for MTC | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155135 | 1263 | - | Adding RMCs for MTC to Annex A | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155136 | 1264 | - | Adding abbreviations and SS HD-FDD testing information | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155139 | 1266 | - | Uncertainties and Test tolerances for Test cases 4.3.2 and 4.3.4.2 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155141 | 1267 | - | Uncertainties and Test tolerances for Test cases 4.4.1 and 4.4.2 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155143 | 1268 | - | Uncertainties and Test tolerances for Test cases 4.5.1.1 and 4.5.2.1 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155145 | 1269 | - | Uncertainties and Test tolerances for Test case 8.20.2A | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155147 | 1270 | - | Uncertainties and Test tolerances for Test case 8.20.4A | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155149 | 1271 | - | Uncertainties and Test tolerances for Test case 8.20.4B | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155197 | 1286 | - | Uncertainties and Test Tolerance for E-UTRAN TDD – UE Timing Advanced Adjustment Accuracy Test for SCell in sTAG | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155270 | 1297 | - | Correction to test case 7.1.2 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155399 | 1304 | - | Addition of cell configuration mapping for newly introduced MTC tests | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155427 | 1305 | - | Uncertainties and Test Tolerances for 3DL FDD RSRP for E-UTRAN in Carrier Aggregation | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155429 | 1307 | - | Update to 3DL PCell in FDD RSRQ for E-UTRAN in Carrier Aggregation | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155437 | 1310 | - | Removable of wait time after activation of Scell | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155574 | 1319 | - | Incorrect Io values for eICIC tests | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155680 | 1328 | - | Adding new RRM test case 8.16.35 for 3DL CA | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155681 | 1329 | - | Adding new RRM test case 8.16.36 for 3DL CA | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155770 | 1330 | - | Introduction of test case 9.1.41.1 - FD-FDD Intra Frequency Absolute RSRP Accuracy for UE category 0 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155771 | 1331 | - | Introduction of test case 9.1.41.2 - FD-FDD Intra Frequency Relative RSRP Accuracy for UE category 0 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155772 | 1332 | - | Introduction of test case 9.1.42.1 - HD-FDD Intra Frequency Absolute RSRP Accuracy for UE category 0 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155773 | 1333 | - | Introduction of test case 9.1.42.2 - HD-FDD Intra Frequency Relative RSRP Accuracy for UE category 0 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155774 | 1334 | - | Introduction of test case 9.1.43.1 - TDD Intra Frequency Absolute RSRP Accuracy for UE category 0 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155775 | 1335 | - | Introduction of test case 9.1.43.2 - TDD Intra Frequency Relative RSRP Accuracy for UE category 0 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155776 | 1336 | - | Introduction of cell configuration mapping for Intra Frequency RSRP Accuracy for UE category 0 Test Cases | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155805 | 1256 | 1 | Update minimum requirements for test case 8.1.1 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155806 | 1265 | 1 | Updating and aligning of RMC tables in TS36.521-3 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155820 | 1250 | 1 | Introduction of new test case 6.1.8 MTC | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155821 | 1252 | 1 | Introduction of new test case 8.1.12 MTC | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155822 | 1253 | 1 | Introduction of new test case 8.1.13 MTC | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155823 | 1254 | 1 | Introduction of new test case 8.1.17 MTC | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155824 | 1255 | 1 | Introduction of new test case 8.1.18 MTC | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155825 | 1259 | 1 | Adding new MTC handover test case 5.1.10 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155826 | 1260 | 1 | Adding new MTC handover test case 5.1.11 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155827 | 1261 | 1 | Adding new MTC handover test case 5.1.12 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155828 | 1272 | 1 | Adding new MTC FD-FDD RLM test case for out-of-sync and definition update | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155829 | 1273 | 1 | Adding new MTC FD-FDD RLM test case for in-sync | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155830 | 1274 | 1 | Adding new MTC FD-FDD RLM test case for out-of-sync in DRX | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155831 | 1275 | 1 | Adding new MTC FD-FDD RLM test case for in-sync in DRX | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155832 | 1276 | 1 | Adding new MTC HD-FDD RLM test case for out-of-sync | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155833 | 1277 | 1 | Adding new MTC HD-FDD RLM test case for in-sync | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155834 | 1278 | 1 | Adding new MTC HD-FDD RLM test case for out-of-sync in DRX | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155835 | 1279 | 1 | Adding new MTC HD-FDD RLM test case for in-sync in DRX | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155836 | 1280 | 1 | Adding new MTC TDD RLM test case for out-of-sync | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155837 | 1281 | 1 | Adding new MTC TDD RLM test case for in-sync | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155838 | 1282 | 1 | Adding new MTC TDD RLM test case for out-of-sync in DRX | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155839 | 1283 | 1 | Adding new MTC TDD RLM test case for in-sync in DRX | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155862 | 1284 | 1 | Update to RSRP accuracy test cases for FDD-TDD CA | 12.7.0 | 12.8.0 |

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|---|---|---|---|---|---|---|---|
| 2015-12 | RAN#70 | R5-155863 | 1299 | 1 | Update of message contents for 3DL CA RSRP and RSRQ Accuracy tests | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155864 | 1309 | 1 | Correction to test applicability in 3DL CA RRM test cases | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155866 | 1315 | 1 | Clarification on frequency swapping and references to connection diagrams in 3DL CA RRM test cases | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155867 | 1316 | 1 | Addition of missing applicability release to test case 9.2.39 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155868 | 1320 | 1 | Introduction of new 3DL CA RRM test case TC 8.16.41 “3 DL FDD CA activation and deactivation of unknown SCell in non-DRX” | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155869 | 1327 | 1 | Updating 3DL CA test cases 8.16.29 and 8.16.30 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155878 | 1288 | 1 | Uncertainties and Test Tolerances for 3DL CA Event Triggered Reporting under Deactivated SCells in Non-DRX Test cases | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155879 | 1290 | 1 | Uncertainties and Test Tolerances for 3DL E-UTRA for Carrier Aggregation RSRQ Accuracy Test cases | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155880 | 1321 | 1 | Uncerts and Test Tools for TCs 8.16.31+8.16.32+8.16.33+8.16.34 36.521-3 CR | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155881 | 1257 | 1 | Updated to TC 7.1.4A | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155888 | 1285 | 1 | Update RSRQ and reported values for serving cell Intra Frequency Absolute RSRQ Accuracy Test cases | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155889 | 1287 | 1 | Uncertainties and Test Tolerances for 2DL CA activation and deactivation of known SCell in non-DRX Test cases | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155896 | 1293 | 1 | Add new LC MTC test case 8.1.19 E-UTRAN FD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps for UE category 0 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155897 | 1294 | 1 | Add new LC MTC test case 8.1.20 E-UTRAN FDD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0. | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155898 | 1295 | 1 | Add new LC MTC test case 8.1.21 E-UTRAN HD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps for UE category 0 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-155899 | 1296 | 1 | Add new LC MTC test case 8.1.22 E-UTRAN HD - FDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-156009 | 1298 | 1 | Update Test applicability for test case 7.1.7 and 7.2.5 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-156015 | 1326 | 1 | Addition of UE transmit timing accuracy test case for PRoSe Direct discovery. | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-156032 | 1289 | 1 | Uncertainties and Test Tolerances for 3DL E-UTRA for Carrier Aggregation RSRP Accuracy Test cases | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-156033 | 1308 | 1 | Update of Annex E for 3DL CA RRM | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-156034 | 1311 | 1 | Correction to 3DL CA RSRP/RSRQ test cases | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-156051 | 1251 | 1 | Introduction of new test case 8.1.11 MTC | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-156053 | 1291 | 1 | Add new LC MTC test case 6.1.6 E-UTRAN FD-FDD Intra-frequency RRC Re-establishment for UE category 0 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-156054 | 1292 | 1 | Add new LC MTC test case 6.1.7 E-UTRAN HD-FDD Intra-frequency RRC Re-establishment for UE category 0category 0 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-156067 | 1322 | 1 | Addition of new SCE-L1 RRM TC 8.22.1 E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-156068 | 1323 | 1 | Addition of new SCE-L1 RRM TC 8.22.2 E-UTRAN TDD-TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells in DRX based on CRS based discovery signal | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-156069 | 1324 | 1 | Addition of new SCE-L1 RRM TC 8.22.3 E-UTRAN FDD-FDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-156070 | 1325 | 1 | Addition of new SCE-L1 RRM TC 8.22.4 E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation conditions in DRX based on CRS based discovery signal | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-156128 | 1313 | 1 | Correction to message contents for RRM 3DL CA TC 9.2.41 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-156129 | 1314 | 1 | Corrections to message contents for RRM 3DL CA TCs 8.16.27+8.16.28 | 12.7.0 | 12.8.0 |
| 2015-12 | RAN#70 | R5-156137 | 1301 | 2 | Update of test applicabilities for R12 RRM cases in 36.521-3 | 12.7.0 | 12.8.0 |
| 2016-03 | RAN#71 | R5-160121 | 1346 | - | Updates and Test Tolerances for 2DL FDD-TDD E-UTRA for Carrier Aggregation RSRP Accuracy Test cases | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160191 | 1349 | - | Updates and Test Tolerances for 2DL E-UTRA for Carrier Aggregation RSRQ Accuracy Test cases | 12.8.0 | 12.9.0 |

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|---|---|---|---|---|---|---|---|
| 2016-03 | RAN#71 | R5-160200 | 1350 | - | Band 66 update of 3DL E-UTRA for CA RSRP Accuracy Test cases | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160204 | 1351 | - | Correction to test case 8.20.2B | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160207 | 1353 | - | Correction to test case 8.16.17 and 8.16.18 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160221 | 1356 | - | Uncertainties and TT for 8.16.23 and 8.16.24 TDD-FDD CA event triggered reporting under deactivated SCell in non-DRX with PCell in FDD/TDD 36.521-3 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160222 | 1357 | - | Uncertainties and TT for 8.16.25 and 8.16.26 TDD-FDD CA event triggered reporting on deactivated SCell with PCell interruption in non-DRX with PCell in FDD/TDD 36.521-3 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160237 | 1358 | - | Condition definitions for discovery measurements and measurement accuracy requirements | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160257 | 1363 | - | Addition of new SCE-L1 RRM TC 8.22.9 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX based on CRS based discovery signal | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160258 | 1364 | - | Addition of new SCE-L1 RRM TC 8.22.10 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX based on CRS based discovery signal | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160261 | 1367 | - | Addition of new SCE-L1 RRM TC 9.1.25 FDD intra-frequency absolute and relative RSRP accuracies in CRS based discovery signal | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160262 | 1368 | - | Addition of new SCE-L1 RRM TC 9.1.26 TDD intra-frequency absolute and relative RSRP accuracies in CRS based discovery signal | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160263 | 1369 | - | Addition of new SCE-L1 RRM TC 9.1.27 FDD—FDD inter-frequency absolute and relative RSRP accuracies in CRS based discovery signal | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160264 | 1370 | - | Addition of new SCE-L1 RRM TC 9.1.28 TDD—TDD inter-frequency absolute and relative RSRP accuracies in CRS based discovery signal | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160265 | 1371 | - | Addition of new SCE-L1 RRM TC 9.1.33 FDD absolute and relative RSRP accuracies for E-UTRAN Carrier Aggregation in CRS based discovery signal | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160266 | 1372 | - | Addition of new SCE-L1 RRM TC 9.1.34 TDD absolute and relative RSRP accuracies for E-UTRAN Carrier Aggregation in CRS based discovery signal | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160267 | 1373 | - | Addition of new SCE-L1 RRM TC 9.2.28 FDD intra-frequency absolute RSRQ accuracy with CRS based discovery signal | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160268 | 1374 | - | Addition of new SCE-L1 RRM TC 9.2.29 TDD intra-frequency absolute RSRQ accuracy with CRS based discovery signal | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160269 | 1375 | - | Addition of new SCE-L1 RRM TC 9.2.30 FDD-FDD inter-frequency absolute and relative RSRQ accuracies with CRS based discovery signal | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160270 | 1376 | - | Addition of new SCE-L1 RRM TC 9.2.31 TDD-TDD inter-frequency absolute and relative RSRQ accuracies with CRS based discovery signal | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160271 | 1377 | - | Addition of new SCE-L1 RRM TC 9.2.32 FDD absolute and relative RSRQ accuracy for E-UTRAN Carrier Aggregation in CRS based discovery signal | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160272 | 1378 | - | Addition of new SCE-L1 RRM TC 9.2.33 TDD absolute and relative RSRQ accuracy for E-UTRAN Carrier Aggregation in CRS based discovery signal | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160443 | 1382 | - | Introduction of new Dual Connectivity test case 8.23.1 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160444 | 1383 | - | Introduction of new Dual Connectivity test case 8.23.2 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160445 | 1384 | - | Introduction of new Dual Connectivity test case 8.23.3 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160446 | 1385 | - | Introduction of new Dual Connectivity test case 8.23.4 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160447 | 1386 | - | Introduction of new Dual Connectivity test case 8.23.5 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160448 | 1387 | - | Introduction of new Dual Connectivity test case 8.23.6 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160450 | 1389 | - | Update to MTC test cases 8.1.11, 8.1.12, 8.1.13, 8.1.17, 8.1.18 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160452 | 1391 | - | Update to test cases 8.16.17 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160497 | 1397 | - | Correction to event triggered tests without measurement gaps | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160498 | 1398 | - | Correction to RLM tests in feICIC | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160556 | 1405 | - | CA RRM: Inconsistent message contents in 3CC event triggered tests | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160558 | 1407 | - | CA RRM: Missing message contents in 3CC TCs 9.1.37 and 9.1.38 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160559 | 1408 | - | RRM: Correction of applicability of TCs 9.3.2 and 9.4.2 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160572 | 1409 | - | Updating TS36.521-3 clause 3.5.1 for band 66 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160574 | 1410 | - | Updating Annex I with band group FDD_B | 12.8.0 | 12.9.0 |

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| 2016-03 | RAN#71 | R5-160610 | 1416 | - | Antenna configuration principle for 36.521-3 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160702 | 1417 | - | Introducing DC RRM test case 7.3.38 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160703 | 1418 | - | Introducing DC RRM test case 7.3.39 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160704 | 1419 | - | Introducing DC RRM test case 7.3.40 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160705 | 1420 | - | Updating References, Definitions and Abbreviations for DC in 36.521-3 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160821 | 1361 | 1 | Addition of new SCE-L1 RRM TC 8.22.6 E-UTRAN TDD-TDD intra-frequency event triggered reporting in DRX based on CSI-RS based discovery signal | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160822 | 1362 | 1 | Addition of new SCE-L1 RRM TC 8.22.8 E-UTRAN TDD-TDD inter-frequency event triggered reporting under fading propagation condition in DRX based on CSI-RS based discovery signal | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160823 | 1365 | 1 | Addition of new SCE-L1 RRM TC 8.22.11 E-UTRAN FDD event triggered reporting under deactivated SCell in non-DRX based on CSI-RS based discovery signal | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160824 | 1366 | 1 | Addition of new SCE-L1 RRM TC 8.22.12 E-UTRAN TDD event triggered reporting under deactivated SCell in non-DRX based on CSI-RS based discovery signal | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160833 | 1406 | 1 | CA RRM: Ambiguous description of relative RSRQ measurements in 3CC tests | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160834 | 1395 | 1 | Correction to test frequency configuration for 3DL CA RRM | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160847 | 1352 | 1 | Correction to test case 7.1.4 and 7.1.4_1 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160872 | 1402 | 1 | Updating Annex F for E-UTRAN 3DL CA Activation and Deactivation of Known SCell in Non-DRX test cases in TS36.521-3 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160876 | 1388 | 1 | Uncertainties and Test Tolerances for E-UTRAN UE Transmit Timing Accuracy Tests for SCell in sTAG in FDD/TDD Test cases | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160883 | 1390 | 1 | Update to RRM test cases 7.1.3, 7.1.3_1, 7.1.4, 7.1.4_1, 7.1.4A for TT and Uncerts | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160884 | 1414 | 1 | Update of HO testcases for MTC | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160886 | 1339 | 2 | Introduction of TC 8.1.15 E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for UE category 0MTC | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160887 | 1338 | 2 | Add new LC MTC test case 8.1.14 E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for UE category 0 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160888 | 1340 | 2 | Introduction of TC 8.1.16 E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells with DRX for UE category 0 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160889 | 1341 | 2 | Introduction of TC 8.2.7 E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps for UE category 0 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160890 | 1342 | 2 | Introduction of TC 8.2.8 E-UTRAN TDD Intra-frequency identification of a new CGI of E-UTRA cell using autonomous gaps with DRX for UE category 0 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160891 | 1343 | 2 | Introduction of TC 9.2.42.1 FD-FDD Intra Frequency Absolute RSRQ Accuracy for UE category 0 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160892 | 1344 | 2 | Introduction of TC 9.2.43.1 HD-FDD Intra Frequency Absolute RSRQ Accuracy for UE category 0 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160893 | 1345 | 2 | Introduction of TC 9.2.44.1 TDD Intra Frequency Absolute RSRQ Accuracy for UE category 0 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-160894 | 1415 | 1 | Update of RLM testcases for MTC | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-161002 | 1394 | 1 | Correction to RRC message exceptions for 3DL CA RSRP/RSRQ | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-161007 | 1403 | 1 | Addition of test cases for transmission timing accuracy and Interruptions for D2D | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-161037 | 1359 | 1 | Addition of new RRM TC 8.22.5 E-UTRAN FDD-FDD intra-frequency event triggered reporting in DRX based on CSI-RS based discovery signal | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-161038 | 1360 | 1 | Addition of new RRM TC 8.22.7 E-UTRAN FDD-FDD inter-frequency event triggered reporting for DRS measurement based on CSI-RS with DRX | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-161051 | 1411 | 1 | Correcting the applicability of test cases 9.5.1 and 9.5.2 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-161068 | 1392 | 1 | Update to test cases 8.16.17A and 8.16.18A | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-161094 | 1393 | 2 | Correction to Event Triggered Reporting test under Deactivated SCells | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-161095 | 1396 | 2 | Correction to 3DL CA with deactivated SCell with PCell and SCell interruptions | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-161096 | 1400 | 1 | Updating RRM test case 8.16.35 | 12.8.0 | 12.9.0 |

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| 2016-03 | RAN#71 | R5-161097 | 1401 | 2 | Updating RRM test case 8.16.36 | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-161098 | 1354 | 2 | New TC 8.16.39 E-UTRA TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in FDD | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-161099 | 1355 | 2 | New TC 8.16.40 E-UTRA TDD-FDD 3DL CA Activation and Deactivation of Unknown SCell in Non-DRX with PCell in TDD | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-161100 | 1380 | 2 | Addition of new test case for 3DL FDD CA activation and deactivation of known cell | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-161106 | 1347 | 2 | Test Tolerances for TC 8.16.38 3DL TDD CA activation and deactivation of known SCell in non-DRX | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-161117 | 1379 | 2 | Updates of FDD 2DL CA activation and deactivation test case | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-161118 | 1399 | 2 | Correction to 9.1.15.1 and 9.1.15.2 RSRP tests for feICIC | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-161122 | 1413 | 3 | Correction of test procedure for 3DL CA RRM test case TC 8.16.41 “3 DL FDD CA activation and deactivation of unknown SCell in non-DRX” | 12.8.0 | 12.9.0 |
| 2016-03 | RAN#71 | R5-161123 | 1412 | 3 | Introduction of new 3DL CA RRM test case TC 8.16.42 “3 DL TDD CA activation and deactivation of unknown SCell in non-DRX” | 12.8.0 | 12.9.0 |
| 2016-06 | RAN#72 | R5-162186 | 1422 | - | Editorial correction uplink absolute power measurement in Annex F.1.2 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162190 | 1423 | - | Update of test tolerance for UE ProSe discovery transmission timing | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162193 | 1424 | - | Update of Test Tolerance analyses for 7.5.1 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162278 | 1425 | - | Correction to Table number for using MAC-MainConfig-RBC message | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162286 | 1430 | - | Update of RRM RLM test cases for Cat 0 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162288 | 1431 | - | Update of Annex F for Cat 0 RLM test cases | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162310 | 1432 | - | RRM: Correction to TCs 7.1.7A and 7.1.7B | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162311 | 1433 | - | RRM: Corrections to ri-ConfigIndex in 3CC test cases | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162312 | 1434 | - | RRM: Missing release applicability of inter-mode 3CC test cases | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162313 | 1435 | - | RRM: Clarification of message content in TC 8.16.36 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162321 | 1437 | - | Addition of test cases for SLSS initiation and cease | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162330 | 1439 | - | Introduction of TDD intra frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162334 | 1442 | - | Correction of 3DL CA RRM test case 8.16.36 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162336 | 1443 | - | Introduction of TDD-TDD inter frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162397 | 1447 | - | Uncertainties and Test tolerances for TS 36.521-3 Test cases 9.1.6.1_1 and 9.1.12.1_1 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162399 | 1448 | - | Uncertainties and Test tolerances for TS 36.521-3 Test cases 9.1.6.2_1 and 9.1.12.2_1 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162430 | 1450 | - | Update of RRM test cases 9.1.1.1, 9.1.2.1, 9.1.16.1 to include band 65, 66 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162433 | 1451 | - | Update of RRM test cases 9.1.1.1_1, 9.1.2.1_1, 9.1.16.1_1 to include band 65, 66 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162436 | 1452 | - | Update of RRM test cases 9.1.1.2, 9.1.2.2, 9.1.16.2 to include band 65, 66 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162439 | 1453 | - | Update of RRM test cases 9.1.3.1, 9.1.4.1, 9.1.17.1 to include band 65, 66 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162600 | 1461 | - | Correction to test applicability in 3DL CA activation and deactivation TCs | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162830 | 1463 | 1 | Correction to 3DL CA activation and deactivation TCs | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162831 | 1454 | 1 | Correction to applicability for EUTRA TDD to UTRA TDD Son test case | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162833 | 1438 | 1 | Introduction of FDD intra frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162834 | 1441 | 1 | Introduction of FDD-FDD inter frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162835 | 1444 | 1 | Introduction of FDD absolute and relative CSI-RSRP accuracies for E-UTRAN Carrier Aggregation in CSI-RS based discovery signal | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162836 | 1445 | 1 | Introduction of TDD absolute and relative CSI-RSRP accuracies for E-UTRAN Carrier Aggregation in CSI-RS based discovery signal | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162881 | 1436 | 1 | Addition of ProSe cell selection test cases | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162891 | 1421 | 1 | Uncertainties and Test Tolerances for MTC RRM TC 8.1.11, 8.1.12, 8.1.13, 8.1.17 and 8.1.18 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162894 | 1429 | 1 | Test Tolerance updates for MTC test cases in clause 6 | 12.9.0 | 12.10.0 |

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| 2016-06 | RAN#72 | R5-162895 | 1457 | 1 | Add Test requiements for TS 36.521-3 Test case 8.22.1 and 8.22.2 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162897 | 1458 | 1 | Add Test requiements for TS 36.521-3 Test case 8.22.3 and 8.22.4 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162957 | 1426 | 1 | Update to Dual Connectivity RRM test case 8.23.4 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162958 | 1427 | 1 | Update to Dual Connectivity RRM test case 8.23.5 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162959 | 1428 | 1 | Update to Dual Connectivity RRM test case 8.23.6 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162960 | 1446 | 1 | Added new clause for Syncronous and Asynchronous DC test principles | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162961 | 1456 | 1 | Update to Dual Connectivity RRM test case 8.23.2 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162962 | 1467 | 1 | Update to Dual Connectivity RRM test case 8.23.3 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162963 | 1468 | 1 | Update of DC RLM test cases | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162964 | 1469 | 1 | Adding new RLM test case 7.3.41 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162965 | 1470 | 1 | Adding new RLM test case 7.3.42 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-162966 | 1471 | 1 | Adding new RLM test case 7.3.43 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-163111 | 1455 | 1 | Update to Dual Connectivity RRM test case 8.23.1 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-163128 | 1459 | 1 | Add Test requiements for TS 36.521-3 Test case 9.1.25 and 9.1.26 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-163197 | 1462 | 2 | Correction to duration of T3 in 3 DL CA Event Triggered Reporting on Deactivated Scell | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-163134 | 1464 | 1 | Clean up on RRM test cases in chapter 7, 8, 9 | 12.9.0 | 12.10.0 |
| 2016-06 | RAN#72 | R5-163135 | 1465 | 1 | Correction to event triggered reporting under deactivated SCell(10MHz+5MHz and 20MHz+10MHz) | 12.9.0 | 12.10.0 |
| 2016-09 | RAN#73 | R5-165130 | 1472 | - | Parameter correction for TC 9.2.16.1 | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165166 | 1477 | - | Update Test Tolerances and Test requirements for TS 36.521-3 Test cases 9.1.3.1_1 and 9.1.4.1_1 | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165168 | 1478 | - | Update Test Tolerances and Test requirements for TS 36.521-3 Test cases 9.1.3.2_1 and 9.1.4.2_1 | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165296 | 1482 | - | Band 66 update of Inter-freq relative RSRP accuracy Test cases | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165389 | 1490 | - | Update of test tolerance for UE ProSe direct communication transmission timing | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165428 | 1494 | - | Modification of measurement report configuration for TC 8.1.8 and 8.2.6 | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165527 | 1496 | - | Add Test requiements for TS 36.521-3 Test case 8.22.9 and 8.22.10 | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165528 | 1497 | - | Add Test requiements for TS 36.521-3 Test case 9.2.28 and 9.2.29 | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165529 | 1498 | - | Add Test requiements for TS 36.521-3 Test case 9.2.30 and 9.2.31 | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165530 | 1499 | - | Add Test requiements for TS 36.521-3 Test case 9.2.32 and 9.2.33 | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165577 | 1507 | - | Update Annex F for TS 36.521-3 Test case 9.1.27, 9.1.28, 9.1.33 and 9.1.34 | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165643 | 1511 | - | Correction to applicability for TDD-UTRAN TDD SON | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165664 | 1513 | - | Correction to test parameters of 3DL CA event triggered report test cases for 8.16.31, 8.16.32, 8.16.33 and 8.16.34 | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165665 | 1514 | - | Clarification of test parameters for RRM asymmetric bandwidth 2CA test cases | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165666 | 1515 | - | Clarification of test applicability for 2DL CA with intra-band non-contiguous | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165667 | 1516 | - | Clean up on CA RRM test cases 8.16.25 and 9.1.40_1 | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165668 | 1517 | - | Update of cell configuration mappings for FDD/TDD 3CA band combination | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165669 | 1518 | - | Correction to the test applicability for test cases 8.20.4, 8.20.4A and 8.20.4B | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165670 | 1519 | - | Clean up on RRM test cases 8.7.2 and 8.11.4 | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165851 | 1537 | - | Removing redundant Editor's note from Clause 9 in 36.521-3 | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165852 | 1538 | - | Adding antenna connection diagram references to RRM DC RLM test cases | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165853 | 1539 | - | Updating Annex E with DC RLM test cases 7.3.38 – 7.3.43 | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165861 | 1540 | - | Update of definitions for IncMon | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-166024 | 1500 | 1 | Correction of triggering event for SCE test case 8.22.5 and 8.22.6 | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-166025 | 1501 | 1 | Introduce Cell configuration mapping for SCE RRM test cases | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-166098 | 1484 | 1 | Update of test tolerance for UE ProSe direct communication SLSS initiation and cease | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-166100 | 1512 | 1 | Correction to 3DL CA Activation and Deactivation test cases | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165575 | 1505 | - | Update Test requiement for TS 36.521-3 Test case 9.1.27and 9.1.28 | 12.10.0 | 12.11.0 |

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|---|---|---|---|---|---|---|---|
| 2016-09 | RAN#73 | R5-165576 | 1506 | - | Update Test requiement for TS 36.521-3 Test case 9.1.33 and 9.1.34 | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-166305 | 1536 | 1 | Adding CQI reporting periodicity to MTC test cases | 12.10.0 | 12.11.0 |
| 2016-09 | RAN3#73 | R5-166306 | 1492 | 1 | Uncertainties and Test Tolerances for intra frequency absolute and relative CSI-RSRP accuracies in CRI-RS based discovery signal Aggregation | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-166308 | 1527 | 1 | Add note to cell mapping table in Annex E for band 66 high range testing | 12.10.0 | 12.11.0 |
| 2016-09 | RAN#73 | R5-165385 | 1488 | - | Introduction of FD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeA | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-165388 | 1489 | - | Introduction of HD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeA | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-165390 | 1491 | - | Introduction of TDD RSRP Intra frequency case for Cat-M1 UE in CEModeA | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-165505 | 1495 | - | Introduction of Band 45 into 36.521-3 | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5=165570 | 1502 | - | Update Test requiement for TS 36.521-3 Test case 6.2.7 and 6.2.8 | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-165571 | 1503 | - | Update Test requiement for TS 36.521-3 Test case 7.2.4 | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-165572 | 1504 | - | Update Annex F for TS 36.521-3 Test case 6.2.7, 6.2.8 and 7.2.4 | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-165739 | 1520 | - | Correction to Band 66 notes in E-UTRA band groups in 36.521-3 | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-165836 | 1528 | - | Adding new eMTC test case 6.1.9 to 36.521-3 | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-165839 | 1529 | - | Adding new eMTC test case 6.1.10 to 36.521-3 | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-165840 | 1530 | - | Adding new eMTC test case 6.1.11 to 36.521-3 | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-165841 | 1531 | - | Adding new eMTC test case 7.1.10 to 36.521-3 | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-165842 | 1532 | - | Adding new eMTC test case 7.1.11 to 36.521-3 | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-165844 | 1533 | - | Adding new eMTC RRM test cases to Annex E of 36.521-3 | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-165845 | 1534 | - | Adding new eMTC RMCs to Annex A of 36.521-3 | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-166089 | 1485 | 1 | Introduction of E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-166090 | 1487 | 1 | Introduction of E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA in DRX | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-166163 | 1474 | 1 | Introduction of eMTC RRM TC 4.2.12 | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-166164 | 1475 | 1 | Introduction of eMTC RRM TC 4.2.13 | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-166167 | 1486 | 1 | Introduction of E-UTRAN FDD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeA | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-166182 | 1543 | 1 | Addition of RLM test cases for FD-FDD CE Mode A | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-166175 | 1523 | 1 | New Test Case: 7.3.50: E-UTRAN FD-FDD Radio Link Monitoring Test for Out-of-Sync in DRX for UE Category M1 Configured in CEMode A | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-166176 | 1524 | 1 | New test case: 7.3.51: E-UTRAN FD-FDD Radio Link Monitoring Test for In-sync in DRX for UE Category M1 Configured in CEMode A | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-166178 | 1526 | 1 | New test case: 7.3.55: E-UTRAN HD-FDD Radio Link Monitoring Test for In-sync in DRX for UE Category M1 configured in CEMode A | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-166183 | 1544 | 1 | Addition of RLM test cases for HD-FDD CE Mode A | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-166316 | 1521 | 2 | New Test Case: 6.2.10: E-UTRAN FDD Contention Based Random Access Test for Cat-M1 UEs in Normal Coverage | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-166317 | 1522 | 2 | New Test Case: 6.2.11: E-UTRAN HD-FDD Contention Based Random Access Test for Cat-M1 UEs in Normal Coverage | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-166318 | 1535 | 1 | Adding new OCNG pattern definitions for eMTC to Annex D of 36.521-3 | 12.11.0 | 13.0.0 |
| 2016-09 | RAN#73 | R5-166333 | 1525 | 2 | New Test Case: 7.3.54: E-UTRAN HD-FDD Radio Link Monitoring Test for Out-of-sync in DRX for UE Category M1 Configured in CEMode A | 12.1.00 | 13.0.0 |
| 2016-12 | RAN#74 | R5-168061 | 1546 | F | New Reference Channel for TS 37.571-1 RSTD tests | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168131 | 1547 | F | Addition of new test case E-UTRAN TDD Radio Link Monitoring Test for Out-of-sync for Cat-M1 UE in CEMode A | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168132 | 1548 | F | Addition of new test case E-UTRAN TDD Radio Link Monitoring Test for In-sync for Cat-M1 UE in CEMode A | 13.0.0 | 13.1.0 |

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|---|---|---|---|---|---|---|---|
| 2016-12 | RAN#74 | R5-168133 | 1549 | F | Addition of new test case E-UTRAN TDD Radio Link Monitoring Test for Out-of-Sync in DRX for UE Category M1 Configured in CEMode A | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168134 | 1550 | F | Addition of new test case E-UTRAN TDD Radio Link Monitoring Test for In-Sync in DRX for UE Category M1 Configured in CEMode A | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168156 | 1551 | F | Introduction of new definitions for NB-IoT in Chapter 3 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168157 | 1552 | F | Introduction of Reference NPRACH Configurations for NB-IoT | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168158 | 1553 | F | Introduction of OCNG pattern for NB-IoT | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168159 | 1554 | F | Correction of NPDCCH RMC for NB-IoT | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168170 | 1558 | F | Uncertainties and Test Tolerances for inter frequency absolute and relative CSI-RSRP accuracies in CSI-RS based discovery signal | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168172 | 1559 | F | Uncertainties and Test Tolerances for FDD/TDD absolute and relative CSI-RSRP accuracies for E-UTRAN Carrier Aggregation in CSI-RS based discovery signal | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168179 | 1565 | F | Uncertainties and Test Tolerances for intra frequency absolute and relative RSRP accuracies of Cat-M1 UE in CEModeA | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168185 | 1568 | F | Introduction of HD-FDD RSRP Intra frequency case for Cat-M1 UE in CEModeB | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168197 | 1569 | F | Updating Annex F by adding clause 6 test cases for eMTC | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168199 | 1570 | F | Updating Annex F by adding clause 7 test cases for eMTC | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168200 | 1571 | F | Updating Annex E by adding test cases 6.2.10 and 6.2.11, eMTC | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168202 | 1572 | F | Adding Reference PRACH Configuration for eMTC into 36.521-3 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168206 | 1576 | F | Updating eMTC RRM test case 7.1.10 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168208 | 1577 | F | Updating eMTC RRM test case 7.1.11 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168248 | 1585 | F | Addition of new eMTC RRM test cases to Annex E of 36.521-3 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168251 | 1586 | F | Introduction of eMTC RRM new test case 4.2.14 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168289 | 1588 | F | HD-FDD Inter-frequency RRC Re-establishment for UE category NB1 In-Band mode under enhanced coverage | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168295 | 1590 | F | Adding new eMTC RRM test cases to Annex E | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168296 | 1591 | F | Introduction of new eMTC RRM test case 8.1.29 for category M1 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168297 | 1592 | F | Introduction of new eMTC RRM test case 8.1.30 for category M1 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168306 | 1594 | F | Updated to DC RRM inter-frequency event triggered reporting test cases | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168475 | 1600 | F | Uncertainties, Test Tolerances and RSRQ absolute accuracy requirements for MTC RRM TC 9.2.42.1, 9.2.43.1 and 9.2.44.1 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168504 | 1604 | F | NB-IoT TA adjustment enhanced coverage | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168505 | 1605 | F | HD-FDD Intra-frequency RRC Re-establishment for UE category NB1 under normal coverage | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168608 | 1619 | F | Update to CE Mode A RLM test cases | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168620 | 1620 | F | Updates to message contents for eMTC RRM TC 4.2.12 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168621 | 1621 | F | Updates to message contents for eMTC RRM TC 4.2.13 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168622 | 1622 | F | Test Tolerances for eMTC RRM TC 4.2.12 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168623 | 1623 | F | Test Tolerances for eMTC RRM TC 4.2.13 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168662 | 1626 | F | Introduction of Intra frequency handover Test case 5.1.13 for Cat-M1 UEs in CEModeA | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168663 | 1627 | F | Introduction of Intra frequency handover Test case 5.1.14 for Cat-M1 UEs in CEModeA | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168664 | 1628 | F | Introduction of Intra frequency handover Test case 5.1.15 for Cat-M1 UEs in CEModeA | 13.0.0 | 13.1.0 |

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|---|---|---|---|---|---|---|---|
| 2016-12 | RAN#74 | R5-168678 | 1629 | F | Clarification of test parameters for RRM asymmetric bandwidth 2CA test cases | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168679 | 1630 | F | Correction to TDD CA Activation and Deactivation of known SCell in Non-DRX test cases | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168680 | 1631 | F | Correction to figure group for timing adjustment 2CA TCs | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168681 | 1632 | F | Correction to RRM TDD FDD 2CA Event Triggered Report test cases | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168682 | 1633 | F | Correction to Information Bit Payload for TDD PDSCH RMC | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168710 | 1636 | F | Adding new test case 7.3.44 DC RLM FDD-TDD | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168711 | 1637 | F | Adding new test case 7.3.45 DC RLM TDD-FDD | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168713 | 1638 | F | Adding new test case 7.3.46 DC RLM TDD-FDD | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168716 | 1639 | F | Adding new test case 7.3.47 DC RLM TDD-FDD | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168718 | 1640 | F | Adding new DC RLM TDD-FDD test cases into Annex E | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168732 | 1644 | F | Adding new test cases for 4 Rx RRM RLM FDD testing to annex E | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168810 | 1646 | F | Updating DC RLM RRM test cases | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168814 | 1648 | F | Updating DC RLM RRM test cases Annex F | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168830 | 1650 | F | E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeA | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168836 | 1652 | F | E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA in DRX | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168848 | 1653 | F | Updating tet cases 9.1.6, 9.1.7, 9.1.12, 9.1.13 To include band group FDD_B | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168854 | 1654 | F | Updating test cases 9.1.6.2, 9.1.7.2, 9.1.12.2, 9.1.13.2 To include band group FDD_B | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168880 | 1656 | F | Updating eMTC RRM test case 6.1.9 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168881 | 1657 | F | Updating eMTC RRM test case 6.1.10 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168882 | 1658 | F | Updating eMTC RRM test case 6.1.11 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168938 | 1659 | F | Uncertainties and Test Tolerances for intra frequency absolute and relative RSRP accuracies for UE category 0 Aggregation | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-168971 | 1661 | F | E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeA | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169517 | 1634 | F | Correction to RRM TDD FDD 2CA measurement accuracy test cases | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169529 | 1556 | F | Introduction of Cell configuration mapping for NB-IoT | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169552 | 1610 | F | Addition of the new TC 9.1.44 4 DL CA PCell in FDD FDD-TDD RSRP for E-UTRAN in Carrier Aggregation | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169553 | 1611 | F | Addition of the new TC 9.2.45 4 DL CA PCell in FDD FDD-TDD RSRQ for E-UTRAN in Carrier Aggregation | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169575 | 1578 | F | New Test Case: 6.2.13: E-UTRAN FDD Contention Based Random Access Test for Cat-M1 UEs in Enhanced Coverage | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169576 | 1579 | F | New Test Case: 6.2.14: E-UTRAN HD-FDD Contention Based Random Access Test for Cat-M1 UEs in Enhanced Coverage | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169577 | 1580 | F | New Test Case: 6.2.15: E-UTRAN TDD Contention Based Random Access Test for Cat-M1 UEs in Enhanced Coverage | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169578 | 1581 | F | Update Annex F Test Tolerance for E-UTRAN FDD-FDD and HD-FDD Radio Link Monitoring Test in DRX for UE Category M1 Configured in CEMode A | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169579 | 1560 | F | Introduction of test cases 7.2.6 in TS36.521-3 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169580 | 1561 | F | Introduction of test cases 7.2.7 in TS36.521-3 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169581 | 1562 | F | Introduction of test cases 7.2.8 in TS36.521-3 | 13.0.0 | 13.1.0 |

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| 2016-12 | RAN#74 | R5-169582 | 1601 | F | New test case: 8.1.33 E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in asynchronous cells for Cat-M1 UE in CEModeB | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169583 | 1602 | F | New test case: 8.1.34 E-UTRAN HD-FDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169584 | 1603 | F | New test case: 8.1.35 E-UTRAN TDD intra-frequency event triggered reporting under fading propagation conditions in synchronous cells for Cat-M1 UE in CEModeB | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169585 | 1624 | F | Introduction of new eMTC RRM TC 4.2.15 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169586 | 1635 | F | eMTC RRM TC 6.2.10 prach-StartingSubframe and numRepetitionPerPreambleAttempt should match | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169587 | 1647 | F | Introduction of E-UTRAN HD - FDD Intra-Frequency Case for Cat-M1 UE in Enhanced Coverage Test Case | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169588 | 1649 | F | Introduction of E-UTRAN TDD - TDD Intra-Frequency Case for Cat-M1 UE in Enhanced Coverage Test Case | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169594 | 1660 | F | Update to E-UTRAN DC intra-frequency event triggered reporting DC Test cases | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169595 | 1609 | F | Update of test tolerance for UE Initiation and Cease of SLSS for ProSe direct communication | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169599 | 1595 | F | Introduction of new eMTC RRM test case 9.1.57 for category M1 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169600 | 1617 | F | Addition of a new test case 9.1.59 TDD RSRP Intra frequency case for Cat-M1 UE in CEModeB | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169601 | 1618 | F | Update Annex F for TS 36.521-3 Test case 9.1.57, 9.1.58 and 9.1.59 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169609 | 1614 | F | Correction of triggering event for SCE test case 8.22.7 and 8.22.8 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169641 | 1641 | F | Adding Antenna connection configuration guide for 4 Rx RRM testing | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169679 | 1642 | F | Adding new test case 7.3.1_1 for 4 Rx RRM RLM FDD testing | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169680 | 1643 | F | Adding new test case 7.3.2_1 for 4 Rx RRM RLM FDD testing | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169695 | 1555 | F | Correction of NPDSCH RMC for NB-IoT | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169698 | 1625 | F | Addition of in-sync RLM test cases for NB-IOT | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169711 | 1587 | F | Introduction of eMTC RRM new test case 6.2.12 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169718 | 1615 | F | Correction of triggering event for SCE test case 8.22.11 and 8.22.12 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169726 | 1606 | F | Uncertainties and Test Tolerances for MTC RRM TCs 8.1.19,8.1.20,8.1.21 and 8.2.22 | 13.0.0 | 13.1.0 |
| 2016-12 | RAN#74 | R5-169727 | 1607 | F | Uncertainties and Test Tolerances for MTC RRM TCs 8.2.7 and 8.2.8 | 13.0.0 | 13.1.0 |

# History

| Document history | | |
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| V13.0.0 | December 2016 | Publication |
| V13.1.0 | March 2017 | Publication |
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